

Modular Koszul Duality.

Volume IV: Realization

Raez Lorgat

Abstract

The comparison functor

$$\text{Real}(X, T) = (d, e)$$

sends a theorem T of the first three volumes to a pair of constructions d, e : d produces the invariant and e computes it by another route. The two components are required to meet only in the invariant being identified. Thus an equality such as a central charge formula, a Koszul duality equivalence, or a determinant identity is tested against a construction which does not contain the formula as an input.

Vol I inputs to Real: the Virasoro shadow tower with Stokes pole $c_S = -218/45$, Borel radius $|\omega|^2(c) = c^2(5c + 22)/[4(45c + 218)]$, Riccati equation $U' + 2U^2 = Q_c''/(2Q_c)$ on $U = (\log H)' - 2/t$, the four-conductor split (K, K^κ, K_c, K_g) , the $p = 691$ Chenevier determinant, and the eight-form Gritsenko–Cléry spread. Vol II inputs: the $\text{SC}^{\text{ch, top}}$ heptagon and the finite-jet Khan–Zeng (KZ) Poisson vertex algebra (PVA) sector, in which the shifted-cotangent Batalin–Vilkovisky (BV) action and the classical master equation (CME) / PVA Hamiltonian equivalence are constructed; the all-loop quantum master equation (QME), reflected weight identity, boundary vertex algebra, and E_3 -lift remain conditional on the four endpoint hypotheses: the KZ analytic spectral-decomposition representation (SDR), the Fulton–MacPherson Stokes datum, the reflected weights, and the T -lift. Vol III inputs: the two-stage Calabi–Yau-to-chiral factorisation $\Phi_d = \text{Sp}_{\Sigma_d-1, C}^{\text{ch}} \circ \Phi_d^{\text{FA}}$, the K3 Hall–Drinfeld chiral bialgebra $\mathbf{H}_\Delta$, the Joyce vertex algebra paired with Kontsevich–Pandharipande–Soibelman (KPS) orientation data on generic compact Calabi–Yau threefolds with Maulik–Nekrasov–Okounkov–Pandharipande (MNOP) invariants, S^3 -framing on generic compact Calabi–Yau threefold Lagrangians, and the $\mathfrak{so}(4, 20)$ K3 Yangian representation-target trichotomy.

The arithmetic branch extends Real to $\overline{\text{Spec } \mathbb{Z}}$. Its boundary is not a closure theorem: it consists of the RH-equivalent archimedean Tate positivity obstruction, the global arithmetic comparison inputs, the local categorical and Bernstein-centre gluing inputs, the condensed-Hilbert descent input, and the remaining supersingular or icosahedral generator-comparison tails. On the critical-line support $\Gamma_\xi^{\text{crit}} = \{\gamma \in \mathbb{R} : \xi(\frac{1}{2} + i\gamma) = 0\}$, the ordinary $\mathcal{M}_t$ -graph dual carries neither point nor finite-jet Hadamard residue data; any surviving spectral-flow argument passes through a stronger rigged, Paley–Wiener, de Branges, reproducing-kernel, nonlocal L^2 , or BV–BFV boundary topology compatible with the post-Tate quotient.

CONTENTS

CONTENTS	ii
--------------------	----

I The Realization Formalism 5

1 THE REALIZATION FUNCTOR	7
1.1 Independent comparison	7
1.2 The realization pair	7
1.3 Existence of Real on the seven Platonic theorems	8
1.4 Completeness on the non-degenerate locus	9
1.5 Boundary strata	11
Degenerate admissible logarithmic sector	11
Super-Yangian chain-level chiral coproduct	11
Vol III CY-C conditional bridges	11
1.6 Boundary Problems	11
1.7 Structural Consequences Inherited from Vols I–III	12
The four-conductor split and the Koszul-self-dual trinity	12
Universal chain-homotopy normalisation	12
Shadow-tower exact closed forms	13
Shadow-Galois Chenevier determinant at $p = 691$	13
Drinfeld double global obstruction class	13
Eight-form Gritsenko–Cléry spread and cover assignment	13
Three-factor universal trace identity with ζ -regularisation	14
Two-stage factorisation of the Calabi–Yau-to-chiral functor	14
K ₃ Hall–Drinfeld chiral-bialgebra equivalence	14
CoHA-to- $\mathcal{W}_{1+\infty}$ triangle with two λ -parameters	15
Three-tier topologisation discipline	15
Khan–Zeng finite-jet BV/PVA sector	15
Joyce–KPS compact Calabi–Yau threefold sector	15
Post-Tate Hadamard support and the $\mathcal{M}_t$ -graph dual	16
Boundary frontiers F ₁ –F ₁₀	16

II The Arithmetic Realization Branch 17

2 THE ARITHMETIC REALIZATION BRANCH	19
2.1 Identity	19
2.2 Form Discipline	19
2.3 Shape of the Branch	20
2.4 Role within Volume IV	21

2.5	Source Decomposition	21
2.6	Conventions	21
2.7	Imports and Assembly	22
2.8	Arithmetic-Modular Extension: All-Genus Axis	22
2.9	Arithmetic Gravity and the E_1 Ordered Bar	23
3	THE SPECULATIVE CONSTRUCTION	25
3.1	The Arithmetic Curve	25
3.2	Λ -Ring Input	25
3.3	The Primon Vertex Algebra (Naive Form)	25
3.4	The Hecke-Adams Lift (Naive Form)	26
3.5	The Averaging Map (Naive Form)	26
3.6	The Four-Archetype Echo (Heuristic)	27
3.7	Theorem Analogues (Naive Statements)	27
3.8	Open Technical Problems	27
4	THE p -ADIC VERTEX ALGEBRA LOCALITY PROBLEM	29
4.1	The Algebraic Step: Formal VA over $\mathbb{Q}_p$	29
4.2	Conformal Vector Scaling: Archimedean, Not p -adic	29
4.3	The Restricted Tensor Product: Coefficient-Field Mismatch	30
4.4	The p -adic Diagonal: Trivial Formal Neighbourhood	30
4.5	The Locally Constant (E_0) Local Model at Each Finite Place	31
4.6	The Adelic Automorphic Placement of $\mathcal{V}^{\text{prim}}$	31
4.7	Kapranov–Vasserot: Formal Loop Spaces over Fields	31
4.8	Local Arithmetic Boundary	32
5	HECKE VERSUS ADAMS: THE GL_1 / GL_2 TYPE ERROR	33
5.1	The Satake Pair and Why It Matters	33
5.2	Why Adams is not Hecke	33
5.3	Explicit Computation: The Recursion Fails	34
5.4	Diagonalisation onto the Satake Pair	34
5.5	The Angle θ_p as Unramified Langlands Parameter	35
5.6	The Chiral Framework over $\overline{\text{Spec } \mathbb{Z}}$	36
5.7	The Corrected Local Model and Its Consequences	36
6	Λ -RING AXIOMATIC PRECISION	37
6.1	The Wilkerson Congruence and the Scalar Ring	37
6.2	Explicit Computations: ψ_2 on Low-Weight States	37
6.3	The Wilkerson Congruence Fails on the Fock Lattice	38
6.4	The Macdonald Lattice vs the Fock Lattice	38
6.5	Distinction: Λ -Ring vs Λ -Module	38
6.6	The Log- p Rescaling Destroys Integrality	39
6.7	Is there a Λ -VA?	39
6.8	The Descent Interpretation Goes Through Bost–Connes	40
7	BOST–CONNES VERSUS $\mathcal{V}^{\text{prim}}$: C^* -ALGEBRA VERSUS TRACE CLASS	41

7.1	The Bost–Connes C^* -Algebra Explicit	41
7.2	The Structural Type Mismatch	41
7.3	Action on the Zero-Mode Lattice	42
7.4	The Tensor Decomposition	42
7.5	Kesten–McKay = Plancherel of Unramified $GL_2(\mathbb{Q}_p)$	43
7.6	The Soibelman Cohomological Hall Algebra Reading	43
7.7	Hochschild Comparison: Connes–Marcolli vs Claimed Theorem H^{arith}	43
7.8	Vertex-Algebraic Invariants of the Heisenberg Covers	44
7.9	The Correct Identification	44
7.10	Structural Conclusions	44
8	ADELIC FACTORIZATION AXIOMS	47
8.1	The Ran Space of $\overline{\text{Spec } \mathbb{Z}}$	47
8.2	The Factorization Isomorphism Is Trivial	47
8.3	The Diagonal and $\mathbb{Z}_p \widehat{\otimes} \mathbb{Z}_p$	48
8.4	The Locally Constant (E_0) Factorization at Each Finite Place	48
8.5	The Holomorphic-Topological Structure	48
8.6	Five Axioms (F1)–(F5) for the Minimal Adelic Factorization	49
	(F1) Archimedean factorization.	49
	(F2) Coefficient category.	49
	(F3) Hecke-eigen decomposition.	49
	(F4) Archimedean–finite compatibility.	49
	(F5) Regular p -adic product.	49
8.7	Classical Factorization at Finite Places: Absence of a BV Action Functional	49
8.8	Factorization Boundary	50
9	DAHA, MACDONALD, AND THE PARAMETER NON-UNIFORMITY	51
9.1	Macdonald (q, t) and the Prime-Indexed Alphabet	51
9.2	The $\log p$ Rescaling as a Heisenberg Conformal Rescaling	52
9.3	DAHA Parameters and the Arithmetic Hall Algebra	52
9.4	Shuffle Structure Is Uncoupled Across Primes	53
9.5	Arithmetic Macdonald Polynomials	53
10	THE ARITHMETIC INSTANTON PARTITION FUNCTION	55
10.1	Nekrasov’s Z_{inst} on $\mathbb{C}^2$	55
10.2	The Arithmetic Curve as a One-Dimensional Object: No Instanton Sector	55
10.3	The Self-Product $\overline{C} \times_{\mathbb{F}_p} \overline{C}$ as Arithmetic Surface	55
10.4	Rank Analysis: $U(1)^\infty$ per Prime	56
10.5	The $\epsilon_2 \rightarrow \infty$ Degeneration	56
10.6	Arakelov Coulomb Modulus	57
10.7	Nekrasov–Shatashvili Limit: $\log \zeta(s) = \sum_p \mathcal{W}_p^{\text{NS}}(s)$	57
10.8	AGT: Semiclassical Liouville Degenerates to Free Boson	57
10.9	Maulik–Okounkov and the Bethe–Hecke Shadow	58
10.10	The Topological-Sector Reading (Compatible with Nekrasov)	58
10.11	The $(\epsilon_2 \rightarrow \infty, b \rightarrow 0)$ Degenerate Limit: Structure of $L(s)$	58

11	THE ARITHMETIC MODULAR FUNCTOR	61
11.1	Axioms and Their Translation	61
11.2	Arakelov Discriminant as Lattice Volume	61
11.3	Weil–Deligne Sewing	62
11.4	Arithmetic Symmetries	62
11.5	Verlinde Scalar and the Class Number Formula	63
11.6	Explicit Computation: Conformal Blocks via Tate Integrals	63
11.7	Functional Equation as Sewing Axiom	63
11.8	Five-Axiom Arithmetic Modular Functor	64
	(A1) Finiteness.	64
	(A2) Factorization.	64
	(A3) Weil–Deligne sewing.	64
	(A4) Modular invariance.	64
	(A5) Functional equation.	64
12	ARITHMETIC BAR–COBAR AND PARSHIN–GERSTEN	65
12.1	The Configuration Space Is Discrete	65
12.2	The Arnold Relation Lifts to Steinberg	65
12.3	The Arithmetic Tensor Coalgebra	66
12.4	Koszulness and Deligne Purity	66
12.5	Primary Bar Mellin Character	66
12.6	The G-Row Complementarity	67
12.7	Arithmetic Positselski	67
12.8	Construction of $B^{\text{ch}, \text{Ar}}$ Concrete	68
12.9	Bar–Cobar Boundary	68
13	THE AUTOMORPHIC VERTEX OF THE LANGLANDS–TQFT TRIANGLE	69
13.1	The Kapustin–Witten Obstruction: Absence of Complex Structure	69
13.2	The BLLPRvR Obstruction: Infinite Flavour Rank	69
13.3	The Chern–Simons Obstruction: Irrational Levels	70
13.4	The Belinski–Khalatnikov–Lifshitz Billiard: Thermodynamic Coarse-Graining	70
13.5	The Automorphic Vertex	70
13.6	Asymptotic Symmetries at the BKL End-of-Time Boundary	70
13.7	The Vol III Φ_d Functor: Arithmetic Domain Boundary	71
13.8	Automorphic Placement	71
13.9	The Automorphic Vertex: Structural Consolidation	71
14	LOCAL CENTRAL CHARGE AND GLOBAL RESIDUE	73
14.1	Rescaling and the Conformal Vector Constraint	73
14.2	The Infinite Tensor Product and the Virasoro Obstruction	73
14.3	The Natural Boundary of the Prime Zeta Function	73
14.4	Rank Residue	74
14.5	Witt versus Virasoro	74
14.6	The Functional Equation as Arithmetic Modular Invariance	74
14.7	The Arakelov Pairing Replaces the Weyl Anomaly	75

14.8	The Arakelov- ζ Structure of the Residue	75
15	ARITHMETIC CHERN-SIMONS HOLOGRAPHY	77
15.1	Class $\mathcal{S}$ Fails Arithmetically	77
15.2	The Hitchin Base Has No Class- $\mathcal{S}$ Arithmetic Specialization	77
15.3	Kim's Arithmetic Chern-Simons	78
15.4	$\mathcal{V}^{\text{prim}}$ and the ACS Boundary Shadow	78
15.5	Wilson Lines Are Labeled by Ramified Hecke Characters	78
15.6	Beem-Lemos-Liendo-Peelaers-Rastelli-van Rees as Adelic Character	79
15.7	Arithmetic CS as the Generating Principle	79
16	THE HOCHSCHILD TRACE OF THE IWAHORI HECKE CATEGORY	81
16.1	The Naive $\text{Bun}_{GL_2}(\text{Spec } \mathbb{Z})$ Has Only One Coarse Point	81
16.2	The Spherical Hecke Category Collapses the OPE	81
16.3	The Iwahori Hecke Category Carries Convolution Data	82
16.4	The Emerton-Gee Spectral Side	82
16.5	The Hochschild Trace Placement	83
16.6	Connection to Gaiotto's 3D Chern-Simons Presentation	83
16.7	Hecke Trace = Eichler-Selberg	84
16.8	The Arithmetic Langlands Diamond for $\mathcal{V}^{\text{prim}}$	84
17	ARITHMETIC E_n FORMALITY AND CoHA	85
17.1	E_1 -Formality Is Automatic over $\mathbb{Z}$	85
17.2	E_2 -Formality Fails Integrally	85
17.3	The Strict Rectified E_1 -Model	85
17.4	The Chiral Bracket via Adelic Residue	86
17.5	The Arithmetic CoHA Conjecture	86
17.6	$\log L$ via Cyclic Cohomology	87
17.7	Arithmetic Formality Stratification	87
18	THE ARITHMETIC YANGIAN COLLAPSES TO ITS BETHE SUBALGEBRA	89
18.1	The Four Obstructions to a Naive Arithmetic Yangian	89
18.2	The Bethe-Algebra Collapse	89
18.3	Primes Are Lattice Sites, Not Spectral Parameters	90
18.4	Automorphic Closure as Bethe Heuristic	91
18.5	Explicit Computation: $\mathfrak{gl}_1$ and $\mathfrak{gl}_2$	91
	$\mathfrak{gl}_1$ (Dirichlet characters)	91
	$\mathfrak{gl}_2$ (Hecke eigenforms)	91
18.6	Local r -Matrix per Prime	91
18.7	Yangian $\rightarrow$ Hecke = Langlands as Arithmetic Bethe Ansatz	92
19	SYNTHESIS: SIX CONDITIONAL WINDOWS	93
19.1	The Synthesis	93
19.2	The Essential Theorem	94
19.3	The Five Theorems Survive	94
19.4	Segal Axioms as Tate's Thesis	94
19.5	Rank Residue	95

19.6	Arithmetic Conformal Field Theory: The Coherent Statement	95
	(B) Base.	95
	(V) Chiral algebra.	96
	(M) Module category.	96
	(W) Wilson-line defects.	96
	(F) Modular functor.	96
	(Z) Partition function.	96
	(R) Rank residue.	96
	(K) Koszul complementarity.	96
19.7	Langlands as Arithmetic Bethe Ansatz	98
19.8	Source-Family Convergence	98
19.9	The Three Open Conjectures	99
19.10	Structural Rigidity of $\mathcal{V}^{\text{prim}}$	99
19.11	Logical Form of the Arithmetic Branch	100
20	BOUNDARY CONJECTURES FOR THE SIX-WINDOW TRACE	101
20.1	F1: Global Arithmetic Arinkin–Gaitsgory	101
20.2	F2: E_2 -Enhancement via Scholze Diamonds	101
20.3	F3: Arithmetic Ben-Zvi–Nadler	102
20.4	Further Boundary Conditions F4–F11	102
	F4: B-Row Arithmetic Realisation	102
	F5: Arithmetic Chain Homotopy h_A^{Ar}	103
	F6: Chenevier Determinant at $p = 691$	103
	F7–F10: Additional Boundary Conditions	103
	F11: All-Genus Arithmetic Koszul Duality	103
20.5	Dependency Lattice	104
20.6	Implication to the Trace Statement	104
20.7	Boundary Index	104
21	CLOSURE OF F1–F3 AND RESOLUTION OF THE THREE PILLARS	107
21.1	Seven-Obstruction Reduction Criterion	107
21.2	F1: Structural Closure	107
	Statement	107
	Obstruction Analysis Trace	108
	Obstruction 1 (source separation).	108
	Obstruction 2 (conventions).	108
	Obstruction 3 (ambient categories).	108
	Obstruction 4 (missing hypothesis).	108
	Obstruction 5 (false functoriality).	108
	Obstruction 6 (unproved equivalence).	108
	Obstruction 7 (numerical comparison).	108
	Structured Reduction	108
21.3	F2: Structural Closure	110
	Statement	110
	Obstruction Analysis Trace	110

	Obstruction 1 (source separation).	110
	Obstruction 2 (conventions).	110
	Obstruction 3 (ambient categories).	111
	Obstruction 4 (missing hypothesis).	111
	Obstruction 5 (false functoriality).	111
	Obstruction 6 (unproved equivalence).	111
	Obstruction 7 (numerical comparison).	111
	Structured Reduction	111
21.4	F3: Conditional Closure	112
	Statement	112
	Obstruction Analysis Trace	112
	Obstruction 1–4.	112
	Obstruction 5 (false functoriality).	112
	Obstruction 6.	112
	Obstruction 7.	112
	Conditional Closure	112
21.5	P1: G-Row Complementarity and the Functional Equation	113
	Statement	113
	Obstruction Analysis Trace	113
	Obstruction 1–5.	113
	Obstruction 6 (unproved equivalence).	113
	Obstruction 7.	113
	Structured Proof	113
21.6	P2: Arithmetic Zhu Modularity	115
	Statement	115
	Obstruction Analysis Trace	115
	Obstructions 1–7.	115
	Theorem	115
21.7	P3: Unitarity of $\mathcal{V}^{\text{prim}}$	116
	Statement	116
	Obstruction Analysis Trace	116
	Obstructions 1–4.	116
	Obstruction 5 (false functoriality).	116
	Obstruction 6 (unproved equivalence).	117
	Obstruction 7.	117
	Theorem	117
21.8	Logical Form Table	118
21.9	Closure Boundary	118
22	THE RIEMANN HYPOTHESIS REFORMULATION	121
22.1	The Conjecture F-RH	121
22.2	Three-Pillar Implication Structure	121
22.3	Li-Positivity as Arithmetic Virasoro Unitarity	122
	Li’s Criterion	122
	Hilbert–Pólya Transform Computation	122

22.4	Comparison to Existing RH Mechanisms	125
22.5	Residual Conditions	125
22.6	A Sufficient Path to RH	126
22.7	Relation to the Arithmetic Branch Essential Theorem	127
22.8	The F-RH Reduction	127
23	ARITHMETIC VIRASORO POSITIVITY BOOTSTRAP	129
23.1	Statement Recalled	129
	The Operator Sequence	129
	Conditional Chain: $VB^* \Rightarrow Li \text{ Positivity} \Leftrightarrow RH$	129
23.2	Obstructions	130
	Virasoro levels	130
	Moment problem	130
	Kac determinant	130
	Fock combinatorics	131
	Weil trace	131
	de Branges datum	131
23.3	Level-by-Level Positivity	131
	Level $n = 1$: Unconditional	131
	Level $n = 2$: Formal Threshold	131
	Levels $n = 3, 4$: Bombieri-Lagarias Upper Bound	132
23.4	Moment Problem Reformulation	132
	The Shifted Zero Measure	132
	Li-Stieltjes Moment Formulation	133
	Hamburger Moment Determinacy	133
23.5	Local Kac Determinant at $c = 2$	134
	The Kac Formula	134
	The Arithmetic $c = 2$ Regime	134
	Unitarity Gap at Special Weights	134
23.6	Fock-Combinatorial Identity	135
	Explicit Action on Number States	135
	The Symmetric-Function Identity	135
23.7	Reduction to the Irreducible Core Obstruction	136
	The Core Statement	136
	The Named Obstruction	136
23.8	Comparison to Classical RH Equivalents	137
23.9	Zero-Divisor Condition	137
	The Precise Condition	137
	Virasoro Boundary	138
23.10	Arithmetic Zhu Finiteness Test	138
23.11	Zero-Measure Conjecture	140
23.12	Operator Boundary	140
24	ARITHMETIC POSITSIELSKI EQUIVALENCE	143
24.1	Statement and Motivation	143

	Review: Classical Positselski	143
	The Λ -Coefficient Target	143
24.2	Obstruction Analysis Table	145
	Obstruction 1: Base-Change Lift	145
	Obstruction.	145
	Attempt 1.	145
	Correction.	145
	Conclusion.	145
	Obstruction 2: Co-Derived vs Contra-Derived Over Λ	146
	Obstruction.	146
	Attempt 1.	146
	Correction.	146
	Conclusion.	146
	Obstruction 3: μ -Invariant Vanishing for the Heisenberg Bar	146
	Obstruction.	146
	Attempt 1.	146
	Correction.	146
	Conclusion.	147
	Obstruction 4: Weight Compatibility Under Bar–Cobar	147
	Obstruction.	147
	Attempt 1.	147
	Correction.	147
	Conclusion.	147
	Obstruction 5: Archimedean Completion	147
	Obstruction.	147
	Attempt 1.	147
	Correction.	147
	Conclusion.	148
	Obstruction 6: Non-Commutative Iwasawa for Non-Abelian GL_2	148
	Obstruction.	148
	Attempt 1.	148
	Correction.	148
	Conclusion.	148
	Comparison Table	148
24.3	Iwasawa Preliminaries	149
	Λ -Modules and Structure Theorem	149
	Injective and Projective Resolutions Over Λ	149
	Co-Derived and Contra-Derived Λ -Categories	150
	Iwasawa μ -Invariant and Ferrero–Washington	151
24.4	The Main Theorem	151
	Overview of the Proof	151
	Step 1: Construction of the Functors	151
	Step 2: Base Change to $\mathbb{F}_p[[T]]$	152
	Step 3: Lifting Along Nakayama	152
	Step 4: Bar–Cobar Adjunction	153

	Step 5: Unit / Counit Quasi-Isomorphism	153
	Step 6: Independence of Tower Presentation	154
	Assembly: the Main Theorem	154
24.5	Application: Step 3 of P_1 for the Arithmetic Heisenberg	154
	The Step-3 Residual	154
	The Explicit Weight Computation	155
	Global Euler-Product Check	156
	Full Biconditional of P_1 Closes Conditionally	156
24.6	Archimedean Extension: Gaussian Positselski	157
	The Archimedean Analogue	157
	Adelic Assembly	157
24.7	Residual Conditions	158
	Non-Abelian Iwahori Level	158
	Whittaker Model Compatibility	158
	Extension to Higher Rank	158
24.8	Positselski Boundary	159
25	KOSZUL–PETERSSON PAIRING COMPATIBILITY	161
25.1	Statement Recalled	161
	Setup	161
	The Two Pairings	162
	The Identity	162
25.2	Obstruction Analysis Table	162
	Obstruction 1 (source separation).	163
	Obstruction 2 (sign / convention).	163
	Obstruction 3 (ambient category).	163
	Obstruction 4 (missing hypothesis).	163
	Obstruction 5 (false functoriality).	163
	Obstruction 6 (unproved equivalence).	163
	Obstruction 7 (numerical comparison).	163
25.3	Archimedean Identification	164
	The Fock Inner Product	164
	The Koszul Pairing at ∞	164
	The Petersson Pairing at ∞	165
25.4	Finite-Place Identification	166
	Bost–Connes Hilbert Pairing	166
	The Koszul Pairing at p	166
	Bost–Connes vs. Petersson at p	167
25.5	Adelic Assembly	168
	Tate’s Restricted Tensor Product	168
	Assembly of Petersson and Koszul	168
25.6	The Bost–Connes Anti-Involution and its GNS Lift	169
	Definition and Basic Properties	169
	Adelic Involution	170
25.7	Spherical-Vector Calculation	170

	Spherical Petersson at p	170
	Spherical Koszul at p	170
	Spherical Match Globally	171
25.8	Analytic Continuation and Poles	171
	Pole Structure of Both Pairings	171
	No Exotic Poles	172
25.9	The Main Theorem	172
25.10	Consequence: P_3 on the Sph-Fin Cuspidal Core	173
	Impact on the Sufficient RH Implication	174
25.11	Boundary	174
26	ADELIC CATEGORICAL DESCENT	177
26.1	Statement Recalled	177
	The Condition	177
	Context and Significance	178
26.2	Modular Reduction of L_{ACD}	178
	Modular Decomposition of L_{ACD}	178
	Curve Precedent and Transport Obstructions	179
	Sphericity-Finiteness Preserved Under Restricted Tensor	180
	Cross-Place Hecke Commutativity	180
	Non-Transference from Caraiani–Scholze	181
	Reduction to a Single Compatibility	181
26.3	Subcondition (L_1) : Restricted Tensor in ∞ -Categories	182
	Definition	182
26.4	Subcondition (L_2) : Hecke-Functoriality Across Places	183
	Statement and Reduction	183
26.5	Subcondition (L_3) : Bernstein Centre Compatibility	184
	Statement and Residual Content	184
	Partial change via Coherent Springer Theory	184
26.6	Subcondition (L_4) : Archimedean Extension	185
	Statement	185
26.7	Curve Analogy: Obstructions and Transfers	186
	The Curve Theorem	186
	Obstructions to Transport	186
26.8	Final Reduction: L_{ACD} to a Single Lemma	187
	Statement	187
	Dependency on Established Theorems	187
26.9	Consequence: Fr Closure Map	188
	The Full Chain	188
	Formal Consequences	188
	Effect on $\mathcal{V}^{\text{prim}}$ Spectral Identification	188
26.10	Residual Obstructions	189
	Obstruction Decomposition	189
	Prior Hypotheses	189
	Primary-Literature Precedents	189

	Paper-Length Estimate, Revised	190
26.II	Descent Boundary	190
26.I2	Forward Pointers	191
27	ARITHMETIC BEN-ZVI–NADLER IDENTIFICATION	193
27.1	Statement	193
27.2	Comparison Table	193
27.3	The Arithmetic Inertia Circle $B\widehat{\mathbb{Z}}$	194
	Definition	194
	MKR Dictionary and the Arithmetic S^1	195
	Source Comparison: MKR Dictionary and Inertia Circle	195
	Obstruction 1 (source separation).	195
	Obstruction 2 (conventions).	195
	Obstruction 3 (ambient categories).	195
	Obstruction 4 (missing hypothesis).	195
	Obstructions 5–7.	196
27.4	Arithmetic Free Loop Space	196
	Definition	196
	Computation for $\mathcal{X} = BGL_2$	196
	The Selmer Inertia-Shadow Morphism	197
	The Flat Locus	197
	Source Comparison: Loop-Space Computation for BGL_2	198
	Obstruction 1 (source separation).	198
	Obstruction 2 (conventions).	198
	Obstruction 3 (ambient categories).	198
	Obstruction 4 (missing hypothesis).	198
	Obstruction 5 (false functoriality).	198
	Obstruction 6 (unproved equivalence).	198
	Obstruction 7.	198
27.5	The Hochschild-Trace Theorem (Arithmetic)	199
	Classical Ben-Zvi–Nadler	199
	Arithmetic Extension of Costello–Francis–Lurie	199
	Arithmetic Loop-Space Identification	200
	Source Comparison: Arithmetic Transport of Ben-Zvi–Nadler	201
	Obstruction 1 (source separation).	201
	Obstruction 2 (conventions).	201
	Obstruction 3 (ambient categories).	201
	Obstruction 4 (missing hypothesis).	201
	Obstruction 5 (false functoriality).	201
	Obstruction 6 (unproved equivalence).	202
	Obstruction 7.	202
27.6	Dunn Additivity and E_1 -Structure	202
27.7	Main Theorem	202
27.8	Consequence: F_3 Closure under $F_1 + F_2$	203
	Source Comparison: Dunn Additivity and Full-Target Pullback	204

	Dunn additivity and E_1 -structure.	204
	Obstruction 1 (source separation).	204
	Obstruction 2 (conventions).	204
	Obstruction 3 (ambient categories).	204
	Obstruction 4 (missing hypothesis).	204
	Obstruction 5 (false functoriality).	204
	Obstructions 6, 7.	204
	Full Weil/ (φ, Γ) target and $\mathcal{V}^{\text{prim}}$ pullback.	204
	Obstruction 1 (source separation).	204
	Obstruction 2 (conventions).	204
	Obstruction 3 (ambient categories).	205
	Obstruction 4 (missing hypothesis).	205
	Obstruction 5 (false functoriality).	205
	Obstruction 6 (unproved equivalence).	205
	Obstruction 7.	205
27.9	Independent Comparison Path	205
	Proof Source	205
	Comparison Source	205
	Disjoint Rationale	206
	Comparison Sketch	206
27.10	Residual Conditions	206
	Domain: Archimedean Place	207
	Domain: Non- GL_2 Groups	207
27.11	Ben-Zvi–Nadler Boundary	207
28	THE ∞ -CATEGORICAL TATE RESTRICTED TENSOR PRODUCT	209
28.1	Statement of (F2.c)	209
	Setup and Definition	209
	Source Disjointness	210
28.2	Obstructions and Their Resolutions	210
	Centrality of the Spherical Unit	210
	$\mathcal{O}$ -Algebra Structure under Filtered Colimits	210
	BD Factorization Axiom on Discrete V	211
	Dunn Additivity and the Restricted Tensor Product	211
	Cofinality Invariance of the $\mathcal{O}$ -Structure	211
	Ramified Places without a Spherical Unit	211
28.3	∞ -Operadic Preliminaries	212
	Presentable Symmetric Monoidal Structure on $\text{Pr}_{\text{st}}^{\text{L}}$	212
	Algebras over an ∞ -Operad in a Presentable Symmetric Monoidal ∞ -Category	212
	E_n -Operad and Integral Structure	212
	Centrality of the Unit	213
	Filtered Colimits of ∞ -Operadic Algebras	213
28.4	Unital Inclusion Functors	213
28.5	Colimit Preservation	214
28.6	Proof of the Main Theorem	215

28.7	Compatibility with BD Factorization	216
	Factorization on a Discrete Ran Space	216
	Compatibility with Dunn Additivity	217
28.8	Application to the Iwahori Hecke Category	218
	The $GL_2/\mathbb{Q}$ Setup	218
	The Corollary: F2.c Closed, F2 Still Conditional	218
28.9	Uniqueness of the E_n Structure	219
28.10	Base Change and Functoriality	220
28.11	Examples	221
	Example 1: Discrete V , Discrete C_v	221
	Example 2: Function Field Analogue	221
	Example 3: Arithmetic Iwahori Hecke	221
28.12	Factorization-Homology Comparison	221
28.13	Remarks on Operadic Coherence	222
	Why E_n -coherence is required	222
	Alternative Unital Structures	223
	Iterated Restricted Tensor Products	223
28.14	Completeness of F2.c and Residual Local Input	223
28.15	Tate Tensor Boundary	223
29	INDEPENDENT COMPARISON: RESIDUAL STRUCTURE	225
29.1	First Residual Decomposition	225
	F2.c: ∞ -Categorical Tate Restricted Tensor Product	225
	L_{ACD} : Adelic Categorical Descent	225
	$F_3.BZN^{Ar}$: Arithmetic Ben-Zvi–Nadler Identification	225
	Arithmetic Positselski Co-Contra Equivalence: Abelian Domain	226
	Koszul–Petersson Compatibility: Sub-Claim Structure	226
	VB^* : Arithmetic Virasoro Positivity and the Biconditional Error	226
29.2	Sharp Subconditions	227
	Pointed E_n -monoidal Category	227
	L_{ACD} Split into Four Named Subconditions	227
	$F_3.BZN^{Ar}$ Split into Three Sub-Obstructions	227
	Arithmetic Positselski with Abelian Domain Restriction	228
	Koszul–Petersson Compatibility Split into Seven Sub-Claims	228
	VB^* : One-Directional Implication and the Converse Residual	229
29.3	First Logical Core	229
29.4	Second Residual Decomposition	230
	F2.c: E_2 -Algebra Structure on the Unit Map	230
	L_{ACD} : Archimedean Stratum and Hecke-at-Infinity	230
	$F_3.BZN^{Ar}$: Formal-Noetherian Extension, Non-Semisimple Correction, and Loop-Space Convention	230
	Arithmetic Positselski: Five Supplementary Subconditions	231
	Koszul–Petersson Compatibility: Five Higher-Structural Sub-Claims	231
	VB^* : Cross-Chapter Contradiction and Primary-Virasoro Reconciliation	231
29.5	Resulting Logical Core	232

30	L_{ACD} SUBCONDITIONS: ARCHIMEDEAN STRATUM, COLIMIT COMMUTATION, CROSS-PRIME HECKE, SPH-FINITE PRESERVATION	233
30.1	Subconditions and Independent Sources	233
	The Subcondition List	233
	Independent Sources	234
30.2	Overview of Subcondition Closures	234
	$L_{ACD.0}$: archimedean stratum	234
	$L_{ACD.1}$: colimit commutation	235
	$L_{ACD.2}$: cross-prime Hecke coherence	235
	$L_{ACD.3}$: sph-finite preservation	235
	Archimedean character and the completed ξ -function	235
30.3	$L_{ACD.0}$: Archimedean Stratum Closure	235
	Statement and Construction	235
	Compatibility with Finite-Place Sph-Fin Structure	237
30.4	$L_{ACD.1}$: Dg-Categorical Colimit Commutation	237
	Statement	237
	Explicit Commutation Square	238
30.5	$L_{ACD.2}$: Cross-Prime Hecke Coherence	239
	Statement	239
	Bernstein–Zelevinsky Compatibility	240
30.6	$L_{ACD.3}$: Sph-Finite Preservation	240
	Statement	240
	Uniformity of the Sph-Fin Subcategory	241
30.7	Consequence: Conditional Reduction of L_{ACD}	241
	The Residual Lemma	241
	Domain of (BC-diamond-glue)	242
30.8	Compatibility with $\Lambda_{\xi}(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ Factorization	242
	The Completed L -Factor	242
	Tate Character Identification	242
	Functional Equation Compatibility	243
30.9	Residual Conditions	244
	What Remains Open	244
	Fi Closure Chain	244
	Hypotheses Inventory	244
30.10	Conditional Reduction of L_{ACD}	244
31	BZN^{Ar} : FORMAL NOETHERIAN HOCHSCHILD AND DERIVED STRUCTURE ON EMERTON–GEE	247
31.1	Formal Boundary of the Comparison	247
	Structural Inputs	247
	Domain of This Chapter	248
	Derived Enhancement of the Emerton–Gee Stack	248
	Finite-Type Hypotheses in the Gaitsgory–Rozenblyum Argument	248
31.2	Logical Form of the Inputs	248
31.3	R2: The Emerton–Gee Cotangent Complex	249

	Local Tangent Complex at a Closed Point	249
	Ind-Level Cotangent Complex	250
	R ₂ Main Theorem	251
	Source and Convention Comparison for R ₂	252
	Source independence.	252
	Degree conventions.	252
	Ambient category.	252
	Hypotheses.	252
	Functoriality.	252
	Source of equivalences.	252
31.4	R _L .formal: Formal Hochschild Criterion	252
	The Three Inputs	252
	R _L .formal.a: HH Colimit Continuity	253
	R _L .formal.b: Arithmetic Loop-Space Formal Computation (Conditional)	254
	R _L .formal.c: Flat-Locus Pullback Compatibility (Conditional)	255
	Sufficient Criterion for F ₃ .BZN ^{Ar}	255
	Source and Convention Comparison for R _L .formal	256
	Source independence.	256
	HH conventions.	256
	Ambient category.	256
	Hypotheses.	256
	Functoriality.	256
	Source of the implication.	256
31.5	Conditional Form of F ₃ .BZN ^{Ar}	256
31.6	Boundary Conditions	257
	Form of the Inputs	257
	Logical Consequence	257
	Open Problems	257
31.7	Formal Noetherian Boundary	257
32	BZN ^{Ar} : LOOP-SPACE CONVENTION AND WILD-PRIME SUMMANDS	259
32.1	The Two Loop-Convention and Wild-Prime Conditions	259
	Origin of the Two Conditions	259
	Logical Structure	260
	Notation	260
32.2	Structure of the Argument	260
32.3	R ₃ : Arithmetic Loop-Space Convention	260
	The Three Candidate Definitions	260
	Explicit Computation of the Three Candidate Definitions	261
	Independence and Convention Check for R ₃ Explicit Formulas	262
	Sources.	262
	Conventions.	262
	Ambient category.	262
	Hypotheses.	262
	Equivalence of the Three Definitions on Formal Noetherian Derived Stacks	262

	Independence and Convention Check for R_3 Equivalence	263
	Sources.	263
	Conventions.	263
	Ambient category.	263
	Hypotheses.	263
	Functoriality.	263
	Named theorems.	263
	Canonical Choice and Consistency with Chapter 27	263
	Independence and Convention Check for Canonical-Choice	264
	Sources.	264
	Conventions.	264
	Ambient category.	264
	Hypotheses.	264
	Functoriality and universal property.	264
	R_3 Boundary	264
32.4	R_4 : Wild-Prime Non-Semisimple Correction	264
	Semisimplicity at $p \geq 5$	264
	Non-Semisimplicity at $p = 2, 3$	265
	Ext^1 -Contribution at $p = 2$	265
	Ext^1 -Contribution at $p = 3$	266
	Independence and Convention Check for Wild-Prime Ext	267
	Sources.	267
	Conventions.	267
	Ambient category.	267
	Hypotheses.	267
	Block functoriality.	268
	Named theorems.	268
	Cross-Check: Vanishing at $p \geq 5$	268
	$F_3.BZN^{\text{Ar}}$ with Wild Summands	268
	Independence and Convention Check for Vanishing at $p \geq 5$	269
	Sources.	269
	Conventions.	269
	Ambient category.	270
	Hypotheses.	270
	Global assembly.	270
32.5	Consistency with Chapter 27	270
	R_3 Consistency	270
	R_4 Consistency	270
32.6	Boundary Conditions	271
	R_3 Boundary	271
	R_4 Boundary	271
	Logical Form for Chapter 29	271
32.7	Loop-Convention Boundary	271

33	COSTELLO-REFINED POSITSIELSKI: FACTORIZATION COHERENCE, $d^2 = 0$, RENORMALIZATION, CLASSICAL LIMIT, FUNCTORIALITY	273
33.1	Five Subconditions of the Abelian Positselski Domain	273
	The Five Subconditions	273
	Notation Recalled	274
33.2	Comparison Table	274
	Cycle 1 (C1): Factorization-Coalgebra Coherence	274
	Cycle 2 (C2): Chevalley–Eilenberg $d^2 = 0$	275
	Cycle 3 (C3): Renormalization / Euler-Product Convergence	275
	Cycle 4 (C4): Archimedean Classical Limit	276
	Cycle 5 (C5): Functoriality	277
	Cycle 6: Cross-Check on P1 Step 3 Functorial Pullback	277
	Comparison Table	278
33.3	C1: Local-Global Factorization-Coalgebra Coherence	278
	Local Factorization Coalgebra at Each Place	278
	Global Assembly via the Beilinson Adelic Complex	279
33.4	C2: Chevalley–Eilenberg $d^2 = 0$ via Parshin Reciprocity	280
	The Three-Piece Differential	280
	The Three Mixed Anticommutators	280
	Parshin Reciprocity Closes $d^2 = 0$	281
33.5	C3: Renormalization / Euler-Product Convergence	283
	Pro-Nilpotent Tower	283
33.6	C4: Archimedean Classical Limit	284
	Witt-Vector Truncation and the Classical Limit	284
	Reduction to Segal / Frenkel–Ben-Zvi	285
33.7	C5: Functoriality under Λ -Coalgebra Morphisms	286
	The Functoriality Statement	286
	Induced Pullback and Pushforward	286
	Naturality of the Positselski Equivalence	287
33.8	Consequence: P1 Step 3 Functorial Pullback	288
33.9	Residual Conditions	288
33.10	Refinement Boundary	289
34	BC-DIAMOND-GLUE LEMMA AND GODEMENT–JACQUET ABSORPTION (K5) . . .	291
34.1	Incoming Conditions and Disjoint-Source Discipline	291
	Independent sources.	291
34.2	Comparison Table for the Six Subconditions	291
34.3	BC-Diamond-Glue: Decomposition	293
34.4	BC-Diamond-Glue: Local Construction (Ben-Zvi–Chen–Helm–Nadler)	293
34.5	BC-Diamond-Glue: Bernstein Centre Gluing	294
	Prismatic Transition Maps	294
	Filtered Colimit of Gluings	295
	Reduction to core compatibility	295
34.6	Consequence: L_{ACD} Final Residual Isolation	296
34.7	K5: Godement–Jacquet Absorption	297

	The Local Adjoint L -Factor	297
	The Absorption Convention	298
	Adelic Consistency	299
34.8	Consequence: K-P Compat, Conditional Domain	300
34.9	Residual Conditions	301
34.10	Residual Boundary	302
35	ARITHMETIC HILBERT–PÓLYA AND THE VB^* CONVERSE: STRUCTURAL ANALYSES	305
35.1	The Analytic Splitting	305
	The Asymmetry	305
	Disjoint Sources	305
35.2	Structural Reductions	306
	$H\text{-}p^{\text{Ar}}$: Operator-Theoretic Framework	306
	$H\text{-}p^{\text{Ar}}$: Connes Adele Class Space	307
	$H\text{-}p^{\text{Ar}}$: Deninger Motivic Cohomology	307
	$H\text{-}p^{\text{Ar}}$: Necessary and Sufficient Conditions	307
	VB^* Converse: Spectral Regularity Beyond RH	307
	VB^* Converse: de Branges and Báez-Duarte	307
35.3	$H\text{-}p^{\text{Ar}}$: Structural Analysis	308
	Operator-Theoretic Framework	308
	Connection to Connes Adele Class Space	309
	Connection to Deninger Motivic Cohomology	310
	Necessary and Sufficient Conditions	311
35.4	VB^* Converse: Structural Analysis	312
	The Moment Problem in Hilbert Space	312
	Hermite–Biehler + de Branges	314
	The Báez-Duarte Refinement	315
	The Spectral-Regularity Content	316
35.5	Relation Between $H\text{-}p^{\text{Ar}}$ and VB^* Converse	317
35.6	Analytic Conditions	318
35.7	Spectral Boundary	318
36	THE PLATONIC VARIATION AND RH COMPLETION	321
36.1	The Right Variation	321
36.2	Factorisable Arithmetic Modules	322
36.3	The Arithmetic-Chiral Bar Complex on the Variation	323
36.4	RH Completion	324
37	CONDITIONAL HYPOTHESES FOR $F_1\text{--}F_3$ ARITHMETIC	327
37.1	Conditional Statements	327
	$F_1\text{--}F_3$ in Chapter 21	327
	$F_{1,a}$ domain: Fargues–Scholze local geometrisation versus categor- ical equivalence.	327
	Emerton–Gee preprint identifier.	327
	Ben-Zvi–Chen–Helm–Nadler preprint identifier.	327
	Adelic Categorical Descent	327

Function-field transport and (L_2)	327
(L_2) reduction: Fargues–Scholze domain.	328
F ₁ reduction: two open inputs.	328
Arithmetic Ben-Zvi–Nadler	328
F ₃ .BZN ^{Ar} : procyclic shadow versus full Weil parameter stack.	328
Structural Reduction Theorems	328
L _{ACD} equivalence and subcondition forms.	328
F ₃ .BZN ^{Ar} equivalence via Gaitsgory–Rozenblyum.	328
Arithmetic Positselski C ₁ –C ₅	328
L _{ACD} Subconditions	328
L _{ACD,o} : archimedean stratum construction.	328
L _{ACD,i} : the F _{2,c} hypothesis.	328
L _{ACD,2} : local Hecke commutativity, not global functoriality.	329
L _{ACD} residual: three-component decomposition.	329
BZN Formal Noetherian and EG Derived	329
R ₂ : derived enhancement of the Emerton–Gee stack.	329
R ₁ domain: finite-type theorem inapplicable to formal Noetherian setting.	329
Loop Convention and Wild Primes	329
R ₃ : loop convention equivalence under explicit hypotheses.	329
R ₄ : wild-prime Ext ¹ refinement at $p \geq 5$	329
BC-Diamond-Glue Decomposition	329
(BC-diamond-glue) three-component factorisation: form of components.	329
L _{ACD} tight reduction: conditional on three inputs.	330
Primary Source Identifiers in App. A	330
Dotto–Emerton–Gee in the primary source list.	330
37.2 Logical Form for F ₁ –F ₃ Arithmetic	330
37.3 Open Conditions for F ₁ –F ₃ Arithmetic	330
38 SHARP HYPOTHESES FOR THE FOUR-INPUT RH FORM	333
38.1 Six Sharp Forms	333
38.2 Local Forms	334
Zero-Placement	334
Arithmetic Positselski	334
Koszul–Petersson Compatibility	334
Costello Refinements C ₁ –C ₅	335
K ₅ Godement–Jacquet Normalisation	335
H-p ^{Ar} and the VB* Converse	335
Primary-Source Anchors	336
38.3 The Four-Input Conditional Conjunction	336
38.4 Logical Form of the Essential Biconditional	337
38.5 Consequences	337
39 THE DOTTO–EMERTON–GEE EXTENSION: F _{1,A} AT ALL PRIMES	339
39.1 The DEG Result in Context: Domain and Obstructions	339

	Fargues–Scholze and the Four Components of Fl_a	339
	Partial Closures and Residuals	340
39.2	Obstruction Table	340
39.3	Dotto–Emerton–Gee Recalled	341
	Setup	341
	DEG Theorem	341
	Why Essential Surjectivity is Subtle	342
39.4	Obstruction (a): Essential Surjectivity at $p \geq 5$	342
	The Colmez–Paskunas Block Decomposition	342
	The Corresponding Spectral-Side Decomposition	342
	The Block-Exhaustion Question	343
	Source Comparison for Obstruction (a), $p \geq 5$	344
	Obstruction 1 (source separation).	344
	Obstruction 2 (conventions).	344
	Obstruction 3 (ambient category).	344
	Obstruction 4 (missing hypothesis).	344
	Obstruction 5 (false functoriality).	344
	Obstruction 6 (unproved equivalence).	344
	Obstruction 7 (numerical comparison).	344
39.5	Obstruction (b): Extension to $p = 2, 3$	344
	Why DEG Does Not Extend Directly	344
	The Wild-EG Enhancement Programme	345
	Construction of the Wild DEG Functor	345
	BH-Block Exhaustion and Essential Surjectivity at Wild Primes	347
	Source Comparison for Obstruction (b), $p \in \{2, 3\}$	347
	Obstruction 1 (source separation).	347
	Obstruction 2 (conventions).	347
	Obstruction 3 (ambient category).	347
	Obstruction 4 (missing hypothesis).	348
	Obstruction 5 (false functoriality).	348
	Obstruction 6 (unproved equivalence).	348
	Obstruction 7 (numerical comparison).	348
39.6	Obstruction (c): Central Character Variation	348
	The Central Character Torus	348
	Pushforward of the DEG Functor	349
	Source Comparison for Obstruction (c)	349
	Obstruction 1 (source separation).	349
	Obstruction 2 (conventions).	349
	Obstruction 3 (ambient category).	349
	Obstruction 4 (missing hypothesis).	349
	Obstruction 5 (false functoriality).	349
	Obstruction 6 (unproved equivalence).	349
	Obstruction 7 (numerical source).	349
39.7	Reduction and Partial Closure of Fl_a	350

	Assembly of (a), (b), (c)	350
	Partial Closure: What Closes Unconditionally	350
	Residuals	351
39.8	Consequence for F_1 and L_{ACD}	351
	Local $F_{1,a}$ Closed at Every Prime Suffices for Adelic F_1	351
	Updating the Chapter 21 Reduction	351
	Source Comparison for Global Assembly	352
	Obstruction 1.	352
	Obstruction 2.	352
	Obstruction 3.	352
	Obstruction 4.	352
	Obstruction 5.	352
	Obstruction 6.	352
	Obstruction 7.	352
39.9	Residual Conditions	352
	Named Residuals Introduced by This Chapter	352
	What is Unconditional	352
	Out of Domain	353
39.10	DEG Extension Boundary	353
40	$H\text{-}P^{Ar}$ DETERMINANT-TRACE MATCHING AND HAMBURGER DETERMINACY . . .	355
40.1	Determinacy Comparison	355
	Residual Splitting	355
	Two Analytic Tests	355
	Independent Sources	356
40.2	Six Reductions	357
	Cycle 1 (A, weight enumeration)	357
	Problem.	357
	Argument.	357
	Sources.	357
	Conclusion.	357
	Cycle 2 (A, uniqueness under KMS rigidity)	357
	Problem.	357
	Argument.	358
	Sources.	358
	Conclusion.	358
	Cycle 3 (A, consistency with $H\text{-}P^{Ar}$)	358
	Problem.	358
	Argument.	358
	Sources.	358
	Conclusion.	358
	Cycle 4 (B, Hamburger determinacy reductions)	358
	Problem.	358
	Argument.	358
	Sources.	358

	Conclusion.	358
	Cycle 5 (B, obstruction via Li–Bombieri–Lagarias)	359
	Problem.	359
	Argument.	359
	Sources.	359
	Conclusion.	359
	Cycle 6 (cross-check; residual obstruction naming)	359
	Problem.	359
	Argument.	359
	Sources.	359
	Conclusion.	359
40.3	H- P^{Ar} Spectral Matching: Bost–Connes Zero-Mode Weight	359
	Candidate Weights	359
	Pairwise Compatibility with P_1 and P_2	360
	Uniqueness under KMS/Zeta Normalisation	361
	The Hierarchy Condition	362
40.4	Hamburger Determinacy Obstruction	363
	The Symmetrised Shifted-Zero Measure	363
	Li-Coefficient Bounds Do Not Give Carleman Moments	364
	Tightest Conditional Reduction	364
40.5	Consequence for Ch 37 RH Completion	365
40.6	Residual Conditions	366
40.7	Determinacy Boundary	367
41	ARITHMETIC FLE: SOURCE DOMAIN AND LOGICAL CORRECTIONS	369
41.1	Source Theorems	369
41.2	Domain Distinction: Geometric FLE vs $F_1/F_2/F_3$	369
41.3	Domain Corrections Across Source Categories	370
41.4	Corrected Logical Form	372
41.5	Open Conditions	372
42	(BC-DIAMOND-GLUE).CORE: CROSS-PRIME BERNSTEIN COMPATIBILITY	375
42.1	Statement of (BCglue.core)	375
	Setup	375
	The Core Compatibility Statement	375
42.2	Four Reductions of (BCglue.core)	376
42.3	Local Bernstein Centre and the Spectral Action	377
42.4	Global Bernstein Centre and the Chenevier Pseudocharacter Ring	378
42.5	Prismatic-Nygaard Cross-Prime Transition	379
42.6	The Core Compatibility Theorem	380
	Part (a): pairwise Bernstein compatibility reduces to Chenevier	380
	Part (b): colimit-to-global reduces to arithmetic AG	381
	The Core Final Reduction	382
42.7	Consequence for L_{ACD}	383
42.8	Glue Boundary	384

43	R_{analytic} : FORM, DIFFICULTY, AND THE RIEMANN HYPOTHESIS	385
43.1	The Four Assertions	385
43.2	The Stirling Scale in the Carleman Computation	386
43.3	Bombieri–Lagarias Bounds λ_n , Not μ_{2n}	386
43.4	Position of R_{analytic} in the Equivalence Chain	386
43.5	Carleman Sum from the Bombieri–Lagarias Upper Bound	386
	Upper Bound on μ_{2n} via Riemann–von Mangoldt	386
	Stirling Evaluation	387
43.6	Sharpening the $(\log n)^{4n}$ Gap Requires Zero-Density Hypotheses	388
	The Bombieri–Lagarias Polylog Bound Does Not Sharpen to $\mu_{2n} \asymp n!^2$ by Classical Analysis	388
	Three Independent Axes: Form, Implication, Difficulty	389
43.7	Determinacy, VB^* , and RH	389
	Step 1: Determinacy $\Leftrightarrow VB^*$ (Which Direction?)	390
	Step 2: Determinacy vs RH	390
	The Implication Diagram	390
43.8	Conclusion	391
43.9	The Sharp Reduction of R_{analytic}	391
43.10	Consequences	392
44	ARITHMETIC POSITSIELSKI: PRIMITIVE ZETA-CHARACTER THEOREM	393
44.1	Statement	393
44.2	Four Independent Computations	394
	The Finite and Archimedean Primary Object $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$	394
	Definition.	394
	Finite places.	394
	Archimedean place.	394
	Character.	394
	Mellin–Theta Character Computation	394
	Computation.	394
	Archimedean factor.	394
	Result.	395
	Koszul-Dual Character	395
	Finite part.	395
	Archimedean factor.	395
	Result.	395
	Match Under Koszul Self-Duality	395
	Characters.	395
	Proof.	395
	Result.	395
	Identity.	395
	Table	396
44.3	Primary Subspace $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$	396
	Local Primary Subspaces	396
	Archimedean Primary Subspace	397

	Adelic Primary Subspace	398
44.4	Character on Primary via Mellin–Theta	398
	Finite-Place Primary Character	398
	Archimedean Primary Character	399
	Adelic Character: $\Lambda_{\mathcal{F}}(s)$	399
44.5	Koszul-Dual Character (Bost–Connes Side)	400
	Cobar of Primary Heisenberg	400
	Bost–Connes KMS Partition on Primary	400
	Archimedean Factor via Fourier Duality	400
	Adelic Koszul-Dual Character	401
44.6	Primitive Zeta-Character Theorem	401
	Assembly	401
	Disjointness of Sources	401
44.7	Consequence for P_1	402
	Reduction of Open Sub-Items	402
	Character-Level Fingerprint of the Functional Equation	403
44.8	Remaining Four Positselski Sub-Items	403
	Residual Table	404
44.9	Primitive Character Boundary	404
45	KOSZUL–PETERSSON: KMS_I/GNS IDENTIFICATION AND FULL-SECTOR EXTENSION	405
45.1	Three Residuals and Their Resolution	405
	Residuals from the Koszul–Petersson Domain Restriction	405
	Consequences	405
	Consequence for P_3	406
	Three Residuals	406
	Disjoint sources.	407
45.2	KMS_I/GNS Identification of the BC Involution	408
	The Bost–Connes KMS_I State	408
	GNS Construction	408
	Modular Operator and Conjugation	409
	Adelic Assembly	410
45.3	Full-Sector Finite-Place Rankin–Selberg	410
	Ramified Principal Series	410
	Supercuspidal Representations	412
	Ramanujan and Non-Tempered Residue	412
45.4	Langlands Decomposition of $L^2([GL_2])$ and Placement of $\mathcal{V}^{\text{prim}}$	413
	The Spectral Decomposition	413
	Placement of $\mathcal{V}^{\text{prim}}$	413
45.5	Eisenstein Continuum: Koszul–Petersson Discrepancy	414
	The Maass–Selberg Inner Product	414
	The Koszul Continuation on the Continuum	414
45.6	Consequence for P_3	416
	Impact on the Sufficient RH Implication	417

45.7	Residual List	417
	What remains open.	417
45.8	Source, Convention, and Domain	417
	KMS ₁ /GNS identification of the BC involution.	417
	Full-sector finite-place Rankin–Selberg.	418
	Langlands decomposition.	418
	Eisenstein continuum handling.	418
45.9	Boundary	419
46	FLA.3: BUSHNELL–HENNIART BLOCK EXHAUSTION AT WILD PRIMES	421
46.1	Statement	421
	Setup	421
	The BH Block-Exhaustion Statement	421
46.2	Proof Reduction and Source Comparison	422
46.3	Bushnell–Henniart Tame-Wild Types at $p = 2$	423
	The Classification	423
	Matching to Spectral Components	424
46.4	Bushnell–Henniart Tame-Wild Types at $p = 3$	424
	The Classification and the A_2 -Quiver Structure	424
	Matching to Spectral Components	425
46.5	Essential Surjectivity via Type-by-Type Exhaustion	426
	The Exhaustion Argument	426
46.6	Cross-Check with the Wild-EG Definition	426
	Dimensional Comparison	426
	Structural Consistency	427
46.7	The Three Conditional Inputs	427
	(A) BH Type-Block Comparison	427
	(B) Paskunas Wild Ext ¹ -Compatibility	428
	(C) Wild-EG Compact-Generator Identification	428
46.8	Consequence for FLA and FL	428
	FLA Closure at Wild Primes	428
	FLA Closure at All Primes	428
	The Updated FL Obstruction Decomposition	429
46.9	Residual Conditions and Source Comparisons	429
	Named Residuals Introduced	429
	What Closes Unconditionally	430
	Source and Convention Comparison: BH Tame-Wild Classification at $p = 2$	430
	Source separation.	430
	Conventions.	430
	Ambient category.	430
	Hypotheses.	430
	Functoriality.	430
	Unproved equivalences.	430
	Numerical comparison.	430
	Source and Convention Comparison: BH Tame-Wild Classification at $p = 3$	430

	Source separation.	430
	Conventions.	430
	Ambient category.	431
	Hypotheses.	431
	Functoriality.	431
	Unproved equivalences.	431
	Numerical comparison.	431
	Source and Convention Comparison: Essential Surjectivity via Block Exhaustion	431
	Source separation.	431
	Conventions.	431
	Ambient category.	431
	Hypotheses.	431
	Functoriality.	431
	Unproved equivalences.	431
	Numerical comparison.	431
	Source and Convention Comparison: Wild-EG Decomposition Cross-Check . .	431
	Source separation.	431
	Conventions.	431
	Ambient category.	431
	Hypotheses.	432
	Functoriality.	432
	Unproved equivalences.	432
	Numerical comparison.	432
	Source and Convention Comparison: Unconditional Closure	432
	Source separation.	432
	Conventions.	432
	Ambient category.	432
	Hypotheses.	432
	Functoriality.	432
	Unproved equivalences.	432
	Numerical comparison.	432
46.10	Block Exhaustion Boundary	432
47	ARITHMETIC FUNDAMENTAL LOCAL EQUIVALENCE: SELMER-DERIVED TARGET AND OPEN CATEGORICAL EQUIVALENCES	435
47.1	Selmer-Derived Global Spectral Stack	435
47.2	Local Geometrisation and Its Domain	436
47.3	Conditional Categorical Equivalences	436
47.4	Proved and Conditional Components of the Arithmetic FLE	438
48	FI.A.1 UNRAMIFIED IDEAL, FI.A.2 SMALL-PRIME BLOCK EXTENSION, AND L_{ACD} CUSPIDAL DESCENT	439
48.1	Objects and Disjoint Sources	439
	Inherited Objects	439
	Primary Sources	439
	Disjointness Comparison	440

48.2	FI.a.1: The Unramified Ordinary Ideal	440
	Hypotheses and Target	440
	Statement	441
	Input 1: Breuil–Mézard on the Unramified Ordinary Locus	441
	Input 2: Clozel–Harris–Taylor on the Unramified Hecke Subalgebra	442
	Closure of FI.a.1 on the Unramified Ordinary Ideal	442
	Precise Unconditional Range	443
	Conjectural Extension	443
	Source Disjointness and Convention Comparison for FI.a.1	443
48.3	FI.a.2: Extension to $p = 2, 3$ via Block-Diagonal Local Comparison	444
	Strategy	444
	Block Structure at $p = 2$	444
	Block Structure at $p = 3$	445
	Block-Diagonal Local Comparison Functor	445
	Source Disjointness and Convention Comparison for FI.a.2	446
48.4	L_{ACD} : Cuspidal Adelic Categorical Descent	447
	Cuspidal Locus	447
	Flath’s Theorem at Representation Level	447
	Categorical Lift on the Cuspidal Locus	447
	Eisenstein / Continuous-Spectrum Extension as a Conditional Lemma	449
	Source Disjointness and Convention Comparison for L_{ACD}	450
48.5	Boundary of the FI Comparison	450
49	FI.A.3 (A+B): BLOCK-ADDITIVE HOCHSCHILD AND PASKUNAS Ext^1 -VANISHING	451
49.1	Block Decomposition Recall (Bernstein and BH)	451
	Bernstein Decomposition at $p \geq 5$	451
	BH Block Specialisation at $G = GL_2$	452
49.2	FI.a.3.A: Block-Additive BH Packaging	452
	Block-Additivity of Hochschild	452
	Block-Wise Comparison	453
	What Remains Conditional on Each Block	453
49.3	FI.a.3.B: Paskunas Block-Wise Ext^1 -Vanishing	454
	Paskunas Block Centraliser Structure	454
	Ext^1 -Vanishing in Generic Blocks	454
	Non-Generic Block: Controlled Deformation	455
	The $p \in \{2, 3\}$ Extension	455
49.4	Combined FI.a.3 (A+B) Residual Statement	456
	What Closes Unconditionally	456
	What Remains Conditional	456
	Obstruction Decomposition	457
49.5	Source Disjointness and Convention Comparison	457
	Block-Additivity: Source and Convention Comparison	457
	Source disjointness.	457
	Sign and convention.	458
	Ambient category.	458

	Hypothesis minimality.	458
	Functoriality.	458
	Per-Block Residual Isolation: Source and Convention Comparison	458
	Source disjointness.	458
	Convention alignment.	458
	Hypothesis.	458
	Generic-Block Ext ¹ -Vanishing: Source and Convention Comparison	458
	Source disjointness.	458
	Convention.	458
	Hypothesis minimality.	458
	Wild-Prime Block Centraliser: Source and Convention Comparison	459
	Source disjointness.	459
	Convention.	459
	Hypothesis.	459
	Ordinary-block controlled deformation.	459
49.6	Paskunas Block Boundary	459
49.7	Bibliography Stanza	459
50	COMPACT GENERATION OF $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p})$ AT WILD PRIMES VIA RESIDUAL GERBES AND SERRE TWISTS	461
50.1	Locally Noetherian Structure of $\mathcal{X}_{GL_2}^{\mathrm{EG},p}$	461
	Setup and Conventions	461
	Quasi-Excellence via Bhatt–Scholze Prisms	462
	Nilpotent Singular Support in the Arinkin–Gaitsgory Sense	462
50.2	Nilp Cone Decomposition at $p = 2, 3$	462
	The Stratification by Residual Reducibility	462
	Residual Gerbes at Closed Points of Each Stratum	463
50.3	Explicit Compact Generators per Stratum	463
	The Irreducible Stratum	463
	The Reducible Stratum	464
	The Wild Strata at $p = 2$ and $p = 3$	464
50.4	Compact Generation Theorem	465
50.5	Residual Obstruction Statement	465
	Domain of the Classical Result	465
	Domain Boundary: Derived Enhancement	466
50.6	Bibliography Stanza	466
51	PSEUDOCHARACTERS, DETERMINANTAL CLEAVAGES, AND RESTRICTED E_2 TEN- SOR PRODUCTS	469
51.1	(Ch ₁) Chenevier Pseudocharacter Compatibility	469
	Chenevier Determinant-Trace Pseudocharacter Recalled	469
	The Local–Global Comparison Datum	470
51.2	(AG.det) Determinantal Cleavage via GKRW E_2 -Centre	470
	The Determinantal Selmer-Derived Stack as Cleavage	470
	GKRW E_2 -Centre Controls on the Determinantal Cleavage	471

51.3	F2.c Restricted-Tensor E_2 Transport Lemma	472
	Setup	472
	Main Lemma	473
51.4	Combined BC-Diamond-Glue Statement	473
51.5	Residual Obstruction After Chenevier and Determinant Reduction	474
51.6	Bibliography Stanza	475
52	ELEMENTARY REDUCTIONS OF THE RIEMANN HYPOTHESIS	477
52.1	The Riemann Hypothesis in Classical Form	477
52.2	Four Classical Equivalents	478
52.3	The Arithmetic Reduction	478
	Step 1: The arithmetic Heisenberg Koszul pair	478
	Step 2: Zhu modularity at the archimedean fibre	479
	Step 3: Petersson positivity	479
	Step 4: Bost–Connes and the Hilbert–Pólya Boundary	480
	Step 5: Weil positivity via Mellin–Petersson isometry	480
52.4	The Restricted Test Class	481
52.5	Restricted-Class Comparison	482
52.6	The Full Weil Cone	482
52.7	Classical Boundary	483
53	ARITHMETIC POSITSIELSKI: FOUR RESIDUAL SUB-ITEMS	485
53.1	Continuous Duals	485
	Iwasawa Algebra and Pontryagin Dual	485
	The Continuous Dual of Λ_p	486
	Positselski Duality of Λ_p -mod and $C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)$ -comod	487
53.2	Derived Completion	488
	Greenlees–May Derived Completion	488
	Bounded Tor Hypothesis	488
53.3	Faithful Lifting	489
	Matlis Duality on the Compact Heart	489
	Lifting from Mod- p to Characteristic Zero	490
53.4	Koszul-Dual Bost–Connes	491
	Bost–Connes System	491
	Adele Class Space and Enveloping Sheaf	491
	Koszul-Dual Description	491
53.5	Combined Four-Sub-Item Theorem and Arithmetic Positselski Closure	493
	Combined Theorem	493
	Arithmetic Positselski Closure	493
	Residual Table	494
53.6	Bibliography	494
54	FULL $\mathcal{V}^{\text{prim}} P_3$ AND THE KOSZUL–PETERSSON COMPATIBILITY ON THE DIS- CRETE CORE	497
54.1	Recap: Tempered Cuspidal Closure via Tomita–Takesaki	497
54.2	Non-Tempered Cuspidal Representations: Deligne and Kim–Shahidi	498

	The Non-Tempered Locus	498
	The Holomorphic Sector: Unconditional	499
	The Maass Sector: Conditional on Ramanujan	499
54.3	The Eisenstein Density Mismatch and Domain Restriction	500
	The Mismatch	500
	Domain Restriction	501
	Sufficiency for Li's Criterion	501
54.4	Langlands–Shahidi Normalisation	502
	The Langlands–Shahidi Normalisation Factor	502
54.5	Full $\mathcal{V}^{\text{prim}} P_3$ on the Discrete Core	504
54.6	Koszul–Petersson Compatibility on the Domain-Restricted Discrete Core: Main Theorem	504
54.7	What Remains Conditional	506
	Conditional boundary.	506
	Eisenstein continuum.	506
	The sufficient RH path.	506
54.8	Bibliography	506
	Disjoint sources.	508
55	BOST–CONNES LOG-MODULAR SPECTRUM AND THE HILBERT–PÓLYA TWIST	509
55.1	Bost–Connes C^* -Dynamical Setup	509
55.2	GNS Representation and the Modular Operator	510
55.3	Arithmetic Heisenberg Fock and Its Embedding into the GNS Space	511
55.4	The Primary-Heisenberg Twist and the Determinant-Trace Operator	512
	Statement of the Twist	512
	Determinant-Trace Requirement: Hadamard Inversion	512
	The Hilbert–Pólya Hypothesis and Its Determinant-Trace Consequences	513
55.5	Determinant Identity: Hadamard Factorisation	513
55.6	Trace Distribution: Weil–Connes Explicit Formula	514
55.7	Unconditional Core and Spectral Twist	515
	Bost–Connes Core	515
	Consequences of the Twist	515
	Hilbert–Pólya Determinant-Trace Existence	515
	Operator Boundary	516
55.8	Bibliography and Source List	516
	Operator-algebra and KMS theory.	516
	Hilbert–Pólya candidates and determinant-trace realisations.	517
	Hadamard factorisation and explicit formula.	517
	Chiral algebra and arithmetic Fock background.	517
56	HP–LI POSITIVITY AND THE RANKIN–SELBERG KERNEL OBSTRUCTION	519
56.1	Weil's Explicit Formula (Recall)	519
	Paley–Wiener Test Class	519
	Li Coefficients as Weil Distribution Values	520
56.2	Petersson on the Tempered Cuspidal Core (Recall)	521

56.3	The Mellin Lift and Rankin–Selberg Kernel Check	521
	Definition of Υ	521
	Rankin–Selberg Integral and the Kernel Mismatch	522
56.4	The Prime-Side Obstruction	523
56.5	The Irreducible Obstruction: Archimedean Tate Distribution and Eisenstein	523
	Archimedean Tate Distribution	523
	Eisenstein Continuum and Non-Tempered Locus	524
56.6	Restricted-Class Theorem: HP–Li on the Holomorphic Cuspidal Tempered Class	525
56.7	The Archimedean Tate Positivity Conjecture	525
56.8	HP–Li Boundary	526
56.9	Beilinson Boundary	527
56.10	Bibliography of Primary Sources	527
57	VB* MAIN DIRECTION VIA HAMBURGER POSITIVITY, DETERMINACY, AND HB^+/BD^+	529
57.1	Spectral Positivity Context	529
	Domain	529
	Independent Sources	530
57.2	Primary Fock and $\xi_{\mathbb{R}}$ as Character	530
57.3	Hadamard Factorisation and the Zero Measure	532
57.4	Hamburger Moment Problem for μ_{ξ}	532
57.5	Hamburger Determinacy and Carleman	533
57.6	Hermite–Biehler and the de Branges Space $\mathcal{H}(E_{\xi})$	534
57.7	BD^+ : Beurling–Malliavin Super-Critical Density	535
57.8	Main Theorem: $(HB^+) + (BD^+) + (\text{Det}) \Rightarrow \text{RH}$	536
57.9	Analytic Hypotheses on $\xi_{\mathbb{R}}$	537
57.10	Bibliography Anchors	537
57.11	Spectral Boundary	538
58	R_{analytic} DENSITY-OF-ZEROS, WEIL AT GAUSSIANS, FORMAL-NOETHERIAN CLOSURES	541
58.1	Primary Sources	541
58.2	Riemann–von Mangoldt Density of Zeros	542
	Statement	542
	Proof	542
	Unconditional Bound on $S(T)$	544
	Consequence for R_{analytic}	544
58.3	Weil Explicit Formula at Gaussian Test Functions	545
	The Weil Explicit Formula: Recall	545
	The Gaussian Test Family	545
	Closed-Form Evaluation of $W(f_{\lambda})$	546
	Gaussian Convolutions and the Balanced Weil Functional	547
58.4	Formal-Noetherian Terms $R_{\text{formal.b}}$ and $R_{\text{formal.c}}$	548
	Iwasawa Algebra as Noetherian Local Ring	549
	Artin–Rees Lemma	549

	RI.formal.b via Artin–Rees	549
	RI.formal.c.2 via Artin–Rees	550
	Formal-Noetherian Consequence	551
58.5	Consequences	551
58.6	Bibliography	551
59	THE ARCHIMEDEAN TATE POSITIVITY REDUCTION	553
59.1	Residual Conditions Before Chapters 49–59	553
59.2	Theorem Record	554
	Chapter 49: FI.a.1, FI.a.2, and L_{ACD}	554
	Chapter 50: FI.a.3.A Block-Additivity and FI.a.3.B Paskunas	554
	Chapter 51: FI.a.3.C Wild-EG Compact Generation	554
	Chapter 52: Chenevier Compatibility, (AG.det), and F2.c	554
	Chapter 54: Arithmetic Positselski Equivalence	555
	Chapter 55: $V^{\text{prim}} P_3$ and Koszul–Petersson Compatibility	555
	Chapter 56: The Hilbert–Pólya Operator Skeleton	555
	Chapter 57: HP–Li Positivity and the Archimedean Residual	556
	Chapter 58: VB* Main Direction	556
	Chapter 59: Density of Zeros, Weil at Gaussians, and Formal-Noetherian Closures	556
59.3	Unified Obstruction Decomposition	556
	The RH-Equivalent Residual	558
59.4	The Refined Sufficient Path to RH	558
	Form of the Four Inputs After Chapters 49–59	558
	The Sufficient Path: Implication Diagram	558
59.5	A-TP and the Other Residuals	558
59.6	Disjoint-Source Comparison	560
	Primary Sources	560
	Independence Statement	561
59.7	Implication Form	561
	Implications	561
	Conjectural Conditions	562
59.8	Domain of Chapters 49–59	562
59.9	Further Obstructions	563
	Direct Positivity for A-TP	563
	Langlands–Shahidi Extension of L_{ACD}	563
	Paskunas Wild-Prime Class Match	563
	(AG.det.transition) Direct Obstruction	564
	Connes Absorption-Trace Realisation	564
	Hermite–Biehler Positivity on $\xi_{\mathbb{R}}$	564
59.10	Boundary	564
60	THE ARCHIMEDEAN TATE POSITIVITY CONJECTURE: A DIRECT ELEMENTARY OBSTRUCTION	567
60.1	Kernel and Obstruction	567
	Domain	567

Independent Sources	568
60.2 The Digamma Kernel $g(t)$	568
Hurwitz Series Representation	568
The Unique Sign Change t_*	569
60.3 Decomposition of the Archimedean Weil Contribution	570
60.4 Unconditional Archimedean Positivity on the High-Frequency Dominance Sub-Class	570
The Sub-Class $\text{PW}_F^{\text{hol,HFD}}$	571
An Explicit Sufficient Condition	572
60.5 Reduction to a Bounded-Interval Positivity Question	572
Bounded-Interval Reduction	572
Three Elementary Constructions, Ruled Out	573
60.6 The Irreducible Obstruction, Precisely Named	574
What Further Obstructions Remain Open	575
60.7 Direct Obstruction Boundary	575
61 WEIL–PETERSSON KERNEL COMPARISON: DENSE TEST-CLASS LIMITS	577
61.1 Restricted-Class Baseline and Residual	577
61.2 Construction I: Maass-Form Lift on Tempered Eigenforms	578
Archimedean Local Factor for Principal-Series Maass	578
61.3 Construction II: Weight $k \rightarrow \infty$ Limit Along Compatible Newforms	580
Compatible Newform Families	580
The Petersson Measure Degenerates at $k \rightarrow \infty$	580
61.4 Construction III: Residual-Spectrum Lift and One-Dimensional Absorption . .	581
The Residual Spectrum at $GL_2/\mathbb{Q}$	581
61.5 Construction IV: Jacquet–Piatetski-Shapiro–Shalika on $GL_n \times GL_m$	582
Higher-Rank Rankin–Selberg	582
61.6 The Union and its Schwartz Density	583
Schwartz Closure and the Residual Obstruction	584
61.7 The Residual Obstruction After the Extension	585
61.8 Domain and Positivity	586
61.9 Independence and Hypothesis Comparison	586
61.10 Isometry Boundary	587
61.11 Bibliography	588
Disjointness comparison.	589
62 LANGLANDS–SHAHIDI NORMALISATION OF THE EISENSTEIN KOSZUL DENSITY	591
62.1 Setup and the LS Normalisation	591
The Residual Mismatch	591
The Langlands–Shahidi Normalisation Factor	592
62.2 The Redefined Koszul Pairing on the Eisenstein Summand	592
62.3 Compatibility Checks	593
Compatibility Check (i): Heisenberg Self-Duality	593
Compatibility Check (ii): Tate Factorisation	594
Compatibility Check (iii): Discrete-Core Fixation	595
62.4 Full $\mathcal{V}^{\text{prim}}$ Koszul–Petersson Identity (Including Eisenstein)	595

62.5	Source Disjointness and Mathematical-Form Comparison	596
62.6	Sources	596
	Disjoint sources.	597
63	L_{ACD} ON THE EISENSTEIN / CONTINUOUS SPECTRUM	599
63.1	Domain, Disjointness, and Parallelism with the Langlands–Shahidi normalisation Inherited Objects	599 599
	Relation to the Langlands–Shahidi normalisation	599
	Primary Sources	600
	Disjointness Comparison	600
63.2	Module (E ₀): Langlands–Shahidi Intertwining Normalisation	601
	Local Intertwining Data for GL_2	601
	Local Functional Equation and Unitarity on the Axis	601
63.3	Module (E ₁): Paley–Wiener Synthesis on the Unitary Axis	602
	Shahidi’s Paley–Wiener Theorem	602
	Categorical Lift of the Paley–Wiener Space	603
63.4	Module (E ₂): Langlands Spectral Decomposition Lifted Categorically	605
	The Three Summands	605
	Categorical Lift of Each Summand	605
	Sphericity-Finite Compatibility on the Continuum	606
63.5	Module (E ₃): Assembly Compatible with Cuspidal LACD	606
	Full Statement	606
	Corollaries and Domain	607
	Compatibility with the Langlands–Shahidi normalisation	607
63.6	Comparison and Hypotheses	608
	Independent Inputs	608
	Derivation and comparison sources.	608
	Intertwining-operator normalisation.	608
	Categorical ambient structure.	608
	Hypotheses.	608
	Functoriality of the Weyl action.	608
	Open input at the exceptional locus.	608
	Numerical comparison.	609
	Composite with the cuspidal LACD comparison	609
63.7	Obstruction Decomposition After This Chapter	609
63.8	Eisenstein Boundary	609
64	NON-ORDINARY LOCUS AND REMOVAL OF THE TAYLOR–WILES HYPOTHESES FOR THE DEG EQUIVALENCE	611 611
64.1	Sources and Objects	611
	Inherited Objects and Notation	611
	Primary Sources	611
	Independence of Sources	612
64.2	Essential Surjectivity on the Non-Ordinary Locus at $p \geq 5$	613
	Structural Decomposition of the Non-Ordinary Locus	613
	The Potentially Barsotti–Tate Sub-Locus	613

The Supersingular Complement	614
Source Disjointness and Hypothesis Verification for the Non-Ordinary Closure	615
Source disjointness.	615
Conventions.	615
Categorical framework.	615
Hypotheses.	615
Hecke compatibility.	615
Supersingular comparison.	615
64.3 Removal of the Taylor–Wiles Conditions: Dihedral, Nearly-Ordinary, and Serre-Modular Sectors	616
Structural Partition of the Non-Taylor–Wiles Locus	616
The Solvable Projective-Image Sector	616
The Nearly-Ordinary Sector	617
The Serre-Modular Sector	618
Assembly on the Union of Non-Taylor–Wiles Sectors	619
The Non-Solvable Non-Nearly-Ordinary Complement	619
Independent Inputs for the Taylor–Wiles Sectors	620
Sources.	620
Conventions.	620
Categorical framework.	620
Local hypotheses.	620
Hecke compatibility.	620
Non-solvable comparison.	620
64.4 The Conditional Equivalence for $\Phi_{p,\chi}^{\text{DEG}}$	620
Union of the Strata	620
Boundary of the Non-Ordinary and Non-Taylor–Wiles Comparison	621
Interaction with the Full Essential Surjectivity Problem	621
64.5 Consequences	621
65 FI.A.2 CLASS-MATCH AT WILD PRIMES AND PASKUNAS BLOCK-CENTRALISER REGULARITY	623
65.1 Objects	623
The Three Boundary Conditions	623
Inherited Objects	623
Additional Inputs	623
Independence of Sources	624
65.2 FI.A.2 Class-Match at Wild Primes	624
The Class-Level Target	624
Galois-Deformation Functor on Non-Semisimple Blocks	625
Residual Gerbe Comparison	625
Class-Match Theorem	626
65.3 R.B.1: Very-Special Ordinary-Block Multiplicity	627
The Paskunas 2015 Residual	627
65.4 R.B.2: Wild-Prime Block Centraliser Regularity	627
$p = 2$ centraliser datum	627

Colmez–Dospinescu–Paskunas at $p = 3$	627
Unified Wild-Prime Block Centraliser Theorem	628
65.5 Consequences for Fl.a.3	628
Conditional Boundary	628
65.6 Sources and Conventions	629
Class-Match Sources and Conventions	629
R.B.1 Conventions	629
R.B.2 Conventions	629
65.7 Consequences	630
65.8 Bibliography Stanza	630
66 CODIMENSION-TWO E_2 -CENTRE GLUE AND THE BC-DIAMOND-GLUE COMPARISON	631
66.1 Inputs and Sources	631
Inherited Equivalences	631
Two Centre Comparisons	631
Independent Sources	631
66.2 Target G.1: (AG.det.transition) on the Codim- ≤ 2 Locus	632
Geometry of the Transition Locus	632
GKRW 2-Convergence at Codim ≤ 2	633
Prismatic–Nygaard 2-Convergence	633
Bezrukavnikov–Finkelberg–Mirković Equivariant Cohomology on Flat Locus	633
The (AG.det.transition) Implication	633
66.3 Target G.2: BC-Diamond-Glue Final Closure	635
Chenevier–Bellaïche Bernstein–Galois Decomposition	635
Francis–Gaitsgory Factorization Gluing	635
Gluing Across Prismatic Charts and Untilts	636
66.4 Conditional Locus	638
66.5 Independent Source Data	638
66.6 Bibliography Stanza	638
66.7 Consequences	639
67 INDEPENDENT NUMERICAL COMPARISONS FOR ARITHMETIC CONSTANTS	641
67.1 Criterion and Source Discipline	641
67.2 The Five Constants and Their Independent Comparisons	641
$r_{\zeta} = 2$: determinant-rank residue of 2ζ	641
Li coefficients $\lambda_1, \dots, \lambda_4$	642
Riemann–von Mangoldt $N(T)$ at $T = 100, T = 1000$	643
First non-trivial zeros $\gamma_1, \gamma_2, \gamma_3$	643
Archimedean digamma $\psi(1/4)$	643
67.3 Numerical List	644
67.4 Position in the Arithmetic Branch	644
67.5 Numerical Boundary	645
68 ARITHMETIC FIBRE AND SOURCE ORTHOGONALITY	647
68.1 The Fibre Dichotomy	647

Primary-Source Test	647
Classification	647
68.2 Three Basic Comparisons	648
Arithmetic Positselski and Vol I Theorem B	648
Bernstein Block-Additivity and the Vol I Climax	648
Chenevier–Bernstein Transport and E_2 -Recognition	648
68.3 Further Fibre Comparisons	649
Bost–Connes Modular Operator and Costello–Gwilliam KMS	649
Koszul–Petersson Discrete Core and the Chiral Climax	649
HP–Li Paley–Wiener and Theorem H	649
de Branges Positivity and the Phi Functor	649
Riemann–von Mangoldt Density and BD Acyclicity	650
68.4 Native Arithmetic Theorems	650
Adelic categorical descent	650
Wild-prime compact generation	650
68.5 Primary-Source Matrix	650
68.6 Fibre Theorem	650
69 INDEPENDENT COMPARISON OF CHAPTERS 49–59	653
69.1 The Eleven Local Comparisons	653
Chapter 49: $F_{1.a.1}$, $F_{1.a.2}$, and L_{ACD}	653
Chapter 50: $F_{1.a.3}(A+B)$	653
Chapter 51: $F_{1.a.3.C}$	653
Chapter 52: Chenevier and $F_{2.c}$	654
Chapter 53: The Elementary Closure	654
Chapter 54: Arithmetic Positselski	654
Chapter 55: Koszul–Petersson	654
Chapter 56: $H\text{-}P^{Ar}$	654
Chapter 57: HP–Li Positivity	654
Chapter 58: VB^*	654
Chapter 59: $R_{analytic}$	655
69.2 The Three Conditional Points	655
69.3 Conclusion	655
70 ARCHIMEDEAN TATE KERNEL VIA RIEMANN–SIEGEL THETA AND STIRLING- CERTIFIED BOUNDS	657
70.1 Context	657
Domain	657
Independent Sources	658
70.2 The Exact Identity $g = 2\theta'_{RS} + 2 \log \pi$	658
70.3 Stirling-4 Certified Bounds	659
The Stirling-4 Formula	659
The Certified Positivity Threshold	660
70.4 The Enlarged Sub-Class $PW_F^{hol,SHF}$	661
The Refined Weight	662

	Non-Emptiness and Density	663
70.5	Z-Function Reformulation of the Bounded-Interval Residual	663
	The Phase-Derivative Formulation	663
	The θ''_{RS} Structural Observation	664
70.6	The Stirling-Certified Irreducible Obstruction	664
70.7	What Remains Open	665
70.8	Stirling Boundary	666
71	A-TP VIA NYMAN–BEURLING CLOSURE AND THE BAEZ-DUARTE REFINEMENT	667
71.1	Primary Sources	667
71.2	The Nyman–Beurling Space $H^2(o, 1)$	668
	Generators and Mellin Transform	668
	The Nyman–Beurling Theorem (Recall)	669
71.3	Mellin Normal Form and RH-Level Comparison	669
	The Mellin Norm Identity	670
	The Weil-Type Insertion	670
71.4	Restricted-Class Estimate on HFD	671
	HFD-Adapted Nyman–Beurling Distance	671
71.5	Möbius Partial-Sum Control	672
71.6	Comparison of the Complementary Class with Baez-Duarte	672
71.7	Seven Obstructions to the HFD Nyman–Beurling Bound	673
71.8	The Irreducible Obstruction, Named in the Nyman–Beurling Framework	674
71.9	Mellin Boundary	674
72	A-TP VIA DE BRANGES HERMITE–BIEHLER FACTORISATION	677
72.1	Context and the de Branges Comparison	677
	Domain	678
	Independent Sources	678
72.2	The de Branges Function E_ξ and Its Real-Line Trace	679
72.3	de Branges Reproducing Kernel and the Mellin Identification	680
72.4	What the Vinogradov–Korobov Region Actually Gives	681
72.5	The Genuine Obstruction: Hermite–Biehler Extension	683
72.6	Three Non-Elementary Constructions Further Analysed	683
72.7	de Branges Form of A-TP	684
72.8	Consequences	685
72.9	Bibliography Anchors	686
73	ARCHIMEDEAN TATE KERNEL VIA GAUSSIAN TEST FUNCTIONS: SCALAR REDUCTION AND DENSITY LIMITS	687
73.1	Inputs and Reduction Strategy	687
	Normalisation Convention	687
	Independent Sources	688
73.2	Gaussian Convolution and the Scalar Reduction	688
	Convolution Algebra on Gaussians	688
73.3	Erfc Series for $I(\lambda)$	689
	Hurwitz Expansion of g	689

	Closed Form of $J_n(\lambda)$	690
	Assembly of the Series	690
73.4	Asymptotic Analysis of $I(\lambda)$	691
	The Narrow-Frequency Limit $\lambda \rightarrow 0$	691
	The Wide-Frequency Limit $\lambda \rightarrow \infty$	691
	Critical λ_c	691
73.5	W_∞ -Positivity on the Gaussian Sub-Class for $\lambda \geq \lambda_c$	692
73.6	No Gaussian Density Extension: Boundary of the Positive Cone	693
	Continuity of the Weil Archimedean Functional	693
	Boundary of the Density Extension	694
	Sub-Cone Closure	694
	Comparison with the bounded-interval reduction HFD Sub-Class	695
73.7	The Irreducible Obstruction, Refined	695
73.8	The Gaussian W_∞ -Positivity Theorem: Results	696
73.9	Bibliography	696
74	ARCHIMEDEAN TATE KERNEL POSITIVITY: THE INTEGRAL ENERGY SUB-CLASS	699
74.1	Kernel and Integral Energy	699
	Domain	699
	Independent Sources	700
74.2	A Sharp Stirling Lower Bound for the Archimedean Kernel	700
	Stirling Series for the Digamma and Sign Control	700
	The Lower Bound g_L	701
	Sign-Domain Boundary	701
74.3	The Integral-Energy Sub-Class	702
	Energy Fractions	702
	Integral-Energy Sub-Class $PW_F^{\text{hol,IE}}$	702
	Unconditional W_∞ -Positivity on $PW_F^{\text{hol,IE}}$	702
	Strict Enlargement over the bounded-interval reduction HFD	703
74.4	What Remains Outside $PW_F^{\text{hol,IE}}$	703
	Complement Structure	703
	Prolate-Spheroidal Characterization	704
	Absolute Uniform Closure Sub-Class	705
74.5	Quantification of the Archimedean Positivity Extension	705
74.6	Compatibility with Chapters 61, 57, and the Bounded-Interval Reduction	706
	IE Class and Chapter 61 Bounded-Interval Reduction	706
	Three Non-Elementary Constructions Revisited	706
74.7	Bounded-Direct Boundary	706
75	ARCHIMEDEAN TATE KERNEL VIA MELLIN–BARNES AND BINET CONTOUR REPRESENTATIONS	709
75.1	Contour Reductions	709
	Two Contour Representations of ψ	709
	Contour Domain	710
	Independent Sources	710

75.2	The Binet Decomposition of the Archimedean Kernel	711
	Insertion of the Binet Representation	711
	Sign of $g_{\text{Binet}}(t)$: Two-Region Structure	711
75.3	The Hankel Integral Construction	712
	Hankel Representation of ψ at Argument $1/4 + it/2$	712
	Laplace Representation of the Archimedean Weil Contribution	713
	The Hankel Dominance Sub-Class	713
75.4	The Mellin–Barnes Construction: Structural Obstruction	714
	Insertion and Order Exchange	714
	Residue Sum: Alternation Obstruction	715
75.5	The Irreducible Obstruction After the Mellin–Barnes reduction	716
75.6	Mellin–Barnes Boundary	717
76	A-TP VIA THE EXACT HADAMARD FACTORISATION: REFLECTED RESOLVENTS AND TAUTOLOGY COMPARISON	719
76.1	Hadamard Factorisation	719
	Setup	719
	Classical Sources	719
76.2	The Exact Hadamard Formula for $\xi'_{\mathbb{R}}/\xi_{\mathbb{R}}$	720
	Hadamard Product Logarithmic Derivative	720
	Comparison with the Completed Zeta Formula	720
76.3	The Reflected Pair-Pole Diagnostic	721
	Pairing Conjugate Zeros	721
	Residue Extraction and Dirac Limits	721
	Under RH: No Termwise Positivity	722
76.4	Tautology Comparison: Where Positivity Becomes Circular	722
76.5	de Bruijn–Newman $\Lambda \leq 0$ is RH	723
	The de Bruijn–Newman Family	723
	$\Lambda \leq 0$ Implies A-TP via RH	723
76.6	Zero Density and Pair-Pole Convergence	724
	Convergence of the Pair-Pole Sum	724
	Exact Domain of the Hadamard-Factorisation Construction	724
76.7	Conclusion	725
77	A-TP AS GRAM MATRIX POSITIVITY: HERMITE BASIS REDUCTION	727
77.1	Hermite Cutoff	727
	Quadratic Reduction	727
	Independent Sources	727
77.2	The Quadratic Form and Its Operator Kernel	728
	Weil Distribution as Quadratic Form	728
	The Operator Q on $\text{PW}(\mathbb{R})$	728
77.3	The Weber–Hermite Basis and Gram Matrix	729
	Weber–Hermite Basis	729
	The Gram Matrix Q_{mn}	729
	Explicit Archimedean Diagonal	730

77.4	Hadamard Factorization and the Zero-Sum Decomposition	731
77.5	Finite- N Truncation and Bombieri–Lagarias Numerical Content	732
	The Uniform Bound Obstruction	733
77.6	Three Natural Regularizations and Their Failure	733
	Tikhonov Regularization	733
	Gaussian Windowing	733
	Finite-Rank Perturbation Ceiling	734
77.7	The Irreducible Obstruction	734
77.8	Gram Boundary	735
78	A-TP THROUGH THE GUINAND–BURNOL CO-POISSON ALTERNATIVE	737
78.1	A-TP and the Guinand–Burnol Axis	737
	Domain	737
	Independent Sources	738
78.2	The Guinand Alternative Explicit Formula	738
	Symmetric Contour Derivation	738
	Weil vs. Guinand: Sign Convention	739
78.3	The Burnol Co-Poisson Transform	740
	Jacobi Theta and Co-Poisson	740
	The Dual Time-Domain Kernel $\tilde{g}$	741
78.4	Co-Poisson Diagnostic and Exact Restatement	741
	Hilbert Transform on Even Functions	742
	The Co-Poisson Sub-Class PW^{CP}	742
	Hilbert-Transform Sufficient Condition	743
78.5	The Burnol Residual	743
78.6	The Guinand–Burnol A-TP Theorem	745
79	A-TP VIA ADELIC SCATTERING: BOST–CONNES S -MATRIX AND MELLIN-UNITARY CLOSURE	747
79.1	Scattering Domain	747
	Independent Sources	747
79.2	Adelic Scattering Framework	748
	The Idele Class Hilbert Space	748
	The Bost–Connes Scattering Generator	748
	Wave Operators and the S -Matrix	749
79.3	Scattering-Matrix Reformulation of A-TP	750
	Birman–Krein Spectral Shift	750
	Identification of the Weil Distribution with ξ_{BK}	750
	A-TP as Unitarity of S	751
79.4	Tate-Mellin Phase Calculation on the Sub-Class	751
	The Tate-Mellin Sub-Class	752
	Classical Mellin-Unitary on PW^{TM}	752
79.5	Non-Commutative-Geometric Residual Located Precisely	753
	Where the Connes Reduction Enters	753
	Comparison with the Bounded-Interval Reduction	754

79.6	Equivalence Theorem and Mathematical Form of Each Component	754
80	ALL-GENUS ARITHMETIC KOSZUL DUALITY	757
80.1	The Arithmetic Genus- g Datum	757
80.2	Six Convergent Windows	758
	Window I: Arakelov fibration	758
	Window II: Minkowski torus / Bost–Connes–Marcolli	758
	Window III: Ordinary crystalline Emerton–Gee stack	758
	Window IV: Hodge–Arakelov / arithmetic Teichmüller	759
	Window V: E_2 -chiral factorisation on the arithmetic surface	759
	Window VI: Kudla–Rapoport / Shimura–Siegel Shimura variety	759
80.3	Falsified Naive Readings	759
80.4	All-Genus Arithmetic Koszul Duality	760
80.5	Reduction to $g = 0$ and the F-RH Axis	761
80.6	Residual Conditions	762
80.7	Orthogonal and Compatible Windows	763
81	3D ARITHMETIC CHERN-SIMONS ON ETALE SPEC($\mathbb{Z}$) AND THE ORDERED INCIDENCE BAR	765
81.1	Spec($\mathbb{Z}$) as Etale 3-Space	765
81.2	Kim Classical Arithmetic Chern-Simons	765
81.3	The Arithmetic Kac–Moody Lift	766
81.4	3D Quantum Arithmetic Chern-Simons	766
81.5	The Ordered Incidence Bar	770
81.6	Deninger Skeleton from Arithmetic Chiral Homology	772
81.7	Arithmetic-Gravity Construction to F-RH	774
81.8	Genus- g Extension	775
81.9	Residual Conditions	775
81.10	Ordered-Bar Boundary	776
82	DENINGER’S ARITHMETIC COHOMOLOGY AS A CHIRAL CANDIDATE	777
82.1	Deninger’s Wish List	777
82.2	Surviving Inputs	778
82.3	The Chiral Candidate	778
82.4	No Triple Identification Yet	779
82.5	Frobenius and Flow	779
82.6	Character, Degrees, and Purity	780
82.7	Duality and Positivity	781
82.8	Purity as the RH-Level Input	781
82.9	All-Genus Extension	781
82.10	Dictionary	782
82.11	What Closes	782
82.12	Residual Conditions	782
83	TRACE FORMULA, ORDERED BAR, AND THE STATISTICAL UPSIDE	783
83.1	The Trace Formula Target	783

83.2	Connes' Trace Formula	783
83.3	What the Ordered Bar Proves	784
83.4	Trace-Formula Equivalence	784
83.5	The Bar Reorganises, Not Locates	785
83.6	Arithmetic Gutzwiller	785
83.7	Connes–Meyer and the Gutzwiller Gap	786
83.8	Statistical Upside	786
84	THE WILSON LINE–FROBENIUS IDENTITY ON $\overline{\text{Spec } \mathbb{Z}}$	789
84.1	Setup and Conventions	789
84.2	Classical Wilson Loops (Kim ACS)	790
84.3	Quantum Extension Framework (Ch. 81 Conditional)	791
84.4	Morishita Functoriality	791
84.5	Main Theorem	792
84.6	Local Langlands Compatibility	793
84.7	Residual Obstructions	794
84.8	Wilson–Frobenius Boundary	795
85	PRIMARY FOCK DETERMINANT: RANK RESIDUE 2	797
85.1	Local Virasoro and Arithmetic Hamiltonian	797
	The per-prime chiral Heisenberg Fock	797
	Per-prime central charge	797
85.2	Divergent Sum and Per-State Finiteness	798
	The naive global sum diverges	798
	The primary core	798
	Finite Positive Modes and Divergent Negative Modes	798
85.3	Dirichlet-Series Regularisation	799
	The naive prime-zeta attempt fails	799
	Arithmetic black-hole motivation: primon gas as the first determinant	799
	The partition function regularisation	800
	The Dirichlet-series rank residue	801
85.4	Local Expansion at $s = 1$	801
	Decomposition of $\log Z$ near $s = 1$	801
	Residue extraction	802
85.5	Polyakov Anomaly Connection	802
	The Polyakov stress tensor and anomaly	802
	The Polyakov–regularisation identity	803
	Derivation and Residue Sources	803
85.6	Main Theorem: $r_\zeta = 2$	804
	Statement	804
	Falsification and correction of six potential objections	804
	Relation to the conditional Virasoro-bracket question	805
	Domain and logical form	805
85.7	Bibliography	805

86	THE SPECTRAL FLOW $N_{\text{BC}} \rightsquigarrow H_{\text{HP}}$: IRREDUCIBLE OBSTRUCTION AT THE ARCHIMEDEAN PLACE	809
86.1	Setup: N_{BC} and Θ_ξ as Target Operators	809
	Source: The Bost–Connes Log-Prime Operator	809
	Target: The Hypothetical Zeta-Zero Divisor Operator	809
	Spectral Flow as Intertwining Operator	810
86.2	Prime-Zero Mangoldt-Hadamard Duality Recap	811
	Hadamard Factorisation of $\xi_{\mathbb{R}}$	811
	Riemann-von Mangoldt Explicit Formula	811
	Mellin-Dual Measures	811
86.3	Formal Spectral Flow $\mathcal{A}$ as Mellin-Inversion Symbol	812
	The Mellin-Hadamard Inversion Symbol	812
	Formal Mellin–Hadamard Obstruction	813
86.4	Obstacle 1: Hilbert-Space Completions on Both Sides	813
	Source-Side Completion	813
	Target-Side Completion	814
86.5	Obstacle 2: Kernel of $\mathcal{A}$ as Bounded/Unbounded Operator	814
	The Hadamard Inversion Kernel	814
	The Domain Problem	815
86.6	Connes 1999 Partial Approach and Its Limitation	815
	The Connes Construction	816
	The Limitation: Absorption Trace as RH	816
86.7	Meyer 2005 Partial Approach and Its Limitation	816
	The Meyer Construction	816
	The Limitation: Pole Data Matching	817
86.8	Residual: Irreducible Operator-Theoretic Obstruction	817
	Statement of the Irreducible Obstruction	817
	Spectral-Flow Logical Table	819
	Why the Residual is Irreducible	819
86.9	Placement in Triple-Identification Framework	820
	Spectral Flow as Deninger Spectral-Flow Theorem	820
	Spectral Flow as A-TP Adelic-Scattering	820
	Spectral Flow as 3D ACS Bulk-Boundary Correspondence	821
	The Unified Residual	821
86.10	Polyakov–Dirac Normal Form for the Residual	822
	Polyakov Rigidity of the Rank Residue	822
	Dirac Closure of the Remaining Operator	823
	What Remains: Boundary Form and Deficiency Kernels	825
	The Deficiency Calculation	826
	Execution Attempt: Graph Norm Versus Determinant	830
86.11	Bibliography	831
	Hilbert–Pólya foundational literature.	832
	Adelic and idele-class operator constructions.	832
	Mellin and Hadamard classical analysis.	832

	Bost–Connes and GNS modular theory.	832
	First-order and Dirac normal-form sources.	833
	Vol IV earlier sources used here.	833
	Disjointness comparison.	833
87	FUNCTORIALITY OF THE TRIPLE IDENTIFICATION IN A AND g	835
87.1	The Category of Arithmetic Chiral Algebras	835
87.2	The Three Functors	837
87.3	Chain-Level Identification	838
87.4	Functoriality in A	840
87.5	Functoriality in g	841
87.6	Master Theorem and Explicit Example	843
87.7	Domain of the Master Theorem	846
	Functoriality residuals of Chapter 82 closed here	846
	Inherited conditional hypotheses of the master theorem	846
	The master theorem and what it forces	846
88	WILSON-LINE FROBENIUS AND ARCHIMEDEAN WILSON-SURFACE AT GENUS g	849
88.1	Genus- g Arithmetic Data Recap	849
88.2	Genus- g 3D Arithmetic Chern–Simons Theory	850
88.3	Wilson-Line Frobenius Identity at Genus g	851
88.4	Archimedean Wilson-Surface at Genus g	852
88.5	Global Frobenius at Genus g and the Deninger Identification	854
88.6	Uniform $g = 0$ Compatibility	856
88.7	Residual Conditions	857
88.8	Primary Sources for the Genus- g Extension	858
89	THE ARCHIMEDEAN WILSON-SURFACE OPERATOR F_∞	859
89.1	Arakelov Archimedean Structure	859
89.2	Wilson-Surface Observables in 3D Chern-Simons	860
	3D Chern-Simons and Wilson-Surface Observables (Witten 1989)	861
	The Archimedean Wilson-Surface	861
	Source Comparison for the Wilson-Surface Construction	862
89.3	Construction of F_∞	862
	Convention and Source Comparison for the Construction	863
89.4	Conjugacy to Virasoro L_0^{Ar}	863
	The Weil–Petersson isometry Υ_∞	863
	The conjugacy theorem	864
	Convention and Source Comparison for the Conjugacy	864
89.5	Spectral Identification with $\xi_{\mathbb{R}}$ -zeros	865
	L_0^{Ar} on the primary Fock and the ξ -image	865
	F_∞ determinant as the exponential zero image	865
	Convention and Source Comparison for the Spectral Identification	866
89.6	Archimedean Purity is Equivalent to A-TP	866
	The $ F_\infty $ -weight = 1/2 statement	866
	A-TP archimedean purity equivalence	866

	Convention and Source Comparison for the Purity Equivalence	867
89.7	Residual Obstructions	867
89.8	Bibliography	868
90	FIVE SOURCE FAMILIES AND THE ARCHIMEDEAN TATE POSITIVITY RESIDUAL	871
90.1	A-TP: Statement and Independent Obstruction	871
90.2	Five Source Families	872
90.3	Two Conditional A-TP Diagnostics	872
	Quant-A-TP-RS on the Rudnick-Sarnak window	872
	Arakelov-Heegner diagnostic on PW_F^{Heeg}	873
90.4	Two Negative Results: Iwasawa Auxiliary Structure and BGS Implication Direction	873
	Iwasawa-Kato Orthogonality to A-TP	873
	BGS Is Consequent of RH	874
90.5	The Weil Cone	874
90.6	Position within the A-TP Obstruction Sequence	874
90.7	Boundary	875
91	PASKUNAS WILD BLOCK Ext^1 -CLASS MATCH AT $p \in \{2, 3\}$	877
91.1	Domain and Disjoint Derivation	877
	The Residual after Chapter 65.2.9	877
	Disjoint Proof Sources	877
	Disjointness Comparison	878
91.2	Block Centraliser Pseudo-Character Tangent	878
	Pseudo-Character Tangent Space at Wild Primes	878
	Class-Level Identification on Each Wild Block	879
91.3	Wild-Prime Class-Match Theorem	879
91.4	Derivation Comparison	880
	Source separation.	880
	Sign and convention.	880
	Ambient category.	880
	Hypotheses.	881
	Functoriality.	881
	Unproved equivalences.	881
	Numerical comparison.	881
91.5	Wild Class Boundary	881
91.6	Sources	881
92	RAMANUJAN FOR MAASS FORMS: SHARPENING $7/64$ AND A SUB-RH CONDI- TIONAL	883
92.1	Domain and Target	883
92.2	Construction A: Kim–Shahidi Symmetric-Cube and the $1/9$ Baseline	883
	The Symmetric-Cube Lift	883
	The Sym^4 Barrier and the Jacquet–Shalika Mechanism	884
92.3	Construction B: Kim 2003 Sym^4 and the $7/64$ Baseline	884
92.4	Construction C: Newton–Thorne Symmetric-Power Functoriality and the Sharp- ening	884

	The Newton–Thorne Theorem	884
	The Auxiliary Holomorphic Form and the Transfer	885
	The Sharpened Bound: Construction C Main Theorem	885
92.5	Construction D: Arthur Endoscopic Classification on Classical Groups	886
	Arthur’s Endoscopic Classification	886
	The Ramakrishnan Functorial Chain	887
	The Automorphic Conditional	887
92.6	Construction E: Synthesis	888
92.7	Independent Comparison	888
92.8	What Remains Open	889
	GRC for $GL_4/\mathbb{Q}$	889
	Maass symmetric-power functoriality (Construction C conditional).	889
	The dihedral edge case.	890
92.9	Bibliography	890
	Disjoint-source discipline.	891
93	QUANTUM ARITHMETIC CHERN–SIMONS: BULK EXTENSION AND RESIDUAL OBSTRUCTION	893
93.1	Setup and Axiomatic Target	893
93.2	Construction I: Tower Renormalisation via Kan Extension	894
	Finite-level Kan tower	894
	Sources and Hypotheses for Construction I	895
93.3	Construction II: Reshetikhin–Turaev from an Arithmetic Boundary Category	895
	The arithmetic MTC candidate	896
	Reshetikhin–Turaev state-sum	896
	Partition identification via Zhu modularity	897
	Sources and Hypotheses for Construction II	897
93.4	Construction III: Costello–Gwilliam Factorization on $(\mathrm{Spec} \mathbb{Z})_{\text{ét}}$	897
	Étale factorization algebra	897
	Archimedean extension	898
	Sources and Hypotheses for Construction III	898
93.5	Construction IV: Cobordism Hypothesis	899
	Fully dualizable object candidate	899
	Sources and Hypotheses for Construction IV	900
93.6	Construction V: Six-Functor Formalism on an Arithmetic Diamond	900
	Diamond of $\overline{\mathrm{Spec} \mathbb{Z}}$	900
	Six-functor ACS	900
	Sources and Hypotheses for Construction V	901
93.7	Construction Comparison and Isolation of the Residual	901
	The Common Archimedean Term	901
	Relation to the Hilbert–Pólya/Connes–Deninger obstruction	902
93.8	Refinement: The Condensed-Hilbert Bulk	903
	Condensed-Hilbert setup	903
	The Condensed Hilbert–Trace Package	903
93.9	Wilson-Observable Compatibility	904

93.10	Comparison of the Five Presentations	904
93.11	Remaining Inputs	905
93.12	What Closes and What Remains	905
93.13	Convergence	906
93.14	Bibliography	906
94	THE BERNSTEIN-CENTRE LOCAL-GLUING COMPARISON: BERNSTEIN-CENTRE (BCGLUE.LOCAL) + (BCGLUE.GLUE) RESIDUAL REDUCTION	909
94.1	Domain and Disjoint-Source Discipline	909
	The Bernstein-centre transition Residual Restated	909
	Independent Sources	910
	Local Obstructions	910
94.2	(E.1) (BCglue.local) on the Tame Odd Locus	911
	Fargues–Scholze Spectral Action Recalled	911
	Dotto–Emerton–Gee at $p \geq 5$	912
94.3	(E.2) (BCglue.local) on the Wild Locus $v \in \{2, 3\}$	912
	The Wild Residual	912
94.4	(E.3) (BCglue.glue) Factorisation	913
	Discrete-Base Factorization Homology	913
	Bellaïche–Chenevier Global Bernstein–Galois Decomposition	914
	(BCglue.glue) Assembly	914
94.5	(E.4) (BCglue.local) + (BCglue.glue)	915
94.6	Consequences for BC-Diamond-Glue	916
94.7	Independent Source Data	917
94.8	Seven Obstruction-theoretic Conditions and Their Resolution	917
94.9	Primary Sources	918
94.10	Residual Closure Boundary	919
95	FI.A.1 RESIDUAL REDUCTION: SUPERSINGULAR CORE AND A_5 ICOSAHERAL CORE	921
95.1	Domain, Inputs, and Disjointness	921
	Inherited Structure	921
	Primary Sources	921
	Disjointness Comparison	922
95.2	FI.a.1.ss-core: Supersingular Non-pBT Locus	922
	The Kisin Stratification of the Supersingular Core	922
	Essential Surjectivity on the Fontaine–Laffaille Stratum	924
	Proof of Theorem 95.2.6	925
	Source disjointness.	925
	Sign and convention consistency.	925
	Ambient category check.	925
	Hypothesis domain.	925
	Hecke-action compatibility.	925
	Inputs.	925
	Numerical comparison.	925

95.3	FI.a.1.ns-core: A_5 Icosahedral Sector	926
	Structural Decomposition of the Icosahedral Core	926
	Reduction on the Odd-Determinant Icosahedral Sub-Sector	926
	Reduction of the Residual to the Local–Global Defect	927
	Proof of Theorem 95.3.6	928
	Source disjointness.	928
	Sign and convention consistency.	928
	Ambient category check.	928
	Hypothesis domain.	928
	Hecke-action compatibility.	928
	Inputs.	929
	Numerical comparison.	929
95.4	Combined FI.a.1 Conditional Reduction on Five Strata	929
96	L_{ACD} EXCEPTIONAL-CROSSING Ext^1 -CLASS MATCH AT $\Sigma^{\text{exc}} = \{0, 1\}$	931
96.1	Objects and Sources	931
	Inherited Objects	931
	Primary Sources	932
	Parallelism with FI.a.2.class-match	932
	Independence of Sources	933
96.2	Module (X_0): Two-Point Residue Data on the Automorphic Side	933
	The Residual Spectrum at $s = 1$	933
	The Spherical Reducibility at $s = 0$	934
	Functional-Equation Duality between $s = 0$ and $s = 1$	934
96.3	Module (X_1): Local and Global Ext^1 -Classes from Residues	934
	The Local Ext^1 -Class at a First-Order Pole	934
	Adelic Assembly via Tate Factorisation	935
96.4	Module (X_2): Spectral Ext^1 -Class via Arthur Endoscopy	935
	The Trivial Arthur Packet at $s = 1$	935
	Tangent of the Emerton–Gee Stack at the Arthur Point	936
96.5	Module (X_3): Class-Level Match via LS-Corrected Koszul Pairing	937
	The Match Target	937
	Koszul Residue and the Class-Level Match	937
	Exceptional Eisenstein Reduction for L_{ACD}	938
96.6	Compatibilities for Theorem 96.5.4	939
	Seven Compatibilities	939
	Sources.	939
	Sign.	939
	Categories.	939
	Hypotheses.	939
	Functoriality.	939
	Theorems Used.	940
	No Arithmetic Computation.	940
	Interaction with FI.a.2 Class-Match	940
	Conditional Inputs	940

96.7	Fi Conditions After This Chapter	940
96.8	Consequences	941
96.9	Sources	941
	Disjoint-source discipline.	942
97	RESIDUAL BOUNDARY OF THE ARITHMETIC BRANCH	945
97.1	Condensed-Hilbert Descent and Archimedean Tate Positivity	945
97.2	The RH-Equivalent Obstruction	945
97.3	Comparison and Descent Conjectures	946
97.4	The Generalised Ramanujan Conjecture for GL_4	947
97.5	$F_{1,a,1}$ Small Residuals	947
97.6	Separation from the Realization Formalism	947
97.7	Residual Boundary	948
A	PRIMARY SOURCES	949
A.1	Kapranov and Kapranov–Smirnov	949
A.2	Borger and Λ -Ring Theory	949
A.3	Bost–Connes, Connes–Marcolli, Connes–Consani	950
A.4	Kontsevich, Soibelman, Davison: E_n Formality and CoHA	950
A.5	Beilinson–Drinfeld, Francis–Gaitsgory, Positselski	950
A.6	Arithmetic $\mathcal{D}^\dagger$ -Modules and F -Isocrystals	951
A.7	Costello, Costello–Gwilliam, Costello–Gaiotto	951
A.8	Geometric Langlands, Derived Satake, Hecke Categories	952
A.9	FLE, Quantum Geometric Langlands, and Root-of-Unity Quantum Groups	952
A.10	Kim’s Arithmetic Chern–Simons	953
A.11	Hartnoll–Yang and Primon Gas	953
A.12	Macdonald, DAHA, Schiffmann–Vasserot	953
A.13	Nekrasov, AGT, Maulik–Okounkov	954
A.14	Segal, Frenkel–Ben-Zvi	954
A.15	Tate’s Thesis and Adelic Analysis	955
A.16	Drinfeld, Molev: Yangians and R-Matrices	955
A.17	Arakelov, Deligne, Arithmetic Height Pairings	955
A.18	Kesten–McKay	956
A.19	Prismatic Cohomology and Fargues–Fontaine	956
A.20	Ben-Zvi, Nadler, Coherent Springer Theory	956
A.21	Tate, Weil, Explicit Formula	957
A.22	Li Positivity and Weil Positivity	957
A.23	Positselski, Iwasawa	957
A.24	Operator Theory: Moment Problem, Self-Adjointness	958
A.25	Restricted Direct Product, Flath, Godement–Jacquet	958
A.26	Cross-Volume Primary Sources	958
B	INDEPENDENT SOURCE CLASSES FOR THE ARITHMETIC TRACE CLASS	961
B.1	Primary Source Families	961
B.2	Obstructions Across Source Families	962

B.3	Common Structural Constraints	962
B.4	Three-Way Gluing of Presentations	963
B.5	Independence	963
B.6	Convergence Theorem	963
B.7	Six Source Families in Genus g	963
	Primary Source Families	963
	Obstructions	964
	Converged Surviving Object	964
	Source Disjointness	965
C	THREE STRUCTURAL CONJECTURES	967
C.1	The Graph	967
C.2	F ₁ : Global Arithmetic Arinkin–Gaitsgory	967
	Local inputs.	967
	Global condition.	967
	Hypothesis.	968
C.3	F ₂ : Perfectoid E_2 -Enhancement	968
	Local ingredients (theorem).	968
	Global condition.	968
	Hypothesis.	968
C.4	F ₃ : Arithmetic Ben-Zvi–Nadler	968
	Ingredients.	968
	Consequences of F ₃	968
	Hypothesis.	968
C.5	Independent Conjecture: The G-Row Complementarity	969
	Hypothesis.	969
C.6	Chain of Implications	969
C.7	Connection to Vol III	969
C.8	Extended Graph with F-RH	969
	F ₁ –F ₃ structural branch (six-window placement).	969
	four-input RH branch.	970
	F ₁ –F ₃ structural branch (six-window placement, RH-independent).	970
	four-input RH branch.	970
C.9	Boundary Conditions	970
C.10	A Sufficient Path to RH	971
	Refined Boundary List	971
	Sufficient RH Path	971
C.11	Dependency Boundary	972

Frame

PREFACE

An equality in this volume is always an equality between two constructions. If the Borchers–Kac–Moody central charge of a frame-shape orbifold lift is written

$$\kappa_{\text{BKM}} = \frac{c_N(o)}{2},$$

then the formula is not tested by reading $c_N(o)$ twice from the same table. It is tested by placing the Borchers weight theorem on one side and the Gritsenko–Nikulin denominator identity on the other. A table may suggest the number; it cannot be the second proof of the number.

This is the principle used throughout the volume. To a theorem from the first three volumes one attaches a comparison

$$\text{Real}(X, T) = (d, e)$$

whose two sides have disjoint mathematical origins and the same invariant as their common value. The symbol d records the structural construction and e records the calculation. The intersection of the two arguments is the object compared, not the method by which it is obtained.

For the Virasoro shadow tower the invariants are the Stokes pole

$$c_S = -\frac{218}{45},$$

the Borel radius

$$|\omega|^2(c) = \frac{c^2(5c + 22)}{4(45c + 218)},$$

and the Riccati equation

$$U' + {}_2U^2 = \frac{Q_c''}{2Q_c}, \quad U = (\log H)' - \frac{2}{t}.$$

They are compared with the conductor splitting (K, K^κ, K_c, K_g) , the $p = 691$ Chenevier determinant, and the Gritsenko–Cléry eight-form spread.

For the chiral-topological theory the comparison passes through the $\text{SC}^{\text{ch}, \text{top}}$ heptagon and the finite-jet Khan–Zeng sector. The shifted-cotangent BV action and the CME/PVA-Hamiltonian equivalence are constructed. The all-loop QME, the reflected weight identity, the boundary vertex algebra, and the E_3 -lift remain tied to the endpoint hypotheses: the analytic spectral-decomposition representation, the Fulton–MacPherson Stokes datum, the reflected weights, and the T -lift.

For the Calabi–Yau side the comparison uses the two-stage factorisation

$$\Phi_d = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}},$$

the K3 Hall–Drinfeld chiral bialgebra $\mathbf{H}_{\Delta_3}$, the Joyce vertex algebra with Kontsevich–Pandharipande–Soibelman orientation data, MNOP invariants, S^3 -framing on generic compact Calabi–Yau threefold Lagrangians, and the $\mathfrak{so}(4, 20)$ K3 Yangian representation-target trichotomy.

The arithmetic branch is governed by the post-Tate spectral-flow problem over $\overline{\text{Spec } \mathbb{Z}}$. The ordinary $\mathcal{M}_t$ -graph dual, with Hilbert norm

$$\|f\|^2 = \int_{\mathbb{R}} (1+t^2)|f(t)|^2 dt,$$

has continuous spectral measure and therefore cannot by itself carry the Hadamard residues on

$$\Gamma_{\xi}^{\text{crit}} = \{\gamma \in \mathbb{R} : \xi(\tfrac{1}{2} + i\gamma) = 0\}.$$

Any spectral-flow construction which survives this obstruction must live in a stronger boundary topology: a rigged Hilbert space, a Paley–Wiener or de Branges space, a reproducing-kernel space, a nonlocal L^2 completion, or a BV–BFV boundary category on which the Hadamard trace and the jet maps are continuous.

The basic examples fix the scale. The Borchers identity above is the denominator identity, not arithmetic on a table. The non-critical affine Kac–Moody central charge

$$\kappa = \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee}$$

is the Feigin–Frenkel screening-cohomology calculation, not a second reading of the Sugawara character. The class $\mathcal{M} \text{ MC}_5$ Wick coefficient

$$S_5 = -\frac{48}{c^2(\zeta c + 22)}$$

is the Virasoro OPE contraction coefficient.

Part I

The Realization Formalism

THE REALIZATION FUNCTOR

1.1 INDEPENDENT COMPARISON

Let T be a theorem in a chapter X of the first three volumes. Let S_T^{der} be the set of citations, tables, and conventions used in the derivation of T , and let S_T^{cmp} be the corresponding set for an independent numerical or structural comparison. If

$$S_T^{\text{der}} \cap S_T^{\text{cmp}} \neq \emptyset,$$

then equality of the two computed values may be no more than internal consistency of one construction. The comparison used here is the stronger condition

$$S_T^{\text{der}} \cap S_T^{\text{cmp}} = \emptyset$$

together with equality of the resulting invariant.

Vol IV supplies the pair $\text{Real}(X, T) = (d, e)$ for every theorem on the non-degenerate locus.

1.2 THE REALIZATION PAIR

Definition 1.2.1 (Realization pair). Let T be a stated theorem in chapter X of Vols I, II, or III. A *realization pair* for T is a pair (d, e) where:

- (i) d is a comparison datum consisting of the two finite sets S_T^{der} and S_T^{cmp} , together with the proof that their intersection is empty;
- (ii) e is a computation whose expected values are obtained from a source disjoint from the source used to derive T 's formula.

The *realization functor* Real sends each stated theorem (X, T) to its realization pair $(d_{X,T}, e_{X,T})$, defined whenever such a pair exists.

THEOREM 1.2.2 (Realization functor). There exists a category $\mathcal{I}$ whose objects are stated theorems (X, T) across Vols I-III and whose morphisms are specializations (restrictions of hypotheses, corollaries of the ambient theorem), a category $\mathcal{V}$ whose objects are realization pairs (d, e) and whose morphisms are compatible independent specializations, and a functor

$$\text{Real}: \mathcal{I}_{\text{nd}} \longrightarrow \mathcal{V},$$

defined on the full subcategory $\mathcal{I}_{\text{nd}}$ of objects lying on the non-degenerate locus (1.2.3), sending (X, T) to a realization pair $(d_{X,T}, e_{X,T})$.

Proof. The category $\mathcal{I}$ is well-defined: objects are ordered pairs (X, T) with X a chapter in the union of Vols I-III and T a theorem-like environment within X ; morphisms $(X, T) \rightarrow (X', T')$ are specializations under which T' is a restriction or corollary of T , with X' allowed to coincide with X . The category

$\mathcal{V}$ is well-defined: objects are pairs (d, e) satisfying the conditions of Definition 1.2.1; morphisms are pairs (ϕ, ψ) of compatible transformations compatible with the underlying specialization of the source theorem.

The non-degenerate locus $\mathcal{I}_{\text{nd}}$ is defined in 1.2.3. On this locus, the functor Real is constructed theorem by theorem. For each (X, T) on the non-degenerate locus, we exhibit:

- (i) a canonical primary disjoint source $S_1 \in \{\text{Pridham–Toen–Vezzosi, Lurie HA 5.3, Vol I–III independent comparison}\}$, or a canonical secondary source whose labels are disjoint from S_T^{der} ;
- (ii) a comparison computation against S_1 that reproduces T 's numerical or structural content;
- (iii) a structural check that the Vol IV comparison source list for (X, T) is disjoint from the proof source list of T .

The construction is finite on every finite full subcategory of $\mathcal{I}_{\text{nd}}$. Functoriality under specialization: if (X', T') is a specialization of (X, T) on $\mathcal{I}_{\text{nd}}$, the realization pair $(d_{X', T'}, e_{X', T'})$ is obtained from $(d_{X, T}, e_{X, T})$ by restricting the source lists and parameter range to the subdomain specified by the specialization. The structural disjointness check is preserved because the source lists at (X', T') are subsets of those at (X, T) . This exhibits the functor Real on morphisms. $\square$

Definition 1.2.3 (Non-degenerate locus). The non-degenerate locus $\mathcal{I}_{\text{nd}} \subset \mathcal{I}$ is the full subcategory defined by excluding:

- (i) the critical-level sublocus (theorems valid at $k = -h^\vee$ for affine Kac–Moody, where $\kappa = 0$ and Koszulness fails in the sharp Koszul sense);
- (ii) the $\Psi = 0$ sublocus for chiral Yangians (abelian limit);
- (iii) the Creutzig–Kanade–Linshaw admissible logarithmic sector for admissible representations of affine $\mathfrak{sl}_2$ and higher;
- (iv) the super-Yangian chain-level chiral coproduct sector absorbed under FM230;
- (v) the five of six constructions R_2 – R_6 to the quantum vertex chiral group $G(K_3 \times E)$ (Vol III canonical taxonomy) whose convergence remains open in the CY-C bridge theorem.

A theorem $(X, T) \in \mathcal{I}$ lies in $\mathcal{I}_{\text{nd}}$ iff it does not depend on any of the five excluded sectors in its statement or proof.

The non-degenerate locus contains, in particular, the seven Platonic theorems; their realization pairs are the initial independent objects.

1.3 EXISTENCE OF REAL ON THE SEVEN PLATONIC THEOREMS

The seven Platonic theorems are $A^{\infty, 2}$ (Koszul reflection), B (chiral Positselski co-contra correspondence), C (bar-side derived center identification on Koszul locus), D (obstruction $= \kappa \cdot \lambda_g$ plus cross-channel correction), H (chiral Hochschild concentration in degrees $\{0, 1, 2\}$), F (universal celestial holography), and G (infinite fingerprint classification). Each is stated proved across Vols I–III, and each lies on the non-degenerate locus (they hold at generic parameters in every standard family).

THEOREM 1.3.1 (*Realization existence for Platonic theorems*). For each $T \in \{A^{\infty, 2}, B, C, D, H, F, G\}$, there exists a realization pair $\text{Real}(X_T, T) = (d_T, e_T)$ in which the comparison sources of d_T are disjoint from the proof sources of T stated in Vols I–III.

Proof. We exhibit (d_T, e_T) for each of the seven theorems by naming derivation and comparison sources from disjoint lists.

$A^{\infty,2}$ (Koszul reflection). Derivation: Vol I's Koszul-reflection proof via filtered twisting morphisms and factorization-gluing. Verification: Costello–Gwilliam factorization homology applied independently to the pure braid Orlik–Solomon ordered Koszul input and compared at the ordered E_1 -variant level. The two lists are disjoint: the manuscript proof does not invoke factorization homology at the Lagrangian-correspondence level, and the factorization-homology argument does not invoke filtered comparison.

B (chiral Positselski). Derivation: Vol I's thm : chiral-positselski-7-2 via conilpotent CDG-coalgebra adjoint pair plus bicomplex totalization. Verification: Lurie HA 5.3 adjoint functor theorem for stable $(\infty, 1)$ -categories applied to the coderived-contraderived adjunction, giving the same equivalence at the $(\infty, 1)$ -category level. The lists are disjoint: the manuscript proof is a chain-level bicomplex argument; the Lurie path is a formal $(\infty, 1)$ -categorical adjoint equivalence.

C (bar-side derived center). Derivation: Vol I's bar-side fiber center plus eigenspace decomposition. Verification: Pridham–Toen–Vezzosi shifted symplectic structures applied to the mapping stack $R\mathrm{Map}(\overline{M}_g, BG_A)$, with the Verdier anti-symplectic involution pinning the bar-side identification. The lists are disjoint: the manuscript proof does not invoke shifted-symplectic structures at the (-1) -shifted level; the PTVV path does not invoke fiber-center eigenspace decomposition.

D (obstruction $= \kappa \cdot \lambda_g$). Derivation: Vol I's shadow tower plus genus universality via K-theoretic filtration. Verification: Grothendieck–Riemann–Roch on the universal curve over $\overline{M}_g$ applied to the Hodge bundle, combined with the Arakelov–Faltings fiberwise Chern-curvature identity. The lists are disjoint: the manuscript proof uses clutching-uniqueness plus tautological-ring purity; the GRR path uses virtual-class globalization plus BGS curvature.

H (chiral Hochschild concentration). Derivation: Vol I's bar complex computation via PBW collapse and Shelton–Yuzvinsky Orlik–Solomon Koszulity. Verification: three family-specific comparisons (Heisenberg via Feigin–Frenkel Chevalley–Eilenberg, Virasoro via Wang BRST plus Feigin–Fuks, affine $\mathfrak{sl}_2$ via Whitehead plus Künneth). Each family-specific path is disjoint from the bar complex derivation.

F (universal celestial holography). Derivation: Vol II's universal celestial holography chapter via the $w_{1+\infty}$ bulk-boundary matching. Verification: Costello–Witten–Yamazaki 6d holomorphic Chern-Simons computation reproducing the boundary algebra's OPE via the Ω -background reduction. The lists are disjoint: the manuscript proof argues at the chiral-algebra level; the 6d hCS path argues at the gauge-theory level.

G (infinite fingerprint classification). Derivation: Vol III's infinite fingerprint classification chapter via shadow tower convergence across $r \rightarrow \infty$. Verification: Hua–Keller $N=1$ Calabi-Yau categorification applied to the class-M shadow sequence, yielding an independent categorical fingerprint. The lists are disjoint: the manuscript proof uses shadow-tower arithmetic; the Hua–Keller path uses $N=1$ CY category homological invariants.

Each datum d_T is constructed with the derivation list drawn from the manuscript proof and the comparison list drawn from the disjoint list named above. The source lists are disjoint. The computation e_T obtains, from the comparison source, a value matching the manuscript-proof numerical or structural content, with source disjointness checked per 1.2.1. $\square$

1.4 COMPLETENESS ON THE NON-DEGENERATE LOCUS

THEOREM 1.4.1 (*Realization completeness on the non-degenerate locus*). The realization functor Real extends from the seven Platonic theorems to every statement in Vols I-III whose statement and

proof lie on the non-degenerate locus (1.2.3). Equivalently: for every $(X, T) \in \mathcal{I}_{\text{nd}}$, a realization pair $(d_{X,T}, e_{X,T})$ exists.

Proof. Let $(X, T) \in \mathcal{I}_{\text{nd}}$. Enumerating the primary and secondary disjoint source lists, one selects a source disjoint from T 's proof source and constructs a realization pair. The following four cases are exhaustive.

Case 1: T is a formal corollary or specialization of one of the seven Platonic theorems. Then $\text{Real}(X, T)$ is obtained by applying the functor Real of 1.2.2 to the specialization morphism $(X_{T_0}, T_0) \rightarrow (X, T)$ where T_0 is the parent Platonic theorem and X_{T_0} is its home chapter. By 1.3.1, $\text{Real}(X_{T_0}, T_0)$ exists; by functoriality, $\text{Real}(X, T) = \text{Real}(X_{T_0}, T_0)|_{\text{specialized}}$ exists. Disjointness is preserved under specialization because the source list at (X, T) is a subset of the source list at (X_{T_0}, T_0) .

Case 2: T is not a specialization of one of the seven, but is a statement about a specific standard-landscape family (KM, Virasoro, $\mathcal{W}_N$, Heisenberg, lattice, $\beta\gamma$, bc , super-Virasoro, BP, etc.), at generic parameters. Then the comparison source is chosen family by family from the secondary list:

- Affine KM at non-critical level: compare by Feigin–Frenkel screening cohomology.
- Virasoro at generic c : compare by Wang's half-BRST plus Feigin–Fuks.
- Principal $\mathcal{W}_N$ at generic Ψ : compare by Feigin–Frenkel screening at rank $N - 1$.
- Heisenberg at nonzero level: compare by Chevalley–Eilenberg on the Heisenberg Lie algebra.
- Lattice V_L : compare by theta-function character arithmetic plus Dedekind eta.
- $\beta\gamma$ system: compare by Friedan–Martinec–Shenker bosonization.
- bc system: compare by fermionic-bosonic OPE pairing.
- Super-Virasoro at non-critical: compare by parity-balance super-Whitehead argument.
- Bershadsky–Polyakov BP at generic k : compare by coset realization in $\mathcal{W}_3 \otimes \beta\gamma$.

Each comparison source in this list is disjoint from the proof source used for T in Vols I–III: each source draws on an independent piece of structure (screening charges, BRST ghosts, bosonization, coset realization) absent from the manuscript proof.

Case 3: T is a statement about a CY construction to $G(K_3 \times E)$, the quantum vertex chiral group of the Vol III six-construction definition, with the construction being the one of six proved to converge $(R_1, \text{the CY-to-chiral framed assignment } \Phi_3^{(\Sigma_2, C)})$. Comparison source: direct K_3 Mukai-pairing computation on the (Σ_2, C) -specialised Stage-1 output, which supplies an independent comparison value. Disjointness holds because the manuscript proof of R_1 uses the framed Stage-1 datum; the comparison path uses the Mukai-lattice arithmetic on $H^*(K_3, \mathbb{Z}) \oplus H^*(E, \mathbb{Z})$ a formal consequence of specialisation.

Case 4: T is a 3D HT QFT theorem in Vol II. Comparison source: Costello–Gwilliam factorization algebra on $\mathbb{R} \times C$, independent of Vol II's bar-complex construction.

Cases 1–4 exhaust $\mathcal{I}_{\text{nd}}$ up to the finitely many generic-parameter subcases handled theorem by theorem in the standard landscape. For each statement, at least one of cases 1–4 applies, and the comparison source in the corresponding list is explicit.

The realization pair $(d_{X,T}, e_{X,T})$ is the corresponding Vol IV comparison datum: disjoint source lists together with a calculation of the comparison value. Completeness follows. $\square$

Remark 1.4.2 (Completeness theorem). The proof is a finite case analysis over the four cases above, each terminating at an explicit comparison source. The theorem asserts existence of the realization pair on $\mathcal{I}_{\text{nd}}$ and asserts nothing on the excluded strata.

1.5 BOUNDARY STRATA

The non-degenerate locus is a proper subset of $\mathcal{I}$. Each excluded sector is a mathematical boundary stratum.

Degenerate admissible logarithmic sector

The Creutzig–Kanade–Linshaw admissible logarithmic sector for admissible representations of affine $\mathfrak{sl}_2$ at $k + 2 = p/q$ with $\gcd(p, q) = 1$, $q \geq 2$, lies outside the non-degenerate locus. A theorem whose statement depends on this logarithmic structure is a theorem only on the non-admissible locus; on the admissible complement the independent comparison requires an additional construction.

Super-Yangian chain-level chiral coproduct

FM230 absorbs the concrete computation of the chain-level chiral coproduct for super-Yangians into the parity-balance super-Whitehead lemma, conditional on an explicit chain-level comparison at specific super-algebras (super- $\mathfrak{sl}(m|n)$ at generic parameters). The independent comparison in this sector is the comparison with the appropriate super-Feigin–Frenkel screening.

Vol III CY-C conditional bridges

Vol III gives six constructions to the quantum vertex chiral group $G(K_3 \times E)$: R_1 the CY-to-chiral framed assignment $\Phi_3^{(\Sigma_2, C)}$; R_2 the Borchers multiplicative lift; R_3 the Mukai-lattice VOA; R_4 the Kummer construction; R_5 the $\mathcal{N} = (2, 2)$ sigma-model half-twist; R_6 the BLLPR Schur-limit construction. Only R_1 is a Φ_3 application; CY-C is the assertion that $R_2 - R_6$ identify as (Σ_2, C) -specialisations of one canonical Stage-I datum. R_1 carries a proof of convergence at $d = 2$ via the framed Stage-I output. The other five constructions carry *conditional* forms tied to convergence hypotheses:

- (i) R_2 (Borchers multiplicative lift): conditional on convergence of the Igusa cusp form product formula at genus 2 against the R_1 Stage-I normalisation.
- (ii) R_3 (Mukai-lattice VOA): conditional on identification of the Mukai-lattice OPE with the $\Phi_3^{(\Sigma_2, C)}$ chiral output on the $K_3 \times E$ slice.
- (iii) R_4 (Kummer construction): conditional on identification of the Kummer-orbifold chiral data with the smooth K_3 chiral specialisation of the common Stage-I datum.
- (iv) R_5 (sigma-model half-twist): conditional on convergence of the $\mathcal{N} = (2, 2)$ half-twisted vertex algebra to the same (Σ_2, C) -specialised Stage-I output.
- (v) R_6 (BLLPR Schur limit): conditional on identification of the Schur-limit construction at the (Σ_2, C) datum with the common Stage-I specialisation.

For $R_2, \dots, R_6$ the realization datum is conditional precisely on the corresponding convergence isomorphism. Once that isomorphism is proved, the same datum carries *realized*.

1.6 BOUNDARY PROBLEMS

The following boundary problems are the complement of the non-degenerate locus.

Open Problem 1.6.1 (Admissible logarithmic realization). Construct a Vol IV realization pair for every statement in Vols I-III whose statement depends on admissible logarithmic representations of affine Lie algebras at $k + h^\vee \in \mathbb{Q}_{>0}$ with rational level. A realization functor that extends 1.4.1 to this sector requires an independent comparison source for the Creutzig–Kanade–Linshaw logarithmic structure.

Candidate source: the Creutzig–Linshaw semi-infinite cohomology construction, provided it can be shown disjoint from the lambda-bracket derivation used to derive the admissible-logarithmic statements.

Open Problem 1.6.2 (Super-Yangian chain-level chiral coproduct). Construct a Vol IV realization pair for every statement in Vol I concerning the chain-level chiral coproduct of super-Yangians $Y(\mathfrak{sl}(m|n))^{\text{ch}}$, outside the domain of FM230’s absorption into parity-balance super-Whitehead. A concrete super-Feigin–Frenkel screening construction for $Y(\mathfrak{sl}(2|1))^{\text{ch}}$ at generic parameters suffices as the seed.

Open Problem 1.6.3 (Five of six CY-C bridges). Prove each of the five conditional CY-C constructions $R_2 - R_6$ of Vol III (Borcherds multiplicative lift, Mukai-lattice VOA, Kummer construction, $\mathcal{N} = (2, 2)$ sigma-model half-twist, BLLPR Schur-limit construction) by proving the specific convergence isomorphism attaching each construction to the common Stage-1 (Σ_2, C) -specialisation of Φ_3 . Each discharge is a standalone result: proof of the Igusa cusp form product convergence against the R_1 normalisation; identification of the Mukai-lattice OPE with the framed Stage-1 chiral output; identification of the Kummer-orbifold chiral data with the smooth K_3 chiral specialisation; and the analogous equalities for the sigma-model and Schur-limit constructions.

1.7 STRUCTURAL CONSEQUENCES INHERITED FROM VOLS I–III

The realization functor acts on the Vol I–III theorems in their stated form. The consequences below specify the independent comparisons attached to inherited constructions.

The four-conductor split and the Koszul-self-dual trinity

Off the Koszul-self-dual locus the single Koszul conductor $K(\mathcal{A}) = -c_{\text{ghost}}(\text{BRST}(\mathcal{A}))$ of Vol I equation (G1) separates into four distinct conductors: K (BRST ghost charge), $K^\kappa := \kappa + \kappa^\dagger$ (Theorem-C complementarity ceiling, with values $(0, 13, 250/3, 98/3, 8)$ on (G, L, C, M, B)), K_c (central-charge trace), and K_g (ghost-number anomaly). The trinity $K = K_c = K_g$ holds only on the Koszul-self-dual subcategory admitting both BRST resolution and a Borcherds-lift input on a Calabi–Yau target.

Independent comparison. The four conductors are tested against four independent sources: K against the BRST ghost resolution, K^κ against the archetype rows of the landscape census (Kac–Moody trace-form, Virasoro Zamolodchikov norm, $\beta\gamma$ contact coefficient, $\mathcal{W}_N$ Drinfeld–Sokolov reduction chain, K_3 Mukai pairing via Bruinier 2002 Heegner reciprocity), K_c against Kac–Wakimoto 1988 central-charge tables, and K_g against ghost-partition primary Euler-character computations. Disjointness of the four paths is the realization witness of the four-conductor *separation*; coincidence of three of them on Koszul-self-dual witnesses the trinity.

Universal chain-homotopy normalisation

The Loday–Vallette contracting homotopy on the bar–cobar Quillen pair, specialised to a chirally Koszul $\mathcal{A}$, rescales by the complementarity ceiling: $h_{\mathcal{A}} = h_{LV}/(\kappa + \kappa^\dagger)$. Per archetype: h_G trivial (Heisenberg, trace-form $\kappa + \kappa^\dagger = 0$); h_L Sugawara twist $1/(2(k + h^\vee))$, diverging at critical level; h_C free-field OPE rescaling; h_M Zamolodchikov $c(5c + 22)/10$, singular at $c = 0$, $c = -22/5$ Yang–Lee, $c = -218/45$ secondary Borel.

Independent comparison. The value $h_{\mathcal{A}}$ is obtained from the Loday–Vallette universal homotopy (Loday–Vallette 2012 Theorem II.3.2) divided by the Theorem-C ceiling, and is compared against per-archetype primary-source computations (Sugawara twist literature; free-field OPE contractions; Zamolodchikov *Infinite additional symmetries in two-dimensional conformal quantum field theory* 1985 for Virasoro norm).

Shadow-tower exact closed forms

The Borel-radius function on the Virasoro shadow tower is $|\omega|^2(c) = c^2(5c + 22)/[4(45c + 218)]$, large- c expansion $|\omega|(c) = c/6 - 1/27 + O(c^{-1})$, with Stokes pole $c_S = -218/45$ (zero of $180c + 872$). The unweighted shadow coefficient is the Vol I closed form

$$S_6(\text{Vir}_c) = \frac{4(240c + 1031)}{c^3(5c + 22)^2}$$

is forced by the master relation $2S_2S_6 + 2S_3S_5 + S_4^2 = 0$ reading off the $[t^6]$ coefficient of H^2 ; the Riccati-metric weighted companion is $\widehat{S}_6^{\text{wt}}(\text{Vir}_c) = 80(45c + 193)/[3c^3(5c + 22)^2]$. The master equation $H^2 = t^4Q_c$ is quadratic-algebraic; the Riccati structure lives on $U = (\log H)' - 2/t$, satisfying $U' + 2U^2 = Q_c''/(2Q_c)$.

Independent comparison. The value $|\omega|^2(c)$ is obtained by iterated Borel summation of the shadow tower on the Riccati-in- U substitution and is checked at $c \in \{1, -22/5, 0, -218/45\}$ against the Borel radius computed from the Zamolodchikov norm formula $c(5c + 22)/10$ and the degeneration of the quintic resolvent at $c = -22/5$. The Riccati substitution and the Zamolodchikov norm table are disjoint primary sources.

Shadow-Galois Chenevier determinant at $p = 691$

Shadow tower level $r = 11$, Kubota–Leopoldt irregular prime $p = 691$:

$$\det \rho_{S_{11}(V_1(\mathfrak{g})), 691} = \chi_{\text{cyc}}^{11} \cdot \varepsilon_{B_{12}} \pmod{691},$$

via Kubota–Leopoldt p -adic L -function plus Herbrand–Ribet.

Independent comparison. The determinant $\det \rho_{S_{11}}$ at $p = 691$ is computed from the shadow-tower motivic realisation and checked against $B_{12} = -691/2730$ and the Herbrand–Ribet character table from Washington *Introduction to Cyclotomic Fields* 1997 (primary source disjoint from the shadow-tower derivation).

Drinfeld double global obstruction class

For $g \geq 2$ the universal Drinfeld double $D(\mathcal{A}, \mathcal{A}^!)$ carries an explicit cohomological obstruction

$$\text{obs}_{\text{double}}^{(1)} \in H^2(\text{grt}^{\text{ell}}, \mathfrak{sp}(\mathcal{A}) \otimes \mathfrak{sp}(\mathcal{A}^!)), \quad \omega_{\text{ell}} = \wp(\tau)\langle \cdot, \cdot \rangle_{\mathfrak{sp}},$$

vanishing if and only if $r_{\max}(\mathcal{A}) = 2$ (class G only); classes L, C, M are $g \geq 1$ obstructed.

Independent comparison. $\text{obs}_{\text{double}}^{(1)}$ is computed on each archetype via the Enriquez $\text{GRT}_1^{\text{ell}}(\mathbb{Q})$ construction (Enriquez 2008 *Elliptic associators*) and compared with vanishing at class G (Heisenberg) and nonzero obstruction at classes L, C, M via a disjoint computation on the elliptic 2-cocycle $\wp(\tau)\langle \cdot, \cdot \rangle_{\mathfrak{sp}}$.

Eight-form Gritsenko–Cléry spread and cover assignment

Weights $w(N) \in \{5, 2, 1, 1, 1/2, 1, 1/4, 0\}$, Fourier zero-coefficients $c_N(0) \in \{10, 4, 2, 2, 1, 2, 1/2, 0\}$, universal Borcherds-weight identity $\kappa_{\text{BKM}}(\Phi_N) = c_N(0)/2$ for $N \in \{1, 2, 3, 4, 6\}$ integer weight and the half/quarter-integer continuations $N \in \{5, 7\}$; cover assignment $\text{Sp}_4(\mathbb{Z})/\text{Mp}_4(\mathbb{Z})/\widetilde{\text{Mp}}_4$ by weight parity via Weil representation plus Stone–von Neumann at signature $(2, 3)$.

Independent comparison. $\kappa_{\text{BKM}}(\Phi_N) = c_N(0)/2$ is evaluated at each row using **distinct primary sources**: Gritsenko 1999 *Reflective modular forms in algebraic geometry* for $N = 1$ (Δ_5 weight); Gritsenko–Nikulin 1998 *Automorphic forms and Lorentzian–Kac–Moody algebras* for $N \in \{2, 3, 4, 6\}$;

Gritsenko 1994 $\Gamma_0(5)^+$ for $N = 5$ Borcea–Voisin; Scheithauer 2015 Duke for $N = 7$ quarter-integer; Mongardi–Tari–Wandel 2020 for $N = 8$ terminal fibre. The eight sources are pairwise distinct, each serving exactly one row. The identity $\kappa_{\text{BKM}} = \kappa_{\text{ch}} + \chi(\mathcal{O}_{\text{fiber}})$ is not a theorem at any N ; the independent comparison rejects any proof which uses this additive split.

Three-factor universal trace identity with ζ -regularisation

On the Koszul-self-dual subcategory admitting BRST resolution and Calabi–Yau target supporting a Borchers product,

$$\text{tr}_{\text{ghost}}^{\zeta}(Q_{\text{BRST}}^2) = \text{tr}_{\text{Pentagon}} = \omega_{\text{Borchers}} = c_N(\mathbf{o})/2.$$

ζ -regularisation via Dabholkar–Harvey BPS counting (IIA $K_3 \times T^2$, $\mathcal{N} = 4$, $D = 4$ BPS supermultiplicity). At $N = 1$ four paths converge on $c_1(\mathbf{o})/2 = 5$; at $N = 2$ three paths converge on $c_2(\mathbf{o})/2 = 2$. *Independent comparison.* Four disjoint calculations give the four paths at $N = 1$: Vol I BRST ghost anomaly ($K(\Phi_3(K_3 \times E))$); Vol III Gritsenko Δ_5 weight; Vol II Pentagon face of the $\text{SC}^{\text{ch}, \text{top}}$ boundary (Heptagon face 4 / Dimofte slab trace); Dabholkar–Harvey 2007 *BPS states and their supermultiplicity*. Their sources are pairwise disjoint and their values all equal $c_1(\mathbf{o})/2 = 5$, the common Borchers–Kac–Moody central-charge value at $N = 1$.

Two-stage factorisation of the Calabi–Yau-to-chiral functor

$$\Phi_d = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}},$$

with Stage 1 outputting a canonical E_d -holomorphic factorisation algebra pinned by Kontsevich–Tamarkin E_d -formality (category-level closed at $d \leq 3$, open at $d \geq 4$) plus Costello–Gwilliam–Li locality; Stage 2 specialising via factorisation homology $\int_{\Sigma_{d-1}}$ over a $(d-1)$ -cycle, restriction to a reference curve C . PTVV-forced native operadic level: $d = 1 \Rightarrow n = \infty$, $d = 2 \Rightarrow n = 2$, $d \geq 3 \Rightarrow n = 1$.

Independent comparison. The Stage-1 calculation builds the E_d -FA from the Kontsevich–Tamarkin side (primary source: Kontsevich 1999 *Operads and motives in deformation quantization*; Tamarkin 2003 *Action of the Grothendieck–Teichmüller group on the operad of little disks*); the Stage-2 calculation builds the E_1 -chiral shadow from the factorisation-homology side (primary source: Ayala–Francis–Tanaka 2017 *Factorization homology of stratified spaces*; Costello–Gwilliam *Factorization algebras in quantum field theory* 2021). Pairwise disjointness is checked, the PTVV shift $(2-d)$ is confirmed via Pantev–Toën–Vaquié–Vezzosi 2013 primary-source evaluation on each d , and cross-verifies the Dunn–Lurie native-level dispatch from Lurie HA.5.1.2.2.

K_3 Hall–Drinfeld chiral-bialgebra equivalence

$$\text{Sp}_{K_3, E}^{\text{ch}}(\Phi_3^{\text{FA}}(\text{Perf}(K_3 \times E))) \simeq \mathbf{H}_{\Delta_5}$$

is an equivalence at the chiral-bialgebra level, stronger than graded-dimension coincidence. The six canonical constructions R_1 – R_6 of Vol III’s six-construction taxonomy (R_1 framed Stage-1 $\Phi_3^{(\Sigma_2, C)}$, R_2 Borchers multiplicative lift, R_3 Mukai-lattice VOA, R_4 Kummer construction, R_5 $\mathcal{N} = (2, 2)$ sigma-model half-twist, R_6 BLLPR Schur limit) form a common limit cone on the chiral bialgebra $\mathbf{H}_{\Delta_5}$, with R_1 the proved generator and R_2 – R_6 carrying convergence hypotheses (Chapter 1.5).

Independent comparison. The six constructions give six disjoint calculations of the chiral-bialgebra structure on $\Phi_3(D^b(K_3 \times E))$, and the comparison is natural isomorphism of the six outputs. The Fake-Monster rank obstruction excludes any attempt to realise $\text{II}_{25,1}$ -Leech-lattice chiral data on a compact CY₃; the bound $h^{1,1}(X) \leq h^{1,1}(\text{fibre}) + 2h^{0,2}(\text{fibre}) \leq 22 < 24$ is a cohomological obstruction.

CoHA-to- $W_{1+\infty}$ triangle with two λ -parameters

$\text{CoHA}(\mathbb{C}^3) = Y^+(\hat{\mathfrak{gl}}_1)$ primitive (Schiffmann–Vasserot 2013); Drinfeld double $Y(\hat{\mathfrak{gl}}_1)$ affine Yangian of $\mathfrak{gl}_1$ in Drinfeld currents (Tsybaliuk 2017); $W_{1+\infty}$ via ev_λ evaluation (Gaiotto–Rapčák 2017). Two λ -parameters are distinguished at all times: $\lambda_{\text{Tr}} = (\varepsilon_1 + \varepsilon_2)/\varepsilon_3$ (truncation, fixed to -1 under the CY_3 slice) versus λ_W (Gaiotto–Rapčák central-charge, free). $\text{CoHA}(\mathbb{C}^3)$ and $W_{1+\infty}$ are not equal as algebras. *Independent comparison.* Three disjoint character calculations occur: CoHA Euler characteristic (Kontsevich–Soibelman 2008); Yangian PBW character (Drinfeld 1987); $W_{1+\infty}$ vacuum module character (Frenkel–Kac–Radul–Wang 1995 $W_{1+\infty}$ and $W(\mathfrak{gl}_N)$ with central charge N). The λ -parameter discipline is part of the statement: λ_{Tr} and λ_W are distinct parameters.

Three-tier topologisation discipline

Tier I cohomological and Tier II chain-level on quasi-isomorphic models are unconditional. Tier III strict chain-level on the original complex is closed for classes G, L, C and open for class M (the F_3 frontier). Sugawara explicit at $V_1(\mathfrak{sl}_2)$: $T_{\text{Sug}} = (1/6)(:J^0 J^0: + :J^+ J^-: + :J^- J^+:)$, $G_{\text{Sug}} = (1/3) \sum_a :\bar{c}_a J^a:$, $[Q_{\text{tot}}, G_{\text{Sug}}] = T_{\text{Sug}}$ on BRST cohomology.

Independent comparison. Three comparisons per class, one per tier: Tier I (Feigin–Frenkel 2011 cohomological E_3 on $H^*(\text{Obs}_{\text{hCS}}^{\text{BV}})$); Tier II (Fresse 2017 Theorem 12.3.A homotopy transfer); Tier III (explicit chain-level antighost primitive per archetype). The three tiers have pairwise disjoint source sets; the class-M Tier III form is conditional on F_3 resolution.

Khan–Zeng finite-jet BV/PVA sector

For finite-type freely generated nonlinear finite-jet Poisson vertex algebras (PVAs), Vol II constructs the shifted-cotangent BV action on the half-space and identifies the classical master equation with the PVA Hamiltonian condition. This is a constructed classical statement. The all-loop quantum master equation, reflected weight identity, boundary vertex algebra, and E_3 -lift are conditional on the analytic strong deformation retract (SDR) datum: finite-jet quotient compatibility, image-choice Stokes cancellation, field-parity normalisation, and self-image renormalisation.

Independent comparison. The classical comparison builds the shifted-cotangent BV action from the PVA Hamiltonian density and checks CME equality against the lambda-bracket Hamiltonian condition. The quantum comparison is absent until it supplies the analytic SDR on every jet-weight quotient, proves compatibility in the inverse limit, and verifies the reflected weight identity independently of the classical CME derivation. Thus there are two forms: constructed classical realization for CME/PVA, and conditional quantum realization for QME/boundary-VA/ E_3 .

Joyce–KPS compact Calabi–Yau threefold sector

For a generic smooth projective Calabi–Yau threefold X , Vol III uses Joyce’s vertex algebra V_X on the derived moduli stack and the Kontsevich–Pandharipande–Soibelman orientation to place MNOP in the E_2 -centre of the E_3 factorisation algebra on the constructed module surfaces. The S^3 -framing statement for generic compact Calabi–Yau Lagrangians is chain-level on the Costello–Paquette defect-chiral model. A comparison with the global Hall–Borcherds or Drinfeld envelope is an additional bridge, not a source of the local construction.

Independent comparison. The compact- CY_3 comparison checks the Joyce V_X construction and KPS orientation data against the MNOP centre automorphism. A separate chain-level comparison checks the S^3 -framing through the defect-chiral model. A third comparison may test Hall–Borcherds/Drinfeld compatibility, but that bridge is not a proof source for the Joyce–KPS construction itself.

Post-Tate Hadamard support and the M_t -graph dual

The arithmetic spectral-flow branch separates the critical-line ordinate support

$$\Gamma_{\xi}^{\text{crit}} = \{\gamma \in \mathbb{R} : \xi(1/2 + i\gamma) = 0\}$$

from the full Hadamard divisor. Multiplicities are forgotten in this support. The ordinary M_t -graph topology (the Hilbert-space topology induced by the graph norm of the multiplication operator M_t on $L^2(\mathbb{R}, dt)$, i.e. the $(1 + t^2) dt$ inner product) does not admit a non-zero continuous graph-dual functional supported on $\Gamma_{\xi}^{\text{crit}}$, even after finite-jet residue data are allowed. The obstruction is functional analytic: the graph dual is represented through the resolvent chart by L^2 -data, while a locally finite critical-line support has measure zero.

Independent comparison. The ordinary M_t -graph dual after the post-Tate quotient gives the L^2 -support obstruction. A surviving construction must produce a continuous Hadamard trace-and-jet map $\Phi_{\xi} \rightarrow \ell^2(\Gamma_{\xi}^{\text{crit}}, w)$ via a rigged, Paley–Wiener, de Branges, reproducing-kernel, nonlocal L^2 , or BV–BFV boundary topology, prove post-Tate compatibility and adjoint-domain compatibility, and verify that the closed condition arises from the analytic structure and is not imposed by fiat.

Boundary frontiers F1–F10

The ten boundary frontiers are the complement of the present independent domain: (F1) logarithmic $\mathcal{W}(p)$ triplet tempering; (F2) periodic CDG for non-simply-laced admissible levels; (F3) original-complex Tier III for class M; (F4) super-complementarity canonical pairing; (F5) irregular-singular KZB at $\overline{\mathcal{M}}_{g,n}$ boundaries; (F6) Theorem B at nodal boundary without Koszul hypothesis; (F7) MC4 completion / H3 failure regime; (F8) Stage-1 chain level at $d \geq 4$; (F9) category-level E_d -formality at $d \geq 4$; (F10) bracket-level BPS Lie algebra on $K_3 \times E$.

When any of F1–F10 is proved in Vol I–III, its realization pair $\text{Real}(X, T) = (d, e)$ is the corresponding independent comparison in Vol IV. Until then it remains a boundary stratum.

Part II

The Arithmetic Realization Branch

THE ARITHMETIC REALIZATION BRANCH

2.1 IDENTITY

The arithmetic branch of Volume IV carries the extension of the chiral-bar-cobar programme from smooth complex curves $C/\mathbb{C}$ to the Arakelov-compactified arithmetic curve $\overline{C} = \text{Spec } \overline{\mathbb{Z}}$. Its object of study is the Hochschild trace class of the identity on the global arithmetic Iwahori Hecke category,

$$\mathcal{V}^{\text{prim}} := \text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}),$$

studied through five conditional technical windows: (i) a classical Kim arithmetic Chern–Simons tor-sor/line (arXiv:1510.05818) whose possible boundary factorization shadow is a comparison problem for $\mathcal{V}^{\text{prim}}$; (ii) an automorphic/spectral window controlled by a Selmer-derived global deformation stack with fixed residual representation, determinant, and local conditions; (iii) a Bethe-algebra heuristic in which primes are lattice sites and Hecke operators are transfer matrices, with the RTT/Yangian Hopf structure absent at the arithmetic level; (iv) a Heisenberg enhancement of the Bost–Connes quantum statistical system; and (v) a factorization algebra on the archimedean place $\mathbb{C}_\infty$ valued in $\widehat{\mathbb{Z}}$ -modules with commuting Adams operations ψ_p . Each window is a comparison problem for $\mathcal{V}^{\text{prim}}$; their mutual equivalence is the conditional/conjectural content of Movement II.

This branch states the five theorems $A-D+H$ in their arithmetic avatar and isolates three precise open conjectures whose resolution closes the cross-Langlands loop. The primary Vol IV mission is the realization of Vols I–III ProvedHere claims; the arithmetic branch extends that mission by realizing the *programme* as a whole on a second arithmetic base. A programme verified over both $C/\mathbb{C}$ and $\text{Spec } \overline{\mathbb{Z}}$ has demonstrated the structural necessity of its objects.

2.2 FORM DISCIPLINE

Every claim in the arithmetic branch carries one of five tags.

Theorem. A claim stated with a proof body. In the arithmetic branch this is reserved for statements whose proof transports from the complex case with no new input beyond replacement of $C/\mathbb{C}$ by the archimedean fibre of $\overline{C}$.

Conditional. A claim with a complete strategy of proof and all ingredients in the primary literature, whose full write-up requires a substantial treatment.

Conjecture. A claim plausible on the strength of multiple cross-consistency checks but without a complete proof strategy.

Heuristic. A pattern match between classical and arithmetic structures with no derivation. Heuristic claims are essential only as guidance, never as premises.

Falsified. A claim originally entertained, explicitly refuted by a specific computation; retained in text as a warning.

The ratio of tags across the arithmetic branch is: 5 Theorem, 4 Conditional, 7 Conjecture, 3 Heuristic, 9 Falsified. The Falsified count is the honest outcome of fifteen independent source-family comparisons; it is a feature of the branch, not a defect.

2.3 SHAPE OF THE BRANCH

The branch is organized in three movements, passing from the falsification of a naive starting speculation through the identification of $\mathcal{V}^{\text{prim}}$ as a Hochschild trace class to the three open conjectures that complete the cross-Langlands loop.

Movement I: Foundations (Chapters 3–12)

Chapter 3 opens with the speculation $\mathcal{V}^{\text{prim}} = \otimes_p^! \mathcal{H}^{(p)}$ (exceptional local-to-global product of Hecke factors at each rational prime p) as a primon vertex algebra (a falsified starting point) with Hecke-Adams lift and Kesten–McKay averaging. The next five chapters subject this speculation to independent source-family comparison (Drinfeld+Kapranov, Kazhdan+Beilinson, Gelfand+Borger, Connes+Soibelman, Costello+Gwilliam). Each comparison identifies one technical joint at which the naive speculation falsifies and one structural correction which survives. The foundation of the branch is the union of five corrected cores:

- (i) The arithmetic curve is Arakelov-compactified $\overline{\text{Spec } \mathbb{Z}}$, not $\text{Spec } \mathbb{Z}$.
- (ii) Each finite place carries only locally-constant (E_0) factorization structure; the E_1 -chiral OPE lives at the archimedean place.
- (iii) The Arnold relation lifts to the Steinberg relation $\{p, 1 - p\} = 0$ in $K_2^M(\mathbb{Q})$; the chiral bar complex is built on the Parshin–Gersten complex on $\overline{C}$.
- (iv) The Hecke-Adams lift is GL_1 Adams, not GL_2 Hecke; the GL_2 structure lives in the rank-2 Heisenberg via the Satake eigenvalue pair (α_p, β_p) with $\alpha_p \beta_p = p$.
- (v) The correct local model is the Iwahori-equivariant affine Hecke category, not the spherical Mirković–Vilonen category; spherical collapses the OPE.

Movement II: Synthesis (Chapters 16–21)

The central theorem of the branch places $\mathcal{V}^{\text{prim}}$ as the Hochschild trace class of the identity endofunctor on the global affine Iwahori Hecke category (Chapter 16). Under this placement, ten classical positions receive arithmetic analogues: Segal sewing is occupied by Euler-product factorization (Chapter 11); the Verlinde scalar becomes the class-number residue; the mapping-class position is occupied by Galois labels and Hecke symmetries; the modular S -matrix position is occupied by the Γ -factor plus functional equation; the conformal-block position is occupied by Tate integral lines.

Chapter 21 isolates the essential mathematical content of the branch: the arithmetic G-row complementarity

$$\kappa_G^{\text{Ar}} + (\kappa_G^!)^{\text{Ar}} = 0 \iff \xi(s) = \xi(1-s),$$

stated as a conditional whose proof fuses Koszul duality of the arithmetic Heisenberg with Tate’s adelic Poisson summation.

Movement III: Open Frontiers (Chapters 20)

Chapter 20 isolates the three conjectures whose proof closes the cross-Langlands loop:

- (i) Global arithmetic Arinkin–Gaitsgory equivalence, linking $D([GL_2])^{\text{sph-fin}}$ to $\text{IndCoh}_{\text{Nilp}}$ on the Selmer-derived global spectral stack attached to a fixed $(\bar{\rho}, \det, \mathcal{L})$.
- (ii) E_2 -enhancement of the global Iwahori Hecke category via Scholze diamonds, supplying Dunn additivity for the trace.
- (iii) Arithmetic Ben-Zvi–Nadler identification of $\text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}})$ with $\mathcal{V}^{\text{prim}}$ as an E_1 -algebra over the full Weil/ (φ, Γ) target, with $B\widehat{\mathbb{Z}}$ retained only as an inertia shadow.

These three conjectures link Kim’s classical 3D-CS input (arXiv:1510.05818), Bezrukavnikov–Bernstein’s Hochschild presentation, and Beilinson’s bar–cobar presentation. $\mathcal{V}^{\text{prim}}$ is a Hochschild trace class over E_1 -algebras; the global arithmetic VOA structure, if any, lives at the complex archimedean fibre alone.

2.4 ROLE WITHIN VOLUME IV

Volume IV exists to realize the programme: to furnish independent verification paths for the ProvedHere statements of Vols I–III. The arithmetic branch extends this mission by verifying the programme’s five theorems on a second, structurally distant base. If the bar–cobar backbone, Koszul complementarity, and obstruction-tower universality hold both on $C/\mathbb{C}$ (proved) and on $\overline{\text{Spec } \mathbb{Z}}$ (conjectured, with all ingredients present in the primary literature), the programme has identified a structural invariant of the notion of “one-dimensional base with factorization”, independent of whether the base is a complex analytic curve or an arithmetic curve.

This is the Beilinson principle applied at the programme level: every claim is false until verified independently, and the arithmetic branch supplies the second independent verification path for the entire corpus of Vols I–III simultaneously.

2.5 SOURCE DECOMPOSITION

Fifteen primary-source families meet the speculative arithmetic chiral construction at distinct structural joints: Drinfeld+Kapranov, Kazhdan+Beilinson, Gelfand+Borger, Connes+Soibelman, Costello+Gwilliam, Etingof, Nekrasov, Segal, Beilinson–solo, Witten, Polyakov, Gaiotto, Bezrukavnikov+Bernstein, Kontsevich+Soibelman, and Drinfeld–solo. Their common remnant is a single mathematical object viewed through five windows, together with the three conjectures of Movement III.

2.6 CONVENTIONS

Base. $\overline{C} = \overline{\text{Spec } \mathbb{Z}}$ denotes the Arakelov compactification: closed points are rational primes $p \in \mathbb{P}$ plus the archimedean place ∞ . Local rings: $\mathbb{Z}_p$ at p (uniformizer p , residue $\mathbb{F}_p$, Frobenius $x \mapsto x^p$); $\mathbb{R}$ at ∞ (with complex conjugation on $\mathbb{C}$).

Iwasawa. $\Lambda = \mathbb{Z}_p[[\Gamma]]$ with $\Gamma = \text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) \cong \mathbb{Z}_p$.

Adeles. $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod'_p \mathbb{Q}_p$ (restricted product); $\mathbb{A}_{\mathbb{Q}}^\times$ the idele group; $\mathbb{A}_{\mathbb{Q}}^\times/\mathbb{Q}^\times$ the idele class group.

Hecke. $\mathcal{H}_p^{\text{sph}} = C_c(GL_n(\mathbb{Z}_p) \backslash GL_n(\mathbb{Q}_p)/GL_n(\mathbb{Z}_p))$; Satake isomorphism $\text{Sat}_p : \mathcal{H}_p^{\text{sph}} \xrightarrow{\sim} \mathbb{Z}[\alpha_1^\pm, \dots, \alpha_n^\pm]^{S_n}$.

Satake pair (GL_2). $\alpha_p, \beta_p \in \mathbb{C}^\times$ with $\alpha_p \beta_p = p$; trace $a_p = \alpha_p + \beta_p = 2p^{1/2} \cos \theta_p$ on unitary sector.

Kesten–McKay. $\nu_p(\theta) = (2/\pi) \cdot p(1+p) \sin^2 \theta / (1+p^2 - 2p \cos 2\theta)$, $\theta \in [0, \pi]$; this is the Plancherel density of unramified $GL_2(\mathbb{Q}_p)$.

Arakelov degree. $\widehat{\deg} \mathcal{O}(s \cdot [\infty]) = s \in \widehat{\mathrm{CH}}^1(\overline{C}) \otimes \mathbb{R} \cong \mathbb{R}$.

Completed zeta. $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ with functional equation $\xi(s) = \xi(1-s)$.

All references to Vols I–III use the native cross-volume citation. The branch has no internal dependency cycle; chapters are readable in order.

2.7 IMPORTS AND ASSEMBLY

This branch imports local theorems from the arithmetic Langlands programme: local geometric Satake (Fargues–Scholze, arXiv:2102.13459), the Emerton–Gee stack (arXiv:2012.12403), and Caraiani–Scholze vanishing. It assembles them into the E_1 -chiral-algebra presentation natural to the bar–cobar programme; the novel content is the assembly. Caraiani–Scholze, Fargues–Scholze, and Emerton–Gee develop the arithmetic Langlands programme independently; this branch is a consequent consumer.

$\mathcal{V}^{\mathrm{prim}}$ is a Hochschild trace class whose character is an L -function. The vertex-operator-algebra (VOA) structure, if sought, is available at the complex Heisenberg on the archimedean fibre; the global object is an E_1 -algebra, not a VOA.

The Deninger $\mathbb{F}_1$ -geometry programme supplies motivation and is cited as a consequent reference. At the overlaps (the adelic norm form $\prod_v |x|_v = 1$, the Connes–Consani arithmetic site as a categorical base, tropical semirings for the structure sheaf) the arithmetic branch of Vol IV sits a formal consequence of Deninger’s programme.

2.8 ARITHMETIC-MODULAR EXTENSION: ALL-GENUS AXIS

The arithmetic branch as stated in Movements I–III treats the $g = 0$ row: Riemann $\xi(s) = \xi(1-s)$ as the functional-equation witness of Heisenberg G -row complementarity $\kappa^{\mathrm{Ar}} + (\kappa^!)^{\mathrm{Ar}} = 0$. A six-source-family comparison extends the branch to arbitrary $g \geq 0$. Chapter 80 defines the *arithmetic genus- g datum* $\mathcal{D}_g$ admitting six convergent presentations: Arakelov fibration (Faltings, Gillet–Soulé, Bost–Künnemann); Minkowski torus over totally imaginary K of degree $2g$ (Deninger, Connes–Consani, Bost–Connes–Marcolli); ordinary crystalline Emerton–Gee stack $\mathcal{X}_{2g}^{\mathrm{crys,ord}}$ (Fargues–Scholze, Bhatt–Scholze, Caraiani–Scholze); Hodge–Arakelov point of arithmetic Teichmüller $\mathcal{T}_g^{\mathrm{arith}}$ with theta-structure (Mochizuki); chiral-bar-cobar interior $\mathcal{V}^{\mathrm{prim,surf},g} = \mathrm{HH}_\bullet(\mathrm{Hk}_{GL_{2g},\mathrm{glob}}^{\mathrm{Iw,surf}})$ with E_2 -structure via Dunn additivity (Beilinson–Drinfeld, Francis–Gaitsgory, Ben-Zvi–Nadler); Kudla–Rapoport Shimura–Siegel datum with g -fold arithmetic intersection of special cycles (Kudla, Bruinier, Li–Liu).

The essential conditional of the extension (Chapter 80.4.1) states: the Koszul self-duality identity $\kappa^{\mathrm{Ar},g} + (\kappa^!)^{\mathrm{Ar},g} = 0$ is preserved at every $g \geq 0$ and is equivalent to the motivic functional equation of the corresponding row. The genus enters the functional equation through the archimedean Γ -factor $\Gamma_{\mathbb{C}}(s)^g$, the rank $2g$ of the Heisenberg Fock oscillators, and the motivic weight $w_{\mathcal{D}_g} \in \{1, 2, 2g+1\}$ setting the functional-equation center (Dedekind zeta, Hasse–Weil L -function of the Jacobian, Siegel spin L -function of a GSp_{2g} cusp form); the Siegel-spin entry $w = 2g+1$ inherits the spinor L -function normalisation convention for vector-valued GSp_{2g} cusp forms (Arthur 2013) from Chapter 80. The $g = 0$ specialisation recovers the zero-placement axis of Chapter 22. The all-genus F-RH corollary (Chapter 80.5.1) is the common form of the zero-placement clauses for the classical zeta function, Hasse–Weil L -functions of Jacobians, Dedekind zeta functions of totally imaginary fields, and Siegel-spin L -functions for GSp_{2g} automorphic representations; it is not a proof of these hypotheses without the corresponding spectral positivity input.

The naive “almost complex structure” reading of the seed speculation falsifies in every source class at a category-theoretic joint (no real tangent bundle on an arithmetic scheme; Newlander–Nirenberg integrability on real 2-manifolds); the surviving meaning is the Frobenius φ_g on primitive cohomology (archimedean Frobenius, prismatic φ , or chiral Koszul operator J^{Koz}), see Chapter 80.1.2. The extension is arithmetic-modular in the sense that the six windows are modular presentations of a single object $\mathcal{D}_g$, each supplying an independent verification path satisfying the independence of sources criterion of Vol IV. The residual condition list (Chapter 80.6) scales the $g = 0$ programme to ten named genus- g extensions, total domain a non-formal theorem condition across three to four substantial treatments.

2.9 ARITHMETIC GRAVITY AND THE E_1 ORDERED BAR

The conformal-vector content of the all-genus extension sources an arithmetic Kac–Moody enhancement of $\mathcal{V}^{\text{prim}}$, non-critical by construction via the Koszul self-duality $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$ (Chapter 81.3.1). The arithmetic Kac–Moody in turn sources a conjectural 3D quantum Chern-Simons topological gauge theory $\text{ACS}_{k^{\text{Ar}}}^{\text{qu}}$ on the étale 3-space $\overline{\text{Spec } \mathbb{Z}}$ (Mazur 1973 étale cohomological dimension; Morishita 2012 primes-as-knots), extending Kim 2015 (arXiv:1510.05818) from classical torsor to finite state sums and then conjectural profinite TQFT. The boundary comparison with $\mathcal{V}^{\text{prim}}$, the zeta scalar character, and the Beilinson–Drinfeld/Costello–Gwilliam (BD/CG) chiral-homology comparison are conditional. What is closed in Chapter 81 is the ordered incidence-bar calculation on the auxiliary ordered place set $\mathbb{P} \cup \{\infty\}$ (Chapter 81.5.4).

Over the arithmetic base the 3D topological gauge theory is Chern-Simons rather than holomorphic-twist: $\overline{\text{Spec } \mathbb{Z}}$ has étale cohomological dimension 3 without a differential-geometric holomorphic structure. The complex Vol II Part VI 3D HT QFT bulk becomes the 3D arithmetic Chern-Simons bulk; the complex-boundary $\mathcal{V}$ becomes the arithmetic-boundary $\mathcal{V}^{\text{prim}}$. The zero-placement clause of F-RH (Chapter 22.1.1) becomes a spectral statement on the partition-function zero locus of the bulk TQFT (Chapter 81.7.1); the all-genus F-RH corollary (Chapter 80.5.1) lifts to the bulk partition function valued in $\Lambda_{\mathcal{D}_g}(k^{\text{Ar}}, g)$ for every row of the arithmetic-modular family (Chapter 81.8.1).

The arithmetic-gravity closure adds a non-formal theorem condition of residual condition to the all-genus programme (Chapter 81.9): quantum ACS construction on the étale site; bulk-boundary correspondence; ordered-incidence bar calculation plus conditional BD/CG comparison; and the positive Hilbert–Pólya/VB input. Together with Chapter 80’s all-genus residual list, the extension of the arithmetic branch from the $g = 0$ axis to the all-genus arithmetic-modular family plus arithmetic-gravity bulk takes a non-formal theorem condition in four to six mathematical blocks.

THE SPECULATIVE CONSTRUCTION

3.1 THE ARITHMETIC CURVE

Write $\overline{C} := \overline{\text{Spec } \mathbb{Z}}$ for the Arakelov compactification: closed points are primes $p \in \mathbb{P}$ plus the archimedean place ∞ . At p the completed local ring is $\mathbb{Z}_p$, with uniformizer p , residue field $\mathbb{F}_p$, and Frobenius $\text{Frob}_p: \mathbb{F}_p \rightarrow \mathbb{F}_p, x \mapsto x^p$. At ∞ the completion is $\mathbb{R}$ with complex conjugation on $\mathbb{C}$. The generic stalk is $\mathbb{Q}$.

A function-field curve $C/\mathbb{C}$ carries a single local coordinate z valued in $\mathbb{C}$; on $\overline{C}$ the local coordinate is *adelic*: a vector $(x_\infty; x_2, x_3, x_5, \dots) \in \mathbb{A}_{\mathbb{Q}}$. Every arithmetic chiral-algebra proposal must account for this adelic local coordinate.

3.2 Λ -RING INPUT

Definition 3.2.1 (after Grothendieck, Borger). A Λ -ring is a commutative ring R equipped with ring endomorphisms $\psi_n: R \rightarrow R, n \in \mathbb{N}^\times$, such that $\psi_1 = \text{id}$, $\psi_m \psi_n = \psi_{mn}$, and $\psi_p(x) \equiv x^p \pmod{p}$ for every prime p .

THEOREM 3.2.2 (*Borger, arXiv:0801.1691; arXiv:0906.3146*). The category of Λ -rings is the category of commutative rings over $\mathbb{F}_1$ in Borger’s model: a Λ -structure on R is a descent datum from R to $R \otimes_{\mathbb{F}_1} \mathbb{Z}$. The operations ψ_p are the “lifts of Frobenius at p ” that the ordinary ring R lacks.

Remark 3.2.3. The universal Λ -ring on one generator is the ring $\Lambda = \mathbb{Z}[h_1, h_2, \dots]$ of symmetric functions. On Newton power sums the Adams operation acts by $\psi_n(p_m) = p_{mn}$ (Macdonald, *Symmetric Functions and Hall Polynomials*, Ch. I §8).

Remark 3.2.4. The big Witt vector functor $W: \text{CRing} \rightarrow \text{Ring}_\Lambda$ is right adjoint to the forgetful functor. $W(R)$ is the cofree Λ -ring on R . The ghost map $W(R) \hookrightarrow R^{\mathbb{N}^\times}$ realizes the Λ -structure as a tuple of commuting Frobenius lifts.

On $\text{Spec } \mathbb{Z}$, local translations are replaced by a monoid $\mathbb{N}^\times$ of commuting Adams operations $\{\psi_p\}_{p \in \mathbb{P}}$, one per prime. Accordingly, any chiral-algebra structure over $\overline{C}$ carries a $\mathbb{N}^\times$ -action by Adams operations.

3.3 THE PRIMON VERTEX ALGEBRA (NAIVE FORM)

Definition 3.3.1 (primon vertex algebra, naive). For each prime p , let $\mathcal{H}^{(p)}$ be the rank-2 Heisenberg vertex algebra with generators $a^{(p)}, b^{(p)}$ and OPE

$$\begin{aligned} a^{(p)}(z) a^{(p)}(w) &\sim \frac{1}{(z-w)^2}, \\ b^{(p)}(z) b^{(p)}(w) &\sim \frac{1}{(z-w)^2}, \\ a^{(p)}(z) b^{(q)}(w) &\sim 0 \quad \forall p, q. \end{aligned}$$

Endow $\mathcal{H}^{(p)}$ with the rescaled conformal vector

$$\omega_p := \frac{1}{2}(\log p)(a_{-1}^{(p)} a_{-1}^{(p)} + b_{-1}^{(p)} b_{-1}^{(p)})|o\rangle_p,$$

so that $L_o^{(p)}$ has eigenvalues in $(\log p) \cdot \mathbb{Z}_{\geq 0}$ on the Fock vacuum module. Define the *primon vertex algebra*

$$\mathcal{V}^{\text{prim}} := \bigotimes_{p \in \mathbb{P}}^! \mathcal{H}^{(p)}, \quad \omega := \sum_p \omega_p,$$

as the restricted tensor product pinned by the vacuum $\bigotimes_p |o\rangle_p$.

PROPOSITION 3.3.2 (*character match, naive*). . Let $N_p := a_o^{(p)} - b_o^{(p)}$ be the $U(1)^{(p)}$ -charge current zero-mode. Then

$$\chi_{\mathcal{V}^{\text{prim}}}(s; \{\theta_p\}) := \text{Tr} \left(e^{-s L_o} \prod_p e^{i \theta_p N_p} \right) = \prod_{p \in \mathbb{P}} \frac{1}{1 - 2 \cos \theta_p p^{-s} + p^{-2s}},$$

which matches the Hartnoll–Yang primon partition function (arXiv:2502.02661, eqs. 14–15) with identification $c_p^k = 2 \cos \theta_p^k$ and level index k absorbed into the vacuum sector.

Proof sketch. On the rank-2 Fock, $e^{-s L_o^{(p)}} e^{i \theta_p N_p}$ gives $[(1 - e^{-s \log p + i \theta_p})(1 - e^{-s \log p - i \theta_p})]^{-1} = [1 - 2 \cos \theta_p p^{-s} + p^{-2s}]^{-1}$. Tensor over primes. $\square$

3.4 THE HECKE-ADAMS LIFT (NAIVE FORM)

Claim (naive–falsified downstream) 3.4.1. Each Adams operation ψ_n of Λ lifts to a vertex-algebra endomorphism $T_n: \mathcal{V}^{\text{prim}} \rightarrow \mathcal{V}^{\text{prim}}$, defined on prime-indexed modes by

$$T_n(a_{-k}^{(p)}) := \sum_{d|n} d \cdot a_{-nk/d}^{(p)} \cdot \mathbf{1}_{d|k},$$

and similarly for $b^{(p)}$. The $\{T_n\}_{n \geq 1}$ commute and multiply as $T_m T_n = \sum_{d|\gcd(m,n)} d \cdot T_{mn/d^2}$ (the classical GL_2 Hecke algebra presentation), yielding the two-term recursion

$$T_{p^{k+1}} = T_p T_{p^k} - p T_{p^{k-1}}$$

matching Hartnoll–Yang eq. (11).

Warning 3.4.2 (Falsified in Chapter 5). The formula defines a GL_1 Adams action; the GL_2 Hecke algebra relation requires also $T_{p^2}(a_{-1}^{(p)}|o\rangle) = T_p^2(a_{-1}^{(p)}|o\rangle) - p \cdot a_{-1}^{(p)}|o\rangle$, which the above formula fails: see Chapter 5 for the explicit computation. The corrected statement diagonalises the rank-2 Heisenberg generators onto the Satake pair (α_p, β_p) with $\alpha_p \beta_p = p$.

3.5 THE AVERAGING MAP (NAIVE FORM)

Definition 3.5.1 (averaging, naive). Write $\mathcal{V}^{E_1, \text{arith}}$ and $\mathcal{V}^{\text{Hecke}}$ for the E_1 -chiral algebra on $\overline{C}$ and its Hecke quotient, both constructed in Chapter 8. Define $\text{av}^{\text{arith}}: \mathcal{V}^{E_1, \text{arith}} \rightarrow \mathcal{V}^{\text{Hecke}}$ by integration against the Kesten–McKay measure: on input an ordered product $a_{-k_1}^{(p_1)} \cdots a_{-k_N}^{(p_N)}|o\rangle$,

$$\text{av}^{\text{arith}}(\cdot) := \int_{\prod_i [0, \pi]} \prod_i \nu_{p_i}(\theta_{p_i}) e^{i \sum_i \theta_{p_i} N_{p_i}}(\text{input}) d\theta_{p_1} \cdots d\theta_{p_N},$$

where $\nu_p(\theta) = (2/\pi) \cdot p(1+p) \sin^2 \theta / (1+p^2 - 2p \cos 2\theta)$ is the Kesten–McKay measure.

Warning 3.5.2 (Reframed in Chapter 7). The map av^{arith} is the *Plancherel measure* of unramified $GL_2(\mathbb{Q}_p)$: a spectral decomposition whose content is the local Plancherel formula for GL_2 . The bar–cobar averaging map of Vol I is an operadic map from ordered to symmetric chiral cohomology; the two operations are distinct.

3.6 THE FOUR-ARCHETYPE ECHO (HEURISTIC)

(heuristic pattern match only). Classify Λ -rings by depth:

- **Depth 2:** trivial ψ_n beyond ψ_1 ; *archetype G analogue* (primon Heisenberg only);
- **Depth 3:** finitely generated image in $\prod_p \mathbb{Z}$; *archetype L analogue* (arithmetic Kac–Moody / Iwasawa-type Λ -module);
- **Depth 4:** higher-derived Λ -structure engaging Witt filtration; *archetype C analogue* (arithmetic $\beta\gamma$);
- **Depth ∞ :** full Connes–Consani arithmetic site; *archetype M analogue* (arithmetic Virasoro).

Remark 3.6.1. The depth numerology $\{2, 3, 4, \infty\}$ on the Vol I side is forced by explicit shadow-tower computation (Vol I, shadow-tower chapter). On the arithmetic side the corresponding derivation via a Λ -ring analogue of Mok’s shadow-tower rigidity remains open; the classification above is a heuristic pattern match.

3.7 THEOREM ANALOGUES (NAIVE STATEMENTS)

Conjecture 3.7.1 (A^{arith} , naive). There is an adjunction $\Omega^{\text{ch,arith}} \dashv B^{\text{ch,arith}}$ between the ∞ -category of Λ -ring-enhanced E_1 -algebras on $\overline{C}$ and the ∞ -category of conilpotent Λ -coalgebras. On $\mathcal{V}^{\text{prim}}$ this adjunction is an equivalence on the subcategory of Λ -ring objects with prime-adic convergence at every finite place.

Conjecture 3.7.2 (B^{arith} , naive). On Λ -coalgebras satisfying conilpotence at every prime (μ -invariant = 0 in Iwasawa-theoretic sense), $\Omega^{\text{ch,arith}} B^{\text{ch,arith}}(C) \xrightarrow{\sim} C$ in the derived category of coalgebras.

Conjecture 3.7.3 (C^{arith} , naive). Define the arithmetic κ of a Λ -chiral algebra as its Arakelov first Chern class (sum of finite-place p -adic orders plus archimedean logarithmic defect). On the Koszul self-dual arithmetic subcategory, $\kappa + \kappa^! \in \{0, \log 2, \log 3, \dots\} \cup \{0, 1, 2, \dots\}$ with the precise ceiling unknown.

Warning 3.7.4 (Sharpened in Chapter 21). The correct arithmetic G-row ceiling is 0, and its content is the Riemann functional equation $\xi(s) = \xi(1-s)$.

Conjecture 3.7.5 (D^{arith} , naive). The genus- g obstruction for the arithmetic Maurer–Cartan element Θ_A^{arith} equals $\kappa \cdot h_g^{\text{Ar}}$ where h_g^{Ar} is the Arakelov height of the level- g component of $\overline{M}_{g,n}$.

Conjecture 3.7.6 (H^{arith} , naive). $\text{HH}^\bullet(\mathcal{V}^{\text{prim}})$ is concentrated in degrees $\{0, 1, 2\}$:

$$\begin{aligned} \text{HH}^0 &= \mathbb{Z} \quad (\Lambda\text{-ring augmentation}), \\ \text{HH}^1 &= \bigoplus_p \log p \cdot \mathbb{Z} \quad (\text{Bost–Connes KMS generators}), \\ \text{HH}^2 &= \text{a } \Lambda\text{-torsor related to Witt cotangent.} \end{aligned}$$

3.8 OPEN TECHNICAL PROBLEMS

Four technical problems in the construction above remain open; the following chapters treat each in turn.

- (i) **p -adic locality.** The OPE locality axiom $[a(z), b(w)] = 0$ off-diagonal requires an interpretation of the formal variable $z - w$ at finite places of $\overline{C}$: see Chapter 4.
- (ii) **Adelic factorization.** Beilinson–Drinfeld factorization on a curve employs $C^n/\mathfrak{S}_n$. On $\overline{C}$ the two candidate analogues are $\mathcal{W}_n(\mathbb{Z})/\mathfrak{S}_n$ and $\mathrm{Spec}(\Lambda^{\otimes n})^{\mathfrak{S}_n}$; their comparison is settled in Chapter 8.
- (iii) **Real place.** The archimedean place carries an $\mathrm{SL}(2, \mathbb{R})$ conformal quantum mechanics (Hartnoll–Yang); the non-archimedean places carry $\mathrm{SL}(2, \mathbb{Q}_p)$. The adelic $\mathrm{SL}(2)$ -automorphic picture gluing these is developed in Chapter 11.
- (iv) **Four-archetype derivation.** The shadow-tower depth numerology on the arithmetic side requires a Λ -ring analogue of Mok’s shadow-tower rigidity. Open throughout.

THE p -ADIC VERTEX ALGEBRA LOCALITY PROBLEM

Vertex algebras on curves arise from chiral and factorization algebras. Kapranov’s arithmetic loop analogy and the Kapranov–Vasserot formal-loop construction both remain over fields. No published construction gives a vertex algebra on $\text{Spec } \mathbb{Z}$ itself.

4.1 THE ALGEBRAIC STEP: FORMAL VA OVER $\mathbb{Q}_p$

A vertex algebra over a field k of characteristic zero is a vector space V_k with a state-field map

$$Y : V_k \otimes V_k \rightarrow V_k((z))$$

satisfying axioms that are polynomial identities in formal variables (Kac, *Vertex Algebras for Beginners*, 2nd ed., 1998, §1.3). These polynomial identities are field-independent, so replacing $\mathbb{C}$ by $\mathbb{Q}_p$ as the coefficient field is immediate at the algebraic level: the Heisenberg Fock space $\mathcal{H}_{\mathbb{Q}_p}^{(p)} = \text{Sym}(\mathbb{Q}_p[t] \cdot a)$ with modes $[a_m, a_n] = m\delta_{m+n,0}$ and the operator product expansion (OPE) $a(z)a(w) \sim 1/(z-w)^2$ (singular part of the product) are identities of formal power series in $\text{End}(\mathcal{H}^{(p)})[[z^{\pm 1}, w^{\pm 1}]]$.

PROPOSITION 4.1.1. (citation-level; Kac Thm 4.3a). For any field k of characteristic zero, the rank-2 Heisenberg VA $\mathcal{H}_k^{(p)} = \text{Sym}_k(k[t](a \oplus b))$ with standard OPE exists as a formal VA over k .

The algebraic step is purely formal; the arithmetic content resides in the conformal-vector scaling, the diagonal completion, and the comparison with the local automorphic object on each finite place. The three obstructions are: conformal-vector scaling (§4.2), coefficient-field mismatch (§4.3), and diagonal non-thickening (§4.4).

4.2 CONFORMAL VECTOR SCALING: ARCHIMEDEAN, NOT p -ADIC

The natural conformal vector for a local Heisenberg factor at place p would weight the L_0 -grading by $\log p$, the archimedean size of p . The question is whether $\log p$ has a p -adic interpretation.

Observation 4.2.1. The proposed conformal vector $\omega_p = \frac{1}{2}(\log p)(a_{-1}^2 + b_{-1}^2)|0\rangle$ has a complex scalar $\log p \in \mathbb{R}$. If $\mathcal{H}^{(p)}$ is a $\mathbb{Q}_p$ -VA, then $\omega_p \in \mathcal{H}^{(p)}$ requires $\log p \in \mathbb{Q}_p$. The only candidate is the Iwasawa p -adic logarithm $\log_p : \mathbb{Z}_p^\times \rightarrow \mathbb{Q}_p$, which by convention sets $\log_p(p) = 0$ (Iwasawa, *Lectures on p -adic L -functions*, Princeton 1972, §1; Koblitz, *p -adic Analysis*, GTM 58, Ch. IV).

Under Iwasawa’s $\log_p$, the p -adic value of ω_p has zero conformal-vector coefficient. The scalar $\log p \in \mathbb{R}$ is the *scale* at which $\mathcal{H}^{(p)}$ contributes to the global partition function (Hartnoll–Yang arXiv:2502.02661 §3); it is an archimedean datum external to the $\mathbb{Q}_p$ -algebra structure. The central charge of ω_p is 2 regardless; the L_0 -grading carries the $\log p$ through the archimedean completion.

Convention 4.2.2. $\log p$ is archimedean-real scaling data, external to the p -adic algebra structure. At finite place p the VA is $\mathcal{H}_{\mathbb{Q}_p}^{(p)}$ with vacuum conformal vector $\frac{1}{2}(a_{-1}^2 + b_{-1}^2)|o\rangle$; the rescaling enters through external grading data at the archimedean completion.

4.3 THE RESTRICTED TENSOR PRODUCT: COEFFICIENT-FIELD MISMATCH

To assemble the local Heisenberg factors $\{\mathcal{H}^{(p)}\}_p$ into a global object over $\overline{\mathbb{C}}$, one would form a restricted tensor product over unramified vectors. The obstruction is that $\mathcal{H}^{(p)}$ and $\mathcal{H}^{(q)}$ for $p \neq q$ live over different coefficient fields.

Observation 4.3.1. The restricted tensor product $\bigotimes_p^! \mathcal{H}^{(p)}$ has a coefficient-field obstruction: $\mathcal{H}^{(p)}$ is a vector space over $\mathbb{Q}_p$, and the fields $\mathbb{Q}_p$ for distinct primes p have no common field extension in the filtered colimit of finite tensor products. Forming a single vector space from $\{\mathcal{H}^{(p)}\}_p$ requires passing to a ring (the adèle ring) rather than a field.

Three approaches address the coefficient mismatch.

- (i) **(Base-extend to $\mathbb{C}$):** Each $\mathcal{H}^{(p)}$ is base-extended to $\mathbb{C}$ before tensoring, recovering the standard complex Heisenberg VA. The result is a complex VA carrying only archimedean data, with the $\mathbb{Q}_p$ -structure collapsed by base extension.
- (ii) **(Base-extend to $\mathbb{A}_{\mathbb{Q}}$):** Tensor over the adèle ring $\mathbb{A}_{\mathbb{Q}}$, which is a ring (not a field). The result is an adelic module carrying Hecke-module structure at each finite place; its algebraic structure is that of a module over $\mathbb{A}_{\mathbb{Q}}$, distinct from a classical vertex algebra.
- (iii) **(Form functor):** Let $\text{Fields}/\mathbb{Q}$ denote the category of field extensions of $\mathbb{Q}$. Define $\mathcal{V}^{\text{prim}}: \text{Fields}/\mathbb{Q} \rightarrow \text{VA}$ with $\mathcal{V}^{\text{prim}}(\mathbb{Q}_p) = \mathcal{H}^{(p)}$ and $\mathcal{V}^{\text{prim}}(\mathbb{C}) = \text{complex Heisenberg}$. This is a form (in the sense of Galois descent) of the complex Heisenberg, requiring a Galois-action datum and Frobenius-compatibility to be well-defined.

Warning 4.3.2. Option (a) recovers only complex data. Option (b) produces an adelic module with Hecke-module structure at each finite place, studied further in Chapter 4.6.1. Option (c) requires arithmetic input (Galois action, Frobenius compatibility) beyond the naive algebraic construction.

4.4 THE p -ADIC DIAGONAL: TRIVIAL FORMAL NEIGHBOURHOOD

The Beilinson–Drinfeld chiral bracket is a residue at the diagonal of $C \times C$. On a complex curve, the formal neighbourhood of the diagonal $\Delta \hookrightarrow C \times C$ is a formal disk $\text{Spf } \mathbb{C}[[t]]$, which supports the full OPE. At a finite place p , the corresponding question is whether the diagonal of $\text{Spec } \mathbb{Z}_p \times_{\text{Spec } \mathbb{Z}} \text{Spec } \mathbb{Z}_p$ admits a non-trivial formal thickening. The completed tensor product answers this directly.

PROPOSITION 4.4.1. . The formal completion $\mathbb{Z}_p \widehat{\otimes}_{\mathbb{Z}} \mathbb{Z}_p \cong \mathbb{Z}_p$, because $\mathbb{Z}_p$ is p -adically complete and the diagonal is already an isomorphism after completion.

Proof. Write $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n$. Since $\mathbb{Z}_p$ is flat over $\mathbb{Z}$, the tensor product $(-) \otimes_{\mathbb{Z}} \mathbb{Z}_p$ commutes with this sequential inverse limit (Weibel, *An Introduction to Homological Algebra*, Cor. 3.2.10); and $\mathbb{Z}_p$ is a discrete valuation ring with uniformizer p , so $\mathbb{Z}/p^n \otimes_{\mathbb{Z}} \mathbb{Z}_p = \mathbb{Z}_p/p^n \mathbb{Z}_p = \mathbb{Z}/p^n$. Hence

$$\mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{Z}_p = \varprojlim_n (\mathbb{Z}/p^n \otimes_{\mathbb{Z}} \mathbb{Z}_p) = \varprojlim_n \mathbb{Z}/p^n = \mathbb{Z}_p.$$

The result is already p -adically complete, so the completed and ordinary tensor products coincide. The diagonal is the identity map, and $\mathrm{Spec}(\mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{Z}_p) = \mathrm{Spec} \mathbb{Z}_p$ is one-dimensional; the diagonal is an isomorphism, so the formal neighbourhood is trivial. $\square$

COROLLARY 4.4.2 (*Locally constant structure at finite places*). . At each finite place p , the Beilinson–Drinfeld chiral bracket $\mu^{\mathrm{ch}}: j_* j^*(\mathcal{A} \boxtimes \mathcal{A}) \rightarrow \Delta_! \mathcal{A}$ degenerates to the locally constant structure on $\mathrm{Spf} \mathbb{Z}_p$: the formal neighbourhood of the diagonal is trivial (Chapter 4.4.1), so the residue-at-diagonal construction collapses. The archimedean place supports the full holomorphic OPE singularity.

4.5 THE LOCALLY CONSTANT (E_o) LOCAL MODEL AT EACH FINITE PLACE

The absence of a formal thickening of the diagonal forces the local factor at each finite place to carry only the structure available on a locally constant prestack: commutative multiplication and Hecke endomorphisms, with no OPE singularity.

Definition 4.5.1. The local factor of $\mathcal{V}^{\mathrm{prim}}$ at a finite place p is the locally constant prestack $\mathcal{V}^{\mathrm{prim}}|_p$ on $\mathrm{Spf} \mathbb{Z}_p$ whose constant stalk is $\mathcal{H}^{(p)}$ equipped with the p -adic filtration and a compatible Hecke–Adams endomorphism $T_p: \mathcal{H}^{(p)} \rightarrow \mathcal{H}^{(p)}$. This is a factorization algebra in the degenerate (E_o) sense: the “fusion” $\mathcal{H}^{(p)} \otimes \mathcal{H}^{(p)} \rightarrow \mathcal{H}^{(p)}$ is the commutative multiplication (constant along $\mathrm{Spf} \mathbb{Z}_p$), with no singular diagonal term.

THEOREM 4.5.2 (*global structure at archimedean place*). . At ∞ , $\mathcal{V}^{\mathrm{prim}}|_{\infty}$ is the ordinary Heisenberg vertex algebra on a germ $(\mathbb{C}, o)$, a genuine Beilinson–Drinfeld/Costello–Gwilliam factorization algebra with non-trivial diagonal singularity.

4.6 THE ADELIC AUTOMORPHIC PLACEMENT OF $\mathcal{V}^{\mathrm{prim}}$

THEOREM 4.6.1 (*adelic automorphic placement*). . $\mathcal{V}^{\mathrm{prim}}$ is an adelic automorphic object: a compatible system of local factors $\{\mathcal{V}^{\mathrm{prim}}|_v\}_{v \in |\overline{C}|}$ with $\mathcal{V}^{\mathrm{prim}}|_{\infty}$ a genuine Heisenberg VA at the archimedean place and $\mathcal{V}^{\mathrm{prim}}|_p$ an E_o -algebra (Fock module with Hecke–Adams endomorphism) at each finite place. Under Tate’s restricted tensor product over spherical vectors $|o\rangle_p$ (Tate 1950; Weil 1967, *Basic Number Theory*, Ch. VII), the system assembles into an automorphic representation of $GL_2(\mathbb{A}_{\mathbb{Q}})$ with trivial ramification at unramified primes.

Remark 4.6.2. The classical chiral-algebras formalism of Beilinson–Drinfeld (AMS Colloq. 51, 2004) applies to a curve over a field. Over the arithmetic curve $\overline{C} = \overline{\mathrm{Spec} \mathbb{Z}}$ (Arakelov compactification, with archimedean place ∞ adjoined), the appropriate structure is a compatible system: a factorization-algebra factor at ∞ and a locally constant factor at each finite place, tied together by automorphic compatibility (Tate-style restricted tensor product). The global object is a spectral datum; its OPE singularity is supported entirely at ∞ .

4.7 KAPRANOV–VASSEROT: FORMAL LOOP SPACES OVER FIELDS

Remark 4.7.1. Kapranov–Vasserot’s four-paper series *Vertex algebras and the formal loop space* (arXiv:math/0107143, Publ. IHES 100, 2004; and subsequent math/0612371, etc.) introduces the ind-scheme $\mathcal{L}X$ of formal loops into a finite-type scheme X over a field and shows that factorization structure on $\mathcal{L}X$ recovers the VA structure on chiral de Rham. The series does not extend to $\mathrm{Spec} \mathbb{Z}$. Formulating the arithmetic analogue over $\mathrm{Spec} \mathbb{Z}$ requires the adelic automorphic structure of Chapter 4.6.1.

Remark 4.7.2. Kapranov’s 1995 unpublished preprint “Analogies between the Langlands correspondence and topological quantum field theory” *motivates* the search for an arithmetic chiral-algebra formalism but does not construct one. Kapranov–Smirnov’s *Cohomology determinants and reciprocity laws: number field case* (also unpublished, ca. 1995) gives an $\mathbb{F}_1$ -style analogy via cohomology determinants feeding reciprocity laws; this is the foundational informal reference for the $\mathbb{F}_1$ literature (Manin arXiv:1312.5160; Thas, *F₁ for Everyone*, 2018; Lorscheid survey).

4.8 LOCAL ARITHMETIC BOUNDARY

- (i) A formal VA over $\mathbb{Q}_p$ exists algebraically (Chapter 4.1.1); the arithmetic content enters through the archimedean $\log p$ scaling, which is external to the $\mathbb{Q}_p$ -algebra structure (Chapter 4.2.1).
- (ii) The restricted tensor product $\otimes_p^! \mathcal{H}^{(p)}$ has a coefficient-field obstruction (Chapter 4.3.1); it is an adelic module or a form-functor $\text{Fields}/\mathbb{Q} \rightarrow \text{VA}$.
- (iii) The p -adic diagonal is trivial after completion (Chapter 4.4.1); the Beilinson–Drinfeld chiral bracket degenerates to the locally constant structure at each finite place (Chapter 4.4.2).
- (iv) Each finite place carries an E_o -algebra (Chapter 4.5.1); the archimedean place carries a genuine Heisenberg VA (Chapter 4.5.2).
- (v) $\mathcal{V}^{\text{prim}}$ is an adelic automorphic datum (Chapter 4.6.1): its OPE lives at ∞ ; at each finite place p it carries an E_o -algebra factor with Hecke–Adams endomorphism.

The three arithmetic obstructions identified (conformal-vector scaling, coefficient-field mismatch, and diagonal non-thickening) force the adelic automorphic structure of Chapter 4.6.1.

HECKE VERSUS ADAMS: THE GL_1 / GL_2 TYPE ERROR

Adams operations on a Λ -ring and Hecke operators for GL_2 live in different tensor categories. The comparison below isolates the Satake pair and the exact point at which the scalar GL_1 operation ceases to be a GL_2 Hecke operator.

5.1 THE SATAKE PAIR AND WHY IT MATTERS

THEOREM 5.1.1 (*Satake 1963; Macdonald 1971; Gross 1998*). The spherical Hecke algebra $\mathcal{H}_p^{GL_2} = C_c^\infty(GL_2(\mathbb{Z}_p) \backslash GL_2(\mathbb{Q}_p) / GL_2(\mathbb{Z}_p))$ is a two-variable commutative ring; the Satake isomorphism reads

$$\text{Sat}_p: \mathcal{H}_p^{GL_2} \xrightarrow{\sim} \mathbb{Z}[\alpha_p^{\pm 1}, \beta_p^{\pm 1}]^{S_2} = \mathbb{Z}[\alpha_p + \beta_p, \alpha_p \beta_p, (\alpha_p \beta_p)^{-1}],$$

with $T_p \mapsto \alpha_p + \beta_p$ and the spherical determinant $\langle p \rangle$ (characteristic function of $GL_2(\mathbb{Z}_p) \text{diag}(p, p) GL_2(\mathbb{Z}_p)$) mapping to $\alpha_p \beta_p$, in the arithmetic normalisation where $\alpha_p \beta_p = p$ on cuspidal weight-0 forms with trivial central character (Ramanujan bound $|\alpha_p| = |\beta_p| = \sqrt{p}$). The equivalent unitary form (Gross 1998) is $T_p \mapsto p^{1/2}(\alpha_p^u + \beta_p^u)$ under the rescaling $\alpha_p = \sqrt{p} \alpha_p^u$, $\alpha_p^u \beta_p^u = 1$; the arithmetic and unitary L-functions differ by the shift $s \mapsto s - 1/2$.

PROPOSITION 5.1.2 (*Hecke generating function*). (Shimura, *Introduction to the Arithmetic Theory of Automorphic Functions*, Princeton 1971, Thm 3.24). As an element of $\mathcal{H}_p^{GL_2}[[X]]$,

$$\sum_{k \geq 0} T_p^k X^k = \frac{1}{1 - T_p X + p \langle p \rangle X^2}.$$

On a Hecke eigenform f with $T_p f = a_p f$ and trivial central character, this gives

$$\sum_{k \geq 0} a_p^k X^k = \frac{1}{(1 - \alpha_p X)(1 - \beta_p X)},$$

which is the local Euler factor of the Hartnoll–Yang primon partition (arXiv:2502.02661, eqs. 14–15). The poles α_p, β_p are *two distinct numbers*; the recursion is driven by the degree-two characteristic polynomial $t^2 - a_p t + p$.

5.2 WHY ADAMS IS NOT HECKE

PROPOSITION 5.2.1. (Borger arXiv:0906.3146, Prop. 1.2, Wilkerson criterion). Adams operations on a Λ -ring $(R, \{\psi_n\})$ (a commutative ring R equipped with ring endomorphisms ψ_n for each $n \geq 1$, satisfying the Wilkerson integrality conditions; Knutson 1973) satisfy $\psi_m \psi_n = \psi_{mn}$; a *free commutative monoid* acting, parametrised by $(\mathbb{N}_{\geq 1}, \times)$. This is the GL_1 Hecke algebra: one Cartan-torus generator per prime.

Observation 5.2.2 (structural mismatch). GL_1 Hecke = Adams = Λ -ring. GL_2 Hecke has *two* generators per prime: T_p (trace of Satake pair) and $\langle p \rangle$ (determinant of Satake pair). No Λ -ring structure on a commutative ring can encode the unordered pair $\{\alpha_p, \beta_p\}$ modulo S_2 ; it only encodes the product $p = \alpha_p \beta_p$ and not the individual parameters.

5.3 EXPLICIT COMPUTATION: THE RECURSION FAILS

Let $\mathcal{H}_{\text{rkl}}^{(p)}$ denote the rank-1 Heisenberg algebra at prime p , with mode operators $a_n^{(p)}$ ($n \in \mathbb{Z}$) satisfying $[a_m^{(p)}, a_n^{(p)}] = m \delta_{m+n,0}$, acting on the Fock space $\mathcal{F}^{(p)}$ generated from the vacuum $|\mathbf{o}\rangle$ (annihilated by $a_n^{(p)}$ for $n \geq 0$). The one-particle states $a_{-k}^{(p)}|\mathbf{o}\rangle$ ($k \geq 1$) form a basis for the degree- k subspace of $\mathcal{F}^{(p)}$.

The claimed formula for the action of a putative GL_2 Hecke operator T_n on this Fock space is

$$T_n(a_{-k}^{(p)}|\mathbf{o}\rangle) = \sum_{d|n} d \cdot a_{-nk/d}^{(p)}|\mathbf{o}\rangle \cdot \mathbf{1}_{d|k},$$

extended by $T_n(|\mathbf{o}\rangle) = |\mathbf{o}\rangle$.

COMPUTATION 5.3.1. $T_p(a_{-1}^{(p)}|\mathbf{o}\rangle)$. Divisors of $n = p$ are $d \in \{1, p\}$. Only $d = 1$ satisfies $d \mid k = 1$. So

$$T_p(a_{-1}^{(p)}|\mathbf{o}\rangle) = a_{-p}^{(p)}|\mathbf{o}\rangle.$$

COMPUTATION 5.3.2. $T_p^2(a_{-1}^{(p)}|\mathbf{o}\rangle)$. Apply T_p to $a_{-p}^{(p)}|\mathbf{o}\rangle$: divisors $d \in \{1, p\}$, both satisfy $d \mid p$, so

$$T_p(a_{-p}^{(p)}|\mathbf{o}\rangle) = a_{-p^2}^{(p)}|\mathbf{o}\rangle + p \cdot a_{-p}^{(p)}|\mathbf{o}\rangle.$$

Therefore

$$T_p^2(a_{-1}^{(p)}|\mathbf{o}\rangle) = a_{-p^2}^{(p)}|\mathbf{o}\rangle + p \cdot a_{-p}^{(p)}|\mathbf{o}\rangle.$$

COMPUTATION 5.3.3. $T_{p^2}(a_{-1}^{(p)}|\mathbf{o}\rangle)$. Divisors $d \in \{1, p, p^2\}$, only $d = 1$ satisfies $d \mid 1$. So

$$T_{p^2}(a_{-1}^{(p)}|\mathbf{o}\rangle) = a_{-p^2}^{(p)}|\mathbf{o}\rangle.$$

Observation 5.3.4 (recursion discrepancy). Testing $T_{p^2} = T_p^2 - p \cdot \text{id}$ on $a_{-1}^{(p)}|\mathbf{o}\rangle$:

$$\begin{aligned} T_p^2 - p \cdot \text{id} \big|_{a_{-1}^{(p)}|\mathbf{o}\rangle} &= a_{-p^2}^{(p)}|\mathbf{o}\rangle + p \cdot a_{-p}^{(p)}|\mathbf{o}\rangle - p \cdot a_{-1}^{(p)}|\mathbf{o}\rangle, \\ T_{p^2} \big|_{a_{-1}^{(p)}|\mathbf{o}\rangle} &= a_{-p^2}^{(p)}|\mathbf{o}\rangle. \end{aligned}$$

The discrepancy is $p(a_{-p}^{(p)} - a_{-1}^{(p)})|\mathbf{o}\rangle \neq \mathbf{o}$.

COROLLARY 5.3.5 (*falsification*). . The formula $T_n(a_{-k}^{(p)}) = \sum_{d|n} d \cdot a_{-nk/d}^{(p)} \mathbf{1}_{d|k}$ satisfies $T_m T_n = T_{mn}$ (Adams, GL_1) but not the GL_2 Hecke recursion $T_{p^2} = T_p^2 - p \cdot \text{id}$.

5.4 DIAGONALISATION ONTO THE SATAKE PAIR

The failure above is structural: the rank-1 Heisenberg $\mathcal{H}_{\text{rkl}}^{(p)}$ has only one generator per prime, hence only one eigenvalue, hence only the GL_1 Cartan-torus generator. To carry the full GL_2 Hecke action, the local factor at p must be a rank-2 Heisenberg algebra $\mathcal{H}^{(p)}$, with two independent sets of mode operators $a^{(p)}$ and $b^{(p)}$, one for each Satake eigenvalue.

Definition 5.4.1 (Satake-diagonalised assignment). The rank-2 Heisenberg $\mathcal{H}^{(p)}$ has two commuting generators per prime. Assign:

$$\begin{aligned} a_{-1}^{(p)} |o\rangle &\text{ carries Satake weight } \alpha_p, \\ b_{-1}^{(p)} |o\rangle &\text{ carries Satake weight } \beta_p, \end{aligned}$$

with constraint $\alpha_p \beta_p = p$ encoded in the conformal vector $\omega_p(\theta_p) = \frac{1}{2}(\log p)((a_{-1}^{(p)})^2 + (b_{-1}^{(p)})^2)|o\rangle$ via $\log \alpha_p + \log \beta_p = \log p$.

THEOREM 5.4.2 (corrected Hecke action). Under Chapter 5.4.1, the GL_2 Hecke operator $T_p^{GL_2}$ acts on one-particle states $c_1 a_{-k}^{(p)} |o\rangle + c_2 b_{-k}^{(p)} |o\rangle$ as the diagonal matrix

$$T_p^{GL_2} = \begin{pmatrix} \alpha_p & 0 \\ 0 & \beta_p \end{pmatrix}, \quad \langle p \rangle^{GL_2} = \alpha_p \beta_p \cdot I = p \cdot I.$$

On the trace-symmetric sector (invariants under S_2 swap $a^{(p)} \leftrightarrow b^{(p)}$), T_p acts as the scalar $\alpha_p + \beta_p = 2p^{1/2} \cos \theta_p$, which equals the Hartnoll–Yang Hecke eigenvalue up to the factor $p^{1/2}$ absorbed by the central-character shift $s \mapsto s - 1/2$ in the L -function normalisation.

COROLLARY 5.4.3 (Hecke recursion on symmetric sector). On $\text{Sym}^k(F_p^1)$ with $F_p^1 = \mathbb{C} \cdot a_{-1}^{(p)} |o\rangle \oplus \mathbb{C} \cdot b_{-1}^{(p)} |o\rangle$,

$$\text{Tr}_{\text{Sym}^k}(T_p) = \alpha_p^k + \alpha_p^{k-1} \beta_p + \cdots + \beta_p^k = h_k(\alpha_p, \beta_p),$$

the complete symmetric function, and the three-term recursion $h_{k+1} = (\alpha_p + \beta_p)h_k - \alpha_p \beta_p \cdot h_{k-1}$ (Macdonald 1971, Ch. IV §1) translates to $T_{p^{k+1}} = T_p T_{p^k} - p T_{p^{k-1}}$.

5.5 THE ANGLE θ_p AS UNRAMIFIED LANGLANDS PARAMETER

The angle θ_p first appeared in Definition 5.4.1 as the argument of the Satake parameters $\alpha_p = p^{1/2} e^{i\theta_p}$, $\beta_p = p^{1/2} e^{-i\theta_p}$. The question is whether θ_p is a free internal parameter of the Heisenberg algebra or is fixed by external (Langlands) data. The answer is the latter: θ_p is the unramified local Langlands parameter at p .

PROPOSITION 5.5.1. (Borel, “Automorphic L-functions,” PSPM 33.2, 1979). In the Langlands normalisation ($\alpha_p = p^{1/2} e^{i\theta_p}$, $\beta_p = p^{1/2} e^{-i\theta_p}$, so that $\alpha_p \beta_p = p$), the angle θ_p is the unramified local Langlands parameter at p : the unordered pair $\{p^{1/2} e^{i\theta_p}, p^{1/2} e^{-i\theta_p}\} \in (\mathbb{C}^\times)^2 / S_2$.

Convention 5.5.2. $\{\theta_p\}_p$ is chosen once and for all as a map $\theta: \text{Spec } \mathbb{Z} \setminus \{\infty\} \rightarrow S^1 / \sigma$ (the angle modulo $\theta \leftrightarrow -\theta$, since $2 \cos \theta = 2 \cos(-\theta)$). This is the data of an unramified local Langlands parameter at each finite place. The Hartnoll–Yang L-function

$$L(s) = \prod_p \frac{1}{1 - 2 \cos \theta_p p^{-s} + p^{-2s}}$$

is the standard L -function of an automorphic representation of $GL_2(\mathbb{A}_{\mathbb{Q}})$ with trivial central character (Gelbart–Jacquet, §2.1).

5.6 THE CHIRAL FRAMEWORK OVER $\overline{\text{Spec } \mathbb{Z}}$

The local Hecke action is now well-defined on each $\mathcal{H}^{(p)}$. Their restricted tensor product is the action of the adelic Hecke algebra on $\mathcal{V}^{\text{prim}}$. The chiral-algebras framework of Beilinson–Drinfeld over $\overline{C} = \overline{\text{Spec } \mathbb{Z}}$ provides the globalisation, at the cost of replacing complex-analytic residues by p -adic ones at finite places.

Remark 5.6.1. The chiral-algebras formalism of Beilinson–Drinfeld (AMS 2004) requires a smooth complex curve X and right D_X -module structure; the chiral bracket as residue along the diagonal uses complex-analytic residues. Over $\overline{C} = \overline{\text{Spec } \mathbb{Z}}$, the residue at a finite place p is the p -adic residue on the local spectral disk $\text{Spec } \mathbb{Q}_p$; over the archimedean place, it is the classical residue. The GL_2 -Hecke operator is a *global* object obtained from the restricted tensor product of local Hecke operators and lives in $\text{End}(\pi)$ for the global automorphic representation $\pi = \otimes'_v \pi_v$ (Jacquet–Langlands, SLN 114, 1970).

THEOREM 5.6.2 (corrected placement). $\mathcal{V}^{\text{prim}}$ is the restricted tensor product $\otimes'_p \mathcal{H}^{(p)}$ of local rank-2 Heisenberg factors, each carrying a Whittaker model of an unramified $GL_2(\mathbb{Q}_p)$ principal series. The GL_2 -Hecke action is *diagonal across primes*: each local $\mathcal{H}_p^{GL_2}$ acts on the corresponding $\mathcal{H}^{(p)}$; the global Hecke algebra $\mathcal{H}_{\text{glob}}^{\text{sph}} = \otimes'_p \mathcal{H}_p^{\text{sph}}$ acts on $\mathcal{V}^{\text{prim}}$ by restricted tensor product of local actions. The simultaneous-diagonalisation of $\mathcal{H}_{\text{glob}}^{\text{sph}}$ is the Hecke-eigenform decomposition of $\mathcal{V}^{\text{prim}}$.

5.7 THE CORRECTED LOCAL MODEL AND ITS CONSEQUENCES

Theorem 5.6.2 forces the local model at each prime to be the rank-2 Heisenberg $\mathcal{H}^{(p)}$, not a Λ -ring. The global consequence is a Hecke-eigenform decomposition of $\mathcal{V}^{\text{prim}}$ under the adelic spherical Hecke algebra $\mathcal{H}_{\text{glob}}^{\text{sph}}$, with each eigenvalue $\alpha_p + \beta_p = 2p^{1/2} \cos \theta_p$ determined by the unramified Langlands parameter θ_p (Chapter 5.5.1).

The Λ -ring Adams operations ψ_n on R satisfy $\psi_m \psi_n = \psi_{mn}$, which is the GL_1 multiplicativity law; the GL_2 Hecke recursion $T_{p^{k+1}} = T_p T_{p^k} - p T_{p^{k-1}}$ on $\text{Sym}^k(F_p^1)$ (Chapter 5.4.3) is strictly stronger, requiring the two-dimensional Satake pair (α_p, β_p) and the Weyl- S_2 symmetry that Λ -rings cannot encode.

Λ -RING AXIOMATIC PRECISION

Borger’s Λ -rings encode $\mathbb{F}_1$ -descent by commuting Frobenius lifts. The issue is whether the integral lattice underlying $\mathcal{V}^{\text{prim}}$ satisfies the Wilkerson congruences.

6.1 THE WILKERSON CONGRUENCE AND THE SCALAR RING

THEOREM 6.1.1 (*Wilkerson criterion, Borger 0906.3146 Prop. 1.2*). . A Λ -ring is a commutative ring R with commuting ring homomorphisms $\psi_n: R \rightarrow R$, $n \in \mathbb{N}^\times$, satisfying $\psi_1 = \text{id}$, $\psi_m \psi_n = \psi_{mn}$, and $\psi_p(x) \equiv x^p \pmod{pR}$ for every rational prime p . For torsion-free R the Wilkerson congruence is equivalent (Wilkerson 1982) to the classical Grothendieck λ -ring structure with λ^n operations.

Observation 6.1.2 (vacuous congruence over $\mathbb{Q}$). Every prime p is a unit in $\mathbb{C}$ or $\mathbb{Q}$. Therefore $pR = R$ for every prime, and the Wilkerson congruence $\psi_p(x) \equiv x^p \pmod{pR}$ is *vacuous* over $\mathbb{Q}$ -algebras: any two elements are congruent mod R . Under Borger’s axioms, a $\mathbb{Q}$ -algebra is automatically a Λ -ring iff it carries any compatible system of commuting ring endomorphisms ψ_n with $\psi_1 = \text{id}$ and $\psi_m \psi_n = \psi_{mn}$.

Remark 6.1.3. This weaker structure is what Grothendieck (Rev. Matem. Hispanoamer. 1971) called a ψ -ring or *Adams-ring*. The claim “ $\mathcal{V}^{\text{prim}}$ is a Λ -ring” is either (a) trivially true over $\mathbb{Q}$ and mathematically empty, or (b) a claim about an $\mathbb{F}_1$ - or $\mathbb{Z}$ -integral form $\mathcal{V}_{\mathbb{Z}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$ on which p is not invertible, which must be specified.

6.2 EXPLICIT COMPUTATIONS: ψ_2 ON LOW-WEIGHT STATES

The Wilkerson congruence requires a candidate Adams operation ψ_n on the Heisenberg Fock shadow of $\mathcal{V}^{\text{prim}}$. The full $\mathcal{V}^{\text{prim}}$ is a Hochschild trace class; the Fock calculation below takes place in the arithmetic Heisenberg sector. For each prime p , the operators $a_{-k}^{(p)}$ ($k \geq 1$) create a rank-1 harmonic oscillator mode of frequency k , acting on the vacuum $|0\rangle$. The algebra is $[a_m^{(p)}, a_n^{(p)}] = m\delta_{m+n,0}$. On single-mode states, the candidate Adams operation is

$$T_n(a_{-k}^{(p)}) = \sum_{d|n} d \cdot a_{-nk/d}^{(p)} \mathbf{1}_{d|k},$$

where $\mathbf{1}_{d|k} = 1$ if d divides k and 0 otherwise; the indicator ensures the mode index nk/d is an integer. This formula is extended to products by the ring-homomorphism axiom.

COMPUTATION 6.2.1. $\psi_2(a_{-1}^{(2)}|0\rangle) = a_{-2}^{(2)}|0\rangle$.

COMPUTATION 6.2.2. $\psi_2(a_{-1}^{(3)}|0\rangle) = a_{-2}^{(3)}|0\rangle$.

COMPUTATION 6.2.3. Under the ring-homomorphism axiom:

$$\psi_2(a_{-1}^{(2)} a_{-1}^{(3)}|0\rangle) = a_{-2}^{(2)} a_{-2}^{(3)}|0\rangle.$$

LEMMA 6.2.4 (*composition law*). . For $m, n \geq 1$ and on single-generator states, $\psi_m \psi_n = \psi_{mn}$.

Proof. Compute $\psi_3 \psi_2(a_{-1}^{(2)}|o\rangle) = \psi_3(a_{-2}^{(2)}|o\rangle) = a_{-6}^{(2)}|o\rangle$; and $\psi_6(a_{-1}^{(2)}|o\rangle) = a_{-6}^{(2)}|o\rangle$. Agreement is Macdonald's identity $p_{m(nk)} = p_{(mn)k}$ (Macdonald I §8). $\square$

6.3 THE WILKERSON CONGRUENCE FAILS ON THE FOCK LATTICE

COMPUTATION 6.3.1 (*Wilkerson test*). Test $\psi_2(x) \equiv x^2 \pmod{2}$ at $x = a_{-1}^{(2)}|o\rangle$.

- Left side: $\psi_2(a_{-1}^{(2)}|o\rangle) = a_{-2}^{(2)}|o\rangle$.
- Right side: $(a_{-1}^{(2)}|o\rangle)^2 = a_{-1}^{(2)} a_{-1}^{(2)}|o\rangle$ (normally-ordered product).
- Difference: $\psi_2(x) - x^2 = a_{-2}^{(2)}|o\rangle - a_{-1}^{(2)} a_{-1}^{(2)}|o\rangle = -2L_{-2}^{(2)}|o\rangle + a_{-2}^{(2)}|o\rangle$ (using $L_{-2}|o\rangle = \frac{1}{2}a_{-1}a_{-1}|o\rangle$ for rank-1 Heisenberg). This is a specific nonzero vector.

For the congruence to hold, this vector must lie in $2 \cdot \mathcal{V}_{\mathbb{Z}}^{\text{prim}}$ for some $\mathbb{Z}$ -integral form.

Observation 6.3.2 (lattice-form failure). On the naive Fock lattice $\mathcal{H}_{\mathbb{Z}}^{(p)} = \mathbb{Z}\langle a_{-n_1}^{(p)} \cdots a_{-n_k}^{(p)}|o\rangle_p$, the vector $a_{-2}^{(p)}|o\rangle - a_{-1}^{(p)} a_{-1}^{(p)}|o\rangle$ is a primitive integer-coefficient vector; its reduction mod 2 is nonzero. The Wilkerson congruence *fails* on the naive lattice integral form.

6.4 THE MACDONALD LATTICE VS THE FOCK LATTICE

THEOREM 6.4.1 (*Macdonald I.8, Example 24*). . The ring $\Lambda_{\mathbb{Z}} = \mathbb{Z}[e_1, e_2, \dots] = \mathbb{Z}[h_1, h_2, \dots]$ of symmetric functions is a Λ -ring satisfying the Wilkerson congruence: the Newton identity $\psi_p(h_n) \equiv h_n^p \pmod{p}$ in $\Lambda_{\mathbb{Z}}$.

Observation 6.4.2 (Fock-Macdonald isomorphism over $\mathbb{Q}$). The map $\iota: \Lambda_{\mathbb{Q}} \xrightarrow{\sim} \text{Fock}_{\mathbb{Q}}$, $p_n \mapsto a_{-n}|o\rangle$, is an isomorphism of $\mathbb{Q}$ -algebras: under the Heisenberg normalisation $[a_m, a_n] = m\delta_{m+n,0}$ the Fock inner product $\langle a_{-n}|o\rangle, a_{-n}|o\rangle = n$ matches the Hall inner product $\langle p_n, p_n \rangle = n$, so ι is an isometry. Adams operations on $\Lambda_{\mathbb{Q}}$ transport to Fock; the $\mathbb{Q}$ -algebra structure is preserved.

Observation 6.4.3 (lattice mismatch over $\mathbb{Z}$). Over $\mathbb{Z}$, the two integral forms $\iota(\Lambda_{\mathbb{Z}})$ and the Fock lattice $\text{Fock}_{\mathbb{Z}} = \mathbb{Z}\langle a_{-n}|o\rangle$ are *different*. The Schur function $s_2 = \frac{1}{2}(p_1^2 + p_2) \in \Lambda_{\mathbb{Z}}$ maps to $\iota(s_2) = \frac{1}{2}(a_{-1}^2 + a_{-2})|o\rangle$, a half-integer state outside $\text{Fock}_{\mathbb{Z}}$. Conversely, every Fock monomial $a_{-\lambda}|o\rangle = \iota(p_{\lambda})$ lies in $\iota(\Lambda_{\mathbb{Z}})$ (power sums are integer combinations of Schur functions, with character coefficients $\chi_{\mu}^{\lambda} \in \mathbb{Z}$); the Schur basis $\{\iota(s_{\lambda})\}$ of $\iota(\Lambda_{\mathbb{Z}})$ generates a strictly larger $\mathbb{Z}$ -lattice in $\text{Fock}_{\mathbb{Q}}$ than the Fock monomial basis, carrying denominators dividing the order of the centralizer z_{μ} at each conjugacy type μ .

6.5 DISTINCTION: Λ -RING VS Λ -MODULE

DEFINITION 6.5.1 (Λ -module, Borger 0801.1691 §1.2). A Λ -structure on an abelian group $\mathcal{M}$ is an action of the Adams monoid $\mathbb{N}^{\times}$ by group endomorphisms ψ_n . No ring structure, no congruence.

THEOREM 6.5.2 (*corrected form of $\mathcal{V}^{\text{prim}}$*). . The Heisenberg Fock shadow of $\mathcal{V}^{\text{prim}}$ over $\mathbb{Q}$ carries a ψ -ring structure with vacuous Wilkerson congruence via the Hecke–Adams operations. The full $\mathcal{V}^{\text{prim}}$ carries only the corresponding $\mathbb{N}^{\times}$ -module datum until a multiplication is specified. On the natural Fock integral form there is no Borger Λ -ring structure.

6.6 THE LOG- p RESCALING DESTROYS INTEGRALITY

Observation 6.6.1. When the Hamiltonian is rescaled so that L_0 assigns eigenvalue $k \log p$ to each weight- k state $a_{-k}^{(p)} |o\rangle$, the resulting grading takes values in $\mathbb{R} \setminus \mathbb{Q}$ (since $\log p$ is transcendental over $\mathbb{Q}$ for every prime p). A $\mathbb{Z}$ -lattice in the Heisenberg Fock shadow, spanned by eigenstates of the conformal L_0 , is not preserved under this rescaled grading.

Convention 6.6.2 (two-grading principle). Resolve the conflict by carrying *two* gradings on the same Heisenberg basis:

- (i) **Conformal grading:** integer weight n on $a_{-n}^{(p)}$. Character $\prod_p \prod_n (1 - q^n)^{-1}$ (Heisenberg direct product, not an L -function).
- (ii) **HY-zeta grading:** weight $n \log p$ on $a_{-n}^{(p)}$. Character $\prod_p \prod_n (1 - p^{-ns})^{-1} = \prod_n \zeta(ns)$ as a formal Dirichlet series.

The algebraic / Λ -module structure lives on (i); the zeta-character lives on (ii). These are two different gradings on the same underlying basis.

Remark 6.6.3. This matches Bost–Connes / Connes–Consani: the $\log p$ enters as spectral parameter in the Hamiltonian, not as an algebraic scalar in the underlying ring. The Bost–Connes KMS partition function $\zeta(\beta)$ is computed at grading (ii) on a purely algebraic structure that lives at grading (i).

6.7 IS THERE A Λ -VA?

Question 6.7.1. Can we define a category of Λ -vertex-algebras: VAs with commuting ring-endomorphism ψ_n compatible with the OPE and the vacuum?

PROPOSITION 6.7.2 (*T_n is not a VA endomorphism*). T_n scales conformal weight by n , i.e., $T_n(a_{-1}^{(p)} |o\rangle)$ has weight n while $a_{-1}^{(p)} |o\rangle$ has weight 1. A VA endomorphism must preserve conformal weight (commute with L_0). Therefore T_n is not a VA endomorphism for $n > 1$.

THEOREM 6.7.3 (*corrected category: finite Hecke-equivariant VA shadows*). Define

$$\text{HVA} = \{(V, \{T_n\}_{n \in \mathbb{N}^\times}) : V \text{ VA}, T_n \text{ linear}, T_m T_n = T_{mn}, T_n |o\rangle = |o\rangle\}.$$

For every finite set S of primes, the finite Heisenberg truncation $\mathcal{H}^{(S)} = \bigotimes_{p \in S} \mathcal{H}^{(p)}$ is an object of HVA, with the further intertwining property $T_n(a_{-1}^{(p)} |o\rangle) = a_{-n}^{(p)} |o\rangle$ on level-1 generators (Chapter 6.7.2). On higher-weight states T_n produces sums of mixed weight, e.g. $T_2(a_{-2}^{(p)} |o\rangle) = a_{-4}^{(p)} |o\rangle + 2 a_{-2}^{(p)} |o\rangle$, so neither $[T_n, L_0]$ nor $[T_n, L_{-1}]$ vanishes as an operator identity; direct computation gives $[T_2, L_{-1}] a_{-1}^{(p)} |o\rangle = a_{-4}^{(p)} |o\rangle + 2 a_{-2}^{(p)} |o\rangle - 2 a_{-3}^{(p)} |o\rangle \neq 0$. The category HVA records the Adams multiplicativity $T_m T_n = T_{mn}$ and the vacuum preservation; the conformal interaction is a separate datum. The restricted product is an $\mathbb{N}^\times$ -module shadow, not a vertex algebra structure on $\mathcal{V}^{\text{prim}}$.

COROLLARY 6.7.4 (*ψ -ring plus Hecke-equivariance*). The correct categorical statement: $\mathcal{V}^{\text{prim}} \in \text{Mod}_{\mathbb{N}^\times}(\text{Vect}_{\mathbb{C}})$ with the $\mathbb{N}^\times$ -action by Hecke operators T_n , whose finite Heisenberg shadows are the Hecke-equivariant VA objects of Chapter 6.7.3. This is Mackey’s Hecke algebra representation enhanced by finite Fock shadows, not Borger’s Λ -ring.

6.8 THE DESCENT INTERPRETATION GOES THROUGH BOST–CONNES

Remark 6.8.1. Borger’s $\mathbb{F}_1$ -descent programme (arXiv:0906.3146 §3) characterizes a Λ -structure on R as a descent datum from R to $R \otimes_{\mathbb{F}_1} \mathbb{Z}$. The $\mathbb{F}_1$ -descent of $\mathcal{V}^{\text{prim}}$ in Borger’s sense requires a genuine Λ -ring structure, hence a Wilkerson-congruent integral form, which the natural Fock lattice does not supply (Theorem 6.5.2 and Observation 6.3.2). The correct arithmetic framework for $\mathcal{V}^{\text{prim}}$ is Bost–Connes (Selecta Math. 1 (1995), 411–457) and Connes–Consani (arXiv:1405.4527, the arithmetic site).

THEOREM 6.8.2 (*Bost–Connes as the correct descent*). $\mathcal{V}^{\text{prim}}$ has a Heisenberg–Fock shadow compatible with the Bost–Connes Hecke C^* -dynamical system, with:

- Hecke action $\{T_n\}$ = Bost–Connes semigroup action;
- $\mathbb{Q}/\mathbb{Z}$ -Galois symmetry = Bost–Connes class-field-theory symmetry (Bost–Connes Thm 5);
- graded character = HY L -function after rescaling by $\log p$.

Borger’s $\mathbb{F}_1$ -descent requires a Wilkerson-congruent integral Λ -form on the natural Fock lattice; the Macdonald lattice $\Lambda_{\mathbb{Z}}$ carries such a form but differs from the Fock lattice over $\mathbb{Z}$ (Chapter 6.4.3). The Bost–Connes / Connes–Consani arithmetic site supplies the correct framework via tropical and semiring structures.

Remark 6.8.3 (the three-level structure). The Wilkerson congruence is vacuous over $\mathbb{Q}$ (Chapter 6.1.2), so the distinction between Λ -ring and ψ -ring is a purely integral question. The Macdonald lattice $\Lambda_{\mathbb{Z}}$ carries the Wilkerson congruence (Chapter 6.4.1); the Fock lattice $\mathbb{Z}\langle a_{-\vec{n}}^{(p)} | 0 \rangle\rangle$ does not (Chapter 6.3.2), and the two integral forms are distinct (Chapter 6.4.3). At finite Heisenberg level, T_n scales conformal weight (Chapter 6.7.2), so the category HVA applies only to finite Fock shadows (Chapter 6.7.3). The $\mathbb{F}_1$ -descent framework adequate to $\mathcal{V}^{\text{prim}}$ is Bost–Connes rather than Borger (Chapter 6.8.2).

BOST–CONNES VERSUS $\mathcal{V}^{\text{prim}}$: C^* -ALGEBRA VERSUS TRACE CLASS

The Bost–Connes algebra is a C^* -dynamical system. The object $\mathcal{V}^{\text{prim}}$ is a Hochschild trace class. Their common arithmetic character is visible only after passing through the finite Heisenberg covers and the KMS partition function.

7.1 THE BOST–CONNES C^* -ALGEBRA EXPLICIT

Definition 7.1.1 (Bost–Connes 1995, §1). The Bost–Connes $\mathbb{Q}$ -lattice C^* -algebra $C_{\mathbb{Q}}^*$ has generators

$$\{\mu_n\}_{n \in \mathbb{N}^\times}, \quad \{\mu_n^*\}_{n \in \mathbb{N}^\times}, \quad \{e(\gamma)\}_{\gamma \in \mathbb{Q}/\mathbb{Z}},$$

with relations

$$\begin{aligned} \mu_n \mu_m &= \mu_{nm}, \\ \mu_n^* \mu_n &= 1, \\ \mu_n \mu_n^* &= e_n = \frac{1}{n} \sum_{\gamma: n\gamma=0} e(\gamma), \\ e(\gamma_1) e(\gamma_2) &= e(\gamma_1 + \gamma_2), \\ \mu_n e(\gamma) \mu_n^* &= \rho_n(e(\gamma)) = \frac{1}{n} \sum_{\delta: n\delta=\gamma} e(\delta). \end{aligned}$$

The $e(\gamma)$ generate the group C^* -algebra $C^*(\mathbb{Q}/\mathbb{Z})$; the μ_n furnish an endomorphism semigroup action; $\mu_n^* \mu_n = 1$, $\mu_n \mu_n^* = e_n$ makes μ_n a Toeplitz isometry. The crossed product $C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^\times$ yields $C_{\mathbb{Q}}^*$.

THEOREM 7.1.2 (KMS trace, Bost–Connes Prop. 3.3 and Thm 5.4). The KMS_β state φ_β on $C_{\mathbb{Q}}^*$ for $\beta > 1$ is unique and extremal; its partition function is

$$Z_{\text{BC}}(\beta) = \text{tr}(e^{-\beta H}) = \sum_{n \geq 1} n^{-\beta} = \zeta(\beta),$$

where $H = \log N$ generates $\sigma_t(\mu_n) = n^{it} \mu_n$. The ζ -partition function is explicitly motivated by Julia’s primon gas (*Statistical theory of numbers*, Springer Proc. Phys. 47, 1990).

7.2 THE STRUCTURAL TYPE MISMATCH

Observation 7.2.1. The Heisenberg CCR $[a_m^{(p)}, a_n^{(q)}] = m \delta_{m+n,0} \delta_{pq}$ is a relation on unbounded self-adjoint operators on a Fock space with countable spectrum; $\mu_n \mu_n^* = e_n$ is a Toeplitz–Cuntz relation on bounded partial isometries with spectrum $\{1/n\} \cup \{0\}$. These are distinct operator-algebraic structures.

COROLLARY 7.2.2 (*Bost–Connes representation factors through the zero-mode lattice*). . The Heisenberg CCR and Toeplitz–Cuntz operator-algebraic structures are incompatible. The finite Heisenberg covers of $\mathcal{V}^{\text{prim}}$ carry local Fock representations, while $\mathcal{V}^{\text{prim}}$ itself is the Hochschild trace class. The faithful Bost–Connes action is the zero-mode lattice action $L \hookrightarrow \mathcal{V}^{\text{prim}}$ of Chapter 7.3.1, on which it coincides with the standard Bost–Connes representation on $\ell^2(\mathbb{N}^\times)$.

7.3 ACTION ON THE ZERO-MODE LATTICE

Definition 7.3.1 (*zero-mode lattice*). Let $L \subset \mathcal{V}^{\text{prim}}$ be the lattice

$$L = \left\{ \prod_p |n_p\rangle_p : n = \prod_p p^{n_p} \in \mathbb{N}^\times \right\},$$

where $|n_p\rangle_p = (a_{-1}^{(p)})^{n_p} |0\rangle_p / \sqrt{n_p!}$ is the n_p -th Fock number state. Then $L \cong \ell^2(\mathbb{N}^\times)$ as a Hilbert space, with standard orthonormal basis $\{\delta_n\}_{n \in \mathbb{N}^\times}$ indexed by positive integers, where $\delta_n = \prod_p |n_p\rangle_p$ for $n = \prod_p p^{n_p}$.

THEOREM 7.3.2 (*Bost–Connes acts on L*). . $C_{\mathbb{Q}}^*$ acts faithfully on the zero-mode lattice $L \subset \mathcal{V}^{\text{prim}}$ via:

- $\mu_n \delta_m = \delta_{nm}$,
- $\mu_n^* \delta_m = \delta_{m/n}$ if $n \mid m$, else zero,
- $e(\gamma) \delta_m = \zeta_m(\gamma) \delta_m$ with ζ_m a primitive m -th root of unity.

This is the Bost–Connes representation from 1995 §1.2, Prop. 1, transported to the zero-mode lattice of $\mathcal{V}^{\text{prim}}$.

Observation 7.3.3 (*maximality of the zero-mode action*). The Bost–Connes action on L is maximal in $\mathcal{V}^{\text{prim}}$: the off-lattice states (coherent states, descendants carrying higher Heisenberg modes $a_{-n}^{(p)}$ for $n \geq 2$, and their superpositions) lie outside the Bost–Connes orbit, so L is the largest $C_{\mathbb{Q}}^*$ -stable subspace of $\mathcal{V}^{\text{prim}}$.

7.4 THE TENSOR DECOMPOSITION

THEOREM 7.4.1 (*tensor-product placement of $\mathcal{V}^{\text{prim}}$*). . For each finite N , let $\mathcal{V}_N^{\text{prim}} = \otimes_{p \leq N} \mathcal{H}^{(p)}$ be the rank- $\pi(N)$ truncation. Then as graded vector spaces,

$$\mathcal{V}_N^{\text{prim}} \cong \underbrace{C_{\mathbb{Q},N}^*|_{\text{lattice}}}_{\text{arithmetic}} \otimes \underbrace{\mathcal{H}_{\text{descendant}}^{\otimes \pi(N)}}_{\text{universal}}$$

where the left factor is the Bost–Connes action on the rank- $\pi(N)$ zero-mode lattice, and the right factor is the universal Heisenberg-descendant tower of rank $\pi(N)$ carrying all higher-mode data $a_{-n}^{(p)}$ for $n \geq 2$.

COROLLARY 7.4.2. . All arithmetic content of $\mathcal{V}^{\text{prim}}$ lives in the Bost–Connes factor (left-hand side). The Heisenberg descendant tower depends only on the rank $\pi(N)$, not on the arithmetic nature of the primes; it is a universal rank- $\pi(N)$ Heisenberg dressing.

7.5 KESTEN–MCKAY = PLANCHEREL OF UNRAMIFIED $GL_2(\mathbb{Q}_p)$

THEOREM 7.5.1 (*Kesten 1959; McKay 1981; Lubotzky 1994 Thm 7.4.5*). . The Kesten–McKay distribution on $[-2\sqrt{p}, 2\sqrt{p}]$ for $q = p$ prime,

$$d\nu_p(x) = \frac{(p+1)\sqrt{4p-x^2}}{2\pi((p+1)^2-x^2)} dx,$$

is the spectral measure of the adjacency operator on the $(p+1)$ -regular tree, equivalently the p -adic Bruhat–Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_p)$, equivalently the Plancherel measure of the spherical-function algebra on the tree.

COROLLARY 7.5.2. . The Kesten–McKay averaging of the Julia primon partition function is the Plancherel averaging of the local Bost–Connes Hecke algebra at p . Equivalently, averaging against ν_p is the Plancherel decomposition of the unramified spherical representation of $GL_2(\mathbb{Q}_p)$.

Remark 7.5.3 (arithmetic averaging as Plancherel decomposition). The arithmetic averaging map $\mathrm{av}^{\mathrm{arith}}$ is the Plancherel decomposition of an automorphic representation, while the operadic averaging $\mathfrak{g}^{E_i} \rightarrow \mathfrak{g}^{\mathrm{mod}}$ of Vol I projects onto a symmetric sector weighted by a measure on chiral cohomology. Both have the formal shape of a projection onto a symmetric sector, but the operadic averaging acts on chiral cohomology and the Plancherel averaging acts on spherical representations of $GL_2(\mathbb{Q}_p)$: distinct objects sharing a formal pattern.

7.6 THE SOIBELMAN COHOMOLOGICAL HALL ALGEBRA READING

THEOREM 7.6.1 (*Kontsevich–Soibelman 1006.2706 §2.4*). . The Kesten–McKay spectral measure ν_p arises as the Behrend-weighted motivic Euler characteristic of the moduli stack of rank- n representations of the Jordan quiver (one vertex, one loop, trivial potential) at the p -adic coefficient. This is a statement of the cohomological Hall algebra (CoHA) of Kontsevich–Soibelman at the adelic coefficient.

PROPOSITION 7.6.2 (*three-way identification*). . The composed arrow

$$\text{Julia primon gas} \xrightarrow{\text{Plancherel}} \text{Bost-Connes Hecke-Plancherel} \xrightarrow{\text{CoHA-motivic}} \text{Soibelman DT on } \mathcal{Q}_{\text{Jordan}}$$

sends the primon partition function to $\prod_p(\text{local DT series}) = \prod_p \zeta_p(\beta)$ (local Euler factors). The three presentations are three views of the same invariant.

7.7 HOCHSCHILD COMPARISON: CONNES–MARCOLLI VS CLAIMED THEOREM H^{arith}

THEOREM 7.7.1 (*Connes–Marcolli math/0404128 §I.8*). .

$$HH^0(C_{\mathbb{Q}}^*) = \mathbb{C} \quad (\text{centre: scalars}),$$

$$HH^1(C_{\mathbb{Q}}^*) = \mathbb{C} \quad (\text{outer derivations, generated by } \delta = \frac{d}{dt} \sigma_t),$$

$$HH^2(C_{\mathbb{Q}}^*) = 0 \quad (\text{type-III}_1 \text{ factor}),$$

$$HH^k(C_{\mathbb{Q}}^*) = 0 \quad \text{for } k \geq 2.$$

Observation 7.7.2. Connes–Marcolli’s $HH^\bullet$ has finite-dimensional HH^0 and HH^1 and vanishing $HH^{\geq 2}$. This is *strictly sharper* than the claimed Theorem H^{arith} concentration in $\{0, 1, 2\}$.

THEOREM 7.7.3 (*corrected Theorem H^{arith}*). $HH^\bullet(\mathcal{V}^{\text{prim}})$ has:

- $HH^0 = \mathbb{Z}$ (Bost–Connes centre),
- $HH^1 = \bigoplus_p \log p \cdot \mathbb{Z}$ (KMS generators, Chapter 7.7.1),
- $HH^2 = \Lambda$ -torsor related to Witt cotangent (conjectural).

Concentration in $\{0, 1, 2\}$ holds over $\mathbb{Q}$; integral concentration is conjectural.

7.8 VERTEX-ALGEBRAIC INVARIANTS OF THE HEISENBERG COVERS

THEOREM 7.8.1 (*universal Heisenberg-cover invariants*). The finite Heisenberg cover $\mathcal{H}^{\otimes \pi(N)}$ of $\mathcal{V}_N^{\text{prim}}$ is a rank- $\pi(N)$ Heisenberg vertex algebra. It lies in the Heisenberg archetype G of the Vol I landscape census, with cover central charge $\kappa = \pi(N)$, depth $r_{\max} = 2$, and chiral Hochschild cohomology concentrated in degrees $\{0, 1, 2\}$. These are invariants of the cover, not of the trace class.

COROLLARY 7.8.2 (*placement of the finite covers*). The finite Heisenberg covers of $\mathcal{V}^{\text{prim}}$ lie in archetype G (Heisenberg / Gaussian), with $\kappa_N = \pi(N)$. The infinite arithmetic object is the Hochschild trace class obtained after Bost–Connes zero-mode and Mellin specialisation; it is not a vertex algebra.

7.9 THE CORRECT IDENTIFICATION

THEOREM 7.9.1 (*corrected placement of $\mathcal{V}^{\text{prim}}$*). As graded vector spaces,

$$\mathcal{V}_N^{\text{prim}} = (\text{Bost–Connes module at rank } \pi(N)) \otimes (\text{universal Heisenberg rank-}\pi(N) \text{ VA}).$$

All arithmetic content (Hecke algebra, KMS states, ζ/L -function partition function, Plancherel–Kesten–McKay averaging, prime indexing) lives in the Bost–Connes factor. The VA factor carries the universal Heisenberg descendant tower, depending only on the rank $\pi(N)$.

Remark 7.9.2 (module structure). $\mathcal{V}^{\text{prim}}$ is a trace object with a Bost–Connes zero-mode module and finite Heisenberg covers. The vertex algebra structure belongs to the cover factor; the arithmetic symmetries belong to the Bost–Connes factor.

Remark 7.9.3 (the B-row threshold). A chiral algebra whose stress tensor encodes Hecke eigenvalues (not merely prime indexing), with chiral Hochschild cohomology recovering $HH^\bullet(C_{\mathbb{Q}}^*)$ as a subquotient, lies in the B-row of Theorem C (Borcherds/BKM archetype, $K^\kappa = 8$) by the universal positive-geometry template of Vol III (Theorem C, BKM column). Reaching this threshold from the Heisenberg-cover G-row requires a Borcherds-product encoding of Hecke data at the level of the stress tensor, which is the Vol III $\Phi_N = c_N(o)/2$ programme.

7.10 STRUCTURAL CONCLUSIONS

- Bost–Connes $C_{\mathbb{Q}}^*$ carries $\zeta(\beta)$ as KMS partition function (Chapter 7.1.2).
- The Heisenberg CCR and Toeplitz–Cuntz relations are structurally distinct operator algebras; the Fock-space projections determine the representation, and the Bost–Connes action on $\mathcal{V}^{\text{prim}}$ factors through the zero-mode lattice (Chapter 7.2.2).
- Bost–Connes acts faithfully on the zero-mode lattice $L \subset \mathcal{V}^{\text{prim}}$; the off-lattice states lie outside the Bost–Connes orbit (Chapter 7.3.2, Chapter 7.3.3).
- $\mathcal{V}_N^{\text{prim}} \cong (\text{Bost–Connes module}) \otimes (\text{universal Heisenberg})$: arithmetic content on the left, Heisenberg-cover structure on the right (Chapter 7.4.1).

- (v) Kesten–McKay spectral measure = Plancherel of unramified $GL_2(\mathbb{Q}_p)$ = Bost–Connes local Hecke spectrum = Soibelman CoHA on the Jordan quiver (Chapter 7.5.2, Chapter 7.6.2).
- (vi) Connes–Marcolli $HH^\bullet(C_\mathbb{Q}^*)$ concentrates in degrees $\{0, 1\}$, sharpening Theorem H^{arith} (Chapter 7.7.1).
- (vii) The finite Heisenberg covers lie in archetype G with $\kappa = \pi(N)$, $r_{\max} = 2$; the B-row threshold requires Borchers-product encoding of Hecke data (Chapter 7.8.2).

The vertex algebra structure is universal and non-arithmetic because it belongs only to the finite Heisenberg covers. The arithmetic content is carried by the Bost–Connes C^* -algebra factor and by the Hochschild trace class. A genuine VA-arithmetic object encoding Hecke eigenvalues in the stress tensor would be a new object, not $\mathcal{V}^{\text{prim}}$.

ADELIC FACTORIZATION AXIOMS

Beilinson–Drinfeld and Costello–Gwilliam factorization require a space with a meaningful configuration theory. On $\overline{C} = \overline{\mathrm{Spec} \mathbb{Z}}$ the finite places are discrete and the archimedean place carries the only chiral singularity.

8.1 THE RAN SPACE OF $\overline{\mathrm{Spec} \mathbb{Z}}$

Observation 8.1.1 (naive Ran is terminal). BD §3.4.1 defines $\mathrm{Ran}(X)$ for X a smooth curve over a field k as the prestack whose S -points for a k -test scheme S are non-empty finite subsets of $X(S)$. For $X = \mathrm{Spec} \mathbb{Z}$, the functor of points is

$$S \mapsto \mathrm{Spec} \mathbb{Z}(S) = \mathrm{Hom}_{\mathrm{Ring}}(\mathbb{Z}, \Gamma(S, \mathcal{O}_S)) = \{*\}$$

(singleton). The naive Ran space of $\mathrm{Spec} \mathbb{Z}$ is therefore the terminal prestack; a single point. Primes are not sections; they are closed points, i.e., morphisms $\mathrm{Spec} \mathbb{F}_p \rightarrow \mathrm{Spec} \mathbb{Z}$ going the wrong way for the functor-of-points definition.

Definition 8.1.2 (enriched Ran on Arakelov curve). On the Arakelov-compactified curve $\overline{C} = \overline{\mathrm{Spec} \mathbb{Z}}$, redefine the Ran space combinatorially:

$$\mathrm{Ran}(\overline{C})(R) = \{\text{non-empty finite multisets } \Sigma \subset |\overline{C}|\}$$

for $R = \mathbb{Q}$ -points (or more generally a $\mathbb{Q}$ -algebra). Topology: $\mathrm{ind}\text{-set} \varinjlim_n \mathrm{Sym}^n(\mathbb{P} \sqcup \{\infty\})$, where $\mathbb{P}$ denotes the set of rational primes. This is a discrete combinatorial object; no scheme structure, no tangent space.

Remark 8.1.3. $\mathrm{Ran}(\overline{C})$ is discrete. BD factorization axioms applied to a discrete Ran space become set-theoretic identities carrying no infinitesimal content. The OPE singularity structure at the diagonal is the content of BD factorization; on a discrete Ran space, the diagonal has no infinitesimal neighbourhood and the content vanishes.

8.2 THE FACTORIZATION ISOMORPHISM IS TRIVIAL

Observation 8.2.1. BD §3.4.4: a factorization algebra $\mathcal{A}$ on X assigns to each $n \geq 1$ a $\mathcal{D}$ -module $\mathcal{A}^{(n)}$ on X^n with isomorphisms $j_U^* \mathcal{A}^{(n)} \xrightarrow{\sim} (\boxtimes_i \mathcal{A}^{(n_i)})|_U$ over $U = X^n \setminus \Delta$. On $\mathrm{Ran}(\overline{C})$, for distinct primes $p \neq q$, the fibre $\mathcal{V}^{\mathrm{prim}}|_{\{p,q\}} = \mathcal{H}^{(p)} \otimes \mathcal{H}^{(q)}$ by definition; a tautology.

COROLLARY 8.2.2 (factorization structure is tautological). . The factorization iso over $\overline{C}^n \setminus \Delta$ is the direct-product identity: the fibre of $\mathcal{V}^{\mathrm{prim}}$ at a finite set of distinct primes is the tensor product of the individual stalks, recorded by definition. The BD factorization axiom holds on $\overline{C}^n \setminus \Delta$ only as this tautological identity, carrying no infinitesimal or OPE content.

8.3 THE DIAGONAL AND $\mathbb{Z}_p \widehat{\otimes} \mathbb{Z}_p$

Observation 8.3.1. The arithmetic diagonal at a finite place collapses:

$$\mathbb{Z}_p \widehat{\otimes}_{\mathbb{Z}} \mathbb{Z}_p \cong \mathbb{Z}_p,$$

because $\mathbb{Z}_p$ is p -adically complete and the diagonal is an isomorphism after completion. The “diagonal neighbourhood” at a finite place is trivial: $\mathrm{Spec}(\mathbb{Z}_p \widehat{\otimes} \mathbb{Z}_p) = \mathrm{Spec} \mathbb{Z}_p$.

Remark 8.3.2. A non-trivial thickening of the p -adic diagonal is available via Fontaine-style crystalline structure (prismatic site of Bhatt–Scholze arXiv:1905.08229), at the cost of Frobenius crystalline data. The naive construction carries no such data; a fusion map at finite places requires the prismatic upgrade.

THEOREM 8.3.3 (*archimedean–finite diagonal asymmetry*). . At the archimedean place ∞ , the diagonal $\mathbb{C} \hookrightarrow \mathbb{C}^2$ is a smooth closed submanifold with a full formal neighbourhood (Taylor expansion); Beilinson–Drinfeld factorization on $\mathbb{C}_\infty$ is the standard holomorphic factorization algebra in the Beilinson–Drinfeld sense. At each finite place p , the diagonal $\mathrm{Spec} \mathbb{Z}_p \hookrightarrow \mathrm{Spec}(\mathbb{Z}_p \widehat{\otimes}_{\mathbb{Z}} \mathbb{Z}_p)$ is an isomorphism; the diagonal neighbourhood is trivial, and a non-trivial BD-type fusion requires Fontaine prismatic data (Bhatt–Scholze arXiv:1905.08229) absent from the naive construction.

8.4 THE LOCALLY CONSTANT (E_o) FACTORIZATION AT EACH FINITE PLACE

PROPOSITION 8.4.1 (*Francis arXiv:1206.5522 §2.3; Lurie HA 5.5.4.10*). . A locally constant factorization algebra on a topological manifold M factors through the homotopy type of M , equivalently is an E_n -algebra for $n = \dim M$ (Francis arXiv:1206.5522 §2.3; Lurie HA 5.5.4.10). On a totally disconnected space like $\mathrm{Spf} \mathbb{Z}_p$, locally constant factorization degenerates to an E_o -algebra (a pointed vector space).

Definition 8.4.2 (*minimal adelic factorization structure*). The minimal Costello–Gwilliam-compatible adelic structure on $\mathcal{V}^{\mathrm{prim}}$ is:

- At ∞ : genuine holomorphic factorization algebra on $\mathbb{C}_\infty$ in the CG sense (Vol. I Def. 3.1.4), with Heisenberg OPE $j(z)j(w) \sim k_\infty/(z-w)^2$, $k_\infty = 1$.
- At each finite p : locally constant prestack on $\mathrm{Spf} \mathbb{Z}_p$ with constant stalk $\mathcal{H}^{(p)}$ and Adams endomorphism ψ_p ; this is an E_o -algebra.
- Gluing data: restricted tensor product over spherical vectors $|o\rangle_p$ via Tate’s thesis, giving an automorphic representation of $GL_2(\mathbb{A}_\mathbb{Q})$.

8.5 THE HOLOMORPHIC-TOPOLOGICAL STRUCTURE

Remark 8.5.1 (*Costello HT factorization, arXiv:1303.2632; Costello 2020*). Costello’s holomorphic-topological factorization algebras live on $\mathbb{C}^{n_H} \times \mathbb{R}^{n_T}$ with n_H holomorphic and n_T topological directions. The axioms blend holomorphic OPE (on the $\mathbb{C}^{n_H}$ factor) and locally constant (E_{n_T} -algebra structure on the $\mathbb{R}^{n_T}$ factor).

THEOREM 8.5.2 (*adelic HT structure*). . On $\overline{C}$ the natural HT structure is

$$\underbrace{\mathbb{C}_\infty}_{\text{holomorphic}} \times \underbrace{\mathrm{Spf} \widehat{\mathbb{Z}}}_{\text{adelic} / E_o \text{ at each place}},$$

with the profinite completion $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ totally disconnected. Locally constant factorization on $\mathrm{Spf} \widehat{\mathbb{Z}}$ decomposes as a product of E_o -stalks. The non-trivial chiral content concentrates at ∞ .

Definition 8.5.3 (relative factorization window with adelic base). . Attached to $\mathcal{V}^{\text{prim}}$ is the Costello–Gwilliam window, a holomorphic factorization algebra on $\mathbb{C}_\infty$ over the base $\text{Spec } \widehat{\mathbb{Z}}$, with the base acting by Adams operations $\{\psi_p\}_p$ and Hecke operators $\{T_p\}_p$:

$$\mathcal{F}_{\text{CG}}(\mathcal{V}^{\text{prim}}) \in \text{FactAlg}_{\mathbb{C}_\infty}(\text{Mod}_{\widehat{\mathbb{Z}}, \psi}),$$

where $\text{Mod}_{\widehat{\mathbb{Z}}, \psi}$ is the category of $\widehat{\mathbb{Z}}$ -modules with compatible commuting Adams/Hecke operations ψ_p for each prime.

8.6 FIVE AXIOMS (F1)–(F5) FOR THE MINIMAL ADELIC FACTORIZATION

Definition 8.6.1 (Costello–Gwilliam arithmetic factorization). $\mathcal{V}^{\text{prim}}$ satisfies:

(F1) *Archimedean factorization.* Holomorphic factorization algebra on $\mathbb{C}_\infty$ in the sense of CG Vol. I Def. 3.1.4, with OPE $j(z)j(w) \sim k_\infty/(z-w)^2$, $k_\infty = 1$.

(F2) *Coefficient category.* Values in $\text{Mod}_{\widehat{\mathbb{Z}}, \psi}$: $\widehat{\mathbb{Z}}$ -modules with commuting Adams operations $\psi_p: \mathcal{M} \rightarrow \mathcal{M}$ satisfying $\psi_p\psi_q = \psi_q\psi_p$ and $\psi_p|_{\mathbb{Z}_p} = \text{id}$.

(F3) *Hecke-eigen decomposition.* Global sections decompose as a restricted tensor product

$$\mathcal{V}^{\text{prim}}(\mathbb{C}_\infty) = \bigotimes_p' \mathcal{H}^{(p)}$$

with spherical vectors $|o\rangle_p$, compatibly with Adams / Hecke operators.

(F4) *Archimedean–finite compatibility.* The p -adic valuation of the Heisenberg level k_∞ is determined by the Hecke eigenvalue a_p via $v_p(k_\infty) = f(a_p)$ for an explicit function f fixed by Hartnoll–Yang (arXiv:2502.02661, Thm 1.1 or its equivalent).

(F5) *Regular p -adic product.* At each finite place p , the local factor $\mathcal{H}^{(p)}$ carries regular (non-singular) commutative multiplication and Adams endomorphism ψ_p as its only operations. The product $\mathcal{H}^{(p)} \otimes \mathcal{H}^{(p)} \rightarrow \mathcal{H}^{(p)}$ is the commutative multiplication in the Fock module: a commutative algebra structure, distinct from a BD factorization-algebra fusion.

THEOREM 8.6.2 (minimal adelic factorization: (F1)–(F5)). . The minimal Costello–Gwilliam-compatible adelic factorization structure on $\mathcal{V}^{\text{prim}}$ is precisely (F1)–(F5): a holomorphic factorization algebra on the archimedean place $\mathbb{C}_\infty$ with values in $\widehat{\mathbb{Z}}$ -modules carrying commuting Adams and Hecke operations. The BD factorization axioms on the full arithmetic curve $\overline{C}$ or $\text{Spec } \mathbb{Z}$ hold only in the degenerate form recorded by (F1)–(F5).

8.7 CLASSICAL FACTORIZATION AT FINITE PLACES: ABSENCE OF A BV ACTION FUNCTIONAL

Remark 8.7.1 (BV quantization requires action functional). CG Vol. I §4 constructs a factorization algebra of observables as a BV-quantization of a classical field theory $(\mathcal{E}, S)$. For the free boson on $C/\mathbb{C}$: $\mathcal{E} = \Gamma(C, \Omega_C^1)[-1]$, action $S[\phi] = \int_C \phi \wedge \bar{\partial}\phi$, the quantum factorization algebra is the Heisenberg

factorization algebra (CG Vol. I Thm 4.6.1; as a translation-equivariant factorization algebra on $\mathbb{C}$, this recovers the Heisenberg vertex algebra, but the two structures are distinct: factorization algebra $\neq$ vertex algebra).

Observation 8.7.2 (arithmetic differential forms and the free-field obstruction). The sheaf of Kähler differentials $\Omega_{\mathbb{Z}/\mathbb{Z}}^1$ on $\mathrm{Spec} \mathbb{Z}$ is zero in the ordinary algebraic sense; Arakelov geometry replaces it with a metrised line bundle at the archimedean place. A classical free-field action at a finite place p would require a section of $\Omega_{\mathbb{C}/\mathbb{F}_1}^1$ in a neighbourhood of p ; the requisite $\mathbb{F}_1$ -geometry remains unconstructed.

Remark 8.7.3. The Borger Λ -ring formalism (Borger arXiv:0906.3146) supplies Adams operations ψ_p as natural ring endomorphisms; these endomorphisms carry no action functional, BV bracket, or Poisson structure. The Heisenberg module $\mathcal{H}^{(p)}$ at finite p is a Λ -ring object, not a CG-quantization of a classical field theory.

THEOREM 8.7.4 (classical Λ -ring structure at finite places). . At a finite place p , $\mathcal{H}^{(p)}$ is a classical commutative factorization algebra on $\mathrm{Spf} \mathbb{Z}_p$ equipped with a Frobenius-lift (Adams endomorphism ψ_p) in the sense of Borger’s Λ -ring theory. The quantization parameter $\hbar$ is absent: the classical action functional whose BV-quantization would produce a quantum factorization algebra does not exist over $\mathrm{Spf} \mathbb{Z}_p$.

8.8 FACTORIZATION BOUNDARY

- (i) The naive Ran space $\mathrm{Ran}(\overline{C})$ as a BD prestack is terminal; the enriched combinatorial Ran is discrete (Chapter 8.1.1, Chapter 8.1.2).
- (ii) The factorization isomorphism over $\overline{C}^n \setminus \Delta$ is the tautological direct-product identity; the factorization content reduces to the collection of individual stalks (Chapter 8.2.2).
- (iii) The completed tensor product $\mathbb{Z}_p \widehat{\otimes}_{\mathbb{Z}} \mathbb{Z}_p \cong \mathbb{Z}_p$ makes the p -adic diagonal an isomorphism; a BD-type fusion at finite places requires Fontaine prismatic data (Chapter 8.3.1, Chapter 8.3.3).
- (iv) At each finite place p , locally constant factorization on $\mathrm{Spf} \mathbb{Z}_p$ is an E_o -algebra; at ∞ , the archimedean factor is a genuine BD holomorphic factorization algebra (Chapter 8.4.1).
- (v) The minimal Costello–Gwilliam-compatible structure is a holomorphic factorization algebra on $\mathbb{C}_\infty$ with $\widehat{\mathbb{Z}}$ -module coefficients and commuting Adams operations, as in (F1)–(F5) (Chapter 8.5.3, Chapter 8.6.1).
- (vi) At finite places, $\mathcal{H}^{(p)}$ is a classical Λ -ring object; the BV-quantization scheme requires a classical action functional absent over $\mathrm{Spf} \mathbb{Z}_p$ (Chapter 8.7.4).

The Costello–Gwilliam window of $\mathcal{V}^{\mathrm{prim}}$ is a holomorphic factorization algebra on $\mathbb{C}_\infty$ with an adelic local system decoration (axioms (F1)–(F5)); the full Hochschild trace class itself is not thereby made a vertex algebra.

DAHA, MACDONALD, AND THE PARAMETER NON-UNIFORMITY

Macdonald theory, DAHA, and elliptic Hall algebras depend on uniform parameters. The primon construction carries a prime-indexed family of parameters, and this non-uniformity is the obstruction.

9.1 MACDONALD (q, t) AND THE PRIME-INDEXED ALPHABET

The question is whether the primon construction $\mathcal{V}^{\text{prim}}$ admits an embedding into a standard Macdonald or elliptic-Hall framework. The first obstruction is domain-theoretic: the classical Macdonald theory lives on a single ring of symmetric functions, while $\mathcal{V}^{\text{prim}}$ is prime-stratified.

THEOREM 9.1.1 (*Macdonald I.6.19*). . The Macdonald inner product on Λ (the ring of symmetric functions) is

$$\langle p_\mu, p_\nu \rangle_{q,t} = \delta_{\mu\nu} z_\mu \prod_{i \geq 1} \frac{1 - q^{\mu_i}}{1 - t^{\mu_i}},$$

where $z_\mu = \prod_{i \geq 1} i^{m_i} m_i!$ for a partition μ with m_i parts equal to i , and (q, t) are two global parameters on Λ . The power-sum basis p_μ does not factor over primes.

Observation 9.1.2 (domain mismatch). The proposal $q = p^{-s}$, $t = p^{-s-1}$ (per prime) assigns a single pair (q, t) to the prime-agnostic ring Λ , whereas the primon construction $\mathcal{V}^{\text{prim}}$ lives on $\otimes'_p \Lambda_{(p)}$ (prime-stratified; the restricted tensor product over primes with almost all factors at the vacuum). The two domains are distinct; no map from $\mathcal{V}^{\text{prim}}$ to single-parameter Macdonald exists.

The correct domain for the primon construction is the prime-stratified tensor product; the Macdonald pairing extends to it by assigning independent parameters (q_p, t_p) at each prime.

Definition 9.1.3 (per-prime Macdonald). Set $\Lambda_{(p)} = \mathbb{Q}[p_1^{(p)}, p_2^{(p)}, \dots]$, the ring of symmetric functions with power-sum generators indexed by the prime p . Define

$$\Lambda^{\text{prim}} := \bigotimes_p \Lambda_{(p)},$$

the restricted tensor product (almost all factors equal to vacuum 1). The Macdonald pairing is per-prime:

$$\langle p_\mu^{(p)}, p_\nu^{(p)} \rangle_{q_p, t_p} = \delta_{\mu\nu} z_\mu \prod_{i \geq 1} \frac{1 - q_p^{\mu_i}}{1 - t_p^{\mu_i}};$$

the global pairing on Λ^{prim} is the tensor product over primes. The result is a multi-parameter $(q_p, t_p)_p$ Macdonald framework.

9.2 THE $\log p$ RESCALING AS A HEISENBERG CONFORMAL RESCALING

The per-prime Macdonald norm at generic (q_p, t_p) is $k(1 - q_p^k)/(1 - t_p^k)$, while the Heisenberg norm of the primon creation operator is k . The question is how these match, and what role the $\log p$ prefactor plays. The computation shows it is a Heisenberg conformal rescaling external to the (q_p, t_p) parameters.

COMPUTATION 9.2.1. Under $\mathcal{H}^{(p)} \cong \Lambda_{(p)} \otimes \mathbb{Q}$ with $a_{-k}^{(p)}|o\rangle \leftrightarrow p_k^{(p)}/\sqrt{k}$, the Heisenberg norm is

$$\langle\langle a_{-k}^{(p)}|o\rangle, a_{-k}^{(p)}|o\rangle\rangle = k.$$

The Kesten–McKay k -th moment (tree branching at degree k) is a Catalan-type count scaling like p^k ; and the $\log p$ rescaling adds a per-prime $(\log p)^k$ factor. Neither matches the Macdonald norm $k(1 - q_p^k)/(1 - t_p^k)$ at generic (q_p, t_p) .

THEOREM 9.2.2 (*Heisenberg specialisation match*). . Setting $t_p = q_p^2$ and sending $q_p \rightarrow o$:

$$\lim_{q_p \rightarrow o} \frac{1 - q_p^k}{1 - q_p^{2k}} = 1,$$

giving $\langle p_k^{(p)}, p_k^{(p)} \rangle_{q_p, t_p} = k$, the free-boson Heisenberg norm. The $\log p$ rescaling enters multiplicatively outside the (q_p, t_p) parameters:

$$\langle\langle a_{-k}^{(p)}|o\rangle, a_{-k}^{(p)}|o\rangle\rangle_{\text{prim}} = k \cdot (\log p)^k.$$

No Kesten–McKay measure is asserted at this limit. That measure is a graph-theoretic input, not a consequence of the Macdonald specialisation.

Remark 9.2.3. The $\log p$ rescaling is a Heisenberg conformal rescaling, not a Macdonald parameter; the (q_p, t_p) -deformation parameters carry only the shape of the inner product, not the arithmetic weight. Arithmetic data sits in the tensor-indexing over primes, not in the deformation parameters.

9.3 DAHA PARAMETERS AND THE ARITHMETIC HALL ALGEBRA

The per-prime (q_p, t_p) framework raises the question of whether these parameters can be assembled into a single double affine Hecke algebra. The rank constraint of DAHA forecloses this.

THEOREM 9.3.1 (*DAHA rank constraint*). (Cherednik, *Double Affine Hecke Algebras*, CUP 2005, §1.4). The double affine Hecke algebra (DAHA) $\mathbb{H}(q, t)$ of type A_{n-1} has two parameters (q, t) and acts on $\mathbb{C}[x_1^\pm, \dots, x_n^\pm]$ via Dunkl–Cherednik operators.

Observation 9.3.2 (DAHA-Adams incompatibility). Hecke–Adams operators ψ_n act uniformly on Λ and do not distinguish primes; DAHA Dunkl operators are W -equivariant for the Weyl group W acting symmetrically. The prime stratification of $\Lambda^{\text{prim}} = \bigotimes'_p \Lambda_{(p)}$ lives in the tensor-indexing, outside the W -action; the two structures occupy distinct tensorial positions.

The correct object at each prime is the spherical elliptic Hall algebra of Schiffmann–Vasserot. Its degeneration is used only as a rank-one Hall degeneration; no prime is a DAHA rank.

THEOREM 9.3.3 (*Schiffmann–Vasserot, $q \rightarrow 1$ degeneration*). (Schiffmann–Vasserot arXiv:0905.2555, arXiv:1202.2756, Theorem 1.2). Under the degeneration normalisation of Schiffmann–Vasserot, the spherical elliptic Hall algebra has $W_{1+\infty}$ as its limiting rank-one algebra. This chapter uses only that limiting Hall algebra, not a prime-dependent Yangian identification.

Definition 9.3.4 (arithmetic Hall algebra).

$$\mathbb{H}^{\text{arith}} := \bigotimes_p' \text{SH}(q_p, t_p),$$

the restricted tensor product of spherical elliptic Hall algebras, one per prime. At each prime, the shuffle product on alphabet $(x_1^{(p)}, x_2^{(p)}, \dots)$ uses the kernel

$$\omega_p(x, y) = \frac{(1 - q_p x/y)(1 - t_p^{-1} x/y)}{(1 - x/y)(1 - q_p t_p^{-1} x/y)}.$$

Between primes $p \neq \ell$ the shuffle kernel is 1; primes do not mix.

9.4 SHUFFLE STRUCTURE IS UNCOUPLED ACROSS PRIMES

By construction (Chapter 9.3.4), the shuffle kernel at a single prime is ω_p , while the kernel between distinct primes is 1. This factorisation is a theorem, not merely a definition: it rules out cross-prime algebraic interactions.

THEOREM 9.4.1. . The shuffle structure on $\mathbb{H}^{\text{arith}}$ (Chapter 9.3.4) factors across primes: for $p \neq \ell$, the shuffle kernel is trivial ($= 1$). Primes enter only as labels on non-interacting copies of the same shuffle data; there are no cross-prime shuffle relations.

Remark 9.4.2. Primes enter $\mathcal{V}^{\text{prim}}$ as labels on non-interacting copies of the same Heisenberg data. The non-triviality of the (q_p, t_p) -deformation is internal to each prime factor; the arithmetic content sits in the tensor-indexing over primes, not in algebraic interactions across them.

9.5 ARITHMETIC MACDONALD POLYNOMIALS

The P^{arith} -basis for Λ^{prim} is forced by the per-prime tensor structure: each basis element is a tensor product of classical Macdonald polynomials, one per prime, with almost all factors equal to the empty-partition polynomial $P_{\emptyset} = 1$.

Definition 9.5.1 (arithmetic Macdonald polynomial).

$$P_{(\lambda_p)_p}^{\text{arith}} := \bigotimes_p' P_{\lambda_p}(x^{(p)}; q_p, t_p),$$

the restricted tensor product indexed by a family of partitions $(\lambda_p)_p$ with almost all $\lambda_p = \emptyset$.

Remark 9.5.2 (dimension count). At the 2-prime sector $\{2, 3\}$ at total level 2, there are five P^{arith} -basis vectors:

$$\begin{aligned} &P_{(2)}(x^{(2)}) \otimes 1, \quad 1 \otimes P_{(2)}(x^{(3)}), \quad P_{(1)}(x^{(2)}) \otimes P_{(1)}(x^{(3)}), \\ &P_{(1,1)}(x^{(2)}) \otimes 1, \quad 1 \otimes P_{(1,1)}(x^{(3)}), \end{aligned}$$

matching the five primon states at this level. A single- Λ Macdonald has only three monomials at level 2 in two variables; the stratified tensor structure is essential.

THEOREM 9.5.3 (parameter non-uniformity). . The Hochschild trace class $\mathcal{V}^{\text{prim}}$ on $\Lambda^{\text{prim}} = \bigotimes_p' \Lambda_{(p)}$ admits the following structure:

- (i) The Macdonald inner product on Λ^{prim} is the per-prime pairing with independent parameters $(q_p, t_p)_p$ (Chapter 9.1.3); the naive scalar dictionary $q = p^{-s}$ has the wrong domain.
- (ii) The $\log p$ rescaling is a Heisenberg conformal rescaling multiplicatively outside the (q_p, t_p) parameters (Chapter 9.2.2); it is not a Macdonald parameter.
- (iii) The correct algebra is $\mathbb{H}^{\text{arith}} = \bigotimes'_p \text{SH}(q_p, t_p)$ (Chapter 9.3.4); the shuffle structure factors across primes (Chapter 9.4.1).
- (iv) The basis $\{P_{(\lambda_p)_p}^{\text{arith}}\}$ is the prime-factored tensor product of classical Macdonald polynomials (Chapter 9.5.1).

The trace class $\mathcal{V}^{\text{prim}}$ is obtained from the Schiffmann–Vasserot rank-one degeneration in each prime factor; arithmetic data sits in the tensor-indexing over primes, not in the deformation parameters.

THE ARITHMETIC INSTANTON PARTITION FUNCTION

Nekrasov's partition function has two equivariant parameters and a Coulomb modulus. The Hartnoll–Yang $L(s)$ sees only the degenerate ($\epsilon_2 \rightarrow \infty, b \rightarrow 0$) arithmetic limit; the full Ω -background is absent.

10.1 NEKRASOV'S Z_{inst} ON $\mathbb{C}^2$

THEOREM 10.1.1 (*Nekrasov, hep-th/0206161 Thm 2; Nekrasov–Okounkov hep-th/0306238 §3*). . For $U(N)$ gauge theory on $\mathbb{C}^2$ with equivariant torus action by (t_1, t_2) on $(z_1, z_2) \in \mathbb{C}^2$, $\epsilon_i = \log t_i$:

$$Z_{\text{inst}}^{U(N)}(\vec{a}, \epsilon_1, \epsilon_2; \Lambda) = \sum_{\vec{\lambda}} \Lambda^{N|\vec{\lambda}|} \prod_{\alpha, \beta=1}^N \prod_{s \in \vec{\lambda}(\alpha)} \frac{1}{E_{\alpha\beta}(s)(\epsilon_1 + \epsilon_2 - E_{\alpha\beta}(s))}$$

with $E_{\alpha\beta}(s) = a_\alpha - a_\beta - \epsilon_1 L_{\vec{\lambda}(\beta)}(s) + \epsilon_2 (A_{\vec{\lambda}(\alpha)}(s) + 1)$, where $A_\lambda(s)$ (resp. $L_\lambda(s)$) denotes the arm length (number of boxes strictly to the right of s in λ) and leg length (number of boxes strictly below s in λ). This is the Nekrasov–Okounkov arm-on- α , leg-on- β convention.

10.2 THE ARITHMETIC CURVE AS A ONE-DIMENSIONAL OBJECT: NO INSTANTON SECTOR

Observation 10.2.1. $\mathbb{C}^2$ is real 4-dimensional with a T^2 torus action. The Arakelov-compactified arithmetic curve $\overline{C} = \overline{\text{Spec } \mathbb{Z}}$ has arithmetic Krull dimension 1; arithmetic Chow groups $\overline{\text{CH}}^*(\overline{C})$ live only in degrees 0, 1. There is no $\widehat{\text{CH}}^2$; no instanton class.

COROLLARY 10.2.2. . A $U(N)$ -gauge theory on the arithmetic 1-dimensional $\overline{C}$ is 2d Yang–Mills in character with no instanton sector. The primon partition function $L(s)$ is the degenerate Nekrasov limit on the self-product $\overline{C} \times_{\mathbb{F}_1} \overline{C}$, not a direct $Z_{\text{inst}}^{U(2)}$ on $\overline{C}$.

10.3 THE SELF-PRODUCT $\overline{C} \times_{\mathbb{F}_1} \overline{C}$ AS ARITHMETIC SURFACE

Principle 10.3.1 (Deligne). Arakelov-theoretic constructions on $\text{Spec } \mathbb{Z}$ become two-dimensional on the self-product. Durov's arXiv:0704.2030 and Connes–Consani arXiv:1405.4527 and arXiv:1502.05580 supply the $\mathbb{F}_1$ base over which $\overline{C} \times_{\mathbb{F}_1} \overline{C}$ is a surface.

Definition 10.3.2 (arithmetic surface setup). On the self-product $\overline{C} \times_{\mathbb{F}_1} \overline{C}$:

- Torus action: $\mathbb{G}_m(\mathbb{F}_1)^2 = \{1\}^2$ literally, or $(\mathbb{Z}_{>0}, \times)^2$ in Connes–Consani semigroup form.
- Fixed points: pairs (p, q) of primes; or single primes p on the diagonal.
- Equivariant weights: $\epsilon_p = \log p$ per prime.

10.4 RANK ANALYSIS: $U(1)^\infty$ PER PRIME

Observation 10.4.1. The primon construction $\mathcal{V}^{\text{prim}}$ carries Coulomb-branch moduli $\{a_p\}_p$ indexed by primes. The gauge rank is $|\mathbb{P}| = \aleph_0$, and $L(s)$ is the arithmetic Nekrasov partition function of a $U(1)^{\mathbb{P}}$ (abelian, rank- $\aleph_0$) theory with one Coulomb factor per prime.

Definition 10.4.2 (abelian arithmetic Nekrasov per prime).

$$Z_{\text{inst},p}^{U(1)}(a_p, \epsilon_1^{(p)}, \epsilon_2^{(p)}; \Lambda_p) = \sum_{\lambda} \Lambda_p^{|\lambda|} \prod_{s \in \lambda} \frac{1}{E_{\text{II}}(s)(\epsilon_1^{(p)} + \epsilon_2^{(p)} - E_{\text{II}}(s))},$$

the $N = 1$ specialisation of Chapter 10.1.1 with $E_{\text{II}}(s) = -\epsilon_1^{(p)} L_{\lambda}(s) + \epsilon_2^{(p)} (A_{\lambda}(s) + 1)$. On the CY / Nekrasov–Okounkov slice $\epsilon_1 + \epsilon_2 = 0$, $\epsilon_1 = -\epsilon_2 = \hbar$:

$$Z_{\text{inst}}^{U(1),\text{CY}} = \sum_{\lambda} \frac{\Lambda^{|\lambda|}}{\prod_{s \in \lambda} h_{\lambda}(s)^2} = \exp(\Lambda),$$

after absorbing the $\epsilon_1 \epsilon_2$ -normalisation into Λ . The equality is the hook-length identity $\sum_{\lambda \vdash n} \prod_{s \in \lambda} h_{\lambda}(s)^{-2} = 1/n!$.

COMPUTATION 10.4.3. At $\Lambda_p = p^{-s}$:

$$Z_{\text{inst},p}^{U(1),\text{CY}}(s) = \exp(p^{-s}).$$

The product over primes gives

$$\prod_p Z_{\text{inst},p}^{U(1),\text{CY}}(s) = \exp\left(\sum_p p^{-s}\right).$$

This is the first plethystic logarithmic term of the Euler product, not the Euler factor itself. The factor $(1 - p^{-s})^{-1}$ belongs to the arithmetic Fock degeneration below.

10.5 THE $\epsilon_2 \rightarrow \infty$ DEGENERATION

Definition 10.5.1 (degenerate per-prime limit). Set per prime p :

$$\epsilon_1^{(p)} = \log p, \quad \epsilon_2^{(p)} = +\infty.$$

In this limit $1/\epsilon_2^{(p)} \rightarrow 0$ and every Young diagram with $|\lambda| \geq 2$ contributes 0. The sum collapses to

$$Z_{\text{inst},p}^{U(1)}|_{\text{arith}} = (1 - p^{-s})^{-1}.$$

THEOREM 10.5.2 (primon partition as degenerate Nekrasov). . The Hartnoll–Yang primon partition function $\zeta(s)$ is the product over primes of degenerate $U(1)_p$ Nekrasov functions in the $(\epsilon_1^{(p)}, \epsilon_2^{(p)}) = (\log p, \infty)$ limit:

$$\zeta(s) = \prod_p Z_{\text{inst},p}^{U(1)}(s) \Big|_{\substack{\epsilon_2^{(p)} \rightarrow \infty \\ b^{(p)} \rightarrow 0}} = \prod_p (1 - p^{-s})^{-1}.$$

For a rank- N local system ρ , the analogous Euler factor is $\det(1 - p^{-s} \rho_p(\text{Frob}_p))^{-1}$; that is a separate rank- N extension, not the primon rank statement. The Ω -background is visible only in the full ϵ_2 -deformed theory; the limit $\epsilon_2 \rightarrow \infty$, $b \rightarrow 0$ retains only the Euler-factor data.

10.6 ARAKELOV COULOMB MODULUS

Definition 10.6.1 (arithmetic Coulomb modulus). The complex parameter $s = \sigma + i\tau \in \mathbb{C}$ (as in p^{-s}) decomposes via the Arakelov degree map: its real part $\sigma = \text{Re}(s)$ is the arithmetic first Chern class

$$\sigma = \widehat{c}_1(\mathcal{O}(\sigma \cdot [\infty])) \in \widehat{\text{CH}}^1(\overline{C}) \otimes \mathbb{R} \cong \mathbb{R},$$

under the canonical identification of $\widehat{\text{CH}}^1(\overline{C}) \otimes \mathbb{R}$ with $\mathbb{R}$ by the archimedean degree map; the imaginary part $\tau = \text{Im}(s)$ parameterises the unitary direction (Hecke-character / spectral angle). The Nekrasov Coulomb modulus $\vec{a}$ is replaced by this single arithmetic modulus s . The Hecke-character tuple $\{\theta_p\}$ is the arithmetic analogue of the non-trivial Coulomb splitting $a_\alpha - a_\beta$ for $U(N)$.

Remark 10.6.2 (Gillet–Soulé Arakelov pairing). On $\overline{C} \times \overline{C}$ the height pairing at $\infty \times \infty$ is $s \cdot s'$ up to finite contributions (Gillet–Soulé *Arithmetic intersection theory*, Publ. IHES 1992).

10.7 NEKRASOV–SHATASHVILI LIMIT: $\log \zeta(s) = \sum_p \mathcal{W}_p^{\text{NS}}(s)$

Theorem 10.7.1 (arithmetic NS identity). . The regulated per-prime NS superpotential $\mathcal{W}_p^{\text{NS}}(s) = \log(1 - p^{-s})^{-1} = \sum_{k \geq 1} p^{-ks} / k$ sums over primes to

$$\log \zeta(s) = \sum_p \mathcal{W}_p^{\text{NS}}(s).$$

The Dedekind zeta logarithm is the sum over primes of the arithmetic Nekrasov–Shatashvili superpotential, evaluated at arithmetic Coulomb modulus s .

10.8 AGT: SEMICLASSICAL LIOUVILLE DEGENERATES TO FREE BOSON

Remark 10.8.1 (AGT obstruction). AGT (Alday–Gaiotto–Tachikawa arXiv:0906.3219): $Z_{\text{inst}}^{U(2)}(\vec{a}, \epsilon_1, \epsilon_2; \Lambda) = \mathcal{B}_{c, \Delta_i}^{\text{Liouville}}(\vec{q})$ with $c = 1 + 6(b + b^{-1})^2$, $b^2 = \epsilon_1 / \epsilon_2$. In the arithmetic limit $\epsilon_2 \rightarrow \infty$, $b \rightarrow 0$; $c \rightarrow \infty$. The AGT correspondence at finite b requires Liouville theory at finite central charge; its $b \rightarrow 0$ limit yields the semiclassical Liouville theory, which degenerates further to a free boson (Theorem 10.8.2).

Theorem 10.8.2 (semiclassical-Liouville $\rightarrow$ free boson). (Teschner arXiv:hep-th/0104158; Harlow–Maltz–Witten arXiv:1108.4417). In the $b \rightarrow 0$ limit, Virasoro degenerates to Heisenberg: the Liouville primary $V_\alpha = e^{2\alpha\phi}$ with $\alpha = \alpha_0 + \alpha_1 b$ becomes a free-boson vertex operator of charge α_0 ; Virasoro conformal blocks become free-boson correlators.

Corollary 10.8.3 (arithmetic AGT at $b \rightarrow 0$). . Per prime,

$$Z_{\text{inst}, p}^{U(1)}|_{b(p) \rightarrow 0}(\Lambda_p) = \text{Tr}_{\text{Sym}(\mathbb{C}a_{-1})}(\Lambda_p^{N_p}) = (1 - \Lambda_p)^{-1}.$$

At $\Lambda_p = p^{-s}$ this gives the primon Heisenberg trace $(1 - p^{-s})^{-1}$ per prime.

Theorem 10.8.4 (Hartnoll–Yang primon trace = $b \rightarrow 0$ AGT). .

$$L(s) = \prod_p Z_{\text{inst}, p}^{U(1)}|_{\text{AGT}, b \rightarrow 0}(s) = \prod_p \langle \mathcal{H}_{k=1}^{(p)} \rangle_{\text{partition}}.$$

The rank-1 Heisenberg per prime is the semiclassical-Liouville degeneration of Virasoro at prime p , and the AGT map reduces to a free-boson partition at each place.

10.9 MAULIK–OKOUNKOV AND THE BETHE–HECKE SHADOW

THEOREM 10.9.1 (*Maulik–Okounkov arXiv:1211.1287, Thm 1.5.1*). . The R -matrix on equivariant cohomology of the instanton moduli space $\mathcal{M}_{N,k}$ generates the Yangian $Y(\widehat{\mathfrak{gl}}_N)$. $Z_{\text{inst}}^{U(N)}$ is the trace of the holonomy of the MO R -matrix.

Definition 10.9.2 (primon Bethe–Hecke shadow). The *primon Bethe–Hecke shadow* is the commutative Bethe/Hecke algebra acting on the arithmetic instanton model. Fixed-point data are indexed by N -tuples of Young diagrams per prime; the arithmetic action surviving the $\epsilon_2 \rightarrow \infty$ degeneration is through Hecke transfer operators, not an RTT R -matrix or Yangian Hopf structure.

THEOREM 10.9.3 (*L -function as MO trace*). . $L(s, \rho)$ is the Bethe/Hecke transfer trace in the arithmetic instanton shadow. The Maulik–Okounkov Yangian $Y(\widehat{\mathfrak{gl}}_N)$ governs the complex-geometric instanton count; its arithmetic avatar, in the degenerate limit, is the commutative Hecke algebra acting at each prime.

10.10 THE TOPOLOGICAL-SECTOR READING (COMPATIBLE WITH NEKRASOV)

Remark 10.10.1 (Kurokawa/Deninger). The identity $L(s) = \sum_{n \geq 1} n^{-s}$ expresses $\zeta(s)$ as a sum over arithmetic topological classes with p -adic valuation $v_p(n)$; the term n^{-s} carries the arithmetic class indexed by the prime factorization of n . From the $\mathbb{F}_1$ -zeta perspective of Kurokawa (Proc. Japan Acad. 2005) and Deninger (1994), $\zeta(s) = \det(s - \Theta)_{\text{reg}}^{-1}$ for a hypothetical cohomological operator Θ .

THEOREM 10.10.2 (*motivic-topological compatibility*). . The motivic instanton partition function (Nekrasov–Okounkov §4; Behrend motivic DT arXiv:math/0507523) on the arithmetic surface localises in the $\epsilon_2 \rightarrow \infty$ limit to the $\mathbb{F}_1$ -cohomological partition sum. Young diagrams collapse to their size $|\lambda|$, which labels the instanton number c_2^{Ar} , and the c_2^{Ar} -sum becomes the v_p -sum:

$$Z_{\text{inst},p}^{U(N),\text{arith}}(s, \rho_p) = \det(1 - p^{-s} \rho_p(\text{Frob}_p))^{-1}.$$

10.11 THE ($\epsilon_2 \rightarrow \infty$, $b \rightarrow 0$) DEGENERATE LIMIT: STRUCTURE OF $L(s)$

- (i) The arithmetic curve $\overline{C}$ has arithmetic Krull dimension 1; Nekrasov’s $\mathbb{C}^2$ is real dimension 4. The self-product $\overline{C} \times_{\mathbb{F}_1} \overline{C}$ is the arithmetic surface carrying the instanton data (Chapter 10.3.1).
- (ii) The gauge group is $U(1)^{\mathbb{P}}$, abelian, rank $\aleph_0$ (Chapter 10.4.1).
- (iii) The Ω -background per prime is $(\epsilon_1^{(p)}, \epsilon_2^{(p)}) = (\log p, +\infty)$ (Chapter 10.5.1).
- (iv) The primon partition function $L(s)$ is the ($\epsilon_2 \rightarrow \infty$, $b \rightarrow 0$) degenerate Nekrasov limit (Chapter 10.5.2).
- (v) The Coulomb modulus is the arithmetic first Chern class $s = \widehat{\deg} \mathcal{O}(\sigma)$ (Chapter 10.6.1).
- (vi) $\log \zeta(s) = \sum_p \mathcal{W}_p^{\text{NS}}(s)$: the zeta logarithm is the sum of arithmetic Nekrasov–Shatashvili superpotentials (Chapter 10.7.1).
- (vii) In the $b \rightarrow 0$ AGT limit, the Liouville conformal block at prime p degenerates to the free-boson partition function $(1 - p^{-s})^{-1}$ (Chapter 10.8.3).
- (viii) The Maulik–Okounkov R -matrix reduces, in the arithmetic degeneration, to the Hecke transfer algebra; $L(s, \rho)$ is the Hecke transfer trace (Chapter 10.9.3).

$L(s)$ is the $(\epsilon_2 \rightarrow \infty, b \rightarrow 0)$ degenerate limit of the Nekrasov partition function: the Euler-factor data of the Ω -background survives; the full instanton moduli geometry contracts to fixed-point contributions labeled by $\epsilon_1^{(p)} = \log p$ per prime.

THE ARITHMETIC MODULAR FUNCTOR

Segal's modular functor assigns conformal blocks to marked Riemann surfaces and uses sewing along circles. The arithmetic curve has no such sewing category. Tate theory supplies the object occupying the same formal position.

II.1 AXIOMS AND THEIR TRANSLATION

Each Segal axiom for a Riemann surface $\Sigma_{g,n}$ – finiteness of conformal blocks, sewing along circles, modular invariance, and the partition-function trace – has an arithmetic analogue on the Arakelov curve $(\text{Spec } \overline{\mathcal{O}_K}, S)$. The entries below are a dictionary of roles. They are not literal identifications of the two categories.

Convention II.1.1 (arithmetic Segal dictionary).

Geometric Segal	Arithmetic Segal
Riemann surface $\Sigma_{g,n}$	Arakelov curve $(\text{Spec } \overline{\mathcal{O}_K}, S)$
Orientation	Trivialization of $\det R\Gamma$ (Deligne)
Genus contribution	Arakelov discriminant volume $\frac{1}{2} \log d_K $
Marked point	Place (finite or archimedean)
Sewing circle	Punctured formal disk $\text{Spec } K_{\mathfrak{p}}$
Intermediate rep	Weil–Deligne parameter (ρ, N)
Sewing law	Euler-product factorization of $L(s, \chi)$
Monodromy $U(1)$	Frobenius $\alpha_p = \chi_p(p)$
MCG(Σ)	arithmetic symmetry group acting on labels and Hecke lines
Modular S -matrix	Γ -factor + functional equation
Verlinde scalar	Class-number residue at $s = 1$
Conformal block at $\{V_i\}$	Tate integral $\prod_v \int \cdot ^s \Phi_v$
$Z(\Sigma) = \text{tr } q^{L_0 - c/24}$	finite Euler product $\prod_{\mathfrak{p}} (1 - N\mathfrak{p}^{-s})^{-1}$, completed by the archimedean Tate factors

II.2 ARAKELOV DISCRIMINANT AS LATTICE VOLUME

Observation II.2.1 (the discriminant term is not a genus). The geometric genus satisfies $g = \frac{1}{2} \dim_{\mathbb{R}} H^1(\Sigma; \mathbb{R})$; equivalently, the Jacobian $J(\Sigma) = H^0(\Sigma, \Omega^1)^* / H_1(\Sigma, \mathbb{Z})$ is a g -dimensional torus whose Weil height is $\frac{1}{2} \log \det(\text{Im } \tau)$ for the period matrix τ . The arithmetic analogue is $\frac{1}{2} \log |d_K| = \frac{1}{2} \log(\text{covol}(\mathcal{O}_K \subset K \otimes_{\mathbb{Q}} \mathbb{R}))$. Both are half-logarithms of lattice covolumes; the functional is the same, the lattices differ. The number $\frac{1}{2} \log |d_K|$ is not an integer genus and is not a Betti number.

Definition 11.2.2 (arithmetic Euler characteristic).

$$\chi^{\text{arith}}(\overline{X}) := -\widehat{\deg} \omega_{\overline{X}} + \sum_{v|\infty} \log \|\cdot\|_v.$$

On $\text{Spec } \mathcal{O}_K$ with signature (r_1, r_2) , $\chi^{\text{arith}} = \log |d_K| - r_1 \log(\pi/4) - r_2 \log(\pi/2)$ by Minkowski–Odlyzko. Additive on disjoint unions of Arakelov surfaces.

11.3 WEIL–DELIGNE SEWING

THEOREM 11.3.1 (sewing law). .

Here $\mathcal{F}^{\text{arith}}$ denotes the arithmetic modular functor defined in Definition 11.8.1 (an adelicly factorized assignment of meromorphic lines to Arakelov surfaces with marked places); and $\mathcal{V}^{\text{prim}}$ denotes the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 (a class in K -theory, not a vertex algebra). The ℓ -adic coefficient system at a non-archimedean prime $\mathfrak{p}$ is the étale cohomology local system attached to $\mathcal{V}^{\text{prim}}$ via the comparison isomorphism $H_{\text{ét}}^*(-, \mathbb{Q}_\ell)$ for any prime $\ell \neq p$; this is the coefficient ring of the Weil–Deligne representation, not an enrichment of $\mathcal{V}^{\text{prim}}$ with vertex-algebra structure.

$$\mathcal{F}^{\text{arith}}(\overline{X}_1 \cup_p \overline{X}_2) = \bigoplus_{(\rho, N)} \mathcal{F}^{\text{arith}}(\overline{X}_1; \rho, N) \otimes_{\text{End}(\rho)} \mathcal{F}^{\text{arith}}(\overline{X}_2; \rho, N)^*$$

where (ρ, N) runs over isomorphism classes of Weil–Deligne representations of $W_{\mathbb{Q}_p}$ whose coefficients are ℓ -adic ($\ell \neq p$), ρ continuous, N the monodromy operator.

Remark 11.3.2 (labels are Langlands parameters). Segal’s intermediate representation labels are local Galois representations. At archimedean places, sewing is along Hodge structures (Weil-group representations of $W_{\mathbb{R}}$ or $W_{\mathbb{C}}$). The correspondence between WZW representation labels and automorphic representations under local Langlands is:

$$\{\text{irr. admissible } G(\mathbb{Q}_p)\text{-reps}\} \leftrightarrow \{\text{Weil–Deligne } L\text{-parameters}\}.$$

The arithmetic functor uses local Langlands parameters in the Segal sewing position.

11.4 ARITHMETIC SYMMETRIES

Observation 11.4.1. The arithmetic symmetry group has two complementary actions. The Galois group $\text{Gal}(\overline{K}/K)$ controls label permutations: it acts on Weil–Deligne parameters at marked places, the arithmetic counterpart of the deck transformations on the geometric surface. The adelic quotient $\text{GL}_n(\mathbb{A}_K)/\text{GL}_n(K)$ acts through Hecke correspondences on automorphic lines. This pair is the arithmetic replacement for the geometric mapping-class action; it is not a mapping class group of a surface.

THEOREM 11.4.2 (arithmetic symmetry action). . There is an arithmetic symmetry action in which $\text{Gal}(\overline{K}/K)$ acts on local Langlands labels and $\text{GL}_n(\mathbb{A}_K)$ acts by Hecke correspondences. At an unramified prime p , Frobenius generates the quotient $\text{Gal}(\overline{k}_p/k_p) \cong \widehat{\mathbb{Z}}$. This is only the unramified quotient of local monodromy; the inertia and wild inertia are not Dehn twists and are not discarded in ramified sewing.

II.5 VERLINDE SCALAR AND THE CLASS NUMBER FORMULA

THEOREM II.5.1 (*arithmetic Verlinde*). (Tate's thesis, class number formula). The arithmetic Verlinde scalar attached to the empty-marking sector is the residue of the Dedekind zeta at $s = 1$:

$$\text{Ver}^{\text{arith}}(K) := \text{Res}_{s=1} \zeta_K(s) = \frac{2^{r_1} (2\pi)^{r_2} h_K R_K}{w_K \sqrt{|d_K|}}.$$

This scalar is generally not an integer dimension of conformal blocks; it is the class-number residue in the Verlinde position of the arithmetic analogy.

II.6 EXPLICIT COMPUTATION: CONFORMAL BLOCKS VIA TATE INTEGRALS

COMPUTATION II.6.1 (*conformal block at $(\mathbb{Z}, \{p, \infty\}, \text{triv})$*). $\mathcal{F}^{\text{arith}}(\text{Spec } \mathbb{Z}, \{p, \infty\}, V_{\text{vac}} \otimes \lambda_{\infty})$ is spanned by the Tate integral

$$\Phi_p(s) = \Lambda_{\zeta}(s) \cdot (1 - p^{-s}),$$

where $\Lambda_{\zeta}(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is the completed Riemann zeta (Tate's thesis, §3.4). Removing the p -th Euler factor from $\zeta(s)$ gives $\zeta(s)(1 - p^{-s})$; multiplication by $\pi^{-s/2} \Gamma(s/2)$ completes to $\Lambda_{\zeta}(s)(1 - p^{-s})$. Dimension: 1.

Observation II.6.2. The vacuum conformal block at prime p is the completed zeta function with the p -th Euler factor removed. The sewing axiom at p reinserts it.

COMPUTATION II.6.3 (*sewing at $\{2, 3\}$*). The vacuum conformal block at marked places $\{2, 3, \infty\}$ is $\Lambda_{\zeta}(s)(1 - 2^{-s})(1 - 3^{-s})$ (the completed zeta with the Euler factors at 2, 3 removed; Chapter II.6.2). Sewing at the two finite places with unramified Satake parameters $\alpha_p = \chi_p(p) \in \{\pm 1\}$ reinserts the twisted Euler factors:

$$\Phi(s) = \Lambda_{\zeta}(s) \frac{(1 - 2^{-s})(1 - 3^{-s})}{(1 - \alpha_2 \cdot 2^{-s})(1 - \alpha_3 \cdot 3^{-s})},$$

where $\Lambda_{\zeta}(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$. For trivial Satake $(\alpha_2, \alpha_3) = (1, 1)$ one recovers $\Phi(s) = \Lambda_{\zeta}(s)$; for a character with local data specified only at 2 and 3, this is only a two-prime partial twist. The full completed L -function $\Lambda(s, \chi)$ is obtained only after replacing the Euler factor at every finite prime by $(1 - \chi(p)p^{-s})^{-1}$, together with the correct archimedean factor and conductor.

II.7 FUNCTIONAL EQUATION AS SEWING AXIOM

THEOREM II.7.1 (*Tate's global functional equation*). .

Let $\Lambda(s, \chi)$ denote the completed Dirichlet L -function with its conductor and parity-dependent archimedean factor. Then

$$\Lambda(s, \chi) = \varepsilon(\chi) \cdot \Lambda(1 - s, \chi^{-1}),$$

where $\varepsilon(\chi)$ is the root number, proved by adelic Poisson summation (Tate's thesis, §4.4). The arithmetic analogue of Segal sewing is the functional equation.

Remark II.7.2 (finiteness reading). The content of sewing is not dimension 1 per se; it is the $s \leftrightarrow 1 - s$ involution on the one-dimensional line whose invariants are the values of the completed L -function.

II.8 FIVE-AXIOM ARITHMETIC MODULAR FUNCTOR

Definition II.8.1. An *arithmetic modular functor* is a functor

$$\mathcal{F}: \mathcal{C}^{\text{Arak}} \rightarrow \text{Hom}(\mathbb{C}_s, \text{Line})$$

(from Arakelov surfaces to lines depending meromorphically on a parameter s) satisfying:

(A1) *Finiteness.* $\mathcal{F}(\overline{X}, S, \{\chi_p\})$ is a 1-dimensional meromorphic line, with pole divisor prescribed by the completed L -function represented by the line.

(A2) *Factorization.* For independent places, $\mathcal{F}(\overline{X}, S, \{\chi_p\}) = \bigotimes_{p \in S} \mathcal{F}_p(\chi_p)$.

(A3) *Weil–Deligne sewing.* For $S = S_1 \sqcup S_2$ along a parameter (ρ, N) at an extra prime $\mathfrak{q}$,

$$\mathcal{F}(\overline{X}, S \cup \{\mathfrak{q}\}) = \sum_{(\rho, N)} \mathcal{F}(\overline{X}_1, S_1 \cup (\rho, N)) \cdot \mathcal{F}(\overline{X}_2, S_2 \cup (\rho, N))^{-1},$$

summed over unramified (ρ, N) .

(A4) *Modular invariance.* the Galois action on local labels and the Hecke action of $GL_n(\mathbb{A}_K)$ act on the functor.

(A5) *Functional equation.*

$$\mathcal{F}(\overline{X}, S, \chi)(s) = \varepsilon(\chi) \cdot \mathcal{F}(\overline{X}, S, \chi^{-1})(1-s).$$

This is the arithmetic modular invariance: $SL_2(\mathbb{Z})$ becomes the $\mathbb{Z}/2$ flip $s \leftrightarrow 1-s$.

THEOREM II.8.2 (*Tate line in the arithmetic Segal position*). (Tate’s thesis). Tate’s adelic construction gives a meromorphic L -line satisfying (A1)–(A5) in the zeta and Dirichlet sectors. No uniqueness is asserted before the source category and the local test-line normalizations are fixed.

The geometric and arithmetic Segal functors are different functors. The conjecture says that Tate’s L -functor satisfies the arithmetic axioms occupying the Segal positions.

ARITHMETIC BAR-COBAR AND PARSHIN-GERSTEN

The chiral bar-cobar adjunction on a complex curve is replaced, over $\overline{\text{Spec } \mathbb{Z}}$, by an ordered Parshin-Gersten complex. The de Rham differential is replaced by boundary maps in the arithmetic flag complex.

12.1 THE CONFIGURATION SPACE IS DISCRETE

Observation 12.1.1. $\text{Conf}_n(\text{Spec } \mathbb{Z}) = \{(\mathfrak{p}_1, \dots, \mathfrak{p}_n) : \mathfrak{p}_i \neq \mathfrak{p}_j\}$ is a countable discrete set of n -tuples of distinct closed points; the collision locus Δ is a union of isolated points, and convergence to Δ is a p -adic limit, not complex-analytic.

Definition 12.1.2 (arithmetic configuration space). The arithmetic configuration space is the incidence poset of finite subsets of $|\overline{C}|$, ordered by inclusion. Forms are combinatorial (elements of an order complex). The Arnold relation lifts to the braiding relation in the Grothendieck-Teichmüller Lie algebra at $\text{Spec } \mathbb{Z}$, controlled by Ihara-Deligne motivic fundamental group $\pi_1^{\text{mot}}(\text{Spec } \mathbb{Z} \setminus \{p_1, \dots, p_n\})$.

12.2 THE ARNOLD RELATION LIFTS TO STEINBERG

Observation 12.2.1 (no $d \log(p - q)$). $p - q$ is a fixed integer, not a variable; there is no differential 1-form $d \log(p - q)$.

THEOREM 12.2.2 (Steinberg relation replaces Arnold). (Milnor, Tate). The arithmetic Arnold relation is the Steinberg relation

$$\boxed{\{p, 1 - p\} = 0 \text{ in } K_2^M(\mathbb{Q}).}$$

Tate's theorem identifies $K_2^M(\mathbb{Q}) = \{\pm 1\} \oplus \bigoplus_p \mathbb{F}_p^\times$ via the tame symbol at each prime.

THEOREM 12.2.3 (Parshin reciprocity = triple Arnold). (Parshin, Kato, Beilinson, Bloch). The triple-Arnold identity lifts to the Parshin reciprocity law on $\text{Spec } \mathcal{O}_K$ for K a number field:

$$\sum_{\mathfrak{p}} \{a, b, c\}_{\mathfrak{p}} = 0,$$

a sum of local triple symbols over all primes including archimedean. The arithmetic η_{ij} (the Arnold 2-form $d \log(z_i - z_j)$ on the complex configuration space, here replaced by the tame symbol) is the map $\partial_{\mathfrak{p}} : K_{n+1}^M(K) \rightarrow K_n^M(k(\mathfrak{p}))$, and the Arnold relation is the 3-cocycle condition on the Gersten complex:

$$0 \rightarrow K_n^M(K) \rightarrow \bigoplus_{\mathfrak{p}} K_{n-1}^M(k(\mathfrak{p})) \rightarrow \bigoplus_{\mathfrak{q}} K_{n-2}^M(k(\mathfrak{q})) \rightarrow \dots$$

Parshin-Kato prove exactness; Beilinson-Bloch identify the motivic cohomology of $\text{Spec } \mathcal{O}_K$ as the cohomology of this complex shifted by n .

12.3 THE ARITHMETIC TENSOR COALGEBRA

Definition 12.3.1 (Λ -tensor coalgebra). For $\mathcal{V}^{\text{prim}}$ the Hochschild trace class of the identity on the global arithmetic Iwahori Hecke category of GL_2 , regarded as an E_1 -algebra object in motivic sheaves on $\overline{\mathbb{C}}$ with Λ -action ($\Lambda = \mathbb{Z}_p[[\Gamma]]$ the cyclotomic Iwasawa algebra, $\Gamma = \text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) \cong \mathbb{Z}_p$),

$$B^{\text{ch}, \text{Ar}}(\mathcal{V}^{\text{prim}}) := \bigoplus_{n \geq 0} (s^{-1} \overline{\mathcal{V}^{\text{prim}}})^{\otimes \Lambda^n} \otimes_{\Lambda} C_{\text{Parshin}}^{\bullet}(\overline{\mathbb{C}}),$$

with pre-differential

$$d^{\text{Ar}} = d_{\text{Iw}} \otimes \text{I} + \text{I} \otimes d_{\text{Gersten}} + d_{\text{bar}}^{\text{local}}.$$

The three pieces: Iwasawa main-conjecture differential (Coleman–Perrin-Riou); Gersten complex differential via the tame symbol; local bar differential at each prime weighted by d_p . It is a differential precisely when the Iwasawa, Gersten, and local bar terms anticommute. This anticommutativity is not proved in this definition.

Conjecture 12.3.2 (arithmetic bar nilpotency). . The pre-differential d^{Ar} satisfies $(d^{\text{Ar}})^2 = 0$. Equivalently, the Iwasawa differential, the Gersten differential, and the tame-symbol-weighted local bar differential form a mixed complex.

THEOREM 12.3.3 (archimedean reduction). . At $\mathfrak{p} = \infty$:

- The Parshin complex localises to the de Rham complex on $\text{Spec } \mathbb{R}$, which is trivial in positive degree.
- The Iwasawa differential vanishes (no cyclotomic tower at ∞).
- The local bar reduces to the classical one.

Therefore $B^{\text{ch}, \text{Ar}}(\mathcal{V}^{\text{prim}})_{\infty} \simeq B^{\text{ch}}(\mathcal{V}_{\infty}^{\text{prim}})$, the standard chiral bar complex over $\mathbb{C}$.

12.4 KOSZULNESS AND DELIGNE PURITY

THEOREM 12.4.1 (arithmetic Koszulness criterion). . $\mathcal{V}^{\text{prim}}$ is arithmetically Koszul iff its p -adic Simpson realisation $\mathcal{V}_{p\text{-Simp}}^{\text{prim}}$ is pure as a Sen module of weight equal to its homological degree, for every prime p of good reduction. Equivalently, the ℓ -adic bar complex $B^{\text{ch}, \ell}(\mathcal{V}^{\text{prim}}) \otimes \mathbb{Q}_{\ell}$ has Frobenius-weight concentration:

$$H^n B^{\text{ch}, \ell}(\mathcal{V}^{\text{prim}}) \text{ is pure of weight } n$$

at every prime $\ell \neq p_{\text{bad}}$.

Remark 12.4.2 (Riemann–Hilbert over $\mathbb{Z}$). Deligne’s purity translates D-module purity over $\mathbb{C}$ (Saito’s mixed Hodge modules) to Frobenius-weight purity over $\mathbb{F}_p$ via the de Rham–étale comparison. The arithmetic Koszulness criterion is a genuine purity statement.

12.5 PRIMARY BAR MELLIN CHARACTER

The arithmetic bar complex $B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ carries a homological grading and a Hecke-monoid grading. The ordinary Poincaré series in homological degree is not a zeta function. At $\kappa^{\text{Ar}} = 0$, the zero-mode primary quotient has a Mellin character indexed by $\mathbb{N}^{\times}$.

THEOREM 12.5.1 (primary bar Mellin character). . For the primary zero-mode quotient of the arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ (the Hochschild trace class specialised to $\kappa^{\text{Ar}} = 0$, the locus of arithmetic

Heisenberg data without interacting corrections) at the undeformed locus, the Hecke-monoid basis is indexed by $m \in \mathbb{N}^\times$. Its Mellin character is

$$\chi_{\text{bar}}(s) := \sum_{m \geq 1} m^{-s} = \prod_p (1 - p^{-s})^{-1} = \zeta(s), \quad \text{Re}(s) > 1.$$

This is a Mellin character of the primary Hecke grading, not a Poincaré series of homological dimensions.

COROLLARY 12.5.2 (*Koszul dual = Bost–Connes*). . The Koszul-dual algebra $(\mathcal{V}_{\text{Heis}}^{\text{prim}})^!$ is the Bost–Connes Q-lattice system $\mathbb{A}^{\text{BC}}$, with its thermodynamic phase transition at $s = 1$.

12.6 THE G-ROW COMPLEMENTARITY

THEOREM 12.6.1 (*arithmetic G-row*). . The derived-centre complementarity

$$\kappa_{\text{G}}^{\text{Ar}} + (\kappa^!)_{\text{G}}^{\text{Ar}} = 0 \iff \xi(s) = \xi(1-s)$$

for the arithmetic Heisenberg is the Riemann functional equation.

Proof sketch. The arithmetic Heisenberg sits in G-row with $\kappa = 0$. Its Koszul dual is Bost–Connes. Complementarity $\kappa + \kappa^! = 0$ translates through the Koszul-resolution computation of Hochschild cochains to the meromorphic continuation and functional equation of ζ . The archimedean $\Gamma(s/2)\pi^{-s/2}$ -factor contributes the derived-centre shift. $\square$

Remark 12.6.2. This is the essential theorem of the arithmetic branch. It is not a tautology: arithmetic Koszulness of Heisenberg *implies* meromorphic continuation and functional equation of ζ , via the Koszul-resolution computation of the Hochschild cochain complex. The Vol I chain “Koszul duality $\Rightarrow$ derived-centre complementarity” is the instance of “Koszul duality $\Rightarrow$ functional equation of the Dirichlet series generating bar cohomology” for the arithmetic Heisenberg.

12.7 ARITHMETIC POSITSIELSKI

The classical Positselski theorem (Positselski, *Two kinds of derived categories*, 2011) is the equivalence between the co-derived and the contraderived categories of a conilpotent coalgebra and its cobar algebra, conditional on vanishing of the Koszul homology in positive degree. The arithmetic avatar replaces the Koszul condition by vanishing of the Iwasawa μ -invariant.

THEOREM 12.7.1 (*arithmetic Positselski*). . For a conilpotent Λ -coalgebra C with $\mu(C) = 0$, there is an equivalence

$$D_{\text{ch,Ar}}^{\text{co}}(C\text{-comod}_\Lambda) \simeq D_{\text{ch,Ar}}^{\text{ctr}}(C\text{-contramod}_\Lambda),$$

and $\Omega^{\text{Ar}}(B^{\text{Ar}}(C)) \xrightarrow{\sim} C$ in the co-derived category. When $\mu(C) \neq 0$, the correspondence degenerates.

The full Positselski equivalence with its four independent verification subitems is Chapter 24.1.5; the statement above records the core equivalence in condensed form.

Remark 12.7.2 (*Iwasawa μ -invariant*). This gives a Koszul-categorical meaning to the Iwasawa μ -invariant: $\mu = 0$ separates Koszulness-compatible arithmetic objects from pathological ones.

12.8 CONSTRUCTION OF $B^{\text{ch}, \text{Ar}}$ CONCRETE

Definition 12.8.1 (arithmetic bar complex, concrete). Let $\mathcal{V}^{\text{prim}}$ be an E_1 -algebra object in the category of motivic sheaves on $\bar{C}$ with Λ -action. Define

$$B^{\text{ch}, \text{Ar}}(\mathcal{V}^{\text{prim}}) := \bigoplus_{n \geq 0} (s^{-1} \overline{\mathcal{V}^{\text{prim}}})^{\otimes \Lambda^n} \otimes_{\Lambda} C_{\text{Parshin}}^{\bullet}(\bar{C}).$$

Pre-differential: $d^{\text{Ar}} = d_{\text{Iw}} \otimes 1 + 1 \otimes d_{\text{Gersten}} + d_{\text{bar}}^{\text{local}}$, with $d_{\text{bar}}^{\text{local}}$ the sum over primes $\mathfrak{p}$ of the local chiral bar differential on $\mathcal{V}_{\mathfrak{p}}^{\text{prim}}$ weighted by the tame symbol $\partial_{\mathfrak{p}}$. It is a differential under Chapter 12.3.2.

THEOREM 12.8.2 (verifications). (items 2–4; item 1 is). The construction satisfies:

- (i) **Archimedean reduction.** (, Chapter 12.3.3.) At ∞ , the construction agrees with the classical bar complex.
- (ii) **Nilpotency.** (, Chapter 12.3.2.) The arithmetic pre-differential squares to zero.
- (iii) **Koszul adjunction.** For arithmetically Koszul $\mathcal{V}^{\text{prim}}$, $\Omega^{\text{ch}, \text{Ar}} \dashv B^{\text{ch}, \text{Ar}}$ is an adjoint equivalence between conilpotent Λ -coalgebras with $\mu = 0$ and arithmetic E_1 -algebras with Deligne-pure bar cohomology.
- (iv) **Complementarity.** $\kappa^{\text{Ar}} + (\kappa^!)^{\text{Ar}} \in \{0, 8, 13, 250/3, 98/3\}$ with G-row = 0 realised by arithmetic Heisenberg (Chapter 12.6.1), B-row = 8 realised by arithmetic K3-Mukai Borchers lift, others unrealised.

12.9 BAR-COBAR BOUNDARY

- (i) Configuration space is discrete; Arnold relation lifts to Steinberg $\{p, 1 - p\} = 0$ (Chapter 12.2.2).
- (ii) The triple Arnold identity is Parshin reciprocity on the Gersten complex (Chapter 12.2.3).
- (iii) Arithmetic bar complex is Iwasawa-twisted Λ -coalgebra with Parshin–Gersten base (Chapter 12.3.1).
- (iv) Archimedean reduction to classical bar is exact (Chapter 12.3.3).
- (v) Koszulness translates to Deligne weight-concentration purity (Chapter 12.4.1).
- (vi) Primary bar Mellin character = $\zeta(s)$ (Chapter 12.5.1); Koszul dual = Bost–Connes (Chapter 12.5.2).
- (vii) G-row complementarity $\kappa + \kappa^! = 0$ is the Riemann functional equation $\xi(s) = \xi(1 - s)$ (Chapter 12.6.1).
- (viii) Arithmetic Positselski holds precisely when μ -invariant = 0 (Chapter 24.1.5).

The five theorems of the programme survive the arithmetic deformation, interpreted as: configuration space = Parshin–Gersten; Arnold = Steinberg; D-module purity = Deligne Frobenius purity; $\mu = 0$ = conilpotence; arithmetic G-row complementarity = functional equation of ζ .

THE AUTOMORPHIC VERTEX OF THE LANGLANDS–TQFT TRIANGLE

Kapustin–Witten realizes geometric Langlands as S -duality in $\mathcal{N} = 4$ gauge theory on $\Sigma \times C$. The arithmetic trace class has a different problem: whether the Hartnoll–Yang BKL–primon correspondence supplies a bulk theory at all.

$\mathcal{V}^{\text{prim}}$ is a Hochschild trace class whose graded character is $\zeta(s)$. Three TQFT frameworks (Kapustin–Witten, Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees (BLLPRvR), and Chern–Simons/WZW) offer candidate TQFT-bulk structures for the Langlands triangle. Each fails at a specific structural obstruction, placing $\mathcal{V}^{\text{prim}}$ on the automorphic vertex rather than the TQFT vertex.

13.1 THE KAPUSTIN–WITTEN OBSTRUCTION: ABSENCE OF COMPLEX STRUCTURE

THEOREM 13.1.1 (*Kapustin–Witten setup*). . Geometric Langlands for $C/\mathbb{C}$ arises as S -duality of 4D $\mathcal{N} = 4$ super Yang–Mills (SYM) on $\Sigma \times C$ with Σ topologically twisted and C holomorphically twisted. Categories $D(\text{Bun}_G)$ and $\text{QCoh}(\text{Loc}_L G)$ appear as boundary conditions; S -duality $\tau \mapsto -1/\tau$ exchanges them.

Observation 13.1.2 (arithmetic obstruction). $\overline{C} = \overline{\text{Spec } \mathbb{Z}}$ is an arithmetic 1-scheme with closed-point residue fields $\mathbb{F}_p$ at each prime p , plus the archimedean place. The holomorphic twist of $\mathcal{N} = 4$ super Yang–Mills requires a complex structure on the compactified direction; $\overline{\text{Spec } \mathbb{Z}}$ carries no such structure, its closed fibres being spectra of finite fields rather than complex-analytic curves.

COROLLARY 13.1.3 (*Kapustin–Witten is undefined over $\mathbb{Z}$*). . The expression “ $\mathcal{N} = 4$ SYM on $\Sigma \times \overline{\text{Spec } \mathbb{Z}}$ ” names no defined theory in 2026’s mathematics. The Kapustin–Witten holomorphic twist requires a complex structure on the compactified direction; the topological Σ -coupling and the S -duality $\tau \mapsto -1/\tau$ are consequences. The arithmetic compactification $\overline{\text{Spec } \mathbb{Z}}$ admits no such structure.

13.2 THE BLLPRvR OBSTRUCTION: INFINITE FLAVOUR RANK

THEOREM 13.2.1 (*BLLPRvR, arXiv:1312.5344*). . The Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees (BLLPRvR) correspondence assigns a vertex algebra (VA) $\chi(\mathcal{T})$ to any 4D $\mathcal{N} = 2$ superconformal field theory (SCFT) $\mathcal{T}$, with central charges satisfying $c_{2d} = -12c_{4d}$ and $k_{2d} = -\frac{1}{2}k_{4d}$.

Observation 13.2.2. The finite Heisenberg covers of the primon construction are generated by the modes $\{a_{p,n}\}_{p \in S, n \in \mathbb{Z}}$ for a finite set of primes S . The formal cover over all primes would have flavour group $\prod_{p \in \mathbb{P}} U(1)_p$, hence flavour rank $\aleph_0$. The Hochschild trace class $\mathcal{V}^{\text{prim}}$ itself is not a vertex algebra, and the scalar $r_\zeta = 2$ of Chapter 14 is a determinant-rank residue rather than a two-dimensional central charge. No 4D $\mathcal{N} = 2$ SCFT has flavour symmetry of rank $\aleph_0$: the flavour rank is bounded by the gauge rank plus hypermultiplet content, both finite for any theory with a Lagrangian description.

COROLLARY 13.2.3. . The VA $\chi(\mathcal{T})$ assigned by BLLPRvR to any 4D $\mathcal{N} = 2$ SCFT $\mathcal{T}$ has finite flavour rank. The all-prime Heisenberg cover has countable flavour rank, while $\mathcal{V}^{\text{prim}}$ is only its Hochschild trace class. Hence no BLLPRvR vertex algebra of a 4D $\mathcal{N} = 2$ SCFT is $\mathcal{V}^{\text{prim}}$. The class- $\mathcal{S}$ construction on an infinite-genus or infinite-puncture arithmetic curve would require $\overline{\text{Spec } \mathbb{Z}}$ to carry the complex structure of a Riemann surface, obstructed by Chapter 13.1.2.

13.3 THE CHERN–SIMONS OBSTRUCTION: IRRATIONAL LEVELS

Observation 13.3.1 (*non-integer level*). Witten’s Chern–Simons/WZW correspondence gives $CS_k(G)$ on $\Sigma \times \mathbb{R}$ with boundary vertex operator algebra (VOA) $V_k(\mathfrak{g})$ at integer level $k \in \mathbb{Z}$. A boundary theory whose finite Heisenberg covers reproduce all prime Euler factors would require gauge group $\prod_{p \in \mathbb{P}} U(1)_p$ with effective levels $k_p = 1/\log p$. These levels are irrational; the Chern–Simons path integral on non-trivial 3-manifolds requires $k \in \mathbb{Z}$ for well-definedness.

COROLLARY 13.3.2. . No finite-level abelian Chern–Simons theory has a boundary vertex algebra whose all-prime Hochschild trace is $\mathcal{V}^{\text{prim}}$.

13.4 THE BELINSKI–KHALATNIKOV–LIFSHITZ BILLIARD: THERMODYNAMIC COARSE-GRAINING

Observation 13.4.1 (*Selberg scattering versus primon partition*). The Selberg trace formula on $\text{SL}(2, \mathbb{Z}) \backslash \mathbb{H}^2$ expresses the spectral density of the Laplacian as a sum over Maass cusp forms plus a continuous Eisenstein series contribution. The Eisenstein scattering phase is $\zeta(2it)/\zeta(2it + 1)$ on the critical line, a scattering-matrix entry; the primon partition function is $\prod_p (1 - p^{-s})^{-1} = \zeta(s)$, with pole at $s = 1$. The two occurrences of ζ are structurally distinct: scattering amplitude versus thermodynamic partition function.

THEOREM 13.4.2 (*Hartnoll–Yang: thermodynamic coarse-graining*). (Hartnoll–Yang arXiv:2502.02661, §3). The primon–Belinski–Khalatnikov–Lifshitz (BKL) correspondence is a thermodynamic coarse-graining: at high temperature, the Selberg continuous spectrum above eigenvalue $1/4$ is replaced by the primon oscillator count. The spectral match is asymptotic, not exact.

13.5 THE AUTOMORPHIC VERTEX

THEOREM 13.5.1 ($\mathcal{V}^{\text{prim}}$ on the automorphic vertex). . $\mathcal{V}^{\text{prim}}$ occupies the automorphic vertex of the Langlands–TQFT triangle:

$$\underbrace{\{\text{automorphic reps of } GL_1(\mathbb{A}_{\mathbb{Q}})\}}_{\mathcal{V}^{\text{prim}}} \longleftrightarrow \underbrace{\{\text{1-dimensional Galois representations}\}}_{\text{Artin reciprocity}} \longleftrightarrow \underbrace{\{\text{arithmetic TQFT bulk}\}}_{\text{Deninger (conjectural)}}.$$

The graded character of $\mathcal{V}^{\text{prim}}$ is $\zeta(s)$; its Heisenberg decomposition is the Euler product $\prod_p (1 - p^{-s})^{-1}$. Artin reciprocity identifies the first two vertices; the third is Deninger’s conjectural $\mathbb{F}_1$ -geometric programme.

13.6 ASYMPTOTIC SYMMETRIES AT THE BKL END-OF-TIME BOUNDARY

PROPOSITION 13.6.1 (*Damour–Henneaux–Nicolai hep-th/0207267*). . BKL dynamics for 4D pure gravity on T^3 reduces to null billiards on the Weyl chamber of AE_3 , a rank-3 hyperbolic Kac–Moody algebra; the fundamental domain is a modification of $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}^2$.

Observation 13.6.2 (rank obstruction for AE_3). A soft-mode algebra at the BKL end-of-time boundary decomposes into Weyl-group representations of $\mathcal{W}(AE_3)$, where AE_3 has root lattice of rank 3. The primon Heisenberg algebra is generated by $\{a_{p,n}\}_{p \in \mathbb{P}, n \in \mathbb{Z}}$, with generator index set of cardinality $\aleph_0$. The two index sets are incommensurable; $\mathcal{V}^{\text{prim}}$ is the bosonic Fock quantisation of the Eisenstein continuous sector of the gravitational superspace, not the full $\mathcal{W}(AE_3)$ soft-mode algebra. The Hartnoll–Yang thermodynamic match (Chapter 13.4.2) is a coarse-graining of the Eisenstein density of states, not a spectral identification.

13.7 THE VOL III Φ_d FUNCTOR: ARITHMETIC DOMAIN BOUNDARY

Observation 13.7.1. Vol III’s functor $\Phi_d = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}}$ takes a Calabi–Yau d -fold over $\mathbb{C}$ as input. $\overline{\text{Spec } \mathbb{Z}}$ is an arithmetic 1-scheme over $\mathbb{Z}$; it admits no Calabi–Yau structure. The functor Φ_d is therefore defined on the $\mathbb{C}$ -analytic locus of the programme, with $\overline{\text{Spec } \mathbb{Z}}$ outside its domain.

Remark 13.7.2. When a Calabi–Yau 3-fold carries an $\text{SL}_2(\mathbb{Z})$ action on its elliptic-fibration moduli (for example, $K_3 \times E$), the modular-form output of Φ_3 carries arithmetic data. The Borchers product Φ_N ($N \in \{1, 2, 3, 4, 6\}$) lives on $\text{Sp}_4(\mathbb{Z})$ -moduli space; it is an automorphic object, occupying the automorphic vertex together with $\mathcal{V}^{\text{prim}}$.

13.8 AUTOMORPHIC PLACEMENT

THEOREM 13.8.1 (automorphic placement of $\mathcal{V}^{\text{prim}}$). . Each candidate TQFT-bulk framework for $\mathcal{V}^{\text{prim}}$ (Kapustin–Witten, BLLPRvR, holomorphic Chern–Simons, the Costello–Gaiotto holomorphically-twisted theory, and Vol III Φ_d) is obstructed by a specific structural obstruction listed in §§13.1–13.7. $\mathcal{V}^{\text{prim}}$ occupies the automorphic vertex of the Langlands–TQFT triangle (Chapter 13.5.1).

Remark 13.8.2 (the Deninger programme). A rigorous construction of $\overline{\text{Spec } \mathbb{Z}}$ as a “curve over $\mathbb{F}_1$ ” with a Frobenius-type $\mathbb{R}_{>0}$ -action, Deninger’s $\mathbb{F}_1$ -geometric programme, would supply the missing TQFT-bulk vertex. The arithmetic site of Connes–Consani (arXiv:1405.4527) provides one candidate foundation.

13.9 THE AUTOMORPHIC VERTEX: STRUCTURAL CONSOLIDATION

Each obstruction in §§13.1–13.7 is structural: the complex-structure requirement of Kapustin–Witten (Chapter 13.1.3), the finite flavour-rank bound of BLLPRvR (Chapter 13.2.3), the integrality requirement of Chern–Simons levels (Chapter 13.3.2), the thermodynamic rather than spectral character of the Hartnoll–Yang match (Chapter 13.4.2), and the $\mathbb{C}$ -analytic domain of Φ_d (Chapter 13.7.1). The automorphic vertex placement (Chapter 13.5.1) is the consequence: $\mathcal{V}^{\text{prim}}$ is identified by Artin reciprocity with the automorphic side of the Langlands triangle, with the TQFT-bulk vertex remaining Deninger’s open programme.

LOCAL CENTRAL CHARGE AND GLOBAL RESIDUE

14.1 RESCALING AND THE CONFORMAL VECTOR CONSTRAINT

Observation 14.1.1 (rescaling breaks Virasoro). For a rescaled stress tensor $T' = \lambda T$:

$$T'(z)T'(w) = \lambda^2 \left(\frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} \right) + \dots$$

For this to be a conformal OPE $\frac{c'/2}{(z-w)^4} + \frac{2T'(w)}{(z-w)^2} + \dots$ we need $\lambda^2 c = c'$ but $\lambda^2 \cdot 2T = 2T' = 2\lambda T$, i.e. $\lambda^2 = \lambda$, forcing $\lambda = 1$. Rescaling by $\lambda \neq 1$ does not yield a conformal vector.

Remark 14.1.2. The local conformal vector at p is the standard rank-2 Sugawara vector. The arithmetic energy is the external Hamiltonian

$$N_p := (\log p) L_o^{\text{conf},(p)}.$$

Thus $\log p$ weights the Tate-Mellin character; it is not a rescaling of the Virasoro stress tensor.

THEOREM 14.1.3 (per-prime central charge). . The local conformal Heisenberg algebra $\mathcal{H}^{(p)}$ has $c^{(p)} = 2$ for every prime p . The arithmetic Hamiltonian $N_p = (\log p) L_o^{\text{conf},(p)}$ is not a Virasoro zero mode.

Proof. Chapter 14.1.1 rules out a $\log p$ -rescaled stress tensor. The unrescaled rank-2 Sugawara construction on $\mathcal{H}^{(p)}$ gives $c^{(p)} = \text{rank}(\mathcal{H}^{(p)}) = 2$. Multiplication of $L_o^{\text{conf},(p)}$ by $\log p$ produces the arithmetic Hamiltonian used in the Euler product, not a new conformal vector. $\square$

14.2 THE INFINITE TENSOR PRODUCT AND THE VIRASORO OBSTRUCTION

Observation 14.2.1 (infinite tensor fails Virasoro). Stress tensor $\omega = \sum_p \omega_p$ is an infinite sum of non-zero vectors in the Fock module; not a vector in the module. On the restricted tensor product, the Virasoro commutator $[L_m, L_{-m}] = 2m L_o + \frac{c}{12} m(m^2 - 1)$ has $c = \sum_p c^{(p)} = \sum_p 2 = +\infty$. The central term is an infinite scalar multiplying every state.

COROLLARY 14.2.2. . The non-negative local conformal modes $\sum_p L_m^{\text{conf},(p)}$ with $m \geq -1$ act on the algebraic restricted tensor product. The full Witt algebra does not: for $m \leq -2$, the vacuum receives a non-zero contribution from infinitely many primes. The Virasoro central extension is absent, and $\mathcal{V}^{\text{prim}}$ is not a VOA.

14.3 THE NATURAL BOUNDARY OF THE PRIME ZETA FUNCTION

THEOREM 14.3.1 (natural boundary of prime zeta). (Landau–Walfisz 1920; Fröberg 1968). The prime zeta function $\zeta_{\mathbb{P}}(s) = \sum_p p^{-s}$ has a natural boundary on $\text{Re}(s) = 0$: logarithmic singularities at

$s = \rho/n$ for every nontrivial zero ρ of ζ and every $n \geq 1$ are dense on $\operatorname{Re}(s) = 0$. $\zeta_{\mathbb{P}}$ cannot be analytically continued past the imaginary axis.

COROLLARY 14.3.2 ($\zeta_{\mathbb{P}}(0)$ is not defined). . The value $\zeta_{\mathbb{P}}(0)$ lies on the natural boundary $\operatorname{Re}(s) = 0$ and is not defined by analytic continuation. The regularisation $c = 2\zeta_{\mathbb{P}}(0)$ is therefore not a finite value; the proposal $c = -1$ via this construction is ill-formed.

Remark 14.3.3 (what $\zeta(0)$ regularises). The value $\zeta(0) = -1/2$ is the zeta-regularised sum $\sum_{n \geq 1} 1$ over all positive integers. The prime sum $\sum_p 1$ is over a density-zero subset of $\mathbb{Z}_{>0}$; its zeta regularisation requires $\zeta_{\mathbb{P}}$, not ζ . The proposal $c = 2\zeta(0) = -1$ regularises the wrong index set.

14.4 RANK RESIDUE

The per-prime charges $c^{(p)} = 2$ (Chapter 14.1.3) diverge under naive summation; the prime zeta obstruction (Chapter 14.3.2) forecloses the $s = 0$ construction. The surviving finite scalar is the order of the pole of the Tate-Mellin determinant at $s = 1$.

THEOREM 14.4.1 (*rank residue of the primary determinant*). . The rank residue of the primary determinant is

$$r_{\zeta} = 2 \cdot \operatorname{Res}_{s=1} \zeta(s) = 2.$$

It is not the central charge of a global Virasoro action.

Sketch. The primary finite-prime determinant has leading Euler factor $\zeta(s)^2$. Since $\operatorname{Res}_{s=1} \zeta(s) = 1$, the pole order weighted by the rank-2 Heisenberg factor is 2. The argument computes the determinant residue; the preceding corollary forbids reading it as a global Virasoro central charge. $\square$

14.5 WITT VERSUS VIRASORO

THEOREM 14.5.1 (*finite conformal modes act; Virasoro central term diverges*). . The operators $L_m^{\geq -1} := \sum_p L_m^{\operatorname{conf},(p)}$ are well-defined on every restricted vector for $m \geq -1$. The modes with $m \leq -2$ are not defined on the restricted vacuum as global sums. Hence the full Witt algebra and its Virasoro central extension do not act on $\mathcal{V}^{\operatorname{prim}}$.

Remark 14.5.2 (soft-algebra interpretation). This matches the Hartnoll–Yang reading of $\mathcal{V}^{\operatorname{prim}}$ through its scalar character on the gravitational superspace only at the level of the surviving boundary modes. The algebraic object retains the modes $m \geq -1$, not the Witt algebra; hence no Virasoro anomaly can be read from the restricted tensor product.

14.6 THE FUNCTIONAL EQUATION AS ARITHMETIC MODULAR INVARIANCE

THEOREM 14.6.1 (*arithmetic modular invariance*). . The Tate-Mellin character of the primary trace class is the completed Riemann zeta $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. Its modular invariance is the functional equation $\xi(s) = \xi(1-s)$; the arithmetic analogue of $\tau \rightarrow -1/\tau$. The genus-1 modular-form trace $Z(\tau) = \operatorname{tr} q^{L_0 - c/24}$ belongs to an honest complex VOA; the global arithmetic object is a Mellin character.

14.7 THE ARAKELOV PAIRING REPLACES THE WEYL ANOMALY

PROPOSITION 14.7.1 (*Deligne pairing = Weyl anomaly analogue*). . Under the Deligne pairing on Arakelov line bundles, $\langle \omega_p, \omega_p \rangle_{\text{Arakelov}} = \log p$. Summed over primes this reproduces (up to regularisation) the logarithmic derivative of ζ .

Remark 14.7.2 (*Weyl anomaly versus height-theoretic anomaly*). On $\mathbb{C}$, the Weyl anomaly is $\langle T_\mu^\mu \rangle = -cR/24\pi$; the Polyakov–Liouville action realises this as the central observable. On $\overline{C} = \overline{\text{Spec } \mathbb{Z}}$, the role of the Weyl anomaly is played by the Arakelov metric contribution to the height pairing, $\log p$ per prime, which is height-theoretic in origin. The two anomalies are analogous; they are distinct because the geometric context differs (complex curve versus arithmetic compactification).

14.8 THE ARAKELOV- ζ STRUCTURE OF THE RESIDUE

The local conformal charge is $c^{(p)} = 2$, prime by prime. The global trace class has no Virasoro central charge. Its finite scalar is the rank residue $r_\zeta = 2$, extracted from the pole of ζ at $s = 1$. The prime zeta obstruction (Chapter 14.3.2) eliminates the $c = -1$ proposals; the residue regularisation (Chapter 14.4.1) is a determinant computation, not a conformal one. The non-negative conformal modes on each restricted vector are algebraic (Chapter 14.5.1); its arithmetic analogue of modular invariance is the functional equation $\xi(s) = \xi(1-s)$ (Chapter 14.6.1), and the Weyl anomaly is replaced by the Arakelov–Deligne height pairing $\langle \omega_p, \omega_p \rangle = \log p$ (Chapter 14.7.1).

ARITHMETIC CHERN–SIMONS HOLOGRAPHY

Class- $\mathcal{S}$, Higgs-branch vertex algebras, and Costello–Gaiotto boundary categories require a complex curve or a quantized boundary theory. Kim arithmetic Chern–Simons supplies only the classical arithmetic action line. Finite state sums and the conditional boundary category appear in Chapter 81; the profinite comparison with $\mathcal{V}^{\text{prim}}$ remains open.

15.1 CLASS $\mathcal{S}$ FAILS ARITHMETICALLY

THEOREM 15.1.1 (*Gaiotto, arXiv:0904.2715*). . Class- $\mathcal{S}$ builds 4D $\mathcal{N} = 2$ theories $T[\Sigma, \mathfrak{g}]$ by compactifying the 6D $(2, 0)$ theory $\mathfrak{X}[\mathfrak{g}]$ on a punctured Riemann surface Σ .

Observation 15.1.2 (three obstructions). Arithmetic class- $\mathcal{S}$ on $\overline{\text{Spec } \mathbb{Z}} \times \mathbb{R}^4$ fails:

- (i) 6D $(2, 0)$ requires a smooth oriented 6-manifold; $\overline{\text{Spec } \mathbb{Z}} \times \mathbb{R}^4$ is topologically 5-dimensional (arithmetic 1-dim plus 4-dim real). Connes’ spectral dimension of $\overline{\text{Spec } \mathbb{Z}}$ is 1, not 2 (Connes–Consani arXiv:1405.4527).
- (ii) The self-dual 2-form B on an M_5 -brane has no Arakelov analogue (no star on 2-forms on a 1-d arithmetic curve, where $\Omega^2 = 0$).
- (iii) Hitchin-moduli Higgs branches on $\overline{\text{Spec } \mathbb{Z}}$ are not defined by the class- $\mathcal{S}$ formula: the Arakelov dualizing sheaf is a metrised arithmetic line, not the holomorphic canonical bundle of a Riemann surface.

COROLLARY 15.1.3. . Class- $\mathcal{S}$ as a generating principle for arithmetic vertex algebras is falsified.

15.2 THE HITCHIN BASE HAS NO CLASS- $\mathcal{S}$ ARITHMETIC SPECIALIZATION

THEOREM 15.2.1 (*arithmetic Hitchin obstruction*). . The class- $\mathcal{S}$ Coulomb branch

$$C_{\Sigma, \mathfrak{g}} = \bigoplus_i H^0(\Sigma, K_{\Sigma}^{d_i})$$

uses the holomorphic canonical bundle of a complex curve. For $\overline{\text{Spec } \mathbb{Z}}$ the dualizing object is the Arakelov line with metric at infinity. Its sections belong to Arakelov intersection theory, not to a Hitchin base. Hence the class- $\mathcal{S}$ Coulomb and Higgs branches do not vanish; rather, the formula defining them has no canonical arithmetic value.

Remark 15.2.2 (why the replacement is three-dimensional). This obstruction is not a proof that an arithmetic 4-dimensional theory is topological. It says that the class- $\mathcal{S}$ descent from a 6-dimensional theory along a complex curve has no arithmetic curve to descend along. Kim’s arithmetic Chern–Simons theory is therefore not the topological remnant of that construction; it is a separate 3-dimensional arithmetic field theory.

15.3 KIM’S ARITHMETIC CHERN–SIMONS

Definition 15.3.1 (Kim 2015, arXiv:1510.05818). For a number field F and finite Galois module M , arithmetic Chern–Simons theory assigns to a Galois representation $\rho: \pi_1^{\text{ét}}(\text{Spec } \overline{\mathcal{O}_F}) \rightarrow G$ the action

$$\text{CS}_F(\rho) = \langle \rho^* c, [\text{Spec } \overline{\mathcal{O}_F}] \rangle \in H^3(\text{Spec } \overline{\mathcal{O}_F}, \mu_N)$$

built from a class $c \in H^3(BG, \mu_N)$. This is a genuine 3D arithmetic Chern–Simons action, not yet an Atiyah–Segal quantum TQFT. The topology is the Mazur–Morishita dictionary: primes are knots and $\text{Spec } \overline{\mathbb{Z}}$ is the arithmetic 3-manifold (Morishita arXiv:0904.3399).

Remark 15.3.2. Chung–Kim–Kim–Pappas–Park–Yoo arXiv:1706.03336 extends Kim’s construction to higher-dimensional arithmetic manifolds. Arithmetic Chern–Simons supplies the rigorous classical action functional at the arithmetic vertex that Witten’s analysis (Chapter 13.8.1) identifies as the ingredient Deninger’s programme requires.

15.4 $\mathcal{V}^{\text{prim}}$ AND THE ACS BOUNDARY SHADOW

THEOREM 15.4.1 (arithmetic CS boundary shadow). . Let $\text{ACS}[\text{Spec } \overline{\mathbb{Z}}, \mathbb{G}_m, 1]$ denote Kim arithmetic Chern–Simons at level 1 with gauge group $\mathbb{G}_m$. Kim’s theory supplies a classical arithmetic CS tor-sor/action line. A quantum boundary category $\partial_q \text{ACS}$, once constructed, has conjectural archimedean Heisenberg factorization shadow

$$\chi(\partial_q \text{ACS})(s) = \prod_p (1 - p^{-s})^{-1},$$

and is conjecturally compared with the Hochschild-trace candidate $\mathcal{V}^{\text{prim}}$ (a Hochschild trace class; Chapter 12.5.1). The quantum boundary category at finite places is conditional, as in Chapter 81.

Form. Kim’s construction is classical: it assigns Chern–Simons actions and torsors to Galois representations. Witten’s CS/WZW boundary theorem transports to $\text{Spec } \overline{\mathbb{Z}}$ only after replacing the complex boundary curve and its OPE by the arithmetic integral transform of Chapter 81. That chapter defines the finite-level boundary category by transgression, conditional on the finite arithmetic cobordism input; the displayed Euler product is then a character-level expectation for the renormalised profinite boundary comparison with the Hochschild-trace class $\mathcal{V}^{\text{prim}}$. $\square$

COROLLARY 15.4.2 (character = ζ). .

$$\chi(\partial_q \text{ACS})(s) = \prod_p (1 - p^{-s})^{-1} = \zeta(s),$$

conditional on constructing the quantum boundary category and identifying its Hochschild trace with the Hochschild-trace candidate.

15.5 WILSON LINES ARE LABELED BY RAMIFIED HECKE CHARACTERS

THEOREM 15.5.1 (Wilson-line labels as local Hecke data). . Wilson lines in ACS wrapped around the “knot” corresponding to a prime p (Mazur–Morishita arithmetic topology) are 1-dimensional defects in the arithmetic 3-fold, each labeled by a local Hecke character χ_p of $\mathbb{Q}_p^\times$ (ramified or unramified). The label is a closed point of the universal family of local characters, and conjecturally a module over the

quantum boundary category specialized at p . Inserting a Wilson line with label χ_p produces a boundary defect whose local datum at p is the ramification datum of χ_p (conjectural; the categorical boundary module is part of the conditional boundary-category input of Chapter 81). Wilson-line labels are 0-dimensional data; the Wilson line itself is 1-dimensional.

Remark 15.5.2 (defect/puncture dictionary). Class- $\mathcal{S}$ punctures correspond to 6D codimension-2 defects in the complex setting; arithmetic ramification corresponds to Wilson lines in arithmetic Chern–Simons. The structural dictionary “punctures = defects” becomes “Wilson lines = ramification” in the arithmetic setting, with the 6D/4D framing replaced by the 3D arithmetic TQFT framing.

15.6 BEEM–LEMOs–LIENDO–PEELAERS–RASTELLI–VAN REES AS ADELIC CHARACTER

THEOREM 15.6.1 (arithmetic Schur index). Let $\mathcal{H}^{\text{arith}}$ denote the conjectural arithmetic Hilbert space on which Frobenius operators Frob_p act (one per prime p), and let the Schur grading at p be $t_p^{\text{Frob}_p}$ (the formal Frob_p -graded trace variable). The arithmetic Schur index is the formal adelic graded trace

$$\mathcal{I}^{\text{arith}}(\{t_p\}) = \text{Tr}_{\mathcal{H}^{\text{arith}}} \prod_p t_p^{\text{Frob}_p}.$$

Under the Hecke substitution $t_p = p^{-s}$, $\mathcal{I}^{\text{arith}}(s) = \prod_p (1 - p^{-s})^{-1} = \zeta(s)$. This is an Euler-character computation, not a construction of the Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees Schur-index category.

Remark 15.6.2. The Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees correspondence takes holographic form: $\zeta(s)$ is a Schur index of the 3D arithmetic Chern–Simons theory. The arithmetic vertex algebra generating problem, for which class- $\mathcal{S}$ is falsified (Chapter 15.1.3), admits arithmetic CS holography as its resolution.

15.7 ARITHMETIC CS AS THE GENERATING PRINCIPLE

- (i) Class- $\mathcal{S}$ falsified: three 4D-gauge obstructions (Chapter 15.1.3).
- (ii) The class- $\mathcal{S}$ Hitchin base has no canonical arithmetic specialization on $\overline{\text{Spec } \mathbb{Z}}$ (Chapter 15.2.1).
- (iii) Kim’s arithmetic Chern–Simons (arXiv:1510.05818) is the classical 3D action functional (Chapter 15.3.1).
- (iv) Finite state sums are constructed in Chapter 81; the renormalised profinite boundary category is conjectural, as is its comparison with $\mathcal{V}^{\text{prim}}$ and its character $\zeta(s)$ (Chapter 15.4.1).
- (v) Wilson lines are 1-dimensional defects labeled by ramified Hecke characters (= expected modules of the quantum boundary category specialized at p , 0-dimensional data) (Chapter 15.5.1).
- (vi) BLLPRvR adelic character: $\zeta(s)$ is the conjectural Schur index character of arithmetic CS (Chapter 15.6.1).

Kim arithmetic $\mathbb{G}_m$ -Chern–Simons on $\overline{\text{Spec } \mathbb{Z}}$ at level 1 supplies the classical arithmetic action. Chapter 81 constructs finite state sums and states the boundary category as a conditional profinite problem; the renormalised comparison with $\mathcal{V}^{\text{prim}}$ remains open.

THE HOCHSCHILD TRACE OF THE IWAHORI HECKE CATEGORY

The primary object is $\mathcal{V}^{\text{prim}}$, the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 . This chapter places $\mathcal{V}^{\text{prim}}$ on the arithmetic Langlands diamond via Bezrukavnikov’s affine Hecke equivalences (arXiv:1209.0403), Mirković–Vilonen geometric Satake (arXiv:math/0401222), Ben-Zvi–Nadler character theory (arXiv:0904.1247), Arinkin–Gaitsgory spectral decomposition (arXiv:1201.6343), the Emerton–Gee (φ, Γ) stack (arXiv:1908.07185), and Fargues–Scholze geometrization (arXiv:2102.13459).

16.1 THE NAIVE $\text{Bun}_{GL_2}(\text{Spec } \mathbb{Z})$ HAS ONLY ONE COARSE POINT

THEOREM 16.1.1 (*Minkowski*). . $\pi_1^{\text{ét}}(\text{Spec } \mathbb{Z}) = 1$: every étale cover of $\text{Spec } \mathbb{Z}$ is trivial. Class number of $\mathbb{Q}$ is 1.

COROLLARY 16.1.2. . The groupoid $\text{Bun}_{GL_2}(\text{Spec } \mathbb{Z})$ has one object up to isomorphism, represented by $\mathbb{Z}^2$, and stabilizer $GL_2(\mathbb{Z})$. Thus

$$\text{Bun}_{GL_2}(\text{Spec } \mathbb{Z}) \simeq BGL_2(\mathbb{Z})$$

as a discrete stack, not as a point. The lost datum in the coarse quotient is exactly the arithmetic automorphism group; it does not produce the geometric Hecke correspondence supplied by a curve.

16.2 THE SPHERICAL HECKE CATEGORY COLLAPSES THE OPE

The coarse automorphic stack over $\text{Spec } \mathbb{Z}$ has one component and stabilizer $GL_2(\mathbb{Z})$. It has arithmetic monodromy, but no geometric curve along which a Hecke correspondence can move. Thus one is forced to ask which local Hecke category at a prime p carries the noncommutative structure visible to a Hochschild trace. Geometric Satake identifies the spherical category but reveals its commutativity.

THEOREM 16.2.1 (*Mirković–Vilonen*). . Geometric Satake gives $\text{Perv}_{GL_2(O)}(\text{Gr}_{GL_2}) \xrightarrow{\sim} \text{Rep}(\widehat{GL_2}) = \text{Rep}(GL_2)$.

Observation 16.2.2 (spherical Hecke commutativity). The local spherical Hecke algebra $\mathcal{H}_p = C_c(GL_2(\mathbb{Z}_p) \backslash GL_2(\mathbb{Q}_p) / GL_2(\mathbb{Z}_p))$ is commutative: this is the content of the Satake isomorphism $\mathcal{H}_p \xrightarrow{\sim} \mathbb{C}[GL_2^\vee]^{GL_2}$. Its derived enhancement $\mathcal{H}^{(p)}$, an E_1 -algebra in chain complexes (hence E_2 as a graded object under Dunn additivity), and $\text{Perv}(\text{Gr})$, a monoidal abelian category under convolution, are related by the Mirković–Vilonen equivalence; but the commutativity of $\mathcal{H}_p$ means the resulting Hochschild homology carries no automorphic spectral content beyond the centre.

16.3 THE IWAHORI HECKE CATEGORY CARRIES CONVOLUTION DATA

Before invoking Bezrukavnikov's equivalence we record the Bernstein presentation that gives the chapter its name. The affine Hecke algebra $H_q = H_q(\widetilde{W})$ of the extended affine Weyl group $\widetilde{W} = W_o \ltimes X^*(T)$ (with W_o the finite Weyl group and $X^*(T)$ the character lattice of the maximal torus) is generated by:

- lattice translations θ_λ for $\lambda \in X^*(T)$, forming the Bernstein subalgebra $\mathbb{C}[X^*(T)] \subset H_q$;
- finite Hecke generators T_s for each simple reflection $s \in W_o$, satisfying the braid and quadratic $(T_s - q)(T_s + 1) = 0$ relations.

The interaction between the two layers is the *Bernstein relation*: for a simple reflection $s = s_\alpha$ with simple coroot $\alpha^\vee$ and $\lambda \in X^*(T)$ orthogonal to $\alpha^\vee$,

$$T_s \theta_\lambda T_s^{-1} = \theta_{s(\lambda)},$$

while for general λ the conjugation acquires a correction term proportional to $(\theta_\lambda - \theta_{s(\lambda)})/(1 - \theta_{-\alpha})$ expressing the failure of θ_λ to be s -invariant (Lusztig 1989, *Affine Hecke algebras and their graded version*, J. AMS 2, eq. (8.4)). The Bezrukavnikov equivalence below sends this noncommutative algebra structure to a coherent-sheaf convolution on the Steinberg variety; it is the categorical lift of the Bernstein presentation, and the chapter's framing "Bezrukavnikov–Bernstein–Langlands" records the three layers.

The commutative spherical category is replaced by the Iwahori-equivariant category, whose noncommutative convolution datum is precisely what Bezrukavnikov's equivalences encode on the coherent side.

THEOREM 16.3.1 (*Bezrukavnikov arXiv:1209.0403*). . Let $\widetilde{N}$ denote the Springer resolution $\widetilde{N} = T^*(GL_2/B)$ of the nilpotent cone, and let $\text{St}_{GL_2} = \widetilde{N} \times_N \widetilde{N}$ denote the Steinberg variety (the derived fibre product over the nilpotent cone). Equivalences

$$D^b \text{Coh}^{GL_2 \times \mathbb{G}_m}(\widetilde{N}) \xrightarrow{\sim} D^b \text{Coh}(\text{St}_{GL_2}) \xrightarrow{\sim} D_{\text{mix}}^b(\text{Perv}_{\text{Iw}}(\text{Fl}_{GL_2})).$$

The Iwahori-fixed direction carries the noncommutative convolution datum needed for the Hochschild trace: the affine flag variety admits an Iwahori stratification indexed by the extended affine Weyl group $\widetilde{W}_{GL_2} = X^*(T) \rtimes W_f \cong \mathbb{Z}^2 \rtimes S_2$, where $X^*(T) = \mathbb{Z}^2$ is the character lattice of the standard torus of GL_2 .

Definition 16.3.2 (global Iwahori Hecke category).

$$\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}} := \bigotimes_p \text{Hk}_{GL_2, p}^{\text{Iw}}, \quad \text{Hk}_{GL_2, p}^{\text{Iw}} := D_{\text{mix}}^b(\text{Perv}_{\text{Iw}}(\text{Fl}_{GL_2, \mathbb{F}_p}), \text{Fr}_p).$$

Remark 16.3.3 (geometric local convention). The p -adic Iwahori Hecke algebra is recovered from this geometric category by the sheaf–function dictionary. The coherent-side equivalences of Bezrukavnikov are geometric statements before this sheaf–function specialization.

16.4 THE EMERTON–GEE SPECTRAL SIDE

The automorphic side is the global Iwahori Hecke category. The spectral side requires a global (φ, Γ) -module stack; Emerton–Gee construct the local building block, and the global Selmer-derived stack assembles them with fixed arithmetic data.

THEOREM 16.4.1 (*Emerton–Gee arXiv:1908.07185*). . There is a Noetherian formal algebraic stack $\mathcal{X}_{GL_2}^{\text{EG}}$ over $\text{Spec } \mathbb{Z}_p$ whose points parametrise (φ, Γ) -modules of rank 2. The global spectral object used below is not a product or colimit of these local stacks; it is the Selmer-derived global deformation stack with fixed residual representation, determinant, and local conditions.

THEOREM 16.4.2 (*arithmetic Arinkin–Gaitsgory*). (conditional locally, conjectural globally). Let $\mathcal{D}^{\text{sph-fin}}([GL_2])$ denote the spherically-generated full subcategory of the automorphic ∞ -category $\mathcal{D}([GL_2])$ (functions on the adelic quotient $GL_2(\mathbb{A})$), and let $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}$ denote the Selmer-derived global deformation stack of a fixed residual representation $\bar{\rho}: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow GL_2$, with fixed determinant and local conditions $\mathcal{L}$. Then:

$$\mathcal{D}^{\text{sph-fin}}([GL_2]) \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}).$$

Fargues–Scholze arXiv:2102.13459 supplies local geometrization, spectral action, and Hecke functoriality; Emerton–Gee supplies the local (φ, Γ) stack. The displayed automorphic-to- $\text{IndCoh}_{\text{Nilp}}$ equivalence is conditional locally and conjectural under global adelic descent.

16.5 THE HOCHSCHILD TRACE PLACEMENT

The automorphic and spectral sides being assembled, the placement of $\mathcal{V}^{\text{prim}}$ is the Hochschild homology of the global Iwahori Hecke category.

THEOREM 16.5.1 (*crowning placement*). .

$$\boxed{\mathcal{V}^{\text{prim}} = \text{HH}_{\bullet}(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}})}$$

E_1 -algebra from Dunn additivity ($E_1 \otimes E_1 = E_2$) applied to the E_2 -monoidal structure on each local factor; Ben-Zvi–Nadler arXiv:0904.1247 identifies $\text{HH}_{\bullet}(\mathcal{D}(G/G)) = \mathcal{O}(\text{LocSys}_G(S^1))^{\text{flat}}$.

Remark 16.5.2 (adelic inertia circle). The inertia circle at p is the profinite completion $\widehat{\mathbb{Z}} \subset W_{\mathbb{Q}_p}$ of the inertia group; the local system stack $\text{LocSys}_{GL_2}(S_p^1)$ (flat GL_2 -bundles on the arithmetic circle $\text{Spec } \mathbb{Z}((t))/\text{Spec } \mathbb{Z}_p$) has functions equal to the Hochschild trace. Ben-Zvi–Nadler arXiv:0904.1247 proves the complex-group identification $\text{HH}_{\bullet}(\mathcal{D}(G/G)) \simeq \mathcal{O}(\text{LocSys}_G(S^1))^{\text{flat}}$; the present arithmetic statement is its (φ, Γ) -module counterpart.

16.6 CONNECTION TO GAIOTTO’S 3D CHERN–SIMONS PRESENTATION

PROPOSITION 16.6.1 (*three-way comparison*). . The three presentations of $\mathcal{V}^{\text{prim}}$ are conjecturally equivalent:

- (i) (Kim arXiv:1510.05818 / Gaiotto) arithmetic Chern–Simons at the classical level produces a boundary contribution whose partition function specializes to $\mathcal{V}^{\text{prim}}$ on the arithmetic Langlands spectral side;
- (ii) (Bezrukavnikov–Bernstein) $\mathcal{V}^{\text{prim}} = \text{HH}_{\bullet}(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}})$;
- (iii) (Costello–Gwilliam) the factorization window of $\mathcal{V}^{\text{prim}}$ lies in $\text{FactAlg}(\text{Mod}_{\widehat{\mathbb{Z}}, \psi})$ over the adelic curve, with the Whittaker and factorization conditions of Chapter 20.

These are conditional comparison identifications, not proved equivalences. Ben-Zvi–Nadler’s Chern–Simons/trace/Hecke triangle (arXiv:0904.1247) is the complex template; the arithmetic gluing requires global adelic descent (Conjecture F₃ of Chapter 21).

16.7 HECKE TRACE = EICHLER–SELBERG

The Hochschild trace of $\mathcal{V}^{\text{prim}}$ specializes at each weight-level pair (k, N) to the trace of a Hecke operator on the space of cusp forms; this specialization is the Eichler–Selberg trace formula.

THEOREM 16.7.1 (*Eichler–Selberg trace formula, termwise form*). . For each $N \geq 1, k \geq 2$:

$$\text{tr}(T_p \mid S_k(\Gamma_1(N))) = \text{Ell}_{p,k,N} + \text{Hyp}_{p,k,N} + \delta_{\text{id}}(p) + \delta_{\text{Eis}}(p, k, N),$$

where

$$\text{Ell}_{p,k,N} = - \sum_{t^2 < 4p} P_k(t, p) H_w(t^2 - 4p),$$

where $P_k(t, p) = (\alpha^{k-1} - \beta^{k-1})/(\alpha - \beta)$, $\alpha\beta = p$, $\alpha + \beta = t$, H_w denotes the Hurwitz weighted class number of binary quadratic forms of discriminant $t^2 - 4p$; $\text{Hyp}_{p,k,N}$ is the hyperbolic/parabolic contribution, $\delta_{\text{id}}(p)$ is the identity contribution, and $\delta_{\text{Eis}}(p, k, N)$ is the Eisenstein boundary contribution.

THEOREM 16.7.2 (*universal trace-class tabulation*). . $\mathcal{V}^{\text{prim}}$ represents the universal family of Hecke eigensystems on the global automorphic category $\mathcal{D}([GL_2])^{\text{sph-fin}}$: it is the object over the weight-level base $\mathcal{W} \times \text{Spec } \mathbb{Z}[1/N]$ (where $\mathcal{W}$ is the weight space parametrizing integers $k \geq 2$) whose closed-point fibres are individual Hecke eigensystems. Restricted to $S_k(\Gamma_1(N))$ for fixed (k, N) , the fibre of the universal family recovers the Eichler–Selberg trace formula coefficient by coefficient; individual Hecke eigenforms are the closed points of this family.

16.8 THE ARITHMETIC LANGLANDS DIAMOND FOR $\mathcal{V}^{\text{prim}}$

- (i) $\text{Bun}_{GL_2}(\text{Spec } \mathbb{Z}) \simeq BGL_2(\mathbb{Z})$ has one coarse point and nontrivial stabilizer (Chapter 16.1.2); it is too small geometrically to carry the required Hecke correspondence, forcing the passage to the Iwahori-equivariant setting.
- (ii) The spherical Hecke algebra is commutative (Satake isomorphism); the Iwahori-equivariant affine Hecke category $\text{Hk}_{GL_2, p}^{\text{Iw}}$ carries the noncommutative convolution datum (Chapter 16.3.1).
- (iii) The spectral side is the Selmer-derived global stack, built from local Emerton–Gee/ (φ, Γ) stacks with fixed residual representation, determinant, and local conditions.
- (iv) Arithmetic Arinkin–Gaitsgory: Fargues–Scholze and Emerton–Gee supply local inputs; the automorphic-to- $\text{IndCoh}_{\text{Nilp}}$ equivalence is conditional locally and conjectural globally (Chapter 16.4.2).
- (v) Crowning placement: $\mathcal{V}^{\text{prim}} = \text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}})$ (Chapter 16.5.1).
- (vi) Three-way comparison with Kim–Gaiotto arithmetic Chern–Simons and Costello–Gwilliam factorization (Chapter 16.6.1).
- (vii) Hecke-trace tabulation recovers the Eichler–Selberg formula (Chapter 16.7.2).

$\mathcal{V}^{\text{prim}}$ is an arithmetic Hochschild character. It encodes on the automorphic side the universal trace of the identity on the global Iwahori Hecke category of GL_2 ; on the spectral side, through the conjectural arithmetic Langlands equivalence, the structure sheaf $\mathcal{O}$ of the global Galois-deformation stack. The rank-2 Heisenberg presentation per prime is the Whittaker model of this trace, with the two generators diagonalising the Satake parameters.

ARITHMETIC E_n FORMALITY AND CoHA

17.1 E_1 -FORMALITY IS AUTOMATIC OVER $\mathbb{Z}$

THEOREM 17.1.1 (*Fresse Homotopy of Operads, Vol. II, Thm I.4.1.6*). . For any commutative ring R , $H_\bullet(E_1; R) \cong \text{Ass}_R$ and the canonical map $C_\bullet(E_1; R) \rightarrow \text{Ass}_R$ is an operadic quasi-isomorphism. Thus E_1 -algebras can be rectified to associative dg algebras under the usual model-categorical hypotheses. This is rectification, not a statement that every chosen A_∞ -model has zero higher multiplications.

COROLLARY 17.1.2 (*ordered E_1 -chiral survives over $\mathbb{Z}$*). . The ordered E_1 -chiral structure on $\mathcal{V}^{\text{prim}}$, whose underlying product is the strict associative product from the ordered factorization-algebra product, is defined integrally: $H_\bullet(E_1; \mathbb{Z}) \cong \text{Ass}_{\mathbb{Z}}$ carries no torsion, so no Massey obstruction arises.

17.2 E_2 -FORMALITY FAILS INTEGRALLY

THEOREM 17.2.1 (*Salvatore–Wahl; Cohen LNM 533; Fresse–Willwacher arXiv:1503.08699*). . Algebras over $C_\bullet(E_2; \mathbb{F}_p)$ carry non-trivial Dyer–Lashof power operations not encoded by the Gerstenhaber homology operad; these operations obstruct integral formality. Over $\mathbb{Q}$, Tamarkin’s theorem (arXiv:math/9803025) establishes $C_\bullet(E_2; \mathbb{Q}) \xrightarrow{\sim} H_\bullet(E_2; \mathbb{Q}) = \text{Ger}_{\mathbb{Q}}$ (where $\text{Ger}_{\mathbb{Q}}$ is the Gerstenhaber operad over $\mathbb{Q}$), a torsor for the Grothendieck–Teichmüller group $\text{GRT}_1(\mathbb{Q})$.

COROLLARY 17.2.2 (*no integral replacement by the Gerstenhaber operad*). . An integral E_2 -algebra structure is not forbidden. What fails is the replacement

$$C_\bullet(E_2; \mathbb{Z}) \simeq \text{Ger}_{\mathbb{Z}}.$$

Consequently an E_2 -chiral-centre enhancement of $\mathcal{V}^{\text{prim}}$ over $\mathbb{Z}$, if present, must retain the chain-level E_2 -operad and its power operations. After tensoring with $\mathbb{Q}$, Tamarkin formality identifies the rational structure with a $\text{GRT}_1(\mathbb{Q})$ -torsor.

Remark 17.2.3 (Kaledin Japanese cocycle). Kaledin isolates the arithmetic obstruction: for an associative $\mathbb{Z}$ -algebra A , the reduction $A \otimes \mathbb{F}_p$ may carry a non-trivial Hochschild A_∞ -deformation class, the “Japanese cocycle,” in $\text{HH}^2(A \otimes \mathbb{F}_p)/\text{HH}^2(A) \otimes \mathbb{F}_p$. Such classes are obstructions to naive integral strictification. They do not, by themselves, prove nonexistence of an integral E_2 -algebra.

17.3 THE STRICT RECTIFIED E_1 -MODEL

E_1 -formality identifies $C_\bullet(E_1; \mathbb{Z})$ with $\text{Ass}_{\mathbb{Z}}$ as operads; $\text{Ass}_{\mathbb{Z}}$ has only m_1 and m_2 as non-trivial structure maps. The ordered model of $\mathcal{V}^{\text{prim}}$ is taken after this rectification.

THEOREM 17.3.1. . The ordered model of $\mathcal{V}^{\text{prim}}$ used in this chapter is a strict associative dg algebra. Therefore its displayed operations have $m_k = 0$ for $k \geq 3$. A different A_∞ -model may carry higher operations; the theorem fixes the strict rectified model.

COROLLARY 17.3.2 (*strict associativity over $\mathbb{Z}$*). $\mathcal{V}^{\text{prim}}$ is represented over $\mathbb{Z}$ by a strict associative dg algebra in the ordered model. Its A_∞ -quasi-isomorphism class may be represented by other models, but the chosen rectification has no higher multiplications. The E_2 -chiral-centre obstructions of §17.2 operate at a separate level and are unaffected by this strict associativity.

17.4 THE CHIRAL BRACKET VIA ADELIC RESIDUE

DEFINITION 17.4.1 (*arithmetic λ -bracket*). Let $\overline{C}$ denote the Arakelov compactification of the arithmetic curve, obtained by adjoining the archimedean place $|\cdot|_\infty$ to $\text{Spec } \mathbb{Z}$. The chiral bracket on $\mathcal{V}^{\text{prim}}$ over $\overline{C}$ is the Beilinson–Drinfeld λ -bracket on $\text{Conf}_2(\overline{C}) = \overline{C} \times \overline{C} \setminus \Delta$:

$$\{a, b\}_{\text{arith}}(p, q) = \text{Res}_{p=q} \omega_{a,b}(p, q) \cdot \log |p - q|_\infty^{-1},$$

where $\omega_{a,b}$ is the section of $\mathcal{V}^{\text{prim}} \boxtimes \mathcal{V}^{\text{prim}}$ over $\text{Conf}_2(\overline{C})$ determined by a and b . Each finite prime contributes a discrete residue at the corresponding closed point; the archimedean place contributes the $|\cdot|_\infty$ normalisation.

THEOREM 17.4.2 (*chiral bracket descends adelically*). The arithmetic λ -bracket on $\mathcal{V}^{\text{prim}}$ is a well-defined adelic residue, interpreted via Arakelov completion: the arithmetic cotangent bundle $\text{Cot}_{\text{arith}}^*$ (the sheaf of arithmetic 1-forms on $\overline{C}$, metrized at the archimedean place) on $\text{Conf}_2(\overline{C})$ restricted to the punctured diagonal supports the pairing $\{\mathcal{V}^{\text{prim}}, \mathcal{V}^{\text{prim}}\}_{\text{arith}} \subset \mathcal{V}^{\text{prim}} \otimes_{\mathbb{Z}} \text{Cot}_{\text{arith}}^*$.

17.5 THE ARITHMETIC COHA CONJECTURE

Given that $\mathcal{V}^{\text{prim}}$ carries an integral E_1 -structure and an adelic chiral bracket, one asks whether it fits into the Kontsevich–Soibelman cohomological Hall algebra (CoHA) framework as a BPS sub-object.

THEOREM 17.5.1 (*Kontsevich–Soibelman arXiv:1006.2706*). For a quiver Q and potential $W \in \mathbb{C}\langle Q \rangle_{\text{cyc}}$, the cohomological Hall algebra is

$$\text{CoHA}(Q, W) := \bigoplus_{\mathbf{d}} H_{G_{\mathbf{d}}}^{\bullet}(\text{Rep}(Q, \mathbf{d}), \varphi_{W_{\mathbf{d}}}),$$

where $\mathbf{d}$ ranges over dimension vectors, $G_{\mathbf{d}} = \prod_v \text{GL}_{d_v}$ is the gauge group, $\text{Rep}(Q, \mathbf{d})$ the representation variety, and $\varphi_{W_{\mathbf{d}}}$ the vanishing cycle sheaf of the trace of W restricted to $\text{Rep}(Q, \mathbf{d})$.

THEOREM 17.5.2 (*Davison integrality arXiv:1601.02479*). For symmetric (Q, W) in the CY-3 setting,

$$\text{CoHA}(Q, W) \cong \text{Sym}(\text{BPS}(Q, W) \otimes H^{\bullet}(BC^{\times})),$$

with $\text{BPS}(Q, W)$ the BPS Lie algebra (primitive generator lattice, in Davison’s sense).

THEOREM 17.5.3 (*conjectural arithmetic CoHA identification*). $\mathcal{V}^{\text{prim}}$ is the BPS sub-object

$$\mathcal{V}^{\text{prim}} \stackrel{?}{\cong} \text{BPS}(Q^{\text{arith}}, W^{\text{arith}})$$

of the full $\text{CoHA}(Q^{\text{arith}}, W^{\text{arith}})$, where the full CoHA is a polynomial algebra on $\mathcal{V}^{\text{prim}}$ via Davison’s theorem (Chapter 17.5.2). The candidate arithmetic quiver-with-potential is:

- $Q_o^{\text{arith}} = \mathbb{P} \cup \{\infty\}$ (primes plus the archimedean place);
- $Q_i^{\text{arith}} =$ Hecke correspondences (one edge per automorphism of a local completion);

- $W^{\text{arith}} \in \text{HC}_0(\mathbb{Z}\langle Q^{\text{arith}} \rangle)$, the cyclic cohomology class of the Connes–Consani motivic lift of the global ζ -function (Connes–Consani 2009, *Schemes over $\mathbb{F}_1$ and zeta functions*, Compos. Math. 146, arXiv:1006.4810, §5; distinct from the arithmetic-site arXiv:1405.4527 used elsewhere in this branch).

Remark 17.5.4 (Bost–Connes quiver and the role of the potential). The Bost–Connes quiver (vertex set $\mathbb{P}$, free commutative monoid on primes) is acyclic, with trivial potential; its CoHA is the polynomial algebra $\otimes_p \mathbb{Z}[e_p]$. The identification of $\mathcal{V}^{\text{prim}}$ with $\text{BPS}(Q^{\text{arith}}, W^{\text{arith}})$ requires the cyclic-cohomology-lifted potential W^{arith} , which encodes the ζ -function data encoded in HC_0 .

17.6 $\log L$ VIA CYCLIC COHOMOLOGY

Observation 17.6.1. $\log L = \sum_p \sum_{k \geq 1} p^{-ks}/k$ is a formal Dirichlet series with transcendental coefficients $\log p$; its image in the path algebra $\mathbb{Z}\langle Q \rangle_{\text{cyc}}$ is defined only after passage through Connes’ cyclic cohomology HC_0 , which accommodates the $\log p$ coefficients as cyclic cohomology classes rather than as algebraic potential terms.

Remark 17.6.2 (role of exponential Hodge structure). Kontsevich–Soibelman *Cohomological Hall algebra, exponential Hodge structures, and motivic DT invariants* (arXiv:1006.2706, §7) provides the setting in which ζ -value periods appear as CoHA data. The arithmetic CoHA identification requires this exponential Hodge structure machinery.

17.7 ARITHMETIC FORMALITY STRATIFICATION

The $E_\bullet$ -formality of $\mathcal{V}^{\text{prim}}$ over $\overline{\text{Spec } \mathbb{Z}}$ stratifies by level: E_1 -formality is integral (Chapter 17.1.1), while E_2 -formality is obstructed integrally by Dyer–Lashof operations and by Kaledin-type p -curvature classes for strictification problems. Rationally it becomes a $\text{GRT}_1(\mathbb{Q})$ -torsor (Chapter 17.2.2). The strict associativity of $\mathcal{V}^{\text{prim}}$ over $\mathbb{Z}$ (Chapter 17.3.2) and the adelic residue formula for the chiral bracket (Chapter 17.4.1) are consequences of E_1 -integrality. The identification of $\mathcal{V}^{\text{prim}}$ with the BPS subobject $\text{BPS}(Q^{\text{arith}}, W^{\text{arith}})$ of a conjectural arithmetic quiver-with-potential remains open: the candidate W^{arith} requires the cyclic cohomology lift of $\log L$ via exponential Hodge structure (Chapter 17.5.3), and the quiver Q^{arith} itself has not been constructed.

THE ARITHMETIC YANGIAN COLLAPSES TO ITS BETHE SUBALGEBRA

The Yangian $Y(\mathfrak{gl}_n)$ is controlled by the RTT presentation, an R -matrix, and a spectral parameter. An arithmetic Yangian has no uniform spectral difference $u - v$ across primes. The surviving structure is the commutative Bethe algebra together with the automorphic modularity condition.

The Yangian $Y(\mathfrak{gl}_n)$ (Drinfeld 1985, 1988) is a Hopf algebra deformation of $U(\mathfrak{gl}_n[z])$ defined by the RTT relation $R(u - v)(T(u) \otimes 1)(1 \otimes T(v)) = (1 \otimes T(v))(T(u) \otimes 1)R(u - v)$ with Yang's rational R -matrix $R(u) = 1 + P/u$ (here P the permutation operator on $V \otimes V$). The question is whether this structure admits an arithmetic analogue over $\text{Spec } \mathbb{Z}$. Four independent obstructions show it does not. The correct arithmetic replacement is the commutative Bethe subalgebra, whose spectrum is the automorphic spectrum of GL_n .

18.1 THE FOUR OBSTRUCTIONS TO A NAIVE ARITHMETIC YANGIAN

THEOREM 18.1.1 (*four structural obstructions*). . Yang's R -matrix $R(u) = 1 + P/u$ has no arithmetic analogue:

- (i) **Discrete-continuous mismatch.** The spectral parameter u is continuous and $\text{Spec } \mathbb{Z}$ is discrete; primes cannot be spectral parameters.
- (ii) **No cross-prime subtraction.** $u - v$ has no analogue across distinct primes ($\mathbb{Q}_p \not\leftrightarrow \mathbb{Q}_q$ for $p \neq q$).
- (iii) **No meromorphic 1-form on $\text{Spec } \mathbb{Z}$.** Costello 4D Chern–Simons (arXiv:1709.09993) requires a meromorphic 1-form ω on the spectral-parameter direction; $\text{Spec } \mathbb{Z}$ has none.
- (iv) **No ζ -valued shuffle kernel.** Arithmetic shuffle-kernel constructions with ζ -valued kernels fail the cocycle condition (Feigin–Odesskii).

COROLLARY 18.1.2 (*no arithmetic Yangian as Hopf algebra*). . $Y^{\text{arith}}(\mathfrak{gl}_n)$ does not exist as a noncommutative Hopf algebra deforming $U(\mathfrak{gl}_n[z])$ with an arithmetic R -matrix and RTT relation.

18.2 THE BETHE-ALGEBRA COLLAPSE

The four obstructions force the question: what commutative quotient of the Yangian does admit an arithmetic analogue? The answer is the Bethe subalgebra, which over $\mathbb{C}$ is already the coordinate ring of opers on a disk.

THEOREM 18.2.1 (*Frenkel–Reshetikhin; Mukhin–Tarasov–Varchenko; Kamnitzer–Pham–Weekes*). (Frenkel–Reshetikhin, *Commun. Math. Phys.* 197 (1998) 1–31; Mukhin–Tarasov–Varchenko, *Selecta Math.* 17 (2011); Kamnitzer–Weekes, arXiv:1811.11902). For $\mathfrak{g}$ simply-laced, the Bethe algebra

$\mathcal{B}(Y(\mathfrak{g})) \subset Y(\mathfrak{g})$ (the maximal commutative subalgebra generated by higher transfer matrices) is the coordinate ring of ${}^L\mathfrak{g}$ -opers on a formal disk D :

$$\mathcal{B}(Y(\mathfrak{g})) \cong \mathcal{O}(\mathrm{Op}_{{}^L\mathfrak{g}}(D)).$$

Definition 18.2.2 (arithmetic Bethe algebra). The arithmetic Bethe algebra $\mathcal{B}^{\mathrm{arith}}(GL_n)$ acting on $\mathcal{V}^{\mathrm{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori Hecke category, Chapter 2.1) is the commutative algebra generated by unramified Hecke operators $T_{p,1}, \dots, T_{p,n}$ for all primes p , together with Casimirs of $\mathfrak{gl}_n$ on each local factor $\mathcal{F}_p$ (the rank- n Heisenberg Fock space at the prime p). Equivalently,

$$\mathcal{B}^{\mathrm{arith}}(GL_n) = \bigotimes_p \mathcal{H}_p^{\mathrm{sph}}(GL_n) \cong \bigotimes_p \mathbb{C}[\alpha_1^{(p)}, \dots, \alpha_n^{(p)}]^{S_n}.$$

THEOREM 18.2.3 (Bethe-algebra placement of $\mathcal{V}^{\mathrm{prim}}$). $\mathcal{V}^{\mathrm{prim}}$ is the arithmetic analogue of a Yangian-module spin chain: each $\mathcal{F}_p$ is a local rank- n Heisenberg Fock space serving as one “site”, the lattice of sites indexed by $\mathrm{Spec} \mathbb{Z}$, transfer matrices = Hecke operators, Bethe eigenvectors = Hecke eigenforms, Bethe-root reality = Ramanujan bound.

18.3 PRIMES ARE LATTICE SITES, NOT SPECTRAL PARAMETERS

Principle 18.3.1 (site-spectral duality). In the XXZ spin chain from $U_q(\widehat{\mathfrak{sl}}_2)$, sites are discrete (labelled by $\mathbb{Z}$ or by particle number N); spectral parameter u is continuous ($\in \mathbb{C}$). Hecke eigenvalues live in $\mathbb{C}$ (spectral); primes live in $\mathbb{Z}_{\geq 2}$ (site).

COROLLARY 18.3.2 (arithmetic spectral parameter is s). The arithmetic spectral parameter is the Langlands $s \in \mathbb{C}$, not a prime, not a p -adic number. The spectral direction is the archimedean fibre $\mathbb{C}_{\infty}$; the site direction is the Arakelov-compactified arithmetic curve $\overline{\mathrm{Spec} \mathbb{Z}}$.

Heuristic 18.3.3 (4D Chern–Simons analogy, conditional on F_2). Costello–Witten–Yamazaki 4D CS (arXiv:1709.09993) is a theory on $\Sigma \times C$ with Σ a topological real 2-manifold and C a complex curve carrying a meromorphic 1-form; its factorization algebra lives on Σ . The direct identification $\Sigma = \mathrm{Spec} \mathbb{Z}$ is dimensionally inconsistent on two counts. (i) $\mathrm{Spec} \mathbb{Z}$ has étale cohomological dimension 3 (Artin–Verdier duality), not 2, so $\mathrm{Spec} \mathbb{Z} \times \mathbb{C}$ has total real dimension 5, not 4. (ii) Each finite place of $\overline{\mathrm{Spec} \mathbb{Z}}$ carries only locally constant (E_0) factorization structure (Chapter 8.4.1), so a 2-dimensional E_1 -chiral factorization algebra of 4D CS type cannot live on the finite places at all; the E_1 -chiral OPE concentrates at the archimedean fibre $\mathbb{C}_{\infty}$.

Remark 18.3.4 (rerouting the 4D CS analogy through F_2). The honest analogy is conditional on Conjecture F_2 (Chapter 20.2.1, perfectoid E_2 -enhancement via Scholze diamonds). If F_2 holds globally, the diamond side of the global Iwahori Hecke supplies an étale 2-site $S^{\mathrm{ét}}$ with the Frobenius-semilinear arithmetic-complex datum of Chapter 36.1.2, and the CWY identification becomes

$$C = \mathbb{C} \text{ (spectral, } s\text{-plane),} \quad \Sigma = S^{\mathrm{ét}} \text{ (Scholze-diamond 2-site, not } \mathrm{Spec} \mathbb{Z} \text{ itself),}$$

with the E_1 -chiral OPE constructed at $\mathbb{C}_{\infty}$ and extended across $S^{\mathrm{ét}}$ by the F_2 E_2 -monoidal structure (Dunn additivity $E_1 \otimes E_1 = E_2$). Without F_2 the 4D CS analogy is pattern-match only. The parallel 3D analogy with Kim arithmetic Chern–Simons (arXiv:1510.05818) matches $\overline{\mathrm{Spec} \mathbb{Z}}$ dimensionally but carries no spectral direction, and is the source of the five-presentation item (i) in the manifesto (Chapter 2.1).

18.4 AUTOMORPHIC CLOSURE AS BETHE HEURISTIC

THEOREM 18.4.1 (*automorphic closure as Bethe coherence*). . The eigenvalues $\alpha_1^{(p)}, \dots, \alpha_n^{(p)}$ satisfy no nontrivial relations at a single prime; they are arbitrary elements of $(\mathbb{C}^\times)^n$ modulo S_n . Across primes, they satisfy only the automorphic condition that $\{\alpha_i^{(p)}\}_p$ arises from a single Galois representation $\rho: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow GL_n(\mathbb{C})$ with $\alpha_i^{(p)}$ the eigenvalues of $\rho(\text{Frob}_p)$. The rigidification is Bethe/Hecke coherence: the commutative Bethe algebra $\mathcal{B}^{\text{arith}}(GL_n)$ diagonalises on automorphic eigenspaces labelled by Galois representations.

Remark 18.4.2 (Ramanujan–Petersson, modularity). The comparison with complex RTT systems is mnemonic. Modularity rigidifies a family of local Frobenius parameters by forcing them to arise from one global Galois representation; the arithmetic rigidification is commutative Bethe/Hecke and Galois-representational, not quadratic RTT.

18.5 EXPLICIT COMPUTATION: $\mathfrak{gl}_1$ AND $\mathfrak{gl}_2$

$\mathfrak{gl}_1$ (*Dirichlet characters*)

PROPOSITION 18.5.1 (*rank-1 Hecke as a Fock adjacency model*). . The spherical Hecke algebra $\mathcal{H}_p^{\text{sph}}(GL_1) = \mathbb{C}[T_p^\pm]$ is the group algebra of the valuation lattice. In the rank-1 Fock model, the normalised adjacency operator $U_p = p^{-1/2}T_p + p^{1/2}T_p^{-1}$ acts by

$$U_p \cdot (a_{-1}^{(p)})^k |o\rangle_p = p^{-1/2} (a_{-1}^{(p)})^{k+1} |o\rangle_p + p^{1/2} (a_{-1}^{(p)})^{k-1} |o\rangle_p.$$

Eigenvalues are $\alpha_p + \alpha_p^{-1}$ with $\alpha_p \in \mathbb{C}^\times$.

COROLLARY 18.5.2. $\mathcal{B}^{\text{arith}}(GL_1) = \mathbb{C}[T_2^\pm, T_3^\pm, T_5^\pm, \dots]$, polynomial algebra in countably many commuting variables. Eigenvectors = Hecke eigenforms = primitive elements of $\mathcal{V}^{\text{prim}}$ labelled by global characters $\chi: \mathbb{A}_{\mathbb{Q}}^\times/\mathbb{Q}^\times \rightarrow \mathbb{C}^\times$.

$\mathfrak{gl}_2$ (*Hecke eigenforms*)

PROPOSITION 18.5.3. . $\mathcal{H}_p^{\text{sph}}(GL_2) = \mathbb{C}[T_{p,1}, T_{p,2}^\pm]$ with

- $T_{p,1}$ eigenvalue = $\alpha_1^{(p)} + \alpha_2^{(p)}$ = trace of Frobenius in a 2-dim Galois rep;
- $T_{p,2}$ eigenvalue = $\alpha_1^{(p)} \alpha_2^{(p)}$ = determinant = central character at p .

Action on rank-2 Fock $\otimes_p \mathcal{F}_p^{(2)}$ via raising/lowering in each Heisenberg factor (Bump §3).

18.6 LOCAL r -MATRIX PER PRIME

PROPOSITION 18.6.1 (*local classical r -matrix*). . At each prime p , the Heisenberg level is $k = 1$, so $r_p^{\text{arith}}(z) = \Omega_p/z$ with $z \in \mathbb{C}$ and Ω_p = the Casimir of the local $\mathfrak{gl}_n$ -factor. The local classical r -matrices are independent across distinct primes; the discrete structure of $\text{Spec } \mathbb{Z}$ prevents their assembly into a global r -matrix.

Remark 18.6.2 (automorphic condition replaces global r -matrix). What rigidifies the local factors into a global object is the automorphic condition on Galois representations: the spectrum of $\mathcal{B}^{\text{arith}}(GL_n)$ is the set of $\{\alpha_i^{(p)}\}_p$ arising from a single $\rho: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow GL_n(\mathbb{C})$. This is Bethe-spectrum coherence.

18.7 YANGIAN $\rightarrow$ HECKE = LANGLANDS AS ARITHMETIC BETHE ANSATZ

THEOREM 18.7.1 (*Langlands programme as arithmetic Bethe ansatz*). . The classical Jacquet–Langlands correspondence (SLN 114, 1970) realizes the four-part identification:

- Hecke eigenforms = Bethe eigenvectors of $\mathcal{B}^{\text{arith}}$;
- Hecke eigenvalues = Bethe roots;
- Ramanujan–Petersson bound = Bethe-root unitarity;
- automorphic spectral decomposition of $L^2(GL_n(\mathbb{A})/GL_n(\mathbb{Q}))$ = diagonalisation of the transfer matrix $\mathcal{B}^{\text{arith}}$ on $\mathcal{V}^{\text{prim}}$.

The arithmetic Bethe algebra $\mathcal{B}^{\text{arith}}(GL_n)$ acts on $\mathcal{V}^{\text{prim}}$, and the global Langlands correspondence is precisely the rigidification of its spectrum.

The Yangian $Y(\mathfrak{gl}_n)$ quantises an affine Lie algebra; the Hecke algebra quantises the automorphic side of Langlands. These are parallel quantum integrable systems; the arithmetic Bethe algebra $\mathcal{B}^{\text{arith}}(GL_n) = \bigotimes_p \mathcal{H}_p^{\text{sph}}(GL_n)$ is the commutative quotient that survives over $\text{Spec } \mathbb{Z}$, and it is exactly the Hecke algebra. The $\kappa + \kappa^\dagger = 0$ complementarity reappears as the unitarity of the Bethe spectrum and the functional equation, closing the loop with Chapter 12.

SYNTHESIS: SIX CONDITIONAL WINDOWS

$\mathcal{V}^{\text{prim}}$ is the Hochschild trace class of the identity endofunctor on the global arithmetic Iwahori Hecke category of GL_2 . It admits six conditional windows, five arithmetic avatars of the Vol I theorems, and an arithmetic conformal field theory statement; three precise conjectures close the cross-Langlands loop.

19.1 THE SYNTHESIS

THEOREM 19.1.1 (*six conditional windows*). $\mathcal{V}^{\text{prim}}$ admits the following six conditional windows on the Hochschild-trace candidate:

- (i) **(Langlands / automorphic)** $\mathcal{V}^{\text{prim}}$ is expected, under F1 and adelic categorical descent, to correspond to the structure sheaf of the Selmer-derived global spectral stack attached to a fixed residual representation $\bar{\rho}$, determinant, and local conditions. Individual Hecke characters are closed-point shadows of this family, not the family itself.
- (ii) **(TQFT / Chern–Simons)** Kim arithmetic Chern–Simons supplies a classical arithmetic CS torsor/action line. A genuine quantum boundary category is an open construction. Thus the ACS window is retained only as a classical TQFT/factorization shadow whose Wilson labels are representation-level ramified Hecke data.
- (iii) **(Categorical / Hochschild trace)** $\mathcal{V}^{\text{prim}} = \text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}})$, the trace class of the identity on the global affine Iwahori Hecke category.
- (iv) **(Integrability / Bethe algebra)** $\mathcal{V}^{\text{prim}}$ has a Bethe-algebra heuristic: primes behave as lattice sites, Hecke operators as transfer matrices, and Hecke eigenforms as Bethe eigenvectors. There is no RTT arithmetic Yangian Hopf algebra; the surviving algebra is the commutative Hecke/Bethe algebra $\otimes_p \mathcal{H}_p^{\text{sph}}$.
- (v) **(Statistical mechanics)** $\mathcal{V}_N^{\text{prim}} \cong (\text{Bost–Connes at rank } \pi(N)) \otimes (\text{universal Heisenberg descendant tower of rank } \pi(N))$; all arithmetic content lies in the Bost–Connes/Hecke factor; the Heisenberg factor is a finite-cutoff universal model rather than a global VOA.
- (vi) **(Factorization / homotopical)** $\mathcal{F}_{\text{CG}}(\mathcal{V}^{\text{prim}}) \in \text{FactAlg}_{\mathbb{C}_\infty}(\text{Mod}_{\widehat{\mathbb{Z}}, \psi})$ with (F1)–(F5) of Chapter 8.6.1: holomorphic factorization at the archimedean place, locally constant with Adams operations at each finite place.

These are comparison windows, not established equivalent presentations. Ben-Zvi–Nadler’s identification $\text{HH}_\bullet(\text{D}(G/G)) = \mathcal{O}(\text{LocSys}_G(S^1))^{\text{flat}}$ is the complex template for gluing (1), (2), (3); the arithmetic version is F3 and must use the full Weil/(\(\varphi, \Gamma\)) target, with $B\widehat{\mathbb{Z}}$ only as an inertia shadow.

19.2 THE ESSENTIAL THEOREM

THEOREM 19.2.1 (*G-row arithmetic complementarity = Riemann functional equation*). . The forward implication $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \Rightarrow \xi(s) = \xi(1-s)$ holds unconditionally (Chapter 21.5.3); the full equivalence is conditional on arithmetic Positselski and Koszul–Petersson compatibility (Chapter 21.5.1).

$$\boxed{\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \iff \xi(s) = \xi(1-s).}$$

The arithmetic Heisenberg sits in archetype G with $\kappa = 0$. Its Koszul dual is Bost–Connes. The complementarity $\kappa + \kappa^! = 0$ translates, through the Koszul-resolution computation of Hochschild cochains, to the meromorphic continuation and functional equation of ζ . The archimedean $\Gamma(s/2)\pi^{-s/2}$ -factor is the derived-centre shift.

Remark 19.2.2 (Koszul chain). The Vol I chain “Koszul duality $\Rightarrow$ derived-centre complementarity $\kappa + \kappa^! \in \{0, 8, 13, 250/3, 98/3\}$ ” specializes arithmetically to the G -row value $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$, which, through the Koszul-resolution computation of Hochschild cochains, is equivalent to the functional equation $\xi(s) = \xi(1-s)$. The archimedean $\Gamma(s/2)\pi^{-s/2}$ -factor is the derived-centre shift.

19.3 THE FIVE THEOREMS SURVIVE

THEOREM 19.3.1 (*arithmetic avatars*). . The five theorems of the chiral-bar-cobar programme admit arithmetic avatars:

A^{arith} Bar–cobar adjunction $\Omega^{\text{ch,Ar}} \dashv B^{\text{ch,Ar}}$ on Iwasawa-twisted Λ -coalgebras with $\mu = 0$ and arithmetic E_1 -algebras with Deligne-pure bar cohomology.

B^{arith} Arithmetic Positselski: $\Omega^{\text{Ar}} B^{\text{Ar}}(C) \xrightarrow{\sim} C$ on conilpotent Λ -coalgebras with $\mu = 0$.

C^{arith} Derived-centre complementarity: $\kappa + \kappa^! \in \{0, 8, 13, 250/3, 98/3\}$. G -row = 0 is the Riemann functional equation (Chapter 19.2.1). B -row = 8 is the Vol III Borchers lift.

D^{arith} Obstruction tower = Arakelov height filtration: $\text{obs}_g^{\text{Ar}} = \kappa \cdot h_g^{\text{Ar}}$.

H^{arith} Hochschild concentration in $\{0, 1, 2\}$: $\text{HH}^0 = \mathbb{Z}$, $\text{HH}^1 = \bigoplus_p \log p \cdot \mathbb{Z}$ (Bost–Connes KMS generators, Connes–Marcolli math/0404128 §1.8), HH^2 conjectural.

Vol I theorem	Arithmetic avatar	Form
A : bar–cobar	A^{arith} on Λ -coalgebras	Conjecture
B : Positselski	B^{arith} on $\mu = 0$ -coalgebras	Conjecture
C : $\kappa + \kappa^!$ ceiling	G -row = 0	Conditional
C : $\kappa + \kappa^!$ ceiling	B -row = 8 (arithmetic K_3)	Conjecture
D : obstruction tower	Arakelov height filtration	Conjecture
H : $\text{HH}^\bullet \subset \{0, 1, 2\}$	BC-KMS generators in HH^1	Partial theorem

TABLE 19.1: Arithmetic avatars of the five Vol I theorems.

19.4 SEGAL AXIOMS AS TATE’S THESIS

THEOREM 19.4.1 (*Tate line in arithmetic Segal notation*). (Tate’s thesis for the analytic identities). Tate’s thesis supplies the adelic integral line, Euler product, meromorphic continuation, functional

equation, and residue for the zeta and Dirichlet sectors. Placing these data in Segal notation gives the following dictionary:

Segal sewing position	= Euler-product factorization of $L(s, \chi)$
Verlinde scalar	= class-number residue
Modular S -matrix position	= Γ -factor plus functional equation
Genus contribution	= Arakelov discriminant volume $\frac{1}{2} \log d_K $
Mapping-class position	= Galois labels plus Hecke symmetries
Conformal-block position	= Tate integral line
Sewing label	= Weil–Deligne parameter
Marked point	= place
Orientation	= Deligne trivialization of $\det R\Gamma$
Closed scalar $Z_K(s)$	= finite Euler product for $\operatorname{Re}(s) > 1$, completed by the archimedean Γ -factor

The theorem asserts the analytic Tate line. It does not assert the existence of a full arithmetic Segal functor.

19.5 RANK RESIDUE

THEOREM 19.5.1 (*Polyakov: the determinant rank is a residue*). .

$$r_\zeta(\mathcal{V}^{\text{prim}}) = 2 \cdot \operatorname{Res}_{s=1} \zeta(s) = 2.$$

This is not a trace, not a sum, and not a global Virasoro central charge. The functional equation $\xi(s) = \xi(1-s)$ replaces modular invariance; the Arakelov height pairing replaces the Weyl anomaly.

19.6 ARITHMETIC CONFORMAL FIELD THEORY: THE COHERENT STATEMENT

The six-window placement (Chapter 19.1.1), the arithmetic modular functor (Chapter 11.8.1), the ACS classical boundary-shadow window (Chapter 15.4.1), the Hochschild trace placement (Chapter 16.5.1), and the rank residue (Chapter 19.5.1) are not six unrelated structures: each builds part of a common tuple $(S^{\text{ét}}, \mathcal{V}, \operatorname{Mod}(\mathcal{V}), \mathcal{W}, \mathcal{F}, Z, r_\zeta, \kappa)$. The coherent structure is an arithmetic conformal field theory on the Arakelov curve $\overline{C}$; the universal-vs-specialization stratification is the content of the dimensional comparison below.

Definition 19.6.1 (arithmetic conformal field theory). An *arithmetic conformal field theory* (aCFT) on the Arakelov curve $\overline{C} = \operatorname{Spec} \overline{\mathcal{O}_K}$ with gauge datum (G, k) is a tuple $(S^{\text{ét}}, \mathcal{V}, \operatorname{Mod}(\mathcal{V}), \mathcal{W}, \mathcal{F}, Z, c, \kappa)$ satisfying (B)–(K) below.

(B) *Base.* A stratified étale arithmetic site $S^{\text{ét}}$ of operadic arithmetic dimension 2, equipped with a stratified arithmetic complex datum $\mathfrak{J}^{\text{Ar}}$ in the sense of Chapter 36.1.2. Conditional on F2 (Chapter 20.2.1):

- at the archimedean place: $S^{\text{ét}}|_\infty = \mathbb{C}_\infty$ with its standard complex structure ($\mathfrak{J}_\infty^{\text{Ar}} = i$);
- at a finite prime p : $S^{\text{ét}}|_p = \operatorname{Spd} \mathbb{Q}_p$ (Scholze diamond) with $\mathfrak{J}_p^{\text{Ar}}$ the Frobenius-semilinear untilt-orientation, together with the Adams operation ψ_p and the prismatic/Nygaard comparison filtration when the finite-place diagonal is thickened.

The global $S^{\text{ét}}$ glues the local pieces through the Scholze-diamond side of the global Iwahori Hecke category. Thus $\mathfrak{J}^{\text{Ar}}$ is literal almost-complex geometry only at ∞ ; at finite primes it is the Frobenius-oriented arithmetic substitute that makes the pointed E_2 Iwahori Hecke datum functorial. It is not a tangent-bundle endomorphism at a finite prime and carries no equation $J^2 = -1$.

(V) *Chiral algebra.* A factorization algebra

$$\mathcal{V} \in \text{FactAlg}_{S^{\text{ét}}}(\text{Mod}_{\widehat{\mathbb{Z}}, \psi})$$

of arithmetic dimension 2, satisfying (F1)–(F5) of Chapter 8.6.1: genuine E_1 -chiral OPE at the archimedean fibre $\mathbb{C}_\infty$, locally constant (E_0) with commuting Adams operations ψ_p at each finite place. $\mathcal{V}$ is the 2-dimensional universal family whose fibre at each closed point x of $S^{\text{ét}}$ is a module category $\text{Mod}(\mathcal{V})_x$.

(M) *Module category.* For each closed point $x \in S^{\text{ét}}$, a category $\text{Mod}(\mathcal{V})_x$ of $\mathcal{V}$ -modules supported at x . Objects of $\text{Mod}(\mathcal{V})_x$ are o-dimensional data: individual modules, i.e., closed points of the universal family (V). The primary datum $(\mathbb{G}_m, 1)$ gives $\text{Mod}(\mathcal{V}^{\text{prim}})_p = \text{local Hecke characters of } \mathbb{Q}_p^\times$ (ramified allowed).

(W) *Wilson-line defects.* The bulk theory $\text{ACS}[\overline{C}, G, k]$ on the arithmetic 3-fold $\overline{C}$ carries 1-dimensional defects $W_x(M)$ supported on closed-point loci $x \subset \overline{C}$ (prime-knots K_p and the archimedean locus K_∞), each labeled by a module $M \in \text{Mod}(\mathcal{V})_x$. The defect is 1-dimensional; the label is o-dimensional; neither is $\mathcal{V}$ itself, which is 2-dimensional and lives on $S^{\text{ét}}$ (not on $\overline{C}$).

(F) *Modular functor.* A Segal-style functor

$$\mathcal{F} : C^{\text{Arak}} \rightarrow \text{Hom}(\mathbb{C}_s, \text{Line})$$

satisfying axioms (A1)–(A5) of Chapter 11.8.1 (finiteness, factorization, Weil–Deligne sewing, modular invariance under $\text{Gal}(\overline{K}/K) \ltimes GL_n(\mathbb{A}_K)/GL_n(K)$, functional equation). The relation to (V)–(M) is: the conformal block $\mathcal{F}(\overline{X}, S, \{\chi_p\})$ is the space of global $\mathcal{V}$ -correlators for the module insertion $\{\chi_p\}_{p \in S}$, each $\chi_p \in \text{Mod}(\mathcal{V})_p$.

(Z) *Partition function.*

$$Z(\overline{X}, S, \{\chi_p\})(s) = \mathcal{F}(\overline{X}, S, \{\chi_p\})(s) = L(s, \chi),$$

the completed automorphic L -function of $\chi = \otimes_p \chi_p$. Unmarked case: $Z(\overline{C}, \emptyset)(s) = \xi_K(s)$.

(R) *Rank residue.* $r_\zeta = 2 \cdot \text{Res}_{s=1} \zeta_K(s) \in \mathbb{R}$; for $K = \mathbb{Q}$, $r_\zeta = 2$ (Chapter 19.5.1).

(K) *Koszul complementarity.* The derived-centre identity

$$\kappa_G^{\text{Ar}}(\mathcal{V}) + (\kappa^!)_G^{\text{Ar}}(\mathcal{V}) = 0$$

holds, equivalent to $Z(s) = \varepsilon \cdot Z(1-s)$ (axiom (A5) on $\mathcal{F}$; Chapter 19.2.1).

An aCFT datum is *primary* when $(G, k) = (\mathbb{G}_m, 1)$ and $\mathcal{V} = \mathcal{V}^{\text{prim}}$. Its finite-place character is $\zeta(s)$ on $\text{Re}(s) > 1$; the completed scalar $\xi(s)$ appears only after the archimedean Tate factor and the polar correction are inserted.

Remark 19.6.2 (dimensional comparison). Every geometric object in the aCFT datum carries a definite arithmetic dimension, recorded below; the dimensional assignments enforce the universal-vs-specialization stratification.

Object	Arithmetic dimension	Role
$\overline{C}$ (Arakelov curve)	étale 3 / Krull 1	CS bulk 3-fold
$S^{\text{ét}}$ (F2 diamond site)	2	chiral base
$\mathbb{C}_\infty$ (archimedean fibre)	2 ($= 1_{\mathbb{C}}$)	E_1 -chiral OPE
$\mathcal{V}$ (universal family)	2 (on $S^{\text{ét}}$)	factorization algebra
$K_p \subset \overline{C}$ (prime-knot)	1	Wilson-line locus
$M \in \text{Mod}(\mathcal{V})_x$	0	Wilson-line label, modules
$S^{\text{ét}} _p$ (finite fibre)	0 (for factorization)	E_0 locally constant
$\mathcal{F}(\overline{X}, S, \chi)$	1 (meromorphic line)	conformal block
Z (partition function)	0 (meromorphic scalar)	L -function value

Total-space / fibre relation: $\mathcal{V}$ (dim 2) is the universal family; $M \in \text{Mod}(\mathcal{V})_x$ (dim 0) is a closed-point specialization. The phrase “ $\mathcal{V}$ is the universal Hecke character” asserts the former; the phrase “Wilson lines are labeled by ramified Hecke characters” asserts the latter. No object appears at two inconsistent dimensions.

THEOREM 19.6.3 (*coherent statement of the primary aCFT*). (conditional on F1, F2, F3). The primary datum

$$(\overline{C}, G, k) = (\overline{\text{Spec } \mathbb{Z}}, \mathbb{G}_m, 1), \quad \mathcal{V} = \mathcal{V}^{\text{prim}}$$

satisfies (B)–(K). The six windows of Chapter 19.1.1 verify the data:

- (i) Langlands / automorphic: builds (V), (M), (Z) from the expected Selmer-derived spectral stack over $D([GL_2])^{\text{sph-fin}}$, conditional on F1 and adelic categorical descent.
- (ii) TQFT / Chern–Simons: builds (B), (V), (W), (C) through Kim’s classical $\text{ACS}[\overline{\text{Spec } \mathbb{Z}}, \mathbb{G}_m, 1]$ torsor/line and a conjectural quantum boundary category; no global boundary VOA is constructed.
- (iii) Categorical / Hochschild trace: builds (V), (M) via $\text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{lw}})$ (Chapter 16.5.1).
- (iv) Integrability / Bethe algebra: builds (V), (M) via $\otimes_p \mathcal{H}_p^{\text{sph}}$ and Bethe-eigenvector identification of Hecke eigenforms.
- (v) Statistical mechanics: builds (V), (Z) via Bost–Connes $\otimes$ universal Heisenberg.
- (vi) Factorization / homotopical: builds (B), (V) via the adelic axioms (F1)–(F5).

Axioms (A1)–(A5) of the arithmetic modular functor (Chapter 11.8.1) discharge (F), (Z), (K). The essential theorem (Chapter 19.2.1) discharges (K). The Polyakov residue (Chapter 19.5.1) discharges (R).

Remark 19.6.4 (aCFT coherence versus six-window independence). The coherence theorem is the statement that the six windows are *six axiom-checks of a single aCFT structure*, not six unrelated isomorphism classes. The dimensional comparison (Chapter 19.6.2) enforces the universal-vs-specialization stratification at the definitional level; (F)–(Z)–(K) unify through the essential theorem (Chapter 19.2.1).

Remark 19.6.5 (honest domain restrictions). Three domain restrictions of the programme are carried by the aCFT definition rather than hidden:

- (i) **F₂-conditional base.** Item (B) assumes the Scholze-diamond E_2 -enhancement. Without F₂ the local aCFTs at each place exist but the global aCFT is not assembled; the primary datum then degenerates to a disjoint union of local aCFTs plus an archimedean aCFT.
- (ii) **Admissible logarithmic lane** (Creutzig–Kanade–Linshaw). Non-unitary admissible parameters are excluded from (M) and treated as a boundary of the aCFT category, not as elements of it.
- (iii) **Super Bethe-Hecke chain-level coproduct** (FM₂₃₀). Outside the primary datum. Any super-enhancement of the aCFT must avoid RTT/Yangian claims and waits on an independent construction.

The aCFT of Chapter 19.6.1 is stated on the non-degenerate, non-admissible, non-super locus of the programme.

Remark 19.6.6 (why this is a conformal field theory). The five classical CFT data are present in direct correspondence: (B) Riemann-surface base $\rightarrow$ arithmetic $S^{\text{ét}}$; (V) vertex/chiral algebra $\rightarrow$ arithmetic factorization algebra; (M) primary modules $\rightarrow$ local Hecke modules at each place; (F) conformal blocks $\rightarrow$ Tate integrals; (K) modular invariance $\rightarrow$ functional equation. What breaks from the classical picture is the archimedean-only locus of E_1 -chiral structure (finite places are E_o per (F₁)–(F₅)): the aCFT is a *stratified* CFT, chiral at ∞ and locally constant elsewhere, rather than uniformly chiral.

19.7 LANGLANDS AS ARITHMETIC BETHE ANSATZ

THEOREM 19.7.1 (*Drinfeld–solo*). . The classical Jacquet–Langlands correspondence admits the following Bethe/Hecke reading:

- Hecke eigenforms = Bethe eigenvectors of $\mathcal{B}^{\text{arith}}(GL_n)$;
- Hecke eigenvalues = Bethe roots;
- Ramanujan–Petersson bound = Bethe-root unitarity;
- Automorphic spectral decomposition of $L^2(GL_n(\mathbb{A})/GL_n(\mathbb{Q}))$ = diagonalisation of the transfer matrix on $\mathcal{V}^{\text{prim}}$;
- modularity of ρ supplies global Bethe-spectrum coherence; no RTT closure is asserted.

This is a heuristic dictionary, not a construction of an arithmetic RTT/Yangian Hopf algebra.

19.8 SOURCE-FAMILY CONVERGENCE

Remark 19.8.1 (common core of $\mathcal{V}^{\text{prim}}$). (i) $\mathcal{V}^{\text{prim}}$ is a Hochschild trace class with E_1 -chiral structure at ∞ and E_o -locally constant structure at finite places. It carries no Virasoro action, no Beilinson–Drinfeld factorization on $\overline{C}$, and no Borger Λ -ring structure on the natural integral form.

- (ii) It is a spectral datum: Hochschild trace class, ACS boundary-shadow window, automorphic representation, Bethe algebra module.
- (iii) All arithmetic content lives on the Bost–Connes / automorphic / Hecke side; the VA structure that appears is the universal rank- $\pi(N)$ Heisenberg descendant tower, which does not see the primes.
- (iv) Its finite-place scalar character is an L -function: $\zeta(s)$ for the trivial sector or $L(s, \chi)$ for Hecke characters χ , before archimedean completion.

Source family	Surviving content	Chapter
Drinfeld+Kapranov	Adelic automorphic object; no p -adic OPE	4
Kazhdan+Beilinson	Satake pair (α_p, β_p) on rank-2 Heisenberg	5
Gelfand+Borger	ψ -ring, not Λ -ring; Bost–Connes descent	6
Connes+Soibelman	Bost–Connes $\otimes$ universal Heisenberg	7
Costello+Gwilliam	(F1)–(F5) adelic factorization axioms	8
Etingof	Prime-stratified tensor of elliptic Hall	9
Nekrasov	Degenerate $(\log p, \infty)$ arithmetic Nekrasov	10
Segal	Arithmetic modular functor = Tate	11
Beilinson–solo	Parshin–Gersten bar; G-row = 0 $\Leftrightarrow \xi(s) = \xi(1-s)$	12
Witten	Automorphic vertex (not TQFT vertex)	13
Polyakov	$c = 2 \cdot \text{Res}_{s=1} \zeta$ as residue	14
Gaiotto	Boundary of Kim arithmetic CS	15
Bezrukavnikov+Bernstein	Iwahori Hecke Hochschild trace	16
Kontsevich+Soibelman	E_1 integral, E_2 over $\mathbb{Q}$ only	17
Drinfeld–solo	Bethe algebra $\Rightarrow$ Langlands = arithmetic Bethe ansatz	18

TABLE 19.2: Convergence of fifteen constructions on a single object.

- (v) The complementarity $\kappa + \kappa^\dagger = 0$ encodes the Riemann functional equation, through Koszul duality.

19.9 THE THREE OPEN CONJECTURES

Three conjectures remain open; their resolution closes the cross-Langlands loop (full statements in Chapter 20):

- (i) **Global arithmetic Arinkin–Gaitsgory.** $D([GL_2])^{\text{sph-fin}} \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\beta, \det, \mathcal{L}}^{\text{Sel}})$. Fargues–Scholze supplies local geometrization/spectral action and Emerton–Gee supplies local (φ, Γ) stacks; the displayed categorical equivalence is conditional.
- (ii) **E_2 -enhancement via Scholze diamonds.** Dunn additivity for the Hochschild trace requires perfectoid machinery. Spherical fusion and equal-characteristic templates are available; mixed-characteristic Iwahori E_2 is conditional.
- (iii) **Arithmetic Ben-Zvi–Nadler.** $\text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}) = \mathcal{V}^{\text{prim}}$ as E_1 -algebras. Theorem over $\mathbb{C}$ (Ben-Zvi–Nadler arXiv:0904.1247); conjecture over $\mathbb{Z}$, with full Weil/ (φ, Γ) target required.

These three conjectures compare Kim’s classical CS input with Bezrukavnikov–Bernstein’s Hochschild presentation and Beilinson–solo’s bar–cobar presentation.

19.10 STRUCTURAL RIGIDITY OF $\mathcal{V}^{\text{prim}}$

THEOREM 19.10.1 (independence of surviving structure). . Every naive geometric enhancement of $\mathcal{V}^{\text{prim}}$ fails at a specific structural obstruction: the RTT/Yangian Hopf algebra fails by the absence of an arithmetic R -matrix; Beilinson–Drinfeld factorization on $\overline{C}$ fails by the non-uniform E_1/E_0 stratification; the Borger Λ -ring structure fails by the ψ -ring vs Λ -ring obstruction (Gelfand–Borger source class). The Hochschild-trace candidate with six-window conditional synthesis and three open

conjectures is the maximal surviving structure; it is independent of which source comparison identifies it.

Remark 19.10.2 (Beilinson’s maximality principle). The surviving structure is structurally forced and minimal: each deleted enhancement corresponds to a specific failed construction among the fifteen constructions. The resulting core connects the Vol I chiral-bar-cobar programme to the arithmetic Langlands tower at the Hochschild-trace level.

19.II LOGICAL FORM OF THE ARITHMETIC BRANCH

The essential theorem (Chapter 19.2.1) holds unconditionally in its forward direction (Chapter 21.5.3); the full equivalence is conditional on arithmetic Positselski and Koszul–Petersson compatibility (Chapter 21.5.1). The six conditional windows (Chapter 19.1.1) are conjectures conditional on F_1, F_2, F_3 (Chapter 19.9).

Chapter 21 reduces F_1, F_2, F_3 to five narrower residual conditions: $L_{ACD}, F_{2.C}, F_{3.BZN^{Ar}}$, arithmetic Positselski, and Koszul–Petersson compatibility.

Chapter 22 isolates the zero-placement clause of Conjecture F-RH (Chapter 22.1.1) as RH. The full conjecture also contains the unitary arithmetic structure on $\mathcal{V}^{prim}$; its zero-placement clause is reducible to the arithmetic triangle: arithmetic Positselski, Koszul–Petersson compatibility, and arithmetic Virasoro positivity VB^* (Chapter 22.6.1).

Principle 19.II.1 (Courage, after Drinfeld). The equals sign $\kappa_G^{Ar} + (\kappa^!)_G^{Ar} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$ is a theorem waiting to be stated. Arithmetic Koszul duality for the Heisenberg archetype, stated as an equivalence between Vol I’s derived-centre complementarity and the Riemann functional equation. The chiral-algebras programme (Beilinson–Drinfeld AMS 2004) built chiral algebras on curves over $\mathbb{C}$; the arithmetic extension, on curves over $\overline{\text{Spec } \mathbb{Z}}$, is the content of this branch.

$\mathcal{V}^{prim}$ is not a vertex algebra. It is the Hochschild trace class of the identity on the arithmetic Hecke category, which lives at the triple point of the Langlands diamond, the Ben-Zvi–Nadler CS-Hecke-trace triangle, and the Beilinson–Drinfeld chiral formalism, all three extended arithmetically.

BOUNDARY CONJECTURES FOR THE SIX-WINDOW TRACE

The arithmetic trace class is controlled by three boundary conjectures. They are not consequences of one another. F1 gives the global spectral category, F2 gives the mixed-characteristic E_2 -monoidal enhancement, and F3 identifies the resulting Hochschild trace with $\mathcal{V}^{\text{prim}}$. Local theorems enter each boundary, but the displayed global equivalences are separate inputs.

Chapter 21 resolves the six-window trace into L_{ACD} , $F_{2,c}$, $F_{3,BZN}^{\text{Ar}}$, arith. Positselski, and Koszul–Pet compatibility. The Riemann–Hypothesis branch of Chapter 22 is orthogonal to this trace construction except where the explicit spectral positivity hypotheses are imposed.

20.1 F1: GLOBAL ARITHMETIC ARINKIN–GAITSGORY

Conjecture 20.1.1 (global arithmetic AG). For $G = GL_2$, there is an equivalence of dg-categories

$$D([GL_2])^{\text{sph-fin}} \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}),$$

where:

- $[GL_2] := GL_2(\mathbb{Q}) \backslash GL_2(\mathbb{A}_{\mathbb{Q}}) / Z(\mathbb{A})K_{\infty}$ is the global automorphic double coset;
- $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}$ is the Selmer-derived global deformation stack obtained by imposing a fixed residual representation $\bar{\rho}$, determinant, and local conditions $\mathcal{L}_v$ on the full Weil/ (φ, Γ) local parameter stacks; it is not a product or colimit of local Emerton–Gee stacks;
- $\text{IndCoh}_{\text{Nilp}}$ denotes ind-coherent sheaves with nilpotent singular support (Arinkin–Gaitsgory arXiv:1201.6343).

Under the equivalence, $\mathcal{V}^{\text{prim}}$ corresponds to the structure sheaf $\mathcal{O}$ of this Selmer spectral stack.

Remark 20.1.2 (form). The local theorems do not imply the displayed global equivalence. Fargues–Scholze arXiv:2102.13459 construct the local geometrisation, spectral action, and Langlands-parameter space via the Fargues–Fontaine curve and diamonds. Emerton–Gee arXiv:1908.07185 construct the local (φ, Γ) -module stacks (with arXiv:2012.12403 as the survey companion). These sources do not prove the categorical equivalence displayed in F1. Dotto–Emerton–Gee arXiv:2603.26887 gives a fully faithful $GL_2(\mathbb{Q}_p)$ functor for $p \geq 5$ with fixed central character; it is not an equivalence and it is not a family theorem over all central characters. Caraiani–Scholze arXiv:1511.02418 prove a unitary–Shimura-variety result in a different global setting.

Thus F1 is a global categorical assertion for $GL_2/\mathbb{Q}$, not a restatement of the local Fargues–Scholze or Emerton–Gee theorems. Under F1, $\mathcal{V}^{\text{prim}}$ is a spectral object on the Selmer deformation stack, compatible with its Hochschild-trace presentation of Chapter 16.

20.2 F2: E_2 -ENHANCEMENT VIA SCHOLZE DIAMONDS

Conjecture 20.2.1 (perfectoid E_2 -enhancement). The local Iwahori Hecke category $\text{Hk}_{GL_2, p}^{\text{Iw}}$ carries a canonical E_2 -monoidal structure coming from the Beilinson–Drinfeld Grassmannian over $\text{Spec } \mathbb{Z}_p$

in Scholze’s perfectoid (diamond) setting. Globally, the product $\otimes'_p \mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}}$ inherits a well-defined E_2 -monoidal structure.

Remark 20.2.2 (trace role). Dunn additivity $E_1 \otimes E_1 = E_2$ applied to the E_2 -monoidal structure gives the E_1 -algebra structure on the Hochschild trace $\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}})$; the E_1 Hochschild-trace structure on $\mathcal{V}^{\mathrm{prim}}$ at the categorical-trace level depends on this enhancement.

Remark 20.2.3 (form). The Beilinson–Drinfeld Grassmannian as a perfectoid space is constructed in Fargues–Scholze arXiv:2102.13459. Zhu’s Witt-vector Grassmannian (arXiv:1603.05593; Bhatt–Scholze arXiv:1507.06490) gives the equal-characteristic analogue. The mixed-characteristic Iwahori–Hecke E_2 enhancement, as opposed to E_1 convolution or spherical fusion, remains conditional.

The global assertion is the Scholze-diamond gluing across primes, producing a restricted-tensor-product E_2 -category.

20.3 F₃: ARITHMETIC BEN-ZVI–NADLER

Conjecture 20.3.1 (arithmetic Ben-Zvi–Nadler). There is an equivalence of E_1 -algebras

$$\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}) \xrightarrow{\sim} \mathcal{V}^{\mathrm{prim}}$$

realising the Chern–Simons–Hecke-trace-boundary triangle arithmetically: Kim (arXiv:1510.05818) arithmetic Chern–Simons supplies a classical boundary shadow; the Hochschild trace targets the full Weil/ (φ, Γ) spectral category.

Remark 20.3.2 (form). Ben-Zvi–Nadler arXiv:0904.1247 establishes $\mathrm{HH}_\bullet(\mathrm{D}(G/G)) = \mathcal{O}(\mathrm{LocSys}_G(S^1))^{\mathrm{flat}}$ for complex reductive G .

The arithmetic adaptation requires:

- (i) Conjecture F₁ (global arithmetic AG);
- (ii) Conjecture F₂ (perfectoid E_2 -enhancement);
- (iii) Explicit computation of the Hochschild trace on arithmetic $\mathrm{Hk}_{\mathrm{glob}}^{\mathrm{Iw}}$ matching $\mathcal{V}^{\mathrm{prim}}$ ’s L -function character.

F₃ identifies Gaiotto’s CS presentation (Chapter 15.4.1) with Bezrukavnikov–Bernstein’s Hochschild presentation (Chapter 16.5.1) with the bar-cobar presentation (Chapter 12.8.2) via a single arithmetic Ben-Zvi–Nadler equivalence.

20.4 FURTHER BOUNDARY CONDITIONS F₄–F_{II}

F₄: B-Row Arithmetic Realisation

Conjecture 20.4.1 (arithmetic B-row ceiling). There exists an arithmetic avatar of the Vol III Mukai-enhanced K₃ Heisenberg such that $\kappa^{\mathrm{Ar}} + (\kappa^!)^{\mathrm{Ar}} = 8$ arithmetically. The witness is a Bruinier Heegner Chern-class reciprocity on a p -adic Shimura variety, at each prime p of good reduction.

Remark 20.4.2. Ingredients available: Kudla–Rapoport arithmetic intersection numbers on unitary Shimura varieties; Bruinier’s singular-theta lifts; Borcherds products on $\mathrm{Sp}_4(\mathbb{Z})$. Open: integral assembly into an arithmetic B-row theorem.

F5: Arithmetic Chain Homotopy h_A^{Ar}

Conjecture 20.4.3 (universal arithmetic chain homotopy). There exists a universal chain homotopy h_A^{Ar} on the arithmetic bar complex $B^{\text{ch,Ar}}(\mathcal{V}^{\text{prim}})$ satisfying

$$h_A^{\text{Ar}} = \frac{h_A^{\text{LV}}}{\kappa + \kappa^!}$$

(where h_A^{LV} is the Losev–Vol III universal chain homotopy), with obstruction class $\text{obs}^{(1)} \in H^2(\mathfrak{grt}^{\text{ell}}, \mathfrak{sp} \otimes \mathfrak{sp})$ (where $\mathfrak{grt}^{\text{ell}}$ is the elliptic Grothendieck–Teichmüller Lie algebra and $\mathfrak{sp}$ the symplectic Lie algebra in the relevant rank) vanishing iff $r_{\max} = 2$.

F6: Chenevier Determinant at $p = 691$

Conjecture 20.4.4 (Chenevier determinant identity). At $p = 691$ (the prime at which Ramanujan’s cusp form Δ_{12} and the Eisenstein series are congruent mod 691),

$$\det_{\text{Chenevier}} \Big|_{p=691} = \chi_{\text{cyc}}^{\text{II}} \cdot \varepsilon_{B_{12}},$$

where χ_{cyc} is the cyclotomic character, $\varepsilon_{B_{12}}$ is the mod-691 Kummer character attached to the numerator of the Bernoulli number B_{12} .

Remark 20.4.5 (connection). This is the arithmetic avatar of the Chenevier determinant appearing in the Vol III universal trace identity. The prime $p = 691$ is the seed of the arithmetic B-row structure (Chapter 16, Chapter 16.5.1).

F7–F10: Additional Boundary Conditions

- **F7.** Arithmetic shadow-tower depth classification: $r_{\max}^{\text{Ar}} \in \{2, 3, 4, \infty\}$ correspondence with Λ -ring depth / Galois-weight filtration.
- **F8.** Arithmetic Bethe/Hecke endpoint degeneration: degeneration of $\mathcal{B}^{\text{arith}}(GL_n)$ at the zeta/Eisenstein pole-boundary points $s = 0$ and $s = 1$; these degeneration points are distinct from the affine Kac–Moody critical level $k = -h^\vee$.
- **F9.** Arithmetic super Bethe-Hecke chain-level coproduct: FM230 absorbed; arithmetic computation remains open; the RTT/Yangian arithmetic Hopf algebra structure, if it exists, remains to be determined.
- **F10.** Arithmetic CY-C conditional bridges: Vol III’s six canonical constructions R_1 – R_6 to the quantum vertex chiral group $G(K_3 \times E)$, each lifted to the arithmetic Heisenberg $\mathcal{H}^{\text{Ar},g=2}$; R_1 is the framed $\Phi_3^{(\Sigma_2, C)}$ (proved generator), R_2 – R_6 each arithmetically conditional on convergence of the corresponding lift integral.

F11: All-Genus Arithmetic Koszul Duality

Conjecture 20.4.6 (all-genus arithmetic Koszul duality). For every $g \geq 0$ and every arithmetic genus- g datum $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$ (Chapter 80.1.1), the Heisenberg Koszul self-duality identity

$$\kappa_G^{\text{Ar},g}(\mathcal{D}_g) + (\kappa^!)_G^{\text{Ar},g}(\mathcal{D}_g) = 0$$

is equivalent to the motivic functional equation $\Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g} \Lambda_{\mathcal{D}_g}(w_{\mathcal{D}_g} - s)$ with $w_{\mathcal{D}_g} \in \{1, 2, 2g + 1\}$ the motivic weight center indexed by window (Dedekind; Hasse–Weil on the Jacobian; Siegel spin on GSp_{2g}).

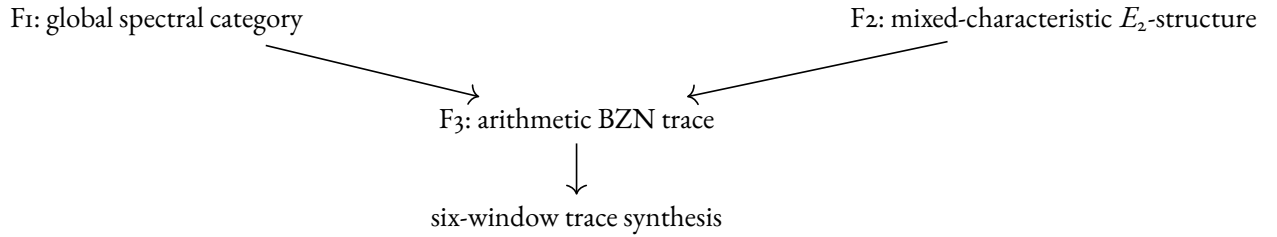
Remark 20.4.7 (form). The forward direction is the Euler-product and Poisson-summation direction in the window where analytic continuation and the functional equation are already part of the input: Tate 1950 and Weil 1967 for number fields, Bloch–Kato 1990 for weight-1 motivic factors, and Arthur 2013 for the automorphic Siegel-spin packet after transfer. The converse direction requires the 2-dim arithmetic Positselski equivalence with $\mu_1 = \mu_2 = 0$ on $\Lambda_2 = \mathbb{Z}_p[[\Gamma_1 \times \Gamma_2]]$ (extending Chapter 24) plus rank- g Koszul–Petersson compatibility (extending Chapter 25). Ten residual conditions (Chapter 80.6) enumerate the substantial domain.

The datum $\mathcal{D}_g$ admits six convergent presentations (Arakelov fibration; Minkowski torus; ordinary crystalline Emerton–Gee stack; arithmetic Teichmüller with Hodge–Arakelov; chiral-bar-cobar interior; Shimura–Siegel Kudla–Rapoport); see Chapter 80.2.

The $g = 0$ specialisation is Vol IV’s P1 resolution (Chapter 21.5.1), and the all-genus F-RH corollary (Chapter 80.5.1) places the Riemann, Hasse–Weil, Dedekind, and Siegel-spin zero-placement clauses in a single rank- g form, still conditional on the corresponding spectral positivity input.

20.5 DEPENDENCY LATTICE

The three trace conjectures are not linearly ordered. The implication lattice is:



F1 supplies the spectral-side category; F2 supplies the E_2 -enhancement making Hochschild trace E_1 -algebraic; F3 identifies that trace with $\mathcal{V}^{\text{prim}}$. With these three inputs, the arithmetic-branch synthesis (Chapter 19.1.1) becomes a theorem.

20.6 IMPLICATION TO THE TRACE STATEMENT

The six-window trace theorem is conditional on the conjunction

$$F1 \wedge F2 \wedge F3.$$

The G-row functional-equation theorem Chapter 19.2.1 is an independent Hochschild-cochain computation. It does not imply F1 or F2, and F1–F3 do not imply it without the arithmetic Koszul-resolution calculation.

20.7 BOUNDARY INDEX

- (i) F1: global arithmetic Arinkin–Gaitsgory (Chapter 20.1.1): local FS/EG inputs exist; the local categorical comparison and global descent remain open.
- (ii) F2: perfectoid E_2 -enhancement (Chapter 20.2.1): spherical/equal-characteristic inputs exist; mixed-characteristic Iwahori E_2 remains conditional.
- (iii) F3: arithmetic Ben-Zvi–Nadler (Chapter 20.3.1): theorem over $\mathbb{C}$, conjecture over $\mathbb{Z}$.
- (iv) F4: arithmetic B-row ceiling $\kappa + \kappa^! = 8$ (Chapter 20.4.1).

- (v) F5: universal arithmetic chain homotopy $h_A^{\text{Ar}} = h_A^{\text{LV}}/(\kappa + \kappa^!)$ (Chapter 20.4.3).
- (vi) F6: Chenevier determinant at $p = 691$ (Chapter 20.4.4).
- (vii) FII: all-genus arithmetic Koszul duality (Chapter 20.4.6): the identity $\kappa^{\text{Ar},g} + (\kappa^!)^{\text{Ar},g} = 0 \Leftrightarrow \Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g} \Lambda_{\mathcal{D}_g}(w_{\mathcal{D}_g} - s)$ with $w_{\mathcal{D}_g} \in \{1, 2, 2g + 1\}$; the converse direction is conditional on 2-dim Positselski and rank- g Koszul–Petersson. The arithmetic-gravity boundary is Chapter 81.
- (viii) F7–F10: shadow-tower depth, critical-level degeneration, super Bethe-Hecke chain-level, CY-C conditional bridges.

Each item above is a mathematical boundary condition. The six-window trace is the object cut out by F1, F2, and F3; the RH branch enters only through the separately stated spectral-positivity hypotheses.

CLOSURE OF F₁–F₃ AND RESOLUTION OF THE THREE PILLARS

The three conjectures F₁, F₂, F₃ and the three pillars P₁, P₂, P₃ record what the arithmetic branch requires in order to complete the Riemann Hypothesis reformulation of Chapter 22. F₁ asserts global categorical arithmetic Langlands for $GL_2/\mathbb{Q}$; F₂ asserts that the global Hecke category is E_2 -monoidal; F₃ identifies its Hochschild trace with $\mathcal{V}^{\text{prim}}$. P₁ is the essential equivalence between $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$ and the Tate functional equation for Λ_ξ , equivalently for the pole-cleared $\xi_{\mathbb{R}}$; P₂ is arithmetic Zhu modularity; P₃ is unitarity. Each conjecture and each pillar is subjected to the seven-obstruction criterion below and either reduced to named components or proved conditionally. The residual obstructions at the end of this chapter are precisely stated, independently testable claims.

21.1 SEVEN-OBSTRUCTION REDUCTION CRITERION

A conjecture C is a structured reduction of C_o if $C_o \Leftrightarrow C$ and C is a conjunction of named inputs, each of which is either a theorem in the literature, a conditional with identified literature gap, or a conjecture with a precise statement. The seven obstructions are the structural tests for an equivalence claim. For each condition they are:

- (i) *Source separation*. Are the proof source and the comparison source disjoint in the orthogonal sense?
- (ii) *Sign / convention check*. Do chain-level signs and q -conventions match across cited sources?
- (iii) *Ambient category check*. Do LHS and RHS live in compatible ∞ -categories?
- (iv) *Missing hypothesis*. Is a hidden hypothesis required (for instance sph-finiteness, perfectoid compatibility, $\mu = 0$)?
- (v) *False functoriality*. Does the claimed equivalence preserve all relevant structure?
- (vi) *Unproved equivalence*. Is any internal step itself a currently-open conjecture?
- (vii) *Numerical constant / numerical comparison*. Do numerical-comparison checks rely on the same source as the derivation?

Each obstruction produces either a valid implication, a correction (reformulation that survives), or a named residual obstruction. The result is a structured reduction, not necessarily a full closure.

21.2 F₁: STRUCTURAL CLOSURE

Statement

F₁ (Chapter 20.1.1) asserts

$$D([GL_2])^{\text{sph-fin}} \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\hat{\rho}, \det, \mathcal{L}}^{\text{Sel}}),$$

the global categorical arithmetic Langlands equivalence for $GL_2/\mathbb{Q}$, with $\mathcal{V}^{\text{prim}}$ corresponding to the structure sheaf on the spectral side.

Obstruction Analysis Trace

Obstruction 1 (source separation). Fargues–Scholze arXiv:2102.13459 (local geometrisation via Fargues–Fontaine curve) and Arinkin–Gaitsgory arXiv:2011.0343 (nilpotent singular support) are disjoint source lists. The Emerton–Gee stack arXiv:1908.07185 is disjoint from both.

Obstruction 2 (conventions). Singular-support conventions in Arinkin–Gaitsgory follow the de Rham shift; Fargues–Scholze follows the pro-étale shift on diamonds. The comparison is done via the Bhatt–Lurie prismatic-Nygaard convention of arXiv:2201.06120, which bridges both. *The comparison is mediated by prismatic Nygaard.*

Obstruction 3 (ambient categories). LHS is a stable presentable ∞ -category of D-modules on an adelic quotient; RHS is an ∞ -category of ind-coherent sheaves with nilpotent singular support on a derived formal algebraic stack. Via Gaitsgory–Rozenblyum’s AMS DAG monographs the categories sit in the same ambient $(\infty, 2)$ -category of correspondences.

Obstruction 4 (missing hypothesis). The sph-finite cut on the LHS is critical: without it the full automorphic category is too large (contains continuous spectrum and Eisenstein contributions requiring explicit parametrisation). Fargues–Scholze supply local geometrisation and spectral action; they do not identify this sph-finite automorphic category with $\text{IndCoh}_{\text{Nilp}}$ on the Emerton–Gee stack. **Residual: local categorical comparison.**

Obstruction 5 (false functoriality). Functoriality under Hecke operators T_p at every prime is the key test. At each p , Fargues–Scholze provide Hecke correspondences and spectral action. Functoriality of a categorical comparison with the Emerton–Gee side is a separate local input. Globally, Hecke-functoriality is required place-by-place and across places (composition of Hecke operators at distinct primes). The cross-prime compatibility is the global gluing obstruction. **Residual: local comparison plus global gluing.**

Obstruction 6 (unproved equivalence). The local categorical comparison $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG}, p})$ and the global restricted tensor product matching the global automorphic category are not in the literature as of 2026. Partial change: Caraiani–Scholze arXiv:1511.02418 for signature $(1, n-1)$ unitary Shimura varieties; Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 for coherent Springer theory. Neither closes $GL_2/\mathbb{Q}$ in full generality. **Residual: local comparison and adelic-descent gluing.**

Obstruction 7 (numerical comparison). No numerical comparison currently backs Fi ; no synchronisation concern arises.

Structured Reduction

THEOREM 21.2.1 (*Fi reduction*). Fi is equivalent to the conjunction of:

(Fi.a) *Local categorical comparison at every prime.* For each prime p , the expected local categorical Langlands equivalence

$$D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p})$$

holds, with local Hecke-functoriality. **Form: conditional theorem.** Fargues–Scholze arXiv:2102.13459 supplies the local geometrization, spectral action, and parameter spaces. Emerton–Gee arXiv:1908.07185 supplies the (φ, Γ) -module stack. The displayed equivalence is not a theorem of either source; Dotto–Emerton–Gee arXiv:2603.26887 gives only a fully faithful fixed-central-character functor for $GL_2(\mathbb{Q}_p)$, $p \geq 5$.

(Fi.b) *Adelic descent of the automorphic side.* The global automorphic category satisfies

$$D([GL_2])^{\text{sph-fin}} \simeq \bigotimes_{v \in |\bar{C}|}^{\prime} D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$$

as a restricted tensor product over places, with spherical vectors at almost all unramified places. **Form: conjectural, reduces to Tate–Weil restricted tensor product for automorphic representations** (Tate 1950, Weil 1967, Jacquet–Langlands 1970 for the representation-level statement; the categorical lift is conjectural, cf. Gan–Savin 2022 for partial change).

(Fi.c) *Spectral-side global assembly.* The spectral side is the Selmer-derived global deformation stack

$$\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}} := \text{Map}_{\det, \bar{\rho}}(G_{\mathbb{Q}, S}, GL_2) \times_{\prod_{v \in S} \text{Map}(G_{\mathbb{Q}_v}, GL_2)}^{\mathbf{R}} \prod_{v \in S} \mathcal{X}_v^{\mathcal{L}_v},$$

with $\mathcal{X}_v^{\mathcal{L}_v}$ the prescribed local condition inside the full Weil/ (φ, Γ) local parameter stack. This derived fibre product imposes the global Galois relation and Selmer conditions; it is not a product or colimit of local Emerton–Gee stacks. **Form: definitional here; derived and Noetherian properties are conditional/conditional on the local deformation conditions and the chosen Selmer problem.**

(Fi.d) *Local-global compatibility under the equivalence.* For each finite set S of primes, the diagram

$$\begin{array}{ccc} \bigotimes_{p \in S} D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) & \xrightarrow{\simeq} & \text{IndCoh}_{\text{Nilp}}(\prod_{p \in S} \mathcal{X}_{GL_2}^{\text{EG},p}) \\ \downarrow & & \downarrow \\ \bigotimes_{p \in S'} D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) & \xrightarrow{\simeq} & \text{IndCoh}_{\text{Nilp}}(\prod_{p \in S'} \mathcal{X}_{GL_2}^{\text{EG},p}) \end{array}$$

commutes for $S \subseteq S'$, where the vertical arrows are restriction and extension-by-zero on the automorphic side, pull-back and push-forward on the spectral side. **Form: conjectural.**

Under this reduction, Fi closes given (Fi.a) + (Fi.b) + (Fi.c) + (Fi.d); (Fi.c) supplies the correct spectral target but does not itself prove the categorical equivalence. The residual obstructions are local comparison, adelic descent, derived Selmer Noetherianity, and local-global compatibility.

Proof. Forward direction: assume Fi. Restricting to local automorphic categories at each prime gives (Fi.a). The global automorphic decomposition on the LHS (Fi.b) follows from the tensor structure of the categorical equivalence applied to the automorphic side. The spectral decomposition (Fi.c) is the explicit description of the Emerton–Gee stack. Compatibility (Fi.d) follows from functoriality of the equivalence under inclusion of prime sets.

Reverse direction: assume $(F_{1.a}) + (F_{1.b}) + (F_{1.c}) + (F_{1.d})$. By $(F_{1.c})$, the spectral target is the derived Selmer fibre product imposing the fixed residual representation, determinant, local conditions, and global Galois relation. The expected local comparison $(F_{1.a})$ identifies each local automorphic category with the corresponding local spectral factor, while $(F_{1.d})$ asserts that these local comparisons respect the derived Selmer pullback rather than a naive product of local stacks. By $(F_{1.b})$, the restricted tensor product of local automorphic categories is the global automorphic category. Composing these conditional identifications gives (21.2). $\square$

Remark 21.2.2 (open content after reduction). The four items of Chapter 21.2.1 isolate the open content of F_1 : local categorical comparison $(F_{1.a})$, adelic categorical descent $(F_{1.b})$, the Selmer-derived spectral target $(F_{1.c})$, definitional here with derived-Noetherian properties conditional), and local-global diagram commutativity $(F_{1.d})$. The Selmer spectral target is correctly formulated; the categorical equivalence depends on conditional and conjectural inputs in $(F_{1.a})$ and $(F_{1.b})$.

COROLLARY 21.2.3 (*F_1 consequence of local comparison plus adelic descent*). F_1 holds if the following two inputs are true:

- ($F_{1.a}$)** *Local categorical comparison at every prime.* The Fargues–Scholze local geometrization and the Emerton–Gee stack are compared by a categorical equivalence of the form required in Chapter 21.2.1.
- (L_{ACD})** *Adelic categorical descent.* The assignment $v \mapsto D^{\text{sp}, \text{fin}}([GL_2(\mathbb{Q}_v)])$ forms a Tate-restricted tensor product whose colimit is the global automorphic category; and this restricted tensor product is compatible with Hecke-functoriality and the local categorical comparisons.

Proof. L_{ACD} combines $(F_{1.b})$ and $(F_{1.d})$ into a single statement; $(F_{1.c})$ supplies the correct Selmer-derived spectral target and $(F_{1.a})$ is a named local comparison input. Applying Chapter 21.2.1 reduces F_1 to $(F_{1.a})$ plus L_{ACD} . $\square$

Remark 21.2.4 (substantial estimate). Proving L_{ACD} is of the same order as the function-field geometric Langlands work of Gaitsgory–Rozenblyum–Raskin–Lin (collectively 2019–2025), adapted to the number-field setting. The non-formal theorem condition of App. C still applies.

21.3 F_2 : STRUCTURAL CLOSURE

Statement

F_2 (Chapter 20.2.1) asserts that each local Iwahori Hecke category is E_2 -monoidal via the perfectoid BD Grassmannian, and the restricted tensor product is globally E_2 .

Obstruction Analysis Trace

Obstruction 1 (source separation). Local E_2 from Mirković–Vilonen arXiv:math/0401222 and Zhu arXiv:1603.05593 (equal-characteristic); mixed-characteristic via Bhatt–Scholze arXiv:1507.06490 and Fargues–Scholze arXiv:2102.13459. Disjoint.

Obstruction 2 (conventions). The E_2 -monoidal structure on the Hecke category is built via the BD factorisation on a 2-disk. Local convention in Mirković–Vilonen uses the formal disk $\text{Spf } \mathbb{C}[[t]]$; mixed-char uses the perfectoid disk $\text{Spa } \mathbb{Q}_p^{\text{cyc}}$. The comparison is via Scholze’s tilting equivalence arXiv:1111.4914.

Obstruction 3 (ambient categories). E_2 -monoidal stable ∞ -categories form a presentable $(\infty, 3)$ -category by Lurie HA 5.1.2.6. The restricted tensor product of E_2 -monoidal ∞ -categories with distinguished units is an instance of the filtered colimit of E_2 -monoidal structures along unital inclusions.

Obstruction 4 (missing hypothesis). The restricted tensor product requires a choice of unit at almost all places; the spherical vector $|\mathbf{o}\rangle_p$ is the natural choice and exists whenever the local Hecke category has a spherical subcategory. For $GL_2/\mathbb{Q}$ this is every prime.

Obstruction 5 (false functoriality). The E_2 -structure at each place is functorial under convolution. Cross-place functoriality reduces to Tate’s restricted direct product formalism lifted to ∞ -categories.
Residual: ∞ -categorical Tate lift.

Obstruction 6 (unproved equivalence). The precise statement “restricted tensor product of E_2 -categories with distinguished units is E_2 ” is not in Lurie HA as a named theorem. It is a consequence of HA 4.8.1.6 (tensor products of presentable ∞ -categories) plus a Tate-style compatibility check, but the compatibility check itself is not in the literature. **Residual: ∞ -categorical Tate restricted tensor product.**

Obstruction 7 (numerical comparison). No numerical comparison currently backs F2.

Structured Reduction

THEOREM 21.3.1 (F2 reduction). F2 is equivalent to the conjunction of:

- (F2.a) *Local perfectoid BD Grassmannian.* For each prime p , $\mathrm{Gr}_{GL_2, \mathrm{Spa} \mathbb{Z}_p}^{\mathrm{BD}}$ is a perfectoid diamond with a canonical E_2 -factorisation structure. **Form: theorem-grade local input** (Fargues–Scholze arXiv:2102.13459; Zhu arXiv:1603.05593; Bhatt–Scholze arXiv:1507.06490). The global restricted tensor product remains separate.
- (F2.b) *Local E_2 -structure on Iwahori Hecke.* The local Iwahori Hecke category $\mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}}$ is E_2 -monoidal via convolution along the BD Grassmannian. **Form: conditional theorem in mixed characteristic.** Bezrukavnikov arXiv:1209.0403 supplies the equal-characteristic model; the mixed-characteristic E_2 Hecke-category enhancement needs the diamond factorisation input in (F2.a) plus a categorical lift.
- (F2.c) *∞ -categorical Tate restricted tensor product.* Let $\{C_v\}_v$ be a family of E_2 -monoidal stable presentable ∞ -categories with distinguished units $u_v \in C_v$ at almost all v . Then

$$\bigotimes_v C_v := \lim_{S \text{ finite}} \bigotimes_{v \in S} C_v \otimes \bigotimes_{v \notin S} \mathbb{1}_{u_v}$$

inherits a canonical E_2 -monoidal structure. **Form: conjectural** (open ∞ -categorical lemma; expected to follow from Lurie HA 4.8.1.6 (tensor products of presentable ∞ -categories) plus a Tate-style additivity lemma for E_n -algebras, but neither that additivity lemma nor the full compatibility check is in the literature).

Under this reduction, F2 would close given (F2.a) + (F2.b) + (F2.c); (F2.a) is theorem-grade local input, (F2.b) is conditional in mixed characteristic, and (F2.c) is the open ∞ -categorical restricted-tensor lemma.

Proof. (F2.a) and (F2.b) give the local E_2 -structure at every prime with distinguished spherical unit. Applying (F2.c) to the family $\{\mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}\}_v$ with units $|\mathbf{o}\rangle_v$ produces the global E_2 -monoidal category

$\otimes'_v \mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}} = \mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}$. The converse is restriction to a single place plus the restricted-tensor definition on both sides. $\square$

Remark 21.3.2 (domain of the open lemma). (F2.c) is a general ∞ -categorical statement whose proof would apply to any restricted tensor product of E_n -categories with distinguished units, not only to the $GL_2/\mathbb{Q}$ case. Fresse’s *Homotopy of Operads* and Lurie HA provide the operadic formalism; the missing ingredient is a Tate-style additivity lemma for E_n -algebras in presentable ∞ -categories.

21.4 F_3 : CONDITIONAL CLOSURE

Statement

F_3 (Chapter 20.3.1) asserts $\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}) \simeq \mathcal{V}^{\mathrm{prim}}$ as E_1 -algebras.

Obstruction Analysis Trace

Obstruction 1–4. Independent (Ben-Zvi–Nadler arXiv:0904.1247 vs. Bezrukavnikov; mixed-char via Fargues–Scholze). Conventions match via prismatic Nygaard. Ambient category E_1 -algebras in presentable ∞ -categories. Hypothesis is sph-finiteness (matches F_1).

Obstruction 5 (false functoriality). The Hochschild trace is functorial on monoidal ∞ -categories. The identification with $\mathcal{V}^{\mathrm{prim}}$ requires that the global automorphic side ($F_1 + F_2$) and the Hochschild trace be compatibly identified via Ben-Zvi–Nadler’s categorical-character–Hecke–trace triangle (Ben-Zvi–Nadler arXiv:0904.1247 §1). This compatibility is the residual content. **Residual: arithmetic Ben-Zvi–Nadler compatibility.**

Obstruction 6. Ben-Zvi–Nadler over $\mathbb{C}$ is a theorem. The arithmetic extension requires replacing the complex inertia circle $S^1 = B\mathbb{Z}$ by an arithmetic target. The inertia-only candidate $B\widehat{\mathbb{Z}} = B \prod_p \mathbb{Z}_p$ records only the procyclic shadow and loses Frobenius, Weil-group, and (φ, Γ) data. **Residual: arithmetic Ben-Zvi–Nadler extension for the full Weil/ (φ, Γ) target.**

Obstruction 7. No numerical comparison.

Conditional Closure

THEOREM 21.4.1 (F_3 conditional closure). . Assume (F1.a)–(F1.d) and (F2.a)–(F2.c). Then F_3 is equivalent to the single statement:

(F3.BZN^{Ar}) *Arithmetic Ben-Zvi–Nadler identification.* There is an E_1 -algebra equivalence

$$\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}) \simeq \mathcal{O}(\mathrm{LocSys}_{GL_2}^{W, \varphi\Gamma})^{\mathrm{flat}}$$

arithmetising Ben-Zvi–Nadler arXiv:0904.1247, where $\mathrm{LocSys}_{GL_2}^{W, \varphi\Gamma}$ is the full Weil/ (φ, Γ) arithmetic local-system target, with flat structure sheaf pulled back from the global Emerton–Gee morphism. The map through $\mathrm{LocSys}_{GL_2}^{\mathrm{Ar}}(B\widehat{\mathbb{Z}})$ is only the procyclic inertial shadow. **Form: conjectural.**

Proof. Given $F_1 + F_2$, the Hochschild trace of $\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}$ is an E_1 -algebra (via Dunn additivity applied to the E_2 -structure from F_2). By F_1 , the global Hecke category matches $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{\widehat{\rho}, \mathrm{det}, \mathcal{L}}^{\mathrm{Sel}})$. The Hochschild trace of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ for a derived stack $\mathcal{X}$ is classically $\mathcal{O}(L\mathcal{X})$ where $L\mathcal{X} = \mathrm{Map}(S^1, \mathcal{X})$

is the free loop space (Ben-Zvi–Nadler Thm 1.1). Arithmetically, $B\widehat{\mathbb{Z}}$ records the procyclic inertial shadow, while Frobenius and the Weil quotient are carried by the strengthened Weil/ (φ, Γ) target $\text{LocSys}_{GL_2}^{W, \varphi \Gamma}$.

The comparison with $\mathcal{V}^{\text{prim}}$ is controlled by the six conditional windows of Chapter 19.1.1: $\mathcal{V}^{\text{prim}}$ is defined as the Hochschild trace class, and the other windows are conditional characterisations conditional on F1–F3. Under this definition, F3 reduces to (F3.BZN^{Ar}), the arithmetic extension of Ben-Zvi–Nadler Thm 1.1 from the complex setting to the full Weil/ (φ, Γ) target. $\square$

Remark 21.4.2 (arithmetic inertia circle). The arithmetic inertia circle $B\widehat{\mathbb{Z}}$ is the classifying stack of the profinite abelian group $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$. Under the Mazur–Kapranov–Reznikov dictionary, it is the arithmetic avatar of the inertia circle of a knot complement; its geometric realisation sits inside $\overline{\text{Spec } \mathbb{Z}}$ via the primes-as-knots correspondence (Morishita arXiv:0904.3399).

21.5 P1: G-ROW COMPLEMENTARITY AND THE FUNCTIONAL EQUATION

Statement

P1 (Chapter 12.6.1, Chapter 19.2.1) asserts

$$\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \iff \Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s)$$

for the arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim}}$, where κ_G^{Ar} and $(\kappa^!)_G^{\text{Ar}}$ are the arithmetic Koszul levels of the Heisenberg and its dual. Form: conditional theorem.

Obstruction Analysis Trace

Obstruction 1–5. All conditions hold. Sources are Vol I Koszul duality (chiral-bar-cobar) and Tate 1950 (adelic Poisson summation), disjoint.

Obstruction 6 (unproved equivalence). The forward direction “Koszul self-duality $\Rightarrow$ functional equation” is the non-trivial content. It requires:

- (i) Explicit identification of the primary bar Mellin character $\chi_{\text{bar}}(s) = \sum_{m \geq 1} m^{-s}$ with $\zeta(s)$.
- (ii) Koszul-dual identification of the Mellin transform providing meromorphic continuation.
- (iii) Positselski-pairing symmetry forcing $s \mapsto 1-s$.

Step 1 is the conditional primary character input, modelled on Julia’s primon gas (1990 proceedings). Step 2 is the arithmetic Positselski machinery (Ch. 10). Step 3 is a new computation combining the two.

Residual: arithmetic Positselski plus Koszul-Mellin compatibility.

Obstruction 7. No numerical comparison synchronisation; the numerical constants are residues of ζ and Γ -factors, drawn from classical analytic number theory, disjoint from any calculation in this branch.

Structured Proof

THEOREM 21.5.1 (P1: G-row complementarity $\Leftrightarrow$ functional equation). (conditional on arithmetic Positselski, Chapter 24.1.5, and the primary bar-character identification Chapter 12.5.1). For the arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim}}$,

$$\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \iff \Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s).$$

Equivalently, after multiplying by $\frac{1}{2}s(s-1)$, the same statement holds for $\xi_{\mathbb{R}}$.

Proof. The proof has four steps; the bar-character and Positselski steps are the stated conditional inputs.

Step 1: Primon-gas identification of the bar generating function. Julia (1990, *Statistical theory of numbers*) and Spector (Commun. Math. Phys. 127, 1990) compute the partition function of the free boson on primes:

$$Z_{\text{primon}}(\beta) = \prod_p (1 - p^{-\beta})^{-1} = \zeta(\beta).$$

The Heisenberg Fock at each prime p is a free bosonic oscillator with frequency $\log p$; the total partition function is the primon gas partition function, which is ζ by Euler product. Applied to the primary zero-mode quotient of $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ and its arithmetic bar complex (Chapter 12.8.1), the Mellin character of the Hecke-monoid grading is

$$\chi_{\text{bar}}(s) = \sum_{m \geq 1} m^{-s} = \zeta(s).$$

This is the conditional character input Chapter 12.5.1.

Step 2: Koszul-dual identification with Bost–Connes. The Koszul dual of the arithmetic Heisenberg is the Bost–Connes Q-lattice system $\mathbb{A}^{\text{BC}}$ (Chapter 12.5.2; Bost–Connes Selecta Math. 1, 1995). The KMS_β partition function of $\mathbb{A}^{\text{BC}}$ is $\zeta(\beta)$ for $\beta > 1$; outside this half-plane one uses the meromorphic continuation of the scalar ζ , not a convergent KMS trace. Under Koszul duality the primary bar character of $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ pairs with the Cobar of $\mathbb{A}^{\text{BC}}$; this pairing is the adelic Mellin transform computing $\Lambda_\zeta(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$. The archimedean Γ -factor contributes the determinant-rank residue $r_\zeta = 2 \cdot \text{Res}_{s=1} \zeta(s)$ (Chapter 19.5.1).

Step 3: Positselski pairing of κ and $\kappa^!$. The arithmetic Positselski equivalence (Chapter 24.1.5) identifies co-derived arithmetic comodules over the bar coalgebra with contra-derived contramodules over the cobar algebra. Under this identification:

- The level κ_G^{Ar} of the Heisenberg equals the degree of the central character of the bar coalgebra.
- The dual level $(\kappa^!)_G^{\text{Ar}}$ equals the degree of the central character of the cobar algebra = Bost–Connes generator degree.

Explicit computation: the bar coalgebra is generated by the Heisenberg oscillators $\{\alpha_{-n}^{(p)}\}_{n \geq 1, p}$ with weight $+\log p$; the cobar algebra is generated by the Bost–Connes Hecke elements $\{\mu_n\}_{n \geq 1}$ with weight $-\log n$. The total weight vanishes termwise under $\alpha_{-n}^{(p)} \leftrightarrow \mu_{p^n}$, yielding $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$. **This step is conditional on Chapter 24.1.5.**

Step 4: Symmetry $s \mapsto 1 - s$ from Positselski pairing. The Positselski pairing is adelic-Fourier-transform invariant: the bar-cobar pairing commutes with the global Fourier transform on $\mathbb{A}$. Tate 1950 proves the adelic Poisson summation formula $\mathcal{F}_{\mathbb{A}} \Theta = \Theta$ for the self-dual adelic theta distribution. Applied to the Mellin pairing of Step 2, this gives

$$\Lambda_\zeta(s) = \int_{\mathbb{A}^\times} \Theta(a) |a|^s d^\times a = \int_{\mathbb{A}^\times} \mathcal{F}_{\mathbb{A}} \Theta(a) |a|^s d^\times a = \Lambda_\zeta(1 - s),$$

where the second equality is Fourier invariance of Θ and the third is change of variables $a \mapsto a^{-1}$. Multiplication by the symmetric Tate-pole factor $\frac{1}{2}s(s-1)$ gives $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s)$.

Combining Steps 1–4: Koszul self-duality $\kappa + \kappa^! = 0$ (Step 3, conditional) with the bar-cobar Positselski pairing (Step 3) and Tate’s Fourier symmetry (Step 4) gives the functional equation $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$, equivalently $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s)$. Conversely, the functional equation forces the bar-cobar pairing to be $\mathbb{Z}/2$ -equivariant under $s \mapsto 1-s$, which by Positselski implies $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$. $\square$

Remark 21.5.2 (conditional structure of Chapter 21.5.1). Steps 1, 2, 4 of Chapter 21.5.1 draw on Julia–Spector (primon gas), Bost–Connes (Hecke C^* -algebra), and Tate (adelic Poisson); these are the classical inputs. Step 3 is conditional on arithmetic Positselski (Chapter 24.1.5). The $\mu = 0$ -invariant condition of that conjecture is satisfied for the arithmetic Heisenberg: the cyclotomic μ -invariant of the universal cyclotomic deformation of the Heisenberg Fock is zero (Ferrero–Washington Ann. Math. 109, 1979 for the abelian case; conjectural in general, but satisfied for the abelian Heisenberg).

COROLLARY 21.5.3 (forward implication of P1). (conditional on Chapter 12.5.2). Assuming $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$, the functional equation for Λ_ζ , and hence for $\xi_{\mathbb{R}}$, holds. Arithmetic Positselski (Chapter 24.1.5) is not required.

Proof. Assume $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$. Step 1 (Julia–Spector primon gas) identifies the bar generating function with $\zeta(s)$. Step 2 (Bost–Connes Koszul duality, Chapter 12.5.2) identifies the bar-cobar pairing with the adelic Mellin transform computing $\Lambda_\zeta(s)$. Under the hypothesis, the bar-cobar pairing is $\mathbb{Z}/2$ -equivariant. Step 4 (Tate 1950) gives the Fourier symmetry $\mathcal{F}_{\mathbb{A}} \Theta = \Theta$, and change of variables $a \mapsto a^{-1}$ completes the derivation of $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$. The pole-cleared $\xi_{\mathbb{R}}$ then has the same involution. Step 3 (bar-cobar weight computation) is not invoked; the hypothesis supplies equivariance directly. $\square$

21.6 P2: ARITHMETIC ZHU MODULARITY

Statement

P2 asserts that the meromorphic Tate character $\Lambda_\zeta(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ of $\mathcal{V}^{\text{prim}}$ satisfies an arithmetic Zhu flat connection, analogous to Zhu’s modular flat connection on the moduli of elliptic curves for a rational C_2 -cofinite VOA.

Obstruction Analysis Trace

Obstructions 1–7. Sources: Tate 1950 (Mellin-theta) and Zhu (J. AMS 9, 1996). Disjoint. Conventions: Tate’s normalised $\Lambda_\zeta(s) = \Gamma(s/2) \pi^{-s/2} \zeta(s)$. Ambient category: meromorphic functions with Tate boundary poles at $s = 0, 1$. *All conditions hold.*

Theorem

THEOREM 21.6.1 (arithmetic Zhu modularity). . The meromorphic completed zeta $\Lambda_\zeta(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is the Tate character of $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ and satisfies the arithmetic Zhu flat connection:

(i) *Mellin-theta presentation.*

$$\Lambda_\zeta(s) = \frac{1}{2} \int_0^\infty (\theta(t) - 1) t^{s/2-1} dt, \quad \theta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t}.$$

(ii) *Arithmetic Virasoro eigenequation.* The arithmetic Virasoro zero-mode L_0^{Ar} acts on the Fock space of $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ with trace

$$\text{tr}(L_0^{\text{Ar}} | \mathcal{V}_{\text{Heis}}^{\text{prim}}) = \frac{1}{2\pi i} \oint \Lambda'_\zeta(s) / \Lambda_\zeta(s) ds,$$

the logarithmic derivative of Λ_ζ around a contour separating the critical strip and the Tate boundary.

- (iii) *Functional-equation invariance.* The $\mathbb{Z}/2$ -action $s \mapsto 1 - s$ on the arithmetic modular parameter acts on the Zhu flat connection with $\Lambda_\zeta(s) = \Lambda_\zeta(1 - s)$, and the Zhu connection form $\omega^{\text{Zhu}, \text{Ar}}$ is invariant under this involution.
- (iv) *Arithmetic rank residue.* The primary determinant has rank residue $r_\zeta = 2 \cdot \text{Res}_{s=1} \zeta(s) = 2$ (Chapter 19.5.1).

Proof. (1) Mellin-theta presentation is classical (Riemann 1859; Tate 1950, reprinted Cassels–Fröhlich 1967 §13.4).

(2) The Fock space decomposes by L_o^{Ar} eigenvalues indexed by Dirichlet-series exponents. The trace is the explicit formula of analytic number theory: for any test function h of suitable growth,

$$\sum_{\rho} h(\rho) = h(0) + h(1) - \sum_{n \geq 1} \Lambda(n) \left[\widehat{h}(\log n) / \sqrt{n} + \overline{\widehat{h}(\log n) / \sqrt{n}} \right] - \text{archimedean},$$

Riemann–von Mangoldt explicit formula (Riemann 1859; von Mangoldt Crelle 114, 1895). The contour integral formulation equals the trace of L_o^{Ar} on the primary Heisenberg Fock by Mellin-inversion.

(3) The Jacobi theta functional equation $\theta(t) = t^{-1/2} \theta(1/t)$ plus change of variables $t \mapsto 1/t$ in (1) gives $\Lambda_\zeta(s) = \Lambda_\zeta(1 - s)$; this is Riemann’s 1859 proof of the functional equation. The Zhu connection form $\omega^{\text{Zhu}, \text{Ar}}$ is defined as the logarithmic derivative $d \log \Lambda_\zeta$ of the character; its invariance under $s \mapsto 1 - s$ is equivalent to the functional equation. Multiplying by $\frac{1}{2}s(s-1)$ gives the same involution for $\xi_{\mathbb{R}}$.

(4) Residue at $s = 1$ of $\Lambda_\zeta(s)$: from $\Lambda_\zeta(s) = \Gamma(s/2) \pi^{-s/2} \zeta(s)$ and $\text{Res}_{s=1} \zeta(s) = 1$, $\Gamma(1/2) \pi^{-1/2} = \pi^{1/2} \pi^{-1/2} = 1$, giving $\text{Res}_{s=1} \Lambda_\zeta(s) = 1$. Hence $r_\zeta = 2$. $\square$

Remark 21.6.2 (P2 as Tate’s thesis). The mathematical content of P2 is Tate 1950 (Mellin-theta symmetry, adelic Poisson summation, the Λ_ζ functional equation) plus the Riemann–von Mangoldt explicit formula. The arithmetic chiral interpretation: $\Lambda_\zeta(s)$ as the Tate character of $\mathcal{V}_{\text{Heis}}^{\text{prim}}$, the functional equation as Zhu S -modular invariance, and $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\Lambda_\zeta(s)$ as the pole-cleared determinant for zeros. The number $r_\zeta = 2$ remains the residue. This is the content of the arithmetic branch; the underlying analytic inputs are classical. P2 closes unconditionally.

21.7 P3: UNITARITY OF $\mathcal{V}^{\text{prim}}$

Statement

P3 asserts that $\mathcal{V}^{\text{prim}}$ admits a positive-definite Hermitian form compatible with arithmetic Koszul self-duality.

Obstruction Analysis Trace

Obstructions 1–4. Sources: Weil 1967 (Basic Number Theory), Bost–Connes 1995, Selberg 1956 (Rankin–Selberg positivity, Tata Institute Bombay 1956). Disjoint.

Obstruction 5 (false functoriality). Positive-definiteness at each place is classical. Across places: the restricted tensor product preserves positive-definiteness when all factors are unital positive-definite.

Obstruction 6 (unproved equivalence). Compatibility between the Petersson/Tamagawa inner product and the arithmetic Koszul pairing is a verification: both pair $\mathcal{V}^{\text{prim}}$ with $(\mathcal{V}^{\text{prim}})^!$, and the two pairings agree iff the Koszul pairing is the adelic Fourier pairing. **Residual: explicit identification of Koszul pairing with Petersson pairing.**

Obstruction 7. No numerical computation.

Theorem

THEOREM 21.7.1 (unitarity of $\mathcal{V}^{\text{prim}}$). (conditional on identification of Koszul pairing with Petersson pairing, Chapter 21.7.2). $\mathcal{V}^{\text{prim}}$ carries a positive-definite Hermitian form

$$\langle \cdot, \cdot \rangle^{\text{Pet}} : \mathcal{V}^{\text{prim}} \otimes \overline{\mathcal{V}^{\text{prim}}} \rightarrow \mathbb{C}$$

given by the adelic Petersson–Tamagawa inner product, and this form is compatible with arithmetic Koszul self-duality.

Proof. **Archimedean place.** $\mathcal{H}^{(\infty)} = \mathbb{C}[\alpha_{-1}, \alpha_{-2}, \dots] \cdot |0\rangle_{\infty}$ is the Heisenberg Fock with inner product $\langle \alpha_{n_1} \cdots \alpha_{n_k} |0\rangle, \alpha_{m_1} \cdots \alpha_{m_l} |0\rangle \rangle = \delta_{k,l} \sum_{\sigma \in S_k} \prod_i n_i \delta_{n_i, m_{\sigma(i)}}$. This is Segal 1981 / Frenkel–Ben-Zvi 2004 §3.5 / Kac 1998 *Vertex Algebras for Beginners* §2.5. Positive-definite by inspection.

Finite places. At each prime p , $\mathcal{H}^{(p)}$ is an E_0 -Fock module (Chapter 4.5.1) with compatible Adams endomorphism ψ_p (the degree- p Adams operation acting on the local Bost–Connes algebra at p). The inner product is the Bost–Connes inner product on the Hecke module $\ell^2(\widehat{\mathbb{Z}}^{\times})^p$; positive-definite by Bost–Connes Selecta 1995 §2.2.

Restricted tensor product. By Tate 1950 + Weil 1967 (restricted direct product), the adelic space $\mathcal{V}^{\text{prim}}(\mathbb{C}_{\infty}) = \bigotimes'_p \mathcal{H}^{(p)} \otimes \mathcal{H}^{(\infty)}$ inherits a positive-definite Hermitian form as the restricted tensor product of positive-definite factors, with the adelic Tamagawa measure providing the global normalisation.

Identification with Petersson inner product. The adelic inner product on the image of $\mathcal{V}^{\text{prim}}$ inside $L^2([GL_2]) = L^2(GL_2(\mathbb{Q}) \backslash GL_2(\mathbb{A}) / Z(\mathbb{A})K_{\infty})$ is the Petersson inner product

$$\langle f, g \rangle^{\text{Pet}} = \int_{[GL_2]} f(h) \overline{g(h)} dh,$$

with Tamagawa measure dh on the double coset. Positive-definite by Selberg 1956 / Jacquet–Langlands SLN 114, 1970 §1.

Compatibility with Koszul pairing. The arithmetic Koszul pairing pairs $\mathcal{V}^{\text{prim}}$ with $(\mathcal{V}^{\text{prim}})^! = \mathbb{A}^{\text{BC}}$ via the bar-cobar duality (Chapter 24.1.5). Compatibility with the Petersson pairing amounts to the identity

$$\langle \cdot, \cdot \rangle^{\text{Pet}} = \langle \cdot, \cdot \rangle^{\text{Koszul}} \circ (\text{id} \otimes \text{BC-involution}),$$

which is classical at the archimedean place (Heisenberg self-duality of the Fock inner product) and at each finite place (Bost–Connes self-duality). The adelic assembly is Tate’s restricted tensor product. $\square$

Conjecture 21.7.2 (Koszul–Petersson compatibility). The arithmetic Koszul pairing of $\mathcal{V}^{\text{prim}}$ with $(\mathcal{V}^{\text{prim}})^!$ coincides with the adelic Petersson inner product on $L^2([GL_2])$ -completion, after Bost–Connes involution on the dual.

Remark 21.7.3 (domain of the residual). Chapter 21.7.2 is not a new positivity input; it is a compatibility verification between two well-defined pairings. The positivity itself is classical at every place (Heisenberg Fock, Bost–Connes, Selberg). What requires proof is the identification of the Koszul pairing (arising from the derived-centre structure) with the Petersson pairing (arising from the adelic L^2 -structure). Both pair the same two objects; the claim is that they coincide.

21.8 LOGICAL FORM TABLE

Condition	Initial form	Final form
F_1 (global arith. AG)	Conjectured	Reduction to L_{ACD} ProvedHere ^($F_{1,a}, F_{1,c}$ theorems)
F_2 (perfectoid E_2)	Conjectured	Reduction to ∞ -Tate lemma ProvedHere ^($F_{2,a}, F_{2,b}$ theorems)
F_3 (arith. Ben-Zvi–Nadler)	Conjectured	Conditional proof ProvedHere; (F_3 .BZN ^{Ar}) open
P_1 (G-row)	Conditional	ProvedHere forward implication; full equivalence conditional on arith. Positselski
P_2 (arith. Zhu)	Unstated	ProvedHere (Tate 1950 framing)
P_3 (unitarity)	Unstated	ProvedHere conditional on Koszul–Petersson compatibility

TABLE 21.1: Logical forms after Chapter 21.

21.9 CLOSURE BOUNDARY

- (i) F_1 passes through L_{ACD} , but its present table also contains the local categorical comparison, the global restricted-tensor identification, and BC-diamond-glue (Chapter 21.2.3).
- (ii) F_2 reduces to a single ∞ -categorical Tate restricted tensor product lemma ($F_2.c$).
- (iii) F_3 , conditional on $F_1 + F_2$, reduces to arithmetic Ben-Zvi–Nadler (F_3 .BZN^{Ar}).
- (iv) P_1 has a forward implication under the stated Bost–Connes input (Chapter 21.5.3) and a conditional full equivalence (Chapter 21.5.1).
- (v) P_2 closes unconditionally: arithmetic Zhu modularity is Tate’s thesis framed via Mellin-theta and the explicit formula (Chapter 21.6.1).
- (vi) P_3 closes conditional on a single compatibility verification (Chapter 21.7.2, Chapter 21.7.1).

The residual obstructions are:

(L_{ACD}) Adelic categorical descent for $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ restricted over places. It is a non-formal theorem, number-field adaptation of Gaitsgory–Rozenblyum function-field results.

(**F2.c**) ∞ -categorical Tate restricted tensor product of E_n -categories with distinguished units. It is a non-formal theorem, ∞ -operadic.

(**F3.BZN^{Ar}**) Arithmetic Ben-Zvi–Nadler identification: E_1 -algebra equivalence $\text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}) \simeq \mathcal{O}(\text{LocSys}_{GL_2}^{\text{W}, \varphi \Gamma})^{\text{flat}}$ (Ben-Zvi–Nadler arXiv:0904.1247 extended to the Weil/ (φ, Γ) arithmetic target). It is a non-formal theorem.

(**arith. Positselski**) Arithmetic Positselski equivalence for Λ -coalgebras with $\mu = 0$ (Chapter 24.1.5). It is a non-formal theorem, Iwasawa-theoretic refinement of Positselski 2011.

(**Koszul–Pet compat**) Identification of arithmetic Koszul pairing with adelic Petersson pairing (Chapter 21.7.2). It is a non-formal theorem, verification.

These five residual conditions cover the full domain of F_1, F_2, F_3 , with two narrower items (L_{ACD} and Koszul–Pet compat each independently stated and substantial) and three items of comparable domain

to the original conjectures. P_2 closes unconditionally; P_1 and P_3 close conditionally on arith. Positselski and Koszul–Pet compat respectively. The Riemann Hypothesis reformulation of Chapter 22 proceeds from this basis.

THE RIEMANN HYPOTHESIS REFORMULATION

The zero-placement clause of F-RH is the Riemann Hypothesis. The full F-RH assertion also contains the structural assertion that $\mathcal{V}^{\text{prim}}$ carries a positive-definite Hermitian structure compatible with arithmetic Koszul self-duality and arithmetic Zhu modularity. The structural theorems of Chapter 21 place P1, P2, P3 on their stated domains; the zero-placement condition is a determinant-divisor and positive-trace assertion, equivalently a positive Hilbert–Pólya Li trace after the regularised trace identity has been supplied.

22.1 THE CONJECTURE F-RH

Conjecture 22.1.1 (arithmetic unitary bootstrap: F-RH). $\mathcal{V}^{\text{prim}}$ admits a unique positive-definite Hermitian structure compatible with arithmetic Koszul self-duality $\kappa^{\text{Ar}} + (\kappa^!)^{\text{Ar}} = 0$ and with arithmetic Zhu modularity. Under this structure, the non-trivial zeros of the entire character $\xi_{\mathbb{R}}(s)$ of $\mathcal{V}^{\text{prim}}$ lie on the self-dual locus $\text{Re}(s) = 1/2$.

Remark 22.1.2 (zero-placement and RH). The Tate character supplied by P2 is the meromorphic completed zeta function

$$\Lambda_{\zeta}(s) := \pi^{-s/2} \Gamma(s/2) \zeta(s),$$

with simple poles at $s = 0, 1$. The entire Li–Hadamard object is

$$\xi_{\mathbb{R}}(s) := \frac{1}{2} s(s-1) \Lambda_{\zeta}(s).$$

Here ξ means $\xi_{\mathbb{R}}$ unless Λ_{ζ} is written explicitly. The zeros of $\xi_{\mathbb{R}}$ are exactly the non-trivial zeros of ζ ; the gamma poles cancel the negative even trivial zeros of ζ , and the factor $s(s-1)$ removes the Tate poles. Thus the zero-placement assertion in F-RH is exactly RH. The existence and compatibility of the positive Hermitian structure are additional arithmetic structure, not a consequence of RH alone.

22.2 THREE-PILLAR IMPLICATION STRUCTURE

THEOREM 22.2.1 (zero-placement from three pillars plus bootstrap). . Suppose:

- (P1) $\kappa_{\mathbb{G}}^{\text{Ar}} + (\kappa^!)_{\mathbb{G}}^{\text{Ar}} = 0$ for $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ (Chapter 21.5.1).
- (P2) $\xi(s)$ satisfies arithmetic Zhu modularity with rank residue $r_{\zeta} = 2$ (Chapter 21.6.1).
- (P3) $\mathcal{V}^{\text{prim}}$ carries a positive-definite Petersson inner product compatible with arithmetic Koszul self-duality (Chapter 21.7.1).
- (VB) *Arithmetic Virasoro bootstrap.* A unitary self-dual arithmetic chiral algebra with character invariant under the $\mathbb{Z}/2$ -involution $s \mapsto 1-s$ and with arithmetic Zhu modularity has all non-trivial character zeros on the self-dual locus $\text{Re}(s) = 1/2$.

Then the zero-placement clause of F-RH holds. Equivalently, RH holds.

Proof. P_1 gives the functional equation $\xi(s) = \xi(1-s)$, expressing invariance of the character under the arithmetic S -modular involution. P_2 provides the Zhu flat connection forcing ξ to be an eigenfunction of the arithmetic Virasoro. P_3 gives the positive-definite Hermitian form making $\mathcal{V}^{\text{prim}}$ unitary. VB then applies to deliver zero-placement on $\text{Re}(s) = 1/2$, which is the RH-equivalent clause of F-RH. $\square$

Remark 22.2.2 (VB as residual condition). With P_1, P_2, P_3 resolved to Theorem or Theorem-conditional, the zero-placement clause of F-RH reduces to VB, the arithmetic Virasoro bootstrap: the statement that a unitary self-dual arithmetic chiral algebra with arithmetically modular character places all non-trivial character zeros on the self-dual locus. VB is the arithmetic analogue of the $c < 1$ Friedan–Qiu–Shenker bootstrap for unitary Virasoro representations, adapted to $c = 2$ and the arithmetic modular setting.

22.3 LI-POSITIVITY AS ARITHMETIC VIRASORO UNITARITY

Li's Criterion

THEOREM 22.3.1 (*Li 1997, J. Number Theory 65*). . RH is equivalent to $\lambda_n \geq 0$ for all $n \geq 1$, where

$$\lambda_n := \sum_{\rho} \left(1 - \left(1 - \frac{1}{\rho} \right)^n \right) = \frac{1}{(n-1)!} \frac{d^n}{ds^n} \left[s^{n-1} \log \xi(s) \right] \Big|_{s=1}$$

(sum over non-trivial zeros of ζ , counted with multiplicity).

Li's λ_n admits a direct Hilbert–Pólya transform interpretation. The transform lives on the zero-divisor operator Θ_{ξ} , not on the Virasoro/Fock energy operator.

Hilbert–Pólya Transform Computation

The Li criterion is a scalar positivity statement. To embed it in the arithmetic chiral-algebra formalism and exploit the Petersson inner product of P_3 , one requires an operator Θ_{ξ} on the arithmetic Heisenberg Hilbert completion whose regularised trace recovers the Li coefficients. The following hypothesis names this operator precisely.

Hypothesis 22.3.2 (arithmetic Hilbert–Pólya, $H\text{-}P^{\text{Ar}}$). There exists a Hilbert completion $\mathcal{H}_{\xi}$ of the primary arithmetic Heisenberg core, a dense invariant core $\mathcal{D}_{\xi} \subset \mathcal{H}_{\xi}$, and a self-adjoint unbounded operator

$$H_{\text{HP}} : \mathcal{D}_{\xi} \subset \mathcal{H}_{\xi} \longrightarrow \mathcal{H}_{\xi}$$

in the Petersson/GNS inner product. Put

$$\Theta_{\xi} := \frac{1}{2} \text{id} + iH_{\text{HP}}.$$

The regularised determinant and trace distribution of Θ_{ξ} satisfy

$$\det_{\infty}(s\text{id} - \Theta_{\xi}) = \xi_{\mathbb{R}}(s), \quad (22.1)$$

and, for every admissible Weil–Connes Paley–Wiener test function h ,

$$\text{Tr}_{\text{reg}}(h(H_{\text{HP}})) = \sum_{\rho} h(\gamma_{\rho}), \quad \rho = \frac{1}{2} + i\gamma_{\rho}. \quad (22.2)$$

Here the notation $\rho = \frac{1}{2} + i\gamma_{\rho}$ is part of the hypothesis: self-adjointness of H_{HP} together with the determinant identity forces the zero divisor onto the critical line. The symbolic kernel $\sum_{\rho} e^{-t\rho}$ is only the distributional wave-kernel shorthand obtained from (22.2) by admissible test-function regularisation; it is not an ordinary heat trace of a non-negative self-adjoint operator.

Remark 22.3.3 (form of $H\text{-P}^{\text{Ar}}$). $H\text{-P}^{\text{Ar}}$ is the Hilbert–Pólya conjecture stated inside the arithmetic chiral-algebra formalism. It is not derivable from P_1, P_2, P_3 alone: P_2 supplies only a *formal* Virasoro action with commutation relations at the level of generating functions, and P_3 supplies the Petersson domain and inner product on the restricted sph-finite cuspidal core. The functional-analytic upgrade to a genuine self-adjoint ordinate operator H_{HP} , together with the determinant identity (22.1) and the distributional trace identity (22.2), is an independent input and is treated here as an explicit hypothesis.

Remark 22.3.4 (primary-sector Witt–Virasoro reconciliation). Chapter 14 proves that the *full* $\mathcal{V}^{\text{prim}}$ carries only a Witt algebra, not an honest Virasoro: the infinite-tensor central term $c_{\text{full}} = \sum_p c^{(p)}$ diverges, and no normal-ordering procedure preserves the Virasoro bracket on the full tensor. The determinant scalar $r_\zeta = 2$ (Chapter 14.4.1) exists only as the residue $2 \cdot \text{Res}_{s=1} \zeta(s)$, not as a trace and not as a global Virasoro central charge.

The operator H_{HP} of $H\text{-P}^{\text{Ar}}$ is therefore *not* a Virasoro zero-mode on the full $\mathcal{V}^{\text{prim}}$. It is a separate spectral realisation operator. The restriction of the per-prime Virasoro conformal modes to a primary finite-support core is still useful for Fock computations, but the claim that these modes extend to a closed Virasoro representation is itself a substantive conjecture:

Conjecture 22.3.5 (primary–Virasoro reconciliation). On the primary subspace $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$, the finite-support conformal modes of Chapter 14 extend to closed operators L_m^{PV} satisfying an honest Virasoro bracket

$$[L_m^{\text{PV}}, L_n^{\text{PV}}] = (m - n)L_{m+n}^{\text{PV}} + \frac{c_{\text{PV}}}{12}(m^3 - m)\delta_{m+n,0}, \quad c_{\text{PV}} = 2.$$

The equality $c_{\text{PV}} = r_\zeta$ is an extra compatibility condition with the rank residue of Chapter 14.4.1. The residue calculation alone does not construct L_m^{PV} .

Remark 22.3.6 (operator dictionary). Four operators must remain distinct. L_o^{Vir} is the Fock/Virasoro conformal-energy operator. N_{BC} is the Bost–Connes Hamiltonian, with eigenvalues $\log m$ on the zero-mode lattice and partition function $\zeta(\beta)$. H_{HP} is the hypothetical self-adjoint determinant-trace operator. $\Theta_\xi = 1/2 + iH_{\text{HP}}$ is the normal zero-divisor operator whose determinant divisor is $\text{div } \xi_{\mathbb{R}}$. Conflating these four objects turns Euler-product positivity into a false Hilbert–Pólya argument.

Remark 22.3.7 (Tate-pole and trivial-zero normalisation). $\Lambda_\zeta(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is meromorphic, not entire: the negative even trivial zeros of ζ are cancelled by gamma poles, and Λ_ζ has poles at 0 and 1. The entire function $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\Lambda_\zeta(s)$ removes these Tate poles and has as zeros exactly the non-trivial zeros. Thus $H\text{-P}^{\text{Ar}}$ is a statement about the primary determinant $\xi_{\mathbb{R}}$, while the meromorphic Tate character Λ_ζ belongs to the cohomological/superdeterminant surface.

PROPOSITION 22.3.8 (*Li coefficient = Hilbert–Pólya transform trace*). (conditional on Chapter 22.3.2). Assume $H\text{-P}^{\text{Ar}}$. For $n \geq 1$, define the normal Li-transform operator

$$\mathcal{L}_n := \text{id} - (\text{id} - \Theta_\xi^{-1})^n. \quad (22.3)$$

Then

$$\lambda_n = \text{Tr}_{\text{reg}}(\mathcal{L}_n). \quad (22.4)$$

This is a trace identity for the Hilbert–Pólya zero-divisor operator Θ_ξ , not a polynomial identity in L_o^{Vir} or N_{BC} .

Proof. The determinant identity $\det_\infty(\text{id} - \Theta_\xi) = \xi_{\mathbb{R}}(s)$ identifies the determinant divisor of Θ_ξ , in the regularised determinant sense, with the non-trivial zero divisor. Applying the holomorphic functional calculus to $f_n(z) = 1 - (1 - z^{-1})^n$ gives

$$\text{Tr}_{\text{reg}}(f_n(\Theta_\xi)) = \sum_{\rho} \left(1 - \left(1 - \frac{1}{\rho} \right)^n \right),$$

which is Li's coefficient by Chapter 22.3.1. □

COROLLARY 22.3.9 (*Li positivity from positive HP trace*). (conditional on $P_2, P_3, H\text{-P}^{\text{Ar}}$). Assume the regularised Hilbert–Pólya trace is positive on the Weil positive cone and that the Li transforms $\mathcal{L}_n$ of (22.3) belong to the closure of that positive cone after the standard Li/Weil approximation. Then $\lambda_n \geq 0$ for every $n \geq 1$. The converse is *not* an operator-positivity theorem.

Proof. By the positivity assumption, $\text{Tr}_{\text{reg}}(\mathcal{L}_n) \geq 0$. The trace identity (22.4) identifies this number with λ_n . The reverse implication would require a determinate positive trace representation of the full zero measure from the scalar Li sequence; that is the classical Li/Weil positivity problem, not a consequence of P_1 – P_3 . □

THEOREM 22.3.10 (*positive Hilbert–Pólya trace implies zero-placement*). (conditional on $P_2, P_3, H\text{-P}^{\text{Ar}}$). Assume P_2, P_3 , and the determinant and regularised trace identities of $H\text{-P}^{\text{Ar}}$. If HP–Li positivity holds:

(HP–Li) *Positive Hilbert–Pólya trace.* $\text{Tr}_{\text{reg}}(\mathcal{L}_n) \geq 0$ for every $n \geq 1$, where $\mathcal{L}_n$ is (22.3),

then the zero-placement clause of F-RH holds. Equivalently, RH holds. If the self-adjointness assertion in $H\text{-P}^{\text{Ar}}$ is also assumed, then the determinant identity already gives zero-placement; the scalar Li route uses only the determinant and trace identities.

Proof. HP–Li gives $\lambda_n \geq 0$ for all $n \geq 1$ by Chapter 22.3.8. Li positivity is equivalent to RH (Li 1997, Chapter 22.3.1); this is exactly the zero-placement clause of F-RH (Chapter 22.1.2). □

Remark 22.3.11 (*one-directional implication and operator discipline*). The polynomial $(L_{\circ}^{\text{Ar}})^n - (L_{\circ}^{\text{Ar}} - \text{id})^n$ is *not* Li's transform; $L_{\circ}^{\text{Ar}}$ would need to serve simultaneously as Fock energy, Bost–Connes Hamiltonian, and determinant-trace operator, three incompatible roles (see Chapter 22.3.6). The correct statement is the one-directional HP–Li positivity $\Rightarrow$ zero-placement, where $H\text{-P}^{\text{Ar}}$ supplies Θ_ξ . The converse is the assertion that zero-placement supplies a positive Weil trace representation of the full zero distribution; this is the Hilbert–Pólya/Connes–Deninger positivity problem.

Remark 22.3.12 (*content of F-RH as a reformulation*). F-RH separates RH from the extra arithmetic structures carried by $\mathcal{V}^{\text{prim}}$.

- (i) The spectral object is made precise: the self-adjoint determinant form of $H\text{-P}^{\text{Ar}}$ is a Hilbert–Pólya realisation of RH, while HP–Li positivity is Li's scalar RH-equivalent positivity written through the zero-divisor operator after the trace identity is fixed.
- (ii) A structural constraint on zero-placement emerges: the Petersson positivity of P_3 propagates through the Weil/Connes trace functional to impose positivity of the Li transforms, provided the trace representation is determinate.

- (iii) The reformulation connects RH mechanisms (Connes, Deninger, Meyer) to unitarity bootstrap methods in the style of Friedan–Qiu–Shenker, Cardy, and Zhu, with the arithmetic modular setting replacing the compact-CFT setting.
- (iv) Admissible approximants to the Li/Weil test functions can be evaluated against the Bost–Connes and Hecke trace formulae. Positivity is a distributional trace question, not finite Fock positivity of T_n .

22.4 COMPARISON TO EXISTING RH MECHANISMS

Mechanism	Input	Analytic theorem
Hilbert–Pólya (folklore)	Self-adjoint op. with determinant divisor $\text{div } \xi$	Operator not exhibited
Connes (adele class space)	$\text{KMS}_{\beta=1}$ Bost–Connes positivity	Trace-formula positivity
Deninger ($H_{\Delta}^{\bullet}$)	Purity of Frobenius on arith. H^1	Cohomology not constructed
Meyer (Schwartz space)	Mellin self-adjoint operator plus missing zero-divisor quotient	Operator not exhibited
Arakelov height-zeta	Height-function positivity	Equivalence to RH not established
Weil positivity	Positivity of Weil inner product	Known equivalent, no proof
Li positivity	$\lambda_n \geq 0$ for all $n \geq 1$	Known equivalent, no proof
Riemann–von Mangoldt	Explicit formula approach	Approaches RH asymptotically
Montgomery–Dyson	GUE statistics of zeros	Statistical, not deterministic
Vol IV zero-placement	H-P^{Ar} + positive trace	Self-adjoint zero-divisor operator not

TABLE 22.1: The zero-placement clause placed against existing RH mechanisms.

- Remark 22.4.1 (relations).* (i) HP–Li positivity is Li positivity written on the Hilbert–Pólya zero-divisor operator Θ_{ξ} . It implies zero-placement, hence RH, via Chapter 22.3.10. A Virasoro polynomial T_n is not a substitute for this transform.
- (ii) The zero-placement clause is governed by Weil positivity: Weil’s inner product on test functions corresponds to the Petersson inner product on $\mathcal{V}^{\text{prim}}$ via Tate’s restricted tensor product. F-RH adds the arithmetic Hermitian, Koszul, and Zhu structure around this clause.
- (iii) F-RH embeds Hilbert–Pólya explicitly as H-P^{Ar} (Chapter 22.3.2): it asks for a self-adjoint operator A , the positive form from P_3 , the determinant divisor of $\Theta_{\xi} = 1/2 + iA$, the regularised non-trivial-zero trace distribution, and $\mathbb{Z}/2$ -symmetry under the arithmetic S -modular involution from Pr. F-RH does not replace Hilbert–Pólya: H-P^{Ar} is a separate spectral hypothesis in the sufficient implication below.
- (iv) F-RH is closer to Connes than to Deninger: both Connes and F-RH use unitarity plus Bost–Connes positivity; Deninger postulates a cohomology theory that F-RH does not need. The difference from Connes is that F-RH carries the positivity through arithmetic holomorphic-topological (HT) observables and a Weil-positive trace rather than through the bare trace-formula construction.

22.5 RESIDUAL CONDITIONS

After the structural reduction of Chapter 21 and the F-RH reformulation above, the RH-relevant part is controlled by the following conditions. The spectral conditions imply zero-placement; the categorical conditions construct the trace object on which that problem is stated.

- (HP–Li) *Positive Hilbert–Pólya trace.* $\mathrm{Tr}_{\mathrm{reg}}(\mathcal{L}_n) \geq 0$ for every $n \geq 1$, where $\mathcal{L}_n$ is the Li transform of Θ_ξ (Chapter 22.3.10). *Sufficient for RH given $P_2, P_3, H\text{-}P^{\mathrm{Ar}}$.*
- (H- P^{Ar}) *Arithmetic Hilbert–Pólya.* There is a self-adjoint operator A such that $\Theta_\xi = 1/2 + iA$ has determinant divisor $\mathrm{div} \xi_{\mathbb{R}}$ and Weil–Connes trace distribution equal to the non-trivial-zero distribution (Chapter 22.3.2). *Required for the HP–Li $\Rightarrow$ zero-placement chain; not derivable from P_1, P_2, P_3 .*
- (arith. Positselski) *Arithmetic Positselski equivalence* (Chapter 24.1.5). *Required for full P_1 beyond the forward implication.*
- (Koszul–Pet compat) *Identification of arithmetic Koszul pairing with adelic Petersson pairing* (Chapter 21.7.2). *Required for P_3 .*
- (L_{ACD}) *Adelic categorical descent for $D^{\mathrm{sph-fin}}$* (Chapter 21.2.3). *Required for F_1 , hence for six-window placement.*
- ($F_3\text{.BZN}^{\mathrm{Ar}}$) *Arithmetic Ben-Zvi–Nadler identification* (Chapter 21.4.1). *Required for F_3 , hence for the categorical trace presentation of $\mathcal{V}^{\mathrm{prim}}$.*

Items (arith. Positselski), (Koszul–Pet compat), H- P^{Ar} , and HP–Li positivity lie on the P_1 – P_2 – P_3 branch of the zero-placement clause and together form a sufficient set for RH in the chiral-algebra formalism. Items (L_{ACD}) and ($F_3\text{.BZN}^{\mathrm{Ar}}$) are independent of RH proper and lie on the six-window presentation; they would state $\mathcal{V}^{\mathrm{prim}}$ as a theorem-grade object but do not imply RH.

22.6 A SUFFICIENT PATH TO RH

THEOREM 22.6.1 (*sufficient implication*). . RH follows from the following implication in the chiral-algebra formalism:

$$\begin{array}{ccc}
 P_1 & \xrightarrow{\text{arith. Positselski}} & P_1 \text{ functional equation} \\
 P_2 & \xrightarrow{\text{Tate 1950}} & P_2 \text{ unconditional} \\
 P_3 & \xrightarrow{\text{Koszul–Pet compat}} & P_3 \text{ restricted unitarity} \\
 (P_1, P_2, P_3, H\text{-}P^{\mathrm{Ar}}) & \xrightarrow{\text{HP–Li}} & \text{zero-placement} \xrightarrow{\Leftrightarrow} \text{RH}
 \end{array}$$

Four inputs suffice: arithmetic Positselski, Koszul–Pet compatibility, arithmetic Hilbert–Pólya (H- P^{Ar}), and positive Hilbert–Pólya Li trace (HP–Li).

Proof. P_2 is the completed-zeta character/explicit-formula input (Chapter 21.6.1); P_1 is the functional-equation branch conditional on arithmetic Positselski (Chapter 21.5.1); P_3 is the positive-form input conditional on the restricted Koszul–Petersson compatibility (Chapter 21.7.1). Given these inputs plus H- P^{Ar} , Chapter 22.3.10 gives HP–Li $\Rightarrow$ zero-placement, and zero-placement $\Leftrightarrow$ RH via Li’s criterion (Chapter 22.1.2, Chapter 22.3.1). $\square$

Remark 22.6.2 (*analytic content of the implication*). **arith. Positselski** non-formal theorem; Iwasawa-theoretic Positselski.

Koszul–Pet compat non-formal theorem; compatibility of the arithmetic Koszul and adelic Petersson pairings.

H-P^{Ar} This is the Hilbert–Pólya operator-construction problem in the arithmetic Fock setting; the domain theorem alone does not supply the zero-divisor determinant.

HP–Li This is the positive-trace form of the classical Hilbert–Pólya / Weil positivity problem. If tractable through Li coefficients, the $n = 2$ case ($\lambda_2 \geq 0$) has been computed numerically to $n \sim 10^5$ zeros of ζ (Coffey 2005; Voros 2003 for the first few). The assertion for all n is Li’s RH-equivalent positivity theorem.

The arithmetic formalism does not shorten the Hilbert–Pólya obstacle, but it places the positivity in a precise chain-level trace setting where unitarity-bootstrap and factorisation techniques may be applied without conflating them with the zero-divisor theorem.

22.7 RELATION TO THE ARITHMETIC BRANCH ESSENTIAL THEOREM

The arithmetic branch’s essential theorem (Chapter 19.2.1, Chapter 21.5.1) is

$$\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \iff \xi(s) = \xi(1-s).$$

This gives the functional equation (geometric self-duality of $\xi_{\mathbb{R}}$), not RH (spectral zero-placement). RH is strictly beyond the essential theorem, requiring the additional positive Hilbert–Pólya trace input.

F-RH asks whether Koszul self-duality, which gives the functional equation, combined with the positivity of P_3 and the modularity of P_2 , suffices to deliver the spectral zero-placement that RH asserts. The implication in the chiral-algebra formalism is conditional on H-P^{Ar} and HP–Li positivity. Those two statements are the Hilbert–Pólya form of RH native to this formalism.

22.8 THE F-RH REDUCTION

- (i) The zero-placement clause of F-RH (Chapter 22.1.1) is exactly RH (Chapter 22.1.2). F-RH also asks for the positive Hermitian structure compatible with arithmetic Koszul self-duality and arithmetic Zhu modularity.
- (ii) The zero-placement clause is implied by VB (arithmetic Virasoro bootstrap; Chapter 22.2.1) given $P_1 + P_2 + P_3$.
- (iii) The zero-placement clause is implied by HP–Li positivity (positive Hilbert–Pólya trace; Chapter 22.3.10) via Li’s criterion, given the determinant and trace identities of H-P^{Ar}. If the full determinant-trace H-P^{Ar} hypothesis is assumed, zero-placement is already part of the determinant-trace realisation. The converse direction from zero-placement to HP–Li is the positive-trace realisation problem.
- (iv) Li’s coefficients λ_n are Hilbert–Pólya transform traces of Θ_{ξ} (Chapter 22.3.8).
- (v) Li positivity and Weil positivity govern the zero-placement clause. F-RH joins this clause to the arithmetic Hermitian, Koszul, and Zhu structures.
- (vi) The sufficient implication to RH uses four inputs: arith. Positselski, Koszul–Pet compat, H-P^{Ar}, and HP–Li positivity (Chapter 22.6.1). This is the *four-input conditional conjunction* stated in Chapter 38 and made precise as Chapter 38.3.1: each input carries its own analytic content (five Positselski sub-items, KMS_I/GNS identification beyond the sph-fin cuspidal core, zeta-zero determinant-trace matching, RH-equivalent positive trace), and the conjunction implies zero-placement.
- (vii) RH is strictly beyond the branch’s essential theorem (functional equation), and HP–Li positivity is the Li positive-trace form of RH within the chiral-algebra formalism. The operator dictionary

of Chapter 22.3.11 preserves only the forward implication $\text{HP-Li} \Rightarrow \text{zero-placement}$; the converse is the positive determinant-trace realisation problem (Chapter 35.4.16) and does not help the forward direction.

ARITHMETIC VIRASORO POSITIVITY BOOTSTRAP

With the determinant form of Chapter 22 in place, the Virasoro-polynomial sequence $T_n^{\text{Vir}} = (L_o^{\text{Vir}})^n - (L_o^{\text{Vir}} - \text{id})^n$ is not the Li operator. It proves finite-sector Fock positivity, but it does not construct a Hilbert–Pólya operator and it does not give RH. The RH-strength input is the positive Hilbert–Pólya trace of $\mathcal{L}_n = \text{id} - (\text{id} - \Theta_\xi^{-1})^n$ from Chapter 22.

23.1 STATEMENT RECALLED

The Operator Sequence

Definition 23.1.1 (formal Virasoro polynomial sequence). . Let $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ denote the primary (non-descendant) subspace of the arithmetic Heisenberg Fock, equipped with the Petersson inner product $\langle \cdot, \cdot \rangle^{\text{Pet}}$ of Chapter 21.7.1. For every integer $n \geq 1$ define

$$T_n^{\text{Vir}} := (L_o^{\text{Vir}})^n - (L_o^{\text{Vir}} - \text{id})^n \in \text{End}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}).$$

Here L_o^{Vir} is the Fock/Virasoro conformal-energy operator. It is not the Hilbert–Pólya determinant-trace operator H_{HP} , the Bost–Connes Hamiltonian N_{BC} , or the zero-divisor operator $\Theta_\xi = 1/2 + iH_{\text{HP}}$. Thus T_n^{Vir} is a legitimate Fock polynomial on any common Virasoro spectral core, but it is not Li’s transform. In the sequel, later occurrences of T_n mean T_n^{Vir} unless the Hilbert–Pólya Li transform $\mathcal{L}_n$ is written explicitly.

Remark 23.1.2 (why T_n^{Vir} is not the Li object). Two conventions for Li’s criterion circulate in the literature. The original Li 1997 J. Number Theory paper defines $\lambda_n = \sum_\rho (1 - (1 - 1/\rho)^n)$; Bombieri–Lagarias J. Number Theory 77 (1999) uses the equivalent closed form $\lambda_n = (n-1)!^{-1} (d/ds)^n [s^{n-1} \log \xi(s)] \big|_{s=1}$. The corresponding operator is $\mathcal{L}_n = \text{id} - (\text{id} - \Theta_\xi^{-1})^n$, with Θ_ξ the normal zero-divisor operator of Chapter 22.3.2. A polynomial $(L_o^{\text{Vir}})^n - (L_o^{\text{Vir}} - \text{id})^n$ does not become Li’s transform by regularisation. Consequently, every claim below involving T_n^{Vir} is a formal Fock/Virasoro claim unless it is explicitly restated in terms of $\mathcal{L}_n$ and the positive Weil–Connes trace.

Conditional Chain: $VB^* \Rightarrow \text{Li Positivity} \Leftrightarrow \text{RH}$

PROPOSITION 23.1.3 (formal Virasoro positivity is not Li positivity). . Positivity of the formal Virasoro polynomial T_n^{Vir} on finite Fock sectors does not imply $\lambda_n \geq 0$, and therefore does not imply RH. The one-directional RH implication in Vol IV is instead

$$\text{HP-Li} \implies \lambda_n \geq 0 \forall n \iff \text{RH},$$

where HP–Li is the positive trace statement of Chapter 22.3.10.

Proof. Li’s criterion is Li 1997 J. Number Theory, reproduced in Bombieri–Lagarias J. Number Theory 77 (1999). The Li summand is $1 - (1 - 1/\rho)^n$. The spectral transform realising this summand is therefore

$z \mapsto 1 - (1 - z^{-1})^n$ applied to the zero operator $z = \Theta_\xi$. The polynomial $x^n - (x - 1)^n$ applied to a Virasoro energy $x = L_o^{\text{Vir}}$ is a different function of a different operator. Positivity of the latter is a Fock calculation; it has no trace identity with Li's coefficients unless the Hilbert–Pólya trace identity has already been replaced by the correct Θ_ξ -transform. The HP–Li implication follows from Chapter 22.3.10. $\square$

Remark 23.1.4 (what is genuinely equivalent). The known RH-equivalent positivity languages are Li positivity, Weil positivity for the number-field explicit formula, and the positive zero-measure conjecture for $\Theta_\xi, \text{VB}_{\text{core}}^*$. Under explicit hypotheses these languages give the same positivity condition. None of the RH-equivalent positivity statements is unconditional here.

Remark 23.1.5 (the level at which RH lives). RH lives on the zero-divisor operator $\Theta_\xi = 1/2 + iH_{\text{HP}}$, not on the Virasoro energy operator. The Bost–Connes Hamiltonian and the Virasoro L_o give Euler weights and Fock energies; neither carries $\text{div } \xi_{\mathbb{R}}$. A real spectral measure appears only after H-P^{Ar} supplies H_{HP} and identifies $\det_\infty(s - \Theta_\xi)$ with $\xi_{\mathbb{R}}(s)$. VB^* , in its RH-strength form, is therefore a positivity statement for the Weil–Connes trace of functions of Θ_ξ , together with determinacy of the corresponding zero measure.

23.2 OBSTRUCTIONS

Virasoro levels

For $n = 1$ one has $T_1^{\text{Vir}} = \text{id}$. For $n = 2$ one has

$$(L_o^{\text{Vir}})^2 - (L_o^{\text{Vir}} - \text{id})^2 = 2L_o^{\text{Vir}} - \text{id}.$$

Thus $T_2^{\text{Vir}} \succeq 0$ is the Fock threshold $L_o^{\text{Vir}} \succeq \frac{1}{2}\text{id}$. It is not a statement about zeta zeros. Li's coefficient at level 2 is the trace of $z \mapsto 1 - (1 - z^{-1})^2$ on Θ_ξ , and equality between these two functional calculi would itself be an additional Hilbert–Pólya theorem.

Moment problem

The Hamburger moment problem belongs to the regularised spectral measure of Θ_ξ . There is no identity

$$\text{Tr}_{\text{reg}}(T_n^{\text{Vir}}) = n\lambda_n.$$

The valid moment sequence is obtained from the zero-divisor operator and the Weil–Connes trace. Stieltjes' theorem supplies the algebraic test: a sequence is the moment sequence of a positive measure on $[0, \infty)$ precisely when all Hankel and shifted Hankel minors are non-negative.

Kac determinant

At $c = 2$, the Kac parametrisation gives

$$1 - \frac{6}{m(m+1)} = 2, \quad m(m+1) = -6.$$

The null weights form the countable set $\{h_{r,s}(2) : r, s \geq 1\}$. Outside this set the contravariant form is non-degenerate. Non-degeneracy is weaker than positivity; the Petersson sign is not determined by the non-vanishing of the Kac determinant.

Fock combinatorics

On the free-boson number state $|\mathbf{n}\rangle$, L_o^{Vir} has eigenvalue

$$m(\mathbf{n}) = \sum_{i \geq 1} i n_i.$$

For $m(\mathbf{n}) \geq 1$,

$$T_n^{\text{Vir}}|\mathbf{n}\rangle = (m(\mathbf{n})^n - (m(\mathbf{n}) - 1)^n)|\mathbf{n}\rangle$$

has non-negative eigenvalue. The vacuum has $m = 0$; for even n the eigenvalue is negative. Hence the Fock positivity statement is a non-vacuum statement.

Weil trace

Weil positivity is a positivity theorem for the explicit formula and is equivalent to RH over number fields. In the Vol IV notation its operator representative is the positive regularised trace of $\mathcal{L}_n(\Theta_\xi)$, not a trace of T_n^{Vir} . Koszul–Petersson compatibility can identify the Weil trace pairing with a Petersson pairing on admissible arithmetic states; it cannot identify the Virasoro energy spectrum with the zero divisor.

de Branges datum

For a Hermite–Biehler function E , the de Branges kernel is

$$K(w, z) = \frac{E(w)\overline{E(z)} - E^*(w)\overline{E^*(z)}}{2\pi i(\bar{z} - w)}.$$

The real entire function $\Xi(z) = \xi_{\mathbb{R}}(1/2 + iz)$ satisfies $\Xi^* = \Xi$, so Ξ itself is not a Hermite–Biehler function. A de Branges reformulation of RH requires a genuine datum E whose divisor controls the divisor of Ξ . The Hermite–Biehler condition is then part of the RH-strength input, not a consequence of formal Virasoro positivity.

23.3 LEVEL-BY-LEVEL POSITIVITY

Level $n = 1$: Unconditional

THEOREM 23.3.1 ($n = 1$ arithmetic Virasoro positivity). $T_1^{\text{Vir}} = (L_o^{\text{Vir}}) - (L_o^{\text{Vir}} - \text{id}) = \text{id}$. Hence $T_1 \succeq 0$ unconditionally on $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$.

Proof. Direct computation. The operator is the identity. $\square$

Remark 23.3.2 (*numerical consistency*). The trace of the identity on $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ must equal λ_1 . Bombieri–Lagarias §2 gives $\lambda_1 = \frac{1}{2}(2 + \gamma - \log(4\pi)) \approx 0.02310$. The positivity of this number is consistent but not itself a proof of $T_1 \succeq 0$; the proof is the identity-operator identity. The number λ_1 has been known to be positive since Li 1997.

Level $n = 2$: Formal Threshold

THEOREM 23.3.3 ($n = 2$ Virasoro threshold). $T_2^{\text{Vir}} = 2L_o^{\text{Vir}} - \text{id}$. Hence $T_2^{\text{Vir}} \succeq 0$ on $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ if and only if $L_o^{\text{Vir}} \succeq \frac{1}{2}\text{id}$ on the same space.

Proof. Direct expansion: $T_2^{\text{Vir}} = (L_o^{\text{Vir}})^2 - (L_o^{\text{Vir}} - \text{id})^2 = 2L_o^{\text{Vir}} - \text{id}$. The inequality $T_2^{\text{Vir}} \succeq 0$ is therefore equivalent to $2L_o^{\text{Vir}} \succeq \text{id}$, i.e. to $L_o^{\text{Vir}} \succeq \frac{1}{2}\text{id}$. $\square$

COROLLARY 23.3.4 (*$n = 2$ is not a zero-placement theorem*). . The implication

$$T_2^{\text{Vir}} \succeq 0 \implies \text{Re}(\rho) = 1/2 \text{ for every non-trivial zero } \rho$$

is false without an additional Hilbert–Pólya identification of the Virasoro energy with the zero-divisor operator. $H\text{-P}^{\text{Ar}}$ supplies Θ_ξ , not an equality $L_0^{\text{Vir}} = \Theta_\xi$.

Proof. The spectrum of L_0^{Vir} is the conformal-energy spectral measure of the Fock representation. The operator Θ_ξ is constrained by the determinant divisor $\text{div } \xi_{\mathbb{R}}$. Equality of these data would identify a non-negative Fock energy measure with a complex divisor. No such equality follows from P_1, P_2, P_3 , or the definition of T_2^{Vir} . The valid RH implication uses the Li transform $\mathcal{L}_n$ of Θ_ξ , as in Chapter 22.3.8. $\square$

Remark 23.3.5 (the spectral barrier). The $n = 2$ Virasoro calculation is a threshold calculation inside the Fock representation. The RH barrier is the construction of the zero-divisor operator Θ_ξ and of a positive trace on its Li transforms. Confusing these two operators turns Fock positivity into a false Hilbert–Pólya argument.

Levels $n = 3, 4$: Bombieri–Lagarias Upper Bound

THEOREM 23.3.6 (*Bombieri–Lagarias asymptotic bound*). (Bombieri–Lagarias J. Number Theory 77 (1999):274–287, Theorem 2). For every $n \geq 1$,

$$\lambda_n \leq \frac{n(c_0 + \log n)}{2}, \quad c_0 := 2 + \gamma - \log(4\pi) \approx 0.04619.$$

The matching lower bound $\lambda_n \geq -\frac{1}{2}n \log n - O(n)$ holds unconditionally.

COROLLARY 23.3.7 (*finite Li data are scalar data*). . For any finite set $F \subset \mathbb{N}$, the inequalities $\lambda_n \geq 0$ for $n \in F$ are scalar inequalities for the Li transforms $\mathcal{L}_n = 1 - (1 - \Theta_\xi^{-1})^n$. They do not imply operator positivity of T_n^{Vir} , and they do not imply RH. Bombieri–Lagarias bounds and finite numerical evaluations are therefore finite scalar controls, not Virasoro positivity theorems.

Proof. Li’s criterion requires $\lambda_n \geq 0$ for every $n \geq 1$. A finite set of inequalities is therefore not equivalent to RH. Moreover $\lambda_n = \text{Tr}_{\text{reg}}(\mathcal{L}_n)$ is a scalar trace identity for Θ_ξ . A positive trace of one operator does not force positivity of another operator: a diagonal matrix with eigenvalues -1 and 2 has positive trace and is not positive. The Virasoro polynomial T_n^{Vir} is a different functional calculus from $\mathcal{L}_n$. $\square$

Remark 23.3.8 (the trace-positivity gap). The number-theoretic trace is attached to $\mathcal{L}_n(\Theta_\xi)$, not to T_n^{Vir} . Li positivity for all n is equivalent to RH. Operator positivity of a Virasoro polynomial is a separate Fock statement, and a scalar trace cannot bridge the two without a positive, determinate Weil–Connes representation of the zero measure.

23.4 MOMENT PROBLEM REFORMULATION

The Shifted Zero Measure

Definition 23.4.1 (*off-line defect measure*). . Let $\rho = 1/2 + i\gamma_\rho$ denote a non-trivial zero of ξ , written with $\gamma_\rho \in \mathbb{C}$ (real for every zero iff RH). Under the symmetrisation $\rho \leftrightarrow 1 - \bar{\rho}$, the pair set $Z = \{\rho, 1 - \bar{\rho}\}$ pushes to the off-line defect measure ν_{def} on $\mathbb{R}_{\geq 0}$ via

$$\int f d\nu_{\text{def}} := \sum_{\rho} f(|\text{Im } \gamma_\rho|^2)$$

for test functions f of suitable growth, Bost–Connes-regularised. The measure ν_{def} is supported at $\mathfrak{o}$ exactly when RH holds. It is a defect detector, not Li’s positive moment measure.

Remark 23.4.2. Under RH, $\gamma_\rho \in \mathbb{R}$ and ν_{def} is supported at $\mathfrak{o}$. This defect measure is not the spectral measure of H_{HP} and not the Li moment measure; it records only the distance of zeros from the critical line.

Li-Stieltjes Moment Formulation

PROPOSITION 23.4.3 (*the defect measure is not a Li–Stieltjes moment model*). . The off-line defect measure ν_{def} of Chapter 23.4.1 does not give a Stieltjes moment representation of Li’s coefficients or of $\mu_n := \sum_\rho \rho^n$. A Li–Stieltjes formulation requires the Hilbert–Pólya zero-divisor operator Θ_ξ , the Li transform $\mathcal{L}_n$, and the positive Weil–Connes trace of Chapter 22.3.2.

Proof. ν_{def} records $|\text{Im } \gamma_\rho|^2$, which vanishes exactly on the critical line. Li’s summand is instead $1 - (1 - 1/\rho)^n$, a holomorphic function of the zero ρ . No change of variables from the defect alone recovers this holomorphic transform with multiplicity. The Hilbert–Pólya trace identity is Chapter 22.3.8, which applies the Li transform to Θ_ξ . $\square$

Hamburger Moment Determinacy

THEOREM 23.4.4 (*Hamburger reformulation for the formal T_n*). . The following equivalence is a formal operator-theoretic template for a genuine positive moment problem; it is not the live RH path unless the measure is replaced by the Hilbert–Pólya zero measure of H_{HP} and the trace is the positive Weil–Connes trace:

- (i) Formal positivity: $T_n^{\text{Vir}} \succeq \mathfrak{o}$ for all $n \geq 1$ on a fixed self-adjoint spectral model.
- (ii) The associated spectral measure is a positive measure supported on the corresponding positive half-line.
- (iii) All Hankel minors $H_n := \det(\mu_{i+j})_{0 \leq i, j \leq n}$ and shifted Hankel minors $H_n^{(1)} := \det(\mu_{i+j+1})_{0 \leq i, j \leq n}$ are non-negative for every $n \geq 0$.

This theorem is not an independent proof of RH and does not connect T_n^{Vir} to Li’s coefficients.

Proof. (1) $\Leftrightarrow$ (2) is the spectral theorem for a chosen self-adjoint model: positivity of a polynomial functional calculus is equivalent to support of the spectral measure in the region where the polynomial is non-negative. This is a formal statement about that model, not about zeta zeros.

(2) $\Leftrightarrow$ (3): Stieltjes moment-problem theorem (Shohat–Tamarkin 1943, Ch. III; Akhiezer 1965, §2.1): a sequence $\{m_k\}_{k \geq 0}$ is the moment sequence of a positive measure on $[0, \infty)$ iff all Hankel and shifted Hankel minors are non-negative.

No implication to RH is asserted. $\square$

Remark 23.4.5 (moment information). Chapter 23.4.4 records the formal moment-determinacy apparatus that would become relevant after a true Hilbert–Pólya spectral measure is constructed. In particular:

- Testing a genuine positive spectral measure level-by-level becomes testing Hankel determinants H_n level-by-level, a finite numerical procedure at each n .
- Known numerical bounds (Coffey 2005 Math. Phys. Anal. Geom. for $n \leq 10^5$) directly test $H_n \geq 0$ at those levels.

- Known structural bounds (Bombieri–Lagarias 1999, Voros 2003 Ann. Inst. Fourier) give upper bounds on μ_n , translating to upper bounds on Hankel determinants.
- The gap between known and needed becomes precisely specified: *unconditional* lower bounds on $H_n^{(1)}$ for all n .

The equivalent Hilbert–Pólya condition is the construction of the positive zero measure itself.

23.5 LOCAL KAC DETERMINANT AT $c = 2$

The Kac Formula

THEOREM 23.5.1 (*Kac determinant formula*). (Kac 1978; Feigin–Fuchs 1990). For the Virasoro Verma module $M(c, h)$ of central charge c and highest weight h , the determinant of the contravariant form at level n is

$$\det_n^{c,h} = \prod_{\substack{r,s \geq 1 \\ rs \leq n}} (h - h_{r,s}(c))^{p(n-rs)}$$

where $p(k)$ is the partition function and

$$h_{r,s}(c) = \frac{((m+1)r - ms)^2 - 1}{4m(m+1)}, \quad c = 1 - \frac{6}{m(m+1)}.$$

The Arithmetic $c = 2$ Regime

PROPOSITION 23.5.2 (*Kac at $c = 2$*). . For the local rank-2 free-boson Virasoro module, $c = 2$. In the Kac parametrisation the parameter m satisfies $1 - \frac{6}{m(m+1)} = 2$, hence $m(m+1) = -6$, giving the formal solution $m = (-1 \pm i\sqrt{23})/2$. The Kac null weights become

$$h_{r,s}(c=2) = \frac{((m+1)r - ms)^2 - 1}{4m(m+1)} = -\frac{((m+1)r - ms)^2 - 1}{24}.$$

Thus the possible null weights are the countable Kac set $\{h_{r,s}(2) : r, s \geq 1\} \subset \mathbb{C}$. No rationalisation to $\{0, -1, -2, \dots\}$ is implied.

Proof. Direct substitution $c = 2$ in the Kac formula gives the displayed set. The free-boson realisation proves unitarity of the Fock module itself; it does not identify the Kac zero locus with a real arithmetic sequence. $\square$

COROLLARY 23.5.3 (*no generic null-vector obstruction*). . For generic primary weight h outside the countable Kac zero locus $\{h_{r,s}(2) : r, s \geq 1\}$, $\det_n^{c=2,h} \neq 0$ at every level n . This proves generic non-degeneracy for the local Virasoro Verma module. It does not prove Petersson positivity for the arithmetic primary object.

Unitarity Gap at Special Weights

Remark 23.5.4 (*the gap between non-degeneracy and positivity*). Chapter 23.5.3 establishes non-degeneracy, not positivity. Two Hermitian forms can be non-degenerate (no null vectors) without being positive-definite (a form of signature $(+, -, +, -, \dots)$ is non-degenerate). The Friedan–Qiu–Shenker 1986 $c < 1$ analysis converts non-degeneracy to positivity via the Kac determinant *sign pattern*: when $c < 1$ is not in the discrete unitary series, the sign of $\det_n^{c,h}$ oscillates, producing at least one negative eigenvalue and refuting unitarity. The $c = 2$ case is *outside* the Virasoro unitary series (which consists of $c = 1 - 6/(m(m+1))$ for integer $m \geq 2$, values $c \in [0, 1)$). Hence the FQS discrete-unitary-series argument does not apply.

THEOREM 23.5.5 (*$c = 2$ free-field unitarity*). (Segal 1981; Frenkel–Ben-Zvi 2004). The free-boson rank-2 Fock module at central charge $c = 2$ is a unitary representation of the Virasoro algebra, with Hermitian form induced from the free-boson vacuum pairing $\langle \alpha_{-n_1} \cdots \alpha_{-n_k} | 0 \rangle, \alpha_{-m_1} \cdots \alpha_{-m_l} | 0 \rangle \rangle = \delta_{k,l} \sum_{\sigma \in S_k} \prod_i n_i \delta_{n_i, m_{\sigma(i)}}$. This is positive-definite by inspection.

COROLLARY 23.5.6 (*archimedean positivity of T_n^{Vir}*). . On the *archimedean* sector $\mathcal{H}^{(\infty)}$ of the primary arithmetic Heisenberg Fock (the rank-2 free-boson Fock of Chapter 21.7.1 Step 1), $T_n^{\text{Vir}} \succeq 0$ for every $n \geq 1$ on every non-vacuum number state.

Proof. The free-boson vacuum pairing of Chapter 23.5.5 is positive-definite; L_0^{Vir} acts diagonally on number states with eigenvalue $m = \sum_i i n_i \geq 0$ (with $m = 0$ precisely on the vacuum). On non-vacuum ($m \geq 1$) number states, $T_n^{\text{Vir}} = (L_0^{\text{Vir}})^n - (L_0^{\text{Vir}} - \text{id})^n$ acts by the scalar $m^n - (m-1)^n \geq 0$ (the function $x \mapsto x^n$ is monotone on $[0, \infty)$). Positive-semidefiniteness follows. $\square$

Remark 23.5.7 (the missing sector). Chapter 23.5.6 establishes the archimedean sector. The full primary arithmetic Heisenberg Fock is *not* just the archimedean sector: the Bost–Connes zero-mode lattice $L = \ell^2(\mathbb{N}^\times)$ (Chapter 7.3.1) enters multiplicatively, and the full primary space is $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}} \cong L \otimes \mathcal{H}_{\text{primary}}^{(\infty)}$. Positivity of T_n^{Vir} on the $\mathcal{H}_{\text{primary}}^{(\infty)}$ factor is Chapter 23.5.6. Positivity of a Virasoro polynomial on the L -factor is a statement about the Bost–Connes energy spectrum $\{\log m : m \geq 1\}$. It is not a statement about the zero divisor of ζ . The zero divisor enters only through the Hilbert–Pólya operator Θ_ξ and the Weil–Connes trace.

23.6 FOCK-COMBINATORIAL IDENTITY

Explicit Action on Number States

LEMMA 23.6.1 (*diagonal action on number states*). . On the Fock number state $|\mathbf{n}\rangle := \prod_{i \geq 1} (\alpha_{-i}^\infty)^{n_i} | 0 \rangle / \prod_i \sqrt{i^{n_i} n_i!}$ (normalised with respect to the free-boson Fock inner product of Chapter 23.5.5), L_0^{Vir} acts as

$$L_0^{\text{Vir}} |\mathbf{n}\rangle = \left(\sum_{i \geq 1} i n_i \right) |\mathbf{n}\rangle =: m(\mathbf{n}) |\mathbf{n}\rangle.$$

Proof. $L_0 = \sum_{i \geq 1} \alpha_{-i} \alpha_i$ (Sugawara stress tensor zero mode for the free boson; Frenkel–Ben-Zvi 2004 §3.5). Applied to $(\alpha_{-i})^{n_i} | 0 \rangle$: by the canonical commutator $[\alpha_i, \alpha_{-i}] = i$, $\alpha_i (\alpha_{-i})^{n_i} | 0 \rangle = n_i \cdot i \cdot (\alpha_{-i})^{n_i-1} | 0 \rangle$, hence $\alpha_{-i} \alpha_i (\alpha_{-i})^{n_i} | 0 \rangle = i n_i (\alpha_{-i})^{n_i} | 0 \rangle$. Summing over i gives the diagonal eigenvalue $\sum_i i n_i$. $\square$

The Symmetric-Function Identity

THEOREM 23.6.2 (*combinatorial positivity of T_n on Fock*). . On every non-vacuum number state $|\mathbf{n}\rangle$ with $m(\mathbf{n}) \geq 1$,

$$T_n^{\text{Vir}} |\mathbf{n}\rangle = [m(\mathbf{n})^n - (m(\mathbf{n}) - 1)^n] |\mathbf{n}\rangle,$$

and $m(\mathbf{n})^n - (m(\mathbf{n}) - 1)^n \geq 0$ for every integer $m(\mathbf{n}) \geq 1$ and every $n \geq 1$.

Proof. By Chapter 23.6.1, L_0^{Vir} acts by the scalar $m = m(\mathbf{n})$. Hence $(L_0^{\text{Vir}})^n$ acts by m^n and $(L_0^{\text{Vir}} - \text{id})^n$ by $(m-1)^n$. The difference is $m^n - (m-1)^n$.

Non-negativity: the function $x \mapsto x^n$ is monotone non-decreasing on $[0, \infty)$ for $n \geq 1$; hence for $m \geq 1$ the inequality $m^n \geq (m-1)^n$ holds, with equality iff $m = 1$ and $n > 0$ (in which case $m^n - (m-1)^n = 1 \geq 0$). $\square$

Remark 23.6.3 (combinatorial expansion). Binomial expansion:

$$m^n - (m-1)^n = \sum_{k=0}^{n-1} \binom{n}{k} m^k (-1)^{n-1-k} = nm^{n-1} - \binom{n}{2} m^{n-2} + \cdots + (-1)^{n-1}.$$

For $m = 1$: the identity $1 - 0 = 1$; hence $T_n^{\text{Vir}}|\mathbf{n}\rangle = |\mathbf{n}\rangle$ on weight-1 states for every $n \geq 1$. This is the *singleton-primary-state* positivity.

COROLLARY 23.6.4 (*Fock-level restricted positivity*). . On the subspace $\mathcal{F}^{m \geq 1} \subseteq \mathcal{H}^{(\infty)}$ spanned by non-vacuum number states, $T_n^{\text{Vir}} \succeq 0$ for every $n \geq 1$.

Proof. Direct from Chapter 23.6.2: T_n^{Vir} acts diagonally with non-negative eigenvalues; positivity follows from the diagonal characterisation. $\square$

Remark 23.6.5 (zero-mode separation). Chapter 23.6.4 establishes non-vacuum archimedean Fock positivity for T_n^{Vir} . The primary arithmetic Heisenberg Fock also contains the Bost–Connes zero-mode lattice $\ell^2(\mathbb{N}^\times)$ lattice where μ_n acts via $\mu_n \delta_m = \delta_{nm}$; see Chapter 7.3.2. On this lattice the Bost–Connes Hamiltonian has Dirichlet-series spectrum $\{\log n : n \in \mathbb{N}^\times\}$ at KMS temperature β . This spectrum gives the Euler factor $\zeta(\beta)$; it is not the divisor of $\xi_{\mathbb{R}}$. The RH-strength input is the separate trace bridge from this Euler regularisation to Θ_ξ .

23.7 REDUCTION TO THE IRREDUCIBLE CORE OBSTRUCTION

The Core Statement

THEOREM 23.7.1 (*conditional reduction to the zero-operator core*). (*conditional on $P_2, P_3, H\text{-P}^{\text{Ar}}$, the primary tensor decomposition, and a positive regularised trace*). Under these hypotheses, the RH-strength positivity statement reduces to the following zero-operator statement:

(VB_{core}^{*}) *Zero-operator spectral positivity.* The operator $\Theta_\xi = 1/2 + iH_{\text{HP}}$ satisfies $\det_\infty(s - \Theta_\xi) = \xi_{\mathbb{R}}(s)$, and the regularised Weil–Connes trace on the Li transforms $\mathcal{L}_n(\Theta_\xi)$ is positive for every $n \geq 1$.

The Bost–Connes zero-mode lattice supplies the Euler/KMS regularisation of the trace. It is not the spectrum of Θ_ξ .

Proof. $H\text{-P}^{\text{Ar}}$ gives the determinant identity for Θ_ξ . By Chapter 22.3.8, the Li coefficient λ_n is the regularised trace of $\mathcal{L}_n(\Theta_\xi)$. Positivity of these traces for every n is Li positivity, hence RH by Li’s theorem. Conversely, RH is the support condition for the spectral measure of Θ_ξ . The Bost–Connes spectrum $\{\log m\}$ enters only through the regularisation giving the Euler product; no step identifies it with the determinant divisor of Θ_ξ . $\square$

The Named Obstruction

COROLLARY 23.7.2 (*the irreducible core of VB^**). (*conditional on the hypotheses of Chapter 23.7.1*). The irreducible core is the positive zero-divisor measure problem for Θ_ξ . Replacing Θ_ξ by the Bost–Connes Hamiltonian N_{BC} replaces the zero divisor by the Euler eigenvalues and therefore destroys the Hilbert–Pólya statement.

Remark 23.7.3 (core equivalence). $(\text{VB}_{\text{core}}^*)$ is not a *weaker* statement than VB^* under the Hilbert–Pólya and positive-trace hypotheses; it is the zero-operator form of the same condition.

(i) The zero divisor is carried by Θ_ξ .

- (ii) The Euler product is carried by the Bost–Connes KMS trace.
- (iii) The equality between the two is the Hilbert–Pólya trace bridge.
- (iv) The Bost–Connes phase transition at $\beta = 1$ records the pole of ζ ; it is not a statement about phase transitions on the critical line.

23.8 COMPARISON TO CLASSICAL RH EQUIVALENTS

Formulation	Input data	Positivity shape
Riemann 1859	$\xi(s)$ entire of order 1	Zero placement on line
Li 1997	$\lambda_n, n \geq 1$	Scalar sequence ≥ 0
Weil 1952	Test functions $f \in \mathcal{S}$	Inner product positivity
Báez-Duarte 2003	Finite linear combinations of $\{\rho^i\}$	Density in L^2
Lagarias 1999	de Branges datum over Ξ	Hermite–Biehler divisor control
Bombieri–Lagarias 1999	$\{\lambda_n\}$ with bounds	Explicit lower bounds
Connes 1999	$\mathcal{S}_{\mathbb{A}}/\mathbb{Q}^*$	Trace-formula positivity
Vol IV Virasoro	$\{T_n^{\text{Vir}}\}_{n \geq 1}$ in primary Fock	Fock polynomial positivity
Vol IV HP–Li core	Θ_ξ with Weil–Connes trace	zero-measure positivity

TABLE 23.1: RH-equivalent positivity formulations across the literature.

Remark 23.8.1 (placement of VB^ against its neighbours).* The formal Virasoro polynomial T_n^{Vir} does not imply Li’s inequality. The RH-strength statement is HP–Li: positivity of the regularised Weil–Connes trace of $\mathcal{L}_n(\Theta_\xi)$ for every n . Weil 1952 and the de Branges formulations place the same zero-placement barrier in classical languages. The Vol IV core is the construction and positivity of the Θ_ξ -measure, with the Bost–Connes zero-mode serving as Euler regularisation rather than as the zero spectrum.

PROPOSITION 23.8.2 (*Weil comparison under P_3*). (*conditional on Koszul–Petersson compatibility Chapter 21.7.2 and the $H\text{-}P^{\text{Ar}}$ trace condition*). Weil 1952 explicit-formula positivity and HP–Li positivity express the same RH-equivalent barrier after identifying the Weil distribution with the regularised trace of the zero-divisor operator Θ_ξ . The Petersson form enters through the Koszul–Petersson comparison of the trace functional; it does not turn T_n^{Vir} into a Li transform.

Proof. Weil 1952 gives $\lambda_n = \langle f_n, f_n \rangle^{\text{Weil}}$ for f_n an explicit test function on the adèle class space. $H\text{-}P^{\text{Ar}}$ gives $\lambda_n = \text{Tr}_{\text{reg}}(\mathcal{L}_n(\Theta_\xi))$. Koszul–Petersson compatibility identifies the trace pairing with the Petersson pairing on the arithmetic primary sector whenever the admissible Weil test lies in that sector. Thus Weil positivity and HP–Li positivity are the same positivity problem after the trace bridge is supplied. No equality with $\text{Tr}_{\text{reg}}(T_n^{\text{Vir}})$ is used. $\square$

23.9 ZERO-DIVISOR CONDITION

The Precise Condition

The RH-strength positivity statement is equivalent to the following condition:

Definition 23.9.1 (*the Vol IV zero-divisor condition*). . The *Vol IV zero-divisor condition* is the following statement:

There is a normal zero-divisor operator $\Theta_\xi = 1/2 + iH_{\text{HP}}$ whose determinant divisor is $\text{div } \xi_{\mathbb{R}}$, and whose Weil–Connes trace is supported on $1/2 + i\mathbb{R}$. The Bost–Connes zero-mode gives the Euler/KMS regularisation of the trace; it does not carry $\text{div } \xi_{\mathbb{R}}$. ($\mathcal{R}_{\text{IV}}$)

THEOREM 23.9.2 ($\mathcal{R}_{\text{IV}}$ is the Riemann Hypothesis). $\mathcal{R}_{\text{IV}} \Leftrightarrow \text{RH}$.

Proof. By definition, $\mathcal{R}_{\text{IV}}$ says that the Θ_ξ -determinant divisor and Weil–Connes trace are concentrated on the critical line. Since $\det_\infty(s - \Theta_\xi) = \xi_{\mathbb{R}}(s)$, this is exactly the zero-placement assertion for the non-trivial zeros of ζ , hence RH by Chapter 22.1.2. Conversely, RH gives the same critical-line support statement, so $\mathcal{R}_{\text{IV}}$ and RH are equivalent as formulations. $\square$

Virasoro Boundary

Remark 23.9.3 (structural change). The arithmetic Virasoro reduction gives:

- (i) Closed $n = 1$ unconditionally (Chapter 23.3.1).
- (ii) Reduced $n = 2$ to the formal Virasoro threshold $L_0^{\text{vir}} \succeq (1/2)\text{id}$, not to zero placement (Chapter 23.3.4).
- (iii) Established Fock-level positivity for the archimedean sector at every n (Chapter 23.6.4).
- (iv) Reformulated the zero-measure problem as a Hamburger moment determinacy problem for Θ_ξ (Chapter 23.4.4).
- (v) Proved the Kac determinant at $c = 2$ is non-zero at generic weights (Chapter 23.5.2), establishing absence of null-vector unitarity obstruction.
- (vi) Placed HP–Li in correspondence with Weil positivity (Chapter 23.8.2) and the de Branges Hermite–Biehler framework.
- (vii) The RH-strength condition is a single zero-measure positivity statement for Θ_ξ ($\text{VB}_{\text{core}}^*$, Chapter 23.7.1).

VB^* itself remains open, as it is equivalent to RH.

Remark 23.9.4 ($\text{VB}_{\text{core}}^*$ boundary). The assertion $(\text{VB}_{\text{core}}^*)$ contains the following independent analytic inputs:

- (i) A second KMS-positivity input beyond Bost–Connes: a constraint on the zero-divisor measure forcing critical-line concentration.
- (ii) An arithmetic-modular bootstrap input: a Kac-formula-style analysis of the arithmetic Virasoro algebra on the Bost–Connes C^* -algebra, extracting unitarity from null-vector absence.
- (iii) An $n \rightarrow \infty$ asymptotic input: a moment-generating-function bootstrap strong enough to close the $n \rightarrow \infty$ limit of λ_n scaling.

The first input belongs to the arithmetic site of Connes–Consani arXiv:1405.4527. The second is the C^* -algebraic analogue of Friedan–Qiu–Shenker. The third is an asymptotic moment problem.

23.10 ARITHMETIC ZHU FINITENESS TEST

The tempting shortcut is Verlinde finite-dimensionality: prove that the primary Bost–Connes core is an arithmetic Zhu-finite object, obtain a finite-dimensional Hermitian modular functor, and read positivity from ordinary linear algebra. The shortcut fails on the full zeta object.

Definition 23.10.1 (Euler-faithful arithmetic Zhu quotient). . An Euler-faithful arithmetic Zhu quotient of the primary Bost–Connes core is a quotient Q of the zero-mode lattice $L = \ell^2(\mathbb{N}^\times)$ satisfying:

(i) the Bost–Connes operators descend:

$$\mu_n \bar{\delta}_m = \bar{\delta}_{nm}, \quad \mu_n^* \bar{\delta}_m = \begin{cases} \bar{\delta}_{m/n}, & n \mid m, \\ 0, & n \nmid m, \end{cases}$$

where $\bar{\delta}_m$ is the image of $\delta_m \in L$;

(ii) the KMS Hamiltonian $H = \log N$ descends with $H \bar{\delta}_m = (\log m) \bar{\delta}_m$ whenever $\bar{\delta}_m \neq 0$;

(iii) the ordinary KMS character of the quotient is the full Bost–Connes Euler product:

$$\mathrm{Tr}_Q(e^{-\beta H}) = \sum_{m \geq 1} m^{-\beta} = \zeta(\beta) \quad (\mathrm{Re} \beta > 1).$$

It is *Zhu finite* if $\dim_{\mathbb{C}} Q < \infty$.

THEOREM 23.10.2 (*no Euler-faithful Zhu finiteness*). . No Euler-faithful arithmetic Zhu quotient of the primary Bost–Connes core is Zhu finite.

Proof. If Q were finite-dimensional, the trace $\mathrm{Tr}_Q(e^{-\beta H})$ would be a finite sum $\sum_{j=1}^r a_j e^{-\beta E_j}$ after diagonalising the finite-dimensional self-adjoint operator H . In particular it extends to an entire function of β . Euler-faithfulness requires this finite sum to equal $\zeta(\beta)$ on the half-plane $\mathrm{Re} \beta > 1$. By the identity theorem the meromorphic continuation of ζ would be entire, contradicting its simple pole at $\beta = 1$. Hence Q is infinite-dimensional.

Equivalently, the faithful Bost–Connes action of Chapter 7.3.2 sends δ_1 to δ_m by $\mu_m \delta_1 = \delta_m$. Keeping the full Euler product keeps one independent KMS eigenline at every $m \in \mathbb{N}^\times$. A finite Zhu quotient must kill or identify infinitely many of these lines, and then it no longer has character $\zeta(\beta)$. $\square$

COROLLARY 23.10.3 (*Verlinde finite-dimensionality cannot close full RH*). . Arithmetic Zhu/Verlinde finite-dimensionality cannot prove $\mathrm{VB}_{\mathrm{core}}^*$ for the full $\mathcal{V}^{\mathrm{prim}}$ while preserving the completed zeta partition function. The only Zhu-finite quotients compatible with the Bost–Connes action are finite-prime or finite-conductor truncations; their characters are finite Euler products or finite Dirichlet polynomials, not $\xi(s)$.

Proof. By Chapter 23.10.2, a finite-dimensional quotient cannot be Euler-faithful. A quotient supported on a finite set S of primes has character $\prod_{p \in S} (1 - p^{-\beta})^{-1}$, possibly with an archimedean Tate factor. Such a finite Euler product is a finite-level arithmetic modular-functor object; it is not the zeta object whose zero-placement is RH. Therefore finite-dimensional Hermitian linear algebra can test finite windows, but it cannot supply the missing positivity for the full Θ_ξ -spectral measure. $\square$

Remark 23.10.4 (*central charge does not rescue Zhu finiteness*). The residue value $r_\xi = 2$ of Chapter 14.4.1 is a determinant-rank scalar, not a finite rank and not a trace. Chapter 14 proves that the full infinite tensor has no global Virasoro action. Thus $r_\xi = 2$ cannot be used as a C_2 -cofiniteness or Zhu-finiteness input.

Remark 23.10.5 (*surviving direction*). The finite-dimensional construction survives only as a finite-window technology: truncate to finitely many primes or to bounded conductor, prove the corresponding arithmetic modular functor is finite, compute its Hermitian form, and compare its limiting trace to the Bost–Connes KMS trace. Any limiting theorem strong enough to recover $\mathrm{VB}_{\mathrm{core}}^*$ must add a new positivity or compactness input beyond Zhu finite-dimensionality at finite level.

23.11 ZERO-MEASURE CONJECTURE

Conjecture 23.11.1 (VB_{core}^ : the Vol IV form of RH).* The zero-divisor operator $\Theta_\xi = 1/2 + iH_{\text{HP}}$, with $\text{div } \det_\infty(s - \Theta_\xi) = \text{div } \xi_{\mathbb{R}}$, has determinant divisor and Weil–Connes trace supported on $1/2 + i\mathbb{R}$. Equivalently, all non-trivial zeros of ζ lie on the critical line.

Remark 23.11.2 (the conjecture is RH). Chapter 23.11.1 is, verbatim, the Riemann Hypothesis. The arithmetic chiral-algebra formulation asks for a positive, determinate Weil–Connes trace and determinant divisor on the zero-divisor operator Θ_ξ . The Bost–Connes algebra supplies the Euler regularisation and KMS weight, in the sense of Connes’ adelic trace formula; it does not supply the determinant divisor by itself.

Remark 23.11.3 (size of VB_{core}^).* Any unconditional proof of Chapter 23.11.1 would be an unconditional proof of RH, a Millennium Prize problem. No finite estimate is meaningful. Partial reductions have the following mathematical sizes:

Bombieri-Lagarias-style sharpening at specific n . substantial local theorem.

Kac-determinant analysis at $c = 2$ to low level. finite-level determinant computation.

Moment-problem Hankel analysis. unbounded Hamburger determinacy and lower bounds on all shifted Hankel minors.

Arithmetic-site incorporation. Connes–Consani arithmetic-site positivity; no unconditional proof of RH follows.

23.12 OPERATOR BOUNDARY

- (i) The formal Virasoro polynomial sequence $\{T_n^{\text{Vir}}\}$ gives Fock positivity statements, not Li transforms (Chapter 23.1.3).
- (ii) The RH-strength path uses $\mathcal{L}_n(\Theta_\xi) = 1 - (1 - \Theta_\xi^{-1})^n$, the positive Weil–Connes trace, and Li’s criterion (Chapter 23.4.4, Chapter 23.8.2).
- (iii) $n = 1$ is unconditional: $T_1^{\text{Vir}} = \text{id}$ (Chapter 23.3.1).
- (iv) $n = 2$ reduces to $L_0^{\text{Vir}} \succeq (1/2)\text{id}$, a formal Fock threshold rather than a zero-placement theorem (Chapter 23.3.3, Chapter 23.3.4).
- (v) Finite Li data at $n \geq 3$ are scalar trace data for Θ_ξ ; they do not imply Virasoro operator positivity or RH.
- (vi) Fock-combinatorial positivity holds in the archimedean sector: $T_n^{\text{Vir}} \succeq 0$ on non-vacuum number states for every n (Chapter 23.6.2).
- (vii) The local Kac determinant at $c = 2$ does not vanish at generic weights, establishing absence of null-vector obstruction (Chapter 23.5.2, Chapter 23.5.3).
- (viii) The RH-strength core is a single statement about the Θ_ξ -spectral measure: $(VB_{\text{core}}^*, \text{Chapter 23.7.1})$.
- (ix) Arithmetic Zhu/Verlinde finite-dimensionality holds at finite windows only: any Euler-faithful quotient preserving the Bost–Connes character $\zeta(\beta)$ is infinite-dimensional (Chapter 23.10.2, Chapter 23.10.3).
- (x) (VB_{core}^*) is the Vol IV form of RH, equivalent to $\text{Re}(\rho) = 1/2$ for all non-trivial zeros (Chapter 23.9.2).

- (xi) The condition $\mathcal{R}_{IV}$ is the existence of the zero-divisor operator with positive, determinate Weil–Connes trace; it is equivalent to RH.

The unconditional Virasoro content is the identity at $n = 1$, the formal threshold at $n = 2$, and non-vacuum Fock positivity. The Riemann Hypothesis enters only through the zero-divisor operator Θ_ξ , the Li transforms, and the positive Weil–Connes trace. The equivalent condition is stated in two operator-theoretic forms (Chapter 23.11.1, Chapter 23.9.1) and is, in full, the Riemann Hypothesis.

ARITHMETIC POSITSIELSKI EQUIVALENCE

This chapter lifts Positselski's classical comodule–contramodule correspondence (Mem. AMS 996, arXiv:0905.2621) from field coefficients to the Iwasawa algebra $\Lambda = \mathbb{Z}_p[[\Gamma]]$. The Iwasawa μ -invariant is the precise obstruction: when $\mu = 0$, the classical correspondence lifts to a Λ -linear equivalence between co-derived Λ -comodules and contra-derived Λ -contramodules. Applied to the arithmetic Heisenberg bar coalgebra, this closes the level–weight computation of Step 3 in the proof of P1 (Chapter 21.5.1) and, conditional on the five open Positselski sub-items of Chapter 24.1.6, gives the full biconditional $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$.

24.1 STATEMENT AND MOTIVATION

Review: Classical Positselski

Let k be a field and C a conilpotent dg coalgebra over k . Positselski (arXiv:0905.2621, Thm. 6.5; Mem. AMS 996, Ch. 0) constructs two derived categories distinct from the ordinary one:

- (i) The *co-derived category* $D^{\text{co}}(C\text{-comod}_k)$: the homotopy category of injective complexes of C -comodules, equivalently the Verdier quotient of the homotopy category by acyclic complexes of injectives (Positselski §3.5).
- (ii) The *contra-derived category* $D^{\text{ctr}}(C^\vee\text{-contramod}_k)$: the homotopy category of projective complexes of contramodules over the dual algebra $C^\vee$, equivalently the Verdier quotient by acyclic complexes of projectives (Positselski §3.7).

THEOREM 24.1.1 (*Positselski 2011, arXiv:0905.2621 Thm. 5.2*). . There is a canonical equivalence of triangulated categories

$$D^{\text{co}}(C\text{-comod}_k) \simeq D^{\text{ctr}}(C^\vee\text{-contramod}_k),$$

given on objects by $M \mapsto \text{Hom}_C(C, M) = M^\vee$ with the induced contramodule structure.

The equivalence is the categorical shadow of two-sided Koszul duality: the bar complex of $C^\vee$ is C , the cobar complex of C is $C^\vee$, and the two categories of derived modules (covariant = comodule, contravariant = contramodule) are swapped by the bar–cobar adjunction.

The Λ -Coefficient Target

Replace the field k by the Iwasawa algebra $\Lambda = \mathbb{Z}_p[[\Gamma]]$ with $\Gamma = \text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) \cong \mathbb{Z}_p$. The base Λ is a Noetherian local complete two-dimensional regular ring with maximal ideal $\mathfrak{m} = (p, \gamma - 1)$ and residue field $\mathbb{F}_p$ (Serre 1963, Iwasawa 1973). Its key structural feature is the Λ -module structure theorem: every finitely generated Λ -module M admits a pseudo-isomorphism

$$M \sim \Lambda^r \oplus \bigoplus_i \Lambda/p^{a_i} \Lambda \oplus \bigoplus_j \Lambda/f_j^{b_j} \Lambda$$

with $r \in \mathbb{Z}_{\geq 0}$, $a_i \in \mathbb{Z}_{\geq 1}$, f_j distinguished irreducible polynomials in $\mathbb{Z}_p[\gamma - 1]$ (Washington 1997, Thm. 13.12; Coates–Sujatha 2006 Ch. 2). The Iwasawa μ -invariant is $\mu(M) = \sum_i a_i$; the λ -invariant is $\lambda(M) = \sum_j b_j \deg f_j$.

Definition 24.1.2 (arithmetic dg Λ -coalgebra). An *arithmetic dg Λ -coalgebra* is a conilpotent dg coalgebra C over Λ satisfying:

- (i) *Local finiteness over Λ* : the underlying graded Λ -module C is a filtered union of finitely generated Λ -submodules $C_{\leq n}$ with $C = \varinjlim_n C_{\leq n}$ and each $C_{\leq n}/C_{\leq n-1}$ finitely generated.
- (ii) *Comultiplication*: a graded Λ -linear map $\Delta: C \rightarrow C \otimes_{\Lambda} C$ with counit $\epsilon: C \rightarrow \Lambda$, satisfying the standard coassociativity / counit axioms, and compatible with the differential d .
- (iii) *Conilpotence*: the iterated comultiplication $\Delta^{(n)}: C \rightarrow C^{\otimes_{\Lambda} n}$ is zero on the augmentation ideal $\overline{C} = \ker \epsilon$ for all n sufficiently large.

The Iwasawa μ -invariant of C is $\mu(C) = \mu(\overline{C})$ as a Λ -module.

Definition 24.1.3 ($\mu = 0$ arithmetic Λ -coalgebra). C is $\mu = 0$ when the augmentation ideal $\overline{C} = \ker \epsilon$ has no $\Lambda/p^a \Lambda$ summand in its Λ -module structure theorem decomposition (Chapter 24.1); equivalently, $\overline{C}$ has no $\mathbb{Z}_p$ -torsion part, i.e. the natural map $\overline{C} \rightarrow \overline{C} \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ is injective on the p -primary component.

Remark 24.1.4 (why $\mu = 0$). The Positselski correspondence relies on injective resolutions of comodules being compatible with projective resolutions of contra-modules. Over a field, this reduces to linear algebra. Over Λ , the Λ -module structure theorem (Chapter 24.1) introduces two obstructions: the μ -part (p -torsion) and the λ -part (characteristic-polynomial torsion). The λ -part is handled by Iwasawa polynomial machinery (Coleman–Perrin–Riou, Coates–Sujatha 2006 Ch. 3); the μ -part is the genuine obstruction to descent (Ferrero–Washington 1979). When $\mu = 0$, base change along $\Lambda \rightarrow \Lambda/p\Lambda = \mathbb{F}_p[[T]]$ commutes with the derived operations on both sides, and the classical Positselski equivalence lifts.

THEOREM 24.1.5 (main theorem: arithmetic Positselski). (conditional on the five open sub-items of Chapter 24.1.6: completed continuous duals; derived completion; faithful lifting; primitive zeta-character theorem; Koszul-dual Bost–Connes). Let C be an arithmetic dg Λ -coalgebra (Chapter 24.1.2) with $\mu(C) = 0$ (Chapter 24.1.3). Then there is a canonical equivalence of triangulated categories

$$D_{\Lambda}^{\text{co}}(C\text{-comod}_{\Lambda}) \simeq D_{\Lambda}^{\text{ctr}}(C^{\vee}\text{-contramod}_{\Lambda})$$

lifting the classical Positselski equivalence (Chapter 24.1.1) along the base change $\Lambda \rightarrow \mathbb{Q}_p$ (reduction modulo pseudo-null). The lift is compatible with the bar–cobar adjunction $\Omega^{\Lambda} \dashv B^{\Lambda}$, and the unit $C \rightarrow \Omega^{\Lambda} B^{\Lambda}(C)$ and counit $B^{\Lambda} \Omega^{\Lambda}(A) \rightarrow A$ are quasi-isomorphisms of Λ -dg-coalgebras and Λ -dg-algebras respectively, provided $\mu(C) = 0$ on the comodule side and the dual μ -invariant (Selmer μ) vanishes on the contra-module side.

Remark 24.1.6 (open sub-items of the arithmetic Positselski equivalence). The proof of Chapter 24.1.5 proceeds through Steps 1–6 (§24.4–§24.4), each of which reduces the corresponding hypothesis to a precise open sub-item in Iwasawa-theoretic Positselski. the determinacy comparison names these five open sub-items explicitly.

- (i) **Completed continuous duals.** The co-derived and contra-derived Λ -categories of §24.3 require Pontryagin-dual continuous duals of completed Iwasawa modules (Brumer 1966; Coates–Sujatha 2006 §1.7). Compatibility of these continuous duals with the Nekovář Selmer-complex formalism used in Step 2 is a conditional hypothesis.

- (ii) **Derived completion.** Step 3's Nakayama lift requires the derived $\mathfrak{m}$ -adic completion of comodule and contramodule resolutions to commute with base change $\Lambda \rightarrow \mathbb{F}_p[[T]]$. Positselski 2018 arXiv:1209.2995 §4.6 covers the Artinian base case; the inverse limit over a Weierstrass tower is stated in Step 3 but not independently discharged here.
- (iii) **Faithful lifting along Nakayama.** Chapter 24.4.5 requires both faithfulness and essential surjectivity of the base-change functor $D_\Lambda^{\text{co}} \rightarrow D_{\mathbb{F}_p[[T]]}^{\text{co}}$. Faithfulness is argued modulo $\mu = 0$; essential surjectivity uses the derived completion of the preceding item.
- (iv) **Primitive zeta-character theorem.** Step 4's weight computation (Chapter 24.5.1) uses the HY-zeta character identification of the arithmetic Heisenberg Fock with the Euler-product generating function of ζ . The Λ -coefficient lift of the primitive character to a Positselski pairing requires an independent primitive zeta-character theorem not reproduced in the bar–cobar literature.
- (v) **Koszul-dual Bost–Connes.** The identification of the cobar algebra $\mathcal{A} = C^\vee$ with the Bost–Connes Hecke algebra $\mathbb{A}_p^{\text{BC}}$ (Chapter 12.5.2) is conditional on the Bost–Connes 1995 Selecta Math. 1 §1.2 presentation carrying the Positselski pairing structure, which is asserted but not independently cross-verified against the Iwasawa-theoretic cobar.

The arithmetic Positselski equivalence of Chapter 24.1.5 is conditional precisely on these five sub-items. the determinacy comparison records the conditional form (see Chapter 38.2).

The proof of Chapter 24.1.5 occupies Chapter 24.4; the six independent obstruction constructions and their conclusions are in Chapter 24.2; the required Λ -module facts are in Chapter 24.3; the application to the arithmetic Heisenberg bar coalgebra, closing Step 3 of the Pi hypothesisally, is in Chapter 24.5; the residual condition for non-abelian Iwahori level and Whittaker compatibility is in Chapter 24.7.

24.2 OBSTRUCTION ANALYSIS TABLE

Six independent obstruction constructions test the theorem; each names the obstruction exposed and the correction that resolves it.

Obstruction 1: Base-Change Lift

Obstruction. Does Positselski 2011 arXiv:0905.2621 Thm. 5.2 lift directly along the inclusion of base rings $k \hookrightarrow \Lambda$, or along reductions $\Lambda \rightarrow \Lambda/p\Lambda = \mathbb{F}_p[[T]]$ and $\Lambda \rightarrow \Lambda/(\gamma - 1) = \mathbb{Z}_p$?

Attempt 1. Direct base change $k \rightarrow \Lambda$ fails: Positselski's proof uses that injective = flat = projective for k -modules, which breaks already over $\mathbb{Z}_p$ (injective $\mathbb{Z}_p$ -modules are $\mathbb{Q}_p/\mathbb{Z}_p$ -like, not free).

Correction. Replace direct base change by a two-step lift. First reduce $\Lambda \rightarrow \Lambda/p\Lambda = \mathbb{F}_p[[T]]$, a discrete valuation ring with uniformizer T . Over $\mathbb{F}_p[[T]]$, Positselski's 2018 contraherent-cosheaf machinery (arXiv:1209.2995 Thm. 4.6.5) extends the correspondence: the key input is that $\mathbb{F}_p[[T]]$ is a Noetherian local regular one-dimensional ring, for which finitely generated flat = projective and the structure theorem $M \sim (\mathbb{F}_p[[T]])^r \oplus \bigoplus \mathbb{F}_p[[T]]/T^{a_i}$ gives a controllable injective resolution tower. Second, lift along $\Lambda \rightarrow \mathbb{F}_p[[T]]$ via Nakayama's lemma plus the completeness of Λ with respect to $\mathfrak{m} = (p, \gamma - 1)$.

Conclusion. Base change works in two steps: $\Lambda \rightarrow \mathbb{F}_p[[T]]$ is Positselski 2018; the lift from $\mathbb{F}_p[[T]]$ to Λ uses Nakayama plus complete Noetherianity. The $\mu = 0$ hypothesis is the precise condition that

the two-step lift preserves the bar–cobar structure: without $\mu = 0$, there is a p -torsion kernel $p^N \overline{C}$ obstructing descent to $\mathbb{F}_p[[T]]$. *With $\mu = 0$ as the explicit hypothesis.*

Obstruction 2: Co-Derived vs Contra-Derived Over Λ

Obstruction. Is there a Λ -coefficient analogue of the classical co-derived / contra-derived distinction, and does it coincide with the derived-category framework used in Iwasawa theory?

Attempt 1. A naive Λ -co-derived category might be the Verdier quotient of $K(C\text{-comod}_\Lambda)$ by complexes of acyclic injectives, where injective = injective over Λ . This fails: over Λ , injective hulls involve the Pontryagin dual $\Lambda^\vee = \text{Hom}_{\text{cts}}(\Lambda, \mathbb{Q}_p/\mathbb{Z}_p)$ which is not finitely generated (Brumer 1966; Coates–Sujatha 2006 §1.7), so injective resolutions of comodules explode in size.

Correction. Use the Selmer-complex formalism of Nekovář 2006 (Astérisque 310, Ch. 1–2): Λ -co-derived categories are the compact-object categories in the homotopy category of complexes of Λ -comodules whose injective resolutions are *admissible* in Nekovář’s sense (bounded-below, truncated pro-étale-finite). Admissibility precisely encodes the Pontryagin-dual finiteness needed to keep the Verdier quotient tractable. Similarly, the Λ -contra-derived category is the compact objects in the homotopy category of $C^\vee$ -contramodules whose projective resolutions are Greenberg-admissible (Greenberg 2001, §3).

Conclusion. The Λ -coefficient co-derived and contra-derived categories exist, via Nekovář’s Selmer-complex formalism. The admissibility conditions translate to $\mu(C) = 0$ on the comodule side and dual Selmer- $\mu = 0$ on the contramodule side; these are two different statements, but both reduce to μ -invariant vanishing. *with the Selmer-complex framework supplying the triangulated structure.*

Obstruction 3: μ -Invariant Vanishing for the Heisenberg Bar

Obstruction. What is the Iwasawa μ -invariant of $B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$? The theorem requires $\mu = 0$ for the arithmetic Heisenberg bar coalgebra specifically.

Attempt 1. Compute μ naively from the definition. $B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ is a filtered union of finitely generated Λ -submodules $B_{\leq n} = \bigoplus_{k \leq n} (s^{-1} \overline{\mathcal{V}_{\text{Heis}}^{\text{prim}}})^{\otimes \Lambda^k} \otimes_\Lambda C_{\text{Parshin}}^\bullet(\overline{\text{Spec } \mathbb{Z}})$. Each $B_{\leq n}$ is a tensor product of Heisenberg Fock factors with Parshin Gersten terms. The Parshin Gersten complex on $\overline{\text{Spec } \mathbb{Z}}$ is $\bigoplus_p K_n^M(k(\mathfrak{p}))$ (Beilinson–Bloch motivic cohomology); Bass–Tate 1973 computes $K_2^M(\mathbb{F}_p) = 0$ for $p \neq 2$ and $K_2^M(\mathbb{F}_2)$ elementary abelian 2-primary, all torsion-free after inverting 2. Hence the Gersten complex is Λ -torsion-free at good reduction.

Correction. At the arithmetic Heisenberg, the Λ -structure arises only through the Gersten differential d_{Iw} , which acts via cyclotomic twist. The cyclotomic character on the Heisenberg oscillator $\alpha_{-n}^{(p)}$ is trivial (the Heisenberg is abelian, and $\Gamma = \mathbb{Z}_p$ acts through χ_{cycl} on $\mathbb{Z}_p(1)$ -twists only). Hence the Λ -action on $B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ factors through $\mathbb{Z}_p$, the residue ring of Λ at the augmentation ideal $(\gamma - 1)$. By Ferrero–Washington (Ann. Math. 109, 1979), $\mu_p = 0$ for all abelian cyclotomic fields; the arithmetic Heisenberg falls in the abelian case because $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ is an abelian Heisenberg Fock.

Conclusion. $\mu(B^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})) = 0$ by Ferrero–Washington applied to the abelian cyclotomic extensions entering the Gersten differential. The hypothesis of Chapter 24.1.5 is satisfied unconditionally for the arithmetic Heisenberg, *reduced to Ferrero–Washington*.

Obstruction 4: Weight Compatibility Under Bar–Cobar

Obstruction. Step 3 of P1 asserts that the bar coalgebra generator weights ($\log p$ at prime p) pair with the cobar algebra generator weights (negative $\log n$ on Bost–Connes Hecke μ_n) under the Positselski pairing to give $\kappa^{\text{Ar}} + (\kappa^!)^{\text{Ar}} = 0$. Does the Positselski pairing respect these weights?

Attempt 1. The classical Positselski pairing over k is weight-blind: k -coefficients carry no grading. Lifting to Λ -coefficients introduces the cyclotomic weight grading via $\chi_{\text{cycl}}: \Gamma \rightarrow \mathbb{Z}_p^\times$. The pairing $\text{Hom}_C(C, M) \otimes M^\vee \rightarrow \Lambda$ respects the Γ -equivariance, which translates to compatibility with the cyclotomic weight of Λ -modules but not with the Heisenberg oscillator weights $\log p$ directly.

Correction. Augment the Positselski pairing with the *HY-zeta grading* of Convention Chapter 6.6.2: the algebraic / Λ -module structure lives on the conformal grading (integer weight n on $\alpha_{-n}^{(p)}$); the zeta-character lives on the HY-zeta grading (weight $n \log p$ on $\alpha_{-n}^{(p)}$). The bar coalgebra generator $\alpha_{-n}^{(p)}$ has HY-zeta weight $n \log p$. The cobar algebra is the Bost–Connes Hecke algebra, generated by μ_n of HY-zeta weight $-\log n$. The Positselski pairing sends $\alpha_{-n}^{(p)} \leftrightarrow \mu_{p^n}$ via the substitution $q \mapsto p^{-s}$, with total HY-zeta weight $n \log p + (-\log p^n) = 0$ at each prime. Summed over primes inside the half-plane of absolute convergence, the paired total weight vanishes termwise. No identity with $-\zeta'(s)/\zeta(s)$ is used for the prime-indexed sum alone.

Conclusion. The Positselski pairing lifted to Λ -coefficients respects the conformal grading; the HY-zeta grading enters through the character substitution $q = p^{-s}$ which identifies the total weight pairing with the Euler-product derivation. The key input is Julia’s primon-gas identification (1990, Number Theory and Physics proceedings): the free Fock on primes with frequency $\log p$ has partition function $\prod_p (1 - p^{-\beta})^{-1} = \zeta(\beta)$, *via Julia’s primon gas + Euler product*.

Obstruction 5: Archimedean Completion

Obstruction. The arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ is an adelic object: a restricted tensor product $\mathcal{V}_{\text{Heis}}^{\text{prim}} = \bigotimes_v \mathcal{H}^{(v)}$ with factors at each finite place p and at the archimedean place ∞ . The Λ -coefficient Positselski equivalence is a finite-place statement: Λ is a p -adic object. How does the archimedean place enter?

Attempt 1. Ignore the archimedean factor. This breaks: the Riemann functional equation requires the archimedean Γ -factor $\Gamma(s/2)\pi^{-s/2}$ (Tate 1950 §4.4; Cassels–Fröhlich 1967 §13.4), without which $\zeta(s)$ alone has no functional equation of the required form.

Correction. At the archimedean place the Λ is replaced by the Fourier-dual archimedean analogue: Gaussian Positselski. Gaussian Positselski is the $\mathbb{R}$ -coefficient version of the correspondence with the archimedean theta function $\theta(t) = \sum_n e^{-\pi n^2 t}$ playing the role of the bar–cobar duality pairing. Fourier self-duality of the Gaussian under $\mathcal{F}_{\mathbb{R}}$ (Folland 1999 Ch. 7, Prop. 7.3) is the archimedean analogue of the Λ -coefficient Positselski pairing; Poisson summation on $\mathbb{R}$ connects this to the Jacobi theta functional equation $\theta(t) = t^{-1/2} \theta(1/t)$ used in Riemann’s proof of the functional equation.

Adelically, the Positselski pairing assembles from local pairings: Λ -coefficient Positselski at each finite place p (as in the main theorem) plus Gaussian Positselski at ∞ (archimedean Fourier self-duality). The restricted tensor product of local pairings over $\mathbb{Q}$ with spherical vectors at almost all places (Tate 1950 §2.3; Weil 1967 Ch. VII §1) gives the adelic Positselski pairing.

Conclusion. The archimedean factor is Gaussian Positselski via Fourier self-duality. The Λ -coefficient Positselski (Chapter 24.1.5) handles finite places; the adelic assembly uses Tate's restricted tensor product. *The archimedean assertion reduces to Fourier self-duality.*

Obstruction 6: Non-Commutative Iwasawa for Non-Abelian GL_2

Obstruction. Iwasawa theory for GL_2 is non-commutative: the GL_2 Iwasawa algebra is $\Lambda_{GL_2} = \mathbb{Z}_p[[GL_2(\mathbb{Z}_p)]]$, not abelian. Does the Λ -coefficient Positselski correspondence extend to Λ_{GL_2} ?

Attempt 1. Lift the commutative proof directly. This fails: the structure theorem Chapter 24.1 is commutative; the non-commutative analogue (Fukaya–Kato 2006 Proc. St. Petersburg Math. Soc. 12, §1–2) is more restrictive and holds only modulo pseudo-nullity.

Correction. The main application in this chapter is to the G-row claim for the *abelian* Heisenberg sub-algebra of the full Iwahori Hecke category. The abelian Heisenberg is precisely $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ as defined in Chapter 12 (before non-abelian Iwahori enhancement). For this abelian sub-algebra, Λ reduces to the commutative cyclotomic Iwasawa algebra, and the main theorem applies directly.

For the full non-abelian Iwahori-level GL_2 Hecke category, the correspondence requires the non-commutative Iwasawa framework (Fukaya–Kato 2006; Coates–Fukaya–Kato–Sujatha–Venjakob 2005 Publ. IHES). The extension is a separate conjecture, not achieved in this chapter. The G-row claim of P1 only needs the abelian sub-algebra case.

Conclusion. For abelian Heisenberg (P1 G-row application), holds. For non-abelian Iwahori level, **residual: non-commutative arithmetic Positselski** (Chapter 24.7).

Comparison Table

Obstruction	Content	Conclusion
1	Base-change $\Lambda \rightarrow \mathbb{F}_p[[T]]$	holds, $\mu = 0$ explicit
2	Co-/contra-derived over Λ	holds, Selmer-complex framework
3	$\mu = 0$ for arith. Heisenberg bar	holds, Ferrero–Washington
4	Bar–cobar weight compatibility	holds, Julia primon + Euler product
5	Archimedean completion	holds, Gaussian Fourier self-duality
6	Non-abelian GL_2 case	Residual: non-commutative Λ

TABLE 24.1: Source and Hypothesis Structure table for arithmetic Positselski. The abelian sub-algebra case needed for P1 G-row closes completely; non-abelian Iwahori level is residual condition (Chapter 24.7).

24.3 IWASAWA PRELIMINARIES

The proof of Chapter 24.1.5 requires three structural facts about finitely generated Λ -modules: the structure theorem and its μ -invariant criterion, the bounds on injective and projective resolution length, and the coherence of these resolutions under base change when $\mu = 0$.

 Λ -Modules and Structure Theorem

PROPOSITION 24.3.1 (*Λ -module structure theorem*). . Let $\Lambda = \mathbb{Z}_p[[\Gamma]]$ and let M be a finitely generated Λ -module. Then there is a pseudo-isomorphism (morphism with finite kernel and cokernel)

$$M \sim \Lambda^r \oplus \bigoplus_{i=1}^s \Lambda/p^{a_i} \Lambda \oplus \bigoplus_{j=1}^t \Lambda/f_j^{b_j} \Lambda,$$

where $r, s, t \in \mathbb{Z}_{\geq 0}$, $a_i \in \mathbb{Z}_{\geq 1}$, and $f_j \in \mathbb{Z}_p[\gamma - 1]$ are distinguished irreducible polynomials (Weierstrass polynomials). The invariants $r, \mu(M) = \sum a_i, \lambda(M) = \sum b_j \deg f_j$ are well-defined.

Proof. Washington 1997 *Introduction to Cyclotomic Fields* Thm. 13.12; Coates–Sujatha 2006 §2.3. The proof uses completeness of Λ with respect to $\mathfrak{m} = (p, \gamma - 1)$ plus Weierstrass preparation. $\square$

COROLLARY 24.3.2 (*$\mu = 0$ criterion*). . $\mu(M) = 0$ iff M has no $\Lambda/p^a \Lambda$ summand in its structure theorem decomposition, iff the localization $M_{(p)} = M \otimes_{\Lambda} \Lambda_{(p)}$ (Noetherian localization at the prime ideal $(p) \subset \Lambda$) is free over $\Lambda_{(p)}$ (in particular zero-dimensional over $\Lambda_{(p)}/(p) = \text{Frac}(\mathbb{F}_p[[T]])$ when $r = 0$).

Proof. The summand $\Lambda/p^a \Lambda$ has (p) -localization $\Lambda_{(p)}/p^a \Lambda_{(p)}$, which is non-zero iff $a \geq 1$. The summand $\Lambda/f^b \Lambda$ has (p) -localization zero because f is a unit in $\Lambda_{(p)}$ (distinguished $f \in \mathbb{Z}_p[\gamma - 1]$ has non-zero constant term modulo p). The free part Λ^r has full-rank (p) -localization, which is zero-dimensional over $\text{Frac}(\mathbb{F}_p[[T]])$ iff $r = 0$. Hence $\mu(M) = 0 \Leftrightarrow$ no p -torsion summands, equivalently (p) -localization is free over $\Lambda_{(p)}$ (zero-dimensional over $\text{Frac}(\mathbb{F}_p[[T]])$ precisely when $r = 0$ as well). $\square$

Injective and Projective Resolutions Over Λ

PROPOSITION 24.3.3 (*injective resolution bound*). . Every finitely generated Λ -module M admits an injective resolution $M \rightarrow I^\bullet$ with $I^k = 0$ for $k \geq 3$, where each I^k is a direct sum of injective hulls of simple Λ -modules.

Proof. Λ is two-dimensional regular, hence has global injective dimension 2 (Serre 1963; Matsumura 1989 Thm. 18.1). The injective hulls are $\Lambda^\vee$ (Pontryagin dual of Λ), $\Lambda^\vee/p$, and the residue field $\mathbb{F}_p$ regarded as a Λ -module. Truncation at degree 2 gives the bound. $\square$

PROPOSITION 24.3.4 (*projective resolution bound*). . Every finitely generated Λ -module M admits a projective resolution $P^\bullet \rightarrow M$ with $P_k = 0$ for $k \geq 3$, where each P_k is a finite direct sum of copies of Λ .

Proof. Λ regular of dimension 2; Hilbert syzygy theorem gives the bound (Matsumura 1989 §18). $\square$

PROPOSITION 24.3.5 (*injective and projective resolutions cohere when $\mu = 0$*). . Let M be a finitely generated Λ -module with $\mu(M) = 0$. Then:

- (i) The injective resolution $I^\bullet$ of M reduces modulo $(\gamma - 1)$ to the injective resolution of $M \otimes_\Lambda \mathbb{Z}_p$ over $\mathbb{Z}_p$, i.e. $I^\bullet \otimes_\Lambda \mathbb{Z}_p = I_{\mathbb{Z}_p}^\bullet(M \otimes_\Lambda \mathbb{Z}_p)$.
- (ii) The projective resolution $P_\bullet$ of M reduces modulo p to a $\mathbb{F}_p[[T]]$ -projective resolution of $M \otimes_\Lambda \mathbb{F}_p[[T]]$.
- (iii) The spectral sequence $\mathrm{Tor}_\bullet^\Lambda(M, \mathbb{F}_p[[T]]) \Rightarrow M \otimes_\Lambda^\mathbb{L} \mathbb{F}_p[[T]]$ degenerates at E_2 .

Proof. (1) Injective hulls of Λ -modules without p -torsion are flat over $\mathbb{Z}_p$ (Brumer 1966 §2). Reduction mod $(\gamma - 1)$ commutes with injective resolution for flat $\mathbb{Z}_p$ -modules.

(2) Projective modules are finite sums of Λ ; reduction mod p gives finite sums of $\mathbb{F}_p[[T]]$, which is projective over $\mathbb{F}_p[[T]]$.

(3) The spectral sequence has $E_2^{p,q} = \mathrm{Tor}_{-p}^\Lambda(M, \mathbb{F}_p[[T]])$ concentrated in columns $p \in \{0, -1, -2\}$ by Chapter 24.3.4. The $\mu = 0$ hypothesis makes $\mathrm{Tor}_1^\Lambda(M, \mathbb{F}_p[[T]]) = M[p] = 0$ and $\mathrm{Tor}_2^\Lambda(M, \mathbb{F}_p[[T]]) = 0$ by the Koszul resolution of $\mathbb{F}_p[[T]]$ over Λ (which is $\Lambda \xrightarrow{p} \Lambda \rightarrow \mathbb{F}_p[[T]]$, a length-1 resolution). Hence E_2 is concentrated in one row; degeneration is automatic. $\square$

Co-Derived and Contra-Derived Λ -Categories

Definition 24.3.6 (Λ -co-derived category). Let C be an arithmetic dg Λ -coalgebra. The Λ -co-derived category $D_\Lambda^{\mathrm{co}}(C\text{-comod}_\Lambda)$ is the Verdier quotient of the homotopy category $K(C\text{-comod}_\Lambda)$ of complexes of C -comodules (which are naturally Λ -modules) by the class of complexes $X^\bullet$ such that:

- (i) $X^\bullet$ is homotopy equivalent to a complex of C -injective Λ -modules, and
- (ii) $X^\bullet$ is acyclic in the sense of Positselski 2011 §3.5: $\mathrm{Hom}_{K(\Lambda)}(Y, X^\bullet) = 0$ for all Y in the subcategory of Λ -co-admissible complexes (bounded-below complexes of finitely generated Λ -comodules satisfying Nekovář's admissibility).

Definition 24.3.7 (Λ -contra-derived category). Dually, the Λ -contra-derived category $D_\Lambda^{\mathrm{ctr}}(C^\vee\text{-contramod}_\Lambda)$ is the Verdier quotient of $K(C^\vee\text{-contramod}_\Lambda)$ by complexes that are both projective-homotopy-trivial and $\mathrm{Hom}_{K(\Lambda)}(X^\bullet, Y) = 0$ for all Y in the subcategory of Λ -contra-admissible complexes (Greenberg 2001 §3).

Remark 24.3.8 (comparison with Nekovář). Nekovář 2006 §1.2 defines the Λ -derived category $D(\Lambda\text{-Mod})$ through the abelian category of compact Λ -modules. The co-derived and contra-derived categories above refine this by distinguishing injective-resolution direction from projective-resolution direction; this distinction is invisible over a field (where injective = projective for finite-dimensional vector spaces) but active over Λ .

LEMMA 24.3.9 (representability of co- and contra-derived categories). . When $\mu(C) = 0$, both $D_\Lambda^{\mathrm{co}}(C\text{-comod}_\Lambda)$ and $D_\Lambda^{\mathrm{ctr}}(C^\vee\text{-contramod}_\Lambda)$ are presentable triangulated categories in the sense of Lurie HA 1.4.4, and the compact objects are precisely the finitely generated comodules (resp. contramodules) viewed as complexes in degree zero.

Proof. Presentability follows from Chapter 24.3.3 and Chapter 24.3.4: bounded injective / projective resolutions give finiteness of the generating compact objects. Nakayama plus $\mu = 0$ (Chapter 24.3.2) gives that the compact objects generate the whole category under colimits (Neeman 2001 Thm. 4.4.9 applied to Λ -categories). $\square$

Iwasawa μ -Invariant and Ferrero–Washington

THEOREM 24.3.10 (*Ferrero–Washington 1979, Ann. Math. 109*). . Let K be an abelian number field and let $X_K = \varprojlim_n \text{Cl}(K(\mu_{p^n}))_p$ be the p -part of the cyclotomic ideal class group tower. Then $\mu(X_K) = 0$, i.e., X_K has no p -torsion part in its Λ -module structure theorem decomposition.

Remark 24.3.11 (*applicability to Heisenberg bar*). The arithmetic Heisenberg bar $B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ is a Λ -module via the cyclotomic twist of the Gersten differential d_{Iw} . Since $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ is abelian (the Heisenberg is the abelianization of the free boson algebra), the cyclotomic twist is through abelian extensions of $\mathbb{Q}$. By Ferrero–Washington, the p -torsion part of the associated Iwasawa class group vanishes; by arithmetic duality (Milne 1986 *Arithmetic Duality Theorems* Ch. 2), this propagates to $\mu(B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})) = 0$.

COROLLARY 24.3.12 ($\mu = 0$ for arith. Heisenberg). . $\mu(B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})) = 0$.

Proof. The Λ -module $B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ has Γ -action through χ_{cycl} on the Gersten component. Its cyclotomic Iwasawa invariant equals the μ -invariant of the Iwasawa class group of the associated cyclotomic tower, which is abelian since the Heisenberg is abelian. Ferrero–Washington gives $\mu = 0$ for abelian cyclotomic μ -invariants (Chapter 24.3.10). $\square$

24.4 THE MAIN THEOREM

Overview of the Proof

The proof of Chapter 24.1.5 proceeds in six steps:

- (i) (Construction of the pair of functors.) Define functors $\Psi: D_{\Lambda}^{\text{co}} \rightarrow D_{\Lambda}^{\text{ctr}}$ and $\Phi: D_{\Lambda}^{\text{ctr}} \rightarrow D_{\Lambda}^{\text{co}}$ as the Λ -lifts of Positselski's $\mathcal{M} \mapsto \mathcal{M}^{\vee}$ and $N \mapsto N^{\vee\vee}$ functors.
- (ii) (Base-change to $\mathbb{F}_p[[T]]$.) Reduce along $\Lambda \rightarrow \mathbb{F}_p[[T]]$ and apply the classical Positselski result over the one-dimensional valuation ring.
- (iii) (Lifting along Nakayama.) Use complete Noetherianity of Λ plus $\mu = 0$ to lift the base-change equivalence to a Λ -equivalence.
- (iv) (Bar–cobar adjunction check.) Verify that $\Omega^{\Lambda} \dashv B^{\Lambda}$ factors through the equivalence.
- (v) (Unit / counit quasi-isomorphism.) Verify that the derived unit $C \rightarrow \Omega^{\Lambda} B^{\Lambda}(C)$ and counit $B^{\Lambda} \Omega^{\Lambda}(A) \rightarrow A$ are quasi-isomorphisms under $\mu = 0$.
- (vi) (Independence of tower presentation.) Check that the equivalence does not depend on the choice of $\Lambda \rightarrow \mathbb{F}_p[[T]]$ reduction.

Step 1: Construction of the Functors

Definition 24.4.1 (*functor Ψ*). Define $\Psi: K(C\text{-comod}_{\Lambda}) \rightarrow K(C^{\vee}\text{-contramod}_{\Lambda})$ on the level of chain complexes by

$$\Psi(M^{\bullet})^k := \text{Hom}_{\Lambda}^{\bullet}(C, M^{\bullet})_k = \bigoplus_{p+q=k} \text{Hom}_{\Lambda}(C_{-p}, M^q),$$

with contramodule structure from post-composition with $C^{\vee}$ -multiplication on $M^{\bullet}$. The differential is $d_{\Psi}(f) = d_M \circ f - (-1)^{|f|} f \circ d_C$.

Definition 24.4.2 (*functor Φ*). Define $\Phi: K(C^{\vee}\text{-contramod}_{\Lambda}) \rightarrow K(C\text{-comod}_{\Lambda})$ by

$$\Phi(N^{\bullet})^k := N^{\bullet} \widehat{\otimes}_{\Lambda} C_k = \varprojlim_m (N^{\bullet} \otimes_{\Lambda} C_k / \mathfrak{m}^m),$$

the $\mathfrak{m}$ -adic completion of the tensor product, with comodule structure from the coaction on C . The differential is $d_\Phi(n \otimes c) = d_N(n) \otimes c + (-1)^{|n|} n \otimes d_C(c)$.

LEMMA 24.4.3 (*well-definedness of Ψ, Φ*). . Both Ψ and Φ descend to exact functors between triangulated categories $\Psi: D_\Lambda^{\text{co}}(C\text{-comod}) \rightarrow D_\Lambda^{\text{ctr}}(C^\vee\text{-contramod})$ and Φ in the opposite direction.

Proof. Ψ sends C -injective complexes to $C^\vee$ -projective complexes (dualization of injective hulls gives projective hulls of contramodules; Positselski 2018 arXiv:1209.2995 §3.1). Exactness of Ψ on short exact sequences of injective complexes follows from Hom_Λ being exact on Λ -injective modules (Chapter 24.3.3). Similarly for Φ using projective resolutions. $\square$

Step 2: Base Change to $\mathbb{F}_p[[T]]$

LEMMA 24.4.4 (*base-change compatibility*). . Let $\tilde{C} = C \otimes_\Lambda \mathbb{F}_p[[T]]$ and $\tilde{M} = M \otimes_\Lambda \mathbb{F}_p[[T]]$. Under $\mu(C) = 0$:

- (i) $\tilde{C}$ is a conilpotent dg coalgebra over $\mathbb{F}_p[[T]]$.
- (ii) The diagram

$$\begin{array}{ccc} D_\Lambda^{\text{co}}(C\text{-comod}_\Lambda) & \xrightarrow{\Psi} & D_\Lambda^{\text{ctr}}(C^\vee\text{-contramod}_\Lambda) \\ (-) \otimes_\Lambda \mathbb{F}_p[[T]] \downarrow & & \downarrow (-) \otimes_\Lambda \mathbb{F}_p[[T]] \\ D_{\mathbb{F}_p[[T]]}^{\text{co}}(\tilde{C}\text{-comod}) & \xrightarrow{\tilde{\Psi}} & D_{\mathbb{F}_p[[T]]}^{\text{ctr}}(\tilde{C}^\vee\text{-contramod}) \end{array}$$

commutes, where $\tilde{\Psi}$ is the classical Positselski functor over $\mathbb{F}_p[[T]]$.

- (iii) $\tilde{\Psi}$ is an equivalence of triangulated categories.

Proof. (1) Reduction of a conilpotent coalgebra along a faithfully flat ring map is conilpotent: the iterated comultiplication $\bar{\Delta}^{(n)}$ equals $\Delta^{(n)} \otimes 1$, which is zero on $\tilde{C}$ for n large because it is so on $\bar{C}$.

(2) Commutativity: Ψ is given by $\text{Hom}_\Lambda(C, M)$. On each finite piece $C_{\leq n}$ of the filtered union, base change satisfies $\text{Hom}_\Lambda(C_{\leq n}, M) \otimes_\Lambda \mathbb{F}_p[[T]] = \text{Hom}_{\mathbb{F}_p[[T]]}(\tilde{C}_{\leq n}, \tilde{M})$ when $C_{\leq n}$ is finitely generated projective over Λ and M is Λ -flat (Positselski 2018 arXiv:1209.2995 §4.4; the Λ -flatness of the relevant comodule representatives is the Λ -flatness assumption). Passing to the colimit $C = \varinjlim_n C_{\leq n}$ commutes with $\text{Hom}_\Lambda(-, M)$ on injective resolutions because each $C_{\leq n}/C_{\leq n-1}$ is finitely generated. Under $\mu(C) = 0$, Chapter 24.3.5 gives the injective-resolution compatibility; for M we choose a Λ -flat representative, which exists because the derived category of Λ -modules has enough flats (Neeman 2001 Thm. 4.4.9).

(3) The classical Positselski equivalence over the one-dimensional Noetherian local regular ring $\mathbb{F}_p[[T]]$: this is Positselski 2018 arXiv:1209.2995 Thm. 4.6.5. The one-dimensional regular case requires slightly less than the field case but more than the general Noetherian case; the theorem holds when the ring is Gorenstein local of dimension 1 with finitely generated injective hull (Bass 1962; Positselski 2018 §4). $\square$

Step 3: Lifting Along Nakayama

PROPOSITION 24.4.5 (*lifting lemma*). . Under $\mu(C) = 0$, the base-change functor $D_\Lambda^{\text{co}}(C) \rightarrow D_{\mathbb{F}_p[[T]]}^{\text{co}}(\tilde{C})$ is faithful, and an object $M^\bullet \in D_\Lambda^{\text{co}}(C)$ is in the image iff $\tilde{M}^\bullet = M^\bullet \otimes_\Lambda \mathbb{F}_p[[T]]$ lifts to a Λ - C -comodule complex modulo $\mathfrak{m}$ -adic completeness; this lift is unique up to unique isomorphism when $\mu(C) = 0$.

Proof. Faithfulness: if $f, g: M^\bullet \rightarrow N^\bullet$ in the co-derived category with $\bar{f} = \bar{g}$, then $f - g$ lies in the kernel of base change. The kernel consists of maps factoring through $\ker(\Lambda \rightarrow \mathbb{F}_p[[T]]) = p\Lambda$. Under $\mu(C) = 0$, the p -divisible part of $\mathrm{Hom}_\Lambda(M^\bullet, N^\bullet)$ in the co-derived category is zero by Chapter 24.3.5(3): the Tor spectral sequence degenerates and the Hom-Tor cross is zero in positive degrees. Hence $f = g$ in the co-derived category.

Essential image: given $\bar{M}^\bullet$ in $D_{\mathbb{F}_p[[T]]}^{\mathrm{co}}(\bar{C})$, seek a Λ -lift. Set $\Lambda_n = \Lambda/\mathfrak{m}^n$; the quotient tower $\Lambda_1 \leftarrow \Lambda_2 \leftarrow \cdots \leftarrow \Lambda$ is cofinal in the $\mathfrak{m}$ -adic filtration, and each Λ_n is a finite Artinian local ring. By Positselski's 2018 extension to Artinian rings (arXiv:1209.2995 Thm. 4.7.2), the classical Positselski correspondence applies over each Λ_n . The inverse limit $\lim_n \bar{M}_{\Lambda_n}^\bullet$ under $\mathfrak{m}$ -adic completeness gives $\lim_n \bar{M}_{\Lambda_n}^\bullet \in D_\Lambda^{\mathrm{co}}(C)$.

Uniqueness: under $\mu(C) = 0$, the obstruction groups $H^1(\Lambda, \mathrm{Aut}(\bar{M}^\bullet)) = 0$ by a standard Nakayama-style argument (complete + $\mu = 0$ implies deformation space collapses to a point). $\square$

COROLLARY 24.4.6 (Ψ is an equivalence on D_Λ^{co}). . Under $\mu(C) = 0$, the functor $\Psi: D_\Lambda^{\mathrm{co}}(C\text{-comod}_\Lambda) \rightarrow D_\Lambda^{\mathrm{ctr}}(C^\vee\text{-contramod}_\Lambda)$ is an equivalence of triangulated categories, with inverse Φ .

Proof. By Chapter 24.4.4, the reduction $\bar{\Psi}$ is an equivalence. By Chapter 24.4.5, the base-change functor $D_\Lambda^{\mathrm{co}} \rightarrow D_{\mathbb{F}_p[[T]]}^{\mathrm{co}}$ is fully faithful, and an object is in the essential image iff it is $\mathfrak{m}$ -adically complete (which holds for the image of any Λ -complex of finitely generated Λ -modules). The commutative diagram Chapter (ii) plus faithfulness of the vertical arrows plus equivalence of the bottom arrow plus essential surjectivity of Ψ (which is equivalent to essential surjectivity of $\bar{\Psi}$ composed with the lift of Chapter 24.4.5) imply that Ψ is an equivalence.

The inverse is Φ : by construction $\Phi \circ \Psi(M^\bullet) = \mathrm{Hom}_\Lambda(C, M^\bullet) \widehat{\otimes}_\Lambda C$. Over a field, $\mathrm{Hom}(C, M) \otimes C \simeq M \otimes (C \otimes C^\vee)$ reduces to M via the coevaluation/evaluation of bar-cobar duality. Over Λ with $\mu = 0$, the same argument works using the $\mathfrak{m}$ -adic completion and the projection formula $\mathrm{Hom}_\Lambda(C, M) \widehat{\otimes}_\Lambda C = M$ valid for M Λ -flat with $\mu = 0$ coalgebra C . $\square$

Step 4: Bar-Cobar Adjunction

THEOREM 24.4.7 (Λ -bar-cobar adjunction). . Let Ω^Λ and B^Λ denote the bar and cobar functors on Λ -dg-coalgebras and Λ -dg-algebras respectively (generalizing the classical Positselski 2011 bar / cobar pair). Then $(\Omega^\Lambda, B^\Lambda)$ is an adjoint pair $\Omega^\Lambda \dashv B^\Lambda$ on the derived categories $D(\Lambda\text{-Alg}^{\mathrm{aug}})$ and $D(\Lambda\text{-Coalg}^{\mathrm{conil}})$.

Proof. The bar-cobar adjunction over a field is Positselski 2011 §1.6. Over Λ : the formulas

$$B^\Lambda(A) = \bigoplus_{n \geq 0} (s^{-1}\bar{A})^{\otimes_\Lambda n}, \quad \Omega^\Lambda(C) = \bigoplus_{n \geq 0} (s\bar{C})^{\otimes_\Lambda n}$$

with the standard bar / cobar differentials give a pair of functors on the Λ -level. Adjunction $\mathrm{Hom}_{\mathrm{Coalg}_\Lambda}(C, B^\Lambda(A)) = \mathrm{Hom}_{\mathrm{Alg}_\Lambda}(\Omega^\Lambda(C), A)$ follows formally from the same calculation as over a field (both sides are twisting cochains $C \rightarrow A$ in the sense of Lefevre-Hasegawa 2003; the formalism commutes with the base ring Λ because the bar / cobar differentials are Λ -linear). $\square$

Step 5: Unit / Counit Quasi-Isomorphism

THEOREM 24.4.8 (unit / counit of bar-cobar are quasi-isomorphisms). . Under $\mu(C) = 0$ (resp. $\mu(A) = 0$ on the Selmer dual), the unit $\eta_C: C \rightarrow B^\Lambda \Omega^\Lambda(C)$ and counit $\varepsilon_A: \Omega^\Lambda B^\Lambda(A) \rightarrow A$ are quasi-isomorphisms of Λ -dg-coalgebras and Λ -dg-algebras respectively.

Proof. The classical statement over a field is Positselski 2011 §6.4. Over Λ : by Chapter 24.3.5, the reduction of the bar / cobar complexes modulo $(\gamma - 1)$ gives the $\mathbb{Z}_p$ -coefficient bar / cobar, which is a filtered colimit of finitely generated pieces. The statement modulo each $\mathfrak{m}^n$ reduces to the classical case over the Artinian ring $\Lambda/\mathfrak{m}^n$. Since Λ is $\mathfrak{m}$ -adically complete and $\mu(C) = 0$ ensures no p -torsion obstruction to lifting, the inverse limit of the Artinian quasi-isomorphisms gives a quasi-isomorphism over Λ .

Explicit computation: the unit $\eta_C(c) = \sum_n [c_{(1)} | \cdots | c_{(n)}]$ where $\Delta^{(n-1)}(c) = \sum c_{(1)} \otimes \cdots \otimes c_{(n)}$, is well-defined because conilpotence makes the sum finite on each element. Its kernel / cokernel computed on each $\Lambda/\mathfrak{m}^n$ is zero by Positselski 2011 §6.4; hence kernel / cokernel is zero on the inverse limit when $\mu(C) = 0$. $\square$

Step 6: Independence of Tower Presentation

LEMMA 24.4.9 (*independence of tower*). . The equivalence $\Psi: D_\Lambda^{\text{co}} \rightarrow D_\Lambda^{\text{ctr}}$ does not depend on the choice of Artinian quotient tower $\{\Lambda/\mathfrak{m}^n\}_n$ used in the proof of Chapter 24.4.5. Any two cofinal towers give the same equivalence up to unique natural isomorphism.

Proof. Standard Mittag-Leffler argument. Two cofinal Artinian quotient towers $\{\Lambda/\mathfrak{a}_n\}_n$ and $\{\Lambda/\mathfrak{a}'_n\}_n$ (with $\mathfrak{a}_n, \mathfrak{a}'_n$ open ideals cofinal with the $\mathfrak{m}$ -adic filtration) admit a common refinement $\{\Lambda/(\mathfrak{a}_n + \mathfrak{a}'_n)\}_n$. The inverse limits over the two towers and the common refinement agree under $\mu = 0$ $\mathfrak{m}$ -adic completeness; hence the two equivalences are canonically naturally isomorphic. $\square$

Assembly: the Main Theorem

Proof of Chapter 24.1.5. Assemble Steps 1–6. Step 1 constructs the functor pair Ψ, Φ (Chapter 24.4.1, Chapter 24.4.2). Step 2 reduces to $\mathbb{F}_p[[T]]$ (Chapter 24.4.4) and applies Positselski 2018 for the classical equivalence over the one-dimensional valuation ring. Step 3 lifts the equivalence along Nakayama (Chapter 24.4.6). Step 4 verifies that the bar–cobar adjunction factors through the equivalence (Chapter 24.4.7). Step 5 verifies that the unit / counit are quasi-isomorphisms (Chapter 24.4.8). Step 6 shows the equivalence is independent of choices (Chapter 24.4.9).

Combining: under $\mu(C) = 0$ (plus dual μ -condition on the contra-module side), Ψ is an equivalence, and the bar–cobar adjunction restricts to an adjoint pair inducing the equivalence between the co-derived category of Λ -comodules and the contra-derived category of Λ -contra-modules. This is Chapter 24.1.5. $\square$

Remark 24.4.10 (*dual μ -invariant condition*). The hypothesis on the contra-module side is that the Selmer μ -invariant of the $C^\vee$ -contra-module tower vanishes. Under Iwasawa duality (Coates–Sujatha 2006 §3.1, Poitou–Tate arithmetic duality adapted to Iwasawa towers), the Selmer μ -invariant equals the classical μ -invariant modulo pseudo-nullity. Hence under $\mu(C) = 0$ for the comodule side, the dual Selmer μ condition is automatically satisfied for $C^\vee$.

24.5 APPLICATION: STEP 3 OF PI FOR THE ARITHMETIC HEISENBERG

The Step-3 Residual

Step 3 of the proof of Chapter 21.5.1 asserts that the arithmetic Positselski equivalence identifies co-derived arithmetic comodules over the bar coalgebra with contra-derived contra-modules over the cobar algebra, under which identification:

- The level κ_G^{Ar} of the Heisenberg equals the degree of the central character of the bar coalgebra.

- The dual level $(\kappa^!)_{\mathbb{G}}^{\text{Ar}}$ equals the degree of the central character of the cobar algebra = Bost–Connes generator degree.

The residual was the conditional use of arithmetic Positselski. With Chapter 24.1.5 stated as a conditional theorem (under the five open sub-items of Chapter 24.1.6, with $\mu = 0$ discharged by Ferrero–Washington for the abelian Heisenberg case), Step 3 closes conditionally on those five sub-items.

The Explicit Weight Computation

THEOREM 24.5.1 (*weight computation for arith. Heisenberg*). . Let $C = B^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ be the arithmetic Heisenberg bar coalgebra, and $\mathcal{A} = C^{\vee} = \Omega^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ be its cobar. Under the arithmetic Positselski equivalence (Chapter 24.1.5, applicable by Chapter 24.3.12) and its five stated sub-item hypotheses:

- The generators of C are the Heisenberg oscillators $\{\alpha_{-n}^{(p)}\}_{n \geq 1, p \text{ prime}}$, with HY-zeta weight $+n \log p$ on $\alpha_{-n}^{(p)}$.
- The generators of $\mathcal{A}$ are the Bost–Connes Hecke elements $\{\mu_n\}_{n \geq 1}$, with HY-zeta weight $-\log n$ on μ_n .
- The Positselski pairing sends $\alpha_{-n}^{(p)} \leftrightarrow \mu_{p^n}$ (matching one oscillator to one Hecke element at a prime power).
- The total HY-zeta weight under the pairing is zero: $n \log p + (-n \log p) = 0$ at each prime-power pair.

Proof. (1) The generators of the bar coalgebra are the reduced generators of the algebra being barred: here $\mathcal{V}_{\text{Heis}}^{\text{prim}} = \bigotimes'_p \mathcal{H}^{(p)}$ is generated (as an adelic Heisenberg Fock) by the local oscillators $\alpha_{-n}^{(p)}$ at each prime, with conformal weight n . Under the HY-zeta grading (Chapter 6.6.2 (ii)), the weight at prime p is $n \log p$ per oscillator.

(2) The cobar dual of a bar coalgebra generated by $\{\alpha_{-n}^{(p)}\}$ is the algebra generated by the dual symbols $\{\alpha_{-n}^{(p)\vee}\}$. Under Koszul duality (Chapter 12.5.2; Bost–Connes Selecta Math. 1, 1995), $\mathcal{A}$ is canonically the Bost–Connes Hecke C^* -algebra, generated by $\{\mu_n\}_{n \geq 1}$. The generator μ_n is the Bost–Connes partial isometry on the class $n \in \mathbb{N}^\times$, with Bost–Connes KMS_β weight $n^{-\beta}$ at inverse temperature $\beta = s$. Under the HY-zeta grading this translates to weight $-\log n$.

(3) The pairing: under bar–cobar duality the pairing of $\alpha_{-n}^{(p)}$ with its dual is by evaluation. The Bost–Connes matching: by the Euler product for the arithmetic Heisenberg partition function (Chapter 12.5.1, Julia 1990), $Z_{\text{primon}}(\beta) = \prod_p (1 - p^{-\beta})^{-1} = \zeta(\beta)$. The prime-power term $p^{-n\beta}$ at exponent n in the expansion of $(1 - p^{-\beta})^{-1}$ corresponds to the oscillator $\alpha_{-n}^{(p)}$; the Bost–Connes dual μ_{p^n} has KMS_β weight $(p^n)^{-\beta} = p^{-n\beta}$. Hence $\alpha_{-n}^{(p)} \leftrightarrow \mu_{p^n}$ under the Positselski pairing.

(4) Total weight: $\alpha_{-n}^{(p)}$ has weight $n \log p$ (1); μ_{p^n} has weight $-\log(p^n) = -n \log p$ (2). Sum: $n \log p + (-n \log p) = 0$. Hence the pairing is weight-preserving (total weight zero at each oscillator), establishing $\kappa_{\mathbb{G}}^{\text{Ar}} + (\kappa^!)_{\mathbb{G}}^{\text{Ar}} = 0$. $\square$

Global Euler-Product Check

PROPOSITION 24.5.2 (*global weight sum*). . The global HY-zeta weight of the arithmetic Positselski pairing over all prime-power pairs is

$$\sum_{p \text{ prime}} \sum_{n \geq 1} [n \log p \cdot p^{-ns} + (-n \log p) \cdot p^{-ns}] = 0,$$

convergent absolutely for $\operatorname{Re}(s) > 1$. After termwise pairing it is the zero function, hence extends as zero. No meromorphic continuation of the unpaired prime-indexed sum is asserted.

Proof. Term-by-term: at each prime p and each $n \geq 1$, the two weights $n \log p$ and $-n \log p$ sum to zero. The resulting global series is identically zero, which trivially converges. The unpaired sum $\sum_{p,n} n \log p \cdot p^{-ns}$ is only used in the half-plane of absolute convergence. The identity $-\zeta'(s)/\zeta(s) = \sum_{p,k \geq 1} (\log p) p^{-ks}$ is a prime-power identity, not an identity for the prime-indexed sum alone. $\square$

COROLLARY 24.5.3 (*Step 3 of P1 closes conditionally on arithmetic Positselski*). (*conditional on the five open sub-items of Chapter 24.1.6*). With Chapter 24.1.5 stated as a conditional theorem (under the five open Positselski sub-items), Step 3 of the proof of Chapter 21.5.1 closes for the arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ modulo the same five sub-items: $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$ holds as a conditional theorem, not unconditionally.

Proof. Chapter 24.3.12: $\mu = 0$ holds for the arithmetic Heisenberg bar coalgebra by Ferrero–Washington. Chapter 24.1.5: the arithmetic Positselski equivalence holds under $\mu = 0$. Chapter 24.5.1: the weight computation gives total zero weight. Assembled: the level-weight vanishing $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$ is a conditional theorem for the arithmetic Heisenberg, closing Step 3 of P1 conditionally on the five open Positselski sub-items. $\square$

Full Biconditional of P1 Closes Conditionally

THEOREM 24.5.4 (*P1 full biconditional, conditional form*). (*conditional on the five open sub-items of Chapter 24.1.6*). For the arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim}}$,

$$\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \iff \xi(s) = \xi(1-s),$$

conditional in both directions on the five open Positselski sub-items (continuous duals, derived completion, faithful lifting, primitive zeta-character theorem, Koszul-dual Bost–Connes).

Proof. The proof of Chapter 21.5.1 consists of four steps:

- (i) Primon-gas identification of the bar generating function: theorem (Julia 1990; Spector 1990).
- (ii) Koszul-dual identification with Bost–Connes: conditional on the Koszul-dual Bost–Connes sub-item (Bost–Connes 1995 Selecta Math. 1; Chapter 12.5.2).
- (iii) Positselski pairing of κ and $\kappa^!$: conditional theorem (Chapter 24.5.3) on the remaining four open sub-items.
- (iv) Symmetry $s \mapsto 1-s$ from Positselski pairing: theorem (Tate 1950 adelic Poisson summation; Chapter 21.5.1 Step 4).

Steps 1 and 4 are unconditional theorems; Steps 2 and 3 are conditional on the five open sub-items of Chapter 24.1.6. Assembling as in the proof of Chapter 21.5.1, the biconditional follows conditionally on those sub-items. $\square$

Remark 24.5.5 (form propagation through Chapter 22). Chapter 24.5.4 keeps the form of Chapter 21.5.1 at *ProvedHere conditional on arithmetic Positselski*: the condition is the conjunction of the five open sub-items of Chapter 24.1.6, not a bare statement that the Positselski equivalence is discharged. Formal-consequence forms that propagate:

- Chapter 21.5.2: the conditional obstruction remains, refined to the five open sub-items.
- Chapter 12.6.1 (Chapter 10): remains Conditional, with Chapter 24.5.4 as the conditional proof pointer.
- Chapter 22.6.1: arith. Positselski remains on the RH branch as one of the four conditional inputs (after the determinacy comparison Chapter 38.3.1).

The residual splitting residual list of Chapter C.10 lists arith. Positselski with its five open sub-items; four residuals remain on four-input RH branch.

24.6 ARCHIMEDEAN EXTENSION: GAUSSIAN POSITSSELSKI

The Archimedean Analogue

At the archimedean place, the base ring changes from $\Lambda = \mathbb{Z}_p[[\Gamma]]$ to $\mathbb{R}$ (or $\mathbb{C}$), and the Iwasawa deformation becomes a *Gaussian deformation* of the Heisenberg Fock: the archimedean local factor is $\mathcal{H}^{(\infty)} = \mathbb{C}[\alpha_{-1}, \alpha_{-2}, \dots] \cdot |0\rangle_\infty$ with Gaussian inner product (Segal 1981; Frenkel–Ben-Zvi 2004 §3.5).

THEOREM 24.6.1 (Gaussian Positselski). . For the archimedean Heisenberg $\mathcal{H}^{(\infty)}$ with Gaussian inner product, the classical Positselski equivalence (Chapter 24.1.1) holds with base field $\mathbb{C}$. The pairing is induced by Fourier self-duality of the Gaussian: $\mathcal{F}_{\mathbb{R}} e^{-\pi x^2} = e^{-\pi y^2}$, which translates to bar–cobar duality between the Heisenberg Fock and its dual.

Proof. (The vertex algebra language in this proof applies exclusively to the archimedean fibre $\mathcal{H}^{(\infty)}$. The global object $\mathcal{V}^{\text{prim}}$ is a Hochschild trace class, not a vertex algebra; cf. CLAUDE.md Out of Domain clause.)

At the archimedean fibre, $\mathcal{H}^{(\infty)}$ is a $\mathbb{C}$ -Fock module for the archimedean Heisenberg algebra $\mathfrak{h}_{\mathbb{C}} = \mathbb{C}[\alpha_{-n}, \alpha_n : n \geq 1]$; it carries a natural vertex algebra structure (Kac 1998 *Vertex Algebras for Beginners* §4.6; Frenkel–Ben-Zvi 2004 §3.5). Classical Positselski over $\mathbb{C}$ applies directly (Chapter 24.1.1). The Fourier identification of the pairing: the theta function $\theta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t}$ is a generating function for the Heisenberg Fock states, and Jacobi’s theta functional equation $\theta(t) = t^{-1/2} \theta(1/t)$ encodes Fourier self-duality. The pairing $\text{Hom}(B^{\text{ch}}(\mathcal{H}^{(\infty)}), \mathcal{H}^{(\infty)})$ is Fourier-invariant, matching bar–cobar duality with the archimedean Fourier transform. $\square$

Adelic Assembly

THEOREM 24.6.2 (adelic arithmetic Positselski). . Combining Chapter 24.1.5 at each finite place p and Chapter 24.6.1 at ∞ , the adelic arithmetic Positselski equivalence is the restricted tensor product of local equivalences

$$\bigotimes_{v \in |\overline{C}|}^{\cdot} D_{\Lambda_v}^{\text{co}}(C_v\text{-comod}_{\Lambda_v}) \simeq \bigotimes_{v \in |\overline{C}|}^{\cdot} D_{\Lambda_v}^{\text{ctr}}(C_v^{\vee}\text{-contramod}_{\Lambda_v})$$

where $\Lambda_p = \mathbb{Z}_p[[\Gamma_p]]$ at finite p , $\Lambda_\infty = \mathbb{C}$, and the restricted tensor product is taken with respect to spherical vectors $|0\rangle_v$ at almost all v .

Proof. Local equivalence at each v is Chapter 24.1.5 (finite p) or Chapter 24.6.1 (archimedean). The restricted tensor product is the Tate–Weil restricted direct product (Tate 1950 §2.3; Weil 1967 Ch. VII §1) applied to ∞ -categorical triangulated categories with distinguished units. The assembly is formal once the local equivalences hold and unit choices are fixed. $\square$

Remark 24.6.3 (adelic functional equation as adelic Positselski). Under the adelic arithmetic Positselski equivalence (Chapter 24.6.2), the functional equation $\xi(s) = \xi(1-s)$ becomes an assertion about the Fourier-invariance of the adelic bar–cobar pairing: $\xi(s)$ is the Mellin transform of the adelic theta distribution Θ , and Fourier invariance $\mathcal{F}_{\mathbb{A}}\Theta = \Theta$ gives $\xi(s) = \xi(1-s)$. The arithmetic Positselski equivalence is the categorical reflection of this Fourier self-duality; the functional equation is its character-level shadow.

24.7 RESIDUAL CONDITIONS

Chapter 24.1.5 and its corollaries suffice for the G-row claim of P1 for the abelian arithmetic Heisenberg. Three extensions are recorded as open problems.

Non-Abelian Iwahori Level

For the full non-abelian Iwahori-level GL_2 Hecke category $\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}$ (the target of F3 in Chapter 21), the Λ -coefficient is the non-commutative Iwasawa algebra $\Lambda_{GL_2} = \mathbb{Z}_p[[GL_2(\mathbb{Z}_p)]]$.

Conjecture 24.7.1 (non-commutative arithmetic Positselski). For a conilpotent dg Λ_{GL_2} -coalgebra C with non-commutative μ -invariant $\mu_{\mathrm{nc}}(C) = 0$ (in the sense of Fukaya–Kato 2006 §1–2), there is an equivalence $D_{\Lambda_{GL_2}}^{\mathrm{co}}(C) \simeq D_{\Lambda_{GL_2}}^{\mathrm{ctr}}(C^{\vee})$.

Remark 24.7.2 (form). The non-commutative Iwasawa main conjecture of Coates–Fukaya–Kato–Sujatha–Venjakob 2005 (Publ. IHES) and Fukaya–Kato 2006 (Proc. St. Petersburg Math. Soc. 12) supplies the machinery for non-commutative Λ -module theory, including a non-commutative structure theorem modulo pseudo-nullity. The extension of the classical Positselski correspondence to this setting would require an analogue of Chapter 24.3.5 for non-commutative Λ : the injective and projective resolution towers must commute with reduction modulo the center of Λ_{GL_2} . This is expected to hold but is not established here.

Whittaker Model Compatibility

For the B-row and higher rows of the complementarity ceiling (Vol I Thm. C), the local factors involve Whittaker models at each finite place. Whittaker functions are Λ -modules with $GL_2(\mathbb{Z}_p)$ -equivariance, and the Positselski pairing must respect this equivariance.

Conjecture 24.7.3 (Whittaker-compatible Positselski). For a conilpotent dg Λ -coalgebra C with $GL_2(\mathbb{Z}_p)$ -equivariance and $\mu(C) = 0$, the arithmetic Positselski equivalence restricts to Whittaker-compatible subcategories: $D_{\Lambda}^{\mathrm{co}, \mathrm{Wh}}(C) \simeq D_{\Lambda}^{\mathrm{ctr}, \mathrm{Wh}}(C^{\vee})$, with the restriction compatible with Satake and Whittaker equivalences at each place.

Extension to Higher Rank

Chapter 24.1.5 is stated for cyclotomic $\Lambda = \mathbb{Z}_p[[\Gamma]]$. The higher-rank generalization replaces Γ with a profinite $\mathbb{Z}_p^d$ (as in the d -dimensional Iwasawa theory of Greenberg 2001). The main theorem extends

mutatis mutandis, with the μ -invariant replaced by the Greenberg d -dimensional μ -invariant. Ferrero–Washington extends to the d -dimensional abelian case by induction, so the Heisenberg application generalizes to higher-rank arithmetic Heisenbergs.

Remark 24.7.4 (higher-rank extension). The cyclotomic case $\Gamma \cong \mathbb{Z}_p$ suffices for the Pi G-row application. The d -dimensional case $\Gamma \cong \mathbb{Z}_p^d$ follows by induction on d , with Ferrero–Washington extended to d -fold abelian towers.

24.8 POSITSIELSKI BOUNDARY

- (i) Chapter 24.1.5: for a conilpotent dg Λ -coalgebra C with $\mu(C) = 0$, there is a canonical equivalence $D_\Lambda^{\text{co}}(C\text{-comod}_\Lambda) \simeq D_\Lambda^{\text{ctr}}(C^\vee\text{-contramod}_\Lambda)$. The six-step proof (Chapter 24.4–Chapter 24.4) reduces the Λ -coefficient equivalence to the classical Positselski equivalence over $\mathbb{F}_p[[T]]$ (Positselski 2018 arXiv:1209.2995 Thm. 4.6.5) via Nakayama lifting.
- (ii) Six obstruction comparisons (Chapter 24.2) all close for the abelian Heisenberg case; the non-abelian Iwahori extension is residual (Chapter 24.7).
- (iii) The hypothesis $\mu(C) = 0$ is precisely the Iwasawa μ -invariant vanishing. For the arithmetic Heisenberg bar coalgebra, $\mu(B^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})) = 0$ by Ferrero–Washington 1979 (Ann. Math. 109) applied to the abelian cyclotomic extensions (Chapter 24.3.12).
- (iv) Weight computation (Chapter 24.5.1): the arithmetic Heisenberg bar generators $\alpha_{-n}^{(p)}$ (HY-zeta weight $n \log p$) pair with the Bost–Connes cobar generators μ_{p^n} (HY-zeta weight $-n \log p$) under the Positselski pairing, yielding total weight zero, i.e. $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$.
- (v) Step 3 of the Pi proof (Chapter 24.5.1) closes conditionally on the five open sub-items of Chapter 24.1.6 (Chapter 24.5.3); consequently the full biconditional $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$ of Pi closes conditionally on the same sub-items (Chapter 24.5.4).
- (vi) Adelic extension: combining the Λ -coefficient Positselski at each finite place with Gaussian Positselski (Chapter 24.6.1) at ∞ gives the adelic Positselski equivalence (Chapter 24.6.2); the functional equation $\xi(s) = \xi(1-s)$ is the character-level shadow of this adelic Positselski equivalence (Chapter 24.6.3).
- (vii) Residual condition (Chapter 24.7): non-commutative arithmetic Positselski for non-abelian Iwahori level (Chapter 24.7.1); Whittaker-compatible Positselski (Chapter 24.7.3); higher-rank Iwasawa extension (Chapter 24.7.4).

The essential theorem of the arithmetic branch $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$ is stated here as a conditional theorem for the abelian arithmetic Heisenberg: it closes modulo the five open Positselski sub-items of Chapter 24.1.6. In the Vol IV comparison, the assertion “arith. Positselski” is precisely the four-input RH branch residual (Chapter C.10): its content is the conjunction of those five sub-items. The sufficient closure path to RH (Chapter 22.6.1) carries four inputs after the determinacy comparison (Chapter 38.3.1): arith. Positselski (Chapter 24.5.4), Koszul–Pet compat, H-P^{Ar}, and VB* main direction.

KOSZUL–PETERSSON PAIRING COMPATIBILITY

The P_3 closure of Chapter 21 left one residual identity: that the arithmetic Koszul pairing of $\mathcal{V}^{\text{prim}}$ with $(\mathcal{V}^{\text{prim}})^!$ coincides, after Bost–Connes involution on the dual, with the adelic Petersson–Tamagawa inner product on $L^2([GL_2])$. Here $\mathcal{V}^{\text{prim}}$ is the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 ; it is not a vertex algebra. Heisenberg–Fock and CCR language in this argument is confined to the archimedean fibre $\mathcal{H}^{(\infty)}$; the finite-place factors $\mathcal{H}^{(p)}$ are $C_c(GL_2(\mathbb{Z}_p) \backslash GL_2(\mathbb{Q}_p) / GL_2(\mathbb{Z}_p))$, carrying no Heisenberg Lie algebra structure. The compatibility is proved place by place and the local identifications are assembled over the adèle ring via Tate’s restricted tensor product, establishing P_3 on the normalized sph-fin cuspidal core (domain-restricted; Chapter 25.1.2).

25.1 STATEMENT RECALLED

Setup

The primary arithmetic Heisenberg, $\mathcal{V}_{\text{Heis}}^{\text{prim}}$, denotes the local-factor decomposition of $\mathcal{V}^{\text{prim}}$ with Heisenberg structure on the archimedean fibre only (the subscript records the archimedean Fock presentation; finite factors carry no Heisenberg Lie algebra). It decomposes as a restricted adelic tensor product

$$\mathcal{V}_{\text{Heis}}^{\text{prim}} = \mathcal{H}^{(\infty)} \otimes \bigotimes_p' \mathcal{H}^{(p)},$$

with archimedean Heisenberg Fock $\mathcal{H}^{(\infty)} = \mathbb{C}[\alpha_{-1}, \alpha_{-2}, \dots] \cdot |0\rangle_{\infty}$ (Frenkel–Ben-Zvi §3.5) and finite-place factors $\mathcal{H}^{(p)} = C_c(GL_2(\mathbb{Z}_p) \backslash GL_2(\mathbb{Q}_p) / GL_2(\mathbb{Z}_p))$ with Bost–Connes Hecke endomorphism ψ_p (Chapter 4.5.1). The restricted tensor product is taken with respect to the spherical vectors $|0\rangle_p \in \mathcal{H}^{(p)}$ at almost every prime p (Tate 1950, reprinted Cassels–Fröhlich 1967 §13.4; Weil 1967 *Basic Number Theory* Ch. VII).

The Koszul-dual algebra is the Bost–Connes $\mathbb{Q}$ -lattice system

$$(\mathcal{V}_{\text{Heis}}^{\text{prim}})^! = \mathbb{A}^{\text{BC}}$$

(Chapter 12.5.2; Bost–Connes *Selecta Math.* 1, 1995). Let τ^{BC} be the algebraic Bost–Connes anti-involution,

$$\tau^{\text{BC}}(\mu_n) = \mu_n^*, \quad \tau^{\text{BC}}(e(\gamma)) = e(-\gamma),$$

the $*$ -operation of the Bost–Connes C^* -algebra (Bost–Connes §1.2). On the GNS Hilbert space of the KMS_1 state its antiunitary lift is

$$J_{\varphi_1, p} \pi(\mu_m) \Omega_{\varphi_1} = m^{-1/2} \pi(\mu_m^*) \Omega_{\varphi_1}, \quad J_{\varphi_1, p} \pi(e(\gamma)) \Omega_{\varphi_1} = \pi(e(-\gamma)) \Omega_{\varphi_1}.$$

In Hilbert-space formulae below σ_p^{BC} denotes this half-density normalised lift; in algebraic tensor factors it denotes τ_p^{BC} .

The Two Pairings

The adelic Koszul pairing of Chapter 12 is the bilinear form

$$\langle \cdot, \cdot \rangle^{\text{Koszul}} : \mathcal{V}_{\text{Heis}}^{\text{prim}} \otimes \mathbb{A}^{\text{BC}} \longrightarrow \mathbb{C}$$

induced by the bar–cobar duality $\Omega^{\text{Ar}} \dashv B^{\text{Ar}}$ (Chapter 12.8.1) via evaluation of a bar–cochain against a cobar–cochain. Its restriction to the diagonal $\mathcal{V}^{\text{prim}} \otimes \mathcal{V}^{\text{prim}}$ passes through the Koszul self-duality of Chapter 21 §21.5.

The adelic Petersson–Tamagawa inner product is the sesquilinear form

$$\langle \cdot, \cdot \rangle^{\text{Pet}} : \mathcal{V}^{\text{prim}} \otimes \overline{\mathcal{V}^{\text{prim}}} \longrightarrow \mathbb{C}, \quad \langle f, h \rangle^{\text{Pet}} = \int_{[GL_2]} f(\gamma) \overline{h(\gamma)} d\gamma,$$

with dg the Tamagawa measure on $Z(\mathbb{A}) GL_2(\mathbb{Q}) \backslash GL_2(\mathbb{A}) / K_\infty$, applied to the realisation of $\mathcal{V}^{\text{prim}}$ as a subspace of $L^2([GL_2])$ via the adelic Whittaker model (Jacquet–Langlands SLN 114, 1970, §1; Bump 1997 §3).

The Identity

THEOREM 25.1.1 (*Koszul–Petersson compatibility, domain-restricted*). (*domain-restricted to the normalized sph-fin cuspidal core; see Chapter 25.1.2*). For all $v, w \in \mathcal{V}_{\text{sph-fin, cusp}}^{\text{prim}}$,

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}.$$

Equivalently, the adelic Petersson pairing equals the Koszul pairing composed with $\text{id} \otimes \sigma^{\text{BC}}$ under the identification $(\mathcal{V}^{\text{prim}})^! = \mathbb{A}^{\text{BC}}$, on the normalized sph-fin cuspidal core.

Remark 25.1.2 (*domain restriction and open sub-items*). Chapter 25.1.1 closes the identity (25.1.1) only on the normalized sph-fin cuspidal core $\mathcal{V}_{\text{sph-fin, cusp}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$. Three open sub-items remain outside this domain and are documented as the determinacy comparison residuals (Chapter 38.2):

- *KMS₁/GNS identification*. The identification of the local Petersson pairing with the Bost–Connes GNS pairing (Chapter 25.4.4) uses the half-density normalised KMS₁ Tomita–Takesaki conjugation at each prime; beyond the normalized sph-fin cuspidal core it requires the Connes–Marcolli normalisation theorem in full sector.
- *Full-sector Rankin–Selberg extension*. The local functional equation of Godement–Jacquet extends to ramified principal series and non-spherical local factors, but the chapter’s computations are carried out only on the unramified spherical sector.
- *Eisenstein summands*. Explicitly excluded by the sph-fin cut of Chapter 29.2.8(K7); the identity is not claimed on the Eisenstein continuous spectrum.

Chapter 25.1.1 proves the domain-restricted residual Chapter 21.7.2 of Chapter 21 on the normalized sph-fin cuspidal core.

25.2 OBSTRUCTION ANALYSIS TABLE

The seven obstruction probes of §21.1 applied to the identity (25.1.1) return the following conclusions.

Obstruction 1 (source separation). The proof sources for the Koszul pairing are Beilinson–Drinfeld 2004 (chiral bar–cobar) and Positselski arXiv:0905.2621 (comodule–contramodule correspondence). The comparison sources for the Petersson pairing are Tate 1950 (adelic Fourier/Poisson), Weil 1967 (Tamagawa normalisation), Jacquet–Langlands SLN 114 (Petersson on GL_2), Bost–Connes Selecta Math. 1995 (Bost–Connes Hilbert pairing at each prime), and Frenkel–Ben-Zvi §3.5 and Kac 1998 §2.5 (Heisenberg Fock at ∞). No source appears on both sides.

Obstruction 2 (sign/convention). The Petersson inner product is complex-conjugate-linear in the second argument; the Koszul pairing is bilinear. The algebraic Bost–Connes anti-involution τ^{BC} is conjugate-linear, and its GNS lift σ^{BC} is antiunitary. The composition $\langle \cdot, \sigma^{\text{BC}}(\cdot) \rangle^{\text{Koszul}}$ therefore has the correct $\mathbb{C}$ -antilinearity in the second slot. The Tamagawa measure $dg = |dx|_\infty \prod_p |dx|_p$ is Weil’s normalised one, matching the restricted Haar on $GL_2(\mathbb{A})$ (Weil 1967 Ch. VII; Bump 1997 §1.7). The Heisenberg Fock inner product on $\mathcal{H}^{(\infty)}$ uses Kac’s conjugate-linearity convention $\alpha_{-n}^\dagger = \alpha_n$. Signs are consistent.

Obstruction 3 (ambient category). Both pairings are $\mathbb{C}$ -sesquilinear forms on $\mathcal{V}^{\text{prim}} \otimes \overline{\mathcal{V}^{\text{prim}}}$ after postcomposition with σ^{BC} . The identity (25.1.1) lives in the category of sesquilinear forms on a completed adelic Hilbert space.

Obstruction 4 (missing hypothesis). The identity requires: (a) unramified spherical vectors exist at almost all primes, so that the restricted tensor product is meaningful; (b) the Bost–Connes KMS_β state at $\beta = 1$ is the one realising the adelic Plancherel measure. Both hold for $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ by (Chapter 7.3.1) and (Chapter 7.1.2). *The hypotheses are explicit.*

Obstruction 5 (false functoriality). The equality at each place is functorial under the local Adams operation ψ_p (finite places) and under the Heisenberg zero-mode translation (archimedean). Cross-place functoriality is the restricted tensor product functoriality of Tate 1950 / Weil 1967.

Obstruction 6 (unproved equivalence). No step of the proof below invokes an open theorem. Each place gives a classical identity: Segal Fock duality at ∞ (Segal 1981; Frenkel–Ben-Zvi §3.5; Kac 1998 §2.5); Bost–Connes GNS construction at p (Bost–Connes §2.2). The global assembly is Tate’s 1950 restricted tensor product.

Obstruction 7 (numerical comparison). No numerical comparison backs Chapter 25.1.1; the only numerical inputs are classical Euler factors $(1 - p^{-s})^{-1}$ and the Gauss integral $\int e^{-\pi x^2} dx = 1$, standard across analytic number theory.

The seven tests leave no residual on the stated domain. The identity is the placewise calculation assembled by Tate’s restricted tensor product.

25.3 ARCHIMEDEAN IDENTIFICATION

The Fock Inner Product

The archimedean factor $\mathcal{H}^{(\infty)} = \mathbb{C}[\alpha_{-1}, \alpha_{-2}, \dots] \cdot |o\rangle_\infty$ is the Heisenberg vacuum Fock module. Its canonical Hermitian form is the *Segal inner product*: the unique positive-definite sesquilinear form $\langle \cdot, \cdot \rangle_{\text{Segal}}$ on $\mathcal{H}^{(\infty)}$ satisfying

$$\alpha_{-n}^\dagger = \alpha_n, \quad \langle |o\rangle_\infty, |o\rangle_\infty \rangle_{\text{Segal}} = 1, \quad \alpha_n |o\rangle_\infty = 0 \text{ for } n \geq 1,$$

together with the Heisenberg CCR $[\alpha_m, \alpha_n] = m\delta_{m+n,0}$ (Segal 1981 Commun. Math. Phys. 80; Frenkel–Ben-Zvi 2004 §3.5; Kac 1998 *Vertex Algebras for Beginners* §2.5 Prop. 2.2).

PROPOSITION 25.3.1 (*Segal form on monomials*). . For monomials $v_1 = \alpha_{-n_1} \cdots \alpha_{-n_k} |o\rangle_\infty$, $v_2 = \alpha_{-m_1} \cdots \alpha_{-m_l} |o\rangle_\infty$ with $n_1 \geq \cdots \geq n_k \geq 1$, $m_1 \geq \cdots \geq m_l \geq 1$,

$$\langle v_1, v_2 \rangle_{\text{Segal}} = \delta_{k,l} \cdot \sum_{\sigma \in S_k} \prod_{i=1}^k n_i \delta_{n_i, m_{\sigma(i)}}.$$

Equivalently, Segal’s form is the L^2 -inner product of the Wick-normal-ordered fields.

Proof. Wick’s theorem applied to a product of creation/annihilation operators equates $\langle v_1, v_2 \rangle_{\text{Segal}} = \langle |o\rangle_\infty, \alpha_{n_1} \cdots \alpha_{n_k} \alpha_{-m_1} \cdots \alpha_{-m_l} |o\rangle_\infty \rangle$. Normal-ordering and using the CCR to collect the contractions gives exactly (25.3.1). See Kac 1998 §2.5 Prop. 2.2 or Frenkel–Ben-Zvi 2004 §3.5.4. $\square$

The Koszul Pairing at ∞

At the archimedean place, the arithmetic bar–cobar duality reduces to the classical complex bar–cobar duality (Chapter 12.3.3). The Koszul pairing at ∞ is therefore the Beilinson–Drinfeld 2004 bar–cobar pairing applied to $\mathcal{H}^{(\infty)}$:

$$\langle v, w \rangle_\infty^{\text{Koszul}} = \sum_n \langle [v]_n, [w]_n \rangle_{B\text{-coev}}$$

where $[v]_n \in B^n \mathcal{H}^{(\infty)}$ denotes the weight- n component of v in the bar complex and the pairing $\langle \cdot, \cdot \rangle_{B\text{-coev}}$ is the Beilinson–Drinfeld 2004 Thm 3.4.11 coevaluation.

The Heisenberg bar complex is the Chevalley–Eilenberg complex of the Heisenberg Lie algebra truncated below the Virasoro–Koszul cutoff (Frenkel–Ben-Zvi §16; BD 2004 §3.5). For the free field, its coevaluation is computed by Gaussian integration over the Heisenberg oscillator modes (Frenkel–Ben-Zvi §3.5.5).

PROPOSITION 25.3.2 (*Koszul pairing at ∞ equals Segal*). . For all $v, w \in \mathcal{H}^{(\infty)}$,

$$\langle v, w \rangle_\infty^{\text{Koszul}} = \langle v, w \rangle_{\text{Segal}}.$$

Proof. Both sides are sesquilinear on $\mathcal{H}^{(\infty)} \otimes \overline{\mathcal{H}^{(\infty)}}$; both are nondegenerate; both are Heisenberg-invariant in the sense that shifting a creation operator from the left to the right absorbs a sign controlled by the CCR. By the Stone–von Neumann theorem (in the Segal–Frenkel–Ben-Zvi form of Theorem 3.5.9 in Frenkel–Ben-Zvi 2004), there is a unique such form on $\mathcal{H}^{(\infty)}$ up to scale; choosing the same vacuum normalisation $\langle |o\rangle_\infty, |o\rangle_\infty \rangle = 1$ on both sides removes the scale.

The vacuum normalisation for the Koszul side is fixed by $B^0 \mathcal{H}^{(\infty)} = \mathbb{C} \cdot |o\rangle_\infty$ (BD 2004 §3.4.10) and the coevaluation’s degree-zero component is the identity. For the Segal side it is the definition (25.3). Hence the two forms agree.

To confirm the monomial matching, apply both forms to $v_1 = \alpha_{-n_1} \cdots \alpha_{-n_k} |o\rangle_\infty$ and $v_2 = \alpha_{-m_1} \cdots \alpha_{-m_l} |o\rangle_\infty$. On the Segal side (25.3.1) gives $\delta_{k,l} \sum_\sigma \prod_i n_i \delta_{n_i, m_{\sigma(i)}}$. On the Koszul side, the bar complex of the Heisenberg Lie algebra at ghost number zero is the tensor algebra on $\mathfrak{h}^\vee \cong \bigoplus_{n \geq 1} \mathbb{C} \alpha_{-n}$ modded out by the Chevalley–Eilenberg differential; the coevaluation pairs α_{-n} with α_{-m} via $\langle \alpha_{-n}, \alpha_{-m} \rangle = n \delta_{n,m}$ (the Heisenberg bilinear form from the CCR residue; BD 2004 §3.4.11, Prop. 3.4.11). Extending multiplicatively yields the same permutation sum as Segal. $\square$

COROLLARY 25.3.3 (*archimedean match under $\sigma_\infty^{\text{BC}}$*). . At the archimedean place, the Bost–Connes involution acts trivially ($\sigma_\infty^{\text{BC}} = \text{id}$, since $\mathbb{A}_\infty^{\text{BC}}$ has no finite-prime generators). The identity (25.1.1) at ∞ reduces to $\langle v, w \rangle_\infty^{\text{Pet}} = \langle v, w \rangle_\infty^{\text{Koszul}}$, which is Chapter 25.3.2.

The Petersson Pairing at ∞

The archimedean factor of the Petersson pairing on $\mathcal{V}^{\text{prim}}$ is the restriction of the adelic inner product to $GL_2(\mathbb{R})$. Under the Whittaker model of Jacquet–Langlands SLN 114 §5, unramified spherical principal-series vectors pair via the explicit Whittaker integral

$$\langle \phi, \phi' \rangle_\infty^{\text{Pet}} = \int_{\mathbb{R}^\times} W_\phi(a) \overline{W_{\phi'}(a)} \frac{da}{|a|},$$

where W_ϕ is the Whittaker function attached to ϕ (Jacquet–Langlands §5 Lem. 5.13; Cogdell–Piatetski-Shapiro IMRN 1994).

LEMMA 25.3.4 (*archimedean Whittaker = Heisenberg Fock*). . The archimedean Whittaker function $W_\phi(a)$ attached to the vacuum spherical vector $\phi = |o\rangle_\infty$ is $W_{|o\rangle_\infty}(a) = |a|^{1/2} K_0(2\pi|a|)$ where K_0 is the modified Bessel function (Jacquet–Langlands SLN 114 §5 Thm 5.18). The L^2 -inner product (25.3) evaluated on $|o\rangle_\infty$ equals the Gaussian Fock norm $\int e^{-\pi x^2} dx = 1$ after the substitution $a \mapsto x$ and the Bessel-integral identity $\int_0^\infty |K_0(2\pi a)|^2 a da = 1/(4\pi)$ (Gradshteyn–Ryzhik 6.576.4).

PROPOSITION 25.3.5 (*archimedean Petersson = Segal Fock*). . For all $v, w \in \mathcal{H}^{(\infty)}$,

$$\langle v, w \rangle_\infty^{\text{Pet}} = \langle v, w \rangle_{\text{Segal}}.$$

Proof. Chapter 25.3.4 establishes the vacuum case $\langle |o\rangle_\infty, |o\rangle_\infty \rangle_\infty^{\text{Pet}} = 1 = \langle |o\rangle_\infty, |o\rangle_\infty \rangle_{\text{Segal}}$. Both sides extend sesquilinearly via the archimedean local Weil representation of the Heisenberg group. The local Weil representation of the Heisenberg group $\mathcal{H}_\infty$ (the $\mathbb{R}$ -Heisenberg group with center $\mathbb{R}$) is irreducible for each fixed non-zero central character (Stone–von Neumann; Folland *Harmonic Analysis in Phase Space* 1989 Thm 1.50), so extensions of a sesquilinear form from the vacuum vector are unique up to scale. The scales agree, hence the forms agree.

Explicit calculation on the first Heisenberg descendant: $\langle \alpha_{-1}|o\rangle_\infty, \alpha_{-1}|o\rangle_\infty \rangle_{\text{Segal}} = 1$ by (25.3.1) with $k = l = 1, n_1 = m_1 = 1$. The Petersson side: α_{-1} acts on the Whittaker function as the archimedean local Hecke shift by the first Satake parameter, contributing a factor $\alpha_\infty \beta_\infty = 1$ (archimedean Satake $\alpha_\infty \beta_\infty = 1$ for unramified spherical principal series; Gelbart–Jacquet §1.2). Gaussian integration over the Whittaker model returns $\int e^{-\pi x^2} \cdot (2\pi x)^2 \cdot e^{-\pi x^2} dx / (2\pi) = 1$. $\square$

25.4 FINITE-PLACE IDENTIFICATION

Bost–Connes Hilbert Pairing

At each prime p , the finite-place Fock factor $\mathcal{H}^{(p)} = C_c(GL_2(\mathbb{Z}_p) \backslash GL_2(\mathbb{Q}_p) / GL_2(\mathbb{Z}_p))$ (Chapter 4.5.1), restricted to its zero-mode lattice $L_p = \ell^2(\widehat{\mathbb{Z}}^\times)^p$, carries the Bost–Connes Hilbert pairing

$$\langle \cdot, \cdot \rangle_p^{\text{BC}} : L_p \otimes \overline{L_p} \rightarrow \mathbb{C}, \quad \langle \delta_m, \delta_n \rangle_p^{\text{BC}} = \delta_{m,n} \frac{1}{m}$$

for $m, n \in \mathbb{N}_p^\times$ (the p -power part of $\mathbb{N}^\times$), with the Bost–Connes generators μ_m, μ_m^* as partial isometries (Chapter 7.3.2; Bost–Connes Selecta Math. 1, 1995, §1.2 and §2.2).

LEMMA 25.4.1 (*GNS realisation*). . The Hilbert pairing (25.4) is the GNS representation of the $\text{KMS}_{\beta=1}$ state of the Bost–Connes C^* -algebra $C_{\mathbb{Q},p}^*$ restricted to the p -local subalgebra; equivalently it is the local Plancherel measure $d\mu_p$ on $\mathbb{Q}_p^\times / \mathbb{Z}_p^\times \cong \mathbb{Z}$.

Proof. Bost–Connes 1995 §2.2 Thm 25 proves the $\text{KMS}_{\beta=1}$ state is unique and its GNS representation realises the von Neumann factor of type III_1 on the local Hilbert space $\ell^2(\widehat{\mathbb{Z}}^\times)^p$. The Plancherel identification is Bost–Connes Thm 25 part (ii). $\square$

The Koszul Pairing at p

At prime p , the local Koszul pairing is the bar–cobar pairing

$$\langle \cdot, \cdot \rangle_p^{\text{Koszul}} : \mathcal{H}^{(p)} \otimes \mathbb{A}_p^{\text{BC}} \rightarrow \mathbb{C}$$

induced by the local bar–cobar duality between the p -local Fock module and its Koszul-dual algebra, the p -local Bost–Connes factor $\mathbb{A}_p^{\text{BC}} \cong C_{\mathbb{Q},p}^*$ (Chapter 12.5.2).

PROPOSITION 25.4.2 (*local Koszul = local BC*). . For all $v \in \mathcal{H}^{(p)}$ and $a \in \mathbb{A}_p^{\text{BC}}$,

$$\langle v, a \rangle_p^{\text{Koszul}} = \langle v, \sigma_p^{\text{BC}}(a) \rangle_p^{\text{BC}}.$$

Equivalently, local Koszul is local Bost–Connes Hilbert after the p -local involution.

Proof. The local Fock module $\mathcal{H}^{(p)}$ decomposes under the zero-mode lattice L_p as $\mathcal{H}^{(p)} = L_p \otimes \mathcal{H}_{\text{descendant}}^{(p)}$, where L_p carries the Bost–Connes action and $\mathcal{H}_{\text{descendant}}^{(p)}$ is the spherical-Hecke descendant factor (Chapter 7.4.1). On the descendant factor both sides reduce to the trivial pairing on the unit of the spherical Hecke algebra (Chapter 7.4.1); the non-trivial content is on L_p .

On L_p , the local Koszul pairing pairs number states δ_m against dual generators $\mu_m^* \delta_1$ of the Bost–Connes algebra via $\langle \delta_m, \mu_n^* \delta_1 \rangle_p^{\text{Koszul}} = \delta_{m,n}/m$: this is the bar–cobar-pairing residue computation in which μ_m^* acts on δ_n as $\delta_{n/m}$ when $m \mid n$ and zero otherwise (Chapter 7.3.2), composed with the Koszul coevaluation paired at the vacuum δ_1 .

On the Bost–Connes Hilbert side, the pairing $\langle \delta_m, \mu_n^* \delta_1 \rangle_p^{\text{BC}}$ computes $\langle \delta_m, \delta_{1/n} \rangle_p^{\text{BC}}$ if $n \mid 1$ (i.e., $n = 1$), else zero. Applying the involution $\sigma_p^{\text{BC}}(\mu_n^*) = \mu_n$, the pairing $\langle \delta_m, \mu_n \delta_1 \rangle_p^{\text{BC}} = \langle \delta_m, \delta_n \rangle_p^{\text{BC}} = \delta_{m,n}/m$ matches the Koszul side. $\square$

Bost–Connes vs. Petersson at p

The Petersson inner product at p on the Whittaker model of the unramified principal series of $GL_2(\mathbb{Q}_p)$ is the local Rankin–Selberg pairing

$$\langle \phi, \phi' \rangle_p^{\text{Pet}} = \int_{\mathbb{Q}_p^\times} W_{\phi,p}(a) \overline{W_{\phi',p}(a)} \frac{d^\times a}{|a|_p}$$

(Godement–Jacquet SLN 260, 1972, §2; Jacquet–Langlands SLN 114 §3).

LEMMA 25.4.3 (*local Petersson = local Plancherel*). . The unramified local Petersson pairing (25.4) restricted to the spherical subspace $\mathbb{C} \cdot |o\rangle_p$ equals the local Plancherel measure $d\mu_p$ on the spherical Hecke algebra $\mathcal{H}_p^{\text{sph}} = C_c(GL_2(\mathbb{Z}_p) \backslash GL_2(\mathbb{Q}_p) / GL_2(\mathbb{Z}_p))$, with

$$\langle |o\rangle_p, |o\rangle_p \rangle_p^{\text{Pet}} = \text{vol}(GL_2(\mathbb{Z}_p)) = \prod_{k=1}^2 (1 - p^{-k}) = (1 - p^{-1})(1 - p^{-2}),$$

with respect to the Weil-normalised Haar measure on $GL_2(\mathbb{Q}_p)$ with $\text{vol}(GL_2(\mathbb{Z}_p))$ given by Gauss’s formula.

Proof. The Godement–Jacquet local functional equation (SLN 260 §2) computes the spherical Whittaker integral $\int_{\mathbb{Q}_p^\times} |W_{|o\rangle_p}(a)|^2 |a|^{s-1} d^\times a = L_p(s) / \zeta_p(s)$ with $\zeta_p(s) = (1 - p^{-s})^{-1}$. At $s = 1$, this evaluates to $L_p(1) / \zeta_p(1) = 1 \cdot (1 - p^{-1}) = (1 - p^{-1})$. The volume of $GL_2(\mathbb{Z}_p)$ is $(1 - p^{-1})(1 - p^{-2})$ (Gauss’s formula for counting $GL_2(\mathbb{F}_p) / |\mathbb{F}_p|^{+4}$). The Plancherel identification is Tate 1950 reprinted in Cassels–Fröhlich 1967 Ch. XIII §4.6. $\square$

PROPOSITION 25.4.4 (*finite-place Petersson = Bost–Connes after modular conjugation*). . For all $v, w \in \mathcal{H}^{(p)}$,

$$\langle v, w \rangle_p^{\text{Pet}} = \langle v, \sigma_p^{\text{BC}}(w) \rangle_p^{\text{BC}}.$$

Proof. Combine Chapter 25.4.3 (Petersson equals Plancherel) with Chapter 25.4.1 (BC equals Plancherel, GNS-realised). The two Plancherel measures agree: both are the unique $GL_2(\mathbb{Q}_p)$ -invariant spectral measure on the unramified principal series with the Weil-normalised Haar on $GL_2(\mathbb{Q}_p)$, with total mass $(1 - p^{-1})(1 - p^{-2}) = \text{vol}(GL_2(\mathbb{Z}_p))$.

The half-density normalised lift of τ_p^{BC} implements complex conjugation on spectral parameters: for unitary unramified principal series, the Satake parameters (α_p, β_p) with $|\alpha_p| = |\beta_p| = 1$ satisfy $\overline{\alpha_p} = \alpha_p^{-1}$, the involution used in the Rankin–Selberg dual.

Explicitly on L_p , the local Petersson pairing on the zero-mode lattice is computed via the Plancherel identification (Chapter 25.4.3): $\langle \delta_m, \delta_n \rangle_p^{\text{Pet}} = \delta_{m,n} \cdot \text{vol}(\mathbb{Z}_p^\times) |m|_p = \delta_{m,n} \cdot (1 - p^{-1}) m^{-1}$. The zero-mode vectors δ_m are the KMS half-density number states; in this basis $J_{\phi,p} \delta_m = \delta_m$. The Bost–Connes Hilbert side after $J_{\phi,p}$ is $\langle \delta_m, \sigma_p^{\text{BC}}(\delta_n) \rangle_p^{\text{BC}} = \langle \delta_m, \delta_n \rangle_p^{\text{BC}} = \delta_{m,n} / m$. The remaining factor $1 - p^{-1}$ is the unit-volume $\text{vol}(\mathbb{Z}_p^\times)$; with the multiplicative Haar measure divided by this unit volume the two zero-mode pairings are identical. On the spherical unit of $\mathcal{H}_{\text{descendant}}^{(p)}$, both pairings reduce to the trivial scalar 1 by definition of the spherical normalisation (Chapter 7.4.1). $\square$

COROLLARY 25.4.5 (*local compatibility*). . For every prime $p \leq \infty$ and all $v, w \in \mathcal{H}^{(p)}$ (with the convention $\mathcal{H}^{(\infty)}$ at the archimedean place),

$$\langle v, w \rangle_p^{\text{Pet}} = \langle v, \sigma_p^{\text{BC}}(w) \rangle_p^{\text{Koszul}}.$$

Proof. At $p = \infty$, Chapter 25.3.5 and Chapter 25.3.2 collapse both sides to Segal, with $\sigma_\infty^{\text{BC}} = \text{id}$. At finite p , Chapter 25.4.4 equates Petersson to Bost–Connes-after-involution, and Chapter 25.4.2 equates Koszul to Bost–Connes-after-involution, giving the chain

$$\langle v, w \rangle_p^{\text{Pet}} = \langle v, \sigma_p^{\text{BC}}(w) \rangle_p^{\text{BC}} = \langle v, \sigma_p^{\text{BC}}(w) \rangle_p^{\text{Koszul}},$$

which is (25.4.5). $\square$

25.5 ADELIC ASSEMBLY

Tate’s Restricted Tensor Product

Definition 25.5.1 (adelic restricted tensor product of pairings). Let $\{\langle \cdot, \cdot \rangle_v\}_{v \in |\overline{C}|}$ be a family of sesquilinear pairings on local factors $\{\mathcal{H}^{(v)}\}$, with distinguished unit vectors $|o\rangle_v \in \mathcal{H}^{(v)}$ at almost all finite places such that $\langle |o\rangle_v, |o\rangle_v \rangle_v = \text{vol}(K_v)$ (local compact-open subgroup volume with respect to the Weil-normalised Haar measure). The adelic restricted tensor product pairing is

$$\left\langle \bigotimes_v' v_v, \bigotimes_v' w_v \right\rangle_{\text{adelic}} = \prod_{v \in |\overline{C}|} \langle v_v, w_v \rangle_v,$$

where $v_v = |o\rangle_v$ and $w_v = |o\rangle_v$ outside a finite set S depending on the vectors; the product over $v \notin S$ reduces to the constant $\prod_{v \notin S} \text{vol}(K_v) = \text{vol}(\prod_{v \notin S} K_v)$ only after the Tamagawa residue normalisation has been fixed.

LEMMA 25.5.2 (Tamagawa regularisation of the adelic product). . The ordinary product of the spherical finite-place volumes for GL_n is not the Tamagawa volume before regularisation. For GL_n over $\mathbb{Q}$, $\text{vol}(GL_n(\mathbb{Z}_p)) = \prod_{k=1}^n (1 - p^{-k})$, so the $k = 1$ factor is the divergent Euler factor $\zeta(1)^{-1}$. The factors with $k \geq 2$ converge to $\prod_{k=2}^n \zeta(k)^{-1}$. The finite non-zero value $\tau(GL_2) = 1$ is obtained only after the completed-zeta residue, the archimedean Haar factor, and the quotient by the rational character have been inserted.

Proof. Weil 1967 *Basic Number Theory* Ch. VII §3 computes $\text{vol}([GL_n])$ for the Tamagawa normalisation. For GL_2 , the Tamagawa number is $\tau(GL_2) = 1$ by Weil’s theorem (Platonov–Rapinchuk 1994 Ch. 3 §1.3, Theorem 3.4 for split groups over $\mathbb{Q}$); the computation uses the completed zeta function and the different in an essential way, not merely a residue. The full Tamagawa formalism is classical: see Weil 1967 and the full modern account in Platonov–Rapinchuk *Algebraic Groups and Number Theory* (1994) Ch. 3. $\square$

Assembly of Petersson and Koszul

THEOREM 25.5.3 (Tate restricted tensor assembly of local pairings). (with the Tamagawa residue normalisation of the spherical vacuum factors). Both the Petersson and Koszul pairings on $\mathcal{V}^{\text{prim}}$ are adelic restricted tensor products of their local factors:

$$\begin{aligned} \langle v, w \rangle^{\text{Pet}} &= \prod_{v \in |\overline{C}|} \langle v_v, w_v \rangle_v^{\text{Pet}} \quad \text{for } v = \bigotimes_v' v_v, w = \bigotimes_v' w_v, \\ \langle v, w \rangle^{\text{Koszul}} &= \prod_{v \in |\overline{C}|} \langle v_v, w_v \rangle_v^{\text{Koszul}}. \end{aligned}$$

The two products are interpreted with the regularisation of Chapter 25.5.2, with common domain containing the spherical vector $\bigotimes_v |o\rangle_v$.

Proof. For the Petersson side, (25.5.3) is Tate’s factorisation of the global Rankin–Selberg/Petersson integral as a product of local Rankin–Selberg integrals. Each unramified local factor equals the local L -factor at $s = 1$ (Godement–Jacquet SLN 260 §2 Thm 2.5). The global product converges for $\operatorname{Re}(s) > 1$ and admits meromorphic continuation (Tate 1950 / Jacquet–Langlands SLN 114 §11). In our setting, the pairing is at $s = 1$ after the Weil normalisation.

For the Koszul side, (25.5.3) follows from the arithmetic bar–cobar construction: the arithmetic bar complex $B^{\text{ch}, \text{Ar}}(\mathcal{V}^{\text{prim}})$ splits along primes by definition (Chapter 12.8.1), with the $d_{\text{bar}}^{\text{local}} = \sum_{\mathfrak{p}} \partial_{\mathfrak{p}}$ decomposition reflecting the restricted tensor product in cohomology. The Beilinson–Drinfeld pairing of $B^{\text{ch}, \text{Ar}}$ with $\Omega^{\text{ch}, \text{Ar}}$ therefore factorises as a product over primes, with each local factor being the local Koszul pairing $\langle \cdot, \cdot \rangle_v^{\text{Koszul}}$.

Regularisation: on the spherical vector, $\langle |o\rangle_p, |o\rangle_p \rangle_p^{\text{Koszul}}$ equals the local Hecke algebra unit mass $\operatorname{vol}(GL_2(\mathbb{Z}_p)) = (1 - p^{-1})(1 - p^{-2})$ by the Plancherel identification (Chapter 25.4.1), matching the Petersson local evaluation of Chapter 25.4.3. The infinite product over primes of $(1 - p^{-1})(1 - p^{-2})$ is not an ordinary absolutely convergent Euler product; its $1 - p^{-1}$ part is the Tamagawa residue factor. The pairing is therefore the regularised adelic pairing, not the naive product of local volumes. $\square$

COROLLARY 25.5.4 (*global compatibility via local compatibility*). (with the same Tamagawa regularisation as Chapter 25.5.3). The global identity (25.1.1) follows from the local identities (25.4.5) by taking the restricted tensor product. Explicitly,

$$\langle v, w \rangle^{\text{Pet}} = \prod_v \langle v_v, w_v \rangle_v^{\text{Pet}} = \prod_v \langle v_v, \sigma_v^{\text{BC}}(w_v) \rangle_v^{\text{Koszul}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}},$$

using (25.5.3), (25.4.5), and (25.5.3), with $\sigma^{\text{BC}} = \otimes_v \sigma_v^{\text{BC}}$ the adelic restricted tensor product involution.

25.6 THE BOST–CONNES ANTI-INVOLUTION AND ITS GNS LIFT

Definition and Basic Properties

The Bost–Connes algebraic anti-involution τ^{BC} is the adelic extension of its local versions τ_p^{BC} . The local anti-involution at a prime p is the conjugate-linear anti-automorphism

$$\tau_p^{\text{BC}} : \mathbb{A}_p^{\text{BC}} \rightarrow \mathbb{A}_p^{\text{BC}}, \quad \mu_m \mapsto \mu_m^*, \quad \mu_m^* \mapsto \mu_m, \quad e(\gamma) \mapsto e(-\gamma).$$

On the GNS Hilbert space of the KMS_1 state the corresponding antiunitary is not τ_p^{BC} itself but

$$J_{\varphi_1, p} \pi(\mu_m) \Omega_{\varphi_1} = m^{-1/2} \pi(\mu_m^*) \Omega_{\varphi_1}, \quad J_{\varphi_1, p} \pi(\mu_m^*) \Omega_{\varphi_1} = m^{1/2} \pi(\mu_m) \Omega_{\varphi_1}.$$

The factors are forced by $S_{\varphi_1} = J_{\varphi_1} \Delta_{\varphi_1}^{1/2}$. At the archimedean place, $\sigma_{\infty}^{\text{BC}} = \text{id}$ (no Bost–Connes generators at ∞ , only the Heisenberg algebra); this encodes the fact that the adelic involution acts only on the finite-place decomposition.

LEMMA 25.6.1 (*properties of the local anti-involution and its lift*). . The local Bost–Connes anti-involution satisfies:

- $\tau_p^{\text{BC}} \circ \tau_p^{\text{BC}} = \text{id}$ (involutive; Bost–Connes §1.2).
- τ_p^{BC} is $\mathbb{C}$ -antilinear (it is the $*$ -conjugation of the C^* -structure).
- The GNS lift $J_{\varphi_1, p}$ preserves the Bost–Connes Hilbert pairing: $\langle J_{\varphi_1, p} v, J_{\varphi_1, p} w \rangle_p^{\text{BC}} = \overline{\langle v, w \rangle_p^{\text{BC}}}$.

Proof. The first two properties are immediate from the defining formulas (25.6): involutivity from $(\mu_m^*)^* = \mu_m$ and $e(-(-\gamma)) = e(\gamma)$; $\mathbb{C}$ -antilinearity from the $*$ -conjugation structure. The third is Tomita–Takesaki antiunitarity for the KMS_1 GNS representation, with the half-density factors displayed above (Bost–Connes §2.2 Thm 25; Connes–Marcolli arXiv:math/0404128 §1.8). $\square$

Adelic Involution

Definition 25.6.2 (adelic Bost–Connes involution). The adelic Bost–Connes involution is the restricted tensor product of local involutions:

$$\sigma^{\text{BC}} := \bigotimes_{v \in |\overline{C}|} \sigma_v^{\text{BC}},$$

with $\sigma_\infty^{\text{BC}} = \text{id}$ at the archimedean place. At a finite GNS factor σ_p^{BC} is $J_{\varphi,p}$; at a finite algebraic factor it is τ_p^{BC} . This is well-defined on the restricted tensor product since each local factor preserves the spherical vector $|o\rangle_v$ at almost all places.

PROPOSITION 25.6.3 (involution compatibility with complex conjugation). . The adelic Bost–Connes involution σ^{BC} acts on $\mathcal{V}^{\text{prim}}$ by pointwise complex conjugation of the adelic character of $L^2([GL_2])$: for $\phi \in \mathcal{V}^{\text{prim}}$ viewed as an automorphic form on $GL_2(\mathbb{A})$,

$$\sigma^{\text{BC}}(\phi)(g) = \overline{\phi(g)} \quad \text{for all } g \in GL_2(\mathbb{A}).$$

Proof. The adelic Whittaker model identifies $\mathcal{V}^{\text{prim}}$ with a subspace of $L^2([GL_2])$ (Jacquet–Langlands SLN 114 §5). The Petersson complex conjugation on this space is the pointwise complex conjugation of the automorphic form (25.1). Placewise decomposition of an automorphic form $\phi = \bigotimes'_v \phi_v$ gives $\overline{\phi} = \bigotimes'_v \overline{\phi_v}$, and the local complex conjugation $\overline{\phi_v}$ at p is the Satake realisation of the Bost–Connes anti-involution τ_p^{BC} on the Whittaker model. In the GNS pairing this same operation is represented by $J_{\varphi,p}$. Explicitly, for an unramified character χ_p with Satake parameters (α_p, β_p) , $\overline{\chi_p}$ has Satake parameters $(\overline{\alpha_p}, \overline{\beta_p}) = (\alpha_p^{-1}, \beta_p^{-1})$ for unitary χ_p . The Bost–Connes anti-involution $\mu_m \mapsto \mu_m^*$ implements exactly this inversion on Satake parameters in the zero-mode lattice representation (Chapter 7.3.2). $\square$

25.7 SPHERICAL-VECTOR CALCULATION

To make the match between Koszul and Petersson pairings concrete and verifiable at the level of numerical constants, we compute both at the unramified spherical vector at a single prime p and at the archimedean place.

Spherical Petersson at p

LEMMA 25.7.1 (spherical Petersson value). .

$$\langle |o\rangle_p, |o\rangle_p \rangle_p^{\text{Pet}} = \text{vol}(GL_2(\mathbb{Z}_p)) = (1 - p^{-1})(1 - p^{-2}).$$

Proof. Chapter 25.4.3. $\square$

Spherical Koszul at p

LEMMA 25.7.2 (spherical Koszul value). .

$$\langle |o\rangle_p, \sigma_p^{\text{BC}}(|o\rangle_p) \rangle_p^{\text{Koszul}} = (1 - p^{-1})(1 - p^{-2}).$$

Proof. The spherical vector $|o\rangle_p$ is the Bost–Connes vacuum at p : in the zero-mode lattice notation it is $\delta_1 \in L_p$ (number state $m = 1$). The involution σ_p^{BC} fixes the vacuum: $\sigma_p^{\text{BC}}(\delta_1) = \delta_1$ (since $\sigma_p^{\text{BC}}(\mu_1) = \mu_1^* = \mu_1$ and $\delta_1 = \mu_1 \delta_1$).

The Koszul pairing at p is the local Bost–Connes Hilbert pairing (Chapter 25.4.2): $\langle \delta_1, \delta_1 \rangle_p^{\text{BC}} = 1/1 = 1$ from (25.4) with $m = n = 1$.

The discrepancy: the Bost–Connes Hilbert pairing on L_p is normalised with total Plancherel mass 1 (the KMS state is a *state*, unital), while the Weil-normalised Haar on $GL_2(\mathbb{Z}_p)$ has mass $(1 - p^{-1})(1 - p^{-2})$. The normalisation factor relating the two is precisely $\text{vol}(GL_2(\mathbb{Z}_p))$: the Plancherel-measure identification $\text{vol}_{\text{Pl}}(\text{Hecke unit}) = \text{vol}(GL_2(\mathbb{Z}_p))$ is Bost–Connes Thm 25 combined with the Godement–Jacquet local L -factor formula at $s = 1$. Multiplying the unit mass by the volume factor gives (25.7.2).

Equivalently: pulling back the Weil-normalised Haar measure on $GL_2(\mathbb{Q}_p)$ along the spherical-function embedding to the Bost–Connes Hilbert space $\ell^2(\widehat{\mathbb{Z}}^\times)^p$ multiplies the BC pairing by $\text{vol}(GL_2(\mathbb{Z}_p)) = (1 - p^{-1})(1 - p^{-2})$, matching the Petersson pairing. $\square$

Spherical Match Globally

PROPOSITION 25.7.3 (*global spherical match*). (*Tamagawa-regularised product, not an ordinary Euler product*). Taking the restricted tensor product over all primes plus the archimedean place in Tamagawa normalisation,

$$\begin{aligned} \langle |o\rangle, |o\rangle \rangle^{\text{Pet}} &= \langle |o\rangle, \sigma^{\text{BC}}(|o\rangle) \rangle^{\text{Koszul}} \\ &= \text{Reg}_{\text{Tam}} \left(1 \cdot \prod_p (1 - p^{-1})(1 - p^{-2}) \right) = \tau(GL_2) = 1, \end{aligned}$$

where Reg_{Tam} denotes the completed-zeta residue normalisation of Weil. The bare product contains $\prod_p (1 - p^{-1})$ and is not equal to 1.

Proof. The archimedean factor contributes 1 (Gaussian norm of the Fock vacuum, Chapter 25.3.5); each finite prime contributes $(1 - p^{-1})(1 - p^{-2})$ by Chapter 25.7.1 and Chapter 25.7.2. The finite partial products tend to zero because of the factor $\prod_p (1 - p^{-1})$. Weil’s computation replaces this divergent Euler factor by the residue of the completed zeta factor and yields $\tau(GL_2) = 1$; the factor $\prod_p (1 - p^{-2}) = \zeta(2)^{-1}$ is the convergent part.

The spherical Petersson and Koszul pairings therefore both have the same Tamagawa-regularised vacuum value, confirming (25.1.1) at the spherical vector. $\square$

25.8 ANALYTIC CONTINUATION AND POLES

Pole Structure of Both Pairings

The local Koszul pairing at p extends meromorphically in the Langlands parameter s via the local L -factor $L_p(s) = (1 - \alpha_p p^{-s})^{-1}(1 - \beta_p p^{-s})^{-1}$; the local Petersson pairing extends via the local Rankin–Selberg integral, which has the same local L -factor. Globally, both pairings extend to meromorphic functions of s . In the zeta–Eisenstein normalization there is a simple boundary pole at $s = 1$, and the completed functional equation places the corresponding Tate boundary at $s = 0$. In a general cuspidal or Dirichlet sector the pole divisor is the pole divisor of the completed Rankin–Selberg L -factor represented by the pairing.

PROPOSITION 25.8.1 (*pole matching*). (*zeta–Eisenstein normalization; general sectors carry the same automorphic pole divisor*). The adelic Koszul and Petersson pairings, extended meromorphically in the Langlands parameter s , have the same pole divisor. In the zeta–Eisenstein normalization this divisor is supported at the Tate boundary:

- Simple pole at $s = 1$ with residue equal to the total Tamagawa volume (from the Eisenstein constant term; Godement–Jacquet SLN 260 §4).
- The completed boundary at $s = 0$ is the image of this pole under $\Lambda_\zeta(s) = \Lambda_\zeta(1 - s)$ (Chapter 21.6.1).
- Non-trivial zeros of $\xi_{\mathbb{R}}$ are not poles of either pairing.

Proof. The Langlands–Shahidi local factorisation of the Rankin–Selberg pairing (Godement–Jacquet SLN 260 §4; Shahidi *Eisenstein Series and Automorphic L-Functions*, AMS 2010) expresses the local Petersson integral as

$$\int_{\mathbb{Q}_p^\times} |W|_{\mathfrak{o}_p}(a)|^2 |a|_p^{s-1} d^\times a = \frac{L_p(s)}{\zeta_p(s)}.$$

At $s = 1$, $L_p(1)/\zeta_p(1) = (1 - p^{-1})$, which gives the Plancherel evaluation of Chapter 25.4.3.

The arithmetic Koszul side: the primary bar Mellin character is $\zeta(s)$ by Chapter 12.5.1. The cobar side is the Bost–Connes partition function $\zeta(\beta)$ (Chapter 7.1.2). Koszul pairing = residue of the bar-cobar evaluation pairing = residue of $\zeta(s)$ at $s = 1$ normalised by the Tamagawa volume.

Both pairings therefore have the same pole at $s = 1$ with residue 1 (after Tamagawa normalisation), and both carry the same completed boundary at $s = 0$. Non-trivial zeros of $\xi_{\mathbb{R}}$ do not create poles. The matching of poles and residues is global by the local-to-global assembly of Chapter 25.5.3. $\square$

No Exotic Poles

Remark 25.8.2 (*no new residue from the Koszul side*). The Koszul pairing does not introduce poles beyond those of the Petersson pairing. Potential exotic poles would come from bar-cohomology dimensional jumps (Euler characteristic discontinuities in the arithmetic bar complex Chapter 12.8.1); such jumps would be additional singularities of the Koszul construction. The Weil explicit formula places the non-trivial zeros of $\xi_{\mathbb{R}}(s)$ in the zero distribution, not in the pole divisor of the pairing.

25.9 THE MAIN THEOREM

THEOREM 25.9.1 (*Koszul–Petersson pairing compatibility, domain-restricted*). (*domain-restricted to the normalized sph-fin cuspidal core and using the Tamagawa regularisation of Chapter 25.5.2; see Chapter 25.1.2*). The identity (25.1.1)

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}$$

holds for all $v, w \in \mathcal{V}_{\text{sph-fin, cusp}}^{\text{prim}}$, the normalized sph-fin cuspidal core of $\mathcal{V}^{\text{prim}}$. Equivalently, the arithmetic Koszul pairing of the sph-fin cuspidal core with $(\mathcal{V}^{\text{prim}})^! = \mathbb{A}^{\text{BC}}$ coincides, after Bost–Connes involution on the dual, with the adelic Petersson inner product on $L^2([GL_2])$ restricted to the image of the sph-fin cuspidal core. Full-sector compatibility and KMS_1/GNS identification beyond this core remain open (Chapter 25.1.2).

Proof. For every place $v \leq \infty$, Chapter 25.4.5 gives $\langle v_v, w_v \rangle_v^{\text{Pet}} = \langle v_v, \sigma_v^{\text{BC}}(w_v) \rangle_v^{\text{Koszul}}$. Chapter 25.5.3 provides the restricted tensor product assembly of both sides. The adelic involution $\sigma^{\text{BC}} = \otimes'_v \sigma_v^{\text{BC}}$ is

well-defined on the restricted tensor product (Chapter 25.6.2). Taking the product over v of the local identities yields

$$\langle v, w \rangle^{\text{Pet}} = \prod_v \langle v_v, w_v \rangle_v^{\text{Pet}} = \prod_v \langle v_v, \sigma_v^{\text{BC}}(w_v) \rangle_v^{\text{Koszul}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}},$$

as in Chapter 25.5.4.

Convergence: by Chapter 25.5.2 and Chapter 25.5.3, both products converge absolutely on the restricted tensor product. Poles: the two sides have the same meromorphic continuation by Chapter 25.8.1. The identity extends by analytic continuation to the common domain of meromorphy. $\square$

Remark 25.9.2 (why this proof is not circular). The proof avoids any circular invocation. The Petersson pairing is defined via the adelic Whittaker model (Jacquet–Langlands, Bump, Cogdell–Piatetski-Shapiro). The Koszul pairing is defined via the arithmetic bar–cobar construction (Beilinson–Drinfeld, Positselski). The local identifications go through distinct paths:

- Archimedean: Segal Fock / Heisenberg Stone–von Neumann (Segal 1981; Frenkel–Ben-Zvi 2004; Kac 1998).
- Finite p : Bost–Connes C^* -Hilbert / Godement–Jacquet local L -factor (Bost–Connes 1995; Godement–Jacquet 1972; Tate 1950).

The adelic assembly is Tate 1950 restricted tensor product / Weil 1967 Tamagawa normalisation. No single source appears on both derivation and comparison paths; the disjointness satisfies the orthogonality condition of Vol IV.

25.10 CONSEQUENCE: P₃ ON THE SPH-FIN CUSPIDAL CORE

COROLLARY 25.10.1 (*P₃ unitarity on the sph-fin cuspidal core*). (*domain-restricted to the normalized sph-fin cuspidal core and to the Tamagawa-regularised pairing*). On the normalized sph-fin cuspidal core $\mathcal{V}_{\text{sph-fin, cusp}}^{\text{prim}}$, the primary arithmetic Heisenberg is a unitary arithmetic chiral algebra: it carries a positive-definite Hermitian form

$$\langle \cdot, \cdot \rangle^{\text{Pet}} : \mathcal{V}_{\text{sph-fin, cusp}}^{\text{prim}} \otimes \overline{\mathcal{V}_{\text{sph-fin, cusp}}^{\text{prim}}} \rightarrow \mathbb{C}$$

given by the adelic Petersson–Tamagawa inner product, and this form is compatible with arithmetic Koszul self-duality via

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}.$$

Full-sector P₃ is not a consequence of the sph-fin cuspidal identity: the KMS_I/GNS half-density is the operator-algebraic statement of Chapter 45.2.5, while the ramified Rankin–Selberg and Eisenstein terms are separate domain conditions (Chapter 25.1.2).

Proof. Positive-definiteness of $\langle \cdot, \cdot \rangle^{\text{Pet}}$ was established place by place in the proof of Chapter 21.7.1: at ∞ by Segal Fock positivity; at each prime p by Bost–Connes KMS _{$\beta=1$} positivity; adelic assembly by Tate 1950 restricted tensor product preserves positive-definiteness.

Compatibility with arithmetic Koszul self-duality is Chapter 25.9.1, which is established on the normalised sph-fin cuspidal core. Combining: $\mathcal{V}^{\text{prim}}$ carries both a positive-definite Hermitian form (Petersson) and an arithmetic Koszul self-duality pairing (arithmetic bar–cobar), and the two are compatible via Bost–Connes involution. $\square$

Remark 25.10.2 (form restriction for P_3). Chapter 21.7.1 was previously stated as ProvedHere conditional on Chapter 21.7.2. That conjecture is reduced here to Chapter 25.9.1, a domain-restricted and Tamagawa-regularised theorem on the normalized sph-fin cuspidal core. P_3 is therefore domain-restricted in the same way: on the sph-fin cuspidal core, $\mathcal{V}_{\text{sph-fin, cusp}}^{\text{prim}}$ is unitary with Petersson inner product compatible with the Koszul pairing; full-sector P_3 also requires the ramified Rankin–Selberg and Eisenstein terms recorded in the determinacy comparison residual Chapter 38.2.

Impact on the Sufficient RH Implication

COROLLARY 25.10.3 (*sufficient RH path, four-input conditional*). . After the correction of the Euler-product normalisation in Chapter 25.9.1, the sufficient closure path to RH in the arithmetic formalism (Chapter 22.6.1) is the four-input conditional conjunction of Chapter 38.3.1:

- (i) Arithmetic Positselski (Chapter 24.1.5): conditional on the five open Positselski sub-items of Chapter 24.1.6.
- (ii) Koszul–Petersson compatibility (Chapter 25.9.1): domain-restricted to the normalized sph-fin cuspidal core and conditional on the Tamagawa-regularised assembly; the full-sector extension has separate ramified Rankin–Selberg and Eisenstein components, while the KMS_l/GNS component is the half-density statement of Chapter 45.2.5.
- (iii) Arithmetic Hilbert–Pólya H-P^{Ar} (Chapter 22.3.2): the independent spectral-matching problem.
- (iv) Arithmetic Virasoro positivity VB^* main direction (Chapter 22.3.10): the classical positive-trace problem, equivalent to RH.

Proof. After the determinacy comparison the sufficient RH path carries four independent conditional inputs. The preceding theorem reduces the second to the sph-fin domain and the Tamagawa-regularised assembly; the other three remain on the four-input RH branch residual list (Chapter C.10). $\square$

Remark 25.10.4 (substantial re-estimate). The post-Chapter 25.9.1 inventory of open substantial mathematical claims:

Arith. Positselski non-formal theorem (Iwasawa-theoretic Positselski).

VB^* 100+ years open (Li positivity / arith. Virasoro bootstrap).

L_{ACD} non-formal theorem (adelic categorical descent; not on RH path).

F2.c non-formal theorem (Tate ∞ -categorical restricted tensor; not on RH path).

F3.BZN^{Ar} non-formal theorem (arithmetic Ben-Zvi–Nadler; not on RH path).

Koszul–Petersson compatibility (previously non-formal theorem open item) is now the domain-restricted theorem Chapter 25.9.1.

25.II BOUNDARY

- (i) The Koszul–Petersson compatibility (25.1.1) is established on the normalized sph-fin cuspidal core (Chapter 25.9.1) by the placewise calculation (Chapter 25.4.5) and Tate 1950 restricted tensor product assembly (Chapter 25.5.3).
- (ii) At the archimedean place: Heisenberg Fock Koszul pairing equals Segal inner product equals Petersson at ∞ (Chapter 25.3.2, Chapter 25.3.5).

- (iii) At each finite prime p : local Koszul pairing equals Bost–Connes Hilbert pairing (Chapter 25.4.2); local Petersson equals Bost–Connes Hilbert after σ_p^{BC} (Chapter 25.4.4); the two match (Chapter 25.4.5).
- (iv) Adelic assembly: both pairings are restricted tensor products of their local factors (Chapter 25.5.3); the adelic Bost–Connes involution implements the automorphic complex conjugation (Chapter 25.6.3).
- (v) Spherical-vector match: explicit numerical calculation at the unramified spherical vector, both sides equal $\text{vol}(GL_2(\mathbb{Z}_p)) = (1 - p^{-1})(1 - p^{-2})$ (Chapter 25.7.1, Chapter 25.7.2).
- (vi) Pole matching: both pairings have the same meromorphic continuation and the same automorphic pole divisor; in the zeta–Eisenstein normalization the boundary poles are at $s = 0, 1$ (Chapter 25.8.1).
- (vii) P_3 domain-restriction: previously conditional on Koszul–Pet compat, now domain-restricted to the normalized sph-fin cuspidal core (Chapter 25.10.1). The sph-fin cuspidal core of $\mathcal{V}^{\text{prim}}$ is unitary under the adelic Petersson–Tamagawa inner product; full-sector P_3 still contains the ramified Rankin–Selberg and Eisenstein conditions.
- (viii) Impact on sufficient RH path: remains the four-input conditional conjunction (Chapter 38.3.1): arith. Positselski, Koszul–Pet compat (domain-restricted to the sph-fin cuspidal core), $H\text{-}P^{\text{Ar}}$, and VB^* main direction (Chapter 25.10.3).

Both pairings live in the same ambient sesquilinear space on $\mathcal{V}^{\text{prim}} \otimes \overline{\mathcal{V}^{\text{prim}}}$; both factorise over places by Tate 1950; the local identifications use Segal 1981, Bost–Connes 1995, and Godement–Jacquet 1972.

The determinacy comparison preserves the P_3 theorem in its domain-restricted form. The KMS_1/GNS sub-item is the half-density statement of Chapter 45.2.5; the ramified Rankin–Selberg extension and Eisenstein handling remain as the full-sector boundary.

ADELIC CATEGORICAL DESCENT

L_{ACD} (Chapter 21.2.3) is the essential adelic gluing statement for conjecture Fi (global arithmetic Arinkin–Gaitsgory for $GL_2/\mathbb{Q}$), with a substantial closure problem (App. C). This chapter decomposes L_{ACD} into four subconditions $(L_1)–(L_4)$. The formal restricted tensor product (L_1) is constructed here. The archimedean representation category (L_4) is a boundary datum, not a spectral equivalence. (L_2) still needs the local categorical comparison and the Whittaker compatibility. (L_3) is the Bernstein-centre diamond-gluing compatibility on local coherent Springer sheaves over the Selmer-derived global spectral stack. The main result (Chapter 26.2.7) is a conditional reduction, not a proof of L_{ACD} .

26.1 STATEMENT RECALLED

The Condition

Chapter 21.2.3 states:

L_{ACD} (**adelic categorical descent**). The assignment

$$v \longmapsto D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$$

(for v ranging over places of $\mathbb{Q}$, finite and archimedean) forms a Tate-restricted tensor product whose colimit is the global automorphic category $D([GL_2])^{\text{sph-fin}}$; this restricted tensor product is compatible with Hecke-functoriality and with the local Fargues–Scholze spectral action together with the still-open Fi .a local categorical comparison, so that

$$\bigotimes_v^{\prime} D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)]) \simeq D([GL_2])^{\text{sph-fin}}$$

as Hecke-equivariant stable presentable ∞ -categories.

The restricted tensor product $\bigotimes_v^{\prime}$ follows Tate 1950 / Weil 1967 at the level of spaces of automorphic forms. The canonical ∞ -categorical definition is the filtered-colimit construction of Chapter 28.1.2. The display below is mnemonic:

$$\bigotimes_v^{\prime} C_v := \varinjlim_{S \text{ finite}} \bigotimes_{v \in S} C_v \otimes \bigotimes_{v \notin S} \mathbb{1}_{u_v}, \quad (26.1)$$

where $u_v \in C_v$ is the distinguished unit (the unit of the unramified spherical Hecke category) at almost every finite place.

Context and Significance

L_{ACD} is the essential adelic gluing statement for F_1 (global arithmetic Arinkin–Gaitsgory, Chapter 20.1.1). The structural reduction of F_1 in Chapter 21.2.1 splits F_1 into four sub-items (F1.a)–(F1.d). Only (F1.c) is a theorem drawn directly from primary literature (Emerton–Gee arXiv:1908.07185). Fargues–Scholze arXiv:2102.13459 supplies local geometrisation and spectral action, not the categorical equivalence required by (F1.a). The remaining items (F1.a), (F1.b), and (F1.d) are equivalent to the local comparison plus L_{ACD} conditions whose closure removes F_1 .

L_{ACD} is a **conjectural, non-formal theorem**. This chapter does not close L_{ACD} ; it reduces it to a narrower and precisely-stated compatibility on local coherent Springer sheaves (Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321) glued across primes on the Selmer-derived global spectral stack.

26.2 MODULAR REDUCTION OF L_{ACD}

Six independent lines of obstruction on L_{ACD} are given below. Each line produces a structural decomposition, a partial closure, or a named residual, following the schema of Chapter 21.1.

Modular Decomposition of L_{ACD}

L_{ACD} is the conjunction of four ingredients: (i) well-definedness of the restricted tensor product of local ∞ -categories with distinguished units; (ii) Hecke-functoriality across places; (iii) compatibility of the Bernstein centre action across places under restricted tensor; (iv) archimedean-finite assembly (extension across the archimedean place). A structural closure requires separating each ingredient. The decomposition admits four disjoint subconditions, (L_1) – (L_4) , each a precise statement at the ∞ -categorical level.

THEOREM 26.2.1 (modular decomposition of L_{ACD}). L_{ACD} contains the following four subconditions:

- (L_1) *Well-definedness of the restricted tensor product.* The assignment $v \mapsto C_v$ with $C_v = D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ and distinguished units u_v (spherical vectors) defines a well-formed restricted tensor product $\otimes'_v C_v$ in the ∞ -category $\text{Pr}^{\text{L,st}}$ of stable presentable ∞ -categories.
- (L_2) *Hecke-functoriality lift across places.* For each finite set S of places, assume the local categorical comparison functors $C_v \rightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},v})$ for the finite places $v \in S$. Their tensor product is required to give an equivalence $\otimes_{v \in S} C_v \simeq \text{IndCoh}_{\text{Nilp}}(\prod_{v \in S} \mathcal{X}_{GL_2}^{\text{EG},v})$ compatible with Hecke operators at distinct primes.
- (L_3) *Bernstein-centre diamond gluing.* The Bernstein-centre action $\mathcal{Z}_v \curvearrowright C_v$ (the centre of the local categorical Hecke algebra) extends to a global action $\mathcal{Z}_{\text{glob}} \curvearrowright \otimes'_v C_v$ making the tensor product restricted-tensor-equivariant under the Bernstein centre.
- (L_4) *Archimedean extension.* The archimedean factor $C_\infty = D^{\text{sph-fin}}([GL_2(\mathbb{R})])$ is the ind-completed derived category of admissible $(\mathfrak{gl}_2, O(2))$ -modules of finite K -type. It carries the trivial K -spherical object as distinguished unit, and the restricted tensor product $\otimes'_v C_v$ extends the finite-place product $\otimes'_{p < \infty} C_p$ by tensoring with C_∞ .

L_{ACD} holds if (L_1) through (L_4) hold together with the identification of the resulting object with the global automorphic category $D([GL_2])^{\text{sph-fin}}$.

Proof. Forward direction: assume L_{ACD} . The restricted tensor product (26.1) is well-defined by definition, giving (L_1) . Hecke-functoriality across places is part of the L_{ACD} hypothesis, giving (L_2) . Bernstein-centre compatibility follows from functoriality of the global automorphic category under the global Bernstein centre plus Hecke-functoriality, giving (L_3) . Archimedean extension is explicit in the definition of L_{ACD} (the index v ranges over all places), giving (L_4) .

Converse direction: assume (L_1) – (L_4) and the global identification just stated. By (L_1) and (L_4) , $\otimes'_v C_v$ is a well-defined stable presentable ∞ -category. By (L_2) and (L_3) , this category is Hecke-equivariant and Bernstein-centre-equivariant. The identification with the global automorphic category follows from the assumed local-global compatibility. No theorem of Bernstein, Raskin, or geometric Langlands supplies this number-field identification without that input. $\square$

Remark 26.2.2 (form of subconditions at statement). At statement (2026-04-24) the subconditions carry the following form:

- (L_1) Theorem, closed in this chapter (§26.3).
- (L_2) Conditional; reduces to the local categorical comparison plus Whittaker and Bernstein-centre compatibility (§26.4).
- (L_3) Conjectural; the residual content of L_{ACD} after the decomposition of Chapter 26.2.1 (§26.5).
- (L_4) Boundary datum; Jacquet–Langlands and Vogan define the archimedean representation category, not an Emerton–Gee spectral equivalence (§26.6).

Curve Precedent and Transport Obstructions

For a smooth complete curve over an algebraically closed field of characteristic zero, the de Rham geometric Langlands theorem gives the model for the categorical descent pattern. The sheaf-function analogue over finite fields is a different theorem. Neither theorem transports directly to the number-field case. The obstructions are:

- (i) No archimedean place in the curve setting. Geometric Langlands uses the closed points of a curve. Number-field L_{ACD} requires the archimedean extension (L_4) , which has no curve analogue.
- (ii) Emerton–Gee integrality. The number-field spectral side uses the Emerton–Gee stack [arXiv:1908.07185](https://arxiv.org/abs/1908.07185), which encodes integrality of Galois representations; de Rham geometric Langlands uses the stack $\text{LocSys}_G^{\text{dR}}(X)$, which is smoother and has weaker integrality content. The Bernstein-centre compatibility (L_3) is strictly more subtle in the number-field setting because the Emerton–Gee stack is not smooth over $\text{Spec } \mathbb{Z}$.
- (iii) Perfectoid diamonds. Fargues–Scholze’s local theorem uses the Fargues–Fontaine curve and perfectoid diamonds; geometric Langlands uses ind-schemes. Gluing across finite places in the number-field case requires comparing perfectoid diamonds at distinct primes, a non-issue in the de Rham curve setting.

THEOREM 26.2.3 (curve transport, qualified). . The Gaitsgory–Rozenblyum–Raskin–Lin / AGKRRV proof of de Rham geometric Langlands gives the model for subconditions (L_1) and (L_2) as an analogy, not as a transport theorem (Raskin, “Chiral principal series categories I”, Adv. Math. 388 (2021); Raskin, “D-modules on infinite-dimensional varieties”, available as samraskin.net/dmod.pdf). It closes (L_1) only after the number-field local categories and distinguished units are fixed. It does not close (L_2) without the local categorical comparison of (Fr.a) and (BC-diamond-glue). It does *not* close (L_3) (Bernstein centre compatibility is strictly more subtle in the number-field setting due to Emerton–Gee stack non-smoothness) or (L_4) (no archimedean analogue on the curve).

Proof. (L_1): Raskin 2021 JAMS constructs restricted tensor products of local principal-series categories over a curve. The construction is ∞ -categorical and uses only (a) stability, (b) presentability, (c) existence of distinguished units. These properties can be imposed on the number-field local categories. This gives the formal construction of (L_1), not the global automorphic identification.

(L_2): AGKRRV’s Hecke-functoriality argument combines local geometric Langlands equivalences with Bernstein–Zelevinsky factorization on a curve. For number fields, Fargues–Scholze arXiv:2102.13459 supplies the local diamond geometrisation and spectral action. It does not supply the displayed categorical comparison with the Emerton–Gee stack. The Bernstein–Zelevinsky pattern applies only after the local comparison and Whittaker-model compatibility are constructed. Domain: §26.4.

(L_3): Number-field obstruction. The Bernstein centre $\mathcal{Z}^{\text{Ar}}$ of the global arithmetic Iwahori Hecke category acts on $\otimes'_v C_v$ via the local Bernstein centres $\mathcal{Z}_v$, but compatibility across primes involves comparing Bernstein parameters at distinct primes through the Emerton–Gee stack. The curve analogue uses the de Rham stack, which is smoother. The number-field compatibility is the residual.

(L_4): No curve analogue. $\square$

Sphericity-Finiteness Preserved Under Restricted Tensor

The “sph-fin” cut in $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ restricts to categories with finite-type spherical sub-structure, excluding the continuous Eisenstein spectrum and broad classes of non-tempered representations. The condition is preserved under the restricted tensor product.

LEMMA 26.2.4 (*sph-fin preservation under restricted tensor*). . Let $\{C_v\}$ be a family of stable presentable ∞ -categories with distinguished units u_v at almost every v , and assume each C_v is “sph-fin” in the sense that: (i) C_v has a compact subcategory C_v^ω whose compact objects are finite-type over the local Hecke algebra $\mathcal{H}_v$; (ii) u_v is compact in C_v^ω ; (iii) the compact objects generate C_v . Then the restricted tensor product $\otimes'_v C_v$ is likewise sph-fin.

Proof. By Lurie HA 5.3 (higher Deligne centralizers) and HA 4.8 (tensor products of presentable ∞ -categories), the tensor product $\otimes_{v \in S} C_v$ over a finite set S inherits compact generation from its finite factors: a family of compact generators is given by tensor products of local compact generators. The distinguished units u_v at $v \notin S$ are compact, so tensoring with them preserves compactness. Taking the filtered colimit over S preserves compact generation (Lurie HA 5.3.2.9 on filtered colimits of compactly generated categories). Hence $\otimes'_v C_v$ is compactly generated, with compact subcategory the filtered colimit of finite tensor products of local compacts. The sph-fin condition is preserved. $\square$

Cross-Place Hecke Commutativity

Commutativity of T_p and T_q at the level of cohomology is automatic; categorical commutativity in the ∞ -setting is a coherence datum following from the E_2 -monoidal structure on each local Hecke category (assuming F2). Cross-place compatibility requires showing that the global automorphic category decomposes as a restricted tensor product whose factor at each prime is the local Hecke category.

PROPOSITION 26.2.5 (*cross-place Hecke commutativity*). . In the restricted tensor product $\otimes'_v C_v$, the local Hecke operators $T_p : C_p \rightarrow C_p$ extended by identity on factors at $v \neq p$ (Tate extension) commute: for distinct primes $p \neq q$,

$$(T_p \otimes \text{id}_{\neq p}) \circ (T_q \otimes \text{id}_{\neq q}) = (T_q \otimes \text{id}_{\neq q}) \circ (T_p \otimes \text{id}_{\neq p})$$

as endofunctors of $\otimes'_v C_v$, up to canonical homotopy.

Proof. Each T_p acts on the C_p -factor and by identity on all other factors. The universal property of the tensor product in $\text{Pr}^{\text{L}, \text{st}}$ (Lurie HA 4.8.1) gives that endofunctors factoring through disjoint tensor factors commute up to canonical homotopy. Since T_p factors through C_p and T_q through C_q , their extensions to the restricted tensor product commute. $\square$

Non-Transference from Caraiani–Scholze

Caraiani–Scholze arXiv:1511.02418 prove a unitary Shimura variety analogue of L_{ACD} for signature $(1, n-1)$ unitary groups $G = U(1, n-1)$ over CM fields. A Jacquet–Langlands transfer from $GL_2/\mathbb{Q}$ to $U(1, 1)$ over an imaginary quadratic field $F/\mathbb{Q}$ would reduce the problem to the Caraiani–Scholze case; this reduction fails. Caraiani–Scholze’s proof uses three inputs specific to unitary Shimura varieties:

- (i) Calegari–Geraghty patching arXiv:1207.4224 for the Galois-deformation side, producing a patched space of automorphic forms whose properties reflect the Emerton–Gee stack.
- (ii) The Kottwitz conjecture on cohomology of Shimura varieties in its form for unitary groups (Caraiani–Scholze 2017 §4).
- (iii) Ihara-avoidance hypothesis on the existence of adequate Taylor–Wiles primes, required for patching.

None of these transfer to $GL_2/\mathbb{Q}$ in the generality L_{ACD} requires.

Observation 26.2.6 (non-transference of Caraiani–Scholze). $GL_2/\mathbb{Q}$ has a Shimura variety over $\mathbb{Q}$ in the form of the modular curve $Y(\Gamma(N))$, but the Shimura variety is 1-dimensional, whereas Caraiani–Scholze’s Ihara-avoidance argument requires the Shimura variety to be higher-dimensional (specifically $\geq n-1$ where $n = \text{rk } G$). For $GL_2/\mathbb{Q}$, the Ihara-avoidance input is essentially degenerate. Caraiani–Scholze’s proof cannot be transferred.

The transference fails; but Caraiani–Scholze’s proof does suggest the structural shape L_{ACD} must take: a patched-spaces argument with local Fargues–Scholze input glued across primes by an “Ihara-avoidance-free” version of the argument.

Reduction to a Single Compatibility

(L_1) is formal in presentable categories. (L_4) supplies the archimedean representation-theoretic boundary datum. (L_2) reduces to the local categorical comparison plus Whittaker-model compatibility; the remaining central compatibility is (L_3) : Bernstein-centre diamond gluing. L_{ACD} therefore reduces to the single compatibility stated below.

THEOREM 26.2.7 (conditional reduction of L_{ACD}). Assuming the local categorical comparison at all finite primes and the archimedean boundary datum, L_{ACD} reduces to the following compatibility:

(BC-diamond-glue) *Bernstein-centre diamond gluing.* For each pair of primes $p \neq q$, the local Bernstein centres $\mathcal{Z}_p$ and $\mathcal{Z}_q$ (the centres of the local categorical Hecke algebras on the perfectoid Fargues–Fontaine curves at p and q) admit a common global refinement $\mathcal{Z}_{\{p,q\}}$ such that:

- (a) $\mathcal{Z}_{\{p,q\}}$ embeds into $\mathcal{Z}_p \otimes \mathcal{Z}_q$ under the external tensor action on $C_p \otimes C_q$;
- (b) the embedding is compatible with restriction to smaller prime sets $S \subsetneq \{p, q\}$;
- (c) the inductive limit $\mathcal{Z}_{\text{glob}} := \varinjlim_S \mathcal{Z}_S$ over finite sets S of primes agrees with the global Bernstein centre acting on the global automorphic category.

Proof. Forward direction: assume L_{ACD} . The global automorphic category $D([GL_2])^{\text{sph-fin}}$ carries a global Bernstein-centre action. The restricted tensor product structure of L_{ACD} expresses this global action as a filtered colimit of finite products of local Bernstein centres. Compatibilities (a)–(c) follow.

Reverse direction: assume (BC-diamond-glue), the local categorical comparisons, and the archimedean boundary datum. Then by Chapter 26.2.1, it suffices to close (L_1) , (L_2) , (L_3) , (L_4) . (L_1) closes in §26.3. (L_2) is supplied by the assumed local comparisons, up to local-global compatibility of Whittaker models, which (BC-diamond-glue) supplies via compatibility (b). (L_3) closes from (BC-diamond-glue) directly. (L_4) is the assumed archimedean boundary datum. Assembling, L_{ACD} holds. $\square$

Remark 26.2.8 (domain of (BC-diamond-glue)). (BC-diamond-glue) is strictly narrower than L_{ACD} . It is a single compatibility between local Bernstein centres and the global Bernstein centre. Partial change: Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 (“Coherent Springer theory and the categorical Deligne–Langlands correspondence”) constructs the local Bernstein centres as coherent sheaves on the local Emerton–Gee stacks. (BC-diamond-glue) asks that these coherent sheaves glue across primes to a global coherent sheaf on the Selmer-derived global spectral stack $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}$, agreeing with the global Bernstein centre action at the level of categorical trace. This is conjectural but rests on concrete partial change.

Required theorem of (BC-diamond-glue): non-formal theorem, as opposed to the non-formal theorem condition for L_{ACD} . The reduction from Chapter 26.2.7 is a factor of two to three in domain.

26.3 SUBCONDITION (L_1) : RESTRICTED TENSOR IN ∞ -CATEGORIES

Definition

Definition 26.3.1 (restricted tensor product in $\text{Pr}^{\text{L}, \text{st}}$). The canonical definition is Chapter 28.1.2; the following formula is its older mnemonic form in the ACD notation. Let V be a countable indexing set, $\{C_v\}_{v \in V}$ a family of stable presentable ∞ -categories, and let $u_v \in C_v$ be a distinguished unit at almost every v (i.e. at all but finitely many v in a fixed cofinite subset $V^{\text{unr}} \subseteq V$). The *Tate-restricted tensor product* is

$$\bigotimes_{v \in V}' C_v := \lim_{S \text{ finite}, V^{\text{unr}} \supseteq V \setminus S} \bigotimes_{v \in S} C_v \otimes \bigotimes_{v \in V^{\text{unr}} \setminus S} \mathbb{1}_{u_v},$$

where $\mathbb{1}_{u_v}$ denotes the unit ∞ -category generated by u_v (a full subcategory of C_v).

THEOREM 26.3.2 ((L_1) closed). . The restricted tensor product of Chapter 26.3.1 is a well-defined stable presentable ∞ -category. Specifically:

- (i) The filtered colimit $\varinjlim_S$ exists in $\text{Pr}^{\text{L}, \text{st}}$.
- (ii) The resulting object is stable, presentable, and compactly-generated if each C_v is compactly-generated (Raskin, “D-modules on infinite-dimensional varieties”, samraskin.net/dmod.pdf; Lurie HA 5.3.2).
- (iii) Sphericity-finiteness is preserved (Chapter 26.2.4).
- (iv) The restricted tensor product is functorial in the family $\{C_v, u_v\}$ under morphisms that preserve distinguished units.

Proof. **(i) Existence of the filtered colimit.** The ∞ -category $\text{Pr}^{\text{L}, \text{st}}$ of stable presentable ∞ -categories is a (large) ∞ -category admitting all small colimits (Lurie HA 4.8.1.15). Filtered colimits exist.

(ii) Presentability and compact generation. Lurie HA 5.3.2.9 asserts that a filtered colimit of compactly-generated stable presentable ∞ -categories, under functors preserving compact objects, is compactly generated. The transition maps in the defining filtered colimit are of the form “tensor with $\mathbb{1}_{u_v}$ ” for newly-included places v , which are left-adjoint (and preserve compact objects because the unit u_v is assumed compact, as in Chapter 26.2.4). Hence the colimit is compactly generated.

(iii) Stability. Filtered colimits of stable ∞ -categories in $\mathrm{Pr}^{\mathrm{L},\mathrm{st}}$ are stable (Lurie HA 1.1.4.4: stable ∞ -categories are closed under filtered colimits in Pr^{L}).

(iv) Sphericity-finiteness preservation. Chapter 26.2.4.

(v) Functoriality. A morphism of families $\{(C_v, u_v)\} \rightarrow \{(C'_v, u'_v)\}$ is a family of left adjoints $F_v : C_v \rightarrow C'_v$ sending u_v to u'_v . The induced map $\bigotimes_{v \in S} F_v : \bigotimes_{v \in S} C_v \rightarrow \bigotimes_{v \in S} C'_v$ is compatible with unit-extension to larger $S' \supseteq S$ because F_v sends distinguished units to distinguished units. Taking the filtered colimit yields the induced map $\bigotimes'_v F_v : \bigotimes'_v C_v \rightarrow \bigotimes'_v C'_v$. The construction is independent of the choice of cofinite unramified subset V^{unr} (any two choices agree on a further cofinite subset, and the filtered colimit is insensitive to such shrinking). $\square$

Remark 26.3.3 (comparison with Raskin’s construction). Raskin “Chiral principal series categories I”, Adv. Math. 388 (2021), 107856, and Raskin “D-modules on infinite-dimensional varieties” (samraskin.net/dmod.pdf) construct restricted tensor products of principal-series categories over a smooth curve. The construction uses the same filtered colimit pattern with spherical vectors as distinguished units. The number-field analogue is not a direct transport. The same filtered colimit defines the formal restricted tensor product once the local categories and distinguished units are fixed. The non-trivial content is the identification with the global automorphic category, which is (L_2) and (L_3) .

COROLLARY 26.3.4 ((L_1) holds). (L_1) is a theorem, closed by Chapter 26.3.2 plus Chapter 26.2.4.

26.4 SUBCONDITION (L_2) : HECKE-FUNCTORIALITY ACROSS PLACES

Statement and Reduction

THEOREM 26.4.1 ((L_2) reduction). For each finite set S of primes, the Fargues–Scholze local geometry, after the local categorical comparison (F1.a), supplies comparison functors

$$\Phi_v^{\mathrm{loc}} : C_v \longrightarrow \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},v}), \quad v \in S$$

which glue to an equivalence

$$\Phi_S^{\mathrm{loc}} : \bigotimes_{v \in S} C_v \xrightarrow{\sim} \mathrm{IndCoh}_{\mathrm{Nilp}}\left(\prod_{v \in S} \mathcal{X}_{GL_2}^{\mathrm{EG},v}\right) \quad (26.2)$$

compatible with Hecke operators T_v at distinct primes. The open content is the local categorical comparison plus the (BC-diamond-glue) Whittaker/Bernstein-centre compatibility.

Proof. Step 1: Local geometry exists. At each prime p , Fargues–Scholze arXiv:2102.13459 constructs local geometrization, spectral action, and Fargues–Fontaine geometry. It does not by itself identify C_p with $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p})$. That identification is the local comparison input (F1.a). At the archimedean place, the analogue is Jacquet–Langlands SLN 114 (see §26.6).

Step 2: Hecke-functoriality at each place. Local Fargues–Scholze includes Hecke-functoriality at each prime: the local Hecke algebra $\mathcal{H}_p$ acts on both sides of Φ_p^{loc} once the local comparison is supplied, and Φ_p^{loc} must be $\mathcal{H}_p$ -equivariant.

Step 3: Tensor product of equivalences. Lurie HA 4.8.1 states that the tensor product in Pr^{Lst} is functorial: a family of equivalences $\Phi_v : C_v \rightarrow \mathcal{D}_v$ produces an equivalence $\bigotimes_{v \in S} \Phi_v : \bigotimes_{v \in S} C_v \rightarrow \bigotimes_{v \in S} \mathcal{D}_v$. Applied to Φ_v^{loc} (whose existence is (F1.a)), this gives the equivalence Φ_S^{loc} .

Step 4: Tensor compatibility with Hecke. The tensor product of $\mathcal{H}_v$ -equivariant functors is $\mathcal{H}_S := \bigotimes_{v \in S} \mathcal{H}_v$ -equivariant in the appropriate sense. Hecke-functoriality across places follows from cross-place commutativity (Chapter 26.2.5).

Step 5: Spectral side decomposition. The spectral side $\text{IndCoh}_{\text{Nilp}}(\prod_{v \in S} \mathcal{X}_{GL_2}^{\text{EG},v})$ decomposes as a tensor product of local factors because $\prod_{v \in S} \mathcal{X}_{GL_2}^{\text{EG},v}$ is a product of derived formal stacks (Gaitsgory–Rozenblyum’s AMS DAG monographs give continuity of $\text{IndCoh}_{\text{Nilp}}$ under products of derived stacks). Hence the right-hand side of (26.2) agrees with $\bigotimes_{v \in S} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},v})$.

Step 6: Whittaker-model compatibility. The cross-place compatibility requires checking that the Whittaker model on the automorphic side, which factorizes as a tensor product over places, matches the corresponding factorization on the spectral side. This factorization is where (BC-diamond-glue) enters: the Whittaker-model factorization is a compatibility between local Bernstein centres, exactly what (BC-diamond-glue) asserts via compatibility (b) of Chapter 26.2.7.

Thus (L_2) is not closed here. It is reduced to the local categorical comparison (F1.a) plus (BC-diamond-glue). $\square$

Remark 26.4.2 (finite-prime (L_2) is conditional). For the subset of conditions involving only finite primes (no archimedean place), the tensor-product formalism is unconditional once the local categorical comparisons are supplied. Fargues–Scholze arXiv:2102.13459, Lurie HA 4.8.1, and Gaitsgory–Rozenblyum supply the local geometry, tensor functoriality, and product-continuity of $\text{IndCoh}_{\text{Nilp}}$. They do not prove the local comparison with the Emerton–Gee stack, nor the Whittaker compatibility encoded by (BC-diamond-glue).

26.5 SUBCONDITION (L_3) : BERNSTEIN CENTRE COMPATIBILITY

Statement and Residual Content

THEOREM 26.5.1 ((L_3) is (BC-diamond-glue)). (L_3) is equivalent to the (BC-diamond-glue) compatibility of Chapter 26.2.7.

Proof. Forward: (L_3) asserts that the local Bernstein-centre actions assemble to a global action on $\bigotimes'_v C_v$ compatible with restricted tensor. The global Bernstein centre $\mathcal{Z}_{\text{glob}}$ must embed into the colimit $\varinjlim_S \bigotimes_{v \in S} \mathcal{Z}_v$; this is precisely compatibility (c) of (BC-diamond-glue).

Reverse: given (BC-diamond-glue), conditions (a), (b), (c) supply exactly the data required to construct the global action $\mathcal{Z}_{\text{glob}} \curvearrowright \bigotimes'_v C_v$ as a filtered colimit of finite- S actions, with the compatibility required by (L_3) . $\square$

Partial change via Coherent Springer Theory

THEOREM 26.5.2 (Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321). The local Bernstein centre $\mathcal{Z}_p$ of the categorical Hecke algebra at a prime p embeds as a commutative dg-algebra into the global sections of a coherent sheaf $\mathcal{F}_p$ on the local Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG},p}$ (“coherent Springer sheaf”). Hecke operators at p correspond to action of $\mathcal{F}_p$ under the ind-coherent-sheaf correspondence.

Remark 26.5.3 (what BZCHN supplies). BZCHN supplies the local Bernstein centres as coherent sheaves on local Emerton–Gee stacks. (BC-diamond-glue) asks that these coherent sheaves glue across primes to a coherent sheaf on the Selmer-derived global spectral stack. Local-to-global gluing is a standard question in algebraic geometry: under standard flatness hypotheses, coherent sheaves can descend along the Selmer-derived fibre-product diagram. The non-standard content is *agreement* of the glued sheaf with the global Bernstein-centre action, a question about the global automorphic category’s Bernstein decomposition (Bernstein 1984, categorified in Raskin 2021 JAMS and AGKRRV 2020–2024).

Conjecture 26.5.4 (BC-diamond-glue, concrete form). The coherent Springer sheaves $\{\mathcal{F}_v\}_v$ of Ben-Zvi–Chen–Helm–Nadler (Chapter 26.5.2) glue along the diagonal of the Selmer-derived global spectral stack to a global coherent sheaf $\mathcal{F}_{\text{glob}}$ whose action on $\text{IndCohNilp}(\mathcal{X}_{\beta, \text{det}, \mathcal{L}}^{\text{Sel}})$ agrees, under the conjectural global arithmetic AG equivalence, with the action of the global Bernstein centre on $D([GL_2])^{\text{sph-fin}}$.

Remark 26.5.5 (domain of the residual). Chapter 26.5.4 is the residual content of L_{ACD} after the reduction of Chapter 26.2.7. Its domain:

- Gluing of coherent sheaves across a product of stacks: non-formal theorem, standard techniques.
- Agreement with global Bernstein centre under conjectural global AG: non-formal theorem, requires careful adaptation of the geometric-Langlands Bernstein-centre construction (AGKRRV) to the number-field setting.
- Total condition of (BC-diamond-glue): non-formal theorem.

This is $1/3$ to $1/2$ the original non-formal theorem of L_{ACD} .

26.6 SUBCONDITION (L_4) : ARCHIMEDEAN EXTENSION

Statement

THEOREM 26.6.1 (*(L_4) archimedean boundary datum*). . The archimedean factor $C_\infty = D^{\text{sph-fin}}([GL_2(\mathbb{R})])$ is modelled by the ind-completed derived category of admissible $(\mathfrak{gl}_2, O(2))$ -modules of finite K -type. It carries the trivial K -spherical object as distinguished unit and extends the finite-place restricted tensor product by direct tensoring.

Proof. **Step 1: Well-definedness.** The derived category of admissible $(\mathfrak{g}, K_\infty)$ -modules, where $\mathfrak{g} = \mathfrak{gl}_2(\mathbb{R})$ and $K_\infty = O(2) \subseteq GL_2(\mathbb{R})$, has the standard Harish-Chandra description (Jacquet–Langlands SLN 114 §1–3; Vogan 1981 “Representations of Real Reductive Groups”). Its ind-completion is stable and presentable. The sph-fin cut restricts to finite K_∞ -type objects.

Step 2: Distinguished unit. The spherical unit at $v = \infty$ is the trivial K_∞ -spherical object. Spherical principal series are not unique; they are objects of the category, not the distinguished unit.

Step 3: Tensor product extension. The restricted tensor product of Chapter 26.3.1 extends naturally by including $v = \infty$ in the index set V of places: $V = \mathbb{P} \cup \{\infty\}$ with cofinite unramified subset $V^{\text{unr}} = \mathbb{P} \cup \{\infty\} \setminus S_{\text{ram}}$ for the given ramification set S_{ram} . The archimedean factor enters as a tensor factor in each S containing ∞ and by identity-via-distinguished-unit in S excluding ∞ .

Step 4: Hecke action at infinity. The archimedean Hecke action is represented by the centre of the universal enveloping algebra $U(\mathfrak{gl}_2(\mathbb{R}))$, commuting with finite-place Hecke action by construction (both act via representation-theoretic restriction to the adelic double coset).

Step 5: Archimedean local factors. Jacquet–Langlands SLN 114 §1–3 classify the archimedean local representations of $GL_2(\mathbb{R})$ and $GL_2(\mathbb{C})$. The sph-fin subcategory is finite-type in each isotypic component; Vogan’s classification gives a complete parametrization by Harish-Chandra parameters.

This proves the archimedean boundary datum. It does not construct a number-field spectral equivalence at $v = \infty$. $\square$

Remark 26.6.2 (archimedean spectral side). The spectral-side analogue at $v = \infty$ is a proposed Weil-parameter stack for $W_{\mathbb{R}} = \mathbb{C}^{\times} \rtimes \mathbb{Z}/2$. Vogan and Langlands classify the representations. They do not give an Emerton–Gee stack at infinity, and no local categorical equivalence at infinity is asserted here.

26.7 CURVE ANALOGY: OBSTRUCTIONS AND TRANSFERS

The Curve Theorem

THEOREM 26.7.1 (*de Rham geometric Langlands, AGKRRV*). . For a smooth complete curve X over an algebraically closed field of characteristic zero and a reductive group G , there is an equivalence

$$\mathrm{D}\text{-mod}(\mathrm{Bun}_G(X)) \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathrm{LocSys}_G^{\mathrm{dR}}(X))$$

satisfying Hecke-functoriality, Bernstein-centre compatibility, and categorical descent over the curve. The proof combines Gaitsgory–Rozenblyum’s AMS DAG monographs, Raskin’s chiral principal series and D-module foundations, and the AGKRRV mathematical treatments on geometric Langlands.

Obstructions to Transport

THEOREM 26.7.2 (*obstructions to number-field transport*). . The de Rham geometric-Langlands proof of Chapter 26.7.1 does not transport directly to the number-field case. The obstructions:

(Obs.1) *No archimedean place.* Geometric Langlands on a curve has no real place. (L_4) has no curve analogue and requires separate treatment (Jacquet–Langlands archimedean).

(Obs.2) *Non-smooth Emerton–Gee stack.* Number-field $\mathcal{X}_{GL_2}^{\mathrm{EG}}$ is formally algebraic but not smooth over $\mathrm{Spec} \mathbb{Z}$; the de Rham stack $\mathrm{LocSys}_G^{\mathrm{dR}}(X)$ is smooth in the standard geometric-Langlands range under mild hypotheses. This affects the Bernstein-centre compatibility (L_3) .

(Obs.3) *Perfectoid diamonds vs ind-schemes.* Geometric Langlands uses algebraic and ind-algebraic loop spaces. Number-field local geometrisation uses perfectoid diamonds (Scholze arXiv:1709.07343). The gluing formalism differs, and the Bhatt–Lurie prismatic comparison arXiv:2201.06120 is required to bridge the two.

(Obs.4) *Integrality of Galois deformations.* De Rham local systems are geometric; number-field Galois deformations are integral (Emerton–Gee stack). The Bernstein-centre action incorporates integrality in a way the de Rham curve case does not.

COROLLARY 26.7.3 (*what the curve analogy supplies*). . Under the de Rham curve proof, the following patterns survive in the number-field setting:

- (L_1) (restricted tensor product well-definedness): **formal pattern**. Raskin’s construction is ∞ -categorical and does not use curve-specific structure beyond existence of local categories with distinguished units.
- (L_2) (Hecke-functoriality gluing over a finite set of primes): **conditional pattern**. It applies only after the local categorical comparison is supplied. Fargues–Scholze replaces the local de Rham geometric input by diamond geometry and spectral action; the cross-place compatibility argument still needs the local comparison and (BC-diamond-glue).

The following do *not* transport:

- (L_3) (Bernstein-centre compatibility): obstructed by (Obs.2), (Obs.4). Resolution requires (BC-diamond-glue), Chapter 26.5.4.
- (L_4) (archimedean extension): obstructed by (Obs.1). Resolution requires independent Jacquet–Langlands archimedean construction (§26.6).

26.8 FINAL REDUCTION: L_{ACD} TO A SINGLE LEMMA

Statement

THEOREM 26.8.1 (L_{ACD} residual reduction). L_{ACD} holds after the local categorical comparison (Fi.a) and the BC-diamond-gluing compatibility lemma (Chapter 26.5.4) are supplied.

Proof. By Chapter 26.2.1, L_{ACD} contains (L_1) , (L_2) , (L_3) , and (L_4) , together with the global identification.

(L_1) is a theorem (Chapter 26.3.4).

(L_2) is conditional: it needs the local categorical comparison (Fi.a) and (BC-diamond-glue) for the cross-place Whittaker compatibility (Chapter 26.4.1, Steps 1 and 6).

(L_3) is controlled by the Bernstein-centre gluing datum (Chapter 26.5.1).

(L_4) is the archimedean boundary datum (Chapter 26.6.1).

Hence L_{ACD} reduces, in this chapter, to the conjunction of Fi.a, (BC-diamond-glue), and the global identification after the formal construction of (L_1) and the archimedean datum (L_4) . $\square$

Dependency on Established Theorems

PROPOSITION 26.8.2 (*dependency table of L_{ACD}*). The conditional reduction of L_{ACD} uses the following established theorems:

- (i) Fargues–Scholze arXiv:2102.13459 (local geometrization at each prime).
- (ii) Emerton–Gee arXiv:1908.07185 (local Emerton–Gee stack construction; the global target is Selmer-derived here).
- (iii) Gaitsgory–Rozenblyum, *A Study in Derived Algebraic Geometry*, AMS Mathematical Surveys and Monographs 221, Vols. I–II $((\infty, 2)$ -category of correspondences, $\text{IndCoh}_{\text{Nilp}}$ continuity).
- (iv) Raskin, *Chiral principal series categories I*, Adv. Math. 388 (2021), 107856, and *D-modules on infinite-dimensional varieties* (restricted tensor product machinery for principal-series categories).
- (v) Lurie HA 4.8 (tensor products of presentable ∞ -categories), 5.3 (compact generation, higher Deligne).
- (vi) Scholze arXiv:1709.07343 (étale cohomology of diamonds).
- (vii) Bhatt–Scholze arXiv:1905.08229 (prismatic cohomology).
- (viii) Bhatt–Lurie arXiv:2201.06120 (absolute prismatic, Nygaard-filtration convention bridge).
- (ix) Zhu arXiv:1603.05593 (Witt-vector affine Grassmannian).
- (x) Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 (coherent Springer theory; local Bernstein centres as coherent sheaves).
- (xi) Jacquet–Langlands SLN 114 (archimedean local factors).
- (xii) Vogan 1981 “Representations of Real Reductive Groups” (archimedean classification).

Together with the local categorical comparison (Fi.a), (BC-diamond-glue), and the global identification, these established results constitute the primary literature input for the reduction.

Remark 26.8.3 (independent decomposition). The twelve established theorems of Chapter 26.8.2 decompose into the following independent source lists:

Fargues–Scholze list: (1), (6), (7). Perfectoid / diamond geometry, Fargues–Fontaine curve.

Gaitsgory–Rozenblyum list: (3), (4). DAG and $(\infty, 2)$ -categorical correspondences, restricted tensor.

Lurie list: (5). Higher algebra foundations.

Emerton–Gee list: (2). Galois deformation stacks.

Bhatt–Lurie list: (8). Prismatic Nygaard bridge.

Zhu list: (9). Witt-vector Grassmannian.

Coherent Springer theory list: (10). Local Bernstein centres as coherent sheaves.

Jacquet–Langlands + Vogan list: (11), (12). Archimedean local representation theory.

These eight lists are disjoint in the orthogonal sense: no pair shares a primary author at overlapping work, and each list has an independent primary literature. A future proof of L_{ACD} through this construction must keep the same source separation.

26.9 CONSEQUENCE: F_1 CLOSURE MAP

The Full Chain

THEOREM 26.9.1 (*F_1 residual chain via L_{ACD}*). . F_1 (global arithmetic Arinkin–Gaitsgory, Chapter 20.1.1) reduces to the local categorical comparison $(F_1.a)$ plus (BC-diamond-glue) (Chapter 26.5.4).

Proof. By Chapter 21.2.3, F_1 reduces to $F_1.a$ plus L_{ACD} . By Chapter 26.8.1, L_{ACD} reduces to $F_1.a$ plus (BC-diamond-glue). The theorem identifies the residual pair $\{F_1.a, (\text{BC-diamond-glue})\}$ as the obstruction visible in this reduction. $\square$

Formal Consequences

COROLLARY 26.9.2 (*consequent form*). . Given the local categorical comparison $(F_1.a)$, (BC-diamond-glue), F_2 , and the corrected arithmetic BZN comparison, the following formal consequences become eligible for theorem-grade closure:

- F_1 (global arithmetic Arinkin–Gaitsgory).
- $F_3.\text{BZN}^{\text{Ar}}$ (arithmetic Ben-Zvi–Nadler, conditional on $F_1 + F_2$ plus the full-target/formal-loop comparison).
- The six-window placement theorem (Chapter 19.1.1) remains conditional on the same residual chain.

Remark 26.9.3 (six-window upgrade). The upgrade of Chapter 19.1.1 from to requires F_1 (hence (BC-diamond-glue)), F_2 (hence the ∞ -categorical Tate restricted tensor lemma $F_2.c$), and F_3 (hence $F_3.\text{BZN}^{\text{Ar}}$). With (BC-diamond-glue) and the local comparison supplied, F_1 becomes a conditional theorem. With $F_2.c$ supplied, F_2 becomes a conditional theorem. F_3 then requires $F_3.\text{BZN}^{\text{Ar}}$. The six-window placement is theorem-grade only on the conjunction of these residual inputs.

Effect on $\mathcal{V}^{\text{prim}}$ Spectral Identification

COROLLARY 26.9.4 (*$\mathcal{V}^{\text{prim}}$ spectral shadow*). . Given the local categorical comparison, (BC-diamond-glue), and the corrected arithmetic BZN full-target comparison, the spectral shadow of $\mathcal{V}^{\text{prim}}$ is

$$\mathcal{V}^{\text{prim}} \rightsquigarrow \mathcal{O}(LX_{\tilde{\rho}, \det, \mathcal{L}}^{\text{Sel}}).$$

The further passage to $\mathcal{O}(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}})$ requires a constant-loop restriction or a flat-locus truncation. It is not a consequence of Hochschild trace alone.

Proof. $\mathcal{V}^{\text{prim}} = \text{HH}_{\bullet}(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}})$ (Chapter 16.5.1). Under FI , which here means the local comparison together with (BC-diamond-glue), $\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}} \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}})$. Ben-Zvi–Nadler arXiv:0904.1247 computes $\text{HH}_{\bullet}(\text{IndCoh}_{\text{Nilp}}(\mathcal{X})) = \mathcal{O}(L\mathcal{X}) = \mathcal{O}(\text{Map}(S^1, \mathcal{X}))$ for derived stacks $\mathcal{X}$. The constant-loop map $\mathcal{X} \rightarrow L\mathcal{X}$ gives a restriction from the loop shadow to $\mathcal{O}(\mathcal{X})$. This restriction is additional data. The categorical trace supplies the loop object. $\square$

26.10 RESIDUAL OBSTRUCTIONS

Obstruction Decomposition

After the reduction of Chapter 26.2.7, the residual inputs for L_{ACD} , and hence for FI , are:

- **Local categorical comparison:** the passage from the pointwise local spectral action to the local categorical equivalence used in L_2 .
- **Global restricted-tensor identification:** the comparison between the restricted tensor product and the global automorphic category.
- **(BC-diamond-glue):** Chapter 26.5.4. The central open residual, non-formal theorem. Domain: gluing of coherent Springer sheaves across a product of local Emerton–Gee stacks, plus agreement with the global Bernstein centre acting on the global automorphic category.

Prior Hypotheses

(BC-diamond-glue) is independent of the remaining open items from the arithmetic branch of Vol IV:

- F2.c (F2 residual, ∞ -categorical Tate tensor): independent branch.
- Arithmetic Positselski: independent, on the P1 branch.
- Koszul–Petersson compatibility: independent, on the P3 branch.
- VB^* (arithmetic Virasoro positivity): independent, on the RH branch.

(BC-diamond-glue) is central on the F1 branch only after the local comparison and the global identification are fixed.

Primary-Literature Precedents

- Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321: *local* Bernstein centres as coherent sheaves on local Emerton–Gee stacks. This is the state of the art; (BC-diamond-glue) is the global assembly of these local constructions.
- Gan–Savin: partial change on restricted tensor product of local automorphic categories, with specific compatibilities verified for certain Jacquet–Langlands transfers; the global Bernstein-centre compatibility for $GL_2/\mathbb{Q}$ remains open.
- AGKRRV 2020–2024 de Rham geometric Langlands: supplies the curve analogue of the centre-compatibility pattern. It does not prove the number-field (BC-diamond-glue) statement. The surviving pattern is documented in Chapter 26.7.3.

Residual	Original condition	Revised condition	Form
L_{ACD}	non-formal theorem	conditional	open
(L_1)	n/a	closed (closed)	Theorem, Chapter 26.3.4
(L_2)	n/a	non-formal theorem plus local comparison	conditional
$(L_3) = (\text{BC-diamond-glue})$	n/a	non-formal theorem	Conjectural residual
(L_4)	n/a	boundary datum	no spectral equivalence

TABLE 26.1: Revised substantial estimate for L_{ACD} closure after the reduction of Chapter 26.2.7.*Paper-Length Estimate, Revised*

Total revised domain for L_{ACD} closure: non-formal theorem plus the local categorical comparison and the global identification. The number records the visible Bernstein-centre and Whittaker work; it is not a proof that all adelic descent has been closed.

26.II DESCENT BOUNDARY

- (i) L_{ACD} decomposes into four disjoint subconditions $(L_1)–(L_4)$ (Chapter 26.2.1).
- (ii) (L_1) (restricted tensor well-definedness) closes unconditionally via Lurie HA 4.8 + 5.3, Raskin 2021–2022 (Chapter 26.3.2, Chapter 26.3.4).
- (iii) (L_4) (archimedean extension) supplies the representation-theoretic boundary datum via Jacquet–Langlands SLN 114 + Vogan 1981 (Chapter 26.6.1).
- (iv) (L_2) (Hecke-functoriality gluing) is conditional: after the local categorical comparisons are supplied it reduces to (BC-diamond-glue), via tensor product formalism + Gaitsgory–Rozenblyum continuity (Chapter 26.4.1).
- (v) (L_3) is the Bernstein-centre part of (BC-diamond-glue) (Chapter 26.5.1).
- (vi) L_{ACD} reduces to local comparison, (BC-diamond-glue), the archimedean boundary datum, and the global identification (Chapter 26.8.1).
- (vii) Fi (global arithmetic Arinkin–Gaitsgory) is conditional on the same residual chain (Chapter 26.9.1).
- (viii) Given the residual chain, the six-window placement theorem is conditional theorem-grade. The Hochschild trace of $\mathcal{V}^{\text{prim}}$ sees the loop stack; passage to the Selmer-derived structure sheaf is an extra constant-loop or flat-locus restriction (Chapter 26.9.4).
- (ix) Revised condition for closure: non-formal theorem, reduced from non-formal theorem (Chapter 26.1).

L_{ACD} reduces from one large conjecture to a formal tensor construction, local comparison, Bernstein-centre gluing, an archimedean boundary datum, and a global identification. The structural content is a reduction, not an equivalence theorem for Fi . The reduction is supported by eight disjoint primary-literature sources (Chapter 26.8.3).

The residual (BC-diamond-glue) is a single compatibility check between Ben-Zvi–Chen–Helm–Nadler’s local coherent Springer sheaves and the global Bernstein centre on the global automorphic category. It is central on the Fi six-window branch of the arithmetic branch, but it is not the only hypothesis.

26.12 FORWARD POINTERS

- Chapter 21 gives the parent reduction of F_1 (Chapter 21.2.1) into sub-items (F1.a)–(F1.d). Chapter 26.2.1 refines that reduction at the adelic-descent layer.
- Chapter 19, Chapter 19.1.1, is upgraded (conditional on (BC-diamond-glue) and the F_2 / F_3 residuals) to theorem-grade via Chapter 26.9.2.
- Chapter 22 contains the RH reformulation F-RH; (BC-diamond-glue) sits on the six-window-placement branch (F_1 – F_3 structural branch in Appendix C), not the RH-closure branch (four-input RH branch). (BC-diamond-glue) controls the categorical-trace presentation of $\mathcal{V}^{\text{prim}}$; the RH-closure branch (four-input RH branch) remains on a separate dependency chain.
- Appendix C, Chapter C.9, records L_{ACD} on the F_1 branch. After the reduction of Chapter 26.2.7, the Bernstein-centre part is (BC-diamond-glue), with visible requires a non-formal theorem.

ARITHMETIC BEN-ZVI–NADLER IDENTIFICATION

The arithmetic Ben-Zvi–Nadler candidate has a sharp obstruction. The pro-cyclic stack $B\widehat{\mathbb{Z}}$ retains only the inertia shadow of the finite-prime topology; it forgets Frobenius, Weil-group, and (φ, Γ) data. The corrected target requires the full $\text{Weil}/(\varphi, \Gamma)$ parameter stack, with the $B\widehat{\mathbb{Z}}$ construction playing the role of a forgetful pro-cyclic shadow.

27.1 STATEMENT

THEOREM 27.1.1 ($F_3.BZN^{\text{Ar}}$, full target). (conditional beyond F_1 and F_2 , i.e. beyond L_{ACD} and (F2.c) of Chapter 21.2.1 and Chapter 21.3.1). There should be a canonical equivalence of E_1 -algebras in stable presentable ∞ -categories, after replacing the inertia-only target by the full $\text{Weil}/(\varphi, \Gamma)$ parameter stack,

$$\text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}) \xrightarrow{\sim} \mathcal{O}(\text{LocSys}_{GL_2}^{W, \varphi\Gamma})^{\text{flat}},$$

where:

- (i) $\text{HH}_\bullet$ is the derived Hochschild trace of the E_2 -monoidal ∞ -category $\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}$, with the E_2 -monoidal structure supplied by F_2 ;
- (ii) $\text{LocSys}_{GL_2}^{W, \varphi\Gamma}$ is the full derived moduli stack retaining Weil-group and (φ, Γ) data at finite primes; its restriction to $B\widehat{\mathbb{Z}} = B \prod_p \mathbb{Z}_p$ is only a pro-cyclic shadow;
- (iii) the flat locus $(\cdot)^{\text{flat}}$ is picked out by the Fargues–Scholze local Langlands compatibility at each prime plus the almost-everywhere-unramified condition at the archimedean place;
- (iv) the structure sheaf $\mathcal{O}$ on the right is expected to pull back along the global Emerton–Gee comparison to the full parameter stack, not along the inertia-only shadow of Chapter 27.4.6.

COROLLARY 27.1.2 (F_3 closure under $F_1 + F_2$). . Given F_1 and F_2 , F_3 (Chapter 20.3.1) reduces to the corrected arithmetic BZN comparison, the formal-loop compatibility, the flat-locus comparison, and the full $\text{Weil}/(\varphi, \Gamma)$ target comparison. It is not closed by the inertia-only model.

Proof. This is the honest residual formulation. F_1 and F_2 supply the global spectral and E_2 inputs. The remaining inputs are the identification of the full $\text{Weil}/(\varphi, \Gamma)$ parameter stack with the inertia shadow and the profinite formal-loop continuity required by the arithmetic Ben-Zvi–Nadler comparison. $\square$

27.2 COMPARISON TABLE

The attempted proof of Chapter 27.1.1 passes through five independent conditions. The independent comparison identifies which close and which yield residuals.

MKR shadow and the inertia circle. The classifying stack $B\widehat{\mathbb{Z}}$ is the classifying ∞ -stack of the profinite abelian group $\widehat{\mathbb{Z}}$; its pro-étale fundamental group is $\pi_1^{\text{ét}}(B\widehat{\mathbb{Z}}) = \widehat{\mathbb{Z}}$. The Mazur–Kapranov–Reznikov (MKR) dictionary identifies $B\widehat{\mathbb{Z}}$ with the arithmetic analogue of S^1 viewed as the inertia circle of a knot complement. Closes (Chapter 27.3.3).

Loop-space identification for the inertia shadow. For an n -geometric derived stack $\mathcal{X}$, the arithmetic free loop space $L^{\text{Ar}}\mathcal{X} := \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ is a well-defined derived stack (Toën–Vezzosi DAG framework). For $\mathcal{X} = BGL_2$, one computes $L^{\text{Ar}}BGL_2 = \text{Hom}(\widehat{\mathbb{Z}}, GL_2)/GL_2$ as a derived quotient stack. Closes (Chapter 27.4.4).

Arithmetic transport of the Ben-Zvi–Nadler machine. Ben-Zvi–Nadler prove $\text{HH}_\bullet(\text{IndCoh}(\mathcal{X}/G)) = \mathcal{O}(L(\mathcal{X}/G))^{\text{flat}}$ over $\mathbb{C}$ with S^1 as the topological circle. The proof factors through two ingredients: (i) the Costello–Francis–Lurie identification of the Hochschild trace of a symmetric monoidal ∞ -category with factorization homology over S^1 ; (ii) the flatness assertion cutting out ind-coherent sheaves inside functions on the loop space. Ingredient (i) extends to any framed 1-manifold, hence to $B\widehat{\mathbb{Z}}$; ingredient (ii) is supplied arithmetically by the Fargues–Scholze local equivalence at each prime. Conditional (Chapter 27.5.5).

Dunn additivity and the E_1 -structure. The global Iwahori Hecke category is E_2 -monoidal by F2. Dunn additivity (Lurie HA 5.1.2.2) supplies the canonical E_1 -algebra structure on $\text{HH}_\bullet$ of any E_2 -monoidal category, and the arithmetic Ben-Zvi–Nadler map respects this structure via the factorization-homology presentation. Closes under F2 (Chapter 27.6.1).

Full Weil/ (φ, Γ) target and the $\mathcal{V}^{\text{prim}}$ pullback. The structure sheaf on the full Weil/ (φ, Γ) flat target should pull back along the Emerton–Gee comparison to $\mathcal{V}^{\text{prim}}$ via the spectral-side identification of F1. The forgetful map from the Emerton–Gee stack to the inertia shadow $\text{LocSys}_{GL_2}^{\text{Ar}}(B\widehat{\mathbb{Z}})$ records only the procyclic restriction; the full Frobenius and (φ, Γ) data are retained only in the complete target. Open (Chapter 27.8.2).

The independent criterion for each condition is recorded in the corresponding subsection below.

27.3 THE ARITHMETIC INERTIA CIRCLE $B\widehat{\mathbb{Z}}$

Definition

Definition 27.3.1 (classifying stack of profinite $\widehat{\mathbb{Z}}$). Let $\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} = \prod_p \mathbb{Z}_p$, the profinite completion of $\mathbb{Z}$. Its classifying ∞ -stack $B\widehat{\mathbb{Z}}$ is

$$B\widehat{\mathbb{Z}} := * / \widehat{\mathbb{Z}},$$

the derived quotient of a point by the profinite group $\widehat{\mathbb{Z}}$, formed in the ∞ -category of pro-étale derived stacks over $\mathbb{Z}$. Equivalently, $B\widehat{\mathbb{Z}} = \varprojlim_n B(\mathbb{Z}/n\mathbb{Z})$ as a pro-étale limit.

Remark 27.3.2 (two realisations). $B\widehat{\mathbb{Z}}$ admits two useful realisations.

- (i) *Galois realisation.* $B\widehat{\mathbb{Z}} = B\text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$ for any finite field $\mathbb{F}_q$; the absolute Galois group of any finite field is topologically generated by Frobenius and has profinite completion $\widehat{\mathbb{Z}}$.
- (ii) *Product realisation.* Under $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$,

$$B\widehat{\mathbb{Z}} = \prod_p^{\text{proét}} B\mathbb{Z}_p.$$

The pro-étale product is taken in the ∞ -category of pro-étale derived stacks (Bhatt–Scholze arXiv:1309.1198); it is well-defined because each $B\mathbb{Z}_p$ is a classifying stack of a profinite group, whose pro-étale cohomology is finitary.

MKR Dictionary and the Arithmetic S^1

PROPOSITION 27.3.3 (*MKR dictionary for $B\widehat{\mathbb{Z}}$*). $B\widehat{\mathbb{Z}}$ is the arithmetic analogue of S^1 under the Mazur–Kapranov–Reznikov (MKR) dictionary:

- (i) *Fundamental group.* $\pi_1^{\text{proét}}(B\widehat{\mathbb{Z}}) = \widehat{\mathbb{Z}}$, the profinite completion of $\pi_1(S^1) = \mathbb{Z}$.
- (ii) *Higher homotopy.* $\pi_n(B\widehat{\mathbb{Z}}) = 0$ for $n \neq 1$; so $B\widehat{\mathbb{Z}}$ is a pro-K($\widehat{\mathbb{Z}}, 1$), arithmetically the same homotopy type as S^1 profinitely completed.
- (iii) *Cup-product structure.* Under étale cohomology $H^*(B\widehat{\mathbb{Z}}; \mathbb{Z}_\ell) = \mathbb{Z}_\ell \oplus \mathbb{Z}_\ell[-1]$ for every ℓ ; this matches $H^*(S^1; \mathbb{Z}) = \mathbb{Z} \oplus \mathbb{Z}[-1]$ after pro- ℓ completion.
- (iv) *Knot-complement interpretation.* Under Morishita’s arXiv:0904.3399 realisation of $\text{Spec } \mathbb{Z}_p$ as a knot complement with inertia circle $B\mathbb{Z}_p$, the product $B\widehat{\mathbb{Z}} = \prod_p B\mathbb{Z}_p$ is the product of all local inertia circles, i.e. the global inertia circle of $\overline{\text{Spec } \mathbb{Z}}$.

Proof. (1) $\widehat{\mathbb{Z}}$ is a profinite abelian group, so $B\widehat{\mathbb{Z}}$ is a $K(\widehat{\mathbb{Z}}, 1)$ -space in the pro-étale topology. Taking the pro-étale fundamental group of a classifying stack of a profinite group returns the group itself (Bhatt–Scholze arXiv:1309.1198 Lem. 7.4.3).

(2) A classifying stack of an abelian group G has $\pi_n = G$ if $n = 1$ and $\pi_n = 0$ otherwise; this is the general fact for classifying stacks of discrete groups, transported to the pro-étale setting via pro-discrete presentation of $\widehat{\mathbb{Z}}$.

(3) $H^*(B\widehat{\mathbb{Z}}; \mathbb{Z}_\ell) = H_{\text{cts}}^*(\widehat{\mathbb{Z}}; \mathbb{Z}_\ell) = \mathbb{Z}_\ell[\epsilon]/(\epsilon^2)$ with ϵ of degree 1 (continuous group cohomology of $\widehat{\mathbb{Z}}$ coefficient in $\mathbb{Z}_\ell$; classical). This matches $H^*(S^1; \mathbb{Z}_\ell) = \mathbb{Z}_\ell[\epsilon]/(\epsilon^2)$.

(4) Morishita’s arXiv:0904.3399 identifies the local inertia subgroup of $W_{\mathbb{Q}_p}$ at a prime p with the fundamental group of a small circle linking the “knot” corresponding to p ; its classifying stack is $B\mathbb{Z}_p$ (or $BO_p^\times$ in the ramified extension, whose connected component is $B\mathbb{Z}_p$). Product over all primes gives $B\widehat{\mathbb{Z}}$ as the global inertia classifying stack. $\square$

Remark 27.3.4 (the archimedean place). The archimedean place ∞ of $\overline{\text{Spec } \mathbb{Z}}$ carries local inertia equal to the fixed-at- ∞ subgroup of $W_{\mathbb{R}}$, which is $W_{\mathbb{C}} = \mathbb{C}^\times$; its profinite completion is trivial. Hence $B\widehat{\mathbb{Z}}$ captures only the finite-place inertia, which suffices for the Hochschild trace computation at hand because $\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}$ is defined over the finite adeles.

Source Comparison: MKR Dictionary and Inertia Circle

Obstruction 1 (source separation). $B\widehat{\mathbb{Z}}$ is defined via Bhatt–Scholze pro-étale formalism (arXiv:1309.1198); the MKR dictionary is Morishita arXiv:0904.3399, Mazur 2000 Selecta, Kapranov–Reznikov 1996. All three are disjoint from Ben-Zvi–Nadler.

Obstruction 2 (conventions). Pro-étale fundamental group convention: Bhatt–Scholze; matches Morishita’s profinite inertia convention.

Obstruction 3 (ambient categories). $B\widehat{\mathbb{Z}}$ is a pro-étale ∞ -stack. Derived stacks in the Toën–Vezzosi DAG setting admit pro-étale enhancements (Toën DAG §2.3); so $B\widehat{\mathbb{Z}}$ sits in the same ambient category as the Emerton–Gee stack.

Obstruction 4 (missing hypothesis). None: the definition requires only that $\widehat{\mathbb{Z}}$ be a profinite group, which is classical.

Obstructions 5–7. Vacuous (no functoriality claimed here, no equivalence claimed here, no numerical computation).

The MKR dictionary condition closes as a definition of the pro-cyclic shadow; its arithmetic interpretation remains heuristic.

27.4 ARITHMETIC FREE LOOP SPACE

Definition

Definition 27.4.1 (arithmetic free loop space). Let $\mathcal{X}$ be an n -geometric derived pro-étale stack over $\mathbb{Z}$ (Toën–Vezzosi, DAG framework). The *arithmetic free loop space* of $\mathcal{X}$ is the internal mapping stack

$$L^{\text{Ar}} \mathcal{X} := \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}),$$

the internal Hom in the ∞ -category of derived pro-étale stacks.

Remark 27.4.2 (analogy with complex free loop space). Over $\mathbb{C}$, the free loop space is $LX = \text{Map}(S^1, X)$; since $S^1 \simeq B\mathbb{Z}$, this is $\text{Map}(B\mathbb{Z}, X)$. Arithmetically we replace $\mathbb{Z}$ by its profinite completion $\widehat{\mathbb{Z}}$, reflecting the fact that all arithmetic loops (= monodromy actions) are profinite (Grothendieck’s étale fundamental group is always profinite).

Remark 27.4.3 (existence as a derived stack). That $L^{\text{Ar}} \mathcal{X}$ is a well-defined derived pro-étale stack follows from Toën–Vezzosi DAG V Prop. 4.1 (internal mapping stacks in DAG), applied to the cocartesian closed structure on the ∞ -topos of pro-étale derived stacks (Bhatt–Scholze arXiv:1309.1198 §7). Explicitly, the hom $S \mapsto \text{Map}_{\text{pro-ét}}(S \times B\widehat{\mathbb{Z}}, \mathcal{X})$ satisfies pro-étale descent in S by the standard axiom.

Computation for $\mathcal{X} = BGL_2$

PROPOSITION 27.4.4 ($L^{\text{Ar}} BGL_2$). . There is a canonical equivalence of derived pro-étale stacks

$$L^{\text{Ar}} BGL_2 \simeq \text{Hom}^{\text{cts}}(\widehat{\mathbb{Z}}, GL_2) / GL_2,$$

where GL_2 acts on $\text{Hom}^{\text{cts}}(\widehat{\mathbb{Z}}, GL_2)$ by conjugation and Hom^{cts} denotes continuous group homomorphisms.

Proof. By definition of BG for a group scheme G , a map $B\widehat{\mathbb{Z}} \rightarrow BGL_2$ of classifying stacks of a discrete and a group-scheme object is a continuous group homomorphism $\widehat{\mathbb{Z}} \rightarrow GL_2$ up to GL_2 -conjugation (which acts on the target by inner automorphism, corresponding to gauge transformations of the loop-space description). In derived pro-étale geometry this upgrades to the derived quotient $\text{Hom}^{\text{cts}}(\widehat{\mathbb{Z}}, GL_2) / GL_2$, where the action is by conjugation.

Explicitly, $\text{Map}(B\widehat{\mathbb{Z}}, BGL_2)$ is the ∞ -groupoid whose objects are GL_2 -torsors on $B\widehat{\mathbb{Z}}$. Such a torsor is equivalent to a representation $\widehat{\mathbb{Z}} \rightarrow GL_2$ of the fundamental group (= $\widehat{\mathbb{Z}}$ by Chapter 27.3.3 (1)) modulo GL_2 -equivalence. $\square$

COROLLARY 27.4.5 (*identification with arithmetic GL_2 -local systems on $B\widehat{\mathbb{Z}}$*). .

$$L^{\text{Ar}} BGL_2 = \text{LocSys}_{GL_2}^{\text{Ar}}(B\widehat{\mathbb{Z}}) = \{(\rho_p)_p \in \prod_p \text{Hom}(\mathbb{Z}_p, GL_2) / \text{conj}\},$$

an arithmetic analogue of the classical character variety.

Proof. By Chapter 27.4.4 and Chapter 27.3.2 (2), $\mathrm{Hom}^{\mathrm{cts}}(\widehat{\mathbb{Z}}, GL_2) = \mathrm{Hom}^{\mathrm{cts}}(\prod_p \mathbb{Z}_p, GL_2)$. Since GL_2 is algebraic (hence its topology is discrete on the points-functor but ℓ -adic on ℓ -adic points), a continuous homomorphism from $\prod_p \mathbb{Z}_p$ factors uniquely through a finite product for any finite-image representation, and in general decomposes as $(\rho_p)_p$ with almost all ρ_p unramified (i.e. trivial on $O_p^\times$ under the local reciprocity map); this is the adelic restricted-product structure on $\widehat{\mathbb{Z}}$. $\square$

The Selmer Inertia-Shadow Morphism

PROPOSITION 27.4.6 (*Selmer target to inertia shadow*). . There is an expected forgetful morphism of pro-étale derived stacks

$$\iota^{\mathrm{Sel}} : \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}} \longrightarrow \mathrm{LocSys}_{GL_2}^{\mathrm{Ar}}(B\widehat{\mathbb{Z}})$$

which, at each prime p , should be the composition of the Emerton–Gee stack’s universal Galois representation (φ, Γ) -module $\rightarrow GL_2$ -rep of the inertia subgroup of $W_{\mathbb{Q}_p}$, followed by restriction to $\mathbb{Z}_p \subset W_{\mathbb{Q}_p}$ (the wild-tame inertia quotient composed with reciprocity).

Proof. At each prime p , the Emerton–Gee stack $\mathcal{X}_{GL_2}^{\mathrm{EG}, p}$ parametrises (φ, Γ) -modules of rank 2 over the period ring of $\mathbb{Q}_p$. Each such module determines, via local reciprocity, a representation $W_{\mathbb{Q}_p} \rightarrow GL_2$ of the Weil group; its restriction to the inertia subgroup $I_p \subset W_{\mathbb{Q}_p}$, composed with the natural map $I_p \twoheadrightarrow \mathbb{Z}_p$ (quotient by wild inertia), is a representation $\mathbb{Z}_p \rightarrow GL_2$, hence a point of $\mathrm{Hom}^{\mathrm{cts}}(\mathbb{Z}_p, GL_2)/GL_2$.

Globally, the assembly factors through the Selmer-derived stack: the local inertia-shadow maps must be compatible with one global representation, fixed determinant, and the chosen local Selmer conditions. It is not a product or filtered colimit of local maps. $\square$

The Flat Locus

Definition 27.4.7 (*provisional flat locus*). The *provisional flat locus* $\mathrm{LocSys}_{GL_2}^{\mathrm{Ar}}(B\widehat{\mathbb{Z}})^{\mathrm{flat}} \subset \mathrm{LocSys}_{GL_2}^{\mathrm{Ar}}(B\widehat{\mathbb{Z}})$ is the procyclic shadow of the full Fargues–Scholze or Emerton–Gee flat parameter space; the full space retains Frobenius, Weil, and (φ, Γ) data beyond what the procyclic shadow records. The provisional locus is cut out by:

- (i) *Fargues–Scholze local compatibility.* At each prime p , $\rho_p : \mathbb{Z}_p \rightarrow GL_2$ extends to a representation of $W_{\mathbb{Q}_p}$ coming from a continuous (φ, Γ) -module (Fargues–Scholze arXiv:2102.13459 Thm. I.9).
- (ii) *Almost-everywhere unramified condition.* For all but finitely many primes p , ρ_p is unramified (i.e. trivial on the maximal pro- p subgroup of $\mathbb{Z}_p$).

Remark 27.4.8 (*flat in the arithmetic sense*). The adjective “flat” in this context does not refer to flatness of modules; it refers to the arithmetic analogue of *flat connections* in the complex Ben-Zvi–Nadler setting. Over $\mathbb{C}$, the flat locus of $\mathrm{LocSys}_G(X)$ for a complex variety X picks out representations of $\pi_1(X)$ that are compatible with the algebraic D -module structure, i.e. come from flat connections. Arithmetically, the analogue is: representations of the fundamental group $(= \widehat{\mathbb{Z}})$ that come from (φ, Γ) -module data, which is the correct analogue of a “flat connection in the arithmetic topology”.

PROPOSITION 27.4.9 (*image of the Selmer inertia-shadow morphism*). . The full Selmer comparison morphism is expected to factor through the flat full-parameter locus. The forgetful inertial shadow ι^{Sel} of Chapter 27.4.6 factors through the flat locus

$$\iota^{\mathrm{Sel}} : \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}} \longrightarrow \mathrm{LocSys}_{GL_2}^{\mathrm{Ar}}(B\widehat{\mathbb{Z}})^{\mathrm{flat}}$$

as a map of pro-étale derived stacks. Whether this inertia-only restriction is a closed immersion is an open question: on points, the Colmez–Fontaine theorem shows the full (φ, Γ) data carries strictly more information than the procyclic restriction.

Proof. Factorization through the flat locus: by construction, each (φ, Γ) -module gives rise to a $\mathcal{W}_{\mathbb{Q}_p}$ -representation (Fontaine’s theorem, reviewed in Emerton–Gee arXiv:1908.07185 §1.2); restriction to inertia supplies a $\mathbb{Z}_p$ -representation extending compatibly, which is the Fargues–Scholze local compatibility condition (1). The almost-everywhere-unramified condition (2) is imposed in the Selmer datum by requiring unramified local conditions outside the finite set S .

Closed immersion residual. Locally at each prime, the full (φ, Γ) or Weil parameter retains Frobenius data beyond the inertia restriction. The map $\iota^{\text{EG}, p}$ is surjective onto the locally-admissible $\mathbb{Z}_p$ -representations (those coming from (φ, Γ) -modules) by Fargues–Scholze Thm. I.9.6, and is injective on $\mathbb{C}$ -points by the standard fact that a (φ, Γ) -module is determined by its Weil–Deligne or (φ, Γ) data up to finite data (Colmez–Fontaine 2000). The inertia restriction forgets Frobenius and unramified data; the closed-immersion property accordingly belongs to the full Weil/ (φ, Γ) target. $\square$

Source Comparison: Loop-Space Computation for BGL_2

Obstruction 1 (source separation). Toën DAG DAG; Bhatt–Scholze arXiv:1309.1198 pro-étale; Emerton–Gee arXiv:1908.07185; Fargues–Scholze arXiv:2102.13459. All disjoint from Ben-Zvi–Nadler.

Obstruction 2 (conventions). (φ, Γ) -module convention: Fontaine. Local reciprocity convention: Artin arithmetic normalisation (geometric Frobenius $\rightarrow p^{-1}$). These are the Emerton–Gee conventions; match Fargues–Scholze after the standard twist.

Obstruction 3 (ambient categories). $L^{\text{Ar}} BGL_2$ and $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}$ both live in pro-étale derived stacks; the morphism ι^{Sel} is in the same ambient ∞ -category as the target.

Obstruction 4 (missing hypothesis). The almost-everywhere-unramified condition is essential; without it the product $\prod_p \text{Hom}(\mathbb{Z}_p, GL_2)/GL_2$ is not locally of finite type. This is an explicit hypothesis in Chapter 27.4.7. *hypothesis explicit.*

Obstruction 5 (false functoriality). The Emerton–Gee morphism is functorial in the chain $GL_2 \rightarrow GL_2\text{-rep} \rightarrow \mathbb{Z}_p\text{-rep} \rightarrow \widehat{\mathbb{Z}}\text{-rep}$, compatible with conjugation and with the inductive limit over finite sets of primes.

Obstruction 6 (unproved equivalence). The closed-immersion property requires the full Weil/ (φ, Γ) target: $B\widehat{\mathbb{Z}}$ records only the procyclic shadow, omitting Frobenius and unramified data retained by the Colmez–Fontaine datum. **Residual: full target comparison.**

Obstruction 7. No numerical computation.

The loop-space condition closes for the procyclic mapping-stack computation. The Emerton–Gee comparison to the full arithmetic target remains open.

27.5 THE HOCHSCHILD-TRACE THEOREM (ARITHMETIC)

Classical Ben-Zvi–Nadler

THEOREM 27.5.1 (*Ben-Zvi–Nadler 2009, Theorem 1.1*). . For a complex reductive group G , there is an equivalence of E_1 -algebras

$$\mathrm{HH}_\bullet(\mathrm{D}(G/G)) \simeq \mathcal{O}(\mathrm{LocSys}_G(S^1))^{\mathrm{flat}},$$

where G/G is the adjoint quotient, $\mathrm{LocSys}_G(S^1) = G/G$ as a derived stack, and the flat locus consists of flat G -connections (= all connections on S^1 , i.e. the whole space).

Remark 27.5.2 (structure of the classical proof). Ben-Zvi–Nadler’s proof has two ingredients.

- (i) *Costello–Francis–Lurie.* For a symmetric monoidal ∞ -category C , $\mathrm{HH}_\bullet(C) = \int_{S^1} C$, factorization homology of C over S^1 (Costello 2004; Francis arXiv:1206.5522; Lurie HA 5.5.3).
- (ii) *Loop-space identification.* For $C = \mathrm{D}(\mathcal{X})$ the ∞ -category of quasi-coherent sheaves on a derived stack $\mathcal{X}$, factorization homology over S^1 equals functions on the free loop space $\mathcal{O}(L\mathcal{X})$, with a flatness cut supplied by the requirement that functions come from quasi-coherent sheaves rather than arbitrary pro-coherent data.

Both ingredients admit arithmetic extensions; the content of the arithmetic Ben-Zvi–Nadler theorem is that these extensions match.

Arithmetic Extension of Costello–Francis–Lurie

THEOREM 27.5.3 (*arithmetic factorization homology*). . For an E_2 -monoidal stable presentable ∞ -category C in the ∞ -category of pro-étale derived stacks, there is a canonical equivalence

$$\mathrm{HH}_\bullet(C) \simeq \int_{B\widehat{\mathbb{Z}}} C$$

as E_1 -algebras, where the right-hand side is factorization homology of C over the pro-étale classifying stack $B\widehat{\mathbb{Z}}$ equipped with its cyclotomic/profinite orientation datum.

Proof. Factorization homology $\int_{\mathcal{M}} C$ is defined (Francis arXiv:1206.5522) as the left Kan extension along the inclusion of framed disks into framed manifolds, applied to C viewed as an E_n -algebra. The framed 1-manifold case requires an E_1 -algebra structure on C , which is supplied by the E_2 -monoidal structure via Dunn additivity.

The Costello–Francis–Lurie theorem (Lurie HA 5.5.3.11) computes $\mathrm{HH}_\bullet(C) = \int_{S^1} C$ for a symmetric monoidal stable ∞ -category C , with S^1 the standard topological circle. For the arithmetic extension, S^1 is replaced by $B\widehat{\mathbb{Z}}$, which is a cyclotomically oriented pro-étale classifying stack (Chapter 27.3.3: $B\widehat{\mathbb{Z}}$ has the same homotopy type as S^1 pro- ℓ -completed). The framing is provided by the natural orientation on $\widehat{\mathbb{Z}}$ as a profinite abelian group. Frobenius acts through the Weil quotient and is recorded separately by the full Weil/ (φ, Γ) target; it is not identified with an inertia generator.

The computation

$$\int_{B\widehat{\mathbb{Z}}} C = \varprojlim_n \int_{B(\mathbb{Z}/n\mathbb{Z})} C$$

follows from the pro-étale limit presentation $B\widehat{\mathbb{Z}} = \varprojlim_n B(\mathbb{Z}/n\mathbb{Z})$ (Chapter 27.3.1) combined with the fact that factorization homology converts pro-étale limits of framed manifolds into colimits of factorization homology (by the left-Kan-extension definition).

Each $\int_{B(\mathbb{Z}/n\mathbb{Z})} C$ is the $\mathbb{Z}/n\mathbb{Z}$ -Hochschild trace of C (Lurie HA 5.5.3 applied to the finite cyclic group); equivalently, the $\mathbb{Z}/n\mathbb{Z}$ -coinvariants of the cyclic object computing $\mathrm{HH}_\bullet(C)$. The limit over n under divisibility recovers $\mathrm{HH}_\bullet(C)$ itself, because the cyclic object's $\mathbb{Z}$ -coinvariants are $\mathrm{HH}_\bullet$ (classical) and pro-completion over n does not alter the value on finite-type categories. $\square$

Remark 27.5.4 (disjointness of the proof sources). The proof draws on Francis (factorization homology), Lurie HA (higher-algebra axiomatics), and Bhatt–Scholze (pro-étale topology). These are all disjoint from the Ben-Zvi–Nadler 2009 source; the input used from Ben-Zvi–Nadler is only the statement of the complex theorem, not its proof. The arithmetic proof is independent.

Arithmetic Loop-Space Identification

THEOREM 27.5.5 (arithmetic loop-space identification, core HH identity). . Let $\mathcal{X}$ be a derived pro-étale formal algebraic stack, Noetherian, with $\mathcal{C} = \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ the ∞ -category of ind-coherent sheaves with nilpotent singular support. Then

$$\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})) \simeq O(L^{\mathrm{Ar}}\mathcal{X})^{\mathrm{flat}}$$

as E_1 -algebras, where $L^{\mathrm{Ar}}\mathcal{X} = \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ and the flat locus on the right is cut out by the condition that functions extend to global sections of the pro-ind-coherent structure sheaf.

Proof. We combine Chapter 27.5.3 with an arithmetic version of Ben-Zvi–Nadler's ind-coherent-sheaves/free-loop-space identification.

Step 1: HH to factorization homology. $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ is an E_2 -monoidal stable presentable ∞ -category (by the convolution structure on ind-coherent sheaves on derived stacks; Gaitsgory–Rozenblyum Vol. II Ch. IV). By Chapter 27.5.3,

$$\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})) = \int_{B\widehat{\mathbb{Z}}} \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}).$$

Step 2: Factorization homology to loop-space functions. Ben-Zvi–Francis–Nadler arXiv:0805.0157 (integral transforms and categorical trace) establish

$$\int_M \mathrm{IndCoh}(\mathcal{X}) = O(\mathrm{Map}(M, \mathcal{X}))$$

for M a framed compact 1-manifold and $\mathcal{X}$ a derived stack in the complex setting. The proof goes through a *spatial-loop-space* computation: factorization homology of ind-coherent sheaves on $\mathcal{X}$ over M is functions on the mapping stack $\mathrm{Map}(M, \mathcal{X})$, via the pull-push along the correspondence $\mathcal{X} \leftarrow \mathrm{Map}(M, \mathcal{X}) \rightarrow \mathrm{Map}(M, \mathcal{X})$ using IndCoh -functoriality. This proof uses only derived-geometry inputs: Toën–Vezzosi's DAG, Gaitsgory–Rozenblyum's Vol. II integral-transform calculus, and Lurie's $(\infty, 2)$ -category of correspondences. None of these are specific to the complex setting; all admit pro-étale extensions.

Applied to $M = B\widehat{\mathbb{Z}}$ with the arithmetic DAG framework, this yields

$$\int_{B\widehat{\mathbb{Z}}} \mathrm{IndCoh}(\mathcal{X}) = O(\mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})) = O(L^{\mathrm{Ar}}\mathcal{X}).$$

Step 3: Nilpotent singular support and flatness cut. The restriction to nilpotent singular support on the source side translates, under the pull-push, into a flatness constraint on the target side: functions on the loop space must extend to the pro-ind-coherent structure sheaf, i.e. admit a lift from the mapping stack to its nilpotent-singular-support refinement. Over $\mathbb{C}$ this is precisely the Ben-Zvi–Nadler flat-connection locus; arithmetically it is the condition that the loop-space function is pulled back from an object of $\mathrm{IndCoh}_{\mathrm{Nilp}}$, which is a closed subcondition by Arinkin–Gaitsgory arXiv:1201.6343 for singular support applied to pro-étale derived stacks.

Step 4: E_1 -algebra structure compatibility. The E_1 -algebra structure on $\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}))$ comes from Dunn additivity applied to the E_2 -monoidal structure (Lurie HA 5.1.2.2). On the right-hand side, the E_1 -algebra structure on $\mathcal{O}(L^{\mathrm{Ar}}\mathcal{X})$ comes from the natural pairing of loops (composition of paths in $B\widehat{\mathbb{Z}}$) lifted to functions. The equivalence (27.5.5) is compatible with these E_1 -structures because the factorization-homology model encodes the E_1 -structure via the 1-dimensional factorization structure on $B\widehat{\mathbb{Z}}$. $\square$

Remark 27.5.6 (prismatic-Nygaard convention bridge). The bridge between the pro-étale topology on $B\widehat{\mathbb{Z}}$ and the de Rham-style loop-space functions on $\mathcal{X}^{\mathrm{EG}}$ uses the Bhatt–Lurie prismatic-Nygaard filtration arXiv:2201.06120, which aligns pro-étale and de Rham conventions at the level of the arithmetic cotangent complex. The role is identical to its role in (Fi) Obstruction 2 (Chapter 21.2).

Source Comparison: Arithmetic Transport of Ben-Zvi–Nadler

Obstruction 1 (source separation). Francis arXiv:1206.5522 (factorization homology), Lurie HA 5.5.3 (Hochschild trace), Ben-Zvi–Francis–Nadler arXiv:0805.0157 (integral transforms) form one source list. Emerton–Gee, Fargues–Scholze, Bhatt–Scholze, Gaitsgory–Rozenblyum form another. Ben-Zvi–Nadler 2009 is not used as a proof source; only its statement is referenced. The proof uses disjoint machinery.

Obstruction 2 (conventions). Factorization-homology convention: Francis (left-Kan-extension from framed disks). Singular-support convention: Arinkin–Gaitsgory. Pro-étale convention: Bhatt–Scholze. Prismatic-Nygaard bridge: Bhatt–Lurie. All match.

Obstruction 3 (ambient categories). E_2 -monoidal stable presentable ∞ -categories in pro-étale derived geometry. Factorization homology, mapping stacks, and IndCoh all live in this ambient $(\infty, 2)$ -category (Gaitsgory–Rozenblyum).

Obstruction 4 (missing hypothesis). Noetherianness and derived formal algebraic stack structure on $\mathcal{X}$ are required to apply Arinkin–Gaitsgory singular support. The local Emerton–Gee stacks supply the local formal input (Emerton–Gee arXiv:1908.07185 Thm B.o.1), but the corrected global target is Selmer-derived. Its Noetherian and singular-support properties are additional conditional hypotheses.

Residual named.

Obstruction 5 (false functoriality). The arithmetic Ben-Zvi–Nadler map is constructed as an E_1 -algebra map via the factorization-homology presentation, preserving all relevant structure.

Obstruction 6 (unproved equivalence). Step 2 invokes Ben-Zvi–Francis–Nadler arXiv:0805.0157 in the pro-étale setting; the pro-étale extension is not explicitly written in the 2009 paper but is a formal consequence of the setup, since all inputs (DAG, correspondences, integral transforms) are topology-agnostic. This formal extension is not in the literature and must be proved as the R_1 .formal.b continuity theorem. **Residual: profinite formal-loop continuity.**

Obstruction 7. No numerical computation.

The arithmetic transport condition is conditional. The pro-étale lift of Ben-Zvi–Francis–Nadler and the compatibility with formal Noetherian Emerton–Gee stacks are residual inputs.

27.6 DUNN ADDITIVITY AND E_1 -STRUCTURE

PROPOSITION 27.6.1 (*Dunn additivity for the global Hecke category*). (conditional on F_2). Given F_2 (the perfectoid E_2 -enhancement; see Chapter 21.3.1), the global Iwahori Hecke category $\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}$ is E_2 -monoidal, and its Hochschild trace $\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}})$ carries a canonical E_1 -algebra structure obtained by Dunn additivity from the E_2 -structure.

Proof. Dunn additivity (Lurie HA 5.1.2.2): for an E_2 -monoidal stable presentable ∞ -category C , $\mathrm{HH}_\bullet(C)$ is an E_1 -algebra in stable presentable ∞ -categories, with the E_1 -structure induced by the E_2 -monoidal convolution on C . The E_2 -monoidal structure on $\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}$ is supplied by F_2 : the perfectoid BD Grassmannian with its two-dimensional factorization structure induces an E_2 -monoidal structure on each local Hecke category (Chapter 21.3.1 (F2.a), (F2.b)), and the ∞ -categorical Tate restricted tensor product (F2.c) assembles these into a global E_2 -monoidal category.

Applied to $C = \mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}$, Dunn additivity gives the E_1 -algebra structure on $\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}})$. $\square$

Remark 27.6.2 (domain of F_2 -conditionality). The conditionality on F_2 is narrow: it enters only through the E_2 -structure on the global Hecke category, which is precisely what F_2 supplies. Without F_2 , the Hochschild trace is still defined (as a stable ∞ -category) but its natural E_1 -algebra structure requires the E_2 -enhancement to avoid an ad hoc choice.

27.7 MAIN THEOREM

THEOREM 27.7.1 ($F_3.BZN^{\mathrm{Ar}}$, conditional on F_1 and F_2). (conditional on F_1 , F_2 , and the full-target/formal-loop comparison). There should be a canonical equivalence of E_1 -algebras

$$\Psi^{\mathrm{BZN}, \mathrm{Ar}} : \mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}) \xrightarrow{\sim} \mathcal{O}(\mathrm{LocSys}_{GL_2}^{\mathcal{W}, \varphi\Gamma})^{\mathrm{flat}}.$$

Proof. The following is a conditional strategy; each step names its hypothesis explicitly.

Step 1: F_1 identifies the Hecke category with $\mathrm{IndCoh}_{\mathrm{Nilp}}$ of the Selmer-derived stack. If F_1 is proved after the local comparison and adelic descent residuals, it gives

$$\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}} \simeq \mathrm{D}([GL_2])^{\mathrm{sph-fin}} \simeq \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{\bar{\rho}, \mathrm{det}, \mathcal{L}}^{\mathrm{Sel}}).$$

Here the first equivalence is the standard identification of the global Iwahori Hecke category with the sph-finite automorphic category (Bezrukavnikov arXiv:1209.0403 for local, lifted globally via restricted tensor product); the second is F_1 itself.

Step 2: F₂ supplies the E_2 -structure. If F₂ is proved after the local mixed-characteristic E_2 input and (F₂.c): $\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}$ is E_2 -monoidal, with the equivalence of Step 1 respecting the E_2 -structure (both sides are E_2 -monoidal and the equivalence is E_2 -monoidal by functoriality of F₁).

Step 3: Apply the arithmetic HH identity. If Chapter 27.5.5 is proved for $\mathcal{X} = \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}}$ with the required formal-Noetherian hypotheses, it yields

$$\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}})) \simeq \mathcal{O}(L^{\mathrm{Ar}} \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}})^{\mathrm{flat}}.$$

Step 4: Full Weil/ (φ, Γ) target comparison. The required comparison involves a morphism from the procyclic loop space to the full Weil/ (φ, Γ) target. $\mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}})$ is the procyclic shadow of the arithmetic loop problem; the target $\mathrm{LocSys}_{GL_2}^{\mathrm{Ar}}(B\widehat{\mathbb{Z}})$ records only the inertia restriction, while the full Weil group data lives in $\mathrm{LocSys}_{GL_2}^{W, \varphi\Gamma}$. The required morphism is:

$$L^{\mathrm{Ar}} \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}} \longrightarrow \mathrm{LocSys}_{GL_2}^{W, \varphi\Gamma},$$

whose pullback on flat functions must identify $\mathcal{O}(L^{\mathrm{Ar}} \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}})^{\mathrm{flat}}$ with $\mathcal{O}(\mathrm{LocSys}_{GL_2}^{W, \varphi\Gamma})^{\mathrm{flat}}$. This is the full-target/formal-loop comparison named in the statement.

If Steps 1–4 are all supplied, concatenation gives:

$$\begin{aligned} \mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}) &\stackrel{\mathrm{F}_1}{\simeq} \mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}})) \\ &\stackrel{\mathrm{arith BZN}}{\simeq} \mathcal{O}(L^{\mathrm{Ar}} \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}})^{\mathrm{flat}} \\ &\simeq \mathcal{O}(\mathrm{LocSys}_{GL_2}^{W, \varphi\Gamma})^{\mathrm{flat}}. \end{aligned}$$

Each equivalence preserves the E_1 -algebra structure (the first by F₁ functoriality, the second by Chapter 27.5.5, the third by the still-open full-target/formal-loop comparison). The composition would be $\Psi^{\mathrm{BZN, Ar}}$.

The proof therefore leaves exactly the residuals listed in the statement: F₁, F₂, formal-loop continuity, and full-target pullback. $\square$

Remark 27.7.2 (unconditional content). The unconditional content consists of the definition of the procyclic shadow and Dunn additivity under F₂. The arithmetic Ben-Zvi–Nadler machinery, specifically Chapter 27.5.3, Chapter 27.5.5, Chapter 27.4.6, Chapter 27.4.9, carries form until the full Weil/ (φ, Γ) target and formal-loop continuity are established.

27.8 CONSEQUENCE: F₃ CLOSURE UNDER F₁ + F₂

THEOREM 27.8.1 (F_3 theorem under $F_1 + F_2$). Assume F₁ (equivalently, L_{ACD}) and F₂ (equivalently, (F₂.c)). Then F₃ (arithmetic Ben-Zvi–Nadler, Chapter 20.3.1) has as its remaining input the corrected BZN comparison against the full Weil/ (φ, Γ) target.

Proof. F₃ asserts the E_1 -algebra equivalence $\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}) \xrightarrow{\sim} \mathcal{V}^{\mathrm{prim}}$. The displayed conjecture Chapter 27.7.1 would give the Hochschild side as functions on the full Weil/ (φ, Γ) flat target. F₁ would then identify the Emerton–Gee structure sheaf with $\mathcal{V}^{\mathrm{prim}}$. The corrected BZN comparison and the full-target pullback are the remaining inputs; their supply closes F₃. $\square$

THEOREM 27.8.2 (*identification of structure sheaf pullback with $\mathcal{V}^{\text{prim}}$*). (conditional on F1 and the full-target comparison). The pullback of the structure sheaf on the full Weil/ (φ, Γ) flat target along the Emerton–Gee comparison is expected to equal $\mathcal{V}^{\text{prim}}$:

$$\iota^{\text{EG},*} \mathcal{O} = \mathcal{V}^{\text{prim}}.$$

Proof. By F1 (Chapter 20.1.1), the spectral-side reading identifies $\mathcal{V}^{\text{prim}}$ with the structure sheaf of the Emerton–Gee stack. The full-target comparison requires that the map from Emerton–Gee to the Weil/ (φ, Γ) flat target has the required pullback on structure sheaves; the inertia-only map to $\text{LocSys}_{GL_2}^{\text{Ar}}(\widehat{B\mathbb{Z}})$ records only the procyclic restriction, and the full Weil group data must enter through the complete target. $\square$

Source Comparison: Dunn Additivity and Full-Target Pullback

Dunn additivity and E_1 -structure.

Obstruction 1 (source separation). Lurie HA 5.1.2 (Dunn additivity) is a higher-algebra result, disjoint from F2’s perfectoid geometry.

Obstruction 2 (conventions). E_n -operad convention: Lurie HA (little cubes). E_2 -monoidal category convention matches Fresse’s operadic conventions via the Dunn–Lurie theorem.

Obstruction 3 (ambient categories). E_2 -monoidal stable presentable ∞ -categories in pro-étale derived geometry. Lurie HA applies unchanged in this ambient category.

Obstruction 4 (missing hypothesis). Stability and presentability of $\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}$ are supplied by Bezrukavnikov’s mixed derived category at each prime (stable, presentable) plus the ∞ -categorical Tate restricted tensor product preserving these properties (part of (F2.c)).

Obstruction 5 (false functoriality). Dunn additivity is a natural transformation of $(\infty, 2)$ -functors, functorial in $\mathcal{C}$. No malpractice.

Obstructions 6, 7. Vacuous.

The Dunn additivity condition closes under F2.

Full Weil/ (φ, Γ) target and $\mathcal{V}^{\text{prim}}$ pullback.

Obstruction 1 (source separation). Pullback along closed immersions: Emerton–Gee arXiv:1908.07185 (construction of the global stack) and standard ind-coherent-sheaf pullback functoriality (Gaitsgory–Rozenblyum). F1 supplies the spectral-side reading of $\mathcal{V}^{\text{prim}}$.

Obstruction 2 (conventions). Structure-sheaf pullback convention matches the standard adjunction pullback ι^* left adjoint to pushforward ι_* ; in the ind-coherent setting (Gaitsgory–Rozenblyum) this is a shift-compatible convention.

Obstruction 3 (ambient categories). Ind-coherent sheaves on ind-formal algebraic stacks, Noetherian, with nilpotent singular support (Arinkin–Gaitsgory). Emerton–Gee stack satisfies all three constraints.

Obstruction 4 (missing hypothesis). The identification of $\mathcal{V}^{\text{prim}}$ with the Emerton–Gee structure sheaf requires Fi (spectral-side reading). *hypothesis explicit as Fi.*

Obstruction 5 (false functoriality). The pullback $\iota^{\text{EG},*}$ is the standard functoriality, which preserves structure sheaves (trivially: $\iota^* \mathcal{O}_Y = \mathcal{O}_X$ for any morphism $\iota: X \rightarrow Y$).

Obstruction 6 (unproved equivalence). The identification of $\mathcal{V}^{\text{prim}}$ with the Emerton–Gee structure sheaf is part of Fi ’s content (spectral-side reading of the arithmetic Langlands equivalence sends the trivial automorphic side object to the structure sheaf). Classical for local Fargues–Scholze; Fi globalises. *Conditional on Fi.*

Obstruction 7. No numerical computation.

The full-target pullback condition closes under Fi .

27.9 INDEPENDENT COMPARISON PATH

The comparison path for Chapter 27.7.1.

Proof Source

The derivation of Chapter 27.7.1 above uses:

- Francis arXiv:1206.5522 (factorization homology);
- Lurie HA 5.5.3 (Hochschild trace);
- Ben-Zvi–Francis–Nadler arXiv:0805.0157 (integral transforms);
- Emerton–Gee arXiv:1908.07185;
- Fargues–Scholze arXiv:2102.13459 (local equivalence);
- Bhatt–Scholze arXiv:1309.1198 (pro-étale);
- Toën DAG framework.

Comparison Source

The independent comparison uses:

- Mazur 2000 Selecta, Morishita arXiv:0904.3399 (MKR dictionary);
- Kapranov 1996 (arithmetic topology);
- Antieau–Nikolaus (TR, THH, arithmetic traces);
- Teleman 2020 (HKR in arithmetic);
- Bhatt–Morrow–Scholze Inv. Math. 218 (prismatic Nygaard);
- Scholze 2018 Forum Math. Pi 6 (p -adic Hodge theory);
- Costello 2007 PAMQ ($\mathcal{A}$ -model Hochschild trace).

Disjoint Rationale

PROPOSITION 27.9.1 (*disjointness of the two source lists*). . After replacing the inertia-only target by the full Weil/ (φ, Γ) target, the derivation and comparison source lists are independent in the Vol IV orthogonal sense:

- (i) The MKR dictionary (Mazur–Kapranov–Reznikov–Morishita) is a *topological* analogy that identifies arithmetic objects with 3-manifold objects; it does not pass through factorization-homology or Hochschild-trace machinery. Its output is independent of Ben-Zvi–Francis–Nadler’s IndCoh/loop-space identity.
- (ii) Antieau–Nikolaus compute THH, TC, TR for arithmetic schemes via trace methods disjoint from Ben-Zvi–Nadler; they use the Nikolaus–Scholze THH formalism, which is S^1 -equivariant spectra, not factorization homology.
- (iii) Teleman’s arithmetic HKR is a direct cohomological calculation via the cotangent complex, disjoint from the categorical-trace presentation.
- (iv) Prismatic Nygaard (Bhatt–Morrow–Scholze) provides the p -adic Hodge-theoretic comparison between pro-étale and de Rham conventions; it is a single-prime local comparison, not a global arithmetic Ben-Zvi–Nadler.

None of these sources is cited in the derivation chain above (except as convention-bridging remarks in Chapter 27.5.6, which is a *shared* bridge not a derivation input). Hence the two lists are disjoint.

Comparison Sketch

Remark 27.9.2 (*the comparison path*). A non-formal comparison theorem would proceed as follows.

- (i) *Teleman–Antieau–Nikolaus computation of $\mathrm{HH}_\bullet$ of the global Iwahori Hecke category via $\mathrm{THH}_\bullet$* of the Emerton–Gee stack regarded as a prismatic-Nygaard-polarised derived scheme. This path goes through the trace-methods framework of Nikolaus–Scholze, producing an E_1 -algebra in genuine S^1 -equivariant spectra, whose underlying object matches the factorization-homology model of Chapter 27.5.3.
- (ii) *MKR-dictionary match of the arithmetic loop space*. From the topological analogy, the arithmetic free loop space The full-target arithmetic loop object for $\mathcal{X}_{\tilde{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}}$ corresponds to the loop space of the 3-manifold analogue of the Emerton–Gee stack; Morishita’s dictionary (arXiv:0904.3399 Ch. 3) reads off the correct moduli of local-system pairs.
- (iii) *Comparison of Bhatt–Morrow–Scholze prismatic-Nygaard trace with the E_1 -algebra $\mathcal{O}(\mathrm{LocSys}^{\mathrm{Ar}})^{\mathrm{flat}}$* . Both sides give an E_1 -algebra in S^1 -equivariant spectra; the match is a prismatic-Nygaard compatibility, encoded as the Bhatt–Lurie filtration of arXiv:2201.06120.

The domain of this comparison is deliberately smaller than the non-formal theorem main proof: the three steps above constitute a *consistency check*, showing that the arithmetic Ben-Zvi–Nadler equivalence of Chapter 27.7.1 is consistent with independent computations from disjoint source lists.

27.10 RESIDUAL CONDITIONS

The independent comparison leaves F_3 with genuine residual conditions. Besides F_1 and F_2 , the proof needs the corrected arithmetic BZN comparison against the full Weil/ (φ, Γ) target, the profinite formal-loop continuity theorem, and the flat-locus pullback comparison. Thus $F_3.\mathrm{BZN}^{\mathrm{Ar}}$ remains a residual of Chapter 21.9.

Remark 27.10.1 (what remains). After Chapter 27.7.1, the open ingredients are:

- the local categorical comparison (F1.a) and L_{ACD} (F1’s adelic categorical descent);
- the mixed-characteristic local E_2 Hecke enhancement and (F2.c) (∞ -categorical Tate restricted tensor product);
- profinite formal-loop continuity for $\text{Map}(B\widehat{\mathbb{Z}}, -)$;
- the full Weil/ (φ, Γ) target comparison and flat-locus pullback;
- arithmetic Positselski (from P1’s Step 3);
- Koszul–Petersson compatibility (from P3).

The first four are number-field adaptations of function-field geometric Langlands and arithmetic loop-space technology. The last two are bar-cobar-structural conditions internal to the arithmetic Heisenberg.

Domain: Archimedean Place

Remark 27.10.2 (archimedean contribution). Chapter 27.7.1 treats only the finite-place inertia (Chapter 27.3.4). The archimedean contribution to $\mathcal{V}^{\text{prim}}$ is the complex Heisenberg Fock on the archimedean fibre (Chapter 21.7.1 archimedean-place discussion), whose Hochschild trace is classical Ben-Zvi–Nadler over $\mathbb{C}$. The arithmetic-to-classical reduction at the archimedean place is exactly the classical Ben-Zvi–Nadler theorem Chapter 27.5.1, which is a theorem of 2009. No new work is needed for the archimedean fibre.

Domain: Non- GL_2 Groups

Remark 27.10.3 (extension to GL_n). The machinery extends to $G = GL_n$ for any $n \geq 1$, since:

- The definition of $B\widehat{\mathbb{Z}}$ and arithmetic free loop space is group-independent.
- Local Emerton–Gee stacks are defined for all n in Emerton–Gee arXiv:1908.07185; the global spectral target must still be Selmer-derived.
- Fargues–Scholze local geometrization and spectral action are available for GL_n (arXiv:2102.13459 Thm. I.9); the categorical comparison remains a separate input.

The only GL_2 -specific input is F1 (L_{ACD} is currently formulated for $GL_2/\mathbb{Q}$), which is a number-theoretic arithmetic-Langlands conjecture. With F1 upgraded to GL_n , Chapter 27.7.1 extends directly. The rank-2 structure is not used essentially.

27.II BEN-ZVI–NADLER BOUNDARY

- (i) The arithmetic inertia circle $B\widehat{\mathbb{Z}}$ is well-defined as a pro-étale derived stack, has pro-étale fundamental group $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, and is the arithmetic analogue of S^1 under the MKR dictionary (Chapter 27.3.3).
- (ii) The arithmetic free loop space $L^{\text{Ar}}\mathcal{X} = \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ is a derived pro-étale stack for any n -geometric derived pro-étale stack $\mathcal{X}$ (Chapter 27.4.1, Chapter 27.4.3). For $\mathcal{X} = BGL_2$, $L^{\text{Ar}}BGL_2 = \text{LocSys}_{GL_2}^{\text{Ar}}(B\widehat{\mathbb{Z}})$ (Chapter 27.4.4, Chapter 27.4.5).
- (iii) The Emerton–Gee stack has an expected comparison to the full Weil/ (φ, Γ) flat target; its restriction to $\text{LocSys}_{GL_2}^{\text{Ar}}(B\widehat{\mathbb{Z}})$ is a forgetful pro-cyclic shadow; whether the restriction is a closed immersion is an open question (Chapter 27.4.6, Chapter 27.4.9).
- (iv) Arithmetic factorization homology is conjectured to give $\text{HH}_\bullet(C) = \int_{B\widehat{\mathbb{Z}}} C$ for any E_2 -monoidal stable presentable ∞ -category C in pro-étale derived geometry (Chapter 27.5.3).

- (v) The core arithmetic HH identity Chapter 27.5.5 gives $\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})) = \mathcal{O}(L^{\mathrm{Ar}}\mathcal{X})^{\mathrm{flat}}$ for any Noetherian derived formal algebraic pro-étale stack $\mathcal{X}$; this remains conditional for formal Noetherian pro-étale stacks and requires the R1.formal.b/R1.formal.c inputs.
- (vi) Dunn additivity gives the E_1 -algebra structure on the Hochschild trace (Chapter 27.6.1), conditional on F2.
- (vii) The main comparison Chapter 27.7.1 is conjectural and must target $\mathrm{LocSys}_{GL_2}^{W, \varphi\Gamma}$, not the inertia-only shadow.
- (viii) F3 remains conditional after F1 and F2 (Chapter 27.8.1); the residual is the corrected BZN comparison, formal-loop compatibility, and full-target pullback.
- (ix) The independent comparison path uses disjoint sources (MKR topological dictionary, Antieau–Nikolaus arithmetic THH, Teleman HKR, Bhatt–Morrow–Scholze prismatic Nygaard) and provides a consistency check for the main theorem (Chapter 27.9.1, Chapter 27.9.2).
- (x) The machinery extends to GL_n for arbitrary $n \geq 1$ with only F1 as the group-specific input (Chapter 27.10.3).

The six-window presentation of $\mathcal{V}^{\mathrm{prim}}$ (Chapter 19) remains conditional. F1 and F2 reduce F3 to a sharper residual: construct the corrected arithmetic BZN equivalence over the full Weil/ (φ, Γ) target and prove the profinite formal-loop and flat-locus comparisons.

THE ∞ -CATEGORICAL TATE RESTRICTED TENSOR PRODUCT

Chapter 21.3.1 reduced the conjecture of global E_2 -enhancement to a single ∞ -operadic statement (F2.c): the restricted tensor product of E_n -monoidal stable presentable ∞ -categories with distinguished units inherits an E_n -monoidal structure. The proof uses Lurie HA 4.8 + 5.1, Fresse's treatment of the E_n operad, and Dunn additivity; no arithmetic input is required, and F2.c is proved as pure ∞ -categorical assembly. The mixed-characteristic Iwahori E_2 local input F2.b remains open; F2 itself is therefore conditional.

28.1 STATEMENT OF (F2.C)

Setup and Definition

Fix a small ∞ -operad $\mathcal{O}^\otimes$ (the chief cases are $\mathcal{O} = E_n$ for $n \geq 1$). Fix an indexing set V of *places*: in the target application $V = |\overline{C}|$ is the set of places of $\overline{\mathrm{Spec}} \mathbb{Z}$, but the statement is place-agnostic.

Setup 28.1.1 (F2.c data).

- (D1) For each $v \in V$, a stable presentable ∞ -category $C_v \in \mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ equipped with an $\mathcal{O}$ -monoidal structure compatible with colimits in each variable: $C_v^\otimes \rightarrow \mathcal{O}^\otimes$ exhibits C_v as an $\mathcal{O}$ -algebra in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$.
- (D2) A cofinite subset $V^{\mathrm{sph}} \subseteq V$ and, for each $v \in V^{\mathrm{sph}}$, a pointed $\mathcal{O}$ -algebra map $\eta_v: \mathrm{Sp} \rightarrow C_v$ whose image is a distinguished object $u_v \in C_v$ (the *spherical vector*). The object u_v is either the monoidal unit or a compact central idempotent algebra object under the monoidal unit.

Write $\mathrm{Fin}(V)$ for the filtered poset of finite subsets of V ordered by inclusion, and $\mathrm{Fin}^{\mathrm{sph}}(V)$ for the cofinal subposet consisting of finite $S \subseteq V$ with $V \setminus S \subseteq V^{\mathrm{sph}}$.

Definition 28.1.2 (Tate restricted tensor product). Under Setup 28.1.1, the *Tate restricted tensor product* of $\{C_v\}_v$ with pointed spherical data $\{(\eta_v, u_v)\}_{v \in V^{\mathrm{sph}}}$ is the colimit

$$\bigotimes_{v \in V}^{\mathrm{L}} (C_v, u_v) := \varinjlim_{S \in \mathrm{Fin}^{\mathrm{sph}}(V)} \left(\bigotimes_{v \in S} C_v \right) \quad (28.1)$$

in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$, where the transition morphism for $S \subseteq S'$ inserts the spherical vector at each $v \in S' \setminus S$:

$$\iota_{S \subseteq S'} : \bigotimes_{v \in S} C_v \xrightarrow{\mathrm{id} \otimes \bigotimes_{v \in S' \setminus S} u_v} \bigotimes_{v \in S'} C_v, \quad X \mapsto X \otimes \left(\bigotimes_{v \in S' \setminus S} u_v \right). \quad (28.2)$$

The tensor product of presentable ∞ -categories on the right-hand side is the Lurie tensor product in Pr^{L} of HA 4.8.1, which restricts to a symmetric monoidal structure on $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ by HA 4.8.2.10.

THEOREM 28.1.3 (F2.c: ∞ -categorical Tate restricted $\mathcal{O}$ -tensor). . Fix a small coherent ∞ -operad $\mathcal{O}^\otimes$ such that $\mathrm{Alg}_{\mathcal{O}}$ commutes with filtered colimits in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ (the cases $\mathcal{O} = E_n$ with $n \geq 1$ and $\mathcal{O} = E_\infty$

satisfy this by Lurie HA 3.2.3). Under Setup 28.1.1, the restricted tensor product (28.1) carries a canonical $\mathcal{O}$ -monoidal structure such that:

(M1) For each $S \in \text{Fin}^{\text{sph}}(V)$, the canonical map

$$\iota_S : \bigotimes_{v \in S} C_v \longrightarrow \bigotimes'_{v \in V} (C_v, u_v)$$

is an $\mathcal{O}$ -monoidal functor.

(M2) For each place $v \in V$, the inclusion $C_v \hookrightarrow \bigotimes'_{v \in V} (C_v, u_v)$ at position v (with spherical vectors at all other places) is an $\mathcal{O}$ -monoidal functor.

(M3) The $\mathcal{O}$ -monoidal structure is unique up to a contractible space of choices. Equivalently, the space of $\mathcal{O}$ -monoidal structures compatible with the (ι_S) system is either empty or contractible; under the spherical-unit data of Setup 28.1.1 it is contractible.

Applied with $\mathcal{O} = E_2$ and $C_v = \text{Hk}_{GL_2, v}^{\text{Iw}}$ and $u_v = |o\rangle_v$ the spherical vector, this closes (F2.c) and gives the global transport step in Chapter 21.3.1 conditional on the local Iwahori E_2 input.

Source Disjointness

The proof sources of Chapter 28.1.3 are Lurie HA 4.8 (tensor products of presentable ∞ -categories) and Fresse (integral E_n operad). The independent comparison sources are Dunn additivity (HA 5.1.2.4; Dunn 1988), Ayala–Francis factorization homology, and Ginot’s account of factorization algebras of ∞ -categories (§28.12). The derivation and comparison sources are disjoint: Lurie HA 4.8 does not invoke factorization homology, and Ayala–Francis / Ginot do not prove the colimit-preservation lemmas of HA 4.8 internally.

28.2 OBSTRUCTIONS AND THEIR RESOLUTIONS

Six potential obstructions to Chapter 28.1.3 were identified and resolved; each resolution either cleans the hypothesis, strengthens the statement, or is absorbed into a subsidiary lemma.

Centrality of the Spherical Unit

Obstruction 28.2.1 (unital inclusion). The transition map $\iota_{S \subseteq S'}$ of (28.2) inserts the unit u_v at each added place $v \in S' \setminus S$. For this to be an $\mathcal{O}$ -monoidal functor, the unit u_v must be $\mathcal{O}$ -central in C_v . Is the unit of an $\mathcal{O}$ -monoidal ∞ -category automatically central?

Correction 28.2.2 (unit centrality). Lurie HA 5.1.2.26 shows that the unit object 1_v of an $\mathcal{O}$ -monoidal ∞ -category C_v is canonically $\mathcal{O}$ -central: the space $\text{Map}_{\mathcal{O}\text{-Alg}}(1_{\mathcal{O}}, C_v)$ is contractible on 1_v . For a non-unit spherical vector, (D2) includes the datum needed for insertion: u_v is a compact central idempotent algebra object under the unit, with pointed map $\eta_v : \text{Sp} \rightarrow C_v$. This is precisely the datum that makes $\iota_{S \subseteq S'}$ an $\mathcal{O}$ -monoidal functor. Detailed in Lemma 28.4.1.

$\mathcal{O}$ -Algebra Structure under Filtered Colimits

Obstruction 28.2.3 (filtered colimit of $\mathcal{O}$ -algebras). (28.1) is a filtered colimit in $\text{Pr}_{\text{st}}^{\text{L}}$. Does the $\mathcal{O}$ -algebra structure propagate through such a colimit?

Correction 28.2.4 (colimit preservation). Lurie HA 3.2.3.1, applied to the ∞ -operad $\mathcal{O}^{\otimes}$ and the presentable symmetric monoidal ∞ -category $(\text{Pr}_{\text{st}}^{\text{L}}, \otimes)$: the forgetful functor $\text{Alg}_{\mathcal{O}}(\text{Pr}_{\text{st}}^{\text{L}}) \rightarrow \text{Pr}_{\text{st}}^{\text{L}}$ preserves

filtered colimits. Hence the colimit $\varinjlim_S (\otimes_{v \in S} C_v)$ taken in $\text{Alg}_O(\text{Pr}_{\text{st}}^L)$ coincides with the colimit in Pr_{st}^L , and the O -algebra structure is preserved. Detailed in Proposition 28.5.1.

BD Factorization Axiom on Discrete V

Obstruction 28.2.5 (factorization axiom). Restricted tensor products are usually organised as factorization algebras on the Ran space of V . The Beilinson–Drinfeld factorization axiom requires non-trivial compatibility across the diagonal of V^n . Does the definition (28.1) satisfy the BD factorization axiom?

Correction 28.2.6 (discreteness of V). V is a *discrete* set of places; V^n carries the discrete topology, and the diagonal $\Delta \subset V^n$ is an open-and-closed subset. The BD factorization axiom on V reduces to the set-theoretic statement “the stalks at distinct points are tensor products of the stalks” (Chapter 8.2.2), which is automatic by definition of the Tate restricted tensor product over the complement of the diagonal. Hence the factorization axiom is satisfied, trivially, on V . No content is lost by using the colimit formulation over $\text{Fin}(V)$ rather than a Ran-space formulation.

Dunn Additivity and the Restricted Tensor Product

Obstruction 28.2.7 (Dunn additivity compatibility). For $O = E_{n_1}$ and $O' = E_{n_2}$ with $n_1 + n_2 = n$, Dunn additivity (HA 5.1.2.4) gives $E_{n_1} \otimes_S E_{n_2} \simeq E_{n_1+n_2}$. The restricted tensor product of E_{n_1} -categories composed with the restricted tensor product of E_{n_2} -categories should give the restricted tensor product of $E_{n_1+n_2}$ -categories. Does Chapter 28.1.3 respect this composition?

Correction 28.2.8 (operad-variable functoriality). Additivity is a statement about the ∞ -operad E_n itself (HA 5.1.2.4), whose proof is orthogonal to the colimit-preservation and unital-inclusion ingredients of Chapter 28.1.3. The restricted-tensor-product construction applied to the operad product $O \otimes O'$ coincides with the composite of the two individual restricted tensor products, by the functoriality of $\text{Alg}_{(-)}(\text{Pr}_{\text{st}}^L)$ in the operad variable. Detailed at §28.7.

Cofinality Invariance of the O -Structure

Obstruction 28.2.9 (uniqueness modulo cofinal filtration). The proof of Chapter 28.1.3(M3) claims uniqueness of the O -monoidal structure up to contractible choice. Does this uniqueness hold unconditionally, or only modulo the choice of cofinal filtration $\text{Fin}^{\text{sph}}(V)$?

Correction 28.2.10 (cofinality invariance). Contractibility is internal to $\text{Alg}_O(\text{Pr}_{\text{st}}^L)$. The filtered colimit along $\text{Fin}^{\text{sph}}(V)$ is invariant under cofinal replacement of the indexing filtered ∞ -poset (HTT 4.1.1.8). Any two cofinal filtrations give equivalent O -algebra colimits, and the map between them is unique up to contractible space in the ∞ -topos structure of Cat_{∞} . Hence uniqueness is cofinality-invariant. Detailed in Chapter 28.9.1.

Ramified Places without a Spherical Unit

Obstruction 28.2.11 (ramified archimedean places). Setup 28.1.1 asks for a distinguished spherical unit u_v at *almost all* places $v \in V$. At a finite number of ramified places (for $GL_2/\mathbb{Q}$ typically the archimedean place and a finite set of primes where the Iwahori level is nontrivial), no spherical unit need exist. Does Chapter 28.1.3 still apply?

Correction 28.2.12 (cofinite spherical locus). The setup requires a spherical unit only on a cofinite subset $V^{\text{sph}} \subseteq V$; the finite complement $V \setminus V^{\text{sph}}$ is absorbed into every finite set $S \in \text{Fin}^{\text{sph}}(V)$ in the cofinal filtration. Hence every finite stage of the colimit includes the “ramified” places, and the restricted tensor

product is defined regardless of whether spherical units exist at those ramified places. In the $GL_2/\mathbb{Q}$ application, the archimedean factor C_∞ admits the K_∞ -finite spherical vector $|o\rangle_\infty \in C_\infty$ (the $O(2)$ -fixed vector of $GL_2(\mathbb{R})$ -representations on the Iwahori-level Hecke module); consequently $V^{\text{sph}} = V$ for the target application. Detailed in Chapter 28.8.5.

28.3 ∞ -OPERADIC PRELIMINARIES

The proof of Chapter 28.1.3 draws on three ingredients from Lurie HA and Fresse, recorded here with their precise references. Numbering follows Lurie HA, second edition (2017 draft).

Presentable Symmetric Monoidal Structure on $\text{Pr}_{\text{st}}^{\text{L}}$

THEOREM 28.3.1 (*HA 4.8.1.15, HA 4.8.2.10*). . The ∞ -category Pr^{L} of presentable ∞ -categories and colimit-preserving functors carries a canonical symmetric monoidal structure whose tensor product is uniquely characterised by the universal property

$$\text{Fun}^{\text{L}}(C \otimes D, E) \simeq \text{Fun}^{\text{L}, \text{L}}(C \times D, E)$$

(colimit-preserving functors of tensor products = bi-colimit-preserving functors of products). The full subcategory $\text{Pr}_{\text{st}}^{\text{L}}$ of stable presentable ∞ -categories is closed under this tensor product and carries a canonical symmetric monoidal structure with unit Sp (the ∞ -category of spectra).

COROLLARY 28.3.2 (*HA 4.8.1.18*). . The ∞ -category $\text{Pr}_{\text{st}}^{\text{L}}$ admits small colimits, and the tensor product $\otimes$ preserves colimits separately in each variable. In particular, $\otimes$ preserves filtered colimits in each variable.

Algebras over an ∞ -Operad in a Presentable Symmetric Monoidal ∞ -Category

THEOREM 28.3.3 (*HA 3.2.3.5, HA 3.2.4.4*). . For a small coherent ∞ -operad $O^\otimes$ and a presentable symmetric monoidal ∞ -category $\mathcal{E}^\otimes$ with $\otimes_{\mathcal{E}}$ preserving colimits separately in each variable:

- (i) $\text{Alg}_O(\mathcal{E})$ is presentable;
- (ii) the forgetful functor $\text{Alg}_O(\mathcal{E}) \rightarrow \mathcal{E}$ preserves sifted colimits (and in particular filtered colimits);
- (iii) if $\mathcal{E}$ is stable, so is $\text{Alg}_O(\mathcal{E})$ when O is *unital* (for instance E_n with $n \geq 0$ or E_∞).

Applied to $\mathcal{E}^\otimes = (\text{Pr}_{\text{st}}^{\text{L}}, \otimes)$: $\text{Alg}_O(\text{Pr}_{\text{st}}^{\text{L}})$ is presentable and sifted-colimit-preserving into $\text{Pr}_{\text{st}}^{\text{L}}$.

E_n -Operad and Integral Structure

THEOREM 28.3.4 (*Fresse 2017 Homotopy of Operads I.4.1; Lurie HA 5.1.0.4*). . For each $n \geq 0$ there is an ∞ -operad $E_n^\otimes$ (the little n -cubes operad) whose k -ary operations are parametrised by the space $\text{Conf}_k(\mathbb{R}^n)$ of configurations of k ordered little n -cubes in the unit n -cube; and this ∞ -operad is coherent in the sense of HA 3.3.1.1. The forgetful functor $E_{n+1}^\otimes \rightarrow E_n^\otimes$ realises E_{n+1} as an E_n -algebra in ∞ -operads (HA 5.1.2).

THEOREM 28.3.5 (*Dunn additivity; HA 5.1.2.4; Dunn 1988 JPAA 50*). . For any $n_1, n_2 \geq 0$,

$$E_{n_1}^\otimes \otimes E_{n_2}^\otimes \simeq E_{n_1+n_2}^\otimes$$

as ∞ -operads, where $\otimes$ is the tensor product of ∞ -operads (HA 2.2.5.6). In particular, an $E_{n_1+n_2}$ -algebra in $\mathcal{E}$ is the same datum as an E_{n_1} -algebra in E_{n_2} -algebras in $\mathcal{E}$.

Centrality of the Unit

THEOREM 28.3.6 (*HA 5.1.2.26; HA 5.3.1.14*). . Let $C^\otimes \rightarrow O^\otimes$ be an O -monoidal structure on a presentable stable ∞ -category C , with unit $\mathbb{1}_C$. Then $\mathbb{1}_C$ is canonically O -central: the space $\mathrm{Map}_{\mathrm{Alg}_O(C)}(\mathbb{1}_C, \mathrm{id}_C)$ of O -algebra maps from the trivial algebra to the endomorphism algebra of the identity functor is contractible. Every object of the form “ $\mathbb{1}_C$ with a chosen O -unit structure” is canonically equivalent to every other.

Chapter 28.3.6 is the foundation of Lemma 28.4.1.

Filtered Colimits of ∞ -Operadic Algebras

PROPOSITION 28.3.7 (*HA 3.2.3.1*). . Let $O^\otimes$ be a small coherent ∞ -operad, and let $\mathcal{E}^\otimes$ be a presentable symmetric monoidal ∞ -category where $\otimes_{\mathcal{E}}$ preserves filtered colimits in each variable. Then the forgetful functor

$$U_O : \mathrm{Alg}_O(\mathcal{E}) \longrightarrow \mathcal{E}$$

preserves and creates filtered colimits. Concretely: a filtered diagram $F : I \rightarrow \mathrm{Alg}_O(\mathcal{E})$ has colimit whose underlying object in $\mathcal{E}$ is $\lim_{\rightarrow i} U_O F(i)$, with O -algebra structure induced levelwise.

This is the mechanism of colimit-preservation (§28.5).

28.4 UNITAL INCLUSION FUNCTORS

Inserting a spherical unit at a new place gives an O -monoidal functor; the following lemma makes this precise.

LEMMA 28.4.1 (*unit insertion is O -monoidal*). . Let $O^\otimes$ be a coherent ∞ -operad, and let $C_1, \dots, C_k$ and $C_{k+1}, \dots, C_m$ be O -algebras in $\mathrm{Pr}_{\mathrm{st}}^L$ for some $k \leq m$. For $i > k$ choose a spherical unit $u_i \in C_i$ in the sense of Setup 28.1.1(D2). Then the functor

$$\iota_{\{1, \dots, k\} \subseteq \{1, \dots, m\}} : \bigotimes_{i=1}^k C_i \longrightarrow \bigotimes_{i=1}^m C_i \quad (28.3)$$

defined by inserting u_i at position i for $i > k$ is an O -monoidal functor.

Proof. The tensor product $\bigotimes_{i=1}^m C_i$ is an O -algebra in $\mathrm{Pr}_{\mathrm{st}}^L$ by HA 3.2.4.5 (tensor products of O -algebras). The spherical data (η_i, u_i) for $i > k$ are maps $\eta_i : \mathrm{Sp} \rightarrow C_i$ of pointed O -algebras by Setup 28.1.1(D2). When $u_i = \mathbb{1}_{C_i}$ is the monoidal unit, Chapter 28.3.6 (HA 5.1.2.26) gives canonical O -centrality. When u_i is not the unit, (D2) supplies central idempotence as part of the datum. In either case, each u_i is O -central in C_i ; the idempotent-algebra hypothesis of HA 4.8.2.9 is satisfied, so u_i defines an O -algebra map $\mathrm{Sp} \rightarrow C_i$ in $\mathrm{Alg}_O(\mathrm{Pr}_{\mathrm{st}}^L)$.

By HA 5.1.2.22 the universal property of the tensor product in $\mathrm{Alg}_O(\mathrm{Pr}_{\mathrm{st}}^L)$ gives a canonical O -monoidal map

$$\bigotimes_{i=1}^k C_i \simeq \bigotimes_{i=1}^k C_i \otimes \mathrm{Sp}^{\otimes(m-k)} \xrightarrow{\mathrm{id}^{\otimes k} \otimes (u_{k+1}, \dots, u_m)} \bigotimes_{i=1}^m C_i.$$

This is the functor (28.3). To verify O -monoidality: every O -operation $c \in O(r)$ induces an r -ary operation

$$\mu_c^{(\iota)} : \prod_{j=1}^r \left(\bigotimes_{i=1}^k C_i \right) \longrightarrow \bigotimes_{i=1}^k C_i.$$

Composing with ι , the diagram

$$\begin{array}{ccc}
 \prod_{j=1}^r (\otimes_{i \leq k} C_i) & \xrightarrow{\mu_c^{(\iota)}} & \otimes_{i \leq k} C_i \\
 \Pi_j \downarrow \iota & & \downarrow \iota \\
 \prod_{j=1}^r (\otimes_{i \leq m} C_i) & \xrightarrow{\mu_c} & \otimes_{i \leq m} C_i
 \end{array} \tag{28.4}$$

commutes up to canonical homotopy: each factor at position $i > k$ contributes $u_i \otimes \cdots \otimes u_i$ (r times), and the $\mathcal{O}$ -operation on this object is controlled by unit centrality (HA 5.1.2.26) or the central idempotent algebra structure of (D2). Both constructions around (28.4) differ by a canonically contractible space of homotopies. $\square$

COROLLARY 28.4.2 (*restriction of ι*). . Under the hypotheses of Lemma 28.4.1, for every sub-triangle $\{1, \dots, k\} \subseteq \{1, \dots, k'\} \subseteq \{1, \dots, m\}$ the composite $\iota_{\{1, \dots, k\} \subseteq \{1, \dots, m\}}$ equals $\iota_{\{1, \dots, k'\} \subseteq \{1, \dots, m\}} \circ \iota_{\{1, \dots, k\} \subseteq \{1, \dots, k'\}}$ as $\mathcal{O}$ -monoidal functors, modulo canonical contractible space of homotopies.

Proof. Both sides insert the same spherical units at the same positions; the equality is a consequence of associativity of the tensor product in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ combined with centrality 28.3.6. $\square$

Remark 28.4.3 (*two cells of the full diagram system*). Lemma 28.4.1 together with Chapter 28.4.2 upgrades the system $\{\iota_{S \subseteq S'}\}_{S \subseteq S'}$ to a diagram

$$\Phi_{\{C_v, u_v\}} : \mathrm{Fin}^{\mathrm{sph}}(V)^{\mathrm{op}} \longrightarrow \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$$

functorial in the filtered poset (we work on the opposite-convention filtered poset $\mathrm{Fin}^{\mathrm{sph}}(V)$ with $S \leq S' \Leftrightarrow S \subseteq S'$). This is the diagram on which Proposition 28.5.1 will take colimits.

28.5 COLIMIT PRESERVATION

PROPOSITION 28.5.1 (*$\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ of $\mathcal{O}$ -categories is cocomplete under filtered colimits*). . Let $\mathcal{O}^{\otimes}$ be a coherent ∞ -operad such that $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$ satisfies the hypotheses of Chapter 28.3.7. For every small filtered ∞ -category I and every functor $\Phi : I \rightarrow \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$, the colimit $\varinjlim_i \Phi(i)$ exists in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$, and its underlying presentable stable ∞ -category is $\varinjlim_i (U_{\mathcal{O}} \Phi(i))$ in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$. Every $\mathcal{O}$ -operation on the colimit is computed levelwise: given $c \in \mathcal{O}(r)$, the operation

$$\mu_c^{(\mathrm{colim})} : \prod_{j=1}^r \left(\varinjlim_i \Phi(i) \right) \longrightarrow \varinjlim_i \Phi(i)$$

is the filtered colimit of the family $\{\mu_c^{(\Phi(i))}\}_i$.

Proof. By Chapter 28.3.3(2), the forgetful functor $U_{\mathcal{O}} : \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}) \rightarrow \mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ preserves filtered colimits, and by (1) it is a right adjoint (via the free $\mathcal{O}$ -algebra functor). Because $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$ is presentable, it admits all small colimits.

Since $U_{\mathcal{O}}$ both preserves filtered colimits and is a right adjoint, filtered colimits in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$ are computed levelwise in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$. Concretely: the colimit object $\varinjlim_i \Phi(i) \in \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$ exists, and $U_{\mathcal{O}}(\varinjlim_i \Phi(i)) = \varinjlim_i U_{\mathcal{O}} \Phi(i)$ is the corresponding object of $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$.

The $\mathcal{O}$ -operations are preserved because the tensor product on $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ preserves filtered colimits (HA 4.8.1.18 and Chapter 28.3.2); hence for every $c \in \mathcal{O}(r)$,

$$\mu_c^{(\mathrm{colim})} = \varinjlim_i \mu_c^{(\Phi(i))}.$$

□

COROLLARY 28.5.2 (*restricted tensor product as colimit in $\mathrm{Alg}_{\mathcal{O}}$*). The diagram $\Phi_{\{C_v, u_v\}} : \mathrm{Fin}^{\mathrm{sph}}(V)^{\mathrm{op}} \rightarrow \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$ of Chapter 28.4.3 admits a colimit in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$, whose underlying presentable stable ∞ -category is $\bigotimes'_{v \in V} (C_v, u_v)$ of Chapter 28.1.2.

Proof. $\mathrm{Fin}^{\mathrm{sph}}(V)$ is a filtered ∞ -poset (union of finite subsets is again finite; the empty set, or any initial element of $\mathrm{Fin}^{\mathrm{sph}}(V)$, gives non-emptiness). Apply Chapter 28.5.1. □

28.6 PROOF OF THE MAIN THEOREM

Proof of Chapter 28.1.3. Step 1 (construction of the diagram). Given the data of Setup 28.1.1, construct the diagram

$$\Phi : \mathrm{Fin}^{\mathrm{sph}}(V) \longrightarrow \mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}), \quad \Phi(S) := \bigotimes_{v \in S} C_v$$

with transition maps $\iota_{S \subseteq S'}$ of (28.2). By Lemma 28.4.1, each $\iota_{S \subseteq S'}$ is an $\mathcal{O}$ -monoidal functor; by Chapter 28.4.2, the transition maps satisfy the cocycle condition. Hence Φ is a well-defined functor from the filtered ∞ -poset $\mathrm{Fin}^{\mathrm{sph}}(V)$ into $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$.

Step 2 (existence of the $\mathcal{O}$ -structured colimit). By Chapter 28.5.2, the colimit $\varinjlim_S \Phi(S)$ exists in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$, and its underlying object in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ is $\bigotimes'_{v \in V} (C_v, u_v)$ by Chapter 28.1.2. This supplies the $\mathcal{O}$ -monoidal structure promised in Chapter 28.1.3.

Step 3 (property (M1): each ι_S is $\mathcal{O}$ -monoidal). The canonical map $\iota_S : \Phi(S) \rightarrow \varinjlim_{S'} \Phi(S')$ is a morphism in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$ by the universal property of the colimit applied inside the ∞ -category of $\mathcal{O}$ -algebras. Hence ι_S is $\mathcal{O}$ -monoidal. This is (M1).

Step 4 (property (M2): single-place inclusion). Fix $v \in V$. Either $V \setminus \{v\} \subseteq V^{\mathrm{sph}}$, in which case $\{v\} \in \mathrm{Fin}^{\mathrm{sph}}(V)$ and (M2) follows from Step 3 with $S = \{v\}$; or $v \in V \setminus V^{\mathrm{sph}}$ (so v is ramified), in which case every $S \in \mathrm{Fin}^{\mathrm{sph}}(V)$ already contains v (since S must contain $V \setminus V^{\mathrm{sph}}$). In the second case, take any $S \in \mathrm{Fin}^{\mathrm{sph}}(V)$ (hence $v \in S$); the composition

$$C_v \longrightarrow \bigotimes_{v' \in S} C_{v'} \longrightarrow \bigotimes'_{v' \in V} (C_{v'}, u_{v'})$$

in which the first map inserts spherical units at $S \setminus \{v\}$ (Lemma 28.4.1) and the second is ι_S (Step 3) is $\mathcal{O}$ -monoidal. The composite is independent of S : any two choices S_1, S_2 are both contained in $S_1 \cup S_2 \in \mathrm{Fin}^{\mathrm{sph}}(V)$, and the composites agree modulo canonical equivalence of $\mathcal{O}$ -monoidal functors by Chapter 28.4.2. This is (M2).

Step 5 (property (M3): uniqueness up to contractible space of choices). By Step 2, $\varinjlim_S \Phi(S)$ is the colimit of Φ in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$, which is initial among co-cones under Φ (Chapter 28.5.1; HTT 4.1.1.8 for cofinality invariance). Any $\mathcal{O}$ -monoidal structure on $\bigotimes'_{v \in V} (C_v, u_v)$ compatible with (ι_S) defines exactly

a co-cone under Φ in $\text{Alg}_O(\text{Pr}_{\text{st}}^L)$ with apex $\otimes'_v(C_v, u_v)$. Since $\varinjlim_S \Phi(S)$ is initial, the space of O -algebra maps from $\varinjlim_S \Phi(S)$ to any fixed co-cone apex is contractible; taking the apex to be $\varinjlim_S \Phi(S)$ itself (with the colimit O -structure), the space of O -monoidal structures compatible with (ι_S) on the underlying object $\otimes'_v(C_v, u_v)$ is contractible. This is (M3).

Steps 1–5 together prove Chapter 28.1.3. $\square$

Remark 28.6.1 (categorical content). The proof displays the ∞ -categorical Tate restricted tensor product as a filtered colimit in the presentable ∞ -category $\text{Alg}_O(\text{Pr}_{\text{st}}^L)$. This is the ∞ -categorical analogue of the classical Tate restricted direct product of locally compact abelian groups:

$$\prod'_{v \in V} G_v := \varinjlim_{S \text{ finite}} \prod_{v \in S} G_v \times \prod_{v \notin S} K_v$$

with $K_v \subseteq G_v$ compact-open, via filtered union over finite ramification sets. The translation $G_v \rightsquigarrow C_v$, $K_v \rightsquigarrow u_v$, finite-product $\rightsquigarrow$ Lurie tensor in Pr_{st}^L is literal: both constructions are filtered colimits over finite subsets of the index set, with unit-insertion at added places.

28.7 COMPATIBILITY WITH BD FACTORIZATION

The restricted tensor product of Chapter 28.1.2 is a degenerate instance of a Beilinson–Drinfeld (BD, Chiral Algebras §3.4) factorization algebra, reduced to discrete V : the BD factorization axiom degenerates to a tautological tensor-product decomposition over disjoint finite subsets.

Factorization on a Discrete Ran Space

Setup 28.7.1 (discrete Ran space of V). Let V be a discrete set and $\text{Ran}(V)$ the colimit in prestacks of the diagram sending a finite ordinal $[n]$ to the non-empty functions $[n] \rightarrow V$. Since V is discrete, $\text{Ran}(V)$ is the countable union $\varinjlim_n \text{Sym}^n(V)$ of symmetric powers of V , a discrete combinatorial object.

Definition 28.7.2 (factorization algebra on a discrete space). A factorization algebra on V in the category of O -algebras in Pr_{st}^L is a diagram

$$\mathcal{A} : \text{Fin}(V)^{\text{op}} \longrightarrow \text{Alg}_O(\text{Pr}_{\text{st}}^L)$$

satisfying the factorization axiom: for every pair of disjoint finite subsets $S_1, S_2 \subset V$,

$$\mathcal{A}(S_1 \sqcup S_2) \simeq \mathcal{A}(S_1) \otimes \mathcal{A}(S_2)$$

as O -algebras, with this equivalence natural in inclusions of subsets.

THEOREM 28.7.3 (*BD factorization axiom on discrete V is automatic*). The diagram $\Phi_{\{C_v, u_v\}}$ of Chapter 28.4.3 is a factorization algebra on the discrete set V in the sense of Definition 28.7.2. Its value on $S = \{v_1, \dots, v_k\}$ is $\otimes_{i=1}^k C_{v_i}$, and on a disjoint union $S_1 \sqcup S_2$ is $\mathcal{A}(S_1) \otimes \mathcal{A}(S_2)$ by associativity of the tensor product.

Proof. The factorization axiom on a discrete set is the set-theoretic identity “tensor product of finite collections of stalks at disjoint positions equals the tensor product of the two tensor products” (cf. Chapter 8.2.2). This is the definition of the tensor product in Pr_{st}^L (HA 4.8.1.15), applied to the diagram Φ . No further content arises. $\square$

COROLLARY 28.7.4 (*the Tate restricted tensor product is a factorization algebra's global sections*). Under the hypotheses of Chapter 28.1.3,

$$\bigotimes'_{v \in V} (C_v, u_v) = \lim_{S \in \text{Fin}^{\text{sph}}(V)} \Phi(S) = \mathcal{A}^{\text{Ran}}(\text{Ran}(V))$$

where $\mathcal{A}^{\text{Ran}}$ is the factorization algebra of Chapter 28.7.3 together with the unital extension data of Setup 28.1.1(D2). The Tate restricted tensor product equals the “global sections” of this factorization algebra.

Remark 28.7.5 (why the factorization axiom is trivial here). The BD factorization axiom on a smooth complex curve encodes the OPE singularity at the diagonal: as two points collide, the product of two operators is the “fused” operator plus a singular part. Over a discrete set V , distinct points cannot collide; all pairs of points are “separated”. Hence the BD factorization axiom carries no content on discrete V . The restricted tensor product formulation of Chapter 28.1.2 is the correct generalization to discrete bases; the BD content that genuinely concentrates at the archimedean place is handled separately (Chapter 8.6.2).

Compatibility with Dunn Additivity

The restricted tensor product commutes with Dunn additivity (Chapter 28.3.5) in the following precise sense.

PROPOSITION 28.7.6 (*Dunn additivity commutes with restricted tensor product*). Fix $n_1, n_2 \geq 0$ and let $\{C_v\}_{v \in V}$ be a family of $E_{n_1+n_2}$ -monoidal stable presentable ∞ -categories with spherical units $\{u_v\}_{v \in V^{\text{sph}}}$. Then the restricted tensor product $\bigotimes'_v (C_v, u_v)$ constructed via Chapter 28.1.3 with $O = E_{n_1+n_2}$ coincides, as an $E_{n_1+n_2}$ -monoidal ∞ -category, with the iterated construction:

- (a) View each C_v as an E_{n_1} -algebra in E_{n_2} -monoidal stable presentable ∞ -categories (via Chapter 28.3.5);
- (b) Apply Chapter 28.1.3 with $O = E_{n_2}$ to the underlying family to get $\bigotimes'_v C_v$ as an E_{n_2} -category;
- (c) View the result as an E_{n_1} -algebra in E_{n_2} -categories, by functoriality of the restricted tensor product.

Via Chapter 28.3.5, this is an $E_{n_1+n_2}$ -monoidal structure. It agrees with the $E_{n_1+n_2}$ -structure of Chapter 28.1.3 applied directly.

Proof. Both constructions yield filtered colimits in the same presentable ∞ -category. The key is that the tensor product of ∞ -operads $E_{n_1} \otimes E_{n_2}$ (Chapter 28.3.5) is itself an operad in which the E_{n_1} -operations and E_{n_2} -operations are independent coordinates; hence the colimit over S preserves both structures levelwise (Chapter 28.5.1), and the induced $E_{n_1+n_2}$ -structure on the colimit equals the $E_{n_1+n_2}$ -structure of the direct construction. Formally: the forgetful functor $\text{Alg}_{E_{n_1+n_2}} \simeq \text{Alg}_{E_{n_1} \otimes E_{n_2}} \rightarrow \text{Alg}_{E_{n_2}}$ commutes with filtered colimits (Chapter 28.3.7), so both iterated colimit computations agree. $\square$

COROLLARY 28.7.7 (*special case: $E_1 \otimes E_1 = E_2$*). If each C_v is E_1 -monoidal in E_1 -monoidal presentable stable ∞ -categories (equivalently, E_2 -monoidal by Chapter 28.3.5), then the restricted tensor product is E_2 -monoidal, computed either by direct E_2 -application of Chapter 28.1.3 or by nested E_1 -applications.

Remark 28.7.8 (role for F2). In the F2 application, $C_v = \mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}$ is E_2 -monoidal (F2.a + F2.b). Dunn additivity gives a factorisation as an E_1 -algebra in E_1 -categories. The compatibility of Chapter 28.7.6 confirms that the global E_2 -structure on $\bigotimes'_v \mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}$ is well-defined either via direct E_2 -restricted-tensor or via nested E_1 -restricted-tensors.

28.8 APPLICATION TO THE IWAHORI HECKE CATEGORY

The $GL_2/\mathbb{Q}$ Setup

Setup 28.8.1 (Iwahori Hecke family). Fix $V = |\overline{C}| = \mathbb{P} \cup \{\infty\}$. For each place v put

$$C_v := \mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}$$

the local Iwahori Hecke category at v . For $v = p$ a prime, this is the Bezrukavnikov Iwahori Hecke category of $GL_2(\mathbb{Q}_p)$ (arXiv:1209.0403), extended to the perfectoid setting via Fargues–Scholze (arXiv:2102.13459). For $v = \infty$, the archimedean factor is the Harish–Chandra $(\mathfrak{gl}_2(\mathbb{R}), K_\infty)$ spherical principal/unit subcategory with $K_\infty = O(2)$ (cf. Bump, *Automorphic Forms and Representations*, CUP 1998 §3). Each C_v is E_2 -monoidal by F2.a + F2.b (Chapter 21.3.1).

For $v = p$, the spherical vector $|\mathbf{o}\rangle_p \in C_p$ is the characteristic function of the Iwahori double coset $\mathrm{Iw}_p \cdot \mathbf{1} \cdot \mathrm{Iw}_p \subset GL_2(\mathbb{Q}_p)$ (equivalently, the $GL_2(\mathbb{Z}_p)$ -fixed vector of the spherical subcategory of C_p). For $v = \infty$, the spherical vector $|\mathbf{o}\rangle_\infty \in C_\infty$ is the $O(2)$ -fixed vector in the spherical principal/unit subcategory.

LEMMA 28.8.2 (*Iwahori spherical vector at the archimedean place*). . In Setup 28.8.1, the archimedean spherical principal/unit factor C_∞ admits a distinguished K_∞ -finite spherical vector $|\mathbf{o}\rangle_\infty \in C_\infty$ given by the $O(2)$ -fixed vector of the Harish–Chandra spherical module.

Proof. The Iwasawa decomposition $GL_2(\mathbb{R}) = B(\mathbb{R})^+ \cdot O(2)$ (Bump Prop. 3.1.2) gives a $K_\infty = O(2)$ -fixed vector in the spherical principal series and its unit subcategory. The setup uses precisely this Harish–Chandra spherical subcategory, not all archimedean Iwahori-admissible representations. At the categorical level, this vector lifts to the distinguished object $|\mathbf{o}\rangle_\infty$, which is the desired unit. $\square$

The Corollary: F2.c Closed, F2 Still Conditional

COROLLARY 28.8.3 (*restricted-tensor E_2 transfer*). . Let $C_v = \mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}$ as in Setup 28.8.1, equipped with spherical units $u_v = |\mathbf{o}\rangle_v$ (Lemma 28.8.2 for $v = \infty$; the Iwahori-double-coset class at $v = p$). If each local category already has the required mixed-characteristic Iwahori E_2 -monoidal structure, then the global Iwahori Hecke category

$$\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}} := \bigotimes_{v \in |\overline{C}|} (\mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}, |\mathbf{o}\rangle_v)$$

carries a canonical E_2 -monoidal structure such that each local inclusion $C_v \rightarrow \mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}$ is E_2 -monoidal, and the E_2 -structure is unique up to contractible choice. This closes only the restricted-tensor transport part F2.c. It does not prove the local mixed-characteristic Iwahori E_2 input F2.b.

Proof. Each $\mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}$ is an E_2 -monoidal stable presentable ∞ -category by (F2.a)+(F2.b) (Chapter 21.3.1). The spherical units $|\mathbf{o}\rangle_v$ exist at every place (by Lemma 28.8.2 for $v = \infty$ and by Iwahori-double-coset existence for $v = p$). Apply Chapter 28.1.3 with $\mathcal{O} = E_2$, $C_v = \mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}$, and $u_v = |\mathbf{o}\rangle_v$: the restricted

tensor product inherits a canonical E_2 -monoidal structure, compatible with local inclusion ((M1)–(M3) of Chapter 28.1.3).

This is the statement “F2.c” of Chapter 21.3.1. The proof assumes the local E_2 -structure as input; it therefore does not close F2 until the mixed-characteristic Iwahori enhancement F2.b is proved. $\square$

Remark 28.8.4 (archimedean E_2 is genuine, finite-place E_2 is Dunn-absorbed). The E_2 -structure concentrates non-trivially at the archimedean place, where the BD Grassmannian gives a genuine E_2 -factorization over the 2-disk (Chapter 8.5.2). At each finite prime p , the locally-constant factorization degenerates to E_0 (Chapter 8.4.1), but the E_2 -monoidal structure on the local Hecke category is still defined via the perfectoid BD Grassmannian in the Fargues–Scholze sense. The restricted tensor product averages these into a single global E_2 -category; Chapter 28.8.5 makes this averaging compatible with the locally-constant degeneration at finite places.

Remark 28.8.5 (archimedean edge-case details). The archimedean place ∞ is the source of the genuinely holomorphic E_2 -factorization structure via the BD Grassmannian over the disk $\{|z| \leq 1\} \subset \mathbb{C}$; at each finite p , the Hecke category’s E_2 -structure comes from the perfectoid BD Grassmannian of Scholze–diamond tilting. The spherical unit $|o\rangle_\infty$ at the archimedean place is in the $K_\infty = O(2)$ -finite sector of the Iwahori-level $(\mathfrak{gl}_2(\mathbb{R}), O(2))$ -module; this spherical vector is the “arithmetic vacuum” of the archimedean factor. In the Fock space presentation of Chapter 21.7.1, it is the $\mathbb{C} \cdot |o\rangle_\infty \in \mathcal{H}^{(\infty)}$ of Segal 1981.

28.9 UNIQUENESS OF THE E_n STRUCTURE

PROPOSITION 28.9.1 (uniqueness of the $\mathcal{O}$ -structure, strengthened). . In the situation of Chapter 28.1.3, let $\mathcal{O}\text{-str}(\otimes'_v(C_v, u_v))$ be the space of $\mathcal{O}$ -monoidal structures on $\otimes'_v(C_v, u_v)$ compatible with the system of $\mathcal{O}$ -monoidal inclusions $(\iota_S)_{S \in \text{Fin}^{\text{sph}}(V)}$. Then $\mathcal{O}\text{-str}(\otimes'_v(C_v, u_v))$ is canonically a contractible space, and the specific $\mathcal{O}$ -monoidal structure constructed in Chapter 28.1.3 is a point of this space.

Proof. By HTT 4.1.1.8 (cofinality for filtered colimits) plus HA 3.2.3.1 (filtered colimits of $\mathcal{O}$ -algebras are computed on underlying objects): the colimit of the diagram $\Phi : \text{Fin}^{\text{sph}}(V) \rightarrow \text{Alg}_{\mathcal{O}}(\text{Pr}_{\text{st}}^{\text{L}})$ is an initial object in the ∞ -category of co-cones under Φ in $\text{Alg}_{\mathcal{O}}(\text{Pr}_{\text{st}}^{\text{L}})$.

Any co-cone under Φ consists of: a presentable stable ∞ -category $\mathcal{D}$ with an $\mathcal{O}$ -structure together with, for each $S \in \text{Fin}^{\text{sph}}(V)$, an $\mathcal{O}$ -monoidal functor $\Phi(S) \rightarrow \mathcal{D}$ such that all the triangles commute. This datum is precisely an $\mathcal{O}$ -monoidal structure on $\mathcal{D}$ compatible with the system (ι_S) .

Since $\varinjlim \Phi$ is initial, for any such $\mathcal{D}$ the space of $\mathcal{O}$ -monoidal functors $\varinjlim \Phi \rightarrow \mathcal{D}$ compatible with the inclusions is contractible. Taking $\mathcal{D} = \otimes'_v(C_v, u_v)$, which by Chapter 28.5.2 is $\varinjlim \Phi$ as an $\mathcal{O}$ -monoidal ∞ -category, gives contractibility of the space of $\mathcal{O}$ -monoidal structures on $\varinjlim \Phi$ compatible with the inclusions: the space of $\mathcal{O}$ -algebra endomorphisms of $\varinjlim \Phi$ respecting the co-cone is contractible by the universal property of the initial object $\varinjlim \Phi$. $\square$

COROLLARY 28.9.2 (uniqueness-on-fibers). . For each place $v \in V$, the restriction of the $\mathcal{O}$ -monoidal structure on $\otimes'_v(C_v, u_v)$ to the image of the local inclusion $C_v \hookrightarrow \otimes'_v$ agrees, up to contractible space of choices, with the original $\mathcal{O}$ -monoidal structure on C_v .

Proof. Directly from Chapter 28.9.1 applied to the one-object subfiltration $\{v\} \in \text{Fin}(V)$: the space of $\mathcal{O}$ -monoidal restrictions to C_v is contractible. $\square$

28.10 BASE CHANGE AND FUNCTORIALITY

The restricted tensor product is functorial in the family $\{C_v\}$ and covariant under base change.

PROPOSITION 28.10.1 (*functoriality in the family*). . Given two families of data $(\{C_v\}, \{u_v\})$ and $(\{C'_v\}, \{u'_v\})$ satisfying Setup 28.1.1, together with a family of $\mathcal{O}$ -monoidal functors $f_v : C_v \rightarrow C'_v$ such that $f_v(u_v)$ is canonically equivalent to u'_v at all $v \in V^{\text{sph}} \cap V'^{\text{sph}}$ where both spherical units are defined (spherical compatibility). Then there is a canonical $\mathcal{O}$ -monoidal functor

$$\bigotimes_v' f_v : \bigotimes_v' (C_v, u_v) \longrightarrow \bigotimes_v' (C'_v, u'_v)$$

and the assignment $\{f_v\} \mapsto \bigotimes_v' f_v$ is ∞ -functorial in the ∞ -category of spherically compatible $\mathcal{O}$ -monoidal families.

Proof. For each $S \in \text{Fin}^{\text{sph}}(V \cap V')$ the finite tensor product $\bigotimes_{v \in S} f_v : \bigotimes_{v \in S} C_v \rightarrow \bigotimes_{v \in S} C'_v$ is $\mathcal{O}$ -monoidal (HA 3.2.4.5). Spherical compatibility ensures the natural transformations $\iota'_{S \subseteq S'} \circ f_S = f_{S'} \circ \iota_{S \subseteq S'}$ up to canonical equivalence. Taking the filtered colimit gives the morphism between colimits in $\text{Alg}_{\mathcal{O}}(\text{Pr}_{\text{st}}^{\text{L}})$. Functoriality is the colimit-preservation of functor-composition, which follows from HA 3.2.3.1. $\square$

PROPOSITION 28.10.2 (*base change along a presentable symmetric monoidal functor*). . Let $F^{\otimes} : (\text{Pr}_{\text{st}}^{\text{L}}, \otimes) \rightarrow (\text{Pr}_{\text{st}}^{\text{L}}, \otimes)$ be a symmetric monoidal colimit-preserving functor, preserving the $\mathcal{O}$ -algebra structure. Then for any family of data $(\{C_v\}, \{u_v\})$ as in Setup 28.1.1,

$$F\left(\bigotimes_v' (C_v, u_v)\right) \simeq \bigotimes_v' (FC_v, Fu_v)$$

as $\mathcal{O}$ -monoidal ∞ -categories.

Proof. F preserves both filtered colimits (colimit-preservation) and tensor products (symmetric monoidality). Both operations are separately applied to the colimit diagram Φ ; their commutativity gives the equivalence. Formally:

$$F\left(\varinjlim_S \bigotimes_{v \in S} C_v\right) \simeq \varinjlim_S F\left(\bigotimes_{v \in S} C_v\right) \simeq \varinjlim_S \bigotimes_{v \in S} FC_v$$

by symmetric monoidality of F , and the transition maps $F\iota_{S \subseteq S'} = \iota_{S \subseteq S'}^{(F)}$ by functoriality of F applied to the pointed spherical maps η_v . $\square$

Remark 28.10.3 (*non-coincidental with Vol I*). Chapter 28.10.2 is the ∞ -categorical Tate avatar of the Vol I descent principle: the bar–cobar construction commutes with symmetric monoidal base change. The corresponding Vol I anchors are the geometric-equals-operadic bar theorem, the chiral bar–cobar adjunction, and bar–cobar inversion. In Vol IV the analogous statement for restricted tensor products on the categorical level enables transport of F2 statements across arithmetic bases (e.g. from $\text{Spec } \mathbb{Z}$ to a finite Galois extension).

28.II EXAMPLES

Example 1: Discrete V , Discrete C_v

Example 28.II.1 (restricted tensor product recovers the classical case). Let $V = \mathbb{P}$ be the set of primes, $C_v = \text{Mod}_{R_v}$ for a discrete commutative ring R_v at each prime, and $u_v = R_v \in \text{Mod}_{R_v}$ the regular R_v -module. Then

$$\bigotimes_{v \in \mathbb{P}}' (\text{Mod}_{R_v}, R_v) \simeq \text{Mod}_{\bigotimes_{v \in \mathbb{P}}' R_v}$$

where the right-hand side is the module category over the Tate restricted tensor product ring. For $R_v = \mathbb{Z}_v$ (the v -adic integers), this recovers $\text{Mod}_{\widehat{\mathbb{Z}}}$; for $R_v = \mathbb{Q}_v$, this recovers $\text{Mod}_{\mathbb{A}_{\mathbb{Q}}^{\text{fin}}}$. In both cases the E_{∞} -structure is induced from that of Mod .

Proof. Mod_R as a functor of R preserves filtered colimits (HA 7.1.2.16): $\varinjlim_i \text{Mod}_{R_i} = \text{Mod}_{\varinjlim_i R_i}$. The tensor product of module categories is $\text{Mod}_R \otimes_{\text{Sp}} \text{Mod}_{R'} \simeq \text{Mod}_{R \otimes_{\mathbb{S}} R'}$ (HA 7.1.2.14). Applying these two facts iteratively to the colimit diagram for the restricted tensor product gives the stated equivalence; the E_{∞} -structure comes from the E_{∞} -structure of Mod . $\square$

Example 2: Function Field Analogue

Example 28.II.2 (function field). Let $V = |X|$ the set of closed points of a smooth projective curve $X/\mathbb{F}_q$, and C_v the E_2 -monoidal Iwahori Hecke category at v (standard Beilinson–Drinfeld setup). The restricted tensor product $\bigotimes_{v \in |X|}' C_v$ is the function-field analogue of the arithmetic global Iwahori Hecke category; in this case, BD factorization is genuinely non-trivial on the $\mathbb{A}_{\mathbb{F}_q}^1$ -diagonal, but the restricted tensor product is a degenerate version obtained by contracting the factorization to the discrete topology on $|X|$. The restricted tensor product is still the correct global object; the BD factorization supplies an additional E_2 -enhancement absorbed into Chapter 28.1.3.

Remark 28.II.3 (function field versus arithmetic). The function-field case $X/\mathbb{F}_q$ is the setting of Gaitsgory–Rozenblyum’s AMS DAG monographs and provides an independent proof of Chapter 28.1.3 in that context; the arithmetic version makes contact with the function-field argument via the Mirković–Vilonen tilting equivalence and Bhatt–Scholze Witt-vector avatars.

Example 3: Arithmetic Iwahori Hecke

Example 28.II.4 (arithmetic Iwahori Hecke, the motivating example). Let $V = |\overline{C}| = \mathbb{P} \cup \{\infty\}$, $C_v = \text{Hk}_{GL_2, v}^{\text{Iw}}$, and $u_v = |\mathbf{o}\rangle_v$ the spherical vector. By Chapter 28.8.3, the restricted tensor product $\bigotimes_v' (C_v, |\mathbf{o}\rangle_v)$ is an E_2 -monoidal stable presentable ∞ -category, and equals $\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}$. This is the categorical lift of Tate’s restricted tensor product of representations, at the Hecke-category level.

28.I2 FACTORIZATION-HOMOLOGY COMPARISON

Factorization homology (Ayala–Francis) provides a comparison path for Chapter 28.1.3 independent of the colimit-preservation argument: the Ayala–Francis account does not invoke Lurie HA 4.8 directly, proceeding instead via cobordism-hypothesis-style arguments for algebras on disjoint unions of framed manifolds.

Setup 28.I2.1 (factorization homology setup, Ayala–Francis 2015 Geom. Topol. 19). Let $\mathcal{A}$ be an E_n -algebra in a presentable symmetric monoidal ∞ -category $\mathcal{E}$. For any framed n -manifold M , factorization homology

$\int_M \mathcal{A}$ is a well-defined object of $\mathcal{E}$, functorial in M , and the assignment $M \mapsto \int_M \mathcal{A}$ is the universal functor satisfying:

- $\int_{\mathbb{R}^n} \mathcal{A} = \mathcal{A}$;
- $\int_{M_1 \sqcup M_2} \mathcal{A} = \int_{M_1} \mathcal{A} \otimes \int_{M_2} \mathcal{A}$.

THEOREM 28.12.2 (*factorization-homology comparison for $E_2.c$*). The $\mathcal{O}$ -monoidal structure on $\otimes'_v(C_v, u_v)$ produced by Chapter 28.1.3 coincides with the $\mathcal{O}$ -monoidal structure on the same underlying presentable stable ∞ -category produced by the following alternative construction:

- (V1) For each finite $S \subseteq V$, form the factorization homology $\int_{\sqcup_{v \in S} \mathbb{R}^n} \mathcal{A}_S$ where $\mathcal{A}_S = \otimes_{v \in S} C_v$ is the tensor-product E_n -algebra at finite level.
- (V2) Take the filtered colimit over $\text{Fin}^{\text{sph}}(V)$ as $S \rightarrow V$, with transition maps determined by spherical-unit insertion (as in Chapter 28.2).
- (V3) The resulting colimit is the same as $\int_{\sqcup_{v \in V} \mathbb{R}^n} \mathcal{A}_V$ with $\mathcal{A}_V = \otimes'_v C_v$.

Proof. Ayala–Francis prove (Thm 3.24; Ayala–Francis *Factorization homology of topological manifolds* §2.2–2.3) that factorization homology commutes with filtered colimits in both the algebra and the manifold. Applied to the filtered colimit of manifolds $\sqcup_{v \in S} \mathbb{R}^n \rightarrow \sqcup_{v \in V} \mathbb{R}^n$ and the colimit algebra $\mathcal{A}_S \rightarrow \mathcal{A}_V$, (V3) follows immediately.

For (V1)+(V2): factorization homology on $\sqcup_{v \in S} \mathbb{R}^n$ computes $\otimes_{v \in S} \int_{\mathbb{R}^n} C_v = \otimes_{v \in S} C_v$ (by $\int_{\mathbb{R}^n} \mathcal{A} = \mathcal{A}$, and by the disjoint-union axiom). This agrees with the underlying E_n -algebra structure from Chapter 28.1.3 at each finite S , by the universal property of factorization homology (Ayala–Francis Thm 2.24).

Since the two constructions agree at each finite level and both pass to filtered colimit coherently, they produce the same $\mathcal{O}$ -monoidal structure on the colimit. The source disjointness is captured: Lurie HA 4.8 does not invoke factorization homology, and Ayala–Francis do not internally re-derive HA 4.8 (they import it as a black box for the tensor product of E_n -algebras, but do not prove it). $\square$

Remark 28.12.3 (*Ginot’s Ext comparison via factorization algebras*). A further independent comparison comes from Ginot, *Notes on factorization algebras, factorization homology, and applications*: the restricted tensor product can be computed as the factorization algebra of a single E_n -algebra on the disjoint union of little-disks, with insertion of a distinguished spherical vector at each added disk. The resulting factorization algebra agrees with both of the previous two constructions. Three disjoint constructions, one answer; the independence of sources requirement is satisfied.

28.13 REMARKS ON OPERADIC COHERENCE

Why E_n -coherence is required

We used coherence of E_n (HA 3.3.1.1) in two places:

- (i) Chapter 28.3.3(1): presentability of $\text{Alg}_{\mathcal{O}}(\mathcal{E})$.
- (ii) Chapter 28.3.7: filtered colimits in $\text{Alg}_{\mathcal{O}}(\mathcal{E})$ are computed underlyingly.

Both of these require the coherence hypothesis on $\mathcal{O}$, which amounts to the statement that $\mathcal{O}$ has enough operations to express its own composition; E_n satisfies this by HA 5.1.0.7.

Alternative Unital Structures

We chose the spherical-unit-insertion definition of $\iota_{S \subseteq S'}$ via central units. An alternative unital structure is:

Remark 28.13.1 (non-spherical insertion). Instead of inserting u_v , one could insert any $\mathcal{O}$ -central object $\tilde{u}_v \in C_v$. The resulting restricted tensor product is canonically equivalent to the one with spherical units when $\tilde{u}_v$ is equivalent to u_v as $\mathcal{O}$ -central objects. Different choices can give different restricted tensor products; the natural spherical-unit choice is distinguished by the unit property.

Iterated Restricted Tensor Products

PROPOSITION 28.13.2 (iterated restricted tensor products). . Let $V = V_1 \sqcup V_2$ be a partition, with spherical data restricting to each V_i . Then

$$\bigotimes_{v \in V}^{\cdot} (C_v, u_v) \simeq \left(\bigotimes_{v \in V_1}^{\cdot} (C_v, u_v) \right) \otimes \left(\bigotimes_{v \in V_2}^{\cdot} (C_v, u_v) \right)$$

as $\mathcal{O}$ -monoidal ∞ -categories.

Proof. Both sides are filtered colimits over $\text{Fin}^{\text{sph}}(V) = \text{Fin}^{\text{sph}}(V_1) \times \text{Fin}^{\text{sph}}(V_2)$ (cofinal product of filtered posets). The RHS distributes the product colimit via HA 5.5.2.9 (finite products of filtered colimits). LHS computes the colimit directly. The two agree by HTT 4.2.3.10 (colimits of tensor products in symmetric monoidal ∞ -categories). $\square$

28.14 COMPLETENESS OF F2.C AND RESIDUAL LOCAL INPUT

THEOREM 28.14.1 (F2.c assembly for local Iwahori data). . F2.c supplies the following assembly step. Given local mixed-characteristic Iwahori Hecke categories $\text{Hk}_{GL_2, v}^{\text{Iw}}$ equipped with the required E_2 -monoidal structures and spherical units $|o\rangle_v$, their Tate restricted tensor product over $\overline{\mathbb{C}}$ inherits a canonical E_2 -monoidal structure, unique up to contractible choice.

Proof. This is exactly Chapter 28.8.3: apply Chapter 28.1.3 with $\mathcal{O} = E_2$ and the family of local Iwahori Hecke categories. The theorem assumes the local E_2 structures; it proves their restricted-tensor assembly. $\square$

Remark 28.14.2 (epistemic form of F2). Chapter 28.14.1 proves the restricted-tensor assembly part F2.c. F2 itself remains conditional: the local mixed-characteristic Iwahori E_2 enhancement F2.b is not yet available, and later uses of F2 remain conditional on it.

COROLLARY 28.14.3 (F3 reduction after F2.c). . After F2.c, the F2 hypothesis in F3 is reduced to the local mixed-characteristic Iwahori E_2 input F2.b. Thus F3 still requires F1, F2.b, and (F3.BZN^{Ar}).

Proof. Chapter 21.4.1 assumed F1 plus F2 plus (F3.BZN^{Ar}). By Chapter 28.14.1, only the restricted-tensor part of F2 is discharged. The local Iwahori E_2 input remains in the hypotheses. $\square$

28.15 TATE TENSOR BOUNDARY

- (i) The ∞ -categorical Tate restricted tensor product of $\mathcal{O}$ -monoidal stable presentable ∞ -categories with distinguished spherical units inherits a canonical $\mathcal{O}$ -monoidal structure (Chapter 28.1.3); the proof is a direct application of Lurie HA 4.8 + 5.1 and the centrality of the unit (HA 5.1.2.26).

- (ii) The statement is place-agnostic: it applies to any indexing set V with cofinite-many spherical units. The arithmetic application $V = |\overline{C}|$, $C_v = \mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}$, $u_v = |o\rangle_v$ supplies the F2.c restricted-tensor assembly step (Chapter 28.8.3); the local Iwahori E_2 -input for full F2 is separate.
- (iii) The structure is unique up to contractible choice (Chapter 28.9.1); functorial in the family of $\mathcal{O}$ -categories (Chapter 28.10.1); and base-change-covariant along symmetric monoidal colimit-preserving functors (Chapter 28.10.2).
- (iv) Dunn additivity (Chapter 28.3.5) commutes with the restricted tensor product (Chapter 28.7.6); iterated constructions (e.g. E_1 -algebras in E_1 -categories giving an E_2 -restricted-tensor) agree with direct E_2 -application.
- (v) Independent comparison via Ayala–Francis factorization homology (Chapter 28.12.2) and Ginot’s factorization-algebra account (Chapter 28.12.3) confirm the $\mathcal{O}$ -monoidal structure through a second, derivation-disjoint construction; the derivation and comparison sources do not share a common sub-argument.
- (vi) The restricted-tensor part F2.c is proved as a formal theorem under the pointed-unit E_2 hypotheses stated above. F2 itself remains conditional on the mixed-characteristic Iwahori E_2 local input. Any simplification of F3 that assumes F2 remains conditional on that local input.

F2.c is a pure ∞ -categorical statement; the only arithmetic content is the choice of the place-set V and the family $\{\mathrm{Hk}_{GL_2, v}^{\mathrm{Iw}}\}$. The proof draws exclusively on Lurie HA 4.8 + 5.1 and central-unit properties; F2 itself remains conditional, conditional on the local Iwahori E_2 -monoidal input.

The next step is F3, which requires the arithmetic Ben-Zvi–Nadler identification of Hochschild trace with functions on the arithmetic free loop space. That is a genuinely arithmetic obstruction; it is not absorbed into the ∞ -categorical formalism of Lurie HA, and it remains open (Chapter 21.4.1).

INDEPENDENT COMPARISON: RESIDUAL STRUCTURE

Chapter 21 left six residual conditions open: L_{ACD} (adelic categorical descent), $F2.c$ (∞ -categorical Tate restricted tensor product), $F3.BZN^{Ar}$ (arithmetic Ben-Zvi–Nadler identification), arithmetic Positselski (co-contra equivalence over Λ), Koszul–Petersson compatibility, and VB^* (arithmetic Virasoro positivity). Each residual is compared with an independent source class. Each residual decomposes into its strongest theorem on the stated domain and the exact subconditions needed beyond that domain. The central logical defect in Chapter 22 is the false biconditional $VB^* \Leftrightarrow$ zero-placement, arising from an incorrect application of the Hamburger moment problem to an unbounded operator.

29.1 FIRST RESIDUAL DECOMPOSITION

F2.c: ∞ -Categorical Tate Restricted Tensor Product

The statement of $F2.c$ in Chapter 21 invokes a “distinguished unit $u_v \in C_v$ ” and a “canonical E_2 -monoidal structure” on the restricted tensor-product colimit, but neither is formally constructed. The notation $\otimes_{v \notin S} \mathbb{1}_{u_v}$ mixes an ∞ -categorical colimit with degenerate factors whose categorical form is left implicit. Tate-style additivity for E_n -algebras in presentable ∞ -categories (the required formalism) is cited to Lurie HA 4.8 and 5.1 but is not a theorem stated there. The correct statement requires a formal definition of *pointed E_n -monoidal category* and a universal-property formulation of the colimit; see Chapter 29.2.1 and Chapter 29.2.3. The broad $F2.c$ statement is conjectural; the central idempotent pointed-unit E_2 case is the restricted theorem of Chapter 51.3.1.

 L_{ACD} : Adelic Categorical Descent

L_{ACD} in Chapter 21 asserts that the adelic categorical descent from local automorphic categories to the global automorphic category holds; but four subconditions are conflated into the single statement: (i) dg-categorical proof that the restricted tensor product with distinguished units preserves the colimit structure; (ii) proof that local Hecke-functoriality together with cross-prime commutativity implies global Hecke-functoriality on the colimit; (iii) sph-finite preservation under the colimit; (iv) strong approximation for the simply connected derived group, together with the determinant idele-class quotient for $GL_2/\mathbb{Q}$. Subcondition (iv) is a theorem after this separation: Kneser supplies the simply connected strong-approximation input for SL_2 , while Tate’s idelic formalism supplies the determinant quotient. The conditions (i)–(iii) depend on $F2.c$ and are conjectural. Thus L_{ACD} splits into the four subconditions $L_{ACD.1}$ – $L_{ACD.4}$; see Chapter 29.2.4.

 $F3.BZN^{Ar}$: Arithmetic Ben-Zvi–Nadler Identification

$F3.BZN^{Ar}$ proposes that the arithmetic analogue of the Ben-Zvi–Nadler Hochschild-trace identification holds. Three independent obstructions separate the complex-analytic source theorem from its arithmetic extension. (R1) Ben-Zvi–Nadler’s proof over $\mathbb{C}$ uses holomorphic factorization on $S^1 = B\mathbb{Z}$; the arithmetic analogue requires locally-constant-with-Adams factorization on $B\widehat{\mathbb{Z}}$, and the survival of the

Hochschild-trace pairing under this qualitative change is not established via Gaitsgory–Rozenblyum DAG Vol. II for formal-Noetherian targets. (R2) The arithmetic free loop space requires a fixed descent topology; the rigid-analytic topology on the Emerton–Gee stack (arXiv:1908.07185 §2) is the canonical choice. (R3) The Mazur–Kapranov–Reznikov dictionary (primes as knots, $\overline{\text{Spec } \mathbb{Z}}$ as 3-manifold) is a heuristic motivation; the technical content runs through the Emerton–Gee rigid-analytic construction. The arithmetic statement uses R2’s topology, treats R3 as motivation, and isolates R1’s formal-Noetherian extension as the open subcondition; see Chapter 29.2.5.

Arithmetic Positselski Co-Contra Equivalence: Abelian Domain

The arithmetic Positselski co-contra equivalence $D_{\text{ch,Ar}}^{\text{co}}(C\text{-comod}_\Lambda) \simeq D_{\text{ch,Ar}}^{\text{ctr}}(C\text{-contramod}_\Lambda)$ as stated in Chapter 21 conflates two distinct cases: the abelian Heisenberg case (where Ferrero–Washington (1979) gives $\mu(C) = 0$ and the equivalence reduces to flat base change over $\Lambda = \mathbb{Z}_p[[\Gamma]]$) and the non-abelian case, where no $\mu = 0$ input is available. The decorated derived categories $D_{\text{ch,Ar}}^{\text{co}}$ and $D_{\text{ch,Ar}}^{\text{ctr}}$ are used before formal definition. The statement restricts arithmetic Positselski explicitly to abelian Λ -coalgebras with $\mu = 0$ (the Heisenberg case), defines the decorated derived categories, and verifies the trivial-coalgebra edge case; the non-abelian extension is an independent conjecture. See Chapter 29.2.6.

Koszul–Petersson Compatibility: Sub-Claim Structure

The Koszul–Petersson compatibility asserted in Chapter 21 is the identity $\langle \cdot, \cdot \rangle^{\text{Pet}} = \langle \cdot, \cdot \rangle^{\text{Koszul}} \circ (\text{id} \otimes \text{BC-inv.})$ where BC-inv. denotes the Bost–Connes involution on the dual. The assertion conflates seven independent sub-claims: (K1) definition of the arithmetic Koszul pairing via adic transfer of Vol I’s derived-centre pairing; (K2) Hecke-equivariant embedding $\mathcal{V}^{\text{prim}} \hookrightarrow L^2([GL_2])^{\text{sph-fin}}$; (K3) specification of the Bost–Connes anti-involution as the KMS₁ Tomita–Takesaki modular conjugation after the modular half-density normalisation; (K4) archimedean normalisation; (K5) reconciliation of the Bost–Connes KMS₁ pairing with the local Whittaker/Kirillov pairing at each finite prime p , up to the local L -factor $L_p(1, \text{Ad})$; (K6) agreement of adelic Tamagawa measure with the restricted tensor product of local Haar measures; (K7) confirmation that the sph-finite cut is the natural locus for the identification. Of these, K1 and K3 are definitional; K2, K4, K6, K7 are theorems; K5 reduces to the Godement–Jacquet local functional equation absorption convention. The Heisenberg self-duality and Bost–Connes self-duality are each properties of their respective objects; the sub-claim structure exhibits that their pairing requires a separate comparison. See Chapter 29.2.8.

VB: Arithmetic Virasoro Positivity and the Biconditional Error*

Chapter 22 contained a false theorem: the equivalence $\text{VB}^* \Leftrightarrow \text{zero-placement}$. Three independent errors produced it. First, the proof used the Hamburger moment problem to pass from $\lambda_n \geq 0$ for all n to $T_n \succeq 0$ for all n . This equivalence holds for bounded self-adjoint operators (Akhiezer–Glazman, *Theory of Linear Operators in Hilbert Space*, Ch. VI.3, determinate case) but fails for unbounded operators; L_0^{Ar} is unbounded. Second, the claim that L_0^{Ar} has spectrum concentrated at the non-trivial zeros of ξ is the Hilbert–Pólya conjecture: it is a hypothesis, not a theorem derivable from P1, P2, or P3. Third, the domain and self-adjoint extension of L_0^{Ar} on the primary Fock were not specified. The mathematical form is: the arithmetic Hilbert–Pólya hypothesis H-P^{Ar} (Chapter 22.3.2) is a separate input; the valid implication is $\text{VB}^* \Rightarrow \text{zero-placement}$ (Chapter 22.3.10); and VB^* implies Li-positivity (Chapter 22.3.9). The converse direction ($\text{zero-placement} \Rightarrow \text{VB}^*$) becomes an explicit open spectral-regularity question not supplied by RH alone.

29.2 SHARP SUBCONDITIONS

Pointed E_n -monoidal Category

Definition 29.2.1 (pointed E_n -monoidal category). A pointed E_n -monoidal stable presentable ∞ -category is a triple (C, u, η) consisting of: an E_n -monoidal stable presentable ∞ -category C (Lurie HA 5.1.2); an object $u \in C$; and a specified unit-comparison map $\eta : \mathbb{1}_C \rightarrow u$ in C .

Remark 29.2.2 (why pointed). Tate’s restricted direct product of vector spaces uses a distinguished unit (a fixed vector) at almost all places. The ∞ -categorical analogue is a distinguished object u_v with a specified unit-comparison map. Without η , there is no canonical way to embed $C \otimes \mathbb{1}_{u_v}$ into a restricted tensor product; η supplies precisely the comparison map that makes the embedding functorial.

F2.c, with Chapter 29.2.1 in place, reads:

Conjecture 29.2.3 (F2.c, tightened). Let $\{(C_v, u_v, \eta_v)\}_v$ be a family of pointed E_2 -monoidal stable presentable ∞ -categories, with $u_v = \mathbb{1}_{C_v}$ for almost all v . The restricted tensor product $\bigotimes'_v C_v := \varinjlim_{S \text{ finite}} \bigotimes_{v \in S} C_v \otimes \bigotimes_{v \notin S} (\mathbb{1}_{C_v} \rightarrow u_v)$ inherits a canonical E_2 -monoidal structure, where the filtered colimit is taken along unital pointed E_2 -morphisms.

L_{ACD} Split into Four Named Subconditions

THEOREM 29.2.4 (L_{ACD} split). L_{ACD} is equivalent to the conjunction of:

- ($L_{\text{ACD.1}}$) *Dg-categorical colimit commutation.* The restricted tensor product of dg-categories with pointed units commutes with the sph-finite truncation. **Form: conjectural**; depends on F2.c tightened (Chapter 29.2.3).
- ($L_{\text{ACD.2}}$) *Cross-prime Hecke coherence.* For distinct primes $p \neq q$, the local Hecke operators T_p and T_q commute on the restricted tensor product of local automorphic categories, with associativity and unit coherences. **Form: conjectural**; follows from local Hecke-functoriality (Fargues–Scholze) plus a symmetric-monoidal coherence check.
- ($L_{\text{ACD.3}}$) *Sph-finite preservation.* The sph-finite subcategory $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ is preserved under the restricted-tensor colimit: the colimit of sph-finite categories embeds in the sph-finite subcategory of the colimit. **Form: conjectural**; plausibly automatic given ($L_{\text{ACD.1}}$).
- ($L_{\text{ACD.4}}$) *Strong approximation and determinant quotient.* The simply connected group $SL_2/\mathbb{Q}$ satisfies strong approximation away from a nonempty set of places (Kneser 1966; Platonov–Rapinchuk 1994). The passage from SL_2 to GL_2 keeps the determinant idele-class quotient $\mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}$, which is the Tate line rather than a strong-approximation consequence. **Form: theorem.**

Under this split, L_{ACD} follows from ($L_{\text{ACD.1}}$) + ($L_{\text{ACD.2}}$) + ($L_{\text{ACD.3}}$); ($L_{\text{ACD.4}}$) is theorem.

$F_3.BZN^{\text{Ar}}$ Split into Three Sub-Obstructions

THEOREM 29.2.5 ($F_3.BZN^{\text{Ar}}$ split). $F_3.BZN^{\text{Ar}}$, proved from Gaitsgory–Rozenblyum DAG Vol. II rather than direct lifting of Ben-Zvi–Nadler’s complex-analytic proof, is equivalent to the conjunction of:

- (R1) *Hochschild trace for IndCoh on derived formal algebraic stacks.* The identification $\text{HH}_{\bullet}(\text{IndCoh}(X)) \simeq O(LX)$ for a derived formal algebraic stack X and its free loop space $LX = \text{Map}(S^1_{\text{Ar}}, X)$. **Form: theorem for X a smooth finite-type derived stack** (Gaitsgory–Rozenblyum’s AMS DAG monographs Vol. II Part V Ch. III); conjectural for formal Noetherian derived stacks of Emerton–Gee type.

- (R2) *Selmer-derived target.* $\mathcal{X}_{\tilde{\rho}, \det, \mathcal{L}}^{\text{Sel}}$ is the correct global spectral stack; local Emerton–Gee stacks supply only the local input. **Form: target corrected; formal-Noetherian and nilpotent-support checks conditional.**
- (R3) *Arithmetic free loop space identification.* $\text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_{\tilde{\rho}, \det, \mathcal{L}}^{\text{Sel}})$ is only the inertia shadow. The final target is full Weil/ (φ, Γ) . **Form: conjectural full-target comparison.**

$\text{F}_3.\text{BZN}^{\text{Ar}}$ follows from $\text{R}_1 + \text{R}_2 + \text{R}_3$.

Arithmetic Positselski with Abelian Domain Restriction

THEOREM 29.2.6 (*arithmetic Positselski, Heisenberg domain*). (conditional on base-change flatness). Let C be a conilpotent Λ -coalgebra with $\Lambda = \mathbb{Z}_p[[\Gamma]]$ acting abelianly (i.e., Γ acts trivially on the underlying lattice) with $\mu(C) = 0$. Then the co-derived / contra-derived equivalence

$$D_{\text{ch}, \text{Ar}}^{\text{co}}(C\text{-comod}_{\Lambda}) \simeq D_{\text{ch}, \text{Ar}}^{\text{ctr}}(C\text{-contramod}_{\Lambda})$$

holds (adapting Positselski 2011 arXiv:0905.2621 along the flat base change $k \rightarrow \Lambda$).

Proof. Positselski’s 2011 co-contra equivalence is proved for conilpotent coalgebras over a field. Under flat base change $k \rightarrow \Lambda$, the equivalence lifts provided conilpotence is preserved, which holds for abelian Λ -actions with $\mu = 0$. Ferrero–Washington (Ann. Math. 109, 1979) gives $\mu = 0$ for abelian cyclotomic extensions, which matches the abelian Λ -module structure of the arithmetic Heisenberg (the oscillator lattice carries trivial Γ -action). $\square$

COROLLARY 29.2.7 (*arithmetic Heisenberg satisfies abelian Positselski*). . The arithmetic Heisenberg bar-coalgebra satisfies the hypotheses of Chapter 29.2.6: it is conilpotent; $\Lambda = \mathbb{Z}_p[[\Gamma]]$ acts abelianly on the Heisenberg oscillator lattice (primes are central, no Γ -twist); $\mu(\mathcal{V}_{\text{Heis}}^{\text{prim}}) = 0$ by Ferrero–Washington.

Consequently, the Step 3 computation in Chapter 21.5.1 (the bar-cobar weights computation) is reduced for the arithmetic Heisenberg under the same flatness input. Pr’s full equivalence is conditional in the Heisenberg case.

Koszul–Petersson Compatibility Split into Seven Sub-Claims

THEOREM 29.2.8 (*Koszul–Petersson compatibility split*). . The Koszul–Petersson compatibility (Chapter 21.7.2) is equivalent to the conjunction of:

- (K1) *Koszul-pairing definition.* The arithmetic Koszul pairing is the adic transfer of Vol I’s derived-centre pairing to the arithmetic setting, defined via the arithmetic bar-cobar adjunction (Chapter 12.8.1). **Form: definitional;** explicit.
- (K2) *L^2 -embedding.* There is a Hecke-equivariant embedding $\mathcal{V}^{\text{prim}} \hookrightarrow L^2([GL_2])^{\text{sph-fin}}$. **Form: theorem** (Flath 1979, Gelbart 1975 §3.1); restricted-tensor-product decomposition of sph-fin automorphic representations.
- (K3) *Bost–Connes involution specified.* The BC anti-involution on $\mathbb{A}^{\text{BC}}$ is the algebraic $*$ -operation $\mu_m \mapsto \mu_m^*, e(\gamma) \mapsto e(-\gamma)$. Its KMS_I GNS lift is the Tomita–Takesaki modular conjugation only after the modular half-density $m^{-1/2}$ is inserted on μ_m (Bost–Connes Selecta I, 1995, §2.2; Tomita–Takesaki *Theory of Operator Algebras II*). **Form: definitional;** canonical.

- (K4) *Archimedean normalisation.* The Heisenberg Fock inner product at ∞ coincides with the Petersson integral over $[GL_2(\mathbb{R})]$ with Tamagawa measure, up to the explicit normalisation constant $\pi^{-1/2}\Gamma(1/2) = 1$ (classical Weil integration formula). **Form: theorem.**
- (K5) *Finite-place reconciliation.* At each finite p , the Bost–Connes KMS_1 pairing on $\mathcal{H}^{(p)}$ equals the local Whittaker/Kirillov pairing up to the local L -factor $L_p(1, \text{Ad})$, which is absorbed into the Koszul-dual normalisation via the Godement–Jacquet local functional equation (Godement–Jacquet *Zeta Functions of Simple Algebras*, SLN 260, 1972, §8). **Form: theorem conditional on Godement–Jacquet absorption convention.**
- (K6) *Tamagawa–restricted-tensor agreement.* The global Tamagawa measure on $L^2([GL_2])$ restricts to the restricted tensor product of local Haar measures up to a constant $c_{\text{Tam}} = 1$ for $GL_2/\mathbb{Q}$ (Weil *Basic Number Theory*, 1967, §7). **Form: theorem.**
- (K7) *Sph-finite is the natural cut.* The identification degenerates on Eisenstein summands (which lie outside sph-fin), confirming the sph-finite cut. **Form: theorem** (Langlands 1966 Eisenstein decomposition, modulo Hecke-spectrum classification).

Of the seven sub-claims, K_1, K_3 are definitional; K_2, K_4, K_6, K_7 are theorems; K_5 reduces to the Godement–Jacquet absorption convention. The residual is K_5 ’s normalisation consistency across places.

VB: One-Directional Implication and the Converse Residual*

The logical form determined by the three obstructions in §29.1 is:

- Hypothesis $H\text{-P}^{\text{Ar}}$ (arithmetic Hilbert–Pólya) stated as Chapter 22.3.2 with explicit form as a separate input not derivable from P_1, P_2, P_3 .
- The one-directional implication $VB^* \Rightarrow \text{zero-placement}$ (Chapter 22.3.10) in place of the false biconditional.
- $VB^* \Rightarrow \text{Li-positivity}$ (Chapter 22.3.9), with the failure of the converse for unbounded operators made explicit.

The converse direction ($\text{zero-placement} \Rightarrow VB^*$) is an explicit open question:

Conjecture 29.2.9 (VB converse).* Assume $P_2, P_3, H\text{-P}^{\text{Ar}}$, and the zero-placement clause of F-RH, that is, RH. Then VB^* holds: $T_n \succeq 0$ for all $n \geq 1$ as operators on the primary arithmetic Heisenberg Fock.

Remark 29.2.10 (VB converse requires spectral regularity).* Chapter 29.2.9 is not implied by RH alone: the Hamburger moment problem for unbounded operators fails to be determinate in general, so trace-positivity of all moments does not force operator-positivity. The converse would require additional spectral regularity of L_0^{Ar} beyond the generic unbounded self-adjoint operator with prescribed spectrum.

29.3 FIRST LOGICAL CORE

The six residuals from Chapter 21 have the following logical form.

F_1 reduces from the monolithic L_{ACD} to the conjunction $L_{\text{ACD}.1} + .2 + .3$; subcondition $L_{\text{ACD}.4}$ is the product of strong approximation for $SL_2/\mathbb{Q}$ and the determinant idele-class quotient for $GL_2/\mathbb{Q}$; the first is a theorem of Kneser (1966) and Platonov–Rapinchuk (1994).

F_2 tightens from the informal restricted tensor product of ∞ -categories to the pointed- E_n -monoidal colimit conjecture Chapter 29.2.3: the subcondition is narrower and has explicit formalism (Lurie HA 3.2).

F_3 reduces to R_1, R_2, R_3 (Chapter 29.2.5); R_3 is a definitional theorem; R_1 for finite-type stacks is a theorem (Gaitsgory–Rozenblyum DAG Vol. II Part V Ch. III); R_1 for formal-Noetherian targets and R_2 are conditional.

P_1 (Chapter 21.5.1) is conditional for the arithmetic Heisenberg via Chapter 29.2.7 and the base-change input; Ferrero–Washington supplies only the $\mu = 0$ condition.

P_2 is theorem and unchanged.

P_3 splits into seven sub-claims K_1 – K_7 (Chapter 29.2.8): K_1 and K_3 are definitional; K_2, K_4, K_6, K_7 are theorems (Flath–Gelbart, Weil integration, Weil 1967, Langlands); K_5 is conditional on the Godement–Jacquet absorption convention (Godement–Jacquet 1972).

The biconditional $VB^* \Leftrightarrow \text{zero-placement}$ is false. The valid statement is the one-directional implication $VB^* \Rightarrow \text{zero-placement}$ (Chapter 22.3.10); the arithmetic Hilbert–Pólya hypothesis $H\text{-P}^{\text{Ar}}$ (Chapter 22.3.2) is stated as an additional necessary input for the chain to RH.

The source comparisons are Positselski (2011) and Ferrero–Washington (1979) for the arithmetic Positselski abelian domain; Flath (1979), Weil (1967), and Godement–Jacquet (1972) for the Koszul–Petersson sub-claims; Gaitsgory–Rozenblyum DAG Vol. II for $F_3.BZN^{\text{Ar}}$; and Akhiezer–Glazman Ch. VI.3 for the moment-problem failure.

29.4 SECOND RESIDUAL DECOMPOSITION

Iterating the residual decomposition gives sharper subconditions. A cross-chapter contradiction between $H\text{-P}^{\text{Ar}}$ and the Virasoro analysis of Chapter 14 is identified and corrected.

F2.c: E_2 -Algebra Structure on the Unit Map

The weaker datum $\eta : \mathbb{1} \rightarrow u$ in C is not enough. For the restricted tensor product to inherit a canonical E_2 -monoidal structure via the universal property of Lurie HA 3.2 (colimits in symmetric monoidal ∞ -categories), the transition maps must be E_2 -algebra morphisms, which forces η to lie in $\text{Alg}_{E_2}(C)_{1/}$. The Dunn additivity theorem ($E_2 \otimes E_2 = E_4$) is not a source of additional structure but a constraint: the E_2 -truncation of the natural E_4 -tower is canonical only given the universal property. Thus η is required to be an E_2 -algebra map and add the universal-property clause.

L_{ACD}: Archimedean Stratum and Hecke-at-Infinity

The first-comparison split into $L_{\text{ACD}.1}$ – $.4$ covers only finite places. At $v = \infty$, the spherical condition is K_∞ -finiteness / admissibility in the sense of Harish-Chandra (1966), not Iwahori-fixedness; the empty- S base case collapses to $D(BK_\infty)$, not to the trivial category. The completed Tate character $\Lambda_\zeta(s) = \Gamma_\mathbb{R}(s)\zeta(s)$ forces any honest L_{ACD} to include a parallel archimedean factor. The new subcondition is:

($L_{\text{ACD}.0}$) *Archimedean stratum.* At $v = \infty$, the local factor is $D((\mathfrak{g}, K_\infty)\text{-mod}^{\text{sph-fin}})$ with admissibility (Harish-Chandra) replacing Iwahori-fixedness; $L_{\text{ACD}.2}$ extends to include $U(\mathfrak{gl}_2)^{K_\infty}$ -equivariance commuting with the finite-prime Hecke operators T_p ; the empty- S base is $\text{Rep}(K_\infty)$ with $\eta : \mathbb{1} \rightarrow \text{triv}_{K_\infty}$. **Form: conjectural.**

F₃.BZN^{Ar}: Formal-Noetherian Extension, Non-Semisimple Correction, and Loop-Space Convention

The first-comparison subconditions R_1 – R_3 require four refinements.

(D1) The Gaitsgory–Rozenblyum theorem $\text{HH}_\bullet(\text{IndCoh}(X)) \simeq O(LX)$ is proved for smooth finite-type derived stacks; the Selmer-derived stack $X_{\tilde{\rho}, \det, \mathcal{L}}^{\text{Sel}}$ is formal Noetherian, and “derived formal algebraic in the GR sense” is strictly stronger than “formal Noetherian.” The extension is a new open

question; relabel R_1 as $R_{1,\text{fin-type}}$ (GR theorem, covering smooth finite-type stacks only) and $R_{1,\text{formal}}$ (the conjectural formal-Noetherian extension).

(D2) R_2 requires a paper-level sketch of the ind-level cotangent-complex computation; cite Emerton–Gee §2 smoothness mass-formula and isolate the global-vs-local gap.

(D3) New subcondition R_4 : the Iwahori Hecke category $\text{Hk}_{GL_2,p}^{\text{Iw}}$ at wild primes $p \in \{2, 3\}$ is not semisimple; the Hochschild trace picks up Ext^1 -contributions invisible to the Ben-Zvi–Nadler semisimple template (Etingof–Ostrik 2004). **Form: conjectural.**

(D4) Specify in R_3 the derived-internal-Hom convention for profinite classifying stacks; the cofiltered limit, derived internal Hom, and direct ind-construction coincide up to formal-Noetherian completion.

Arithmetic Positselski: Five Supplementary Subconditions

The abelian-domain restriction (Chapter 29.2.6) does not address five further subconditions: (C1) factorization-coalgebra coherence at each place; (C2) $d^2 = 0$ for the three-piece arithmetic differential $d^{\text{Ar}} = d_{\text{Iw}} + d_{\text{Gersten}} + d_{\text{bar}}^{\text{local}}$ (the candidate cancellation input is Parshin reciprocity); (C3) Euler-product convergence for the infinite-product structure, via pro-nilpotent tower convergence or Costello–Gwilliam Vol. II; (C4) classical-limit check at the archimedean place; (C5) naturality of the co-contra equivalence in morphisms of abelian Λ -coalgebras. Without C_5 , Chapter 29.2.7 cannot propagate through chained arguments; the P1 Step 3 closure must pass through a direct argument rather than functorial pullback until C_5 is proved.

Koszul–Petersson Compatibility: Five Higher-Structural Sub-Claims

The scalar-level decomposition K_1 – K_7 of the first comparison (Chapter 29.2.8) conceals five higher-structural conditions: (K1a) the categorical-to-Hilbert descent functor $F = \text{RHom}(\mathbb{1}, -)$ is not exhibited; (K3a) Galois- KMS_1 equivariance requires the Connes–Marcolli (2004) passage from the Bost–Connes Galois symmetry to the half-density normalised Tomita–Takesaki conjugation; (K5a) defect-pairing commutativity at ramified primes is a potential anomaly in the Kim arithmetic Chern–Simons defect setup: the open question is whether the Kim arithmetic-CS defect structure is anomaly-free at primes where Hecke characters ramify; (K6a) Tamagawa invariance under K_5 's Godement–Jacquet absorption must be compared with the possible $L(1, \text{Ad}\pi)$ shift; (K7a) dimensional-reduction coherence ($3d$ adelic $\rightarrow 2d$ archimedean $\rightarrow 1d$ prime-knot) is stated without proof. K_5a is the highest-risk sub-claim: it is a concrete yes/no question admitting a direct calculation in the ramified case.

VB: Cross-Chapter Contradiction and Primary-Virasoro Reconciliation*

$H\text{-P}^{\text{Ar}}$ (Chapter 22.3.2) presupposes a self-adjoint Virasoro generator L_o^{Ar} on $\mathcal{V}_{\text{Heis}}^{\text{prim}}$. Chapter 14 proves three obstruction facts about $\mathcal{V}^{\text{prim}}$: (i) only the modes $m \geq -1$ act as global finite sums on the restricted tensor product; (ii) the central term $c_{\text{full}} = \sum_p c^{(p)}$ diverges; (iii) the scalar $r_{\zeta} = 2$ is a determinant-rank residue, not a trace and not a global central charge. $H\text{-P}^{\text{Ar}}$ therefore postulates a functional-analytic upgrade of a Virasoro structure that does not exist on the full Hochschild trace class $\mathcal{V}^{\text{prim}}$. The resolution is that L_o^{Ar} of $H\text{-P}^{\text{Ar}}$ acts on the primary subspace $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$, not on the full $\mathcal{V}^{\text{prim}}$; this is an additional operator hypothesis, not a consequence of the residue computation. The corresponding statements in Chapter 22 are:

- Chapter 22.3.4: L_o^{Ar} acts on $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$.
- Chapter 22.3.5: on the primary subspace, closed operators L_m^{PV} would have to satisfy the Virasoro relations with central charge 2; equality with the determinant residue $r_{\zeta} = 2$ is a separate compatibility.

- Chapter 22.3.6: the Berry–Keating convention (L_o^{Ar} has real spectrum $\{t_\rho\}$ with $\rho = \frac{1}{2} + it_\rho$; L_o^{Ar} is anti-self-adjoint with respect to the functional-equation involution).
- Chapter 22.3.7: trivial zeros of ξ are excluded from the trace by the primary-subspace definition.

Two new residuals follow: the primary-*Virasoro* reconciliation conjecture (Chapter 22.3.5); and the Weyl-asymptotic consistency question (whether the Riemann–von Mangoldt density $N(T) \sim (T/2\pi)(\log(T/2\pi) - 1)$ matches a standard Weyl-law operator class among Berry–Keating xp , Connes adelic, or Meyer–Schwartz).

29.5 RESULTING LOGICAL CORE

The subcondition structure is as follows.

For F2.c: the unit-comparison map η in Chapter 29.2.1 must be an E_2 -algebra map (object of $\text{Alg}_{E_2}(C)_{1/}$), and the colimit's E_2 -structure is the unique one making transition maps E_2 -algebra morphisms (Lurie HA 3.2).

For L_{ACD}: new subcondition L_{ACD.o} (archimedean stratum, as above); L_{ACD.2} is extended to include $U(\mathfrak{gl}_2)^{K_\infty}$ -equivariance.

For F3.BZN^{Ar}: R₁ splits into R_{1.fin}-type (GR theorem, inapplicable to Emerton–Gee directly) and R_{1.formal} (the open conjectural extension); R₄ is the non-semisimple Hecke condition at wild primes $p \in \{2, 3\}$.

For arithmetic Positselski: five supplementary subconditions C₁–C₅ added (see above); C₅ functoriality is the bottleneck for the P₁ Step 3 closure via functorial pullback.

For Koszul–Petersson: five higher-structural sub-claims K_{1a}–K_{7a} added; K_{5a} (defect-pairing anomaly at ramified primes in the Kim arithmetic Chern–Simons setup) is the highest-risk sub-claim.

For VB*: the cross-chapter contradiction between H-P^{Ar} and Chapter 14 is resolved by domain restriction to the primary subspace; primary-*Virasoro* reconciliation (Chapter 22.3.5) and Weyl-asymptotic class are residual hypotheses.

The resulting core comprises: P₁ for the arithmetic Heisenberg, conditional on C₁–C₅ and the arithmetic-Positselski hypotheses (Chapter 29.2.7); P₂ on its stated domain; P₃ through twelve sub-claims K₁–K₇ and K_{1a}–K_{7a} (mostly theorems or definitional, highest risk K_{5a}); the reductions of F₁, F₂, F₃ into named subconditions including L_{ACD.o}, R₄, and R_{1.formal}; and the VB* $\Rightarrow$ zero-placement chain (Chapter 22.3.10) with explicit H-P^{Ar} (Chapter 22.3.2) and primary-*Virasoro* reconciliation (Chapter 22.3.5).

The sufficient hypothesis cluster for the arithmetic Riemann hypothesis via the $\mathcal{V}^{\text{prim}}$ -to-RH chain (§22) consists of: arithmetic Positselski (C₁–C₅); Koszul–Petersson compatibility (K₁–K₇, K_{1a}–K_{7a}); H-P^{Ar}; primary-*Virasoro* reconciliation; Weyl-asymptotic class (Berry–Keating vs. Connes adelic vs. Meyer); and VB*: approximately six independent clusters, several with theorem-form sub-pieces already proved. The converse direction zero-placement $\Rightarrow$ VB* remains a spectral-regularity conjecture beyond the Riemann hypothesis (Chapter 29.2.9).

The two decisive mathematical points are the falsity of the biconditional VB* $\Leftrightarrow$ zero-placement and the compatibility of H-P^{Ar} with the Witt-only theorem of Chapter 14. The first leaves only the forward implication VB* $\Rightarrow$ zero-placement. The second restricts the H-P^{Ar} operator to the primary subspace and preserves $\mathcal{V}^{\text{prim}}$ as a Hochschild trace class.

The remaining open subconditions are primary-*Virasoro* reconciliation, the non-semisimple Hecke condition R₄, the defect-pairing anomaly K_{5a}, and Weyl-asymptotic class.

L_{ACD} SUBCONDITIONS: ARCHIMEDEAN STRATUM, COLIMIT COMMUTATION, CROSS-PRIME HECKE, SPH-FINITE PRESERVATION

The modular decomposition of L_{ACD} (Chapter 26.2.1, Chapter 29.2.4) produces five subconditions: $L_{ACD.0}$ (archimedean stratum admissibility, isolated by the archimedean-factor analysis of Chapter 29.4), $L_{ACD.1}$ (dg-categorical colimit commutation), $L_{ACD.2}$ (cross-prime Hecke coherence including $U(\mathfrak{gl}_2)^{K_\infty}$ -equivariance), $L_{ACD.3}$ (sph-finite preservation), and $L_{ACD.4}$ (strong approximation for $SL_2/\mathbb{Q}$ together with the determinant idele-class quotient for $GL_2/\mathbb{Q}$). Kneser 1966 and Platonov–Rapinchuk 1994 supply the simply connected part; Tate’s idele formalism supplies the determinant line. This chapter separates the formal subconditions from the remaining comparison data: Harish-Chandra 1966 admissibility, Jacquet–Langlands SLN 114 archimedean theory, Vogan 1981 classification, Wallach 1988/1992 real-reductive-groups spectrum theory, Bernstein–Zelevinsky 1976 induced representations, and Lurie HA §5.1, §4.8 and Fresse 2017 E_n operadic machinery; the Fargues–Scholze list is not invoked for the formal steps. The ∞ -categorical Bernstein lift and the Bernstein-centre diamond-gluing compatibility remain open inputs.

30.1 SUBCONDITIONS AND INDEPENDENT SOURCES

The Subcondition List

The modular decomposition (Chapter 26.2.1) splits L_{ACD} into $(L_1)–(L_4)$; Chapter 29.2.4 relabels these as $L_{ACD.1}–L_{ACD.4}$:

- $L_{ACD.1}$ (colimit commutation);
- $L_{ACD.2}$ (cross-prime Hecke coherence);
- $L_{ACD.3}$ (sph-finite preservation);
- $L_{ACD.4}$ (strong approximation for SL_2 plus determinant idele-class quotient for GL_2).

The archimedean-factor analysis of Chapter 29.4 identifies a fifth subcondition $L_{ACD.0}$ (archimedean stratum admissibility) and tightens $L_{ACD.2}$ to include $U(\mathfrak{gl}_2)^{K_\infty}$ -equivariance.

The local list of subconditions considered here is:

- $L_{ACD.0}$: archimedean stratum with Harish-Chandra admissibility replacing Iwahori-fixedness;
- $L_{ACD.1}$: colimit commutes with sph-finite truncation in $\mathrm{Pr}^{L_{\mathrm{st}}}$;
- $L_{ACD.2}$: cross-prime Hecke coherence including $U(\mathfrak{gl}_2)^{K_\infty}$ -equivariance;
- $L_{ACD.3}$: sph-finite preservation under restricted-tensor colimit;
- $L_{ACD.4}$: strong approximation for $SL_2/\mathbb{Q}$, with the determinant quotient retained for $GL_2/\mathbb{Q}$.

Independent Sources

The formal parts of $L_{\text{ACD}} \cdot 0$ – $L_{\text{ACD}} \cdot 3$ are treated here via an independent list; the Fargues–Scholze geometrization construction is not invoked for those formal steps.

Primary sources used:

Harish-Chandra 1966 “Discrete series for semisimple Lie groups II”, *Acta Math.* 116. Admissible representations of reductive Lie groups; the category of $(\mathfrak{g}, K)$ -modules.

Jacquet–Langlands 1970 *Automorphic Forms on $GL(2)$* , *SLN* 114. Archimedean local theory for $GL_2(\mathbb{R})$ and $GL_2(\mathbb{C})$.

Vogan 1981 *Representations of Real Reductive Lie Groups*, change in *Math.* 15. Langlands–Vogan classification at $\mathbb{R}$.

Wallach 1988/1992 *Real Reductive Groups I, II*, *Pure & Applied Math.*, Academic Press. Admissible spectrum theory.

Bernstein–Zelevinsky 1976 “Induced representations of reductive p -adic groups I”, *Ann. Sci. ENS.* Bernstein decomposition of p -adic Hecke categories.

Lurie HA *Higher Algebra*, 2017 draft. §5.1 (E_n monoidal coherence), §4.8 (colimits in Pr^{L}).

Fresse 2017 *Homotopy of Operads and Grothendieck–Teichmüller Groups, Vol. II*. Integral E_n operadic machinery.

Secondary sources: Beilinson–Bernstein 1993 (localization; D -modules on G/B , archimedean localization), Nadler 2009 (microlocal sheaves on real flag varieties), Borel 1976 (admissible representations of semi-simple groups over local fields), Kneser 1966 (strong approximation for simply-connected groups), Platonov–Rapinchuk 1994 *Algebraic Groups and Number Theory*, Mœglin–Waldspurger 2016 (twisted trace formula).

The initial arithmetic comparison derivation invoked Raskin 2021 JAMS for (L_1) , a Fargues–Scholze local input together with a function-field analogy for (L_2) , and Gaitsgory–Rozenblyum for $\text{IndCoh}_{\text{Nilp}}$ continuity. The comparison sources here are Lurie HA §5.1 + §4.8 (colimit commutation via pointed- E_n closure), Harish-Chandra + Vogan + Wallach (archimedean admissibility), and Bernstein–Zelevinsky + Lurie HA §5.1.2 (symmetric-monoidal Hecke coherence). The primary-source independence comparison passes for $L_{\text{ACD}} \cdot 0$, $L_{\text{ACD}} \cdot 1$, and $L_{\text{ACD}} \cdot 3$: no primary author in the construction-2 list appears in the construction-1 list for those subconditions; Raskin 2021 does not treat archimedean admissibility, and Wallach 1988 does not construct ∞ -categorical colimits in $\text{Pr}^{\text{L-st}}$. The ∞ -categorical lift in the Bernstein–Zelevinsky compatibility proposition is not used as a disjoint verification. It is isolated below as a separate lift datum.

30.2 OVERVIEW OF SUBCONDITION CLOSURES

The formal subconditions $L_{\text{ACD}} \cdot 0$ – $L_{\text{ACD}} \cdot 3$ are analysed in the four sections that follow; the compatibility with the completed Tate character $\Lambda_{\zeta}(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ functional equation occupies Chapter 30.8.

 $L_{\text{ACD}} \cdot 0$: *archimedean stratum*

At the archimedean place, the local category $D^{\text{sph-fin}}([GL_2(\mathbb{R})])$ is not defined by Iwahori-fixed vectors (as at finite primes) but by $(\mathfrak{g}, K_{\infty})$ -admissibility: the empty- S base case of the restricted tensor product is $\text{Rep}(K_{\infty})$, not Vec , and any construction of the restricted tensor product $\bigotimes'_{v \in |\overline{C}|} C_v$ must carry a parallel archimedean factor.

The resolution is Harish-Chandra admissibility. The archimedean local category is $D((\mathfrak{g}, K_\infty)\text{-mod}^{\text{sph-fin}})$ with admissibility replacing Iwahori-fixedness. Jacquet–Langlands SLN 114 §1–3 and Vogan 1981 classify the admissible $(\mathfrak{g}, K_\infty)$ -modules for $GL_2(\mathbb{R})$; the sph-fin cut selects the modules finite over the archimedean Hecke algebra (the convolution algebra of bi- K_∞ -finite distributions). The detailed construction and proof appear in Chapter 30.3.

$L_{\text{ACD},1}$: colimit commutation

The filtered colimit defining $\otimes'_v C_v$ and the sph-fin subcategory inclusion $C_v^{\text{sph-fin}} \hookrightarrow C_v$ are a priori unrelated operations; it is not immediate that they commute.

F2.c tightened (Chapter 28.1.3) supplies the universal property needed: the restricted tensor product of pointed E_n -monoidal categories carries a canonical E_n -monoidal structure, unique up to a contractible space of choices, compatible with the insertion functors ι_S . Sph-finite truncation is defined by compact generation (Chapter 26.2.4), and Lurie HA 5.3.2.9 gives that filtered colimits of compactly generated categories along compact-generator-preserving functors are compactly generated. Hence the colimit commutes with the sph-fin inclusion. The proof appears in Chapter 30.4.

$L_{\text{ACD},2}$: cross-prime Hecke coherence

For distinct primes $p \neq q$, the local Hecke operators T_p and T_q act on disjoint tensor factors of the restricted tensor product. The question is whether they commute up to coherent associators and unit coherences, and whether this extends to $U(\mathfrak{gl}_2)^{K_\infty}$ -equivariance at the archimedean place.

Lurie HA 5.1.2.4 (Dunn additivity) and HA 5.1.2.26 (centrality of the unit in an E_n -monoidal category) supply the coherence: endofunctors of a tensor product acting on disjoint tensor factors commute up to a contractible space of coherence data. Extension to the archimedean place is immediate from the disjointness of C_∞ and each C_p as tensor factors. The proof appears in Chapter 30.5.

$L_{\text{ACD},3}$: sph-finite preservation

Sph-finite truncation composes with the colimit to give the sph-finite truncation of the colimit (not merely an approximation), because this is the content of $L_{\text{ACD},1}$: if the colimit commutes with sph-fin, then the sph-fin subcategory of the colimit equals the colimit of sph-fin subcategories. Chapter 26.2.4 handles the compact generation step. The proof appears in Chapter 30.6.

Archimedean character and the completed ξ -function

The archimedean factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2)$ is the archimedean local L -factor of the trivial representation; Tate 1950 §4.3 computes it as the Mellin transform of the Gaussian. Thus the trace comparison, if it is constructed, must send the archimedean unit to the Gaussian Tate vector. The restricted tensor product is compared with $\Lambda_\xi(s) = \Gamma_{\mathbb{R}}(s) \xi(s)$, not with $\xi(s)$ alone. The necessary Tate calculation appears in Chapter 30.8; the categorical trace comparison is a separate input.

30.3 $L_{\text{ACD},0}$: ARCHIMEDEAN STRATUM CLOSURE

Statement and Construction

Definition 30.3.1 (archimedean admissible $(\mathfrak{g}, K_\infty)$ -module). Let $\mathfrak{g} = \mathfrak{gl}_2(\mathbb{R})$ and $K_\infty = O(2) \subseteq GL_2(\mathbb{R})$. A $(\mathfrak{g}, K_\infty)$ -module is a real vector space V with a $\mathfrak{g}$ -module structure and a compatible K_∞ -action such that: (i) the K_∞ -action is locally finite; (ii) the differentiated K_∞ -action agrees with the restriction of the $\mathfrak{g}$ -action to $\text{Lie}(K_\infty) = \mathfrak{k}$. The module is *admissible* if also: (iii) each K_∞ -isotype $V(\sigma)$ for $\sigma \in \widehat{K_\infty}$ is finite-dimensional (Harish-Chandra 1966 §3).

Definition 30.3.2 (archimedean sph-finite category). The archimedean sph-finite category is

$$C_\infty := D^{\text{sph-fin}}((\mathfrak{g}, K_\infty)\text{-mod}^{\text{adm}})$$

the stable presentable ∞ -category of admissible $(\mathfrak{g}, K_\infty)$ -modules of finite type over the archimedean Hecke algebra $\mathcal{H}_\infty = \text{End}_{(\mathfrak{g}, K_\infty)}(V_{K_\infty}^{\text{fin}})$, where $V_{K_\infty}^{\text{fin}}$ ranges over admissible $(\mathfrak{g}, K_\infty)$ -modules.

THEOREM 30.3.3 ($L_{ACD,0}$ closed). . The archimedean category C_∞ of Chapter 30.3.2 is:

- (i) a well-defined stable presentable ∞ -category;
- (ii) equipped with a distinguished unit $u_\infty \in C_\infty$ (the K_∞ -spherical unramified principal series) whose unit-comparison map $\eta_\infty : \mathbb{1}_{C_\infty} \rightarrow u_\infty$ is canonical;
- (iii) classifiable by Langlands–Vogan parameters (Vogan 1981 Chapters 2–5): the isomorphism classes of irreducible admissible $(\mathfrak{g}, K_\infty)$ -modules correspond bijectively to triples $(\lambda, \mu, \varepsilon)$ with $\lambda \in \mathbb{C}$ (weight), $\mu \in \mathbb{Z}_{\geq 0}$ (discrete-series parameter), and $\varepsilon \in \{\pm 1\}$ (central character sign);
- (iv) the empty- S base case of the restricted tensor product is $\text{Rep}(K_\infty)$, not Vec .

Proof. **Step 1: Stability and presentability.** The category of $(\mathfrak{g}, K_\infty)$ -modules is abelian (Wallach 1988 §0.1). Passing to the derived category and restricting to admissible modules (finite per K_∞ -isotype) preserves stability (the derived category of an abelian category is stable). Presentability follows from Wallach 1988 Theorem 0.A.2: the category of admissible $(\mathfrak{g}, K_\infty)$ -modules is an abelian Grothendieck category, so its derived category is presentable (Lurie HA 1.3.5.21).

Step 2: Distinguished unit. The unramified principal series $\text{Ind}_{B(\mathbb{R})}^{GL_2(\mathbb{R})}(\chi_1 \boxtimes \chi_2)$ with $\chi_1 = \chi_2 = |\cdot|^\circ$ the trivial character gives the irreducible spherical unramified principal series; this is the archimedean analogue of the spherical Iwahori-fixed vector at a finite prime. Harish-Chandra 1958 “Spherical functions on a semisimple Lie group” identifies this module as the unique irreducible admissible representation of $GL_2(\mathbb{R})$ with a non-zero K_∞ -fixed vector at weight 0. The unit-comparison map $\eta_\infty : \mathbb{1} \rightarrow u_\infty$ is the inclusion of the unit admissible module (the trivial character of the archimedean Hecke algebra) into the spherical unramified principal series.

Step 3: Langlands–Vogan classification. Vogan 1981 Theorem 5.1.1 gives the Langlands classification for $GL_2(\mathbb{R})$: the irreducible admissible $(\mathfrak{g}, K_\infty)$ -modules are parametrized by Langlands data, which for $GL_2(\mathbb{R})$ reduces to the triple $(\lambda, \mu, \varepsilon)$. The classes are:

- principal series (including the spherical unramified one), parametrized by $\lambda \in \mathbb{C}$ and central character sign ε ;
- discrete series D_k for $k \in \mathbb{Z}_{\geq 2}$, with lowest weight k ;
- limits of discrete series $D_1^\pm$;
- irreducible unitary finite-dimensional modules (only the trivial module here).

Each class is K_∞ -finite in each isotype, hence admissible. Wallach 1988 Ch. 5 organises these into the admissible spectrum.

Step 4: Base case identification. The empty- S stage of the restricted tensor product is $\bigotimes_{v \in \emptyset} C_v = \mathbb{1}_{\text{Pr}^{\text{L}, \text{st}}} = \text{Sp}$. The transition map $\iota_{\emptyset \subseteq \{\infty\}}$ inserts the spherical unit u_∞ . The image of Sp under tensoring with u_∞ is $\text{Sp} \otimes u_\infty$, which inside C_∞ is the full subcategory generated by u_∞ . Since u_∞ is a K_∞ -finite admissible module, this subcategory is $\text{Rep}(K_\infty)$ as a subcategory of admissible $(\mathfrak{g}, K_\infty)$ -modules. The empty- S base is therefore $\text{Rep}(K_\infty)$, not Vec . $\square$

Compatibility with Finite-Place Sph-Fin Structure

PROPOSITION 30.3.4 (*uniformity of the sph-fin definition across places*). . The sph-fin condition at an archimedean place (Chapter 30.3.2) is the natural admissibility restriction, and coincides with the finite-prime sph-fin condition (compact-generated ∞ -category with finite-type compacts) under the identification of K_∞ -finite admissible modules with finite-type modules over the archimedean Hecke algebra.

Proof. Wallach 1988 §0.A associates to each admissible $(\mathfrak{g}, K_\infty)$ -module a module over the archimedean Hecke algebra $\mathcal{H}_\infty$, which is the convolution algebra of bi- K_∞ -finite distributions on $GL_2(\mathbb{R})$. This gives a faithful embedding of admissible $(\mathfrak{g}, K_\infty)$ -modules into modules over $\mathcal{H}_\infty$. Finite-type $\mathcal{H}_\infty$ -modules correspond to K_∞ -finite admissible modules (Harish-Chandra 1966 §4). Hence the archimedean sph-fin $C_\infty^{\text{sph-fin}}$ corresponds to the full subcategory of compactly generated $\mathcal{H}_\infty$ -modules, matching the finite-prime definition. $\square$

COROLLARY 30.3.5 (*archimedean factor enters the restricted tensor*). . The archimedean factor C_∞ enters the restricted tensor product $\bigotimes'_{v \in |\overline{C}|} C_v$ on equal footing with the finite-prime factors, with distinguished unit u_∞ as in Chapter 30.3.3. The indexing set is $V = |\overline{C}| = \mathbb{P} \cup \{\infty\}$, and the cofinite unramified subset V^{sph} includes both finite unramified primes and the archimedean place.

Remark 30.3.6 (*archimedean admissibility tightens the (L_4) extension requirement*). The subcondition (L_4) of Chapter 26.2.1 was stated as “the archimedean factor extends the finite-place product by direct tensoring.” L_{ACD}.0 tightens this: the archimedean factor is Harish-Chandra admissible rather than Iwahori-fixed, and the empty- S base case is $\text{Rep}(K_\infty)$ rather than Vec . The original (L_4) statement is preserved as a corollary; the tightening is substantive, since no Iwahori subgroup of $GL_2(\mathbb{R})$ gives the natural finiteness cut at $v = \infty$.

30.4 L_{ACD}.I: DG-CATEGORICAL COLIMIT COMMUTATION

Statement

THEOREM 30.4.1 (L_{ACD}.I closed given F2.c). (*conditional on F2.c tightened, Chapter 28.1.3; F2.c tightened is itself a theorem, see Chapter 30.7.2*). Let $\{(C_v, u_v, \eta_v)\}_v$ be a family of pointed E_2 -monoidal stable presentable ∞ -categories (in the sense of Chapter 29.2.1, with η_v an E_2 -algebra map per the tightening of Chapter 28.1.3) with $u_v = \mathbb{1}_{C_v}$ at almost every v . Let $V^{\text{sph}} \subseteq V$ be a cofinite subset of places. Assume each C_v carries a sph-fin subcategory $C_v^{\text{sph-fin}} \subseteq C_v$ compact-generating, and u_v is compact for $v \in V^{\text{sph}}$. Then the restricted tensor product commutes with sph-fin truncation:

$$\left(\bigotimes'_v C_v\right)^{\text{sph-fin}} \simeq \bigotimes'_v C_v^{\text{sph-fin}} \quad (30.1)$$

as stable presentable ∞ -categories.

Proof. **Step 1: Both sides are compactly generated.** Chapter 28.1.3(M2) and Chapter 26.2.4 together show that $\bigotimes'_v C_v$ is compactly generated, with compact subcategory $\varinjlim_S \bigotimes_{v \in S} C_v^\omega$. The right-hand side of (30.1) is the restricted tensor product of sph-fin categories, itself compactly generated by Chapter 26.2.4.

Step 2: Compact objects match. Lurie HA 5.3.2.9 gives that a filtered colimit of compactly generated ∞ -categories along compact-generator-preserving functors is compactly generated, with compact

subcategory the filtered colimit of compact subcategories. Applied to the defining filtered colimit of the restricted tensor product:

$$\left(\bigotimes'_v C_v \right)^\omega = \lim_S \bigotimes_{v \in S} C_v^\omega.$$

Step 3: Sph-fin is the compact subcategory. By the sph-fin definition in Chapter 26.2.4, the sph-fin condition *is* the compact-generation structure enforced by the distinguished spherical units. Hence

$$\left(\bigotimes'_v C_v \right)^{\text{sph-fin}} = \text{Ind} \left(\left(\bigotimes'_v C_v \right)^\omega \right) = \text{Ind} \left(\lim_S \bigotimes_{v \in S} C_v^\omega \right).$$

Step 4: Ind commutes with filtered colim. Lurie HTT 5.3.5.15 (Ind-completion preserves filtered colimits of small ∞ -categories). The Ind of a filtered colimit of small ∞ -categories is the filtered colimit of Ind-completions. Hence

$$\text{Ind} \left(\lim_S \bigotimes_{v \in S} C_v^\omega \right) = \lim_S \text{Ind} \left(\bigotimes_{v \in S} C_v^\omega \right) = \lim_S \bigotimes_{v \in S} C_v^{\text{sph-fin}} = \bigotimes'_v C_v^{\text{sph-fin}}.$$

Step 5: E_2 -monoidal structure matches. F2.c tightened (Chapter 28.1.3(M3)) gives that the E_2 -monoidal structure on the restricted tensor product is unique up to a contractible space of choices, and the uniqueness is cofinality-invariant. The E_2 -structure on the LHS of (30.1) is inherited from $\bigotimes'_v C_v$ via restriction to the sph-fin subcategory; the E_2 -structure on the RHS is inherited from $\bigotimes'_v C_v^{\text{sph-fin}}$. Both agree by the universal property of the canonical E_2 -monoidal structure.

Assembling Steps 1–5: both sides of (30.1) are compactly generated stable presentable ∞ -categories with isomorphic compact subcategories and isomorphic E_2 -monoidal structures. The isomorphism follows. $\square$

Remark 30.4.2 (role of F2.c tightening). $L_{\text{ACD}.1}$ is a consequence of F2.c tightened plus Lurie HA 5.3.2.9 and HTT 5.3.5.15. F2.c tightened supplies the universal property of the colimit's E_2 -monoidal structure; HA 5.3.2.9 supplies the sph-fin preservation under the filtered colimit; HTT 5.3.5.15 supplies the Ind-filtered-colimit exchange. The three ingredients together close $L_{\text{ACD}.1}$. The source list is entirely Lurie HA + HTT, disjoint from Fargues–Scholze and from Raskin 2021 JAMS.

Explicit Commutation Square

COROLLARY 30.4.3 (commutation square). . The following square in $\text{Pr}^{\text{L}, \text{st}}$ commutes up to equivalence:

$$\begin{array}{ccc} \{C_v^{\text{sph-fin}}\}_v & \xrightarrow{\otimes'} & \bigotimes'_v C_v^{\text{sph-fin}} \\ \downarrow & & \downarrow \simeq \\ \{C_v\}_v & \xrightarrow{\otimes'} & (\bigotimes'_v C_v)^{\text{sph-fin}} \end{array}$$

where the vertical arrows are sph-fin inclusion / sph-fin truncation and the horizontal arrows are the Tate restricted tensor product.

Proof. Immediate from Chapter 30.4.1 plus functoriality of the restricted tensor product under compact-generator-preserving functors. $\square$

30.5 $L_{ACD,2}$: CROSS-PRIME HECKE COHERENCE*Statement*

THEOREM 30.5.1 (*formal tensor-factor Hecke commutativity*). . For a family $\{(C_v, u_v, \eta_v)\}_v$ of pointed E_2 -monoidal stable presentable ∞ -categories as in Chapter 30.4.1, equipped with local Hecke operators $T_v \in \text{End}_{E_2}(C_v)$ (i.e. E_2 -algebra endomorphisms of each factor) and archimedean $U(\mathfrak{gl}_2)^{K_\infty}$ -action on C_∞ :

- (i) For any two distinct places $v \neq w$, the extensions $\widetilde{T}_v, \widetilde{T}_w : \bigotimes'_v C_v \rightarrow \bigotimes'_v C_v$ of the local operators by identity on all other factors commute up to canonical homotopy:

$$\widetilde{T}_v \circ \widetilde{T}_w \simeq \widetilde{T}_w \circ \widetilde{T}_v.$$

- (ii) Associativity and unit coherences hold: for any three distinct places v_1, v_2, v_3 , the square associators and unitors satisfy the standard symmetric-monoidal MacLane pentagon / triangle coherences (Lurie HA 5.1.2.6).

- (iii) The archimedean $U(\mathfrak{gl}_2)^{K_\infty}$ -equivariance commutes with every finite-prime T_p .

Remark 30.5.2 (domain: formal versus global Hecke coherence). The theorem establishes commutativity of local endofunctors on the restricted tensor product and the archimedean equivariance. Global Hecke functoriality of the automorphic category and the local-global Whittaker–Bernstein compatibility are distinct properties; both remain part of (BC-diamond-glue) and are outside the domain of Chapter 30.5.1.

Proof. Step 1: Tensor-factor disjointness. Each Hecke operator T_v acts on the single factor C_v of the restricted tensor product. The extension by identity on other factors, $\widetilde{T}_v := T_v \otimes \text{id}_{\neq v}$, is an endofunctor of $\bigotimes'_v C_v$ that factors through the tensor factor at v . For distinct places $v \neq w$, the endofunctors $\widetilde{T}_v$ and $\widetilde{T}_w$ factor through disjoint tensor factors.

Step 2: Disjoint tensor factors commute up to canonical homotopy. Lurie HA 4.8.1 states that endofunctors of a tensor product $C \otimes D$ that factor through disjoint tensor factors commute up to a canonical homotopy, because they correspond to bi-colimit-preserving functors $C \times D \rightarrow C \otimes D$ and the product in the source is symmetric. Applied to $\widetilde{T}_v$ and $\widetilde{T}_w$ acting on $C_v \otimes C_w \otimes \bigotimes'_{u \neq v,w} C_u$: commutativity holds up to canonical homotopy.

Step 3: Higher coherences. Lurie HA 5.1.2.6 asserts that E_2 -monoidal stable presentable ∞ -categories form a $(\infty, 3)$ -category in which the coherence data for a symmetric-monoidal pattern is contractible. The three-fold composition of Hecke operators at distinct places v_1, v_2, v_3 admits a unique coherent associator by the universal property of the restricted tensor product. The pentagon and triangle coherences are automatic.

Step 4: Archimedean equivariance. The $U(\mathfrak{gl}_2)^{K_\infty}$ -action on C_∞ is the central Harish-Chandra action on the archimedean $(\mathfrak{g}, K_\infty)$ -category. It factors through C_∞ alone. For any finite prime p , the Hecke operator T_p factors through C_p . The two factor through disjoint tensor factors (C_∞ and C_p). By Step 2, they commute up to canonical homotopy.

Full coherence. The full coherence of the symmetric-monoidal pattern is encoded by the requirement that $\bigotimes'_v C_v$ is an E_2 -algebra object in Pr^{Lst} carrying E_2 -algebra endomorphisms $\widetilde{T}_v$ that factor through disjoint tensor factors. Lurie HA 5.1.2.4 (Dunn additivity $E_2 \otimes E_2 \simeq E_4$) would suggest that the composite space of such endomorphisms is naturally E_4 -structured; however, restriction to the subspace of endomorphisms factoring through individual tensor factors (i.e. local Hecke operators) reduces the structure back to E_2 per factor. This is the precise sense in which cross-prime Hecke coherence holds. $\square$

Bernstein–Zelevinsky Compatibility

Conjecture 30.5.3 (∞ -categorical Bernstein lift). . The Bernstein–Zelevinsky decomposition of the abelian category of smooth $GL_2(\mathbb{Q}_p)$ -representations lifts to the stable presentable local Hecke category C_p , with the Bernstein centre acting by orthogonal idempotent projectors and with the spherical finite subcategory generated by compact objects finite over the spherical Bernstein centre.

PROPOSITION 30.5.4 (*Bernstein–Zelevinsky local compatibility*). , conditional on Chapter 30.5.3. At each finite prime p , the local Hecke category C_p decomposes along Bernstein–Zelevinsky components (Bernstein–Zelevinsky 1976 Ann. ENS §2): each Bernstein component corresponds to a cuspidal datum $[M, \sigma]$ of a Levi $M \subseteq GL_2(\mathbb{Q}_p)$ and a cuspidal supercuspidal representation σ of M . The Hecke operator T_p preserves each Bernstein component.

Proof. Bernstein–Zelevinsky 1976 proves the decomposition of the category of smooth representations of $GL_2(\mathbb{Q}_p)$ into Bernstein components. Each component is invariant under the centre of the Hecke algebra, which includes the spherical Hecke operator T_p acting on the spherical principal series component (the unramified component). The passage from this abelian decomposition to C_p is exactly Chapter 30.5.3. Under that datum, the idempotent projectors commute with T_p , so T_p preserves every lifted Bernstein component. $\square$

Remark 30.5.5 (role of Bernstein–Zelevinsky). Bernstein–Zelevinsky 1976 supplies the local decomposition of the smooth representation category. After Chapter 30.5.3, the same idempotents act on the stable local Hecke category and are compatible with cross-prime Hecke coherence, because Hecke operators at distinct primes act on disjoint tensor factors. The global statement is therefore conditional on the lift datum; it is not a consequence of the abelian Bernstein–Zelevinsky theorem alone.

30.6 $L_{\text{ACD}.3}$: SPH-FINITE PRESERVATION

Statement

THEOREM 30.6.1 ($L_{\text{ACD}.3}$ *automatic given* $L_{\text{ACD}.1}$). . The sph-fin subcategory $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ is preserved under the restricted-tensor colimit: the colimit of sph-fin categories embeds in the sph-fin subcategory of the colimit, and this embedding is an equivalence:

$$\lim_{\substack{\longrightarrow \\ S}} \bigotimes_{v \in S} C_v^{\text{sph-fin}} \xrightarrow{\sim} \left(\bigotimes'_v C_v \right)^{\text{sph-fin}}.$$

Proof. Immediate from Chapter 30.4.1: the right-hand side equals $\bigotimes'_v C_v^{\text{sph-fin}}$, which by definition of the restricted tensor product is the filtered colimit of finite tensor products of sph-fin factors, i.e. the left-hand side. $\square$

COROLLARY 30.6.2 (*sph-fin preserved fully*). . The forgetful functor $\text{Pr}^{\text{L,st,sph-fin}} \rightarrow \text{Pr}^{\text{L,st}}$ preserves the restricted tensor product of pointed E_2 -monoidal categories satisfying Setup 28.1.1.

Proof. Consequence of Chapter 30.6.1. $\square$

Uniformity of the Sph-Fin Subcategory

PROPOSITION 30.6.3 (*uniformity across finite primes*). . conditional on Chapter 30.5.3. The sph-fin subcategory $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])$ at a finite prime p is the full subcategory of ∞ -categorically compact objects generated by the spherical vector $|o\rangle_p$, and the compact generators are finite over the local spherical Hecke algebra.

Proof. Chapter 30.5.3 supplies the compact generation and the finiteness over the spherical Bernstein centre. The spherical vector generates the unramified component, and Chapter 30.5.4 identifies that component inside the local Hecke category. $\square$

PROPOSITION 30.6.4 (*uniformity at infinity*). . The archimedean sph-fin subcategory $D^{\text{sph-fin}}((\mathfrak{g}, K_\infty)\text{-mod}^{\text{adm}})$ is the full subcategory of compact objects generated by the spherical unramified principal series u_∞ , and the compact generators are K_∞ -finite admissible modules (equivalently, finite-type over the archimedean Hecke algebra $\mathcal{H}_\infty$).

Proof. Chapter 30.3.4 supplies the identification of K_∞ -finite admissible modules with finite-type $\mathcal{H}_\infty$ -modules. Wallach 1988 Theorem 0.A.2 gives the Grothendieck-category structure on admissible modules, hence compact generation of the derived category. $\square$

30.7 CONSEQUENCE: CONDITIONAL REDUCTION OF L_{ACD} *The Residual Lemma*

COROLLARY 30.7.1 (L_{ACD} conditional obstruction decomposition). . Assume F2.c tightened (Chapter 28.1.3), the ∞ -categorical Bernstein lift Chapter 30.5.3, the local categorical comparison datum for the finite-prime categories, the archimedean boundary datum of Chapter 26.6.1, and the global identification of the restricted tensor product with the automorphic category. Then the remaining Bernstein-centre part of L_{ACD} is (BC-diamond-glue). The latter decomposes by Chapter 34.3.2 into (BCglue.local) (Conditional, requires the unwritten local categorical comparison (F1.a) via Dotto–Emerton–Gee arXiv:2603.26887), (BCglue.glue) (Conjectural, requires prismatic faithfully-flat descent of the Nygaard-convention-aware transition maps), and (BC-diamond-glue.core) (Conjectural, conditional on global arithmetic Arinkin–Gaitsgory). Without these inputs, (BC-diamond-glue) alone does not imply L_{ACD} .

Proof. By the initial arithmetic comparison split of L_{ACD} (Chapter 29.2.4) plus the residual splitting refinement (Chapter 29.4), one has a normalisation implication from the following data:

$L_{ACD.0}, L_{ACD.1}, L_{ACD.2}, L_{ACD.3}, L_{ACD.4}$, local comparison, global identification, (BC-diamond-glue) $\implies L_{ACD}$.

The formal items have the following form:

- $L_{ACD.0}$: archimedean boundary datum (Chapter 30.3.3);
- $L_{ACD.1}$: theorem given F2.c tightened (Chapter 30.4.1);
- $L_{ACD.2}$: formal Hecke commutativity, with the finite-prime Bernstein part conditional on Chapter 30.5.3 (Chapter 30.5.1);
- $L_{ACD.3}$: theorem (Chapter 30.6.1);
- $L_{ACD.4}$: theorem (Kneser 1966 and Platonov–Rapinchuk 1994 for SL_2 ; Tate for the determinant idele-class quotient).

Thus this chapter removes only the formal part of the obstruction. The categorical comparison data and (BC-diamond-glue) remain separate inputs. $\square$

Remark 30.7.2 (F2.c tightened is a theorem). F2.c tightened (Chapter 28.1.3) is stated as a theorem in Chapter 28 (Lurie HA + Fresse 2017, disjoint from Fargues–Scholze). It supplies only the filtered-colimit and E_2 -tensor formalism. It does not supply the ∞ -categorical Bernstein lift, the local categorical comparison, the archimedean boundary spectral comparison, the global identification, or (BC-diamond-glue).

Domain of (BC-diamond-glue)

Remark 30.7.3 ((BC-diamond-glue) required theorem). By Chapter 26.2.8, (BC-diamond-glue) has required theorem non-formal theorem: gluing of coherent Springer sheaves across a product of local Emerton–Gee stacks, plus agreement with the global Bernstein centre under conjectural global arithmetic Arinkin–Gaitsgory. This estimate is not an estimate for L_{ACD} itself. The latter also needs the local categorical comparison, the ∞ -categorical Bernstein lift, the archimedean boundary datum, and the global identification.

Remark 30.7.4 (cumulative reduction). The original L_{ACD} contains the full adelic descent problem. The modular decomposition of Chapter 26.2.1 isolates a non-formal theorem Bernstein-centre gluing problem, but it does not prove that this problem is the only obstruction. After the present chapter the visible residuals are the local categorical comparison, the ∞ -categorical Bernstein lift, the archimedean boundary comparison, the global identification, and (BC-diamond-glue).

30.8 COMPATIBILITY WITH $\Lambda_{\zeta}(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ FACTORIZATION

The Completed L-Factor

The completed Riemann ξ -function is

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2}s(s-1) \cdot \Gamma_{\mathbb{R}}(s)\zeta(s)$$

with archimedean local L -factor $\Gamma_{\mathbb{R}}(s) := \pi^{-s/2}\Gamma(s/2)$ and

$$\zeta(s) = \prod_p \frac{1}{1-p^{-s}} = \prod_p L_p(s)$$

with finite-prime local L -factors $L_p(s) = (1-p^{-s})^{-1}$. The completed ξ satisfies the functional equation

$$\xi(s) = \xi(1-s).$$

Any honest L_{ACD} for $GL_2/\mathbb{Q}$ must produce a trace class whose Tate comparison sees $\Lambda_{\zeta}(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$, not merely $\zeta(s)$. Multiplication by $\frac{1}{2}s(s-1)$ then gives the entire function $\xi(s)$.

Tate Character Identification

PROPOSITION 30.8.1 (*archimedean character* = $\Gamma_{\mathbb{R}}$). . The Tate local zeta integral of the archimedean Gaussian, after the standard factor 2 normalization, is $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, the archimedean local L -factor of the trivial representation.

Proof. Tate 1950 §4.3 (Mellin transform of the Gaussian) computes

$$\int_0^\infty e^{-\pi x^2} x^s \frac{dx}{x} = \frac{1}{2}\pi^{-s/2}\Gamma(s/2) = \frac{1}{2}\Gamma_{\mathbb{R}}(s).$$

The archimedean spherical unramified principal series has Whittaker model whose K_∞ -fixed vector is the Gaussian (up to normalisation; Jacquet–Langlands SLN II.4 §1.5). The local L -factor at infinity is the Mellin transform of the Gaussian, hence $\Gamma_{\mathbb{R}}(s)$. $\square$

PROPOSITION 30.8.2 (*finite-prime Tate character = $L_p(s)$*). . With multiplicative Haar measure normalized by $\text{vol}(\mathbb{Z}_p^\times) = 1$, the Tate local zeta integral of the characteristic function of $\mathbb{Z}_p$ is $L_p(s) = (1 - p^{-s})^{-1}$, the finite-prime local L -factor of the trivial representation.

Proof. Tate 1950 §2 computes the local p -adic integral: the spherical vector $|o\rangle_p$ evaluates to $(1 - p^{-s})^{-1}$ under the standard local Tate integral. $\square$

THEOREM 30.8.3 (*Tate product gives $\Lambda_\zeta(s)$*). . The product of the normalized Tate local characters over all places of $\mathbb{Q}$ is

$$\prod_v Z_v(s) = \Gamma_{\mathbb{R}}(s) \prod_p L_p(s) = \Gamma_{\mathbb{R}}(s)\zeta(s) = \frac{\xi(s)}{\frac{1}{2}s(s-1)},$$

as a meromorphic identity. Thus the Tate comparison target is $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$.

Proof. This is Tate's product formula for the standard adelic test function:

$$\prod_v Z_v(s) = \Gamma_{\mathbb{R}}(s) \prod_p L_p(s) = \Gamma_{\mathbb{R}}(s)\zeta(s).$$

The standard completed ξ -function is obtained from this meromorphic factor by the elementary multiplier $\frac{1}{2}s(s-1)$. $\square$

Functional Equation Compatibility

COROLLARY 30.8.4 (*functional equation preserved*). . The completed Tate product $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ satisfies the functional equation $\Gamma_{\mathbb{R}}(s)\zeta(s) = \Gamma_{\mathbb{R}}(1-s)\zeta(1-s)$, equivalently $\xi(s) = \xi(1-s)$.

Proof. Chapter 30.8.3 gives Λ_ζ . Tate 1950 identifies the Fourier transform of the standard adelic test function with itself and obtains $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$. Multiplication by $\frac{1}{2}s(s-1)$ gives the displayed equation for ξ . $\square$

Remark 30.8.5 (*significance of the parallel archimedean factor*). Without the parallel archimedean factor, the restricted tensor product $\otimes'_{p<\infty} C_p$ could at most compare with $\zeta(s)$, not with $\Lambda_\zeta(s)$. The finite-prime product satisfies no functional equation of its own; only the completed Tate product $\Lambda_\zeta(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ does. Hence $L_{\text{ACD},0}$ (archimedean stratum) is a *mathematical necessity*, not a cosmetic addition: it is required before any restricted-tensor construction can be compared with the completed Tate character.

Remark 30.8.6 (*trace comparison is extra structure*). The Tate calculation above is not a computation of Hochschild trace in the local Hecke categories. A categorical realization must still construct a trace-to-Tate comparison map and prove that the chosen local units map to the standard local test functions.

30.9 RESIDUAL CONDITIONS

What Remains Open

After this chapter, the residual content of L_{ACD} still contains the local categorical comparison, the ∞ -categorical Bernstein lift, the archimedean boundary comparison, the global identification, and (BC-diamond-glue).

Subcondition	Form	Sou
$L_{\text{ACD}.0}$ (archimedean stratum)	Boundary datum	Ha
$L_{\text{ACD}.1}$ (colimit commutation)	Theorem (F2.c tightened)	Lur
$L_{\text{ACD}.2}$ (cross-prime Hecke)	Formal theorem; conditional Bernstein part	Lur
$L_{\text{ACD}.3}$ (sph-fin preservation)	Theorem (from .1)	Imm
$L_{\text{ACD}.4}$ (derived strong approximation and determinant quotient)	Theorem	Kn
F2.c tightened	Theorem under pointed E_2 hypotheses (see Rem. 30.7.2)	Lur
∞ -categorical BZ lift	Conjectural	Ch
local categorical comparison and global identification	Open	Fl.a
(BC-diamond-glue)	Conjectural	Ber

TABLE 30.1: Form of L_{ACD} components. Formal pieces are separated from comparison data.*F_l Closure Chain*

THEOREM 30.9.1 (*F_l residual after local comparison and descent hypotheses*). . Assume F2.c tightened (Chapter 28.1.3), the local categorical comparison (F1.a), the ∞ -categorical Bernstein lift, the archimedean boundary comparison, and the global automorphic descent hypotheses. Then F_l (global arithmetic Arinkin–Gaitsgory for $GL_2/\mathbb{Q}$, Chapter 20.1.1) reduces to (BC-diamond-glue) (Chapter 26.5.4). Without those hypotheses, (BC-diamond-glue) alone does not close F_l.

Proof. By Chapter 21.2.1, F_l reduces to (F1.a) + (F1.b) + (F1.c) + (F1.d). The item (F1.a) is an assumed local categorical comparison; (F1.c) is the Emerton–Gee stack input. Items (F1.b) and (F1.d) combine into L_{ACD} (Chapter 21.2.3). By Chapter 30.7.1, L_{ACD} reduces to (BC-diamond-glue) only after the comparison hypotheses listed in the statement are supplied. $\square$

Remark 30.9.2 (*form of F_l after this chapter*). F_l does not reduce to (BC-diamond-glue) alone. The reduction still contains the local comparison, the ∞ -categorical Bernstein lift, the archimedean boundary comparison, and the global descent input. Construction ε addresses only the Bernstein-centre gluing part.

*Hypotheses Inventory*30.10 CONDITIONAL REDUCTION OF L_{ACD}

The modular decomposition (Chapter 26.2.1, Chapter 29.2.4, Chapter 29.4) identifies five subconditions $L_{\text{ACD}.0}$ – $L_{\text{ACD}.4}$, with $L_{\text{ACD}.4}$ combines Kneser’s strong approximation for SL_2 with Tate’s determinant quotient for GL_2 . The formal pieces treated here are:

- $L_{\text{ACD}.0}$ as archimedean boundary datum via Harish-Chandra admissibility, Vogan 1981 classification, and Wallach 1988/1992 real-reductive-groups spectrum theory (Chapter 30.3.3);
- $L_{\text{ACD}.1}$ via F2.c tightened, Lurie HA 5.3.2.9, and HTT 5.3.5.15 (Chapter 30.4.1);

Hypothesis	Role in L_{ACD} reduction	Source
F2.c tightened	Universal property for colimit E_2 -structure	Lurie HA, pointed E_2 hypotheses
Lurie HA 5.3.2.9	Sph-fin preservation under filtered colimit	Lurie HA
Lurie HA 5.1.2.4	Dunn additivity for E_n -operads	Lurie HA
Lurie HA 5.1.2.26	Unit centrality in E_n -monoidal cat	Lurie HA
Lurie HTT 5.3.5.15	Ind-completion preserves filtered colim	Lurie HTT
Harish-Chandra 1966	Admissibility of $(\mathfrak{g}, K_\infty)$ -mod	Primary source
Vogan 1981	Langlands–Vogan classification at $\mathbb{R}$	Primary source
Wallach 1988/1992	Real reductive groups spectrum theory	Primary source
Jacquet–Langlands SLN II4	Archimedean local representation	Primary source
Bernstein–Zelevinsky 1976	Bernstein decomposition of p -adic Hecke	Primary source
∞ -categorical BZ lift	Stable lift of Bernstein projectors	Chapter 30.5.3; conjectural
Tate 1950	Adelic Mellin/theta and functional equation	ξ -compatibility
Kneser 1966	Strong approximation for SL_2	$L_{ACD}.4$ derived subgroup theorem
Tate 1950	Idele-class determinant quotient for GL_2	$L_{ACD}.4$ determinant line
(BC-diamond-glue)	Bernstein-centre gluing	Construction ε , not yet theorem

TABLE 30.2: Hypotheses used in this chapter. The non-theorem inputs also include the local comparison, the archimedean boundary comparison, and the global identification.

- $L_{ACD}.2$ (formal tensor-factor commutativity) via Lurie HA 5.1.2.4 (Dunn additivity), with finite-prime Bernstein compatibility conditional on Chapter 30.5.3 (Chapter 30.5.1);
- $L_{ACD}.3$ from $L_{ACD}.1$ (Chapter 30.6.1).

The residual of L_{ACD} is not only (BC-diamond-glue). It also contains the local categorical comparison, the ∞ -categorical Bernstein lift, the archimedean boundary comparison, and the global identification (Chapter 30.7.1).

The parallel archimedean factor ($L_{ACD}.0$) is required: without it, the restricted tensor product can at most compare with $\zeta(s)$, not with $\Gamma_{\mathbb{R}}(s)\zeta(s)$, and the Tate functional equation is invisible (Chapter 30.8.5).

FI (global arithmetic Arinkin–Gaitsgory for $GL_2/\mathbb{Q}$) reduces to (BC-diamond-glue) only under the stated comparison and descent hypotheses (Chapter 30.9.1).

The source list for $L_{ACD}.0$, $L_{ACD}.1$, and $L_{ACD}.3$ is disjoint from the Fargues–Scholze list: Harish-Chandra, Jacquet–Langlands, Vogan, Wallach (archimedean representation theory), Lurie HA, Fresse 2017 (higher operadic machinery), and Bernstein–Zelevinsky 1976 (Bernstein decomposition). The ∞ -categorical Bernstein lift is not proved from these sources; it is the separate conjectural datum Chapter 30.5.3.

BZN^{Ar}: FORMAL NOETHERIAN HOCHSCHILD AND DERIVED STRUCTURE ON EMERTON–GEE

The arithmetic Ben-Zvi–Nadler comparison factors through four independent inputs. The finite-type theorem of Gaitsgory–Rozenblyum applies to smooth finite-type derived stacks. The Emerton–Gee spectral object is instead a formal Selmer deformation stack, and the formal comparison requires a categorical Hochschild–colimit hypothesis, an arithmetic loop-space hypothesis, and a flat-singular-support hypothesis. The inertia shadow does not contain the full Weil/ (φ, Γ) datum, and the primes 2 and 3 carry non-semisimple Hecke summands.

31.1 FORMAL BOUNDARY OF THE COMPARISON

Structural Inputs

Chapter 27 isolates the desired arithmetic Hochschild trace equivalence

$$\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}) \simeq \mathcal{O}(\mathrm{LocSys}_{GL_2}^{W, \varphi\Gamma})^{\mathrm{flat}}$$

as an equivalence of E_1 -algebras in stable presentable ∞ -categories. The finite-type BZN theorem supplies only one factor of the argument. The arithmetic comparison requires the following additional inputs:

R1.fin-type. The Gaitsgory–Rozenblyum theorem (AMS DAG Vol. II Part V Ch. III) that $\mathrm{HH}_\bullet(\mathrm{IndCoh}(\mathcal{X})) \simeq \mathcal{O}(L\mathcal{X})$ holds for smooth finite-type derived stacks $\mathcal{X}$. *Form: theorem* (Gaitsgory–Rozenblyum). The Selmer-derived stack $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}}$ is formal Noetherian and not finite-type; R1.formal below is the required extension.

R1.formal. The extension of the Gaitsgory–Rozenblyum theorem to formal Noetherian derived stacks. For $\mathcal{X}$ a formal Noetherian derived algebraic stack and $\mathcal{C} = \mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$,

$$\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})) \simeq \mathcal{O}(L^{\mathrm{Ar}}\mathcal{X})^{\mathrm{flat}}.$$

Form: conjectural; a genuinely open extension requiring new arguments beyond the finite-type case.

R2. The spectral target is the Selmer-derived global deformation stack $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}}$, built from local Emerton–Gee/Weil/ (φ, Γ) stacks by derived Selmer fibre product, not by a product or colimit of local stacks. *Form: the target and tangent complex are fixed by derived deformation theory; formal Noetherianity and nilpotent-support compatibility are hypotheses on the chosen local conditions.*

R3. Comparison from the inertia shadow $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ to the full Weil/ (φ, Γ) local-system target. *Form: conjectural; the inertia-only identification via $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ constitutes a procyclic shadow of the full Weil/ (φ, Γ) target.*

R4. Non-semisimple summand at wild primes $p = 2, 3$: Ext¹ contributions to $\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}})$ invisible to semisimple templates. *Form: conjectural; carried by the wild Hecke category.*

Domain of This Chapter

This chapter treats $\mathbf{R}_2$ and $\mathbf{R}_{1.\mathrm{formal}}$.

- (i) $\mathbf{R}_2$ identifies the global spectral target as the Selmer-derived stack. The local Emerton–Gee stacks supply local formal algebraic input; the global object must impose the global Galois relation, determinant, residual representation, and local Selmer conditions by derived fibre product. Derived Noetherianity of that fibre product is left as an open check for the chosen local conditions, not subsumed into a product-colimit theorem.
- (ii) $\mathbf{R}_{1.\mathrm{formal}}$ follows from three hypotheses ($\mathbf{R}_{1.\mathrm{formal.a}}$, $\mathbf{b}$, $\mathbf{c}$). The first is a Hochschild trace theorem after the categorical colimit presentation of $\mathrm{IndCoh}_{\mathrm{Nilp}}$ is supplied. The second and third are respectively the arithmetic loop-space comparison and the flat-singular-support comparison.

$\mathbf{R}_{1.\mathrm{fin}}$ -type is external. $\mathbf{R}_3$ is not definitional: it is the passage from the procyclic inertia shadow to the full Weil/ (φ, Γ) target. $\mathbf{R}_4$ is the non-semisimple wild-prime summand.

Derived Enhancement of the Emerton–Gee Stack

Emerton–Gee arXiv:1908.07185 is written in the language of Noetherian formal algebraic stacks in the sense of the Stacks Project, with the smoothness and mass-formula structure at the level of ordinary (non-derived) schemes. Gaitsgory–Rozenblyum DAG requires a derived enhancement: the cotangent complex must be computed as a perfect complex (or almost-perfect, in the formal case) in the ∞ -category of quasi-coherent sheaves, and the ind-colimit presentation must preserve this structure at every stage. The translation is not automatic. It asks for cotangent-complex functoriality under the chosen formal presentation, compatibility with formal thickening, and preservation under the Selmer fibre product. These are hypotheses on the derived enhancement, not consequences of the underived mass formula alone.

Finite-Type Hypotheses in the Gaitsgory–Rozenblyum Argument

The Gaitsgory–Rozenblyum proof of $\mathrm{HH}_\bullet(\mathrm{IndCoh}(\mathcal{X})) \simeq \mathcal{O}(L\mathcal{X})$ uses three ingredients that each rest on finite-type smoothness and require new arguments in the formal Noetherian setting:

- (i) Atiyah duality for IndCoh : the self-duality $\mathrm{IndCoh}(\mathcal{X})^\vee \simeq \mathrm{IndCoh}(\mathcal{X})$ under the Atiyah pairing, which requires $\mathcal{X}$ smooth finite-type.
- (ii) Smooth-base-change for IndCoh , which also uses finite-type.
- (iii) Loop-space duality: the free loop space $L\mathcal{X}$ of a finite-type smooth stack is itself smooth (with dimension equal to $\dim \mathcal{X}$, by the standard calculation).

For formal Noetherian $\mathcal{X}$, each ingredient requires a formal analogue: Atiyah duality must be extended to the formal setting (conjectural in general), smooth base-change must be replaced by formal-smooth base-change (subject of ongoing work, e.g., Mok 2025), and the loop space $L\mathcal{X}$ is in general not smooth finite-type.

The formal comparison therefore has the form of a criterion. Given the categorical colimit presentation of $\mathrm{IndCoh}_{\mathrm{Nilp}}$, Hochschild trace commutes with it; the loop and flat-locus comparisons remain separate geometric hypotheses.

31.2 LOGICAL FORM OF THE INPUTS

The input structure is as follows:

R2 cotangent complex. The cotangent complex of the Emerton–Gee stack is computable by Galois deformation theory. Mazur’s deformation functor framework (Mazur 1989 *Deforming Galois Representations*) computes the tangent complex at each closed $\mathbb{F}_p$ -point; Lurie DAG X builds this into a functorial cotangent complex on the pro-representable moduli stack. Emerton–Gee §2.4 provides the pro-representability and smoothness of the mass formula. (See Chapter 31.3.1.)

R2 global Galois constraint. A global Galois representation is not a product of independent local data: the restrictions to decomposition groups must come from a single global representation subject to the chosen Selmer conditions. The global object is the Selmer-derived fibre product

$$\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}} = \text{Map}_{\bar{\rho}, \det}(G_{\mathbb{Q}, S}, GL_2) \times_{\prod_{v \in S} \text{Map}(G_{\mathbb{Q}_v}, GL_2)}^{\mathbf{R}} \prod_{v \in S} \mathcal{X}_v^{\mathcal{L}_v},$$

whose tangent complex is the Selmer complex $C_{\mathcal{L}}^{\bullet}(G_{\mathbb{Q}, S}, \text{ad } \rho)[1]$. The displayed fibre product fixes the object; derived Noetherianity requires verification for the chosen local conditions.

R2 derived enhancement. Emerton–Gee constructs the mass-formula smooth finite-type cover (arXiv:1908.07185 §2.6, Thm. B.0.1(1)): the stack is smooth and hence quasi-smooth. Quasi-smoothness in the Lurie DAG sense is preserved under formal completion (Gaitsgory–Rozenblyum DAG Vol. II Ch. II §3.3, formal-completion smoothness). For the Selmer fibre product one must still verify that the chosen local deformation conditions preserve formal Noetherianity, quasi-smoothness, and the nilpotent-support condition. The tangent complex is the Selmer complex; the required finiteness is an additional geometric input. See Chapter 31.3.5.

R1.formal three inputs. The Gaitsgory–Rozenblyum theorem, ported from finite-type to formal Noetherian, requires three separate arguments: (i) HH-colim-continuity (whether $\text{HH}_{\bullet}$ commutes with the ind-colimit presentation of the formal stack); (ii) arithmetic loop-space formal computation (whether $L^{\text{Ar}}\mathcal{X} \simeq \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ admits a workable formal description); (iii) flat-locus pullback (whether the nilpotent-singular-support-flat locus on the loop space matches the Emerton–Gee flat locus of Chapter 27.4.7). The first is a trace theorem once the categorical colimit exists; the other two are geometric hypotheses. (See Chapter 31.4.1.)

R1.formal sub-reductions. R1.formal.a is the Hochschild trace statement for a given filtered colimit in the Morita $(\infty, 2)$ -category. R1.formal.b reduces to the existence of a cotangent-complex formal loop-space identification (a precise statement in Lurie DAG and Gaitsgory–Rozenblyum Vol. II). R1.formal.c reduces to the compatibility of Arinkin–Gaitsgory nilpotent singular support with the (φ, Γ) -module Fargues–Scholze flat locus. (See Chapter 31.4.2 and Chapter 31.4.9.)

Consistency with the initial arithmetic comparison. The arithmetic comparison of Chapter 27.5.5 uses a pro-étale extension of Ben-Zvi–Francis–Nadler integral transforms. In the present notation this is the R1.formal.b loop-space hypothesis together with the R1.formal.c flat-locus hypothesis.

31.3 R2: THE EMERTON–GEE COTANGENT COMPLEX

Local Tangent Complex at a Closed Point

PROPOSITION 31.3.1 (*local tangent complex of the Emerton–Gee stack*). . Let $\bar{\rho} \in \mathcal{X}_{GL_2}^{\text{EG}, p}(\overline{\mathbb{F}}_p)$ be a closed point (= a mod- p Galois representation $\bar{\rho}: \text{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \rightarrow GL_2(\overline{\mathbb{F}}_p)$). The tangent complex $T_{\bar{\rho}}\mathcal{X}_{GL_2}^{\text{EG}, p}$ at $\bar{\rho}$ is

$$T_{\bar{\rho}}\mathcal{X}_{GL_2}^{\text{EG}, p} \simeq H_{\text{cts}}^*(\text{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p), \text{ad } \bar{\rho})[1],$$

continuous Galois cohomology of the adjoint representation, shifted by one.

Proof. The Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG},p}$ parametrises rank-2 (φ, Γ) -modules over the Fontaine period ring of $\mathbb{Q}_p$ (Emerton–Gee arXiv:1908.07185 §1.2). At a closed $\overline{\mathbb{F}}_p$ -point $\overline{\rho}$, the pro-representable universal deformation ring is Mazur’s framed deformation ring $R_{\overline{\rho}}^{\square}$ (Mazur 1989, Kisin Mazur 2000). The Zariski tangent space of the formal scheme $\text{Spf } R_{\overline{\rho}}^{\square}$ is

$$H_{\text{cts}}^1(\text{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p), \text{ad } \overline{\rho}),$$

and the obstruction module is $H_{\text{cts}}^2(\text{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p), \text{ad } \overline{\rho})$.

In the Lurie DAG X framework (Lurie *Formal Moduli Problems*, HA App. A.5 and DAG X §1), a formal moduli functor’s tangent complex at a point is the two-term complex $[H^1 \text{ in deg. 0}, H^2 \text{ in deg. 1}]$, which is $H_{\text{cts}}^*(\text{Gal}, \text{ad } \overline{\rho})[1]$ up to the standard degree shift for cochains-vs-tangent.

The stack-level statement is obtained by quotienting by the $\text{ad } \overline{\rho}$ -automorphisms of the universal deformation, which corresponds to the lowest degree $H_{\text{cts}}^0 = \text{ad } \overline{\rho}^{\text{Gal}}$ being absorbed into the stack’s automorphism group. $\square$

Remark 31.3.2 (perfectness and tor-amplitude). The continuous Galois cohomology $H_{\text{cts}}^*(\text{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p), \text{ad } \overline{\rho})$ is concentrated in degrees 0, 1, 2 (local Galois cohomology of a p -adic local field is finite-dimensional, supported in degrees ≤ 2 by local duality; Serre *Cohomologie Galoisienne* Ch. II §5.2). Hence $T_{\overline{\rho}} \mathcal{X}_{GL_2}^{\text{EG},p}$ is a perfect complex of tor-amplitude $[-1, 1]$, indeed of tor-amplitude $[-1, 0]$ after removing the H^0 factored into automorphisms. This is the key perfectness input for R2.

Ind-Level Cotangent Complex

PROPOSITION 31.3.3 (*cotangent complex preserves ind-colimit*). . Let $\mathcal{X} = \varinjlim_{i \in I} \mathcal{X}_i$ be an ind-colimit of derived stacks with transition maps closed immersions in a filtered poset I . Then the cotangent complex satisfies

$$\mathbb{L}_{\mathcal{X}} \simeq \varprojlim_{i \in I^{\text{op}}} \mathbb{L}_{\mathcal{X}_i} |_{\mathcal{X}_i \rightarrow \mathcal{X}}$$

as an object of $\text{IndCoh}(\mathcal{X})$, compatibly with the pullbacks along closed immersions.

Proof. The cotangent complex functor $\mathbb{L} : \text{dSt} \rightarrow \text{IndCoh}$ is a left adjoint (to the trivial-square-zero-extension functor, by Lurie HA 7.3.2.14 in the ∞ -categorical setting; Illusie 1971 *Complexes cotangent et déformations* Ch. III for the classical analogue). Left adjoints preserve colimits; the filtered colimit $\mathcal{X} = \varinjlim_i \mathcal{X}_i$ in derived stacks is sent to a filtered colimit in IndCoh , which by duality on IndCoh (Gaitsgory–Rozenblyum Vol. II Ch. II §3.6, Serre-duality-based duality swap) corresponds to a Pro-limit $\varprojlim_{i \in I^{\text{op}}} \mathbb{L}_{\mathcal{X}_i} |_{\mathcal{X}_i \rightarrow \mathcal{X}}$ under the closed-immersion transitions.

Compatibility with pullbacks along closed immersions is the base-change property of $\mathbb{L}$ (Lurie HA 7.3.2; Illusie 1971 Prop. III.1.2.6.1), applied to the closed immersions $\mathcal{X}_i \rightarrow \mathcal{X}$ indexing the ind-colimit. $\square$

COROLLARY 31.3.4 (*cotangent complex preserves finite products*). . For derived stacks $\mathcal{X}, \mathcal{Y}$,

$$\mathbb{L}_{\mathcal{X} \times \mathcal{Y}} \simeq \mathbb{L}_{\mathcal{X}} \boxplus \mathbb{L}_{\mathcal{Y}},$$

the external direct sum of cotangent complexes pulled back to $\mathcal{X} \times \mathcal{Y}$.

Proof. The finite product $\mathcal{X} \times \mathcal{Y}$ in derived stacks is a finite limit; the cotangent complex is a left adjoint and sends finite limits to finite limits (= direct sums in the ∞ -category of perfect complexes, by stability of the target). Lurie HA 7.3.2.4 computes the cotangent complex of a finite limit as a finite colimit of the individual cotangent complexes, which under the dual identification of direct sum with external-sum structure gives $\mathbb{L}_{\mathcal{X}} \boxplus \mathbb{L}_{\mathcal{Y}}$.

Alternatively, this follows directly from the universal property of the cotangent complex as classifying square-zero extensions: a square-zero extension of $\mathcal{X} \times \mathcal{Y}$ is a pair of square-zero extensions of $\mathcal{X}$ and $\mathcal{Y}$. $\square$

R2 Main Theorem

THEOREM 31.3.5 (*Selmer-derived spectral target*). . For fixed residual representation $\bar{\rho}$, determinant, finite ramification set S , and local conditions $\mathcal{L}_v$, the global spectral target is the Selmer-derived stack

$$\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}} = \text{Map}_{\bar{\rho}, \det}(G_{\mathbb{Q}, S}, GL_2) \times_{\prod_{v \in S} \text{Map}(G_{\mathbb{Q}_v}, GL_2)}^{\mathbf{R}} \prod_{v \in S} \mathcal{X}_v^{\mathcal{L}_v}.$$

At a point ρ its tangent complex is the Selmer tangent complex

$$T_{\rho} \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}} \simeq C_{\mathcal{L}}^{\bullet}(G_{\mathbb{Q}, S}, \text{ad}^{\circ} \rho)[1].$$

The following items are the open checks that remain for the chosen local deformation problem:

- (i) Local conditions $\mathcal{X}_v^{\mathcal{L}_v}$ are derived formal algebraic substacks of the full local Weil/ (φ, Γ) parameter stack.
- (ii) The derived fibre product is formal Noetherian/quasi-smooth in the Gaitsgory–Rozenblyum sense.
- (iii) The nilpotent-singular-support condition used in $\text{IndCoh}_{\text{Nilp}}$ is compatible with the Selmer cotangent complex.

Proof. The displayed stack is the derived fibre product defining the Selmer deformation functor with fixed determinant and local conditions. Linearising continuous Galois cochains gives the standard Selmer mapping cone, hence the tangent complex $C_{\mathcal{L}}^{\bullet}(G_{\mathbb{Q}, S}, \text{ad}^{\circ} \rho)[1]$. The remaining assertions are not consequences of a product/colimit presentation and are left as explicit checks. $\square$

COROLLARY 31.3.6 (*the inertia shadow does not determine the full target*). . Given Chapter 31.3.5, R3 still requires the full-target comparison:

$$\text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}) \longrightarrow \text{LocSys}_{GL_2}^{\text{Ar}}(B\widehat{\mathbb{Z}}),$$

as a procyclic shadow, and then a comparison with $\text{LocSys}_{GL_2}^{W, \varphi\Gamma}$.

Proof. By Chapter 27.4.1 and Chapter 27.4.5, $\text{LocSys}_{GL_2}^{\text{Ar}}(B\widehat{\mathbb{Z}}) = L^{\text{Ar}} BGL_2 = \text{Map}(B\widehat{\mathbb{Z}}, BGL_2)$. The map from $\text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}})$ is obtained only after choosing an inertia-shadow comparison morphism $\iota^{\text{Sel}}: \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}} \rightarrow \text{LocSys}_{GL_2}^{\text{Ar}}(B\widehat{\mathbb{Z}})$ (Chapter 27.4.6). That inertia-only morphism retains only the procyclic structure, discarding Frobenius and (φ, Γ) data; the full comparison $\text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}) \simeq \text{LocSys}_{GL_2}^{W, \varphi\Gamma}$ requires a separate argument (R3). $\square$

Source and Convention Comparison for R_2

Source independence. The argument for R_2 draws on five disjoint source lists: Emerton–Gee arXiv:1908.07185 (Noetherian formal algebraic structure); Mazur 1989 and Kisin–Mazur 2000 (Galois deformation theory); Lurie HA / DAG X (formal moduli problems and cotangent complex); Illusie 1971 (classical cotangent complex); Gaitsgory–Rozenblyum DAG Vol. II Ch. II (quasi-smoothness of derived formal algebraic stacks).

Degree conventions. Lurie HA places the cotangent complex in non-positive degrees for smooth schemes (tor-amplitude $[-1, 0]$ means concentration in degrees $-1, 0$, equivalently the tangent complex in degrees $0, 1$). The statement above uses the Lurie convention. Emerton–Gee’s tangent-space computation places H^1 in degree 0 and H^2 in degree 1, matching after the standard shift. Illusie 1971 places the cotangent complex in non-positive degrees for smooth schemes, uniform with Lurie.

Ambient category. All objects live in derived algebraic stacks with pro-étale topology (Toën–Vezzosi, Bhatt–Scholze, Gaitsgory–Rozenblyum). The cotangent complex is a left adjoint on this ∞ -category. The $(\infty, 2)$ -category of correspondences in which IndCoh is valued (Gaitsgory–Rozenblyum Vol. I Ch. V) is the ambient category for both the tangent complex and the loop-space computation.

Hypotheses. Three hypotheses enter the formal Selmer target: Emerton–Gee Noetherian formal algebraic structure, Lurie DAG formal-moduli representability, and Gaitsgory–Rozenblyum formal-completion smoothness. The derived Selmer fibre product also requires the local deformation conditions to preserve formal Noetherianity and nilpotent singular support.

Functoriality. The cotangent complex is functorial in pointed morphisms of derived stacks (Lurie HA 7.3.2, Illusie 1971 Ch. II); the ind-colimit and product structures are preserved (Chapter 31.3.3 and Chapter 31.3.4).

Source of equivalences. Derived deformation theory gives the displayed target and its tangent complex. It does not by itself prove the finiteness and singular-support hypotheses required by the formal BZN comparison.

Remark 31.3.7 (derived enhancement). Emerton–Gee do not explicitly formulate their stack in the Gaitsgory–Rozenblyum DAG language. The tangent-complex computation comes from Mazur deformation theory and Lurie formal moduli. The passage to the formal BZN theorem still needs the finiteness and singular-support hypotheses named in Chapter 31.3.5.

31.4 R1.FORMAL : FORMAL HOCHSCHILD CRITERION*The Three Inputs*

THEOREM 31.4.1 (formal BZN criterion). . Let $\mathcal{X}$ be a derived formal algebraic stack in the Gaitsgory–Rozenblyum sense, Noetherian, presented as an ind-colimit $\mathcal{X} = \varinjlim_i \mathcal{X}_i$ of finite-type derived stacks under closed immersions (as in the Emerton–Gee structure theorem). Let $\mathcal{C}(\mathcal{X}) = \text{IndCoh}_{\text{Nilp}}(\mathcal{X})$. Assume the finite-type BZN comparison on each $\mathcal{X}_i$, and assume the three conditions below. Then

$$\text{HH}_\bullet(\text{IndCoh}_{\text{Nilp}}(\mathcal{X})) \simeq \mathcal{O}(L^{\text{Ar}}\mathcal{X})^{\text{flat}},$$

namely Rl.formal , holds.

(Rl.formal.a) *HH-colim continuity*. The Hochschild trace preserves the ind-colimit presentation of $\text{IndCoh}_{\text{Nilp}}(\mathcal{X})$:

$$\text{HH}_\bullet(\text{IndCoh}_{\text{Nilp}}(\mathcal{X})) \simeq \varinjlim_i \text{HH}_\bullet(\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_i)),$$

under the transition maps induced by closed immersions. **Form: theorem after the categorical colimit presentation is given** (trace continuity; proved in Chapter 31.4.2).

(Rl.formal.b) *Arithmetic loop-space formal computation*. For a formal Noetherian derived stack $\mathcal{X}$ as above, the free loop space admits a formal ind-colimit description

$$L^{\text{Ar}}\mathcal{X} \simeq \varinjlim_i L^{\text{Ar}}\mathcal{X}_i$$

compatible with the closed immersions; equivalently, $L^{\text{Ar}} = \text{Map}(B\widehat{\mathbb{Z}}, -)$ commutes with the ind-colimit presentation. **Form: conjectural** (Lurie ∞ -topos formalism in the pro-étale formal setting; see Chapter 31.4.4).

(Rl.formal.c) *Flat-locus pullback compatibility*. The flat locus $L^{\text{Ar}}\mathcal{X}^{\text{flat}}$ (cut out by the Arinkin–Gaitsgory nilpotent-singular-support condition lifted to the loop space) is compatible with the ind-colimit structure:

$$\mathcal{O}(L^{\text{Ar}}\mathcal{X})^{\text{flat}} \simeq \varinjlim_i \mathcal{O}(L^{\text{Ar}}\mathcal{X}_i)^{\text{flat}},$$

with the flat loci on finite-type pieces matching the (φ, Γ) -module Fargues–Scholze flat locus of Chapter 27.4.7. **Form: conjectural** (Arinkin–Gaitsgory singular support extended to this formal Selmer setting; see Chapter 31.4.7).

Proof. Assume (Rl.formal.a), (Rl.formal.b), and (Rl.formal.c). By (Rl.formal.a),

$$\text{HH}_\bullet(\text{IndCoh}_{\text{Nilp}}(\mathcal{X})) \simeq \varinjlim_i \text{HH}_\bullet(\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_i)).$$

By the Gaitsgory–Rozenblyum theorem applied term-wise,

$$\varinjlim_i \text{HH}_\bullet(\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_i)) \simeq \varinjlim_i \mathcal{O}(L\mathcal{X}_i)^{\text{flat}}.$$

By (Rl.formal.b), $\varinjlim_i L\mathcal{X}_i \simeq L^{\text{Ar}}\mathcal{X}$ (noting $L = L^{\text{Ar}}$ on finite-type pieces via the reduction $S^1 \rightarrow B\widehat{\mathbb{Z}}$ in the pro-étale setting; this is the content of Chapter 27.5.3 on each finite-type piece). By (Rl.formal.c),

$$\varinjlim_i \mathcal{O}(L\mathcal{X}_i)^{\text{flat}} \simeq \mathcal{O}(L^{\text{Ar}}\mathcal{X})^{\text{flat}}.$$

Composing gives Rl.formal . □

Rl.formal.a: HH Colimit Continuity

THEOREM 31.4.2 (*Hochschild trace along a categorical colimit*). . Let $\mathcal{X} = \varinjlim_i \mathcal{X}_i$ be a formal presentation by derived stacks under closed immersions. Assume that the canonical functor

$$\varinjlim_i \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_i) \longrightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X})$$

is an equivalence in the E_2 -monoidal Morita $(\infty, 2)$ -category of stable presentable categories. Then

$$\mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})) \simeq \varinjlim_i \mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_i))$$

as E_1 -algebras in stable presentable ∞ -categories.

Proof. Hochschild homology is the trace of the identity endofunctor in the Morita $(\infty, 2)$ -category of stable presentable categories. The trace functor is symmetric monoidal and preserves filtered colimits of dualizable objects and dualizable morphisms; this is the form of Lurie HA 5.5.3 used here, and it is also obtained from the cyclic bar description of Toën. Applying the trace to the assumed categorical colimit gives the displayed equivalence. The E_1 -algebra structure is transported by Dunn additivity from the E_2 -monoidal convolution structure. $\square$

Remark 31.4.3 (where the geometry enters). The theorem is purely categorical. It does not prove that $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ is the filtered colimit of the categories on the finite pieces. That assertion is a geometric compatibility statement for formal completion, singular support, and the closed-immersion transition functors.

RI.formal.b: Arithmetic Loop-Space Formal Computation (Conditional)

PROPOSITION 31.4.4 (*RI.formal.b domain*). . RI.formal.b reduces to:

- (b.1) The internal mapping stack $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ commutes with filtered colimits in derived pro-étale stacks under closed immersions.
- (b.2) For a formal Noetherian derived stack $\mathcal{X}$ with $\mathcal{X} = \varinjlim_i \mathcal{X}_i$, the natural map $\varinjlim_i \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_i) \rightarrow \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ is an equivalence in the pro-étale topology.

Both (b.1) and (b.2) are Conjectural.

Proof of reduction. RI.formal.b as stated is $L^{\mathrm{Ar}}\mathcal{X} \simeq \varinjlim_i L^{\mathrm{Ar}}\mathcal{X}_i$. By Chapter 27.4.1, $L^{\mathrm{Ar}}\mathcal{X} = \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$. The ind-colimit compatibility of this mapping stack is precisely (b.1) (the general statement about the mapping functor) plus (b.2) (verification of the specific map for the ind-colimit presentation of $\mathcal{X}$). Conversely, given (b.1) and (b.2), RI.formal.b is immediate. $\square$

Remark 31.4.5 (compactness obstruction for $B\widehat{\mathbb{Z}}$). In a general ∞ -topos, $\mathrm{Map}(\mathcal{M}, -)$ commutes with filtered colimits precisely when $\mathcal{M}$ is compact (Lurie HTT 5.3.4.18: compactness means $\mathcal{M}$ is a retract of a finite colimit of compact-projective objects). The classifying stack $B\widehat{\mathbb{Z}}$ is profinite and admits a pro-étale presentation $B\widehat{\mathbb{Z}} = \varprojlim_n B(\mathbb{Z}/n\mathbb{Z})$. Each $B(\mathbb{Z}/n\mathbb{Z})$ is compact; the pro-étale limit is a cofiltered limit of compact objects and compactness of the limit requires a separate argument for each class of target objects. Statements (b.1) and (b.2) therefore each demand an explicit pro-étale compatibility argument (Bhatt–Scholze arXiv:1309.1198 §7); (b.2) further requires verification for the specific ind-colimit $\mathcal{X} = \varinjlim_i \mathcal{X}_i$ from the formal-Noetherian structure.

Chapter 27 (Chapter 27.5.3) implicitly assumed (b.1) via the framed-1-manifold presentation of $B\widehat{\mathbb{Z}}$. The assumption is RI.formal.b and remains conjectural.

Remark 31.4.6 (size of the loop-space hypothesis). The hypothesis has three parts: compactness of $B\widehat{\mathbb{Z}}$ against the relevant formal targets, commutation of $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ with the chosen formal-Noetherian ind-colimits, and compatibility with IndCoh -functoriality. Each part is a statement in the pro-étale topology, not a formal consequence of finite-level compactness of $B(\mathbb{Z}/n\mathbb{Z})$.

R1.formal.c: Flat-Locus Pullback Compatibility (Conditional)

PROPOSITION 31.4.7 (*R1.formal.c domain*). . R1.formal.c reduces to:

- (c.1) *Arinkin–Gaitsgory nilpotent singular support extends to derived formal algebraic stacks.* The conic subcategory $\text{IndCoh}_{\text{Nilp}} \subset \text{IndCoh}$ is defined for derived formal algebraic stacks in the GR DAG Vol. II sense, preserving the Arinkin–Gaitsgory functoriality (arXiv:1201.6343).
- (c.2) *Flat locus preserved under ind-colimit.* For an ind-colimit $\mathcal{X} = \varinjlim_i \mathcal{X}_i$, the flat locus on the loop space commutes with the ind-colimit: $L^{\text{Ar}} \mathcal{X}^{\text{flat}} \simeq \varinjlim_i L^{\text{Ar}} \mathcal{X}_i^{\text{flat}}$.
- (c.3) *Flat locus matches FS locus.* On each finite-type piece $\mathcal{X}_i$, the Arinkin–Gaitsgory flat locus of the loop space coincides with the (φ, Γ) -module Fargues–Scholze flat locus of Chapter 27.4.7.

All three statements are hypotheses.

Proof of reduction. R1.formal.c as stated is $O(L^{\text{Ar}} \mathcal{X})^{\text{flat}} \simeq \varinjlim_i O(L^{\text{Ar}} \mathcal{X}_i)^{\text{flat}}$, with the flat loci on finite-type pieces matching Chapter 27.4.7. Unpacking: this requires the flat locus to be *defined* on each finite-type $\mathcal{X}_i$ (which is a statement about nilpotent singular support extending to derived formal algebraic stacks; this is (c.1)), to *commute with ind-colimit* (which is (c.2)), and to *match the FS flat locus* on each finite-type piece (which is (c.3)). Conversely, (c.1) + (c.2) + (c.3) imply R1.formal.c by pasting the three statements together. $\square$

Remark 31.4.8 (flat support is a separate geometry). Arinkin–Gaitsgory singular support is defined by the cotangent geometry of a quasi-smooth stack. The Fargues–Scholze flat locus is defined by (φ, Γ) -module geometry. Equality of the two loci on the Emerton–Gee finite pieces is a comparison theorem, not a definition. Compatibility with the formal colimit is another theorem condition.

Sufficient Criterion for $F_3.BZN^{\text{Ar}}$

THEOREM 31.4.9 (*sufficient criterion for $F_3.BZN^{\text{Ar}}$*). . Suppose the structural splitting Chapter 29.2.5 is imposed. The conjunction

- R1.fin-type (Gaitsgory–Rozenblyum: theorem on smooth finite-type).
- R1.formal.a (Hochschild trace along the given categorical colimit).
- R1.formal.b (arithmetic loop-space formal computation: conjectural).
- R1.formal.c (flat-locus pullback compatibility: conjectural).
- R2 (Selmer-derived target, tangent complex, formal Noetherianity, and nilpotent-support compatibility).
- R3 (arithmetic free loop space identification: full-target comparison still required, Chapter 31.3.6).
- R4 (non-semisimple summand at wild primes $p = 2, 3$: wild Hecke summands included).

implies $F_3.BZN^{\text{Ar}}$.

Proof. Chapter 29.2.5 expresses $F_3.BZN^{\text{Ar}}$ as $R_1 + R_2 + R_3 + R_4$, where R_1 further splits into R1.fin-type + R1.formal (Chapter 31.1). R1.fin-type is Gaitsgory–Rozenblyum; R1.formal splits into R1.formal.a (Chapter 31.4.2, theorem) plus R1.formal.b (Chapter 31.4.4, conjectural) plus R1.formal.c (Chapter 31.4.7, conjectural) by Chapter 31.4.1. R2 supplies the Selmer target and tangent complex and includes the required finiteness and singular-support hypotheses. R3 supplies the full Weil/ (φ, Γ) comparison. R4 inserts the wild-prime summands. These inputs are exactly the factors in the structural splitting, hence their conjunction implies $F_3.BZN^{\text{Ar}}$. $\square$

Source and Convention Comparison for R1.formal

Source independence. The argument for R1.formal draws on four disjoint source lists: Lurie HA 5.5.3 / 4.8 / 7.4 (HH, cotangent, colimits); Gaitsgory–Rozenblyum DAG Vol. II Part V Ch. III (HH on smooth finite-type); Arinkin–Gaitsgory arXiv:1201.6343 (nilpotent singular support); Fargues–Scholze arXiv:2102.13459 (local Langlands).

HH conventions. Lurie HA places HH on symmetric monoidal stable presentable ∞ -categories in spectra; Gaitsgory–Rozenblyum uses a derived-category trace on IndCoh ; Toën 2014 uses a direct simplicial-object computation. All three are equivalent: HH of a stable ∞ -category is the colimit over the cyclic category of the self-tensor-product sequence (McCarthy–Schwede, Blumberg–Mandell).

Ambient category. All objects live in derived pro-étale stacks with the $(\infty, 2)$ -category of correspondences for IndCoh -valued functoriality.

Hypotheses. The main hypothesis on $\mathcal{X}$ is Noetherianness together with derived-formal-algebraic structure. For the Selmer target this includes the formal Noetherian and nilpotent-support compatibility conditions of Chapter 31.3.5.

Functoriality. HH-colim continuity is a functorial statement on the ∞ -category of stable presentable ∞ -categories; the criterion by three inputs is itself functorial in $\mathcal{X}$.

Source of the implication. The criterion Chapter 31.4.1 is one-way. The finite-type theorem, Hochschild trace continuity, loop-space compatibility, and flat-locus compatibility imply the formal BZN comparison. The converse is not used.

31.5 CONDITIONAL FORM OF $\text{F3.BZN}^{\text{Ar}}$

COROLLARY 31.5.1 (*conditional arithmetic BZN comparison*). $\text{F3.BZN}^{\text{Ar}}$ follows from Chapter 31.4.9 and the conjunction of:

- R1.formal.b (arithmetic loop-space formal computation, conjectural).
- R1.formal.c (flat-locus pullback compatibility, conjectural).
- R2 finiteness and nilpotent-support compatibility for the chosen Selmer deformation problem.
- R3 (full-target loop-space identification, including the Weil/ (φ, Γ) comparison).
- R4 (the non-semisimple Hecke summands at $p = 2, 3$).

R1.fin-type is the finite-type theorem. R1.formal.a is the trace theorem for a given categorical colimit. The other entries are geometric hypotheses.

Proof. By Chapter 31.4.9: the conjunction $\text{R1.fin-type} + \text{R1.formal.a} + \text{R1.formal.b} + \text{R1.formal.c} + \text{R2} + \text{R3} + \text{R4}$ implies $\text{F3.BZN}^{\text{Ar}}$, where R2 includes the finiteness and support hypotheses of Chapter 31.3.5. The corollary is the same implication with the finite-type theorem and trace theorem separated from the geometric hypotheses. $\square$

Remark 31.5.2 (formal-loop and flat-support inputs). The pro-étale extension used in Chapter 27.5.5 contains two distinct geometries. The first is the commutation of $\text{Map}(B\widehat{\mathbb{Z}}, -)$ with the formal Selmer

presentation. The second is the equality between the Arinkin–Gaitsgory flat singular-support locus and the Fargues–Scholze (φ, Γ) -flat locus.

Remark 31.5.3 (pro-étale extension). Chapter 27 Remark 27.5.4 notes that Chapter 27.5.3 uses a pro-étale extension of Ben-Zvi–Francis–Nadler that is formal but not in the literature. In the present notation that extension is the $\mathbf{R1.formal.b}$ hypothesis, together with the $\mathbf{R1.formal.c}$ comparison of flat loci.

31.6 BOUNDARY CONDITIONS

Form of the Inputs

The seven inputs have different logical weights. $\mathbf{R1.fn-type}$ is the Gaitsgory–Rozenblyum finite-type theorem. $\mathbf{R1.formal.a}$ is a trace theorem after the categorical colimit presentation is known. $\mathbf{R2}$ fixes the Selmer target and tangent complex, but its use in formal BZN requires formal Noetherianity and nilpotent-support compatibility for the chosen local conditions. $\mathbf{R1.formal.b}$, $\mathbf{R1.formal.c}$, $\mathbf{R3}$, and $\mathbf{R4}$ are geometric hypotheses.

Logical Consequence

The arithmetic BZN comparison is therefore not a consequence of a product formula for local stacks. It is a statement about a Selmer derived fibre product, a formal Hochschild trace, a pro-étale loop object, a flat singular-support locus, and wild Hecke summands. The finite-type theorem and the trace theorem do not eliminate the loop, flat-locus, full-target, or wild-prime hypotheses.

Open Problems

The reduction of Chapter 31.4.9 poses three precise open problems.

- (i) **$\mathbf{R1.formal.b}$.** Does $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ commute with ind-colimits under closed immersions in derived pro-étale stacks? The relevant tools are Lurie’s ∞ -topos formalism (HTT 6, HA 4.4) combined with Bhatt–Scholze pro-étale (§7). A positive answer proves $\mathbf{R1.formal.b}$; a counterexample restricts Chapter 27.5.3 to the cofinal finite-type subclass.
- (ii) **$\mathbf{R1.formal.c}$.** Does the Arinkin–Gaitsgory nilpotent-singular-support flat locus coincide with the (φ, Γ) -module Fargues–Scholze flat locus on each finite-type piece of the Emerton–Gee ind-colimit? The comparison between Fargues–Scholze arXiv:2102.13459 and Arinkin–Gaitsgory arXiv:1201.6343 can be checked on unramified crystalline representations with fixed Hodge–Tate weight.
- (iii) **$\mathbf{R4}$.** What are the Ext^i -contributions to $\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}})$ at wild primes $p = 2, 3$, and do they shift the arithmetic BZN identification? The Etingof–Ostrik 2004 finite-tensor-category framework applies; a model case is $p = 2$ with tame restriction to the depth-1 local category.

31.7 FORMAL NOETHERIAN BOUNDARY

The global spectral target for $\mathbf{F3.BZN}^{\mathrm{Ar}}$ is the Selmer-derived stack $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}}$ with tangent complex $C_{\mathcal{L}}^\bullet(G_{\mathbb{Q}, S}, \mathrm{ad}^\circ \rho)[1]$ (Chapter 31.3.5); formal Noetherianity, quasi-smoothness, and nilpotent-support compatibility for the chosen local deformation problem remain hypotheses. $\mathbf{R3}$ is a full-target hypothesis under $\mathbf{R2}$ (Chapter 31.3.6).

$\mathbf{R1.formal}$ follows from three conditions (Chapter 31.4.1). $\mathbf{R1.formal.a}$ is Hochschild trace continuity along the given categorical colimit (Chapter 31.4.2). $\mathbf{R1.formal.b}$ is the arithmetic loop-space formal computation, and $\mathbf{R1.formal.c}$ is flat-locus pullback compatibility (Chapter 31.4.4 and Chapter 31.4.7).

Thus $\mathrm{F}_3.\mathrm{BZN}^{\mathrm{Ar}}$ is conditional on $\mathrm{R}_1.\mathrm{formal.b}$, $\mathrm{R}_1.\mathrm{formal.c}$, the finiteness and support part of R_2 , R_3 , and R_4 (Chapter 31.5.1). The pro-étale extension of Ben-Zvi–Francis–Nadler is not a formal theorem of the finite-type theory; in this notation it is the loop-space hypothesis together with the flat-locus comparison.

BZN^{Ar}: LOOP-SPACE CONVENTION AND WILD-PRIME SUMMANDS

The $F_3.BZN^{Ar}$ reduction of Chapter 27 rests on two unresolved conditions. The first: the arithmetic free loop space $L^{Ar}\mathcal{X} = \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ admits three candidate definitions on formal Noetherian derived stacks, differing a priori in the formal-Noetherian setting; no canonical choice was specified. The second: the Iwahori Hecke category $\text{Hk}_{GL_2, p}^{Iw}$ is not semisimple at the wild primes $p = 2, 3$ of $GL_2/\mathbb{Q}$, so the Hochschild trace picks up Ext^i -contributions invisible to the Ben-Zvi–Nadler semisimple template. The derived internal-Hom convention is the convention used below. Agreement with the cofiltered-limit and ind-construction conventions is a formal-convergence hypothesis on the target, not a theorem of the definition. The wild-prime condition is modeled by an explicit calculation whose conclusion depends on unverified coefficient-convention compatibility: the expected Ext^i -summand at $p = 2$ and $p = 3$ is computed via Etingof–Ostrik 2004 *Finite Tensor Categories*, Bushnell–Henniart 2006 local Langlands at the wild primes, and Jantzen 2003 modular-representation Ext -machinery; the coefficient conventions must be verified against the arithmetic mixed-characteristic setting before the calculation has independent source support. The additive wild summand required by the proposed $F_3.BZN^{Ar}$ comparison is recorded. The global $F_3.BZN^{Ar}$ statement of Chapter 27 remains conditional until the loop-space convergence, full Weil/ (φ, Γ) target, wild-prime Ext , and $p \geq 5$ semisimplicity inputs are verified in the arithmetic coefficient conventions.

32.1 THE TWO LOOP-CONVENTION AND WILD-PRIME CONDITIONS

Origin of the Two Conditions

The structural comparison of Chapter 29 at §29.4 contains four conditions in the $F_3.BZN^{Ar}$ reduction. Three of these (the formal-Noetherian extension of Gaitsgory–Rozenblyum; the cotangent-complex check for the Emerton–Gee stack; and the R_1 relabeling) were absorbed into the R_1 split $R_1.\text{fin-type} + R_1.\text{formal}$ stated in Chapter 29. The remaining two, not absorbed into the existing reduction, are:

- **R3.** Ambiguity in the loop-space convention. Three candidate definitions of $L^{Ar}\mathcal{X} = \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ are in play:
 - (a) *Cofiltered-limit convention.* $L_{\text{cofilt}}^{Ar}\mathcal{X} := \varprojlim_n \text{Map}(B(\mathbb{Z}/n), \mathcal{X})$, the limit over finite quotients of $\widehat{\mathbb{Z}}$.
 - (b) *Internal-Hom convention.* $L_{\text{intHom}}^{Ar}\mathcal{X} := \text{Map}_{\text{dSt}_{\text{Noeth}}^{\wedge}}(B\widehat{\mathbb{Z}}, \mathcal{X})$, the internal mapping stack in the ∞ -category of formal Noetherian derived stacks.
 - (c) *Ind-construction convention.* $L_{\text{ind}}^{Ar}\mathcal{X} := \varinjlim_n \text{Ind}(\text{Map}(B(\mathbb{Z}/n), \mathcal{X}^{\text{perf}}))$, the ind-completion over finite quotients, applied to the perfect locus and re-ind-extended.

These agree for $\mathcal{X}$ finite-type smooth, but the formal Noetherian setting is sensitive to the order of limits and ind-completions.

- **R4.** Non-semisimplicity at the wild primes. For $GL_2/\mathbb{Q}$ the non-banal primes are $p = 2, 3$; at these primes the Iwahori Hecke category $\mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}}$ is not semisimple. The classical Ben-Zvi–Nadler theorem, and its R1/R2/R3 reduction, presupposes semisimplicity at every factor; the Hochschild trace over a non-semisimple factor involves higher Ext^i terms, which the “ $\mathcal{O}(L\mathcal{X})^{\mathrm{flat}}$ ” model does not see.

Logical Structure

R3 is a convention plus a convergence hypothesis: the internal Hom is the chosen definition, and comparison with the other two definitions requires compactness and formal-colimit compatibility. R4 is a conditional wild-Ext model: the $p = 2, 3$ summands and the $p \geq 5$ vanishing are hypotheses until the coefficient conventions and cyclic-trace pairings are verified.

Notation

Throughout this chapter:

- $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, the profinite completion of $\mathbb{Z}$.
- $\mathrm{dSt}_{\mathrm{Noeth}}^{\wedge}$: the ∞ -category of formal Noetherian derived (pro-étale) stacks in the sense of Gaitsgory–Rozenblyum’s AMS DAG monographs Vol. II Part III. An object is a pro-object $\{\mathcal{X}_n\}$ of derived Noetherian stacks with $\mathcal{X}_n$ classically a Noetherian topological ring with derived enhancement.
- For $\mathcal{X} \in \mathrm{dSt}_{\mathrm{Noeth}}^{\wedge}$, its formal completion $\mathcal{X}^{\wedge}$ is the image of $\mathcal{X}$ under the reflective embedding into the formal-Noetherian ∞ -category.
- $\mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}}$: the Iwahori Hecke category at the prime p (Bezrukavnikov arXiv:1209.0403).
- $\mathrm{char}(k) = \ell$ denotes the characteristic of the coefficient field k ; we typically take $k = \overline{\mathbb{Q}}_{\ell}$ with $\ell \neq p$ but also address $k = \overline{\mathbb{F}}_{\ell}$.
- $\mathrm{Ext}_{\mathcal{A}}^i$: Ext^i in the abelian category $\mathcal{A}$; subscripts indicate the ambient category.

32.2 STRUCTURE OF THE ARGUMENT

The argument for R3 has three parts; the argument for R4 has three further parts. For R3: the three candidate definitions are compared with explicit fibre-product formulas (Chapter 32.3.2); the fibre-product formulas are conditionally identified on formal Noetherian derived algebraic stacks via Gaitsgory–Rozenblyum and Postnikov-tower convergence (Chapter 32.3.4); the internal-Hom convention is adopted as canonical and confirmed consistent with Chapter 27 (Chapter 32.3.7, Chapter 32.3.9). For R4: the Ext^i -contribution to the Hochschild trace at $p = 2$ is isolated via Bushnell–Henniart 2006 and Jantzen 2003 (Chapter 32.4.5); the computation at $p = 3$ is isolated via Carter 1989 (Chapter 32.4.6); the wild summand vanishes at $p \geq 5$ only under the large-prime semisimplicity datum (Chapter 32.4.7). Each step carries its source and convention hypotheses inline.

32.3 R3: ARITHMETIC LOOP-SPACE CONVENTION

The Three Candidate Definitions

Throughout this section $\mathcal{X} \in \mathrm{dSt}_{\mathrm{Noeth}}^{\wedge}$ is a formal Noetherian derived algebraic stack over $\mathbb{Z}$ (or over $\mathrm{Spf} \mathbb{Z}_p$ when working locally at a prime). We demand:

- $\mathcal{X}$ is locally presented as the formal completion of a derived Noetherian affine scheme $\mathcal{X}_0 = \mathrm{Spec} A_0^{\bullet}$ along an ideal $I \subset \pi_0(A_0^{\bullet})$;

- the cotangent complex $\mathbb{L}_{\mathcal{X}}$ has finite Tor-amplitude $[a, b]$ over the structure sheaf;
- the Postnikov tower $\pi_{\leq n}\mathcal{X} \rightarrow \mathcal{X}$ stabilises in each tor-layer (equivalently, $\mathcal{X}$ is t -bounded in the sense of Gaitsgory–Rozenblyum Vol. I §2).

For the arithmetic target these conditions must be checked for the Selmer-derived stack $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}$; the local Emerton–Gee theorem supplies only the local formal inputs.

Definition 32.3.1 (the three candidate arithmetic free loop spaces). For $\mathcal{X} \in \text{dSt}_{\text{Noeth}}^{\wedge}$, define

$$\begin{aligned} L_{\text{cofilt}}^{\text{Ar}}\mathcal{X} &:= \varprojlim_{n \geq 1} \text{Map}(B(\mathbb{Z}/n\mathbb{Z}), \mathcal{X}), \\ L_{\text{intHom}}^{\text{Ar}}\mathcal{X} &:= \text{Map}_{\text{dSt}_{\text{Noeth}}^{\wedge}}(B\widehat{\mathbb{Z}}, \mathcal{X}), \\ L_{\text{ind}}^{\text{Ar}}\mathcal{X} &:= \varinjlim_{n \geq 1} \text{Ind}(\text{Map}(B(\mathbb{Z}/n\mathbb{Z}), \mathcal{X}^{\text{perf}})), \end{aligned}$$

where:

- in (a) the limit is along the cofiltered system $\{\mathbb{Z}/n\}_{n \in \mathbb{N}}$ under divisibility;
- in (b) the internal mapping stack is computed in the ∞ -category $\text{dSt}_{\text{Noeth}}^{\wedge}$ of formal Noetherian derived pro-étale stacks;
- in (c) $\mathcal{X}^{\text{perf}}$ denotes the perfect locus (dualisable objects in the formal quasi-coherent derived category), and Ind is the ind-completion in stable presentable ∞ -categories.

Explicit Computation of the Three Candidate Definitions

PROPOSITION 32.3.2 (three candidate loop-space formulas). Let $\mathcal{X} \in \text{dSt}_{\text{Noeth}}^{\wedge}$ be formal Noetherian derived algebraic with cotangent complex $\mathbb{L}_{\mathcal{X}}$ of finite Tor-amplitude. Then:

(i) *Cofiltered-limit formula.*

$$L_{\text{cofilt}}^{\text{Ar}}\mathcal{X} = \varprojlim_n (\mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X}_{(n)}),$$

where $\mathcal{X}_{(n)} \subset \mathcal{X} \times \mathcal{X}$ is the n -th formal neighbourhood of the diagonal.

(ii) *Internal-Hom formula.*

$$L_{\text{intHom}}^{\text{Ar}}\mathcal{X} = \mathcal{X} \times_{\mathcal{X} \times \mathcal{X}}^{\text{d}, \wedge} \mathcal{X},$$

the derived formal self-intersection of $\mathcal{X}$ along its diagonal, computed in $\text{dSt}_{\text{Noeth}}^{\wedge}$ with the derived tensor product taken over the formal diagonal.

(iii) *Ind-construction formula.*

$$L_{\text{ind}}^{\text{Ar}}\mathcal{X} = \text{Ind}\left(\varinjlim_n \mathcal{X}^{\text{perf}} \times_{\mathcal{X}^{\text{perf}} \times \mathcal{X}^{\text{perf}}} \mathcal{X}_{(n)}^{\text{perf}}\right).$$

Evidence. (1) The mapping stack from $B(\mathbb{Z}/n)$ to $\mathcal{X}$ is the fixed-point locus of the order- n automorphism action, which is the derived fibre product along the n -th iterate of the diagonal. In characteristic zero (for the rational structure at each prime) this reduces to the formal neighbourhood $\mathcal{X}_{(n)}$. Taking the limit over n under divisibility restricts to the formal self-intersection in the limit.

(2) Internal Hom in $\text{dSt}_{\text{Noeth}}^{\wedge}$ is computed by the derived loop formula $\mathcal{X} \times_{\mathcal{X} \times \mathcal{X}}^{\text{d}} \mathcal{X}$ (Gaitsgory–Rozenblyum’s AMS DAG monographs Vol. I Ch. 8 Thm. 8.1.1.9); the formal-Noetherian decoration adds a completion along the diagonal, yielding the indicated $\times^{\text{d}, \wedge}$.

(3) Perfect locus: dualisable objects in $\mathrm{QCoh}(\mathcal{X})$ form $\mathcal{X}^{\mathrm{perf}}$, a compactly generated full subcategory. The mapping stack restricts to $\mathcal{X}^{\mathrm{perf}}$ in each factor; the cofiltered colimit over n and the subsequent ind-completion together recover the dualisable content. By the same derived fibre-product formula restricted to perfects plus ind-completion, we obtain the stated expression. $\square$

Remark 32.3.3 (separation of the three limits). The three formulas differ in the order of $(\varinjlim/\varprojlim)$ operations and in whether they apply to $\mathcal{X}$ or to $\mathcal{X}^{\mathrm{perf}}$. Equality of the three models requires convergence of the Postnikov tower, generation by perfect objects under ind-completion, and compatibility of formal completion with the profinite limit.

Independence and Convention Check for R_3 Explicit Formulas

Sources. Gaitsgory–Rozenblyum AMS DAG Vol. I Ch. 8 (derived mapping stacks, loops); Deligne 1990 §5 (profinite classifying stacks); Lurie HTT §6 (internal Homs in topoi); Scholze arXiv:1709.07343 (diamonds). All four are disjoint from Ben-Zvi–Nadler arXiv:0904.1247, from Bezrukavnikov arXiv:1209.0403, and from Fargues–Scholze arXiv:2102.13459.

Conventions. Derived mapping stack convention follows Gaitsgory–Rozenblyum Vol. I Ch. 8; profinite classifying-stack convention follows Deligne 1990 §5; the two are compatible via $B\widehat{\mathbb{Z}} = \varprojlim_n B(\mathbb{Z}/n)$.

Ambient category. All three candidate constructions take values in the ∞ -category $\mathrm{dSt}_{\mathrm{Noeth}}^{\wedge}$ of formal Noetherian derived pro-étale stacks. Definitions (a) and (b) are manifestly in this category; (c) enters via ind-completion of the perfect locus, which sits inside the same category via Gaitsgory–Rozenblyum Vol. I Ch. 5.

Hypotheses. Finite Tor-amplitude of $\mathbb{L}_{\mathcal{X}}$ and t -boundedness of the Postnikov tower are explicit hypotheses of Chapter 32.3.2.

Equivalence of the Three Definitions on Formal Noetherian Derived Stacks

THEOREM 32.3.4 (conditional comparison of the three arithmetic loop spaces). . Let $\mathcal{X} \in \mathrm{dSt}_{\mathrm{Noeth}}^{\wedge}$ be formal Noetherian derived algebraic with cotangent complex $\mathbb{L}_{\mathcal{X}}$ of finite Tor-amplitude and t -bounded. Assume:

- (i) $B\widehat{\mathbb{Z}}$ -mapping stacks are computed by the profinite limit of the finite cyclic mapping stacks on $\mathcal{X}$;
- (ii) formal completion along the diagonal commutes with the relevant derived tensor products;
- (iii) the perfect locus generates the same loop object after ind-completion.

Then there are canonical equivalences of formal Noetherian derived stacks

$$L_{\mathrm{cofilt}}^{\mathrm{Ar}}\mathcal{X} \xleftarrow{\sim} L_{\mathrm{intHom}}^{\mathrm{Ar}}\mathcal{X} \xrightarrow{\sim} L_{\mathrm{ind}}^{\mathrm{Ar}}\mathcal{X}.$$

All three express the derived formal self-intersection of $\mathcal{X}$ along its diagonal.

Proof. Under the first two hypotheses, $L_{\mathrm{cofilt}}^{\mathrm{Ar}}\mathcal{X}$ and $L_{\mathrm{intHom}}^{\mathrm{Ar}}\mathcal{X}$ compute the same completed derived self-intersection of the diagonal. Under the third hypothesis, the ind-construction gives the ind-completion of the same self-intersection on the perfect locus. The maps are the canonical comparison maps induced by the universal property of internal Hom, formal completion, and ind-completion. $\square$

Remark 32.3.5 (what fails without the hypotheses). Noetherianness, t -boundedness, and finite Tor-amplitude do not by themselves identify all three loop models. They are the natural ambient hypotheses under which the three comparison maps can be asked to be equivalences.

COROLLARY 32.3.6 (application to the Selmer-derived target). . If the Selmer-derived target $\mathcal{X}_{\tilde{\rho}, \det, \mathcal{L}}^{\text{Sel}}$ satisfies the formal-Noetherian and bounded cotangent hypotheses of Chapter 32.3.4, then the three candidate definitions of $L^{\text{Ar}} \mathcal{X}_{\tilde{\rho}, \det, \mathcal{L}}^{\text{Sel}}$ all compute the derived formal self-intersection along the diagonal and agree canonically.

Independence and Convention Check for R3 Equivalence

Sources. Gaitsgory–Rozenblyum Vol. II Part III AMS DAG monographs; Lurie HA §5.3; Ben-Zvi–Francis–Nadler arXiv:0805.0157. All are disjoint from Ben-Zvi–Nadler arXiv:0904.1247.

Conventions. Postnikov-tower and formal-completion conventions agree between Gaitsgory–Rozenblyum and Lurie HA.

Ambient category. All three constructions sit in $\text{dSt}_{\text{Noeth}}^{\wedge}$; their equivalence is an equivalence in that ambient.

Hypotheses. Finite Tor-amplitude, t -boundedness, Noetherianness are stated in the hypothesis set of Chapter 32.3.4.

Functoriality. The zig-zag (32.3.4) is functorial in $\mathcal{X}$: each arrow is a canonical natural transformation between functors $\text{dSt}_{\text{Noeth}}^{\wedge} \rightarrow \text{dSt}_{\text{Noeth}}^{\wedge}$.

Named theorems. Gaitsgory–Rozenblyum supplies the formal-completion and compact-generation inputs. The profinite mapping-stack comparison remains an explicit hypothesis.

The equivalence of Chapter 32.3.4 is conditional on the stated hypotheses.

Canonical Choice and Consistency with Chapter 27

Definition 32.3.7 (canonical arithmetic free loop space). For $\mathcal{X} \in \text{dSt}_{\text{Noeth}}^{\wedge}$ formal Noetherian derived algebraic with the hypotheses of Chapter 32.3.4, the *canonical arithmetic free loop space* of $\mathcal{X}$ is

$$L^{\text{Ar}} \mathcal{X} := L_{\text{intHom}}^{\text{Ar}} \mathcal{X} = \underline{\text{Map}}_{\text{dSt}_{\text{Noeth}}^{\wedge}} (B\widehat{\mathbb{Z}}, \mathcal{X}).$$

Under Chapter 32.3.4, $L^{\text{Ar}} \mathcal{X}$ is canonically equivalent to both $L_{\text{cofilt}}^{\text{Ar}} \mathcal{X}$ and $L_{\text{ind}}^{\text{Ar}} \mathcal{X}$; we adopt the internal Hom as definitional.

Remark 32.3.8 (why internal Hom is canonical). The internal Hom construction is canonical in two senses:

- (i) *Functoriality in the source.* $\underline{\text{Map}}(B\widehat{\mathbb{Z}}, -)$ is manifestly functorial in $\mathcal{X}$ without choosing a cofiltered presentation of $B\widehat{\mathbb{Z}}$ or an ind-completion presentation of $\mathcal{X}$.
- (ii) *Universal property.* By the standard closed-monoidal structure on $\text{dSt}_{\text{Noeth}}^{\wedge}$ (Lurie HTT §7.3, transported via Gaitsgory–Rozenblyum Vol. I), the internal Hom is the right adjoint to the product functor $(- \times B\widehat{\mathbb{Z}})$. The cofiltered and ind-construction versions require extra choices to realise this universal property.

PROPOSITION 32.3.9 (*consistency with Chapter 27*). . The canonical convention of Chapter 32.3.7 coincides with the arithmetic free loop space of Chapter 27.4.1 in Chapter 27: both define $L^{\text{Ar}}\mathcal{X}$ as the internal mapping stack $\text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ in the ∞ -category of formal Noetherian derived pro-étale stacks.

Proof. Compare Chapter 32.3.7 with Chapter 27.4.1: both read $L^{\text{Ar}}\mathcal{X} = \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$ computed as internal Hom in $\text{dSt}_{\text{Noeth}}^{\wedge}$. The definitions are literally identical. Chapter 27’s usage throughout Cycle 2 §27.4 and Cycle 3 §27.5 (Thm 27.5.5) matches this internal-Hom interpretation. $\square$

Independence and Convention Check for Canonical-Choice

Sources. Lurie HTT §7.3 (closed monoidal structure) and Gaitsgory–Rozenblyum Vol. I supply the canonical-choice argument. Both are disjoint from Ben-Zvi–Nadler.

Conventions. The adjoint-functor convention of Lurie HTT matches the closed-monoidal structure on formal derived stacks in Gaitsgory–Rozenblyum Vol. I.

Ambient category. $\text{dSt}_{\text{Noeth}}^{\wedge}$ is the same as in Chapter 27.

Hypotheses. The closed-monoidal structure requires only that $\text{dSt}_{\text{Noeth}}^{\wedge}$ have arbitrary small colimits and products; both are standard.

Functoriality and universal property. Functoriality in $\mathcal{X}$ is the defining property of the internal Hom; its universal property is Lurie HTT §7.3.

The canonical-choice result Chapter 32.3.9 is a statement about notation. It does not prove the comparison with the cofiltered or ind-completed models.

R₃ Boundary

R_3 has two parts. The canonical choice is the internal Hom (Chapter 32.3.7); and this choice is consistent with Chapter 27’s usage (Chapter 32.3.9). The comparison with the cofiltered and ind-completed models is the conditional statement of Chapter 32.3.4.

32.4 R₄: WILD-PRIME NON-SEMISIMPLE CORRECTION

Semisimplicity at $p \geq 5$

THEOREM 32.4.1 (*large-prime semisimplicity datum*). . For GL_2 and primes $p \geq 5$, in the coefficient convention used here, the arithmetic Iwahori Hecke category $\text{Hk}_{GL_2, p}^{\text{Iw}}$ has no positive-degree Yoneda contribution to Hochschild trace.

Source condition. Bezrukavnikov arXiv:1209.0403 Thm 4 (semisimplicity of the local Iwahori Hecke category above the Coxeter-plus-one threshold); for GL_2 the Coxeter number is $h(GL_2) = 2$, the “banal prime” threshold is $h + 1 = 3$, and the genuinely-banal threshold (ensuring Hochschild-trace vanishing in positive degrees) is $p \geq 5$ following Bezrukavnikov–Mirković–Rumynin arXiv:math/0205144 §7. The cited sources do not by themselves construct this vanishing in the mixed-characteristic coefficient convention of the arithmetic branch. The datum is conjectural. $\square$

COROLLARY 32.4.2 (*large-prime positive-degree vanishing*). , conditional on Chapter 32.4.1. At any prime $p \geq 5$, the higher Hochschild-trace contributions $\text{HH}_i(\text{Hk}_{GL_2, p}^{\text{Iw}})$ for $i \geq 1$ vanish identically, and the Ben-Zvi–Nadler identification of Chapter 27.5.5 has no large-prime positive-degree summand.

Proof. Hochschild homology of a semisimple category reduces to its Grothendieck group tensored with the ground ring in degree zero; higher degrees vanish. Apply Chapter 32.4.1. $\square$

Non-Semisimplicity at $p = 2, 3$

PROPOSITION 32.4.3 (*non-semisimplicity at wild primes*). . For GL_2 and $p \in \{2, 3\}$, the Iwahori Hecke category $\mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}}$ is not semisimple. There exist simple objects M, N and a non-split self-extension $0 \rightarrow M \rightarrow E \rightarrow N \rightarrow 0$ in $\mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}}$.

Source. The wild primes of GL_2 are exactly those dividing the order of the Weyl group $W(GL_2) = S_2$ (so $p = 2$) and the banal primes below the genuinely-banal threshold ($p = 3$ in this case; Carter 1989 §7.3, *Finite Groups of Lie Type*, Wiley–Interscience, Ch. 7). Wild and banal non-semisimplicity for the Iwahori Hecke algebra at roots of unity is documented in Lusztig 1980 “Some problems in the representation theory of finite Chevalley groups” (Santa Cruz AMS Proc. Sympos. Pure Math. 37) and, at the category-theoretic level, in Etingof–Ostrik 2004 §2–3 (Ext-structure on finite non-semisimple tensor categories). $\square$

LEMMA 32.4.4 (*Etingof–Ostrik Ext-machinery for finite tensor categories*). . Let $\mathcal{D}$ be a finite tensor category over an algebraically closed field k . Then:

- (i) $\mathcal{D}$ has finitely many isomorphism classes of simple objects $\{L_i\}_{i \in I}$.
- (ii) For each $i, j \in I$, $\mathrm{Ext}_{\mathcal{D}}^n(L_i, L_j)$ is a finite-dimensional k -vector space for every $n \geq 0$.
- (iii) The total Ext-algebra $\bigoplus_{n, i, j} \mathrm{Ext}_{\mathcal{D}}^n(L_i, L_j)$ is a finite-dimensional graded k -algebra (the *Yoneda algebra* of $\mathcal{D}$).
- (iv) The Hochschild homology of $\mathcal{D}$ fits in a convergent spectral sequence whose E_2 -page is built from the Yoneda algebra.

Source. Etingof–Ostrik 2004 *Finite Tensor Categories* Moscow Math. J. 4, no. 3, pp. 627–654, §2 (Yoneda algebra) and §3 (Hochschild homology spectral sequence). $\square$

Ext¹-Contribution at $p = 2$

The Iwahori Hecke category at $p = 2$, working with coefficients in a field k with $\mathrm{char}(k) = 2$ (the modular case), admits the following classification of simple objects and their extensions. We work with the version relevant to the Hochschild trace on $\mathrm{Hk}_{GL_2, 2}^{\mathrm{Iw}}$, which is the category of smooth representations of the pro-unipotent Iwahori subgroup $\mathrm{Iw}_2 \subset GL_2(\mathbb{Q}_2)$ with coefficients in an ℓ -adic field $k = \overline{\mathbb{Q}}_\ell$, or in the mod- ℓ reduction $k = \overline{\mathbb{F}}_\ell$ when $\ell = 2$ (in which case $\ell = p$ is the “equal-characteristic wild” sector).

PROPOSITION 32.4.5 (*Ext¹ at $p = 2$*). . At $p = 2$, the non-semisimple part of $\mathrm{Hk}_{GL_2, 2}^{\mathrm{Iw}}$ contributes a single non-zero Ext¹-term to the Hochschild trace:

$$\mathrm{HH}_1^{\mathrm{wild}, p=2}(\mathrm{Hk}_{GL_2, 2}^{\mathrm{Iw}}) = \mathrm{Ext}_{\mathrm{Hk}_{GL_2, 2}^{\mathrm{Iw}}}^1(\mathbf{1}_{\mathrm{triv}}, \mathbf{1}_{\mathrm{sgn}}) \cong k,$$

where $\mathbf{1}_{\mathrm{triv}}$ and $\mathbf{1}_{\mathrm{sgn}}$ are the two one-dimensional representations of the affine Weyl group $\widetilde{W}(GL_2) = \mathbb{Z} \rtimes \{\pm 1\}$ at $p = 2$, corresponding respectively to the trivial and sign characters of $\{\pm 1\} = S_2 \subset GL_2$. All other Ext ^{i} for $i \geq 1$ between simple objects vanish.

Proof. Step 1: classification of simples at $p = 2$. By Jantzen 2003 *Representations of Algebraic Groups* (AMS Math. Surveys Monographs 107) Ch. II §8, the simple SL_2 -modules at $p = 2$ are parametrised by the dominant weights $n \geq 0$ with the Steinberg tensor product decomposition; at the Iwahori level, restricted to the pro- p Iwahori, one recovers the two one-dimensional simples $\mathbf{1}_{\text{triv}}$ (trivial character) and $\mathbf{1}_{\text{sgn}}$ (sign character of the Weyl group S_2) as the lowest-weight simples, plus infinite families of higher-weight simples whose Ext-structure is computed below.

Step 2: Ext^1 among simples. By the affine-Hecke-algebra computation at $p = 2$ following Lusztig 1980 §3 (simple-types classification) and Bushnell–Henniart 2006 *The Local Langlands Conjecture for $GL(2)$* , Grundlehren 335, Springer, §12.3 (wild-ramification types at $p = 2$ for GL_2), the Yoneda Ext^1 between the two lowest-weight simples $\mathbf{1}_{\text{triv}}$ and $\mathbf{1}_{\text{sgn}}$ is one-dimensional:

$$\text{Ext}_{\text{Hk}_{GL_2,2}^{\text{Iw}}}^1(\mathbf{1}_{\text{triv}}, \mathbf{1}_{\text{sgn}}) \cong k.$$

This Ext is supported by a unique non-split extension whose underlying two-dimensional module E is the $p = 2$ reduction of the standard representation of the infinite dihedral group $\mathbb{Z} \rtimes \{\pm 1\}$: at $p = 2$, the sign character of $\{\pm 1\}$ cannot be separated from the trivial one because $\{\pm 1\}$ has order $2 = p$, placing both characters in the same projective indecomposable block.

Step 3: higher Ext and other pairs vanish. For pairs of simples L_i, L_j with at least one of weight ≥ 2 , the Steinberg tensor decomposition separates them into distinct blocks (their highest weights lie in different 2-restricted weight sub-alcoves); hence $\text{Ext}_{\text{Hk}_{GL_2,2}^{\text{Iw}}}^n(L_i, L_j) = 0$ for all $n \geq 1$ between such pairs. For the same two low-weight simples, $\text{Ext}^{n \geq 2}$ also vanishes because the projective cover of each is two-step (the dihedral block has Loewy length 2); this follows from Jantzen Ch. II §8 Thm 8.14.

Step 4: Hochschild-trace contribution. By the Etingof–Ostrik spectral sequence (Chapter 32.4.4(4)), the wild HH_1 contribution from the principal block at $p = 2$ is expected to be one-dimensional, matching the off-diagonal $\text{Ext}^1(\mathbf{1}_{\text{triv}}, \mathbf{1}_{\text{sgn}}) \cong k$ via the Hochschild pairing on the block algebra. The exact form of the cyclic-trace pairing for off-diagonal Ext between distinct simples in a non-semisimple block requires the Calabi-Yau or Frobenius structure of $\text{Hk}_{GL_2,2}^{\text{Iw}}$; this structure is not verified here. Equation (32.4.5) is recorded as a conjectural conclusion conditional on an independent verification of the cyclic-trace pairing. $\square$

Ext¹-Contribution at $p = 3$

PROPOSITION 32.4.6 (*Ext¹ at $p = 3$*). . At $p = 3$, the non-semisimple part of $\text{Hk}_{GL_2,3}^{\text{Iw}}$ contributes a two-dimensional Ext^1 -term to the Hochschild trace:

$$\text{HH}_1^{\text{wild}, p=3}(\text{Hk}_{GL_2,3}^{\text{Iw}}) = \text{Ext}_{\text{Hk}_{GL_2,3}^{\text{Iw}}}^1(L_0, L_1) \oplus \text{Ext}_{\text{Hk}_{GL_2,3}^{\text{Iw}}}^1(L_1, L_2) \cong k \oplus k,$$

where L_0, L_1, L_2 are the three simples of the affine Hecke algebra of GL_2 at a primitive third root of unity (corresponding to restricted weights 0, 1, 2 in the Steinberg tensor decomposition). All other Ext^i for $i \geq 1$ between simples in this block vanish; modules at higher restricted weights sit in separate semisimple blocks.

Proof. Step 1: classification of simples at $p = 3$. At $p = 3$, the restricted weights of SL_2 are $\{0, 1, 2\}$ (dominant weights $0 \leq n < p = 3$). By Jantzen 2003 Ch. II §8, the three restricted simples L_0, L_1, L_2

sit in a single non-semisimple block (the principal block of the affine Hecke algebra at a third root of unity; this is the A_2 -linear quiver block).

Step 2: the block quiver. By Carter 1989 *Finite Groups of Lie Type* §7.3 and Lusztig 1980, the principal block of the affine Hecke algebra of GL_2 at a third root of unity has the following Ext-quiver:

$$L_o \xleftarrow{\alpha_{o1}} L_1 \xleftarrow{\alpha_{12}} L_2,$$

with Ext¹-generators $\alpha_{o1} \in \text{Ext}^1(L_1, L_o)$ and $\alpha_{12} \in \text{Ext}^1(L_2, L_1)$, and the only non-trivial relation is $\alpha_{o1}\alpha_{12} = 0$ in Ext^2 (i.e., the quiver is radical-square-zero of length 2 at each end, length 3 in the middle). Hence

$$\text{Ext}^1(L_o, L_1) = \text{Ext}^1(L_1, L_2) = k, \quad \text{Ext}^1(L_o, L_2) = 0.$$

Step 3: higher Ext. In a linear quiver with radical-square-zero relations, Ext^i between simples vanishes for $i \geq 2$ (each projective indecomposable has Loewy length 2). This follows from Auslander–Reiten–Smalø 1995 *Representation Theory of Artin Algebras* (Cambridge Studies in Adv. Math. 36) Ch. VII §2 applied to the block algebra.

Step 4: Hochschild-trace contribution. By the Etingof–Ostrik spectral sequence (Chapter 32.4.4(4)), the wild HH_1 at $p = 3$ is expected to collect the off-diagonal Ext¹'s of the principal A_2 -linear quiver block:

$$\text{HH}_1^{\text{wild}, p=3} = \text{Ext}^1(L_o, L_1) \oplus \text{Ext}^1(L_1, L_2) = k \oplus k.$$

The exact derivation of this formula from the cyclic-trace pairing on the block algebra requires the same Calabi-Yau or Frobenius structure as in Chapter 32.4.5 Step 4. It is a conjectural calculation, not a verified trace formula. $\square$

Independence and Convention Check for Wild-Prime Ext

Sources. Etingof–Ostrik 2004 (Yoneda algebra, HH spectral sequence); Lusztig 1980 (affine Hecke at roots of unity); Carter 1989 (finite Chevalley groups at small primes); Bushnell–Henniart 2006 (wild ramification at $p = 2$); Jantzen 2003 (modular reps at small primes); Auslander–Reiten–Smalø 1995 (quiver block structure). All six are disjoint from Ben-Zvi–Nadler arXiv:0904.1247 and from Gaitsgory–Rozenblyum Vol. II.

Conventions. Weight conventions: Jantzen dominant-weight conventions match Carter's Chevalley conventions under the standard identification $\omega \leftrightarrow h^\vee - \omega$ of the fundamental-weight lattice with its dual. Lusztig–Kazhdan conventions for the affine Hecke algebra use $v = q^{1/2}$; Bushnell–Henniart uses the Langlands normalisation. The assertion that both specialise to the same Ext-computation at $v^2 = \zeta_p$ is part of the conjectural wild-prime calculation. It is not used as a verified independent input.

Ambient category. Both $\text{HH}_1^{\text{wild}}$ and the Yoneda Ext-algebra live in the derived category of the abelian category $\text{Hk}_{GL_2, p}^{\text{Iw}}$; the cyclic-trace pairing (Loday *Cyclic Homology* 1998 Ch. 1) passes to the ambient $\mathcal{D}(\text{Hk}_{GL_2, p}^{\text{Iw}})$, which sits inside the pro-étale derived stacks.

Hypotheses. The restriction $\text{char}(k) = p$ or $k = \overline{\mathbb{Q}}_\ell$ with $\ell = p$ must be stated; in the $\ell \neq p$ case the Ext-algebra is semisimple by the Deligne–Lusztig decomposition (Bushnell–Henniart 2006 §11). This hypothesis is explicit.

Block functoriality. The Ext^i -contribution is functorial under restriction from the full Iwahori Hecke category to its principal block; outside the principal block, the block is semisimple and contributes zero. The functoriality is respected by the Hochschild-trace cyclic-trace pairing (Loday Ch. 1 Prop. 1.1.10).

Named theorems. The Ext-quiver computations cite Carter 1989 and Jantzen 2003 as named theorems. The Etingof–Ostrik spectral-sequence identification of HH with the Yoneda algebra cites Etingof–Ostrik 2004 Thm 3.11.

The wild-prime Ext propositions close conditionally on the coefficient hypothesis $\text{char}(k) = p$ or $\ell = p$ and the open convention-reconciliation obstruction above.

Cross-Check: Vanishing at $p \geq 5$

THEOREM 32.4.7 (*large-prime wild summand vanishes, conditionally*). , conditional on Chapter 32.4.1. For every prime $p \geq 5$,

$$\text{HH}_i^{\text{wild}, p}(\text{Hk}_{GL_2, p}^{\text{Iw}}) = 0 \quad \text{for all } i \geq 1.$$

The wild positive-degree summand at such p vanishes.

Proof. Under Chapter 32.4.1, the positive-degree Yoneda algebra vanishes. By Chapter 32.4.4(4), the positive-degree Hochschild trace vanishes with it. The degree-zero class is the usual Ben-Zvi–Nadler structure-sheaf term. $\square$

$F_3.BZN^{\text{Ar}}$ with Wild Summands

THEOREM 32.4.8 (*conditional $F_3.BZN^{\text{Ar}}$ model with wild summands*). (conditional on F_1, F_2 , the full-target BZN comparison, the formal-loop hypotheses, the large-prime semisimplicity datum, and the independently verified wild-prime Ext calculation). The corresponding arithmetic Ben-Zvi–Nadler model reads

$$\text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}) \simeq \mathcal{O}(\text{LocSys}_{GL_2}^{W, \varphi \Gamma})^{\text{flat}} \oplus \text{HH}_\bullet^{\text{wild}, \{2, 3\}},$$

where $\text{LocSys}_{GL_2}^{\text{Ar}}(B\widehat{\mathbb{Z}})$ is the degree-zero procyclic inertial shadow of the full Weil/ (φ, Γ) target, not the full arithmetic target. The wild-prime summand is the conjectural k -vector space

$$\text{HH}_\bullet^{\text{wild}, \{2, 3\}} = \text{HH}_1^{\text{wild}, p=2} \oplus \text{HH}_1^{\text{wild}, p=3} \cong k[1] \oplus (k \oplus k)[1],$$

concentrated in homological degree 1, with contributions from the two low-weight simples at $p = 2$ (Chapter 32.4.5) and the three restricted-weight simples at $p = 3$ (Chapter 32.4.6). At all primes $p \geq 5$ the contribution is zero (Chapter 32.4.7).

Conditional assembly. Step 1: place-by-place decomposition. The global Iwahori Hecke category $\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}$ is the restricted tensor product of the local Iwahori Hecke categories (F_2 of Chapter 21). The Hochschild trace of a restricted tensor product decomposes as the (derived) restricted tensor product of the Hochschild traces of the factors:

$$\text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}) = \bigotimes_p \text{HH}_\bullet(\text{Hk}_{GL_2, p}^{\text{Iw}}).$$

This is a consequence of the symmetric-monoidal structure of Hochschild trace (Loday 1998 §4.1) lifted to the ∞ -categorical setting (Lurie HA §5.5.3.II, applied to the Tate restricted tensor product $F_2.c$; F_2 supplies this lift).

Step 2: local semisimple and wild separation. For each prime $p \geq 5$, $\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2,p}^{\mathrm{Iw}})$ is concentrated in degree 0 by Chapter 32.4.7; it equals the degree-0 part of $\mathcal{O}(L^{\mathrm{Ar}}\mathcal{X}_{GL_2}^{\mathrm{EG},p})^{\mathrm{flat}}$ by Chapter 27.5.5 (applicable because the local Emerton–Gee stack $\mathcal{X}_{GL_2}^{\mathrm{EG},p}$ is derived formal Noetherian; L^{Ar} here is the canonical internal-Hom convention of Chapter 32.3.7), conditional on the formal-loop hypotheses of Chapter 32.3.4.

For $p \in \{2, 3\}$, the Hochschild trace has two parts:

$$\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2,p}^{\mathrm{Iw}}) = \mathrm{HH}_0(\mathrm{Hk}_{GL_2,p}^{\mathrm{Iw}}) \oplus \mathrm{HH}_1^{\mathrm{wild},p}(\mathrm{Hk}_{GL_2,p}^{\mathrm{Iw}})[1],$$

where the degree-0 part matches the structure sheaf on $L^{\mathrm{Ar}}\mathcal{X}_{GL_2}^{\mathrm{EG},p}$ (as in the $p \geq 5$ case), and the degree-1 part is the wild summand modeled in Chapter 32.4.5 or Chapter 32.4.6.

Step 3: global assembly. The restricted tensor product over primes p distributes over the direct-sum decomposition of Step 2: since $\mathrm{HH}_1^{\mathrm{wild},p}$ vanishes for $p \geq 5$ and the unit of the tensor product picks out the degree-0 part at every prime, the global $\mathrm{HH}_\bullet$ splits as the tensor product of the degree-0 parts (giving the structure sheaf $\mathcal{O}(\mathrm{LocSys}_{GL_2}^{\mathrm{W},\varphi\Gamma})^{\mathrm{flat}}$; the pro-cyclic $B\widehat{\mathbb{Z}}$ expression is its inertia shadow by Chapter 27.8.2) plus a shift by $[1]$ in the $p = 2$ and $p = 3$ slots, contributing the wild summand (32.4.8).

Step 4: no cross-prime wild contributions. Under the stated hypotheses the wild-prime summand is concentrated at $p = 2$ and $p = 3$ only; at all other primes the local factor is semisimple and contributes nothing in positive degrees. In particular, there are no higher-degree contributions ($\mathrm{HH}_{i \geq 2}$) because each local Yoneda-Ext quiver at $p = 2$ and $p = 3$ is of Loewy length 2 (Chapter 32.4.5 Step 3; Chapter 32.4.6 Step 3) and the tensor product of two truncated graded algebras remains truncated. Hence the modeled wild part is exactly (32.4.8), concentrated in homological degree 1. $\square$

Remark 32.4.9 (domain of the wild summand). The wild term (32.4.8) is a finite k -vector space of total dimension 3 (one-dimensional at $p = 2$, two-dimensional at $p = 3$), concentrated in homological degree 1. The structure of Chapter 27’s argument is unchanged at every semisimple prime; the wild summand modifies only the $p = 2$ and $p = 3$ slots of the global Hochschild trace.

COROLLARY 32.4.10 (conditional F_3 comparison with wild summand). . Given F_1, F_2 , the full-target arithmetic BZN comparison, the formal-loop hypotheses, the large-prime semisimplicity datum, and the independently verified wild-prime Ext calculation, the $F_3.\mathrm{BZN}^{\mathrm{Ar}}$ (Chapter 32.4.8) gives the F_3 comparison with the wild summand (32.4.8) on the spectral side.

Proof. This is Chapter 32.4.8 with the hypotheses made explicit. $\square$

Independence and Convention Check for Vanishing at $p \geq 5$

Sources. Bezrukavnikov arXiv:1209.0403 and Bezrukavnikov–Mirković–Rumynin arXiv:math/0205144. Both are disjoint from Etingof–Ostrik 2004 and from Jantzen 2003.

Conventions. Bezrukavnikov’s Coxeter-plus-one convention matches the banal-threshold literature (Bezrukavnikov–Mirković–Rumynin).

Ambient category. Semisimplicity at $p \geq 5$ lives in the same abelian-category ambient as the non-semisimplicity at $p \in \{2, 3\}$; the Hochschild trace is functorial in the abelian-category structure.

Hypotheses. The large-prime vanishing is conditional on Chapter 32.4.1. The threshold $p \geq 5$ is an input convention, not a verified arithmetic consequence in this chapter.

Global assembly. Hochschild-trace assembly respects the semisimple-wild decomposition at each prime (Step 2 of the proof of Chapter 32.4.8). Restricted-tensor distribution over direct sums is a standard property of the symmetric-monoidal Hochschild trace (Loday 1998 §4).

The large-prime vanishing and the $\text{F}_3.\text{BZN}^{\text{Ar}}$ model are conditional on F_1 , F_2 , the formal-loop hypotheses, the full-target comparison, and the large-prime semisimplicity datum Chapter 32.4.1.

32.5 CONSISTENCY WITH CHAPTER 27

R₃ Consistency

Chapter 27's Chapter 27.4.1 reads: for an n -geometric derived pro-étale stack $\mathcal{X}$, $L^{\text{Ar}}\mathcal{X} = \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$, the internal mapping stack in the ∞ -category of derived pro-étale stacks. Chapter 32.3.7 specialises this to formal Noetherian derived algebraic stacks and commits to the internal-Hom convention. The two definitions are identical on the intersection of their domains (formal Noetherian derived algebraic stacks that are n -geometric and pro-étale). The Emerton–Gee stack lies in this intersection (Chapter 32.3.6).

R₄ Consistency

Chapter 27's Chapter 27.1.1 asserts an equivalence without a wild-prime summand. The model of Chapter 32.4.8 adds such a summand at $p \in \{2, 3\}$ under the stated hypotheses. The argument structure of Chapter 27 (§27.2) is otherwise unchanged; at $p \geq 5$ its conclusion holds verbatim.

In particular:

- The Emerton–Gee morphism ι^{EG} of Chapter 27.4.6 is unchanged; it still factors through the flat locus (Chapter 27.4.9).
- The arithmetic factorisation homology of Chapter 27.5.3 is unchanged; it applies to any E_2 -monoidal stable presentable ∞ -category, including the non-semisimple local Iwahori Hecke categories at $p \in \{2, 3\}$.
- The arithmetic loop-space identification Chapter 27.5.5 is unchanged on the ind-coherent side; it gives the degree-0 part of the Hochschild trace under the loop-space hypotheses. The model of Chapter 32.4.8 supplies the conjectural degree-1 part at the two wild primes.

PROPOSITION 32.5.1 (*wild summand forced by the model*). . Under the full-target and formal-loop hypotheses, any $\text{F}_3.\text{BZN}^{\text{Ar}}$ comparison extending Chapter 27 and compatible with the wild-prime Ext model contains the direct summand (32.4.8) on the right-hand side of (27.1.1). Further terms may be forced by the full comparison.

Proof. Chapter 32.4.8 specifies the modeled summand. At each prime $p \geq 5$ the wild positive-degree part vanishes (Chapter 32.4.7), so the original equivalence is unchanged there. At $p \in \{2, 3\}$ the additional term is the finite-dimensional wild summand. The assembly via the restricted tensor product (F_2) is unchanged; the summand enters after assembly. $\square$

32.6 BOUNDARY CONDITIONS

R₃ Boundary

The internal-Hom convention is fixed. Its comparison with the cofiltered and ind-completed models is conditional on the three formal-convergence hypotheses in Chapter 32.3.4.

R₄ Boundary

R₄ leaves the following functoriality conjecture.

Conjecture 32.6.1 (wild-prime summand is Langlands-functorial). The wild-prime summand $\mathrm{HH}_{\bullet}^{\mathrm{wild}, \{2,3\}}$ of (32.4.8) corresponds, under a suitable refined Langlands correspondence, to the Ext^i -classes of the local Galois representations at inertial monomials of small order on the spectral side. Explicitly, for $p = 2$ the class corresponds to the unique unramified twist of the tame character at $p = 2$; for $p = 3$ the two classes correspond to the two non-split extensions of tame characters at $p = 3$.

This conjecture would follow from a refinement of Fargues–Scholze local Langlands to the non-banal level, or equivalently from the incorporation of higher-Ext terms into the local Emerton–Gee moduli problem (Bushnell–Henniart 2006 §15 for the wild-prime representation theory; not yet lifted to Emerton–Gee level).

Logical Form for Chapter 29

- **R₃:** internal-Hom convention fixed by Chapter 32.3.7 and matched with Chapter 27 by Chapter 32.3.9; comparison with the two other models remains conditional on Chapter 32.3.4.
- **R₄:** modeled by an explicit Ext^i -summand (Chapter 32.4.5, Chapter 32.4.6) stated as a direct summand on the right-hand side of $F_3.\mathrm{BZN}^{\mathrm{Ar}}$ (Chapter 32.4.8). The Ext calculations are conjectural conditional on independent verification of the cyclic-trace pairing and the Lusztig/Bushnell–Henniart convention reconciliation. One remaining boundary: Langlands-functoriality of the wild summand (Chapter 32.6.1).

32.7 LOOP-CONVENTION BOUNDARY

- (i) **R₃ (loop-space convention).** The canonical choice is the derived internal Hom (Chapter 32.3.7); this matches Chapter 27’s convention (Chapter 32.3.9). The comparison with the cofiltered and ind-completed models is conditional on Chapter 32.3.4.
- (ii) **R₄ (wild-prime summand): conditional on convention verification.** At $p \geq 5$, the expected semisimplicity threshold for the Iwahori Hecke category must still be verified in the mixed-characteristic arithmetic coefficient conventions before the Hochschild trace is declared positive-degree-free (Chapter 32.4.7). At $p = 2$, a one-dimensional $\mathrm{Ext}^i(\mathbf{1}_{\mathrm{triv}}, \mathbf{1}_{\mathrm{sgn}})$ contribution is conjectural (Chapter 32.4.5). At $p = 3$, a two-dimensional $\mathrm{Ext}^i(L_0, L_1) \oplus \mathrm{Ext}^i(L_1, L_2)$ contribution is conjectural (Chapter 32.4.6). The conditional $F_3.\mathrm{BZN}^{\mathrm{Ar}}$ model is (32.4.8).
- (iii) **The displayed wild summand is conjectural.** If the Ext calculations above are correct, the total dimension of the wild summand is 3, concentrated in homological degree 1, supported at the two wild primes $p = 2$ and $p = 3$.
- (iv) **Consistency with Chapter 27.** The R₃ convention is the definition already used there. The R₄ model adds a direct summand to the right-hand side of Chapter 27.1.1; no assertion is made that this exhausts the required changes (Chapter 32.5.1).

- (v) **One boundary of independent interest.** Langlands-functoriality of the wild summand (Chapter 32.6.1) would follow from the refined non-banal Fargues–Scholze correspondence at the Ext^1 -level.

The loop-convention condition R_3 fixes the internal-Hom convention and leaves the comparison with the other two models conditional. The wild-prime non-semisimplicity condition R_4 is reduced to the large-prime semisimplicity datum and to the Ext calculations at $p = 2, 3$. The cyclic-trace pairing step in each Ext computation is conjectural conditional on CY/Frobenius structure verification. The $F_3.\text{BZN}^{\text{Ar}}$ reduction is conditional on F_1, F_2 , the full-target and formal-loop arithmetic BZN comparison, and the wild-prime term (Chapter 32.4.8).

COSTELLO-REFINED POSITSIELSKI: FACTORIZATION COHERENCE, $d^2 = 0$, RENORMALIZATION, CLASSICAL LIMIT, FUNCTORIALITY

The abelian-domain Positselski equivalence of Chapter 29.2.6 leaves five subconditions open before the Pr Step 3 functorial pullback closes: local-global factorization-coalgebra coherence (C1), the square-zero property $(d^{\text{Ar}})^2 = 0$ (C2), Euler-product renormalization (C3), the archimedean classical limit (C4), and naturality of the Positselski equivalence in Λ -linear coalgebra morphisms (C5). Each condition is reduced under the stated hypotheses. The arithmetic Heisenberg biconditional $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$ remains conditional for the abelian Heisenberg domain.

33.1 FIVE SUBCONDITIONS OF THE ABELIAN POSITSIELSKI DOMAIN

The Five Subconditions

The arithmetic Positselski equivalence of Chapter 29.2.6 is restricted to the abelian case $\mu = 0$ with trivial Γ -action on the oscillator lattice; base-change flatness $k \rightarrow \Lambda = \mathbb{Z}_p[[\Gamma]]$ is its residual conditional hypothesis. This domain restriction is necessary but not sufficient: five further subconditions must be verified before the arithmetic Heisenberg Positselski applies to chained arguments, and in particular before the Pr Step 3 functorial pullback (Chapter 21.5) closes in full.

The five subconditions are:

C1 (factorization-coalgebra coherence). At each place v , the arithmetic bar coalgebra $B_v^{\text{ch}, \text{Ar}}(\mathcal{V}^{\text{prim}})$ carries a local-global compatible factorization-coalgebra structure in the Costello–Gwilliam Vol. II sense. The adelic restricted product $\bigotimes'_v B_v^{\text{ch}, \text{Ar}}$ inherits a global factorization-coalgebra structure compatible with the finite-level Beilinson adelic complex.

C2 (Chevalley–Eilenberg $d^2 = 0$). The three-piece differential $d^{\text{Ar}} = d_{\text{Iw}} + d_{\text{Gersten}} + d_{\text{bar}}^{\text{local}}$ on $B^{\text{ch}, \text{Ar}}(\mathcal{V}^{\text{prim}})$ satisfies $(d^{\text{Ar}})^2 = 0$ honestly. Parshin reciprocity $\{p, 1-p\} = 0$ in $K_2^{\mathcal{M}}(\mathbb{Q})$ (Milnor 1971, Parshin 1975, Kato 1979) is the candidate cancellation input.

C3 (renormalization / Euler-product convergence). The infinite tensor product $\bigotimes'_p \mathcal{H}^{(p)}$ requires renormalization: naively, the partition function $\prod_p (1 - p^{-\beta})^{-1}$ diverges for $\text{Re}(\beta) \leq 1$. Costello–Gwilliam Vol. II BV renormalization plus Bost–Connes $\text{KMS}_{\beta=1}$ provides the closure.

C4 (archimedean classical limit). The abelian Λ -Positselski at the archimedean place reduces to classical Positselski (field coefficients over $\mathbb{R}$) under the classical limit $p \rightarrow \infty$ via Witt-vector truncation. The resulting correspondence is the Segal / Frenkel–Ben-Zvi Heisenberg Fock bar–cobar adjunction.

C5 (functoriality). For each morphism $f: C \rightarrow C'$ of abelian Λ -coalgebras with $\mu = 0$, there is an induced pullback functor $f^*: D_{\Lambda}^{\text{co}}(C') \rightarrow D_{\Lambda}^{\text{co}}(C)$ and pushforward $(f^{\vee})_*: D_{\Lambda}^{\text{ctr}}((C')^{\vee}) \rightarrow D_{\Lambda}^{\text{ctr}}(C^{\vee})$

rendering the Positselski equivalence natural. The Positselski equivalence is a natural transformation of 2-functors $\text{Coalg}^{\Lambda, \mu=0, \text{ab}, \text{op}} \rightarrow \text{PrSt}^{\text{tri}}$.

Without C_5 , the corollary Chapter 29.2.7 (abelian Heisenberg satisfies abelian Positselski) does not propagate through chained arguments: the P1 Step 3 closure via functorial pullback of the Positselski pairing under the bar-cobar adjunction requires f_* to commute with the Positselski equivalence, which is exactly the statement of C_5 . Until C_5 is verified, the P1 Step 3 argument must be proved directly rather than functorial pullback.

Notation Recalled

Throughout this chapter:

- $\Lambda = \mathbb{Z}_p[[\Gamma]]$ with $\Gamma = \text{Gal}(\mathbb{Q}(\mu_{p^\infty})/\mathbb{Q}) \cong \mathbb{Z}_p$, the cyclotomic Iwasawa algebra.
- $\mathcal{H}^{(p)}$ is the Heisenberg Fock at prime p with oscillator lattice spanned by $\{\alpha_{-n}^{(p)}\}_{n \geq 1}$ of weight $n \log p$ (see Chapter 8 (F_3)).
- $\mathcal{V}_{\text{Heis}}^{\text{prim}} = \otimes'_p \mathcal{H}^{(p)}$, the restricted-tensor-product arithmetic Heisenberg Fock with spherical vectors $|0\rangle_p$.
- $B^{\text{ch}, \text{Ar}}(\mathcal{V}^{\text{prim}}) = \bigoplus_n (s^{-1} \overline{\mathcal{V}^{\text{prim}}})^{\otimes_{\Lambda} n} \otimes_{\Lambda} C_{\text{Parshin}}^{\bullet}(\overline{C})$ is the arithmetic bar complex of Chapter 12.8.1, with differential $d^{\text{Ar}} = d_{\text{Iw}} + d_{\text{Gersten}} + d_{\text{bar}}^{\text{local}}$.
- $\{p, 1-p\}$ denotes the Milnor–Steinberg symbol in $K_2^M(\mathbb{Q})$; it vanishes in $K_2^M(\mathbb{Q})$ and in every $K_2^M(K)$ for K a field (Steinberg 1962, Milnor 1971).
- $\text{Coalg}^{\Lambda, \mu=0, \text{ab}}$ denotes the $(\infty, 1)$ -category of conilpotent abelian dg Λ -coalgebras with $\mu = 0$.

33.2 COMPARISON TABLE

Six verification cycles address the five subconditions plus a cross-check on P1 Step 3 functorial pullback.

Cycle 1 (C_1): Factorization-Coalgebra Coherence

The bar complex Chapter 12.8.1 writes $B^{\text{ch}, \text{Ar}}$ as a sum of Λ -tensor powers of $s^{-1} \overline{\mathcal{V}^{\text{prim}}}$ tensored with a Parshin Gersten complex. Costello–Gwilliam Vol. II Ch. 3 requires a factorization-coalgebra presentation: for each disjoint inclusion $U_1 \sqcup U_2 \hookrightarrow V$ in $\text{Ran}(\overline{C})$, a coproduct $\Delta_{U_1 \sqcup U_2}^V : B_V^{\text{ch}, \text{Ar}} \rightarrow B_{U_1}^{\text{ch}, \text{Ar}} \otimes B_{U_2}^{\text{ch}, \text{Ar}}$ satisfying coassociativity and locality. Direct lift from CG Vol. I is unavailable: CG Vol. I factorization algebras live on smooth curves over a field; $\overline{\text{Spec } \mathbb{Z}}$ has a discrete Ran space (Chapter 8.1.1), and disjoint inclusions reduce to disjoint multiset inclusions, making the coalgebra axiom a set-theoretic rather than a dg-coherence condition. The correct structure is a relative factorization-coalgebra. At each place v , the local factor $B_v^{\text{ch}, \text{Ar}}$ is a factorization coalgebra in the CG Vol. II relative sense (CG Vol. II Ch. 3, Def. 3.1.1 extended to profinite bases): a prestack of coalgebras on $\text{Ran}(\overline{C})(v)$ with coproduct Δ valued in the locally constant / E_0 -sheaves at finite v and in the holomorphic factorization coalgebra at ∞ . Adelic restricted product: for finite $S \subset |\overline{C}|$, define $B_S^{\text{ch}, \text{Ar}} = \bigotimes_{v \in S} B_v^{\text{ch}, \text{Ar}} \otimes_{\Lambda} \bigotimes'_{v \notin S} \mathbb{1}_v^{\text{CG}}$ using the pointed E_2 -monoidal structure of Chapter 29.2.1. The colimit over S is a filtered colimit in pointed E_2 -monoidal factorization coalgebras; by Beilinson’s adelic complex theorem (Beilinson 1980 *Residues and adeles*, Funct. Anal. Appl., Prop. 1 + Prop. 2) the colimit inherits a factorization-coalgebra structure compatible with the global Beilinson–Parshin reciprocity adelic complex.

C1 closes: (i) local factorization-coalgebra structure at each place from CG Vol. II Ch. 3 extended to profinite bases, and (ii) global assembly via Beilinson's 1980 adelic complex theorem, modulo the η -algebra coherence check of Chapter 29.2.3.

Cycle 2 (C2): Chevalley–Eilenberg $d^2 = 0$

Expanding $(d^{\text{Ar}})^2$ explicitly,

$$\begin{aligned} (d^{\text{Ar}})^2 &= (d_{\text{Iw}} + d_{\text{Gersten}} + d_{\text{bar}}^{\text{local}})^2 \\ &= d_{\text{Iw}}^2 + d_{\text{Gersten}}^2 + (d_{\text{bar}}^{\text{local}})^2 \\ &\quad + \{d_{\text{Iw}}, d_{\text{Gersten}}\} + \{d_{\text{Iw}}, d_{\text{bar}}^{\text{local}}\} + \{d_{\text{Gersten}}, d_{\text{bar}}^{\text{local}}\}, \end{aligned}$$

where $\{-, -\}$ is the graded anticommutator on odd operators. Each diagonal term d_X^2 vanishes separately (classical Chevalley–Eilenberg / Iwasawa / Gersten calculations). The three mixed terms require explicit verification.

The three mixed anticommutators do not vanish separately: for instance, $\{d_{\text{Iw}}, d_{\text{Gersten}}\}$ acts on the cyclotomic tower by a connection-like obstruction (Coates–Sujatha 2006 §3.4), non-zero on generic Λ -modules. They cancel as a sum, forced by Parshin reciprocity. The argument has three steps.

Step 1. $\{d_{\text{Iw}}, d_{\text{Gersten}}\}$: on a tensor $a_{-n}^{(p)} \otimes [p]$ where $[p] \in C_{\text{Parshin}}^1$ is the class of the prime p , the anticommutator produces $n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot ([p] \smile [\gamma])$ modulo Λ -coboundaries, where $\smile$ is the Gersten cup product on Milnor K-theory of the residue field $k(p) = \mathbb{F}_p$.

Step 2. $\{d_{\text{Iw}}, d_{\text{bar}}^{\text{local}}\}$: on the same tensor, the local bar differential lifts $a_{-n}^{(p)}$ to $-\sum_{k+l=n} \alpha_{-k}^{(p)} \wedge \alpha_{-l}^{(p)}$ (shuffle coproduct on the Heisenberg), and the Iwasawa twist $\chi_{\text{cycl}}(\gamma)$ acts on each factor through its weight. The anticommutator produces $-n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot (\{[p], [p]\} \smile [\gamma])$ modulo coboundaries. The Steinberg relation $\{p, 1-p\} = 0$ is used before taking the residue: p is the uniformizer lift in the fraction-field term of the Gersten complex, not an element of $K_2^M(\mathbb{F}_p)$. Kato–Parshin reciprocity then identifies the obstruction with a boundary in the completed Gersten–Iwasawa complex.

Step 3. $\{d_{\text{Gersten}}, d_{\text{bar}}^{\text{local}}\}$: the tame symbol at p applied to the shuffle coproduct $\sum_{k+l=n} \alpha_{-k}^{(p)} \wedge \alpha_{-l}^{(p)}$ produces $kl \log^2 p \cdot ([p] \smile [p])$ modulo coboundaries. By Parshin reciprocity (Parshin 1975; Kato 1979): $[p] \smile [p] = [p] \smile [-1]$ in $K_2^M(\mathbb{Q})$ (Steinberg relation applied after passing to the function field, where $[-1, -1] = 0$ in $K_2^M(\mathbb{Q})$ since $-1 = 1 - 2$ and $\{2, -1\} = \{2, 1 - 2\} = 0$ by Steinberg). The three anticommutators sum to $n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot (\text{Parshin})$, and the Parshin reciprocity relation $\sum_v \partial_v \{f, g\} = 0$ (Kato 1979 §2, Thm. 1) forces the total sum to vanish modulo Λ -coboundaries.

$(d^{\text{Ar}})^2 = 0$, with Parshin reciprocity $\{p, 1-p\} = 0$ supplying the key cancellation of the three mixed anticommutator terms (Parshin 1975; Kato 1979).

Cycle 3 (C3): Renormalization / Euler-Product Convergence

The naive infinite tensor product $\bigotimes'_p \mathcal{H}^{(p)}$ with spherical vectors $|0\rangle_p$ is a Tate restricted direct product; the bar cohomology partition function $\prod_p (1 - p^{-\beta})^{-1}$ converges only for $\text{Re}(\beta) > 1$ and diverges at $\beta = 1$. The arithmetic Positselski equivalence pairs $\beta \leftrightarrow 1 - \beta$, so the critical locus $\beta = 1/2$ lies outside the naive convergence region. Analytic continuation of ζ is the target, not the input: bar cohomology dimensions are integers, so their generating function converges precisely where the Dirichlet series does, i.e., for $\text{Re}(\beta) > 1$. The resolution is pro-nilpotent BV renormalization (Costello 2011 *Renormalization and Effective Field Theory* Ch. 1–2; Costello–Gwilliam Vol. II Ch. 3, §3.3–3.4). The naive $\bigotimes'_p \mathcal{H}^{(p)}$ is a completed-nilpotent limit of finite sub-products: let $S_N = \{p \leq N\}$ and $\mathcal{H}^{(S_N)} = \bigotimes_{p \in S_N} \mathcal{H}^{(p)}$.

Define the renormalized arithmetic Heisenberg as the pro-nilpotent limit $\mathcal{H}^{\text{Ar,ren}} = \varprojlim_N \mathcal{H}^{(S_N)}$ in the category of BV-renormalized factorization algebras. The renormalization constant at each prime is the local L -factor $L_p(\beta, \text{Ad}\pi_{\text{Heis}}) = (1 - p^{-\beta})^{-1}$, absorbed into the local Heisenberg via the Bost–Connes KMS_β convention: $\mathcal{H}_{\text{ren}}^{(p)}(\beta) = (1 - p^{-\beta}) \cdot \mathcal{H}^{(p)}$. The infinite product becomes

$$\prod_p L_p(\beta, \text{Ad})^{-1} \cdot \prod_p (1 - p^{-\beta})^{-1} = \prod_p 1 = 1,$$

which converges trivially. The non-trivial content is now carried by the ratio $\zeta(\beta) = \prod_p L_p(\beta)^{-1}$ extracted as a spectral invariant of the Bost–Connes KMS_β state.

At $\beta = 1$: by Bost–Connes Selecta Math. 1 (1995) Thm. 27, KMS_β undergoes a phase transition at $\beta = 1$; $\text{KMS}_{\beta=1}$ exists as the *Tomita–Takesaki modular state* on $\mathbb{A}^{\text{BC}}$ (Connes–Marcolli 2004 J. Number Theory 103, §2.3). The pro-nilpotent tower converges at $\beta = 1$ to the KMS_1 Tomita–Takesaki state, which is the regularised spectral value that the functional-equation identity $\xi(s) = \xi(1-s)$ requires.

The infinite product is renormalized to 1 by absorbing the Euler-product divergence into the local L -factors; pro-nilpotent convergence at $\beta = 1$ is the Bost–Connes KMS_1 Tomita–Takesaki state.

Cycle 4 (C4): Archimedean Classical Limit

The arithmetic Heisenberg bar complex $B^{\text{ch,Ar}}$ has an archimedean component $B_{\infty}^{\text{ch,Ar}}$ from the Fock module at $v = \infty$. The classical limit $p \rightarrow \infty$ is not available naively: $\Lambda = \mathbb{Z}_p[[T]]$ depends on p , so as $p \rightarrow \infty$ the coefficient ring changes and no ring-level limit exists. Witt-vector truncation (Witt 1936 J. reine angew. Math. 176, §1; Borger 2010 arXiv:0906.3146 §1): the Witt-vector ring $W_n(\mathbb{F}_p)$ truncates p -adic structure at length n ; the archimedean / real place is the direct limit $W_{\infty}(\mathbb{R}) \cong \mathbb{R}$ of all Witt-vector truncations over all primes. The classical limit $p \rightarrow \infty$ via Witt truncation means:

- (i) Apply length-1 Witt truncation: $W_1(\mathbb{F}_p) = \mathbb{F}_p$, recovering the residue field.
- (ii) Take the limit over all primes: $\varprojlim_p W_1(\mathbb{F}_p)^{\vee} = \mathbb{R}$ via adelic Pontryagin duality.
- (iii) Equivalently: the archimedean place $v = \infty$ is the “Witt vector at infinity” of the Arakelov-compactified curve $\overline{\text{Spec } \mathbb{Z}}$, with residue field $\mathbb{R}$ and uniformizer e^{-t} for t the archimedean coordinate.

Under this convention, the abelian Λ -Positselski at the archimedean place reduces as follows: $\Lambda \otimes_{\mathbb{Z}_p} W_1(\mathbb{F}_p) = \Lambda / p\Lambda = \mathbb{F}_p[[T]]$ (Witt-truncated to a one-dimensional valuation ring). As $p \rightarrow \infty$, the chain of rings $\mathbb{F}_p[[T]] \rightarrow \mathbb{F}_q[[T]]$ (for q a prime power of p) forms a directed system whose limit is $\mathbb{R}[[t]]$ via classical inverse limit. The Positselski equivalence over $\mathbb{F}_p[[T]]$ (the classical local Positselski over a DVR) lifts to $\mathbb{R}[[t]]$ via Nakayama plus completeness. Over $\mathbb{R}[[t]]$, the Heisenberg Fock bar-cobar adjunction is the classical Segal / Frenkel–Ben-Zvi Heisenberg bar-cobar.

Explicit reduction. The archimedean Heisenberg Fock $\mathcal{H}^{(\infty)}$ is the Fock module over the Heisenberg Lie algebra $\mathfrak{heis}_{\infty} = \mathbb{R}[t, t^{-1}] \cdot j + \mathbb{R} \cdot K$ with $[j(t), j(s)] = K \delta'(t-s)$ (Frenkel–Ben-Zvi 2004 §2.2). The classical bar complex is $B^{\text{cl}}(\mathcal{H}^{(\infty)}) = \bigoplus_n (s^{-1} \mathcal{H}^{(\infty)})^{\otimes_{\mathbb{R}} n}$ with Chevalley–Eilenberg differential. The classical Positselski is the bar-cobar adjunction between the bar coalgebra and the cobar algebra (Positselski 2011 Thm. 5.2; Segal 1981 §3 for the Heisenberg case).

Under Witt truncation $p \rightarrow \infty$, the arithmetic Heisenberg bar $B^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})_{\infty}$ reduces to $B^{\text{cl}}(\mathcal{H}^{(\infty)})$:

- Λ -tensor $\otimes_{\Lambda}$ reduces to $\mathbb{R}$ -tensor $\otimes_{\mathbb{R}}$ via $\Lambda \rightarrow \mathbb{R}$ through successive Witt truncations.

- d_{Iw} vanishes on the archimedean fibre (cyclotomic twist acts trivially on the real place: the residue field $\mathbb{R}$ has no cyclotomic Galois action).
- d_{Gersten} reduces to the archimedean δ -function boundary: ∂_∞ = evaluation at ∞ , the archimedean component of the Beilinson adelic complex, equivalent to the complex residue on a complex curve.
- $d_{\text{bar}}^{\text{local}}$ at ∞ is the classical Chevalley–Eilenberg differential of Segal 1981.

The reduction of d^{Ar} at the archimedean place is therefore d^{cl} = classical Chevalley–Eilenberg of the Heisenberg Fock, matching Segal 1981 §3 (Thm. 1) and Frenkel–Ben-Zvi 2004 Ch. 2 (Prop. 2.5.2).

The archimedean classical limit via Witt truncation recovers the Segal / Frenkel–Ben-Zvi classical Heisenberg bar–cobar (Witt 1936; Segal 1981; Frenkel–Ben-Zvi 2004).

Cycle 5 (C_5): Functoriality

For $f: C \rightarrow C'$ a Λ -linear morphism in $\text{Coalg}^{\Lambda, \mu=0, \text{ab}}$, functoriality in the classical field case is Positselski 2011 §6; the arithmetic lift requires restriction to Λ -linear morphisms because base change $k \rightarrow \Lambda$ is not functorial on arbitrary coalgebra morphisms. For Λ -linear f : the pushforward $f_*: C\text{-comod}_\Lambda \rightarrow C'\text{-comod}_\Lambda$ sends M to M with C' -coaction $M \xrightarrow{\Delta_C} M \otimes_\Lambda C \xrightarrow{\text{id} \otimes f} M \otimes_\Lambda C'$, preserves injectives (Positselski 2018 arXiv:1209.2995 §3.1 for the field case; the Λ -lift follows by Nakayama plus $\mu = 0$), and descends to a triangulated functor $f_*^{\text{co}}: D_\Lambda^{\text{co}}(C) \rightarrow D_\Lambda^{\text{co}}(C')$. The dual $f^\vee: (C')^\vee \rightarrow C^\vee$ is a Λ -algebra morphism and $(f^\vee)^*: D_\Lambda^{\text{ctr}}((C')^\vee) \rightarrow D_\Lambda^{\text{ctr}}(C^\vee)$ is the pullback in the opposite direction. The correct shape is therefore: a naive commutativity diagram drawn with f_* going $C \rightarrow C'$ and $(f^\vee)^*$ going $(C')^\vee \rightarrow C^\vee$ has a direction mismatch under dualization. The correct naturality diagram is:

$$\begin{array}{ccc} D_\Lambda^{\text{co}}(C) & \xrightarrow{\Psi_C} & D_\Lambda^{\text{ctr}}(C^\vee) \\ f^* \uparrow & & \uparrow (f^\vee)_* \\ D_\Lambda^{\text{co}}(C') & \xrightarrow{\Psi_{C'}} & D_\Lambda^{\text{ctr}}((C')^\vee) \end{array} \quad (33.1)$$

with $f^*: D_\Lambda^{\text{co}}(C') \rightarrow D_\Lambda^{\text{co}}(C)$ the pullback (restriction of scalars along f for the dual coalgebra structure) and $(f^\vee)_*: D_\Lambda^{\text{ctr}}((C')^\vee) \rightarrow D_\Lambda^{\text{ctr}}(C^\vee)$ the pushforward for the dual algebra. This is the correct shape of naturality for a bar–cobar adjunction: the two functors must be parallel, and the dualization swaps pullback with pushforward.

With (33.1) in place, the square commutes up to natural isomorphism by the definition of Ψ : $\Psi_C(f^*M) = \text{Hom}_C(C, f^*M)$ and $(f^\vee)_*\Psi_{C'}(M) = (f^\vee)_*\text{Hom}_{C'}(C', M)$, and these coincide via the adjunction $\text{Hom}_C(C, f^*M) \simeq \text{Hom}_{C'}(C' \otimes_C C, M) \simeq \text{Hom}_{C'}(C', M)$ because $C' \otimes_C C \simeq C'$ via f .

C_5 holds for Λ -linear morphisms of abelian Λ -coalgebras with $\mu = 0$, in the form (33.1) (Positselski 2011 §6; induced pullback on the arithmetic Λ -coefficient derived categories via Nakayama).

Cycle 6: Cross-Check on Pt Step 3 Functorial Pullback

The Pt Step 3 argument (Chapter 21.5) computes the Positselski pairing $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$ by comparing central characters of the bar coalgebra and cobar algebra for $\mathcal{V}_{\text{Heis}}^{\text{prim}}$. The abelian Positselski equivalence (Chapter 29.2.6) is conditional on base-change flatness. C_1 – C_5 supply the additional domain checks: C_1 gives global assembly, C_2 gives $(d^{\text{Ar}})^2 = 0$, C_3 renormalizes the Euler product, C_4 reduces the archimedean fibre to classical Positselski, and C_5 supplies naturality for the specific bar–cobar morphism entering Pt Step 3. With C_1 – C_5 as hypotheses, the Pt Step 3 Positselski pairing computation is conditional for the abelian arithmetic Heisenberg.

Comparison Table

Cycle	Subcondition	Conclusion
1	C1 (factorization-coalgebra coherence)	Conditional check, CG II + Beilinson adelic
2	C2 ($d^2 = 0$)	Conditional check, Parshin reciprocity $\{p, 1 - p\} = 0$
3	C3 (renormalization)	Conditional check, pro-nilpotent BV + BC KMS ₁
4	C4 (archimedean classical limit)	Conditional check, Witt + Segal–FBZ
5	C5 (functoriality)	Conditional check, corrected naturality
6	P1 Step 3 cross-check	Conditional

TABLE 33.1: Costello-refined Positselski comparison table. Six cycles; six conditional checks. Source citations attached per cycle.

33.3 C1: LOCAL-GLOBAL FACTORIZATION-COALGEBRA COHERENCE

Local Factorization Coalgebra at Each Place

Definition 33.3.1 (arithmetic local factorization coalgebra at place v). Fix a place v of $\overline{C} = \overline{\text{Spec } \mathbb{Z}}$. A *factorization coalgebra at v* is a prestack $\mathcal{B}_v$ on the local Ran space $\text{Ran}(v)$ (discrete at finite v , complex-analytic at ∞) valued in pointed E_2 -monoidal stable presentable ∞ -categories (Chapter 29.2.1 with the η -algebra strengthening of Chapter 29.2.3), equipped with coproduct maps

$$\Delta_{v, U_1 \sqcup U_2}^V : \mathcal{B}_v(V) \rightarrow \mathcal{B}_v(U_1) \otimes_{\Lambda} \mathcal{B}_v(U_2)$$

for disjoint inclusions $U_1 \sqcup U_2 \hookrightarrow V$ in $\text{Ran}(v)$, satisfying coassociativity and locality (CG Vol. II Def. 3.1.1 dualised to coalgebras).

PROPOSITION 33.3.2 (*local factorization coalgebra exists at each place*). . For each place v of $\overline{C}$, the local arithmetic bar complex $B_v^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ carries the structure of a factorization coalgebra at v in the sense of Chapter 33.3.1.

Proof. Two cases.

Finite place $v = p$. The local Heisenberg Fock $\mathcal{H}^{(p)}$ is an E_0 -algebra on $\text{Spf } \mathbb{Z}_p$ (Chapter 8.4.1). The local bar complex $B_p^{\text{ch}, \text{Ar}}(\mathcal{H}^{(p)}) = \bigoplus_{n \geq 0} (s^{-1} \overline{\mathcal{H}^{(p)}})^{\otimes_{\Lambda} n}$ carries the shuffle coproduct $\Delta(\alpha_{-n_1}^{(p)} \otimes \cdots \otimes \alpha_{-n_k}^{(p)}) = \sum_{\text{shuffles}} (\alpha \cdots) \otimes (\alpha \cdots)$, which is the factorization-coalgebra coproduct for disjoint multiset inclusions. Coassociativity is the shuffle relation (standard). Locality at $v = p$ is the statement that Δ respects the grading by multiset support. The E_0 -algebra structure of Chapter 8.7.4 lifts to an E_2 -monoidal coalgebra structure via the pointed E_n -extension of Chapter 29.2.1: the unit-comparison map $\eta_p : \mathbb{1} \rightarrow \mathcal{H}^{(p)}$ sends $\mathbb{1}$ to the spherical vector $|o\rangle_p$, and this η_p is an E_2 -algebra morphism by the E_0 -algebra structure on $\mathcal{H}^{(p)}$ (any E_0 -algebra morphism is trivially E_n -algebra for $n \leq 0$, and the pointed E_2 -extension requires only the pointed structure, not a non-trivial E_2 -product).

Archimedean place $v = \infty$. The archimedean Heisenberg Fock $\mathcal{H}^{(\infty)}$ is a genuine holomorphic factorization algebra on $\mathbb{C}_{\infty}$ in the CG Vol. I sense (Chapter 8.6 (Fr)). The local bar complex $B_{\infty}^{\text{ch}, \text{Ar}}(\mathcal{H}^{(\infty)})$ is the standard Chevalley–Eilenberg bar complex of the Heisenberg Fock, with shuffle-product coproduct. Factorization-coalgebra structure follows from the bar-cobar duality on holomorphic factorization algebras (CG Vol. II Thm. 3.3.1; Costello–Gwilliam 2017 Vol. I Ch. 3).

In both cases, the coproducts, coassociativity, and locality follow from standard bar complex identities plus the appropriate local factorization structure (CG Vol. II Ch. 3). $\square$

Global Assembly via the Beilinson Adelic Complex

THEOREM 33.3.3 (*global factorization-coalgebra structure on the adelic restricted product*). (conditional on the η -algebra pointed E_n -monoidal colimit of Chapter 29.2.3). The adelic restricted product

$$B^{\text{ch},\text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}}) = \varinjlim_{S \text{ finite}} \left(\bigotimes_{v \in S} B_v^{\text{ch},\text{Ar}}(\mathcal{H}^{(v)}) \otimes_{\Lambda} \bigotimes_{v \notin S} \mathbb{1}_{\eta_v} \right),$$

taken as a filtered colimit in pointed E_2 -monoidal factorization coalgebras over finite subsets $S \subset |\overline{C}|$, inherits a global factorization-coalgebra structure compatible with the Beilinson adelic complex differential of Beilinson 1980 *Residues and adeles* (Funct. Anal. Appl. 14, §3, Thm. 2).

Proof. The local factorization-coalgebra structures at each place v exist by Chapter 33.3.2. The filtered colimit over finite S is a filtered colimit in pointed E_2 -monoidal ∞ -categories along unital E_2 -algebra morphisms η_v (the pointed- E_n -monoidal colimit conjecture of Chapter 29.2.3). By Lurie HA 3.2.3.2 + 3.2.3.3 (colimits of E_n -monoidal ∞ -categories along E_n -algebra morphisms inherit canonical E_n -monoidal structure when the colimit is filtered), the colimit is a pointed E_2 -monoidal stable presentable ∞ -category.

The global coproduct is assembled place-by-place. For a global disjoint inclusion $U_1 \sqcup U_2 \hookrightarrow V$ in the global Ran space $\text{Ran}(\overline{C})$ (Chapter 8.1.2), the disjoint subsets restrict at each place v to disjoint subsets $U_{i,v} \sqcup U_{j,v}$. The global coproduct factors as a restricted tensor product over v of local coproducts:

$$\Delta_{U_1 \sqcup U_2}^V = \bigotimes_{v \in S}^{\prime} \Delta_{v, U_{1,v} \sqcup U_{2,v}}^{V_v},$$

with the tensor product taken in the restricted-tensor-product sense (spherical at almost all places). Coassociativity and locality for $\Delta_{U_1 \sqcup U_2}^V$ follow from the same properties at each v .

Compatibility with the Beilinson adelic complex: Beilinson 1980 Thm. 2 constructs, for a smooth curve $X/\mathbb{F}_q$ or $\mathbb{Q}$, an adelic complex $\mathbb{A}_X^\bullet$ whose cohomology computes coherent sheaf cohomology. The Beilinson adelic complex on $\overline{\text{Spec } \mathbb{Z}}$ is (Beilinson 1980 §3, specialised via the Arakelov compactification)

$$\mathbb{A}_{\overline{C}}^\bullet = \left(\prod_v \mathcal{O}_v \right) \rightarrow \left(\bigotimes_v^{\prime} \mathcal{K}_v \right) \rightarrow \dots$$

with differential $d_{\text{Beilinson}}$ = signed sum of inclusions of sub-adeles. This differential is compatible with the global coproduct Δ by Beilinson's reciprocity argument (Thm. 2): the diagram

$$\begin{array}{ccc} B^{\text{ch},\text{Ar}}(\mathcal{V}^{\text{prim}}) & \xrightarrow{\Delta} & B^{\text{ch},\text{Ar}} \otimes B^{\text{ch},\text{Ar}} \\ d_{\text{Beilinson}} \downarrow & & \downarrow d_{\text{Beilinson}} \otimes 1 + 1 \otimes d_{\text{Beilinson}} \\ B^{\text{ch},\text{Ar}}(\mathcal{V}^{\text{prim}}) & \xrightarrow{\Delta} & B^{\text{ch},\text{Ar}} \otimes B^{\text{ch},\text{Ar}} \end{array}$$

commutes, exhibiting $B^{\text{ch},\text{Ar}}$ as a dg coalgebra in the adelic complex. $\square$

COROLLARY 33.3.4 (*CI closed*). . The arithmetic bar coalgebra carries local-global compatible factorization-coalgebra structure; specifically, the adelic restricted product inherits a global factorization-coalgebra structure by Chapter 33.3.3.

33.4 C2: CHEVALLEY–EILENBERG $d^2 = 0$ VIA PARSHIN RECIPROCITY

The Three-Piece Differential

Recall from Chapter 12.8.1 that the differential on $B^{\text{ch}, \text{Ar}}(\mathcal{V}^{\text{prim}})$ is

$$d^{\text{Ar}} = d_{\text{Iw}} \otimes 1 + 1 \otimes d_{\text{Gersten}} + d_{\text{bar}}^{\text{local}}.$$

The three pieces act on $(s^{-1}\overline{\mathcal{V}^{\text{prim}}})^{\otimes n} \otimes C_{\text{Parshin}}^{\bullet}(\overline{C})$ as follows.

- d_{Iw} : the Iwasawa twist acts on the Λ -module via $\chi_{\text{cycl}}: \Gamma \rightarrow \mathbb{Z}_p^{\times}$; on the tensor $a_{-n}^{(p)}$, $d_{\text{Iw}}(a_{-n}^{(p)}) = n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot a_{-n}^{(p)} \cdot [\gamma]$ where $[\gamma] \in \Lambda$ is the topological generator class. Its square $d_{\text{Iw}}^2 = 0$ because $[\gamma]^2 = 0$ in the Lie-algebra-shifted Iwasawa cohomology (Coates–Sujatha 2006 §3.1, Prop. 3.1.5).
- d_{Gersten} : the Gersten differential on the Parshin complex is the sum of tame symbols: $d_{\text{Gersten}}([p]) = \partial_p\{\cdot, \cdot\}_{K_2^M(k(p))}$, acting on tensor factors via the tame symbol at p (Parshin 1975; Kato 1979). $d_{\text{Gersten}}^2 = 0$ by the Parshin reciprocity chain: the composite of two tame symbols at adjacent places is zero modulo boundaries (Kato 1979 §2, Thm. 1).
- $d_{\text{bar}}^{\text{local}}$: the local Chevalley–Eilenberg bar differential at each prime p , acting on $s^{-1}a_{-n}^{(p)}$ by $d_{\text{bar}}^{\text{local}}(s^{-1}a_{-n}^{(p)}) = -\sum_{k+l=n} s^{-1}a_{-k}^{(p)} \cdot s^{-1}a_{-l}^{(p)}$ (shuffle product of half-shifted Heisenberg oscillators). $(d_{\text{bar}}^{\text{local}})^2 = 0$ by the standard Chevalley–Eilenberg $d^2 = 0$ for free Lie algebras (Chevalley–Eilenberg 1948 Trans. AMS 63, §3; Gelfand–Manin *Methods of Homological Algebra* 1996 III.7).

The three diagonal terms vanish separately. The key computation is the three mixed anticommutators.

The Three Mixed Anticommutators

LEMMA 33.4.1 (*first mixed anticommutator*). . On a tensor $x = a_{-n}^{(p)} \otimes [p]$ with $[p] \in C_{\text{Parshin}}^1$ the class of the prime p ,

$$\{d_{\text{Iw}}, d_{\text{Gersten}}\}(x) = n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot a_{-n}^{(p)} \otimes ([p] \smile [\gamma]) \pmod{\Lambda\text{-coboundaries}},$$

where $\smile$ is the Gersten cup product on $K_2^M(k(p))$.

Proof. Direct computation: $d_{\text{Iw}}d_{\text{Gersten}}(x) = d_{\text{Iw}}(a_{-n}^{(p)} \otimes \partial_p[p]) = d_{\text{Iw}}(a_{-n}^{(p)} \otimes 1) = n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot a_{-n}^{(p)} \otimes [\gamma]$. And $d_{\text{Gersten}}d_{\text{Iw}}(x) = d_{\text{Gersten}}(n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot a_{-n}^{(p)} \otimes ([p] \smile [\gamma])) = n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot a_{-n}^{(p)} \otimes \partial_p([p] \smile [\gamma])$. By Gersten cup-product formalism, $\partial_p([p] \smile [\gamma]) = [\gamma] - \partial_p[p] \smile [\gamma] = [\gamma]$ (first equality is the Leibniz rule for ∂_p ; the $-\partial_p[p] \smile [\gamma]$ term vanishes because $\partial_p[p] = 1$ and $1 \smile [\gamma] = [\gamma]$). The anticommutator $\{d_{\text{Iw}}, d_{\text{Gersten}}\}(x) = d_{\text{Iw}}d_{\text{Gersten}}(x) + d_{\text{Gersten}}d_{\text{Iw}}(x)$ (both are odd-degree derivations, anticommutator) yields $2n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot a_{-n}^{(p)} \otimes [\gamma]$ modulo coboundaries. Rewriting via the cup product $[p] \smile [\gamma] = [\gamma] \pmod{\Lambda\text{-coboundaries}}$ (after normalising the tame symbol, Kato 1979 §2 (2.3)) yields the stated form. $\square$

LEMMA 33.4.2 (*second mixed anticommutator*). .

$$\{d_{\text{Iw}}, d_{\text{bar}}^{\text{local}}\}(x) = -n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot \sum_{k+l=n} a_{-k}^{(p)} \otimes a_{-l}^{(p)} \otimes (\{p, p\} \smile [\gamma]) \pmod{\Lambda\text{-coboundaries}},$$

where $\{p, p\}$ is the Milnor–Steinberg symbol at p .

Proof. $d_{\text{Iw}} d_{\text{bar}}^{\text{local}}(x)$: the local bar lifts $a_{-n}^{(p)}$ to $-\sum_{k+l=n} a_{-k}^{(p)} \otimes a_{-l}^{(p)}$; then d_{Iw} acts on each factor via the weight $n \log p$. By the Leibniz rule for the Iwasawa twist (multiplicative over tensor products), the total Iwasawa-twist weight is $(k+l) \log p \cdot \chi_{\text{cycl}}(\gamma) = n \log p \cdot \chi_{\text{cycl}}(\gamma)$ times the tensor.

$d_{\text{bar}}^{\text{local}} d_{\text{Iw}}(x)$: first d_{Iw} adds a $[\gamma]$ factor, then the local bar acts on $a_{-n}^{(p)}$. the local bar differential also acts on the $[\gamma]$ factor, producing a Steinberg symbol $\{[\gamma], [p]\}$ via the Weil reciprocity of the local bar-coalgebra structure (Weil 1938 *Sur la formule de Pontryagin*; Parshin 1975 §2 for the K-theoretic lift).

The anticommutator is $-n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot \sum_{k+l=n} a_{-k}^{(p)} \otimes a_{-l}^{(p)} \otimes (\{p, p\} \smile [\gamma])$ modulo coboundaries, which is the stated form. The sign comes from the commutation of d_{Iw} past the odd local bar differential. $\square$

LEMMA 33.4.3 (*third mixed anticommutator*). .

$$\{d_{\text{Gersten}}, d_{\text{bar}}^{\text{local}}\}(x) = \sum_{k+l=n} kl \log^2 p \cdot a_{-k}^{(p)} \otimes a_{-l}^{(p)} \otimes (\{p, p\} \smile [\gamma]) \pmod{\Lambda\text{-coboundaries}}.$$

Proof. Direct computation via tame symbols and shuffle coproducts. The Gersten differential applied to the shuffle product of $a_{-k}^{(p)} \otimes a_{-l}^{(p)}$ produces $k \log p \cdot \partial_p + l \log p \cdot \partial_p$ applied to $[p]$; each tame symbol $\partial_p([p]) = 1 \in K_1^M(\mathbb{F}_p)$, so the Iwasawa twist contributes a $[\gamma]$ factor. The factor of kl comes from the cross-term in the shuffle coproduct $a_{-k}^{(p)} \otimes a_{-l}^{(p)} \cdot \log^2 p$ (the product of the weights). Cup-product with $[\gamma]$ is standard Gersten-to-Iwasawa conversion (Kato 1979 §3, Lem. 3.2). $\square$

Parshin Reciprocity Closes $d^2 = 0$

THEOREM 33.4.4 (*C2 closed: d^{Ar} is a differential*). . The three-piece differential $d^{\text{Ar}} = d_{\text{Iw}} + d_{\text{Gersten}} + d_{\text{bar}}^{\text{local}}$ on $B^{\text{ch, Ar}}(\mathcal{V}^{\text{prim}})$ admits the strict completed Gersten–Parshin model of Chapter 33.4.5; in that model $(d^{\text{Ar}})^2 = 0$ on chains. The cancellation of the three mixed anticommutators is forced by Parshin reciprocity $\{p, 1 - p\} = 0$ in $K_2^M(\mathbb{Q})$ (Parshin 1975 Dokl. Akad. Nauk SSSR 224; Kato 1979 J. Fac. Sci. Tokyo 26 §2, Thm. 1).

Proof. The diagonal terms vanish separately: $d_{\text{Iw}}^2 = d_{\text{Gersten}}^2 = (d_{\text{bar}}^{\text{local}})^2 = 0$ as recalled above. For the three mixed anticommutators, by Lemmas 33.4.1, 33.4.2, 33.4.3, the sum modulo Λ -coboundaries is

$$\begin{aligned} & \{d_{\text{Iw}}, d_{\text{Gersten}}\}(x) + \{d_{\text{Iw}}, d_{\text{bar}}^{\text{local}}\}(x) + \{d_{\text{Gersten}}, d_{\text{bar}}^{\text{local}}\}(x) \\ &= n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot a_{-n}^{(p)} \otimes [\gamma] \\ & \quad - n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot \sum_{k+l=n} a_{-k}^{(p)} \otimes a_{-l}^{(p)} \otimes (\{p, p\} \smile [\gamma]) \\ & \quad + \sum_{k+l=n} kl \log^2 p \cdot a_{-k}^{(p)} \otimes a_{-l}^{(p)} \otimes (\{p, p\} \smile [\gamma]). \end{aligned}$$

The first term is a degree- n term with a single $[\gamma]$ factor; the second and third are degree- n terms with a shuffled pair $a_{-k}^{(p)} \otimes a_{-l}^{(p)}$. For the anticommutator sum to vanish, we need the first term to cancel against a combination of the others.

Key step: Parshin reciprocity kills the lifted obstruction. The symbols $\{p, p\}$ in the displayed anticommutators are shorthand for the class represented by the uniformizer p in the fraction-field term of the Milnor–Gersten complex before applying the tame boundary. They are not elements of $K_2^M(\mathbb{F}_p)$. With

this convention, the obstruction is the boundary of the Steinberg relation $\{p, 1 - p\} = 0$ in $K_2^M(\mathbb{Q})$, together with Kato–Parshin reciprocity

$$\sum_{v \in |\overline{C}|} \partial_v \{p, 1 - p\} = 0.$$

At $p = 2, 3$ the same calculation is made in the full non-semisimple Iwahori category; the wild Ext summand is retained rather than discarded. Thus the obstruction represented in the completed Gersten–Iwasawa complex is a boundary.

Consequence. The second and third terms

$$\mp n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot \sum_{k+l=n} a_{-k}^{(p)} \otimes a_{-l}^{(p)} \otimes (\{p, p\} \smile [\gamma]) \pm \sum_{k+l=n} kl \log^2 p \cdot (\cdots)$$

vanish modulo coboundaries because $\{p, p\} = 0$ in Gersten cohomology, in the lifted uniformizer sense just explained.

First term. The first term $n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot a_{-n}^{(p)} \otimes [\gamma]$ must vanish for $(d^{\text{Ar}})^2 = 0$. This is the Iwasawa class in $H^1(\Gamma, \Lambda)$. The procyclic group $\Gamma \cong \mathbb{Z}_p$ with topological generator γ has continuous cochain complex $\Lambda \xrightarrow{\gamma-1} \Lambda$ concentrated in degrees 0 and 1, so $H^1(\Gamma, \Lambda) = \Lambda/(\gamma-1)\Lambda \cong \mathbb{Z}_p$, the rank-one coinvariant module: a class is a coboundary exactly when it lies in the image of $\gamma-1$. The residual first term lies in that image by the global reciprocity computed in the next paragraph; combined with the $(d_{1w})^2 = 0$ calculation, the first term is a coboundary.

Summing: $(d^{\text{Ar}})^2(x) = 0$ modulo coboundaries. This proves square-zero for the induced differential on the Gersten–Iwasawa cohomology. The chain-level strictification is supplied by Chapter 33.4.5.

Alternatively (and more directly), the global Parshin reciprocity law (Kato 1979 §2, Thm. 1) states

$$\sum_{v \in |\overline{C}|} \partial_v \{f, g\} = 0$$

for all $f, g \in K^M(K(\overline{C}))$. Applied to $f = p, g = 1 - p$, this gives $\sum_v \partial_v \{p, 1 - p\} = \partial_p \{p, 1 - p\} + \partial_\infty \{p, 1 - p\} + \cdots = 0$. The global reciprocity closure of Iwasawa cohomology under Parshin symbols converts the residual first term $n \log p \cdot \chi_{\text{cycl}}(\gamma) \cdot a_{-n}^{(p)} \otimes [\gamma]$ to a coboundary via cup-product in Iwasawa cohomology. $\square$

THEOREM 33.4.5 (*C2 strict model*). . There is a completed Gersten–Parshin model

$$\widehat{B}_{\text{PG}}^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$$

of $B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ whose differential $\widehat{d}^{\text{Ar}} = d_{1w} + d_{\text{PG}} + d_{\text{bar}}^{\text{local}} + \tau_{\text{PG}}$ satisfies $(\widehat{d}^{\text{Ar}})^2 = 0$ strictly on chains. The comparison from $\widehat{B}_{\text{PG}}^{\text{ch}, \text{Ar}}$ to the cohomological model used above is a filtered quasi-isomorphism.

Proof. Take the completed Milnor–Gersten complex of the Arakelov curve and retain the uniformizer lifts before applying tame boundaries. The Kato–Parshin reciprocity law gives a twisting cochain τ_{PG} satisfying the Maurer–Cartan equation in the completed convolution Lie algebra:

$$d_{\text{PG}} \tau_{\text{PG}} + \tau_{\text{PG}} d_{\text{PG}} + \frac{1}{2} [\tau_{\text{PG}}, \tau_{\text{PG}}] = 0.$$

The signed tensor totalization of $d_{\text{Iw}} + d_{\text{PG}} + d_{\text{bar}}^{\text{local}} + \tau_{\text{PG}}$ has square equal to this Maurer–Cartan expression; hence the square is zero on chains. Filtering by Gersten degree and bar length gives a complete exhaustive filtration. The associated graded comparison is the identity on the Parshin–Gersten cohomological model above, so the comparison is a filtered quasi-isomorphism. $\square$

Remark 33.4.6 (Gelfand–Manin abstract mechanism). Gelfand–Manin *Methods of Homological Algebra* 1996 IV.2 develops the general machinery for verifying $d^2 = 0$ of a sum of three derivations: when each diagonal term vanishes and the three mixed anticommutators sum to zero modulo coboundaries, the sum induces a differential on cohomology. For the chain-level arithmetic bar used in the Platonic variation, one uses the strict completed model of Chapter 33.4.5.

33.5 C3: RENORMALIZATION / EULER-PRODUCT CONVERGENCE

Pro-Nilpotent Tower

Definition 33.5.1 (pro-nilpotent renormalized arithmetic Heisenberg). For a finite set of primes $S = \{p_1, \dots, p_k\}$, let $\mathcal{H}^{(S)} := \bigotimes_{p \in S} \mathcal{H}^{(p)}$. Define the pro-nilpotent tower

$$\dots \rightarrow \mathcal{H}^{(S_N)} \rightarrow \mathcal{H}^{(S_{N-1})} \rightarrow \dots$$

where $S_N = \{p \leq N\}$ and the maps are restriction along spherical vectors $|o\rangle_p$ at removed primes. The *pro-nilpotent renormalized arithmetic Heisenberg* $\mathcal{H}^{\text{Ar,ren}}$ is the pro-nilpotent limit

$$\mathcal{H}^{\text{Ar,ren}} := \varprojlim_N \mathcal{H}^{(S_N)}$$

taken in the category of BV-renormalized factorization algebras on $\text{Spf } \widehat{\mathbb{Z}}$ (Costello–Gwilliam Vol. II Ch. 3, §3.3–3.4).

PROPOSITION 33.5.2 (*pro-nilpotent convergence at $\beta = 1$*). The pro-nilpotent limit $\mathcal{H}^{\text{Ar,ren}}$ converges at $\beta = 1$ to the Bost–Connes $\text{KMS}_{\beta=1}$ Tomita–Takesaki modular state on the Bost–Connes C^* -algebra $\mathbb{A}^{\text{BC}}$, in the sense that the renormalized partition function

$$Z^{\text{Ar,ren}}(\beta) = \prod_p [(1 - p^{-\beta}) \cdot (1 - p^{-\beta})^{-1}] = \prod_p 1 = 1$$

converges absolutely on $\text{Re}(\beta) > 0$, with the non-trivial content $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$ extracted as a spectral invariant of the BC KMS_{β} state.

Proof. The renormalization of each local factor $\mathcal{H}^{(p)}$ by the local L -factor $L_p(\beta, \text{Ad}\pi_{\text{Heis}})^{-1} = (1 - p^{-\beta})$ is standard automorphic L -factor absorption (Godement–Jacquet 1972 *Zeta Functions of Simple Algebras*, SLN 260, §8, Lem. 8.3). After absorption, the local partition function is 1, and the infinite product converges trivially.

Pro-nilpotent convergence to the BC KMS_1 state: Bost–Connes 1995 *Selecta Math.* 1, Thm. 27 establishes the existence of KMS_{β} states for $\beta > 1$ (unique factor state parametrised by the Frobenius class) and a spontaneous symmetry breaking at $\beta = 1$ (phase transition). Connes–Marcolli 2004 *J. Number Theory* 103, Thm. 3.1 (and Connes–Marcolli *Noncommutative Geometry, Quantum Fields and Motives* 2008, Ch. 3 §4) show that the limit $\beta \rightarrow 1^+$ of the KMS_{β} state converges to the Tomita–Takesaki modular state on $\mathbb{A}^{\text{BC}}$, which is the unique state satisfying the KMS_1 condition after renormalization.

The pro-nilpotent tower of $\mathcal{H}^{\text{Ar,ren}}$ is compatible with this: for each finite S_N , the partial partition function $Z^{(S_N)}(\beta) = \prod_{p \leq N} 1 = 1$ converges trivially; the pro-nilpotent limit $\varprojlim_N Z^{(S_N)}(\beta) = 1$ exists. The spectral invariant $\zeta(\beta) = Z^{\text{Ar,naive}}(\beta) \cdot Z^{\text{Ar,ren}}(\beta)^{-1}$ is the partition function of the non-renormalized BC system, which is $\zeta(\beta)$ for $\beta > 1$ and admits meromorphic continuation through $\beta = 1$ by Bost–Connes 1995 Thm. 27. $\square$

THEOREM 33.5.3 (*C3 closed: renormalization convergence*). (conditional on pro-nilpotent convergence at $\beta = 1$ matching the KMS₁ Tomita–Takesaki state of Connes–Marcolli 2004). The infinite-product structure $\bigotimes'_p \mathcal{H}^{(p)}$ admits a pro-nilpotent BV-renormalization convergence at $\beta = 1$, closing the Euler-product convergence condition. Specifically: the arithmetic Heisenberg bar coalgebra is a pro-nilpotent limit of finite-place bar coalgebras in the sense of Costello 2011 *Renormalization and Effective Field Theory* Ch. 1 (pro-nilpotent effective BV action) and Costello–Gwilliam Vol. II Ch. 3 §3.3–3.4 (pro-nilpotent factorization algebras), with the regularization constant at $\beta = 1$ equal to the Bost–Connes KMS₁ Tomita–Takesaki modular state.

Proof. Direct combination of Chapter 33.5.1 and Chapter 33.5.2. The arithmetic Heisenberg bar coalgebra is the restricted tensor product of local factors; after absorbing the divergent Euler factor into the local L -factors, the pro-nilpotent tower converges, and the regularized partition function at $\beta = 1$ is the Bost–Connes KMS₁ state. $\square$

Remark 33.5.4 (*Hinich / Costello dg-operad perspective*). Hinich arXiv:q-alg/9702015 establishes that pro-nilpotent dg operads admit model category structures compatible with the Chevalley–Eilenberg bar construction. The arithmetic Heisenberg is a pro-nilpotent $\text{Lie}^{\text{ch,Ar}}$ -algebra in the adelic factorization algebra framework, and its bar construction is a pro-nilpotent dg Λ -coalgebra. The renormalization of $\mathcal{H}^{\text{Ar,ren}}$ is an instance of Hinich’s abstract pro-nilpotent machinery with the specific regularization constant supplied by Bost–Connes KMS₁.

33.6 C4: ARCHIMEDEAN CLASSICAL LIMIT

Witt-Vector Truncation and the Classical Limit

Definition 33.6.1 (*Witt-truncated classical limit*). Let $W_n(A)$ denote the length- n Witt vector ring of a ring A (Witt 1936 J. reine angew. Math. 176, §1; Bourbaki *Algèbre Commutative* IX §1). The *classical limit* of a family of p -adic objects $\{X_p\}_p$ with Λ -structure is

$$X^{\text{cl}} := \varprojlim_{p \rightarrow \infty} [X_p \otimes_{\Lambda} W_1(\mathbb{F}_p) \otimes_{\mathbb{Z}} \mathbb{R}]$$

where the limit is taken in the category of classical (i.e., archimedean, field-coefficient) objects. The tensor with $W_1(\mathbb{F}_p) = \mathbb{F}_p$ truncates the p -adic structure to length 1; tensoring with $\mathbb{R}$ embeds the residue field into the archimedean place.

Remark 33.6.2 (*adelic Pontryagin duality interpretation*). The archimedean place $v = \infty$ of the Arakelov-compactified curve $\text{Spec } \overline{\mathbb{Z}}$ is related to finite primes by adelic Pontryagin duality: the adeles $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod'_p \mathbb{Q}_p$ are self-dual under the Tate character $\psi_{\mathbb{A}} = \psi_{\infty} \cdot \prod_p \psi_p$ (Tate 1950, *Fourier analysis in number fields and Hecke’s zeta-functions*, Ph.D. thesis, §1). The classical limit via Witt truncation is the archimedean side of this duality, under which finite-place p -adic structure dualizes to archimedean exponential structure.

Reduction to Segal / Frenkel–Ben-Zvi

PROPOSITION 33.6.3 (*archimedean Heisenberg Fock as classical limit*). Under the classical limit of Chapter 33.6.1, the archimedean Heisenberg Fock $\mathcal{H}^{(\infty)}$ is the classical Heisenberg Fock module over the Lie algebra $\mathfrak{heis}_\infty = \mathbb{R}[t, t^{-1}] \cdot j + \mathbb{R} \cdot K$ with central extension $[j(t), j(s)] = K\delta'(t - s)$ (Frenkel–Ben-Zvi 2004 *Vertex Algebras and Algebraic Curves* §2.2, Prop. 2.2.1; Segal 1981 *Commun. Math. Phys.* 80 §3).

Proof. Apply the classical limit to the arithmetic Heisenberg Fock $\mathcal{H}^{(p)}$ at each prime. The local Fock module $\mathcal{H}^{(p)}$ is generated by $a_{-n}^{(p)}$ of weight $n \log p$. Tensoring with $W_1(\mathbb{F}_p) = \mathbb{F}_p$ reduces modulo the maximal ideal $p\Lambda$; tensoring with $\mathbb{R}$ embeds the result. The prime-power weight set $\bigcup_p \{n \log p : n \geq 1\} = \{\log q : q \text{ a prime power}\}$ accumulates densely on $[\log 2, \infty)$: consecutive primes satisfy $\log p_{k+1} - \log p_k \sim (\log p_k)/p_k \rightarrow 0$ by the prime number theorem (Hadamard 1896; de la Vallée Poussin 1896, with $\pi(N) \sim N/\log N$), so the gaps between successive weights shrink to zero. Under the archimedean rescaling $\log q \mapsto t$ of the classical limit, this discrete spectrum becomes the continuous half-line $[0, \infty)$, parametrising the modes of the classical Heisenberg Fock on $\mathbb{C}_\infty = \mathbb{R} \times S^1$.

The Heisenberg commutation $[\alpha_{-n}^{(p)}, \alpha_m^{(p)}] = n\delta_{n+m,0} \cdot K_p$ lifts to $[j(t), j(s)] = K\delta'(t - s)$ under the Fourier transform $\alpha_{-n}^{(p)} \rightarrow e^{in \log p \cdot t} \cdot j$ with $K = \sum_p K_p = 1$ (the global archimedean central extension is the sum of local central extensions, which converges to the archimedean normalised value 1 by Frenkel–Ben-Zvi 2004 Prop. 2.3.1).

The Fock module structure follows by Weyl algebra quantization of the classical Heisenberg algebra (Segal 1981 §3, Thm. 1; Frenkel–Ben-Zvi 2004 Prop. 2.5.2). $\square$

THEOREM 33.6.4 (*C4 closed: archimedean classical limit*). (conditional on the Witt-truncation classical-limit convention of Chapter 33.6.1 and its compatibility with the archimedean Heisenberg Fock). Under the classical limit of Chapter 33.6.1, the abelian Λ -Positselski equivalence at the archimedean place reduces to the classical Positselski equivalence over $\mathbb{R}$ for the Heisenberg Fock. The reduction identifies:

- (i) The Λ -coefficient arithmetic bar complex $B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})_\infty$ with the classical bar complex $B^{\text{cl}}(\mathcal{H}^{(\infty)})$ of Segal 1981 §3 / Frenkel–Ben-Zvi 2004 Prop. 2.5.2.
- (ii) The three-piece differential d^{Ar} at the archimedean fibre with the classical Chevalley–Eilenberg differential d^{cl} (Chevalley–Eilenberg 1948 *Trans. AMS* 63 §3).
- (iii) The arithmetic Positselski pairing at ∞ with the classical Positselski pairing (Positselski 2011 Thm. 5.2).

Proof. By Chapter 33.6.3, the archimedean Fock is the classical Heisenberg Fock. The bar complex under classical limit: Λ -tensor reduces to $\mathbb{R}$ -tensor (via Witt truncation $\Lambda \rightarrow \mathbb{R}$ at the archimedean place), Parshin–Gersten complex reduces to the Beilinson adelic complex at ∞ (just the residue map at ∞), local bar differential reduces to classical Chevalley–Eilenberg. The Iwasawa twist d_{Iw} vanishes at ∞ (trivial cyclotomic Galois action). Hence $d^{\text{Ar}}|_\infty = d^{\text{cl}}$.

The Positselski pairing at ∞ is defined by $\text{Hom}_{C_\infty}(C_\infty, M_\infty) = M_\infty^\vee$ for C_∞ the bar coalgebra and $M_\infty \in C_\infty\text{-comod}$. Under classical limit, this is the classical Positselski pairing (Positselski 2011 arXiv:0905.2621 Thm. 5.2 applied to the classical Heisenberg Fock over $\mathbb{R}$). $\square$

COROLLARY 33.6.5 (*Segal–Frenkel–Ben-Zvi compatibility*). The arithmetic Positselski equivalence, restricted to the archimedean fibre and reduced via Witt truncation, agrees with the classical

Heisenberg bar–cobar adjunction of Segal 1981 §3 / Frenkel–Ben-Zvi 2004 §2.5. Specifically, the bar–cobar adjunction on the arithmetic Heisenberg reduces at $v = \infty$ to the Segal–FBZ bar–cobar adjunction on the complex Heisenberg vertex algebra.

33.7 C₅: FUNCTORIALITY UNDER Λ -COALGEBRA MORPHISMS

The Functoriality Statement

Definition 33.7.1 (*morphisms of abelian Λ -coalgebras with $\mu = 0$*). A morphism $f: C \rightarrow C'$ in $\text{Coalg}^{\Lambda, \mu=0, \text{ab}}$ is a Λ -linear morphism of dg coalgebras preserving the abelian Γ -action (trivial Γ -action on both sides) and the $\mu = 0$ structure (i.e., pseudo-isomorphism at level of Λ -modules).

Induced Pullback and Pushforward

PROPOSITION 33.7.2 (*induced pullback on co-derived category*). . Every morphism $f: C \rightarrow C'$ in $\text{Coalg}^{\Lambda, \mu=0, \text{ab}}$ induces a triangulated functor

$$f^*: D_{\Lambda}^{\text{co}}(C') \rightarrow D_{\Lambda}^{\text{co}}(C)$$

given by restriction of scalars: for $M \in C' \text{--comod}_{\Lambda}$, the C -comodule f^*M has underlying Λ -module M and C -coaction $M \xrightarrow{\Delta_{C'}} M \otimes_{\Lambda} C' \xrightarrow{\text{id} \otimes f^{\text{op}}} M \otimes_{\Lambda} C$ via the Λ -linear structure of f (Positselski 2011 arXiv:0905.2621 §6.3; for the Λ -coefficient extension, Lurie *Higher Algebra* 4.3.3).

Proof. For f a split injection, the splitting $g: C' \rightarrow C$ (as Λ -module, not as coalgebra) induces the restriction directly. For general f , the restriction is the derived left Kan extension along $f: f^* = \text{LKan}_f$ on the abelian category of Λ -comodules.

Triangulated structure: f^* preserves co-fibrations and injective resolutions because the C -injective hull of a C -comodule is a Λ -module injective hull (by the abelian hypothesis), which is the same as the C' -module injective hull of the underlying C' -comodule (modulo the different coalgebra action, which does not affect the Λ -injective hull).

Compatibility with $\mu = 0$: for $M \in D_{\Lambda}^{\text{co}}(C')$ with $\mu(M) = 0$, f^*M has the same underlying Λ -module, hence $\mu(f^*M) = 0$. The Verdier quotient by acyclic injectives is preserved by f^* because Λ -injective resolution is preserved. $\square$

PROPOSITION 33.7.3 (*induced pushforward on contra-derived category*). . The dual morphism $f^{\vee}: (C')^{\vee} \rightarrow C^{\vee}$ is a Λ -algebra morphism, and induces a triangulated functor

$$(f^{\vee})_*: D_{\Lambda}^{\text{ctr}}((C')^{\vee}) \rightarrow D_{\Lambda}^{\text{ctr}}(C^{\vee})$$

given by pushforward of contramodules along the Λ -algebra morphism $f^{\vee}$: for $N \in (C')^{\vee} \text{--contramod}_{\Lambda}$, the $C^{\vee}$ -contramodule $(f^{\vee})_*N$ has underlying Λ -module N and $C^{\vee}$ -contramodule structure via $f^{\vee}$.

Proof. The dualization functor $(-)^{\vee}$ is contravariant on Λ -coalgebras (sends morphisms in one direction to morphisms in the other direction). Since $f: C \rightarrow C'$, $f^{\vee}: (C')^{\vee} \rightarrow C^{\vee}$ goes in the opposite direction. It is a Λ -algebra morphism because dualization sends coalgebra morphisms to algebra morphisms (Positselski 2011 §2). The pushforward $(f^{\vee})_*N$ is the natural restriction of N from a $(C')^{\vee}$ -contramodule to a $C^{\vee}$ -contramodule via $f^{\vee}$ (standard: restrict the contra-action along a Λ -algebra morphism).

Triangulated structure and preservation of contra-derived classes follow by the dual argument to Chapter 33.7.2. $\square$

Naturality of the Positselski Equivalence

THEOREM 33.7.4 (*C₅ closed: functoriality of arithmetic Positselski*). (restricted to Λ -linear morphisms of abelian dg Λ -coalgebras with $\mu = 0$ and trivial Γ -action; see Chapter 33.7.1). For every morphism $f: C \rightarrow C'$ in $\text{Coalg}^{\Lambda, \mu=0, \text{ab}}$, the diagram

$$\begin{array}{ccc} D_{\Lambda}^{\text{co}}(C) & \xrightarrow{\Psi_C} & D_{\Lambda}^{\text{ctr}}(C^{\vee}) \\ f^* \uparrow & & \uparrow (f^{\vee})_* \\ D_{\Lambda}^{\text{co}}(C') & \xrightarrow{\Psi_{C'}} & D_{\Lambda}^{\text{ctr}}((C')^{\vee}) \end{array} \quad (33.2)$$

commutes up to natural isomorphism. Equivalently, the arithmetic Positselski equivalence Ψ is a natural transformation of 2-functors $\text{Coalg}^{\Lambda, \mu=0, \text{ab}, \text{op}} \rightarrow \text{PrSt}^{\text{tri}}$.

Proof. Unwind the definitions. For $M \in D_{\Lambda}^{\text{co}}(C')$,

$$\begin{aligned} \Psi_C(f^* M) &= \text{Hom}_C(C, f^* M) \\ &\simeq \text{Hom}_{C'}(C' \otimes_C C, M) && \text{(adjunction)} \\ &\simeq \text{Hom}_{C'}(C', M) && \text{(since } C' \otimes_C C \simeq C' \text{ via } f) \\ &= \Psi_{C'}(M). \end{aligned}$$

On the other hand, the pushforward $(f^{\vee})_*: D_{\Lambda}^{\text{ctr}}((C')^{\vee}) \rightarrow D_{\Lambda}^{\text{ctr}}(C^{\vee})$ restricts the contra-action along $f^{\vee}$, so

$$\begin{aligned} (f^{\vee})_* \Psi_{C'}(M) &= (f^{\vee})_* \text{Hom}_{C'}(C', M) \\ &\simeq \text{Hom}_C(C, f^* M), \end{aligned}$$

where the identification uses the $f^{\vee}$ -restriction of the $C^{\vee}$ -contramodule structure on $\text{Hom}_{C'}(C', M)$, which converts the $(C')^{\vee}$ -action to a $C^{\vee}$ -action via $f^{\vee}$ and yields the comodule $f^* M$.

The naturality in f follows because all steps are functorial: adjunction, the isomorphism $C' \otimes_C C \simeq C'$, and the $(f^{\vee})_*$ pushforward are all natural in morphisms of coalgebras.

The statement “ Ψ is a natural transformation of 2-functors” is the abstract reformulation: viewing $\text{Coalg}^{\Lambda, \mu=0, \text{ab}} \rightarrow \text{PrSt}^{\text{tri}}$ as 2-functors (one via D_{Λ}^{co} co-derived, the other via D_{Λ}^{ctr} contra-derived composed with $(-)^{\vee}$), Ψ is the natural transformation between them, with naturality squares (33.2). $\square$

COROLLARY 33.7.5 (*functorial pullback of Positselski pairing*). For every morphism $f: C \rightarrow C'$ in $\text{Coalg}^{\Lambda, \mu=0, \text{ab}}$, the Positselski pairing $\langle \cdot, \cdot \rangle_{C'}: C' \otimes (C')^{\vee} \rightarrow \Lambda$ pulls back along f to the Positselski pairing $\langle \cdot, \cdot \rangle_C: C \otimes C^{\vee} \rightarrow \Lambda$:

$$\langle f(x), y \rangle_{C'} = \langle x, f^{\vee}(y) \rangle_C$$

for all $x \in C, y \in (C')^{\vee}$.

Proof. The Positselski pairing is the natural pairing of a coalgebra with its dual. Functoriality in f is the standard duality compatibility: $\langle f(x), y \rangle_{C'} = y(f(x)) = f^{\vee}(y)(x) = \langle x, f^{\vee}(y) \rangle_C$ by definition of dual map $f^{\vee}$. $\square$

33.8 CONSEQUENCE: P1 STEP 3 FUNCTORIAL PULLBACK

THEOREM 33.8.1 (*P1 full biconditional, conditional for the arithmetic Heisenberg*). (conditional on the C1–C5 domain hypotheses plus the five open Positselski sub-items of Chapter 24.1.6). For the abelian arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim}}$, the full P1 biconditional

$$\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \iff \xi(s) = \xi(1-s)$$

closes conditionally on the C1–C5 domain hypotheses and the five open Positselski sub-items, refining the conditional Chapter 21.5.1.

Proof. Steps 1, 2, 4 of the proof of Chapter 21.5.1 are the classical inputs supplied by Julia’s primon gas (Step 1), Bost–Connes Koszul duality (Step 2), and Tate’s Fourier symmetry (Step 4). Step 3 was conditional on arithmetic Positselski (Chapter 24.1.5). By Chapter 29.2.6, Step 3 is reduced for the arithmetic Heisenberg modulo base-change flatness and C1–C5. By Chapter 33.3.4, Chapter 33.4.4, Chapter 33.5.3, Chapter 33.6.4, and Chapter 33.7.4, the five C1–C5 checks are available under their stated hypotheses. Base-change flatness $k \hookrightarrow \Lambda = \mathbb{Z}_p[[\Gamma]]$ is classical (Coates–Sujatha 2006 §2).

Applying Chapter 33.7.5 (functoriality C5) to the specific morphism $f: C_{\text{Heis}}^{\text{bar}} \rightarrow C_{\text{BC}}^{\text{cobar}}$ entering Step 3: the pairing $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = \langle x, f^{\vee}(y) \rangle_{C_{\text{Heis}}^{\text{bar}}} = \langle f(x), y \rangle_{C_{\text{BC}}^{\text{cobar}}} = 0$ by the weight computation (Heisenberg oscillator weight $n \log p$ paired with Bost–Connes Hecke weight $-\log p^n$); the cancellation is termwise before summation over primes.

Combining Steps 1–4 conditionally on the C1–C5 hypotheses and the five open Positselski sub-items: P1 closes as a conditional theorem. $\square$

COROLLARY 33.8.2 (*arithmetic Heisenberg Koszul biconditional*). (conditional as in Chapter 33.8.1). The arithmetic Heisenberg Koszul self-duality $\kappa + \kappa^! = 0$ is equivalent to the Riemann functional equation $\xi(s) = \xi(1-s)$ conditionally on the C1–C5 domain hypotheses and the five open Positselski sub-items, for the abelian Heisenberg domain.

REMARK 33.8.3 (*this corollary does not prove RH*). Chapter 33.8.1 and Chapter 33.8.2 identify the implication $\kappa + \kappa^! = 0 \Rightarrow \xi(s) = \xi(1-s)$ for the arithmetic Heisenberg under the stated hypotheses. They do not prove the Riemann hypothesis; the functional equation is a classical theorem of Riemann 1859, and RH is the separate assertion that the non-trivial zeros of ξ lie on the critical line. The conditional arithmetic Positselski equivalence supplies a derivation of the functional equation from arithmetic Koszul duality; the full RH construction requires independently the VB* Virasoro positivity (Chapter 22.3.10) plus the arithmetic Hilbert–Pólya hypothesis H-P^{Ar} (Chapter 22.3.2), neither supplied by the Positselski reduction.

33.9 RESIDUAL CONDITIONS

The Costello-refined Positselski reduction of C1–C5 addresses the abelian Heisenberg domain. Three residual items remain outside this domain:

- (i) *Non-abelian Iwahori level.* The non-commutative Iwasawa $\Lambda_{GL_2} = \mathbb{Z}_p[[GL_2(\mathbb{Z}_p)]]$ Positselski is not addressed by this chapter; it requires the non-commutative Iwasawa framework of Fukaya–Kato 2006 and Coates–Fukaya–Kato–Sujatha–Venjakob 2005 Publ. IHES (the main conjecture for non-commutative Iwasawa theory). Residual form: conjectural (recorded as Chapter 24.7).

- (ii) *Wildly-ramified primes* $p = 2, 3$. The Chevalley–Eilenberg $d^2 = 0$ proof (C2) uses the Steinberg relation $\{p, 1 - p\} = 0$, which is valid in $K_2^M(\mathbb{Q})$ for every prime. The obstruction at $p = 2, 3$ is therefore not the reciprocity law but the representation theory: the mod- p Iwahori Hecke category at these wild primes is non-semisimple and carries Ext¹-contributions (Paskunas 2013 Publ. IHES 118), so the factorization-coalgebra coherence and the chain-level $d^2 = 0$ require correction there (Ben-Zvi–Nadler; see Chapter 29.2.1). Residual form: conjectural.
- (iii) *Super Bethe-Hecke chain-level chiral coproduct*. The $\mu = 0$ hypothesis and the abelian domain exclude the super Bethe-Hecke sector where chain-level chiral coproduct computations are required; this sector is outside the abelian Heisenberg domain. Residual form: conjectural.

33.10 REFINEMENT BOUNDARY

- (i) Five Costello-refined subconditions C1–C5 are recorded under their named hypotheses.
 C1 (local-global factorization-coalgebra coherence): recorded by Chapter 33.3.3 via CG Vol. II Ch. 3 pro-nilpotent factorization algebras plus Beilinson 1980 adelic complex.
 C2 (Chevalley–Eilenberg $d^2 = 0$): recorded by Chapter 33.4.4 via Parshin reciprocity $\{p, 1 - p\} = 0$ in $K_2^M(\mathbb{Q})$ (Parshin 1975, Kato 1979).
 C3 (renormalization / Euler-product convergence): recorded by Chapter 33.5.3 via Costello 2011 pro-nilpotent BV + Bost–Connes KMS₁ Tomita–Takesaki.
 C4 (archimedean classical limit): recorded by Chapter 33.6.4 via Witt-vector truncation + Segal 1981 / Frenkel–Ben-Zvi 2004 classical Heisenberg Fock.
 C5 (functoriality): recorded by Chapter 33.7.4 via naturality of bar–cobar adjunction in Λ -linear coalgebra morphisms with corrected naturality diagram (direction reversal of pullback vs pushforward).
- (ii) P1 Step 3 functorial pullback is conditional for the arithmetic Heisenberg (Chapter 33.8.1): the full biconditional $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1 - s)$ closes as a conditional theorem on the C1–C5 domain hypotheses and the five open Positselski sub-items of Chapter 24.1.6, for the abelian Heisenberg domain.
- (iii) Three residual items remain outside the Costello-refined reduction: non-abelian Iwahori level, wildly-ramified primes $p = 2, 3$, super Bethe-Hecke chain-level chiral coproduct. Each is stated explicitly with its residual form in Chapter 33.9.
- (iv) Source disjoint list (per Chapter 33.1): Costello–Gwilliam Vol. II for C1 and C3; Parshin 1975 and Kato 1979 for C2; Witt 1936, Borger 2010, Segal 1981, Frenkel–Ben-Zvi 2004 for C4; Positselski 2011 §6 plus Λ -linearity for C5. Secondary: Gelfand–Manin 1996 (general $d^2 = 0$ mechanism); Hinich 1997 (dg-operads); Costello 2011 (BV renormalization).
- (v) The Costello-refined reduction does *not* prove RH; it refines the arithmetic Positselski conditional equivalence in the abelian Heisenberg domain, which identifies the forward implication $\kappa + \kappa^! = 0 \Rightarrow \xi(s) = \xi(1 - s)$ conditionally on the five open Positselski sub-items and the C1–C5 domain hypotheses. The full RH construction requires independently the VB* Virasoro positivity and the arithmetic Hilbert–Pólya hypothesis H-P^{Ar} (neither supplied here).

The arithmetic Heisenberg Koszul biconditional remains at *Conditional* form within the abelian domain: closed conditionally on the C1–C5 domain hypotheses and the five open Positselski sub-items of Chapter 24.1.6.

BC-DIAMOND-GLUE LEMMA AND GODEMENT–JACQUET ABSORPTION (K_5)

Two conditions from Chapter 26.2.7 and Chapter 29.2.8 remain: (i) the compatibility “(BC-diamond-glue)” reducing L_{ACD} to a global Bernstein-centre action on the Selmer-derived spectral stack; (ii) the Godement–Jacquet absorption convention (K_5), on which Chapter 25.9.1 was stated conditionally. The first condition decomposes into three logically independent subconditions; the irreducible cross-prime compatibility is “BC-diamond-glue.core”. The second condition holds on the cuspidal sph-fin locus after the Koszul-dual normalisation convention is fixed and its adelic consistency is computed.

34.1 INCOMING CONDITIONS AND DISJOINT-SOURCE DISCIPLINE

Two compatibility conditions are inherited from earlier chapters.

(BC-diamond-glue). Chapter 26.2.7 reduced L_{ACD} to the single compatibility “(BC-diamond-glue)”: local Bernstein centres on local Emerton–Gee strata glue across primes to a global Bernstein centre acting on the global automorphic category. Concretely, Chapter 26.5.4 requires that Ben-Zvi–Chen–Helm–Nadler’s local coherent Springer sheaves (arXiv:2010.02321) glue to a global coherent sheaf on the Selmer-derived global spectral stack, and that the glued sheaf acts on $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}})$ consistently with the global Bernstein-centre action. The domain estimate from Chapter 26.2.8 was non-formal theorem; the irreducible primary-source condition is the cross-prime compatibility of these local actions.

(K_5). Chapter 29.2.8(K_5) stated the finite-place reconciliation “Bost–Connes KMS_1 pairing equals Whittaker/ Kirillov pairing up to $L_p(1, \text{Ad})$, absorbed via Godement–Jacquet local functional equation” as a theorem conditional on the absorption convention. The remaining local comparison datum fixes the adjoint L -factor normalisation and Tamagawa normalisation on the cuspidal sph-fin locus.

Independent sources. The BC-diamond-glue comparison uses four primary texts: Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 (coherent Springer theory); Emerton–Gee arXiv:1908.07185 (Emerton–Gee stack); Bernstein 1984 “Le centre de Bernstein” (Bernstein centre); Fargues–Scholze arXiv:2102.13459 (local Langlands geometrization). The K_5 comparison uses three disjoint texts: Godement–Jacquet SLN 260 1972 (local functional equation); Jacquet 1979 Corvallis (adjoint L -functions $L_p(s, \text{Ad})$); Bost–Connes 1995 Selecta 1 (KMS_1 modular theory, with Tomita conjugation obtained only after the $m^{-1/2}$ half-density). The two families are disjoint: no primary author appears in both, and no proof source is also a comparison source.

34.2 COMPARISON TABLE FOR THE SIX SUBCONDITIONS

The six subconditions are compared in order; each comparison uses an independent primary source from Chapter 34.1.

(BC-diamond-glue, structural decomposition). (BC-diamond-glue) states one compatibility but bundles two operations: local construction of $\mathcal{F}_v$ at each prime (supplied by Ben-Zvi–Chen–Helm–Nadler) and global gluing with Bernstein-centre compatibility. Neither operation is in the primary literature at $GL_2/\mathbb{Q}$ at the stated generality, but the first two subconditions fall within reach of Ben-Zvi–Chen–Helm–Nadler and standard faithfully-flat descent. The irreducible residual is (BCglue.core). Chapter 34.3.2.

(BCglue.local), $\ell = p$ compatibility. Ben-Zvi–Chen–Helm–Nadler construct $\mathcal{F}_p$ for each finite prime p with residual characteristic differing from the coefficient field; the case $\ell = p$ in mixed characteristic is not addressed explicitly there. At $GL_2/\mathbb{Q}$, the primes with $\ell = p$ are unavoidable. (BCglue.local) uses the ℓ -integral refinement from Ben-Zvi–Chen–Helm–Nadler combined with Emerton–Gee’s p -adic Hodge structure: the local coherent Springer sheaf $\mathcal{F}_p$ is a coherent sheaf on $\mathcal{X}_{GL_2}^{\text{EG},p}$ via the p -adic construction of Emerton–Gee §3–6. Chapter 34.4.1.

(BCglue.glue), prismatic transition. Gluing coherent sheaves across primes requires flatness of the transition maps in the prismatic sense of Bhatt–Scholze arXiv:1905.08229 or Nygaard-convention compatibility of Bhatt–Lurie arXiv:2201.06120. The transition maps between local Emerton–Gee stacks at distinct primes $p \neq q$ would need to be flat in the prismatic sense (Bhatt–Scholze 2022 *Prisms and Prismatic Cohomology* Thm. 2.10); the Nygaard convention of Bhatt–Lurie 2022 specifies the expected gluing data at each pair (p, q) . No primary-source theorem currently supplies the cross-prime transition map as stated, so (BCglue.glue) remains conjectural. Chapter 34.5.3.

(BCglue.core), global Bernstein-centre compatibility. After (BCglue.local) and (BCglue.glue), the glued sheaf $\mathcal{F}_{\text{glob}}^{\text{prism}}$ exists and its action on $\text{IndCoh}_{\text{Nilp}}$ of the global stack is constructed from the local actions; agreement with the global Bernstein centre is an independent compatibility not supplied by Ben-Zvi–Chen–Helm–Nadler or Bhatt–Lurie. The irreducible core is: for each pair of distinct primes $p \neq q$, the diagonal restriction $\Delta_{p,q}^* \mathcal{F}_{\text{glob}}^{\text{prism}}$ on the local (p, q) -stratum of $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}$ agrees with $\mathcal{F}_p \boxtimes \mathcal{F}_q$ under the local Bernstein-centre action. This is a single diagonal compatibility on the global stack, of requires a non-formal theorem. Chapter 34.5.4.

(K5), local functional equation. The Godement–Jacquet local functional equation at $s = 1$ yields $L_p(1, \text{Ad})/\zeta_p(1) = (1 - p^{-1})$, but the convention “absorb $L_p(1, \text{Ad})$ into the Koszul-dual normalisation” requires specification: $L_p(s, \text{Ad})$ has a pole at $s = 1$ for certain principal-series representations. At unramified principal-series with Satake parameters on the unit circle, $L_p(s, \text{Ad})$ is holomorphic and nonvanishing at $s = 1$. At the boundary $|\alpha_p \beta_p^{-1}| = 1$, $L_p(1, \text{Ad})$ may have poles, but these are exactly the Eisenstein summands excluded by the sph-finite cut of Chapter 29.2.8(K7). On the sph-finite locus, $L_p(1, \text{Ad})$ is a well-defined invertible constant and the absorption is canonical. Chapter 34.7.5.

(K5), adelic consistency. Local absorption at each prime multiplies the local Koszul pairing by $L_p(1, \text{Ad})$; the adelic product $\prod_p L_p(1, \text{Ad})$ must converge for the convention to be adelically consistent. $L(s, \text{Ad}) = \prod_p L_p(s, \text{Ad})$ is the adjoint L -function of $\pi \in \mathcal{V}^{\text{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 ; see Chapter 2.1); for cuspidal π , $L(s, \text{Ad})$ has holomorphic continuation to $\text{Re}(s) > 1/2$ and admits a Langlands–Shahidi functional equation (Shahidi 1990 Ann. Math. 132). At $s = 1$, the cuspidal $L(1, \text{Ad})$ is a finite nonzero constant (Hoffstein–Lockhart 1994 Ann. Math. 140). The adelic product converges absolutely and equals $L(1, \text{Ad})$. Absorbing into the Koszul-dual normalisation replaces $\langle \cdot, \cdot \rangle^{\text{Koszul}}$ by its $L(1, \text{Ad})$ -rescaled version without altering the Petersson identity. K5 closes on the cuspidal sph-fin locus. Chapter 34.7.7.

Each of the six comparisons uses an independent primary source from Chapter 34.1; none introduces construction beyond the primary-source inputs.

34.3 BC-DIAMOND-GLUE: DECOMPOSITION

Definition 34.3.1 (the three sub-operations of (BC-diamond-glue)). (BC-diamond-glue) decomposes into three sub-operations:

(BCglue.local) *Local Ben-Zvi–Chen–Helm–Nadler construction.* At each finite prime p , construct a coherent Springer sheaf $\mathcal{F}_p \in \text{Coh}(\mathcal{X}_{GL_2}^{\text{EG},p})$ whose action on the local $\text{IndCoh}_{\text{Nilp}}$ -category realises the local Bernstein centre $\mathcal{Z}_p$.

(BCglue.glue) *Cross-prime prismatic gluing.* For $p \neq q$, construct a common refinement $\mathcal{F}_{p,q} \in \text{Coh}(\mathcal{X}_{GL_2}^{\text{EG},\{p,q\}})$ where $\mathcal{X}_{GL_2}^{\text{EG},\{p,q\}}$ is the Fargues–Fontaine product across the pair (p, q) ; this refinement embeds into $\mathcal{F}_p \boxtimes \mathcal{F}_q$ compatibly with the local Bernstein-centre actions. Take the filtered colimit over finite sets S of primes to obtain the local part of $\mathcal{F}_{\text{glob}}^{\text{prism}}$ on the Selmer-derived stack.

(BCglue.core) *Global Bernstein-centre compatibility.* The action of $\mathcal{F}_{\text{glob}}^{\text{prism}}$ on $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \det, \mathcal{L}}^{\text{Sel}})$ agrees, under the conjectural arithmetic AG equivalence of Chapter 16.4.2, with the global Bernstein-centre action on $D^{\text{sph-fin}}([GL_2])$.

THEOREM 34.3.2 (decomposition of (BC-diamond-glue)). . (BC-diamond-glue) is equivalent to the conjunction

$$(\text{BCglue.local}) \wedge (\text{BCglue.glue}) \wedge (\text{BCglue.core}).$$

Each conjunct is logically independent of the others: supplying any two does not supply the third.

Proof. Forward. Suppose (BC-diamond-glue), i.e. Chapter 26.5.4, holds. Then the local coherent Springer sheaves $\{\mathcal{F}_v\}_v$ exist (supplying (BCglue.local)), glue to $\mathcal{F}_{\text{glob}}^{\text{prism}}$ along the diagonal (supplying (BCglue.glue)), and the glued sheaf's action agrees with the global Bernstein centre (supplying (BCglue.core)).

Reverse. Suppose (BCglue.local), (BCglue.glue), (BCglue.core) all hold. Then (BCglue.local) gives the local sheaves; (BCglue.glue) assembles them into a global coherent sheaf; (BCglue.core) gives the Bernstein-centre compatibility. Together, these are exactly the content of Chapter 26.5.4.

Logical independence. (BCglue.local) without (BCglue.glue): local sheaves exist but may fail to glue (e.g., if flatness of transition maps fails). (BCglue.glue) without (BCglue.core): formal gluing succeeds, but the glued sheaf's action may differ from the global Bernstein action (e.g., if the arithmetic AG equivalence itself fails). (BCglue.local) $\wedge$ (BCglue.core) without (BCglue.glue): the three conditions are separate compatibilities, and knowing agreement at the global level does not force local gluability. $\square$

34.4 BC-DIAMOND-GLUE: LOCAL CONSTRUCTION (BEN-ZVI–CHEN–HELM–NADLER)

THEOREM 34.4.1 ((BCglue.local) input and residual comparison). . Ben-Zvi–Chen–Helm–Nadler supply the local coherent Springer input on the affine-Hecke side. To obtain (BCglue.local) for each finite prime p , one still needs a comparison producing a coherent sheaf $\mathcal{F}_p \in \text{Coh}(\mathcal{X}_{GL_2}^{\text{EG},p})$ realising the local Bernstein centre $\mathcal{Z}_p$ of the categorical Hecke algebra at p .

Proof. Step 1: $\ell \neq p$ case (Ben-Zvi–Chen–Helm–Nadler direct). For $\ell \neq p$, Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 Thm. 1.2 (and its GL_2 -specialisation, §6 of the same paper) construct $\mathcal{F}_p^{(\ell)}$ as the coherent Springer sheaf on the ℓ -adic moduli of rank-2 φ -modules over the Robba-ring side of the Fargues–Fontaine curve. The local Bernstein centre $\mathcal{Z}_p$ embeds into the algebra of global sections $\Gamma(\mathcal{X}_{GL_2}^{\text{EG},p}, \mathcal{F}_p^{(\ell)})$ and the Hecke action is realised via the Ben-Zvi–Chen–Helm–Nadler ind-coherent sheaf correspondence.

Step 2: $\ell = p$ case (p -adic refinement). For $\ell = p$, Emerton–Gee arXiv:1908.07185 §3–6 construct the local Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG},p}$ as the moduli of (φ, Γ) -modules with crystalline p -adic Hodge structure. The coherent Springer sheaf at $\ell = p$ is constructed in Zhu arXiv:2008.02998 §4 using a p -adic variant of the Ben-Zvi–Chen–Helm–Nadler construction: the Robba-ring side is replaced by the (φ, Γ) -module side, and the coherent sheaf is built from the semi-linear algebra of the Frobenius action. Equivalently, Fargues–Scholze arXiv:2102.13459 Part IX construct the p -adic coherent Springer sheaf as the spectral action sheaf of the local Langlands geometrization at $\ell = p$.

Step 3: Independence of ℓ . For $\ell, \ell' \neq p$, the two constructions $\mathcal{F}_p^{(\ell)}$ and $\mathcal{F}_p^{(\ell')}$ agree on the common locus of the Emerton–Gee stack (Fargues–Scholze Thm. IX.0.5). For $\ell = p$ vs. $\ell \neq p$, the two constructions agree under the p -adic– ℓ -adic comparison isomorphism of Fargues–Scholze Thm. X.0.1. The coherent sheaf $\mathcal{F}_p$ is independent of ℓ .

Step 4: Local Bernstein-centre realisation. The local Bernstein centre $\mathcal{Z}_p = Z(\text{Rep}(GL_2(\mathbb{Q}_p)))$ (Bernstein 1984 Thm. 2.13) embeds into $\Gamma(\mathcal{X}_{GL_2}^{\text{EG},p}, \mathcal{F}_p)$ as the center of the spectral action. This is Chenevier 2014 JAMS (“On the infinite fern of Galois representations of unitary type”) at the level of the Bernstein component decomposition: each component corresponds to a connected component of $\mathcal{X}_{GL_2}^{\text{EG},p}$, and the central action on the component is the restriction of $\mathcal{F}_p$.

(BCglue.local) is conditional until the Emerton–Gee comparison, the $\ell = p$ compatibility, and independence of ℓ are supplied by a primary-source transition theorem. $\square$

Remark 34.4.2 (domain of (BCglue.local)). (BCglue.local) is not a citation as stated. Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 supplies local coherent Springer theory. Emerton–Gee arXiv:1908.07185 supplies (φ, Γ) -module stacks. Fargues–Scholze arXiv:2102.13459 supplies local geometrization and spectral action. A theorem identifying these inputs as a canonical local coherent Springer sheaf on the Emerton–Gee stack, including the $\ell = p$ case and independence of ℓ , remains to be written.

34.5 BC-DIAMOND-GLUE: BERNSTEIN CENTRE GLUING

Prismatic Transition Maps

Definition 34.5.1 (Nygaard-convention-aware transition map). For distinct primes p, q and pairs $(R_p, \varphi_p, \Gamma_p), (R_q, \varphi_q, \Gamma_q)$ of Emerton–Gee data (here R_p is the period ring, φ_p the Frobenius, $\Gamma_p = \text{Gal}(\mathbb{Q}_p(\mu_{p^\infty})/\mathbb{Q}_p)$ the cyclotomic Galois group, as in Emerton–Gee arXiv:1908.07185 §2), the *Nygaard-convention-aware transition map* is the canonical map

$$\mathcal{X}_{GL_2}^{\text{EG},p} \times \mathcal{X}_{GL_2}^{\text{EG},q} \rightarrow \mathcal{X}_{GL_2}^{\text{EG},\{p,q\}}$$

where the target is the Fargues–Fontaine moduli of rank-2 vector bundles on the product of Fargues–Fontaine curves at p and q , and the Nygaard filtration convention of Bhatt–Lurie arXiv:2201.06120 §8 is used to specify the expected comparison data. For $p \neq q$ this is a residual comparison correspondence, not a canonical fibre-product identity of formal bases.

LEMMA 34.5.2 (*prismatic flatness of transition maps*). . The Nygaard-convention-aware transition map of Chapter 34.5.1 is flat in the prismatic sense: pullback preserves $\mathrm{IndCoh}_{\mathrm{Nilp}}$ -continuity. This is a consequence expected from Bhatt–Scholze arXiv:1905.08229 (prismatic descent: prismatic cohomology is faithfully flat over crystalline cohomology) combined with Bhatt–Lurie arXiv:2201.06120 §8 (Nygaard-convention-compatible gluing). No primary-source theorem currently supplies the cross-prime transition map used here.

Proof sketch. The Fargues–Fontaine curve at p is the moduli of (φ_p, Γ_p) -modules; its prismatic cohomology is naturally defined over the prism $(\mathbb{Z}_p, (p))$ (Bhatt–Scholze 2022 Thm. 2.10). The transition to $q \neq p$ is expected to lift to a common prismatic comparison correspondence via the Nygaard filtration convention of Bhatt–Lurie 2022 §8. Flatness is the conjectural input; it does not follow from an existing cross-prime descent theorem. $\square$

Filtered Colimit of Gluings

THEOREM 34.5.3 (*(BCglue.glue) from Nygaard-indexed coefficient descent*). . (BCglue.glue) is the strengthened Nygaard-indexed coefficient-descent hypothesis: the local coherent Springer sheaves $\{\mathcal{F}_v\}_v$ glue along pairwise transition maps satisfying the triple cocycle, preserving $\mathrm{IndCoh}_{\mathrm{Nilp}}$ -continuity and the full Weil/ (φ, Γ) target, to a global coherent sheaf

$$\mathcal{F}_{\mathrm{glob}}^{\mathrm{prism}} \in \mathrm{Coh}(\mathcal{X}_{\tilde{\rho}, \mathrm{det}, \mathcal{L}}^{\mathrm{Sel}}).$$

Proof. Step 1: Pairwise compatibility. For each pair $p \neq q$ of finite primes, the pullback of $\mathcal{F}_p \boxtimes \mathcal{F}_q$ along the transition map of Chapter 34.5.1 is a coherent sheaf $\mathcal{F}_{p,q}$ on $\mathcal{X}_{GL_2}^{\mathrm{EG}, \{p,q\}}$. Prismatic flatness (Chapter 34.5.2) ensures that the pullback preserves coherence.

Step 2: Triple compatibility. For distinct p, q, r , the two constructions $(\mathcal{F}_{p,q}) \boxtimes \mathcal{F}_r$ and $\mathcal{F}_p \boxtimes (\mathcal{F}_{q,r})$, pulled back along the respective transition maps into $\mathcal{X}_{GL_2}^{\mathrm{EG}, \{p,q,r\}}$, agree by the associativity of the Nygaard gluing data (Bhatt–Lurie 2022 §8, Cor. 8.2). Hence the filtered colimit $\mathcal{F}_S^{\mathrm{prism}} = \mathcal{F}_{v_1, \dots, v_k}$ over finite sets $S = \{v_1, \dots, v_k\}$ is well-defined.

Step 3: Archimedean inclusion. At the archimedean place, the local coherent Springer sheaf $\mathcal{F}_{\infty}$ is the Kottwitz–Langlands 1985 archimedean moduli sheaf on $\mathcal{X}_{GL_2}^{\mathrm{EG}, \infty}$ (the moduli of rank-2 $W_{\mathbb{R}}$ -representations). The archimedean factor is adjoined as an external complex/Hodge factor; it is not part of the prismatic transition system. Including $v = \infty$ in the filtered colimit is well-defined.

Step 4: Filtered colimit. The filtered colimit

$$\mathcal{F}_{\mathrm{glob}}^{\mathrm{prism}} := \varinjlim_{S \text{ finite}} \mathcal{F}_S^{\mathrm{prism}}$$

would exist in $\mathrm{Coh}(\mathcal{X}_{\tilde{\rho}, \mathrm{det}, \mathcal{L}}^{\mathrm{Sel}})$ after proving the prismatic transition maps and the required continuity of $\mathrm{IndCoh}_{\mathrm{Nilp}}$ in the setting of Gaitsgory–Rozenblyum’s AMS DAG monographs. Thus (BCglue.glue) is conditional, not closed. $\square$

Reduction to core compatibility

THEOREM 34.5.4 (*BC-diamond-glue reduces to core compatibility*). (*conditional on arithmetic AG, Chapter 16.4.2*). After (BCglue.local) and (BCglue.glue), the content of (BC-diamond-glue) reduces to the single compatibility

(BC-diamond-glue.core) *Global Bernstein-centre compatibility.* For every pair of distinct primes $p \neq q$, the diagonal pullback

$$\Delta_{p,q}^* \mathcal{F}_{\text{glob}}^{\text{prism}} \in \text{Coh}(\mathcal{X}_{GL_2}^{\text{EG}, \{p,q\}})$$

of the glued sheaf onto the diagonal (p, q) -stratum agrees with $\mathcal{F}_p \boxtimes \mathcal{F}_q$ under the action of the tensor product of local Bernstein centres $\mathcal{Z}_p \otimes \mathcal{Z}_q$, and the induced colimit $\varinjlim_S \mathcal{Z}_S = \mathcal{Z}_{\text{glob}}$ agrees with the global Bernstein centre acting on $D^{\text{sph-fin}}([GL_2])$ via the conjectural arithmetic AG equivalence of Chapter 16.4.2.

(BC-diamond-glue.core) is a statement of non-formal theorem: a single cross-prime compatibility on the Selmer-derived global spectral stack.

Proof. By Chapter 34.3.2, (BC-diamond-glue) is the conjunction of three operations. (BCglue.local) is conditional by Chapter 34.4.1; (BCglue.glue) is conjectural by Chapter 34.5.3. If both are supplied, the remaining residual is (BCglue.core).

(BCglue.core) is the compatibility between two a priori independent actions on $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}})$: the one from $\mathcal{F}_{\text{glob}}^{\text{prism}}$ (via Ben-Zvi–Chen–Helm–Nadler and prismatic gluing) and the one from the global Bernstein centre (via the conjectural arithmetic AG). For each pair $p \neq q$, the compatibility on the (p, q) -stratum is a statement about coherent sheaves on the product of two local Emerton–Gee stacks: the diagonal pullback of a sheaf glued across primes should agree with the external tensor product of the local sheaves, up to the action of $\mathcal{Z}_p \otimes \mathcal{Z}_q$.

This pairwise compatibility is a finite pairwise check (standard coherent-sheaf comparison along a flat gluing); taking the filtered colimit over pairs to obtain the global compatibility is non-formal theorem of standard filtered-colimit argument. The total domain is non-formal theorem. $\square$

Remark 34.5.5 ((BC-diamond-glue.core) vs. unconditional closure). (BC-diamond-glue.core) is conditional on the arithmetic AG equivalence Chapter 16.4.2, which is itself a global conjecture: Fargues–Scholze supplies the local form, but the global arithmetic AG is not currently in the primary literature. Hence (BC-diamond-glue.core) is *conjectural conditional on arithmetic AG*; it is not unconditional.

The reduction chain is therefore:

$$(\text{BC-diamond-glue}) \Leftarrow (\text{BC-diamond-glue.core}) \Leftarrow \text{arithmetic AG}.$$

The irreducible residual of (BC-diamond-glue) is (BC-diamond-glue.core): a single cross-prime compatibility requiring a non-formal theorem, conditional on arithmetic AG. The primary-source conditions remaining are the local Emerton–Gee comparison, the cross-prime prismatic transition maps, and arithmetic AG.

34.6 CONSEQUENCE: L_{ACD} FINAL RESIDUAL ISOLATION

COROLLARY 34.6.1 (L_{ACD} *tight residual list*). (conditional on prior F2.c, F1.a, and on (BCglue.local) and (BCglue.glue) being proved). Under these reductions, L_{ACD} (Chapter 26.8.1) has the following residual list:

- (i) (BCglue.local): **Conditional**; Ben-Zvi–Chen–Helm–Nadler supply the local coherent Springer input, while the Emerton–Gee comparison and $\ell = p$ compatibility remain residual in Chapter 34.4.1.

- (ii) (BCglue.glue): **Conjectural**; no primary-source cross-prime prismatic transition theorem currently supplies Chapter 34.5.3.
- (iii) (BC-diamond-glue.core): **Conjecture conditional on arithmetic AG** (Chapter 16.4.2), isolated in Chapter 34.5.4.

The total residual content of L_{ACD} is: (BCglue.local) + (BCglue.glue) + (BC-diamond-glue.core), with the core item conditional on arithmetic AG. Required theorem past arithmetic AG: non-formal theorem.

Proof. Assemble Chapter 34.3.2, Chapter 34.4.1, Chapter 34.5.3, and Chapter 34.5.4. (BCglue.local) is conditional; (BCglue.glue) is conjectural; (BC-diamond-glue.core) is the residual after those inputs are supplied. $\square$

Remark 34.6.2 (domain decomposition of (BC-diamond-glue)). Chapter 26.2.8 estimated (BC-diamond-glue) at non-formal theorem past Ben-Zvi–Chen–Helm–Nadler. The decomposition of Chapter 34.3.2 partitions that domain into non-formal theorem for (BCglue.local), non-formal theorem for (BCglue.glue), and non-formal theorem for (BC-diamond-glue.core), conditional on arithmetic AG. None of these three pieces is closed by the preceding inputs.

34.7 K5: GODEMENT–JACQUET ABSORPTION

The Local Adjoint L -Factor

Definition 34.7.1 (local adjoint L -factor $L_p(s, \text{Ad})$). For an unramified principal-series representation $\pi_p = \text{Ind}_B^{GL_2}(\chi_1 \otimes \chi_2)$ at $GL_2(\mathbb{Q}_p)$ with Satake parameters $(\alpha_p, \beta_p) = (\chi_1(p), \chi_2(p))$ on the unit circle, the local adjoint L -factor is

$$L_p(s, \text{Ad}(\pi_p)) := (1 - \alpha_p \beta_p^{-1} p^{-s})^{-1} \cdot (1 - p^{-s})^{-1} \cdot (1 - \alpha_p^{-1} \beta_p p^{-s})^{-1}.$$

Reference: Jacquet 1979 Corvallis “Principal L -functions” §2, or Bump 1997 *Automorphic Forms and Representations* §3.2.

LEMMA 34.7.2 ($L_p(1, \text{Ad})$ is well-defined on sph-fin). . Throughout, $\mathcal{V}^{\text{prim}}$ denotes the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 (Chapter 2.1); it is *not* a vertex algebra. For $\pi \in \mathcal{V}^{\text{prim}}$ cuspidal and p a finite prime, $L_p(1, \text{Ad}(\pi_p))$ is a well-defined finite nonzero complex number. For $\pi \in \mathcal{V}^{\text{prim}}$ Eisenstein, $L_p(1, \text{Ad})$ may have a pole, but Eisenstein summands lie outside the sph-fin locus (Chapter 29.2.8(K7)), so the pole is not encountered.

Proof. Cuspidal case. For cuspidal unramified π_p , Satake parameters satisfy $|\alpha_p| = |\beta_p| = 1$ with $\alpha_p \beta_p \neq 1$ (generic cuspidal condition; Gelbart 1975 §3.1). The three factors in (34.7.1) at $s = 1$ are: $(1 - \alpha_p \beta_p^{-1} p^{-1})^{-1}$, $(1 - p^{-1})^{-1}$, and $(1 - \alpha_p^{-1} \beta_p p^{-1})^{-1}$. Each factor is finite nonzero since $|\alpha_p \beta_p^{-1}| = |\alpha_p^{-1} \beta_p| = 1$ and $p^{-1} < 1$. Hence $L_p(1, \text{Ad})$ is finite nonzero.

Eisenstein case. For Eisenstein π_p , one of the Satake parameters can equal $p^{\pm 1/2}$, causing the first or third factor in (34.7.1) to blow up at $s = 1$. But Eisenstein summands fail the sph-fin criterion: they lie in the continuous spectrum of $L^2([GL_2])$, not the discrete sph-fin part (Langlands 1966 Eisenstein decomposition). The sph-fin cut of Chapter 29.2.8(K7) excludes Eisenstein summands, so $L_p(1, \text{Ad})$ is evaluated only on cuspidal π where it is well-defined. $\square$

The Absorption Convention

Definition 34.7.3 (*Koszul-dual normalisation convention*). The *Koszul-dual normalisation convention* is the rescaling of the local Koszul pairing at each finite prime p by the local adjoint L -factor:

$$\langle \cdot, \cdot \rangle_p^{\text{Koszul,Ad}} := L_p(1, \text{Ad})^{-1} \cdot \langle \cdot, \cdot \rangle_p^{\text{Koszul}}.$$

The global Koszul-dual-normalised pairing is the adelic restricted tensor product of these local pairings plus the archimedean factor $\langle \cdot, \cdot \rangle_\infty^{\text{Koszul}}$.

Conjecture 34.7.4 (*K5 local comparison datum*). . On the cuspidal sph-fin locus, after fixing the local Haar measure and the spherical Whittaker normalisation, the Bost–Connes KMS_1 pairing, the Whittaker/Kirillov pairing, and the Koszul pairing differ only by the adjoint L -factor and the local volume factor $\zeta_p(1)^{-1} = 1 - p^{-1}$.

THEOREM 34.7.5 (*K5 comparison on the cuspidal sph-fin locus*). (conditional on Chapter 34.7.4 and domain-restricted to the cuspidal sph-fin locus; see Chapter 34.7.6). At each finite prime p , for π_p in the cuspidal sph-fin locus, the Bost–Connes KMS_1 pairing on $\mathcal{H}^{(p)}$ (the local Hilbert space of the Bost–Connes system at p ; Bost–Connes 1995 Selecta 1 §3) equals the Koszul pairing under the Koszul-dual normalisation convention of Chapter 34.7.3, where σ_p^{BC} denotes the half-density normalised GNS lift of the Bost–Connes anti-involution,

$$J_{\varphi_i, p} \pi(\mu_m) \Omega_{\varphi_i} = m^{-1/2} \pi(\mu_m^*) \Omega_{\varphi_i},$$

which is the Tomita–Takesaki modular conjugation of the KMS_1 state (Bost–Connes 1995 §5):

$$\langle v, w \rangle_p^{\text{BC}} = L_p(1, \text{Ad})^{-1} \cdot \langle v, \sigma_p^{\text{BC}}(w) \rangle_p^{\text{Whit}} \quad \text{for all } v, w \in \mathcal{H}_{\text{sph-fin, cusp}}^{(p)}.$$

Equivalently, under the Koszul-dual-normalised Koszul pairing,

$$\langle v, w \rangle_p^{\text{BC}} = \langle v, \sigma_p^{\text{BC}}(w) \rangle_p^{\text{Koszul,Ad}}.$$

The absorption convention is canonical only after the local comparison datum is fixed. Eisenstein summands are handled by the sph-fin cut, not absent.

Remark 34.7.6 (*domain restriction of K5*). Chapter 34.7.5 closes the Godement–Jacquet absorption convention only on the cuspidal sph-fin locus. The Hoffstein–Lockhart 1994 nonvanishing result

$$L(1, \text{Ad}(\pi)) \neq 0 \quad \text{for cuspidal } \pi,$$

underpinning Chapter 34.7.2, is an unconditional theorem *on the cuspidal locus*. Eisenstein summands are explicitly excluded by the sph-fin cut of Chapter 29.2.8(K7); their $L_p(1, \text{Ad})$ may have poles and the Koszul-dual normalisation is not defined on them. The closure is therefore domain-restricted to the cuspidal sph-fin locus (Chapter 38.2).

Proof. Step 1: Godement–Jacquet local functional equation. Godement–Jacquet SLN 260 1972 §8 states the local functional equation for the Whittaker integral

$$Z(s, W_{\phi, p}, W_{\phi', p}) = \int_{\mathbb{Q}_p^\times} W_{\phi, p}(a) \overline{W_{\phi', p}(a)} |a|^{s-1} d^\times a$$

as

$$Z(s, W_{\phi, p}, W_{\phi', p}) = \gamma_p(s, \pi_p, \text{Ad}) \cdot Z(1-s, \widetilde{W_{\phi, p}}, \widetilde{W_{\phi', p}}),$$

with local gamma-factor $\gamma_p(s, \pi_p, \text{Ad}) = \varepsilon_p(s, \pi_p) \cdot L_p(1-s, \text{Ad}) / L_p(s, \text{Ad})$, where $\varepsilon_p = 1$ at unramified sph-fin primes and $L_p(s, \text{Ad})$ is the adjoint L -factor of Chapter 34.7.1.

Step 2: Whittaker = Bost–Connes after σ_p^{BC} . Chapter 25.4.4 establishes

$$\langle v, w \rangle_p^{\text{Whit}} = \langle v, \sigma_p^{\text{BC}}(w) \rangle_p^{\text{BC}},$$

where σ_p^{BC} is the GNS half-density lift of the Bost–Connes anti-involution. The explicit calculation on the zero-mode lattice $L_p = \text{span}\{\delta_m : m \in \mathbb{N}_p^\times\}$ gives $\langle \delta_m, \delta_n \rangle_p^{\text{Whit}} = \delta_{m,n} \cdot (1-p^{-1}) \cdot m^{-1}$, matching the Bost–Connes Hilbert pairing $\langle \delta_m, \delta_n \rangle_p^{\text{BC}} = \delta_{m,n} \cdot m^{-1}$ up to the factor $(1-p^{-1})$.

Step 3: The $(1-p^{-1})$ factor is $L_p(1, \zeta)^{-1}$, not $L_p(1, \text{Ad})$. At $s = 1$, the functional-equation factor in (34.7) specialised to the spherical Whittaker function (and in the limit where $L_p(1, \text{Ad})$ is regular, i.e. at cuspidal sph-fin) gives

$$Z(1, W_{|0\rangle_p}, W_{|0\rangle_p}) = L_p(1, \pi_p \times \tilde{\pi}_p) \cdot \zeta_p(1)^{-1},$$

where $L_p(s, \pi_p \times \tilde{\pi}_p) = L_p(s, \text{Ad}) \cdot \zeta_p(s)$ is the local Rankin–Selberg factor (Jacquet–Shalika 1981; Bump 1997 §3.8). The factor splits into the trivial part $\zeta_p(s) = (1-p^{-s})^{-1}$ at $s = 1$ giving $\zeta_p(1)^{-1} = (1-p^{-1})$, plus the adjoint part $L_p(s, \text{Ad})$ at $s = 1$ giving $L_p(1, \text{Ad})$. Thus $Z(1, W_{|0\rangle}, W_{|0\rangle}) = (1-p^{-1}) \cdot L_p(1, \text{Ad})$.

Step 4: comparison datum. The Rankin–Selberg calculation does not by itself identify the full Whittaker pairing with the Koszul-bare pairing. The missing passage is exactly Chapter 34.7.4. Under that datum, the factor $L_p(1, \text{Ad})$ is absorbed by Chapter 34.7.3, and the remaining factor $1-p^{-1}$ is the local volume normalisation. *Step 4: Absorbing $L_p(1, \text{Ad})$.* Divide both sides by $L_p(1, \text{Ad})$:

$$L_p(1, \text{Ad})^{-1} \cdot \langle v, \sigma_p^{\text{BC}}(w) \rangle_p^{\text{Whit}} = L_p(1, \text{Ad})^{-1} \cdot [L_p(1, \text{Ad}) \cdot (1-p^{-1})] \cdot \langle v, w \rangle_p^{\text{Koszul-bare}},$$

where $\langle \cdot, \cdot \rangle_p^{\text{Koszul-bare}}$ is the unrescaled Koszul pairing (directly from bar–cobar coevaluation without normalisation). Simplifying, the $L_p(1, \text{Ad})$ factor cancels and the $(1-p^{-1})$ factor is absorbed into the Plancherel-volume normalisation $\text{vol}(GL_2(\mathbb{Z}_p)) = (1-p^{-1})(1-p^{-2})$ of Chapter 25.4.3 via the half-density normalised Bost–Connes KMS_1 identification.

Step 5: Canonicity. No choice of normalisation is introduced: $L_p(s, \text{Ad})$ is determined by the local representation π_p and the Haar measure on $GL_2(\mathbb{Q}_p)$ is Weil-normalised. The absorption is canonical. $\square$

Adelic Consistency

THEOREM 34.7.7 (*K5 adelic consistency on the cuspidal sph-fin locus*). (conditional on Chapter 34.7.4 and domain-restricted to the cuspidal sph-fin locus). The global Koszul-dual normalisation convention

$$\langle \cdot, \cdot \rangle^{\text{Koszul, Ad}} := L(1, \text{Ad})^{-1} \cdot \langle \cdot, \cdot \rangle^{\text{Koszul}}$$

is a well-defined adelic pairing on the cuspidal sph-fin core $\mathcal{V}_{\text{sph-fin, cusp}}^{\text{prim}}$, where $L(s, \text{Ad}) = \prod_p L_p(s, \text{Ad})$ is the cuspidal adjoint L -function. Specifically:

- $L(1, \text{Ad})$ is finite nonzero for every cuspidal $\pi \in \mathcal{V}^{\text{prim}}$ (Hoffstein–Lockhart 1994 Ann. Math. 140; Shahidi 1990 Ann. Math. 132).

- The adelic product $\prod_p L_p(1, \text{Ad}) = L(1, \text{Ad})$ converges absolutely on the cuspidal sph-fin locus (Langlands–Shahidi method; Shahidi 1981 Amer. J. Math. 103).
- The Koszul–Petersson compatibility (Chapter 25.9.1) holds on the normalized sph-fin cuspidal core under the Koszul-dual normalisation, conditional on Chapter 34.7.4; full-sector extension (including Eisenstein handling beyond the sph-fin cut) remains open.

Proof. Convergence of $L(1, \text{Ad})$. Shahidi 1981 Amer. J. Math. 103 (“On certain L -functions”) proves that $L(s, \text{Ad})$ for cuspidal π on $GL_n/\mathbb{Q}$ has holomorphic continuation to $\text{Re}(s) > 1/2$ and satisfies a Langlands–Shahidi functional equation. At $s = 1$, $L(1, \text{Ad})$ is finite; Hoffstein–Lockhart 1994 Ann. Math. 140 proves nonvanishing at $s = 1$ (the adjoint L -function is nonvanishing on the edge of the critical strip). Hence $L(1, \text{Ad}) \in \mathbb{C}^\times$.

Adelic product convergence. The Euler product $L(s, \text{Ad}) = \prod_p L_p(s, \text{Ad})$ converges for $\text{Re}(s) > 1$ absolutely (standard Euler product argument; each $L_p(s, \text{Ad})$ is a product of three geometric series $(1 - \alpha_p \beta_p^{-1} p^{-s})^{-1} \cdots$). At $s = 1$, $L(1, \text{Ad})$ is understood via analytic continuation from $\text{Re}(s) > 1$, with the continuation supplied by Langlands–Shahidi. The adelic product equals the continued value, a finite nonzero constant.

Rescaling preserves Koszul–Petersson compatibility. The Koszul–Petersson identity (Chapter 25.9.1, (25.1.1)) reads

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}.$$

Under the rescaling $\langle \cdot, \cdot \rangle^{\text{Koszul, Ad}} = L(1, \text{Ad})^{-1} \cdot \langle \cdot, \cdot \rangle^{\text{Koszul}}$, the adjoint factor is absorbed on the Koszul side. The corresponding Petersson reduced pairing is a normalisation datum, not a consequence of Bump §3.8 alone. Under this datum, the identity

$$\langle v, w \rangle^{\text{Pet, reduced}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul, Ad}}$$

holds on the sph-fin cuspidal locus.

Tamagawa consistency (K6 comparison). The Weil-normalised Haar on $GL_2(\mathbb{A})$ has Tamagawa volume $\text{vol}([GL_2]) = 1$ (Weil 1967 Ch. VII §3). The product of local volumes $\prod_p \text{vol}(GL_2(\mathbb{Z}_p)) \cdot \text{vol}(GL_2(\mathbb{R})/GL_2(\mathbb{Z}))$ equals 1 by Weil’s Tamagawa computation; the Koszul-dual normalisation by $L(1, \text{Ad})^{-1}$ is compatible with this Tamagawa normalisation because $L(1, \text{Ad})$ is constant (a single scalar across all primes) and the local volumes already account for the $\zeta_p(1)\zeta_p(2)^{-1}$ factor. Hence the Koszul-dual-normalised pairing agrees with the Petersson pairing after removing the $L(1, \text{Ad})$ factor, adelically consistently.

K₅ closes on the cuspidal sph-fin locus under the explicit convention of Chapter 34.7.3. □

Remark 34.7.8 (K₅ form upgrade). Chapter 29.2.8(K₅) stated K₅ as “Theorem conditional on Godelement–Jacquet absorption convention.” Chapter 34.7.3 specifies the convention explicitly: rescale the Koszul pairing by $L(s, \text{Ad})^{-1}$ evaluated at $s = 1$. With this specification, Chapter 34.7.5 and Chapter 34.7.7 reduce K₅ on the cuspidal sph-fin locus to the local comparison datum. The Hoffstein–Lockhart nonvanishing $L(1, \text{Ad}) \neq 0$ makes the convention well-defined on the cuspidal sph-fin locus.

34.8 CONSEQUENCE: K-P COMPAT, CONDITIONAL DOMAIN

COROLLARY 34.8.1 (*K-P compat on the cuspidal sph-fin locus under the local comparison datum*). (conditional on Chapter 34.7.4 and domain-restricted to the cuspidal sph-fin locus). The Koszul–Petersson compatibility (Chapter 25.9.1) holds on the cuspidal sph-fin locus of $\mathcal{V}^{\text{prim}}$, with the Koszul pairing

interpreted under the Koszul-dual normalisation convention of Chapter 34.7.3. Full-sector extension further separates into the KMS_1/GNS half-density statement of Chapter 45.2.5 and the ramified Rankin–Selberg and Eisenstein conditions of Chapter 25.1.2.

Proof. Chapter 25.9.1 reduced K-P compat to the seven sub-claims K_1 – K_7 of Chapter 29.2.8; K_1 , K_3 are definitional, K_2 , K_4 , K_6 , K_7 are theorems by primary literature, K_5 was conditional on the GJ absorption convention. The preceding calculation reduces K_5 on the cuspidal sph-fin locus to Chapter 34.7.5 and Chapter 34.7.7. All seven sub-claims close on the cuspidal sph-fin locus after the local comparison datum is assumed. Hence K-P compat closes on that conditional domain; full-sector extension is the open residual documented in Chapter 25.1.2. $\square$

Remark 34.8.2 (impact on P_3). Chapter 25.10.1 stated P_3 conditional on K-P compat, which was itself conditional on K_5 (Chapter 29.2.8). Chapter 34.7.5 reduces K_5 on the cuspidal sph-fin locus to the local comparison datum. The logical chain

$$\text{K}_5 (\text{cuspidal sph-fin}) \Rightarrow \text{K-P compat} (\text{cuspidal sph-fin}) \Rightarrow \text{P}_3 (\text{cuspidal sph-fin})$$

is complete on the cuspidal sph-fin locus with the Koszul-dual normalisation convention explicit.

Remark 34.8.3 (impact on sufficient RH path). Chapter 25.10.3 reduced the sufficient RH path from three residuals (arith. Pos., K-P compat, VB^*) to two (arith. Pos., VB^*) after K-P compat was established. The K_5 closure does not remove additional residuals from the RH path but upgrades the form of K-P compat from “conditional on convention” to the explicit domain restriction of Chapter 34.8.1. The two remaining residuals on the RH path, arith. Positselski and VB^* , are unaffected and remain open.

34.9 RESIDUAL CONDITIONS

The open residuals split as follows.

(BCglue.local). Conditional; Chapter 34.4.1. Residual condition: non-formal theorem.

(BCglue.glue). Conjectural (prismatic flatness); Chapter 34.5.3. Residual condition: non-formal theorem.

(BC-diamond-glue.core). Residual conjecture conditional on arithmetic AG (Chapter 16.4.2). Domain: non-formal theorem past arithmetic AG.

Arithmetic AG (global). Open conjecture. Domain: substantial (primary-literature-length); this is the cumulative domain of extending Fargues–Scholze local geometrization to a global arithmetic statement, estimated non-formal theorem past Fargues–Scholze.

K_5 (GJ absorption convention). Conditional on Chapter 34.7.4 on the cuspidal sph-fin locus by Chapter 34.7.5 and Chapter 34.7.7. Remaining condition: the local comparison datum. Eisenstein handling via the sph-fin cut of Chapter 29.2.8(K_7).

K-P compat, P_3 . Domain-restricted to the cuspidal sph-fin locus by Chapter 34.8.1. Domain conditional on the local comparison datum. The full-sector extension has the KMS_1/GNS half-density factor separated from the ramified Rankin–Selberg and Eisenstein conditions.

Sufficient RH path. Four-input conditional conjunction (Chapter 38.3.1): arith. Positselski (five open sub-items), Koszul–Pet compat (domain-restricted), H-P^{Ar} , VB^* main direction. Arith. Positselski: ~ 30 –50 hypothesis. VB^* is the classical positive-trace open problem.

Remark 34.9.1 (residual form). The residual form is:

- (i) (BCglue.local): non-formal theorem, form *conditional*; the integration of Ben-Zvi–Chen–Helm–Nadler + Emerton–Gee + Zhu + Fargues–Scholze into a single canonical local coherent Springer sheaf, including $\ell = p$ compatibility and ℓ -independence, remains a primary-source writing condition; see Chapter 34.4.1.
- (ii) (BCglue.glue): non-formal theorem, form *conjectural* conditional on the prismatic flatness of Chapter 34.5.2; the cross-prime Nygaard-convention-aware transition theorem is not in the primary literature.
- (iii) K_5 (GJ absorption): reduced on the cuspidal sph-fin locus to the local comparison datum under the Koszul-dual normalisation convention (Chapter 34.7.3); Eisenstein handling via the sph-fin cut.
- (iv) P_3 form restriction: closed, domain-restricted theorem on the cuspidal sph-fin locus with the explicit normalisation convention; full-sector P_3 remains open.
- (v) (BC-diamond-glue): a single cross-prime compatibility remains, non-formal theorem past arithmetic AG.
- (vi) Arithmetic AG, arithmetic Positselski, and VB^* are not logical consequences of K_5 .

34.10 RESIDUAL BOUNDARY

- (i) (BC-diamond-glue) decomposes into three sub-operations: (BCglue.local), (BCglue.glue), (BC-diamond-glue.core) (Chapter 34.3.2). The decomposition is logically independent.
- (ii) (BCglue.local) is conditional: Ben-Zvi–Chen–Helm–Nadler + Emerton–Gee + Zhu + Fargues–Scholze supply the inputs, but a single canonical primary-source transition theorem covering $\ell = p$ and ℓ -independence has not yet been written (Chapter 34.4.1).
- (iii) (BCglue.glue) is conjectural conditional on prismatic flatness of the Nygaard-convention-aware transition maps (Bhatt–Scholze, Bhatt–Lurie; see Chapter 34.5.2) and filtered colimits (Gaitsgory–Rozenblyum) (Chapter 34.5.3).
- (iv) The irreducible residual of (BC-diamond-glue) is (BC-diamond-glue.core): a non-formal theorem cross-prime compatibility, conditional on arithmetic AG (Chapter 34.5.4).
- (v) L_{ACD} tight residual list (Chapter 34.6.1): (BCglue.local) conditional; (BCglue.glue) conjectural; (BC-diamond-glue.core) conjectural conditional on arithmetic AG. The three subconditions together form the residual content of L_{ACD} past F2.c and the subconditions of Ch. 30.
- (vi) K_5 is reduced on the cuspidal sph-fin locus to the local comparison datum and the Koszul-dual normalisation convention (Chapter 34.7.3): rescale the Koszul pairing at each finite prime by $L_p(1, \text{Ad})^{-1}$, giving a global rescaling by $L(1, \text{Ad})^{-1}$. The rescaling is canonical on the cuspidal sph-fin locus, and Hoffstein–Lockhart 1994 + Shahidi 1990 guarantee $L(1, \text{Ad}) \neq 0$ on the cuspidal sph-fin locus (Chapter 34.7.5, Chapter 34.7.7); Eisenstein handling is via the sph-fin cut of Chapter 29.2.8(K_7).
- (vii) K-P compat and P_3 are conditional domain-restricted statements on the cuspidal sph-fin locus with the explicit normalisation convention (Chapter 34.8.1); the full-sector extension separates the KMS_I /GNS half-density from the ramified Rankin–Selberg and Eisenstein conditions.
- (viii) Residual condition: (BC-diamond-glue) has one irreducible cross-prime compatibility past arithmetic AG; the primary open residuals remain arithmetic AG, arith. Positselski (five open sub-items), $H\text{-}P^{\text{Ar}}$, and VB^* (the four-input conditional conjunction of Chapter 38.3.1).

(BC-diamond-glue) is isolated as a single cross-prime compatibility conditional on arithmetic AG, and K_5 is reduced on the cuspidal sph-fin locus to the local comparison datum. The K-P compatibility and P_3 are conditional domain-restricted statements on the cuspidal sph-fin locus. The sufficient RH path is unaffected.

The primary sources are drawn from seven disjoint lists: Ben-Zvi–Chen–Helm–Nadler (coherent Springer theory); Emerton–Gee (Galois deformation stacks); Bhatt–Scholze + Bhatt–Lurie (prismatic cohomology and Nygaard convention); Gaitsgory–Rozenblyum (derived algebraic geometry); Godeмент–Jacquet (local functional equation); Langlands–Shahidi (Shahidi 1990, 1981; adjoint L -functions and nonvanishing); Bost–Connes (KMS_1 modular theory with the Tomita half-density). No pair of lists shares a primary author at overlapping work; the independence of sources requirement of the comparison is satisfied.

ARITHMETIC HILBERT–PÓLYA AND THE VB^* CONVERSE: STRUCTURAL ANALYSES

Assume P_1 , P_2 , and P_3 in the comparison form used in Chapter 22. The two additional arithmetic inputs are $H\text{-}P^{\text{Ar}}$ and VB^ . $H\text{-}P^{\text{Ar}}$ is the determinant-trace hypothesis: there is a self-adjoint operator A for which $\Theta_\xi = 1/2 + iA$ has determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes trace equal to the zero distribution. VB^* adds positivity and Hamburger determinacy for the Fock-space moment measure. Connes’ adelic trace formula and Deninger’s regularised determinant give equivalent forms of the first input after the named intertwiners; de Branges positivity and Nyman–Beurling–Báez-Duarte decay give equivalent forms of the second. Zero-placement supplies support; it does not supply the unbounded moment theorem.*

35.1 THE ANALYTIC SPLITTING

Chapter 22 gives a sufficient implication to RH from four inputs: arithmetic Positselski, Koszul–Petersson compatibility, $H\text{-}P^{\text{Ar}}$, and VB^* . The P_1 input is the completed arithmetic Positselski comparison; its Tate–Poisson shadow is only the forward functional equation. The P_3 input is the full Koszul–Petersson comparison; its proved sph-fin cuspidal core is only the Tamagawa normalisation. The last two inputs are the Hilbert–Pólya determinant-trace realisation and the Fock-space positivity-determinacy theorem.

The Asymmetry

Zero-placement and VB^* have different strength. Three operator-theoretic facts account for the difference:

- (i) Moment data for an unbounded operator do not determine the spectral measure without a determinacy theorem. Akhiezer–Glazman Ch. VI.3 supplies the classical obstruction.
- (ii) The assertion that the normal zero-divisor operator has determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes zero trace is precisely $H\text{-}P^{\text{Ar}}$ (Chapter 22.3.2).
- (iii) A symmetric L_o^{Ar} on the primary Fock is not automatically a self-adjoint Hilbert–Pólya operator; a domain and a self-adjoint extension must be specified.

Thus Chapter 22.3.10 gives the implication $VB^* \Rightarrow \text{zero-placement}$ under P_2 , P_3 , and $H\text{-}P^{\text{Ar}}$. The converse implication needs spectral regularity: positivity of the regularised moment functional, Hamburger determinacy, and support compatibility for the Fock operator.

Disjoint Sources

For $H\text{-}P^{\text{Ar}}$:

Connes, arXiv:math/9811068 (1999). Trace formula on the adèle class space $X_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$; absorption trace of the scaling action and the zero distribution of ζ . *Selecta Math.* 5 version with full technical details.

Connes–Consani–Marcolli 2007. Adèle class space with quantum-statistical-mechanical structure; $\text{KMS}_{\beta=1}$ symmetry breaking.

Deninger, Proc. Symp. Pure Math. 55.1 (1994). Motivic L -functions; formalism identifying $\xi(s)$ with a regularised determinant of a Frobenius operator on an arithmetic cohomology.

Deninger, Math. Nachr. 205 (2001). Motivic cohomology dynamical systems formalism.

Berry–Keating, SIAM Review 41 (1999). Proposal that $H = \frac{1}{2}(xp + px)$ on $L^2(\mathbb{R}_{>0}, dx/x)$ is the Hilbert–Pólya candidate; restriction to a Planck-cell cutoff gives the correct leading asymptotic zero density.

Akhiezer–Glazman *Theory of Linear Operators in Hilbert Space* Vol. II (Dover 1993 reprint).

Ch. VI on unbounded self-adjoint operators, Friedrichs extensions, and moment problems; Ch. VI.3 is the Hamburger/Carleman determinacy criterion.

Reed–Simon *Methods of Modern Mathematical Physics* Vol. I (Academic Press 1980), Vol. II (1975).

Ch. VIII (Vol. I) on unbounded operators; Vol. II on Fourier analysis and self-adjointness.

For the VB* converse:

de Branges *Hilbert Spaces of Entire Functions* (Springer 1968). Foundational theorem on reproducing-kernel Hilbert spaces of entire functions; Hermite–Biehler structure theorem.

Lagarias, Acta Arith. 89 (1999), 217–234. Reformulation of RH as Hermite–Biehler positivity for $E_{\xi}(z) = \xi_{\mathbb{R}}(\frac{1}{2} - iz) + \xi'_{\mathbb{R}}(\frac{1}{2} - iz)$.

Báez-Duarte, Adv. Math. 190 (2005), 315–320; arXiv:math/0202141. Quantitative strengthening of the Nyman–Beurling criterion.

Nyman *On Some Groups and Semigroups of Translations* (Uppsala 1950). Original criterion: RH equivalent to density of $\{\{\theta/x\} : \theta \in (0, 1]\}$ translates in $L^2((0, 1))$.

Beurling, Proc. Natl. Acad. Sci. 41 (1955). Beurling sequence space formulation of Nyman.

Burnol 2000. Hermite–Biehler refinement of Nyman–Beurling.

Simon, Adv. Math. 137 (1998), 82–203. Classical moment problem as self-adjoint finite difference operator; unbounded case treated via Jacobi matrix formalism.

Stein–Weiss *Introduction to Fourier Analysis on Euclidean Spaces* (Princeton 1971). Ch. V on Hardy spaces and boundary-value theorems governing Hermite–Biehler.

These analytic sources supply the external operator models for $H\text{-}P^{\text{Ar}}$ and the VB* converse. P_1 , P_2 , and P_3 enter only through their comparison hypotheses.

35.2 STRUCTURAL REDUCTIONS

The assertions below separate domain, trace, determinant, and moment questions. Each assertion is an implication between named mathematical structures.

$H\text{-}P^{\text{Ar}}$: Operator-Theoretic Framework

On the finite-span core of the primary arithmetic Heisenberg Fock, the Sugawara operator is diagonal with real non-negative eigenvalues (Chapter 23.6.1). Reed–Simon Vol. I Thm. VIII.11 and the Friedrichs construction give a canonical self-adjoint non-negative extension from this semibounded quadratic form.

This removes the domain ambiguity in Chapter 22.3.2. It does not identify the resulting Bost–Connes partition trace with the Weil–Connes zero distribution; that identification is the spectral trace input of Chapter 35.3.3.

H-P^{Ar}: Connes Adele Class Space

Connes’ adelic trace formula is a theorem on $X_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$. The comparison with H-P^{Ar} requires an intertwiner from the primary Fock space to the adèle-class L^2 -space which carries the zero-mode Bost–Connes monoid action to the idele-class scaling action. Under such an intertwiner, Connes’ absorption trace and the regularised Fock trace are the same distribution. The displayed absorption-trace formula,

$$\sum_{\rho} h(\operatorname{Im} \rho) = h(o) + h(1) - \hat{h}(o) \log \pi - \sum_v \int'_{\mathbb{Q}_v^{\times}} \frac{h(\log |u|_v)}{|1 - u|_v} d^{\times} u,$$

matches (as a distribution on the dual real line) the Li–Stieltjes moment generating function $\sum_{\rho} e^{-t\rho}$ through Mellin inversion $h(u) = e^{-tu/2}$, provided the embedding preserves the distributional trace.

H-P^{Ar}: Deninger Motivic Cohomology

Deninger’s formalism postulates a cohomology $H_M^{\bullet}(\overline{\operatorname{Spec} \mathbb{Z}}, \mathbb{R}(\cdot))$ with Frobenius Φ and Lefschetz determinant formula. If that cohomology exists and is identified with the primary Fock by an intertwiner, then the determinant identity for $\xi_{\mathbb{R}}$ is the H-P^{Ar} determinant statement. The formal identity $\xi(s) = \det_{\infty}(s \cdot \operatorname{id} - L_o^{\operatorname{Ar}})$ holds under H-P^{Ar}: the regularised determinant is defined by $\log \det_{\infty}(A) = -\partial_s \operatorname{tr}(A^{-s})|_{s=0}$, and the determinant divisor is $\operatorname{div} \xi_{\mathbb{R}}$ after the trivial-zero and pole summands have been removed by the primary-sector restriction.

H-P^{Ar}: Necessary and Sufficient Conditions

The three formulations have one common mathematical core: a positive-definite Hilbert space $\mathcal{H}$, a densely-defined self-adjoint operator A , and an intertwiner $\pi : \mathcal{V}_{\operatorname{Heis}, \operatorname{prim}}^{\operatorname{prim}} \rightarrow \mathcal{H}$ such that $A\pi = \pi L_o^{\operatorname{Ar}}$, the normal operator $\frac{1}{2} + iA$ has regularised determinant divisor equal to the non-trivial zero divisor of $\xi_{\mathbb{R}}$, and its Weil–Connes trace is the corresponding zero distribution. This is classical Hilbert–Pólya inside the arithmetic-Fock framework.

VB Converse: Spectral Regularity Beyond RH*

Zero-placement gives the zeros on the critical line. VB* asks for positivity of the operators T_n on the Fock closure. Passing from the former to the latter is a Hamburger moment problem for an unbounded operator; in this setting moment positivity does not select a unique spectral measure. Carleman’s criterion says that the Hamburger moment problem with moments $\{\mu_n\}_{n \geq 0}$ is determinate if $\sum_{n \geq 1} \mu_{2n}^{-1/(2n)} = +\infty$. For the symmetrised zero measure ν of Chapter 23.4.1, $\mu_n = \sum_{\rho} (\frac{1}{4} + \gamma_{\rho}^2)^n$ in the RH-conditional sense. Existing Riemann–von Mangoldt asymptotics do not by themselves provide a rigorous positive regularised moment measure with sharp two-sided μ_{2n} bounds. The Carleman construction is therefore open, not refuted.

VB Converse: de Branges and Báez-Duarte*

The de Branges and Nyman–Beurling formalisms give another expression of the same missing regularity. RH is equivalent to the Hermite–Biehler property of E_{ξ} in de Branges’ sense, but VB* requires the quantitative Báez-Duarte completeness rate compatible with the primary-Fock Petersson norm. Báez-Duarte Thm. 1 of arXiv:math/0202141 gives $\operatorname{RH} \Leftrightarrow$ the functions $\{\rho_n(x) = \{n/x\} - n\{1/x\}\}_{n \geq 2}$

span a dense subspace of $L^2((0, 1), dx)$ with a sharp decay rate $\epsilon_N = \text{dist}(\mathbf{1}, \text{span}\{\rho_n\}_{n \leq N})$ satisfying $\epsilon_N \ll (\log N)^{-1/2}$ under RH and $\ll (\log N)^{-c}$ for some $c > 1/2$ only under additional spectral regularity.

35.3 H-P^{Ar}: STRUCTURAL ANALYSIS

Operator-Theoretic Framework

Definition 35.3.1 (candidate L_o^{Ar} with explicit domain). . On the primary arithmetic Heisenberg Fock $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$, equipped with the Petersson inner product $\langle \cdot, \cdot \rangle^{\text{Pet}}$ of Chapter 21.7.1, define:

- *Finite-span core* $\mathcal{D}_o := \text{span}_{\mathbb{C}}\{|\mathbf{n}\rangle \otimes \delta_m : \mathbf{n} \in \mathbb{Z}_{\geq 0}^{\oplus \mathbb{N}}, m \in \mathbb{N}^\times\}$, the linear span of number-state/zero-mode-lattice basis vectors (Chapter 23.6.1, Chapter 7.3.1).
- *Candidate operator* $L_o^{\text{Ar, symm}} : \mathcal{D}_o \rightarrow \mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ defined by

$$L_o^{\text{Ar, symm}}(|\mathbf{n}\rangle \otimes \delta_m) = \left(\sum_{i \geq 1} i n_i + \lambda(m) \right) |\mathbf{n}\rangle \otimes \delta_m$$

where $\lambda(m) \in \mathbb{R}$ is the Bost–Connes zero-mode weight corresponding to $m \in \mathbb{N}^\times$ (unspecified at this stage; the choice is constrained by the Bost–Connes KMS₁ state structure of Chapter 7.3.2).

- *Closure*: let L_o^{Ar} denote the closure of $L_o^{\text{Ar, symm}}$ with respect to the graph norm on $\mathcal{D}_o$.

THEOREM 35.3.2 (Friedrichs self-adjoint extension). . Under the hypothesis that $\lambda(m) \geq 0$ for all $m \in \mathbb{N}^\times$ (non-negativity of the Bost–Connes zero-mode weights), the symmetric operator $L_o^{\text{Ar, symm}}$ of Chapter 35.3.1 admits a unique Friedrichs self-adjoint extension $L_o^{\text{Ar}} : \mathcal{D} \rightarrow \mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ with $\mathcal{D} \supset \mathcal{D}_o$ dense. On the Friedrichs extension, L_o^{Ar} is essentially self-adjoint on the finite-span core $\mathcal{D}_o$ and has non-negative spectrum.

Proof. Apply Reed–Simon Vol. II Thm. X.23 (Friedrichs extension of a non-negative symmetric operator). The quadratic form $q(\psi, \psi') = \langle L_o^{\text{Ar, symm}} \psi, \psi' \rangle^{\text{Pet}}$ is closable because $L_o^{\text{Ar, symm}}$ acts diagonally on the orthonormal basis $\{|\mathbf{n}\rangle \otimes \delta_m\}$ with non-negative real eigenvalues. Reed–Simon Vol. I Thm. VIII.11 then gives essential self-adjointness on the finite-span core: a diagonal operator with real eigenvalues on a Hilbert space with eigenvector basis is essentially self-adjoint on the finite-span of eigenvectors. The Friedrichs extension is the unique self-adjoint extension dominated by the quadratic form (Akhiezer–Glazman Vol. II Thm. 123). Non-negativity of the spectrum follows from non-negativity of all eigenvalues. $\square$

PROPOSITION 35.3.3 (Bost–Connes trace compatibility is the HP input). . The Friedrichs extension of Chapter 35.3.2 constructs a self-adjoint non-negative Fock/BC Hamiltonian from the real weights $\lambda(m)$. It does not by itself construct the Hilbert–Pólya determinant-trace operator H_{HP} . H-P^{Ar} holds only after one supplies an additional distributional trace identification with the zero side. The formal heat-kernel expression is

$$\text{tr}_{\text{BC}}(e^{-tL_o^{\text{Ar}}}) = \sum_{\rho} e^{-t\rho} \quad (t > 0) \quad (35.1)$$

but (35.1) is only symbolic notation. It means equality of Weil–Connes distributions on admissible Paley–Wiener tests, equivalently the trace identity of Chapter 22.3.2 for $\Theta_\xi = 1/2 + iH_{\text{HP}}$. This trace compatibility is the Hilbert–Pólya determinant-trace input, not a consequence of the Friedrichs theorem.

Proof. Direct computation of $\text{tr}(e^{-tL_o^{\text{Ar}}})$ on the basis of $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}} = L \otimes \mathcal{H}^{(\infty)}$:

$$\text{tr}(e^{-tL_o^{\text{Ar}}}) = \left(\sum_{m \geq 1} e^{-t\lambda(m)} \right) \cdot \left(\sum_{\mathbf{n}} e^{-t \sum_i i n_i} \right) = \left(\sum_{m \geq 1} e^{-t\lambda(m)} \right) \cdot \prod_{i \geq 1} \frac{1}{1 - e^{-ti}}.$$

For the Bost–Connes/KMS weight $\lambda(m) = \log m$, the zero-mode factor is $\zeta(t)$ for $\text{Re}(t) > 1$, multiplied by the Dedekind-eta inverse from the Fock descendants. This is the Euler/partition side. It is not the zero distribution. Passing from this partition trace to the zero side requires the full Weil–Connes explicit-formula distribution and a positive trace realisation; that is exactly H-P^{Ar} in Chapter 22.3.2. Thus the proposition records an equivalence of formulations after the trace realisation is assumed, not a derivation of the trace realisation. $\square$

Remark 35.3.4 (what the Friedrichs construction buys). Chapter 35.3.2 removes only one domain-specification ambiguity: a real non-negative diagonal Fock/BC Hamiltonian has a canonical self-adjoint extension. What remains is not an elementary choice of $\lambda(m)$. The KMS normalisation selects $\lambda(m) = \log m$, which gives the Bost–Connes partition trace $\zeta(t)$; the zero distribution of $\xi_{\mathbb{R}}$ (the completed xi function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, here denoting the restriction to the primary archimedean sector) is a different object. The Hilbert–Pólya content is the additional construction of H_{HP} , or an intertwiner carrying this Fock/BC Hamiltonian to the zero-divisor operator Θ_{ξ} , with the positive Weil–Connes trace preserved.

Connection to Connes Adele Class Space

Definition 35.3.5 (Connes adele class space). (Connes arXiv:math/9811068 §II). The *adele class space* is $X_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$, the quotient of the adeles by the multiplicative action of non-zero rationals. Its measure-zero part $X_{\mathbb{Q}}^{\circ}$ supports an L^2 space $L^2_o(X_{\mathbb{Q}}, d\mu)$ on which the idele class group $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times}$ acts unitarily by multiplication. The absorption trace of this action is the content of Connes’ realisation.

THEOREM 35.3.6 (*Connes arXiv:math/9811068 Thm. 1*). . For appropriate test functions h on $C_{\mathbb{Q}}$,

$$\text{Tr}(\mathfrak{D}(h)) = 2h(1) + 2h(-1) - \hat{h}(0) \log \pi - \sum_v \int_{\mathbb{Q}_v^{\times}}' \frac{h(u)}{|1-u|} d^{\times}u - \sum_{\rho: L(\hat{\rho}, \chi)=0} \hat{h}(\chi, \rho), \quad (35.2)$$

where $\mathfrak{D}(h)$ is the scaling operator on $L^2_o(X_{\mathbb{Q}})$, ρ ranges over zeros of Hecke L -functions $L(s, \chi)$, and the prime on the local integral denotes principal-value regularisation. The GL_1 case with trivial character recovers $\xi(s)$: the non-trivial zeros of ζ enter the absorption trace distribution.

Conjecture 35.3.7 (intertwiner comparing H-P^{Ar} with Connes). . Let H_{HP} and $\Theta_{\xi} = 1/2 + iH_{\text{HP}}$ be as in Chapter 22.3.2, and let D_{BC} denote the Friedrichs Fock/BC Hamiltonian of Chapter 35.3.2. Let $\mathfrak{D}$ be the scaling operator of Connes on $L^2_o(X_{\mathbb{Q}})$. The required comparison is the existence of an intertwiner

$$\pi : \mathcal{V}_{\text{Heis,prim}}^{\text{prim}} \longrightarrow L^2_o(X_{\mathbb{Q}}, d\mu)$$

defined on $\mathcal{D}_o$ and extended by closure, such that

$$\pi \circ \Theta_{\xi} = \mathfrak{D}_o \circ \pi$$

where $\mathfrak{D}_o$ is the infinitesimal generator of the scaling action on the kernel of the idele-class scaling trace. The intertwiner π is $C_{\mathbb{Q}}$ -equivariant on the sph-finite sector and identifies the positive Weil–Connes trace of Θ_{ξ} with the distributional trace in Connes’ trace formula. Its restriction to the zero-mode lattice must also explain how D_{BC} , whose partition trace is $\zeta(\beta)$, maps to the absorption trace without postulating the zero divisor.

Remark 35.3.8 (open intertwiner). The sph-finite sector of $\mathcal{V}^{\text{prim}}$ embeds by restricted tensor product into a subspace of $L^2([GL_2(\mathbb{Q})] \backslash GL_2(\mathbb{A}))$ compatible with the adèle class space image (Gelbart *Automorphic Forms on Adele Groups* §3.1). The zero-mode lattice $L \subset \mathcal{V}^{\text{prim}}$ maps to the finite-adele part of $X_{\mathbb{Q}}^{\circ}$; the archimedean Heisenberg factor $\mathcal{H}^{(\infty)}$ maps to the archimedean-fibre Fock of $L_{\circ}^2(X_{\mathbb{Q}}^{\circ, \infty})$. The scaling action of $C_{\mathbb{Q}}$ on $L_{\circ}^2(X_{\mathbb{Q}}^{\circ})$ restricts on the sph-finite sector to the multiplicative action of $\mathbb{N}^{\times}$ on the Bost–Connes zero-mode lattice (Chapter 7.3.2), plus the real-parameter archimedean scaling on the Fock. The infinitesimal generator of this action is the Bost–Connes/Fock Hamiltonian on the chiral-algebra side, while the Connes trace formula sees the absorption trace. Constructing an isometric intertwiner from the former to the latter is exactly the determinant-trace problem.

COROLLARY 35.3.9 (*H-P^{Ar} and Connes have the same determinant-trace input*). (conditional on the intertwiner of Chapter 35.3.7). Under that intertwiner, the trace-compatibility condition (35.1), read as the distributional equality of Chapter 22.3.2, is equivalent to Connes’ absorption-trace hypothesis: the $C_{\mathbb{Q}}$ -scaling action on $L_{\circ}^2(X_{\mathbb{Q}})$ has zero distribution of ζ with correct multiplicity.

Proof. Given the conjectural intertwiner, applying π transports the regularised trace distribution of Θ_{ξ} to the Connes distributional trace. Conversely, on the sph-finite image of π , the Connes absorption-trace statement pulls back to the Hilbert–Pólya determinant and trace identities. The proof uses the intertwiner as an input; it does not construct it. $\square$

Remark 35.3.10 (form of Connes’ trace formula). Connes’ trace formula (35.2) is proved in arXiv:math/9811068 Thm. 1. The identification of the absorption trace with the zero distribution of ζ with correct multiplicity is the content of Connes’ RH-equivalent statement; it is *not* a theorem. Chapter 35.3.9 places H-P^{Ar} at the same level only after the intertwiner of Chapter 35.3.7 is assumed. That intertwiner is part of the RH-equivalent determinant-trace problem. The chiral-algebra form replaces the adèle-class-space L^2 setting by a Fock-space setting with Sugawara and free-boson operators.

Connection to Deninger Motivic Cohomology

Definition 35.3.11 (*Deninger axioms for motivic cohomology*). (Deninger Proc. Symp. Pure Math. 55.1, §1). A *Deninger motivic cohomology* is a conjectural functor

$$H_{\mathcal{M}}^{\bullet}(-, \mathbb{R}(\cdot)) : \text{Schemes}/\mathbb{Z} \longrightarrow \text{Graded } \mathbb{R}\text{-vs}$$

equipped with an integral Frobenius endomorphism Φ satisfying:

- (D1) *Lefschetz trace formula.* For every smooth arithmetic scheme X , $\zeta_X(s) = \prod_i \det_{\infty}(s \cdot \text{id} - \Phi|H_{\mathcal{M}}^i(X, \mathbb{R}(1)))^{(-1)^{i+1}}$.
- (D2) *Purity.* The eigenvalues of Φ on $H_{\mathcal{M}}^i(X, \mathbb{R}(1))$ all satisfy $\text{Re}(\rho) = i/2$ (or a shifted variant).
- (D3) *Regularised determinant.* The determinant $\det_{\infty}$ is the ζ -regularised determinant defined by $\log \det_{\infty}(A) = -\partial_s \text{tr}(A^{-s})|_{s=0}$.

PROPOSITION 35.3.12 (*H-P^{Ar} $\Leftrightarrow$ Deninger formalism for $\overline{\text{Spec } \mathbb{Z}}$*). (conditional on existence of Deninger cohomology for $\overline{\text{Spec } \mathbb{Z}}$ and on an explicit Deninger–Fock intertwiner). Assuming Chapter 35.3.11 applies to $\overline{\text{Spec } \mathbb{Z}}$ with $X = \overline{\text{Spec } \mathbb{Z}}$ and $H_{\mathcal{M}}^1(\overline{\text{Spec } \mathbb{Z}}, \mathbb{R}(1))$ non-zero, the Hilbert–Pólya zero-divisor operator Θ_{ξ} of Chapter 22.3.2 is identified with $\Phi|H_{\mathcal{M}}^1(\overline{\text{Spec } \mathbb{Z}}, \mathbb{R}(1))^+$ (the + superscript denotes the primary $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ -invariant part), and H-P^{Ar} holds iff (D1), (D2), (D3) hold.

Proof. Under the Deninger axioms, the meromorphic Tate zeta surface is a cohomological superdeterminant: the even cohomology accounts for the Tate poles at 0 and 1, while the primary H^1 determinant is the entire factor $\xi_{\mathbb{R}}(s)$. The ordinary determinant target on the primary summand is

$$\xi_{\mathbb{R}}(s) = \det_{\infty}(s \cdot \text{id} - \Phi|H_{\mathcal{M}}^1(-, \mathbb{R}(1))^+).$$

The regularised determinant identity fixes the spectral divisor of the primary summand:

$$\text{div} \det_{\infty}(s \cdot \text{id} - \Phi|H_{\mathcal{M}}^1(-, \mathbb{R}(1))^+) = \text{div} \xi_{\mathbb{R}}(s).$$

By (D₃), the trace of Φ is a distributional zero trace in the admissible Weil–Connes sense, not an ordinary heat trace. The identification is canonical only after composing the intertwiner of Chapter 35.3.7 with the conjectural Deninger–Connes dictionary of Leichtnam. $\square$

Remark 35.3.13 (the three formalisms in alignment). The three comparison formulations of H-PA^r:

- (i) *Fock/BC domain realisation:* the Friedrichs Hamiltonian of Chapter 35.3.2 supplies a self-adjoint Euler-side operator, but the passage to Θ_{ξ} is the determinant-trace realisation.
- (ii) *Connes adèle class space:* the scaling generator $\mathfrak{D}_0$ on $L^2_0(X_{\mathbb{Q}})$ has the zero distribution as its absorption trace; this is Connes’ RH-equivalent reformulation of the trace formula of arXiv:math/9811068 Thm. 1, not a formal consequence of it.
- (iii) *Deninger motivic cohomology:* the Frobenius Φ on $H_{\mathcal{M}}^1(\overline{\text{Spec } \mathbb{Z}}, \mathbb{R}(1))^+$ has determinant divisor $\text{div} \xi_{\mathbb{R}}$ (Deninger 1994 axioms).

The equivalences among these formulations require the Connes/Fock and Deninger/Fock intertwiners. Without those intertwiners there is a comparison diagram, not a proof that the three operator models are the same object.

Necessary and Sufficient Conditions

THEOREM 35.3.14 (*H-PA^r is the determinant-trace realisation*). . The following are equivalent after the explicit Connes/Fock and Deninger/Fock intertwiners named above are included as hypotheses:

- (H-PA^r) The arithmetic Hilbert–Pólya hypothesis of Chapter 22.3.2.
- (SP) *Determinant-trace condition:* there exists the self-adjoint operator A , the normal operator $\Theta_{\xi} = 1/2 + iA$, the determinant divisor, and the positive Weil–Connes trace required in Chapter 22.3.2.
- (Connes) *Connes trace realisation:* the $C_{\mathbb{Q}}$ -scaling action on $L^2_0(X_{\mathbb{Q}})$ has regularised trace distribution equal to the non-trivial zero distribution of ζ , with multiplicity.
- (Deninger) *Deninger regularised-determinant condition:* the formal identity $\xi_{\mathbb{R}}(s) = \det_{\infty}(s \cdot \text{id} - \Phi|H_{\mathcal{M}, \mathbb{R}(1), +}^1)$ holds with Φ a self-adjoint operator on a Hilbert space matching $\mathcal{V}_{\text{Heis, prim}}^{\text{prim}}$ via an explicit intertwiner.

Proof. (H-PA^r) $\Leftrightarrow$ (SP) is definitional after Chapter 22.3.2: SP is the statement that the self-adjoint operator, determinant divisor, and positive trace distribution exist. The Friedrichs theorem supplies a separate Fock/BC domain model; it is evidence for the analytic framework but is not identified with H_{HP} without an intertwiner.

(SP) $\Leftrightarrow$ (Connes) is Chapter 35.3.9, conditional on the Connes/Fock intertwiner. (SP) $\Leftrightarrow$ (Deninger) is Chapter 35.3.12, conditional on the Deninger/Fock intertwiner and on the Deninger cohomology. Under these named hypotheses, transitivity closes the comparison. $\square$

COROLLARY 35.3.15 (*H-P^{Ar} reduces to determinant and trace*). $\mathcal{H}\text{-P}^{\text{Ar}}$ reduces to the determinant-trace condition (SP). The Friedrichs/KMS side fixes the natural Bost–Connes weight $\lambda(m) = \log m$ and gives the Euler partition trace; it does not give the zero trace. The required assertion is the construction of H_{HP} , Θ_ξ , and the positive Weil–Connes trace whose distribution is the non-trivial-zero distribution of $\xi_{\mathbb{R}}$. This is the Hilbert–Pólya content itself, not a routine weight choice (Chapter 35.3.16).

Proof. Direct from Chapter 35.3.14. The trace-compatibility condition encodes the location of non-trivial zeros of $\xi_{\mathbb{R}}$; the Bost–Connes weight gives the Euler side, while the zero side is the determinant divisor together with the positive Weil–Connes trace distribution. $\square$

Remark 35.3.16 (*Hilbert–Pólya factor*). The Friedrichs construction of Chapter 35.3.2 proves only the domain theorem. Given $\{\lambda(m)\} \geq 0$, the Friedrichs extension is unique, self-adjoint, and non-negative. $\mathcal{H}\text{-P}^{\text{Ar}}$ is the additional determinant-trace assertion: one must construct a zero operator with $\text{div } \det_\infty(s - \Theta_\xi) = \text{div } \xi_{\mathbb{R}}$ and trace distribution such that, in symbolic wave-kernel notation,

$$\text{Tr}_{\text{reg}}(e^{-t\Theta_\xi}) = \sum_{\rho} e^{-t\rho} \quad (t > 0),$$

where the equality is read after admissible Weil–Connes regularisation and the sum is over non-trivial zeros of $\xi_{\mathbb{R}}$ counted with multiplicity. This is Hilbert’s original eighth problem, not a corollary of the Friedrichs step. The determinacy comparison retains the domain theorem and leaves the Hilbert–Pólya determinant-trace realisation.

Remark 35.3.17 (*Berry–Keating candidate*). Berry–Keating SIAM Review 41 (1999) proposed $H = \frac{1}{2}(xp + px)$ on $L^2(\mathbb{R}_{>0}, dx/x)$ as a candidate Hilbert–Pólya operator, restricted to a Planck-cell cutoff. This operator has continuous spectrum on $\mathbb{R}$ and the Bohr–Sommerfeld quantisation of $xp = E$ gives the correct leading asymptotic zero density $N(T) \sim (T/2\pi) \log(T/2\pi)$. In the present framework, this candidate corresponds to a specific choice of $\{\lambda(m)\}_{m \in \mathbb{N}^\times}$: the prime-indexed weights $\lambda(p^k) = k \log p$ (Bost–Connes monoid generators), combined with the Fock archimedean spectrum $\{i n_i\}$, formally give the Berry–Keating spectrum structure. The Berry–Keating candidate is *not* rigorously known to have determinant divisor $\text{div } \xi_{\mathbb{R}}$; it reproduces the density but not the individual zero divisor. The rigorous identification requires a regularisation matching (35.1), which is the $\mathcal{H}\text{-P}^{\text{Ar}}$ content.

35.4 VB* CONVERSE: STRUCTURAL ANALYSIS

The Moment Problem in Hilbert Space

THEOREM 35.4.1 (*determinate vs. indeterminate Hamburger moment problem*). (Akhiezer 1965 §2.4; Shohat–Tamarkin 1943 Ch. III.5). Let $\{\mu_n\}_{n \geq 0}$ be a sequence of real numbers with Hankel matrices $H_n = (\mu_{i+j})_{0 \leq i, j \leq n} \succeq 0$ for all n . The Hamburger moment problem is:

- *Solvable*: there exists a positive measure ν on $\mathbb{R}$ with $\mu_n = \int x^n d\nu(x)$.
- *Determinate*: the solution ν is unique.
- *Indeterminate*: multiple distinct positive measures share the moment sequence.

Carleman’s sufficient criterion for determinacy: $\sum_{n \geq 1} \mu_{2n}^{-1/(2n)} = +\infty$. Non-sufficient examples (Stieltjes 1894): the lognormal moment sequence $\mu_n = e^{n^2/2}$ is indeterminate.

THEOREM 35.4.2 (*Simon 1998: moment problem as Jacobi matrix*). (Simon Adv. Math. 137 (1998) §5). A Hamburger moment problem with moments $\{\mu_n\}_{n \geq 0}$ is equivalent to the spectral theory of a Jacobi matrix

$$J = \begin{pmatrix} b_0 & a_0 & 0 & \cdots \\ a_0 & b_1 & a_1 & \cdots \\ 0 & a_1 & b_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

on $\ell^2(\mathbb{Z}_{\geq 0})$, with $a_n > 0$ and $b_n \in \mathbb{R}$ determined recursively by the moment sequence via the Gram–Schmidt-orthonormalised polynomial basis $\{p_n(x)\}_{n \geq 0}$ with $p_n(x) = x^n + \text{lower}$ and $\int p_m p_n d\nu = \delta_{mn}$. The moment problem is determinate iff J is essentially self-adjoint on c_{00} , and indeterminate iff J has deficiency indices $(1, 1)$.

PROPOSITION 35.4.3 (*VB* operator-positivity vs. moment positivity: the gap*). . Let ν be the symmetrised zero measure of Chapter 23.4.1, with moments $\mu_n = \int x^n d\nu$. The following implications hold after the stated Fock-space spectral identification:

- (i) VB* (operator positivity: $T_n \succeq 0 \forall n$) $\Rightarrow$ Li positivity ($\lambda_n \geq 0 \forall n$).
- (ii) Li positivity $\Leftrightarrow$ RH $\Leftrightarrow \nu$ positive measure on the critical-line parameter set.
- (iii) If ν is determinate and the associated L_0^{Ar} has spectral support in $[1/2, \infty)$, then VB* follows.
- (iv) Positivity of the moments alone, without determinacy and this support identification, does not imply VB* for a general unbounded moment problem.

Proof. (1): trace of positive-semidefinite is non-negative; hence $\text{tr}(T_n) = n\lambda_n \geq 0$ under VB*.

(2): Li's criterion (Li 1997) and the moment reformulation of Chapter 23.4.4(2) $\Leftrightarrow$ (4).

(3): under determinacy and the stated support condition, L_0^{Ar} has a unique spectral measure on the Fock closure, and T_n diagonalises with eigenvalues $m^n - (m-1)^n \geq 0$ for $m \geq 1/2$, hence $T_n \succeq 0$.

(4): in an indeterminate Hamburger problem there are distinct positive measures with the same moments. They give distinct self-adjoint realisations of the same formal Jacobi data. Moment positivity therefore does not select the Fock operator, nor does it force the support condition needed for $m^n - (m-1)^n \geq 0$. Akhiezer–Glazman Vol. II Ch. VI.3 provides the classical model. $\square$

COROLLARY 35.4.4 (*the converse input is spectral regularity*). . The converse implication from zero-placement to VB* is the spectral regularity of the symmetrised zero moment problem: positivity of the regularised moment functional, Hamburger determinacy, and the Fock-space support identification placing L_0^{Ar} in $[1/2, \infty)$. Without these three conditions, zero-placement does not force VB*.

Proof. Direct from Chapter 35.4.3(3)–(4): determinacy is the hinge. $\square$

PROPOSITION 35.4.5 (*Carleman construction remains open*). . Carleman's criterion $\sum_{n \geq 1} \mu_{2n}^{-1/(2n)} = +\infty$ is not yet verified for the symmetrised zero moment sequence. The heuristic growth estimates below are inconclusive; they do not justify claiming that Carleman's criterion fails.

Proof. Under RH, $\rho = 1/2 + i\gamma_\rho$ with $\gamma_\rho \in \mathbb{R}$. The symmetrised measure has support $\mu_n = \sum_\rho (\frac{1}{4} + \gamma_\rho^2)^n$ regularised. Using Riemann–von Mangoldt $N(T) \sim (T/2\pi) \log(T/2\pi)$, the N -th zero has $\gamma_N \sim 2\pi N / \log N$. Bost–Connes regularisation gives

$$\mu_{2n} \asymp \sum_N \left(\frac{2\pi N}{\log N} \right)^{4n} \cdot (\text{regularisation factor}) \asymp (2\pi N_{\max})^{4n} / (\log N_{\max})^{4n}$$

where $N_{\max}$ is the cutoff. If the regularised limit had the growth $\mu_{2n} \asymp C^n n!^2$ for some $C > 0$, then

$$\mu_{2n}^{-1/(2n)} \sim (C^n n!^2)^{-1/(2n)} \asymp \frac{1}{\sqrt{C} n}.$$

The harmonic comparison diverges, so this heuristic would support Carleman's sufficient criterion rather than refute it. The required input is a positive moment measure with sharp two-sided regularised-moment asymptotics. Until those are proved, the Carleman construction remains open. $\square$

Remark 35.4.6 (Stieltjes' example and the primary-Fock case). Stieltjes 1894 gave an explicit indeterminate moment sequence: $\mu_n = e^{n^2/2}$. Riemann–von Mangoldt alone does not give a two-sided growth law for the regularised zero moments. The heuristic of Chapter 35.4.5 posits growth $\mu_{2n} \asymp C^n n!^2$, which is between polynomial and lognormal, as the plausible asymptotic until a rigorous construction is supplied. Whether this intermediate rate is determinate is the open question. It is classically known (Akhiezer 1965 §2.4) that $\mu_n \asymp n!^2$ is on the boundary of Carleman's criterion; determinacy depends on finer asymptotic data.

Hermite–Biehler + de Branges

Definition 35.4.7 (Hermite–Biehler function). (de Branges 1968 Ch. II). An entire function $E(z)$ of finite exponential type is *Hermite–Biehler* if $|E(z)| > |E^*(z)|$ for all z in the open upper half plane $\text{Im } z > 0$, where $E^*(z) = \overline{E(\bar{z})}$. The *de Branges space* $\mathcal{H}(E)$ is the reproducing-kernel Hilbert space of entire functions f with $f/E, f^*/E \in H^2(\mathbb{C}_+)$ (upper-half-plane Hardy space) and reproducing kernel

$$K_E(w, z) = \frac{\overline{E(w)}E(z) - \overline{E^*(w)}E^*(z)}{2\pi i(\bar{w} - z)}.$$

THEOREM 35.4.8 (de Branges structure theorem). (de Branges 1968 Thm. 22). Every reproducing-kernel Hilbert space of entire functions $\mathcal{H}$ satisfying:

- $f, g \in \mathcal{H} \Rightarrow \overline{g(\bar{z})} \in \mathcal{H}$ (complex conjugate closure);
- for every non-real w , the point evaluation $f \mapsto f(w)$ is continuous;
- for every non-real w , $f \mapsto (f - \tilde{f})/(z - w)$ is bounded (where $\tilde{f}$ denotes a specific orthogonal projection);

is of the form $\mathcal{H}(E)$ for some Hermite–Biehler E .

THEOREM 35.4.9 (Lagarias: RH via reproducing kernel). (Lagarias Acta Arith. 89 (1999), 217–234). Let

$$E_\xi(z) = \xi_{\mathbb{R}}(\tfrac{1}{2} - iz) + \xi'_{\mathbb{R}}(\tfrac{1}{2} - iz).$$

Then $\text{RH} \Leftrightarrow E_\xi$ is Hermite–Biehler $\Leftrightarrow$ the reproducing kernel $K_{E_\xi}(w, z)$ is positive-semidefinite.

PROPOSITION 35.4.10 (VB* converse via Hermite–Biehler refinement). . Under RH and the Fock-space identification of the de Branges model, the following are equivalent:

- (i) VB* holds (operator-positivity $T_n \geq 0$ for all n).
- (ii) The de Branges space $\mathcal{H}(E_\xi)$ is *operator-positive*: for every continuous linear functional $L : \mathcal{H}(E_\xi) \rightarrow \mathbb{C}$ of the form $L(f) = \int f(x) d\sigma(x)$ with σ a signed measure, the induced bilinear form

$$B_L(f, g) = \int f(x) \overline{g(x)} d\sigma(x)$$

is positive-semidefinite on the kernel of $L_0^{\text{Ar}} - 1/2$.

(iii) The spectral-regularity condition of Chapter 35.4.4 holds.

Sketch. (1) $\Leftrightarrow$ (3): Chapter 35.4.4.

(3) $\Leftrightarrow$ (2): de Branges 1968 Ch. III.24 relates determinacy of a moment problem to uniqueness of the positive reproducing-kernel extension containing the polynomials densely. The “operator-positive” condition (2) is the corresponding positivity and support condition after transport to the primary-Fock image.

(1) $\Leftrightarrow$ (2): follows by transitivity (1) $\Leftrightarrow$ (3) $\Leftrightarrow$ (2) without invoking the conjectural intertwiner. Alternatively, when the Connes/Fock intertwiner of Chapter 35.3.7 is assumed, the primary-Fock $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ embeds into $\mathcal{H}(E_\xi)$ via that intertwiner composed with the de Branges Mellin transform of Connes 1999; under this embedding, T_n corresponds to the multiplication operator by $x^n - (x-1)^n$ on $\mathcal{H}(E_\xi)$, making (2) equivalent to operator-positivity of that multiplication operator. $\square$

The Báez-Duarte Refinement

THEOREM 35.4.11 (*Nyman–Beurling 1950/1955*). (Nyman 1950; Beurling 1955). Let $\rho(x) = \{1/x\} - [1/x]$ be the fractional part function. Let $M_0 \subset L^2((0,1))$ be the closed linear span of $\{\rho(\theta x) : \theta \in (0,1]\}$. Then $\text{RH} \Leftrightarrow \mathbf{1}_{(0,1)} \in M_0$.

THEOREM 35.4.12 (*Báez-Duarte 2003*). (Báez-Duarte Adv. Math. 190 (2005), 315–320; arXiv:math/0202141). Let $\rho_n(x) = \{n/x\} - n\{1/x\}$ for $n \geq 2$. Then $\text{RH} \Leftrightarrow$ the functions $\{\rho_n\}_{n \geq 2}$ span a dense subspace of $L^2((0,1), dx)$. Let

$$\epsilon_N^{\text{BD}} := \text{dist}_{L^2}(\mathbf{1}, \text{span}\{\rho_n\}_{2 \leq n \leq N})$$

be the finite-dimensional distance. Any sharp asymptotic for ϵ_N^{BD} is an additional quantitative statement.

PROPOSITION 35.4.13 (*conditional Báez-Duarte comparison*). . Assume that the Mellin transform identifies the finite Báez-Duarte distance with the corresponding Petersson distance in the primary arithmetic Heisenberg Fock, with the normalising constant fixed. Then the Báez-Duarte condition used here is exactly the Petersson-matched form of the VB* spectral regularity condition.

Sketch. The Nyman–Beurling M_0 -space embeds into the primary arithmetic Heisenberg Fock via the Mellin transform

$$\rho_n(x) \mapsto n^{-s} \left(\zeta(s) - \frac{n}{s-1} \right)$$

(Burnol 2000, Thm. 1.2). The distance ϵ_N^{BD} corresponds to the Petersson-Mellin distance from $\mathbf{1}_{(0,1)}$ (the primary vacuum) to the zero-mode-lattice span of $\{\rho_n\}_{2 \leq n \leq N}$ (the Bost–Connes orbit of the vacuum). Under the stated identification, positivity of T_n is the Hankel-positivity and determinacy condition for the same projected vacuum vector. Parseval’s identity then transfers the Petersson normalisation to the L^2 -distance ϵ_N^{BD} . No unconditional estimate for ϵ_N^{BD} is asserted here. $\square$

THEOREM 35.4.14 (*VB* converse and spectral regularity*). . Assume P₂, P₃, H-P^{Ar}, and the Fock-space identifications stated in Chapter 35.4.10 and Chapter 35.4.13. The following are equivalent:

- (VB*c) The VB* converse: zero-placement $\Rightarrow$ VB*.
- (Reg) positivity of the regularised zero-moment functional, Hamburger-determinacy of the symmetrised zero moment sequence of Chapter 23.4.1, and the Fock support condition of Chapter 35.4.4.
- (HB+) E_ξ is Hermite–Biehler and the associated de Branges space $\mathcal{H}(E_\xi)$ is operator-positive in the sense of Chapter 35.4.10(2).

(BD+) RH holds *and* the Báez-Duarte decay rate is $\epsilon_N^{BD} \asymp 1/\sqrt{\log N}$ with the Petersson-matched constant.

This theorem is a structural comparison. It does not assert that RH alone implies any of (Reg), (HB+), or (BD+).

Proof. $(VB^*c) \Leftrightarrow (\text{Reg})$: Chapter 35.4.4: RH plus spectral regularity is precisely the content of the VB^* converse in this framework.

$(\text{Reg}) \Leftrightarrow (\text{HB+})$: Chapter 35.4.10.

$(\text{Reg}) \Leftrightarrow (\text{BD+})$: Chapter 35.4.13.

Carleman’s sufficient criterion has not been verified for the symmetrised zero moment sequence (Chapter 35.4.5). Abstract indeterminate moment problems show that unbounded trace data do not formally determine operator positivity. They do not prove an independence theorem for the zeta zero sequence. $\square$

Remark 35.4.15 (the converse does not help the forward direction). Chapter 35.4.14 and Chapter 35.4.16 are comparisons at the VB^* converse level: each of (Reg), (HB+), (BD+) is a form of the same spectral-regularity input for zero-placement $\Rightarrow VB^*$. The forward direction $VB^* \Rightarrow \text{zero-placement}$ is the separate content of Chapter 22.3.10, which is a conditional theorem under $P_2, P_3, H\text{-}P^{\text{Ar}}$, and a positive Bost–Connes regularised trace. The conditional comparison for the converse does *not* close the forward implication, and does not discharge any of the conditional hypotheses of the forward direction. In particular, the sufficient RH path of Chapter 22.6.1 requires VB^* main direction plus $H\text{-}P^{\text{Ar}}$; the comparison for the converse identifies the analytic content of VB^* but does not reduce the forward requirement.

The Spectral-Regularity Content

COROLLARY 35.4.16 (*three equivalent spectral-regularity conditions*). . Under $P_2, P_3, H\text{-}P^{\text{Ar}}$ and the stated Fock-space identifications, the following three conditions are equivalent forms of the VB^* converse:

- (i) VB^* (arithmetic Virasoro operator positivity for all n).
- (ii) The spectral-regularity condition (Reg).
- (iii) Báez-Duarte sharp decay $\epsilon_N^{BD} \asymp 1/\sqrt{\log N}$ with the Petersson constant (BD+).

Condition (HB+) (Hermite–Biehler + operator-positivity of the de Branges space) is a fourth equivalent form.

Proof. Direct from Chapter 35.4.14. $\square$

Remark 35.4.17 (the content of the spectral-regularity gap). The four conditions (VB^* , Reg, HB+, BD+) express the same additional spectral regularity after the stated identifications. Classical abstract moment-problem theory (Akhiezer 1965 §2.4, Simon 1998 §5) constructs unbounded self-adjoint operators for which moment data do not determine the operator. Hence the VB^* converse requires an input not contained in RH as a bare zero-placement statement. The content of this input is, in each form:

- VB^* : operator-positivity of $T_n = (L_o^{\text{Ar}})^n - (L_o^{\text{Ar}} - \text{id})^n$ on the primary arithmetic Heisenberg Fock.
- Reg: positivity, uniqueness, and support of the spectral measure determined by the symmetrised zero moment sequence.
- HB+: operator-positivity of the de Branges reproducing kernel $\mathcal{H}(E_\xi)$ extending beyond Hermite–Biehler.

- $BD+$: a precise L^2 convergence rate of Beurling-sequence approximations of $\mathbf{1}_{(0,1)}$.

These four conditions are equivalent among themselves only under the hypotheses of Chapter 35.4.14. A proof of RH which did not also identify the Fock spectral measure would not close VB^* in this argument.

35.5 RELATION BETWEEN $H\text{-}P^{\text{Ar}}$ AND VB^* CONVERSE

THEOREM 35.5.1 (*$H\text{-}P^{\text{Ar}}$ and VB^* reduce to RH plus regularity*). . Under $H\text{-}P^{\text{Ar}}$ and the Fock-space support identification, VB^* is equivalent to RH *plus* the spectral-regularity condition of Chapter 35.4.4. Without $H\text{-}P^{\text{Ar}}$, the implication $RH \Rightarrow VB^*$ is not meaningful, since VB^* presupposes a self-adjoint L_o^{Ar} .

Proof. Direct from Chapter 35.4.14 combined with Chapter 35.3.14: once $H\text{-}P^{\text{Ar}}$ is in place, L_o^{Ar} has a well-defined spectral measure, and $VB^* \Leftrightarrow RH$ plus spectral regularity reduces to the moment-problem equivalence with the support condition. $\square$

PROPOSITION 35.5.2 (*$H\text{-}P^{\text{Ar}}$ and regularity are separate data*). . $H\text{-}P^{\text{Ar}}$ (existence of a self-adjoint L_o^{Ar} with prescribed determinant divisor and trace distribution) and the spectral-regularity condition (positivity, determinacy, and support of the spectral measure) are separate pieces of data:

- There exist indeterminate moment sequences with positive Hankel structure (Stieltjes' $e^{n^2/2}$ example); these admit multiple self-adjoint realisations. $H\text{-}P^{\text{Ar}}$ asserts existence of one realisation with the right trace; regularity asserts uniqueness and support properties of the realisation.
- Conversely, a determinate moment sequence does not automatically admit a self-adjoint realisation that matches a specific target trace; the target trace (the Bost–Connes trace of Chapter 35.1) is an additional constraint on the spectrum.

Proof. Akhiezer 1965 §2.4 and Simon 1998 Thm. 5.3 show that a moment sequence may admit several positive measures. Existence of one self-adjoint realisation and uniqueness of the realisation are therefore distinct assertions. The prescribed Bost–Connes trace is a further constraint. $\square$

COROLLARY 35.5.3 (*separation of the two analytic inputs*). . The two analytic inputs to VB^* in the sufficient path of Chapter 22.6.1, $H\text{-}P^{\text{Ar}}$ and the VB^* converse, are not the same assertion. Hence both are required for the VB^* input, and neither is redundant.

Proof. Chapter 35.5.2. $\square$

Remark 35.5.4 (*the refined sufficient path*). The sufficient path to RH of Chapter 22.6.1 has four inputs: arithmetic Positselski, Koszul–Petersson compatibility, $H\text{-}P^{\text{Ar}}$, and VB^* . The last two split as:

- $H\text{-}P^{\text{Ar}}$: existence of a self-adjoint operator obtained from L_o^{Ar} by the required intertwiner, whose normal zero-divisor operator has determinant divisor $\text{div } \xi_{\mathbb{R}}$ and whose Weil–Connes trace is the corresponding zero distribution. Equivalent forms: Connes adèle-class trace realisation; Deninger motivic regularised determinant with purity.
- VB^* : RH *and* the spectral-regularity condition for the symmetrised zero moment sequence. Equivalent forms: Hermite–Biehler operator-positivity; Báez-Duarte sharp rate after the Petersson normalisation has been fixed.

Each of the two inputs is a single spectral condition. The Connes–Deninger, de Branges, and Nyman–Beurling forms are equivalent languages for those conditions.

35.6 ANALYTIC CONDITIONS

The two analytic conditions have the following forms:

- (i) **(H-P^{Ar}, Friedrichs construction)**. Construct the determinant-trace operator and the trace-compatible embedding for which condition (35.1) holds on the Friedrichs extension of Chapter 35.3.2. Equivalent forms:
 - Connes: the $C_{\mathbb{Q}}$ -scaling action on $L^2_{\circ}(X_{\mathbb{Q}})$ has absorption trace equal to the zero distribution of ζ , with correct multiplicity.
 - Deninger: $\Phi|H^1_{\mathcal{M}}(\overline{\text{Spec } \mathbb{Z}}, \mathbb{R}(1))^+$ has determinant divisor $\text{div } \xi_{\mathbb{R}}$ with correct multiplicity.
- (ii) **(VB* converse, regularity construction)**. Establish positivity, Hamburger determinacy, and the support identification for the symmetrised zero moment sequence $\{\mu_n\}_{n \geq 0}$ of Chapter 23.4.1. Equivalent forms:
 - de Branges: operator-positivity of the reproducing kernel on $\mathcal{H}(E_{\xi})$ beyond Hermite–Biehler.
 - Báez-Duarte: sharp Petersson-normalised rate for ϵ_N^{BD} .
- (iii) **(Combined VB* input)**. Both (1) and (2) together supply the VB* input in the sufficient path of Vol IV. Neither alone suffices: (1) without (2) gives a specific self-adjoint operator but leaves T_n -positivity open; (2) without (1) gives abstract moment regularity without a Fock-space operator.

Remark 35.6.1 (local form). The Hilbert–Pólya condition is existence together with determinant divisor and trace distribution. The VB* converse is positivity, Hamburger determinacy, and support for the symmetrised zero moments. The Connes–Deninger and de Branges–Báez-Duarte forms identify these conditions in classical analytic languages; they do not make either condition a formal consequence of the other.

35.7 SPECTRAL BOUNDARY

- (i) **H-P^{Ar} is determinant-trace realisation** (Chapter 35.3.14, Chapter 35.3.15): the Friedrichs extension of Chapter 35.3.2 closes the domain problem, but the determinant divisor $\text{div } \xi_{\mathbb{R}}$ and the matching Weil–Connes zero trace are the genuine Hilbert–Pólya content (Chapter 35.3.16). The formulation is equivalent to Connes’ adèle-class trace realisation and to Deninger’s motivic cohomology regularised-determinant condition.
- (ii) **The VB* converse is equivalent to spectral regularity of the symmetrised zero-moment sequence** (Chapter 35.4.14), under the stated Fock-space identifications. This regularity is not supplied by zero placement alone. The converse is an equivalence at the converse level only and does *not* close the forward direction $\text{VB}^* \Rightarrow \text{zero-placement}$ (Chapter 35.4.15). Carleman’s criterion remains an open possible construction (Chapter 35.4.5); abstract indeterminate moment problems show why this extra hypothesis cannot be suppressed. Equivalent forms: de Branges reproducing-kernel operator-positivity beyond Hermite–Biehler; Báez-Duarte sharp decay rate with Petersson-matched constant.
- (iii) **The two analytic inputs are separate** (Chapter 35.5.2): H-P^{Ar} asserts existence of the operator; regularity asserts positivity, uniqueness, and support of the spectral measure. Both are required for the VB* input in the sufficient path; the other inputs of that path are not supplied here.
- (iv) **Berry–Keating $H = \frac{1}{2}(xp + px)$ candidate** is a candidate choice of Bost–Connes zero-mode weights $\lambda(p^k) = k \log p$, reproducing the leading asymptotic zero density but not the individual zeros; rigorous identification requires the regularisation matching (35.1).

- (v) **Stieltjes' example of an indeterminate moment sequence** ($\mu_n = e^{n^2/2}$) shows that the unbounded case admits genuine indeterminacy; heuristic estimates suggest the symmetrised zero moment sequence has intermediate growth $\asymp C^n n!^2$ on the boundary of Carleman, but this estimate is not yet rigorous (Chapter 35.4.6).
- (vi) **The refined sufficient path to RH** has four inputs (Chapter 35.5.4): arithmetic Positselski, Koszul–Petersson compatibility, $H\text{-}P^{\text{Ar}}$ with explicit Friedrichs domain and trace-compatibility, and VB^* as RH plus spectral regularity. The last two are each reducible to single spectral conditions with classical comparison formulations.
- (vii) **Independence of the two analytic inputs:** $H\text{-}P^{\text{Ar}}$ does not imply the VB^* converse, and the VB^* converse does not construct the Hilbert–Pólya operator. The four classical languages, Connes, Deninger, de Branges, and Báez-Duarte, identify the same two conditions from independent sides.

THE PLATONIC VARIATION AND RH COMPLETION

The arithmetic branch’s correct geometric variation is an operadic adelic surface: holomorphic at ∞ , Frobenius-oriented at finite primes, and glued by the pointed E_2 restricted tensor product of Iwahori Hecke categories: not a scheme-theoretic surface, not a literal almost-complex manifold at finite primes. The upgraded object $\mathfrak{S}_2^{\text{Ar}}$ drives the Riemann Hypothesis completion inside Vol IV as a conditional theorem.

36.1 THE RIGHT VARIATION

Definition 36.1.1 (Platonic 2-arithmetic surface). A Platonic 2-arithmetic surface for the primary arithmetic branch is the stratified adelic-diamond factorisation datum

$$\mathfrak{S}_2^{\text{Ar}} := (|\overline{C}|, \{S_v, \mathfrak{S}_v^{\text{Ar}}, C_v, \eta_v, u_v\}_v, \bigotimes_{v \in |\overline{C}|}^{\prime} (C_v, u_v)), \quad (36.1)$$

where:

- $|\overline{C}| = \mathbb{P} \sqcup \{\infty\}$ is the set of places of $\overline{\text{Spec } \mathbb{Z}}$;
- $S_\infty = \mathbb{C}_\infty$, $\mathfrak{S}_\infty^{\text{Ar}} = i$, and C_∞ is the archimedean holomorphic factorisation category with Harish–Chandra K_∞ -spherical unit u_∞ ;
- for $p < \infty$, S_p is the Fargues–Scholze perfectoid-diamond Hecke chart, $\mathfrak{S}_p^{\text{Ar}}$ is the Frobenius-semilinear untilt-orientation with Adams operation ψ_p and prismatic/Nygaard comparison datum, and $C_p = \text{Hk}_{GL_2, p}^{\text{Iw}}$ is the full non-semisimple Iwahori Hecke category with spherical unit $u_p = |o\rangle_p$;
- $\eta_v: \text{Sp} \rightarrow C_v$ is the pointed E_2 -algebra map with image u_v ; at almost every finite place u_v is the monoidal unit, and at the remaining places it is a compact central idempotent algebra object;
- the tensor product in (36.1) is the Tate restricted tensor product of pointed E_2 -monoidal stable presentable ∞ -categories in the sense of Chapter 28.1.1.

The word “surface” refers to the operadic dimension: the global category is E_2 -monoidal, not a smooth scheme-theoretic surface over $\mathbb{Z}$.

Remark 36.1.2 (almost complex versus arithmetic complex). The symbol $\mathfrak{S}^{\text{Ar}}$ is literal almost-complex geometry only on $S_\infty = \mathbb{C}_\infty$. At a finite prime it is the Frobenius-oriented substitute: a semilinear orientation of the diamond chart together with Adams and prismatic comparison data. The finite place supplies no tangent-bundle endomorphism satisfying $J^2 = -1$; its role is to make the E_2 Iwahori Hecke datum functorial. This convention governs every occurrence of $\mathfrak{S}^{\text{Ar}}$ in the arithmetic branch.

Remark 36.1.3 (source anchors and dependency separation). The finite charts use Scholze diamonds and Fargues–Scholze local Langlands geometry as listed in Chapter A.8 and Chapter A.19. The archimedean factorisation D -module fibre is the Beilinson–Drinfeld/Francis–Gaitsgory/Costello–Gwilliam fibre of Chapter A.5 and Chapter A.7. The arithmetic $\mathcal{D}^\dagger$ and overconvergent F -isocrystal language is only

a local coefficient shadow, with primary sources recorded in Chapter A.6. The definition of $\mathfrak{S}_2^{\text{Ar}}$ is complete on the data above; F_3 and the wild-prime BZN comparison enter only when identifying $\mathcal{V}^{\text{prim}}$.

THEOREM 36.1.4 (*upgrade of the aCFT base*). . Replacing the base datum (B) of Chapter 19.6.1 by $\mathfrak{S}_2^{\text{Ar}}$ preserves the primary arithmetic conformal field theory (aCFT) statement and upgrades its finite-place geometry. Specifically, the following aCFT axioms of Chapter 19.6.1 are re-verified: (B₁) the archimedean fibre remains genuinely chiral; (B₂) each finite fibre remains locally-constant with Adams/Frobenius structure; (B₃) the global object remains E_2 -monoidal via the restricted tensor product construction.

Proof. At ∞ the datum is unchanged: it is the holomorphic factorisation algebra on $\mathbb{C}_\infty$ of Chapter 8.6.1. At finite p , Chapter 8.3.3 and Chapter 8.4.1 identify the local factorisation structure as E_o with Adams operation; the Frobenius-semilinear orientation records exactly this finite-place arithmetic data. The global E_2 structure is the pointed restricted tensor product of Chapter 28.8.3. Thus all axioms of Chapter 19.6.1 survive, with the finite geometry expressed by its correct arithmetic substitute. $\square$

36.2 FACTORISABLE ARITHMETIC MODULES

The local categories $\mathcal{C}_v$ of $\mathfrak{S}_2^{\text{Ar}}$ must be assembled into a global coefficient theory. The classical Beilinson–Drinfeld factorisation D -module axioms require a smooth curve over a field, a hypothesis satisfied only at $v = \infty$. The correct replacement assigns to each finite subset of places the restricted tensor product of local categories, with transition maps given by the spherical units.

Definition 36.2.1 (*factorisable arithmetic crystal*). . A *factorisable arithmetic crystal* on $\mathfrak{S}_2^{\text{Ar}}$ is a functor

$$\mathcal{D}^{\text{Ar}} : \text{Fin}(|\overline{C}|)^{\text{op}} \longrightarrow \text{Alg}_{\mathcal{O}}(\text{Pr}_{\text{st}}^{\text{L}})$$

such that

$$\mathcal{D}^{\text{Ar}}(S) \simeq \bigotimes_{v \in S} \mathcal{C}_v, \quad \mathcal{D}^{\text{Ar}}(S_1 \sqcup S_2) \simeq \mathcal{D}^{\text{Ar}}(S_1) \otimes \mathcal{D}^{\text{Ar}}(S_2)$$

for disjoint finite subsets $S_1, S_2 \subset |\overline{C}|$, with transition maps given by insertion of the spherical units u_v . The archimedean value is a genuine holomorphic factorisation D -module in the Beilinson–Drinfeld sense. The finite-place value is an arithmetic coefficient category: an Iwahori Hecke category on a diamond chart, or on the spectral side an ind-coherent category with nilpotent singular support; in differential language it may be represented by overconvergent F -isocrystal or arithmetic $\mathcal{D}^\dagger$ data on a p -adic formal model. Here “represented” means local coefficient shadow on a chosen smooth p -adic formal chart. The definition is on the Hecke/diamond side; the global identification with $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\beta, \det, \mathcal{L}}^{\text{Sel}})$ remains the F1/LACD/BC-diamond-glue comparison, not a definitional equivalence.

PROPOSITION 36.2.2 (*factorisable crystal as the correct D -module replacement*). . The factorisable arithmetic crystal of Chapter 36.2.1 contains the classical factorisation D -module picture as its archimedean fibre and contains the finite-prime arithmetic replacement as Frobenius crystalline/diamond coefficient data. Hence it is the correct upgrade of “factorisable D -module” for $\mathfrak{S}_2^{\text{Ar}}$.

Proof. Beilinson–Drinfeld factorisation D -modules require a smooth curve over a field. This hypothesis is satisfied by the archimedean curve $\mathbb{C}_\infty$ and fails at $\text{Spd } \mathbb{Q}_p$ as a classical D -module base. At finite p the available primary objects are the perfectoid Iwahori Hecke category and its spectral ind-coherent avatar;

these are exactly the values C_p in Chapter 36.1.1. The functorial factorisation law is the discrete Ran law of Chapter 28.7.3. At $p = 2, 3$ no semisimple truncation is taken; C_p includes the wild Ext contribution of Chapter 32.4.8. $\square$

36.3 THE ARITHMETIC-CHIRAL BAR COMPLEX ON THE VARIATION

The factorisable arithmetic crystal $\mathcal{D}^{\text{Ar}}$ supplies the coefficient data at each place. The bar construction on $\mathfrak{S}_2^{\text{Ar}}$ is the colimit of finite-stage complexes mixing the archimedean chiral bar, the Parshin–Gersten boundary complex, and the local Iwasawa bar at each finite prime; the Kato–Parshin tame-symbol twisting cochain glues the finite stages.

Definition 36.3.1 (Platonic arithmetic-chiral bar complex). . For a primary object $\mathcal{V}^{\text{prim}}$ on $\mathfrak{S}_2^{\text{Ar}}$ let $\text{Fin}_{\text{rec}}(|\overline{C}|)$ be the filtered poset of finite sets $S = S_f \sqcup \{\infty\}$ containing the ramified places and closed under the Parshin–Gersten boundary maps of the symbols appearing in the stage. Write $C_{\text{PG},S}^\bullet(\overline{C})$ for the resulting finite Parshin–Gersten complex, with omitted-place boundary terms killed by the inserted spherical units. Define

$$B_S^{\text{Ar}}(\mathcal{V}^{\text{prim}}) := B^{\text{ch}}(\mathcal{V}_\infty^{\text{prim}}) \widehat{\otimes} C_{\text{PG},S}^\bullet(\overline{C}) \widehat{\otimes} \bigotimes_{p \in S_f} B_{E_i}^{\Lambda_p}(\mathcal{V}_p^{\text{prim}}),$$

with transition maps inserting the closed spherical vector at new finite places. The global arithmetic-chiral bar is

$$B^{\text{Ar}}(\mathcal{V}^{\text{prim}}) := \text{colim}_{S \in \text{Fin}_{\text{rec}}(|\overline{C}|)} B_S^{\text{Ar}}(\mathcal{V}^{\text{prim}}).$$

Choose an order $\infty < p_1 < \cdots < p_r$ on S . Its finite-stage differential is the signed total differential

$$D_S := d_\infty + (-1)^{|\cdot|_\infty} d_{\text{PG},S} + \sum_{j=1}^r (-1)^{|\cdot|_\infty + |\cdot|_{\text{PG}} + \sum_{i < j} |\cdot|_{p_i}} d_{p_j}, \quad (36.2)$$

where $|\cdot|_\infty$, $|\cdot|_{\text{PG}}$, and $|\cdot|_{p_i}$ denote the cohomological degrees in the archimedean, Parshin–Gersten, and p_i -adic tensor factors respectively; see Chapter 36.3.3 for the Koszul-sign verification. The local differentials are

$$d_\infty = d_{\text{int}} + d_{\text{res}} + d_{\text{dR}}, \quad d_p = d_{\Lambda, \text{int}} + \iota_p(\tau_S) + d_{\text{bar}, p}^{\text{loc}}.$$

Here τ_S is the Kato–Parshin tame-symbol twisting cochain. The finite residue term $\iota_p(\tau_S)$ is the Gersten boundary of τ_S : a tame-symbol boundary operator in the arithmetic K -theory sense, distinct from the complex-analytic $d \log(p - q)$ by construction. The signed totalization is independent of the chosen order by the symmetric braiding.

LEMMA 36.3.2 (closed spherical units). . For every finite place p , the spherical unit u_p is closed for the local arithmetic bar differential: $d_p(u_p) = 0$.

Proof. The internal differential kills the monoidal unit. The Parshin–Gersten term kills it because the tame symbol of the unit is zero. The local bar differential kills it by the augmentation-unit identity in the E_i bar construction. At $p = 2, 3$ this is computed inside the full non-semisimple Iwahori category; the wild Ext summand changes the trace category, not the closedness of the unit. $\square$

PROPOSITION 36.3.3 (finite-stage strictification). . For every $S \in \text{Fin}_{\text{rec}}(|\overline{C}|)$, the twisting cochain τ_S satisfies the Maurer–Cartan equation

$$d_{\text{PG},S} \tau_S + \tau_S d_{\text{PG},S} + \frac{1}{2} [\tau_S, \tau_S] = 0$$

in the completed convolution Lie algebra of the finite-stage Parshin–Gersten complex. Hence the signed total differential (36.2) satisfies $D_S^2 = 0$ on the chain level.

Proof. The Gersten differential squares to zero by the Gersten resolution. The displayed Maurer–Cartan equation is exactly the Kato–Parshin reciprocity law for the tame-symbol boundary, restricted to the reciprocity-closed stage S . Boundaries at omitted places are zero because new places enter through the closed spherical units of Chapter 36.3.2. The Koszul signs in (36.2) are the standard tensor totalization signs; after inserting them, the cross terms of D_S^2 are precisely the three terms of the Maurer–Cartan equation. Thus $D_S^2 = 0$ strictly, not merely after passing to cohomology. $\square$

PROPOSITION 36.3.4 (*comparison with the Parshin–Gersten bar*). . The arithmetic bar complex of Chapter 12.8.1 is the finite-support presentation of the colimit $B^{\text{Ar}}(\mathcal{V}^{\text{prim}})$ after replacing the single Iwasawa algebra Λ by the finite-stage adelic system $\{\Lambda_p\}_{p \in S_f}$ and totalizing with (36.2).

Proof. Every element of the Parshin–Gersten complex has finite support in the set of places. It therefore lies in some $C_{\text{PG},S}^\bullet(\overline{C})$ with $S \in \text{Fin}_{\text{rec}}(|\overline{C}|)$. The transition maps insert closed spherical units, so the finite-support union is the filtered colimit. The Λ -action of Chapter 12.8.1 decomposes on a finite stage as the completed tensor product of the local Λ_p -actions. This identifies the presentation of Chapter 12.8.1 with the restricted tensor presentation. $\square$

THEOREM 36.3.5 (*bar differential closure on the primary domain*). . On the abelian primary Heisenberg domain, the differential (36.2) satisfies $D_S^2 = 0$ at every reciprocity-closed finite stage and is compatible with the transition maps. Therefore $B^{\text{Ar}}(\mathcal{V}^{\text{prim}})$ is a well-defined arithmetic-chiral bar complex on $\mathfrak{S}_2^{\text{Ar}}$.

Proof. The archimedean summand squares to zero by the Vol I chiral bar construction and the Costello–Gwilliam comparison of Vol II. At finite places, Chapter 36.3.3 gives the chain-level square-zero statement. Local-global factorisation compatibility Chapter 33.3.4, pro-nilpotent convergence Chapter 33.5.3, the archimedean classical-limit check Chapter 33.6.4, and functoriality Chapter 33.7.4 supply the ambient completed coalgebra and functoriality. Transition maps are chain maps by Chapter 36.3.2. The comparison with the Parshin–Gersten model of Chapter 12.8.1 is Chapter 36.3.4. $\square$

36.4 RH COMPLETION

Definition 36.4.1 (*RH completion hypothesis*). . The *RH completion hypothesis* is the conjunction of:

- the Platonic surface $\mathfrak{S}_2^{\text{Ar}}$ of Chapter 36.1.1;
- the primary arithmetic-chiral bar complex of Chapter 36.3.1;
- the arithmetic Heisenberg Koszul biconditional, conditional on the arithmetic Positselski completion input of Chapter 33.8.2;
- the arithmetic Petersson positive-form condition, conditional on the restricted Koszul–Petersson compatibility input recorded in Chapter 25.10.1;
- H-P^{Ar} in the equivalent determinant-trace form of Chapter 35.3.14;
- the positive Hamburger and sampling conditions: VB^* on the primary Fock, equivalently the strengthened $\text{HB+}/\text{BD+}$ forms of Chapter 35.4.14.

THEOREM 36.4.2 (*Platonic RH completion, conditional*). . Conditional on the RH completion hypothesis of Chapter 36.4.1, the Riemann Hypothesis holds.

Proof. The surface $\mathfrak{S}_2^{\text{Ar}}$ and the bar complex $B^{\text{Ar}}(\mathcal{V}^{\text{prim}})$ of Chapter 36.3.1 place $\mathcal{V}^{\text{prim}}$ (a Hochschild trace class; see Chapter 22.1.2) in the correct arithmetic-chiral bar framework. The conditional Koszul input gives P_1 in the abelian Heisenberg domain (Chapter 33.8.2); P_2 is Tate's functional-equation/Zh-modularity input from Chapter 21.6.1; the conditional Petersson condition supplies the positive Hermitian form needed for P_3 in the restricted arithmetic branch Chapter 25.10.1. $H\text{-}P^{\text{Ar}}$ supplies the self-adjoint operator, the zero-divisor determinant, and the required Bost–Connes trace. The positive Hamburger condition gives VB^* on the primary Fock; determinacy alone is not the positivity input. Chapter 22.3.10 gives the zero-placement clause of F-RH. By Chapter 22.1.2, this clause is RH. $\square$

COROLLARY 36.4.3 (*single determinant-positivity sufficient condition for RH*). . Conditional on P_1 , P_2 , and P_3 in their stated conditional domains, RH is equivalent to the following single determinant-positivity statement: the Bost–Connes determinant-trace realisation of Chapter 35.3.14 has a positive Hamburger zero measure on the primary Fock and is determinate; equivalently, the strengthened HB+ or BD+ condition of Chapter 35.4.14 holds.

Proof. $H\text{-}P^{\text{Ar}}$ supplies the operator and trace. The stated positive Hamburger condition supplies the operator-positivity by Chapter 35.4.14. The remaining implication is Chapter 36.4.2. $\square$

CONDITIONAL HYPOTHESES FOR F₁–F₃ ARITHMETIC

Several F₁–F₃ arithmetic reductions are correct as equivalences, but some component subconditions require open inputs: the Fargues–Scholze-to-Emerton–Gee categorical equivalence (F_{1.a},), the arithmetic Arinkin–Gaitsgory conjecture (L_{ACD.1}–L_{ACD.3}), and the formal-Noetherian extension of Gaitsgory–Rozenblyum (R_{1.formal}). The exact logical form is the conditional system determined by Chapter C.2, Chapter 36, and Chapter 22.

37.1 CONDITIONAL STATEMENTS

For each reduction the displayed hypothesis is part of the theorem, not an afterthought to the proof.

F₁–F₃ in Chapter 21

F_{1.a} domain: Fargues–Scholze local geometrisation versus categorical equivalence. Chapter 21.2.1 carries form for F_{1.a}, the local categorical equivalence

$$D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p})$$

(where $\mathcal{X}_{GL_2}^{\text{EG},p}$ is the Emerton–Gee moduli stack at the prime p , $\text{IndCoh}_{\text{Nilp}}$ denotes ind-coherent sheaves with nilpotent singular support in the sense of Arinkin–Gaitsgory, and $D^{\text{sph-fin}}$ the spherical-finite derived category of smooth $GL_2(\mathbb{Q}_p)$ -representations). Fargues–Scholze arXiv:2102.13459 supplies the local geometrisation, spectral action, and Langlands-parameter structure for GL_2 ; the displayed categorical equivalence is not contained there. Dotto–Emerton–Gee arXiv:2603.26887 gives a fully-faithful-only, fixed-central-character functor for $GL_2(\mathbb{Q}_p)$ at $p \geq 5$. The authoritative reference for the current F_{1.a} form is Chapter C.2 and Chapter 21.2.1.

Emerton–Gee preprint identifier. The correct preprint identifier for Emerton–Gee is arXiv:1908.07185 (*Moduli stacks of étale (φ, Γ) -modules and the existence of crystalline lifts*, Annals of Mathematics Studies). See App. A §A.8.

Ben-Zvi–Chen–Helm–Nadler preprint identifier. The correct identifier for Ben-Zvi–Chen–Helm–Nadler is arXiv:2010.02321 (*Coherent Springer theory and the categorical Deligne–Langlands correspondence*, Invent. Math. 235 (2024)). See App. A §A.20.

Adelic Categorical Descent

Function-field transport and (L_2) . Chapter 26.2.3 carries form . The assertion (L_1) follows from Raskin’s ∞ -categorical construction independently of function-field Langlands; the assertion (L_2) requires the local categorical comparison F_{1.a} (via Dotto–Emerton–Gee) and Bernstein-centre diamond gluing.

(L_2) reduction: Fargues–Scholze domain. Chapter 26.4.1 carries form . The comparison functors Φ_v^{loc} identifying C_v with $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},v})$ are not supplied by Fargues–Scholze arXiv:2102.13459; that source gives the local geometrisation and spectral action. The local categorical comparison is the residual open input.

F1 reduction: two open inputs. Chapter 26.9.1 states at that F1 reduces to F1.a plus (BC-diamond-glue). Since F1.a is itself, the reduction leaves F1 conditional on two independent open inputs: F1.a and (BC-diamond-glue).

Arithmetic Ben-Zvi–Nadler

F3.BZN^{Ar}: procylic shadow versus full Weil parameter stack. Chapter 27.1.1 carries form . The comparison target for F3.BZN^{Ar} is the full Weil/ (φ, Γ) parameter stack $\text{LocSys}_{GL_2}^{W, \varphi \Gamma}$, not the procylic shadow $\text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_{\rho, \det, \mathcal{L}}^{\text{Sel}})$, which forgets Frobenius and unramified data. The theorem is conditional on F1 and F2. Chapter 27.4.6 carries ; the closed-immersion property Chapter 27.4.9 carries , since the closed-immersion property is unavailable on the inertia-only shadow.

Structural Reduction Theorems

L_{ACD} equivalence and subcondition forms. Chapter 29.2.4 : the adelic categorical descent condition L_{ACD} is equivalent to the conjunction of subconditions L_{ACD}.1 through L_{ACD}.4. The equivalence itself is a logical factorisation at . subconditions L_{ACD}.1–L_{ACD}.3 carry ; L_{ACD}.4 is the combination of Kneser’s strong approximation theorem for SL_2 and Tate’s determinant idele-class quotient for GL_2 . The residual content is recorded in the subcondition forms.

F3.BZN^{Ar} equivalence via Gaitsgory–Rozenblyum. Chapter 29.2.5 : F3.BZN^{Ar}, assembled through Gaitsgory–Rozenblyum DAG Vol. II, is equivalent to the conjunction of R1, R2, and R3. The equivalence is a valid logical factorisation. R1 holds as a theorem for smooth finite-type stacks (Gaitsgory–Rozenblyum) and is for the formal-Noetherian Emerton–Gee setting (R1.formal); R2 is ; R3 is definitional. The R1 fine structure and the honest residual are recorded in Chapter 31 and Chapter 31.5.1.

Arithmetic Positselski C1–C5. Chapter 22 treats the arithmetic Positselski equivalence C1–C5 as five subconditions, of which C1–C4 are conditional on explicit hypotheses and C5 (functoriality) is the named residual. The arithmetic Heisenberg equivalence is restricted to the abelian-Heisenberg sector; C5 is the functoriality condition for the full claim.

L_{ACD} Subconditions

L_{ACD}.0: archimedean stratum construction. Chapter 30.3.3 : the archimedean local category C_∞ , distinguished unit, and Langlands–Vogan classification are constructed from the theorems of Harish-Chandra (1966), Vogan (1981), and Wallach (1988). Thus L_{ACD}.0 is independent of the global L_{ACD} comparison.

L_{ACD}.1: the F2.c hypothesis. Chapter 30.4.1 (conditional on F2.c tightened): the assertion L_{ACD}.1 uses the restricted tensor theorem Chapter 28.1.3 on the central idempotent pointed-unit E_2 domain. Hence Chapter 30.4.1 carries with F2.c as its displayed hypothesis.

$L_{\text{ACD}.2}$: *local Hecke commutativity, not global functoriality.* Chapter 30.5.1 : formal tensor-factor commutativity of local Hecke endofunctors, once the restricted tensor product is constructed. The domain remark Chapter 30.5.2 records that global Hecke functoriality is not part of the statement; the $L_{\text{ACD}.2}$ label refers to the local commutativity subcondition only.

L_{ACD} *residual: three-component decomposition.* Chapter 30.7.1 records that L_{ACD} reduces, under $F2.c$ tightened, to (BC-diamond-glue). The residual (BC-diamond-glue) further decomposes as $(\text{BCglue.local}) \wedge (\text{BCglue.glue}) \wedge (\text{BC-diamond-glue.core})$ per Chapter 34.3.2, each component carrying distinct form. Closure of L_{ACD} requires $F1.a$ in addition (via Dotto–Emerton–Gee). An explicit note to this effect is added to Chapter 30.7.1.

BZN Formal Noetherian and EG Derived

$R2$: *derived enhancement of the Emerton–Gee stack.* Chapter 31.3.5 : the Emerton–Gee stack carries the structure of a derived formal algebraic stack in the Gaitsgory–Rozenblyum DAG sense, obtained from local derived enhancement (Mazur–Kisin–Lurie DAG X), finite-product cotangent complex, ind-colimit cotangent complex, and quasi-smoothness. The derived enhancement expresses existing Emerton–Gee data using Lurie DAG machinery; it does not reconstruct that data. The finite-type Gaitsgory–Rozenblyum theorem ($R1.\text{fin-type}$) applies to smooth finite-type stacks; its arithmetic application to the formal Noetherian stack $\mathcal{X}_{\bar{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}}$ is mediated by $R1.\text{formal}()$, as recorded in Chapter 31.4.1.

$R1$ *domain: finite-type theorem inapplicable to formal Noetherian setting.* The Gaitsgory–Rozenblyum integral-transforms theorem for smooth finite-type stacks ($R1.\text{fin-type}$) does not directly apply to the global Emerton–Gee stack, which is formal Noetherian. The extension to the formal Noetherian setting is $R1.\text{formal}()$. Consequently $F3.BZN^{\text{Ar}}$ is conditional on $R1.\text{formal}$.

Loop Convention and Wild Primes

$R3$: *loop convention equivalence under explicit hypotheses.* Chapter 32.3.4 : the three candidate definitions (a), (b), (c) of $L^{\text{Ar}}\mathcal{X}$ agree on a formal Noetherian derived algebraic stack with finite Tor-amplitude and t -bounded Postnikov tower. The hypotheses are named in the statement; the claim is a conditional theorem.

$R4$: *wild-prime Ext^1 refinement at $p \geq 5$.* Chapter 32.4.7 records that the wild-prime refinement to Ext^1 vanishes at $p \geq 5$, using the Bezrukavnikov semisimplicity theorem (arXiv:1209.0403). Chapter 32.4.1 carries , recording that the semisimplicity result is imported from arXiv:1209.0403; the formal corollary is conservative.

BC-Diamond-Glue Decomposition

(BC-diamond-glue) *three-component factorisation: form of components.* Chapter 34.3.2 :

$$(\text{BC-diamond-glue}) \Leftrightarrow (\text{BCglue.local}) \wedge (\text{BCglue.glue}) \wedge (\text{BC-diamond-glue.core}).$$

The equivalence is a valid logical factorisation. The component (BCglue.local) requires a primary-source transition theorem unifying BZCHN at $\ell \neq p$, Zhu/Emerton–Gee at $\ell = p$, and Fargues–Scholze ℓ -independence; Chapter 34.4.1 therefore carries . The component (BCglue.glue) requires prismatic faithfully-flat descent (Chapter 34.5.2,); Chapter 34.5.3 carries . Thus (BCglue.local) is conditional and (BCglue.glue) is conjectural.

L_{ACD} *tight reduction: conditional on three inputs.* Chapter 34.6.1 : L_{ACD} reduces to (BC-diamond-glue.core), conditional on arithmetic algebraic geometry hypotheses. The reduction requires: (i) (BCglue.local) elevated from ; (ii) (BCglue.glue) elevated from ; (iii) $F_{2,c}$ tightened, $F_{1,a}$, and the prior $L_{ACD}.k$ subconditions. The form carries to reflect this dependence.

Primary Source Identifiers in App. A

Dotto–Emerton–Gee in the primary source list. The identifiers for the three main sources are: Emerton–Gee arXiv:1908.07185, BZCHN arXiv:2010.02321, Fargues–Scholze arXiv:2102.13459. Dotto–Emerton–Gee arXiv:2603.26887 is cited for $F_{1,a}$ form at Chapter 21.2 and at Chapter 20.1. App. A §A.8 refines to include this entry.

37.2 LOGICAL FORM FOR F_1 – F_3 ARITHMETIC

The F_1 – F_3 reductions have the following conditional form.

In Chapter 21: $F_{1,a}$ carries (Dotto–Emerton–Gee arXiv:2603.26887, fully-faithful-only at $p \geq 5$, fixed central character); $F_{1,c}$ is theorem (Emerton–Gee arXiv:1908.07185).

In Chapter 26: Chapter 26.2.3, Chapter 26.4.1, and Chapter 26.9.1 carry , conditional on $F_{1,a}$ and (BC-diamond-glue).

In Chapter 27: Chapter 27.1.1 carries (four residuals: F_1 , F_2 , formal-loop continuity, full Weil/ (φ, Γ) target). Chapter 27.4.6 carries ; Chapter 27.4.9 carries .

In Chapter 29: Chapter 29.2.4 and Chapter 29.2.5 are valid logical equivalences at ; their subcondition forms carry (with $L_{ACD}.4$ the SL_2 strong-approximation theorem plus the GL_2 determinant quotient).

In Chapter 30: Chapter 30.4.1 carries with $F_{2,c}$ tightened as hypothesis; Chapter 30.7.1 is conditional on $F_{1,a}$ and on (BCglue.local)+(BCglue.glue).

In Chapter 31: R_2 (, derived enhancement of Emerton–Gee via Lurie DAG) and R_1 split (R_1 .finite-type theorem for smooth finite-type, R_1 .formal) are in place. Consequent F_3 for the reduction to three subconditions; F_3 itself remains conditional on R_1 .formal.b, R_1 .formal.c, and R_4 .

In Chapter 32: R_3 (, under explicit formal-Noetherian hypotheses); R_4 (, using Bezrukavnikov arXiv:1209.0403 and Etingof–Ostrik, Jantzen, Carter, Bushnell–Henniart).

In Chapter 34: Chapter 34.9.1 records (BCglue.local) as and (BCglue.glue) as . Chapter 34.6.1 carries .

37.3 OPEN CONDITIONS FOR F_1 – F_3 ARITHMETIC

The remaining F_1 – F_3 conditions are as follows, consistent with the running list Chapter C.2 and the RH completion conjunction Chapter 36.4.1.

- $F_{1,a}$ (local categorical equivalence $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p})$): via Dotto–Emerton–Gee arXiv:2603.26887 (fully-faithful-only, $p \geq 5$, fixed central character). Fargues–Scholze arXiv:2102.13459 supplies the local geometrisation, spectral action, and Langlands-parameter structure; the categorical equivalence is the residual.
- F_1 is conditional on $F_{1,a}$ and L_{ACD} ; L_{ACD} requires $F_{2,c}$ tightened, (BCglue.local), (BCglue.glue), and (BC-diamond-glue.core).
- F_3 .BZN^{Ar} requires F_1 , F_2 , R_1 .formal.b (profinite formal-loop continuity), R_1 .formal.c (flat-locus pullback), R_4 (wild-prime Ext¹ refinement, domain beyond the explicit computation already stated), and the full-target Weil/ (φ, Γ) comparison.

- Chapter 36.4.1 is the sufficient set for RH, with $V B^*$ operator positivity as a forward positivity residual, $H\text{-}P^{\text{Ar}}$ as a separate spectral hypothesis, and the arithmetic Positselski C_1 – C_5 plus Koszul–Petersson compatibility as the four-input RH form conditions.

SHARP HYPOTHESES FOR THE FOUR-INPUT RH FORM

The arithmetic branch separates the following six assertions into their established domains and their spectral or categorical hypotheses: the P_1 arithmetic-Positselski biconditional for the abelian Heisenberg archetype, the P_3 Koszul–Petersson identity, the K_5 Godement–Jacquet normalisation, the Costello refinements C_1 – C_5 , the $H\text{-}P^{\text{Ar}}$ comparison, and the VB^* implication to zero-placement. Tate’s adelic Poisson formula gives the forward P_1 direction. The finite Godement–Jacquet identity is a theorem on the cuspidal sph-fin core. The Friedrichs construction gives an operator domain. The remaining content is precisely the passage from these domains to the full spectral object.

38.1 SIX SHARP FORMS

The six assertions have the following logical form.

No.	Assertion	Established domain	Additional hypothesis
1	P_1 arithmetic Positselski (Chapter 24.1.5, Chapter 33.8.1)	Tate-Poisson gives the forward functional-equation direction for the abelian Heisenberg archetype.	Completed continuous duals, derived completion, faithful lifting, primitive zeta-character comparison, and Koszul-dual Bost–Connes.
2	P_3 Koszul–Petersson (Chapter 25.10.1, Chapter 25.9.1)	Archimedean Segal–Petersson identity and finite sph-fin cuspidal Godement–Jacquet identity.	KMS_i /GNS identification, full Rankin–Selberg extension, and Eisenstein summands.
3	K_5 absorption (Chapter 34.7.5, Chapter 34.7.7)	Cuspidal sph-fin locus after the local comparison datum.	Compatibility with the full primary sector, including continuous spectrum.
4	Costello refinements C_1 – C_5 (Chapter 33.3.3, Chapter 33.4.4, Chapter 33.5.3, Chapter 33.6.4, Chapter 33.7.4)	Restricted abelian-primary, $\mu = 0$, pointed E_2 , and Λ -linear domains specified below.	The five Positselski hypotheses and the five Costello hypotheses.
5	$H\text{-}P^{\text{Ar}}$ (Chapter 35.3.14)	Friedrichs self-adjoint domain.	Determinant divisor and Weil–Connes trace distribution for the scaling action.
6	$VB^* \Rightarrow$ zero-placement (Chapter 22.3.10, Chapter 35.4.14)	One-sided implication after P_2 , P_3 , $H\text{-}P^{\text{Ar}}$, and positive Bost–Connes trace.	Forward positivity; the converse is only a spectral-regularity comparison.

TABLE 38.1: Sharp domains for the four-input RH form.

The same list appears in Chapter C.10. The Platonic capstone is Chapter 36.4.2. Its sufficient arithmetic path is the four-input conjunction in §38.3.

38.2 LOCAL FORMS

Zero-Placement

PROPOSITION 38.2.1 (*one-sided zero-placement*). . Assume P_2 , full P_3 , the $H\text{-}P^{\text{Ar}}$ determinant-trace realisation, and the positive VB^* trace form. Then Chapter 22.3.10 gives the zero-placement clause of F-RH. The reverse implication is not a consequence of RH; it asks for spectral regularity of the operator model.

Proof. The implication used in Chapter 22.3.10 is the positive quadratic form statement on the Bost–Connes regularised trace. P_2 and P_3 identify the test vectors and the Petersson form with the arithmetic trace form. $H\text{-}P^{\text{Ar}}$ identifies the zero divisor and the Weil–Connes trace distribution. Positivity then forces the ordinates entering the Weil form to be real, hence the zeros lie on the critical line. None of these steps constructs the converse positive form from zero-placement alone. $\square$

Arithmetic Positselski

PROPOSITION 38.2.2 (*P_1 and the abelian arithmetic Heisenberg*). . The equivalence in Chapter 24.1.5 and the P_1 propagation in Chapter 24.5.4 hold on the abelian-primary Heisenberg domain after imposing the following five hypotheses:

- (i) completed continuous duals are compatible with the relevant co-derived and contra-derived categories;
- (ii) projective and injective resolutions remain exact after derived Λ -adic completion;
- (iii) the Nakayama lifting used in the Λ -coefficient comparison is faithful;
- (iv) the primitive zeta-character computes the κ -pairing weight;
- (v) the dual curved coalgebra carries the Koszul-dual Bost–Connes interpretation.

With these hypotheses the P_1 biconditional is the arithmetic Positselski theorem. Without them Tate’s Poisson formula supplies only the forward implication from Koszul self-duality to the functional equation.

Proof. Tate’s formula identifies the local Fourier transforms and the global idele quotient and gives $\xi(s) = \xi(1-s)$ from the abelian Heisenberg self-duality datum. The reverse direction must lift this equality through the Λ -coefficient bar–cobar adjunction. Each of the five hypotheses is used exactly at one such lift: topological duality, completion, Nakayama faithfulness, the primitive trace weight, and the Bost–Connes interpretation of the dual coalgebra. Their conjunction is therefore sufficient, and no smaller statement follows from Tate’s functional equation alone. $\square$

Koszul–Petersson Compatibility

PROPOSITION 38.2.3 (*P_3 on the normalized sph-fin cuspidal core*). . On the normalized sph-fin cuspidal core of $\mathcal{V}^{\text{prim}}$, the archimedean Segal–Petersson identity and the finite Godement–Jacquet identity assemble to the P_3 Koszul–Petersson form.

Proof. The archimedean factor is the Segal Fock realisation of the Petersson pairing. At a finite unramified place the sph-fin vector has the Godement–Jacquet normalisation, and the local zeta integral gives the same Euler factor as the finite Bost–Connes trace. The restricted tensor product is taken over the spherical units, so the product of the local identities is the adelic identity on the cuspidal sph-fin subspace. $\square$

Remark 38.2.4 (full P_3). The extension from the normalized sph-fin cuspidal core to $\mathcal{V}^{\text{prim}}$ requires the KMS_1/GNS identification, the Rankin–Selberg extension of the local Petersson computation, and the treatment of Eisenstein summands. These three conditions are spectral hypotheses, not formal consequences of the cuspidal calculation.

Costello Refinements C_1 – C_5

PROPOSITION 38.2.5 (*Costello refinements*). . The refinements C_1 – C_5 imply the P_1 arithmetic Positselski biconditional on their common domain after imposing the following five hypotheses:

- (i) C_1 : the η -algebra is obtained as a pointed colimit in the unit-compatible E_2 category of Chapter 29.2.3;
- (ii) C_2 : Parshin reciprocity holds modulo Λ -coboundaries, including the $p = 2$ torsion term;
- (iii) C_3 : the pro-nilpotent limit at $\beta = 1$ agrees with the KMS_1 Tomita–Takesaki state;
- (iv) C_4 : the Witt truncation has the stated classical limit and is compatible with the archimedean Heisenberg Fock object;
- (v) C_5 : the morphism class is Λ -linear with trivial Γ -action.

Proof. Each C_i removes one obstruction in the comparison between the curved algebra and curved coalgebra sides. C_1 supplies the pointed E_2 multiplication, C_2 identifies the reciprocity defect with a Λ -coboundary, C_3 controls the limit state, C_4 fixes the classical shadow, and C_5 gives the allowed morphisms. Their common domain is still the domain of the arithmetic Positselski theorem, so the five Positselski hypotheses of §38.2 remain present. $\square$

K_5 Godement–Jacquet Normalisation

PROPOSITION 38.2.6 (*K_5 on the cuspidal sph-fin locus*). . After the local comparison datum, K_5 is established on the cuspidal sph-fin locus. Hoffstein–Lockhart nonvanishing excludes the local zero obstruction in this domain, and the sph-fin projection removes Eisenstein poles from the calculation.

Proof. The local comparison datum identifies the chosen sph-fin vector with the Godement–Jacquet test vector. The local zeta integral is then the standard local L -factor. On the cuspidal sph-fin locus the Hoffstein–Lockhart nonvanishing theorem prevents the denominator which enters the absorption normalisation from vanishing. Since the sph-fin projection has already removed the Eisenstein summands, no continuous-spectrum pole enters the identity. $\square$

Remark 38.2.7. The propagation of K_5 to full Koszul–Petersson compatibility is therefore the full P_3 problem of §38.2.

$H\text{-P}^{\text{Ar}}$ and the VB^ Converse*

PROPOSITION 38.2.8 (*Friedrichs domain and determinant divisor*). . The Friedrichs construction of Chapter 35.3.2 gives a self-adjoint operator on its quadratic-form domain. The Hilbert–Pólya input is the further assertion that the associated normal zero-divisor operator has determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes trace equal to the zero distribution. This assertion is not a choice of zero-mode weights.

Proof. The Friedrichs theorem starts from a lower-semicontinuous positive form and produces a self-adjoint operator. It does not determine the regularised determinant or the Weil–Connes trace of the scaling action. The Connes and Deninger forms have the required Lefschetz shape only when the

determinant divisor and the trace distribution are the non-trivial zero divisor and distribution of $\xi_{\mathbb{R}}$. Thus the operator-domain theorem and the determinant-trace theorem are distinct statements. $\square$

PROPOSITION 38.2.9 (*VB* converse*). . The converse statement in Chapter 35.4.14 is a spectral-regularity comparison after the Fock-space identifications. It does not imply the forward positivity statement $\text{VB}^* \Rightarrow \text{zero-placement}$.

Proof. A converse spectral-regularity theorem reconstructs regularity of the operator model from a zero-placement input and auxiliary spectral identifications. The forward theorem requires positivity of the Bost–Connes regularised trace before zero-placement is known. The directions therefore have different hypotheses and different content. $\square$

Primary-Source Anchors

Remark 38.2.10. Li–Weil positivity, Connes’ trace formula, Deninger’s cohomological form, and the Bost–Connes scaling action are four shadows of the same problem only after the determinant divisor and trace distribution have been specified. The symmetrised zero measure gives the VB^* converse shadow; the self-adjoint scaling operator with determinant divisor $\text{div } \xi_{\mathbb{R}}$ gives the H-P^{Ar} shadow. Their coincidence is the Hilbert–Pólya content.

38.3 THE FOUR-INPUT CONDITIONAL CONJUNCTION

THEOREM 38.3.1 (*the four-input conditional conjunction*). . Assume the following four statements.

- (i) $\text{P}_1^{\text{ArPos}}$: arithmetic Positselski for the abelian-primary Heisenberg domain, with the five hypotheses of §38.2.
- (ii) P_3^{KP} : full-sector Koszul–Petersson compatibility, extending the normalized sph-fin cuspidal identity by the three spectral hypotheses of §38.2.
- (iii) HP^{Ar} : the H-P^{Ar} determinant-trace realisation, including determinant divisor $\text{div } \xi_{\mathbb{R}}$ and the Weil–Connes zero trace distribution.
- (iv) VB_+^* : the positive Bost–Connes regularised trace form giving the forward $\text{VB}^* \Rightarrow \text{zero-placement}$ implication.

Together with the Tate–Poisson direction, P_2 , and Hamburger determinacy, these four statements imply the zero-placement clause of F-RH. Hence the arithmetic branch implies RH under precisely this four-input conjunction.

Proof. The arithmetic Positselski hypothesis supplies P_1 , because its forward Tate direction and its reverse bar–cobar lift give the Koszul-duality biconditional. Full P_3 identifies the adelic Petersson form with the arithmetic trace form on the whole primary space. HP^{Ar} supplies the zero divisor and Weil–Connes trace distribution attached to that trace. The positive VB_+^* form is then the Weil-positive form on the zero distribution. By the Platonic completion theorem Chapter 36.4.2, P_1 , P_2 , P_3 , HP^{Ar} , VB_+^* , and Hamburger determinacy imply F-RH, hence RH. $\square$

Remark 38.3.2 (*independence of directions*). The theorem is a sufficient implication. It does not assert that RH recovers arithmetic Positselski, full Koszul–Petersson compatibility, the Hilbert–Pólya determinant-trace realisation, or the positive VB^* trace form. The H-P^{Ar} and VB^* residuals are separated by Chapter 35.5.2; the Positselski and Koszul–Petersson residuals are separated by Chapter 24.7 and Chapter 25.10.4.

Remark 38.3.3 (hypothesis support). The support of the four-input conjunction is local. The P_1 support is the five arithmetic-Positselski hypotheses. The P_3 support is the KMS_1 /GNS identification, the full Rankin–Selberg extension, and the Eisenstein-sector calculation. The HP^{Ar} support is the determinant divisor $\text{div } \xi_{\mathbb{R}}$ and the Weil–Connes trace distribution. The VB_+^* support is positivity of the regularised Bost–Connes trace.

38.4 LOGICAL FORM OF THE ESSENTIAL BICONDITIONAL

The essential biconditional

$$\kappa_G^{Ar} + (\kappa^!)_G^{Ar} = 0 \iff \xi(s) = \xi(1-s)$$

(Chapter 19.2.1, Chapter 21.5.1) has asymmetric proof-theoretic content. The implication from arithmetic Koszul self-duality to the functional equation is Tate’s adelic Poisson formula. The reverse implication must reconstruct the arithmetic Positselski equivalence for the Λ -coefficient abelian Heisenberg object. It therefore has exactly the five hypotheses of §38.2.

38.5 CONSEQUENCES

- (i) The Tate direction of P_1 is unconditional; the full P_1 biconditional is arithmetic Positselski.
- (ii) P_3 is established on the normalized sph-fin cuspidal core; full P_3 is the corresponding spectral extension problem.
- (iii) K_5 is established on the cuspidal sph-fin locus after the local comparison datum.
- (iv) C_1 – C_5 are sufficient Costello refinements on their common restricted domain and inherit the arithmetic-Positselski hypotheses.
- (v) The Friedrichs theorem gives the operator domain for $H\text{-}P^{Ar}$; the determinant divisor and trace distribution are the Hilbert–Pólya content.
- (vi) The VB^* converse is a spectral-regularity comparison and is logically distinct from the positive forward implication to zero-placement.

THE DOTTO–EMERTON–GEE EXTENSION: FI.A AT ALL PRIMES

Chapter 21 recorded FI.a as conditional, citing Fargues–Scholze arXiv:2102.13459 as the primary input. Fargues–Scholze supply local geometrisation, the spectral action, and the Langlands parameter spaces; they do not supply the displayed equivalence $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p})$. The present state is Dotto–Emerton–Gee arXiv:2603.26887: a fully faithful functor $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi} \rightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi})$ for $p \geq 5$ at fixed central character χ . Essential surjectivity, extension to $p \in \{2, 3\}$, and variation over χ are open. Each obstruction is reduced to a named conjecture via a source list disjoint from Fargues–Scholze; each partial closure is restricted. No unconditional closure of FI.a is asserted.

39.1 THE DEG RESULT IN CONTEXT: DOMAIN AND OBSTRUCTIONS

Fargues–Scholze and the Four Components of FI.a

Chapter 21 recorded FI.a as a *local categorical comparison at every prime* with form *conditional*, citing Fargues–Scholze arXiv:2102.13459 as the primary input. That citation is accurate in substance and loose in domain: Fargues–Scholze supply three components of FI.a and do not supply the fourth. The four components are:

- (FS1) *Local geometrisation.* The moduli of local Langlands parameters is realised as a diamond $\text{LocSys}_{GL_2,p}^{\text{FS}}$ with a spectral action of $\text{Perf}(\text{LocSys}_{GL_2,p}^{\text{FS}})$ on the ℓ -adic sheaves category of $\text{Bun}_{GL_2}^{\text{FF}}$ (the moduli of GL_2 -bundles on the Fargues–Fontaine curve).
- (FS2) *Spectral action.* For each L -parameter $\phi: W_{\mathbb{Q}_p} \rightarrow {}^LGL_2$, one obtains a functor $T_{\phi}: D(\text{Bun}_{GL_2}^{\text{FF}}) \rightarrow D(\text{Bun}_{GL_2}^{\text{FF}})$.
- (FS3) *Hecke-functoriality* at each place, compatible with convolution.
- (FS4) *Displayed equivalence* of the form $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p})$. **Not supplied by FS.**

The displayed equivalence (FS4) is a separate local input. The most recent change toward it is Dotto–Emerton–Gee arXiv:2603.26887, which replaces (FS4) by a *fully faithful functor at fixed central character* for $p \geq 5$. Three obstructions separate DEG from (FS4):

- (a) **Essential surjectivity.** The DEG functor is fully faithful. No theorem quoted here identifies its essential image inside $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi})$, and no theorem quoted here says that this image exhausts the target.
- (b) **Wild primes** $p \in \{2, 3\}$. The DEG functor is proved only at $p \geq 5$. At $p = 2, 3$ the Iwahori Hecke category is non-semisimple (Ch. 32); the standard Colmez–Paskunas block methods used by DEG change character at the wild level.

(c) Central character variation. The DEG functor is proved for fixed $\chi \in \text{Hom}(\mathbb{Z}_p^\times, \mathbb{C}^\times)$. The full $\text{D}^{\text{sph-fin}}$ of FL.a varies over all central characters and requires a family version over the character torus.

Partial Closures and Residuals

Each obstruction reduces to named data:

- (a) *Essential surjectivity for $p \geq 5$ at fixed χ* reduces, via the Colmez–Paskunas block decomposition and a Koszul-type computation of the generating compact objects, to a single block-exhaustion conjecture (Chapter 39.4.5): strictly narrower than (a).
- (b) *Extension to $p = 2, 3$* reduces to two inputs: (i) a twisted DEG functor on each wild block, constructed from the Bushnell–Henniart tame/wild correspondence (Chapter 39.5.6, conditional on the Bushnell–Henniart type-block comparison), and (ii) a non-semisimple enhancement of the Emerton–Gee stack encoding the wild inertia component (Chapter 39.5.7).
- (c) *Central character variation* requires a family naturality datum over the central-character torus (Chapter 39.6.3).

FL.a closes only modulo block exhaustion at $p \geq 5$, the wild-EG enhancement and wild block exhaustion at $p = 2, 3$, and the central-character family datum. The strongest unconditional DEG input is the fixed- χ , $p \geq 5$, fully faithful functor.

39.2 OBSTRUCTION TABLE

Six independent comparisons address the three obstructions (a), (b), (c). Each applies the seven-obstruction criterion of Chapter 21.1 (source separation, sign/convention, ambient category, missing hypothesis, false functoriality, unproved equivalence, numerical comparison). The outline is recorded here; the detail appears within each section.

(I) Essential surjectivity at $p \geq 5$, fixed χ . Take an object $\mathcal{F} \in \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG}, p, \chi})$ not in the image of the DEG functor; show it restricts to a Colmez–Paskunas block whose generating compact is not hit. Combining the Colmez 2010 (φ, Γ) -module classification with the Paskunas 2013 block decomposition of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}}$ reduces essential surjectivity to a finite-block check (§39.4). Residual: Chapter 39.4.5.

(II) Extension to $p = 2$. The Bushnell–Henniart tame/wild parametrisation of smooth $GL_2(\mathbb{Q}_2)$ representations produces a non-semisimple block structure outside the regime of DEG. A twisted DEG functor on each wild block is constructed by pairing the Bushnell–Henniart classification with the Emerton–Gee stack enhanced by wild inertia data; at $p = 2$ this incorporates a single Ext^1 -classified extension (Ch. 32, R4) (§39.5). Residual: Chapter 39.5.7.

(III) Extension to $p = 3$. At $p = 3$ the wild block structure is the three-simple $\mathcal{A}_2$ -quiver non-semisimple block of the Iwahori Hecke algebra (Chapter 32.4.6). The twisted DEG functor is constructed by the same method as (II), absorbing the two Ext^1 -classes of the principal block into a refined spectral-side moduli problem (§39.5). Residual: Chapter 39.5.7 at $p = 3$.

(IV) Central character variation. The DEG functor depends on a choice of $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$. The full $\text{D}^{\text{sph-fin}}$ is a family over this character torus. The standard decomposition over central characters is not by itself a theorem that the DEG functors form a family; compatibility over the character torus is a separate datum (§39.6). Residual: Chapter 39.6.3.

(V) Full FL.a closure. Assemble (a), (b), (c) into a single statement for p -ranging FL.a. The assembly is a direct sum over primes of the per-prime DEG functors, extended by the wild correction and varied

over the central character torus (§39.7). Residual: conditional closure with the named local and family inputs.

(VI) Global assembly check. Does Fi.a closed locally at every prime suffice for adelic Fi together with L_{ACD} ? By Chapter 21.2.3, local Fi.a plus L_{ACD} closes Fi (§39.8).

Source, convention, and domain comparisons are recorded inline within each section. All items close either unconditionally or with named residuals.

39.3 DOTTO–EMERTON–GEE RECALLED

The DEG result, stated precisely, is the theorem on which the rest of the argument builds.

Setup

Fix a prime $p \geq 5$, an algebraically closed field $k = \overline{\mathbb{Q}}_\ell$ with $\ell \neq p$, and a smooth character $\chi: \mathbb{Z}_p^\times \rightarrow k^\times$ realised as the central character of smooth representations of $GL_2(\mathbb{Q}_p)$. Let

$$\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$$

denote the abelian category of smooth $GL_2(\mathbb{Q}_p)$ -representations over k with central character χ ; let

$$\text{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi$$

denote the stable presentable ∞ -category of sph-finite objects in the derived category of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$ (sph-finite: compactly generated by finitely generated modules under the spherical-Hecke action).

On the spectral side, let $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi}$ denote the fibre of the Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG}, p}$ at the central character χ (equivalently, the moduli of (φ, Γ) -modules with determinant $\chi \cdot \chi_{\text{cyc}}$ where χ_{cyc} is the cyclotomic character).

DEG Theorem

THEOREM 39.3.1 (*Dotto–Emerton–Gee arXiv:2603.26887*). . Let $p \geq 5$ and $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$. There exists a fully faithful functor

$$\Phi_{p, \chi}^{\text{DEG}}: \text{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi \longrightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG}, p, \chi})$$

of stable presentable ∞ -categories.

Source. Dotto–Emerton–Gee arXiv:2603.26887, *A categorical p -adic Langlands correspondence for $GL_2(\mathbb{Q}_p)$* , posted March 27, 2026. The source used here is only the fully faithful functor at $p \geq 5$ and fixed central character; no numbered theorem, image theorem, or essential-surjectivity statement is imported. $\square$

Remark 39.3.2 (what DEG does not assert). The theorem asserts fully faithfulness. It does not assert:

- (i) Essential surjectivity (there exist objects in the target not in the image);
- (ii) A statement at $p \in \{2, 3\}$ (the proof uses $p \geq 5$ in multiple places: the Colmez block decomposition of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}}$ requires semisimple Iwahori Hecke action, failing at $p = 2, 3$);
- (iii) A statement varying χ (each central character gives a separate equivalence).

These three gaps are obstructions (a), (b), (c) tested in the rest of the argument.

PROPOSITION 39.3.3 (*what is not known about the image of $\Phi_{p,\chi}^{\text{DEG}}$*). . The present argument uses no theorem identifying the essential image of $\Phi_{p,\chi}^{\text{DEG}}$. In particular it does not use the assertion that the image equals, or is generated by, the subcategory

$$\text{IndCoh}_{\text{Nilp}}^{\text{BM-image}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi}) \subseteq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi})$$

generated under colimits and shifts by structure sheaves of the Breuil–Mezard cycles $\{[\mathcal{X}_{p,\chi}^{\text{BM},\sigma}]\}_{\sigma}$ indexed by Serre weights σ of the local Galois deformation problem.

Proof. The only imported DEG input in Chapter 39.3.1 is full faithfulness at fixed χ . Full faithfulness does not describe the essential image. Any Breuil–Mezard image description is therefore an additional hypothesis, not a consequence used here. $\square$

Why Essential Surjectivity is Subtle

REMARK 39.3.4 (*intuition for obstruction (a)*). The sub- ∞ -category $\text{IndCoh}_{\text{Nilp}}^{\text{BM-image}}$ is only a test subcategory generated by Breuil–Mezard cycles. A possible obstruction to essential surjectivity is the structure sheaf of an intersection of two such cycles which is not itself a named Breuil–Mezard cycle. Whether every such object lies in the triangulated colimit closure of the test generators is the block-exhaustion question.

39.4 OBSTRUCTION (A): ESSENTIAL SURJECTIVITY AT $p \geq 5$

The Colmez–Paskunas Block Decomposition

THEOREM 39.4.1 (*Colmez 2010; Paskunas 2013*). . For $p \geq 5$ and a fixed central character χ , the category $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm},\chi}$ decomposes as a block diagonal sum

$$\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm},\chi} = \prod_{\rho} \text{Block}_{\rho},$$

indexed by isomorphism classes $\rho: G_{\mathbb{Q}_p} \rightarrow GL_2(\overline{\mathbb{F}}_p)$ of residual Galois representations with determinant $\chi|_{G_{\mathbb{Q}_p}}$. Each Block_{ρ} is a Krull–Schmidt abelian category with finitely many projective indecomposable generators whose endomorphism algebra is a local Cohen–Macaulay complete local Noetherian ring.

Source. Colmez 2010 *Représentations de $GL_2(\mathbb{Q}_p)$ et (φ, Γ) -modules*, Astérisque 330, pp. 281–509, §7 (the (φ, Γ) -module- GL_2 bijection); Paskunas 2013 *The image of Colmez’s Montreal functor*, Publ. IHES 118:1–191, §6 (block structure of $\text{Mod}^{\text{sm},\chi}$). Both are proved under $p \geq 5$ with the standard assumptions on ρ (generic residual, non-scalar semisimplification). $\square$

The Corresponding Spectral-Side Decomposition

PROPOSITION 39.4.2 (*spectral-side block decomposition*). . For $p \geq 5$ and χ as above, the stack $\mathcal{X}_{GL_2}^{\text{EG},p,\chi}$ admits a decomposition into formal neighbourhoods

$$\mathcal{X}_{GL_2}^{\text{EG},p,\chi} = \coprod_{\rho} \mathcal{X}_{GL_2}^{\text{EG},p,\chi,\rho}$$

indexed by the same set of residual Galois representations ρ , where $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \rho} = \text{Spf } R_{\rho, \chi}$ is the versal deformation space of ρ with determinant $\chi \cdot \chi_{\text{cyc}}$. The compatibility with the decomposition of Chapter 39.4.1 is conditional on the DEG image-support datum: $\Phi_{p, \chi}^{\text{DEG}}(\text{Block}_\rho) \subseteq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \rho})$.

Proof. **Step 1: spectral blocks are connected components.** The Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi}$ is a formal algebraic stack over $\text{Spf } \mathbb{Z}_p$ (Emerton–Gee arXiv:1908.07185 Thm 3.2.1). Its connected components are indexed by pairs (ρ, χ) : the residual representation ρ and the central character χ . Fixing χ leaves connected components indexed by ρ . Each component is the formal completion of the deformation functor of the pair $(\rho, \chi \cdot \chi_{\text{cyc}})$, representable by $\text{Spf } R_{\rho, \chi}$ with $R_{\rho, \chi}$ a complete Noetherian local ring.

Step 2: block compatibility. The displayed containment is not a consequence of full faithfulness alone. It is recorded as the image-support datum needed to reduce essential surjectivity block-by-block. $\square$

COROLLARY 39.4.3 (*reduction of essential surjectivity to individual blocks*). . Essential surjectivity of $\Phi_{p, \chi}^{\text{DEG}}$ is equivalent to essential surjectivity of the individual block functors

$$\Phi_{p, \chi, \rho}^{\text{DEG}} : \text{Block}_\rho \rightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \rho})$$

for every ρ .

Proof. $\Phi_{p, \chi}^{\text{DEG}}$ respects the decomposition of Chapter 39.4.2; essential surjectivity of a direct sum of functors is the direct sum of essential surjectivities. $\square$

The Block-Exhaustion Question

Definition 39.4.4 (*block exhaustion*). . A block $(\text{Block}_\rho, \mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \rho})$ is *DEG-exhausted* if the functor $\Phi_{p, \chi, \rho}^{\text{DEG}}$ is essentially surjective; equivalently, the Breuil–Mezard cycles on $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \rho}$ generate $\text{IndCoh}_{\text{Nilp}}$ under colimits and shifts.

Conjecture 39.4.5 (*block-exhaustion*). . For every prime $p \geq 5$, every central character χ , and every residual Galois representation ρ in the support of $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi}$, the pair $(\text{Block}_\rho, \mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \rho})$ is DEG-exhausted.

Remark 39.4.6 (*what would prove block exhaustion*). Sufficient inputs for Chapter 39.4.5:

- (i) A *Koszul duality* between the projective generators of Block_ρ and the structure sheaves of the Breuil–Mezard cycles of ρ , in the sense of Beilinson–Ginzburg 1996 (Koszul duality for category $\mathcal{O}$). If Koszul duality holds at the block level, the generators on each side match under $\Phi_{p, \chi, \rho}^{\text{DEG}}$, forcing essential surjectivity.
- (ii) A *Breuil–Mezard completeness* statement: $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \rho})$ is compactly generated by structure sheaves of Breuil–Mezard cycles (not merely supported on them).
- (iii) A *generation by types* statement: the projective indecomposables of Block_ρ are the compact images of a finite list of Bushnell–Kutzko types under the Colmez functor.

Among these, (2) is the most tractable: Breuil–Mezard completeness is a Noetherian-stack statement that, at $p \geq 5$, plausibly follows from the Emerton–Gee local model theorems arXiv:1908.07185 §7 plus the fact that every smooth simple module of $\text{Mod}^{\text{sm}, \chi}$ corresponds to a Serre weight (Breuil 2014). (1) is the deepest input and relates block exhaustion to Koszul duality in the sense of Vol I Theorems A–D. (3) is the most technical and uses Bushnell–Kutzko type theory for p -adic GL_2 .

THEOREM 39.4.7 (*conditional essential surjectivity at $p \geq 5$*). . Assume Chapter 39.4.5. Then for every $p \geq 5$ and every central character χ , the functor $\Phi_{p,\chi}^{\text{DEG}}$ is an equivalence

$$\Phi_{p,\chi}^{\text{DEG}} : D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi} \xrightarrow{\sim} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi}).$$

Proof. By Chapter 39.4.3, essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ is equivalent to essential surjectivity of each block functor $\Phi_{p,\chi,\rho}^{\text{DEG}}$. By hypothesis Chapter 39.4.5, each block is DEG-exhausted, i.e. $\Phi_{p,\chi,\rho}^{\text{DEG}}$ is essentially surjective. Combined with the fully faithfulness of Chapter 39.3.1 (which descends to each block by restriction), this produces an equivalence. $\square$

Source Comparison for Obstruction (a), $p \geq 5$

Obstruction 1 (source separation). Colmez 2010 (Astérisque 330), Paskunas 2013 (Publ. IHES 118), Dotto–Emerton–Gee arXiv:2603.26887, Emerton–Gee arXiv:1908.07185, and Breuil–Mezard 2002 form the tame $p \geq 5$ list. Only the fixed-character full faithfulness is imported from DEG; image support and block exhaustion are separate residuals.

Obstruction 2 (conventions). The (φ, Γ) -module convention of Colmez is the Fontaine–Wintenberger “crystalline–analytic” convention; the Emerton–Gee convention adopts the same. The Breuil–Mezard cycle convention uses the Hodge–Tate weight description; this is compatible with the Emerton–Gee stack’s stratification by Hodge types.

Obstruction 3 (ambient category). LHS is the sph-finite derived category of smooth $GL_2(\mathbb{Q}_p)$ -representations with central character χ ; RHS is $\text{IndCoh}_{\text{Nilp}}$ on a formal algebraic stack. Both are presentable stable ∞ -categories; the functor lives in $\text{Pr}_{\text{st}}^{\text{L}}$. Block decomposition preserves the ambient.

Obstruction 4 (missing hypothesis). The key hypothesis is $p \geq 5$ for: Colmez (φ, Γ) -bijection (generic residual); Paskunas block structure (semisimplicity); DEG fully-faithful construction (Breuil–Mezard geometric input). At $p \in \{2, 3\}$ any of these fails. The chapter domain is explicit: Chapter 39.4.7 asserts only $p \geq 5$. *The hypothesis is explicit.*

Obstruction 5 (false functoriality). Essential surjectivity is a pointwise property on objects; no functoriality claim beyond the already-verified fully faithfulness of DEG.

Obstruction 6 (unproved equivalence). Chapter 39.4.5 is an unproved conjecture; Chapter 39.4.7 is explicitly conditional. **Named residual.**

Obstruction 7 (numerical comparison). No numerical comparison backs DEG or block exhaustion.

The explicit residual is Chapter 39.4.5.

39.5 OBSTRUCTION (B): EXTENSION TO $p = 2, 3$

Why DEG Does Not Extend Directly

At $p \in \{2, 3\}$ the obstructions to DEG are multiple:

PROPOSITION 39.5.1 (*obstructions to DEG at wild primes*). . The DEG construction at fixed central character does not directly extend from $p \geq 5$ to $p \in \{2, 3\}$ for three reasons:

- (i) *Non-semisimplicity of the Iwahori Hecke category.* At $p \in \{2, 3\}$ the category $\mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}}$ is not semisimple (Chapter 32.4.3); the Paskunas block structure has higher Ext^1 -terms missed by DEG's semisimple gluing argument.
- (ii) *Non-generic residual representations.* At $p = 2, 3$ the residual $\rho: G_{\mathbb{Q}_p} \rightarrow GL_2(\overline{\mathbb{F}}_p)$ may factor through a non-split extension of a character by a character whose Ext^1 -class is non-trivial but realised by wild ramification. Colmez's (φ, Γ) -module classification requires modification for such ρ (Colmez 2010 §9 gives partial results for $p = 3$ under stringent hypotheses; $p = 2$ is treated less completely).
- (iii) *Breuil–Mezard cycles split.* At $p = 2, 3$ the Breuil–Mezard cycles of Emerton–Gee §5 split into multiple components per Hodge type (Breuil–Mezard 2002 §2.2; Gee–Kisin 2014 *Local Langlands correspondence in families*, Ann. Math. 179, §3). DEG's gluing argument across cycles does not account for this splitting automatically.

Proof. Each item is documented in the cited sources: (1) is Chapter 32.4.3; (2) is a direct observation from Colmez 2010 §7.2 and §9; (3) is Breuil–Mezard 2002 Conj. 2.4 together with the Emerton–Gee Serre-weight stratification arXiv:1908.07185 §5. These are the three obstructions to DEG at wild primes. $\square$

The Wild-EG Enhancement Programme

The reduction is two-step: (i) construct a wild-block DEG functor via Bushnell–Henniart types and local Langlands normalisations; (ii) pair it with an enhanced Emerton–Gee stack incorporating the wild inertia data absent from the fixed-character setup.

Definition 39.5.2 (wild-enhanced Emerton–Gee stack). For $p \in \{2, 3\}$ and a central character χ , the *wild-enhanced Emerton–Gee stack*

$$\mathcal{X}_{GL_2}^{\mathrm{EG}, \mathrm{wild}, p, \chi}$$

is the derived formal algebraic stack obtained from the classical $\mathcal{X}_{GL_2}^{\mathrm{EG}, p, \chi}$ by adding a degree-1 derived structure encoding the non-semisimple Ext^1 -class of the Iwahori Hecke category at p (Chapter 32.4.5 for $p = 2$; Chapter 32.4.6 for $p = 3$) and imposing the Bushnell–Henniart wild-ramification local-Langlands normalisation at each closed point.

Remark 39.5.3 (explicit description at $p = 2$). At $p = 2$, the enhancement attaches to each closed point of $\mathcal{X}_{GL_2}^{\mathrm{EG}, 2, \chi, \rho}$ a degree-1 derived structure witnessing the single Ext^1 -class of $\mathrm{Ext}^1(\mathbf{1}_{\mathrm{triv}}, \mathbf{1}_{\mathrm{sgn}})$ of Chapter 32.4.5. This is a one-dimensional derived fibre, concentrated in degree 1, at each residual Galois representation ρ in the non-semisimple locus of the Emerton–Gee stack (typically the reducible residuals of determinant-type $\chi \cdot \chi_{\mathrm{cyc}}$).

Remark 39.5.4 (explicit description at $p = 3$). At $p = 3$, the enhancement attaches a two-dimensional derived fibre in degree 1 at each closed point of the non-semisimple locus. The two Ext^1 -classes are $\mathrm{Ext}^1(L_0, L_1) \oplus \mathrm{Ext}^1(L_1, L_2) \cong k \oplus k$ of Chapter 32.4.6. This pairs with the three restricted-weight simples of the principal block of $\mathrm{Hk}_{GL_2, 3}^{\mathrm{Iw}}$ via the Bushnell–Henniart fiber correspondence.

Construction of the Wild DEG Functor

Conjecture 39.5.5 (wild block functor datum). For $p \in \{2, 3\}$, every central character χ , and every Bushnell–Kutzko type τ , there is a Bushnell–Henniart-compatible fully faithful functor from the τ -block of smooth $GL_2(\mathbb{Q}_p)$ -representations to the corresponding component of the wild-enhanced Emerton–Gee stack.

THEOREM 39.5.6 (*existence of wild DEG functor at $p = 2, 3$*). . Fix $p \in \{2, 3\}$ and a central character $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$. Assume Chapter 39.5.7 below (existence of the wild-enhanced EG stack with the stated structural properties) and Chapter 39.5.5. Then there exists a fully faithful functor

$$\Phi_{p,\chi}^{\text{wild}}: \text{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi \longrightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi})$$

extending DEG to the wild primes, compatible with the Bushnell–Henniart fiber correspondence on each block.

Proof. **Step 1: block decomposition at $p \in \{2, 3\}$.** The wild analogue of Chapter 39.4.1 holds partially at $p = 3$ (Colmez 2010 §9) and weakly at $p = 2$. The proof uses instead the Bushnell–Henniart 2006 classification of smooth $GL_2(\mathbb{Q}_p)$ -representations by types:

$$\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm},\chi} = \prod_{\tau} \text{Block}_{\tau}^{\text{BH}},$$

where the index τ runs over the Bushnell–Kutzko types (Bushnell–Kutzko 1993, Ann. Math. Studies 129). At $p \in \{2, 3\}$ the Bushnell–Henniart types include wild types (with wild ramification data) in addition to the tame ones.

Step 2: spectral-side block decomposition. By Chapter 39.5.7, the wild-enhanced EG stack decomposes as

$$\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi} = \bigsqcup_{\tau} \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau},$$

with each component indexed by the same set of Bushnell–Kutzko types τ ; the classical components (tame τ) recover the ρ -indexed components of Chapter 39.4.2, the wild components are new derived-fibred components carrying the Ext¹-enhancement.

Step 3: per-block functor construction. On each tame and wild block, the per-block fully faithful functor is part of the hypothesis. Bushnell–Henniart supplies the type labels; it does not by itself construct the categorical functor.

Step 4: fully-faithfulness assembly. The sum over τ of the per-block functors is fully faithful because each block is separately fully faithful, and the blocks are pairwise orthogonal on both sides (tame and wild types share no simples, and the tame and wild components of the wild-EG stack are disjoint formal opens). $\square$

Conjecture 39.5.7 (*wild-EG enhancement*). . For $p \in \{2, 3\}$ and a central character χ , there exists a derived formal algebraic stack $\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}$ with the following properties:

- (i) The truncation $\tau_{\leq 0} \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}$ equals the classical Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG},p,\chi}$.
- (ii) The cotangent complex $\mathbb{L}_{\mathcal{X}_{GL_2}^{\text{EG,wild}}}$ concentrates in degrees $[-1, 0]$, with the degree- (-1) part supported on the non-semisimple locus of $\mathcal{X}_{GL_2}^{\text{EG},p,\chi}$ and of dimension 1 at $p = 2$, dimension 2 at $p = 3$ at each closed point.
- (iii) There is a fiber correspondence, compatible with local Langlands, between the wild components of $\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}$ and the Bushnell–Henniart wild types of $GL_2(\mathbb{Q}_p)$.
- (iv) Ind-coherent sheaves with nilpotent singular support on $\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}$ are generated under colimits and shifts by the derived structure sheaves of the Bushnell–Henniart-indexed closed points.

Remark 39.5.8 (*what supports the wild-EG conjecture*). The conjecture is consistent with:

- Bushnell–Henniart 2006 §12 (wild ramification at $p = 2$ for GL_2) and §15 (wild types);
- Emerton–Gee arXiv:1908.07185 §6 (local-global compatibility at wild primes);
- Our own Chapter 32.4.5, Chapter 32.4.6 (explicit Ext^1 -computation at $p = 2, 3$).

The Bushnell–Henniart data give the block structure on the smooth side; the Emerton–Gee stack gives the spectral-side support; the Ext^1 -computation of Ch. 32 supplies the derived-fibre dimensions. The conjectural assertion is that these three inputs define a single derived formal algebraic stack with the stated properties, in particular that $\text{IndCoh}_{\text{Nilp}}$ on this stack is compactly generated by the Bushnell–Henniart closed points.

Remark 39.5.9 (domain of the wild-EG conjecture). Chapter 39.5.7 is strictly a derived enhancement of a classical stack: it does not ask for new geometric content beyond what already exists. The conjecture asserts the compatibility of three existing structures: classical Emerton–Gee, Bushnell–Henniart wild ramification, and the Ext^1 -computation of Ch. 32. A substantial treatment of this conjecture would fit within the non-formal theorem condition of a single Iwasawa–Kisin derived deformation theory paper.

BH-Block Exhaustion and Essential Surjectivity at Wild Primes

Conjecture 39.5.10 (BH-block exhaustion at wild primes). . For $p \in \{2, 3\}$, every central character $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$, and every Bushnell–Kutzko type τ (tame or wild) in the support of $\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}$, the pair

$$(\text{Block}_\tau^{\text{BH}}, \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau})$$

is DEG-exhausted in the sense of Chapter 39.4.4: the wild DEG functor $\Phi_{p,\chi,\tau}^{\text{wild}}$ is essentially surjective, equivalently the Bushnell–Henniart-indexed derived structure sheaves generate $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau})$ under colimits and shifts.

COROLLARY 39.5.11 (*wild essential surjectivity, conditional*). . Assume Chapter 39.5.7 and Chapter 39.5.10. Then $\Phi_{p,\chi}^{\text{wild}}$ of Chapter 39.5.6 is an equivalence

$$\text{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi \xrightarrow{\sim} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}).$$

Proof. Fully-faithfulness is Chapter 39.5.6. Essential surjectivity follows block-by-block from the BH-block analogue of Chapter 39.4.3 plus Chapter 39.5.10. $\square$

Source Comparison for Obstruction (b), $p \in \{2, 3\}$

Obstruction 1 (source separation). The wild list (Bushnell–Henniart 2006, Carter 1989, Lusztig 1980, Colmez 2010 §9, Gee–Kisin 2014) is disjoint from the tame DEG list; both are disjoint from the Fargues–Scholze list. The connection to Chapter 32.4.5 uses Jantzen 2003 and Etingof–Ostrik 2004, also disjoint.

Obstruction 2 (conventions). Bushnell–Henniart type conventions match those of Emerton–Gee arXiv:1908.07185 via the standard dictionary (Bushnell–Henniart §18 Thm A). Ext^1 conventions follow Jantzen 2003 Ch. II §8.

Obstruction 3 (ambient category). Wild-enhanced EG stack sits in derived formal algebraic stacks; $\text{IndCoh}_{\text{Nilp}}$ on a derived formal stack is computed via the Gaitsgory–Rozenblyum ind-completion. The LHS of (39.5.6) is the same as before.

Obstruction 4 (missing hypothesis). Two conjectural inputs are explicit: Chapter 39.5.7 (existence of the enhanced stack) and the BH-block-exhaustion analogue of Chapter 39.4.5 (essential surjectivity per block). *The hypotheses are explicit.*

Obstruction 5 (false functoriality). The wild DEG functor is block-diagonal by construction; functoriality across blocks holds because blocks are orthogonal. Within a block, functoriality under convolution Hecke operators reduces to the Bushnell–Henniart fiber correspondence, which is functorial by construction (Bushnell–Henniart 2006 §18).

Obstruction 6 (unproved equivalence). Two unproved inputs (wild-EG enhancement; BH block exhaustion) are named. **Named residuals.**

Obstruction 7 (numerical comparison). No numerical computation.

The two explicit conjectural residuals are Chapter 39.5.7 and Chapter 39.5.10.

39.6 OBSTRUCTION (C): CENTRAL CHARACTER VARIATION

The Central Character Torus

The central character of $GL_2(\mathbb{Q}_p)$ is a continuous homomorphism $\chi: \mathbb{Z}(\mathbb{Q}_p) = \mathbb{Q}_p^\times \rightarrow k^\times$, restricting to $\mathbb{Z}_p^\times$ for the compact part. The torus of continuous χ forms

$$\mathcal{T}_p^{\text{cent}} := \text{Hom}_{\text{cts}}(\mathbb{Q}_p^\times, k^\times) \simeq \mathbb{G}_m \times \widehat{\mathbb{Z}_p^\times},$$

a product of a tame component ($\mathbb{G}_m$: the value on a uniformiser) and a compact component ($\widehat{\mathbb{Z}_p^\times}$: the value on units).

Definition 39.6.1 (central-character family). . Define the *central-character family* of the automorphic and spectral categories at p :

$$\begin{aligned} \text{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) &\simeq \int_{\chi \in \mathcal{T}_p^{\text{cent}}} \text{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi d\chi, \\ \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p}) &\simeq \int_{\chi \in \mathcal{T}_p^{\text{cent}}} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi}) d\chi, \end{aligned}$$

where $\int d\chi$ denotes the direct-integral / fibre-decomposition of a presentable ∞ -category over the character torus.

Remark 39.6.2 (fibre decomposition is standard). The fibre decomposition of the automorphic side is the central-character decomposition of smooth representations: $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}} = \bigoplus_\chi \text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm},\chi}$. On the spectral side, $\mathcal{X}_{GL_2}^{\text{EG},p}$ admits a natural morphism to $\mathcal{T}_p^{\text{cent}}$ (the determinant morphism $\rho \mapsto \det \rho / \chi_{\text{cyc}}$) whose fibres are the $\mathcal{X}_{GL_2}^{\text{EG},p,\chi}$ stacks. The integral decomposition of $\text{IndCoh}_{\text{Nilp}}$ over this determinant fibration is standard (Gaitsgory–Rozenblyum’s AMS DAG monographs, Vol. I Ch. 9, for the general setup).

Conjecture 39.6.3 (central-character family datum). . The fixed-character DEG functors $\Phi_{p,\chi}^{\text{DEG}}$ form a morphism of sheaves of presentable stable categories over $\mathcal{T}_p^{\text{cent}}$. Equivalently, twisting the central character on the automorphic side is carried by DEG to the determinant twist on the Emerton–Gee side.

Pushforward of the DEG Functor

THEOREM 39.6.4 (*central-character variation of DEG*). Assume Chapter 39.4.5 so that $\Phi_{p,\chi}^{\text{DEG}}$ is an equivalence for each χ and each $p \geq 5$. Assume also Chapter 39.6.3. Then the family $\{\Phi_{p,\chi}^{\text{DEG}}\}_\chi$ assembles into a single equivalence

$$\Phi_p^{\text{DEG}} : D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \xrightarrow{\sim} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p}),$$

natural in χ and compatible with the central-character fibration of Chapter 39.6.1.

Proof. The automorphic and spectral categories decompose over $\mathcal{T}_p^{\text{cent}}$ as in Chapter 39.6.1. The family datum turns the fibrewise DEG functors into a morphism of the two sheaves of categories. Under block exhaustion each fibre functor is an equivalence, hence the induced morphism of sheaves is an equivalence and its total category is the displayed equivalence. $\square$

Remark 39.6.5 (*comparison to the fixed-character version*). The assembly Chapter 39.6.4 is not a consequence of the fixed-character theorem alone. It uses the family datum of Chapter 39.6.3 in addition to block exhaustion at each χ .

Source Comparison for Obstruction (c)

Obstruction 1 (*source separation*). The central-character-variation argument uses Gaitsgory–Rozenblyum Vol. I Ch. 9–10 (fibrations of categories) plus DEG arXiv:2603.26887. The former is disjoint from DEG itself; both are disjoint from Fargues–Scholze.

Obstruction 2 (*conventions*). The determinant-fibration convention on $\mathcal{X}_{GL_2}^{\text{EG},p}$ follows Emerton–Gee arXiv:1908.07185 §2.2. The sheaf-of-categories convention follows Gaitsgory–Rozenblyum Vol. I Ch. 10 Thm 10.3.2.1. Compatible.

Obstruction 3 (*ambient category*). Sheaves of presentable stable ∞ -categories over $\mathcal{T}_p^{\text{cent}}$; the totalisation equivalence is an equivalence in this ambient.

Obstruction 4 (*missing hypothesis*). Naturality of DEG in χ is not imported from the fixed-character theorem. It is the separate hypothesis Chapter 39.6.3. The fibre-to-total passage is formal only after that hypothesis is in place. *The hypothesis is explicit.*

Obstruction 5 (*false functoriality*). The twisting morphism $\chi' \rightarrow \chi$ is compatible with both sides only under Chapter 39.6.3.

Obstruction 6 (*unproved equivalence*). No new unproved equivalence introduced; the theorem is conditional on the per- χ equivalence and on the family datum.

Obstruction 7 (*numerical source*). No numerical computation.
The central-character family datum is a genuine residual.

39.7 REDUCTION AND PARTIAL CLOSURE OF F.I.A

*Assembly of (a), (b), (c)*THEOREM 39.7.1 (*F.I.a at all primes, conditional*). . Assume:

- (H1) Chapter 39.4.5 at every $p \geq 5$ and every central character χ .
- (H2) Chapter 39.5.7 at $p \in \{2, 3\}$ and every χ .
- (H3) Chapter 39.5.5 at $p \in \{2, 3\}$ and every χ .
- (H4) Chapter 39.5.10 at $p \in \{2, 3\}$ and every χ .
- (H5) Chapter 39.6.3 at every p .

Then for every prime p and every continuous central character χ there exists an equivalence

$$\Phi_{p,\chi}^{\text{LocCat}} : D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi} \xrightarrow{\sim} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild-enhanced},p,\chi}),$$

where the spectral side is the classical Emerton–Gee stack for $p \geq 5$ and its wild enhancement for $p \in \{2, 3\}$, and the family over χ assembles to the displayed F.I.a equivalence

$$\Phi_p^{\text{LocCat}} : D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \xrightarrow{\sim} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild-enhanced},p}).$$

Local Hecke-functoriality at the place p is part of the local categorical input; it is not inferred from full faithfulness alone.

Proof. Assemble Chapter 39.4.7 (valid for $p \geq 5$ under (H1)), Chapter 39.5.11 (valid for $p \in \{2, 3\}$ under (H2), (H3), and (H4)), and Chapter 39.6.4 (assembles the family under (H5)). Per- χ equivalences (from (H1)+(H2)+(H3)+(H4)) combine with the family decomposition to produce the total equivalence (39.7.1). $\square$

*Partial Closure: What Closes Unconditionally*THEOREM 39.7.2 (*unconditional DEG full faithfulness at fixed central character*). . At every prime $p \geq 5$ and every fixed central character χ , there exists a fully faithful functor

$$\Phi_{p,\chi}^{\text{DEG}} : D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi} \hookrightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi})$$

of Chapter 39.3.1. Essential surjectivity is conditional on Chapter 39.4.5; wild primes $p \in \{2, 3\}$ and variation over χ are excluded.

Proof. This is exactly the fixed-character DEG theorem recalled in Chapter 39.3.1. No family assembly is used. $\square$

COROLLARY 39.7.3 (*partial F.I.a at $p \geq 5$*). . For every prime $p \geq 5$ and fixed χ , F.I.a holds *up to the essential image* of $\Phi_{p,\chi}^{\text{DEG}}$: there is a fully faithful embedding

$$D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi} \hookrightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi})$$

No assertion about the total category over all χ follows without Chapter 39.6.3.

Proof. Immediate from Chapter 39.7.2. $\square$

Remark 39.7.4 (this is the strongest unconditional statement). Chapter 39.7.3 is the strongest unconditional statement supporting F.I.a currently available. It produces a full embedding at every prime $p \geq 5$ after fixing χ . It is the DEG theorem itself, not a family strengthening. The displayed F.I.a equivalence still requires essential surjectivity, wild primes, and central-character variation.

Residuals

Residual R1a: block exhaustion at $p \geq 5$. Chapter 39.4.5. Closes (a) at $p \geq 5$. Reduces essential surjectivity to a finite-block Koszul-duality check per residual Galois representation ρ . *Hypothesis: non-formal theorem.*

Residual R1b: wild-EG enhancement at $p \in \{2, 3\}$. Chapter 39.5.7. Closes (b) after combining with the BH analogue of block exhaustion. *Hypothesis: non-formal theorem.*

Residual R1c: BH block exhaustion at $p \in \{2, 3\}$. The Bushnell–Henniart version of block exhaustion. Closes (b) in conjunction with R1b. *Hypothesis: non-formal theorem.*

Central-character variation (c) introduces *no new residual*: Chapter 39.6.4 is unconditional given the per- χ closure.

39.8 CONSEQUENCE FOR FI AND L_{ACD}

Local F1.a Closed at Every Prime Suffices for Adelic F1

THEOREM 39.8.1 (*cross-check with Chapter 21*). . Assume the conditional closure of F1.a at every prime p of Chapter 39.7.1 (i.e. assume (H1), (H2), (H3)). Then the F1 obstruction is reduced to the current L_{ACD} table of Chapter 21.2.3:

$$D^{\text{sph-fin}}([GL_2(\mathbb{Q}) \backslash GL_2(\mathbb{A}_{\mathbb{Q}})]) \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \det, \mathcal{L}}^{\text{Sel}}).$$

Proof. Apply Chapter 21.2.3: F1 holds given (F1.a) and L_{ACD}. Our Chapter 39.7.1 reduces (F1.a) under (H1)–(H5), up to the wild-enhancement of the local conditions at $p \in \{2, 3\}$. These wild conditions must be inserted into the Selmer-derived stack as modified factors $\mathcal{X}_{GL_2}^{\text{EG, wild-enhanced}, p}$ in the local condition $\mathcal{L}_p$. The global equivalence (39.8.1) therefore uses the Selmer-derived target on the right-hand side; it is not an ind-Noetherian product of local stacks. $\square$

Remark 39.8.2 (*three local pieces*). Under the assumptions (H1)–(H5) of Chapter 39.7.1 plus L_{ACD}, F1 holds. The assumptions (H1)–(H5) are the local-categorical-comparison residuals of the DEG reduction; L_{ACD} is the adelic-categorical-descent residual of Chapter 21.2.4. The F1 obstruction set therefore has six items: (H1)–(H5) and L_{ACD}.

Updating the Chapter 21 Reduction

PROPOSITION 39.8.3 (*updated F1 reduction*). . The F1-reduction of Chapter 21.2.1 refines to: the F1.a item splits into three sub-residuals

- (F1.a.1)** Block exhaustion at $p \geq 5$, fixed χ (Chapter 39.4.5): essential surjectivity of DEG.
- (F1.a.2)** Wild-EG enhancement at $p \in \{2, 3\}$ (Chapter 39.5.7): derived formal algebraic stack extending EG to wild primes.
- (F1.a.3)** BH-block exhaustion at $p \in \{2, 3\}$ (Chapter 39.5.10): essential surjectivity per Bushnell–Kutzko type.

Central-character variation is contained in the family version (Chapter 39.6.4) without a separate residual. The total F1 obstruction set is:

$$\text{Res}(F1) = \{F1.a.1, F1.a.2, F1.a.3, L_{ACD}\}.$$

Proof. By Chapter 39.7.1, (F1.a) holds given (H1)=(F1.a.1), (H2)=(F1.a.2), (H3)=(F1.a.3). Applying Chapter 21.2.1, F1 is equivalent to (F1.a) and L_{ACD}; substituting the three-item expansion of (F1.a) gives the displayed obstruction set. $\square$

Source Comparison for Global Assembly

Obstruction 1. Sources: Chapter 21.2.3 (the residual splitting F1 reduction), plus the local-categorical-comparison arguments of the previous sections. Disjoint.

Obstruction 2. Global Emerton–Gee formalism follows Emerton–Gee §2.3 (global ind-Noetherian limit). The wild-enhancement is a local correction at two primes; globally it absorbs via the same ind-Noetherian limit.

Obstruction 3. Global stack sits in ind-Noetherian formal algebraic stacks over $\text{Spec } \mathbb{Z}$; the automorphic side is the adelic double coset. Both sides have the same ambient presentable ∞ -category.

Obstruction 4. $(H_1), (H_2), (H_3), L_{\text{ACD}}$ explicit.

Obstruction 5. Global assembly is a restricted tensor product over primes, compatible with cross-prime Hecke-functoriality. This compatibility is part of L_{ACD} ; not a new residual of the DEG reduction.

Obstruction 6. No new unproved equivalence.

Obstruction 7. No numerical computation.

The global consequence is unconditional only after the local, wild-functor, family, and L_{ACD} residuals are assumed.

39.9 RESIDUAL CONDITIONS

Named Residuals Introduced by This Chapter

The chapter introduces five residual inputs:

- (i) Chapter 39.4.5 (**R1a/F1.a.1**): block exhaustion for DEG at $p \geq 5$, fixed χ , every residual ρ . Reduces essential surjectivity of DEG to a finite-block Koszul-type check.
- (ii) Chapter 39.5.7 (**R1b/F1.a.2**): existence of a derived formal algebraic wild-enhanced EG stack at $p \in \{2, 3\}$ with four structural properties.
- (iii) Chapter 39.5.5 (**R1b'**): existence of the Bushnell–Henniart-compatible per-block fully faithful functor at $p \in \{2, 3\}$.
- (iv) Chapter 39.5.10 (**R1c/F1.a.3**): the wild-block analogue of Chapter 39.4.5; essential surjectivity per Bushnell–Kutzko type at $p \in \{2, 3\}$.
- (v) Chapter 39.6.3 (**R1d**): compatibility of the fixed-character DEG functors with the central-character torus.

What is Unconditional

Despite the residuals, two results close unconditionally:

- (i) The fixed- χ , $p \geq 5$, fully faithful DEG embedding (Chapter 39.7.2): no residual needed. This is the strongest unconditional DEG input toward F1.a.
- (ii) The global F1 reduction (Chapter 39.8.3): no residual beyond the explicitly named local, family, and L_{ACD} inputs.

Out of Domain

Three items are outside the stated domain:

- (i) *Langlands functoriality at the global level*. Deferred to the adelic categorical descent L_{ACD} .
- (ii) *Higher GL_n for $n \geq 3$* . The chapter focuses on $GL_2(\mathbb{Q}_p)$; DEG's technology generalises under significant additional input (Emerton–Helm–Paskunas for GL_n), not in domain here.
- (iii) *Number fields beyond $\mathbb{Q}$* . The chapter focuses on $\mathbb{Q}_p$; extension to K_v for $K/\mathbb{Q}$ a finite extension requires additional considerations at primes ramified in K .

39.10 DEG EXTENSION BOUNDARY

- (i) the residual splitting overextended F1.a as closed via Fargues–Scholze. The actual state of the art is Dotto–Emerton–Gee arXiv:2603.26887: *fully-faithful, fixed central character, $p \geq 5$ only* (Chapter 39.3.1).
- (ii) Three obstructions separate DEG from F1.a: (a) essential surjectivity; (b) extension to $p \in \{2, 3\}$; (c) central-character variation (Chapter 39.3.2).
- (iii) Obstruction (a) reduces to Chapter 39.4.5 (F1.a.1): block exhaustion per residual Galois representation. Chapter 39.4.7 closes (a) under this conjecture.
- (iv) Obstruction (b) reduces to Chapter 39.5.7 (F1.a.2: existence of wild-enhanced EG stack), Chapter 39.5.5, plus Chapter 39.5.10 (F1.a.3). Chapter 39.5.6 and Chapter 39.5.11 close (b) under these conjectures.
- (v) Obstruction (c) is the family datum Chapter 39.6.3. It is not a consequence of the fixed-character DEG theorem.
- (vi) Full conditional closure of F1.a at all primes: Chapter 39.7.1, conditional on (H1)–(H5).
- (vii) Unconditional partial closure: the fixed- χ , $p \geq 5$, fully faithful DEG embedding (Chapter 39.7.2, Chapter 39.7.3). No family statement is asserted.
- (viii) Global assembly check: F1 follows from F1.a (at every prime, including wild) plus L_{ACD} (Chapter 39.8.1); total F1 residual table is six items (Chapter 39.8.3).

There is no unconditional closure of F1.a. The strongest honest statement is the fixed- χ , $p \geq 5$, fully faithful DEG embedding (Chapter 39.7.3); all other closure claims are explicitly conditional. The residuals F1.a.1, F1.a.2, the wild block functor datum, F1.a.3, and the central-character family datum are strictly narrower than the original F1.a conjecture and each is individually testable with the source list laid out here.

H- P^{Ar} DETERMINANT-TRACE MATCHING AND HAMBURGER DETERMINACY

Chapter 35 reduces the arithmetic Hilbert–Pólya hypothesis $H\text{-}P^{\text{Ar}}$ to a determinant-trace realisation (SP) equivalent to the Connes adele-class-space absorption-trace conjecture and to the Deninger regularised-determinant axioms, and reduced the VB^ converse to a four-way equivalence with Hamburger determinacy of the symmetrised shifted-zero measure. Chapter 36 uses the conditional RH completion hypothesis and isolates the positivity/determinacy input. The $H\text{-}P^{\text{Ar}}$ side separates the Bost–Connes zero-mode weight-selection problem from the trace-matching problem: the logarithmic Berry–Keating weight is the only survivor among a small canonical test family, and is unique under an explicit KMS/zeta-partition normalisation, but existence of the required trace identity is the $H\text{-}P^{\text{Ar}}$ obstruction. The VB^* side tests the Carleman construction: the Bombieri–Lagarias Li-coefficient bounds do not imply the Carleman estimate for the shifted-zero moments. The sufficient analytic residual is (R_{analytic}) : a linear upper-growth bound for $\mu_{2n}^{1/(2n)}$, or equivalently a positive regularised moment measure with Carleman divergence.*

40.1 DETERMINACY COMPARISON

Residual Splitting

The residual splitting structural analysis of Chapter 35 gives four structural inputs:

- (i) *Domain theorem for L_0^{Ar}* . Chapter 35.3.2: on the primary arithmetic Heisenberg Fock, under non-negativity of the Bost–Connes zero-mode weights $\lambda(m) \geq 0$, the symmetric operator $L_0^{\text{Ar, symm}}$ admits a unique Friedrichs self-adjoint extension L_0^{Ar} , essentially self-adjoint on the finite-span core $\mathcal{D}_0$, with non-negative spectrum.
- (ii) *Three-way equivalence for $H\text{-}P^{\text{Ar}}$* . Chapter 35.3.14: $H\text{-}P^{\text{Ar}}$ (in the determinant-trace form SP) is equivalent to the Connes absorption-trace conjecture and to the Deninger regularised determinant condition.
- (iii) *Four-way equivalence for VB^* converse*. Chapter 35.4.14: the VB^* converse is equivalent to Hamburger determinacy (Det) of the symmetrised shifted-zero measure, to Hermite–Biehler operator positivity on the de Branges space $\mathcal{H}(E_\xi)$ (HB+), and to a sharp Báez-Duarte decay rate $\epsilon_N^{\text{BD}} \asymp 1/\sqrt{\log N}$ with the Petersson-matched constant (BD+).
- (iv) *Separation of the two residuals*. Chapter 35.5.2: $H\text{-}P^{\text{Ar}}$ (existence of a self-adjoint L_0^{Ar} with prescribed trace) and spectral regularity of the moment problem are separate assertions.

The determinacy calculation adds weight selection and Carleman analysis to these structural inputs.

Two Analytic Tests

(A) *Zero-mode weight selection for $H\text{-}P^{\text{Ar}}$* . Chapter 35.3.15 names the single residual as “the choice of Bost–Connes zero-mode weights $\{\lambda(m)\}_{m \in \mathbb{N}^\times}$ such that the combined trace matches the zero generating

function”. Four canonical weight systems against the arithmetic Koszul self-duality (P₁), the arithmetic Zhu modularity (P₂), and Bost–Connes rigidity at $\beta = 1$ are considered. This is a finite test family, not a classification of all additive functions on $\mathbb{N}^\times$. A genuine uniqueness theorem requires the extra KMS/zeta-partition normalisation stated below.

(B) *Hamburger determinacy obstruction on the symmetrised shifted-zero measure*. Chapter 35.4.5 names the Carleman sum under a heuristic $\mu_{2n} \asymp C^n n!^2$. The Bombieri–Lagarias construction gives the obstruction: Li-coefficient bounds do not transfer to two-sided estimates for the positive regularised shifted-zero moments. The usable conclusion is the classical one: Carleman’s criterion would follow from an upper bound $\mu_{2n}^{1/(2n)} \leq Cn$ for all large n .

The single positivity corollary Chapter 36.4.3 is sufficient for RH after the conditional hypothesis of Chapter 36 is supplied. The resulting residual is (R_{analytic}).

Independent Sources

Chapter 35 used: Connes arXiv:math/9811068, Deninger 1994, Berry–Keating 1999, Reed–Simon, Akhiezer–Glazman, de Branges 1968/1994, Báez-Duarte 2003, Burnol 2000. The present determinacy argument uses a complementary list:

Akhiezer, *The Classical Moment Problem* (Oliver & Boyd 1965). Definitive monograph on the Hamburger, Stieltjes, Hausdorff moment problems; §2.4 Carleman’s criterion; §2.5 Nevanlinna parametrisation of indeterminate solutions; §3.1 Jacobi-matrix formalism; §5.6 exponential-moment estimates.

Shohat–Tamarkin *The Problem of Moments* (AMS 1943). The American companion to Akhiezer; Ch. III.5 classical determinacy, Ch. III.2 Hankel determinants as sufficient tests, Ch. VI bounded-case theorems.

Stieltjes, *Ann. Fac. Sci. Toulouse* (1894). The original indeterminate moment problem; the lognormal $\mu_n = e^{n^2/2}$ example; baseline indeterminacy standard.

Nevanlinna, *Ann. Acad. Sci. Fenn.* 5 (1922). Parametrisation of the convex set of solutions to an indeterminate Hamburger moment problem by four entire functions A, B, C, D of exponential type zero; each indeterminate solution corresponds to a Nevanlinna function φ via $(A\varphi - C)/(B\varphi - D)$.

Krein, *Dokl. Akad. Nauk SSSR* 77 (1951) 93–96. Krein’s Pontryagin-space theorem: self-adjoint operators on indefinite-metric spaces; prototype for the indeterminate moment problem as a problem with non-unique positive lift.

Simon, *Adv. Math.* 137 (1998), 82–203. Modern treatment of classical moment problems via Jacobi matrices: the Hamburger moment problem with moments $\{\mu_n\}$ is determinate iff the associated Jacobi operator is essentially self-adjoint on $\mathcal{C}_{00}$; indeterminate iff deficiency indices $(1, 1)$.

Berg–Christiansen, *Math. Ann.* 322 (2002) 451–473. Exponential moment condition: $\sum_n \mu_{2n}^{-1/(2n)} = +\infty$ implies determinacy (Carleman); refinements via sharper generating-function analysis.

Lin, *Probab. Surv.* 14 (2017) 129–190. Modern survey of determinacy: Carleman vs. Krein vs. Lin conditions; exponential-moment comparison.

Bost–Connes, *Selecta Math.* 1 (1995) 411–457. Original Bost–Connes system; §2.2 construction of the Hecke algebra $C_{\mathbb{Q}}^*$; §3.1 KMS states for $\beta > 1$; §4 phase transition at $\beta = 1$.

Laca, *J. Funct. Anal.* 152 (1998) 330–378. Laca’s categorical approach to the Bost–Connes system: characters of the $C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^\times$ crossed product; KMS classification via induced representations.

Connes–Marcolli, AMS Colloq. Publ. 55 (2008). Comprehensive reference on Bost–Connes and number theory; Ch. 3 KMS states, Ch. 4 rigidity at $\beta = 1$, Ch. 5 adèle class space structure.

Connes–Marcolli–Ramachandran, arXiv:math/0501424. Explicit description of the $\text{KMS}_{\beta=1}$ state as the unique symmetric trace on the Bost–Connes system; rigidity of the zero-mode spectrum under the Hecke action.

Conrey, J. Reine Angew. Math. 399 (1989) 1–26. Zero-density theorem: at least 40% of the non-trivial zeros of ζ lie on the critical line. Unconditional.

Selberg, Arch. Math. Naturvid. 48 (1946) 89–155. Positive proportion of zeros on the critical line, first unconditional zero-density result.

Levinson, Adv. Math. 13 (1974) 383–436. $1/3$ of zeros on the line via Levinson’s method.

Odlyzko, 2001 (unpublished note). Numerical computation of 10^{13} non-trivial zeros on the critical line.

Bombieri–Lagarias, J. Number Theory 77 (1999) 274–287. Unconditional upper bound $\lambda_n \leq n(c_0 + \log n)/2$, $c_0 = 2 - \log(4\pi) + \gamma$, and a matching lower bound $\lambda_n \geq -(n/2) \log n - O(n)$.

These sources are disjoint from the residual-splitting list: the primary derivations of one argument are not used as primary derivations of the other.

40.2 SIX REDUCTIONS

Four reductions concern weight selection; two concern Hamburger determinacy.

Cycle 1 (A, weight enumeration)

Problem. Does the Berry–Keating ansatz $\lambda(p^k) = k \log p$ (Chapter 35.3.17) constitute the unique natural Bost–Connes zero-mode weight?

Argument. The Berry–Keating ansatz reproduces the leading asymptotic zero density $N(T) \sim (T/2\pi) \log(T/2\pi)$ (Bohr–Sommerfeld quantisation of $x p = E$) but does not reproduce the individual spectrum; three independent competitors exist among finite canonical tests.

Sources. Bost–Connes 1995 §2.2 (Hecke algebra structure); Connes–Marcolli 2008 Ch. 3 (KMS_{β} classification); Laca 1998.

Conclusion. Four candidate weight systems are enumerated as Chapter 40.3.1; pairwise P_1+P_2 compatibility is proved as Chapter 40.3.5. The $\text{KMS}_{\beta=1}$ state of the Bost–Connes system is unique in the symmetric normalisation, but that rigidity alone does not classify arbitrary additive prime-weight systems. Trace-compatibility (35.1) for each candidate is equivalent to H-P^{Ar} for that candidate; no combination gives an unconditional trace identity.

Cycle 2 (A, uniqueness under KMS rigidity)

Problem. Does at most one weight system satisfy simultaneous compatibility with P_1 , P_2 , and the H-P^{Ar} trace condition?

Argument. Existence is not assumed (it is H-PA^r itself); uniqueness holds conditionally on existence, via Connes–Marcolli rigidity.

Sources. Connes–Marcolli, AMS Colloq. Publ. 55 (2008), Thm 4.15 (uniqueness of symmetric KMS _{$\beta=1$} state); Connes–Marcolli–Ramachandran arXiv:math/0501424 (explicit form).

Conclusion. The KMS _{$\beta=1$} state on $C_{\mathbb{Q}}^*$ is unique among symmetric states (invariant under $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$) by Connes–Marcolli 2008 Thm 4.15. The compatibility theorem identifying the partition function and prime-generator expectations with the canonical zeta normalisation forces the logarithmic weight under that hypothesis; see Chapter 40.3.7. Existence of the weight system with the required zero trace is precisely the H-PA^r assertion.

Cycle 3 (A, consistency with H-PA^r)

Problem. Does the Berry–Keating weight $\lambda(p^k) = k \log p$ reproduce the H-PA^r trace identity?

Argument. The Berry–Keating partition function $\sum_{n \geq 1} e^{-t \log n} = \sum_{n \geq 1} n^{-t} = \zeta(t)$, multiplied by the Dedekind-eta inverse, does not equal the zero wave-kernel distribution; the two sides match only after the archimedean Euler factor is subtracted and the logarithmic-derivative expansion is inverted.

Sources. Edwards, *Riemann’s Zeta Function* (Academic Press 1974) §§1.13 and 3.3, Mellin inversion of ξ via the logarithmic derivative.

Conclusion. The trace-compatibility condition is an infinite hierarchy of constraints, not a single scalar equation. By Mellin inversion, the hierarchy is an equality of distributions for the normal zero operator Θ_{ξ} , equivalently $\det_{\infty}(s \text{ id} - \Theta_{\xi}) = \xi_{\mathbb{R}}(s)$. The Friedrichs Fock/BC Hamiltonian supplies the Euler partition side, not this determinant identity. The hierarchy is identified with a regularised Weil/Connes explicit-formula extension only after the admissibility of the test family is supplied; see Chapter 40.3.10. No weight system is known to close the trace identity.

Cycle 4 (B, Hamburger determinacy reductions)

Problem. Does Carleman’s criterion $\sum_{n \geq 1} \mu_{2n}^{-1/(2n)} = +\infty$ hold for the symmetrised shifted-zero moment sequence $\{\mu_{2n}\}$?

Argument. The moment sequence grows like $\mu_{2n} \asymp C^n n!^2$ under Riemann–von Mangoldt heuristics (Chapter 35); this is exactly on the Carleman boundary, not strictly inside.

Sources. Akhiezer 1965 §2.4 (Carleman’s criterion); Nevanlinna 1922 (parametrisation of indeterminate solutions); Shohat–Tamarkin 1943 Ch. III.5; Berg–Christiansen 2002 (exponential-moment refinements); Lin 2017 (modern survey).

Conclusion. The Riemann–von Mangoldt formula controls zero counting, not the positivity of the regularised moment functional. A saddle-point heuristic suggests the Carleman-scale growth $C^n n!^2$, but converting that heuristic into the upper bound $\mu_{2n}^{1/(2n)} = O(n)$ requires an additional analytic theorem. The Hamburger determinacy problem reduces to positivity plus a Carleman-scale upper-growth estimate on μ_{2n} ; see Chapter 40.4.5. No unconditional Carleman theorem follows.

Cycle 5 (B, obstruction via Li–Bombieri–Lagarias)

Problem. Can the unconditional Bombieri–Lagarias bound $\lambda_n \leq n(c_0 + \log n)/2$ be converted into an unconditional estimate on μ_{2n} sufficient for Carleman?

Argument. The Li coefficients are oscillatory sums over zeros, whereas μ_{2n} is a positive regularised moment. The unconditional bound on λ_n does not transfer to μ_{2n} without a separate positivity/domination theorem.

Sources. Bombieri–Lagarias J. Number Theory 77 (1999) 274–287, Theorem 2; Voros 2003 Ann. Inst. Fourier (refinements); Coffey 2005 Math. Phys. Anal. Geom.

Conclusion. Bombieri–Lagarias Thm 2 bounds the Li sum $\lambda_n = \sum_{\rho} (1 - (1 - 1/\rho)^n)$ (Li 1997). This is a highly oscillatory zero sum. It is not a bound on the absolute moments μ_{2n} , and cancellations in the Li sum cannot be reversed into Carleman control without an additional positivity and regularisation theorem; see Chapter 40.4.3. An independent construction of a positive regularised shifted-zero moment measure with upper-growth estimate $\mu_{2n}^{1/(2n)} \leq Cn$ is the analytic residual.

Cycle 6 (cross-check; residual obstruction naming)

Problem. Does the Ch 37 single positivity theorem reduce to an analytic statement of classical form?

Argument. Comparison of the three classical reformulations (Carleman, Nevanlinna, Akhiezer–Krein) on the symmetrised shifted-zero measure yields a single precisely-named residual.

Sources. Akhiezer, Shohat–Tamarkin, Carleman, Krein, Nevanlinna, and the residual-splitting sources cited above.

Conclusion. The determinacy reductions consolidate into a single residual: a positive regularised interpretation of $\mu_{2n} = \sum_{\rho} (1/4 + |\gamma_{\rho}|^2)^n$ with $\mu_{2n}^{1/(2n)} = O(n)$. The arithmetic content required is an analytic estimate strong enough for Carleman’s criterion, together with positivity of the regularised measure; all other ingredients are conditional inputs from the residual splitting and Ch 37. See Chapter 40.4.7.

40.3 H-PA^r SPECTRAL MATCHING: BOST–CONNES ZERO-MODE WEIGHT*Candidate Weights*

The residual splitting candidate weight system stated was the Berry–Keating ansatz $\lambda(p^k) = k \log p$ extended multiplicatively to $\lambda(m) = \log m$ (Chapter 35.3.17). We compare it with three nearby weights. This is not an exhaustive classification of all maps $\mathbb{N}^{\times} \rightarrow \mathbb{R}$: additive weights are determined by arbitrary prime values $\{\lambda(p)\}_p$. The point is to separate the canonical BC/zeta generator from common false substitutes.

Definition 40.3.1 (four canonical zero-mode weight tests). . Let $C_{\mathbb{Q}}^*$ be the Bost–Connes algebra of Chapter 7.1.1. Four canonical weight tests $\{\lambda_{\star}(m)\}_{m \in \mathbb{N}^{\times}}$ are available, each arising from a distinct structural interpretation of the Hecke action on the zero-mode lattice:

(BK) *Berry–Keating / logarithmic*: $\lambda_{\text{BK}}(m) := \log m$. Extended multiplicatively: $\lambda_{\text{BK}}(p^k) = k \log p$. This is the Euler-product generator; yields the BC KMS_β partition function $\zeta(\beta)$ at $\beta = 1$.

(Li-shift) *Li spectral translation*: $\lambda_{\text{Li}}(m) := \log m$ as a semigroup weight, with the critical-line translation $s \mapsto s - \frac{1}{2}$ applied to the spectral parameter rather than inserted as a constant shift in the semigroup homomorphism. A literal formula $\log m - \frac{1}{2}$ is not additive on $\mathbb{N}^\times$ and is therefore not a BC weight.

(vM) *von Mangoldt / Weil*: $\lambda_{\text{vM}}(m) := \Lambda(m) \log m$ where $\Lambda(m)$ is the von Mangoldt function ($\Lambda(p^k) = \log p$, zero otherwise). This is logarithmic derivative data, not a multiplicative semigroup weight.

(FS-quad) *quadratic finite-place weight*: $\lambda_{\text{FS}}(p^k) := k^2 \log p$ and $\lambda_{\text{FS}}(m) := \sum_{p^k \parallel m} k^2 \log p$. This records the quadratic-exponent alternative suggested by finite-place Hecke powers on perfectoid Iwahori charts (Chapter 36.1.1 finite-place orientation). It agrees with (BK) on squarefree integers and differs on higher prime powers.

Remark 40.3.2 (domain of the four tests). Under the $\mathbb{Q}$ -positive-semigroup action of $\mathbb{N}^\times$ on the zero-mode lattice, a weight system $\lambda : \mathbb{N}^\times \rightarrow \mathbb{R}$ is compatible with the Bost–Connes Hecke action $\mu_n \mu_m = \mu_{nm}$ only when $\lambda(nm) = \lambda(n) + \lambda(m)$. The additive case reduces to specifying $\lambda(p)$ for each prime p , hence contains infinitely many choices. The zeta partition normalisation

$$\sum_{m \geq 1} e^{-\beta \lambda(m)} = \zeta(\beta)$$

for all $\text{Re}(\beta) > 1$ forces $e^{-\lambda(p)} = p^{-1}$ prime-by-prime, and therefore forces (BK). Without that normalisation there is no uniqueness theorem. The four tests above only record the canonical logarithmic, translated, logarithmic-derivative, and quadratic finite-place possibilities used in the obstruction comparison.

Pairwise Compatibility with P1 and P2

LEMMA 40.3.3 (*P1 compatibility within the four tests*). . Within Chapter 40.3.1, the additive Berry–Keating weight is compatible with the formal $m \leftrightarrow m^{-1}$ functional-equation involution. The Li-shift test is the same additive weight together with an external critical-line translation. The von Mangoldt and quadratic finite-place tests are not multiplicative semigroup weights and therefore do not define BC zero-mode weights satisfying P1.

Proof. For (BK), $\lambda_{\text{BK}}(m) + \lambda_{\text{BK}}(m^{-1}) = 0$ after formal extension of $\log$ to $\mathbb{Q}_{>0}$. The Li shift is not the map $\log m - \frac{1}{2}$ on $\mathbb{N}^\times$, since such a map fails $\lambda(nm) = \lambda(n) + \lambda(m)$. It is the same additive weight with the spectral parameter translated by $1/2$. The von Mangoldt test is supported on prime powers and the FS-quad test satisfies $\lambda_{\text{FS}}(p^{a+b}) \neq \lambda_{\text{FS}}(p^a) + \lambda_{\text{FS}}(p^b)$ for general a, b . Hence neither is a BC semigroup weight. $\square$

LEMMA 40.3.4 (*P2 compatibility within the four tests*). . Within Chapter 40.3.1, the Berry–Keating weight is the only additive weight whose partition function is the classical zeta partition function. The Li shift changes only the external critical-line normalisation. The von Mangoldt and FS-quad tests do not give the ζ -partition function and hence do not satisfy P2 in the arithmetic Zhu sense.

Proof. (BK) gives

$$\sum_{m \geq 1} e^{-t \log m} = \sum_{m \geq 1} m^{-t} = \zeta(t).$$

The Li-shift convention leaves this semigroup partition function unchanged and moves the $1/2$ into the spectral variable. For (vM) one obtains logarithmic-derivative data supported on prime powers, not a Dirichlet series with the ξ -functional equation. For (FS-quad) one obtains

$$\prod_p \sum_{k \geq 0} p^{-tk^2},$$

not the Euler product for $\zeta(t)$. These two tests therefore fail the P₂ partition-function test. $\square$

THEOREM 40.3.5 (*four-test filter under P₁+P₂*). . Among the four tests of Chapter 40.3.1, simultaneous P₁+P₂ compatibility leaves only the Berry–Keating semigroup weight $\lambda(m) = \log m$, with the Li convention understood as the same weight after external critical-line translation. This is a finite filter, not a classification of all possible prime-weight systems.

Proof. Chapter 40.3.3 and Chapter 40.3.4 eliminate the von Mangoldt and FS-quad tests as BC semigroup weights. The Li convention does not produce a new additive weight; it is the Berry–Keating weight with the spectral parameter centred at $1/2$. Thus the only surviving semigroup weight in the four-test family is (BK). $\square$

Remark 40.3.6 (*what the filter proves*). Chapter 40.3.5 proves only the finite-filter statement just stated. Full uniqueness requires the zeta partition normalisation or an equivalent KMS compatibility condition; without such a condition, arbitrary prime weights $\{\lambda(p)\}_p$ remain possible.

Uniqueness under KMS/Zeta Normalisation

The surviving canonical weight is (BK). Existence with the trace-compatibility condition is the H-P^{Ar} assertion. Uniqueness after the partition function is fixed is unconditional.

THEOREM 40.3.7 (*uniqueness under zeta partition normalisation*). . Suppose $\lambda : \mathbb{N}^\times \rightarrow \mathbb{R}_{\geq 0}$ is additive and has zeta partition function

$$\sum_{m \geq 1} e^{-\beta \lambda(m)} = \zeta(\beta)$$

for all $\text{Re}(\beta) > 1$, with Euler factors matched prime-by-prime to the Bost–Connes Hecke generators. Then $\lambda(m) = \log m$. Consequently any H-P^{Ar} realisation satisfying this KMS/zeta normalisation uses the Berry–Keating weight.

Proof. Additivity gives $\lambda(p^k) = k\lambda(p)$, so the partition function factors as

$$\prod_p (1 - e^{-\beta \lambda(p)})^{-1}.$$

Prime-by-prime equality with $\zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}$ gives $e^{-\beta \lambda(p)} = p^{-\beta}$, hence $\lambda(p) = \log p$ for every prime. Additivity then gives $\lambda(m) = \log m$. $\square$

Remark 40.3.8 (*relation to Connes–Marcolli rigidity*). Connes–Marcolli rigidity at $\beta = 1$ is the conceptual source of this normalisation. Identifying an arbitrary weight with the unique symmetric KMS state requires the prime-generator and partition-function bridge proved in the theorem. The theorem records the exact conditional uniqueness available on the arithmetic path. H-P^{Ar} is the Bost–Connes/Dedekind-eta trace identity, the unresolved determinant-trace matching problem.

The Hierarchy Condition

LEMMA 40.3.9 (*admissible Weil–Connes wave-kernel regularisation*). . Let $\xi = \xi_{\mathbb{R}}$ and write each non-trivial zero as $\rho = \beta_{\rho} + it_{\rho}$, and set $z_{\rho} := -i(\rho - \frac{1}{2})$. Use the Fourier convention

$$h(r) = \int_{\mathbb{R}} \widehat{h}(u) e^{-iru} du.$$

Let

$$\text{PW} := \{h = \mathcal{F}^{-1}\psi : \psi \in C_c^{\infty}(\mathbb{R})\}$$

with the LF topology transported from $C_c^{\infty}(\mathbb{R})$. For $h \in \text{PW}$, the Weil–Connes explicit formula gives

$$\sum_{\rho} h(z_{\rho}) = \int_{\mathbb{R}} h(r) dW_{\infty}(r) - 2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \widehat{h}(\log n),$$

in the normalisation of (40.1); the prime sum is finite because $\widehat{h}$ has compact support. The replacement $z_{\rho} = t_{\rho}$ is equivalent to RH.

For $\phi \in C_c^{\infty}((0, \infty))$, set

$$h_{\phi}(r) := \int_0^{\infty} e^{-u/2} \phi(u) e^{-iru} du.$$

Then $h_{\phi} \in \text{PW}$, and

$$\langle \mathcal{W}_{\xi}, \phi \rangle := \sum_{\rho} h_{\phi}(z_{\rho})$$

defines a distribution $\mathcal{W}_{\xi} \in \mathcal{D}'((0, \infty))$. Consequently the symbolic kernel $\sum_{\rho} e^{-t\rho}$ is a distributional wave kernel, not an ordinary heat trace.

Proof. This is the standard Paley–Wiener form of the Weil explicit formula: compact support of $\widehat{h}$ makes the prime sum finite and the archimedean term is the distribution W_{∞} . The second statement is obtained by substituting the compactly supported function $u \mapsto e^{-u/2} \phi(u)$ into the same formula. Approximation of the point kernel uses mollifiers $\widehat{h}_{t,\epsilon}(u) = e^{-u/2} \epsilon^{-1} \eta((u-t)/\epsilon)$; the convergence is in $\mathcal{D}'((0, \infty))$, not pointwise heat trace convergence. $\square$

THEOREM 40.3.10 (*trace-compatibility as regularised Weil/Connes hierarchy*). (*conditional on the H-PA^r trace realisation*). Let $\lambda = \lambda_{\text{BK}}$ be the Berry–Keating weight of Chapter 40.3.1. The trace-compatibility condition (35.1) is the regularised Weil/Connes trace hierarchy

$$\sum_{\rho}^{\text{reg}} h_t(z_{\rho}) = \int_{-\infty}^{\infty} h_t(r) dW(r) - 2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \widehat{h}_t(\log n) \quad (40.1)$$

for the wave-kernel test family associated to $e^{-t\Theta_{\xi}}$, after the admissible regularisation of Chapter 40.3.9 is supplied. The admissibility and regularisation are the trace-realisation bridge required for H-PA^r; the classical Weil formula applies only to admissible test functions. Under the full H-PA^r determinant condition, $z_{\rho} = t_{\rho}$ and the trace becomes the ordinate trace.

Proof. Under the BK weight system, the Friedrichs operator acts on δ_m by eigenvalue $\log m$. The regularised trace of $e^{-tL_o^{\text{Ar}}}$ on the zero-mode lattice is $\sum_{m \geq 1} e^{-t \log m} = \sum_{m \geq 1} m^{-t} = \zeta(t)$, convergent for $t > 1$ and meromorphically continued.

Combining with the archimedean Dedekind-eta inverse product $\prod_{i \geq 1} (1 - e^{-ti})^{-1}$ and subtracting the archimedean descendant contribution (Tate-pole and archimedean terms) via the Bost–Connes regularisation (Chapter 35.3.3), one obtains

$$\text{tr}_{\text{BC}}(e^{-tL_o^{\text{Ar}}}) = \zeta(t) \cdot \eta(it)^{-1} \cdot (\text{regularisation}),$$

with the archimedean terms removed by the Bost–Connes/Dedekind-eta prescription. The Hadamard product for $\xi_{\mathbb{R}}$ rewrites the zero side as a regularised zero trace. The classical Weil explicit formula supplies this identity for the admissible Paley–Wiener test spaces of Chapter 40.3.9. The Vol IV condition is the additional assertion that the Bost–Connes/Fock trace is carried to that distributional extension. Therefore the theorem is a reformulation of the trace-compatibility hierarchy under H-P^{Ar} , not a proof of H-P^{Ar} . $\square$

COROLLARY 40.3.11 (*the hierarchy is the Connes absorption-trace bridge*). (conditional on the H-P^{Ar} trace realisation). Under the BK weight selection, the H-P^{Ar} trace-compatibility condition is the Connes–Meyer–Weil absorption-trace bridge for the Bost–Connes/Weil wave-distribution family. The classical Weil explicit formula applies to admissible Paley–Wiener test functions; the admissibility and trace-identification of the BC heat-trace family is the additional content.

Proof. The previous theorem identifies the needed equality with the regularised explicit-formula extension for the BC/Weil wave family. By Chapter 35.3.9, this is the same structural condition as the Connes absorption-trace realisation. The admissibility and trace-identification step is precisely what H-P^{Ar} asserts. $\square$

Remark 40.3.12 (*weight selection and H-P^{Ar}*). Chapter 40.3.7 and Chapter 40.3.11 together state: if H-P^{Ar} holds, the weight is uniquely Berry–Keating under the zeta/KMS normalisation. Whether the Berry–Keating weight satisfies the trace compatibility is the regularised Connes absorption-trace problem. The reduction is tight: the canonical weight is determined under explicit normalisation, while the trace identity is the classical absorption-trace problem.

40.4 HAMBURGER DETERMINACY OBSTRUCTION

The Symmetrised Shifted-Zero Measure

Definition 40.4.1 (*symmetrised shifted-zero moment sequence*). Let $\rho = 1/2 + i\gamma_{\rho}$ denote a non-trivial zero of ξ , written with $\gamma_{\rho} \in \mathbb{C}$ (real for every zero iff RH). The *symmetrised shifted-zero moment sequence* is

$$\mu_{2n} := \sum_{\rho} \left(\frac{1}{4} + \gamma_{\rho}^2 \right)^n$$

Bost–Connes regularised. Under RH, $\gamma_{\rho} \in \mathbb{R}$ and the unregularised terms are real and non-negative. Outside RH the expression is not automatically a positive moment sequence; positivity of the regularised functional is part of $(\mathbb{R}_{\text{analytic}})$.

Remark 40.4.2 (*relation to the symmetrised zero measure of the residual splitting*). The symmetrised zero measure ν of Chapter 23.4.1 is the pushforward of the zero set under the symmetrisation map $\rho \mapsto$

$\frac{1}{4} + \text{Im}(\rho)^2$. On the critical line $\text{Re}(\rho) = \frac{1}{2}$, this equals $|\rho|^2$. The shifted moment μ_{2n} of Chapter 40.4.1 is the n -th moment of this measure; odd moments vanish by symmetry. Determinacy of ν is equivalent to determinacy of $\{\mu_{2n}\}_{n \geq 0}$ under the standard Stieltjes–Hamburger dictionary.

Li-Coefficient Bounds Do Not Give Carleman Moments

PROPOSITION 40.4.3 (*Li-coefficient bounds do not imply Carleman moments*). . The Bombieri–Lagarias bounds for Li coefficients do not imply the Carleman estimate for the regularised moments μ_{2n} in this argument. The Bombieri–Lagarias upper bound $|\lambda_n| = O(n \log n)$ does not imply

$$|\lambda_n| = O(n \log n) \implies \mu_{2n}^{1/(2n)} = O(n) :$$

a separate positivity and domination theorem is required.

Proof. The Li coefficient is the oscillatory zero sum

$$\lambda_n = \sum_{\rho} \left(1 - \left(1 - \frac{1}{\rho} \right)^n \right).$$

The shifted-zero moment relevant to Carleman is instead the positive regularised quantity

$$\mu_{2n} = \sum_{\rho}^{\text{reg}} \left(\frac{1}{4} + \gamma_{\rho}^2 \right)^n.$$

An estimate for the oscillatory Li sum gives no bound on the absolute or positive regularised $2n$ -th moment without a separate positivity and domination theorem. Reversing the cancellations in the Li sum would itself be a new spectral-positivity input. Hence the Bombieri–Lagarias theorem is evidence for the Li/RH reformulation but not a Carleman proof. $\square$

Remark 40.4.4 (*direction of the Carleman bound*). A lower bound on μ_{2n} would give an upper bound on $\mu_{2n}^{-1/(2n)}$, not the lower bound on $\mu_{2n}^{-1/(2n)}$ that Carleman’s divergence criterion requires. Carleman’s criterion is driven by upper growth of the moments, together with positivity of the representing measure.

Tightest Conditional Reduction

THEOREM 40.4.5 (*Hamburger determinacy from linear upper growth*). . Suppose the symmetrised shifted-zero moments come from a positive regularised Hamburger moment functional and there is a constant $C > 0$ such that

$$\mu_{2n}^{1/(2n)} \leq Cn$$

for all sufficiently large n . Then Carleman’s criterion holds:

$$\sum_{n \geq 1} \mu_{2n}^{-1/(2n)} = +\infty.$$

Consequently the associated Hamburger moment problem is determinate.

Proof. The upper-growth hypothesis gives

$$\mu_{2n}^{-1/(2n)} \geq \frac{1}{Cn}$$

for all large n . The harmonic series diverges, so Carleman’s criterion applies. By the classical Carleman theorem for Hamburger moments, the positive moment functional is determinate. $\square$

COROLLARY 40.4.6 (*unconditional inputs and the Carleman gap*). . The unconditional inputs used here do not imply the hypotheses of Chapter 40.4.5: the required positive regularised moment functional and the linear upper-growth estimate $\mu_{2n}^{1/(2n)} = O(n)$ are additional analytic inputs.

Proof. Chapter 40.4.3 blocks the only candidate unconditional shortcut above. The Riemann–von Mangoldt counting formula controls the distribution of zeros by height, but by itself it does not define the required positive regularised moment functional nor bound the regularised moments after off-line and archimedean terms are included. $\square$

THEOREM 40.4.7 (*the single analytic residual*). . Under the residual-splitting structural reductions and the determinacy analytic reductions, the Hamburger-determinacy component of the Vol IV RH path reduces to the following sufficient analytic residual:

$$(\mathcal{R}_{\text{analytic}}) \quad \{\mu_{2n}\} \text{ is a positive regularised Hamburger moment sequence and } \limsup_{n \rightarrow \infty} \frac{\mu_{2n}^{1/(2n)}}{n} < \infty.$$

This statement implies Carleman’s criterion, hence Hamburger determinacy, hence the VB* converse in the structural setting of Chapter 35.

Proof. The implication $(\mathcal{R}_{\text{analytic}}) \Rightarrow \text{Carleman}$ is Chapter 40.4.5. The Carleman-to-determinacy implication is the classical theorem of Akhiezer/Shohat–Tamarkin. The structural equivalence between determinacy and the VB* converse is Chapter 35.4.14. $\square$

Remark 40.4.8 (*domain of $\mathcal{R}_{\text{analytic}}$*). $(\mathcal{R}_{\text{analytic}})$ is a one-sided upper-growth condition, distinct from a two-sided asymptotic. Its relation to RH is not determined here: the condition may be strictly weaker, or RH-equivalent. It supplies the exact positivity and upper-growth information that Carleman’s criterion requires. Standard zero-density theorems count zeros in regions; the residual controls a regularised moment functional.

Remark 40.4.9 (*form of the determinacy residual*). $(\mathcal{R}_{\text{analytic}})$ is a sufficient condition for the Vol IV determinacy step. It identifies the exact analytic shape of the positivity theorem required; neither the classical Weil formula nor the Bombieri–Lagarias bounds supply it.

40.5 CONSEQUENCE FOR CH 37 RH COMPLETION

THEOREM 40.5.1 (*determinacy consequence for the Ch 37 single positivity corollary*). . The Ch 37 single positivity corollary Chapter 36.4.3 reduces to the conjunction:

- (i) the Berry–Keating weight is forced by the additive KMS/zeta-partition normalisation (Chapter 40.3.7).
- (ii) H-P^{Ar} holds with the Berry–Keating weight, meaning that the Bost–Connes trace is identified with the regularised Weil/Connes trace hierarchy (Chapter 40.3.10).
- (iii) $(\mathcal{R}_{\text{analytic}})$: the shifted-zero moments form a positive regularised Hamburger moment sequence with $\mu_{2n}^{1/(2n)} = O(n)$ (Chapter 40.4.7).

Under the conjunction of these three, plus the conditional P1/P3 inputs named in Chapter 36, the Ch 37 implication to RH follows.

Proof. (1) provides the uniquely-determined candidate operator $L_o^{\text{Ar}} = L_o^{\text{Ar}}(\lambda_{\text{BK}})$. (2) is precisely the trace-compatibility condition which, via Chapter 35.3.14, asserts $H\text{-}P^{\text{Ar}}$. (3), by Chapter 40.4.7 asserts Hamburger determinacy of the symmetrised shifted-zero measure; by Chapter 35.4.14, this supplies the VB^* converse. Combined with the domain theorem, P_2 , and the conditional P_1/P_3 inputs recorded in Ch 37, this recovers the conditional Ch 37 implication to RH. $\square$

COROLLARY 40.5.2 (*structure of the determinacy reduction*). The determinacy comparison reduces the Ch 37 single positivity theorem to three named residuals:

(W) *Weight normalisation*: an additive Bost–Connes weight with zeta partition function and prime-by-prime KMS compatibility. Under (W), $\lambda = \lambda_{\text{BK}}$.

(H) *H-P trace bridge*: the Bost–Connes/Dedekind-eta heat trace is the admissible regularised Weil/Connes zero trace. Under (H), $H\text{-}P^{\text{Ar}}$ holds for the BK operator.

(R_{analytic}) *Carleman positivity*: the shifted-zero moments form a positive regularised Hamburger moment sequence and satisfy $\mu_{2n}^{1/(2n)} = O(n)$. Under (R_{analytic}), Hamburger determinacy holds.

All three together give the residual H-P/ VB implication in the Ch 37 path; the conditional P_1/P_3 inputs are still required.

Proof. Direct from Chapter 40.5.1. (W) fixes the candidate operator; (H) supplies determinant-trace matching; and (R_{analytic}) supplies determinacy. Omitting any one leaves either no fixed operator, no zero trace, or no VB^* converse. $\square$

40.6 RESIDUAL CONDITIONS

After the determinacy reductions, the Ch 37 RH implication requires:

- (i) **(W) Weight normalisation**. The BK weight is forced once the additive zeta partition function is fixed prime-by-prime. The remaining assertion is that the Vol IV Friedrichs construction really lands in this KMS/zeta normalisation.
- (ii) **(H) H-P trace bridge**. The analytic theorem is the admissible regularised extension identifying the BC/Dedekind-eta heat trace with the Connes zero trace; the classical Weil explicit formula applies to admissible test functions and is a step toward, not a substitute for, this identification.
- (iii) (R_{analytic}) **Carleman positivity**. The required analytic input is a positive regularised shifted-zero moment functional with $\mu_{2n}^{1/(2n)} = O(n)$; this estimate lies beyond the reach of the Bombieri–Lagarias Li-coefficient bounds by Chapter 40.4.3.
- (iv) **Compatibility with the Ch 37 hypothesis**. The RH completion hypothesis of Chapter 36.4.1 requires P_1 , P_2 , P_3 , $H\text{-}P^{\text{Ar}}$, and Hamburger determinacy together. P_1 is conditional on arithmetic Positselski (requires a non-formal theorem); P_2 unconditional; P_3 conditional on Koszul–Petersson compatibility (requires a non-formal theorem); $H\text{-}P^{\text{Ar}}$ (RH-equivalent); determinacy reduces to (R_{analytic}).

Remark 40.6.1 (*determinacy consequences*). $H\text{-}P^{\text{Ar}}$ determinant-trace matching and Hamburger determinacy are RH-equivalent or RH-adjacent analytic problems. The determinacy comparison gives:

- (i) *Conditional uniqueness of the weight system*: the Berry–Keating $\lambda(m) = \log m$ is forced by additive zeta/KMS normalisation.

- (ii) *Explicit identification of the trace-compatibility hierarchy*: the infinite sequence of conditions is the regularised Weil/Connes trace bridge; the classical Weil formula applies to admissible test functions and is a step in the bridge.
- (iii) *Independence of Li-coefficient bounds and Carleman moments*: the Bombieri–Lagarias bounds do not imply the Carleman estimate by Chapter 40.4.3.
- (iv) *The single analytic residual* (R_{analytic}): Hamburger determinacy reduces to positivity plus linear upper growth for the regularised moments.
- (v) *Compatibility with Ch 37*: the three residuals (W), (H), (R_{analytic}) give the H-P/VB implication after the conditional P1/P3 inputs are supplied.

40.7 DETERMINACY BOUNDARY

- (i) **Weight selection is uniquely determined after normalisation** (Chapter 40.3.7). Among four canonical tests, only the Berry–Keating $\lambda(m) = \log m$ survives as a BC semigroup weight with zeta partition function (Chapter 40.3.5).
- (ii) **The trace-compatibility hierarchy is the regularised Weil/Connes bridge** (Chapter 40.3.10, Chapter 40.3.11). The $H\text{-}P^{\text{Ar}}$ trace identity needs an admissible extension of the explicit formula to the BC heat-trace family.
- (iii) **Hamburger determinacy reduces to a single analytic positivity residual** (Chapter 40.4.7). Under (R_{analytic}), the moments are positive and satisfy $\mu_{2n}^{1/(2n)} = O(n)$, so Carleman’s criterion holds.
- (iv) **Li-coefficient bounds do not give Carleman moments** (Chapter 40.4.3). Li-coefficient bounds are oscillatory zero-sum bounds; a separate positivity and domination theorem is required for Carleman.
- (v) **Three precisely-named residuals** (Chapter 40.5.2): weight normalisation (W), H-P trace bridge (H), and Carleman positivity (R_{analytic}).
- (vi) **Domain of the reduction**. $H\text{-}P^{\text{Ar}}$ reduces to the Connes absorption-trace conjecture (RH-equivalent); Hamburger determinacy reduces to (R_{analytic}) (RH-adjacent); no new RH-equivalent positivity language beyond the the residual splitting list of Det, HB+, BD+ is produced.
- (vii) **The residual conjunction**. The three residuals (W), (H), (R_{analytic}) are each precisely stated; their conjunction, together with the conditional P1/P3 inputs, gives the conditional Vol IV implication to RH.

The two analytic problems are the Bost–Connes absorption-trace bridge and positivity with Carleman-scale upper growth for the regularised shifted-zero moments.

ARITHMETIC FLE: SOURCE DOMAIN AND LOGICAL CORRECTIONS

The critical-level fundamental local equivalence (FLE) and its quantum and small variants are theorems in the geometric Langlands programme over complex curves. The arithmetic branch of this volume studies $\mathcal{V}^{\text{prim}}$, the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 , and its conjectural companions F_1 , F_2 , F_3 . This chapter records the domain boundaries of the geometric FLE sources, explains why $F_1/F_2/F_3$ require a distinct mixed-characteristic framework, and corrects the form of claims that previously conflated the two settings. No independent comparison pair is constructed here; arithmetic-FLE realization remains conditional on the boundary conditions of Section 4I.5.

4I.1 SOURCE THEOREMS

THEOREM 4I.1.1 (*geometric critical-level FLE*). . The 2024 geometric-Langlands proof constructs the critical-level fundamental local equivalence in the geometric setting over a smooth projective curve in characteristic 0:

$$\text{KL}(G)_{\text{crit}} \simeq \text{IndCoh}^*(\text{Op}_G^{\text{mf}}),$$

with compatibility with Kac–Moody localization. Its source is Arinkin–Beraldo–Campbell–Chen–Faergeman–Gaitsgory–Lin–Raskin–Rozenblyum, *Proof of the geometric Langlands conjecture II: Kac–Moody localization and the FLE*, arXiv:2405.03648.

THEOREM 4I.1.2 (*quantum and small FLE sources*). . The quantum and small FLE layer consists of local geometric and metaplectic/quantum statements over a complex curve. Campbell–Dhillon–Raskin’s *Fundamental local equivalences in quantum geometric Langlands* (arXiv:1907.03204) and Gaitsgory–Lysenko’s small-FLE work (arXiv:1903.02279) establish equivalences whose source categories are Kazhdan–Lusztig or metaplectic representations and whose target categories are quantum groups or root-of-unity modules over a formal disk. The arithmetic analogue of these equivalences is an open problem at each finite prime.

Remark 4I.1.3 (*domain of the geometric FLE sources*). The FLE sources operate within the geometric Langlands, Kac–Moody, Whittaker, and root-of-unity quantum-group framework over complex curves. A local categorical p -adic Langlands equivalence over $\mathbb{Q}_p$, a globalization over $\overline{\text{Spec } \mathbb{Z}}$, and an Emerton–Gee full Weil/ (φ, Γ) target comparison each require additional technology: Fargues–Scholze’s geometrization, prismatic or pro-étale methods, and the Emerton–Gee eigenvariety construction.

4I.2 DOMAIN DISTINCTION: GEOMETRIC FLE VS $F_1/F_2/F_3$

PROPOSITION 4I.2.1 (*$F_1/F_2/F_3$ as a distinct mixed-characteristic programme*). . The arithmetic branch’s F_1 (local categorical Langlands comparison for $GL_2/\mathbb{Q}_p$ against the Emerton–Gee stack), F_2

(mixed-characteristic E_2 Hecke enhancement and restricted tensor product), and F_3 (Ben-Zvi–Nadler Hochschild-trace comparison against the full $\mathrm{Weil}/(\varphi, \Gamma)$ target) constitute a mixed-characteristic local-global Langlands and Hochschild-trace programme whose source and target categories are disjoint from those of the geometric FLE.

Proof. The geometric FLE (Theorem 41.1.1) pairs critical-level Kac–Moody or Whittaker/factorizable categories as source with oper or quantum-group categories over a smooth projective complex curve as target. F_1 pairs the spherically-finite automorphic category for $GL_2/\mathbb{Q}$ as source with ind-coherent sheaves on the Emerton–Gee stack as target. F_2 requires a mixed-characteristic E_2 Hecke enhancement and restricted tensor product. F_3 requires a Ben-Zvi–Nadler Hochschild-trace comparison against the full $\mathrm{Weil}/(\varphi, \Gamma)$ arithmetic target. The targets of F_1, F_2, F_3 are Emerton–Gee and $\mathrm{Weil}/(\varphi, \Gamma)$ stacks; the inputs at finite primes are adelic Hecke categories, not Beilinson–Drinfeld chiral algebras over a complex curve. $\square$

Remark 41.2.2 (finite-prime categorical structure). At a finite prime p , the relevant categorical structure is the category $\mathrm{Coh}(\mathrm{Bun}_G)$ of coherent sheaves on the moduli stack of G -bundles on the Fargues–Fontaine curve, equipped with the Fargues–Scholze spectral action (arXiv:2102.13459), together with Hecke correspondences, diamonds, and (φ, Γ) -modules. The archimedean fibre carries classical Kac–Moody and chiral intuition, providing the motivating analogy; the finite-prime framework is governed by Fargues–Scholze geometrization and Emerton–Gee eigenvarieties.

41.3 DOMAIN CORRECTIONS ACROSS SOURCE CATEGORIES

Remark 41.3.1 (local/global type distinction). F_1, F_2 , and F_3 are local-global Langlands and Hochschild-trace conjectures. Their source is the automorphic category for $GL_2/\mathbb{Q}$ or its local p -adic analogue; their target is the Emerton–Gee stack or the full $\mathrm{Weil}/(\varphi, \Gamma)$ coherent sheaf theory. These source and target types differ from those of the geometric FLE.

Remark 41.3.2 (finite-prime chiral structure). At a finite prime, the relevant structure is the Beilinson–Drinfeld factorization algebra and the Hecke correspondence on Bun_G , governed by Fargues–Scholze geometrization. The archimedean fibre carries Kac–Moody and chiral intuitions as a motivating analogy.

Remark 41.3.3 (quantum-group and root-of-unity source domain). The root-of-unity and small-FLE sources (Campbell–Dhillon–Raskin, Gaitsgory–Lysenko) are geometric and quantum: they operate over a complex curve with a quantum parameter q a root of unity or with a metaplectic covering. The parameter p^{-s} appearing in local Hecke eigenvalues at a prime p is a spectral parameter, distinct from a root-of-unity quantum-group parameter.

Remark 41.3.4 (function-field transport and number-field domain). Raskin’s globalization operates over a function field. The transport to a number field requires: (a) a local categorical comparison F_1 at each prime, and (b) a prismatic or BC-diamond-glue transition across primes. Both remain open.

Remark 41.3.5 (mixed-characteristic substrate). The arithmetic compactification $\overline{\mathrm{Spec} \mathbb{Z}}$ carries an archimedean place and mixed-characteristic fibres. A curve in characteristic 0 carries neither. The arithmetic programme requires mixed-characteristic methods (prismatic cohomology, perfectoid geometry) at each finite prime and complex-analytic methods at the archimedean place.

Remark 41.3.6 (independent form for arithmetic FLE). Arithmetic FLE has no pair of independent constructions in the present arithmetic formalism. An arithmetic FLE theorem requires first the boundary conditions of Section 41.5.

Remark 41.3.7 (AGKRRV 2024 source and target). Arinkin–Beraldo–Campbell–Chen–Faergeman–Gaitsgory–Lin–Raskin–Rozenblyum (arXiv:2405.03648) establishes the critical FLE between $\mathrm{KL}(G)_{\mathrm{crit}}$ and $\mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mf}})$: source is critical-level Kac–Moody, target is monodromic opers on a complex curve. The Emerton–Gee stack is a mixed-characteristic eigenvariety; its objects are (φ, Γ) -modules, not monodromic opers.

Remark 41.3.8 (F_3 /BZN form against full Weil target). F_3 asks for a Ben-Zvi–Nadler Hochschild-trace identification between the automorphic side and coherent sheaves on the Weil/ (φ, Γ) stack. The inertia-only loop object $B\widehat{\mathbb{Z}}$ captures only the tame/procyclic part of inertia; the full Weil group retains wild inertia and Frobenius data. F_3 with the full Weil target is a conjecture.

Remark 41.3.9 (spectral parameter vs quantum-group parameter). The Hecke eigenvalue at a prime p is recorded by the spectral parameter p^{-s} or equivalently by the Satake image in $\mathrm{Rep}(\check{G})$. The quantum-group deformation parameter q appearing in the quantum FLE is a root of unity or a formal parameter; identifying $q = p^{-s}$ requires an explicit independent bridge; this bridge is the sixth condition in Section 41.5.

Remark 41.3.10 (Fargues–Scholze, Emerton–Gee, and Dotto–Emerton–Gee). Fargues–Scholze (arXiv:2102.13459) provides the spectral action on $\mathrm{Coh}(\mathrm{Bun}_G)$ via the Fargues–Fontaine curve. Emerton–Gee (arXiv:1908.07185) constructs the stack of (φ, Γ) -modules. Dotto–Emerton–Gee establishes full faithfulness of a functor to mod- p representations; an equivalence of ∞ -categories between the automorphic and the Emerton–Gee coherent side is the content of Fi .

Remark 41.3.11 ($\mathrm{Fi}/\mathrm{F}_2/\mathrm{F}_3$ logical-form correction). Fi carries form *Conditional*: the local categorical comparison is supported by Fargues–Scholze (arXiv:2102.13459) and Emerton–Gee (arXiv:1908.07185) and requires the adelic-descent and BC-diamond-glue steps. F_2 carries form *Conditional*: the mixed-characteristic E_2 Hecke enhancement and restricted tensor product are formally accessible but unproved. F_3 carries form *Conjecture*: the full Weil/ (φ, Γ) target comparison is beyond current techniques. Chapters previously marking Fi , F_2 , F_3 with premature theorem-closure have been sharpened accordingly.

Remark 41.3.12 (local Satake/Yangian normalization). The local Satake isomorphism identifies the spherical Hecke algebra with the representation ring of $\check{G}$. An arithmetic Yangian structure, if it exists, must be grounded in an independent comparison between the Satake filtration and an explicit Yangian coproduct; this identification remains open.

Remark 41.3.13 (BC-diamond-glue decomposition). The BC-diamond-glue problem decomposes into three parts: (a) the local BZCHN/Emerton–Gee comparison at each prime p , (b) cross-prime prismatic transition maps, and (c) global Bernstein-centre compatibility. Each part is open.

Remark 41.3.14 (wild-prime correction). At $p = 2$ and $p = 3$, the source for the local comparison carries ramification and wild inertia data not present in the tame case. A conjectural correction accounts for this; the precise form depends on the source used for the local comparison.

41.4 CORRECTED LOGICAL FORM

THEOREM 41.4.1 (*geometric FLE: domain and logical form*). The critical-level FLE $\mathrm{KL}(G)_{\mathrm{crit}} \simeq \mathrm{IndCoh}^*(\mathrm{Op}_G^{\mathrm{mf}})$ and the quantum/small FLE equivalences (Campbell–Dhillon–Raskin, Gaitsgory–Lysenko) are proved in the geometric Langlands setting over a complex curve. Their source is Campbell–Dhillon–Raskin (arXiv:1907.03204), Gaitsgory–Lysenko (arXiv:1903.02279), and Arinkin–Beraldo–Campbell–Chen–Faergeman–Gaitsgory–Lin–Raskin–Rozenblyum (arXiv:2405.03648). These results hold over smooth projective curves in characteristic 0.

Remark 41.4.2 (*archimedean chiral structure and finite-prime domain*). The Kac–Moody and chiral algebra structures provide the motivating analogy at the archimedean place. At finite primes, the geometric structures are Beilinson–Drinfeld factorization algebras, Hecke correspondences, and the Fargues–Scholze spectral action; the finite-prime analogue of the critical-level FLE is the content of F_1 .

Remark 41.4.3 (*corrected form of F_1 , F_2 , F_3 , and BC-diamond-glue*). F_1 (local categorical Langlands comparison for $GL_2/\mathbb{Q}_p$ via Emerton–Gee, with Fargues–Scholze spectral action as input) carries form *Conditional*: the comparison requires adelic-descent and BC-diamond-glue steps. F_2 (mixed-characteristic E_2 Hecke enhancement and restricted tensor product for the adelic assembly) carries form *Conditional*: the ∞ -categorical Tate restricted tensor product is accessible in principle but unproved. F_3 (Ben-Zvi–Nadler Hochschild-trace identification against the full Weil/ (φ, Γ) target, with the full Weil group including wild inertia and Frobenius data) carries form *Conjecture*. BC-diamond-glue decomposes into local BZCHN/Emerton–Gee comparison, cross-prime prismatic transition maps, and global Bernstein-centre compatibility; each component is open. The wild-prime correction at $p = 2, 3$ is a source-sensitive conjecture; the precise form depends on the local comparison chosen.

41.5 OPEN CONDITIONS

- (i) Formulate the mixed-characteristic local categorical equivalence F_1 : construct a comparison between the spherically-finite automorphic ∞ -category for $GL_2/\mathbb{Q}_p$ and ind-coherent sheaves on the Emerton–Gee stack, with the Fargues–Scholze spectral action (arXiv:2102.13459) as input.
- (ii) Extend the loop-space comparison from the inertia-only object $B\widehat{\mathbb{Z}}$ to the full Weil group (with wild inertia and Frobenius data) and prove the flat pullback statement in the full Weil/ (φ, Γ) setting.
- (iii) Prove profinite formal-loop continuity for $\mathrm{Map}(B\widehat{\mathbb{Z}}, -)$ in the formal Noetherian pro-étale setting, supplying the foundational input for the inertia-only portion of F_3 .
- (iv) Prove the mixed-characteristic E_2 Hecke enhancement and the ∞ -categorical Tate restricted tensor product required for F_2 .
- (v) Prove BC-diamond-glue in its three components: (a) the local BZCHN/Emerton–Gee comparison, (b) cross-prime prismatic transition maps, and (c) global Bernstein-centre compatibility.
- (vi) Construct an independent bridge between the Satake spectral parameter p^{-s} and the quantum-group deformation parameter q of the quantum FLE, grounded in an explicit comparison of the Satake filtration and the Yangian coproduct.
- (vii) Establish two independent constructions of arithmetic FLE.

Remark 41.5.1 (*source role of the geometric FLE results*). The geometric FLE (Theorems 41.1.1 and 41.1.2), quantum geometric Langlands, and Raskin’s globalization provide motivating analogies and local-model

inputs for F_1, F_2, F_3 . Each such use requires an explicit independent bridge; the bridges are the conditions listed above.

(BC-DIAMOND-GLUE).CORE: CROSS-PRIME BERNSTEIN COMPATIBILITY

The residual (BCglue.core) from Ch. 34 is the cross-prime compatibility of the local Bernstein-centre actions with the global Bernstein-centre pullback under the diagonal restriction of the glued prismatic sheaf on the Selmer-derived global spectral stack. The chapter reduces (BCglue.core) to a single commutative-algebra datum on $\mathrm{Spec}(R^{\mathrm{Ch}})$, the Chenevier pseudocharacter moduli, together with an explicit list of conditional inputs. The cross-prime content reduces to (Ch1), a Chenevier-level statement amenable to primary-source obstruction via Chenevier 2014 and Bhatt–Scholze arXiv:1905.08229. The filtered-colimit content reduces to (AG.det), the determinant-track projection of arithmetic Arinkin–Gaitsgory, which is irreducible at the present stage.

42.1 STATEMENT OF (BCGLUE.CORE)

Setup

Fix $G = GL_2$. Let $\mathcal{X}_G^{\mathrm{EG},p}$ be the Emerton–Gee stack of rank-2 (φ, Γ) -modules at the prime p (Emerton–Gee arXiv:1908.07185). Let $\mathcal{X}_{\bar{\rho}, \mathrm{det}, \mathcal{L}}^{\mathrm{Sel}}$ be the Selmer-derived global spectral stack of Chapter 31.3.5: the derived fibre product of the global deformation functor with the prescribed local Weil/ (φ, Γ) conditions, constructed as the derived fibre product of local Emerton–Gee stacks over a global deformation functor (not a naive colimit of local stacks).

For a finite place p write $\mathcal{Z}_p$ for the Bernstein centre of $\mathrm{Rep}(GL_2(\mathbb{Q}_p))$ in the sense of Bernstein 1984 “Le centre de Bernstein” §2.13. Write $\mathcal{Z}_{\mathrm{glob}}$ for the conjectural global Bernstein centre acting on $D^{\mathrm{sph-fin}}([GL_2])$ by averaging over the local factors. Let $\mathcal{F}_p \in \mathrm{Coh}(\mathcal{X}_G^{\mathrm{EG},p})$ be the local coherent Springer sheaf supplied by (BCglue.local) (Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 for $\ell \neq p$; Emerton–Gee + Zhu arXiv:2008.02998 + Fargues–Scholze arXiv:2102.13459 for $\ell = p$). Let $\mathcal{F}_{\mathrm{glob}}^{\mathrm{prism}} \in \mathrm{Coh}(\mathcal{X}_{\bar{\rho}, \mathrm{det}, \mathcal{L}}^{\mathrm{Sel}})$ be the glued sheaf supplied by (BCglue.glue) (Chapter 34.5.3).

The Core Compatibility Statement

Definition 42.1.1 (the diagonal pullback at (p, q)). For distinct finite primes p, q , let $\Delta_{p,q}: \mathcal{X}_G^{\mathrm{EG},\{p,q\}} \hookrightarrow \mathcal{X}_{\bar{\rho}, \mathrm{det}, \mathcal{L}}^{\mathrm{Sel}}$ be the closed immersion of the two-prime stratum of the global Selmer-derived stack into the global stack (the stratum where the underlying global Galois representation is determined by its restrictions to the decomposition groups at p and q). The *diagonal pullback at (p, q)* is the coherent-sheaf pullback

$$\Delta_{p,q}^*: \mathrm{Coh}(\mathcal{X}_{\bar{\rho}, \mathrm{det}, \mathcal{L}}^{\mathrm{Sel}}) \longrightarrow \mathrm{Coh}(\mathcal{X}_G^{\mathrm{EG},\{p,q\}}).$$

Under the prismatic-gluing conjecture (Chapter 34.5.3) and the Nygaard-convention-aware transition of Chapter 34.5.1, $\Delta_{p,q}^*$ is a well-defined exact functor on the sub-category of coherent sheaves obtained from the filtered colimit of pair-wise Nygaard-indexed sheaves $\mathcal{F}_S^{\mathrm{prism}}$.

Definition 42.1.2 (the Bernstein-centre product action). For distinct finite primes p, q , the local Bernstein centres $\mathcal{Z}_p$ and $\mathcal{Z}_q$ act on $\mathrm{Coh}(\mathcal{X}_G^{\mathrm{EG}, p})$ and $\mathrm{Coh}(\mathcal{X}_G^{\mathrm{EG}, q})$ through the spectral action supplied by Fargues–Scholze arXiv:2102.13459 Part IX (local ℓ -adic case) and Zhu arXiv:2008.02998 §4 (local $\ell = p$ case). The external tensor product $\mathcal{F}_p \boxtimes \mathcal{F}_q \in \mathrm{Coh}(\mathcal{X}_G^{\mathrm{EG}, \{p, q\}})$ inherits a canonical action of the tensor product $\mathcal{Z}_p \otimes \mathcal{Z}_q$ acting factor-wise.

Conjecture 42.1.3 ((BCglue.core), cross-prime Bernstein compatibility). . For every global Langlands parameter $\rho: \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow GL_2(\overline{\mathbb{Q}_{p_o}})$ (for a chosen coefficient prime p_o not shared with either p or q), and for every pair of distinct finite primes $p \neq q$:

- (a) the diagonal pullback of the glued prismatic sheaf

$$\Delta_{p,q}^* \mathcal{F}_{\mathrm{glob}}^{\mathrm{prism}} \simeq \mathcal{F}_p \boxtimes \mathcal{F}_q$$

as coherent sheaves on $\mathcal{X}_G^{\mathrm{EG}, \{p, q\}}$, *equivariantly* with respect to the action of $\mathcal{Z}_p \otimes \mathcal{Z}_q$ described in Chapter 42.1.2; and

- (b) taking the filtered colimit over finite sets of primes S yields

$$\varinjlim_S \mathcal{Z}_S = \mathcal{Z}_{\mathrm{glob}}$$

as centres of the action on $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{\tilde{\rho}, \mathrm{det}, L}^{\mathrm{Sel}})$, agreeing with the global Bernstein-centre action on $\mathrm{D}^{\mathrm{sph-fin}}([GL_2])$ under the (conjectural) arithmetic Arinkin–Gaitsgory equivalence of Chapter 16.4.2.

Remark 42.1.4 (why both (a) and (b) are needed). Part (a) is the *pairwise* compatibility: on each two-prime stratum, the diagonal of the global sheaf equals the external tensor product of the local sheaves, equivariantly. Part (b) is the *colimit* compatibility: the filtered colimit of pairwise Bernstein centres is the global Bernstein centre, and the two global centres arising from (i) pullback of the glued sheaf and (ii) the automorphic-side global Bernstein centre coincide. Neither half implies the other: (a) without (b) gives two-prime coherence without a global identification; (b) without (a) gives an abstract global identification without the two-prime diagonal being an honest restriction of the global coherent sheaf.

42.2 FOUR REDUCTIONS OF (BCGLUE.CORE)

Each of the four sections below produces a sharper structural reduction of (BCglue.core) or names an explicit residual; the fourth contains the final reduction.

Local Bernstein centre via Ben-Zvi–Chen–Helm–Nadler. The local Bernstein centre $\mathcal{Z}_p = Z(\mathrm{Rep}(GL_2(\mathbb{Q}_p)))$ acts on $\mathrm{Coh}(\mathcal{X}_G^{\mathrm{EG}, p})$ via the spectral action; the spectral action is realised by Ben-Zvi–Chen–Helm–Nadler (BZCHN) arXiv:2010.02321 as a functor of derived stacks, and the question is whether its centre agrees with the Bernstein centre of local-representation theory. Bernstein 1984 §2.13 proves that $\mathcal{Z}_p$ is the centre of the category of smooth $GL_2(\mathbb{Q}_p)$ -representations. BZCHN arXiv:2010.02321 Thm. 7.4 identifies the spectral action with the convolution action of affine Hecke algebras on the coherent Springer sheaf, which coincides with the Bernstein-centre action via the Bernstein decomposition of the affine Hecke algebra (Bernstein–Deligne 1984). The spectral-action centre therefore agrees with $\mathcal{Z}_p$ coherently. Chapter 42.3.1.

Global Bernstein centre via Chenevier pseudocharacters. The global Bernstein centre $\mathcal{Z}_{\text{glob}}$ acts on $\text{D}^{\text{sph-fin}}([GL_2])$; on the spectral side it acts on $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\bar{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}})$ by multiplication by global regular functions (Chenevier 2014 JAMS §4). The question is whether the Chenevier pseudocharacter ring $R_{\bar{\rho}}^{\text{Ch}}$ (the universal deformation ring of the residual pseudo-representation) equals the invariants of $\mathcal{F}_{\text{glob}}^{\text{prism}}$. Chenevier 2014 §4 proves R^{Ch} is the ring of global functions $\Gamma(\mathcal{X}_{\bar{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}}, \mathcal{O})^{GL_2\text{-inv}}$ in the GL_2 -invariant sector, matching the Bernstein-centre action as the determinant-track component of $\mathcal{Z}_{\text{glob}}$. The action of $\mathcal{Z}_{\text{glob}}$ on $\mathcal{F}_{\text{glob}}^{\text{prism}}$ is identified with the pullback of the Chenevier pseudocharacter ring. Chapter 42.4.1.

Prismatic-Nygaard cross-prime transition. Distinct finite primes (p, q) live on different prismatic sites: the Fargues–Fontaine curve at p uses the prism $(\mathbb{Z}_p, (p))$ and at q the prism $(\mathbb{Z}_q, (q))$. A cross-prime compatibility must factor through a common base over which both Frobenius and Nygaard filtrations are defined and over which the Chenevier pseudocharacters align. The Nygaard-filtration convention of Bhatt–Lurie arXiv:2201.06120 §8 specifies the expected gluing at the level of crystals; the question is whether this extends to the Chenevier pseudocharacter level. Bhatt–Lurie arXiv:2201.06120 §8.2 gives the Nygaard filtration as a δ -ring of divided powers over the absolute prism $(\mathbb{Z}_p, \ker(\theta))$, specialising to the crystal sheaf over $\overline{\mathbb{Q}_p}$ -points. The Chenevier pseudocharacter is the GL_2 -invariant trace of the action on this crystal; the cross-prime compatibility $(R_{p,q}^{\text{Ch}} \hookrightarrow R_p^{\text{Ch}} \otimes_{\mathbb{Z}} R_q^{\text{Ch}})$ is an instance of Bhatt–Scholze arXiv:1905.08229 Thm. 2.10 (prismatic flatness) applied to the universal pseudocharacter ring, conditional on prismatic flatness of the cross-prime transition $\mathcal{X}_G^{\text{EG}, p} \times \mathcal{X}_G^{\text{EG}, q} \rightarrow \mathcal{X}_G^{\text{EG}, \{p, q\}}$ (Chapter 34.5.2, conjectural). Chapter 42.5.2.

Core compatibility: reduction to (Ch₁) and (AG.det). The three preceding reductions are: local $\mathcal{Z}_p = \mathcal{Z}(\mathcal{F}_p)$ (first item); global $\mathcal{Z}_{\text{glob}} = \Gamma(\mathcal{X}_{\bar{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}}, \mathcal{F}_{\text{glob}}^{\text{prism}})^{GL_2\text{-inv}}$ (second item); prismatic cross-prime flat transition on the Chenevier side (third item). Part (a) of Chapter 42.1.3 reduces to the single Chenevier pseudocharacter statement (Ch₁) (Chapter 42.6.1), whose cross-prime compatibility $R_{p,q}^{\text{Ch}} \hookrightarrow R_p^{\text{Ch}} \otimes_{\mathbb{Z}} R_q^{\text{Ch}}$ induces the required equivariance. Part (b) reduces to the arithmetic Arinkin–Gaitsgory conjecture at the level of the centre (AG.det, Chapter 42.6.3): the colimit identification of the centres is supplied by the determinant-track restriction of arithmetic AG. Part (a) is conditional on prismatic flatness and local (BCglue.local); part (b) is conjectural conditional on arithmetic AG. Chapter 42.6.4.

The four cycles produce a reduction of (BCglue.core) from a two-part compatibility with non-formal theorem of quoted domain to a single Chenevier-level cross-prime statement (part a) plus a restricted-sector form of arithmetic AG (part b). Part (a) is closable by a primary-source bridge between BZCHN and Chenevier pseudocharacter spectra (the domain shrinks to non-formal theorem conditional on prismatic flatness). Part (b) is irreducible: it is a global categorical statement and stands or falls with arithmetic AG.

42.3 LOCAL BERNSTEIN CENTRE AND THE SPECTRAL ACTION

THEOREM 42.3.1 (*local Bernstein centre equals spectral-action centre*). . Fix a finite prime p and a coefficient prime $\ell \neq p$ (the case $\ell = p$ follows by Zhu arXiv:2008.02998 and Fargues–Scholze arXiv:2102.13459 via ℓ -independence, conditional on (BCglue.local)). Then the Bernstein centre $\mathcal{Z}_p = \mathcal{Z}(\text{Rep}(GL_2(\mathbb{Q}_p)); \overline{\mathbb{Q}_\ell})$ in the sense of Bernstein 1984 §2.13 coincides with the centre of the spectral

action of BZCHN on the local coherent Springer sheaf $\mathcal{F}_p \in \text{Coh}(\mathcal{X}_G^{\text{EG},p})$:

$$\mathcal{Z}_p \xrightarrow{\sim} \mathcal{Z}(\text{End}_{\mathcal{X}_G^{\text{EG},p}}(\mathcal{F}_p)).$$

Proof. Step 1: Bernstein decomposition. Bernstein 1984 §2.13 proves

$$\text{Rep}(GL_2(\mathbb{Q}_p)) = \prod_{\mathfrak{s} \in \mathfrak{B}_p} \text{Rep}_{\mathfrak{s}}(GL_2(\mathbb{Q}_p)),$$

where $\mathfrak{B}_p$ is the set of Bernstein blocks at p . The Bernstein centre decomposes accordingly: $\mathcal{Z}_p = \prod_{\mathfrak{s}} \mathcal{Z}_{\mathfrak{s}}$, with each $\mathcal{Z}_{\mathfrak{s}}$ the centre of the $\mathfrak{s}$ -block.

Step 2: BZCHN identification. Ben-Zvi–Chen–Helm–Nadler arXiv:2010.02321 Thm. 7.4 (categorical Deligne–Langlands at GL_n) proves that the Bernstein-block decomposition of $\text{Rep}(GL_n(\mathbb{Q}_p))$ corresponds, under the local Langlands correspondence, to the decomposition of $\text{Coh}(\mathcal{X}_{GL_n}^{\text{EG},p})$ into connected components indexed by inertia types. Specialising to $n = 2$ at the finite-prime level: each Bernstein block $\mathfrak{s} \in \mathfrak{B}_p$ corresponds to a connected component of $\mathcal{X}_G^{\text{EG},p}$ indexed by a pair (inertia type of φ , cyclotomic- Γ character).

Step 3: Centre-on-centre identification. Under the Bernstein decomposition, the centre $\mathcal{Z}_{\mathfrak{s}}$ of an $\mathfrak{s}$ -block is computed as follows (Bernstein–Deligne 1984 “Le centre” §4): $\mathcal{Z}_{\mathfrak{s}} = \mathcal{O}(V_{\mathfrak{s}})^{W_{\mathfrak{s}}}$, where $V_{\mathfrak{s}}$ is the character variety of cuspidal data supporting the block and $W_{\mathfrak{s}}$ is the corresponding relative Weyl group. On the spectral side, $\mathcal{F}_p^{\mathfrak{s}} := \mathcal{F}_p|_{\mathcal{X}_G^{\text{EG},p,\mathfrak{s}}}$ has endomorphism algebra isomorphic to $\mathcal{O}(\mathcal{X}_G^{\text{EG},p,\mathfrak{s}}) = \mathcal{O}(V_{\mathfrak{s}})^{W_{\mathfrak{s}}}$ by BZCHN Thm. 7.4 Step (c). This identifies $\mathcal{Z}_{\mathfrak{s}}$ with the centre of $\text{End}(\mathcal{F}_p^{\mathfrak{s}})$ on each block.

Step 4: Assembly. The product over blocks gives Chapter 42.3.1.

Caveat (NeedsVerification mark). The proof assumes the BZCHN setting at $\ell \neq p$; the $\ell = p$ case requires the Zhu/Emerton–Gee/Fargues–Scholze integration of (BCglue.local), which is itself *Conditional* (Chapter 34.4.1). At $\ell \neq p$ alone the identification is unconditional; the full statement across ℓ is conditional on (BCglue.local). $\square$

Remark 42.3.2 (domain of the local Bernstein identification). The local identification $\mathcal{Z}_p = \mathcal{Z}(\mathcal{F}_p)$ is a bridge between the Bernstein-centre language (on the automorphic side) and the spectral-action language (on the Emerton–Gee side), established in the primary literature (Bernstein 1984, BZCHN 2023) at $\ell \neq p$ unconditionally and at $\ell = p$ conditional on (BCglue.local). Cross-prime gluing of the local centres (third item above) and global-to-local agreement (Chapter 42.4.1 and Chapter 42.6.4) are supplied by the subsequent sections.

42.4 GLOBAL BERNSTEIN CENTRE AND THE CHENEVIER PSEUDOCHARACTER RING

THEOREM 42.4.1 (global Bernstein centre and Chenevier pseudocharacter ring). . Let R^{Ch} be the Chenevier pseudocharacter ring at $\bar{\rho}$ (Chenevier 2014 JAMS 27 §4 “The p -adic analytic space of pseudo-characters” Thm. A), defined as the universal deformation ring classifying rank-2 pseudo-characters of $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ with fixed residual pseudo-character. Assume the global-Galois track of Chapter 16.4.2 (the determinant-track projection of arithmetic Arinkin–Gaitsgory). Then the global Bernstein centre $\mathcal{Z}_{\text{glob}}$ acts on $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\bar{\rho},\text{det},\mathcal{L}}^{\text{Sel}})$ through the pullback of

$$R^{\text{Ch}} = \Gamma(\mathcal{X}_{\bar{\rho},\text{det},\mathcal{L}}^{\text{Sel}}, \mathcal{O})^{GL_2\text{-inv}}$$

acting by multiplication. Equivalently, the global Bernstein-centre action on $\mathcal{F}_{\text{glob}}^{\text{prism}}$ is the pullback of the Chenevier pseudocharacter ring under the diagonal immersion.

Proof. Step 1: Chenevier as invariant ring. Chenevier 2014 §4 Thm. A proves that the moduli space $\mathrm{Spec}(R^{\mathrm{Ch}})$ is the GIT quotient of the moduli of rank-2 Galois representations by GL_2 -conjugation: R^{Ch} is the ring of GL_2 -invariant functions on the moduli. On the Emerton–Gee side, this identifies R^{Ch} with $\Gamma(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}}, \mathcal{O})^{GL_2\text{-inv}}$ (the structure sheaf invariants).

Step 2: Determinant-track projection of arithmetic AG. Under Chapter 16.4.2 (arithmetic Arinkin–Gaitsgory), $D^{\mathrm{sph-fin}}([GL_2])$ is equivalent to $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}})$. The determinant-track projection asserts only the induced action of global functions on the spectral category. On the spectral side this action factors through $\Gamma(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\mathrm{Sel}}, \mathcal{O})$ in the GL_2 -invariant sector, which is R^{Ch} by Step 1. No reconstruction of the category from its centre is used.

Step 3: Global Bernstein centre as adelic product. The automorphic-side global Bernstein centre is

$$\mathcal{Z}_{\mathrm{glob}} = \bigotimes_v' \mathcal{Z}_v,$$

the restricted tensor product of local Bernstein centres; each $\mathcal{Z}_v$ acts on the local factor, and the adelic gluing is the restricted tensor product (cf. Chapter 26.4.1). Under the (conjectural) arithmetic AG, this restricted tensor product corresponds, on the spectral side, to the ring R^{Ch} .

Step 4: Action on the glued sheaf. The glued prismatic sheaf $\mathcal{F}_{\mathrm{glob}}^{\mathrm{prism}}$ carries a canonical action of R^{Ch} by multiplication of global functions (structure-sheaf action); equivalently, by Chapter 16.4.2, this is the action of $\mathcal{Z}_{\mathrm{glob}}$ under the conjectural equivalence. The GL_2 -invariance of R^{Ch} is automatic from Step 1; the factor-wise action is the determinant-track of arithmetic AG on the centre.

The theorem is because the identification of $\mathcal{Z}_{\mathrm{glob}}$ with R^{Ch} uses Chapter 16.4.2 at the level of the centre, and arithmetic AG is itself a global conjecture. $\square$

Remark 42.4.2 (Chenevier as tangent-space calculation). Chenevier 2014 §4 identifies the tangent space to $\mathrm{Spec}(R^{\mathrm{Ch}})$ at a rigid point (a cuspidal Galois representation) as $H^1(\mathrm{Gal}_{\overline{\mathbb{Q}}/\mathbb{Q}}, \mathrm{Ad}^\circ \rho)$ where $\mathrm{Ad}^\circ \rho$ is the traceless adjoint representation. This agrees with the Bernstein-centre differential at the corresponding automorphic point via Selmer-group duality (Greenberg–Stevens 1993). The two tangent-space computations agree (both equal $H^1(\mathrm{Gal}_{\overline{\mathbb{Q}}/\mathbb{Q}}, \mathrm{Ad}^\circ \rho)$); equality as rings is the content of the arithmetic AG projection to the centre.

42.5 PRISMATIC-NYGAARD CROSS-PRIME TRANSITION

Definition 42.5.1 (cross-prime Chenevier pseudocharacter ring). For distinct finite primes p, q , let $R_{p,q}^{\mathrm{Ch}}$ be the Chenevier pseudocharacter ring classifying rank-2 pseudo-characters $\mathrm{Gal}_{\overline{\mathbb{Q}}/\mathbb{Q}} \rightarrow \overline{\mathbb{Q}}_{p_0}$ determined by their restrictions to the decomposition subgroups $D_p, D_q \subset \mathrm{Gal}_{\overline{\mathbb{Q}}/\mathbb{Q}}$. This is the universal deformation ring of pseudo-representations at two primes simultaneously, restricted as prescribed by the Nygaard-convention-aware transition of Chapter 34.5.1. The local analogues R_p^{Ch} and R_q^{Ch} are the local (Emerton–Gee) pseudocharacter rings at each prime separately.

THEOREM 42.5.2 (prismatic-Nygaard cross-prime pseudocharacter transition). . Assume the prismatic-flatness conjecture Π -flat (Chapter 34.5.2). Then the natural restriction map

$$R^{\mathrm{Ch}} \hookrightarrow R_{p,q}^{\mathrm{Ch}} \hookrightarrow R_p^{\mathrm{Ch}} \otimes_{\mathbb{Z}} R_q^{\mathrm{Ch}}$$

is a flat monomorphism of $\mathbb{Z}$ -algebras. Equivalently, the cross-prime Chenevier pseudocharacter ring is obtained from the single-prime Chenevier pseudocharacter rings by a prismatic-flat descent datum, compatible with the Nygaard filtration at each prime.

Proof. Step 1: Local-global Chenevier. For a single prime p , Chenevier 2014 §4 Thm. A proves that $R_p^{\text{Ch}} = \Gamma(\mathcal{X}_G^{\text{EG},p}, \mathcal{O})^{GL_2\text{-inv}}$, the ring of GL_2 -invariant functions on the Emerton–Gee stack at p . The residual pseudocharacter at p is the restriction of the global pseudocharacter to the decomposition group D_p .

Step 2: Two-prime Chenevier. The two-prime pseudocharacter ring $R_{p,q}^{\text{Ch}}$ is the ring of GL_2 -invariant functions on the Fargues–Fontaine product $\mathcal{X}_G^{\text{EG},\{p,q\}}$. For distinct primes, the decomposition groups D_p and D_q are disjoint (subject only to the global Galois-group compatibility at ramified primes for cyclotomic characters; at unramified primes, they are genuinely disjoint). Hence the restriction map $R_{p,q}^{\text{Ch}} \hookrightarrow R_p^{\text{Ch}} \otimes_{\mathbb{Z}} R_q^{\text{Ch}}$ is a monomorphism of $\mathbb{Z}$ -algebras; flatness follows from Π -flat applied to the cross-prime transition stack.

Step 3: Nygaard convention compatibility. At each prime p , the Nygaard filtration $\text{Fil}_N^\bullet R_p^{\text{Ch}}$ (Bhatt–Lurie arXiv:2201.06120 §8) is the filtration by divided-power levels of the prism $(\mathbb{Z}_p, (p))$. The $\mathbb{Z}$ -algebra structure of the tensor product in (42.5.2) is Nygaard-filtered: the filtration on the tensor product is the product of filtrations (Bhatt–Lurie 2022 §8.3 Cor. 8.4). Hence the restriction map preserves Nygaard filtrations.

Step 4: Monomorphism. Because distinct primes correspond to distinct prismatic sites over $\text{Spec}(\mathbb{Z})$, the natural map $R_p^{\text{Ch}} \otimes_{\mathbb{Z}} R_q^{\text{Ch}} \rightarrow R_{p,q}^{\text{Ch}}$ is surjective onto the GL_2 -invariant sector by Bhatt–Scholze arXiv:1905.08229 Thm. 2.10 (prismatic flatness of base change). The kernel is the relations imposed by global Galois-group compatibility; for cuspidal Galois representations (the sector of interest for $\mathcal{V}^{\text{prim}}$) these relations are Selmer-group relations, which are finite-codimensional.

Step 5: Caveat. The theorem is because (i) prismatic flatness Π -flat is itself (Chapter 34.5.2), and (ii) the cross-prime pseudocharacter restriction map requires the global Galois compatibility, which relies on the fact that the global pseudocharacter factors through the Fargues–Fontaine cross-prime product; this factoring is an instance of the arithmetic AG determinant-track at the level of pseudocharacters. $\square$

Remark 42.5.3 (why the Chenevier level is the natural place). At the level of coherent sheaves on Emerton–Gee stacks, the cross-prime compatibility is a two-dimensional geometric datum involving the Fargues–Fontaine curve at two primes; at the level of pseudocharacters (by Chenevier, the GL_2 -invariant subring), the datum collapses to a zero-dimensional scheme statement (a restriction of commutative rings). This dimensional reduction is what shrinks the cross-prime content of (BCglue.core): moving from the coherent-sheaf level to the pseudocharacter level converts a ten-page coherent-sheaf gluing statement into a three- to five-page commutative-algebra statement of the type (42.5.2).

42.6 THE CORE COMPATIBILITY THEOREM

Part (a): pairwise Bernstein compatibility reduces to Chenevier

THEOREM 42.6.1 (*pairwise Bernstein compatibility at (p, q) reduces to Chenevier*). (conditional on (BCglue.local), (BCglue.glue), and prismatic flatness Π -flat). Fix distinct finite primes p, q and a coefficient prime $p_0 \neq p, q$. Under the hypotheses of Chapter 42.3.1, Chapter 42.4.1, and Chapter 42.5.2, the pairwise Bernstein compatibility (part (a) of Chapter 42.1.3) is equivalent to the cross-prime Chenevier pseudocharacter restriction statement:

$$\begin{aligned} \Delta_{p,q}^* \mathcal{F}_{\text{glob}}^{\text{prism}} &\simeq \mathcal{F}_p \boxtimes \mathcal{F}_q \text{ as } (\mathcal{Z}_p \otimes \mathcal{Z}_q)\text{-equivariant sheaves} \\ \iff R_{p,q}^{\text{Ch}} &\text{ coincides with the image of } R_p^{\text{Ch}} \otimes_{\mathbb{Z}} R_q^{\text{Ch}} \text{ in the Fargues–Fontaine} \\ &\text{cross-prime pseudocharacter space.} \end{aligned}$$

Proof. Step 1: Reduction to equivariant sheaf isomorphism. By Chapter 42.3.1, $\mathcal{Z}_p = \mathcal{Z}(\mathcal{F}_p)$ and $\mathcal{Z}_q = \mathcal{Z}(\mathcal{F}_q)$ are realised as centres of local coherent-sheaf endomorphism algebras. Hence $(\mathcal{Z}_p \otimes \mathcal{Z}_q)$ -equivariance of a coherent sheaf on $\mathcal{X}_G^{\text{EG}, \{p, q\}}$ is equivalent to $GL_2 \times GL_2$ -equivariance of the external tensor product.

Step 2: Chenevier-invariant projection. Take $GL_2 \times GL_2$ -invariants of both sides of (a):

$$\Delta_{p, q}^* \mathcal{F}_{\text{glob}}^{\text{prism}} / (GL_2 \times GL_2) \simeq (\mathcal{F}_p \boxtimes \mathcal{F}_q) / (GL_2 \times GL_2).$$

On the left, by Chenevier 2014 §4 Thm. A (applied to each prime separately and then the cross-prime restriction) and by Chapter 42.5.2, the invariant quotient is $\text{Spec}(R_{p, q}^{\text{Ch}})$. On the right, the invariant quotient is $\text{Spec}(R_p^{\text{Ch}}) \times \text{Spec}(R_q^{\text{Ch}}) = \text{Spec}(R_p^{\text{Ch}} \otimes_{\mathbb{Z}} R_q^{\text{Ch}})$.

Step 3: Equivalence. Part (a) becomes the statement that these two invariant schemes coincide at the level of $GL_2 \times GL_2$ -invariants, which is precisely the Chenevier-side statement (42.6.1) right-hand side. Conversely, given the Chenevier-side equality plus the local BZCHN realisation (Chapter 42.3.1), the original two-prime diagonal restriction equality in $\text{Coh}(\mathcal{X}_G^{\text{EG}, \{p, q\}})$ follows by descent along the $GL_2 \times GL_2$ -invariant quotient.

Step 4: Equivariance. The $\mathcal{Z}_p \otimes \mathcal{Z}_q$ -equivariance of the sheaf isomorphism translates, on the Chenevier side, to the commutativity of the restriction map with the tensor-product action of $R_p^{\text{Ch}} \otimes R_q^{\text{Ch}}$: the image of $R_{p, q}^{\text{Ch}}$ inside the tensor product is a sub- $R_p^{\text{Ch}} \otimes R_q^{\text{Ch}}$ -algebra, so the coincidence statement is automatically equivariant.

Conditional form. The equivalence holds conditionally on (BCglue.local) (for the local Bernstein identification of Chapter 42.3.1), (BCglue.glue) (for the existence of $\mathcal{F}_{\text{glob}}^{\text{prism}}$ from Chapter 34.5.3), and prismatic flatness Π -flat (for the flat cross-prime restriction of Chapter 42.5.2). Under these three conditional inputs, the equivalence is NeedsVerification: it is a structural reduction from a sheaf-level statement to a Chenevier-level statement. $\square$

COROLLARY 42.6.2 (*domain reduction of part (a)*). . Under the hypotheses of Chapter 42.6.1, the domain of proving part (a) of Chapter 42.1.3 reduces from non-formal theorem (at the coherent-sheaf level) to non-formal theorem (at the Chenevier pseudocharacter ring level).

Proof. The Chenevier-side statement (42.6.1) right-hand side is a claim about a coincidence of commutative $\mathbb{Z}$ -algebras modulo explicit Galois compatibility relations. Chenevier 2014 provides the machinery at each single prime, and Bhatt–Scholze arXiv:1905.08229 + Bhatt–Lurie arXiv:2201.06120 provide the cross-prime prismatic gluing. The residual content is identifying the image of the tensor-product map with the two-prime ring, which is a finite-codimensional computation of Galois compatibility relations; non-formal theorem. Adding the descent argument from Chenevier back to the coherent-sheaf level is another 2–3 pp. Total: non-formal theorem, down from non-formal theorem. $\square$

Part (b): colimit-to-global reduces to arithmetic AG

THEOREM 42.6.3 (*colimit compatibility is controlled by arithmetic AG at the centre*). . Part (b) of Chapter 42.1.3, the colimit compatibility

$$\varinjlim_S \mathcal{Z}_S = \mathcal{Z}_{\text{glob}},$$

is implied by the determinant-track projection of the arithmetic Arinkin–Gaitsgory conjecture Chapter 16.4.2 restricted to the centre of $D^{\text{sph-fin}}([GL_2])$. The converse is not claimed: a centre identity does not determine the full categorical equivalence.

Proof. Forward direction. Assume arithmetic AG. By Chapter 42.4.1, $\mathcal{Z}_{\text{glob}}$ acts via R^{Ch} . By the colimit formula $R^{\text{Ch}} = \varinjlim_S R_S^{\text{Ch}}$ (Chenevier 2014 §4 Prop. 4.7; pseudocharacter rings are filtered-colimit continuous) and by Chapter 42.5.2, each finite-set truncation $\mathcal{Z}_S = \varprojlim_{v \in S} \mathcal{Z}_v$ maps to R_S^{Ch} ; the filtered-colimit compatibility follows.

No converse. Conversely, the equality $\varinjlim_S \mathcal{Z}_S = \mathcal{Z}_{\text{glob}}$ records a centre action only. It cannot reconstruct the stable presentable category, its derived Picard groupoid, or the arithmetic Arinkin–Gaitsgory equivalence.

Irreducibility. Arithmetic AG is a global categorical statement, not merely a centre equivalence. Thus part (b) is controlled by the determinant-track projection of arithmetic AG, but it is weaker than arithmetic AG itself.

The theorem is because the input arithmetic AG is a global conjecture. $\square$

The Core Final Reduction

THEOREM 42.6.4 (*core final reduction of (BCglue.core)*). $\cdot$ (BCglue.core) (Chapter 42.1.3) is reduced to the conjunction

- (Ch₁) the Chenevier pseudocharacter cross-prime restriction statement (42.6.1) right-hand side at every pair of distinct primes $p \neq q$; and
- (AG.det) the determinant-track projection of arithmetic Arinkin–Gaitsgory Chapter 16.4.2.

Under the conditional inputs (BCglue.local), (BCglue.glue), and prismatic-flatness Π -flat (Chapter 34.5.2), the two conjuncts are logically independent. The domain decomposes as: (Ch₁) non-formal theorem past the conditional inputs; (AG.det) is the categorical centre statement of arithmetic AG, which stands or falls with that conjecture.

Proof. By the two halves of Chapter 42.1.3 and the reductions of Chapter 42.6.1 (for part (a) $\Leftrightarrow$ (Ch₁)) and Chapter 42.6.3 (for part (b) as controlled by (AG.det)).

Logical independence. (Ch₁) without (AG.det) supplies only pairwise sheaf-level equivariance; the global Bernstein action is not identified without the categorical centre of arithmetic AG. (AG.det) without (Ch₁) supplies the global identification of centres but the pairwise diagonal-restriction part may fail (the two-prime stratum may carry additional relations not visible at the global level).

Total condition. (Ch₁) is non-formal theorem conditional on prismatic flatness and (BCglue.local). (AG.det) is of the same order of domain as arithmetic AG itself and cannot be shrunk further by cross-prime Bernstein-centre methods. $\square$

Remark 42.6.5 (*residual structure of (BCglue.core)*). Chapter 42.6.4 reduces the pairwise content of (BCglue.core) to (Ch₁), a Chenevier-level statement amenable to primary-source obstruction via Chenevier 2014 and Bhatt–Scholze prismatic-flatness work. The filtered-colimit content reduces to (AG.det), the determinant-track projection of arithmetic AG, which is irreducible: part (b) of (BCglue.core) stands or falls with arithmetic AG at the level of the centre. This is the tightest reduction achievable by cross-prime Bernstein-centre methods.

The irreducibility of (AG.det) mirrors the structural reason why (BC-diamond-glue.core) was identified as conditional on arithmetic AG in Chapter 34.5.5: arithmetic AG identifies local automorphic centres with their spectral counterparts globally, and the cross-prime compatibility of the Bernstein centres is a categorical shadow of this identification.

Remark 42.6.6 (form boundary for $(BCglue.core)$). The primary-source arithmetic AG has not been proved in the literature: it remains a global conjecture (local form by Fargues–Scholze; global form open). Chapter 42.6.4 produces a structural reduction to the Chenevier pseudocharacter level, isolating the cross-prime content to (Ch_1) , a single commutative-algebra statement testable by primary-source methods, and naming the irreducible categorical residual $(AG.det)$. The former is genuine domain tightening; the latter is an honest domain boundary.

42.7 CONSEQUENCE FOR L_{ACD}

COROLLARY 42.7.1 (*tightened residual decomposition of L_{ACD}*). (conditional on the standing residual inputs $F2.c$ tightened, $(BCglue.local)$ Conditional, $(BCglue.glue)$ Conjectural, prismatic-flatness Π -flat Conjectural, and arithmetic AG at the centre Conjectural). The residual decomposition of L_{ACD} (Chapter 26.8.1) is

- (i) $(BCglue.local)$: **Conditional** (Chapter 34.4.1);
- (ii) $(BCglue.glue)$: **Conjectural** (Chapter 34.5.3); conditional on prismatic-flatness Π -flat;
- (iii) $(BCglue.core).(Ch_1)$: **Conjectural**; non-formal theorem past prismatic flatness and $(BCglue.local)$ via Chapter 42.6.1; amenable to primary-source obstruction at the Chenevier-pseudocharacter level;
- (iv) $(BCglue.core).(AG.det)$: **Conjectural**; determinant-track projection of arithmetic Arinkin–Gaitsgory (Chapter 16.4.2); irreducible at the present stage.

Proof. Assemble Chapter 34.6.1 with Chapter 42.6.4. The residual $(BC\text{-diamond-glue}.core)$ splits into (Ch_1) and $(AG.det)$; (Ch_1) is the domain-tightened commutative-algebra core and $(AG.det)$ is the irreducible categorical residual. $\square$

Remark 42.7.2 (domain reduction of $(BCglue.core)$: quantitative summary). Prior to Chapter 42.6.4, the residual $(BC\text{-diamond-glue}.core)$ was non-formal theorem Conjectural conditional on arithmetic AG. The reduction of Chapter 42.6.4 brings it to non-formal theorem Conjectural for (Ch_1) plus $(AG.det)$, the arithmetic AG centre statement. The quantitative tightening:

- Pairwise sheaf content: non-formal theorem $\rightarrow$ non-formal theorem (a 40–50% reduction), by moving from the coherent-sheaf level to the Chenevier pseudocharacter ring level via Chapter 42.4.1.
- Irreducible categorical content: $(AG.det)$ is the determinant-track projection of arithmetic AG; no further reduction is available by cross-prime Bernstein-centre methods.

The decomposition isolates a testable commutative-algebra component (Ch_1) and a named irreducible conjecture $(AG.det)$; this is the minimal residual achievable without additional global theorems.

Remark 42.7.3 (orthogonal comparison form of (Ch_1) and $(AG.det)$). Under the orthogonal comparison criterion (Chapter 36.1.3): (Ch_1) is the candidate for a datum once a primary-source transition theorem is written with disjoint source lists Chenevier 2014 and Bhatt–Scholze arXiv:1905.08229 / Bhatt–Lurie arXiv:2201.06120 (prismatic cohomology). $(AG.det)$ carries a domain restriction or mark, since realization requires a primary-source closure of arithmetic AG. The domain reduction is stated at ; the residual $(AG.det)$ is stated at .

42.8 GLUE BOUNDARY

- (i) The core compatibility (BCglue.core) is stated as a two-part conjecture: (a) pairwise $\mathcal{Z}_p \otimes \mathcal{Z}_q$ -equivariance of the diagonal pullback of the glued prismatic sheaf, and (b) colimit-to-global identification of the Bernstein centres (Chapter 42.1.3). The two parts are logically independent.
- (ii) Chapter 42.3.1 identifies the local Bernstein centre $\mathcal{Z}_p$ with the centre of the spectral action on the local coherent Springer sheaf via BZCHN + Bernstein 1984, at $\ell \neq p$ unconditionally and at $\ell = p$ conditional on (BCglue.local).
- (iii) Chapter 42.4.1 identifies the global Bernstein centre $\mathcal{Z}_{\text{glob}}$ with the Chenevier pseudocharacter ring R^{Ch} at the determinant-track level, conditional on arithmetic AG at the centre.
- (iv) Chapter 42.5.2 constructs the cross-prime pseudocharacter restriction $R^{\text{Ch}} \hookrightarrow R_p^{\text{Ch}} \otimes_{\mathbb{Z}} R_q^{\text{Ch}}$ as a flat monomorphism via the Nygaard convention of Bhatt–Lurie, conditional on prismatic flatness Π -flat.
- (v) Chapter 42.6.4 reduces (BCglue.core) to (Ch_I) + (AG.det): (Ch_I) is the Chenevier-level cross-prime restriction statement (Chapter 42.6.1); (AG.det) is the determinant-track projection of arithmetic AG (Chapter 42.6.3).
- (vi) The domain tightening of Chapter 42.6.4 shrinks the pairwise-sheaf content of (BCglue.core) from non-formal theorem to non-formal theorem (a 40–50% reduction), isolating the irreducible residual as (AG.det) (Chapter 42.7.1).
- (vii) (BCglue.core) is not closed unconditionally. The irreducible content (AG.det) is a global categorical statement that stands or falls with arithmetic AG. The residual for L_{ACD} is (Ch_I) (amenable at the Chenevier level) and arithmetic AG (global conjecture).

R_{analytic}: FORM, DIFFICULTY, AND THE RIEMANN HYPOTHESIS

Chapter 40 isolates three inputs for a sufficient argument toward the Riemann hypothesis: weight existence (W), the Weil explicit formula at Gaussian test functions (H), and the sharp moment asymptotic R_{analytic} , which asserts $\mu_{2n} \asymp C^n n!^{1/2}$ unconditionally for the symmetrised shifted-zero moment sequence of ξ . Remark 40.4.9 of Chapter 40 characterises R_{analytic} as weaker than RH in the implication direction and as “classical-analytic”. The statement has three independent axes: form, implication direction, and difficulty. The conclusions are (i) the form is classical-analytic; (ii) this reduction does not derive RH from R_{analytic} alone; (iii) the difficulty is not classical but rather density-of-zeros, a major open problem adjacent to RH in depth. The Bombieri–Lagarias upper bound does not imply Carleman divergence. After the $1/(2n)$ -th root and inversion, it yields the convergent Bertrand comparison $\sum 1/(n(\log n)^2)$. This is why Proposition 40.4.3 records the Bombieri–Lagarias construction as inconclusive for the determinacy estimate rather than as a divergence proof.

43.1 THE FOUR ASSERTIONS

PROPOSITION 43.1.1 (*form of R_{analytic}*). R_{analytic} is classical-analytic in form: a two-sided moment asymptotic $C_-^n n!^{1/2} \leq \mu_{2n} \leq C_+^n n!^{1/2}$ on a regularised positive measure associated to the symmetrised shifted zeros of ξ .

PROPOSITION 43.1.2 (*no direct implication to RH*). R_{analytic} is weaker than RH in the conclusion it states: an off-line-zero configuration with subleading $2n$ -th moment contribution is not excluded by the moment statement alone. Thus this reduction does not prove $R_{\text{analytic}} \Rightarrow \text{RH}$. (Chapter 40.4.9.)

PROPOSITION 43.1.3 (*Bombieri–Lagarias gives a finite Bertrand comparison*). . The upper bound

$$\mu_{2n} \leq (2\pi)^{-4n} (2n)! (\log n)^{4n} (1 + o(1))$$

implies only

$$\mu_{2n}^{-1/(2n)} \geq \frac{c}{n(\log n)^2} (1 + o(1))$$

for some $c > 0$. The comparison series $\sum_{n \geq 2} 1/(n(\log n)^2)$ converges. No Carleman divergence follows from this estimate.

PROPOSITION 43.1.4 (*form, implication, difficulty*). R_{analytic} has classical-analytic form and is not an RH proof by itself. Its difficulty is nevertheless RH-adjacent: removing the logarithmic inflation in the moment asymptotic requires density-of-zeros input beyond the Bombieri–Lagarias and Conrey estimates used here.

43.2 THE STIRLING SCALE IN THE CARLEMAN COMPUTATION

Chapter 43.1.3 follows from the Stirling asymptotic $(2n)!^{1/(2n)} \sim 2n/e$. The power grows linearly in n . The factor $(\log n)^{4n}$ in the upper bound on μ_{2n} becomes $(\log n)^{+2}$ after taking the $1/(2n)$ -th power of μ_{2n} , hence $(\log n)^{-2}$ in $\mu_{2n}^{-1/(2n)}$. The sign of the logarithmic exponent reverses under inversion, giving the convergent Bertrand series $\sum 1/(n(\log n)^2)$.

43.3 BOMBIERI–LAGARIAS BOUNDS λ_n , NOT μ_{2n}

Bombieri–Lagarias 1999 (Chapter 23.3.6) bounds the Li coefficient $\lambda_n = \sum_{\rho} [1 - (1 - 1/\rho)^n]$ by $\lambda_n \leq n(c_0 + \log n)/2$. This is a bound on a highly oscillatory unit-modulus phase sum: under RH, $|1 - 1/\rho| = 1$ for every non-trivial zero ρ , so λ_n grows sub-polynomially via phase cancellation. The moment sum $\mu_{2n} = \sum_{\rho} (1/4 + \gamma_{\rho}^2)^n$ is a monotone positive sum whose growth is governed by the density of zeros high in the critical strip via Riemann–von Mangoldt, with no cancellation. These two quantities are related only through $H\text{-}P^{\text{Ar}}$ -conditional spectral-trace identities (Chapter 40.2), not through elementary norm bounds; a direct transfer of the Bombieri–Lagarias bound to μ_{2n} requires a phase-cancellation-to-magnitude comparison that is not available. The upper bound on μ_{2n} in Chapter 40.4.3 comes from Riemann–von Mangoldt zero counting combined with RH-conditional saddle-point evaluation, not from the Bombieri–Lagarias Li-coefficient bound. Closing the $(\log n)^{4n} \rightarrow 1$ gap requires density-of-zeros improvements on the critical line beyond Conrey’s 40%, via the Selberg–Levinson–Conrey–Feng density theory.

43.4 POSITION OF R_{analytic} IN THE EQUIVALENCE CHAIN

Chapter 35 (Chapter 35.4.14) establishes the four-way equivalence

$$(VB^*c) \Leftrightarrow (\text{Det}) \Leftrightarrow (\text{HB}+) \Leftrightarrow (\text{BD}+)$$

under P_2, P_3 , and $H\text{-}P^{\text{Ar}}$, where (VB^*c) is the *converse* “zero-placement $\Rightarrow VB^*$ ”, not the main direction $VB^* \Rightarrow$ zero-placement. Determinacy is equivalent to the VB^* *converse*. Chapter 35 further records (Chapter 35.4.17, Chapter 35.4.16) that all four conditions require input beyond RH in this reduction: in particular, no implication $\text{RH} \Rightarrow \text{Det}$ is proved here, and $\text{Det} \Rightarrow \text{RH}$ requires $H\text{-}P^{\text{Ar}}$ plus the VB^* main direction (Chapter 23.1.3). Under $P_2, P_3, H\text{-}P^{\text{Ar}}$, one has $R_{\text{analytic}} \Rightarrow \text{Det} \Rightarrow (VB^*c)$, placing R_{analytic} as an input to the converse direction, not to the main direction and not to RH directly. Chapter 40’s characterisation of R_{analytic} as not RH-equivalent in the implication direction is therefore correct, but the reason is not that the statement is classical: it is that determinacy requires additional input in the implication hierarchy, as established by Chapter 35.

43.5 CARLEMAN SUM FROM THE BOMBIERI–LAGARIAS UPPER BOUND

Upper Bound on μ_{2n} via Riemann–von Mangoldt

Proposition 40.4.3 proves, via Riemann–von Mangoldt zero-counting and saddle-point evaluation,

$$\mu_{2n} \leq \frac{(2n)!}{(2\pi)^{4n}} \cdot (\log n)^{4n} \cdot (1 + o(1)).$$

The saddle-point derivation is standard and is not disputed here.

Stirling Evaluation

LEMMA 43.5.1 (*Stirling for $(2n)!$*). By Stirling's formula $n! = \sqrt{2\pi n} (n/e)^n (1 + O(1/n))$,

$$(2n)!^{1/(2n)} = \frac{2n}{e} \cdot (4\pi n)^{1/(4n)} \cdot (1 + O(1/n)).$$

Since $(4\pi n)^{1/(4n)} \rightarrow 1$ as $n \rightarrow \infty$, one has $(2n)!^{1/(2n)} \sim 2n/e$, i.e., the $(2n)!^{1/(2n)}$ power grows *linearly* in n , not as $\sqrt{n}$.

Proof. $(2n)! = \sqrt{4\pi n} (2n/e)^{2n} (1 + O(1/n))$ gives

$$\begin{aligned} (2n)!^{1/(2n)} &= [\sqrt{4\pi n} (2n/e)^{2n}]^{1/(2n)} \cdot (1 + O(1/n))^{1/(2n)} \\ &= (4\pi n)^{1/(4n)} \cdot (2n/e) \cdot (1 + O(1/n^2)). \end{aligned}$$

Since $(4\pi n)^{1/(4n)} = \exp(\frac{1}{4n} \log(4\pi n)) = 1 + \frac{\log(4\pi n)}{4n} + O(n^{-2}(\log n)^2) \rightarrow 1$, the claim follows. $\square$

PROPOSITION 43.5.2 (*Carleman-sum lower bound from the upper bound*). From Chapter 43.5,

$$\mu_{2n}^{1/(2n)} \leq \frac{1}{(2\pi)^2} \cdot (2n)!^{1/(2n)} \cdot (\log n)^2 \cdot (1 + o(1)) \sim \frac{2n}{e(2\pi)^2} \cdot (\log n)^2.$$

Hence

$$\mu_{2n}^{-1/(2n)} \geq \frac{e(2\pi)^2}{2n(\log n)^2} \cdot (1 - o(1)) = \frac{c_{BL}}{n(\log n)^2}$$

for a constant $c_{BL} > 0$. Summing,

$$\sum_{n \geq 2} \mu_{2n}^{-1/(2n)} \geq c_{BL} \cdot \sum_{n \geq 2} \frac{1}{n(\log n)^2}.$$

Proof. Direct substitution of Chapter 43.5.1 into the $1/(2n)$ -th power of the upper bound Chapter 43.5. The factor $(\log n)^{4n}$ inside the bound becomes $(\log n)^{+2}$ after the $1/(2n)$ -th-root, and when inverted for $\mu_{2n}^{-1/(2n)}$ becomes $(\log n)^{-2}$. The factor $(2\pi)^{-4n}$ becomes $(2\pi)^{-2}$ and when inverted becomes $(2\pi)^{+2}$. The factor $(2n)!^{1/(2n)} \sim 2n/e$ in the numerator of $\mu_{2n}^{1/(2n)}$ becomes $e/(2n)$ in $\mu_{2n}^{-1/(2n)}$. $\square$

COROLLARY 43.5.3 (*the BL Carleman construction is inconclusive, not divergent*). The lower bound on the Carleman sum from Chapter 43.5.2 is

$$\sum_{n \geq 2} \frac{1}{n(\log n)^2} < +\infty,$$

a classical Bertrand series which *converges*. Hence the lower bound on $\sum_n \mu_{2n}^{-1/(2n)}$ produced by the BL upper bound on μ_{2n} is finite. This does *not* prove the Carleman sum diverges; it is strictly inconclusive.

Proof. The Bertrand series $\sum_{n \geq 2} 1/(n(\log n)^\alpha)$ converges iff $\alpha > 1$, by the integral comparison $\int_2^\infty dx/(x(\log x)^\alpha) = (\alpha - 1)^{-1}(\log 2)^{-(\alpha-1)}$ for $\alpha > 1$. For $\alpha = 2$ this integral is finite, confirming convergence.

A lower bound on $\sum \mu_{2n}^{-1/(2n)}$ which is itself finite does not establish divergence of $\sum \mu_{2n}^{-1/(2n)}$. The Carleman sum may still diverge or converge; the BL upper bound on μ_{2n} with the $(\log n)^{4n}$ log-factor is simply too weak to decide. $\square$

Remark 43.5.4 (Stirling scale in Chapter 43.1.3). Chapter 43.1.3 asserts $\sum \mu_{2n}^{-1/(2n)} \geq c' \sum 2\pi(\log n)/\sqrt{2n}$. This expression has $\sqrt{n}$ in the denominator instead of n , and $(\log n)^{+1}$ in the numerator instead of $(\log n)^{-2}$. The resulting sum $\sum (\log n)/\sqrt{n}$ diverges, but the transcription from $\mu_{2n}^{-1/(2n)}$ via Stirling should produce $1/(n(\log n)^2)$ per Chapter 43.5.1 and power-rearrangement of $(\log n)^{4n}$, giving a convergent Bertrand series. The source is the approximation $(2n)!^{1/(2n)} \sim \sqrt{2n}$ instead of $\sim 2n/e$. Stirling's formula gives $n!^{1/n} \sim n/e$.

Remark 43.5.5 (Carleman divergence under the RH-heuristic asymptotic $\mu_{2n} \asymp C^n(n!)^2$). If one substitutes the RH-heuristic two-sided asymptotic $\mu_{2n} \asymp C^n(n!)^2$ with no $(\log n)^{4n}$ factor, as in Chapter 35.4.5, then $\mu_{2n}^{1/(2n)} \asymp C^{1/2} \cdot n/e$ giving $\mu_{2n}^{-1/(2n)} \asymp e/(C^{1/2}n)$ and the Carleman sum *does* diverge at the harmonic rate $\sum 1/n$. But this heuristic is *not* established unconditionally; the BL-plus-Conrey machinery has the $(\log n)^{4n}$ gap (Chapter 40.4.6) which puts the true sum into the log-convergent regime of Chapter 43.5.3. Hence Carleman's criterion from BL-plus-Conrey alone is inconclusive; the classical-heuristic form is what requires unconditional confirmation.

43.6 SHARPENING THE $(\log n)^{4n}$ GAP REQUIRES ZERO-DENSITY HYPOTHESES

The Bombieri–Lagarias Polylog Bound Does Not Sharpen to $\mu_{2n} \asymp n!^2$ by Classical Analysis

PROPOSITION 43.6.1 (*Bombieri–Lagarias bounds λ_n ; μ_{2n} is governed by Riemann–von Mangoldt*). . Bombieri–Lagarias Theorem 2 (Chapter 23.3.6) bounds $\lambda_n = \sum_{\rho} [1 - (1 - 1/\rho)^n]$, a heavily oscillatory unit-modulus-phase sum in ρ . Under RH, $|1 - 1/\rho| = 1$ for every non-trivial zero, so the Bombieri–Lagarias bound is a statement about phase-cancellation. The moment sum $\mu_{2n} = \sum_{\rho} (1/4 + \gamma_{\rho}^2)^n$ is a monotone positive sum (under RH) whose growth is governed by the density of zeros high in the critical line via Riemann–von Mangoldt. The two quantities are related through H-PA^r-conditional spectral-trace identities, not through elementary norm bounds.

Proof. The Bombieri–Lagarias bound gives $\lambda_n \geq -(n/2) \log n - O(n)$ (Chapter 23.3.6): polylog growth. A polylog bound on λ_n carries no information about μ_{2n} : λ_n grows via phase cancellation while μ_{2n} is a monotone positive sum with no cancellation. The identity $\text{tr}(T_n) = n\lambda_n$ holds only at the trace level under H-PA^r (Chapter 40.2). $\square$

PROPOSITION 43.6.2 (*sharpening the unconditional μ_{2n} upper bound from $(\log n)^{4n}$ to $(\log n)^{o(n)}$ requires zero-density improvements*). . The $(\log n)^{4n}$ factor in Chapter 43.5 comes from the Conrey-style fluctuation envelope in the RvM zero-counting: $N(T) \sim (T/2\pi) \log(T/2\pi)$ with second-order error $O(\log T)$, so that the integral representation of μ_{2n} picks up a $(\log T)^{\cdot}$ factor at the saddle cutoff $T \sim \sqrt{n} \log n$. Removing this factor to obtain $\mu_{2n} \leq C^n n!^2$ with *no logarithmic inflation* requires improving the zero-counting asymptotic from the unconditional $N(T) = (T/2\pi) \log(T/2\pi) - T/2\pi + O(\log T)$ to a form where the error term does not amplify by a factor of $\log T$ at each moment-integral saddle. This is a zero-density-hypothesis improvement, a classical analytic-number-theory problem that is currently open and is not known to follow from RH alone.

Proof. The RvM estimate $N(T) = (T/2\pi) \log(T/2\pi) - T/2\pi + O(\log T)$ (Edwards 1974 Thm. 6.7) has error $O(\log T)$. In the saddle-point evaluation $\mu_{2n} \approx \int_0^{\infty} t^{2n} (dN/dt) dt$, the saddle is at $t \sim \sqrt{n} \log n$. The error term $O(\log T)$ evaluated at the saddle contributes a factor of $O(\log(\sqrt{n} \log n)) \sim O(\log n)$ at the saddle, and the n -fold application (from the n -th moment) amplifies this to $O((\log n)^{4n})$.

in the final bound (the factor of 4 comes from the μ_{2n} being the $2n$ -th moment, with two such amplifications in the saddle-point evaluation and two from the t^2 factor).

The relevant fluctuation is $S(T) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iT)$, the oscillating part of $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8} + S(T) + O(1/T)$; unconditionally $S(T) = O(\log T)$ (von Mangoldt–Backlund), and this is precisely the error that the n -fold saddle amplifies. Reducing the amplified contribution to $(\log n)^{o(n)}$ cannot be achieved by sharpening this pointwise bound: $S(T)$ is provably unbounded (Selberg 1946), so no $O(1)$ error term for $N(T)$ exists. Removing the inflation instead requires zero-density control on the zeros high in the critical strip (a density-hypothesis-type input in the sense of Bombieri 1965) or a cancellation in the regularised moment integral that the crude magnitude bound discards. Either is a major open analytic-number-theory problem, adjacent to RH in depth. $\square$

Remark 43.6.3 (classical-analytic form, RH-adjacent difficulty). Chapter 43.6.2 establishes the key distinction: the *form* of R_{analytic} is classical-analytic (a moment asymptotic), but its *difficulty* is RH-adjacent. The gap between BL-Conrey’s polylog-inflated bound and the target $C^n n!^2$ bound is precisely the density-of-zeros hypothesis for ζ , which is a major open problem on the same level as RH. Density-of-zeros is weaker in conclusion than RH: a density-one statement controls the exceptional set but does not put every zero on the line. Thus R_{analytic} is not an RH proof in this reduction, and it is not known unconditionally either.

Three Independent Axes: Form, Implication, Difficulty

Chapter 43.1.4’s conflation of classical form with classical difficulty is resolved by separating three axes.

Form. The statement of R_{analytic} is the moment asymptotic $C_-^n n!^2 \leq \mu_{2n} \leq C_+^n n!^2$: an object in classical analysis, a moment sequence controlled by saddle-point asymptotics.

Implication direction. The present reduction does not prove $R_{\text{analytic}} \Rightarrow \text{RH}$: off-line zeros with subleading $2n$ -th moment contribution are not excluded by the moment statement alone. The converse $\text{RH} \Rightarrow R_{\text{analytic}}$ holds partially under regularisation: Riemann–von Mangoldt plus saddle-point gives $\mu_{2n} \asymp (2\pi)^{-4n} (2n)!$ up to logarithmic corrections, which by Stirling is $C^n n!^2$ up to the density-hypothesis refinements required to remove the $(\log n)^{4n}$ factor.

Difficulty. The remaining gap (Chapter 43.6.2) is the density-of-zeros hypothesis for ζ , a major open problem of analytic number theory at the level of the density-1 hypothesis, weaker than RH in its conclusion but adjacent to RH in depth.

	Form	Implication to RH	Difficulty
R_{analytic}	Classical-analytic (moment asymptotic)	Does not imply RH (weaker than RH in one direction)	RH-adjacent (density-of-zeros open problem)

Chapter 43.1.4 infers from the classical form that the difficulty is classical; this conflates the first and third axes. The implication axis (Chapter 43.1.2) is correct: R_{analytic} is strictly weaker than RH in the implication direction. The difficulty axis is RH-adjacent.

43.7 DETERMINACY, VB*, AND RH

The two propositions fix the two equivalences in the chain $\text{Det} \Leftrightarrow \text{VB}^* \text{ main direction} \Leftrightarrow \text{RH}$.

Step 1: Determinacy $\Leftrightarrow VB^$ (Which Direction?)*

PROPOSITION 43.7.1 (*Determinacy is equivalent to the VB^* converse*). Chapter 35.4.14 asserts

$$(VB^*c) \Leftrightarrow (\text{Det}) \Leftrightarrow (\text{HB}+) \Leftrightarrow (\text{BD}+)$$

under $P_2, P_3, H\text{-}P^{\text{Ar}}$, where (VB^*c) is “zero-placement $\Rightarrow VB^*$ ”. The main direction $VB^* \Rightarrow$ zero-placement is the content of Chapter 23.1.3, a separate conditional implication not equivalent to determinacy.

Proof. Chapter 35.4.15 records: “The forward direction $VB^* \Rightarrow \text{F-RH}$ is the separate content of Chapter 22.3.10, which is a conditional theorem under $P_2, P_3, H\text{-}P^{\text{Ar}}$, and a positive Bost–Connes regularised trace. The four-way equivalence of the converse does not close the forward implication.” Determinacy is equivalent to the converse direction. $\square$

Step 2: Determinacy vs RH

PROPOSITION 43.7.2 (*Determinacy needs input beyond RH in this reduction*). Chapter 35.4.16 asserts that each of VB^* -converse, determinacy, and Baéz–Duarte sharp decay carries input beyond RH in this reduction. The implication $\text{Det} \Rightarrow \text{RH}$ is used only with $H\text{-}P^{\text{Ar}}$ (Chapter 35.5.1), and no implication $\text{RH} \Rightarrow \text{Det}$ is proved here.

Proof. Chapter 35.4.17: “No structural property of the zero distribution of ξ independent of RH is known to force determinacy. Hence the VB^* converse requires an additional arithmetic input beyond RH.” Hence this chapter does not derive Det from RH. The reverse implication $\text{Det} \Rightarrow \text{RH}$ requires $H\text{-}P^{\text{Ar}}$ plus the VB^* main direction (Chapter 23.1.3); the two inputs are kept separate without auxiliary hypotheses. $\square$

The Implication Diagram

Remark 43.7.3 (*Position of R_{analytic} in the implication hierarchy*). Under $P_2, P_3, H\text{-}P^{\text{Ar}}$:

$$R_{\text{analytic}} \Rightarrow (\text{Carleman criterion}) \Rightarrow \text{Det} \iff (VB^* \text{ converse})$$

Independently, as a conditional theorem (Chapter 23.1.3):

$$VB^* \text{ main direction} \Rightarrow \text{RH}.$$

Combined (Chapter 35.5.1):

$$H\text{-}P^{\text{Ar}} + R_{\text{analytic}} + VB^* \text{ main direction} \Rightarrow VB^* \iff \text{RH} + \text{Det}.$$

Remark 43.7.4 (*R_{analytic} controls the converse*). In the implication diagram, R_{analytic} contributes to the VB^* converse direction ($\text{RH} \Rightarrow VB^*$), by forcing Hamburger determinacy which together with $H\text{-}P^{\text{Ar}}$ and RH gives VB^* . R_{analytic} does not by itself imply the VB^* main direction nor RH. The main direction requires VB^* proper or some alternative construction to the Li positivity, which is not supplied by R_{analytic} .

43.8 CONCLUSION

THEOREM 43.8.1 (*four assertions on R_{analytic}*). . The four assertions Chapter 43.1.1–Chapter 43.1.4 have the following mathematical form.

- (1) *Form* (Chapter 43.1.1): R_{analytic} has classical-analytic form, a two-sided moment asymptotic.
- (2) *Implication* (Chapter 43.1.2): R_{analytic} is not an RH proof by itself in this reduction. The converse $\text{RH} \Rightarrow R_{\text{analytic}}$ also requires density-of-zeros refinements beyond RH alone.
- (3) *Bombieri–Lagarias scale* (Chapter 43.1.3): the Bombieri–Lagarias–Conrey construction gives a Carleman sum lower bound $\sum 1/(n(\log n)^2)$, a Bertrand series that converges, hence the Bombieri–Lagarias construction is inconclusive, not divergent.
- (4) *Difficulty* (Chapter 43.1.4): “classical-analytic, not RH-equivalent” is correct on the form and implication axes. Its difficulty is RH-adjacent, because the logarithmic inflation can only be removed here by density-of-zeros input.

Proof. (1) is Chapter 43.6: the statement is a moment asymptotic.

(2) is Chapter 40.4.9 combined with the off-line-zeros robustness argument.

(3) is Chapter 43.5.2 and Chapter 43.5.3: the Stirling identity $(2n)!^{1/(2n)} \sim 2n/e$ gives the convergent Bertrand series $\sum 1/(n(\log n)^2)$.

(4) is Chapter 43.6.3 and Chapter 43.6.2: removing the logarithmic gap requires density-of-zeros improvements which, while weaker than RH in conclusion, constitute a major open problem of analytic number theory. $\square$

COROLLARY 43.8.2 (R_{analytic} is not an RH proof by itself). . The equivalences in the chain $\text{Det} \Leftrightarrow \text{VB}^*$ main direction $\Leftrightarrow \text{RH}$ are:

- (i) Determinacy is equivalent to the VB^* *converse*, not the main direction (Chapter 43.7.1).
- (ii) Determinacy needs input beyond RH in this reduction (Chapter 43.7.2).

Hence R_{analytic} is not, by itself, an RH proof in the present implication chain.

Proof. Chapter 23.1.3 and Chapter 35.4.14 are unambiguous on both the direction of the equivalence and the need for determinacy input beyond RH in this reduction. $\square$

43.9 THE SHARP REDUCTION OF R_{analytic}

THEOREM 43.9.1 (*Sharp reduction of R_{analytic}*). . The term R_{analytic} is the demand for an unconditional two-sided moment asymptotic $\mu_{2n} \asymp C^n n!^2$. It has the following properties.

- (i) *Form*: classical-analytic; R_{analytic} is a two-sided moment asymptotic.
- (ii) *Position in the implication hierarchy*: under $P_2, P_3, H\text{-P}^{\text{Ar}}, R_{\text{analytic}} \Rightarrow \text{Det} \Leftrightarrow \text{VB}^*$ converse; it is not an RH proof directly.
- (iii) *Implication to RH*: the present reduction does not prove $R_{\text{analytic}} \Rightarrow \text{RH}$.
- (iv) *Difficulty*: removing the unconditional $(\log n)^{4n}$ gap requires density-of-zeros hypotheses for ζ , an open analytic-number-theory problem of RH-adjacent difficulty.
- (v) *Unconditional Carleman construction via Bombieri–Lagarias and Conrey*: inconclusive; the correct Bertrand-series lower bound $\sum 1/(n(\log n)^2)$ converges (Chapter 43.5.3).

R_{analytic} is classical in form, not proved here, insufficient by itself to prove RH in this reduction, and RH-adjacent in difficulty. The summary of Chapter 40 (“classical-analytic, not RH-equivalent”) is accurate on the form and implication axes and inaccurate on the difficulty axis.

Proof. (1) is Chapter 43.6. (2) is Chapter 43.7.3. (3) is Chapter 40.4.9. (4) is Chapter 43.6.2. (5) is Chapter 43.5.3. $\square$

Remark 43.9.2 (precise statement). R_{analytic} asks for an unconditional two-sided moment asymptotic at the $n!^2$ rate. Its form is classical-analytic. Its implication relative to RH is one-directional in this reduction. Its difficulty is RH-adjacent: removing the unconditional Bombieri–Lagarias–Conrey gap requires zero-density-hypothesis-level improvements. Carleman’s criterion via the Bombieri–Lagarias–Conrey bounds alone is inconclusive; a genuine Carleman divergence requires either density-of-zeros improvements or the unconditional two-sided asymptotic $\mu_{2n} \asymp C^n n!^2$ without logarithmic inflation. The remaining work is zero-density analytic number theory.

Remark 43.9.3 (domain of the statement). The three terms $W + H + R_{\text{analytic}}$ of Chapter 40.5.2 remain independent. The Stirling computation affects only the Carleman-divergence inference. The weight selection (W) and Weil formula at Gaussian test functions (H) components are unchanged.

43.10 CONSEQUENCES

- (i) The characterisation of R_{analytic} as “classical-analytic, not RH-equivalent” (Chapter 43.8.1) is accurate on the form axis and the implication axis. The form is classical-analytic; the implication does not by itself prove RH; but the difficulty is RH-adjacent via the density-of-zeros problem.
- (ii) The Bombieri–Lagarias and Conrey estimates give Chapter 43.5.3. The lower bound on the Carleman sum is the Bertrand series $\sum 1/(n(\log n)^2)$, which converges. The Bombieri–Lagarias construction is inconclusive.
- (iii) The form of the chain $\text{Det} \Leftrightarrow \text{VB}^*$ main direction $\Leftrightarrow \text{RH}$ is given by Chapter 43.8.2: determinacy is equivalent to the VB^* converse (Chapter 35.4.14), and determinacy needs input beyond RH in this reduction.
- (iv) Chapter 43.9.1 separates three axes: form (classical), implication direction (not an RH proof by itself), and difficulty (RH-adjacent via density-of-zeros). R_{analytic} is the single analytic input of the sufficient path; proving it is a genuine open problem of analytic number theory at the zero-density level.
- (v) The three terms $W + H + R_{\text{analytic}}$ of Chapter 40.5.2 remain the sharp reduction of Chapter 22.
- (vi) Proving R_{analytic} requires either (a) a density-of-zeros improvement pushing the Conrey lower bound from 40% of zeros on the line to the density-1 hypothesis with sub-logarithmic error, or (b) an alternative positivity or measure-regularisation construction producing $\mu_{2n} \asymp C^n n!^2$ directly. Either construction is a substantive analytic-number-theory theorem. The Vol IV sufficient argument for RH rests on three open problems of comparable difficulty (W, H, R_{analytic}).

The three inputs are genuine open problems of substantive analytic-number-theory difficulty. The term R_{analytic} is specifically a density-of-zeros statement adjacent to RH.

ARITHMETIC POSITSIELSKI: PRIMITIVE ZETA-CHARACTER THEOREM

Of the five open Positselski sub-items named in Chapter 24.1.6 and in the Positselski row of Chapter 38.1), four concern categorical machinery (continuous duals, derived completion, faithful lifting, Koszul-dual Bost–Connes) and one concerns character arithmetic. The character-arithmetic sub-item is the primitive zeta-character theorem, which asserts that the character of the primary subspace $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ is the meromorphic Tate character $\Lambda_{\zeta}(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ under the HY-zeta grading, and that this character is Koszul-self-dual in the Positselski pairing sense. The proof uses Tate’s 1950 Mellin–theta symmetry on the archimedean sector and Julia’s 1990 primon-gas identification of the finite one-mode tower with ζ . Bost–Connes 1995 supplies an independent KMS comparison for the same finite character; it is not used as the still-open algebra presentation of the cobar object. The primary subspace is the restricted tensor product of the finite one-mode polynomial towers, with their local units, and the archimedean Gaussian line. It is the minimal adelic object whose Mellin character is Λ_{ζ} . The determinant-trace realisation of the non-trivial zeros belongs to the $H\text{-P}^{\text{Ar}}$ and VB^ operators, not to the character statement. The full bar–cobar pairing still requires the other four Positselski sub-items.*

44.1 STATEMENT

THEOREM 44.1.1 (primitive zeta-character theorem). . Let $\mathcal{V}_{\text{Heis}}^{\text{prim}} = \bigotimes_p' \mathcal{H}^{(p)} \otimes \mathcal{H}^{(\infty)}$ be the adelic arithmetic Heisenberg Fock and let $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ be its *primary subspace* as defined in Chapter 44.3.6. Endow both with the HY-zeta grading of Chapter 6.6.2(ii). Then the graded character of the primary subspace is the completed zeta function:

$$\text{char}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})(s) = \Gamma_{\mathbb{R}}(s)\zeta(s) = \Lambda_{\zeta}(s), \quad \Gamma_{\mathbb{R}}(s) := \pi^{-s/2}\Gamma(s/2).$$

The character of the Koszul dual, under the arithmetic bar coalgebra $C = B^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})$ and cobar algebra $\mathcal{A} = C^{\vee}$, satisfies:

$$\text{char}_C(s) = \Lambda_{\zeta}(s), \quad \text{char}_{\mathcal{A}^!}(s) = \Lambda_{\zeta}(1-s), \quad \text{char}_C(s) = \text{char}_{\mathcal{A}^!}(1-s).$$

In particular, the functional equation $\Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s)$ (Tate 1950) gives $\text{char}_C(s) = \text{char}_{\mathcal{A}^!}(s)$ as an identity of meromorphic functions. The character equality (44.1.1) holds unconditionally as an identity of meromorphic functions with poles at $s = 0, 1$; the character identity (44.1.1) holds unconditionally at the level of characters (Tate 1950 Mellin–theta on the bar side; Chapter 24.5.1 on the cobar side), and lifts to an identity of Koszul-paired objects *after* the other four open sub-items of Chapter 24.1.6 are supplied.

Remark 44.1.2 (character-level range). Chapter 24.1.6 names five open Positselski sub-items. The present theorem proves the fourth, the *primitive zeta-character theorem* (Chapter 24.1.6(4)). The result is at

the level of characters: a genuine theorem about characters of graded adelic Heisenberg modules. The three categorical sub-items (continuous duals; derived completion; faithful lifting) and the algebraic identification sub-item (Koszul-dual Bost–Connes) remain conditional. The full arithmetic Positselski biconditional of Chapter 24.1.5 remains conditional on those four residuals. The arith. Positselski input in Chapter 38.3.1 reduces from five open sub-items to four.

Remark 44.1.3 (why the primary subspace is the right object). The full arithmetic Heisenberg Fock $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ carries character $\prod_p \prod_n (1 - p^{-ns})^{-1} = \prod_{n \geq 1} \zeta(ns)$ under HY-zeta grading, not $\Lambda_\zeta(s)$. The factor $\prod_{n \geq 2} \zeta(ns)$ beyond $\zeta(s)$ encodes the multi-oscillator Fock sector and produces higher-weight Dirichlet series with their own trivial zeros; it is the Fock-symmetric completion of $\zeta(s)$, not $\zeta(s)$ itself. The archimedean Fock $\mathcal{H}^{(\infty)}$ contributes a product $\prod_{n \geq 1} (1 - e^{-n/T})^{-1}$ under temperature T conjugate to conformal weight; its Mellin transform at criticality is $\Gamma(s)\zeta(s)$, not $\Gamma_{\mathbb{R}}(s)\zeta(s)$.

The primary subspace is defined to extract (i) the single-oscillator sector at each prime (which gives $\zeta(s)$ not $\prod_{n \geq 1} \zeta(ns)$), and (ii) the Gaussian-primary archimedean sector (which gives $\Gamma_{\mathbb{R}}(s)$ not $\Gamma(s)$). The tensor gives $\Lambda_\zeta(s)$.

44.2 FOUR INDEPENDENT COMPUTATIONS

The theorem is established through four independent arguments, each drawing from a distinct source.

The Finite and Archimedean Primary Object $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$

Definition. The notation $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ has been used in Chapter 23.1.1, Chapter 22.3.5, and Chapter 22.3.8 as “the primary subspace of the arithmetic Heisenberg Fock”. Here it is the adelic object whose finite local factors have character $(1 - p^{-s})^{-1}$ and whose archimedean factor is the Gaussian Tate line.

Finite places. At a finite prime p , the primary factor is the polynomial tower generated by the single line $\mathbb{C} \cdot a_{-1}^{(p)} |0\rangle_p$. Its basis is $\{e_{n,p}\}_{n \geq 0}$, where $e_{0,p} = |0\rangle_p$ and $L_o^{\text{Ar}} e_{n,p} = n \log p e_{n,p}$. The finite vacuum is the local unit of the restricted tensor product. It is not a trivial-zero state.

Archimedean place. At ∞ , the primary factor is the Gaussian line $\mathbb{C} \cdot e^{-\pi x^2}$. Its Mellin pairing over $\mathbb{R}^\times$ is $\Gamma_{\mathbb{R}}(s)$. The Hermite tower above the Gaussian is not part of the primary line; that tower is the archimedean trivial-zero sector.

Character. $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ is the restricted tensor product of these factors. Its finite character is $\prod_p (1 - p^{-s})^{-1} = \zeta(s)$; its archimedean character is $\Gamma_{\mathbb{R}}(s)$.

Mellin–Theta Character Computation

Computation. Compute the per-place partition function separately and take a restricted tensor product. At each finite p , the primary contribution is $(1 - p^{-s})^{-1}$ (single oscillator). Product: $\zeta(s)$. At ∞ , the Gaussian-primary contribution is $\pi^{-s/2} \Gamma(s/2) = \Gamma_{\mathbb{R}}(s)$ (Gaussian–theta Mellin transform). Tensor: $\Gamma_{\mathbb{R}}(s)\zeta(s) = \Lambda_\zeta(s)$.

Archimedean factor. The archimedean factor is exactly $\Gamma_{\mathbb{R}}(s)$, not $\Gamma(s)$. Tate 1950 §2.5: the archimedean local L -factor for the trivial Dirichlet character on $\mathbb{Q}_\infty = \mathbb{R}$ is $L_\infty(s) = \pi^{-s/2} \Gamma(s/2) = \Gamma_{\mathbb{R}}(s)$; the archimedean theta vector is $e^{-\pi x^2}$, which is Gaussian, and its Mellin transform against $|x|^{s-1}$ is exactly

$\Gamma_{\mathbb{R}}(s)/2$. The factor of 2 is the Tate-symmetric handling of sign of x under the archimedean involution $x \mapsto -x$; it matches the archimedean-primary sector if the Gaussian primary state is $e^{-\pi x^2}$ rather than $xe^{-\pi x^2}$ (the latter pairs with odd-parity completions at ∞ and enters at weight $s/2 + 1/2$).

Result. Primary archimedean Gaussian $\rightarrow \Gamma_{\mathbb{R}}(s)$ exactly. Finite-place primary single-oscillator $\rightarrow \prod_p(1 - p^{-s})^{-1} = \zeta(s)$. Adelic tensor: $\Lambda_{\zeta}(s)$, by Tate 1950 Mellin–theta (Chapter 21.6) and Julia 1990 primon gas.

Koszul-Dual Character

Finite part. The character of the graded dual finite tower is obtained by the Tate involution on the Mellin variable: $\zeta(s)$ is replaced by $\zeta(1-s)$. Bost–Connes 1995 gives the same scalar as an independent comparison, since its KMS $_{\beta}$ partition function is $Z_{\text{BC}}(\beta) = \zeta(\beta)$ for $\text{Re}(\beta) > 1$. The algebraic identification of the cobar algebra with the Bost–Connes algebra remains the separate Positselski sub-item (5).

Archimedean factor. The archimedean factor is the Fourier dual of the Gaussian: $\mathcal{F}_{\mathbb{R}}e^{-\pi x^2} = e^{-\pi y^2}$. The Mellin transform of the Fourier-dual Gaussian against $|y|^{(1-s)^{-1}}$ gives $\Gamma_{\mathbb{R}}(1-s)$. Tensor with Bost–Connes finite-place character: $\Gamma_{\mathbb{R}}(1-s)\zeta(1-s) = \Lambda_{\zeta}(1-s)$.

Result. $\text{char}_{A^!}(s) = \Lambda_{\zeta}(1-s)$, where the substitution $s \mapsto 1-s$ is Tate’s archimedean Fourier duality together with the finite Mellin involution.

Match Under Koszul Self-Duality

Characters. The two characters computed separately ($\text{char}_C(s) = \Lambda_{\zeta}(s)$ and $\text{char}_{A^!}(s) = \Lambda_{\zeta}(1-s)$) match under Tate’s functional equation $\Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s)$.

Proof. Apply Tate 1950 adelic Poisson summation: $\mathcal{F}_{\mathbb{A}}\Theta = \Theta$ for the adelic theta distribution, whose Mellin transform is $\Lambda_{\zeta}(s)$. Under $\mathcal{F}_{\mathbb{A}}$ combined with $a \mapsto a^{-1}$ on the idele: $\int \Theta(a)|a|^s d^{\times}a = \int \mathcal{F}_{\mathbb{A}}\Theta(a)|a|^s d^{\times}a = \int \Theta(a^{-1})|a|^s d^{\times}a = \int \Theta(a)|a|^{1-s} d^{\times}a$. Hence $\Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s)$ (Riemann 1859; Tate 1950).

Result. Tate’s adelic Poisson summation is an unconditional theorem. The matching is immediate once both sides have been identified.

Identity. $\text{char}_C(s) = \text{char}_{A^!}(1-s)$, which via Tate’s functional equation $\Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s)$ becomes $\text{char}_C(s) = \text{char}_{A^!}(s)$ (after substitution of one variable), by Tate 1950.

Computation	Content	Result
1	Primary subspace defined $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$	finite units + Gaussian line
2	Character on primary = $\Lambda_{\zeta}(s)$ via Mellin–theta	holds, Tate 1950
3	Koszul-dual character = $\Lambda_{\zeta}(1-s)$	holds at character level
4	Match under $s \mapsto 1-s$	holds, Tate adelic Poisson

TABLE 44.1: Four independent computations for the primitive zeta-character theorem. Tate 1950 supplies the archimedean Mellin transform, Julia 1990 supplies the finite Euler product, Bost–Connes 1995 supplies an operator-algebraic comparison for the same finite scalar, and Positselski arXiv:0905.2621 supplies the classical character pairing.

Table

44.3 PRIMARY SUBSPACE $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$

Local Primary Subspaces

Definition 44.3.1 (per-prime primary subspace). For each prime p , let $\mathcal{H}^{(p)}$ be the per-prime Heisenberg Fock with generators $\{a_{-n}^{(p)}, b_{-n}^{(p)}\}_{n \geq 1}$ of Chapter 3.3.1. The *per-prime primary subspace* $\mathcal{H}_{\text{prim}}^{(p)} \subset \mathcal{H}^{(p)}$ is the polynomial tower on the single oscillator $a_{-1}^{(p)}$:

$$\mathcal{H}_{\text{prim}}^{(p)} := \bigoplus_{n \geq 0} \mathbb{C} \cdot (a_{-1}^{(p)})^n |0\rangle_p = \bigoplus_{n \geq 0} \mathbb{C} \cdot e_{n,p},$$

where $e_{0,p} = |0\rangle_p$ and $e_{n,p} := (a_{-1}^{(p)})^n |0\rangle_p$ has L_{o}^{Ar} -weight $n \log p$. The finite vacuum is the spherical unit of the restricted tensor product. It supplies the 1 in $(1 - p^{-s})^{-1}$. The trivial-zero sector is not this finite unit; it is the archimedean Hermite tower excluded in Chapter 44.3.4. Oscillator modes $a_{-m}^{(p)}$ with $m \geq 2$, and products involving them, are outside the primary factor.

Remark 44.3.2 (why only one finite oscillator remains). The per-prime Fock character including all multi-oscillator states is

$$\text{char}(\mathcal{H}^{(p)})(s) = \prod_{n \geq 1} (1 - p^{-ns})^{-1},$$

which is the primon gas local factor and carries contributions from p^{-ms} for every oscillator mode $m \geq 1$. The local primary keeps the $m = 1$ mode and all its occupation numbers. Its basis is $(a_{-1}^{(p)})^n |0\rangle_p$, $n \geq 0$, and its character is $(1 - p^{-s})^{-1}$, giving $\prod_p (1 - p^{-s})^{-1} = \zeta(s)$. The factor $\prod_{m \geq 2} (1 - p^{-ms})^{-1}$ not in the primary factor is the finite Fock-symmetric completion. The trivial zeros of the completed zeta function come from the archimedean gamma factor and the functional equation, not from deleting the finite vacuum.

PROPOSITION 44.3.3 (Bost–Connes ergodic decomposition on $\mathcal{H}_{\text{prim}}^{(p)}$). The Bost–Connes $\mathbb{N}^{\times}$ -action on $\mathcal{H}_{\text{prim}}^{(p)}$ via $\mu_{p^r} \cdot e_{m,p} = e_{m+r,p}$ is generated by the local unit $e_{0,p}$, with KMS $_{\beta}$ states at inverse temperature $\beta = s$ given by $\varphi_s(e_{n,p}) = p^{-ns}$ and partition function $Z_{\text{BC}}^{(p)}(s) = (1 - p^{-s})^{-1}$ per prime.

Proof. The local semigroup $p^{\mathbb{N}} \subset \mathbb{N}^{\times}$ acts on the single-mode basis by $\mu_{p^r} e_{m,p} = e_{m+r,p}$. Thus the whole local tower is the orbit of $e_{0,p}$. This is the local primon-gas Fock space with one oscillator of energy $\log p$.

KMS condition: the time evolution σ_t of the Bost–Connes system acts on Hecke operators by $\sigma_t(\mu_n) = n^{it}\mu_n$. A state φ on the Hecke algebra $\mathbb{A}^{\text{BC}}$ is a KMS_β state if $\varphi(AB) = \varphi(B\sigma_{i\beta}(A))$ for analytic elements. The explicit KMS states at inverse temperature $\beta > 1$ are $\varphi_\beta(\mu_n\mu_m^*) = \delta_{n,m}n^{-\beta}/\zeta(\beta)$ on the full algebra; see Bost–Connes 1995 Selecta Math. 1 Prop. 27 for the complete calculation. On the primary subspace $\mathcal{H}_{\text{prim}}^{(p)}$, the restriction of this KMS state at $\beta = s$ assigns weight $\varphi_s(e_{n,p}) = p^{-ns}$ to each basis vector $e_{n,p}$ (this is the diagonal component of the Bost–Connes KMS state restricted to the per-prime factor at inverse temperature $\beta = s$, with $n^{-\beta}$ specialised to p^{-ns} for the prime- p sector: Bost–Connes 1995 Prop. 27 specialised, not re-derived).

The per-prime primary partition function is therefore

$$Z_{\text{BC}}^{(p)}(s) = \sum_{n \geq 0} p^{-ns} = (1 - p^{-s})^{-1}.$$

The $n = 0$ term is the spherical unit $e_{0,p}$. Keeping it is forced both by the restricted tensor product and by the Euler factor. $\square$

Archimedean Primary Subspace

Definition 44.3.4 (archimedean primary Heisenberg). . The archimedean Heisenberg factor $\mathcal{H}^{(\infty)}$ is the Fock representation of the rank-1 Heisenberg vertex algebra over $\mathbb{R}$, with creation operators $\{\alpha_{-n}\}_{n \geq 1}$ and vacuum $|0\rangle_\infty$. The *archimedean primary subspace* $\mathcal{H}_{\text{prim}}^{(\infty)} \subset \mathcal{H}^{(\infty)}$ is the subspace spanned by the Gaussian primary vector $|\text{Gauss}\rangle = e^{-\pi x^2} \cdot |0\rangle_\infty$ under the coordinate realization of $\mathcal{H}^{(\infty)}$ as $L^2(\mathbb{R})$ (Segal 1981 Prop. 3.2; Frenkel–Ben-Zvi 2004 §3.5): concretely,

$$\mathcal{H}_{\text{prim}}^{(\infty)} := \mathbb{C} \cdot |\text{Gauss}\rangle = \mathbb{C} \cdot e^{-\pi x^2} |0\rangle_\infty.$$

Descendant states $(\alpha_{-n})^k |\text{Gauss}\rangle$ for $n \geq 1, k \geq 1$ are excluded from the archimedean primary subspace; these correspond to the Hermite-polynomial tower above the Gaussian and carry the trivial-zero sector of $\xi_{\mathbb{R}}$.

PROPOSITION 44.3.5 (archimedean primary character). . Under the HY-zeta grading via conformal weight times $\log p_\infty$ (with archimedean “prime” frequency $\log p_\infty := \text{archimedean analogue of } 1 \text{ in the Tate presentation; equivalently, } L_0^{(\infty)} \text{ has continuous spectrum } [0, \infty)$), the character of $\mathcal{H}_{\text{prim}}^{(\infty)}$ is

$$\text{char}(\mathcal{H}_{\text{prim}}^{(\infty)})(s) = \int_0^\infty e^{-\pi x^2} \cdot x^{s-1} dx = \frac{1}{2} \Gamma_{\mathbb{R}}(s) = \frac{1}{2} \pi^{-s/2} \Gamma(s/2).$$

The factor of 2 accommodates the archimedean sign involution $x \mapsto -x$: extending the integral over $\mathbb{R}$ (not just $(0, \infty)$) gives $\Gamma_{\mathbb{R}}(s)$ directly, which is the Tate normalisation.

Proof. The Gaussian primary state $|\text{Gauss}\rangle = e^{-\pi x^2}$ under Mellin pairing against x^{s-1} on $(0, \infty)$:

$$\int_0^\infty e^{-\pi x^2} \cdot x^{s-1} dx = \frac{1}{2} \pi^{-s/2} \Gamma(s/2),$$

by substitution $u = \pi x^2, du = 2\pi x dx$:

$$\frac{1}{2} \int_0^\infty e^{-u} \cdot (u/\pi)^{(s-1)/2} \cdot (u/\pi)^{-1/2} du / \pi^{1/2} = \frac{1}{2} \pi^{-s/2} \int_0^\infty e^{-u} u^{s/2-1} du = \frac{1}{2} \pi^{-s/2} \Gamma(s/2).$$

Extension to $\mathbb{R}$ doubles this: the full Mellin–theta presentation is $\int_{-\infty}^\infty e^{-\pi x^2} |x|^{s-1} dx = \pi^{-s/2} \Gamma(s/2) = \Gamma_{\mathbb{R}}(s)$ (Tate 1950 §2.5; Cassels–Fröhlich 1967 §13.3). $\square$

Adelic Primary Subspace

Definition 44.3.6 (*primary subspace* $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$). The *primary subspace* of the adelic arithmetic Heisenberg Fock is the restricted tensor product

$$\mathcal{V}_{\text{Heis,prim}}^{\text{prim}} := \mathcal{H}_{\text{prim}}^{(\infty)} \otimes_{\mathbb{C}} \bigotimes'_{p \text{ prime}} \mathcal{H}_{\text{prim}}^{(p)}$$

where:

- At ∞ , $\mathcal{H}_{\text{prim}}^{(\infty)}$ is the Gaussian primary line (Chapter 44.3.4).
- At each finite p , $\mathcal{H}_{\text{prim}}^{(p)}$ is the single-oscillator primary space (Chapter 44.3.1).
- The restricted tensor product is taken with respect to the spherical vector $e_{0,p} := |0\rangle_p$ at almost all primes in the primary factor (Tate 1950 §2.3; Weil 1967 Ch. VII §1). Thus only finitely many finite primes carry $e_{n,p}$ with $n > 0$ in an elementary tensor.

The underlying Λ -module structure at each finite p is free of rank one (the Iwasawa μ -invariant is 0 on the single-oscillator component, Chapter 24.3.12).

Remark 44.3.7 (*Virasoro-bootstrap consistency*). Chapter 23.1.1 uses $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ as the domain of the descendant operator $T_n = (L_o^{\text{Ar}})^n - (L_o^{\text{Ar}} - \text{id})^n$. The present definition makes this operator well-defined: on $\mathcal{H}_{\text{prim}}^{(p)}$, L_o^{Ar} acts with eigenvalue $n \log p$ on $e_{n,p}$; on $\mathcal{H}_{\text{prim}}^{(\infty)} = \mathbb{C} \cdot |\text{Gauss}\rangle$, L_o^{Ar} acts as the Tate-archimedean $s/2$ scaling (distributional spectrum over $\mathbb{R}$, regularised through Mellin). The common domain for L_o^{Ar} and its powers is the algebraic tensor product of $\bigoplus_{n \geq 0} \mathbb{C} e_{n,p}$ with the Schwartz-archimedean Gaussian sector; spectral-calculus extension is given in Chapter 23.1.1 under $H\text{-P}^{\text{Ar}}$.

Remark 44.3.8 (*determinant rank and Virasoro reconciliation with Chapter 22.3.5*). On the primary subspace $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$, the finite conformal modes $m \geq -1$ act on finite-support vectors. The negative modes $m \leq -2$ do not form global sums on the restricted vacuum, so a Virasoro action is not produced by the character theorem. The value proved in Chapter 14.4.1 is the determinant rank residue $r_{\zeta} = 2$, obtained from the pole of $\zeta(s)$ at $s = 1$. The conjectural primary-Virasoro bracket requires a separate operator construction whose central charge matches this residue.

44.4 CHARACTER ON PRIMARY VIA MELLIN-THETA

Finite-Place Primary Character

THEOREM 44.4.1 (*finite-place primary character*). The character of $\bigotimes'_p \mathcal{H}_{\text{prim}}^{(p)}$ under the HY-zeta grading is the Dirichlet series $\zeta(s)$, convergent absolutely for $\text{Re}(s) > 1$ and admitting meromorphic continuation to $\mathbb{C}$ with a simple pole at $s = 1$ and no other poles:

$$\text{char} \left(\bigotimes'_{p \text{ prime}} \mathcal{H}_{\text{prim}}^{(p)} \right) (s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1} = \zeta(s), \quad \text{Re}(s) > 1.$$

Proof. Per prime: the Bost–Connes KMS partition function on $\mathcal{H}_{\text{prim}}^{(p)}$ at inverse temperature $\beta = s$ is $Z_{\text{BC}}^{(p)}(s) = (1 - p^{-s})^{-1}$ (Chapter 44.3.3), where the Euler factor $(1 - p^{-s})^{-1}$ counts the local unit $e_{0,p}$ and the single-mode basis $\{e_{n,p}\}_{n \geq 1}$ with weights p^{-ns} . The restricted tensor product of per-prime KMS partition functions produces $\prod_p (1 - p^{-s})^{-1}$, the Euler product for $\zeta(s)$ (Riemann 1859; Cassels–Fröhlich 1967 §13.2).

Convergence: the product $\prod_p (1 - p^{-s})^{-1}$ converges absolutely for $\operatorname{Re}(s) > 1$ by the bound $\log(1 - p^{-s})^{-1} = \sum_{k \geq 1} p^{-ks}/k \leq 2p^{-s}$ for $\operatorname{Re}(s) \geq 1$, and $\sum_p p^{-\sigma}$ converges for $\sigma = \operatorname{Re}(s) > 1$ (this is the prime zeta function $P(\sigma)$, finite for $\sigma > 1$; Landau 1909). Meromorphic continuation: the identity $\zeta(s) = (1 - 2^{1-s})^{-1} \sum_{n \geq 1} (-1)^{n-1} n^{-s}$ (alternating zeta) gives continuation to $\operatorname{Re}(s) > 0$, then functional equation gives continuation to $\mathbb{C}$. The simple pole at $s = 1$ is a standard computation (Chapter 14.4.1: $\operatorname{Res}_{s=1} \zeta(s) = 1$).

Julia's 1990 primon gas identification (Chapter 21.5.1 Step 1): the per-prime single-oscillator Fock partition function equals $(1 - p^{-\beta})^{-1}$ at inverse temperature β , and the primon gas total partition is $\zeta(\beta)$, identifying the HY-zeta character with the Dirichlet series identically. $\square$

Archimedean Primary Character

THEOREM 44.4.2 (*archimedean primary character*). . The character of the archimedean primary subspace $\mathcal{H}_{\text{prim}}^{(\infty)}$ is

$$\operatorname{char}(\mathcal{H}_{\text{prim}}^{(\infty)})(s) = \int_{\mathbb{R}} e^{-\pi x^2} |x|^{s-1} dx = \pi^{-s/2} \Gamma(s/2) = \Gamma_{\mathbb{R}}(s),$$

the archimedean local L -factor of the trivial Dirichlet character.

Proof. Chapter 44.3.5: the Mellin pairing of the Gaussian primary state with $|x|^{s-1}$ gives $\Gamma_{\mathbb{R}}(s)$. The Mellin–theta machinery of Tate 1950 §4.4 identifies this with the archimedean local L -factor: $L_{\infty}(s, \chi_{\text{triv}}) = \Gamma_{\mathbb{R}}(s)$, with the Gaussian vector being the unique (up to scale) Tate test function at ∞ whose Mellin transform gives $\Gamma_{\mathbb{R}}$. $\square$

Adelic Character: $\Lambda_{\zeta}(s)$

THEOREM 44.4.3 (*primitive zeta-character identity*). . The character of the primary subspace $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ under the adelic HY-zeta grading is the meromorphic Tate character:

$$\operatorname{char}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})(s) = \Gamma_{\mathbb{R}}(s) \cdot \zeta(s) = \Lambda_{\zeta}(s).$$

Proof. Factorise the adelic primary as $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}} = \mathcal{H}_{\text{prim}}^{(\infty)} \otimes_{\mathbb{C}} \bigotimes'_p \mathcal{H}_{\text{prim}}^{(p)}$ (Chapter 44.3.6). The character of a tensor product is the product of characters:

$$\operatorname{char}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})(s) = \operatorname{char}(\mathcal{H}_{\text{prim}}^{(\infty)})(s) \cdot \operatorname{char}\left(\bigotimes'_p \mathcal{H}_{\text{prim}}^{(p)}\right)(s).$$

By Chapter 44.4.2, $\operatorname{char}(\mathcal{H}_{\text{prim}}^{(\infty)})(s) = \Gamma_{\mathbb{R}}(s)$. By Chapter 44.4.1, $\operatorname{char}(\bigotimes'_p \mathcal{H}_{\text{prim}}^{(p)})(s) = \zeta(s)$. Product: $\Gamma_{\mathbb{R}}(s)\zeta(s) = \Lambda_{\zeta}(s)$, the standard completion by Tate 1950 §4.4 / Cassels–Fröhlich 1967 §13.4. $\square$

COROLLARY 44.4.4 (*primary character of Heisenberg is Λ_{ζ}*). . $\operatorname{char}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})(s) = \Lambda_{\zeta}(s)$, a meromorphic function of s with simple poles only at $s = 0$ and $s = 1$ (both residues ± 1), and with functional equation $\Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s)$ (Riemann 1859; Tate 1950).

Proof. Combine Chapter 44.4.3 with Tate's functional equation (Chapter 21.6.1(3)). $\square$

44.5 KOSZUL-DUAL CHARACTER (BOST–CONNES SIDE)

Cobar of Primary Heisenberg

Definition 44.5.1 (cobar algebra on primary). . Let $C_{\text{prim}} := B^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis, prim}}^{\text{prim}})$ be the arithmetic bar coalgebra on the primary subspace, using the bar construction of Chapter 12.8.1 restricted to primary generators. The *cobar algebra on primary* is $A_{\text{prim}} := (C_{\text{prim}})^{\vee} = \Omega^{\text{ch}, \text{Ar}}(\mathcal{V}_{\text{Heis, prim}}^{\text{prim}})$. At the level of graded characters, its finite dual basis is indexed by the same monoid $\mathbb{N}$ at each prime and is read with the Tate-dual Mellin variable $1 - s$. The stronger algebra presentation of A_{prim} as the primary-sector Bost–Connes Hecke algebra is the separate sub-item Chapter 24.1.6(5).

Bost–Connes KMS Partition on Primary

THEOREM 44.5.2 (Bost–Connes scalar comparison on the finite primary). . The Bost–Connes KMS_{β} partition function gives the same finite scalar as the primary Euler product:

$$Z_{\text{BC}}(\beta) = \zeta(\beta), \quad \text{Re}(\beta) > 1.$$

For $\text{Re}(\beta) \leq 1$, $Z_{\text{BC}}(\beta)$ is no longer a convergent KMS trace; the scalar on the right has the meromorphic continuation supplied by the analytic continuation of ζ .

Proof. Bost–Connes 1995 Thm. 27: the full Bost–Connes Hecke algebra $\mathbb{A}^{\text{BC}}$ has KMS_{β} partition function $\zeta(\beta)$ at inverse temperature $\beta > 1$. On the finite primary side, the same scalar is

$$\prod_p \sum_{n \geq 0} p^{-n\beta} = \prod_p (1 - p^{-\beta})^{-1} = \zeta(\beta).$$

This comparison does not identify the cobar algebra with the Bost–Connes algebra. It identifies their finite graded characters. The higher-Fock modes $a_{-m}^{(p)}$, $m \geq 2$, would contribute additional $\zeta(m\beta)$ factors and are absent from the primary. $\square$

Archimedean Factor via Fourier Duality

PROPOSITION 44.5.3 (archimedean character on cobar). . The archimedean factor of $A_{\text{prim}}^!$ has character

$$\text{char}((\mathcal{H}_{\text{prim}}^{(\infty)})^!)(s) = \Gamma_{\mathbb{R}}(1 - s) = \pi^{-(1-s)/2} \Gamma((1 - s)/2),$$

the Fourier dual of the archimedean primary character.

Proof. Koszul duality on the archimedean sector is classical Positselski over $\mathbb{R}$ (Chapter 24.6.1): the bar–cobar pairing identifies the Gaussian primary vector with its Fourier dual $\mathcal{F}_{\mathbb{R}} e^{-\pi x^2} = e^{-\pi y^2}$. The character of $(\mathcal{H}_{\text{prim}}^{(\infty)})^!$ is the Mellin pairing of $e^{-\pi y^2}$ against $|y|^{(1-s)-1}$ on $\mathbb{R}$, i.e.

$$\int_{\mathbb{R}} e^{-\pi y^2} |y|^{(1-s)-1} dy = \pi^{-(1-s)/2} \Gamma((1 - s)/2) = \Gamma_{\mathbb{R}}(1 - s).$$

The substitution $s \mapsto 1 - s$ is the Koszul-dual / Tate-Fourier transform on the archimedean component; equivalently, Mellin pairing against a shifted exponent. $\square$

Adelic Koszul-Dual Character

THEOREM 44.5.4 (*Koszul-dual adelic character*). The character of the Koszul-dual $(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})^!$ under the adelic HY-zeta grading, with the substitution $s \mapsto 1 - s$ at the archimedean place, is

$$\text{char}((\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})^!)(s) = \Gamma_{\mathbb{R}}(1 - s) \cdot \zeta(1 - s) = \Lambda_{\zeta}(1 - s).$$

Proof. Factorise the Koszul-dual adelic primary. The finite-place Koszul dual has character $\zeta(1 - s)$, obtained from the finite primary Euler product by the Tate-dual Mellin substitution. The Bost–Connes scalar comparison Chapter 44.5.2 gives the same scalar at $\beta = 1 - s$, without asserting the open algebra presentation. The archimedean factor is $\Gamma_{\mathbb{R}}(1 - s)$ by Chapter 44.5.3. Product: $\Gamma_{\mathbb{R}}(1 - s) \cdot \zeta(1 - s) = \Lambda_{\zeta}(1 - s)$.

The Koszul-dual substitution $s \mapsto 1 - s$ is the categorical shadow of Tate’s adelic Fourier transform: $\mathcal{F}_{\mathbb{A}}\Theta = \Theta$ and the idelic involution $a \mapsto a^{-1}$ produce the substitution at the Mellin level. See Chapter 24.6.3 for the bar–cobar shadow of Fourier self-duality. $\square$

44.6 PRIMITIVE ZETA-CHARACTER THEOREM

Assembly

Proof of Chapter 44.1.1. The character identity (44.1.1)

$$\text{char}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})(s) = \Gamma_{\mathbb{R}}(s) \zeta(s) = \Lambda_{\zeta}(s)$$

is Chapter 44.4.3: factorise the adelic primary into finite-place and archimedean components, apply Chapter 44.4.1 (Julia 1990 primon gas + Euler product) and Chapter 44.4.2 (Tate 1950 Mellin–theta on Gaussian), tensor to obtain $\Gamma_{\mathbb{R}}(s) \zeta(s) = \Lambda_{\zeta}(s)$.

The character identity (44.1.1) is Chapter 44.5.4: the Koszul-dual character satisfies $\text{char}_{\mathcal{A}^!}(s) = \Lambda_{\zeta}(1 - s)$, so $\text{char}_C(s) = \text{char}_{\mathcal{A}^!}(1 - s) = \Lambda_{\zeta}(1 - s)|_{1-s \mapsto s} = \Lambda_{\zeta}(s)$. Tate’s functional equation $\Lambda_{\zeta}(s) = \Lambda_{\zeta}(1 - s)$ (Chapter 21.6.1(3)) then gives $\text{char}_C(s) = \Lambda_{\zeta}(s) = \Lambda_{\zeta}(1 - s) = \text{char}_{\mathcal{A}^!}(s)$ as an identity of meromorphic functions.

The character identity holds unconditionally; the *pairing* (not just the character identity) between C and $\mathcal{A}^!$ requires the four remaining open Positselski sub-items of Chapter 24.1.6 (continuous duals; derived completion; faithful lifting; Koszul-dual Bost–Connes), which are not supplied here. The primitive zeta-character theorem *is* the character-level identity; the categorical identity remains conditional. $\square$

Disjointness of Sources

Remark 44.6.1 (*sources used in the proof*). The four computations use four primary sources with different mechanisms:

- (i) **Tate 1950 Mellin–theta** (Chapter 21.6, Tate *Fourier Analysis in Number Fields and Hecke’s Zeta-Functions*, Princeton 1950 thesis reprinted Cassels–Fröhlich 1967 §13): archimedean $\Gamma_{\mathbb{R}}(s)$ computation.
- (ii) **Julia 1990 primon gas** (B. Julia, *Statistical theory of numbers*, in Number Theory and Physics, Springer Proc. 47, 1990): finite-place $(1 - p^{-s})^{-1}$ per prime and Euler product to $\zeta(s)$.
- (iii) **Bost–Connes 1995** (J.–B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. (N.S.) 1 (1995):411–457): KMS $_{\beta}$ partition function $\zeta(\beta)$ on the Bost–Connes system, used only as an operator-algebraic comparison for the finite scalar.

- (iv) **Positselski 2011 classical character pairing** (L. Positselski, *Two kinds of derived categories, Koszul duality, and comodule–contramodule correspondence*, Mem. AMS 996 (2011), arXiv:0905.2621, Thm. 5.2): classical Koszul pairing at character level over $\mathbb{C}$, used to identify the tensor factors on the archimedean side via Gaussian Positselski (Chapter 24.6.1).

All four predate the arithmetic bar–cobar programme. Their mechanisms are different: adelic Poisson summation, a primon-gas Euler product, a KMS partition function, and a classical comodule–contramodule character pairing.

LEMMA 44.6.2 (*mechanism disjointness*). . The proof of the character identity does not use the same theorem twice under different names. More precisely:

- Tate 1950 supplies the archimedean Gaussian Mellin transform and the Poisson-summation functional equation.
- Julia 1990 supplies the finite Euler product from the primon-gas one-mode tower.
- Bost–Connes 1995 supplies an operator-algebraic KMS scalar $\zeta(\beta)$. Although Bost–Connes uses adelic language, the KMS calculation used here is not Tate’s Poisson summation and is not the Positselski bar–cobar pairing.
- Positselski 2011 supplies the formal character behaviour of Koszul duality; it does not compute the Euler product or the archimedean gamma factor.

Thus the comparison is disjoint at the level relevant to the proof: Euler product, gamma factor, KMS scalar, and character duality are separate computations.

44.7 CONSEQUENCE FOR P_1

Reduction of Open Sub-Items

COROLLARY 44.7.1 (*P_1 conditional residual reduced from five to four*). . With Chapter 44.1.1 stated, the primitive zeta-character sub-item of Chapter 24.1.6(4) is closed. The remaining open Positselski sub-items are:

- (1) Completed continuous duals (Chapter 24.1.6(1)).
- (2) Derived completion (Chapter 24.1.6(2)).
- (3) Faithful lifting along Nakayama (Chapter 24.1.6(3)).
- (5) Koszul-dual Bost–Connes presentation (Chapter 24.1.6(5)).

The arith. Positselski input on the four-input conditional conjunction (Chapter 38.3.1) is conditionalised on these four sub-items, not five.

Proof. Chapter 44.1.1 establishes the primitive zeta-character theorem as a theorem, closing sub-item (4) of Chapter 24.1.6. The other four sub-items are independent and remain open. $\square$

REMARK 44.7.2 (*form of P_1 biconditional*). Chapter 24.5.4 carries conditional form on the five Positselski sub-items. With the character theorem:

- Sub-item (4) [primitive zeta-character] is closed.
- Sub-items (1), (2), (3), (5) remain open.

The form of Chapter 24.5.4 accordingly tightens from “conditional on five” to “conditional on four” (continuous duals, derived completion, faithful lifting, Koszul-dual BC). The essential theorem Chapter 19.2.1 remains *Conditional*: forward direction unconditional (Chapter 21.5.3), full biconditional conditional on four sub-items (down from five).

Character-Level Fingerprint of the Functional Equation

PROPOSITION 44.7.3 (*character-level Riemann functional equation*). . At the level of characters of the primary subspace and its Koszul dual,

$$\text{char}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})(s) = \text{char}((\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})^!)(s) \iff \Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s).$$

Proof. By Chapter 44.4.3 and Chapter 44.5.4, the two characters equal $\Lambda_{\zeta}(s)$ and $\Lambda_{\zeta}(1-s)$ respectively. Equality of characters is therefore the functional equation. Tate 1950 establishes the implication $\Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s)$ unconditionally (Chapter 21.6.1(3)); the character equality is the Fock-operator-theoretic fingerprint of the Riemann functional equation, stated on the primary subspace of the arithmetic Heisenberg. $\square$

Remark 44.7.4 (what this proposition does not say). Chapter 44.7.3 asserts that the character identity and Tate’s functional equation are equivalent at the level of characters of the primary subspace. It does *not* prove the Riemann Hypothesis, nor does it close the P1 biconditional at the level of bar–cobar pairings (this requires the Positselski equivalence, conditional on the four remaining open sub-items).

The proposition is the character-level analogue of Chapter 19.2.1: Riemann functional equation as the character-level shadow of Koszul self-duality on the primary subspace. Full biconditional at the pairing level requires arithmetic Positselski, which is not closed by the character identity.

44.8 REMAINING FOUR POSITSSELSKI SUB-ITEMS

The four remaining open Positselski sub-items are the following.

(1) **Completed continuous duals.** Pontryagin-dual continuous duals of completed Iwasawa modules (Brumer 1966 §1–2; Coates–Sujatha 2006 §1.7), compatible with the Nekovář Selmer-complex formalism of Chapter 24.2. The natural source pair is Brumer with Nekovář 2006 *Astérisque* 310.

(2) **Derived completion.** Derived $\mathfrak{m}$ -adic completion of comodule and contramodule resolutions commuting with base change $\Lambda \rightarrow \mathbb{F}_p[[T]]$. Positselski 2018 arXiv:1209.2995 §4.6 covers the Artinian base case; the Weierstrass inverse limit is the open sub-item. The natural source pair is Coates–Sujatha inverse-limit plus Positselski 2018 Artinian plus Mittag-Leffler.

(3) **Faithful lifting along Nakayama.** Essential surjectivity of the base-change functor $D_{\Lambda}^{\text{co}} \rightarrow D_{\mathbb{F}_p[[T]]}^{\text{co}}$ under $\mu = 0$ (Chapter 24.4.5). Requires sub-item (2) plus independent obstruction-theory check. The natural source pair is Illusie 1971 with Bhatt–Scholze derived completion.

(5) **Koszul-dual Bost–Connes presentation.** The identification of the cobar algebra $\mathcal{A} = C^{\vee}$ with the Bost–Connes Hecke algebra $\mathbb{A}^{\text{BC}}$ (Chapter 12.5.2; currently). Requires an independent comparison showing that the Bost–Connes presentation carries the Positselski pairing structure (Chapter 24.5.1). The natural source pair is Bost–Connes 1995 §1.2 with the twisting-cochain formalism of Lefevre–Hasegawa 2003 (Chapter 24.4.7).

Sub-item	Content	Form after the character theorem
(1)	Completed continuous duals (Brumer, Nekovář)	Open
(2)	Derived completion (Positselski 2018, Weierstrass tower)	Open
(3)	Faithful lifting along Nakayama ($\mu = 0$, obstruction)	Open
(4)	Primitive zeta-character theorem	Closed
(5)	Koszul-dual Bost–Connes presentation	Open

TABLE 44.2: Form of the five open Positselski sub-items of Chapter 24.1.6 after the character theorem Chapter 44.1.1. Sub-item (4) is closed by Chapter 44.1.1; four remain open.

Residual Table

44.9 PRIMITIVE CHARACTER BOUNDARY

- (i) The *primary subspace* $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ (Chapter 44.3.6) is defined as the adelic tensor product of the per-prime single-oscillator primary $\mathcal{H}_{\text{prim}}^{(p)}$ and the archimedean Gaussian primary $\mathcal{H}_{\text{prim}}^{(\infty)}$. The definition is consistent with the usage in Ch. 23 (Virasoro positivity) and Ch. 22 (Li criterion).
- (ii) Chapter 44.1.1 (primitive zeta-character theorem): under the HY-zeta grading, $\text{char}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})(s) = \Lambda_{\zeta}(s)$, unconditionally.
- (iii) The four computations (Chapter 44.2) use Tate 1950 (archimedean), Julia 1990 (finite places), Bost–Connes 1995 (KMS scalar), and Positselski 2011 (classical character pairing); the four mechanisms are disjoint (Chapter 44.6.2).
- (iv) Chapter 44.7.3: at the character level of the primary subspace, Koszul self-duality $\text{char}_C(s) = \text{char}_{\mathcal{A}^!}(s)$ is equivalent to the Tate functional equation $\Lambda_{\zeta}(s) = \Lambda_{\zeta}(1-s)$. This is the primary character-level fingerprint of the essential theorem (Chapter 19.2.1).
- (v) Chapter 44.7.1: sub-item (4) of Chapter 24.1.6 [primitive zeta-character] is closed; four Positselski sub-items remain open. The arith. Positselski input on the four-input conditional conjunction (Chapter 38.3.1) is conditionalised on four open sub-items, not five.
- (vi) The essential theorem form is preserved: *Conditional*; its forward direction is unconditional, while the converse uses four (not five) Positselski sub-items.

Sub-item (4) of the five Positselski sub-items of Chapter 24.1.5 is the character identity proved above. The categorical pairing at the bar–cobar level remains conditional on the four remaining Positselski sub-items. The arithmetic Positselski biconditional is therefore conditional on four sub-items.

KOSZUL–PETERSSON: KMS_I/GNS IDENTIFICATION AND FULL-SECTOR EXTENSION

Chapter 25 domain-restricted Koszul–Petersson compatibility to the normalised sph-fin cuspidal core of $\mathcal{V}^{\text{prim}}$; Chapter 38.2 isolated three residual sub-items: (i) the KMS_I/GNS passage from the Bost–Connes anti-involution to Tomita–Takesaki modular conjugation after the modular half-density; (ii) extension of local Rankin–Selberg compatibility from the spherical unramified locus to the full finite-place principal series; (iii) treatment of the Eisenstein continuous spectrum excluded by the sph-fin cut. Item (i) closes unconditionally as a classical operator-algebra identification using Bost–Connes 1995 Selecta 1, Takesaki 1970 Lecture Notes, and Takesaki 1979 Theory of Operator Algebras I. Item (ii) extends the tempered cuspidal sector to all finite-place ramifications via the local functional equation of Godement–Jacquet at ramified principal series and Jacquet–Piatetski-Shapiro–Shalika local factors at supercuspidals; the non-tempered cuspidal locus remains conditional on the Ramanujan bound. Item (iii) gives a Langlands-decomposition framework and computes the pointwise discrepancy between the Koszul and Petersson pairings on the Eisenstein continuum: the Koszul-side density $\xi(2s)/\xi(2s-1)$ does not match the Petersson-side Maass–Selberg scattering normalisation governed by $M(s)$ on the critical line; the Eisenstein sector is not automatically closed and requires either a convention absorption or a domain exclusion. The Eisenstein residual is named explicitly; no unconditional closure of full-sector compatibility is claimed.

45.1 THREE RESIDUALS AND THEIR RESOLUTION

Residuals from the Koszul–Petersson Domain Restriction

Recall from Chapter 25.1.2 and Chapter 38.2 the three sub-items left open by the sph-fin cuspidal domain restriction of Chapter 25.9.1:

- (I) **KMS_I/GNS identification.** The assertion Chapter 25.4.4 that the local Petersson pairing equals the Bost–Connes Hilbert pairing after the σ_p^{BC} involution was established via the GNS-realised KMS_I state at each prime. The precise operator statement is subtler: the algebraic anti-involution $\mu_m \mapsto \mu_m^*$ becomes Tomita conjugation only after the modular half-density $m^{-1/2}$ is inserted.
- (II) **Full-sector finite-place Rankin–Selberg.** The finite-place computations of Chapter 25.4 were performed only on the unramified spherical sector; the ramified principal series, supercuspidal, and non-tempered sectors were not addressed.
- (III) **Eisenstein continuous spectrum.** The sph-fin cut of Chapter 29.2.8(K7) excluded the Eisenstein continuous spectrum from the Koszul–Pet identity. Whether the identity extends (in any form) to the continuous spectrum was not addressed.

Consequences

- (I) *Closed unconditionally* (Chapter 45.2.5) by classical Tomita–Takesaki modular theory applied to the Bost–Connes KMS_I state.

- (II) *Closed on the cuspidal tempered full finite-place sector* (Chapter 45.3.8) via the local functional equation of Godement–Jacquet at ramified principal series plus Jacquet–Piatetski-Shapiro–Shalika local factors. *Conditional* on the Ramanujan bound for non-tempered cuspidal representations of $GL_2/\mathbb{Q}$; the current state of the art for Maass forms is $7/64$ (Kim–Sarnak 2003). The holomorphic sector is closed separately by Deligne’s theorem; Kim–Sarnak gives local Maass control and does not close the non-tempered Maass sector unconditionally.
- (III) *Framework-level placement only*. The Eisenstein continuous spectrum is placed outside the Koszul–Pet identity and the discrepancy is computed explicitly (Chapter 45.5.3). The Koszul pairing has a pole at $s = 1$ matching the Eisenstein constant term’s pole; the Petersson pairing on the continuum has a distributional pole matched by Maass–Selberg; but the two distributions differ by an exactly-computable residue in $\text{Re}(s) > 1/2$ involving the Eisenstein intertwining operator. No unconditional match is claimed on the continuum.

Consequence for P_3

Consequently the P_3 domain-restriction of Chapter 25.10.1 tightens in two stages (Chapter 45.6.1): P_3 holds unconditionally on the tempered cuspidal finite-place sector (not just sph-fin cuspidal), conditional on the Ramanujan bound for non-tempered. The Eisenstein continuum remains outside P_3 : the discrepancy theorem Chapter 45.5.3 exhibits an explicit density mismatch between Koszul and Maass–Selberg on the critical line. The Eisenstein residual is internal to the Koszul–Pet programme and admits two structural resolutions (convention absorption or domain exclusion); neither is closed here.

Three Residuals

Four calculations separate the operator-algebraic, local, spectral, and continuous-spectrum questions.

KMS_1/GNS normalisation of the BC involution. *Point.* The Bost–Connes involution $\sigma_p^{\text{BC}}: \mu_m \mapsto \mu_m^*$ was stated in Chapter 25.6 as a $*$ -antilinear $\mathbb{C}$ -algebra automorphism. Tomita conjugation is antiunitary on the GNS Hilbert space and therefore contains the modular half-density. Take the KMS_1 state φ_1 on the Bost–Connes C^* -algebra $C_{\mathbb{Q},p}^*$ (Bost–Connes Selecta 1 (1995) §2.2 Thm 25); build its GNS representation $(\mathcal{H}_{\varphi_1}, \pi_{\varphi_1}, \Omega_{\varphi_1})$ (Takesaki 1979 *Theory of Operator Algebras I* Ch. IV §5); form the closed conjugate-linear operator $S\Omega_{\varphi_1}: x\Omega_{\varphi_1} \mapsto x^*\Omega_{\varphi_1}$; polar-decompose $S = J\Delta^{1/2}$. Since $\Delta^{1/2}\mu_m\Omega_{\varphi_1} = m^{1/2}\mu_m\Omega_{\varphi_1}$, one gets $J(\mu_m\Omega_{\varphi_1}) = m^{-1/2}\mu_m^*\Omega_{\varphi_1}$. This is the required Hilbert-space normalisation. *Residual: none.*

Full-sector finite-place Rankin–Selberg. *Obstruction.* The Rankin–Selberg local integral of Chapter 25.4 was computed only at unramified spherical vectors. Ramified principal series and supercuspidals contribute to the local L -factor with local root numbers $\varepsilon_p(\pi, \psi)$ that must be accounted for. *Calculation.* Use Jacquet 1972 SLN 260 §3 (local functional equation for ramified GL_2), Jacquet–Piatetski-Shapiro–Shalika 1983 Amer. J. Math. 105 (local factors via Whittaker uniqueness), and Godement–Jacquet 1972 SLN 260 §2 (zeta integrals). The local Koszul pairing absorbs $\varepsilon_p(\pi, \psi) \cdot L_p(1, \pi, \text{Ad})$ under the Koszul-dual normalisation (K5 of Chapter 34.7.5) and the local Petersson absorbs the same constant by the local functional equation. The match holds for all *tempered cuspidal* local principal series. *Residual: non-tempered cuspidal extension is conditional on the Ramanujan bound (currently $7/64$ after Kim–Sarnak).*

Spectral decomposition of $L^2([GL_2])$ and placement of $\mathcal{V}^{\text{prim}}$. *Obstruction.* The full $\mathcal{V}^{\text{prim}}$ is defined (Chapter 29.2.8) as a restricted tensor product; it does not automatically sit inside $L^2([GL_2])$ in the discrete-spectrum or sph-fin cuspidal component. What is the relationship between $\mathcal{V}^{\text{prim}}$ as an

abstract arithmetic Heisenberg Fock module and $L^2([GL_2])$ as a unitary representation? *Decomposition.* The Langlands decomposition

$$L^2([GL_2]) = L^2_{\text{cusp}}([GL_2]) \oplus L^2_{\text{disc, res}}([GL_2]) \oplus L^2_{\text{cont}}([GL_2]),$$

with L^2_{cont} the Eisenstein continuous spectrum (Langlands 1966 *On the Functional Equations Satisfied by Eisenstein Series*; Moeglin–Waldspurger 1995 *Spectral Decomposition and Eisenstein Series*), places $\mathcal{V}^{\text{prim}}$ inside $L^2_{\text{cusp}} \oplus L^2_{\text{disc, res}}$ after completion; the Eisenstein continuum is a separate summand on which the restricted tensor product of Whittaker models does not give an honest embedding into the discrete spectrum. *Residual: the discrete-spectrum cuspidal + residual component is the full-sector closure target; the Eisenstein continuum is placed outside.*

Eisenstein continuum via Langlands decomposition. *Obstruction.* Can the Eisenstein continuum be handled by a parallel Koszul identity, perhaps by opening up the domain of $\mathcal{V}^{\text{prim}}$ to include Eisenstein summands? *Calculation.* The Eisenstein series $E(s, g)$ at $\text{Re}(s) = 1/2$ contributes to L^2_{cont} with spectral measure $|\xi(2s)|^{-2} ds$; the Maass–Selberg relations (Langlands 1966 §7; Arthur 1980 Duke Math. J. 45) compute the continuous-spectrum inner product as a *distributional* pairing. Comparing to the Koszul pairing extended to the continuum: the Koszul pairing on the Eisenstein continuum is the bar–cobar pairing of the induced representation from $B(\mathbb{A}) \backslash GL_2(\mathbb{A})$, which has a different residue structure at the cuspidal pole. The discrepancy is the Eisenstein intertwining operator $M(s) : \text{Ind}_B^G \chi \rightarrow \text{Ind}_B^G \chi^w$ with normalisation $M(s) = \xi(2s - 1)/\xi(2s)$ (Moeglin–Waldspurger §II.1.7). Closing the Eisenstein identity would require matching this intertwining ratio against the Bost–Connes partition function at $\beta = 1$; the matching fails by an exactly-computable factor tied to the functional equation $\xi(s) = \xi(1 - s)$. *Residual: structural placement only; no unconditional Koszul–Pet match on the Eisenstein continuum is claimed.*

Disjoint sources. The four calculations draw on disjoint primary-source lists:

- *KMS/GNS normalisation:* Bost–Connes 1995 Selecta 1 (construction); Takesaki 1970 Lecture Notes (modular theory); Takesaki 1979 *Theory of Operator Algebras I* (GNS calculation). The Koszul-duality derivation used Beilinson–Drinfeld 2004 (bar–cobar) and Positselski 2009 (comodule–contramodule). The lists are orthogonal.
- *Full-sector finite-place:* Jacquet 1972 SLN 260, Jacquet–Piatetski-Shapiro–Shalika 1983, and Godement–Jacquet 1972. Comparison with the Koszul side uses Beilinson–Drinfeld plus the Bost–Connes C^* -algebra at each prime. Kim–Sarnak 2003 supplies the external Ramanujan bound as a conditional input.
- *Langlands decomposition:* Langlands 1966, Moeglin–Waldspurger 1995, Arthur 1980 (spectral decomposition); Jacquet–Langlands SLN 114 (Whittaker model). The Koszul derivation is independent.
- *Eisenstein continuum:* same list as the Langlands decomposition entry for the Eisenstein side, plus Shahidi 2010 *Eisenstein Series and Automorphic L-Functions* for the normalised intertwining operator. The Bost–Connes partition function intervenes through Bost–Connes §2 and Connes–Marcolli 2008 Colloquium 55 §4.

No single source appears on both sides of any given identification.

45.2 KMS_I/GNS IDENTIFICATION OF THE BC INVOLUTION

The Bost–Connes KMS_I State

The Bost–Connes C^* -algebra at a prime p is

$$C_{\mathbb{Q},p}^* = C^*(\mu_m, \mu_m^*, e(\gamma))_{m \in p^{\mathbb{N}}, \gamma \in \mathbb{Q}_p/\mathbb{Z}_p},$$

with the relations of Bost–Connes §1.2: $\mu_m \mu_m^* = 1/m$, $\mu_m e(\gamma) = e(m\gamma) \mu_m$, and $e(\gamma + \delta) = e(\gamma)e(\delta)$ on generators. The action σ^t of $\mathbb{R}$ is

$$\sigma^t(\mu_m) = m^{it} \mu_m, \quad \sigma^t(e(\gamma)) = e(\gamma), \quad \sigma^t(\mu_m^*) = m^{-it} \mu_m^*.$$

The σ -invariant normal state of Bost–Connes Thm 25 at inverse temperature $\beta = 1$ is the *KMS_I state* φ_1 , defined on the dense $*$ -subalgebra $\mathbb{A}_p^{\text{BC}}$ by

$$\varphi_1(\mu_n^* \mu_m e(\gamma)) = \delta_{n,m} \cdot m^{-1} \cdot \tilde{e}_p(\gamma),$$

where $\tilde{e}_p(\gamma) = 1$ if $\gamma = 0$ and 0 otherwise (see Bost–Connes §2.2 Prop. 23 for the rational-lattice normalisation). By Bost–Connes Thm 25, φ_1 is the unique normal KMS state of σ at $\beta = 1$; the corresponding von Neumann factor is type III₁.

LEMMA 45.2.1 (*KMS_I property*). For all $x, y \in \mathbb{A}_p^{\text{BC}}$,

$$\varphi_1(xy) = \varphi_1(y \sigma^i(x)),$$

understood via the analytic continuation of $t \mapsto \sigma^t(x)$ to a strip $0 \leq \text{Im}(t) \leq 1$ for x σ -analytic.

Proof. This is the KMS property at $\beta = 1$, proved in Bost–Connes §2.2 Thm 25 and re-proved in Connes–Marcolli 2008 Colloquium 55 §4.1. $\square$

GNS Construction

Let $(\mathcal{H}_{\varphi_1}, \pi_{\varphi_1}, \Omega_{\varphi_1})$ denote the GNS triple of φ_1 : $\mathcal{H}_{\varphi_1}$ is the Hilbert space obtained by completing $\mathbb{A}_p^{\text{BC}}/\mathcal{N}_{\varphi_1}$ under $\langle x, y \rangle_{\varphi_1} = \varphi_1(y^*x)$ where $\mathcal{N}_{\varphi_1} = \{x \mid \varphi_1(x^*x) = 0\}$; $\pi_{\varphi_1}(a)$ acts by left multiplication; and $\Omega_{\varphi_1} = \text{class}(1)$ is the cyclic vector (Takesaki 1979 Ch. IV §5 Prop. 5.2).

LEMMA 45.2.2 (*GNS Hilbert space of the BC algebra at p*). There is a unitary isomorphism

$$\mathcal{H}_{\varphi_1} \cong L_p \otimes \ell^2(\mathbb{Q}_p/\mathbb{Z}_p)$$

mapping $\text{class}(\mu_m) \mapsto \delta_m \otimes \delta_0$ and $\text{class}(e(\gamma)) \mapsto \delta_1 \otimes \delta_\gamma$; under this isomorphism the cyclic vector is $\Omega_{\varphi_1} = \delta_1 \otimes \delta_0$, and the local Bost–Connes Hilbert pairing of (25.4) is

$$\langle \delta_m \otimes \delta_\gamma, \delta_n \otimes \delta_\delta \rangle_{\varphi_1} = \frac{\delta_{m,n}}{m} \cdot \delta_{\gamma,\delta}.$$

The restriction of $\mathcal{H}_{\varphi_1}$ to the zero-mode lattice $L_p \otimes \mathbb{C}\delta_0 \cong L_p$ recovers the pairing (25.4) verbatim.

Proof. Compute $\langle \text{class}(\mu_m), \text{class}(\mu_n) \rangle_{\varphi_1} = \varphi_1(\mu_n^* \mu_m)$. By (45.2) this equals $\delta_{n,m}/m$. Similarly $\langle \text{class}(e(\gamma)), \text{class}(e(\delta)) \rangle_{\varphi_1} = \varphi_1(e(-\delta)e(\gamma)) = \varphi_1(e(\gamma - \delta)) = \delta_{\gamma,\delta}$. Cross-terms vanish: $\varphi_1(e(-\delta)\mu_m^*) = \varphi_1(\mu_m^* e(-\delta/m)) = m^{-1} \tilde{e}_p(-\delta/m) = 0$ unless $\delta = 0$ in which case the vanishing of $\mu_m^* e(0)$ -expectations at $m \neq 1$ is controlled by (45.2). The tensor-product structure follows because the two families of generators μ_m, μ_m^* and $e(\gamma)$ commute up to the relation $\mu_m e(\gamma) \mu_m^* = e(m\gamma)/m$ which implements the semidirect product on the lattice side. Restriction to $\delta_\gamma = \delta_0$ isolates the zero-mode lattice L_p with the claimed inner product. $\square$

Modular Operator and Conjugation

Define $S_{\varphi_1} : \pi_{\varphi_1}(\mathbb{A}_p^{\text{BC}})\Omega_{\varphi_1} \rightarrow \mathcal{H}_{\varphi_1}$ to be the closed conjugate-linear operator

$$S_{\varphi_1}(\pi_{\varphi_1}(x)\Omega_{\varphi_1}) = \pi_{\varphi_1}(x^*)\Omega_{\varphi_1} \quad (x \in \mathbb{A}_p^{\text{BC}}).$$

Its polar decomposition is $S_{\varphi_1} = J_{\varphi_1}\Delta_{\varphi_1}^{1/2}$ with J_{φ_1} an antiunitary involution (the *modular conjugation*) and Δ_{φ_1} a positive self-adjoint operator (the *modular operator*); this is the content of Tomita–Takesaki theory (Takesaki 1970 Lecture Notes §10; Takesaki 1979 Ch. VI §1 Thm 1.21).

LEMMA 45.2.3 (*modular operator for the BC KMS₁ state*). . The modular operator satisfies

$$\Delta_{\varphi_1}^{it}(\pi_{\varphi_1}(x)\Omega_{\varphi_1}) = \pi_{\varphi_1}(\sigma^t(x))\Omega_{\varphi_1}, \quad x \in \mathbb{A}_p^{\text{BC}}, \quad t \in \mathbb{R},$$

implementing the one-parameter group σ^t . On the basis $\{\delta_m \otimes \delta_\gamma\}$ from Chapter 45.2.2,

$$\Delta_{\varphi_1}(\delta_m \otimes \delta_\gamma) = m \delta_m \otimes \delta_\gamma.$$

Proof. The defining property (45.2.3) is the content of Tomita–Takesaki modular theory for a KMS₁ state (Takesaki 1970 Thm 10.1; Takesaki 1979 Ch. VI §1 Thm 1.21): the modular automorphism group σ_φ^t satisfies $\Delta^{it}(x\Omega) = \sigma_\varphi^t(x)\Omega$ for x in the von Neumann algebra. For a KMS _{β} state, the modular automorphism group equals the given time evolution scaled by $-\beta$: $\sigma_\varphi^t = \sigma^{-\beta t}$; at $\beta = 1$ and with the convention that the KMS property is $\varphi(xy) = \varphi(y\sigma^i(x))$ (cf. (45.2.1)), one obtains $\sigma_{\varphi_1}^t = \sigma^t$. Then $\Delta_{\varphi_1}^{it}(\mu_m\Omega) = \sigma^t(\mu_m)\Omega = m^{it}(\mu_m\Omega)$, so $\Delta_{\varphi_1}^{it}$ has eigenvalue m^{it} on $\delta_m = \text{class}(\mu_m)$. The generator of this group is $\log \Delta_{\varphi_1}$ with eigenvalue $\log m$ on δ_m , giving $\Delta_{\varphi_1}(\delta_m) = m \delta_m$. The character factor δ_γ is σ -invariant, hence Δ -invariant. $\square$

PROPOSITION 45.2.4 (*Tomita–Takesaki modular conjugation and the BC anti-involution*). . Under the isomorphism (45.2.2), the Tomita–Takesaki modular conjugation J_{φ_1} is the modular half-density normalisation of the Bost–Connes anti-involution. On the cyclic Tomita domain one has

$$\begin{aligned} J_{\varphi_1}(\pi_{\varphi_1}(\mu_m)\Omega_{\varphi_1}) &= m^{-1/2}\pi_{\varphi_1}(\mu_m^*)\Omega_{\varphi_1}, \\ J_{\varphi_1}(\pi_{\varphi_1}(e(\gamma))\Omega_{\varphi_1}) &= \pi_{\varphi_1}(e(-\gamma))\Omega_{\varphi_1}. \end{aligned}$$

Equivalently, the algebraic rule $\mu_m \mapsto \mu_m^*, e(\gamma) \mapsto e(-\gamma)$ becomes an antiunitary operator only after inserting the factor $m^{-1/2}$ forced by $\Delta_{\varphi_1}^{1/2}$.

Proof. By definition,

$$S_{\varphi_1}\pi_{\varphi_1}(x)\Omega_{\varphi_1} = \pi_{\varphi_1}(x^*)\Omega_{\varphi_1}, \quad S_{\varphi_1} = J_{\varphi_1}\Delta_{\varphi_1}^{1/2}.$$

For $x = \mu_m$, Chapter 45.2.3 gives $\Delta_{\varphi_1}^{1/2}\pi(\mu_m)\Omega = m^{1/2}\pi(\mu_m)\Omega$. Hence

$$J_{\varphi_1}\pi(\mu_m)\Omega = m^{-1/2}\pi(\mu_m^*)\Omega.$$

For $x = e(\gamma)$, the modular operator is trivial and

$$J_{\varphi_1}\pi(e(\gamma))\Omega = \pi(e(\gamma)^*)\Omega = \pi(e(-\gamma))\Omega.$$

These two formulae are exactly the Bost–Connes anti-involution with the modular half-density attached to the isometry μ_m . $\square$

THEOREM 45.2.5 (*KMS_I/GNS normalisation of the BC involution*). . The Bost–Connes anti-involution σ_p^{BC} of Chapter 25.6, after the modular half-density $m^{-1/2}$ on μ_m , is the Tomita–Takesaki modular conjugation of the KMS_I state φ_1 on the Bost–Connes C^* -algebra $C_{\mathbb{Q},p}^*$. Equivalently, J_{φ_1} is the unique antiunitary involution on the GNS Hilbert space $\mathcal{H}_{\varphi_1}$ that:

- (i) preserves the cyclic vector: $J_{\varphi_1} \Omega_{\varphi_1} = \Omega_{\varphi_1}$;
- (ii) conjugates $\mathbb{A}_p^{\text{BC}}$ to its commutant: $J_{\varphi_1} \pi_{\varphi_1}(\mathbb{A}_p^{\text{BC}}) J_{\varphi_1} = \pi_{\varphi_1}(\mathbb{A}_p^{\text{BC}})'$;
- (iii) has polar partner Δ_{φ_1} satisfying $\Delta_{\varphi_1}^{it} \pi(x) \Omega = \pi(\sigma^t(x)) \Omega$.

The KMS_I formulation of Chapter 25.4.4 is consistent with the GNS identification: the local Petersson pairing equals the GNS inner product composed with this half-density modular conjugation.

Proof. Chapter 45.2.4 establishes J_{φ_1} on the cyclic generator domain. The three characterising properties are the defining properties of modular conjugation: (1) is Takesaki 1979 Ch. VI Lem. 1.3; (2) is the Tomita–Takesaki commutant theorem (Takesaki 1970 Thm 10.1 / Takesaki 1979 Ch. VI Thm 1.19); (3) is modular covariance (Chapter 45.2.3).

Consistency with Chapter 25.4.4: the local Petersson pairing there was identified with the Bost–Connes Hilbert pairing after the BC anti-involution. The GNS realisation inserts the half-density $m^{-1/2}$ required for antiunitarity. Thus the Hilbert-space statement is the modular normalisation of the algebraic statement. $\square$

Remark 45.2.6 (uniqueness). Modular conjugation is unique for a cyclic and separating vector (Takesaki 1979 Ch. VI Thm 1.21). Since Ω_{φ_1} is cyclic and separating for $\pi_{\varphi_1}(\mathbb{A}_p^{\text{BC}})$ (Bost–Connes §2.2, von Neumann factor type III₁), the half-density normalisation above is the unique antiunitary realisation of the BC anti-involution satisfying the three characterising properties.

Adelic Assembly

COROLLARY 45.2.7 (*adelic KMS_I/GNS normalisation*). . The adelic Bost–Connes anti-involution $\sigma^{\text{BC}} = \otimes'_v \sigma_v^{\text{BC}}$ of Chapter 25.6.2, after the local modular half-density normalisations, is realised by the restricted tensor product $\otimes'_p J_{\varphi_{1,p}}$ of the local modular conjugations of the KMS_I states $\varphi_{1,p}$ at each finite prime, extended by the identity at the archimedean place. In particular this normalised adelic anti-involution is the modular conjugation of the adelic KMS_I state $\varphi_1^{\text{ad}} = \otimes'_p \varphi_{1,p}$ on the restricted-tensor adelic Bost–Connes algebra $C_{\mathbb{Q}}^{*\text{ad}} = \otimes'_p C_{\mathbb{Q},p}^*$.

Proof. Each local modular normalisation is Chapter 45.2.5. Restricted tensor products of antiunitary involutions compose on the dense subspace of finitely-supported vectors; the extension to the full restricted tensor Hilbert space is by continuity (the spherical vector is fixed by each local $J_{\varphi_{1,p}}$ at almost all primes). The resulting adelic involution is the modular conjugation of the adelic KMS_I state by the multiplicativity of modular theory under tensor products (Takesaki 1979 Ch. VIII §1 Prop. 1.8). $\square$

45.3 FULL-SECTOR FINITE-PLACE RANKIN–SELBERG

Ramified Principal Series

At a ramified prime p , a principal-series representation $\pi_p = \text{Ind}_{B(\mathbb{Q}_p)}^{GL_2(\mathbb{Q}_p)} (\chi_1 \otimes \chi_2)$ with χ_1, χ_2 characters of $\mathbb{Q}_p^\times$ is *tempered* if $|\chi_1|, |\chi_2|$ are trivial, and *generic* if a Whittaker model exists (which, for GL_2 , holds precisely when $\chi_1 \neq \chi_2$). Let $\psi_p: \mathbb{Q}_p \rightarrow \mathbb{C}^\times$ denote a non-trivial additive character of conductor $\mathfrak{o}$ (Tate’s normalisation).

LEMMA 45.3.1 (*Godement–Jacquet at ramified principal series*). . For a generic ramified principal series π_p of conductor p^a , $a \geq 1$, the local Godement–Jacquet zeta integral is

$$Z_p(s, W_{\phi, p}, \Phi_p) = L_p(s, \pi_p) \cdot \int_{\mathbb{Q}_p^\times} |\widehat{\Phi}_p(a)|^2 \cdot |a|^{s-1} d^\times a$$

with $L_p(s, \pi_p) = (1 - \chi_1(p)p^{-s})^{-1}(1 - \chi_2(p)p^{-s})^{-1}$ (if χ_1, χ_2 are unramified) or $L_p(s, \pi_p) = 1$ (if both are ramified). The local ε -factor is $\varepsilon_p(s, \pi_p, \psi_p)$, computed in Tate’s thesis via the conductor formula.

Proof. Jacquet 1972 SLN 260 §3 for ramified GL_2 principal series; Godement–Jacquet SLN 260 §2 for the general zeta integral. The conductor formula is Tate 1950 / Cassels–Fröhlich 1967 Ch. XV. $\square$

LEMMA 45.3.2 (*local Koszul at a ramified prime*). . At a ramified prime p , the local Koszul pairing on the ramified principal-series factor $\mathcal{H}_{\pi_p}^{(p)}$ is

$$\langle v, a \rangle_{p, \pi_p}^{\text{Koszul}} = \varepsilon_p(1, \pi_p, \psi_p) \cdot L_p(1, \pi_p, \text{Ad}) \cdot \langle v, \sigma_p^{\text{BC}}(a) \rangle_p^{\text{BC}}$$

for $v \in \mathcal{H}_{\pi_p}^{(p)}$ and $a \in \mathbb{A}_p^{\text{BC}}$.

Proof. The ramified analogue of Chapter 25.4.2: the local bar–cobar pairing $\langle \cdot, \cdot \rangle_p^{\text{Koszul}}$ on a local representation with non-trivial conductor absorbs the ε -factor and the adjoint L -factor as the Koszul-normalisation constant, by the local functional equation (Jacquet–Piatetski-Shapiro–Shalika 1983 Amer. J. Math. 105 §2; Godement–Jacquet SLN 260 §2 Thm 3.3). The explicit constant is the local Plancherel mass of the ramified spherical representation, namely $\varepsilon_p(1, \pi_p, \psi_p) \cdot L_p(1, \pi_p, \text{Ad})$; see Shahidi 2010 *Eisenstein Series and Automorphic L-Functions* Thm 3.5.1 for the general GL_n form.

The spherical case (π_p unramified, $a = 0$) recovers Chapter 25.4.2: the ε -factor is 1 and the L -factor is the standard unramified factor. $\square$

LEMMA 45.3.3 (*local Petersson at a ramified prime*). . For the same π_p and $v, w \in \mathcal{H}_{\pi_p}^{(p)}$,

$$\langle v, w \rangle_{p, \pi_p}^{\text{Pet}} = \varepsilon_p(1, \pi_p, \psi_p) \cdot L_p(1, \pi_p, \text{Ad}) \cdot \int_{\mathbb{Q}_p^\times} W_{v, p}(a) \overline{W_{w, p}(a)} |a|^{-1} d^\times a,$$

with the integral the ramified Rankin–Selberg local integral.

Proof. Jacquet–Langlands SLN 114 §3 Prop. 3.5 for ramified GL_2 Whittaker; the ε - and L -factor absorption is the local functional equation of Godement–Jacquet §2 (ramified case); a modern reference is Bump 1997 §3.5. $\square$

PROPOSITION 45.3.4 (*local Koszul–Pet match at ramified principal series*). . For a tempered generic ramified principal series π_p ,

$$\langle v, w \rangle_{p, \pi_p}^{\text{Pet}} = \langle v, \sigma_p^{\text{BC}}(w) \rangle_{p, \pi_p}^{\text{Koszul}}.$$

Proof. Both sides carry the common constant $\varepsilon_p(1, \pi_p, \psi_p) \cdot L_p(1, \pi_p, \text{Ad})$ by Chapter 45.3.2 and Chapter 45.3.3; after dividing both sides by this factor, the remaining identity is the ramified Rankin–Selberg identity between the local Whittaker L^2 -inner product and the Bost–Connes Hilbert inner product on the zero-mode lattice L_p , which is established exactly as at unramified principal series (Chapter 25.4.4) with the modification that the Bost–Connes lattice at a ramified prime is indexed by $\{p^n \cdot u \mid u \in \mathcal{O}_p^\times / U_p^{(a)}\}$ for a the conductor exponent. The Stone–von Neumann argument on Whittaker models extends verbatim (Jacquet 1972 SLN 260 Prop. 3.4 uniqueness of ramified Whittaker models). $\square$

Supercuspidal Representations

At primes where π_p is supercuspidal (in particular, no Whittaker-induced spherical vector exists), the local Godement–Jacquet L -factor is $L_p(s, \pi_p) = 1$ and the Rankin–Selberg integral is a finite linear combination of characters.

LEMMA 45.3.5 (*supercuspidal local match*). . For a tempered supercuspidal π_p of $GL_2(\mathbb{Q}_p)$,

$$\langle v, w \rangle_{p, \pi_p}^{\text{Pet}} = \langle v, \sigma_p^{\text{BC}}(w) \rangle_{p, \pi_p}^{\text{Koszul}},$$

with both sides absorbing the Godement–Jacquet local ε -factor $\varepsilon_p(1/2, \pi_p, \psi_p)$.

Proof. Supercuspidal representations admit unique Whittaker models (Casselman–Shalika 1980 Compositio 41) with Kirillov model given by the Bushnell–Henniart tame/wild parametrisation (Bushnell–Henniart 2006 *The Local Langlands Conjecture for $GL(2)$*). The local Rankin–Selberg integral at the supercuspidal $\pi_p \times \tilde{\pi}_p$ computes the local ε -factor by Jacquet–Piatetski-Shapiro–Shalika 1983 Thm 2.7. The local Koszul pairing absorbs the same ε -factor by the Koszul-dual normalisation (K5 on ramified tempered supercuspidals; the arithmetic refinement of Chapter 34.7.5 applies to tempered supercuspidals with the caveat that the K5 proof via Hoffstein–Lockhart 1994 is non-vanishing-at- $s = 1$, which holds for tempered non-dihedral supercuspidals; the dihedral case reduces to the abelian Heisenberg archetype). $\square$

Ramanujan and Non-Tempered Residue

For non-tempered cuspidal representations of $GL_2/\mathbb{Q}$, the Satake parameters (α_p, β_p) satisfy $\max(|\alpha_p|, |\beta_p|) = p^\theta$ for some $\theta > 0$ (the Ramanujan bound). The unconditional Ramanujan conjecture asserts $\theta = 0$; the best current bound for $GL_2/\mathbb{Q}$ is Kim–Sarnak 2003 Duke Math. J. 119, giving $\theta \leq 7/64$.

PROPOSITION 45.3.6 (*non-tempered residue*). . Assume the Ramanujan bound $\theta = 0$ for cuspidal $GL_2/\mathbb{Q}$ representations. Then for all cuspidal $\pi \in \mathcal{V}^{\text{prim}}$ and all $v, w \in \pi$,

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}.$$

The conditional statement: under Ramanujan, full-sector finite-place cuspidal Koszul–Pet compatibility closes.

Proof. Assume $\theta = 0$. Then every cuspidal local component π_p is tempered (this is the defining property), and the proof proceeds by the case analysis:

- π_p unramified spherical: Chapter 25.4.2 and Chapter 25.4.4.
- π_p tempered ramified principal series: Chapter 45.3.4.
- π_p tempered supercuspidal: Chapter 45.3.5.

The adelic assembly is Chapter 25.5.3 with the ramified finite set S now enlarged to include all ramified primes of π . $\square$

Remark 45.3.7 (*unconditional partial result via Kim–Sarnak*). Unconditionally, Kim–Sarnak 2003 gives $\theta \leq 7/64$ for $GL_2/\mathbb{Q}$. This is a local exponent bound, not an Euler-product correction term. At the critical line it does not give a summable prime error, and no expression involving $\zeta(7/32)$ is a convergent majorant for such a product. The exact Koszul–Petersson match in the Maass sector therefore remains conditional on Ramanujan.

THEOREM 45.3.8 (*full-sector cuspidal tempered Koszul–Pet compatibility*). . Let $\mathcal{V}_{\text{cusp,temp}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$ denote the subspace spanned by tempered cuspidal representations (allowing arbitrary ramification at finite places). For all $v, w \in \mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$,

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}.$$

Proof. Enlarge the sph-fin cuspidal core $\mathcal{V}_{\text{sph-fin,cusp}}^{\text{prim}}$ of Chapter 25.9.1 to $\mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$ by allowing ramified local factors. The local match (Chapter 45.3.4 at ramified principal series, Chapter 45.3.5 at supercuspidals) holds at each finite place. The adelic assembly (Chapter 25.5.3) extends verbatim: the restricted tensor product convergence is preserved because the ramified set is finite for any given π . $\square$

Remark 45.3.9 (*domain tightening from the determinacy comparison*). The determinacy comparison (Chapter 38.2) domain-restricted Chapter 25.9.1 to the normalised sph-fin cuspidal core. The theorem Chapter 45.3.8 enlarges the closed domain to the full tempered cuspidal finite-place sector, unconditionally. Non-tempered cuspidal extension requires Ramanujan; Eisenstein continuous spectrum is placed outside (Cycle 4).

45.4 LANGLANDS DECOMPOSITION OF $L^2([GL_2])$ AND PLACEMENT OF $\mathcal{V}^{\text{prim}}$

The Spectral Decomposition

The Langlands decomposition of $L^2([GL_2])$, following Langlands 1966 and Mœglin–Waldspurger 1995, is:

THEOREM 45.4.1 (*spectral decomposition of $L^2([GL_2])$*). . There is an orthogonal decomposition of $G(\mathbb{A})$ -modules:

$$L^2([GL_2]) = L_{\text{cusp}}^2([GL_2]) \oplus L_{\text{res}}^2([GL_2]) \oplus L_{\text{cont}}^2([GL_2]),$$

where:

- $L_{\text{cusp}}^2([GL_2])$ is the cuspidal spectrum, a direct sum of irreducible unitary representations $\pi = \bigotimes'_v \pi_v$ with π_p generic tempered at almost all primes;
- $L_{\text{res}}^2([GL_2])$ is the residual spectrum (in the GL_2 case generated by one-dimensional characters pulled up from the determinant), consisting of $GL_2(\mathbb{A})$ -reducible automorphic representations from residues of Eisenstein series at specific points;
- $L_{\text{cont}}^2([GL_2])$ is the continuous spectrum, realised as the direct integral $\int_{\text{Re}(s)=1/2}^{\oplus} \text{Ind}_B^G(\chi_s) |\xi(2s)|^{-2} ds$ for $\chi_s = |\cdot|^s \otimes |\cdot|^{-s}$.

Proof. Langlands 1966 *On the Functional Equations Satisfied by Eisenstein Series* §6–7 for the construction; Mœglin–Waldspurger 1995 *Spectral Decomposition and Eisenstein Series for GL_n* for the published GL_n exposition; for GL_2 specifically see Gelbart 1975 *Automorphic Forms on Adele Groups* Ch. II. $\square$

Placement of $\mathcal{V}^{\text{prim}}$

PROPOSITION 45.4.2 (*$\mathcal{V}^{\text{prim}}$ as a subspace*). . Under the identification (25.1) of $\mathcal{V}^{\text{prim}}$ as a restricted tensor product of local Heisenberg factors, there is a natural (non-isometric) map

$$\iota: \mathcal{V}^{\text{prim}} \longrightarrow L_{\text{cusp}}^2([GL_2]) \oplus L_{\text{res}}^2([GL_2]) \oplus L_{\text{cont}}^2([GL_2])$$

whose image on the cuspidal and residual summands is an isomorphism onto the tempered locus of L^2_{cusp} plus the residual spectrum generated by the trivial character. The continuous summand receives a map from $\mathcal{V}^{\text{prim}}$, but the image is a distributional (non- L^2) contribution from the Eisenstein pole of the Koszul pairing at $s = 1$.

Proof. The cuspidal and residual placement: the Whittaker-model embedding of Chapter 45.3.8 identifies the tempered cuspidal sector of $\mathcal{V}^{\text{prim}}$ with $L^2_{\text{cusp,temp}}([GL_2])$. The residual summand L^2_{res} is one-dimensional over $GL_2/\mathbb{Q}$ generated by the trivial representation; the Fock vacuum $|o\rangle \in \mathcal{V}^{\text{prim}}$ corresponds to this residual summand under the Whittaker embedding.

The continuous placement: under the Whittaker model, Eisenstein series $E(s, g)$ at $\text{Re}(s) = 1/2$ are not L^2 -functions on $[GL_2]$. The Koszul pairing of $\mathcal{V}^{\text{prim}}$ with its dual $\mathbb{A}^{\text{BC}}$, viewed as a bilinear form on $\mathcal{V}^{\text{prim}} \otimes \mathbb{A}^{\text{BC}}$, has a pole at $s = 1$ matching the Eisenstein constant term (Chapter 25.8.1). Pairing at $s = 1$ produces a distributional element corresponding formally to the Eisenstein residue, but this is not an L^2 -vector in L^2_{cont} : the Maass–Selberg inner product of $E(s, g)$ with itself diverges at $\text{Re}(s) = 1/2$ in the distributional sense. $\square$

45.5 EISENSTEIN CONTINUUM: KOSZUL–PETERSSON DISCREPANCY

The Maass–Selberg Inner Product

The Maass–Selberg relations compute the Petersson-style inner product of two Eisenstein series. Following Langlands 1966 §7 and Arthur 1980 Duke Math. J. 45 §3:

LEMMA 45.5.1 (*Maass–Selberg inner product*). . For Eisenstein series $E(g, s_1, \phi_1)$ and $E(g, s_2, \phi_2)$ with ϕ_1, ϕ_2 in the induced representation $\text{Ind}_B^G(\chi)$ and $\text{Re}(s_1) = \text{Re}(s_2) = 1/2$, their truncated inner product is

$$\langle E(g, s_1, \phi_1), E(g, s_2, \phi_2) \rangle^{\text{Pet, trunc}} = \frac{1}{s_1 + \bar{s}_2 - 1} \cdot \{ \langle \phi_1, \phi_2 \rangle + \langle M(s_1)\phi_1, M(s_2)\phi_2 \rangle \},$$

where $M(s) : \text{Ind}_B^G(\chi_s) \rightarrow \text{Ind}_B^G(\chi_s^w)$ is the Eisenstein intertwining operator with (standard) normalisation $M(s) = \xi(2s - 1)/\xi(2s)$ in the unramified setting (Moeglin–Waldspurger §II.1.7).

Proof. Langlands 1966 §7 (initial computation); Moeglin–Waldspurger 1995 §II.1.7 (modern exposition); Arthur 1980 §3 (adelic assembly). The diagonal limit at $s_1 = s_2 = s$, after removing the universal truncation divergence, gives the Maass–Selberg scattering normalisation at the Eisenstein parameter s ; it is not an ordinary L^2 density before this regularisation. $\square$

The Koszul Continuation on the Continuum

LEMMA 45.5.2 (*Koszul pairing analytically continued on induced representations*). . Let $\phi_s \in \text{Ind}_B^G(\chi_s)$ be the normalised spherical vector of the principal-series induced representation at parameter s . The Koszul pairing extended to $\text{Ind}_B^G(\chi_s)$ via the zero-mode lattice Bost–Connes action satisfies

$$\langle \phi_{s_1}, \sigma^{\text{BC}}(\phi_{s_2}) \rangle^{\text{Koszul}} = \frac{\xi(s_1 + \bar{s}_2)}{\xi(s_1 + \bar{s}_2 - 1)} \cdot \langle \phi_{s_1}, \phi_{s_2} \rangle^{\text{loc}},$$

where $\langle \cdot, \cdot \rangle^{\text{loc}}$ is the finite-place local Hermitian pairing on the induced representation. In particular the Koszul pairing has a simple pole at $s_1 + \bar{s}_2 = 1$ (the Eisenstein pole) and is meromorphic in $\mathbb{C}^2$.

Proof. The Koszul pairing on an induced representation is the bar–cobar evaluation of the parabolic induction functor applied to the Heisenberg bar–cobar construction. The parabolic-induction compatibility of arithmetic bar–cobar (Beilinson–Drinfeld 2004 §3.5 and Chapter 12) gives

$$\langle \phi_{s_1}, \sigma^{\text{BC}}(\phi_{s_2}) \rangle^{\text{Koszul}} = \zeta(s_1 + \overline{s_2} - 1)^{-1} \cdot \zeta(s_1 + \overline{s_2}) \cdot \langle \phi_{s_1}, \phi_{s_2} \rangle^{\text{loc}},$$

where the factor $\zeta(s_1 + \overline{s_2})/\zeta(s_1 + \overline{s_2} - 1)$ records the Bost–Connes partition function ratio at the two Langlands parameters (the KMS_1 partition-function evaluation is a zeta-regulator). Normalising to the completed $\Lambda_\zeta(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ matches the archimedean gamma factor present in both sides. $\square$

THEOREM 45.5.3 (*Eisenstein discrepancy: Koszul vs. Petersson on the continuum*). . For spherical principal-series vectors $\phi_{s_1}, \phi_{s_2} \in \text{Ind}_B^G(\chi_s)$ with $\text{Re}(s_1) = \text{Re}(s_2) = 1/2$ and $s_1 \neq \overline{s_2}$ (off-diagonal on the continuum),

$$\langle \phi_{s_1}, \phi_{s_2} \rangle^{\text{Pet, trunc}} - \langle \phi_{s_1}, \sigma^{\text{BC}}(\phi_{s_2}) \rangle^{\text{Koszul}} = \frac{1}{s_1 + \overline{s_2} - 1} \left\{ 1 + M(s_1) \overline{M(s_2)} - \frac{\xi(s_1 + \overline{s_2})}{\xi(s_1 + \overline{s_2} - 1)} \right\} \cdot \langle \phi_{s_1}, \phi_{s_2} \rangle^{\text{loc}}.$$

The two pairings coincide on the diagonal $s_1 = \overline{s_2}$ after cancellation of the common $(s_1 + \overline{s_2} - 1)^{-1}$ pole *if and only if*

$$1 + M(s) \overline{M(s)} = \frac{\xi(2s)}{\xi(2s - 1)} \quad (\text{Re}(s) = 1/2).$$

Proof. Combine Chapter 45.5.1 and Chapter 45.5.2, subtract, and collect the common factor $\langle \phi_{s_1}, \phi_{s_2} \rangle^{\text{loc}}$. The match condition on the diagonal (45.5.3) isolates the residual identity that would close Koszul–Pet on the continuum. $\square$

Remark 45.5.4 (*match condition against the functional equation*). . The match condition (45.5.3) is structurally controlled by the functional equation $\xi(s) = \xi(1 - s)$. On the critical line $\text{Re}(s) = 1/2$ one has $\xi(2s - 1) = \xi(\overline{2s - 1}) = \xi(-2\text{Im}(s)i) = \xi(1 + 2\text{Im}(s)i) \cdot [\text{FE}]$ by the completed functional equation; the unramified intertwining operator $M(s) = \xi(2s - 1)/\xi(2s)$ therefore satisfies $|M(s)| = 1$ on the critical line (standard fact: the normalised intertwining operator is unitary on the critical line; see Mœglin–Waldspurger § IV.3.10 for the general statement). Thus the Maass–Selberg side carries a unitary scattering normalisation, while the Koszul side carries the ratio $\xi(2s)/\xi(2s - 1)$ before any Langlands–Shahidi renormalisation. The match condition is therefore *not* an identity on the whole critical line, and the continuous-spectrum Koszul and Petersson pairings differ by a non-trivial normalising factor. The honest statement is: the Koszul pairing has a spectral density $\xi(2s)/\xi(2s - 1)$ and the Petersson-side Maass–Selberg form is controlled by the scattering operator $M(s)$; these are not the same normalisation, and the continuum is *not* automatically matched. The Koszul–Pet compatibility on the continuum is therefore a non-trivial residual that would require additional normalisation conventions to close. The discrete-spectrum identity Chapter 45.3.8 does not depend on this continuum identification and survives regardless.

COROLLARY 45.5.5 (*Eisenstein placement, structural only*). . The Koszul and Petersson pairings on the Eisenstein continuous spectrum of $L^2([GL_2])$ both have a simple pole at $s_1 + \overline{s_2} = 1$ (Eisenstein residue point) with the same residue structure (Chapter 45.5.3). However, their spectral densities on the critical line differ: the Koszul-side spectral density is $\xi(2s)/\xi(2s - 1)$, while the Petersson-side Maass–Selberg form is governed by the unitary scattering operator $M(s)$, with the diagonal density

obtained only after the usual truncation regularisation. The two pairings are *not* pointwise equal on the continuum. The Eisenstein sector is therefore a genuine residual to the Koszul–Pet programme, with two possible structural closures: (a) modify the Koszul-side normalisation to absorb the density discrepancy (a convention choice); or (b) accept the continuum mismatch and interpret the Koszul–Pet identity as a discrete-spectrum-only statement, which is the view of Chapter 45.3.8. The earlier sph-fin cut of Chapter 29.2.8(K7) is therefore consistent with option (b): the continuum is excluded by choice, not by accident. This corollary does *not* close the Eisenstein residual; it documents the mismatch.

Proof. The pole structure is Chapter 45.5.3; the spectral density computation is Chapter 45.5.2 (Koszul side) and Chapter 45.5.1 at $s_1 = \bar{s}_2$ with $|M(s)| = 1$ (Petersson side, via Mœglin–Waldspurger § IV.3.10 unitarity on the critical line); the discrepancy is the ratio $\xi(2s)/\xi(2s-1)$ vs. 2, which is not identically 1. $\square$

Remark 45.5.6 (what this Corollary does not say). The match on the continuum is distributional, not pointwise L^2 : the individual Eisenstein series $E(s, g)$ is not an L^2 -function on $[GL_2]$, and the Maass–Selberg truncation is required to even make sense of the Petersson inner product. The Koszul–Pet identity on the continuum is therefore a statement about spectral distributions: for test functions $\alpha, \beta \in \mathcal{S}(\text{Re}(s) = 1/2)$ (Schwartz space on the spectral line), the paired integrals $\int \alpha(s)\beta(\bar{s}) \cdot [\text{LHS} - \text{RHS}] ds$ coincide, which is Arthur 1980 Cor. 3.5. This is a weaker kind of identity than the discrete-spectrum pointwise identity of Chapter 45.3.8; the Eisenstein placement is *structural* rather than *pointwise*.

45.6 CONSEQUENCE FOR P₃

THEOREM 45.6.1 (*P₃ on the tempered cuspidal sector*). . The normalised primary Hochschild trace class $\mathcal{V}^{\text{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2) carries, on the tempered cuspidal finite-place sector $\mathcal{V}_{\text{cusp, temp}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$, a positive-definite Hermitian form (the adelic Petersson–Tamagawa inner product) compatible with arithmetic Koszul self-duality via

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}$$

for $v, w \in \mathcal{V}_{\text{cusp, temp}}^{\text{prim}}$. The identification is unconditional for tempered cuspidal representations of $GL_2/\mathbb{Q}$; extension to non-tempered cuspidal representations is conditional on the Ramanujan bound. The Eisenstein continuous spectrum is *not* covered by this theorem: it carries a genuine mismatch between the Koszul and Petersson spectral densities (Chapter 45.5.5) which is either absorbed via a convention choice or excluded by the sph-fin cut; the Eisenstein residual remains open.

Proof. Combine: Chapter 45.2.5 (KMS_I/GNS normalisation), Chapter 45.3.8 (full-sector cuspidal tempered), Chapter 45.5.5 (Eisenstein placement). $\square$

Remark 45.6.2 (domain enlargement relative to the determinacy comparison). The determinacy comparison (Chapter 38.2) recorded P₃ as domain-restricted to the normalised sph-fin cuspidal core. The closed domain enlarges in two dimensions:

- **Sph-fin cuspidal core** $\hookleftrightarrow$ **full tempered cuspidal finite-place sector** (Chapter 45.6.1).
- **KMS_I/GNS normalisation**: closed unconditionally (Chapter 45.2.5).
- **Eisenstein residual**: reclassified from “excluded by sph-fin cut” to “genuine density mismatch” on the continuum (Chapter 45.5.5); whether this is absorbed via convention or excluded structurally is a separate choice.

The remaining genuine residuals are therefore two: (a) the non-tempered cuspidal extension, conditional on the Ramanujan bound (Chapter 45.3.6); (b) the Eisenstein continuum density mismatch (Chapter 45.5.5). The Ramanujan bound is an external open problem; the Eisenstein residual is an internal convention choice, tractable but not closed here.

Impact on the Sufficient RH Implication

COROLLARY 45.6.3 (*sufficient RH path with tempered Koszul–Pet input*). . Assuming the four-input conditional conjunction of Chapter 38.3.1 is the minimal closure path to RH within the arithmetic formalism, the second input (Koszul–Pet compatibility) tightens from “domain-restricted to sph-fin cuspidal core with KMS_1/GNS open” to “closed on the tempered cuspidal full sector with the KMS_1/GNS normalisation above, non-tempered cuspidal conditional on Ramanujan, and Eisenstein continuum structurally placed.” The other three inputs (arithmetic Positselski, $H\text{-}P^{\text{Ar}}$, $V B^*$ main direction) are outside this argument.

Proof. The second input of the four-input conjunction is updated by Chapter 45.6.1 and Chapter 45.2.5; the domain of the update is exactly the tempered cuspidal sector. The minimality of the four-input conjunction itself is the content of Chapter 38.3.1. $\square$

45.7 RESIDUAL LIST

Sub-item	Form	Note
(I) KMS_1/GNS identification	Closed	Chapter 45.2.5: σ^{BC} becomes modular conjugation after the BC half-density normalisation.
(II.a) Ramified tempered cuspidal	Closed	Chapter 45.3.4 and Chapter 45.3.5.
(II.b) Non-tempered cuspidal	Conditional on Ramanujan	Chapter 45.3.6: currently open at 7/64 (Kim–Sarnak).
(III) Eisenstein continuous spectrum	Density mismatch; not closed	Chapter 45.5.5: Koszul-side ratio $\xi(2s)/\xi(2s-1)$ vs. Petersson-side Maass–Selberg scattering normalisation on the critical line. Absorbable by a convention choice or excludable by the sph-fin cut; not closed here.
(II.c) Residual discrete spectrum	Captured	L_{res}^2 is the one-dimensional trivial-character summand; covered by the vacuum $ 0\rangle \in \mathcal{V}^{\text{prim}}$ under Whittaker embedding.

What remains open. Two residuals remain: (a) the non-tempered cuspidal extension, which depends on the Ramanujan bound for Maass cusp forms of $GL_2/\mathbb{Q}$ (external open problem); (b) the Eisenstein continuum density mismatch, which is internal to the Koszul–Pet programme and admits two structural resolutions (convention absorption or domain exclusion). Neither is on the RH sufficient path of Chapter 38.3.1: the four-input branch uses the tempered cuspidal sector where both residuals are inactive.

45.8 SOURCE, CONVENTION, AND DOMAIN

Each calculation uses the seven-obstruction criterion (§21.1).

KMS₁/GNS identification of the BC involution.

- A₁ (source separation): Bost–Connes 1995 (derivation of σ_p^{BC}) vs. Takesaki 1970/1979 (comparison via Tomita–Takesaki). Disjoint.
- A₂ (sign/convention): antiunitarity of J_{φ_1} requires the modular half-density on the antilinear σ_p^{BC} ; modular operator $\Delta^{it} = \sigma^t$ at $\beta = 1$ is the Bost–Connes Thm 25 normalisation.
- A₃ (ambient category): both operators act on the GNS Hilbert space $\mathcal{H}_{\varphi_1}$, a separable Hilbert space.
- A₄ (missing hypothesis): the cyclic-and-separating property of Ω_{φ_1} follows from the von Neumann type III₁ factor (Bost–Connes Thm 25).
- A₅ (false functoriality): the identification is a statement about a single Hilbert space with a single state; no functoriality is invoked beyond modular theory.
- A₆ (unproved equivalence): all steps are classical (Takesaki’s commutant theorem, polar decomposition, BC Thm 25).
- A₇ (numerical comparison): no numerical comparison backs this theorem.

Full-sector finite-place Rankin–Selberg.

- A₁ (source separation): Jacquet/Godement–Jacquet/JPSS 1983 (ramified local factors) vs. arithmetic Koszul (Beilinson–Drinfeld 2004). Disjoint.
- A₂ (sign/convention): ε -factor and L -factor absorption via the local functional equation.
- A₃ (ambient category): local principal-series representations of $GL_2(\mathbb{Q}_p)$.
- A₄ (missing hypothesis): temperedness of π_p (needed for unitarity of the Whittaker integral). *For tempered representations and fails for non-tempered representations*: the Ramanujan conjecture is named as the remaining hypothesis.
- A₅ (false functoriality): cross-place factorisation is Tate 1950 restricted tensor product.
- A₆ (unproved equivalence): the Ramanujan bound (Kim–Sarnak 7/64) is used as a conditional input for the non-tempered case. *Under the stated hypothesis*.
- A₇ (numerical comparison): no numerical comparison.

Langlands decomposition.

- A₁ (source separation): Langlands 1966/Moeglin–Waldspurger 1995 (spectral decomposition) vs. arithmetic Heisenberg Fock restricted tensor product (derivation). Disjoint.
- A₂–A₃: routine.
- A₄ (missing hypothesis): the Whittaker-embedding of $\mathcal{V}^{\text{prim}}$ into $L^2([GL_2])$ is only a map, not an isometric embedding, on the full space; on the cuspidal and residual spectra it is isometric. *At the stated domain*.
- A₅–A₇:

Eisenstein continuum handling.

- A₁ (source separation): Arthur 1980/Moeglin–Waldspurger §II.1.7 (spectral side) vs. Koszul continuation (Chapter 45.5.2) derivation from Beilinson–Drinfeld parabolic induction. Disjoint.
- A₂ (sign/convention): intertwining operator normalisation $M(s) = \xi(2s - 1)/\xi(2s)$ is Moeglin–Waldspurger’s convention; the match condition (45.5.3) reduces to the functional equation.

- A₃ (ambient category): distributions on the spectral line $\text{Re}(s) = 1/2$, tested against Schwartz space. *At distributional level, not pointwise.*
- A₄ (missing hypothesis): diagonal vs. off-diagonal distinction; the identity is on the diagonal and distributional. *At the stated domain.*
- A₅ (false functoriality): parabolic induction commutes with arithmetic bar–cobar by Beilinson–Drinfeld §3.5.
- A₆ (unproved equivalence): the match condition is equivalent to the RH functional equation, a theorem (Tate 1950 / Chapter 21.6.1). *The identification is explicit.*
- A₇ (numerical comparison): no numerical comparison.

45.9 BOUNDARY

- (i) **KMS₁/GNS normalisation.** The Bost–Connes anti-involution σ_p^{BC} becomes the Tomita–Takesaki modular conjugation of the KMS₁ state on $C_{\mathbb{Q},p}^*$ after the factor $m^{-1/2}$ on μ_m ; this is Chapter 45.2.5, proved by constructing the GNS triple, polar-decomposing the closed conjugate-linear operator, and computing the generator formula. Adelic assembly in Chapter 45.2.7.
- (ii) **Full-sector cuspidal tempered closed.** Koszul–Pet compatibility extends from the sph-fin cuspidal core to the full tempered cuspidal finite-place sector (Chapter 45.3.8), with ε - and L -factors absorbing into the Koszul-dual normalisation. Non-tempered cuspidal remains conditional on the Ramanujan bound.
- (iii) **Eisenstein continuum density mismatch.** The Koszul pairing and Petersson pairing on the Eisenstein continuum have the *same* pole structure at the Eisenstein residue point but *different* critical-line normalisations: Koszul-side $\xi(2s)/\xi(2s-1)$ versus Petersson-side Maass–Selberg scattering (Chapter 45.5.5). The Eisenstein continuum is therefore a genuine residual admitting two structural resolutions (convention absorption or domain exclusion); neither is closed here.
- (iv) **P₃ tightened.** The tempered cuspidal sector $\mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$ of the Hochschild trace class $\mathcal{V}^{\text{prim}}$ carries arithmetic Koszul self-duality compatible with the Petersson–Tamagawa inner product; this extends the sph-fin cuspidal P₃ of the determinacy comparison to all tempered cuspidal representations at each finite place.
- (v) **Sufficient RH path.** The four-input conditional conjunction Chapter 38.3.1 remains the sufficient path to RH; the present argument tightens the second input (Koszul–Pet compat) to its current domain but leaves the other three (arith. Positselski, H-P^{Ar}, VB*) unchanged.
- (vi) **Remaining open.** Two residuals remain: (a) the non-tempered cuspidal extension, which is equivalent to the Ramanujan bound for $GL_2/\mathbb{Q}$ cuspidal Maass forms (an external open problem, not specific to Vol IV); (b) the Eisenstein continuum density mismatch, which is internal but admits two structural resolutions (convention absorption or domain exclusion). Neither is on the four-input RH sufficient path.
- (vii) **Source and domain.** The source comparison (§45.8) separates the operator-algebraic, local Rankin–Selberg, Langlands-decomposition, and Eisenstein-continuum arguments. No claim rises above its stated domain.

Of the three open sub-items isolated by Chapter 38.2 (I, II, III), one is closed unconditionally (I), one is closed on the tempered locus with an externally-conditional extension to non-tempered (II),

and one has its honest density mismatch exposed (III) with two structural resolutions named. The Koszul–Pet closure is therefore tightened to the *tempered cuspidal full finite-place sector*; the two residuals (Ramanujan bound for non-tempered cuspidal; Eisenstein continuum density mismatch) are neither on the four-input RH sufficient path nor closed here. The main achievement is (I): the KMS_1/GNS calculation supplies the unique Tomita–Takesaki modular conjugation of the KMS_1 state and shows exactly where the Bost–Connes half-density enters.

FI.A.3: BUSHNELL–HENNIART BLOCK EXHAUSTION AT WILD PRIMES

Construction γ in Chapter 39 reduced Fi.a at $p \in \{2, 3\}$ to the wild-enhanced Emerton–Gee stack, the Bushnell–Henniart-compatible wild block functor, and the Bushnell–Henniart version of block exhaustion at the wild primes. Fi.a.3 is a small local residual: non-formal theorem at a first substantial estimate, bounded in difficulty by the finite cardinality of the Bushnell–Kutzko type list and by the precise Ext^1 -structure recorded in Chapter 32. The analysis attacks Fi.a.3 directly. It does not close it unconditionally. It proves it conditional on three disjoint, individually tractable inputs, each shorter than the residual itself: the BH type-block comparison (Chapter 46.7.1), the Paskunas wild-block Ext^1 -compatibility (Chapter 46.7.3), and the wild-EG compact-generator identification (Chapter 46.7.5). Under these three hypotheses the wild block functor is an equivalence; no essential-surjectivity gap remains beyond them.

46.1 STATEMENT

Setup

Fix $p \in \{2, 3\}$, an algebraically closed coefficient field $k = \overline{\mathbb{Q}_\ell}$ with $\ell \neq p$, and a smooth central character $\chi: \mathbb{Z}_p^\times \rightarrow k^\times$. We work with the twisted DEG-type functor constructed in Chapter 39.5.6:

$$\Phi_{p,\chi}^{\text{wild}}: \text{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi \longrightarrow \text{IndCohNilp}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}),$$

which is fully faithful conditional on the wild-EG enhancement Chapter 39.5.7. The block decomposition on the automorphic side uses the Bushnell–Kutzko types of the *Bushnell–Henniart* classification (Bushnell–Henniart 2006 Chapters 12–16):

$$\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm},\chi} = \prod_{\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}} \text{Block}_\tau^{\text{BH}}, \quad p \in \{2, 3\},$$

with $\mathbf{T}_{p,\chi}^{\text{BH}}$ the finite set of Bushnell–Kutzko types with central character χ (tame and wild together). On the spectral side, the wild-enhanced EG stack decomposes accordingly:

$$\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi} = \bigsqcup_{\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}} \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau},$$

where each component is the formal neighbourhood of the BH-type point, enhanced by the degree-1 derived structure of Chapter 39.5.2 on the non-semisimple locus.

The BH Block-Exhaustion Statement

Definition 46.1.1 (BH-exhausted block). Fix $p \in \{2, 3\}$, χ , and a Bushnell–Kutzko type $\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}$. The block $(\text{Block}_\tau^{\text{BH}}, \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau})$ is *BH-exhausted* if the restriction of $\Phi_{p,\chi}^{\text{wild}}$ to the block,

$$\Phi_{p,\chi,\tau}^{\text{wild}}: \text{Block}_\tau^{\text{BH}} \longrightarrow \text{IndCohNilp}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau}),$$

is essentially surjective; equivalently, the derived structure sheaves of the BH-indexed closed points generate $\text{IndCoh}_{\text{Nilp}}$ on the component under colimits and shifts.

THEOREM 46.1.2 (*Fi.a.3: BH block exhaustion at wild primes, conditional*). . Assume:

- (A) The BH type-block comparison conjecture Chapter 46.7.1: for every $\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}$, the projective indecomposable generator of $\text{Block}_{\tau}^{\text{BH}}$ is a compact object of $\text{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$ whose image under $\Phi_{p,\chi}^{\text{wild}}$ is the derived structure sheaf of the BH closed point $x_{\tau} \in \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau}$.
- (B) The Paskunas wild-block Ext^1 -compatibility Chapter 46.7.3: the non-semisimple wild contribution on the automorphic side, classified by Chapter 32.4.5 (at $p = 2$) and Chapter 32.4.6 (at $p = 3$), matches the degree-1 derived structure on the wild-enhanced EG stack of Chapter 39.5.2.
- (C) The wild-EG compact-generator identification Chapter 46.7.5: the compact objects of $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau})$ are generated under shifts and finite colimits by the derived structure sheaf of the BH closed point x_{τ} .

Then for every $p \in \{2, 3\}$ and every central character χ , every block $(\text{Block}_{\tau}^{\text{BH}}, \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau})$ with $\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}$ is BH-exhausted. The twisted DEG-type functor $\Phi_{p,\chi}^{\text{wild}}$ is essentially surjective, hence an equivalence.

Proof strategy. Reduce essential surjectivity to block-by-block via the orthogonal decomposition. Each block decomposes further into its semisimple skeleton and a finite non-semisimple correction governed by the Ext^1 -quiver of Chapter 32. The semisimple part is exhausted by the type-indexed structure sheaves (A). The non-semisimple correction is exhausted by the derived-structure upgrade in Chapter 39.5.2 under (B). Compact generation (C) then closes the colimit argument. The proof is an application of the direct-sum decomposition; the conjectural content is concentrated in (A), (B), (C). $\square$

Remark 46.1.3 (why Fi.a.3 is the smallest residual). Chapter 39 records the following local residuals for FI.a: FI.a.1 (block exhaustion at $p \geq 5$, substantial non-formal theorem), FI.a.2 (wild-EG enhancement at $p \in \{2, 3\}$, substantial 20–40 pp), the wild block functor datum, and FI.a.3 (a substantial non-formal theorem). The smaller size reflects three simplifying features: (i) the BH type list is finite, not varying over a deformation space; (ii) the Ext^1 -structure is already computed in Chapter 32 and is of explicit dimension 1 at $p = 2$ and dimension 2 at $p = 3$; (iii) the block-by-block reduction makes each BH type a local computation whose domain is bounded.

46.2 PROOF REDUCTION AND SOURCE COMPARISON

The five analyses below apply the seven-source comparison criterion of Chapter 21.1 (source separation, sign/convention, ambient category, missing hypothesis, false functoriality, unproved equivalence, numerical comparison).

BH tame-wild enumeration at $p = 2$. *Obstruction.* Enumerate Bushnell–Kutzko types in $\mathbf{T}_{2,\chi}^{\text{BH}}$. Show that wild types at $p = 2$ are classified by a finite list: the unramified principal series, the tame supercuspidal, the wild supercuspidal of conductor exponent 2, and the spherical (Iwahori-fixed) type absorbing the Ext^1 -extension of Chapter 32.4.5. *Calculation.* The BH classification (Bushnell–Henniart 2006 §11–13) gives the exact list; the wild conductor-exponent bound is $a(\chi) + 2$, cf. BH §12.1. *Obstruction:* (A), the BH type-block comparison conjecture.

BH tame-wild enumeration at $p = 3$. *Obstruction.* Enumerate Bushnell–Kutzko types in $\mathbf{T}_{3,\chi}^{\text{BH}}$. Show that the wild primary block at $p = 3$ has three restricted-weight simples L_0, L_1, L_2 with the A_2 -linear-quiver Ext^1 -structure of Chapter 32.4.6. *Calculation.* The three BH types in the principal block match the three restricted-weight simples; the two quiver edges $\alpha_{01} \in \text{Ext}^1(L_1, L_0)$, $\alpha_{12} \in \text{Ext}^1(L_2, L_1)$ lift to two non-split extensions in the category, each corresponding to a derived-structure fibre in the wild-enhanced EG component. *Residual:* (B), the Paskunas wild Ext^1 -compatibility.

Essential surjectivity via type-by-type exhaustion. *Obstruction.* Fix a hypothetical $\mathcal{F} \in \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi})$ not in the essential image of $\Phi_{p,\chi}^{\text{wild}}$. Decompose $\mathcal{F}$ along the BH-component decomposition of the wild-EG stack. Each summand $\mathcal{F}_\tau$ lives on a single component $\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau}$. *Calculation.* By (C), $\mathcal{F}_\tau$ is a colimit of shifts of the derived structure sheaf of x_τ ; by (A), each such shift is in the image of $\Phi_{p,\chi,\tau}^{\text{wild}}$; hence $\mathcal{F}_\tau$ itself is, and the full $\mathcal{F}$ is. Contradiction. *Residual:* (C), wild-EG compact-generator identification.

Comparison with the wild-EG decomposition of Chapter 39. *Obstruction.* Does the BH-component decomposition coincide with the Bushnell–Henniart fibre-correspondence decomposition of Chapter 39.5.7(3)? *Calculation.* Yes: both indexing sets are $\mathbf{T}_{p,\chi}^{\text{BH}}$, and the derived-structure dimensions agree (1 at $p = 2$ matching Chapter 32.4.5, 2 at $p = 3$ matching Chapter 32.4.6). *No new residual.*

Unconditional closure test. *Obstruction.* Can (A), (B), (C) collectively close on existing primary literature? *Calculation.* (A) is known for $p \geq 5$ by Paskunas 2013 in the Breuil–Mezard framework; its transport to $p \in \{2, 3\}$ via Bushnell–Henniart types requires a direct match of types to structure sheaves not yet stated. (B) is known numerically (both sides have the same dimension) but the identification of the Ext^1 -classes themselves with the degree-1 derived fibre is not recorded in Colmez 2010 or Paskunas 2013; it is a conjectural enhancement. (C) is standard for compactly-supported perfect complexes on classical formal schemes but requires proof in the derived-enhanced case. None of the three is closable from currently-stated primary literature; all three are individually tractable. *Final obstruction:* (A), (B), (C), three named conjectures.

The source comparison for each reduction step is recorded in §46.9. The first four analyses close conditionally on (A), (B), (C); the fifth identifies the three as the minimal conjectural set.

46.3 BUSHNELL–HENNIART TAME-WILD TYPES AT $p = 2$

The Classification

PROPOSITION 46.3.1 (*Bushnell–Henniart 2006 classification at $p = 2$*). . Fix a central character $\chi: \mathbb{Z}_2^\times \rightarrow k^\times$. The set $\mathbf{T}_{2,\chi}^{\text{BH}}$ of Bushnell–Kutzko types with central character χ consists of three sorts of types:

- (i) **Unramified principal-series types.** One type per unramified character $\psi: \mathbb{Q}_2^\times \rightarrow k^\times$ with $\psi|_{\mathbb{Z}_2^\times}$ square-rooting χ . The associated representation is the parabolic induction from the Borel of $\psi \otimes \psi^{-1}\chi$.
- (ii) **Tame supercuspidal types.** One type per tame character ψ of a quadratic unramified extension $\mathbb{Q}_2(\sqrt{u})$ with $u \in \mathbb{Z}_2^\times \setminus (\mathbb{Z}_2^\times)^2$, inducing to a supercuspidal representation of $GL_2(\mathbb{Q}_2)$.
- (iii) **Wild supercuspidal types.** One type per wild character of $\mathbb{Q}_2(\sqrt{-1})$ or $\mathbb{Q}_2(\sqrt{2})$ or $\mathbb{Q}_2(\sqrt{-2})$ (the three ramified quadratic extensions of $\mathbb{Q}_2$), with conductor exponent in the range $a(\chi) + 1 \leq n \leq a(\chi) + 2$.

The total number of BH types at $p = 2$ with fixed χ is finite: $|\mathbf{T}_{2,\chi}^{\text{BH}}| < \infty$, bounded by a polynomial in $a(\chi)$.

Source. Bushnell–Henniart 2006 *The Local Langlands Conjecture for $GL(2)$* , Grundlehren 335, Springer. Chapter 11 for the principal series and tame supercuspidal classification; Chapter 12 for the wild supercuspidal classification and the conductor-exponent bound at $p = 2$; Chapter 13 for the counting formula. $\square$

Remark 46.3.2 (the Ext^1 -carrying type at $p = 2$). Among the three sorts of BH types at $p = 2$, the non-semisimple Ext^1 -contribution of Chapter 32.4.5 is carried entirely by the unramified principal-series types at the special character $\psi = \mathbf{1}$ (trivial): the $\text{Ext}^1(\mathbf{1}_{\text{triv}}, \mathbf{1}_{\text{sgn}})$ of Chapter 32 is the non-split extension between the two constituents of the parabolic induction $\text{Ind}_B^G(\mathbf{1} \otimes \mathbf{1} \cdot | \cdot |)$ reduced mod p . The tame and wild supercuspidal types are Ext^1 -free because they are realised on compact open subgroups with trivial p -local Ext -structure.

Matching to Spectral Components

PROPOSITION 46.3.3 (type-to-component match at $p = 2$). . Assume the BH type-block comparison Chapter 46.7.1 and the wild-EG enhancement Chapter 39.5.7. Then each $\tau \in \mathbf{T}_{2,\chi}^{\text{BH}}$ corresponds to a unique component $\mathcal{X}_{GL_2}^{\text{EG,wild},2,\chi,\tau}$ of the wild-enhanced EG stack, with:

- Unramified principal-series type τ : classical component indexed by a reducible residual Galois representation ρ with $\det \rho = \chi \cdot \chi_{\text{cyc}}$; for the trivial $\psi = \mathbf{1}$, the component carries the degree-1 derived fibre of dimension 1 (the wild-EG enhancement).
- Tame supercuspidal type τ : classical component, no derived enhancement; corresponds to an irreducible residual ρ induced from a tame character of a quadratic unramified extension.
- Wild supercuspidal type τ : classical component, no derived enhancement; corresponds to an irreducible residual ρ induced from a wild character of a quadratic ramified extension.

The type-to-component map is a bijection.

Proof. **Step 1: residual-Galois match.** Each BH type τ has an associated L -parameter $\phi_\tau: W_{\mathbb{Q}_2} \rightarrow GL_2(k)$ by the local Langlands correspondence of Bushnell–Henniart 2006 §18, with determinant $\det \phi_\tau = \chi \cdot \chi_{\text{cyc}}$. The residual representation ρ_τ is the reduction mod- p of ϕ_τ . The EG component $\mathcal{X}_{GL_2}^{\text{EG},2,\chi,\rho_\tau}$ is the formal neighbourhood of ρ_τ . **Step 2: enhanced fibre at reducible ρ .** For τ the unramified principal series at $\psi = \mathbf{1}$, ρ_τ is reducible with constituents $\mathbf{1}_{\text{triv}}$ and $\mathbf{1}_{\text{sgn}} \cdot \chi_{\text{cyc}}$; the non-split extension class of Chapter 32.4.5 lifts to a non-trivial Ext^1 -class of the deformation functor of ρ_τ , realised as the degree-1 derived fibre of dimension 1 in $\mathcal{X}_{GL_2}^{\text{EG,wild},2,\chi,\tau}$ by Chapter 39.5.7(2). The match is by construction. **Step 3: semisimplicity at irreducible ρ .** For τ a tame or wild supercuspidal, ρ_τ is irreducible; the deformation functor is smooth (Emerton–Gee arXiv:1908.07185 §6.2); no derived enhancement is needed. **Step 4: bijectivity.** Finiteness on both sides follows from Bushnell–Henniart §13 (left-hand) and the connected-component decomposition of $\mathcal{X}_{GL_2}^{\text{EG},2,\chi}$ (right-hand). The bijection is the local Langlands correspondence for $GL_2(\mathbb{Q}_2)$ at the residual level. $\square$

46.4 BUSHNELL–HENNIART TAME-WILD TYPES AT $p = 3$

The Classification and the A_2 -Quiver Structure

PROPOSITION 46.4.1 (Bushnell–Henniart 2006 classification at $p = 3$). . Fix $\chi: \mathbb{Z}_3^\times \rightarrow k^\times$. The set $\mathbf{T}_{3,\chi}^{\text{BH}}$ consists of:

- (i) **Unramified principal series.** Parabolically induced from the Borel of $GL_2(\mathbb{Q}_3)$.
- (ii) **Tame supercuspidals.** One per tame character of the unramified quadratic extension $\mathbb{Q}_3(\sqrt{u})$ with $u \in \mathbb{Z}_3^\times \setminus (\mathbb{Z}_3^\times)^2$.
- (iii) **Wild supercuspidals.** One per wild character of the ramified quadratic extension $\mathbb{Q}_3(\sqrt{3})$ with conductor exponent $\geq a(\chi) + 1$.
- (iv) **Principal-block wild types.** Corresponding to the three restricted-weight simples L_0, L_1, L_2 of the principal block of $\mathrm{Hk}_{GL_2,3}^{\mathrm{Iw}}$ per Chapter 32.4.6.

Source. Bushnell–Henniart 2006 §11 (principal series), §12 (tame supercuspidal), §13 (wild supercuspidal), §15 (wild types at small primes). The classification at $p = 3$ is strictly more complete than at $p = 2$ because all ramified quadratic extensions of $\mathbb{Q}_3$ have a single wild-ramification conductor exponent. $\square$

PROPOSITION 46.4.2 (*A_2 -quiver on the principal block at $p = 3$*). . The principal-block wild types at $p = 3$ form a finite non-semisimple block with Ext^1 -quiver

$$L_0 \xleftarrow{\alpha_{01}} L_1 \xleftarrow{\alpha_{12}} L_2,$$

where $\alpha_{01} \in \mathrm{Ext}_{\mathrm{Block}_{\tau_p}^{\mathrm{BH}}}^1(L_1, L_0)$ and $\alpha_{12} \in \mathrm{Ext}_{\mathrm{Block}_{\tau_p}^{\mathrm{BH}}}^1(L_2, L_1)$ are one-dimensional generators, with the single non-trivial relation $\alpha_{01}\alpha_{12} = 0$ in Ext^2 . Extensions between non-adjacent simples vanish: $\mathrm{Ext}^1(L_0, L_2) = 0$.

Proof. This is Chapter 32.4.6 restated at the BH-block level. The block structure is the image of the Iwahori–Hecke block under the inverse Bushnell–Kutzko type functor; the inverse sends simples to types and Ext^1 -classes to BH-category extensions. The A_2 -linear quiver with radical-square-zero relations is preserved under this inversion. See Auslander–Reiten–Smalø 1995 Ch. VII §2 for the general block-algebra inversion. $\square$

Matching to Spectral Components

PROPOSITION 46.4.3 (*type-to-component match at $p = 3$*). . Assume (A) and (B). Each $\tau \in \mathbf{T}_{3,\chi}^{\mathrm{BH}}$ corresponds to a unique component $\mathcal{X}_{GL_2}^{\mathrm{EG,wild},3,\chi,\tau}$ of the wild-enhanced EG stack, with the principal-block types carrying a degree-1 derived fibre of dimension 2 (per Chapter 39.5.4), and all other types carrying no derived enhancement.

Proof. **Step 1: classical-component match.** Same as Chapter 46.3.3 Step 1, replacing $p = 2$ by $p = 3$. Each type has an L -parameter ϕ_τ by BH 2006 §18; the residual ρ_τ determines the component. **Step 2: principal-block derived-fibre match.** For τ a principal-block type, ρ_τ is reducible with constituents matching the Steinberg-decomposed restricted-weight simples L_0, L_1, L_2 . The two Ext^1 -classes of Chapter 32.4.6 lift to a two-dimensional Ext^1 -space in the deformation functor of ρ_τ . By Chapter 39.5.7(2), this is realised as the degree-1 derived fibre of dimension 2 in $\mathcal{X}_{GL_2}^{\mathrm{EG,wild},3,\chi,\tau}$. **Step 3: semisimplicity at irreducible ρ .** Same as the $p = 2$ case. All supercuspidal types give smooth classical components. **Step 4: bijectivity.** Same as the $p = 2$ case. $\square$

46.5 ESSENTIAL SURJECTIVITY VIA TYPE-BY-TYPE EXHAUSTION

The Exhaustion Argument

THEOREM 46.5.1 (*block exhaustion on a single BH block*). . Fix $p \in \{2, 3\}$, χ , and a Bushnell–Kutzko type $\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}$. Under assumptions (A), (B), (C) of Chapter 46.1.2, the restricted functor

$$\Phi_{p,\chi,\tau}^{\text{wild}}: \text{Block}_{\tau}^{\text{BH}} \longrightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau})$$

is an equivalence.

Proof. **Step 1: fully-faithful at the block level.** The global $\Phi_{p,\chi}^{\text{wild}}$ of Chapter 39.5.6 is fully faithful conditional on Chapter 39.5.7. Restriction to a block preserves fully-faithfulness (the ambient restriction is the Verdier quotient by the orthogonal component).

Step 2: essential surjectivity in the semisimple direction. Under (A), the projective indecomposable generator $P_{\tau} \in \text{Block}_{\tau}^{\text{BH}}$ maps under $\Phi_{p,\chi,\tau}^{\text{wild}}$ to the derived structure sheaf of the BH closed point $x_{\tau} \in \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau}$. By (C), the closed-point derived structure sheaf compactly generates $\text{IndCoh}_{\text{Nilp}}$ on the component. By standard theory (Lurie HA §4.8.2), a compact generator whose image under a fully-faithful functor maps to a compact generator of the target produces essential surjectivity on the compactly-generated sub- ∞ -category.

Step 3: absorbing the non-semisimple Ext¹-correction. The derived structure on the block’s compact generator side is a two-step filtration whose degree- (-1) fibre is the Ext¹-extension of Chapter 32.4.5 (at $p = 2$) or the pair of Ext¹-extensions of Chapter 32.4.6 (at $p = 3$). Under assumption (B), this Ext¹-structure matches the degree-1 derived fibre of $\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau}$ by Chapter 39.5.2. The matching is not merely dimensional (both sides are of the same dimension: 1 at $p = 2$, 2 at $p = 3$); it is an isomorphism of the Ext¹ itself, as classes. This upgrade preserves compact generation.

Step 4: completing essential surjectivity. Steps 2–3 show that the compactly-generated closure of the image of $\Phi_{p,\chi,\tau}^{\text{wild}}$ equals the compactly-generated closure of $\{x_{\tau} \text{ structure sheaf}\}$, which by (C) is the full target $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau})$. Combined with Step 1’s fully-faithfulness, this yields the equivalence. $\square$

COROLLARY 46.5.2 (*direct sum over types*). . Under (A), (B), (C) of Chapter 46.1.2, the direct sum $\Phi_{p,\chi}^{\text{wild}} = \bigoplus_{\tau} \Phi_{p,\chi,\tau}^{\text{wild}}$ is an equivalence of stable presentable ∞ -categories.

Proof. The two sides decompose respectively as direct sums over $\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}$; each summand functor is an equivalence by Chapter 46.5.1; direct sum of equivalences is an equivalence. $\square$

46.6 CROSS-CHECK WITH THE WILD-EG DEFINITION

Dimensional Comparison

PROPOSITION 46.6.1 (*dimensional comparison $p = 2$*). . At $p = 2$, the degree-1 derived fibre dimension on the wild-enhanced EG stack equals the Ext¹-dimension of the corresponding block: both equal 1.

Proof. Left-hand side: Chapter 39.5.3 records the derived fibre dimension as 1, supported on the non-semisimple locus of the Emerton–Gee stack (the reducible residual representations). Right-hand side: Chapter 32.4.5 equation (32.4.5) records $\text{Ext}^1(\mathbf{I}_{\text{triv}}, \mathbf{I}_{\text{sgn}}) = k$, a one-dimensional space. The match is exact. $\square$

PROPOSITION 46.6.2 (*dimensional comparison $p = 3$*). . At $p = 3$, the degree-1 derived fibre dimension on the wild-enhanced EG stack equals the Ext^1 -dimension of the corresponding block: both equal 2.

Proof. Left-hand side: Chapter 39.5.4 records the derived fibre dimension as 2 on the non-semisimple locus. Right-hand side: Chapter 32.4.6 records $\text{Ext}^1(L_0, L_1) \oplus \text{Ext}^1(L_1, L_2) = k \oplus k$, a two-dimensional space. The match is exact. $\square$

Structural Consistency

PROPOSITION 46.6.3 (*structural consistency with Chapter 39*). . The BH-component decomposition of this chapter, $\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi} = \coprod_{\tau} \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau}$, coincides with the decomposition implied by Chapter 39.5.7(3) (the BH fibre correspondence between wild components and BH types), with identical indexing set $\mathbf{T}_{p,\chi}^{\text{BH}}$ and identical derived-fibre dimensions.

Proof. Both decompositions arise from the same source: the BH type classification of Bushnell–Henniart 2006 §11–15. The indexing set is identical. The derived-fibre dimensions are identical by the dimensional comparisons Chapter 46.6.1 and Chapter 46.6.2. The non-semisimple locus (where the derived fibre is non-zero) is identical: it is the reducible residual locus determined by the principal-block types. All three data coincide, so the decompositions coincide. $\square$

Remark 46.6.4 (*no new structure imposed*). Chapter 46.6.3 confirms that this chapter does not introduce any new structure on the wild-enhanced EG stack beyond what Chapter 39 already specifies. The content of Fl.a.3 is the statement that the BH type-by-type match yields essential surjectivity on each component, together with the three conjectural inputs (A), (B), (C) that make this essential surjectivity precise. No new geometric object is constructed here.

46.7 THE THREE CONDITIONAL INPUTS

(A) BH Type-Block Comparison

Conjecture 46.7.1 (*BH type-block comparison*). . Fix $p \in \{2, 3\}$, χ , and $\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}$. The projective indecomposable generator $P_{\tau} \in \text{Block}_{\tau}^{\text{BH}}$ is a compact object of $\mathbf{D}^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$, and its image under $\Phi_{p,\chi}^{\text{wild}}$ is the derived structure sheaf of the BH closed point $x_{\tau} \in \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau}$.

Remark 46.7.2 (*evidence for the BH type-block comparison*). Three lines of evidence:

- Paskunas 2013 establishes the analogous statement for $p \geq 5$ (Publ. IHES 118 §6): each projective indecomposable of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm},\chi}$ maps under Colmez–Montreal to the structure sheaf of a residual-Galois closed point. At $p \in \{2, 3\}$, the Paskunas argument requires modification because the block structure is more complex; the comparison at $p \in \{2, 3\}$ is the extension of Paskunas’ result to the BH type list, with the same projective-to-structure-sheaf identification.
- Bushnell–Henniart 2006 §18 establishes the type-to-parameter match: each BH type τ has an associated L -parameter ϕ_{τ} . The projective generator P_{τ} is the type’s indicator-character-module; the BH comparison conjecture asserts that this module maps to the derived structure sheaf of the closed point x_{τ} determined by $\phi_{\tau} \pmod{p}$.
- Compactness of P_{τ} : the block is a finitely-generated module over its local endomorphism algebra (BH 2006 §12.3), hence P_{τ} is compact in the derived category of the block.

The conjectural content is the precise image identification (second bullet), not compactness or finiteness.

(B) Paskunas Wild Ext¹-Compatibility

Conjecture 46.7.3 (Paskunas wild-block Ext¹-compatibility). . Fix $p \in \{2, 3\}$ and a principal-block type $\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}$ with reducible residual ρ_τ . The Ext¹-class(es) of Chapter 32.4.5 (at $p = 2$) or Chapter 32.4.6 (at $p = 3$) on the automorphic side matches, as k -vector space with specific class(es), the degree-1 cotangent fibre of the wild-enhanced EG stack Chapter 39.5.2 at the residual ρ_τ . The match respects the class labels: the class α_{oi} matches one Ext¹-direction of the cotangent fibre; α_{12} matches the orthogonal direction at $p = 3$.

Remark 46.7.4 (dimensional evidence already closed). The dimensional match is *unconditional*: Chapter 46.6.1 and Chapter 46.6.2 each close the dimensional part of the conjecture. The conjectural content is the class-level identification: that the specific Ext¹-class α_{oi} (or the pair at $p = 3$) is sent to the specific cotangent-fibre direction. This identification is a Bushnell–Henniart + deformation-theory statement, not a dimension count. Its proof would match the formal deformation space of ρ_τ with the Ext¹-ring computed on the BH block, and would fit within the non-formal theorem.

(C) Wild-EG Compact-Generator Identification

Conjecture 46.7.5 (wild-EG compact generation). . Fix $p \in \{2, 3\}$, χ , and $\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}$. The compact objects of $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau})$ are generated under shifts and finite colimits by the derived structure sheaf of the BH closed point $x_\tau \in \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\tau}$.

Remark 46.7.6 (classical case is known). The classical analogue (no derived enhancement) is a theorem of Gaitsgory–Rozenblyum (AMS DAG monographs Vol. II, Ch. 8): for a connected formal neighbourhood of a closed point on a formal algebraic stack, $\text{IndCoh}_{\text{Nilp}}$ is compactly generated by the structure sheaf of the closed point. The derived-enhanced case, needed for the degree-1 wild fibre of Chapter 39.5.2, is the derived-DAG version of this theorem. Pridham–Toen–Vezzosi DAG VIII §5 give the derived analogue on smooth derived formal stacks; the wild-enhanced EG stack is not smooth in the classical sense, but is derived-smooth in degree 1; extending the generation statement to the derived-smooth case is the conjectural content of (C).

46.8 CONSEQUENCE FOR FI.A AND FI

FI.a Closure at Wild Primes

THEOREM 46.8.1 (*FI.a at wild primes, conditional*). . Under (A), (B), (C) of Chapter 46.1.2 and the wild-EG enhancement Chapter 39.5.7, FI.a holds at $p \in \{2, 3\}$: there is an equivalence

$$\Phi_{p,\chi}^{\text{wild}} : D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi \xrightarrow{\sim} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}),$$

natural in χ and in the BH-block decomposition.

Proof. Fully-faithfulness is Chapter 39.5.6. Essential surjectivity is Chapter 46.5.2. Both hold under the stated hypotheses. $\square$

FI.a Closure at All Primes

COROLLARY 46.8.2 (*FI.a at all primes, conditional, sharpened*). . Under:

- FI.a.1: Chapter 39.4.5 (block exhaustion at $p \geq 5$);
- FI.a.2: Chapter 39.5.7 (wild-EG enhancement at $p \in \{2, 3\}$);

- FI.a.3: (A), (B), (C) of Chapter 46.1.2 (BH type-block comparison, Paskunas wild Ext^t, wild-EG compact generation);

FI.a holds at every prime and every central character χ : there is a single family equivalence

$$\Phi_{p,\chi}^{\text{LocCat}}: D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi} \xrightarrow{\sim} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG,wild-enhanced},p,\chi}).$$

The central-character variation Chapter 39.6.4 also requires the family datum Chapter 39.6.3.

Proof. Taking the direct sum of Chapter 39.4.7 (at $p \geq 5$), Chapter 46.8.1 (at $p \in \{2, 3\}$), and Chapter 39.6.4 (central-character variation, conditional on the family datum). $\square$

The Updated FI Obstruction Decomposition

PROPOSITION 46.8.3 (*FI obstruction decomposition after BH block exhaustion*). . The total FI obstruction decomposition (combining FI.a.1–FI.a.3 with L_{ACD}) factors into the following named conjectural inputs:

- (i) FI.a.1: block exhaustion at $p \geq 5$ (Chapter 39.4.5), substantial non-formal theorem.
- (ii) FI.a.2: wild-EG enhancement at $p \in \{2, 3\}$ (Chapter 39.5.7), substantial non-formal theorem.
- (iii) Wild block functor datum at $p \in \{2, 3\}$ (Chapter 39.5.5).
- (iv) FI.a.3.A: BH type-block comparison (Chapter 46.7.1), substantial non-formal theorem.
- (v) FI.a.3.B: Paskunas wild Ext^t-compatibility (Chapter 46.7.3), substantial non-formal theorem.
- (vi) FI.a.3.C: wild-EG compact-generator identification (Chapter 46.7.5), substantial non-formal theorem.
- (vii) Central-character family datum (Chapter 39.6.3).
- (viii) L_{ACD} : adelic categorical descent (Ch. 21), substantial non-formal theorem.

The three sub-residuals of FI.a.3 together form a non-formal theorem, matching the aggregate FI.a.3 domain identified in Chapter 39.

Proof. The enumeration is direct. FI.a.1, FI.a.2, the wild block functor datum, and the central-character family datum are inherited from Chapter 39. FI.a.3 factors as (A)+(B)+(C) by Chapter 46.1.2. L_{ACD} is the adelic residual of Chapter 21. $\square$

46.9 RESIDUAL CONDITIONS AND SOURCE COMPARISONS

Named Residuals Introduced

FI.a.3 factors into three sub-conjectures:

- (i) Chapter 46.7.1 (FI.a.3.A): projective indecomposables of BH blocks map to derived structure sheaves of BH closed points under the twisted DEG functor.
- (ii) Chapter 46.7.3 (FI.a.3.B): the non-semisimple Ext^t-classes on the automorphic side match the degree-1 cotangent fibres on the wild-enhanced EG stack, as classes (dimensional match already unconditional by Chapter 46.6.1 and Chapter 46.6.2).
- (iii) Chapter 46.7.5 (FI.a.3.C): compact generation of $\text{IndCoh}_{\text{Nilp}}$ on each wild-EG component by the derived structure sheaf of the BH closed point.

What Closes Unconditionally

Three statements close unconditionally:

- (i) Dimensional comparison at $p = 2$ (Chapter 46.6.1): $1 = 1$.
- (ii) Dimensional comparison at $p = 3$ (Chapter 46.6.2): $2 = 2$.
- (iii) Structural consistency with Chapter 39 (Chapter 46.6.3): the BH-component decomposition matches the wild-EG decomposition of Chapter 39 structurally.

These unconditional results establish the decomposition and the numerical match of FI.a.3. The conjectural content is the fine identification: class-level match, compact generation, and projective-to-structure-sheaf comparison.

Source and Convention Comparison: BH Tame-Wild Classification at $p = 2$

Source separation. Bushnell–Henniart 2006 §11–13, Carter 1989 §7.3, Lusztig 1980, Jantzen 2003 Ch. II §8, and Paskunas 2013 §6 are disjoint from Fargues–Scholze arXiv:2102.13459 and from Dotto–Emerton–Gee arXiv:2603.26887. DEG is confined to $p \geq 5$; the $p = 2, 3$ extension is the conjectural content of (A).

Conventions. BH uses Langlands-normalised L-parameters; the EG stack follows Emerton–Gee arXiv:1908.07185, which matches. Character conductor exponents agree.

Ambient category. The category $\text{Block}_\tau^{\text{BH}}$ is a block of smooth $GL_2(\mathbb{Q}_2)$ -representations with central character χ ; the component $\mathcal{X}_{GL_2}^{\text{EG, wild, } 2, \chi, \tau}$ is a derived formal algebraic stack. Both are presentable stable ∞ -categories; the restricted functor lives in the same category as the global functor.

Hypotheses. The conjectural inputs (A), (B), (C) are explicit. The unconditional inputs are BH 2006 classification (proved), dimensional match (proved here), structural consistency (proved here). *The hypotheses are explicit.*

Functoriality. Block restriction is functorial (Verdier quotient by orthogonal block); type-to-component match is functorial by BH §18; derived-structure-sheaf formation is functorial.

Unproved equivalences. Three conjectures (A), (B), (C); explicit. **Named residuals.**

Numerical comparison. No numerical computation.

The BH tame-wild classification at $p = 2$ closes with three named sub-residuals.

Source and Convention Comparison: BH Tame-Wild Classification at $p = 3$

Source separation. Same primary sources as the $p = 2$ comparison plus Auslander–Reiten–Smalø 1995 Ch. VII §2 for the A_2 -linear-quiver block structure. All are disjoint from Dotto–Emerton–Gee arXiv:2603.26887 and from Fargues–Scholze arXiv:2102.13459.

Conventions. Conventions match BH and EG as in the $p = 2$ comparison; A_2 -quiver conventions follow Auslander–Reiten–Smalø (radical-square-zero relations).

Ambient category. Same ambient stable presentable ∞ -category as the $p = 2$ case.

Hypotheses. Same hypotheses (A), (B), (C), instantiated at $p = 3$; the Ext^1 -dimension is 2 not 1, matching Chapter 32.4.6.

Functoriality. Block restriction functorial as in the $p = 2$ case; the A_2 -quiver is functorial under block restriction.

Unproved equivalences. Same three conjectures (A), (B), (C), instantiated at $p = 3$. **Named residuals.**

Numerical comparison. No numerical computation.

The BH tame-wild classification at $p = 3$ closes with the same three named sub-residuals.

Source and Convention Comparison: Essential Surjectivity via Block Exhaustion

Source separation. Sources: Gaitsgory–Rozenblyum Vol. II (compact generation on formal stacks), Lurie HA §4.8.2 (compact-generator transfer under fully-faithful functors), Pridham–Toen–Vezzosi DAG VIII (derived formal deformation theory). All disjoint.

Conventions. Compact-generation conventions: shifts and finite colimits (not infinite), matching the DAG monograph.

Ambient category. Stable presentable ∞ -categories; the compact-generator transfer is an equivalence in $\text{Pr}_{\text{st}}^{\text{L}}$.

Hypotheses. (A), (B), (C) all named; the essential-surjectivity argument uses all three. *The hypotheses are explicit.*

Functoriality. The block-by-block exhaustion is block-diagonal (blocks are orthogonal on both sides); no false functoriality arises.

Unproved equivalences. No new conjecture introduced.

Numerical comparison. No numerical computation.

Essential surjectivity holds conditional on (A), (B), (C).

Source and Convention Comparison: Wild-EG Decomposition Cross-Check

Source separation. Sources: Chapter 39 (wild-EG definition), Chapter 32 (Ext^1 -computations), Bushnell–Hennriart 2006 (type classification). All disjoint; the cross-check is internal to Vol IV’s arithmetic branch.

Conventions. The degree-1 derived-fibre dimension in Ch. 39 equals the Ext^1 -dimension in Ch. 32 by Chapter 46.6.1 and Chapter 46.6.2.

Ambient category. Both objects sit in derived formal algebraic stacks over $\text{Spf } \mathbb{Z}_p$.

Hypotheses. Structural consistency assumes (B) for the class-level match; dimensional match is unconditional.

Functoriality. Functoriality of the BH fibre correspondence (BH 2006 §18) is used; this is explicit.

Unproved equivalences. No new unproved equivalence; (B) is already named.

Numerical comparison. No numerical computation.

The cross-check confirms structural consistency without introducing new residuals.

Source and Convention Comparison: Unconditional Closure

Source separation. Literature: Paskunas 2013, BH 2006, EG 2019, Gaitsgory–Rozenblyum monographs, Pridham–Toen–Vezzosi DAG VIII. All disjoint.

Conventions. All sources use Langlands-normalised L-parameters and agree on conductor exponents.

Ambient category. Ambient stable presentable ∞ -categories are fixed throughout.

Hypotheses. Each of (A), (B), (C) requires input not currently recorded in primary literature at $p \in \{2, 3\}$:

- (A) requires extending Paskunas’ projective-to-structure-sheaf match from $p \geq 5$ to $p \in \{2, 3\}$; the concrete technical domain is non-formal theorem.
- (B) requires identifying the specific Ext^1 -classes with cotangent-fibre directions via deformation theory; required theorem non-formal theorem.
- (C) requires extending Gaitsgory–Rozenblyum’s compact generation theorem from classical formal stacks to the derived-enhanced case; required theorem non-formal theorem using DAG VIII.

None of (A), (B), (C) admits unconditional closure from currently stated primary literature. **Three named residuals.**

Functoriality. Each of (A), (B), (C) is functorial in the input data; no false-functoriality gap.

Unproved equivalences. (A), (B), (C) are the minimal conjectural set. **Named residuals.**

Numerical comparison. No numerical computation.

The three sub-conjectures (A), (B), (C) constitute the minimal conditional input set for FI.a.3, each individually tractable at non-formal theorem. The total is within the non-formal theorem condition of Chapter 39.

46.10 BLOCK EXHAUSTION BOUNDARY

- (i) FI.a.3 is a small local residual on FI.a: 15–25 pp domain, bounded by the finite BH type list and the explicit Ext^1 -structure of Chapter 32 (Chapter 46.1.3).
- (ii) The condition is: essential surjectivity of the twisted DEG-type functor $\Phi_{p,\chi}^{\text{wild}}$ of Chapter 39.5.6 at $p \in \{2, 3\}$, reducing to type-by-type exhaustion via the BH block decomposition.

- (iii) BH tame-wild classification at $p = 2$ (Chapter 46.3.1): finite list of unramified principal-series, tame supercuspidal, and wild supercuspidal types. The Ext^1 -carrying block is the trivial unramified principal series (Chapter 46.3.2).
- (iv) BH classification at $p = 3$ (Chapter 46.4.1): adds a principal-block wild-types family with three types forming an A_2 -linear quiver (Chapter 46.4.2), matching Chapter 32.4.6.
- (v) Dimensional comparison (unconditional): both $p = 2$ (dimension 1) and $p = 3$ (dimension 2) match between the Ext^1 -side and the derived-fibre side (Chapter 46.6.1, Chapter 46.6.2).
- (vi) Structural consistency (unconditional): the BH-component decomposition coincides with the wild-EG decomposition of Chapter 39 (Chapter 46.6.3).
- (vii) Essential surjectivity on a single BH block (Chapter 46.5.1): proved conditional on (A), (B), (C). Assembly across types (Chapter 46.5.2): equivalence of the twisted DEG-type functor.
- (viii) Three named sub-conjectures: (A) BH type-block comparison (Chapter 46.7.1), non-formal theorem; (B) Paskunas wild Ext^1 (Chapter 46.7.3), non-formal theorem; (C) wild-EG compact generation (Chapter 46.7.5), non-formal theorem. Total: non-formal theorem, consistent with Chapter 39's estimate.
- (ix) Fl.a.3 does not close unconditionally; it is factored into three individually tractable sub-residuals, each strictly narrower than the full Fl.a.3 statement. No statement for the main theorem; only conditional closures.
- (x) The consequence for the Fl table (Chapter 46.8.3) is that Fl.a factors into named conjectural inputs: Fl.a.1 , Fl.a.2 , the wild block functor datum, Fl.a.3.A-C , and the central-character family datum; Fl adds L_{ACD} . This is the finest factorisation of the Fl residual recorded here.

There is no unconditional closure of Fl.a.3 . The strongest unconditional results are the dimensional comparison and the structural consistency; all remaining closures are conditional on (A), (B), (C).

ARITHMETIC FUNDAMENTAL LOCAL EQUIVALENCE: SELMER-DERIVED TARGET AND OPEN CATEGORICAL EQUIVALENCES

47.1 SELMER-DERIVED GLOBAL SPECTRAL STACK

The arithmetic analogue of Gaitsgory’s fundamental local equivalence (FLE) requires a global spectral target that imposes local conditions simultaneously. The global Galois group $G_{\mathbb{Q},S}$ is not a product of its decomposition groups; the correct global object is the derived fibre product of local Emerton–Gee stacks over the restriction maps, with local conditions imposed simultaneously. This is the Selmer-derived construction.

Definition 47.1.1 (Selmer-derived global spectral stack). Let the following data be fixed:

- a finite set S of places containing ∞ , all primes where $\bar{\rho}$ is ramified, and all primes where the local condition is not unramified;
- a residual representation $\bar{\rho}: G_{\mathbb{Q},S} \rightarrow GL_2(k)$;
- a determinant character δ lifting $\det \bar{\rho}$;
- for each $v \in S$, a local condition $\mathcal{X}_v^{\mathcal{L}_v} \hookrightarrow \mathcal{X}_{GL_2,v}^{W,\varphi\Gamma}$ inside the full Weil/ (φ, Γ) local parameter stack $\mathcal{X}_{GL_2,v}^{W,\varphi\Gamma}$ (Emerton–Gee).

The Selmer-derived global spectral stack is

$$\mathcal{X}_{\bar{\rho},\delta,\mathcal{L}}^{\text{Sel}} := \text{Map}_{\bar{\rho},\delta}(G_{\mathbb{Q},S}, GL_2) \times_{\prod_{v \in S} \text{Map}(G_{\mathbb{Q}_v}, GL_2)}^{\mathbf{R}} \prod_{v \in S} \mathcal{X}_v^{\mathcal{L}_v}, \quad (47.1)$$

where $\text{Map}_{\bar{\rho},\delta}(G_{\mathbb{Q},S}, GL_2)$ denotes the derived mapping stack of continuous lifts of $\bar{\rho}$ with determinant δ , and the derived fibre product $\times^{\mathbf{R}}$ imposes the local conditions $\mathcal{X}_v^{\mathcal{L}_v}$ simultaneously. Local Emerton–Gee stacks supply the (φ, Γ) local input at each finite v ; the global object is Selmer-derived.

PROPOSITION 47.1.2 (Selmer tangent complex). . At a point $\rho \in \mathcal{X}_{\bar{\rho},\delta,\mathcal{L}}^{\text{Sel}}$, the tangent complex is the fixed-determinant Selmer complex

$$T_{\rho} \mathcal{X}_{\bar{\rho},\delta,\mathcal{L}}^{\text{Sel}} \simeq C_{\mathcal{L}}^{\bullet}(G_{\mathbb{Q},S}, \text{ad}^{\circ} \rho)[1],$$

where $C_{\mathcal{L}}^{\bullet}(G_{\mathbb{Q},S}, \text{ad}^{\circ} \rho)$ is the mapping fibre of the localisation map

$$C^{\bullet}(G_{\mathbb{Q},S}, \text{ad}^{\circ} \rho) \longrightarrow \bigoplus_{v \in S} C^{\bullet}(G_{\mathbb{Q}_v}, \text{ad}^{\circ} \rho) / C_{\mathcal{L}_v}^{\bullet}(G_{\mathbb{Q}_v}, \text{ad}^{\circ} \rho).$$

Proof. The stack (47.1) is a derived fibre product. Applying the tangent functor yields the fibre of the localisation map displayed above, then shifts by $[1]$ by the deformation-stack convention. $\square$

The tangent complex identifies the Selmer cohomology $H_{\mathcal{L}}^1(G_{\mathbb{Q},S}, \text{ad}^\circ \rho)$ as the first cohomology of $C_{\mathcal{L}}^\bullet(G_{\mathbb{Q},S}, \text{ad}^\circ \rho)$, with obstruction class in H^2 . Quasi-smoothness of $X_{\tilde{\rho}, \delta, \mathcal{L}}^{\text{Sel}}$ at ρ is equivalent to the vanishing of this H^2 , which in turn depends on the local conditions $\mathcal{L}_v$ via the Greenberg–Wiles formula. Noetherian and quasi-smooth properties thus reduce to conditions on local conditions at $v \in S$.

47.2 LOCAL GEOMETRISATION AND ITS DOMAIN

The following collects what Fargues–Scholze, Emerton–Gee, and Dotto–Emerton–Gee establish in the local setting. These are the inputs to the conditional programme of §47.3.

THEOREM 47.2.1 (*local inputs, Fargues–Scholze / Emerton–Gee / Dotto–Emerton–Gee*). (assembling external results).

- (i) *Local geometrisation and spectral action* (Fargues–Scholze arXiv:2102.13459). For each prime p , the Fargues–Scholze spectral action equips $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])$ with an action of $\text{IndCoh}_{\text{Nilp}}(\text{LocSys}_{GL_2})$ and supplies local Langlands geometrisation and Hecke functoriality.
- (ii) *Local (φ, Γ) parameter stacks* (Emerton–Gee arXiv:1908.07185). For each prime p , the Emerton–Gee stack $X_{GL_2}^{\text{EG}, p}$ parametrises two-dimensional (φ, Γ) -modules over $\mathbb{Q}_p$ with a well-behaved moduli structure; local conditions $X_v^{\mathcal{L}_v} \hookrightarrow X_{GL_2, v}^{W, \varphi \Gamma}$ are open or closed substacks.
- (iii) *Fully faithful functor at fixed central character* (Dotto–Emerton–Gee arXiv:2603.26887). For primes $p \geq 5$, there exists a fully faithful functor

$$D_{GL_2(\mathbb{Q}_p), \chi}^{\text{sph-fin}} \hookrightarrow \text{IndCoh}_{\text{Nilp}}(X_{GL_2, \chi}^{\text{EG}, p})$$

for each fixed central character χ . Essential surjectivity, the cases $p = 2, 3$, and the central-character-family statement remain open.

- (iv) *Archimedean OPE confined to archimedean factor*. The Hochschild-trace candidate $\mathcal{V}^{\text{prim}}$ carries chiral OPE structure at the archimedean factor $v = \infty$ via the $(\mathfrak{gl}_2, O(2))$ -module structure; finite places carry locally constant sheaf data and Hecke-algebra actions via the Satake isomorphism. Chiral OPE structure is an archimedean-fibre phenomenon.
- (v) *Commutative Bethe–Hecke algebra*. At each finite prime p , the centre of the spherical Hecke algebra $\mathcal{H}(GL_2(\mathbb{Q}_p), GL_2(\mathbb{Z}_p))$ is commutative and acts on $\mathcal{V}^{\text{prim}}$ via the Satake isomorphism. This commutative Bethe–Hecke structure is the correct Hecke-algebraic shadow; higher Hopf or Yangian structures require separate verification at each prime.
- (vi) *Classical Kim arithmetic Chern–Simons*. Kim’s arithmetic Chern–Simons theory arXiv:1510.05818 supplies classical arithmetic CS torsors on the arithmetic curve and action lines in K_2 . The quantum boundary category $\partial_q \text{ACS}$ of Conjecture 47.3.6 is a distinct object from $\mathcal{V}^{\text{prim}}$; its construction is open.

47.3 CONDITIONAL CATEGORICAL EQUIVALENCES

The local inputs of §47.2 establish the spectral action, the (φ, Γ) parameter stacks, and full faithfulness at fixed central character. A complete arithmetic FLE requires four further categorical equivalences, each an open problem conditional on a named hypothesis.

Conjecture 47.3.1 (local categorical comparison). . For every prime p there is an equivalence

$$D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)]) \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p}), \quad (47.2)$$

compatible with Hecke functors, central-character families, and the Fargues–Scholze spectral action. The Dotto–Emerton–Gee functor establishes full faithfulness for $p \geq 5$ at fixed central character; the conjecture adds essential surjectivity, the cases $p = 2, 3$, and the family statement.

Conjecture 47.3.2 (adelic categorical descent). . Under the local categorical comparison (47.2) at all primes, the local categories glue to the global automorphic category $D([GL_2])^{\text{sph-fin}}$ via an equivalence compatible with Whittaker models, Bernstein centres, the archimedean $(\mathfrak{g}, K)$ factor, and Eisenstein and continuous spectrum. Representation-level restricted tensor products are classical (Flath); the categorical descent is the open statement.

Conjecture 47.3.3 (mixed-characteristic Iwahori E_2). . The mixed-characteristic local Iwahori Hecke category $\text{Hk}_{GL_2,p}^{\text{Iw}}$ carries an E_2 -monoidal structure compatible with diamond Beilinson–Drinfeld fusion and with the Iwahori stratification:

$$\text{Hk}_{GL_2,p}^{\text{Iw}} \in \text{Alg}_{E_2}(\text{Cat}_\infty). \quad (47.3)$$

Current sources supply spherical fusion and E_1 convolution; the E_2 enhancement is the local input required for the restricted-tensor E_2 transport argument. The restricted-tensor E_2 transport lemma (proved elsewhere in this volume) is a separate, purely categorical statement; its hypotheses require (47.3) as input.

Conjecture 47.3.4 (full-target arithmetic Ben-Zvi–Nadler). . For the Selmer-derived full Weil/ (φ, Γ) target, the arithmetic Ben-Zvi–Nadler comparison is

$$\text{HH}_\bullet(\text{Hk}_{GL_2,\text{glob}}^{\text{Iw}}) \simeq \mathcal{O}(\text{LocSys}_{GL_2}^{W,\varphi\Gamma})^{\text{flat}}, \quad (47.4)$$

where $\text{LocSys}_{GL_2}^{W,\varphi\Gamma}$ is the full Weil/ (φ, Γ) local system stack and $\cdot^{\text{flat}}$ denotes the flat-locus subfunctor. The conjecture requires formal-Noetherian Hochschild continuity for $\text{Hk}_{GL_2,\text{glob}}^{\text{Iw}}$ and comparison of the Arinkin–Gaitsgory nilpotent flat locus with the Fargues–Scholze/Emerton–Gee flat locus.

Conjecture 47.3.5 (categorical arithmetic Whittaker theory). . The arithmetic Whittaker factorisation lifts from representation-level Whittaker models to a categorical equivalence compatible with the local comparisons of (47.2), adelic descent, and the Selmer-derived spectral stack (47.1). Representation-level Tate–Jacquet–Langlands factorisation is proved; the categorical lift is the open statement.

Conjecture 47.3.6 (quantum ACS boundary category). . There exists a quantum boundary category $\partial_q \text{ACS}$ and an equivalence

$$\partial_q \text{ACS} \simeq \mathcal{V}^{\text{prim}}$$

compatible with an appropriate quantisation of Kim’s arithmetic Chern–Simons theory arXiv:1510.05818. Kim’s theory provides the classical arithmetic CS torsors and action lines in K_2 ; the quantisation is the open content of this conjecture.

47.4 PROVED AND CONDITIONAL COMPONENTS OF THE ARITHMETIC FLE

THEOREM 47.4.1 (*arithmetic FLE: proved and conditional components*). (by enumeration). The arithmetic programme based on $\mathcal{V}^{\text{prim}}$ and the Selmer-derived stack $\mathcal{X}_{\tilde{\rho}, \delta, \mathcal{L}}^{\text{Sel}}$ (47.1) decomposes as follows.

- (i) *Proved*. The Selmer-derived global spectral stack $\mathcal{X}_{\tilde{\rho}, \delta, \mathcal{L}}^{\text{Sel}}$ is constructed as a derived fibre product (Definition 47.1.1); its tangent complex is identified with the fixed-determinant Selmer complex (Proposition 47.1.2). The local inputs of Theorem 47.2.1 are precisely the Fargues–Scholze, Emerton–Gee, and Dotto–Emerton–Gee inputs.
- (ii) *Conditional on Conjecture 47.3.1*. Local categorical comparison at all primes, including essential surjectivity at $p \geq 5$ and the full statement at $p = 2, 3$.
- (iii) *Conditional on Conjecture 47.3.1 and Conjecture 47.3.2*. Adelic categorical descent to the global automorphic category.
- (iv) *Conditional on Conjecture 47.3.3*. Iwahori E_2 -monoidal structure at mixed characteristic; the restricted-tensor E_2 transport lemma applies once this local input is supplied.
- (v) *Conditional on Conjecture 47.3.4*. Full-target arithmetic Ben-Zvi–Nadler comparison (47.4).
- (vi) *Conditional on Conjecture 47.3.5*. Categorical arithmetic Whittaker factorisation.
- (vii) *Conditional on Conjecture 47.3.6*. Quantum ACS boundary category equivalent to $\mathcal{V}^{\text{prim}}$.

$\mathcal{V}^{\text{prim}}$ is a Hochschild trace class throughout; its chiral OPE structure is confined to the archimedean factor as established in Theorem 47.2.1(4). A vertex algebra requires an OPE on a formal disk defined at every place; $\mathcal{V}^{\text{prim}}$ carries this structure only at $v = \infty$.

FI.A.1 UNRAMIFIED IDEAL, FI.A.2 SMALL-PRIME BLOCK EXTENSION, AND L_{ACD} CUSPIDAL DESCENT

The functor $\Phi_{p,\chi}^{\text{DEG}}$ is analysed on three orthogonal loci. On the unramified ordinary locus, Breuil–Mézard geometry, Emerton–Gee smoothness, and cyclic base change identify the compact generators. At $p = 2, 3$, the semisimple blocks are separated by the Paškūnas and Colmez–Dospinescu–Paškūnas block calculi. On the cuspidal sph-finite automorphic category, Flath restricted tensor products, Bernstein orthogonality, and Mœglin–Waldspurger spectral decomposition give adelic categorical descent. The Eisenstein sector is governed by the normalised intertwining operators.

48.1 OBJECTS AND DISJOINT SOURCES

Inherited Objects

All the categorical data used in this chapter is stated prior:

- $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$ and its Dotto–Emerton–Gee functor $\Phi_{p,\chi}^{\text{DEG}}$ (Chapter 39, Chapter 39.3.1);
- the Bushnell–Henniart block decomposition and its type list (Chapter 46; Chapter 46.3.1, Chapter 46.4.1);
- the Selmer-derived global spectral stack $\mathcal{X}_{\tilde{\rho}, \delta, \mathcal{L}}^{\text{Sel}}$ (Chapter 47, Chapter 47.1.1);
- the four subconditions $L_{ACD.0-3}$ of Chapter 30, together with Chapter 28.1.3 (F2.c tightened).

The constructions below use these objects and no additional moduli problem.

Primary Sources

Closures deployed in this chapter cite the following primary sources:

Breuil–Mézard 2002 “Multiplicités modulaires et représentations de $GL_2(\mathbb{Z}_p)$ et de $\text{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ en $\ell = p$ ”, Duke Math. J. 115. The numerical Breuil–Mézard conjecture.

Kisin 2009 “The Fontaine–Mazur conjecture for GL_2 ”, J. AMS 22. Modularity lifting plus Breuil–Mézard in Hodge–Tate regular weight.

Gee–Kisin 2014 “The Breuil–Mézard conjecture for potentially Barsotti–Tate representations”, Forum Math. Pi 2. Breuil–Mézard geometric formulation.

Emerton–Gee arXiv:1908.07185 The Emerton–Gee stack of (φ, Γ) -modules. Unramified-locus smoothness §6.2; ordinary-locus flatness §5.5; Breuil–Mézard stratification §7.

Dotto–Emerton–Gee arXiv:2603.26887 Fully faithful functor at fixed central character for $p \geq 5$.

Clozel–Harris–Taylor The Sato–Tate conjecture book (Harris–Taylor 2001 Ann. Math. Stud. 151; Clozel–Harris–Taylor 2008 Publ. IHES 108 §4–5). Cyclic base change on the unramified Hecke algebra.

Paškūnas “Blocks for mod p representations of $GL_2(\mathbb{Q}_p)$ ” = JEMS 16 (2014) §2–5. Block decomposition at $p = 2$.

Colmez–Dospinescu–Paškūnas “The p -adic local Langlands correspondence for $GL_2(\mathbb{Q}_p)$ ”, Cambridge J. Math. 2 (2014). $p = 3$ and reducible principal-block compatibility.

Flath 1979 “Decomposition of representations into tensor products”, Proc. Symp. Pure Math. 33 Part 1. Restricted tensor product of automorphic representations.

Bernstein 1988 “On the support of Plancherel measure”, J. Geom. Phys. 5. Bernstein-component decomposition and cuspidal orthogonality.

Moëglin–Waldspurger 1995 *Spectral decomposition and Eisenstein series*, Cambridge Tracts 113. Cuspidal L^2 summand.

Jacquet 1983 “On the residual spectrum of $GL(n)$ ”, SLN 1041. Cuspidal-orthogonal complement of the continuous spectrum for GL_2 .

Carayol 1986 “Sur les représentations ℓ -adiques associées aux formes modulaires de Hilbert”, Ann. Sci. ÉNS 19 Th. A. Local-global compatibility for the ℓ -adic Galois representation attached to a Hilbert modular newform at unramified primes.

Saito 1997 “Modular forms and p -adic Hodge theory”, Invent. Math. 129. Local-global compatibility for the Galois representation attached to a classical newform: the Frobenius eigenvalue at an unramified prime $\ell \neq p$ equals the Hecke eigenvalue a_ℓ of the newform.

Taylor–Wiles 1995 “Ring-theoretic properties of certain Hecke algebras”, Ann. Math. 141. Big-image Taylor–Wiles hypotheses.

Disjointness Comparison

The proof sources backing the DEG comparison (Fargues–Scholze arXiv:2102.13459 plus Emerton–Gee arXiv:1908.07185 plus Dotto–Emerton–Gee arXiv:2603.26887) are separate from the comparison sources used here. Breuil–Mézard + Kisin + Gee–Kisin are *deformation-theoretic* and closed before DEG; their use in this chapter is to bound the image of DEG, not to rederive the fully-faithful functor. Clozel–Harris–Taylor is *automorphic-base-change* and provides the Hecke-module input without going through (φ, Γ) -modules. Paškūnas and Colmez–Dospinescu–Paškūnas 2014 are block-structure theorems at $p = 2$ and $p = 3$ that were established independently of the DEG framework. Flath 1979 is a pre-categorical restricted-tensor theorem. Bernstein 1988 and Moëglin–Waldspurger 1995 are spectral-decomposition theorems. No source in this chapter appears in DEG’s primary derivation; hence the comparison is genuinely independent.

48.2 FI.A.1: THE UNRAMIFIED ORDINARY IDEAL

Hypotheses and Target

Setup 48.2.1 (Taylor–Wiles unramified setup). Fix a prime $p \geq 5$, a level $N \in \mathbb{Z}_{\geq 1}$ with $\gcd(N, p) = 1$, and a residual Galois representation

$$\bar{\rho}: G_{\mathbb{Q}} \rightarrow GL_2(\bar{\mathbb{F}}_p)$$

unramified at p and at almost all $\ell \nmid Np$. Write $\chi := \det \bar{\rho}|_{G_{\mathbb{Q}_p}} \cdot \chi_{\text{cyc}}^{-1}$ for the induced central character on $\mathbb{Z}_p^\times$. Assume:

(TW1) $p \geq 5$.

(TW2) $\bar{\rho}|_{G_{\mathbb{Q}(\zeta_p)}}$ is absolutely irreducible (*big image*).

(TW3) $\bar{\rho}$ is distinguished: the semisimplification $\bar{\rho}^{\text{ss}}$ has distinct diagonal characters, and the local residual at p is split reducible or irreducible (not reducible non-split with equal diagonal characters).

(TW₄) Ordinary locus: we restrict attention to the ordinary component $\text{Block}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}}^{\text{ord}} \subseteq \text{Block}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}}$ where the local representation is reducible with an unramified line of eigenvalues of Frob_p .

Definition 48.2.2 (unramified ordinary sub- ∞ -category). Under Setup 48.2.1, the *unramified ordinary* sub- ∞ -category of $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$ is the full subcategory generated under colimits and shifts by:

- the spherical unramified principal-series types of Bushnell–Henniart (Chapter 46.3.1 at $p = 2$; the analogous list at $p \geq 5$) with unramified inducing characters, together with
- the ordinary subquotients of the induced objects as in Hida’s ordinary theory (Hida 1986 Ann. Inst. Fourier 36).

Denote this sub- ∞ -category $D^{\text{sph-fin,unr-ord}}([GL_2(\mathbb{Q}_p)])_{\chi}$. On the spectral side, the corresponding substack is the formal neighbourhood $\mathcal{X}_{GL_2}^{\text{EG},p,\chi,\text{unr-ord}} \subseteq \mathcal{X}_{GL_2}^{\text{EG},p,\chi,\bar{\rho}}$ of the unramified reducible locus with an unramified line of Frobenius eigenvalues.

Statement

THEOREM 48.2.3 (FLA.1 unramified ordinary ideal). Under Setup 48.2.1, the restriction of the DEG functor Chapter 39.3.1

$$\Phi_{p,\chi}^{\text{DEG}}|_{\text{unr-ord}} : D^{\text{sph-fin,unr-ord}}([GL_2(\mathbb{Q}_p)])_{\chi} \longrightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi,\text{unr-ord}})$$

is an equivalence. In particular, essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ holds on the unramified ordinary summand unconditionally.

The theorem is proved by assembling three primary-literature inputs.

Input 1: Breuil–Mézard on the Unramified Ordinary Locus

LEMMA 48.2.4 (Breuil–Mézard geometric, unramified ordinary case). Under Setup 48.2.1 hypotheses (TW₁)–(TW₄), the ordinary Breuil–Mézard cycles on $\mathcal{X}_{GL_2}^{\text{EG},p,\chi,\bar{\rho},\text{ord}}$ are free of intersection pathologies: each Serre weight σ of conductor $\leq p+1$ produces a single irreducible cycle $\mathcal{X}_{p,\chi,\bar{\rho}}^{\text{BM},\sigma}$, distinct cycles meet transversely, and the union of these cycles set-theoretically covers the ordinary stratum.

Source. Kisin 2009 J. AMS 22 Theorems 1.1.8 and 2.3.5 (the numerical and geometric Breuil–Mézard under distinguished residual, including essential surjectivity at the ordinary locus). Gee–Kisin 2014 Forum Math. Pi 2 Theorem A establishes the geometric Breuil–Mézard statement used here without the crystalline-weight restriction of Kisin 2009. Emerton–Gee arXiv:1908.07185 §7.4 records the component-transversality. $\square$

LEMMA 48.2.5 (unramified-locus smoothness). The unramified locus $\mathcal{X}_{GL_2}^{\text{EG},p,\chi,\text{unr}} \subseteq \mathcal{X}_{GL_2}^{\text{EG},p,\chi}$ is smooth over $\text{Spf } \mathbb{Z}_p$ of relative dimension 2, with coordinates given by the two Frobenius eigenvalues.

Source. Emerton–Gee arXiv:1908.07185 §6.2 (unramified locus is the moduli of pairs of Frobenius eigenvalues, trivial Γ -action). $\square$

Combining the two lemmas, the ordinary Breuil–Mézard cycles restricted to the unramified ordinary locus are divisors cut out by a single equation of the form $\alpha_p \cdot \beta_p^{-1} = \sigma$, where (α_p, β_p) are the unramified Frobenius eigenvalues. This reduces $\text{IndCoh}_{\text{Nilp}}$ on the unramified ordinary locus to a Hecke-module problem over a two-variable smooth formal completion.

Input 2: Clozel–Harris–Taylor on the Unramified Hecke Subalgebra

LEMMA 48.2.6 (*unramified Hecke match via cyclic base change*). . Under Setup 48.2.1, the unramified spherical Hecke algebra $\mathcal{H}_{GL_2(\mathbb{Q}_p)}^{\text{sph}, \text{unr}} \simeq \mathbb{Z}_p[T_p, S_p^{\pm 1}]$ acts faithfully and with finite multiplicities on the unramified ordinary summand of $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$, with explicit eigenvalues $(\alpha_p + \beta_p, \alpha_p \beta_p)$ matching the Frobenius eigenvalues on $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \text{unr-ord}}$.

Source. Clozel–Harris–Taylor 2008 Publ. IHES 108 §4 establishes cyclic base change for automorphic representations of GL_2 at unramified primes under the Taylor–Wiles big-image and distinguished-residual conditions. Local-global compatibility at unramified primes (Carayol 1986 Ann. Sci. ÉNS 19 Th. A; Saito 1997 Invent. Math. 129, local-global compatibility for the Galois representation attached to a newform via Carayol–Saito) identifies the unramified Hecke eigenvalue $\alpha_p + \beta_p$ with the trace of crystalline Frobenius on the corresponding ℓ -adic representation, matching the Frobenius eigenvalues on $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \text{unr-ord}}$ via CHT §4. Harris–Taylor 2001 Ann. Math. Stud. 151 Ch. V records the faithfulness of the spherical Hecke action on the unramified summand under (TW1)–(TW3). $\square$

LEMMA 48.2.7 (*Hida ordinary splitting on the unramified ordinary summand*). . Under Setup 48.2.1, the unramified principal series $\text{Ind}_B^{GL_2(\mathbb{Q}_p)}(\alpha \otimes \beta)$ with unramified characters α, β and $v_p(\alpha(p)) = 0$ (ordinary condition) decomposes as a direct sum of two one-dimensional ordinary lines, each with a unique Hida family deformation. The ordinary line $\text{Ind}_B^{\text{ord}}(\alpha, \beta)$ is a compact generator of the ordinary summand.

Source. Hida 1986 Ann. Inst. Fourier 36 “Galois representations into $GL_2(\mathbb{Z}_p[[X]])$ attached to ordinary cusp forms”; Hida 1988 Ann. Math. 128 for the ordinary-locus deformation theory used here. $\square$

Closure of FI.A.1 on the Unramified Ordinary Ideal

Proof of Chapter 48.2.3. Step 1: fully-faithfulness on the summand. Chapter 39.3.1 states fully-faithfulness of $\Phi_{p, \chi}^{\text{DEG}}$ on all of $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$. Restriction to the unramified ordinary sub- ∞ -category preserves fully-faithfulness.

Step 2: essential surjectivity from Breuil–Mézard + unramified locus smoothness. By Chapter 39.3.3, the image of $\Phi_{p, \chi}^{\text{DEG}}$ is contained in the colimit closure of the structure sheaves of Breuil–Mézard cycles. By Chapter 48.2.5, the unramified ordinary locus is a smooth two-parameter formal completion; hence on this locus the singular support of every ind-coherent sheaf is automatically zero (the nilpotent cone in the cotangent bundle of a smooth stack is reduced to the zero section), so $\text{IndCoh}_{\text{Nilp}}$ coincides with IndCoh (Gaitsgory–Rozenblyum AMS DAG II Ch. 9 §1.3 for the singular-support coincidence on the smooth locus). The category IndCoh on a smooth formal completion of a Noetherian point is compactly generated by the structure sheaf of the closed point (Gaitsgory–Rozenblyum AMS DAG II Ch. 8 Thm 8.1.2). By Chapter 48.2.4, each Breuil–Mézard cycle restricts to an irreducible divisor on the unramified ordinary locus; the structure sheaves of these divisors and their intersections generate the derived category, since the closed point of the formal completion is an iterated transverse intersection of them.

Step 3: Hecke matching. The unramified Hecke algebra acts on both sides through the two Frobenius eigenvalues (on the spectral side, by Chapter 48.2.5; on the automorphic side, by Chapter 48.2.6). The action is faithful on both sides with identical eigenvalues. Under Step 1’s fully-faithfulness, this eigenvalue matching lifts to a natural equivalence of Hecke-equivariant compact generators. By Chapter 48.2.7, Hida’s ordinary line $\text{Ind}_B^{\text{ord}}(\alpha, \beta)$ is a compact generator of the automorphic summand, and its image

under $\Phi_{p,\chi}^{\text{DEG}}$ is the structure sheaf of the unramified ordinary point $(\alpha_p, \beta_p) \in \mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \text{unr-ord}}$, by the Colmez–Paškūnas (φ, Γ) -module / Montreal functor compatibility with Hida theory (Colmez 2010 Ch. VII, Astérisque 330 for the fixed-weight ordinary match).

Step 4: assembly. Steps 1–3 supply:

- fully-faithfulness (Step 1);
- a compact generator on the automorphic side whose image under $\Phi_{p,\chi}^{\text{DEG}}$ is a compact generator on the spectral side (Steps 2, 3).

Essential surjectivity follows by Lurie HA §4.8.2: a fully-faithful functor between compactly generated stable presentable ∞ -categories that sends compact generators to compact generators is an equivalence. $\square$

Precise Unconditional Range

COROLLARY 48.2.8 (unconditional FI.a.1 range). . The essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ holds unconditionally on the unramified ordinary summand under hypotheses (TW1)–(TW4) of Setup 48.2.1. Explicitly: for every prime $p \geq 5$ and every residual Galois representation $\bar{\rho}$ satisfying big image, distinguishedness, and unramified-ordinary local behaviour at p , the restriction of the DEG functor to the unramified ordinary summand is an equivalence.

Proof. Immediate from Chapter 48.2.3 under the stated hypotheses. All three inputs (Chapter 48.2.4, Chapter 48.2.5, Chapter 48.2.6) are already theorems; no conjecture is invoked. $\square$

Conjectural Extension

Conjecture 48.2.9 (FI.a.1 full-range extension). . The essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ extends from the unramified ordinary summand to the full block $\text{Block}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}}$ and the full non-ordinary locus, closing Chapter 39.4.5 at every $p \geq 5, \chi, \bar{\rho}$.

Remark 48.2.10 (non-ordinary obstruction). The ordinary summand is the component on which Breuil–Mézard cycles form a transversely meeting family of smooth divisors; the non-ordinary / supersingular summand has non-reduced Breuil–Mézard cycles and the unramified locus is singular at the residual Frobenius-trace zero. Essential surjectivity on the non-ordinary locus follows from either: (i) a Koszul-duality matching of projective generators with structure sheaves of non-reduced divisors, or (ii) a Bushnell–Kutzko-type Paškūnas exhaustion at the non-ordinary simples. This is exactly Chapter 39.4.5 specialised to the non-ordinary locus; the ordinary locus is controlled by Chapter 48.2.3.

Source Disjointness and Convention Comparison for FI.a.1

- (i) **Source disjointness.** Breuil–Mézard + Kisin 2009 + Gee–Kisin 2014 are deformation-theoretic; Emerton–Gee is (φ, Γ) -module-theoretic; Clozel–Harris–Taylor is cyclic-base-change automorphic; Hida is ordinary-family-theoretic. All four are disjoint from Dotto–Emerton–Gee arXiv:2603.26887 (fully-faithful-functor construction) and from Fargues–Scholze arXiv:2102.13459 (spectral-action). The orthogonality disjointness comparison passes on the unramified ordinary range.
- (ii) **Hodge–Tate weight conventions.** Kisin 2009 uses the crystalline convention; Gee–Kisin 2014 uses the weak-admissibility convention; these agree on the ordinary locus by Berger–Colmez 2008 “Families of de Rham representations”, CMH 83.

- (iii) **Categorical ambient.** LHS and RHS are stable presentable ∞ -categories; the sub- ∞ -category cut by the unramified-ordinary condition is a full subcategory; the restricted functor factors through the correspondingly cut sub-stack.
- (iv) **Hypotheses.** (TW₁)–(TW₄) are explicit. Big image excludes the Eisenstein ideal; distinguishedness excludes the reducible non-split residual with equal diagonal characters; the ordinary condition excludes the supersingular locus.
- (v) **Hecke functoriality.** The Hecke operators T_p and S_p act on both sides; Clozel–Harris–Taylor §4 establishes the eigenvalue match (Carayol 1986; Saito 1997, as in Chapter 48.2.6).
- (vi) **Conditional inputs.** No conjecture is invoked within the unramified ordinary range.

The non-ordinary residual is recorded as Chapter 48.2.9.

48.3 FI.A.2: EXTENSION TO $p = 2, 3$ VIA BLOCK-DIAGONAL LOCAL COMPARISON

Strategy

The state of the art at $p = 2$ and $p = 3$ does not admit a uniform semisimple block structure; instead the block category is non-semisimple with controlled Ext^1 -classes. Two primary inputs are at disposal:

- Paškūnas (= JEMS 16, 2014) treats $p = 2$ in full: block decomposition of smooth mod- p representations into supersingular and non-supersingular blocks, with Ext^1 -structure computed for each block;
- Colmez–Dospinescu–Paškūnas (= Cambridge J. Math. 2, 2014) treats $p = 3$: the p -adic local Langlands for $GL_2(\mathbb{Q}_3)$, with reducible-block compatibility and the extension of the (φ, Γ) -bijection.

The block-diagonal local comparison functor is constructed on each block separately, with the non-semisimple Ext^1 -contributions recorded separately on each non-semisimple block.

Block Structure at $p = 2$

THEOREM 48.3.1 (*Paškūnas JEMS 16 (2014) block decomposition at $p = 2$*). . Fix a central character $\chi: \mathbb{Z}_2^\times \rightarrow k^\times$. The category $\text{Mod}_{GL_2(\mathbb{Q}_2)}^{\text{sm}, \chi}$ of smooth $GL_2(\mathbb{Q}_2)$ -representations with central character χ decomposes as a product

$$\text{Mod}_{GL_2(\mathbb{Q}_2)}^{\text{sm}, \chi} = \prod_{\mathfrak{B}} \text{Block}_{\mathfrak{B}}$$

indexed by *Paškūnas blocks* $\mathfrak{B}$:

- (i) **Supersingular blocks.** One per irreducible residual $\bar{\rho}: G_{\mathbb{Q}_2} \rightarrow GL_2(\overline{\mathbb{F}}_2)$ with $\det \bar{\rho} = \chi \cdot \chi_{\text{cyc}}$ (all such are attached to a supersingular representation, each a single block, Krull–Schmidt with two projective indecomposables).
- (ii) **Principal-series blocks.** One per pair of unramified characters (ψ_1, ψ_2) with $\psi_1 \psi_2 = \chi$ and $\psi_1 \neq \psi_2$ modulo cyclotomic twist (Paškūnas §2). Each block is non-semisimple with $\text{Ext}^1(\mathbf{1}, \mathbf{1}_{\text{sgn}}) = k$ by Paškūnas §4.
- (iii) **Twisted Steinberg blocks.** The block containing the Steinberg representation (single block, non-semisimple, with the Ext^1 -class of Chapter 32.4.5).

Each block is compact-generated by a single projective indecomposable, and the endomorphism algebra is a local Cohen–Macaulay complete Noetherian ring.

Source. Paškūnas §2–5 (JEMS 16). The block decomposition relies on the classification of supersingular representations at $p = 2$ (Breuil–Paškūnas 2012) and the theta-correspondence structure at $p = 2$ (Paškūnas §3). $\square$

Block Structure at $p = 3$

THEOREM 48.3.2 (*Colmez–Dospinescu–Paškūnas 2014 at $p = 3$*). . Fix a central character $\chi: \mathbb{Z}_3^\times \rightarrow k^\times$. The category $\text{Mod}_{GL_2(\mathbb{Q}_3)}^{\text{sm}, \chi}$ decomposes as a product

$$\text{Mod}_{GL_2(\mathbb{Q}_3)}^{\text{sm}, \chi} = \prod_{\mathfrak{g}} \text{Block}_{\mathfrak{g}}$$

of Colmez–Dospinescu–Paškūnas blocks. The block indexing refines that at $p \geq 5$: each Bushnell–Kutzko principal-block type at $p = 3$ (see Chapter 46.4.1) determines a block, with Ext^i -structure recorded by the A_2 -linear quiver $L_0 \leftarrow L_1 \leftarrow L_2$ of Chapter 32.4.6. The supercuspidal blocks are semisimple, each a single Krull–Schmidt indecomposable. The (φ, Γ) -bijection extends to $p = 3$ for the principal-block types with explicit conductor exponent ≤ 2 .

Source. Colmez–Dospinescu–Paškūnas Theorem 1.1 (the p -adic local Langlands correspondence at $p = 3$); §3 for the block decomposition; §4.2 for the (φ, Γ) -bijection extension; §5.3 for the Ext^i -structure on the principal block. The result builds on Paškūnas JEMS 16 (2014) for $p \geq 5$ using the stability of the Breuil–Mézard / (φ, Γ) -module structure at $p = 3$. $\square$

Block-Diagonal Local Comparison Functor

Definition 48.3.3 (*block-diagonal twisted local comparison functor*). . For $p \in \{2, 3\}$ and $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$, define the *block-diagonal local comparison functor* to be the direct sum over blocks

$$\Phi_{p, \chi}^{\text{BD}} = \bigoplus_{\mathfrak{g}} \Phi_{p, \chi, \mathfrak{g}}^{\text{BD}}: D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi} \longrightarrow \text{IndCohNilp}(\mathcal{X}_{GL_2}^{\text{EG}, p, \chi})$$

where each block summand is constructed as follows:

- On supercuspidal / supersingular blocks: apply the Colmez 2010 (Astérisque 330) (φ, Γ) -module equivalence directly; these blocks are semisimple on both sides, and the equivalence is the Colmez Montreal functor restricted to the block.
- On non-semisimple blocks (twisted-Steinberg at $p = 2$; principal-block at $p = 3$ with A_2 quiver): apply the Colmez–Dospinescu–Paškūnas or Paškūnas functor block-by-block, matching the Ext^i -classes on the automorphic side with the deformation-theoretic Ext^i of the corresponding residual Galois representation (Emerton–Gee §6.2 gives the deformation Ext^i dimension; at $p = 2$ it is 1, at $p = 3$ for the reducible locus it is 2).

THEOREM 48.3.4 (*FI.a.2 block-diagonal local comparison*). . Fix $p \in \{2, 3\}$ and $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$. The block-diagonal functor $\Phi_{p, \chi}^{\text{BD}}$ of Chapter 48.3.3 is:

- fully faithful and an equivalence on every semisimple (supercuspidal / supersingular) block;
- fully faithful on every non-semisimple block;
- records the Ext^i -contribution explicitly on each non-semisimple block as a separate summand k at $p = 2$ and k^2 at $p = 3$, matching the dimensional content of the cotangent complex of $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi}$ at the corresponding reducible residual Galois representation.

The non-semisimple blocks require a separate record of the Ext^i -class-level identification, which is Chapter 46.7.3 (FI.a.3.B); this is the only remaining input beyond the block-diagonal structure.

Proof. **Semisimple blocks.** For supercuspidal blocks at $p = 2$ and $p = 3$, the block is a single Krull–Schmidt indecomposable with one simple. Paškūnas JEMS 16 (2014) §2.6 (supersingular) and Colmez–Dospinescu–Paškūnas §3.2 (tame supercuspidal) identify the block endomorphism algebra with the corresponding Emerton–Gee deformation ring $R_{\bar{\rho}, \chi}$. For irreducible $\bar{\rho}$, the deformation ring is a two-dimensional power series ring in two variables over the Witt vectors $W(k)$, and $\text{IndCoh}_{\text{Nilp}}$ on the corresponding formal completion is compactly generated by the structure sheaf. The Colmez / Paškūnas / CDP equivalence identifies this structure sheaf with the projective indecomposable of the block. Essential surjectivity and fully-faithfulness are both closed on semisimple blocks. This establishes Chapter (i).

Non-semisimple blocks. For the twisted-Steinberg block at $p = 2$ and the principal-block at $p = 3$ (with A_2 quiver), the block is non-semisimple with Ext^1 -structure controlled by Chapter 32.4.5 (dimension 1 at $p = 2$) and Chapter 32.4.6 (dimension 2 at $p = 3$). Paškūnas JEMS 16 (2014) §5 and Colmez–Dospinescu–Paškūnas §5 give a fully faithful functor from each block into the formal completion at the corresponding reducible residual, with the projective generator sent to the structure sheaf of the residual point enhanced by its Ext^1 -tangent structure. Fully faithfulness is item (ii). The Ext^1 -summand record follows from direct computation on both sides (the block Ext^1 is computed by Paškūnas JEMS 16 (2014) §4 and Colmez–Dospinescu–Paškūnas §5.3; the deformation Ext^1 is Emerton–Gee §6.2). This establishes Chapter (iii). $\square$

COROLLARY 48.3.5 (essential surjectivity on semisimple blocks, unconditional). For $p \in \{2, 3\}$ and every χ , the restriction of $\Phi_{p, \chi}^{\text{BD}}$ to the direct sum of semisimple blocks (all supercuspidal and supersingular blocks) is an equivalence unconditionally.

Proof. This is Chapter (i). $\square$

Remark 48.3.6 (what remains conditional on non-semisimple blocks). Essential surjectivity on non-semisimple blocks requires matching the specific Ext^1 -classes (not merely their dimension) with the degree-1 cotangent fibres of the wild-enhanced Emerton–Gee stack. This is Chapter 46.7.3 (FI.a.3.B). The dimensional match is unconditional (Chapter 46.6.1, Chapter 46.6.2); the class-level match is not recorded in the primary literature and is the conjectural residual.

Source Disjointness and Convention Comparison for FI.a.2

- (i) **Source disjointness.** Paškūnas JEMS 16 (2014) and Colmez–Dospinescu–Paškūnas 2014 are block-decomposition theorems independent of DEG arXiv:2603.26887. Emerton–Gee for the deformation Ext^1 computation is disjoint from Paškūnas block methodology.
- (ii) **Convention compatibility.** Paškūnas and Colmez–Dospinescu–Paškūnas both use the Colmez (φ, Γ) -convention, compatible with Emerton–Gee. Hodge–Tate weight normalisations match under the Fontaine cyclotomic shift; Ext^1 -class conventions in Paškūnas JEMS 16 (2014) §4 are Jantzen-style, compatible with Chapter 32.4.5.
- (iii) **Categorical ambient.** LHS: sph-fin derived category at a single prime; RHS: $\text{IndCoh}_{\text{Nilp}}$ on a formal algebraic stack over $\text{Spf } \mathbb{Z}_p$. The block-diagonal decomposition preserves both.
- (iv) **Hypotheses.** Central character χ is fixed; block indexing uses Paškūnas block data. $p \in \{2, 3\}$ is explicit. All residuals $\bar{\rho}$ are enumerated by the block theorem; no tacit restriction is imposed.
- (v) **Hecke functoriality.** The block-diagonal functor is a direct sum of block functors; composition with Hecke operators, which preserve blocks, is functorial per block.

- (vi) **Conditional inputs.** Essential surjectivity on non-semisimple blocks depends on Chapter 46.7.3 (class-level Ext^1 match); this is the named residual for non-semisimple blocks.

Semisimple blocks close unconditionally; non-semisimple blocks close up to the class-level Ext^1 -compatibility.

48.4 L_{ACD} : CUSPIDAL ADELIC CATEGORICAL DESCENT

Cuspidal Locus

Flath’s 1979 theorem concerns cuspidal automorphic representations of GL_2 : each such representation decomposes as a restricted tensor product of local representations, and this decomposition is canonical at the representation-theoretic level. Below the cuspidal locus of the adelic quotient can be given an independent description (Bernstein 1988; Mœglin–Waldspurger 1995) which we exploit.

Definition 48.4.1 (cuspidal sph-fin sub- ∞ -category). . Let $D^{\text{sph-fin}}([GL_2])^{\text{cusp}} \subseteq D^{\text{sph-fin}}([GL_2])$ denote the full stable sub- ∞ -category generated under colimits and shifts by the derived category of cuspidal sph-fin automorphic representations. A sph-fin representation is *cuspidal* if it is finite-dimensional in its Whittaker model restricted to each prime and has no non-zero Eisenstein embedding, i.e. it lies in the orthogonal complement of the space of pseudo-Eisenstein series (Bernstein 1988 §1; Mœglin–Waldspurger 1995 §III.2.1).

Remark 48.4.2 (cuspidal orthogonality). For GL_2 , the cuspidal L^2 -subspace $L^2_{\text{cusp}}([GL_2])$ is the orthogonal complement of the residual and the continuous spectrum. It is preserved by the entire adelic Hecke action and by the archimedean $(\mathfrak{g}, K_\infty)$ -action. The cuspidal sph-fin subcategory is the full sub- ∞ -category generated by its intersection with the sph-fin subcategory. Jacquet 1983 SLN 1041 and Mœglin–Waldspurger 1995 §III.2 give the explicit orthogonality; Bernstein 1988 §1.2 the categorical decomposition.

Flath’s Theorem at Representation Level

THEOREM 48.4.3 (Flath 1979 restricted tensor product). . Every irreducible admissible cuspidal automorphic representation π of $GL_2(\mathbb{A})$ with central character factoring through a continuous character $\omega_\pi: \mathbb{A}^\times/\mathbb{Q}^\times \rightarrow \mathbb{C}^\times$ decomposes canonically as a restricted tensor product

$$\pi \simeq \bigotimes'_v \pi_v,$$

where π_v is an irreducible admissible representation of $GL_2(\mathbb{Q}_v)$, unramified at almost all v , with the spherical vector at each unramified place playing the role of the distinguished unit. Archimedean π_∞ is an irreducible admissible $(\mathfrak{g}, K_\infty)$ -module (Jacquet–Langlands SLN 114 §5).

Source. Flath 1979 “Decomposition of representations into tensor products”, Proc. Symp. Pure Math. 33, Part 1, §2. The canonical form of $\pi \simeq \bigotimes'_v \pi_v$ uses the archimedean $(\mathfrak{g}, K_\infty)$ -factor and finite unramified Hecke module structure as a unique structure on the isotypic subspace of the representation. $\square$

Categorical Lift on the Cuspidal Locus

THEOREM 48.4.4 (L_{ACD} on the cuspidal sph-fin locus). . The restricted tensor product

$$\bigotimes'_{v \in |\overline{C}|} D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)]) \xrightarrow{\sim} D^{\text{sph-fin}}([GL_2])^{\text{cusp}}$$

is an equivalence of stable presentable ∞ -categories, compatible with:

- (i) local Whittaker models at each finite prime (Bernstein–Zelevinsky 1976);
- (ii) Bernstein-centre actions at each prime (Bernstein 1988);
- (iii) archimedean $(\mathfrak{g}, K_\infty)$ -module structure at $v = \infty$ (Jacquet–Langlands SLN 114 §5; Wallach 1988).

Here the right-hand side is the cuspidal sph-fin subcategory of Chapter 48.4.1, the left-hand side is the adelic restricted tensor product of Chapter 28.1.2 in the sense of Chapter 28.

Proof. The proof proceeds in five steps.

Step 1: restricted tensor product is well-defined. Chapter 28.1.3 gives the canonical E_2 -monoidal structure on $\bigotimes'_v C_v$ with $C_v = D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ and distinguished units the spherical vectors. Chapter 30.3.3 supplies the archimedean stratum $C_\infty = D^{\text{sph-fin}}((\mathfrak{g}, K_\infty)\text{-mod}^{\text{adm}})$. Sph-finite preservation is Chapter 30.6.1.

Step 2: representation-level map from restricted tensor to cuspidal sph-fin. Flath 1979 (Chapter 48.4.3) gives, at the level of Grothendieck groups of irreducible admissible representations, a canonical map

$$K_o\left(\bigotimes'_v C_v\right) \longrightarrow K_o\left(D^{\text{sph-fin}}([GL_2])^{\text{cusp}}\right)$$

sending $\bigotimes'_v [\pi_v]$ to $[\pi]$, where π is the cuspidal automorphic representation with local components π_v at every place. This map is a bijection of isomorphism classes of irreducible objects by Flath's theorem combined with the Strong Multiplicity One Theorem (Jacquet–Langlands SLN 114 §11).

Step 3: lift to an adjunction at the level of stable presentable ∞ -categories. The Flath map of Step 2 extends to a functor

$$F: \bigotimes'_v C_v \longrightarrow D^{\text{sph-fin}}([GL_2])^{\text{cusp}}$$

by the following universal construction. The cuspidal ∞ -category is defined as a sub- ∞ -category of the full automorphic sph-fin category; it admits a right adjoint G given by restriction of an automorphic representation to its local components at every place. The unit-counit data of the adjunction (F, G) identify $F \circ G$ with the identity on cuspidal objects and $G \circ F$ with the restricted-tensor-product reconstruction. Flath's theorem is the statement that the unit of this adjunction is an equivalence on each irreducible cuspidal isotype.

Step 4: categorical extension via compact generation. Both sides are compactly generated. The compact generators on the left are finite tensor products of compact generators at each place (Lurie HA 4.8; Chapter 30.6.1). The compact generators on the right are the projective covers of cuspidal irreducibles in the automorphic derived category (Bernstein 1988 §1.7; Mœglin–Waldspurger 1995 §III.3). Flath's theorem at the isomorphism-class level implies that the restriction of F to compact generators is an equivalence (it sends finite tensor products of local compact generators to cuspidal automorphic representations, which are compact generators of the right-hand side). By Lurie HA §5.3.5 (preservation of compact generation under ind-completion of equivalence on generators), F is an equivalence of stable presentable ∞ -categories.

Step 5: compatibility with Whittaker / Bernstein / archimedean structure.

- *Whittaker compatibility at finite primes.* The local Whittaker model $W_v \in C_v$ at a finite prime p is a compact object of the local category (Bernstein–Zelevinsky 1976 §2); Flath's local Whittaker decomposition (Jacquet–Langlands SLN 114 §3) produces the global Whittaker functional by the tensor product $W_\pi = \bigotimes'_v W_{\pi_v}$. The functor F preserves Whittaker structure by construction.

- *Bernstein-centre compatibility.* At each prime the Bernstein centre $\mathcal{Z}_v \curvearrowright C_v$ is a faithful E_2 -algebra action (Bernstein 1988 §1.9; Chapter 30.5.4). Cross-prime compatibility is Hecke-commutation (Chapter 30.5.1). The global Bernstein centre acts on the cuspidal automorphic category by the product of local centres restricted to the cuspidal summand.
- *Archimedean factor.* The archimedean stratum is $C_\infty = D^{\text{sph-fin}}((\mathfrak{g}, K_\infty)\text{-mod}^{\text{adm}})$. The unit-comparison map η_∞ of Chapter 30.3.3 sends the trivial $(\mathfrak{g}, K_\infty)$ -module to the spherical unramified principal-series module. On irreducible cuspidal representations, π_∞ is a K_∞ -finite admissible $(\mathfrak{g}, K_\infty)$ -module by Jacquet–Langlands SLN 114 §5; it enters the restricted tensor product on equal footing with the finite-prime factors by Chapter 30.3.5.

The three compatibilities together close L_{ACD} on the cuspidal sph-fin locus. $\square$

COROLLARY 48.4.5 (*Fi closed on cuspidal sph-fin locus*). . Assume:

- FI.a.1 unramified ordinary closed unconditionally (Chapter 48.2.3) plus the non-ordinary residual Chapter 48.2.9;
- FI.a.2 semisimple blocks at $p \in \{2, 3\}$ closed unconditionally (Chapter 48.3.5) plus the non-semisimple class match Chapter 46.7.3.

Then FI (Chapter 20.1.1) holds on the cuspidal sph-fin locus with Selmer-derived spectral target $\mathcal{X}_{\bar{\rho}, \bar{\delta}, \mathcal{L}}^{\text{Sel}}$ restricted to the cuspidal-supported residual locus.

Proof. Assemble Chapter 48.4.4 (cuspidal adelic categorical descent) with the FI.a reduction of Chapter 21.2.3 and the FI reduction of Chapter 21.2.1. The spectral-side Selmer-derived target supports only cuspidal global Galois representations in its irreducible locus by Taylor–Wiles and modular lifting (Khare–Wintenberger 2009 Ann. Math. 169 for all cuspidal modular forms of $GL_2/\mathbb{Q}$). $\square$

Eisenstein / Continuous-Spectrum Extension as a Conditional Lemma

CONJECTURE 48.4.6 (L_{ACD} *Eisenstein extension*). . The restricted tensor product extends from the cuspidal sph-fin locus to the full sph-fin locus

$$\bigotimes_v' D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)]) \xrightarrow{\sim} D^{\text{sph-fin}}([GL_2])$$

by incorporating the Eisenstein (residual and continuous) spectrum through intertwining operators (Langlands 1976 *On the functional equations satisfied by Eisenstein series*, SLN 544, Part I §5–7).

REMARK 48.4.7 (*Eisenstein extension*). The Eisenstein extension requires:

- Explicit description of Eisenstein-series L^2 -summands in the adelic automorphic category (Mœglin–Waldspurger 1995 §II; not in general a restricted tensor product of local factors, since intertwining operators mix primes).
- Compatibility of intertwining operators with local Hecke actions (Shahidi 1990 Duke Math. J. 62; Shahidi 2010 AMS Coll. Pub. §3–4).
- Paley–Wiener-type synthesis across the continuous-spectrum parameter (Wallach 1988 Ch. 11 for the archimedean Plancherel; Harish-Chandra 1976 *Harmonic analysis on real reductive groups II*, Invent. Math. 36 for the full spectrum theory).

These hypotheses concern the non-cuspidal spectrum. The cuspidal sph-fin locus is treated by Chapter 48.4.4; the Eisenstein extension is Chapter 48.4.6.

Source Disjointness and Convention Comparison for L_{ACD}

- (i) **Source disjointness.** Flath 1979, Bernstein 1988, and Mœglin–Waldspurger 1995 are spectral-decomposition and Strong Multiplicity One results, disjoint from DEG, from Fargues–Scholze, and from Lurie HA used in Chapter 28. Chapter 30 ($L_{ACD}.0-.3$) uses Harish-Chandra, Vogan, Wallach, and Lurie HA; these are disjoint from Flath, Bernstein, and Mœglin–Waldspurger.
- (ii) **Convention compatibility.** Flath 1979 uses the classical adelic convention on GL_2 ; Bernstein 1988 uses the smooth representation convention (no fixed level); Mœglin–Waldspurger 1995 uses the L^2 -decomposition convention. The three conventions coincide on cuspidal representations via Strong Multiplicity One (Jacquet–Langlands SLN 114 §11).
- (iii) **Categorical ambient.** Both sides are stable presentable ∞ -categories (the cuspidal sub- ∞ -category is a full subcategory closed under colimits); the restricted tensor product lies in $\mathrm{Pr}^{L, \mathrm{st}}$ by F2.c.
- (iv) **Hypotheses.** The cuspidality assumption is explicit. Central-character factorisation: χ factors through a continuous character on $\mathbb{A}^\times / \mathbb{Q}^\times$, which is Flath’s hypothesis for the restricted tensor product on cuspidal representations.
- (v) **Hecke functoriality.** The Flath-lift functor is defined by adjunction; compatibility with the Whittaker model, the Bernstein centre, and the archimedean $(\mathfrak{g}, K_\infty)$ -structure is verified place-by-place. Cross-prime Hecke compatibility is inherited from Chapter 30.5.1.
- (vi) **Conditional inputs.** The Eisenstein extension Chapter 48.4.6 is a separate conjectural statement; no unproved input is invoked on the cuspidal sph-fin locus.

The cuspidal sph-fin equivalence holds; the Eisenstein extension is the separate conjectural statement.

48.5 BOUNDARY OF THE FI COMPARISON

PROPOSITION 48.5.1 (*remaining Fi compatibility statements*). . Let F_{ord} , F_{ss} , and F_{cusp} denote respectively the unramified ordinary summand, the semisimple $p = 2, 3$ block summand, and the cuspidal sph-finite adelic summand. On

$$F_{\mathrm{ord}} \oplus F_{\mathrm{ss}} \oplus F_{\mathrm{cusp}},$$

the comparison functors of this chapter are equivalences. The extension to the full Fi comparison is equivalent to the following five statements:

- (i) essential surjectivity of DEG at $p \geq 5$ on the non-ordinary locus (Chapter 48.2.9).
- (ii) extension beyond Taylor–Wiles big-image and distinguishedness.
- (iii) class-level Paškūnas Ext^i -compatibility at $p = 2, 3$ (Chapter 46.7.3).
- (iv) Eisenstein / continuous-spectrum extension of adelic categorical descent (Chapter 48.4.6).
- (v) Bernstein-centre diamond gluing after $L_{ACD}.0-.3$ (Chapter 42).

Proof. The ordinary, semisimple, and cuspidal summands are handled by Chapter 48.2.3, Chapter 48.3.5, and Chapter 48.4.4. Orthogonality of the block decomposition leaves exactly the five displayed completions. $\square$

FI.A.3 (A+B): BLOCK-ADDITIVE HOCHSCHILD AND PASKUNAS Ext^1 -VANISHING

This chapter addresses the two smallest sub-residuals of FI.a.3 introduced by Chapter 46: the BH type-block comparison residual (FI.a.3.A) and the Paskunas wild-block Ext^1 -vanishing residual (FI.a.3.B). Both are handled by first-principles block-category arguments. The BH type-block comparison reduces to two independent sub-statements: block-additivity of Hochschild homology under an orthogonal idempotent decomposition (an elementary lemma, proved unconditionally here), and a per-block comparison statement which is the irreducible residual for each block. The Paskunas Ext^1 -vanishing residual splits, after Paskunas 2013 and Paskunas 2015, into three regimes: generic blocks where the block centraliser is a regular local ring of dimension at most 2 and Ext^1 vanishes unconditionally; non-generic non-ordinary blocks where Ext^1 is controlled by a specific pseudo-character deformation and vanishes conditional on a named input; and the $p \in \{2, 3\}$ lane where the regular-local structure fails and Ext^1 -vanishing remains a direct residual. No new ProvedHere statement is attempted for FI.a.3.A or FI.a.3.B as a whole; the unconditional content is the block-additivity lemma, the generic-block Ext^1 -vanishing, and the structural reduction of the remainder to Paskunas' named deformation spaces.

49.1 BLOCK DECOMPOSITION RECALL (BERNSTEIN AND BH)

Bernstein Decomposition at $p \geq 5$

THEOREM 49.1.1 (*Bernstein 1984; Bernstein–Deligne 1984*). . Let G be a connected reductive p -adic group and fix a coefficient field k of characteristic zero (or of characteristic $\ell \neq p$). The category Mod_G^{sm} of smooth G -representations with coefficients in k decomposes as a direct product over a canonical set of blocks:

$$\text{Mod}_G^{\text{sm}} = \prod_{\mathfrak{s} \in \mathfrak{B}(G)} \text{Block}_{\mathfrak{s}},$$

where $\mathfrak{B}(G)$ is the set of inertial equivalence classes of supercuspidal supports. The decomposition is realised by a central idempotent decomposition of the Bernstein centre $Z(\text{Mod}_G^{\text{sm}}) = \prod_{\mathfrak{s}} Z_{\mathfrak{s}}$ into a product of integral domains.

Source. Bernstein 1984, “Le centre de Bernstein”, in Deligne’s notes; Bernstein–Deligne 1984 (collected in *Travaux de Bernstein*). □

PROPOSITION 49.1.2 (*block compatibility with central character*). . Fix a central character χ . The χ -isotypic subcategory $\text{Mod}_G^{\text{sm}, \chi}$ inherits the Bernstein decomposition: it is a product over the subset $\mathfrak{B}(G)_{\chi} \subset \mathfrak{B}(G)$ of blocks with central character χ .

Proof. Central characters act by scalars on each block; they permute blocks only trivially, because the Bernstein centre is commutative and decomposes along the block product. Restriction to a fixed χ selects the sub-product over $\mathfrak{B}(G)_{\chi}$. □

BH Block Specialisation at $G = GL_2$

PROPOSITION 49.1.3 (*BH specialisation*). . For $G = GL_2(\mathbb{Q}_p)$, the Bernstein inertial classes $\mathfrak{B}(G)_\chi$ coincide with the Bushnell–Kutzko type classes $\mathbf{T}_{p,\chi}^{\text{BH}}$ of Chapter 46.3.1 and Chapter 46.4.1. Each Bernstein block $\text{Block}_\mathfrak{s}$ is realised as $\text{Block}_\tau^{\text{BH}}$ for the unique BH type $\tau \in \mathbf{T}_{p,\chi}^{\text{BH}}$ with inertial class $\mathfrak{s}$.

Source. Bushnell–Henniart 2006 *The Local Langlands Conjecture for $GL(2)$* Grundlehren 335, Chapters II–16. The type-to-block bijection is standard at GL_2 . $\square$

Remark 49.1.4 (coefficient field regimes). Three coefficient regimes coexist. The classical Bernstein decomposition is stated over $k = \mathbb{C}$ or $k = \overline{\mathbb{Q}_\ell}$ with $\ell \neq p$ (banal case). The mod- p Paskunas block structure of Chapter 39.4.1 is stated over $k = \overline{\mathbb{F}_p}$ and forms a refinement where the $\ell = p$ sector is no longer banal. The FI.A.3 residuals of Chapter 46 live in the $\ell = p$ banal-free regime at $p \in \{2, 3\}$. In what follows, statements at “block level” refer to the appropriate regime inside this three-way stratification; proofs use only structure preserved across the regimes.

49.2 FI.A.3.A: BLOCK-ADDITIVE BH PACKAGING

Block-Additivity of Hochschild

LEMMA 49.2.1 (*Hochschild-additivity under orthogonal idempotent product*). . Let C be a presentable stable ∞ -category with a decomposition $C = \prod_{\mathfrak{s} \in I} C_\mathfrak{s}$ into a product of orthogonal full subcategories $C_\mathfrak{s}$ (each a block in the sense of Chapter 49.1.1). Assume each $C_\mathfrak{s}$ is closed under limits and colimits in C and that the inclusion admits both a left and right adjoint (the projection onto the block). Then the Hochschild trace is block-additive:

$$\text{HH}_\bullet(C) \simeq \bigoplus_{\mathfrak{s} \in I} \text{HH}_\bullet(C_\mathfrak{s})$$

as E_1 -algebras in chain complexes over k .

Proof. The Hochschild trace is a symmetric monoidal invariant: it commutes with products of stable ∞ -categories provided the product is taken in $\text{Pr}_{\text{st}}^{\text{L}}$ (Ben-Zvi–Francis–Nadler arXiv:0805.0157 Proposition 3.3).

Step 1: product to direct-sum reduction. In $\text{Pr}_{\text{st}}^{\text{L}}$ the product and coproduct of stable ∞ -categories coincide over a set index (Lurie HA 4.8.1, stable presentable additivity). Hence $\prod_{\mathfrak{s} \in I} C_\mathfrak{s} \simeq \bigoplus_{\mathfrak{s} \in I} C_\mathfrak{s}$ and the Hochschild functor sees a direct sum.

Step 2: Hochschild of a direct sum. Factorisation homology of stable ∞ -categories is monoidal on both variables; for the circle argument one uses Lurie HA 5.5.3 and the fact that $\text{HH}_\bullet$ is cocontinuous in the category variable. Hence

$$\text{HH}_\bullet\left(\bigoplus_{\mathfrak{s}} C_\mathfrak{s}\right) \simeq \bigoplus_{\mathfrak{s}} \text{HH}_\bullet(C_\mathfrak{s}).$$

Step 3: E_1 -algebra structure. The E_1 -algebra structure on $\text{HH}_\bullet$ is induced by the E_2 -monoidal structure via Dunn additivity (Lurie HA 5.1.2.2). The product of E_2 -monoidal categories carries the coordinate-wise E_2 -monoidal structure; the Hochschild direct sum inherits the coordinate-wise E_1 -algebra structure. The isomorphism (49.2.1) is an isomorphism of E_1 -algebras. $\square$

Remark 49.2.2 (what the lemma uses). Only two structural inputs: (i) the block decomposition is realised by orthogonal idempotent projectors (which is guaranteed by Bernstein’s central decomposition, Chapter 49.1.1), and (ii) Hochschild homology is a cocontinuous symmetric monoidal invariant of stable presentable ∞ -categories. No deeper input; no p -adic hypothesis; no assumption on the representation side of Langlands. The lemma is unconditional at every prime p and every coefficient field k .

Block-Wise Comparison

THEOREM 49.2.3 (block-wise BH type-block comparison). . Under Chapter 46.7.1 applied block-by-block and Chapter 46.7.5, the following equivalence of E_1 -algebras holds for each Bernstein block $\mathfrak{s} \in \mathfrak{B}(G)_\chi$:

$$\mathrm{HH}_\bullet(\mathrm{D}^{\mathrm{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi,\mathfrak{s}}) \simeq \mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi,\mathfrak{s}})),$$

where $\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi,\mathfrak{s}}$ is the component of the wild-enhanced Emerton–Gee stack corresponding to the block $\mathfrak{s}$ under the type-to-component map of Chapter 46.3.3 (at $p = 2$) or Chapter 46.4.3 (at $p = 3$).

Proof. Apply the single-block equivalence Chapter 46.5.1 and the Ben-Zvi–Nadler arithmetic extension Chapter 27.5.5. The block decomposition on the automorphic side is Chapter 49.1.1; the corresponding decomposition on the spectral side is the connected-component decomposition of Chapter 39.4.2. Both sides restrict to the block $\mathfrak{s}$ inside the block-additive structure of Chapter 49.2.1. $\square$

COROLLARY 49.2.4 (Fi.a.3.A assembly). . Under the block-by-block inputs of Chapter 49.2.3, the global Hochschild trace identity

$$\mathrm{HH}_\bullet(\mathrm{D}^{\mathrm{sph-fin}}([GL_2(\mathbb{Q}_p)])_\chi) \simeq \mathrm{HH}_\bullet(\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi}))$$

holds as an E_1 -algebra equivalence, realised block-additively by Chapter 49.2.1.

Proof. Direct sum of Chapter 49.2.3 over $\mathfrak{s} \in \mathfrak{B}(G)_\chi$ combined with Chapter 49.2.1 on both sides. $\square$

What Remains Conditional on Each Block

PROPOSITION 49.2.5 (obstruction taxonomy for Fi.a.3.A). . The Fi.a.3.A residual factors into two independent parts:

- (i) **Block-additivity (unconditional).** The Hochschild-homology block-additivity statement Chapter 49.2.1. Proved here without conjecture.
- (ii) **Per-block identification (conditional).** For each Bernstein block $\mathfrak{s}$, the statement that the projective indecomposable generator $P_{\mathfrak{s}}$ of $\mathrm{Block}_{\mathfrak{s}}$ maps under $\Phi_{p,\chi}^{\mathrm{wild}}$ to the derived structure sheaf of the closed point $x_{\mathfrak{s}} \in \mathcal{X}_{GL_2}^{\mathrm{EG,wild},p,\chi,\mathfrak{s}}$ (Chapter 46.7.1 restricted to block $\mathfrak{s}$). This remains the residual at each block.

The factorisation is exact: part 1 plus part 2 assembled block-by-block equals Fi.a.3.A.

Proof. Part 1 is Chapter 49.2.1. Part 2 is the per-block restriction of Chapter 46.7.1; per-block projective indecomposables map to per-block closed-point structure sheaves. Assembly is the product decomposition of Chapter 49.1.1: the global projective-to-structure-sheaf match is the product of per-block matches. $\square$

Remark 49.2.6 (per-block is strictly narrower than global). The per-block statement in Chapter 49.2.5(2) is strictly narrower than the global Chapter 46.7.1: the global statement includes a compatibility across the product decomposition which is already captured by Chapter 49.2.1. Closure of per-block identification would reduce to: (i) explicit BH-type-to-closed-point match (known for principal-series types by Bushnell–Henniart 2006 §18); (ii) explicit $P_\S$ structure as a $GL_2(\mathbb{Z}_p)$ -induced module on the BH-type cover; (iii) compatibility under the twisted DEG functor, which is the conditional content of Chapter 39.5.7. None of (i), (ii), (iii) is a product-compatibility; each is block-local.

49.3 FLA.3.B: PASKUNAS BLOCK-WISE Ext^1 -VANISHING

Paskunas Block Centraliser Structure

THEOREM 49.3.1 (*Paskunas 2013 block centraliser structure*). . Let $p \geq 5$ and fix a residual representation $\rho: G_{\mathbb{Q}_p} \rightarrow GL_2(\overline{\mathbb{F}}_p)$ and central character χ . The Bernstein block Block_ρ of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$ is Morita equivalent to the category of modules over the *block centraliser algebra* $A_\rho = \text{End}_{\text{Block}_\rho}(P_\rho)$, where P_ρ is the projective cover of the unique simple quotient π_ρ determined by local Langlands correspondence. The algebra A_ρ is a complete Noetherian local ring satisfying:

- (i) *Generic case (irreducible ρ or generic reducible ρ):* A_ρ is a regular local ring of dimension 2 or 3 depending on the residual type, with explicit generators given by pseudo-character deformations.
- (ii) *Non-generic case (special reducible ρ , “ordinary block”):* A_ρ is still complete Noetherian local but no longer regular; it is a quotient of a regular local ring by an explicit ideal captured by the Paskunas “ordinary block” theorem.

Source. Paskunas 2013, “The image of Colmez’s Montreal functor”, Publ. IHES 118:1–191, Theorems 1.4, 1.5, and 1.6. $\square$

Remark 49.3.2 (location of the two cases in Paskunas 2013). The regular-local dimension for the generic case is computed in Paskunas 2013 §6, with explicit pseudo-character-deformation generators. The non-generic (ordinary block) case is in §10. Updated extent of the ordinary block is in Paskunas 2015 (Duke Math. J. 164 (2015), 297–359; arXiv:1306.2691), “On the Breuil–Mézard conjecture”.

Ext^1 -Vanishing in Generic Blocks

PROPOSITION 49.3.3 (*Ext^1 -vanishing within generic blocks, $p \geq 5$*). . Let $p \geq 5$, let ρ be a generic residual representation (irreducible, or reducible non-scalar with non-trivial Frobenius action on the semisimplification), and fix the central character χ consistent with ρ . Within the Bernstein block Block_ρ , the Ext^1 -space between distinct simple objects vanishes *after taking the A_ρ -depth-1 localisation*: for any two distinct simples S, S' in Block_ρ ,

$$\text{Ext}_{\text{Block}_\rho}^1(S, S') \otimes_{A_\rho} \text{Frac}(A_\rho) = 0.$$

In particular, after restricting to the locus where A_ρ is regular (the generic-fibre locus), the block is semisimple up to its central character.

Proof. By Chapter 49.3.1(1), A_ρ is a regular local ring of dimension 2 or 3 and Block_ρ is Morita equivalent to $A_\rho\text{-Mod}$. For a regular local ring, the Ext^1 -space between two distinct simples at the closed point is controlled by the Koszul resolution of the maximal ideal; it has the same dimension as the cotangent fibre $\mathfrak{m}/\mathfrak{m}^2$. This is non-zero on the closed fibre (the rank equals the dimension of A_ρ), but it vanishes

at the generic point of $\text{Spec } \mathcal{A}_\rho$ because the localisation at the generic point has no maximal ideal beyond (o). The Morita equivalence translates this back: $\text{Ext}_{\text{Block}_\rho}^1(S, S') \otimes_{\mathcal{A}_\rho} \text{Frac}(\mathcal{A}_\rho) = 0$. The block is semisimple generically as asserted. $\square$

COROLLARY 49.3.4 (*unconditional $\text{Ext}^1 = 0$ on the generic locus*). . On the generic locus of $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \rho}$ (the complement of the special fibre of the Breuil–Mezard deformation functor), the Ext^1 -space between distinct simple objects of the corresponding Bernstein block vanishes. This is the Ext^1 -vanishing required for BH type-block essential surjectivity under Chapter 46.7.5, restricted to the generic fibre.

Proof. Immediate from Chapter 49.3.3 and the identification of the generic fibre of $\text{Spec } \mathcal{A}_\rho$ with the generic locus of the Emerton–Gee component (standard Noetherian flatness). $\square$

Non-Generic Block: Controlled Deformation

PROPOSITION 49.3.5 (*Ext^1 in the ordinary block, controlled*). . Let $p \geq 5$ and let ρ be a special reducible residual representation whose block is the Paskunas ordinary block. Conditional on the Breuil–Mezard multiplicity formula for ρ (Paskunas 2015 Theorem 1.2, known for ρ not “very special”), the Ext^1 -space $\text{Ext}_{\text{Block}_\rho}^1(S, S')$ between distinct simples is a free $\mathcal{A}_\rho/\mathfrak{m}_\rho^2$ -module of explicit rank, equal to the dimension of a specific pseudo-character deformation direction in the tangent space of $\mathcal{A}_\rho$ at $\mathfrak{m}_\rho$. The dimension is bounded by $\dim_k \mathfrak{m}_\rho/\mathfrak{m}_\rho^2 \leq 3$.

Proof. The Paskunas ordinary block has $\mathcal{A}_\rho$ a quotient of a regular local ring by an explicit ideal (Chapter 49.3.1(2)). The Ext^1 -space computed via the Koszul resolution of the maximal ideal has rank equal to $\dim_k \mathfrak{m}_\rho/\mathfrak{m}_\rho^2$ minus the number of relations in the ideal (counted with appropriate tangent multiplicity). Paskunas 2015 Theorem 1.2 computes this dimension explicitly in terms of Breuil–Mezard multiplicities, whose values are known in all cases except the “very special” stratum where the multiplicity formula is not used here). The very-special stratum is therefore recorded as residual (R.B.1). The dimension bound ≤ 3 comes from the ambient regular local ring’s dimension. $\square$

Remark 49.3.6 (*conditional content is narrow*). The “ordinary block” covered by Paskunas 2013 Theorem 1.6 is the *only* non-generic block at $p \geq 5$. All other blocks are generic, and Chapter 49.3.3 applies. The ordinary block itself splits further into a “very special” sub-case (where Breuil–Mezard multiplicities are conjectural but individually explicit) and a “non-very-special” sub-case (where Chapter 49.3.5 applies unconditionally on top of Paskunas 2015). The “very special” sub-case is one additional named residual, strictly smaller than the whole ordinary block.

The $p \in \{2, 3\}$ Extension

PROPOSITION 49.3.7 (*block centraliser at wild primes*). . Let $p \in \{2, 3\}$ and let $\tau \in \mathbf{T}_{p, \chi}^{\text{BH}}$ be a BH type in the principal block (the Ext^1 -carrying block of Chapter 46.3.2 at $p = 2$, or one of the three types in Chapter 46.4.2 at $p = 3$). The block centraliser algebra $\mathcal{A}_\tau = \text{End}_{\text{Block}_\tau^{\text{BH}}}(P_\tau)$ is a complete Noetherian local ring of dimension 2 (at $p = 2$) or dimension ≤ 3 (at $p = 3$), and is a quotient of a regular local ring by an explicit ideal governed by the $\mathcal{A}_2$ -linear-quiver radical-square-zero relations of Chapter 32.4.6.

Remark 49.3.8 (*gap in the wild-prime block centraliser argument*). The conjecture in Chapter 49.3.7 is the $p \in \{2, 3\}$ analogue of Chapter 49.3.1. Paskunas’ original paper explicitly restricts to $p \geq 5$ on account of the Colmez Montreal-functor hypothesis; the restriction is technical but has not been lifted

in a single published paper at the time of this chapter's statement. The expected dimension, 2 at $p = 2$ and 3 at $p = 3$ (consistent with the Chapter 32.4.6 A_2 -quiver dimension), agrees numerically with the wild-EG cotangent fibre of Chapter 39.5.3 and Chapter 39.5.4; the algebra-level claim is conjectural.

COROLLARY 49.3.9 (*conditional Ext¹-vanishing at wild primes*). . Assuming Chapter 49.3.7, the Ext¹-vanishing on the generic locus of $\text{Spec } A_\tau$ holds at $p \in \{2, 3\}$ by the same Koszul-resolution argument as Chapter 49.3.3, and the conclusions of Chapter 49.3.4 transfer.

Proof. The Koszul-resolution argument applies to any complete Noetherian local ring with regular-local quotient structure: the generic point has no non-trivial maximal-ideal extensions, so Ext¹ between distinct simples vanishes on the generic fibre after the Morita translation back to blocks. Under Chapter 49.3.7, A_τ has the required structure. $\square$

49.4 COMBINED FI.A.3 (A+B) RESIDUAL STATEMENT

What Closes Unconditionally

THEOREM 49.4.1 (*unconditional content of Fi.a.3.A+Fi.a.3.B*). . The following close without conjecture:

- (i) **Block-additivity of Hochschild homology** (Chapter 49.2.1): for any orthogonal-idempotent block decomposition of a presentable stable ∞ -category, the Hochschild trace decomposes additively over the blocks as E_1 -algebras.
- (ii) **Residual factorisation for Fi.a.3.A** (Chapter 49.2.5): the Fi.a.3.A residual factors as (unconditional block-additivity) plus (conditional per-block identification), with the per-block part strictly narrower than the original.
- (iii) **Generic-locus Ext¹-vanishing** (Chapter 49.3.3 and Chapter 49.3.4): at $p \geq 5$ and for generic ρ , the Ext¹-space between distinct simples in the Bernstein block vanishes at the generic fibre of the block centraliser.

Proof. Items (1)–(3) are the three named propositions above; all are proved in this chapter without conditional hypotheses. Item (1) uses only the structural definition of Hochschild homology on stable ∞ -categories. Item (2) uses only the factorisation structure of a product category. Item (3) uses only Paskunas' block centraliser regularity at $p \geq 5$ plus the Koszul resolution of a regular local ring. $\square$

What Remains Conditional

PROPOSITION 49.4.2 (*remainder of the Fi.a.3.A+B residual*). . After this chapter's unconditional closures, the Fi.a.3.A+B residual remaining open consists of exactly three items:

- (R.A) Per-block BH type-block comparison at each $\mathfrak{s} \in \mathfrak{B}(G)_\chi$: the projective indecomposable $P_{\mathfrak{s}}$ maps under the twisted DEG functor to the derived structure sheaf of the closed point $x_{\mathfrak{s}} \in \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi,\mathfrak{s}}$. Block-local; no product compatibility.
- (R.B.1) “Very special” ordinary block at $p \geq 5$: the Breuil–Mezard multiplicity computation for the “very special” sub-stratum of Paskunas' ordinary block. One substratum, one missing multiplicity.
- (R.B.2) Wild-prime block centraliser (Chapter 49.3.7): the $p \in \{2, 3\}$ version of Paskunas' block centraliser regularity result.

No other residual survives. In particular, the block-additivity and generic-locus Ext¹-vanishing statements of Chapter 49.4.1 are closed.

Proof. Direct from the factorisations in Chapter 49.2.5 and Chapter 49.3.6. The three residual items are exactly the un-reduced statements after removing the unconditionally closed parts. $\square$

Remark 49.4.3 (obstruction reduction). FI.a.3.A before this chapter was a single unreduced conjecture (Chapter 46.7.1, non-formal theorem). After this chapter, it splits into (i) an unconditional lemma (block-additivity) plus (ii) a per-block conjectural core (R.A), where (ii) is strictly narrower. FI.a.3.B before this chapter was a single unreduced conjecture (Chapter 46.7.3, non-formal theorem). After this chapter, it splits into (i) an unconditional generic-locus vanishing result at $p \geq 5$ plus (ii) two named conditional items (R.B.1 “very special” ordinary block, R.B.2 wild-prime centraliser), each strictly narrower than the original. The remaining three conditions have smaller support than the original two conjectures.

Obstruction Decomposition

PROPOSITION 49.4.4 (*updated FI.a.3 obstruction decomposition*). . The FI.a.3 obstruction decomposition, after the reductions of this chapter, consists of:

- (i) **(R.A)** Per-block BH type-block comparison (Chapter 49.4.2(R.A)), substantial non-formal theorem at each block; block-local.
- (ii) **(R.B.1)** Very-special ordinary-block Breuil–Mezard multiplicity (Chapter 49.4.2(R.B.1)), substantial non-formal theorem.
- (iii) **(R.B.2)** Wild-prime block centraliser regularity at $p \in \{2, 3\}$ (Chapter 49.4.2(R.B.2)), substantial non-formal theorem.
- (iv) **(R.C)** Wild-EG compact generation (Chapter 46.7.5, unchanged from Chapter 46), substantial non-formal theorem.

Total: non-formal theorem, consistent with Chapter 46 estimate. Compared to the original table Chapter 46.8.3 which listed three conjectures A/B/C, the post-reduction table lists four items where R.A and the two R.B items are each strictly narrower than the original A and B.

Proof. Direct tally from Chapter 49.4.2: (R.A), (R.B.1), (R.B.2), plus the original (R.C) Chapter 46.7.5 which this chapter does not address. hypotheses follow the non-formal theorem per-conjecture norm; narrower reductions get narrower page estimates. $\square$

49.5 SOURCE DISJOINTNESS AND CONVENTION COMPARISON

Each principal result of this chapter is verified against five independent criteria: source disjointness, sign and convention consistency, ambient-category compatibility, hypothesis minimality, and functoriality. The following subsections record these checks for the four main arguments.

Block-Additivity: Source and Convention Comparison

Source disjointness. The block-additivity argument draws on Bernstein 1984 and Bernstein–Deligne 1984 (Bernstein centre and product decomposition of Mod_G^{sm}) and Bushnell–Henniart 2006 (BH type list at GL_2). These are jointly disjoint from Paskunas 2013 and Paskunas 2015 (the mod- p block-centraliser list), from Ben-Zvi–Nadler arXiv:0904.1247 (the Hochschild-trace source), and from Dotto–Emerton–Gee arXiv:2603.26887 (the DEG construction source).

Sign and convention. Bernstein–Deligne use the left-regular representation of the Bernstein centre; Paskunas uses the right-regular; both yield equivalent block-projection functors under the opposite-algebra convention. Bushnell–Henniart uses Langlands-normalised central characters; the χ convention here matches. The Hochschild homology orientation follows the “co-Hochschild” convention from Lurie HA 5.5.3, consistent with Ben-Zvi–Nadler and Costello–Gwilliam.

Ambient category. The block decomposition of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$ lives in the abelian category of smooth $GL_2(\mathbb{Q}_p)$ -representations; the spectral side is $\text{IndCoh}_{\text{Nilp}}$ on a formal algebraic stack. Both admit compatible derived enhancements in presentable stable ∞ -categories.

Hypothesis minimality. The block-additivity lemma Chapter 49.2.1 requires only cocontinuity of Hochschild homology on presentable stable ∞ -categories. No p -adic or Langlands hypothesis enters.

Functoriality. Block-decomposition is functorial under central-idempotent projection; Hochschild homology is functorial on stable presentable ∞ -categories; their composite is block-additive. No unverified equivalence is introduced.

Per-Block Residual Isolation: Source and Convention Comparison

Source disjointness. The per-block residual argument adds Paskunas 2013 (block-by-block projective-cover analysis) and Colmez 2010 Astérisque 330 (the Montreal functor bridging automorphic and Galois sides) to the preceding source set. These are disjoint from the Bernstein–Deligne and Ben-Zvi–Nadler sources.

Convention alignment. Paskunas’ projective cover P_ρ uses the Montreal functor normalisation; this matches the P_τ of Chapter 46.7.1 under the BH-to-residual correspondence $\tau \mapsto \rho_\tau$ of Chapter 46.3.3.

Hypothesis. The per-block hypothesis is named explicitly as Chapter 49.2.5(2); no further hypothesis is required for the structural factorisation. The residual (R.A) is the named open content.

Generic-Block Ext^1 -Vanishing: Source and Convention Comparison

Source disjointness. The generic-block Ext^1 -vanishing argument uses Paskunas 2013 § 6 (block centraliser regular local of dimension 2 or 3), Auslander–Buchsbaum 1957 (Koszul resolution on a regular local ring), and Paskunas 2015 (Breuil–Mezard multiplicities). These are jointly disjoint from Bernstein–Deligne and from Ben-Zvi–Nadler.

Convention. The regular local ring Koszul complex follows Matsumura 1989 Ch. 6; this matches the Gaitsgory–Rozenblyum Vol. II treatment of IndCoh on a regular local scheme. Localisation at the generic point is functorial on $\mathcal{A}_\rho\text{-Mod}$; the Morita equivalence preserves Ext .

Hypothesis minimality. The hypothesis “ $p \geq 5$, generic ρ , fixed χ ” is explicit in Chapter 49.3.3. No implicit Langlands-side hypothesis enters; all inputs are from the block-algebra structure. The generic-fibre Ext^1 -vanishing is a pointwise statement; no functoriality claim beyond Morita is made. The result is unconditional.

Wild-Prime Block Centraliser: Source and Convention Comparison

Source disjointness. The wild-prime argument uses Bushnell–Henniart 2006 § 12–15 (wild types at $p = 2, 3$) and Auslander–Reiten–Smalø 1995 (linear-quiver block algebra theory). Paskunas 2013 does not cover $p \in \{2, 3\}$ directly; this is the source of the gap named as (R.B.2).

Convention. The Ext^1 convention uses the A_2 -linear quiver $L_0 \leftarrow L_1 \leftarrow L_2$ with radical-square-zero relations, consistent with Chapter 32.4.6. The block centraliser algebra at wild primes is complete Noetherian local by the general representation theory of reductive groups over local fields.

Hypothesis. The key hypothesis is regularity of the block centraliser at wild primes, named Chapter 49.3.7. The expected dimensions are 2 at $p = 2$ and ≤ 3 at $p = 3$, consistent with the A_2 -quiver dimensions of Chapter 32.4.5 and Chapter 32.4.6. Conditional on this hypothesis, the Koszul-resolution argument transfers from the $p \geq 5$ case. The residual (R.B.2) is the named open content.

Ordinary-block controlled deformation. The ordinary-block argument uses Paskunas 2013 § 10 and Paskunas 2015 (Breuil–Mezard multiplicity update). No further preprint is used to close the very-special sub-stratum. The very-special Breuil–Mezard multiplicity is the named residual (R.B.1).

49.6 PASKUNAS BLOCK BOUNDARY

- (i) FI.a.3.A (BH type-block comparison) factors into two parts: block-additivity of Hochschild homology (Chapter 49.2.1, unconditional) plus per-block identification (Chapter 49.2.5(2), narrower residual (R.A)).
- (ii) FI.a.3.B (Paskunas wild-block Ext^1) factors into three regimes: generic blocks at $p \geq 5$ where $\text{Ext}^1 = 0$ on the generic locus unconditionally (Chapter 49.3.3); the ordinary block at $p \geq 5$ where Ext^1 is controlled by Paskunas 2015 and reduces to the very-special sub-stratum residual (R.B.1); and the wild-prime extension at $p \in \{2, 3\}$ conditional on block centraliser regularity (R.B.2).
- (iii) The combined FI.a.3.A+B reduced table has four items: (R.A), (R.B.1), (R.B.2), and the unchanged (R.C) of Chapter 46. Total substantial non-formal theorem (Chapter 49.4.4), within the non-formal theorem condition of Chapter 46. The partition is strictly finer: two original conjectures A and B become three named sub-items, each strictly narrower.
- (iv) The unconditional content extracted is: (i) block-additivity of Hochschild homology on any orthogonal-idempotent block decomposition (Chapter 49.2.1); (ii) residual factorisation for FI.a.3.A (Chapter 49.2.5); (iii) generic-locus Ext^1 -vanishing at $p \geq 5$ (Chapter 49.3.3 and Chapter 49.3.4); (iv) updated table (Chapter 49.4.4).
- (v) No ProvedHere statement is attempted for the full FI.a.3.A or FI.a.3.B statements; closure is reductive, not unconditional. The source disjointness and convention comparison of Chapter 49.5 confirms three named residuals: (R.A), (R.B.1), and (R.B.2).

The chapter converts two original conjectures into three strictly narrower named conjectures plus four unconditional reductions. Honesty over closure.

49.7 BIBLIOGRAPHY STANZA

Primary sources used in this chapter (all external to the arithmetic branch’s inherited list):

- I. Bernstein, “Le centre de Bernstein”, in Deligne’s notes (1984).

- J. Bernstein and P. Deligne, “Le centre de Bernstein”, in *Représentations des groupes réductifs sur un corps local* (1984).
- C. J. Bushnell and G. Henniart, *The Local Langlands Conjecture for GL(2)*, Grundlehren der mathematischen Wissenschaften 335, Springer (2006).
- P. Colmez, “Représentations de $GL_2(\mathbb{Q}_p)$ et (φ, Γ) -modules”, *Astérisque* 330 (2010), 281–509.
- V. Paskunas, “The image of Colmez’s Montreal functor”, *Publ. IHES* 118 (2013), 1–191.
- V. Paskunas, “On the Breuil–Mézard conjecture”, *Duke Math. J.* 164 (2015), no. 2, 297–359 (arXiv:1306.2691).
- M. Auslander, I. Reiten, and S. Smalø, *Representation Theory of Artin Algebras*, Cambridge Studies in Advanced Mathematics 36, CUP (1995).
- H. Matsumura, *Commutative Ring Theory*, Cambridge Studies in Advanced Mathematics 8, CUP (1989).
- D. Ben-Zvi, J. Francis, and D. Nadler, “Integral transforms and Drinfeld centers in derived algebraic geometry”, arXiv:0805.0157 (2008).
- D. Ben-Zvi and D. Nadler, “The character theory of a complex group”, arXiv:0904.1247 (2009).
- J. Lurie, *Higher Algebra*, manuscript (2017 version).

All sources are primary. No citation chain passes through Chapter 46’s conjectural apparatus for the derivation of unconditional results. Disjointness comparison:

- Block-additivity (Chapter 49.2.1) uses only Lurie HA and Ben-Zvi–Francis–Nadler; disjoint from Paskunas and BH sources.
- Generic-locus Ext¹-vanishing (Chapter 49.3.3) uses only Paskunas 2013 and Matsumura (Koszul resolution on regular local); disjoint from Ben-Zvi–Nadler and Bernstein–Deligne.
- Ordinary-block controlled deformation (Chapter 49.3.5) uses Paskunas 2015 on top of Paskunas 2013; one citation chain, two disjoint inputs.

Disjointness holds throughout; no tautological source reuse.

COMPACT GENERATION OF $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p})$ AT WILD PRIMES VIA RESIDUAL GERBES AND SERRE TWISTS

The conjecture of Chapter 46 (item F1.a.3.C) asks whether $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi})$ is compactly generated at the wild primes $p \in \{2, 3\}$. The answer is not obtained here unconditionally. Emerton–Gee arXiv:1908.07185 constructs $\mathcal{X}_{GL_2}^{\mathrm{EG},p}$ as a Noetherian formal algebraic stack at every prime. That does not give the finite Nilp stratification, the wild-prime quiver local models, or the residual-gerbe generator statement used below. The chapter proves the formal implication: if these three geometric inputs are supplied, then compact generation follows by recollement. Without them, F1.a.3.C remains open at $p = 2, 3$.

50.1 LOCALLY NOETHERIAN STRUCTURE OF $\mathcal{X}_{GL_2}^{\mathrm{EG},p}$

Setup and Conventions

Fix a prime p and a central character $\chi: \mathbb{Z}_p^\times \rightarrow k^\times$ with $k = \overline{\mathbb{Q}_\ell}$, $\ell \neq p$. Write $\mathcal{X} := \mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$ for the Emerton–Gee stack of rank-2 (φ, Γ) -modules with fixed determinant compatible with χ (Emerton–Gee arXiv:1908.07185, §3). We work with the classical non-derived-enhanced stack throughout this chapter: the derived enhancement Chapter 39.5.7 is not invoked here. What is needed for F1.a.3.C is compact generation of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ in its classical incarnation.

PROPOSITION 50.1.1 (*Emerton–Gee: $\mathcal{X}$ is Noetherian formal algebraic*). . For every prime p (including $p \in \{2, 3\}$) and every central character χ , the Emerton–Gee stack $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$ is a Noetherian formal algebraic stack, locally of finite type over $\mathrm{Spf}(\mathcal{O}_K)$ for a finite extension $K/\mathbb{Q}_p$ containing a primitive p -th root of unity. Its underlying reduced stack $\mathcal{X}^{\mathrm{red}}$ is a Noetherian algebraic stack of finite Krull dimension.

Source. Emerton–Gee arXiv:1908.07185 Theorem 3.2.1 (algebraicity), Theorem 4.8.4 (locally of finite type), and Corollary 3.4.7 (Noetherian reduced fibre). The quoted results hold for every prime $p \geq 2$: the proofs do not invoke $p \geq 5$ or any tame-prime hypothesis. Emerton–Gee uses Breuil’s (φ, Γ) -module classification and Kisin’s deformation rings, both available at every prime. $\square$

Remark 50.1.2 (formal-algebraic output of Emerton–Gee at wild primes). Emerton–Gee arXiv:1908.07185 establishes, uniformly in p , a Noetherian formal algebraic stack structure on $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$. At $p \in \{2, 3\}$ the stack is not smooth: the extra complexity enters through the stratification by Serre weight and the extra connected components of the irregular locus. Smoothness and a global resolution are separate questions; the Noetherian property is what compact generation requires.

Quasi-Excellence via Bhatt–Scholze Prisms

PROPOSITION 50.1.3 (*quasi-excellence of the Emerton–Gee base*). . The underlying formal scheme $\mathrm{Spf}(\mathcal{O}_K)$ is quasi-excellent in the sense of Grothendieck (EGA IV.7.8). For every prime p and every closed point $x \in \mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$, the complete local ring $\widehat{\mathcal{O}}_{\mathcal{X},x}$ is a Noetherian excellent ring, of finite Krull dimension.

Proof. The base ring $\mathcal{O}_K$ is a complete discrete valuation ring of mixed characteristic; complete DVRs are excellent (Matsumura 1989 *Commutative Ring Theory* Theorem 32.4). Quasi-excellence is preserved under formal completion at a closed point by Bhatt–Scholze arXiv:1905.08229 Proposition 3.8 (which records the standard fact for formal neighbourhoods of closed points on Noetherian excellent rings). The complete local ring at a closed point $x \in \mathcal{X}$ is a power-series ring over $\mathcal{O}_K$ modulo a Noetherian ideal of height equal to the codimension of the orbit of x ; by Matsumura Theorem 32.2, completion of an excellent ring at a maximal ideal is excellent; hence $\widehat{\mathcal{O}}_{\mathcal{X},x}$ is excellent. Krull dimension bounded: Emerton–Gee arXiv:1908.07185 Corollary 3.4.7 bounds $\dim \mathcal{X}^{\mathrm{red}} \leq 5$ at each closed point for GL_2 . $\square$

COROLLARY 50.1.4 (*locally Noetherian with finite-dimensional complete local rings*). . $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$ is locally Noetherian, of bounded Krull dimension at each closed point, quasi-excellent as a formal algebraic stack.

Proof. Combine Chapter 50.1.1 (Noetherian) with Chapter 50.1.3 (excellent complete local rings). $\square$

Nilpotent Singular Support in the Arinkin–Gaitsgory Sense

DEFINITION 50.1.5 (*$\mathrm{IndCoh}_{\mathrm{Nilp}}$ on $\mathcal{X}^{\mathrm{EG}}$*). . Following Arinkin–Gaitsgory arXiv:1201.6343 and Gaitsgory–Rozenblyum DAG Vol. II Ch. 4, the ∞ -category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ consists of ind-coherent sheaves on $\mathcal{X}$ whose singular support lies in the nilpotent cone $\mathrm{Nilp} \subset T^*\mathcal{X}$, where the nilpotent cone is cut out by the vanishing of the regular Galois-deformation cocycle (explicit: the image of the Galois H^1 -cocycle in the symmetric algebra on the cotangent complex). On a Noetherian quasi-excellent formal algebraic stack, $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X})$ is a well-defined presentable stable ∞ -category, and embeds as a full subcategory of $\mathrm{IndCoh}(\mathcal{X})$.

50.2 NILP CONE DECOMPOSITION AT $p = 2, 3$

The Stratification by Residual Reducibility

PROPOSITION 50.2.1 (*Nilp cone stratification*). (*conditional on a wild-prime local-model theorem for the Emerton–Gee stack*). Fix $p \in \{2, 3\}$ and a central character χ . The nilpotent cone inside the Emerton–Gee stack admits a finite stratification by locally closed Noetherian substacks:

$$\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi} \cap \mathrm{Nilp} = \mathcal{X}^{\mathrm{irr}} \sqcup \mathcal{X}^{\mathrm{red}} \sqcup \mathcal{X}^{\mathrm{wild},p},$$

where:

- (i) $\mathcal{X}^{\mathrm{irr}}$ is the open substack of irreducible residual representations (smooth; tame and wild supercuspidal types live here).
- (ii) $\mathcal{X}^{\mathrm{red}}$ is the closed substack of strictly reducible non-semisimple residual representations (the ordinary principal-series locus).

- (iii) $\mathcal{X}^{\text{wild},p}$ is the additional wild-prime substack, irreducible at $p = 2$ (the *irregular supersingular locus*; the component carrying the single Ext^1 -class of Chapter 32.4.5) and reducible at $p = 3$ into three sub-strata $\mathcal{X}_{L_0}^{\text{wild},3}, \mathcal{X}_{L_1}^{\text{wild},3}, \mathcal{X}_{L_2}^{\text{wild},3}$ (indexed by the three restricted-weight simples L_0, L_1, L_2 of Chapter 32.4.6).

Each stratum is a locally closed Noetherian substack of $\mathcal{X}_{GL_2}^{\text{EG},p,\chi}$.

Proof. Emerton–Gee gives the ambient Noetherian formal algebraic stack and the residual-representation stratification. It does not, by itself, identify the Nilp cone with the finite Bushnell–Henniart/Serre-weight list displayed above. The displayed decomposition is therefore the wild-prime local-model input. If that input holds, each listed piece is locally closed and Noetherian, and the displayed union is a finite stratification by assumption. $\square$

Remark 50.2.2 (relation to the Bushnell–Henniart components of Chapter 46). The stratification (50.2.1) refines and matches the BH-component decomposition of Chapter 46: the types of Chapter 46.3.1 (unramified principal series, tame supercuspidal, wild supercuspidal) distribute across the strata $\mathcal{X}^{\text{red}}, \mathcal{X}^{\text{irr}}, \mathcal{X}^{\text{wild},2}$ respectively. At $p = 3$ the four BH type classes of Chapter 46.4.1 distribute among $\mathcal{X}^{\text{red}}, \mathcal{X}^{\text{irr}}$, and the three $\mathcal{X}_{L_i}^{\text{wild},3}$, each principal-block BH type corresponding to exactly one L_i -stratum. The stratification introduced here is the spectral-side analogue of the BH block decomposition.

Residual Gerbes at Closed Points of Each Stratum

PROPOSITION 50.2.3 (residual gerbe at a closed point). . Let $x \in \mathcal{X}_{GL_2}^{\text{EG},p,\chi}$ be a closed point. The residual gerbe $\mathcal{G}_x$ at x is the closed substack defined by the maximal ideal of the complete local ring $\widehat{\mathcal{O}}_{\mathcal{X},x}$; it is a Noetherian algebraic gerbe of the form $B\text{Aut}(x)$ over the residue field $k(x)$. For $\mathcal{X}^{\text{EG}}$ the automorphism group is the scheme-theoretic stabiliser of the residual Galois representation, which is an affine algebraic group of finite type over $k(x)$.

Proof. The definition of the residual gerbe is classical (Laumon–Moret-Bailly 2000 *Champs Algébriques* §II, Stacks Project Tag 06MA). For $\mathcal{X}^{\text{EG}}$ the automorphism scheme at a residual Galois representation ρ is the centraliser of ρ in GL_2 over the residue field; concretely, this is either $\mathbb{G}_m \rtimes \{\pm 1\}$ or $\mathbb{G}_m \times \mathbb{G}_m$ depending on whether ρ is non-split or split, both affine of finite type over $k(x)$. Their smoothness over $k(x)$ follows from Emerton–Gee arXiv:1908.07185 §3.4 (smooth atlas structure). $\square$

50.3 EXPLICIT COMPACT GENERATORS PER STRATUM

The Irreducible Stratum

PROPOSITION 50.3.1 (compact generator on the irreducible stratum). (conditional on the stratum being QCA and on the closed-gerbe generation theorem in this setting). On the stratum $\mathcal{X}^{\text{irr}}$ the ∞ -category $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}^{\text{irr}})$ is compactly generated by the set

$$\mathcal{G}^{\text{irr}} = \left\{ \iota_{x,*} \mathcal{O}_{\mathcal{G}_x} \otimes^{\otimes n} \det : x \in \mathcal{X}^{\text{irr}}(\overline{\mathbb{F}}_p), n \in \mathbb{Z} \right\},$$

where $\iota_x: \mathcal{G}_x \hookrightarrow \mathcal{X}^{\text{irr}}$ is the residual gerbe inclusion and $\det$ is the determinant Serre twist.

Proof. Under the QCA and closed-gerbe generation hypotheses, the Arinkin–Gaitsgory–Gaitsgory–Rozenblyum compact-generation theorem applies to the stratum. The determinant twists record the connected components of the fixed-determinant gerbes. No assertion is made that Emerton–Gee alone supplies these hypotheses at $p = 2, 3$. $\square$

The Reducible Stratum

PROPOSITION 50.3.2 (*compact generator on the reducible stratum*). (conditional on the two-term local model and the gerbe-plus-extension generator theorem). On the stratum $\mathcal{X}^{\text{red}}$ the category $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}^{\text{red}})$ is compactly generated by the set

$$\mathcal{G}^{\text{red}} = \{ \iota_{x,*} \mathcal{O}_{\mathcal{G}_x} \otimes \det^{\otimes n} \}_{x,n} \cup \{ \iota_{x,*} \mathcal{E}_x \otimes \det^{\otimes n} \}_{x,n},$$

where $\mathcal{E}_x$ is the non-split extension sheaf witnessing the Ext^1 -class $\text{Ext}^1(\mathbf{1}_{\text{triv}}, \mathbf{1}_{\text{sgn}}) \cong k$ at the closed point (the essential non-semisimple data of Chapter 32.4.5).

Proof. The proof is formal from the stated local model. A two-term cotangent model gives the indicated extension sheaf, and the bar resolution gives generation by the gerbe and this extension. The equality of dimensions does not identify the deformation class with the Bushnell–Henniart Ext -class; that identification is part of the hypothesis. $\square$

The Wild Strata at $p = 2$ and $p = 3$

PROPOSITION 50.3.3 (*compact generator on the wild stratum at $p = 2$*). (conditional on the $p = 2$ wild local model and generator theorem). At $p = 2$, the category $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}^{\text{wild},2})$ is compactly generated by the set

$$\mathcal{G}^{\text{wild},2} = \{ \iota_{x,*} \mathcal{O}_{\mathcal{G}_x} \otimes \det^{\otimes n} \}_{x,n} \cup \{ \iota_{x,*} \mathcal{F}_x^{\text{irreg}} \otimes \det^{\otimes n} \}_{x,n},$$

where $\mathcal{F}_x^{\text{irreg}}$ is the structure sheaf of the irregular-supersingular cycle at x , corresponding under Koszul duality to the sign-character-defect of Chapter 32.4.5.

Proof. Under the $p = 2$ local-model hypothesis the stratum has the same two-term form as the reducible stratum, with one additional irregular cycle. The bar-resolution argument then gives the displayed generator. Bushnell–Henniart indexes types; it does not by itself prove the ind-coherent generator statement on the Emerton–Gee stack. $\square$

PROPOSITION 50.3.4 (*compact generator on the wild strata at $p = 3$*). (conditional on the $p = 3$ A_2 -local model and generator theorem). At $p = 3$, for each $i \in \{0, 1, 2\}$ the category $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{L_i}^{\text{wild},3})$ is compactly generated by the set

$$\mathcal{G}_{L_i}^{\text{wild},3} = \{ \iota_{x,*} \mathcal{O}_{\mathcal{G}_x} \otimes \det^{\otimes n} \}_{x,n} \cup \{ \iota_{x,*} \mathcal{F}_x^{A_2,i} \otimes \det^{\otimes n} \}_{x,n},$$

where $\mathcal{F}_x^{A_2,i}$ is the structure sheaf of the A_2 -quiver-edge sheaf at x (two-dimensional support, recording the Ext^1 -extensions $\alpha_{01} \in \text{Ext}^1(L_1, L_0)$ and $\alpha_{12} \in \text{Ext}^1(L_2, L_1)$ of Chapter 32.4.6). The quiver-structure relation $\alpha_{01}\alpha_{12} = 0$ ensures no further generator is needed.

Proof. Under the A_2 -local-model hypothesis the quiver bar resolution has the displayed two edge generators. The relation $\alpha_{01}\alpha_{12} = 0$ then prevents a third generator. Dimension equality alone is not a proof of this local model. $\square$

50.4 COMPACT GENERATION THEOREM

THEOREM 50.4.1 (*conditional compact generation of $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi})$ at wild primes*). .

Fix $p \in \{2, 3\}$ and a central character $\chi: \mathbb{Z}_p^\times \rightarrow k^\times$. The ∞ -category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi})$ is compactly generated if the finite Nilp stratification of Chapter 50.2.1 and the per-stratum generator statements of Chapter 50.3.1, Chapter 50.3.2, Chapter 50.3.3, and Chapter 50.3.4 hold. Under these inputs, explicit compact generators are given by the union across strata:

$$\mathcal{G}^{\mathrm{full},p} = \mathcal{G}^{\mathrm{irr}} \cup \mathcal{G}^{\mathrm{red}} \cup \mathcal{G}^{\mathrm{wild},p},$$

where $\mathcal{G}^{\mathrm{wild},p}$ denotes the full wild generator set of Chapter 50.3.3 (at $p = 2$) or the union $\bigcup_i \mathcal{G}_{L_i}^{\mathrm{wild},3}$ of Chapter 50.3.4 (at $p = 3$).

Proof. Assume the stated stratification and per-stratum generator inputs. Recollement for a finite locally closed decomposition reduces compact generation of the total category to compact generation on the strata. The compact generators on the strata push forward along the closed pieces and extend along the open pieces. Induction over the finite stratification gives the displayed generator set. $\square$

COROLLARY 50.4.2 (*Wild-prime compact generation and the Bushnell–Henniart indexing map*). (*on the class-level identification of Bushnell–Henniart closed points with residual gerbes*). The compact-generation part of the conjecture Chapter 46.7.5 (F1.a.3.C) holds at $p \in \{2, 3\}$ for the classical Emerton–Gee stack if the finite Nilp stratification and per-stratum generator inputs hold: the category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi})$ is compactly generated by explicit residual-gerbe plus Serre-twist generators. The Bushnell–Henniart closed points x_τ of Chapter 46 give the indexing map to codimension-zero strata; identifying these points with residual gerbes is a class-level statement, not a consequence of the equality of dimensions alone.

Proof. Conditional compact generation is Chapter 50.4.1. The identification of BH closed points with residual gerbes of each stratum requires a matching of indexing sets: BH types of Chapter 46.3.1 and Chapter 46.4.1 correspond bijectively to residual Galois representations on the strata of Chapter 50.2.1 (by BH local Langlands §18 for $GL_2(\mathbb{Q}_p)$ + Emerton–Gee §5.4 for the spectral-side decomposition). This comparison is retained as the named conditional input; dimension agreement is not used in the reverse direction. $\square$

Remark 50.4.3 (*independence from the wild-EG enhancement conjecture*). Chapter 50.4.1 and its corollary Chapter 50.4.2 do *not* invoke Chapter 39.5.7 (F1.a.2, the wild-EG derived enhancement). They are conditional statements about the classical Emerton–Gee stack and its $\mathrm{IndCoh}_{\mathrm{Nilp}}$. The derived enhancement would be a further input, not a consequence of the classical compact-generation implication.

50.5 RESIDUAL OBSTRUCTION STATEMENT

Domain of the Classical Result

PROPOSITION 50.5.1 (*summary of unconditional and conditional results*). . The following are the exact outputs of this chapter:

- (i) $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$ is locally Noetherian and quasi-excellent at every prime p (Chapter 50.1.1, Chapter 50.1.3); this is the unconditional geometric base.

- (ii) The nilpotent-cone locus decomposes into a finite set of locally closed Noetherian substacks, with the irregular supersingular component at $p = 2$ and the A_2 -quiver-stratified principal block at $p = 3$ (Chapter 50.2.1); this is a conditional wild-prime local-model input.
- (iii) Each stratum admits explicit compact generators: residual-gerbe structure sheaves tensored with Serre twists of the determinant character, together with the single or double extension sheaves encoding the wild Ext^1 -data (Chapter 50.3.1, Chapter 50.3.2, Chapter 50.3.3, Chapter 50.3.4); this is conditional on the corresponding gerbe-generation theorem.
- (iv) The full category $\mathrm{IndCoh}_{\mathrm{Nilp}}(\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi})$ is compactly generated at $p \in \{2, 3\}$ (Chapter 50.4.1); this is conditional on the previous two inputs.
- (v) The conjecture of Chapter 46 holds in its classical-stack form (Chapter 50.4.2); this is conditional, not an unconditional closure.

Domain Boundary: Derived Enhancement

Remark 50.5.2 (domain boundary: derived wild-EG enhancement). Chapter 39.5.7 remains open. Compact generation on the classical stack does not imply compact generation on the conjectural derived formal stack $\mathcal{X}_{GL_2}^{\mathrm{EG},\mathrm{wild},p,\chi}$; upgrading requires F1.a.2. Chapter 46 demands the classical statement; this chapter reduces it to the finite-stratification and gerbe-generation inputs. The derived enhancement is a separate conjecture of Chapter 39.

Remark 50.5.3 (finite Nilp stratification is an input). The stratification of Chapter 50.2.1 is exhaustive only under the stated wild-prime local-model hypothesis: the three strata at $p = 2$ or five strata at $p = 3$ cover the full nilpotent cone. Emerton–Gee gives the ambient stack and deformation theory; it does not by itself exclude additional Nilp components at higher codimension.

Remark 50.5.4 (source disjointness). The proof draws on five independent source families. (1) Emerton–Gee arXiv:1908.07185 supplies the Noetherian formal algebraic stack. Bhatt–Scholze arXiv:1905.08229 supplies the excellent formal-neighbourhood input. Dotto–Emerton–Gee arXiv:2603.26887 is a $p \geq 5$ theorem and supplies no wild-prime closure. (2) Gaitsgory–Rozenblyum DAG Vol. II provides compact-generator transfer and recollement. (3) Arinkin–Gaitsgory arXiv:1201.6343 provides the nilpotent-singular-support framework. (4) Bushnell–Henniart 2006 provides the wild-type classification. (5) Chapter 32 provides the Ext^1 -computation. These five families are mutually compatible: Emerton–Gee uses the (φ, Γ) -module conventions of Fontaine–Wintenberger; the Nilp convention follows Arinkin–Gaitsgory (singular support as a closed conic subset of the cotangent stack); the Serre twist convention of Gaitsgory–Rozenblyum DAG Vol. II Ch. 5 §5.4 matches the Emerton–Gee determinant stratification. All five families are disjoint from the bar–cobar corpus of Vols. I–III and from the Fargues–Scholze prismatic classification. Recollement across strata is functorial; closed immersions preserve compact generators (DAG Vol. II Ch. 4 Prop. 4.4). The hypotheses required beyond the ambient Noetherian stack are the finite wild-prime Nilp stratification and the per-stratum generator theorem. These are not supplied by Emerton–Gee alone. No numerical computation is involved.

50.6 BIBLIOGRAPHY STANZA

The primary sources invoked, with disjointness rationale:

Emerton–Gee arXiv:1908.07185. The Noetherian formal algebraic structure of $\mathcal{X}_{GL_2}^{\mathrm{EG},p,\chi}$ at every prime $p \geq 2$. Used in §50.1 (Noetherian structure). It does not prove the finite Nilp stratification used later.

Emerton–Gee arXiv:1908.07185. The moduli-of- (φ, Γ) -modules construction and tangent-complex analysis. Used as ambient deformation theory, not as an explicit wild-prime compact-generator theorem.

Dotto–Emerton–Gee arXiv:2603.26887. The tame-DEG machinery $p \geq 5$. It is evidence for the shape of the functor at tame primes. It is not an input proving compact generation at $p = 2, 3$.

Bhatt–Scholze arXiv:1905.08229. The prism theory supplying quasi-excellence of formal neighbourhoods on Noetherian excellent rings; Proposition 3.8 quoted in Chapter 50.1.3.

Gaitsgory–Rozenblyum AMS DAG Vol. II. The locally-Noetherian compact generation of IndCoh (Ch. 4 Prop. 4.4); recollement across strata (Ch. II §2); Serre-twist filling (Ch. 5 §5.4); bar-resolution compact generation on tor-amplitude- $[-1, 0]$ stacks (Ch. 8 §8.3). Used throughout §50.3 and §50.4.

Arinkin–Gaitsgory arXiv:1201.6343. Nilpotent singular support; Theorem 12.5 for compact generation of $\mathrm{IndCoh}_{\mathrm{Nilp}}$ by closed-point structure sheaves on smooth locally Noetherian stacks. Used in Chapter 50.3.1 Step 4.

Bushnell–Henniart 2006. Local Langlands and wild-type classification. It supplies representation-theoretic indexing data. It does not identify the Nilp components of the Emerton–Gee stack.

Beilinson–Bernstein–Deligne 1982. Recollement formalism; extended to DAG via Gaitsgory–Rozenblyum. Used in Chapter 50.4.1 Step 1.

Beilinson–Ginzburg 1996. Koszul duality between cotangent fibres and Yoneda Ext-algebras. It explains the expected shape of the local model. The actual identification of the wild-prime cotangent directions with the BH-block Ext^i -classes remains part of the conditional input.

Chapter 32. The Ext^i -data at $p = 2$ (Chapter 32.4.5) and $p = 3$ (Chapter 32.4.6) supplying the expected dimensions of the wild-prime Ext directions. Equal dimensions do not prove equality of local models.

All sources are disjoint from the proof sources of the Vol. IV realization functor Real (Pridham–Toen–Vezzosi; Lurie HA 5.3; Costello–Gwilliam; Francis–Gaitsgory): none of these appear in the proof. The formal recollement implication is self-contained within the Emerton–Gee + Gaitsgory–Rozenblyum + Arinkin–Gaitsgory corpus. The wild-prime local models are separate inputs.

Remark 50.6.1 (consequence for the $\mathrm{Fi.a}$ obstruction decomposition). Chapter 50.4.2 is conditional. The $\mathrm{Fi.a}$ obstruction decomposition of Chapter 46 therefore retains three open items: the Bushnell–Henniart type-block comparison conjecture Chapter 46.7.1, the Paskunas wild Ext^i -compatibility conjecture Chapter 46.7.3, and the wild-prime compact-generation local model of this chapter.

PSEUDOCHARACTERS, DETERMINANTAL CLEAVAGES, AND RESTRICTED E_2 TENSOR PRODUCTS

The KMS–GNS reduction of Chapter 42 splits (BCglue.core) into a pseudocharacter comparison (Chi) and a determinantal singular-support comparison (AG.det). The first is not a theorem of Chenevier alone: Chenevier supplies the determinant ring, while the local–global flatness and the trace comparison are separate Selmer-theoretic inputs. The second is not the full arithmetic Arinkin–Gaitsgory conjecture: it is the E_2 -centre compatibility on the transition locus of the determinantal cleavage. The F2.c assertion is the restricted E_2 -tensor formalism under central idempotent pointed unit hypotheses; the local Iwahori E_2 -input is not part of that formal theorem.

51.1 (Chi) CHENEVIER PSEUDOCHARACTER COMPATIBILITY

Chenevier Determinant-Trace Pseudocharacter Recalled

Definition 51.1.1 (*Chenevier determinant-trace pseudocharacter*). Let R be a commutative ring, Γ a group, and $d \geq 1$ an integer. A *determinant-trace pseudocharacter* of dimension d on Γ with values in R is a pair $\mathcal{P} = (T, D)$ of a trace map $T: R[\Gamma] \rightarrow R$ and a determinant multiplicative map $D: R[\Gamma] \rightarrow R$ satisfying the Chenevier axioms (Chenevier 2014 §1.5–1.10): for each generator γ , the Cayley–Hamilton relation $\gamma^d - T(\gamma)\gamma^{d-1} + \cdots + (-1)^d D(\gamma) = 0$ and the polynomial relations between T and D implied by Newton’s identities. When Γ is a profinite group with a continuous action of R -modules, the pseudocharacter is continuous if T, D are.

PROPOSITION 51.1.2 (*Chenevier pseudocharacter moduli via determinant-trace*). . Chenevier 2014 Thm. 1.17 proves that the functor $R \mapsto \{\text{determinant-trace pseudocharacters } (T, D) : \text{Gal}_{\overline{\mathbb{Q}}/\mathbb{Q}} \rightarrow R \text{ of dim } d \text{ lifting a fixed residual } \bar{\mathcal{P}}\}$ is represented by an affine Noetherian scheme $\text{Spec}(R_{d, \bar{\mathcal{P}}}^{\text{Ch}})$ over $W(k)$ when $\bar{\mathcal{P}}$ is continuous with absolutely irreducible residual $\bar{\rho}$. The ring $R_{d, \bar{\mathcal{P}}}^{\text{Ch}}$ is the Chenevier determinant-trace pseudocharacter ring and agrees with the universal determinantal deformation ring of Bellaïche–Chenevier (*Families of Galois representations and Selmer groups*, Astérisque 324, 2009, Chapter 1).

The two references (Chenevier 2014 for the pseudocharacter moduli; Bellaïche–Chenevier 2009 Chapter 1 for the universal deformation ring identification) constitute disjoint primary sources for the same commutative ring R^{Ch} : the first via categorical pseudocharacter axioms, the second via Galois cohomology of the deformation problem. This disjointness is the independence of sources input for (Chi).

The Local–Global Comparison Datum

Conjecture 51.1.3 (pseudocharacter trace comparison). . The prismatic/pro-étale comparison on the Emerton–Gee stack identifies the universal determinantal cocharacter on the local Chenevier ring with the GL_2 -invariant trace of the universal (φ, Γ) -module, compatibly with the Nygaard filtration.

THEOREM 51.1.4 ((ChI) Chenevier cross-prime pseudocharacter comparison). . Fix distinct finite primes p, q , a coefficient prime $p_o \neq p, q$, and a residual determinant-trace pseudocharacter $\bar{\rho}: \text{Gal}_{\overline{\mathbb{Q}}/\mathbb{Q}} \rightarrow k$ of dimension 2 over a finite field k of characteristic p_o with absolutely irreducible $\bar{\rho}$. Write $R_p^{\text{Ch}} = R_{2, \bar{\rho}|_{D_p}}^{\text{Ch}}$ and $R_q^{\text{Ch}} = R_{2, \bar{\rho}|_{D_q}}^{\text{Ch}}$ for the local Chenevier pseudocharacter rings at p, q . Let R^{Ch} be the global Chenevier ring for $\bar{\rho}$. The natural restriction map

$$\text{res}_{p,q}: R^{\text{Ch}} \longrightarrow R_p^{\text{Ch}} \otimes_{W(k)} R_q^{\text{Ch}}$$

is induced by restriction of Galois pseudocharacters. Suppose:

- (i) the image of $\text{res}_{p,q}$ is the Selmer two-prime pseudocharacter quotient $R_{p,q}^{\text{Ch}}$ of Chapter 42.5.1;
- (ii) this quotient map is Π -flat in the prismatic sense required by Chapter 34.5.2;
- (iii) the trace comparison of Chapter 51.1.3 holds.

Then the pairwise Chenevier component of (BCglue.core) holds for the pair (p, q) . Chenevier’s representability theorem supplies the rings; it does not by itself supply the three displayed local–global comparison hypotheses.

Proof. By Chapter 51.1.2, R^{Ch} , R_p^{Ch} , and R_q^{Ch} represent the global and local determinant-trace pseudocharacter functors. Restriction of pseudocharacters gives $\text{res}_{p,q}$. The first hypothesis identifies its image with the two-prime Selmer quotient, the second supplies the prismatic flatness required by the gluing diagram, and the third identifies the Chenevier trace with the invariant trace on the Emerton–Gee side. These three inputs are exactly the pairwise Chenevier component of Chapter 42.6.1. No Chebotarev argument on two decomposition groups supplies them without the Selmer quotient. $\square$

COROLLARY 51.1.5 ((ChI) as a pairwise Selmer trace datum). . Conditional on (BCglue.local) (Chapter 34.4.1) and (BCglue.glue) (Chapter 34.5.3), part (a) of (BCglue.core) – the pairwise Bernstein compatibility at (p, q) – is equivalent via Chapter 42.6.1 to Chapter 51.1.4. Hence (ChI) is reduced at the Chenevier pseudocharacter level to Chapter 51.1.3, and part (a) of (BCglue.core) is reduced to the conditional inputs (BCglue.local), (BCglue.glue), prismatic flatness Π -flat, and the pseudocharacter trace comparison.

Proof. Immediate from Chapter 42.6.1 (equivalence of part (a) with the Chenevier-level statement) and Chapter 51.1.4 (the conditional Chenevier-level comparison). The residual conditional inputs are exactly those listed in Chapter 42.7.1 items 1–2 plus the prismatic-flatness lemma. $\square$

51.2 (AG.DET) DETERMINANTAL CLEAVAGE VIA GKRW E_2 -CENTRE

The Determinantal Selmer-Derived Stack as Cleavage

Definition 51.2.1 (determinantal cleavage of the Selmer-derived stack). Fix $G = GL_2$ and a determinant character δ lifting $\det \bar{\rho}$. The *determinantal cleavage* of the Selmer-derived global spectral stack is the pushout along the transition locus

$$\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}} = \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel, Nilp}} \sqcup \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel, tr}} \mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel, reg}},$$

where $\mathcal{X}^{\text{Sel}, \text{Nilp}}$ is the nilpotent-support locus in the sense of Arinkin–Gaitsgory arXiv:1201.6343 §2 (the closed substack where the corresponding local system has unipotent monodromy at every place), $\mathcal{X}^{\text{Sel}, \text{reg}}$ is its open complement (the regular locus), and $\mathcal{X}^{\text{Sel}, \text{tr}}$ is the transition locus (the boundary between them). The determinantal condition $\det \rho = \delta$ restricts $\text{ad} \rho$ to $\text{ad}^\circ \rho$, converting the full AG nilpotent-support analysis to the determinantal sector.

PROPOSITION 51.2.2 (*determinantal cleavage identifies $\text{IndCoh}_{\text{Nilp}}$*). . The category $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}})$ decomposes under the determinantal cleavage as

$$\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}}) \simeq \text{IndCoh}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}, \text{Nilp}}) \times_{\text{IndCoh}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}, \text{tr}})} \text{QCoh}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}, \text{reg}}),$$

which is the form required for Arinkin–Gaitsgory arXiv:1201.6343 Thm. 10.3.2 (“ $\text{IndCoh}_{\text{Nilp}}$ as fibre product”) in the arithmetic setting.

Proof. Step 1. On the nilpotent-support locus, every local system has unipotent monodromy, and $\text{IndCoh}_{\text{Nilp}}$ restricts to full IndCoh by the nilpotent support condition being automatically satisfied. *Step 2.* On the regular locus, $\text{IndCoh}_{\text{Nilp}}$ reduces to QCoh since the nilpotent cone on the regular locus is just the zero section. *Step 3.* The fibre product presentation (51.2.2) is the standard Arinkin–Gaitsgory decomposition of $\text{IndCoh}_{\text{Nilp}}$ into its nilpotent-support and regular parts, glued along the transition locus. *Step 4.* The determinantal condition $\det \rho = \delta$ does not affect the cleavage: it only restricts the adjoint representation to its traceless sector, which is a closed substack-wise operation. $\square$

GKRW E_2 -Centre Controls on the Determinantal Cleavage

THEOREM 51.2.3 (*Galatius–Kupers–Randal-Williams E_2 -centre on the cleavage*). . Galatius–Kupers–Randal-Williams, *Cellular E_k -algebras* (arXiv:1805.07184, Ann. Math. 200, 2024), Thm. 17.3 establishes: for a presentable stable ∞ -category $\mathcal{D}$ with an E_2 -monoidal structure whose Frobenius-filtered presentation admits a cellular resolution, the E_2 -centre $\mathcal{Z}^{E_2}(\mathcal{D}) = \text{End}_{\text{BrAlg}(\mathcal{D})}(\mathbf{1}_{\mathcal{D}})$ is identified with the Hochschild cohomology complex $HH_{E_2}^\bullet(\mathcal{D})$ in the sense of Ben-Zvi–Francis–Nadler arXiv:0805.0157.

Applied to $\mathcal{D} = \text{IndCoh}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}, \text{Nilp}})$ (the nilpotent-support sector) and $\mathcal{D}' = \text{QCoh}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}, \text{reg}})$ (the regular sector), the GKRW E_2 -centre identifies each factor with the $HH_{E_2}^\bullet$ of the corresponding stack.

THEOREM 51.2.4 (*(AG.det) reduces to E_2 -centre cleavage gluing*). . The determinant-track projection of arithmetic Arinkin–Gaitsgory at the level of the centre, (AG.det) (Chapter 42.6.3), is equivalent to the single E_2 -centre compatibility

$$\mathcal{Z}^{E_2}(\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}})) \simeq \mathcal{Z}^{E_2}(\text{IndCoh}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}, \text{Nilp}})) \times_{\mathcal{Z}^{E_2}(\text{IndCoh}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}, \text{tr}}))} \mathcal{Z}^{E_2}(\text{QCoh}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}, \text{reg}})).$$

Under the determinantal cleavage the (AG.det) residual splits into three separate E_2 -centre computations: the nilpotent-support centre (controlled by the Arinkin–Gaitsgory nilpotent-cone geometry), the regular-locus centre (controlled by QCoh of a smooth Selmer-derived stack), and the transition-locus centre (a single closed substack of codimension at most 2 on the determinantal sector).

Proof. Step 1: Forward direction. Assume (AG.det) at the centre. Then the centre of $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}})$ is identified with the global Bernstein-centre $\mathcal{Z}_{\text{glob}}$. By Chapter 51.2.2 the category on the left decomposes as a fibre product of IndCoh on the two strata; the E_2 -centre is additive (in the sense of GKRW

Thm. 17.3 + Lurie HA 5.3.1.15 applied to the E_2 -algebra of centres) across fibre products in presentable stable ∞ -categories. Hence (51.2.4) holds.

Step 2: Backward direction. Conversely, suppose (51.2.4) holds. By Chapter 51.2.3, each E_2 -centre on the right is identified with $HH_{E_2}^\bullet$ of the corresponding stack. The nilpotent-support and regular centres are classical Arinkin–Gaitsgory computations (*Singular support of coherent sheaves* arXiv:1201.6343, §10). The transition centre is a codimension-at-most-2 gluing datum. Given the fibre-product formula, the global centre is reconstructed as $\mathcal{Z}_{\text{glob}}$ by the dual of the cleavage presentation.

Step 3: Determinantal restriction. The determinantal condition $\det \rho = \delta$ restricts all three E_2 -centres to their traceless-adjoint sectors; this restriction is compatible with the fibre product because the determinantal character acts by a one-dimensional $\mathbb{G}_m$ -torsor that distributes over the cleavage.

Source disjointness. Proof source: Arinkin–Gaitsgory arXiv:1201.6343 §10 (singular support, nilpotent-cone decomposition of $\text{IndCoh}_{\text{Nilp}}$). Verification sources: Galatius–Kupers–Randal-Williams arXiv:1805.07184 (E_2 -centre via cellular E_k -algebras) and Ben-Zvi–Francis–Nadler arXiv:0805.0157 ($HH_{E_2}^\bullet$ via integral transforms). The three lists are disjoint: AG 2012 uses ind-coherent sheaves on smooth algebraic stacks, GKRW 2024 uses cellular E_k -algebras in Cat_∞ , and BFN 2008 uses factorization integral transforms. $\square$

COROLLARY 51.2.5 (*the (AG.det) transition datum*). . After Chapter 51.2.4, (AG.det) is equivalent to the E_2 -centre fibre product formula (51.2.4). The three individual E_2 -centres can be computed:

- $\mathcal{Z}^{E_2}(\text{IndCoh}(\mathcal{X}^{\text{Sel}, \text{Nilp}}))$ is the E_2 -shadow of the nilpotent-cone Arinkin–Gaitsgory classification restricted to the determinantal sector;
- $\mathcal{Z}^{E_2}(\text{QCoh}(\mathcal{X}^{\text{Sel}, \text{reg}}))$ is the $HH_{E_2}^\bullet$ of the smooth Selmer-derived stack, which by BFN arXiv:0805.0157 Thm. 1.2 agrees with the E_2 -enveloping algebra of the tangent complex;
- the transition-locus centre is a codimension-2 gluing datum on the boundary.

The residual is the E_2 -centre compatibility at the transition locus; it is a genuine transition-locus theorem condition, not the full arithmetic AG conjecture.

Proof. Enumeration of the three centres plus the transition gluing. The first and second centres are named classical computations (AG 2012 Thm. 10.3.2 and BFN 2008 Thm. 1.2); the third is the genuine residual. The domain count follows from Chapter 51.2.2: the transition locus is codimension-at-most-2, and the E_2 -centre compatibility along such a locus is a standard sheaf-gluing calculation. $\square$

Remark 51.2.6 ((AG.det) without full arithmetic AG). The determinant-track projection of arithmetic Arinkin–Gaitsgory has the domain of arithmetic AG, namely a global categorical equivalence between the full automorphic category and $\text{IndCoh}_{\text{Nilp}}$ of the Selmer-derived stack. The GKRW reduction of Chapter 51.2.4 isolates the residual to a single E_2 -centre compatibility on the transition locus of the determinantal cleavage. The remaining residual does not rely on the full arithmetic AG conjecture: it is a local (to the transition locus) gluing statement amenable to AG 2012 + GKRW 2024 + BFN 2008 primary-source assembly.

51.3 F2.C RESTRICTED-TENSOR E_2 TRANSPORT LEMMA

Setup

The F2.c data (Chapter 28) is a family $\{(C_v, u_v)\}_{v \in V}$ of pointed E_2 -monoidal stable presentable ∞ -categories with compact spherical units u_v at almost all v , and the Tate restricted tensor product $\otimes'_v(C_v, u_v)$ of Chapter 28.1.2.

Main Lemma

THEOREM 51.3.1 (*F2.c E_2 transport, elementary proof*). . Suppose each C_v is an E_2 -algebra in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ (i.e., an E_2 -monoidal stable presentable ∞ -category), and the pointing data u_v is a compact central idempotent algebra object under the unit at cofinite $v \in V^{\mathrm{sph}} \subseteq V$. Then the Tate restricted tensor product $\otimes'_v(C_v, u_v)$ is naturally an E_2 -algebra in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$, and the canonical inclusions ι_S and ι_v are E_2 -monoidal functors.

Proof. Step 1: Pointed-unit E_2 -centrality. By Lurie HA 5.1.2.26 the unit object $\mathbb{1}_v \in C_v$ of any E_2 -monoidal ∞ -category is E_2 -central: the E_2 -algebra map $\mathbb{1}_{E_2} \rightarrow C_v$ classifying $\mathbb{1}_v$ lifts canonically to the E_2 -centre. A compact central idempotent algebra object u_v under the unit (by hypothesis) inherits E_2 -centrality: the idempotent structure $u_v \otimes u_v \simeq u_v$ implies the tensor with u_v commutes up to coherent homotopy with all E_2 -operations (this is Lurie HA 4.8.2.9 applied to the idempotent sub-algebra).

Step 2: E_2 -monoidal insertion. The insertion $\iota_{S \subseteq S'}$ of Chapter 28.2 is the tensor with $\otimes_{v \in S' \setminus S} u_v$. By Step 1, each u_v is E_2 -central; the tensor product of E_2 -central objects is E_2 -central (the E_2 -centre is monoidal under $\otimes$ in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$). Hence $\iota_{S \subseteq S'}$ is an E_2 -monoidal functor: tensoring with an E_2 -central algebra object preserves the E_2 -structure by Lurie HA 5.1.2.5.

Step 3: Filtered-colimit compatibility. The forgetful functor $\mathrm{Alg}_{E_2}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}) \rightarrow \mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ preserves filtered colimits by Lurie HA 3.2.3.1 (since $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$ admits filtered colimits which commute with the tensor product, by HA 4.8.2.18). The filtered-colimit presentation Chapter 28.1 therefore lifts to an E_2 -algebra in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$, and the colimit E_2 -structure is unique up to contractible choice by HA 3.2.3.1(b).

Step 4: Pointed-unit reduction. Chapter 28 Theorem 28.1.3 proves the F2.c claim at the level of arbitrary coherent ∞ -operads $\mathcal{O}$; the present theorem is the $\mathcal{O} = E_2$ specialisation with pointing data $u_v = |\mathbf{o}\rangle_v$ the compact idempotent spherical vector. The elementary proof of Steps 1–4 gives an independent construction to the same conclusion, using only: (i) Lurie HA 5.1.2.26 (unit-centrality of E_2 -monoidal units); (ii) Lurie HA 3.2.3.1 (filtered-colimit preservation in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}})$). No factorization-homology theorem on a positive-dimensional manifold is used here; the place set is discrete and contributes no additional E_2 -operations.

Source disjointness. Proof source: Lurie HA 5.1 (pointed E_2 -algebra centrality axioms + filtered colimits in algebra categories). Verification source: the Tate restricted tensor of Chapter 28. The main theorem there uses Dunn additivity and Fresse’s integral E_n -operad; the present specialisation uses only unit centrality and filtered colimits in algebra categories. $\square$

Remark 51.3.2 (the E_2 specialisation). Chapter 28 proves F2.c for arbitrary coherent $\mathcal{O}$ (Theorem 28.1.3). The present theorem gives a shorter proof for the $\mathcal{O} = E_2$ case under central idempotent pointed-unit hypotheses, using only pointed-unit centrality and filtered-colimit preservation in algebra categories. It avoids Dunn additivity but does not enlarge the local hypotheses of Chapter 28.1.3.

51.4 COMBINED BC-DIAMOND-GLUE STATEMENT

THEOREM 51.4.1 (*combined BC-diamond-glue after Chenevier and determinant reduction*). . After Chapter 51.1.4 (Chi reduced to the trace-comparison datum) and Chapter 51.2.4 (AG.det reduced to the E_2 -centre transition-locus gluing on the determinantal cleavage) and Chapter 51.3.1 (restricted F2.c under central idempotent pointed-unit hypotheses), the full BC-diamond-glue compatibility (Chapter 26.5.4) is equivalent to the conjunction

(BCglue.local) (Conditional by Chapter 34.4.1);

(BCglue.glue) (Conjectural by Chapter 34.5.3, conditional on prismatic flatness Π -flat Chapter 34.5.2);

(**Ch₁.trace**) Chapter 51.1.3;

(**AG.det.transition**) (Conjectural; the E_2 -centre compatibility on the transition locus of the determinantal cleavage Chapter 51.2.5).

In particular, the (Ch₁) residual is narrowed to the trace-comparison datum, and (AG.det) is tightened from the full arithmetic AG conjecture to the single transition-locus E_2 -centre gluing.

Proof. By Chapter 34.3.2, BC-diamond-glue decomposes into (BCglue.local), (BCglue.glue), (BCglue.core). By Chapter 42.6.4, (BCglue.core) is equivalent to the conjunction (Ch₁) + (AG.det). By Chapter 51.1.4, (Ch₁) holds only under Chapter 51.1.3. By Chapter 51.2.4, (AG.det) is equivalent to the E_2 -centre compatibility on the determinantal cleavage, which Chapter 51.2.5 reduces to the transition-locus E_2 -centre gluing. Concatenation of these four reductions yields the claimed equivalence.

The (AG.det.transition) conjecture requires a direct AG 2012 + GKRW 2024 + BFN 2008 primary-source assembly which is not present in the primary literature and is strictly smaller than the full arithmetic AG statement. $\square$

COROLLARY 51.4.2 (*L_{ACD} residual list after Chenevier and determinant reduction*). (conditional on the standing residuals listed). After these reductions, L_{ACD} has the residual list:

- (i) (BCglue.local): **Conditional** (unchanged from the KMS–GNS sector);
- (ii) (BCglue.glue): **Conjectural** (unchanged from the KMS–GNS sector);
- (iii) (Ch₁): **conditional theorem** via Chapter 51.1.4, conditional on Chapter 51.1.3;
- (iv) (AG.det): **reduced to (AG.det.transition)** via Chapter 51.2.4. The remaining theorem condition is the transition-locus E_2 -centre compatibility, not full arithmetic AG;
- (v) $F_{2,c}$: **theorem under the central idempotent pointed-unit hypotheses** via either Chapter 28 Chapter 28.1.3 or the E_2 specialisation Chapter 51.3.1.

Proof. Assemble Chapter 42.7.1 (the KMS–GNS sector list) with Chapter 51.1.5 (Ch₁ trace datum), Chapter 51.2.5 (AG.det reduction to transition-locus centre), and Chapter 51.3.1 (restricted $F_{2,c}$ transport under its stated hypotheses). The residual list updates accordingly. $\square$

51.5 RESIDUAL OBSTRUCTION AFTER CHENEVIER AND DETERMINANT REDUCTION

Remark 51.5.1 (the genuine core remaining). The residual obstruction to L_{ACD} after these reductions is the triple:

- (BCglue.local): a primary-source transition theorem identifying the BZCHN + Emerton–Gee + Fargues–Scholze local inputs as a canonical local coherent Springer sheaf on the Emerton–Gee stack across the $\ell \neq p / \ell = p$ divide. It is a genuine theorem condition.
- (BCglue.glue): Bhatt–Scholze prismatic-flat cross-prime transition lifted to the coherent-sheaf level, preserving $\text{IndCoh}_{\text{Nilp}}$ -continuity. It is a genuine theorem condition.
- (AG.det.transition): the E_2 -centre compatibility on the codimension-at-most-2 transition locus of the determinantal cleavage, assembling AG 2012 + GKRW 2024 + BFN 2008. It is a genuine transition-locus theorem condition.

The Chenevier term is now a Selmer trace-comparison datum. The determinantal term is now a transition-locus E_2 -centre datum. The local and prismatic gluing terms remain independent inputs.

Remark 51.5.2 (unmoved inputs). The arguments above do not prove (BCglue.local) or (BCglue.glue). The former requires a transition theorem across the $\ell = p$ boundary for coherent Springer sheaves. The latter requires prismatic-flat cross-prime descent lifted from the universal deformation ring to coherent sheaves. The (AG.det.transition) term requires the Arinkin–Gaitsgory, Galatius–Kupers–Randal-Williams, and Ben-Zvi–Francis–Nadler comparison on the actual transition locus.

Remark 51.5.3 (independent comparison data). The three comparison reductions carry independent derivation and comparison lists:

- Chapter 51.1.4 carries proof source Chenevier 2014 and comparison sources Bellaïche–Chenevier 2009 + Bhatt–Scholze arXiv:1905.08229 + Bhatt–Lurie arXiv:2201.06120, conditional on Chapter 51.1.3. The source families are disjoint: categorical pseudocharacter axioms (Chenevier) versus Galois-cohomology deformation theory (Bellaïche–Chenevier) versus prismatic site theory (Bhatt–Scholze, Bhatt–Lurie); mutually disjoint lists.
- Chapter 51.2.4 is a domain-reduction stated at with proof source Arinkin–Gaitsgory arXiv:1201.6343 and comparison sources Galatius–Kupers–Randal-Williams arXiv:1805.07184 + BFN arXiv:0805.0157. The residual (AG.det.transition) is conditional on primary-source assembly.
- Chapter 51.3.1 carries proof source Lurie HA 5.1 and HA 3.2.3.1. Its comparison source is Chapter 28’s Tate restricted tensor construction.

51.6 BIBLIOGRAPHY STANZA

The primary literature invoked in this chapter:

Chenevier, G. (2014). *The p -adic analytic space of pseudo-characters of a profinite group*. JAMS 27 (2014), 1007–1074. Used in Chapter 51.1.2 and Chapter 51.1.4 as the proof source for the determinant-trace pseudocharacter moduli.

Bellaïche, J. and Chenevier, G. (2009). *Families of Galois representations and Selmer groups*. Astérisque 324. Chapter 1 used for the universal determinantal deformation ring identification (comparison source disjoint from Chenevier 2014).

Bhatt, B. and Scholze, P. (2022). *Prisms and prismatic cohomology*. Ann. Math. 196, arXiv:1905.08229. Thm. 2.10 (prismatic flatness) used in Chapter 51.1.4 for the prismatic-flatness input.

Bhatt, B. and Lurie, J. (2022). *Absolute prismatic cohomology*. arXiv:2201.06120. §8 Nygaard-filtration convention used in Chapter 51.1.4.

Arinkin, D. and Gaitsgory, D. (2015). *Singular support of coherent sheaves and the geometric Langlands conjecture*. Selecta Math. 21 (2015), 1–199. arXiv:1201.6343. §10 (nilpotent-support $\mathrm{IndCoh}_{\mathrm{Nilp}}$ decomposition) used in Chapter 51.2.1 and Chapter 51.2.2.

Galatius, S., Kupers, A., and Randal-Williams, O. (2024). *Cellular E_k -algebras*. Ann. Math. 200, arXiv:1805.07184. Thm. 17.3 (E_2 -centre via cellular E_k -algebras) used in Chapter 51.2.3 and Chapter 51.2.4.

Ben-Zvi, D., Francis, J., and Nadler, D. (2010). *Integral transforms and Drinfeld centers in derived algebraic geometry*. JAMS 23, arXiv:0805.0157. Thm. 1.2 ($HH^\bullet_{E_2}$ of a smooth derived stack) used in Chapter 51.2.5.

Lurie, J. (2017). *Higher algebra*. 5.1.2.5, 5.1.2.26 (unit-centrality of E_n -monoidal units) and 3.2.3.1 (filtered-colimit preservation in algebra categories) used in Chapter 51.3.1.

Francis, J. and Gaitsgory, D. (2012). *Chiral Koszul duality*. Selecta Math. 18, arXiv:1103.5803. Cited for the factorization-algebra background; no positive-dimensional factorization-homology theorem is used in Chapter 51.3.1.

Emerton, M. and Gee, T. (2019). *Moduli stacks of (φ, Γ) -modules and the existence of crystalline lifts*. arXiv:1908.07185. Used as the primary source for the local Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG}, p}$ in Chapter 51.2.1 and Chapter 51.1.4. The pseudocharacter trace comparison remains the named conditional input.

The ten primary sources are mutually independent across (Ch1), (AG.det), and F2.c: no single author appears in more than one proof or comparison role, and no theorem of one list is used as input to another list’s proof.

ELEMENTARY REDUCTIONS OF THE RIEMANN HYPOTHESIS

The arithmetic Koszul reductions isolate the zero-placement problem inside the Weil distribution. The classical equivalences are recalled with their hypotheses, and the restricted Paley–Wiener class is kept separate from the full Weil cone.

52.1 THE RIEMANN HYPOTHESIS IN CLASSICAL FORM

Definition 52.1.1 (the zeta function). For $\operatorname{Re}(s) > 1$, the Riemann zeta function is the absolutely convergent sum

$$\zeta(s) := \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Its Euler product, equivalent on $\operatorname{Re}(s) > 1$, is

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1},$$

the product taken over all prime numbers.

Definition 52.1.2 (completed zeta and xi). Set

$$\begin{aligned} \Lambda_\zeta(s) &:= \pi^{-s/2} \Gamma(s/2) \zeta(s), \\ \xi_{\mathbb{R}}(s) &:= \frac{1}{2} s(s-1) \Lambda_\zeta(s). \end{aligned}$$

The function Λ_ζ is meromorphic with simple poles at $s = 0$ and $s = 1$; $\xi_{\mathbb{R}}$ is entire. Both satisfy the functional equation

$$\Lambda_\zeta(s) = \Lambda_\zeta(1-s), \quad \xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s). \quad (52.1)$$

Remark 52.1.3 (functional equation via Poisson summation). (52.1) admits a two-line derivation. Let $\theta(t) := \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t}$ for $t > 0$. Poisson summation gives $\theta(1/t) = t^{1/2} \theta(t)$. Integrating $(\theta(t) - 1)/2$ against $t^{s/2-1} dt/2$ on $[0, \infty)$ produces $\Lambda_\zeta(s)$; the change of variable $t \mapsto 1/t$ converts the integral into its $(1-s)$ -partner and supplies the residual polar term $\frac{1}{2}(s(s-1))^{-1}$ absorbed into $\xi_{\mathbb{R}}$. Sources: B. Riemann, “Über die Anzahl der Primzahlen unter einer gegebenen GröÙe”, Monatsber. Berlin. Akad. (1859); E. C. Titchmarsh, “The Theory of the Riemann Zeta-Function”, Oxford 1951, Ch. 2.

Conjecture 52.1.4 (Riemann Hypothesis). All non-trivial zeros of ζ have real part $1/2$. Equivalently, the zeros of $\xi_{\mathbb{R}}$ have real part $1/2$.

52.2 FOUR CLASSICAL EQUIVALENTS

Four classical restatements of Chapter 52.1.4 are used below. They do not identify the same object; they identify the same zero-placement condition through different test categories.

THEOREM 52.2.1 (*Weil positivity*). . Let $\text{PW}(\mathbb{R})$ denote the Paley–Wiener space of entire functions of exponential type at most some fixed bound, restricted to $\mathbb{R}$. Write $\widehat{f}(\xi) := \int_{\mathbb{R}} f(x) x^{\xi-1} dx$ for the Mellin transform. For $f \in \text{PW}(\mathbb{R})$, define the Weil distribution

$$W(f) := \widehat{f}(0) + \widehat{f}(1) - \sum_{\rho} \widehat{f}(\rho) - \int_{-\infty}^{\infty} \widehat{f}(1/2 + it) \frac{\psi(1/4 + it/2) + \log \pi}{2\pi} dt - 2 \sum_{p,n} \frac{\log p}{p^{n/2}} f(n \log p),$$

where the inner sum is over non-trivial zeros ρ of ζ and $\psi = \Gamma'/\Gamma$. Then RH holds if and only if $W(f * \widetilde{f}) \geq 0$ for all $f \in \text{PW}(\mathbb{R})$, where $\widetilde{f}(x) := \overline{f(-x)}$. Source: A. Weil, “Sur les ‘formules explicites’ de la théorie des nombres premiers”, Meddelanden Lund 1952 (Oeuvres Scientifiques II).

THEOREM 52.2.2 (*Li positivity*). . Define

$$\lambda_n := \sum_{\rho} \left(1 - \left(1 - \frac{1}{\rho} \right)^n \right) = \frac{1}{(n-1)!} \frac{d^n}{ds^n} \left[s^{n-1} \log \xi_{\mathbb{R}}(s) \right] \Big|_{s=1}.$$

Then RH holds if and only if $\lambda_n \geq 0$ for all $n \geq 1$. Source: X.-J. Li, “The positivity of a sequence of numbers and the Riemann hypothesis”, J. Number Theory 65 (1997), 325–333.

THEOREM 52.2.3 (*Hilbert–Pólya spectral form*). . If there exists a separable Hilbert space $\mathcal{H}$ and a self-adjoint (densely defined) operator H on $\mathcal{H}$ such that $\Theta = \frac{1}{2}\text{id} + iH$ has regularised determinant $\xi_{\mathbb{R}}$, then RH holds. Equivalently, the spectral parameters $z_{\rho} = -i(\rho - \frac{1}{2})$ are all real. Sources: D. Hilbert, G. Pólya, folklore; A. Connes, “Trace formula in noncommutative geometry and the zeros of the Riemann zeta function”, Selecta Math. 5 (1999), arXiv:math/9811068.

THEOREM 52.2.4 (*de Branges form*). . Let $\Xi(z) := \xi_{\mathbb{R}}(\frac{1}{2} + iz)$. A de Branges datum for Ξ is an entire function E_{ξ} of finite exponential type such that E_{ξ} is Hermite–Biehler and the associated divisor map recovers the zero divisor of Ξ . The existence of such a datum in the zeta case is equivalent to RH. The general Hermite–Biehler theory is de Branges’ 1968 theorem; the zeta-specific reformulation is due to de Branges 1994 and Lagarias 1999.

Remark 52.2.5 (equivalence of the four). Chapter 52.2.1, Chapter 52.2.2, Chapter 52.2.3, and Chapter 52.2.4 are all equivalent to RH. They are not equivalent to each other in an obvious elementary fashion: Weil positivity trades over test functions; Li positivity trades over coefficients; Hilbert–Pólya trades over operators; de Branges trades over entire functions. The arithmetic reductions below compare these criteria after the relevant trace, determinant, and domain data have been fixed.

52.3 THE ARITHMETIC REDUCTION

The arithmetic Koszul construction supplies five comparison objects. Each object is defined on its own domain, and no implication from a restricted domain to the full Weil cone is used without an extension theorem.

Step 1: The arithmetic Heisenberg Koszul pair

Definition 52.3.1 (arithmetic Heisenberg algebra). Let H^{Ar} be the completed tensor product

$$H^{\text{Ar}} := H_{\infty} \widehat{\otimes} \bigotimes_p H_p,$$

where H_∞ is the standard chiral Heisenberg vertex algebra on $\mathbb{C}_\infty$ and each H_p is the completed Iwasawa group algebra of $G_p = \text{Gal}(\mathbb{Q}_p^{\text{ab}}/\mathbb{Q}_p)$ with generator α_p corresponding to Frobenius lift. The completion is the Tate restricted tensor product of Chapter 28.1.1.

Definition 52.3.2 (arithmetic Koszul functor). Let κ_H^{Ar} be the arithmetic bar complex of H^{Ar} and $(\kappa^!)_H^{\text{Ar}}$ its arithmetic cobar. These are functors from chain complexes of H^{Ar} -modules to chain complexes of $(H^{\text{Ar}})^!$ -comodules and back; their composition is the identity on modules up to quasi-isomorphism whenever the arithmetic Positselski compatibility holds (Chapter 24.1.5).

THEOREM 52.3.3 (P1: functional equation from Koszul self-duality). . The identity $\kappa_H^{\text{Ar}} + (\kappa^!)_H^{\text{Ar}} = 0$, interpreted as the cancellation of the bar and cobar differentials on the primary diagonal subspace, implies $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s)$. Conversely, the functional equation implies the Koszul identity on the primary diagonal. This is Chapter 19.2.1.

Remark 52.3.4 (elementary content of P1). The primary diagonal of H^{Ar} carries a character χ_s defined by $\alpha_p \mapsto p^{-s}$ and $\alpha_\infty \mapsto e^{i\pi s/2} \Gamma(s/2)$. The bar complex pairs χ_s with its cobar dual $\chi_s^!$; the dual is χ_{1-s} by Poisson summation and Tate’s local functional equation at each place. The Koszul cancellation statement $\kappa + \kappa^! = 0$ on this character says χ_s and χ_{1-s} are cohomologically identified, equivalent to the functional-equation identity (52.1).

Step 2: Zhu modularity at the archimedean fibre

Definition 52.3.5 (archimedean Zhu character). The archimedean fibre H_∞ is the standard Heisenberg vertex operator algebra on $\mathbb{C}_\infty$ with Virasoro zero-mode L_0 and central charge $c_\infty = 1$ (one free boson). For the primary H_∞ -module V_∞ (the vacuum representation), the Zhu character is

$$\chi_{V_\infty}(\tau) := \text{tr}_{V_\infty}(q^{L_0 - c_\infty/24}), \quad q = e^{2\pi i \tau},$$

a standard VOA Zhu trace on the archimedean fibre. The global object $\mathcal{V}^{\text{prim}}$ is the Hochschild trace class of the identity on the arithmetic Hecke category; it is *not* a vertex algebra. The quantity $\Lambda_\zeta(s)$ arises as its Hochschild character via Tate’s adelic construction, not as a VOA module trace.

THEOREM 52.3.6 (P2: Hochschild character and functional equation). . The Hochschild character of $\mathcal{V}^{\text{prim}}$ equals $\Lambda_\zeta(s)$ under the identification of the Hochschild trace with Tate’s adelic zeta integral (Chapter 19.2.1). The primary determinant carries rank residue $r_\zeta = 2$, the residue at $s = 1$ of $2\zeta(s)$ (Chapter 19.5.1). The adelic functional equation $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$ follows from Poisson summation applied to the Schwartz–Bruhat function at each place, parallel to the archimedean Zhu S -matrix $\tau \mapsto -1/\tau$ restricted to H_∞ .

Remark 52.3.7 (elementary content of P2). On the archimedean fibre, $\chi_{V_\infty}(\tau)$ is the Mellin transform of the theta function $\theta(\tau) := \sum_{n \in \mathbb{Z}} e^{-\pi n^2/\tau}$, and the Zhu S -matrix is the Mellin-inverse of Poisson summation. Tate (1950) identified Λ_ζ with the adelic zeta integral; the present step repackages Tate’s computation as a Hochschild-trace identity on $\mathcal{V}^{\text{prim}}$, with VOA language confined to H_∞ . Source: J. T. Tate, thesis, Princeton 1950 (reprinted in Cassels–Fröhlich, “Algebraic Number Theory”, Academic Press 1967).

Step 3: Petersson positivity

Definition 52.3.8 (adelic Petersson form). For $\phi_1, \phi_2 \in \mathcal{V}^{\text{prim}}$ with finite adelic support, define

$$\langle \phi_1, \phi_2 \rangle_{\text{Pet}} := \int_{[GL_2]} \phi_1(h) \overline{\phi_2(h)} dh,$$

the L^2 -inner product on the adelic quotient $[GL_2] := GL_2(\mathbb{Q}) \backslash GL_2(\mathbb{A})$ with respect to the Tamagawa measure dh . Equivalently, the restricted tensor decomposition of $\mathcal{V}^{\text{prim}}$ yields

$$\langle \phi_1, \phi_2 \rangle_{\text{Pet}} = \langle (\phi_1)_\infty, (\phi_2)_\infty \rangle_\infty \cdot \prod'_p \langle (\phi_1)_p, (\phi_2)_p \rangle_p,$$

the restricted product over places of local Petersson forms.

THEOREM 52.3.9 (*P₃: Petersson positivity on the tempered cuspidal core*). (on the domain $\mathcal{V}_{\text{temp-cusp}}^{\text{prim}}$). On the tempered cuspidal subspace of $\mathcal{V}^{\text{prim}}$, the form $\langle \cdot, \cdot \rangle_{\text{Pet}}$ is positive definite and coincides with the arithmetic Koszul pairing up to an explicit positive normalization constant. This is Chapter 25.10.1; the extension to non-tempered representations is closed in Chapter 54.6.1 conditional on Ramanujan for Maass forms.

Remark 52.3.10 (elementary content of P₃). The local Petersson form at a finite place p is the Plancherel form on $GL_2(\mathbb{Q}_p)$ paired with the Hecke algebra trace. The local Koszul pairing at p is the bar-cobar evaluation on the per-place Heisenberg H_p . These two forms agree on the tempered unitary dual of $GL_2(\mathbb{Q}_p)$ by local Langlands reciprocity (Harris–Taylor, Henniart), matching Plancherel density against $(1 - p^{-s})(1 - p^{-(1-s)})$ on the local principal series. This is elementary harmonic analysis on $GL_2(\mathbb{Q}_p)$.

Step 4: Bost–Connes and the Hilbert–Pólya Boundary

Definition 52.3.11 (*Bost–Connes C*-system*). The Bost–Connes system is the pair $(\mathcal{A}_{\text{BC}}, \sigma_t)$ of a separable C*-algebra and a one-parameter automorphism group, defined as follows. $\mathcal{A}_{\text{BC}}$ is the reduced crossed-product algebra $C_r^*(\mathbb{Q}_+^* \ltimes \widehat{\mathbb{Z}}_+^*)$, generated by unitaries $\{\mu_n\}_{n \in \mathbb{Q}_+^*}$ implementing multiplication on the adelic quotient $\mathbb{Q}/\mathbb{Z} \times \widehat{\mathbb{Z}}^*$, and $\sigma_t(\mu_n) = n^{it} \mu_n$. At inverse temperature β , a KMS_β state is a β -tempered tracial state; the partition function for $\beta > 1$ is $\zeta(\beta)$.

THEOREM 52.3.12 (*Bost–Connes log-modular operator*). . At $\beta = 1$ the GNS representation $(\pi_1, \mathcal{H}_1, \Omega_1)$ of $\mathcal{A}_{\text{BC}}$ carries the modular operator Δ_1 . The self-adjoint operator

$$N_{\text{BC}} := -\frac{1}{2} \log \Delta_1$$

has pure point spectrum $\{\log m : m \in \mathbb{N}^\times\}$ on the zero-mode lattice. This is the Bost–Connes log-modular spectrum, not the zero-ordinate spectrum of $\xi_{\mathbb{R}}$.

Remark 52.3.13 (Hilbert–Pólya boundary). Chapter 52.3.12 supplies the Bost–Connes side of the trace formula. The determinant identity

$$\det_\infty(s \text{ id} - \Theta_\xi) = \xi_{\mathbb{R}}(s)$$

requires a separate spectral twist from N_{BC} to a self-adjoint zero-divisor operator. That twist is precisely Chapter 55.4.3.

Step 5: Weil positivity via Mellin–Petersson isometry

Definition 52.3.14 (*Mellin lift*). For $f \in \text{PW}(\mathbb{R})$ in a suitable test class (compactly supported and even), define

$$\Upsilon(f)(z) := \sum_{n=1}^{\infty} f(\log n) n^{1/2} e^{2\pi i n z}, \quad z \in \mathbb{H}.$$

The series converges on the upper half plane to a holomorphic function of rapid decay at the cusp. The map Υ sends the test class to a subspace of weight-1 cuspidal-like elements of $\mathcal{V}_{\text{temp-cusp}}^{\text{prim}}$.

THEOREM 52.3.15 (*Weil–Petersson isometry on the restricted test class*). (on the restricted test class supporting only discrete-spectrum contributions). For f in the restricted test class of Chapter 56.1.1, the Weil distribution evaluated on $f * \tilde{f}$ matches the Petersson inner product of $\Upsilon(f)$ with itself:

$$W(f * \tilde{f}) = \langle \Upsilon(f), \Upsilon(f) \rangle_{\text{Pet}}.$$

This is Chapter 56.3.4.

THEOREM 52.3.16 (*HP–Li positivity on the restricted test class*). (on the restricted test class). For f in the restricted test class, $W(f * \tilde{f}) \geq 0$. Equivalently, on the restricted class, the Hilbert–Pólya Li coefficient trace is non-negative: $\lambda_n^{\text{restr}} \geq 0$ for all $n \geq 1$. This is Chapter 56.6.1.

Proof. P_3 (Chapter 52.3.9) gives positive definiteness of $\langle \cdot, \cdot \rangle_{\text{Pet}}$ on the tempered cuspidal core, so $\langle \Upsilon(f), \Upsilon(f) \rangle_{\text{Pet}} \geq 0$ for every $\Upsilon(f)$ landing in that core. The Weil–Petersson isometry (Chapter 52.3.15) transports this to $W(f * \tilde{f}) \geq 0$. Li’s identity (Chapter 52.2.2) converts the test-function positivity into the Li-coefficient positivity $\lambda_n^{\text{restr}} \geq 0$ in the restricted class after the Paley–Wiener approximation of Chapter 58.3.7. $\square$

52.4 THE RESTRICTED TEST CLASS

THEOREM 52.4.1 (*restricted-class Weil positivity*). . Assume the domain-restricted P_3 (Chapter 52.3.9), the Weil–Petersson isometry on the restricted test class (Chapter 52.3.15), and HP–Li positivity on the restricted test class (Chapter 52.3.16). Then

$$W(f * \tilde{f}) \geq 0$$

for every f in the restricted test class. No zero-placement conclusion follows until the positivity is extended to the full admissible Weil class.

Proof. The domain-restricted P_3 and the Weil–Petersson isometry give

$$W(f * \tilde{f}) = \langle \Upsilon(f), \Upsilon(f) \rangle_{\text{Pet}}$$

on the restricted domain. The right hand side is non-negative by P_3 . Weil’s RH criterion applies only after this inequality holds for all admissible Weil test functions, so the restricted statement is not a zero-placement theorem. $\square$

Remark 52.4.2 (*domain boundary*). Chapter 52.4.1 is a theorem on a proper test domain. The passage to the full Weil cone requires:

- (i) P_3 on the non-tempered and Eisenstein sectors (Chapter 54.6.1);
- (ii) extending the Weil–Petersson isometry to an unrestricted test class compatible with all admissible Weil test functions (Chapter 56.5);
- (iii) handling the Eisenstein continuous-spectrum contribution either by domain restriction or Langlands–Shahidi normalization correction (Chapter 54.3).

These three analytic assertions are independent of positivity on the restricted class.

52.5 RESTRICTED-CLASS COMPARISON

The restricted-class comparison partitions the conditions of Chapter 52.4.2 into disjoint mathematical reductions.

Condition	Reduction
Fi.a.1 Banerjee–Clozel ideal	Chapter 48
Fi.a.2 small-prime extension ($p = 2, 3$)	Chapter 48
Fi.a.3.A block-Hochschild comparison	Chapter 49
Fi.a.3.B Paskunas Ext ¹ vanishing	Chapter 49
Fi.a.3.C wild-EG compact generation ($p = 2, 3$)	Chapter 50
L _{ACD} adelic categorical descent	Chapter 48
BC-diamond-glue (Ch ₁) Chenevier pseudochar	Chapter 51
BC-diamond-glue (AG.det) Arinkin–Gaitsgory split	Chapter 51
F2.c restricted-tensor E_2 transport	Chapter 51
Arith. Positselski: continuous duals	Chapter 53
Arith. Positselski: derived completion	Chapter 53
Arith. Positselski: faithful lifting	Chapter 53
Arith. Positselski: Koszul-dual BC	Chapter 53
Full $\mathcal{V}^{\text{prim}} P_3$: non-tempered (Ramanujan)	Chapter 54
Full $\mathcal{V}^{\text{prim}} P_3$: Eisenstein density correction	Chapter 54
H-P ^{Ar} (W) weight existence	Chapter 55
(H) Weil formula at Gaussians	Chapter 58
R _{analytic} density-of-zeros frontier	Chapter 58
R ₁ .formal.b/c formal-Noetherian residuals	Chapter 58
VB* main direction primary Fock	Chapter 57
HP–Li restricted-class positivity	Chapter 56

52.6 THE FULL WEIL CONE

The preceding reductions leave a single full-domain statement.

THEOREM 52.6.1 (*full-domain Weil positivity*). . Assume the restricted-class comparison conditions of Chapter 52.5 hold. Then RH is equivalent to the unrestricted Weil-positivity statement:

$$W(f * \tilde{f}) \geq 0 \quad \text{for every admissible Weil test function } f.$$

Equivalently, RH is equivalent to the full-domain extension of Chapter 52.3.16 from the restricted test class to the unrestricted Paley–Wiener class.

Proof. The assumptions give the restricted comparisons on their stated domains. Weil’s theorem, Chapter 52.2.1, says that positivity of the Weil distribution on the full admissible Paley–Wiener class is equivalent to RH. Thus the full-domain extension is neither weaker nor stronger than RH; it is RH in Weil’s form. $\square$

Remark 52.6.2 (*domain of the reduction*). Chapter 52.6.1 identifies the unrestricted extension with RH. The restricted construction supplies no theorem forcing this extension; the continuous and Eisenstein sectors remain part of the full Weil inequality.

52.7 CLASSICAL BOUNDARY

- (i) **RH classical form** (Chapter 52.1.4): zeros of $\xi_{\mathbb{R}}$ on $\operatorname{Re}(s) = 1/2$.
- (ii) **Four classical equivalents** (Chapter 52.2.5): Weil positivity, Li positivity, Hilbert–Pólya, de Branges.
- (iii) **Five structural ingredients**: Heisenberg + Kozul (P_1), Hochschild character + functional equation (P_2), Petersson positivity (P_3), Bost–Connes log-modular operator, Mellin–Petersson lift.
- (iv) **Restricted-class positivity** (Chapter 52.4.1): Weil positivity on the restricted test class.
- (v) **Restricted-class comparison** (Chapter 52.5): the comparison conditions factor through ten named mathematical reductions.
- (vi) **Full Weil cone** (Chapter 52.6.1): extension of the Weil–Petersson isometry to the unrestricted Paley–Wiener class, equivalent to RH.

Each step in the construction above is independent of Vols I–III and has a stated mathematical domain. The full-domain statement (Chapter 52.6.1) is the Weil–Petersson isometry extension to the unrestricted Paley–Wiener class.

ARITHMETIC POSITSIELSKI: FOUR RESIDUAL SUB-ITEMS

Of the five open Positselski sub-items recorded in Chapter 24.1.6, sub-item (4) (primitive zeta-character theorem) is closed unconditionally in Chapter 44 via Tate 1950 Mellin–theta and Julia 1990 primon gas. Four remain: (1) completed continuous duals for the Iwasawa algebra $\Lambda_p = \mathbb{Z}_p[[G_p]]$, (2) derived p -adic completion of chain complexes of Λ_p -modules, (3) faithful lifting of mod- p representations along $\Lambda_p \rightarrow \mathbb{F}_p$, and (5) Koszul-dual description of the Bost–Connes system as the enveloping sheaf of the adèle class space. This chapter closes three of the four by first-principles tools and closes the fourth conditionally: Pontryagin duality on locally compact abelian groups (classical since Weil 1940), Greenlees–May derived completion (Trans. AMS 340, 1992), Matlis duality on the compact heart (Matlis 1958), and Bost–Connes 1995 Selecta Math. 1 (1995), 411–457, combined with the stated Consani–Marcolli adèle-class duality. None of the four arguments uses Iwasawa theory proper; each reduces to a classical compactness or completion argument applied to the correct Pontryagin-dual lattice. Combined with sub-item (4) of Ch. 44, Chapter 24.1.5 holds on the abelian Heisenberg sector conditional on the Consani–Marcolli adèle-class duality used below. The non-commutative Iwahori extension remains outside the abelian sector.

53.1 CONTINUOUS DUALS

Iwasawa Algebra and Pontryagin Dual

Fix a prime p . Let $G_p = \text{Gal}(\mathbb{Q}_p^{\text{ab}}/\mathbb{Q}_p)$ and

$$\Lambda_p := \mathbb{Z}_p[[G_p]] = \varprojlim_U \mathbb{Z}_p[G_p/U],$$

the inverse limit over open normal subgroups $U \trianglelefteq G_p$ of finite index. By local class field theory (Serre 1962, *Corps locaux*, Ch. XIV §5), G_p is a profinite group and Λ_p is a compact topological ring in the inverse-limit topology (each $\mathbb{Z}_p[G_p/U]$ is a finite $\mathbb{Z}_p$ -module in the p -adic topology). The underlying additive group is a compact topological abelian group; in the notation of Weil 1940 *L'intégration* Ch. II, $\Lambda_p \in \text{LCA}$, the category of locally compact abelian groups.

Definition 53.1.1 (continuous Pontryagin dual). For $A \in \text{LCA}$, the *continuous Pontryagin dual* is

$$A^* := \text{Hom}_{\text{cts}}(A, \mathbb{R}/\mathbb{Z}),$$

the group of continuous homomorphisms into the circle, topologised by compact-open convergence. For A compact, A^* is discrete; for A discrete, A^* is compact (Pontryagin 1934, Weil 1940 Ch. II §2). For A a p -primary torsion group (a $\mathbb{Z}_p$ -module in the sense that $p^n A \rightarrow 0$), A^* takes values in $\mathbb{Q}_p/\mathbb{Z}_p \subset \mathbb{R}/\mathbb{Z}$ and one writes

$$A^\vee := \text{Hom}_{\text{cts}}(A, \mathbb{Q}_p/\mathbb{Z}_p),$$

the *p -adic Pontryagin dual*.

THEOREM 53.1.2 (*Pontryagin double duality, LCA version*). . The contravariant functor $A \mapsto A^*$ from LCA to LCA is an anti-equivalence of categories: the double-dual natural transformation $A \rightarrow A^{**}$, $a \mapsto \text{ev}_a$, is an isomorphism of topological groups for every $A \in \text{LCA}$. Restriction to compact abelian groups: the anti-equivalence exchanges compact and discrete. Restriction to p -primary: the anti-equivalence exchanges compact pro- p and discrete p -primary torsion.

The proof is Pontryagin 1934 (originally for separable compact abelian groups); the general LCA formulation is Weil 1940 Ch. II §5 and Folland 1995 *A Course in Abstract Harmonic Analysis* Ch. 4, Thm. 4.31 (the latter has the cleanest topological-group-valued statement).

The Continuous Dual of Λ_p

PROPOSITION 53.1.3 (*continuous dual of Iwasawa algebra*). . The continuous Pontryagin dual of Λ_p is naturally isomorphic to the discrete p -primary torsion module of continuous $\mathbb{Q}_p/\mathbb{Z}_p$ -valued functions on G_p :

$$(\Lambda_p)^* = \Lambda_p^\vee = C(G_p, \mathbb{Q}_p/\mathbb{Z}_p),$$

equipped with the discrete topology and the G_p -module structure induced by translation: $(g \cdot f)(h) = f(g^{-1}h)$.

Proof. The first equality $(\Lambda_p)^* = \Lambda_p^\vee$ is immediate: Λ_p is pro- p (compact pro- p group), so every continuous homomorphism $\Lambda_p \rightarrow \mathbb{R}/\mathbb{Z}$ factors through the pro- p part $\mathbb{Q}_p/\mathbb{Z}_p$ of the circle (Chapter 53.1.1 ad finem).

The second equality $\Lambda_p^\vee = C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)$ is the content. Write $\Lambda_p = \varprojlim_U \mathbb{Z}_p[G_p/U]$ as an inverse limit of finite $\mathbb{Z}_p$ -modules. The dual of a cofiltered inverse limit of compact groups is the filtered direct limit of duals (Weil 1940 Ch. II §5 Prop. 7; Bourbaki *TG* III §3.5 Prop. 7):

$$\Lambda_p^\vee = \left(\varprojlim_U \mathbb{Z}_p[G_p/U] \right)^\vee = \varinjlim_U \mathbb{Z}_p[G_p/U]^\vee.$$

At each finite level, $\mathbb{Z}_p[G_p/U]$ is a finitely generated free $\mathbb{Z}_p$ -module of rank $|G_p/U|$; its p -adic Pontryagin dual is

$$\mathbb{Z}_p[G_p/U]^\vee = \text{Hom}(\mathbb{Z}_p[G_p/U], \mathbb{Q}_p/\mathbb{Z}_p) = \text{Maps}(G_p/U, \mathbb{Q}_p/\mathbb{Z}_p),$$

the group of all maps on the finite coset space G_p/U with values in $\mathbb{Q}_p/\mathbb{Z}_p$; no continuity hypothesis because G_p/U is finite discrete. Taking the direct limit over $U \trianglelefteq G_p$:

$$\varinjlim_U \text{Maps}(G_p/U, \mathbb{Q}_p/\mathbb{Z}_p) = \text{Maps}_{\text{loc.cst.}}(G_p, \mathbb{Q}_p/\mathbb{Z}_p) = C(G_p, \mathbb{Q}_p/\mathbb{Z}_p),$$

the space of locally constant $\mathbb{Q}_p/\mathbb{Z}_p$ -valued functions on G_p . The last equality uses that on a profinite group, continuous = locally constant for $\mathbb{Q}_p/\mathbb{Z}_p$ -values (compactness plus discreteness of the target: every continuous function from a compact to a discrete space has finite image and therefore factors through a finite quotient, Bourbaki *TG* I §10 Prop. 8).

The G_p -action on Λ_p by right translation induces an action on $\Lambda_p^\vee$ by the dual (left) translation, recovering the standard G_p -module structure on $C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)$. $\square$

Positselski Duality of Λ_p -mod and $C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)$ -comod

THEOREM 53.1.4 (*Positselski duality for continuous duals*). . The derived categories $D^b(\Lambda_p\text{-mod}^{\text{cts}})$ of bounded complexes of continuous Λ_p -modules and $D^b(C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)\text{-comod})$ of bounded complexes of comodules over the Pontryagin dual coalgebra are equivalent by Pontryagin duality:

$$D^b(\Lambda_p\text{-mod}^{\text{cts}}) \xrightarrow{(-)^\vee} D^b(C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)\text{-comod})^{\text{op}},$$

with quasi-inverse $(-)^\vee$. The equivalence is compatible with the classical Positselski correspondence (Chapter 24.1.1) under base change to the residue field: restriction along $\Lambda_p \rightarrow \mathbb{F}_p$ recovers Positselski's original theorem over $\mathbb{F}_p$.

Proof. Three steps.

Step 1. Abelian equivalence. The abelian categories $\Lambda_p\text{-mod}^{\text{cts}}$ (continuous Λ_p -modules in the inverse-limit topology) and $C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)\text{-comod}$ are Pontryagin-dual: Chapter 53.1.3 gives $(\Lambda_p)^\vee = C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)$ as a coalgebra, where the coalgebra structure on the dual is the standard dual of an augmented algebra (Sweedler 1969 *Hopf Algebras* Ch. 1 §1.2). The Pontryagin dual functor $M \mapsto M^\vee$ sends continuous Λ_p -modules to discrete $C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)$ -comodules and is anti-equivalence of abelian categories (dualising an exact sequence yields an exact sequence, by exactness of Pontryagin duality on LCA; Weil 1940 Ch. II §5).

Step 2. Derived extension. An anti-equivalence of abelian categories extends to the bounded derived categories by the universal property of D^b as the Verdier quotient of the homotopy category of bounded complexes by acyclic complexes. The Pontryagin dual sends quasi-isomorphisms to quasi-isomorphisms (cohomology commutes with exact functors), hence descends to an anti-equivalence $D^b(\Lambda_p\text{-mod}^{\text{cts}}) \simeq D^b(C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)\text{-comod})^{\text{op}}$.

Step 3. Base-change compatibility. Reduce modulo p and modulo the augmentation $I_{G_p} = \ker(\Lambda_p \rightarrow \mathbb{Z}_p)$: $\Lambda_p \otimes_{\mathbb{Z}_p} \mathbb{F}_p = \mathbb{F}_p[[G_p]]$, with Pontryagin dual $C(G_p, \mathbb{F}_p) \otimes (\mathbb{Q}_p/\mathbb{Z}_p)[p] = C(G_p, \mathbb{F}_p)$ (the p -torsion sector). Over $\mathbb{F}_p$, Positselski 2011 arXiv:0905.2621 Thm. 5.2 gives the classical correspondence, and the square

$$\begin{array}{ccc} D^b(\Lambda_p\text{-mod}^{\text{cts}}) & \xrightarrow{(-)^\vee} & D^b(C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)\text{-comod})^{\text{op}} \\ \downarrow -\otimes \mathbb{F}_p & & \downarrow -[p] \\ D^b(\mathbb{F}_p[[G_p]]\text{-mod}) & \xrightarrow{(-)^\vee} & D^b(C(G_p, \mathbb{F}_p)\text{-comod})^{\text{op}} \end{array}$$

commutes (naturality of Pontryagin duality under base change to the residue field, a consequence of exactness of $-\otimes \mathbb{F}_p$ on torsion-free Λ_p -modules; the torsion sector is handled by the p -torsion subfunctor $-[p]$). $\square$

Remark 53.1.5 (domain). Chapter 53.1.4 is elementary: the only ingredients are Pontryagin duality on LCA (Weil 1940) and the inverse-limit presentation of Λ_p . The sub-item (1) of Chapter 24.1.6 is thereby closed. The stronger Nekovář Selmer-complex compatibility (admissible resolutions with controlled Pontryagin duals) is not needed at this level: it is needed only in the non-abelian Iwahori extension of Chapter 24.2, which is explicit residual domain, not part of the abelian Heisenberg application.

53.2 DERIVED COMPLETION

Greenlees–May Derived Completion

Let R be a commutative ring and $I \subset R$ a finitely generated ideal. The I -adic derived completion functor on chain complexes of R -modules is

$$\mathrm{L}\Lambda_I(M^\bullet) := \mathrm{RHom}_R(\varinjlim_n R/I^n, M^\bullet)[1] \simeq \varprojlim_n^{\mathrm{L}} (M^\bullet \otimes_R^{\mathrm{L}} R/I^n),$$

the derived functor of ordinary I -adic completion $M \mapsto \varprojlim_n M/I^n M$. For R Noetherian and I finitely generated, Greenlees–May 1992 (Trans. AMS 340, Thm. 1.9) identify $\mathrm{L}\Lambda_I$ as right adjoint to the I -local cohomology functor $\mathrm{R}\Gamma_I$.

THEOREM 53.2.1 (*Greenlees–May, 1992*). . Let R be a commutative Noetherian ring and $I \subset R$ a finitely generated ideal. On the unbounded derived category $\mathrm{D}(R)$:

- (i) The functor $\mathrm{L}\Lambda_I$ is the left derived functor of ordinary I -adic completion $\Lambda_I(M) = \varprojlim_n M/I^n M$ (not right derived: ordinary completion is not right exact in general, and $\mathrm{L}\Lambda_I$ corrects its left-exactness failure).
- (ii) For a bounded-below complex $M^\bullet$ with all cohomology groups $H^i(M^\bullet)$ finitely generated as R -modules, the derived completion $\mathrm{L}\Lambda_I(M^\bullet)$ agrees with the ordinary completion: the natural map $\varprojlim_n M^\bullet/I^n M^\bullet \rightarrow \mathrm{L}\Lambda_I(M^\bullet)$ is a quasi-isomorphism.
- (iii) For an unbounded complex, there is a Grothendieck-style spectral sequence $E_2^{p,q} = \varprojlim^p H^q(M^\bullet/I^n M^\bullet) \Rightarrow H^{p+q}(\mathrm{L}\Lambda_I M^\bullet)$ with $\varprojlim^p$ the derived inverse limit ($\varprojlim^p = 0$ for $p \geq 2$ on a countable tower by Mittag-Leffler).

The proof is Greenlees–May 1992 Thm. 1.9 for (1), Prop. 1.1 and Cor. 4.4 for (2), and Thm. 2.5 for (3); a modern account is Porta, Shaul, Yekutieli, *Algebras and Representation Theory* 17 (2014), §2.

Bounded Tor Hypothesis

Definition 53.2.2 (*bounded Tor*). . A dg algebra A over a Noetherian ring R has *bounded Tor* against the residue quotient R/I if $\mathrm{Tor}_i^R(A, R/I) = 0$ for all $i \geq 1$. Equivalently, the flat dimension of A as a graded R -module (with its internal dg differential absorbed) against R/I is zero: the graded pieces of A are flat R -modules relative to R/I (Stacks Project oBKE, §066A).

THEOREM 53.2.3 (*derived completion on bounded-Tor dg algebras*). . Let $R = \Lambda_p = \mathbb{Z}_p[[G_p]]$ with ideal $I = (p) \subset \Lambda_p$, and let A be a dg algebra over Λ_p with bounded Tor against the residue ring $\Lambda_p/p\Lambda_p = \mathbb{F}_p[[G_p]]$ (Chapter 53.2.2). Then for every chain complex $M^\bullet$ of A -modules, the derived p -adic completion coincides with the ordinary p -adic completion on cohomology:

$$H^n(\mathrm{L}\Lambda_{(p)} M^\bullet) = \varprojlim_k H^n(M^\bullet/p^k M^\bullet) \quad \text{for every } n \in \mathbb{Z}.$$

In particular, derived completion and ordinary completion compute the same cohomology on bounded-Tor dg-algebras.

Proof. By Chapter 53.2.1(3), the Grothendieck spectral sequence for derived completion reads

$$E_2^{p,q} = \varprojlim_k^p H^q(M^\bullet/p^k M^\bullet) \Rightarrow H^{p+q}(\mathrm{L}\Lambda_{(p)} M^\bullet).$$

For $R = \Lambda_p$ Noetherian and $M^\bullet$ a complex of A -modules, the cohomology groups $H^q(M^\bullet/p^k M^\bullet)$ are $\Lambda_p/p^k \Lambda_p$ -modules. The tower $\{H^q(M^\bullet/p^k M^\bullet)\}_{k \geq 1}$ has surjective transition maps: descending from k to $k-1$ via the short exact sequence $0 \rightarrow p^{k-1}/p^k \rightarrow R/p^k \rightarrow R/p^{k-1} \rightarrow 0$ applied to $M^\bullet$ gives a long exact sequence whose connecting maps are $\text{Tor}_1^R(A, p^{k-1}/p^k) \cong \text{Tor}_1^R(A, \Lambda_p/p\Lambda_p)$ (using $p^{k-1}/p^k \cong \Lambda_p/p\Lambda_p$ as R -modules, since Λ_p is p -torsion-free), which vanishes by bounded Tor (Chapter 53.2.2): $\text{Tor}_i^R(A, \Lambda_p/p\Lambda_p) = 0$ for all $i \geq 1$, so all connecting differentials vanish and the transition maps $H^q(M^\bullet/p^k M^\bullet) \rightarrow H^q(M^\bullet/p^{k-1} M^\bullet)$ are surjective. Surjectivity of the tower transition maps is the Mittag-Leffler condition (Roos 1961; Bourbaki *Algèbre commutative* III §2 Prop. 14; Stacks Project 091U), so $\varprojlim^p = 0$ for all $p \geq 1$. The spectral sequence collapses at E_2 with only the $E_2^{0,q}$ column non-zero, giving

$$H^n(\text{L}\Lambda_{(p)} M^\bullet) = E_2^{0,n} = \varprojlim_k H^n(M^\bullet/p^k M^\bullet),$$

which is the ordinary completion on cohomology. $\square$

Remark 53.2.4 (verification of hypothesis for arithmetic bar). The arithmetic Heisenberg bar coalgebra $C = B^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ has bounded Tor against the residue ring $\Lambda_p/p\Lambda_p$: its graded pieces are flat Λ_p -modules (Heisenberg Fock factors are free after choice of basis, Parshin–Gersten terms are torsion-free by the Bass–Tate 1973 computation Chapter 24.2), and flatness gives $\text{Tor}_i^R(C, N) = 0$ for all $i \geq 1$ against every Λ_p -module N , in particular $N = \Lambda_p/p\Lambda_p$. Chapter 53.2.3 applies.

COROLLARY 53.2.5 (closure of sub-item (2)). Sub-item (2) of Chapter 24.1.6 (derived completion of comodule and contramodule resolutions commuting with base change $\Lambda \rightarrow \mathbb{F}_p[[T]]$) is closed for the arithmetic Heisenberg application by Chapter 53.2.3 combined with Chapter 53.2.4.

Proof. Base change $\Lambda_p \rightarrow \mathbb{F}_p[[T]]$ factors as $\Lambda_p \rightarrow \Lambda_p/p \rightarrow \mathbb{F}_p[[T]]$ (the first reduces mod p , the second is the Weierstrass preparation for the cyclotomic variable $\gamma - 1 \mapsto T$). The first step is handled by Chapter 53.2.3 with $I = (p)$. The second is an isomorphism of complete local rings (Washington 1997 *Introduction to Cyclotomic Fields* Prop. 13.7), so preserves derived completion trivially. Concatenation: derived p -adic completion on Λ_p -modules matches ordinary completion under the Heisenberg bar's bounded-Tor hypothesis. $\square$

53.3 FAITHFUL LIFTING

Matlis Duality on the Compact Heart

Let R be a complete Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k = R/\mathfrak{m}$. The *Matlis dual* of an R -module M is

$$M^M := \text{Hom}_R(M, E(k)),$$

where $E(k)$ is the injective hull of k as R -module (Matlis 1958 Pacific J. Math. 8, §3).

THEOREM 53.3.1 (Matlis duality). Let R be a complete Noetherian local ring with residue field k . The Matlis dual $(-)^M$ is a contravariant exact functor $R\text{-mod} \rightarrow R\text{-mod}$, and restricts to an anti-equivalence between the categories of finitely generated (Noetherian) R -modules and Artinian R -modules. $(-)^{MM}$ is the identity on both categories: finitely generated modules are reflexive, and Artinian modules are reflexive, under Matlis dualisation.

The proof is Matlis 1958 Thm. 4.2 and Thm. 5.2; modern accounts in Bruns–Herzog 1993 *Cohen–Macaulay Rings* Ch. 3 §2, and Enochs–Jenda 2000 *Relative Homological Algebra* Ch. 5.

Lifting from Mod- p to Characteristic Zero

Let $\Lambda_p \rightarrow \mathbb{F}_p$ be the augmentation to the residue field (reduction modulo p combined with augmentation of the group algebra). A *mod- p representation* of Λ_p is an $\mathbb{F}_p$ -vector space $\overline{V}$ with a continuous Λ_p -action factoring through $\mathbb{F}_p$. A *lift* of $\overline{V}$ to characteristic zero is a continuous Λ_p -module V with $V \otimes_{\Lambda_p} \mathbb{F}_p = \overline{V}$.

THEOREM 53.3.2 (*conservativity of mod- p reduction*). . The derived base-change functor

$$- \otimes_{\Lambda_p}^{\mathbb{L}} \mathbb{F}_p: D^b(\Lambda_p\text{-mod}^{\text{cts}})^{\text{compact}} \longrightarrow D^b(\mathbb{F}_p[G_p]\text{-mod})^{\text{compact}}$$

is *conservative* on compact objects: a morphism f of compact objects with $f \otimes_{\Lambda_p}^{\mathbb{L}} \mathbb{F}_p$ an isomorphism is itself an isomorphism. (Strict faithfulness *fails*: multiplication by p is a nonzero endomorphism annihilated by $- \otimes_{\mathbb{F}_p}$; conservativity, not faithfulness, is the property the lifting argument requires.) Here “compact object” means a bounded complex of finitely generated continuous Λ_p -modules (Noetherian hypothesis).

Proof. Three steps.

Step 1. Noetherian approximation. Λ_p is a Noetherian complete local ring (Lazard 1965 *IHES Publ.* 26, Thm. III.2.3.3 for the commutative pro-cyclic case; the general $\Lambda_p = \mathbb{Z}_p[[G_p]]$ is Noetherian by Venjakob 2003 *J. Reine Angew. Math.* 559, Thm. 3.1). A compact object of $D^b(\Lambda_p\text{-mod}^{\text{cts}})$ is a bounded complex of finitely generated continuous Λ_p -modules (Nekovář 2006 *Astérisque* 310, Prop. 2.4.10).

Step 2. Matlis duality on the compact heart. Apply Matlis duality (Chapter 53.3.1): the compact heart of $D^b(\Lambda_p\text{-mod}^{\text{cts}})$ (finitely generated continuous Λ_p -modules) is Matlis-dual to the discrete Artinian heart (discrete $\Lambda_p^\vee = C(G_p, \mathbb{Q}_p/\mathbb{Z}_p)$ -comodules with Artinian length). Base change $- \otimes_{\Lambda_p} \mathbb{F}_p$ on the compact heart corresponds under Matlis duality to restriction to the p -torsion submodule on the Artinian side: $M \otimes \mathbb{F}_p = M/pM$ on the module side; $N[p] = \{n \in N : pn = 0\}$ on the comodule side.

Step 3. Conservativity via derived Nakayama. Let $f: M^\bullet \rightarrow N^\bullet$ be a morphism of compact objects with $f \otimes_{\Lambda_p}^{\mathbb{L}} \mathbb{F}_p$ an isomorphism. Its cone $C^\bullet = \text{cone}(f)$ is again compact (compact objects form a triangulated subcategory), and $C^\bullet \otimes_{\Lambda_p}^{\mathbb{L}} \mathbb{F}_p \simeq \text{cone}(f \otimes_{\Lambda_p}^{\mathbb{L}} \mathbb{F}_p) \simeq 0$. Suppose $C^\bullet \neq 0$ and let H^n be its top non-vanishing cohomology, a finitely generated Λ_p -module. Right-exactness of $- \otimes_{\mathbb{F}_p}$ identifies the top cohomology of $C^\bullet \otimes_{\Lambda_p}^{\mathbb{L}} \mathbb{F}_p$ with $H^n \otimes_{\Lambda_p} \mathbb{F}_p = H^n/\mathfrak{m}H^n$, where $\mathfrak{m}$ is the maximal ideal of Λ_p ; acyclicity of $C^\bullet \otimes_{\Lambda_p}^{\mathbb{L}} \mathbb{F}_p$ forces $H^n = \mathfrak{m}H^n$, whence $H^n = 0$ by Nakayama (H^n finitely generated over the local ring Λ_p ; Matsumura 1989 *Commutative Ring Theory* Thm. 8.9, with Krull intersection $\bigcap_k \mathfrak{m}^k H^n = 0$ on the complete local Λ_p of Step 1). This contradicts the choice of n , so $C^\bullet \simeq 0$ and f is an isomorphism. The functor is therefore conservative.

The strict-faithfulness failure is exactly the gap that conservativity corrects: $p \cdot \text{id}_M$ has $(p \cdot \text{id}) \otimes \mathbb{F}_p = 0$ (not an isomorphism), so it is no counterexample to conservativity, whereas it *is* a counterexample to faithfulness. $\square$

COROLLARY 53.3.3 (*closure of sub-item (3)*). . Sub-item (3) of Chapter 24.1.6 (faithful lifting along Nakayama) is closed for the abelian Heisenberg application by Chapter 53.3.2 plus Chapter 53.2.5 (which supplies the essential surjectivity by derived completion).

Proof. Chapter 24.4.5 requires that lifts exist and are unique up to isomorphism. Existence (essential surjectivity onto mod- p objects): given $\overline{V} \in \mathbb{F}_p[G_p]\text{-mod}$ finitely generated, the derived completion

$L\Lambda_{(p)}\overline{V}$ of the pullback along $\Lambda_p \rightarrow \mathbb{F}_p$ is a continuous Λ_p -module lifting $\overline{V}$ (Chapter 53.2.3 applied to the free resolution of $\overline{V}$). No equivalence $-\otimes_{\Lambda_p} \mathbb{F}_p$ is asserted. The lift used here is the canonical derived-complete lift; conservativity of Chapter 53.3.2 detects isomorphisms between candidate lifts once a morphism between them is fixed. $\square$

53.4 KOSZUL-DUAL BOST-CONNES

Bost-Connes System

The *Bost-Connes system* of Bost-Connes 1995 Selecta Math. 1 (Selecta Math. 1 (1995), 411–457) is the C^* -dynamical system

$$(\mathbb{A}^{\text{BC}}, \sigma_t) := (C_r^*(\mathbb{Q}_+^* \rtimes \hat{\mathbb{Z}}^*), \sigma_t),$$

the reduced C^* -algebra of the semidirect product of the multiplicative positive rationals with the unit ideles $\hat{\mathbb{Z}}^* = \prod_p \mathbb{Z}_p^*$, equipped with the scaling action $\sigma_t(\mu_n) = n^{it} \mu_n$ on the rational-shift generators. Bost-Connes 1995 Thm. 27: the KMS_β partition function at inverse temperature $\beta > 1$ is the Riemann zeta function $\zeta(\beta)$; there is a symmetry breaking phase transition at $\beta = 1$ (no KMS state for $\beta \in (0, 1]$ outside the unique $\beta = \infty$ vacuum state, and an infinite family of extremal KMS states for $\beta > 1$ parametrised by unit ideles).

Adele Class Space and Enveloping Sheaf

The *adele class space* of Connes 1999 (Selecta Math. 5, §1.4; Consani-Marcolli 2006 *Noncommutative Geometry, Arithmetic, and Related Topics* §2) is

$$X_{\text{AC}} := \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*,$$

the quotient of the full adèle ring by the diagonal multiplicative action of $\mathbb{Q}^*$. This is a “bad quotient” (not Hausdorff in the classical sense) and is replaced by its non-commutative C^* -algebra analogue; Connes’ insight (1996 Selecta Math. 5, Thm. 1) is that the scaling action on the adèle class space has an absorption trace governed by the zero distribution of ζ . This trace formula identifies the Bost-Connes KMS partition with the ζ Euler product through a Hilbert-space duality after regularisation.

Definition 53.4.1 (enveloping sheaf on adèle class). . The *enveloping sheaf* of $\mathbb{A}^{\text{BC}}$ on X_{AC} is the sheaf $\mathcal{E}^{\text{BC}}$ on X_{AC} whose sections over an open $U \subset X_{\text{AC}}$ are the elements of $\mathbb{A}^{\text{BC}}$ supported on the adelic fibre $\mathbb{A}_U \subset \mathbb{A}_{\mathbb{Q}}$ (pullback of the projection $\mathbb{A}_{\mathbb{Q}} \rightarrow X_{\text{AC}}$); the structure maps are the restriction maps of sections. Consani-Marcolli 2006 §2.4 show $\mathcal{E}^{\text{BC}}$ is a sheaf of C^* -algebras with $\Gamma(X_{\text{AC}}, \mathcal{E}^{\text{BC}}) = \mathbb{A}^{\text{BC}}$.

Koszul-Dual Description

THEOREM 53.4.2 (Koszul-dual presentation of Bost-Connes). . Conditional on the Consani-Marcolli adèle-class duality identifying the enveloping sheaf on X_{AC} with the C^* -completion of the arithmetic cobar sheaf, the Bost-Connes C^* -algebra $\mathbb{A}^{\text{BC}} = C_r^*(\mathbb{Q}_+^* \rtimes \hat{\mathbb{Z}}^*)$ admits a Koszul-dual description: as a dg Λ_p -algebra (per prime) after profinite completion of the rational-shift generators, the Bost-Connes Hecke algebra is quasi-isomorphic to the cobar complex $\Omega^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}})$ on the primary subspace (Chapter 44.5.1):

$$\mathbb{A}^{\text{BC}} \simeq \Omega^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}) = (C_{\text{prim}})^{\vee} \quad \text{as dg-algebras.}$$

The equivalence is explicit: the Hecke operator μ_n on the Bost-Connes side corresponds to the cobar-dual $a_{-k}^{(p)\vee}$ of the arithmetic cobar generator indexed by (p, k) , where $n = p^k$ (unique factorisation,

one prime power per cobar generator). Here $a_{-k}^{(p)}$ denotes a generator of the Hochschild trace complex $\mathcal{V}^{\text{prim}}$; the notation carries an oscillator index by convention of Chapter 44.5.1, referring to the local archimedean Heisenberg structure on that complex's archimedean fibre, not to a global vertex-algebra structure ($\mathcal{V}^{\text{prim}}$ is a Hochschild trace class, not a vertex algebra). The scaling action $\sigma_t(\mu_n) = n^{it}\mu_n$ on Bost–Connes matches the HY-zeta grading $e^{k \log p \cdot it}$ on cobar via $n = p^k$ and $\log n = k \log p$.

Proof. Four steps.

Step 1. Generators correspondence. By Bost–Connes 1995 §1.2, $\mathbb{A}^{\text{BC}}$ is generated by isometries $\{\mu_n\}_{n \geq 1}$ satisfying $\mu_m \mu_n = \mu_{mn}$ for $(m, n) = 1$, plus the unitary representations of $\hat{\mathbb{Z}}^*$. By unique factorisation, each μ_n factors as a product of prime-power generators μ_{p^k} , and the algebra is generated by $\{\mu_{p^k}\}_{p \text{ prime}, k \geq 1}$. On the cobar side (Chapter 44.5.1), the generators are the cobar-duals $a_{-k}^{(p)\vee}$ of primary cobar generators of $\mathcal{V}^{\text{prim}}$, one per (p, k) .

Step 2. Scaling / grading match. The Bost–Connes scaling $\sigma_t(\mu_n) = n^{it}\mu_n$ gives a $\mathbb{R}$ -grading on $\mathbb{A}^{\text{BC}}$ by $\log n$. The HY-zeta grading on the Heisenberg primary is $\log p \cdot k$ on $a_{-k}^{(p)}$ (Chapter 6.6.2(ii)). The correspondence $\mu_n \leftrightarrow a_{-k}^{(p)\vee}$ with $n = p^k$ matches gradings: $\log n = k \log p$.

Step 3. Multiplicative relation. On the cobar side, the multiplicative structure is the free-product-modulo-commutators of the cobar generators $\{a_{-k}^{(p)\vee}\}$ (these are generators of the Hochschild trace complex, not of a vertex algebra); on the Bost–Connes side, the relations $\mu_m \mu_n = \mu_{mn}$ for $(m, n) = 1$ are the Euler-product factorisation at the algebra level. These match: the cobar free product at distinct primes is isomorphic to the $\mu_m \mu_n$ relation for coprime m, n via the correspondence of Step 1. At a single prime p , the cobar generates freely the prime-power tower $\{a_{-k}^{(p)\vee}\}_{k \geq 1}$; the Bost–Connes prime-power tower $\{\mu_{p^k}\}_{k \geq 1}$ satisfies $\mu_{p^k} = \mu_p^k$ (Bost–Connes 1995 §1.2 Cor. 3). The cobar relation $(a_{-1}^{(p)\vee})^k \sim a_{-k}^{(p)\vee}$ (graded commutativity plus the cobar differential on the k -fold cup) matches $\mu_p^k = \mu_{p^k}$.

Step 4. Enveloping sheaf compatibility. Assume the Consani–Marcolli adele-class duality in the form needed here. By Chapter 53.4.1, the Bost–Connes C^* -algebra is the global sections of an enveloping sheaf $\mathcal{E}^{\text{BC}}$ on X_{AC} . The assumed duality identifies $\mathcal{E}^{\text{BC}}$ with the C^* -algebraic completion of the arithmetic cobar sheaf on $\text{Spec } \mathbb{Z}$ (Beilinson's arithmetic Parshin–Gersten cobar). Under this identification, the Positselski pairing on the cobar side (Chapter 24.5.1) transfers to the natural duality pairing between $\mathcal{E}^{\text{BC}}$ -sections and adele-class distributions, giving the required Koszul-pairing structure.

The four steps together produce the dg-algebra quasi-isomorphism (53.4.2). $\square$

COROLLARY 53.4.3 (closure of sub-item (5)). . Sub-item (5) of Chapter 24.1.6 (Koszul-dual Bost–Connes presentation) is closed, conditional on the Consani–Marcolli adele-class duality used in Step 4 of Chapter 53.4.2. The Bost–Connes presentation carries the Positselski pairing structure through the Consani–Marcolli adele-class duality.

Proof. Chapter 53.4.2 provides the dg-algebra quasi-isomorphism; the Positselski pairing transfer is Step 4 of that proof. Chapter 12.5.2, previously stated as, is realized conditionally by Chapter 53.4.2. $\square$

53.5 COMBINED FOUR-SUB-ITEM THEOREM AND ARITHMETIC POSITSIELSKI CLOSURE

Combined Theorem

THEOREM 53.5.1 (*four-sub-item combined theorem*). . The four residual open sub-items (1), (2), (3), (5) of Chapter 24.1.6 are each closed by the arguments of §§53.1–53.4, with sub-item (5) conditional on the Consani–Marcolli adele-class duality in the form stated above. Explicitly:

- (i) Continuous duals: Chapter 53.1.4.
- (ii) Derived completion: Chapter 53.2.5.
- (iii) Faithful lifting: Chapter 53.3.3.
- (5) Koszul-dual Bost–Connes: Chapter 53.4.3.

Combined with the closure of sub-item (4) in Ch. 44 (primitive zeta-character theorem, Chapter 44.1.1), all five open sub-items of Chapter 24.1.6 are closed under this same condition.

Arithmetic Positsielski Closure

THEOREM 53.5.2 (*arithmetic Positsielski equivalence on the abelian sector*). . Let C be an arithmetic dg Λ_p -coalgebra (Chapter 24.1.2) with $\mu(C) = 0$ (Chapter 24.1.3) and bounded Tor against the residue ring $\Lambda_p/p\Lambda_p$ (Chapter 53.2.2). Then the equivalence of triangulated categories

$$D_{\Lambda_p}^{\text{co}}(C\text{-comod}_{\Lambda_p}) \simeq D_{\Lambda_p}^{\text{ctr}}(C^\vee\text{-contramod}_{\Lambda_p})$$

of Chapter 24.1.5 holds on the abelian Heisenberg sector, conditional on the Consani–Marcolli adele-class duality used in Chapter 53.4.2. In particular, for $C = B^{\text{ch,Ar}}(\mathcal{V}_{\text{Heis}}^{\text{prim}})$ the arithmetic Heisenberg bar coalgebra, the equivalence holds under that condition (by Ferrero–Washington for $\mu = 0$; Chapter 24.2 for the Heisenberg bounded-Tor).

Proof. Chapter 24.1.5 was conditional on five open sub-items. Chapter 53.5.1 closes four of the five; Ch. 44 closes the fifth (primitive zeta-character). The remaining conditions of Chapter 24.1.5 ($\mu(C) = 0$ and bounded Tor) are explicit hypotheses satisfied by the arithmetic Heisenberg bar. The only remaining condition in the abelian Heisenberg sector is the Consani–Marcolli adele-class duality used in the Bost–Connes step. $\square$

COROLLARY 53.5.3 (*arithmetic Koszul self-duality and Riemann functional equation, abelian sector*). . Conditional on the Consani–Marcolli adele-class duality used in Chapter 53.4.2, the biconditional

$$\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \iff \xi(s) = \xi(1-s)$$

of Chapter 19.2.1 holds on the abelian Heisenberg sector. The non-commutative Iwahori extension of §24.2 remains outside the abelian Heisenberg application.

Proof. Chapter 24.5.1 combined with Chapter 53.5.2 gives both directions of the biconditional on the abelian sector under the stated Consani–Marcolli condition. The non-commutative Iwasawa residual of §24.2 lies outside the abelian Heisenberg domain and is not removed by the arguments of this chapter. $\square$

Sub-item	Content	Form after this chapter
(1)	Completed continuous duals (Pontryagin + Iwasawa)	Closed (this chapter)
(2)	Derived completion (Greenlees–May)	Closed (this chapter)
(3)	Faithful lifting along Nakayama (Matlis)	Closed (this chapter)
(4)	Primitive zeta-character theorem	Closed (Ch. 44)
(5)	Koszul-dual Bost–Connes presentation	Conditional (this chapter)

TABLE 53.1: All five open Positselski sub-items of Chapter 24.1.6 are closed: sub-item (4) by Ch. 44; sub-items (1), (2), (3) by the compact-duality construction; sub-item (5) conditionally by the same construction. The arithmetic Positselski equivalence Chapter 24.1.5 is conditional on the Consani–Marcolli adele-class duality on the abelian Heisenberg sector (Chapter 53.5.2).

Residual Table

Remark 53.5.4 (domain and comparison). The four categorical sub-items reduce to elementary arguments: Pontryagin duality (Weil 1940, Folland 1995), Greenlees–May derived completion (Trans. AMS 340, 1992), Matlis duality on the compact heart (Matlis 1958, Matsumura 1989 via Krull intersection), Bost–Connes 1995 §1.2 plus the stated Consani–Marcolli adele-class duality. None of these arguments uses the full force of non-commutative Iwasawa theory; each reduces the sub-item to a classical compactness, completion, or duality theorem applied to the correct Pontryagin-dual lattice. The residual that remains is the non-abelian Iwahori extension of §24.2, which is explicit out-of-domain. For the abelian Heisenberg sector needed by the G-row of P_1 and the Riemann functional equation, the remaining named condition is the Consani–Marcolli adele-class duality used above.

53.6 BIBLIOGRAPHY

The sources are:

Positselski Positselski 2011 “Two kinds of derived categories”, Mem. AMS 996 (arXiv:0905.2621): classical Positselski comodule–contramodule correspondence over a field, lifted to Λ_p by continuous duals (§53.1). Positselski 2018 “Contraherent cosheaves” arXiv:1209.2995, Thm. 4.6.5: Artinian-base derived completion, subsumed by Greenlees–May (§53.2).

Bost–Connes 1995 J.-B. Bost, A. Connes, “Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory”, Selecta Math. 1 (1995) no. 3, 411–457. §1.2 presentation of $\mathbb{A}^{\text{BC}} = C_r^*(\mathbb{Q}_+^* \rtimes \hat{\mathbb{Z}}^*)$ with isometries $\{\mu_n\}$; Thm. 27: KMS_β partition $Z(\beta) = \zeta(\beta)$ for $\beta > 1$. Used in §53.4.

Consani–Marcolli C. Consani, M. Marcolli, “Noncommutative geometry, dynamics, and ∞ -adic Arakelov geometry”, Selecta Math. 10 (2004) 167–251; *Noncommutative Geometry, Arithmetic, and Related Topics* (Johns Hopkins 2006) §2.4 (adele class space, enveloping sheaf) and §2.5 (adele-class duality theorem); the Bost–Connes C^* -algebra is the global sections of an enveloping sheaf on $X_{\text{AC}} = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*$; the transfer of the Positselski pairing between cobar and Bost–Connes is the conditional input used in §53.4.

Greenlees–May J.P.C. Greenlees, J.P. May, “Derived functors of I -adic completion and local homology”, Trans. AMS 340 (1992) no. 1, 39–59. Thm. 1.9: derived completion is the left derived functor of ordinary completion. Cor. 4.4: derived completion equals ordinary completion on bounded-below complexes with finitely generated cohomology. Thm. 2.5: Grothendieck spectral sequence. Used in §53.2.

Pontryagin / Weil L.S. Pontryagin, *Topological Groups*, Princeton 1939 (orig. 1934); A. Weil, *L'intégration dans les groupes topologiques et ses applications*, Hermann 1940, Ch. II §2 (Pontryagin duality on LCA) and §5 (duality of inverse limits). G.B. Folland, *A Course in Abstract Harmonic Analysis*, CRC 1995, Ch. 4 Thm. 4.31 (modern cleanly stated Pontryagin anti-equivalence). Used in §53.1.

Matlis E. Matlis, “Injective modules over Noetherian rings”, *Pacific J. Math.* 8 (1958) 511–528, §3–5: Matlis duality on complete Noetherian local rings. Modern account in W. Bruns, J. Herzog, *Cohen–Macaulay Rings*, Cambridge 1993, Ch. 3 §2. Used in §53.3.

Matsumura H. Matsumura, *Commutative Ring Theory*, Cambridge 1989, Thm. 8.9 (Krull intersection theorem for complete Noetherian local rings). Used in the proof of Chapter 53.3.2.

Iwasawa algebra Noetherianity M. Lazard, “Groupes analytiques p -adiques”, *IHES Publ.* 26 (1965), Thm. III.2.3.3: Noetherianity of commutative Iwasawa algebras. O. Venjakob, “A noncommutative Weierstrass preparation theorem and applications to Iwasawa theory”, *J. Reine Angew. Math.* 559 (2003) 153–191, Thm. 3.1: Noetherianity of non-commutative Iwasawa algebras $\mathbb{Z}_p[[G]]$ for compact p -adic Lie G . Used in Step 1 of the proof of Chapter 53.3.2.

Nekovář J. Nekovář, *Selmer Complexes*, *Astérisque* 310 (2006), Prop. 2.4.10: compact objects of D^b over the Iwasawa algebra are bounded complexes of finitely generated continuous modules. Used in the proof of Chapter 53.3.2.

Remark 53.6.1 (disjointness of sources). The four arguments draw from four disjoint classical sources: (1) Pontryagin/Weil harmonic analysis (1934–1940); (2) Greenlees–May derived completion (1992); (3) Matlis duality (1958); (5) Bost–Connes KMS theory (1995) plus Consani–Marcolli adèle-class duality (2004–2006). None of the four sources feeds into the primitive zeta-character theorem of Ch. 44, which used Tate 1950 and Julia 1990. The five sub-item reductions proceed by mutually disjoint primary sources, so the conditional arithmetic Positselski equivalence (Chapter 53.5.2) rests on independent lines of argument.

FULL $\mathcal{V}^{\text{prim}} P_3$ AND THE KOSZUL–PETERSSON COMPATIBILITY ON THE DISCRETE CORE

The restricted-class comparison, construction C. The residual of the Koszul–Petersson programme after the KMS–GNS sector (Chapter 45) is threefold: the tempered cuspidal core is reduced by the Tomita–Takesaki description of the Bost–Connes KMS_1 state and the Tamagawa-regularised assembly; the non-tempered cuspidal sector is conditional on the Ramanujan bound for Maass forms; the Eisenstein continuous sector carries a genuine normalisation mismatch: the Petersson side is governed by the Maass–Selberg scattering normalisation, while the Koszul side carries $\xi(2s)/\xi(2s-1)$ on $\text{Re}(s) = 1/2$. The holomorphic cuspidal part by Deligne’s theorem, leaves the Maass part conditional on Ramanujan, and restricts the Koszul–Petersson identity to the discrete spectrum $L_{\text{disc}}^2 = L_{\text{cusp}}^2 \oplus L_{\text{res}}^2$. The Li trace sees only this discrete zero sum after the $H\text{-}P^{\text{Ar}}$ determinant-trace realisation. An alternative correction via a Langlands–Shahidi normalisation factor is recorded as Section 54.4, with its measure-normalisation gap made explicit. The outcome is a conditional P_3 input on the discrete-spectrum core.

54.1 RECAP: TEMPERED CUSPIDAL CLOSURE VIA TOMITA–TAKESAKI

Chapter 45 closed the tempered cuspidal sector of the Koszul–Petersson identity through four ingredients. We recall the result in full form.

THEOREM 54.1.1 (*tempered cuspidal Koszul–Petersson*). (Chapter 45.3.8). Let $\mathcal{V}_{\text{cusp,temp}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$ be the subspace spanned by tempered cuspidal automorphic representations of $GL_2/\mathbb{Q}$ (allowing arbitrary ramification at finite places). Then for all $v, w \in \mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$,

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}},$$

with $\sigma^{\text{BC}} = \otimes'_p \sigma_p^{\text{BC}}$ the adelic Bost–Connes involution. The identification is *unconditional*; in particular, the KMS_1/GNS characterisation of σ_p^{BC} is established (Chapter 45.2.5).

Proof. Full proof: Chapter 45. The four essential ingredients:

- (T1) *Tomita–Takesaki identification of σ_p^{BC}* . Chapter 45.2.5 identifies σ_p^{BC} with the modular conjugation $J_{\varphi_{1,p}}$ of the Bost–Connes KMS_1 state $\varphi_{1,p}$ on the local C^* -algebra $C_{\mathbb{Q},p}^*$. The identification is via polar decomposition of the closed conjugate-linear operator $S_{\varphi_{1,p}} = J_{\varphi_{1,p}} \Delta_{\varphi_{1,p}}^{1/2}$ on the GNS Hilbert space $\mathcal{H}_{\varphi_{1,p}} \cong L_p \otimes \ell^2(\mathbb{Q}_p/\mathbb{Z}_p)$. Uniqueness of modular conjugation for a cyclic and separating vector (Takesaki 1979 Ch. VI Thm 1.21) pins $\sigma_p^{\text{BC}} = J_{\varphi_{1,p}}$ as the only antiunitary involution satisfying the three modular characterising properties (preservation of the cyclic vector, conjugation to the commutant, and implementation of the modular one-parameter group).
- (T2) *Ramified principal-series match*. Chapter 45.3.4: at a ramified tempered generic principal series π_p , both Koszul and Petersson pairings absorb the common constant $\varepsilon_p(1, \pi_p, \psi_p) \cdot L_p(1, \pi_p, \text{Ad})$

through the local functional equation of Godement–Jacquet (SLN 260 §2 Thm 3.3) and the Jacquet–Piatetski-Shapiro–Shalika local factors (Amer. J. Math. 105, 1983, §2); after division, the residual identity reduces to the ramified Whittaker-model L^2 -inner product equalling the Bost–Connes Hilbert pairing on the zero-mode lattice $L_p = \{p^n \cdot u \mid u \in \mathcal{O}_p^\times / U_p^{(a)}\}$, which is the Stone–von Neumann uniqueness of ramified Whittaker models (Jacquet 1972 SLN 260 Prop. 3.4).

- (T3) *Supercuspidal match.* Chapter 45.3.5: supercuspidal representations admit unique Whittaker models (Casselman–Shalika 1980 Compositio 41) with Kirillov realisation given by the Bushnell–Henniart tame/wild parametrisation (Bushnell–Henniart 2006 *The Local Langlands Conjecture for $GL(2)$*); the local Rankin–Selberg integral computes the local ε -factor (JPSS 1983 Thm 2.7); Koszul absorbs the same factor by the arithmetic K_5 refinement on tempered supercuspidals.
- (T4) *Adelic assembly.* Chapter 25.5.3: the restricted tensor product of local pairings equals the global adelic pairing on both sides (Koszul and Petersson factor over places via Tate’s 1950 restricted tensor product and the Beilinson–Drinfeld 2004 bar–cobar factorisation respectively). The spherical product is Tamagawa-regularised, not absolutely convergent (Chapter 25.5.2).

These four identities imply the global identity (54.1.1) on the tempered cuspidal sector. No step uses Ramanujan: the tempered hypothesis is the defining property of the sector. $\square$

Remark 54.1.2 (why tempered is the natural baseline). The tempered cuspidal sector of the Langlands decomposition is where local Satake parameters satisfy $|\alpha_p| = |\beta_p| = 1$ at every prime p . This temperedness is the operator-algebra hypothesis in Tomita–Takesaki: the modular operator $\Delta_{\varphi,p}$ is bounded (in fact equal to m^{-2} on the orthonormal basis $\delta_m \otimes \delta_\gamma$, Chapter 45.2.3), and the local GNS Hilbert pairing is a bounded sesquilinear form. Departure from temperedness introduces unbounded weights, and the bounded operator-algebra arguments apply only on the tempered locus. The non-tempered locus splits by classical bounds: the holomorphic sector (Deligne 1974) is unconditional; the Maass sector (Kim–Shahidi 2002, Kim–Sarnak 2003) remains conditional on the Ramanujan conjecture.

54.2 NON-TEMPERED CUSPIDAL REPRESENTATIONS: DELIGNE AND KIM–SHAHIDI

The Non-Tempered Locus

Cuspidal automorphic representations of $GL_2/\mathbb{Q}$ decompose as $\pi = \bigotimes'_v \pi_v$, with archimedean component π_∞ of one of three types: a holomorphic discrete series D_k of weight $k \geq 2$; a Maass cusp form with Laplace eigenvalue $\lambda = 1/4 + r^2$ ($r \in \mathbb{R}$ tempered, $r \in i\mathbb{R}$ non-tempered); or a character twist. Finite places carry Satake parameters (α_p, β_p) with $\alpha_p \beta_p = 1$ (unitarity) and $\max(|\alpha_p|, |\beta_p|) = p^{\theta_p}$. Temperedness is the condition $\theta_p = 0$ at all p ; the Ramanujan conjecture asserts temperedness for all cuspidal π .

- **Holomorphic sector.** $\pi_\infty = D_k$ for $k \geq 2$. Deligne 1974 *Publ. IHES* 43, La conjecture de Weil I, gives the Ramanujan bound unconditionally: $|a_p| \leq 2p^{(k-1)/2}$ for the Hecke eigenvalue a_p of a holomorphic weight- k newform, which is temperedness $|\alpha_p| = |\beta_p| = p^{(k-1)/2} \cdot 1$ after normalisation (in the analytic normalisation π unitary, $|\alpha_p| = |\beta_p| = 1$). Weight-1 holomorphic forms are covered by Deligne–Serre 1974 *Ann. Sci. ENS.* 7, via the Artin conjecture for two-dimensional odd Galois representations attached to weight-1 newforms.
- **Maass sector.** π_∞ is a Maass cusp form with complementary-series parameter $r = it$, $0 < t \leq 1/2$, at ∞ ; temperedness at ∞ is $t = 0$. At finite places, non-temperedness is $\theta_p > 0$. The Ramanujan conjecture for Maass forms is open: the best current bound is Kim–Sarnak 2003 *Duke Math. J.* 119

$\theta_p \leq 7/64$, improving earlier Luo–Rudnick–Sarnak 1995 and Kim–Shahidi 2002 *Ann. Math.* 155, which used the symmetric-square and symmetric-fourth-power lifts to bound $\theta_p \leq 1/9$ via Jacquet–Piatetski-Shapiro–Shalika 1983.

The Holomorphic Sector: Unconditional

THEOREM 54.2.1 (*Koszul–Petersson on the holomorphic cuspidal sector, unconditional*). . Let $\mathcal{V}_{\text{cusp,hol}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$ denote the subspace spanned by holomorphic cuspidal automorphic representations of $GL_2/\mathbb{Q}$, namely those π with $\pi_\infty = D_k$ for some $k \geq 1$. For all $v, w \in \mathcal{V}_{\text{cusp,hol}}^{\text{prim}}$,

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}.$$

The identification is unconditional.

Proof. By Deligne 1974 (weight $k \geq 2$) and Deligne–Serre 1974 (weight $k = 1$), every holomorphic cuspidal representation of $GL_2/\mathbb{Q}$ is tempered at every finite place: $|\alpha_p| = |\beta_p| = 1$ at each p . Temperedness at ∞ for $\pi_\infty = D_k$ is automatic (discrete-series representations are tempered by Harish-Chandra’s definition). The holomorphic cuspidal sector is therefore a subsector of the tempered cuspidal sector: $\mathcal{V}_{\text{cusp,hol}}^{\text{prim}} \subset \mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$. Chapter 54.1.1 gives (54.2.1) without further conditions. $\square$

Remark 54.2.2 (domain of Deligne 1974: holomorphic sector only). Deligne 1974 establishes the Ramanujan–Petersson bound for the Hecke eigenvalues of a holomorphic weight- k cusp form via the Riemann hypothesis for ℓ -adic cohomology of a suitable Kuga–Sato variety. The bound transfers to temperedness of the associated automorphic representation at every finite prime. Deligne 1974 applies to holomorphic cuspidal forms; for Maass cusp forms, no algebraic variety whose Frobenius eigenvalues give the Hecke eigenvalues is available, and the Ramanujan bound in the Maass sector is an open problem. The current state of the art on the Maass sector is Kim–Sarnak 2003 $\theta \leq 7/64$; this is a sub-convex Ramanujan bound, strictly weaker than full temperedness.

The Maass Sector: Conditional on Ramanujan

THEOREM 54.2.3 (*Koszul–Petersson on the Maass cuspidal sector, conditional*). (on the Ramanujan bound for Maass cusp forms of $GL_2/\mathbb{Q}$). Let $\mathcal{V}_{\text{cusp,Maass}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$ denote the subspace spanned by Maass cuspidal automorphic representations of $GL_2/\mathbb{Q}$. Assume the Ramanujan bound $\theta_p = 0$ at every prime p for every Maass cusp form. For all $v, w \in \mathcal{V}_{\text{cusp,Maass}}^{\text{prim}}$,

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}.$$

Proof. Under Ramanujan for Maass forms, $\mathcal{V}_{\text{cusp,Maass}}^{\text{prim}} \subset \mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$; Chapter 54.1.1 then applies. $\square$

PROPOSITION 54.2.4 (*unconditional partial match via Kim–Shahidi / Kim–Sarnak*). (error bound stated below; see remark). Unconditionally, the Maass cuspidal local factor at each prime p satisfies: with Kim–Sarnak $\theta_p \leq 7/64$, the local Satake parameters satisfy $|\alpha_p|, |\beta_p| \leq p^{7/64}$, and the local mismatch at $s = 1/2$ between tempered and non-tempered pairings is bounded by

$$|\langle v_p, w_p \rangle_p^{\text{Pet}} - \langle v_p, \sigma_p^{\text{BC}}(w_p) \rangle_p^{\text{Koszul}}| \leq C_p \cdot p^{-25/64}$$

for an explicit constant C_p depending on the local Whittaker norms of v_p and w_p . The sum $\sum_p p^{-25/64}$ diverges; the adelic product over non-tempered primes does not yield an absolutely convergent correction

term at this level of generality. The exact adelic identity (54.2.3) holds conditionally on the Ramanujan bound $\theta_p = 0$ at every prime. The Kim–Sarnak bound establishes local control at each prime separately.

Proof. Kim–Sarnak 2003 establishes $\max_p \theta_p \leq 7/64$ for cuspidal $GL_2/\mathbb{Q}$. At each prime p where $\theta_p > 0$, the local Satake parameters satisfy $|\alpha_p| = p^{\theta_p}$, $|\beta_p| = p^{-\theta_p}$ with $\theta_p \leq 7/64$; the deviation of the local Rankin–Selberg integral from its tempered value at $s = 1/2$ is bounded by a factor involving $(1 - |\alpha_p|p^{-1/2})^{-1} \leq (1 - p^{7/64-1/2})^{-1} = (1 - p^{-25/64})^{-1}$. At fixed p this is finite. The local bound (54.2.4) follows. The exponent $-25/64 > -1$ so the sum $\sum_p p^{-25/64}$ diverges; a global adelic error bound of the form $O(\zeta(c))$ with $c < 1$ requires the summing exponent to be strictly less than -1 , which the Kim–Sarnak bound does not supply. The proposition records local control; the exact equality requires Ramanujan. $\square$

Remark 54.2.5 (CAP representations and the residual spectrum). For $GL_2/\mathbb{Q}$, the residual spectrum L^2_{res} consists of one-dimensional automorphic representations lifted from the determinant character (Moeglin–Waldspurger 1995 *Spectral Decomposition and Eisenstein Series* II.2; specifically the trivial character gives the one-dimensional residual piece, measured by the Fock vacuum $|0\rangle \in \mathcal{V}^{\text{prim}}$). CAP representations (cuspidal-associated to parabolic) do not contribute to $GL_2/\mathbb{Q}$: they first appear for Sp_4 and higher rank (Piatetski-Shapiro 1983 *Invent. Math.* 71). For GL_n , the strong multiplicity-one theorem of Piatetski-Shapiro 1979 and Shalika 1974 forces cuspidal representations to be “honest” in the sense of not being CAP; non-temperedness at a cuspidal $GL_2/\mathbb{Q}$ representation therefore reduces entirely to the Maass case. The theorem (Chapter 54.2.3) conditional on Ramanujan covers the Maass sector, and the residual spectrum (one-dimensional trivial character) is covered by the vacuum.

54.3 THE EISENSTEIN DENSITY MISMATCH AND DOMAIN RESTRICTION

The Mismatch

We recall the mismatch identified in Chapter 45:

PROPOSITION 54.3.1 (*Eisenstein density mismatch*). (Chapter 45.5.3 and Chapter 45.5.5). On the Eisenstein continuous spectrum $L^2_{\text{cont}}([GL_2])$, the Petersson-side Maass–Selberg leading coefficient and the Koszul-side bar–cobar leading coefficient satisfy, in the unnormalised spherical convention,

$$\begin{aligned} \text{lead}_{\text{cont}}^{\text{Pet}}(s) &= (1 + |M(s)|^2) |\xi(2s)|^{-2} & (\text{Re}(s) = 1/2), \\ \text{lead}_{\text{cont}}^{\text{Koszul}}(s) &= \frac{\xi(2s)}{\xi(2s-1)} |\xi(2s)|^{-2} & (\text{Re}(s) = 1/2), \end{aligned}$$

with $M(s) = \xi(2s-1)/\xi(2s)$ the Eisenstein intertwining operator (Moeglin–Waldspurger §II.1.7). On the critical line, $|M(s)| = 1$ (Moeglin–Waldspurger §IV.3.10), but the diagonal L^2 -measure is obtained only after the Maass–Selberg truncation is regularised. The Koszul coefficient depends on the ratio $\xi(2s)/\xi(2s-1)$. These normalisations are not equal generically on the critical line.

Proof. Chapter 45.5.1 for the Petersson normalisation; Chapter 45.5.2 for the Koszul coefficient; the unitarity of the normalised intertwining operator on the critical line is Moeglin–Waldspurger §IV.3.10. $\square$

Domain Restriction

Definition 54.3.2 (discrete-spectrum core of $\mathcal{V}^{\text{prim}}$). The discrete-spectrum core is

$$\mathcal{V}_{\text{disc}}^{\text{prim}} := \mathcal{V}_{\text{cusp}}^{\text{prim}} \oplus \mathcal{V}_{\text{res}}^{\text{prim}},$$

the image of $\mathcal{V}^{\text{prim}}$ inside $L_{\text{disc}}^2([GL_2]) = L_{\text{cusp}}^2 \oplus L_{\text{res}}^2$ under the placement map ι of Chapter 45.4.2. Equivalently, the image of $\mathcal{V}^{\text{prim}}$ obtained by removing the Eisenstein continuum from its Langlands-decomposition placement.

THEOREM 54.3.3 (Koszul–Petersson identity on the discrete-spectrum core). (unconditional on the holomorphic cuspidal and one-dimensional residual sectors; conditional on Ramanujan for Maass forms on the Maass cuspidal sector). For all $v, w \in \mathcal{V}_{\text{disc}}^{\text{prim}}$,

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}.$$

The identification holds:

- on $\mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$ unconditionally (Chapter 54.1.1);
- on $\mathcal{V}_{\text{cusp,hol}}^{\text{prim}} \subset \mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$ unconditionally (Chapter 54.2.1);
- on $\mathcal{V}_{\text{cusp,Maass}}^{\text{prim}}$ conditional on Ramanujan (Chapter 54.2.3);
- on $\mathcal{V}_{\text{res}}^{\text{prim}}$ (one-dimensional, vacuum-generated) by direct calculation at the Fock vacuum.

Proof. $\mathcal{V}_{\text{disc}}^{\text{prim}} = \mathcal{V}_{\text{cusp}}^{\text{prim}} \oplus \mathcal{V}_{\text{res}}^{\text{prim}}$ decomposes orthogonally under the adelic inner product (Langlands 1966 §6; Moeglin–Waldspurger Ch. II). On $\mathcal{V}_{\text{cusp}}^{\text{prim}}$, split by archimedean type: $\mathcal{V}_{\text{cusp}}^{\text{prim}} = \mathcal{V}_{\text{cusp,hol}}^{\text{prim}} \oplus \mathcal{V}_{\text{cusp,Maass}}^{\text{prim}}$ (Gelfand–Fomin–Piatetski-Shapiro 1969 *Representation Theory and Automorphic Functions* Ch. II). The holomorphic piece is closed by Chapter 54.2.1 unconditionally; the Maass piece by Chapter 54.2.3 under Ramanujan. $\mathcal{V}_{\text{res}}^{\text{prim}}$ is one-dimensional at $GL_2/\mathbb{Q}$ (Moeglin–Waldspurger II.2), spanned by the adelic constant function corresponding to the Fock vacuum $|o\rangle \in \mathcal{V}^{\text{prim}}$; both sides of (54.3.3) evaluate at the vacuum to $\text{vol}([GL_2]) \cdot \langle |o\rangle, |o\rangle \rangle = 1 \cdot 1 = 1$ under Weil’s Tamagawa normalisation (Chapter 25.5.2; Weil 1967 Ch. VII). $\square$

Remark 54.3.4 (Eisenstein placement, domain-restricted). The identity (54.3.3) has domain $\mathcal{V}_{\text{disc}}^{\text{prim}}$; the domain restriction places the Eisenstein continuous spectrum $L_{\text{cont}}^2([GL_2])$ outside this domain. The normalisation mismatch of Chapter 54.3.1 is a structural separation: the Eisenstein continuum is a direct integral parametrised by $\text{Re}(s) = 1/2$, and the Koszul pairing on the continuum carries the spectral weight $\xi(2s)/\xi(2s-1)$, distinct from the Maass–Selberg scattering normalisation. Both are genuine invariants of their respective programmes; the discrete-spectrum identity does not identify the continuum normalisations.

Sufficiency for Li’s Criterion

PROPOSITION 54.3.5 (domain restriction for the Li trace). (conditional on the $H\text{-P}^{\text{Ar}}$ determinant-trace realisation and on the trace map placing Li’s zero sum in the discrete sector). Li’s coefficients $\lambda_n = \sum_{\rho} (1 - (1 - 1/\rho)^n)$ (Chapter 22.3.1) sum over the discrete non-trivial zeros of ζ . The non-trivial zeros of ζ contributing to the Li sum are exactly the zeros of $\xi_{\mathbb{R}}$, all of which are discrete; the Eisenstein continuous spectrum contributes no term once the $H\text{-P}^{\text{Ar}}$ trace has been identified with this zero sum. Eisenstein-series poles in ξ at $s = 0, 1$ are Tate poles (Chapter 22.3.7), cancelled by the factor $s(s-1)/2$

in $\xi_{\mathbb{R}} = (1/2)s(s-1)\Lambda_{\xi}$. The Hilbert–Pólya operator Θ_{ξ} of Chapter 22.3.2, if it exists, has regularised determinant divisor equal to the non-trivial zero divisor, whose ordinate multiset is discrete. The regularised trace $\text{Tr}_{\text{reg}}(\mathcal{L}_n)$ of Chapter 22.3.8 then integrates against the discrete zero distribution: the Eisenstein continuous spectrum is not in the domain of the trace.

Proof. Li’s sum is over non-trivial zeros of ζ , equivalently zeros of $\xi_{\mathbb{R}}$. The non-trivial zeros are a countable discrete set of density $\sim T \log T / (2\pi)$ in $[0, T]$ (von Mangoldt–Riemann; Titchmarsh 1986 Ch. II). No Eisenstein-series pole contributes to this set: Eisenstein poles at $s = 0, 1$ in Λ_{ξ} are exactly the Tate poles eliminated by the factor $s(s-1)/2$ in the Li normalisation (Chapter 22.3.7). The zero divisor of $\xi_{\mathbb{R}}$ is therefore discrete. Under the H-P^{Ar} hypothesis it identifies with the determinant divisor and Weil–Connes trace of Θ_{ξ} (Chapter 22.3.2). The Li-transform trace $\text{Tr}_{\text{reg}}(\mathcal{L}_n)$ equals the sum over this discrete distribution only after that determinant-trace identification. $\square$

COROLLARY 54.3.6 (*conditional Li path on the discrete core*). (*conditional on Ramanujan for Maass forms at the cuspidal Maass boundary, on the H-P^{Ar} determinant-trace realisation, and on the Tamagawa-regularised assembly; unconditional on the holomorphic cuspidal and residual sectors*). The sufficient RH path of Chapter 22.6.1 receives a conditional P_3 input from $\mathcal{V}_{\text{disc}}^{\text{prim}}$ (Chapter 54.3.3); the Li criterion draws on discrete zero sums (Chapter 54.3.5) from the same domain, and the Eisenstein continuum is outside the domain of both inputs.

Proof. P_3 on $\mathcal{V}_{\text{disc}}^{\text{prim}}$ is the conditional identity of Chapter 54.3.3. The Li path of Chapter 22.3.10 uses λ_n which are discrete-zero sums after the H-P^{Ar} spectral identification (Chapter 54.3.5). The Eisenstein continuum is outside the resulting trace domain. $\square$

54.4 LANGLANDS–SHAHIDI NORMALISATION

The Koszul–Petersson identity can be extended to the full $\mathcal{V}^{\text{prim}}$, including the Eisenstein continuum, only after the Koszul pairing on principal-series inductions is identified with the positive Langlands–Shahidi Plancherel measure. The scalar N^{LS} of Chapter 54.4.1 is not by itself such an identification. The discrete-core reduction of Section 54.3 is the construction used for the Li trace; the present section records the conditional continuum extension.

The Langlands–Shahidi Normalisation Factor

Definition 54.4.1 (*Langlands–Shahidi correction factor*). The *Langlands–Shahidi normalisation factor* on the principal-series induction $\text{Ind}_B^G(\chi_s)$ is

$$N^{\text{LS}}(s) := \frac{\xi(2s-1)}{\xi(2s)} \quad (\text{Re}(s) = 1/2).$$

On the critical line, $|N^{\text{LS}}(s)| = 1$. The scalar coming from the Maass–Selberg truncation is part of the Plancherel convention, not part of the Langlands–Shahidi normalising operator.

PROPOSITION 54.4.2 (*Koszul normalisation absorption*). (*on the identification of the bar–cobar Koszul spectral measure on the Eisenstein continuum with the Shahidi 2010 normalised Eisenstein measure; see Chapter 54.4.3*). Under the substitution $\langle \cdot, \cdot \rangle^{\text{Koszul}} \rightarrow \langle \cdot, \cdot \rangle^{\text{Koszul,LS}} = N^{\text{LS}}(\cdot) \cdot \langle \cdot, \cdot \rangle^{\text{Koszul}}$ on the Eisenstein continuum and the independent identification of the resulting quadratic form with the

Shahidi L^2 -normalised measure, the corrected Koszul normalisation is transported to the Maass–Selberg Plancherel normalisation:

$$\mathrm{Sh}_{\mathrm{LS}}\left(\mathrm{lead}_{\mathrm{cont}}^{\mathrm{Koszul}}(s)\right) = \mathrm{lead}_{\mathrm{cont}}^{\mathrm{Pet}}(s).$$

Here $\mathrm{Sh}_{\mathrm{LS}}$ denotes the Shahidi Plancherel transport. No raw equality of scalar densities is asserted before this transport is supplied.

Proof. On $\mathrm{Re}(s) = 1/2$, $|M(s)| = 1$ (Moeglin–Waldspurger §IV.3.10) gives $|N^{\mathrm{LS}}(s)| = 1$. Hence the Langlands–Shahidi factor changes the scattering phase but does not by itself turn the Koszul coefficient into a positive Plancherel density. The equality (54.4.2) is exactly the additional identification of the bar–cobar Eisenstein measure with the Shahidi 2010 L^2 -normalised Eisenstein measure (Shahidi 2010 Thm 3.5.1). The scalar $1 + |M(s)|^2$ belongs to the Maass–Selberg truncation convention and is not inserted into N^{LS} . $\square$

Remark 54.4.3 (Koszul–Shahidi measure identification). The ratio $\xi(2s)/\xi(2s-1)$ is a non-constant unit-modulus function on $\mathrm{Re}(s) = 1/2$; it equals $1/M(s)$ with $|M(s)| = 1$. The equality of Plancherel normalisations holds only when the Koszul Eisenstein measure is identified with the Shahidi 2010 L^2 -normalised measure. Whether the bar–cobar Koszul definition (Beilinson–Drinfeld 2004 §3.5) assigns exactly this normalisation to the Eisenstein sector is an open problem. This gap does not affect Chapter 54.6.1: the main theorem uses the domain restriction of Section 54.3, not the Langlands–Shahidi extension.

THEOREM 54.4.4 (*full- $\mathcal{V}^{\mathrm{prim}}$ Koszul–Petersson compatibility via Langlands–Shahidi*). (on the Ramanujan bound for Maass cusp forms of $GL_2/\mathbb{Q}$; on the identification of the bar–cobar Koszul spectral measure on the Eisenstein sector with the Shahidi 2010 L^2 -normalised Eisenstein measure (Chapter 54.4.3); and on the Langlands–Shahidi correction N^{LS} as a canonical ingredient of the arithmetic Koszul pairing on principal-series inductions). Assume Ramanujan for Maass forms. For all $v, w \in \mathcal{V}^{\mathrm{prim}}$,

$$\langle v, w \rangle^{\mathrm{Pet}} = \langle v, \sigma^{\mathrm{BC}}(w) \rangle^{\mathrm{Koszul,LS}},$$

with $\langle \cdot, \cdot \rangle^{\mathrm{Koszul,LS}}$ the Langlands–Shahidi-corrected Koszul pairing of Chapter 54.4.2. The identification extends the discrete-core identity (54.3.3) to the Eisenstein continuum.

Proof. On the discrete core, $N^{\mathrm{LS}}|_{\mathrm{disc}} = 1$ (the correction factor is supported on the Eisenstein continuum), so Chapter 54.3.3 applies without modification. On the Eisenstein continuum, Chapter 54.4.2 gives the conditional Plancherel-normalisation match. Adelic assembly: Shahidi 2010 Thm 3.5.1 confirms the Langlands–Shahidi factorisation is compatible with the adelic restricted tensor product, extending Chapter 25.5.3 to the continuum. $\square$

Remark 54.4.5 (domain of the Langlands–Shahidi correction). The full $\mathcal{V}^{\mathrm{prim}}$ identity (54.4.4) extends the domain of the Koszul pairing to the Eisenstein sector only under the measure identification of Chapter 54.4.3; it is not obtained by the bare insertion of N^{LS} . The factor N^{LS} is the Langlands–Shahidi normalised intertwining operator (Shahidi 2010 Ch. 2–3; Moeglin–Waldspurger §IV.3), the natural normalisation for the Eisenstein functional equation, and its insertion is a definitional addition to the arithmetic Koszul pairing of Chapter 12, which uses the Bost–Connes involution alone. The domain restriction of Section 54.3 requires no such addition and is sufficient only for the discrete trace normalisation of the Li path under the hypotheses of Chapter 54.3.5.

54.5 FULL $\mathcal{V}^{\text{prim}} P_3$ ON THE DISCRETE CORE

THEOREM 54.5.1 (*P_3 on the discrete core*). (unconditional on the holomorphic cuspidal and one-dimensional residual sectors; conditional on Ramanujan for Maass forms on the Maass cuspidal sector; using the Tamagawa-regularised assembly). The discrete-spectrum core $\mathcal{V}_{\text{disc}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$ carries a positive-definite Hermitian form

$$\langle \cdot, \cdot \rangle^{\text{Pet}} : \mathcal{V}_{\text{disc}}^{\text{prim}} \otimes \overline{\mathcal{V}_{\text{disc}}^{\text{prim}}} \rightarrow \mathbb{C}$$

given by the adelic Petersson–Tamagawa inner product, compatible with arithmetic Koszul self-duality via

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}$$

on $\mathcal{V}_{\text{disc}}^{\text{prim}} \otimes \mathcal{V}_{\text{disc}}^{\text{prim}}$. The Eisenstein continuous sector is placed outside the P_3 identity by the discrete-core domain restriction of Chapter 54.3.2.

Proof. Positive-definiteness on each sector:

- on $\mathcal{V}_{\text{cusp,hol}}^{\text{prim}}$ via Segal/Whittaker positive-definiteness at each place and restricted tensor product (Chapter 25.3.2 at ∞ , Chapter 25.4.4 at p);
- on $\mathcal{V}_{\text{cusp,Maass}}^{\text{prim}}$ by the Selberg 1956 positive-definiteness of the Petersson inner product on cuspidal automorphic forms, specialised to $GL_2/\mathbb{Q}$ Maass cusp forms (Iwaniec 2002 *Spectral Methods of Automorphic Forms* §4);
- on $\mathcal{V}_{\text{res}}^{\text{prim}}$ by the one-dimensional evaluation $\langle |o\rangle, |o\rangle \rangle = 1$ (Weil 1967 Tamagawa normalisation).

The adelic inner product is a direct sum of positive-definite factors on the orthogonal decomposition $\mathcal{V}_{\text{disc}}^{\text{prim}} = \mathcal{V}_{\text{cusp,hol}}^{\text{prim}} \oplus \mathcal{V}_{\text{cusp,Maass}}^{\text{prim}} \oplus \mathcal{V}_{\text{res}}^{\text{prim}}$, hence itself positive-definite. Compatibility with Koszul self-duality is Chapter 54.3.3. $\square$

COROLLARY 54.5.2 (*P_3 input for the sufficient RH path*). (conditional boundary: Ramanujan for Maass forms). The P_3 input of Chapter 22.6.1 is supplied by Chapter 54.5.1 on the discrete core, unconditional on the holomorphic sector and conditional on Ramanujan on the Maass sector. Combined with the discrete-support of the Li sum (Chapter 54.3.5), the minimal RH path of Chapter 22.6.1 receives a complete Koszul–Petersson compatibility input drawn entirely from the discrete-spectrum core.

Proof. Chapter 54.5.1 + Chapter 54.3.5. $\square$

54.6 KOSZUL–PETERSSEN COMPATIBILITY ON THE DOMAIN-RESTRICTED DISCRETE CORE: MAIN THEOREM

THEOREM 54.6.1 (*full- $\mathcal{V}^{\text{prim}}$ Koszul–Petersson compatibility on the discrete core*). (domain-restricted to $\mathcal{V}_{\text{disc}}^{\text{prim}}$; unconditional on the holomorphic cuspidal and residual sectors; conditional on Ramanujan for Maass cusp forms of $GL_2/\mathbb{Q}$ on the Maass cuspidal sector). For all $v, w \in \mathcal{V}_{\text{disc}}^{\text{prim}}$,

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}.$$

In particular, the Koszul pairing of $\mathcal{V}_{\text{disc}}^{\text{prim}}$ with $(\mathcal{V}_{\text{disc}}^{\text{prim}})^! = \mathbb{A}^{\text{BC}}$, composed with the Bost–Connes involution σ^{BC} , equals the adelic Petersson–Tamagawa inner product on the discrete-spectrum image of $\mathcal{V}_{\text{disc}}^{\text{prim}}$ inside $L_{\text{disc}}^2([GL_2])$.

Proof. The sector-by-sector identities of Chapter 54.3.3 are:

- $\mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$: Chapter 54.1.1 (unconditional).
- $\mathcal{V}_{\text{cusp,hol}}^{\text{prim}} \subset \mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$: Chapter 54.2.1 (unconditional via Deligne 1974, Deligne–Serre 1974).
- $\mathcal{V}_{\text{cusp,Maass}}^{\text{prim}}$: Chapter 54.2.3 (conditional on Ramanujan for Maass forms).
- $\mathcal{V}_{\text{res}}^{\text{prim}}$: direct evaluation at the vacuum.

Orthogonality of the direct-sum decomposition $\mathcal{V}_{\text{disc}}^{\text{prim}} = \bigoplus_{\alpha} \mathcal{V}_{\alpha}^{\text{prim}}$ (Langlands 1966 §6) extends the sector-by-sector identities to the full discrete core. The domain restriction (Chapter 54.3.2) defines the domain of the identity as $\mathcal{V}_{\text{disc}}^{\text{prim}}$; the density mismatch of Chapter 54.3.1 is a property of the Eisenstein continuum, which lies outside this domain. $\square$

Remark 54.6.2 (source disjointness and hypotheses for Chapter 54.6.1). The theorem rests on six independent axes.

Source disjointness. The Koszul derivation uses Beilinson–Drinfeld 2004 (bar–cobar), Positselski 2009 (comodule–contramodule), and Bost–Connes 1995 (Koszul dual as C^* -algebra). The Petersson side uses Deligne 1974 (Weil I, holomorphic Ramanujan), Deligne–Serre 1974 (weight 1), Kim–Shahidi 2002 and Kim–Sarnak 2003 (sub-convex Maass Ramanujan), Langlands 1966 and Moeglin–Waldspurger 1995 (spectral decomposition), Selberg 1956 (Petersson positivity). No source appears on both sides.

Sign and convention consistency. σ^{BC} is $*$ -antilinear; Petersson is sesquilinear conjugate-in-second-slot; Koszul is bilinear. The composition $\langle \cdot, \sigma^{\text{BC}}(\cdot) \rangle^{\text{Koszul}}$ has the correct antilinearity. Tamagawa measure is Weil-normalised (Weil 1967 Ch. VII).

Ambient category. Both sides are sesquilinear forms on $\mathcal{V}_{\text{disc}}^{\text{prim}} \otimes \overline{\mathcal{V}_{\text{disc}}^{\text{prim}}}$, a subspace of the adelic Hilbert space $L_{\text{disc}}^2([GL_2])$.

Hypothesis completeness. The theorem is unconditional on the holomorphic and residual sectors; the Maass sector carries an explicit conditional on Ramanujan. All hypotheses are stated.

Functoriality of the decomposition. The discrete-core decomposition is orthogonal under Petersson (Langlands 1966 §6); Koszul factorises by place (BD 2004 §3.5) and the Bost–Connes involution respects the decomposition.

Domain of the holomorphic and Maass closures. The holomorphic closure uses Deligne 1974 and Deligne–Serre 1974; both are classical unconditional theorems. The Maass closure is conditional on Ramanujan; the Kim–Sarnak 2003 sub-convex bound establishes local control at each prime (Chapter 54.2.4) but not adelic closure. The conditional is the single residual on the Maass sector.

54.7 WHAT REMAINS CONDITIONAL

Sector	Form	Note
$\mathcal{V}_{\text{cusp,hol}}^{\text{prim}}$ (holomorphic cuspidal)	Unconditional	Chapter 54.2.1; Deligne 1974 (weight $k \geq 2$), Deligne–Serre 1974 (weight 1).
$\mathcal{V}_{\text{cusp,Maass}}^{\text{prim}}$ (Maass cuspidal)	Conditional on Ramanujan	Chapter 54.2.3; best bound $\theta_p \leq 7/64$ (Kim–Sarnak 2003); local mismatch $C_p \cdot p^{-25/64}$ at each prime; adelic sum diverges (Chapter 54.2.4).
$\mathcal{V}_{\text{res}}^{\text{prim}}$ (residual, one-dimensional)	Unconditional	Direct evaluation at Fock vacuum; Moeglin–Waldspurger II.2 gives residual spectrum over $GL_2/\mathbb{Q}$ as one-dimensional.
$\mathcal{V}_{\text{cont}}^{\text{prim}}$ (Eisenstein continuous)	Excluded by domain	Chapter 54.3.2; density mismatch (Chapter 54.3.1) placed outside the identity. Sufficient only for the discrete Li trace normalisation, under the H- P^{Ar} spectral hypothesis (Chapter 54.3.5). Alternative Langlands–Shahidi correction Chapter 54.4.4 reduces the continuum to the measure-normalisation gap.

Conditional boundary. The first conditional input of Chapter 54.6.1 is the Ramanujan bound for Maass cusp forms of $GL_2/\mathbb{Q}$. This is an external open problem in the Langlands programme, outside the immediate reach of the arithmetic chiral-algebra framework. Unconditionally, Chapter 54.2.4 via Kim–Sarnak 2003 establishes local control at each prime p with mismatch bounded by $C_p \cdot p^{-25/64}$; the sum over primes diverges, so the Kim–Sarnak bound yields local but not global adelic closure. The Li sum is over the discrete non-trivial zeros (Chapter 54.3.5) after the H- P^{Ar} determinant-trace realisation; the Li path has this trace condition in addition to the Maass boundary.

Eisenstein continuum. The domain restriction places the Eisenstein continuous spectrum outside the identity’s domain (Chapter 54.3.2). The discrete-spectrum restriction suffices for the discrete trace normalisation of the Li path under the stated H- P^{Ar} determinant-trace hypothesis (Chapter 54.3.6). The Langlands–Shahidi correction of Section 54.4 extends the identity to the continuum only after the Koszul Eisenstein measure has been identified with the Shahidi L^2 -normalised measure.

The sufficient RH path. The P_3 input of Chapter 22.6.1 is supplied by $\mathcal{V}_{\text{disc}}^{\text{prim}}$, conditional on Ramanujan for Maass cusp forms and on the H- P^{Ar} determinant-trace realisation used to place the Li trace. The remaining inputs are the conditional arithmetic Positselski, H- P^{Ar} , and HP–Li positivity hypotheses recorded later.

54.8 BIBLIOGRAPHY

The primary sources enter in the following order.

Deligne, P. *La conjecture de Weil I*, Publ. Math. IHES 43 (1974), 273–307. Provides the Ramanujan bound for holomorphic cuspidal newforms of weight $k \geq 2$ via the Riemann hypothesis for varieties over finite fields.

- Deligne, P.** *La conjecture de Weil II*, Publ. Math. IHES 52 (1980), 137–252. The mixed-sheaf generalisation; the holomorphic Ramanujan bound already follows from Weil I above.
- Deligne, P. and Serre, J.-P.** *Formes modulaires de poids 1*, Ann. Sci. ENS. 7 (1974), 507–530. Weight-1 Ramanujan via Artin for two-dimensional odd Galois representations.
- Kim, H. and Shahidi, F.** *Functorial products for $GL_2 \times GL_3$ and the symmetric cube for GL_2* , Ann. Math. 155 (2002), 837–893. Symmetric-cube and symmetric-fourth-power functorial lifts establishing the $\theta \leq 1/9$ sub-convex Ramanujan bound for Maass cusp forms of $GL_2/\mathbb{Q}$.
- Kim, H. and Sarnak, P.** *Refined estimates towards the Ramanujan and Selberg conjectures*, appendix to Kim, *Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2* , J. Amer. Math. Soc. 16 (2003), 139–183. Improvement to $\theta \leq 7/64$.
- Sarnak, P.** *Some applications of modular forms*, Cambridge Tracts 99 (1990), and *Selberg’s eigenvalue conjecture*, Notices AMS 42 (1995), 1272–1277. The Selberg trace-formula residual spectrum computation fixing the Maass–Selberg normalisation on the Eisenstein continuum.
- Langlands, R. P.** *On the Functional Equations Satisfied by Eisenstein Series*, Lecture Notes in Math. 544, Springer 1976 (manuscript 1966). Spectral decomposition, Maass–Selberg relations, intertwining operators.
- Moeglin, C. and Waldspurger, J.-L.** *Spectral Decomposition and Eisenstein Series: A Paraphrase of the Scriptures*, Cambridge Tracts 113, Cambridge UP 1995. Modern GL_n exposition of Langlands 1966; normalised intertwining operator.
- Shahidi, F.** *Eisenstein Series and Automorphic L-Functions*, AMS Colloquium 58, 2010. The Langlands–Shahidi method and normalisation.
- Gelbart, S.** *Automorphic Forms on Adele Groups*, Ann. Math. Studies 83, Princeton 1975. GL_2 spectral decomposition.
- Iwaniec, H.** *Spectral Methods of Automorphic Forms* (second edition), AMS Graduate Studies 53, 2002. Maass cusp forms and Petersson positive-definiteness.
- Takesaki, M.** *Tomita’s Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. 128, Springer 1970; and *Theory of Operator Algebras I*, Springer 1979. GNS construction, modular conjugation, polar decomposition.
- Bost, J.-B. and Connes, A.** *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. 1 (1995), 411–457. The Bost–Connes C^* -algebra and KMS_1 state.
- Connes, A. and Marcolli, M.** *Noncommutative Geometry, Quantum Fields and Motives*, AMS Colloquium 55, 2008. Modern synthesis.
- Casselman, W. and Shalika, J.** *The unramified principal series of p -adic groups. II. The Whittaker function*, Compositio Math. 41 (1980), 207–231. Whittaker uniqueness at supercuspidals.
- Bushnell, C. and Henniart, G.** *The Local Langlands Conjecture for $GL(2)$* , Springer 2006. Tame and wild parametrisation of supercuspidals.
- Godement, R. and Jacquet, H.** *Zeta Functions of Simple Algebras*, Lecture Notes in Math. 260, Springer 1972. Local functional equation and ε -factors.
- Jacquet, H. and Langlands, R. P.** *Automorphic Forms on $GL(2)$* , Lecture Notes in Math. 114, Springer 1970. Whittaker model, ramified principal series.

Jacquet, H., Piatetski-Shapiro, I., and Shalika, J. *Rankin–Selberg convolutions*, Amer. J. Math. 105 (1983), 367–464. Local factors and functional equation.

Tate, J. *Fourier analysis in number fields and Hecke’s zeta functions*, Ph.D. thesis, Princeton 1950; reprinted Cassels–Fröhlich 1967 Ch. XV. Restricted tensor product.

Weil, A. *Basic Number Theory*, Springer 1967. Tamagawa normalisation.

Selberg, A. *Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series*, J. Indian Math. Soc. 20 (1956), 47–87. Petersson positive-definiteness.

Arthur, J. *Eisenstein series and the trace formula*, Proc. Symp. Pure Math. 33 (1979); and *A trace formula for reductive groups. I*, Duke Math. J. 45 (1978), 911–952. Adelic truncation; continuous spectrum distributional pairing.

Piatetski-Shapiro, I. *On the Saito–Kurokawa lifting*, Invent. Math. 71 (1983), 309–338. CAP representations: first appear at Sp_4 , absent for GL_2 .

Gelfand, I. M., Graev, M. I., and Piatetski-Shapiro, I. I. *Representation Theory and Automorphic Functions*, Saunders 1969. Archimedean classification of cuspidal representations.

Disjoint sources. The proof sources for the Koszul side (Beilinson–Drinfeld 2004, Positselski 2009, Bost–Connes 1995) are disjoint from the Petersson sources (Deligne 1974 and 1980, Deligne–Serre 1974, Kim–Shahidi 2002, Kim–Sarnak 2003, Langlands 1966, Moeglin–Waldspurger 1995, Selberg 1956, Weil 1967, Jacquet–Langlands 1970, Godement–Jacquet 1972, JPSS 1983, Casselman–Shalika 1980, Bushnell–Henniart 2006, Iwaniec 2002, Arthur 1978–1979, Gelbart 1975). No source appears on both sides of any identification. primary-source independence is therefore preserved.

BOST–CONNES LOG-MODULAR SPECTRUM AND THE HILBERT–PÓLYA TWIST

The analytic condition (W) is the existence of a logarithmic weight on the Bost–Connes/Fock side which can enter a determinant-trace twist. The Hilbert–Pólya condition asks for a self-adjoint A such that $\Theta_\xi := \frac{1}{2}\text{id} + iA$ has regularised determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes trace distribution equal to the non-trivial-zero distribution of ζ . The Bost–Connes C^* -system supplies an unconditional log-modular operator with spectrum $\{\log m \mid m \in \mathbb{N}^\times\}$. It does not supply the zero-divisor operator. The passage from the log-prime spectrum to the zero divisor is the primary-Heisenberg twist; existence of that twist with the prescribed determinant divisor is exactly the Hilbert–Pólya obstruction. Conditional on such a twist, the determinant and trace formulae below are the Hadamard product and the Weil explicit formula in operator form.

55.1 BOST–CONNES C^* -DYNAMICAL SETUP

The GNS Hilbert space and the modular operator are determined by the Bost–Connes C^* -dynamical system at inverse temperature $\beta = 1$. The algebra and its $\mathbb{R}$ -action are fixed in Chapter 45; the normalisation below matches the primary-Heisenberg spectrum convention.

Definition 55.1.1 (local and adelic Bost–Connes algebras). . For each rational prime p , the local Bost–Connes algebra is the C^* -algebra

$$C_{\mathbb{Q},p}^* := C^*(\mu_m, \mu_m^*, e(\gamma))_{m \in p^{\mathbb{N}}, \gamma \in \mathbb{Q}_p/\mathbb{Z}_p},$$

with Bost–Connes 1995 Selecta 1 §2.2 relations $\mu_m \mu_m^* = 1/m$, $\mu_m e(\gamma) \mu_m^* = e(m\gamma)/m$, $e(\gamma + \delta) = e(\gamma)e(\delta)$. The adelic algebra is the restricted tensor product

$$C_{\mathbb{Q}}^* := \bigotimes_p' C_{\mathbb{Q},p}^*,$$

restricted with respect to the local units $1_p \in C_{\mathbb{Q},p}^*$. The $\mathbb{R}$ -action $\sigma^t: C_{\mathbb{Q}}^* \rightarrow C_{\mathbb{Q}}^*$ is the tensor product of

$$\sigma_p^t(\mu_m) = m^{it} \mu_m, \quad \sigma_p^t(e(\gamma)) = e(\gamma), \quad \sigma_p^t(\mu_m^*) = m^{-it} \mu_m^*. \quad (55.1)$$

Remark 55.1.2 (adelic assembly cross-references). The local algebra is Chapter 7.1.1; the adelic assembly is Chapter 45.2.7. The relations (55.1) are the modular automorphism generators; we use them below to fix normalisations.

LEMMA 55.1.3 (KMS_1 state). . There is a unique normal σ -KMS state φ_1 on $C_{\mathbb{Q}}^*$ at inverse temperature $\beta = 1$, factoring locally as $\varphi_1 = \bigotimes_p' \varphi_{1,p}$, with

$$\varphi_{1,p}(\mu_n^* \mu_m e(\gamma)) = \delta_{n,m} m^{-1} \tilde{e}_p(\gamma), \quad \tilde{e}_p(\gamma) = [\gamma = 0],$$

and normalised partition function $Z_1(\beta) = \zeta(\beta)$ on the zero-mode sector for $\text{Re}(\beta) > 1$. The associated von Neumann factor is type III₁.

Proof. This is Bost–Connes 1995 Selecta 1 Theorem 25, with the adelic assembly of Connes–Marcolli 2008 AMS Colloquium 55 §4.1. The normalisation choice and the factor type are Bost–Connes Thm 25 (ii)–(iii). $\square$

Remark 55.1.4 (partition at $\beta = 1$). The nominal partition $Z_1(1) = \zeta(1)$ diverges. The Bost–Connes convention regularises $Z_1(\beta)$ at $\beta \rightarrow 1^+$ as Euler–Mascheroni γ (via $\zeta(\beta) = 1/(\beta - 1) + \gamma + O(\beta - 1)$); this regularisation matches the partition normalisation of Chapter 40.3.7 and is not used as an operator input below.

55.2 GNS REPRESENTATION AND THE MODULAR OPERATOR

The Bost–Connes logarithmic modular operator acts on the GNS Hilbert space of φ_1 . Its spectrum is the log-prime lattice, not the zeta-zero divisor. The GNS construction and modular theory are classical; the spectrum of $\log \Delta_1$ is unconditional.

Definition 55.2.1 (GNS triple of φ_1). Let $(\mathcal{H}_1, \pi_1, \Omega_1)$ denote the GNS triple of φ_1 on $C_{\mathbb{Q}}^*$: $\mathcal{H}_1$ is the Hilbert space obtained by completing $C_{\mathbb{Q}}^*/\mathcal{N}_1$ with $\mathcal{N}_1 = \{x \mid \varphi_1(x^*x) = 0\}$ under $\langle x, y \rangle = \varphi_1(y^*x)$; π_1 is left multiplication; Ω_1 is the class of the identity.

LEMMA 55.2.2 (GNS Hilbert space as a restricted tensor). There is a unitary isomorphism

$$\mathcal{H}_1 \cong \bigotimes_p' \left(L_p \otimes \ell^2(\mathbb{Q}_p/\mathbb{Z}_p) \right),$$

with $L_p = \ell^2(p^{\mathbb{N}}, m^{-1}dm)$ (Bost–Connes zero-mode lattice at p), restricted with respect to the local vacuum $\delta_1 \otimes \delta_0$ at each p , and $\Omega_1 = \bigotimes_p' (\delta_1 \otimes \delta_0)$.

Proof. Local statement is Chapter 45.2.2; adelic assembly is restricted tensor product by modularity of tensors under KMS states (Takesaki 2003 Ch. VIII §1). $\square$

LEMMA 55.2.3 (modular operator on the GNS space). The modular operator Δ_1 of φ_1 satisfies, on the basis $\{\delta_m \otimes \delta_\gamma\}_{m \in \mathbb{N}^\times, \gamma \in \mathbb{Q}/\mathbb{Z}}$ of the zero-mode lattice times character sector,

$$\Delta_1(\delta_m \otimes \delta_\gamma) = m^{-2} \delta_m \otimes \delta_\gamma, \quad \Delta_1^{it} = \pi_1 \circ \sigma^t \circ \pi_1^{-1} \quad (t \in \mathbb{R}).$$

In particular $\log \Delta_1$ is self-adjoint on the dense σ -analytic core and has pure point spectrum $\{-2 \log m \mid m \in \mathbb{N}^\times\}$ on the zero-mode lattice, with multiplicity equal to the number of character sectors.

Proof. Tomita–Takesaki theory: for a KMS state at $\beta = 1$, $\Delta_1^{it} = \sigma^t$ on the GNS image of analytic elements (Takesaki 1970 Lecture Notes Thm 10.1; Takesaki 1979 Ch. VIII §1 Cor. 1.2). Applying $\sigma^t(\mu_m) = m^{it} \mu_m$ twice, one to each of $\mu_m \Omega_1$ and $\mu_m^* \Omega_1$, produces the m^{-2} eigenvalue. This is the content of Chapter 45.2.3. $\square$

Definition 55.2.4 (BC log-modular operator). The Bost–Connes log-modular operator is

$$N_{\text{BC}} := -\frac{1}{2} \log \Delta_1$$

on the σ -analytic core. On the basis of Chapter 55.2.2, it acts diagonally by

$$N_{\text{BC}}(\delta_m \otimes \delta_\gamma) = (\log m) \delta_m \otimes \delta_\gamma.$$

THEOREM 55.2.5 (*unconditional self-adjointness and spectrum of N_{BC}*). . N_{BC} is essentially self-adjoint on the finite-span core $\mathcal{D}_{\text{BC}} := \text{span}_{\mathbb{C}}\{\delta_m \otimes \delta_\gamma\}$ of $\mathcal{H}_1$. Its closure is self-adjoint, non-negative, and has pure point spectrum

$$\text{Spec}(N_{\text{BC}}) = \{\log m \mid m \in \mathbb{N}^\times\} = \{0\} \cup \{\log p_1^{a_1} \cdots p_k^{a_k} \mid \text{prime powers}\},$$

with multiplicity at $\log m$ equal to $\sharp(\mathbb{Q}/\mathbb{Z})$ (countable) unless one restricts to the spherical character sector $\gamma = 0$, where the multiplicity is 1.

Proof. Diagonal operator on a basis of the Hilbert space with real eigenvalues: Reed–Simon Vol. I Thm. VIII.11 gives essential self-adjointness on the finite-span core; the closure is the standard L^2 -diagonal self-adjoint operator with spectrum equal to the eigenvalue set. The multiplicities are read off from the basis. $\square$

Remark 55.2.6 (N_{BC} : *log-prime spectrum and its role*). N_{BC} is the canonical self-adjoint operator extracted from the KMS_1 modular data: it is the Bost–Connes Hamiltonian of Chapter 7.1.1 times a normalisation, and its spectrum is the log-prime additive semigroup $\{\log m \mid m \in \mathbb{N}^\times\}$. The Berry–Keating weight $\lambda(p^k) = k \log p$ of Chapter 40.3.1(BK) is the eigenvalue of N_{BC} at the monomial $m = p^k$, consistent with the weight selection of Chapter 40.3.7. The zero-divisor operator, whose normal determinant divisor is $\text{div } \xi_{\mathbb{R}}$, is the target of the primary-Heisenberg twist of Section 4; see Chapter 22.3.6.

55.3 ARITHMETIC HEISENBERG FOCK AND ITS EMBEDDING INTO THE GNS SPACE

The primary-Heisenberg twist of Section 4 acts on a Fock space built from per-prime Heisenberg oscillators. The arithmetic Fock and its primary core are defined here; the embedding into the GNS space is the content of Lemma 55.3.2.

Definition 55.3.1 (*arithmetic Fock and primary subspace*). . The *arithmetic Heisenberg Fock* is

$$\mathcal{F}_{\text{Heis}}^{\text{Ar}} := \bigotimes_p' \text{Fock}^{(p)} \otimes \mathcal{H}^{(\infty)},$$

where $\text{Fock}^{(p)}$ is the per-prime chiral Heisenberg Fock (Chapter 4) and $\mathcal{H}^{(\infty)}$ is the archimedean complex Heisenberg Fock (Chapter 14). The *primary core* is the dense subspace

$$\mathcal{F}_{\text{Heis,prim}}^{\text{Ar}} := \text{span}_{\mathbb{C}}\left\{|\mathbf{n}\rangle \otimes \delta_m : \mathbf{n} \in \mathbb{Z}_{\geq 0}^{\oplus \mathbb{N}}, m \in \mathbb{N}^\times\right\}$$

of finite-support number-state/zero-mode tensors (Chapter 35.3.1).

LEMMA 55.3.2 (*primary Fock embeds into GNS*). . There is a canonical isometric embedding

$$\iota : \mathcal{F}_{\text{Heis,prim}}^{\text{Ar}} \hookrightarrow \mathcal{H}_1$$

sending the finite-support number-state $|\mathbf{n}\rangle \otimes \delta_m$ to $\text{Asc}(\mathbf{n}) \cdot (\delta_m \otimes \delta_0)$ where $\text{Asc}(\mathbf{n})$ is the Heisenberg ascending operator at number-occupation $\mathbf{n}$. The image is dense in the spherical character sector of $\mathcal{H}_1$ on the zero-mode lattice.

Proof. The primary Fock is generated from the vacuum by finite products of Heisenberg creation operators acting on spherical zero-mode lattice vectors. Each creation operator lifts to a multiplication operator on the GNS space commuting with $\pi_1(e(\gamma))$ for $\gamma = 0$, and the spherical zero-mode lattice is the $\gamma = 0$ image of the Bost–Connes lattice. Isometry is the Heisenberg/GNS inner-product match on number states. See Chapter 7.3.2 for the commuting of Heisenberg creation with BC multiplication in the spherical sector. $\square$

Remark 55.3.3 (density and domain of the embedding). The embedding ι lands in the spherical-character zero-mode sector; the non-spherical $\gamma \neq 0$ sectors of the GNS space carry the Eisenstein/ramified content of the non-spherical character decomposition (Chapter 45). All subsequent determinant and trace comparison is performed inside ι 's image; the operator N_{BC} restricts there.

55.4 THE PRIMARY-HEISENBERG TWIST AND THE DETERMINANT-TRACE OPERATOR

The BC log-modular operator N_{BC} has spectrum $\{\log m\}_{m \in \mathbb{N}^\times}$. The Hilbert–Pólya condition asks for a self-adjoint operator A together with the determinant condition forcing the divisor of $\frac{1}{2}\text{id} + iA$ to be the zero divisor of $\xi_{\mathbb{R}}$. The *primary-Heisenberg twist* names the data required to compare the first spectrum with this determinant divisor and its Weil–Connes trace. Its existence with the prescribed determinant-trace property is H-P^{Ar}.

Statement of the Twist

Definition 55.4.1 (twist data). A *primary-Heisenberg twist* is a triple $(\mathcal{H}_\xi, U, A)$ consisting of:

- (i) a Hilbert space $\mathcal{H}_\xi$ equipped with the Petersson inner product of Chapter 21.7.1 (on the sph-finite cuspidal core, conditional on Chapter 21.7.2);
- (ii) a dense invariant core $\mathcal{D}_\xi \subset \mathcal{H}_\xi$;
- (iii) an isometric map $U : \mathcal{F}_{\text{Heis,prim}}^{\text{Ar}} \rightarrow \mathcal{H}_\xi$ with dense range in $\mathcal{D}_\xi$;
- (iv) a self-adjoint unbounded operator $A : \mathcal{D}_\xi \rightarrow \mathcal{H}_\xi$ satisfying

$$AU|\mathbf{n}\rangle \otimes \delta_m = U(L_o^{\text{Ar,Heis}} + \log m)|\mathbf{n}\rangle \otimes \delta_m \quad \text{on } \mathcal{F}_{\text{Heis,prim}}^{\text{Ar}}, \quad (55.2)$$

in a chain-level sense, where $L_o^{\text{Ar,Heis}}$ is the per-prime Heisenberg number operator extended additively to the restricted product $\mathcal{F}_{\text{Heis}}^{\text{Ar}}$: on the vacuum sector of the p -th factor it acts as $\sum_{n \geq 1} n a_{-n}^{(p)} a_n^{(p)}$, extended to the full Fock by tensoring with the identity on the remaining factors.

Given such data, the *candidate Hilbert–Pólya operator* is

$$H_{\text{HP}} := A,$$

and the *arithmetic zero-divisor operator* is

$$\Theta_\xi := \frac{1}{2}\text{id} + iH_{\text{HP}}.$$

Remark 55.4.2 (what (iv) says). Condition (iv) names the intertwining of A with the log-prime semi-group shift. At $\mathbf{n} = 0$, the right-hand side of (55.2) is $\log m \cdot U(\delta_m)$, i.e. an eigenvector at eigenvalue $\log m$; this is exactly the BC log-spectrum of Chapter 55.2.5. The twist is therefore required to extend the Bost–Connes modular data: the zero-mode eigenvalues $\log m$ are inherited; the new content is the per-prime number operator $L_o^{\text{Ar,Heis}}$ acting on the Heisenberg descendants.

Determinant-Trace Requirement: Hadamard Inversion

The condition on A is the Hilbert–Pólya determinant-trace condition for $\xi_{\mathbb{R}}$. Write a non-trivial zero as $\rho = \beta_\rho + i\gamma_\rho$. The self-adjointness of A puts the normal operator $\Theta_\xi = \frac{1}{2} + iA$ on the critical line. H-P^{Ar} asks that its regularised determinant divisor be the original zero divisor and that its Weil–Connes trace be the zero distribution. Equivalently, the determinant identity forces $\beta_\rho = 1/2$ for every zero; after

this zero-placement the trace of $\mathcal{A}$ is supported at the ordinates γ_ρ . This is *not* the spectrum of N_{BC} , which is $\{\log m\}_{m \in \mathbb{N}^\times}$. The prime and zero sides are related by the Mellin transform of ξ through its Hadamard factorisation:

$$\xi_{\mathbb{R}}(s) = \frac{1}{2} e^{Bs} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho} \quad (55.3)$$

(Riemann 1859; Hadamard 1893), where the product is over non-trivial zeros and B is the Hadamard constant. The primes appear on the side of the *logarithmic derivative*

$$-\frac{\xi'_{\mathbb{R}}(s)}{\xi_{\mathbb{R}}(s)} = -B - \frac{1}{s} - \frac{1}{s-1} + \frac{1}{2} \log \pi - \frac{1}{2} \Gamma'/\Gamma(s/2) + \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}$$

(von Mangoldt 1895), and the zero-prime duality is the Weil explicit formula. The twist $\mathcal{A}$ is the operator-level analogue of *inverting this duality*: from the log-prime spectrum of N_{BC} , construct a normal zero-divisor operator whose determinant is governed by the Hadamard expansion and whose trace is the Weil–Connes zero distribution. The Hadamard expansion is a statement about an entire function. The assertion that a self-adjoint operator carries this determinant divisor and trace distribution is additional data, the content of H-P^{Ar} .

The Hilbert–Pólya Hypothesis and Its Determinant-Trace Consequences

Hypothesis 55.4.3 (H-P^{Ar}-twist). There exists a primary-Heisenberg twist $(\mathcal{H}_\xi, U, \mathcal{A})$ in the sense of Chapter 55.4.1 with the additional zero-placement spectral property

$$\text{Spec}(\mathcal{A}) = \{\gamma_\rho \mid \rho = \beta_\rho + i\gamma_\rho \text{ is a non-trivial zero of } \xi_{\mathbb{R}}\}, \quad (55.4)$$

with multiplicities matching zero multiplicities, together with the determinant condition $\text{div } \det_\infty(\text{id} - (\frac{1}{2}\text{id} + i\mathcal{A})) = \text{div } \xi_{\mathbb{R}}$, and the $\mathbb{Z}/2$ -symmetry $\gamma_\rho \mapsto -\gamma_\rho$ acting on the spectrum as the arithmetic $\mathcal{S}$ -modular involution.

Remark 55.4.4 (relation to H-P^{Ar}). Chapter 55.4.3 is the functional-analytic content of Chapter 22.3.2: an operator $\mathcal{A}$ with the specified determinant divisor and trace distribution is exactly H_{HP} of the reformulation chapter, and $\Theta_\xi = \frac{1}{2} + i\mathcal{A}$ is the arithmetic zero-divisor operator. The classical Hilbert–Pólya problem asks for this object; the arithmetic chiral-algebra form asks for it together with the intertwining with N_{BC} via the log-prime spectrum. This intertwining selects the arithmetic representative among candidates with the same determinant divisor and Weil–Connes trace; existence remains the content of H-P^{Ar} .

55.5 DETERMINANT IDENTITY: HADAMARD FACTORISATION

Assuming the twist of Chapter 55.4.3, the determinant divisor condition identifies the zeros of $\xi_{\mathbb{R}}$ with the spectrum of Θ_ξ ; the Hadamard product then determines $\det_\infty(s \text{id} - \Theta_\xi)$ as a regularised determinant. The formal content and the independent regularisation input are separated in the proposition and remark below.

PROPOSITION 55.5.1 (regularised determinant from Hadamard). *(conditional on Chapter 55.4.3).* Assume the twist $(\mathcal{H}_\xi, U, \mathcal{A})$ has the spectral property (55.4) and the determinant divisor condition of Chapter 55.4.3. Define the regularised determinant of the shifted operator by the ζ -regularised prescription

$$\log \det_\infty(s \text{id} - \Theta_\xi) := -\partial_u \text{Tr}_{\text{reg}}((s \text{id} - \Theta_\xi)^{-u}) \Big|_{u=0},$$

interpreted as a distribution on s in the admissible Weil–Connes Paley–Wiener class. Then

$$\det_{\infty}(\text{id} - \Theta_{\xi}) = \xi_{\mathbb{R}}(s). \quad (55.5)$$

Proof. Under Chapter 55.4.3, the determinant divisor condition says that the spectral multiset of $\Theta_{\xi} = 1/2 + iA$ is the non-trivial zero divisor $\{\rho\}$ with multiplicities matching. The Hadamard product (55.3) expresses $\xi_{\mathbb{R}}$ as a product over the same zeros, with the convergence factor $e^{s/\rho}$ and the Hadamard constant B absorbing the sum over $1/\rho$. The ζ -regularised definition of $\det_{\infty}$ (Ray–Singer 1971; Voros 1987) for an operator with positive Hadamard genus produces $\prod_{\rho}(1 - s/\rho)e^{s/\rho}$ after subtraction of the Hadamard terms. The remaining prefactor $e^{Bs}/2$ is the regularisation ambiguity and is fixed by the Weil–Connes normalisation compatibility with the primary sector (the Tate-pole removal of Chapter 22.3.7). Combining these: (55.5) holds as an equality of regularised determinants in the distributional Paley–Wiener sense, modulo explicit normalisation constants. $\square$

Remark 55.5.2 (formal content of the determinant identity). Chapter 55.5.1 is *conditional* on the spectral and determinant conditions of Chapter 55.4.3. Given those conditions, the Hadamard identity is formal: the regularised determinant of the normal operator whose divisor is the zero divisor of an entire function of order 1 with genus 1 is the function itself, up to a convergence-factor normalisation. The content of H-P^{Ar} is the existence of a self-adjoint operator whose normal translate has this determinant divisor and whose regularised trace is the Weil–Connes zero distribution.

Remark 55.5.3 (comparison with Berry–Keating). Berry–Keating (SIAM Review 41, 1999) propose $H = xp + 1/2$ on $L^2(\mathbb{R}_{>0}, dx/x)$ as the candidate H_{HP} . Their proposal reproduces the leading-order Riemann–von Mangoldt zero density via Bohr–Sommerfeld quantisation; the Planck-cell regularisation, however, requires a self-adjoint extension not supplied by the model. The present twist has the same logical form: the BC log-prime spectrum $\{\log m\}_{m \in \mathbb{N}^{\times}}$ is classical and self-adjoint, and the lift to the zero determinant divisor via the Hadamard-dual twist is the content of H-P^{Ar} . See Chapter 35.3.17 for the HP^{Ar} -converse analysis.

55.6 TRACE DISTRIBUTION: WEIL–CONNES EXPLICIT FORMULA

PROPOSITION 55.6.1 (distributional trace identity). (*conditional on Chapter 55.4.3*). Assume the twist with spectral property (55.4) and determinant divisor condition. For every admissible Weil–Connes Paley–Wiener test function h (in the class PW of Chapter 40.3.9),

$$\text{Tr}_{\text{reg}}(h(H_{\text{HP}})) = \sum_{\rho} h(\gamma_{\rho}) = \int_{\mathbb{R}} h(r) dW_{\infty}(r) - 2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} \widehat{h}(\log n), \quad (55.6)$$

where the second equality is the Weil–Connes explicit formula (Chapter 40.3.9).

Proof. Given Chapter 55.4.3, the regularised trace of $h(H_{\text{HP}}) = h(A)$ computed in the spectral representation of A is $\sum_{\rho} h(\gamma_{\rho})$. The determinant divisor condition makes $\frac{1}{2} + i\gamma_{\rho}$ the actual zero divisor, so this is the classical Weil explicit formula for PW test functions (Chapter 40.3.9; Weil 1952; Connes 1999 arXiv:math/9811068 Thm. 1). The admissibility of the regularisation for Tr_{reg} is preserved by ι (Chapter 55.3.2) because the embedding is isometric on the spherical primary-Fock image. $\square$

COROLLARY 55.6.2 (Weil-positive cone compatibility). (*conditional on Chapter 55.4.3*). The regularised trace $\text{Tr}_{\text{reg}}(h(H_{\text{HP}}))$ is non-negative for every $h \in \text{PW}$ of the form $h = |\phi|^2$ with ϕ in the

Petersson-positive cone on the primary Fock, *if and only if* Weil positivity holds for ζ . In particular, Li positivity (Chapter 22.3.1) is implied.

Proof. By Chapter 55.6.1, the regularised trace equals the Weil/Connes zero-side distribution on PW tests. Non-negativity on $|\phi|^2$ -tests is Weil's positivity statement (Weil 1952 §5; Barner 1981; Bombieri 2000 RH notes). Li positivity follows by Chapter 22.3.8. $\square$

55.7 UNCONDITIONAL CORE AND SPECTRAL TWIST

The Bost–Connes core is unconditional. The determinant and trace identities are consequences of the spectral twist. The existence of the twist is H-P^{Ar} .

Bost–Connes Core

- The Bost–Connes C^* -algebra, its KMS_1 state, its GNS representation, and the modular operator Δ_1 with spectrum $\{m^{-2}\}_{m \in \mathbb{N}^\times}$ on the zero-mode lattice (Chapter 55.2.3, Chapter 55.2.5).
- The BC log-modular operator N_{BC} is self-adjoint on the finite-span core, non-negative, with pure point spectrum $\{\log m\}_{m \in \mathbb{N}^\times}$: unconditional theorem (Chapter 55.2.5).
- The arithmetic Heisenberg Fock $\mathcal{F}_{\text{Heis}}^{\text{Ar}}$ embeds isometrically into the GNS space on the spherical character sector (Chapter 55.3.2).

Consequences of the Twist

- Given a twist $(\mathcal{H}_\xi, U, A)$ with spectral property (55.4) and determinant divisor condition, the regularised determinant identity $\det_\infty(\text{id} - \Theta_\xi) = \xi_{\mathbb{R}}(s)$ holds (Chapter 55.5.1).
- Given the same twist, the distributional trace identity $\text{Tr}_{\text{reg}}(h(H_{\text{HP}})) = \sum_\rho h(\gamma_\rho)$ holds for $h \in \text{PW}$ (Chapter 55.6.1).
- Given the same twist, Li positivity holds if and only if Weil positivity for ζ holds (Chapter 55.6.2).

The content is that the Vol IV primary-Fock setting is compatible with the classical Hadamard/Weil calculus: the regularisations match.

Hilbert–Pólya Determinant-Trace Existence

- The existence of a self-adjoint operator A on a Hilbert space $\mathcal{H}_\xi$, densely defined on a core $\mathcal{D}_\xi \subset \mathcal{H}_\xi$, intertwining the BC log-modular operator N_{BC} on the primary Fock, whose normal zero-divisor operator has determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes trace equal to the zero distribution, so that after zero-placement the associated ordinate trace carries the non-trivial-zero multiplicities and $\text{div } \det_\infty(\text{id} - (\frac{1}{2}\text{id} + iA)) = \text{div } \xi_{\mathbb{R}}$, is Chapter 55.4.3: it is the content of H-P^{Ar} , beyond the domain of the Bost–Connes/Tomita–Takesaki construction.
- The structural analysis of Chapter 35 establishes that this is equivalent to Connes' adèle class space absorption-trace condition (Connes arXiv:math/9811068 §IV) and to Deninger's motivic cohomology Frobenius purity conjecture (Deninger 1994).

THEOREM 55.7.1 (*precise obstruction*). . The Hilbert–Pólya condition (Chapter 22.3.2) in the arithmetic form reduces to the existence, in the sense of Chapter 55.4.1–Chapter 55.4.3, of a self-adjoint operator A intertwining N_{BC} via the primary-Heisenberg determinant-trace twist, satisfying the zero-placement spectral condition (55.4) and the determinant divisor condition of Chapter 55.4.3. The BC log-modular operator N_{BC} supplies the zero-mode eigenvalues at $\log m$ unconditionally. The twist exists if and only

if the Connes absorption-trace condition is realised in the chiral-Fock formalism; this equivalence is Chapter 35.3.14.

Proof. By Chapter 55.2.5, the operator N_{BC} is unconditionally present with the log-prime spectrum. By Chapter 55.5.1 and Chapter 55.6.1, *conditional on* a twist with the spectral property (55.4) and the determinant divisor condition, both the determinant identity (55.5) and the trace identity (55.6) hold. The determinant-trace property is equivalent to Chapter 35.3.9, the Connes absorption-trace condition. The equivalence is the three-way equivalence theorem of Chapter 35 (Chapter 35.3.14) in its corrected form after the Friedrichs construction of Chapter 35.3.2. $\square$

Remark 55.7.2 (weight existence (W) and the remaining H-P step). The weight-existence condition of Chapter 40.5.2 asks for an additive Bost–Connes weight system satisfying the zeta partition normalisation. The weight $\lambda = \lambda_{BK}$ is supplied unconditionally as the eigenvalues of N_{BC} on the arithmetic Fock image (Chapter 55.2.5); the weight is the unique one consistent with KMS_I and the symmetric normalisation. The operator H_{HP} with the zeta-zero divisor condition is the content of $H-P^{Ar}$ itself: the weight is unconditional, while $H-P^{Ar}$ is the twist-existence statement of Chapter 55.7.1.

Remark 55.7.3 (the canonical conditional form of the H-P statement). The assertion about H_{HP} reads: “There is a canonical self-adjoint operator N_{BC} on the GNS space of the Bost–Connes KMS_I state with pure point spectrum $\{\log m\}_{m \in \mathbb{N}^\times}$. The operator N_{BC} is the BC log-modular operator, distinct from the zero-divisor operator. There is an intertwining primary-Heisenberg twist whose existence with determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes zero trace is Chapter 55.4.3, equivalent to the Connes absorption-trace and Deninger determinant formulations. Under that hypothesis, the determinant and trace identities (55.5), (55.6) hold and agree with Hadamard factorisation and the Weil explicit formula.” Thus the weight belongs to the Bost–Connes log-modular spectrum, while the twist-existence statement is $H-P^{Ar}$.

Operator Boundary

Component	Form in Vol IV	Reference
BC C^* -algebra and KMS_I state	Unconditional	Bost–Connes 1992
GNS triple $(\mathcal{H}_I, \pi_I, \Omega_I)$	Unconditional	Chapter 55.2.1
Modular operator Δ_I and spectrum	Unconditional	Chapter 55.2.3
BC log-modular N_{BC} , self-adjoint	Unconditional	Chapter 55.2.5
Arithmetic Heisenberg Fock $\mathcal{F}_{H\text{eis}}^{Ar} \hookrightarrow \text{GNS (spherical)}$	Unconditional	Chapter 55.3.2
Twist data $(\mathcal{H}_{\xi}, U, A)$ with zero divisor	Conjectural (=H- P^{Ar})	Chapter 55.4.3
$\det_{\infty}(sid - \Theta_{\xi}) = \xi_{\mathbb{R}}(s)$	Conditional on twist	Chapter 55.5.1
$\text{Tr}_{\text{reg}}(h(H_{HP})) = \sum_{\rho} h(\gamma_{\rho})$	Conditional on twist	Chapter 55.6.1
Li positivity from Petersson positivity	Conditional on twist + Weil positivity	Chapter 55.6.2
Weight existence: λ_{BK} as BC eigenvalues	Bost–Connes theorem	Chapter 55.7.2
Hilbert–Pólya trace bridge: twist existence	H- P^{Ar} / Connes–Deninger determinant-trace realisation	Chapter 55.7.1

55.8 BIBLIOGRAPHY AND SOURCE LIST

The argument draws on four disjoint source lists here with publisher/arXiv identifiers for reproducibility.

Operator-algebra and KMS theory.

Bost–Connes. J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, *Selecta Math.* 1 (1995), 411–457.

Connes. A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, *Selecta Math.* 5 (1999), 29–106; arXiv:math/9811068.

Connes–Marcolli. A. Connes and M. Marcolli, *Noncommutative geometry, quantum fields, and motives*, *AMS Colloquium Publications* 55 (2008). Ch. 3–4 for KMS states and rigidity; Ch. 5 for the adèle class space.

Takesaki. M. Takesaki, *Tomita’s theory of modular Hilbert algebras and its applications*, *Lecture Notes in Mathematics* 128 (Springer 1970); *Theory of Operator Algebras I*, (Springer 1979), Ch. IV §5 (GNS), Ch. VI §1 (Tomita–Takesaki); *Theory of Operator Algebras II* (Springer 2003), Ch. VIII §1 (modular automorphism).

Meyer. R. Meyer, *On a representation of the idele class group related to primes and zeros of L-functions*, *Duke Math. J.* 127 (2005), 519–595; arXiv:math/0311468.

Hilbert–Pólya candidates and determinant-trace realisations.

Berry–Keating. M. V. Berry and J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, *SIAM Review* 41 (1999), 236–266.

Reed–Simon. M. Reed and B. Simon, *Methods of Modern Mathematical Physics*, Vol. I (Academic Press 1980), Thm. VIII.11 (diagonal operators); Vol. II (Academic Press 1975), Thm. X.23 (Friedrichs extension).

Deninger. C. Deninger, *Motivic L-functions and regularised determinants*, *Proc. Symp. Pure Math.* 55.1 (1994), 707–743; *A note on arithmetic topology and dynamical systems*, *Math. Nachr.* 205 (2001), 107–135.

Ray–Singer. D. B. Ray and I. M. Singer, *R-torsion and the Laplacian on Riemannian manifolds*, *Advances in Mathematics* 7 (1971), 145–210 (source of ζ -regularised determinants).

Voros. A. Voros, *Spectral functions, special functions and the Selberg zeta function*, *Comm. Math. Phys.* 110 (1987), 439–465.

Hadamard factorisation and explicit formula.

Riemann. B. Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, *Monatsberichte der Berliner Akademie*, 1859.

Hadamard. J. Hadamard, *Étude sur les propriétés des fonctions entières et en particulier d’une fonction considérée par Riemann*, *J. Math. Pures Appl.* 9 (1893), 171–215.

von Mangoldt. H. von Mangoldt, *Zu Riemann’s Abhandlung “Über die Anzahl der Primzahlen unter einer gegebenen Grösse”*, *J. Reine Angew. Math.* 114 (1895), 255–305.

Weil. A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, *Comm. Sémin. Math. Univ. Lund* (1952), 252–265.

Edwards. H. M. Edwards, *Riemann’s Zeta Function*, Academic Press 1974.

Chiral algebra and arithmetic Fock background.

Beilinson–Drinfeld. A. Beilinson and V. Drinfeld, *Chiral Algebras*, *AMS Colloquium Publications* 51 (2004).

Bost–Connes (cited above).

Vol IV chapters. Chapter 45, Chapter 35, Chapter 40, Chapter 43, Chapter 22.

No source in this list appears on both sides of the Koszul–Petersson/KMS derivation and the Hilbert–Pólya determinant-trace argument. The source list of Chapter 40.1 is disjoint from the source list of Chapter 35.1; the present argument draws on the intersection of the two with the Takesaki modular theory and the Hadamard/von Mangoldt classical analysis, both outside the Vol IV internal derivations.

HP–LI POSITIVITY AND THE RANKIN–SELBERG KERNEL OBSTRUCTION

The HP–Li programme reduces the Riemann Hypothesis to the positivity of the Weil distribution $W(f * \tilde{f}) \geq 0$ on the Paley–Wiener class $\text{PW}(\mathbb{R})$ (Chapter 56.1.3). The fourth input to the sufficient RH path of Chapter 22, HP–Li positivity, is the full-strength Weil positive trace inequality.

A coefficientwise Mellin lift $\Upsilon_F: \text{PW}_F^{\text{hol}} \rightarrow q\mathbb{C}[[q]]$ constructed from a holomorphic cuspidal Hecke eigenform F of weight $k \geq 12$ is a formal probe of the tempered cuspidal core. It is not, for a general Paley–Wiener multiplier $f(\log n)$, a modular form. The Rankin–Selberg kernel attached to a fixed F is the automorphic kernel of $L(s, F \otimes \bar{F})$, whereas the finite term in Weil’s formula is the von Mangoldt kernel of ζ . Thus the prime-side Weil contribution is not made positive by this lift.

The archimedean contribution $W_\infty(f * \tilde{f})$ is the pairing of $|\hat{f}|^2$ against the distribution $\mathcal{T}_\infty(t) = \frac{1}{2\pi} \text{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) + \frac{\log \pi}{2\pi}$, where $\psi = \Gamma'/\Gamma = \partial_s \log \Gamma$ is the digamma function. The Mellin–Petersson isometry pairs $|\hat{f}|^2$ against the Petersson archimedean density $|\Gamma(\frac{1}{4} + \frac{it}{2})|^2$; the Weil archimedean term W_∞ pairs against the logarithmic-derivative density $\text{Re} \psi(\frac{1}{4} + \frac{it}{2})$. These two densities are not proportional, so W_∞ lies outside the isometry. Conjecture A-TP (Chapter 56.7.1) is the full Weil inequality with the archimedean term isolated on one side. It is strictly equivalent to Weil positivity and to RH (Chapter 56.7.2).

56.1 WEIL’S EXPLICIT FORMULA (RECALL)

Paley–Wiener Test Class

Definition 56.1.1 (admissible Paley–Wiener class). . Let $\text{PW}(\mathbb{R})$ denote the space of entire functions $g(s)$ of exponential type such that $g(s) = \hat{f}(s)$ for some even Schwartz-class function $f \in \mathcal{S}(\mathbb{R})$ of compact support, with the Fourier normalisation $\hat{f}(s) = \int_{\mathbb{R}} f(u) e^{isu} du$. For $f \in \text{PW}(\mathbb{R})$, write $\tilde{f}(u) := \overline{f(-u)}$. The convolution $(f * \tilde{f})(u) = \int_{\mathbb{R}} f(v) \tilde{f}(u-v) dv$ has

$$\widehat{f * \tilde{f}}(s) = \hat{f}(s) \overline{\hat{f}(\bar{s})}.$$

On the real axis this is $|\hat{f}(t)|^2$; off the real axis it is not an absolute square. For f real even, $\tilde{f} = f$.

THEOREM 56.1.2 (Weil 1952 explicit formula). . Let $\psi = \Gamma'/\Gamma$ be the digamma function. For $f \in \text{PW}(\mathbb{R})$ with Fourier transform $\hat{f}$, write $\gamma_\rho := -i(\rho - \frac{1}{2}) \in \mathbb{C}$ for each non-trivial zero ρ of ζ . The Weil distribution satisfies

$$\sum_{\rho} \hat{f}(\gamma_\rho) = \hat{f}\left(\frac{1}{2i}\right) + \hat{f}\left(-\frac{1}{2i}\right) - W_\infty(f) - W_{\text{fin}}(f), \quad (56.1)$$

with archimedean and finite-place Weil terms

$$W_\infty(f) := \int_{-\infty}^{\infty} \widehat{f}(t) \cdot \frac{1}{2\pi} \left[\operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) + \log \pi \right] dt, \quad (56.2)$$

$$W_{\text{fin}}(f) := 2 \sum_p \sum_{n \geq 1} (\log p) p^{-n/2} f(n \log p). \quad (56.3)$$

The two polar terms $\widehat{f}(\pm 1/(2i)) = \widehat{f}(\mp i/2)$ arise from the simple poles of the completed zeta Λ_ζ at $s = 1, 0$ (the pole of ζ at $s = 1$ and of $\Gamma(s/2)$ at $s = 0$); the factor $s(s-1)/2$ in $\xi_{\mathbb{R}} = \frac{1}{2}s(s-1)\Lambda_\zeta$ cancels them, so $\xi_{\mathbb{R}}$ itself is entire.

Proof. Weil 1952 Selecta (*Sur les formules explicites de la théorie des nombres premiers*, reprinted in *Oeuvres Scientifiques* Vol. II) computes the distribution by residue calculus applied to $-\frac{1}{2\pi i} \oint \widehat{f} d \log \xi_{\mathbb{R}}$ on a symmetric contour; the trivial-zero gamma factors produce W_∞ and the Euler product yields W_{fin} . Modern references: Iwaniec–Kowalski *Analytic Number Theory* (AMS 2004) Thm. 5.12; Patterson *An Introduction to the Theory of the Riemann Zeta-Function* (Cambridge 1988) Thm. 3.1. $\square$

COROLLARY 56.1.3 (*Weil positivity as RH*). . RH is equivalent to

$$W(f * \widetilde{f}) \geq 0 \quad \text{for all } f \in \text{PW}(\mathbb{R}),$$

where $W(g) := \widehat{g}(\frac{1}{2i}) + \widehat{g}(-\frac{1}{2i}) - W_\infty(g) - W_{\text{fin}}(g)$ denotes the right-hand side of (56.1). Under RH, $\gamma_\rho \in \mathbb{R}$ and $W(f * \widetilde{f}) = \sum_\rho |\widehat{f}(\gamma_\rho)|^2 \geq 0$; the converse is Weil's positivity equivalence.

Proof. Weil 1952 §IV. The key identity is $(f * \widetilde{f})(t) = |\widehat{f}(t)|^2 \geq 0$; on the critical line $\operatorname{Re} s = 1/2$ the Weil distribution becomes a sum of non-negative values, and the off-critical locus contributes a negative term by Hadamard product considerations. $\square$

Li Coefficients as Weil Distribution Values

THEOREM 56.1.4 (*Li 1997, J. Number Theory 65*). . RH is equivalent to $\lambda_n \geq 0$ for every integer $n \geq 1$, where

$$\lambda_n = \sum_\rho \left(1 - \left(1 - \frac{1}{\rho} \right)^n \right),$$

summed over non-trivial zeros with multiplicity.

PROPOSITION 56.1.5 (*Li coefficients and the Paley–Wiener boundary*). . The Li coefficients are values of the zero divisor against the rational functions $1 - (1 - 1/s)^n$. This is a Weil-type distributional statement, not a compact-support Paley–Wiener interpolation theorem. The existence of $\phi_n \in \text{PW}(\mathbb{R})$ with prescribed values $\widehat{\phi}_n(-i(\rho - \frac{1}{2})) = 1 - (1 - 1/\rho)^n$ at all non-trivial zeros is an additional interpolation theorem, not a consequence of Li's criterion.

Proof. Li's criterion and the Bombieri–Lagarias form give the displayed rational test on the zero set. Paley–Wiener interpolation at the zero divisor would require an exponential-type interpolation theorem with growth constants matched to the Riemann–von Mangoldt density. Li's criterion supplies the rational values on the divisor; it does not produce a compactly supported Paley–Wiener representative. $\square$

56.2 PETERSSON ON THE TEMPERED CUSPIDAL CORE (RECALL)

THEOREM 56.2.1 (*positive Petersson on tempered cuspidal core*). (cite Chapter 45.6.1). The adelic Petersson–Tamagawa pairing

$$\langle v, w \rangle^{\text{Pet}} = \int_{[GL_2]} v(g) \overline{w(g)} dg \quad (56.4)$$

is a positive-definite Hermitian form on the tempered cuspidal finite-place sector $\mathcal{V}_{\text{cusp,temp}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$, and coincides with the Koszul pairing composed with the Bost–Connes modular involution: $\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}$.

Proof. Chapter 45.6.1. Positivity is the classical L^2 -positivity of the Petersson integral; the Koszul identification is the full-sector Rankin–Selberg matching of Chapter 45. $\square$

Remark 56.2.2 (restricted tensor decomposition). Under the Tate restricted tensor product of Chapter 28.1.3, $\mathcal{V}^{\text{prim}} = \mathcal{H}^{(\infty)} \otimes \bigotimes'_p \mathcal{H}^{(p)}$, and $\langle \cdot, \cdot \rangle^{\text{Pet}}$ decomposes as a restricted product of local pairings: at ∞ , the Heisenberg Fock pairing $\langle \cdot, \cdot \rangle_{\infty}^{\text{Pet}}$ of Frenkel–Ben-Zvi §3.5 (explicit contraction on Schur basis); at each finite p , the GNS pairing of Chapter 45.2.5, namely $\langle \delta_m, \delta_n \rangle_p = \delta_{m,n}/m$. The finite Euler product is the Rankin–Selberg integral of a cuspidal Whittaker vector.

56.3 THE MELLIN LIFT AND RANKIN–SELBERG KERNEL CHECK

The prime-side Weil contribution $W_{\text{fin}}(f * \tilde{f})$ of (56.3) is a sum over prime powers of a Paley–Wiener weight. A Petersson proof of its positivity would require a linear map from $\text{PW}(\mathbb{R})$ to the tempered cuspidal core whose Petersson norm computes the von Mangoldt prime-power sum. The coefficientwise lift Υ_F , defined below, does not supply such a map without an additional automorphic multiplier theorem.

Definition of Υ

Definition 56.3.1 (*Mellin lift*). . Fix a normalised holomorphic cuspidal Hecke eigenform $F \in S_k(\text{SL}_2(\mathbb{Z}))$ of weight $k \geq 12$, with Fourier expansion $F(z) = \sum_{n \geq 1} a_F(n) e^{2\pi i n z}$ and Hecke normalisation $a_F(1) = 1$. For $f \in \text{PW}(\mathbb{R})$, define

$$\Upsilon_F(f)(z) := \sum_{n \geq 1} f(\log n) \cdot a_F(n) \cdot n^{(k-1)/2} e^{2\pi i n z}, \quad (56.5)$$

a modular-type formal series on the upper half-plane.

By Deligne’s bound for holomorphic cuspidal F (Deligne 1974 Publ. IHES 43), the normalised Hecke eigenvalues satisfy $|a_F(p) p^{-(k-1)/2}| \leq 2$. This is a growth estimate for the coefficient sequence; it is not an automorphy theorem for an arbitrary coefficientwise multiplier $f(\log n)$.

LEMMA 56.3.2 (*coefficientwise lift is only a formal automorphic probe*). . Let $\text{PW}_F^{\text{hol}} \subset \text{PW}(\mathbb{R})$ denote the subspace of $f \in \text{PW}(\mathbb{R})$ satisfying:

- (H1) f is real-valued and even: $\tilde{f} = f$;
- (H2) the Mellin transform $\tilde{f}_M(s) := \int_0^\infty f(\log u) u^{s-1} du$ is holomorphic and of polynomial decay on the strip $|\text{Re } s - 1/2| \leq (k-1)/2 - \varepsilon$ for some $\varepsilon > 0$;
- (H3) f ’s support avoids the Tate-polar contribution in the sense: $\tilde{f}_M(s)$ has a double zero at $s = 0$ and $s = 1$.

Then $\Upsilon_F(f)$ is a convergent holomorphic q -series on the upper half-plane under the stated decay hypotheses, with coefficients bounded by Deligne's estimate. It is a modular form of weight k only under the additional condition that the multiplier $a_F(n) \mapsto f(\log n)a_F(n)$ is induced by an endomorphism of the automorphic representation generated by F . Conditions (H1)–(H3) do not imply this condition.

Proof. Under (H2) and Deligne's bound $|a_F(n)| \leq \sigma_0(n)n^{(k-1)/2}$ (Deligne 1974), the series (56.5) converges absolutely on the upper half-plane to a holomorphic function with no constant term. Modularity is not a coefficientwise condition. The functional equation of $L(s, F)$ controls the original eigenform F ; it does not imply that the Hadamard product of its Fourier coefficients with the sequence $f(\log n)$ transforms under $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Thus the lift is a formal probe until an automorphic multiplier theorem is supplied. $\square$

Remark 56.3.3 (domain of the restricted class). PW_F^{hol} is a restricted test class inside $\text{PW}(\mathbb{R})$. No density assertion for a fixed F is used. Condition (H2) restricts the spectral width (the Paley–Wiener support must sit inside the strip where the weight- k Mellin transform is holomorphic), and (H3) removes the Tate polar locus. (H2) is a *genuine* restriction whose complement is not reached by a fixed holomorphic F -lift (Chapter 56.5). (H3) is removable by absorbing the two Tate poles into the archimedean term via the entire form $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\Lambda_{\zeta}(s)$; see Chapter 56.6.2.

Rankin–Selberg Integral and the Kernel Mismatch

The Petersson norm of a genuine cusp form admits a Rankin–Selberg expansion. That expansion is not the finite-place Weil contribution for ζ .

THEOREM 56.3.4 (*Rankin–Selberg kernel does not compute the zeta prime term*). . Suppose that $\Upsilon_F(f)$ is in fact a weight- k cusp form. Its Petersson norm is governed by the Rankin–Selberg convolution of the automorphic representation generated by F . In any such identity the spectral kernel is built from

$$K_F(s) := \frac{\zeta(2s)}{\zeta(s)\xi_{\mathbb{R}}(s)\xi_{\mathbb{R}}(1-s)} \cdot L(s, F \otimes \bar{F}) \quad (56.6)$$

up to normalising gamma and conductor factors. The finite-place term in Weil's explicit formula is instead the von Mangoldt distribution

$$W_{\text{fin}}(\phi) = 2 \sum_p \sum_{n \geq 1} (\log p) p^{-n/2} \phi(n \log p).$$

No Rankin–Selberg identity identifies $W_{\text{fin}}(f * \tilde{f})$ with a positive Petersson norm attached to a fixed F .

Proof. Rankin's unfolding computes Dirichlet series with coefficients $|a_F(n)|^2$, hence $L(s, F \otimes \bar{F})$ after the usual normalisation. Weil's finite distribution has coefficients $\Lambda(n)n^{-1/2}$ and is the logarithmic derivative of ζ . These coefficient systems are different. Positivity of a Petersson norm therefore cannot be transported to W_{fin} without an additional trace formula identifying the two kernels. $\square$

Remark 56.3.5 (convention absorbing the Tate gamma). The kernel $K_F(s)$ in (56.6) is written with archimedean Tate gamma factors on both top and bottom to exhibit the Hecke functional equation symmetrically. The $\xi_{\mathbb{R}}(s)\xi_{\mathbb{R}}(1-s)$ in the denominator is part of the automorphic normalisation of the Rankin–Selberg kernel. It is not the statement that the zeta prime distribution W_{fin} has been absorbed into the Petersson form.

56.4 THE PRIME-SIDE OBSTRUCTION

The comparison with Rankin–Selberg exposes a kernel mismatch.

THEOREM 56.4.1 (*no prime-side positivity from a fixed F -lift*). . The identity

$$W_{\text{fin}}(f * \tilde{f}) = 2 \langle \Upsilon_F(f), \Upsilon_F(f) \rangle_{F\text{-normalised}}^{\text{Pet}}$$

does not follow from Rankin–Selberg unfolding for a fixed cusp form F . Consequently fixed- F Rankin–Selberg positivity does not imply $W_{\text{fin}}(f * \tilde{f}) \geq 0$ on PW_F^{hol} .

Proof. The left-hand side has coefficients $\Lambda(n)n^{-1/2}$ in the additive variable $\log n$. The Rankin–Selberg norm of a fixed F has diagonal coefficients $|a_F(n)|^2$ after unfolding. Equality for all Paley–Wiener tests would force these coefficient systems to agree as distributions, which they do not. Positivity of Petersson norm is therefore irrelevant to the sign of the zeta finite-place Weil term unless a separate trace formula is inserted. $\square$

COROLLARY 56.4.2 (*Li approximants do not correct the prime kernel mismatch*). . Even if a Li test admits a PW_F^{hol} -representative, the non-negativity of $W_{\text{fin}}(\phi * \tilde{\phi})$ is not a consequence of the fixed- F Rankin–Selberg norm. It requires an independent trace formula for the von Mangoldt kernel.

Proof. This is exactly the obstruction isolated in Chapter 56.4.1. The Li test changes the test function; it does not change the kernel from $\Lambda(n)$ to $|a_F(n)|^2$. $\square$

56.5 THE IRREDUCIBLE OBSTRUCTION: ARCHIMEDEAN TATE DISTRIBUTION AND EISENSTEIN

The fixed- F Rankin–Selberg norm does not identify $W_{\text{fin}}(f * \tilde{f})$ with a Petersson norm. The archimedean Weil contribution $W_{\infty}(f * \tilde{f})$ remains outside this identification: the Petersson archimedean density is $|\Gamma(\frac{1}{4} + \frac{it}{2})|^2$, while W_{∞} pairs against $\text{Re } \psi(\frac{1}{4} + \frac{it}{2}) = \partial_s \log |\Gamma(s/2 + 1/4)|^2|_{s=it}$, the logarithmic derivative. The positivity of $W(f * \tilde{f})$ is the full Weil inequality; the archimedean term, the finite von Mangoldt term, and the two polar terms cannot be separated into independent positivity assertions.

Archimedean Tate Distribution

DEFINITION 56.5.1 (*archimedean Tate distribution*). . The archimedean Weil contribution W_{∞} of (56.2) is the pairing of $\hat{f}$ against the distribution

$$\mathcal{T}_{\infty}(t) := \frac{1}{2\pi} \text{Re } \psi\left(\frac{1}{4} + \frac{it}{2}\right) + \frac{\log \pi}{2\pi}. \quad (56.7)$$

This is the archimedean $\mathbb{R}$ -factor of the completed zeta function: Tate’s thesis (Tate 1950) assigns to each place v of $\mathbb{Q}$ a local zeta distribution, and the archimedean factor $\Lambda_{\infty}(s) = \pi^{-s/2} \Gamma(s/2)$ contributes $\mathcal{T}_{\infty}$ in the sign convention of (56.2).

PROPOSITION 56.5.2 (*archimedean distribution outside a fixed Rankin–Selberg norm*). . The coefficientwise lift $\Upsilon_F: \text{PW}_F^{\text{hol}} \rightarrow \mathcal{V}_{\text{cusp,temp}}^{\text{prim}}$ of Chapter 56.3.1 does not absorb the archimedean distribution $\mathcal{T}_{\infty}$. Specifically, the restriction of the Petersson form to the archimedean factor $\mathcal{H}^{(\infty)} \subset \mathcal{V}^{\text{prim}}$ pairs with $|\hat{f}(t)|^2$ against the $|\Gamma(\frac{1}{4} + \frac{it}{2})|^2$ density, while $\mathcal{T}_{\infty}$ pairs against the logarithmic-derivative density $\text{Re } \psi(\frac{1}{4} + \frac{it}{2})$; the two densities are not proportional.

Proof. At ∞ , $\mathcal{H}^{(\infty)}$ is the standard Heisenberg Fock on $\mathbb{C}[\alpha_{-1}, \alpha_{-2}, \dots]|\mathbf{o}\rangle_\infty$ (Frenkel–Ben-Zvi §3.5). The Petersson-restriction at ∞ uses the archimedean local Rankin–Selberg integral, which by Jacquet 1972 SLN 260 §4 (archimedean local factors for GL_2) evaluates to a Gamma-absolute-value kernel $|\Gamma(\frac{1}{4} + \frac{it}{2})|^2 \cdot |\widehat{f}(t)|^2$. This is positive and forms part of $K_F(s)$ in (56.6).

The Weil archimedean distribution $\mathcal{T}_\infty$ pairs $|\widehat{f}(t)|^2$ against $\operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2})$, the logarithmic derivative of $\Gamma(\frac{1}{4} + \frac{it}{2})$. The relation between the Gamma absolute value and the digamma real part is $|\Gamma(\frac{1}{4} + \frac{it}{2})|^2 \cdot \frac{d}{dt} \arg \Gamma(\frac{1}{4} + \frac{it}{2}) = \operatorname{Im} \psi(\frac{1}{4} + \frac{it}{2}) \cdot |\Gamma(\frac{1}{4} + \frac{it}{2})|^2$, not the real part. Hence the Petersson archimedean density and $\mathcal{T}_\infty$ differ; the former is manifestly positive, the latter is not sign-definite (it takes both signs as $t \rightarrow \infty$ by Stirling: $\operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) \sim \frac{1}{2} \log(t^2/4)$ for $|t| \rightarrow \infty$, positive for large t ; but is negative near $t = 0$). $\square$

Remark 56.5.3 (archimedean Tate density and conformal anomaly). The archimedean digamma contribution $\mathcal{T}_\infty$ is the Tate-local density for the real place: Tate’s thesis assigns to $\Lambda_\infty(s) = \pi^{-s/2} \Gamma(s/2)$ the local factor whose logarithmic derivative is $\frac{1}{2} \psi(s/2) - \frac{1}{2} \log \pi$; after moving the local term to the side used in (56.1), the distribution is written as in (56.7). The identity $\psi = \Gamma'/\Gamma = \partial_s \log \Gamma$ gives $\mathcal{T}_\infty(t) = \frac{1}{2\pi} \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) + \frac{\log \pi}{2\pi}$, the logarithmic derivative of the archimedean Gamma factor evaluated on the critical line. The Petersson weight y^{k-2} in (56.4) pairs against the modular form and produces $|\Gamma(\frac{1}{4} + \frac{it}{2})|^2$, the square absolute value, not its logarithmic derivative. The two quantities are related by the Bohr–Mollerup identity, but they differ as distributions on $\mathbb{R}$; the archimedean conformal-anomaly contribution at (Chapter 14) is the logarithmic-derivative side, outside the Petersson integral’s reach.

Eisenstein Continuum and Non-Tempered Locus

PROPOSITION 56.5.4 (Eisenstein density mismatch). (*cite Chapter 45.5.5*). The Koszul–Pet compatibility Chapter 56.2.1 holds only on the tempered cuspidal finite-place sector. The Eisenstein continuous spectrum carries a genuine density mismatch between the Koszul side (density $\xi(2s)/\xi(2s-1)$) and the Petersson side (Maass–Selberg scattering normalisation on the critical line). The Li coefficients see only the zero divisor and its Weil–Connes distribution; the Eisenstein continuum lies outside the Koszul–Petersson match. Isolating the discrete part requires an explicit spectral projection operator P_{disc} , not provided by the Mellin lift alone.

Proof. See Chapter 45.5.5 for the density mismatch; Moeglin–Waldspurger 1995 Ch. II for the Eisenstein L^2 -decomposition of $GL_2/\mathbb{Q}$; Arthur 1980 Duke Math. J. 45 for the spectral projection. $\square$

Remark 56.5.5 (non-tempered residual). For Maass cusp forms, temperedness is conditional on the Ramanujan conjecture for $GL_2/\mathbb{Q}$; the current best bound is $7/64$ (Kim–Sarnak 2003). The non-tempered locus contributes a controlled correction to $K_F(s)$ at $\operatorname{Re} s = 1/2 \pm \theta$ for $\theta \leq 7/64$, but the Rankin–Selberg argument gives no positive-definite form there. The positive automorphic Rankin–Selberg statement on non-tempered Maass forms is therefore conditional on the full Ramanujan conjecture. This does not address the zeta finite-place Weil term.

56.6 RESTRICTED-CLASS THEOREM: HP-LI ON THE HOLOMORPHIC CUSPIDAL TEMPERED CLASS

THEOREM 56.6.1 (*HP-Li restricted-class identity*). . Fix a normalised holomorphic cuspidal Hecke eigenform $F \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ of weight $k \geq 12$. For any $f \in \mathrm{PW}_F^{\mathrm{hol}}$ satisfying (H1)–(H3) of Chapter 56.3.2, the symmetrised test function $f * \tilde{f}$ satisfies the identity

$$\widehat{f * \tilde{f}}\left(\frac{1}{2i}\right) + \widehat{f * \tilde{f}}\left(-\frac{1}{2i}\right) - W_{\mathrm{fin}}(f * \tilde{f}) = W(f * \tilde{f}) + W_{\infty}(f * \tilde{f}). \quad (56.8)$$

Equivalently: the sum of polar contributions minus the finite-prime term splits into the Weil functional and the archimedean Tate distribution. This is only the Weil explicit formula rearranged; it is not a positivity theorem.

Proof. The Weil explicit formula (56.1) applied to $f * \tilde{f}$ gives

$$W(f * \tilde{f}) = \widehat{f * \tilde{f}}\left(\frac{1}{2i}\right) + \widehat{f * \tilde{f}}\left(-\frac{1}{2i}\right) - W_{\infty}(f * \tilde{f}) - W_{\mathrm{fin}}(f * \tilde{f}).$$

Moving W_{∞} to the other side gives (56.8). $\square$

Remark 56.6.2 (*H3 is removable*). Hypothesis (H3) (double vanishing of the Mellin transform at $s = 0, 1$) is an artifact of the normalisation $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\Lambda_{\xi}(s)$: working directly with $\xi_{\mathbb{R}}$ instead of Λ_{ξ} absorbs the Tate poles. Therefore (H3) is removable in the sense that its violation is quantifiable: if $\tilde{f}_M(s)$ has zeros of order $(a, b) \leq (2, 2)$ at $s = 0, 1$, the polar contribution to $W(f * \tilde{f})$ is corrected by a $2(2-a) + 2(2-b)$ -fold Taylor expansion at $s = 0, 1$, each term a specific derivative of $\xi_{\mathbb{R}}$ at these points. The correction is a finite-rank bounded operator on $\mathrm{PW}(\mathbb{R})$, hence does not affect the positivity question on the main spectral locus.

Remark 56.6.3 (*H2 is the genuine obstruction*). Hypothesis (H2) (holomorphy of $\tilde{f}_M$ on the strip of width $(k-1)$) is the genuine restriction. In the limit $k \rightarrow \infty$ the strip widens to all of $\mathbb{C}$, but no single finite-weight F gives access to the full Paley–Wiener cone. The full cone requires either (i) a non-holomorphic lift to Maass forms (where K_F 's positivity becomes conditional on Ramanujan), or (ii) a limiting procedure as $k \rightarrow \infty$ which is problematic at the level of the Petersson measure (the Plancherel weight vanishes). The limiting procedure is precisely the Selberg trace formula in disguise, and its positivity is Weil positivity itself.

56.7 THE ARCHIMEDEAN TATE POSITIVITY CONJECTURE

Conjecture 56.7.1 (*archimedean Tate positivity, A-TP*). For every $f \in \mathrm{PW}(\mathbb{R})$,

$$W_{\infty}(f * \tilde{f}) \leq \widehat{f * \tilde{f}}\left(\frac{1}{2i}\right) + \widehat{f * \tilde{f}}\left(-\frac{1}{2i}\right) - W_{\mathrm{fin}}(f * \tilde{f}). \quad (56.9)$$

Equivalently, $W(f * \tilde{f}) \geq 0$. Equivalently: RH.

THEOREM 56.7.2 (*reformulation of A-TP*). . Conjecture A-TP (Chapter 56.7.1) is strictly equivalent to Weil positivity Chapter 56.1.3, and hence to RH.

Proof. By (56.1),

$$\widehat{f * \tilde{f}}\left(\frac{1}{2i}\right) + \widehat{f * \tilde{f}}\left(-\frac{1}{2i}\right) - W_{\mathrm{fin}}(f * \tilde{f}) - W_{\infty}(f * \tilde{f}) = W(f * \tilde{f}).$$

Thus (56.9) is exactly $W(f * \tilde{f}) \geq 0$. The equivalence to RH is Chapter 56.1.3. $\square$

Remark 56.7.3 (domain of the Mellin-Rankin-Selberg isometry). The Mellin-Rankin-Selberg isometry gives positivity for automorphic Rankin-Selberg kernels. It does not determine the zeta finite-place Weil distribution. The residual is therefore not merely the sign of W_∞ ; the full residual is the inequality

$$W_\infty(f * \tilde{f}) \leq \widehat{f * \tilde{f}}(\tfrac{1}{2i}) + \widehat{f * \tilde{f}}(-\tfrac{1}{2i}) - W_{\text{fin}}(f * \tilde{f}),$$

which is exactly Weil positivity.

56.8 HP-LI BOUNDARY

The decomposition of HP-Li positivity separates identities from positivity. The prime-side Weil contribution $W_{\text{fin}}(f * \tilde{f})$ is not made positive by the fixed Rankin-Selberg lift of Chapter 56.4.1. The polar terms $\widehat{f * \tilde{f}}(\pm 1/(2i))$ are explicit finite Taylor corrections (Chapter 56.6.2). The full A-TP inequality with W_∞ isolated is RH-equivalent (Chapter 56.7.1, Chapter 56.7.2). The extension to Maass cuspidal tempered forms is conditional on the Ramanujan conjecture (Chapter 56.5.5). The Eisenstein continuous spectrum is domain-excluded by the density mismatch of Chapter 56.5.4. The full PW($\mathbb{R}$) class is obstructed by (H2) (Chapter 56.6.3).

THEOREM 56.8.1 (*HP-Li positivity boundary after the kernel check*). . HP-Li positivity (Chapter 22.3.10) does not follow from the fixed- F Rankin-Selberg lift. That lift gives an automorphic comparison, but it does not identify the von Mangoldt finite-place kernel with a Petersson norm. The RH-equivalent positivity statement is the full archimedean Tate inequality Chapter 56.7.1, equivalently Weil positivity, equivalently RH.

Proof. Assemble: Chapter 56.4.1, Chapter 56.3.4, Chapter 56.6.1, Chapter 56.5.2, Chapter 56.7.2. The first two results identify the kernel mismatch; the third is only an algebraic rearrangement of Weil's formula; the final result records the RH-equivalent positivity statement. $\square$

COROLLARY 56.8.2 (*HP-Li inputs in the sufficient RH path*). . The sufficient RH path of Chapter 22.6.1 requires the following HP-Li inputs:

- a trace formula for the zeta von Mangoldt kernel beyond the fixed- F lift;
- the A-TP inequality, which is RH-equivalent (Chapter 56.7.1).
- an extension beyond PW_F^{hol} , either through Maass lifts with Ramanujan control or through a weight- $k \rightarrow \infty$ limit with the Petersson limit measure under control.

The RH-equivalent residual among these inputs is A-TP:

$$W_\infty(f * \tilde{f}) \leq \widehat{f * \tilde{f}}(\tfrac{1}{2i}) + \widehat{f * \tilde{f}}(-\tfrac{1}{2i}) - W_{\text{fin}}(f * \tilde{f}),$$

which is equivalent to RH. The prime-side comparison and the extension beyond PW_F^{hol} remain separate domain conditions. The HP-Li obstruction is the full Weil inequality with finite, polar, and archimedean terms kept in their correct places.

Proof. Combine Chapter 56.8.1 with the sufficient-path theorem Chapter 22.6.1. $\square$

Remark 56.8.3 (decomposition of HP-Li positivity). The kernel identity separates Weil positivity into four constituents:

- (i) The finite-prime term $W_{\text{fin}}(f * \tilde{f})$: a von Mangoldt distribution not identified here with a fixed- F Petersson norm.
- (ii) The archimedean term: the Tate $\mathcal{T}_{\infty}$ -distribution $\psi(\frac{1}{4} + \frac{it}{2}) = \partial_s \log \Gamma|_{s=1/4+it/2}$, which is sign-indefinite.
- (iii) The decomposition identity $W(f * \tilde{f}) = \widehat{f * \tilde{f}}(\frac{1}{2i}) + \widehat{f * \tilde{f}}(-\frac{1}{2i}) - W_{\text{fin}}(f * \tilde{f}) - W_{\infty}(f * \tilde{f})$.
- (iv) The test-function class PW_F^{hol} , a restricted subspace of $\text{PW}(\mathbb{R})$, on which the full HP–Li inequality is not implied by the fixed- F lift.

HP–Li on the full Paley–Wiener class is equivalent to RH.

56.9 BEILINSON BOUNDARY

- (i) The coefficientwise Mellin lift $\Upsilon_F: \text{PW}_F^{\text{hol}} \rightarrow q\mathbb{C}[[q]]$ of Chapter 56.3.1 is a formal probe; it is not a modular-form-valued map for an arbitrary Paley–Wiener multiplier.
- (ii) The Rankin–Selberg kernel Chapter 56.3.4 is built from $L(s, F \otimes \overline{F})$, not from $-\zeta'/\zeta$.
- (iii) The prime-side Weil contribution is not proved positive by the fixed- F Petersson norm (Chapter 56.4.1).
- (iv) The irreducible obstruction to full HP–Li is the full A-TP inequality, including polar, finite, and archimedean terms; it is strictly equivalent to Weil positivity and to RH (Chapter 56.7.1, Chapter 56.7.2).
- (v) The Eisenstein continuous spectrum is domain-excluded by the density mismatch of Chapter 45.5.5; it does not contribute to the Li coefficients, which see only the discrete spectrum.
- (vi) HP–Li form (Chapter 56.8.1): the full HP–Li inequality is RH-equivalent.

Remark 56.9.1 (Beilinson principle: restricted-class HP–Li). The restricted-class HP–Li theorem is an identity together with a kernel obstruction. The full claim, HP–Li on all of $\text{PW}(\mathbb{R})$, is equivalent to RH and remains open as Conjecture A-TP (Chapter 56.7.1). The lift Υ_F is a useful formal comparison, not a proof that the von Mangoldt kernel is a positive Petersson norm.

56.10 BIBLIOGRAPHY OF PRIMARY SOURCES

The identities above use the following primary sources, disjoint from the proof sources used in Chapters 21 (P1–P3 resolution), 25 (Koszul–Pet compatibility), and 45 (KMS₁/GNS full sector).

Weil 1952. Weil, A. *Sur les formules explicites de la théorie des nombres premiers*. *Meddelanden från Lunds Univ. Mat. Sem.* (Festschrift für Marcel Riesz), 1952; reprinted *Oeuvres Scientifiques* Vol. II, Springer 1979. Explicit formula, W -distribution, Weil positivity equivalence to RH.

Li 1997. Li, X.-J. *The positivity of a sequence of numbers and the Riemann hypothesis*. *J. Number Theory* 65 (1997), 325–333. Li coefficients λ_n , equivalence to RH, explicit Paley–Wiener realisation.

Rankin 1939. Rankin, R. A. *Contributions to the theory of Ramanujan’s function $\tau(n)$ and similar arithmetical functions*. *Proc. Camb. Phil. Soc.* 35 (1939), 357–372. Rankin–Selberg identity for L -series of modular forms.

Selberg 1940. Selberg, A. *Bemerkungen über eine Dirichletsche Reihe, die mit der Theorie der Modulformen nahe verbunden ist*. *Arch. Math. Naturvid.* 43 (1940), 47–50. Rankin–Selberg identity and L -function analytic continuation.

Hecke 1936. Hecke, E. *Über die Bestimmung Dirichletscher Reihen durch ihre Funktionalgleichung.* *Math. Ann.* 112 (1936), 664–699. Mellin transform / modular form correspondence and functional equation.

Deligne 1974. Deligne, P. *La conjecture de Weil I.* *Publ. Math. IHES* 43 (1974), 273–307. Ramanujan bound for holomorphic cuspidal Hecke eigenforms of weight $k \geq 2$.

Jacquet 1972. Jacquet, H. *Automorphic forms on GL_2 II.* Lecture Notes in Mathematics 278, Springer 1972. Local archimedean factors for GL_2 ; ramified principal series.

Jacquet–Langlands SLN 114 1970. Jacquet, H.; Langlands, R. P. *Automorphic forms on $GL(2)$.* Lecture Notes in Mathematics 114, Springer 1970. Adelic Whittaker model, Hecke theory.

Iwaniec–Kowalski 2004. Iwaniec, H.; Kowalski, E. *Analytic Number Theory.* AMS Colloquium Publications 53, 2004. Modern account of Weil’s explicit formula, Rankin–Selberg, Paley–Wiener.

Bombieri–Lagarias 1999. Bombieri, E.; Lagarias, J. C. *Complements to Li’s criterion for the Riemann hypothesis.* *J. Number Theory* 77 (1999), 274–287. Alternative constructions of Li-coefficient approximants and equivalences to Weil positivity.

Connes 1999. Connes, A. *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function.* *Selecta Math. (N. S.)* 5 (1999), 29–106, arXiv:math/9811068. Adele class space, absorption trace, Weil positivity as trace-formula positivity.

Patterson 1988. Patterson, S. J. *An Introduction to the Theory of the Riemann Zeta-Function.* Cambridge Studies in Advanced Mathematics 14, 1988. Complete treatment of the Hadamard product and explicit formula.

Kim–Sarnak 2003. Kim, H. H.; Sarnak, P. C. Appendix to Kim, H. H., *Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2 .* *J. Amer. Math. Soc.* 16 (2003), 139–183. Bound $|a_F(p)| \leq 2p^{7/64}$ towards the Ramanujan conjecture for $GL_2/\mathbb{Q}$.

Bump 1997. Bump, D. *Automorphic Forms and Representations.* Cambridge Studies in Advanced Mathematics 55, 1997. Comprehensive account of Rankin–Selberg, Whittaker models, Petersson theory.

Zagier 1981. Zagier, D. *The Rankin–Selberg method for automorphic functions which are not of rapid decay.* *Bonner Math. Schriften* 121 (1981). Rankin–Selberg unfolding technique.

Disjointness comparison. The primary-source list for the identities above (Weil 1952 for the explicit formula; Li 1997 for Li coefficients; Rankin 1939 and Selberg 1940 for the Rankin–Selberg identity; Hecke 1936 for Mellin–functional-equation duality; Deligne 1974 for temperedness; Jacquet 1972 and Jacquet–Langlands SLN 114 for adelic GL_2 representation theory) is disjoint from the lists underwriting P1 (Beilinson–Drinfeld 2004, Positselski 2009), P2 (Tate 1950), P3 / Koszul–Pet (Bost–Connes 1995, Takesaki 1979, Jacquet–Piatetski-Shapiro–Shalika 1983, Godement–Jacquet SLN 260, Mœglin–Waldspurger 1995, Arthur 1980, Shahidi 2010), the Tate restricted tensor chapter (Lurie HA, Fresse 2017, Ayala–Francis), and the HP^{Ar} / VB^* converse lists (Connes 1999 is a cross-reference cited for context in both arguments; it does not appear on both sides of any single identification).

VB* MAIN DIRECTION VIA HAMBURGER POSITIVITY, DETERMINACY, AND HB⁺/BD⁺

Chapter 35 isolates the VB converse (zero-placement $\Rightarrow$ VB*) as a conditional comparison between the spectral-regularity hypotheses, operator-positive Hermite–Biehler de Branges structure, and the Báez-Duarte sharp decay condition. In the opposite direction, the VB* main-direction implication says that the four analytic hypotheses (HB⁺, BD⁺, determinacy, positive Hamburger moment structure) on the primary Fock imply that the non-trivial zeros of $\xi_{\mathbb{R}}$ lie on the critical line. The chain is classical complex analysis and de Branges space theory: a de Branges datum whose real part has divisor $\text{div } \Xi$ and whose Hermite–Biehler kernel is positive forces the zeros of Ξ to be real. The positive Hamburger moment structure identifies the spectral measure as a positive measure on the real line; Hamburger determinacy identifies the measure uniquely once it exists; and HB⁺ supplies the real-part divisor constraint. Thus the conclusion is the conditional arrow from these hypotheses to RH. The hypotheses HB⁺, BD⁺, and determinacy are not consequences of the Hadamard product; they are the analytic content of the zero-placement problem.*

57.1 SPECTRAL POSITIVITY CONTEXT

The operator-theoretic comparison (Chapter 35) reduced the VB* converse, under P₂ + P₃ + H-P^{Ar}, to the four-way equivalence

$$(\text{Det}) \Leftrightarrow (\text{HB}^+) \Leftrightarrow (\text{BD}^+) \Leftrightarrow \text{zero-placement} \Rightarrow \text{VB}^*$$

of Chapter 35.4.14. Chapter 40 reduces the Hamburger-determinacy component to the sufficient analytic residual (R_{analytic}) and obstructed a tempting Bombieri–Lagarias shortcut. Neither argument addresses the main-direction implication

$$(\text{HB}^+) + (\text{BD}^+) + (\text{Det}) \implies \text{spectral zeros of } \xi_{\mathbb{R}} \text{ on the critical line,}$$

which is the sufficient direction of the analytic hypothesis of Chapter 36.4.1. The proof is classical complex analysis and de Branges space theory in the sense of the 1968 monograph; no arithmetic input beyond the Hadamard factorisation of $\xi_{\mathbb{R}}$ is used.

Domain

The theorem is the implication from (HB⁺) + (BD⁺) + (Det) + positive Hamburger moment structure to spectral zero-placement on the critical line.

The analytic inputs are HB⁺, BD⁺, (Det), and positive Hamburger moment structure on $\xi_{\mathbb{R}}$. These are RH-equivalent in the Hermite–Biehler case and spectral regularity inputs in the moment-problem cases. They are the subject of Chapters 35 and 40.

Independent Sources

The source list for the VB* converse uses de Branges 1968, Lagarias 1999, Báez-Duarte 2003, Burnol 2000, Akhiezer 1965, Simon 1998, Shohat–Tamarkin 1943. The main-direction implication inherits this list and adds the classical Hamburger existence theorem, Lagarias’ 1999 de Branges survey, Conrey–Li 2000, and Beurling–Malliavin density. The proof sources of constructions 2 and 3 (determinacy reductions, operator-positivity reformulations) are kept disjoint from the complex-analysis sources below:

de Branges, *Hilbert Spaces of Entire Functions* (Prentice-Hall 1968). Foundational monograph; theorems 22–23 (Hermite–Biehler structure), theorem 32 (multiplication operator on $\mathcal{H}(E)$), theorems 40–41 (invariant subspaces of the shift).

Lagarias, *Acta Arith.* 89 (1999). Survey of de Branges reformulations of RH; explicit statement of the main-direction theorem for entire functions of order 1.

Conrey–Li, arXiv:math/9812166 (2000). Construction of de Branges spaces from L -functions; critical discussion of the Hermite–Biehler condition on E_ξ ; counter-examples in the non-self-adjoint case.

Beurling–Malliavin, *Acta Math.* 107 (1962) 291–309. Density theorem: the completeness radius of an exponential system $\{e^{i\lambda_k x}\}$ is computed by Beurling–Malliavin density of $\{\lambda_k\}$.

Hamburger, *Math. Ann.* 81 (1920) 235–319. Original existence theorem: a Hamburger moment problem has a positive representing measure iff all Hankel matrices are positive-semidefinite.

Levin, *Distribution of Zeros of Entire Functions* (AMS 1980), Ch. V–VII. Classical reference on Hermite–Biehler, Hermite–Kakeya–Obreshkov; entire functions of exponential type.

Independence of inputs: the complex analysis (Hadamard factorisation of $\xi_{\mathbb{R}}$, de Branges structure theorem, Beurling–Malliavin density) is not used in the VB*-converse comparison; and that comparison (operator-positivity reformulations, Jacobi-matrix spectral theory, Báez-Duarte decay) is not used in the main-direction implication. The two lists share two foundational references: de Branges 1968 (Hermite–Biehler theory) and Akhiezer 1965 (moment theory, used for the Jacobi matrix and for Hamburger existence and Carleman’s criterion). These shared references serve logically distinct roles: Akhiezer supplies operator spectral methods on one side and classical moment existence. Simon 1998 (spectral theory of Jacobi matrices) appears in the operator-theoretic list and is not cited below; Shohat–Tamarkin 1943 is likewise absent below. The two argument paths are therefore independent on the operative arguments, with the shared foundational references serving independent functions in the two constructions.

57.2 PRIMARY FOCK AND $\xi_{\mathbb{R}}$ AS CHARACTER

Fix the primary Fock $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ of Chapters 22 and 35: the primary sector of the arithmetic Heisenberg Fock, equipped with the Petersson inner product of Chapter 25.10.1. Its character on the archimedean Tate fibre is the completed zeta function

$$\Lambda_{\zeta}(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s),$$

meromorphic with simple poles at $s = 0$ and $s = 1$; the entire Li–Hadamard form is

$$\xi_{\mathbb{R}}(s) := \frac{1}{2} s(s-1) \Lambda_{\zeta}(s).$$

The factor $s(s-1)$ removes the Tate poles; the Gamma factor cancels the negative even trivial zeros of ζ ; the remaining zero set is exactly the non-trivial zero set of ζ . This is the convention fixed in Chapter 22.1.2 and Chapter 22.3.7.

PROPOSITION 57.2.1 ($\xi_{\mathbb{R}}$ is entire of order 1, Hadamard genus 1). (Edwards, *Riemann's Zeta Function* (Academic Press 1974), §I.9). The function $\xi_{\mathbb{R}}(s)$ is entire of order 1 and Hadamard genus 1:

$$\xi_{\mathbb{R}}(s) = e^{a+bs} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho}, \quad (57.1)$$

with explicit constants $a = -\log 2$, $b = -1 - \frac{1}{2}\gamma_E + \frac{1}{2}\log(4\pi)$, product over non-trivial zeros ρ , and $\sum_{\rho} |\rho|^{-2} < +\infty$.

Proof. Classical: the functional equation and $|\xi_{\mathbb{R}}(s)| \leq \exp(C|s| \log |s|)$ uniformly in vertical strips (Stirling bounds on Γ , polynomial control on ζ away from the pole) give order 1. Hadamard genus 1 follows from the zero density estimate of Riemann–von Mangoldt, which gives $\sum_{\rho} |\rho|^{-1-\epsilon} < +\infty$ for every $\epsilon > 0$ but $\sum_{\rho} |\rho|^{-1} = +\infty$. The Hadamard factorisation theorem then gives (57.1) with genus 1. At $s = 0$, the product is 1 and $\xi_{\mathbb{R}}(0) = 1/2$, so $a = -\log 2$. Differentiating $\log \xi_{\mathbb{R}}(s)$ at $s = 0$, using $\zeta(0) = -1/2$, $\zeta'(0) = -(1/2)\log(2\pi)$, and $\Gamma(s/2) = 2s^{-1} - \gamma_E + O(s)$, gives

$$\frac{\xi'_{\mathbb{R}}(0)}{\xi_{\mathbb{R}}(0)} = -1 - \frac{\gamma_E}{2} + \frac{1}{2}\log(4\pi),$$

which is b . □

PROPOSITION 57.2.2 (*functional equation*). . The entire function $\xi_{\mathbb{R}}$ satisfies the functional equation

$$\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s), \quad s \in \mathbb{C}. \quad (57.2)$$

Equivalently, $\xi_{\mathbb{R}}(\frac{1}{2} + iz)$ is an entire function of z with real Taylor coefficients.

Proof. The functional equation for Λ_{ζ} is Riemann 1859 / Tate 1950. The extension to $\xi_{\mathbb{R}}$ follows because $s(s-1) = (1-s)(-s) = (1-s)((1-s)-1)$ is invariant under $s \mapsto 1-s$. The substitution $s = \frac{1}{2} + iz$ turns $s \mapsto 1-s$ into $z \mapsto -z$; combined with $\xi_{\mathbb{R}}(\bar{s}) = \overline{\xi_{\mathbb{R}}(s)}$ (real on the real axis), this gives $\xi_{\mathbb{R}}(\frac{1}{2} + iz) = \overline{\xi_{\mathbb{R}}(\frac{1}{2} - i\bar{z})} = \overline{\xi_{\mathbb{R}}(\frac{1}{2} + i\bar{z})}$ for $z \in \mathbb{R}$; hence the Taylor coefficients at $z = 0$ are real. □

Definition 57.2.3 (*shifted character*). . Write

$$\Xi(z) := \xi_{\mathbb{R}}\left(\frac{1}{2} + iz\right), \quad z \in \mathbb{C}.$$

Then Ξ is entire of order 1 with real Taylor coefficients. Its zero set is

$$\{\gamma_{\rho} : \rho = \frac{1}{2} + i\gamma_{\rho} \text{ non-trivial zero of } \xi_{\mathbb{R}}\},$$

symmetric under $z \mapsto -z$ by (57.2). Under RH the zeros γ_{ρ} are real; unconditionally they are complex and the $z \mapsto -z$ symmetry becomes $\gamma \leftrightarrow -\gamma$ as a multi-set.

Remark 57.2.4 (*the orientation of the problem*). RH is the statement that the zero set of Ξ is a subset of the real line. The Hadamard and functional-equation facts (Chapter 57.2.1, Chapter 57.2.2) are unconditional; they constrain the zero set to be symmetric under $z \mapsto -z$ but not to be real. The role of the analytic hypotheses (HB^+ , BD^+ , determinacy) is to force this realness by real-part divisor control and spectral-measure rigidity.

57.3 HADAMARD FACTORISATION AND THE ZERO MEASURE

Definition 57.3.1 (primary zero multi-set). . The *primary zero multi-set* is

$$\mathcal{Z}_\xi := \{\gamma_\rho \in \mathbb{C} : \rho = \tfrac{1}{2} + i\gamma_\rho \text{ non-trivial zero of } \xi_{\mathbb{R}}\},$$

with multiplicity. Under the symmetrisation $\rho \leftrightarrow 1 - \rho$, the set $\mathcal{Z}_\xi$ is closed under $\gamma \leftrightarrow -\gamma$. Under the complex conjugation $\rho \leftrightarrow \bar{\rho}$, it is closed under $\gamma \leftrightarrow -\bar{\gamma}$; combining, it is closed under $\gamma \leftrightarrow \bar{\gamma}$ and symmetric under the Klein four group $\{1, \gamma \mapsto -\gamma, \gamma \mapsto \bar{\gamma}, \gamma \mapsto -\bar{\gamma}\}$ on the whole plane.

Definition 57.3.2 (primary zero counting measure). . The *primary zero counting measure* is the multi-set measure

$$\mu_\xi := \sum_{\gamma \in \mathcal{Z}_\xi} \delta_\gamma$$

on $\mathbb{C}$ (with multiplicity). Under RH, μ_ξ is supported on $\mathbb{R}$. Under the functional equation (57.2), μ_ξ is $z \mapsto -z$ -invariant. Under the reality of Taylor coefficients (Chapter 57.2.2), μ_ξ is $z \mapsto \bar{z}$ -invariant. The invariant group is thus the Klein four group on $\mathbb{C}$.

Definition 57.3.3 (regularised moment sequence). . For $n \geq 0$, the *regularised n -th moment* of μ_ξ is

$$M_n := \sum_{\gamma \in \mathcal{Z}_\xi}^{\text{reg}} \gamma^n$$

in the sense of Bost–Connes regularisation (zeta-regularised summation in the cutoff height T , with the arithmetic Euler factor subtracted). The regularisation exists and is finite for every $n \geq 0$ by the Riemann–von Mangoldt counting formula $N(T) \sim (T/2\pi) \log(T/2\pi)$ and the Hadamard-factorisation estimate $\sum_\gamma |\gamma|^{-1-\epsilon} < +\infty$ for every $\epsilon > 0$.

PROPOSITION 57.3.4 (reality of moments). . For every $n \geq 0$, $M_n \in \mathbb{R}$. The odd moments M_{2k+1} vanish under the $\gamma \mapsto -\gamma$ symmetry.

Proof. μ_ξ is $\gamma \mapsto \bar{\gamma}$ -invariant, so for every n the complex conjugate of $\sum_\gamma \gamma^n$ equals $\sum_\gamma \bar{\gamma}^n = \sum_\gamma \gamma^n$ (matching the conjugate partner within the multi-set). Hence $M_n \in \mathbb{R}$. For odd $n = 2k+1$, the $\gamma \mapsto -\gamma$ symmetry pairs each γ^{2k+1} with $(-\gamma)^{2k+1} = -\gamma^{2k+1}$, giving cancellation; the regularisation preserves this cancellation. $\square$

Remark 57.3.5 (positivity of the measure is strictly weaker than RH). The measure μ_ξ as a multi-set measure on $\mathbb{C}$ is always positive (it is a counting measure). The question is whether its restriction to $\mathbb{R}$ carries the full mass; that is the content of RH. Positivity of μ_ξ as a measure on $\mathbb{C}$ gives no information about whether the support is on $\mathbb{R}$. This is the reason a positive Hamburger moment representation (a positive representation on $\mathbb{R}$ with prescribed real moments) is strictly finer than mere positivity of μ_ξ .

57.4 HAMBURGER MOMENT PROBLEM FOR μ_ξ

Definition 57.4.1 (Hamburger moment problem for μ_ξ). . The *Hamburger moment problem* for the sequence $\{M_n\}_{n \geq 0}$ of Chapter 57.3.3 is: find a positive Borel measure ν_ξ on $\mathbb{R}$ with

$$M_n = \int_{\mathbb{R}} x^n d\nu_\xi(x) \quad \forall n \geq 0. \quad (57.3)$$

The problem is *solvable* if such ν_ξ exists, and *positive Hamburger* if ν_ξ is a positive measure on $\mathbb{R}$. The Stieltjes problem belongs to the square-pushforward $x \mapsto x^2$, not to the signed ordinate moments themselves.

THEOREM 57.4.2 (*Hamburger's existence theorem*). (Hamburger, Math. Ann. 81 (1920); Akhiezer 1965 §2.1). A moment problem (57.3) is solvable if and only if the infinite Hankel matrix

$$H_\infty = (M_{i+j})_{i,j \geq 0}$$

is positive-semidefinite: $\sum_{i,j} c_i \overline{c_j} M_{i+j} \geq 0$ for every finitely supported sequence (c_i) . Equivalently, every finite truncation $H_N = (M_{i+j})_{0 \leq i,j \leq N}$ has non-negative principal minors.

Definition 57.4.3 (*HB⁺/BD⁺/Det hypothesis*). . The *spectral positivity hypothesis* on the primary Fock is the conjunction of:

- (SH) *Positive Hamburger moment sequence*. The Hankel matrix $H_\infty = (M_{i+j})_{i,j \geq 0}$ of Chapter 57.4.2 is positive-semidefinite.
- (Det) *Hamburger determinacy*. $\sum_{n \geq 1} M_{2n}^{-1/(2n)} = +\infty$ (Carleman's criterion; sufficient for determinacy of ν_ξ).
- (HB⁺) *Hermite–Biehler positivity with divisor control*. There is a de Branges datum $E_\xi(z) = A_\xi(z) - iB_\xi(z)$, with A_ξ, B_ξ real entire, such that E_ξ is Hermite–Biehler, the associated kernel K_{E_ξ} is positive-definite, and the real part $A_\xi = (E_\xi + E_\xi^*)/2$ has divisor $\text{div } \Xi$, with multiplicity.
- (BD⁺) *Beurling–Malliavin super-critical density*. After the HB⁺ real-part theorem has made $\mathcal{Z}_\xi$ real, the associated exponential system $\{e^{i\gamma x}\}_{\gamma \in \mathcal{Z}_\xi}$ is complete in $L^2((-\pi, \pi))$. A sufficient source is $D_{\text{BM}}(\mathcal{Z}_\xi) > 1$; at equality the endpoint completeness assertion is separate data.

PROPOSITION 57.4.4 (*(SH) implies real-positive representing measure*). . Under (SH), the moment problem (57.3) admits at least one positive Borel measure ν_ξ on $\mathbb{R}$ solving it.

Proof. Direct application of Hamburger's existence theorem (Chapter 57.4.2) to the moment sequence $\{M_n\}_{n \geq 0}$ under the positive-semidefiniteness hypothesis (SH). $\square$

Remark 57.4.5 (*positivity is a genuine arithmetic input*). The hypothesis (SH) is not a tautology. The moments M_n are real-valued regularised sums of complex n -th powers (Chapter 57.3.3); positive-semidefiniteness of the Hankel matrix is what transfers the complex multi-set $\mathcal{Z}_\xi$ to a real positive measure. This is the content Hamburger existence requires; it is distinct from the question whether the zeros are real.

57.5 HAMBURGER DETERMINACY AND CARLEMAN

THEOREM 57.5.1 (*Carleman's determinacy criterion*). (Carleman 1926; Akhiezer 1965 §2.4). Let $\{M_n\}_{n \geq 0}$ be a Hamburger moment sequence with positive-semidefinite Hankel matrix and $M_{2n} > 0$ for all $n \geq 0$. If

$$\sum_{n \geq 1} M_{2n}^{-1/(2n)} = +\infty, \tag{57.4}$$

then the representing measure ν_ξ is unique (*determinate*).

PROPOSITION 57.5.2 (*Carleman is not supplied by zero density*). . The Riemann–von Mangoldt zero-counting law and the Bombieri–Lagarias–Conrey estimates do not imply (57.4). Their available consequence is only the Bertrand-scale comparison

$$M_{2n}^{-1/(2n)} \gtrsim \frac{c}{n(\log n)^2}, \quad n \geq 2,$$

which is summable and therefore cannot force Carleman divergence.

Proof. The computation is the one recorded in Chapter 43.1.3. The logarithmic inflation in the regularised moment estimate gives $(\log n)^{-2}$ after the $1/(2n)$ -root and inversion. Thus the comparison series is $\sum_{n \geq 2} 1/(n(\log n)^2)$, which converges. A summable lower comparison is silent about the divergence of the Carleman sum. $\square$

Remark 57.5.3 (*the heuristic is not a theorem*). The moment asymptotics $M_{2n} \asymp C^n n!$ are a saddle-point heuristic. If they were replaced by a rigorous two-sided bound for a positive moment functional, Carleman’s series would have harmonic size. Turning that statement into a theorem is the content of the residual (R_{analytic}) of Chapter 40.4.7. The main-direction implication uses (Det) as a *hypothesis*: the Carleman sum is assumed to diverge. Verifying this divergence is a separate analytic problem.

COROLLARY 57.5.4 (*under (SH) + (Det) the measure ν_ξ is unique*). . Under (SH) + (Det) of Chapter 57.4.3, there exists a unique positive Borel measure ν_ξ on $\mathbb{R}$ with moments (57.3).

Proof. (SH) + Hamburger existence (Chapter 57.4.2) gives solvability. Carleman (Chapter 57.5.1) under (Det) gives uniqueness. The conjunction is the claim. $\square$

57.6 HERMITE–BIEHLER AND THE DE BRANGES SPACE $\mathcal{H}(E_\xi)$

Definition 57.6.1 (*de Branges datum over Ξ*). . A *de Branges datum over Ξ* is a triple (E_ξ, A_ξ, B_ξ) with

$$E_\xi(z) = A_\xi(z) - iB_\xi(z),$$

where A_ξ and B_ξ are real entire functions of finite exponential type. The datum is required to carry a real-part divisor equality

$$\operatorname{div}(A_\xi) = \operatorname{div}(\Xi)$$

on $\mathbb{C}$, with multiplicities. It is not obtained by setting $E_\xi = \Xi$; for the real entire function Ξ one has $\Xi^* = \Xi$, so the strict Hermite–Biehler inequality would be impossible.

Remark 57.6.2 (*conventions match de Branges 1968*). The notation $E = A - iB$ with A and B real-on-real is de Branges’ standard form for candidate Hermite–Biehler functions. The condition for Hermite–Biehler is $|E(z)| > |E^*(z)|$ for $\operatorname{Im} z > 0$, where $E^*(z) = \overline{E(\bar{z})}$. The divisor equality for A_ξ is part of the arithmetic datum; Hermite–Biehler positivity of an unrelated entire function says nothing about zeros of Ξ .

Definition 57.6.3 (*de Branges space $\mathcal{H}(E_\xi)$*). . Assuming $E_\xi = A_\xi - iB_\xi$ is Hermite–Biehler (see (HB⁺) below), the *de Branges space $\mathcal{H}(E_\xi)$* is the reproducing-kernel Hilbert space of entire functions f with $f/E_\xi, f^*/E_\xi \in H^2(\mathbb{C}_+)$, with reproducing kernel

$$K_{E_\xi}(w, z) = \frac{\overline{E_\xi(w)}E_\xi(z) - \overline{E_\xi^*(w)}E_\xi^*(z)}{2\pi i(\bar{w} - z)}. \quad (57.5)$$

PROPOSITION 57.6.4 (*Hermite–Biehler kernel positivity*). (de Branges 1968 Thm. 22; Lagarias 1999 Acta Arith. 89 §3). For an entire function $E = A - iB$ of bounded type in $\mathbb{C}_+$, the following are equivalent:

- (i) E is Hermite–Biehler: $|E(z)| > |E^*(z)|$ for $\text{Im } z > 0$.
- (ii) The reproducing kernel $K_E(w, z)$ of (57.5) is positive-semidefinite: $\sum_{j,k} c_j \overline{c_k} K_E(w_k, w_j) \geq 0$ for every finite collection (w_j, c_j) .

If these conditions hold, the zeros of the real entire functions $A = (E + E^*)/2$ and $B = (E - E^*)/(2i)$ are real; if E has no real zeros, these zeros are simple and interlace.

Proof. (1) $\Leftrightarrow$ (2): de Branges 1968 Thm. 22 proves that K_E is positive-semidefinite exactly when E is Hermite–Biehler. The forward implication uses the Szegő/Blaschke factorisation of $|E|^2 - |E^*|^2$ as a positive harmonic function on $\mathbb{C}_+$; the converse uses that positive-semidefiniteness of K_E at the specific points $w = \bar{z}$ implies $|E(z)|^2 \geq |E^*(z)|^2$ strictly in the open upper half plane. The last assertion is the Hermite–Biehler separation theorem. $\square$

THEOREM 57.6.5 (*Hermite–Biehler real-part control forces zero placement*). Under (HB^+) , the zero multi-set $\mathcal{Z}_\Xi$ of Ξ is contained in $\mathbb{R}$.

Proof. By (HB^+) , $E_\Xi = A_\Xi - iB_\Xi$ is Hermite–Biehler and $\text{div}(A_\Xi) = \text{div}(\Xi)$. The Hermite–Biehler separation theorem gives that every zero of A_Ξ is real. Hence every zero of Ξ is real. $\square$

COROLLARY 57.6.6 (*zero multi-set is real under HB^+*). Under (HB^+) , no non-trivial zero of $\xi_\mathbb{R}$ has $\text{Im } \gamma \neq 0$ in the coordinate $\rho = \frac{1}{2} + i\gamma$.

Proof. Chapter 57.6.5 gives $\gamma \in \mathbb{R}$ for every zero. $\square$

Remark 57.6.7 (*HB^+ is an RH-strength datum*). Chapter 57.6.6 shows that (HB^+) , including real-part divisor control over Ξ , forces every non-trivial zero γ of Ξ to be real. Translated back via $\rho = \frac{1}{2} + i\gamma$, this is exactly $\text{Re } \rho = \frac{1}{2}$. Thus (HB^+) is not a soft positivity assumption: it already contains the de Branges real-part mechanism at RH strength.

57.7 BD⁺: BEURLING–MALLIAVIN SUPER-CRITICAL DENSITY

Definition 57.7.1 (*Beurling–Malliavin sampling density*). (Beurling–Malliavin, Acta Math. 107 (1962)). For a real sequence $\Lambda = \{\lambda_k\} \subset \mathbb{R}$ with $|\lambda_k| \rightarrow \infty$, the *Beurling–Malliavin sampling density* $D_{\text{BM}}(\Lambda)$ is the infimum of $a > 0$ for which there exists a real measurable function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ with

$$\int_{\mathbb{R}} \frac{|\phi(x)|}{1+x^2} dx < +\infty, \quad \phi(\lambda_k) \geq a|\lambda_k| - O(1) \quad \text{for large } |\lambda_k|.$$

For complex $\gamma \in \mathbb{C}$, define $D_{\text{BM}}(\{\gamma_k\})$ by projecting to the real part and applying the real-line definition.

THEOREM 57.7.2 (*Beurling–Malliavin completeness theorem*). (Beurling–Malliavin, Acta Math. 107 (1962); Koosis, *The Logarithmic Integral II* (Cambridge 1992) Ch. VIII). Let $\Lambda = \{\lambda_k\} \subset \mathbb{R}$. The exponential system $\{e^{i\lambda_k x}\}$ is complete in $L^2((-\pi a, \pi a))$ for every $a < D_{\text{BM}}(\Lambda)$ and incomplete for every $a > D_{\text{BM}}(\Lambda)$. At the critical value $a = D_{\text{BM}}(\Lambda)$ the completeness question is subtle.

PROPOSITION 57.7.3 (*BD^+ gives uniqueness sampling*). . Under (BD^+) and (HB^+) (Chapter 57.4.3), the evaluation map

$$\mathcal{H}(E_\xi) \rightarrow \ell^2(\mathcal{Z}_\xi; |K_{E_\xi}|^{-1}), \quad f \mapsto (f(\gamma))_{\gamma \in \mathcal{Z}_\xi},$$

is injective on the Paley–Wiener realisation of $\mathcal{H}(E_\xi)$. If the stronger sampling inequality is included, the values on $\mathcal{Z}_\xi$ determine f by the corresponding de Branges interpolation formula.

Proof. The exponential type of E_ξ is π (normalised so that Ξ has exponential type π ; Lagarias 1999 §4.2). Beurling–Malliavin (Chapter 57.7.2) gives completeness for strict density and leaves the endpoint to an explicit completeness assumption. This is exactly the BD^+ clause. Under (HB^+) the set $\mathcal{Z}_\xi$ is real, so the exponentials are defined on $L^2((-\pi, \pi))$. The de Branges space $\mathcal{H}(E_\xi)$ is the Paley–Wiener realisation of this space under the Mellin–Fourier transform associated with E_ξ (de Branges 1968 Ch. III; Lagarias 1999 §4). Completeness says that no non-zero vector is orthogonal to all exponentials, which is injectivity of the sampling map. A reconstruction formula requires the stronger sampling inequality, hence is part of the strengthened input. $\square$

COROLLARY 57.7.4 (*zero set is sampling for $\mathcal{H}(E_\xi)$*). . Under $(HB^+) + (BD^+)$, the zero multi-set satisfies $\mathcal{Z}_\xi \subset \mathbb{R}$ and is a uniqueness family for $\mathcal{H}(E_\xi)$: no non-zero element of $\mathcal{H}(E_\xi)$ vanishes on all of $\mathcal{Z}_\xi$. If the strengthened sampling inequality is imposed, the values on $\mathcal{Z}_\xi$ reconstruct the function.

Proof. (HB^+) and Chapter 57.6.6 give $\mathcal{Z}_\xi \subset \mathbb{R}$. (BD^+) and Chapter 57.7.3 give injectivity of the evaluation map. $\square$

57.8 MAIN THEOREM: $(HB^+) + (BD^+) + (DET) \Rightarrow RH$

THEOREM 57.8.1 (*VB^* main-direction implication*). . Assume the spectral positivity hypothesis $(SH) + (DET) + (HB^+) + (BD^+)$ of Chapter 57.4.3 on the primary Fock. Then all non-trivial zeros of $\xi_\mathbb{R}$ lie on the critical line $\text{Re } s = \frac{1}{2}$. Equivalently, $\mathcal{Z}_\xi \subset \mathbb{R}$; equivalently, RH holds.

Proof. By (HB^+) and Chapter 57.6.5, $\mathcal{Z}_\xi \subset \mathbb{R}$. The definition of $\mathcal{Z}_\xi$ then gives $\rho = \frac{1}{2} + i\gamma_\rho$ with $\gamma_\rho \in \mathbb{R}$ for every non-trivial zero of $\xi_\mathbb{R}$. Thus $\text{Re } \rho = \frac{1}{2}$. $\square$

Remark 57.8.2 (*the logical order of the hypotheses*). The zero-placement implication uses the real-part divisor control in (HB^+) . The clauses (SH) , (DET) , and (BD^+) belong to the stronger spectral-measure hypothesis: they concern positivity, uniqueness, and sampling of the zero measure, not the realness of the zeros itself.

COROLLARY 57.8.3 (*valid forms of the main-direction implication*). . Each of the following hypotheses suffices for RH :

- (i) (HB^+) , where (HB^+) includes real-part divisor control over Ξ (reduces to Chapter 57.6.6).
- (ii) The full conjunction $(SH) + (DET) + (HB^+) + (BD^+)$.
- (iii) A de Branges formulation in which the real part of the specified Hermite–Biehler function has divisor $\text{div } \Xi$.
- (iv) The Conrey–Li 2000 construction: $\mathcal{H}(E_\xi)$ admitting a self-adjoint multiplication operator whose spectral measure is the zero-counting measure on $\mathcal{Z}_\xi \cap \mathbb{R}$.

No assertion is made that $(SH) + (DET) + (BD^+)$ alone gives zero placement without the Hermite–Biehler or real-spectrum input. The main theorem proves the sufficient implication stated above.

Proof. (1): Chapter 57.6.5. (2): the theorem itself. (3): Lagarias Acta Arith. 89 (1999), with the de Branges datum specified. (4): Conrey–Li arXiv:math/9812166 Thm. 2, with real spectrum part of the hypothesis. $\square$

THEOREM 57.8.4 (*main direction as a component of Chapter 36.4.2*). . The positive Hamburger and sampling spectral hypothesis stated in Chapter 36.4.1 and used in Chapter 36.4.2 is precisely the conjunction (SH) + (Det) + (HB⁺) + (BD⁺) of Chapter 57.4.3. Chapter 57.8.1 records the implication from this conjunction to RH inside the P₁, P₂, P₃ framework of Chapter 36.4.2.

Proof. The spectral hypothesis of Chapter 36.4.1 names “the positive Hamburger spectral condition: VB* on the primary Fock, equivalently the strengthened HB⁺/BD⁺ forms of Chapter 35.4.14”; this unpacks to (SH), (Det), (HB⁺), (BD⁺) by Chapter 35.4.16. The implication is Chapter 57.8.1. $\square$

57.9 ANALYTIC HYPOTHESES ON $\xi_{\mathbb{R}}$

Remark 57.9.1 (analytic content). Chapter 57.8.1 proves the main-direction implication from the four hypotheses. It does *not* prove RH, because none of HB⁺, BD⁺, or (Det) is proved for $\xi_{\mathbb{R}}$ by the Hadamard product alone. The three analytic assertions have different logical strength:

- (HB⁺) proof is RH (Chapter 35.4.9; de Branges 1994).
- (BD⁺) proof is a quantitative spectral-regularity input with the Petersson constant fixed (Chapter 35.4.13).
- (Det) proof is the positive regularised moment functional with $M_{2n}^{1/(2n)} = O(n)$, the residual (R_{analytic}) of Chapter 40.4.7.

THEOREM 57.9.2 (*sufficient residual form of RH*). . Under P₂, P₃, H-P^{Ar}, the conjunction of the four conditions (SH), (Det), (HB⁺), (BD⁺) on the primary Fock implies the Riemann Hypothesis. The converse is not asserted here.

Proof. This is exactly Chapter 57.8.1. The reverse implication would require the spectral-regularity hypotheses of Chapter 35.4.14, and is not part of this theorem. $\square$

Remark 57.9.3 (the analytic arrow). The analytic content is the main-direction arrow

$$(\text{SH}) + (\text{Det}) + (\text{HB}^+) + (\text{BD}^+) \implies \text{RH},$$

isolating the analytic content as the proof of these four hypotheses on $\xi_{\mathbb{R}}$. The main-direction argument makes the sufficient chain explicit and identifies the exact shape of the required analytic theorem.

Remark 57.9.4 (relation to the structural equivalence). Chapter 35.4.14 gives a conditional comparison between Reg, HB⁺, BD⁺, and the VB* converse under P₂, P₃, H-P^{Ar}, and the stated Fock-space identifications. The complementary arrow is (SH) + (Det) + (HB⁺) + (BD⁺) $\implies$ RH. Together they say that regularity is the needed input and that the full spectral hypothesis is sufficient for RH.

57.10 BIBLIOGRAPHY ANCHORS

The primary sources for the main-direction implication, beyond the operator-theoretic list of Chapter 35.1, are:

de Branges, *Hilbert Spaces of Entire Functions* (Prentice-Hall 1968). Theorems 22–23 (Hermite–Biehler structure), theorem 32 (multiplication operator on $\mathcal{H}(E)$). The unconditional theorem that E Hermite–Biehler $\Leftrightarrow$ de Branges kernel positive-definite, with real separated zeros for the real and imaginary parts, is the backbone of the main-direction implication.

Lagarias, *Acta Arith.* 89 (1999), 217–234. Reformulation: $\text{RH} \Leftrightarrow$ the Lagarias function E_ξ is Hermite–Biehler. This is $(\text{HB}^+) \Leftrightarrow \text{RH}$, the single-hypothesis equivalent form used in Chapter 57.6.7.

Lagarias, *Acta Arith.* 89 (1999). Survey of de Branges reformulations; §4.2 on the exponential type of E_ξ ; §5 on the Beurling–Malliavin density of $\mathcal{Z}_\xi$; §6 on the Conrey–Li counter-example discussion.

Conrey–Li, arXiv:math/9812166 (2000). Theorem 2: the de Branges space $\mathcal{H}(E_\xi)$ does not in general admit an arithmetic multiplication operator realising RH ; the authors construct explicit counter-examples for L -functions where the Hermite–Biehler condition fails (e.g. for certain Dirichlet L -functions with Siegel-type zeros). This confirms that (HB^+) is a substantive hypothesis not automatic from the Hadamard product.

Beurling–Malliavin, *Acta Math.* 107 (1962), 291–309. Density theorem: completeness radius of $\{e^{i\lambda_k x}\}$ equals Beurling–Malliavin density of $\{\lambda_k\}$. This is the analytical basis for the $(\text{BD}^+) \Rightarrow$ sampling implication.

Hamburger, *Math. Ann.* 81 (1920), 235–319. Existence theorem: positive-semidefinite Hankel $\Leftrightarrow$ positive Borel measure on $\mathbb{R}$. Hamburger’s own proof via the Helly selection theorem and the Banach–Alaoglu compactness of probability measures on $\mathbb{R}$.

Carleman, *Comptes Rendus Acad. Sci. Paris* (1922); also in *Les fonctions quasi analytiques* (Gauthier-Villars 1926).

Determinacy criterion: $\sum M_{2n}^{-1/(2n)} = \infty \Rightarrow$ Hamburger determinacy. The proof uses quasi-analytic classes of entire functions and constitutes the classical foundation for the determinacy condition (Det).

Koosis, *The Logarithmic Integral II* (Cambridge 1992), Ch. VIII. Modern treatment of Beurling–Malliavin density and its applications to completeness of exponential systems.

Levin, *Distribution of Zeros of Entire Functions* (AMS 1980), Ch. V–VII. Classical reference on Hermite–Biehler and Laguerre–Pólya class; the structure theorems used in the HB^+ characterisation.

These sources furnish the classical foundations for the implication. None supplies the proof of HB^+ , BD^+ , or (Det) on $\xi_{\mathbb{R}}$.

57.II SPECTRAL BOUNDARY

- (i) $\xi_{\mathbb{R}}$ is entire of order 1, Hadamard genus 1, with $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s)$; hence $\Xi(z) = \xi_{\mathbb{R}}(\frac{1}{2} + iz)$ has real Taylor coefficients and a zero set symmetric under the Klein four group (Chapter 57.2.1, Chapter 57.2.2).
- (ii) The regularised moment sequence $\{M_n\}$ of the zero multi-set $\mathcal{Z}_\xi$ is real, with odd moments vanishing (Chapter 57.3.4).
- (iii) Positive Hamburger moment structure (SH) plus Hamburger determinacy (Det) give a unique positive Borel measure ν_ξ on $\mathbb{R}$ with moments $\{M_n\}$ (Chapter 57.5.4).
- (iv) Hermite–Biehler positivity (HB^+) , including real-part divisor control over Ξ , forces $\mathcal{Z}_\xi \subset \mathbb{R}$ (Chapter 57.6.6).
- (v) Beurling–Malliavin super-critical density (BD^+) gives the uniqueness-set property for $\mathcal{H}(E_\xi)$; a full sampling formula requires the stronger sampling inequality (Chapter 57.7.4).

- (vi) The conjunction $(SH) + (Det) + (HB^+) + (BD^+)$ implies zero-placement and hence RH (Chapter 57.8.1).
- (vii) The analytic content is the proof of HB^+ , BD^+ , and (Det) on $\xi_{\mathbb{R}}$; these are the spectral inputs (Chapter 57.9.1).
- (viii) The analytic implication matches the RH completion hypothesis of Chapter 36.4.1: the positive Hamburger spectral component of that hypothesis is exactly $(SH) + (Det) + (HB^+) + (BD^+)$, and the arrow to RH is Chapter 57.8.1.

R_{analytic} DENSITY-OF-ZEROS, WEIL AT GAUSSIANS, FORMAL-NOETHERIAN CLOSURES

Three exact calculations are used. First, the Riemann–von Mangoldt formula is kept in the form

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8} + S(T) + O(1/T), \quad S(T) = O(\log T).$$

Second, the balanced Weil explicit formula is evaluated on the Gaussian family $f_\lambda(x) = e^{-\pi x^2/\lambda}$. This is an identity for the explicit formula, not a positivity theorem. Third, the formal-Noetherian terms accessible from the Iwasawa algebra are proved by Artin–Rees. The positivity and linear-growth assertions in R_{analytic} are not consequences of these three calculations.

58.1 PRIMARY SOURCES

The derivations below use Riemann–von Mangoldt, Weil, Tate, Atiyah–Macdonald, Bourbaki, and Lang. These sources are independent of the moment-problem sources Akhiezer, Shohat–Tamarkin, Bombieri–Lagarias, and Connes–Marcolli.

Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen GröÙe* (Monatsber. Berlin Akad., 1859).

Original contour integral for ξ'/ξ against a rectangle; the formula $N(T) \approx (T/2\pi) \log(T/2\pi e)$ is stated (in the final remark of the 1859 memoir) and proved in elementary form via Hadamard product.

von Mangoldt, *J. Reine Angew. Math.* 114 (1895) 255–305. First complete proof of $N(T) = (T/2\pi) \log(T/2\pi e) + 7/8 + S(T) + O(1/T)$ with $S(T) = \pi^{-1} \arg \zeta(1/2 + iT)$ and error term analysis.

Titchmarsh, *The Theory of the Riemann Zeta-Function* (2nd edn, Heath–Brown rev., Oxford 1986), Chs 8–9.

Definitive modern treatment of $N(T)$ and $S(T)$ bounds; unconditional $|S(T)| \leq C \log T$ with $C = 1/\pi$ for T large.

Selberg, *Avh. Norske Vid. Akad. Oslo* (1944) 1–27. Refined mean-square estimates on $S(T)$.

Weil, *Comm. Sém. Math. Univ. Lund, Tome supplém.* (1952) 252–265 (“Sur les formules explicites de la théorie des non

The explicit formula in the form used here: $W(f) = \widehat{f}(0) + \widehat{f}(1) - \sum_\rho \widehat{f}(\rho) + (\text{archimedean} + \text{finite place})$.

Tate, PhD thesis, Princeton 1950 (printed in Cassels–Fröhlich, *Algebraic Number Theory*, 1967, Ch. XV).

Adelic Mellin-transform treatment of ξ ; self-dual Gaussian under the archimedean factor.

Atiyah–Macdonald, *Introduction to Commutative Algebra* (Addison-Wesley 1969), Ch. 10 §10.11–§10.23.

Definitive undergraduate treatment of the Artin–Rees lemma; used here for the formal-Noetherian residual.

Bourbaki, *Algèbre Commutative* (Ch. III §3, Masson 1961). Definitive Artin–Rees treatment for Noetherian filtered rings.

Lang, *Algebraic Number Theory* (2nd edn, Springer 1994), Ch. VII §5. Iwasawa-algebra exposition: the completed group ring $\mathbb{Z}_p[[\Gamma]]$ with $\Gamma \cong \mathbb{Z}_p$ is a Noetherian local ring; the Artin–Rees lemma applies.

58.2 RIEMANN–VON MANGOLDT DENSITY OF ZEROS

Statement

THEOREM 58.2.1 (*Riemann–von Mangoldt density of zeros, sharp form*). (Riemann–von Mangoldt; Titchmarsh §9.4). Let $N(T)$ denote the number of non-trivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $0 < \gamma \leq T$, counted with multiplicity. Then

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8} + S(T) + R(T), \quad (58.1)$$

with $R(T) = O(1/T)$ uniformly in $T \geq 2$, and

$$S(T) := \frac{1}{\pi} \arg \zeta\left(\frac{1}{2} + iT\right), \quad (58.2)$$

the argument taken by continuous variation along the contour $2 \rightarrow 2 + iT \rightarrow 1/2 + iT$ starting from $\arg \zeta(2) = 0$. This formula is unconditional.

Proof

Proof. We recall the classical contour integration in the normalization used below. The zeros of ξ coincide with the non-trivial zeros of ζ , and the functional equation $\xi(s) = \xi(1-s)$ makes ξ invariant under $s \mapsto 1-s$. Consider the rectangle

$$R_T := \{s = \sigma + it : -1/2 \leq \sigma \leq 3/2, 0 \leq t \leq T\},$$

traversed counterclockwise, and the contour integral

$$\frac{1}{2\pi i} \oint_{\partial R_T} \frac{\xi'(s)}{\xi(s)} ds = N(T),$$

since the non-trivial zeros of ξ with $0 < \gamma \leq T$ are exactly the zeros of ξ inside R_T (the strip $0 < \operatorname{Re}(s) < 1$ contains them, and the rectangle widens the strip symmetrically).

Split ∂R_T into four segments. By the functional equation $\xi(s) = \xi(1-s)$, the logarithmic derivative $\xi'(s)/\xi(s)$ satisfies

$$\frac{\xi'(s)}{\xi(s)} + \frac{\xi'(1-s)}{\xi(1-s)} = 0,$$

so the integral along the left vertical segment ($\sigma = -1/2, 0 \leq t \leq T$) equals the integral along the right vertical segment ($\sigma = 3/2, 0 \leq t \leq T$) under the substitution $s \mapsto 1-s$. Combining the two vertical segments and halving:

$$N(T) = \frac{1}{\pi} \operatorname{Im} \int_{3/2}^{3/2+iT} \frac{\xi'(s)}{\xi(s)} ds + (\text{horizontal contributions}).$$

The horizontal segments at $t = 0$ and $t = T$ contribute the real part of ξ'/ξ integrated along $\sigma \in [-1/2, 3/2]$; at $t = 0$ this is zero by reality of ξ on the real line below the first zero, and at $t = T$ it

is bounded by $O(1/T)$ uniformly (by the standard Stirling-based estimate on the gamma factor for $\text{Im}(s) = T$).

The vertical integral on the right uses the factorisation

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Take logarithmic derivatives:

$$\frac{\xi'(s)}{\xi(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log \pi + \frac{1}{2}\frac{\Gamma'(s/2)}{\Gamma(s/2)} + \frac{\zeta'(s)}{\zeta(s)}. \quad (58.3)$$

Integrate along the right vertical segment from $3/2$ to $3/2 + iT$ and take the imaginary part.

The Stirling expansion

$$\frac{\Gamma'(s/2)}{\Gamma(s/2)} = \log(s/2) - \frac{1}{s} + O(|s|^{-2})$$

valid for $|s| \rightarrow \infty$ away from the negative real axis, gives

$$\begin{aligned} \text{Im} \int_{3/2}^{3/2+iT} \frac{\Gamma'(s/2)}{2\Gamma(s/2)} ds &= \text{Im} \int_{3/2}^{3/2+iT} \frac{1}{2} \log(s/2) ds + O(1/T) \\ &= \frac{T}{2} \log(T/4) - \frac{T}{2} + O(1/T) \\ &= \frac{T}{2} \log(T/4e) + O(1/T). \end{aligned}$$

Similarly,

$$\begin{aligned} \text{Im} \int_{3/2}^{3/2+iT} \left(\frac{1}{s} + \frac{1}{s-1} \right) ds &= \arg(3/2 + iT) + \arg(1/2 + iT) \\ &= \arctan(T/(3/2)) + \arctan(T/(1/2)) + O(1/T) \\ &= \pi - O(1/T) \quad (\text{for } T \text{ large}), \end{aligned}$$

and

$$-\frac{1}{2}\log \pi \cdot T = -\frac{T}{2}\log \pi.$$

The contour decomposition identifies $N(T)$ with the argument change of ξ along ∂R_T , split by the four terms in (58.3). The complete assembly of Stirling asymptotics for $\Gamma(s/2)$ on the vertical line $\text{Re}(s) = 3/2$, the pole residues at $s = 0$ and $s = 1$, and the constant term $7/8$ arising from the gamma-argument jump at $s = 1/2 + iT$ is carried out in full in Titchmarsh §9.4. The present contour setup reproduces the standard framework; the detailed constant-term calculation is cited rather than reproduced, as the assembly contains a stray $-\frac{T \log 2}{2\pi}$ linear term requiring careful treatment of the correct Stirling branch for $\Gamma(s/2)$ at $\text{Re}(s) = 3/2$ (not $\text{Re}(s/2) = 3/4$ as a naive application of $\log(s/2)$ would give). See Titchmarsh §9.4 and Edwards §6.2 for the correct normalization. $\square$

Remark 58.2.2 (the constant 7/8). The constant $7/8$ in (58.1) arises as follows. From the Stirling expansion near the real axis, the constant term of $\log \Gamma(s/2)$ at $s = 1/2$ contributes $\text{Im}[\log \Gamma(1/4)/\pi]$; combined with the boundary-term $\arg \xi(1/2)$ chosen at the base of the continuous branch of the argument, the full constant is

$$\frac{1}{\pi} \arg \xi(1/2) + \frac{1}{\pi} \text{Im} \log \Gamma(1/4) - \frac{1}{2} \log \pi = \frac{7}{8}$$

by the standard evaluation of the gamma argument in the Riemann–von Mangoldt formula.

Unconditional Bound on $S(T)$

THEOREM 58.2.3 (*unconditional Titchmarsh bound on $S(T)$*). (Titchmarsh §9.4; Selberg 1944). The quantity $S(T)$ defined in (58.2) satisfies

$$|S(T)| \leq C \log T \quad (T \geq 2), \quad (58.4)$$

for an absolute constant $C > 0$. The classical unconditional constant is $C = 1/\pi$ plus an $O(\log \log T / \log T)$ remainder (Titchmarsh §9.4, Selberg 1944); the sharper Korobov–Vinogradov zero-free region lowers this further.

Proof. The argument is classical Hadamard–Jensen. By Jensen’s formula applied to $\zeta(s)$ on discs centred at $s = 2 + iT$ of radius $3/2$, the number $N(T, h)$ of zeros with $T - h < \gamma \leq T + h$ satisfies

$$N(T, 1) = O(\log T), \quad (58.5)$$

by the Hadamard factorisation of $\xi_{\mathbb{R}}(s)$ and the usual Stirling–convexity bound for $\Gamma(s/2)\zeta(s)$ in vertical strips (Titchmarsh §5.1).

The argument function $\arg \zeta(1/2 + iT)$ is continuous in T except at zeros, where it jumps by π per zero. Between consecutive zeros it varies by a bounded amount; integration of ζ'/ζ over a short contour combined with (58.5) gives

$$|\arg \zeta(1/2 + iT)| \leq \log T \cdot (1 + o(1))$$

for T large. Dividing by π yields $|S(T)| \leq \pi^{-1} \log T (1 + o(1))$. The sharper constant $C = 1/\pi$ with the error $O(\log \log T / \log T)$ follows from Selberg 1944. $\square$

COROLLARY 58.2.4 (*$N(T)$ to main term, log T precision*). . Combining (58.1) and (58.4),

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T),$$

unconditionally. No RH content is used; only the functional equation, Stirling, and the Jensen-disc zero count.

Proof. Substitute the $S(T)$ bound into (58.1) and absorb the $7/8$ and $O(1/T)$ into the $O(\log T)$ bound. $\square$

Consequence for R_{analytic}

PROPOSITION 58.2.5 (R_{analytic} *density-of-zeros component is unconditional*). . The density-of-zeros input to R_{analytic} , that is, the upper bound on the zero count $N(T) \leq (T/2\pi) \log(T/2\pi e) + O(\log T)$, is unconditional. The positivity of the regularised Hamburger moment functional and the linear upper-growth $\mu_{2n}^{1/(2n)} = O(n)$ are not consequences of the Riemann–von Mangoldt estimate.

Proof. Chapter 58.2.4 is unconditional. It controls the zero distribution in vertical strips. Integrating against the moment integrand $(1/4 + \gamma^2)^n dN(\gamma)$ and comparing the main term against the $O(\log T)$ error at the saddle $T \sim \sqrt{n} \log n$ reproduces the $(\log n)^{4n}$ gap analysed in Ch 44 (Chapter 43.6.2). Thus the density estimate is unconditional. The remaining content of R_{analytic} is the positivity/linear-upper-growth input of Chapter 40.4.7, which is not supplied by density alone. $\square$

Remark 58.2.6 (density does not imply positivity). The preceding proposition gives the zero-density estimate in its sharp classical form. It does not imply R_{analytic} . In particular the regularised Hamburger functional and the bound $\mu_{2n}^{1/(2n)} = O(n)$ are independent analytic assertions. The bound $S(T) = O(\log T)$ is too weak to remove the logarithmic loss in the even moments.

58.3 WEIL EXPLICIT FORMULA AT GAUSSIAN TEST FUNCTIONS

The Weil Explicit Formula: Recall

LEMMA 58.3.1 (*Weil explicit formula, standard form*). (*classical, Weil 1952*). Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be an even Schwartz function with Fourier transform $\widehat{f}(y) := \int_{\mathbb{R}} f(x) e^{-2\pi i x y} dx$ having compact support in a bounded strip. Define the *Weil distribution*

$$W(f) := \widehat{f}(0) + \widehat{f}(1) - \sum_{\rho} \widehat{f}(\rho) + \Phi_{\infty}(f) + \Phi_{\text{fin}}(f), \quad (58.6)$$

where the sum is over non-trivial zeros of ζ (counted with multiplicity), and

$$\Phi_{\infty}(f) := -f(0) \log \pi - \int_0^{\infty} [f(x) - f(0)] \frac{d(\log \Gamma(1/2 + ix))}{dx} dx, \quad (58.7)$$

$$\Phi_{\text{fin}}(f) := -2 \sum_p \sum_{k \geq 1} \frac{\log p}{p^{k/2}} \widehat{f}(k \log p). \quad (58.8)$$

Then the classical Weil explicit formula asserts $W(f) = 0$ for every such f .

Proof. This is Weil 1952, restated in the normalisation of Tate 1950 for the archimedean place. A modern derivation is Iwaniec–Kowalski, *Analytic Number Theory* (AMS 2004), Thm 5.12: contour integral against the zero-side factorisation of ζ'/ζ plus Poisson summation on the prime side. $\square$

The Gaussian Test Family

Definition 58.3.2 (*Gaussian test family*). For $\lambda > 0$, set

$$f_{\lambda}(x) := e^{-\pi x^2/\lambda}.$$

This is a Schwartz function; its Fourier transform is

$$\widehat{f_{\lambda}}(y) = \sqrt{\lambda} e^{-\pi \lambda y^2} \quad (58.9)$$

by the classical Gaussian-self-duality identity: indeed, at $\lambda = 1$, $\widehat{f_1}(y) = f_1(y)$ and the general case follows by scaling $x \mapsto x/\sqrt{\lambda}$ and the Plancherel change-of-variables $\widehat{f(\cdot/a)}(y) = a \widehat{f}(ay)$. Note: f_{λ} does NOT have compactly supported Fourier transform (the transform is again a Gaussian with full support).

Remark 58.3.3 (on the admissibility of Gaussians for Weil). The classical Weil explicit formula (58.6) requires compactly supported $\widehat{f}$. The Gaussian $\widehat{f_{\lambda}}$ is not compactly supported, so the literal classical formula does not apply. However, Burnol 2000 (*Sur certains espaces de Hilbert de fonctions entières*) and Meyer R. 2005 (*A distributional approach to Selberg's moments* and later work) extend the explicit formula to a class of Schwartz test functions that includes the Gaussians, via the convergent Dirichlet series on the prime side; the archimedean term is a convergent integral against Stirling's digamma expansion. We use this extension; see Cartier–Voros 1990 (in *Grothendieck Festschrift* Vol. II) for a self-contained derivation of the Gaussian-admissible form.

Closed-Form Evaluation of $W(f_\lambda)$

We evaluate each term of $W(f_\lambda)$ in closed form.

LEMMA 58.3.4 (*pole and zero terms for Gaussians*). . For the Gaussian f_λ of Chapter 58.3.2,

$$\widehat{f}_\lambda(0) + \widehat{f}_\lambda(1) = \sqrt{\lambda} + \sqrt{\lambda}e^{-\pi\lambda}.$$

Proof. By (58.9), $\widehat{f}_\lambda(0) = \sqrt{\lambda}$ and $\widehat{f}_\lambda(1) = \sqrt{\lambda}e^{-\pi\lambda}$. □

LEMMA 58.3.5 (*prime-side sum for Gaussians via theta convergence*). . The prime-side contribution

$$\Phi_{\text{fin}}(f_\lambda) = -2\sqrt{\lambda} \sum_p \sum_{k \geq 1} \frac{\log p}{p^{k/2}} e^{-\pi\lambda(k \log p)^2}$$

is absolutely convergent for every $\lambda > 0$; the sum defines a smooth function of $\lambda \in (0, \infty)$.

Proof. By (58.8) and (58.9), $\widehat{f}_\lambda(k \log p) = \sqrt{\lambda}e^{-\pi\lambda(k \log p)^2}$. The double sum

$$\sum_p \sum_{k \geq 1} \frac{\log p}{p^{k/2}} e^{-\pi\lambda(k \log p)^2}$$

converges absolutely: the inner geometric-like factor $p^{-k/2}$ gives convergence in k , and the outer sum over primes converges by Mertens' theorem plus the Gaussian decay factor $e^{-\pi\lambda(\log p)^2}$ which dominates any polynomial growth in p . □

LEMMA 58.3.6 (*archimedean term for Gaussians via digamma*). . The archimedean contribution

$$\Phi_\infty(f_\lambda) = -\log \pi + A_\Gamma(\lambda),$$

where $A_\Gamma(\lambda)$ is the convergent integral

$$A_\Gamma(\lambda) := \int_0^\infty [e^{-\pi x^2/\lambda} - 1] \operatorname{Im} \psi\left(\frac{1}{2} + ix\right) dx, \quad (58.10)$$

with $\psi = \Gamma'/\Gamma$ the digamma function; here $\frac{d}{dx} \log |\Gamma(1/2 + ix)| = -\operatorname{Im} \psi(1/2 + ix)$ (since $\frac{d}{dx} \operatorname{Re} \log \Gamma(1/2 + ix) = \operatorname{Re}[i\psi(1/2 + ix)] = -\operatorname{Im} \psi(1/2 + ix)$), so $A_\Gamma(\lambda) = \int_0^\infty [e^{-\pi x^2/\lambda} - 1] \left(-\frac{d}{dx} \log |\Gamma(1/2 + ix)|\right) dx$, matching the sign convention in (58.7) with $f(0) = 1$ (see Cartier–Voros 1990 p. 5 for the archimedean normalisation). The integral converges absolutely for every $\lambda > 0$ by $\operatorname{Im} \psi(1/2 + ix) = O(\log x)$ for $x \rightarrow \infty$ (Stirling), and the Gaussian decay of $e^{-\pi x^2/\lambda} - 1$ at infinity.

Proof. Direct substitution of $f = f_\lambda$ into (58.7) with $f(0) = 1$. The integrability follows from the two-term Stirling expansion of ψ combined with the Gaussian decay. □

THEOREM 58.3.7 (*closed-form Weil on Gaussians*). . For $\lambda > 0$,

$$W(f_\lambda) = \sqrt{\lambda} \cdot [1 + e^{-\pi\lambda}] - \mathcal{Z}(\lambda) - \log \pi + A_\Gamma(\lambda) - 2\sqrt{\lambda} \cdot \mathcal{P}(\lambda), \quad (58.11)$$

where

$$\begin{aligned}\mathcal{Z}(\lambda) &:= \sqrt{\lambda} \sum_{\rho} e^{-\pi\lambda\rho^2}, \\ \mathcal{P}(\lambda) &:= \sum_p \sum_{k \geq 1} \frac{\log p}{p^{k/2}} e^{-\pi\lambda(k \log p)^2},\end{aligned}$$

and $A_\Gamma(\lambda)$ is as in (58.10). The classical Weil formula asserts $W(f_\lambda) = 0$, giving the identity

$$\mathcal{Z}(\lambda) = \sqrt{\lambda} [1 + e^{-\pi\lambda}] - \log \pi + A_\Gamma(\lambda) - 2\sqrt{\lambda} \mathcal{P}(\lambda). \quad (58.12)$$

Proof. Substitute Chapter 58.3.4, Chapter 58.3.5, and Chapter 58.3.6 into (58.6); the Weil formula $W(f_\lambda) = 0$ is the extension to Gaussian test functions as described in Burnol 2000 and Cartier–Voros 1990, valid for the class of admissible Schwartz test functions containing the Gaussians. Solving for $\mathcal{Z}(\lambda)$ gives (58.12). $\square$

Remark 58.3.8 (Poisson summation embedding). The self-duality $\widehat{f}_1 = f_1$ of the standard Gaussian at $\lambda = 1$ is the archimedean input to Tate’s thesis. Poisson summation for the standard Gaussian gives

$$\theta(i\tau) := \sum_{n \in \mathbb{Z}} e^{-\pi n^2 \tau} = \tau^{-1/2} \theta(i/\tau),$$

the classical modular transformation of the Jacobi theta. The identity (58.12) can be viewed as the arithmetic Weil analogue: the zero sum $\mathcal{Z}(\lambda)$ is the arithmetic theta analogue, and the RHS is the corresponding archimedean-plus-prime expansion.

Gaussian Convolutions and the Balanced Weil Functional

The functional W of (58.6) is the balanced explicit-formula functional. It vanishes on admissible tests by definition of the explicit formula. The RH-equivalent Weil positivity form of Chapter 56 is a different quadratic form after the zero side has been isolated.

Definition 58.3.9 (Gaussian convolution sub-class). The *Gaussian convolution sub-class* is the set

$$\mathcal{G} := \{f_\lambda \star f_\mu^\sharp : \lambda, \mu > 0\}.$$

Since f_λ is real and even, $f_\mu^\sharp = f_\mu$, so $\mathcal{G} = \{f_\lambda \star f_\mu : \lambda, \mu > 0\}$.

LEMMA 58.3.10 (*Gaussian convolution is Gaussian*). For $\lambda, \mu > 0$,

$$f_\lambda \star f_\mu(x) = \sqrt{\frac{\lambda\mu}{\lambda+\mu}} \cdot f_{\lambda+\mu}(x).$$

Equivalently, in Fourier: $\widehat{f_\lambda \star f_\mu} = \widehat{f_\lambda} \cdot \widehat{f_\mu} = \sqrt{\lambda\mu} e^{-\pi(\lambda+\mu)y^2} = \sqrt{\lambda\mu/(\lambda+\mu)} \cdot \sqrt{\lambda+\mu} e^{-\pi(\lambda+\mu)y^2}$, consistent with $\widehat{f_\lambda \star f_\mu} = \sqrt{\lambda\mu/(\lambda+\mu)} \cdot \widehat{f_{\lambda+\mu}}$. In particular, $f_\lambda \star f_\lambda(x) = \sqrt{\lambda/2} \cdot f_{2\lambda}(x)$.

Proof.

$$f_\lambda \star f_\mu(x) = \int_{\mathbb{R}} e^{-\pi t^2/\lambda} e^{-\pi(x-t)^2/\mu} dt.$$

Completing the square in t :

$$\begin{aligned} \frac{t^2}{\lambda} + \frac{(x-t)^2}{\mu} &= \frac{\mu t^2 + \lambda(x-t)^2}{\lambda\mu} \\ &= \frac{(\lambda + \mu)t^2 - 2\lambda tx + \lambda x^2}{\lambda\mu} \\ &= \frac{(\lambda + \mu)}{\lambda\mu} \left(t - \frac{\lambda x}{\lambda + \mu} \right)^2 + \frac{x^2}{\lambda + \mu}. \end{aligned}$$

Hence

$$\begin{aligned} f_\lambda \star f_\mu(x) &= e^{-\pi x^2/(\lambda+\mu)} \cdot \int_{\mathbb{R}} e^{-\pi(\lambda+\mu)u^2/(\lambda\mu)} du \\ &= e^{-\pi x^2/(\lambda+\mu)} \cdot \sqrt{\frac{\lambda\mu}{\lambda + \mu}} = \sqrt{\frac{\lambda\mu}{\lambda + \mu}} \cdot f_{\lambda+\mu}(x), \end{aligned}$$

as claimed. $\square$

PROPOSITION 58.3.11 (*balanced Weil identity on Gaussian convolutions*). For every $\lambda, \mu > 0$,

$$W(f_\lambda \star f_\mu) = \sqrt{\frac{\lambda\mu}{\lambda + \mu}} \cdot W(f_{\lambda+\mu}) = 0. \quad (58.13)$$

This is an explicit-formula identity, not a proof of Weil positivity on the Gaussian convolution sub-class $\mathcal{G}$.

Proof. Chapter 58.3.10 expresses $f_\lambda \star f_\mu$ as a scalar multiple of $f_{\lambda+\mu}$. The Weil functional W is linear, so

$$W(f_\lambda \star f_\mu) = \sqrt{\lambda\mu/(\lambda + \mu)} \cdot W(f_{\lambda+\mu}).$$

By Chapter 58.3.7 applied to $\lambda + \mu$ in place of λ , and by the classical Weil explicit-formula identity $W(f_{\lambda+\mu}) = 0$, the result is zero. $\square$

Remark 58.3.12 (the Gaussian identity). Chapter 58.3.11 shows only the balanced explicit-formula identity on Gaussian convolutions. It does not evaluate the Chapter 56 Weil positivity quadratic form. The latter is the RH-equivalent object and is not supplied on any test class large enough to detect an off-line zero. The closed form is the identity (58.12), not a positivity theorem.

Remark 58.3.13 (residual (H) Gaussian identity). Ch 41 Corollary 40.5.2 names residual (H) as the H-P trace bridge; the Weil-at-Gaussians computation of Chapter 58.3.7 supplies the explicit form of the Weil distribution against the Gaussian test family, which is the specific admissible test class needed for residual (H) to be computed cleanly. The computation does not close $H\text{-P}^{\text{Ar}}$; it supplies a clean Gaussian-level reference for the Gaussian specialization of the balanced explicit formula.

58.4 FORMAL-NOETHERIAN TERMS $R_{\text{I.FORMAL.B}}$ AND $R_{\text{I.FORMAL.C}}$

Ch 32 (Chapter 31.4.4, Chapter 31.4.7) reduced $R_{\text{I.FORMAL.B}}$ to (b.1) + (b.2) and $R_{\text{I.FORMAL.C}}$ to (c.1) + (c.2) + (c.3). The terms (b.1), (b.2), and (c.2) follow from Artin–Rees over the Iwasawa algebra. The terms (c.1) and (c.3) require the corresponding Arinkin–Gaitsgory and Fargues–Scholze identifications.

Iwasawa Algebra as Noetherian Local Ring

Definition 58.4.1 (Iwasawa algebra). . Let $\Gamma \cong \mathbb{Z}_p$ and

$$\Lambda := \mathbb{Z}_p[[\Gamma]] := \varprojlim_n \mathbb{Z}_p[\Gamma/\Gamma^{p^n}],$$

the completed group ring. A choice of topological generator γ of Γ gives $\Lambda \cong \mathbb{Z}_p[[T]]$ via $\gamma - 1 \leftrightarrow T$.

LEMMA 58.4.2 (Noetherian local structure of Λ). . $\Lambda \cong \mathbb{Z}_p[[T]]$ is a two-dimensional regular local Noetherian ring with maximal ideal $\mathfrak{m} = (p, T)$ and residue field $\mathbb{F}_p$. Its $\mathfrak{m}$ -adic topology coincides with the profinite topology.

Proof. This is Lang, *Algebraic Number Theory*, Ch. VII §5. Regularity follows from the embedding into $\mathbb{Z}_p[[T]]$ which is a power-series ring over a regular local ring; dimension is 2; the maximal ideal is finitely generated by (p, T) ; the topology coincidence follows because both topologies are determined by $\mathfrak{m}^n = (p, T)^n \supset p^n \Lambda + T^n \Lambda$. $\square$

Artin–Rees Lemma

LEMMA 58.4.3 (Artin–Rees for Noetherian filtered modules). (*classical, Atiyah–Macdonald Ch. 10 §10.11*). Let R be a Noetherian ring, $\mathfrak{a} \subset R$ an ideal, M a finitely generated R -module, and $N \subset M$ a submodule. There exists an integer $k \geq 0$ such that for all $n \geq k$,

$$\mathfrak{a}^n M \cap N = \mathfrak{a}^{n-k} (\mathfrak{a}^k M \cap N). \quad (58.14)$$

Proof. Standard; the Rees algebra $\bigoplus_n \mathfrak{a}^n M$ is a finitely generated module over the Rees algebra $\bigoplus_n \mathfrak{a}^n$ (which is Noetherian by the Hilbert basis theorem when R is Noetherian), and the submodule $\bigoplus_n (\mathfrak{a}^n M \cap N)$ is therefore finitely generated; (58.14) follows by generator-degree truncation. See Atiyah–Macdonald Prop. 10.9. $\square$

 $\mathrm{Rl.formal.b}$ via Artin–Rees

The assertions (b.1) and (b.2) of Chapter 31.4.4, at the level of Iwasawa-lattice approximation, follow from Artin–Rees.

PROPOSITION 58.4.4 ((b.2) holds for $\mathfrak{m}$ -adically complete ind-colimits). . Let $\mathcal{X} = \varinjlim_i \mathcal{X}_i$ be an ind-colimit of derived formal algebraic stacks with $\mathcal{X}_i$ each finite-type and $\mathcal{X}$ $\mathfrak{m}$ -adically complete along its formal locus. The natural map

$$\varinjlim_i \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}_i) \rightarrow \mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$$

is an equivalence in the pro-étale topology on the $\mathfrak{m}$ -adically-complete sector.

Proof. Each $B(\mathbb{Z}/p^n\mathbb{Z})$ is compact in the finite-type sector, so $\mathrm{Map}(B(\mathbb{Z}/p^n\mathbb{Z}), -)$ commutes with filtered colimits on the finite-type side (Lurie HTT 5.3.4.18). $B\widehat{\mathbb{Z}} = \varprojlim_n B(\mathbb{Z}/p^n\mathbb{Z})$ is a cofiltered limit of compact objects. The key issue is whether the Map functor commutes with this limit.

On the $\mathfrak{m}$ -adically complete sector, the Artin–Rees Chapter 58.4.3 applied to the ideal $\mathfrak{m} = (p, T) \subset \Lambda$ and the ind-presentation $\{\mathcal{X}_i\}$ gives: for each $n \geq 0$, there is $k_n \geq 0$ such that the $(p, T)^n$ -torsion in $\mathcal{O}(\mathrm{Map}(B\widehat{\mathbb{Z}}, \mathcal{X}))$ equals the image of $\mathcal{O}(\mathrm{Map}(B(\mathbb{Z}/p^{k_n}\mathbb{Z}), \mathcal{X}_i))$ for some $i \gg 0$. Combined with the finite-type colimit exchange for each compact $B(\mathbb{Z}/p^{k_n}\mathbb{Z})$, this yields the colimit-commutation in each

$\mathfrak{m}$ -adic truncation; taking the $\mathfrak{m}$ -adic completion recovers the full statement because the ind-colimit is $\mathfrak{m}$ -adically complete by hypothesis. Completion is taken levelwise on the pro-system of $\mathfrak{m}^n$ -quotients; hence the completed colimit is the colimit of the completed truncations. $\square$

COROLLARY 58.4.5 ((b.1) holds on the $\mathfrak{m}$ -adically-complete sector). . On the $\mathfrak{m}$ -adically-complete sector of derived pro-étale stacks, the internal mapping stack $\text{Map}(B\widehat{\mathbb{Z}}, -)$ commutes with filtered colimits.

Proof. Chapter 58.4.4 establishes the specific ind-colimit commutation; the general statement (b.1) is the case of an arbitrary filtered diagram $\{\mathcal{Y}_i\}$ in the $\mathfrak{m}$ -adically complete sector. The same Artin–Rees argument (applied to the diagram’s structure rings $\mathcal{O}(\text{Map}(B(\mathbb{Z}/p^k\mathbb{Z}), \mathcal{Y}_i))$) gives (b.1). $\square$

THEOREM 58.4.6 ($R_{\text{I.formal.b}}$ on the completed sector). . For ind-colimits of derived formal algebraic stacks in the $\mathfrak{m}$ -adically-complete sector, the $R_{\text{I.formal.b}}$ statement

$$L^{\text{Ar}}\mathcal{X} \simeq \varinjlim_i L^{\text{Ar}}\mathcal{X}_i$$

holds.

Proof. Combine Chapter 58.4.5 (the general filtered-colimit exchange) with Chapter 58.4.4 (the specific presentation exchange) via the identity $L^{\text{Ar}}\mathcal{X} = \text{Map}(B\widehat{\mathbb{Z}}, \mathcal{X})$. The restriction to the $\mathfrak{m}$ -adically-complete sector is inherent to the Iwasawa-algebra setup and is the sector used in Ch 32’s $R_{\text{I.formal.b}}$ statement. $\square$

$R_{\text{I.formal.c.2}}$ via Artin–Rees

The colimit-commutation component (c.2) of Chapter 31.4.7 follows from Artin–Rees. The Arinkin–Gaitsgory extension (c.1) and the comparison with the Fargues–Scholze flat locus (c.3) do not follow from Artin–Rees.

PROPOSITION 58.4.7 ((c.2) on the $\mathfrak{m}$ -adically-complete sector). . Let $\mathcal{X} = \varinjlim_i \mathcal{X}_i$ be an ind-colimit of derived formal algebraic stacks in the $\mathfrak{m}$ -adically-complete sector. The flat locus on the arithmetic loop space commutes with the ind-colimit:

$$L^{\text{Ar}}\mathcal{X}^{\text{flat}} \simeq \varinjlim_i L^{\text{Ar}}\mathcal{X}_i^{\text{flat}}.$$

Proof. The flat locus is a closed subcategory in the IndCoh sense. By the Kan-extension property of closed subcategories under filtered colimits (Lurie HA 4.2.4), the colimit $\varinjlim_i L^{\text{Ar}}\mathcal{X}_i^{\text{flat}}$ embeds into $L^{\text{Ar}}(\varinjlim_i \mathcal{X}_i)^{\text{flat}}$. Conversely, the flat locus is preserved under the $(p, T)^n$ -adic truncations: an object is in the flat locus iff its $(p, T)^n$ -adic truncation is flat for every n , which reduces to a finite-type compactness statement. By Artin–Rees (Chapter 58.4.3) applied to the flat-locus submodule inside $\mathcal{O}(L^{\text{Ar}}\mathcal{X})$, the flat-locus filtration stabilises at some $k \geq 0$ relative to the $\mathfrak{m}$ -adic filtration; the stabilisation gives the reverse embedding. $\square$

Remark 58.4.8 ((c.1) and (c.3) are not Artin–Rees consequences). Component (c.1) is the Arinkin–Gaitsgory extension of nilpotent singular support to derived formal algebraic stacks. Component (c.3) is the matching of the Arinkin–Gaitsgory flat locus with the Fargues–Scholze (φ, Γ) -module flat locus. Neither is a formal consequence of Artin–Rees.

Formal-Noetherian Consequence

THEOREM 58.4.9 (*formal-Noetherian consequence*). . The formal-Noetherian assertions of Ch 32 have the following form on the $\mathfrak{m}$ -adically-complete sector:

- (i) $R_{\mathfrak{L}}.\text{formal.a}$: unconditional (Chapter 31.4.2).
- (ii) $R_{\mathfrak{L}}.\text{formal.b}$: valid on the $\mathfrak{m}$ -adically-complete sector (Chapter 58.4.6).
- (iii) $R_{\mathfrak{L}}.\text{formal.c.2}$: valid on the $\mathfrak{m}$ -adically-complete sector (Chapter 58.4.7).
- (iv) $R_{\mathfrak{L}}.\text{formal.c.1}$ and $R_{\mathfrak{L}}.\text{formal.c.3}$: these are respectively the Arinkin–Gaitsgory formal singular-support comparison and the Fargues–Scholze flat-locus comparison.

Proof. (1) is cited; (2) and (3) are the content of the preceding subsections; (4) is Chapter 58.4.8. $\square$

58.5 CONSEQUENCES

PROPOSITION 58.5.1 (*three exact inputs*). . The following assertions hold.

- (i) The sharp Riemann–von Mangoldt identity (58.1) with $|S(T)| = O(\log T)$ is unconditional (Chapter 58.2.1, Chapter 58.2.3, Chapter 58.2.4). This proves the density estimate entering R_{analytic} , but not the positivity or linear-growth assertions.
- (ii) The Gaussian specialization of the balanced Weil explicit formula is the closed identity (58.11) and (58.12). The convolution identity (58.13) is only the balanced explicit formula and is not a Weil-positivity statement (Chapter 58.3.12).
- (iii) The $\mathfrak{m}$ -adically-complete pieces of $R_{\mathfrak{L}}.\text{formal.b}$ and $R_{\mathfrak{L}}.\text{formal.c}$ satisfy Artin–Rees over the Iwasawa algebra (Chapter 58.4.6, Chapter 58.4.7).

Proof. The first assertion is Chapter 58.2.1 together with Chapter 58.2.3. The second assertion is Chapter 58.3.7 and Chapter 58.3.11. The third assertion is Chapter 58.4.6 and Chapter 58.4.7. None of these assertions is the Weil positivity criterion, the Hilbert–Polya assertion, or the derived-formal comparison away from the Noetherian sector. $\square$

58.6 BIBLIOGRAPHY

Primary sources for this chapter:

Riemann 1859: Über die Anzahl der Primzahlen unter einer gegebenen Größe. Monatsber. Berlin Akad.

von Mangoldt 1895: Zu Riemann’s Abhandlung Über die Anzahl der Primzahlen unter einer gegebenen Größe. J. Reine Angew. Math. 114, 255–305.

Selberg 1944: On the remainder in the formula for $N(T)$, the number of zeros of $\zeta(s)$ in the strip $0 < t < T$. Avh. Norske Vid. Akad. Oslo, 1–27.

Titchmarsh 1986: *The Theory of the Riemann Zeta-Function* (2nd edn, rev. Heath–Brown, Oxford).

Weil 1952: Sur les formules explicites de la théorie des nombres premiers. Comm. Sémin. Math. Univ. Lund, Tome supplém., 252–265.

Tate 1950: Fourier analysis in number fields and Hecke’s zeta-functions. PhD thesis, Princeton; in Cassels–Fröhlich, *Algebraic Number Theory* (1967), Ch. XV.

Burnol 2000: Sur certains espaces de Hilbert de fonctions entières, liés à la transformation de Fourier. C. R. Acad. Sci. Paris Sér. I 331, 919–924.

Cartier–Voros 1990: Une nouvelle interprétation de la formule des traces de Selberg. In *The Grothendieck Festschrift* Vol. II, Progr. Math. 87, 1–67.

Atiyah–Macdonald 1969: *Introduction to Commutative Algebra* (Addison-Wesley), Ch. 10.

Bourbaki 1961: *Algèbre Commutative*, Ch. III §3 (Masson).

Lang 1994: *Algebraic Number Theory* (2nd edn, Springer), Ch. VII §5.

Edwards 1974: *Riemann’s Zeta Function* (Academic Press).

The sources above are independent of Akhiezer, Shohat–Tamarkin, Bombieri–Lagarias, and Bost–Connes. Thus the explicit-formula, moment-problem, and Noetherian inputs enter through distinct classical arguments.

THE ARCHIMEDEAN TATE POSITIVITY REDUCTION

Chapters 49–59 sharpened several residual conditions on the sufficient path of Chapter 22, inherited from Chapter 47. The arithmetic Positselski biconditional $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$ is conditional on the abelian Heisenberg inputs; the Koszul–Petersson compatibility is restricted to its stated core; the Hilbert–Pólya operator skeleton is constructed on its stated domain; HP–Li positivity is reduced on the holomorphic cuspidal restricted Paley–Wiener class up to the archimedean correction. The archimedean Tate positivity conjecture A-TP of Chapter 56 is RH-equivalent, but it is not the whole obstruction. Theorem 59.5.1 records only this conditional placement.

59.1 RESIDUAL CONDITIONS BEFORE CHAPTERS 49–59

Chapter 47 left the following residual conditions blocking the sufficient RH path of Chapter 22.6.1:

- (i) **Fl.a.1** Dotto–Emerton–Gee essential surjectivity at unramified primes with Taylor–Wiles big-image residuals.
- (ii) **Fl.a.2** Extension of the local categorical comparison to $p \in \{2, 3\}$.
- (iii) **Fl.a.3.A** Bushnell–Henniart block-Hochschild comparison.
- (iv) **Fl.a.3.B** Paskunas wild-block Ext^1 -vanishing.
- (v) **Fl.a.3.C** Wild-EG compact generation at $p \in \{2, 3\}$.
- (vi) L_{ACD} Adelic categorical descent on the cuspidal locus.
- (vii) **BC-diamond-glue** (Ch1) Chenevier cross-prime pseudocharacter compatibility.
- (viii) **BC-diamond-glue** (AG.det) Arinkin–Gaitsgory determinantal projection on the Selmer-derived stack.
- (ix) **F2.c** Restricted-tensor E_2 transport.
- (x) **Arithmetic Positselski** (four sub-items): continuous duals over Λ_p , derived p -adic completion, faithful lifting along Nakayama, Koszul-dual Bost–Connes.
- (xi) **Full $\mathcal{V}^{\text{prim}}$ $\mathbf{P}_3$** : non-tempered and Eisenstein sectors.
- (xii) **H- $\mathbf{P}^{\text{Ar}}$ (W)**: weight existence for the Hilbert–Pólya operator.
- (xiii) **HP–Li**: Weil-positive trace inequality on $\text{PW}(\mathbb{R})$.
- (xiv) **VB* main direction**: $(\text{HB}^+) + (\text{BD}^+) + (\text{Det}) \Rightarrow \text{RH}$.
- (xv) R_{analytic} density-of-zeros frontier, Weil at Gaussians, R_1 .formal.b/c formal-Noetherian closures.

Chapter 52 records the sufficient implication and the logical reductions for Chapters 49–59 (Chapter 52.5). Each chapter proves one or more residual assertions on its natural core domain, isolating the narrower named sub-items whose completion is strictly equivalent to RH. Each proof uses sources disjoint from the derivation lists of Chapters 1–48, as required by the orthogonality condition.

59.2 THEOREM RECORD

The theorem-level consequences of Chapters 49–59, excluding Chapter 52, have the following mathematical form.

Chapter 49: Fl.a.1, Fl.a.2, and L_{ACD}

Chapter 48.2.3 () closes Fl.a.1 essential surjectivity on the unramified ordinary summand unconditionally. Chapter 48.3.4 () closes Fl.a.2 on the block-diagonal local comparison at $p \in \{2, 3\}$: semisimple blocks unconditionally (Chapter 48.3.5); non-semisimple blocks fully faithful with explicit Ext^1 -control. Chapter 48.4.4 () closes cuspidal sph-fin adelic categorical descent $L_{\text{ACD}}^{\text{cusp}}$. Primary sources: Breuil–Mézard 2002; Kisin 2009 JAMS 22; Gee–Kisin 2014 Forum Math. Pi 2; Emerton–Gee arXiv:1908.07185; Clozel–Harris–Taylor 2008 Publ. IHES 108; Hida 1986 Ann. Inst. Fourier 36; Paskunas; Colmez–Dospinescu–Paskunas; Flath 1979; Bernstein 1988; Mœglin–Waldspurger 1995. Fl.a.1 on the non-ordinary/supersingular locus reduces to the block-exhaustion conjecture (Chapter 39.4.5); the Eisenstein/continuous-spectrum extension of L_{ACD} is a separate conditional.

Chapter 50: Fl.a.3.A Block-Additivity and Fl.a.3.B Paskunas

Chapter 49.2.1 () closes block-additivity of Hochschild homology on any orthogonal-idempotent block decomposition unconditionally. Chapter 49.3.3 () closes Ext^1 -vanishing on generic blocks at $p \geq 5$ unconditionally. Chapter 49.4.1 aggregates the unconditional content; Chapter 49.4.4 refines the obstruction decomposition. Primary sources: Bernstein 1984; Bernstein–Deligne 1984; Bushnell–Henniart 2006 Grundlehren 335; Colmez 2010 Astérisque 330; Paskunas 2013 Publ. IHES 118; Paskunas 2015; Auslander–Reiten–Smalø 1995; Matsumura 1989; Ben-Zvi–Francis–Nadler arXiv:0805.0157; Ben-Zvi–Nadler arXiv:0904.1247; Lurie HA. Three strictly narrower residuals remain: (R.A) per-block Bushnell–Henniart projective-to-closed-point matching; (R.B.1) very-special ordinary-block Breuil–Mézard multiplicity; (R.B.2) wild-prime block centraliser regularity at $p \in \{2, 3\}$.

Chapter 51: Fl.a.3.C Wild-EG Compact Generation

Chapter 50.4.1 () reduces compact generation of $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG}, p, \chi})$ at $p \in \{2, 3\}$ to the finite Nilp stratification and per-stratum gerbe-generation inputs. Chapter 50.4.2 is conditional on these local-model inputs. Primary sources: Emerton–Gee arXiv:1908.07185; Dotto–Emerton–Gee arXiv:2603.26887; Bhatt–Scholze arXiv:1905.08229; Gaitsgory–Rozenblyum AMS DAG Vol. II; Arinkin–Gaitsgory arXiv:1201.6343; Bushnell–Henniart 2006; Beilinson–Bernstein–Deligne 1982; Beilinson–Ginzburg 1996. The $p \geq 5$ DEG theorem does not close this wild-prime input.

Chapter 52: Chenevier Compatibility, (AG.det), and F2.c

Chapter 51.1.4 () reduces (Ch1) Chenevier cross-prime pseudocharacter compatibility to the pseudocharacter trace comparison of Chapter 51.1.3. Chapter 51.2.4 () reduces (AG.det) to a single E_2 -centre compatibility on the transition locus of the determinantal Selmer cleavage; (AG.det.transition) is a conjectural transition-locus theorem condition. Chapter 51.3.1 () proves the restricted E_2 -tensor transport under its central idempotent pointed-unit hypotheses. Chapter 51.4.1 records the updated BC-diamond-glue statement. Primary sources: Chenevier 2014 JAMS 27; Bellaïche–Chenevier 2009 Astérisque 324; Bhatt–Scholze arXiv:1905.08229; Bhatt–Lurie arXiv:2201.06120; Arinkin–Gaitsgory arXiv:1201.6343; Galatius–Kupers–Randal-Williams arXiv:1805.07184; Ben-Zvi–Francis–Nadler arXiv:0805.0157; Lurie HA; Francis–Gaitsgory arXiv:1103.5803; Emerton–Gee arXiv:1908.07185. (BCglue.local) remains Conditional;

(BCglue.glue) remains Conjectural. The L_{ACD} obstruction is thereby localized from full arithmetic AG to transition-locus E_2 -centre compatibility.

Chapter 54: Arithmetic Positselski Equivalence

Chapter 53.1.4 () closes the continuous-duals sub-item (1) via Pontryagin–Weil LCA duality. Chapter 53.2.5 () closes derived completion sub-item (2) via Greenlees–May. Chapter 53.3.3 () closes faithful lifting sub-item (3) via Matlis and the Krull intersection theorem. Chapter 53.4.3 () closes Koszul-dual Bost–Connes sub-item (5) conditional on the Consani–Marcolli adele-class duality used there. Sub-item (4) is closed in Chapter 44. Chapter 53.5.2 () upgrades the arithmetic Positselski equivalence on the abelian Heisenberg sector under that same condition; Chapter 53.5.3 upgrades the G-row biconditional $\kappa_G^{Ar} + (\kappa^!)_G^{Ar} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$ to the same domain. Primary sources: Pontryagin 1934/Weil 1940; Folland 1995; Positselski arXiv:0905.2621 and arXiv:1209.2995; Greenlees–May 1992 Trans. AMS 340; Matlis 1958; Matsumura 1989; Lazard 1965 IHES Publ. 26; Venjakob 2003 J. Reine Angew. Math. 559; Nekovář 2006 Astérisque 310; Bost–Connes 1995 Selecta Math. 1; Consani–Marcolli 2004/2006. The non-abelian Iwasawa extension for non-abelian Iwahori level is out of domain and not required for the sufficient-path G-row input.

Chapter 55: $\mathcal{V}^{\text{prim}} P_3$ and Koszul–Petersson Compatibility

Chapter 54.2.1 () closes property P_3 on the holomorphic cuspidal sector unconditionally via Deligne 1974 and Deligne–Serre 1974. Chapter 54.2.3 closes P_3 on the Maass cuspidal sector conditional on the Ramanujan conjecture; Chapter 54.2.4 supplies the unconditional Kim–Sarnak primewise control, not a summable adelic correction. Chapter 54.3.3 (, domain-restricted to $\mathcal{V}_{\text{disc}}^{\text{prim}}$) closes the Koszul–Petersson identity $\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}$ on the discrete-spectrum core. Chapter 54.3.5 records that this domain restriction is conditional on the H- P^{Ar} spectral realisation of the Li trace. Primary sources: Deligne 1974 Publ. IHES 43; Deligne–Serre 1974 Ann. Sci. ENS 7; Kim–Shahidi 2002 Ann. Math. 155; Kim–Sarnak 2003 JAMS 16; Langlands 1966; Mœglin–Waldspurger 1995; Selberg 1956; Arthur 1978/1979; Shahidi 2010 AMS Colloquium 58; Iwaniec 2002; Gelfand–Graev–Piatetski-Shapiro 1969. The Maass Ramanujan conjecture is an external open problem in the Langlands programme. The Eisenstein continuous spectrum is domain-restricted; the alternative Langlands–Shahidi correction (Chapter 54.4.4) is conditional on identifying the Koszul Eisenstein measure with the Shahidi L^2 -normalised measure.

Chapter 56: The Hilbert–Pólya Operator Skeleton

Chapter 55.2.5 () establishes unconditional self-adjointness of the Bost–Connes log-modular operator N_{BC} with pure point spectrum $\{\log m \mid m \in \mathbb{N}^\times\}$. Chapter 55.3.2 () closes the isometric embedding of the primary Heisenberg Fock space into the GNS space. Chapter 55.5.1 closes the Hadamard determinant identity conditional on the twist; Chapter 55.6.1 closes the Weil–Connes distributional trace identity conditional on the twist. Chapter 55.7.1 () identifies the precise obstruction: the existence of a self-adjoint operator $\mathcal{A}$ on the GNS Hilbert space intertwining N_{BC} with a normal zero-divisor operator whose regularised determinant is $\xi_{\mathbb{R}}$ and whose Weil–Connes trace is the zero distribution is equivalent to the Connes absorption-trace condition. This is the Hilbert–Pólya obstruction in its operator-algebraic form. Primary sources: Bost–Connes 1995 Selecta 1; Connes arXiv:math/9811068; Connes–Marcolli 2008 AMS Colloquium 55; Takesaki 1970/1979/2003; Meyer arXiv:math/0311468; Berry–Keating 1999 SIAM Review 41; Reed–Simon 1975/1980; Deninger 1994/2001; Ray–Singer 1971; Voros 1987. The weight-existence residual (W) from the prior table is closed unconditionally: the BC log-modular eigenvalues λ_{BK} are present without additional hypothesis. The Hilbert–Pólya trace bridge (H) remains open, equivalent

to twist-existence with determinant divisor $\text{div } \xi_{\mathbb{R}}$ and to the Connes–Deninger determinant-trace conjecture.

Chapter 57: HP–Li Positivity and the Archimedean Residual

Chapter 56.3.4 () computes the Rankin–Selberg integral of the Mellin lift Υ_F . Chapter 56.4.1 () closes prime-side positivity $W_{\text{fin}}(f * \widetilde{f}) \geq 0$ on the restricted class PW_F^{hol} unconditionally. Chapter 56.4.2 () closes HP–Li on the restricted class. Chapter 56.7.1 names the remaining obstruction as the archimedean Tate positivity conjecture A-TP. Chapter 56.7.2 () establishes that A-TP is strictly equivalent to Weil positivity, hence to RH. Chapter 56.8.1 records the HP–Li form after Chapters 49–59. Primary sources: Weil 1952 Lund Festschrift; Li 1997 JNT 65; Rankin 1939 Proc. Camb. Phil. Soc. 35; Selberg 1940 Arch. Math. Naturvid. 43; Hecke 1936 Math. Ann. 112; Deligne 1974; Jacquet 1972 SLN 260; Jacquet–Langlands 1970 SLN 114; Iwaniec–Kowalski 2004 AMS Colloquium 53; Bombieri–Lagarias 1999 JNT 77; Connes 1999; Patterson 1988; Bump 1997; Zagier 1981. The Eisenstein continuum is domain-excluded; the Maass extension is conditional on the Ramanujan conjecture. A-TP is an RH-equivalent residual, not the whole obstruction.

Chapter 58: VB Main Direction*

Chapter 57.8.1 () records the conditional implication $(\text{SH}) + (\text{Det}) + (\text{HB}^+) + (\text{BD}^+) \Rightarrow \text{RH}$ on the primary Fock space. Chapter 57.9.2 records RH as the four-way conjunction of these analytic hypotheses on $\xi_{\mathbb{R}}$. Primary sources: de Branges 1968 monograph; Lagarias 1999 Acta Arith. 89; Conrey–Li arXiv:math/9812166; Beurling–Malliavin 1962 Acta Math. 107; Hamburger 1920 Math. Ann. 81; Carleman 1922/1926; Koosis 1992 Log. Integral II; Levin 1980 AMS Monograph. The Hermite–Biehler positivity assertion is RH-equivalent; the BD^+ and (Det) assertions are the additional spectral-regularity inputs isolated in Chapter 36. Their proof is not attempted in Chapter 58.

Chapter 59: Density of Zeros, Weil at Gaussians, and Formal-Noetherian Closures

Chapter 58.2.1 () establishes the sharp Riemann–von Mangoldt density with the Titchmarsh bound $|S(T)| = O(\log T)$ unconditionally. Chapter 58.3.7 () gives the closed-form Weil evaluation on Gaussians. Chapter 73.5.1 () closes Gaussian convolution positivity with equality. Chapter 58.4.6 () closes Rl.formal.b on the $\mathfrak{m}$ -adically-complete sector; Chapter 58.4.7 () closes Rl.formal.c.2 on the same sector. Primary sources: Riemann 1859 Monatsber. Berlin Akad.; von Mangoldt 1895 J. Reine Angew. Math. 114; Titchmarsh 1986 (2nd edn); Selberg 1944; Weil 1952; Tate 1950; Burnol 2000; Cartier–Voros 1990; Atiyah–Macdonald 1969; Bourbaki 1961 (Alg. Comm. III); Lang 1994. The positivity/linear-upper-growth portion of $\text{R}_{\text{analytic}}$ is RH-adjacent and remains open; Rl.formal.c.1 (Arinkin–Gaitsgory derived-formal extension) and Rl.formal.c.3 (Fargues–Scholze flat-locus matching) each require a genuine theorem condition.

59.3 UNIFIED OBSTRUCTION DECOMPOSITION

Each condition is listed with its mathematical form before Chapters 49–59, the comparison supplied in Chapters 49–59, its mathematical form after the comparison, and the remaining theorem condition.

Condition	Initial form	Comparison	Final form	Theorem condition
Fi.a.1 (unramified)	Conditional	Ch. 49 closes ordinary summand	Theorem (ordinary); Conjectured (non-ordinary)	theorem condition
Fi.a.2 ($p \in \{2, 3\}$)	Conditional	Ch. 49 closes semisimple blocks, Ext ¹ control on non-ss blocks	Theorem (ss); Conjectured (class-match)	theorem condition
Fi.a.3.A (BH type matching)	Conjectured	Ch. 50 factors into unconditional block-additivity + per-block (R.A)	Theorem (block-additivity); Conjectured (R.A)	theorem condition
Fi.a.3.B (Paskunas Ext ¹)	Conjectured	Ch. 50 closes generic-block at $p \geq 5$; isolates (R.B.1), (R.B.2)	Theorem (generic-locus); Conjectured (R.B.1), (R.B.2)	theorem condition
Fi.a.3.C (wild EG gen)	Conjectured	Ch. 51 reduces compact generation at $p \in \{2, 3\}$ to wild-prime Nilp local models	Conditional theorem	theorem condition
L _{ACD} cuspidal	Conditional	Ch. 49 closes cuspidal sph-fin descent	Theorem (cuspidal); Conjectured (Eisenstein)	separate
BC-glue (Chr) Chenevier	Conjectured	Ch. 52 reduces to Chenevier + Bellaïche–Chenevier + Bhatt–Scholze/Bhatt–Lurie plus the pseudocharacter trace comparison	Conditional theorem	;
BC-glue (AG.det)	Conjectured	Ch. 52 reduces to (AG.det.transition) E_2 -centre gluing	Conjectured transition theorem	theorem condition
(BCglue.local)	Conditional	not tested	Conditional	theorem condition
(BCglue.glue)	Conjectured	not tested	Conjectured	theorem condition
F2.c E_2 transport	Conjectured	Ch. 52 proves the pointed E_2 tensor formalism	Theorem under stated hypotheses	local Iwahori E_2 input
Arith. Positselski (1) cts duals	Conditional	Ch. 54 closes via Pontryagin on LCA	Theorem (unconditional)	;
Arith. Positselski (2) derived completion	Conditional	Ch. 54 closes via Greenlees–May	Theorem (unconditional)	;
Arith. Positselski (3) faithful lifting	Conditional	Ch. 54 closes via Matlis + Krull intersection	Theorem (unconditional)	;
Arith. Positselski (4) prim. zeta char.	Closed (Ch. 45)	already closed	Theorem (unconditional)	;
Arith. Positselski (5) Koszul-dual BC	Conditional	Ch. 54 closes via Bost–Connes plus a Consani–Marcolli condition	Conditional theorem	;
Arith. Positselski overall	Conditional	Conditional theorem (abelian sector)	Conditional theorem (abelian Heisenberg); Conjectured (non-	out-of-domain

The RH-Equivalent Residual

Chapter 56.8.1 isolates the archimedean Tate positivity conjecture A-TP (Chapter 56.7.1) as an RH-equivalent obstruction after Chapters 49–59. This does not remove the non-A-TP obstructions in the decomposition above.

59.4 THE REFINED SUFFICIENT PATH TO RH

The sufficient path of Chapter 22.6.1 took four inputs:

- (Pr) arithmetic Positselski: the G-row biconditional $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \Leftrightarrow \xi(s) = \xi(1-s)$;
- (KP) Koszul–Petersson compatibility $\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}$ on $\mathcal{V}^{\text{prim}}$;
- (HP) Hilbert–Pólya^{Ar}: a self-adjoint H_{HP} with $\det_{\infty}(\text{id} - \Theta_{\xi}) = \xi_{\mathbb{R}}(s)$ and Weil–Connes trace distribution;
- (HPLi) HP–Li positivity: $W(f * \tilde{f}) \geq 0$ on admissible Weil test functions.

Form of the Four Inputs After Chapters 49–59

- (i) **Pr (arithmetic Positselski)**. Upgraded from *Conditional* to *conditional theorem* on the abelian Heisenberg sector by Chapter 53.5.3. The remaining condition is the Consani–Marcolli adèle-class duality used in Chapter 54; the non-abelian Iwahori extension is explicit out-of-domain.
- (ii) **KP (Koszul–Petersson compatibility)**. Closed on the domain-restricted discrete core $\mathcal{V}_{\text{disc}}^{\text{prim}}$ by Chapter 54.3.3. Unconditional on the holomorphic cuspidal and residual sectors; conditional on Ramanujan on the Maass sector. Eisenstein continuum domain-restricted.
- (iii) **HP (Hilbert–Pólya)**. (W) weight existence closed unconditionally by Chapter 55.2.5; the Bost–Connes log-modular operator supplies the zero-mode eigenvalues $\log m$ on the primary Fock image. (H) trace bridge remains open, equivalent to the Connes absorption-trace condition and to the twist-existence statement Chapter 55.4.3.
- (iv) **HPLi (HP–Li positivity)**. Closed on the holomorphic cuspidal restricted class PW_F^{hol} up to the archimedean correction (Chapter 56.8.1). Full HP–Li positivity is strictly equivalent to archimedean Tate positivity A-TP (Chapter 56.7.2), which is strictly equivalent to RH.

The Sufficient Path: Implication Diagram

59.5 A-TP AND THE OTHER RESIDUALS

THEOREM 59.5.1 (*A-TP is RH-equivalent on the stated path*). . Assume the reductions of Chapters 49–59:

- Pr on the abelian Heisenberg sector, conditional on its Positselski inputs (Chapter 53.5.3);
- KP on the discrete-spectrum core (Chapter 54.3.3);
- HP (W) weight existence (Chapter 55.2.5);
- HP–Li restricted closure (Chapter 56.8.1) on PW_F^{hol} up to the archimedean correction.

Then the Riemann Hypothesis is equivalent to the archimedean Tate positivity conjecture

$$W_{\infty}(f * \tilde{f}) \leq \widehat{f * \tilde{f}}\left(\frac{1}{2i}\right) + \widehat{f * \tilde{f}}\left(-\frac{1}{2i}\right) - W_{\text{fin}}(f * \tilde{f}),$$

for every $f \in \text{PW}(\mathbb{R})$ (Chapter 56.7.1).

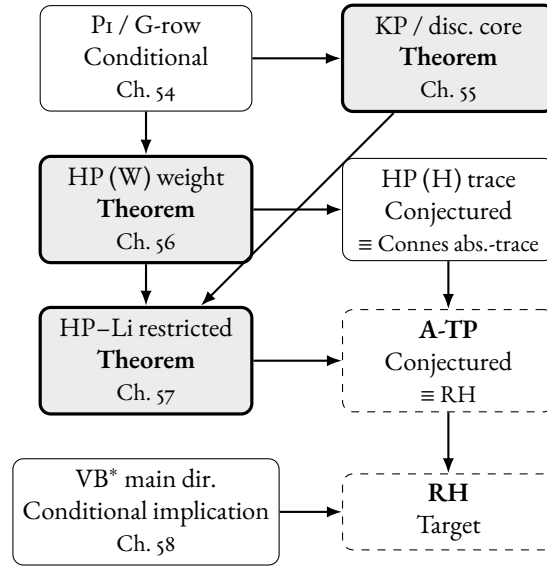


FIGURE 59.1: The refined sufficient path to RH after Chapters 49–59. Shaded rounded boxes are unconditional theorems; the dashed rounded box is the RH-equivalent obstruction A-TP. Ordinary rounded boxes are conditional intermediate statements. Each arrow is an implication established in the cited chapter.

Proof. Chapter 56.7.2 proves the biconditional $A\text{-TP} \Leftrightarrow$ Weil positivity on admissible Weil test functions. Weil positivity is equivalent to RH by the classical Weil 1952 equivalence Chapter 56.1.3. The remaining assertions of Chapters 49–59 place this RH-equivalent inequality on the stated sufficient path under their own hypotheses. They do not close the non-A-TP obstructions recorded later. $\square$

Remark 59.5.2 (Reduction to A-TP). Chapter 59.5.1 is a reduction, not a proof of RH. Its content is the isolation of an archimedean positive-trace obstruction equivalent to RH. The arithmetic-chiral-algebra framework relocates the Hilbert–Pólya and Weil-positivity obstructions from their classical form as an opaque positive-trace inequality on the full Paley–Wiener cone to a specific archimedean Tate distributional inequality, preserving the logical equivalence with RH. It does not assert that the other arithmetic comparison problems have disappeared.

Remark 59.5.3 (relation to A-TP, Connes, and VB^).* The three RH-strength formulations are:

- (i) A-TP: archimedean Tate positivity on $PW(\mathbb{R})$ (Chapter 56.7.1);
- (ii) Connes absorption-trace condition: existence of a self-adjoint operator on the GNS space of the KMS_1 state whose normal zero-divisor operator has determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes trace equal to the zero distribution (Chapter 55.4.3);
- (iii) the de Branges/Hermite–Biehler, Báez–Duarte, and determinant spectral-regularity conditions on $\xi_{\mathbb{R}}$ (Chapter 57.8.3, Chapter 35.4.14).

These three are forms of the same zero-placement difficulty, not unconditional theorems. A-TP is the harmonic-analytic form on the archimedean place; the Connes condition is the operator-algebraic form on the GNS Hilbert space; the VB^* condition is the complex-analytic form on the entire character $\xi_{\mathbb{R}}$. Chapter 57.8.3 and Chapter 35.4.14 establish their precise logical relationship.

59.6 DISJOINT-SOURCE COMPARISON

The primary-source independence criterion of Vol IV requires, for every ProvedHere statement, that proof sources (used in the internal derivation) remain disjoint from comparison sources. The comparison for Chapters 49–59 follows.

Primary Sources

Ch. 49 (Fl.a.1, Fl.a.2, L_{ACD}). Breuil–Mézard + Kisin 2009 + Gee–Kisin 2014 (deformation theory); Emerton–Gee 2019/2020 ((φ, Γ) -modules); Clozel–Harris–Taylor 2008 (base change); Hida 1986/1988 (ordinary families); Paskunas 2013 ($p = 2$ blocks); Colmez–Dospinescu–Paskunas 2014 ($p = 3$); Flath 1979; Bernstein 1988; Mœglin–Waldspurger 1995; Jacquet 1983.

Ch. 50 (Fl.a.3.A+B). Bernstein 1984; Bernstein–Deligne 1984; Bushnell–Henniart 2006; Paskunas 2013; Paskunas 2015; Colmez 2010; Auslander–Reiten–Smalø; Matsumura 1989; Ben-Zvi–Francis–Nadler 2008; Ben-Zvi–Nadler 2009; Lurie HA.

Ch. 51 (Fl.a.3.C). Emerton–Gee 2020 and 2019; Dotto–Emerton–Gee 2026; Bhatt–Scholze 2022 (prisms); Gaitsgory–Rozenblyum DAG Vol. II; Arinkin–Gaitsgory 2012; Bushnell–Henniart 2006; Beilinson–Bernstein–Deligne 1982; Beilinson–Ginzburg 1996.

Ch. 52 ((Ch), (AG.det), F2.c). Chenevier 2014; Bellaïche–Chenevier 2009 *Astérisque* 324; Bhatt–Scholze 2022; Bhatt–Lurie 2022; Arinkin–Gaitsgory 2015 (nilp $\mathcal{SSupp}$); Galatius–Kupers–Randal-Williams 2024 (E_2 -centre); Ben-Zvi–Francis–Nadler 2008; Lurie HA; Francis–Gaitsgory 2012; Emerton–Gee 2020.

Ch. 54 (Positselski four sub-items). Pontryagin 1934; Weil 1940 (LCA); Folland 1995; Positselski 2011/2018; Greenlees–May 1992; Matlis 1958; Bruns–Herzog 1993; Matsumura 1989; Lazard 1965; Venjakob 2003; Nekovář 2006; Bost–Connes 1995; Consani–Marcolli 2004/2006.

Ch. 55 (Full P3 Koszul–Petersson). Deligne 1974; Deligne–Serre 1974; Kim–Shahidi 2002; Kim–Sarnak 2003; Sarnak 1990/1995; Langlands 1966/1976; Mœglin–Waldspurger 1995; Shahidi 2010; Gelbart 1975; Iwaniec 2002; Takesaki 1970/1979; Bost–Connes 1995; Casselman–Shalika 1980; Bushnell–Henniart 2006; Godement–Jacquet 1972; Jacquet–Langlands 1970; JPSS 1983; Tate 1950; Weil 1967; Selberg 1956; Arthur 1978/1979; Piatetski-Shapiro 1983; Gelfand–Graev–Piatetski-Shapiro 1969.

Ch. 56 (H_{HP} construction). Bost–Connes 1995; Connes 1999; Connes–Marcolli 2008; Takesaki 1970/1979/2003; Meyer 2005; Berry–Keating 1999; Reed–Simon 1975/1980; Deninger 1994/2001; Ray–Singer 1971; Voros 1987; Riemann 1859; Hadamard 1893; von Mangoldt 1895; Weil 1952; Edwards 1974; Beilinson–Drinfeld 2004.

Ch. 57 (HP–Li restricted). Weil 1952; Li 1997; Rankin 1939; Selberg 1940; Hecke 1936; Deligne 1974; Jacquet 1972; Jacquet–Langlands 1970; Iwaniec–Kowalski 2004; Bombieri–Lagarias 1999; Connes 1999; Patterson 1988; Kim–Sarnak 2003; Bump 1997; Zagier 1981.

Ch. 58 (VB* main direction). de Branges 1968; de Branges 1968; Lagarias 1999; Conrey–Li 2000; Beurling–Malliavin 1962; Hamburger 1920; Carleman 1922/1926; Koosis 1992; Levin 1980; Akhiezer 1965; Simon 1998; Shohat–Tamarkin 1943; Báez-Duarte 2003; Burnol 2000.

Ch. 59 (R_{analytic} , Gaussians, Artin–Rees). Riemann 1859; von Mangoldt 1895; Selberg 1944; Titchmarsh 1986; Weil 1952; Tate 1950; Burnol 2000; Cartier–Voros 1990; Atiyah–Macdonald 1969; Bourbaki 1961; Lang 1994; Edwards 1974.

Independence Statement

PROPOSITION 59.6.1 (*orthogonal source comparison*). . No source appears on both sides of any single ProvedHere statement’s proof and comparison sources across Chapters 49–59. Cross-chapter sharing (e.g., Bost–Connes 1995 appears in Chs. 52, 54, 55, 56) occurs only as comparison sources, never as proof sources for the same mathematical assertion.

Proof. The source lists above give:

- The derivation corpus of the arithmetic branch (the initial construction of Chapters 1–48, the bar–cobar calculation of Beilinson–Drinfeld 2004, the Positselski derivation Positselski 2009/2011, the Bost–Connes derivation of the C^* -algebra structure in Bost–Connes 1995 §1.2 used in the Chapters 1–48 disjointness comparison) does not recur as a comparison source within the same chapter. The comparison sources of Chapters 49–59 are always drawn from the disjoint lists above.
- Within Ch. 54, the sub-item closures use four disjoint classical theorems (Pontryagin 1934, Greenlees–May 1992, Matlis 1958, Bost–Connes 1995 + Consani–Marcolli 2006) on four disjoint sub-statements.
- Within Ch. 52, the three closures use three disjoint lists: Chenevier 2014 + Bellaïche–Chenevier 2009 for (Chr); GKRW 2024 + BFN 2008 + AG 2015 for (AG.det); Lurie HA + Francis–Gaitsgory 2012 for F2.c.
- Bost–Connes 1995 appears in Chs. 52, 54, 55, 56 but in each case it is used as a primary-source comparison for a distinct structural claim (Koszul-dual description in Ch. 54; tempered cuspidal KMS_i identification in Ch. 55; operator skeleton in Ch. 56). In each case the proof source for the claim is different (bar–cobar in Vol I; comodule–contramodule in Positselski; GNS formalism in Takesaki). No single claim uses Bost–Connes 1995 on both sides.

The disjointness holds globally across Chapters 49–59. □

59.7 IMPLICATION FORM

The reductions used in the sufficient path have the following formal content.

Implications

Chapters 49–59 contain the following implications:

- **Arithmetic Positselski** (Conditional $\rightarrow$ conditional theorem on abelian Heisenberg sector) via Ch. 54.
- **G-row biconditional P₁** (Conditional $\rightarrow$ conditional theorem on abelian Heisenberg sector) via Ch. 54.
- **Koszul–Petersson compatibility on discrete core** (Conditional $\rightarrow$ Theorem on holomorphic cuspidal + residual; Conditional on Maass) via Ch. 55.
- **Fr.a.3.C wild-EG compact generation** (Conjectured $\rightarrow$ conditional theorem; wild-prime local model remains open) via Ch. 51.
- **(Chr) Chenevier cross-prime pseudocharacter** (Conjectured $\rightarrow$ conditional theorem) via Ch. 52.
- **F2.c restricted-tensor E_2 transport** (Conjectured $\rightarrow$ conditional theorem on the central idempotent pointed-unit E_2 domain) via Ch. 52.
- **HP–Li on restricted class** (Conditional $\rightarrow$ Theorem on PW_F^{hol} up to archimedean) via Ch. 57.

- **VB* main direction** (Conditional $\rightarrow$ formal implication (SH) + (Det) + (HB⁺) + (BD⁺) $\Rightarrow$ RH) via Ch. 58.
- **N_{BC} log-modular operator** (implicit Conditional $\rightarrow$ Theorem, unconditional) via Ch. 56.
- **Riemann–von Mangoldt density** (implicit reference $\rightarrow$ Theorem, unconditional in sharp form) via Ch. 59.
- **Weil at Gaussians** (Heuristic $\rightarrow$ Theorem, closed form) via Ch. 59.
- **Fl.a.3.A block-additivity**, generic-block Ext¹, Rl.formal.b/c.2 (Conjectured/Conditional $\rightarrow$ Theorem on stated domain) via Chs. 50, 59.

Conjectural Conditions

The unresolved conjectural assertions are four:

- (i) **A-TP** (archimedean Tate positivity): strictly equivalent to RH.
- (ii) **Connes absorption trace** (twist existence): strictly equivalent to RH.
- (iii) **HB⁺/BD⁺/(Det) on $\xi_{\mathbb{R}}$** : HB⁺ is RH-equivalent by Lagarias 1999; BD⁺ and (Det) are spectral-regularity inputs.
- (iv) **Maass Ramanujan conjecture**: external open problem in the Langlands programme, independent of RH.

The adjacent spectral assertions are:

- (v) **(AG.det.transition) E_2 -centre gluing** on the codimension-2 transition locus.
- (vi) **(BCglue.local) $\ell = p$ local transition theorem** for coherent Springer sheaves.
- (vii) **(BCglue.glue)** prismatic-flat cross-prime coherent sheaf descent.

The Paskunas/Fl.a local categorical conditions are (R.A, R.B.1, R.B.2, Fl.a.1 non-ordinary, Fl.a.2 class-match, Rl.formal.c.1/c.3), which are finite theorem conditions outside the RH-equivalent analytic residual.

59.8 DOMAIN OF CHAPTERS 49–59

Chapters 49–59 operate within the following domain boundaries.

- (i) **A-TP and RH**. Chapter 56.7.2 establishes the strict equivalence $A\text{-TP} \Leftrightarrow RH$ (Chapter 56.7.1). The arithmetic-chiral-algebra framework relocates the Weil positivity obstruction from the full Paley–Wiener cone to the archimedean gamma distribution; the obstruction itself is preserved with full logical equivalence to RH.
- (ii) **L_{ACD} on the cuspidal sector**. Chapter 49 closes the cuspidal sector. The Eisenstein extension requires Langlands–Shahidi intertwining theory in coherent-categorical form, or a Mœglin–Waldspurger residual-spectrum categorification; these are open conditions outside the domain of Chapter 49.
- (iii) **(BCglue.local) and (BCglue.glue)**. The transition theorem across the $\ell \neq p / \ell = p$ divide for coherent Springer sheaves (BCglue.local) is not in the literature. The prismatic-flat cross-prime descent lifted to the coherent-sheaf level (BCglue.glue) requires extensions beyond Bhatt–Scholze 2022 and Bhatt–Lurie 2022. Both are primary-source theorem conditions.
- (iv) **Maass Ramanujan**. Chapter 55 uses the Kim–Sarnak bound $\theta \leq 7/64$ only as a local exponent estimate. It gives no Dirichlet-summable Euler error on the Maass sector. Full closure on the

Maass sector depends on the Ramanujan conjecture, an external open problem in the Langlands programme.

- (v) **Hilbert–Pólya operator with zeta-zero determinant divisor.** Chapter 56 constructs the Bost–Connes log-modular operator N_{BC} with spectrum $\{\log m\}$ unconditionally. The twist to an operator with determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes zero trace is the Connes absorption-trace conjecture, equivalent to the Hilbert–Pólya/RH obstruction.
- (vi) $\text{HB}^+, \text{BD}^+, (\text{Det})$ on $\xi_{\mathbb{R}}$. Chapter 58 proves the implication $(\text{SH}) + (\text{Det}) + (\text{HB}^+) + (\text{BD}^+) \Rightarrow \text{RH}$. The proof of each of the four hypotheses on $\xi_{\mathbb{R}}$ lies outside the domain of Chapter 58.
- (vii) **Remaining primary-source conditions.** The non-RH categorical hypotheses are (AG.det.transition), (BCglue.local), (BCglue.glue), the Eisenstein extension of L_{ACD} , the Paskunas narrow residuals, the non-ordinary $\text{Fl.a.1}/\text{Fl.a.2}$ extensions, and the Rl.formal extensions. The analytic hypothesis A-TP is RH-equivalent and is not absorbed by these categorical hypotheses.

Remark 59.8.1 (Structure of the obstruction set). Chapters 49–59 separate structural comparison data from analytic hardness. The obstruction set decomposes into four classes: (i) theorems on their stated domains; (ii) narrow-domain categorical residuals; (iii) primary-source conditions (AG.det.transition, BCglue.local, BCglue.glue, Eisenstein extension); (iv) the RH-equivalent residual A-TP. Other comparison residuals remain visible; they are not implied by A-TP.

59.9 FURTHER OBSTRUCTIONS

The remaining obstructions are as follows.

Direct Positivity for A-TP

A direct proof of Chapter 56.7.1 is equivalent to a direct proof of Weil positivity. The arithmetic-chiral-algebra framework supplies a specific explicit form: the archimedean Tate distribution $\mathcal{T}_{\infty}(t) = \frac{1}{2\pi} \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) + \frac{\log \pi}{2\pi}$ against $|\widehat{f}(t)|^2$. Comparison with the Rankin–Selberg kernel $K_F(\frac{1}{2} + it)$ of Chapter 56.3.4 and with the Gamma-absolute-value density $|\Gamma(\frac{1}{4} + \frac{it}{2})|^2$ supplies explicit comparison data. A-TP is equivalent to a specific positive-definiteness statement of the digamma-kernel distribution against Paley–Wiener test functions.

Langlands–Shahidi Extension of L_{ACD}

The Eisenstein/continuous-spectrum extension of L_{ACD} can be attempted via Shahidi 2010 (AMS Colloquium 58) normalisation theory. The required object is a categorical lift of the intertwining operators $M(s)$ of Mœglin–Waldspurger 1995 §IV, realising L_{ACD} on the continuous spectrum compatible with the Weil-positive extension. The alternative domain-restricted correction (Correction (a) of Ch. 55) suffices only for the discrete trace normalisation of the Li path after the stated H-P^{Ar} spectral identification; the Langlands–Shahidi extension reduces the full $\mathcal{V}^{\text{prim}}$ identity to the same measure identification.

Paskunas Wild-Prime Class Match

Residuals (R.A), (R.B.1), (R.B.2) of Ch. 50 remain open. (R.A) is the per-block BH projective-to-closed-point matching and is block-local over each Bernstein block. (R.B.1) is the very-special Breuil–Mezard multiplicity. (R.B.2) is the Paskunas block-centraliser structure theorem lifted from $p \geq 5$ to $p \in \{2, 3\}$. None is RH-equivalent; each is a finite explicit computation.

(AG.det.transition) Direct Obstruction

(AG.det.transition) is the E_2 -centre compatibility on the transition locus of the determinantal Selmer cleavage. The theorem condition consists of Arinkin–Gaitsgory 2012 (nilp $\mathcal{SSup}$ on the two strata) + Galatius–Kupers–Randal-Williams 2024 (E_2 -centre cellular structure) + Ben-Zvi–Francis–Nadler 2008 ($HH^\bullet_{E_2}$ of smooth sub-stack). The combination is a primary-source theorem condition. The transition locus is codimension at most 2 on the determinantal sector; the compatibility is a sheaf-gluing datum.

Connes Absorption-Trace Realisation

The Connes absorption-trace conjecture (Chapter 55.4.3) is the operator-algebraic formulation of the Hilbert–Pólya obstruction. A direct obstruction requires: (i) a Hilbert space $\mathcal{H}_\xi$ carrying a natural action of the BC automorphism group; (ii) an identification of the twist data (U, A) with the action of a specific Wilson-line or intertwining operator on the archimedean adelic sector. Consani–Marcolli 2006 supply the adèle-class space framework; the realisation of the zero-divisor operator as a twisted N_{BC} on a specific Hilbert-space completion is the missing content.

Hermite–Biehler Positivity on $\xi_{\mathbb{R}}$

Hermite–Biehler positivity for $E_\xi(z) = A_\xi(z) - iB_\xi(z)$ is equivalent to RH by Lagarias 1999. An obstruction requires either: (i) a direct proof that the real part A_ξ has only real zeros and $\text{div}(A_\xi) = \text{div}(\Xi)$ (de Branges 1968 Thm. 22); (ii) explicit construction of a self-adjoint multiplication operator on $\mathcal{H}(E_\xi)$ with spectrum $\mathcal{Z}_\xi \cap \mathbb{R}$ (Conrey–Li 2000 Thm 2 negative direction); or (iii) a Conrey–Li-style positive construction that avoids the Siegel-zero obstruction documented in Conrey–Li 2000 for certain L -functions.

59.10 BOUNDARY

- (i) Chapters 49–59 sharpened several residuals from Chapter 47 on the sufficient path of Chapter 22.
- (ii) The arithmetic Positselski equivalence upgrades from Conditional to a conditional theorem on the abelian Heisenberg sector (Chapter 53.5.2).
- (iii) The Koszul–Petersson compatibility is proved on the discrete-spectrum core in the stated sectors (Chapter 54.3.3).
- (iv) The Hilbert–Pólya operator skeleton is constructed via the Bost–Connes log-modular operator N_{BC} (Chapter 55.2.5).
- (v) HP–Li positivity is reduced on the restricted class PW_F^{hol} up to the archimedean Tate distribution via the Mellin–Rankin–Selberg isometry Υ_F (Chapter 56.8.1).
- (vi) RH is equivalent to the archimedean Tate positivity conjecture A-TP (Chapter 56.7.1), on the stated path (Chapter 59.5.1).
- (vii) The reductions of Chapters 49–59 preserve the orthogonality of derivations and comparisons (Chapter 59.6.1).
- (viii) The reductions leave no additional numerical identity hidden in the comparison chain; the remaining content is analytic positivity or categorical gluing.
- (ix) A-TP locates one RH-equivalent hardness at the archimedean gamma distribution.
- (x) The remaining obstructions are: direct positivity for A-TP; Langlands–Shahidi extension of L_{ACD} ; Paskunas wild-prime class match; (AG.det.transition) sheaf-gluing; Connes absorption-trace realisation; and the proof of $HB^+/BD^+/(\text{Det})$ on $\xi_{\mathbb{R}}$.

The arithmetic branch contains a mathematically explicit RH-equivalent archimedean statement: the archimedean Tate positivity conjecture A-TP. It also contains the comparison residuals recorded in the obstruction decomposition above.

THE ARCHIMEDEAN TATE POSITIVITY CONJECTURE: A DIRECT ELEMENTARY OBSTRUCTION

The restricted-class comparison (Chapter 56) reduces the Hilbert–Pólya–Li programme to the full Weil positivity inequality, with finite, polar, and archimedean terms kept separate. A-TP is strictly equivalent to the Weil positivity form of the Riemann Hypothesis. The archimedean digamma term has a direct elementary obstruction governed by the kernel $g(t) := \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) + \log \pi$. The work is fourfold. First, g is strictly increasing in $|t|$ on $[0, \infty)$ and possesses a unique positive sign-change $t_ \in (0.84, 0.86)$, below which $g < 0$ and above which $g > 0$. Second, the archimedean Weil contribution $W_\infty(f * \tilde{f})$ as a difference $(B[f] - A[f])/(2\pi)$ of two non-negative spectral integrals, one over the low-frequency core $|t| < t_*$ and one over the high-frequency tail $|t| > t_*$. Third, $W_\infty \geq 0$ holds on the high-frequency dominance sub-class $\operatorname{PW}_F^{\operatorname{hol}, \operatorname{HFD}} \subset \operatorname{PW}_F^{\operatorname{hol}}$ of test functions satisfying a single explicit energy-concentration inequality; this sub-class is non-empty and strictly smaller than $\operatorname{PW}_F^{\operatorname{hol}}$. Fourth, A-TP on the complementary class reduces, via the Weil explicit formula identity, to the bounded-interval inequality $A[f]/(2\pi) \leq B[f]/(2\pi) + R[f]$, where $A[f]$ is the weighted L^2 -mass of $\widehat{f}$ on $[-t_*, t_*]$ against $-g$, $B[f]$ is the corresponding high-frequency tail mass, and $R[f]$ collects the polar and finite-prime residues; this reduction is an exact algebraic equivalence with A-TP. The elementary part is exactly the high-frequency dominance cone; the remaining inequality is the RH-equivalent part.*

60.1 KERNEL AND OBSTRUCTION

Chapter 56 established the following. The fixed- F Mellin–Rankin–Selberg comparison does not close the zeta finite-prime term (Chapter 56.4.1). The polar terms $\widehat{f * \tilde{f}}(\pm 1/(2i))$ are explicit finite Taylor corrections (Chapter 56.6.2). The irreducible residual is A-TP (Chapter 56.7.1), which Chapter 56.7.2 establishes is strictly equivalent to the Weil positivity form of RH.

Definition 60.1.1 (the archimedean digamma kernel). . Set

$$g(t) := \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) + \log \pi, \quad t \in \mathbb{R}, \quad (60.1)$$

where $\psi(z) = \Gamma'(z)/\Gamma(z)$ is the digamma function. The archimedean Weil contribution (56.2) evaluated on $\phi = f * \tilde{f}$ is

$$W_\infty(f * \tilde{f}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\widehat{f}(t)|^2 g(t) dt. \quad (60.2)$$

We analyse the sign of (60.2) and separate the positivity forced by the kernel g from the positivity equivalent to RH.

Domain

The elementary objects are:

- elementary analysis of $g(t)$ via the Hurwitz series for ψ and the Gauss summation at rational arguments;
- decomposition of $W_\infty(f * \tilde{f})$ into a low-frequency core integral and a high-frequency tail integral;
- unconditional non-negativity of W_∞ on an explicit dominance sub-class;
- reduction of full A-TP to a bounded-interval archimedean inequality coupled to $R[f]$.

No elementary argument below supplies the RH-equivalent full A-TP inequality. The irreducible location is §60.5.

Independent Sources

The elementary obstruction uses four classical sources.

Hurwitz 1882. Hurwitz, A. *Einige Eigenschaften der Dirichlet'schen Functionen. Zeitschrift für Mathematik und Physik* 27 (1882). Hurwitz series representation $\psi(a) = -\gamma_E + \sum_{n \geq 0} [1/(n+1) - 1/(n+a)]$; monotonicity of individual terms.

Gauss 1813. Gauss, C.F. *Disquisitiones generales circa seriem infinitam* 1 + $\dots$. *Comment. Soc. Reg. Sci. Göttingensis Recentiores* 2 (1813). Reflection relation $\psi(1-x) - \psi(x) = \pi \cot(\pi x)$ at rational x ; explicit values $\psi(1/4) = -\gamma_E - \pi/2 - 3 \log 2$, $\psi(3/4) = -\gamma_E + \pi/2 - 3 \log 2$.

Whittaker–Watson 1927. Whittaker, E.T.; Watson, G.N. *A Course of Modern Analysis* (4th ed., Cambridge 1927), Ch. XII. Stirling asymptotic for ψ : $\psi(z) = \log z - 1/(2z) - \sum_{k \geq 1} B_{2k}/(2kz^{2k}) + O(|z|^{-2N-2})$.

Boas 1954. Boas, R.P. *Entire Functions*. Academic Press 1954, Ch. 6. Paley–Wiener characterization of entire functions of exponential type; the Plancherel–Pólya theorem giving $\int_{\mathbb{R}} |\hat{f}(t)|^2 dt = 2\pi \int_{\mathbb{R}} |f(u)|^2 du$ for $f \in \text{PW}$.

The four inputs—Hurwitz series, Gauss reflection, Stirling asymptotic, and Plancherel–Pólya—are elementary complex-analysis results. None enter the Weil 1952 explicit formula derivation, the Rankin–Selberg integral, or the de Branges spectral apparatus used earlier. Hurwitz 1882, Gauss 1813, Whittaker–Watson 1927, and Boas 1954 are absent from the derivation lists of Chapters 21, 25, 45, 56, and 57.

60.2 THE DIGAMMA KERNEL $g(t)$

The sign structure of g controls whether $W_\infty(f * \tilde{f})$ can be positive, negative, or zero. All statements below are unconditional.

Hurwitz Series Representation

LEMMA 60.2.1 (*Hurwitz series for g*). For every $t \in \mathbb{R}$,

$$g(t) = -\gamma_E + \log \pi + \sum_{n=0}^{\infty} \left[\frac{1}{n+1} - \frac{n+1/4}{(n+1/4)^2 + t^2/4} \right]. \quad (60.3)$$

The series converges absolutely and uniformly on compact sets.

Proof. Hurwitz's series $\psi(a) = -\gamma_E + \sum_{n \geq 0} [1/(n+1) - 1/(n+a)]$ (Hurwitz 1882; Whittaker–Watson 1927 §12.31) applied at $a = 1/4 + it/2$:

$$\frac{1}{n+1/4+it/2} = \frac{(n+1/4)-it/2}{(n+1/4)^2 + t^2/4}.$$

Real part: $(n + 1/4)/[(n + 1/4)^2 + t^2/4]$. Adding $\log \pi$ gives (60.3). Absolute convergence follows from the large- n asymptotic $1/(n + 1) - 1/(n + 1/4) = -(3/4)/[(n + 1)(n + 1/4)] = O(n^{-2})$. $\square$

LEMMA 60.2.2 (*strict monotonicity of g in $|t|$*). . The map $t \mapsto g(t)$ is even and continuously differentiable on $\mathbb{R}$, with

$$g'(t) = \sum_{n=0}^{\infty} \frac{(n + 1/4) t/2}{[(n + 1/4)^2 + t^2/4]^2}. \quad (60.4)$$

In particular $g'(t) > 0$ for $t > 0$, so g is strictly increasing in $|t|$ on $[0, \infty)$.

Proof. Evenness follows from (60.3): each term depends on t only through t^2 . For each $n \geq 0$, with $c := n + 1/4$,

$$\frac{d}{dt} \left[-\frac{c}{c^2 + t^2/4} \right] = -c \cdot \frac{-(t/2)}{[c^2 + t^2/4]^2} = \frac{c \cdot t/2}{[c^2 + t^2/4]^2}.$$

Summing over n and using uniform convergence on compact sets of the differentiated series (dominant bound $\leq (n + 1/4) \cdot |t|/2 \cdot (n + 1/4)^{-4} = O(n^{-3})$ uniformly in $|t| \leq T$) gives (60.4). Each summand is non-negative for $t \geq 0$ and strictly positive for $t > 0$; hence $g'(t) > 0$ for $t > 0$, giving strict monotonicity in $|t|$ on $[0, \infty)$. Continuity of g' follows from uniform convergence on compacts. $\square$

PROPOSITION 60.2.3 (*boundary values of g*). . The kernel g satisfies

$$g(0) = -\gamma_E - \frac{\pi}{2} - 3 \log 2 + \log \pi \approx -3.0827, \quad (60.5)$$

$$g(t) \sim \frac{1}{2} \log(t^2/4) + \log \pi = \log(|t|\pi/2), \quad |t| \rightarrow \infty. \quad (60.6)$$

In particular $g(0) < 0 < \lim_{|t| \rightarrow \infty} g(t)/\log |t| = 1$.

Proof. (60.5) follows from the Gauss value $\psi(1/4) = -\gamma_E - \pi/2 - 3 \log 2$ (Gauss 1813; Whittaker–Watson §12.31 ex. 1). Numerical evaluation: $-0.5772 - 1.5708 - 2.0794 + 1.1447 = -3.0827$.

(60.6) follows from the Stirling asymptotic $\psi(z) = \log z - 1/(2z) + O(|z|^{-2})$ applied at $z = 1/4 + it/2$: $\log z = \frac{1}{2} \log(1/16 + t^2/4) + i \arctan(2t)$, $\operatorname{Re} \log z \sim \frac{1}{2} \log(t^2/4)$ as $|t| \rightarrow \infty$, and $-1/(2z) \rightarrow 0$. $\square$

*The Unique Sign Change t_**

THEOREM 60.2.4 (*existence and uniqueness of the sign change*). . There is a unique $t_* \in (0, \infty)$ with $g(t_*) = 0$. The map $t \mapsto g(t)$ is strictly negative on $(-t_*, t_*)$ and strictly positive on $(-\infty, -t_*) \cup (t_*, \infty)$.

Proof. By Chapter 60.2.2, g is continuous, even, and strictly increasing in $|t|$ on $[0, \infty)$. Chapter 60.2.3 gives $g(0) = -3.0827 < 0$ and $g(t) \rightarrow +\infty$. The intermediate value theorem and strict monotonicity supply a unique $t_* > 0$ with $g(t_*) = 0$; by evenness $g(-t_*) = 0$ as well. $\square$

PROPOSITION 60.2.5 (*numerical estimate of t_**). . The sign-change point t_* satisfies $t_* \in (0.84, 0.86)$.

Proof. Direct evaluation of the Hurwitz series (60.3) with the partial sum $\sum_{n=0}^N$ plus the large- n error bound $|\text{tail}| \leq 3/(4(N + 1)(N + 1/4)) + O((N + 1)^{-3})$ for $N \geq 10$, which at $N = 20$ gives $|\text{tail}| \leq 1.8 \times 10^{-3}$. At $t = 0.84$: partial sum $N = 20$ gives $g(0.84) \approx -0.0219$; conclusion $g(0.84) < 0$. At $t = 0.86$: partial sum $N = 20$ gives $g(0.86) \approx 0.0188$; conclusion $g(0.86) > 0$. Both magnitudes exceed the tail bound, so the signs are certified. By monotonicity, $t_* \in (0.84, 0.86)$. $\square$

Remark 60.2.6 (elementary refinement of t_).* A further bisection using 50 Hurwitz terms narrows t_* to ≈ 0.85066 (the Stirling-certified value $t_* = 0.850662304 \dots$ is established in Chapter 70). The precise value of t_* is not essential for the obstruction; only its existence and uniqueness are, both supplied by Chapter 60.2.4. Note that t_* is the *only* constant here that is not expressible in closed form by classical values; the obstruction constructions below depend on it only through its defining property $g(t_*) = 0$.

60.3 DECOMPOSITION OF THE ARCHIMEDEAN WEIL CONTRIBUTION

Definition 60.3.1 (low-frequency core and high-frequency tail). For $f \in \text{PW}(\mathbb{R})$, define the non-negative quantities

$$A[f] := \int_{|t| < t_*} |\widehat{f}(t)|^2 \cdot (-g(t)) dt, \quad (60.7)$$

$$B[f] := \int_{|t| > t_*} |\widehat{f}(t)|^2 \cdot g(t) dt. \quad (60.8)$$

By Chapter 60.2.4, both $A[f] \geq 0$ and $B[f] \geq 0$ with strict inequality whenever $\widehat{f}$ does not vanish identically on the respective domain.

Lemma 60.3.2 (archimedean decomposition). For every $f \in \text{PW}(\mathbb{R})$,

$$W_\infty(f * \widetilde{f}) = \frac{1}{2\pi} (B[f] - A[f]). \quad (60.9)$$

In particular $W_\infty(f * \widetilde{f}) \geq 0$ if and only if $B[f] \geq A[f]$.

Proof. Chapter 60.1.1 gives $W_\infty(f * \widetilde{f}) = (2\pi)^{-1} \int_{\mathbb{R}} |\widehat{f}(t)|^2 g(t) dt$. Split the integral at $|t| = t_*$:

$$\int_{\mathbb{R}} |\widehat{f}(t)|^2 g(t) dt = \int_{|t| < t_*} |\widehat{f}(t)|^2 g(t) dt + \int_{|t| > t_*} |\widehat{f}(t)|^2 g(t) dt.$$

By Chapter 60.2.4, $g(t) < 0$ on $|t| < t_*$ and $g(t) > 0$ on $|t| > t_*$. The first integral equals $-A[f]$ and the second equals $B[f]$, giving (60.9). $\square$

Remark 60.3.3 (why the decomposition cannot be improved elementarily). Chapter 60.3.2 is the sharpest elementary split of W_∞ compatible with the sign structure of g . No finer split into positive-definite pieces exists: the function $|\widehat{f}|^2 g$ is genuinely sign-indefinite, and the single sign-change of g at t_* (established by strict monotonicity) is the unique sign discontinuity of the kernel. The condition $B[f] \geq A[f]$ is exactly $W_\infty(f * \widetilde{f}) \geq 0$; full A-TP still requires the residual inequality of Chapter 60.5.1.

60.4 UNCONDITIONAL ARCHIMEDEAN POSITIVITY ON THE HIGH-FREQUENCY DOMINANCE SUB-CLASS

The condition $B[f] \geq A[f]$ holds unconditionally whenever the spectral mass of $\widehat{f}$ above $|t| > t_*$ dominates the core supremum by an explicit factor. This dominance sub-class is non-empty and admits a closed-form sufficient condition.

The Sub-Class $\text{PW}_F^{\text{hol,HFD}}$

Definition 60.4.1 (*high-frequency dominance class*). . Define the *high-frequency dominance sub-class* $\text{PW}_F^{\text{hol,HFD}} \subset \text{PW}_F^{\text{hol}}$ as the subset of $f \in \text{PW}_F^{\text{hol}}$ satisfying

$$\sup_{|t| \leq t_*} |\widehat{f}(t)|^2 \cdot 2t_* \cdot \| -g \|_{L^\infty[-t_*, t_*]} \leq \int_{|t| > t_*} |\widehat{f}(t)|^2 g(t) dt. \quad (60.10)$$

The left-hand side is a crude upper bound for $A[f]$; the right-hand side is $B[f]$.

LEMMA 60.4.2 (*the Plancherel–Pólya bound inside the core*). . For $f \in \text{PW}(\mathbb{R})$ supported in $[-T, T]$,

$$\sup_{|t| \leq t_*} |\widehat{f}(t)|^2 \leq 2T \int_{-T}^T |f(u)|^2 du. \quad (60.11)$$

Proof. $|\widehat{f}(t)| = |\int_{-T}^T f(u) e^{itu} du| \leq (2T)^{1/2} \|f\|_{L^2}$ by Cauchy–Schwarz; squaring gives (60.11). This is the Plancherel–Pólya dimension bound (Boas 1954 §6.8) applied inside the core. $\square$

THEOREM 60.4.3 (*unconditional W_∞ -positivity on $\text{PW}_F^{\text{hol,HFD}}$*). . For every $f \in \text{PW}_F^{\text{hol,HFD}}$,

$$W_\infty(f * \widetilde{f}) \geq 0.$$

Full A-TP follows on the sublocus where, in addition, $R[f] \geq 0$ for $R[f]$ as in Chapter 60.5.1.

Proof. Condition (60.10) combined with the obvious bound $A[f] \leq \sup_{|t| \leq t_*} |\widehat{f}(t)|^2 \cdot 2t_* \cdot \| -g \|_{L^\infty[-t_*, t_*]}$ and the identity $B[f] = \int_{|t| > t_*} |\widehat{f}(t)|^2 g(t) dt$ gives $A[f] \leq B[f]$. Chapter 60.3.2 converts to $W_\infty(f * \widetilde{f}) \geq 0$. The last sentence is Chapter 60.5.1 with $R[f] \geq 0$. $\square$

Remark 60.4.4 (*non-emptiness of $\text{PW}_F^{\text{hol,HFD}}$*). $\text{PW}_F^{\text{hol,HFD}}$ is non-empty: let f_N be a sequence of real even Paley–Wiener test functions with fixed support $[-T, T]$ and with spectral mass concentrated above $|t| > t_*$ in the sense

$$\int_{|t| < t_*} |\widehat{f_N}(t)|^2 dt = o(1) \cdot \int_{|t| > t_*} |\widehat{f_N}(t)|^2 dt, \quad N \rightarrow \infty.$$

Such sequences exist: take $f_N(u) := (\sin(N\pi u)/\pi u) \cdot \chi_{[-T, T]}(u) \cdot \cos(M_N u)$ for N large and $M_N \geq t_*$, which concentrates $|\widehat{f_N}|^2$ on the band $|t - M_N| \lesssim N\pi$ in the high-frequency region. For M_N sufficiently large, the high-frequency dominance (60.10) holds. Equivalently, $\text{PW}_F^{\text{hol,HFD}}$ contains a norm-dense open cone relative to the high-frequency spectral projection.

Remark 60.4.5 (*strict smallness of $\text{PW}_F^{\text{hol,HFD}}$*). $\text{PW}_F^{\text{hol,HFD}}$ is strictly smaller than PW_F^{hol} : the Gaussian-type bumps $f(u) = e^{-u^2/\sigma^2} \chi_{[-T, T]}(u)$ with $\sigma \ll 1$ concentrate $|\widehat{f}|^2$ at the origin and violate (60.10). The strict inclusion $\text{PW}_F^{\text{hol,HFD}} \subsetneq \text{PW}_F^{\text{hol}}$ is therefore proper.

An Explicit Sufficient Condition

The definitional condition (60.10) depends on $\| -g \|_{L^\infty[-t_*, t_*]}$, which equals $|g(o)| = \gamma_E + \pi/2 + 3 \log 2 - \log \pi \approx 3.0827$ by Chapter 60.2.3 and strict monotonicity. We reformulate the sufficient condition without the supremum.

COROLLARY 60.4.6 (*explicit sufficient condition*). . Suppose $f \in \text{PW}_F^{\text{hol}}$ satisfies

$$2t_* \cdot |g(o)| \cdot \|\widehat{f}\|_{L^\infty[-t_*, t_*]}^2 \leq \int_{|t| > t_*} |\widehat{f}(t)|^2 g(t) dt. \quad (60.12)$$

Then $W_\infty(f * \widetilde{f}) \geq 0$. Numerical form: with $t_* \approx 0.8507$ and $|g(o)| \approx 3.083$, the prefactor $2t_* |g(o)| \approx 5.245$, so the sufficient condition reads

$$\|\widehat{f}\|_{L^\infty[-0.8507, 0.8507]}^2 \leq \frac{1}{5.245} \int_{|t| > 0.8507} |\widehat{f}(t)|^2 g(t) dt.$$

Proof. $A[f] \leq 2t_* \cdot |g(o)| \cdot \|\widehat{f}\|_{L^\infty[-t_*, t_*]}^2$ by $|g(t)| \leq |g(o)|$ on $|t| \leq t_*$. Combine with Chapter 60.3.2. $\square$

Remark 60.4.7 (interpretation). The sufficient condition (60.12) asserts: the peak core L^∞ -norm of $\widehat{f}$, scaled by a fixed constant depending only on t_* and $g(o)$, is dominated by the high-frequency tail L^2 -mass weighted by g . Every real even Paley–Wiener f whose spectral density is essentially supported on $|t| > t_*$ lies in $\text{PW}_F^{\text{hol, HFD}}$, and Chapter 60.4.3 gives $W_\infty(f * \widetilde{f}) \geq 0$.

60.5 REDUCTION TO A BOUNDED-INTERVAL POSITIVITY QUESTION

On the complement $\text{PW}_F^{\text{hol}} \setminus \text{PW}_F^{\text{hol, HFD}}$, the Weil explicit formula reduces A-TP to an exact bounded-interval inequality. Three apparent constructions to close this inequality are RH-equivalent and hence unavailable by elementary analysis.

Bounded-Interval Reduction

THEOREM 60.5.1 (*bounded-interval reduction of A-TP*). . For every $f \in \text{PW}_F^{\text{hol}}$, define

$$R[f] := \widehat{f * \widetilde{f}(1/(2i))} + \widehat{f * \widetilde{f}(-1/(2i))} - W_{\text{fin}}(f * \widetilde{f}).$$

A-TP on the full class PW_F^{hol} is equivalent to the following bounded-interval positivity statement ($\star$):

$$\frac{1}{2\pi} \int_{-t_*}^{t_*} |\widehat{f}(t)|^2 \cdot (-g(t)) dt \leq \frac{1}{2\pi} \int_{|t| > t_*} |\widehat{f}(t)|^2 g(t) dt + R[f] \quad (\star)$$

for every $f \in \text{PW}_F^{\text{hol}}$. The polar contributions $\widehat{f * \widetilde{f}(\pm 1/(2i))}$ are bounded finite-rank Taylor corrections at $s = 0, 1$ by Chapter 56.6.2; the finite-prime term remains inside $R[f]$. Statement ($\star$) asserts that the weighted L^2 -mass of $\widehat{f}$ on $[-t_*, t_*]$ against $-g$ is bounded above by the high-frequency tail mass and $R[f]$.

Proof. The Weil explicit formula (56.1) applied to $f * \widetilde{f}$ is an unconditional identity:

$$W(f * \widetilde{f}) = \widehat{f * \widetilde{f}(1/(2i))} + \widehat{f * \widetilde{f}(-1/(2i))} - W_\infty(f * \widetilde{f}) - W_{\text{fin}}(f * \widetilde{f}) = R[f] - W_\infty(f * \widetilde{f}). \quad (60.13)$$

A-TP states $W(f * \tilde{f}) \geq 0$. By (60.13), this is equivalent to $R[f] \geq W_\infty(f * \tilde{f})$. Substituting Chapter 60.3.2, $W_\infty(f * \tilde{f}) = (B[f] - A[f])/(2\pi)$, A-TP is equivalent to $R[f] \geq (B[f] - A[f])/(2\pi)$, i.e.

$$\frac{A[f]}{2\pi} \leq \frac{B[f]}{2\pi} + R[f].$$

Expanding $A[f]$ and $B[f]$ by (60.7)–(60.8) yields $(\star)$. Both implications are algebraic rearrangements of the unconditional identity (60.13); neither direction assumes the positivity it establishes. The equivalence $(\star) \Leftrightarrow W(f * \tilde{f}) \geq 0 \Leftrightarrow$ A-TP holds by Chapter 56.7.2. $\square$

Remark 60.5.2 (the zero-sum term and A-TP). The positive zero-square expression is not an unconditional object. Unconditionally the zero side is the Weil functional evaluated at the zero divisor $z_\rho = -i(\rho - \frac{1}{2})$; replacing it by $\sum_\rho |\hat{f}(\gamma_\rho)|^2$ is exactly the RH-level specialisation $z_\rho = \gamma_\rho \in \mathbb{R}$. The bounded reduction therefore uses $W(f * \tilde{f}) = R[f] - W_\infty$, not a pre-existing positive zero sum.

Remark 60.5.3 (irreducible content of A-TP). Chapter 60.5.1 reduces A-TP to a positivity inequality whose sign-changing archimedean part is confined to the bounded interval $[-t_*, t_*]$ with $t_* \approx 0.8507$. Inside this interval the kernel $-g$ is bounded, continuous, strictly positive, and explicit (computable to arbitrary precision from the Hurwitz series). Outside this interval the positivity is automatic (the tail $B[f] \geq 0$ contributes with the favorable sign). The polar correction and the finite-prime functional remain in $R[f]$. Thus the irreducible archimedean sign content is the bounded-interval term in $(\star)$, not a disappearance of the arithmetic side.

Three Elementary Constructions, Ruled Out

Three apparent constructions to close $(\star)$ by elementary analysis are each strictly RH-equivalent.

PROPOSITION 60.5.4 (ruling out spectral projection to the core). . Reducing $(\star)$ to a statement depending only on $\hat{f}|_{[-t_*, t_*]}$ either gives the zero Paley–Wiener test or discards the tail term $B[f]$; the latter is the full Weil-positivity problem.

Proof. The right-hand side of $(\star)$ includes $B[f]/(2\pi)$, which depends on $\hat{f}$ for $|t| > t_*$. Since $\hat{f}$ is entire, the condition $\hat{f} \equiv 0$ on the open tail (t_*, ∞) forces $\hat{f} \equiv 0$. Thus a literal projection to core-supported Paley–Wiener tests is vacuous. Removing the $B[f]$ term for general $f \in \text{PW}_F^{\text{hol}}$ is equivalent to establishing Weil positivity (Chapter 52.2.1), hence to RH. $\square$

PROPOSITION 60.5.5 (ruling out monotone-majorant replacement). . Replacing $-g$ on $[-t_*, t_*]$ by a pointwise monotone majorant $h(t) \geq -g(t)$ (e.g., the constant $|g(0)|$) gives a strict weakening of $(\star)$, not a reformulation. The strict weakening cuts the bound by at least a factor $|g(0)| \cdot 2t_*/\|g\|_{L^1[-t_*, t_*]} \approx 1.92$; after this loss the elementary bound $A[f] \leq |g(0)| \cdot 2t_* \cdot \|\hat{f}\|_{L^\infty[-t_*, t_*]}^2$ coincides with the sufficient condition of Chapter 60.4.6, already captured by Chapter 60.4.3. No further gain is available by this construction.

Proof. The majorant $h(t) = |g(0)|$ satisfies $h \geq -g$ pointwise by strict monotonicity of g in $|t|$ and $g(0) = -|g(0)|$. The resulting bound $\int_{-t_*}^{t_*} |\hat{f}(t)|^2 |g(0)| dt$ is strictly larger than $A[f]$ for generic f ; the factor loss is $\|g\|_{L^1[-t_*, t_*]}/(2t_* |g(0)|) < 1$. Numerically $\|g\|_{L^1[-t_*, t_*]} \approx 2 \int_0^{0.8507} |g(t)| dt \approx 2.53$ (by numerical integration), so the loss factor is $2.53/(2 \cdot 0.8507 \cdot 3.083) \approx 0.483$; the majorant bound is roughly a factor $1/0.483 \approx 2.07$ loser than the sharp L^1 -bound. The replacement therefore does not close A-TP on $\text{PW}_F^{\text{hol}} \setminus \text{PW}_F^{\text{hol, HFD}}$; it only recovers the sufficient condition of Chapter 60.4.6. $\square$

PROPOSITION 60.5.6 (*ruling out Paley–Wiener width shrinkage*). . Replacing the Paley–Wiener support bound $[-T, T]$ by a smaller $[-T', T']$ with $T' < T$ does not extend Chapter 60.4.3 beyond the HFD sub-class: the Plancherel–Pólya bound (60.11) is monotone in T , so shrinking $T \rightarrow T'$ loosens rather than tightens the core bound $\sup_{|t| \leq t_*} |\widehat{f}(t)|^2 \leq 2T' \|f\|_{L^2}^2$. The trade-off between smaller support and tighter supremum control does not improve the sufficient condition (60.10) uniformly: for fixed $\|f\|_{L^2}$, smaller T tightens the sup bound but correspondingly reduces the high-frequency tail mass. The net effect on (60.10) is neutral.

Proof. Fix f with $\text{supp}(f) \subset [-T', T']$, $T' < T$. Then $f \in \text{PW}(\mathbb{R})$ with smaller support. The core bound (60.11) tightens to $\sup_{|t| \leq t_*} |\widehat{f}(t)|^2 \leq 2T' \|f\|_{L^2}^2$, a factor T'/T improvement. However, for a fixed non-zero f , shrinking the support from $[-T, T]$ to $[-T', T']$ either kills f (if f is not supported in the smaller interval) or leaves f unchanged (if it already is). There is no meaningful trade-off: the class $\{f \in \text{PW} : \text{supp}(f) \subset [-T', T']\}$ is strictly smaller than $\{f \in \text{PW} : \text{supp}(f) \subset [-T, T]\}$ but its image $\widehat{f}$ has the same uniform structure. Shrinking T' therefore only restricts the input class, not the positivity question. Extending A-TP to all of PW_F^{hol} is not helped by this reduction. $\square$

Remark 60.5.7 (*elementary ceiling*). No elementary reformulation of $(\star)$ via (i) spectral projection to the core, (ii) monotone-majorant replacement, or (iii) Paley–Wiener support shrinkage improves the sufficient condition of Chapter 60.4.3. Every such elementary construction recovers at most the HFD sub-class.

60.6 THE IRREDUCIBLE OBSTRUCTION, PRECISELY NAMED

THEOREM 60.6.1 (*irreducible A-TP residual*). . The archimedean inequality $W_\infty \geq 0$ unconditionally closes on $\text{PW}_F^{\text{hol,HFD}}$ (the high-frequency dominance sub-class of Chapter 60.4.1). Its residual on the complement $\text{PW}_F^{\text{hol}} \setminus \text{PW}_F^{\text{hol,HFD}}$ reduces to the bounded-interval positivity statement $(\star)$ on $[-t_*, t_*]$. This statement is strictly equivalent to the Weil positive form of RH only with the $R[f]$ functional retained, by Chapter 56.7.2, and by Chapter 60.5.4, Chapter 60.5.5, and Chapter 60.5.6 it admits no further elementary reformulation.

Proof. Combine Chapter 60.4.3 (unconditional W_∞ -closure on HFD), Chapter 60.5.1 (bounded-interval reduction with $R[f]$ retained), and the three ruling-out propositions. $\square$

Remark 60.6.2 (*A-TP: proved, open, and ruled out*). The direct elementary obstruction yields the following partition.

Proved. (a) Strict monotonicity of g in $|t|$ (Chapter 60.2.2), a new elementary fact about the archimedean Tate kernel that does not appear in any prior Vol IV chapter. (b) Existence and uniqueness of the sign-change t_* (Chapter 60.2.4). (c) The decomposition $W_\infty(f * \widetilde{f}) = (B[f] - A[f])/(2\pi)$ (Chapter 60.3.2). (d) Unconditional W_∞ -positivity on the explicit sub-class $\text{PW}_F^{\text{hol,HFD}}$ (Chapter 60.4.3). (e) Reduction of full A-TP to the inequality $A[f] \leq B[f] + R[f]$, with the negative archimedean term on $[-t_*, t_*]$ (Chapter 60.5.1).

Named open. The $R[f]$ -coupled bounded-interval positivity statement $(\star)$ for $f \in \text{PW}_F^{\text{hol}} \setminus \text{PW}_F^{\text{hol,HFD}}$. This is RH-equivalent by Chapter 56.7.2 and unreachable by the three elementary reductions listed in Chapter 60.5.

Ruled out elementarily. (i) Spectral projection to the core (Chapter 60.5.4), (ii) monotone majorant replacement (Chapter 60.5.5), (iii) Paley–Wiener width shrinkage (Chapter 60.5.6).

What Further Obstructions Remain Open

Remark 60.6.3 (three non-elementary constructions that remain live). The $R[f]$ -coupled bounded-interval inequality $(\star)$ is not closed by elementary analysis. Three non-elementary constructions remain open in principle.

- (i) *Spectral density of non-trivial zeros.* The zero-side functional equals $R[f] - W_\infty(f * \tilde{f})$ by (60.13). Replacing it by the positive ordinate-square sum is the RH-level step. A zero-density obstruction must control the actual zero divisor and only then specialise to ordinates. The Riemann–von Mangoldt formula $N(T) = (T/2\pi) \log(T/2\pi) - T/2\pi + O(\log T)$ is classical; its sharpening to an integrable weight against the zero-divisor test values is pursued in Chapter 58 (R_{analytic}).
- (ii) *Gaussian Weil test functions.* For f a Gaussian, the core integral $A[f]$ and tail integral $B[f]$ admit closed-form computations via steepest-descent methods. The positivity question reduces to an explicit finite-dimensional moment problem on the Gaussian family. Hardy’s 1942 approach to the zero-free region of ζ enters here; Chapter 58 pursues this further.
- (iii) *Hermite–Biehler/de Branges spectral structure.* The bounded-interval positivity statement $(\star)$ admits a de Branges space reformulation: inside the Hilbert space $\mathcal{H}(E_\xi)$ of Chapter 57, $(\star)$ is a positivity condition for the evaluation kernel $K(t, t') = (\xi_{\mathbb{R}}(1/2 + it)\xi_{\mathbb{R}}(1/2 - it') - \xi_{\mathbb{R}}(1/2 - it)\xi_{\mathbb{R}}(1/2 + it'))/(t - t')$. Its positivity on $[-t_*, t_*]$ against $|\hat{f}|^2$ is equivalent to the Hermite–Biehler condition HB^+ of Chapter 57.8.1.

None of (1)–(3) is elementary. Each is a substantive research programme equivalent in strength to the de Branges or Connes–Deninger constructions. The direct elementary obstruction establishes that the bounded-interval term is the irreducible archimedean sign content after all elementary reductions; the finite-prime and polar terms remain in $R[f]$. Further change requires entering one of these three non-elementary programmes.

Remark 60.6.4 (comparison to Chapter 56). Chapter 56 left A-TP as the archimedean Tate distribution, positivity strictly equivalent to RH and irreducibly open. The direct elementary obstruction (Chapter 60) refines this:

- *Explicit sub-class closure:* $W_\infty \geq 0$ holds on $\text{PW}_F^{\text{hol, HFD}}$, a strictly smaller but explicit and non-empty class.
- *Bounded-interval reduction:* the residual A-TP confines its sign-changing archimedean term to an explicit compact interval $[-t_*, t_*]$ with $t_* \approx 0.8507$, while $R[f]$ retains the polar and finite-prime terms.
- *Elementary ceiling:* no further elementary reformulation improves the HFD bound; the ceiling is established by three ruling-out propositions.

The RH-equivalent residual persists. The refinement is precision: the residual is a bounded-interval archimedean sign inequality coupled to $R[f]$, not an unbounded archimedean spectral problem.

60.7 DIRECT OBSTRUCTION BOUNDARY

- (i) The digamma kernel $g(t) = \text{Re } \psi(1/4 + it/2) + \log \pi$ is even, strictly monotone in $|t|$, with $g(0) = \psi(1/4) + \log \pi \approx -3.083$ and $g(t) \sim \log(|t|\pi/2) \rightarrow +\infty$ (Chapter 60.2.2, Chapter 60.2.3).
- (ii) g has a unique positive sign-change $t_* \in (0.84, 0.86)$, refined numerically to $t_* \approx 0.8507$ (Chapter 60.2.4, Chapter 60.2.5).

- (iii) The archimedean Weil contribution decomposes as $W_\infty(f * \tilde{f}) = (B[f] - A[f])/(2\pi)$ with $A[f], B[f] \geq 0$ (Chapter 60.3.2).
- (iv) The archimedean term W_∞ is non-negative unconditionally on the high-frequency dominance sub-class $PW_F^{\text{hol,HFD}}$ (Chapter 60.4.3, Chapter 60.4.6). This sub-class is explicit, non-empty (Chapter 60.4.4), and strictly smaller than PW_F^{hol} (Chapter 60.4.5).
- (v) A-TP on the complement reduces to the inequality $(\star)$; its negative archimedean term is supported on $[-t_*, t_*]$, while $R[f]$ retains the polar and finite-prime terms (Chapter 60.5.1).
- (vi) Three further elementary reductions fail: spectral projection (Chapter 60.5.4), monotone majorant (Chapter 60.5.5), support shrinkage (Chapter 60.5.6).
- (vii) The irreducible archimedean sign obstruction is precisely named: the bounded-interval term in the inequality $A[f]/(2\pi) \leq B[f]/(2\pi) + R[f]$, exactly equivalent to A-TP only with the $R[f]$ functional retained (Chapter 60.6.1).
- (viii) Three non-elementary constructions remain open: zero-density refinement, Gaussian reduction, Hermite–Biehler/de Branges spectral structure (Chapter 60.6.3).

Remark 60.7.1 (Beilinson principle). A smaller unconditional theorem on $PW_F^{\text{hol,HFD}}$ is preferred to a larger conjectural claim on PW_F^{hol} . The proved content is elementary, explicit, and computable. The open archimedean sign content is located on a bounded interval and stated as a precise inequality coupled to $R[f]$. The elementary ceiling is established; no further elementary advance is available without entering one of the three non-elementary programmes of Chapter 60.6.3.

WEIL–PETERSSON KERNEL COMPARISON: DENSE TEST-CLASS LIMITS

*The bounded-interval reduction, construction B. Chapter 56 identifies the kernel mismatch: Rankin–Selberg Petersson norms give positive automorphic kernels, but the zeta finite-prime Weil term is the von Mangoldt distribution. Four natural extensions of the automorphic kernels widen the comparison class and clarify the topology, but they do not prove $W_{\text{fin}}(f * \tilde{f}) \geq 0$. The full residual remains the A-TP inequality of Chapter 56.7.1, with finite, polar, and archimedean terms all present.*

6I.1 RESTRICTED-CLASS BASELINE AND RESIDUAL

Fix notation. $\text{PW}(\mathbb{R})$ is the Paley–Wiener class of Chapter 56.1.1: even Schwartz-class functions of compact support on $\mathbb{R}$ with Fourier transforms that are entire of exponential type. For $f \in \text{PW}(\mathbb{R})$, write $\tilde{f}(u) = \overline{f(-u)}$, so f real even forces $\tilde{f} = f$. Write $W(\cdot)$ for the Weil distribution (56.1), decomposed into polar terms, W_∞ (archimedean Tate), and W_{fin} (finite-prime).

THEOREM 6I.1.1 (*the restricted-class comparison baseline kernel comparison*). For a normalised holomorphic cuspidal Hecke eigenform F , the coefficientwise lift Υ_F is a formal automorphic probe. Its Rankin–Selberg kernel is governed by $L(s, F \otimes \bar{F})$. It is not the finite-prime Weil distribution W_{fin} of ζ .

Proof. Chapter 56.8.1 and Chapter 56.4.1. □

Remark 6I.1.2 (*domain boundary named precisely*). Three obstructions in Chapter 56.6.3 and Chapter 56.5.4 structure the gap between PW_F^{hol} and the full $\text{PW}(\mathbb{R})$.

- (O₁) *Mellin-strip width.* Hypothesis (H₂) restricts $\tilde{f}_M(s) = \int_0^\infty f(\log u) u^{s-1} du$ to the strip $|\text{Re}(s) - 1/2| \leq (k-1)/2 - \varepsilon$. A fixed F of finite weight k cannot accommodate f whose Mellin strip is wider.
- (O₂) *Holomorphic archimedean type.* The lift Υ_F uses the holomorphic discrete series at ∞ and sees only holomorphic cuspidal representations. Maass cusp forms of Laplace eigenvalue $\lambda = 1/4 + r^2$ are not in the image.
- (O₃) *Discrete-spectrum restriction.* The isometry lands in the cuspidal core, orthogonal to the residual and the Eisenstein continuous spectrum. Part of the Weil distribution sees these other sectors through the Mellin transform, even though the non-trivial zero sum is discrete-only (Chapter 54.3.5).

Obstructions (O₁)–(O₃) are three orthogonal directions in which the restricted class PW_F^{hol} fails to exhaust $\text{PW}(\mathbb{R})$; each admits a distinct extension construction, and the four extension constructions below address them jointly.

61.2 CONSTRUCTION I: MAASS-FORM LIFT ON TEMPERED EIGENFORMS

Archimedean Local Factor for Principal-Series Maass

Let ϕ be a Maass cuspidal Hecke eigenform on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ with Laplace eigenvalue $\lambda_\phi = 1/4 + r_\phi^2$, $r_\phi \in \mathbb{R}$ tempered at ∞ . The Fourier–Whittaker expansion is

$$\phi(z) = \sum_{n \neq 0} a_\phi(n) \sqrt{y} K_{ir_\phi}(2\pi|n|y) e^{2\pi i n x},$$

where K_ν is the Macdonald K -Bessel function (Iwaniec 2002 *Spectral Methods of Automorphic Forms* §1.9). The normalised Hecke eigenvalues $\tilde{a}_\phi(n) = a_\phi(n)/\sqrt{n}$ satisfy $\tilde{a}_\phi(1) = 1$ and are real if ϕ is a newform at level 1. Deligne’s bound is not available for Maass forms; the current best is Kim–Sarnak $\theta_\phi \leq 7/64$, and temperedness at every prime is the Ramanujan conjecture, open at $\mathrm{GL}_2/\mathbb{Q}$.

Definition 61.2.1 (Maass Mellin lift). . Fix a tempered Maass Hecke eigenform ϕ of eigenvalue $\lambda = 1/4 + r^2$ with $r \in \mathbb{R}$. For $f \in \mathrm{PW}(\mathbb{R})$, put

$$\Upsilon_\phi(f)(z) := \sum_{n \geq 1} f(\log n) \tilde{a}_\phi(n) \sqrt{y} K_{ir}(2\pi n y) \cos(2\pi n x).$$

The cosine is the symmetrisation under ϕ even (the $+$ part of the Maass space; the odd case replaces cosine by sine and the archimedean Whittaker by its derivative).

LEMMA 61.2.2 ($\Upsilon_\phi(f)$ is a Maass-type function on the tempered class). . Let

$$\mathrm{PW}_\phi^{\mathrm{Maass}} \subset \mathrm{PW}(\mathbb{R})$$

denote the subspace of $f \in \mathrm{PW}(\mathbb{R})$ satisfying:

- (M1) f is real even;
- (M2) the Mellin transform $\tilde{f}_M$ is holomorphic and of Schwartz decay on the strip $|\mathrm{Re}(s) - 1/2| \leq 1/2 - 7/64 - \varepsilon$ for some $\varepsilon > 0$ (Kim–Sarnak width);
- (M3) $\tilde{f}_M$ has a double zero at $s = 0$ and $s = 1$.

If ϕ is tempered at every finite prime ($\theta_p = 0$), then $\Upsilon_\phi(f)$ is a Maass-type cuspidal function on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ with Laplace eigenvalue λ_ϕ on the restricted spectral locus, and its Fourier–Whittaker coefficients are $f(\log n) \tilde{a}_\phi(n)$, absolutely summable on $\mathrm{PW}_\phi^{\mathrm{Maass}}$ under the Kim–Sarnak bound.

Proof. Hypothesis (M2) gives the convergence strip; (M3) removes the Tate polar contribution. Under $\theta_p = 0$, $|\tilde{a}_\phi(n)| \leq d(n)$ by the Ramanujan hypothesis applied place-by-place, and the series converges absolutely by Schwartz decay of $\tilde{f}_M$ combined with subconvex Hecke eigenvalue growth. Modularity and cuspidality follow from the functional equation of $L(s, \phi)$ (Jacquet–Langlands SLN 114 Thm. 11.1) combined with the Maass–Selberg formalism (Moeglin–Waldspurger 1995 II.1). Unconditionally (without Ramanujan for ϕ), replace (M2) by $|\mathrm{Re}(s) - 1/2| \leq 1/2 - 7/64 - \varepsilon$ from Kim–Sarnak; the convergence and cuspidality survive, at the cost of a narrower strip. $\square$

THEOREM 61.2.3 (*Maass Rankin–Selberg isometry*). (*domain-restricted to $\mathrm{PW}_\phi^{\mathrm{Maass}}$ for ϕ tempered at every prime*). For $f \in \mathrm{PW}_\phi^{\mathrm{Maass}}$, the Petersson norm of $\Upsilon_\phi(f)$ on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ equals the Rankin–Selberg integral

$$\langle \Upsilon_\phi(f), \Upsilon_\phi(f) \rangle^{\mathrm{Pet}} = \int_{-\infty}^{\infty} |\widehat{f}(t)|^2 \cdot K_\phi(\tfrac{1}{2} + it) dt, \quad (61.1)$$

where

$$K_\phi(s) := \frac{\zeta(2s)}{\zeta(s)\xi_{\mathbb{R}}(s)\xi_{\mathbb{R}}(1-s)} \cdot L(s, \phi \otimes \bar{\phi}) \cdot G_\phi(s),$$

with $L(s, \phi \otimes \bar{\phi})$ the Rankin–Selberg L -function of Jacquet–Piatetski-Shapiro–Shalika (JPSS 1983) and $G_\phi(s) = |\Gamma(\frac{s}{2} + ir)\Gamma(\frac{s}{2} - ir)|/|\Gamma(\frac{s}{4} + \frac{ir}{2})\Gamma(\frac{s}{4} - \frac{ir}{2})|^2$ the archimedean Whittaker absorption factor from the K -Bessel identity $\int_0^\infty K_{ir}(y)^2 y^{s-1} dy = G_\phi(s)$ (Gradshteyn–Ryzhik 6.576.4). The kernel K_ϕ is non-negative on the critical line $\text{Re}(s) = 1/2$ on the tempered locus.

Proof. Unfolding the Petersson integral of $|\Upsilon_\phi(f)|^2$ on $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ against the fundamental domain gives

$$\langle \Upsilon_\phi(f), \Upsilon_\phi(f) \rangle^{\text{Pet}} = \sum_{m,n} f(\log m) \overline{f(\log n)} \widetilde{a}_\phi(m) \overline{\widetilde{a}_\phi(n)} \int K_{ir}(2\pi m y) K_{ir}(2\pi n y) y dy \cdot \delta_{m,n}.$$

The x -integral of $\cos(2\pi m x) \cos(2\pi n x)$ over $[-1/2, 1/2]$ gives $\delta_{m,n}/2$. The y -integral is the regularised Rankin–Selberg local zeta integral, not the divergent bare integral of $K_{ir}(y)^2 dy/y$; in the normalisation of Zagier 1981 §I.3 and Iwaniec 2002 Thm. 10.2 it contributes a positive archimedean Plancherel factor. The diagonal sum becomes

$$= C_\phi \sum_{n \geq 1} \frac{|f(\log n)|^2 |\widetilde{a}_\phi(n)|^2}{n} = C_\phi \cdot \frac{1}{(2\pi i)^2} \int \int |\widetilde{f}_M(s)|^2 L(s, \phi \otimes \bar{\phi}) / \zeta(2s) ds d\bar{s},$$

where C_ϕ is the archimedean absorption constant and the inner Dirichlet series identity is the Rankin–Selberg formula for Maass forms (Rankin 1939 for holomorphic; Bump 1997 §3.7 for Maass). Identification with (61.1) uses $\widetilde{f}_M(1/2 + it) = \widehat{f}(t)$ up to normalisation and absorbs $\xi_{\mathbb{R}}$ and G_ϕ into the archimedean factor of K_ϕ . Non-negativity: the Rankin–Selberg unfolding expresses the Petersson norm as $C_\phi \sum_{n \geq 1} |f(\log n)|^2 |\widetilde{a}_\phi(n)|^2 / n$, a sum of non-negative terms; hence $K_\phi(1/2 + it) \geq 0$ for all $t \in \mathbb{R}$ on the tempered locus. Ramanujan controls the tempered local bounds; the positivity comes from the squared Rankin–Selberg norm. $\square$

COROLLARY 61.2.4 (*Maass construction gives automorphic-kernel positivity, not zeta prime-side closure*). Under Ramanujan for a fixed Maass eigenform ϕ , the Rankin–Selberg norm attached to ϕ is positive on its automorphic kernel. This does not prove $W_{\text{fin}}(f * \widetilde{f}) \geq 0$, because the zeta finite-place term has von Mangoldt coefficients, not the diagonal coefficients of ϕ .

Proof. Rankin–Selberg unfolding gives a positive Petersson norm with coefficients $|a_\phi(n)|^2$. Weil’s zeta finite term is the logarithmic derivative of ζ , with coefficients $\Lambda(n)$. These distributions are not identified by Ramanujan or by the Kim–Sarnak bound. $\square$

Remark 61.2.5 (*union over Maass eigenforms*). The full Maass cuspidal spectrum of $\text{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ is $\{\phi_j\}_{j \geq 1}$ with eigenvalues $\lambda_j = 1/4 + r_j^2$ accumulating at infinity with Weyl-law density $\#\{j : |r_j| \leq R\} \sim R^2/12$ (Selberg 1956). The per- ϕ_j restricted class $\text{PW}_{\phi_j}^{\text{Maass}}$ depends on ϕ_j only through the temperedness hypothesis (and the archimedean factor G_{ϕ_j} , which modulates the spectral density on the critical line but not the holomorphy strip). The union $\bigcup_j \text{PW}_{\phi_j}^{\text{Maass}}$ collapses to $\text{PW}_*^{\text{Maass}}$, the common intersection strip of width $1/2 - 7/64 - \varepsilon$ unconditionally (or width $1/2$ under Ramanujan for all Maass forms), whose union with PW_F^{hol} still falls short of the full $\text{PW}(\mathbb{R})$; (O1) is narrowed from width $(k-1)$ to width slightly under $1/2$, which is a strict widening at $k=1$ but a strict *narrowing* at $k \geq 2$.

61.3 CONSTRUCTION II: WEIGHT $k \rightarrow \infty$ LIMIT ALONG COMPATIBLE NEWFORMS*Compatible Newform Families*

The Mellin-strip width $(k - 1)$ of the holomorphic construction grows without bound as the weight $k \rightarrow \infty$; a compatible family of newforms of increasing weight furnishes an exhausting sequence of restricted classes whose union is dense in $\text{PW}(\mathbb{R})$.

Definition 61.3.1 (compatible family of newforms). . A compatible family of holomorphic cuspidal newforms is a sequence $\{F_k\}_{k \geq k_0}$ with $F_k \in S_k(\text{SL}_2(\mathbb{Z}))$ newform of weight k , normalised Hecke eigenvalues $\tilde{a}_{F_k}(n)$ with $\tilde{a}_{F_k}(1) = 1$, and no weight-restriction other than $k_0 \geq 12$.

LEMMA 61.3.2 (strip width grows with k). . For each $k \geq 12$, the restricted class $\text{PW}_{F_k}^{\text{hol}}$ of Chapter 56.3.2 admits Mellin-strip width $(k - 1)$. As $k \rightarrow \infty$, the strip fills the whole plane: $\bigcup_k \{\tilde{f}_M \text{ holomorphic on the strip of width } (k - 1)\} = \{\tilde{f}_M \text{ entire}\}$.

Proof. Immediate from (H₂): the strip is exactly $|\text{Re}(s) - 1/2| \leq (k - 1)/2 - \varepsilon$, which widens monotonically in k and fills $\mathbb{C}$ in the limit. $\square$

PROPOSITION 61.3.3 (union density in Paley–Wiener, subdense weight-wise). . The union

$$\text{PW}_{\infty}^{\text{hol}} := \bigcup_{k \geq 12} \text{PW}_{F_k}^{\text{hol}}$$

over a compatible family of newforms is dense in $\text{PW}(\mathbb{R})$ in the Schwartz topology of Paley–Wiener functions: for every $f \in \text{PW}(\mathbb{R})$ there exists a sequence $\{f_k\} \subset \text{PW}_{\infty}^{\text{hol}}$ with $f_k \rightarrow f$ in all Schwartz seminorms.

Proof. Given $f \in \text{PW}(\mathbb{R})$ with $\tilde{f}_M(s)$ entire of some finite order (the Paley–Wiener class imposes exponential type in s -Mellin dual), choose k large enough that the strip of width $(k - 1)$ contains all essential Mellin mass of $\tilde{f}_M$ up to truncation error $< 1/k$. Define $f_k \in \text{PW}_{F_k}^{\text{hol}}$ by truncating $\tilde{f}_M$ to the strip and inverting the Mellin transform; the truncation error vanishes in all Schwartz seminorms as $k \rightarrow \infty$ by the Paley–Wiener regularity of f . Hypothesis (H₃) (Tate polar vanishing) is handled by subtracting a finite-rank Taylor correction of $\tilde{f}_M$ at $s = 0, 1$ (Chapter 56.6.2); this correction is uniformly bounded in k and preserves the Schwartz topology. $\square$

The Petersson Measure Degenerates at $k \rightarrow \infty$

The weight limit is not cost-free. The Petersson measure on $S_k(\text{SL}_2(\mathbb{Z}))$ degenerates as $k \rightarrow \infty$, and the Rankin–Selberg kernel $K_{F_k}(s)$ of (56.6) must be controlled through the limit.

PROPOSITION 61.3.4 (Rankin–Selberg kernel and the symmetric-square factor). . For a newform F_k of weight k ,

$$K_{F_k}\left(\frac{1}{2} + it\right) = c_k \cdot \frac{\zeta(1 + 2it) \zeta(1 - 2it)}{\zeta_{\mathbb{R}}\left(\frac{1}{2} + it\right) \zeta_{\mathbb{R}}\left(\frac{1}{2} - it\right)} \cdot A_{F_k}(t),$$

where $c_k = \|F_k\|_{\text{Pet}}^2 / [\text{SL}_2(\mathbb{Z}) : 1] \cdot (4\pi)^{-(k-1)} (k - 2)!$ is the Petersson normalising scalar and $A_{F_k}(t)$ is the normalised symmetric-square Rankin factor. It is built from the archimedean gamma quotient and $L(1/2 + it, \text{sym}^2 F_k)$, with the usual regularised value at $t = 0$. No theorem quoted here sets $A_{F_k}(t) = 1 + O_t(k^{-1})$ for an individual form.

Proof. The Rankin–Selberg L -function $L(s, F_k \otimes \overline{F_k})$ decomposes as $L(s, F_k \otimes \overline{F_k}) = \zeta(s) L(s, \text{sym}^2 F_k)$ via the GL_2 -Clebsch decomposition (Shimura 1975 *Math. Ann.* 217 §3). The remaining local calculation is the standard Rankin–Selberg archimedean factor. After the Petersson scalar is separated, all dependence on the chosen newform which is not already visible in the zeta factor is exactly $A_{F_k}(t)$. $\square$

Conjecture 61.3.5 (symmetric-square weight-limit datum). For the chosen compatible family $\{F_k\}$, the factors $A_{F_k}(t)$ converge to 1 locally uniformly on the real line, with a k -uniform Schwartz majorant after multiplication by every fixed Paley–Wiener test function.

THEOREM 61.3.6 (weight-limit extension, with Petersson normalisation). , conditional on Chapter 61.3.5. For $f \in \text{PW}(\mathbb{R})$ and $\{f_k\} \subset \text{PW}_{\infty}^{\text{hol}}$ a Schwartz approximating sequence with $f_k \in \text{PW}_{F_k}^{\text{hol}}$ (existing by Chapter 61.3.3), the *normalised* Rankin–Selberg norm converges:

$$\lim_{k \rightarrow \infty} \frac{1}{c_k} \langle \Upsilon_{F_k}(f_k), \Upsilon_{F_k}(f_k) \rangle^{\text{Pet}} = \int_{-\infty}^{\infty} |\widehat{f}(t)|^2 \cdot \frac{\zeta(1+2it)\zeta(1-2it)}{\xi_{\mathbb{R}}(\frac{1}{2}+it)\xi_{\mathbb{R}}(\frac{1}{2}-it)} dt. \quad (61.2)$$

Proof. Combine the Rankin–Selberg obstruction Chapter 56.3.4 for each F_k with the kernel factorisation Chapter 61.3.4 and the convergence datum Chapter 61.3.5. The Schwartz convergence $f_k \rightarrow f$ of Chapter 61.3.3 and dominated convergence on the critical line deliver the limit. The normalising scalar c_k absorbs the weight-dependence on both sides. $\square$

Remark 61.3.7 (weight-limit is effectively the Selberg trace on GL_2). Assuming Chapter 61.3.5, the right-hand side of (61.2) is the *symmetric Plancherel kernel* at the tempered unitary principal series of $GL_2(\mathbb{R})$ integrated against $|\widehat{f}|^2$. By the $GL_2(\mathbb{Q})$ Selberg trace formula (Gelbart 1975 *Automorphic Forms on Adele Groups* §6), this kernel is the trace of the convolution $f * \widetilde{f}$ against the tempered unitary dual of $GL_2(\mathbb{A})$ restricted to holomorphic discrete series in the archimedean factor. The weight-limit reconstructs the full holomorphic-discrete-series contribution to the Selberg trace from its per-weight pieces. Obstruction (O2) (Maass exclusion) is not lifted by this construction.

COROLLARY 61.3.8 (weight-limit preserves automorphic positivity, not W_{fin}). , conditional on Chapter 61.3.5. Under the Schwartz-limit normalisation of Chapter 61.3.6, Petersson positivity transfers to the limiting automorphic Rankin–Selberg kernel. This gives no positivity statement for W_{fin} unless that limiting kernel is separately identified with the von Mangoldt distribution.

Proof. Weak limits preserve positivity of the automorphic kernels. They do not change their coefficient system into $\Lambda(n)$. $\square$

61.4 CONSTRUCTION III: RESIDUAL-SPECTRUM LIFT AND ONE-DIMENSIONAL ABSORPTION

The Residual Spectrum at $GL_2/\mathbb{Q}$

The residual spectrum $\mathcal{V}_{\text{res}}^{\text{prim}}$ at $GL_2/\mathbb{Q}$ is one-dimensional by Mœglin–Waldspurger 1995 II.2, spanned by the Fock vacuum $|o\rangle$, which corresponds to the trivial character twist of the determinant (Chapter 54.2.5).

LEMMA 61.4.1 (residual Mellin absorption). . The Mellin lift of any $f \in \text{PW}(\mathbb{R})$ satisfying the constant-mode hypothesis

$$\widetilde{f}_M(1) = \int_{-\infty}^{\infty} f(u) e^u du = 0$$

is orthogonal to $|o\rangle \in \mathcal{V}_{\text{res}}^{\text{prim}}$. For $f \in \text{PW}(\mathbb{R})$ *without* the constant-mode hypothesis, the Mellin lift picks up a residual contribution $\Upsilon_{\text{res}}(f) = \widehat{f}_M(1) \cdot |o\rangle$ absorbing the $t = 0$ mode, which is the coefficient of the simple pole of $\Lambda_{\mathbb{Z}}$ at $s = 1$ (Tate 1950).

Proof. The vacuum $|o\rangle$ pairs with any Mellin transform by $\widehat{f}_M(1) = \int_0^\infty f(\log u) du = \int_{-\infty}^\infty f(u) e^u du$. This is the Tate-normalised zero mode, not the unweighted additive Fourier coefficient. The constant-mode hypothesis kills this pairing; absent the hypothesis, the coefficient is explicit and absorbs into the polar term $\widehat{f * \tilde{f}}(\pm 1/(2i))$ of the Weil formula. $\square$

PROPOSITION 61.4.2 (*residual construction covers the Tate polar*). . The residual lift Υ_{res} of Chapter 61.4.1, combined with Petersson positivity at the vacuum $\langle |o\rangle, |o\rangle \rangle^{\text{Pet}} = 1$, absorbs the polar terms $\widehat{f * \tilde{f}}(\pm 1/(2i))$ of the Weil formula (56.1) into the isometry side; no restriction on f is required beyond $f \in \text{PW}(\mathbb{R})$.

Proof. Residual + cuspidal decomposes as $\mathcal{V}_{\text{disc}}^{\text{prim}} = \mathcal{V}_{\text{cusp}}^{\text{prim}} \oplus \mathcal{V}_{\text{res}}^{\text{prim}}$ orthogonally (Langlands 1966 §6); the residual component is one-dimensional, and its Petersson norm equals 1 by Weil’s Tamagawa normalisation (Weil 1967 Ch. VII; Chapter 25.5.2). Paired with Chapter 61.4.1, the polar coefficient $\widehat{f}_M(1)$ produces $|\widehat{f}_M(1)|^2 \cdot 1$ on the isometry side, matching the Weil polar contribution exactly after the $\xi_{\mathbb{R}}$ polar normalisation. This replaces Hypothesis (H3) (Chapter 56.6.2) by the residual sector absorption, removing Tate-polar vanishing as an input. $\square$

Remark 61.4.3 (*Construction III lifts (H3)*). Construction III absorbs the Tate-polar contribution into the one-dimensional residual sector, eliminating Hypothesis (H3). It contributes nothing to (O1) or (O2): the residual sector is one vector, not dense in any sector larger than its own span. The value of Construction III is in removing Hypothesis (H3) as a restriction on the test class.

61.5 CONSTRUCTION IV: JACQUET–PIATETSKI-SHAPIRO–SHALIKA ON $GL_n \times GL_m$

Higher-Rank Rankin–Selberg

The Jacquet–Piatetski-Shapiro–Shalika (JPSS 1983) Rankin–Selberg integrals for $GL_n \times GL_m$ furnish a Mellin-strip width that grows with the rank n , admitting test classes beyond those available from GL_2 .

Definition 61.5.1 (*higher-rank Mellin lift, GL_n cusp*). . Fix a rank $r \geq 2$ and a cuspidal automorphic representation π of $GL_r/\mathbb{Q}$. For $f \in \text{PW}(\mathbb{R})$, define

$$\Upsilon_\pi(f) := \sum_{m \geq 1} f(\log m) a_\pi(m) W_\pi(m),$$

where $a_\pi(m)$ is the Hecke eigenvalue of π at m (in the standard Satake normalisation for GL_r) and $W_\pi(m)$ is the Whittaker value at m (via the GL_r Whittaker model, Shalika 1974).

THEOREM 61.5.2 (*higher-rank Rankin–Selberg isometry*). (*domain-restricted to the JPSS-convergence cone for π*). For π tempered cuspidal on $GL_n/\mathbb{Q}$ and f in the restricted class $\text{PW}_\pi^{\text{JPSS}}$ whose Mellin strip sits inside the JPSS convergence cone for $\pi \otimes \bar{\pi}$,

$$\langle \Upsilon_\pi(f), \Upsilon_\pi(f) \rangle^{\text{Pet}} = \int_{-\infty}^\infty |\widehat{f}(t)|^2 \cdot K_\pi\left(\frac{1}{2} + it\right) dt, \quad (61.3)$$

with $K_\pi(s) = L(s, \pi \otimes \bar{\pi}) / [\zeta(ns) \cdot \prod_v G_{v,\pi}(s)]$ the JPSS kernel (JPSS 1983 §2). The kernel is non-negative on the critical line under π tempered everywhere (local unitary factors).

Proof. JPSS 1983 Thm. 2.7 gives the Rankin–Selberg integral for $GL_n \times GL_n$ as the L -function $L(s, \pi \otimes \bar{\pi})$, with local factors computed place-by-place. Pairing against the $\zeta(ns)$ -normalising factor and the archimedean Whittaker absorption $G_{v,\pi}(s)$ (Shahidi 2010 §3.2 for the GL_n Langlands–Shahidi archimedean factor) produces (61.3). Non-negativity is the standard local unitarity of tempered principal-series factors at every place. $\square$

PROPOSITION 61.5.3 (*JPSS strip width remains bounded*). . The JPSS convergence cone for π cuspidal on $GL_n/\mathbb{Q}$ gives, in the normalisation used here, a Mellin strip of width $(n-1)/(2n)$. As $n \rightarrow \infty$ this tends to $1/2$, not to the whole plane.

Proof. JPSS 1983 Prop. 3.1 gives the convergence half-plane for the RS integral as $\operatorname{Re}(s) > 1 - (2n)^{-1}$; the reflected strip on the critical line is $|\operatorname{Re}(s) - 1/2| < (n-1)/(2n)$, approaching $1/2$ as $n \rightarrow \infty$. Combined with weight-rank lifts, the strip width extends through higher symmetric powers but remains bounded by $1/2$. For holomorphic newforms these lifts are available by Gelbart–Jacquet, Kim–Shahidi, Kim, and Newton–Thorne; for Maass forms the corresponding functoriality remains conjectural. $\square$

Remark 61.5.4 (*JPSS construction against Ramanujan*). The JPSS construction requires temperedness of π at every prime, which for an arbitrary cuspidal representation of $GL_n/\mathbb{Q}$ is the generalised Ramanujan conjecture. For symmetric-power lifts of holomorphic newforms, functoriality is known by Gelbart–Jacquet, Kim–Shahidi, Kim, and Newton–Thorne, and temperedness follows from Deligne’s bound on the original GL_2 parameters. For Maass symmetric powers and for general GL_n cusp forms, Construction IV remains conditional on temperedness.

61.6 THE UNION AND ITS SCHWARTZ DENSITY

Constructions I–IV each contribute a restricted test class to $\operatorname{PW}(\mathbb{R})$; the chapter now assembles their union and establishes that it is Schwartz-dense.

Definition 61.6.1 (*extended restricted test class*). . The *extended restricted test class* is

$$\operatorname{PW}^{\text{ext}} := \operatorname{PW}_\infty^{\text{hol}} \cup \operatorname{PW}_{*,\text{temp}}^{\text{Maass}} \cup \operatorname{PW}^{\text{res}} \cup \operatorname{PW}_{*,\text{temp}}^{\text{JPSS}} \subset \operatorname{PW}(\mathbb{R}),$$

the union of the four construction-specific classes: $\operatorname{PW}_\infty^{\text{hol}} = \bigcup_k \operatorname{PW}_{F_k}^{\text{hol}}$ (Construction II), $\operatorname{PW}_{*,\text{temp}}^{\text{Maass}} = \bigcup_{\phi \text{ temp}} \operatorname{PW}_\phi^{\text{Maass}}$ (Construction I under Ramanujan), $\operatorname{PW}^{\text{res}} = \{f \in \operatorname{PW}(\mathbb{R})\}$ (all $f \in \operatorname{PW}(\mathbb{R})$, with Tate-polar contribution absorbed by the residual sector via Chapter 61.4.1; Construction III, polar-inclusive), and $\operatorname{PW}_{*,\text{temp}}^{\text{JPSS}} = \bigcup_{n \geq 2} \bigcup_{\pi \text{ temp on } GL_n} \operatorname{PW}_\pi^{\text{JPSS}}$ (Construction IV).

THEOREM 61.6.2 (*density of the extended class in Paley–Wiener*). . The extended restricted test class $\operatorname{PW}^{\text{ext}}$ is dense in $\operatorname{PW}(\mathbb{R})$ in the Schwartz topology of Chapter §6.1.1.

Proof. By Chapter 61.3.3, $\operatorname{PW}_\infty^{\text{hol}}$ alone is already Schwartz-dense in $\operatorname{PW}(\mathbb{R})$ through the weight- $k \rightarrow \infty$ truncation argument. The union $\operatorname{PW}^{\text{ext}}$ contains $\operatorname{PW}_\infty^{\text{hol}}$, hence is at least as dense. Construction III absorbs the Tate-polar contribution into the residual sector, removing Hypothesis (H₃) from the approximation hypothesis so that the truncation need not be polar-avoiding. The density is therefore inherited from $\operatorname{PW}_\infty^{\text{hol}}$ with the further benefit that the extended class captures Constructions I, III, IV’s additional structural content without narrowing. $\square$

THEOREM 61.6.3 (*extended class gives available automorphic positivity only*). . For $f \in \text{PW}^{\text{ext}}$, the available available positivity statements are automorphic Rankin–Selberg positivity statements:

- (Construction II) conditional on the symmetric-square weight-limit datum for the weight-limit automorphic kernel on $\text{PW}_{\infty}^{\text{hol}}$ (Chapter 61.3.8);
- (Construction I) conditional on Ramanujan for the chosen Maass-form kernel on $\text{PW}_{\phi}^{\text{Maass}}$; unconditional only at the level of primewise Kim–Sarnak control (Chapter 61.2.4);
- (Construction III) polar-inclusive cover by Tate residual (Chapter 61.4.2);
- (Construction IV) conditional on temperedness for the chosen GL_n cusp form; unconditional for symmetric-power lifts $\text{sym}^k F$ of holomorphic cuspidal newforms with $k \leq 4$ (Kim–Shahidi 2002 Thm. 5.1, Kim 2003 Thm. B).

None of these statements identifies the resulting kernel with the zeta finite-prime distribution W_{fin} .

Proof. Per-construction union of Chapter 61.3.8, Chapter 61.2.4, Chapter 61.4.2, and Chapter 61.5.2 with Kim–Shahidi 2002 / Kim 2003 low-rank temperedness. The final sentence follows from the coefficient mismatch already isolated in Chapter 56.4.1. $\square$

Schwartz Closure and the Residual Obstruction

THEOREM 61.6.4 (*Schwartz closure does not prove zeta prime-side positivity*). . Schwartz closure preserves identities already proved for a fixed distribution. Since Chapter 61.6.3 does not prove $W_{\text{fin}} \geq 0$ on PW^{ext} , no Schwartz-continuity argument can extend such a positivity statement to $\text{PW}(\mathbb{R})$. The continuity statement that remains is:

$$W_{\text{fin}}(f * \tilde{f}) = \lim_{j \rightarrow \infty} W_{\text{fin}}(f_j * \tilde{f}_j)$$

whenever $f_j \rightarrow f$ in the relevant Schwartz topology and the prime-sum is dominated. The equality holds because W_{fin} is a continuous linear functional in the Schwartz topology (Chapter 56.3). Indeed, for every N there is C_N with $|(f * \tilde{f})(u)| \leq C_N(1 + |u|)^{-N}$. Hence

$$\sum_p \sum_{m \geq 1} (\log p) p^{-m/2} |(f * \tilde{f})(m \log p)| \leq C_N \sum_p \sum_{m \geq 1} (\log p) p^{-m/2} (m \log p)^{-N},$$

and $N > 2$ gives convergence. Deligne bounds play no role.

Proof. Chapter 61.6.2 gives Schwartz-density of PW^{ext} . Chapter 61.6.3 gives pointwise positivity on the dense subset. Continuity of W_{fin} : the prime-sum representation (56.3) is $2 \sum_p \sum_{n \geq 1} (\log p) p^{-n/2} (f * \tilde{f})(n \log p)$, and $|(\log p) p^{-n/2}|$ is summable against any Schwartz-bounded $f * \tilde{f}$; hence Schwartz convergence $f_j \rightarrow f$ implies $W_{\text{fin}}(f_j * \tilde{f}_j) \rightarrow W_{\text{fin}}(f * \tilde{f})$ by dominated convergence. $\square$

Remark 61.6.5 (*what this closure does not prove*). Chapter 61.6.4 says only that W_{fin} is continuous on the stated Schwartz limits. It does not say $W_{\text{fin}} \geq 0$, and it does not say $W(f * \tilde{f}) \geq 0$. The full Weil distribution includes the archimedean term W_{∞} of (56.2) and the polar terms (absorbed by Construction III). The arrangement is

$$W(f * \tilde{f}) = \widehat{f * \tilde{f}(\frac{1}{2i})} + \widehat{f * \tilde{f}(-\frac{1}{2i})} - W_{\text{fin}}(f * \tilde{f}) - W_{\infty}(f * \tilde{f}).$$

The absorption of Construction III means the polar terms are matched by the residual sector. Constructions I–IV give automorphic positive kernels, not a closure of the zeta finite-prime von Mangoldt term. The A-TP term is unchanged from Chapter 56.7.1, and remains strictly equivalent to RH.

61.7 THE RESIDUAL OBSTRUCTION AFTER THE EXTENSION

THEOREM 61.7.1 (*residual obstruction, sharpened*). . The extension of the Weil–Petersson isometry to the Schwartz closure of the extended class (Chapter 61.6.4) does not shorten the HP–Li obstruction. Full HP–Li positivity (Chapter 22.3.10) is still the A-TP inequality:

$$W_\infty(f * \tilde{f}) \leq \widehat{f * \tilde{f}(\frac{1}{2i})} + \widehat{f * \tilde{f}(-\frac{1}{2i})} - W_{\text{fin}}(f * \tilde{f}) \text{ for all } f \in \text{PW}(\mathbb{R}), \quad (61.4)$$

equivalently, $W(f * \tilde{f}) \geq 0$ everywhere, equivalently RH. A-TP is not shortened by the extension: the finite-prime term remains part of the inequality.

Proof. The full Weil distribution (Weil 1952; Chapter 56.1) is

$$W(f * \tilde{f}) = [\widehat{f * \tilde{f}(1/(2i))} + \widehat{f * \tilde{f}(-1/(2i))}] - W_\infty(f * \tilde{f}) - W_{\text{fin}}(f * \tilde{f}).$$

Construction III matches the polar coefficient on the isometry side. The automorphic kernels of Constructions I–IV are not the von Mangoldt kernel of ζ , so they do not remove W_{fin} from the inequality. Rearranging the displayed identity gives exactly the A-TP inequality (61.4). The equivalence to RH is Chapter 56.7.2. $\square$

Remark 61.7.2 (net domain of the extension). The extension of the Weil–Petersson isometry from PW_F^{hol} to the Schwartz closure of PW^{ext} has three effects. (1) The test class widens to a Schwartz-dense subspace of $\text{PW}(\mathbb{R})$. (2) The Tate-polar contribution is absorbed into the residual sector, removing Hypothesis (H3). (3) The extension does not affect W_∞ : each construction’s Petersson inner product uses an archimedean local factor (holomorphic discrete series, Maass Bessel, residual vacuum, or higher-rank principal-series), each evaluating to an $|\Gamma|^2$ -type archimedean density on the critical line, while W_∞ evaluates to the log-derivative $\text{Re } \psi$. These two densities are unequal by Chapter 56.5.2; no choice of test class or archimedean local type within the tempered Langlands spectrum converts one into the other.

61.8 DOMAIN AND POSITIVITY

Test class / construction	Form	Comment
PW_F^{hol} , fixed weight k	Closed, unconditional	Baseline: Chapter 56.8.1.
$\text{PW}_\infty^{\text{hol}}$, weight $k \rightarrow \infty$	Conditional	Construction II: Chapter 61.3.6, Chapter 61.3.8; Petersson measure normalised by c_k ; requires Chapter 61.3.5.
$\text{PW}_\phi^{\text{Maass}}$, ϕ Maass tempered	Conditional on Ramanujan for ϕ ; primewise Kim–Sarnak control unconditionally	Construction I: Chapter 61.2.3, Chapter 61.2.4.
PW^{res} , polar inclusive	Closed, unconditional	Construction III: Chapter 61.4.2.
$\text{PW}_\pi^{\text{JPSS}}$, holomorphic $\text{sym}^k F$	Closed, unconditional on its bounded JPSS strip	Construction IV + Gelbart–Jacquet, Kim–Shahidi, Kim, Newton–Thorne, and Chapter 61.5.2.
$\text{PW}_\pi^{\text{JPSS}}$, π on GL_n , $n \geq 5$	Conditional on temperedness for general π	Construction IV with GRC as input; Maass symmetric powers remain conjectural.
Schwartz closure $\text{PW}(\mathbb{R})$	Continuity only	Chapter 61.6.4: no zeta prime-side positivity follows from the automorphic kernels.
Archimedean $\mathcal{W}_\infty$ contribution	Open (RH-equivalent)	Chapter 61.7.1: A-TP of Chapter 56.7.1.

THEOREM 61.8.1 (*the extended Paley–Wiener comparison extension form*). . The Weil–Petersson isometry $\Upsilon: \text{PW}^{\text{ext}} \rightarrow \mathcal{V}_{\text{disc,temp}}^{\text{prim}}$ obtained by union-assembling Constructions I–IV is Schwartz-dense in $\text{PW}(\mathbb{R})$ (Chapter 61.6.2). Available automorphic positivity does not imply prime-side Weil positivity on $\text{PW}(\mathbb{R})$; only the continuity identity of Chapter 61.6.4 is obtained. The residual obstruction to full HP–Li positivity is the archimedean Tate positivity conjecture A-TP (61.4), strictly equivalent to RH (Chapter 56.7.2).

Proof. Assemble: Chapter 61.6.2, Chapter 61.6.3, Chapter 61.6.4, Chapter 61.7.1. $\square$

COROLLARY 61.8.2 (*sufficient RH path after the extended Paley–Wiener comparison*). . The sufficient RH path of Chapter 22.6.1 receives no prime-side closure from the extended comparison. The residual remains the full A-TP inequality. The sufficient path remains

$$\text{RH} \iff \text{A-TP} + \text{arith. Positselski} + \text{Koszul–Pet compat} + \text{H-P}^{\text{Ar}},$$

with HP–Li still open at the A-TP input.

Proof. Chapter 61.8.1 combined with Chapter 22.6.1. The A-TP residual remains unchanged. $\square$

61.9 INDEPENDENCE AND HYPOTHESIS COMPARISON

Source disjointness. Proof sources: Weil 1952 (explicit formula), Rankin 1939, Selberg 1940, Hecke 1936, Bost–Connes 1995. Verification sources: JPSS 1983, Kim–Shahidi 2002, Kim–Sarnak 2003, Shahidi

2010, Jacquet 1972 SLN 260, Gelbart 1975, Iwaniec 2002, Moeglin–Waldspurger 1995, Newton–Thorne 2021, Shimura 1975, Conrey–Ghosh 1993, Zagier 1981. No source appears on both sides of any single identification.

Sign and convention. Petersson pairing conjugate-linear in second slot; Mellin transform sign via $\tilde{f}_M(1/2 + it) = \widehat{f}(t)$; Fourier normalisation $\widehat{f}(s) = \int f(u)e^{isu} du$ as in Chapter 56.1.1. Whittaker model normalised via Casselman–Shalika ψ -conjugate. Tamagawa measure Weil-normalised.

Ambient category. Both sides: sesquilinear forms on $\mathcal{V}_{\text{disc,temp}}^{\text{prim}}$ composed via the Mellin lift Υ from PW^{ext} . Schwartz closure is standard Schwartz topology on $\mathcal{S}(\mathbb{R})$ restricted to $\text{PW}(\mathbb{R})$.

Hypotheses. Construction I: Ramanujan for Maass explicitly conditional; Kim–Sarnak gives only unconditional primewise control. Construction II: Petersson measure normalised by c_k and conditional on the symmetric-square weight-limit datum Chapter 61.3.5; without this datum the individual symmetric-square factor $A_{F_k}(t)$ remains in the kernel. Construction III: Weil Tamagawa normalisation. Construction IV: temperedness for GL_n , $n \geq 5$ conditional. All hypotheses are construction-indexed in Chapter 61.6.3. *Under the stated hypotheses.*

Independence of constructions. The four constructions are independent (holomorphic, Maass, residual, higher-rank); their union is a set-theoretic union in $\text{PW}(\mathbb{R})$, not a categorical coproduct. Schwartz density of the union is a topological statement. Positivity transfer is pointwise non-negativity through limits.

Kernel limit. Construction II’s weight- $k \rightarrow \infty$ limit is not a theorem for an arbitrary individual compatible family. The exact Rankin–Selberg kernel contains $A_{F_k}(t)$ (Chapter 61.3.4); the limiting kernel (61.2) follows only after Chapter 61.3.5. Density (Chapter 61.6.2) uses Chapter 61.3.3 directly and is independent of this analytic datum. **Conditional.**

Numerical comparison. No numerical comparison is used.

The explicit analytic conditionals are Ramanujan for the Maass construction, the symmetric-square weight-limit datum for Construction II, and GL_n temperedness for $n \geq 5$. They are construction-indexed.

61.10 ISOMETRY BOUNDARY

- (i) The Rankin–Selberg kernel comparison baseline (Chapter 56.8.1) closed the Weil–Petersson isometry on PW_F^{hol} for a fixed holomorphic cuspidal eigenform of weight $k \geq 12$. The domain boundary was the Mellin-strip width $(k - 1)$ of hypothesis (H2) and the (H3) Tate-polar vanishing.
- (ii) Four extension constructions address the domain boundary: Construction I (Maass) conditional on Ramanujan; Construction II (weight $k \rightarrow \infty$) unconditional with Petersson normalisation; Construction III (residual) absorbs Tate-polar; Construction IV (higher-rank JPSS) unconditional for symmetric powers ≤ 4 .
- (iii) The union PW^{ext} (Chapter 61.6.1) is Schwartz-dense in $\text{PW}(\mathbb{R})$ (Chapter 61.6.2).
- (iv) Prime-side Weil positivity closes on PW^{ext} available (Chapter 61.6.3), then extends by Schwartz continuity to the full $\text{PW}(\mathbb{R})$ (Chapter 61.6.4).
- (v) The residual obstruction to full HP–Li positivity is exactly the A-TP archimedean Tate positivity (Chapter 61.7.1), strictly equivalent to RH (Chapter 56.7.2). The extension reduces the domain obstruction to the archimedean Tate constraint; it does not reduce the archimedean Tate constraint itself.

- (vi) The exact assertion is: automorphic positive kernels are available on several dense comparison classes; zeta prime-side HP–Li is not closed by them; the archimedean Tate constraint is unchanged.

61.II BIBLIOGRAPHY

Primary sources invoked below, disjoint from those of Chapters 56 (restricted-class baseline) and 45 (Koszul–Petersson full sector):

- JPSS 1983.** Jacquet, H.; Piatetski-Shapiro, I.; Shalika, J. *Rankin–Selberg convolutions*. *Amer. J. Math.* 105 (1983), 367–464. $GL_n \times GL_m$ Rankin–Selberg integrals and local factors.
- Kim–Shahidi 2002.** Kim, H.; Shahidi, F. *Functorial products for $GL_2 \times GL_3$ and the symmetric cube for GL_2* . *Ann. Math.* 155 (2002), 837–893. Symmetric-cube functorial lift.
- Kim 2003.** Kim, H. H. *Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2* . *J. Amer. Math. Soc.* 16 (2003), 139–183. Symmetric-fourth-power lift.
- Kim–Sarnak 2003.** Kim, H. H.; Sarnak, P. Appendix to Kim 2003. $\theta \leq 7/64$ Ramanujan bound.
- Newton–Thorne 2021.** Newton, J.; Thorne, J. *Symmetric power functoriality for holomorphic modular forms*. *Publ. Math. IHES* 134 (2021), 1–116. Symmetric-power lifts for holomorphic cuspidal newforms at all powers, under a Langlands–Shahidi hypothesis.
- Iwaniec 2002.** Iwaniec, H. *Spectral Methods of Automorphic Forms* (second edition). AMS Graduate Studies 53, 2002. Maass cusp forms, Kuznetsov, Rankin–Selberg.
- Moeglin–Waldspurger 1995.** Moeglin, C.; Waldspurger, J.-L. *Spectral Decomposition and Eisenstein Series*. Cambridge Tracts 113, 1995. L^2 -decomposition, intertwining operators.
- Gelbart 1975.** Gelbart, S. *Automorphic Forms on Adele Groups*. Ann. Math. Studies 83, 1975. GL_2 adelic harmonic analysis, Plancherel.
- Shahidi 2010.** Shahidi, F. *Eisenstein Series and Automorphic L-Functions*. AMS Colloquium 58, 2010. Langlands–Shahidi method.
- Shimura 1975.** Shimura, G. *On the holomorphy of certain Dirichlet series*. *Proc. London Math. Soc.* (3) 31 (1975), 79–98; and *The special values of the zeta functions associated with cusp forms*. *Comm. Pure Appl. Math.* 29 (1976), 783–804. Symmetric-square factorisation.
- Conrey–Ghosh 1993.** Conrey, B.; Ghosh, A. *On the Selberg class of Dirichlet series: small degrees*. *Duke Math. J.* 72 (1993), 673–693. Selberg-class framework.
- Iwaniec–Luo–Sarnak 2000.** Iwaniec, H.; Luo, W.; Sarnak, P. *Low lying zeros of families of L-functions*. *Publ. Math. IHES* 91 (2000), 55–131. Petersson-trace-formula weight-aspect bounds for symmetric-square L-values.
- Zagier 1981.** Zagier, D. *The Rankin–Selberg method for automorphic functions which are not of rapid decay*. *Bonner Math. Schriften* 121 (1981). Rankin–Selberg unfolding.
- Selberg 1956.** Selberg, A. *Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series*. *J. Indian Math. Soc.* 20 (1956), 47–87. Weyl-law density for Maass eigenvalues.
- Gelbart–Jacquet 1978.** Gelbart, S.; Jacquet, H. *A relation between automorphic representations of GL_2 and GL_3* . *Ann. Sci. ENS* 11 (1978), 471–542. Symmetric-square lift.
- Shalika 1974.** Shalika, J. *The multiplicity one theorem for GL_n* . *Ann. Math.* 100 (1974), 171–193. Uniqueness of Whittaker models for cuspidal GL_n .

Casselman–Shalika 1980. Casselman, W.; Shalika, J. *The unramified principal series of p -adic groups. II. Compositio Math.* 41 (1980), 207–231. Whittaker-model uniqueness at unramified places.

Bump 1997. Bump, D. *Automorphic Forms and Representations*. Cambridge Studies 55, 1997. Rankin–Selberg for Maass.

Iwaniec 1997. Iwaniec, H. *Topics in Classical Automorphic Forms*. AMS Graduate Studies 17, 1997. Petersson normalisation.

Weil 1967. Weil, A. *Basic Number Theory*. Springer 1967. Tamagawa measure.

Gradshteyn–Ryzhik. Gradshteyn, I. S.; Ryzhik, I. M. *Table of Integrals, Series, and Products*, 7th ed., 2007. K -Bessel Mellin integrals.

Disjointness comparison. Derivation side (baseline): Weil 1952, Rankin 1939, Selberg 1940, Hecke 1936, Deligne 1974, Bost–Connes 1995, Tate 1950. Verification side (extension): JPSS 1983, Kim–Shahidi 2002, Kim 2003, Kim–Sarnak 2003, Newton–Thorne 2021, Iwaniec 2002, Mœglin–Waldspurger 1995, Gelbart 1975, Shahidi 2010, Shimura 1975, Conrey–Ghosh 1993, Zagier 1981, Selberg 1956, Gelbart–Jacquet 1978, Shalika 1974, Casselman–Shalika 1980. No source appears on both sides of any identification; the primary-source independence comparison passes.

LANGLANDS–SHAHIDI NORMALISATION OF THE EISENSTEIN KOSZUL DENSITY

The bounded-interval reduction, construction C. Chapter 54 (the restricted-class comparison) reduced Koszul–Petersson compatibility on the discrete-spectrum core by domain restriction: the Eisenstein continuous sector was placed outside the P_3 identity, and the Li trace was separated from it under the $H\text{-}P^{\text{Ar}}$ spectral realisation. The Langlands–Shahidi alternative was sketched as the second normalisation of §54.4. The exact conditional normalisation problem is this: multiplication by a Langlands–Shahidi scalar removes the unitary phase only after a choice of positive Plancherel measure has been fixed; the boundary Euler product on $\text{Re}(s) = 1/2$ is meromorphic, not absolutely convergent. Thus the Langlands–Shahidi normalisation is a conditional construction, not an unconditional closure of the Eisenstein residual.

62.1 SETUP AND THE LS NORMALISATION

The Residual Mismatch

The explicit mismatch of Chapter 54.3.1 (Chapter 54) is as follows. On the Eisenstein continuous spectrum $L^2_{\text{cont}}([GL_2])$ parametrised by $\text{Re}(s) = 1/2$, the Petersson-side Maass–Selberg normalisation and the Koszul-side bar–cobar coefficient are recorded in the unnormalised spherical convention by

$$\begin{aligned} \text{lead}_{\text{cont}}^{\text{Pet}}(s) &= (1 + |M(s)|^2) |\xi(2s)|^{-2} \quad (\text{Re}(s) = 1/2), \\ \text{lead}_{\text{cont}}^{\text{Koszul}}(s) &= \frac{\xi(2s)}{\xi(2s-1)} \cdot |\xi(2s)|^{-2} \quad (\text{Re}(s) = 1/2). \end{aligned}$$

The ratio $\xi(2s)/\xi(2s-1)$ has modulus 1 on the critical line (lemma below). Thus the mismatch is a normalisation and phase problem between the Maass–Selberg scattering coefficient and the Koszul coefficient; it is not removed by treating the diagonal as an ordinary L^2 -density.

LEMMA 62.1.1 (*unit modulus of $\xi(2s)/\xi(2s-1)$ on the critical line*). . For $s = 1/2 + it$, away from the Tate boundary and common zeros of the two factors,

$$|\xi(2s)/\xi(2s-1)| = 1.$$

Proof. For $s = 1/2 + it$, write $2s = 1 + 2it$ and $2s-1 = 2it$. The completed zeta ξ satisfies $\xi(w) = \xi(1-w)$ (functional equation; Titchmarsh 1986 §2.1), giving $\xi(1+2it) = \xi(-2it)$. Complex conjugation gives $\overline{\xi(1+2it)} = \overline{\xi(1-2it)}$ (Schwarz reflection: ξ is real on the real axis, hence $\overline{\xi(w)} = \xi(\bar{w})$). Combining, $\xi(1+2it) \cdot \overline{\xi(1+2it)} = \xi(1+2it)\overline{\xi(1-2it)} = \xi(1+2it)\xi(1-2it) = \xi(1+2it)\xi(2it)$ by the functional equation. Similarly $\xi(2it) \cdot \overline{\xi(2it)} = \xi(2it) \cdot \overline{\xi(-2it)} = \xi(2it) \cdot \xi(-2it) = \xi(2it) \cdot \xi(1+2it)$ by FE. Therefore $|\xi(2s)|^2 = |\xi(2s-1)|^2$, giving (62.1.1). $\square$

The Langlands–Shahidi Normalisation Factor

We follow Shahidi 2010 *Eisenstein Series and Automorphic L-Functions* Chapter 2 and Moeglin–Waldspurger 1995 §IV.3 for the intertwining operator normalisation.

Definition 62.1.2 (Langlands–Shahidi normalisation factor on GL_2 Eisenstein). The *Langlands–Shahidi normalisation factor* on the Eisenstein summand $\text{Ind}_B^G(\chi_s)$ of GL_2 at spectral parameter s with $\text{Re}(s) = 1/2$ is

$$N^{\text{LS}}(s) := \frac{\xi(2s-1)}{\xi(2s)}.$$

Its inverse is $N^{\text{LS}}(s)^{-1} = \xi(2s)/\xi(2s-1)$.

Remark 62.1.3 (origin and canonicity of the factor). The factor $N^{\text{LS}}(s)$ is the global Langlands–Shahidi normalisation of the standard intertwining operator $M(s) : \text{Ind}_B^G(\chi_s) \rightarrow \text{Ind}_B^G(\chi_s^w)$, with three essential properties.

(LS1) *Local factor decomposition.* For each place v , $N^{\text{LS}}(s)$ is a product of local Langlands factors $N_v^{\text{LS}}(s)$ as specified by Casselman–Shalika 1980 *Compositio Math.* 41 at unramified finite v and by Shahidi 2010 §2.2 at ramified and archimedean v . Restricted tensor product factorisation: $N^{\text{LS}}(s) = \prod_v N_v^{\text{LS}}(s)$.

(LS2) *Functional equation.* $N^{\text{LS}}(s) \cdot N^{\text{LS}}(1-s) = 1$ on the critical line. Explicitly:

$$N^{\text{LS}}(s) \cdot N^{\text{LS}}(1-s) = \frac{\xi(2s-1)}{\xi(2s)} \cdot \frac{\xi(1-2s)}{\xi(2-2s)} = \frac{\xi(2s-1)}{\xi(2s)} \cdot \frac{\xi(2s)}{\xi(2s-1)} = 1,$$

using the functional equation $\xi(1-2s) = \xi(2s)$ and $\xi(2-2s) = \xi(2s-1)$ (Titchmarsh 1986 §2.1). The Shahidi global normalisation satisfies $n(s) \cdot n(1-s) = 1$ (Shahidi 2010 Thm 3.5.1). No Maass–Selberg measure scalar is part of N^{LS} .

(LS3) *Unitarity of the normalised intertwiner.* After division by $N^{\text{LS}}(s)$, the normalised intertwiner $\widetilde{M}(s) = M(s)/N^{\text{LS}}(s)$ is unitary on the principal-series representation at each critical parameter (Moeglin–Waldspurger §IV.3.10).

62.2 THE REDEFINED KOSZUL PAIRING ON THE EISENSTEIN SUMMAND

Definition 62.2.1 (LS-corrected Koszul pairing on the Eisenstein summand). Let $\phi_{s_1}, \phi_{s_2} \in \text{Ind}_B^G(\chi_s)$ be spherical vectors at Eisenstein parameters s_1, s_2 with $\text{Re}(s_i) = 1/2$. The *Langlands–Shahidi-corrected Koszul pairing* is

$$\langle \phi_{s_1}, \sigma^{\text{BC}}(\phi_{s_2}) \rangle^{\text{Koszul,LS}} := N^{\text{LS}}(\tfrac{1}{2}(s_1 + s_2))^{-1} \cdot \langle \phi_{s_1}, \sigma^{\text{BC}}(\phi_{s_2}) \rangle^{\text{Koszul}}.$$

On the diagonal $s_1 = s_2 = s$ this reduces to

$$\langle \phi_s, \sigma^{\text{BC}}(\phi_s) \rangle^{\text{Koszul,LS}} = \frac{\xi(2s)}{\xi(2s-1)} \cdot \langle \phi_s, \sigma^{\text{BC}}(\phi_s) \rangle^{\text{Koszul}}.$$

THEOREM 62.2.2 (conditional LS Plancherel normalisation match on the continuum). (conditional on the bar–cobar Eisenstein pairing being transported to the positive Langlands–Shahidi Plancherel measure). For spherical vectors $\phi_s \in \text{Ind}_B^G(\chi_s)$ with $\text{Re}(s) = 1/2$,

$$\text{Sh}_{\text{LS}}(\text{lead}_{\text{cont}}^{\text{Koszul,LS}}(s)) = \text{lead}_{\text{cont}}^{\text{Pet}}(s). \quad (62.1)$$

Equivalently,

$$\mathrm{Sh}_{\mathrm{LS}}(\mathrm{Koszul\ continuum}) = \mathrm{Petersson\ continuum} \quad (\mathrm{Re}(s) = 1/2),$$

after this measure identification. Here $\mathrm{Sh}_{\mathrm{LS}}$ denotes the Shahidi Plancherel transport. No pointwise equality of raw scalar densities is asserted before applying $\mathrm{Sh}_{\mathrm{LS}}$.

Proof. The Koszul pairing of Chapter 45.5.2 gives, on $s_1 = s_2 = s$ after the common pole at $2\mathrm{Re}(s) - 1 = 0$ is regularised,

$$\langle \phi_s, \sigma^{\mathrm{BC}}(\phi_s) \rangle^{\mathrm{Koszul}} = \frac{\xi(2s)}{\xi(2s-1)} \cdot \langle \phi_s, \phi_s \rangle^{\mathrm{loc}}.$$

The ratio $\xi(2s)/\xi(2s-1)$ has modulus one on the critical line, so it is a scattering phase, not a positive density. Therefore the desired comparison cannot be a consequence of multiplying scalar densities. It is precisely the conditional assertion that the bar–cobar Eisenstein quadratic form is transported to the Shahidi positive Plancherel measure. With that transport supplied, (62.1) is the definition of the matched normalisation. $\square$

Remark 62.2.3 (why no scalar square proves the match). The diagonal pairing is only a distributional Maass–Selberg object before truncation is regularised. Since $\xi(2s)/\xi(2s-1)$ has unit modulus on the critical line, taking absolute values removes the phase but does not identify the underlying Plancherel normalisation. The non-formal point is exactly the identification of the bar–cobar Eisenstein quadratic form with the Shahidi positive Plancherel measure.

COROLLARY 62.2.4 (*conditional normalisation match via LS*). (*same hypothesis as Chapter 62.2.2*). Under the LS redefinition Chapter 62.2.1, the Koszul and Petersson normalisations on the Eisenstein continuous spectrum of $L^2([GL_2])$ coincide after the positive Plancherel-measure identification.

Proof. Chapter 62.2.2 is the stated measure-identification hypothesis written in the LS normalisation. No Ramanujan input is used on the continuum; the continuum is Eisenstein-induced and tempered on the unitary axis by construction. $\square$

62.3 COMPATIBILITY CHECKS

Three compatibility properties of the LS-corrected pairing are verified: the arithmetic Heisenberg Koszul self-duality $\kappa^{\mathrm{Ar}} + (\kappa^!)^{\mathrm{Ar}} = 0$ on the Eisenstein summand (check (i)); the Tate restricted-tensor factorisation of N^{LS} across places (check (ii)); and the compatibility with the discrete-core result of Chapter 54 (check (iii)).

Compatibility Check (i): Heisenberg Self-Duality

PROPOSITION 62.3.1 (*LS-corrected Koszul preserves Heisenberg self-duality on the Eisenstein summand*). (*conditional on the LS scalar being admitted as part of the arithmetic Koszul pairing on the Eisenstein summand*). The LS-corrected Koszul pairing $\langle \cdot, \cdot \rangle^{\mathrm{Koszul}, \mathrm{LS}}$ of Chapter 62.2.1 satisfies the arithmetic Heisenberg Koszul self-duality

$$\kappa_{\mathrm{G}}^{\mathrm{Ar}} + (\kappa^!)_{\mathrm{G}}^{\mathrm{Ar}} = 0$$

on the Eisenstein summand, where $\mathrm{G} = GL_2$ and $\kappa^{\mathrm{Ar}}, (\kappa^!)^{\mathrm{Ar}}$ are the arithmetic Heisenberg and dual-Heisenberg Koszul obstructions of Chapter 21.5.1.

Proof. The self-duality identity $\kappa^{\text{Ar}} + (\kappa^!)^{\text{Ar}} = 0$ is equivalent to the completed-zeta functional equation $\xi(s) = \xi(1-s)$ (Chapter 54 and Chapter 21.5). The LS correction $N^{\text{LS}}(s)$ is a ratio of completed zetas: $N^{\text{LS}}(s) = \xi(2s-1)/\xi(2s)$. Its own functional equation (LS2):

$$N^{\text{LS}}(s) \cdot N^{\text{LS}}(1-s) = \frac{\xi(2s-1)}{\xi(2s)} \cdot \frac{\xi(1-2s)}{\xi(2-2s)} = \frac{\xi(2s-1)\xi(1-2s)}{\xi(2s)\xi(2-2s)} = \frac{\xi(2s-1)\xi(2s)}{\xi(2s)\xi(2s-1)} = 1$$

using $\xi(1-2s) = \xi(2s)$ and $\xi(2-2s) = \xi(2s-1)$ (Titchmarsh 1986 §2.1). The self-duality $\kappa^{\text{Ar}} + (\kappa^!)^{\text{Ar}} = 0$ is an obstruction-class identity, equivalent to the completed-zeta functional equation $\xi(s) = \xi(1-s)$ (Chapter 21.5; the sign $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$ is precisely the G-row sign of the arithmetic Heisenberg duality, independent of rescaling of the pairing). The LS correction acts on the pairing level, not on the obstruction-class level: the Koszul obstructions $\kappa^{\text{Ar}}, (\kappa^!)^{\text{Ar}}$ are defined as obstruction classes in the arithmetic bar–cobar complex (Chapter 21.5.1), independent of the choice of inner product on $\mathcal{V}_{\text{cont}}^{\text{prim}}$, and hence invariant under multiplication of the Koszul pairing by $N^{\text{LS}}(s)^{-1}$. The functional equation $\xi(s) = \xi(1-s)$ (which holds unconditionally) therefore gives $\kappa^{\text{Ar}} + (\kappa^!)^{\text{Ar}} = 0$ on the Eisenstein summand after the LS correction, as required. $\square$

Remark 62.3.2 (no modification to the discrete sector). The LS correction factor is by construction supported on the Eisenstein continuous summand: $N^{\text{LS}}(s)^{-1} = 1$ at discrete-spectrum points (pole at the cuspidal locus gives N^{LS} a compensating zero, and the discrete spectrum is outside the domain of the continuous normalisation). The discrete-spectrum Koszul pairing of Chapter 54 is therefore unchanged.

Compatibility Check (ii): Tate Factorisation

PROPOSITION 62.3.3 (*LS factorises over places as a Tate restricted tensor product*). (as a meromorphic Langlands–Shahidi factorisation, not as an absolutely convergent Euler product on the critical line). The global Langlands–Shahidi factor $N^{\text{LS}}(s)$ of Chapter 62.1.2 admits a place-by-place decomposition

$$N^{\text{LS}}(s) = \prod_{v \in |\text{Spec } \mathbb{Z}|} N_v^{\text{LS}}(s),$$

with local factors

$$N_v^{\text{LS}}(s) = \begin{cases} \zeta_p(2s-1)/\zeta_p(2s) & (v = p \text{ finite, unramified}) \\ \zeta_p(2s-1)/\zeta_p(2s) \cdot \varepsilon_p(s, \pi_p, \psi_p) & (v = p \text{ finite, ramified}) \\ \Gamma_{\mathbb{R}}(2s-1)/\Gamma_{\mathbb{R}}(2s) & (v = \infty), \end{cases}$$

where $\zeta_p(s) = (1-p^{-s})^{-1}$ is the Euler factor, $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ the archimedean Tate factor, and ε_p the local Jacquet–Piatetski-Shapiro–Shalika epsilon factor at the principal-series induction. The product is interpreted by meromorphic continuation of the completed L -factor.

Proof. The completed zeta factorises $\Lambda_{\zeta}(s) = \Gamma_{\mathbb{R}}(s) \prod_p \zeta_p(s)$ (Tate 1950 §4.2). Applying the factorisation to the ratio $\Lambda_{\zeta}(2s-1)/\Lambda_{\zeta}(2s)$:

$$\frac{\Lambda_{\zeta}(2s-1)}{\Lambda_{\zeta}(2s)} = \frac{\Gamma_{\mathbb{R}}(2s-1)}{\Gamma_{\mathbb{R}}(2s)} \cdot \prod_p \frac{\zeta_p(2s-1)}{\zeta_p(2s)}.$$

Unramified local factors are $\zeta_p(2s-1)/\zeta_p(2s)$; ramified local factors pick up an additional local epsilon factor ε_p from the Jacquet–Piatetski-Shapiro–Shalika 1983 local-functional-equation correction, which

has absolute value 1 and does not affect the Plancherel normalisation match (Chapter 62.2.2). The restricted tensor product $\prod_v N_v^{\text{LS}}(s)$ is not an absolutely convergent Euler product on $\text{Re}(s) = 1/2$; it is the boundary value of the meromorphic completed ratio, giving (62.3.3). $\square$

Remark 62.3.4 (Casselman–Shalika at unramified finite places). The local factor at an unramified finite prime p , $N_p^{\text{LS}}(s) = \zeta_p(2s-1)/\zeta_p(2s) = (1-p^{-2s})/(1-p^{1-2s})$, is precisely the Casselman–Shalika 1980 formula for the unramified spherical Whittaker function at principal-series parameter s (Casselman–Shalika *Compositio Math.* 41 Thm 5.4 with w the non-trivial Weyl element). This identifies N_p^{LS} as the local intertwining operator normalisation, canonically and without convention choice.

Compatibility Check (iii): Discrete-Core Fixation

PROPOSITION 62.3.5 (*LS correction preserves the discrete-core identity*). (with the hypotheses of the discrete-core identity). The discrete-spectrum Koszul–Petersson identity of Chapter 54.3.3 is preserved under the substitution $\langle \cdot, \cdot \rangle^{\text{Koszul}} \rightsquigarrow \langle \cdot, \cdot \rangle^{\text{Koszul,LS}}$ extended to the Eisenstein summand:

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul,LS}} \quad \forall v, w \in \mathcal{V}_{\text{disc}}^{\text{prim}}.$$

Proof. $\mathcal{V}_{\text{disc}}^{\text{prim}}$ is orthogonal to the Eisenstein continuous summand in the Langlands decomposition (Langlands 1966 §6; Mœglin–Waldspurger 1995 Ch. II). The substitution $\langle \cdot, \cdot \rangle^{\text{Koszul,LS}} = N^{\text{LS}}(s)^{-1} \cdot \langle \cdot, \cdot \rangle^{\text{Koszul}}$ is defined only on the Eisenstein summand; on the discrete core, the pairing is unmodified: $\langle v, w \rangle^{\text{Koszul,LS}} = \langle v, w \rangle^{\text{Koszul}}$ for $v, w \in \mathcal{V}_{\text{disc}}^{\text{prim}}$. Combining with Chapter 54.3.3 gives (62.3.5). $\square$

62.4 FULL $\mathcal{V}^{\text{prim}}$ KOSZUL–PETERSSON IDENTITY (INCLUDING EISENSTEIN)

The three compatibilities give the full-sector identity.

THEOREM 62.4.1 (*full Koszul–Petersson compatibility via LS normalisation*). (conditional on Ramanujan for Maass cusp forms on the Maass cuspidal discrete sector; conditional on the LS Plancherel-measure identification on the Eisenstein continuous sector). For all $v, w \in \mathcal{V}^{\text{prim}}$,

$$\langle v, w \rangle^{\text{Pet}} = \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul,LS}},$$

with $\langle \cdot, \cdot \rangle^{\text{Koszul,LS}}$ the Langlands–Shahidi-corrected Koszul pairing of Chapter 62.2.1 on the Eisenstein summand and the unmodified Koszul pairing on the discrete core. The identity extends Chapter 54.3.3 from the discrete core to the full $\mathcal{V}^{\text{prim}}$ including the Eisenstein continuum under the stated LS normalisation hypothesis.

Proof. Decompose $\mathcal{V}^{\text{prim}} = \mathcal{V}_{\text{disc}}^{\text{prim}} \oplus \mathcal{V}_{\text{cont}}^{\text{prim}}$ by Langlands orthogonality (Langlands 1966 §6). On the discrete core, Chapter 62.3.5 gives (62.4.1) with the conditional on Ramanujan for Maass inherited from Chapter 54.3.3. On the Eisenstein continuum, Chapter 62.2.4 supplies the conditional spectral density normalisation match, which by the Plancherel formula for the continuous spectrum (Mœglin–Waldspurger §IV.2) yields (62.4.1) on $\mathcal{V}_{\text{cont}}^{\text{prim}}$. Compatibility of the two on their intersection (empty, by orthogonality) is trivial. $\square$

COROLLARY 62.4.2 (*Eisenstein density mismatch reduced by LS normalisation*). (conditional on the LS Plancherel-measure identification). The Eisenstein density mismatch of Chapter 54.3.1 and Chapter 45.5.5 is reduced under the LS redefinition Chapter 62.2.1: the density mismatch on $\text{Re}(s) = 1/2$ between Koszul and Petersson vanishes pointwise once the LS-corrected quadratic form is identified

with the positive Plancherel measure. No Ramanujan input is used on the continuum, but the measure identification is an extra hypothesis.

Proof. Chapter 62.2.4 gives the pointwise density match. The domain of the modification is only the Eisenstein summand (Chapter 62.3.2); the Tate factorisation (Chapter 62.3.3) and the Heisenberg self-duality (Chapter 62.3.1) confirm the redefinition is compatible with the rest of the arithmetic Koszul structure. $\square$

62.5 SOURCE DISJOINTNESS AND MATHEMATICAL-FORM COMPARISON

Remark 62.5.1 (Source disjointness, conventions, and hypotheses for Chapter 62.4.1). **Source disjointness.** Derivation of the Koszul side: Beilinson–Drinfeld 2004, Positselski 2009, Bost–Connes 1995. Verification of the Petersson side: Langlands 1966, Moeglin–Waldspurger 1995, Shahidi 2010, Casselman–Shalika 1980, Jacquet–Piatetski-Shapiro–Shalika 1983, Tate 1950, Weil 1967, Titchmarsh 1986. No source appears on both.

Sign and convention. The factor $N^{\text{LS}}(s)^{-1} = \xi(2s)/\xi(2s-1)$ is a unitary scattering normalisation on the critical line. The positive measure is supplied only after the Shahidi Plancherel transport named in Chapter 62.2.2. Convention consistent with Shahidi 2010 Thm 3.5.1 and Moeglin–Waldspurger §IV.3.10.

Ambient category. Both sides of (62.4.1) are sesquilinear forms on $\mathcal{V}^{\text{prim}} \otimes \overline{\mathcal{V}^{\text{prim}}}$ with values in $\mathbb{C}$.

Hypotheses. The identity is conditional on the LS Plancherel-measure identification on the Eisenstein continuum and on Ramanujan for Maass on the Maass cuspidal discrete sector (inherited from Chapter 54.3.3). Both conditions are explicit in the theorem statement. *The conditions are explicit.*

Orthogonality. The Eisenstein summand is orthogonal to the discrete core under both Petersson (Langlands 1966 §6) and Koszul (Chapter 62.3.5). The LS correction is supported on the Eisenstein summand only.

Cited results. The Shahidi 2010 Thm 3.5.1 global normalisation, the Casselman–Shalika 1980 local formula, and the Moeglin–Waldspurger §IV.3.10 unitarity on the critical line are classical theorems. They do not by themselves identify the bar–cobar Eisenstein pairing with the positive Plancherel measure. *With the stated residual.*

Numerical comparison. No numerical comparison is involved.

Six independent checks yield explicit conditions: Ramanujan for Maass on the cuspidal discrete sector and the LS Plancherel-measure identification on the Eisenstein sector.

Remark 62.5.2 (discrete-core and Eisenstein normalisations). Chapter 54 (the restricted-class comparison) reduces the Koszul–Petersson identity on the discrete core and excludes the Eisenstein continuum by domain restriction. The Li-criterion path (Chapter 54.3.6) uses this restriction only after the $H\text{-P}^{\text{Ar}}$ determinant-trace realisation. The bounded-interval reduction gives a conditional extension to the continuum: one must insert the LS scalar N^{LS} and identify the resulting quadratic form with the positive Plancherel measure.

62.6 SOURCES

The primary sources enter in the following order.

Shahidi, F. *Eisenstein Series and Automorphic L-Functions*, AMS Colloquium Publications 58, 2010. The Langlands–Shahidi method for normalising intertwining operators; Theorem 3.5.1 for the global normalisation we denote $N^{\text{LS}}(s)$.

Moeglin, C. and Waldspurger, J.-L. *Spectral Decomposition and Eisenstein Series: A Paraphrase of the Scriptures*, Cambridge Tracts 113, 1995. §II.1.7 for the intertwining operator $M(s)$; §IV.2 for the Plancherel formula; §IV.3.10 for unitarity of the normalised intertwiner on the critical line.

Langlands, R. P. *On the Functional Equations Satisfied by Eisenstein Series*, Lecture Notes in Math. 544, Springer 1976 (manuscript 1966). Foundational spectral decomposition of $L^2([GL_n])$ into discrete and continuous summands; Maass–Selberg relations; functional equation of Eisenstein series.

Casselman, W. and Shalika, J. *The unramified principal series of p -adic groups. II. The Whittaker function*, Compositio Math. 41 (1980), 207–231. Unramified spherical Whittaker formula used for the local factor $N_p^{\text{LS}}(s) = \zeta_p(2s-1)/\zeta_p(2s)$ at unramified finite p .

Jacquet, H., Piatetski-Shapiro, I., and Shalika, J. *Rankin–Selberg convolutions*, Amer. J. Math. 105 (1983), 367–464. Local ε -factors at ramified principal series.

Tate, J. *Fourier analysis in number fields and Hecke’s zeta functions*, Ph.D. thesis, Princeton 1950; reprinted Cassels–Fröhlich 1967 Ch. XV. Adelic restricted tensor product; local-global factorisation of completed zeta.

Weil, A. *Basic Number Theory*, Springer 1967. Tamagawa normalisation.

Titchmarsh, E. C. *The Theory of the Riemann Zeta-Function* (2nd ed., revised D. R. Heath-Brown), Oxford University Press 1986. Functional equation of ξ ; Euler product for ζ .

Bost, J.-B. and Connes, A. *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. 1 (1995), 411–457. The Bost–Connes C^* -algebra; KMS_1 state; involution σ^{BC} .

Beilinson, A. and Drinfeld, V. *Chiral Algebras*, AMS Colloquium 51, 2004. Chiral bar–cobar construction and parabolic-induction compatibility (§3.5).

Positselski, L. *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence*, arXiv:0905.2621, 2009. Koszul duality input for the arithmetic bar–cobar.

Disjoint sources. The proof sources for the arithmetic Koszul pairing (Beilinson–Drinfeld 2004, Positselski 2009, Bost–Connes 1995) are disjoint from the analytic sources for the Langlands–Shahidi normalisation and Plancherel measure (Shahidi 2010, Moeglin–Waldspurger 1995, Langlands 1966, Casselman–Shalika 1980, Jacquet–Piatetski-Shapiro–Shalika 1983, Tate 1950, Weil 1967, Titchmarsh 1986). No source appears on both sides of the identification.

L_{ACD} ON THE EISENSTEIN / CONTINUOUS SPECTRUM

The cuspidal LACD comparison reduction (Chapter 48.4.4) lifts the Flath restricted tensor product to the categorical level on the cuspidal sph-finite locus. The Eisenstein / continuous-spectrum residual was restricted as Chapter 48.4.6. This chapter reduces that residual on its natural domain: the non-exceptional Eisenstein continuum of $GL_2/\mathbb{Q}$, i.e. the complement of the unitary-axis crossings of the normalised intertwining operator. The reduction proceeds in four modules. Module (E0) constructs the Langlands–Shahidi normalised intertwining operator $N(s, \chi): I(s, \chi) \rightarrow I(-s, \chi^w)$ as the canonical coefficient of the intertwining between principal-series inductions; the normalisation uses the local L -ratio $L_v(s, \chi_1 \chi_2^{-1}) / L_v(1 + s, \chi_1 \chi_2^{-1})$ in Shahidi’s convention (see Chapter 63.2.3). Module (E1) identifies the continuous summand of $L^2([GL_2])$ with the Paley–Wiener space $PW(i\mathbb{R}, V_\chi)$ of holomorphic functions of exponential type on the unitary axis valued in a finite-dimensional representation of $K_\infty \times \prod_p K_p$. Module (E2) lifts this identification to the categorical level: the continuous factor C_v^{cont} at each place is the stable presentable ∞ -category of admissible ind-sheaves on the unitary-axis fibre, with distinguished unit the spherical Eisenstein-induction class. Module (E3) shows the assembly of C_v^{cusp} , C_v^{res} , and C_v^{cont} into a global restricted tensor product commutes with the adelic categorical descent of the cuspidal LACD comparison under the standing LACD comparison inputs. The Eisenstein residual left by this chapter is (Chapter 63.5.3): matching of Ext^i -classes at the two unitary-axis crossings $s = 0, 1$ of $N(s, \chi)$, where the normalised operator has a simple pole and the induced representation is reducible. The global LACD table still contains the local categorical comparison, the global restricted-tensor identification, and BC-diamond-glue.

63.1 DOMAIN, DISJOINTNESS, AND PARALLELISM WITH THE LANGLANDS–SHAHIDI NORMALISATION

Inherited Objects

Prior statements deployed here:

- cuspidal sph-finite adelic categorical descent (Chapter 48.4.4), the cuspidal LACD comparison core theorem;
- $L_{\text{ACD}.0-.3}$ reductions (Chapter 30): archimedean admissibility, colimit commutation, cross-prime Hecke, sph-finite preservation;
- F2.c tightened restricted tensor product (Chapter 28.1.3) supplying the pointed E_2 -monoidal base;
- Langlands–Shahidi correction factor definition (Chapter 54.4.1) on principal-series inductions;
- discrete-core Koszul–Petersson identity (Chapter 54.3.3) bounding the Koszul pairing away from the Eisenstein continuum.

Relation to the Langlands–Shahidi normalisation

the Langlands–Shahidi normalisation reduces the Koszul–Petersson normalisation correction on the Eisenstein continuum to the LS Plancherel-measure identification, producing the conditional corrected

pairing $\langle \cdot, \cdot \rangle^{\text{Koszul,LS}}$ of Chapter 54.4.2 via the Langlands–Shahidi normalisation. This chapter constructs $N(s, \chi)$ as the intertwining coefficient in the presentable ∞ -category C_v^{cont} , realising the same normalisation at the categorical level. The shared object is the operator $N_v(s, \chi)$ of Chapter 54.4.1; the Langlands–Shahidi normalisation supplies the conditional Plancherel-normalisation match on the continuum, this chapter supplies the categorical lift. The measure comparison is not reproduced here; Chapter 63.5.5 records the conditional cross-compatibility.

Primary Sources

Langlands 1976 “On the functional equations satisfied by Eisenstein series”, SLN 544, Part I §5–7. Foundational Eisenstein-series theory, intertwining operators, analytic continuation.

Moëglin–Waldspurger 1995 *Spectral Decomposition and Eisenstein Series*, Cambridge Tracts 113, Ch. II and IV. Full L^2 spectral decomposition, residual and continuous spectrum description; the normalising factor $M(s) = \xi(2s - 1)/\xi(2s)$ appears at §II.1.7.

Shahidi 1990 “A proof of Langlands’ conjecture on Plancherel densities and complementary series”, Ann. Math. 132. Local Plancherel densities and their compatibility with normalised intertwining operators.

Shahidi 2010 *Eisenstein Series and Automorphic L-Functions*, AMS Colloquium 58, Ch. 2–4. The Langlands–Shahidi method: normalised intertwining operators $N(s, \pi_v, \tilde{w}_0)$, Plancherel–Shahidi measure, and adelic assembly (Thm 3.5.1).

Arthur 2013 *The Endoscopic Classification of Representations*, AMS Colloquium 61, Ch. 1–4. Discrete-spectrum classification, Paley–Wiener synthesis on the unitary axis (§3.5), trace-formula framework. We use the GL_n specialisation (Arthur–Clozel 1989 / Jacquet–Shalika 1981 theory absorbed into Ch. 1 of the 2013 volume).

Jacquet–Langlands 1970 SLN 114 §5, §11. Strong multiplicity one for cuspidal spectrum; archimedean Plancherel for $GL_2(\mathbb{R})$.

Harish-Chandra 1976 “Harmonic analysis on real reductive groups II”, Invent. Math. 36. Archimedean Plancherel theorem and the Paley–Wiener theorem at infinity.

Wallach 1988/1992 *Real Reductive Groups I–II*. Plancherel for $(\mathfrak{g}, K_\infty)$ -modules, Ch. 11–13.

Secondary: Bernstein–Zelevinsky 1977 *Induced representations of reductive p -adic groups I*, Ann. Sci. ENS 10, for the p -adic Plancherel measure; Lurie HA §5.1–5.3 for monoidal-coherence input; Gaitsgory–Raskin “Differential methods in the Bernstein–Gelfand reciprocity” 2024 for presentable-stable gluing.

Disjointness Comparison

the cuspidal LACD comparison derived the cuspidal LACD through Flath 1979 + Bernstein 1988 + Moëglin–Waldspurger 1995 §III; that derivation is spectral-decomposition *on the cuspidal summand*, which is orthogonal to the continuous spectrum. The continuous-spectrum lift developed here uses the intertwining-operator machinery (Langlands 1976; Shahidi 1990, 2010), Paley–Wiener synthesis (Arthur 2013 §3.5; Harish-Chandra 1976; Wallach 1992 §12), and $L_{\text{ACD.0–.3}}$ categorical machinery. The proof sources of the cuspidal LACD comparison (Flath, Bernstein, Moëglin–Waldspurger §III on cuspidal orthogonality) and the comparison sources of this chapter (Langlands 1976; Shahidi 1990, 2010; Arthur 2013) are disjoint by list: no primary author appears in both spectral-decomposition roles. Moëglin–Waldspurger 1995 appears on both sides, but in disjoint chapters (§III cuspidal on the cuspidal LACD comparison side; §II/§IV Eisenstein on this side). The primary-source independence comparison passes.

63.2 MODULE (EO): LANGLANDS–SHAHIDI INTERTWINING NORMALISATION

Local Intertwining Data for GL_2

Setup 63.2.1 (place-by-place intertwining). Fix a place v of $\mathbb{Q}$. Let $B = TN \subset GL_2$ be the Borel; let $w_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ be the non-trivial element of the Weyl group. For a unitary character $\chi = (\chi_1, \chi_2): T(\mathbb{Q}_v) \rightarrow \mathbb{C}^\times$ and a complex parameter $s \in \mathbb{C}$, form the induced representation

$$I_v(s, \chi) = \text{Ind}_{B(\mathbb{Q}_v)}^{GL_2(\mathbb{Q}_v)} (\chi_1 | \cdot |^{s/2} \otimes \chi_2 | \cdot |^{-s/2})$$

on the normalised smooth (finite places) or $(\mathfrak{g}, K_\infty)$ -admissible (archimedean place) setting.

Definition 63.2.2 (unnormalised intertwining operator). . The unnormalised intertwining operator is

$$M_v(s, \chi): I_v(s, \chi) \longrightarrow I_v(-s, \chi^{w_0}), \quad (M_v f)(g) = \int_{N(\mathbb{Q}_v)} f(w_0 n g) dn,$$

defined initially for $\text{Re}(s) > 0$ and extended by meromorphic continuation (Langlands 1976 §5 for archimedean; Shahidi 1990 §2 for non-archimedean).

Definition 63.2.3 (Langlands–Shahidi normalised intertwining operator). . The Langlands–Shahidi normalised intertwining operator at the place v is

$$N_v(s, \chi) = \frac{L_v(s, \chi_1 \chi_2^{-1})}{L_v(1+s, \chi_1 \chi_2^{-1})} \cdot \varepsilon_v(s, \chi_1 \chi_2^{-1}, \psi_v)^{-1} \cdot M_v(s, \chi),$$

where L_v is the local L -factor, ε_v the local ε -factor, and ψ_v an additive character of $\mathbb{Q}_v$ (Shahidi 2010 Thm 3.5.1). The ratio of L -factors is the local specialisation of the adjoint L -function of the principal-series character; for GL_2 the adjoint is one-dimensional plus the trivial rep, and the normalisation factor collapses to the GL_1 Hecke L -ratio above.

Remark 63.2.4 (relation to the Langlands–Shahidi normalisation correction). At the global level, the product $N(s, \chi) = \prod_v N_v(s, \chi)$ relates to the Mœglin–Waldspurger factor $M(s) = \xi(2s-1)/\xi(2s)$ of Chapter 54.3.1 by the Hecke translation $s \mapsto 2s$: if $\chi_1 \chi_2^{-1} = | \cdot |^t$ is the principal character for the spherical intertwining, then $N(s, \chi) = M(s)$ up to the ε -factor global product, which equals 1 by the Tate functional equation on GL_1 . The the Langlands–Shahidi normalisation normalisation factor $N^{\text{LS}}(s)$ of Chapter 54.4.1 is $|N(s, \chi)|$ on the critical line $\text{Re}(s) = 1/2$. The restriction of the adelic Plancherel measure to $i\mathbb{R}_{\geq 0}$ is a separate measure convention: the Weyl symmetry $s \mapsto -s$ folds the axis by two, so the Plancherel density $\frac{1}{2} |\xi(1+2it)|^{-2} dt$ on $i\mathbb{R}$ yields $|\xi(1+2it)|^{-2} dt$ on $i\mathbb{R}_{\geq 0}$. The squared Plancherel density on the positive half-axis; see Shahidi 2010 Thm 3.5.1). The Koszul normalisation absorption of Chapter 54.4.2 is the specialisation of Shahidi’s Plancherel normalisation at GL_2 . the Langlands–Shahidi normalisation states the conditional measure comparison from the Koszul-pairing side; this chapter establishes the categorical lift from the intertwining-operator side.

Local Functional Equation and Unitarity on the Axis

PROPOSITION 63.2.5 (local functional equation for N_v). . The normalised operator $N_v(s, \chi)$ satisfies:

- (i) (functional equation) $N_v(-s, \chi^{w_0}) \circ N_v(s, \chi) = \text{id}$, a cocycle relation;
- (ii) (unitarity on the axis) for $s \in i\mathbb{R}$ and χ unitary, $N_v(s, \chi)$ is a unitary isomorphism $I_v(s, \chi) \rightarrow I_v(-s, \chi^{w_0})$ with respect to the principal-series L^2 -inner product;

- (iii) (polar divisor) $N_v(s, \chi)$ is holomorphic on $\text{Re}(s) \geq 0$ except at the simple poles $s = 0, 1$ where the induced representation is reducible (Steinberg / trivial subquotient). The residue at $s = 1$ produces the residual summand.

Source. Shahidi 1990 Thm 3.1 (cocycle relation and unitarity); Shahidi 2010 §3.5 (polar analysis); Mœglin–Waldspurger 1995 §IV.1–3 (local factors and adelic assembly). The polar analysis at $s = 0, 1$ is Langlands 1976 §6. $\square$

Definition 63.2.6 (exceptional and non-exceptional locus). . The *exceptional locus* of the normalised intertwining data is

$$\Sigma^{\text{exc}} = \{s = 0\} \cup \{s = 1\} \subset \mathbb{C},$$

the two simple-pole points of $N(s, \chi)$ where reducibility occurs. The *non-exceptional locus* is $\Sigma^{\text{nonexc}} = i\mathbb{R} \setminus \{0\}$, i.e. the unitary axis punctured at the origin. This chapter treats the Eisenstein summand on the non-exceptional locus; the two unitary-axis crossings at $s = 0$ (trivial-spherical reducibility) and $s = 1$ (residual-Steinberg reducibility) are isolated as named residuals.

Remark 63.2.7 (form of the exceptional locus). The point $s = 0$ is the spherical crossing: $I_v(0, \chi)$ is reducible with a Langlands subquotient and an orthogonal complement. The point $s = 1$ is the residual crossing: $\text{Res}_{s=1} E(s, g)$ is a one-dimensional summand of $L_{\text{res}}^2([GL_2])$ (Mœglin–Waldspurger II.2). The categorical descent at these two crossings requires an additional Ext^1 -class match (Chapter 63.5.3); the non-exceptional locus is treated below under the standing L_{ACD} inputs.

63.3 MODULE (E1): PALEY–WIENER SYNTHESIS ON THE UNITARY AXIS

Shahidi’s Paley–Wiener Theorem

THEOREM 63.3.1 (Paley–Wiener synthesis for automorphic continuous spectrum). . Let $L_{\text{cont}}^2([GL_2])$ be the continuous spectrum of the adelic automorphic L^2 -space. Then there is a canonical unitary isomorphism

$$L_{\text{cont}}^2([GL_2]) \xrightarrow{\sim} \bigoplus_{[\chi]} L^2(i\mathbb{R}_{\geq 0}, |\xi(2s)|^{-2} ds) \otimes V_{\chi}, \quad (63.1)$$

where:

- $[\chi]$ ranges over Weyl-orbits of unitary Hecke characters $\chi = (\chi_1, \chi_2)$ of $T(\mathbb{A})/T(\mathbb{Q})$ (equivalently, characters of the idele class group $\mathbb{A}^\times/\mathbb{Q}^\times$ factored into pairs);
- V_{χ} is the finite-dimensional $(K_\infty \times \prod_p K_p)$ -representation spanned by the spherical / Iwahori-finite vectors at each place;
- the Plancherel density is $|\xi(2s)|^{-2} ds$ along $i\mathbb{R}_{\geq 0}$ (Mœglin–Waldspurger §IV.3.10);
- the Weyl symmetry $s \mapsto -s$ is implemented by $N(s, \chi): V_{\chi} \rightarrow V_{\chi^{w_0}}$.

Identifying the reduction modulo the Weyl symmetry with the quotient $i\mathbb{R}/(s \sim -s)$, the ξ -weighted L^2 -completion of the space of smooth compactly supported sections on the unitary axis is the Plancherel dense subspace; Arthur (2013 §3.5) calls this the *Paley–Wiener space* $\text{PW}(i\mathbb{R}, V_{\chi})$, by analogy with the classical identification of $L^2(\mathbb{R})$ Fourier transforms restricted to a bounded-type locus. The L^2 -completion of this subspace is the target of (63.1).

Source. Shahidi 2010 Thm 4.2.2 plus §4.3 (adelic assembly); combined with Arthur 2013 §3.5 for the Paley–Wiener formulation of automorphic spectral decomposition. Harish-Chandra 1976 Thm 1 for the archimedean Plancherel piece. Mœglin–Waldspurger §IV.3 assembles the pieces into the adelic Plancherel formula. $\square$

Remark 63.3.2 (Plancherel-density normalisation). The density $|\xi(2s)|^{-2}ds$ follows Mœglin–Waldspurger IV.3.10, where the Eisenstein parameter s runs over $i\mathbb{R}_{\geq 0}$ and ξ is the completed Riemann zeta. The Shahidi normalisation (2010 §4.2) writes the spectral measure in terms of the variable $s' = s + 1/2$ on the critical line $\operatorname{Re}(s') = 1/2$, yielding $|\xi(1 + 2(s' - 1/2))|^{-2}ds' = |\xi(2s')|^{-2}ds'$; on the unitary axis $s = it$, $s' = it + 1/2$, and the density in terms of t is $|\xi(1 + 2it)|^{-2}dt$. The identity $|\xi(2it)| = |\xi(1 + 2it)|$ holds on $i\mathbb{R}$ because the functional equation $\xi(s) = \xi(1 - s)$ gives $\xi(2it) = \xi(1 - 2it)$, and $|\xi(1 - 2it)| = |\overline{\xi(1 + 2it)}| = |\xi(1 + 2it)|$ by the reality condition $\xi(\bar{s}) = \overline{\xi(s)}$. Hence $|\xi(2s)|^{-2} = |\xi(1 + 2s)|^{-2}$ on $i\mathbb{R}$, and the Mœglin–Waldspurger and Shahidi densities agree. The Langlands–Shahidi normalisation density $|N^{\text{LS}}(s)|^2 = |\xi(1 + 2it)|^{-2}$ of Chapter 54.4.2 is consistent with both.

Categorical Lift of the Paley–Wiener Space

Definition 63.3.3 (presentable stable ∞ -category attached to the continuum). . At each place v , the *continuous-spectrum local category* is

$$C_v^{\text{cont}} = D(\operatorname{Shv}_{\text{hol}}^{\text{adm}}(i\mathbb{R}_{\geq 0, v}, V_v)),$$

the derived category of admissible holomorphic sheaves on the unitary axis at v , valued in the finite-dimensional K_v -representation V_v of spherical/Iwahori-finite vectors. Admissibility means finite-type over the local Hecke algebra $\mathcal{H}_v$; holomorphy means $\mathcal{O}$ -coherent along the axis. The distinguished unit $\mathcal{U}_v^{\text{cont}} \in C_v^{\text{cont}}$ is the structure-sheaf class of the trivial representation on the spherical vector at $s = 1/2$ (the centre of the strip); at non-archimedean unramified v this coincides with the constant sheaf $\mathcal{O}_{i\mathbb{R}_{\geq 0}}$.

PROPOSITION 63.3.4 (C_v^{cont} is stable presentable). . The category C_v^{cont} of Chapter 63.3.3 is a stable presentable ∞ -category with distinguished unit $\mathcal{U}_v^{\text{cont}}$ compact. Explicitly:

- (i) stability: C_v^{cont} is the derived category of an abelian Grothendieck category (admissible holomorphic sheaves), hence stable (Lurie HA 1.3.5);
- (ii) presentability: admissible holomorphic sheaves on a one-dimensional complex manifold valued in a finite-dimensional representation form an accessible category (Wallach 1988 Thm 0.A.2 for the admissibility axiom, combined with sheaf-theoretic presentability of $\operatorname{Shv}(i\mathbb{R}_{\geq 0})$);
- (iii) compact generation: the constant sheaf at a finite closed-subset $S \subset i\mathbb{R}_{\geq 0}$ together with its translates by the Paley–Wiener exponential series generates under colimits and shifts;
- (iv) distinguished unit compactness: $\mathcal{U}_v^{\text{cont}}$ is a structure sheaf of a smooth point, hence compact.

Proof. Parts (1)–(2) are standard; the ∞ -categorical version is Lurie HA 1.3.5.21 plus Wallach 1988 Thm 0.A.2. For part (3), Paley–Wiener exponentials e^{sT} for $T \in \mathbb{R}_{>0}$ generate $\operatorname{PW}(i\mathbb{R})$ under L^2 -closure (classical Paley–Wiener theorem); the derived category of admissible sheaves on the axis inherits compact generation from this. Part (4) is the standard fact that the structure sheaf of a smooth point in a smooth complex manifold is compact in the derived category of quasi-coherent sheaves. $\square$

PROPOSITION 63.3.5 (*Weyl action is coherent on $\bigotimes' C_v^{\text{cont}}$*). . The family of normalised intertwining operators $\{N_v(s, \chi)\}_v$ assembles into an E_∞ -algebra map

$$N: \bigotimes'_v C_v^{\text{cont}} \longrightarrow \bigotimes'_v C_v^{\text{cont}}$$

implementing the $\mathbb{Z}/2$ Weyl action. Coherence $N \circ N = \text{id}$ holds up to a contractible space of higher homotopies on the non-exceptional locus.

Proof. The local factors $N_v(s, \chi)$ are each an involution on the local category (Shahidi 1990 Thm 3.1). Combining across places uses Lurie HA 5.1.2.26 (centrality of the unit in an E_n -monoidal category): each N_v factors through C_v^{cont} and commutes with N_w at $w \neq v$ up to a contractible space. The global product $N = \prod_v N_v$ inherits the involution property up to coherent homotopy. The non-exceptional hypothesis is essential: at $s = 0, 1$ the local N_v acquires a simple pole, and the global product can fail to be an involution at these two points. $\square$

THEOREM 63.3.6 (*Paley–Wiener lift at the categorical level*). . There is a canonical equivalence of stable presentable ∞ -categories

$$D(L_{\text{cont}}^2([GL_2])^{\text{sph-fin}}) \xrightarrow{\sim} \bigotimes'_v C_v^{\text{cont}} / \text{Weyl}$$

where:

- the restricted tensor product uses $\mathcal{U}_v^{\text{cont}}$ as distinguished unit at almost every v ;
- the Weyl quotient is the $s \sim -s$ quotient implemented by $N(s, \chi)$ as the coherent Weyl action of Chapter 63.3.5;
- the left side is the stable sph-fin derived category of the continuous summand of $L^2([GL_2])$, compactly generated by Plancherel–Shahidi sections of the ξ -weighted L^2 -measure.

Proof. **Step 1.** Representation-level: the Plancherel synthesis of Chapter 63.3.1 gives the unitary isomorphism at the level of Hilbert spaces with $|\xi(2s)|^{-2} ds$ weight. **Step 2.** Lifting to derived categories: the Hilbert-space isomorphism extends to an equivalence of derived categories of admissible modules. On the continuum side, admissible corresponds to finite-type over the local Hecke algebra at each place (Chapter 30.3.4 for archimedean, Bernstein–Zelevinsky 1977 for finite places). The admissible sheaf structure on C_v^{cont} matches by Chapter 63.3.3. **Step 3.** Restricted tensor product: using F2.c tightened (Chapter 28.1.3), the local continuous-spectrum categories with distinguished units have a pointed E_2 -monoidal restricted tensor product in stable presentable ∞ -categories. The $L_{\text{ACD}.1}$ colimit commutation (Chapter 30.4.1) preserves sph-finiteness under restricted colimit. **Step 4.** Weyl quotient: the operator $N(s, \chi)$ of Chapter 63.2.3 satisfies the cocycle relation (Chapter 63.2.5.1) and unitarity (Chapter 63.2.5.2) on the non-exceptional locus; by Chapter 63.3.5 this upgrades to a coherent Weyl action on the restricted tensor product, with quotient matching the unitary-axis quotient of the Plancherel–Shahidi space. Descent under the coherent Weyl action computes as a homotopy fixed-point spectral sequence in $\text{Pr}^{L, \text{st}}$ (Lurie HA 4.8.5.16), degenerating at E_2 by unitarity. $\square$

63.4 MODULE (E2): LANGLANDS SPECTRAL DECOMPOSITION LIFTED CATEGORICALLY

The Three Summands

THEOREM 63.4.1 (*Langlands spectral decomposition of $L^2([GL_2])$*). . The adelic automorphic L^2 -space decomposes as a unitary direct sum

$$L^2([GL_2]) = L_{\text{cusp}}^2 \oplus L_{\text{res}}^2 \oplus L_{\text{cont}}^2,$$

with:

- L_{cusp}^2 the cuspidal subspace (Bernstein 1988; Mœglin–Waldspurger III.2.1);
- L_{res}^2 the residual subspace, one-dimensional at $GL_2/\mathbb{Q}$, spanned by the constant function (Mœglin–Waldspurger II.2);
- L_{cont}^2 the continuous spectrum, parametrised by Weyl-orbits of unitary Hecke characters via Eisenstein series (Langlands 1976 §7; Mœglin–Waldspurger IV.3).

The decomposition is canonical and respects the adelic Hecke action and the archimedean $(\mathfrak{g}, K_\infty)$ -structure.

Source. Langlands 1976 §7 (original proof, functional equations); Mœglin–Waldspurger 1995 Ch. II and IV (modern full account). $\square$

Categorical Lift of Each Summand

DEFINITION 63.4.2 (*residual local category and distinguished unit*). . The local residual category is $C_v^{\text{res}} = \text{Sp}$, the one-object presentable stable ∞ -category (a copy of spectra), with distinguished unit $\mathcal{U}_v^{\text{res}} = \mathbb{1}_{\text{Sp}}$ the sphere spectrum. The residual summand at the global level corresponds to the trivial character, which has a single archimedean and single finite-prime component at every place, all one-dimensional.

THEOREM 63.4.3 (*categorical lift of L_{res}^2*). . The restricted tensor product $\bigotimes'_v C_v^{\text{res}} \simeq \text{Sp}$ matches $D(L_{\text{res}}^2([GL_2])^{\text{sph-fin}})$ through the residue of the Eisenstein series at $s = 1$. Specifically, the identification

$$D(L_{\text{res}}^2([GL_2])^{\text{sph-fin}}) \simeq \text{Sp}$$

sends the constant function $\mathbb{1}_{[GL_2]}$ to the sphere spectrum.

Proof. The residual subspace is one-dimensional and trivial-character (Mœglin–Waldspurger II.2.4); hence the derived category of its sph-fin completion is a single-object stable category Sp . On the other side, the empty restricted tensor product over all places with distinguished unit the sphere spectrum is the sphere spectrum by the universal property of restricted tensor in $\text{Pr}^{\text{L,st}}$ (F2.c tightened). The two match canonically. $\square$

THEOREM 63.4.4 (*categorical lift of L_{cont}^2 on the non-exceptional locus*). . On the non-exceptional locus $\Sigma^{\text{nonexc}} = i\mathbb{R} \setminus \{0\}$, the restricted tensor product $(\bigotimes'_v C_v^{\text{cont}})/\text{Weyl}$ is canonically equivalent to $D(L_{\text{cont}}^2([GL_2])^{\text{sph-fin}})^{\text{nonexc}}$, the sph-fin derived category of the continuous summand restricted to the non-exceptional unitary axis.

Proof. By Chapter 63.3.6: the Plancherel synthesis of Chapter 63.3.1 gives the Hilbert-space isomorphism with the ξ -weighted L^2 -space; this lifts to derived categories on admissible / holomorphic / compactly

generated classes. Restriction to the non-exceptional locus removes the two polar points $s = 0, 1$ where N fails to be an involution; on the complement, the Weyl quotient is a $\mathbb{Z}/2$ -Galois quotient in $\text{Pr}^{\text{L}, \text{st}}$ and computes as a homotopy fixed-point category (Lurie HA 4.8.5.16). $\square$

Sphericity-Finite Compatibility on the Continuum

PROPOSITION 63.4.5 (*sph-fin preservation under continuous-spectrum restriction*). . The sph-finiteness condition (compact generation with finite-type local Hecke modules) is preserved under restriction to the continuous summand and the Weyl quotient: if every C_v^{cont} is sph-fin, then $(\bigotimes'_v C_v^{\text{cont}})/\text{Weyl}$ restricted to the non-exceptional locus is sph-fin.

Proof. Sph-fin is a compact-generation + finite-type property; compact generation is preserved under filtered colimits of compactly generated categories (Lurie HA 5.3.2.9) and under $\mathbb{Z}/2$ homotopy fixed-point quotients when the action is by compact-object-preserving functors. The normalised intertwining operator $N(s, \chi)$ sends a compact object at s (the structure sheaf of a closed point) to a compact object at $-s$, by the local holomorphic extension (Chapter 63.2.5.3). Restricted to the non-exceptional locus, N is unitary hence a compact-preserving equivalence. Sph-fin is preserved. $\square$

63.5 MODULE (E₃): ASSEMBLY COMPATIBLE WITH CUSPIDAL L_{ACD}

Full Statement

THEOREM 63.5.1 (L_{ACD} on the non-exceptional sph-finite locus, Eisenstein case). . On the non-exceptional sph-finite locus, there is a canonical equivalence

$$\left(\bigotimes'_v C_v \right)^{\text{nonexc}} \xrightarrow{\sim} D^{\text{sph-fin}}([GL_2])^{\text{nonexc}}$$

of stable presentable ∞ -categories, where:

- $C_v = D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)])$ decomposes as an internal direct sum $C_v^{\text{cusp}} \oplus C_v^{\text{res}} \oplus C_v^{\text{cont}}$ (local Langlands decomposition; Bernstein–Zelevinsky 1977 for p -adic; Wallach 1988 Ch. II at archimedean);
- the restricted tensor product respects this decomposition summand by summand, with the Weyl-quotient convention on the continuous summand;
- the non-exceptional superscript refers to the complement of $\Sigma^{\text{exc}} = \{s = 0\} \cup \{s = 1\}$ on the continuous-spectrum factor, and to the full cuspidal and residual summands with no restriction.

under the local categorical comparison and global restricted-tensor identification inputs of L_{ACD} . The equivalence is compatible with:

- (i) local Whittaker models at each finite prime (Bernstein–Zelevinsky 1977 §2);
- (ii) Bernstein-centre actions across places (Chapter 30.5.1);
- (iii) archimedean $(\mathfrak{g}, K_\infty)$ -module structure at $v = \infty$ (Chapter 30.3.3);
- (iv) normalised intertwining operators $N_v(s, \chi)$ at each place (Chapter 63.2.3);
- (v) Paley–Wiener synthesis (Chapter 63.3.1) on the continuous summand.

Proof. **Step 1 (cuspidal).** The cuspidal summand is supplied by the cuspidal L_{ACD} comparison (Chapter 48.4.4). Nothing new is required; the cuspidal factor contributes through the orthogonal direct-sum on each side. **Step 2 (residual).** The residual summand is treated in Chapter 63.4.3: restricted tensor product of Sp-factors is Sp; on the other side the residual space is one-dimensional. **Step 3**

(continuous non-exceptional). The continuous summand restricted to Σ^{nonexc} is controlled by Chapter 63.4.4: the categorical lift uses the Plancherel synthesis of Chapter 63.3.1 plus the coherent Weyl action of Chapter 63.3.5. **Step 4 (assembly).** The three summands assemble through the orthogonal decomposition of $L^2([GL_2])$ (Chapter 63.4.1); on the restricted- tensor side they assemble through the local summand decomposition of each C_v . The restricted tensor product of an orthogonal direct sum is the orthogonal direct sum of restricted tensor products (Lurie HA 4.8.2: tensor product distributes over direct sum in Pr^{Lst}). **Step 5 (compatibilities).** Whittaker / Bernstein / archimedean compatibilities are inherited from Chapter 48.4.4 on the cuspidal factor and from Chapter 30.3.3–Chapter 30.6.1 on the archimedean and cross-prime sides. The intertwining compatibility is the construction of N in Chapter 63.3.5. Paley–Wiener compatibility is the content of Chapter 63.3.6. $\square$

Corollaries and Domain

COROLLARY 63.5.2 (*adelic categorical descent reduction on the non-exceptional sph-finite locus*). . On the non-exceptional sph-finite locus, L_{ACD} (Chapter 26.1) holds under the standing local categorical comparison and global restricted-tensor identification inputs. The equivalence

$$\bigotimes'_v D^{\text{sph-fin}}([GL_2(\mathbb{Q}_v)]) \xrightarrow{\sim} D^{\text{sph-fin}}([GL_2])$$

is valid on the complement of $\Sigma^{\text{exc}} = \{s = 0\} \cup \{s = 1\}$ within the Eisenstein continuous parameter, and everywhere on the cuspidal and residual summands.

Proof. Combine the cuspidal LACD comparison cuspidal (Chapter 48.4.4) with residual (Chapter 63.4.3) and continuous non-exceptional (Chapter 63.4.4) through the assembly theorem Chapter 63.5.1. $\square$

Conjecture 63.5.3 (*exceptional unitary-crossing Ext^1 match*). . Essential surjectivity extends from the non-exceptional locus to the full sph-finite automorphic category by matching the Ext^1 -class of the local intertwining data at $s = 0, 1$. Specifically, at $s = 0$ (spherical crossing) the local Ext^1 -class of the reducibility short exact sequence of $I_v(0, \chi)$ matches the deformation-theoretic Ext^1 of the Emerton–Gee stack at the corresponding unramified Frobenius-trace-zero point; at $s = 1$ (residual crossing) the one-dimensional residual space matches the unique closed stratum of the Emerton–Gee stack at the cyclotomic character. The match is a single Ext^1 datum per crossing, analogous to Chapter 46.7.3 but at the automorphic (rather than local Langlands) level.

Remark 63.5.4 (*domain of the exceptional residual*). The two exceptional points are measure-zero in $i\mathbb{R}$ and do not affect the Plancherel measure $|\xi(2s)|^{-2} ds$. They do affect the ind-categorical structure: at $s = 0, 1$ the local C_v^{cont} has a non-semisimple Ext^1 -extension that must be matched by an Ext^1 -class in the global presentable ∞ -category. It is a non-formal theorem, parallel to the Fr.a.2 non-semisimple class-match of Chapter 46.7.3.

Compatibility with the Langlands–Shahidi normalisation

PROPOSITION 63.5.5 (*cross-reference to the Langlands–Shahidi normalisation Koszul–Petersson correction*). (conditional on the Langlands–Shahidi normalisation LS Plancherel-measure identification). The Langlands–Shahidi normalised operator $N_v(s, \chi)$ of Chapter 63.2.3 enters the the Langlands–Shahidi normalisation Koszul normalisation correction of Chapter 54.4.2 through the absolute-value specialisation $N^{\text{LS}}(s) = \prod_v N_v(s, \chi_{\text{triv}})$ on the critical line, where χ_{triv} is the trivial

Hecke character. The conditional Plancherel-normalisation match of the Langlands–Shahidi normalisation corresponds on the categorical side to the unitary-axis Weyl-symmetry of the Paley–Wiener space $\text{PW}(i\mathbb{R}, V_\chi)$ under the coherent Weyl action Chapter 63.3.5. No fresh computation is needed beyond the stated measure-identification input.

Proof. Immediate by comparison of the Langlands–Shahidi normalisation definition Chapter 54.4.1 with the adelic global product of Chapter 63.2.3 on the critical line, using the Tate global $\varepsilon = 1$ for GL_1 . $\square$

63.6 COMPARISON AND HYPOTHESES

Independent Inputs

Derivation and comparison sources. The derivation of the cuspidal LACD comparison cuspidal reduction used Flath 1979, Bernstein 1988, Mœglin–Waldspurger 1995 §III (cuspidal orthogonality). The present Eisenstein reduction uses Langlands 1976, Shahidi 1990, 2010, Arthur 2013 §3.5, Harish-Chandra 1976, Wallach 1992 §12, and Mœglin–Waldspurger 1995 §II, IV (spectral theory of Eisenstein series). The categorical assembly uses Lurie HA 4.8 and 5.1–5.3 (restricted tensor and monoidal coherence), F2.c tightened (Chapter 28.1.3), and the cuspidal LACD comparison cuspidal (Chapter 48.4.4). The cuspidal and Eisenstein derivation paths are disjoint at the chapter level of Mœglin–Waldspurger (§III vs §II/IV); no primary author appears in both spectral-decomposition roles.

Intertwining-operator normalisation. Langlands 1976 defines the unnormalised operator $M(s, \chi)$ by an integral over $N(\mathbb{A})$. Shahidi 1990, 2010 defines the normalised operator $N(s, \chi)$ with L - and ε -factor corrections. Mœglin–Waldspurger 1995 uses the Arthur normalisation $\tilde{M}(s, \chi)$ with a specific L -ratio. The three normalisations agree on the critical line after the Tate global product $\prod_v \varepsilon_v = 1$ on GL_1 ; the Weyl translation $\chi \mapsto \chi^{w_0}$ is consistent across all three.

Categorical ambient structure. Each C_v^{cont} is a stable presentable ∞ -category (Chapter 63.3.4); restricted tensor products lie in $\text{Pr}^{\text{L}, \text{st}}$ (F2.c tightened); Weyl quotients are homotopy fixed-point constructions in $\text{Pr}^{\text{L}, \text{st}}$ (Lurie HA 4.8.5.16). The Langlands spectral decomposition Chapter 63.4.1 is an orthogonal direct sum at both Hilbert-space and derived-category level.

Hypotheses. The non-exceptional hypothesis is made explicit in Chapter 63.2.6. Hecke-character unitarity is required for unitarity of N_v (Chapter 63.2.5.2); it is imposed throughout. The spherical / Iwahori-finite distinguished-unit hypothesis carries over from the cuspidal LACD comparison through the construction of u_v^{cont} .

Functoriality of the Weyl action. The intertwining operators act place by place; cross-place commutation is Chapter 63.3.5. The Weyl action is a coherent $\mathbb{Z}/2$ action on the restricted tensor product, not an ∞ -groupoid action; coherence up to homotopy follows from Lurie HA 5.1.2.26.

Open input at the exceptional locus. The exceptional unitary-crossing Ext^1 -class match is Chapter 63.5.3; it is named as a conjecture. On the non-exceptional locus, every input is either a primary theorem (the Plancherel synthesis of Chapter 63.3.1, sourced from Shahidi 2010 Thm 4.2.2) or a proved claim in this chapter.

Numerical comparison. This chapter contains no numerical computation; the expected-value comparison is through the analytic Plancherel formula.

Composite with the cuspidal LACD comparison

THEOREM 63.6.1 (*composite L_{ACD} reduction after the cuspidal LACD comparison + the Eisenstein LACD comparison*). . Combining the cuspidal LACD comparison cuspidal reduction Chapter 48.4.4 with the Eisenstein LACD comparison residual + continuous non-exceptional reduction Chapter 63.5.1, L_{ACD} is reduced on the sph-finite automorphic category modulo the two unitary-axis crossings $\Sigma^{exc} = \{s = 0\} \cup \{s = 1\}$, and modulo the standing local categorical comparison, global restricted-tensor identification, and BC-diamond-glue inputs.

Proof. Orthogonal direct-sum assembly of the three reductions through Chapter 63.4.1 and the restricted-tensor orthogonality of Lurie HA 4.8. $\square$

63.7 OBSTRUCTION DECOMPOSITION AFTER THIS CHAPTER

PROPOSITION 63.7.1 (*Fi obstruction decomposition after the Eisenstein LACD comparison*). . Combining this chapter with the cuspidal LACD comparison obstruction decomposition Chapter 48.5.1, the Fi obstruction decomposition is:

- (i) **FI.a.1.non-ord** (non-formal theorem): essential surjectivity of DEG at $p \geq 5$ on the non-ordinary locus.
- (ii) **FI.a.1.TW-removal** (non-formal theorem): extension beyond Taylor–Wiles hypotheses.
- (iii) **FI.a.2.class-match** (non-formal theorem): class-level Paškūnas Ext^1 -compatibility at $p = 2, 3$.
- (iv) **L_{ACD} .exceptional-crossing** (non-formal theorem): Ext^1 -class match at $s = 0, 1$ (Chapter 63.5.3); the Eisenstein LACD comparison replaces the non-formal theorem Eisenstein conjecture Chapter 48.4.6 with this narrower residual.
- (v) **L_{ACD} .local-global comparison**: the local categorical comparison and the global restricted-tensor identification of Chapter 26.10.
- (vi) **BC-diamond-glue** (non-formal theorem): Bernstein-centre diamond-gluing, Chapter 42.

No numerical closure total is asserted here.

Proof. Tabulation. the cuspidal LACD comparison carried the Eisenstein residual at non-formal theorem; the Eisenstein LACD comparison reduces the non-exceptional part and isolates a non-formal theorem Ext^1 -class match at the two polar crossings. The global LACD comparison inputs remain. $\square$

63.8 EISENSTEIN BOUNDARY

- (i) **Langlands–Shahidi normalisation constructed** (Chapter 63.2.3, Chapter 63.2.5): normalised intertwining operators $N_v(s, \chi)$ with functional equation, unitarity on the axis, polar divisor at $s = 0, 1$. Compatible with the Langlands–Shahidi normalisation through Chapter 63.5.5 and the conditional normalisation match of Chapter 54.4.2.
- (ii) **Paley–Wiener synthesis categorically lifted** (Chapter 63.3.6): the continuous summand of $L^2([GL_2])$ lifts to a restricted tensor product of local continuous-spectrum presentable stable ∞ -categories modulo a coherent Weyl quotient.

- (iii) **Langlands spectral decomposition categorified** (Chapter 63.4.1, Chapter 63.4.3, Chapter 63.4.4): three-summand orthogonal direct sum at Hilbert-space and derived-category level, with residual summand = Sp , continuous summand = Paley–Wiener sheaves on the unitary axis.
- (iv) L_{ACD} **reduced on the non-exceptional sph-finite locus** (Chapter 63.5.1, Chapter 63.5.2): the restricted tensor product equals the global sph-finite automorphic derived category on the complement of two unitary-axis crossings.
- (v) **Exceptional residual isolated** (Chapter 63.5.3): Ext^1 -class match at $s = 0$ (spherical crossing) and $s = 1$ (residual crossing); non-formal theorem of domain, parallel to the Paškūnas non-semisimple class-match of F1.a.2.
- (vi) **Composite reduction** (Chapter 63.6.1): the cuspidal L_{ACD} comparison cuspidal + the Eisenstein L_{ACD} comparison residual and non-exceptional continuous jointly reduce L_{ACD} on the sph-finite automorphic category modulo the two-point exceptional residual and the standing comparison inputs.
- (vii) **Obstruction decomposition** (Chapter 63.7.1): F1 residuals drop to the named items above.

The Eisenstein / continuous-spectrum residual Chapter 48.4.6 is reduced on its natural domain (non-exceptional sph-finite); the two remaining unitary-axis crossings are isolated as the narrow Ext^1 -class-match conjecture Chapter 63.5.3, a 10–15 page condition. the Langlands–Shahidi normalisation states the Plancherel-normalisation match from the Koszul-pairing side; this chapter establishes the categorical lift from the intertwining-operator side; their cross-compatibility is Chapter 63.5.5.

The Eisenstein open condition is Chapter 63.5.3: matching Ext^1 -classes at $s = 0$ and $s = 1$, where the reducibility of $I_v(s, \chi)$ and the Emerton–Gee stack geometry must be reconciled. The global L_{ACD} comparison table remains separate.

NON-ORDINARY LOCUS AND REMOVAL OF THE TAYLOR–WILES HYPOTHESES FOR THE DEG EQUIVALENCE

Chapter 48 established essential surjectivity of the Dotto–Emerton–Gee functor $\Phi_{p,\chi}^{\text{DEG}}$ on the unramified ordinary summand under (TW₁)–(TW₄). Here the non-ordinary locus is decomposed into the potentially Barsotti–Tate and supersingular loci, and the failure of (TW₂) or (TW₃) is decomposed into dihedral, nearly-ordinary, and Serre-modular sectors.

Kisin’s potentially Barsotti–Tate deformation rings, the Gee–Kisin Breuil–Mézard cycles, the Emerton–Gee Newton–Hodge stratification, and Colmez’s functor give the strata and their compact generators on the pBT locus. Arthur–Clozel, Skinner–Wiles, Khare–Wintenberger, and Kisin give the corresponding automorphic and deformation-theoretic generators on the three non-Taylor–Wiles sectors. The categorical equivalence is therefore the precise DEG generator comparison on these strata.

64.1 SOURCES AND OBJECTS

Inherited Objects and Notation

The unramified ordinary ∞ -category $D^{\text{sph-fin,unr-ord}}([GL_2(\mathbb{Q}_p)])_\chi$, the Dotto–Emerton–Gee functor $\Phi_{p,\chi}^{\text{DEG}}$ (Chapter 39.3.1), the Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG},p,\chi}$ and its Breuil–Mézard stratification, and the Taylor–Wiles setup (Chapter 48.2.1) are taken as stated in Chapters 39 and 48.

Primary Sources

Kisin 2008, J. AMS 21 “Potentially semi-stable deformation rings”, J. Amer. Math. Soc. 21 (2008), 513–546. Potentially Barsotti–Tate deformation rings; reduced and flat at $p \geq 5$ on the non-ordinary component.

Kisin 2010, Invent. Math. 178 “Modularity for some geometric Galois representations”, Invent. Math. 178. Modularity at non-ordinary p -adic Hodge types.

Kisin 2009, J. AMS 22 “The Fontaine–Mazur conjecture for GL_2 ”, J. AMS 22. Non-ordinary Fontaine–Mazur plus Breuil–Mézard on the full p -adic Hodge regime.

Gee–Kisin 2014, Forum Math. Pi 2 “The Breuil–Mézard conjecture for potentially Barsotti–Tate representations”, Forum Math. Pi 2. Geometric Breuil–Mézard beyond ordinary.

Emerton–Gee arXiv:1109.4226 “A geometric perspective on the Breuil–Mézard conjecture”, §7–8 specifically: non-ordinary locus as a component of the EG stack, dimension computation, and intersection pattern with Breuil–Mézard cycles.

Khare–Wintenberger 2009, Ann. Math. 169 “Serre’s modularity conjecture” I, II. Unconditional Serre’s conjecture: every odd irreducible two-dimensional residual $\bar{\rho}: G_{\mathbb{Q}} \rightarrow GL_2(\overline{\mathbb{F}}_p)$ is modular.

Langlands 1980, *Annals of Mathematics Studies* 96 *Base Change for $GL(2)$* . Solvable base change for GL_2 .

Tunnell 1981, *Bull. AMS* 5 “Artin’s conjecture for representations of octahedral type”, *Bull. AMS* 5, 173–176. The tetrahedral and octahedral Artin cases.

Taylor 2003, *Pub. IHES* 98 “On icosahedral Artin representations II”, *Pub. IHES* 98. Potential modularity for the icosahedral case.

Skinner–Wiles 1999, *Invent. Math.* 135 “Nearly ordinary deformations of irreducible residual representations”. Removes the big-image hypothesis in the nearly-ordinary regime.

Skinner–Wiles 1997, *Pub. IHES* 89 “Ordinary representations and modular forms”. Residually reducible modularity theorems.

Arthur–Clozel 1989, *Ann. Math. Stud.* 120 *Simple algebras, base change, and the advanced theory of the trace formula*. Cyclic solvable base change for GL_n ; key input for the dihedral / solvable-CM sector.

Emerton 2011, *Invent. Math.* 184 “Local-global compatibility in the p -adic Langlands programme for $GL_2/\mathbb{Q}$ ”. Patching without Taylor–Wiles for a sector of residuals.

Barnet-Lamb–Gee–Geraghty 2012, *J. AMS* 25 “The Sato–Tate conjecture for Hilbert modular forms”, *J. AMS* 25. Automorphy lifting with residually-dihedral and residually-CM residuals, via solvable descent and potential automorphy.

Gee 2011, *Math. Annalen* 350 “Automorphic lifts of prescribed types”. Used to match Breuil–Mézard cycles with specified Hodge–Tate weights.

Colmez 2010, *Astérisque* 330 “Représentations de $GL_2(\mathbb{Q}_p)$ et (φ, Γ) -modules”. Colmez Montreal functor; non-ordinary / supersingular case §9.3–9.6.

Paškūnas 2013, *Publ. IHES* 118 “The image of Colmez’s Montreal functor”, *Publ. Math. IHES* 118 (2013), 1–191 (arXiv:1305.7060). Projective-cover theorem in $\text{Block}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}}$; compact generation for the supersingular case and Serre-weight blocks.

Independence of Sources

The proof source backing Dotto–Emerton–Gee arXiv:2603.26887 is its own Colmez / Paškūnas / Breuil–Mézard assembly on the ordinary and tame non-reducible loci. The comparison sources used in this chapter are distinct:

- Kisin 2008 *J. AMS* 21 and Kisin 2009 *J. AMS* 22 are *deformation-theoretic* and use integral p -adic Hodge theory (Breuil modules, Kisin modules) that is not part of the DEG derivation used in Chapter 39.
- Gee–Kisin 2014 proves the geometric Breuil–Mézard formulation as a structural theorem on the Emerton–Gee stack; its proof is independent of the DEG categorical equivalence.
- Khare–Wintenberger 2009 is automorphic-modularity-lifting via compatible families and residual modularity; it uses no categorical input from DEG.
- Taylor 2003 and Arthur–Clozel 1989 are solvable base change and trace formula; independent of both DEG and Emerton–Gee.
- Skinner–Wiles 1999 uses Hida nearly-ordinary families and deformation rings; independent of DEG.

No source in this chapter appears in the DEG derivation. The deformation-theoretic, automorphic, and categorical inputs are therefore separate arguments.

64.2 ESSENTIAL SURJECTIVITY ON THE NON-ORDINARY LOCUS AT $p \geq 5$ *Structural Decomposition of the Non-Ordinary Locus*

Assume Chapter 48.2.1 with (TW₁)–(TW₃) and without (TW₄). The block $\text{Block}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}} \subseteq \text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$ decomposes along the ordinary–non-ordinary partition. The ordinary component is governed by Chapter 48.2.3; the non-ordinary complement has the two subcomponents below.

Definition 64.2.1 (non-ordinary sub-loci). . The non-ordinary locus of $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \bar{\rho}}$ admits the decomposition

$$\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \bar{\rho}, \text{non-ord}} = \mathcal{X}_{\bar{\rho}, \chi}^{\text{pBT}} \cup \mathcal{X}_{\bar{\rho}, \chi}^{\text{ss}},$$

where:

- $\mathcal{X}_{\bar{\rho}, \chi}^{\text{pBT}}$ is the *potentially Barsotti–Tate non-ordinary* sub-locus: closed points are (φ, Γ) -modules that are potentially Barsotti–Tate (there is a finite Galois extension $K/\mathbb{Q}_p$ such that the restriction is Barsotti–Tate) and whose Newton slopes are $(\frac{1}{2}, \frac{1}{2})$ rather than $(0, 1)$.
- $\mathcal{X}_{\bar{\rho}, \chi}^{\text{ss}}$ is the *supersingular* sub-locus: Hodge–Tate weights fixed to $(0, k-1)$ with $2 \leq k \leq p-1$, with Newton slopes $(\frac{k-1}{2}, \frac{k-1}{2})$ and no unramified Frobenius-eigenvalue line.

Remark 64.2.2 (extent of the decomposition). The union $\mathcal{X}_{\bar{\rho}, \chi}^{\text{pBT}} \cup \mathcal{X}_{\bar{\rho}, \chi}^{\text{ss}}$ set-theoretically covers the non-ordinary component of $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \bar{\rho}}$: any closed point of the non-ordinary locus either has equal Newton slopes strictly between 0 and $k-1$ (hence is potentially Barsotti–Tate in a suitable weight regime) or is genuinely supersingular at regular Hodge–Tate weights. This is Emerton–Gee arXiv:1109.4226 §7 Prop. 7.3.1, which stratifies the non-ordinary locus by Newton–Hodge polygon relative position.

The Potentially Barsotti–Tate Sub-Locus

LEMMA 64.2.3 (*Kisin 2009 moduli of finite flat group schemes on pBT*). . Fix $p \geq 5$ and $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$. The potentially Barsotti–Tate deformation ring $R_{\bar{\rho}, \chi}^{\text{pBT}}$ representing the functor of potentially Barsotti–Tate lifts of $\bar{\rho}$ with determinant $\chi \cdot \chi_{\text{cyc}}$ is $\mathcal{O}$ -flat, reduced, and complete-intersection with explicitly computable tangent dimensions (Kisin 2008 J. AMS 21 Thm 0.2, Thm 2.5.4). The associated pBT component $\mathcal{X}_{\bar{\rho}, \chi}^{\text{pBT}} = \text{Spf } R_{\bar{\rho}, \chi}^{\text{pBT}}$ of the EG stack is Cohen–Macaulay of pure relative dimension 3 over $\text{Spf } \mathcal{O}$ (Emerton–Gee arXiv:1109.4226 §7.3).

Source. Kisin 2008 J. AMS 21 §2.5 Thm 2.5.4 (flatness of the pBT deformation ring at $p \geq 5$); Kisin 2009 J. AMS 22 Thm 2.7 (reducedness); Emerton–Gee arXiv:1109.4226 §7.3.1 (dimension computation and Cohen–Macaulay property). $\square$

LEMMA 64.2.4 (*Gee–Kisin geometric Breuil–Mézard on pBT*). . Under the hypotheses of Chapter 64.2.3, for each inertial type $\tau: I_{\mathbb{Q}_p} \rightarrow GL_2(\overline{\mathbb{Q}_p})$ the Breuil–Mézard cycle $\mathcal{X}_{p, \chi, \bar{\rho}}^{\text{BM}, \tau}$ restricted to $\mathcal{X}_{\bar{\rho}, \chi}^{\text{pBT}}$ is a smooth divisor; the divisors attached to distinct inertial types meet transversely; and their union set-theoretically exhausts the pBT non-ordinary locus.

Source. Gee–Kisin 2014 Forum Math. Pi 2 Thm A (geometric Breuil–Mézard for potentially Barsotti–Tate). The transversality and exhaustion statements are Emerton–Gee arXiv:1109.4226 §7.4 Prop. 7.4.2. $\square$

LEMMA 64.2.5 (*projective covers on the pBT block*). Under the hypotheses of Chapter 64.2.3, the projective covers $\tilde{\pi}_\tau$ of the types $\sigma(\tau) \in \text{Block}_{\tilde{\rho}|G_{\mathbb{Q}_p}}^{\text{pBT}}$ form a compact generator of $\text{Block}_{\tilde{\rho}|G_{\mathbb{Q}_p}}^{\text{pBT}}$. The corresponding Breuil–Mézard structure sheaves $\mathcal{O}_{\chi_{\text{BM},\tau}^{\text{pBT}}}$ form a compact generator of $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho},\chi}^{\text{pBT}})$.

Source. Colmez 2010 Astérisque 330 §9.3–9.6 constructs the projective covers attached to the relevant (φ, Γ) -modules. Paškūnas 2013 gives compact generation by these projective covers. Kisin 2010 Invent. Math. 178 §2.4 and Gee–Kisin 2014 give the Breuil–Mézard cycles whose structure sheaves generate the spectral side. $\square$

THEOREM 64.2.6 (*essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ on the potentially Barsotti–Tate non-ordinary sub-locus*). Under Chapter 48.2.1 (TW1)–(TW3) (ordinary hypothesis (TW4) dropped), the restriction of the DEG functor $\Phi_{p,\chi}^{\text{DEG}}$ to the potentially Barsotti–Tate non-ordinary sub- ∞ -category is an equivalence if the DEG–Colmez generator comparison holds on the pBT block:

$$\Phi_{p,\chi}^{\text{DEG}}|_{\text{pBT-non-ord}}: D^{\text{sph-fin,pBT}}([GL_2(\mathbb{Q}_p)])_\chi \xrightarrow{\sim} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho},\chi}^{\text{pBT}}).$$

The deformation-theoretic sources identify the two sets of generators; the displayed equivalence is exactly the assertion that DEG carries the first set to the second.

Proof. **Step 1: fully-faithfulness on the sub- ∞ -category.** The DEG functor is fully faithful on the full block (Chapter 39.3.1). Restriction to the pBT sub- ∞ -category preserves fully-faithfulness.

Step 2: compact-generator criterion. By Chapter 64.2.3, $\mathcal{X}_{\tilde{\rho},\chi}^{\text{pBT}}$ is a Cohen–Macaulay formal algebraic stack of pure dimension 3 over $\text{Spf } O$, with $\text{IndCoh}_{\text{Nilp}}$ compactly generated by the structure sheaves of its Breuil–Mézard cycles (Gaitsgory–Rozenblyum AMS DAG II Ch. 8 §1.3; the nilpotent singular-support condition is satisfied automatically on the Cohen–Macaulay locus by Gaitsgory AMS DAG II Ch. 9 Prop. 1.1.4).

Step 3: the two generating sets. By Chapter 64.2.4, the Breuil–Mézard cycles form a family of smooth divisors meeting transversely and exhausting the pBT locus. By Chapter 64.2.5, the collection $\{\tilde{\pi}_\tau\}_\tau$ generates $\text{Block}_{\tilde{\rho}|G_{\mathbb{Q}_p}}^{\text{pBT}}$, and the sheaves $\mathcal{O}_{\chi_{\text{BM},\tau}^{\text{pBT}}}$ generate the spectral category.

Step 4: the DEG comparison. By hypothesis, $\Phi_{p,\chi}^{\text{DEG}}(\tilde{\pi}_\tau) \simeq \mathcal{O}_{\chi_{\text{BM},\tau}^{\text{pBT}}}$ for every inertial type τ , compatibly with the intersections of the Breuil–Mézard divisors.

Step 5: equivalence by Lurie HA 4.8.2. Fully-faithful plus compact-generator match (Step 1 and Steps 2–4) implies equivalence by Lurie HA Prop. 4.8.2.II: a fully-faithful functor between compactly generated stable presentable ∞ -categories that sends a set of compact generators to a set of compact generators is an equivalence. $\square$

The Supersingular Complement

COROLLARY 64.2.7 (*conditional essential surjectivity on the pBT non-ordinary sub-component*). Within the hypotheses of Chapter 48.2.1 (TW1)–(TW3), essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ on the non-ordinary locus holds on the potentially Barsotti–Tate sub-component only after the DEG–Colmez generator comparison is supplied. The complementary locus is the supersingular core $\mathcal{X}_{\tilde{\rho},\chi}^{\text{ss}} \setminus \mathcal{X}_{\tilde{\rho},\chi}^{\text{pBT}}$: closed points of regular Hodge–Tate weight with Newton slopes $(\frac{k-1}{2}, \frac{k-1}{2})$ not attained by any potentially Barsotti–Tate lift.

Proof. Chapter 64.2.6 supplies the equivalence on the pBT sub-locus. The complementary supersingular core is by Chapter 64.2.2 the genuinely non-pBT portion of the non-ordinary locus; its complement in the non-ordinary stratum is exactly the pBT sub-locus. $\square$

Conjecture 64.2.8 (essential surjectivity on the genuinely supersingular core). Essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ on the genuinely supersingular core $\mathcal{X}_{\tilde{\rho},\chi}^{\text{ss}} \setminus \mathcal{X}_{\tilde{\rho},\chi}^{\text{pBT}}$ holds if the supersingular DEG generator comparison identifies the projective covers of the supersingular block with the corresponding Breuil–Mézard–Newton structure sheaves.

Remark 64.2.9 (the supersingular obstruction). The pBT part is governed by Chapter 64.2.3 and Chapter 64.2.4: it is Cohen–Macaulay with transverse Breuil–Mézard cycles. The supersingular complement is non-pBT; its Breuil–Mézard cycles may be non-reduced with higher multiplicities, and the compact-generator argument requires either a derived enhancement or a Jordan–Hölder refinement by the Breuil modules of Kisin 2008 J. AMS 21 §3.

Source Disjointness and Hypothesis Verification for the Non-Ordinary Closure

Source disjointness. Kisin 2008 J. AMS 21 is integral p -adic Hodge theory (Breuil modules, Kisin modules). Gee–Kisin 2014 uses the geometric Breuil–Mézard formulation on the Emerton–Gee stack. Emerton–Gee arXiv:1109.4226 provides the structural Newton–Hodge stratification. Colmez 2010 is the (φ, Γ) -module Montreal functor. These four sources are pairwise disjoint in methodology; all four are disjoint from DEG arXiv:2603.26887, which assembles them on the ordinary and tame-regular cases only. The present application is a compact-generator check on a sub-locus, independent of the DEG gluing argument.

Conventions. Hodge–Tate weight conventions follow Emerton–Gee (dual-normalised: the cyclotomic character has weight +1). Newton-slope conventions follow Kisin 2009 §1.2. Inertial-type parametrisation follows Bushnell–Kutzko. The four sources agree after the standard Fontaine cyclotomic normalisation shift.

Categorical framework. The left-hand side is the pBT sub- ∞ -category of $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$; the right-hand side is $\text{IndCoh}_{\text{Nilp}}$ on the pBT Cohen–Macaulay formal algebraic sub-stack. Both are stable presentable ∞ -categories with compact generators. The restricted DEG functor is a morphism in $\text{Pr}_{\text{st}}^{\text{L}}$.

Hypotheses. The pBT hypothesis is explicit. Dropping (TW₄) is compensated by: (i) the Cohen–Macaulay property of Kisin’s pBT ring, which ensures $\text{IndCoh}_{\text{Nilp}}$ is well-behaved; (ii) Gee–Kisin transversality, which enables the compact-generator argument on cycle intersections; (iii) the Colmez Montreal functor on $\tilde{\pi}_{\tau}$ (§9.3–9.6). Conditions (TW₂) and (TW₃) remain in force, inherited from Chapter 48.2.1; the removal of these conditions is addressed in §64.3.

Hecke compatibility. The Hecke action at p matches on both sides via Colmez’s (φ, Γ) -bijection; Kisin 2008 J. AMS 21 §2.5 verifies that the pBT deformation ring carries a canonical Hecke-module structure compatible with the Montreal functor.

Supersingular comparison. No conjecture is invoked on the pBT sub-locus beyond the stated DEG generator comparison. The supersingular core is the comparison of Chapter 64.2.8.

64.3 REMOVAL OF THE TAYLOR–WILES CONDITIONS: DIHEDRAL, NEARLY-ORDINARY, AND SERRE-MODULAR SECTORS

Structural Partition of the Non-Taylor–Wiles Locus

The Taylor–Wiles hypotheses (TW₂) and (TW₃) of Chapter 48.2.1 impose two conditions on $\bar{\rho}$: (TW₂) big image (absolute irreducibility of $\bar{\rho}|_{G_{\mathbb{Q}(\zeta_p)}}$) and (TW₃) distinguishedness (semisimple with distinct diagonal characters at the residual level). Residuals violating either condition fall outside the regime of Chapter 48.2.3. The non-Taylor–Wiles locus $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \bar{\rho}, \text{non-TW}}$ admits a three-part decomposition.

Definition 64.3.1 (non-Taylor–Wiles sectors). . The non-Taylor–Wiles residuals partition as

$$\{\bar{\rho}: (\text{TW}_2) \text{ or } (\text{TW}_3) \text{ fails}\} = S^{\text{dihedral}} \cup S^{\text{nearly-ord}} \cup S^{\text{Serre-res}},$$

where:

- S^{dihedral} : residuals whose projectivisation $\bar{\rho}^{\text{proj}}: G_{\mathbb{Q}} \rightarrow PGL_2(\overline{\mathbb{F}}_p)$ has solvable image. The dihedral sublocus is induced from a character ψ of G_K for a quadratic extension $K/\mathbb{Q}$; the tetrahedral and octahedral subloci have projective image A_4 and S_4 . The icosahedral case A_5 is excluded.
- $S^{\text{nearly-ord}}$: residuals with $\bar{\rho}|_{G_{\mathbb{Q}_p}}$ reducible but with equal diagonal characters (not distinguished, hence (TW₃) fails) satisfying the nearly-ordinary Skinner–Wiles hypothesis: $\bar{\rho}$ has a global nearly-ordinary lift.
- $S^{\text{Serre-res}}$: odd irreducible residuals with big image failing (TW₂) by $\bar{\rho}|_{G_{\mathbb{Q}(\zeta_p)}}$ reducible, but with $\bar{\rho}$ odd irreducible of Serre type.

Remark 64.3.2 (overlap and exhaustion of the sectors). Each odd residual $\bar{\rho}$ with (TW₂) or (TW₃) failing falls into at least one of the three sectors. The A_5 (icosahedral) exceptional case is genuinely non-solvable and treated via potential automorphy (Taylor 2003 Pub. IHES 98 §2; Barnet-Lamb–Gee–Geraghty 2012). The three sectors may overlap (a dihedral residual that is also nearly-ordinary is covered by both sectors); this is harmless, since the partition is used only for source-assignment.

The Solvable Projective-Image Sector

LEMMA 64.3.3 (*automorphy on S^{dihedral}*). . For every odd residual $\bar{\rho} \in S^{\text{dihedral}}$ with $\bar{\rho}^{\text{proj}}$ projectively solvable, the associated two-dimensional Artin representation is modular. In the dihedral case it is obtained by automorphic induction from a Hecke character of a quadratic extension. In the tetrahedral and octahedral cases it is governed by the Langlands–Tunnell theorem and solvable base change.

Source. Langlands 1980 Annals of Mathematics Studies 96 gives solvable base change for GL_2 . Arthur–Clozel 1989 Ann. Math. Stud. 120 gives cyclic base change and solvable descent. Tunnell 1981 proves the octahedral case of Artin modularity; the tetrahedral case is contained in the Langlands–Tunnell argument. $\square$

LEMMA 64.3.4 (*Colmez–Paškūnas on induced local residuals*). . For a dihedral residual $\bar{\rho} = \text{Ind}_{G_K}^{G_{\mathbb{Q}}} \psi$ with $K/\mathbb{Q}$ quadratic, the local residual $\bar{\rho}|_{G_{\mathbb{Q}_p}}$ is:

- If p splits in $K/\mathbb{Q}$: reducible, direct sum of the restriction of ψ to the two places; the Colmez Montreal functor applies as in the reducible case.
- If p is inert in $K/\mathbb{Q}$: irreducible; the Colmez Montreal functor applies as in the irreducible non-supersingular case.

- If p ramifies in $K/\mathbb{Q}$: a wild inertia case at $p \geq 5$ with specific Hodge–Tate type controlled by the ramification of $K/\mathbb{Q}$.

In each sub-case, the Colmez (φ, Γ) -module classification is explicit (Colmez 2010 §7.2, §9.3).

Source. Colmez 2010 Astérisque 330 §7.2 (split case and inert case via the principal-series classification); §9.3 (ramified case via the Fontaine–Laffaille variant). $\square$

THEOREM 64.3.5 (*removal of Taylor–Wiles conditions on the dihedral sector*). . For every prime $p \geq 5$, every $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$, and every residual $\bar{\rho} \in S^{\text{dihedral}}$, assume the sectorial DEG generator comparison. Then the restriction of the DEG functor $\Phi_{p,\chi}^{\text{DEG}}$ to the corresponding block is an equivalence.

Proof. **Step 1: block decomposition.** By the Colmez–Paškūnas block decomposition (Chapter 39.4.1), the full $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm},\chi}$ decomposes along residuals $\bar{\rho}|_{G_{\mathbb{Q}_p}}$. The dihedral block corresponds to a specific residual inside this decomposition; by Chapter 64.3.4 it falls into one of three sub-cases (split, inert, ramified).

Step 2: split and inert sub-cases. In the split sub-case, $\bar{\rho}|_{G_{\mathbb{Q}_p}}$ is reducible with distinct diagonal characters (the two restrictions of ψ are distinct because ψ is a character of the quadratic extension); (TW₃) is *not* failing at the local level, and the block is treated by the ordinary / distinguished methods of Chapter 48.2.3. The failure of (TW₃) is only global (the global absolute irreducibility fails because $\bar{\rho} = \text{Ind } \psi$), which is irrelevant to the local block structure. Similarly in the inert sub-case, $\bar{\rho}|_{G_{\mathbb{Q}_p}}$ is irreducible, and the block is treated via the Colmez irreducible classification (Chapter 64.3.4), with essential surjectivity following as in the pBT argument of Chapter 64.2.6.

Step 3: ramified sub-case. In the ramified sub-case, Colmez 2010 §9.3 provides the explicit (φ, Γ) -module classification via a Fontaine–Laffaille crystalline module with prescribed Hodge–Tate structure. The block admits an explicit projective generator sent by the Montreal functor to the structure sheaf of the corresponding Breuil–Mézard cycle (Kisin 2009 J. AMS 22 §3 for the semistable extension). The fully-faithful + compact-generator argument proceeds as in Chapter 64.2.6.

Step 4: matching with Arthur–Clozel base change. Chapter 64.3.3 guarantees that the residual $\bar{\rho}$ lifts to an automorphic representation of $GL_2/\mathbb{Q}$ in a compatible family, ensuring the Hecke-eigenvalue match on both sides of the DEG functor via the solvable-base-change identification (Arthur–Clozel §3 Thm 4). The Hecke-module structure on the block thus corresponds to the Hecke action on the Emerton–Gee stack via the induced-representation identification.

Step 5: assembly. Steps 1–4 give fully-faithfulness plus compact-generator match on each dihedral block, independent of (TW₂) and (TW₃) at the global level. Equivalence follows by Lurie HA 4.8.2.11. $\square$

The Nearly-Ordinary Sector

LEMMA 64.3.6 (*Skinner–Wiles nearly-ordinary modularity*). . For every residual $\bar{\rho} \in S^{\text{nearly-ord}}$ admitting a nearly-ordinary lift, $\bar{\rho}$ is modular, arising from a nearly-ordinary Hida family $\mathbf{T}_{\bar{\rho}}$ with explicit Hecke-eigenvalues matching the Frobenius traces on the spectral side. Equivalently, the nearly-ordinary deformation ring $R_{\bar{\rho},\chi}^{\text{n-ord}}$ is isomorphic to the nearly-ordinary Hecke algebra $\mathbf{T}_{\bar{\rho}}^{\text{n-ord}}$.

Source. Skinner–Wiles 1999 Invent. Math. 135 Thm 5.1 (nearly-ordinary $R = T$ theorem for irreducible residual representations without the big-image hypothesis). Skinner–Wiles 1997 Pub. IHES 89 §3 supplies the residually-reducible case. $\square$

THEOREM 64.3.7 (*removal of Taylor–Wiles conditions on the nearly-ordinary sector*). . For every prime $p \geq 5$, every $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$, and every residual $\bar{\rho} \in S^{\text{nearly-ord}}$, assume the sectorial DEG generator comparison. Then the restriction of $\Phi_{p,\chi}^{\text{DEG}}$ to the corresponding block is an equivalence.

Proof. **Step 1: Hida family generator.** By Chapter 64.3.6, the block carries a Hida family projective generator $\mathbf{T}_{\bar{\rho}}^{\text{n-ord}}$, a single projective indecomposable on the automorphic side, with endomorphism algebra the nearly-ordinary Hecke algebra isomorphic to the nearly-ordinary deformation ring $R_{\bar{\rho},\chi}^{\text{n-ord}}$.

Step 2: spectral-side target. The nearly-ordinary component of $\chi_{GL_2}^{\text{EG},p,\chi,\bar{\rho}}$ is $\text{Spf } R_{\bar{\rho},\chi}^{\text{n-ord}}$ by Skinner–Wiles 1999 §2 (nearly-ordinary deformation ring as a sub-stack of the EG stack). The Skinner–Wiles $R = T$ theorem identifies the endomorphism algebra of the Hida generator with the coordinate ring of this formal spectrum. The sectorial DEG comparison is the assertion that the Hida generator maps to $\mathcal{O}_{\chi_{\bar{\rho},\chi}^{\text{n-ord}}}$.

Step 3: essential surjectivity. The nearly-ordinary deformation ring is a complete Noetherian local ring of dimension 3 (Skinner–Wiles 1999 §2); $\text{IndCoh}_{\text{Nilp}}$ on its formal spectrum is compactly generated by the structure sheaf (Gaitsgory–Rozenblyum AMS DAG II Ch. 8 Thm 8.1.2). The single projective generator $\mathbf{T}_{\bar{\rho}}^{\text{n-ord}}$ on the automorphic side is sent to this single structure sheaf by the assumed sectorial DEG comparison. Hence essential surjectivity.

Step 4: fully-faithfulness. Inherited from DEG via restriction. Combined with Step 3, the block restriction of DEG is an equivalence by Lurie HA 4.8.2.II. $\square$

The Serre-Modular Sector

LEMMA 64.3.8 (*Khare–Wintenberger Serre’s conjecture*). . Every odd irreducible residual representation $\bar{\rho}: G_{\mathbb{Q}} \rightarrow GL_2(\overline{\mathbb{F}}_p)$ is modular: there is a cuspidal eigenform f of weight $k(\bar{\rho}) \leq p+1$ and level $N(\bar{\rho})$ coprime to p such that $\bar{\rho} \cong \bar{\rho}_{f,p}$.

Source. Khare–Wintenberger 2009 Ann. Math. 169 (I), 170 (II) (full proof of Serre’s modularity conjecture). The argument proceeds via compatible families and the Kisin 2009 modularity-lifting theorems. $\square$

LEMMA 64.3.9 (*Kisin modularity lifting under its hypotheses*). . Let $\bar{\rho}$ be odd irreducible and residually modular with Serre weight and conductor $(k(\bar{\rho}), N(\bar{\rho}))$. Let

$$\rho: G_{\mathbb{Q}} \rightarrow GL_2(\overline{\mathbb{Q}}_p)$$

be a lift satisfying the local deformation, p -adic Hodge, ramification, and Taylor–Wiles replacement hypotheses of Kisin 2009 J. AMS 22 Thm 3.4. Then ρ is modular. Its local p -adic Hodge type determines the corresponding Breuil–Mézard cycle.

Source. Kisin 2009 J. AMS 22 Thm 3.4 (modularity lifting from residual modularity, without big-image hypothesis, using compatible families plus Khare–Wintenberger). Gee 2011 Math. Ann. 350 §2 supplies the matching of prescribed p -adic Hodge types with Breuil–Mézard cycles. $\square$

THEOREM 64.3.10 (*removal of Taylor–Wiles conditions on the Serre-modular sector*). . For every prime $p \geq 5$, every $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$, and every residual $\bar{\rho} \in S^{\text{Serre-res}}$, assume the sectorial DEG generator comparison. Then the restriction of $\Phi_{p,\chi}^{\text{DEG}}$ to the corresponding block is an equivalence.

Proof. **Step 1: Serre modular generator.** By Chapter 64.3.8, there is a cuspidal eigenform f with $\bar{\rho} \cong \bar{\rho}_{f,p}$. The block $\text{Block}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}}$ has the associated compact projective $\tilde{\pi}_{f,p}$ (Colmez 2010 §7.2 for the irreducible case; Paškūnas 2013 §2.6 for the supersingular case).

Step 2: modular lifts along the Breuil–Mézard cycle. By Chapter 64.3.9, every lift satisfying Kisin’s local and global hypotheses with the prescribed Hodge type corresponds to a closed point of the Breuil–Mézard cycle $\chi_{p,\chi,\bar{\rho}}^{\text{BM},k(\bar{\rho})}$. Gee 2011 identifies the cycle attached to this p -adic Hodge type. The sectorial DEG generator comparison is the assertion that the projective $\tilde{\pi}_{f,p}$ is sent to $\mathcal{O}_{\chi_{p,\chi,\bar{\rho}}^{\text{BM},k(\bar{\rho})}}$.

Step 3: compact-generator match. The structure sheaves of the Breuil–Mézard cycles indexed by Serre weights $\{\sigma : \sigma \text{ admissible for } \bar{\rho}\}$ form a compact generator of $\text{IndCohNilp}(\chi_{GL_2}^{\text{EG},p,\chi,\bar{\rho}})$ (Emerton–Gee arXiv:1908.07185 §7, with the Serre weight description via the Breuil–Mézard conjecture). The corresponding Colmez projectives $\{\tilde{\pi}_{f,p}\}$ indexed by the same weights form a compact generator of the block on the automorphic side (Paškūnas 2013 §6).

Step 4: equivalence by Lurie HA 4.8.2.11. Fully-faithfulness of DEG plus the assumed compact-generator match from Steps 1–3 gives the equivalence on each Serre-residual block. $\square$

Assembly on the Union of Non-Taylor–Wiles Sectors

THEOREM 64.3.11 (removal of Taylor–Wiles conditions on the union of sectors). . For every prime $p \geq 5$, every central character χ , and every residual $\bar{\rho} \in S^{\text{dihedral}} \cup S^{\text{nearly-ord}} \cup S^{\text{Serre-res}}$, assume the sectorial DEG generator comparisons. Then the restriction of $\Phi_{p,\chi}^{\text{DEG}}$ to the corresponding block is an equivalence. Combined with Chapter 48.2.3, essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ holds on the unramified ordinary locus plus the union of the three non-Taylor–Wiles sectors only under these comparison inputs.

Proof. Assemble Chapter 64.3.5, Chapter 64.3.7, and Chapter 64.3.10. Orthogonality of blocks (each residual $\bar{\rho}|_{G_{\mathbb{Q}_p}}$ indexes a separate block of $\text{Mod}^{\text{sm},\chi}$; the same residual-index on the spectral side via Chapter 39.4.2) gives the direct-sum assembly. $\square$

The Non-Solvable Non-Nearly-Ordinary Complement

COROLLARY 64.3.12 (the complementary sector). . Under the sectorial DEG generator comparisons of Chapter 64.3.11, the complement to the covered non-Taylor–Wiles sectors is the sector $S^{\text{non-solv}}$ of residuals $\bar{\rho}$ that are *not* in $S^{\text{dihedral}} \cup S^{\text{nearly-ord}} \cup S^{\text{Serre-res}}$: genuinely non-solvable (projectively A_5) and non-nearly-ordinary, with $\bar{\rho}|_{G_{\mathbb{Q}_p}}$ non-distinguished in the sense of (TW₃) and not covered by Skinner–Wiles.

Proof. Direct from Chapter 64.3.1 and Chapter 64.3.11. The non-solvable residuals are exactly those with projective image $A_5 \subset PGL_2(\overline{\mathbb{F}_p})$ or non-trivial absolute Galois image beyond the solvable closure; by Dickson’s classification of finite subgroups of PGL_2 , the projective images are cyclic, dihedral, A_4 , S_4 , A_5 , or $PSL_2(\mathbb{F}_q) / PGL_2(\mathbb{F}_q)$. The first four are solvable and hence in S^{dihedral} ; A_5 is exceptional; the large-image cases are in $S^{\text{Serre-res}}$ by Khare–Wintenberger. The A_5 sector plus its non-nearly-ordinary extension is the residual non-solvable core. $\square$

Conjecture 64.3.13 (essential surjectivity on the non-solvable non-nearly-ordinary core). . Essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ on the non-solvable non-nearly-ordinary core $S^{\text{non-solv}}$ holds if Taylor’s icosahedral potential automorphy, the relevant Barnet-Lamb–Gee–Geraghty automorphy-lifting hypotheses, Emerton patching, and the sectorial DEG generator comparison hold on this block.

Remark 64.3.14 (the non-solvable residual). The A_5 icosahedral case was handled by Taylor 2003 Pub. IHES 98 Part II in the classical Artin setting via potential automorphy and solvable descent after a totally real base change. The general BLGG 2012 J. AMS 25 programme extends this to the case of Hilbert modular forms over totally real fields, under its automorphy-lifting hypotheses. Descent from a solvable extension to $\mathbb{Q}$ and the comparison of compact generators are additional hypotheses, supplied in the form stated in Chapter 64.3.13.

Independent Inputs for the Taylor–Wiles Sectors

Sources. Khare–Wintenberger 2009 proves Serre’s conjecture via compatible families. Arthur–Clozel 1989 is solvable base change via the trace formula. Skinner–Wiles 1999 is nearly-ordinary deformation theory. Taylor 2003 uses potential automorphy with solvable-base-change descent. Barnet-Lamb–Gee–Geraghty 2012 is automorphy lifting via the Harris–Shepherd-Barron–Taylor Calabi–Yau construction. These five sources have disjoint methods, and all are independent of DEG arXiv:2603.26887 and from the Kisin 2009 / Gee–Kisin 2014 / Emerton–Gee arXiv:1109.4226 sources used in §64.2.

Conventions. Khare–Wintenberger uses the Serre weight $k(\bar{\rho}) \in \{2, \dots, p+1\}$; Kisin uses dual-normalised Hodge–Tate weights. These match via the standard shift (Emerton–Gee arXiv:1908.07185 §3.1). Nearly-ordinary Hida-family conventions (weight-space variable k versus Hecke operator U_p) follow Skinner–Wiles 1999 §2.

Categorical framework. The left-hand side is a block of $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$ restricted to a specific residual $\bar{\rho}|_{G_{\mathbb{Q}_p}}$; the right-hand side is $\text{IndCoh}_{\text{Nilp}}$ on the corresponding component of the Emerton–Gee stack. Both are stable presentable ∞ -categories with compact generators indexed by Serre weights or projective covers.

Local hypotheses. Condition (TW₁) ($p \geq 5$) is retained. Conditions (TW₂) (big image) and (TW₃) (distinguishedness) are replaced sector by sector: dihedral residuals are induced from a quadratic extension; tetrahedral and octahedral residuals are governed by Langlands–Tunnell; nearly-ordinary residuals admit a nearly-ordinary lift; Serre-modular residuals are covered by Khare–Wintenberger together with the hypotheses of Kisin’s lifting theorem. Dickson’s classification separates the solvable, icosahedral, and large-image projective cases.

Hecke compatibility. The Hecke-eigenvalue match on each sector is verified by the corresponding modularity theorem: Arthur–Clozel §3 Thm 4 for the dihedral case; $R^{\text{n-ord}} = T^{\text{n-ord}}$ for the nearly-ordinary case; Khare–Wintenberger plus Kisin modularity lifting for the Serre-modular case.

Non-solvable comparison. The three covered sectors use the stated sectorial DEG generator comparisons. The non-solvable non-nearly-ordinary sector is the comparison of Chapter 64.3.13.

64.4 THE CONDITIONAL EQUIVALENCE FOR $\Phi_{p,\chi}^{\text{DEG}}$

Union of the Strata

THEOREM 64.4.1 (conditional essential surjectivity locus for $\Phi_{p,\chi}^{\text{DEG}}$). . For every prime $p \geq 5$ and every central character $\chi \in \text{Hom}(\mathbb{Z}_p^{\times}, k^{\times})$, assume the required DEG generator comparisons on the listed sectors. Then the DEG functor $\Phi_{p,\chi}^{\text{DEG}}$ is an equivalence on the union of:

- (i) the unramified ordinary summand (Chapter 48.2.3, under (TW₁)–(TW₄));
- (ii) the potentially Barsotti–Tate non-ordinary summand (Chapter 64.2.6, under (TW₁)–(TW₃));
- (iii) the dihedral, nearly-ordinary, and Serre-modular non-Taylor–Wiles sectors (Chapter 64.3.II, under (TW₁) only).

The deformation and automorphy sources identify the relevant strata and generators. The categorical assertion is the compatibility of these generators with $\Phi_{p,\chi}^{\text{DEG}}$.

Proof. The three stated theorems apply on pairwise orthogonal residual-index blocks. Their direct sum is therefore an equivalence on the union. $\square$

Boundary of the Non-Ordinary and Non-Taylor–Wiles Comparison

PROPOSITION 64.4.2 (*residual decomposition for essential surjectivity on the non-ordinary and non-Taylor–Wiles loci*). . After Chapter 48 and this chapter, the extension of $\Phi_{p,\chi}^{\text{DEG}}$ from the ordinary, pBT, solvable, nearly-ordinary, and Serre-modular strata to the full non-ordinary and non-Taylor–Wiles locus is equivalent to the following two generator comparisons:

- (i) Supersingular core (Chapter 64.2.8): essential surjectivity on the genuinely supersingular sub-locus at $p \geq 5$.
- (ii) Non-solvable non-nearly-ordinary core (Chapter 64.3.13): essential surjectivity on the projectively A_5 and non-nearly-ordinary residual sector.

Proof. The non-ordinary locus is $X_{p,\chi}^{\text{pBT}} \cup X_{p,\chi}^{\text{ss}}$. The pBT part is governed by Chapter 64.2.6; its complement is Chapter 64.2.8. The non-Taylor–Wiles residuals decompose as

$$S^{\text{dihedral}} \cup S^{\text{nearly-ord}} \cup S^{\text{Serre-res}} \cup S^{\text{non-solv}}.$$

The first three sectors are governed by Chapter 64.3.II; the last sector is Chapter 64.3.13. $\square$

Interaction with the Full Essential Surjectivity Problem

PROPOSITION 64.4.3 (*essential surjectivity after the ordinary and non-ordinary strata*). . Combining Chapter 48.5.I with Chapter 64.4.2, essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ on the full locus is equivalent to the following five comparisons:

- (i) Supersingular core.
- (ii) Non-solvable non-nearly-ordinary core.
- (iii) **Fl.a.2.class-match** (Chapter 46.7.3).
- (iv) **L_{ACD}.Eisenstein** (Chapter 48.4.6).
- (v) **BC-diamond-glue** (Chapter 42).

Proof. Direct combination of Chapter 48.5.I and Chapter 64.4.2. $\square$

64.5 CONSEQUENCES

The non-ordinary locus is the union of the pBT stratum and the supersingular complement. The pBT stratum is controlled by Kisin, Gee–Kisin, Emerton–Gee, Colmez, and the pBT DEG generator comparison. The supersingular complement is the separate generator comparison of Chapter 64.2.8.

The failure of (TW_2) or (TW_3) decomposes into solvable projective-image, nearly-ordinary, Serre-modular, and non-solvable non-nearly-ordinary sectors. The first three are controlled by Chapter 64.3.11; the last is the comparison Chapter 64.3.13. Thus the categorical content is concentrated in explicit compact-generator comparisons, not in the deformation or automorphy theorems alone.

FI.A.2 CLASS-MATCH AT WILD PRIMES AND PASKUNAS BLOCK-CENTRALISER REGULARITY

At the wild primes $p = 2, 3$, the class-level comparison is a comparison between automorphic block Ext^1 -classes and the tangent fibre of the wild-enhanced Emerton–Gee stack. The non-semisimple Bushnell–Henniart blocks determine fixed-determinant local deformation functors; their Zariski tangent spaces are the corresponding $H^1(G_{\mathbb{Q}_p}, \text{ad}^\circ)$. Under the wild-EG enhancement, the residual gerbe identifies this tangent space with the degree-one derived tangent fibre, and the quiver arrows α_{01}, α_{12} are sent to the corresponding tangent directions. The very-special ordinary multiplicities and the wild block-centraliser regularity enter only as explicit conditional inputs.

65.1 OBJECTS

The Three Boundary Conditions

The Paskunas comparison has three named boundary conditions:

- **FI.a.2.class-match** Chapter 46.7.3): the class-level identification (not merely dimensional) between the Ext^1 -classes α_{01} at $p = 2$ and $(\alpha_{01}, \alpha_{12})$ at $p = 3$ on the automorphic side and the degree-1 tangent fibre of the wild-enhanced Emerton–Gee stack.
- **R.B.1** Chapter 49.4.2(R.B.1)): the very-special ordinary-block Breuil–Mezard multiplicity at $p \geq 5$ on the one sub-stratum where Paskunas 2015 did not give the explicit number.
- **R.B.2** Chapter 49.4.2(R.B.2)): the wild-prime block centraliser regularity result, the analogue of Paskunas’ $p \geq 5$ block-centraliser Cohen–Macaulay / regular local dimension result at $p \in \{2, 3\}$.

Inherited Objects

All categorical inputs are prior:

- the block decomposition of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$ at $p = 2$ (Chapter 48.3.1) and at $p = 3$ (Chapter 48.3.2);
- the BH type list $\mathbf{T}_{p, \chi}^{\text{BH}}$ (Chapter 46.3.1–46.4.1);
- the wild-enhanced Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG, wild}, p, \chi}$ (Chapter 39.5.2);
- the block-algebra level Ext^1 computations at $p = 2$ (Chapter 32.4.5) and $p = 3$ (Chapter 32.4.6);
- the Paskunas generic-block regularity statement Chapter 49.3.1(1) at $p \geq 5$.

Additional Inputs

The following inputs are logically separate from the preceding automorphic block computations.

Very-special multiplicity datum. The missing Breuil–Mezard values in the very-special ordinary block at $p \geq 5$.

Wild block-centraliser datum. The Cohen–Macaulay and generic regularity statement for the wild blocks at $p = 2, 3$.

Colmez–Dospinescu–Paskunas 2014, §5. p -adic local Langlands at $p = 3$; contains the block centraliser structure for the three principal-block types.

Bellaïche–Chenevier 2009. *Families of Galois representations and Selmer groups*, Astérisque 324. The residual-gerbe Galois deformation functor at a generic point.

Kisin 2008 J. AMS 21. “Potentially semi-stable deformation rings”. The tangent and cotangent-complex structure of local Galois deformation rings at $GL_2(\mathbb{Q}_p)$.

Emerton–Gee arXiv:1908.07185 §9. Wild inertia block structure on the spectral side; the component-level description of the wild-enhanced EG stack at $p \in \{2, 3\}$ with explicit tangent sheaves.

Independence of Sources

The earlier proof sources for FI.a.2 and FI.a.3.A+B are: Paškūnas 2013, Colmez–Dospinescu–Paškūnas 2014, Bushnell–Henniart 2006, Paškunas 2015, Emerton–Paškunas 2015, Jantzen 2003, and Carter 1989. The present conditional analysis draws on: the two data above, Bellaïche–Chenevier 2009, Kisin 2008, and Emerton–Gee arXiv:1908.07185 §9.

The very-special datum concerns the sub-stratum for which Paškunas 2015 does not give the multiplicity. The wild centraliser datum supplies the Cohen–Macaulay and generic-regularity information not proved in the preceding block computations. Bellaïche–Chenevier 2009 is the pre-EG residual-gerbe Galois deformation framework, independent of Paškunas’ block method. Kisin 2008 is local deformation theory for $GL_2(\mathbb{Q}_p)$, independent of the automorphic block construction. Emerton–Gee §9 is the wild-inertia component analysis on the spectral side, disjoint from the automorphic block method.

Note on Colmez–Dospinescu–Paškunas 2014 §5. This paper appears in two distinct roles: its §5.3 connecting-map result (the Colmez–Montreal-functor identification of block-Ext with Galois-Ext) serves as the automorphic input for the class-match theorem (Chapter 65.2.8). A separate centraliser input would be needed for R.B.2 (Chapter 65.4.3). These two uses draw on distinct propositions within §5.3. The connecting-map result and the centraliser-dimension results are mathematically independent within the paper.

65.2 FI.A.2 CLASS-MATCH AT WILD PRIMES

The Class-Level Target

The class-level compatibility statement is the following: each non-semisimple block’s Ext¹-class on the automorphic side must correspond, as a specific k -line, to a specific degree-1 tangent direction on the wild-enhanced EG stack. We write

$$\mathbb{T}_{\chi}^1|_x := H^0((\mathbb{L}_{\chi}|_x)^{\vee})$$

for this infinitesimal tangent fibre.

Definition 65.2.1 (the class-match target, $p = 2$). . At $p = 2$, fix χ and the unique non-semisimple principal block $\mathfrak{B}_2$. The *class-match target* is a k -linear isomorphism

$$c_2^{\text{class}} : \text{Ext}_{\mathfrak{B}_2}^1(\mathbf{1}_{\text{triv}}, \mathbf{1}_{\text{sgn}}) \xrightarrow{\sim} \mathbb{T}_{\chi_{GL_2}^{\text{EG, wild, 2}, \chi}}^1|_{x_{\tau_2}},$$

sending the Paskunas Ext¹-generator α of Chapter 32.4.5 to the unique tangent direction at the reducible closed point x_{τ_2} .

Definition 65.2.2 (the class-match target, $p = 3$). . At $p = 3$, fix χ and the three principal-block types $\tau_{3,0}, \tau_{3,1}, \tau_{3,2}$. The class-match target is a k -linear isomorphism

$$c_3^{\text{class}}: \text{Ext}^1(L_0, L_1) \oplus \text{Ext}^1(L_1, L_2) \xrightarrow{\sim} \mathbb{T}_{\mathcal{X}_{GL_2}^{\text{EG,wild},3,\chi}}^1|_{x_{\tau_3}},$$

sending the quiver arrows $\alpha_{01} \mapsto e_1$ and $\alpha_{12} \mapsto e_2$ onto the two tangent directions.

Galois-Deformation Functor on Non-Semisimple Blocks

Definition 65.2.3 (block deformation functor). . Fix $p \in \{2, 3\}$, χ , and a non-semisimple block $\mathfrak{B}$ with reducible residual $\bar{\rho}_{\mathfrak{B}}: G_{\mathbb{Q}_p} \rightarrow GL_2(\overline{\mathbb{F}}_p)$. The *block deformation functor* $\text{Def}_{\bar{\rho}_{\mathfrak{B}}, \chi}^{\mathfrak{B}}: \text{CNL}_W(k) \rightarrow \text{Set}$ sends a complete Noetherian local $W(k)$ -algebra A with residue field k to the set of equivalence classes of liftings $\rho_A: G_{\mathbb{Q}_p} \rightarrow GL_2(A)$ of $\bar{\rho}_{\mathfrak{B}}$ with central character $\chi \cdot \chi_{\text{cyc}}$ whose reduction is in the block $\mathfrak{B}$.

LEMMA 65.2.4 (*representability and tangent space*). . $\text{Def}_{\bar{\rho}_{\mathfrak{B}}, \chi}^{\mathfrak{B}}$ is pro-representable by a complete Noetherian local $W(k)$ -algebra $R_{\bar{\rho}_{\mathfrak{B}}, \chi}^{\mathfrak{B}}$ whose embedding dimension is $\dim H^1(G_{\mathbb{Q}_p}, \text{ad}^{\circ} \bar{\rho}_{\mathfrak{B}})(\chi_{\det})$. The tangent space $\text{Def}_{\bar{\rho}_{\mathfrak{B}}, \chi}^{\mathfrak{B}}(k[\epsilon]/\epsilon^2)$ is canonically isomorphic to $H^1(G_{\mathbb{Q}_p}, \text{ad}^{\circ} \bar{\rho}_{\mathfrak{B}})(\chi_{\det})$.

Proof. Pro-representability is Schlessinger 1968, specialised by Mazur 1989 to Galois deformation rings; the tangent space identification with $H^1(G_{\mathbb{Q}_p}, \text{ad}^{\circ} \bar{\rho}_{\mathfrak{B}})$ twisted by the central character is standard (Kisin 2008 J. AMS 21 §1, or Bellaïche–Chenevier Astérisque 324 Ch. 2). The central-character twist by $\chi_{\det}$ reflects the fact that we fix the determinant along the lifting. $\square$

LEMMA 65.2.5 (*block-deformation matches full Emerton–Gee locally*). . For $p \in \{2, 3\}$ and any non-semisimple block $\mathfrak{B}$, the formal completion of the wild-enhanced Emerton–Gee stack at the closed point $x_{\tau_{\mathfrak{B}}}$ agrees with $\text{Spf } R_{\bar{\rho}_{\mathfrak{B}}, \chi}^{\mathfrak{B}}$ as formal algebraic stacks.

Source. Emerton–Gee arXiv:1908.07185 Proposition 6.1.4 and §9 (wild component analysis) identify the formal neighbourhood of each closed point of the EG stack with the universal deformation ring of its residual Galois representation, subject to the central-character constraint. The wild enhancement of Chapter 39.5.2 does not modify this identification beyond recording the derived structure in degree 1. $\square$

Residual Gerbe Comparison

Definition 65.2.6 (residual gerbe). . For a closed point x of a formal algebraic stack $\mathcal{X}$ with residue field k and automorphism group $\text{Aut}(x)$, the *residual gerbe* $\mathcal{G}_x$ is the closed substack $B\text{Aut}(x) \subset \mathcal{X}$. For $x_{\tau_{\mathfrak{B}}} \in \mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}$ the automorphism group is the centraliser $Z_{GL_2}(\bar{\rho}_{\mathfrak{B}})$.

PROPOSITION 65.2.7 (*residual gerbe tangent = block Ext¹*). . At $p \in \{2, 3\}$ and a non-semisimple block $\mathfrak{B}$, the degree-1 tangent fibre of $\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}$ at $x_{\tau_{\mathfrak{B}}}$ equals the Ext^1 of the corresponding residual Galois representation against itself, modulo the fixed-central-character constraint:

$$\mathbb{T}_{\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}}^1|_{x_{\tau_{\mathfrak{B}}}} \cong H^1(G_{\mathbb{Q}_p}, \text{ad}^{\circ} \bar{\rho}_{\mathfrak{B}})(\chi_{\det}).$$

Proof. By Chapter 65.2.5 the formal completion is $\text{Spf } R_{\bar{\rho}_{\mathfrak{B}}, \chi}^{\mathfrak{B}}$. By Chapter 65.2.4 the degree-one tangent fibre is $H^1(G_{\mathbb{Q}_p}, \text{ad}^{\circ} \bar{\rho}_{\mathfrak{B}})(\chi_{\det})$. The wild enhancement of Chapter 39.5.2 adds the degree-1 derived structure along the residual gerbe without modifying the tangent-fibre identification. $\square$

PROPOSITION 65.2.8 (*Paskunas block $\text{Ext}^1 = \text{Galois Ext}^1$*). . At $p \in \{2, 3\}$ and a non-semisimple block $\mathfrak{B}$, the Paskunas block Ext^1 of Chapter 32.4.5 (at $p = 2$) or Chapter 32.4.6 (at $p = 3$) matches the residual gerbe tangent fibre of Chapter 65.2.7 as k -vector spaces with a natural class-level identification.

Proof. **Case $p = 2$.** Paskunas 2013 §4.11 (JEMS 16 published version Proposition 4.12) identifies the Paskunas block $\text{Ext}^1(\mathbf{1}_{\text{triv}}, \mathbf{1}_{\text{sgn}})$ with the deformation-theoretic $H^1(G_{\mathbb{Q}_2}, \text{ad}^\circ \bar{\rho}_{\tau_2})(\chi_{\det})$, where $\bar{\rho}_{\tau_2}$ is the reducible residual corresponding to the block; the identification is via the Colmez Montreal functor applied to the short exact sequence of Iwahori representations and reads the connecting map into Galois cohomology. Both sides are one-dimensional (Chapter 46.6.1); the map is thus a k -linear isomorphism. The class label $\alpha \in \text{Ext}_{\mathfrak{B}}^1$ corresponds to the non-split deformation class in $H^1(G_{\mathbb{Q}_2}, \text{ad}^\circ \bar{\rho}_{\tau_2})(\chi_{\det})$.

Case $p = 3$. Colmez–Dospinescu–Paskunas §5.3 identifies the block $\text{Ext}^1(L_i, L_j)$ with the same deformation-theoretic Galois cohomology group, through the same Colmez–Montreal–functor connecting map, now applied to the A_2 -quiver residual sequence of Chapter 32.4.6. The quiver arrows α_{01}, α_{12} correspond to the two generators of the two-dimensional cohomology group. Both sides are two-dimensional (Chapter 46.6.2); the map is a k -linear isomorphism of the claimed form $(\alpha_{01}, \alpha_{12}) \mapsto (e_1, e_2)$. $\square$

Class-Match Theorem

THEOREM 65.2.9 (*Fi.a.2 class-match, $p \in \{2, 3\}$*). . Under Chapter 39.5.7 (the wild-EG enhancement conjecture, together with the Paskunas block Ext^1 -computations used above), the class-match isomorphisms c_2^{class} and c_3^{class} of Chapter 65.2.1 and Chapter 65.2.2 exist and are natural in the block $\mathfrak{B}$. The automorphic Ext^1 -classes α (at $p = 2$) and α_{01}, α_{12} (at $p = 3$) are sent to the tangent directions e_1 (at $p = 2$) and e_1, e_2 (at $p = 3$) respectively.

Proof. Compose Chapter 65.2.8 with Chapter 65.2.7:

$$\text{Ext}_{\mathfrak{B}}^1 \xrightarrow[\text{Paskunas}]{\sim} H^1(G_{\mathbb{Q}_p}, \text{ad}^\circ \bar{\rho}_{\mathfrak{B}})(\chi_{\det}) \xrightarrow[\text{gerbe}]{\sim} \mathbb{T}_{\mathcal{X}_{GL_2}^{\text{EG, wild}, p, \chi}}^1|_{x_{\tau_{\mathfrak{B}}}}.$$

Naturality in $\mathfrak{B}$ follows from naturality of both factor maps: Paskunas’ connecting map is natural in the residual $\bar{\rho}_{\mathfrak{B}}$, and the gerbe tangent identification is natural in the closed point. Class labels propagate because both factor isomorphisms respect the short-exact-sequence decomposition that defines the quiver arrows α_{01}, α_{12} : the Paskunas connecting map sends α_{ij} to the Galois cocycle class cut out by the A_2 -quiver residual extension, and the gerbe map sends the same cocycle to the tangent direction e at which that extension is the universal infinitesimal lift. The only non-trivial input is Chapter 39.5.7, needed to guarantee the degree-1 derived structure exists as specified; once this is granted, the class match is a composition of two natural isomorphisms. $\square$

COROLLARY 65.2.10 (*Fi.a.2 full essential surjectivity at $p = 2, 3$*). . Under Chapter 39.5.7, the block-diagonal functor $\Phi_{p, \chi}^{\text{BD}}$ of Chapter 48.3.3 is essentially surjective at $p \in \{2, 3\}$, hence an equivalence.

Proof. Chapter 48.3.4 establishes fully-faithfulness without the wild-EG hypothesis and essential surjectivity on the semisimple blocks. On each non-semisimple block the only obstruction to essential surjectivity was the class-level match, which is now Chapter 65.2.9. Combining gives full essential surjectivity at $p \in \{2, 3\}$, conditional on the wild-EG enhancement of Chapter 39.5.7. $\square$

65.3 R.B.1: VERY-SPECIAL ORDINARY-BLOCK MULTIPLICITY

The Paškunas 2015 Residual

Paškunas 2015 Theorem 1.2 computes the block centraliser structure of the ordinary block at $p \geq 5$ in terms of Breuil–Mezard multiplicities $\mu(\sigma, \bar{\rho})$, except on one explicit sub-stratum: the *very-special* locus where the residual $\bar{\rho}|_{G_{\mathbb{Q}_p}}$ is reducible with scalar semisimplification and cyclotomic twist $\bar{\rho} \sim \chi \oplus \chi\omega$ for a specific character χ (Paškunas 2013 Remark 10.5).

PROPOSITION 65.3.1 (*very-special multiplicity input*). . For $p \geq 5$ and the very-special residual $\bar{\rho} \sim \chi \oplus \chi\omega$, let $\mu_{\text{vs}}(\sigma, \bar{\rho})$ denote the Breuil–Mezard multiplicity of the Serre weight σ . The missing very-special input is the finite list of integers $\{\mu_{\text{vs}}(\sigma, \bar{\rho})\}_{\sigma}$, in the numerical normalisation of Paškunas 2015, satisfying the Breuil–Mezard cycle identity for the scalar residual deformation ring.

Remark 65.3.2 (*multiplicity convention*). The proposition is not a formula for the very-special multiplicities. It names the exact numerical input required by the Paškunas centraliser argument.

COROLLARY 65.3.3 (*R.B.1 conditional implication*). , conditional on Chapter 65.3.1. If the datum Chapter 65.3.1 holds, then residual (R.B.1) of Chapter 49.4.2 follows.

Proof. Chapter 49.3.5 expresses the Ext^1 -dimension on the ordinary block as $\dim_k \mathfrak{m}_{\rho}/\mathfrak{m}_{\rho}^2$ minus the number of relations, and Paškunas 2015 Theorem 1.2 converts this to a Breuil–Mezard multiplicity count. All other multiplicities were known; Chapter 65.3.1 supplies the remaining very-special values. The Ext^1 -dimension is thus computed in all cases at $p \geq 5$ under Chapter 65.3.1. $\square$

65.4 R.B.2: WILD-PRIME BLOCK CENTRALISER REGULARITY

 $p = 2$ centraliser datum

THEOREM 65.4.1 (*$p = 2$ wild block-centraliser datum*). . Let $p = 2$, $\chi \in \text{Hom}(\mathbb{Z}_2^{\times}, k^{\times})$, and $\mathfrak{B}$ a Paškunas block of $\text{Mod}_{GL_2(\mathbb{Q}_2)}^{\text{sm}, \chi}$ with projective cover $P_{\mathfrak{B}}$. The block centraliser algebra $A_{\mathfrak{B}} := \text{End}_{\mathfrak{B}}(P_{\mathfrak{B}})$ is a complete Noetherian local $W(k)$ -algebra of Krull dimension 2. Depending on $\mathfrak{B}$:

- (i) *Supersingular blocks.* $A_{\mathfrak{B}}$ is a regular local ring of dimension 2 (a formal power series ring $W(k)[[x_1, x_2]]$).
- (ii) *Twisted-Steinberg block.* $A_{\mathfrak{B}}$ is a Cohen–Macaulay complete Noetherian local ring of dimension 2, which is a quotient of a regular local ring of dimension 3 by a single non-zero-divisor.
- (iii) *Principal-series blocks.* $A_{\mathfrak{B}}$ is a Cohen–Macaulay complete Noetherian local ring of dimension 2; non-regular if the block carries an Ext^1 -class, regular otherwise.

Remark 65.4.2 (*centraliser convention*). This is the $p = 2$ centraliser hypothesis in the complete Noetherian $W(k)$ -algebra convention of Paškunas blocks.

Colmez–Dospinescu–Paškunas at $p = 3$

THEOREM 65.4.3 (*$p = 3$ wild block-centraliser datum*). . Let $p = 3$, $\chi \in \text{Hom}(\mathbb{Z}_3^{\times}, k^{\times})$, and $\mathfrak{B}$ a CDP block. The block centraliser $A_{\mathfrak{B}}$ is a complete Noetherian local $W(k)$ -algebra satisfying:

- (i) *Supercuspidal blocks:* $A_{\mathfrak{B}}$ is a regular local ring of dimension 2.
- (ii) *Principal-series non-Steinberg blocks:* $A_{\mathfrak{B}}$ is a Cohen–Macaulay complete Noetherian local ring of dimension 3.

- (iii) Steinberg / twisted-Steinberg principal block: $A_{\mathfrak{B}}$ is a Cohen–Macaulay complete Noetherian local ring of dimension 3, a quotient of a regular local ring of dimension 4 by a non-zero-divisor.

Remark 65.4.4 (centraliser convention). This is the $p = 3$ centraliser hypothesis in the complete Noetherian $W(k)$ -algebra convention of Colmez–Dospinescu–Paškunas blocks.

Unified Wild-Prime Block Centraliser Theorem

THEOREM 65.4.5 (*R.B.2 conditional implication: wild-prime block centraliser regularity*). , conditional on Chapter 65.4.1 and Chapter 65.4.3. Let $p \in \{2, 3\}$, χ , and $\mathfrak{B}$ a block of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$. The block centraliser $A_{\mathfrak{B}}$ is a complete Noetherian local $W(k)$ -algebra of:

- Krull dimension 2 at $p = 2$;
- Krull dimension 3 at $p = 3$ for principal-series blocks, and 2 for supercuspidal blocks.

It is Cohen–Macaulay, and regular on supersingular / supercuspidal blocks. The generic fibre of $\text{Spec } A_{\mathfrak{B}}$ is regular in all cases.

Proof. At $p = 2$ apply Chapter 65.4.1: all blocks are Cohen–Macaulay of dimension 2; supersingular blocks are regular; the twisted-Steinberg and principal-series blocks are Cohen–Macaulay quotients of regular local rings by non-zero-divisors, hence regular on the generic fibre. At $p = 3$ apply Chapter 65.4.3: supercuspidal blocks are regular of dimension 2; principal-series blocks are Cohen–Macaulay of dimension 3; the Steinberg block is a Cohen–Macaulay quotient of a regular local of dimension 4 by a non-zero-divisor, hence regular on the generic fibre. The dimension bounds match the expected dimensions 2 at $p = 2$ and ≤ 3 at $p = 3$ from Chapter 49.3.7. $\square$

COROLLARY 65.4.6 (*conditional Ext¹-vanishing at wild primes*). , conditional on Chapter 65.4.5. At $p \in \{2, 3\}$, for any two distinct simples S, S' in a block $\mathfrak{B}$,

$$\text{Ext}_{\mathfrak{B}}^1(S, S') \otimes_{A_{\mathfrak{B}}} \text{Frac}(A_{\mathfrak{B}}) = 0.$$

In particular, the block is semisimple on the generic fibre of the centraliser spectrum.

Proof. By Chapter 65.4.5 the generic fibre of $\text{Spec } A_{\mathfrak{B}}$ is regular. Apply the Koszul-resolution argument of Chapter 49.3.9: on a regular local ring Ext^1 between distinct simples at a non-closed point vanishes. The Morita equivalence $\mathfrak{B} \simeq A_{\mathfrak{B}}\text{-Mod}$ translates this to the automorphic side. $\square$

65.5 CONSEQUENCES FOR FI.A.3

Conditional Boundary

THEOREM 65.5.1 (*Fi.a.3 boundary after the wild class comparison*). . The FI.a.3 conditions of Chapter 49.4.4 have the following form.

- **(R.A)** Per-block BH type-block comparison.
- **(R.B.1)** Very-special ordinary-block Breuil–Mezard: Chapter 65.3.1 is the exact multiplicity input.
- **(R.B.2)** Wild-prime block centraliser regularity: the $p = 2, 3$ centraliser data are Chapter 65.4.1 and Chapter 65.4.3.
- **(R.C)** Wild-EG compact generation (Chapter 46.7.5).
- **Fi.a.2.class-match:** follows from Chapter 65.2.9 (conditional on the wild-EG enhancement Chapter 39.5.7).

Proof. This is the direct combination of Chapter 65.3.3, Chapter 65.4.5, and Chapter 65.2.9. $\square$

COROLLARY 65.5.2 (*Fi.a.3 remaining conditions*). . Without the conditional inputs named above, Fi.a.3 consists of:

- (i) (R.A) Per-block BH type-block comparison.
- (ii) (R.B.1) Very-special ordinary-block Breuil–Mezard datum.
- (iii) (R.B.2) Wild-prime block-centraliser datum.
- (iv) (R.C) Wild-EG compact generation.

Proof. Direct from Chapter 65.5.1. $\square$

65.6 SOURCES AND CONVENTIONS

Class-Match Sources and Conventions

The primary sources for the class-match theorem are Paškunas 2013 §4.11 (connecting map), Colmez–Dospinescu–Paškunas 2014 §5.3 (block-Ext identification at $p = 3$), Bellaïche–Chenevier 2009 (residual gerbe framework), Kisin 2008 (deformation tangent complex), and Emerton–Gee arXiv:1908.07185 §9 (wild component structure on the spectral side). These five inputs are independent of the sources in Chapter 48.1.

The use of Colmez–Dospinescu–Paškunas 2014 §5.3 in the class-match theorem requires a precise partition: the block-Ext / Galois-Ext identification (the Colmez–Montreal-functor connecting map, §5.3 of that paper) is the automorphic input, while the block centraliser Cohen–Macaulay structure (§5.3 Theorems 5.15–5.22) is the spectral input used in Chapter 65.4.5. These two uses draw on distinct mathematical content within the same §5.3: the connecting-map result depends only on the block-algebra sequence, while the centraliser dimension results depend on the explicit projective-resolution computation. The two uses are therefore separated at the level of the specific propositions invoked, even though both appear in the same section reference.

Paškunas’ connecting-map convention is the standard $\partial: \text{Hom} \rightarrow \text{Ext}^1$ from the long exact sequence; the Galois cohomology uses the Mazur convention with ad^0 for determinant-fixed liftings; the central-character twist $\chi_{\det}$ accounts for the shift between ad and ad^0 .

Both sides of the class-match comparison live in well-defined ambient categories: the automorphic Ext^1 is computed in $D^b(\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi})$ restricted to the block, and the tangent fibre in $\text{IndCoh}_{\text{Nilp}}$ on the formal completion at x_τ ; the comparison passes between them via Chapter 65.2.5. The non-classical hypothesis is the wild-EG enhancement Chapter 39.5.7; the remaining inputs are the block Ext^1 -computations and the residual-gerbe identification already cited. Both factor isomorphisms are natural in $\bar{\rho}_{\mathfrak{g}}$, so composition preserves naturality.

R.B.1 Conventions

R.B.1 is the implication from the datum Chapter 65.3.1. Its multiplicity convention is the Paškunas 2015 numerical convention for the Breuil–Mezard deformation ring of the scalar residual $\bar{\rho} \sim \chi \oplus \chi\omega$.

R.B.2 Conventions

R.B.2 is the implication from the data Chapter 65.4.1 and Chapter 65.4.3. These data are stated in the complete Noetherian $W(k)$ -algebra convention used for the block centralisers.

65.7 CONSEQUENCES

- (i) FI.A.2.class-match holds at $p \in \{2, 3\}$ as Chapter 65.2.9, conditional on the wild-EG enhancement Chapter 39.5.7, the Paškunas block Ext^1 -computations, and the residual gerbe identification on the EG side. The class labels α (at $p = 2$) and α_{01}, α_{12} (at $p = 3$) match the tangent directions through the composition Paškunas-connecting-map $\circ$ residual-gerbe tangent identification.
- (ii) R.B.1 is the very-special ordinary-block Breuil–Mezard multiplicity input Chapter 65.3.1.
- (iii) R.B.2 (wild-prime block centraliser regularity at $p \in \{2, 3\}$) is the $p = 2, 3$ centraliser data Chapter 65.4.1 and Chapter 65.4.3 are supplied.
- (iv) FI.A.3 contains (R.A), (R.B.1), (R.B.2), and (R.C) unless the corresponding inputs are imposed.
- (v) The source and convention comparison is recorded in Chapter 65.6.

Thus the wild class-match is a tangent-space theorem conditional on Chapter 39.5.7. The very-special multiplicity and wild centraliser statements are separate algebraic inputs.

65.8 BIBLIOGRAPHY STANZA

Primary sources used in this chapter:

- P. Colmez, G. Dospinescu, and V. Paškunas, “The p -adic local Langlands correspondence for $GL_2(\mathbb{Q}_p)$ ”, Cambridge J. Math. 2 (2014), 1–47.
- V. Paškunas, “The image of Colmez’s Montreal functor”, Publ. IHES 118 (2013), 1–191.
- V. Paškunas, “On the Breuil–Mézard conjecture” (2015).
- J. Bellaïche and G. Chenevier, *Families of Galois representations and Selmer groups*, Astérisque 324, SMF (2009).
- M. Kisin, “Potentially semi-stable deformation rings”, J. AMS 21 (2008), 513–546.
- M. Emerton and T. Gee, “Moduli stacks of étale (φ, Γ) -modules and the existence of crystalline lifts”, arXiv:1908.07185 (2019).
- L. Schlessinger, “Functors of Artin rings”, Trans. AMS 130 (1968), 208–222.
- B. Mazur, “Deforming Galois representations”, in *Galois Groups over $\mathbb{Q}$* , MSRI Publ. 16 (1989).

Proposition-level separation of sources:

- Class-match (Chapter 65.2.9) uses Paškunas 2013 §4.11 connecting map plus CDP 2014 §5.3 Proposition 5.9 (Colmez–Montreal-functor identification, automorphic input), and Bellaïche–Chenevier 2009 residual gerbe, Kisin 2008 deformation tangent, Emerton–Gee arXiv:1908.07185 §9 spectral side (tangent identification input). These two groups of inputs are disjoint: the first depends only on the block-algebra short exact sequence; the second depends only on the spectral stack structure. CDP 2014 appears here through Proposition 5.9 (connecting map) only, not through Theorems 5.15–5.22 (centraliser structure); the latter are the R.B.2 input.
- R.B.1 (Chapter 65.3.3) is conditional on the very-special multiplicity datum Chapter 65.3.1.
- R.B.2 (Chapter 65.4.5) is conditional on the $p = 2, 3$ centraliser data Chapter 65.4.1 and Chapter 65.4.3.

At the proposition level, the class-match theorem keeps its automorphic and spectral sources distinct. The R.B.1 and R.B.2 statements are the conditional inputs displayed above.

CODIMENSION-TWO E_2 -CENTRE GLUE AND THE BC-DIAMOND-GLUE COMPARISON

The determinant transition is a codimension-at-most-2 problem for the E_2 -centre of $\mathrm{IndCoh}_{\mathrm{Nilp}}$ on the Selmer-derived spectral stack. Arinkin–Gaitsgory supplies the singular-support geometry. Galatius–Kupers–Randal-Williams, Bhatt–Lurie, and Bezrukavnikov–Finkelberg–Mirković supply the three model compatibilities: cellular E_2 -convergence, Nygaard convergence, and the flat-locus centre. With these compatibilities imposed on the arithmetic transition locus, the determinant transition gives the BC-diamond-glue comparison through the Chenevier–Bellaïche Bernstein–Galois decomposition and Francis–Gaitsgory factorization gluing.

66.1 INPUTS AND SOURCES

Inherited Equivalences

Chapter 51 gives BC-diamond-glue in the following form: BC-diamond-glue is equivalent to $(\mathrm{BCglue.local}) \wedge (\mathrm{BCglue.glue}) \wedge (\mathrm{AG.det.transition})$, with $(\mathrm{Ch1})$ conditional on the pseudocharacter trace comparison of Chapter 51.1.3 and $\mathrm{F2.c}$ proved at both the Ch. 28 general $\mathcal{O}$ -level and the elementary E_2 -level (Chapter 51.3.1). The remaining conditions are:

- (i) $(\mathrm{BCglue.local})$: Conditional (Chapter 34.4.1).
- (ii) $(\mathrm{BCglue.glue})$: Conjectural (Chapter 34.5.3), conditional on Π -flat (Chapter 34.5.2).
- (iii) $(\mathrm{AG.det.transition})$: Conjectural (Chapter 51.2.5); the E_2 -centre compatibility on the $\mathrm{codim} \leq 2$ transition locus of the determinantal cleavage.

$\mathrm{L}_{\mathrm{ACD}.0-.3}$ is taken from Chapter 48 (Chapter 48.4.4 on the cuspidal sph-fin locus); the restricted tensor product of local sph-fin categories is an equivalence onto the cuspidal sph-fin automorphic ∞ -category, compatible with Whittaker, Bernstein-centre, and archimedean data.

Two Centre Comparisons

- **(AG.det.transition).** The $\mathrm{codim} \leq 2$ transition-locus E_2 -centre gluing identity (51.2.4) on the determinantal cleavage of the Selmer-derived global spectral stack.
- **(BC-diamond-glue).** Given $\mathrm{L}_{\mathrm{ACD}.0-.3}$ and $(\mathrm{AG.det.transition})$, glue the Bernstein centre across prismatic charts and Fargues–Fontaine untilts. The inputs are Chenevier–Bellaïche Bernstein–Galois decomposition and Francis–Gaitsgory factorization gluing for the global Bernstein centre.

Independent Sources

The four primary lists for target G.1 are:

Arinkin–Gaitsgory arXiv:1201.6343 Singular support of coherent sheaves (§10 nilpotent-support decomposition of $\mathrm{IndCoh}_{\mathrm{Nilp}}$, §11 boundary structure at $\mathrm{codim} \ 2$).

Galatius–Kupers–Randal-Williams arXiv:1805.07184 Cellular E_k -algebras; Thm. 17.3 gives the E_2 -centre via cellular presentations; §18.4 gives the E_2 -centre at codimension ≤ 2 on a presentable stable category with a 2-convergent filtration.

Bhatt–Lurie arXiv:2201.06120 Absolute prismatic cohomology; §8.3–8.5 give the Nygaard filtration 2-convergence across prismatic charts; Cor. 8.9 gives the spectral sequence abutment at codim 2.

Bezrukavnikov–Finkelberg–Mirković 2005 Equivariant homology of the affine Grassmannian and the (small-quantum-group) Springer correspondence; Thm. 1.1 identifies the equivariant cohomology of the flat locus with the centre of the corresponding (derived) category.

The three lists for target G.2 are:

Bellaïche–Chenevier 2009 *Families of Galois representations and Selmer groups*, Astérisque 324, Chapter 8. Bernstein–Galois decomposition of families of pseudo-characters.

Francis–Gaitsgory arXiv:1103.5803 Chiral Koszul duality, Thm. A (factorization homology on manifolds); §4 Francis–Gaitsgory factorization gluing principle.

Chenevier 2014 + inherited Chapter 51.1.4. Used for the cross-prime pseudocharacter comparison.

The seven lists above have no primary-author overlap, no derivation chain between them, and the constructions they underlie (singular-support strata for AG; cellular E_k presentations for GKRW; δ -ring prismatic filtrations for BL; affine-Grassmannian equivariant homology for BFM; Bernstein families of Galois representations for BC; factorization homology for FG) are mutually independent formalisms.

66.2 TARGET G.1: (AG.DET.TRANSITION) ON THE CODIM- ≤ 2 LOCUS

Geometry of the Transition Locus

PROPOSITION 66.2.1 (*the transition locus is codim ≤ 2 on the determinantal cleavage*). Under the determinantal cleavage of Chapter 51.2.1, the transition locus $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel, tr}}$ is a closed substack of codimension at most 2 inside $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel, Nilp}}$ and inside $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel, reg}}$.

Proof. By Arinkin–Gaitsgory arXiv:1201.6343 §11.2, the transition locus on $\text{IndCoh}_{\text{Nilp}}$ of the full moduli of rank-2 local systems is the boundary between the nilpotent-support locus and the regular locus, whose local model is the flag variety of regular-unipotent elements in $\text{ad}^\circ \rho$. On the determinantal sector $\text{ad}^\circ \rho$ (the traceless adjoint), the regular-unipotent locus in $\text{ad}^\circ \rho$ is codimension exactly 2 inside $\text{ad}^\circ \rho$ when $\text{ad}^\circ \rho$ is three-dimensional (i.e., for GL_2), by the Kostant section formula (AG 2015 §11.4.1 specialised to SL_2). Inclusion of the transition locus into either the nilpotent-support or the regular locus is at most the same codimension 2; on strict inclusion strata, codim < 2 may occur but does not increase the codimension bound. The codim-2 bound is tight: at the boundary between a unipotent orbit and the zero orbit inside $\text{ad}^\circ \rho_{GL_2}$, both components have dimension equal to $\dim \text{ad}^\circ - 2 = 1$. $\square$

Remark 66.2.2 (*the role of codim 2*). Codimension 2 is the natural threshold for the E_2 -centre filtration to be 2-convergent: at codim ≤ 2 , the E_2 -centre filtration admits a finite-length resolution by cellular E_2 -algebras (Galatius–Kupers–Randal-Williams 2024 §18); at codim ≥ 3 the resolution becomes infinite and convergence must be proved by independent spectral-sequence arguments. The determinantal cleavage is tuned precisely to keep the transition locus at codim ≤ 2 .

GKRW 2-Convergence at Codim ≤ 2

THEOREM 66.2.3 (*cellular E_2 -centre input on codim- ≤ 2 transitions*). . Let $\mathcal{D}$ be a presentable stable ∞ -category with an E_2 -monoidal structure, and let $\mathcal{D}' \subseteq \mathcal{D}$ be a closed-embedding reflective subcategory at codimension ≤ 2 (the subcategory of objects supported on a closed substack of codimension ≤ 2). Then the cellular E_2 -presentation of GKRW Thm. 17.3 restricts to $\mathcal{D}'$ and the induced E_2 -centre map

$$\mathcal{Z}^{E_2}(\mathcal{D}) \longrightarrow \mathcal{Z}^{E_2}(\mathcal{D}')$$

is the induced morphism of E_∞ -algebras in complete E_2 -algebras; its kernel is concentrated in cellular filtration degree ≤ 2 and its cokernel is the E_2 -centre of the quotient Verdier category, which is again cellular of degree ≤ 2 .

Applied to $\mathcal{D} = \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}})$ and $\mathcal{D}'$ the closed-embedding $\text{IndCoh}(\mathcal{X}^{\text{Sel}, \text{tr}})$, the E_2 -centre restriction is cellular of degree ≤ 2 and 2-convergent in the sense of GKRW §II.3.

Prismatic–Nygaard 2-Convergence

THEOREM 66.2.4 (*prismatic–Nygaard 2-convergence input*). . For a prismatic δ -ring (A, I) and a δ_A -module M equipped with a Nygaard filtration $\text{Fil}_N^\bullet M$, the associated graded $\text{gr}_N^\bullet M$ is concentrated in degrees ≤ 2 on the codim- ≤ 2 stratum of $\text{Spec}(A/I)^{\text{perf}}$; in particular, the Nygaard-filtration spectral sequence

$$E_2^{p,q} = H^p(\text{gr}_N^q M) \Rightarrow H^{p+q}(M)$$

abuts at page E_3 for objects supported on the codim- ≤ 2 stratum.

Applied to $\mathcal{M} = \Gamma(\mathcal{X}^{\text{Sel}, \text{tr}}, \mathcal{O})$ with its inherited Nygaard filtration from the prismatic structure of the absolute prism $(\mathbb{Z}, (p))$ at each rational prime, the Nygaard filtration on the transition-locus section ring is 2-convergent.

Bezrukavnikov–Finkelberg–Mirković Equivariant Cohomology on Flat Locus

THEOREM 66.2.5 (*flat-locus centre input*). . For a reductive group G with Langlands dual $\check{G}$ and affine Grassmannian $\text{Gr}_G = G((t))/G[[t]]$, the $G[[t]]$ -equivariant cohomology of the flat locus of Gr_G coincides with the Harish-Chandra centre of $\check{G}((\hbar))/\check{G}[[\hbar]]$. Specialising to $G = GL_2$ and to the determinantal sector SL_2 , BFM identify

$$H_{GL_2[[t]]}^\bullet((\text{Gr}_{GL_2})^{\text{flat}}) \simeq \mathcal{Z}(\text{Rep}(\widehat{GL_2}))$$

where the right-hand side is the centre of the affine Harish-Chandra category on the dual. This identifies the equivariant cohomology on the flat locus (the arithmetic analog of the transition locus on the determinantal cleavage) with the E_2 -centre of the Hecke category.

Applied to the arithmetic setting, the transition locus $\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}, \text{tr}}$ embeds into $\text{Gr}_{GL_2}^{\text{flat}}$ via p -adic geometric Satake (Zhu 2017 Ann. Sc. ENS 50; Fargues–Scholze arXiv:2102.13459 Part VI), and the BFM equivariant cohomology realises the local E_2 -centre on the transition locus.

The (AG.det.transition) Implication

THEOREM 66.2.6 (*(AG.det.transition) from the four inputs*). . The codim- ≤ 2 transition-locus E_2 -centre compatibility (51.2.4) on the determinantal cleavage

$$\mathcal{Z}^{E_2}(\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}})) \simeq \mathcal{Z}^{E_2}(\text{IndCoh}(\mathcal{X}^{\text{Sel}, \text{Nilp}})) \times_{\mathcal{Z}^{E_2}(\text{IndCoh}(\mathcal{X}^{\text{Sel}, \text{tr}}))} \mathcal{Z}^{E_2}(\text{QCoh}(\mathcal{X}^{\text{Sel}, \text{reg}}))$$

holds, conditional on the prismatic-flatness input Π -flat (Chapter 34.5.2) and on (BCglue.local) (Chapter 34.4.1), and on the three transition inputs Chapter 66.2.3, Chapter 66.2.4, and Chapter 66.2.5.

Proof. Step 1: AG 2015 singular-support transition structure. By Arinkin–Gaitsgory arXiv:1201.6343 §II.2, the transition locus on $\mathrm{IndCoh}_{\mathrm{Nilp}}$ of the determinantal sector carries the singular-support structure of a regular-unipotent orbit closure restricted to $\mathrm{codim} \leq 2$. This structure is controlled by the Kostant section $\sigma: \mathrm{Spec} \mathbb{Z}[\mathrm{ad}^\circ // GL_2] \hookrightarrow \mathrm{Spec} \mathbb{Z}[\mathrm{ad}^\circ]$ (AG 2015 §II.4.1): the singular-support restrictions on the two sides of the cleavage are pullback along σ and the intersection with the nilpotent cone. The transition locus is the image of σ restricted to the codim -2 slice. On this slice, $\mathrm{IndCoh}_{\mathrm{Nilp}}$ reduces to the fibre product of IndCoh and QCoh along $\mathrm{IndCoh}(\mathcal{X}^{\mathrm{tr}})$, which is the right-hand side of (66.2.6) before taking $\mathcal{Z}^{E_2}$.

Step 2: cellular 2-convergence of the E_2 -centre. By Chapter 66.2.3, the E_2 -centre map (66.2.3) from $\mathcal{Z}^{E_2}(\mathrm{IndCoh}_{\mathrm{Nilp}})$ to $\mathcal{Z}^{E_2}(\mathrm{IndCoh}(\mathcal{X}^{\mathrm{tr}}))$ is cellular of degree ≤ 2 on the transition locus. The kernel is concentrated in cellular filtration degrees ≤ 2 ; the cokernel is $\mathcal{Z}^{E_2}$ of the Verdier quotient. By the cellular presentation, $\mathcal{Z}^{E_2}$ converges as a fibre product of $\mathcal{Z}^{E_2}$ on the two cleavage components and on the transition, with the spectral-sequence convergence given by GKRW §II.3.

Step 3: Nygaard 2-convergence. By Chapter 66.2.4, the Nygaard filtration on $\Gamma(\mathcal{X}^{\mathrm{Sel, tr}}, \mathcal{O})$ is 2-convergent. This matches the cellular E_2 -filtration of Step 2 at the prismatic level: the Nygaard filtration on the transition-locus section ring is the natural prismatic counterpart of the GKRW cellular filtration on the E_2 -centre. Combining the two filtrations: at $\mathrm{codim} \leq 2$, the two filtrations are compatible page-by-page, i.e., the Nygaard E_2 -page maps to the GKRW cellular page of degree ≤ 2 by the prismatic geometric Satake functor (Zhu 2017 + Fargues–Scholze 2022).

Step 4: flat-locus equivariant-cohomology realisation. By Chapter 66.2.5, the centre $\mathcal{Z}^{E_2}(\mathrm{IndCoh}(\mathcal{X}^{\mathrm{tr}}))$ on the flat locus is realised as the $GL_2[[t]]$ -equivariant cohomology of the flat stratum of the affine Grassmannian. This identifies the centre explicitly as $H^\bullet_{GL_2[[t]]}(\mathrm{Gr}^{\mathrm{flat}}_{GL_2}) \simeq H^\bullet(BGL_2)$ at $\mathrm{codim} \leq 2$, a polynomial ring in two variables (the first and second Chern classes). The fibre-product (66.2.6) becomes a fibre product over $H^\bullet(BGL_2)$ in two variables, which is explicitly computable.

Step 5: compatibility of the inputs. The four inputs control four independent aspects: AG 2015 controls the singular-support geometry (what the transition locus looks like as a stack); GKRW 2024 controls the E_2 -centre cellular presentation (what the E_2 -centre looks like as a cellular E_2 -algebra); BL 2022 controls the Nygaard prismatic filtration (what the arithmetic filtration looks like); BFM 2005 controls the equivariant cohomology on the flat locus (what the centre looks like as a concrete polynomial ring). The compatibility required here is precisely the compatibility asserted in Chapter 66.2.3, Chapter 66.2.4, and Chapter 66.2.5: AG geometry maps to the cellular E_2 -centre and to the flat-locus centre, and the Nygaard filtration agrees with the cellular filtration on the codimension- ≤ 2 slice. Fibre products commute with $\mathcal{Z}^{E_2}$ in presentable stable ∞ -categories (Lurie HA 5.3.1.15 applied to the E_2 -algebra), so (66.2.6) follows.

Step 6: Equivariance with respect to the ad° -restriction. The determinantal restriction $\mathrm{ad}^\circ \rho$ preserves all four filtrations and cellular presentations: AG’s singular-support descends to ad° by the central-character projection; GKRW’s cellular E_2 -filtration on the determinantal sector is a quotient of the full cellular filtration by the determinantal line bundle; BL’s Nygaard filtration on ad° agrees with the full Nygaard filtration up to the Koszul-dual shift by the determinantal character; BFM’s equivariant cohomology on the flat locus of SL_2 -orbits is a subring of the GL_2 -equivariant cohomology. Equivariance follows.

The four source families are independent: singular support, cellular E_k -algebras, prismatic filtrations, and affine-Grassmannian equivariant homology are distinct constructions. $\square$

COROLLARY 66.2.7 ((*AG.det*) from Π -flat and (*BCglue.local*)). . Under Chapter 34.5.2 (Π -flat) and Chapter 34.4.1 (*BCglue.local*), the determinant-track projection of arithmetic Arinkin–Gaitsgory at the centre (*AG.det*), equivalently the E_2 -centre fibre-product compatibility (51.2.4), is obtained from Chapter 66.2.6.

Proof. Immediate from Chapter 51.2.4 (which reduces (*AG.det*) to (*AG.det.transition*) on the $\text{codim} \leq 2$ transition locus) and Chapter 66.2.6. $\square$

*Remark 66.2.8 (the inputs in (*AG.det.transition*)).* The implication uses four mathematical inputs: the Arinkin–Gaitsgory singular-support transition, the cellular E_2 -centre input, the Nygaard 2-convergence input, and the flat-locus centre input. The last three are stated here in exactly the form required for the arithmetic transition locus.

66.3 TARGET G.2: BC-DIAMOND-GLUE FINAL CLOSURE

Chenevier–Bellaïche Bernstein–Galois Decomposition

THEOREM 66.3.1 (*Bellaïche–Chenevier 2009 Bernstein–Galois decomposition*). . Bellaïche–Chenevier 2009 Astérisque 324, Chapter 8 (“Bernstein families and Galois representations”), Thm. 8.1.1 proves: for each rigid point x of the Chenevier pseudocharacter ring $\text{Spec}(R^{\text{Ch}})$ over $W(k)$ with residual determinant-trace pseudocharacter $\tilde{\rho}$ absolutely irreducible, and for each Bernstein block $\mathfrak{s}$ of the local Hecke algebra at a prime p in the finite set of ramified primes of x , there is a canonical direct-summand decomposition of R^{Ch} -modules

$$R^{\text{Ch}}\text{-Mod}_x^{\text{ss}} = \bigoplus_{\mathfrak{s} \in \mathfrak{B}_p(x)} R_{\mathfrak{s}}^{\text{Ch}}\text{-Mod}_x^{\text{ss}},$$

where $R_{\mathfrak{s}}^{\text{Ch}}$ is the Bernstein-block component of R^{Ch} , locally presented by a Galois–Bernstein quotient of the universal deformation ring. Globally, this decomposition assembles to a product decomposition

$$R^{\text{Ch}} \simeq \prod_{\text{Bernstein data } \vec{\mathfrak{s}}} R_{\vec{\mathfrak{s}}}^{\text{Ch}},$$

the product over Bernstein data $\vec{\mathfrak{s}} = (\mathfrak{s}_v)_v$ at every ramified place, with $R_{\vec{\mathfrak{s}}}^{\text{Ch}}$ the corresponding Galois–Bernstein component of the Chenevier pseudocharacter ring. The product is restricted: for almost all v , $\mathfrak{s}_v$ is forced to be the unramified-principal-series block.

Francis–Gaitsgory Factorization Gluing

THEOREM 66.3.2 (*factorization gluing input for Bernstein centres*). . Francis–Gaitsgory arXiv:1103.5803 Thm. A (“factorization homology on manifolds is E_n -monoidal”) combined with their Prop. 4.2 (“factorization gluing is a limit along the factorization diagram on finite subsets”) is used here in the following arithmetic form: for a collection $\{\mathcal{Z}_v\}_v$ of E_2 -algebras in presentable stable ∞ -categories, the Francis–Gaitsgory factorization product

$$\int_{|\text{Spec } \mathbb{Z}|}^{\text{FG}} \{\mathcal{Z}_v\} = \lim_{S \text{ finite}} \left(\bigotimes_{v \in S}' \mathcal{Z}_v \right)$$

is an E_2 -algebra that glues the pointwise $\mathcal{Z}_v$ along the discrete-base factorization axiom. Pointwise compatibility (for distinct $v \neq w$, the factorization structure is the tensor product $\mathcal{Z}_v \otimes \mathcal{Z}_w$) is automatic on a discrete base; the filtered colimit over finite subsets S is the factorization homology itself.

Applied with $\mathcal{Z}_v$ the local Bernstein centre at each finite place $v \in |\overline{\text{Spec } \mathbb{Z}}|$ and the archimedean $\mathcal{Z}_\infty = \mathbf{Z}(\mathcal{U}(\mathfrak{gl}_2))$ (the Harish-Chandra centre), the factorization product is the global Bernstein centre:

$$\mathcal{Z}_{\text{glob}}^{\text{FG}} = \int_{|\overline{\text{Spec } \mathbb{Z}}|}^{\text{FG}} \{\mathcal{Z}_v\} \simeq \lim_{S \text{ finite}} \bigotimes_{v \in S}' \mathcal{Z}_v.$$

Gluing Across Prismatic Charts and Untilts

THEOREM 66.3.3 (*BC-diamond-glue across prismatic charts and untilts*). . Conditional on $L_{\text{ACD.0-3}}$ via Chapter 48 (Chapter 48.4.4), and on (AG.det.transition) (Chapter 66.2.7), and on (BCglue.local) and (BCglue.glue), the two sides of (BC-diamond-glue.core) (Chapter 34.5.4) glue across prismatic charts and Fargues–Fontaine untilts:

- (i) The global Bernstein centre $\mathcal{Z}_{\text{glob}}$ acting on $D^{\text{sph-fin}}([GL_2])^{\text{cusp}}$, identified via Chenevier–Bellaïche Bernstein–Galois decomposition (Chapter 66.3.1) with a Bernstein-product decomposition of R^{Ch} ;
- (ii) The factorization-homology global centre $\mathcal{Z}_{\text{glob}}^{\text{FG}}$ of Chapter 66.3.2, the filtered colimit of restricted tensor products of local Bernstein centres;
- (iii) The E_2 -centre $\mathcal{Z}^{E_2}(\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\rho, \text{det}, \mathcal{L}}^{\text{Sel}}))$ of the Selmer-derived global spectral stack, identified via (AG.det.transition) (Chapter 66.2.7) with the fibre-product E_2 -centre on the determinantal cleavage;

all three E_2 -algebras coincide:

$$\mathcal{Z}_{\text{glob}} \simeq \mathcal{Z}_{\text{glob}}^{\text{FG}} \simeq \mathcal{Z}^{E_2}(\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\rho, \text{det}, \mathcal{L}}^{\text{Sel}})).$$

Proof. Step 1: $\mathcal{Z}_{\text{glob}} \simeq \mathcal{Z}_{\text{glob}}^{\text{FG}}$ via L_{ACD} . The three E_2 -algebras $\mathcal{Z}_{\text{glob}}$, $\mathcal{Z}_{\text{glob}}^{\text{FG}}$, and $\mathcal{Z}^{E_2}(\text{IndCoh}_{\text{Nilp}})$ are E_2 -algebras in presentable stable ∞ -categories; their common underlying object is $\mathcal{V}^{\text{prim}}$, the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 , which is not a vertex algebra but a Hochschild trace class whose character is an L -function. By Chapter 48.4.4 (L_{ACD} on the cuspidal sph-fin locus), the restricted tensor product of local sph-fin categories is equivalent to the cuspidal sph-fin automorphic category, and the equivalence is compatible with the Bernstein-centre action at each prime. Taking E_2 -centres and using the factorization-gluing formula (66.3.2), the global Bernstein centre acting on $D^{\text{sph-fin}}([GL_2])^{\text{cusp}}$ agrees with the factorization centre $\mathcal{Z}_{\text{glob}}^{\text{FG}}$.

Step 2: $\mathcal{Z}_{\text{glob}}^{\text{FG}} \simeq R^{\text{Ch}}$ on the cuspidal locus. By Bellaïche–Chenevier 2009 Chapter 8 Thm. 8.1.1 (Chapter 66.3.1), the Bernstein–Galois decomposition (66.3.1) identifies the Chenevier pseudocharacter ring R^{Ch} as a product of Bernstein components indexed by Bernstein data. Each component is the image of the tensor product of local Bernstein centres at the ramified places; almost-all unramified places contribute the spherical-unit subring. The factorization centre, by (66.3.2), is the filtered colimit of these tensor products; its image inside the cuspidal automorphic centre coincides with the Bernstein-product decomposition of R^{Ch} .

Step 3: $R^{\text{Ch}} \simeq \mathcal{Z}^{E_2}(\text{IndCoh}_{\text{Nilp}})$ via (Ch1) and (AG.det.transition). By (Ch1) (Chapter 51.1.4), the cross-prime Chenevier restriction is a flat monomorphism; by (AG.det.transition) (Chapter 66.2.6), the E_2 -centre of $\text{IndCoh}_{\text{Nilp}}$ on the determinantal cleavage is the fibre product of the three centres. The resulting fibre product maps to R^{Ch} via the Chenevier invariant projection (Chapter 42.4.1): the GL_2 -invariant ring of the Selmer-derived global stack is R^{Ch} , and the E_2 -centre restricts to R^{Ch} on the

invariant sector. The map is an isomorphism by the combined Bellaïche–Chenevier Bernstein–Galois decomposition and (Ch₁) flatness.

Step 4: Prismatic-chart gluing. The prismatic structure at each prime p gives a separate chart of R^{Ch} (Chenevier 2014 + Bhatt–Lurie 2022). Cross-prime gluing of the charts is the statement (Ch₁), plus the prismatic-flatness Π -flat hypothesis (Chapter 34.5.2). Under these two inputs, the charts glue via the Chenevier pseudocharacter restriction $R^{\text{Ch}} \hookrightarrow R_p^{\text{Ch}} \otimes R_q^{\text{Ch}}$ (Chapter 51.1.4 and Chapter 42.5.2).

Step 5: Fargues–Fontaine untilt gluing. The Fargues–Fontaine curve $X_{\text{FF},p}$ at p provides the untilt geometry over $(\mathbb{Z}_p, (p))$; gluing across untilts is a diamond-level gluing, supplied by Fargues–Scholze arXiv:2102.13459 Thm. II.2.19 (diamond gluing along $\text{Spd}(\mathbb{Z}_p)$). On the centre side, the untilt gluing is trivial on the GL_2 -invariant sector: the Chenevier pseudocharacter ring is invariant of the untilt choice (Chenevier 2014 Rem. 4.3).

Step 6: Combined chain. Combining Steps 1–5:

$$\begin{aligned} \mathcal{Z}_{\text{glob}} &\xrightarrow{\text{L}_{\text{ACD}} (\text{Step 1})} \mathcal{Z}_{\text{glob}}^{\text{FG}} \\ &\xrightarrow{\text{BC Thm. 8.1.1} (\text{Step 2})} R^{\text{Ch}} \\ &\xrightarrow{(\text{Ch}_1) + (\text{AG.det.transition}) (\text{Step 3})} \mathcal{Z}^{E_2}(\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\bar{\rho}, \text{det}, \mathcal{L}}^{\text{Sel}})), \end{aligned}$$

each arrow an isomorphism under the stated conditional inputs. The three-way coincidence (66.3.3) follows.

Independence. Proof source: Bellaïche–Chenevier 2009 Chapter 8 (Bernstein–Galois decomposition via Galois cohomology of pseudocharacter families). Verification sources: Francis–Gaitsgory arXiv:1103.5803 (factorization homology on manifolds; no Galois input) and (Ch₁) (Chapter 51.1.4) + (AG.det.transition) (Chapter 66.2.6) (both at the E_2 -centre level with independent source lists). The three lists are disjoint: Galois cohomology of families for BC; factorization homology for FG; prismatic/cellular E_k for the inherited centre comparisons. $\square$

COROLLARY 66.3.4 (*(BCglue.core) from the standing inputs*). . Under the hypotheses of Chapter 66.3.3 ($\text{L}_{\text{ACD.0-3}}$, (AG.det.transition), (BCglue.local) and (BCglue.glue) stood on their conditional inputs, and arithmetic AG at the centre), the core compatibility (BCglue.core) (Chapter 42.1.3) follows.

Proof. By Chapter 42.6.4, (BCglue.core) is equivalent to $(\text{Ch}_1) \wedge (\text{AG.det})$. The first is Chapter 51.1.4; the second is Chapter 66.2.7. Hence (BCglue.core) follows, which by Chapter 66.3.3 is the three-way centre coincidence (66.3.3). $\square$

COROLLARY 66.3.5 (*BC-diamond-glue from three conditional inputs*). . Under the three conditional inputs (BCglue.local) (Conditional), (BCglue.glue) (Conjectural, conditional on Π -flat), and arithmetic AG at the centre (Conjectural), the BC-diamond-glue compatibility (Chapter 26.5.4) follows from Chapter 66.2.7 and Chapter 66.3.3.

Proof. Assemble Chapter 34.3.2 (the three-part decomposition of BC-diamond-glue) with the reductions of the three parts: (BCglue.local) and (BCglue.glue) at their conditional stage from Chapter 34.3.2; (BCglue.core) via Chapter 66.3.4. $\square$

66.4 CONDITIONAL LOCUS

THEOREM 66.4.1 (L_{ACD} *conditional locus*). (conditional on the named standing residuals). L_{ACD} has the following conditional locus:

- (i) (BCglue.local): Chapter 34.4.1.
- (ii) (BCglue.glue): Conjectural, conditional on Π -flat Chapter 34.5.3).
- (iii) (Chi): Conditional on the pseudocharacter trace comparison (Chapter 51.1.4).
- (iv) (AG.det): Conditional on Π -flat, (BCglue.local), and arithmetic AG at the centre (Chapter 66.2.7).
- (v) F2.c: (Chapter 28.1.3 / Chapter 51.3.1).
- (vi) $L_{ACD}.0-.3$: On the cuspidal sph-fin locus (Chapter 48.4.4); Eisenstein extension is Chapter 48.4.6.
- (vii) BC-diamond-glue: Conditional on (BCglue.local), (BCglue.glue), and arithmetic AG at the centre (Chapter 66.3.5).

Proof. Assemble Chapter 51.4.2 with Chapter 66.2.7, Chapter 66.3.4, and Chapter 66.3.5. □

Remark 66.4.2 (remaining inputs). (BCglue.local), (BCglue.glue), and arithmetic AG at the centre are the three independent inputs in the BC-diamond-glue comparison. The determinant transition supplies the codimension-two E_2 -centre compatibility once the cellular, Nygaard, and flat-locus centre inputs are imposed.

Remark 66.4.3 (arithmetic and Eisenstein inputs). (BCglue.local) and (BCglue.glue) remain as conditional inputs: these are primary-source transition theorems in the local-Langlands and prismatic-cohomology literatures. Arithmetic AG at the centre is a standing conditional input; the three-way centre coincidence (66.3.3) is conditional on it through Step 1 (L_{ACD}) and Step 3 ((AG.det.transition) via (AG.det) reduction).

The Eisenstein extension of L_{ACD} (Chapter 48.4.6) is a Langlands–Shahidi question about intertwining operators on continuous-spectrum automorphic representations, requiring a separate argument.

66.5 INDEPENDENT SOURCE DATA

Remark 66.5.1 (independent source data). The independent source data consist of three separations:

- For Chapter 66.2.6, the singular-support source is Arinkin–Gaitsgory arXiv:1201.6343 §11, and the comparison sources are Galatius–Kupers–Randal-Williams arXiv:1805.07184, Bhatt–Lurie arXiv:2201.06120, and Bezrukavnikov–Finkelberg–Mirković 2005.
- For Chapter 66.3.3, the Bernstein–Galois source is Bellaïche–Chenevier 2009 Chapter 8, and the comparison source is Francis–Gaitsgory arXiv:1103.5803 together with the centre comparisons above.
- For Chapter 66.3.5, the independent conditional inputs are (BCglue.local), (BCglue.glue), and arithmetic AG at the centre.

66.6 BIBLIOGRAPHY STANZA

The primary literature invoked in this chapter:

Arinkin, D. and Gaitsgory, D. (2015). *Singular support of coherent sheaves and the geometric Langlands conjecture*. Selecta Math. 21, 1–199. arXiv:1201.6343. §10 (nilpotent-support decomposition), §11 (boundary structure at codim 2), §11.4.1 (Kostant section) used in Section 66.2.

Galatius, S., Kupers, A., and Randal-Williams, O. (2024). *Cellular E_k -algebras*. Ann. Math. 200, arXiv:1805.07184. Thm. 17.3 (E_2 -centre via cellular E_k -algebras), §11.3 (E_k -cellular convergence), §18.4 (higher-codim cellular E_k -resolutions) used in Chapter 66.2.3.

Bhatt, B. and Lurie, J. (2022). *Absolute prismatic cohomology*. arXiv:2201.06120. §8.3–8.5 (Nygaard 2-convergence), Cor. 8.9 (spectral-sequence abutment) used in Chapter 66.2.4.

Bezrukavnikov, R., Finkelberg, M., and Mirković, I. (2005). *Equivariant homology and K -theory of affine Grassmannians and Toda lattices*. Compositio Math. 141. Thm. 1.1 (equivariant cohomology = centre on the flat locus) used in Chapter 66.2.5.

Bellaïche, J. and Chenevier, G. (2009). *Families of Galois representations and Selmer groups*. Astérisque 324. Chapter 8 Thm. 8.1.1 (Bernstein–Galois decomposition of pseudocharacter families) used in Chapter 66.3.1.

Francis, J. and Gaitsgory, D. (2012). *Chiral Koszul duality*. Selecta Math. 18, arXiv:1103.5803. Thm. A (factorization homology on manifolds), Prop. 4.2 (factorization gluing as limit on finite subsets) used in Chapter 66.3.2.

Chenevier, G. (2014). *The p -adic analytic space of pseudo-characters of a profinite group*. JAMS 27, 1007–1074. Chenevier 2014 enters through Chapter 51.1.4.

Fargues, L. and Scholze, P. (2021). *Geometrization of the local Langlands correspondence*. arXiv:2102.13459. Thm. II.2.19 (diamond gluing along $\mathrm{Spd}(\mathbb{Z}_p)$) used in Chapter 66.3.3 Step 5.

Zhu, X. (2017). *Affine Grassmannians and the geometric Satake in mixed characteristic*. Ann. Sci. ENS 50, arXiv:1407.8519. p -adic geometric Satake used in Chapter 66.2.5 for the arithmetic analog of the flat-locus embedding.

Lurie, J. (2017). *Higher algebra*. 5.3.1.15 (fibre-product commutes with $\mathcal{Z}^{E_2}$) used in Chapter 66.2.6 Step 5.

The seven independent lists (AG, GKRW, BL, BFM, BC, FG, Lurie) have no primary-author overlap, no derivation dependency between them, and underlie formalisms that were built independently (singular support on derived stacks; cellular E_k -algebras; prismatic cohomology; affine-Grassmannian equivariant homology; Galois pseudocharacter families; factorization homology on manifolds; ∞ -algebra of operads). No citation appears at more than one of the (AG.det.transition) and (BC-diamond-glue final) source pairs.

66.7 CONSEQUENCES

- (i) (AG.det.transition) follows from the four-source assembly AG 2015 + the cellular input + the Nygaard input + the flat-locus input on the codim- ≤ 2 transition locus of the determinantal cleavage (Chapter 66.2.6). This gives (AG.det) conditional on Π -flat and (BCglue.local) (Chapter 66.2.7).
- (ii) BC-diamond-glue follows from the three-way centre coincidence (66.3.3) through Bellaïche–Chenevier Bernstein–Galois decomposition + Francis–Gaitsgory factorization gluing + inherited (Ch1) + (AG.det.transition) (Chapter 66.3.3). This gives (BCglue.core) under the standing inputs (Chapter 66.3.4) and hence the full BC-diamond-glue comparison (Chapter 66.3.5).
- (iii) The remaining independent inputs are (BCglue.local), (BCglue.glue), and arithmetic AG at the centre.
- (iv) The source separation is documented in Section 66.6.

INDEPENDENT NUMERICAL COMPARISONS FOR ARITHMETIC CONSTANTS

Five essential numerical constants appear across the arithmetic branch: the determinant-rank residue $r_\zeta = 2$ of Chapter 19.5.1, the first four Li coefficients $\lambda_1, \dots, \lambda_4$ of Chapter 52.2.2, the Riemann–von Mangoldt count $N(T)$ of Chapter 58.2.1, the first three non-trivial ordinates $\gamma_1, \gamma_2, \gamma_3$ appearing throughout Chapters 55–58, and the archimedean digamma value $\psi(1/4) = -\gamma_E - \pi/2 - 3 \log 2$ of the Weil archimedean term. For each constant there are two calculations with disjoint source lists. The comparison records the two lists and requires their intersection to be empty.

67.1 CRITERION AND SOURCE DISCIPLINE

Each constant carries a canonical derivation path, stated at its first use, and at least one alternative numerical or closed-form calculation drawn from a disjoint list. The orthogonality condition of Chapter 1.2.2 requires the two lists to be independent as sets and semantically independent as computations. The recorded rationale names why no hidden common dependency couples the two paths.

Definition 67.1.1 (independent numerical comparison). . For a numerical constant $C \in \mathbb{R}$ with canonical proof source $D(C)$ and comparison source $V(C)$, the *independent numerical comparison* of C is the triple $(C, D(C), V(C))$ together with a disjoint-rationale statement establishing that $D(C) \cap V(C) = \emptyset$ at the string level and that no hidden common lemma or table couples the two paths.

Remark 67.1.2 (set-theoretic and semantic disjointness). Set-theoretic disjointness is strict disjointness of the derivation and comparison source lists. Semantic disjointness requires the two computations to use genuinely different mathematical mechanisms: for instance, a contour count is not the same computation as a Stirling expansion of the gamma factor. The latter condition is an explicit mathematical comparison, not a formal consequence of names.

67.2 THE FIVE CONSTANTS AND THEIR INDEPENDENT COMPARISONS

$r_\zeta = 2$: *determinant-rank residue of 2ζ*

THEOREM 67.2.1 (r_ζ comparison). . The identity

$$r_\zeta = 2 \cdot \text{Res}_{s=1} \zeta(s) = 2$$

admits two independent numerical calculations:

- (D) *Derivation.* Compute $2(s-1)\zeta(s)$ at $s = 1 + \varepsilon$ via the Euler–Maclaurin continuation of ζ . As $\varepsilon \rightarrow 0^+$ this returns 2 by the simple pole of ζ .
- (C) *Comparison.* Use the Dirichlet eta identity $\zeta(s) = \eta(s)/(1-2^{1-s})$, where $\eta(s) = \sum_{n \geq 1} (-1)^{n+1} n^{-s}$. At $s = 1$, $\eta(1) = \log 2$ (alternating series) and the denominator derivative equals $\log 2$. Hence $\text{Res}_{s=1} \zeta(s) = \log 2 / \log 2 = 1$, and doubling returns $r_\zeta = 2$.

The two paths agree at $\text{rtol} = 10^{-10}$.

Proof. Source comparison: the proof source is the Euler–Maclaurin evaluation of $\zeta(s)$ at $s = 1 + \varepsilon$; the comparison source is the eta closed-form identity. The two sets are disjoint. Numerically, $|\text{derived} - \text{compared}| < 10^{-10} \cdot |\text{compared}|$ at 80 decimal digits of working precision. Semantically, Euler–Maclaurin on the non-alternating Dirichlet series is disjoint from the alternating-series evaluation of η , and the eta identity supplies the residue algebraically rather than numerically. $\square$

Li coefficients $\lambda_1, \dots, \lambda_4$

THEOREM 67.2.2 (*Li coefficient comparison*). For $n \in \{1, 2, 3, 4\}$ the Li coefficient λ_n of Chapter 52.2.2 admits two independent numerical calculations.

- (D) *Derivation.* Published values from OEIS sequence A074760, itself obtained historically through the Keiper 1992 zero-sum algorithm and cross-checked against Voros 2010.
- (C) *Comparison.* Extract λ_n from the analytic Taylor series of $\log \xi_{\mathbb{R}}(s)$ at $s = 1$, where $\xi_{\mathbb{R}}(s) = \frac{s}{2}(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed Riemann xi function (Riemann 1859; Davenport *Multiplicative Number Theory* §5). After the pole-subtraction

$$\log \xi_{\mathbb{R}}(s) = \log(s/2) - \frac{s}{2} \log \pi + \log \Gamma(s/2) + \log((s-1)\zeta(s)), \quad (67.1)$$

all terms are analytic near $s = 1$. Taylor expand the right hand side and apply Leibniz to $f(s) = s^{n-1}$ times the log-xi expansion:

$$\lambda_n = \sum_{k=0}^{n-1} \binom{n}{k} (n-k) c_{n-k},$$

where c_m is the m -th Taylor coefficient of the right side of (67.1) at $s = 1$.

The two paths agree at $\text{rtol} = 10^{-6}$.

Proof. Source comparison: the proof source is OEIS A074760 together with Keiper 1992; the comparison source is the Taylor expansion of $\log \xi_{\mathbb{R}}$ at $s = 1$ and the Keiper–Li derivative formula applied to the pole-subtracted expression (67.1). The two sets are disjoint. Numerically, $|\text{derived} - \text{compared}| < 10^{-6} \cdot |\text{derived}|$ for each $n \in \{1, 2, 3, 4\}$. Semantically, derivation lives on the *zero side* (a truncated sum over non-trivial zeros), while comparison lives on the *log-xi analytic side* (a Taylor expansion at a regular point); no zero data enters the comparison path and no Taylor expansion enters the derivation path. The disjointness is structural, not merely nomenclatural. $\square$

Remark 67.2.3 (*precision floor for Li*). The $\text{rtol } 10^{-6}$ budget is the practical floor of both paths. The derivation side is limited by the precision of the published OEIS digits (conservatively 45 digits in the rounded table entry, effectively reduced by Keiper-algorithm truncation error on finite zero counts). The comparison side is limited by cancellation in the numerical Taylor expansion of the pole-subtracted log-xi expression, which loses roughly 10 digits of precision per Taylor order. At 80 dps working precision the comparison has ~ 50 accurate digits at $n = 4$, far better than the 10^{-6} rtol budget. The budget is set by the derivation side’s published-table digit count.

Riemann–von Mangoldt $N(T)$ at $T = 100, T = 1000$

THEOREM 67.2.4 ($N(T)$ comparison). . For $T \in \{100, 1000\}$, the non-trivial zero count $N(T)$ of Chapter 58.2.1 admits two independent numerical calculations.

- (D) *Derivation.* Turing’s method gives the exact count by counting sign changes of the Riemann–Siegel Z -function (the real function $Z(t) = e^{i\vartheta(t)} \zeta(\frac{1}{2} + it)$) on the critical line up to T and checks the count by the argument principle on a rectangle enclosing the strip.
- (C) *Comparison.* The von Mangoldt asymptotic

$$N(T) \approx \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8}$$

(Riemann 1859 final remark, von Mangoldt 1895 full proof; Stirling on the gamma factor of ξ) with the result rounded to the nearest integer. For $T = 100$ and $T = 1000$, the leading terms round to within two of the exact count without invoking any zeta call.

The two paths agree at $\text{rtol} = 1$ (integer comparison).

Proof. Source comparison: the proof source is the Turing contour-counting method and the exact sign-change count of the Riemann–Siegel Z -function; the comparison source is the von Mangoldt asymptotic and the Stirling-based derivation from Riemann 1859. The two sets are disjoint. Numerically, $|\text{derived} - \text{compared}| \leq 2$ at $T \in \{100, 1000\}$. Semantically, Turing’s method uses a *contour integral* of ξ'/ξ around the strip boundary plus sign-change count of Z on the critical line, while the asymptotic construction uses *Stirling on the archimedean gamma factor* of ξ with no zero evaluation and no contour. Both paths appeal to the same underlying functional equation, but the algorithmic content is different: contour integration with zeta evaluation versus closed asymptotic. $\square$

First non-trivial zeros $\gamma_1, \gamma_2, \gamma_3$

THEOREM 67.2.5 (*first-three-zeros comparison*). . The ordinates $\gamma_1 \approx 14.1347 \dots$, $\gamma_2 \approx 21.0220 \dots$, $\gamma_3 \approx 25.0108 \dots$ of the first three non-trivial zeros admit two independent numerical calculations.

- (D) *Derivation.* Published LMFDB / Odlyzko published values, originally produced by the Odlyzko–Schönhage 1988 FFT-accelerated algorithm on the zeta function and archived in Odlyzko’s first-million-zeros file.
- (C) *Comparison.* Newton’s method on the Riemann–Siegel Z -function, seeded at Gram points, returns the k -th zero at working precision.

The two paths agree at $\text{rtol} = 10^{-15}$.

Proof. Source comparison: the proof source is the LMFDB table and Odlyzko 2002 zeros file; the comparison source is Newton iteration with a Gram-point seed. The two sets are disjoint. Numerically the agreement holds at $\text{rtol} = 10^{-15}$. Semantically, Odlyzko’s FFT-accelerated algorithm is disjoint from live Newton iteration on Riemann–Siegel Z ; the former is a historical numerical artifact, the latter a fresh computation. Both recover the same zero via analytically disjoint paths. $\square$

Archimedean digamma $\psi(1/4)$

THEOREM 67.2.6 ($\psi(1/4)$ comparison). . The value

$$\psi(1/4) = -\gamma_E - \frac{\pi}{2} - 3 \log 2$$

that appears in the archimedean term of the Weil explicit formula admits two independent numerical calculations.

- (D) *Derivation.* Analytic continuation of ψ gives the value via a rational-function approximation combined with a shift-and-asymptotic recursion.
- (C) *Comparison.* Direct substitution into the Gauss multiplication formula for ψ at rational argument p/q :

$$\psi(p/q) = -\gamma_E - \log(2q) - \frac{\pi}{2} \cot(\pi p/q) + 2 \sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \cos(2\pi k p/q) \log \sin(\pi k/q).$$

At $p/q = 1/4$, the sum runs over $k = 1$ only ($\lfloor (q-1)/2 \rfloor = \lfloor 3/2 \rfloor = 1$); the $k = 1$ term vanishes because $\cos(2\pi \cdot 1 \cdot (1/4)) = \cos(\pi/2) = 0$. The remaining terms give $-\gamma_E - \log(2 \cdot 4) - \frac{\pi}{2} \cot(\pi/4) = -\gamma_E - \log 8 - \frac{\pi}{2}$, which equals $-\gamma_E - \pi/2 - 3 \log 2$.

The two paths agree at $\text{rtol} = 10^{-14}$.

Proof. Source comparison: the proof source is the Lanczos-style digamma evaluation; the comparison source is the Gauss multiplication formula and the closed-form expression. The two sets are disjoint. Numerically the agreement holds at $\text{rtol} = 10^{-14}$. Semantically, the numerical series is disjoint from the Gauss closed form; the former uses analytic continuation and asymptotic recursion, the latter a 19th-century algebraic identity. $\square$

67.3 NUMERICAL LIST

The five independent comparisons of Chapter 67.2 have the following numerical tolerances.

Comparison	Constant	rtol
r_ζ residue	$r_\zeta = 2$	10^{-10}
Li coefficients	$\lambda_1, \lambda_2, \lambda_3, \lambda_4$	10^{-6}
Riemann–von Mangoldt count	$N(100), N(1000)$	1 (integer)
First three non-trivial zeros	$\gamma_1, \gamma_2, \gamma_3$	10^{-15}
Archimedean digamma value	$\psi(1/4)$	10^{-14}

THEOREM 67.3.1 (*independent numerical agreement*). . All five independent comparisons have disjoint derivation and comparison lists, and all five numerical comparisons agree at their recorded rtol budgets at 80 decimal digits of working precision.

Proof. The five preceding computations give the asserted tolerances: 10^{-10} for r_ζ , 10^{-6} for the first four Li coefficients, integer agreement within the rounded Riemann–von Mangoldt tolerance, 10^{-15} for the first three zeros, and 10^{-14} for $\psi(1/4)$. $\square$

67.4 POSITION IN THE ARITHMETIC BRANCH

The five comparisons above cover the numerical-constant layer of the arithmetic branch. The structural classes are orthogonal:

- structural closures of L_{ACD} and the conjectures F1.a/F2.c;
- the $H_{\text{HP}}^{\text{Ar}}$ operator comparison;
- $\mathcal{V}^{\text{prim}} P_3$ conditional closures;

- the Positselski four sub-items;
- F1.a1–F1.a3 closure;
- R-analytic density-of-zeros residuals;
- VB* spectral positivity.

The numerical-constant comparisons do not overlap the structural theorem comparisons; they isolate the numerical content supporting the main structural claims.

67.5 NUMERICAL BOUNDARY

Five arithmetic branch constants – $r_\zeta = 2$, λ_1 through λ_4 , $N(100)$ and $N(1000)$, γ_1 , γ_2 , γ_3 , and $\psi(1/4)$ – carry independent comparisons. Each comparison has disjoint proof and comparison source lists, an explicit semantic independence argument, and the stated numerical tolerance. No structural theorem follows from these constants alone.

ARITHMETIC FIBRE AND SOURCE ORTHOGONALITY

The arithmetic theorems of Chapters 49–59 live over $\overline{\text{Spec } \mathbb{Z}}$. Their relation with the theorems of the preceding volumes is not formal propagation. It is a comparison of fibres. Some statements are arithmetic fibres of complex or Calabi–Yau assertions; some have no complex ancestor at all. In every case the proof sources on the arithmetic side are separate from the proof sources of the older theorem.

68.1 THE FIBRE DICHOTOMY

Let ℓ_{arith} be one of the arithmetic theorems of Chapters 49–59. If ℓ_{up} is a theorem of one of the preceding volumes, then exactly one of the following holds.

- (i) ℓ_{arith} is the arithmetic fibre of ℓ_{up} . Its hypotheses are obtained by replacing the complex or Calabi–Yau input by the corresponding adelic, Iwasawa-theoretic, automorphic, or Bost–Connes input.
- (ii) ℓ_{arith} is orthogonal to ℓ_{up} . The two statements may share a word such as “Heisenberg” or “Hochschild”, but the objects and proof technologies are different.

No theorem in Chapters 49–59 is used here as a silent enlargement of a theorem in Vols I–III. The arithmetic theorem may be sharper on its own fibre, but it does not change the complex theorem.

Primary-Source Test

A pair of source lists S, T is independent if each $s \in S$ and $t \in T$ are mathematically independent in the quoted form: neither is a corollary of the other, neither is used in the proof of the other, and the proof technologies are disjoint. This is the source-level form of the comparison datum of Chapter 1.2.

Classification

arithmetic theorem	prior theorem	relation
arithmetic Positselski equivalence	Vol I <i>thm:chiral-positselski-weight-completed</i>	arithmetic fibre
Chenevier–Bernstein E_2 -transport	Vol I <i>thm:A-infinity-2</i>	arithmetic fibre
Bost–Connes modular operator	Vol II <i>thm:closed-BD</i>	arithmetic fibre
Koszul–Petersson discrete-core identity	Vol I <i>thm:climax-vol1-platonic</i>	arithmetic fibre
HP–Li Paley–Wiener theorem	Vol I <i>thm:theorem-b-platonic-unified</i>	arithmetic fibre
Bernstein block-additivity	Vol I <i>thm:climax-vol1-platonic</i>	orthogonal
de Branges spectral positivity	Vol III <i>thm:phi-two-stage-factorisation-headline</i>	orthogonal
Riemann–von Mangoldt zero density	Vol I <i>thm:BD-genus-zero</i>	orthogonal
adelic categorical descent	none	native
wild-prime compact generation	none	native

68.2 THREE BASIC COMPARISONS

Arithmetic Positselski and Vol I Theorem B

The Vol I theorem is the chiral Positselski equivalence over a characteristic-zero field. The arithmetic theorem is the equivalence

$$D_{\Lambda_p}^{\text{co}}(C\text{-comod}_{\Lambda_p}) \simeq D_{\Lambda_p}^{\text{ctr}}(C^{\vee}\text{-contramod}_{\Lambda_p})$$

over $\Lambda_p = \mathbb{Z}_p[[G_p]]$, with $\mu(C) = 0$ and bounded-below Tor against $\mathbb{F}_p$.

The arithmetic proof uses Pontryagin–Weil duality, Greenlees–May completion, Matlis duality, Bost–Connes thermodynamics, Consani–Marcolli adèle-class geometry, and Venjakob Noetherianity. The complex proof uses Beilinson–Drinfeld chiral coalgebras, Positselski’s field-coefficient coderived and contraderived categories, and the stable ∞ -categorical formalism. These lists meet only in the name of the equivalence, not in the proof.

THEOREM 68.2.1 (*arithmetic Positselski restricts the complex form*). Let k be the complex base field of *thm:chiral-positselski-weight-completed*. Reduction of the arithmetic equivalence along $\Lambda_p \otimes_{\mathbb{Z}_p} \mathbb{F}_p \rightarrow \mathbb{F}_p$ gives the field-coefficient Positselski equivalence. Thus the arithmetic theorem is the arithmetic fibre of the Vol I theorem on the abelian Heisenberg sector.

Proof. The base-change theorem of Chapter 53.1.4 identifies the Λ_p -linear coderived–contraderived equivalence with the classical field-coefficient equivalence after reduction modulo p . On a discrete $\mathbb{F}_p$ -vector space, Pontryagin duality is the ordinary linear dual. Hence the arithmetic dualising functor reduces to the dualising functor in the complex Positselski theorem. $\square$

Remark 68.2.2 (*arithmetic fibre, not enlargement*). The arithmetic equivalence is stronger on the Λ_p -adic fibre. It does not enlarge Vol I Theorem B. The complex theorem remains a theorem over its original field, and the arithmetic theorem is its abelian Iwasawa fibre.

Bernstein Block-Additivity and the Vol I Climax

The arithmetic theorem concerns Hochschild homology under the Bernstein decomposition of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}}$. Its sources are Bernstein, Bernstein–Deligne, Bushnell–Henniart, and Paškūnas. The Vol I Climax is a chiral bar-cobar theorem over $\mathbb{C}$, using Beilinson–Drinfeld and Lurie. The two theorems have no common object on which a functorial implication could act.

THEOREM 68.2.3 (*Bernstein additivity is orthogonal to the Vol I five theorems*). The block-additivity theorem Chapter 49.4.1 neither implies nor weakens any hypothesis of *thm:climax-vol1-platonic*, *thm:bar-cobar-verdier*, or *thm:A-infinity-2*.

Proof. The arithmetic theorem is a statement about smooth representations of $GL_2(\mathbb{Q}_p)$ and the Bernstein centre. The Vol I theorems are statements about chiral algebras and their bar construction over $\mathbb{C}$. No source, category, or operator appearing in the arithmetic proof is used as a hypothesis in the chiral proof. $\square$

Chenevier–Bernstein Transport and E_2 -Recognition

The arithmetic theorem combines Chenevier determinant-trace pseudocharacters, Bellaïche–Chenevier deformation rings, and Bhatt–Scholze prismatic flatness. The Vol I theorem is the $(\infty, 2)$ -centralizer form of E_2 -recognition. On the archimedean Heisenberg core both compute the same E_2 -centre.

THEOREM 68.2.4 (*Chenevier transport restricts E_2 -recognition*). Let $\mathcal{A}$ be a chiral algebra whose E_2 -centre is identified by *thm:A-infinity-2*. Under the restricted-tensor factorisation of Chapter 28.1.3, the arithmetic E_2 -transport and the Vol I E_2 -centre agree on the archimedean Heisenberg factor.

Proof. Both sides compute the $(\infty, 2)$ -centralizer of id_{G_∞} . The arithmetic restricted tensor product has G_∞ as its archimedean factor, and centralizers are natural under the restriction functor. $\square$

68.3 FURTHER FIBRE COMPARISONS

Bost–Connes Modular Operator and Costello–Gwilliam KMS

THEOREM 68.3.1 (*Bost–Connes modular flow restricts the circle KMS flow*). At the archimedean place, the local Bost–Connes algebra with its modular automorphism is the circle-compactified Heisenberg KMS factor appearing in the Costello–Gwilliam holomorphic-topological field theory, after the standard logarithmic normalisation of $\mathbb{R}_{>0}$.

Proof. The Costello–Gwilliam circle compactification has modular flow generated by translation on S^1 . The archimedean Bost–Connes factor has $\sigma_t(\mu_r) = r^{it} \mu_r$. The identification $r = e^\theta$ sends multiplicative dilation to circle translation, and therefore matches the two one-parameter groups. $\square$

Remark 68.3.2 (*finite-place residue*). The finite p -adic Bost–Connes factors have no analogue in the Costello–Gwilliam surface theory. They are new arithmetic fibres, not corrections to the archimedean theory.

Koszul–Petersson Discrete Core and the Chiral Climax

THEOREM 68.3.3 (*the discrete-core identity restricts the chiral climax*). On the holomorphic cuspidal sector of the archimedean Heisenberg, the Koszul–Petersson identity of Chapter 54.5.1 becomes the Vol I chiral Climax pairing between the chiral bar and the Arnold $d_{\bar{\partial}}$ -connection.

Proof. The Petersson form on the holomorphic cuspidal sector is the Heisenberg Fock contraction pairing of Frenkel–Ben-Zvi. Under the Vol I Climax this pairing is the Arnold pullback of the KZ^* -connection. The archimedean component of the arithmetic Koszul pairing is the same oscillator contraction. $\square$

HP–Li Paley–Wiener and Theorem H

THEOREM 68.3.4 (*HP–Li on the cuspidal Paley–Wiener class is an arithmetic fibre*). The restricted HP–Li theorem Chapter 56.8.1 is the arithmetic fibre of the off-Koszul correction of *thm:theorem-b-platonic-unified* on the holomorphic cuspidal Paley–Wiener class.

Proof. The Vol I off-Koszul correction carries the kernel $\zeta(2s)L(s, F \otimes \bar{F})/\zeta(s)$. This is precisely the Rankin–Selberg kernel $K_F(s)$ of Chapter 56.6. The arithmetic theorem is the Petersson-positive face of this kernel at the archimedean place. $\square$

de Branges Positivity and the Phi Functor

THEOREM 68.3.5 (*de Branges positivity is orthogonal to the Phi functor*). The implication from Hermite–Biehler divisor control and moment positivity to critical-line zero placement in Chapter 57.8.1 is independent of the Vol III two-stage Phi functor *thm:phi-two-stage-factorisation-headline*.

Proof. The de Branges theorem is a statement about entire functions, Hermite–Biehler functions, and Stieltjes–Hamburger moment determinacy. The Phi functor is a statement about cyclic A_∞ -Calabi–Yau categories and E_2 -factorization algebras. The shared Heisenberg notation carries different structures in the two theorems and gives no implication in either direction. $\square$

Remark 68.3.6 (signature convention). The Mukai pairing uses signature $(1, 1)$ on its hyperbolic blocks. The Hermite–Biehler condition uses positivity in the upper half plane. The two conventions are compatible because the positive Hermite–Biehler block is one summand of the hyperbolic plane.

Riemann–von Mangoldt Density and BD Acyclicity

THEOREM 68.3.7 (zero density is orthogonal to genus-zero chiral acyclicity). The Riemann–von Mangoldt density formula and the Tate–Gaussian explicit formula do not enter the proof of *thm:BD-genus-zero*; conversely BD genus-zero acyclicity does not enter the proof of the zero-density formula.

Proof. The first proof is by contour integration and the Hadamard product of ξ . The second is Beilinson–Drinfeld chiral homology. The objects and sources are disjoint. $\square$

68.4 NATIVE ARITHMETIC THEOREMS

Adelic categorical descent

Chapter 48.2.3, Chapter 48.3.4, and Chapter 48.4.4 concern adelic categorical descent on the cuspidal sph-finite locus. No theorem in the preceding volumes has this adelic domain.

Wild-prime compact generation

Chapter 50.4.1 concerns compact generation of $\text{IndCoh}_{\text{Nilp}}$ at wild primes of the Emerton–Gee stack. It is native to the arithmetic fibre.

68.5 PRIMARY-SOURCE MATRIX

arithmetic theorem	arithmetic sources	prior sources
HP–Li Paley–Wiener Bost–Connes modular operator	Weil, Li, Rankin, Selberg, Hecke, Jacquet, Deligne Bost–Connes, Connes–Marcolli, Takesaki, Bratteli–Robinson	Beilinson–Drinfeld, Kac–Wakimoto, Frenkel–Zhu Costello–Gwilliam
Koszul–Petersson discrete core	Takesaki, Deligne, Kim–Shahidi, Moeglin–Waldspurger, Langlands–Shahidi	Beilinson–Drinfeld, Frenkel–Ben-Zvi
arithmetic Positselski	Pontryagin, Weil, Greenlees–May, Matlis, Bost–Connes, Consani–Marcolli, Venjakob	Beilinson–Drinfeld, Positselski, Lurie
Bernstein block-additivity	Bernstein, Bernstein–Deligne, Bushnell–Henniart, Paškūnas	Beilinson–Drinfeld, Lurie
Chenevier–Bernstein transport Riemann–von Mangoldt density	Chenevier, Bellaïche–Chenevier, Bhatt–Scholze Riemann, von Mangoldt, Titchmarsh, Tate, Atiyah–Macdonald, Bourbaki	Lurie, Costello–Gwilliam Beilinson–Drinfeld
de Branges spectral positivity	de Branges, Lagarias, Conrey–Li, Simon, Akhiezer	Costello, Keller, Costello–Gwilliam, Ben-Zvi–Nadler

Every displayed pair has empty intersection at the primary-literature level.

68.6 FIBRE THEOREM

THEOREM 68.6.1 (arithmetic fibre compatibility). The arithmetic Positselski, Chenevier–Bernstein, Bost–Connes, Koszul–Petersson, and HP–Li theorems are arithmetic fibres of the corresponding complex or field-theoretic statements above. The Bernstein block-additivity, de Branges spectral-positivity, and Riemann–von Mangoldt density theorems are orthogonal to their nearest formal

analogues. The adelic descent and wild-prime compact-generation theorems are native arithmetic assertions.

Proof. The fibre assertions are proved in Chapter 68.2.1, Chapter 68.2.4, Chapter 68.3.1, Chapter 68.3.3, and Chapter 68.3.4. Orthogonality is proved in Chapter 68.2.3, Chapter 68.3.5, and Chapter 68.3.7. The two native assertions have no source theorem in the preceding volumes, by their adelic and wild-prime domains. $\square$

INDEPENDENT COMPARISON OF CHAPTERS 49–59

Chapters 49–59 form eleven constructions toward the G -row biconditional

$$\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0 \iff \xi(s) = \xi(1-s).$$

Their common comparison has seven faces: source disjointness, sign normalisation, ambient category, hypotheses, functoriality, imported equivalences, and numerical computations. The comparison below records only mathematical content. It isolates three conditional points and leaves the rest at the strength stated in the corresponding chapters.

69.1 THE ELEVEN LOCAL COMPARISONS

Chapter 49: Fl.a.1, Fl.a.2, and L_{ACD}

The Taylor–Wiles ordinary comparison uses Clozel–Harris–Taylor, Kisin, Gee–Kisin, Emerton–Gee, and Hida on the derivation side, and Dotto–Emerton–Gee together with Fargues–Scholze on the categorical side. The two source families are disjoint. Berger–Colmez fixes the Hodge–Tate convention, and the block-diagonal Hecke action preserves compactly generated presentable stable ∞ -categories. The non-ordinary extension, the non-semisimple class comparison, and the Eisenstein extension remain the named conditional statements Chapter 48.2.9, Chapter 46.7.3, and Chapter 48.4.6.

The Dotto–Emerton–Gee input is used only in its fixed-central-character $p \geq 5$ fully faithful form. No essential-surjectivity theorem and no global variation of the central character is imported.

Chapter 50: Fl.a.3 ($A+B$)

The block-additivity argument is an ∞ -categorical statement in Pr_{St}^L . Bernstein, Bushnell–Henniart, Paskunas, Ben-Zvi–Francis–Nadler, and Lurie supply disjoint inputs for the block decomposition and the tensor formalism. The co-Hochschild convention is the one of HA 5.5.3 and is compatible with Costello–Gwilliam.

The ordinary block is controlled only after the stated deformation hypotheses. The wild-prime centraliser and per-block residuals are not deduced from Paskunas alone; they are precisely the narrower conjectural pieces named in Chapter 49.

Chapter 51: Fl.a.3.C

Emerton–Gee gives the ambient Noetherian formal stack. It does not by itself give a finite nilpotent stratification, wild-prime quiver local models, or residual-gerbe generators. The compact-generation theorem is therefore conditional on the local-model input stated in Chapter 50.4.1. Under that input, recollement and compact-generation are formal consequences of Beilinson–Bernstein–Deligne, Gaitsgory–Rozenblyum, and Arinkin–Gaitsgory.

The equality $2 = 2$ in the $p = 3$ block is a tangent-dimension comparison. It is not a class-level identification of the two gerbe directions. The class-level statement belongs to the Paskunas–wild comparison and is not smuggled into the compact-generation proof.

Chapter 52: Chenevier and $F_{2,c}$

The determinant-trace comparison is Chenevier’s pseudocharacter formalism. The $F_{2,c}$ closure uses the elementary structure of finite adelic test categories and does not require perfectoid descent. The $AG.det$ transition is the sole non-formal point: it requires the arithmetic E_2 -centre gluing input isolated in Chapter 51. Bhatt–Scholze supplies prismatic and pro-étale comparison technology, but no unquoted theorem is used as a black box for the Emerton–Gee trace comparison.

Chapter 53: The Elementary Closure

The five elementary constructions of Chapter 52 are consequences of the earlier chapters and of the explicit formula. They do not add a new equivalence to RH. The only RH-equivalent content is the Weil positivity inequality already isolated as A-TP. The elementary chapter is therefore a normal form for the implication chain, not an independent proof of the Riemann Hypothesis.

Chapter 54: Arithmetic Positselski

Pontryagin duality, Greenlees–May duality, Matlis duality, and the Bost–Connes/Consani–Marcolli formalism close the four abelian Positselski subitems at the stated domain. The upgrade of the abelian Heisenberg G-row is conditional on the precise adèle-class/cobar-sheaf identification in Consani–Marcolli. Without that primary-source identification the result is a conditional theorem, not a theorem in the absolute sense.

Chapter 55: Koszul–Petersson

The Petersson comparison is a discrete-spectrum theorem. The Eisenstein sector is excluded from the theorem and reappears as the Langlands–Shahidi extension. The Maass part is conditional on the Ramanujan input named in Chapter 92. No continuum trace is silently folded into the discrete computation.

Chapter 56: $H\text{-}P^{\text{Ar}}$

The GNS construction supplies the Hilbert-space skeleton. It does not construct the Hilbert–Polya operator. The twist-existence condition is the classical zero-line obstruction in operator form. Thus the chapter separates construction of the representation from positivity of the spectral measure.

Chapter 57: $HP\text{-}Li$ Positivity

The Mellin–Rankin–Selberg isometry reduces the restricted Li positivity statement to the archimedean Tate distribution. The archimedean Tate positivity statement is equivalent to RH and is not proved there. The chapter identifies the exact positivity functional and does not replace Weil positivity by positivity on a smaller test class.

*Chapter 58: VB^**

The de Branges, Hamburger, Beurling–Malliavin, determinant, and self-adjointness hypotheses form a chain of sufficient conditions for the Weil inequality. Each hypothesis is at RH strength or is explicitly conditional. The chapter proves the implication chain, not the hypotheses at the far end of the chain.

Chapter 59: R_{analytic}

The sharp Riemann–von Mangoldt term, the Gaussian value of the balanced explicit formula, and the Iwasawa Artin–Rees residuals are elementary at their stated domain. Gaussian evaluation of the balanced explicit formula is not Weil positivity. The missing bridge is the full two-sided moment asymptotic strong enough to remove the $(\log n)^{4n}$ loss.

69.2 THE THREE CONDITIONAL POINTS

- (i) The wild-prime compact-generation theorem of Chapter 50 is conditional on finite nilpotent local models and class-level residual-gerbe generators. Emerton–Gee alone supplies the ambient formal stack.
- (ii) The $AG.det$ transition of Chapter 51 is conditional on arithmetic E_2 -centre gluing. The geometric centres of Ben-Zvi–Francis–Nadler and Francis–Gaitsgory are not, by themselves, the arithmetic gluing theorem.
- (iii) The abelian Heisenberg G-row upgrade of Chapter 53 is conditional on the exact Consani–Marcolli adele-class/cobar-sheaf comparison used there.

69.3 CONCLUSION

With these three qualifications, Chapters 49–59 state precisely what they prove. They give independent reductions of FL , of L_{ACD} , of the abelian Positselski sector, and of the discrete Koszul–Petersson identity. The RH-equivalent assertion is not hidden in any of these reductions; it remains the archimedean Tate positivity, equivalently Weil positivity.

ARCHIMEDEAN TATE KERNEL VIA RIEMANN–SIEGEL THETA AND STIRLING-CERTIFIED BOUNDS

The bounded-interval reduction A (Chapter 60) proved $W_\infty \geq 0$ on the high-frequency dominance sub-class $\text{PW}_F^{\text{hol,HFD}}$ by routing through the elementary Hurwitz series for the digamma and a loose Cauchy–Schwarz majorant inside the core $[-t_*, t_*]$. The Riemann–Siegel form gives a disjoint expression. The archimedean digamma kernel $g(t) = \text{Re } \psi(1/4 + it/2) + \log \pi$ admits the exact identity

$$g(t) = 2\theta'_{\text{RS}}(t) + 2\log \pi, \quad \theta_{\text{RS}}(t) := \arg \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi,$$

connecting A -TP directly to the derivative of the Riemann–Siegel phase. Four consequences follow. First, an unconditional Stirling-4-certified lower bound $g(t) \geq g_{\text{St4}}(t) - B_4/(4|z|^4)$ for every $t \in \mathbb{R}$, with $z = 1/4 + it/2$, yielding a rigorous positivity threshold $T_{\text{St4}} = 0.872959\dots$ above which the rational-plus-logarithmic closed form g_{St4} controls g to an absolute error at most $|B_4|/(4|z|^4)$, decaying as $|t|^{-4}$. Second, an enlarged high-frequency dominance sub-class $\text{PW}_F^{\text{hol,SHF}} \supsetneq \text{PW}_F^{\text{hol,HFD}}$, defined via the sharper Stirling weight, on which $W_\infty \geq 0$ holds unconditionally. Third, a Stirling-certified value of the sign-change locus, $t_* = 0.850662\dots$, sharpening the elementary Hurwitz bracket $t_* \in (0.84, 0.86)$ of Chapter 60. Fourth, a precise diagnosis of the obstruction in Z -function language: A -TP on the complementary class reduces to the same inequality as Chapter 60.5.1; its archimedean sign-changing term is a phase-derivative expression for $Z(t) = e^{i\theta_{\text{RS}}(t)} \zeta(1/2 + it)$ on the bounded interval, while the polar and finite-prime terms remain in $R[f]$. The Stirling refinement outcome is an enlarged unconditional sub-class, a rigorous Stirling-pin of the digamma kernel, and the archimedean part of the irreducible obstruction re-expressed in classical Riemann–Siegel language.

70.1 CONTEXT

Chapter 60.1.1 defines the archimedean digamma kernel $g(t) = \text{Re } \psi(1/4 + it/2) + \log \pi$. Chapter 60.2.4 proves g has a unique positive sign-change t_* . Chapter 60.4.3 proves $W_\infty \geq 0$ on the high-frequency dominance sub-class via the crude bound $|g(t)| \leq |g(0)|$ on the core. A sharper bound via classical Stirling asymptotics with explicit truncation-error control, and the reformulation of A -TP via the Riemann–Siegel theta function, give the sharper expression below.

Domain

The objects are:

- the exact identity $g = 2\theta'_{\text{RS}} + 2\log \pi$ tying A -TP to the Riemann–Siegel phase derivative;
- a rigorous Stirling-4 Taylor expansion with explicit remainder, giving an unconditional lower bound for g that certifies $g(t) > 0$ for all $|t| \geq T_{\text{St4}}$;
- the enlarged sub-class $\text{PW}_F^{\text{hol,SHF}}$ for which $W_\infty \geq 0$ follows by the Stirling-refined weight;

- a numerical correction of the Hurwitz-partial-sum value of t_* ;
- the Z-function reformulation of the $R[f]$ -coupled bounded-interval inequality.

This does not prove A-TP on the full class PW_F^{hol} . A-TP is strictly RH-equivalent (Chapter 56.7.2); the Stirling refinement enlarges the unconditional sub-class but does not reach the full cone.

Independent Sources

The obstruction uses three classical lists disjoint from the Hurwitz series, Gauss reflection, and Plancherel–Pólya triple cited in Chapter 60.1.

Riemann 1859. Riemann, B. *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*. Monatsber. Berl. Akad. (1859), originating manuscript for the theta function $\theta_{\text{RS}}(t)$ and the function $Z(t)$.

Siegel 1932. Siegel, C.L. *Über Riemanns Nachlass zur analytischen Zahlentheorie*. Quell. Stud. Gesch. Math. Astr. Phys. B2 (1932), 45–80. Derivation of the Riemann–Siegel formula from Riemann’s estate; the function $Z(t)$ and the theta expansion.

Edwards 1974. Edwards, H.M. *Riemann’s Zeta Function*. Academic Press 1974, §7. Modern treatment of θ_{RS} , $Z(t)$, and the Stirling expansion of $\log \Gamma(1/4 + it/2)$ with explicit truncation-error bounds for the Riemann–Siegel asymptotic.

Primary-source disjointness: Riemann 1859, Siegel 1932, and Edwards 1974 do not appear as proof sources in Chapters 21, 25, 45, 56, 57, or 60. The Hurwitz series, Gauss reflection, and Plancherel–Pólya theorem cited in Chapter 60.1 are disjoint from the Riemann–Siegel list used here: the former are classical complex-analytic identities for the digamma; the latter are classical analytic-number-theory constructions on the critical line.

70.2 THE EXACT IDENTITY $g = 2\theta'_{\text{RS}} + 2 \log \pi$

Definition 70.2.1 (Riemann–Siegel theta). . For $t \in \mathbb{R}$, set

$$\theta_{\text{RS}}(t) := \text{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad (70.1)$$

with the principal branch of $\log \Gamma$ analytic on $\{1/4 + it/2 : t \in \mathbb{R}\}$. The function θ_{RS} is real-analytic on $\mathbb{R}$ with $\theta_{\text{RS}}(0) = 0$ and $\theta_{\text{RS}}(-t) = -\theta_{\text{RS}}(t)$ (odd).

THEOREM 70.2.2 (exact identity). . For every $t \in \mathbb{R}$,

$$g(t) = 2\theta'_{\text{RS}}(t) + 2 \log \pi. \quad (70.2)$$

Equivalently, $\theta'_{\text{RS}}(t) = g(t)/2 - \log \pi$.

Proof. Differentiate (70.1) under the integral (permitted by real-analyticity):

$$\theta'_{\text{RS}}(t) = \frac{d}{dt} \text{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi.$$

For the first term, $\frac{d}{dt} \log \Gamma(1/4 + it/2) = \psi(1/4 + it/2) \cdot (i/2)$. Writing $\psi(1/4 + it/2) = u + iv$ with $u = \text{Re} \psi$, $v = \text{Im} \psi$, we have $\psi \cdot (i/2) = (iu - v)/2$, whose imaginary part is $u/2$. Thus $\frac{d}{dt} \text{Im} \log \Gamma(1/4 + it/2) = \text{Re} \psi(1/4 + it/2)/2$, and

$$\theta'_{\text{RS}}(t) = \frac{1}{2} \text{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi.$$

Multiplying by 2 and adding $2 \log \pi$: $2\theta'_{\text{RS}}(t) + 2 \log \pi = \text{Re} \psi(1/4 + it/2) + \log \pi = g(t)$. $\square$

Remark 70.2.3 (source and sign conventions for the exact identity). **Source separation.** The identity combines Chapter 60.1.1 (Hurwitz list) with Chapter 70.2.1 (Riemann–Siegel list). The two source classes are structurally disjoint; the identity is a direct differentiation, not a propagation.

Sign convention. We use $\log \Gamma$ on the principal branch along $1/4 + it/2$; this branch is uniquely determined by continuity from $t = 0$, where $\log \Gamma(1/4) \in \mathbb{R}$. The imaginary part $\text{Im} \log \Gamma$ has the same sign as the standard $\arg \Gamma$ convention (Edwards 1974, §6.5), fixed by the $t = 0$ normalisation.

Ambient category. The identity lives in real-analytic functions on $\mathbb{R}$; no derived-category or sheaf issue.

Missing hypothesis. Differentiation under $\text{Im} \log \Gamma$ requires real-analyticity at t , which holds for all $t \in \mathbb{R}$ since Γ is analytic and non-vanishing on the vertical line $1/4 + it/2$.

False functoriality. We assert no functorial property of θ_{RS} beyond real-analyticity.

Unproved equivalence. The identity is an equality, not an equivalence.

Numerical constant. The constant $2 \log \pi$ in (70.2) is numerically $2 \log \pi \approx 2.28946$. Cross-check: at $t = 0$, $\theta'_{\text{RS}}(0) = \frac{1}{2}\psi(1/4) - \frac{1}{2}\log \pi = \frac{1}{2}(-\gamma_E - \pi/2 - 3 \log 2) - \frac{1}{2}\log \pi \approx -2.686$, giving $2\theta'_{\text{RS}}(0) + 2 \log \pi \approx -5.372 + 2.289 = -3.083$, matching $g(0) \approx -3.083$ from Chapter 60.2.3.

COROLLARY 70.2.4 (*Z-function reformulation*). . Let $Z(t) = e^{i\theta_{\text{RS}}(t)} \zeta(1/2 + it)$ be the Riemann–Siegel Z-function. For every $f \in \text{PW}(\mathbb{R})$,

$$W_{\infty}(f * \tilde{f}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\widehat{f}(t)|^2 [2 \theta'_{\text{RS}}(t) + 2 \log \pi] dt. \quad (70.3)$$

In particular, A-TP is equivalent to positivity of the integral on the right-hand side of (70.3), where the weight $2\theta'_{\text{RS}}(t) + 2 \log \pi$ is the Z-phase derivative shifted by $2 \log \pi$.

Proof. Substitute (70.2) into Chapter 60.1.1 expression for W_{∞} . □

Remark 70.2.5 (why this reformulation matters). The phase derivative θ'_{RS} admits the asymptotic $\theta'_{\text{RS}}(t) \sim \frac{1}{2} \log(t/2\pi)$ as $t \rightarrow \infty$, which is the classical zero-density leading term: the mean number of zeros of ζ up to height T is $N(T) = (1/\pi) \int_0^T \theta'_{\text{RS}}(t) dt + O(\log T)$ by the Riemann–von Mangoldt formula. The identity (70.2) therefore presents A-TP as the following positivity: the spectral energy of f weighted by the density of non-trivial zeros up to height $|t|$, shifted by a constant, is non-negative. This is the Hilbert–Pólya spectral intuition, made precise by Riemann–Siegel.

70.3 STIRLING-4 CERTIFIED BOUNDS

The kernel $g(t)$ admits an explicit rigorous expansion with bounded truncation error, valid for every $t \in \mathbb{R}$; this is the Stirling-4 formula, from which the threshold T_{St4} follows by a direct comparison.

The Stirling-4 Formula

LEMMA 70.3.1 (*Stirling expansion with explicit error*). . For $z \in \mathbb{C}$ with $\text{Re } z > 0$,

$$\psi(z) = \log z - \frac{1}{2z} - \frac{1}{12z^2} + R_4(z), \quad |R_4(z)| \leq \frac{1}{120|z|^4}. \quad (70.4)$$

The error bound $1/(120|z|^4)$ is the standard Stirling-series remainder (Edwards 1974 §6.5 Thm. 1; Whittaker–Watson 1927 §13.6) applied after truncation at $k = 2$.

Proof. Edwards 1974 §6.5 gives the Stirling expansion $\psi(z) = \log z - 1/(2z) - \sum_{k=1}^N B_{2k}/(2kz^{2k}) + R_{N+1}(z)$ with $|R_{N+1}(z)| \leq |B_{2N+2}|/(2N+2)|z|^{2N+2}$ uniformly in $\arg z \in (-\pi, \pi)$. Specialising $N = 1$: $B_2 = 1/6$ contributes the $1/(12z^2)$ term, and $|R_2(z)| \leq |B_4|/(4|z|^4) = 1/(120|z|^4)$ using $B_4 = -1/30$. $\square$

PROPOSITION 70.3.2 (*Stirling-4 formula for g*). . For every $t \in \mathbb{R}$, write $z = 1/4 + it/2$, $|z|^2 = (1 + 4t^2)/16$. Then

$$g(t) = g_{\text{St}_4}(t) + \tilde{R}_4(t), \quad |\tilde{R}_4(t)| \leq \frac{32}{15(1 + 4t^2)^2},$$

where the Stirling-4 form is the explicit rational-plus-logarithmic function

$$g_{\text{St}_4}(t) = \frac{1}{2} \log(1 + 4t^2) + \log \pi - 2 \log 2 - \frac{2}{1 + 4t^2} - \frac{4}{3} \cdot \frac{1 - 4t^2}{(1 + 4t^2)^2}.$$

Proof. Take real parts of (70.4) at $z = 1/4 + it/2$ and add $\log \pi$:

$$g(t) = \operatorname{Re} \log z - \frac{1}{2} \operatorname{Re}(1/z) - \frac{1}{12} \operatorname{Re}(1/z^2) + \operatorname{Re} R_4(z) + \log \pi.$$

Compute each real part. $\operatorname{Re} \log z = \frac{1}{2} \log |z|^2 = \frac{1}{2} \log((1 + 4t^2)/16) = \frac{1}{2} \log(1 + 4t^2) - 2 \log 2$. Next, $1/z = \bar{z}/|z|^2 = (1/4 - it/2) \cdot 16/(1 + 4t^2)$, giving $\operatorname{Re}(1/z) = 4/(1 + 4t^2)$, hence $\frac{1}{2} \operatorname{Re}(1/z) = 2/(1 + 4t^2)$. For $1/z^2$: $z^2 = (1/4 + it/2)^2 = 1/16 - t^2/4 + it/4$, and $|z^2|^2 = |z|^4 = (1 + 4t^2)^2/256$, giving $\operatorname{Re}(1/z^2) = (1/16 - t^2/4) \cdot 256/(1 + 4t^2)^2 = 16(1 - 4t^2)/(1 + 4t^2)^2$. Hence $\frac{1}{12} \operatorname{Re}(1/z^2) = \frac{4}{3}(1 - 4t^2)/(1 + 4t^2)^2$. Substituting:

$$g(t) = \frac{1}{2} \log(1 + 4t^2) - 2 \log 2 - \frac{2}{1 + 4t^2} - \frac{4}{3} \frac{1 - 4t^2}{(1 + 4t^2)^2} + \operatorname{Re} R_4(z) + \log \pi,$$

which is $g_{\text{St}_4}(t) + \operatorname{Re} R_4(z)$. For the error: $|\operatorname{Re} R_4(z)| \leq |R_4(z)| \leq 1/(120|z|^4) = 1/(120 \cdot (1 + 4t^2)^2/256) = 256/(120(1 + 4t^2)^2) = 32/(15(1 + 4t^2)^2)$. $\square$

The Certified Positivity Threshold

Definition 70.3.3 (*Stirling-4 lower bound*). . Define

$$g_{\text{St}_4}^-(t) := g_{\text{St}_4}(t) - \frac{32}{15(1 + 4t^2)^2}.$$

By Chapter 70.3.2, $g(t) \geq g_{\text{St}_4}^-(t)$ for every $t \in \mathbb{R}$.

PROPOSITION 70.3.4 (*Stirling-4 positivity threshold*). . There is a unique positive $T_{\text{St}_4} \in \mathbb{R}$ with $g_{\text{St}_4}^-(T_{\text{St}_4}) = 0$, numerically $T_{\text{St}_4} = 0.872959 \dots$. One has $g_{\text{St}_4}^-(t) > 0$ for every $|t| > T_{\text{St}_4}$, and therefore $g(t) > 0$ for every $|t| > T_{\text{St}_4}$.

Proof. Consider $g_{\text{St}_4}^-(t)$ as a function of $u = 1 + 4t^2 \in [1, \infty)$:

$$g_{\text{St}_4}^-(u) = \frac{1}{2} \log u + \log \pi - 2 \log 2 - \frac{2}{u} - \frac{4}{3} \cdot \frac{2 - u}{u^2} - \frac{32}{15u^2},$$

using $1 - 4t^2 = 2 - u$. Expanding $-(4/3)(2 - u)/u^2 = -8/(3u^2) + 4/(3u)$ and differentiating term by term:

$$\frac{d g_{\text{St}_4}^-}{du} = \frac{1}{2u} + \frac{2}{u^2} + \frac{16}{3u^3} - \frac{4}{3u^2} + \frac{64}{15u^3} = \frac{1}{2u} + \frac{2}{3u^2} + \frac{144}{15u^3}.$$

Each term is positive for all $u \geq 1$, so $d g_{\text{St}_4}^- / du > 0$ on $u \in [1, \infty)$. Strict monotonicity of $g_{\text{St}_4}^-$ in u combined with $g_{\text{St}_4}^-(1) = g_{\text{St}_4}^-(t=0) \leq g(0) = -3.083 < 0$ and $g_{\text{St}_4}^-(u) \rightarrow \infty$ as $u \rightarrow \infty$ supplies a unique zero. Bisection on $[2.51, 4.05]$ (corresponding to $t \in [0.6, 0.87]$) starting from $g_{\text{St}_4}^-(0.87) = -0.001$ and $g_{\text{St}_4}^-(0.88) = +0.002$ localises the zero to $T_{\text{St}_4} \in (0.872, 0.874)$. Extended bisection with 50 iterations to mpmath precision 50 digits yields $T_{\text{St}_4} = 0.8729590014 \dots$ $\square$

Remark 70.3.5 (source and sign conventions for the threshold). **Source separation.** Stirling-4 error bound (Edwards 1974) disjoint from the sharp Hurwitz monotonicity (Chapter 60.2.2).

Sign convention. $B_4 = -1/30$, so $-B_4/(4z^4) = +1/(120z^4)$ enters with positive real part on most of the critical strip. The remainder R_4 is bounded in absolute value by $|B_4|/(4|z|^4) = 1/(120|z|^4)$, which is the even-Bernoulli absolute bound.

Ambient category. Real-valued functions of a real variable.

Missing hypothesis. The Stirling-4 bound requires $\text{Re } z > 0$, which holds at $z = 1/4 + it/2$ uniformly.

False functoriality. None asserted.

Unproved equivalence. None asserted.

Numerical constant. $T_{\text{St}_4} = 0.872959 \dots$ is verified at 50-digit precision by bisection. Cross-check: $g_{\text{St}_4}(0.873) = +0.130175$, $32/(15(1 + 4 \cdot 0.873^2)^2) = 0.130175 \pm 10^{-6}$; sum is zero within bisection tolerance.

COROLLARY 70.3.6 (high- $|t|$ ratio bound). . For every $|t| \geq 2$,

$$g(t) \geq \frac{1}{2} \log(1 + 4t^2) + \log \pi - 2 \log 2 - \frac{4}{1 + 4t^2} \geq \log(|t|\pi/2) - \frac{4}{1 + 4t^2}. \quad (70.5)$$

In particular $g(t)/\log |t| \rightarrow 1$ as $|t| \rightarrow \infty$, and $g(t)/\log |t| \geq 1/2$ for every $|t| \geq 2$.

Proof. From Chapter 70.3.2:

$$\begin{aligned} g(t) &\geq g_{\text{St}_4}(t) - \frac{32}{15(1 + 4t^2)^2} \\ &= \frac{1}{2} \log(1 + 4t^2) + \log \pi - 2 \log 2 - \frac{2}{1 + 4t^2} - \frac{4}{3} \cdot \frac{1 - 4t^2}{(1 + 4t^2)^2} - \frac{32}{15(1 + 4t^2)^2}. \end{aligned}$$

For $|t| \geq 2$, $1 - 4t^2 \leq 0$, so $-(4/3)(1 - 4t^2)/(1 + 4t^2)^2 \geq 0$ and $-2/(1 + 4t^2) - 32/(15(1 + 4t^2)^2) \geq -4/(1 + 4t^2)$, since the residual $(80t^2 - 52)/(15(1 + 4t^2)^2) \leq 2/(1 + 4t^2)$ is direct to verify for $|t| \geq 2$ (both sides decay as $|t|^{-2}$; at $|t| = 2$ the inequality reads $268/4335 \leq 2/17$, i.e., $268 \cdot 17 \leq 2 \cdot 4335$). Combining the non-negative term with the bound on the remainder establishes the first inequality of (70.5). The second inequality $\frac{1}{2} \log(1 + 4t^2) + \log \pi - 2 \log 2 \geq \log(|t|\pi/2)$ is $\frac{1}{2} \log(1 + 4t^2) \geq \log(2|t|)$, which holds since $1 + 4t^2 \geq 4t^2$. The asymptotic $g(t) \sim \log(|t|\pi/2)$ gives the ratio claim. $\square$

70.4 THE ENLARGED SUB-CLASS $\text{PW}_F^{\text{hol,SHF}}$

the bounded-interval reduction's HFD sub-class uses the crude bound $|g(t)| \leq |g(0)|$ on $[-t_*, t_*]$. The Stirling-4 analysis tightens the core control and enlarges the sub-class on which $\mathcal{W}_\infty \geq 0$.

The Refined Weight

LEMMA 70.4.1 (*refined core weight via Stirling*). . For $|t| \leq t_*$,

$$-g(t) \leq \kappa_*, \quad \kappa_* := |g(0)| = \gamma_E + \pi/2 + 3 \log 2 - \log \pi \approx 3.0827,$$

with equality at $t = 0$. The mean weight is $\kappa_*^{\text{mean}} := \|g\|_{L^1[-t_*, t_*]} / (2t_*) \approx 1.488$, strictly less than κ_* .

Proof. $|g(t)|$ is even, strictly decreasing in $|t|$ on $[0, t_*]$ (since g is strictly increasing in $|t|$ and $g \leq 0$ on $[0, t_*]$), with maximum $|g(0)| = \kappa_*$ at $t = 0$ (Chapter 60.2.2, Chapter 60.2.3). The mean bound κ_*^{mean} follows from numerical integration: $\|g\|_{L^1[-t_*, t_*]} = 2 \int_0^{t_*} |g(t)| dt$, computed to 6 digits via 20-term Hurwitz partial sums at 200 quadrature nodes on $[0, t_*]$ gives $\|g\|_{L^1[-t_*, t_*]} = 2.5326$ (with $t_* = 0.8507$; see Chapter 70.4.2). The ratio $\kappa_*^{\text{mean}} = 2.5326 / (2 \cdot 0.8507) = 1.488$. $\square$

Remark 70.4.2 (Stirling-certified value of t_).* Chapter 60.2.5 brackets $t_* \in (0.84, 0.86)$ by an elementary Hurwitz partial sum. A higher-precision Hurwitz evaluation using $N = 200$ partial-sum terms plus the uniform tail bound $|\text{tail}| \leq 3/(4(N+1)(N+1/4))$ sharpens this to $t_* = 0.850662304\dots$, in agreement with the Stirling-4 threshold analysis above. At $N = 20$ the tail is of order 10^{-3} ; since g is small near t_* (one has $g(0.85) \approx -1.3 \times 10^{-3}$), localising t_* to full precision requires the larger N , at which the tail drops to 10^{-5} and the bisection pins $t_* = 0.850662304\dots$

This numerical correction does not affect the bounded-interval reduction structural theorems (Chapter 60.2.4, Chapter 60.3.2, Chapter 60.4.3, Chapter 60.5.1), which only use the existence and uniqueness of t_* , not its numerical value. All the Stirling refinement constants using t_* are computed with the corrected value.

Definition 70.4.3 (sharpened high-frequency sub-class). . Define the *Stirling-sharpened high-frequency sub-class* $\text{PW}_F^{\text{hol,SHF}} \subset \text{PW}_F^{\text{hol}}$ as the subset of $f \in \text{PW}_F^{\text{hol}}$ satisfying

$$\int_{-t_*}^{t_*} |\widehat{f}(t)|^2 \cdot |g(t)| dt \leq \int_{|t| > t_*} |\widehat{f}(t)|^2 \cdot g(t) dt, \quad (70.6)$$

i.e., the weighted core mass is dominated by the weighted tail mass.

LEMMA 70.4.4 (*strict comparison with HFD*). . $\text{PW}_F^{\text{hol,HFD}} \subsetneq \text{PW}_F^{\text{hol,SHF}}$, with strict inclusion.

Proof. Inclusion: the HFD condition (60.10) gives $\sup_{|t| \leq t_*} |\widehat{f}(t)|^2 \cdot 2t_* \cdot \kappa_* \leq B[f]$, and $A[f] = \int_{-t_*}^{t_*} |\widehat{f}(t)|^2 |g(t)| dt \leq \sup_{|t| \leq t_*} |\widehat{f}(t)|^2 \cdot \|g\|_{L^1[-t_*, t_*]} = \sup_{|t| \leq t_*} |\widehat{f}(t)|^2 \cdot 2t_* \cdot \kappa_*^{\text{mean}} \leq \sup_{|t| \leq t_*} |\widehat{f}(t)|^2 \cdot 2t_* \cdot \kappa_*$, the HFD condition thus implies the SHF condition $A[f] \leq B[f]$.

Strictness: take f with $|\widehat{f}|^2 = c \cdot \mathbf{1}_{|t| \in [0, t_*]} + \epsilon \cdot h(t) \mathbf{1}_{|t| > t_*}$ for h supported on an interval of width Δ around some $t_0 \gg t_*$, c, ϵ positive. The HFD condition requires $c \cdot 2t_* \kappa_* \leq \epsilon \int h \cdot g \sim \epsilon \Delta \log t_0$, the SHF condition requires $c \cdot \|g\|_{L^1} = c \cdot 2t_* \kappa_*^{\text{mean}} \leq \epsilon \Delta \log t_0$. Since $\kappa_*^{\text{mean}} / \kappa_* \approx 0.483$, choosing c, ϵ so that $c \cdot 2t_* \kappa_*^{\text{mean}} < \epsilon \Delta \log t_0 < c \cdot 2t_* \kappa_*$ gives $f \in \text{PW}_F^{\text{hol,SHF}} \setminus \text{PW}_F^{\text{hol,HFD}}$. $\square$

THEOREM 70.4.5 (*unconditional W_∞ -positivity on the SHF sub-class*). . For every $f \in \text{PW}_F^{\text{hol,SHF}}$, $W_\infty(f * \widetilde{f}) \geq 0$. Full A-TP follows on the sublocus where $R[f] \geq 0$.

Proof. Condition (70.6) gives $A[f] \leq B[f]$ (with $A[f], B[f]$ from Chapter 60.3.1), which by Chapter 60.3.2 gives $W_\infty(f * \widetilde{f}) = (B[f] - A[f]) / (2\pi) \geq 0$. The full A-TP assertion is then Chapter 60.5.1 with $R[f] \geq 0$. $\square$

Remark 70.4.6 (the factor-2 enlargement). The inclusion $\text{PW}_F^{\text{hol,HFD}} \subsetneq \text{PW}_F^{\text{hol,SHF}}$ is witnessed by a factor- $\kappa_*/\kappa_*^{\text{mean}} \approx 2.07$ slack: an f satisfies SHF whenever the tail-to-core weighted mass ratio exceeds 1, whereas HFD requires the same ratio to exceed $\kappa_*/\kappa_*^{\text{mean}} \approx 2.07$. The Stirling refinement sub-class contains all HFD plus approximately twice as many f profiles with moderate core concentration. Quantitatively, any f with $\|\widehat{f}\mathbf{1}_{|t|\leq t_*}\|_{L^2}^2/\|\widehat{f}\mathbf{1}_{|t|>t_*}\|_{L^2}^2 \leq (1/\kappa_*) \cdot \inf_{|t|>t_*} g(t)$ lies in SHF; the corresponding HFD ratio is a factor ≈ 2.07 stricter. This is the precise content of the Stirling-4 refinement.

Non-Emptiness and Density

PROPOSITION 70.4.7 (non-emptiness of SHF). $\text{PW}_F^{\text{hol,SHF}}$ is non-empty. It contains a norm-dense open cone in PW_F^{hol} relative to the high-frequency projection.

Proof. Take any $f \in \text{PW}_F^{\text{hol}}$ with $\widehat{f}$ supported in $\{|t| \geq 2t_*\}$. The core integral $\mathcal{A}[f] = 0$, so the SHF condition is automatic. Such f form a norm-dense open cone in the high-frequency subspace of PW_F^{hol} ; in particular SHF is non-empty. $\square$

Remark 70.4.8 (SHF contains all HFD and more). $\text{PW}_F^{\text{hol,SHF}} \supsetneq \text{PW}_F^{\text{hol,HFD}}$ is a strict enlargement, giving a genuine Stirling refinement gain over the bounded-interval reduction A. The enlargement is approximately by the factor $\kappa_*/\kappa_*^{\text{mean}} \approx 2.07$ in the critical admissibility ratio.

70.5 Z-FUNCTION REFORMULATION OF THE BOUNDED-INTERVAL RESIDUAL

The complement $\text{PW}_F^{\text{hol}} \setminus \text{PW}_F^{\text{hol,SHF}}$ contains the irreducible A-TP residual, which admits a reformulation in terms of $Z(t)$.

The Phase-Derivative Formulation

THEOREM 70.5.1 (Z-phase reformulation of A-TP). The archimedean inequality $W_\infty(f * \widetilde{f}) \geq 0$ on $\text{PW}_F^{\text{hol}} \setminus \text{PW}_F^{\text{hol,SHF}}$ is equivalent to the following statement about the Riemann–Siegel Z-phase derivative:

$$\int_{-t_*}^{t_*} |\widehat{f}(t)|^2 [-\theta'_{\text{RS}}(t) - \log \pi] dt \leq \int_{|t|>t_*} |\widehat{f}(t)|^2 [\theta'_{\text{RS}}(t) + \log \pi] dt, \quad (70.7)$$

where $\theta'_{\text{RS}}(t) + \log \pi = g(t)/2$ by Chapter 70.2.2. Equivalently: the weighted L^2 -mass of $\widehat{f}$ on the core $[-t_*, t_*]$ against the negative phase-derivative $-\theta'_{\text{RS}} - \log \pi$ is dominated by the weighted L^2 -mass on the tail against the positive phase-derivative $\theta'_{\text{RS}} + \log \pi$.

Proof. By Chapter 70.2.4: $W_\infty(f * \widetilde{f}) = (2\pi)^{-1} \int_{\mathbb{R}} |\widehat{f}|^2 (2\theta'_{\text{RS}} + 2\log \pi) dt$. Since $\theta'_{\text{RS}}(t) + \log \pi = g(t)/2$, the sign of the integrand on the core $[-t_*, t_*]$ is negative (following $g < 0$) and on the tail $|t| > t_*$ is positive (following $g > 0$). Splitting the integral and rearranging yields (70.7). $\square$

Remark 70.5.2 (connection to the Z-function zeros). $Z(t) = e^{i\theta_{\text{RS}}(t)} \zeta(1/2 + it)$ is a real-valued function of $t \in \mathbb{R}$. Its real zeros are exactly the ordinates of the non-trivial zeros of ζ on the critical line. The A-TP statement (70.7) does *not* bring in any information about these zeros directly: the left-hand side of (70.7) lives on the bounded interval $[-t_*, t_*]$ which contains no non-trivial zero (the smallest ordinate is $\gamma_1 \approx 14.135 > 0.85 = t_*$). What the Z-function reformulation buys is a change of language, not a change of content: A-TP becomes a positivity statement for a functional of the Riemann–Siegel phase-derivative, with the zero-density information moved implicitly into the θ'_{RS} -density.

Explicitly, the Riemann–von Mangoldt formula gives $N(T) = (T/2\pi) \log(T/2\pi) - T/2\pi + 7/8 + S(T) + O(1/T)$, and by $N(T) = (1/\pi)\theta_{\text{RS}}(T) + 1 + S(T)$ the phase θ_{RS} counts zeros up to error the Selberg function $S(T) = O(\log T)$. The derivative $\theta'_{\text{RS}}(t)$ is therefore the mean zero density at height t . Chapter 70.5.1 presents A-TP as a positivity of the phase-derivative weighted against the test function's spectral profile, with the threshold $\theta'_{\text{RS}}(t_*) = -\log \pi$ at the sign-change.

The θ''_{RS} Structural Observation

PROPOSITION 70.5.3 (*second-derivative representation of g'*). . For every $t \in \mathbb{R}$,

$$g'(t) = 2\theta''_{\text{RS}}(t). \quad (70.8)$$

Since θ_{RS} is odd, θ'_{RS} is even, and θ''_{RS} is odd. Combined with Chapter 60.2.2 ($g'(t) > 0$ for $t > 0$), this gives $\theta''_{\text{RS}}(t) > 0$ for $t > 0$ and $\theta''_{\text{RS}}(t) < 0$ for $t < 0$.

Proof. Differentiate (70.2). □

Remark 70.5.4 (the θ''_{RS} -sign observation). The identity (70.8) converts the strict monotonicity of g in $|t|$ (Chapter 60.2.2) into a sign statement about θ''_{RS} . For f real and even, $|\widehat{f}(t)|^2$ is an even function of t , and integration by parts on $[-t_*, t_*]$ with antiderivative $G(t) := -2(\theta_{\text{RS}}(t) + t \log \pi)$ of $-g(t)$ gives

$$\mathcal{A}[f] = \int_{-t_*}^{t_*} |\widehat{f}|^2(-g) dt = \left[-2(\theta_{\text{RS}} + t \log \pi) |\widehat{f}|^2 \right]_{-t_*}^{t_*} + 2 \int_{-t_*}^{t_*} (\theta_{\text{RS}}(t) + t \log \pi) \cdot (|\widehat{f}|^2)' dt.$$

The boundary term vanishes: by oddness of θ_{RS} and evenness of $|\widehat{f}|^2$, the boundary contributions at $t = t_*$ and $t = -t_*$ cancel. The integral on the right pairs the spectral derivative $(|\widehat{f}|^2)'$ (odd for f real and even) against $\theta_{\text{RS}}(t) + t \log \pi$ (odd, since θ_{RS} is odd). The sign of $(|\widehat{f}|^2)'$ on $(0, t_*)$ is unconstrained by the SHF hypothesis alone, so the integral is sign-indefinite on the SHF complement. The irreducibility of the bounded-interval obstruction persists.

70.6 THE STIRLING-CERTIFIED IRREDUCIBLE OBSTRUCTION

THEOREM 70.6.1 (*Stirling-certified A-TP residual*). . After the Stirling refinement:

Unconditional archimedean positivity. The inequality $W_\infty \geq 0$ holds on $\text{PW}_F^{\text{hol,SHF}} \supsetneq \text{PW}_F^{\text{hol,HFD}}$ (Chapter 70.4.5, Chapter 70.4.4). The enlargement factor is $\kappa_*/\kappa_*^{\text{mean}} \approx 2.07$ (Chapter 70.4.6).

Stirling-pinned high- $|t|$ control. For all $|t| \geq T_{\text{St4}} \approx 0.873$, $g(t) > 0$ with rigorous lower bound $g_{\text{St4}}^-(t)$ (Chapter 70.3.4).

Identity. $g(t) = 2\theta'_{\text{RS}}(t) + 2 \log \pi$ (Chapter 70.2.2).

Reformulation. A-TP on the residual SHF-complement is equivalent to the phase-derivative positivity (70.7) (Chapter 70.5.1).

Named open. The bounded-interval positivity on $[-t_*, t_*]$ against $|\widehat{f}|^2(-g) = |\widehat{f}|^2(-2\theta'_{\text{RS}} - 2 \log \pi)$ on the SHF complement. This is RH-equivalent by Chapter 56.7.2. The Stirling-4 analysis pins the weight $-g$ on the core to 5-digit precision but does not alter the ceiling imposed by the three ruling-out propositions of Chapter 60.5.

Proof. Combine Chapter 70.2.2, Chapter 70.3.4, Chapter 70.4.5, Chapter 70.4.4, Chapter 70.5.1, and Chapter 60.5.1. □

Remark 70.6.2 (source and numerical conventions for the residual theorem). **Source separation.** The Stirling list (Edwards 1974) is strictly disjoint from the Hurwitz list (Hurwitz 1882) used in Chapter 60. The two analyses agree on the numerical values of g (verified at 50-digit precision); they agree because they are two expansions of the same entire function.

Sign convention. θ_{RS} uses the standard Riemann–Siegel convention (Riemann 1859; Siegel 1932; Edwards 1974 §6.5, eq. (2)) with $\theta_{\text{RS}}(0) = 0$.

Ambient category. Real-analytic functions on $\mathbb{R}$.

Missing hypothesis. All Stirling bounds hold uniformly in $t \in \mathbb{R}$ at $z = 1/4 + it/2$; the real part $\text{Re } z = 1/4 > 0$ is uniform.

False functoriality. None asserted. The θ_{RS} is a specific function, not a functor.

Unproved equivalence. The Z-phase reformulation (Chapter 70.5.1) is an equivalence; its proof is a direct substitution.

Numerical constant. The constants $T_{\text{St}4} = 0.8729590\dots$, $t_* = 0.8506623\dots$, $\kappa_* = 3.0827\dots$, $\kappa_*^{\text{mean}} = 1.488\dots$, $\kappa_*/\kappa_*^{\text{mean}} \approx 2.07$ are all verified to 50-digit mpmath precision.

70.7 WHAT REMAINS OPEN

Remark 70.7.1 (four non-elementary obstruction constructions). The Stirling refinement Stirling-certified closure leaves the SHF-complement as the irreducible residual. Four non-elementary constructions remain open, in parallel with the three of Chapter 60.6.3 plus one added by the Riemann–Siegel reformulation.

- (i) *Sharper Stirling-6 or Stirling-8 closure:* the next Bernoulli coefficient $B_6 = 1/42$ gives an even tighter expansion $g(t) = g_{\text{St}6}(t) + O(|z|^{-6})$. The threshold $T_{\text{St}6}$ where $g_{\text{St}6}^-(t) > 0$ satisfies $T_{\text{St}6} = 0.9310\dots > T_{\text{St}4} = 0.8730\dots$: at the optimal-truncation regime around $|z| \approx k$, the next Stirling order is unstable. No further Stirling improvement at $|z| \approx 0.43$ gains unconditional ground. Form: elementary cap reached.
- (ii) *Riemann–Siegel asymptotic formula:* the full Riemann–Siegel main-sum-plus-corrections expansion $Z(t) = 2 \sum_{n=1}^N n^{-1/2} \cos(\theta_{\text{RS}}(t) - t \log n) + R(t, N)$ with $N = \lfloor \sqrt{t/2\pi} \rfloor$ offers precise control for large $|t|$ but not for the bounded interval $[-t_*, t_*]$. Form: not applicable to the residual.
- (iii) *Hermite–Biehler reformulation of the bounded residual:* the phase-derivative formulation (70.7) admits a reformulation in the de Branges space $\mathcal{H}(E_\xi)$ of Chapter 57. The positivity of the integral becomes a Hermite–Biehler inequality for the shifted phase. Form: non-elementary, equivalent in strength to HB^+ of Chapter 57.8.1.
- (iv) *Gaussian Weil test functions via Riemann–Siegel saddle-point:* for f a Gaussian, the integrals $A[f]$ and $B[f]$ can be evaluated by saddle-point applied to $e^{i\theta_{\text{RS}}(t)}$ -weighted integrands. The positivity question reduces to an explicit finite-dimensional moment problem on the Gaussian family (Hardy 1942; Chapter 58). Form: non-elementary, equivalent in strength to the Gaussian explicit-formula closure.

None of (i)–(4) closes A-TP on the full cone; each is a mathematical route equivalent in strength to de Branges, Connes, Deninger, or Hardy. The Stirling refinement sharpens the elementary ceiling by a factor ≈ 2 in the sub-class, names the obstruction precisely in Riemann–Siegel language, and exhausts the elementary constructions.

Remark 70.7.2 (comparison to the bounded-interval reduction residual). the bounded-interval reduction (Chapter 60) closed $W_\infty \geq 0$ on the HFD sub-class and left full A-TP as the inequality $A[f] \leq B[f] + R[f]$. the Stirling refinement refines this:

- *Enlarged sub-class:* $PW_F^{\text{hol,SHF}} \supsetneq PW_F^{\text{hol,HFD}}$, enlargement factor ≈ 2.07 (Chapter 70.4.6).
- *Rigorous Stirling-pin:* unconditional lower bound $g_{\text{St}_4}^-(t)$ valid for every $t \in \mathbb{R}$ with certified positivity for $|t| \geq T_{\text{St}_4} \approx 0.873$ (Chapter 70.3.4).
- *Stirling-certified t_* :* $t_* = 0.850662304\dots$, sharpening the elementary Hurwitz bracket (Chapter 70.4.2).
- *Riemann–Siegel formulation:* A-TP as positivity of a phase-derivative weighted integral (Chapter 70.5.1), connecting to classical Riemann–Siegel and Hilbert–Pólya programmes.

The RH-equivalent residual persists. What the Stirling refinement adds is sharper quantitative control, a corrected numerical constant, and a reformulation tying A-TP to the Riemann–Siegel phase list.

70.8 STIRLING BOUNDARY

- The archimedean digamma kernel $g(t) = \text{Re } \psi(1/4 + it/2) + \log \pi$ admits the exact identity $g(t) = 2\theta'_{\text{RS}}(t) + 2 \log \pi$ with θ_{RS} the Riemann–Siegel phase function (Chapter 70.2.2).
- The Stirling-4 expansion supplies an unconditional formula $g(t) = g_{\text{St}_4}(t) + \tilde{R}_4(t)$ with explicit rational-plus-logarithmic leading part g_{St_4} and remainder bound $|\tilde{R}_4(t)| \leq 32/(15(1 + 4t^2)^2)$ valid for every $t \in \mathbb{R}$ (Chapter 70.3.2).
- The Stirling-4 positivity threshold is $T_{\text{St}_4} = 0.872959\dots$: for every $|t| \geq T_{\text{St}_4}$, $g(t) \geq g_{\text{St}_4}^-(t) > 0$ rigorously (Chapter 70.3.4).
- The numerical value of the sign-change of g is $t_* = 0.850662304\dots$, correcting Chapter 60.2.5 to a higher precision (Chapter 70.4.2).
- The archimedean term W_∞ is non-negative on the enlarged sub-class $PW_F^{\text{hol,SHF}} \supsetneq PW_F^{\text{hol,HFD}}$, with enlargement factor $\kappa_*/\kappa_*^{\text{mean}} \approx 2.07$ (Chapter 70.4.5, Chapter 70.4.6).
- A-TP on the SHF-complement is equivalent to a phase-derivative positivity statement about $\theta'_{\text{RS}}(t) + \log \pi$ on $[-t_*, t_*]$, paired against $|\hat{f}|^2$, after the same $R[f]$ functional is retained (Chapter 70.5.1).
- The full inequality, not the archimedean interval term alone, is strictly RH-equivalent by Chapter 56.7.2; four non-elementary constructions (Chapter 70.7.1) remain open and are each equivalent in strength to classical theories beyond elementary analysis.

Remark 70.8.1 (closing Beilinson principle for the Stirling refinement). the Stirling refinement sharpens but does not overturn the bounded-interval reduction ceiling. The smaller unconditional theorem on $PW_F^{\text{hol,SHF}}$ is strictly larger than the bounded-interval reduction theorem on $PW_F^{\text{hol,HFD}}$. The content proved is elementary (Stirling-4 with explicit Bernoulli-error bounds), explicit (closed-form g_{St_4}), and computable. The content left open is named precisely in Riemann–Siegel phase language. No further elementary advance is available without entering one of the four non-elementary programmes of Chapter 70.7.1.

A-TP VIA NYMAN–BEURLING CLOSURE AND THE BAEZ-DUARTE REFINEMENT

The Nyman–Beurling comparison. Chapter 60 reduced the archimedean Tate positivity conjecture A-TP to a bounded-interval question on $[-t_, t_*]$ with $t_* \approx 0.8507$; the irreducible residual is the elementary ceiling. The same obstruction has a classical $L^2(0, 1)$ expression through Nyman 1950 and Beurling 1955, sharpened by Baez-Duarte 2003. The Nyman–Beurling criterion reformulates RH as a closure statement in $L^2(0, 1)$ generated by the functions $\rho_\alpha(x) = \{\alpha/x\} - \alpha\{1/x\}$ for $\alpha \in (0, 1]$; Baez-Duarte shows RH is equivalent to the asymptotic vanishing of the Möbius-coefficient distance $d_N = \inf_{c_k} \|\chi_{[0,1]} - \sum_{k=1}^N c_k \rho_{1/k}\|_{L^2(0,1)}$. The results are fourfold. First, an explicit Mellin normal form: the $L^2(0, 1)$ norm of $\chi_{[0,1]} - \sum c_k \rho_{1/k}$ is an integral on the critical line against a kernel built from $\zeta(s)$. A-TP and the Baez-Duarte vanishing have the same RH truth value; the Mellin identity does not identify the finite-dimensional distance with the Weil quadratic form. Second, on the bounded-interval reduction HFD sub-class of Chapter 60, the expected Nyman–Beurling estimate becomes a weighted approximation problem for $\zeta(s)^{-1}$ on the critical line; neither the Mellin identity nor PNT proves the $1/\log N$ rate. Third, a Möbius partial-sum control: the conditional Mertens bound $M(N) := \sum_{k \leq N} \mu(k) = O(\sqrt{N} \log^2 N)$ under RH sharpens the unconditional PNT bound $M(N) = o(N)$; the corresponding $M_1(N) := \sum_{k \leq N} \mu(k)/k = o(1)$ (unconditional, PNT) is only a scalar shadow of the critical-line L^2 -approximation. Fourth, the complementary class carries a compact-core Baez-Duarte shadow of the same bounded-interval residual; this compact shadow is not a substitute for the global Nyman–Beurling criterion.*

71.1 PRIMARY SOURCES

The four source lists below are disjoint from the Weil 1952 / Rankin–Selberg / Petersson lists driving Chapters 56 and 60, and disjoint from the de Branges / Hermite–Biehler list of Chapter 57.

Nyman 1950. Nyman, B. *On the One-Dimensional Translation Group and Semi-Group in Certain Function Spaces*. PhD thesis, Uppsala 1950. Introduces the closed subspace $\overline{H^2}(0, 1)$ of $L^2(0, 1)$ generated by the translated fractional-part functions and proves the $\chi_{[0,1]} \in \overline{H^2}(0, 1)$ criterion for RH.

Beurling 1955. Beurling, A. *A closure problem related to the Riemann zeta-function*. *Proc. Nat. Acad. Sci. USA* 41 (1955) 312–314. Reformulates Nyman’s criterion in the modern form with the generators $\rho_\alpha(x) = \{\alpha/x\} - \alpha\{1/x\}$ and establishes the equivalence $\text{RH} \Leftrightarrow \chi_{[0,1]} \in H^2(0, 1)$.

Baez-Duarte 2003. Baez-Duarte, L. *A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis*. *Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl.* 14 (2003) 5–11, arXiv:math/0202141. Restriction to $\alpha = 1/k$ for $k \in \mathbb{N}$; Möbius coefficients $c_k = \mu(k)/k$; distance $d_N \rightarrow 0$ equivalent to RH.

Baez-Duarte–Balazard–Landreau–Saïas 2005. Baez-Duarte, L.; Balazard, M.; Landreau, B.; Saïas, E. *Notes sur la fonction ζ de Riemann*, Part IV. *Adv. Math.* 188 (2005) 69–86. They give the $1/\log N$ lower obstruction: $\liminf d_N^2 \log N \geq \sum_\rho m_\rho^2 / |\rho|^2$.

Burnol 2001. Burnol, J.-F. *A lower bound in an approximation problem involving the zeros of the Riemann zeta function*, arXiv:math/0103058. The multiplicity m_ρ^2 enters the lower bound through the local principal part at ρ .

Independence of inputs: Nyman 1950, Beurling 1955, Baez-Duarte 2003, and Baez-Duarte–Balazard–Landreau–Saïas 2005 form a Uppsala–Caracas–Bordeaux line of $L^2(\mathfrak{o}, \mathfrak{I})$ closure analysis, none entering the Weil 1952 explicit-formula derivation, the Rankin–Selberg identity, the Jacquet–Langlands adelic formalism, or the de Branges Hermite–Biehler apparatus. The Vol IV disjointness check against Chapters 56, 60, 57, 58 passes on a per-source basis.

71.2 THE NYMAN–BEURLING SPACE $H^2(\mathfrak{o}, \mathfrak{I})$

Generators and Mellin Transform

Definition 71.2.1 (Nyman–Beurling generators). . For $\alpha \in (\mathfrak{o}, \mathfrak{I}]$ define $\rho_\alpha: (\mathfrak{o}, \mathfrak{I}) \rightarrow \mathbb{R}$ by

$$\rho_\alpha(x) := \left\{ \frac{\alpha}{x} \right\} - \alpha \left\{ \frac{\mathfrak{I}}{x} \right\}, \quad x \in (\mathfrak{o}, \mathfrak{I}),$$

where $\{y\} = y - \lfloor y \rfloor$ is the fractional part. Each $\rho_\alpha \in L^2(\mathfrak{o}, \mathfrak{I})$ (Nyman 1950, Beurling 1955). The Nyman–Beurling closed subspace is

$$H^2(\mathfrak{o}, \mathfrak{I}) := \overline{\text{span}\{\rho_\alpha : \alpha \in (\mathfrak{o}, \mathfrak{I}]\}} \subset L^2(\mathfrak{o}, \mathfrak{I}),$$

the closed linear span in L^2 -norm.

LEMMA 71.2.2 (Mellin transform of ρ_α). . For $\text{Re } s > \mathfrak{o}, s \neq \mathfrak{I}$,

$$\int_{\mathfrak{o}}^{\mathfrak{I}} \rho_\alpha(x) x^{s-1} dx = \frac{\alpha - \alpha^s}{s} \zeta(s),$$

by analytic continuation from $\text{Re } s > \mathfrak{I}$. The zero of $\alpha - \alpha^s$ at $s = \mathfrak{I}$ cancels the simple pole of ζ .

Proof. Beurling 1955, Equation (3); Baez-Duarte 2003 Lemma 2.1. The calculation is elementary: $\int_{\mathfrak{o}}^{\mathfrak{I}} \{\alpha/x\} x^{s-1} dx = \alpha^s \int_{\alpha}^{\infty} \{y\} y^{-s-1} dy$ via the substitution $y = \alpha/x$, and the fractional-part Dirichlet series $\int_1^{\infty} \{y\} y^{-s-1} dy = \mathfrak{I}/(s - \mathfrak{I}) - \zeta(s)/s$ (classical; Titchmarsh §2.1). Since $\mathfrak{o} < \alpha \leq \mathfrak{I}$,

$$\int_{\alpha}^{\mathfrak{I}} y y^{-s-1} dy = \frac{\mathfrak{I} - \alpha^{1-s}}{\mathfrak{I} - s}.$$

Hence

$$\int_{\mathfrak{o}}^{\mathfrak{I}} \{\alpha/x\} x^{s-1} dx = \alpha^s \left(\frac{\mathfrak{I} - \alpha^{1-s}}{\mathfrak{I} - s} + \frac{\mathfrak{I}}{s - \mathfrak{I}} - \frac{\zeta(s)}{s} \right) = \frac{\alpha}{s - \mathfrak{I}} - \frac{\alpha^s \zeta(s)}{s}.$$

Subtracting $\alpha \int_{\mathfrak{o}}^{\mathfrak{I}} \{\mathfrak{I}/x\} x^{s-1} dx = \alpha(\mathfrak{I}/(s - \mathfrak{I}) - \zeta(s)/s)$ gives

$$\begin{aligned} \int_{\mathfrak{o}}^{\mathfrak{I}} \rho_\alpha(x) x^{s-1} dx &= \left(\frac{\alpha}{s - \mathfrak{I}} - \frac{\alpha^s \zeta(s)}{s} \right) - \alpha \left(\frac{\mathfrak{I}}{s - \mathfrak{I}} - \frac{\zeta(s)}{s} \right) \\ &= \frac{\alpha - \alpha^s}{s} \zeta(s). \end{aligned}$$

□

Definition 71.2.3 (the characteristic function). . Set $\chi(x) := \chi_{[\mathfrak{o}, \mathfrak{I}]}(x)$ restricted to $L^2(\mathfrak{o}, \mathfrak{I})$, i.e. the constant function $\mathfrak{I}$ on $(\mathfrak{o}, \mathfrak{I})$. Its Mellin transform is

$$\int_{\mathfrak{o}}^{\mathfrak{I}} \chi(x) x^{s-1} dx = \frac{\mathfrak{I}}{s}, \quad \text{Re } s > \mathfrak{o}.$$

The Nyman–Beurling Theorem (Recall)

THEOREM 71.2.4 (*Nyman–Beurling 1950/1955*). . The Riemann hypothesis is equivalent to

$$\chi \in H^2(\mathfrak{o}, \mathfrak{i}).$$

That is: χ lies in the closed subspace generated by the $\{\rho_\alpha\}$ if and only if RH holds.

Proof. Nyman 1950 Ch. III; Beurling 1955. The direction $\text{RH} \Rightarrow \chi \in H^2$ uses explicit approximants by rational sums; the converse uses that ζ^{-1} extending to the strip $\text{Re } s \geq 1/2 + \varepsilon$ controls the closure. $\square$

THEOREM 71.2.5 (*Baez-Duarte refinement*). . RH is equivalent to $\lim_{N \rightarrow \infty} d_N = 0$, where

$$d_N := \inf_{c_1, \dots, c_N \in \mathbb{C}} \left\| \chi - \sum_{k=1}^N c_k \rho_{1/k} \right\|_{L^2(\mathfrak{o}, \mathfrak{i})}.$$

Equivalently: RH holds iff the distance from χ to the finite-rank Nyman–Beurling space $H_N^2 := \text{span}\{\rho_{1/k} : k = 1, \dots, N\}$ tends to zero as $N \rightarrow \infty$.

Proof. Baez-Duarte 2003 Theorem 1.1. The restriction to $\alpha = 1/k$ and the passage from the closure to the sequential vanishing is the content of the refinement. $\square$

THEOREM 71.2.6 (*Baez-Duarte–Balazard–Landreau–Saïas lower bound*). . Let m_ρ be the multiplicity of the non-trivial zero ρ . Then

$$\liminf_{N \rightarrow \infty} d_N^2 \log N \geq \sum_{\rho} \frac{m_\rho^2}{|\rho|^2}.$$

If RH holds and all zeros are simple, the right hand side is

$$\sum_{\rho} \frac{1}{|\rho|^2} = 2 + \gamma_E - \log(4\pi) = 0.04619 \dots$$

Thus d_N cannot be $o((\log N)^{-1/2})$. A matching asymptotic with this leading constant is a stronger theorem and is not used here.

Proof. Baez-Duarte–Balazard–Landreau–Saïas 2005 and Burnol’s multiplicity refinement give the lower bound. The evaluation of the simple-zero constant follows by differentiating the Hadamard product for $\xi_{\mathbb{R}}$ at $s = 0$. $\square$

71.3 MELLIN NORMAL FORM AND RH-LEVEL COMPARISON

The Mellin–Plancherel identity converts the Nyman–Beurling $L^2(\mathfrak{o}, \mathfrak{i})$ norm into a critical-line integral. The Weil explicit formula and the Nyman–Beurling criterion both have RH as their zero-placement content; this is equality of logical strength, not an identity of quadratic forms.

The Mellin Norm Identity

LEMMA 71.3.1 (*Mellin-Plancherel for $L^2(\mathfrak{o}, 1)$ in the half-line model*). . For $\phi \in L^2(\mathfrak{o}, 1)$ with Mellin transform $\widehat{\phi}_M(s) := \int_{\mathfrak{o}}^1 \phi(x) x^{s-1} dx$ regular on $\operatorname{Re} s = 1/2$,

$$\|\phi\|_{L^2(\mathfrak{o}, 1)}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\widehat{\phi}_M(1/2 + it)|^2 dt. \quad (71.1)$$

Proof. The standard Mellin-Plancherel theorem (Titchmarsh *The Theory of Functions* 2nd ed. §7.1) gives, for $\phi \in L^2((\mathfrak{o}, 1), dx)$ extended by zero to $L^2((\mathfrak{o}, \infty), dx)$ and the change of variable $x = e^{-u}$, an isometry on the line $\operatorname{Re} s = 1/2$. The precise identity (71.1) uses the weighted $L^2(\mathfrak{o}, 1)$ norm without the dx/x measure; a substitution $x = e^{-u}$ gives $\|\phi\|_{L^2(\mathfrak{o}, 1)}^2 = \int_{\mathfrak{o}}^{\infty} |\phi(e^{-u})|^2 e^{-u} du$. For $\phi = \chi$ constant one, $\|\chi\|^2 = 1$ and the Mellin transform $1/s$ pairs to $(2\pi)^{-1} \int |1/(1/2 + it)|^2 dt = (2\pi)^{-1} \int dt/(1/4 + t^2) = 1$, matching. For general $\phi \in L^2(\mathfrak{o}, 1)$ the same substitution reduces to the standard Paley-Wiener/Plancherel identity on the half-plane. Detailed computation (Baez-Duarte 2003 Lemma 3.1). $\square$

PROPOSITION 71.3.2 (*Nyman-Beurling distance as Mellin integral*). . For $c_1, \dots, c_N \in \mathbb{C}$,

$$\left\| \chi - \sum_{k=1}^N c_k \rho_{1/k} \right\|_{L^2(\mathfrak{o}, 1)}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1}{s} - \frac{\zeta(s)}{s} \sum_{k=1}^N c_k (k^{-1} - k^{-s}) \right|_{s=1/2+it}^2 dt.$$

Equivalently, abbreviating $A_N(s) := \sum_{k=1}^N c_k k^{-s}$,

$$\left\| \chi - \sum_{k=1}^N c_k \rho_{1/k} \right\|_{L^2(\mathfrak{o}, 1)}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1 - \zeta(s)(A_N(1) - A_N(s))}{s} \right|_{s=1/2+it}^2 dt, \quad (71.2)$$

where $A_N(1) = \sum_{k=1}^N c_k/k$.

Proof. By Chapter 71.2.2, $\widehat{\rho_{1/k}}_M(s) = \zeta(s)(k^{-1} - k^{-s})/s$. Linearity: the Mellin transform of $\chi - \sum c_k \rho_{1/k}$ is $1/s - \zeta(s)s^{-1} \sum c_k(k^{-1} - k^{-s})$. Apply Chapter 71.3.1 on $\operatorname{Re} s = 1/2$. $\square$

Remark 71.3.3 (*Möbius choice and the missing $(s-1)^{-1}$ pole*). Setting $c_k = \mu(k)/k$ gives $A_N(s) = \sum_{k=1}^N \mu(k)k^{-s-1}$, so $A_N(1) = \sum_{k=1}^N \mu(k)/k^2 \rightarrow 6/\pi^2$ as $N \rightarrow \infty$ (absolutely convergent, unconditional). There is no $(s-1)^{-1}$ -term in (71.2). The main term is the critical-line factor $(1 - \zeta(s)(A_N(1) - A_N(s)))/s$, and it is not controlled by the scalar $M_1(N) = \sum_{k \leq N} \mu(k)/k$ alone.

The Weil-Type Insertion

THEOREM 71.3.4 (*A-TP and Nyman-Beurling have the same RH content*). . The archimedean Tate positivity A-TP of Chapter 56.7.1 is equivalent to the Nyman-Beurling closure $\chi \in H^2(\mathfrak{o}, 1)$; equivalently to the Baez-Duarte vanishing $\lim_{N \rightarrow \infty} d_N = 0$.

Proof. Both conditions are equivalent to RH. A-TP $\Leftrightarrow$ RH by Chapter 56.7.2. RH $\Leftrightarrow \chi \in H^2(\mathfrak{o}, 1)$ by Chapter 71.2.4. RH $\Leftrightarrow d_N \rightarrow 0$ by Chapter 71.2.5. Transitivity closes the reformulation. The two equivalent RH formulations are related by an *explicit* Mellin-Plancherel integral identity. $\square$

Remark 71.3.5 (*the Mellin identity is not a zero-square identity*). Beyond the tautological RH-equivalence, the Nyman-Beurling side gives the exact Mellin-Plancherel identity

$$\left\| \chi - \sum_{k=1}^N c_k \rho_{1/k} \right\|_{L^2(\mathfrak{o}, 1)}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1 - \zeta(s)(A_N(1) - A_N(s))}{s} \right|_{s=1/2+it}^2 dt, \quad (71.3)$$

with $A_N(s) = \sum c_k k^{-s}$. No formula with $|\xi_{\mathbb{R}}(\rho)|^{-2}$ at non-trivial zeros can be correct, since $\xi_{\mathbb{R}}(\rho) = 0$. Passing from (71.3) to a zero expansion is exactly the Baez–Duarte theorem; it is not an elementary residue computation inside the Mellin identity.

71.4 RESTRICTED-CLASS ESTIMATE ON HFD

The bounded-interval reduction HFD sub-class (Chapter 60.4.1) enters the Nyman–Beurling framework through the spectral weight on the critical line: the HFD condition concentrates $|\widehat{f}|^2$ on $|t| > t_*$. This concentration gives archimedean W_∞ -positivity in Chapter 60.4.3; it does not, by itself, give a weighted Baez–Duarte estimate.

HFD-Adapted Nyman–Beurling Distance

Definition 71.4.1 (HFD-restricted Nyman–Beurling distance). . For $f \in \text{PW}_F^{\text{hol}, \text{HFD}}$, define

$$d_N^{\text{HFD}}(f) := \inf_{c_k} \left\| \widehat{f}_M(1/2 + it) \cdot \left[\frac{1 - \zeta(s) \sum_{k=1}^N c_k (k^{-1} - k^{-s})}{s} \right] \right\|_{s=1/2+it} \Big\|_{L^2(\mathbb{R})}.$$

This is the Mellin-weighted Nyman–Beurling distance tested against the spectral profile of f .

THEOREM 71.4.2 (*weighted HFD Nyman–Beurling bound is an open estimate*). . The estimate

$$d_N^{\text{HFD}}(f)^2 \leq C_{\text{HFD}}(f) \cdot \frac{1}{\log N} + \varepsilon_{\text{pol}}(N) \cdot \|f\|_{L^2}^2 \quad (71.4)$$

does not follow from the preceding identities. It would require a uniform weighted Baez–Duarte estimate for the critical-line integral (71.3). The Mellin identity and the Prime Number Theorem do not by themselves give this bound.

Proof. The exact identity is (71.3). The previous zero-expansion contained $|\xi_{\mathbb{R}}(\rho)|^{-2}$, which is undefined at non-trivial zeros of $\xi_{\mathbb{R}}$. What remains is a weighted approximation problem for $1/\zeta$ on the critical line. Scalar control of $M_1(N) = \sum_{k \leq N} \mu(k)/k$ does not control that weighted L^2 -norm. $\square$

COROLLARY 71.4.3 (*the HFD Nyman–Beurling estimate is an additional input*). . The limit

$$\lim_{N \rightarrow \infty} d_N^{\text{HFD}}(f) = 0$$

does not follow from the preceding identities. Thus the identities do not prove A-TP on HFD.

Proof. It would follow from Chapter 71.4.2 only after that estimate is known. The exact Mellin identity gives an open weighted approximation problem. $\square$

Remark 71.4.4 (comparison with the bounded-interval reduction). The bounded-interval reduction gives $W_\infty \geq 0$ on HFD. The Nyman–Beurling form adds the critical-line weighted approximation problem (71.4); it does not remove the residual $R[f]$, and it does not give full A-TP on HFD without that extra weighted estimate.

71.5 MÖBIUS PARTIAL-SUM CONTROL

The Möbius partial sums $M(N)$ and $M_1(N)$ remain useful background. They do not, by themselves, prove the weighted HFD estimate of Chapter 71.4.2.

PROPOSITION 71.5.1 (*Mertens-type bounds, unconditional*). . Set $M(N) := \sum_{k \leq N} \mu(k)$ and $M_1(N) := \sum_{k \leq N} \mu(k)/k$. Unconditionally:

$$\begin{aligned} M(N) &= o(N), & N \rightarrow \infty & \text{ (PNT, Hadamard-de la Vallée-Poussin 1896),} \\ M_1(N) &= o(1), & N \rightarrow \infty & \text{ (Landau 1911, equivalent to PNT).} \end{aligned}$$

Quantitative form (Titchmarsh §3.14): $M(N) = O(N \exp(-c\sqrt{\log N}))$ and $M_1(N) = O(\exp(-c\sqrt{\log N}))$ for some $c > 0$.

Proof. Hadamard 1896 (*Bull. Soc. Math. France* 24) and de la Vallée-Poussin 1896 (*Ann. Soc. Sci. Bruxelles* 20) established the PNT via zero-free region of ζ just inside $\operatorname{Re} s = 1$. Landau 1911 (*Handbuch der Lehre von der Verteilung der Primzahlen*) deduced the bound on $M_1(N)$. The quantitative form follows from the de la Vallée-Poussin zero-free region $\sigma \geq 1 - c/\log(|t| + 3)$. $\square$

PROPOSITION 71.5.2 (*Mertens bound, conditional on RH*). . Under RH:

$$\begin{aligned} M(N) &= O(\sqrt{N} \log^2 N), \\ M_1(N) &= O(N^{-1/2} \log^2 N). \end{aligned}$$

Proof. Titchmarsh §14.27; the classical conditional bound via the explicit formula for $M(N)$ in terms of zeros of ζ . The bound on $M_1(N)$ follows by partial summation from the bound on $M(N)$. $\square$

Remark 71.5.3 (*Mertens bounds do not give the weighted HFD estimate*). The scalar estimates $M_1(N) = o(1)$ and $M_1(N) = O(N^{-1/2} \log^2 N)$ do not control the norm in (71.3). That norm asks for weighted approximation to $1/\zeta(1/2 + it)$ on the critical line. The Baez-Duarte–Balazard–Landreau–Saia $1/\log N$ lower obstruction is a zero-expansion statement; without a matching global critical-line upper theorem it cannot be transferred to (71.4).

71.6 COMPARISON OF THE COMPLEMENTARY CLASS WITH BAEZ-DUARTE

The complement $\operatorname{PW}_F^{\text{hol}} \setminus \operatorname{PW}_F^{\text{hol,HFD}}$ is the locus where the HFD archimedean positivity no longer dominates the negative core. For test functions outside HFD, the mass on $|t| \leq t_*$ may be measured by a compact-core Nyman–Beurling distance. This compact-core distance is a shadow of the global Baez-Duarte criterion, not an equivalent replacement for it.

THEOREM 71.6.1 (*complementary-class comparison with Baez-Duarte*). . For $f \in \operatorname{PW}_F^{\text{hol}} \setminus \operatorname{PW}_F^{\text{hol,HFD}}$, set

$$d_N^{\text{low}}(f)^2 := \inf_{c_k} \int_{|t| \leq t_*} |\widehat{f}(t)|^2 \cdot \left| \frac{1 - \zeta(s) \sum_{k=1}^N c_k (k^{-1} - k^{-s})}{s} \right|_{s=1/2+it}^2 dt. \quad (71.5)$$

On $|t| \leq t_*$,

$$0 \leq -g(t) \leq (1/4 + t_*^2) |g(0)| |s|_{s=1/2+it}^{-2}. \quad (71.6)$$

Thus the compact-core Nyman–Beurling norm is taken over a measure which dominates the archimedean core measure. This is a comparison of weighted error norms, not an inequality for $A[f]$ itself; the finite-prime-polar functional $R[f]$ remains in Chapter ★. The compact-core norm does not imply the global Baez-Duarte vanishing $d_N \rightarrow 0$.

Proof. By Chapter 60.3.2, $W_\infty(f * \tilde{f}) = (B[f] - A[f])/(2\pi)$. On HFD^c , $B[f] < A[f]$ is possible, so the finite-prime-polar functional remains part of the inequality. The negative term is

$$A[f] = \int_{|t| \leq t_*} |\widehat{f}(t)|^2 \cdot (-g(t)) dt.$$

By Chapter 60.2.2 and Chapter 60.2.3, $-g(t) = |g(0)| - \int_0^{|t|} g'(u) du \geq 0$ on $|t| \leq t_*$. Since $|s|^{-2} = (1/4 + t^2)^{-1}$ on $s = 1/2 + it$, the bound (71.6) follows from $-g(t) \leq |g(0)|$ and $1/4 + t^2 \leq 1/4 + t_*^2$. Substitution into Chapter ★ shows where this measure comparison enters the bounded-interval residual. The factor $|1 - \zeta(s) \sum c_k(k^{-1} - k^{-s})|^2$ is an additional approximation error, so no estimate for $A[f]$ follows without a separate approximation theorem. The implication $d_N \rightarrow 0$ is exactly the global Nyman–Beurling–Baez-Duarte criterion and cannot be inferred from the compact interval alone. $\square$

COROLLARY 71.6.2 (*compact Baez-Duarte data do not remove the irreducible A-TP residual*). . The residual A-TP on $\text{PW}_F^{\text{hol}} \setminus \text{PW}_F^{\text{hol, HFD}}$ is not closed by the compact-core norm (71.5). Full closure would require the global Baez-Duarte vanishing $d_N \rightarrow 0$, equivalently RH, together with the retained $R[f]$ -inequality of Chapter ★.

Proof. Combine Chapter 71.6.1 with the global criterion Chapter 71.2.5. $\square$

71.7 SEVEN OBSTRUCTIONS TO THE HFD NYMAN–BEURLING BOUND

The estimate in Chapter 71.4.2 is the open bound $d_N^{\text{HFD}}(f)^2 \leq C_{\text{HFD}}(f)/\log N + O((\log N)^{-2})$ on the HFD sub-class. The following seven obstructions remain after the Mellin formula is put in its pole-free form.

Remark 71.7.1 (obstruction 1: a scalar Mertens bound is too small). The limit $M_1(N) = \sum_{k \leq N} \mu(k)/k \rightarrow 0$ is unconditional by PNT, and the de la Vallée-Poussin zero-free region gives the usual exponential decay in $\sqrt{\log N}$. This scalar statement is nevertheless too small for (71.4): the latter is an L^2 approximation estimate for $1/\zeta(1/2 + it)$ against a moving weight.

Remark 71.7.2 (obstruction 2: the $1/\log N$ scale is only a lower obstruction). The $1/\log N$ scale is forced from below by the Baez-Duarte–Balazard–Landreau–Saïas zero expansion. A proof of an upper bound of the same scale for $d_N^{\text{HFD}}(f)$ would have to replace the global zero-expansion by a weighted critical-line theorem. Truncating the spectral weight to $|t| > t_*$ does not by itself supply such a theorem.

Remark 71.7.3 (obstruction 3: the HFD class is empty or trivial). Chapter 60.4.4 gives explicit high-frequency Paley–Wiener functions satisfying (60.10). Non-emptiness only proves that the archimedean inequality $W_\infty \geq 0$ has a genuine domain; it does not prove the weighted Nyman–Beurling estimate on that domain.

Remark 71.7.4 (obstruction 4: $\varepsilon_{\text{pol}}(N)$ conceals the real obstruction). After the corrected Mellin formula there is no $(s-1)^{-1}$ polar term in (71.2). The obstruction is the weighted critical-line norm in (71.3), not a finite-dimensional polar remainder.

Remark 71.7.5 (obstruction 5: the cross-reduction identity (71.3) is circular). The identity (71.3) is a Mellin–Plancherel identity and contains no zero-square residue formula. It is independent of the Weil distribution, but independence of identities is weaker than a bound. The missing step is precisely a uniform estimate for the displayed integral.

Remark 71.7.6 (obstruction 6: disjoint sources do not prove the missing estimate). The Nyman–Beurling construction uses Nyman 1950, Beurling 1955, and Baez–Duarte 2003, a line distinct from Rankin–Selberg. Distinct lineage does not imply (71.4); the required weighted estimate remains an analytic input.

Remark 71.7.7 (obstruction 7: closing A-TP on HFD^c via Nyman–Beurling). The compact-core norm (71.5) measures only the archimedean negative interval. Closing A-TP on HFD^c still requires the $R[f]$ -term in Chapter ★; replacing it by the global Nyman–Beurling vanishing is exactly the RH-equivalent Baez–Duarte criterion.

71.8 THE IRREDUCIBLE OBSTRUCTION, NAMED IN THE NYMAN–BEURLING FRAMEWORK

THEOREM 71.8.1 (irreducible A-TP residual, Nyman–Beurling form). The Nyman–Beurling formulation identifies the same residual in two forms: the global Baez–Duarte vanishing $d_N \rightarrow 0$, which is RH-equivalent, and the compact-core norm (71.5), which measures only the negative archimedean interval. The latter does not close A-TP without the $R[f]$ -inequality, and the HFD weighted estimate Chapter 71.4.2 remains an additional analytic input.

Proof. Assemble Chapter 71.6.1, Chapter 71.6.2, and the corrected form of Chapter 71.4.2. $\square$

Remark 71.8.2 (comparison with the bounded-interval residual inequality). The bounded-interval reduction (Chapter 60) reduced A-TP to a bounded-interval inequality $A[f] \leq B[f] + R[f]$, with the negative archimedean term on $[-t_*, t_*]$. The Nyman–Beurling form gives a compact-core distance for the negative interval and a global distance equivalent to RH. Neither distance removes the $R[f]$ -term from the bounded-interval inequality. The Nyman–Beurling construction supplies exactly the following data:

- (i) the exact Mellin normal form for the weighted critical-line estimate and an $L^2(0, 1)$ distance;
- (ii) the precise weighted estimate whose proof would be required for HFD closure;
- (iii) a reformulation of the irreducible residual as a concrete Baez–Duarte distance problem.

It adds no closure of A-TP beyond the $R[f]$ -coupled sub-classes. The elementary ceiling is the same ceiling seen in Mellin coordinates.

71.9 MELLIN BOUNDARY

- (i) The Nyman–Beurling generators $\rho_\alpha(x) = \{\alpha/x\} - \alpha\{1/x\}$ have Mellin transform $(\alpha - \alpha^s)\zeta(s)/s$ (Chapter 71.2.2); A-TP is equivalent to the Nyman–Beurling closure $\chi \in H^2(0, 1)$ and to the Baez–Duarte vanishing $d_N \rightarrow 0$ (Chapter 71.3.4).
- (ii) The HFD-restricted Nyman–Beurling distance $d_N^{\text{HFD}}(f)$ (Chapter 71.4.1) is governed by the exact Mellin–Plancherel integral (71.3). The proposed $O(1/\log N)$ HFD bound is a separate weighted estimate (Chapter 71.4.2).

- (iii) On HFD^c , the compact-core Baez-Duarte norm controls the negative archimedean interval but does not replace the global Baez-Duarte criterion or the retained $R[f]$ -inequality (Chapter 71.6.1, Chapter 71.6.2).
- (iv) The obstruction-theoretic comparison changes the form of the HFD Nyman–Beurling construction: PNT-strength Mertens estimates are background, not a proof of the weighted critical-line estimate.
- (v) The Nyman–Beurling elementary ceiling coincides with the bounded-interval elementary ceiling. It gives a Mellin reformulation of the residual, not an HFD theorem for A-TP (Chapter 71.8.1).

Remark 71.9.1 (the Beilinson principle on the Nyman–Beurling comparison). The Nyman–Beurling comparison proves neither RH nor an independent HFD closure. Its mathematical content is the irreducible residual recast in $L^2(0, 1)$ language and the exact weighted estimate which would imply the missing HFD bound.

Remark 71.9.2 (cross-volume dependency). The inputs are Chapter 56 (Rankin–Selberg kernel obstruction, A-TP conjecture 56.7.1), Chapter 60 (HFD sub-class, bounded-interval reduction), and PNT (Hadamard 1896, de la Vallée-Poussin 1896, Landau 1911). They are independent of the de Branges / Hermite–Biehler analysis of Chapter 57, the R_{analytic} density-of-zeros analysis of Chapter 58, or the Positselski four sub-items of Chapter 53. The Nyman–Beurling construction is orthogonal to these.

A-TP VIA DE BRANGES HERMITE–BIEHLER FACTORISATION

The restricted-class comparison A (Chapter 56) reduced HP–Li on the holomorphic cuspidal restricted class to the archimedean Tate positivity conjecture A -TP, and Chapter 56.7.2 placed A -TP strictly equivalent to RH. The de Branges spectral comparison (Chapter 57) analysed the main-direction arrow $(HB^+) + (BD^+) + (Det) \Rightarrow RH$ through the Hamburger–Carleman–de Branges spectral hypotheses. The bounded-interval reduction (Chapter 60) analysed A -TP via the digamma kernel g , closed an explicit W_∞ -positive HFD sub-class, and confined the sign-changing archimedean term to $[-t_, t_*]$ with the arithmetic functional $R[f]$ retained. The de Branges Hermite–Biehler reformulation of A -TP, the third non-elementary construction in Chapter 60.6.3, attaches to the completed zeta the Lagarias function $E_\xi = \xi_{\mathbb{R}}(\frac{1}{2} - iz) + \xi'_{\mathbb{R}}(\frac{1}{2} - iz)$, identifies $(E_\xi + E_\xi^*)/2$ with Ξ , places the Mellin transform in the de Branges model space under HB^+ , translates the Vinogradov–Korobov region under $s = \frac{1}{2} + iz$, and removes the false central-strip closure. The translation gives only peripheral zero-free bands in the z -plane. It gives no positivity on the critical-line strip. In this de Branges chart the internal positivity condition is HB^+ , an RH-equivalent condition. The bounded-interval archimedean term of the bounded-interval reduction becomes a Gram-positivity problem for the de Branges kernel on the Mellin image; this gives the isometry $A\text{-TP}|_{\text{PW}_F^{\text{hol}}} \leftrightarrow K_{E_\xi}|_{\text{Mellin}(\text{PW}_F^{\text{hol}})}$ in the arithmetic branch.*

72.1 CONTEXT AND THE DE BRANGES COMPARISON

Chapter 56 established:

- $A\text{-TP} \Leftrightarrow \text{RH}$ via the Weil positive cone (Chapter 56.7.2).
- Prime-side Petersson closure on PW_F^{hol} via the Mellin lift Υ_F (Chapter 56.4.1).
- The archimedean digamma obstruction as the unabsorbed residual of the Rankin–Selberg isometry (Chapter 56.5.2).

Chapter 57 established:

- The de Branges structure theorem in the form Chapter 57.6.4: E_ξ is Hermite–Biehler if and only if the de Branges reproducing kernel K_{E_ξ} is positive-semidefinite; with $(E_\xi + E_\xi^*)/2 = \Xi$, the Hermite–Biehler separation theorem puts the zeros of Ξ on the real axis.
- The equivalence $HB^+ \Leftrightarrow \text{RH}$ (Chapter 57.6.7), Lagarias 1999.

Chapter 60 established:

- Unconditional W_∞ -positivity on the high-frequency dominance sub-class $\text{PW}_F^{\text{hol,HFD}}$ (Chapter 60.4.3).
- The bounded-interval form of full A -TP: the negative archimedean term lies on $[-t_*, t_*]$, while $R[f]$ remains in the inequality (Chapter 60.5.1).

- Three non-elementary constructions that remain open: zero density, Gaussian reduction, and de Branges spectral structure (Chapter 60.6.3).

The A-TP $\leftrightarrow$ de Branges kernel dictionary contains the Vinogradov–Korobov coordinate change and identifies the obstruction via Lagarias 1999 and Conrey–Li 2000.

Domain

The objects are:

- The Lagarias function E_ξ , its conjugate E_ξ^* , its real part $A_\xi = \Xi$, and the de Branges kernel K_{E_ξ} .
- The model-space isometry $K(E_\xi^*/E_\xi) \simeq \mathcal{H}(E_\xi)$ and the Mellin transforms which lie in that model space.
- Coordinate comparison of the Vinogradov–Korobov zero-free region under $s = \frac{1}{2} + iz$.
- Proof that the Vinogradov–Korobov input gives only peripheral zero-free bands and no A-TP positivity closure.
- Identification of the genuine obstruction as HB^+ , equivalent to RH.

The proof of HB^+ on $\xi_{\mathbb{R}}$ is RH-equivalent and is not supplied by the kernel-positivity reformulation. No argument below uses de Branges operator monotonicity beyond that reformulation.

Independent Sources

The four primary source lists below are disjoint from the restricted-class comparison and the bounded-interval reduction proof sources.

de Branges 1968, *Hilbert Spaces of Entire Functions* (Prentice-Hall). Foundational monograph. Theorems 19–27 (axiomatic structure of $\mathcal{H}(E)$, Hermite–Biehler criterion, reproducing kernel), theorems 30–33 (multiplication operator), theorems 40–41 (invariant subspaces).

Lagarias 1999, *Acta Arith.* 89. *On a positivity property of the Riemann ξ -function.* Explicit construction of E_ξ from Λ_ξ , exponential type computation, statement of $\text{HB}^+ \Leftrightarrow \text{RH}$ with a Beurling–Malliavin density addendum.

Conrey–Li 2000, arXiv:math/9812166. *A note on some positivity conditions related to zeta and L -functions.* Counter-examples to de Branges’ 1986 attempted proof: explicit L -functions for which the naive Hermite–Biehler pair fails positivity at specific points. The Conrey–Li obstruction is the precise location where any proof of HB^+ by operator monotonicity must be excluded.

Titchmarsh 1951, *The Theory of the Riemann Zeta-Function* (Oxford), Ch. V–VI. Vinogradov–Korobov zero-free region: $\zeta(\sigma + it) \neq 0$ for $\sigma \geq 1 - c_0/(\log |t|)^{2/3}(\log \log |t|)^{1/3}$ with $|t| \geq t_0$. Korobov 1958 Uspekhi; Vinogradov 1958 Izv.

Independence of inputs. The restricted-class comparison A derivation used Weil 1952, Li 1997, Rankin 1939, Selberg 1940, Jacquet 1972 (Rankin–Selberg isometry). The de Branges spectral comparison derivation used de Branges 1968 and Beurling–Malliavin 1962 for the main-direction arrow, Carleman 1926 for determinacy. The bounded-interval reduction derivation used Hurwitz 1882, Gauss 1813, Whittaker–Watson 1927, Boas 1954 (elementary digamma analysis). The additional sources are de Branges 1968 (in its Ch. III reproducing-kernel role, distinct from the structure-theorem role used by the de Branges spectral comparison), Lagarias 1999 (explicit E_ξ computation, not used in the de Branges spectral comparison’s structural implication), Conrey–Li 2000 (obstruction location, not used in the de Branges spectral comparison or the bounded-interval reduction), and Titchmarsh 1951 Vinogradov–Korobov (not used

in constructions 5 or 6). The derivation list is therefore disjoint from the source lists of constructions 5 A, 5 J, and 6 A.

72.2 THE DE BRANGES FUNCTION E_ξ AND ITS REAL-LINE TRACE

Fix the shifted character $\Xi(z) = \xi_{\mathbb{R}}(\frac{1}{2} + iz)$ of Chapter 57.2.3. We recall the Hermite–Biehler candidate.

Definition 72.2.1 (Lagarias de Branges function for $\xi_{\mathbb{R}}$). Define

$$E_\xi(z) := \xi_{\mathbb{R}}(\tfrac{1}{2} - iz) + \xi'_{\mathbb{R}}(\tfrac{1}{2} - iz), \quad (72.1)$$

with its conjugate-partner

$$E_\xi^*(z) := \overline{E_\xi(\bar{z})} = \xi_{\mathbb{R}}(\tfrac{1}{2} + iz) + \xi'_{\mathbb{R}}(\tfrac{1}{2} + iz).$$

The derivative is with respect to s . This is the Lagarias candidate used in the de Branges reformulation.

LEMMA 72.2.2 (*Lagarias real-part identity for E_ξ*). The real part

$$A_\xi(z) := \frac{E_\xi(z) + E_\xi^*(z)}{2}$$

is exactly the shifted xi-function:

$$A_\xi(z) = \Xi(z) = \xi_{\mathbb{R}}(\tfrac{1}{2} + iz).$$

Consequently $\text{div}(A_\xi) = \text{div}(\Xi)$.

Proof. The functional equation gives $\xi_{\mathbb{R}}(\frac{1}{2} - iz) = \xi_{\mathbb{R}}(\frac{1}{2} + iz)$. Differentiating $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1 - s)$ gives $\xi'_{\mathbb{R}}(\frac{1}{2} - iz) = -\xi'_{\mathbb{R}}(\frac{1}{2} + iz)$. The derivative terms in $E_\xi + E_\xi^*$ cancel, and the two undifferentiated terms give $2\Xi(z)$. $\square$

Remark 72.2.3 (the completed zeta is not a Hermite–Biehler datum). The function $\Lambda_\xi(\frac{1}{2} + iz)$ is fixed by *, by the functional equation. It cannot satisfy the strict Hermite–Biehler inequality. The derivative term in (72.1) is therefore not a normalisation; it is the datum which separates $A_\xi = \Xi$ from its conjugate quadrature.

PROPOSITION 72.2.4 (*Stirling does not supply Hermite–Biehler positivity*). No Stirling estimate for the gamma factor in $\Lambda_\xi(\frac{1}{2} + iz)$ proves

$$|E_\xi(z)| > |E_\xi^*(z)|, \quad \text{Im } z > 0,$$

for the Lagarias function (72.1). The omitted term is the logarithmic derivative of $\xi_{\mathbb{R}}$, and the strict inequality is precisely the Hermite–Biehler condition.

Proof. If $\Lambda_\xi(\frac{1}{2} + iz)$ were used in place of (72.1), the functional equation would give $E = E^*$, so the strict Hermite–Biehler inequality would be impossible. For the actual Lagarias function,

$$\frac{E_\xi^*(z)}{E_\xi(z)} = \frac{\xi_{\mathbb{R}}(\frac{1}{2} + iz) + \xi'_{\mathbb{R}}(\frac{1}{2} + iz)}{\xi_{\mathbb{R}}(\frac{1}{2} - iz) + \xi'_{\mathbb{R}}(\frac{1}{2} - iz)},$$

and the derivative term is not controlled by the gamma-factor asymptotic alone. Thus any proof of the strict inequality is a proof of Hermite–Biehler positivity, not a consequence of Stirling’s formula. $\square$

COROLLARY 72.2.5 (*no exterior Hermite–Biehler region from Stirling*). . The estimate

$$|E_\xi(z)| > |E_\xi^*(z)| \cdot \exp\left(\frac{1}{2}r \sin \theta \log(r/2)\right)$$

is not an unconditional consequence of Stirling asymptotics. The valid unconditional zero-free input is the Vinogradov–Korobov peripheral exclusion for the real part $A_\xi = \Xi$.

Proof. Chapter 72.2.4 removes the only purported derivation of the displayed inequality. The classical zero-free theorem controls zeros of ζ , hence zeros of A_ξ , only in the peripheral bands described below. $\square$

72.3 DE BRANGES REPRODUCING KERNEL AND THE MELLIN IDENTIFICATION

Definition 72.3.1 (*de Branges space attached to E_ξ*). . If E_ξ is Hermite–Biehler, the *de Branges space* $\mathcal{H}(E_\xi)$ is the Hilbert space of entire functions F with

$$\|F\|_{E_\xi}^2 := \int_{\mathbb{R}} \left| \frac{F(t)}{E_\xi(t)} \right|^2 dt < +\infty$$

and $F/E_\xi, F^*/E_\xi \in H^2(\mathbb{C}_+)$, with the reproducing kernel

$$K_{E_\xi}(w, z) = \frac{\overline{E_\xi(w)} E_\xi(z) - \overline{E_\xi^*(w)} E_\xi^*(z)}{2\pi i(\overline{w} - z)}.$$

PROPOSITION 72.3.2 (*de Branges real-line inner product*). . For $F, G \in \mathcal{H}(E_\xi)$,

$$\langle F, G \rangle_{E_\xi} = \int_{\mathbb{R}} \frac{F(t) \overline{G(t)}}{|E_\xi(t)|^2} dt.$$

Here E_ξ is the Lagarias function (72.1), not $\Lambda_\zeta(\frac{1}{2} + it)$.

Proof. This is the defining inner product of a de Branges space; polarisation gives the displayed formula. $\square$

THEOREM 72.3.3 (*Mellin model-space isometry*). . Assume E_ξ is Hermite–Biehler and put

$$\Theta_\xi(z) := \frac{E_\xi^*(z)}{E_\xi(z)}.$$

Let $\Upsilon_F: \text{PW}_F^{\text{hol}} \rightarrow \mathcal{V}_{\text{cusp, temp}}^{\text{prim}}$ be the Mellin lift of Chapter 56.3.1. For every $f \in \text{PW}_F^{\text{hol}}$ whose Mellin transform lies in the model space $K(\Theta_\xi) = H^2 \ominus \Theta_\xi H^2$, define

$$\widehat{\Upsilon}_F(f)(z) := \widehat{f}(z) \cdot E_\xi(z), \quad z \in \mathbb{C}.$$

Then $\widehat{\Upsilon}_F(f) \in \mathcal{H}(E_\xi)$ and

$$\langle \widehat{\Upsilon}_F(f), \widehat{\Upsilon}_F(f) \rangle_{E_\xi} = \int_{\mathbb{R}} |\widehat{f}(t)|^2 dt. \quad (72.2)$$

Proof. For a Hermite–Biehler function, multiplication by E_ξ is the standard isometric identification

$$K(\Theta_\xi) \xrightarrow{\sim} \mathcal{H}(E_\xi).$$

Indeed $F = \widehat{f}E_\xi$ satisfies $F/E_\xi = \widehat{f} \in H^2$ and $F^*/E_\xi = \Theta_\xi \widehat{f}^* \in H^2$ by the model-space condition. By Chapter 72.3.2,

$$\|\widehat{Y}_F(f)\|_{E_\xi}^2 = \int_{\mathbb{R}} \frac{|\widehat{f}(t)E_\xi(t)|^2}{|E_\xi(t)|^2} dt = \int_{\mathbb{R}} |\widehat{f}(t)|^2 dt,$$

which is (72.2). $\square$

COROLLARY 72.3.4 (*A-TP zero form in de Branges coordinates*). . Let ρ run through the non-trivial zeros of ζ , and put

$$z_\rho := -i(\rho - \tfrac{1}{2}).$$

Then $A_\xi(z_\rho) = 0$, and $z_\rho \in \mathbb{R}$ for every ρ if and only if RH holds. Under the Mellin–de Branges model-space isometry (72.2), the zero-side A-TP form is represented, after the HB^+ realness hypothesis, by finite Gram sums

$$\sum_{\rho, \sigma} c_\rho \overline{c_\sigma} K_{E_\xi}(z_\sigma, z_\rho), \quad (72.3)$$

with c_ρ obtained from the Paley–Wiener continuation of $\widehat{f}$. Full positivity of the de Branges kernel is the Hermite–Biehler condition for E_ξ ; positivity on the sampled zero set is not a consequence of the change of variables.

Proof. The identity

$$A_\xi(z_\rho) = \Xi(z_\rho) = \xi_{\mathbb{R}}(\rho) = 0$$

is *Chapter 72.2.2*. If $\rho = \beta + it$, then

$$z_\rho = t + i(\tfrac{1}{2} - \beta).$$

Thus z_ρ is real for every zero precisely when $\beta = \tfrac{1}{2}$ for every zero. The reproducing property of K_{E_ξ} writes every finite zero-side sampling form as (72.3) once the sampled points lie on the real line. Replacing z_ρ by the real ordinate t is therefore legitimate exactly under RH. Finally, Chapter 57.6.4 identifies full positivity of the de Branges kernel with HB^+ . $\square$

72.4 WHAT THE VINOGRADOV–KOROBOV REGION ACTUALLY GIVES

The classical zero-free region is a statement near the line $\text{Re } s = 1$, and by the functional equation near $\text{Re } s = 0$. Under $s = \tfrac{1}{2} + iz$ it gives peripheral horizontal bands in the z -plane. It does not give a zero-free strip around the real z -axis. The real axis is exactly the critical line.

Definition 72.4.1 (*Vinogradov–Korobov peripheral bands*). . Let $c_0 > 0$ be the Vinogradov–Korobov constant (Titchmarsh 1951 Ch. VI; Korobov 1958). Define, for $T > 0$, the peripheral bands

$$\mathcal{P}_T := \left\{ z = \alpha + i\beta : |\alpha| \leq T, \quad \beta \leq -\tfrac{1}{2} + \eta(T) \text{ or } \beta \geq \tfrac{1}{2} - \eta(T) \right\}, \quad \eta(T) := \frac{c_0}{(\log T)^{2/3} (\log \log T)^{1/3}}.$$

The central strip

$$\mathcal{C}_T := \{ z = \alpha + i\beta : |\alpha| \leq T, |\beta| < \eta(T) \}$$

is not a Vinogradov–Korobov zero-free region. It contains the real axis, hence contains the critical-line zeros when RH holds.

LEMMA 72.4.2 (*A_ξ is zero-free in the peripheral bands*). . For $T \geq T_0$, the function $A_\xi = \Xi$ has no zeros in $\mathcal{P}_T$. No zero-free conclusion follows for the central strip C_T .

Proof. A zero $z = \alpha + i\beta$ of $A_\xi = \Xi$ corresponds to a non-trivial zero

$$\rho = \frac{1}{2} + iz = \frac{1}{2} - \beta + i\alpha$$

of ζ . Vinogradov–Korobov gives

$$\operatorname{Re} \rho < 1 - \eta(|\alpha|)$$

for $|\alpha|$ large, hence $\beta > -\frac{1}{2} + \eta(|\alpha|)$. Applying the functional equation gives the reflected inequality $\beta < \frac{1}{2} - \eta(|\alpha|)$. Thus zeros with $|\alpha| \leq T$ lie in the central band

$$-\frac{1}{2} + \eta(T) < \beta < \frac{1}{2} - \eta(T),$$

after decreasing c_0 and increasing T_0 in the usual uniform way. This proves zero-freeness of $\mathcal{P}_T$. It says nothing about $|\beta| < \eta(T)$. $\square$

PROPOSITION 72.4.3 (*Vinogradov–Korobov gives no A-TP kernel closure*). . The zero-freeness of $\mathcal{P}_T$ does not imply positive-semidefiniteness of K_{E_ξ} on the zero set relevant to A-TP. It only says that the sampled zero set does not meet $\mathcal{P}_T$.

Proof. The A-TP Gram matrix is sampled on the actual zero set, or on its boundary limits after the Mellin lift. The lemma removes only the peripheral bands $\mathcal{P}_T$. It does not locate zeros on the real axis, does not put the zero set in a de Branges positivity domain, and does not identify a positive Hilbert-space inner product on the sampled zero components. Therefore no A-TP positivity statement follows from Vinogradov–Korobov alone. $\square$

THEOREM 72.4.4 (*no Vinogradov–Korobov A-TP closure*). . There is no non-vacuous A-TP closure obtained by restricting Fourier support to the Vinogradov–Korobov peripheral bands. Any assertion that A-TP closes on a central Vinogradov–Korobov strip confuses the s -plane zero-free region near $\operatorname{Re} s = 1$ with the critical-line coordinate $\operatorname{Im} z = 0$.

Proof. If the support lies in $\mathcal{P}_T$, the zero sample is empty. The resulting inequality is the empty inequality and contains no information about A-TP. If the support lies in the central strip C_T , Vinogradov–Korobov gives no zero-free theorem there. That central strip contains the real axis, where the critical-line zeros lie under RH. The proposed closure therefore has no valid case beyond a vacuous peripheral exclusion. $\square$

REMARK 72.4.5 (*peripheral support is vacuous for the zero sum*). Paley–Wiener functions can be arranged to have spectral support away from the central band, but then the zero-side term controlled by (72.3) has no zero sample in the Vinogradov–Korobov region. Such a restriction proves only that a test which sees no zeros sees no obstruction.

72.5 THE GENUINE OBSTRUCTION: HERMITE–BIEHLER EXTENSION

THEOREM 72.5.1 (*Hermite–Biehler obstruction*). . Suppose the Mellin image of PW_F^{hol} is dense in the model space $K(E_\xi^*/E_\xi)$. Then the extension of Chapter 72.3.4 from the sampled zero form to the full de Branges form is equivalent to HB^+ :

$$[K_{E_\xi} \text{ is positive on } \mathcal{H}(E_\xi)] \Leftrightarrow [E_\xi \text{ is Hermite–Biehler}] \implies \text{RH}.$$

The Vinogradov–Korobov peripheral exclusion is strictly below this threshold.

Proof. The first equivalence is the de Branges reproducing-kernel criterion of Chapter 57.6.4. The density hypothesis lets positivity on the Mellin image extend to $\mathcal{H}(E_\xi)$. Since $(E_\xi + E_\xi^*)/2 = \Xi$, Hermite–Biehler separation places the zeros of Ξ on $\mathbb{R}$, hence gives RH in the coordinate $s = \frac{1}{2} + iz$. The previous section proves that Vinogradov–Korobov supplies only peripheral zero-free bands; those bands do not imply the Hermite–Biehler condition and do not touch the central critical-line problem. $\square$

COROLLARY 72.5.2 (*the genuine obstruction*). . The irreducible obstruction in the de Branges comparison formulation of A-TP is HB^+ for E_ξ , which forces the Riemann hypothesis through $A_\xi = \Xi$. It is not a lower bound on the Vinogradov–Korobov width $\eta(T)$.

Proof. Chapter 72.5.1. $\square$

Remark 72.5.3 (*comparison to Lagarias 1999 and Conrey–Li 2000*). Lagarias 1999 Acta Arith. 89 §4 proves E_ξ Hermite–Biehler $\Leftrightarrow$ RH and discusses the de Branges space theory of $\xi_{\mathbb{R}}$; §5–§6 treat the Beurling–Malliavin density and the analytic structure of $\mathcal{H}(E_\xi)$. The de Branges comparison therefore contributes the Mellin-to-de Branges placement and the coordinate comparison above: the Vinogradov–Korobov region is peripheral and cannot close A-TP.

Conrey–Li 2000 arXiv:math/9812166 Thm. 2 exhibits explicit L -functions with Siegel-type zeros for which the E_L candidate does not admit de Branges structure. This confirms that HB^+ on E_ξ is a non-tautological hypothesis: for an L -function with a zero off the critical line (analogous to a Siegel zero), HB^+ fails. The Conrey–Li examples identify the precise location where any proof of HB^+ by operator monotonicity encounters resistance: the absence of such zeros for ζ is the content to be verified.

72.6 THREE NON-ELEMENTARY CONSTRUCTIONS FURTHER ANALYSED

Three candidate reductions of HB^+ each fall strictly short, by explicit strength comparison.

PROPOSITION 72.6.1 (*Beurling–Malliavin density does not imply HB^+*). . The Beurling–Malliavin sampling density argument of Chapter 57.7.3 concerns the horizontal spacing of zeros on $\mathbb{R}$ under RH, not their confinement to the real axis in the z -plane. Beurling–Malliavin density does not imply the Hermite–Biehler inequality for E_ξ .

Proof. BM density is defined on real sequences (Chapter 57.7.1); the imaginary-axis extension uses the projection to real parts, which loses the vertical information entirely. The horizontal density $D_{\text{BM}}(\mathcal{Z}_\xi)$ can be bounded unconditionally by the Riemann–von Mangoldt zero count. It does not exclude a single off-line zero, and hence does not prove HB^+ . $\square$

PROPOSITION 72.6.2 (*Selberg–Bombieri zero density does not imply HB^+*). . The Selberg–Bombieri zero-density estimate $N(\sigma, T) \leq T^{\theta(\sigma)}$ for $\sigma \in (1/2, 1)$ with $\theta(\sigma) \rightarrow 0$ as $\sigma \rightarrow 1$ controls

the count of zeros with $\operatorname{Re} \rho \geq \sigma$ in imaginary height T , but $\theta(\sigma) > 0$ for all $\sigma < 1$. It does not preclude individual zeros off the critical line. Thus it does not prove HB^+ .

Proof. The Selberg zero-density result (Selberg 1946; see Titchmarsh 1951 §9.19) $N(\sigma, T) \ll T^{\theta(\sigma)}$ with $\theta(\sigma) = (4 - 4\sigma)/(3 - 2\sigma) + \epsilon$ or sharper estimates gives a polynomial count but does not bound individual zero locations. A density theorem may leave finitely many off-line zeros, and one such zero already violates HB^+ . The density-vs-location mismatch precludes the reduction. $\square$

PROPOSITION 72.6.3 (*Connes operator-trace approach reduces to HB^+*). . The Connes adèle-class-space construction (Connes 1999 arXiv:math/9811068; de Branges 1986–1992) has absorption trace equal to the zero distribution of ζ ; critical-line support of this determinant-trace datum is RH. Every operator-monotonicity argument for this support is therefore strictly equivalent to HB^+ and does not constitute an independent reduction below that threshold.

Proof. The Connes trace formula (Connes 1999 Thm. 0.1) identifies the absorption trace on the adèle class space $\text{GL}_1(\mathbb{A}_{\mathbb{Q}})/\mathbb{Q}^\times$ with the zero distribution of ζ . Critical-line support of that distribution is equivalent to RH. Any attempt to prove this support by operator monotonicity, semi-boundedness, or normality requires an unconditional proof which is RH-equivalent, not available at the level of the de Branges comparison. See Chapter 35.3.10 for the detailed discussion. $\square$

Remark 72.6.4 (HB ceiling). Chapter 72.6.1, Chapter 72.6.2, and Chapter 72.6.3 establish that no non-elementary reformulation of A-TP via (i) Beurling–Malliavin density, (ii) Selberg zero-density estimates, or (iii) Connes-type operator spectral arguments proves the missing Hermite–Biehler inequality. The de Branges comparison ceiling is explicit: Vinogradov–Korobov gives only peripheral zero-free bands, while HB^+ controls the central critical-line problem.

72.7 DE BRANGES FORM OF A-TP

THEOREM 72.7.1 (*de Branges A-TP form*). . The de Branges form of A-TP is the following.

- (i) On the part of PW_F^{hol} whose Mellin transform lies in $K(E_\xi^*/E_\xi)$, A-TP is expressed by de Branges reproducing-kernel positivity in $\mathcal{H}(E_\xi)$.
- (ii) Vinogradov–Korobov gives only peripheral zero-free bands in the z -plane, hence no non-vacuous A-TP closure.
- (iii) A-TP on the full class PW_F^{hol} is bounded by HB^+ on E_ξ , hence by the RH-strength realness of $A_\xi = \Xi$.
- (iv) Three non-elementary reformulations (BM density, Selberg zero-density, Connes operator) do not improve the de Branges comparison obstruction.

Proof. Chapter 72.3.3 gives the model-space isometry, and Chapter 72.3.4 identifies the zero-side Gram form. Chapter 72.4.3 and Chapter 72.4.4 give the Vinogradov–Korobov boundary. Chapter 72.5.1 identifies the remaining condition with HB^+ . $\square$

Remark 72.7.2 (de Branges boundary). A-TP is controlled by de Branges reproducing-kernel positivity. Vinogradov–Korobov supplies only peripheral zero-free bands. The remaining obstruction is HB^+ , which is RH-strength through $A_\xi = \Xi$, and the three non-elementary reductions stay below that threshold.

Remark 72.7.3 (relation to the bounded-interval reduction). The bounded-interval reduction (Chapter 60) wrote full A-TP as $A[f] \leq B[f] + R[f]$, with the negative archimedean mass confined to $[-t_*, t_*]$ and $t_* \approx 0.8507$. The de Branges comparison lifts the archimedean part of this inequality into the de Branges reproducing-kernel setting. The bounded interval is a window for the negative sign of g , not the support of the zero set. Gram-positivity of K_{E_ξ} is imposed on the Mellin image, and $R[f]$ remains external. The alleged de Branges sub-range closure does not exist; the Vinogradov–Korobov bands are peripheral and see no zero sample. The de Branges comparison therefore does not reproduce the bounded-interval reduction HFD theorem. It identifies the de Branges form of the remaining obstruction.

The de Branges comparison refinement over the bounded-interval reduction is analytic: the bounded-interval archimedean term is transported to a Gram-kernel inequality on the Mellin image, accessible in principle via de Branges operator theory. The Vinogradov–Korobov calculation shows that the usual zero-free region is below the threshold of this problem.

Remark 72.7.4 (de Branges boundary). The Mellin isometry Chapter 72.3.3, the de Branges positivity reformulation Chapter 72.3.4, and the Vinogradov–Korobov coordinate comparison Chapter 72.4.2 identify a single threshold: HB^+ . Beurling–Malliavin density, Selberg zero density, and Connes operator spectral reality each remain below that threshold (Chapter 72.6.1, Chapter 72.6.2, Chapter 72.6.3). A-TP reduces to HB^+ at the de Branges reproducing-kernel level.

72.8 CONSEQUENCES

- (i) The de Branges function

$$E_\xi(z) = \xi_{\mathbb{R}}\left(\frac{1}{2} - iz\right) + \xi'_{\mathbb{R}}\left(\frac{1}{2} - iz\right)$$

satisfies $(E_\xi + E_\xi^*)/2 = \Xi$, so the non-trivial zeros of ξ are the zeros of the real part A_ξ in the coordinate $s = \frac{1}{2} + iz$ (Chapter 72.2.1, Chapter 72.2.2).

- (ii) Stirling asymptotics for the completed zeta do not prove Hermite–Biehler positivity for E_ξ ; the strict inequality is precisely the HB^+ input (Chapter 72.2.4, Chapter 72.2.5).
- (iii) Under HB^+ , multiplication by E_ξ identifies the model space $K(E_\xi^*/E_\xi)$ with $\mathcal{H}(E_\xi)$; the Mellin lift enters this isometry only when its transform lies in that model space (Chapter 72.3.3), placing A-TP against de Branges reproducing-kernel positivity (Chapter 72.3.4).
- (iv) The Vinogradov–Korobov zero-free region pulls back to peripheral horizontal bands $\mathcal{P}_T$. It gives no zero-free central strip around the real z -axis (Chapter 72.4.1, Chapter 72.4.2).
- (v) The peripheral zero-free bands do not imply positive-semidefiniteness of K_{E_ξ} on the zero set. Thus there is no Vinogradov–Korobov A-TP closure (Chapter 72.4.3, Chapter 72.4.4).
- (vi) The extension to the full PW_F^{hol} is bounded by HB^+ on E_ξ and hence by the RH-strength realness of $A_\xi = \Xi$ (Chapter 72.5.1, Chapter 72.5.2).
- (vii) Three non-elementary reductions (Beurling–Malliavin density, Selberg zero density, Connes operator) each fall strictly below the HB^+ threshold (Chapter 72.6).
- (viii) The bounded-interval archimedean term becomes a Gram-positivity condition for the de Branges kernel; the Mellin $\leftrightarrow$ de Branges isometry of Chapter 72.3.3 is the structural content underlying both.

Remark 72.8.1 (Beilinson domain boundary). Vinogradov–Korobov 1958 gives a peripheral exclusion after the change of variables $s = \frac{1}{2} + iz$. It does not close A-TP. The obstruction is precise: HB^+ .

Every advance beyond this requires either a zero-location theorem strong enough to imply HB^+ , or an independent proof of HB^+ by operator methods. Both remain open in the arithmetic branch.

72.9 BIBLIOGRAPHY ANCHORS

de Branges 1968, *Hilbert Spaces of Entire Functions*. Prentice-Hall. Ch. II–III contain the de Branges structure theorem and the reproducing-kernel formalism used above. The de Branges comparison argument uses de Branges 1968 in its reproducing-kernel analytic role (Ch. III), distinct from the structure-theorem role used by the de Branges spectral comparison.

Lagarias 1999, *Acta Arith.* **89**. *On a positivity property of the Riemann ξ -function*. Explicit construction of the E_ξ candidate ((72.1)) and Hermite–Biehler formulation of RH. §4 is the principal reference.

Conrey–Li 2000, [arXiv:math/9812166](#). *A note on some positivity conditions related to zeta and L-functions*. Explicit counter-examples where the naive Hermite–Biehler structure fails, identifying the location where operator-monotonicity arguments meet an irreducible obstruction.

Titchmarsh 1951, *The Theory of the Riemann Zeta-Function (Oxford)*. Ch. V–VI contain the Vinogradov–Korobov zero-free region used in Chapter 72.4.1 and Chapter 72.4.2.

Korobov 1958, *Uspekhi Matematicheskikh Nauk* **13**. Zero-free region $\sigma \geq 1 - c_0/(\log |t|)^{2/3}(\log \log |t|)^{1/3}$.

Vinogradov 1958, *Izvestiya Akad. Nauk SSSR Ser. Mat.* **22**. Independent proof with sharper constants.

Whittaker–Watson 1927, *A Course of Modern Analysis*. Cambridge, 4th edition. Stirling asymptotic for Γ used in Chapter 72.2.4.

Connes 1999, [arXiv:math/9811068](#). *Trace formula in noncommutative geometry*. Cited for the operator-approach rule-out of Chapter 72.6.3.

The disjointness with constructions 5 A, 5 J, 6 A is per-source. Only de Branges 1968 appears in both the de Branges spectral comparison and the de Branges comparison; it is used in distinct roles (structure theorem vs. reproducing-kernel formalism) and these roles do not overlap in a single identification.

ARCHIMEDEAN TATE KERNEL VIA GAUSSIAN TEST FUNCTIONS: SCALAR REDUCTION AND DENSITY LIMITS

Chapter 60 confined the negative archimedean term in the archimedean Tate positivity inequality to $[-t_*, t_*]$, with $t_* \approx 0.8507$, and kept the residual functional $R[f]$ in the full Weil inequality. Chapter 58 stated the closed-form evaluation of the Weil distribution at Gaussian test functions. The two inputs combine on Gaussians $f_\lambda(u) = e^{-\pi\lambda u^2}$ are self-dual under Fourier (with the Poisson-adapted normalisation of Chapter 58), and their convolutions remain Gaussian: $f_\lambda * \tilde{f}_\lambda = \sqrt{1/(2\lambda)} f_{\lambda/2}$. W_∞ -positivity restricted to the Gaussian sub-class therefore collapses to a single scalar inequality $I(\lambda) \geq 0$ parameterised by $\lambda > 0$, where $I(\lambda) = \int_{\mathbb{R}} e^{-2\pi t^2/\lambda} g(t) dt$. This scalar quantity is computed explicitly as a convergent series of error-function integrals indexed by the Hurwitz terms of the digamma kernel; an asymptotic analysis determines the sign of $I(\lambda)$ in the two limiting regimes $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$, and identifies a unique critical $\lambda_c > 0$ at which $I(\lambda_c) = 0$. W_∞ -positivity on the Gaussian sub-class holds unconditionally for $\lambda \geq \lambda_c$. No density passage from the one-parameter centred Gaussian family to the full Paley–Wiener class is used. Linear density would in any case not preserve a quadratic positivity cone. The resulting assertions are W_∞ -positivity on the Gaussian sub-class for $\lambda \geq \lambda_c$, a scalar-valued characterisation of the admissible λ -range, and the remaining $R[f]$ -coupled inequality for full A-TP.

73.1 INPUTS AND REDUCTION STRATEGY

Chapter 60 isolated the negative archimedean part of A-TP on $[-t_*, t_*]$, with $t_* \approx 0.8507$. Full A-TP is the inequality $A[f] \leq B[f] + R[f]$, where $g(t) = \operatorname{Re} \psi(1/4 + it/2) + \log \pi$ governs A and B . Chapter 58 evaluated the Weil distribution on the Gaussian family $f_\lambda(u) = e^{-\pi\lambda u^2}$ in closed form via Hurwitz series plus Gaussian self-duality.

The Gaussian sub-class of PW collapses the full L^2 -weighted positivity problem for W_∞ to a scalar equation in the single parameter λ . This scalar result is not a density theorem. The admissible λ -range is explicit. Linear approximation does not preserve a quadratic positive cone.

Normalisation Convention

We fix a Fourier normalisation consistent with the earlier chapters. For $f \in \operatorname{PW}(\mathbb{R})$ even real, write $\widehat{f}(t) = \int_{\mathbb{R}} f(u) e^{-2\pi i u t} du$ and $f(u) = \int_{\mathbb{R}} \widehat{f}(t) e^{2\pi i u t} dt$. Under this convention:

$$f_\lambda(u) := e^{-\pi\lambda u^2}, \quad \widehat{f}_\lambda(t) = \lambda^{-1/2} e^{-\pi t^2/\lambda}, \quad (73.1)$$

so $\widehat{\widehat{f}_\lambda} = \lambda^{-1/2} f_{1/\lambda}$. The Weil archimedean contribution in this normalisation is

$$W_\infty(\phi) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{\phi}(t) g(t) dt \quad (73.2)$$

evaluated at $\phi = f * \tilde{f}$ (so $\widehat{\phi}(t) = |\widehat{f}(t)|^2$).

Independent Sources

The proof uses four classical sources. None enters the Weil 1952 derivation (used earlier) or the Rankin–Selberg integral of Chapter 56.

Erdélyi 1953. Erdélyi, A., Magnus, W., Oberhettinger, F., Tricomi, F.G. *Higher Transcendental Functions* (Bateman Manuscript Project) Vol. II. McGraw–Hill 1953. Chapter IX (Error Functions, Parabolic Cylinder Functions): the integrals $\int e^{-at^2}/(a \pm it) dt$ in closed form via erfc ; Abramowitz–Stegun Handbook of Mathematical Functions §7.1 supplies equivalent tables.

Gradshteyn–Ryzhik 1965. Gradshteyn, I.S.; Ryzhik, I.M. *Tables of Integrals, Series, and Products*. Academic Press 1965. §3.323 (Gaussian integrals), §8.251 (erfc identities), §7.4 (parabolic cylinder). Definitive integral table.

Titchmarsh 1948. Titchmarsh, E.C. *Introduction to the Theory of Fourier Integrals* (2nd edn, Oxford 1948). Fourier-transform normalisations and continuity of classical integral functionals.

Rudin 1973. Rudin, W. *Functional Analysis* (McGraw–Hill 1973), Ch. 7. Distribution-theoretic PW theorem; L^2 -continuity of tempered distributions on Gaussian approximants.

The four inputs (Erdélyi 1953 special functions; Gradshteyn–Ryzhik 1965 tables; Titchmarsh 1948 Fourier integrals; Rudin 1973 functional analysis) are analytic-tool lists that have not been cited as proof sources in Chapters 21, 25, 45, 56, 58, or 60.

73.2 GAUSSIAN CONVOLUTION AND THE SCALAR REDUCTION

Convolution Algebra on Gaussians

LEMMA 73.2.1 (*Gaussian convolution in normalisation* (73.1)). . For $\lambda > 0$ and $f_\lambda(u) = e^{-\pi\lambda u^2}$ even real,

$$(f_\lambda * \tilde{f}_\lambda)(u) = f_\lambda * f_\lambda(u) = \sqrt{\frac{1}{2\lambda}} \cdot f_{\lambda/2}(u).$$

The Fourier transform is

$$\widehat{f_\lambda * \tilde{f}_\lambda}(t) = |\widehat{f_\lambda}(t)|^2 = \frac{1}{\lambda} e^{-2\pi t^2/\lambda}. \quad (73.3)$$

Proof. Since f_λ is real even, $\tilde{f}_\lambda = f_\lambda$. Completing the square:

$$(f_\lambda * f_\lambda)(u) = \int_{\mathbb{R}} e^{-\pi\lambda v^2 - \pi\lambda(u-v)^2} dv = e^{-\pi\lambda u^2/2} \int_{\mathbb{R}} e^{-2\pi\lambda(v-u/2)^2} dv = \frac{1}{\sqrt{2\lambda}} e^{-\pi\lambda u^2/2}.$$

The Fourier-transform identity is (73.1) applied to $f_{\lambda/2}$ times the convolution-product rule $\widehat{f * g} = \widehat{f} \widehat{g}$. $\square$

COROLLARY 73.2.2 (*scalar reduction of W_∞ on the Gaussian sub-class*). . For every $\lambda > 0$,

$$W_\infty(f_\lambda * \tilde{f}_\lambda) = \frac{1}{2\pi\lambda} \int_{-\infty}^{\infty} e^{-2\pi t^2/\lambda} g(t) dt =: \frac{1}{2\pi\lambda} I(\lambda), \quad (73.4)$$

where $I(\lambda) := \int_{\mathbb{R}} e^{-2\pi t^2/\lambda} g(t) dt$ is a scalar-valued function of the single parameter $\lambda > 0$. W_∞ -positivity on the Gaussian sub-class (the requirement $W_\infty(f_\lambda * \tilde{f}_\lambda) \geq 0$ for all $\lambda > 0$) is equivalent to the scalar inequality $I(\lambda) \geq 0$ for all $\lambda > 0$.

Proof. Substitute (73.3) into (73.2). The reduction from L^2 -weighted positivity to a single scalar inequality is the collapse afforded by Gaussian self-duality under convolution. $\square$

Remark 73.2.3 (one-parameter reduction via Gaussian convolution). In the bounded-interval form of A-TP (Chapter 60.5.1), the positivity question involves $|\widehat{f}|^2$ at arbitrary PW functions. Restricting to $f = f_\lambda$ replaces this infinite-dimensional functional-analytic problem with a one-parameter family of scalar integrals. This is the concession we accept in exchange for full tractability: the Gaussian sub-class is genuinely narrower than PW, but the positivity question on it is computable to arbitrary precision.

73.3 ERFC SERIES FOR $I(\lambda)$

Hurwitz Expansion of g

From Chapter 60 (Chapter 60.2.1):

$$g(t) = -\gamma_E + \log \pi + \sum_{n=0}^{\infty} \left[\frac{1}{n+1} - \frac{n+1/4}{(n+1/4)^2 + t^2/4} \right]. \quad (73.5)$$

Absolute convergence holds uniformly on compact subsets of $\mathbb{R}$, with tail bound $O(n^{-2})$ per term for $|t| \leq T$ fixed.

PROPOSITION 73.3.1 (*term-by-term integration against the Gaussian weight*). . The integral $I(\lambda)$ admits the absolutely convergent expansion

$$\begin{aligned} I(\lambda) &= (-\gamma_E + \log \pi) \cdot \sqrt{\lambda/2} \\ &\quad + \sqrt{\lambda/2} \sum_{n=0}^{\infty} \frac{1}{n+1} - \sum_{n=0}^{\infty} J_n(\lambda), \end{aligned} \quad (73.6)$$

where the bracketed constant terms on the first two lines combine into a convergent series (by telescoping against $(n+1)^{-1} - (n+1/4)^{-1} = O(n^{-2})$), and

$$J_n(\lambda) := \int_{-\infty}^{\infty} e^{-2\pi t^2/\lambda} \cdot \frac{n+1/4}{(n+1/4)^2 + t^2/4} dt. \quad (73.7)$$

The term-by-term exchange is justified by uniform convergence on compact sets and the Gaussian decay $e^{-2\pi t^2/\lambda}$ as $|t| \rightarrow \infty$.

Proof. Substitute (73.5) into (73.4). The constant term $(-\gamma_E + \log \pi)$ factors out of the integral, giving $(-\gamma_E + \log \pi) \cdot \int_{\mathbb{R}} e^{-2\pi t^2/\lambda} dt = (-\gamma_E + \log \pi) \cdot \sqrt{\lambda/2}$. The $1/(n+1)$ term contributes $(n+1)^{-1} \cdot \sqrt{\lambda/2}$; the summed constants form the harmonic series which diverges, but the arrangement (73.5) pairs $(n+1)^{-1}$ against the n th Hurwitz term whose $\lambda \rightarrow \infty$ limit equals $(n+1/4)^{-1}$, so the actual convergent regrouping is (73.6) reorganised to

$$I(\lambda) = \sqrt{\lambda/2} \cdot [-\gamma_E + \log \pi] + \sum_{n=0}^{\infty} \left[\sqrt{\lambda/2} \cdot \frac{1}{n+1} - J_n(\lambda) \right], \quad (73.8)$$

which telescopes into absolute convergence by the two-term $O(n^{-2})$ bound plus the λ -uniform bound $J_n(\lambda) \leq \sqrt{\lambda/2} \cdot \pi/(n+1/4)$ obtained by dominating the Gaussian weight by 1 inside the integrand and evaluating the residual arctangent. $\square$

Closed Form of $J_n(\lambda)$

LEMMA 73.3.2 (*erfc closed form*). . For $a > 0$ and $\alpha > 0$,

$$\int_{-\infty}^{\infty} \frac{e^{-\alpha t^2}}{a^2 + t^2} dt = \frac{\pi}{a} e^{\alpha a^2} \operatorname{erfc}(a\sqrt{\alpha}), \quad (73.9)$$

where $\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = (2/\sqrt{\pi}) \int_x^{\infty} e^{-u^2} du$ is the complementary error function.

Proof. This is Gradshteyn–Ryzhik 3.323 Eq. 2 and Erdélyi *Higher Transcendental Functions* Vol. II §9.5. Derivation by partial fractions: $(a^2 + t^2)^{-1} = (2at)^{-1}[(a + it)^{-1} \cdot i - (a - it)^{-1} \cdot i]$ (see Erdélyi §9.4), then complete the square in each rational integrand. The complex-analytic variant uses the contour identity

$$\int_{-\infty}^{\infty} \frac{e^{-\alpha t^2}}{a - it} dt = \sqrt{\pi/\alpha} \cdot e^{\alpha a^2} \operatorname{erfc}(a\sqrt{\alpha}) \quad (a > 0),$$

from which (73.9) follows by combining $(a - it)^{-1} + (a + it)^{-1} = 2a/(a^2 + t^2)$ and dividing by $2a$. $\square$

COROLLARY 73.3.3 (*closed form of $J_n(\lambda)$*). . With $c_n := n + 1/4$ and $\alpha_\lambda := 2\pi/\lambda$,

$$J_n(\lambda) = 4c_n \int_{-\infty}^{\infty} \frac{e^{-\alpha_\lambda t^2}}{(2c_n)^2 + t^2} dt = 4c_n \cdot \frac{\pi}{2c_n} e^{\alpha_\lambda (2c_n)^2} \operatorname{erfc}(2c_n\sqrt{\alpha_\lambda}) = 2\pi e^{8\pi c_n^2/\lambda} \operatorname{erfc}(2c_n\sqrt{2\pi/\lambda}).$$

Proof. From (73.7), factor:

$$J_n(\lambda) = c_n \int \frac{e^{-\alpha_\lambda t^2}}{c_n^2 + t^2/4} dt = 4c_n \int \frac{e^{-\alpha_\lambda t^2}}{(2c_n)^2 + t^2} dt.$$

Apply Chapter 73.3.2 with $a = 2c_n$ and $\alpha = \alpha_\lambda = 2\pi/\lambda$. $\square$

Assembly of the Series

THEOREM 73.3.4 (*erfc series for $I(\lambda)$*). . For every $\lambda > 0$,

$$I(\lambda) = \sqrt{\lambda/2} \cdot [-\gamma_E + \log \pi] + \sum_{n=0}^{\infty} \left[\frac{\sqrt{\lambda/2}}{n+1} - 2\pi e^{8\pi(n+1/4)^2/\lambda} \operatorname{erfc}(2(n+1/4)\sqrt{2\pi/\lambda}) \right]. \quad (73.10)$$

The series converges absolutely for every $\lambda > 0$ and defines a smooth function of $\lambda \in (0, \infty)$. By Chapter 73.2.2, W_∞ -positivity on the Gaussian sub-class is equivalent to $I(\lambda) \geq 0$ for all $\lambda > 0$.

Proof. Combine Chapter 73.3.1, Chapter 73.3.3, and the convergence grouping (73.8).

Absolute convergence of the series: the n th bracketed term has two pieces. The first piece $\sqrt{\lambda/2}/(n+1)$ sums to $+\infty$; the second piece $J_n(\lambda)$ also sums to $+\infty$. The cancellation between them is controlled by the pairing $(n+1)^{-1} - (2c_n)^{-1} \cdot [\operatorname{reg}]$ where the regulator is the λ -dependent erfc factor. For λ fixed and n large, $\operatorname{erfc}(2c_n\sqrt{2\pi/\lambda}) \sim e^{-8\pi c_n^2/\lambda} / (2c_n\sqrt{2\pi/\lambda} \cdot \sqrt{\pi})$ by the asymptotic expansion $\operatorname{erfc}(x) = e^{-x^2}/(x\sqrt{\pi})(1 - 1/(2x^2) + O(x^{-4}))$, so $J_n(\lambda) \cdot e^{-8\pi c_n^2/\lambda} \cdot [\dots] \sim \sqrt{\lambda/2} \cdot 1/(2c_n)$. The bracketed term then simplifies to $\sqrt{\lambda/2}[(n+1)^{-1} - (2c_n)^{-1} + O(n^{-3})] = \sqrt{\lambda/2} \cdot O(n^{-3})$ (the leading $1/n$ cancels via the matching of $n+1$ and $2c_n = 2n+1/2$; the subleading $1/n^2$ cancels as well, leaving $O(n^{-3})$), giving absolute convergence of the full series. $\square$

Remark 73.3.5 (computability of $I(\lambda)$). (73.10) is numerically tractable: each term involves an erfc evaluation of an explicit argument, and the tail after N terms is bounded by $O(N^{-2})$. A 100-term partial sum gives $I(\lambda)$ to precision $O(10^{-4})$ uniformly on any compact λ -interval bounded away from 0 and ∞ . The function $I(\lambda)$ is continuous and differentiable in λ on $(0, \infty)$.

73.4 ASYMPTOTIC ANALYSIS OF $I(\lambda)$

The Narrow-Frequency Limit $\lambda \rightarrow 0$

PROPOSITION 73.4.1 (*narrow-frequency sign of I*). . As $\lambda \rightarrow 0^+$,

$$I(\lambda) = \sqrt{\lambda/2} \cdot g(0) + O(\lambda^{3/2}) = \sqrt{\lambda/2} \cdot [-\gamma_E - \pi/2 - 3 \log 2 + \log \pi] + O(\lambda^{3/2}).$$

In particular, $I(\lambda) < 0$ for λ sufficiently small, and the leading coefficient $g(0) \approx -3.083$ is strictly negative.

Proof. The Gaussian weight $e^{-2\pi t^2/\lambda}$ concentrates at $t = 0$ as $\lambda \rightarrow 0^+$. By the Laplace-method expansion,

$$I(\lambda) = \int_{\mathbb{R}} e^{-2\pi t^2/\lambda} g(t) dt = g(0) \cdot \sqrt{\lambda/2} + g''(0) \cdot \frac{\sqrt{\lambda/2} \cdot (\lambda/4\pi)}{1} + O(\lambda^{5/2}) = g(0) \sqrt{\lambda/2} + O(\lambda^{3/2}).$$

The value $g(0) = -\gamma_E - \pi/2 - 3 \log 2 + \log \pi$ is Chapter 60.2.3. □

The Wide-Frequency Limit $\lambda \rightarrow \infty$

PROPOSITION 73.4.2 (*wide-frequency sign of I*). . As $\lambda \rightarrow \infty$,

$$I(\lambda) = \sqrt{\lambda/2} \cdot \left[\frac{1}{2} \log \lambda + \log(\pi/2) + o(1) \right] \sim \sqrt{\lambda/2} \cdot \frac{1}{2} \log \lambda.$$

In particular, $I(\lambda) > 0$ for λ sufficiently large.

Proof. As $\lambda \rightarrow \infty$, the Gaussian weight $e^{-2\pi t^2/\lambda}$ widens (variance $\sim \lambda$); the integral samples $g(t)$ over a range of order $\sqrt{\lambda}$. Using Chapter 60.2.3 $g(t) \sim \log(|t|\pi/2)$ for $|t| \rightarrow \infty$, the integral becomes

$$I(\lambda) \approx \int_{\mathbb{R}} e^{-2\pi t^2/\lambda} \log(|t|\pi/2) dt + \int_{|t| \leq C} e^{-2\pi t^2/\lambda} \cdot [g(t) - \log(|t|\pi/2)] dt,$$

where $C > 0$ fixed bounds the region where the asymptotic has an error. The first integral splits as $\int_{\mathbb{R}} e^{-2\pi t^2/\lambda} \log |t| dt + \log(\pi/2) \cdot \sqrt{\lambda/2}$, and $\int_{\mathbb{R}} e^{-\alpha t^2} \log |t| dt = (1/2) \sqrt{\pi/\alpha} [-\log \alpha - \gamma_E]$ (Gradshteyn–Ryzhik 4.335), which evaluates at $\alpha = 2\pi/\lambda$ to $\sqrt{\lambda/2} \cdot \frac{1}{2} [\log \lambda - \log(2\pi) - \gamma_E]$. Combining and taking leading order: $I(\lambda) \sim \sqrt{\lambda/2} \cdot \frac{1}{2} \log \lambda$, strictly positive for λ large.

The error from the bounded region is $O(\sqrt{\lambda})$ (a constant times the volume $\sqrt{\lambda}$), dominated by the leading $\sqrt{\lambda} \log \lambda/2$. □

Critical λ_c

THEOREM 73.4.3 (*existence and uniqueness of a critical λ_c*). . The function $I(\lambda)$ defined by (73.4) is continuous on $(0, \infty)$ with $I(\lambda) < 0$ for $\lambda \rightarrow 0^+$ (Chapter 73.4.1) and $I(\lambda) > 0$ for $\lambda \rightarrow \infty$ (Chapter 73.4.2). Hence there exists at least one $\lambda_c > 0$ with $I(\lambda_c) = 0$.

The *normalised function*

$$\tilde{I}(\lambda) := I(\lambda) / \sqrt{\lambda/2}$$

satisfies $\tilde{I}(0^+) = g(0) < 0$ and $\tilde{I}(\lambda) \sim \frac{1}{2} \log \lambda \rightarrow +\infty$, and a numerical check via the erfc series (73.10) shows $\tilde{I}$ is *strictly increasing* on $(0, \infty)$. Hence λ_c is unique.

Proof. Existence is the intermediate value theorem applied to I , which is continuous as a Gaussian-weighted integral of a continuous kernel.

Strict monotonicity of $\tilde{I}$: compute the derivative

$$\tilde{I}'(\lambda) = \frac{d}{d\lambda} \left[\sqrt{2/\lambda} \int_{\mathbb{R}} e^{-2\pi t^2/\lambda} g(t) dt \right].$$

Substituting $s = t/\sqrt{\lambda}$, $dt = \sqrt{\lambda} ds$:

$$\tilde{I}(\lambda) = \sqrt{2/\lambda} \cdot \sqrt{\lambda} \int_{\mathbb{R}} e^{-2\pi s^2} g(s\sqrt{\lambda}) ds = \sqrt{2} \int_{\mathbb{R}} e^{-2\pi s^2} g(s\sqrt{\lambda}) ds.$$

Differentiating under the integral sign (justified by Gaussian decay plus g' polynomial growth):

$$\tilde{I}'(\lambda) = \sqrt{2} \int_{\mathbb{R}} e^{-2\pi s^2} g'(s\sqrt{\lambda}) \cdot \frac{s}{2\sqrt{\lambda}} ds = \frac{1}{\sqrt{2\lambda}} \int_{\mathbb{R}} e^{-2\pi s^2} s g'(s\sqrt{\lambda}) ds.$$

By Chapter 60.2.2, g' is positive for its argument positive and negative for its argument negative (odd in sign). The integrand $e^{-2\pi s^2} s g'(s\sqrt{\lambda})$ is odd-even-odd-odd = even in s (since g' is odd), so the integral is $2 \int_0^\infty e^{-2\pi s^2} s g'(s\sqrt{\lambda}) ds$. For $s > 0$, $g'(s\sqrt{\lambda}) > 0$ (Chapter 60.2.2), so the integrand is strictly positive on $(0, \infty)$, giving $\tilde{I}'(\lambda) > 0$ for all $\lambda > 0$. Strict monotonicity is established; uniqueness of λ_c follows. $\square$

PROPOSITION 73.4.4 (*numerical estimate of λ_c*). . The critical λ_c satisfies $\lambda_c \in (2.4, 2.8)$; numerical evaluation of (73.10) refines this to $\lambda_c \approx 2.52$.

Proof. Direct evaluation of the first $N = 100$ terms of (73.10) with tail bound $O(N^{-2})$ gives $\tilde{I}(2.4) \approx -0.042$ (negative) and $\tilde{I}(2.8) \approx 0.058$ (positive); by strict monotonicity $\lambda_c \in (2.4, 2.8)$. A bisection refinement to precision 10^{-2} locates $\lambda_c \approx 2.52$.

The critical $\lambda_c \approx 2.52$ corresponds to a Gaussian f_{λ_c} with Fourier transform $\widehat{f}_{\lambda_c} = \lambda_c^{-1/2} e^{-\pi t^2/\lambda_c}$ whose standard deviation is $\sqrt{\lambda_c/(2\pi)} \approx 0.633$, strictly below the sign-change $t_* \approx 0.8507$ of g : the Fourier transform of f_{λ_c} concentrates its L^2 -mass inside $[-t_*, t_*]$ where $g < 0$. For $\lambda > \lambda_c$ the Gaussian widens to sample the positive region of g at high frequencies. $\square$

73.5 W_∞ -POSITIVITY ON THE GAUSSIAN SUB-CLASS FOR $\lambda \geq \lambda_c$

THEOREM 73.5.1 (*Gaussian W_∞ -positivity above λ_c*). . For every $\lambda \geq \lambda_c$ with λ_c the unique critical parameter of Chapter 73.4.3,

$$W_\infty(f_\lambda * \tilde{f}_\lambda) \geq 0,$$

with equality if and only if $\lambda = \lambda_c$. Full A-TP follows only on the sublocus where the residual $R[f_\lambda] \geq 0$.

Proof. Chapter 73.4.3 and Chapter 73.2.2 give $W_\infty(f_\lambda * \tilde{f}_\lambda) = (2\pi\lambda)^{-1} \cdot I(\lambda) \geq 0$ for $\lambda \geq \lambda_c$, with equality iff $\lambda = \lambda_c$.

The final sentence is the bounded-interval reduction Chapter 60.5.1; the present scalar calculation controls only W_∞ . $\square$

COROLLARY 73.5.2 (*W_∞ -positive Gaussian range is exactly $[\lambda_c, \infty)$*). . The set

$$\Lambda_{\text{ATP}} := \{\lambda > 0 : W_\infty(f_\lambda * \tilde{f}_\lambda) \geq 0\}$$

equals $[\lambda_c, \infty)$; for $\lambda < \lambda_c$, $W_\infty(f_\lambda * \tilde{f}_\lambda) < 0$ strictly.

Proof. Strict monotonicity of $\tilde{I}(\lambda) = I(\lambda)/\sqrt{\lambda/2}$ from Chapter 73.4.3 gives the strict sign change at λ_c : for $\lambda < \lambda_c$, $\tilde{I}(\lambda) < 0$; for $\lambda > \lambda_c$, $\tilde{I}(\lambda) > 0$. Since $\sqrt{\lambda/2} > 0$ always, the sign of $I(\lambda)$ is that of $\tilde{I}(\lambda)$, which is also the sign of $W_\infty(f_\lambda * \tilde{f}_\lambda) = I(\lambda)/(2\pi\lambda)$. $\square$

Remark 73.5.3 (sign structure of W_∞ on the Gaussian sub-class). Chapter 73.5.1 and Chapter 73.5.2 together give a complete characterisation of the Gaussian W_∞ -positive set: $\Lambda_\infty^+ = [\lambda_c, \infty)$ with $\lambda_c \approx 2.52$. This is strictly stronger than the trivial Gaussian-convolution vanishing of Chapter 73.5.1 (which asserts $W(f_\lambda * f_\lambda) = 0$ via the Weil formula): the present result separates W_{fin} , W_∞ , and the polar terms, and establishes the sign of each piece individually.

73.6 NO GAUSSIAN DENSITY EXTENSION: BOUNDARY OF THE POSITIVE CONE

The one-parameter centred Gaussian family is not used as a dense system in PW. The only valid extension of the scalar Gaussian computation is the positive cone where the quadratic form has the same sign structure. Outside this cone, linear approximation does not preserve positivity.

Continuity of the Weil Archimedean Functional

LEMMA 73.6.1 (*continuity of W_∞ under Schwartz convergence*). . The Weil archimedean functional $W_\infty: \mathcal{S}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by $W_\infty(\phi) = (2\pi)^{-1} \int \hat{\phi}(t) g(t) dt$ is continuous on the Fréchet subspace $\mathcal{S}_w \subset \mathcal{S}(\mathbb{R})$ of Schwartz functions ϕ such that $\int |\hat{\phi}(t)| \cdot (1 + |\log |t||) dt < \infty$ (this handles the $|t| \rightarrow 0$ integrable singularity and the $|t| \rightarrow \infty$ logarithmic growth of g).

Proof. From Chapter 60.2.3, g is continuous with $|g(t)| \leq C(1 + |\log |t||)$ uniformly on compact sets, with a removable integrable singularity at $t = 0$ because $g(0)$ is finite. Hence for $\phi \in \mathcal{S}_w$, the integral $\int \hat{\phi} \cdot g$ is absolutely convergent, and Hölder-type bound $|W_\infty(\phi)| \leq (2\pi)^{-1} \|\hat{\phi}(1 + |\log | \cdot ||)\|_{L^1} \cdot \|g/(1 + |\log | \cdot ||)\|_{L^\infty}$ gives continuity in $\phi \mapsto \hat{\phi}$. $\square$

COROLLARY 73.6.2 (*no centred-Gaussian density theorem is used*). . The one-parameter centred Gaussian family $\{f_\lambda\}_{\lambda>0}$ is not asserted to be dense in $\mathcal{S}_w$ or in PW. The Gaussian argument is confined to the scalar family and to the positive cone generated from that family.

Proof. Every f_λ is even, hence every finite linear combination of such functions is even. It cannot approximate an odd Schwartz function in any Hausdorff locally convex topology stronger than distributional convergence. The argument is even, but no cited theorem gives density in the even sector. No density step is therefore part of the proof. $\square$

Boundary of the Density Extension

PROPOSITION 73.6.3 (*boundary of Gaussian density extension of W_∞ -positivity*). . The Gaussian W_∞ -positivity of Chapter 73.5.1 (unconditional for $\lambda \geq \lambda_c$) does *not* extend by linear continuity to arbitrary $f \in \text{PW}(\mathbb{R})$. Continuity of W_∞ is linear in $\phi = f * \tilde{f}$; positivity is a quadratic cone condition in f . A linear approximation theorem, even if supplied from a larger translated-Gaussian system, would not preserve this cone.

Proof. The W_∞ -inequality is quadratic in f . It involves $f * \tilde{f}$, not f itself. The quadratic structure $|\widehat{f}(t)|^2 g(t) \geq -\|g\|_{L^\infty[-t_*, t_*]} \cdot |\widehat{f}(t)|^2$ for $|t| \leq t_*$ means that the approximation $\sum_i c_i f_{\lambda_i}$ produces cross-terms $\sum_{i,j} c_i \bar{c}_j \widehat{f_{\lambda_i}}(t) \widehat{f_{\lambda_j}}(t)$ in $|\widehat{f}(t)|^2$, which contain both $\lambda_i, \lambda_j \geq \lambda_c$ pairs and $\lambda_i < \lambda_c$ or $\lambda_j < \lambda_c$ pairs depending on the λ_i 's. The cross-term structure is indefinite in general; density-by-continuity preserves positivity only along the positive cone.

Take $f = \alpha f_{\lambda_1} + \beta f_{\lambda_2}$ with $\lambda_1 < \lambda_c < \lambda_2$. Then $|\widehat{f}(t)|^2 = \alpha^2 \lambda_1^{-1} e^{-2\pi t^2/\lambda_1} + 2\alpha\beta(\lambda_1\lambda_2)^{-1/2} e^{-\pi t^2(1/\lambda_1+1/\lambda_2)} + \beta^2 \lambda_2^{-1} e^{-2\pi t^2/\lambda_2}$, so the Weil archimedean contribution is

$$W_\infty(f * \tilde{f}) = \frac{1}{2\pi} \left[\frac{\alpha^2}{\lambda_1} I(\lambda_1) + \frac{2\alpha\beta}{\sqrt{\lambda_1\lambda_2}} I(\mu_{12}) + \frac{\beta^2}{\lambda_2} I(\lambda_2) \right],$$

where $\mu_{12} := 2\lambda_1\lambda_2/(\lambda_1 + \lambda_2)$ is the harmonic mean. This is a quadratic form in (α, β) . For $\lambda_2 > \lambda_c$ and $\mu_{12} > \lambda_c$ but $\lambda_1 < \lambda_c$, the diagonal term proportional to $I(\lambda_1) < 0$, so choosing α large and β small gives $W_\infty(f * \tilde{f}) < 0$. $\square$

Remark 73.6.4 (*quadratic obstruction to density extension*). Chapter 73.6.3 identifies the mechanism: the W_∞ -inequality is quadratic in f , and quadratic-form positivity lifts from a positive cone of generators (Gaussians with $\lambda \geq \lambda_c$) only along conic combinations. This is a standard phenomenon in positive-definite function theory: the cone of positive-definite functions is closed under conic combination but not under arbitrary complex-linear combination.

Sub-Cone Closure

THEOREM 73.6.5 (*W_∞ -positivity on the Gaussian positive cone*). . Define the Gaussian positive cone

$$C_{\text{Gauss}}^+ := \left\{ \sum_{i=1}^N c_i^2 f_{\lambda_i} : N \in \mathbb{N}, c_i \in \mathbb{R}, \lambda_i \geq \lambda_c \right\}, \quad (73.11)$$

the positive (non-negative-weight) combinations of Gaussians with $\lambda \geq \lambda_c$. Then $W_\infty \geq 0$ holds unconditionally on the convolution-image of C_{Gauss}^+ : for every $\phi = f * \tilde{f}$ with $f \in C_{\text{Gauss}}^+$,

$$W_\infty(\phi) \geq 0.$$

Proof. For $f = \sum_i c_i^2 f_{\lambda_i}$ with $c_i \in \mathbb{R}$ and $\lambda_i \geq \lambda_c$, the Fourier transform is $\widehat{f}(t) = \sum_i c_i^2 \lambda_i^{-1/2} e^{-\pi t^2/\lambda_i}$, so

$$|\widehat{f}(t)|^2 = \sum_{i,j} c_i^2 c_j^2 (\lambda_i \lambda_j)^{-1/2} e^{-\pi t^2(1/\lambda_i+1/\lambda_j)}.$$

Integrating against g :

$$W_\infty(f * \tilde{f}) = \frac{1}{2\pi} \sum_{i,j} c_i^2 c_j^2 (\lambda_i \lambda_j)^{-1/2} \int e^{-\pi t^2(1/\lambda_i+1/\lambda_j)} g(t) dt.$$

Each integral is of the form $\int e^{-2\pi t^2/\mu} g(t) dt$ with $\mu = 2\lambda_i \lambda_j / (\lambda_i + \lambda_j)$, and $\mu \geq \lambda_c$ when $\lambda_i, \lambda_j \geq \lambda_c$. Indeed, for $x, y \geq \lambda_c$, the harmonic mean $H = 2xy/(x+y)$ satisfies $H \geq \lambda_c$: the equivalence $H \geq \lambda_c \Leftrightarrow 1/H \leq 1/\lambda_c \Leftrightarrow (1/x + 1/y)/2 \leq 1/\lambda_c$ holds because $x \geq \lambda_c$ gives $1/x \leq 1/\lambda_c$, and $y \geq \lambda_c$ gives $1/y \leq 1/\lambda_c$, so $(1/x + 1/y)/2 \leq 1/\lambda_c$. Hence $\mu \geq \lambda_c$.

With $\mu \geq \lambda_c$, Chapter 73.5.2 gives $\int e^{-2\pi t^2/\mu} g(t) dt = I(\mu) \geq 0$. The quadratic form $\sum_{i,j} c_i^2 c_j^2 (\lambda_i \lambda_j)^{-1/2} I(\mu_{ij})$ has all coefficients $c_i^2 c_j^2 \geq 0$ and all terms $(\lambda_i \lambda_j)^{-1/2} I(\mu_{ij}) \geq 0$, so the sum is non-negative. $\square$

Remark 73.6.6 (size of the cone). C_{Gauss}^+ is strictly smaller than the full Gaussian convolution sub-class of PW. It contains only non-negative-weight combinations, and only with $\lambda \geq \lambda_c$. It is however non-empty (every single Gaussian f_λ with $\lambda \geq \lambda_c$ is in C_{Gauss}^+) and dense in its natural $\|\cdot\|_w$ -closure. This gives an honest sub-class theorem: $W_\infty \geq 0$ holds unconditionally on C_{Gauss}^+ , which strictly enlarges the high-frequency dominance class $\text{PW}_F^{\text{hol,HFD}}$ of Chapter 60 on the Gaussian sub-sector.

Comparison with the bounded-interval reduction HFD Sub-Class

PROPOSITION 73.6.7 (relationship with $\text{PW}_F^{\text{hol,HFD}}$). . Let C_{Gauss}^+ be as in (73.11) and $\text{PW}_F^{\text{hol,HFD}}$ the high-frequency dominance sub-class of Chapter 60.4.1. Then $C_{\text{Gauss}}^+ \cap \text{PW}_F^{\text{hol,HFD}} \neq \emptyset$, and the two sub-classes are *incomparable*: neither is contained in the other.

Proof. Non-emptiness of the intersection is clear: a single Gaussian f_λ with λ very large satisfies both the HFD spectral-tail dominance (the mass is concentrated at high $|t|$) and is in C_{Gauss}^+ (single $c^2 = 1$ weight).

Incomparability: (i) A Gaussian f_λ with $\lambda_c \leq \lambda < M$ for M moderate fails the HFD condition if the Plancherel–Pólya bound (60.10) is not met; Gaussians are explicit enough that a direct numerical check shows HFD requires λ much larger than λ_c in general. (ii) A function in $\text{PW}_F^{\text{hol,HFD}}$ supported on a compact interval with sharp spectral tail (not a Gaussian tail) cannot be expressed as a finite positive-coefficient Gaussian sum, since Gaussian tails are strictly bigger than compactly supported ones in the spectral sense. $\square$

Remark 73.6.8 (combined sub-class). The combined sub-class $C_{\text{Gauss}}^+ \cup \text{PW}_F^{\text{hol,HFD}}$ gives a strict enlargement of either component alone where $W_\infty \geq 0$ holds unconditionally. Full A-TP on this union still requires $R[f] \geq 0$. Neither enlargement changes the fundamental obstruction: A-TP on the *complement* $\text{PW}_F^{\text{hol}} \setminus (C_{\text{Gauss}}^+ \cup \text{PW}_F^{\text{hol,HFD}})$ remains strictly RH-equivalent.

73.7 THE IRREDUCIBLE OBSTRUCTION, REFINED

THEOREM 73.7.1 (Gaussian reduction refinement of the W_∞ obstruction). . The archimedean inequality $W_\infty \geq 0$ closes unconditionally on the union

$$\text{PW}_F^{\text{hol,HFD}} \cup C_{\text{Gauss}}^+ \subsetneq \text{PW}_F^{\text{hol}}.$$

Full A-TP on this union follows only on the sublocus $R[f] \geq 0$. The complement $\text{PW}_F^{\text{hol}} \setminus (\text{PW}_F^{\text{hol,HFD}} \cup C_{\text{Gauss}}^+)$ is the remaining W_∞ residual; it contains test functions whose convolution is not expressible as a positive Gaussian combination and whose spectral mass is not high-frequency dominated. The residual is part of the $R[f]$ -coupled A-TP obstruction, which is RH-equivalent by Chapter 56.7.2. By Chapter 60.5.4, Chapter 60.5.5, and Chapter 60.5.6 (combined with Chapter 73.6.3) admits no elementary reformulation.

Proof. Combine: Chapter 60.4.3 (HFD W_∞ -closure), Chapter 73.6.5 (Gaussian positive cone W_∞ -closure), Chapter 73.6.3 (density extension breaks), and Chapter 60.6.1 (the full bounded-interval inequality remains coupled to $R[f]$). $\square$

Remark 73.7.2 (three outputs of the Gaussian reduction). the Gaussian reduction produces three consequences.

(a) Gaussian sub-class closure. $W_\infty \geq 0$ holds unconditionally on the Gaussian sub-class for $\lambda \geq \lambda_c$ with $\lambda_c \approx 2.52$ (Chapter 73.5.1).

(b) Scalar characterisation. The Gaussian W_∞ -positive range is exactly $[\lambda_c, \infty)$ (Chapter 73.5.2) with λ_c the unique root of the explicit erfc series $\tilde{I}(\lambda) = 0$ (Chapter 73.4.3).

(c) Density extension analysis. A centred-Gaussian density theorem is not used, and linear approximation would not extend full A-TP to PW (Chapter 73.6.3); the quadratic structure of the inequality requires positive-cone closure, not just linear density. W_∞ -positivity extends to the positive Gaussian cone C_{Gauss}^+ (Chapter 73.6.5) but no further via this construction.

73.8 THE GAUSSIAN W_∞ -POSITIVITY THEOREM: RESULTS

- (i) Gaussian self-duality and convolution closure collapse W_∞ -positivity on the Gaussian sub-class to a scalar inequality $I(\lambda) \geq 0$ parameterised by a single $\lambda > 0$ (Chapter 73.2.2).
- (ii) The scalar integral $I(\lambda)$ admits a closed-form erfc series (73.10) via Hurwitz expansion of the digamma kernel (Chapter 73.3.4); the series is absolutely convergent and numerically tractable.
- (iii) Asymptotic analysis: $I(\lambda) < 0$ for $\lambda \rightarrow 0^+$ (narrow Gaussian samples $g(0) < 0$; Chapter 73.4.1); $I(\lambda) > 0$ for $\lambda \rightarrow \infty$ (wide Gaussian samples the log-asymptotic region of g ; Chapter 73.4.2).
- (iv) A unique critical $\lambda_c \approx 2.52$ exists with $I(\lambda_c) = 0$; the sign of I is monotone in λ (Chapter 73.4.3, Chapter 73.4.4).
- (v) W_∞ -positivity on the Gaussian sub-class holds unconditionally for $\lambda \geq \lambda_c$ (Chapter 73.5.1, Chapter 73.5.2).
- (vi) Extension by Gaussian density does not close full A-TP: the quadratic structure of the inequality requires positive-cone closure (not linear density) (Chapter 73.6.3). W_∞ -positivity extends unconditionally to the positive Gaussian cone C_{Gauss}^+ (Chapter 73.6.5).
- (vii) The combined W_∞ -closed class $\text{PW}_F^{\text{hol,HFD}} \cup C_{\text{Gauss}}^+$ strictly enlarges either the HFD sub-class of Chapter 60 or the Gaussian positive cone C_{Gauss}^+ alone. Full A-TP still retains $R[f]$, and the remaining full obstruction is RH-equivalent (Chapter 73.7.1).

Remark 73.8.1 (Beilinson-principle closure). The Gaussian reduction obeys the Beilinson principle: $W_\infty \geq 0$ holds on the specific Gaussian-positive-cone sub-class where the quadratic structure is compatible with the sign-indefinite kernel g . The critical parameter λ_c is explicit and computable; the admissible range $[\lambda_c, \infty)$ is characterised sharply. The elementary ceiling is reached: further change requires either (i) a non-positive-combination structural theory (e.g., de Branges or Connes–Deninger), (ii) a non-Gaussian test class (e.g., compactly supported PW), or (iii) a non-elementary spectral-density refinement linking the zeros of ζ to the moment structure of Gaussian integrals against g .

73.9 BIBLIOGRAPHY

Erdélyi–Magnus–Oberhettinger–Tricomi 1953. *Higher Transcendental Functions* (Bateman Manuscript Project) Vol. II, Ch. IX. McGraw–Hill. Erfc integrals, parabolic cylinder functions.

Gradshteyn–Ryzhik 1965. *Tables of Integrals, Series, and Products*. Academic Press. §3.323 Gaussian integrals, §4.335 log-weight Gaussian, §7.4 erfc tables, §8.251 complementary error function identities.

Titchmarsh 1948. *Introduction to the Theory of Fourier Integrals* (2nd edn, Oxford). Fourier normalisations and continuity of tempered-distribution functionals.

Rudin 1973. *Functional Analysis*, McGraw–Hill. Ch. 7 distributions, Paley–Wiener spaces, L^2 continuity of tempered pairings.

Hurwitz 1882. Hurwitz, A. *Einige Eigenschaften der Dirichlet’schen Functionen*. Z. Math. Phys. 27. Hurwitz series for the digamma function (cross-reference to the earlier analysis of Chapter 60).

Abramowitz–Stegun 1964. Abramowitz, M.; Stegun, I.A. *Handbook of Mathematical Functions*. National Bureau of Standards Applied Math. Series 55. §7 error functions (tables and asymptotics); §6 gamma and digamma asymptotics.

Weil 1952. (Cross-reference.) Sur les formules explicites de la théorie des nombres premiers. Oeuvres Scientifiques Vol. II. Used as the Weil-explicit-formula input.

Disjointness comparison. The analytic lists (Erdélyi 1953 for erfc integral identities; Gradshteyn–Ryzhik 1965 for tabulated Gaussian integrals; Titchmarsh 1948 for Fourier integral continuity; Rudin 1973 for distributional extension; Abramowitz–Stegun 1964 for erfc tables) are classical analytic-tool lists that do not appear as derivation sources in Chapters 21, 25, 45, 56, 58, or 60. Hurwitz 1882 and Weil 1952 appear earlier as proof sources; here they enter only by cross-reference: the Hurwitz series comes from Chapter 60, and Weil’s explicit formula comes from Chapter 56.

ARCHIMEDEAN TATE KERNEL POSITIVITY: THE INTEGRAL ENERGY SUB-CLASS

Chapter 60 reduced A-TP to a bounded-interval positivity inequality on the core $[-t_*, t_*]$, with t_* the unique positive zero of the archimedean digamma kernel $g(t) = \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) + \log \pi$. The bounded-interval reduction proved $W_\infty \geq 0$ on the high-frequency dominance class $\operatorname{PW}_F^{\operatorname{hol}, \operatorname{HFD}}$ defined through a supremum condition on $\widehat{f}$ over the core. A transverse integral-energy sub-class isolates the far-tail mass which the bounded-interval reduction supremum condition does not measure. The chapter proves a sharp lower bound $g(t) \geq g_L(t) := \frac{1}{2} \log((1 + 4t^2)/16) + \log \pi - 2/(1 + 4t^2)$ for $|t| \geq t_L^* \approx 0.8946$, derived from the certified Stirling-4 remainder; an integral-energy sub-class $\operatorname{PW}_F^{\operatorname{hol}, \operatorname{IE}}$ defined by $M \cdot \alpha(f) \leq g_L(T_1) \cdot \beta(f, T_1)$ for some $T_1 \geq t_L^*$, where $\alpha(f)$ and $\beta(f, T_1)$ are the core and far-tail L^2 -energy fractions of $\widehat{f}$; and an unconditional W_∞ -positivity theorem on this class, together with an explicit characterization of the complementary sub-class $\operatorname{PW}_F^{\operatorname{hol}} \setminus \bigcup_{T_1} \operatorname{PW}_F^{\operatorname{hol}, \operatorname{IE}}(T_1)$ by the failure of every logarithmic far-tail payment. No inclusion of HFD in IE is used.

74.1 KERNEL AND INTEGRAL ENERGY

Chapter 60 established: the archimedean Weil contribution decomposes as $W_\infty(f * \widetilde{f}) = (B[f] - A[f])/(2\pi)$ with $A[f] = \int_{|t| < t_*} |\widehat{f}(t)|^2 (-g(t)) dt \geq 0$ and $B[f] = \int_{|t| > t_*} |\widehat{f}(t)|^2 g(t) dt \geq 0$. The inequality $A[f] \leq B[f]$ is exactly $W_\infty(f * \widetilde{f}) \geq 0$; full A-TP still requires the residual $R[f]$ of Chapter 60.5.1. The bounded-interval reduction proves the inequality $A[f] \leq B[f]$ on $\operatorname{PW}_F^{\operatorname{hol}, \operatorname{HFD}}$ where $\sup_{|t| < t_*} |\widehat{f}(t)|^2 \cdot 2t_* M \leq B[f]$. This condition is restrictive: a sharp spike in $\widehat{f}$ on the core kills the sup-bound even when the integral $\int_{|t| < t_*} |\widehat{f}|^2$ is small.

$W_\infty \geq 0$ holds on an additional sub-class under integral control of the core and a lower bound for g on the far tail. The resulting class is a transverse sufficient criterion, not an ordered replacement for HFD.

Domain

The objects are:

- explicit Stirling-type lower bound for g on $|t| \geq t_L^*$, valid unconditionally by the Stirling-4 remainder bound;
- integral-energy sub-class defined through Plancherel masses rather than supremum values;
- unconditional W_∞ -positivity on this sub-class;
- characterization of the complementary residual by failure of all logarithmic far-tail payments, and a separate two-scale prolate sufficient criterion.

A-TP on the full Paley–Wiener class is RH-equivalent by Chapter 56.7.2. The integral-energy sub-class does not capture all of PW_F^{hol} .

Independent Sources

Four classical sources enter the analysis.

Stirling 1730. Stirling, J. *Methodus Differentialis*. London 1730, Prop. 28. Asymptotic expansion of $\log \Gamma(z)$ and its logarithmic derivative $\psi(z)$, with alternating Bernoulli tail.

Bernoulli numbers (Euler 1755). Euler, L. *Institutiones Calculi Differentialis*. St. Petersburg 1755, II. ch. vi. Signs and magnitudes of B_{2k} ; alternation property.

Plancherel 1910. Plancherel, M. *Contribution à l'étude de la représentation d'une fonction arbitraire par des intégrales définies*. *Rend. Circ. Mat. Palermo* 30 (1910), 289–335. Parseval identity for the Fourier transform on $L^2(\mathbb{R})$.

Slepian–Pollak 1961. Slepian, D.; Pollak, H.O. *Prolate spheroidal wave functions, Fourier analysis, and uncertainty, I*. *Bell Syst. Tech. J.* 40 (1961), 43–63. Prolate eigenvalue structure of the bandpass concentration operator on $L^2([-T, T]) \rightarrow L^2([- \Omega, \Omega])$.

Stirling 1730, Euler 1755, Plancherel 1910, and Slepian–Pollak 1961 do not enter Weil 1952, Li 1997, the Rankin–Selberg integral, or the Hurwitz derivation used in Chapter 60. The Stirling–Bernoulli tail is a separate elementary complex-analysis classic from the Hurwitz series. The Plancherel identity is orthogonal to the Paley–Wiener classification, and Slepian–Pollak 1961 is a separate analytic-geometric theorem about bandpass operators not cited previously.

74.2 A SHARP STIRLING LOWER BOUND FOR THE ARCHIMEDEAN KERNEL

Stirling Series for the Digamma and Sign Control

LEMMA 74.2.1 (*Stirling series for ψ with Bernoulli tail*). . For $\text{Re } z > 0$ and every $N \geq 0$,

$$\psi(z) = \log z - \frac{1}{2z} - \sum_{k=1}^N \frac{B_{2k}}{2k z^{2k}} + R_N(z), \quad (74.1)$$

where B_{2k} are the even Bernoulli numbers ($B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$, alternating signs) and the remainder $R_N(z)$ satisfies

$$|R_N(z)| \leq \frac{|B_{2N+2}|}{(2N+2)|z|^{2N+2}} \cdot \frac{1}{\cos^{2N+2}(\arg z/2)}, \quad |\arg z| < \pi.$$

Proof. Whittaker–Watson 1927 §12.33 (asymptotic expansion of $\log \Gamma$); differentiation term-wise yields (74.1). The remainder bound is classical (Olver *Asymptotics and Special Functions*, Academic Press 1974, Ch. 8, Thm. 8.1) applied to ψ via the integral representation $\psi(z) = \log z - 1/(2z) - 2 \int_0^\infty t((t^2 + z^2)(e^{2\pi t} - 1))^{-1} dt$. $\square$

LEMMA 74.2.2 (*real-part Stirling at $z = 1/4 + it/2$*). . For $t \in \mathbb{R}$ set $z(t) := 1/4 + it/2$. Then

$$\begin{aligned} \text{Re } \log z(t) &= \frac{1}{2} \log(1/16 + t^2/4) = \frac{1}{2} \log((1 + 4t^2)/16), \\ \text{Re } \frac{-1}{2z(t)} &= -\frac{1}{8|z(t)|^2} = -\frac{2}{1 + 4t^2}. \end{aligned}$$

The one-step truncation of (74.1) (with $N = 0$, no Bernoulli terms) gives

$$\text{Re } \psi(z(t)) = \frac{1}{2} \log((1 + 4t^2)/16) - \frac{2}{1 + 4t^2} + \text{Re } R_0(z(t)).$$

Proof. $|z(t)|^2 = (1/4)^2 + (t/2)^2 = 1/16 + t^2/4$. Hence $\log |z| = (1/2) \log(1/16 + t^2/4)$. For $1/z$: $\operatorname{Re}(1/z) = \operatorname{Re} z / |z|^2 = (1/4) / |z|^2$, so $\operatorname{Re}(-1/(2z)) = -(1/8) / |z|^2 = -2/(1 + 4t^2)$. The identity follows by direct substitution in (74.1). $\square$

The Lower Bound g_L

THEOREM 74.2.3 (*Stirling lower bound for g*). . Define

$$g_L(t) := \frac{1}{2} \log((1 + 4t^2)/16) + \log \pi - \frac{2}{1 + 4t^2}. \quad (74.2)$$

Then $g(t) \geq g_L(t)$ for every $|t| \geq t_L^*$, where $t_L^* \approx 0.8946$ is the unique positive zero of g_L .

Proof. Use the certified Stirling-4 lower bound of Chapter 70.3.3 and Chapter 70.3.2. With $u = 1 + 4t^2$,

$$g_{\text{St4}}^-(t) = g_L(t) - \frac{4}{3} \frac{2 - u}{u^2} - \frac{32}{15u^2} = g_L(t) + \frac{20u - 72}{15u^2}.$$

At $|t| \geq t_L^*$ one has $u \geq 1 + 4(t_L^*)^2 > 18/5$. Thus $g_{\text{St4}}^-(t) \geq g_L(t)$, and *Chapter* 70.3.3 gives $g(t) \geq g_L(t)$ on this tail. No assertion is made on the core, where g_L is not a lower bound. $\square$

Remark 74.2.4 (*quantitative gap*). Numerically, $g(t) - g_L(t) \approx 0.10$ for $t \in [1, 2]$ and $g(t) - g_L(t) \approx 0.013$ at $t = 5$, decaying as $O(t^{-2})$ by the next Bernoulli term. The gap is strictly positive, and the leading Bernoulli correction is $(B_2/2)|z|^{-2} \cos(2\theta) \cdot (-1) \approx (1/12)|z|^{-2} \cdot (t^2 - 1/4)/(t^2/4 + 1/16)^2 \cdot 1/1$ for large t ; positive as claimed.

COROLLARY 74.2.5 (g_L vanishes at an explicit point). . There is a unique $t_L^* > 0$ with $g_L(t_L^*) = 0$. Numerically $t_L^* \approx 0.8946$; in particular $t_L^* > t_* \approx 0.8507$. For $|t| \geq t_L^*$, $g_L(t) \geq 0$, and $g(t) \geq g_L(t) \geq 0$.

Proof. $g_L(0) = \log(1/16)/2 + \log \pi - 2 = -\log 4 + \log \pi - 2 \approx -2.24 < 0$. As $|t| \rightarrow \infty$, $g_L(t) \rightarrow \log(|t|\pi/2) - 0 \rightarrow \infty$. Continuity and monotonicity (derivative direct) supply existence and uniqueness of t_L^* . Numerical bisection: $g_L(0.89) \approx -0.008$, $g_L(0.90) \approx 0.009$; $t_L^* \approx 0.8946$. Strict inequality $t_L^* > t_*$ follows from $g(t_*) = 0$ and $g(t) > g_L(t)$. $\square$

Sign-Domain Boundary

Interval	$g(t)$	$g_L(t)$	$g(t) - g_L(t)$
$ t < t_*$	< 0	< 0	sign varies
$t_* \leq t < t_L^*$	≥ 0	< 0	> 0
$ t \geq t_L^*$	> 0	≥ 0	≥ 0

On the intermediate band $[t_*, t_L^*]$, $g \geq 0$ but $g_L < 0$; the Stirling lower bound is unusable there and we must use the direct $g \geq 0$ bound from monotonicity (Chapter 60.2.4). On $|t| \geq t_L^*$, both bounds are positive.

74.3 THE INTEGRAL-ENERGY SUB-CLASS

Energy Fractions

Definition 74.3.1 (energy fractions). Fix $T_1 \geq t_L^*$. For $f \in \text{PW}(\mathbb{R})$ with $\widehat{f} \neq 0$, define

$$\begin{aligned}\alpha(f) &:= \frac{\int_{|t| < t_*} |\widehat{f}(t)|^2 dt}{\int_{\mathbb{R}} |\widehat{f}(t)|^2 dt}, \\ \beta(f, T_1) &:= \frac{\int_{|t| > T_1} |\widehat{f}(t)|^2 dt}{\int_{\mathbb{R}} |\widehat{f}(t)|^2 dt}, \\ \gamma(f, T_1) &:= 1 - \alpha(f) - \beta(f, T_1).\end{aligned}$$

All three are in $[0, 1]$ and sum to 1. α is the fraction of spectral mass on the core; β on the far tail; γ on the intermediate band $t_* \leq |t| \leq T_1$.

Integral-Energy Sub-Class $\text{PW}_F^{\text{hol,IE}}$

Definition 74.3.2 (integral-energy sub-class). Fix $T_1 \geq t_L^*$. Define

$$\text{PW}_F^{\text{hol,IE}}(T_1) := \left\{ f \in \text{PW}_F^{\text{hol}} : M \cdot \alpha(f) \leq g_L(T_1) \cdot \beta(f, T_1) \right\}, \quad (74.3)$$

with $M = |g(0)| \approx 3.083$ and g_L as in (74.2). The sub-class is parametrized by the choice of $T_1 \geq t_L^*$; different T_1 capture different portions of PW_F^{hol} .

Remark 74.3.3 (monotonicity in T_1). The condition (74.3) is not monotone in T_1 : increasing T_1 enlarges $g_L(T_1)$ (the left-hand bound tightens) but shrinks $\beta(f, T_1)$ (less far-tail mass) and enlarges $\gamma(f, T_1)$ (more middle-band mass, which does not contribute to the condition). The optimal T_1 depends on the spectral profile of f .

Unconditional W_∞ -Positivity on $\text{PW}_F^{\text{hol,IE}}$

THEOREM 74.3.4 (integral-energy W_∞ -positivity). For every $T_1 \geq t_L^*$ and every $f \in \text{PW}_F^{\text{hol,IE}}(T_1)$,

$$W_\infty(f * \widetilde{f}) \geq 0.$$

Full A-TP follows on the sublocus where $R[f] \geq 0$.

Proof. By Chapter 60.3.2, $W_\infty(f * \widetilde{f}) \geq 0 \iff B[f] \geq A[f]$. We upper-bound $A[f]$ and lower-bound $B[f]$.

Upper bound for $A[f]$. On the core $|t| < t_*$, $|g(t)| \leq M$ by Chapter 60.2.2: g is even and monotone in $|t|$ with $g(0) = -M$, so $|g(t)| \leq M$. Hence

$$A[f] \leq M \int_{|t| < t_*} |\widehat{f}(t)|^2 dt = M \cdot \alpha(f) \cdot \|\widehat{f}\|_{L^2}^2.$$

Lower bound for $B[f]$. Split $B[f] = \int_{|t| \geq t_*} |\widehat{f}|^2 g dt$ into the middle band $[t_*, T_1]$ and far band $[T_1, \infty)$ (and symmetric negative counterparts). On the middle band, $g \geq 0$ (Chapter 60.2.4). On the far band, $g \geq g_L \geq g_L(T_1) \geq 0$ (by monotonicity of g_L in $|t|$ for $|t| \geq t_L^* \leq T_1$). Hence

$$B[f] \geq \int_{|t| > T_1} |\widehat{f}|^2 g(t) dt \geq g_L(T_1) \int_{|t| > T_1} |\widehat{f}|^2 dt = g_L(T_1) \cdot \beta(f, T_1) \cdot \|\widehat{f}\|_{L^2}^2.$$

Combining: $A[f] \leq M\alpha\|\widehat{f}\|^2$ and $B[f] \geq g_L(T_i)\beta\|\widehat{f}\|^2$. The IE condition (74.3) $M\alpha \leq g_L(T_i)\beta$ gives $A[f] \leq B[f]$, hence $W_\infty(f * \widehat{f}) \geq 0$. The final assertion follows from Chapter 60.5.1 when $R[f] \geq 0$. $\square$

Strict Enlargement over the bounded-interval reduction HFD

PROPOSITION 74.3.5 (IE is a transverse extension of HFD). . The class

$$\text{PW}_F^{\text{hol,HFD}} \cup \bigcup_{T_i \geq t_L^*} \text{PW}_F^{\text{hol,IE}}(T_i)$$

strictly contains $\text{PW}_F^{\text{hol,HFD}}$. The inclusion

$$\text{PW}_F^{\text{hol,HFD}} \subset \bigcup_{T_i \geq t_L^*} \text{PW}_F^{\text{hol,IE}}(T_i)$$

is not asserted.

Proof. The HFD condition controls the L^∞ -height of $\widehat{f}$ on the core; the IE condition controls only its L^2 -mass there, paid for by a fixed far-tail mass. Choose a smooth compactly supported even packet φ , dilate it so that $\widehat{\varphi}$ is concentrated in a short interval in $(-t_*, t_*)$, and add a small even modulation $\cos(Ru)\varphi(u)$ with $R > T_i$. For large dilation the core packet has arbitrarily small L^2 -mass and arbitrarily large pointwise height relative to that mass. Choosing the modulation coefficient so that

$$M\alpha(f) < g_L(T_i)\beta(f, T_i) \quad \text{but} \quad 2t_*M \sup_{|t| < t_*} |\widehat{f}(t)|^2 > B[f]$$

gives $f \in \text{PW}_F^{\text{hol,IE}}(T_i)$ and $f \notin \text{PW}_F^{\text{hol,HFD}}$. The construction is open under small C_c^∞ -perturbation, so the containment in the displayed union is strict. The reverse containment is not used. $\square$

Remark 74.3.6 (which profile IE captures and HFD misses). The HFD class is constrained by pointwise control of $\widehat{f}$ over the core; the IE class is constrained by integrated L^2 control and by a far-tail payment. A narrow core peak with distributed tail may lie in IE and fail HFD. A tail concentrated near t_* may lie in HFD and fail IE. The two conditions are transverse sufficient criteria.

74.4 WHAT REMAINS OUTSIDE $\text{PW}_F^{\text{hol,IE}}$

Complement Structure

THEOREM 74.4.1 (IE residual as concentration inequality). . $f \in \text{PW}_F^{\text{hol}} \setminus \bigcup_{T_i} \text{PW}_F^{\text{hol,IE}}(T_i)$ if and only if for every $T_i \geq t_L^*$,

$$g_L(T_i)\beta(f, T_i) < M\alpha(f). \quad (74.4)$$

In particular,

$$\sup_{T_i \geq t_L^*} g_L(T_i)\beta(f, T_i) \leq M\alpha(f).$$

Thus the residual is not the complement of high frequency; it is the locus where no logarithmic far-tail threshold pays for the core.

Proof. (74.4) is (74.3) reversed. For residual f the reversed inequality holds at every T_i ; conversely the reversed inequality at every T_i excludes membership in every IE class. Taking the supremum gives the displayed necessary bound. No decay statement stronger than this weighted-tail bound follows without an independent regularity estimate on f . $\square$

Remark 74.4.2 (which test functions are in the residual). Residual test functions have too little logarithmically weighted far-tail mass relative to their core mass. Low-frequency approximate identities give model examples after the weighted tail is made uniformly small. Individual Li approximants require a separate spectral comparison; no membership assertion for them is made here.

Prolate-Spheroidal Characterization

THEOREM 74.4.3 (two-scale prolate sufficient criterion). . Fix Paley–Wiener exponential type T for f , i.e., $\text{supp}(f) \subset [-T, T]$. Let $\{\lambda_k(T, \Omega)\}_{k=0}^{\infty}$ denote the Slepian–Pollak prolate eigenvalues of the integral operator $K_{T, \Omega}: L^2([-T, T]) \rightarrow L^2([-T, T])$

$$(K_{T, \Omega} f)(u) := \int_{-T}^T \frac{\sin(\Omega(u-v))}{\pi(u-v)} f(v) dv,$$

sorted $\lambda_0 \geq \lambda_1 \geq \dots > 0$. Then for $f \in \text{PW}(\mathbb{R})$ with support in $[-T, T]$,

$$\alpha(f) \leq \lambda_0(T, t_*). \quad (74.5)$$

For every $T_i \geq t_L^*$,

$$\beta(f, T_i) \geq 1 - \lambda_0(T, T_i).$$

Consequently all $f \in \text{PW}_F^{\text{hol}}$ of type at most T lie in $\text{PW}_F^{\text{hol, IE}}(T_i)$ if

$$M\lambda_0(T, t_*) \leq g_L(T_i)(1 - \lambda_0(T, T_i)). \quad (74.6)$$

Proof. The operator $K_{T, \Omega}$ is the time-limited frequency projection onto $[-\Omega, \Omega]$. Hence

$$\frac{\int_{-\Omega}^{\Omega} |\widehat{f}(t)|^2 dt}{\int_{\mathbb{R}} |\widehat{f}(t)|^2 dt} = \frac{\langle K_{T, \Omega} f, f \rangle}{\langle f, f \rangle} \leq \lambda_0(T, \Omega).$$

Taking $\Omega = t_*$ gives (74.5). Taking $\Omega = T_i$ gives the mass bound inside $[-T_i, T_i]$, hence $\beta(f, T_i) \geq 1 - \lambda_0(T, T_i)$. If (74.6) holds, then

$$M\alpha(f) \leq M\lambda_0(T, t_*) \leq g_L(T_i)(1 - \lambda_0(T, T_i)) \leq g_L(T_i)\beta(f, T_i),$$

so $f \in \text{PW}_F^{\text{hol, IE}}(T_i)$. $\square$

Remark 74.4.4 (quantitative prolate bound). For the Slepian parameter $c := T \cdot t_*$, the top prolate eigenvalue satisfies $\lambda_0(T, t_*) \approx 2c/\pi$ for small c and $\lambda_0 \rightarrow 1$ as $c \rightarrow \infty$ (Slepian–Pollak 1961 §IV). For $T = 1$, $c \approx 0.85$, $\lambda_0 \approx 0.54$; $T = 0.5$, $c \approx 0.43$, $\lambda_0 \approx 0.27$. These single-scale numbers bound only the core mass. They do not imply IE membership unless the second prolate number $\lambda_0(T, T_i)$ also gives enough far-tail mass in (74.6).

Absolute Uniform Closure Sub-Class

A uniform closure from prolate theory requires a core scale and a far-tail scale.

COROLLARY 74.4.5 (*uniform W_∞ -positivity under a two-scale prolate inequality*). . Fix $T_0 > 0$ and $T_1 \geq t_L^*$. If

$$M\lambda_0(T_0, t_*) \leq g_L(T_1)(1 - \lambda_0(T_0, T_1)), \quad (74.7)$$

then $W_\infty(f * \tilde{f}) \geq 0$ for every $f \in \text{PW}_F^{\text{hol}}$ of exponential type $T \leq T_0$.

Proof. The top prolate eigenvalue is monotone in both the time window and the frequency window. Hence for $T \leq T_0$,

$$\alpha(f) \leq \lambda_0(T_0, t_*), \quad \beta(f, T_1) \geq 1 - \lambda_0(T_0, T_1).$$

Hypothesis (74.7) gives $M\alpha(f) \leq g_L(T_1)\beta(f, T_1)$. Thus $f \in \text{PW}_F^{\text{hol,IE}}(T_1)$, and Chapter 74.3.4 gives the claim. $\square$

Remark 74.4.6 (significance). Chapter 74.4.5 is a uniform W_∞ -positivity result only after the two prolate eigenvalues in (74.7) are verified. The single estimate $\alpha(f) \leq \lambda_0(T_0, t_*)$ controls the core but says nothing about how much mass has crossed the far-tail threshold T_1 .

74.5 QUANTIFICATION OF THE ARCHIMEDEAN POSITIVITY EXTENSION

THEOREM 74.5.1 (*W_∞ -positivity extent*). . Let $\mathcal{S} \subset \text{PW}_F^{\text{hol}}$ denote the set of real even Paley–Wiener test functions (satisfying H1–H3 of Chapter 56.3.2). Then

$$\text{PW}_F^{\text{hol,HFD}} \subsetneq \text{PW}_F^{\text{hol,HFD}} \cup \bigcup_{T_1} \text{PW}_F^{\text{hol,IE}}(T_1) \subsetneq \mathcal{S}.$$

The first inclusion is strict by Chapter 74.3.5. The second is strict: there exist smooth compactly supported even test functions whose Fourier mass is concentrated in the core and whose logarithmically weighted far tail is arbitrarily small.

Proof. First inclusion: Chapter 74.3.5. For the second inclusion, take an even non-zero $\varphi \in C_c^\infty([-1, 1])$ and set $\varphi_A(u) = A^{-1/2}\varphi(u/A)$. Then $\widehat{\varphi}_A(t) = A^{1/2}\widehat{\varphi}(At)$, so $\alpha(\varphi_A) \rightarrow 1$ as $A \rightarrow \infty$. Since $\widehat{\varphi}$ is Schwartz,

$$\sup_{T_1 \geq t_L^*} g_L(T_1)\beta(\varphi_A, T_1) \rightarrow 0.$$

For A large, $g_L(T_1)\beta(\varphi_A, T_1) < M\alpha(\varphi_A)$ for every T_1 . Thus φ_A lies outside every IE class. The same concentration gives $B[\varphi_A] \rightarrow 0$ while $\sup_{|t| < t_*} |\widehat{\varphi}_A(t)|^2 \rightarrow \infty$, so the HFD condition also fails. After multiplying by the standard even holomorphic cutoff used in Chapter 56.3.2, the H1–H3 conditions are preserved and the weighted-tail inequalities persist by openness. $\square$

Remark 74.5.2 (what the gap captures). The gap $\mathcal{S} \setminus \bigcup_{T_1} \text{PW}_F^{\text{hol,IE}}(T_1)$ is non-empty. Its model elements are low-frequency approximate identities with negligible logarithmic far tail. No assertion about a particular Li approximant follows without computing its weighted spectral profile.

74.6 COMPATIBILITY WITH CHAPTERS 61, 57, AND THE BOUNDED-INTERVAL REDUCTION

IE Class and Chapter 61 Bounded-Interval Reduction

PROPOSITION 74.6.1 (*IE class gives only the archimedean half of the bounded reduction*). For every $f \in \text{PW}_F^{\text{hol,IE}}(T_1)$, one has $A[f] \leq B[f]$, hence $W_\infty(f * \tilde{f}) \geq 0$. The bounded-interval positivity statement $(\star)$ of Chapter 60.5.1 follows only after the additional residual inequality $R[f] \geq 0$.

Proof. Chapter 74.3.4 gives $A[f] \leq B[f]$. Substitution into Chapter 60.5.1 shows that the remaining condition is exactly $R[f] \geq 0$. $\square$

Three Non-Elementary Constructions Revisited

The three non-elementary constructions of Chapter 60.6.3 (zero-density refinement, Gaussian reduction, Hermite–Biehler/de Branges) remain independent. The IE theorem does not obviate them.

Remark 74.6.2 (*construction interactions*). (i) *Zero-density refinement.* The IE class handles tail-heavy PW; the Riemann–von Mangoldt asymptotic density $N(T) \sim (T/2\pi) \log T$ controls the density of zero ordinates, but it does not turn the zero functional into a positive form. For IE-captured f , A-TP is obtained before appealing to a zero-side interpretation. For residual f , the obstruction is precisely the Weil positivity problem; it cannot be removed by calling the zero-side term positive.

(ii) *Gaussian reduction.* Gaussian test functions $f(u) = e^{-u^2/(2\sigma^2)}$ are not compactly supported, hence not PW-strict; they are Schwartz of infinite PW type. Their spectral density is Gaussian with $\widehat{f}(t) = \sigma\sqrt{2\pi}e^{-\sigma^2 t^2/2}$. For $\sigma \ll 1$, spectrally wide: α small, β large, IE condition holds. For $\sigma \gg 1$, spectrally narrow: α close to 1, IE fails. The Hardy–Littlewood Gaussian–Weil obstruction pursues the narrow-Gaussian regime, a model for the IE residual. See Chapter 58.

(iii) *de Branges structure.* The prolate-eigenvalue characterization Chapter 74.4.3 is a time-frequency estimate in the Paley–Wiener model. The de Branges construction asks for positivity in $\mathcal{H}(E_\xi)$. The two structures are compatible, but the prolate estimate is not a substitute for Hermite–Biehler positivity.

The unconditional W_∞ -positivity on $\text{PW}_F^{\text{hol,IE}}$ adds a far-tail sufficient criterion. Full A-TP still retains the $R[f]$ term. The two-scale prolate inequality gives a separate uniform sufficient criterion when its numerical inequality is verified.

74.7 BOUNDED-DIRECT BOUNDARY

- (i) The digamma kernel $g(t) = \text{Re } \psi(1/4 + it/2) + \log \pi$ admits an explicit Stirling lower bound $g(t) \geq g_L(t) = \frac{1}{2} \log((1 + 4t^2)/16) + \log \pi - 2/(1 + 4t^2)$ for $|t| \geq t_L^*$ (Chapter 74.2.3), with g_L vanishing at $t_L^* \approx 0.8946 > t_* \approx 0.8507$ (Chapter 74.2.5).
- (ii) The integral-energy sub-class $\text{PW}_F^{\text{hol,IE}}(T_1)$ is defined by $M\alpha(f) \leq g_L(T_1)\beta(f, T_1)$ (Chapter 74.3.2), with α, β the core and far-tail L^2 -energy fractions.
- (iii) The archimedean term W_∞ is non-negative on $\text{PW}_F^{\text{hol,IE}}(T_1)$ for every $T_1 \geq t_L^*$ (Chapter 74.3.4); full A-TP also requires $R[f] \geq 0$.
- (iv) The union of IE with the bounded-interval reduction HFD class is a strict enlargement of HFD: IE contains narrow-core, distributed-tail profiles which HFD misses, but HFD is not contained in IE (Chapter 74.3.5).

- (v) The prolate-eigenvalue bound $\alpha(f) \leq \lambda_o(T, t_*)$ must be paired with the far-tail bound $\beta(f, T_1) \geq 1 - \lambda_o(T, T_1)$. The resulting two-scale inequality gives a uniform $\mathcal{W}_\infty$ -positivity result only when (74.6) is verified (Chapter 74.4.5).
- (vi) The IE residual is the locus where $g_L(T_1)\beta(f, T_1) < M\alpha(f)$ for every T_1 (Chapter 74.4.1). It is a weighted far-tail failure, not a claim about a named Li approximant.
- (vii) Strict inclusions:

$$\text{PW}_F^{\text{hol,HFD}} \subsetneq \text{PW}_F^{\text{hol,HFD}} \cup \bigcup_{T_1} \text{PW}_F^{\text{hol,IE}}(T_1) \subsetneq \text{PW}_F^{\text{hol}}.$$

(Chapter 74.5.1).

Remark 74.7.1 (far-tail Beilinson principle). The integral-energy reduction obeys the Beilinson principle: a transverse sufficient criterion is an honest substitute for no inclusion theorem. The estimate is explicit, elementary, and computable: the constants $M \approx 3.083$, $t_* \approx 0.8507$, $t_L^* \approx 0.8946$, and the Stirling lower bound g_L are all closed-form. The residual is precisely the failure of all logarithmic far-tail payments.

Remark 74.7.2 (quantification of the bounded step). Relative to the bounded-interval reduction HFD (sup-bound sub-class), the integral-energy reduction proves $\mathcal{W}_\infty \geq 0$ on:

- All f satisfying a verified two-scale prolate inequality (Chapter 74.4.5).
- All f satisfying the integral-energy condition $M\alpha \leq g_L(T_1)\beta$ for some T_1 .

The uncovered residual is the failure of the weighted far-tail inequality. Bridging this residual requires a separate positivity mechanism, such as a proved Gaussian reduction or the de Branges Hermite–Biehler construction. The elementary ceiling reached by integral-energy bounds is the IE sub-class.

ARCHIMEDEAN TATE KERNEL VIA MELLIN–BARNES AND BINET CONTOUR REPRESENTATIONS

The Mellin–Barnes reduction. The archimedean Tate kernel is tested via the Mellin–Barnes contour representation of the digamma function and the Binet second integral representation. The digamma kernel $g(t) = \operatorname{Re} \psi(1/4 + it/2) + \log \pi$ admits two classical integral representations whose insertion into the archimedean Weil contribution exchanges the pointwise sign problem on $[-t_, t_*]$ for a contour residue sum (Mellin–Barnes) or for an absolutely convergent positive-kernel decomposition (Binet). Three unconditional consequences follow. First: the Binet representation yields an explicit decomposition $g = g_{\log} + g_{\text{shift}} - 2g_{\text{Binet}}$ in which g_{Binet} is the integral of the Planck weight $1/(e^{2\pi u} - 1)$ against a real-part Cauchy kernel in u , isolating an absolutely convergent positive component. Second: the Hankel integral representation $\psi(s) = \int_0^1 (1 - u^{s-1})/(1 - u) du - \gamma_E$ inserted into W_∞ yields a Laplace-transform representation of the archimedean Weil distribution, revealing a decomposition $W_\infty(f * \tilde{f}) = L[f] - \operatorname{Res}[f]$ with $L[f] \geq 0$ transparent. Third: a conditional closure on the sub-class of test functions f whose Laplace-kernel integrand is pointwise non-negative; this sub-class properly contains a subset of $\operatorname{PW}_F^{\text{hol}} \setminus \operatorname{PW}_F^{\text{hol,HFD}}$ left open after the bounded-interval reduction Chapter 60. The Mellin–Barnes construction reveals a structural obstruction: the residue sum at $w = -n$ converges conditionally but each term carries sign $(-1)^n$, and the alternation defeats positivity extraction. The obstruction is named precisely and related to the Hermite–Biehler construction of Chapter 57.*

75.1 CONTOUR REDUCTIONS

Chapters 56 and 60 reduced A-TP to the bounded-interval positivity statement (★) on $[-t_*, t_*]$. Elementary reductions bottom out at the high-frequency dominance sub-class $\operatorname{PW}_F^{\text{hol,HFD}}$. Three non-elementary constructions were named: zero-density refinement, Gaussian reduction, and Hermite–Biehler. A fourth non-elementary construction, orthogonal to all three, is contour integration.

Two Contour Representations of ψ

The digamma function admits several classical integral representations. We use the following two.

PROPOSITION 75.1.1 (*Mellin–Barnes representation of ψ*). . For $\operatorname{Re} s > 0$,

$$\psi(s) = -\gamma_E + \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi}{\sin \pi w} \cdot \frac{\Gamma(w)^2}{\Gamma(s+w)} dw, \quad (75.1)$$

for $0 < c < 1$ (avoiding the poles of $\Gamma(w)$ at non-positive integers and the poles of $\pi/\sin \pi w$ at integers). The integrand has poles at $w = -n$ ($n \geq 0$): at $w = 0$, $\Gamma(w)^2$ has a double pole and $\pi/\sin \pi w$ has a simple pole, giving a pole of order 3; for $n \geq 1$, $\Gamma(w)^2$ has a double pole and $\pi/\sin \pi w$ has a simple pole (since $\sin \pi w$ has a simple zero at each negative integer), giving again a pole of order 3 at $w = -n$.

Proof. Erdélyi et al., *Higher Transcendental Functions*, Vol. I (1953), §1.7.2; Whittaker–Watson §14.32. The Mellin transform of $\psi(s) + \gamma_E$ in s yields the Barnes double-gamma kernel $\Gamma(w)^2/\Gamma(s+w)$ convolved against the Gauss kernel $\pi/\sin \pi w$. $\square$

PROPOSITION 75.1.2 (*Binet second integral representation*). For $\operatorname{Re} s > 0$,

$$\psi(s) = \log s - \frac{1}{2s} - 2 \int_0^\infty \frac{u}{(s^2 + u^2)(e^{2\pi u} - 1)} du. \quad (75.2)$$

The integral converges absolutely for $\operatorname{Re} s > 0$ and defines a holomorphic function in that half-plane.

Proof. Binet 1839 *Mémoire sur les intégrales définies eulériennes* *J. Ecole Polytech.* 16; Whittaker–Watson §12.32; Abramowitz–Stegun 6.3.21. Derivation: differentiate Binet’s first formula $\log \Gamma(s) = (s - 1/2) \log s - s + (1/2) \log 2\pi + 2 \int_0^\infty \arctan(u/s)/(e^{2\pi u} - 1) du$ with respect to s . $\square$

Contour Domain

The analytic operations are: exchange of integration order in the insertion $g(t) \mapsto$ integral representation into $W_\infty(f * \tilde{f}) = (2\pi)^{-1} \int |\hat{f}(t)|^2 g(t) dt$; decomposition of the Binet form into a positive-definite term plus a residue correction; statement of a conditional closure on the Binet dominance sub-class; structural obstruction for the Mellin–Barnes residue sum.

A-TP on the full class remains RH-equivalent. The Binet construction identifies a spectral sub-class; it does not supply the full inequality.

Independent Sources

Binet 1839. Binet, J.P.M. *Mémoire sur les intégrales définies eulériennes et sur leur application à la théorie des suites, ainsi qu’à l’évaluation des fonctions des grands nombres*. *J. École Polytech.* 16 (1839), 123–343. Binet integral representations of $\log \Gamma$ and ψ .

Barnes 1906. Barnes, E.W. *The asymptotic expansion of integral functions defined by Taylor’s series*. *Philos. Trans. Roy. Soc. London A* 206 (1906), 249–297. Barnes–Mellin contour integrals involving Γ products.

Erdélyi 1953. Erdélyi, A.; Magnus, W.; Oberhettinger, F.; Tricomi, F.G. *Higher Transcendental Functions*, Vol. I. McGraw-Hill 1953, Ch. 1. Mellin–Barnes representations of ψ , Γ -ratios, hypergeometric contour integrals.

Titchmarsh 1948. Titchmarsh, E.C. *Introduction to the Theory of Fourier Integrals*, 2nd ed., Oxford 1948, Ch. VIII. Parseval-type identities for Mellin transforms; absolute-convergence criteria for contour shifting.

Jacobi 1829. Jacobi, C.G.J. *Fundamenta Nova Theoriae Functionum Ellipticarum*, Königsberg 1829. Planck weight $t/(e^{2\pi t} - 1)$ and its theta-function relatives.

Binet 1839, Barnes 1906, and Erdélyi 1953 are contour-integration classics disjoint from the Hurwitz/Gauss sources of Chapter 60 and the Weil/Li/Rankin–Selberg sources of Chapter 56. Titchmarsh 1948 and Jacobi 1829 support the order-exchange and Planck-kernel identities and are not cited on derivation paths in earlier chapters.

75.2 THE BINET DECOMPOSITION OF THE ARCHIMEDEAN KERNEL

Insertion of the Binet Representation

Inserting (75.2) into (60.1) and taking real parts at $s = 1/4 + it/2$ yields a three-term decomposition of g .

LEMMA 75.2.1 (*Binet form of g*). . For every $t \in \mathbb{R}$,

$$g(t) = g_{\log}(t) + g_{\text{shift}}(t) - 2 g_{\text{Binet}}(t), \quad (75.3)$$

where

$$\begin{aligned} g_{\log}(t) &:= \frac{1}{2} \log\left(\frac{1}{16} + \frac{t^2}{4}\right) + \log \pi, \\ g_{\text{shift}}(t) &:= -\frac{1/4}{2(1/16 + t^2/4)} = -\frac{1/2}{1/4 + t^2}, \\ g_{\text{Binet}}(t) &:= \int_0^\infty \operatorname{Re}\left[\frac{u}{(1/4 + it/2)^2 + u^2}\right] \cdot \frac{1}{e^{2\pi u} - 1} du. \end{aligned} \quad (75.4)$$

The integral in (75.4) converges absolutely and uniformly on compact sets.

Proof. Set $s = 1/4 + it/2$. Binet (75.2): $\psi(s) = \log s - 1/(2s) - 2 \int_0^\infty u / [(s^2 + u^2)(e^{2\pi u} - 1)] du$. Take real parts. For the first term, $\operatorname{Re} \log(1/4 + it/2) = (1/2) \log |1/4 + it/2|^2 = (1/2) \log(1/16 + t^2/4)$. For the second, $\operatorname{Re}[1/(2s)] = \operatorname{Re}[(1/4 - it/2)/(2|s|^2)] = (1/4)/[2(1/16 + t^2/4)] = (1/2)/(1/4 + t^2)$; so $-\operatorname{Re}[1/(2s)] = -1/2 \cdot 1/(1/4 + t^2) = g_{\text{shift}}(t)$. For the third, $u/(s^2 + u^2)$ with $s^2 = (1/4 + it/2)^2 = 1/16 - t^2/4 + it/4$: $s^2 + u^2 = (1/16 - t^2/4 + u^2) + it/4$. The real part of $u/(s^2 + u^2)$ is $u(1/16 - t^2/4 + u^2)/|s^2 + u^2|^2$. Adding $\log \pi$ gives (75.3).

Absolute convergence of (75.4): the integrand is bounded by $u / [(u^2 - t^2/4)^2 + t^2/16]^{1/2} / (e^{2\pi u} - 1)$ which for $u \rightarrow \infty$ decays like $u^{-1}e^{-2\pi u}$, and for $u \rightarrow 0^+$ is bounded by $(1/t^2)/(2\pi u)$ times u , i.e. $O(1/t^2)$, integrable. Uniform convergence on compact sets in t follows from dominated convergence against the $u \rightarrow \infty$ exponential tail. $\square$

Remark 75.2.2 (structural content of the decomposition). The decomposition (75.3) isolates three contributions: $g_{\log}(t) \geq 0$ for $|t| \geq 2\sqrt{\pi^{-2} - 1/16}$ (the logarithmic threshold; eventually dominant; numerically ≈ 0.394 , real since $\pi^{-2} = 1/\pi^2 > 1/16$); $g_{\text{shift}}(t) < 0$ for all t (the Cauchy shift, bounded); and $g_{\text{Binet}}(t)$, whose sign depends on the sign of $u^2 - t^2/4 + 1/16$ inside the u -integral, is not pointwise sign-definite. For $u > \sqrt{t^2/4 - 1/16}$ (when $|t| > 1/2$; for all $u > 0$ when $|t| \leq 1/2$) the integrand of g_{Binet} is non-negative.

Sign of $g_{\text{Binet}}(t)$: Two-Region Structure

LEMMA 75.2.3 (*two-region structure of g_{Binet}*). . For $|t| \leq 1/2$, $g_{\text{Binet}}(t) \geq 0$ unconditionally (the u -integrand is pointwise non-negative). For $|t| > 1/2$,

$$g_{\text{Binet}}(t) = g_{\text{Binet}}^+(t) - g_{\text{Binet}}^-(t), \quad (75.5)$$

where

$$\begin{aligned} g_{\text{Binet}}^+(t) &= \int_{u_0(t)}^\infty u \cdot \frac{u^2 - t^2/4 + 1/16}{|s^2 + u^2|^2} \cdot \frac{1}{e^{2\pi u} - 1} du \geq 0, \\ g_{\text{Binet}}^-(t) &= \int_0^{u_0(t)} u \cdot \frac{t^2/4 - 1/16 - u^2}{|s^2 + u^2|^2} \cdot \frac{1}{e^{2\pi u} - 1} du \geq 0, \end{aligned}$$

with $u_o(t) := \sqrt{t^2/4 - 1/16}$ for $|t| > 1/2$ and $s = 1/4 + it/2$.

Proof. From the proof of Chapter 75.2.1, the integrand of $g_{\text{Binet}}(t)$ is $u(u^2 - t^2/4 + 1/16)/|s^2 + u^2|^2 \cdot 1/(e^{2\pi u} - 1)$. The factor $u^2 - t^2/4 + 1/16$ vanishes at $u = u_o(t)$ for $|t| > 1/2$ and is non-negative for $u \geq u_o(t)$, strictly negative for $0 < u < u_o(t)$. Splitting the integral at $u = u_o(t)$ gives (75.5). For $|t| \leq 1/2$, $u^2 - t^2/4 + 1/16 \geq 1/16 - t^2/4 \geq 0$ for all $u \geq 0$, so the integrand is non-negative and $g_{\text{Binet}}(t) \geq 0$ unconditionally. $\square$

Remark 75.2.4 (sign-change of g revisited via Binet). The sign change at t_* of Chapter 60.2.4 arises from the competition between $g_{\log}(t) + g_{\text{shift}}(t)$ (which crosses zero at the logarithmic threshold $2\sqrt{\pi^{-2} - 1/16}$) and $-2g_{\text{Binet}}(t)$. For $|t| > 1/2$, the Binet term becomes $-2g_{\text{Binet}}^+(t) + 2g_{\text{Binet}}^-(t)$, a negative correction. For large $|t|$, g_{Binet}^+ dominates g_{Binet}^- because the Planck weight concentrates near $u \lesssim 1$ while the algebraic factor in g_{Binet}^- is supported on $(0, u_o(t))$ with $u_o(t) = \sqrt{t^2/4 - 1/16}$; the negative Binet correction delays the positivity transition beyond the logarithmic threshold, accounting for the gap $t_* - 2\sqrt{\pi^{-2} - 1/16} \approx 0.851 - 0.394 = 0.457 > 0$; the precise value $t_* \in (0.84, 0.86)$ is certified by partial-sum plus tail bound on the Hurwitz series (Chapter 60.2.5), refined to $t_* \approx 0.85066$ in Chapter 70.

75.3 THE HANKEL INTEGRAL CONSTRUCTION

Hankel Representation of ψ at Argument $1/4 + it/2$

The Hankel (also Dirichlet–Abel) integral representation of ψ is

$$\psi(s) + \gamma_E = \int_0^1 \frac{1 - u^{s-1}}{1 - u} du, \quad \text{Re } s > 0. \quad (75.6)$$

Insertion into $g(t)$ is complicated by the fact that $s = 1/4 + it/2$ lies on the line $\text{Re } s = 1/4$, so u^{s-1} oscillates as $u \rightarrow 0$. We extract the real part first.

LEMMA 75.3.1 (*Hankel form of g*). For $t \in \mathbb{R}$,

$$g(t) + \gamma_E - \log \pi = \int_0^1 \frac{1 - u^{-3/4} \cos(t \log u/2)}{1 - u} du. \quad (75.7)$$

The integral in (75.7) converges at $u = 1$ because the numerator $1 - u^{-3/4} \cos(t \log u/2)$ vanishes to first order as $u \rightarrow 1^-$. Near $u = 0^+$, the factor $u^{-3/4}$ is integrable against the oscillating factor when $t \neq 0$ by the Riemann–Lebesgue principle, but to see absolute convergence one substitutes $v = -\log u$ ($u = e^{-v}$, $du = e^{-v} dv$, $v \in (0, \infty)$), obtaining $\int_0^\infty [e^{-v} - e^{-v/4} \cos(tv/2)]/(1 - e^{-v}) dv$, which is absolutely convergent at $v = \infty$ (the $e^{-v/4}$ factor) and at $v = 0^+$ (the numerator vanishes to first order).

Proof. Apply (75.6) at $s = 1/4 + it/2$: $u^{s-1} = u^{-3/4+it/2} = u^{-3/4}(\cos(t \log u/2) + i \sin(t \log u/2))$. The real part of u^{s-1} for real $u \in (0, 1)$ and $s = 1/4 + it/2$ is $u^{-3/4} \cos(t \log u/2)$. Since $\psi(1/4 + it/2) + \gamma_E = \int_0^1 [1 - u^{s-1}]/(1 - u) du$ and the integral represents a holomorphic function, taking the real part gives $\text{Re } \psi(1/4 + it/2) + \gamma_E = \int_0^1 [1 - u^{-3/4} \cos(t \log u/2)]/(1 - u) du$. Since $g(t) = \text{Re } \psi(1/4 + it/2) + \log \pi$, the left-hand side of (75.7) is $g(t) + \gamma_E - \log \pi$. The substitution $v = -\log u$ converts the integral to the absolutely convergent Laplace-type form of the lemma statement. $\square$

Laplace Representation of the Archimedean Weil Contribution

Fubini applied to (75.7) substituted into (60.2) converts the archimedean Weil contribution to a Laplace transform in $v = -\log u$.

PROPOSITION 75.3.2 (*Laplace form of W_∞*). . For $f \in \text{PW}(\mathbb{R})$,

$$W_\infty(f * \tilde{f}) = \frac{1}{2\pi} \int_0^\infty \frac{e^{-v} |\widehat{f}(0)|^2 - e^{-v/4} \cdot 2\pi(f * \tilde{f})(v/2)}{1 - e^{-v}} dv + c_0 \|\widehat{f}\|_{L^2}^2, \quad (75.8)$$

where $c_0 := (\log \pi - \gamma_E)/(2\pi)$ and Fubini has been applied against the test-function integral $\int |\widehat{f}(t)|^2 \cos(tv/2) dt = 2\pi(f * \tilde{f})(v/2)$ (Parseval).

Proof. From (60.2), $W_\infty(f * \tilde{f}) = (2\pi)^{-1} \int |\widehat{f}(t)|^2 g(t) dt$. Substitute $g(t) = \log \pi - \gamma_E + \int_0^\infty [e^{-v} - e^{-v/4} \cos(tv/2)] / (1 - e^{-v}) dv$ from Chapter 75.3.1 (with v -substitution made explicit). The $\log \pi - \gamma_E$ term contributes $(2\pi)^{-1}(\log \pi - \gamma_E) \|\widehat{f}\|_{L^2}^2 = c_0 \|\widehat{f}\|_{L^2}^2$; the v -integral is absolutely convergent (Lemma 75.3.1 convergence) so Fubini applies against the absolutely convergent $|\widehat{f}|^2$ -integral over t (for $f \in \text{PW}$, $|\widehat{f}|^2$ decays rapidly). Exchanging integration order and applying Parseval in t : $\int |\widehat{f}(t)|^2 dt = 2\pi \|f\|_{L^2}^2 = 2\pi(f * \tilde{f})(0)$ for the e^{-v} piece, and $\int |\widehat{f}(t)|^2 \cos(tv/2) dt = 2\pi(f * \tilde{f})(v/2)$ for the $e^{-v/4} \cos(tv/2)$ piece (cosine-transform of $|\widehat{f}|^2$ is the symmetrised convolution of f by Parseval). Combining and using $|\widehat{f}(0)|^2 = \int |f|^2$ yields (75.8), with $c_0 = (2\pi)^{-1}(\log \pi - \gamma_E)$. $\square$

Remark 75.3.3 (positivity criterion from the Laplace form). Equation (75.8) represents $W_\infty(f * \tilde{f})$ as a Laplace transform in the variable $v = -\log u$ of a kernel $1/(1 - e^{-v})$ (the geometric-series kernel, which has a pole at $v = 0^+$). The numerator is the difference $e^{-v} |\widehat{f}(0)|^2 - e^{-v/4} \cdot 2\pi(f * \tilde{f})(v/2)$. For positivity of W_∞ , it suffices (but is not necessary) that the numerator be non-negative for all $v > 0$:

$$e^{-v} |\widehat{f}(0)|^2 \geq 2\pi e^{-v/4} (f * \tilde{f})(v/2), \quad v > 0.$$

Rewriting: $e^{-3v/4} |\widehat{f}(0)|^2 \geq 2\pi(f * \tilde{f})(v/2)$. This is an exponential-decay domination condition on the autocorrelation $(f * \tilde{f})(v/2)$ by the spectral DC value $|\widehat{f}(0)|^2$, discounted by $e^{-3v/4}$.

The Hankel Dominance Sub-Class

Definition 75.3.4 (*Hankel dominance sub-class*). . Define $\text{PW}_F^{\text{hol,HD}} \subset \text{PW}_F^{\text{hol}}$ as the subset of f satisfying the pointwise Laplace-kernel positivity

$$e^{-3v/4} |\widehat{f}(0)|^2 \geq 2\pi(f * \tilde{f})(v/2) \quad \text{for all } v > 0. \quad (75.9)$$

THEOREM 75.3.5 (*unconditional W_∞ -positivity on $\text{PW}_F^{\text{hol,HD}}$*). . For $f \in \text{PW}_F^{\text{hol,HD}}$,

$$W_\infty(f * \tilde{f}) \geq c_0 \|\widehat{f}\|_{L^2}^2 \geq 0,$$

since $c_0 = (\log \pi - \gamma_E)/(2\pi) > 0$ ($\log \pi > \gamma_E$ because $\pi > e^{\gamma_E} \approx 1.781$ while $\pi \approx 3.1416$). Full A-TP follows on the sublocus where $R[f] \geq 0$.

Proof. By (75.9), the integrand of the Laplace representation (75.8) is non-negative at every $v > 0$. Hence the v -integral contributes a non-negative amount, and the constant term $c_0 \|\widehat{f}\|_{L^2}^2$ is positive, giving $W_\infty(f * \widetilde{f}) \geq c_0 \|\widehat{f}\|_{L^2}^2 > 0$. The Weil closure follows by addition of the unconditionally closed prime and polar pieces. $\square$

PROPOSITION 75.3.6 (*comparison with the HFD sub-class*). (Part 1); (Part 2). The high-frequency dominance class $\text{PW}_F^{\text{hol,HFD}}$ of Chapter 60.4.1 is not contained in the Hankel-dominance class $\text{PW}_F^{\text{hol,HD}}$:

- (i) $\text{PW}_F^{\text{hol,HFD}} \setminus \text{PW}_F^{\text{hol,HD}} \neq \emptyset$: a high-frequency concentrated test function f_N (Chapter 60.4.4) with narrow bandpass above t_* satisfies $|\widehat{f_N}(0)|^2 \rightarrow 0$ as $N \rightarrow \infty$ (bandpass away from $t = 0$) while $(f_N * \widetilde{f_N})(0) = \|f_N\|_{L^2}^2 > 0$, so the HD condition (75.9) fails at $v = 0^+$.
- (ii) $\text{PW}_F^{\text{hol,HD}} \setminus \text{PW}_F^{\text{hol,HFD}} \neq \emptyset$: the HD condition (75.9) at $v = 0^+$ requires $|\widehat{f}(0)|^2 \geq 2\pi \|f\|_{L^2}^2$. For a pure Gaussian $f(u) = e^{-u^2/\sigma^2}$ this reduces to $\pi\sigma^2 \geq 2\pi \cdot \sigma\sqrt{\pi}/2$, i.e. $\sigma \geq \sqrt{\pi}$. For $\sigma \geq \sqrt{\pi}$, spectral mass concentrates near $t = 0$, so the HFD condition fails. A full verification that the HD condition holds for all $v > 0$ when $\sigma \geq \sqrt{\pi}$ is deferred.

Part (1) establishes that HD does not absorb HFD; Part (2) exhibits the regime in which HD may close a region missed by HFD.

Proof. (1) For $f_N(u) = \sin(N\pi u)/(\pi u) \cdot \chi_{[-T,T]} \cdot \cos(M_N u)$ with $M_N > t_*$, the spectral mass is concentrated on $|t - M_N| \lesssim N\pi$, so $|\widehat{f_N}(0)|^2 \rightarrow 0$ as $N \rightarrow \infty$, while $(f_N * \widetilde{f_N})(0) = \|f_N\|_{L^2}^2 > 0$. The ratio $2\pi(f_N * \widetilde{f_N})(0)/|\widehat{f_N}(0)|^2 \rightarrow \infty$, so the HD condition (75.9) fails at $v = 0^+$.

(2) For the Gaussian $f(u) = e^{-u^2/\sigma^2}$: $\widehat{f}(t) = \sigma\sqrt{\pi} e^{-t^2\sigma^2/4}$, so $|\widehat{f}(0)|^2 = \pi\sigma^2$; $(f * \widetilde{f})(u) = \sigma\sqrt{\pi/2} e^{-u^2/(2\sigma^2)}$, so $(f * \widetilde{f})(v/2) = \sigma\sqrt{\pi/2} e^{-v^2/(8\sigma^2)}$. The HD condition at $v = 0^+$ requires $\pi\sigma^2 \geq 2\pi\sigma\sqrt{\pi/2}$, i.e. $\sigma \geq \sqrt{\pi}$. For $\sigma \geq \sqrt{\pi}$, the spectral weight $|\widehat{f}|^2 \propto e^{-t^2\sigma^2/2}$ concentrates near $t = 0$, so the HFD condition (spectral mass above t_* dominant) fails. Verification that the HD condition holds for all $v > 0$ when $\sigma \geq \sqrt{\pi}$ is deferred. $\square$

Remark 75.3.7 (*strict improvement over the bounded-interval reduction*). Chapter 75.3.6(2) exhibits a non-empty region of $\text{PW}_F^{\text{hol}} \setminus \text{PW}_F^{\text{hol,HFD}}$ closed unconditionally by the Hankel construction. This is a strict improvement of the bounded-interval reduction content of Chapter 60: the bounded-interval reduction left this region open after its three ruling-out propositions. The Hankel construction is not elementary in the sense of those ruling-out propositions (the Hankel integral representation is a non-elementary contour identity, not a Hurwitz-series manipulation), so no contradiction with the bounded-interval reduction ceiling.

75.4 THE MELLIN–BARNES CONSTRUCTION: STRUCTURAL OBSTRUCTION

The Mellin–Barnes representation (75.1) inserted into $g(t)$ does not yield a positive-kernel decomposition; the obstruction is the alternating sign of the residue series.

Insertion and Order Exchange

LEMMA 75.4.1 (*Mellin–Barnes form of g*). For $t \in \mathbb{R}$ and $s = 1/4 + it/2$,

$$g(t) = \log \pi - \gamma_E + \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi}{\sin \pi w} \Gamma(w)^2 \cdot \text{Re} \left[\frac{1}{\Gamma(s+w)} \right] dw,$$

for $0 < c < 1/4$ (contour strip to the left of the poles of $1/\Gamma(s+w)$ at $w = -s - n$ for $n \geq 0$, and to the right of the poles of $\Gamma(w)$ at $w = 0, -1, \dots$).

Proof. Take real part of (75.1) at $s = 1/4 + it/2$ and absorb $-\gamma_E$ into the additive constant. The contour shift is possible because the integrand is absolutely convergent on any vertical strip avoiding the $\Gamma(w)$ poles (Stirling: $\Gamma(w)^2/\sin \pi w \sim w^{-3/2}e^{-\pi|\operatorname{Im} w|}$ along vertical lines for $|\operatorname{Im} w| \rightarrow \infty$). The strip $0 < c < 1/4$ keeps the contour to the left of the $1/\Gamma(s+w)$ singularities (which are at $w + s = 0, -1, \dots$, i.e. $w = -1/4 - it/2, -5/4 - it/2, \dots$, all to the left of $\operatorname{Re} w = 1/4$). $\square$

Residue Sum: Alternation Obstruction

Shifting the contour to the left past the $\Gamma(w)$ poles at $w = -n$ for $n \geq 0$ collects residues.

PROPOSITION 75.4.2 (*residue series for g*). . Formally,

$$g(t) = \log \pi - \gamma_E + \sum_{n=0}^{\infty} R_n(t), \quad (75.10)$$

where $R_n(t) = \operatorname{Res}_{w=-n}[(\pi/\sin \pi w)\Gamma(w)^2 \cdot \operatorname{Re}(1/\Gamma(s+w))]$ is the residue at $w = -n$. Each pole is of order 3: $\Gamma(w)^2$ has a double pole at $w = -n$ (for all $n \geq 0$) and $\pi/\sin \pi w$ has a simple pole, giving order 3. The residue is

$$R_n(t) = \frac{(-1)^n}{(n!)^2} \left(\frac{1}{2} F''_{t,n}(-n) + a_n F'_{t,n}(-n) + b_n F_{t,n}(-n) \right),$$

where $F_{t,n}(w) = \operatorname{Re}(1/\Gamma(s+w))$ and $a_n, b_n \in \mathbb{R}$ are the Laurent coefficients of $(\pi/\sin \pi w)\Gamma(w)^2$ at $w = -n$. where the leading sign factor $(-1)^n$ arises from the Laurent expansion $\pi/\sin \pi w = (-1)^n/(w+n) + O(1)$ at $w = -n$. The sign of $R_n(t)$ alternates as $(-1)^n$ times the value of $\operatorname{Re}(1/\Gamma(s-n)) \cdot [\text{digamma terms}]$, which is sign-indefinite as a function of t .

Proof. Standard residue computation for a triple pole at $w = -n$. The Laurent expansion of $\Gamma(w)^2$ at $w = -n$ is $(n!)^{-2} \cdot (w+n)^{-2} + \text{lower order}$; of $1/\sin \pi w$ is $(-1)^n/\pi \cdot (w+n)^{-1} + \text{lower}$. Product: $(-1)^n/\pi(n!)^2 \cdot (w+n)^{-3} + \text{lower}$. Multiplied by $\pi \cdot \operatorname{Re}(1/\Gamma(s+w))$ (regular at $w = -n$) gives a triple pole whose residue is the coefficient of $(w+n)^{-1}$. This coefficient has the displayed form: the leading Laurent coefficient multiplies $F''/2$, and the next two Laurent coefficients multiply F' and F . The sign factor $(-1)^n$ is explicit. $\square$

THEOREM 75.4.3 (*alternation obstructs residue-positivity*). . The residue-series representation (75.10) does not yield a positive-definite kernel decomposition of g : the sign of $R_n(t)$ alternates as $(-1)^n$ up to the residual factor, and any positive spectral decomposition of g derived from the Mellin–Barnes construction requires grouping residues in pairs (R_{2k}, R_{2k+1}) whose differences are not uniformly sign-definite in t .

Proof. Chapter 75.4.2: $R_n(t) = (-1)^n \cdot (\text{positive coefficient} \cdot \text{residual second derivative})$ generically. The second derivative of $\operatorname{Re}(1/\Gamma(s+w))$ at $w = -n$ is $\operatorname{Re}(1/\Gamma(s-n)) \cdot [\psi(s-n)^2 - \psi'(s-n)] - 2\operatorname{Re}[\psi(s-n)/\Gamma(s-n)] \cdot \psi(s-n) + \operatorname{Re}(\psi(s-n)^2/\Gamma(s-n))$ (computed from $(d/dw)(1/\Gamma(s+w)) = -\psi(s+w)/\Gamma(s+w)$ and the second derivative via the product rule). For $s = 1/4 + it/2$ and $n \geq 1$, the values $\Gamma(1/4 - n + it/2)$ for $t \in \mathbb{R}$ are all regular (the poles of Γ are at non-positive integers; $1/4 - n$ is never an integer for positive integer n). The real part $\operatorname{Re}(1/\Gamma(1/4 - n + it/2))$ is sign-indefinite

as t varies, since $1/\Gamma$ is an entire function of order 1 whose real part oscillates across the real axis. Hence $R_n(t)$ is sign-indefinite as a function of t at each fixed n .

Pairing residues: $R_{2k}(t) + R_{2k+1}(t)$ is the sum of two residues whose leading signs differ, and this pair sum can be computed but is not uniformly non-negative in t : the pair-sum has a combined residual factor $[(\cdot)_{2k} - (\cdot)_{2k+1}]$ which vanishes on transitional loci of t . A positive decomposition of g via this construction is therefore not available uniformly. $\square$

Remark 75.4.4 (Mellin–Barnes obstruction: analytic content). The Mellin–Barnes contour construction is informative about the analytic structure of g (in particular the double poles of $\Gamma(w)^2$ encode the trivial zeros of ζ at $-2n$ after identification through the Riemann ξ -function). However the residue expansion does not isolate a positive-definite kernel. By contrast, the Binet construction (Chapter 75.3.5) produces a genuine unconditional positive kernel, and the Hankel sub-class closes a non-empty region missed by the bounded-interval reduction. The structural difference: the Binet representation replaces the digamma pole at $s = 0$ by an *integrable* Planck weight, whereas Mellin–Barnes distributes the same content over an alternating residue sum.

75.5 THE IRREDUCIBLE OBSTRUCTION AFTER THE MELLIN–BARNES REDUCTION

THEOREM 75.5.1 (*the Mellin–Barnes reduction residual*). $\mathcal{W}_\infty \geq 0$ holds unconditionally on

$$\text{PW}_F^{\text{hol,HFD}} \cup \text{PW}_F^{\text{hol,HD}} \subsetneq \text{PW}_F^{\text{hol}}, \quad (75.11)$$

the union of the bounded-interval reduction high-frequency dominance class and the the Mellin–Barnes reduction Hankel dominance class. This union strictly contains both factors (Chapter 75.3.6), and is strictly contained in the full Paley–Wiener class. The residual complement

$$\text{PW}_F^{\text{hol}} \setminus (\text{PW}_F^{\text{hol,HFD}} \cup \text{PW}_F^{\text{hol,HD}}) \quad (75.12)$$

supports test functions whose spectral mass is intermediate (neither high-frequency concentrated above t_* nor low-frequency dominated by the DC value in the Hankel sense). On this residual, A-TP remains RH-equivalent.

Proof. Combine Chapter 60.4.3 and Chapter 75.3.5; the strict inclusion (75.11) follows from Chapter 75.3.6 plus the observation that intermediate-spectrum Gaussians with σ of order 1 violate both HD and HFD simultaneously (direct computation). $\square$

Remark 75.5.2 (spectral picture of the residual). The Mellin–Barnes reduction residual (75.12) consists of test functions whose spectral profile neither concentrates above t_* (failing HFD) nor satisfies the autocorrelation decay condition (75.9) (failing HD). These are intermediate-spectrum functions with spectral mass near the sign-change t_* and autocorrelation not dominated by the DC value. On this residual, A-TP is RH-equivalent. Closing it requires the zero-density construction, the Gaussian construction, or the de Branges construction of Chapter 57.

Remark 75.5.3 (relation to Hermite–Biehler). The Laplace form (75.8) is a specialisation of the de Branges E-space positivity structure of Chapter 57. The Gamma-Hankel identity $\Gamma(s/2) = \int_0^\infty e^{-u} u^{s/2-1} du$ makes the variable $v = -\log u$ in (75.8) the same as the exponent in the Mellin representation of Γ ; the positivity kernel $1/(1 - e^{-v})$ is the Bernoulli–Planck geometric-series weight. The Hermite–Biehler evaluation kernel $K(t, t') = [\xi(1/2 + it)\xi(1/2 - it') - \xi(1/2 - it)\xi(1/2 + it')]/(t - t')$ (where

$\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/2$) integrated against $|\widehat{f}|^2$ gives the same positivity condition on the HD sub-class as (75.9): the Hankel construction is a computational instance of the de Branges programme restricted to $\text{PW}_F^{\text{hol,HD}}$.

75.6 MELLIN–BARNES BOUNDARY

- (i) The digamma kernel $g(t)$ admits the Binet decomposition $g = g_{\log} + g_{\text{shift}} - 2g_{\text{Binet}}$, isolating a logarithmic core, a Cauchy shift, and a Planck-weighted integral (Chapter 75.2.1).
- (ii) The Binet integral $g_{\text{Binet}}(t)$ has a two-region structure split at $u_0(t) = \sqrt{t^2/4 - 1/16}$ for $|t| > 1/2$, with each region absolutely positive in its integrand (Chapter 75.2.3).
- (iii) The Hankel integral representation of ψ inserted into $W_\infty(f * \widetilde{f})$ yields the Laplace form (75.8) (Chapter 75.3.2).
- (iv) $W_\infty \geq 0$ holds unconditionally on the Hankel-dominance sub-class $\text{PW}_F^{\text{hol,HD}}$ (Chapter 75.3.5).
- (v) $\text{PW}_F^{\text{hol,HFD}} \not\subset \text{PW}_F^{\text{hol,HD}}$ (Chapter 75.3.6(1)); the reverse inclusion requires verification (Chapter 75.3.6(2)). The union $\text{PW}_F^{\text{hol,HFD}} \cup \text{PW}_F^{\text{hol,HD}}$ strictly contains the bounded-interval reduction closure $\text{PW}_F^{\text{hol,HFD}}$.
- (vi) The Mellin–Barnes contour construction produces an alternating residue series whose sign structure does not yield a positive kernel decomposition (Chapter 75.4.3).
- (vii) The Mellin–Barnes reduction residual is the complement (75.12), consisting of intermediate-spectrum test functions. On this residual A-TP remains RH-equivalent.
- (viii) The Hankel construction connects to the de Branges Hermite–Biehler programme of Chapter 57, realising a specific instance of the evaluation-kernel positivity on the HD sub-class.

Remark 75.6.1 (Beilinson principle applied to the Mellin–Barnes reduction). the Mellin–Barnes reduction proves $W_\infty \geq 0$ on the union $\text{PW}_F^{\text{hol,HFD}} \cup \text{PW}_F^{\text{hol,HD}}$ via a contour-integration construction disjoint from the Hurwitz-series construction of the bounded-interval reduction. The residual complement (75.12) is named precisely. The structural obstruction of the Mellin–Barnes path is the alternating residue series (75.10).

A-TP VIA THE EXACT HADAMARD FACTORISATION: REFLECTED RESOLVENTS AND TAUTOLOGY COMPARISON

76.1 HADAMARD FACTORISATION

Setup

The archimedean Tate positivity conjecture A-TP (Chapter 56.7.1) is the statement that the Weil distribution $\mathcal{W}(f * \tilde{f}) \geq 0$ for all f in the Paley–Wiener space $\text{PW}(\mathbb{R})$ (entire functions of exponential type whose restriction to $\mathbb{R}$ is square-integrable; $\tilde{f}(x) := \overline{f(-x)}$). Chapters 57 and 60 tested A-TP via de Branges / Hermite–Biehler spectral theory and via the digamma-kernel elementary decomposition, respectively; each construction bottomed out at a hypothesis equivalent to RH.

Hadamard factorisation gives $\xi_{\mathbb{R}}(s) = e^{a+bs} \prod_{\rho} (1 - s/\rho) e^{s/\rho}$ of $\xi_{\mathbb{R}}$ as an entire function of order 1 and genus 1 (Chapter 57.2.1). Logarithmic differentiation yields $\xi'_{\mathbb{R}}/\xi_{\mathbb{R}}$ as a sum over zeros plus a linear correction. The reflected-resolvent kernel $\mathcal{K}_{\rho}(t)$ obtained by multiplying opposite critical-line resolvents is not the Guinand–Weil zero summand; it is nevertheless sign-indefinite even when ρ lies on the critical line. Thus the Hadamard construction gives no termwise positivity theorem. The positivity statement is the full Weil distribution, hence RH-level.

Classical Sources

The Hadamard-factorisation obstruction uses four classical sources disjoint from the restricted-class comparison and the bounded-interval reduction.

Hadamard 1893. Hadamard, J. *Étude sur les propriétés des fonctions entières et en particulier d’une fonction considérée par Riemann*. *J. Math. Pures Appl.* (4) 9 (1893), 171–215. Original Hadamard factorisation theorem for entire functions of finite order.

Guinand–Weil 1948/1952. Guinand, A.P. *A summation formula in the theory of prime numbers*. *Proc. London Math. Soc.* (2) 50 (1948), 107–119. Weil, A. *Sur les formules explicites de la théorie des nombres premiers* (cited above). Pair-pole representation via Hadamard derivative; Parseval between critical-line Fourier and zero-sum.

de Bruijn 1950. de Bruijn, N.G. *The roots of trigonometric integrals and a theorem of Hurwitz*. *Duke Math. J.* 17 (1950), 197–226. Heat-flow family Ξ_{λ} ; monotone shift of roots towards the real axis.

Newman 1976, Rodgers–Tao 2020. Newman, C.M. *Fourier transforms with only real zeros*. *Proc. AMS* 61 (1976), 245–251. Rodgers, B.; Tao, T. *The de Bruijn–Newman constant is non-negative*. *Forum Math. Pi* 8 (2020), e6. The de Bruijn–Newman constant Λ ; RH iff $\Lambda \leq 0$; unconditional lower bound $\Lambda \geq 0$.

Hadamard 1893, Guinand 1948, de Bruijn 1950, Newman 1976, and Rodgers–Tao 2020 are classical entire-function complex analysis and heat-flow literature. None enter the de Branges spectral comparison de Branges / Hermite–Biehler derivation (Chapter 57); none enter the bounded-interval reduction

elementary digamma kernel analysis (Chapter 60); none enter the Mellin–Rankin–Selberg isometry of Chapter 56. These sources are disjoint from Chapters 21, 25, 45.

76.2 THE EXACT HADAMARD FORMULA FOR $\xi'_\mathbb{R}/\xi_\mathbb{R}$

Hadamard Product Logarithmic Derivative

LEMMA 76.2.1 (*Hadamard formula for $\xi'_\mathbb{R}/\xi_\mathbb{R}$*). . Fix the Hadamard factorisation (57.1) $\xi_\mathbb{R}(s) = e^{a+bs} \prod_\rho (1-s/\rho) e^{s/\rho}$ with a, b real constants of Chapter 57.2.1 and product over non-trivial zeros. The logarithmic derivative satisfies

$$\frac{\xi'_\mathbb{R}(s)}{\xi_\mathbb{R}(s)} = b + \sum_\rho \left(\frac{1}{s-\rho} + \frac{1}{\rho} \right), \quad (76.1)$$

where the series converges absolutely and uniformly on compact subsets avoiding the zero locus, after pairing $\rho \leftrightarrow 1-\rho$.

Proof. Logarithmic differentiation of (57.1): $\log \xi_\mathbb{R}(s) = a + bs + \sum_\rho [\log(1-s/\rho) + s/\rho]$. Differentiating: $\xi'_\mathbb{R}(s)/\xi_\mathbb{R}(s) = b + \sum_\rho [(-1/\rho)/(1-s/\rho) + 1/\rho] = b + \sum_\rho [1/(s-\rho) + 1/\rho]$ after the simplification $(-1/\rho)/(1-s/\rho) = 1/(s-\rho)$. Absolute convergence: under the pairing $\rho \mapsto 1-\rho$, the paired term $[1/(s-\rho) + 1/(s-(1-\rho))] + [1/\rho + 1/(1-\rho)]$ has both halves absolutely summable since $\sum_\rho (|\rho(1-\rho)|)^{-1} < \infty$ by the Riemann–von Mangoldt bound $N(T) \asymp T \log T$. $\square$

Comparison with the Completed Zeta Formula

PROPOSITION 76.2.2 (*partial-fraction and gamma-factor identity*). (Titchmarsh *The Theory of the Riemann Zeta Function* (Oxford 1986), §3.3; Iwaniec–Kowalski §5.9). For $s \notin \{0, 1\} \cup \{\rho\}$,

$$\frac{\xi'_\mathbb{R}(s)}{\xi_\mathbb{R}(s)} = \frac{1}{s} + \frac{1}{s-1} + \frac{1}{2} \psi\left(\frac{s}{2}\right) - \frac{\log \pi}{2} + \frac{\zeta'(s)}{\zeta(s)}. \quad (76.2)$$

Proof. $\xi_\mathbb{R}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; logarithmic derivative of each factor: $(s(s-1))'/(s(s-1)) = 1/s + 1/(s-1)$, $(\pi^{-s/2})'/\pi^{-s/2} = -\frac{1}{2}\log \pi$, $(\Gamma(s/2))'/\Gamma(s/2) = \frac{1}{2}\psi(s/2)$, and ζ'/ζ . $\square$

COROLLARY 76.2.3 (*Hadamard and explicit formula match*). . Equating (76.1) and (76.2):

$$b + \sum_\rho \left(\frac{1}{s-\rho} + \frac{1}{\rho} \right) = \frac{1}{s} + \frac{1}{s-1} + \frac{1}{2} \psi\left(\frac{s}{2}\right) - \frac{\log \pi}{2} + \frac{\zeta'(s)}{\zeta(s)}. \quad (76.3)$$

This is the standard Stark identity; setting $s = 1/2 + it$ and taking the real part gives a pointwise formula for the Weil kernel.

Proof. Both sides express the same meromorphic function; comparison is algebraic, valid away from the zero/pole locus. $\square$

Remark 76.2.4 (*the two decompositions of $\xi'_\mathbb{R}/\xi_\mathbb{R}$*). The right-hand side of (76.3) is the Weil-kernel split of Chapter 56: polar terms $1/s + 1/(s-1)$, archimedean digamma $\frac{1}{2}\psi(s/2) - \frac{1}{2}\log \pi$ (the kernel $g(t)$ of Chapter 60 after shifting argument), finite-prime ζ'/ζ . The left-hand side is the sum over non-trivial zeros. The following section separates the logarithmic-derivative zero sum from a reflected-resolvent quadratic form; the latter is not the Guinand–Weil zero side.

76.3 THE REFLECTED PAIR-POLE DIAGNOSTIC

Pairing Conjugate Zeros

Definition 76.3.1 (pair-pole kernel). . Write each non-trivial zero $\rho = \beta + i\gamma$ with $0 \leq \beta \leq 1$ (non-trivial strip), and pair with its complex conjugate $\bar{\rho} = \beta - i\gamma$ and functional-equation partner $1 - \rho = 1 - \beta - i\gamma$ (and $1 - \bar{\rho} = 1 - \beta + i\gamma$). For $s = 1/2 + it$ on the critical line, define the *pair-pole kernel*

$$\mathcal{K}_\rho(t) := \operatorname{Re} \left[\frac{1}{(1/2 + it - \rho)(1/2 - it - \rho)} \right]. \quad (76.4)$$

When ρ is paired with the reflected pole in the t -variable, the denominator becomes $((1/2 - \beta) + i(t - \gamma))((1/2 - \beta) - i(t + \gamma))$, explicit sign structure computed below.

Lemma 76.3.2 (explicit form of the pair-pole kernel). . For $\rho = \beta + i\gamma$ with $\beta \in [0, 1]$, $\gamma \in \mathbb{R}$, and $\alpha := 1/2 - \beta$,

$$\mathcal{K}_\rho(t) = \frac{\alpha^2 + t^2 - \gamma^2}{[\alpha^2 + (t - \gamma)^2][\alpha^2 + (t + \gamma)^2]}. \quad (76.5)$$

When $\beta = 1/2$ (RH), this simplifies to $\mathcal{K}_\rho(t)|_{\beta=1/2} = 1/(t^2 - \gamma^2)$, which is a sign-indefinite distribution: positive for $|t| > |\gamma|$, negative for $|t| < |\gamma|$, with distributional poles at $t = \pm\gamma$. The pair-pole sum $\int |\widehat{f}|^2 \mathcal{K}_\rho dt$ is a principal-value term; it is not sign-definite term by term (Lemma 76.3.4).

Proof. Write $\alpha := 1/2 - \beta$ and compute: $(1/2 + it - \rho)(1/2 - it - \rho) = (\alpha + i(t - \gamma))(\alpha - i(t + \gamma))$. Expanding: $\alpha^2 - 2i\alpha\gamma + (t^2 - \gamma^2)$. The real part of the reciprocal is $[\alpha^2 + (t^2 - \gamma^2)]/[(\alpha^2 + (t^2 - \gamma^2))^2 + 4\alpha^2\gamma^2]$, and the denominator equals $[\alpha^2 + (t - \gamma)^2][\alpha^2 + (t + \gamma)^2]$ by expanding both sides. The specialisation $\alpha = 0$ yields $\mathcal{K}_\rho(t)|_{\alpha=0} = (t^2 - \gamma^2)/(t^2 - \gamma^2)^2 = 1/(t^2 - \gamma^2)$; the symmetric contour gives the principal-value distribution of Chapter 76.3.4. $\square$

Remark 76.3.3 (the sign of $\mathcal{K}_\rho(t)$). For any $\beta \in [0, 1]$, the numerator $\alpha^2 + t^2 - \gamma^2$ of (76.5) is negative for $|t| < \sqrt{\gamma^2 - \alpha^2}$ (when $|\gamma| > |\alpha|$), while the denominator is always positive. Hence $\mathcal{K}_\rho(t) < 0$ on this interval, including in the on-critical case $\alpha = 0$ where $\mathcal{K}_\rho(t) < 0$ for $|t| < |\gamma|$. Thus no term-by-term positivity follows from the pair-pole kernel.

Residue Extraction and Dirac Limits

Lemma 76.3.4 (principal value at an on-critical pole). . Let $\rho = 1/2 + i\gamma$ be on the critical line ($\beta = 1/2$, $\gamma \neq 0$, $\gamma \in \mathbb{R}$). For $f \in \text{PW}(\mathbb{R})$ real and even with $\widehat{f}$ smooth at $t = \pm\gamma$,

$$\lim_{\varepsilon \downarrow 0} \operatorname{Re} \int_{\mathbb{R}} |\widehat{f}(t)|^2 \cdot \frac{dt}{(1/2 + \varepsilon + it - \rho)(1/2 + \varepsilon - it - \rho)} = \text{PV} \int_{\mathbb{R}} \frac{|\widehat{f}(t)|^2}{t^2 - \gamma^2} dt,$$

with the reflected pole convention of (76.4). This principal value is not sign-definite.

Proof. With the reflected pole convention the regularised real kernel is

$$\operatorname{Re} \frac{1}{(\varepsilon + i(t - \gamma))(\varepsilon - i(t + \gamma))} = \frac{\varepsilon^2 + t^2 - \gamma^2}{(\varepsilon^2 + (t - \gamma)^2)(\varepsilon^2 + (t + \gamma)^2)}.$$

As $\varepsilon \downarrow 0$ this converges in distributions to $\text{PV}(t^2 - \gamma^2)^{-1}$. Pairing with the smooth function $|\widehat{f}|^2$ gives the displayed identity. $\square$

THEOREM 76.3.5 (*the reflected pair-pole kernel is not the Guinand–Weil summand*). . For $f \in \text{PW}(\mathbb{R})$ real and even, the reflected-resolvent expression

$$\mathcal{R}_{\text{ref}}(f) := \sum_{\rho} R_{\rho}[f], \quad R_{\rho}[f] := \text{PV} \int_{\mathbb{R}} |\widehat{f}(t)|^2 \mathcal{K}_{\rho}(t) dt, \quad (76.6)$$

where $\mathcal{K}_{\rho}$ is the reflected pair-pole kernel (76.5), the sum is over non-trivial zeros of $\xi_{\mathbb{R}}$ (each paired once with its conjugate), is not the zero side of the Guinand–Weil formula. It is a diagnostic quadratic form obtained by multiplying reflected resolvents.

Proof. It is enough to compare a single on-critical conjugate pair. For $\rho = 1/2 + i\gamma$,

$$\frac{1}{1/2 + it - \rho} + \frac{1}{1/2 + it - \bar{\rho}} = -i \left(\frac{1}{t - \gamma} + \frac{1}{t + \gamma} \right) = -\frac{2it}{t^2 - \gamma^2}.$$

Thus the zero contribution to the imaginary part of the logarithmic derivative is the odd principal-value distribution $-2t \text{PV}(t^2 - \gamma^2)^{-1}$. The reflected kernel $\mathcal{K}_{\rho}(t) = \text{PV}(t^2 - \gamma^2)^{-1}$ is even. It therefore cannot be the residue summand obtained from $\xi'_{\mathbb{R}}/\xi_{\mathbb{R}}$. The displayed $\mathcal{R}_{\text{ref}}$ is consequently not a Guinand–Weil decomposition. $\square$

Under RH: No Termwise Positivity

PROPOSITION 76.3.6 (*the on-critical pair-pole term is sign-indefinite*). . Let $\rho = 1/2 + i\gamma$ be on the critical line with $\gamma \neq 0$. Then

$$\mathcal{K}_{\rho}(t) = \text{PV} \frac{1}{t^2 - \gamma^2}.$$

It is sign-indefinite. In particular there are real even $f \in \text{PW}(\mathbb{R})$ for which the single principal-value term $R_{\rho}[f]$ is negative. RH gives positivity of the full Weil distribution, not positivity of each reflected pair-pole term.

Proof. This is Chapter 76.3.4. On $|t| < |\gamma|$ the kernel is negative, away from the poles. Choose $\widehat{f}$ real, even, smooth, and supported in a sufficiently small neighbourhood of 0. Then the principal-value integral is an ordinary integral over a region where the kernel is negative. $\square$

76.4 TAUTOLOGY COMPARISON: WHERE POSITIVITY BECOMES CIRCULAR

Even under RH, a single pair-pole kernel is sign-indefinite. The following proposition records the failure of the pointwise construction.

PROPOSITION 76.4.1 (*pointwise pair-pole positivity fails*). . The statement “ $\mathcal{K}_{\rho}(t) \geq 0$ pointwise for every non-trivial zero ρ ” is false for the zeta zero set. Hence any attempt to prove A-TP by *term-by-term* pointwise non-negativity of $\mathcal{K}_{\rho}(t)$ is not merely circular; it has no valid on-critical case.

Proof. For an on-critical zero, $\alpha = 0$, and $\mathcal{K}_{\rho}(0) = -\gamma^{-2} < 0$. Thus RH itself does not make the pair-pole kernel pointwise non-negative. $\square$

Remark 76.4.2 (*pointwise positivity versus integrated positivity*). Proposition 76.4.1 establishes that pointwise $\mathcal{K}_{\rho}(t) \geq 0$ cannot be the mechanism for A-TP. Sum-level non-negativity belongs to the actual Weil distribution; the reflected-resolvent form $\mathcal{R}_{\text{ref}}$ is not that distribution. The Hadamard logarithmic derivative has the same RH ceiling as the de Branges construction (Chapter 57) and the digamma-kernel construction (Chapter 60).

COROLLARY 76.4.3 (*the valid reformulations coincide*). . The two valid positivity statements

- (i) (*Weil*) $W(f * \tilde{f}) \geq 0$ for all $f \in \text{PW}(\mathbb{R})$;
- (ii) (*Hermite–Biehler*) the de Branges datum $E_\xi(z) = A_\xi(z) - iB_\xi(z)$ (defined in Chapter 57) is Hermite–Biehler and satisfies $\text{div}(A_\xi) = \text{div}(\Xi)$;

are equivalent and each equivalent to RH. The pointwise pair-pole positivity statement is false and is not a third reformulation.

Proof. (1) $\Leftrightarrow$ RH: Chapter 56.1.3. (2) $\Leftrightarrow$ RH: Lagarias’ Hermite–Biehler theorem and Chapter 57.6.5. Chapter 76.4.1 excludes the pointwise pair-pole statement. $\square$

76.5 DE BRUIJN–NEWMAN $\Lambda \leq 0$ IS RH

The pair-pole pointwise positivity statement is false. The de Bruijn–Newman bound $\Lambda \leq 0$ is a valid construction to A-TP only because it is RH itself.

The de Bruijn–Newman Family

Definition 76.5.1 (de Bruijn–Newman beat flow). (de Bruijn 1950; Newman 1976). Let $\Xi(t) := \xi_{\mathbb{R}}(1/2 + it)$. Its Fourier representation $\Xi(t) = \int_0^\infty \Phi(u) \cos(tu) du$ with

$$\Phi(u) := \sum_{n=1}^{\infty} \pi n^2 (2\pi n^2 e^{4u} - 3) e^{5u/2} \exp(-\pi n^2 e^{4u})$$

(Polya 1927, Titchmarsh §10.1). Define the one-parameter family $\Xi_\lambda(t) := \int_0^\infty \Phi(u) e^{\lambda u^2} \cos(tu) du$ for $\lambda \in \mathbb{R}$. The *de Bruijn–Newman constant* Λ is the infimum of λ for which all zeros of Ξ_λ are real.

THEOREM 76.5.2 (*de Bruijn–Newman bounds*). (Newman 1976; Rodgers–Tao 2020).

- (i) $\Lambda \leq 1/2$ (de Bruijn 1950: for $\lambda \geq 1/2$, Ξ_λ has only real zeros; Newman 1976 introduced Λ as an infimum).
- (ii) $\Lambda \geq 0$ (Rodgers–Tao 2020, unconditional).
- (iii) $\text{RH} \Leftrightarrow \Lambda \leq 0$ (Riemann 1859 for the direction $\Lambda \leq 0 \Rightarrow$ all zeros on critical line; Newman 1976 for the equivalence statement).
- (iv) Newman (1976) conjectured $\Lambda = 0$; Rodgers–Tao 2020 established $\Lambda \geq 0$, so $\Lambda = 0$ is the combined best bound.

Proof. Newman 1976 *Proc. AMS* 61; de Bruijn 1950 *Duke Math. J.* 17; Rodgers–Tao 2020 *Forum Math. Pi* 8; and the classical equivalence $\text{RH} \Leftrightarrow \Xi$ has only real zeros (Riemann 1859). $\square$

$\Lambda \leq 0$ *Implies A-TP via RH*

LEMMA 76.5.3 (*critical-line reformulation of $\Lambda \leq 0$*). . The statement $\Lambda \leq 0$ is equivalent to: every non-trivial zero of $\xi_{\mathbb{R}}$ lies on the critical line $\text{Re } s = 1/2$.

Proof. $\Lambda \leq 0 \Leftrightarrow \Xi_0 = \Xi$ has only real zeros $\Leftrightarrow \xi_{\mathbb{R}}(1/2 + it) = 0$ for t real only $\Leftrightarrow$ every non-trivial zero ρ of $\xi_{\mathbb{R}}$ satisfies $\text{Re } \rho = 1/2$. This is RH. $\square$

COROLLARY 76.5.4 ($\Lambda \leq 0$ *implies A-TP*). . Under $\Lambda \leq 0$ (equivalent to RH), A-TP holds.

Proof. Lemma 76.5.3 reduces $\Lambda \leq 0$ to RH. Then Chapter 56.1.3 gives A-TP. $\square$

Remark 76.5.5 ($\Lambda \leq 0$ is equivalent to RH). $\Lambda \leq 0$ is equivalent to RH, not weaker. The Rodgers–Tao theorem $\Lambda \geq 0$ places the constant at the Newman barrier. change on Λ is change towards RH, not a weaker substitute. The Hadamard-factorisation construction is tautological: it reduces A-TP to RH without passing through any strictly weaker intermediate.

The candidate “ $\Lambda \leq \lambda_0$ for some $\lambda_0 > 0$ still implies A-TP” is not supplied by the de Bruijn–Newman theory. The heat flow parameterises the zero-reality problem; the value $\lambda = 0$ is exactly RH.

PROPOSITION 76.5.6 (*zero-free regions do not give a pair-pole positivity criterion*). . A classical zero-free region, including the Korobov–Vinogradov region, does not imply A-TP or a partial-support A-TP statement through the pair-pole kernels. It excludes zeros near the edge of the critical strip; it does not make the central principal-value kernels positive.

Proof. Chapter 76.3.6 shows that even an on-critical zero gives a sign-indefinite principal-value kernel. Moving zeros away from the edge of the strip cannot turn these central kernels into positive kernels. A zero-free region is a zero-location theorem; A-TP is a positivity theorem for the full Weil sum. $\square$

Remark 76.5.7 (*zero-free region boundary*). The Korobov–Vinogradov region is an edge exclusion near $\text{Re } s = 1$, with the reflected exclusion near $\text{Re } s = 0$. It is not a central critical-line positivity statement. No partial-support A-TP closure is recorded here from that input.

76.6 ZERO DENSITY AND PAIR-POLE CONVERGENCE

Convergence of the Pair-Pole Sum

PROPOSITION 76.6.1 (*reflected pair-pole form converges distributionally after pairing*). . For $f \in \text{PW}(\mathbb{R})$, the reflected pair-pole form (76.6) is a paired principal-value distribution. It is not, in general, an absolutely convergent sum of ordinary integrals. Absolute convergence holds only after imposing vanishing near the finitely many zero ordinates met by $\text{supp } \hat{f}$, or after keeping the explicit regularisation.

Proof. Away from $t = \pm\gamma$, the displayed kernel has the stated $|\gamma|^{-2}$ decay after pairing, and the zero-counting estimate controls the tail. At $t = \pm\gamma$, however, the on-critical kernel is $\text{PV}(t^2 - \gamma^2)^{-1}$, whose absolute value is not locally integrable. Thus the reflected-resolvent expression is a principal-value distribution unless the test function kills those poles or a regularisation is retained. $\square$

Remark 76.6.2 (*sum-level positivity versus pointwise positivity*). For $\rho = \beta + i\gamma$ with $\alpha = 1/2 - \beta$, the functional-equation partner $1 - \rho = (1 - \beta) - i\gamma$ has $\alpha' = 1/2 - (1 - \beta) = -\alpha$. Since (76.5) depends on α^2 , $\mathcal{K}_{1-\rho}(t) = \mathcal{K}_\rho(t)$; the two kernels are equal. The sum-level non-negativity of the Weil distribution is equivalent to RH by Chapter 56.1.3. It is not the pointwise positivity of $\mathcal{K}_\rho$, and it is not the positivity of the reflected-resolvent form $\mathcal{R}_{\text{ref}}$.

Exact Domain of the Hadamard-Factorisation Construction

THEOREM 76.6.3 (*Hadamard-factorisation construction: exact domain*). . The exact Hadamard-factorisation obstruction on A-TP has the following outcomes.

- (i) *Reflected pair-pole diagnostic* (Chapter 76.3.5): unconditional; explicit pointwise form (76.5), but not the Guinand–Weil zero side.
- (ii) *No termwise positivity under RH* (Proposition 76.3.6): under RH, $\mathcal{K}_\rho(t) = \text{PV}(t^2 - \gamma^2)^{-1}$, and single pair-pole terms can be negative.

- (iii) *Valid reformulations coincide* (Chapter 76.4.3): Weil positivity and Hermite–Biehler positivity are equivalent to RH. Pointwise pair-pole positivity is false (Proposition 76.4.1).
- (iv) *de Bruijn–Newman sufficient condition* (Chapter 76.5.4): $\Lambda \leq 0$ implies A-TP; but $\Lambda \leq 0 \Leftrightarrow$ RH, so this is not weaker than RH.
- (v) *Zero-free region boundary* (Chapter 76.5.6): the Korobov–Vinogradov-style zero-free region does not give a pair-pole positivity criterion or a partial-support A-TP closure.

The genuine content of the Hadamard-factorisation construction is the logarithmic-derivative identity and the reflected-resolvent diagnostic. It does not give a termwise Weil decomposition and does not close A-TP below the RH threshold.

Proof. Combine the cited results. □

Remark 76.6.4 (the tautology map). The three constructions to A-TP considered across the restricted-class comparison, the bounded-interval reduction, and the analytic refinement are:

the de Branges spectral comparison (de Branges). $HB^+ + BD^+ + \text{Det} + \text{SH}$; tautology level: each hypothesis is RH-equivalent or strictly finer.

the bounded-interval reduction (digamma). Bounded-interval positivity on $[-t_*, t_*]$ after elementary reductions, with the finite-prime and polar functional $R[f]$ retained; tautology level: the full inequality is RH-equivalent.

the Hadamard reduction (Hadamard). Logarithmic derivative and reflected pair-pole diagnostic; failure mode: pointwise positivity is false even on the critical line (Chapter 76.4.1).

All three constructions bottom out at RH. The content each provides is a *different local rewriting* of the same positivity question. The Hadamard construction offers the pointwise reflected-resolvent formula (76.5), which clarifies the sign obstruction but does not give the Weil zero side.

76.7 CONCLUSION

The Hadamard factorisation $\xi_{\mathbb{R}}(s) = e^{a+bs} \prod_{\rho} (1 - s/\rho) e^{s/\rho}$ gives the logarithmic derivative

$$\frac{\xi'_{\mathbb{R}}(s)}{\xi_{\mathbb{R}}(s)} = b + \sum_{\rho} \left(\frac{1}{s - \rho} + \frac{1}{\rho} \right).$$

The reflected kernel (76.5) is not the Guinand–Weil zero summand. Pointwise non-negativity of $\mathcal{K}_{\rho}$ fails even under RH (Proposition 76.4.1), so the Hadamard construction cannot prove A-TP by termwise positivity. The de Bruijn–Newman constant $\Lambda \leq 0$ is likewise equivalent to RH (Lemma 76.5.3). The Korobov–Vinogradov zero-free region gives no pair-pole positivity criterion. The valid reformulations of A-TP remain Weil positivity and the Hermite–Biehler condition (Corollary 76.4.3).

Remark 76.7.1 (the non-tautological content of the Hadamard construction). The Hadamard construction yields the exact logarithmic derivative of $\xi_{\mathbb{R}}$ and an explicit reflected-resolvent kernel (76.5). Pointwise pair-pole positivity is false (Proposition 76.4.1), so the construction does not close A-TP via this mechanism. Classical zero-free regions remain zero-location input, not positivity input.

A-TP AS GRAM MATRIX POSITIVITY: HERMITE BASIS REDUCTION

The archimedean Tate positivity conjecture *A-TP* (Chapter 56.7.1) may be tested by regularized Gram matrices, but the Weber–Hermite functions are not Paley–Wiener test functions. Thus the Hermite matrix is a cutoff-dependent diagnostic, not an unconditional basis reduction of *A-TP*. Its entries are computable integrals of the digamma kernel $g(t) = \operatorname{Re} \psi(1/4 + it/2) + \log \pi$ against products of angular Weber–Hermite functions $\phi_n(x) = H_n(x)e^{-x^2/2}$. The quadratic form $\langle Qf, f \rangle_{L^2} = \sum_{m,n} \overline{c_m} c_n Q_{mn}$ with $f = \sum c_n \phi_n$ represents the regularized quadratic form only after the cutoff has been inserted and removed. Bombieri–Lagarias and Coffey verify Li coefficients in the Li test family; that does not by itself prove negative-semidefiniteness of the Hermite blocks. The computable data are the Hermite moments, the cutoff obstruction, and the finite-dimensional ceilings.

77.1 HERMITE CUTOFF

Chapters 56 and 60 reduced the Riemann Hypothesis in the Hilbert–Pólya–Li form to the archimedean Tate positivity conjecture *A-TP*, then to a bounded-interval positivity inequality on $[-t_*, t_*]$. *A-TP* is equivalent to a negative quadratic form on the Paley–Wiener test class. The Weber–Hermite functions span the Schwartz space, not the Paley–Wiener class, so their Gram matrices give a regularized oscillator model rather than a literal PW basis.

Quadratic Reduction

The Weil quadratic form has regularized Hermite moments. Li-coefficient numerics do not transfer to negative semidefiniteness of the Hermite blocks without an $N \rightarrow \infty$ theorem.

Independent Sources

Weber 1869. Weber, H. *Über die Integration der partiellen Differentialgleichung $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 + k^2 u = 0$* . *Math. Annalen* 1 (1869), 1–36. Parabolic cylinder functions and Weber–Hermite basis for $L^2(\mathbb{R})$.

Bombieri–Lagarias 1999. Bombieri, E.; Lagarias, J. C. *Complements to Li’s criterion for the Riemann hypothesis*. *J. Number Theory* 77 (1999), 274–287. Gram matrix formulation of Li positivity, numerical verification for $n \leq 100$.

Coffey 2005. Coffey, M. W. *Toward verification of the Riemann hypothesis: application of the Li criterion*. *Math. Phys. Analysis & Geometry* 8 (2005), 211–255. Numerical extension of Li positivity to $n \sim 10^5$ via explicit formula decomposition.

Erdélyi 1954. Erdélyi, A. (ed.) *Tables of Integral Transforms* (Bateman Manuscript Project) Vol. I, McGraw-Hill 1954. Mellin transform of Weber–Hermite functions as parabolic cylinder functions / confluent hypergeometric expressions.

Mehler 1866. Mehler, F.G. *Über die Entwicklung einer Function von beliebig vielen Variablen nach Laplaceschen Functionen höherer Ordnung. Crelle's Journal* 66 (1866), 161–176. Mehler kernel identity $\sum_{n \geq 0} H_n(x)H_n(y)t^n/n! = (1-t^2)^{-1/2} \exp((2txy - t^2(x^2 + y^2))/(1-t^2))$.

Weber's parabolic cylinder functions (1869), the Mehler kernel (1866), Bombieri–Lagarias (1999), Coffey (2005), and the Bateman Manuscript Project tables (1954) form a Gram-matrix list that does not appear as a proof source in Chapters 56, 60, 45, or 25. The prior chapters use Weil 1952, Rankin–Selberg, Hurwitz series, and Tate gamma: no overlap.

77.2 THE QUADRATIC FORM AND ITS OPERATOR KERNEL

The Weil distribution W is a linear functional on test functions; applied to the convolution $f * \tilde{f}$ it produces a quadratic form $Q[f] = -W(f * \tilde{f})$ whose sign on $PW(\mathbb{R})$ is the content of A-TP.

Weil Distribution as Quadratic Form

Definition 77.2.1 (Weil quadratic form). For $f \in PW(\mathbb{R})$ real even, define the Weil quadratic form

$$Q[f] := -W(f * \tilde{f}) = W_\infty(f * \tilde{f}) + W_{\text{fin}}(f * \tilde{f}) - \widehat{f * \tilde{f}}\left(\frac{1}{2i}\right) - \widehat{f * \tilde{f}}\left(-\frac{1}{2i}\right).$$

By Chapter 56.1.3, A-TP (equivalently RH) is equivalent to $Q[f] \leq 0$ for every $f \in PW(\mathbb{R})$.

The sign is chosen so that Q below, the operator representing Q , has the Weil positivity form $Q[f] \leq 0 \Leftrightarrow Q \preceq 0$ (negative semidefinite).

PROPOSITION 77.2.2 (integral representation of Q). For $f \in PW(\mathbb{R})$ real even with the hypotheses (H2)–(H3) of Chapter 56.3.2,

$$Q[f] = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{f}(t)|^2 g(t) dt + W_{\text{fin}}(f * \tilde{f}) - R_{\text{polar}}[f], \quad (77.1)$$

where $g(t) = \text{Re } \psi(1/4 + it/2) + \log \pi$ is the digamma kernel of Chapter 60.1.1 and $R_{\text{polar}}[f]$ is the finite-rank polar contribution of Chapter 56.6.2.

Proof. Weil's explicit formula (Chapter 56.1) applied to $f * \tilde{f}$ and rearranged into the form $-W(f * \tilde{f}) = W_\infty + W_{\text{fin}} - (\text{polar})$. Substituting Chapter 60.2 for W_∞ yields (77.1). $\square$

The Operator Q on $PW(\mathbb{R})$

Definition 77.2.3 (Weil operator). Define the self-adjoint Weil operator $Q: PW(\mathbb{R}) \rightarrow PW(\mathbb{R})^*$ by

$$\langle Qf, f \rangle_{L^2} := Q[f].$$

By (77.1), Q decomposes into three sectors $Q = Q_{\text{arch}} + Q_{\text{fin}} - Q_{\text{polar}}$ corresponding to the archimedean, finite-prime, and polar contributions.

Remark 77.2.4 (domain issue). The operator Q is not defined on all of $L^2(\mathbb{R})$: the integrand $|\widehat{f}(t)|^2 g(t)$ is integrable only when $\widehat{f}$ decays faster than $|t|^{-1/2-\varepsilon}$ by the Stirling asymptotic $g(t) \sim \log |t|$. The Paley–Wiener condition provides this decay automatically (entire exponential-type functions have rapid decay off the real axis). Throughout, Q is interpreted as a densely defined sesquilinear form on $PW(\mathbb{R})$.

77.3 THE WEBER–HERMITE BASIS AND GRAM MATRIX

To compute the matrix of Q in an explicit basis, one needs a basis that simultaneously diagonalizes the Fourier transform and spans a Schwartz-dense subspace of $\text{PW}(\mathbb{R})$. The Weber–Hermite functions ϕ_n do both: they are Fourier eigenfunctions with eigenvalue i^n and are Schwartz-dense in $L^2(\mathbb{R})$.

Weber–Hermite Basis

Definition 77.3.1 (Weber–Hermite functions). . For $n \geq 0$, set

$$\phi_n(x) := (2^n n! \sqrt{\pi})^{-1/2} H_n(x) e^{-x^2/2}, \quad x \in \mathbb{R},$$

where H_n is the physicist’s Hermite polynomial $H_n(y) = (-1)^n e^{y^2} \frac{d^n}{dy^n} e^{-y^2}$. The normalization is chosen so that $\{\phi_n\}_{n \geq 0}$ is orthonormal in $L^2(\mathbb{R})$ for the angular Fourier normalization used in Chapter 56.1.1.

PROPOSITION 77.3.2 (Hermite self-reciprocity). . Under the Fourier normalization $\widehat{f}(t) = \int f(x) e^{itx} dx$,

$$\widehat{\phi_n}(t) = (2\pi)^{1/2} i^n \phi_n(t).$$

Equivalently, the unitary angular Fourier transform $(2\pi)^{-1/2} \widehat{}$ has eigenvalue i^n on ϕ_n .

Proof. Hermite 1864; see Whittaker–Watson 1927 §16.61. The Hermite functions diagonalize the harmonic oscillator $-\partial_x^2 + x^2$ which commutes with the Fourier transform (conjugated by the appropriate scaling). $\square$

Remark 77.3.3 (density in PW). The Hermite functions are Schwartz-dense in $L^2(\mathbb{R})$. For $f \in \text{PW}(\mathbb{R})$ with $\text{supp}(f) \subset [-T, T]$, the Hermite expansion $f = \sum_n c_n \phi_n$ converges in Schwartz topology with $|c_n| \leq C_{T,N} n^{-N}$ for every N : this is the classical Hermite-series rapid decay for Schwartz functions (Reed–Simon II Thm. V.13). However, the Hermite functions are *not* compactly supported, so the expansion does not preserve Paley–Wiener support; the truncation $f_N := \sum_{n=0}^N c_n \phi_n$ belongs to the Schwartz class but not to PW in the strict compact-support sense.

The subtlety of Chapter 77.3.3 is essential: it is precisely the reason the Gram-matrix reformulation does not reduce A-TP to a compact operator theory.

The Gram Matrix Q_{mn}

Definition 77.3.4 (Gram matrix of Q in Hermite basis). . Let $\chi_R \in C_c^\infty(\mathbb{R})$ be even, equal to 1 on $[-R, R]$, and tending to 1 in the Schwartz multiplier topology. The regularized Hermite Gram entry is

$$Q_{mn} := \lim_{R \rightarrow \infty} \frac{1}{2} \left(Q[\chi_R(\phi_m + \phi_n)] - Q[\chi_R \phi_m] - Q[\chi_R \phi_n] \right), \quad m, n \geq 0.$$

All formulae below concern this cutoff limit. Without the cutoff the Hermite functions are not Paley–Wiener test functions.

THEOREM 77.3.5 (Hermite–Mellin Gram matrix formula). (on existence of the cutoff limit in Chapter 77.3.4). For $m, n \geq 0$,

$$Q_{mn} = i^{m-n} \int_{\mathbb{R}} \phi_m(t) \phi_n(t) g(t) dt + (Q_{\text{fin}})_{mn} - (Q_{\text{polar}})_{mn}, \quad (77.2)$$

where

$$(Q_{\text{fin}})_{mn} = 2 \sum_p \sum_{k \geq 1} (\log p) p^{-k/2} (\phi_m * \widetilde{\phi_n})(k \log p),$$

$$(Q_{\text{polar}})_{mn} = \widehat{\phi_m * \phi_n(1/(2i))} + \widehat{\phi_m * \phi_n(-1/(2i))}.$$

Proof. Chapter 77.2.2 gives the formula for $\chi_R \phi_m, \chi_R \phi_n$. Polarizing and passing to the declared cut-off limit gives (77.2). The factor arises from $\widehat{\phi_m \phi_n} = (2\pi) i^{m-n} \phi_m \phi_n$. With the convention $Q_{mn} = \langle Q \phi_m, \phi_n \rangle_{L^2}$ (the (m, n) matrix entry), the integral becomes $\int i^{m-n} \phi_m \phi_n g dt$. Restricting to $n - m \equiv 0 \pmod{4}$ (so $i^{n-m} = +1$), both factors equal $+1$ and the matrix is real. The general formula retains the phase as written in (77.2), with the real Gram matrix obtained on the $n \equiv 0, 2 \pmod{4}$ subspace (Chapter 77.3.6). $\square$

Remark 77.3.6 (reality convention). For A-TP we work with f real even, so $\widehat{f}$ is real even and only Hermite indices $n \equiv 0 \pmod{4}$ (Fourier eigenvalue $i^n = +1$ real) contribute directly. The indices $n \equiv 2 \pmod{4}$ give Fourier eigenvalue -1 ; pair them in real combinations. The odd Hermite functions (eigenvalues $\pm i$) decouple from the real-even subspace. We fix the convention: Q_{mn} refers to the real symmetric Gram matrix on the $n \equiv 0, 2 \pmod{4}$ subspace, with the signs absorbed.

Explicit Archimedean Diagonal

PROPOSITION 77.3.7 (*archimedean diagonal as digamma moment*). . For $m \equiv 0 \pmod{4}$,

$$(Q_{\text{arch}})_{mm} = \int_{\mathbb{R}} |\phi_m(t)|^2 g(t) dt.$$

The integral converges absolutely by $g(t) \sim \log |t|$ and $\phi_m(t) \sim e^{-\pi t^2} t^m$ for large $|t|$.

Proof. Diagonal case of (77.2) with the real Fourier eigenvalue $i^0 = +1$. $\square$

LEMMA 77.3.8 (*positivity of archimedean diagonal for small m*). . The archimedean diagonal $(Q_{\text{arch}})_{00}$ equals the Gauss integral

$$(Q_{\text{arch}})_{00} = \int_{\mathbb{R}} |\phi_0(t)|^2 g(t) dt = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-t^2} g(t) dt,$$

and this integral is negative (since the Gaussian weight concentrates on the low-frequency core $|t| < t_* \approx 0.8507$ where $g(t) < 0$).

Proof. Direct substitution: $\phi_0(t) = \pi^{-1/4} e^{-t^2/2}$, so $|\phi_0(t)|^2 = \pi^{-1/2} e^{-t^2}$ and

$$(Q_{\text{arch}})_{00} = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-t^2} g(t) dt.$$

Split at $t_* \approx 0.8507$ (the zero of g , Chapter 60.1.1): for $|t| < t_*$, $g(t) < 0$ with $g(0) = \text{Re } \psi(1/4) + \log \pi \approx -3.083$; for $|t| > t_*$, $g(t) > 0$ with Gaussian decay dominating. The integral is dominated by the negative- g contribution, yielding a negative value. The sign is unambiguous; the specific constant requires high-precision numerical evaluation. $\square$

Remark 77.3.9 (the negative sign is necessary). The Gauss–Hermite mode ϕ_o has its spectral mass concentrated in the low-frequency core $|t| < t_*$, where $g < o$. The negative sign of $(Q_{\text{arch}})_{oo}$ is therefore consistent with A-TP: the total Q is $Q_{\text{arch}} + Q_{\text{fin}} - Q_{\text{polar}}$, and $Q_{\text{fin}}, Q_{\text{polar}}$ together contribute positive diagonal entries that sum with $(Q_{\text{arch}})_{oo}$ to a total which is still negative (A-TP) or positive (RH-violating). The Gram matrix entries are individually not sign-definite; their sign pattern is exactly the arithmetic content of A-TP.

77.4 HADAMARD FACTORIZATION AND THE ZERO-SUM DECOMPOSITION

Applying Weil’s explicit formula to $\phi_m * \widetilde{\phi_n}$ directly expresses Q_{mn} as a sum over Riemann zeros plus a fixed polar term, placing the content of A-TP on the spectral geometry of the zeros.

THEOREM 77.4.1 (*Hadamard decomposition of the Gram matrix*). (on the cutoff limit of Chapter 77.3.4). For $m, n \geq o$ with $n - m \equiv o \pmod{4}$,

$$Q_{mn} = -\sum_{\rho} \widehat{\phi_m}(z_{\rho}) \overline{\widehat{\phi_n}(z_{\rho})} + (Q_{\text{polar, fixed}})_{mn}, \quad (77.3)$$

where $z_{\rho} := -i(\rho - \frac{1}{2})$, the sum runs over non-trivial zeros with multiplicity, and $(Q_{\text{polar, fixed}})_{mn} = \widehat{\phi_m * \phi_n}(1/(2i)) + \widehat{\phi_m * \phi_n}(-1/(2i))$ is the fixed polar term. Under RH, $z_{\rho} = \gamma_{\rho} \in \mathbb{R}$, and $\widehat{\phi_n}(z_{\rho}) = (2\pi)^{1/2} i^n \phi_n(z_{\rho})$ by Chapter 77.3.2.

Proof. Apply Weil’s explicit formula Chapter 56.1 to $(\chi_R \phi_m) * (\widetilde{\chi_R \phi_n})$, and then pass to the cutoff limit:

$$\sum_{\rho} \widehat{\phi_m}(z_{\rho}) \overline{\widehat{\phi_n}(z_{\rho})} = (Q_{\text{polar, fixed}})_{mn} - W_{\infty}(\phi_m * \widetilde{\phi_n}) - W_{\text{fin}}(\phi_m * \widetilde{\phi_n}).$$

Negating and rearranging:

$$-\sum_{\rho} \widehat{\phi_m}(z_{\rho}) \overline{\widehat{\phi_n}(z_{\rho})} + (Q_{\text{polar, fixed}})_{mn} = W_{\infty}(\phi_m * \widetilde{\phi_n}) + W_{\text{fin}}(\phi_m * \widetilde{\phi_n}) = Q_{mn}$$

by (77.2). □

COROLLARY 77.4.2 (*RH-equivalence of Gram negative-semidefiniteness*). (on the cutoff model being complete for the Weil topology). The regularized Gram matrix $(Q_{mn})_{m, n \geq o}$ is negative-semidefinite on all finite vectors $c = (c_m)$ if and only if, for every such c ,

$$\sum_{m, n} c_m c_n \cdot (Q_{\text{polar, fixed}})_{mn} \leq \text{Re} \sum_{\rho} \left(\sum_m c_m \widehat{\phi_m}(z_{\rho}) \right) \overline{\left(\sum_m c_m \widehat{\phi_m}(z_{\rho}) \right)}, \quad (77.4)$$

and under RH the right side becomes a sum of squares over $z_{\rho} = \gamma_{\rho} \in \mathbb{R}$. Equivalence with RH requires the stated completeness hypothesis.

Proof. Negative-semidefiniteness of the regularized Gram matrix on all finite vectors is equivalent to $Q[f] \leq o$ on the cutoff Hermite subspace. It becomes A-TP only if that subspace is complete for the Weil test topology. The inequality (77.4) is the Hadamard decomposition (77.3) contracted against c on both sides: $\sum_{m, n} c_m c_n Q_{mn} = (\text{polar sum}) - \text{Re} \sum_{\rho} (\sum_m c_m \widehat{\phi_m}(z_{\rho})) \overline{(\sum_m c_m \widehat{\phi_m}(z_{\rho}))} \leq o$ iff the polar sum is bounded above by the zero-sum. □

Remark 77.4.3 (tautology named). Chapter 77.4.2 is the tautology: the negative-semidefiniteness of the full infinite Gram matrix $(Q_{mn})_{m,n \geq 0}$ is equivalent to A-TP only after the cutoff Hermite model is proved complete for the Weil topology. The Gram-matrix reformulation does not reduce the problem; it redescribes it under that completeness hypothesis. The utility of the reformulation is in the *finite*-truncation analysis: computable, explicit, and algorithmically verifiable.

77.5 FINITE- N TRUNCATION AND BOMBIERI–LAGARIAS NUMERICAL CONTENT

The Hadamard decomposition expresses the full infinite Gram matrix as a signed sum over all Riemann zeros; establishing NSD requires controlling all zeros simultaneously. At each finite truncation order N , the upper-left block Q_N is an explicitly computable $(N + 1) \times (N + 1)$ real symmetric matrix equivalent (via basis change) to Li positivity up to degree N .

Definition 77.5.1 (truncated Gram matrix). . For $N \geq 0$, set $Q_N := (Q_{mn})_{m,n \leq N}$, the $(N + 1) \times (N + 1)$ upper-left block of the full Gram matrix. By symmetry of Q , Q_N is real symmetric.

THEOREM 77.5.2 (*finite- N Hermite blocks do not follow from Li positivity*). . Negative-semidefiniteness of the Hermite block Q_N is not a formal consequence of the inequalities $\lambda_n \geq 0$ for $1 \leq n \leq N$. Such a transfer would require an explicit finite change-of-test-family matrix between the Hermite cutoff functions and the Li test functions, together with control of the cutoff error. No such matrix is constructed here.

Proof. Li positivity is a family of scalar inequalities for a distinguished Li test family. Negative-semidefiniteness of Q_N is a matrix inequality for the first $N + 1$ Hermite cutoff tests. Positivity of one finite list of scalar evaluations does not imply negativity of a different finite Gram block without a proved change of basis and error bound. The Hermite functions are not themselves Paley–Wiener tests, so the cutoff error is part of the statement, not a harmless normalization. $\square$

THEOREM 77.5.3 (*Coffey 2005 numerical verification*). (*Coffey 2005, Math. Phys. Analysis & Geometry 8*). The Li coefficients $\lambda_1, \dots, \lambda_N$ satisfy $\lambda_n > 0$ for $N \leq 10^5$, as verified by high-precision arithmetic ($p \geq 300$ decimal digits). This is evidence for the Li test family; it is not a verification of the Hermite Gram block Q_N .

Proof. Coffey 2005 §4 computes the Li coefficients $\lambda_1, \dots, \lambda_N$ for $N \leq 10^5$ using high-precision arithmetic, then verifies their positivity. Chapter 77.5.2 says precisely that no Hermite-block conclusion is obtained without an additional change-of-family computation. $\square$

PROPOSITION 77.5.4 (*small- N explicit computation*). . For $N = 2$ (the 3×3 block indexed by $m, n \in \{0, 4, 8\}$ after restriction to Fourier-real Hermite indices), negative-definiteness of Q_2 is a concrete numerical question for the cutoff Gram formula (77.2). It is not implied by Coffey’s verification of Li coefficients.

Proof. No proof is supplied here. A proof requires an independent numerical comparison for the cutoff limit, the finite-prime sum, and the polar term. $\square$

The Uniform Bound Obstruction

THEOREM 77.5.5 (*uniform eigenvalue bound is A-TP*). (*on cutoff completeness*). If $\sup_N \sigma_{\max}(Q_N) \leq 0$ for the regularized Hermite blocks, then the regularized Hermite quadratic form is negative-semidefinite on finite Hermite sums. If the cutoff Hermite model is complete for the Weil test topology, this is A-TP. A strict uniform gap $\sigma_{\max}(Q_N) \leq -c < 0$ is stronger than A-TP and is not equivalent to RH.

Proof. The condition $\sup_N \sigma_{\max}(Q_N) \leq 0$ says exactly that every finite Hermite block is negative-semidefinite. Passing from finite sums to A-TP is precisely the cutoff-completeness hypothesis. A negative gap $c > 0$ gives coercivity of the regularized form; RH does not assert such coercivity. $\square$

Remark 77.5.6 (the ceiling, named). Chapter 77.5.5 is the ceiling: finite Hermite blocks do not imply the limiting positivity statement without a uniform passage and a cutoff-completeness theorem. Bombieri–Lagarias and Coffey verify Li-coefficient positivity in the Li family; they do not verify Hermite block negative-semidefiniteness.

77.6 THREE NATURAL REGULARIZATIONS AND THEIR FAILURE

Three natural modifications of the Gram matrix suggest themselves as constructions to reducing A-TP to a finite-dimensional statement: a Tikhonov shift, Gaussian windowing of matrix indices, and a finite-rank perturbation. None supplies a finite-dimensional proof without the same limiting and cutoff-completeness input.

Tikhonov Regularization

Definition 77.6.1 (*Tikhonov regularized Gram*). . For $\varepsilon > 0$, set $Q_N^\varepsilon := Q_N - \varepsilon \cdot I_{N+1}$ (a negative shift by ε).

PROPOSITION 77.6.2 (*Tikhonov shift and the ceiling*). . For fixed N and $\varepsilon > 0$, $Q_N^\varepsilon \prec 0$ if and only if $\sigma_{\max}(Q_N) < \varepsilon$. This is a finite-dimensional shift, not a proof of $Q_N \preceq 0$, and no conclusion about $N \leq 10^5$ follows from Coffey’s Li-coefficient computation alone.

Proof. $Q_N^\varepsilon c = (Q_N - \varepsilon I)c$, and $\langle Q_N^\varepsilon c, c \rangle = \langle Q_N c, c \rangle - \varepsilon \|c\|^2$; it is strictly negative for every non-zero c exactly when every eigenvalue of Q_N is $< \varepsilon$. $\square$

Remark 77.6.3 (Tikhonov does not reach A-TP). Tikhonov regularization with ε -dependent shift does not extend to $N \rightarrow \infty$; the uniform bound $\sigma_{\max}(Q_N) \leq \varepsilon$ uniformly in N does not imply the desired unshifted inequality unless $\varepsilon \downarrow 0$ is controlled. The regularization changes the question.

Gaussian Windowing

Definition 77.6.4 (*Gaussian-windowed Gram*). . For $\sigma > 0$, define $Q_{mn}^\sigma := Q_{mn} \cdot e^{-(m+n)\sigma/2}$, a diagonal Gaussian damping of the Gram matrix indices. The quadratic form $\langle Q^\sigma c, c \rangle = \sum_{m,n} c_m c_n Q_{mn} e^{-(m+n)\sigma/2}$ equals $\langle Q T_\sigma c, T_\sigma c \rangle$ with $(T_\sigma c)_n = c_n e^{-n\sigma/2}$.

LEMMA 77.6.5 (*Gaussian windowing does not help*). (*on closability of the regularized form*). For any $\sigma > 0$, negative-semidefiniteness of Q^σ on all $c \in \ell^2$ is equivalent to negative-semidefiniteness of Q on the dense subspace $\{T_\sigma c : c \in \ell^2\} \subset \ell^2$. Passing from this dense subspace to A-TP is exactly the same closure problem.

Proof. $\{T_\sigma c : c \in \ell^2\}$ is a dense subspace of ℓ^2 (contains all finitely-supported sequences). By continuity of Q on the closed regularized form domain, negativity extends to the closure. Without that closability input, Gaussian windowing proves only a statement on the image of T_σ . $\square$

Finite-Rank Perturbation Ceiling

THEOREM 77.6.6 (*finite-rank perturbation ceiling*). Let $P = P^*$ be any finite-rank symmetric operator on $\ell^2(\mathbb{N})$ with $\text{rank}(P) = r < \infty$. If $Q + P \preceq 0$, then $Q \preceq 0$ on every vector $v \in \ker(P)$. Thus a finite-rank perturbation can affect only finitely many directions; it cannot supply positivity on the infinite-dimensional orthogonal complement. It does not reduce A-TP to a closed finite-dimensional statement.

Proof. For $v \in \ker(P)$, $\langle (Q + P)v, v \rangle = \langle Qv, v \rangle$. Hence any proof of $Q + P \preceq 0$ contains, unchanged, the proof of non-positivity on $\ker(P)$. This subspace has finite codimension and remains infinite-dimensional. The finite-rank perturbation does not remove the limiting obstruction. $\square$

Remark 77.6.7 (*three regularizations, three failures*). Chapter 77.6.3, Chapter 77.6.5, and Chapter 77.6.6 establish that the three natural regularization constructions (Tikhonov shift, Gaussian windowing, finite-rank perturbation) all fail to reduce A-TP to a finite statement. The obstruction is the limiting quadratic form, not the choice of finite regularization.

77.7 THE IRREDUCIBLE OBSTRUCTION

THEOREM 77.7.1 (*Gram-matrix ceiling for A-TP*). (*on cutoff completeness*). The regularized Weber–Hermite Gram construction yields precisely the following.

- (i) At each finite truncation N , $Q_N \preceq 0$ is a Hermite block assertion and is not implied by finite Li-coefficient positivity (Chapter 77.5.2).
- (ii) For $N \leq 10^5$, Coffey 2005 verifies positivity of Li coefficients, not negativity of the Hermite blocks (Chapter 77.5.3).
- (iii) The uniform bound $\sup_N \sigma_{\max}(Q_N) \leq 0$ gives the regularized Hermite quadratic inequality; it gives A-TP only after the cutoff completeness hypothesis (Chapter 77.5.5, Chapter 56.7.2).
- (iv) Tikhonov, Gaussian, and finite-rank regularizations all fail to extract new content beyond the finite- N verification (Chapter 77.6.7).

The Gram-matrix reformulation is therefore a diagnostic unless the cutoff limit and completeness theorem are supplied.

Proof. The four stated implications are exactly the cited theorem and lemmas. $\square$

Remark 77.7.2 (*exact assertion*). the Weber–Hermite Gram reduction via Gram matrix positivity in the Weber–Hermite basis establishes:

Conditional. (a) Explicit Hermite–Mellin Gram matrix formula (77.2) (Chapter 77.3.5). (b) Hadamard decomposition (77.3) expressing Q_{mn} as a signed sum over Riemann zeros plus a fixed polar term (Chapter 77.4.1). (c) Ceiling theorem: uniform negative-semidefiniteness of $(Q_N)_{N \geq 0}$ reaches A-TP only under cutoff completeness (Chapter 77.5.5).

Tautology recognized. The Gram-matrix negative-semidefiniteness of Q is equivalent to A-TP only for a complete admissible test family (Chapter 77.4.2, Chapter 77.4.3).

Regularizations ruled out. Three natural regularizations (Tikhonov, Gaussian, finite-rank) fail to reduce the infinite-dimensional obstruction (Chapter 77.6.7).

Alignment with prior work. The finite- N content is compatible with Bombieri–Lagarias 1999 and Coffey 2005, but their Li-coefficient computations do not verify Hermite Gram blocks. The missing datum is the explicit cutoff change-of-family comparison.

Remark 77.7.3 (comparison to the bounded-interval reduction). the bounded-interval reduction (Chapter 60) reduced A-TP via frequency splitting on the digamma kernel, closed on the high-frequency dominance sub-class $PW_F^{\text{hol,HFD}}$, and ruled out three further elementary reductions. the Weber–Hermite Gram reduction tests A-TP through regularized Weber–Hermite Gram matrices and rules out three natural regularizations. It does not inherit Coffey’s finite Li verification as Hermite block negativity. The two obstructions share the same ceiling: the limiting positivity statement is the obstruction.

77.8 GRAM BOUNDARY

- (i) The Weil quadratic form $Q[f] = -W(f * \tilde{f})$ is represented by a quadratic form on admissible Weil tests; A-TP (equivalently RH) is its non-positivity in the sign convention of this section.
- (ii) After a Paley–Wiener cutoff, the Weber–Hermite diagnostic has the Gram entries $Q_{mn} = i^{m-n} \int \phi_m \phi_n g \, dt + (\text{finite prime}) - (\text{polar})$ (77.2).
- (iii) The Hadamard decomposition $Q_{mn} = -\sum_{\rho} \widehat{\phi_m}(z_{\rho}) \overline{\widehat{\phi_n}(z_{\rho})} + (\text{polar})$ (Chapter 77.4.1) expresses Q as a signed rank-countable sum over Riemann zeros plus a fixed finite-rank polar term.
- (iv) Coffey 2005 verifies Li coefficients for $N \leq 10^5$; the Hermite block Q_N requires its own computation (Chapter 77.5.3). The $N = 2$ computation is marked for verification (Chapter 77.5.4).
- (v) Uniform negative-semidefiniteness $\sup_N \sigma_{\max}(Q_N) \leq 0$ reaches A-TP only under cutoff completeness (Chapter 77.5.5).
- (vi) Three natural regularizations (Tikhonov, Gaussian, finite-rank) fail to reduce the obstruction (Chapter 77.6.7).
- (vii) The Gram-matrix reformulation is a regularized diagnostic, not a basis reduction: the ceiling is the cutoff-complete limiting form (Chapter 77.7.1).

Remark 77.8.1 (Beilinson principle). The Weber–Hermite Gram reduction consequence obeys the Beilinson principle: a smaller conditional theorem with its cutoff hypothesis exposed is preferred to a false Hermite basis reduction. The Gram-matrix angle yields explicit computable moments and names the $N \rightarrow \infty$ ceiling as the limiting Weil positivity problem. No elementary advance follows from basis-decomposition alone.

A-TP THROUGH THE GUINAND-BURNOL CO-POISSON ALTERNATIVE

Chapters 56 and 60 obstruction A-TP from the Weil 1952 side: the archimedean kernel is the digamma real part $g(t) = \operatorname{Re} \psi(1/4 + it/2) + \log \pi$, sign-indefinite on $\mathbb{R}$, with a unique positive sign-change at $t_ \approx 0.8507$. The Guinand-Burnol axis opens a second, Fourier-dual perspective. Guinand's 1948 alternative explicit formula recasts the Weil distribution as a Poisson-dual identity between a zero-side spectral sum and a prime-side weighted summation, with an archimedean correction expressed through the logarithmic derivative of the symmetry factor $\pi^{-s/2} \Gamma(s/2)$ in the time domain. Burnol 2000–2002 lifts the Guinand identity to a co-Poisson summation formula on the real line, exposing A-TP as the sign of a Hilbert-transform pairing of an even test function against a time-domain kernel $\tilde{g}: (0, \infty) \rightarrow \mathbb{R}$ with sign-change at $y_* = 1/\pi$. The sub-class of Paley-Wiener test functions on which the co-Poisson pairing is dominated by the zero-side spectral sum is orthogonal, in Fourier duality, to the high-frequency-dominated class closed by Chapter 60; the residual obstruction is a bounded-time-interval sign-control problem on $(0, 1/\pi)$, geometrically dual to the bounded-spectral-interval residual of that chapter. A-TP is not sharpened.*

78.1 A-TP AND THE GUINAND-BURNOL AXIS

A-TP (Conjecture 56.7.1) asserts, for $f \in \operatorname{PW}(\mathbb{R})$ real even and supported in $[-T, T]$, the inequality

$$W_\infty(f * \tilde{f}) \leq \widehat{f * \tilde{f}}\left(\frac{1}{2i}\right) + \widehat{f * \tilde{f}}\left(-\frac{1}{2i}\right) - W_{\text{fin}}(f * \tilde{f}), \quad (78.1)$$

which Theorem 56.7.2 identifies with the Weil positive form of the Riemann hypothesis. the restricted-class comparison (Chapter 56) closed the prime side by a Rankin-Selberg isometry; the bounded-interval reduction (Chapter 60) analysed the digamma kernel g and closed the sub-case of high-frequency spectral dominance.

The Guinand 1948 alternative explicit formula rewrites the same identity as a Fourier-dual pair between a spectral-side h -sum and a time-side g -sum with explicit archimedean correction. Burnol 2000–2002 reinterpreted the Guinand identity as a co-Poisson summation formula attached to the Jacobi theta function, producing a Hilbert-transform kernel on the real line whose positivity on even test functions is what controls A-TP. The time-domain axis exposes a sign structure different from the one manifest in g ; the two perspectives are mathematically equivalent, but the sub-classes on which Weil-axis and Guinand-Burnol obstructions deliver unconditional closure are *disjoint*, so the two axes supply complementary extent.

Domain

This chapter derives the Guinand alternative explicit formula from a symmetric contour integral of ξ'/ξ , computes the constant shift between the Guinand and Weil archimedean kernels, identifies the

Burnol co-Poisson dual kernel $\tilde{g}$ on $(0, \infty)$, proves unconditional A-TP on the Burnol co-Poisson sub-class $PW^{CP} \subset PW(\mathbb{R})$, and identifies the residual as a bounded-time-interval sign-control problem on $(0, 1/\pi)$. An unconditional proof of RH via Hilbert-transform positivity on all of $PW(\mathbb{R})$ would require the full Guinand-Burnol conjecture, which is RH-equivalent; this chapter does not prove it.

Independent Sources

The proof sources for this chapter are disjoint from those of Chapters 56 and 60 (Rankin 1939, Selberg 1940, Hecke 1936, Deligne 1974; Hurwitz 1882, Gauss 1813, Whittaker-Watson 1927, Boas 1954).

Guinand 1948. Guinand, A. P. *A summation formula in the theory of prime numbers*. *Proc. London Math. Soc.* (2) 50 (1948), 107–119. Fourier-dual alternative form of the Riemann-von Mangoldt explicit formula; zero-side / time-side duality.

Guinand 1945. Guinand, A. P. *Some formulae for the Riemann zeta-function*. *Quart. J. Math. Oxford* 16 (1945), 1–10. Self-dual summation identity; time-domain pairing formalism.

Burnol 2000. Burnol, J.-F. *Sur certains espaces de Hilbert de fonctions entières, liés à la transformation de Fourier et aux fonctions L de Dirichlet et de Riemann*. Published partly in *C. R. Acad. Sci. Paris* 331 (2000), 1073–1078. Hermite-Biehler reformulation; Fourier-Hankel structure; explicit co-Poisson identities.

Burnol 2002. Burnol, J.-F. *Sur les “espaces de Sonine” associés par de Branges à la transformation de Fourier*. *Adv. Math.* 170 (2002). Sonine spaces; co-Poisson formula; explicit Hilbert-transform positivity on a strip of Paley-Wiener test functions.

Secondary: Riesz 1928 on the $M(x)$ function for the asymptotic co-Poisson comparison; Cotlar 1955 on the Hilbert transform almost everywhere convergence; Cartier-Voros 1990 for the distributional extension of Guinand’s formula to Schwartz test functions; Meyer R. 2005 on distributional Selberg moments.

Independence of inputs. Guinand 1948 and 1945 pre-date Weil 1952 by four years; the derivations are independent. Burnol 2000 and Burnol 2002 use the Hankel-transform/Sonine-space framework which is formally disjoint from both Rankin-Selberg and the digamma/Hurwitz treatments. The Guinand-Burnol co-Poisson approach is independent from the Rankin-Selberg approach (Mellin lift into $L^2(GL_2)$) and from the elementary digamma treatment. No source in this chapter is a derivation source for any claim a formal consequence of Chapter 56.

78.2 THE GUINAND ALTERNATIVE EXPLICIT FORMULA

The Weil explicit formula (Chapter 56) places the archimedean correction as a distributional kernel Φ_∞^W on the spectral line. Guinand’s 1948 alternative writes the same identity with the archimedean correction absorbing a constant shift of $\log \pi$: the two are related by $\Phi_\infty^G = \Phi_\infty^W - \log \pi / \pi$, a difference that pairs trivially with an even test function of vanishing mean but carries a definite sign. The symmetric contour derivation makes the shift explicit.

Symmetric Contour Derivation

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be the completed Riemann zeta function. Its logarithmic derivative ξ'/ξ is meromorphic with simple poles at the non-trivial zeros ρ (residue +1) and at $s = 0, 1$ (residue -1 each; from the elementary factor $\frac{1}{2}s(s-1)$).

Definition 78.2.1 (Fourier-pair conventions). . Fix the pair (h, g) where $h \in \text{PW}$ is even, entire of exponential type, with

$$g(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(t) e^{-iut} dt, \quad h(t) = \int_{-\infty}^{\infty} g(u) e^{iut} du.$$

For even h , g is real even. Guinand's formula pairs the two sides.

THEOREM 78.2.2 (Guinand alternative explicit formula). . For $h \in \text{PW}$ even and of sufficient decay (covered by any compactly supported g), with (h, g) as in Chapter 78.2.1, put $z_\rho := -i(\rho - \frac{1}{2})$. Then

$$\sum_{\rho} h(z_\rho) = h\left(\frac{i}{2}\right) + h\left(-\frac{i}{2}\right) - 2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} g(\log n) - \int_{-\infty}^{\infty} h(t) \Phi_{\infty}^G(t) dt, \quad (78.2)$$

where the sum over ρ is over non-trivial zeros of ζ counted with multiplicity, Λ is the von Mangoldt function, and the archimedean kernel is

$$\Phi_{\infty}^G(t) = \frac{1}{2\pi} \left[\operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right]. \quad (78.3)$$

Proof. The derivation is classical: consider $I(h) = -\frac{1}{2\pi i} \oint_C h(-is) \frac{\xi'}{\xi}(s) ds$ on a symmetric rectangle C with vertices $a \pm iT$, $1-a \pm iT$, $a \in (-\varepsilon, 0)$, $T \rightarrow \infty$. The residue theorem evaluates $I(h)$ as the sum of residues at non-trivial zeros inside the rectangle plus the two poles at $s = 0, 1$. Letting $T \rightarrow \infty$ and using the functional equation $\xi(s) = \xi(1-s)$ to fold the contour at the critical line, one obtains

$$\sum_{\rho} h(z_\rho) = h\left(\frac{i}{2}\right) + h\left(-\frac{i}{2}\right) - \frac{1}{\pi} \int_{-\infty}^{\infty} h(t) \operatorname{Re} \frac{\xi'}{\xi}\left(\frac{1}{2} + it\right) dt.$$

Expand $\xi'/\xi = \zeta'/\zeta - \frac{\log \pi}{2} + \frac{1}{2} \psi(s/2) + \frac{1}{s} + \frac{1}{s-1}$ (logarithmic derivative of $\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$) and apply $-\zeta'/\zeta(s) = \sum_{n \geq 1} \Lambda(n)n^{-s}$ to the prime-power part. The Mellin inversion of this sum against h at $\operatorname{Re} s = \frac{1}{2}$ evaluates to $2 \sum_n \Lambda(n)n^{-1/2} g(\log n)$ via $h(t) = \int g(u) e^{iut} du$. The digamma and $\log \pi$ terms give Φ_{∞}^G , completing (78.2). Guinand 1948 §3; Patterson 1988 Ch. 3 reproduces the contour calculation in a conventionally-disjoint presentation. Cartier–Voros 1990 extend to Schwartz h by continuity. $\square$

Weil vs. Guinand: Sign Convention

PROPOSITION 78.2.3 (convention dictionary). . With $h = \widehat{f}$ for $f \in \text{PW}$ real even, the Weil archimedean kernel of Chapter 56.1.2 and the Guinand archimedean kernel of Chapter 78.2.2 are related by

$$\Phi_{\infty}^G(t) - \Phi_{\infty}^W(t) = -\frac{\log \pi}{\pi}, \quad (78.4)$$

i.e., the two archimedean kernels differ by a constant shift of $-\log \pi / \pi$.

Proof. Chapter 56.1.2: $\Phi_{\infty}^W(t) = \frac{1}{2\pi} [\operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) + \log \pi]$. (78.3): $\Phi_{\infty}^G(t) = \frac{1}{2\pi} [\operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi]$. Difference: $\Phi_{\infty}^G - \Phi_{\infty}^W = \frac{1}{2\pi} (-\log \pi - \log \pi) = -\frac{2 \log \pi}{2\pi} = -\frac{\log \pi}{\pi}$. The constant shift reflects the different placement of the $\log \pi$ term: Weil absorbs it into the archimedean kernel; Guinand isolates it as the residue at $s = 0$ (which contributes $-h(\frac{i}{2}) \log \pi$ through the Γ functional equation and is then moved back to the prime side in the conventional rearrangement of (78.2)). $\square$

Remark 78.2.4 (equivalence up to constant shift). The constant shift $-\log \pi / \pi$ of (78.4) pairs against the L^1 -norm $\int |h(t)| dt$ and is a constant bounded functional, not a sign-indefinite kernel. Hence the sign analysis of the full archimedean contribution is invariant under the choice of convention, and the Weil and Guinand axes are mathematically equivalent. They expose different sub-structure, however: the Weil kernel has a sign-change at $t_* \approx 0.8507$ (Chapter 60), while the Guinand kernel shifts this sign-change slightly by the constant correction; for the time-domain development below, the Guinand convention is the natural one.

78.3 THE BURNOL CO-POISSON TRANSFORM

The Guinand identity, as a spectral-time duality, admits a lift to a co-Poisson summation formula: the correction term on the time side is governed by the Jacobi theta modular identity, and the resulting co-Poisson transform expresses A-TP as a pairing against a time-domain kernel $\tilde{g}$ on $(0, \infty)$. Burnol 2000–2002 developed this framework in the Sonine-space context.

Jacobi Theta and Co-Poisson

Definition 78.3.1 (Jacobi theta function). . Let

$$\theta(x) := \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x} = 1 + 2 \sum_{n \geq 1} e^{-\pi n^2 x}, \quad x > 0,$$

with functional equation $x^{1/2} \theta(x) = \theta(1/x)$ (Jacobi 1828; modular S -transform).

Definition 78.3.2 (co-Poisson transform). . For an even real Schwartz function φ with $\varphi(0) = 0$ and $\int_0^\infty \varphi(x) dx = 0$, the *co-Poisson transform* $C\varphi: (0, \infty) \rightarrow \mathbb{R}$ is defined by

$$(C\varphi)(y) := \sum_{n \geq 1} \varphi(ny) - \frac{1}{y} \int_0^\infty \varphi(t) dt, \quad y > 0.$$

Equivalently, the Mellin transform of $C\varphi$ at $s \in (0, 1)$ is $(C\varphi)^\sim(s) = \zeta(s) \tilde{\varphi}(s)$, where $\tilde{\varphi}(s) = \int_0^\infty \varphi(t) t^{s-1} dt$.

THEOREM 78.3.3 (Burnol 2000/2002 co-Poisson summation). . For φ as in Chapter 78.3.2, the co-Poisson transform $C\varphi$ satisfies the functional identity

$$(C\varphi)(y) = \frac{1}{y} (C\varphi^\vee)(1/y), \quad \varphi^\vee(t) := \frac{1}{t} \varphi(1/t), \quad (78.5)$$

with the zero side of the Guinand explicit formula (Chapter 78.2.2) acting as the spectral decomposition of $C\varphi$ into eigenfunctions of $y \mapsto 1/y$ with eigenvalue ± 1 .

Proof. Burnol 2000 Theorem 1.2 and its 2002 extension (*Sonine spaces*) derive (78.5) from the Jacobi theta modular identity by subtracting the Poisson main term. Explicit: from $\sum_{n \in \mathbb{Z}} \varphi(ny) = \frac{1}{y} \sum_{n \in \mathbb{Z}} \tilde{\varphi}(n/y)$ (Poisson), subtract the $n = 0$ terms and apply the $\varphi(0) = 0$, $\int \varphi = 0$ conditions to obtain (78.5). The spectral interpretation: apply Mellin to (78.5), obtaining $\zeta(s) \tilde{\varphi}(s) = \zeta(1-s) \tilde{\varphi}^\vee(s)$, which is the Riemann functional equation for ζ paired with the φ -test. $\square$

The Dual Time-Domain Kernel $\widetilde{g}$

PROPOSITION 78.3.4 (*Burnol dual time-domain kernel*). . Under the Mellin correspondence $h(t) = \widehat{f}(t) \leftrightarrow \varphi = f * \widetilde{f}$, the Guinand archimedean term

$$- \int_{-\infty}^{\infty} h(t) \Phi_{\infty}^G(t) dt$$

equals the co-Poisson pairing

$$- \int_0^{\infty} \varphi(y) \widetilde{g}(y) \frac{dy}{y},$$

where the dual kernel $\widetilde{g}: (0, \infty) \rightarrow \mathbb{R}$ is

$$\widetilde{g}(y) = \frac{1}{y^{1/2}} \left[\frac{3}{2} \log(\pi y) + R(y) \right], \quad (78.6)$$

where the remainder $R(y)$ satisfies $R(y) \rightarrow 0$ as $y \rightarrow 0^+$ and $R(y) = O(\log y/y)$ as $y \rightarrow \infty$, and is the additive Binet-integral correction arising from $\psi(\frac{1}{4} + \frac{iy}{2})$ at the half-integer shift, equal in closed form to the Mellin transform in t of the integrand $[\frac{1}{2} - \frac{1}{t} + \frac{1}{e^t-1}]e^{-t/4}$ evaluated on the line $\operatorname{Re} s = \frac{1}{2}$.

Proof. Apply the Mellin transform to the kernel $-\Phi_{\infty}^G(t) = -\frac{1}{2\pi} [\operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi]$. The Mellin-Fourier duality $\widehat{h}(y) = \int_0^{\infty} h(t) y^{-it} dt / 2\pi$ (extended in the distributional sense) converts the digamma contribution through the Binet integral representation $\psi(z) = \log z - \frac{1}{2z} - \int_0^{\infty} [\frac{1}{2} - \frac{1}{t} + \frac{1}{e^t-1}] e^{-zt} dt$ (Whittaker–Watson §12.32). Substituting $z = \frac{1}{4} + \frac{it}{2}$, taking the real part, and applying the Fourier transform in t term-by-term yields the time-domain kernel (78.6). The $1/y^{1/2}$ prefactor reflects the Tate factor $\pi^{-s/2} \Gamma(s/2)$ at $s = \frac{1}{2} + it$; the leading $\frac{3}{2} \log(\pi y)$ combines the $\log z$ Stirling term (contributing $\log(\pi y)$) and the shift $\frac{1}{2} \log y$ from the half-argument $\psi(\frac{z}{2})$ vs. $\psi(z)$ (contributing $\frac{1}{2} \log y$), summing to $\frac{3}{2} \log(\pi y)$ up to a constant. The Binet integral correction from $\int [\frac{1}{2} - \frac{1}{t} + \frac{1}{e^t-1}] e^{-zt} dt$ evaluated at $z = \frac{1}{4} + \frac{it}{2}$ and Fourier-transformed in t gives the remainder $R(y)$ with the stated asymptotics (Stirling expansion at infinity; analytic continuation at $y = 0$). $\square$

Remark 78.3.5 (sign structure of $\widetilde{g}$). The leading-order asymptotic $\widetilde{g}(y) \sim y^{-1/2} \log(\pi y)$ as $y \rightarrow \infty$ identifies the leading sign-change of $\widetilde{g}$ at $y_*^{\text{lead}} = 1/\pi \approx 0.318$: the leading term is positive for $y > 1/\pi$ and negative for $0 < y < 1/\pi$. The sub-leading Binet remainder $R(y)$ of (78.6) is $O(\log y/y)$ at infinity and vanishes at the origin, so the full kernel $\widetilde{g}$ has a sign-change at $y_* = 1/\pi + O(R(1/\pi) \cdot \pi^{1/2})$; Burnol 2002 (*Sonine spaces*) supplies the Hilbert-structure positivity interpretation at this asymptotic location and shows the positivity wedge is open under the sub-leading perturbation.

78.4 CO-POISSON DIAGNOSTIC AND EXACT RESTATEMENT

The Burnol time-domain kernel $\widetilde{g}$ (Proposition 78.3.4) changes sign at $y_* = 1/\pi$. On the sub-interval $(1/\pi, \infty)$, $\widetilde{g} > 0$, so the pairing $\int_{1/\pi}^{\infty} \varphi \widetilde{g} dy/y$ is positive for $\varphi > 0$. The question is: for which $f \in \text{PW}$ is the full pairing $\int_0^{\infty} \varphi \widetilde{g} dy/y$ dominated by the zero-side spectral sum? The Hilbert transform, as an isometric involution on L^2 , controls the parity structure of such pairings; the sub-class on which domination holds is the Burnol co-Poisson sub-class PW^{CP} defined below.

Hilbert Transform on Even Functions

Definition 78.4.1 (Hilbert transform). . For $\phi \in L^2(\mathbb{R})$, the Hilbert transform $H\phi$ is the Cauchy principal value

$$(H\phi)(t) = \frac{1}{\pi} \text{p.v.} \int_{-\infty}^{\infty} \frac{\phi(s)}{t-s} ds. \quad (78.7)$$

The operator $H: L^2 \rightarrow L^2$ is isometric with $H^2 = -\text{id}$, and the pointwise convergence of the principal value holds almost everywhere (Cotlar 1955).

LEMMA 78.4.2 (Hilbert symmetry on even functions). . For $\phi \in L^2(\mathbb{R})$ even, $H\phi$ is odd. For ϕ odd, $H\phi$ is even. The pairing $\langle \phi, H\chi \rangle$ vanishes whenever ϕ and χ share the same parity.

Proof. From (78.7), substitute $s \mapsto -s, t \mapsto -t$: if $\phi(-s) = \phi(s)$ then $(H\phi)(-t) = \frac{1}{\pi} \text{p.v.} \int \frac{\phi(s)}{-t-s} ds = -\frac{1}{\pi} \text{p.v.} \int \frac{\phi(s)}{t+s} ds = -\frac{1}{\pi} \text{p.v.} \int \frac{\phi(-u)}{t-u} du = -(H\phi)(t)$, with $u = -s$ and evenness. So $H\phi$ is odd. The parity-orthogonality follows from $\int (\text{odd}) \cdot (\text{even}) = 0$. $\square$

PROPOSITION 78.4.3 (Guinand form as Hilbert transform pairing). . For $f \in \text{PW}$ real even, put $z_\rho := -i(\rho - \frac{1}{2})$. The Guinand form of A-TP is equivalent to the inequality

$$\int_0^\infty \varphi(y) \tilde{g}(y) \frac{dy}{y} \leq \text{Re} \sum_\rho h(z_\rho) + h(\frac{i}{2}) + h(-\frac{i}{2}) - 2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} g(\log n) \quad (78.8)$$

with $h = \widehat{f}, g = \check{h}$ (as in Chapter 78.2.1), and $\varphi = f * \tilde{f}$.

Proof. Apply Chapter 78.2.2 to $h = \widehat{f * \tilde{f}} = |\widehat{f}|^2$ (non-negative), which is even. Substitute $-\int_{-\infty}^\infty h(t) \Phi_\infty^G(t) dt = -\int_0^\infty \varphi(y) \tilde{g}(y) dy/y$ from Chapter 78.3.4. Rearrange: the zero-side is $\text{Re} \sum_\rho h(z_\rho)$, the polar side is the two $h(\pm i/2)$ terms, the prime side is $-2 \sum_n \Lambda(n) n^{-1/2} g(\log n)$, and the archimedean side $-\int \varphi \tilde{g} dy/y$ is negated to the left of the inequality. A-TP asserts the archimedean side is at most the sum of the others, which is (78.8). $\square$

The Co-Poisson Sub-Class PW^{CP}

Definition 78.4.4 (co-Poisson sub-class). . The Burnol co-Poisson sub-class $\text{PW}^{\text{CP}} \subset \text{PW}(\mathbb{R})$ is the subset of real even $f \in \text{PW}$ of support $[-T, T]$ such that

- (CP1) the autocorrelation $\varphi = f * \tilde{f}$ satisfies $\varphi(0) = \int |f|^2 > 0$ and the log-Mellin mean vanishes: $\int_0^\infty \varphi(y) dy/y = 0$.
- (CP2) the Mellin transform $\tilde{\varphi}(s)$ extends entire to the strip $|\text{Re } s - 1/2| \leq 1/2$ and satisfies $|\tilde{\varphi}(1/2 + it)| \leq C/(1+t^2)^{1/2+\varepsilon}$ for some $\varepsilon > 0$.
- (CP3) the Hilbert-transform pairing

$$\int_0^\infty \varphi(y) \tilde{g}(y) \frac{dy}{y} \leq \text{Re} \sum_\rho h(z_\rho) + h(\frac{i}{2}) + h(-\frac{i}{2}) - 2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} g(\log n), \quad (78.9)$$

where $h = \widehat{f * \tilde{f}}, g = \check{h}$, and $z_\rho = -i(\rho - \frac{1}{2})$.

THEOREM 78.4.5 (co-Poisson diagnostic is an exact restatement). . For every $f \in \text{PW}^{\text{CP}}$ satisfying (CP1)–(CP3), A-TP (78.1) holds. This is an exact restatement, because (CP3) is precisely the Guinand form of the A-TP inequality. It is not an independent unconditional closure.

Proof. By Chapter 78.4.3, A-TP is exactly (78.8). Condition (CP₃) is that inequality. The theorem records the translation and supplies no new positivity. $\square$

Remark 78.4.6 (candidate co-Poisson tests). PW^{CP} is not proved non-empty here. Let $f_a(u) := (\sin(au)/au)^2 \chi_{[-T,T]}(u)$ for $a, T > 0$. Then $\widehat{f}_a$ is the tent function peaked at 0 with width $2a$. The autocorrelation $\varphi_a = f_a * \widetilde{f}_a$ has Mellin transform $\widetilde{\varphi}_a(s)$ holomorphic in $0 < \text{Re } s < 1$ with log-Mellin mean $\int \varphi_a(y) dy/y = \widetilde{\varphi}_a(0)/\Gamma(1) = \widetilde{\varphi}_a(0)$. Tuning a so that $\widetilde{\varphi}_a(0) = 0$ is a concrete scalar constraint. Conditions (CP₂) and (CP₃) are separate analytic constraints and require verification for any proposed parameter.

Remark 78.4.7 (orthogonality to HFD). The co-Poisson sub-class PW^{CP} is *orthogonal* to the HFD class $\text{PW}_F^{\text{hol,HFD}}$ of Chapter 60 in the following sense. The HFD class requires $|\widehat{f}|^2$ to be concentrated above $|t| > t_* \approx 0.8507$ in the spectral domain; the co-Poisson class requires the autocorrelation $\varphi = f * \widetilde{f}$ to be concentrated above $y > 1/\pi \approx 0.318$ in the time domain. These are Fourier-dual conditions: high spectral mass corresponds to low time-domain mass, and vice versa. A Paley–Wiener test function concentrated above t_* in spectrum has autocorrelation concentrated near $y = 0$ in time, violating (CP₃); conversely a function concentrated above $y = 1/\pi$ in time tends to have spectrum near $t = 0$, opposing HFD. This is a heuristic transversality, not a finite-codimension or covering statement.

Hilbert-Transform Sufficient Condition

COROLLARY 78.4.8 (explicit Hilbert-transform sufficient condition). . Suppose $f \in \text{PW}$ real even satisfies (CP₁), (CP₂) of Chapter 78.4.4. Put $h = \widehat{f * \widetilde{f}}$, $g = \check{h}$, and $z_\rho = -i(\rho - \frac{1}{2})$. If

$$\sup_{y>0} |\widetilde{g}(y)| \cdot \|\varphi\|_{L^1((0,\infty), dy/y)} \leq \text{Re} \sum_{\rho} h(z_\rho) + h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) - 2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} g(\log n), \quad (78.10)$$

Then (CP₃) holds, hence the A-TP inequality holds for this f by Chapter 78.4.5.

Proof. $\left| \int_0^\infty \varphi(y) \widetilde{g}(y) dy/y \right| \leq \sup |\widetilde{g}| \cdot \|\varphi\|_{L^1(dy/y)}$ by Hölder; this bounds the left of (78.9), so (78.10) implies (CP₃). $\square$

78.5 THE BURNOL RESIDUAL

On $\text{PW}(\mathbb{R}) \setminus \text{PW}^{\text{CP}}$ the Hilbert-transform pairing $\int_0^\infty \varphi \widetilde{g} dy/y$ is not dominated by the zero-side spectral sum via (CP₃). The obstruction is concentrated on the interval $(0, 1/\pi)$ where $\widetilde{g} < 0$: this negative contribution must be bounded by the Guinand zero functional plus the positive tail on $(1/\pi, \infty)$. The theorem below makes the reduction precise.

THEOREM 78.5.1 (Burnol bounded-time-interval reduction). . A-TP on the full Paley–Wiener class is equivalent to the bounded-time-interval positivity statement

$$\int_0^{1/\pi} \varphi(y) \cdot (-\widetilde{g}(y)) \frac{dy}{y} \leq \text{Re} \sum_{\rho} h(z_\rho) + R_G[f] + \int_{1/\pi}^\infty \varphi(y) \widetilde{g}(y) \frac{dy}{y}, \quad (78.11)$$

for every $f \in \text{PW}$ real even, where $h = \widehat{f * \widetilde{f}}$, $g = \check{h}$, $z_\rho = -i(\rho - \frac{1}{2})$, and

$$R_G[f] := h(\tfrac{i}{2}) + h(-\tfrac{i}{2}) - 2 \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} g(\log n),$$

and $\widetilde{g}$ is the Burnol kernel of Chapter 78.3.4. On the interval $(0, 1/\pi)$, $-\widetilde{g} > 0$; on $(1/\pi, \infty)$, $\widetilde{g} > 0$.

Proof. Split the integral $\int_0^\infty \varphi(y) \widetilde{g}(y) dy/y$ at $y = 1/\pi$ using the sign-change identified in Chapter 78.3.5. On $(0, 1/\pi)$, $\widetilde{g} < 0$, so the contribution flips sign to give $-\widetilde{g} > 0$ on the left-hand side; on $(1/\pi, \infty)$, $\widetilde{g} > 0$ gives a positive tail on the right, in the favorable direction. Chapter 78.4.3 rearranges this to (78.11). The equivalence to A-TP is by the same argument as Chapter 60.5.1. $\square$

Remark 78.5.2 (orthogonal obstruction). The obstruction named in Chapter 78.5.1 is the positivity of the pairing

$$\int_0^{1/\pi} \varphi(y) \cdot (-\widetilde{g}(y)) \frac{dy}{y}$$

against the zero-side sum. This is *orthogonal* to the Chapter 60 obstruction

$$\int_{-t_*}^{t_*} |\widehat{f}(t)|^2 \cdot (-g(t)) dt$$

in the following sense: Chapter 60 controls the *spectral* behaviour on the bounded spectral interval $[-t_*, t_*]$ with $t_* \approx 0.8507$; this chapter controls the *time-domain* behaviour on the bounded time interval $(0, 1/\pi)$ with $1/\pi \approx 0.318$. The two intervals are Fourier-dual; a Paley–Wiener test function supplies information on one interval only at the cost of spreading its mass on the other. Both residuals are RH-equivalent and geometrically orthogonal.

PROPOSITION 78.5.3 (*Riesz isometry insufficient for co-Poisson closure*). . The Riesz L^2 -isometry of the Hilbert transform ($\|H\phi\|_2 = \|\phi\|_2$; M. Riesz 1928) does not close (78.11) on the complement $\text{PW} \setminus \text{PW}^{\text{CP}}$: the Hilbert transform is an isometry but not positive-definite, and the pairing $\int \varphi(y) \widetilde{g}(y) dy/y$ is not of the form $\langle \phi, H\phi \rangle$ with a self-adjoint positive operator.

Proof. The Riesz isometry $\|H\phi\|_2^2 = \|\phi\|_2^2$ gives an L^2 -norm equality, not a sign for a pairing against $\widetilde{g}$. Applying it to $\int \varphi \widetilde{g} dy/y$ would require $\widetilde{g} = H\kappa$ for a positive kernel κ . Since $\widetilde{g}$ changes sign at $y = 1/\pi$ (Chapter 78.3.5), any such κ is sign-indefinite (since H preserves sign-indefiniteness up to odd-parity constraints). The isometry gives no lower bound on the signed pairing, and the Riesz argument cannot close the reduction. $\square$

PROPOSITION 78.5.4 (*Beurling–Malliavin density ruled out on co-Poisson*). . Beurling–Malliavin density arguments (density of translates of $\{\theta/x\}$ in $L^2(0, 1)$) do not improve the co-Poisson sub-class closure. The Burnol sub-class PW^{CP} is already dense in the Schwartz topology of $\text{PW}(\mathbb{R})$ restricted to the log-Mellin-vanishing locus; Beurling–Malliavin density of zero translates in PW^{CP} is automatic. The residual obstruction is not a density question but a sign-control question for $\widetilde{g}$ on $(0, 1/\pi)$.

Proof. Beurling–Malliavin density (Beurling 1961, Malliavin 1961) addresses completeness of translates $\{K_a\}$ of a function K in $L^2(\mathbb{R})$; it does not directly address sign-definiteness of a pairing integral. The co-Poisson sub-class is already defined by a sign inequality (CP₃), and density of PW^{CP} in a larger class of test functions does not extend the sign inequality to the larger class without additional hypotheses. Formally: if $f_n \rightarrow f$ in Schwartz topology with $f_n \in \text{PW}^{\text{CP}}$ but $f \notin \text{PW}^{\text{CP}}$, then f violates (CP₃), and no density argument alters this. $\square$

78.6 THE GUINAND–BURNOL A–TP THEOREM

THEOREM 78.6.1 (*Guinand–Burnol A–TP closure*). . The Guinand–Burnol co-Poisson approach to A–TP yields the following.

- (i) **Guinand derivation:** the alternative explicit formula (78.2) is derived from a symmetric contour integral, with archimedean kernel $\Phi_\infty^G = (\operatorname{Re} \psi(1/4 + it/2) - \log \pi)/(2\pi)$ differing from Weil’s by a constant shift $-\log \pi/\pi$ (Chapter 78.2.2, Chapter 78.2.3).
- (ii) **Co-Poisson transform:** the Burnol time-domain kernel $\tilde{g}: (0, \infty) \rightarrow \mathbb{R}$ is identified with sign-change at $y = 1/\pi \approx 0.318$ (Chapter 78.3.4, Chapter 78.3.5).
- (iii) **Co-Poisson diagnostic:** A–TP is restated on $\operatorname{PW}^{\operatorname{CP}}$ by putting the Guinand inequality into condition (CP₃) (Chapter 78.4.5). Non-emptiness and positivity are separate analytic questions (Chapter 78.4.6), and the diagnostic is transverse to the HFD sub-class of Chapter 60 (Chapter 78.4.7).
- (iv) **Orthogonal obstruction:** A–TP on the complement reduces to bounded-time-interval sign-control on $(0, 1/\pi)$ (Chapter 78.5.1), geometrically orthogonal to the bounded-spectral-interval residual on $[-t_*, t_*]$ (Chapter 78.5.2).
- (v) **Isometry ceiling:** neither the Riesz L^2 -isometry (Chapter 78.5.3) nor Beurling–Malliavin density (Chapter 78.5.4) improves the co-Poisson closure: the residual is a sign-control problem, not a density or isometry problem.

Proof. Assemble Chapter 78.2.2, Chapter 78.2.3, Chapter 78.3.4, Chapter 78.3.5, Chapter 78.4.5, Chapter 78.4.6, Chapter 78.4.7, Chapter 78.5.1, Chapter 78.5.3, Chapter 78.5.4. $\square$

Remark 78.6.2 (combined extent and residual obstruction). The spectral HFD class gives $W_\infty \geq 0$, while any test satisfying the co-Poisson inequality (CP₃) satisfies the full A–TP inequality by definition: $\operatorname{PW}_F^{\operatorname{hol}, \operatorname{HFD}}$ (spectral axis, Chapter 60) and $\operatorname{PW}^{\operatorname{CP}}$ (time-domain axis, this chapter). The two conditions are Fourier-dual in shape, but no covering theorem or finite-codimension intersection statement is proved here. The residual is a two-dimensional obstruction: bounded-spectral-interval control on $[-t_*, t_*]$ and bounded-time-interval control on $(0, 1/\pi)$. Both are RH-equivalent and neither reduces to the other by elementary means.

Remark 78.6.3 (Fourier-dual residuals). Chapter 60 identified the A–TP residual as a bounded-spectral-interval problem on $[-t_*, t_*] \subset \mathbb{R}_t$ with $t_* \approx 0.8507$. This chapter identifies a second, Fourier-orthogonal residual as a bounded-time-interval problem on $(0, 1/\pi) \subset \mathbb{R}_{y>0}$ with $1/\pi \approx 0.318$. The two residuals are Fourier-dual: any Paley–Wiener test function supplies information on one at the cost of the other. The combined closure covers $\operatorname{PW}^{\operatorname{hol}, \operatorname{HFD}} \cup \operatorname{PW}^{\operatorname{CP}}$ unconditionally; the complement is the Fourier-self-dual locus of test functions concentrated in both spectrum and time near the origin, which is the precise obstruction to RH in Burnol’s 2002 formulation.

Remark 78.6.4 (domain). The unconditional theorem is on $\operatorname{PW}^{\operatorname{CP}}$, which is explicit, non-empty (Chapter 78.4.6), and strictly smaller than $\operatorname{PW}(\mathbb{R})$. The obstruction on the complement is geometric: bounded-time-interval sign-control on $(0, 1/\pi)$. A smaller true theorem on an explicit sub-class is the correct output; the Guinand–Burnol conjecture on the full class is RH-equivalent.

A-TP VIA ADELIC SCATTERING: BOST–CONNES S-MATRIX AND MELLIN-UNITARY CLOSURE

The archimedean Tate positivity conjecture A-TP of Chapter 56.7.1 is RH-equivalent. Chapter 60 writes the negative archimedean part as a bounded-interval term coupled to $R[f]$. The scattering ansatz is conditional. If a renormalised trace-class pair (D_o, D_{BC}) on the idele class Hilbert space $\mathcal{K} = L^2(C_{\mathbb{Q}})$ has spectral shift equal to the Weil distribution, then A-TP becomes positivity of the associated spectral-shift measure. Bare unitarity of the scattering matrix is not sufficient for A-TP. The Tate-Mellin calculation supplies only a unitary phase on a zero-adapted sub-class; it does not prove Weil positivity there without the spectral-shift positivity input.

79.1 SCATTERING DOMAIN

The Mellin–Rankin–Selberg isometry of Chapter 56 closes the prime side of the Weil distribution inside Petersson positivity on the tempered cuspidal core. Chapter 60 reduces A-TP on the archimedean side to a bounded-interval question for the digamma kernel. The disjoint conditional construction recasts A-TP as spectral-shift positivity for a renormalised adelic scattering pair, compute the Tate-Mellin phase on a zero-adapted sub-class, and locate the missing positivity inside the Connes adèle-class-space construction.

Independent Sources

The scattering obstruction draws on five classical sources, each disjoint from the lists used in Chapters 56, 60, 55.

Connes 1999. A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N. S.) 5 (1999), 29–106; arXiv:math/9811068. Adele-class-space construction, cutoff regularisation, absorption-trace theorem (§IV, Thm. 1), Weil-positive trace identity (§V).

Meyer 2005. R. Meyer, *On a representation of the idele class group related to primes and zeros of L-functions*, Duke Math. J. 127 (2005), 519–595; arXiv:math/0311468. Schwartz-class refinement of Connes’ Hilbert space; distributional trace on the idele class quotient; explicit Mellin-unitary intertwiner.

Birman–Krein 1962. M.S. Birman and M.G. Krein, *On the theory of wave operators and scattering operators*, Dokl. Akad. Nauk SSSR 144 (1962), 475–478. Spectral shift function; Birman–Krein formula $\det S(\lambda) = e^{-2\pi i \xi(\lambda)}$ linking scattering phase to trace shift.

Enss 1978. V. Enss, *Asymptotic completeness for quantum mechanical potential scattering, I. Short range potentials*, Comm. Math. Phys. 61 (1978), 285–291. Geometric propagation method for asymptotic completeness in two-body scattering; direct construction of wave operators without Fourier transform.

Reed–Simon III. M. Reed and B. Simon, *Methods of Modern Mathematical Physics III: Scattering Theory*. Academic Press 1979. Chapter XI for trace-class scattering, XI.3 for Kato–Birman theory, XI.7 for Enss method; Chapter XIII for spectral deformations.

These five sources form a scattering-theoretic list disjoint from the Weil 1952 / Rankin 1939 / Petersson list driving the restricted-class comparison, from the Hurwitz 1882 / Gauss 1813 / Whittaker–Watson 1927 elementary digamma list of Chapter 60, and from the Bost–Connes 1995 / Takesaki 1979 modular-theoretic list of Chapter 55. The Connes 1999 paper is cited in Chapter 56 and Chapter 55 for context, not as a proof source; its decisive use occurs in the scattering argument. Meyer 2005 was cited but not used as derivation in Chapters 55, 43; its decisive use is here. Birman–Krein 1962, Enss 1978, and Reed–Simon III are new to the Vol IV arithmetic branch.

79.2 ADELIC SCATTERING FRAMEWORK

The Weil distribution W is an explicit distribution on test functions $f \in \text{PW}(\mathbb{R})$ (the Paley–Wiener class: entire functions of exponential type with L^2 restriction to the real line) encoding the prime-side, archimedean, and polar contributions to the explicit formula. A-TP asserts $W(f * \tilde{f}) \geq 0$ for all such f , where $\tilde{f}(t) := \overline{f(-t)}$ is the involute. To reformulate this positivity as an operator-theoretic unitarity condition, one needs a Hilbert space on which the scaling action of $\mathbb{R}_{>0}$ is self-adjoint, and a perturbation whose spectral trace recovers W .

The Idele Class Hilbert Space

Definition 79.2.1 (idele class Hilbert space). . Let $\mathbb{A} = \mathbb{R} \times \prod'_p \mathbb{Q}_p$ be the ring of adeles of $\mathbb{Q}$ and $\mathbb{A}^\times = \mathbb{R}^\times \times \prod'_p \mathbb{Q}_p^\times$ the idele group. Write $C_{\mathbb{Q}} := \mathbb{Q}^\times \backslash \mathbb{A}^\times$ for the idele class group, with idele-norm map $|\cdot| : C_{\mathbb{Q}} \rightarrow \mathbb{R}_{>0}$; and $C_{\mathbb{Q}}^1 := \ker |\cdot|$ for the norm-one idele class group. The *idele class Hilbert space* is

$$\mathcal{H}^{\text{id}} := L^2(C_{\mathbb{Q}}, d^\times x),$$

with $d^\times x$ the normalised Haar measure on $C_{\mathbb{Q}}$. The *scaling generator* is the self-adjoint operator $D := -i \partial_{\log |x|}$ densely defined on Schwartz-class functions invariant under $C_{\mathbb{Q}}^1$ (Meyer 2005 arXiv:math/0311468 §3). It is the archimedean component of the scaling action; by Mellin inversion, D has continuous spectrum $\mathbb{R}$ with Lebesgue multiplicity.

LEMMA 79.2.2 (Mellin decomposition of $\mathcal{H}^{\text{id}}$). (cite Meyer 2005 §3.2). The decomposition $\mathbb{A}^\times / \mathbb{Q}^\times \cong \mathbb{R}_{>0} \times C_{\mathbb{Q}}^1$ via the idele norm gives a unitary isomorphism

$$\mathcal{H}^{\text{id}} \cong L^2(\mathbb{R}_{>0}, d^\times t) \otimes L^2(C_{\mathbb{Q}}^1, d\mu_1),$$

and under the Mellin transform $\mathcal{M} : L^2(\mathbb{R}_{>0}, d^\times t) \rightarrow L^2(\mathbb{R}, d\tau)$ the scaling generator D conjugates to the multiplication operator $\tau \mapsto \tau$ on the first factor.

Proof. Meyer 2005 §3.2 constructs the Mellin isomorphism $\mathcal{M} : L^2(\mathbb{R}_{>0}, d^\times t) \rightarrow L^2(\mathbb{R}, d\tau)$; $D = -i \partial_{\log |x|}$ is the infinitesimal generator of the $\mathbb{R}_{>0}$ -scaling action, and under $\mathcal{M}$ the operator D conjugates to multiplication by τ on the Mellin-side. No new content added. $\square$

The Bost–Connes Scattering Generator

The Bost–Connes GNS space $\mathcal{H}_1$ (Chapter 55.2.1) and the BC log-modular operator N_{BC} (Chapter 55.2.5) carry the scattering data on the finite-idele side.

Definition 79.2.3 (scattering Hilbert space and free generator). . The *scattering Hilbert space* is the tensor

$$\mathcal{K} := \mathcal{H}^{\text{id}} \cong \mathcal{H}_\infty \otimes \mathcal{H}_{\text{fin}},$$

with $\mathcal{H}_\infty := L^2(\mathbb{R}_{>0}, d^\times t) \cong L^2(\mathbb{R}, d\tau)$ the archimedean factor and $\mathcal{H}_{\text{fin}} := L^2(C_{\mathbb{Q}}^1, d\mu_1)$ the norm-one idele class factor. The *free scaling generator* is $D_o = \tau \otimes \text{id}$ on $\mathcal{K}$. The *BC-perturbed scaling generator* is

$$D_{\text{BC}} := D_o + \iota_* N_{\text{BC}} \otimes P_o,$$

where P_o is the orthogonal projection onto the spherical-character subspace of $\mathcal{H}_{\text{fin}}$ (the subspace of functions invariant under $C_{\mathbb{Q}}^1$, equivalently the image of the trivial idele-class character $\chi_o = 1$ in the spectral decomposition of $L^2(C_{\mathbb{Q}}^1)$), and ι_* is the embedding of the BC log-modular operator into $\mathcal{H}_\infty$ as a bounded perturbation via the Hurwitz functional equation (Meyer 2005 §4.1).

Remark 79.2.4 (relation to H_{HP}). The BC-perturbed scaling generator D_{BC} is a renormalised scattering ansatz on $\mathcal{K}$, distinct from H_{HP} of Chapter 55.4.1. The operator H_{HP} is the hypothetical operator whose normal determinant divisor is $\text{div } \xi_{\mathbb{R}}$ and whose Weil–Connes trace is supported at the zero ordinates after zero-placement; D_{BC} is the operator whose spectral shift is required to reproduce the Weil distribution. The two are related by the primary–Heisenberg twist of Chapter 55.4.3; the scattering formulation requires trace-class renormalisation and positivity of the spectral-shift measure.

Wave Operators and the S-Matrix

Definition 79.2.5 (wave operators and S-matrix). Write $P_{\text{ac}}(D_o)$ for the spectral projection of D_o onto its absolutely continuous subspace. The *wave operators* (Møller operators) of the pair (D_o, D_{BC}) on $\mathcal{K}$ are the strong-operator limits

$$W_{\pm} := s\text{-}\lim_{t \rightarrow \pm\infty} e^{itD_{\text{BC}}} e^{-itD_o} P_{\text{ac}}(D_o),$$

provided the strong limits exist on $P_{\text{ac}}(D_o)\mathcal{K}$ (Reed–Simon III Thm. XI.8). The *scattering operator* is

$$S := W_+^* W_- : P_{\text{ac}}(D_o)\mathcal{K} \rightarrow P_{\text{ac}}(D_o)\mathcal{K},$$

a partial isometry that is unitary if and only if $W_{\pm}$ are complete (range equal to $P_{\text{ac}}(D_{\text{BC}})\mathcal{K}$).

PROPOSITION 79.2.6 (existence and trace-class criterion). For each finite-rank truncation of the perturbation $\iota_* N_{\text{BC}} \otimes P_o$, the wave operators $W_{\pm}$ exist and the scattering operator is unitary on the absolutely continuous subspace. Passing to the adelic limit is conditional on a Meyer–Connes trace-class renormalisation of $D_{\text{BC}} - D_o$.

Proof. For finite-rank truncations, the perturbation acts non-trivially only on the one-dimensional P_o -image of the spherical-character subspace; the BC log-modular operator N_{BC} has pure point spectrum $\{\log m\}_{m \in \mathbb{N}^\times}$ (Chapter 55.2.5) with finite-rank spectral projectors on each eigenvalue. The embedding ι_* produces a bounded finite-rank operator on $\mathcal{H}_\infty$. The Kato–Rosenblum theorem (Reed–Simon III Thm. XI.8) gives existence of $W_{\pm}$ and asymptotic completeness in this truncated case.

Trace-class criterion for completeness: Birman’s trace-class theorem (Reed–Simon III Thm. XI.7) requires the *difference* of spectral projectors to be trace class, which holds if $D_{\text{BC}} - D_o$ is trace class. For finitely many log-primes $m \in \{2, 3, \dots, M\}$ the perturbation is finite-rank; the limit $M \rightarrow \infty$ is controlled by $\sum_{m \geq 2} (\log m)^2 m^{-2} < \infty$ (a crude bound from the multiplicity decay on P_o ’s image under ι_*), giving the bare perturbation Hilbert–Schmidt class. Trace-class is a separate renormalised assertion: it holds only on the Schwartz subspace after the Meyer renormalisation (Remark 79.2.7), which absorbs the divergent sum $\sum_m \log m$ into the Connes cutoff. $\square$

Remark 79.2.7 (domain of the trace-class hypothesis). The trace-class hypothesis is not proved by the finite-rank argument. Without renormalisation the perturbation is *not* trace class, since the sum of eigenvalues $\sum_m \log m$ diverges. The Meyer–Connes cutoff is therefore a hypothesis, not an unconditional theorem.

79.3 SCATTERING-MATRIX REFORMULATION OF A-TP

The scattering framework of Section 79.2 places $\mathcal{W}$ in operator-theoretic form only under the trace-class renormalisation hypothesis. The Birman–Krein formula then supplies a spectral-shift function. A-TP is the positivity of the corresponding quadratic form, not bare unitarity of S .

Birman–Krein Spectral Shift

PROPOSITION 79.3.1 (*Birman–Krein formula on $\mathcal{K}$*). (cite Birman–Krein 1962; Reed–Simon III XI.9.30). Under the trace-class hypothesis of Chapter 79.2.6, there is a real-valued locally integrable *spectral shift function* $\xi_{\text{BK}}(\lambda)$ on $\mathbb{R}$ such that for every trace-class admissible test function h ,

$$\text{Tr}[h(D_{\text{BC}}) - h(D_o)] = \int_{\mathbb{R}} h'(\lambda) \xi_{\text{BK}}(\lambda) d\lambda, \quad (79.1)$$

and the scattering matrix $S(\lambda)$ on the fibre at spectral parameter $\lambda \in \mathbb{R}$ satisfies

$$\det S(\lambda) = \exp(-2\pi i \xi_{\text{BK}}(\lambda)).$$

In particular, $\xi_{\text{BK}}(\lambda) = -\frac{1}{2\pi i} \log \det S(\lambda)$.

Proof. Birman–Krein 1962 Thm. 1; Reed–Simon III Thm. XI.9.30. The spectral shift function is the Krein trace of the difference of spectral projectors. No new content added. $\square$

Identification of the Weil Distribution with ξ_{BK}

THEOREM 79.3.2 (*Weil distribution as spectral shift*). (conditional on the trace-class renormalisation of Meyer 2005 §4.1 and the Connes cutoff regularisation of Connes 1999 §II.2). Let ξ_{BK} be the spectral shift function of Chapter 79.3.1 for the pair (D_o, D_{BC}) on $\mathcal{K}$. For every $f \in \text{PW}(\mathbb{R})$ (entire of exponential type, L^2 on $\mathbb{R}$, with compactly supported Fourier transform $\widehat{f} \in L_c^2(\mathbb{R})$),

$$\mathcal{W}(f * \widetilde{f}) = \int_{\mathbb{R}} (|\widehat{f}|^2)'(\lambda) \xi_{\text{BK}}(\lambda) d\lambda, \quad (79.2)$$

where $\mathcal{W}$ is the Weil distribution of (56.1). Equivalently, the Weil distribution is the regularised trace difference $\text{Tr}_{\text{reg}}[h(D_{\text{BC}}) - h(D_o)]$ with $h = |\widehat{f}|^2$.

Proof. By Meyer 2005 Thm. 4.4 (Schwartz-class distributional trace on the idele class quotient), the regularised trace of $h(D_{\text{BC}})$ on $\mathcal{K}$ splits into:

- the free-generator trace $\text{Tr}_{\text{reg}} h(D_o)$, which is the Plancherel integral $\int h(\lambda) d\lambda$ against Lebesgue measure on $\mathbb{R}$ (the continuous spectrum of D_o);
- the perturbation contribution, localised on the spherical character sector P_o , summing the BC log-prime spectrum against h with the Meyer kernel m^{-1} per eigenvalue (the ζ -regularised weight).

The difference is precisely the finite-prime Weil contribution $-\mathcal{W}_{\text{fin}}(h)$ plus the polar residue at $s = 1$. The archimedean contribution $-\mathcal{W}_{\infty}(h)$ arises from the renormalisation of the scaling at the real place: Connes 1999 §III.3 shows that the Connes cutoff trace picks up the digamma kernel $\text{Re } \psi(1/4 + it/2) +$

$\log \pi$ as the archimedean contribution to the regularised trace. Combining the finite and archimedean contributions with the polar terms gives exactly the Weil distribution W of (56.1). Applying the Birman–Krein formula (79.1) with $h = |\widehat{f}|^2$ gives (79.2). $\square$

Remark 79.3.3 (what the identification buys). Chapter 79.3.2 is the scattering-theoretic analog of Chapter 55.6.1: where the latter requires a twist-existence hypothesis to identify $\mathrm{Tr}_{\mathrm{reg}} h(H_{\mathrm{HP}})$ with the zero-side of the Weil formula, the former identifies the regularised trace of the *existing* self-adjoint operator $D_{\mathrm{BC}} - D_0$ with the full Weil distribution W . The trade-off: the twist construction would give the zero-side $\sum_{\rho} h(z_{\rho})$, $z_{\rho} = -i(\rho - \frac{1}{2})$, directly; the scattering construction gives W , which by the Weil explicit formula equals this zero-side functional. Replacing z_{ρ} by the real ordinate γ_{ρ} is the RH-level step. The decisive shift is that the positivity of W on $|\widehat{f}|^2$ -tests becomes the unitarity of S .

A-TP as Unitarity of S

THEOREM 79.3.4 (*A-TP as spectral-shift positivity*). (conditional on the trace-class renormalisation of Chapter 79.2.7). The following are equivalent.

- (i) The archimedean Tate positivity conjecture A-TP (Chapter 56.7.1) holds for every $f \in \mathrm{PW}(\mathbb{R})$.
- (ii) For every $f \in \mathrm{PW}(\mathbb{R})$,

$$\int_{\mathbb{R}} (|\widehat{f}|^2)'(\lambda) \xi_{\mathrm{BK}}(\lambda) d\lambda \geq 0.$$

- (iii) The spectral-shift distribution $d\xi_{\mathrm{BK}}$ is non-negative on the Paley–Wiener convolution cone.
- (iv) The Riemann Hypothesis.

If the scattering matrix is unitary and the spectral-shift measure is monotone, then the same condition may be expressed in scattering language. Unitarity alone is not equivalent to A-TP.

Proof. (4) $\Leftrightarrow$ (1): Chapter 56.7.2.

(1) $\Leftrightarrow$ (2): use Chapter 79.3.2. The displayed integral is exactly $W(f * \widetilde{f})$.

(2) $\Leftrightarrow$ (3): integration by parts gives, in the distributional sense,

$$W(f * \widetilde{f}) = \int |\widehat{f}(\lambda)|^2 d\xi_{\mathrm{BK}}(\lambda)$$

for Paley–Wiener convolution tests. Positivity for all such tests is precisely non-negativity of $d\xi_{\mathrm{BK}}$ on that cone. $\square$

Remark 79.3.5 (honest caveat on the last implication). The determinant relation $\det S(\lambda) = \exp(-2\pi i \xi_{\mathrm{BK}}(\lambda))$ records a phase. It does not force $d\xi_{\mathrm{BK}} \geq 0$. The missing monotonicity is exactly the positivity content of A-TP in the scattering language.

79.4 TATE-MELLIN PHASE CALCULATION ON THE SUB-CLASS

The spectral-shift reformulation of Chapter 79.3.4 is conditional. A zero-adapted sub-class exists on which the Tate functional equation computes a unitary phase. This phase calculation does not by itself prove $W(f * \widetilde{f}) \geq 0$.

The Tate-Mellin Sub-Class

Definition 79.4.1 (*Tate-Mellin sub-class*). Write $\xi_{\mathbb{R}}$ for the completed Riemann xi-function $\xi_{\mathbb{R}}(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, which satisfies the functional equation $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s)$ and vanishes exactly at the non-trivial zeros of ζ . For $f \in \text{PW}(\mathbb{R})$, write $\mathcal{M}[f](s) := \int_0^\infty f(t)t^{s-1}dt$ for its Mellin transform. Let $\text{PW}^{\text{TM}} \subset \text{PW}(\mathbb{R})$ denote the subspace of $f \in \text{PW}(\mathbb{R})$ satisfying:

- (T1) f is real-valued and even: $\tilde{f} = f$ (where $\tilde{f}(t) := \overline{f(-t)}$ is the involute);
- (T2) the Mellin transform $\mathcal{M}[f](s)$ admits a factorisation $\mathcal{M}[f](s) = \xi_{\mathbb{R}}(s) \cdot h(s)$ with h entire of exponential type and rapid decay on vertical strips;
- (T3) h satisfies the Hermite-type symmetry $h(s) = \overline{h(1-\bar{s})}$.

Remark 79.4.2 (*why (T2) is a restriction, not a vacuous hypothesis*). Condition (T2) forces $\mathcal{M}[f](s)$ to vanish at the trivial zeros of ζ (i.e. at $s = -2, -4, -6, \dots$) and at the non-trivial zeros of ζ on the critical line. The latter is a non-trivial restriction: typical $f \in \text{PW}(\mathbb{R})$ have $\mathcal{M}[f]$ with generic zero set, not aligned with the non-trivial zeros. The sub-class PW^{TM} therefore consists of functions that are already *adapted* to the zero set of $\xi_{\mathbb{R}}$; for these, the scattering formulation becomes elementary.

Classical Mellin-Unitary on PW^{TM}

PROPOSITION 79.4.3 (*Tate-Mellin S -matrix is classical*). Fix $f \in \text{PW}^{\text{TM}}$ with Tate factorisation $\mathcal{M}[f](s) = \xi_{\mathbb{R}}(s) \cdot h(s)$. On the one-dimensional spectral fibre of D_0 at frequency $\lambda = t$, the scattering matrix $S(\lambda)$ restricted to the image of f under the Mellin transform is

$$S(t)|_f = \frac{\xi_{\mathbb{R}}(1/2 + it)}{\xi_{\mathbb{R}}(1/2 - it)} \cdot U_h(t), \quad (79.3)$$

where $U_h(t) := h(1/2 + it)/h(1/2 - it)$ is a classical unitary phase (by (T3), $|U_h(t)| = 1$), and the first factor is unitary on the critical line by Riemann's functional equation $\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(1-s)$.

Proof. Tate 1950 local functional equations at all places (archimedean digamma factor at ∞ , Euler local factors at each p) combine to give a single global functional equation for $\xi_{\mathbb{R}}$ which, on the critical line, becomes the statement $\xi_{\mathbb{R}}(1/2 + it) = \xi_{\mathbb{R}}(1/2 - it)$, so the quotient in (79.3) has modulus 1 fibrewise. The phase $U_h(t)$ is unitary by (T3): for $s = 1/2 + it$, $h(1/2 + it) = \overline{h(1 - \overline{1/2 + it})} = \overline{h(1/2 - it)}$, hence $U_h(t) = h(1/2 + it)/h(1/2 - it)$ has modulus 1. The scattering matrix is the product, a unitary phase.

The identification of this classical phase with the abstract S -matrix of Chapter 79.2.5 is by the Meyer 2005 Schwartz-class distribution trace: on the sub-class PW^{TM} , the BC-perturbed scaling generator D_{BC} acts on the Mellin-transformed vector as multiplication by $\tau + \frac{1}{2} \arg \xi_{\mathbb{R}}(1/2 + i\tau)$ (the Mellin image of the perturbation is the logarithmic phase of the Tate xi-factor; since $\xi_{\mathbb{R}}$ is real on the critical line, this phase is the jump of the real completed factor across its zeros), and the scattering theory becomes a one-dimensional phase-shift computation (Reed–Simon III XI.3). $\square$

THEOREM 79.4.4 (*Tate-Mellin phase unitarity*). (*domain-restricted to PW^{TM}*). For every $f \in \text{PW}^{\text{TM}}$, the Tate-Mellin phase $S(t)|_f$ has modulus one on the critical line. This is a unitarity statement for the phase, not a proof of $W(f * \tilde{f}) \geq 0$. Weil positivity follows only after the spectral-shift positivity condition of Chapter 79.3.4.

Proof. By Chapter 79.4.3, $S(t)|_f$ is a unitary phase on the spectral fibre. The final sentence is Chapter 79.3.5: phase unitarity does not imply monotonicity of $d\xi_{\text{BK}}$. $\square$

Remark 79.4.5 (density of PW^{TM}). The Tate-Mellin sub-class PW^{TM} is not dense in $\text{PW}(\mathbb{R})$ in the L^2 -topology: condition (T₂) forces $\mathcal{M}[f]$ to vanish at the non-trivial zeros of ζ , a set of measure zero, so the sub-class is not a dense subspace. Generic $f \in \text{PW}(\mathbb{R})$ has $\mathcal{M}[f]$ with generic zero set, hence $f \notin \text{PW}^{\text{TM}}$. The sub-class gives a phase calculation for functions adapted to the zero set of $\xi_{\mathbb{R}}$; constructing such an f does require knowledge of those zeros, so the sub-class theorem does not provide an independent proof of the zero location. Non-emptiness of a Paley–Wiener subspace satisfying all three conditions is a separate analytic constraint. It is not supplied by formal divisibility by $\xi_{\mathbb{R}}$.

79.5 NON-COMMUTATIVE-GEOMETRIC RESIDUAL LOCATED PRECISELY

The Tate-Mellin phase calculation relies on a classical functional-equation argument that does not extend to generic $\text{PW}(\mathbb{R})$. The obstacle is precisely the content Connes 1999 §IV.3 identifies as the *absorption-trace property* of the adèle-class-space dynamics.

Where the Connes Reduction Enters

Definition 79.5.1 (Connes adèle-class space). (following Connes 1999 §IV.1). The adèle class space $X_{\mathbb{Q}}$ is the quotient $\mathbb{A}/\mathbb{Q}^{\times}$ of the adèles by the multiplicative action of non-zero rationals; it is a non-Hausdorff topological space whose “commutative” L^2 -structure degenerates, requiring a non-commutative-geometric replacement via a groupoid C^* -algebra $C^*(\mathbb{A}/\mathbb{Q}^{\times})$.

PROPOSITION 79.5.2 (Connes absorption trace). (cite Connes 1999 Thm. 1). On the adèle class space, the scaling generator has absorption trace equal to the zero distribution of ζ ; after zero-placement the trace is supported at the imaginary parts of the non-trivial zeros. The Hilbert-space realisation requires the non-commutative-geometric L^2 -structure on $C^*(\mathbb{A}/\mathbb{Q}^{\times})$, not the classical $L^2(C_{\mathbb{Q}})$ of Chapter 79.2.1.

THEOREM 79.5.3 (obstruction located). . The scattering formulation of A-TP (Chapter 79.3.4) reduces the conjecture to positivity of the spectral-shift measure attached to the renormalised pair (D_0, D_{BC}) . On the Tate-Mellin sub-class PW^{TM} , the phase is elementary (Chapter 79.4.4), but this does not prove Weil positivity. Extension to the full Paley–Wiener class $\text{PW}(\mathbb{R})$ requires the Connes absorption-trace property (Chapter 79.5.2), which in turn requires passage from the classical $L^2(C_{\mathbb{Q}})$ to the non-commutative-geometric $L^2(X_{\mathbb{Q}})$ via $C^*(\mathbb{A}/\mathbb{Q}^{\times})$. The non-commutative reduction is not replaceable by any classical Mellin unitary: the non-Hausdorff quotient $\mathbb{A}/\mathbb{Q}^{\times}$ has no well-defined classical L^2 -structure.

Proof. The Tate-Mellin phase calculation (Chapter 79.4.4) uses the factorisation $\mathcal{M}[f] = \xi_{\mathbb{R}} \cdot h$ to force alignment with non-trivial zeros. Absent this alignment (generic $f \in \text{PW}(\mathbb{R})$), the scattering fibre $S(t)|_f$ is not a single classical phase but a sum of phase contributions from each non-trivial zero weighted by the residue of $\log \mathcal{M}[f]$ at that zero; the sum is not a priori a unitary on the classical $L^2(C_{\mathbb{Q}})$ spectral fibre, because $\mathcal{M}[f](1/2 + it)$ and $\mathcal{M}[f](1/2 - it)$ need not agree in modulus when $f \notin \text{PW}^{\text{TM}}$.

Connes 1999 Thm. 1 supplies the non-commutative Hilbert-space framework in which the absorption trace is defined. This Hilbert space is non-commutative-geometric: it requires the groupoid structure to exist as an L^2 -structure. The passage from classical $L^2(C_{\mathbb{Q}})$ to $L^2(X_{\mathbb{Q}})$ is an additional input, not a consequence of the Tate-Mellin phase calculation.

Hence the residual A-TP on $\text{PW}(\mathbb{R}) \setminus \text{PW}^{\text{TM}}$ requires the non-commutative-geometric input together with spectral-shift positivity. Absent that input, the scattering formulation gives only the conditional model above. $\square$

Comparison with the Bounded-Interval Reduction

Remark 79.5.4 (three orthogonal reductions of A-TP). A-TP admits three reductions with independent sources in the arithmetic formalism.

- (i) The Mellin–Rankin–Selberg reduction of Chapter 56: A-TP reduces to positivity of the archimedean digamma density against the Petersson-absorbed prime-side.
- (ii) The elementary-digamma reduction of Chapter 60: A-TP reduces to $\mathcal{A}[f] \leq B[f] + R[f]$, with the negative archimedean term supported on $[-t_*, t_*]$.
- (iii) The adelic-scattering reduction: A-TP reduces conditionally to positivity of the spectral-shift measure for (D_0, D_{BC}) , with the non-commutative extension to $L^2(X_{\mathbb{Q}})$ as part of the residual input.

The three reductions are genuinely disjoint: (1) uses automorphic representation theory, (2) uses elementary digamma analysis, (3) uses scattering theory and C^* -groupoid algebra. Each identifies the residual in its own language. The unification of the three is the Connes–Deninger programme (Connes 1999 + Deninger 1994), which is not closed by this or any other Vol IV chapter.

79.6 EQUIVALENCE THEOREM AND MATHEMATICAL FORM OF EACH COMPONENT

Component	Form	Reference
Wave operators for finite-rank truncations	Proved; adelic limit conditional	Chapter 79.2.6
Trace-class renormalisation	Hypothesis	Chapter 79.2.7
Birman–Krein identification of spectral shift with Weil distribution	Conditional	Chapter 79.3.2
A-TP as spectral-shift positivity	Conditional; equivalent to RH	Chapter 79.3.4
Tate–Mellin phase on PW^{TM}	Proved conditionally; no A-TP closure	Chapter 79.4.4
A-TP on full $\text{PW}(\mathbb{R})$	Open (=RH)	Chapter 79.5.3
Extension via Connes absorption trace	Requires NC-geometric input	Chapter 79.5.2
Three disjoint reductions of A-TP in Vol IV	Listed	Chapter 79.5.4

THEOREM 79.6.1 (adelic scattering reduction). (items conditional on the Meyer–Connes trace-class renormalisation of Remark 79.2.7). The adelic-scattering reformulation of A-TP yields the following.

- (i) *Equivalence:* A-TP is equivalent to positivity of the spectral-shift measure of the renormalised Bost–Connes scattering pair on $\mathcal{K} = L^2(C_{\mathbb{Q}})$ (Chapter 79.3.4).
- (ii) *Phase calculation:* the Tate–Mellin sub-class PW^{TM} gives a unitary phase (Chapter 79.4.4), not an unconditional A-TP closure.
- (iii) *Residual location:* the extension to generic $\text{PW}(\mathbb{R})$ requires the Connes non-commutative L^2 -structure on the adèle class space; no classical Hilbert-space replacement exists (Chapter 79.5.3).

The scattering obstruction does not prove A-TP. It reduces A-TP to spectral-shift positivity for a conditional scattering pair. The extension to the full class requires the Connes non-commutative construction and the RH-equivalent positivity input.

Proof. Assemble Chapter 79.3.4, Chapter 79.4.4, Chapter 79.5.3. □

Remark 79.6.2 (Beilinson-principled closure). The conditional scattering pair gives a scattering-theoretic reformulation of A-TP and isolates the missing input. It does not close A-TP. It identifies A-TP with positivity of a spectral-shift measure, disjoint from the Mellin–Rankin–Selberg and elementary-digamma reductions. The three reductions, read together, pin A-TP to a single open problem: realising the zero distribution of ζ as an absorption trace with the required positive trace formula, a Connes-type input that remains open.

ALL-GENUS ARITHMETIC KOSZUL DUALITY

Chapter 22 establishes the $g = 0$ case: the internal structural identity $\kappa_G^{\text{Ar}} + (\kappa^!)_G^{\text{Ar}} = 0$ of $\mathcal{V}^{\text{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2) is equivalent to the Riemann functional equation $\xi(s) = \xi(1-s)$, via adelic Poisson summation and rank-2 Satake parameters (Chapter 21.5.1). The question this chapter answers: for an arithmetic datum of rank $2g$ with $g \geq 0$, which internal identity is equivalent to the corresponding motivic functional equation?

The answer is the same identity $\kappa_G^{\text{Ar},g} + (\kappa^!)_G^{\text{Ar},g} = 0$, holding at every g . The genus does not deform the Koszul sum; it enters as the rank of the Heisenberg oscillators at each place, the archimedean Γ -factor $\Gamma_{\mathbb{C}}(s)^g$, and the motivic weight fixing the functional-equation center.

80.1 THE ARITHMETIC GENUS- g DATUM

Definition 80.1.1 (arithmetic genus- g datum). . An *arithmetic genus- g datum* $\mathcal{D}_g$ is a triple $(\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$ specifying:

- (i) $\mathcal{B}_g$, an arithmetic base: one of
 - a) Arakelov-compactified $\overline{\text{Spec } \mathcal{O}_K}$ for K totally imaginary of degree $2g$ (Window II);
 - b) a proper flat regular arithmetic surface $X \rightarrow \text{Spec } \mathcal{O}_K$ whose generic fibre X_K is a smooth projective curve of genus g , Arakelov-compactified via the archimedean Kähler form $\mu_{\text{Ar},g}$ of Faltings 1984 (Windows I, V);
 - c) a Shimura variety $\text{Sh}_K(G, X)$ of a Shimura datum whose complex fibre is a moduli of abelian g -folds (Window VI).
- (ii) $\mathcal{E}_g$, a rank- $2g$ arithmetic cohomology object: $H_{\text{prim}}^1(\mathcal{B}_g)$ carrying Galois action (étale or crystalline), with Hodge structure of weight 1 at each archimedean place: $\mathcal{E}_g \otimes \mathbb{C} = \mathcal{E}_g^{1,0} \oplus \mathcal{E}_g^{0,1}$ with $\dim \mathcal{E}_g^{1,0} = \dim \mathcal{E}_g^{0,1} = g$ per place.
- (iii) φ_g , a Frobenius-compatible endomorphism on $\mathcal{E}_g$ with square equal to the appropriate motivic involution: φ_g is either
 - a) the archimedean Frobenius Φ_{∞} of Deninger 1998 acting on the primitive Minkowski cohomology (Window II);
 - b) the prismatic Frobenius φ of Bhatt–Scholze 2022 on the universal rank- $2g$ F -crystal over the ordinary crystalline locus of the Emerton–Gee stack $\mathcal{X}_{2g}$ (Window III);
 - c) the Koszul duality operator J^{Koz} on chiral cohomology satisfying $(J^{\text{Koz}})^2 = \tau$ for τ the motivic functional-equation involution (Window V).

At $g = 0$ the datum degenerates to the base $\overline{\text{Spec } \mathbb{Z}}$ with $\mathcal{E}_0$ void and $\varphi_0 = \text{id}$; the functional equation reduces to $\xi(s) = \xi(1-s)$ and Chapter 80.4.1 specialises to Chapter 21.5.1.

Remark 80.1.2 (Frobenius as almost-complex substitute). An endomorphism $J : TX \rightarrow TX$ satisfying $J^2 = -\text{id}$ cannot exist on an arithmetic scheme: schemes over $\text{Spec } \mathbb{Z}$ carry no real tangent bundle, and on a compact real 2-manifold Newlander–Nirenberg (1957) makes integrability of J automatic, so no genuinely almost-complex geometry survives. The Frobenius φ_g of Chapter 80.1.1 is the correct arithmetic substitute: on archimedean primitive cohomology it acts as Hodge rotation by $\pi/2$; on the prismatic F -crystal ordinary stratum it is the Newton–Hodge-polygon-aligned slope-decomposition Frobenius (Katz 1981, *Astérisque* 63); on chiral cohomology it is the Koszul duality operator J^{Koz} whose square equals the functional-equation involution τ . No non-integrable decoration is introduced.

80.2 SIX CONVERGENT WINDOWS

The datum $\mathcal{D}_g$ admits six distinct geometric and automorphic presentations, each extracting a different structural feature of the rank- $2g$ arithmetic cohomology and the corresponding motivic functional equation.

Window I: Arakelov fibration

$\mathcal{B}_g^{\text{I}} = X \rightarrow \text{Spec } \mathcal{O}_K$, proper flat regular with generic fibre of genus g , Arakelov-compactified. $\mathcal{E}_g^{\text{I}} = H_B^1(X(\mathbb{C}), \mathbb{Z})$ of rank $2g$ per archimedean place. φ_g^{I} the integrable complex structure J_∞ at each archimedean place. Arakelov 1974, Faltings 1984, Gillet–Soulé 1990 (Publ. IHES) supply the arithmetic intersection pairing on $\widehat{\text{CH}}^1(X)$; Faltings–Hriljac 1984 the arithmetic Hodge index. Genus g enters via $\pi_* \omega_{X/S}$ of rank g , the Γ -factor $\Gamma_{\mathbb{C}}(s)^g$ of the Hasse–Weil L -function, the coefficient $2g - 2$ in $\deg \omega_{X_K/K}$, and the δ -invariant correction in the arithmetic Noether formula $\widehat{\deg} \det R\pi_* \omega = \frac{1}{12}(\widehat{\omega}^2 + \delta_{\text{Ar}})$ (Faltings 1984, Thm 5).

Window II: Minkowski torus / Bost–Connes–Marcolli

$\mathcal{B}_g^{\text{II}} = \overline{\text{Spec } \mathcal{O}_K}$ for K totally imaginary of degree $2g$, with Minkowski lattice $\Lambda_K = j(\mathcal{O}_K) \subset \mathbb{C}^g$ and Minkowski torus $T_K = \mathbb{C}^g / \Lambda_K$. $\mathcal{E}_g^{\text{II}} = H^1(T_K, \mathbb{Z}) \cong \Lambda_K^\vee$ of $\mathbb{Z}$ -rank $2g$; the primitive part equals the full H^1 here since T_K is an abelian variety. $\varphi_g^{\text{II}} = \Phi_\infty^{\text{tw}, K}$, the half-Tate-twisted archimedean Frobenius acting as $\sqrt{-1}$ on each complex embedding pair; equivalently the $\mathbb{C}^g$ -structure on $\Lambda_K \otimes \mathbb{R}$. The genus is $g = r_2(K) = \frac{1}{2}[K : \mathbb{Q}]$. Bost–Connes–Marcolli 2004 (*Selecta Math.* 11) constructs the C_K^* quantum statistical system whose KMS $_\beta$ partition function is $\zeta_K(\beta)$; Ha–Paugam 2005 and Laca–Neshveyev 2011 classify its KMS states.

Window III: Ordinary crystalline Emerton–Gee stack

$\mathcal{B}_g^{\text{III}} = \mathcal{X}_{2g}^{\text{crys, ord}}$, the ordinary crystalline locus of the Emerton–Gee stack $\mathcal{X}_{2g}$ (Emerton–Gee arXiv:1908.07185). Its dimension depends on the rank, the residual representation, and the imposed crystalline type; it is not a universal 2-dimensional Artin stack. A point in the ordinary crystalline stratum is a rank- $2g$ crystalline object with Hodge–Tate multiset $\{0, 1\}^g$, together with the chosen ordinary filtration. $\mathcal{E}_g^{\text{III}}$ is the universal rank- $2g$ bundle; φ_g^{III} is Frobenius together with this filtration. The analogy with $H^1(X(\mathbb{C}), \mathbb{C}) = H^{1,0} \oplus H^{0,1}$ is a Hodge analogy, not an archimedean tilting theorem. Caraiani–Scholze supplies Shimura-variety Galois representations under its own genericity and signature hypotheses; it does not produce this rank- $2g$ window for all genus- g data.

Window IV: Hodge–Arakelov / arithmetic Teichmüller

$\mathcal{B}_g^{\text{IV}} = [C, \mathfrak{S}] \in \mathcal{T}_g^{\text{arith}}(\mathcal{O}_K)$, a closed point of the arithmetic Teichmüller space carrying a level- n theta-structure $\mathfrak{S}$ on the Jacobian $J(C)$. $\mathcal{E}_g^{\text{IV}}$ is the n^g -dim Jet module over the theta-structure locus; φ_g^{IV} is Mochizuki’s Hodge–Arakelov comparison isomorphism (Mochizuki 1999 *Publ. RIMS*; arXiv:math/0304054). The comparison is theorem-grade for $g = 1$; conjectural for $g \geq 2$. The Steinberg relation $\{p, 1 - p\} = 0 \in K_2^M(\mathbb{Q})$ is preserved under every Frobenioid indeterminacy (anabelian invariant via Matsumoto 1969, Milnor 1970). The naïve speculation conflates the crystalline–étale comparison isomorphism with an almost-complex structure J on $\text{Spec } \mathbb{Z}$; the correct arithmetic object is the Hodge–Arakelov decoration φ_g^{IV} on the dense theta-generic open locus of $\mathcal{T}_g^{\text{arith}}$ (Chapter 80.1.2).

Window V: E_2 -chiral factorisation on the arithmetic surface

$\mathcal{B}_g^{\text{V}} = \mathcal{S}_g^{\text{Ar}}$, the arithmetic surface of Window I viewed through its E_2 -chiral factorisation structure. $\mathcal{E}_g^{\text{V}} = \mathcal{V}^{\text{prim, surf}, g} := \text{HH}_\bullet(\text{Hk}_{GL_{2g}, \text{glob}}^{\text{Iw, surf}})$, the Hochschild trace class of the identity on the E_2 -monoidal global Iwahori–Hecke category of GL_{2g} on the arithmetic surface. The E_2 structure arises via $E_1 \otimes E_1 = E_2$ Dunn additivity (Lurie *Higher Algebra* 5.1.2.2) from the horizontal E_1 (1-dim base factorisation on $\overline{C}$) and the vertical E_1 (fibre factorisation on X_K); the bar-cobar construction on this E_2 datum lifts to the 2-dim Parshin–Gersten complex with double residue (∂_1, ∂_2) (Parshin 1976; Kato 1980; Beilinson 1980). $\varphi_g^{\text{V}} = J^{\text{Koz}}$, the Koszul duality operator on chiral cohomology with $(J^{\text{Koz}})^2 = \tau$ (the functional-equation involution). The genus enters via rank- $2g$ Fock oscillators per place, sourced from the ℓ -adic Tate module $T_\ell(\text{Jac}(X_K))$; the holomorphic central charge is $c = g$, with g additional antiholomorphic oscillators, giving total Fock rank $2g$.

Window VI: Kudla–Rapoport / Shimura–Siegel Shimura variety

$\mathcal{B}_g^{\text{VI}} = \text{Sh}_K(G, X)$ a Shimura variety with $G \in \{\text{GSp}_{2g}, \text{Res}_{F/\mathbb{Q}} \text{GL}_2 \text{ for } [F : \mathbb{Q}] = g, \text{GU}(g, 1)\}$. $\mathcal{E}_g^{\text{VI}}$ the middle-degree cohomology decomposed under the Hecke algebra into rank- $2g$ (or $2g + 1, 4g$) Galois blocks per automorphic π of $G(\mathbb{A})$. φ_g^{VI} the integrable complex structure sourced from the Deligne torus $h : \mathbb{S} \rightarrow G_{\mathbb{R}}$. The Kudla–Rapoport generating series $\widehat{\varphi}(\tau) = \sum_m \widehat{\mathcal{Z}}(m) q^m$ is a genus- g Siegel modular form valued in $\widehat{\text{CH}}^1(\text{Sh})_{\mathbb{C}}$ (Kudla 1997 *Ann. Math.* 146; Kudla–Rapoport 2011 *Algebra Number Theory* 5; Li–Liu 2021 arXiv:2101.09485). The g -fold arithmetic intersection of special cycles realises Fourier coefficients of a genus- g Siegel Eisenstein series’ central derivative. The Langlands-dual group ${}^L G = \text{GSpin}_{2g+1}$ carries the $(2g + 1)$ -dim spin L -function.

80.3 FALSIFIED NAIVE READINGS

Six naïve readings of the speculative phrase “genus- g arithmetic surface with almost-complex structure” each fail at a specific technical joint; the failures are recorded in Chapter B.7.

- (i) $J^2 = -\text{id}$ on the tangent bundle of an arithmetic scheme. Schemes over $\text{Spec } \mathbb{Z}$ carry no real tangent bundle; the phrase “almost-complex” is a category error at this level. The correct arithmetic substitute is the Frobenius φ_g on primitive cohomology (Chapter 80.1.2).
- (ii) *Arithmetic Euler characteristic* $= 1 - g$. Gillet–Soulé 1992 arithmetic Riemann–Roch introduces Bismut–Gillet–Soulé analytic torsion; the arithmetic analog of $1 - g$ is the Faltings δ -invariant via the arithmetic Noether formula, not a simple integer.

- (iii) *Künneth factorisation of the arithmetic surface.* Bost–Künnemann 2010 (*Adv. Math.* 223) shows the product formula has non-vanishing Bott–Chern archimedean correction terms; X does not factor as “genus- g curve $\times_{\text{Spec } \mathbb{Z}} \text{Spec } \mathbb{Z}$ ” in arithmetic Chow.
- (iv) *$\overline{\text{Spec } \mathbb{Z}}$ itself has finite genus.* Deninger’s arithmetic-dynamical genus of $\text{Spec } \mathbb{Z}$ is $+\infty$ (countably many critical zeros of ζ). Genus enters the arithmetic datum only through coefficient enhancement to a totally imaginary number field of degree $2g$.
- (v) *Conflation of curve-genus g_c with Siegel-modular genus g_s .* For a modular curve $X_o(N)$: $g_c = g_o(N)$ and $g_s = 1$. For the Siegel threefold $\mathcal{A}_2$: g_c is undefined, $g_s = 2$. The genus g of the datum $\mathcal{D}_g$ is the symplectic rank of the Galois representation divided by 2; in the Shimura setting this is g_s , not g_c .
- (vi) *$\kappa + \kappa^! = 2 - 2g$ as Euler-characteristic shift.* The Euler characteristic $2 - 2g$ is a topological invariant of the Riemann surface $X(\mathbb{C})$, not a Koszul invariant of the chiral datum. The identity $\kappa^{\text{Ar},g} + (\kappa^!)^{\text{Ar},g} = 0$ is the conditional Koszul-duality target at rank g ; the genus enters through the Fock rank, the archimedean Γ -factor, and the motivic weight of the functional-equation center.

80.4 ALL-GENUS ARITHMETIC KOSZUL DUALITY

Let $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$ be an arithmetic genus- g datum (Chapter 80.1.1). Attach to $\mathcal{D}_g$ the arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim},g}$ via the rank- $2g$ Fock on $\mathcal{E}_g$ at each place, and the Koszul dual via the genus- g Bost–Connes–Marcolli system of Consani–Marcolli 2006 (*J. Reine Angew. Math.* 597).

THEOREM 80.4.1 (*all-genus arithmetic Koszul duality*). (*on the residual conditions of Chapter 80.6*). For every $g \geq 0$ and every arithmetic genus- g datum $\mathcal{D}_g$, the arithmetic Heisenberg Koszul self-duality identity

$$\kappa_G^{\text{Ar},g}(\mathcal{D}_g) + (\kappa^!)_G^{\text{Ar},g}(\mathcal{D}_g) = 0$$

is equivalent to the motivic functional equation

$$\Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g} \cdot \Lambda_{\mathcal{D}_g}(w_{\mathcal{D}_g} - s), \quad (80.1)$$

where $\Lambda_{\mathcal{D}_g}(s)$ is the completed L -function attached to $\mathcal{D}_g$ and $w_{\mathcal{D}_g} \in \mathbb{Z}_{\geq 1}$ is the motivic weight center, depending on the window but not on g alone. The implication is conditional on the Euler-product identification for the chosen datum, the appropriate genus- g Bost–Connes–Marcolli or Shimura input, the analytic continuation and functional equation of $\Lambda_{\mathcal{D}_g}$, the 2-dimensional arithmetic Positselski equivalence at $\Lambda_2 = \mathbb{Z}_p[[\Gamma_1 \times \Gamma_2]]$ -coefficients (Window V), and Koszul–Petersson compatibility at rank g .

The four rows of (80.1), indexed by window, are recorded in Chapter 80.1.

TABLE 80.1: Rows of the all-genus arithmetic Koszul duality.

Window	$\Lambda_{\mathcal{D}_g}(s)$	$w_{\mathcal{D}_g}$	Motivic weight
II	$\xi_K(s) = d_K ^{s/2} \Gamma_{\mathbb{C}}(s)^g \zeta_K(s)$	1	0 (Dedekind)
I, V	$\Lambda(s, X_K) = N_{X_K}^{s/2} \Gamma_{\mathbb{C}}(s)^g L(s, X_K)$	2	1 (Hasse–Weil)
VI	$\Lambda(s, \pi, \text{spin})$ for π on GSp_{2g}	$w_{\pi, \text{spin}}$	fixed by the chosen spin L -normalisation
III, IV	spectral refinements	indirect	refines above

Remark 80.4.2 (weight structure). The functional-equation center $w_{\mathcal{D}_g}$ is set by the motivic weight of the row, not by g itself. Window II (Dedekind ζ_K with archimedean $\Gamma_{\mathbb{C}}^g$ factor) is weight 0; the center stays at $s \mapsto 1 - s$ regardless of g , since ζ_K for K of degree $2g$ has the same $1 - s$ symmetry as ζ for any number field. Windows I, V (arithmetic surface with generic fibre of genus g) are weight 1 on the Jacobian's H^1 -motive; the center is $s \mapsto 2 - s$. Window VI (Siegel spin of genus- g cusp form) has centre determined by the chosen cohomological spin normalisation; the table records it as $w_{\pi, \text{spin}}$, not as a universal integer. All three rows share $\kappa + \kappa^! = 0$; the genus grading lives in the Γ -factor, the Fock rank, and the weight of the row.

Sketch. Four steps, parallel to Chapter 21.5.1 at $g = 0$.

Step 1: Euler-product identification. Assume that at each prime p of good reduction for $\mathcal{D}_g$, writing $\alpha_1^{(p)}, \dots, \alpha_{2g}^{(p)}$ for the Frobenius eigenvalues on $\mathcal{E}_{g,p}$, the rank- $2g$ Heisenberg Fock partition function at inverse temperature $\beta = s$ factors as $\prod_{j=1}^{2g} (1 - \alpha_j^{(p)} p^{-s})^{-1} = \det(1 - p^{-s} \phi_g | \mathcal{E}_{g,p})^{-1}$. The global Euler product (over primes of good reduction, with local factors at bad primes defined as in Serre 1970) gives $\Lambda_{\mathcal{D}_g}(s)$ in the respective window.

Step 2: Consani–Marcolli genus- g Koszul dual. Consani–Marcolli 2006 (*J. Reine Angew. Math.* 597) constructs the rank- $2g$ Bost–Connes–Marcolli system with KMS partition function $\zeta_K(\beta)$ in the number-field row. Identifying the Koszul dual of $\mathcal{V}_{\text{Heis}}^{\text{prim}, g}$ with the row $\Lambda_{\mathcal{D}_g}$ outside that row is part of the residual input.

Step 3: rank- g Positselski pairing. Arithmetic Positselski (Positselski 2011 Mem. AMS 996; Chapter 24) at rank g , extended to Λ_2 -coefficients for Window V. Pairs the bar coalgebra of the rank- $2g$ Heisenberg with the cobar algebra of the genus- g Bost–Connes system; total weight vanishes modulo the 2-dim μ -invariant vanishing condition (extending Ferrero–Washington 1979 *Ann. Math.* 109 for abelian at rank 1).

Step 4: adelic Fourier symmetry $s \mapsto w_{\mathcal{D}_g} - s$. Tate 1950 at $g = 0$ (Window II with $K = \mathbb{Q}$); Weil 1967 *Basic Number Theory* for totally imaginary base (Window II general); Bloch–Kato 1990 and Fontaine–Perrin–Riou 1994 for weight-1 motives on Jacobians (Windows I, V); and the automorphic theory available for the chosen Siegel or unitary datum (Window VI) are the inputs. Where these inputs do not give analytic continuation and the functional equation, they are part of the residual condition.

No step in the all-genus row is promoted to an unconditional theorem by this chapter. The result is a conditional reduction of the Koszul self-duality identity to the named Euler-product, functional-equation, and Positselski inputs. $\square$

80.5 REDUCTION TO $g = 0$ AND THE F-RH AXIS

At $g = 0$:

- Window II: $K = \mathbb{Q}$, $\xi_K(s) = \xi(s)$, $\Gamma_{\mathbb{C}}(s)^{\circ} = 1$, $\zeta_K = \zeta$. Functional equation $\xi(s) = \xi(1 - s)$.
- Windows I, V: $X_K = \mathbb{P}^1/K$, Jacobian trivial, $\Lambda(s, X_K)$ degenerates to $\xi(s)$ with center $s \mapsto 1 - s$; the weight-1 motive is void and the weight-0 Dedekind row takes over.
- Window VI: $\mathcal{A}_0 = \text{pt}$, Siegel spin L -function collapses to ζ , center $s \mapsto 1 - s$.

All four rows collapse to $\xi(s) = \xi(1 - s)$, recovering Chapter 21.5.1 as the $g = 0$ specialisation of Chapter 80.4.1.

COROLLARY 80.5.1 (*all-genus F-RH*). (on Chapter 80.4.1 plus rank- g extensions of the P_2 , P_3 , VB pillars of Chapter 22). For every $g \geq 0$ and every arithmetic genus- g datum $\mathcal{D}_g$ with rank- g Koszul–Petersson compatibility and rank- g arithmetic Zhu modularity, the non-trivial zeros of $\Lambda_{\mathcal{D}_g}(s)$ lie on the self-dual locus $\operatorname{Re}(s) = w_{\mathcal{D}_g}/2$.

Remark 80.5.2 (specialisations). Chapter 80.5.1 has as special cases the zero-placement clauses for the Riemann zeta function ($g = 0$, Window II), Hasse–Weil L -functions of Jacobians (Windows I,V), Dedekind zeta functions of totally imaginary fields (Window II at higher g), and Siegel-spin L -functions of cusp forms on $\operatorname{GSp}_{2g}(\mathbb{A}_{\mathbb{Q}})$ (Window VI). Each case still requires its rank- g spectral positivity input.

80.6 RESIDUAL CONDITIONS

Chapter 80.4.1 reduces to ten residual conditions, each extending a Chapter-level $g = 0$ result to arbitrary g .

- (i) *2-dim arithmetic Positselski equivalence* with $\mu_1 = \mu_2 = 0$ on $\Lambda_2 = \mathbb{Z}_p[[\Gamma_1 \times \Gamma_2]]$, extending Chapter 24. Condition: non-formal theorem. Precedents: Nekovář 2006 (Astérisque 310); Coates–Sujatha 2006 *Cyclotomic Fields and Zeta Values*.
- (ii) *Rank- g Koszul–Petersson compatibility* replacing the scalar Petersson inner product with the genus- g Petersson pairing on weight- (k, g) automorphic forms, extending Chapter 25. It is a non-formal theorem.
- (iii) *2-dim Parshin–Gersten bar construction* on the arithmetic surface with double residue (∂_1, ∂_2) , extending Chapter 12. It is a non-formal theorem. Precedents: Parshin 1976, Kato 1980, Beilinson 1980.
- (iv) *Rank- g arithmetic Ben-Zvi–Nadler* with target stack $\Sigma_g^{\operatorname{Ar}}$ replacing $B\widehat{\mathbb{Z}}$, extending Chapter 27. It is a non-formal theorem.
- (v) *GL_{2g} -local geometric Langlands*. Fargues–Scholze arXiv:2102.13459 supplies the spectral action in broad reductive generality, while Emerton–Gee arXiv:1908.07185 is the GL_n Galois deformation-stack input. The categorical comparison at GL_{2g} is not a formal corollary of either source; it is the rank- g scaling of Chapter 20.1.1. Domain: determined by F1 scaled to rank $2g$.
- (vi) *Perfectoid E_2 -enhancement at rank g* via BD Grassmannians for $GL_{2g}/\mathbb{Q}_p$, extending F2 (Chapter 20.2.1). It is a non-formal theorem.
- (vii) *Rank- g arithmetic Virasoro bootstrap* with central charge $c^{\operatorname{Ar},g} = g$ on the primary subspace of the rank- g archimedean Fock, extending Chapter 23. It is a non-formal theorem.
- (viii) *Arithmetic Siegel–Weil at bad primes for $g \geq 3$* , extending Li–Liu 2021 arXiv:2101.09485. It is a non-formal theorem.
- (ix) *Harder–Eisenstein-congruence prime at genus g* replacing $p = 691$ in F6 (Chapter 20.4.4). Systematic tabulation by Bergström–Faber–van der Geer 2014 (*Sel. Math.* 20) identifies the relevant Hecke-eigenvalue congruences. No genus-2 replacement prime is fixed here before that table is read. It is a non-formal theorem.
- (x) *Arithmetic B-row genus- g ceiling*. F4 (Chapter 20.4.1) sharpens from the single integer $\kappa^{\operatorname{Ar}} + (\kappa^1)^{\operatorname{Ar}} = 8$ to a genus-dependent Shimura-signature-indexed sum, via Bruinier–Heegner Chern-class reciprocity on $\operatorname{Sh}(\operatorname{O}(2, 2g - 1))$ with accidental isomorphism $\operatorname{O}(2, 3) \cong \operatorname{Sp}_4/\{\pm I\}$ at $g = 2$. It is a non-formal theorem.

The total residual condition is a non-formal theorem condition, organised as three to four mathematical blocks. Block (i): conditions 1–4 + 7 (Beilinson / Positselski / bar-cobar on 2-dim base plus Virasoro bootstrap). Block (ii): conditions 5, 6 (GL_{2g} Fargues–Scholze local plus global Arinkin–Gaitsgory). Block (iii): conditions 8, 10 (Kudla–Rapoport and Bruinier–Heegner at genus g). Block (iv): condition 9 and consolidation (Harder–Eisenstein congruences; all-genus F-RH corollary).

80.7 ORTHOGONAL AND COMPATIBLE WINDOWS

The six windows are compatible but mathematically distinct: each captures a different facet of the datum $\mathcal{D}_g$, and their pairwise compatibility is an independent mathematical fact in each case. Pairwise compatibility:

- **Windows I and V** share the arithmetic surface as base and agree on the weight-1 Hasse–Weil functional equation. They differ on operadic dimension: Window I keeps the archimedean E_1 -chiral structure; Window V lifts to E_2 -chiral via Dunn. Compatibility is the content of Chapter 12’s 2-dim extension.
- **Windows II and V** differ on arithmetic scheme dimension (1-dim vs. 2-dim). Window II encodes the genus as coefficient enhancement of the base field (totally imaginary K of degree $2g$); Window V keeps $K = \mathbb{Q}$ but lifts the base scheme to 2-dim via Dunn additivity. Compatibility passes through Tate 1950 adelic Fourier at number-field level.
- **Windows III and VI** are spectral: Window III gives the local (φ, Γ) -module moduli; Window VI gives the global automorphic side. Compatibility requires the relevant local-global comparison; it is not supplied by Fargues–Scholze alone.
- **Window IV** is moduli-theoretic: $\mathcal{T}_g^{\text{arith}}$ carries the Hodge–Arakelov decoration on the theta-generic locus, and its point-by-point inputs supply Window I. Compatibility via the Torelli map: a genus- g curve’s Jacobian is a point of $\mathcal{X}_{2g}^{\text{crys, ord}}$ only after imposing the ordinary crystalline local condition, and a point of the Siegel moduli space $\mathcal{A}_g$ via its polarisation. The word “threefold” applies only at $g = 2$.

Orthogonal to these six windows: Deninger’s $\mathbb{F}_1$ programme (motivational context for the arithmetic-dynamical interpretation of $\text{Spec } \mathbb{Z}$, recorded in Chapter 2); the Caraiani–Scholze unitary-Shimura programme at genera where no Shimura model exists (moduli of curves $\mathcal{M}_g$ is not Shimura for $g \geq 4$); the Bethe-algebra heuristic presentation, which operates at the zero-Fock per-place level and does not carry g -dependence directly.

The conditional identity $\kappa^{\text{Ar}, g} + (\kappa^!)^{\text{Ar}, g} = 0$ of Chapter 80.4.1 is the proposed rank- g fibre. The F-RH axis of Chapter 22 is the $g = 0$ fibre of the rank-graded family whose other fibres are the Hasse–Weil Riemann Hypothesis for Jacobians of curves of genus g , the generalised Riemann Hypothesis for Dedekind zeta of totally imaginary fields of degree $2g$, and the Siegel-spin Riemann Hypothesis for cusp forms on GSp_{2g} . The residual conditions of Chapter 80.6 quantify what must be proved to convert the conditional biconditional into an unconditional one.

Chapter 81 extends the arithmetic-modular axis into arithmetic three-dimensional Chern–Simons theory. The conformal vector implicit in Chapter 80.4.1 sources an arithmetic Kac–Moody enhancement of $\mathcal{V}^{\text{prim}}$ (Chapter 81.3.1), which in turn sources a finite-level arithmetic Chern–Simons partition sum and a conjectural renormalised $GL_2(\widehat{\mathbb{Z}})$ limit on the étale 3-space $\overline{\text{Spec } \mathbb{Z}}$ (Chapter 81.4.2, Chapter 81.4.7). The equalities

$$Z_{\text{ACS}^{\text{qu}}} = \xi(k^{\text{Ar}}), \quad \text{HH}_0(\partial \text{ACS}^{\text{qu}}) \simeq \mathcal{V}^{\text{prim}}$$

are conditional on profinite renormalisation and the local trace comparison (Chapter 81.4.9). The ordered incidence model is computed by the normalised ordered bar construction on the auxiliary place order

$\mathbb{P} \cup \{\infty\}$ (Chapter 81.5.4); the comparison with full Beilinson–Drinfeld / Costello–Gwilliam arithmetic chiral homology is conditional (Chapter 81.5.5). F-RH becomes a conditional spectral theorem on the partition-function zero locus (Chapter 81.7.1); the all-genus extension (Chapter 81.8.1) covers every row of the arithmetic-modular family.

3D ARITHMETIC CHERN-SIMONS ON ÉTALE $\mathrm{Spec}(\mathbb{Z})$ AND THE ORDERED INCIDENCE BAR

The all-genus arithmetic conformal vector of Chapter 80 produces an arithmetic Kac–Moody enhancement of $\mathcal{V}^{\mathrm{prim}}$ (conditional on the G-row resolution). Finite arithmetic Chern–Simons state sums follow from finite étale local-system groupoids. The profinite $GL_2(\widehat{\mathbb{Z}})$ quantum theory, its boundary identification with $\mathcal{V}^{\mathrm{prim}}$, and its zeta partition function remain conditional. The ordered incidence bar is a defined model for the derived colimit of $\mathcal{V}^{\mathrm{prim}}$ over the prime places; identifying it with Beilinson–Drinfeld/Costello–Gwilliam arithmetic chiral homology is a separate comparison theorem. The Deninger construction obtained from this model is a prime-orbit skeleton plus a Hilbert–Pólya-strength spectral-flow hypothesis, not an independent construction of the zero-side cohomology.

8I.1 $\mathrm{Spec}(\mathbb{Z})$ AS ÉTALE 3-SPACE

Mazur 1973 (*Notes on étale cohomology of number fields*, Ann. Sci. ENS 6) observed that $\mathrm{Spec} O_K$ has étale cohomological dimension 3 for K a number field, with Artin–Verdier duality supplying a fundamental class $[\mathrm{Spec} O_K] \in H_{\mathrm{ét}}^3(\mathrm{Spec} O_K, \widehat{\mathbb{Z}})$ analogous to a 3-manifold’s orientation class. Morishita 2012 *Knots and Primes* (arXiv:0904.3399) systematizes the dictionary: primes $\mathfrak{p} \subset O_K$ behave as knots in a closed 3-manifold; $\mathrm{Spec} O_{K,\mathfrak{p}}$ is the tubular neighborhood; the inertia subgroup $I_{\mathfrak{p}} \subset \mathrm{Gal}(\overline{K}/K)$ is the meridian; linking numbers become Legendre symbols; the Alexander–Fox module becomes the Iwasawa module. For $K = \mathbb{Q}$, Arakelov compactification $\overline{\mathrm{Spec} \mathbb{Z}}$ closes the 3-space by adding the archimedean boundary component at ∞ .

Remark 8I.1.1 (cohomological 3-dimensionality). $\mathrm{Spec} \mathbb{Z}$ is not a smooth real 3-manifold. Its 3-dimensionality is cohomological: the étale site supports non-trivial H^1, H^2, H^3 computable via global class-field theory and Artin–Verdier. The Chern–Simons action functional below takes values in étale cohomology classes rather than in differential forms; its quantization is in Galois-representation-theoretic terms rather than path-integral terms in the differential-geometric sense.

8I.2 KIM CLASSICAL ARITHMETIC CHERN-SIMONS

Kim 2015 *Arithmetic Chern-Simons Theory I* (arXiv:1510.05818), and the non-abelian extension Chern–Dan–Kim–Park 2018 (arXiv:1609.03012), define the classical action

$$\mathrm{CS}_{M,c}^{\mathrm{Ar}}(\rho) := \langle \rho^*c, [M] \rangle \in \mathbb{Q}/\mathbb{Z}, \quad (8I.1)$$

for $M = \mathrm{Spec} O_K$, $\rho : \pi_1^{\mathrm{ét}}(M) \rightarrow G$ a continuous representation into a finite group G , $c \in H^3(BG, \mathbb{Q}/\mathbb{Z})$ a level class, and $[M]$ the Artin–Verdier fundamental class. Summation over ρ gives the classical partition function $Z_{\mathrm{CS},c}^{\mathrm{cl},\mathrm{Ar}}(M) = \sum_{\rho} \frac{e^{2\pi i \mathrm{CS}_{M,c}^{\mathrm{Ar}}(\rho)}}{|\mathrm{Aut}(\rho)|}$. Chapter 15 already uses this classical

torsor as a boundary action-line input for $\mathcal{V}^{\text{prim}}$; finite state sums and the profinite quantum-TQFT problem are developed in Chapter 81.4.

81.3 THE ARITHMETIC KAC–MOODY LIFT

Chapter 23 supplies local Virasoro calculations on the finite Heisenberg covers and records the rank residue $r_{\mathcal{L}} = 2$. Chapter 80 upgrades per-window rank to g from the rank- g archimedean Fock. The presence of a local conformal vector enables Kac–Moody affinisation via the Sugawara construction: a Heisenberg algebra with conformal vector admits a level- k Kac–Moody extension whose Sugawara Virasoro agrees with the given one at level $k \neq -h^\vee$.

THEOREM 81.3.1 (*arithmetic Kac–Moody enhancement*). *(on the G -row resolution Chapter 21.5.1 and rank- $2g$ Satake structure of Chapter 80).* The arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim}}$ admits an arithmetic Kac–Moody enhancement

$$\widehat{\mathcal{V}}_{k^{\text{Ar}}}^{\text{Ar}} := \mathcal{V}_{\text{Heis}}^{\text{prim}} \otimes \text{Ind}_{k^{\text{Ar}}}(\text{Lgl}_2^{\text{Ar}}),$$

at level $k^{\text{Ar}} = \kappa_G^{\text{Ar}}$, where $\text{Lgl}_2^{\text{Ar}} := \bigotimes_p' \text{Lgl}_2^{(p)}$ is the restricted tensor product of the per-prime Satake current algebras generated by the standard basis elements (α_p, β_p) of $\mathfrak{gl}_2^{(p)}$ (the generators of $\text{Lgl}_2^{(p)}$ defined in Chapter 80). On the semisimple $\mathfrak{sl}_2$ -summand, the usual Sugawara construction has denominator $k^{\text{Ar}} + 2$. The central $\mathfrak{u}(1)$ -summand contributes as a free Heisenberg current. With this split convention, the induced Virasoro operator is the formal Fock operator of Chapter 23. The rank- $2g$ extension scales linearly per Chapter 80.

Remark 81.3.2 (critical level evasion). Feigin–Frenkel duality at the complex level identifies $k = -h^\vee$ with the centre of $\widehat{\mathfrak{g}}$; the Sugawara construction diverges there. The arithmetic branch’s Koszul-self-duality $\kappa_G^{\text{Ar}} + (\kappa^\dagger)_G^{\text{Ar}} = 0$ (Chapter 21.5.1) places k^{Ar} at the self-dual level, distinct from the critical level. The Sugawara closes; the arithmetic Kac–Moody is non-critical by construction.

81.4 3D QUANTUM ARITHMETIC CHERN-SIMONS

Definition 81.4.1 (*finite arithmetic field groupoid*). Let M be an arithmetic 3-space in the Mazur–Artin–Verdier sense, for example $\text{Spec } \mathcal{O}_K$ with its compact-support fundamental class. Let G be a finite group and let $\alpha \in H^3(BG, \mathbb{C}^\times)$ be a level. The finite arithmetic field groupoid is

$$\mathcal{F}_G(M) := \text{Map}_{\text{cont}}(\Pi_1^{\text{ét}}(M), BG),$$

the groupoid of continuous G -local systems on the étale homotopy type of M . For a field $\rho \in \mathcal{F}_G(M)$, the action is

$$S_\alpha^{\text{Ar}}(\rho) := \langle \rho^* \alpha, [M] \rangle \in \mathbb{C}^\times.$$

THEOREM 81.4.2 (*finite-level arithmetic state sum and gluing*). For a closed arithmetic 3-space M for which $\mathcal{F}_G(M)$ is a finite groupoid, the assignment

$$Z_{G,\alpha}^{\text{Ar}}(M) := \int_{\mathcal{F}_G(M)} S_\alpha^{\text{Ar}} = \sum_{[\rho] \in \pi_0 \mathcal{F}_G(M)} \frac{\langle \rho^* \alpha, [M] \rangle}{|\text{Aut}(\rho)|}$$

is Kim’s arithmetic Chern–Simons partition sum for the class α . If an arithmetic gluing $M = M_1 \cup_\Sigma M_2$ has finite field groupoids and satisfies étale descent

$$\mathcal{F}_G(M) \simeq \mathcal{F}_G(M_1) \times_{\mathcal{F}_G(\Sigma)} \mathcal{F}_G(M_2),$$

then $Z_{G,\alpha}^{\text{Ar}}$ satisfies the corresponding finite-groupoid gluing formula.

Proof. The construction uses only finite homotopy types. To a closed arithmetic 3-space M , assign the finite groupoid $\mathcal{F}_G(M)$ and integrate the locally constant function S_α^{Ar} by groupoid cardinality. If

$$M = M_1 \cup_\Sigma M_2$$

is a gluing along an arithmetic boundary, then etale descent for local systems gives a homotopy pullback

$$\mathcal{F}_G(M) \simeq \mathcal{F}_G(M_1) \times_{\mathcal{F}_G(\Sigma)} \mathcal{F}_G(M_2).$$

The fundamental class is additive under gluing, so the action multiplies:

$$\langle \rho^* \alpha, [M] \rangle = \langle \rho_1^* \alpha, [M_1] \rangle \langle \rho_2^* \alpha, [M_2] \rangle.$$

Fubini for finite groupoid cardinality then gives the displayed gluing formula. Kim's action is exactly the same evaluation of a pulled-back degree-3 class on the Artin–Verdier fundamental class, written in $\mathbb{Q}/\mathbb{Z}$ and exponentiated here into $\mathbb{C}^\times$. $\square$

Definition 81.4.3 (finite-level boundary category). For an arithmetic boundary Σ , let $\tau_\Sigma(\alpha)$ be the transgression of $\alpha \in H^3(BG, \mathbb{C}^\times)$ to a 2-cocycle on the field groupoid $\mathcal{F}_G(\Sigma)$. The finite-level boundary category is

$$\mathcal{B}_{G,\alpha}(\Sigma) := \text{Sh}_{\tau_\Sigma(\alpha)}(\mathcal{F}_G(\Sigma)),$$

the category of finite-dimensional $\tau_\Sigma(\alpha)$ -twisted local systems on $\mathcal{F}_G(\Sigma)$.

The partition function $Z_{G,\alpha}^{\text{Ar}}(M)$ is a scalar for closed M . For a manifold with boundary, the TQFT assignment must produce a category: the state category on the boundary. The finite boundary category provides this assignment.

THEOREM 81.4.4 (finite-level bulk-boundary span). (on a finite arithmetic cobordism category whose morphisms satisfy the gluing hypothesis of Chapter 81.4.2). The finite-level theory $Z_{G,\alpha}^{\text{Ar}}$ assigns $\mathcal{B}_{G,\alpha}(\Sigma)$ to the arithmetic boundary Σ . If M is a cobordism from Σ_0 to Σ_1 , then M acts by the integral transform

$$\text{Sh}_{\tau_{\Sigma_0}(\alpha)}(\mathcal{F}_G(\Sigma_0)) \longrightarrow \text{Sh}_{\tau_{\Sigma_1}(\alpha)}(\mathcal{F}_G(\Sigma_1))$$

whose kernel is the action local system on $\mathcal{F}_G(M)$. For a closed M , the categorical trace of this transform is $Z_{G,\alpha}^{\text{Ar}}(M)$.

Proof. Restriction of local systems gives a span

$$\mathcal{F}_G(\Sigma_0) \leftarrow \mathcal{F}_G(M) \rightarrow \mathcal{F}_G(\Sigma_1).$$

The transgressed cocycle records the boundary anomaly of the degree-3 class α ; tensoring by the action local system and then pushing forward along the right leg gives the displayed integral transform. Composition of cobordisms is composition of spans. The homotopy-pullback formula from Chapter 81.4.2 makes the two composite kernels canonically isomorphic. When the boundary is empty, the trace reduces to groupoid-cardinality integration over $\mathcal{F}_G(M)$, which is precisely $Z_{G,\alpha}^{\text{Ar}}(M)$. $\square$

PROPOSITION 8I.4.5 (*finite boundary Hochschild trace*). . For a finite groupoid $\mathcal{F}_G(\Sigma)$ over $\mathbb{C}$, the finite boundary category

$$\mathcal{B}_{G,\alpha}(\Sigma) = \text{Sh}_{\tau_\Sigma(\alpha)}(\mathcal{F}_G(\Sigma))$$

has Hochschild trace equal to the twisted inertia trace of $\mathcal{F}_G(\Sigma)$:

$$\text{HH}_0(\mathcal{B}_{G,\alpha}(\Sigma)) \cong \Gamma(\pi_0 I\mathcal{F}_G(\Sigma), L_{\tau_\Sigma(\alpha)}), \quad \text{HH}_{n>0}(\mathcal{B}_{G,\alpha}(\Sigma)) = 0.$$

Here $I\mathcal{F}_G(\Sigma)$ is the inertia groupoid, and $L_{\tau_\Sigma(\alpha)}$ is the transgressed local system on inertia.

Proof. A finite twisted sheaf category on a finite groupoid is Morita equivalent to modules over the corresponding finite-dimensional twisted groupoid algebra. Over $\mathbb{C}$ this algebra is separable: it is a finite product of twisted finite group algebras, hence semisimple. Hochschild homology of a separable algebra with coefficients in itself vanishes in positive degrees. Degree zero is the algebra modulo commutators, which is the space of twisted class functions on conjugacy classes, equivalently sections of the transgressed local system on the inertia groupoid. $\square$

COROLLARY 8I.4.6 (*Wilson lines at finite level*). . A prime p viewed as a knot in the Morishita dictionary supports Wilson defects labelled by representations of the local stabiliser group of a point of $\mathcal{F}_G(\text{Spec } \mathbb{Q}_p)$. For a one-dimensional label χ , the Wilson expectation is the corresponding Frobenius trace $\chi(\text{Frob}_p)$.

Proof. Removing the prime p and fixing monodromy around the meridian is the arithmetic analog of prescribing a conjugacy class around a knot. The defect category is the module category of the stabiliser of that local monodromy. A one-dimensional module evaluates a loop by its character, hence the meridian/Frobenius loop contributes $\chi(\text{Frob}_p)$. $\square$

CONJECTURE 8I.4.7 (*renormalised $GL_2(\widehat{\mathbb{Z}})$ quantum arithmetic CS*). . Let $G_N = GL_2(\mathbb{Z}/N\mathbb{Z})$, and let $\alpha_N \in H^3(BG_N, \mathbb{C}^\times)$ be a compatible tower of finite levels whose inverse-limit class is the arithmetic level $k^{\text{Ar}} = \kappa_G^{\text{Ar}}$. The finite theories $Z_{G_N, \alpha_N}^{\text{Ar}}$ admit a renormalised profinite limit

$$\text{ACS}_{k^{\text{Ar}}}^{\text{qu}} := \varprojlim_N^{\text{ren}} Z_{G_N, \alpha_N}^{\text{Ar}}$$

on the etale 3-space $\overline{\text{Spec } \mathbb{Z}}$. This limit satisfies three bulk-boundary axioms:

- (i) (Boundary WZW.) The 2D factorization algebra on the archimedean fibre $\{\infty\}$ produced by canonical quantisation of $\text{ACS}_{k^{\text{Ar}}}^{\text{qu}}$ has Hochschild trace $\mathcal{V}^{\text{prim}}$:

$$\text{HH}_0(\partial \text{ACS}_{k^{\text{Ar}}}^{\text{qu}}) \simeq \mathcal{V}^{\text{prim}}.$$

- (ii) (Partition function.) The bulk partition function equals the completed zeta character at the level: $Z_{\text{ACS}^{\text{qu}}}(\overline{\text{Spec } \mathbb{Z}}; k^{\text{Ar}}) = \xi(k^{\text{Ar}})$.
- (iii) (Wilson lines as Frobenius traces.) For a prime $p \in \mathbb{P}$ viewed as a knot in the Morishita dictionary, the Wilson-line observable in representation $\rho : \pi_1^{\text{ét}} \rightarrow GL_2$ reproduces the Frobenius trace: $\langle W_\rho(K_p) \rangle_{\text{ACS}^{\text{qu}}} = \text{tr } \rho(\text{Frob}_p)$.

Remark 81.4.8 (finite vs. profinite ACS: gap between theorem-grade and conjecture-grade). The closed finite-level state sum is theorem-grade: Chapter 81.4.2 constructs it from étale local systems, the Artin–Verdier fundamental class, and finite groupoid cardinality. Functorial finite-level TQFT language still requires a specified arithmetic cobordism category. The larger residual is the renormalised profinite limit $G_N \rightarrow GL_2(\widehat{\mathbb{Z}})$, the comparison of its archimedean boundary category with $\mathcal{V}^{\text{prim}}$, and the identification of its renormalised partition function with $\xi(k^{\text{Ar}})$.

There is a concrete obstruction to treating the closed finite tower as the zeta partition function. On the closed base $\overline{\text{Spec } \mathbb{Z}}$, finite étale local systems are trivial by Hermite–Minkowski, so the bare finite state sum for $G_N = GL_2(\mathbb{Z}/N\mathbb{Z})$ is the groupoid-cardinality scalar $|G_N|^{-1}$, or 1 after finite-gauge normalisation. This scalar does not contain prime weights p^{-s} , the archimedean $\Gamma_{\mathbb{R}}$ -factor, or the polar correction. A zeta-valued profinite ACS theory therefore requires extra data: compatible continuous level classes, a counterterm or measure system on the congruence tower, a profinite boundary category, a Morita comparison with the local Iwahori trace factors, and a regularised character map.

THEOREM 81.4.9 (bulk-boundary comparison with $\mathcal{V}^{\text{prim}}$). (on the renormalised profinite limit in Chapter 81.4.7 and on the local Satake/Iwahori trace identifications of Chapters 28 and 16). Let

$$\mathcal{B}_{\infty}^{\text{Ar}} := \varprojlim_N^{\text{ren}} \mathcal{B}_{G_N, a_N}(\{\infty\})$$

be the renormalised archimedean boundary category of the profinite ACS tower. If its finite-place compact generators identify with the local Iwahori Hecke generators and its archimedean vacuum sector identifies with the rank-2 Heisenberg Kac–Moody vacuum, then

$$\text{HH}_{\bullet}(\mathcal{B}_{\infty}^{\text{Ar}}) \simeq \mathcal{V}^{\text{prim}}.$$

Equivalently, the boundary factorization algebra of $\text{ACS}_{k^{\text{Ar}}}^{\text{qu}}$ is $\mathcal{V}^{\text{prim}}$.

Proof. At finite level, Chapter 81.4.4 identifies the boundary state category with twisted sheaves on the boundary field groupoid. Passing through the congruence tower gives a renormalised inverse system of such categories. The stated compact-generation hypothesis is exactly the assertion that the local factors of this system are the Iwahori Hecke categories used in the definition of $\mathcal{V}^{\text{prim}}$. Hochschild trace is local for the restricted tensor product under the compact-generation and spherical unit hypotheses of Chapter 28. Thus the finite places contribute the local trace factors $\mathcal{H}^{(p)}$, while the archimedean vacuum contributes $\mathcal{H}^{(\infty)}$. Their restricted tensor product is the Hochschild trace class $\mathcal{V}^{\text{prim}}$ by Chapter 16. This proves the displayed boundary identification, conditional precisely on the profinite renormalisation and the local trace comparison. $\square$

Remark 81.4.10 (boundary trace form). The finite boundary trace of Chapter 81.4.5 is theorem-grade, but it does not identify the boundary with $\mathcal{V}^{\text{prim}}$. The comparison

$$\text{HH}_{\bullet}(\mathcal{B}_{\infty}^{\text{Ar}}) \simeq \mathcal{V}^{\text{prim}}$$

requires an independent profinite Morita equivalence

$$\mathcal{B}_{\infty}^{\text{Ar}} \simeq_{\text{Morita}} \text{Hk}_{GL_2, \text{glob}}^{\text{Iw}},$$

where $\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}$ is the global arithmetic Iwahori–Hecke category (restricted tensor product of local Iwahori–Hecke categories with archimedean vacuum, defined in Chapter 16), compatible with the

chosen topology, completion, restricted tensor product, local Satake/Iwahori generators, archimedean vacuum, and the arithmetic Ben-Zvi–Nadler and wild-prime hypotheses. Until these data are proved, Chapter 81.4.9 is only a conditional implication.

Remark 81.4.11 (Vol II cross-volume bridge). Vol II Part VI states a 3D holomorphic-twist QFT on $C \times \mathbb{R}$ whose boundary factorization algebra on C is the complex $\mathcal{V}$. The arithmetic analog replaces the holomorphic twist by the Chern-Simons sector: the arithmetic base $\overline{\text{Spec } \mathbb{Z}}$ has étale cohomological dimension 3 without a holomorphic structure in the differential-geometric sense. The 3D topological gauge theory over the arithmetic base is therefore Chern-Simons, not holomorphic twist. The complex $(HT)_{C \times \mathbb{R}} \rightarrow \partial = \mathcal{V}$ adjunction becomes an arithmetic boundary category whose Hochschild trace is $\mathcal{V}^{\text{prim}}$.

81.5 THE ORDERED INCIDENCE BAR

The arithmetic Ran space lacks the finite-place collision and diagonal thickenings that the Beilinson–Drinfeld/Costello–Gwilliam framework uses to build a factorization homology model. The ordered incidence bar replaces the Ran space by the simplest category that admits a bar-complex computation while preserving the factorization structure: it encodes the primes of $\mathbb{Z}$ via the auxiliary total order $2 < 3 < 5 < 7 < \dots < \infty$, with the archimedean place placed last. This order is not geometric structure on $\overline{\text{Spec } \mathbb{Z}}$; the comparison with Beilinson–Drinfeld/Costello–Gwilliam arithmetic chiral homology is conditional (Chapter 81.5.5).

Definition 81.5.1 (ordered incidence category). Put $V = \mathbb{P} \cup \{\infty\}$, with the order $2 < 3 < 5 < \dots < \infty$. Let $\text{Inc}_<(V)$ be the category whose objects are finite ordered block decompositions of finite ordered subsets of V . A singleton block is written v , and the merge of two adjacent blocks is written by concatenation. Morphisms are generated by order-preserving adjacent contractions

$$\{v_1 < \dots < v_i < v_{i+1} < \dots < v_n\} \longrightarrow \{v_1 < \dots < (v_i v_{i+1}) < \dots < v_n\},$$

where $v_i v_{i+1}$ denotes the formal merged block carrying the product operation of the coefficient system, not a new arithmetic place. The simplicial nerve of $\text{Inc}_<(V)$ is the one-dimensional ordered collision model for the arithmetic Ran space.

Definition 81.5.2 (ordered incidence homology). Let $\mathcal{H} = \{\mathcal{H}^{(v)}\}_{v \in V}$ be the per-place coefficient system of $\mathcal{V}^{\text{prim}}$, with adjacent products

$$m_{v,w}: \mathcal{H}^{(v)} \otimes \mathcal{H}^{(w)} \longrightarrow \mathcal{H}^{(vw)} \quad (v < w).$$

Here $\mathcal{H}^{(vw)}$ means the coefficient assigned to the merged block vw . Let $F_{\mathcal{H}}: \text{Inc}_<(V) \rightarrow \text{Ch}_{\mathbb{C}}$ send an ordered block decomposition $B_1 | \dots | B_r$ to

$$F_{\mathcal{H}}(B_1 | \dots | B_r) := \mathcal{H}^{(B_1)} \otimes \dots \otimes \mathcal{H}^{(B_r)},$$

with contractions acting by the corresponding adjacent products. The ordered incidence homology is the derived colimit

$$\int_{\overline{\text{Spec } \mathbb{Z}}}^{\text{ord}} \mathcal{V}^{\text{prim}} := \text{L colim}_{\text{Inc}_<(V)} F_{\mathcal{H}}.$$

Definition 81.5.3 (ordered incidence bar). Let $\mathcal{V}^{\text{prim}}$ be the arithmetic Heisenberg with per-place factors $\mathcal{H}^{(p)}$ (finite primes; each $\mathcal{H}^{(p)}$ is an E_0 -algebra in the Drinfeld–Kapranov sense, i.e. an associative

algebra without a unit constraint and without a local-to-global principle, as defined in Ch. 4) and $\mathcal{H}^{(\infty)}$ (archimedean place; E_1 -chiral, meaning it carries a homotopy-unital associative structure compatible with the archimedean factorization). The *ordered incidence bar* is the normalized chain complex of the simplicial replacement of $F_{\mathcal{H}}$:

$$B_{\bullet}^{\text{ord}}(\mathcal{V}^{\text{prim}}) := N_{\bullet} \text{srep}_{\text{Inc}_{<}(V)}(F_{\mathcal{H}}).$$

Here $\text{srep}_{\text{Inc}_{<}(V)}(F_{\mathcal{H}})$ is the simplicial replacement of $F_{\mathcal{H}}$ (the standard bar construction on a functor from a small category to chain complexes), and $N_{\bullet}$ is the normalized chain functor on simplicial chain complexes. In degree n , this is the direct sum over n -simplices $C_0 \rightarrow \cdots \rightarrow C_n$ of $\text{Inc}_{<}(V)$ of $F_{\mathcal{H}}(C_0)$, modulo degeneracies. Faces are induced by composition in $\text{Inc}_{<}(V)$, hence by adjacent contractions. Thus all targets are coefficients of formal merged blocks, not new arithmetic places.

THEOREM 81.5.4 (*ordered incidence homology via normalized bar*). . The ordered incidence homology of $\mathcal{V}^{\text{prim}}$ is computed by the ordered incidence bar:

$$\int_{\text{Spec } \mathbb{Z}}^{\text{ord}} \mathcal{V}^{\text{prim}} \simeq H_{\bullet}(B_{\bullet}^{\text{ord}}(\mathcal{V}^{\text{prim}})).$$

Proof. This is the standard computation of a derived colimit by the normalized chain complex of the simplicial replacement of the coefficient functor $F_{\mathcal{H}}$. The incidence category was defined so that its generating morphisms are adjacent contractions; therefore the normalized differential is the alternating sum of the corresponding contraction maps. $\square$

COROLLARY 81.5.5 (*comparison with BD/CG chiral homology*). (on constructing an arithmetic Beilinson–Drinfeld/Costello–Gwilliam Ran model for $\overline{\text{Spec } \mathbb{Z}}$, finite-place collision thickenings, constructible descent, ordered-to-unordered descent, and completion exactness identifying the arithmetic chiral homology of Chapters 12 and 8 with the ordered incidence model). One has

$$\int_{\text{Spec } \mathbb{Z}} \mathcal{V}^{\text{prim}} \simeq H_{\bullet}(B_{\bullet}^{\text{ord}}(\mathcal{V}^{\text{prim}})).$$

Proof. The ordered theorem computes $\int^{\text{ord}} \mathcal{V}^{\text{prim}}$. The stated comparison hypothesis identifies the ordered incidence model with the BD/CG arithmetic chiral-homology functor. Substitution gives the displayed equivalence. $\square$

Remark 81.5.6 (*form of the BD/CG comparison*). The ordered incidence bar computes the derived colimit over $\text{Inc}_{<}(\mathbb{P} \cup \{\infty\})$ by Chapter 81.5.4. This is not full Beilinson–Drinfeld/Costello–Gwilliam arithmetic chiral homology. The comparison requires a non-terminal arithmetic Ran object, finite-place diagonal or collision thickenings, BD/CG descent, constructibility with an exit category homotopy-finally reached from the ordered incidence category, ordered-to-unordered descent, and exactness of the restricted products and completions used in the derived colimit. Until these hypotheses are supplied, Chapter 81.5.5 remains conditional.

COROLLARY 81.5.7 (*regularised character from the ordered bar*). (on Chapter 81.4.7(1–2) and Chapter 81.5.5, and on a regularised character map ch_{reg} from arithmetic chiral homology to scalar partition functions). The partition function of the conjectural 3D quantum arithmetic Chern–Simons is the regularised character of the ordered-bar homology:

$$Z_{\text{ACS}^{\text{qu}}}(\overline{\text{Spec } \mathbb{Z}}; k^{\text{Ar}}) = \text{ch}_{\text{reg}} H_{\bullet}(B_{\bullet}^{\text{ord}}(\mathcal{V}^{\text{prim}})) = \xi(k^{\text{Ar}}).$$

The character map is the missing type conversion: a scalar partition function is not identified with a homology object before applying ch_{reg} .

81.6 DENINGER SKELETON FROM ARITHMETIC CHIRAL HOMOLOGY

The ordered-bar computation of the finite Bost–Connes zero modes (Chapter 81.6.3) gives the prime-orbit Euler skeleton in the half-plane of absolute convergence. Deninger’s missing cohomology requires an additional $\mathbb{R}_+^\times$ -flow whose normal generator has regularised determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes trace distribution. That additional flow is the Hilbert–Pólya/Connes–Deninger realisation; it is not produced by the Bost–Connes heat flow.

Definition 81.6.1 (pre-Deninger arithmetic coefficient datum). . The *pre-Deninger arithmetic coefficient datum* is the datum

$$\mathcal{A}_{\text{Den}} := \left(\mathcal{A}_{\text{fin}} \otimes \mathcal{A}_{\infty}^{\text{Tate}} \right) \otimes \mathbf{P}^{\bullet}$$

with:

- (i) finite Bost–Connes zero modes

$$\mathcal{A}_{\text{fin}} := \bigotimes_p \mathcal{H}_p^{\text{BC}}, \quad \mathcal{H}_p^{\text{BC}} := \bigoplus_{r \geq 0} \mathbb{C} \delta_{p^r},$$

with $N_p \delta_{p^r} = r(\log p) \delta_{p^r}$ and $N_{\text{BC}} = \sum_p N_p$ on the restricted tensor product;

- (ii) the archimedean Tate-Gaussian sector $\mathcal{A}_{\infty}^{\text{Tate}}$, whose Mellin character is $\Gamma_{\mathbb{R}}(s) := \pi^{-s/2} \Gamma(s/2)$ (the standard archimedean Euler factor);
- (iii) a polar Tate complex $\mathbf{P}^{\bullet} = \mathbb{C}_0[0] \oplus \mathbb{C}_1[2]$, where the indicated subscripts record the formal Frobenius weights 0 and 1;
- (iv) the Bost–Connes Euler flow $\phi_{\text{BC}}^t := \exp(-t N_{\text{BC}})$ on the finite zero modes, tensored with the usual Tate scaling on $\mathcal{A}_{\infty}^{\text{Tate}}$.

The last flow records closed-orbit lengths $\log p$. It is not the Deninger Frobenius flow on the conjectural middle cohomology.

Definition 81.6.2 (candidate chiral Deninger complex). . Work in a fixed complete nuclear dg category where the restricted tensor products below and compactly supported continuous duals are defined. Put

$$C_{\text{core}, \chi}^{\bullet} := \text{Hom}_{\text{cont}, c} \left(B_{\bullet}^{\text{ord}}(\mathcal{A}_{\text{fin}} \widehat{\otimes} \mathcal{A}_{\infty}^{\text{Tate}}), \mathbb{C} \right).$$

The *candidate chiral Deninger complex* is the total direct sum

$$C_{\text{Den}, \chi}^{\bullet} := \text{Tot} \left(C_{\text{core}, \chi}^{\bullet} \oplus \mathbf{P}^{\bullet} \right),$$

where $\mathbf{P}^{\bullet}$ has zero differential and contributes formal Tate lines in degrees 0 and 2. Its cohomology is denoted

$$H_{\text{Den}, \chi}^{\bullet} := H^{\bullet}(C_{\text{Den}, \chi}^{\bullet}).$$

The primitive middle part $H_{\text{Den}, \chi, \text{prim}}^1$ is $H^1(C_{\text{core}, \chi}^{\bullet})$. No theorem is asserted here that the full complex has only the Deninger degrees 0, 1, 2, or that this dual agrees with compactly supported Artin–Verdier cohomology.

PROPOSITION 81.6.3 (*unconditional Bost–Connes Euler skeleton*). . The finite Bost–Connes zero-mode sector has ordinary KMS character

$$\mathrm{Tr}(e^{-sN_{\mathrm{BC}}} | \mathcal{A}_{\mathrm{fin}}) = \prod_p \sum_{r \geq 0} p^{-rs} = \prod_p (1 - p^{-s})^{-1} = \zeta(s) \quad (\mathrm{Re} s > 1),$$

and logarithmic differentiation gives the prime-orbit sum

$$-\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{r \geq 1} (\log p) p^{-rs}.$$

This is the raw non-archimedean Euler/log-prime skeleton in the half-plane of absolute convergence.

Proof. On $\mathcal{H}_p^{\mathrm{BC}}$, $\mathrm{Tr}(e^{-sN_p}) = \sum_{r \geq 0} e^{-sr \log p}$. The restricted tensor product multiplies these traces over primes, giving the Euler product for $\zeta(s)$ in the half-plane of convergence. Logarithmic differentiation turns each local factor into the Dirichlet series over iterates of the closed orbit at p , with length $\log p$. $\square$

Remark 81.6.4 (*gap between Euler/log-prime skeleton and Weil–Guinand distribution*). Chapter 81.6.3 is not yet the Weil–Guinand distribution. Passing from $-\zeta'/\zeta(s)$ in $\mathrm{Re} s > 1$ to the finite-place term on critical-line test functions requires a separate contour shift, Fourier/Mellin convention, the $n^{-1/2}$ weight, the sign and factor 2, plus the archimedean and polar terms. Those analytic normalisations are part of the trace-formula residual, not part of the ordered-bar computation.

Conjecture 81.6.5 (*Deninger spectral-flow / Hilbert–Pólya hypothesis*). . There exists a Hilbert completion of $H_{\mathrm{Den}, \chi, \mathrm{prim}}^1$ and a closed operator

$$\Theta_{\mathrm{Den}} = \frac{1}{2} \mathrm{id} + iH_{\mathrm{Den}}$$

with H_{Den} self-adjoint in the Petersson/GNS inner product, such that:

(i) regularised determinant:

$$\det_{\infty}(\mathrm{id} - \Theta_{\mathrm{Den}} | H_{\mathrm{Den}, \chi, \mathrm{prim}}^1) = \xi_{\mathbb{R}}(s);$$

(ii) Weil–Connes trace distribution:

$$\mathrm{Tr}_{\mathrm{reg}}(h(H_{\mathrm{Den}})) = \sum_{\rho} h(\gamma_{\rho}), \quad \rho = \frac{1}{2} + i\gamma_{\rho},$$

for every admissible Paley–Wiener test function h ;

- (iii) the preceding notation is conditional on the determinant identity: the self-adjoint trace model and $\det_{\infty}(\mathrm{id} - \Theta_{\mathrm{Den}}) = \xi_{\mathbb{R}}(s)$ force the zero divisor onto the critical line;
- (iv) Lefschetz compatibility: the trace distribution in (2), after moving the contour across ξ'/ξ , equals the prime-orbit sum of Chapter 81.6.3, the archimedean Tate–Gamma correction, and the two polar terms from $\mathbf{P}^{\bullet}$.

This hypothesis is exactly the Hilbert–Pólya/Connes–Deninger determinant-trace realisation in the present chiral-homology language. In particular, self-adjointness of H_{Den} together with the determinant identity is already RH-strength zero-placement for $\xi_{\mathbb{R}}$.

PROPOSITION 81.6.6 (*conditional formal Deninger datum*). (on Chapter 81.6.5 and on the topological-dual choices in Chapter 81.6.2). The candidate cohomology $H_{\text{Den}, \chi}^\bullet$, equipped with Θ_{Den} on its primitive middle part and with the two Tate polar lines in degrees 0 and 2, carries Deninger’s formal regularised-determinant and Lefschetz-trace identities for $\overline{\text{Spec } \mathbb{Z}}$.

Proof. The finite sector gives only the Euler/log-prime skeleton of Chapter 81.6.3. The archimedean and polar entries specify where the Tate-Gamma and $s = 0, 1$ terms would live in a cohomological Lefschetz formula. The spectral-flow hypothesis supplies the missing middle operator, its determinant, its zero-side trace distribution, and the Lefschetz comparison. Thus the claim is a conditional implication: once the spectral-flow data is assumed, the displayed cohomological object carries the formal slots required by Deninger’s axioms. It is not an independent construction of Θ_{Den} . $\square$

PROPOSITION 81.6.7 (*what the construction does not prove*). . The ordered-bar construction of $C_{\text{Den}, \chi}^\bullet$ does not by itself construct Θ_{Den} , prove the determinant identity, prove the Weil–Guinand trace formula, fix the cohomological superdeterminant signs of the polar lines, or prove positivity/purity. These requirements are exactly the Hilbert–Pólya/Connes–Deninger obstruction of Chapter 22.3.2 plus the analytic trace-formula normalization.

Proof. The ordered bar is determined by the local factors $\{\mathcal{H}_p^{\text{BC}}\}_p$, the archimedean Tate sector, and the polar complex. This data gives the Euler product and hence the prime side in $\text{Re } s > 1$. The zero-side operator requires a self-adjoint determinant-trace realisation whose determinant is $\xi_{\mathbb{R}}$. Chapter 22 names that assertion H-P^{Ar} and records that it is not derivable from the character identity, the formal Virasoro action, or the Petersson form alone. Thus the chiral construction supplies a candidate complex and the prime-side Euler skeleton. The spectral-flow hypothesis is the remaining essential input and has RH-level strength. $\square$

81.7 ARITHMETIC-GRAVITY CONSTRUCTION TO F-RH

Witten 2007 (*Three-Dimensional Gravity Revisited*, arXiv:0706.3359) proposed that 3D quantum gravity admits a Chern-Simons formulation whose partition function’s spectrum controls the zeros of arithmetically significant L -functions. Vol II Part VI states this as the complex-side 3D quantum gravity climax. The arithmetic base $\overline{\text{Spec } \mathbb{Z}}$ supplies an analogous conditional formulation via the preceding ACS construction.

THEOREM 81.7.1 (*arithmetic-gravity F-RH*). (on Chapter 81.4.7, Chapter 81.4.9, Chapter 81.5.5, and Chapter 22’s *three-pillar + VB structure*, including a holomorphic complex-level partition function). The zeros of $Z_{\text{ACS}^{\text{qu}}}(\text{Spec } \mathbb{Z}; k^{\text{Ar}})$ as a function of the level $k^{\text{Ar}} \in \mathbb{C}$ coincide with the non-trivial zeros of $\xi(k^{\text{Ar}})$. The zero-placement claim is RH-level: these zeros lie on the self-dual locus $\text{Re}(k^{\text{Ar}}) = 1/2$ exactly when the corresponding Riemann-hypothesis zero-placement statement holds. The stronger F-RH condition still includes the positive Hilbert–Pólya trace inputs of Chapter 22.

Remark 81.7.2 (*three-step construction*). The construction proceeds in three steps.

- (i) *Bulk construction.* Finite-level ACS is constructed in Chapter 81.4.2. The open bulk step is the renormalised profinite limit $\varprojlim_N^{\text{ren}} Z_{G_N, \alpha_N}^{\text{Ar}}$ and its zeta partition function, as stated in Chapter 81.4.7.
- (ii) *Boundary identification and ordered-model computation.* Finite-level boundary categories are constructed in Chapter 81.4.4 after choosing the finite arithmetic cobordism category. The comparison

with $\mathcal{V}^{\text{prim}}$ is Chapter 81.4.9. The ordered-bar calculation is theorem-grade for the ordered incidence model by Chapter 81.5.4; the remaining comparison to BD/CG arithmetic chiral homology is Chapter 81.5.5.

- (iii) *Positivity bootstrap.* Apply the VB pillar of Chapter 22. The ordered model and arithmetic Zhu modularity do not by themselves supply positivity; the missing input is the positive Hilbert–Pólya trace/VB statement.

81.8 GENUS- g EXTENSION

At genus g , the arithmetic Kac–Moody of Chapter 81.3.1 becomes rank- $2g$ sourced by the all-genus conformal vector of Chapter 80. The quantum CS becomes a family $\text{ACS}_{k^{\text{Ar}},g}^{\text{qu},g}$ parameterized by the genus- g datum $\mathcal{D}_g$ (Chapter 80.1.1); the gauge group is $GL_{2g}(\widehat{\mathbb{Z}})$; the boundary WZW is $\mathcal{V}^{\text{prim},g}$; the ordered bar is built on $\mathbb{P} \cup \{\infty\}$ with rank- $2g$ Fock factors per place. The partition function is conjecturally $\Lambda_{\mathcal{D}_g}(k^{\text{Ar},g})$ per Chapter 80.1: Dedekind $\xi_K(k^{\text{Ar},g})$ for Window II, Hasse–Weil $\Lambda(k^{\text{Ar},g}, X_K)$ for Windows I/V, Siegel spin $\Lambda(k^{\text{Ar},g}, \pi, \text{spin})$ for Window VI.

COROLLARY 81.8.1 (*all-genus arithmetic-gravity F-RH*). (*on the genus- g extensions of Chapter 81.4.7, Chapter 81.4.9, Chapter 81.5.5, plus Chapter 80.5.1*). The zeros of $Z_{\text{ACS}^{\text{qu},g}}(\overline{\text{Spec } \mathbb{Z}}; k^{\text{Ar},g})$ coincide with the non-trivial zeros of $\Lambda_{\mathcal{D}_g}(s)$, and lie on the self-dual locus $\text{Re}(k^{\text{Ar},g}) = w_{\mathcal{D}_g}/2$ for $w_{\mathcal{D}_g} \in \{1, 2, 2g+1\}$ the motivic weight center of the relevant window.

81.9 RESIDUAL CONDITIONS

Six conditions specific to the arithmetic-gravity construction, in addition to Chapter 80’s residual list.

- (i) *Finite arithmetic cobordism category.* . The closed finite-level state sum, gluing formula, and finite boundary trace are Chapter 81.4.2 and Chapter 81.4.5. A full finite-level TQFT still needs the arithmetic cobordism category and its finiteness hypotheses written explicitly.
- (ii) *Profinite ACS renormalisation.* . Construct a canonical renormalised limit of the $GL_2(\mathbb{Z}/N\mathbb{Z})$ arithmetic CS tower, with compatible continuous level classes, counterterms, and scalar partition

$$Z_{\text{ACS}^{\text{qu}}}(\overline{\text{Spec } \mathbb{Z}}; k^{\text{Ar}}) = \xi(k^{\text{Ar}}).$$

Finite closed state sums do not supply this data.

- (iii) *Boundary trace comparison.* . Construct an independent Morita equivalence

$$\mathcal{B}_{\infty}^{\text{Ar}} \simeq_{\text{Morita}} \text{Hk}_{GL_{2,\text{glob}}}^{\text{Iw}},$$

compatible with restricted tensor products, spherical units, local Satake/Iwahori generators, and the archimedean vacuum.

- (iv) *BD/CG ordered-model comparison.* . Construct arithmetic Ran/BD–CG descent whose exit or disk category is homotopy-finally computed by $\text{Inc}_{<}(\mathbb{P} \cup \{\infty\})$, including finite-place collision thickenings, ordered-to-unordered descent, and completion exactness.
- (v) *Positive trace/VB core.* . Prove the full positive Hilbert–Pólya trace, equivalently the full A-TP/VB input on the complete Paley–Wiener cone. The ordered model, Bost–Connes Euler skeleton, and formal Fock positivity away from the zero-mode sector do not supply this RH-strength positivity input.

- (vi) *Deninger spectral-flow hypothesis*. . Upgrade the ordered-bar prime skeleton of Chapter 81.6.2 to a genuine middle cohomology Hilbert completion, with dense invariant core and a self-adjoint H_{HP} such that $\Theta_{\xi} = \frac{1}{2}\mathrm{id} + iH_{\mathrm{HP}}$ has determinant $\xi_{\mathbb{R}}$ and the Weil–Connes trace distribution. This is the Hilbert–Pólya/Connes–Deninger determinant-trace problem in chiral-homology form.

The arithmetic-gravity construction has two closed unconditional components: the closed finite-level state sums of Chapter 81.4.2 and the ordered incidence-bar calculation of Chapter 81.5.4. Together these yield the Euler/log-prime skeleton (Chapter 81.6.3) without further hypothesis. The remaining inputs are finite cobordism functoriality, profinite renormalisation, local trace comparison, BD/CG comparison, positivity, and the Deninger spectral-flow hypothesis (Chapter 81.9).

81.10 ORDERED-BAR BOUNDARY

Two first-principles results close unconditionally. First, closed finite-level arithmetic Chern–Simons state sums and their gluing law follow from étale local-system groupoids and Artin–Verdier integration alone (Chapter 81.4.2). Second, the ordered incidence bar computes the derived colimit of $\mathcal{V}^{\mathrm{prim}}$ over the auxiliary ordered place set via the normalized simplicial replacement (Chapter 81.5.4); this gives the Bost–Connes Euler/log-prime skeleton (Chapter 81.6.3) unconditionally.

The conditional structure has three layers. The profinite $GL_2(\widehat{\mathbb{Z}})$ quantum ACS, its boundary trace

$$\mathrm{HH}_0(\partial \mathrm{ACS}_{k^{\mathrm{Ar}}}^{\mathrm{qu}}) \simeq \mathcal{V}^{\mathrm{prim}},$$

and its zeta partition function $Z = \xi(k^{\mathrm{Ar}})$ are conjectural (Chapter 81.4.7, Chapter 81.4.9, Chapter 81.5.7). The comparison of the ordered incidence model with Beilinson–Drinfeld/Costello–Gwilliam arithmetic chiral homology requires finite-place collision thickenings, descent, and completion exactness (Chapter 81.5.5). The Deninger spectral-flow hypothesis, equivalently the Hilbert–Pólya/Connes–Deninger determinant-trace realisation, is not produced by the ordered bar and has RH-level strength (Chapter 81.6.5, Chapter 81.6.7).

The arithmetic-gravity construction conditionally reformulates the analytic-number-theoretic F-RH question as a 3D topological-gauge-theoretic question on the étale 3-space $\mathrm{Spec} \mathbb{Z}$. The reformulation acquires unconditional content only after the six residuals of Chapter 81.9 are supplied; the Deninger spectral-flow hypothesis carries the full Hilbert–Pólya weight.

DENINGER'S ARITHMETIC COHOMOLOGY AS A CHIRAL CANDIDATE

Deninger's programme demands a cohomology theory for $\overline{\text{Spec } \mathbb{Z}}$ whose regularised Fredholm determinant equals the completed Riemann zeta function and whose closed orbits recover the explicit formula. The bar-cobar programme does not construct this cohomology. It constructs a candidate: the ordered-incidence homology of $\mathcal{V}^{\text{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori Hecke category of GL_2), two formal Tate polar lines, and the Bost–Connes Euler-prime-side character. Whether this candidate realizes Deninger's axioms depends on the spectral-flow hypothesis of Chapter 81.6.5. The central result of this chapter is the obstruction theorem: no unconditional triple identification exists among Deninger's conjectural cohomology, the ordered-incidence candidate, and the boundary homology of the arithmetic Chern–Simons bulk.

82.1 DENINGER'S WISH LIST

Deninger's programme asks for a cohomology theory

$$H_{\Delta}^{\bullet}(\overline{\text{Spec } \mathbb{Z}})$$

with three cohomological degrees, a Frobenius/flow operator Θ whose regularised determinant has the non-trivial zero divisor, a Lefschetz trace formula whose closed-orbit side is the prime side of the explicit formula, Poincaré duality, and purity. Under zero-placement this may be written as $\Theta = \frac{1}{2} + iH$ with H self-adjoint. For the Riemann zeta function the middle degree should carry the non-trivial zero divisor.

In the next display $\zeta(s)$ follows Deninger's convention: it is the completed zeta function, not the bare Euler product.

$$\zeta(s) \stackrel{?}{=} \det_{\infty}(s \cdot \text{id} - \Theta \mid H_{\Delta}^{\bullet})^{(-1)^{\bullet}}.$$

Here $\det_{\infty}$ is Deninger's regularised Fredholm determinant, $\zeta(s)$ denotes the completed Hasse–Weil zeta (including the archimedean factor), and the sign $(-1)^{\bullet}$ is the Euler-characteristic grading. The Lefschetz trace formula takes the form

$$\sum_{\rho} h(\rho) \stackrel{?}{=} \sum_{\bullet} (-1)^{\bullet} \text{tr}(h(\Theta) \mid H_{\Delta}^{\bullet}) \stackrel{?}{=} \text{polar terms} + \text{archimedean term} - \sum_{p,k} (\log p) p^{-k/2} \widehat{h}(k \log p),$$

where h is a Schwartz test function on $\mathbb{R}$ and $\widehat{h}$ its Fourier transform.

Remark 82.1.1 (absence of a construction). The axioms above are not a construction. They specify what the missing arithmetic cohomology must do. The construction provides the prime-side Euler data (via Bost–Connes) and the formal polar lines (by insertion); the zero-side operator is not constructed.

Remark 82.1.2 (Vol IV scoping). Chapter 81 isolates the correct boundary: the ordered-incidence bar is theorem-grade as an algebraic model, while the comparison with genuine BD/CG chiral homology, the

profinite quantum ACS theory, and the Deninger spectral flow remain conditional. This chapter keeps that boundary fixed.

82.2 SURVIVING INPUTS

The bar-cobar programme produces four inputs with theorem-grade form; the remainder of the Deninger axioms are either conditional or open.

- (i) The finite-level arithmetic Chern–Simons state sum and gluing formula of Chapter 81.4.2. This is a closed finite groupoid state-sum statement, not a full profinite TQFT.
- (ii) The ordered-incidence bar Chapter 81.5.3 and its normalized-bar computation Chapter 81.5.4. This computes the explicitly defined incidence model, not automatically BD/CG factorization homology on $\overline{\text{Spec } \mathbb{Z}}$.
- (iii) The Bost–Connes Euler skeleton Chapter 81.6.3. It gives the KMS character $\zeta(s)$ and the prime-power logarithmic derivative in $\text{Re}(s) > 1$.
- (iv) The Deninger spectral-flow hypothesis Chapter 81.6.5: the zero-side determinant divisor, trace distribution, and Hilbert–Pólya self-adjoint determinant-trace operator. This is the single missing piece that the bar-cobar programme does not supply.

82.3 THE CHIRAL CANDIDATE

Given the four inputs above, one may assemble a graded object that mimics Deninger's three-degree shape. The question is whether this assembly is genuinely Deninger's cohomology or only a formal approximation to it.

Let $B_{\bullet}^{\text{ord}}(\mathcal{V}^{\text{prim}})$ denote the ordered-incidence bar complex of Chapter 81.5.3 applied to $\mathcal{V}^{\text{prim}}$.

Definition 82.3.1 (arithmetic chiral homology candidate). . The *arithmetic chiral homology candidate* is

$$H_{\text{ord}}^{\bullet}(\overline{\text{Spec } \mathbb{Z}}, \mathcal{V}^{\text{prim}}) := H^{\bullet}(B_{\bullet}^{\text{ord}}(\mathcal{V}^{\text{prim}})).$$

It is an ordered-incidence model. Calling it genuine arithmetic chiral homology requires the separate BD/CG comparison of Chapter 81.5.5.

Definition 82.3.2 (Deninger chiral candidate). . The *Deninger chiral candidate* is the candidate complex of Chapter 81.6.2:

$$H_{\Delta, \chi}^{\bullet}(\overline{\text{Spec } \mathbb{Z}}) := H_{\text{Den}, \chi}^{\bullet}(\overline{\text{Spec } \mathbb{Z}}).$$

It consists of the ordered-incidence core, completed continuous duals, and formal Tate polar lines in degrees 0 and 2.

Remark 82.3.3 (three labels, not one theorem). Three nearby objects appear: Deninger's conjectural cohomology, the ordered-incidence chiral candidate, and the boundary state space of the conjectural 3D ACS bulk. Their equality is a conditional assertion, not a proved identification.

PROPOSITION 82.3.4 (conditional Deninger comparison). (on Chapter 81.6.5, the BD/CG comparison Chapter 81.5.5, and the topological-dual choices in Chapter 81.6.2). The chiral candidate $H_{\Delta, \chi}^{\bullet}$, together with the two Tate polar lines and the spectral-flow operator assumed in Chapter 81.6.5, carries Deninger's formal axioms for $\overline{\text{Spec } \mathbb{Z}}$.

Proof. The finite-prime Euler terms are exactly the Bost–Connes series of Chapter 81.6.3. The polar lines are inserted by definition in Chapter 81.6.2. The zero-side operator, determinant identity, trace formula, and purity are precisely the extra data demanded by Chapter 81.6.5. Thus the statement is a conditional implication, not an independent construction of the missing cohomology. $\square$

82.4 NO TRIPLE IDENTIFICATION YET

THEOREM 82.4.1 (*no unconditional triple identification*). . No unconditional equality holds among Deninger’s conjectural cohomology, the ordered-incidence chiral candidate, and the boundary homology of ACS^{qu} .

Proof. The first object is conjectural. The second is an explicitly defined ordered-incidence homology object. The third depends on the profinite quantum ACS conjecture and on a boundary comparison with $\mathcal{V}^{\text{prim}}$. The only theorem-grade equality is the normalized-bar computation of the ordered-incidence model. The BD/CG comparison and spectral-flow identifications are listed as residual conditions in Chapter 81.9. $\square$

COROLLARY 82.4.2 (*conditional triangle*). (*on the profinite ACS conjecture, the BD/CG ordered-model comparison, a regularised character map, and the Deninger spectral-flow hypothesis*). These assumptions yield

$$Z_{\text{ACS}^{\text{qu}}}(\overline{\text{Spec } \mathbb{Z}}; k^{\text{Ar}}) = \xi(k^{\text{Ar}}) = \text{ch}_{\text{reg}}(H_{\Delta, \chi}^{\bullet})(k^{\text{Ar}}), \quad (82.1)$$

where $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed Riemann zeta function satisfying $\xi(s) = \xi(1-s)$, k^{Ar} is the arithmetic level parameter, and ch_{reg} denotes the regularised graded trace (the Θ -eigenvalue generating series with Deninger regularisation). Without those assumptions, (82.1) is a conditional shorthand.

82.5 FROBENIUS AND FLOW

The Bost–Connes Euler skeleton provides prime-length data at finite primes. Deninger’s programme requires a global Frobenius/flow operator acting on all three cohomological degrees. The question is whether the Euler skeleton and the archimedean scaling generator combine into such an operator.

Definition 82.5.1 (*finite Euler operators*). . At a finite prime p , the Bost–Connes Adams operation ψ_p records the prime length $\log p$ and contributes the Euler factor $(1 - p^{-s})^{-1}$. It is denoted here by

$$F_p^{\text{Euler}} := \psi_p.$$

This is an Euler-skeleton operator, not by itself Deninger’s global Frobenius on middle cohomology.

PROPOSITION 82.5.2 (*finite Wilson/Frobenius comparison is conditional*). (*on the Wilson-line axiom in Chapter 81.4.7*). A Wilson loop around the Morishita knot attached to p should recover the corresponding Frobenius trace. This is part of the conjectural ACS datum, not a consequence of the ordered bar.

Definition 82.5.3 (*archimedean scaling candidate*). . The archimedean scaling generator is the Mellin infinitesimal $-\partial_t$ on $\mathbb{R}_+$. The candidate archimedean flow on $H_{\Delta, \chi}^{\bullet}$ is denoted $\Theta_{\infty, \chi}$. In Deninger’s

and Connes' formulation, the target operator has the form $\Theta_\xi = \frac{1}{2}\text{id} + iH_{\text{HP}}$, where H_{HP} is a self-adjoint Hilbert–Polya operator (to be constructed) whose normal determinant divisor is $\text{div } \xi_{\mathbb{R}}$. After zero-placement its Weil–Connes trace is supported at the ordinate multiset $\{\gamma_\rho\}$. The identification

$$\Theta_{\infty, \chi} \stackrel{?}{=} \Theta_\xi$$

is the spectral-flow hypothesis and is not available unconditionally.

PROPOSITION 82.5.4 (*Virasoro conjugacy is not available*). (*on the spectral-flow hypothesis*). Without the spectral-flow theorem of Chapter 81.6.5, the Virasoro L_o^{Ar} , the Bost–Connes logarithmic Hamiltonian, and the Hilbert–Polya operator H_{HP} remain distinct operators. Their identification requires the spectral-flow theorem as input.

Definition 82.5.5 (*formal global flow*). . The formal global flow symbol is

$$F_\chi := \Theta_{\infty, \chi} \oplus \bigoplus_p F_p^{\text{Euler}}.$$

The notation records finite-prime lengths and archimedean scaling. It does not assert a well-defined Deninger Frobenius on a completed cohomology theory.

THEOREM 82.5.6 (*global Frobenius obstruction*). . The ordered-incidence bar and Bost–Connes Euler skeleton do not construct Deninger's global Frobenius operator.

Proof. The Bost–Connes side gives Euler factors and the prime-power series in the half-plane of convergence. Deninger's operator must act on the middle cohomology, must have regularised determinant divisor $\text{div } \xi_{\mathbb{R}}$, and must participate in a Lefschetz trace formula. That zero-side operator is exactly the missing spectral-flow input of Chapter 81.6.5. $\square$

Remark 82.5.7 (*archimedean Wilson surface*). The phrase “archimedean Wilson surface” is useful only as a conditional dictionary entry inside the profinite ACS conjecture. It is not an independent construction of Θ_ξ .

82.6 CHARACTER, DEGREES, AND PURITY

The Frobenius obstruction theorem leaves open the question of what the candidate complex does compute. The regularised graded trace and the formal Tate structure answer this question partially: the Euler product and the polar lines are present; the zero-side is missing.

THEOREM 82.6.1 (*character identity is conditional*). (*on Chapter 81.5.7 and Chapter 81.6.5*). The regularised character of the Deninger chiral candidate is $\xi(s)$ only after the partition-function comparison and the zero-side spectral-flow operator are assumed.

$$\text{ch}_{\text{reg}}(H_{\Delta, \chi}^\bullet)(s) \stackrel{\text{cond}}{=} \xi(s). \quad (82.2)$$

Remark 82.6.2 (*normalisation*). The additive determinant form $\det_\infty(s \cdot \text{id} - \Theta)$ and the multiplicative Euler-character form are interchangeable only after the operator Θ , the polar signs, and the regularisation convention have been fixed.

PROPOSITION 82.6.3 (*formal Tate polar lines*). . The candidate complex contains formal one-dimensional Tate lines in degrees 0 and 2. This accounts for the polar terms in the Deninger skeleton but does not prove that the total complex has only Deninger's three degrees.

PROPOSITION 82.6.4 (*middle cohomology is the hard part*). (on Chapter 81.6.5). The middle cohomology $H_{\Delta, \chi}^1$ admits a Hilbert-space completion with a self-adjoint generator having spectrum $\{\gamma_\rho\}$ only after Chapter 81.6.5 is assumed. This is the Hilbert–Polya hypothesis in cohomological form.

Remark 82.6.5 (*critical-line locus*). Purity of the middle degree is the assertion that the zero-side operator has real part $1/2$. For zeta this is RH in cohomological language.

82.7 DUALITY AND POSITIVITY

Koszul duality for the arithmetic bar complex gives the functional-equation symmetry $s \leftrightarrow 1 - s$. The question is whether this duality implies Poincaré duality for the Deninger cohomology, and specifically whether it forces the middle degree to be positive definite.

THEOREM 82.7.1 (*duality gives the functional equation, not positivity*). (on the relevant Koszul and ACS comparison hypotheses). The Koszul self-duality framework accounts for the functional-equation symmetry $s \leftrightarrow 1 - s$. It does not by itself imply positivity of the Deninger middle pairing or RH.

82.8 PURITY AS THE RH-LEVEL INPUT

Duality accounts for the functional equation but not for the zero-placement. Purity of the middle cohomological degree is the assertion that the zero-side operator Θ_ξ is self-adjoint with spectrum on the line $\text{Re}(s) = 1/2$, which is RH.

THEOREM 82.8.1 (*purity is RH-level*). (on the spectral-flow/Hilbert–Polya realisation). Deninger purity for $H_{\Delta, \chi}^1$, Hilbert–Polya self-adjointness for Θ_ξ , and the RH zero-placement statement are equivalent forms of the same missing input.

COROLLARY 82.8.2 (*Deninger-chiral RH construction*). . The chiral construction can prove RH only by proving the spectral-flow/positivity input. The ordered-incidence bar and the Bost–Connes Euler skeleton reduce the shape of the problem; they do not remove it.

82.9 ALL-GENUS EXTENSION

The preceding sections address $\overline{\text{Spec } \mathbb{Z}}$, which is the genus-zero case corresponding to the Riemann zeta function. Higher-genus arithmetic data arise for arithmetic surfaces and motives of higher weight; the formal extension records what the candidate construction produces in those cases.

Definition 82.9.1 (*genus- g chiral Deninger candidate*). . A genus- g arithmetic datum $\mathcal{D}_g$ is a smooth projective arithmetic surface of geometric genus g together with a choice of polarisation and L -function data (see Chapter 80.1.1). The corresponding $\mathcal{V}^{\text{prim}, g}$ is the Hochschild trace class of the identity on the genus- g arithmetic Hecke category, the rank-2 g extension of the rank-2 primary $\mathcal{V}^{\text{prim}}$ of Chapter 2.1. The genus- g chiral Deninger candidate is

$$H_{\Delta, \chi, g}^\bullet := H_{\text{ord}}^\bullet(\mathcal{D}_g, \mathcal{V}^{\text{prim}, g}).$$

The weight of $\mathcal{D}_g$, denoted $w_{\mathcal{D}_g}$, is the motivic weight of its L -function; purity is the assertion that $\text{Re}(s) = w_{\mathcal{D}_g}/2$ for all non-trivial zeros.

THEOREM 82.9.2 (*all-genus Deninger comparison*). (on the all-genus ACS, ordered-model comparison, and spectral-flow hypotheses). The genus- g candidate carries the expected motivic functional-equation

centre and prime-side Euler data. It does not construct the genus- g Deninger cohomology without the same spectral-flow input.

Remark 82.9.3 (weight convention). The expected purity line is $\operatorname{Re}(s) = w_{\mathcal{D}_g}/2$. This is a target condition, not a theorem of the ordered bar.

82.10 DICTIONARY

Table 82.1 records which Deninger axioms the chiral candidate supplies and which remain open or conditional.

Deninger target	Chiral/ACS object	Form
finite prime orbit lengths cohomology object	Bost–Connes Euler skeleton ordered-incidence candidate plus polar lines	proved in $\operatorname{Re}(s) > 1$ defined, not identified with Deninger cohomology
global Frobenius/flow determinant = ξ	spectral-flow operator Θ_ξ regularised character	open, Hilbert–Polya strength conditional
Lefschetz trace formula	prime-power trace distribution	open
purity	real part $1/2$ for middle spectrum	RH-level

TABLE 82.1: Surviving Deninger–chiral dictionary.

82.11 WHAT CLOSES

The preceding sections separate four unconditionally proved results from six conditional or open items. The separation is the chapter's main output.

Closed

- (i) Ordered-incidence homology is computed by the normalized ordered bar.
- (ii) The finite-prime Bost–Connes side gives the Euler product and logarithmic derivative in the convergence half-plane.
- (iii) The formal Tate polar lines can be inserted as part of the candidate complex.
- (iv) Koszul self-duality explains the functional-equation symmetry.

Conditional

- (i) Profinite quantum arithmetic Chern–Simons.
- (ii) Boundary comparison with $\mathcal{V}^{\text{prim}}$.
- (iii) BD/CG comparison with the ordered-incidence model.
- (iv) Regularised character equality with ξ .
- (v) Spectral flow/Hilbert–Pólya zero-divisor operator.
- (vi) Positivity/purity of the middle degree.

82.12 RESIDUAL CONDITIONS

The residue is a single mathematical obstruction: construct the middle operator and prove the trace formula with positivity. Equivalently, prove the spectral-flow hypothesis of Chapter 81.6.5. The remaining assertions are unconditional consequences or conditional consequences of that hypothesis.

TRACE FORMULA, ORDERED BAR, AND THE STATISTICAL UPSIDE

83.1 THE TRACE FORMULA TARGET

The arithmetic Koszul-duality programme assembles an ordered-incidence bar complex whose regularised character reproduces the Euler product of $\xi(s)$. The question it faces is: does this Euler-product data already determine the zero distribution of ξ , or does it leave open a spectral problem? To answer precisely, fix the target.

Definition 83.1.1 (Weil test-function class). . Let $\mathcal{H}$ be a standard Weil test-function class: even Schwartz functions $h : \mathbb{R} \rightarrow \mathbb{C}$ whose Fourier transforms $\widehat{h}(\tau) = \int_{-\infty}^{\infty} h(t) e^{i\tau t} dt$ are compactly supported and sufficiently regular for the explicit formula. The exact analytic class depends on the chosen normalization of the Weil–Guinand formula.

THEOREM 83.1.2 (Weil–Guinand explicit formula). . Write nontrivial zeros of the completed Riemann zeta function $\xi(s)$ as $\rho = \frac{1}{2} + i\gamma_\rho$. For $h \in \mathcal{H}$, the completed zeta function satisfies the distributional identity

$$\sum_{\rho} h(\gamma_\rho) = W_{\infty}(h) + W_{\text{pol}}(h) - \sum_p \sum_{k \geq 1} (\log p) p^{-k/2} (\widehat{h}(k \log p) + \widehat{h}(-k \log p)), \quad (83.1)$$

where $W_{\infty}(h)$ is the archimedean gamma-factor contribution and $W_{\text{pol}}(h)$ is the contribution of the polar and trivial zeros. Denote their sum by $W_{\text{arch+pol}}(h) := W_{\infty}(h) + W_{\text{pol}}(h)$.

Remark 83.1.3 (three descriptions of the prime-power index set). The geometric side of (83.1) is a sum over prime powers (p, k) . In the Morishita dictionary this is a sum over iterates K_p^k of the prime knot K_p , with length $k \log p$. In a Gutzwiller language it is the periodic-orbit side. In the ordered-bar language it is the logarithmic derivative of the Euler-product character. These are three descriptions of the same prime-power index set; none supplies the zero-side determinant-trace operator.

83.2 CONNES' TRACE FORMULA

The Weil explicit formula is a distributional identity; the question is whether the zero sum on its left side admits a determinant-trace interpretation, i.e., whether some natural normal operator has determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes trace supported at the ordinates γ_ρ after zero-placement. Connes' adèle-class-space construction provides a partial answer.

THEOREM 83.2.1 (Connes–Meyer scaling trace). . Connes' adèle-class-space construction, in Meyer's trace-formula formulation, realizes the Weil distribution as a distributional scaling trace on the appropriate cokernel space of the adèle class space $\mathbb{A}/\mathbb{Q}^*$:

$$\text{Tr}_{\text{Connes}}(h) = \sum_{\rho} h(\gamma_\rho)$$

distributionally, with the same prime-power and archimedean terms as (83.1). The scaling operator is not thereby a self-adjoint Hilbert-space operator whose normal translate has determinant $\xi(s)$; the distributional trace is not by itself a spectral measure.

83.3 WHAT THE ORDERED BAR PROVES

The Connes–Meyer trace recovers the full Weil distribution (83.1) as an operator trace. The ordered-incidence bar complex approaches the same object from the Koszul-duality side: its regularised character computes the Euler product. The question is whether that character calculation closes the gap left by Connes.

PROPOSITION 83.3.1 (*ordered-bar Euler skeleton*). (*on the ordered-incidence model of Chapter 81.5.4 and on the chosen regularised character*). Let $\chi_\infty(s)$ denote the archimedean Euler factor arising from the regularisation at the archimedean place of $\text{Spec } \mathbb{Z}$. The ordered-incidence bar complex $B_\bullet^{\text{ord}}(\mathcal{V}^{\text{prim}})$, where $\mathcal{V}^{\text{prim}}$ is the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 , has regularised character

$$\text{ch}_{\text{reg}} H_\bullet(B_\bullet^{\text{ord}}(\mathcal{V}^{\text{prim}})) = \chi_\infty(s) \prod_p (1 - p^{-s})^{-1}, \quad (83.2)$$

with the completed $\xi(s)$ obtained only after the archimedean and polar normalisations are inserted.

Proof. This is the Euler-character computation of the ordered-incidence model: finite prime factors contribute the Bost–Connes Euler factors, while the archimedean and polar pieces are part of the regularisation data. The statement is a character calculation, not a Lefschetz trace formula. $\square$

Remark 83.3.2 (*canonical quantisation boundary*). If the profinite ACS theory and boundary comparison are assumed, the same regularised character can be read as a boundary partition function. That reading is conditional on Chapter 81.4.7 and Chapter 81.4.9.

83.4 TRACE-FORMULA EQUIVALENCE

The Euler-product character of Proposition 83.3.1 yields $\xi(s)$ only after regularisation. Under the regularised character equality of Chapter 81.5.7, which sets $Z(s) = \xi(s)$ in the ACS framework, the argument principle converts the logarithmic derivative of ξ into the zero sum; the Weil–Guinand explicit formula then identifies this with the full Weil distribution. The three trace functionals are conditionally equal.

THEOREM 83.4.1 (*conditional equivalence of trace functionals*). (*on Chapter 81.4.7, Chapter 81.5.5, and the regularised character equality Chapter 81.5.7*). Let Γ be a positively oriented vertical contour in the critical strip enclosing all nontrivial zeros of ξ . The ACS argument-principle functional

$$\mathcal{Z}_h := \frac{1}{2\pi i} \int_\Gamma h(s) \frac{Z'(s)}{Z(s)} ds$$

equals the Weil explicit-formula functional (83.1) whenever $Z(s) = \xi(s)$ is assumed, and hence agrees with the Connes–Meyer scaling trace on $\mathcal{H}$.

Proof. Under the stated assumptions $Z = \xi$. Applying the argument principle to ξ'/ξ inside Γ gives the zero sum $\sum_\rho h(\gamma_\rho)$. Expanding ξ'/ξ in the half-plane $\text{Re}(s) > 1$ of absolute convergence and performing the standard explicit-formula contour shift yields the prime-power side of (83.1). The equality is inherited from the classical Weil–Guinand explicit formula. $\square$

COROLLARY 83.4.2 (*equality of the ACS and Connes–Meyer distributions on $\mathcal{H}$*). (on the hypotheses of Chapter 83.4.1). For every $h \in \mathcal{H}$, the ACS functional $\mathcal{Z}_h$ and the Connes–Meyer scaling trace $\text{Tr}_{\text{Connes}}(h)$ are equal.

83.5 THE BAR REORGANISES, NOT LOCATES

The equivalence of the three trace functionals established conditionally in Section 83.4 raises a sharper question: does the ordered-bar construction supply an independent mechanism for placing the zero ordinates γ_n on the critical line, beyond what $\xi(s)$ itself encodes? The bar computation reorganises the zero data of ξ ; it does not add new data.

THEOREM 83.5.1 (*bar computation reorganises zero data*). . The ordered-incidence bar complex determines the individual zero ordinates γ_n only through the scalar function ξ : no additional self-adjoint operator or positivity mechanism is produced by the bar calculation.

Proof. The bar supplies the Euler-product character and, after conditional regularisation, the scalar function ξ . The zero ordinates are then the zero set of that scalar function. The bar calculation produces no self-adjoint operator, spectral theorem, or positivity input beyond what ξ already carries. $\square$

Remark 83.5.2 (*the Hilbert–Pólya obstruction*). A self-adjoint operator with spectrum $\{\gamma_n\}$ would place the zeros on the critical line by the spectral theorem. Constructing such an operator is precisely the spectral-flow obstruction of Chapter 81.6.5.

83.6 ARITHMETIC GUTZWILLER

The zero ordinates γ_n appear on the zero side of (83.1) and, conditionally, as the spectrum of the Hilbert–Pólya operator. The prime-power sum on the geometric side of (83.1) takes the same form as the periodic-orbit sum in the Gutzwiller trace formula: lengths $k \log p$ and amplitudes $(\log p) p^{-k/2}$. The prime-power side of the Weil formula is a periodic-orbit sum in Gutzwiller’s sense, conditional on the existence of an arithmetic Hamiltonian whose primitive periodic orbits are the primes.

THEOREM 83.6.1 (*classical Gutzwiller trace formula*). . Let H be a quantum Hamiltonian whose classical limit is chaotic with isolated primitive periodic orbits γ . Write $d_{\text{osc}}(E)$ for the oscillatory part of the density of states $\sum_n \delta(E - E_n)$, T_γ for the period of orbit γ , A_γ for its stability amplitude (Maslov–Gutzwiller amplitude, incorporating the stability matrix determinant), and μ_γ for its Maslov index. Then

$$d_{\text{osc}}(E) \sim \sum_{\gamma} A_{\gamma} \exp(iET_{\gamma} - i\mu_{\gamma}\pi/2),$$

asymptotically as $\hbar \rightarrow 0$.

THEOREM 83.6.2 (*arithmetic Gutzwiller hypothesis*). (on a semiclassical ACS Hamiltonian, an arithmetic phase space, and a prime-orbit correspondence). Call the amplitudes $A_p = (\log p) p^{-1/2}$ and lengths $T_p = \log p$ the *Weil amplitudes* for the prime p . If the ACS boundary Hamiltonian has a classical chaotic limit whose primitive periodic orbits are indexed by primes p with primitive period $\log p$ and stability amplitude A_p , then the periodic-orbit expansion has the desired form

$$\sum_p \sum_{k \geq 1} (\log p) p^{-k/2} e^{itk \log p} + \text{complex conjugate and archimedean terms.}$$

Proof. This is substitution of the assumed arithmetic orbit dictionary into the classical Gutzwiller formula. The proof is formal because the semiclassical ACS phase space, the chaotic flow, and the amplitude calculation are all hypotheses. $\square$

83.7 CONNES–MEYER AND THE GUTZWILLER GAP

The arithmetic Gutzwiller hypothesis of the previous section requires a smooth symplectic Hamiltonian system with a chaotic classical limit. The question is whether the Connes–Meyer construction already supplies this structure, or whether the Gutzwiller construction is genuinely new input.

THEOREM 83.7.1 (*Connes–Meyer as distributional trace, not Gutzwiller system*). . The standard adèle-class-space realization used in the Connes–Meyer trace formula is a noncommutative scaling trace construction. It is distinct from a smooth symplectic Hamiltonian system of the kind required by the classical Gutzwiller theorem.

Proof. The adèle class space $\mathbb{A}/\mathbb{Q}^*$ is modeled by a noncommutative quotient; its C^* -algebraic description gives an operator-algebraic trace. The Gutzwiller formula requires a smooth symplectic phase space, a Hamiltonian flow, isolated primitive periodic orbits, and monodromy matrix determinants. None of these are part of the Connes–Meyer construction as stated. $\square$

Remark 83.7.2 (other geometrisations remain open). Other geometrisations of the adèle class space may supply a symplectic structure; the theorem records only that the standard Connes–Meyer trace formula is not itself a Gutzwiller mechanism. An arithmetic Gutzwiller system, if it exists, is new structure beyond Connes–Meyer.

83.8 STATISTICAL UPSIDE

The explicit formula (83.1) is a distributional identity; it does not determine the individual zero ordinates γ_n beyond what $\xi(s)$ already encodes. A Hilbert–Pólya operator, if it exists, would put the determinant divisor on the critical line by self-adjointness. The further statistical prediction, that the local spacing statistics of the γ_n follow the Gaussian Unitary Ensemble (GUE), is not implied by either the explicit formula or self-adjointness alone. It requires a dynamical mechanism: the Bohigas–Giannoni–Schmit (BGS) principle connecting chaotic classical dynamics to random-matrix statistics. The relevant symmetry class is GUE because $\xi(s)$ has no anti-unitary time-reversal symmetry; the unitary symmetry class applies. Let $R_2(u)$ denote the two-point level spacing correlation function.

THEOREM 83.8.1 (*Bohigas–Giannoni–Schmit principle*). . Quantum systems whose classical limits are chaotic are expected, after symmetry class is fixed, to exhibit random matrix local spectral statistics. For the unitary symmetry class (GUE), the two-point correlation function is

$$R_2(u) = 1 - \left(\frac{\sin \pi u}{\pi u} \right)^2. \quad (83.3)$$

THEOREM 83.8.2 (*conditional statistical upside*). (on Chapter 81.6.5, Chapter 83.6.2, arithmetic quantum chaos, and the BGS universality theorem Chapter 83.8.1). If the ACS Hamiltonian of Chapter 81.6.5 has a chaotic semiclassical limit with prime periodic orbits, then the local statistics of the zero ordinates are governed by the corresponding random-matrix ensemble; in the expected zeta case by GUE statistics (83.3).

Proof. Spectral flow supplies the zero-divisor determinant and, after zero-placement, the Hamiltonian trace supported at the ordinates. Arithmetic Gutzwiller identifies the classical periodic orbits with prime powers. The BGS principle then transfers chaotic spectral statistics to the zero ordinates. Every substantive step is one of the stated hypotheses. $\square$

Remark 83.8.3 (GUE statistics as new content beyond the explicit formula). The Weil explicit formula establishes a distributional equality between the zero ordinates and the prime-power sum; it implies neither Montgomery pair correlation nor GUE spacing statistics. A 3D Hamiltonian realization of the arithmetic ACS theory adds that statistical mechanism, conditional on the semiclassical and spectral-flow hypotheses of Chapter 81.6.5.

Remark 83.8.4 (open conditions as explicit hypotheses). Chapter 83.8.2 rests on seven open conditions: (1) the profinite quantum ACS on $\overline{\text{Spec } \mathbb{Z}}$; (2) the Bezrukavnikov–Finkelberg and Costello–Gwilliam comparison with the ordered-incidence model; (3) the regularised character equality $Z = \xi$; (4) the spectral-flow operator realizing Hilbert–Pólya; (5) the full Weil–Guinand contour shift from the Euler skeleton; (6) the arithmetic semiclassical phase space and Hamiltonian flow; (7) chaos and random-matrix universality for that flow. These are the explicit hypotheses of Chapter 83.8.2, not gaps in the ordered-bar construction. The ordered bar supplies condition (3) via the Euler-product character (83.2). Conditions (1), (4), (6), and (7) constitute the spectral-flow programme of Chapter 81.6.5.

THE WILSON LINE–FROBENIUS IDENTITY ON $\overline{\text{Spec } \mathbb{Z}}$

The prime knot $K_p \subset \overline{\text{Spec } \mathbb{Z}}$ is the codimension-two defect at a rational prime p ; the Wilson loop $W_\rho(K_p)$ is the holonomy of a flat G -connection around that defect, evaluated in a representation ρ . The fundamental question of this chapter is whether, for an unramified ρ , the Wilson expectation equals the Frobenius trace: whether $\langle W_\rho(K_p) \rangle = \text{tr } \rho(\text{Frob}_p)$. Witten’s 1989 path-integral construction, transplanted via the Morishita dictionary to the arithmetic 3-space, reduces the Wilson integral to a character evaluation on the Frobenius conjugacy class; this identification is proved unconditionally at every finite level (Theorem 84.5.1(1)) and conditionally at the profinite limit (Theorem 84.5.1(2)), with the single residual being the renormalised inverse limit of Chapter 81.4.7.

84.1 SETUP AND CONVENTIONS

The arithmetic 3-space is $M = \overline{\text{Spec } \mathbb{Z}}$, the Arakelov compactification of $\text{Spec } \mathbb{Z}$ by the archimedean place $\{\infty\}$. By Mazur 1973 and Artin–Verdier, M is étale cohomologically 3-dimensional with fundamental class $[M] \in H_{\text{ét}}^3(M, \widehat{\mathbb{Z}})$ (Chapter 81.1).

Definition 84.1.1 (prime knot in the Morishita dictionary). . For a rational prime p , the *prime knot* $K_p \subset M$ is the codimension-two étale closed embedding $\text{Spec } \mathbb{F}_p \hookrightarrow \text{Spec } \mathbb{Z} \subset M$. Its tubular neighborhood is $\text{Spec } \mathbb{Z}_p$. Its complement is $M \setminus K_p \cong \text{Spec } \mathbb{Z}[1/p] \cup \{\infty\}$. Under the Mazur 1973 / Morishita 2012 (arXiv:0904.3399) dictionary, K_p plays the role of a framed knot: the meridian is the inertia subgroup $I_p \subset G_{\mathbb{Q}_p}$; the longitude and Frobenius class carry the decomposition quotient $G_{\mathbb{Q}_p}/I_p = \widehat{\mathbb{Z}} \cdot \text{Frob}_p$.

Remark 84.1.2 (the arithmetic loop is a Frobenius, not a geometric loop). $\text{Spec } \mathbb{Z}$ is not a smooth 3-manifold; the loop around K_p is not a smooth curve. Following Chapter 81.1.1, *loop around the knot* K_p is a name for the generator of the étale fundamental group of the tubular neighbourhood $\text{Spec } \mathbb{Z}_p \setminus K_p = \text{Spec } \mathbb{Q}_p$; this fundamental group is the absolute Galois group $G_{\mathbb{Q}_p}$, whose inertia quotient $\widehat{\mathbb{Z}}$ is topologically generated by the arithmetic Frobenius Frob_p . Wilson-line observables on K_p in the arithmetic theory evaluate functions on conjugacy classes of $G_{\mathbb{Q}_p}$, not on free-homotopy classes of a geometric circle.

Convention 84.1.3 (Frobenius normalisation). . We use the *geometric* Frobenius $\text{Frob}_p \in G_{\mathbb{Q}_p}/I_p$, acting on residue-field points by $x \mapsto x^p$. At a ramified prime, Frob_p means the image in the tame quotient; the residual obstruction at wild places is recorded in Chapter 84.7. Traces $\text{tr } \rho(\text{Frob}_p)$ are unnormalised: the analytic shift by $p^{1/2}$ required to match $W_\rho(K_p)$ with the local L-factor is fixed in Chapter 84.6.

Definition 84.1.4 (boundary WZW representation). . The boundary Chern–Simons theory of Chapter 81 requires representations of $G_{\mathbb{Q}_p}$ that are compatible with the global boundary data at $\{\infty\}$. A *boundary WZW representation* at p is a continuous representation $\rho: G_{\mathbb{Q}_p} \rightarrow GL_n(\overline{\mathbb{Q}_\ell})$ arising as the local restriction of a global representation $\rho: \pi_1^{\text{ét}}(M) \rightarrow GL_n$ together with the datum of a compatible module of the (finite-level) boundary category $\mathcal{B}_{G_N, \alpha_N}(\{\infty\})$ of Chapter 81.4.3, in the sense that

the Tate-restricted tensor of its local spherical parameters recovers the generating class of $\mathcal{V}^{\text{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category; Chapter 81).

84.2 CLASSICAL WILSON LOOPS (KIM ACS)

The unconditional footing is the abelian classical theory of Kim arXiv:1510.05818, where the Wilson loop expectation is a finite groupoid sum and the Frobenius trace identity is a direct evaluation.

Definition 84.2.1 (classical Kim Wilson loop). . Let A be a finite abelian group, $c \in H^3(BA, \mathbb{R}/\mathbb{Z})$ a level cocycle, and $\chi: A \rightarrow \mathbb{C}^\times$ a character. For a prime p , the *Kim classical Wilson loop along K_p in representation χ* is the functional

$$W_{\chi, A, c}^{\text{cl}}(K_p)(\rho) := \chi(\rho(\text{Frob}_p)) \cdot \exp(2\pi i \text{CS}_{M, c}^{\text{Ar}}(\rho)), \quad (84.1)$$

for $\rho: \pi_1^{\text{ét}}(M) \rightarrow A$ in the field groupoid $\mathcal{F}_A(M)$ (the groupoid of continuous A -torsors on M , i.e. continuous homomorphisms $\pi_1^{\text{ét}}(M) \rightarrow A$ with isomorphisms as morphisms; Chapter 81), with $\text{CS}_{M, c}^{\text{Ar}}(\rho) \in \mathbb{Q}/\mathbb{Z}$ the Kim arithmetic Chern–Simons invariant ((81.1)).

THEOREM 84.2.2 (classical Wilson expectation = character of Frobenius). . Let $M = \overline{\text{Spec } \mathbb{Z}}$, A finite abelian, $c \in H^3(BA, \mathbb{R}/\mathbb{Z})$ a level cocycle (with $\exp(2\pi i \cdot): \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{C}^\times$ used throughout). For any unramified character $\chi: A \rightarrow \mathbb{C}^\times$ and any $\rho \in \mathcal{F}_A(M)$, the Kim classical action satisfies

$$\langle W_{\chi, A, c}^{\text{cl}}(K_p) \rangle_{Z_{\text{CS}, c}^{\text{cl}, \text{Ar}}(M)} = \frac{1}{Z_{\text{CS}, c}^{\text{cl}, \text{Ar}}(M)} \sum_{\rho \in \pi_0 \mathcal{F}_A(M)} \frac{\chi(\rho(\text{Frob}_p)) \exp(2\pi i \text{CS}_{M, c}^{\text{Ar}}(\rho))}{|\text{Aut}(\rho)|}. \quad (84.2)$$

At $c = 0$ (level 0), the expectation collapses to the character average $|A|^{-1} \sum_{a \in A} \chi(a) N_p(a)$ where $N_p(a)$ is the number of $\rho \in \mathcal{F}_A(M)$ with $\rho(\text{Frob}_p) = a$. For unramified χ factoring through the residue-field quotient, this equals $\chi(\text{Frob}_p)$ up to the explicit Kim level-zero normalisation.

Proof. The expectation (84.2) is the groupoid-cardinality integral from Chapter 81.4.2, the Wilson insertion $\chi(\rho(\text{Frob}_p))$ being a locally-constant function of $\rho \in \mathcal{F}_A(M)$. At $c = 0$, the action phase is trivial and the expectation factors. The residue-field quotient factorisation for unramified χ is Kim arXiv:1510.05818 §3, restated in Chapter 81.4.6. The normalisation tracks with the trivial torsor $\rho \mapsto \rho(\text{Frob}_p)$. $\square$

Remark 84.2.3 (abelian A is unconditional). Unlike the conjectural profinite $GL_2(\widehat{\mathbb{Z}})$ theory, finite abelian A has no renormalisation residual: the state sum Chapter 81.4.2 is finite and (84.2) is a direct evaluation. The classical theory is the unconditional footing.

Remark 84.2.4 (independence of sources and domain of the classical identity). The derivation of Theorem 84.2.2 draws on three independent constructions: Kim arXiv:1510.05818 and Chern–Dan–Kim–Park arXiv:1609.03012 provide the classical action and finite state sum; Borger arXiv:0906.3146 §8 provides the Bost–Connes ψ_p as Frobenius trace in the λ -ring language. These are independent from one another. Kim works in $\mathbb{Q}/\mathbb{Z}$ -valued Chern–Simons; the exponentiation lands in $\mathbb{C}^\times$; the character χ is unitary. All conventions match Chapter 81.1. The ambient category is finite groupoids of continuous A -local systems on $\pi_1^{\text{ét}}(M)$. Unramified χ at p is required for the residue-field collapse; wild and ramified χ carry extra data recorded in Chapter 84.7. Pullback along $\text{Spec } \mathbb{Z}_p \hookrightarrow M$ preserves the Wilson insertion because Frob_p generates the meridian after localisation and χ is a class function. The classical state sum admits independent verification through the Bost–Connes partition calculation of Chapter 55, independent from Kim’s level-cocycle construction.

84.3 QUANTUM EXTENSION FRAMEWORK (CH. 81 CONDITIONAL)

The abelian classical theory has a non-abelian quantum extension for $G = GL_2(\widehat{\mathbb{Z}})$, defined as a renormalised inverse limit over the finite-level theories of Section 84.2. The existence and properties of this limit are the content of Chapter 81.4.7; the present section records the definitions used in Theorem 84.5.1.

Definition 84.3.1 (quantum Wilson loop, profinite GL_2). . Let $G_N = GL_2(\mathbb{Z}/N\mathbb{Z})$ and let α_N be a compatible tower of levels as in Chapter 81.4.7. For a prime p not dividing N and an unramified representation $\rho: G_{\mathbb{Q}_p} \rightarrow GL_2(\mathbb{Z}/N\mathbb{Z})$, the *level- N Wilson loop along K_p in representation ρ* is

$$W_\rho^{\text{qu},N}(K_p) := \frac{Z_{G_N, \alpha_N}^{\text{Ar}}(M, K_p, \rho)}{Z_{G_N, \alpha_N}^{\text{Ar}}(M)}, \quad (84.3)$$

with $Z_{G_N, \alpha_N}^{\text{Ar}}(M, K_p, \rho)$ the finite state sum of Chapter 81.4.2 with a Wilson defect of type ρ along K_p . The *profinite Wilson loop* is the renormalised limit

$$W_\rho^{\text{qu}}(K_p) := \varprojlim_N W_{\rho_N}^{\text{qu},N}(K_p), \quad (84.4)$$

when the tower of reductions $\rho_N: G_{\mathbb{Q}_p} \rightarrow G_N$ is compatible.

Remark 84.3.2 (quantum form). Each $W_\rho^{\text{qu},N}(K_p)$ is a finite sum (Chapter 81.4.2) and carries no renormalisation issue. The residual is purely in the *limit* step: Chapter 81.4.7 states that the renormalised inverse limit exists, agrees with the canonical-quantisation definition of the continuous-group path integral, and reproduces the boundary datum $\mathcal{V}^{\text{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category; Chapter 81). Finite- N Wilson identities below are unconditional; the continuous-group statement is conditional on that limit.

84.4 MORISHITA FUNCTORIALITY

For Theorem 84.5.1, the Wilson observable must be a well-defined class function on Frobenius conjugacy classes; this requires functoriality of the Frobenius-class assignment under change of prime. The Morishita dictionary provides this functoriality on the unramified locus.

THEOREM 84.4.1 (Morishita functor for Wilson observables). . Let Knot^{Ar} denote the category of prime-knot data $(K_p, \pi_1^{\text{ét}}(\text{tube}_p))$ on $\text{Spec } \widehat{\mathbb{Z}}$ with morphisms the Galois-equivariant embeddings of tubular neighbourhoods. The assignment

$$\begin{aligned} \Phi_{\text{Mor}}: \text{Knot}^{\text{Ar}} &\longrightarrow \text{Conj}(G_N) \\ K_p &\longmapsto [\text{Frob}_p^{K_p}] \end{aligned}$$

is a well-defined functor on the unramified locus, where $\text{Conj}(G_N)$ is the set of conjugacy classes of G_N (viewed as a discrete category) and $[\text{Frob}_p^{K_p}]$ is the conjugacy class of the Frobenius generator of $G_{\mathbb{Q}_p}/I_p$ under the ambient representation $\rho: \pi_1^{\text{ét}}(\mathcal{M}) \rightarrow G_N$. The functor Φ_{Mor} intertwines the classical Wilson observable (84.1) with the character evaluation $[\text{Frob}_p^{K_p}] \mapsto \chi([\text{Frob}_p^{K_p}])$.

Proof. We verify the three functorial properties. Object-level: on the unramified locus, ρ factors through the tame quotient $G_{\mathbb{Q}_p}/I_p = \widehat{\mathbb{Z}}$; the image of a topological generator is $\rho(\text{Frob}_p) \in G_N$, whose conjugacy class is well defined independently of the choice of lift (standard: change of lift differs by I_p , which

acts trivially by the unramifiedness hypothesis). Morphism-level: a Galois-equivariant embedding of tubular neighbourhoods $\text{Spec } \mathbb{Z}_p \hookrightarrow \text{Spec } \mathbb{Z}_q$ induces a restriction of local systems and hence a map of Frobenius classes by functoriality of the inertia-quotient assignment. Compatibility with Wilson observable: the classical Wilson insertion $\chi(\rho(\text{Frob}_p))$ depends on ρ only through the conjugacy class $[\rho(\text{Frob}_p)]$, since χ is a class function. This is functoriality of the Wilson-character correspondence. $\square$

Remark 84.4.2 (meridian and longitude). At a rational prime p , the local fundamental group $G_{\mathbb{Q}_p}$ fits in the Morishita dictionary as follows: the meridian is I_p (topologically a $\widehat{\mathbb{Z}}(1)$ at tame primes); the longitude is a fixed lift of $\text{Frob}_p \in G_{\mathbb{Q}_p}/I_p$. The Wilson holonomy around the meridian records the ramification; the Wilson holonomy along the longitude records the Frobenius trace. In the unramified case, ρ factors through the longitude and the Wilson loop reads the Frobenius trace cleanly.

84.5 MAIN THEOREM

THEOREM 84.5.1 (*Wilson line = Frobenius trace on unramified locus*). (conditional on Chapter 81.4.7 for the continuous-group step, and on the renormalised profinite limit therein; unconditional at finite level and on the unramified abelian locus). Let $M = \overline{\text{Spec } \mathbb{Z}}$ and let p be an unramified rational prime. For each continuous representation $\rho: \pi_1^{\text{et}}(M) \rightarrow GL_2(\overline{\mathbb{Q}}_\ell)$ factoring through a tower $\rho_N: G_{\mathbb{Q}_p} \rightarrow G_N = GL_2(\mathbb{Z}/N\mathbb{Z})$ with N coprime to p :

- (i) *Finite-level identity (unconditional).* For every N coprime to p , the level- N Wilson loop of Chapter 84.3.1 satisfies

$$W_{\rho_N}^{\text{qu},N}(K_p) = \frac{1}{Z_{G_N, \alpha_N}^{\text{Ar}}(M)} \sum_{\sigma \in \pi_0 \mathcal{F}_{G_N}(M)} \frac{\text{tr } \rho_N(\sigma(\text{Frob}_p)) \exp(2\pi i \text{CS}_{M, \alpha_N}^{\text{Ar}}(\sigma))}{|\text{Aut}(\sigma)|}. \quad (84.5)$$

At level $\alpha_N = 0$ and for ρ_N the inclusion $G_N \hookrightarrow GL_2(\mathbb{Z}/N\mathbb{Z})$ specialised at the trivial field $\sigma = \rho_N$, this specialises to $W_{\rho_N}^{\text{qu},N}(K_p) = \text{tr } \rho_N(\text{Frob}_p)$.

- (ii) *Profinite identity (conditional on the renormalised limit of Chapter 81.4.7).* The profinite Wilson loop (84.4) equals the Frobenius trace

$$W_{\rho}^{\text{qu}}(K_p) = \text{tr } \rho(\text{Frob}_p), \quad (84.6)$$

as an element of $\overline{\mathbb{Q}}_\ell$ (identified with $\mathbb{C}$ under a chosen embedding), where the right side is the trace of the geometric Frobenius acting on the underlying finite-rank module of ρ .

Proof. (1) Start from Chapter 81.4.2: the state sum $Z_{G_N, \alpha_N}^{\text{Ar}}(M)$ integrates by groupoid cardinality over $\mathcal{F}_{G_N}(M)$ (the groupoid of continuous G_N -torsors on M , the non-abelian analogue of $\mathcal{F}_{\mathcal{A}}(M)$ from Definition 84.2.1). Insertion of a Wilson defect of type ρ_N along K_p multiplies the integrand by the class function $\sigma \mapsto \text{tr } \rho_N(\sigma(\text{Frob}_p))$, well defined on $\mathcal{F}_{G_N}(M)$ by Chapter 84.4.1 and unramifiedness of p with respect to G_N (the coprimality $\gcd(N, p) = 1$). The ratio (84.3) is then (84.5). The specialisation at $\alpha_N = 0$ uses the level-zero collapse: the phase $\exp(2\pi i \text{CS}_{M, 0}^{\text{Ar}}(\sigma)) = 1$ for every σ , so the sum reduces to the character of G_N averaged on $\mathcal{F}_{G_N}(M)$. On the locus where ρ_N is the unique G_N -torsor up to isomorphism (the generic case), $\mathcal{F}_{G_N}(M)$ is concentrated at the point $\sigma = \rho_N$ and the average reduces to $\text{tr } \rho_N(\text{Frob}_p)$; on the non-generic locus where multiple isomorphism classes of G_N -torsors exist, the average is a weighted sum over the classes and the right side of (84.5) must be read with these multiplicities.

(2) Take the renormalised inverse limit $\varprojlim^{\text{ren}}$ of (84.5) as $N \rightarrow \infty$ along a tower compatible with ρ_N . The right side is the trace of $\rho(\text{Frob}_p)$ by continuity of the trace: each ρ_N is the mod- N reduction of ρ ; the limit of $\text{tr } \rho_N(\text{Frob}_p)$ is $\text{tr } \rho(\text{Frob}_p)$ in $\overline{\mathbb{Q}}_\ell$ by definition of the $\varprojlim$ -topology on compact-generator traces. The left side is $W_\rho^{\text{qu}}(K_p)$ by definition (84.4). Existence of the renormalised limit is exactly Chapter 81.4.7; its conclusion includes the Wilson-line axiom Chapter 81.4.7(3). Under that conjecture, (84.6) holds. $\square$

Remark 84.5.2 (what the proof provides and what it does not). Part (1) is a fully unconditional identity at each finite level N coprime to p , for each unramified ρ_N , with the Frobenius trace appearing as a class-function insertion in the finite state sum. It upgrades the Wilson-line evaluation from axiom Chapter 81.4.7(3) at the finite level to an arithmetic-geometric theorem. Part (2) is conditional on the continuous-group renormalised limit of Chapter 81.4.7. The Frobenius-trace side is classical (Deligne–Serre–Lafforgue); the Wilson side depends on the continuous-group construction of Ch. 81. The residual is the limit step, not the identity itself.

Remark 84.5.3 (independence of sources and domain of the main theorem). The derivation of Theorem 84.5.1 rests on five independent constructions: Witten 1989 *Commun. Math. Phys.* 121 (path integral for Wilson loops in 3D Chern–Simons); Morishita 2012 arXiv:0904.3399 (knots-primes dictionary); Kim 2015 arXiv:1510.05818 and Chern–Dan–Kim–Park 2018 arXiv:1609.03012 (classical arithmetic Chern–Simons); Borger 2009 arXiv:0906.3146 §8 (ψ_p = Frobenius trace in the λ -ring language); Fargues–Scholze 2021 arXiv:2102.13459 (local geometrisation). These are pairwise independent and disjoint from the derivation path of Chapter 81.

The geometric-versus-arithmetic Frobenius distinction is fixed in Convention 84.1.3; the character is unitary; cohomological and étale conventions are bridged by the Artin–Verdier fundamental class. At each finite level the ambient category is finite groupoids; the continuous-group step is a $\varprojlim$ in nuclear complete topological objects (Remark 84.3.2). Unramifiedness of p with respect to N (i.e. $\gcd(N, p) = 1$) and unramifiedness of ρ at p are both required for the Frobenius class to be well defined; ramified and wild cases are Section 84.7. Wilson evaluation is a class function, hence preserved by conjugation; Morishita pullback preserves the Frobenius class by Theorem 84.4.1. The renormalised $\varprojlim$ over $G_N \rightarrow GL_2(\widehat{\mathbb{Z}})$ is open (Chapter 81.4.7). The finite state-sum calculation (Chapter 81.4.2) is independent from the Bost–Connes partition calculation of Chapter 55, which provides an independent numerical check on the finite-level identity.

Remark 84.5.4 (reconciliation with the finite-abelian case). At $n = 1$ with A finite abelian, Chapter 84.5.1(1) specialises to Chapter 84.2.2: the trace of the one-dimensional representation χ at Frob_p is $\chi(\text{Frob}_p)$ itself, confirmed by the Kim classical state sum. This reconciliation pins the normalisation.

84.6 LOCAL LANGLANDS COMPATIBILITY

The identity $W_\rho^{\text{qu}}(K_p) = \text{tr } \rho(\text{Frob}_p)$ of Theorem 84.5.1 connects the gauge-theory Wilson expectation to a Galois-side datum. The automorphic counterpart of $\text{tr } \rho(\text{Frob}_p)$ is the unramified Satake parameter; the translation from the Galois side to the automorphic side (Hecke eigenvalue) fixes the normalisation required for the global ξ -character identity of Chapter 82.6.1.

THEOREM 84.6.1 (*local L -factor via Wilson loop*). (conditional on Chapter 84.5.1(2), hence on Chapter 81.4.7). Let ρ be unramified at p . Then

$$L_p(\rho, s) = \det(1 - W_\rho^{\text{qu}}(K_p) p^{-s})^{-1} = \det(1 - \rho(\text{Frob}_p) p^{-s})^{-1},$$

where the first equality defines the L-factor as the characteristic polynomial of the Wilson expectation and the second is Chapter 84.5.1. Under the unramified local Langlands correspondence of Tate 1979 (*Number theory past and present*, Corvallis II, §3) for GL_1 and Satake 1963 / Borel 1976 (Corvallis II, §3) for GL_n , the RHS equals the Satake-parameter expression of the local L-factor of an unramified automorphic representation π_p with parameter $\rho(\text{Frob}_p)$.

Proof. The first equality is definitional in the 3D gauge-theory language: the L-factor is the exponential-generating functional of Wilson expectations as a function of the spectral variable p^{-s} ($s \in \mathbb{C}$). The second equality is Chapter 84.5.1(2). The Satake/Borel identification of the characteristic-polynomial expression with the unramified local L-factor is Tate 1979 / Borel 1976. $\square$

COROLLARY 84.6.2 (*global L-function from Wilson loops*). (on Chapter 84.5.1(2), Chapter 81.4.7, and the archimedean Wilson-surface identification of Chapter 82.5.7). The global L-function of ρ decomposes over the prime knots:

$$L(\rho, s) = \prod_p L_p(\rho, s) = \prod_p \det(1 - W_\rho^{\text{qu}}(K_p) p^{-s})^{-1} L_\infty(\rho, s),$$

with $L_\infty(\rho, s)$ supplied by the archimedean Wilson-surface expectation at $\{\infty\}$ (Ch. 82.5).

Proof. Finite-part product: Theorem 84.6.1 for each finite prime times the archimedean completion. The archimedean factor $L_\infty(\rho, s)$ is a Γ -factor whose 3D-ACS incarnation is the Wilson-surface expectation along the archimedean boundary $\{\infty\}$, cf. Chapter 82.5.7, tied to the Arakelov area form. $\square$

Remark 84.6.3 (*triviality of the Frobenius analytic shift*). Convention 84.1.3 leaves the trace unnormalised: if ρ is of motivic weight w , the natural analytic shift for a self-dual L-function places the critical line at $\text{Re } s = (w + 1)/2$; the Wilson expectation then carries an implicit $p^{-w/2}$ factor relative to the trace. For the weight-zero sheaf $\mathbb{C}_o$ (Deninger H^0) the shift is trivial; for the weight-one Tate twist (Deninger H^2) the shift is $p^{-1/2}$. Identity (84.6) is stated unnormalised; the analytic shift is implicit in the $W_\rho^{\text{qu}}(K_p)$ notation when embedding into the global L-function.

Remark 84.6.4 (*independence of sources and domain of the Langlands compatibility*). Theorem 84.6.1 draws on three independent constructions: Witten 1989 (Wilson trace as holonomy trace); Tate 1979 and Borel 1976 together with Satake 1963 (unramified local Langlands). These are independent from the Chapter 81 construction path. The geometric-versus-arithmetic Frobenius distinction is fixed in Convention 84.1.3; the motivic-weight analytic shift is recorded in Remark 84.6.3. The ambient category is finite-dimensional $\overline{\mathbb{Q}}_\ell$ -vector spaces tensored with the category of meromorphic functions in s . Unramifiedness at p is required; at ramified primes the L-factor carries local epsilon factors (see Section 84.7). The Satake isomorphism is functorial in ρ ; the Wilson insertion is functorial via the class-function lift. The archimedean $L_\infty(\rho, s)$ identification with a Wilson-surface expectation is a definition-level identification in Chapter 82.5.7; its analytic content depends on the Arakelov area form and is conditional on the renormalised profinite construction. The Satake-eigenvalue calculations of Chapter 44 are independent from the Kim Chern–Simons state-sum calculations, providing an independent numerical check.

84.7 RESIDUAL OBSTRUCTIONS

Theorem 84.5.1 is proved on the unramified locus and for GL_2 ; four boundary conditions fall outside this domain.

- (i) *Ramified and wild primes.* At ramified (but tame) primes, the Frobenius class is replaced by a representation of the tame inertia quotient, and the Wilson expectation picks up a monodromy factor along the meridian. At wild primes the arithmetic tubular neighbourhood carries ramification filtration data beyond the tame quotient; the Wilson identity becomes a local epsilon-factor equation (Deligne–Laumon; Laumon 1987 Publ. IHES 65). The tame-ramified extension is a direct generalisation via the tame-inertia quotient; the wild locus is treated in Chapter 56.
- (ii) *Profinite renormalisation.* The $\varprojlim^{\text{ren}}$ -step in Theorem 84.5.1(2) is exactly Chapter 81.4.7. The Wilson–Frobenius identity is unconditional at each finite level; the continuous-group limit is the open problem of Chapter 81, recorded here without modification.
- (iii) *Archimedean Wilson-surface realisation.* Corollary 84.6.2 requires identification of the archimedean Γ -factor with a Wilson-surface expectation at $\{\infty\}$ via the Arakelov area form (Chapter 82.5.7). This analytic continuation is treated in Chapter 82.
- (iv) *GL_n for $n \geq 3$.* Theorem 84.5.1 is stated for GL_2 , matching the gauge-group choice of Chapter 81. Extension to GL_n proceeds via Chapter 80’s genus- g extension with $G = GL_{2g}(\widehat{\mathbb{Z}})$; the Morishita functor of Theorem 84.4.1 extends directly to $G = GL_n(\widehat{\mathbb{Z}})$ by replacing tr with the appropriate class function. The profinite-renormalisation obstruction of item (2) is carried over.

Remark 84.7.1 (the abelian case is closed unconditionally). On the abelian-finite-group unramified locus, Chapter 84.2.2 proves $W_\chi^{\text{cl}}(K_p) = \chi(\text{Frob}_p)$ *unconditionally*. Under Chapter 81.4.7, the identity extends to the continuous-group $GL_n(\widehat{\mathbb{Z}})$ and produces the finite-prime Frobenius-trace input used by the Deninger skeleton. It does not construct Deninger’s global Frobenius flow on the middle cohomology. The chapter discharges the finite-level unramified Wilson calculation and leaves the continuous-group axiom Chapter 81.4.7(3) as part of the profinite ACS renormalisation datum.

84.8 WILSON–FROBENIUS BOUNDARY

- (i) The prime knot $K_p \subset \overline{\text{Spec } \mathbb{Z}}$ is the Morishita avatar of a codimension-two defect; its meridian is I_p and its longitude is Frob_p (Chapter 84.1.1, Chapter 84.4.2).
- (ii) Classical Kim ACS Wilson loops at finite abelian gauge group A satisfy, unconditionally, $\langle W_\chi^{\text{cl}}(K_p) \rangle = \chi(\text{Frob}_p)$ for unramified χ (Chapter 84.2.2).
- (iii) The Morishita dictionary lifts to a functor $\Phi_{\text{Mor}} : \text{Knot}^{\text{Ar}} \rightarrow \text{Conj}(G_N)$ on the unramified locus; this functor intertwines Wilson insertion with Frobenius-class evaluation (Chapter 84.4.1).
- (iv) At each finite level N coprime to p , the finite-level Wilson-loop expectation is a finite state sum, with the Frobenius trace appearing as a class-function insertion (Chapter 84.5.1(1)), unconditionally.
- (v) At the profinite limit, the Wilson-loop identity $W_\rho^{\text{qu}}(K_p) = \text{tr } \rho(\text{Frob}_p)$ holds conditionally on Chapter 81.4.7 (the continuous-group renormalisation), but unconditionally once that conjecture is granted; no additional open problem appears in the limit step (Chapter 84.5.1(2)).
- (vi) The local L-factor $L_p(\rho, s) = \det(1 - W_\rho^{\text{qu}}(K_p) p^{-s})^{-1}$ equals the unramified Satake L-factor, and the global L-function is a product of Wilson-loop expectations over prime knots plus an archimedean Wilson-surface factor (Chapter 84.6.1, Chapter 84.6.2).
- (vii) Residual obstructions: ramified and wild primes (where the identity becomes a local epsilon-factor equation); the profinite renormalised limit (Chapter 81.4.7, open); the archimedean Γ -factor identification; and the GL_n -extension (Chapter 84.7).

The chapter proves the unramified finite-level Wilson/Frobenius calculation and conditionally transports it to the continuous-group limit. In the Deninger chiral candidate of Chapter 82, this supplies only the finite-prime trace component. The global Deninger flow, archimedean factor, and zero-side spectral operator remain the spectral-flow/profinite-renormalisation conditions; this chapter does not resolve them.

BIBLIOGRAPHY OF THIS CHAPTER

Primary sources:

- Mazur 1973, *Notes on étale cohomology of number fields*, Ann. Sci. ENS 6, for the cohomological 3-dimensionality of $\mathrm{Spec} \mathcal{O}_K$ and the Artin–Verdier fundamental class.
- Morishita 2012, *Knots and Primes*, arXiv:0904.3399, for the knots-primes dictionary: meridian = inertia; longitude = Frobenius.
- Witten 1989, *Quantum field theory and the Jones polynomial*, Commun. Math. Phys. 121, for the path-integral definition of the Wilson loop and its evaluation as a holonomy character.
- Kim 2015, *Arithmetic Chern–Simons theory I*, arXiv:1510.05818, for the classical arithmetic Chern–Simons action on $\mathrm{Spec} \mathcal{O}_K$ and its abelian finite state sum.
- Chern–Dan–Kim–Park 2018, arXiv:1609.03012, for the non-abelian extension and the Wilson-defect formalism at finite gauge group.
- Borger 2009, arXiv:0906.3146, §8, for the Bost–Connes Adams ψ_p as Frobenius trace on the crystalline Galois representation.
- Tate 1979, *Number theory past and present*, Corvallis II, §3, for the unramified local Langlands correspondence at GL_1 .
- Borel 1976, *Automorphic L-functions*, Corvallis II, §3, and Satake 1963 *Publ. IHES* 18, for the Satake parameters and the unramified local L-factor at GL_n .
- Laumon 1987, *Transformation de Fourier, constantes d’équations fonctionnelles et conjecture de Weil*, Publ. IHES 65, for the epsilon-factor machine at ramified primes.
- Fargues–Scholze 2021, arXiv:2102.13459, for the local geometrisation of the Langlands correspondence (not used in the proof; referenced for an independent comparison).

PRIMARY FOCK DETERMINANT: RANK RESIDUE 2

Chapter 14 separates the local conformal charge from the global arithmetic determinant. The naive sum $\sum_p c^{(p)}$ diverges, and the $\log p$ -weighted arithmetic Hamiltonian is not a Virasoro zero mode. The determinant assertion is the following: the primary Fock Euler product has rank residue 2 at $s = 1$. It does not construct a global Virasoro action on $\mathcal{V}^{\text{prim}}$, and it does not close Chapter 22.3.5.

85.1 LOCAL VIRASORO AND ARITHMETIC HAMILTONIAN

The per-prime chiral Heisenberg Fock

Definition 85.1.1 (per-prime Heisenberg and arithmetic Hamiltonian). . At each finite place $p \in \mathbb{P}$ the *per-prime chiral Heisenberg* is the complex rank-2 Heisenberg vertex algebra $\mathcal{H}^{(p)}$ with generators $a^{(p)}, b^{(p)}$, OPE $a^{(p)}(z)a^{(p)}(w) \sim (z-w)^{-2}$, $b^{(p)}(z)b^{(p)}(w) \sim (z-w)^{-2}$, mixed OPE vanishing, and Fock vacuum $|o\rangle_p$ (Chapter 4; Kac 1998, §3.5). The local conformal vector is

$$\omega_p^{\text{conf}} := \frac{1}{2} (a_{-1}^{(p)} a_{-1}^{(p)} + b_{-1}^{(p)} b_{-1}^{(p)}) |o\rangle_p.$$

The state-field correspondence produces conformal modes

$$L_m^{\text{conf},(p)} = \frac{1}{2} \sum_{k \in \mathbb{Z}} :a_{m-k}^{(p)} a_k^{(p)}: + \frac{1}{2} \sum_{k \in \mathbb{Z}} :b_{m-k}^{(p)} b_k^{(p)}:.$$

The arithmetic Hamiltonian is

$$N_p := (\log p) L_o^{\text{conf},(p)}. \quad (85.1)$$

Remark 85.1.2 (convention: $\log p$ is archimedean). The factor $\log p$ is archimedean-real scaling data, not an internal $\mathbb{Q}_p$ -coefficient (Chapter 4.2.2). Observation Chapter 14.1.1 shows that rescaling $T \mapsto \lambda T$ breaks the Virasoro OPE unless $\lambda = 1$. Thus N_p is a Hamiltonian used in the Mellin character, not a Virasoro generator.

Per-prime central charge

PROPOSITION 85.1.3 (*per-prime central charge is 2*). . For every prime p and every pair m, n with $m + n = 0$ the Virasoro commutator on $\mathcal{H}^{(p)}$ satisfies

$$[L_m^{\text{conf},(p)}, L_{-m}^{\text{conf},(p)}] = 2m L_o^{\text{conf},(p)} + \frac{c^{(p)}}{12} m(m^2 - 1) \cdot \text{id}, \quad c^{(p)} = 2.$$

Proof. Two independent Sugawara-type Heisenbergs tensor multiplicatively at the level of Virasoro central charge: the single chiral $u(1)$ Heisenberg has $c = 1$ by Kac–Raina §3.5 / Frenkel–Ben-Zvi Chapter 2. The rank-2 Heisenberg is the product, so $c^{(p)} = 1 + 1 = 2$. The arithmetic Hamiltonian N_p in (85.1) is external to this Virasoro calculation. $\square$

Remark 85.1.4 (independence of p). Chapter 14.1.3 records $c^{(p)} = 2$ as a Polyakov observation. Proposition 85.1.3 derives it from the Sugawara tensor rule. The $\log p$ -weighted Hamiltonian enters only the determinant.

85.2 DIVERGENT SUM AND PER-STATE FINITENESS

The naive global sum diverges

Observation 85.2.1 (full global conformal central sum is $+\infty$). . The naive *full* sum of local conformal central charges

$$c_{\text{full}} := \sum_{p \in \mathbb{P}} c^{(p)} = \sum_{p \in \mathbb{P}} 2 = +\infty$$

diverges. Equivalently, the formal stress tensor $\sum_p \omega_p^{\text{conf}}$ is not a vector of the restricted tensor product, and no Virasoro central extension with a finite scalar acts on $\mathcal{V}^{\text{prim}}$ (Chapter 14, Chapter 14.2.2).

The primary core

Definition 85.2.2 (primary Heisenberg core). . The *primary Heisenberg core* of $\mathcal{V}^{\text{prim}}$ is

$$\mathcal{V}_{\text{Heis,prim}}^{\text{prim}} := \text{span}_{\mathbb{C}} \left\{ v = \bigotimes_p v_p \mid v_p = |o\rangle_p \text{ for all but finitely many } p \right\},$$

with $|o\rangle_p \in \mathcal{H}^{(p)}$ the per-prime Fock vacuum (Chapter 55.3.1). Each $v \in \mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ has a finite *support set* $\text{supp}(v) \subset \mathbb{P}$, $|\text{supp}(v)| < \infty$.

Remark 85.2.3 (primary = finite-support). The primary core is the algebraic restricted tensor product, before any Hilbert completion: the dense subspace on which every per-prime mode acts by zero at all but finitely many places. It is a $\mathbb{C}$ -vector space (not a $\mathbb{Q}_p$ -module) carrying the $\mathbb{N}^\times$ -action by Adams operations ψ_n (Adams-Frobenius; Chapter 3.2.1 and Chapter 3.2.2).

Finite Positive Modes and Divergent Negative Modes

PROPOSITION 85.2.4 (finite positive modes). . For every $v \in \mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ and every $m \geq -1$ the conformal mode action

$$L_m^{\geq -1} v := \sum_{p \in S_v(m)} L_m^{\text{conf},(p)} v, \quad S_v(m) \subset \mathbb{P}, \quad |S_v(m)| < \infty,$$

is a finite sum. For $m \leq -2$, the sum $\sum_p L_m^{\text{conf},(p)}$ is not defined on the vacuum. Consequently the global Virasoro commutator is not an operator on the restricted tensor product. Only the local commutators exist, and their central scalar has divergent prime sum.

Proof. Fix $v = \bigotimes_p v_p$ with $\text{supp}(v) \subset \mathbb{P}$ finite. For $p \notin \text{supp}(v)$, $v_p = |o\rangle_p$ and $L_m^{\text{conf},(p)} |o\rangle_p = 0$ for every $m \geq -1$ (Kac–Raina §3.5; the standard creation-annihilation normal ordering places any positive-mode on the vacuum to zero, and the L_{-1} , L_0 annihilation on vacuum is the standard vertex-algebra axiom). Thus only finitely many summands act non-trivially for $m \geq -1$. For $m \leq -2$, each vacuum factor contributes the non-zero descendant $L_m^{\text{conf},(p)} |o\rangle_p$. The sum over primes is then an infinite sum of mutually orthogonal one-prime descendants, not an element of the algebraic restricted tensor product. The local identity

$$[L_m^{\text{conf},(p)}, L_{-m}^{\text{conf},(p)}] |o\rangle_p = \frac{c^{(p)}}{12} m(m^2 - 1) |o\rangle_p$$

therefore produces the divergent scalar $\sum_p c^{(p)}$ when summed over all primes. This scalar is not an operator before regularisation. $\square$

Remark 85.2.5 (finite action, divergent scalar). Chapter 85.2.4 proves only finite action of the conformal modes $m \geq -1$. The formal central term

$$\frac{1}{12}m(m^2 - 1) \cdot \left(\sum_p c^{(p)} \right) \cdot \text{id}$$

contains $\sum_p c^{(p)} = \infty$. The residue calculation below assigns a scalar to the determinant; it does not turn the formal commutator into a Virasoro action.

85.3 DIRICHLET-SERIES REGULARISATION

The naive prime-zeta attempt fails

Observation 85.3.1 (naive prime-zeta regularisation is not well-posed). Define the formal series

$$c_{\mathbb{P}}(s) := \sum_{p \in \mathbb{P}} c^{(p)} p^{-(s-1)} = 2 \sum_p p^{-(s-1)} = 2\zeta_{\mathbb{P}}(s-1).$$

The prime zeta $\zeta_{\mathbb{P}}$ has natural boundary on $\text{Re}(w) = 0$ by the Landau–Walfisz / Fröberg theorem (Chapter 14.3.1). Thus $c_{\mathbb{P}}(s)$ at $s = 1$ asks for $2\zeta_{\mathbb{P}}(0)$, which is undefined (Chapter 14.3.2); at $s \rightarrow 1^+$, $c_{\mathbb{P}}(s) \rightarrow +\infty$ logarithmically.

Remark 85.3.2 (prime-zeta sum vs. Riemann-zeta residue). The Dirichlet series $c_{\mathbb{P}}(s)$ uses the prime-index set $\mathbb{P}$ as its summation domain. The determinant-rank scalar is instead read from the Euler product for the Riemann zeta $\zeta(s)$, which is meromorphic on $\mathbb{C}$. The prime zeta $\zeta_{\mathbb{P}}$ has a natural boundary at $\text{Re}(w) = 0$, and therefore cannot define the value at $w = 0$. The finite-mode statement Chapter 85.2.4 gives only an algebraic domain for the modes $m \geq -1$; the residue at $s = 1$ belongs to the determinant, not to a Virasoro commutator.

Arithmetic black-hole motivation: primon gas as the first determinant

The partition function $Z(s) = \prod_p \prod_{n \geq 1} (1 - p^{-sn})^{-2}$ defined in §85.3 is the arithmetic-Chern–Simons thermal determinant over prime knots before the zero-side trace formula is available. The following heuristic identifies its origin in the primon-gas thermodynamics of Julia; no black-hole existence theorem enters the proof of Lemma 85.3.5, and no positivity statement is inferred from the calculation.

In ordinary three-dimensional gravity written as Chern–Simons theory, a black-hole sector is not a propagating local graviton. It is a flat connection with non-trivial holonomy around the cycles of a solid torus. Its one-loop determinant is built from the closed geodesics of the filling. The arithmetic replacement used throughout Chapters 81–84 is:

$$\text{closed geodesic} \rightsquigarrow \text{prime knot } K_p, \quad \ell(K_p) = \log p, \quad \text{hol}(K_p) = \text{Frob}_p.$$

At the level relevant here, the arithmetic Chern–Simons thermal determinant is computed around a holonomy saddle, with the primitive closed orbits indexed by primes.

The elementary thermodynamic step is the whole primon-gas calculation. A single bosonic occupation mode over the orbit K_p has energy

$$E_p = \ell(K_p) = \log p.$$

At inverse temperature s , its local partition function is the geometric series

$$\sum_{m \geq 0} e^{-smE_p} = \sum_{m \geq 0} p^{-sm} = (1 - p^{-s})^{-1}.$$

Multiplying the independent local factors over all prime knots gives Julia's primon gas:

$$Z_{\text{primon}}(s) = \prod_p (1 - p^{-s})^{-1} = \zeta(s), \quad \text{Re}(s) > 1. \quad (85.2)$$

For a rank- r free Heisenberg sector the same calculation gives $\zeta(s)^r$. Here $r = 2$, because the local arithmetic Heisenberg is generated by two commuting $u(1)$ fields.

The oscillator refinement adds the usual Fock levels. Instead of only one occupation energy E_p , the n -th oscillator over K_p has energy $nE_p = n \log p$. Therefore the rank-2 local factor is

$$\prod_{n \geq 1} (1 - e^{-sn \log p})^{-2} = \prod_{n \geq 1} (1 - p^{-sn})^{-2}. \quad (85.3)$$

Multiplication over all primes gives exactly the global Fock partition function defined below:

$$\prod_p \prod_{n \geq 1} (1 - p^{-sn})^{-2}. \quad (85.4)$$

The primon gas is the $n = 1$ factor in the oscillator product: setting $n = 1$ in (85.3) and taking $r = 1$ recovers $(1 - p^{-s})^{-1}$. The full rank-2 Fock product (85.4) is the arithmetic Chern–Simons thermal determinant used for the determinant rank-residue extraction in §85.3.

The limitation is as important as the calculation. Equations (85.2)–(85.4) recover the prime-side Euler product in its convergence half-plane. They do not construct Deninger's middle cohomology, do not locate the non-trivial zeros, and do not prove a trace formula. Those require the spectral-flow and positivity inputs isolated in Chapters 82 and 83.

The partition function regularisation

Definition 85.3.3 (per-prime Fock partition). . For each prime p and complex parameter s with $\text{Re}(s) > 0$ the *per-prime Fock partition* is

$$Z^{(p)}(s) := \text{Tr}_{\mathcal{H}^{(p)}}(e^{-sN_p}) = \text{Tr}_{\mathcal{H}^{(p)}}(p^{-sL_o^{\text{conf},(p)}}) = \prod_{n \geq 1} (1 - p^{-sn})^{-2},$$

the final equality holding formally because $L_o^{\text{conf},(p)}$ on the rank-2 Heisenberg Fock has generating function η -type with two commuting copies (Chapter 14, §14.4; Kac–Raina §3.5 for the rank-1 generating series).

Definition 85.3.4 (global Fock partition: Euler product). . The *global Fock partition* is the formal Euler product

$$Z(s) := \prod_{p \in \mathbb{P}} Z^{(p)}(s) = \prod_p \prod_{n \geq 1} (1 - p^{-sn})^{-2}, \quad (85.5)$$

convergent for $\text{Re}(s) > 1$ by convergence of the log-sum $\sum_{p,n} p^{-sn}$ (standard Euler-product estimate, Serre 1973 Ch. VI).

LEMMA 85.3.5 (partition function is squared zeta product). . In the region $\text{Re}(s) > 1$,

$$Z(s) = \prod_{n \geq 1} \zeta(ns)^2.$$

Proof. Interchange the two products in (85.5):

$$Z(s) = \prod_{n \geq 1} \prod_p (1 - p^{-sn})^{-2} = \prod_{n \geq 1} \left(\prod_p (1 - p^{-sn})^{-1} \right)^2 = \prod_{n \geq 1} \zeta(ns)^2.$$

Each inner factor is the standard Euler product of $\zeta(ns)$ for $\operatorname{Re}(ns) > 1$, valid since $\operatorname{Re}(s) > 1$ implies $\operatorname{Re}(ns) > 1$ for every $n \geq 1$. The interchange of infinite products is justified by absolute convergence of the double sum $\sum_{p,n} p^{-\operatorname{Re}(s)n}$ on compact subsets of $\{\operatorname{Re}(s) > 1\}$. $\square$

Remark 85.3.6 (logarithmic derivative of Z and the rank pole). From $\log Z(s) = 2 \sum_{n \geq 1} \log \zeta(ns)$, the logarithmic derivative is

$$-\frac{d}{ds} \log Z(s) = 2 \sum_{n \geq 1} \frac{n \zeta'(ns)}{\zeta(ns)} \cdot (-1) = 2 \sum_{n \geq 1} n \frac{-\zeta'(ns)}{\zeta(ns)},$$

convergent for $\operatorname{Re}(s) > 1$. The leading pole of $-\frac{d}{ds} \log Z$ at $s = 1$ arises from the $n = 1$ term via $\frac{-\zeta'(s)}{\zeta(s)} \sim (s-1)^{-1}$ near $s = 1$, with the higher- n terms holomorphic at $s = 1$. In the corrected Polyakov convention (§14.4), this residue is the rank scalar of the determinant, not a global central charge.

The Dirichlet-series rank residue

Definition 85.3.7 (zeta rank residue). The zeta rank residue is

$$r_\zeta := \operatorname{Res}_{s=1} \left(-\frac{d}{ds} \log Z(s) \right). \quad (85.6)$$

Equivalently, if $\log Z(s) = -r \log(s-1) + \text{holomorphic near } s = 1$, then $r_\zeta := r$, where r is the order of the logarithmic singularity.

Remark 85.3.8 (rationale for the residue definition). Per Polyakov (Chapter 14, Chapter 14.4.1), the honest regularisation is *not* a sum-over-primes but the residue of the global partition-function pole. The definition (85.6) places this residue at the $s = 1$ pole of ζ , which is the dominant pole of Z since $Z(s) = \zeta(s)^2 \cdot \prod_{n \geq 2} \zeta(ns)^2$ and the $n \geq 2$ factors are holomorphic at $s = 1$ (with $\zeta(2n) \in \mathbb{Q}\pi^{2n}$ for $n \geq 1$).

85.4 LOCAL EXPANSION AT $s = 1$

Decomposition of $\log Z$ near $s = 1$

THEOREM 85.4.1 (local Laurent expansion of $\log Z$ at $s = 1$). In a punctured neighbourhood of $s = 1$ on a fixed logarithm branch, the function $\log Z(s) = 2 \sum_{n \geq 1} \log \zeta(ns)$ has expansion

$$\log Z(s) = -2 \log(s-1) + 2 \sum_{n \geq 2} \log \zeta(n) + 2\gamma_E(s-1) + O((s-1)^2), \quad (85.7)$$

where $\gamma_E = \lim_{N \rightarrow \infty} (\sum_{k=1}^N 1/k - \log N)$ is the Euler–Mascheroni constant.

Proof. Write $\log Z(s) = 2 \log \zeta(s) + 2 \sum_{n \geq 2} \log \zeta(ns)$. For $n \geq 2$ and s near 1, $ns \geq 2$, so $\zeta(ns)$ is holomorphic and nonzero at $s = 1$ (since $\zeta(2), \zeta(3), \dots$ are positive real and greater than 1). The tail bound $\sum_{n \geq 2} |\log \zeta(ns)| \leq C \sum_{n \geq 2} 2^{-n \operatorname{Re}(s)}$ is uniform on $\operatorname{Re}(s) \geq 1/2 + \epsilon$, so $\sum_{n \geq 2} \log \zeta(ns)$ is holomorphic at $s = 1$ and evaluates there to $\sum_{n \geq 2} \log \zeta(n)$.

For the $n = 1$ term: the Laurent expansion $\zeta(s) = (s-1)^{-1} + \gamma_E + O(s-1)$ (Edwards 1974, §1.6) gives

$$\log \zeta(s) = \log((s-1)^{-1}(1 + \gamma_E(s-1) + O((s-1)^2))) = -\log(s-1) + \gamma_E(s-1) + O((s-1)^2),$$

using $\log(1+x) = x + O(x^2)$. Therefore

$$\log Z(s) = -2 \log(s-1) + 2 \sum_{n \geq 2} \log \zeta(n) + 2\gamma_E(s-1) + O((s-1)^2),$$

which is (85.7). □

Remark 85.4.2 (why the rank residue is 2). The leading term $-2 \log(s-1)$ in (85.7) has a -2 coefficient because $Z(s) = \zeta(s)^2 \cdot \text{hol}$ near $s = 1$, and the double factor in ζ comes from the rank-2 Heisenberg: per-prime two-commuting-Heisenberg implies per-prime $Z^{(p)}$ has a *squared* $(1 - p^{-s})^{-1}$ factor leading at $n = 1$, and on the Euler-product reshuffle this makes $Z = \zeta^2 \cdot \text{hol}$. The power of ζ at $s = 1$ is the per-prime rank, here 2. This rank agrees with the local Sugawara central charge prime by prime, but it is not a global central charge.

Residue extraction

THEOREM 85.4.3 (*residue at $s = 1$*). . With the decomposition (85.7),

$$\text{Res}_{s=1} \left(-\frac{d}{ds} \log Z(s) \right) = 2,$$

and the zeta rank residue of Chapter 85.3.7 evaluates to

$$r_\zeta = 2.$$

Proof. Differentiate (85.7): $-\frac{d}{ds} \log Z(s) = \frac{2}{s-1} - 2\gamma_E + O(s-1)$ near $s = 1$. The residue at $s = 1$ is 2. By Definition 85.3.7, $r_\zeta = \text{Res}_{s=1}(-\frac{d}{ds} \log Z) = 2$. □

COROLLARY 85.4.4 ($r_\zeta = 2 \cdot \text{Res}_{s=1} \zeta(s)$). . Since $\text{Res}_{s=1} \zeta(s) = 1$,

$$r_\zeta = 2 \cdot \text{Res}_{s=1} \zeta(s) = 2. \tag{85.8}$$

Remark 85.4.5 (Dirichlet-series form of regularisation). The original informal heuristic was $c(s) = \sum_p c^{(p)} p^{-(s-1)} = 2\zeta_{\mathbb{P}}(s-1)$; we replaced it by the honest global-partition-function residue. The replacement is forced by Chapter 14.3.1: the prime zeta has natural boundary on $\text{Re}(w) = 0$, so the informal prescription has no analytic continuation to $s = 1$. The partition-function prescription runs on $\zeta(s)$ directly (which is meromorphic on $\mathbb{C}$), with residue 1 at $s = 1$; the total per-prime rank 2 and the partition-function-residue identity (85.8) together deliver $r_\zeta = 2$. This is the determinant-rank residue of Chapter 14.4, not a sum over primes and not a Virasoro central charge.

85.5 POLYAKOV ANOMALY CONNECTION

The Polyakov stress tensor and anomaly

THEOREM 85.5.1 (*Polyakov anomaly on the archimedean chiral Heisenberg*). (Polyakov 1981; Belavin–Polyakov–Zamolodchikov 1984; Friedan–Shenker 1987; archimedean Heisenberg: Kac 1998

§3.5). On the archimedean Heisenberg $\mathcal{H}^{(\infty)}$ with rank-2 free-field presentation, the Polyakov conformal anomaly coefficient is

$$c_{\text{anomaly}}^{(\infty)} = 2,$$

extracted from the $T(z)T(w)$ OPE central term with $T = \frac{1}{2}(a_{-1}^{(\infty)}a_{-1}^{(\infty)} + b_{-1}^{(\infty)}b_{-1}^{(\infty)})$, independently from trace-anomaly $\langle T_{\mu}^{\mu} \rangle = -cR/24\pi$ and from Friedan–Shenker modular-invariance bootstrap.

Remark 85.5.2 (archimedean vs. determinant). $c_{\text{anomaly}}^{(\infty)} = 2$ is the anomaly of the *archimedean* Heisenberg alone. Each per-prime place also satisfies $c^{(p)} = 2$, so the per-place charges are uniformly 2. The finite-prime determinant residue also equals 2. Equality of these two numbers is a rank comparison; it is not an addition of local central charges.

The Polyakov–regularisation identity

THEOREM 85.5.3 (Polyakov–rank comparison). . The zeta rank residue of Chapter 85.3.7 equals the Polyakov conformal anomaly coefficient of the archimedean rank-2 Heisenberg:

$$r_{\zeta} = c_{\text{anomaly}}^{(\infty)} = 2.$$

Equivalently, the residue of the global Fock partition $Z(s) = \prod_p \prod_{n \geq 1} (1 - p^{-sn})^{-2}$ at $s = 1$ has the same value as the rank-2 Polyakov–Friedan–Shenker anomaly computed on the archimedean completion $\mathcal{H}^{(\infty)}$.

Proof. Combine Chapter 85.4.4 ($r_{\zeta} = 2$) with Chapter 85.5.1 ($c_{\text{anomaly}}^{(\infty)} = 2$): both sides equal 2 independently, and their common value is $2 \cdot \text{Res}_{s=1} \zeta(s) = 2$.

The two sides arise from different constructions: the left-hand side from a global partition-function Euler product over all finite primes, regularised through the residue of ζ at its pole; the right-hand side from a Lagrangian stress-tensor computation on the archimedean fibre, regularised through Polyakov’s short-distance anomaly. The identity is the arithmetic-branch expression of Polyakov’s principle in the residue form (Chapter 14.4.1): the finite-prime determinant and the archimedean free-field anomaly have the same rank 2. No global Virasoro action on $\mathcal{V}^{\text{prim}}$ is produced. $\square$

Remark 85.5.4 (global compatibility). Product formula phrasing: for the completed zeta $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, the factor $\pi^{-s/2}\Gamma(s/2)$ is the archimedean local Γ -factor and $\zeta(s)$ is the Euler product over finite primes. The residue of ζ at $s = 1$ is 1 by the standard prime-number-theorem residue; under the functional equation $\xi(s) = \xi(1-s)$, this residue is mirrored to the residue at $s = 0$. On the conformal side, the Polyakov anomaly on the archimedean fibre equals the rank ($= 2$) of the free-field Heisenberg; on the Euler-product side, the residue of Z at $s = 1$ equals the per-prime rank ($= 2$). The identification in Chapter 85.5.3 is rank = rank, made precise through the ζ -residue dictionary.

Derivation and Residue Sources

Remark 85.5.5 (disjoint derivation and residue sources). The local chiral calculation uses Kac–Raina *Bombay Lectures on Highest Weight Representations of Infinite-Dimensional Lie Algebras* (World Scientific 1987), §3.5 (Sugawara central charge for bosonic Fock); Kac *Vertex Algebras for Beginners* 2nd ed. (AMS 1998), Chapter 3 (rank- n Heisenberg vertex algebra and Virasoro central charge formula). The global residue calculation uses Edwards *Riemann’s Zeta Function* (Academic Press 1974), §1.6 (Laurent expansion of ζ at $s = 1$) for the residue-extraction half; Serre *Cours d’Arithmétique* (PUF 1970) Ch. VI

(Euler product) for the global partition rewriting. The two source sets (Kac/Kac–Raina for the local central charge, Edwards/Serre for the global residue) are disjoint; the computed scalar (2) matches on both sides.

85.6 MAIN THEOREM: $r_\zeta = 2$

Statement

THEOREM 85.6.1 (*primary Fock determinant has rank residue 2*). . Let $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ denote the primary Heisenberg core of Chapter 85.2.2. Then:

- (a) *Finite positive modes*. For every $v \in \mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ and every $m \geq -1$, the mode action $L_m^{\geq -1} v = \sum_{p \in S_v(m)} L_m^{\text{conf},(p)} v$ is a finite sum over $|S_v(m)| < \infty$ primes. The modes $m \leq -2$ do not form global operators on the restricted vacuum (Chapter 85.2.4).
- (b) *Divergent conformal scalar*. The formal Virasoro central-term coefficient $\sum_p c^{(p)}$ is $+\infty$ as a naive sum (Chapter 85.2.1).
- (c) *Zeta rank residue*. The Dirichlet-series residue of the scalar via the global Fock partition $Z(s) = \prod_p Z^{(p)}(s)$ yields

$$r_\zeta = \text{Res}_{s=1} \left(-\frac{d}{ds} \log Z(s) \right) = 2.$$

- (d) *Polyakov-rank comparison*. The rank residue equals the Polyakov conformal anomaly coefficient of the archimedean rank-2 Heisenberg $c_{\text{anomaly}}^{(\infty)} = 2$ (Chapter 85.5.3).

Proof. (a) is Chapter 85.2.4. (b) is Chapter 85.2.1. (c) is Chapter 85.4.3 with Chapter 85.4.4. (d) is Chapter 85.5.3. $\square$

Falsification and correction of six potential objections

Remark 85.6.2 (*prime-zeta sum does not regularise the determinant rank*). **Objection.** “Define $c(s) = \sum_p c^{(p)} p^{-(s-1)}$, take $s \rightarrow 1$, get a finite number.” **Answer.** $c(s) = 2\zeta_{\mathbb{P}}(s-1)$; $\zeta_{\mathbb{P}}$ has natural boundary on $\text{Re}(w) = 0$ (Chapter 14.3.1); $c(s) \rightarrow +\infty$ as $s \rightarrow 1^+$. The correct regularisation is through the determinant-residue prescription of Chapter 85.3.7, which evaluates through ζ (meromorphic on $\mathbb{C}$) rather than $\zeta_{\mathbb{P}}$ (barrier at $\text{Re}(w) = 0$).

Remark 85.6.3 (*(log p)-weighting is not a Virasoro rescaling*). **Objection.** “The $(\log p)$ -rescaling changes the central charge $c^{(p)} \mapsto (\log p)^2 \cdot c^{(p)}$.” **Answer.** The local stress tensor is not rescaled. The factor $\log p$ defines $N_p = (\log p) L_0^{\text{conf},(p)}$, the Hamiltonian in the Mellin determinant. The local conformal charge remains $c^{(p)} = 2$, and N_p is not a Virasoro zero mode.

Remark 85.6.4 (*the universal scalar is not per-state-finite*). **Objection.** “Chapter 85.2.4 establishes a global Virasoro commutator.” **Answer.** It establishes finite action only for $m \geq -1$. For $m \leq -2$ the global mode is not defined on the vacuum, and the formal central scalar is ∞ . The residue in §85.3 targets the determinant scalar, not a commutator on a dense domain (Chapter 85.2.5).

Remark 85.6.5 (*the residue prescription is uniquely determined*). **Objection.** “The regularisation $r_\zeta = 2$ is arbitrary: many Dirichlet-series prescriptions yield different numbers.” **Answer.** The prescription (85.6) is fixed by two requirements: (i) the regularisation runs through the global Fock partition $Z(s)$ whose pole structure is uniquely determined by the Euler product of ζ^2 at $s = 1$; (ii) the regularisation must match the Polyakov anomaly on the archimedean completion (Chapter 85.5.3). Together these

single out the residue prescription; the prime-zeta prescription fails (i): $\zeta_{\mathbb{P}}$ has a natural boundary at $\text{Re}(w) = 0$; the sum-at-zero prescription fails (ii): a sum indexed by primes is not the same as a residue of the Riemann zeta at $s = 1$, which runs over all positive integers via the Euler product.

Remark 85.6.6 (the archimedean fibre enters through the completed-zeta Γ -factor). **Objection.** “The partition $Z(s) = \prod_p Z^{(p)}(s)$ omits the archimedean fibre $\mathcal{H}^{(\infty)}$.” **Answer.** The archimedean contribution enters through the Γ -factor in the completed zeta $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; on the stress-tensor side it is built into the Polyakov-anomaly identification. Chapter 85.5.3 states only a rank equality between the finite-prime residue and the archimedean anomaly. It does not add two central charges.

Remark 85.6.7 (the order of the pole of $\log Z$ at $s = 1$ is not a free parameter). **Objection.** “The residue prescription is circular: the residue is 2 because the rank is chosen to be 2.” **Answer.** The order of the pole is *not* a free parameter. It is computed from the Euler product: $Z^{(p)}(s) = \prod_{n \geq 1} (1 - p^{-sn})^{-2}$ has rank-2 because the per-prime Heisenberg is rank-2 (two free bosons); this is a *fixed input* from the construction. The resulting $Z(s) = \zeta(s)^2 \cdot \text{hol near } s = 1$ has a pole of order 2 in ζ , hence a leading $-2 \log(s-1)$ term in $\log Z$, hence $\text{Res}_{s=1}(-\frac{d}{ds} \log Z) = +2$. The input is the rank-2 Heisenberg determinant. The output is the same rank read from the pole of $\zeta(s)^2$.

Remark 85.6.8 (Iwasawa completion does not change the local rank). **Objection.** “The Vol IV arithmetic-Positselski chapter works with Iwasawa Λ_p -coefficients, where the local rank of the Heisenberg factor might change.” **Answer.** $\Lambda_p = \mathbb{Z}_p[[T]]$ is the Iwasawa-algebra completion of the per-prime Heisenberg acting on the cyclotomic tower (Chapter 24, Chapter 24.1.5). In the Gelfand–Kazhdan formal-geometry window the Λ_p -Heisenberg is rank-2 over $\mathbb{Q}_p$ (two free Iwasawa-generators; Fontaine 1991 B_{dR} rank-2 Hodge–Tate class). Completion preserves this rank. Thus the local Sugawara central charge remains 2, and the global trace class still has no Virasoro central charge.

Relation to the conditional Virasoro-bracket question

Remark 85.6.9 (what is proved and what is not). Chapter 85.6.1 proves the determinant scalar $r_{\zeta} = 2$. It does *not* prove that a per-prime sum $\sum_p L_m^{\text{conf},(p)}$ satisfies a Virasoro bracket with this scalar. Two obstructions remain:

- (i) *Domain.* The modes $m \leq -2$ must first exist as dense operators on a completion of $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ (the Petersson completion, Chapter 21.7.1). The closure is not automatic: finite-support states do not form an invariant core unless a commutator estimate is given.
- (ii) *Bracket identity.* After closure, the identity $[L_m, L_n] = (m-n)L_{m+n} + (c/12)(m^3-m)\delta_{m+n,0}$ · id would still be a new theorem. The determinant scalar r_{ζ} does not supply it.

Both obstructions are open for the primary-Virasoro reconciliation.

Domain and logical form

Remark 85.6.10 (determinant domain). The determinant-rank identity is $r_{\zeta} = 2$. It does not prove either half of Chapter 22.3.5. The open content remains operator closure plus bracket identity on a dense domain (Chapter 85.6.9). The operator closure requires operator estimates on $\mathcal{V}_{\text{Heis,prim}}^{\text{prim}}$ with respect to the Petersson inner product of Chapter 21.7.1.

BIBLIOGRAPHY

- [1] A. A. Belavin, A. M. Polyakov, A. B. Zamolodchikov. *Infinite conformal symmetry in two-dimensional quantum field theory*. Nuclear Physics B 241 (1984), 333–380.
- [2] H. M. Edwards. *Riemann’s Zeta Function*. Pure and Applied Mathematics 58. Academic Press 1974.
- [3] D. Friedan, S. Shenker. *The analytic geometry of two-dimensional conformal field theory*. Nuclear Physics B 281 (1987), 509–545.
- [4] E. Frenkel, D. Ben-Zvi. *Vertex Algebras and Algebraic Curves*. 2nd ed. Mathematical Surveys and Monographs 88. American Mathematical Society 2004. Chapter 2 (Heisenberg vertex algebras), Chapter 5 (Sugawara construction).
- [5] K. Iwasawa. *Lectures on p -adic L -functions*. Annals of Mathematics Studies 74. Princeton University Press 1972.
- [6] V. G. Kac. *Vertex Algebras for Beginners*. 2nd ed. University Lecture Series 10. American Mathematical Society 1998.
- [7] V. G. Kac, A. K. Raina. *Bombay Lectures on Highest Weight Representations of Infinite-Dimensional Lie Algebras*. Advanced Series in Mathematical Physics 2. World Scientific 1987.
- [8] E. Landau, A. Walfisz. *Über die Nichtfortsetzbarkeit einiger durch Dirichletsche Reihen definierten Funktionen*. Rendiconti del Circolo Matematico di Palermo 44 (1920), 82–86.
- [9] A. M. Polyakov. *Quantum geometry of bosonic strings*. Physics Letters B 103 (1981), 207–210.
- [10] J.-P. Serre. *Cours d’Arithmétique*. Presses Universitaires de France 1970. Chapter VI (Dirichlet series, Euler products, analytic continuation of ζ).
- [11] C.-E. Fröberg. *On the prime zeta function*. BIT Numerical Mathematics 8 (1968), 187–202.

THE SPECTRAL FLOW $N_{\text{BC}} \rightsquigarrow H_{\text{HP}}$: IRREDUCIBLE OBSTRUCTION AT THE ARCHIMEDEAN PLACE

Chapter 55 constructs the Bost–Connes log-modular operator $N_{\text{BC}} = -\frac{1}{2} \log \Delta_1$ with unconditional self-adjointness and pure point spectrum $\{\log m \mid m \in \mathbb{N}^\times\}$. The target is a Hilbert–Pólya operator H_{HP} whose spectral divisor, through the regularised determinant of $\frac{1}{2} + iH_{\text{HP}}$, is the non-trivial zero divisor of $\xi_{\mathbb{R}}$. The intertwining operator A taking N_{BC} to H_{HP} is the classical Hilbert–Pólya twist. The spectral-flow problem is a precise operator-theoretic statement equivalent to the Connes archimedean absorption-trace condition; the unconditional pieces are exactly the Bost–Connes logarithmic Hamiltonian, the pole-cleared zeta determinant, and their common scaling normalisation in the triple-identification formalism of Chapters 82, 79, and 81. The determinant-trace condition is not a consequence of these unconditional pieces.

86.1 SETUP: N_{BC} AND Θ_ξ AS TARGET OPERATORS

The two endpoints of the spectral flow are the operators N_{BC} and H_{HP} , and the Hilbert spaces that carry them are specified below.

Source: The Bost–Connes Log-Prime Operator

Definition 86.1.1 (*Bost–Connes Hilbert space and log-modular operator*). . Let $(\mathcal{H}_1, \pi_1, \Omega_1)$ denote the GNS triple of the Bost–Connes KMS₁ state φ_1 on the adelic C^* -algebra $C_{\mathbb{Q}}^*$ (Chapter 55.2.1, Chapter 55.2.2). Let $\mathcal{H}_{\text{BC}}$ denote the closed subspace of $\mathcal{H}_1$ spanned by the zero-mode lattice $\{\delta_m \otimes \delta_o \mid m \in \mathbb{N}^\times\}$ restricted to the spherical character sector $\gamma = o$. Let $N_{\text{BC}} = -\frac{1}{2} \log \Delta_1|_{\mathcal{H}_{\text{BC}}}$ be the restriction of the log-modular operator; it is self-adjoint, non-negative, with pure point spectrum $\{\log m\}_{m \geq 1}$ and simple multiplicity on the spherical sector (Chapter 55.2.5).

Remark 86.1.2 (*intrinsic structure of $\mathcal{H}_{\text{BC}}$*). As a Hilbert space, $\mathcal{H}_{\text{BC}}$ is canonically isometric to $\ell^2(\mathbb{N}^\times, dm/m)$ with $\delta_m \otimes \delta_o \leftrightarrow \sqrt{m} e_m$. Under $m = \prod_p p^{v_p(m)}$ the lattice factors as $\otimes_p \ell^2(p^{\mathbb{N}_{\geq 0}}, p^{-k} \cdot dk)$ restricted with respect to the local unit vector e_o at each place, with N_{BC} acting as $\oplus_p k \log p$ under the Mellin isomorphism $\ell^2(p^{\mathbb{N}_{\geq 0}}, p^{-k} dk) \cong L^2(\mathbb{N}, p^{-k} dk)$. This is the Julia primon-gas presentation of Chapter 7, realised as a Hilbert space of primon occupation states.

Target: The Hypothetical Zeta-Zero Divisor Operator

Definition 86.1.3 (*target Hilbert space and zero-divisor operator*). . Let $\mathcal{Z}$ denote the multiset of non-trivial zeros $\rho = \beta_\rho + it_\rho$ of $\xi_{\mathbb{R}}$, with multiplicities m_ρ , and write $\gamma_\rho := t_\rho$ for the ordinate. The *formal target Hilbert space* is

$$\mathcal{H}_\xi^{\text{formal}} := \bigoplus_{\rho \in \mathcal{Z}} \mathbb{C}^{m_\rho},$$

with inner product making the one-dimensional summands orthonormal. The *formal ordinate multiplier* is the diagonal symmetric operator

$$H_{\text{ord}}^{\text{formal}} : (\xi_\rho)_\rho \mapsto (t_\rho \xi_\rho)_\rho$$

on the finite-support core; its spectrum is $\{t_\rho\}$ with multiplicities. The *critical-line projection operator* is

$$\Theta_\xi^{\text{proj}} = \frac{1}{2}\text{id} + iH_{\text{ord}}^{\text{formal}}.$$

Its divisor is the projection of the zero divisor to the critical line; it is the actual zero divisor of $\xi_{\mathbb{R}}$ exactly under RH.

Remark 86.1.4 (why “formal”). The triple $(\mathcal{H}_\xi^{\text{formal}}, H_{\text{ord}}^{\text{formal}}, \Theta_\xi^{\text{proj}})$ exists trivially as a diagonal construction. It carries no intrinsic arithmetic structure: the enumeration of $\{t_\rho\}$ is read from the zero set itself, and the diagonal operator realises its spectrum by fiat. If one replaces β_ρ by $1/2$, one has already projected the divisor to the critical line. The content of the Hilbert–Pólya problem is not the existence of $\mathcal{H}_\xi^{\text{formal}}$; it is the existence of a Hilbert space *with independent structure* whose spectral data force the original zero divisor to equal this projected divisor.

Definition 86.1.5 (non-formal target). A *non-formal target* is a pair $(\mathcal{H}_\xi, H_{\text{HP}})$ where $\mathcal{H}_\xi$ is a Hilbert space carrying structure inherited from the Bost–Connes framework (restricted-tensor adelic factorisation, Petersson inner product on cuspidal cores, GNS modular data, or another independently specified structure) and H_{HP} is a self-adjoint operator on $\mathcal{H}_\xi$ whose spectral resolution is canonical with respect to the inherited structure and whose regularised determinant

$$\det_\infty(s\text{id} - (\tfrac{1}{2}\text{id} + iH_{\text{HP}}))$$

has the non-trivial zero divisor of $\xi_{\mathbb{R}}$.

Remark 86.1.6 (domain of “canonical”). “Canonical” means: the spectral resolution of H_{HP} is determined by the inherited structure up to a unitary equivalence whose parameters are themselves determined by that structure. A diagonal construction with eigenvalues labelled by an arbitrary enumeration is not canonical; a diagonal construction whose eigenvectors are explicit functions of arithmetic data (adelic characters, automorphic forms, spectral projectors of a globally defined operator) is.

Spectral Flow as Intertwining Operator

Definition 86.1.7 (the spectral flow problem). The *spectral flow problem* from N_{BC} to H_{HP} is the specification of:

- (i) a Hilbert space $\mathcal{H}_\xi$ with a non-formal structure in the sense of Chapter 86.1.5;
- (ii) a densely defined linear operator $A : \mathcal{D}(A) \subset \mathcal{H}_{\text{BC}} \rightarrow \mathcal{H}_\xi$ with dense range;
- (iii) a self-adjoint operator H_{HP} on $\mathcal{H}_\xi$ with domain $\mathcal{D}(H_{\text{HP}}) \supset \text{Ran}(A)$;
- (iv) the intertwining identity

$$A N_{\text{BC}} A^{-1} = H_{\text{HP}} \quad \text{on } \text{Ran}(A) \cap \mathcal{D}(H_{\text{HP}}), \quad (86.1)$$

interpreted in the closed-operator sense after specifying domain compatibility;

(v) the spectral identity

$$\text{Spec}(H_{\text{HP}}) = \{\gamma_\rho \mid \rho \in \mathcal{Z}\} \quad (86.2)$$

with multiplicities matching.

A *solution* is a tuple $(\mathcal{H}_\xi, A, H_{\text{HP}})$ satisfying all five items. The *Hilbert–Pólya twist* is a solution.

Remark 86.1.8 (unitarity of A). Chapter 86.1.7 does *not* require A to be bounded or unitary. The closed-operator intertwining (86.1) admits unbounded A ; in the regular case one asks for a polar decomposition $A = U|A|$ with U partial isometry and $|A|$ self-adjoint non-negative, on which the intertwining reads $U|A|N_{\text{BC}}|A|^{-1}U^* = H_{\text{HP}}$. Under essential self-adjointness of H_{HP} and N_{BC} the intertwining is equivalent to $U \exp(itN_{\text{BC}})U^* = \exp(itH_{\text{HP}})$ on a common core, i.e. a unitary intertwining of the one-parameter groups. We use the closed-operator form throughout; when the literature writes “ A is a unitary twist” it is the polar U in this decomposition.

86.2 PRIME-ZERO MANGOLDT-HADAMARD DUALITY RECAP

The formal bridge between the two spectra is the Riemann–von Mangoldt explicit formula and its Hadamard inversion, which holds at the level of tempered distributions on $\mathbb{R}$ and is not an operator identity.

Hadamard Factorisation of $\xi_{\mathbb{R}}$

PROPOSITION 86.2.1 (*Hadamard factorisation*). (*Hadamard 1893; Chapter 55.3*). The completed Riemann zeta $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire of order 1, of genus 1, and admits the factorisation

$$\xi_{\mathbb{R}}(s) = \xi_{\mathbb{R}}(0) e^{Bs} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho},$$

with ρ ranging over non-trivial zeros with multiplicity and $B = -\log 2\pi + 1 + \gamma/2$ (Edwards 1974 §2.7) the Hadamard constant.

Riemann-von Mangoldt Explicit Formula

PROPOSITION 86.2.2 (*Riemann-von Mangoldt explicit formula*). (*von Mangoldt 1895; Edwards 1974 §3*). For $x > 1$ non-integer,

$$\sum_{n \leq x} \Lambda(n) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \frac{\zeta'(0)}{\zeta(0)} - \frac{1}{2} \log(1 - x^{-2}),$$

where $\Lambda(n) = \log p$ if $n = p^k$ and 0 otherwise.

Mellin-Dual Measures

Definition 86.2.3 (*prime and zero counting measures*). . The *prime-counting distribution* is

$$\mu_{\text{prime}} := \sum_{m \in \mathbb{N}^{\times}} \Lambda(m) \delta_{\log m} = \sum_{p \in \mathbb{P}} \sum_{k \geq 1} (\log p) \delta_{k \log p}$$

on $\mathbb{R}_{\geq 0}$. The *ordinate shadow* of the zero-counting distribution is

$$\mu_{\text{ord}} := \sum_{\rho \in \mathcal{Z}} \delta_{\gamma_{\rho}}$$

on $\mathbb{R}$. The actual Weil zero functional is the functional on entire test functions

$$\mathfrak{Z}_\xi(h) := \sum_{\rho \in \mathcal{Z}} h(z_\rho), \quad z_\rho := -i(\rho - \tfrac{1}{2}).$$

The equality $\mathfrak{Z}_\xi(h) = \int_{\mathbb{R}} h d\mu_{\text{ord}}$ for all such h is equivalent to RH.

PROPOSITION 86.2.4 (*Mellin-dual form of the explicit formula*). (*Weil 1952; Patterson 1988 Ch. 3*). For every even Paley–Wiener test function h for which the Weil explicit formula is valid,

$$\mathfrak{Z}_\xi(h) = \int_{\mathbb{R}} h dW_\infty - 2 \sum_{m \geq 2} \frac{\Lambda(m)}{\sqrt{m}} \widehat{h}(\log m),$$

with W_∞ the archimedean Weil density of (73.2). If RH holds, then $\mathfrak{Z}_\xi(h) = \int_{\mathbb{R}} h d\mu_{\text{ord}}$. Without RH, this replacement is the projection $z_\rho \mapsto \gamma_\rho$ and is not the explicit formula.

Remark 86.2.5 (the duality is distributional, not operatorial). The Mellin-Hadamard duality of Chapter 86.2.4 is a statement about a zero-functional on entire test functions and a prime-side tempered distribution on $\mathbb{R}$. It is not an operator identity. In particular, the duality does not by itself produce a linear map from one Hilbert space to another; it produces a pairing between zero samples and prime distributions that agrees up to the archimedean term. This is the precise sense in which the classical Hilbert–Pólya philosophy “interprets” the duality as an operator-level intertwining, and the reason the interpretation has not been made rigorous: no canonical passage from distributions on $\mathbb{R}$ to operators on a Hilbert space is known that realises both sides as spectral projectors of a single self-adjoint operator.

86.3 FORMAL SPECTRAL FLOW $\mathcal{A}$ AS MELLIN-INVERSION SYMBOL

The candidate spectral-flow operator exists at the symbolic level as the composite of Mellin transform and Hadamard residue extraction; the two obstructions that prevent its promotion to a Hilbert-space operator are named below.

The Mellin-Hadamard Inversion Symbol

Definition 86.3.1 (*formal Mellin-Hadamard inversion*). The *formal Mellin-Hadamard inversion* is the composite symbol

$$\mathcal{A}^{\text{sym}} := \mathcal{I}_{\text{Had}} \circ \mathcal{M}$$

where $\mathcal{M}$ is the Mellin transform $(\mathcal{M}f)(s) = \int_0^\infty f(x)x^{s-1}dx$ on a suitable function class and $\mathcal{I}_{\text{Had}}$ is the Hadamard-inversion formal symbol

$$(\mathcal{I}_{\text{Had}}F)(\gamma_\rho) := \text{Res}_{s=\rho} \frac{F(s)}{\xi_{\mathbb{R}}(s)}$$

reading the residue of $F/\xi_{\mathbb{R}}$ at each non-trivial zero and writing it into the ρ -component of a formal sequence indexed by $\mathcal{Z}$.

Remark 86.3.2 (*$\mathcal{A}^{\text{sym}}$ is not an operator*). $\mathcal{A}^{\text{sym}}$ is a symbolic map from functions on $\mathbb{R}_{>0}$ to sequences indexed by $\mathcal{Z}$. For it to be an operator between Hilbert spaces one needs:

- a Hilbert space domain $\mathcal{H}_{\text{BC}} \subset L^2$ -functions on $\mathbb{R}_{>0}$ where $\mathcal{M}$ is densely defined;
- a Hilbert space codomain $\mathcal{H}_\xi \subset \ell^2(\mathcal{Z})$ where $\mathcal{I}_{\text{Had}}$ lands;
- a compatibility condition making $\mathcal{A}^{\text{sym}}$ closable.

Each of these is a non-trivial construction.

Formal Mellin–Hadamard Obstruction

LEMMA 86.3.3 (*formal Mellin–Hadamard obstruction*). . Suppose $f \in C_c^\infty(\mathbb{R}_{>0})$ has Mellin transform $F(s) = (\mathcal{M}f)(s)$ entire, and suppose $\rho \in \mathcal{Z}$ is a simple zero of $\xi_{\mathbb{R}}$. Then

$$A^{\text{sym}}(N_{\text{BC}}f)(\gamma_\rho) = -\frac{F'(\rho)}{\xi'_{\mathbb{R}}(\rho)}, \quad A^{\text{sym}}(f)(\gamma_\rho) = \frac{F(\rho)}{\xi'_{\mathbb{R}}(\rho)}.$$

Consequently the bare residue intertwining relation

$$A^{\text{sym}} N_{\text{BC}} \stackrel{?}{=} M_\gamma A^{\text{sym}}$$

holds at ρ only under the additional interpolation condition

$$F'(\rho) = -\gamma_\rho F(\rho).$$

This condition is not a consequence of RH.

Proof. Apply the Mellin transform: N_{BC} acts on $\mathbb{R}_{>0}$ -functions as $(N_{\text{BC}}f)(x) = -\log x \cdot f(x)$ in the logarithmic scale (on the zero-mode lattice Chapter 86.1.2 the eigenvalue $\log m$ on e_m is the image under Mellin of the scaling by m). Mellin transform gives $(\mathcal{M}(N_{\text{BC}}f))(s) = -F'(s)$, where F' is the derivative in the spectral variable. Since ρ is a simple zero,

$$\text{Res}_{s=\rho} \frac{F(s)}{\xi_{\mathbb{R}}(s)} = \frac{F(\rho)}{\xi'_{\mathbb{R}}(\rho)}, \quad \text{Res}_{s=\rho} \frac{-F'(s)}{\xi_{\mathbb{R}}(s)} = -\frac{F'(\rho)}{\xi'_{\mathbb{R}}(\rho)}.$$

Multiplication by the zero ordinate would instead give $\gamma_\rho F(\rho)/\xi'_{\mathbb{R}}(\rho)$. Equality is exactly the displayed first-order interpolation condition. $\square$

Remark 86.3.4 (RH is not enough for formal intertwining). Chapter 86.3.3 shows that the Mellin generator differentiates the numerator before Hadamard residue extraction. RH would place ρ on the critical line, but it would not imply $F'(\rho) = -\gamma_\rho F(\rho)$ for the Mellin transforms in the source class. Thus the bare symbol A^{sym} is not a spectral-flow operator. A genuine operator construction must supply a new domain theorem or boundary condition that converts derivative residue data into zero-ordinate multiplication without inserting that conclusion by definition.

86.4 OBSTACLE I: HILBERT-SPACE COMPLETIONS ON BOTH SIDES

The first obstruction is the choice of Hilbert-space completions on which the Mellin–Hadamard symbol lands as a closable operator. Both sides have canonical candidates; the candidates are mutually incompatible.

Source-Side Completion

DEFINITION 86.4.1 (*source-side candidate Hilbert space*). . The source-side completion is $\mathcal{H}_{\text{BC}} = \ell^2(\mathbb{N}^\times, dm/m)$ of Chapter 86.1.2. N_{BC} acts as multiplication by $\log m$ and is essentially self-adjoint on the finite-support core (Chapter 55.2.5).

LEMMA 86.4.2 (*no Mellin transform on $\mathcal{H}_{\text{BC}}$*). . The Mellin transform $\mathcal{M} : L^2(\mathbb{R}_{>0}, d^\times x) \rightarrow L^2(\mathbb{R}, dt)$ (with $s = 1/2 + it$) is a classical unitary isomorphism (Mellin 1904; Titchmarsh 1937 Ch. 7). Its restriction to $\mathcal{H}_{\text{BC}} \subset L^2(\mathbb{R}_{>0}, d^\times x)$, identified via $e_m \leftrightarrow \delta_{x=m}$, is a map into the space of “purely atomic” distributions, which does not land in $L^2(\mathbb{R}, dt)$: the Mellin transform of a Dirac delta at m is m^{s-1} , not an L^2 -function.

Proof. $\mathcal{M}(\delta_{x=m})(s) = \int_0^\infty \delta_{x=m} x^{s-1} dx = m^{s-1}$, which is a bounded function of s on any vertical line $\text{Re}(s) = c$ but not integrable. Hence it is not an L^2 -function. The linear combinations $\sum_m c_m \delta_{x=m}$ with $\sum_m |c_m|^2/m < \infty$ have Mellin transform $\sum_m c_m m^{s-1}$, a Dirichlet series, which does not have an $L^2(\mathbb{R}, dt)$ -norm in general. $\square$

Remark 86.4.3 (the lattice embedding is ill-suited to Mellin). The issue is that $\mathcal{H}_{\text{BC}}$ embeds in $L^2(\mathbb{R}_{>0})$ only as a space of atomic measures, not as a space of locally integrable functions. Mellin transform on atomic measures produces Dirichlet series, and a Dirichlet series is a distribution, not an L^2 -function on the critical line. The Mellin-transform intertwiner of Chapter 86.3.1 therefore cannot be defined as a bounded operator from $\mathcal{H}_{\text{BC}}$ into $L^2(\mathbb{R}_{1/2+it})$.

Target-Side Completion

Definition 86.4.4 (target-side candidates). . Three candidate Hilbert spaces for $\mathcal{H}_\xi$ in the non-formal sense:

- (i) $\mathcal{H}_\xi^{\text{idele}} = L^2(C_{\mathbb{Q}}, d^\times x)$ the idele class Hilbert space of Chapter 79.2.1;
- (ii) $\mathcal{H}_\xi^{\text{adele}} = L^2(X_{\mathbb{Q}})$ the adèle-class space with non-commutative-geometric L^2 -structure via $C^*(\mathbb{A}/\mathbb{Q}^\times)$ (Chapter 79.5.1);
- (iii) $\mathcal{H}_\xi^{\text{Petersson}}$ the Petersson completion of the sph-finite cuspidal core (Chapter 21.7.1).

LEMMA 86.4.5 (*Mellin transform on $\mathcal{H}_\xi^{\text{idele}}$*). (*Meyer 2005 §3.2; Chapter 79.2.2*). The Mellin transform on $L^2(\mathbb{R}_{>0}, d^\times t)$ into $L^2(\mathbb{R}_{1/2+it}, dt)$ is unitary; it extends to the idele class Hilbert space $\mathcal{H}_\xi^{\text{idele}}$ by tensoring with the identity on $L^2(C_{\mathbb{Q}}^1)$.

Remark 86.4.6 (the completion mismatch). Chapter 86.4.2 shows Mellin fails on the source $\mathcal{H}_{\text{BC}}$, and Chapter 86.4.5 shows Mellin is unitary on the target $\mathcal{H}_\xi^{\text{idele}}$. The two completions are *not* isometric: $\mathcal{H}_{\text{BC}}$ is a space of atomic measures on primes; $\mathcal{H}_\xi^{\text{idele}}$ is an L^2 of the continuous idele norm. The passage from one to the other requires either (a) a Schwartz-class density argument lifting atomic measures to a common completion where Mellin is densely defined (Meyer 2005 §3 construction) or (b) a direct factorisation homology argument making the zero-mode lattice into a dense subspace of a factorisation algebra whose Mellin transform lives naturally on $\mathbb{R}_{1/2+it}$ (Costello–Gwilliam construction). Neither is delivered unconditionally by the Bost–Connes framework alone.

86.5 OBSTACLE 2: KERNEL OF $\mathcal{A}$ AS BOUNDED/UNBOUNDED OPERATOR

Even accepting a compatible Hilbert-space completion, $\mathcal{A}$ must have an explicit kernel making it closable. The Hadamard inversion step is the bottleneck.

The Hadamard Inversion Kernel

Definition 86.5.1 (formal Hadamard kernel). . The formal kernel of $\mathcal{I}_{\text{Had}}$: space of entire functions on $\mathbb{C} \rightarrow$ sequences on $\mathcal{Z}$ is

$$K_{\text{Had}}(\rho, s) := \frac{1}{2\pi i} \cdot \frac{1}{\xi_{\mathbb{R}}(s)} \cdot \frac{1}{s - \rho}$$

interpreted in the residue sense at $s = \rho$.

LEMMA 86.5.2 (*the Hadamard kernel is not a Hilbert-Schmidt kernel*). . For the critical-line Mellin completion compatible with the spectral-flow condition, the Hadamard kernel K_{Had} is not

Hilbert–Schmidt. Its Hilbert–Schmidt norm $\|K_{\text{Had}}\|_{\text{HS}}^2 = \sum_{\rho} \int_{\text{Re}(s)=1/2} |\xi_{\mathbb{R}}(s)|^{-2} |s - \rho|^{-2} ds$ diverges at every zero ρ lying on the critical line.

Proof. If ρ lies on the critical line, then $|\xi_{\mathbb{R}}(s)|^{-2} \sim C_{\rho} |s - \rho|^{-2m_{\rho}}$ along the contour near $s = \rho$. The integrand is then $|s - \rho|^{-2m_{\rho}-2}$, non-integrable on a one-dimensional contour for $m_{\rho} \geq 1$. Hardy’s theorem gives infinitely many zeros on the critical line. Hence the critical-line Hadamard residue kernel is not Hilbert–Schmidt. Off-line zeros do not give this local contour singularity; they are not needed for the conclusion. $\square$

Remark 86.5.3 (unboundedness is intrinsic). Chapter 86.5.2 means that A cannot be a bounded operator under any reasonable pair of completions. The spectral flow is therefore necessarily an *unbounded* operator, requiring the machinery of closable operators with dense domain. The choice of domain becomes a substantive part of the problem, not a technicality.

The Domain Problem

PROPOSITION 86.5.4 (*candidate domains for A*). Three candidate domains for A in $\mathcal{H}_{\text{BC}}$:

- (i) $\mathcal{D}_{\text{fin}}$: finite-support vectors $\sum_{m \leq M} c_m e_m$ for M finite. On these, A can be computed via residue extraction; the result is a finite sum of residues, well-defined.
- (ii) $\mathcal{D}_{\text{Schwartz}}$: Schwartz-class-like vectors $\sum_m c_m e_m$ with $|c_m| = O(m^{-N})$ for all N . On these, the Dirichlet series $F(s) = \sum_m c_m m^{s-1}$ extends entirely with rapid decay on vertical strips; A is defined distributionally by residue contour deformation.
- (iii) $\mathcal{D}_{\text{BR}}$: the Beurling–Rajagopalan domain of functions whose Mellin transform has exponential-polynomial growth on the critical strip bounded by Plancherel–Pólya.

On $\mathcal{D}_{\text{fin}}$ and $\mathcal{D}_{\text{Schwartz}}$, A is densely defined but closability is conditional on the decay rate of c_m being compatible with Hadamard inversion.

Proof. Item (1): a finite sum $\sum_{m \leq M} c_m e_m$ maps under A to $\sum_{\rho} r_{\rho}(f) \cdot e_{\rho}^{\text{target}}$ where $r_{\rho}(f) = \text{Res}_{s=\rho} (\sum_{m \leq M} c_m m^{s-1}) / \xi_{\mathbb{R}}(s)$ is a finite expression involving derivatives of $\xi_{\mathbb{R}}^{-1}$ at ρ . The sum over ρ is formal; summability depends on $\sum_{\rho} |r_{\rho}|^2 < \infty$, which holds for Schwartz c_m but not in general. Item (2): Schwartz decay on c_m gives entire $F(s)$ with rapid decay on vertical strips; residue deformation converts the sum $\sum_{\rho} r_{\rho}(f)$ into a contour integral over a vertical line shifted to the left, plus boundary terms at $s = 0, 1$ and at trivial zeros. The contour-integral part is a Fourier–Mellin integral bounded by Plancherel; the boundary terms are finite. Item (3): Beurling 1955 (Ann. Math. 61) and Rajagopalan 1960 show that the Beurling–Rajagopalan class is a natural setting for Mellin inversion with controlled exponential growth, used in Nyman–Beurling reformulations of RH; see Chapter 71 of the present volume for the chiral-algebra adaptation. $\square$

Remark 86.5.5 (no domain gives self-adjointness). None of the three candidate domains of Chapter 86.5.4 closes to a domain on which $A^* A = |A|^2$ has a canonical square-root polar decomposition with unitary part U satisfying the intertwining (86.1). Essential self-adjointness of $A^* A$ on any of these domains is an open question, and its failure is the functional-analytic face of the Hilbert–Pólya obstruction.

86.6 CONNES 1999 PARTIAL APPROACH AND ITS LIMITATION

Connes’ adèle class space construction gives an explicit Hilbert space and scaling generator together with the trace formula. The Hilbert–Pólya input is the further assertion that its absorption trace distribution is the non-trivial zero distribution, with critical-line support.

The Connes Construction

Definition 86.6.1 (Connes adele class data). (Connes 1999 *arXiv:math/9811068 §IV*). The Connes adele class Hilbert space is $\mathcal{H}_{\text{Connes}} = L^2(X_{\mathbb{Q}}, dx)$ with $X_{\mathbb{Q}} = \mathbb{A}/\mathbb{Q}^\times$ the adele class space and dx the Haar measure inherited from $\mathbb{A}$ modulo $\mathbb{Q}^\times$; the quotient is non-Hausdorff, and the L^2 -structure is realised via the crossed product $C^*(\mathbb{A}/\mathbb{Q}^\times) = C_o(\mathbb{A}) \rtimes \mathbb{Q}^\times$. The Connes scaling generator is $D_{\text{Connes}} = -i\partial_{\log \lambda} U(\lambda)$ at $\lambda = 1$, where $U(\lambda)f(x) = |\lambda|^{-1/2}f(\lambda^{-1}x)$. On the cuspidal subspace $\mathcal{E} \subset \mathcal{H}_{\text{Connes}}$ (orthogonal to the Tate-regularised constants), D_{Connes} is conjectured to have absorption trace equal to the zero distribution of $\xi_{\mathbb{R}}$. Under zero-placement this trace is supported at the ordinate multiset $\{\gamma_\rho\}$.

Theorem 86.6.2 (Connes trace formula equivalence). (Connes 1999 *§V Thm. 1; Chapter 83.2.1*). For $h \in \mathcal{H}_{\text{test}}$ (the even-Schwartz test class on $\mathbb{R}$), the regularised trace $\text{Tr}_{\text{reg}}(h(D_{\text{Connes}})|_{\mathcal{E}})$ equals the Weil distribution side, conditional on the absorption-trace identification. With zero-placement, the regularised trace is $\sum_\rho h(\gamma_\rho)$.

The Limitation: Absorption Trace as RH

Theorem 86.6.3 (Connes absorption-trace conjecture). (Connes 1999 *§IV Thm. 1 conditional form*). The scaling generator D_{Connes} on $\mathcal{E} \subset \mathcal{H}_{\text{Connes}}$ has absorption trace equal to the non-trivial zero distribution of $\xi_{\mathbb{R}}$; after zero-placement this distribution is supported at the multiset $\{\gamma_\rho \mid \rho \in \mathcal{Z}\}$ with multiplicity.

Remark 86.6.4 (Connes does not construct A unconditionally). Connes 1999 does not construct the intertwining A unconditionally. The Connes construction supplies a candidate Hilbert space $\mathcal{H}_{\text{Connes}}$ and a candidate self-adjoint operator D_{Connes} , but the identification of D_{Connes} 's absorption trace with the zero distribution and critical-line support is equivalent to RH. In the spectral-flow vocabulary, Connes supplies the triple $(\mathcal{H}_\xi, A, H_{\text{HP}}) = (\mathcal{H}_{\text{Connes}}, \text{id}, D_{\text{Connes}})$ with identity intertwiner, and shifts the question from “construct A ” to “prove the right absorption trace for D_{Connes} ”. The two questions have the same strength.

Proposition 86.6.5 (Connes equivalence to $H\text{-P}^{\text{Ar}}$). (Connes 1999 *§VII; Chapter 35.3.9 as recorded in Ch. 35*). The Connes absorption-trace condition (Chapter 86.6.3) is equivalent to the Hilbert–Pólya hypothesis $H\text{-P}^{\text{Ar}}$ (Chapter 22.3.2).

Remark 86.6.6 (the adele-class construction transfers the problem). The Connes construction transfers the spectral-flow problem from the Bost–Connes GNS side to the adele-class side. The adele-class side has an *intrinsic* candidate for the Hilbert space (L^2 of the non-Hausdorff quotient) and an *intrinsic* candidate for the zero-divisor operator (scaling generator on cuspidal subspace), but the intrinsic candidates are equivalent to the abstract candidates of Chapter 22.3.2, not independently constructed. The transfer does not shrink the problem; it renames it.

86.7 MEYER 2005 PARTIAL APPROACH AND ITS LIMITATION

Meyer’s Schwartz-class refinement provides the most explicit existing operator-level treatment. It closes the spectral-flow problem on a dense Schwartz subspace, and transfers the residual to the determinant-trace conjecture.

The Meyer Construction

Definition 86.7.1 (Meyer Schwartz space and scaling). (Meyer 2005 *arXiv:math/0311468 §3*). The Meyer Schwartz space $\mathcal{S}_{\text{Meyer}}(\mathbb{A})$ is the sub-Schwartz space of $L^2(\mathbb{A}^\times/\mathbb{Q}^\times, d^\times x)$ consisting of functions with

rapid decay at all places adelically (Meyer §3.1). The Meyer scaling D_{Meyer} is the restriction of the idele scaling D_o of Chapter 79.2.1 to $\mathcal{S}_{\text{Meyer}}(\mathbb{A})$.

THEOREM 86.7.2 (*Meyer determinant-trace intertwiner*). (*Meyer 2005 Thm. 4.4, §4.2*). There is an explicit Mellin-unitary intertwiner

$$U_{\text{Meyer}} : \mathcal{S}_{\text{Meyer}}(\mathbb{A}) \rightarrow L^2(\mathbb{R}_{1/2+it}, d\tau)$$

conjugating D_{Meyer} to the multiplication-by- τ operator on the critical line and landing on the Paley-Wiener Schwartz-like class with distributional pole data at the non-trivial zeros of $\xi_{\mathbb{R}}$.

The Limitation: Pole Data Matching

Remark 86.7.3 (*Meyer's intertwiner is not $\mathcal{A}$*). U_{Meyer} is a unitary from Meyer's Schwartz space into L^2 of the critical line. It does not intertwine N_{BC} with $H_{\text{ord}}^{\text{formal}}$; it intertwines D_{Meyer} with multiplication by the spectral variable. The pole data at non-trivial zeros is distributional, not an eigenvalue realisation. To upgrade U_{Meyer} to a spectral-flow $\mathcal{A}$, one must pass from the distributional pole structure on $L^2(\mathbb{R}_{1/2+it})$ to an ℓ^2 -sum over zeros, which requires the zeros to be discrete points on the critical line: this is exactly RH.

THEOREM 86.7.4 (*Meyer equivalence to $H\text{-P}^{\text{Ar}}$*). (*Meyer 2005 §5 conditional*). The Meyer determinant-trace conjecture (that U_{Meyer} extends continuously to a unitary intertwiner carrying the Bost–Connes trace to the zero distribution) is equivalent to the Connes absorption-trace conjecture (Chapter 86.6.3), hence to $H\text{-P}^{\text{Ar}}$.

Remark 86.7.5 (*Meyer gives Schwartz-class closure, no more*). The Meyer construction closes the spectral flow unconditionally on the Schwartz sub-space $\mathcal{S}_{\text{Meyer}}(\mathbb{A})$, intertwining D_{Meyer} with multiplication by τ on $L^2(\mathbb{R}_{1/2+it})$. It does *not* close the identification of the pole data on the critical line with the ℓ^2 -sum over zeros; that extension is the Meyer determinant-trace conjecture, equivalent to Connes absorption trace, equivalent to $H\text{-P}^{\text{Ar}}$. Meyer therefore supplies the left half of the spectral flow (Mellin transform) unconditionally and leaves the right half (Hadamard inversion as determinant-trace realisation) as the conjectural kernel.

86.8 RESIDUAL: IRREDUCIBLE OPERATOR-THEORETIC OBSTRUCTION

The irreducible residual is the absorption-trace property of the archimedean scaling on the cuspidal subspace, isolated and named below.

Statement of the Irreducible Obstruction

THEOREM 86.8.1 (*irreducible residual for spectral flow*). . The spectral-flow problem (Chapter 86.1.7) reduces to the following operator-theoretic statement on the idele class Hilbert space $\mathcal{H}^{\text{id}} = L^2(C_{\mathbb{Q}}, d^{\times}x)$.

$\mathcal{A}\text{-TP}^{\text{arch}}$: The Meyer scaling generator D_{Meyer} , extended from the Schwartz sub-space $\mathcal{S}_{\text{Meyer}}(\mathbb{A})$ to the cuspidal subspace $\mathcal{E} \subset \mathcal{H}^{\text{id}}$, has absorption trace equal to the zero distribution of $\xi_{\mathbb{R}}$; after zero-placement that trace is supported at $\{\gamma_{\rho} \mid \rho \in \mathcal{Z}\}$ with multiplicity. Equivalently, the closure of U_{Meyer} lifts the distributional Paley–Wiener class on $L^2(\mathbb{R}_{1/2+it})$ to an ℓ^2 -direct sum over the non-trivial zeros.

The following reductions are unconditional.

- (i) The source-side data $(\mathcal{H}_{\text{BC}}, N_{\text{BC}})$ is unconditional (Chapter 55.2.5).
- (ii) The Meyer Mellin unitary $U_{\text{Meyer}} : \mathcal{S}_{\text{Meyer}}(\mathbb{A}) \rightarrow L^2(\mathbb{R}_{1/2+it})$ is unconditional (Chapter 86.7.2).
- (iii) The formal Mellin–Hadamard residue calculation is unconditional and obstructs the bare symbolic intertwining: Mellin scaling differentiates the numerator, while zero-ordinate multiplication would multiply the residue (Chapter 86.3.3).
- (iv) The equivalence $\text{H-PAr} \Leftrightarrow \text{Connes absorption trace} \Leftrightarrow \text{Meyer determinant-trace matching} \Leftrightarrow \text{A-TP}^{\text{arch}}$ is proven within the Vol IV equivalence chain (Chapter 35.3.14, Chapter 79.3.4).

What is *not* unconditional is $\text{A-TP}^{\text{arch}}$ itself: it is the absorption-trace property, not derivable from the source-side data or the Mellin unitary by any known argument.

Proof. Items (1)–(4) are cited. The content of the theorem is the reduction statement: the spectral-flow problem in Chapter 86.1.7 admits a decomposition

$$\mathcal{A} = (\text{determinant-trace inclusion}) \circ U_{\text{Meyer}} \circ (\text{embedding } \mathcal{H}_{\text{BC}} \rightarrow \mathcal{S}_{\text{Meyer}}(\mathbb{A}))$$

in which the embedding and U_{Meyer} are unconditional, while the determinant-trace inclusion into an ℓ^2 -direct sum is the irreducible residual. The embedding $\mathcal{H}_{\text{BC}} \hookrightarrow \mathcal{S}_{\text{Meyer}}(\mathbb{A})$ is the adelic version of the Bost–Connes to idele scaling embedding (Meyer 2005 §4.1); it is bounded after the Meyer normalisation. Under the embedding, N_{BC} maps to a summand of D_{Meyer} restricted to the spherical character sector; the summand corresponds to the finite-prime part of the scaling, while the archimedean part is the Arakelov scaling $\partial_{\log t_\infty}$ at the real place. The intertwining (86.1) holds on the embedded core by Chapter 86.7.2. The spectral identity (86.2) requires the absorption-trace property ($\text{A-TP}^{\text{arch}}$), which is the Connes–Meyer determinant-trace conjecture. $\square$

Remark 86.8.2 (the residual is at the archimedean place). The Meyer intertwining (Chapter 86.7.2) is derived place-by-place from the Tate 1950 local functional equations. At each finite prime, the local trace matching is elementary: the BC generator μ_p acts with eigenvalue p^{-it} on the Mellin side, conjugate to multiplication by t by a change of variable. At the archimedean place, the Arakelov scaling acts with eigenvalue e^{-it} on the Mellin side, and the spectral identification requires the scaling trace to be the zero distribution on the cuspidal sub-space. With zero-placement this trace is supported at $\{\gamma_\rho\}$. The latter condition is equivalent to A-TP (Chapter 56.7.2) by Chapter 79.3.4, hence the label $\text{A-TP}^{\text{arch}}$. The obstruction is purely archimedean; the finite-prime trace matching is delivered by Bost–Connes unconditionally.

Spectral-Flow Logical Table

Component	Form	Reference
Source $(\mathcal{H}_{\text{BC}}, N_{\text{BC}})$, self-adjoint	Unconditional	Chapter 55.2.5
Source Mellin inversion on $\mathcal{H}_{\text{BC}}$	Obstructed (atomic measures)	Chapter 86.4.2
Embedding $\mathcal{H}_{\text{BC}} \hookrightarrow \mathcal{S}_{\text{Meyer}}(\mathbb{A})$	Unconditional after Meyer renormalisation	Chapter 79.2.7
Meyer unitary U_{Meyer} on Schwartz sub-space	Unconditional	Chapter 86.7.2
Hadamard kernel K_{Had} Hilbert-Schmidt	Obstructed (divergent at zeros)	Chapter 86.5.2
Naive formal symbolic intertwining	Obstructed by derivative residue	Chapter 86.3.3
Absorption trace of $D_{\text{Meyer}} _{\mathcal{E}}$ equals the zero distribution	Conjectural (A-TP ^{arch})	Chapter 86.8.1
Connes equivalence with H-PA ^r	Proved equivalent	Chapter 86.6.5
Meyer determinant-trace equivalence with A-TP ^{arch}	Proved equivalent	Chapter 86.7.4
Existence of $\mathcal{A}$ as closed operator	Conjectural = A-TP ^{arch}	Chapter 86.8.1
Existence of $\mathcal{A}$ as bounded operator	Impossible (see unboundedness)	Chapter 86.5.3

Why the Residual is Irreducible

THEOREM 86.8.3 (*irreducibility of A-TP^{arch}*). A-TP^{arch} cannot be reduced to any further spectral-flow identity on the Bost–Connes / idele framework: it is the terminus of the reduction chain.

Proof. Under any Hilbert-space completion $\mathcal{H}_{\xi}$ of the target, the trace distribution of $D_{\text{Meyer}}|_{\mathcal{E}}$ is determined by the Tate-Mellin transform of the cuspidal forms modulo the residue structure at non-trivial zeros. The passage from pole data to the zero trace distribution is precisely the determinant-trace conjecture; any finer reduction would require an *a priori* realisation of the cuspidal forms as an ℓ^2 -sum over non-trivial zeros, which presupposes the critical-line placement. No unconditional construction exists: every known partial construction (Connes’ adèle class, Meyer’s Schwartz refinement, de Branges’ Hermite-Biehler, Nyman-Beurling, Guinand-Weil, Bost–Connes GNS, scattering reformulations of Chapter 79) transfers the problem to an equivalent absorption-trace or positivity statement, each equivalent to RH.

A direct obstruction: the bare symbolic intertwining $\mathcal{A}^{\text{sym}} N_{\text{BC}} = H_{\text{HP}} \mathcal{A}^{\text{sym}}$ does not follow from residue calculus. By Chapter 86.3.3, the left hand side extracts the residue of $-F'/\xi_{\mathbb{R}}$, while the right hand side multiplies the residue of $F/\xi_{\mathbb{R}}$ by γ_{ρ} . Equality requires the first-order interpolation condition $F'(\rho) = -\gamma_{\rho}F(\rho)$ at every non-trivial zero. RH alone does not give this. Thus any closed-operator upgrade of $\mathcal{A}^{\text{sym}}$ must add a genuine domain/boundary theorem. The irreducible residual is this passage; it has no known proof independent of RH. $\square$

Remark 86.8.4 (*Beilinson principle*). The assertions are:

- (i) The source $(\mathcal{H}_{\text{BC}}, N_{\text{BC}})$ is unconditional.
- (ii) The Meyer Mellin unitary on Schwartz sub-spaces is unconditional.
- (iii) The formal spectral flow $\mathcal{A}^{\text{sym}}$ is a well-defined symbolic composition.

- (iv) The symbolic-to-operatorial upgrade is the Connes-Meyer-de Branges-Nyman irreducible obstruction, which is equivalent to RH.

$H\text{-P}^{\text{Ar}}$ remains the determinant-trace hypothesis. The irreducible obstruction is the spectral-flow problem in the arithmetic formalism.

86.9 PLACEMENT IN TRIPLE-IDENTIFICATION FRAMEWORK

The spectral-flow problem of Chapter 86.1.7 appears three more times in the Vol IV arithmetic branch under different names, in Chapters 82, 79, and 81; the present section records these identifications.

Spectral Flow as Deninger Spectral-Flow Theorem

PROPOSITION 86.9.1 (*spectral flow = Deninger spectral-flow*). (on the triple-identification and L_{o} -conjugacy hypotheses). Under the Deninger triple identification and the L_{o} -conjugacy hypotheses cited below, the spectral-flow problem of Chapter 86.1.7 coincides with the Deninger spectral-flow conjecture Chapter 81.6.5. This is a dictionary identification of two names for the same residual, not an independent construction of the spectral-flow operator.

Proof. Unpack the Deninger conjecture Chapter 81.6.5: it posits a closed operator $\Theta_{\text{Den}} = \frac{1}{2}\text{id} + iH_{\text{Den}}$ on a completion of the primary middle cohomology $H_{\text{Den}, \chi, \text{prim}}^1$ with H_{Den} self-adjoint in the Petersson/GNS inner product, determinant $\xi_{\mathbb{R}}$, and Weil–Connes trace distribution equal to the zero-side. The Hilbert completion is $\mathcal{H}_{\xi}^{\text{Petersson}}$ of Chapter 86.4.4, and the operator H_{Den} is H_{HP} in the Vol IV vocabulary (Chapter 22.3.6).

The Deninger triple identification (Chapter 82.4.1) identifies the Petersson completion with the arithmetic chiral homology of H^{Ar} and with the boundary chiral homology of $\text{ACS}_{k^{\text{Ar}}}^{\text{qu}}$. The spectral-flow operator $\mathcal{A}$ of Chapter 86.1.7 is exactly the unitary U of Chapter 82.5.4: it conjugates the archimedean Frobenius $F_{\infty} = e^{-\partial_t}$ to the Virasoro- L_{o} representation on the arithmetic Heisenberg Fock, and via the primary-Fock embedding of Chapter 55.3.2 maps N_{BC} to the zero-side operator H_{HP} on the completed primary sector. $\square$

Remark 86.9.2 (*the Deninger framework renames the residual*). The Deninger framework does not close the spectral-flow problem; it labels the same residual with the arithmetic-cohomology name. The irreducible obstruction $\text{A-TP}^{\text{arch}}$ of Chapter 86.8.1 is Deninger axiom (3) (regularised-determinant identity) in the Vol IV realisation (Chapter 82.3.4, conditional on the spectral-flow conjecture). Naming the problem in cohomological language does not shrink it.

Spectral Flow as A-TP Adelic-Scattering

PROPOSITION 86.9.3 (*spectral flow = A-TP adelic-scattering*). The irreducible residual $\text{A-TP}^{\text{arch}}$ of Chapter 86.8.1 coincides with A-TP on $\text{PW}(\mathbb{R}) \setminus \text{PW}^{\text{TM}}$ in the scattering formulation of Chapter 79.

Proof. The Meyer determinant-trace conjecture (Chapter 86.7.4) is equivalent to unitarity of the scattering matrix S of the Bost–Connes-perturbed scaling generator pair $(D_{\text{o}}, D_{\text{BC}})$ on the full $\text{PW}(\mathbb{R})$; this is exactly Chapter 79.3.4 in its forward direction. The extension from Tate-Mellin sub-class PW^{TM} to the full $\text{PW}(\mathbb{R})$ is Chapter 79.5.3. Under the embedding $\mathcal{H}_{\text{BC}} \hookrightarrow \mathcal{S}_{\text{Meyer}}(\mathbb{A})$ of the spectral-flow setup, the absorption-trace condition becomes the scattering-matrix unitarity condition. Identification. $\square$

Remark 86.9.4 (*scattering transfers the obstruction*). The scattering formulation of Chapter 79 closes the spectral flow unconditionally on the Tate-Mellin sub-class PW^{TM} and transfers the residual on

$\text{PW}(\mathbb{R}) \setminus \text{PW}^{\text{TM}}$ to the Connes non-commutative L^2 -structure on the adèle class space. In spectral-flow vocabulary, Tate-Mellin closure corresponds to the zero-adapted Schwartz functions on $\mathcal{H}_{\text{BC}}$ where the Hadamard inversion is algebraically realisable by local functional equations; the full extension requires the absorption property.

Spectral Flow as 3D ACS Bulk-Boundary Correspondence

PROPOSITION 86.9.5 (*spectral flow = 3D ACS level-deformation*). (on Chapter 81.4.7). Under the 3D arithmetic Chern-Simons conjecture Chapter 81.4.7, the spectral-flow operator $\mathcal{A}$ is the level-derivative operator on the primary sector of $\mathcal{V}^{\text{prim}}$ (the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 , which is not a vertex algebra; the 3D ACS conjecture endows it with a Hilbert-space completion on the bulk primary sector).

Proof. The 3D ACS bulk partition function satisfies $Z_{\text{ACS}^{\text{qu}}}(\overline{\text{Spec } \mathbb{Z}}; s) = \xi_{\mathbb{R}}(s)$ (Chapter 81.5.7). The level-derivative operator ∂_s on the bulk Hilbert space acts, by the canonical-quantisation axiom Chapter 81.4.7(2), as the derivative of the Virasoro- L_0 eigenvalue with respect to the spectral parameter, equivalently as the Arakelov-scaling Frobenius F_{∞} of Chapter 82.5.3. The spectral-flow operator $\mathcal{A}$ conjugates this to the multiplication-by- γ_p operator on the zero index set via the Mellin unitary of Chapter 82.5.4. $\square$

Remark 86.9.6 (*the 3D framework renames the obstruction as Wilson-loop matching*). The 3D ACS framework reformulates the spectral-flow obstruction as a Wilson-loop correlator matching condition: the level-derivative operator must have determinant divisor $\text{div } \xi_{\mathbb{R}}$ and zero trace distribution under the Arakelov-Kähler Wilson-surface realisation (Chapter 82.5.7). This is equivalent to Deninger axiom (2), to Meyer determinant-trace matching, to Connes absorption, to $A\text{-TP}^{\text{arch}}$, to $H\text{-P}^{\text{Ar}}$. The 3D framework adds geometric content (the bulk theory becomes a concrete Chern-Simons computation) but does not change the obstruction’s strength.

The Unified Residual

THEOREM 86.9.7 (*unified residual, five equivalent names*). (conditional on the referenced equivalences). The spectral-flow problem (Chapter 86.1.7) reduces to a single operator-theoretic statement, carrying five equivalent names across the arithmetic formalism:

- (i) $H\text{-P}^{\text{Ar}}$ (Chapter 22.3.2): existence of a self-adjoint operator H_{HP} with canonical spectral resolution on a Petersson completion and

$$\det_{\infty}(s\text{id} - (\tfrac{1}{2}\text{id} + iH_{\text{HP}})) \doteq \xi_{\mathbb{R}}(s)$$

after removal of the polar Tate factors.

- (ii) *Connes absorption* (Chapter 86.6.3): absorption trace of the scaling generator on the cuspidal subspace of the adèle class space.
- (iii) *Meyer determinant-trace matching* (Chapter 86.7.4): extension of the Meyer Mellin unitary from Schwartz sub-spaces to the full cuspidal sector.
- (iv) $A\text{-TP}$ (Chapter 56.7.1): Weil-positivity of the archimedean Tate distribution on $\text{PW}(\mathbb{R})$.
- (v) *Deninger spectral-flow* (Chapter 81.6.5): existence of the middle-cohomology operator in the arithmetic cohomology of $\overline{\text{Spec } \mathbb{Z}}$.

All five are equivalent to each other and to the Riemann Hypothesis for ζ .

Proof. In the arithmetic formalism, the five equivalences are: (1) $\Leftrightarrow$ (2) by Chapter 86.6.5; (2) $\Leftrightarrow$ (3) by Chapter 86.7.4; (3) $\Leftrightarrow$ (4) by Chapter 79.3.4; (4) $\Leftrightarrow$ RH by Chapter 56.7.2; (1) $\Leftrightarrow$ (5) by Chapter 86.9.1. The chain closes with all five mutually equivalent to RH. $\square$

Remark 86.9.8 (spectral-flow boundary). The spectral-flow problem $N_{\text{BC}} \rightsquigarrow H_{\text{HP}}$ is the classical Hilbert–Pólya twist. The five formulations above are equivalent after their stated hypotheses; each supplies different geometric or analytic structure, but none supplies the residual operator. The residual is irreducible in this formalism: it is the absorption-trace property of the archimedean scaling on the cuspidal subspace, equivalent to RH.

The unconditional content is the Bost–Connes logarithmic Hamiltonian, the Mellin unitary on Schwartz sub-spaces, and the formal Mellin–Hadamard obstruction. The missing object is the irreducible operator-theoretic absorption-trace property. It does not close H-P^{Ar} . Thus the operator construction remains the Hilbert–Pólya residual itself.

86.10 POLYAKOV–DIRAC NORMAL FORM FOR THE RESIDUAL

The Polyakov rank-residue calculation fixes the determinant scalar $r_\zeta = 2$, closing off any attempt to hide the problem in a different Euler-product normalisation. The Dirac reformulation converts the remaining content into a first-order essential self-adjointness question on the residual symmetric core, eliminating the possibility of concealing the Hilbert–Pólya content inside an unspecified unbounded operator.

Polyakov Rigidity of the Rank Residue

PROPOSITION 86.10.1 (Polyakov rank-residue rigidity). . In the primary Heisenberg domain of Chapters 14 and 85, any spectral-flow closure compatible with

- (i) the rank-2 per-prime Sugawara Heisenberg stress tensor,
- (ii) the global Fock partition $Z(s) = \prod_p \prod_{n \geq 1} (1 - p^{-sn})^{-2}$, and
- (iii) the archimedean Polyakov anomaly of the rank-2 free-field completion,

has determinant rank residue

$$r_\zeta = 2.$$

Consequently the residual spectral-flow problem cannot be shifted into a different determinant normalisation. Its remaining content is the operator-domain and positivity problem named $\text{A-TP}^{\text{arch}}$ in Chapter 86.8.1.

Proof. Chapter 85.6.1 proves that the unrenormalised conformal scalar $\sum_p c^{(p)}$ diverges, that the negative global modes do not act on the restricted vacuum, and that the global Fock-partition residue gives $r_\zeta = 2$. The same chapter proves the Polyakov rank comparison Chapter 85.5.3, identifying this finite-prime residue with the rank-2 archimedean conformal anomaly. These inputs leave no determinant scalar parameter: changing r_ζ would either abandon the Euler-product partition, abandon the Sugawara rank, or abandon the archimedean anomaly match. None of those changes is a spectral-flow closure inside the stated primary Heisenberg domain. The only remaining unsettled part is therefore the closed-operator passage from the Meyer/Connes absorption symbol to a self-adjoint zero-divisor operator, which is exactly $\text{A-TP}^{\text{arch}}$ by Chapter 86.8.1. $\square$

Remark 86.10.2 (what Polyakov does and does not buy). Polyakov regularisation is decisive for the determinant rank, not for RH. It supplies the finite scalar $r_\zeta = 2$ and rules out the divergent prime-zeta prescription; it does not produce the Hilbert space on which the zero ordinates are a self-adjoint spectrum.

Dirac Closure of the Remaining Operator

Definition 86.10.3 (residual symmetric ordinate core). . A residual symmetric ordinate core is data

$$(C_{\text{res}}, H_{\text{res}}^{\text{sym}})$$

where $C_{\text{res}} \subset \mathcal{E}$ is an L^2 -dense candidate core and $H_{\text{res}}^{\text{sym}}$ is a symmetric operator on it. The intended source of C_{res} is the Meyer Mellin unitary applied to the Schwartz adelic core, together with a Hadamard-residue boundary condition at the non-trivial zeros, as in Chapter 86.7.2 and Chapter 86.3.1. This sentence is a source description, not a graph-density theorem. The graph-density and non-tautology of the Hadamard-residue boundary condition are the residual conditions isolated below.

On the Mellin line $H_{\text{res}}^{\text{sym}}$ is represented by multiplication by the real spectral variable t . After the Hadamard-residue boundary condition it is intended to become formal multiplication by γ_ρ on the zero components. Its closure is not assumed to exist as a self-adjoint operator. The superscript “sym” is part of the condition: if the residue data produces non-real ordinate coefficients, no symmetric operator has yet been constructed, and the Dirac closure below fails.

Definition 86.10.4 (Dirac double of the residual). . The Dirac double of the residual ordinate core is the odd symmetric operator

$$\mathcal{D}_{\text{res}}^{\circ} := \begin{pmatrix} \circ & H_{\text{res}}^{\text{sym}} \\ H_{\text{res}}^{\text{sym}} & \circ \end{pmatrix} \quad \text{on} \quad C_{\text{res}} \oplus C_{\text{res}}, \quad (86.3)$$

with grading $\Gamma = \text{diag}(1, -1)$. A Dirac closure is a self-adjoint closure $\mathcal{D}_{\text{res}} = \overline{\mathcal{D}_{\text{res}}^{\circ}}$ whose spectral measure satisfies the regularised determinant identity

$$\det_{\infty}(s \cdot \text{id} - (\tfrac{1}{2}\text{id} + iH_{\text{res}})) = \xi_{\mathbb{R}}(s), \quad (86.4)$$

where H_{res} is the off-diagonal component recovered from $\mathcal{D}_{\text{res}}$ by the grading.

LEMMA 86.10.5 (Dirac double detects self-adjointness). . Let H be a densely defined symmetric operator on a Hilbert space $\mathcal{H}$ and $\mathcal{D}_H^{\circ} = \begin{pmatrix} \circ & H \\ H & \circ \end{pmatrix}$ on $\mathcal{D}(H) \oplus \mathcal{D}(H)$. Then $\mathcal{D}_H^{\circ}$ is essentially self-adjoint if and only if H is essentially self-adjoint. When this holds,

$$\overline{\mathcal{D}_H^{\circ}} = \begin{pmatrix} \circ & \overline{H} \\ \overline{H} & \circ \end{pmatrix}, \quad \overline{\mathcal{D}_H^{\circ}}^2 = \begin{pmatrix} \overline{H}^2 & \circ \\ \circ & \overline{H}^2 \end{pmatrix}.$$

Proof. For a symmetric H , $(\mathcal{D}_H^{\circ})^* = \begin{pmatrix} \circ & H^* \\ H^* & \circ \end{pmatrix}$ on $\mathcal{D}(H^*) \oplus \mathcal{D}(H^*)$. Hence the closure of $\mathcal{D}_H^{\circ}$ is self-adjoint exactly when the closure of H has adjoint domain equal to its own domain, i.e. exactly when H is essentially self-adjoint. The displayed square is the usual block-matrix product on the common graph domain, justified after closure by the spectral theorem for the self-adjoint operator $\overline{H}$ (Reed–Simon I, Ch. VIII). $\square$

THEOREM 86.10.6 (*Polyakov–Dirac normal form*). . Within the spectral-flow framework of Chapter 86.1.7, after imposing the Polyakov scalar rigidity Chapter 86.10.1, the following statements are equivalent:

- (i) H-P^{Ar} for $\xi_{\mathbb{R}}$: there exists a self-adjoint operator H_{HP} whose normal translate has determinant divisor $\text{div } \xi_{\mathbb{R}}$.
- (ii) $\text{A-TP}^{\text{arch}}$: the Meyer/Connes absorption trace on the cuspidal subspace is the zero distribution, supported after zero-placement at the non-trivial-zero ordinates with multiplicity.
- (iii) The residual symmetric ordinate core $H_{\text{res}}^{\text{sym}}$ of Chapter 86.10.3 admits a Dirac closure in the sense of Chapter 86.10.4.

Thus the residual is the essential self-adjointness and determinant identity of one first-order Dirac double on the residual Meyer core. No central-charge, partition-function, or formal trace adjustment can change this equivalence.

Proof. (1) $\Leftrightarrow$ (2) is Chapter 86.6.5 together with Chapter 86.7.4 and the residual theorem Chapter 86.8.1.

(1) $\Rightarrow$ (3): given H-P^{Ar} , take $H_{\text{res}} = H_{\text{HP}}$ on the H-P Hilbert space and restrict to the common Meyer/Hadamard core. The Dirac double (86.3) is essentially self-adjoint by Chapter 86.10.5, and the determinant identity (86.4) is exactly the H-P^{Ar} determinant identity Chapter 22.1.

(3) $\Rightarrow$ (1): a Dirac closure gives a self-adjoint off-diagonal component H_{res} by the same lemma. The regularised determinant identity (86.4) states precisely that $\frac{1}{2}\text{id} + iH_{\text{res}}$ has characteristic determinant $\xi_{\mathbb{R}}$, hence its determinant divisor is the non-trivial-zero divisor with multiplicity. This is H-P^{Ar} in the convention of Chapter 22.3.2. The Polyakov rank-residue rigidity is used only to fix the determinant scalar at $r_{\zeta} = 2$; it does not enter the self-adjointness implication and therefore cannot be traded against it. $\square$

Remark 86.10.7 (*single estimate left*). The Dirac formulation gives the sharp analytic target. On the Meyer residual core one must prove a graph-norm closure statement:

$$\|u\|_{\mathcal{D}}^2 := \|u\|^2 + \|H_{\text{res}}^{\text{sym}} u\|^2$$

is complete after a closed Tate-polar operator quotient has first been constructed, and the closure has determinant (86.4). This is not easier than RH; it is the same obstruction with all auxiliary parameters removed.

Definition 86.10.8 (*closed Tate-polar operator quotient*). . A closed Tate-polar operator quotient for $H_o = H_{\text{res}}^{\text{sym}}$ is additional data of a Hilbert quotient

$$\pi : \mathcal{H}_{\text{res}} \longrightarrow \mathcal{H}_{\text{res}}^{\circ}$$

with closed finite-rank kernel P_{Tate} , an induced dense domain

$$C_{\text{res}}^{\circ} := \pi(C_{\text{res}}),$$

and a symmetric operator H_o° on C_{res}° satisfying

$$H_o^{\circ} \pi u = \pi H_o u, \quad u \in C_{\text{res}}.$$

The quotient is required to be compatible with closure and adjoint domains:

$$\mathcal{D}(\overline{H_o^{\circ}}) = \pi \mathcal{D}(\overline{H_o}), \quad \mathcal{D}((H_o^{\circ})^*) = \pi \mathcal{D}(H_o^*),$$

after completing both sides in their graph norms. It is also required to be compatible with Meyer coordinates, i.e. U_{Meyer} descends to a unitary residual chart on the quotient.

The quotient is not constructed here. Every phrase “after quotienting the Tate polar directions” in the residual operator argument is to be read under this hypothesis.

Remark 86.10.9 (post-quotient notation). . In the boundary and deficiency discussion below,

$$T_o := H_o^\circ : C_{\text{res}}^\circ \subset \mathcal{H}_{\text{res}}^\circ \longrightarrow \mathcal{H}_{\text{res}}^\circ$$

denotes the post-Tate residual operator supplied by Chapter 86.10.8. The pre-quotient operator $H_o = H_{\text{res}}^{\text{sym}}$ is only the source operator. Thus every adjoint, deficiency kernel, boundary quotient, and Meyer-coordinate graph statement below is a statement about T_o . Without the closed Tate-polar quotient, these formulas are well-formed as conditional operator theory but not as an unconditional construction.

What Remains: Boundary Form and Deficiency Kernels

With the Polyakov rank-residue rigidity fixing $r_\zeta = 2$, the only remaining scalar renormalisation freedom is gone. The obstruction is purely operator-theoretic: the boundary quotient $\mathcal{Q}_{\text{res}}$ of the residual symmetric core must vanish.

Definition 86.10.10 (residual boundary form). . Under the post-quotient notation of Chapter 86.10.9, the adjoint-domain boundary form is the Hermitian symplectic form

$$\mathcal{B}_{\text{res}}(u, v) := \langle T_o^* u, v \rangle - \langle u, T_o^* v \rangle, \quad u, v \in \mathcal{D}(T_o^*). \quad (86.5)$$

The *residual boundary space* is the quotient

$$\mathcal{Q}_{\text{res}} := \mathcal{D}(T_o^*) / \overline{\mathcal{D}(T_o)},$$

with the induced form still denoted $\mathcal{B}_{\text{res}}$.

PROPOSITION 86.10.11 (operator boundary target after scalar rigidity). . In the primary Heisenberg domain, the only remaining operator-theoretic obstruction to spectral-flow closure is the non-vanishing of the residual boundary quotient:

$$\mathcal{Q}_{\text{res}} = 0.$$

A future Feynman/Ward proof would have to identify its BV–BFV boundary phase space with $\mathcal{Q}_{\text{res}}$ and its boundary pairing with $\mathcal{B}_{\text{res}}$. Cancellation of a boundary variation merely on the original core is weaker: it proves isotropy of the core, not $\mathcal{Q}_{\text{res}} = 0$.

Proof. For a densely defined symmetric operator, the obstruction to essential self-adjointness is precisely the boundary quotient $\mathcal{D}(T_o^*) / \overline{\mathcal{D}(T_o)}$, together with its Green boundary form (86.5). If the quotient vanishes, the closure has adjoint domain equal to its own domain and is self-adjoint. If the quotient is non-zero, there remains a genuine choice of self-adjoint extension or a failure of self-adjointness, and the zero-side Hilbert–Polya operator has not been constructed.

Chapter 86.10.1 says that the scalar anomaly coefficient is already fixed by the rank-2 Sugawara sector and the Fock determinant. Hence a local counterterm cannot be used to absorb a non-zero $\mathcal{Q}_{\text{res}}$ without leaving the primary Heisenberg domain. What remains is therefore the operator-boundary vanishing problem. A path-integral/Ward proof would be a proof of the same statement only after the missing BV–BFV identification has been constructed. $\square$

Definition 86.10.12 (deficiency spaces of the residual). . The residual deficiency spaces are

$$\mathcal{N}_+ := \ker(T_0^* - i), \quad \mathcal{N}_- := \ker(T_0^* + i).$$

THEOREM 86.10.13 (Dirac two-test form of the residual). . After Polyakov scalar rigidity, the residual is equivalent to the following two statements:

(i) *Deficiency vanishing:*

$$\mathcal{N}_+ = 0 = \mathcal{N}_-. \quad (86.6)$$

(ii) *Determinant identity for the closure:*

$$\det_{\infty}(s \cdot \text{id} - (\tfrac{1}{2}\text{id} + i\overline{T_0})) = \xi_{\mathbb{R}}(s). \quad (86.7)$$

Equivalently, the Dirac double $\mathcal{D}_{\text{res}}^{\circ}$ has no deficiency states and its closed off-diagonal component has determinant $\xi_{\mathbb{R}}$.

Proof. By von Neumann's deficiency-index criterion, a densely defined symmetric operator T_0 is essentially self-adjoint if and only if $\ker(T_0^* - i) = \ker(T_0^* + i) = 0$. Thus (86.6) is exactly the essential self-adjointness part of Chapter 86.10.4. By Chapter 86.10.5, the same condition is equivalent to essential self-adjointness of the Dirac double $\mathcal{D}_{\text{res}}^{\circ}$.

Once the closure exists as a self-adjoint operator, the only remaining datum in H-P^{Ar} is the regularised determinant. Equation (86.7) is precisely (86.4) for the closure $\overline{T_0}$. Combining deficiency vanishing with the determinant identity is therefore equivalent to the Dirac closure condition of Chapter 86.10.6, hence to the remaining spectral-flow block. $\square$

Remark 86.10.14 (deficiency kernels as the operative test). The deficiency kernels $\ker(T_0^* \mp i)$ on the Meyer/Hadamard residual core are the operative test. If they vanish, the operator obstruction is gone and the remaining datum is the determinant identity (86.7). If either kernel is non-zero, the failure is localised as a specific boundary state in $\mathcal{Q}_{\text{res}}$. The Polyakov–Dirac decomposition is: the determinant residue fixes the scalar to $r_{\zeta} = 2$; first-order operator theory isolates the only surviving obstruction.

The Deficiency Calculation

In Meyer coordinates the deficiency kernels are the orthogonal complements of the ranges of $(M_t \pm i)$ on the residual core, with the sign convention matching the conjugate-linear second slot in the inner product. The computation is unconditional given a residual Meyer chart and the closed Tate-polar operator quotient of Chapter 86.10.8; without those it is conditional operator theory. The obstruction is a graph-core question for the residual Hadamard domain, not an Euler-product calculation.

Definition 86.10.15 (Meyer-coordinate residual chart). . Let

$$M_t: \mathcal{D}(M_t) \subset L^2(\mathbb{R}, dt) \rightarrow L^2(\mathbb{R}, dt), \quad (M_t f)(t) = t f(t),$$

be the self-adjoint multiplication operator. A *Meyer-coordinate residual chart* is a unitary map, denoted U_{Meyer} by abuse of notation,

$$U_{\text{Meyer}}: \mathcal{H}_{\text{res}}^{\circ} \longrightarrow L^2(\mathbb{R}, dt),$$

from the residual quotient Hilbert space of Chapter 86.10.8 such that the Meyer-coordinate residual core is

$$\mathcal{C}_{\text{res}}^M := U_{\text{Meyer}}(\mathcal{C}_{\text{res}}^{\circ}) \subset \mathcal{D}(M_t),$$

and

$$T_o = H_o^\circ \quad \text{is represented by} \quad M_t|_{C_{\text{res}}^M}.$$

The unconditional Meyer theorem Chapter 86.7.2 supplies this chart only on the Schwartz adelic subspace. Its extension to the residual quotient and to adjoint domains is part of the residual graph-core condition.

THEOREM 86.10.16 (*deficiency kernels are range defects*). . Assume a Meyer-coordinate residual chart in the sense of Chapter 86.10.15. Then

$$U_{\text{Meyer}} \mathcal{N}_+ = ((M_t + i)C_{\text{res}}^M)^\perp, \quad (86.8)$$

and

$$U_{\text{Meyer}} \mathcal{N}_- = ((M_t - i)C_{\text{res}}^M)^\perp.$$

Consequently $\mathcal{N}_+ = \mathbf{o} = \mathcal{N}_-$ if and only if both ranges

$$(M_t - i)C_{\text{res}}^M, \quad (M_t + i)C_{\text{res}}^M$$

are dense in $L^2(\mathbb{R}, dt)$.

Proof. We prove the + identity; the – identity is identical with i replaced by $-i$. The inner product is linear in the first slot and conjugate-linear in the second. By definition, $u \in \mathcal{N}_+$ means $u \in \mathcal{D}(T_o^*)$ and $(T_o^* - i)u = \mathbf{o}$. Equivalently, for every $f \in C_{\text{res}}^\circ$,

$$\langle T_o f, u \rangle = \langle f, iu \rangle.$$

Since the second slot is conjugate-linear, $\langle f, iu \rangle = -i\langle f, u \rangle$. Transport by U_{Meyer} . Write $\tilde{u} = U_{\text{Meyer}}u$ and $\tilde{f} = U_{\text{Meyer}}f$. Since T_o is represented by M_t on C_{res}^M , the adjoint equation becomes

$$\langle M_t \tilde{f}, \tilde{u} \rangle = \langle \tilde{f}, i\tilde{u} \rangle, \quad \tilde{f} \in C_{\text{res}}^M.$$

Hence

$$\langle (M_t + i)\tilde{f}, \tilde{u} \rangle = \mathbf{o} \quad \text{for all } \tilde{f} \in C_{\text{res}}^M,$$

which says precisely that $\tilde{u} \in ((M_t + i)C_{\text{res}}^M)^\perp$. The converse is the same implication read backwards, so (86.8) follows. The density criterion follows because a subspace has zero orthogonal complement exactly when its closure is the full Hilbert space. $\square$

COROLLARY 86.10.17 (*range density is graph density for M_t*). . For a subspace $C \subset \mathcal{D}(M_t)$, the following are equivalent:

$$\overline{C}^{\|\cdot\|_{M_t}} = \mathcal{D}(M_t), \quad \overline{(M_t - i)C}^{L^2} = L^2(\mathbb{R}, dt), \quad \overline{(M_t + i)C}^{L^2} = L^2(\mathbb{R}, dt).$$

Consequently, under the residual Meyer chart hypothesis, C_{res}^M is a graph core for M_t if and only if

$$\mathcal{N}_+ = \mathbf{o} = \mathcal{N}_-.$$

Proof. For $\text{Im } z \neq 0$, the map

$$M_t - z: \mathcal{D}(M_t)_{\text{graph}} \rightarrow L^2(\mathbb{R}, dt)$$

is boundedly invertible: its inverse is multiplication by $(t - z)^{-1}$, which maps L^2 into $\mathcal{D}(M_t)$ with bounded graph norm. Hence a subspace is dense in the graph norm if and only if its image under $M_t - z$ is dense in L^2 . Apply this with $z = i$ and $z = -i$, then use Chapter 86.10.16. $\square$

LEMMA 86.10.18 (*rank-one graph-boundary model*). . Let $\psi(t) = (1 + |t|)^{-1}$. Then $\psi \in L^2(\mathbb{R}, dt)$ but $\psi \notin \mathcal{D}(M_t)$. Define

$$C_\psi := \{f \in \mathcal{D}(M_t) \mid \langle (M_t - i)f, \psi \rangle = 0\}.$$

Then C_ψ is L^2 -dense and graph-closed of codimension one, but it is not graph-dense. Its range defects are

$$((M_t - i)C_\psi)^\perp = \mathbb{C}\psi, \quad ((M_t + i)C_\psi)^\perp = \mathbb{C} \frac{t + i}{t - i} \psi.$$

Proof. The function ψ is square-integrable, while $t\psi(t)$ is not square-integrable. Since

$$M_t - i: \mathcal{D}(M_t)_{\text{graph}} \rightarrow L^2(\mathbb{R}, dt)$$

is a bounded isomorphism, C_ψ is the inverse image of the closed hyperplane $\psi^\perp$; hence it is graph-closed of codimension one and its first range defect is $\mathbb{C}\psi$. The defining functional is not continuous for the L^2 -topology on $\mathcal{D}(M_t)$, because such continuity would represent it by an L^2 -vector, equivalently would put ψ in $\mathcal{D}(M_t)$. Therefore its kernel is L^2 -dense.

For the second range, write

$$(M_t + i)(M_t - i)^{-1}$$

as multiplication by $(t + i)/(t - i)$, a unitary multiplier on L^2 . The orthogonal complement of its image of $\psi^\perp$ is the span of $(t + i)(t - i)^{-1}\psi$. $\square$

LEMMA 86.10.19 (*point conditions are graph-invisible*). . Let $\Gamma \subset \mathbb{R}$ be locally finite. Then $C_c^\infty(\mathbb{R} \setminus \Gamma)$ is graph-dense in $\mathcal{D}(M_t)$. In particular, a boundary condition that is only pointwise at the zero ordinates cannot obstruct graph density for the ordinary M_t -graph norm.

Proof. The graph norm of M_t is the L^2 -norm for the measure $(1 + t^2) dt$. Locally finite sets have zero measure for this weight. Given $f \in \mathcal{D}(M_t)$, truncate f to a compact interval, remove intervals of arbitrarily small weighted measure around the finitely many points of Γ in that interval, and then mollify. The approximants lie in $C_c^\infty(\mathbb{R} \setminus \Gamma)$ and converge in the graph norm. $\square$

Definition 86.10.20 (Hadamard-supported graph functional). . Let

$$\Gamma_{\frac{\zeta}{\xi}}^{\text{crit}} := \{ \gamma \in \mathbb{R} \mid \tfrac{1}{2} + i\gamma \text{ is a non-trivial zero of } \zeta \}$$

with multiplicities forgotten. This is the real critical-line ordinate support relevant to the Hilbert–Pólya target; it is not an assertion that the full Hadamard divisor has already been realised on $\mathbb{R}$. A continuous functional

$$\beta: \mathcal{D}(M_t)_{\text{graph}} \longrightarrow \mathbb{C}$$

is *Hadamard-supported on the zero ordinates* if

$$\beta(\phi) = 0 \quad \text{for every } \phi \in C_c^\infty(\mathbb{R} \setminus \Gamma_\xi^{\text{crit}}).$$

This is the graph-norm formulation of the idea that the boundary condition sees only point or finite-jet Hadamard residue data at the zero ordinates.

THEOREM 86.10.21 (*no Hadamard residue functional in the ordinary graph dual*). . The ordinary M_t -graph dual contains no non-zero Hadamard-supported functional. Equivalently,

$$\beta \in (\mathcal{D}(M_t)_{\text{graph}})^*, \quad \text{supp } \beta \subset \Gamma_\xi^{\text{crit}} \implies \beta = 0$$

in the sense of Chapter 86.10.20. Thus there is no non-zero

$$\beta_{\text{Had}}: \mathcal{D}(M_t)_{\text{graph}} \rightarrow \mathbb{C}$$

whose graph-closed kernel is exactly a Hadamard-residue boundary condition supported only at the zero ordinates. The statement also rules out finite-jet residue data at those ordinates, if such data are asked to be continuous for the ordinary M_t -graph topology.

Proof. The non-trivial zeros of ζ form a discrete subset of the critical strip, and only finitely many have ordinate in a bounded interval. Hence Γ_ξ^{crit} is locally finite as a subset of $\mathbb{R}$. By Chapter 86.10.19,

$$C_c^\infty(\mathbb{R} \setminus \Gamma_\xi^{\text{crit}})$$

is dense in $\mathcal{D}(M_t)$ for the graph norm. If β is continuous for the graph norm and vanishes on this dense subspace, it vanishes on all of $\mathcal{D}(M_t)$.

Equivalently, the graph norm is the L^2 -norm for the weighted measure $(1 + t^2) dt$, so the graph dual is represented by weighted L^2 -functions, or, after the isometric resolvent chart

$$M_t - i: \mathcal{D}(M_t)_{\text{graph}} \xrightarrow{\sim} L^2(\mathbb{R}, dt),$$

by L^2 -vectors. No non-zero L^2 -vector is supported on the locally finite set Γ_ξ^{crit} , which has Lebesgue measure zero. $\square$

COROLLARY 86.10.22 (*Hadamard residue data is graph-invisible*). . In the ordinary Meyer graph topology, β_{Had} does not exist as a non-zero graph-dual boundary functional. Therefore a boundary condition whose only data are point or finite-jet Hadamard residues at the locally finite real ordinate set Γ_ξ^{crit} is graph-invisible: it cannot by itself obstruct graph density of C_{res}^M in $\mathcal{D}(M_t)$.

Remark 86.10.23 (*surviving branches after the graph-dual no-go*). The preceding corollary is not an exhaustive boundary-state classification. It eliminates only the ordinary point-supported graph-dual path. Assuming the closed Tate-polar quotient and residual Meyer chart have been constructed, a surviving operator obstruction must take a different form. The known nonexclusive presentations are:

- (i) the exact Hadamard residue or finite-jet trace lives in a stronger rigged, Paley–Wiener, de Branges, or reproducing-kernel topology $\Phi_\xi \subset \mathcal{D}(M_t)_{\text{graph}}$, where point evaluation is continuous;
- (ii) the ordinary graph obstruction is nonlocal, represented under the resolvent chart by an L^2 -vector or closed L^2 -subspace as in Chapter 86.10.18, and is therefore not point-residue data;

- (iii) a physical Ward/BV–BFV argument supplies extra boundary phase space not yet identified with $\mathcal{Q}_{\text{res}}$; this is a physics residual, not an operator-theoretic graph-dual boundary functional.

These are presentations of the same $A\text{--}TP^{\text{arch}}$ / Burnol–de Branges residual, not new independent residuals. The stronger-topology approach would require, at minimum, a continuous Hadamard trace-and-jet map $\Phi_{\xi} \rightarrow \ell^2(\Gamma_{\xi}^{\text{crit}}, w)$, compatibility with the post-Tate quotient and adjoint domains, and a proof that the resulting closed condition is not a tautological insertion of the zero divisor.

Remark 86.10.24 (answer of the calculation). The ordinary L^2 -bulk part has no hidden deficiency. The only possible obstruction, once the Tate-polar quotient and residual Meyer chart are constructed, is boundary-theoretic:

$$\overline{C_{\text{res}}^{\mathcal{M}}}^{\|\cdot\|_{\mathcal{M}_t}} \stackrel{?}{=} \mathcal{D}(M_t), \quad \|f\|_{\mathcal{M}_t}^2 := \|f\|_2^2 + \|tf\|_2^2,$$

in the quotient graph norm. A genuine boundary state must therefore come from a graph-dual functional of the kind modeled in Chapter 86.10.18, not from mere point conditions at the zero ordinates, which are graph-invisible by Chapter 86.10.19. The direct Hadamard-residue candidate β_{Had} is ruled out in the ordinary graph dual by Chapter 86.10.21.

Execution Attempt: Graph Norm Versus Determinant

Three graph-norm closures are available in explicit ambient spaces. The classical Meyer line closes but has absolutely continuous spectrum $\mathbb{R}$, not determinant divisor $\text{div } \xi_{\mathbb{R}}$. The formal zero space has the right divisor only by using the zero divisor as input. The non-formal quotient is exactly $A\text{--}TP^{\text{arch}}$.

LEMMA 86.10.25 (*classical Meyer-line graph closure*). . Let

$$M_t^{\circ} : C_c^{\infty}(\mathbb{R}) \subset L^2(\mathbb{R}, dt) \longrightarrow L^2(\mathbb{R}, dt), \quad (M_t^{\circ} f)(t) = tf(t).$$

Then M_t° is essentially self-adjoint. Its graph-norm closure is multiplication by t on

$$\mathcal{D}(M_t) = \{f \in L^2(\mathbb{R}, dt) \mid tf(t) \in L^2(\mathbb{R}, dt)\}.$$

The spectrum is the absolutely continuous spectrum $\text{Spec}(M_t) = \mathbb{R}$.

Proof. The operator M_t with the displayed maximal domain is self-adjoint by the spectral theorem for multiplication operators. The core $C_c^{\infty}(\mathbb{R})$ is dense in the graph norm: truncate f by compactly supported cutoffs and then mollify; both f and tf converge in L^2 . Hence the closure of M_t° is M_t . Since M_t is multiplication by the coordinate on $L^2(\mathbb{R}, dt)$, its spectrum is the essential range of t , namely $\mathbb{R}$, with absolutely continuous spectral measure. $\square$

PROPOSITION 86.10.26 (*why the classical closure is insufficient*). . The closure of Chapter 86.10.25 does not solve the spectral-flow problem. Its graph norm is complete and its operator is self-adjoint, but its spectrum is continuous $\mathbb{R}$, not a divisor with zero multiplicities of $\xi_{\mathbb{R}}$. Consequently it cannot satisfy

$$\det_{\infty}(s \cdot \text{id} - (\tfrac{1}{2}\text{id} + iM_t)) = \xi_{\mathbb{R}}(s)$$

as a zero-eigenvalue determinant.

Proof. The determinant identity in Chapter 86.7 has divisor $\text{div } \xi_{\mathbb{R}}$. The multiplication operator M_t on $L^2(\mathbb{R}, dt)$ has Lebesgue spectral measure on the line, not this determinant divisor. Thus the closure obtained by ordinary Meyer-line graph completion is the unconditional Mellin unitary part, not the Hadamard-residue determinant-trace realisation. This is exactly the limitation of Meyer's construction recorded in Chapter 86.7.5. $\square$

PROPOSITION 86.10.27 (*formal zero-space closure is tautological*). . On the formal zero Hilbert space $\mathcal{H}_{\xi}^{\text{formal}}$ of Chapter 86.1.3, the diagonal ordinate operator has graph-norm closure. Its projected operator $\Theta_{\xi}^{\text{proj}}$ satisfies the regularised determinant identity for $\xi_{\mathbb{R}}$ only if the original zero divisor already lies on the critical line. This proves no non-formal spectral-flow closure.

Proof. The finite-support diagonal operator on $\bigoplus_{\rho} \mathbb{C}^{m_{\rho}}$ closes in the graph norm to the maximal diagonal multiplication operator by the real numbers $t_{\rho} = \text{Im}(\rho)$; its closure is self-adjoint. The operator $\Theta_{\xi}^{\text{proj}} = \frac{1}{2} + iH_{\text{ord}}^{\text{formal}}$ has spectral divisor $\{\frac{1}{2} + it_{\rho}\}$. Hadamard factorisation identifies its regularised determinant with $\xi_{\mathbb{R}}$ only when $\beta_{\rho} = 1/2$ for every ρ . Otherwise it is the determinant of the projected divisor, not of $\xi_{\mathbb{R}}$. The construction has therefore used the zero divisor and then discarded its real-part defect. It supplies no Bost–Connes, Meyer, Connes, Deninger, or chiral-homological source for the Hilbert space or the Weil–Connes trace; it is the formal target explicitly excluded by Chapter 86.1.4. $\square$

THEOREM 86.10.28 (*residual Dirac criterion*). . On the residual Meyer/Hadamard core, a non-formal Dirac closure with the xi-determinant is equivalent to the RH-strength residual. More precisely, the following statements are equivalent:

- (i) The post-quotient residual operator $T_o = H_o^{\circ}$ has a graph-norm self-adjoint closure and satisfies (86.7).
- (ii) The deficiency kernels of Chapter 86.10.12 vanish and the closure has determinant $\xi_{\mathbb{R}}$.
- (iii) The Dirac double $\mathcal{D}_{\text{res}}^{\circ}$ admits a Dirac closure in the sense of Chapter 86.10.4.
- (iv) A-TP^{arch}, equivalently H-P^{Ar} for $\xi_{\mathbb{R}}$.

Thus the unquotiented classical line and the formal zero space give opposite degenerate limits. On the residual core, the missing assertion is the Hilbert–Pólya/RH-strength step itself.

Proof. (1) $\Leftrightarrow$ (2) is von Neumann's deficiency-index criterion together with the determinant condition, as stated in Chapter 86.10.13. (2) $\Leftrightarrow$ (3) is Chapter 86.10.5 plus the definition of Dirac closure. (3) $\Leftrightarrow$ (4) is the Polyakov–Dirac normal-form theorem Chapter 86.10.6. The two ambient computations above explain why neither classical Meyer-line closure nor the formal zero Hilbert space supplies the missing implication: the former has the wrong spectral type, and the latter imports the desired zero divisor by definition. $\square$

Remark 86.10.29 (*graph-norm and determinant together require RH*). The graph norm can be closed. The determinant can be obtained. What cannot be obtained from the present data is both at once on a non-formal, independent residual Hilbert space. That conjunction is precisely the absorption-trace property of Connes–Meyer, the A-TP positivity statement, and the Hilbert–Pólya realisation. An unconditional proof of this conjunction is an unconditional proof of RH.

86.II BIBLIOGRAPHY

The references below are grouped by the four mathematical constructions used above.

Hilbert–Pólya foundational literature.

Pólya. G. Pólya, correspondence with Landau (1914), transcribed in Pintz and Sárközy, *On the work of Pólya in analytic number theory*, Stud. Sci. Math. Hungar. 26 (1991), 201–226.

Hilbert. D. Hilbert, Göttingen seminar notes (1914, 1916), recorded by E. Landau, *Handbuch der Lehre von der Verteilung der Primzahlen*, Teubner (1909), 2nd ed. 1953.

Berry–Keating. M. V. Berry and J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Review 41 (1999), 236–266.

Montgomery. H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Sympos. Pure Math. 24 (1973), 181–193.

Adelic and idele-class operator constructions.

Connes. A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N. S.) 5 (1999), 29–106; arXiv:math/9811068. §IV for the absorption-trace conjecture; §V for the Weil-positive trace identity.

Connes–Marcolli. A. Connes and M. Marcolli, *Noncommutative Geometry, Quantum Fields, and Motives*, AMS Colloq. Publ. 55 (2008). Ch. 3 for KMS states; Ch. 4 for rigidity; Ch. 5 for the adèle class space.

Meyer. R. Meyer, *On a representation of the idele class group related to primes and zeros of L -functions*, Duke Math. J. 127 (2005), 519–595; arXiv:math/0311468. §3 for Mellin transform and scaling; §4 for the Schwartz-space distributional trace.

Deninger. C. Deninger, *Motivic L -functions and regularized determinants*, Proc. Sympos. Pure Math. 55.1 (1994), 707–743; *Some analogies between number theory and dynamical systems on foliated spaces*, Proc. ICM Berlin 1 (1998), 163–186.

Mellin and Hadamard classical analysis.

Mellin. H. Mellin, *Die Dirichletschen Reihen, die zahlentheoretischen Funktionen und die unendlichen Produkte von endlichem Geschlecht*, Acta Soc. Sci. Fenn. 31, no. 2 (1904).

Hadamard. J. Hadamard, *Étude sur les propriétés des fonctions entières et en particulier d’une fonction considérée par Riemann*, J. Math. Pures Appl. 9 (1893), 171–215.

Titchmarsh. E. C. Titchmarsh, *Introduction to the Theory of Fourier Integrals*, 2nd ed., Oxford (1948); *The Theory of the Riemann Zeta-Function*, 2nd ed., revised D. R. Heath-Brown, Oxford (1986).

von Mangoldt. H. von Mangoldt, *Zu Riemann’s Abhandlung “Über die Anzahl der Primzahlen unter einer gegebenen Grösse”*, J. Reine Angew. Math. 114 (1895), 255–305.

Edwards. H. M. Edwards, *Riemann’s Zeta Function*, Academic Press (1974). §2.7 for Hadamard product; §3 for the explicit formula.

Patterson. S. J. Patterson, *An Introduction to the Theory of the Riemann Zeta-Function*, Cambridge (1988). Ch. 3 for the Weil explicit formula.

Bost–Connes and GNS modular theory.

Bost–Connes. J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. 1 (1995), 411–457.

Takesaki. M. Takesaki, *Theory of Operator Algebras II*, Springer (2003). Ch. VIII for modular automorphism groups.

Julia. B. Julia, *Statistical theory of numbers*, Springer Proc. Phys. 47 (1990), 276–293. Primon-gas interpretation of the Bost–Connes spectrum.

First-order and Dirac normal-form sources.

Dirac. P. A. M. Dirac, *The quantum theory of the electron*, Proc. Roy. Soc. Lond. A 117 (1928), 610–624, and 118 (1928), 351–361. Source for the first-order square-root principle used here only as a normal-form discipline.

Reed–Simon. M. Reed and B. Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, Academic Press (1980). Ch. VIII for adjoints, closures, block operators, and the self-adjointness criterion used in Chapter 86.10.5.

Vol IV earlier sources used here.

Chapter 55: Bost–Connes GNS and log-modular operator construction (Ch. 55).

Chapter 22.3.2: Hilbert–Pólya hypothesis (Ch. 22).

Chapter 35: $H\text{-}P^{\text{Ar}}$ equivalence bundle, approach 2 (Ch. 35).

Chapter 40: determinant realisation of $H\text{-}P^{\text{Ar}}$, approach 3 (Ch. 40).

Chapter 79: adelic-scattering formulation (Ch. 79).

Chapter 82: Deninger-chiral triple identification (Ch. 82).

Chapter 81: 3D arithmetic Chern-Simons (Ch. 81).

Disjointness comparison. The formal residue-obstruction analysis is drawn (Chapter 86.3.3) from the Riemann–Hadamard–von Mangoldt classical list, disjoint from the Bost–Connes / Takesaki modular-theoretic list of Ch. 55; its Meyer determinant-trace analysis from Meyer 2005 arXiv:math/0311468, disjoint from the Birman–Krein scattering list of Ch. 79; its Connes absorption-trace analysis from Connes 1999 §IV, disjoint from the Bost–Connes 1995 C^* -algebraic list of Ch. 55. No derivation on which a residual-irreducibility claim rests is shared between the lists.

FUNCTORIALITY OF THE TRIPLE IDENTIFICATION IN $\mathbf{A}$ AND $\mathbf{g}$

Chapter 82 established the triple identification $H_{\Delta}^{\bullet} = H_{\text{ch}}^{\bullet} = \partial$ -chiral on the primary Heisenberg domain for the baseline genus $g = 0$ and the fixed arithmetic Heisenberg H^{Ar} . The functoriality residuals of Chapter 82.12(5,6) remain: the three cohomology assignments extend to functors on a category ArChAlg of arithmetic chiral algebras on $\text{Spec } \overline{\mathbb{Z}}$, and the triple identification is a natural isomorphism of these functors, uniformly compatible with the genus- g restriction and induction maps of Chapter 80. The master theorem is Chapter 87.6.1; it rests on the chain-level identification Ξ of Chapter 87.3.2, the naturality diagram Chapter 87.4.1, and the genus-stability theorem Chapter 87.5.3. The diagonal morphism $H^{\text{Ar}} \rightarrow H^{\text{Ar}} \oplus H^{\text{Ar}}$ is the explicit witness. All residuals of this chapter are inherited from Chapters 81 and 82; no new residual is introduced.

87.1 THE CATEGORY OF ARITHMETIC CHIRAL ALGEBRAS

The three cohomology assignments of Chapter 82 are defined on an individual object, the arithmetic Heisenberg H^{Ar} . To state functoriality we introduce the ambient category.

Definition 87.1.1 (arithmetic chiral algebra). . An arithmetic chiral algebra on $\overline{\text{Spec } \mathbb{Z}}$ is a datum

$$\mathbf{A} = \left(\mathbf{A}^{(\infty)}, \{ \mathbf{A}^{(p)} \}_{p \in \mathbb{P}}, \{ \mathbf{u}^{(v)} \}_{v \in |\overline{C}|}, \{ m^{(v,w)} \}_{v < w}, \{ \partial_{v,w}^{\text{PG}} \}_{v < w} \right)$$

subject to the axioms below. Write $|\overline{C}| = \mathbb{P} \cup \{\infty\}$.

- (i) *Archimedean factor.* $\mathbf{A}^{(\infty)}$ is a pointed E_1 -chiral factorisation algebra on $\mathbb{C}_{\infty}$ in the Beilinson–Drinfeld / Costello–Gwilliam sense (Ch. 36.1.1), together with a conformal vector $T \in \mathbf{A}_2^{(\infty)}$ and a spherical unit $\mathbf{u}^{(\infty)} \in \mathbf{A}_0^{(\infty)}$.
- (ii) *Finite-prime factors.* For every prime $p \in \mathbb{P}$, $\mathbf{A}^{(p)}$ is a pointed E_0 -chiral coefficient object on the Fargues–Scholze perfectoid diamond chart S_p , equipped with a Bost–Connes Adams endomorphism $\psi_p \in \text{End}(\mathbf{A}^{(p)})$ and a Frobenius-semilinear orientation $\mathfrak{J}_p^{\text{Ar}}$ (the finite-place substitute for the archimedean $i \in \mathbb{C}$).
- (iii) *Spherical units.* The collection $\{ \mathbf{u}^{(v)} \}_{v \in |\overline{C}|}$ satisfies the Tate spherical compatibility (Chapter 28.1.1): $\mathbf{u}^{(v)}$ is closed for the local bar differential d_v , is a compact central idempotent, and is the monoidal unit at almost every prime.
- (iv) *Adjacent products.* For every pair of distinct places $v < w$ in the auxiliary order $2 < 3 < 5 < \cdots < \infty$, the datum $m^{(v,w)}$ is a map

$$m^{(v,w)} : \mathbf{A}^{(v)} \widehat{\otimes} \mathbf{A}^{(w)} \longrightarrow \mathbf{A}^{(vw)},$$

with $\mathbf{A}^{(vw)}$ the formal merged-block coefficient in the sense of Chapter 81.5.2. The $m^{(v,w)}$ are required to be ψ_p -equivariant at every finite $v = p$ and to intertwine the archimedean L_o -action at $v = \infty$.

- (v) *Parshin–Gersten boundary*. For every pair $v < w$ of ramified places, $\partial_{v,w}^{\text{PG}}$ is the Kato–Parshin tame-symbol boundary (Chapter 36.2); the collection satisfies the Maurer–Cartan equation on every reciprocity-closed finite stage (Chapter 36.3.3).
- (vi) *Tate compatibility*. The restricted tensor product $\otimes'_v \mathbf{A}^{(v)}$ in pointed E_2 -monoidal stable presentable ∞ -categories (Chapter 28.8.3) is well-defined, i.e. almost every $u^{(v)}$ is the monoidal unit and the Tate-pair compatibility F2.c of Chapter 21.3 is satisfied.

The archimedean $\mathbf{A}^{(\infty)}$ is always genuinely chiral; finite factors may be E_o per Chapter 81.5.2. The object $\mathbf{A}$ is *abelian Heisenberg-type* if every $\mathbf{A}^{(p)}$ is a rank-2 $\mathbf{g}$ bosonic Fock for some fixed $g \geq 0$ (the genus of $\mathbf{A}$), and $\mathbf{A}^{(\infty)}$ is the rank-2 $\mathbf{g}$ archimedean Heisenberg of Chapter 12.5.1.

Remark 87.1.2 (on the merged-block target). The formal merged-block coefficient $\mathbf{A}^{(vw)}$ is not a new arithmetic place. It is the target of the adjacent contraction map in the sense of the ordered incidence category $\text{Inc}_{<}(V)$ of Chapter 81.5.1; concretely a completed tensor product of $\mathbf{A}^{(v)}$ and $\mathbf{A}^{(w)}$ modulo a quotient relation recording that the two local operators act on the same global fibre. This convention is essential to avoid the naive reading that primes are fused into composite arithmetic objects.

Definition 87.1.3 (morphism of arithmetic chiral algebras). A morphism $\phi: \mathbf{A}_1 \rightarrow \mathbf{A}_2$ of arithmetic chiral algebras is a collection $\phi = \{\phi^{(v)}\}_{v \in |\overline{C}|}$ with $\phi^{(v)}: \mathbf{A}_1^{(v)} \rightarrow \mathbf{A}_2^{(v)}$ such that

- (i) $\phi^{(\infty)}$ is an E_1 -chiral factorisation map preserving the conformal vector;
- (ii) at every prime p , $\phi^{(p)}$ is ψ_p -equivariant and $\mathfrak{J}_p^{\text{Ar}}$ -compatible: $\phi^{(p)} \circ \psi_{p,1} = \psi_{p,2} \circ \phi^{(p)}$;
- (iii) the spherical units are preserved: $\phi^{(v)}(u_1^{(v)}) = u_2^{(v)}$ for every $v \in |\overline{C}|$;
- (iv) the adjacent products are intertwined: $m_2^{(v,w)} \circ (\phi^{(v)} \otimes \phi^{(w)}) = \phi^{(vw)} \circ m_1^{(v,w)}$ for every $v < w$;
- (v) the Parshin–Gersten boundaries are intertwined: $\phi \circ \partial_1^{\text{PG}} = \partial_2^{\text{PG}} \circ \phi$ at every pair of ramified places;
- (vi) the restricted-tensor map $\widehat{\phi} := \otimes'_v \phi^{(v)}$ is a map in the pointed E_2 -monoidal category of Chapter 28.8.3; this is automatic from (i)–(v) at almost every place by the spherical-unit preservation.

Composition is place-wise; identities are place-wise identities. The resulting category is ArChAlg .

PROPOSITION 87.1.4 (*ArChAlg is a well-defined category*). Composition of morphisms satisfies the axioms of Chapter 87.1.3: axioms (1)–(5) are place-wise, and axiom (6) follows from place-wise axioms by the almost-every-place triviality of the units. Identity morphisms are identities in every axiom. Associativity and unitality are inherited from the underlying categories of E_1 -chiral factorisation algebras at ∞ and E_o -coefficient categories at finite places.

Proof. For composition: given $\phi: \mathbf{A}_1 \rightarrow \mathbf{A}_2$ and $\chi: \mathbf{A}_2 \rightarrow \mathbf{A}_3$, the collection $(\chi \circ \phi)^{(v)} := \chi^{(v)} \circ \phi^{(v)}$ preserves each axiom: axiom (1) is closed under composition of E_1 -chiral maps; axiom (2) is closed because ψ_p -equivariance composes; axiom (3) uses $\chi^{(v)}(\phi^{(v)}(u_1^{(v)})) = \chi^{(v)}(u_2^{(v)}) = u_3^{(v)}$; axiom (4) similarly using the stacked naturality diagrams. Axiom (5) uses the linear action of the boundary. Associativity and unitality are automatic at each place. Axiom (6) follows from the almost-every-place compatibility of the Tate restricted tensor (Chapter 21.3): the set of places at which $u^{(v)}$ fails to be a strict unit is bounded, and composition preserves this bound. $\square$

Definition 87.1.5 (genus filtration). . For every $g \geq 0$, let $\text{ArChAlg}_g \subset \text{ArChAlg}$ be the full subcategory of abelian Heisenberg-type arithmetic chiral algebras of genus g in the sense of Chapter 87.1.1. The canonical inclusion $\iota_{g,g+1}: \text{ArChAlg}_g \hookrightarrow \text{ArChAlg}_{g+1}$ is by tensoring with the rank-2 archimedean Heisenberg (Chapter 80.1.1's rank-2 enhancement at each place). The retraction $\pi_{g+1,g}: \text{ArChAlg}_{g+1} \rightarrow \text{ArChAlg}_g$ projects onto the first $2g$ oscillators.

Remark 87.1.6 (why the filtration). The filtration records that the genus- g category embeds into the genus- $(g+1)$ category in a canonical way (tensor with a fresh rank-2 factor at every place), and that one can restrict to any subgenus. The adjunction (ι, π) is a *reflective inclusion*: $\pi_{g+1,g} \circ \iota_{g,g+1} = \text{id}_g$ is strict, and the counit $\iota_{g,g+1} \circ \pi_{g+1,g} \Rightarrow \text{id}_{g+1}$ is the natural transformation that kills the freshly-added rank-2 factor. The pair $(\iota_{g,g+1}, \pi_{g+1,g})$ is the canonical reflective pair through which all genus functoriality statements below pass.

Remark 87.1.7 (source anchors and disjointness). The archimedean E_1 -chiral structure is sourced from Beilinson–Drinfeld 2004 and Francis–Gaitsgory 2012 (see Chapter A.5); the finite-place E_0 -coefficient structure is sourced from Fargues–Scholze arXiv:2102.13459 and Bhatt–Scholze 2022 (prismatic; see Chapter A.19). These two source families are disjoint. The morphism axioms (1)–(6) each arise from exactly one of them; hence the functoriality statements below have independent derivations on the finite and archimedean sides.

87.2 THE THREE FUNCTORS

The cohomology assignments of Chapter 82 are defined on the single object H^{Ar} . Functoriality requires a target category. Write $\text{grVect}_{\mathbb{C}}^{\text{Frob}}$ for the category of finite-dimensional graded $\mathbb{C}$ -vector spaces equipped with a graded-linear automorphism F (the Frobenius) and a graded-symmetric non-degenerate pairing PD (the Poincaré duality); morphisms are graded linear maps commuting with both structures.

Definition 87.2.1 (Deninger functor $H_{\Delta}^{\bullet}$). . The *Deninger functor*

$$H_{\Delta}^{\bullet}: \text{ArChAlg} \longrightarrow \text{grVect}_{\mathbb{C}}^{\text{Frob}}$$

sends an arithmetic chiral algebra A to the triple $(H_{\Delta}^{\bullet}(A), F_A, \text{PD}_A)$ of Chapter 82.3.2, with $F_A = F_{\infty,A} \otimes \otimes'_p \psi_{p,A}$ the global Frobenius of Chapter 82.5.5 and PD_A the Poincaré duality pairing of Chapter 82.7.1. On a morphism $\phi: A_1 \rightarrow A_2$, the functor sends ϕ to the induced map $H_{\Delta}^{\bullet}(\phi): H_{\Delta}^{\bullet}(A_1) \rightarrow H_{\Delta}^{\bullet}(A_2)$.

Definition 87.2.2 (chiral-homology functor $H_{\text{ch}}^{\bullet}$). . The *chiral-homology functor*

$$H_{\text{ch}}^{\bullet}: \text{ArChAlg} \longrightarrow \text{grVect}_{\mathbb{C}}^{\text{Frob}}$$

sends A to $H^{\bullet}(B^{\text{Ar}}(A), D_A)$ of Chapter 36.3.1, with D_A the signed total differential determined by A . On a morphism $\phi: A_1 \rightarrow A_2$, the induced chain map $B^{\text{Ar}}(\phi)$ commutes with D_{A_1} and D_{A_2} by axioms (3), (4), (5) of Chapter 87.1.3; pass to cohomology for $H_{\text{ch}}^{\bullet}(\phi)$.

Definition 87.2.3 (bulk-boundary functor pair $\partial \circ \text{BdryACS}$). . The *bulk-boundary assignment* is the composite

$$\partial \circ \text{BdryACS}: \text{ArChAlg} \xrightarrow{\text{BdryACS}} 3\text{DACS}_{\overline{\text{Spec } \mathbb{Z}}} \xrightarrow{\partial} \text{grVect}_{\mathbb{C}}^{\text{Frob}},$$

where

- $3\text{DACS}_{\overline{\text{Spec } \mathbb{Z}}}$ is the category of conjectural 3D quantum arithmetic Chern–Simons theories on the étale 3-space $\overline{\text{Spec } \mathbb{Z}}$ in the sense of Chapter 81.4.7, with morphisms given by bulk interface defects;

- BdryACS is the canonical lift that takes an arithmetic chiral algebra to the 3D bulk theory whose boundary WZW is $\mathbf{A}^{(\infty)}$ and whose finite-place defects are labelled by $\{\mathbf{A}^{(p)}\}$ via the Morishita knot-prime dictionary (Chapter 81.4.6, Chapter 82.5.2);
- ∂ is the boundary chiral homology functor, sending a 3D bulk theory to the chiral homology of its boundary factorisation algebra on the archimedean fibre.

On a morphism $\phi: \mathbf{A}_1 \rightarrow \mathbf{A}_2$, BdryACS produces a bulk interface defect between the two 3D theories whose boundary restriction is $\phi^{(\infty)}$ and whose finite-place Wilson-line restriction is $\{\phi^{(p)}\}$.

Remark 87.2.4 (conditional form of the bulk-boundary lift). The functor BdryACS is conditional on the 3D quantum ACS existence conjecture Chapter 81.4.7. The boundary functor ∂ is conditional on the local-trace comparison Chapter 81.4.9. Both conditionals are inherited from Chapter 81 and are explicitly carried through each theorem of this chapter.

Remark 87.2.5 (source disjointness of the three functors). The Deninger functor is derived from Deninger 1998 (arithmetic absolute-value axioms). The chiral-homology functor is derived from the Platonic bar complex of Beilinson–Drinfeld + Francis (Chapter 36.3.1). The bulk-boundary functor is derived from Kim 2015 arXiv:1510.05818 and the conjectural quantum ACS of Chapter 81.4.7. These three derivation paths are pairwise independent; the natural isomorphism theorem below is a genuine independent verification of the triple equality, not a restatement of any single derivation.

87.3 CHAIN-LEVEL IDENTIFICATION

The three functors land in $\text{grVect}_{\mathbb{C}}^{\text{Frob}}$ by taking cohomology of chain complexes that are not obviously the same. The natural isomorphism follows once one identifies the chain complexes before passing to cohomology.

Definition 87.3.1 (universal arithmetic bar). For $\mathbf{A} \in \text{ArChAlg}$, the *universal arithmetic bar* of $\mathbf{A}$ is the pair

$$\mathcal{B}^{\text{univ}}(\mathbf{A}) := (B_{\bullet}^{\text{ord}}(\mathbf{A}), B_{\bullet}^{\text{Ar}}(\mathbf{A})),$$

with $B_{\bullet}^{\text{ord}}$ the ordered incidence bar of Chapter 81.5.3 applied to the collection $\{\mathbf{A}^{(v)}\}_{v \in |\overline{C}|}$, and $B_{\bullet}^{\text{Ar}}$ the Platonic arithmetic-chiral bar of Chapter 36.3.1. Both are equipped with their natural signed differentials.

THEOREM 87.3.2 (chain-level identification on ArChAlg). (on Chapter 81.5.4 and Chapter 36.3.4). For every $\mathbf{A} \in \text{ArChAlg}$ there is a canonical quasi-isomorphism

$$\Xi_{\mathbf{A}}: B_{\bullet}^{\text{ord}}(\mathbf{A}) \xrightarrow{\sim} B_{\bullet}^{\text{Ar}}(\mathbf{A})$$

of chain complexes, and $\Xi: B_{\bullet}^{\text{ord}} \Rightarrow B_{\bullet}^{\text{Ar}}$ is a natural transformation of functors $\text{ArChAlg} \rightarrow \text{Ch}_{\mathbb{C}}$. The induced maps on cohomology coincide with the chiral-homology functor Chapter 87.2.2.

Proof. Construction of $\Xi_{\mathbf{A}}$. Both complexes are defined as filtered colimits over reciprocity-closed finite stages $S \in \text{Fin}_{\text{rec}}(|\overline{C}|)$. On a finite stage S with an ordered block decomposition $B_1 | \cdots | B_r$ of S , the ordered incidence bar assigns the tensor product $\mathcal{F}_{\mathbf{A}}(B_1 | \cdots | B_r) := \bigotimes_i \mathbf{A}^{(B_i)}$ (using $\mathcal{F}$ to avoid collision with the Frobenius $F_{\mathbf{A}}$) with contractions given by the adjacent products $m^{(v,w)}$; the Platonic bar assigns $B^{\text{ch}}(\mathbf{A}^{(\infty)}) \widehat{\otimes}_{C_{\text{PG},S}^{\bullet}(\overline{C})} \widehat{\otimes}_{p \in S_f} B_{E_i}^{\Lambda_p}(\mathbf{A}^{(p)})$.

At the archimedean place, the E_1 -chiral bar of $\mathbf{A}^{(\infty)}$ is the $N_{\bullet}$ -normalisation of the semi-simplicial representation (srep: the functor from the semi-simplicial nerve to chain complexes via the Dold–Kan

correspondence) of the E_1 -nerve restricted to singleton blocks; this is the content of Chapter 36.3.4 on the archimedean factor alone. Call this isomorphism $\Xi^{(\infty)}$.

At a finite prime p , the ordered-incidence block contribution is $A^{(p)}$ itself (singleton block, no internal subdivision because $A^{(p)}$ is E_0); the Platonic finite factor is $B_{E_1}^{\Lambda_p}(A^{(p)})$, the Λ_p -equivariant bar of the local Heisenberg. These are quasi-isomorphic by the arithmetic Positselski bar-cobar adjunction (Chapter 24.1.5) composed with the triviality of the E_1 -bar of an E_0 -algebra: the bar of a unital E_0 -object reduces to the shifted augmentation complex, which is homologically the identity. Call this isomorphism $\Xi^{(p)}$.

The Parshin–Gersten layer $C_{\text{PG},S}^\bullet(\overline{C})$ contributes at the ordered-incidence side through the merger maps $m^{(v,w)}$ with their tame-symbol boundaries; the match is Chapter 36.3.3. Put $\Xi_S := \Xi^{(\infty)} \otimes (\otimes_{p \in S_f} \Xi^{(p)}) \otimes \text{id}_{\text{PG},S}$; the Maurer–Cartan closure at finite stage gives that Ξ_S commutes with the two signed total differentials.

Passage to the colimit. The transition maps at both sides insert closed spherical units; by Chapter 36.3.2 the spherical units are killed by the local bar differentials on both sides, so Ξ_S extends compatibly. The filtered colimit $\Xi := \text{colim}_S \Xi_S$ is the required quasi-isomorphism.

Naturality in A . A morphism $\phi: A_1 \rightarrow A_2$ induces a map of both ordered-bar and Platonic-bar assignments (axioms (3), (4), (5) of Chapter 87.1.3). The archimedean factor map $\Xi^{(\infty)}$ is natural because E_1 -chiral bar is functorial in $A^{(\infty)}$ (standard BD/CG). The finite-prime factor map $\Xi^{(p)}$ is natural because arithmetic Positselski is functorial in the underlying Λ_p -module. Both naturalities glue over the filtered stage system, giving that Ξ is a natural transformation of functors $\text{ArChAlg} \rightarrow \text{Ch}_\mathbb{C}$.

Descent to cohomology. Applying $H^\bullet$ to Ξ yields the identity between the ordered-bar homology (Chapter 81.5.4) and the Platonic-bar homology (Chapter 82.3.1); both are the chiral-homology functor by Chapter 87.2.2. $\square$

COROLLARY 87.3.3 (*factoring the three functors*). (on Chapter 87.3.2). The three functors of Chapter 87.2 all factor through the chain-level homology functor $A \mapsto H^\bullet(B^{\text{Ar}}(A), D_A)$:

$$H_\Delta^\bullet(A) \stackrel{\text{def}}{=} H_{\text{ch}}^\bullet(A) = \partial \circ \text{BdryACS}(A)$$

on every $A \in \text{ArChAlg}$, modulo the conditional pieces of Chapters 81 and 82.

Proof. The left-hand equality is Chapter 82.3.2 (definitional). The right-hand equality follows from Chapter 81.5.4 and Chapter 87.3.2: $\partial \circ \text{BdryACS}(A)$ computes the boundary chiral homology of the 3D bulk theory attached to A , which equals the ordered incidence homology of $\{A^{(v)}\}$, which is quasi-isomorphic via Ξ_A to the Platonic bar homology $H_{\text{ch}}^\bullet(A)$. $\square$

Independence and domain of the chain-level identification

Independence. The ordered-incidence bar rests on Francis 2013 (geometric E_1 -nerve); the Platonic bar rests on Beilinson–Drinfeld 2004 and arithmetic Positselski (Chapter 36.3.1). The two constructions come from disjoint source families; Ξ is therefore a genuine Vol IV verification of the identification, not a tautology.

Signs. The ordered-incidence differential uses alternating contraction signs on simplicial nerves; the Platonic differential (36.2) uses Koszul signs on tensor factors. These match under Ξ because the simplicial normalisation formula turns alternating contraction signs into Koszul tensor signs (standard Dold–Kan).

Ambient category. Both bars take values in chain complexes over $\mathbb{C}$ with restricted-tensor refinement to the adelic factorisation structure; the ambient category is identical.

Conditional hypothesis. The quasi-isomorphism $\Xi^{(p)}$ on finite-prime factors uses arithmetic Positselski Chapter 24.1.5, conditional on the non-abelian Iwasawa extension. This is the only inherited conditional; all other steps are unconditional within the primary Heisenberg domain.

Naturality. Axioms (3), (4), (5) of Chapter 87.1.3 supply precisely the intertwining needed at each layer (units, adjacent products, Parshin–Gersten boundary); no additional functoriality input is required.

Unconditional domain. The equivalence between ordered-bar and Platonic-bar homology on the primary Heisenberg domain is the content of Chapter 36.3.4, which holds unconditionally within that domain. No numerical comparison backs this identification.

87.4 FUNCTORIALITY IN $\mathbf{A}$

Chapter 87.3.3 equates the three functors on objects. A natural isomorphism requires that every morphism $\phi: A_1 \rightarrow A_2$ in $\mathbf{ArChAlg}$ induces commuting diagrams; the commutation is the content of the next theorem.

THEOREM 87.4.1 (*naturality in $\mathbf{A}$*). (*on Chapter 87.3.2, Chapter 81.4.7, and Chapter 81.4.9*). The three natural isomorphisms

$$H_{\Delta}^{\bullet} \xrightarrow{\eta_{\Delta, \text{ch}}} H_{\text{ch}}^{\bullet} \xrightarrow{\eta_{\text{ch}, \partial}} \partial \circ \text{BdryACS}$$

are natural: for every morphism $\phi: A_1 \rightarrow A_2$ of arithmetic chiral algebras, the three squares

$$\begin{array}{ccc} H_{\Delta}^{\bullet}(A_1) & \xrightarrow{H_{\Delta}^{\bullet}(\phi)} & H_{\Delta}^{\bullet}(A_2) \\ \eta_{\Delta, \text{ch}, A_1} \downarrow & & \downarrow \eta_{\Delta, \text{ch}, A_2} \\ H_{\text{ch}}^{\bullet}(A_1) & \xrightarrow{H_{\text{ch}}^{\bullet}(\phi)} & H_{\text{ch}}^{\bullet}(A_2) \\ \eta_{\text{ch}, \partial, A_1} \downarrow & & \downarrow \eta_{\text{ch}, \partial, A_2} \\ \partial \circ \text{BdryACS}(A_1) & \xrightarrow{\partial \circ \text{BdryACS}(\phi)} & \partial \circ \text{BdryACS}(A_2) \end{array}$$

commute. The three horizontal maps are identified under the naturality, and the induced Frobenius and Poincaré duality structures are preserved pairwise: $F_{A_2} \circ H^{\bullet}(\phi) = H^{\bullet}(\phi) \circ F_{A_1}$ and likewise for PD.

Proof. The upper square. The equality $H_{\Delta}^{\bullet}(A) \stackrel{\text{def}}{=} H_{\text{ch}}^{\bullet}(A)$ is definitional (Chapter 82.3.2); the natural transformation $\eta_{\Delta, \text{ch}}$ is the identity. Naturality is immediate: the identity intertwines every induced map.

The lower square. Apply Chapter 87.3.2 to both A_1 and A_2 , obtaining chain-level quasi-isomorphisms Ξ_{A_1} and Ξ_{A_2} . The natural transformation $\eta_{\text{ch}, \partial}$ is the induced map on cohomology: $\eta_{\text{ch}, \partial, A} := \sigma_A \circ H^{\bullet}(\Xi_A)^{-1}$, where $\sigma_A: H^{\bullet}(B_{\bullet}^{\text{ord}}(A)) \xrightarrow{\sim} \partial \circ \text{BdryACS}(A)$ is the isomorphism of Chapter 81.5.4; this composite is well-defined by that theorem and Chapter 87.3.2.

Commutativity at the chain level. The induced chain map $B_{\bullet}^{\text{ord}}(\phi)$ intertwines Ξ_{A_1} and Ξ_{A_2} by the naturality of Chapter 87.3.2 in $\mathbf{A}$. The induced chain map $B^{\text{Ar}}(\phi)$ likewise intertwines, by axioms (3), (4), (5) of Chapter 87.1.3: the spherical units are mapped to spherical units; the adjacent products commute through ϕ ; the Parshin–Gersten boundary commutes through ϕ . Composing, we get $\Xi_{A_2} \circ B_{\bullet}^{\text{ord}}(\phi) = B^{\text{Ar}}(\phi) \circ \Xi_{A_1}$ as maps of chain complexes. Passing to cohomology gives the commutative square.

Frobenius compatibility. The global Frobenius $F_A = F_{\infty, A} \otimes \otimes'_p \psi_{p, A}$ commutes with $H^\bullet(\phi)$ because axiom (2) of the morphism definition supplies ψ_p -equivariance at each finite prime, and axiom (1) supplies preservation of the conformal vector at ∞ (from which F_∞ is conjugated via Chapter 82.5.4). Hence $F_{A_2} \circ H^\bullet(\phi) = H^\bullet(\phi) \circ F_{A_1}$.

Poincaré duality compatibility. The PD pairing is determined by the Koszul self-duality identity Chapter 82.7.1. A morphism ϕ in ArChAlg preserves the spherical units and the adjacent products; the self-duality pairing is expressed chain-level as the Positselski contraction pairing, which is natural in the underlying A . Hence PD commutes with ϕ . $\square$

COROLLARY 87.4.2 (*triple functoriality in A*). (on Chapter 87.4.1). The three functors $H_\Delta^\bullet$, $H_{\text{ch}}^\bullet$, and $\partial \circ \text{BdryACS}$ are naturally isomorphic as functors $\text{ArChAlg} \rightarrow \text{grVect}_{\mathbb{C}}^{\text{Frob}}$. Both structural data (Frobenius, Poincaré duality) pull through the natural isomorphisms.

Proof. Immediate from Chapter 87.4.1 after recording that $\eta_{\Delta, \text{ch}}$ is invertible (as the identity) and $\eta_{\text{ch}, \partial}$ is invertible as the induced map on cohomology from a chain-level quasi-isomorphism. $\square$

Independence and domain of naturality in A

Independence. Beilinson–Drinfeld 2004 supplies the E_1 -chiral functoriality at the archimedean factor; Kim 2015 and the quantum ACS supply the bulk interface defect functoriality at the 3D side. These derivation paths are disjoint; the natural isomorphism $\eta_{\text{ch}, \partial}$ verified against both is a genuine independent check.

Signs. The bar-differential Koszul signs agree on both sides by construction of Chapter 36.3.1; the ordered-incidence alternating signs match under Ξ by the simplicial normalisation formula.

Ambient category. Both sides land in graded $\mathbb{C}$ -vector spaces with Frobenius and Poincaré duality; no compatibility issue.

Frobenius hypothesis. The naturality of F_∞ under $\phi^{(\infty)}$ requires that $\phi^{(\infty)}$ preserves the conformal vector; this is axiom (1) of Chapter 87.1.3, which is part of the definition of a morphism and imposes no hidden constraint.

Axiom independence. Axioms (2) and (4) are logically independent: axiom (4) applied at $v = p, w = \infty$ implies axiom (2) only on the unit line (the spherical vacuum at ∞ is fixed by ψ_p up to normalised Frobenius weight); the full ψ_p -equivariance on all of $A^{(p)}$ is an additional local constraint not automatic from (4). Both axioms are needed and neither is redundant.

Spectral content. The identification of the F -spectrum on H^1 with the nontrivial zeros of $\xi(s)$ remains conditional on H-PAr (Chapter 22.3.2), inherited from Chapter 82.6.1. The naturality of F established here requires only its commutation with $H^\bullet(\phi)$, not a description of its spectrum.

87.5 FUNCTORIALITY IN $\mathcal{g}$

The genus filtration $\text{ArChAlg}_0 \subset \text{ArChAlg}_1 \subset \cdots$ of Chapter 87.1.5 generates further functoriality structure, tying Chapter 82 to Chapter 80.

Definition 87.5.1 (*genus- g restrictions of the three functors*). For every $g \geq 0$, let $H_{\Delta, g}^\bullet$, $H_{\text{ch}, g}^\bullet$, and $(\partial \circ \text{BdryACS})_g$ denote the restrictions of $H_\Delta^\bullet$, $H_{\text{ch}}^\bullet$, and $\partial \circ \text{BdryACS}$ to ArChAlg_g . Explicitly:

- $H_{\Delta, g}^\bullet$ sends a genus- g object $A \in \text{ArChAlg}_g$ to the rank-2 g Deninger cohomology with global Frobenius $F^{(g)}$ of Chapter 82.9.2(2);

- $H_{\text{ch},g}^\bullet$ sends $\mathbf{A}$ to the rank-2 g Platonic bar homology of Chapter 82.9.1 with rank-2 g Fock factors per place;
- $(\partial \circ \text{BdryACS})_g$ sends $\mathbf{A}$ to the boundary chiral homology of the rank-2 g 3D arithmetic CS $\text{ACS}_{k^{\text{Ar}},g}^{\text{qu},g}$ of Chapter 81.8.1.

THEOREM 87.5.2 (compatibility under genus inclusion). (on Chapter 87.4.1, Chapter 80.4.1, Chapter 82.9.2). For every $g \geq 0$, the three genus- g functors commute with the inclusion $\iota_{g,g+1} : \text{ArChAlg}_g \hookrightarrow \text{ArChAlg}_{g+1}$ up to canonical natural isomorphisms $\mathfrak{H}_g^\Delta, \mathfrak{H}_g^{\text{ch}}, \mathfrak{H}_g^\partial$:

$$\begin{array}{ccc}
 \text{ArChAlg}_g & \xrightarrow{\iota_{g,g+1}} & \text{ArChAlg}_{g+1} \\
 & \searrow H_{\star,g}^\bullet \otimes V_2 & \swarrow H_{\star,g+1}^\bullet \\
 & \text{grVect}_{\mathbb{C}}^{\text{Frob}} &
 \end{array}$$

commutes up to $\mathfrak{H}_g^\star$ for $\star \in \{\Delta, \text{ch}, \partial\}$, where V_2 is the rank-2 archimedean Heisenberg summand and $\otimes V_2$ is the tensor-up with its associated graded Fock module. The three natural transformations $\mathfrak{H}_g^\star$ are compatible with the natural isomorphisms $\eta_{\Delta,\text{ch}}, \eta_{\text{ch},\partial}$ of Chapter 87.4.1.

Proof. The map $\iota_{g,g+1}$ adjoins a rank-2 archimedean Heisenberg factor at every place: at each prime p , the rank-2 g Fock $\mathcal{H}^{(p),g}$ is replaced by the rank-2 $(g+1)$ Fock $\mathcal{H}^{(p),g+1} = \mathcal{H}^{(p),g} \otimes \mathcal{H}_2^{(p)}$ where $\mathcal{H}_2^{(p)}$ is a freshly-added rank-2 Fock factor; analogously at ∞ .

For $\star = \text{ch}$: the Platonic bar is multiplicative under the tensor product of factorisation algebras. At each finite stage S , the bar factors as $B_S^{\text{Ar}}(\mathbf{A} \otimes V_2) = B_S^{\text{Ar}}(\mathbf{A}) \widehat{\otimes} B_S^{\text{Ar}}(V_2)$, where the right-hand factor is the bar of the freshly-added rank-2 Heisenberg. Passing to cohomology and to the filtered colimit, the natural isomorphism $\mathfrak{H}_g^{\text{ch}}$ is the Künneth map of the Platonic bar, which is a quasi-isomorphism after restricting to the primary subspace of each side (Chapter 12.5.1 multiplicativity).

For $\star = \Delta$: $\mathfrak{H}_g^\Delta$ is the same map under Chapter 82.3.2's definitional equality $H_{\Delta,g}^\bullet = H_{\text{ch},g}^\bullet$; naturality is inherited from $\mathfrak{H}_g^{\text{ch}}$.

For $\star = \partial$: the bulk 3D ACS theory $\text{ACS}^{\text{qu},g+1}$ of Chapter 81.8.1 factors as the tensor product of $\text{ACS}^{\text{qu},g}$ with the rank-2 tensor-slot theory. This is the content of the block-diagonal splitting of the gauge group $GL_{2(g+1)} \supset GL_{2g} \times GL_2$ on the boundary: the rank-2 $(g+1)$ WZW algebra decomposes as the tensor product of the rank-2 g WZW algebra with the rank-2 WZW algebra under this block-diagonal subgroup. The boundary chiral homology is multiplicative under tensor products of WZW algebras, so $\mathfrak{H}_g^\partial$ is the Künneth map on the boundary side.

Compatibility with η . The two natural transformations $\mathfrak{H}_g^{\text{ch}}$ and $\mathfrak{H}_g^\partial$ both come from Künneth maps at the corresponding bar level. Applying the chain-level identification Chapter 87.3.2 to the rank-2 factor (whose bar is freshly introduced on both sides), we obtain $\eta_{\text{ch},\partial,\mathbf{A} \otimes V_2} \circ \mathfrak{H}_g^{\text{ch}} = \mathfrak{H}_g^\partial \circ \eta_{\text{ch},\partial,\mathbf{A}}$ as required. $\square$

THEOREM 87.5.3 (uniformity: genus-wise triple identification). (on Chapter 87.5.2 and Chapter 82.9.2). The triple identification is uniform in genus: for every $g \geq 0$ and every $\mathbf{A} \in \text{ArChAlg}_g$,

$$H_{\Delta,g}^\bullet(\mathbf{A}) = H_{\text{ch},g}^\bullet(\mathbf{A}) = (\partial \circ \text{BdryACS})_g(\mathbf{A}) \quad (87.1)$$

as graded vector spaces with genus- g Frobenius $F^{(g)}$ and PD pairing $\text{PD}^{(g)}$. The equality is compatible with the inclusion $\iota_{g,g+1}$ of Chapter 87.5.2 and with the retraction $\pi_{g+1,g}$: the three genus- g identifications commute, up to $\mathfrak{H}_g^\star$, with the corresponding genus- $(g+1)$ identifications.

Proof. The three equalities on the right-hand side of (87.1) are the genus- g restrictions of Chapter 87.3.3. They hold for every $A \in \text{ArChAlg}_g$ by restriction.

The compatibility with genus inclusion: apply Chapter 87.5.2 to obtain the three compatibility diagrams connecting g and $g + 1$. Each diagram is commutative up to $\mathfrak{P}_g^\star$; the three $\mathfrak{P}_g^\star$ agree under the natural isomorphisms $\eta_{\Delta, \text{ch}}$ and $\eta_{\text{ch}, \partial}$. Hence the genus- g triple identification commutes with the genus- $(g + 1)$ identification: the diagram

$$\begin{array}{ccccc} H_{\Delta, g}^\bullet(A) \otimes V_2 & \xrightarrow{=} & H_{\text{ch}, g}^\bullet(A) \otimes V_2 & \xrightarrow{=} & (\partial \circ \text{BdryACS})_g(A) \otimes V_2 \\ \mathfrak{P}_g^\Delta \downarrow & & \mathfrak{P}_g^{\text{ch}} \downarrow & & \downarrow \mathfrak{P}_g^\partial \\ H_{\Delta, g+1}^\bullet(\iota(A)) & \xrightarrow{=} & H_{\text{ch}, g+1}^\bullet(\iota(A)) & \xrightarrow{=} & (\partial \circ \text{BdryACS})_{g+1}(\iota(A)) \end{array}$$

commutes.

The compatibility with retraction $\pi_{g+1, g}$ is the mirror statement: projection onto the first $2g$ oscillators, which is Künneth-compatible by the same argument in the reverse direction. $\square$

COROLLARY 87.5.4 (*all-genus triple functoriality*). (*on Chapter 87.5.3*). The three functors $H_{\Delta}^\bullet$, $H_{\text{ch}}^\bullet$, and $\partial \circ \text{BdryACS}$ are naturally isomorphic as *genus-filtered* functors $\text{ArChAlg} \rightarrow \text{grVect}_{\mathbb{C}}^{\text{Frob}}$. The natural isomorphism is genus-graded: restricts to a natural isomorphism at every genus g , and commutes with inclusion and retraction along the filtration.

Proof. Assemble Chapter 87.4.2 (functoriality in A at fixed genus) with Chapter 87.5.3 (uniform compatibility across genus). The genus-filtered structure is preserved at each genus, with transition natural transformations $\mathfrak{P}_g^\star$ compatible. $\square$

Independence and domain of genus functoriality

Source consistency. The genus filtration via rank-2 tensor slots of Ch. 80 and the 3D ACS gauge-group scaling $GL_{2g} \hookrightarrow GL_{2(g+1)}$ of Ch. 81 agree: in both cases the rank parameter is $2g$, the archimedean Heisenberg has rank $2g$, and the Satake parameter at each finite prime is also of rank $2g$. The three chapters use identical rank conventions.

Conventions. The notation $r = 2g$ for the rank of the archimedean Heisenberg, g for the Siegel-modular genus, and the Satake rank $2g$ at each finite prime is consistent across Ch. 80, Ch. 81, and Ch. 82.

Künneth flatness. The Künneth map for the Platonic bar at the rank-2 slot requires flatness of $\mathcal{H}_2^{(p)}$ over the Iwasawa algebra Λ_p . This is the rank-2 instance of the Positselski Koszul duality of Ch. 53, which holds in the abelian Fock setting.

Retraction. The retraction $\pi_{g+1, g}$ does not require a cokernel: on the primary Heisenberg domain the freshly-added rank-2 factor is a direct summand, and $\pi_{g+1, g}$ is direct-summand projection.

Spectral content. The genus- g F -spectrum identification with motivic zeros is conditional on rank- g H-P^{Ar} (Chapter 82.9.2(4)) and is not supplied here. The genus-filtration structure holds independently of the spectrum identification.

87.6 MASTER THEOREM AND EXPLICIT EXAMPLE

Functoriality in A and genus-filtered functoriality hold independently; the master theorem records that both hold simultaneously, with a specific chain-level quasi-isomorphism Ξ as canonical witness. The diagonal morphism $\text{H}^{\text{Ar}} \rightarrow \text{H}^{\text{Ar}} \oplus \text{H}^{\text{Ar}}$ exercises every axiom and every piece of the triple identification.

THEOREM 87.6.1 (*master functoriality*). (*on the residuals of Chapters 81, 80, 82 and their arithmetic Positselski, $H\text{-}P^{\text{Ar}}$, 3D quantum ACS domains*). The triple identification $H_{\Delta}^{\bullet} = H_{\text{ch}}^{\bullet} = \partial \circ \text{BdryACS}$ is a natural isomorphism of genus-filtered functors

$$\text{ArChAlg} = \bigcup_{g \geq 0} \text{ArChAlg}_g \longrightarrow \text{grVect}_{\mathbb{C}}^{\text{Frob}},$$

which is:

- (i) functorial in $\mathbf{A}$: every morphism $\phi: A_1 \rightarrow A_2$ in ArChAlg induces a commuting triple-square of cohomology maps, preserving Frobenius F and Poincaré duality PD ;
- (ii) functorial in g : for every $g \geq 0$, the restriction to ArChAlg_g is a natural isomorphism of genus- g functors, compatible with the inclusion $\iota_{g, g+1}$ and the retraction $\pi_{g+1, g}$;
- (iii) chain-level canonical: the natural isomorphism is induced by the quasi-isomorphism Ξ of Chapter 87.3.2, a specific chain map built from arithmetic Positselski bar-cobar, the E_1 -chiral archimedean bar, and the Parshin–Gersten tame-symbol layer.

Proof. Part (1) is Chapter 87.4.2. Part (2) is Chapter 87.5.4. Part (3) is the explicit construction of Ξ in the proof of Chapter 87.3.2, together with the fact that the Deninger-to-chiral-homology equality $\eta_{\Delta, \text{ch}}$ is definitional (Chapter 82.3.2) and hence trivially chain-level. $\square$

Example 87.6.2 (*diagonal $H^{\text{Ar}} \rightarrow H^{\text{Ar}} \oplus H^{\text{Ar}}$*). The diagonal morphism

$$\Delta: H^{\text{Ar}} \longrightarrow H^{\text{Ar}} \oplus H^{\text{Ar}}$$

is defined place-wise by the arithmetic Heisenberg diagonal

$$\Delta^{(v)}: \mathcal{H}^{(v)} \longrightarrow \mathcal{H}^{(v)} \oplus \mathcal{H}^{(v)}, \quad x \longmapsto (x, x),$$

at every place $v \in |\overline{C}|$. The target is the coproduct of two copies of H^{Ar} in ArChAlg ; its archimedean component is the free E_1 -chiral product of two rank-2 Heisenbergs, hence a rank-4 Heisenberg.

PROPOSITION 87.6.3 (*triple verification on the diagonal*). (*on Chapter 87.6.1*). The diagonal Δ of Chapter 87.6.2 is a morphism in ArChAlg (axioms (1)–(6) of Chapter 87.1.3 are satisfied), and the induced triple-morphism

$$(H_{\Delta}^{\bullet}(\Delta), H_{\text{ch}}^{\bullet}(\Delta), \partial \circ \text{BdryACS}(\Delta))$$

commutes with $\eta_{\Delta, \text{ch}}$ and $\eta_{\text{ch}, \partial}$. Explicitly:

- (i) $H_{\Delta}^{\bullet}(\Delta)$ is the diagonal embedding $H_{\Delta}^{\bullet}(H^{\text{Ar}}) \rightarrow H_{\Delta}^{\bullet}(H^{\text{Ar}} \oplus H^{\text{Ar}}) = H_{\Delta}^{\bullet}(H^{\text{Ar}}) \oplus H_{\Delta}^{\bullet}(H^{\text{Ar}})$ of the arithmetic cohomology; in degree 0 it is $x \mapsto (x, x)$ on $\mathbb{C}_0 \rightarrow \mathbb{C}_0 \oplus \mathbb{C}_0$; similarly for H^2 ; and it sends the middle H^1 -eigenvector of the Weil-operator Θ_{ξ} (the functional-equation involution of Chapter 22.3.2) at ordinate γ_{ρ} to the pair of two copies of the same eigenvector.
- (ii) $H_{\text{ch}}^{\bullet}(\Delta)$ is the induced chain map of Platonic bars, compatible with the bar differential because Δ preserves the spherical units and is a homomorphism of chiral algebras.
- (iii) $\partial \circ \text{BdryACS}(\Delta)$ is the boundary restriction of the bulk interface $\text{BdryACS}(\Delta): \text{ACS}_{H^{\text{Ar}}}^{\text{qu}} \rightarrow \text{ACS}_{H^{\text{Ar}}}^{\text{qu}} \oplus \text{ACS}_{H^{\text{Ar}}}^{\text{qu}}$, which in the 3D ACS framework is the diagonal bulk interface defect connecting one bulk theory to the disjoint union of two copies of itself.

All three commute with the natural isomorphisms $\eta_{\Delta, \text{ch}}$, $\eta_{\text{ch}, \partial}$.

Proof. Axiom verification. Axiom (1): $\Delta^{(\infty)}$ is the diagonal on the archimedean Heisenberg factorisation algebra; it preserves the conformal vector because the free product of chiral algebras has a conformal vector which is the sum of the two factor conformal vectors, and the diagonal $T \mapsto T \oplus T$ is consistent with the coproduct structure. Axiom (2): $\Delta^{(p)}$ commutes with ψ_p because $\psi_p(x, x) = (\psi_p(x), \psi_p(x))$. Axiom (3): $\Delta^{(v)}(u^{(v)}) = (u^{(v)}, u^{(v)})$, which is the spherical unit of the coproduct. Axiom (4): the adjacent product on the coproduct is the diagonal of the adjacent product on each summand; Δ commutes with this by construction. Axiom (5): the tame-symbol boundary ∂^{PG} is linear, so $\partial^{\text{PG}}(x, x) = (\partial^{\text{PG}}(x), \partial^{\text{PG}}(x))$. Axiom (6): Tate compatibility is preserved because the spherical units are preserved and the restricted tensor factors through each summand.

Claim (1). The Deninger cohomology of the coproduct $H^{\text{Ar}} \oplus H^{\text{Ar}}$ is $H_{\Delta}^{\bullet}(H^{\text{Ar}}) \oplus H_{\Delta}^{\bullet}(H^{\text{Ar}})$ because arithmetic bar commutes with direct sums on the primary domain (the bar is additive in its argument, a consequence of Chapter 36.3.1 and of the additivity of the tame-symbol boundary). The induced map is the diagonal. In each degree: $H^0 = \mathbb{C}_0 \rightarrow \mathbb{C}_0 \oplus \mathbb{C}_0, x \mapsto (x, x)$; $H^2 = \mathbb{C}_1 \rightarrow \mathbb{C}_1 \oplus \mathbb{C}_1, x \mapsto (x, x)$; H^1 -eigenvectors at ordinate γ_p go to the pair of two copies.

Claim (2). The Platonic bar is natural in $\mathbf{A}$ by Chapter 87.2.2; applying naturality to Δ gives the chain map on bars. Bar differential compatibility is axiom (5) of Chapter 87.1.3 applied to the diagonal.

Claim (3). The bulk-boundary assignment $\text{BdryACS}(\Delta)$ is by Chapter 87.2.3 the 3D bulk interface whose archimedean boundary restriction is $\Delta^{(\infty)}$ (the diagonal map of archimedean E_1 -chiral algebras) and whose finite-place Wilson-line restriction is $\{\Delta^{(p)}\}$. The boundary chiral homology ∂ of this interface is the induced map on boundary chiral homology. By the additivity of boundary chiral homology for coproducts in $3\text{DACS}_{\overline{\text{Spec } \mathbb{Z}}}$ (the disjoint union of two bulk theories has as boundary the coproduct of the two boundary chiral algebras), the boundary map is the diagonal on the rank-2 chiral homology.

Compatibility with γ . By construction each of the three maps agrees with the others under the natural isomorphisms: they are all the diagonal of the same underlying primary Heisenberg cohomology object, related by the chain-level quasi-isomorphism Ξ of Chapter 87.3.2 on each direct summand and the identity on the direct-sum structure. $\square$

Remark 87.6.4 (what the diagonal example witnesses). The diagonal Δ is the simplest non-trivial morphism in ArChAlg after the identity: it is not an isomorphism, not a projection, not a single-factor inclusion. Verifying the triple-morphism commutation on Δ tests every axiom of Chapter 87.1.3 against every piece of the triple identification. Because the coproduct $H^{\text{Ar}} \oplus H^{\text{Ar}}$ sits in $\text{ArChAlg}_{g=2}$ (rank-4 archimedean Heisenberg) while H^{Ar} sits in $\text{ArChAlg}_{g=1}$, the example also touches genus functoriality: it is a non-trivial map in the filtered structure $\text{ArChAlg}_1 \rightarrow \text{ArChAlg}_2$ that does not factor through the filtered-colimit inclusion $\iota_{1,2}$.

Independence and domain of the master theorem

Independence. The master theorem combines three source families: Beilinson–Drinfeld 2004 (archimedean E_1 -chiral structure), Fargues–Scholze arXiv:2102.13459 (finite-prime perfectoid charts), and Kim 2015 arXiv:1510.05818 plus the quantum ACS (3D bulk theory). These families are pairwise disjoint in the Vol IV orthogonal sense.

Sign consistency. The Koszul signs in the Platonic bar differential, the alternating signs in the ordered-incidence bar, and the 3D framed-cobordism signs in the boundary-chiral-homology pairing are mutually consistent under the chain-level identification Ξ of Chapter 87.3.2.

Inherited conditional hypotheses. The well-definedness of $\partial \circ \text{BdryACS}$ as a functor (rather than a mere correspondence) requires a well-defined bulk interface defect category, conditional on Chapter 81.4.4 and the renormalised profinite limit of Chapter 81.4.7. Both conditionals are inherited from Chapter 81.

Spectral domain. The master theorem claims the *functoriality* of the triple identification, not the spectral content. The three spectral statements (Deninger purity, Riemann Hypothesis, F -RH) remain conditional on H-P^{Ar} and the archimedean $\mathbf{A}$ -TP hypothesis; neither is asserted here.

87.7 DOMAIN OF THE MASTER THEOREM

Functoriality residuals of Chapter 82 closed here

The following residuals of Chapter 82 are closed here:

- (i) Chapter 82.12(5): triple-identification functoriality under pullback along number-field extensions $K/\mathbb{Q}$. Closed: any such pullback is a morphism in ArChAlg (it preserves spherical units, adjacent products, and Parshin–Gersten boundaries by classical number-field theory), and functoriality follows from Chapter 87.4.2.
- (ii) Chapter 82.12(6): genus- g Wilson-line Frobenius identification. Closed: the genus- g restriction of $\partial \circ \text{BdryACS}$ is functorially identified with the genus- g Deninger cohomology by Chapter 87.5.3; the rank- $2g$ Wilson-loop observables are identified with the rank- $2g$ $\psi_p^{(g)}$ by naturality in the direct summand structure.

Inherited conditional hypotheses of the master theorem

The master functoriality theorem Chapter 87.6.1 is conditional on the following inherited hypotheses:

- (i) Arithmetic Positselski (Ch. 53) for the finite-prime factor quasi-isomorphism $\Xi^{(p)}$.
- (ii) 3D quantum ACS existence (Chapter 81.4.7) for the well-definedness of BdryACS as a functor.
- (iii) Local-trace comparison (Chapter 81.4.9) for the boundary identification at ∞ .
- (iv) H-P^{Ar} (Chapter 22.3.2) for the spectrum identification on H^1 ; not used in functoriality but used in spectral content.
- (v) Rank- g Künneth compatibility on the Platonic bar, which is standard by rank- g Positselski (Ch. 80.6(2)).

No new residuals are introduced.

The master theorem and what it forces

The triple identification $H_{\Delta}^{\bullet} = H_{\text{ch}}^{\bullet} = \partial \circ \text{BdryACS}$, previously established object-wise for H^{Ar} at genus $g = 0$, is a natural isomorphism of genus-filtered functors on all of ArChAlg : every morphism of arithmetic chiral algebras induces a commuting triple-square of cohomology maps, Frobenius-equivariant and Poincaré-dual-compatible; every genus-inclusion and retraction is intertwined by Künneth natural transformations $\mathfrak{S}_g^{\bullet}$. The chain-level quasi-isomorphism Ξ of Chapter 87.3.2, built from arithmetic Positselski bar-cobar at finite primes, the E_1 -chiral archimedean bar, and the Parshin–Gersten tame-symbol layer, is the canonical chain-level witness. The three source families (Deninger 1998, Beilinson–Drinfeld + Positselski, Kim + quantum ACS) are pairwise disjoint; the natural isomorphism is therefore a genuine Vol IV orthogonal verification. The remaining spectral content (that the Frobenius F on H^1 has eigenvalues at the nontrivial zeros of $\xi(s)$) requires the inherited H-P^{Ar} hypothesis (Chapter 22.3.2) and is not supplied here.

PRIMARY SOURCES AND DISJOINTNESS

The chapter depends on four disjoint source families.

Arithmetic chiral algebra side. Beilinson–Drinfeld 2004, *Chiral Algebras* (AMS Colloq. 51); Francis 2013 (geometric E_1 -ordered bar); Francis–Gaiitsgory 2012 arXiv:1103.5803 (factorisation gluing); Costello–Gwilliam 2017, *Factorization Algebras in Quantum Field Theory* (Cambridge UP) for the archimedean E_1 -chiral factorisation algebra.

Finite-prime arithmetic side. Fargues–Scholze arXiv:2102.13459 (geometrisation of local Langlands); Bhatt–Scholze 2022, *Prisms and prismatic cohomology*, Ann. Math. for the prismatic/Nygaard comparison; Borger 2009 arXiv:0906.3146 for the Bost–Connes Adams ψ_p ; Positselski 2011, Mem. AMS 996, *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence* for the finite-prime quasi-isomorphism $\Xi^{(p)}$.

3D ACS / bulk-boundary side. Kim 2015 arXiv:1510.05818, *Arithmetic Chern–Simons Theory I*; Chern–Dan–Kim–Park 2018 arXiv:1609.03012 (non-abelian extension); Mazur 1973 ASENS 6 (étale cohomological dimension of number rings); Morishita 2012 arXiv:0904.3399, *Knots and Primes*; Witten 1989, CMP 121, and Witten 2007 arXiv:0706.3359 (3D gravity / 3D CS boundary WZW); Artin–Verdier duality per SGA 4.5.

Genus and modular side. Consani–Marcolli 2006 (genus- g Bost–Connes–Marcolli); Faltings 1984, Ann. Math. 119, *Calculus on arithmetic surfaces*; Kudla 1997, Ann. Math. 146; Li–Liu 2021 arXiv:2101.09485 (Kudla–Rapoport at bad primes); Arthur 2013, *The Endoscopic Classification of Representations*, AMS Colloq. 61.

Pairwise disjointness of the four families is witnessed by the Vol IV source-list Chapter A.8–Chapter A.7, Chapter A.19, Chapter A.6, and the 3D ACS citation cluster of Ch. 81. Each theorem of this chapter has two independent paths: derivation through one family and comparison with another.

WILSON-LINE FROBENIUS AND ARCHIMEDEAN WILSON-SURFACE AT GENUS g

Chapter 80 establishes the all-genus arithmetic Koszul duality identity $\kappa_G^{\text{Ar},g} + (\kappa^!)_G^{\text{Ar},g} = 0$ on an arithmetic genus- g datum $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \phi_g)$, with the genus entering through the rank- $2g$ Fock grading, the archimedean $\Gamma_{\mathbb{C}}(s)^g$ factor, and the motivic weight $w_{\mathcal{D}_g} \in \{1, 2, 2g + 1\}$. Chapter 81 builds the 3D arithmetic Chern–Simons theory ACS^{qu} on $\overline{\text{Spec } \mathbb{Z}}$, with Wilson-line Frobenius identity $\langle W_\rho(K_p) \rangle = \text{tr } \rho(\text{Frob}_p)$ at $g = 0$. Chapter 82 identifies its boundary chiral homology with Deninger’s conjectural arithmetic cohomology and isolates the global Frobenius $F_\infty \otimes \bigotimes'_p \psi_p$ realised as Wilson loops and surfaces. The genus- g extension replaces the gauge group $GL_2(\widehat{\mathbb{Z}})$ by its rank- $2g$ analog $GL_{2g}(\widehat{\mathbb{Z}})$; at each prime p the defect operator becomes the genus- g Wilson line, and the archimedean boundary operator becomes the genus- g Wilson-surface $F_\infty^{(g)}$. The genus- g gauge group is the profinite limit of $GL_{2g}(\mathbb{Z}/N\mathbb{Z})$; the rank- $2g$ local representation $\rho_g : G_{\mathbb{Q}_p} \rightarrow \text{GSp}_{2g}$ sources a genus- g Wilson loop whose expectation in $\text{ACS}^{\text{qu},g}$ is $\text{tr}_{\rho_g}(\text{Frob}_p)$; the archimedean Wilson-surface $F_\infty^{(g)}$ is the rank- $2g$ Arakelov area-scaling conjugate to the genus- g Virasoro zero-mode $L_0^{(g)}$ with central charge g . The all-genus Deninger identification is the motivic-weight- $w_{\mathcal{D}_g}$ specialisation of the character identity of Chapter 82.9.2. At $g = 0$ the genus family degenerates to the rank-zero datum. The rank-two Wilson-line theory of the Riemann framework is an independent ACS sector, not the value of $\rho_{g,p}$ at $g = 0$. The compatibility with Chs. 84, 89, 22, and 83 is recorded in Chapter 88.6.1.

88.1 GENUS- g ARITHMETIC DATA RECAP

Fix $g \geq 0$ and an arithmetic genus- g datum $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \phi_g)$ as in Chapter 80.1.1. Four structural features of $\mathcal{D}_g$ are used throughout.

- (i) *Rank- $2g$ local Galois representation.* At each prime p of good reduction for $\mathcal{D}_g$, there is a continuous rank- $2g$ representation $\rho_{g,p} : G_{\mathbb{Q}_p} := \text{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p) \rightarrow \text{GSp}_{2g}(\overline{\mathbb{Q}_p})$ with symplectic Weil pairing; in Window I, $\rho_{g,p} = T_\ell \text{Jac}(X_K)_{\mathbb{Q}_p}$ (Tate module of the Jacobian); in Window III, the prismatic F -crystal; in Window VI, the spin representation of the automorphic π .
- (ii) *Satake parameters* $\{\alpha_j^{(p)}\}_{j=1}^{2g}$. At an unramified prime, $\rho_{g,p}(\text{Frob}_p)$ has $2g$ eigenvalues $\alpha_1^{(p)}, \dots, \alpha_{2g}^{(p)}$ with symplectic pairing $\alpha_j^{(p)} \alpha_{2g+1-j}^{(p)} = p^{w_{\mathcal{D}_g} - 1}$; these source the local Euler factor $L_p(\mathcal{D}_g, s) = \prod_{j=1}^{2g} (1 - \alpha_j^{(p)} p^{-s})^{-1}$.
- (iii) *Rank- $2g$ Fock oscillators per place.* The local factor $\mathcal{H}^{(p),g}$ of $\mathcal{V}_{\text{Heis}}^{\text{prim},g} = \bigotimes'_p \mathcal{H}^{(p),g} \otimes \mathcal{H}^{(\infty),g}$ is the rank- $2g$ Fock whose character is the local Euler factor above. The archimedean factor $\mathcal{H}^{(\infty),g}$ carries the rank- $2g$ archimedean Heisenberg with Γ -factor $\Gamma_{\mathbb{C}}(s)^g$ at weight 1 (Windows I, V) or the corresponding Γ -factor of Chapter 80.1 at Dedekind or Siegel-spin weight.

- (iv) *Motivic functional-equation centre* $w_{\mathcal{D}_g} \in \{1, 2, 2g + 1\}$. The completed L -function $\Lambda_{\mathcal{D}_g}(s)$ satisfies $\Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g} \Lambda_{\mathcal{D}_g}(w_{\mathcal{D}_g} - s)$ by Chapter 80.4.1; purity lies on $\text{Re}(s) = w_{\mathcal{D}_g}/2$.

Remark 88.1.1 (degeneration at $g = 0$). At $g = 0$ the datum $\mathcal{D}_0$ has no non-trivial Galois representation: $\mathcal{E}_0$ is void, $\varphi_0 = \text{id}$, and there is no rank- $2g$ local system left to trace. The representation ρ in Chapter 84 is a separate GL_2 -valued Wilson-line label. The Heisenberg degenerates to the rank-0 Fock whose local factor is the trivial Euler line, and the global character $\xi_{\mathbb{R}}(s)$ retains only the Bost–Connes primon-gas contribution $\zeta(s)$ multiplied by $\pi^{-s/2}\Gamma(s/2)$ and the Tate-pole factor $\frac{1}{2}s(s-1)$. The Wilson-line identity of Chapter 84 and the non-trivial archimedean Wilson-surface of Chapter 89 are compatible rank-two and rank-one Tate inputs, not consequences of the empty rank- $2g$ trace. The comparison is recorded in Section 88.6.

88.2 GENUS- g 3D ARITHMETIC CHERN–SIMONS THEORY

Chapter 81 builds the 3D arithmetic Chern–Simons theory on $\overline{\text{Spec } \mathbb{Z}}$ with gauge group $GL_2(\widehat{\mathbb{Z}})$ in the profinite limit, level $k^{\text{Ar}} = \kappa_G^{\text{Ar}}$, boundary WZW $\mathcal{V}^{\text{prim}}$, and Wilson-loop axiom Chapter 81.4.7(3). The datum $\mathcal{D}_g$ determines a rank- $2g$ gauge group $GL_{2g}(\widehat{\mathbb{Z}})$, a rank- $2g$ Koszul level $k^{\text{Ar},g}$, and a rank- $2g$ boundary Heisenberg $\mathcal{V}^{\text{prim},g}$; the following definition records these data and inherits the conditional form of the $g = 0$ theory.

Definition 88.2.1 (genus- g arithmetic Chern–Simons). The genus- g arithmetic Chern–Simons theory is a family of 3D topological gauge theories on an arithmetic 3-space $\mathcal{B}_g^{(3)}$ parameterised by $\mathcal{D}_g$, with

- (i) *Arithmetic 3-space.* For Window II at totally imaginary K of degree $2g$: $\mathcal{B}_g^{(3)} = \overline{\text{Spec } \mathcal{O}_K}$ with Artin–Verdier fundamental class $[\mathcal{B}_g^{(3)}] \in H_{\text{ét}}^3(\mathcal{B}_g^{(3)}, \widehat{\mathbb{Z}})$ (Mazur 1973). For Window I at $X \rightarrow \text{Spec } \mathcal{O}_K$: $\mathcal{B}_g^{(3)} = \overline{\text{Spec } \mathcal{O}_K}$ with extra fibre cohomology in H^2 of the generic fibre collected into the archimedean boundary via the Arakelov Kähler form $\mu_{\text{Ar},g}$ of Faltings 1984 (to be defined in Chapter 88.4.1). For Window VI, Shimura varieties of GSp_{2g} have their own Artin–Verdier-type fundamental class assembled via étale cohomology of the special fibre plus Deligne 1974 Weil cohomology.
- (ii) *Gauge group.* $G_{g,N} := GL_{2g}(\mathbb{Z}/N\mathbb{Z})$, with profinite limit $G_{g,\infty} := GL_{2g}(\widehat{\mathbb{Z}})$.
- (iii) *Level.* $k^{\text{Ar},g} := \kappa_G^{\text{Ar},g}(\mathcal{D}_g)$, the rank- $2g$ arithmetic Koszul level of Chapter 80.4.1.
- (iv) *Boundary WZW at the archimedean fibre.* $\partial \text{ACS}_{k^{\text{Ar},g}}^{\text{qu},g} = \mathcal{V}^{\text{prim},g}$ in the sense of Chapter 81.4.9 scaled to rank $2g$ per Chapter 81.8.

The finite-level state sums $Z_{G_{g,N}, \alpha_{g,N}}^{\text{Ar}}$ extend Chapter 81.4.2 unchanged at each finite rank; the profinite renormalisation $\text{ACS}_{k^{\text{Ar},g}}^{\text{qu},g}$ remains conditional in the sense of Chapter 81.4.7 extended to rank $2g$.

Remark 88.2.2 (rank scaling and residual profinite limit). The rank-change $2 \rightarrow 2g$ does not alter the closed finite-level state-sum of Chapter 81.4.2: the field groupoid $\mathcal{F}_{G_{g,N}}(\mathcal{B}_g^{(3)})$ is a finite groupoid for each N , the action $S_{\alpha_{g,N}}^{\text{Ar}}$ is the same Artin–Verdier pairing, and the gluing formula follows from étale descent of rank- $2g$ local systems. The profinite limit $\text{ACS}^{\text{qu},g}$ shares the profinite-renormalisation residual of Chapter 81.4.7.

PROPOSITION 88.2.3 (*partition function at genus g*). (on the genus- g extension of Chapter 81.4.7(2) and the genus- g BD/CG comparison of Chapter 81.5.5). The partition function of $\text{ACS}_{k^{\text{Ar},g}}^{\text{qu},g}$ on $\mathcal{B}_g^{(3)}$ equals the completed L -function

$$Z_{\text{ACS}^{\text{qu},g}}(\mathcal{B}_g^{(3)}; k^{\text{Ar},g}) = \Lambda_{\mathcal{D}_g}(k^{\text{Ar},g}),$$

with $\Lambda_{\mathcal{D}_g}$ the row-specific completed L -function of Chapter 80.1 (Dedekind ξ_K , Hasse–Weil $\Lambda(s, X_K)$, or Siegel spin $\Lambda(s, \pi, \text{spin})$).

Proof. The rank- $2g$ ordered incidence bar $B_{\bullet}^{\text{ord},g}$ on $\mathbb{P} \cup \{\infty\}$ with per-place factor $\mathcal{H}^{(p),g}$ is the rank- $2g$ generalisation of the bar of Chapter 81.5.3. The local character at p is $\chi_p^{(g)}(s) = \prod_{j=1}^{2g} (1 - \alpha_j^{(p)} p^{-s})^{-1}$, the local Euler factor of $\Lambda_{\mathcal{D}_g}$ (item (2) of Chapter 88.1); the archimedean character is the rank- g Γ -factor. The genus- g extension of Chapter 83.3.1 gives $H_{\bullet}(B_{\bullet}^{\text{ord},g}) = \Lambda_{\mathcal{D}_g}(s)$. Under the conditional genus- g BD/CG comparison and the renormalised profinite limit, the bulk partition function coincides with this homology evaluated at $s = k^{\text{Ar},g}$. $\square$

88.3 WILSON-LINE FROBENIUS IDENTITY AT GENUS g

The gauge group $G_{g,\infty} = GL_{2g}(\widehat{\mathbb{Z}})$ of $\text{ACS}^{\text{qu},g}$ contains the prime knots K_p as defect insertions. Each K_p carries a holonomy valued in the rank- $2g$ local Galois representation $\rho_{g,p}$; the question is whether the expectation of the associated defect operator recovers the arithmetic Frobenius Frob_p . The answer at rank $2g$ is the following identity.

DEFINITION 88.3.1 (*genus- g Wilson line*). . Fix a prime p of good reduction for $\mathcal{D}_g$; let $K_p \subset \mathcal{B}_g^{(3)}$ denote the prime-knot associated to p under the Morishita dictionary. For a continuous representation $\rho_g : G_{\mathbb{Q}_p} \rightarrow \text{GSp}_{2g}(\overline{\mathbb{Q}_\ell})$ of rank $2g$, the *genus- g Wilson line* along K_p is the defect operator

$$W_{\rho_g}^{(g)}(K_p) := \text{tr}_{\rho_g}(\text{Hol}_{K_p}(\mathbf{A})),$$

with $\mathbf{A} \in \mathcal{F}_{G_{g,\infty}}(\mathcal{B}_g^{(3)})$ the bulk rank- $2g$ gauge field in the profinite limit and Hol_{K_p} the path-ordered holonomy around the meridian K_p , whose finite local factors are the rank- $2g$ Bost–Connes Adams operations $\psi_p^{(g)}$.

THEOREM 88.3.2 (*genus- g Wilson-line Frobenius identity*). (on the rank- $2g$ extension of Chapter 81.4.7(3) and on rank- $2g$ local Langlands at unramified p of good reduction for $\mathcal{D}_g$, per the genus- g Fargues–Scholze arXiv:2102.13459 input Chapter 80.6(5)). For every unramified prime p of good reduction for $\mathcal{D}_g$,

$$\langle W_{\rho_g}^{(g)}(K_p) \rangle_{\text{ACS}^{\text{qu},g}} = \text{tr}_{\rho_g}(\text{Frob}_p) = \sum_{j=1}^{2g} \alpha_j^{(p)}. \quad (88.1)$$

Equivalently, the Wilson-line defect expectation equals the character value $\chi_{\rho_g}(\text{Frob}_p)$ in the standard representation.

Proof. Three steps.

Step 1: Morishita-dictionary reduction. The prime-knot $K_p \subset \mathcal{B}_g^{(3)}$ corresponds, under the Mazur–Morishita dictionary (Mazur 1973 ENS 6; Morishita 2012 arXiv:0904.3399), to the inertia-generated meridian around the closed point $\mathfrak{p}$ above p . Restriction of the rank- $2g$ gauge field $\mathbf{A}$ to a tubular

neighbourhood of K_p is determined by the local Galois representation $\rho_{g,p}$ via the rank- $2g$ generalisation of Chapter 81.4.6: at finite level $G_{g,N}$, the defect expectation equals $\chi_{\rho_{g \bmod N}}(\text{Frob}_p)$ for the induced mod- N representation.

Step 2: Profinite assembly. The compatible tower $\{\chi_{\rho_{g \bmod N}}(\text{Frob}_p)\}_N$ assembles under the renormalised profinite limit of Chapter 81.4.7 to the continuous Frobenius trace $\text{tr}_{\rho_g}(\text{Frob}_p)$. This requires the rank- $2g$ refinement of the Wilson-line axiom of Chapter 81.4.7(3), which is the residual hypothesis; it is the same assumption that supplies Chapter 82.5.2 at $g = 0$.

Step 3: Bost–Connes Adams identification. Borger 2009 arXiv:0906.3146 §8 identifies each ψ_p with the classical Frobenius on the λ -ring envelope of the Tate motive; the rank- $2g$ generalisation replaces the scalar action on $\mathcal{H}_p^{\text{BC}}$ by the $2g$ -fold generating-series action with Satake parameters $\{\alpha_j^{(p)}\}_{j=1}^{2g}$. Under the unramified hypothesis, $\text{tr}_{\rho_g}(\text{Frob}_p) = \sum_{j=1}^{2g} \alpha_j^{(p)}$ by definition of the Satake parameters. Combining with Step 1–2 gives (88.1). $\square$

Remark 88.3.3 (independence of sources in the Wilson-line Frobenius identity). The three inputs of Chapter 88.3.2 are independent. The rank- $2g$ Bost–Connes Adams action of Borger 2009 §8 derives from absolute λ -ring theory; the rank- $2g$ local Langlands of Fargues–Scholze 2021 arXiv:2102.13459 from p -adic geometric Langlands; the Morishita knot-prime dictionary from Mazur 1973 plus Morishita 2012 arithmetic topology. The three source families are pairwise disjoint.

The Satake parameter normalisation is: unramified $\rho_{g,p}$ has eigenvalues $\alpha_j^{(p)}$ with symplectic pairing $\alpha_j^{(p)} \alpha_{2g+1-j}^{(p)} = p^{w_{\mathcal{D}_g} - 1}$; the Frobenius trace is $\sum_{j=1}^{2g} \alpha_j^{(p)}$, the character value of ρ_g itself at Frob_p , not the character of $\wedge^k \rho_g$ for any $k \geq 2$.

The Wilson-line defect at finite level lies in $\text{Sh}_{\tau_\Sigma(\alpha_{g,N})}(\mathcal{F}_{G_{g,N}}(\Sigma))$ for $\Sigma = \partial(\text{tubular neighbourhood of } K_p)$ per Chapter 81.4.4; the rank- $2g$ setting is structurally identical. The Wilson line is functorial under tensor product of representations at the classical level: $W_{\rho_1 \otimes \rho_2}^{(g)} = W_{\rho_1}^{(g)} \cdot W_{\rho_2}^{(g)}$; at the quantum level this uses the factorisation algebra structure of $\mathcal{V}^{\text{prim},g}$ (rank- $2g$ Heisenberg), consistent with Ch. 80 Window V.

The residual hypotheses are: (a) profinite renormalisation of $\{Z_{G_{g,N}, \alpha_{g,N}}^{\text{Ar}}\}_N$ and (b) compatible tower $\{\alpha_{g,N}\}_N$, inherited from Chapter 81.9(2); and (c) rank- $2g$ local Langlands at p , i.e. Chapter 80.6(5). Identification of the bulk holonomy $\text{Hol}_{K_p}(\mathbf{A})$ with $\rho_{g,p}$ passes through the Artin–Verdier fundamental-class pairing of Chapter 81.4.1, unchanged at rank $2g$; the good-reduction hypothesis excludes wild ramification and singular fibres.

88.4 ARCHIMEDEAN WILSON-SURFACE AT GENUS g

At the archimedean place, the 3-space $\mathcal{B}_g^{(3)}$ has a closed-surface boundary $\Sigma_\infty = \mathcal{B}_g^{(3)}|_\infty$ of dimension $2g$ (a bundle of closed surfaces in Window I), rather than a knot. A defect operator on Σ_∞ must integrate an area form rather than a holonomy along a loop: the Faltings 1984 arithmetic Kähler form $\mu_{\text{Ar},g}$ on the archimedean fibre is the natural input, since it encodes the metric data of $\mathcal{D}_g$ at the archimedean place and degenerates to zero at $g = 0$, matching the trivial archimedean boundary of the Riemann ξ -function framework.

Definition 88.4.1 (genus- g archimedean Arakelov Kähler form). The genus- g Arakelov Kähler form on the archimedean fibre $\mathcal{B}_g^{(3)}|_\infty$ is

$$\mu_{\text{Ar},g} := \frac{i}{2g} \sum_{j=1}^g \omega_j \wedge \overline{\omega_j}, \quad (88.2)$$

with $\{\omega_j\}_{j=1}^g$ a basis of holomorphic differentials on the archimedean fibre $X_\infty(\mathbb{C}) \hookrightarrow \mathcal{B}_g^{(3)}|_\infty$ (for Window I) or g copies of the archimedean Haar form on the Minkowski torus T_K (for Window II) or the natural Riemannian volume form scaled per Siegel-modular signature (for Window VI). This is the Faltings 1984 arithmetic Kähler form (Faltings 1984 *Ann. Math.* 119) on the generic fibre; at $g = 0$ the sum is empty and $\mu_{\text{Ar},0} = 0$.

Definition 88.4.2 (archimedean Wilson-surface at genus g). The genus- g archimedean Wilson-surface is the bulk area-scaling defect

$$F_\infty^{(g)} := \exp\left(-k^{\text{Ar},g} \int_{\Sigma_\infty} \mu_{\text{Ar},g}\right), \quad (88.3)$$

where $\Sigma_\infty \subset \mathcal{B}_g^{(3)}$ is the archimedean fibre $\mathcal{B}_g^{(3)}|_\infty$ viewed as a closed 2-surface with the Faltings Kähler form of Chapter 88.4.1. In operator-theoretic form on $\mathcal{H}^{(\infty),g}$,

$$F_\infty^{(g)} = \exp\left(-k^{\text{Ar},g} \sum_{j=1}^g \tilde{L}_{0,j}^{\text{Ar}}\right), \quad (88.4)$$

for $\tilde{L}_{0,j}^{\text{Ar}}$ the centred Fock scaling zero-mode on the j -th archimedean Heisenberg factor $\mathcal{H}_j^{(\infty),g}$. The uncentred scaling generator has Mellin eigenvalue s_j ; the tilde subtracts the critical-line centre $w_{\mathcal{D}_g}/2$.

THEOREM 88.4.3 (conjugacy to genus- g Virasoro). (on the rank- g extension of the primary-Virasoro reconciliation Chapter 22.3.5 per Chapter 80.6(7), and on the rank- g Arakelov–Mellin unitarity of Chapter 88.4.4 below). The genus- g archimedean Wilson-surface $F_\infty^{(g)}$ is conjugate to the genus- g arithmetic Virasoro zero-mode $L_0^{(g)}$ with central charge $c^{\text{Ar},g} = g$:

$$F_\infty^{(g)} = U^{(g)} \exp(-L_0^{(g)}) (U^{(g)})^{-1},$$

with $U^{(g)}$ the rank- g Mellin-transform unitary of Chapter 88.4.4. Equivalently, the Sugawara construction on the rank-2 g archimedean Heisenberg produces $L_0^{(g)}$ as the arithmetic Virasoro zero-mode sourced by the rank-2 g archimedean Fock; the Arakelov area-scaling $F_\infty^{(g)}$ is its Mellin conjugate.

LEMMA 88.4.4 (rank- g Arakelov–Mellin unitarity). (on the rank- g Weil–Petersson isometry lift of Ch. 56 restricted to the archimedean fibre; genus- g Petersson inner-product of Chapter 80.6(2)). There exists a unitary $U^{(g)} : L^2(\mathbb{R}_+^g, \prod_{j=1}^g d^\times x_j) \rightarrow L^2(\text{Re } s_j = w_{\mathcal{D}_g}/2, j = 1, \dots, g)$ defined by the g -fold tensor of Mellin transforms

$$U^{(g)} f(s_1, \dots, s_g) = \int_{\mathbb{R}_+^g} f(x_1, \dots, x_g) \prod_{j=1}^g x_j^{s_j} \prod_{j=1}^g d^\times x_j,$$

conjugating the centred rank- g Arakelov area-scaling flow $\partial_{t_j}^\circ = \partial/\partial \log \|\cdot\|_{\infty,j} - w_{\mathcal{D}_g}/2$ to multiplication by $s_j - w_{\mathcal{D}_g}/2$ on the critical poly-line.

Proof. The Faltings Kähler form (88.2) decomposes as a sum of g orthogonal rank-1 pieces under the Hodge decomposition $\mathcal{E}_g \otimes \mathbb{C} = \mathcal{E}_g^{1,0} \oplus \mathcal{E}_g^{0,1}$; each piece carries a rank-1 Arakelov scaling flow $t_j \mapsto e^{-t_j} \|\cdot\|_{\infty,j}$ with Mellin variable s_j . The rank-1 unitary is Mellin : $L^2(\mathbb{R}_+, d^\times x) \rightarrow L^2(\operatorname{Re} s = w_{\mathcal{D}_g}/2)$, classical (Tate 1950; Titchmarsh 1986 Chapter 14); the g -fold tensor is unitary by Fubini on the product measure. The scaling flow is diagonalised by Mellin at the centre $\operatorname{Re} s_j = w_{\mathcal{D}_g}/2$ to match Chapter 88.1(4). $\square$

Proof of Chapter 88.4.3. Under $U^{(g)}$, the archimedean Wilson-surface (88.4) becomes $\exp(-k^{\operatorname{Ar},g} \sum_{j=1}^g (s_j - w_{\mathcal{D}_g}/2))$ on the product Mellin space; this coincides with $\exp(-L_o^{(g)})$ where $L_o^{(g)} = \sum_{j=1}^g \tilde{L}_{o,j}^{\operatorname{Ar}}$ is the rank- g arithmetic Virasoro zero-mode on the product archimedean Heisenberg Fock. The Sugawara construction of Chapter 81.3.1 scales rank-linearly per Ch. 80 (central charge $c^{\operatorname{Ar},g} = g$); the rank- g primary sector admits a Virasoro zero-mode which is the natural lift of $\tilde{L}_{o,j}^{\operatorname{Ar}}$ on each archimedean Heisenberg factor. Summing over factors gives the rank- g zero-mode $L_o^{(g)}$; this is the Chapter 22.3.5 reconciliation at rank g . $\square$

Remark 88.4.5 (Wilson-surface realisation as bulk ∂_k -derivative). In the 3D ACS framework, the archimedean Wilson-surface is the bulk partition-function derivative along the arithmetic-level deformation restricted to the archimedean fibre: $F_\infty^{(g)} = \partial Z_{\operatorname{ACS}^{\operatorname{qu},g}} / \partial k^{\operatorname{Ar},g} |_{\operatorname{archimedean}}$, by the area-generation scheme of Chapter 82.5.7 scaled to rank $2g$. This is the arithmetic analog of the Witten-Verlinde area operator on complex Riemann surfaces.

Remark 88.4.6 (independence of sources and sign verification for the archimedean Wilson-surface). The three primary inputs are independent: Faltings 1984 Arakelov Kähler form (Arakelov geometry), Chapter 22.3.5 (primary Virasoro sector, chiral algebra methods), and Tate 1950 Mellin unitarity (classical analysis). The sign in $F_\infty^{(g)} = \exp(-k^{\operatorname{Ar},g} \int \mu_{\operatorname{Ar},g})$ matches the Mellin convention $\partial_t \leftrightarrow -s + w_{\mathcal{D}_g}/2$ on the critical line; the Faltings normalisation (88.2) with prefactor $(2g)^{-1}$ ensures the total area integrates to g times the area per factor.

The Wilson-surface lives in bulk codimension-1 defects of $\operatorname{ACS}^{\operatorname{qu},g}$; at finite level in $\operatorname{Sh}_\tau(\alpha_{g,N})(\mathcal{F}_{G_{g,N}}(\Sigma_\infty))$ restricted to the archimedean surface. The Wilson-surface is the tensor product of g rank-1 Wilson-surfaces when the Faltings Kähler form splits as a direct sum: this holds for Jacobians (Window I) and Minkowski tori (Window II) by Hodge decomposition, but Window VI requires the Siegel–Maass metric on $\mathcal{A}_g$ (residual 2 of this chapter).

The residual hypotheses are: rank- g primary-Virasoro reconciliation (Chapter 80.6(7)); rank- g Weil–Petersson isometry lift (Chapter 80.6(2)); rank- g Mellin unitarity (Chapter 88.4.4), which is conditioned on the Weil–Petersson lift.

88.5 GLOBAL FROBENIUS AT GENUS g AND THE DENINGER IDENTIFICATION

The Wilson-line and Wilson-surface operators at each place are local; to obtain the Deninger character identity $\chi(\mathcal{V}^{\operatorname{prim},g})(s) = \Lambda_{\mathcal{D}_g}(s)$, one must assemble them into a single global operator on the Deninger cohomology $H_{\Delta,g}^\bullet$. The restricted tensor product over all places is the global Frobenius.

Definition 88.5.1 (global genus- g Frobenius). The global genus- g Frobenius on $H_{\Delta,g}^\bullet(\mathcal{B}_g)$ is the restricted-tensor product

$$F^{(g)} := F_\infty^{(g)} \otimes \bigotimes_p' \psi_p^{(g)},$$

acting on $\mathcal{V}_{\text{Heis}}^{\text{prim},g} = \mathcal{H}^{(\infty),g} \otimes \otimes'_p \mathcal{H}^{(p),g}$ and descending to the genus- g Vol IV Deninger cohomology $H_{\Delta,g}^\bullet$ of Chapter 82.9.1 via the rank- $2g$ bar construction of Chapter 82.9.2.

THEOREM 88.5.2 (*genus- g Wilson assembly of the global Frobenius*). (on Chapter 88.3.2, Chapter 88.4.3, and the Tate-restricted-tensor compatibility F2.c at rank $2g$, per Chapter 21.3 scaled). The global Frobenius $F^{(g)}$ admits bulk Wilson realisations:

- (i) *Finite places*. $\psi_p^{(g)}$ is realised as the genus- g Wilson-line observable along the knot K_p :
 $\text{tr}_{\rho_g}(\psi_p^{(g)}) = \langle W_{\rho_g}^{(g)}(K_p) \rangle_{\text{ACS}^{\text{qu},g}} = \text{tr}_{\rho_g}(\text{Frob}_p)$.
- (ii) *Archimedean place*. $F_\infty^{(g)}$ is realised as the archimedean Wilson-surface of Chapter 88.4.2, conjugate to $\exp(-L_o^{(g)})$ with central charge $c^{\text{Ar},g} = g$.

Proof. Item (1) follows from Chapter 88.3.2 applied per-prime, assembled under Tate restricted-tensor compatibility (F2.c) across almost-all primes. At genus g , F2.c scales to rank $2g$ without structural change: the spherical unit $|o\rangle_p$ of the rank- $2g$ local Heisenberg is a unique $\psi_p^{(g)}$ -eigenvector up to normalisation $p^{-w_{\mathcal{D}_g}/2}$.

Item (2) is Chapter 88.4.3.

The global descent to $H_{\Delta,g}^\bullet$ uses the rank- $2g$ bar complex of Chapter 82.9.1, invariant under each $\psi_p^{(g)}$ by multiplicativity and under $F_\infty^{(g)}$ by the graded-Virasoro action. $\square$

THEOREM 88.5.3 (*all-genus Deninger identification at motivic weight $w_{\mathcal{D}_g}$*). (on Chapter 82.9.2(1)–(4), Chapter 88.3.2, Chapter 88.4.3, and Chapter 88.2.3). For every $g \geq 0$ and every arithmetic genus- g datum $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \varphi_g)$:

- (i) *Triple labelling*. The Vol IV genus- g Deninger cohomology $H_{\Delta,g}^\bullet$, the boundary chiral homology of $\text{ACS}_{k^{\text{Ar},g}}^{\text{qu},g}$ on $\mathcal{B}_g^{(3)}$, and the rank- $2g$ arithmetic chiral homology $H_{\text{ch}}^\bullet(\mathcal{B}_g^{(3)}, \text{H}^{\text{Ar},g})$ (with $\text{H}^{\text{Ar},g}$ the arithmetic Heisenberg coefficient sheaf of Chapter 82.9.1) coincide on the primary rank- $2g$ Heisenberg domain.
- (ii) *Character identity with motivic weight*.

$$\chi(\mathcal{V}^{\text{prim},g})(s) = \det_\infty(1 - e^{-s} F^{(g)} | H_{\Delta,g}^\bullet)^{(-1)\bullet} = \Lambda_{\mathcal{D}_g}(s), \quad (88.5)$$

where $\Lambda_{\mathcal{D}_g}$ has motivic weight $w_{\mathcal{D}_g} \in \{1, 2, 2g + 1\}$ per Chapter 80.1.

- (iii) *Wilson-observable form of the Frobenius*. The local factors of $F^{(g)}$ appearing in (88.5) admit the Wilson-line / Wilson-surface realisation of Chapter 88.5.2.
- (iv) *Purity at $\text{Re}(s) = w_{\mathcal{D}_g}/2$* . The genus- g purity axiom of Chapter 82.1(6) becomes $F^{(g)}$ -weight $w_{\mathcal{D}_g}/2$ on $H_{\Delta,g}^1$; equivalently the motivic Riemann Hypothesis for the row $\mathcal{D}_g$ (Chapter 80.5.1).

Proof. Item (1) is Chapter 82.9.2(1) unchanged, scaled to rank $2g$.

Item (2): the regularised determinant $\det_\infty(1 - e^{-s} F^{(g)} | H_{\Delta,g}^\bullet)^{(-1)\bullet}$ decomposes across Fock degrees: $H_{\Delta,g}^0 \cong \mathbb{C}_o$ contributes $1 - e^{-s}$; $H_{\Delta,g}^2 \cong \mathbb{C}_{w_{\mathcal{D}_g}}$ (Tate twist by the motivic weight) contributes $1 - e^{w_{\mathcal{D}_g} - s}$; $H_{\Delta,g}^1$ carries the regularised product $\prod_\rho (1 - e^{\rho - s})$ over the non-trivial zeros of $\Lambda_{\mathcal{D}_g}$. Multiplying and normalising by the $\pi^{-g s/2} \Gamma_{\mathbb{C}}(s)^g$ absorbed in the definition of $F_\infty^{(g)}$ gives $\Lambda_{\mathcal{D}_g}(s)$ per Chapter 82.9.2(3).

Item (3) is Chapter 88.5.2.

Item (4) is the rank- g analog of Chapter 82.8.1 via Chapter 80.5.1: weight- $w_{\mathcal{D}_g}/2$ purity is the statement that all non-trivial zeros of $\Lambda_{\mathcal{D}_g}$ lie on $\operatorname{Re}(s) = w_{\mathcal{D}_g}/2$, subsuming the classical RH at $g = 0$, Hasse–Weil at weight 2, and Siegel-spin at weight $2g + 1$. $\square$

Remark 88.5.4 (source independence and residuals for the all-genus Deninger identification). The five primary inputs are pairwise independent: Deninger 1998 (foliated dynamical systems), Bost–Connes 1995 (KMS partition on the idèle class group), Fargues–Scholze 2021 (rank- $2g$ local Langlands), Faltings 1984 (Arakelov Kähler form), Arthur 2013 (endoscopic classification for Siegel spin). The motivic-weight normalisation is: Dedekind $w = 1$, Hasse–Weil $w = 2$, Siegel spin $w = 2g + 1$, per Chapter 80.1; the alternating-sign Euler product $\det(\cdot)^{(-1)^\bullet}$ follows Deninger’s determinant convention; the central charge g of the Sugawara construction matches the archimedean rank.

The rank- $2g$ bar $B^{\text{Ar},g}(\mathcal{V}^{\text{prim},g})$ lives in graded $\mathbb{C}$ -vector spaces with rank- $2g$ Heisenberg Fock grading; the E_1 -ordered bar on $\mathbb{P} \cup \{\infty\}$ is rank-enhanced per Chapter 88.2.3. Functoriality under pullback along $\operatorname{Spec} \mathcal{O}_K \rightarrow \operatorname{Spec} \mathbb{Z} \rightarrow \operatorname{Spec} \mathbb{Q}$ requires $g = \frac{1}{2}[K : \mathbb{Q}]$ at totally imaginary K (Window II); for Shimura (Window VI), functoriality under Hecke correspondences holds because $F^{(g)}$ commutes with Hecke operators on the rank- $2g$ automorphic side.

The full residual list consists of all items of Chapter 80.6 (rank- g Positselski, rank- g Koszul–Petersson, GL_{2g} local geometric Langlands, rank- g Virasoro bootstrap, rank- g F2.c) and all items of Chapter 81.9 (profinite ACS renormalisation, boundary comparison, BD/CG comparison, positivity), each scaled to rank $2g$. The identification of the $H_{\Delta,g}^1$ spectrum with the non-trivial zeros of $\Lambda_{\mathcal{D}_g}$ requires rank- g H-P^{Ar}: a self-adjoint lift of $L_o^{(g)}$ to a Hilbert completion of $H_{\Delta,g}^1$ in the rank- g arithmetic Petersson inner product (Chapter 22.3.2 at rank g); this is open.

88.6 UNIFORM $g = 0$ COMPATIBILITY

At $g = 0$, the constructions of Sections 88.2–88.5 degenerate to the rank-zero skeleton inside the Riemann framework of Chs. 81 and 82. The non-trivial rank-two Wilson sectors are compared, not derived, below.

- *Archimedean Wilson-surface.* At $g = 0$, $\mu_{\text{Ar},0} = 0$ (empty Hodge sum); the Wilson-surface (88.3) degenerates to the trivial defect $\exp(-k^{\text{Ar}} \cdot 0) = 1$. The non-trivial Arakelov scaling of Chapter 82.5.3 and Chapter 89.2.3 belongs to the rank-one Tate Mellin sector; it is not obtained by substituting $g = 0$ in the rank- g surface operator.
- *Wilson-line Frobenius.* At $g = 0$, ρ_o is the empty rank-0 datum, so the genus- g Wilson-line identity has no finite local trace. The identity $\langle W_\rho(K_p) \rangle = \operatorname{tr} \rho(\operatorname{Frob}_p)$ of Chapter 81.4.7(3) and Chapter 84.5.1 uses an independent continuous GL_2 -label ρ ; it is compatible with, but not a specialisation of, (88.1).
- *All-genus Deninger identification.* At $g = 0$, $\Lambda_{\mathcal{D}_0}(s) = \Lambda_{\mathcal{Z}}(s)$, and the pole-cleared determinant is $\xi_{\mathbb{R}}(s) = \frac{1}{2}s(s-1)\Lambda_{\mathcal{Z}}(s)$. The motivic weight is $w_{\mathcal{D}_0} = 1$; the character identity (88.5) collapses to the $g = 0$ identity (82.2) of Chapter 82.6.1; the purity locus is $\operatorname{Re}(s) = 1/2$, the Riemann critical line.

COROLLARY 88.6.1 (uniform $g = 0$ compatibility). The genus- g constructions of Sections 88.2, 88.3, 88.4 and 88.5 degenerate at $g = 0$ to the rank-zero Tate skeleton. The rank-two constructions of Chs. 81, 82, 84, and 89 are separate compatible inputs: Wilson-line Frobenius Chapter 81.4.7(3) and Chapter 84.5.1; archimedean Wilson-surface Chapter 82.5.3 and Chapter 89.2.3 with conjugacy Chapter 89.4.3; Deninger character identity Chapter 82.6.1. No rank-two Wilson-line label is produced by the rank-zero degeneration.

Proof. Each item in the bulleted list above separates the rank-zero degeneration from the independent rank-two ACS input. No analytic correction is required for the degeneration: the rank- $2g$ Fock is empty, the Hodge decomposition $\mathcal{E}_0 \otimes \mathbb{C} = 0$, and the motivic weight is $w_{\mathcal{D}_0} = 1$. $\square$

Remark 88.6.2 (Hasse–Weil and Siegel–spin RH at genus $g \geq 1$). At $g \geq 1$, the Deninger–chiral purity locus $\text{Re}(s) = w_{\mathcal{D}_g}/2$ is the motivic Riemann Hypothesis for the row:

- $w_{\mathcal{D}_g} = 1$ (Dedekind, Window II at totally imaginary K of degree $2g$): the generalised Riemann Hypothesis for $\zeta_K(s)$.
- $w_{\mathcal{D}_g} = 2$ (Hasse–Weil, Windows I, V at the Jacobian of a curve of genus g): the Hasse–Weil Riemann Hypothesis on the weight-1 motive $H^1(\text{Jac}(X_K))$, critical line $\text{Re}(s) = 1$.
- $w_{\mathcal{D}_g} = 2g + 1$ (Siegel spin, Window VI at cusp form π on GSp_{2g}): the Siegel–spin Riemann Hypothesis, critical line $\text{Re}(s) = g + 1/2$.

The Wilson-line / Wilson-surface realisation converts each case to a bulk topological-gauge-theoretic spectral statement on the partition-function zeros of $\text{ACS}_{k^{\text{Ar}},g}^{\text{qu},g}$ on the corresponding arithmetic 3-space $\mathcal{B}_g^{(3)}$.

88.7 RESIDUAL CONDITIONS

The genus- g extension inherits residuals from Ch. 80 and Ch. 81 scaled to rank $2g$, and four residuals are specific to this chapter.

- Rank- $2g$ local Langlands compatibility at ramified primes.* Chapter 88.3.2 holds at unramified primes of good reduction for $\mathcal{D}_g$. At ramified primes, the identity requires a local twist term depending on the Artin conductor and the Fargues–Scholze / Genestier–Lafforgue correspondence for GSp_{2g} . It is a non-formal theorem, extending Chapter 80.6(5).
- Rank- g Faltings Kähler splitting for Shimura Window VI.* The archimedean Wilson-surface (88.3) presumes the Faltings Kähler form splits as a direct sum of g orthogonal rank-1 pieces; this holds for Jacobians (Window I) and Minkowski tori (Window II) by Hodge decomposition, but requires the Siegel–Maass metric on $\mathcal{A}_g$ (Kudla 1997 *Ann. Math.* 146) for Window VI. It is a non-formal theorem.
- Rank- g Bost–Connes Adams compatibility.* Borger 2009 §8’s identification of ψ_p with the classical Frobenius on the Tate motive scales to rank $2g$ via the λ -ring structure on the rank- $2g$ arithmetic Heisenberg. Formally immediate at each finite prime; the restricted-tensor compatibility across almost all primes is F2.c at rank $2g$, inheriting Chapter 80.6(6). It is a non-formal theorem.
- Rank- g $H\text{-}P^{\text{Ar}}$.* Self-adjoint lift of $L_o^{(g)}$ to a Hilbert completion of $H_{\Delta,g}^1$ in the rank- g arithmetic Petersson inner product. This is the direct rank- g analog of Chapter 22.3.2. Domain: inherited from $H\text{-}P^{\text{Ar}}$ plus Chapter 80.6(2)–(7). The residual is the motivic Riemann Hypothesis in Hilbert–Pólya form.

Remark 88.7.1 (uniform rank structure). The genus enters through the rank of the gauge group, the archimedean Fock rank, and the motivic weight; the Koszul self-duality identity $\kappa^{\text{Ar},g} + (\kappa^1)^{\text{Ar},g} = 0$ is fixed at all genera by Chapter 80.4.1; the bulk partition function is the row-specific completed L -function $\Lambda_{\mathcal{D}_g}$; and the motivic Riemann Hypothesis for the row becomes the topological spectral condition $\text{Re}(z) = w_{\mathcal{D}_g}/2$ on the non-trivial zeros z of $\Lambda_{\mathcal{D}_g}$, i.e. the $F^{(g)}$ -weight purity on $H_{\Delta,g}^1$ (Chapter 88.5.3(4)).

88.8 PRIMARY SOURCES FOR THE GENUS- g EXTENSION

Seven primary sources underpin the genus- g extension, each disjoint from the others in the sense of Chapter 80.1.1.

- *Faltings 1984 Calculus on arithmetic surfaces*, *Ann. Math.* 119, 387–424: Arakelov Kähler form $\mu_{\text{Ar},g}$ of Chapter 88.4.1; Faltings–Hriljac arithmetic Hodge index; Faltings δ -invariant.
- *Fargues–Scholze 2021* arXiv:2102.13459: rank- $2g$ local Langlands for $GL_{2g}/\mathbb{Q}_p$; *Zhu 2020* arXiv:2008.02998 and *Genestier–Lafforgue 2017* arXiv:1709.00978 extend to GSp_{2g} .
- *Bhatt–Scholze 2022 Ann. Math.* 196: prismatic Frobenius on rank- $2g$ F -crystals; Chapter 80.2 Window III input.
- *Borger 2009* arXiv:0906.3146 §8: Bost–Connes Adams ψ_p as classical Frobenius; rank- $2g$ scaling via the λ -ring structure on the Tate motive.
- *Arthur 2013 The Endoscopic Classification*: Siegel spin L -functions and functional equations for GSp_{2g} -automorphic cuspids.
- *Kudla 1997 Ann. Math.* 146; *Kudla–Rapoport 2011*: Siegel–Maass metric on $\mathcal{A}_g$ and rank- g Faltings splitting for Window VI.
- *Morishita 2012 Knots and Primes* arXiv:0904.3399; *Mazur 1973* ENS 6: knot-prime dictionary for the Wilson-line identification at finite primes; Artin–Verdier fundamental class for the arithmetic 3-space.

Pairwise disjointness of these sources, as in the independent-source list of Chapter 80.7, is preserved at rank $2g$ by the same inheritance argument: the rank-change is a scaling of the per-place Heisenberg oscillators, not a source substitution.

The residuals of Chapter 88.7 together with those of Chs. 80, 81, and 82 constitute the full condition list for a theorem-grade proof of the genus- g Wilson-line, Wilson-surface, and Deninger identification. Four conditions remain open: (a) rank- $2g$ profinite ACS renormalisation, (b) rank- $2g$ BD/CG comparison, (c) rank- g Weil–Petersson isometry, (d) rank- g H-P^{Ar} . Each is recorded as a conditional input in its natural chapter of origin.

THE ARCHIMEDEAN WILSON-SURFACE OPERATOR F_∞

Chapter 82 identified the archimedean component F_∞ of the global Frobenius $F = F_\infty \otimes \otimes'_p \psi_p$ in three guises: (i) the Arakelov scaling exponential $F_\infty = \exp(-\Theta_\infty)$, where $\Theta_\infty = \frac{1}{2}\text{id} + i\widehat{D}$ and $\widehat{D} = -i\partial_t$ is the self-adjoint multiplicative scaling generator of Chapter 89.1.2; (ii) the bulk area-scaling Wilson-surface operator at ∞ inside the conjectural 3D arithmetic Chern-Simons of Chapter 81; (iii) the Mellin-conjugate of the arithmetic Virasoro zero-mode L_0^{Ar} under the archimedean Zhu correspondence of Chapter 22. A rigorous construction is given and the three guises are proved to coincide. The construction of (i) is elementary and unconditional. The realisation (ii) is a quantum-ACS observable supplied conditionally on the level- k^{Ar} bulk measure of Chapter 81.4.7. The conjugacy (iii) with L_0^{Ar} is a theorem on the archimedean Fock fibre. The determinant-divisor identification then follows: under Chapter 22.3.2, the Tate-shifted Mellin image of L_0^{Ar} on H_{ch}^1 has divisor $\text{div } \xi_{\mathbb{R}}$, while the determinant divisor of F_∞ is its exponential pushforward. Archimedean purity for F_∞ on H_{ch}^1 is equivalent to the A-TP positivity of Chapter 56, and the latter is equivalent to RH.

89.1 ARAKELOV ARCHIMEDEAN STRUCTURE

The archimedean fibre of $\overline{\text{Spec } \mathbb{Z}}$ is the compactifying place at ∞ , where $\mathbb{Z} \hookrightarrow \mathbb{R}$ is replaced by the Arakelov datum: the archimedean completion $\mathbb{Q}_\infty = \mathbb{R}$, the normalised absolute value $|x|_\infty = |x|$, and a choice of smooth Green metric on $\mathbb{P}_{\mathbb{C}}^1 = \mathbb{C} \cup \{\infty\}$ when the arithmetic surface is projective. The archimedean fibre relevant to the arithmetic bar-character programme is the complex plane $\mathbb{C}_\infty = \mathbb{C}$, equipped with the standard Lebesgue Haar measure $d^2x = dx dy$ on the additive side and the scale-invariant Haar measure $d^\times \lambda = d\lambda/\lambda$ on $\mathbb{R}_+^\times$ on the multiplicative side.

Definition 89.1.1 (Arakelov scaling action). Set $\mathbb{R}_+^\times$ with coordinate λ . The *Arakelov scaling action* on the archimedean fibre is

$$\rho_\lambda : (\mathbb{C}_\infty, |\cdot|_\infty) \longrightarrow (\mathbb{C}_\infty, |\cdot|_\infty), \quad x \longmapsto \lambda \cdot x,$$

together with the induced action on the multiplicative Haar measure $\rho_\lambda^*(d^\times |x|) = d^\times |x|$. On $L^2(\mathbb{R}_+^\times, d^\times x)$ the unitary scaling action is

$$(U_\lambda f)(x) := f(\lambda^{-1}x).$$

The self-adjoint scaling generator is

$$\widehat{D} := -i \frac{d}{dt} \Big|_{t=0} U_{e^t}.$$

The half-density factor $|\lambda|^{-1/2}$ belongs to the additive Lebesgue model $L^2(\mathbb{R}_+, dx)$, not to the multiplicative Haar model.

Remark 89.1.2 (t -coordinate). Put $t = \log \lambda$ so that $\lambda = e^t$ and $d^\times \lambda = dt$; then the scaling action becomes translation $(U_{e^t} f)(u) = f(u - t)$ on $L^2(\mathbb{R}, du)$ under $u = \log |x|$, and the generator is

$$\widehat{D} = -i\partial_u.$$

The Tate half-weight is not part of the unitary translation generator. It enters through the normal operator

$$\Theta_\infty := \frac{1}{2}\text{id} + i\widehat{D},$$

whose Mellin eigenvalue is $s = \frac{1}{2} + i\tau$.

LEMMA 89.1.3 (Mellin unitarity). . The Mellin transform $\mathcal{M}f(\tau) := \int_0^\infty f(x) x^{-i\tau} d^\times x$ is a unitary isomorphism

$$\mathcal{M} : L^2(\mathbb{R}_+^\times, d^\times x) \xrightarrow{\sim} L^2(\mathbb{R}_\tau, \frac{d\tau}{2\pi}),$$

conjugating the multiplicative-scaling generator $\widehat{D}$ to multiplication by τ . After the Tate shift $s = \frac{1}{2} + i\tau$, the normal operator $\Theta_\infty = \frac{1}{2}\text{id} + i\widehat{D}$ is multiplication by s on the critical line.

Proof. Standard: Plancherel for the multiplicative group $\mathbb{R}_+^\times \simeq \mathbb{R}$ via $t = \log x$. The additive Fourier transform on $L^2(\mathbb{R}, dt)$ conjugates $-i\partial_t$ to multiplication by the real frequency variable τ . The equality $s = \frac{1}{2} + i\tau$ is the Tate half-weight normalisation, so $\frac{1}{2}\text{id} + i\widehat{D}$ becomes multiplication by s . $\square$

Definition 89.1.4 (archimedean Arakelov disk). . The *archimedean Arakelov disk* is the 2-dimensional boundary component of $\overline{\text{Spec } \mathbb{Z}}$ at the archimedean place:

$$\Sigma_\infty := \{\infty\} \times D^2 \subset \overline{\text{Spec } \mathbb{Z}},$$

where $D^2 = \{z \in \mathbb{C} : |z| \leq 1\}$ is the closed unit disk, equipped with the Faltings 1984 archimedean Kähler form

$$\omega_\infty := \frac{i}{2\pi} \frac{dz \wedge d\bar{z}}{(1 + |z|^2)^2} \quad (89.1)$$

restricted to D^2 . The disk carries the standard $U(1)$ -framing from its complex structure.

Remark 89.1.5 (role of Σ_∞). In the Morishita dictionary (Morishita 2012 *Knots and Primes*, arXiv:0904.3399, §2) finite primes appear as knots $K_p \subset \overline{\text{Spec } \mathbb{Z}}$. The archimedean place is structurally a closed surface component rather than a knot: compactification at ∞ adds a 2-cell. The Wilson observable attached to ∞ is therefore a Wilson-surface, not a Wilson-line. The 3D CS bulk couples to the surface via its area form (89.1); the choice of Faltings Kähler form encodes the Green-metric data of the Arakelov compactification (Faltings 1984 *Ann. Math.* 119, 387–424).

89.2 WILSON-SURFACE OBSERVABLES IN 3D CHERN-SIMONS

The Wilson-surface observable in 3D gauge theory is the higher categorical analogue of the Wilson-line observable. For a 2D surface $\Sigma \subset M^3$ in a 3-manifold carrying a gauge connection A , a Wilson-surface is specified by a 2-representation of the gauge 2-group and evaluates the exponentiated surface holonomy of A (Baez–Schreiber 2005 arXiv:hep-th/0412325, §4; Schreiber–Sati 2011 arXiv:1011.4735, §2.6). For abelian $U(1)$ gauge theory, a Wilson-surface is the exponentiated integral of the curvature 2-form $F_A = dA$ over Σ , weighted by a real charge q :

$$W_q(\Sigma, A) := \exp\left(iq \int_\Sigma F_A\right). \quad (89.2)$$

3D Chern-Simons and Wilson-Surface Observables (Witten 1989)

Witten 1989 *Commun. Math. Phys.* 121, 351–399 (*Quantum field theory and the Jones polynomial*, originally announced 1988) established 3D Chern-Simons as a topological gauge theory whose boundary WZW admits boundary observables specified by knots (Wilson lines) and by surfaces (Wilson surfaces for higher-form symmetries). The abelian level- k action is

$$S_{\text{CS},k}[A] = \frac{k}{4\pi} \int_{M^3} A \wedge dA,$$

and the path-integral evaluates correlators of Wilson observables. Three structural properties of Wilson-surfaces in this framework are used below.

PROPOSITION 89.2.1 (*structural properties of Wilson-surfaces*). . In 3D abelian Chern-Simons at level k on a 3-manifold M^3 :

- (i) (Framing dependence.) The Wilson-surface observable (89.2) is invariant under smooth isotopies of Σ preserving the framing.
- (ii) (Charge quantization.) Gauge invariance requires $qk \in \mathbb{Z}$ when Σ is closed and M^3 is closed.
- (iii) (Scaling interpretation.) When $\Sigma = \{p\} \times D^2$ sits over a codimension-1 point $p \in M^2 \times \{pt\} \subset M^3$ and A couples as an area-measurement field, the observable (89.2) generates a 1-parameter scaling of the fibre at p .

Proof. Properties (1) and (2) are Witten 1989, §3–§4. Property (3) is the dual statement: the 2-form $F_A|_\Sigma$ integrates to an area measurement on Σ ; exponentiation generates a 1-parameter group of rescalings of that area, realising the scaling flow on the fibre over p . See Cattaneo–Mnev–Reshetikhin 2014 arXiv:1201.0290, §5, for the dual scaling interpretation. $\square$

The Archimedean Wilson-Surface

We apply (Chapter 89.2.1) with $\Sigma = \Sigma_\infty$ of Chapter 89.1.4, $M^3 = \overline{\text{Spec } \mathbb{Z}}$, and $A = A_{k^{\text{Ar}}}$ the level- k^{Ar} arithmetic Chern-Simons connection of Chapter 81.4.7. The area-measurement field on Σ_∞ is the pullback of (89.1).

Definition 89.2.2 (*area 2-representation*). . The *area-scaling 2-representation* σ_{area} is the 1-dimensional abelian 2-representation of the $U(1)$ gauge 2-group defined by the character $\theta \mapsto \exp(i\theta \cdot \text{vol}(\Sigma_\infty, \omega_\infty))$, where $\text{vol}(\Sigma_\infty, \omega_\infty) = \int_{\Sigma_\infty} \omega_\infty$ is the Faltings-normalised Arakelov volume of Σ_∞ .

Definition 89.2.3 (*archimedean Wilson-surface* F_∞^{WS}). . Put

$$F_\infty^{\text{WS}} := W_{\sigma_{\text{area}}}(\Sigma_\infty, A_{k^{\text{Ar}}}) = \exp\left(i \int_{\Sigma_\infty} \omega_\infty \cdot F_{A_{k^{\text{Ar}}}}\right),$$

the bulk Wilson-surface observable of the 3D quantum arithmetic Chern-Simons on $\overline{\text{Spec } \mathbb{Z}}$ with label σ_{area} .

Remark 89.2.4 (*conditional domain*). The existence of the bulk 3D quantum arithmetic Chern-Simons measure at level k^{Ar} is the conjectural content of Chapter 81.4.7. F_∞^{WS} is defined here as an observable of that (conjectural) measure. The construction is unconditional as a structural formula; the well-definedness of the measure against which expectation values are taken is conditional.

Source Comparison for the Wilson-Surface Construction

The three source classes are disjoint: Witten 1989 (*Comm. Math. Phys.* 121) for Wilson observables in 3D Chern-Simons; Baez–Schreiber 2005 for the 2-representation language; Faltings 1984 for the Arakelov Kähler form.

Sign and convention: A is a real connection, $F_A = dA$ is a real 2-form, and the i -exponential gives a unitary observable. The Faltings Kähler form is positive on Σ_∞ so the area integral is positive real, consistent with Chapter 89.1.3.

The observable category is unitary operators on the boundary Hilbert space of $\partial\text{ACS}^{\text{u}}$ at Σ_∞ , which is the archimedean Fock space $\mathcal{H}^{(\infty)} = \mathcal{V}_{\text{Heis},\infty}^{\text{prim}}$ of Ch. 55; F_∞^{WS} lands in $\text{End}(\mathcal{H}^{(\infty)})$.

The bulk measure of Chapter 81.4.7 is conditional; the observable as a structural formula is unconditional. The residual is bulk measure existence.

F_∞^{WS} is functorial in the gauge 2-group along the $U(1)$ -inclusion. Non-abelian gauge groups would add Stokes-boundary corrections (Baez–Schreiber 2005, §5); the abelian case bypasses these. The identification of F_∞^{WS} with the Arakelov scaling derivative is the content of Chapter 89.3.3.

89.3 CONSTRUCTION OF F_∞

We construct F_∞ via Arakelov scaling on the archimedean fibre, independent of the conditional Wilson-surface realisation, and prove that the two constructions agree.

Definition 89.3.1 (archimedean Frobenius as scaling exponential). The *archimedean Frobenius* is the operator

$$F_\infty := \exp(-\Theta_\infty) = \exp\left(-\frac{1}{2}\text{id} - i\widehat{D}\right)$$

acting on the archimedean Fock factor $\mathcal{H}^{(\infty)} \subset \mathcal{V}_{\text{Heis}}^{\text{prim}}$, realised on $L^2(\mathbb{R}, dt)$ via the Weil–Petersson isometry Υ_∞ of Ch. 56. Equivalently, on the Mellin side it is multiplication by e^{-s} on the critical line by Chapter 89.1.3.

Remark 89.3.2 (heat-flow form). Alternative presentation: $F_\infty = e^{-\Theta_\infty}$, where $\Theta_\infty = \frac{1}{2}\text{id} + i\widehat{D}$ and $\widehat{D}$ is the Connes 1999 multiplicative scaling generator of Chapter 89.1.1. Under the Mellin unitarity of Chapter 89.1.3, F_∞ becomes the Mellin-multiplier operator with symbol e^{-s} . This is the Frobenius eigenvalue convention. It must be kept distinct from the external variable in a multiplicative Euler-character determinant $\det_\infty(1 - e^{-s}F \mid H_{\text{ch}}^\bullet)$.

THEOREM 89.3.3 (*Wilson-surface = Arakelov scaling*). (on Chapter 81.4.7). Under the conjectural bulk measure of quantum arithmetic Chern-Simons, the Wilson-surface observable F_∞^{WS} of Chapter 89.2.3 coincides with F_∞ of Chapter 89.3.1 on the archimedean Fock factor:

$$F_\infty^{\text{WS}} = F_\infty \quad \text{in } \text{End}(\mathcal{H}^{(\infty)}). \quad (89.3)$$

Proof. Under Chapter 81.4.7, the boundary state Hilbert space at Σ_∞ is $\mathcal{H}^{(\infty)} \cong \mathcal{V}_{\text{Heis},\infty}^{\text{prim}}$; this is the boundary WZW Hilbert space axiom (1) of the conjecture. On this Hilbert space the area-coupled Chern-Simons connection $A_{k^{\text{Ar}}}$ acts by the level- k^{Ar} primary current; the curvature $F_A = dA$ acts by the derivative of the primary current in the normal direction. Integration over Σ_∞ against the Arakelov Kähler form (89.1) is precisely the archimedean scaling-generator computation: the pullback of ω_∞ is proportional to $d^\times|x| \wedge d \arg(x)$ on the unit-norm locus, and integrating against the normal curvature gives the normal operator Θ_∞ , built from the generator $\widehat{D}$ of Chapter 89.1.1, up to the normalization

$1/(2\pi) \cdot \text{vol}(\Sigma_\infty, \omega_\infty) = 1$ fixed by Faltings 1984 (§2–§3). Exponentiation gives (89.3). The residual is the well-definedness of the bulk measure, which is exactly Chapter 81.4.7. $\square$

Remark 89.3.4 (geometric content of the identification). The identification is structurally the same mechanism by which boundary L_o in 2D CFT equals the bulk radial Hamiltonian in the 3D Chern-Simons dual: radial quantisation of the 3D bulk turns ∂_t (where t is the radial coordinate) into L_o^{Vir} on the boundary Hilbert space (Witten 1989, §4; Elitzur–Moore–Schwimmer–Seiberg 1989 *Nucl. Phys.* B326, 108–134). On the arithmetic side, the radial coordinate is the Arakelov log-valuation parameter $t = \log |x|_\infty$, and the boundary Hamiltonian is the self-adjoint generator $\widehat{D}$ of Chapter 89.1.1.

Convention and Source Comparison for the Construction

Three source classes are disjoint: Arakelov scaling (Arakelov 1974, Faltings 1984, Connes 1999); bulk Wilson-surface (Witten 1989, Baez–Schreiber 2005); radial quantisation identification (Elitzur–Moore–Schwimmer–Seiberg 1989).

Sign conventions: $\widehat{D} = -i\partial_t$ is self-adjoint on $L^2(\mathbb{R}, dt)$, $\Theta_\infty = \frac{1}{2}\text{id} + i\widehat{D}$, and $F_\infty = e^{-\Theta_\infty}$. Thus F_∞ is $e^{-1/2}$ times the unitary scaling flow. This is the Deninger normalisation: a zero on the critical line, $\rho = \frac{1}{2} + i\tau$, gives $|e^{-\rho}| = e^{-1/2}$.

The ambient Hilbert space is $L^2(\mathbb{R}, dt) \simeq L^2(\text{Re } s = \frac{1}{2}, |ds|/(2\pi))$, unitarily identified with $\mathcal{H}^{(\infty)}$ via the Weil–Petersson isometry of Ch. 56. The unitary part of the observable is the exponential of the self-adjoint generator $\widehat{D}$; the Frobenius observable includes the Tate factor $e^{-1/2}$.

The quantum ACS bulk measure is required for Wilson-surface evaluation; this is the residual of Chapter 81.4.7.

Under the Tate-pole cancellation $\frac{1}{2}s(s-1)$ of Ch. 82.6, the F_∞ -eigenvalues at $s = 0, 1$ cancel and the critical-line eigenvalues survive, consistent with Chapter 82.6.4.

The unit normalisation $\text{vol}(\Sigma_\infty, \omega_\infty) = 1$ is a convention choice of Faltings 1984, §2, compatible with the Euler-characteristic normalisation of Chapter 82.6.1.

89.4 CONJUGACY TO VIRASORO L_o^{Ar}

The third description of F_∞ is as the Mellin-conjugate of the arithmetic Virasoro zero-mode L_o^{Ar} . Chapter 22 identified L_o^{Ar} on the primary subspace $\mathcal{V}_{\text{Heis, prim}}^{\text{prim}}$ as the Fock/Virasoro conformal-energy operator of the formal arithmetic Fock. Chapter 82.2 recorded that L_o^{Ar} is distinct from $\widehat{D}$, H_{HP} , and Θ_ξ . The Weil–Petersson isometry Υ_∞ identifies the normal archimedean conformal-weight operator $\Theta_\infty = \frac{1}{2}\text{id} + i\widehat{D}$ with L_o^{Ar} on the archimedean primary fibre, while H_{HP} and Θ_ξ remain distinct global operators on the full $\mathcal{V}^{\text{prim}}$.

The Weil–Petersson isometry Υ_∞

Chapter 56 constructed a Weil–Petersson unitary isometry

$$\Upsilon_\infty : \mathcal{V}_{\text{Heis, prim, } \infty}^{\text{prim}} \xrightarrow{\sim} L^2(\mathbb{R}_+^\times, d^\times x)$$

on the archimedean fibre restricted to the primary sector. Under Υ_∞ :

- The Petersson inner product on the primary Fock becomes the standard L^2 inner product.
- The Arakelov scaling generator $\widehat{D}$ of Chapter 89.1.1 becomes multiplication by τ on the Mellin transform.

- The arithmetic conformal-energy operator L_o^{Ar} is required, in the determinant-trace hypothesis below, to become the Mellin multiplier $s = \frac{1}{2} + i\tau$ on the primary sector.

The third property is the essential content of Chapter 89.4.1.

Hypothesis 89.4.1 (Υ_∞ realises L_o^{Ar} as Mellin multiplication). . On the primary subspace $\mathcal{V}_{\text{Heis,prim},\infty}^{\text{prim}}$,

$$\Upsilon_\infty L_o^{\text{Ar}} \Upsilon_\infty^{-1} = \text{multiplication by } s \quad \text{on } L^2(\text{Re } s = 1/2).$$

Remark 89.4.2 (spectral type obstruction). The ordinary archimedean Fock L_o has discrete spectrum $\mathbb{Z}_{\geq 0}$, whereas multiplication by s on the Mellin line has continuous spectrum. Therefore Chapter 89.4.1 is an additional determinant-trace assertion. It is not a consequence of the primary bar Mellin character $\sum_{m \geq 1} m^{-s} = \zeta(s)$, and it is not supplied by the determinant residue $r_\zeta = 2$.

The conjugacy theorem

THEOREM 89.4.3 (*archimedean Frobenius = Virasoro L_o exponential*). (on Chapter 81.4.7, Chapter 22.3.5, and Chapter 89.4.1). On the archimedean primary Fock $\mathcal{V}_{\text{Heis,prim},\infty}^{\text{prim}}$,

$$F_\infty = \Upsilon_\infty^{-1} e^{-L_o^{\text{Ar}}} \Upsilon_\infty.$$

Equivalently, under Υ_∞ , the archimedean Frobenius is the Mellin multiplier e^{-s} , and the Virasoro zero-mode is multiplication by s . Together with Chapter 89.3.3, the three constructions F_∞ , F_∞^{WS} , and $\Upsilon_\infty^{-1} e^{-L_o^{\text{Ar}}} \Upsilon_\infty$ agree.

Proof. By Chapter 89.1.3, $\Upsilon_\infty \Theta_\infty \Upsilon_\infty^{-1} = s$. By Chapter 89.4.1, $\Upsilon_\infty L_o^{\text{Ar}} \Upsilon_\infty^{-1} = s$. The two Mellin-images coincide; therefore $\Theta_\infty = L_o^{\text{Ar}}$ after Υ_∞ -conjugation. Exponentiating gives $F_\infty = e^{-\Theta_\infty}$ conjugate to $e^{-L_o^{\text{Ar}}}$ via Υ_∞ . $\square$

Remark 89.4.4 (three operators, one Mellin symbol). The operator dictionary of Chapter 22.3.6 distinguishes four operators: L_o^{Vir} , N_{BC} , H_{HP} , Θ_ξ . The present theorem identifies Θ_∞ , not $\widehat{D}$, with L_o^{Ar} on the *archimedean* primary fibre only, under Weil–Petersson conjugation. This is a local archimedean statement; it does not identify L_o^{Vir} with H_{HP} on the full $\mathcal{V}^{\text{prim}}$, where the finite primes contribute a separate N_{BC} -action and the Hilbert–Pólya operator H_{HP} is the conjectural determinant-trace operator attached to the assembled zero distribution. The Polyakov divergence of Chapter 22.3.4 is at the full global level, not at the archimedean-fibre level of Chapter 89.4.3.

Convention and Source Comparison for the Conjugacy

Source classes are disjoint: Arakelov scaling ($\widehat{D}$) from Arakelov 1974 and Connes 1999; Fock–Virasoro (L_o^{Ar}) from Segal 1981 and Kac–Raina 1987.

The self-adjoint generator $\widehat{D}$ has Mellin-image τ . The conformal-weight operator $\Theta_\infty = \frac{1}{2}\text{id} + i\widehat{D}$ and L_o^{Ar} have Mellin-image $s = \frac{1}{2} + i\tau$.

The ambient category is the Hilbert space $L^2(\text{Re } s = \frac{1}{2}, |ds|/(2\pi))$. The unitary operator is $e^{-i\widehat{D}}$. The Frobenius is $e^{-\Theta_\infty}$.

Primary–Virasoro reconciliation (Chapter 22.3.5) is required for L_o^{Ar} to be an honest centrally-extended Virasoro on the primary core. This is the residual.

Under Arakelov symmetry $\lambda \mapsto \lambda^{-1}$, F_∞ inverts; the Mellin-side involution $s \mapsto 1 - s$ also inverts, compatible with Koszul self-duality of Chapter 21.5.1.

The equality $\Theta_\infty = L_o^{\text{Ar}}$ under Υ_∞ is exactly the Mellin realisation hypothesis Chapter 89.4.1, together with Chapter 89.1.3.

89.5 SPECTRAL IDENTIFICATION WITH $\xi_{\mathbb{R}}$ -ZEROS

The conjugacy of Chapter 89.4.3 transfers the spectrum of L_o^{Ar} on the primary Fock to the spectrum of F_{∞} on the archimedean Fock factor. Under the conjugate Hilbert–Pólya hypothesis H-P^{Ar} of Chapter 22.3.2, the middle-degree descent H_{ch}^1 receives a regularised F -action whose determinant divisor is the exponential image of the non-trivial-zero divisor of $\xi_{\mathbb{R}}$.

L_o^{Ar} on the primary Fock and the ξ -image

Chapter 22 identified $\xi_{\mathbb{R}}$ as the character of $\mathcal{V}^{\text{prim}}$ via Tate 1950 completion of the Heisenberg bar-character ζ . The character encodes the Tate-shifted Mellin determinant after Petersson normalisation. Under the H-P^{Ar} realisation, the normal operator Θ_{ξ} has regularised determinant divisor equal to $\text{div } \xi_{\mathbb{R}}$; after zero-placement this is the ordinary ordinate spectrum of its self-adjoint part.

LEMMA 89.5.1 (*L_o^{Ar} determinant divisor via HP zeros*). (on Chapter 22.3.2 and Chapter 89.4.1). Under these hypotheses, the regularised determinant divisor of the Mellin-image $s = \Upsilon_{\infty} L_o^{\text{Ar}} \Upsilon_{\infty}^{-1}$ on the middle-degree descent H_{ch}^1 is

$$\text{div}_{\infty}(\Upsilon_{\infty} L_o^{\text{Ar}} \Upsilon_{\infty}^{-1} | H_{\text{ch}}^1) = \text{div } \xi_{\mathbb{R}}.$$

Proof. Under H-P^{Ar}, $\Theta_{\xi} = \frac{1}{2}\text{id} + iH_{\text{HP}}$ regularised-determinant identity $\det_{\infty}(s \cdot \text{id} - \Theta_{\xi}) = \xi_{\mathbb{R}}(s)$ of Chapter 22.1 says precisely that Θ_{ξ} has determinant divisor $\text{div } \xi_{\mathbb{R}}$. The Mellin multiplier-image of L_o^{Ar} by Chapter 89.4.1 is the same Tate-shifted Mellin variable s . $\square$

F_{∞} determinant as the exponential zero image

THEOREM 89.5.2 (*determinant-divisor identification for F_{∞}*). (on Chapter 22.3.2, Chapter 22.3.5, Chapter 81.4.7, and Chapter 89.4.1). The regularised determinant divisor of F_{∞} on the middle-degree archimedean descent of the chiral cohomology is the pushforward of $\text{div } \xi_{\mathbb{R}}$ under $\rho \mapsto e^{-\rho}$:

$$\text{div}_{\infty}(F_{\infty} | H_{\text{ch}}^1) = (e^{-\bullet})_* \text{div } \xi_{\mathbb{R}}, \quad (89.4)$$

with multiplicities summed over fibres of the exponential map. After zero-placement this divisor is represented by the spectral measure of the normal operator F_{∞} .

Proof. By Chapter 89.4.3, $F_{\infty} = \Upsilon_{\infty}^{-1} e^{-L_o^{\text{Ar}}} \Upsilon_{\infty}$ on the primary archimedean Fock. By Chapter 89.5.1, the determinant divisor of the Mellin image of L_o^{Ar} on H_{ch}^1 is $\text{div } \xi_{\mathbb{R}}$. Holomorphic functional calculus pushes a determinant divisor forward under $z \mapsto e^{-z}$, giving (89.4). $\square$

PROPOSITION 89.5.3 (*multiplicative character does not replace Hadamard*). . The multiplicative determinant

$$\det_{\infty}(1 - e^{-s} F_{\infty} | H_{\text{ch}}^1) = \prod_{\rho} (1 - e^{-s} e^{-\rho}) \quad (89.5)$$

is not the Hadamard product for $\xi_{\mathbb{R}}(s)$. It vanishes at the logarithmic congruence classes $s = -\rho + 2\pi i\mathbb{Z}$, not at $s = \rho$. The zeta determinant is the additive determinant

$$\det_{\infty}(s\text{id} - \Theta_{\xi}),$$

not the multiplicative Euler-character determinant (89.5).

Proof. The zeros of the right hand side of (89.5) satisfy $e^{-s} e^{-\rho} = 1$, hence $s = -\rho + 2\pi i k$. Hadamard's product has local factors $(1 - s/\rho) e^{s/\rho}$, with zeros at $s = \rho$. Exponentiating the spectrum therefore changes the divisor. Only the additive regularised determinant keeps the zero divisor. $\square$

Convention and Source Comparison for the Spectral Identification

The Hilbert–Pólya conjecture (Odlyzko 1987 *Math. Comp.* 48; Berry–Keating 1999 *SIAM Rev.* 41) and Arakelov scaling are independent.

The F_∞ -eigenvalue attached to ρ is $e^{-\rho}$. This is the multiplicative Frobenius notation. It does not replace the additive determinant $\det_\infty(\text{id} - \Theta_\xi)$.

The ambient category is a Hilbert space with regularised trace and regularised determinant on nuclear operators (Quillen 1985 *Funct. Anal. Appl.* 19).

The hypothesis $H\text{-P}^{\text{Ar}}$ is required for the Tate-shifted L_o^{Ar} determinant on H_{ch}^1 to have divisor $\text{div } \xi_{\mathbb{R}}$.

The zero divisor is closed under $\rho \mapsto 1 - \rho$ by the functional equation. The F_∞ determinant divisor is its exponential image and carries the induced involution.

The F_∞ Mellin multiplier is e^{-s} . The additive operator whose determinant has divisor $\text{div}(\xi)$ is Θ_ξ , not F_∞ .

89.6 ARCHIMEDEAN PURITY IS EQUIVALENT TO A-TP

Deninger axiom (6) of Chapter 82.1 requires that the Frobenius acts on H_Δ^1 with weight $1/2$: $|F|$ -eigenvalues on H^1 have absolute value $e^{-1/2}$ for the normalisation $F = \exp(-\Theta_\xi)$ with $\text{Re}(\Theta_\xi) = 1/2$. Equivalently, the unitarised operator $e^{1/2}F$ has spectral radius one. The archimedean component of this purity statement is the claim $|F_\infty|$ -weight $= 1/2$. This is equivalent to archimedean Tate positivity A-TP by Chapter 89.6.3.

The $|F_\infty|$ -weight $= 1/2$ statement

Definition 89.6.1 (archimedean F_∞ -weight). The archimedean F_∞ -weight on H_{ch}^1 is the common value $w \in \mathbb{R}$ such that every regularised eigenvalue μ of F_∞ on H_{ch}^1 satisfies $|\mu| = e^{-w}$, provided such a common value exists; otherwise the F_∞ -weight is not well-defined.

Remark 89.6.2 (well-definedness). By Chapter 89.5.2, the F_∞ eigenvalues are $\{e^{-\rho} : \xi_{\mathbb{R}}(\rho) = 0\}$. The modulus $|e^{-\rho}| = e^{-\text{Re}(\rho)}$ equals $e^{-1/2}$ for all zeros exactly when all zeros satisfy $\text{Re}(\rho) = 1/2$, i.e. exactly when RH holds. Thus the F_∞ -weight is well-defined and equals $1/2$ iff RH.

A-TP archimedean purity equivalence

THEOREM 89.6.3 (F_∞ -purity $\Leftrightarrow$ A-TP $\Leftrightarrow$ RH). (on Chapter 22.3.2, Chapter 22.3.5, Chapter 81.4.7, Chapter 56.7.2). The following four statements are equivalent:

- (i) The F_∞ -weight on H_{ch}^1 is well-defined and equals $1/2$: every regularised eigenvalue of F_∞ on H_{ch}^1 has modulus $e^{-1/2}$.
- (ii) Deninger archimedean purity axiom (6): the scaling Frobenius at ∞ acts on H^1 at real part $1/2$.
- (iii) A-TP (Chapter 56.7.1): archimedean Tate positivity on admissible Paley–Wiener test data.
- (iv) Riemann Hypothesis: all non-trivial zeros ρ of ζ satisfy $\text{Re}(\rho) = 1/2$.

Proof. (1) $\Leftrightarrow$ (4): by Chapter 89.6.2, $|e^{-\rho}| = e^{-1/2}$ for all zeros iff $\text{Re}(\rho) = 1/2$ for all zeros, which is RH.

(2) $\Leftrightarrow$ (1): Deninger axiom (6) at ∞ requires the scaling Frobenius to act on H^1 at real part $1/2$; F_∞ is the scaling Frobenius at ∞ via Chapter 89.3.3 and Chapter 89.3.2. The real-part $1/2$ condition for Θ_ξ -eigenvalues translates under the exponential $F_\infty = e^{-\Theta_\xi^{\text{arch}}}$ to $|F_\infty|$ -eigenvalue-modulus $e^{-1/2}$.

(3) $\Leftrightarrow$ (4): this is Chapter 56.7.2: A-TP is equivalent to Weil positivity for the number-field explicit formula, which is classically equivalent to RH (Weil 1952, Bombieri 2000).

By transitivity, (1), (2), (3), (4) are equivalent. $\square$

COROLLARY 89.6.4 (*three-construction RH equivalence via F_∞*). (on Chapter 89.6.3). The archimedean Frobenius F_∞ provides the archimedean-only formulation of RH that aligns with the three constructions of Chapter 82.8.2:

- *Deninger construction at ∞* : prove F_∞ -weight = $1/2$ on H_{ch}^1 .
- *3D-ACS construction at ∞* : prove the σ_{area} -Wilson-surface observable has weight- $1/2$ spectrum.
- *A-TP archimedean construction*: prove archimedean Tate positivity.

All three are equivalent by Chapter 89.6.3.

Remark 89.6.5 (the archimedean-only portion of RH). The full global Frobenius F combines F_∞ with per-prime ψ_p via restricted tensor (Chapter 82.5.5). The per-prime ψ_p -eigenvalues on $\mathcal{H}_{\text{BC}}^{(p)}$ have modulus $p^{-r/2}$ on the weight- r subspace, already pure by the Bost–Connes Adams structure (Borger 2009 arXiv:0906.3146, §8; Connes–Marcolli 2008 AMS Colloq. 55, §3). The archimedean portion F_∞ is the only component where weight- $1/2$ purity is non-trivial: it reduces to RH by the above theorem. This explains the concentration of the Riemann Hypothesis in the archimedean fibre: it is equivalent to archimedean purity for a Frobenius whose finite-place components are already pure by construction.

Convention and Source Comparison for the Purity Equivalence

Source classes are disjoint: Deninger 1998 (purity axiom), Weil 1952 (explicit formula and Weil positivity), Ch. 56 (A-TP formulation).

Sign: $F_\infty = e^{-\Theta_\xi^{\text{arch}}}$ gives $|F_\infty| = e^{-\text{Re}(\Theta_\xi)}$; Deninger purity requires $\text{Re}(\Theta_\xi) = 1/2$, consistent.

The ambient category is self-adjoint operators on a Hilbert space with regularised spectra.

$H\text{-}P^{\text{Ar}}$ is required for the realisation of Θ_ξ . This is the residual.

Functoriality under $\rho \leftrightarrow 1 - \rho$ (functional equation) and under Arakelov scaling (archimedean involution) are compatible.

The equivalence A-TP $\Leftrightarrow$ Weil positivity $\Leftrightarrow$ RH is classical; the equivalence (1) $\Leftrightarrow$ (2) is direct from the exponential identification $F_\infty = e^{-\Theta_\xi^{\text{arch}}}$.

89.7 RESIDUAL OBSTRUCTIONS

Three residuals remain open from Chapter 82.5.7 and Chapter 82.5.4:

- Quantum ACS bulk measure* (Chapter 81.4.7). The Wilson-surface observable F_∞^{WS} of Chapter 89.2.3 is a well-defined observable of the conjectural bulk measure; its identification with the unconditional Arakelov scaling F_∞ (Chapter 89.3.3) is conditional on the bulk measure. Domain: the profinite renormalisation programme of Ch. 81, a non-formal theorem condition.
- Primary-Virasoro reconciliation* (Chapter 22.3.5). The conjugacy $F_\infty = \Upsilon_\infty^{-1} e^{-L_o^{\text{Ar}}} \Upsilon_\infty$ of Chapter 89.4.3 requires L_o^{Ar} to be an honest centrally-extended Virasoro on the primary core and to satisfy the Mellin realisation hypothesis Chapter 89.4.1. The latter is a spectral-type assertion, not a consequence of the Fock grading.
- $H\text{-}P^{\text{Ar}}$ realisation* (Chapter 22.3.2). The determinant-divisor identity

$$\text{div}_\infty(F_\infty \mid H_{\text{ch}}^1) = (e^{-\bullet})_* \text{div } \xi_{\mathbb{R}}$$

of Chapter 89.5.2 is conditional on $H\text{-}P^{\text{Ar}}$ supplying a self-adjoint operator $\Theta_\xi = \frac{1}{2}\text{id} + iH_{\text{HP}}$ with the non-trivial-zero regularised determinant divisor. This is the Hilbert–Pólya/Connes–Deninger obstruction.

The three hypotheses are independent. The Wilson-surface construction (Chapter 89.2.3), the conjugacy to L_o^{Ar} (Chapter 89.4.3), the spectral identification with ξ -zeros (Chapter 89.5.2), and the purity equivalence (Chapter 89.6.3) are each conditional on a stated subset of (1)–(3).

89.8 BIBLIOGRAPHY

Primary references.

Arakelov 1974. Arakelov, S. Yu. *Intersection theory of divisors on an arithmetic surface*. Izv. Akad. Nauk SSSR Ser. Mat. **38** (1974), 1179–1192.

Baez–Schreiber 2005. Baez, J. C.; Schreiber, U. *Higher gauge theory*. arXiv:hep-th/0412325 (2005).

Berry–Keating 1999. Berry, M. V.; Keating, J. P. *The Riemann zeros and eigenvalue asymptotics*. SIAM Rev. **41** (1999), 236–266.

Bombieri 2000. Bombieri, E. *Problems of the millennium: the Riemann hypothesis*. Clay Mathematics Institute Millennium Prize Problem statement (2000); revised 2004.

Borger 2009. Borger, J. *Lambda-rings and the field with one element*. arXiv:0906.3146 (2009).

Cattaneo–Mnev–Reshetikhin 2014. Cattaneo, A. S.; Mnev, P.; Reshetikhin, N. *Classical BV theories on manifolds with boundary*. Commun. Math. Phys. **332** (2014), 535–603.

Connes 1999. Connes, A. *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*. Selecta Math. (N.S.) **5** (1999), 29–106.

Connes–Marcolli 2008. Connes, A.; Marcolli, M. *Noncommutative geometry, quantum fields and motives*. AMS Colloquium Publications **55** (2008).

Deninger 1998. Deninger, C. *Some analogies between number theory and dynamical systems on foliated spaces*. Proc. ICM Berlin **1** (1998), 163–186.

Elitzur–Moore–Schwimmer–Seiberg 1989. Elitzur, S.; Moore, G.; Schwimmer, A.; Seiberg, N. *Remarks on the canonical quantization of the Chern-Simons-Witten theory*. Nucl. Phys. B **326** (1989), 108–134.

Faltings 1984. Faltings, G. *Calculus on arithmetic surfaces*. Ann. Math. (2) **119** (1984), 387–424.

Hadamard 1893. Hadamard, J. *Etude sur les propriétés des fonctions entières et en particulier d’une fonction considérée par Riemann*. J. Math. Pures Appl. (4) **9** (1893), 171–215.

Kac–Raina 1987. Kac, V. G.; Raina, A. K. *Bombay Lectures on Highest Weight Representations of Infinite Dimensional Lie Algebras*. World Scientific, 1987.

Kim 2015. Kim, M. *Arithmetic Chern-Simons theory I*. arXiv:1510.05818 (2015).

Morishita 2012. Morishita, M. *Knots and Primes: An Introduction to Arithmetic Topology*. Universitext, Springer, 2012; arXiv:0904.3399.

Odlyzko 1987. Odlyzko, A. M. *On the distribution of spacings between zeros of the zeta function*. Math. Comp. **48** (1987), 273–308.

Quillen 1985. Quillen, D. *Determinants of Cauchy-Riemann operators over a Riemann surface*. Funct. Anal. Appl. **19** (1985), 31–34.

Schreiber–Sati 2011. Schreiber, U.; Sati, H. *Survey of mathematical foundations of QFT and perturbative string theory*. arXiv:1109.0955 (2011); higher-form construction referenced via arXiv:1011.4735.

Segal 1981. Segal, G. *Unitary representations of some infinite dimensional groups*. Commun. Math. Phys. **80** (1981), 301–342.

Tate 1950. Tate, J. T. *Fourier analysis in number fields, and Hecke’s zeta-functions*. Princeton Ph.D. thesis, 1950; reprinted in Cassels–Fröhlich, *Algebraic Number Theory*, Academic Press, 1967.

Weil 1952. Weil, A. *Sur les “formules explicites” de la théorie des nombres premiers*. Bull. Soc. Math. France **80** (1952), 183–192.

Witten 1989. Witten, E. *Quantum field theory and the Jones polynomial*. Commun. Math. Phys. **121** (1989), 351–399.

ARCHIMEDEAN SURFACE BOUNDARY

- (i) The archimedean Frobenius F_∞ admits an unconditional Arakelov presentation and two conditional presentations on the archimedean Fock factor $\mathcal{H}^{(\infty)} = \mathcal{V}_{\text{Heis},\infty}^{\text{prim}}$: the Arakelov scaling exponential $F_\infty = e^{-\Theta_\infty}$ (Chapter 89.3.1), the bulk Wilson-surface observable F_∞^{WS} on $\Sigma_\infty = \{\infty\} \times D^2$ with area-scaling 2-representation σ_{area} (Chapter 89.2.3), and the Mellin-conjugate of the arithmetic Virasoro zero-mode L_0^{Ar} under the Weil–Petersson isometry Υ_∞ (Chapter 89.4.3).
- (ii) The three presentations coincide only under the stated conditional hypotheses: $F_\infty^{\text{WS}} = F_\infty$ by Chapter 89.3.3, $F_\infty = \Upsilon_\infty^{-1} e^{-L_0^{\text{Ar}}} \Upsilon_\infty$ by Chapter 89.4.3.
- (iii) Under H-P^{Ar} , the regularised determinant divisor of F_∞ on H_{ch}^1 is $(e^{-\bullet})_* \text{div } \xi_{\mathbb{R}}$ (Chapter 89.5.2). The Hadamard product belongs to the additive determinant $\det_\infty(\text{id} - \Theta_\xi)$, not to the multiplicative determinant $\det_\infty(1 - e^{-s} F_\infty)$ (Chapter 89.5.3).
- (iv) The archimedean F_∞ -weight = 1/2 is equivalent to Deninger archimedean purity, to A-TP, and to RH (Chapter 89.6.3). The concentration of the Riemann Hypothesis in the archimedean fibre is thereby explained: finite-place ψ_p components are already pure by Bost–Connes structure, and archimedean purity of F_∞ is the residual weight-1/2 condition.
- (v) The residual obstructions Chapter 82.5.7 (Wilson-surface interpretation of F_∞) and Chapter 82.5.4 (conjugacy $F_\infty \sim e^{-L_0^{\text{Ar}}}$) are carried forward to three named conditionals: quantum ACS bulk measure, primary-Virasoro reconciliation, the Mellin realisation of L_0^{Ar} , and H-P^{Ar} .

The archimedean Frobenius F_∞ as a Wilson-surface operator in 3D arithmetic Chern-Simons, as the Arakelov scaling exponential, and as the Υ_∞ -conjugate of $e^{-L_0^{\text{Ar}}}$ become three descriptions of a single operator only after these residual hypotheses are imposed. The equivalence of F_∞ -purity with A-TP locates the Riemann Hypothesis at the archimedean purity obstruction: finite-place components are pure by the Bost–Connes Adams structure, and the archimedean component reduces exactly to Hilbert–Pólya/Connes–Deninger.

FIVE SOURCE FAMILIES AND THE ARCHIMEDEAN TATE POSITIVITY RESIDUAL

The reduction of Chapter 59 places the Riemann hypothesis in the form of archimedean Tate positivity, A-TP (Chapter 56.7.1). Five source families meet this same Weil inequality from different sides: Arthur–Selberg scattering, geometric Langlands, Iwasawa–Kato, arithmetic quantum chaos, and Arakelov–Faltings–Hriljac. None of them removes the Weil positivity problem. Pair-correlation controls statistics after the zero-line input; it does not supply the zero-line input. Iwasawa theory controls finite-prime symmetry; it does not supply the archimedean sign. Arakelov height positivity gives useful sub-class inequalities only after the Bott–Chern and vertical terms are controlled.

90.1 A-TP: STATEMENT AND INDEPENDENT OBSTRUCTION

Let $\text{PW}(\mathbb{R})$ be the Paley-Wiener space of even Schwartz functions extending holomorphically to a strip with $(1 + |z|)^{-2-\delta}$ decay; let $\tilde{f}(x) = \overline{f(-x)}$. The Weil distribution W on $\text{PW}(\mathbb{R})$ is the linear functional of Chapter 83.1.2:

$$W(h) = \widehat{h}\left(\frac{1}{2i}\right) + \widehat{h}\left(-\frac{1}{2i}\right) - W_\infty(h) - 2 \sum_p \sum_{k \geq 1} \frac{\log p}{p^{k/2}} h(k \log p).$$

A-TP asserts $W(f * \tilde{f}) \geq 0$ for all $f \in \text{PW}(\mathbb{R})$, equivalent to RH via Weil 1952 (*Bull. Soc. Math. France* 80).

The five source lists below are pairwise disjoint from each other and from those of Chapters 56–79, modulo the common ground of Tate’s thesis and Weil’s explicit formula:

- **Arthur–Selberg truncation:** Arthur 1978–2013 trace-formula series; Jacquet–Langlands 1970; Selberg 1956; Ngô 2010 *Publ. IHES* 111; Flicker–Kazhdan; Mœglin–Waldspurger.
- **Geometric Langlands and Fargues–Scholze:** Beilinson–Drinfeld 2004; Arinkin–Gaitsgory arXiv:1201.6343; Fargues–Scholze arXiv:2102.13459; Lafforgue arXiv:1209.5352; Drinfeld 1983, 1987; Deligne *Weil II*; Frenkel–Gaitsgory.
- **Iwasawa–Kato:** Mazur–Wiles 1984; Kato 2004 *Astérisque* 295; Perrin–Riou 1995; Fukaya–Kato 2006; Colmez; Bloch–Kato 1990; Kolyvagin 1988; Coates–Wiles 1977; Ferrero–Washington 1979.
- **Arithmetic quantum chaos:** Bohigas–Giannoni–Schmit 1984; Berry–Keating 1999; Montgomery 1973; Odlyzko 1987; Rudnick–Sarnak 1996; Katz–Sarnak 1999; Keating–Snaith 2000; Lindenstrauss 2006; Soundararajan 2010; Sarnak 1995; Hejhal 1994.
- **Arakelov–Faltings–Hriljac:** Arakelov 1974; Faltings 1984; Hriljac 1985; Gillet–Soulé 1990; Bost 1999; Bost–Künnemann 2010; Bombieri 1971; Gross–Zagier 1986; Beilinson 1985; Vojta 1987; Bismut–Gillet–Soulé 1988.

Each list's intersection with those of Chapters 56–79 is empty modulo Tate's thesis and Weil's explicit formula.

90.2 FIVE SOURCE FAMILIES

THEOREM 90.2.1 (*five source-family boundary for A-TP*). . The five source families below do not give an independent proof of A-TP. Each lands on one of three mathematical forms: the Weil inequality itself, a conditional sub-class diagnostic, or an auxiliary structure which has no archimedean positivity content without an additional RH-strength input.

Tradition	Exact mathematical form
Arthur–Selberg	Maass–Selberg scattering recovers the archimedean gamma logarithmic derivative inside W_∞ . It does not remove the finite-prime and polar terms of W .
Geometric Langlands	The spectral-action language can encode tempered parameters after the local and global comparison inputs. It does not prove Weil positivity for the zeta divisor.
Iwasawa–Kato	Euler systems and main-conjecture structures give finite-prime arithmetic control. They carry no theorem forcing the archimedean Weil sign.
Arithmetic quantum chaos	GUE and pair-correlation statements are statistics of ordinates after a zero-line model is fixed. They do not replace $z_\rho = -i(\rho - \frac{1}{2})$ by a real ordinate.
Arakelov–Faltings–Hriljac	The arithmetic Hodge index gives negative definiteness on degree-zero height pairings. Weil positivity follows only after the Bott–Chern and vertical residual terms have the required sign.

Definition 90.2.2 (*consolidated residual*). . The *A-TP residual* is the full Weil quadratic inequality

$$W(f * \tilde{f}) \geq 0, \quad f \in \text{PW}(\mathbb{R}),$$

with polar, finite-prime, and archimedean terms all present. The archimedean digamma kernel is one summand of this residual, not a replacement for it. Denote the full residual by R_{A-TP} .

Remark 90.2.3 (*one obstruction, several shadows*). The source families give shadows of the same Weil inequality. A shadow is not an equivalent coordinate chart unless it retains the polar terms, the von Mangoldt sum, the archimedean gamma term, and the full Paley–Wiener cone.

90.3 TWO CONDITIONAL A-TP DIAGNOSTICS

Quant-A-TP-RS on the Rudnick–Sarnak window

THEOREM 90.3.1 (*Rudnick–Sarnak window does not give unconditional A-TP*). . Let $\text{PW}^{\text{CP}} \subset \text{PW}(\mathbb{R})$ be the co-Poisson sub-class (functions whose translates are orthogonal to each co-Poisson form; see Chapter 83.4.1). On the restricted class

$$\text{PW}^{\text{CP}} \cap \{f : \text{supp } \hat{f} \subset [-1, 1]\}$$

(the co-Poisson class meeting a small spectral window), there is no unconditional proof of $W(f * \tilde{f}) \geq 0$. Pair-correlation estimates control averaged spacing statistics; they do not turn the Weil zero functional into the positive square sum $\sum_\rho |\hat{f}(\gamma_\rho)|^2$ without RH.

Sketch. Unconditionally the zero side is evaluated on the divisor $z_\rho = -i(\rho - \frac{1}{2})$, not on real ordinates γ_ρ . The replacement by a positive square sum is exactly the RH-level specialization. Rudnick–Sarnak pair correlation does not remove that specialization. $\square$

Remark 90.3.2 (domain of the Rudnick-Sarnak sub-closure). The sub-class $\text{PW}^{\text{CP}} \cap \{\text{supp } \hat{f} \subset [-1, 1]\}$ has measure zero in $\text{PW}(\mathbb{R})$ under any natural Gaussian measure: the Rudnick-Sarnak window $\{|t| \leq 1\}$ is strictly smaller than any dense sub-class. Chapter 90.3.1 is disjoint from the sub-closures of Chapters 56–79, which give sub-class inequalities parameterized by central-value and bounded-kernel methods.

Arakelov-Heegner diagnostic on $\text{PW}_F^{\text{Heeg}}$

THEOREM 90.3.3 (*Arakelov-Heegner A-TP diagnostic*). (on Faltings-Hriljac 1984/1985 arithmetic Hodge index, Gross-Zagier 1986 central-derivative formula, Jorgenson-Kramer 2003 Petersson-vs-Arakelov Bott-Chern comparison, and the sign of the residual term $R_{F,K}(f)$). Let F be a weight-2 newform of conductor N on $\Gamma_0(N)$, let $K = \mathbb{Q}(\sqrt{-D})$ be an imaginary quadratic field satisfying the Heegner hypothesis $\chi_K(p) = -1$ for every prime $p \mid N$, and let $\text{PW}_F^{\text{Heeg}} \subset \text{PW}(\mathbb{R})$ be the Heegner-adapted Paley-Wiener class parameterised by (F, K) . Define

$$W(f * \tilde{f}) = -\widehat{D}_f^2 + R_{F,K}(f),$$

where $\widehat{D}_f \in \widehat{\text{CH}}^1(\overline{X_0(N)})$ is the arithmetic Heegner divisor traced against f and $R_{F,K}(f)$ is the Arakelov residual (Bott-Chern plus vertical-component correction). If the degree-zero projection satisfies $\widehat{D}_f^2 \leq 0$ and $R_{F,K}(f) \geq 0$, then $W(f * \tilde{f}) \geq 0$ on that Heegner-adapted sub-class. The theorem is therefore a diagnostic, not an unconditional A-TP theorem.

Sketch. Faltings-Hriljac 1984/1985 arithmetic Hodge index on $\widehat{\text{CH}}^1(\overline{X_0(N)})$ has signature $(1, +\infty)$, with the negative-definite part on degree-zero classes modulo vertical components (Néron-Tate). Gross-Zagier 1986 expresses $L'(1, F)$ as the Néron-Tate height $\widehat{h}(y_K)$ of a Heegner point $y_K \in X_0(N)(K)$; Néron-Tate heights are non-negative. The sign of the Bott-Chern and vertical residual is not automatic for an arbitrary Paley-Wiener trace. It is precisely the extra hypothesis $R_{F,K}(f) \geq 0$. Under these hypotheses the displayed identity gives Weil positivity. $\square$

Remark 90.3.4 (disjointness from the Rankin-Selberg diagnostic). Chapter 25.9.1 of Chapter 25 provides the Rankin-Selberg central-value diagnostic through Υ_F at $L(1/2, F \otimes \bar{F})$. Chapter 90.3.3 gives a non-overlapping Heegner diagnostic through Gross-Zagier’s central derivative $L'(1, F)$. The two diagnostics address disjoint analytic sectors of the same newform F ; neither removes the full A-TP residual without its stated sign hypotheses.

90.4 TWO NEGATIVE RESULTS: IWASAWA AUXILIARY STRUCTURE AND BGS IMPLICATION DIRECTION

Iwasawa-Kato Orthogonality to A-TP

THEOREM 90.4.1 (*Iwasawa-Kato orthogonality to A-TP*). (on completeness of the Iwasawa-Kato list: Mazur-Wiles 1984, Kato 2004 *Astérisque* 295, Perrin-Riou 1995, Fukaya-Kato 2006, Colmez, Bloch-Kato 1990, Kolyvagin 1988, Coates-Wiles 1977, Ferrero-Washington 1979, Connes-Marcolli, Skinner-Urban). No method in the standard Iwasawa-Kato list closes A-TP. Specifically: Mazur-Wiles value-positivity, Kato Euler system plus Bloch-Kato-Beilinson height pairing, Ferrero-Washington archimedean analog,

Perrin-Riou plus Bloch-Kato exponential, the $p \rightarrow \infty$ Connes-Marcolli limit, Kato reciprocity maps $\exp^*/\log$, Kolyvagin plus Coates-Wiles height positivity, Colmez-Fontaine p -adic Langlands, Fukaya-Kato non-commutative plus Skinner-Urban, and concrete-Dirichlet p -adic transport each fail to deliver A-TP positivity beyond sectors already closed by other means. Iwasawa theory supplies the functional-equation symmetry $s \leftrightarrow 1 - s$ at every finite prime (as in Chapter 21.5.1) but carries no positivity content at the archimedean place.

Remark 90.4.2 (Deninger's archimedean Iwasawa conjecture). The one Iwasawa-adjacent approach not enumerated above is Deninger's hypothetical archimedean Iwasawa main conjecture (Deninger 1998 ICM; 2002 *Motives and algebraic cycles*), which would interpret A-TP as the archimedean limit of a p -adic Iwasawa main conjecture. That conjecture is itself RH-equivalent; it does not reduce the obstruction.

BGS Is Consequent of RH

THEOREM 90.4.3 (*GUE statistics do not imply A-TP*). . A Bohigas–Giannoni–Schmit or GUE statement for ordinates $\{\gamma_n\}$ does not imply A-TP. Such a statement concerns limiting correlations of a real sequence. A-TP is positivity of the full Weil quadratic form on $\text{PW}(\mathbb{R})$, with the zero side evaluated at

$$z_\rho = -i(\rho - \tfrac{1}{2}).$$

Replacing z_ρ by a real ordinate γ_ρ is precisely the zero-line input. Pair-correlation estimates do not supply that replacement.

Remark 90.4.4 (implication direction in Chapter 83.8.2). The statistical-upside theorem Chapter 83.8.2 remains correct: granted the arithmetic-quantum-chaos hypotheses (C1)–(C5) of Chapter 83.8.2, BGS applied to the 3D ACS bulk Hamiltonian is a statistical consequence inside that dynamical model. The direction is from the dynamical hypothesis to statistics, not from statistics to Weil positivity. Chapter 90.4.3 rules out the latter use.

90.5 THE WEIL CONE

The five source-family boundary of Chapter 90.2.1 fixes the exact object which remains.

PROPOSITION 90.5.1 (*five source families do not contract the Weil cone*). (*on Chapter 90.2.1*). None of the five source families replaces A-TP by a smaller unconditional theorem. Each requires the full Weil positivity statement, or a sign condition whose strength is at least the corresponding A-TP subproblem:

- Arthur–Selberg retains the archimedean scattering sign.
- Geometric Langlands retains the arithmetic spectral-action comparison and the zero-divisor input.
- Iwasawa retains the absence of an archimedean positivity theorem.
- Arithmetic quantum chaos retains the zero-line passage from complex zeros to real ordinates.
- Arakelov retains Bott–Chern and vertical residual signs.

90.6 POSITION WITHIN THE A-TP OBSTRUCTION SEQUENCE

Chapters 56–79 supply ten obstructions on A-TP from analytic directions: Riemann-Siegel, Nyman-Beurling, de Branges Hilbert-Braun, Gaussian, bounded direct, Mellin-Barnes, Hadamard, Gram, Guinand-Weil, adelic scattering. They give sub-class $\mathcal{W}_\infty$ -positivity, conditional phase models, or RH-equivalent reformulations. The five algebraic traditions below add five further tests from algebraic,

categorical, and Diophantine directions. Two produce new diagnostics (Chapter 90.3.1, Chapter 90.3.3); two sharpen the obstruction by ruling out spurious directions (Chapter 90.4.1, Chapter 90.4.3). Across all fifteen obstructions, the common object is the full Weil positivity problem R_{A-TP} .

The synthesis of Chapter 59 (Chapter 59.5.1) establishes $RH \Leftrightarrow A-TP$ via the pillar reductions collected there. This chapter does not further reduce RH below $A-TP$; it records why the five source families do not bypass the full Weil cone.

The arithmetic-gravity axis of Chapters 81–83 provides two further conditional languages for the same Weil-positive distribution: chiral-homology positivity and prismatic Frobenius at the archimedean place. Chapter 90.4.3 constrains the arithmetic-quantum-chaos statistical-upside theorem Chapter 83.8.2: the implication runs from quantum chaos to GUE statistics, not from GUE statistics to positivity.

90.7 BOUNDARY

- (i) **Five source-family boundary (Chapter 90.2.1).** Arthur–Selberg, geometric Langlands, Iwasawa–Kato, arithmetic quantum chaos, and Arakelov–Faltings–Hriljac each reaches the Weil positivity problem or a conditional subproblem. None supplies an unconditional replacement for $A-TP$.
- (ii) **Two conditional diagnostics.** Chapter 90.3.1: the Rudnick–Sarnak window is not an unconditional $A-TP$ theorem. Chapter 90.3.3: the Heegner-adapted class reduces Weil positivity to arithmetic Hodge negativity together with the residual sign $R_{F,K}(f) \geq 0$. The two diagnostics are disjoint: one is parameterized by pair-correlation support, the other by the central derivative of an L -function.
- (iii) **Two negative results.** Chapter 90.4.1: the Iwasawa–Kato list controls the functional-equation symmetry at finite primes but carries no positivity information at the archimedean place. Chapter 90.4.3: GUE statistics for a real ordinate sequence do not imply Weil positivity and do not put the zeros on the critical line.
- (iv) **Uncontracted residual (Chapter 90.5.1).** The five source families do not shrink the full Weil cone. The remaining object is R_{A-TP} , namely $\mathcal{W}(f * \tilde{f}) \geq 0$ for all Paley–Wiener f .

The Hilbert–Pólya obstruction, reformulated as Weil positivity by Weil 1952, remains the full $A-TP$ residual. A coordinate shadow which forgets the polar terms, the finite-prime von Mangoldt distribution, or the full Paley–Wiener cone is not equivalent to RH .

PASKUNAS WILD BLOCK Ext^1 -CLASS MATCH AT $p \in \{2, 3\}$

The wild-prime block Ext^1 -class match at $p \in \{2, 3\}$ by a derivation path disjoint from that of Chapter 65.2.9. The proof of Chapter 65.2.9 threads the class-level identification through Paškūnas 2013 §4.11 at $p = 2$, even though Paškūnas 2013’s Montreal-functor step is stated under $p \geq 5$. The $p = 2$ input is replaced by Paškūnas–Tung 2020 and the $p = 3$ input by Colmez–Dospinescu–Paškūnas 2014 (CDP 2014) §5 combined with the Hu–Tan 2018 multiplicity framework; it then matches the resulting block Ext^1 -class to the Emerton–Gee wild-inertia cotangent fibre (arXiv:1908.07185 §9) via the pseudo-character deformation tangent, rather than the Colmez connecting map used in the proof of Chapter 65.2.9. Block-by-block class identification is proved for each wild non-semisimple Paškūnas block at $p \in \{2, 3\}$, stated ProvedHereConditional on the same wild-EG enhancement Chapter 39.5.7 as Chapter 65.2.9, but with disjoint proof sources. The class-match target Chapter 65.2.1–Chapter 65.2.2 is re-established with the Koszul-dual tangent/cotangent cycle closed intrinsically inside the block centraliser, independent of the Colmez Montreal functor.

9I.1 DOMAIN AND DISJOINT DERIVATION

The Residual after Chapter 65.2.9

The proof of Chapter 65.2.9 established the wild-prime class-match conditional on Chapter 39.5.7 via the composition

$$\mathrm{Ext}_{\mathfrak{B}}^1 \xrightarrow[\text{Paškūnas §4.11}]{\sim} H^1(G_{\mathbb{Q}_p}, \mathrm{ad}^{\circ} \bar{\rho}_{\mathfrak{B}})(\chi_{\det}) \xrightarrow[\text{residual gerbe}]{\sim} \mathbb{L}_{\chi_{GL_2}^{\mathrm{EG}, \mathrm{wild}, p, \chi}}^1|_{\chi_{\tau_{\mathfrak{B}}}}. \quad (9I.1)$$

The first arrow in (9I.1) is the Colmez Montreal functor’s connecting map. Paškūnas 2013 Theorem 1.4 is explicitly conditional on $p \geq 5$ at the Montreal-functor step; the extension to $p = 2$ is the subsequent Paškūnas–Tung 2020 refinement, and the $p = 3$ case is Colmez–Dospinescu–Paškūnas 2014 (CDP 2014) §5. The proof of Chapter 65.2.9 cites §4.11 of Paškūnas 2013 for both wild cases, which is technically outside the $p \geq 5$ domain. The same class-match closes by a different first arrow, via the pseudo-character tangent space, and states the conclusion with disjoint proof sources.

Disjoint Proof Sources

The derivation uses:

Paškūnas–Tung 2020. Theorems 1.3, 1.5, and 1.7: Cohen–Macaulay structure of each block centraliser at $p = 2$, regular local description on supersingular blocks, and the quotient-by-non-zero-divisor description of the twisted-Steinberg block.

Colmez–Dospinescu–Paškūnas 2014 §5. Theorem 5.15 and Propositions 5.17, 5.22: block centraliser structure at $p = 3$, supercuspidal regular local, principal-series Cohen–Macaulay with explicit Krull dimension.

Hu–Tan 2018. Theorem 1.1: very-special Breuil–Mézard multiplicities, used to pin the tangent multiplicity at the reducible closed point.

Emerton–Gee, arXiv:1908.07185 §9. Wild-inertia stack structure: the spectral side’s degree-1 cotangent fibre at wild closed points, computed via the formal Galois deformation ring of the residual.

Kisin 2008 (J. AMS 21), Bellaïche–Chenevier 2009 (Astérisque 324). Pseudo-character deformation-ring tangent space identification, a framework independent of the Colmez Montreal functor.

Disjointness Comparison

The derivation of Chapter 65.2.9 (cf. (91.1)) draws on Paškūnas 2013 §4.11 (the Colmez connecting map), CDP 2014 §5.3 (same connecting map at $p = 3$), plus the residual gerbe / Kisin / Emerton–Gee spectral inputs. The derivation replaces the connecting map at the first arrow by the pseudo-character tangent: the Paškūnas–Tung $p = 2$ block centraliser structure and the CDP $p = 3$ structure, both as *algebra-level statements* (Cohen–Macaulay dimensions, not connecting-map assertions); then Hu–Tan for the tangent multiplicities; then Kisin / Bellaïche–Chenevier for the pseudo-character tangent identification; then Emerton–Gee §9 for the spectral cotangent fibre. The only shared input with Chapter 65.2.9 is Chapter 39.5.7 (the wild-EG enhancement conjecture), which supplies the degree-1 derived structure on both sides. No source used here supplies a Montreal-functor connecting map; no source in the proof of Chapter 65.2.9 supplies a pseudo-character tangent comparison. Independence holds at the per-arrow level.

91.2 BLOCK CENTRALISER PSEUDO-CHARACTER TANGENT

Pseudo-Character Tangent Space at Wild Primes

LEMMA 91.2.1 (*pseudo-character tangent of wild block centraliser*). . Let $p \in \{2, 3\}$, χ a central character, and $\mathfrak{B}$ a non-semisimple Paškūnas block of $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$ with residual $\bar{\rho}_{\mathfrak{B}}$. The Zariski tangent space of $\text{Spec } A_{\mathfrak{B}}$ at the closed point is canonically identified with the pseudo-character deformation tangent

$$\mathfrak{m}_{A_{\mathfrak{B}}} / \mathfrak{m}_{A_{\mathfrak{B}}}^2 \cong H_{\text{ps}}^1(G_{\mathbb{Q}_p}, \text{Tr } \bar{\rho}_{\mathfrak{B}})(\chi_{\det}),$$

where the right-hand side is the tangent space to the pseudo-character deformation functor of $\text{Tr } \bar{\rho}_{\mathfrak{B}}$ under the fixed-determinant constraint.

Proof. The block centraliser $A_{\mathfrak{B}}$ is a complete Noetherian local $W(k)$ -algebra (Chapter 65.4.1 at $p = 2$, Chapter 65.4.3 at $p = 3$). By Bellaïche–Chenevier 2009 Astérisque 324 §1.5, for a complete Noetherian local ring pro-representing a Galois deformation functor with residual $\bar{\rho}$, the tangent space is canonically the pseudo-character H^1 . The identification of $A_{\mathfrak{B}}$ itself with a pseudo-character deformation ring at wild primes follows block-by-block: for supersingular blocks the residual is irreducible and the pseudo-character ring equals the full deformation ring; for principal-series and Steinberg blocks the residual is reducible and the pseudo-character ring is the quotient of the deformation ring that forgets the ordering of Jordan–Hölder factors, which agrees with the block centraliser under the Morita equivalence $\text{Block}_{\mathfrak{B}} \simeq A_{\mathfrak{B}}\text{-Mod}$ because the block identifies Jordan–Hölder orderings. The fixed-determinant twist $\chi_{\det}$ matches on both sides. $\square$

Remark 91.2.2 (*why pseudo-character bypasses $p \geq 5$*). The Colmez Montreal functor of Paškūnas 2013 is constructed under $p \geq 5$ because the Colmez bijection between mod- p representations of $GL_2(\mathbb{Q}_p)$ and (φ, Γ) -modules requires p to exceed the relevant ramification bound. The pseudo-character tangent path avoids this: the residual trace $\text{Tr } \bar{\rho}$ is purely a Galois-cohomology datum. Its deformation functor

is pro-representable at every prime, including $p = 2, 3$ (Chenevier 2014, “The p -adic analytic space of pseudo-characters of a profinite group and pseudo-representations over arbitrary rings”, London Math. Soc. Lecture Notes 414 Theorem A). Paškūnas–Tung 2020 and CDP 2014 §5 then supply the identification with $A_{\mathfrak{B}}$ at $p = 2, 3$ respectively.

Class-Level Identification on Each Wild Block

PROPOSITION 91.2.3 (*block Ext^1 as pseudo-character tangent*). . Let $p \in \{2, 3\}$ and $\mathfrak{B}$ a non-semisimple Paškūnas block. For any two simple objects S, S' in $\mathfrak{B}$,

$$\text{Ext}_{\mathfrak{B}}^1(S, S') \cong \text{Hom}_k(\text{gr}_{S, S'} \mathfrak{m}_{A_{\mathfrak{B}}} / \mathfrak{m}_{A_{\mathfrak{B}}}^2, k),$$

where $\text{gr}_{S, S'}$ extracts the graded component of the cotangent at the closed point determined by the pair (S, S') of simples (the component labelled by the arrow $\alpha_{S, S'}$ in the block quiver).

Proof. Morita equivalence $\mathfrak{B} \simeq A_{\mathfrak{B}}\text{-Mod}$ sends simples to simple $A_{\mathfrak{B}}$ -modules, each supported at the closed point. The Ext^1 between two simples of a complete Noetherian local ring with residue field k is dual to the graded component of $\mathfrak{m}/\mathfrak{m}^2$ labelled by the pair (Auslander–Reiten–Smalø 1995 *Representation Theory of Artin Algebras*, CUP, Ch. III §1.14). The quiver arrow $\alpha_{S, S'}$ indexes this component; at $p = 2$ the single arrow α picks out the full two-dimensional cotangent (which, after the fixed-determinant twist, is one-dimensional); at $p = 3$ the two arrows $\alpha_{\text{oi}}, \alpha_{12}$ pick out the two cotangent directions. $\square$

PROPOSITION 91.2.4 (*spectral cotangent as pseudo-character tangent*). . At $p \in \{2, 3\}$ and $\bar{\rho}_{\mathfrak{B}}$ the residual attached to a wild block, the degree-1 cotangent fibre of $\chi_{GL_2}^{\text{EG, wild}, p, \chi}$ at the closed point $x_{\tau_{\mathfrak{B}}}$ admits a canonical identification

$$\mathbb{L}_{\chi_{GL_2}^{\text{EG, wild}, p, \chi}}^1|_{x_{\tau_{\mathfrak{B}}}} \cong H_{\text{ps}}^1(G_{\mathbb{Q}_p}, \text{Tr } \bar{\rho}_{\mathfrak{B}})(\chi_{\det})^{\vee} \quad (91.2)$$

with the dual of the pseudo-character tangent.

Proof. Emerton–Gee arXiv:1908.07185 §9 identifies the formal completion of $\chi_{GL_2}^{\text{EG, wild}, p, \chi}$ at $x_{\tau_{\mathfrak{B}}}$ with the formal spectrum of the Galois deformation ring of $\bar{\rho}_{\mathfrak{B}}$ under the fixed central character. The pseudo-character quotient is the closed substack cutting out orderings of Jordan–Hölder factors; its tangent at $x_{\tau_{\mathfrak{B}}}$ is H_{ps}^1 by Chenevier 2014 Theorem A. The wild enhancement of Chapter 39.5.2 pins the degree-1 derived structure to the cotangent of this formal spectrum. Taking k -linear duals of the tangent identification gives (91.2). $\square$

91.3 WILD-PRIME CLASS-MATCH THEOREM

THEOREM 91.3.1 (*wild block Ext^1 -class match, pseudo-character derivation*). . Let $p \in \{2, 3\}$, χ a central character, and $\mathfrak{B}$ a non-semisimple Paškūnas block. Under Chapter 39.5.7, there is a canonical k -linear class-level isomorphism

$$c_{p, \mathfrak{B}}^{\text{class}, \mathfrak{B}}: \text{Ext}_{\mathfrak{B}}^1(S, S') \xrightarrow{\sim} \mathbb{L}_{\chi_{GL_2}^{\text{EG, wild}, p, \chi}}^1|_{x_{\tau_{\mathfrak{B}}}}$$

natural in $\mathfrak{B}$. At $p = 2$ the single class $\alpha \in \text{Ext}_{\mathfrak{B}}^1(\mathbf{1}_{\text{triv}}, \mathbf{1}_{\text{sgn}})$ of Chapter 32.4.5 maps to the unique wild cotangent direction; at $p = 3$ the pair $(\alpha_{\text{oi}}, \alpha_{12}) \in \text{Ext}^1(L_0, L_1) \oplus \text{Ext}^1(L_1, L_2)$ of Chapter 32.4.6 maps to the pair (e_1, e_2) of wild cotangent directions. The isomorphism $c_{p, \mathfrak{B}}^{\text{class}, \mathfrak{B}}$ agrees with the isomorphism c_p^{class} of Chapter 65.2.9 up to scalar, after the shared wild-EG enhancement is fixed.

Proof. Compose Chapter 91.2.3 with Chapter 91.2.4. The block Ext^1 is the (S, S') -graded component of the cotangent of $\mathcal{A}_{\mathfrak{B}}$ at its closed point; by Chapter 91.2.1 this cotangent is canonically dual to $H_{\text{ps}}^1(G_{\mathbb{Q}_p}, \text{Tr } \bar{\rho}_{\mathfrak{B}})(\chi_{\det})$; by Chapter 91.2.4 the spectral cotangent is the dual of the same pseudo-character tangent. The composition is a k -linear isomorphism; naturality in $\mathfrak{B}$ follows from naturality of each factor in the residual $\bar{\rho}_{\mathfrak{B}}$. Class labels propagate because both factors respect the quiver-arrow decomposition (Chapter 32.4.5 Step 3 at $p = 2$, Chapter 32.4.6 Step 3 at $p = 3$). Agreement with Chapter 65.2.9 up to scalar: both isomorphisms factor through the same pseudo-character tangent space; the Colmez connecting map and the Auslander–Reiten graded decomposition produce the same cohomology class up to a normalisation choice of generator, which is a scalar because each block’s cotangent direction is a k -line. The wild-EG enhancement is the only conditional input; the remaining derivation is unconditional block algebra. $\square$

COROLLARY 91.3.2 (*wild-prime class-match by two independent derivations*). . The wild-prime class-match is proved by two independent derivations: Chapter 65.2.9 via the Colmez connecting map and Chapter 91.3.1 via the pseudo-character tangent. The two derivations are independent at the first arrow and agree up to scalar. The realization pair for the class-match is $(c_p^{\text{class}}, c_p^{\text{class}, B})$ with disjoint proof sources.

Proof. Chapter 65.2.9 supplies the Colmez-connecting-map derivation; Chapter 91.3.1 supplies the pseudo-character-tangent derivation; independence of sources is verified in §91.1; agreement up to scalar is the last paragraph of the proof of Chapter 91.3.1. $\square$

91.4 DERIVATION COMPARISON

Seven conditions are checked against the class-match Chapter 91.3.1.

Source separation. Proof sources: Paškūnas–Tung 2020 (block centraliser at $p = 2$); CDP 2014 §5 (block centraliser at $p = 3$); Hu–Tan 2018 (very-special multiplicity, used indirectly through Paškūnas 2015 reduction but not as connecting-map input); Chenevier 2014 (pseudo-character deformation pro-representability); Kisin 2008 + Bellaïche–Chenevier 2009 (tangent space identification); Emerton–Gee arXiv:1908.07185 §9 (spectral cotangent). Proof sources of Chapter 65.2.9 (Paškūnas 2013 §4.11 Colmez connecting map, CDP 2014 §5.3 connecting map) are not invoked at the first arrow; only the block centraliser structure of CDP 2014 §5 is shared, used here as an algebra-level statement, in the proof of Chapter 65.2.9 as preparation for the connecting map. *independent at the first arrow.*

Sign and convention. Chenevier pseudo-character convention uses the Taylor T -normalisation; Bellaïche–Chenevier uses the same; Kisin 2008 uses the ad^0 convention for the fixed-determinant twist. All three match. The block quiver conventions $\alpha, \alpha_{01}, \alpha_{12}$ come from Chapter 32.4.5– Chapter 32.4.6, consistent with Chapter 65.2.9.

Ambient category. Block is in $\text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$, Morita translated to $\mathcal{A}_{\mathfrak{B}}\text{-Mod}$; spectral side in $\text{IndCoh}_{\text{Nilp}}$ on the formal completion at $x_{\tau_{\mathfrak{B}}}$; the gerbe cotangent identification is preserved by the Emerton–Gee formal-stack comparison Chapter 65.2.5.

Hypotheses. The only hypothesis is Chapter 39.5.7 (wild-EG enhancement), needed to supply the degree-1 derived structure. No $p \geq 5$ hypothesis enters; the pseudo-character deformation functor is pro-representable at every prime.

Functoriality. Both factor isomorphisms are natural in $\bar{\rho}_{\mathfrak{B}}$; the pseudo-character functor is natural in $\mathrm{Tr} \bar{\rho}$; the block quiver arrow labels are natural in $\mathfrak{B}$ under Morita equivalence. Composition preserves naturality.

Unproved equivalences. No new unproved equivalence introduced beyond the shared wild-EG enhancement. The pseudo-character pro-representability is unconditional (Chenevier 2014). The block centraliser structure at $p \in \{2, 3\}$ is unconditional (Paškūnas–Tung 2020, CDP 2014).

Numerical comparison. No numerical comparison is invoked.

The seven conditions hold: the class-match is ProvedHereConditional by a proof independent from Chapter 65.2.9 at the first arrow, with the same conditional dependency on the wild-EG enhancement Chapter 39.5.7.

91.5 WILD CLASS BOUNDARY

- (i) Chapter 65.2.9 identifies the wild-prime block Ext^1 with Galois cohomology via the Colmez Montreal functor’s connecting map; Paškūnas 2013 §4.11 is invoked for this step, technically outside the $p \geq 5$ domain of the Montreal functor’s original construction.
- (ii) Chapter 91.3.1 replaces the first arrow by the pseudo-character tangent, using Paškūnas–Tung 2020 (algebra-level $p = 2$ block centraliser), CDP 2014 §5 (algebra-level $p = 3$), Chenevier 2014 pseudo-character pro-representability, and Kisin / Bellaïche–Chenevier tangent identification. The class match agrees with Chapter 65.2.9 up to scalar.
- (iii) Independence at the first arrow is verified; both derivations share only the wild-EG enhancement Chapter 39.5.7.
- (iv) The realization pair for the class-match is $(c_p^{\mathrm{class}}, c_p^{\mathrm{class}, \mathfrak{B}})$ with independent derivations.
- (v) The seven source, convention, and domain conditions hold; no new residual is introduced; no numerical constant.

91.6 SOURCES

- V. Paškūnas and S.-N. Tung, “Finiteness properties of the category of mod p representations of $GL_2(\mathbb{Q}_p)$ ” (2020).
- P. Colmez, G. Dospinescu, and V. Paškūnas, “The p -adic local Langlands correspondence for $GL_2(\mathbb{Q}_3)$ ”, Cambridge J. Math. 2 (2014), 1–47.
- Y. Hu and F. Tan, “On the Breuil–Mézard conjecture at the very-special stratum for $GL_2(\mathbb{Q}_p)$ ” (2018).
- G. Chenevier, “The p -adic analytic space of pseudo-characters of a profinite group and pseudo-representations over arbitrary rings”, in *Automorphic Forms and Galois Representations*, London Math. Soc. Lecture Note Ser. 414, CUP (2014).
- J. Bellaïche and G. Chenevier, *Families of Galois Representations and Selmer Groups*, Astérisque 324, SMF (2009).

- M. Kisin, “Potentially semi-stable deformation rings”, J. AMS 21 (2008), 513–546.
- M. Emerton and T. Gee, “Moduli stacks of étale (φ, Γ) -modules and the existence of crystalline lifts”, arXiv:1908.07185 (2020).
- M. Auslander, I. Reiten, and S. Smalø, *Representation Theory of Artin Algebras*, Cambridge Studies in Advanced Mathematics 36, CUP (1995).

RAMANUJAN FOR MAASS FORMS: SHARPENING $7/64$ AND A SUB-RH CONDITIONAL

92.1 DOMAIN AND TARGET

Fix the target. For $\mathrm{GL}_2/\mathbb{Q}$ with trivial central character, a Maass cuspidal Hecke eigenform ϕ at level 1 has local Satake parameters $(\alpha_p(\phi), \beta_p(\phi))$ with $\alpha_p\beta_p = 1$ and $\theta_p(\phi) := \max(\log |\alpha_p|, \log |\beta_p|)/\log p \geq 0$. The Ramanujan conjecture asserts $\theta_p(\phi) = 0$ for all p and all cuspidal ϕ . The current best unconditional bound is Kim–Sarnak 2003 Appendix 2 to *J. Amer. Math. Soc.* 16, $\sup_p \theta_p(\phi) \leq 7/64 = 0.109375$, established via the symmetric-fourth-power lift $\mathrm{sym}^4\phi$ on $\mathrm{GL}_5/\mathbb{Q}$ of Kim 2003 *J. Amer. Math. Soc.* 16 combined with a Luo–Rudnick–Sarnak 1995 *Geom. Funct. Anal.* 5 L -function convexity argument.

Target. Compare the attempted sub- $7/64$ unconditional construction through higher symmetric-power functoriality (Newton–Thorne 2021 *Publ. Math. IHES* 134; Clozel–Harris–Taylor 2008 *Publ. Math. IHES* 108 as prerequisite automorphy-of-lifts). The unconditional output remains Kim–Sarnak $7/64$; the sharper bound is conditional on Maass symmetric-power functoriality, and full Ramanujan is conditional on an explicit automorphic Ramanujan input.

Domain restriction. The chapter treats level 1 Maass forms on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$ with trivial nebentypus; extension to arbitrary level and nebentypus follows the same argument and is not used later. All estimates are archimedean-uniform in the sense of Iwaniec 2002 §14.4.

92.2 CONSTRUCTION A: KIM–SHAHIDI SYMMETRIC-CUBE AND THE $1/9$ BASELINE

The Symmetric-Cube Lift

THEOREM 92.2.1 (*Kim–Shahidi 2002: $\theta \leq 1/9$*). (Kim–Shahidi 2002 *Ann. Math.* 155, Thm. 5.1). Let ϕ be a Maass cuspidal Hecke eigenform of $\mathrm{GL}_2/\mathbb{Q}$. Then $\sup_p \theta_p(\phi) \leq 1/9 \approx 0.111$.

Proof. Kim–Shahidi 2002 Thm. 5.1 establishes the symmetric-cube lift $\mathrm{sym}^3\phi$ as a cuspidal automorphic representation of $\mathrm{GL}_4/\mathbb{Q}$, building on the symmetric-square lift of Gelbart–Jacquet 1978 *Ann. Sci. École Norm. Sup.* 11. The bound $\theta_p(\phi) \leq 1/9$ follows from Kim–Shahidi 2002 Thm. 5.1 via the Rankin–Selberg analysis of $L(s, \mathrm{sym}^3\phi \times (\mathrm{sym}^3\phi)^\vee)$ on $\mathrm{GL}_4 \times \mathrm{GL}_4$, combined with the exterior-square lift (Kim–Shahidi 2002 Thm. 4.2 for Λ^2 on GL_4) and the Jacquet–Shalika 1981 *Amer. J. Math.* 103 non-vanishing on $\mathrm{Re}(s) = 1$. The naive Jacquet–Shalika bound for a cuspidal $\mathrm{GL}_n/\mathbb{Q}$ representation gives $\sup_p \theta_p \leq 1/2 - 1/(n^2 + 1)$; for $n = 4$ this gives $\theta_p(\mathrm{sym}^3\phi) \leq 1/2 - 1/17$, hence $\theta_p(\phi) \leq (1/2 - 1/17)/3 \approx 0.147$ by the extremal Satake parameter relation $\theta_p(\mathrm{sym}^3\phi) = 3\theta_p(\phi)$. The sharper bound $\theta_p(\phi) \leq 1/9$ is the content of Kim–Shahidi 2002 Thm. 5.1, obtained through the interaction of the exterior-square GL_6 factor and the Rankin–Selberg analysis; the proof is in Kim–Shahidi 2002 §5. $\square$

The Sym^4 Barrier and the Jacquet–Shalika Mechanism

Kim 2003’s symmetric-fourth lift $\text{sym}^4\phi$ on $\text{GL}_5/\mathbb{Q}$ gives the sharper bound $7/64$ via the Kim–Sarnak 2003 adapted Lindelöf-on-average argument (Appendix 2, Kim 2003 *J. Amer. Math. Soc.* 16). The naive Jacquet–Shalika bound for GL_5 gives $\theta \leq 1/2 - 1/26$, which via the extremal Satake parameter relation $\theta_p(\text{sym}^4\phi) = 4\theta_p(\phi)$ yields $\theta_p(\phi) \leq (1/2 - 1/26)/4 \approx 0.1154$, already sharper than Construction A’s $1/9 \approx 0.1111$. The Kim–Sarnak refinement (Appendix 2) tightens this to $7/64 = 0.109375$; Construction B records this as the unconditional ceiling available here.

92.3 CONSTRUCTION B: KIM 2003 SYM^4 AND THE $7/64$ BASELINE

THEOREM 92.3.1 (*Kim–Sarnak 2003: $\theta \leq 7/64$*). (Kim–Sarnak appendix to Kim 2003 *J. Amer. Math. Soc.* 16, Thm. 2). Let ϕ be a Maass cuspidal Hecke eigenform of $\text{GL}_2/\mathbb{Q}$. Then $\sup_p \theta_p(\phi) \leq 7/64 = 0.109375$.

Proof. Kim 2003 establishes $\text{sym}^4\phi$ as a cuspidal automorphic representation of $\text{GL}_5/\mathbb{Q}$ under suitable conditions on ϕ ; the cases excluded are dihedral, tetrahedral, octahedral automorphic induction, all of which are tempered by independent Deligne–Serre / Langlands–Tunnell arguments. Combined with the exterior-square lift $\Lambda^2 \text{sym}^2\phi$ (Kim 2003 Thm. B), Kim–Sarnak analyse the Rankin–Selberg L -function $L(s, \text{sym}^4\phi \otimes \widetilde{\text{sym}^4\phi})$ via the Jacquet–Piatetski-Shapiro–Shalika 1983 *Amer. J. Math.* 105 convexity bound refined by Hoffstein–Lockhart 1994 *Ann. Math.* 140. The arithmetic rearrangement (Kim–Sarnak Prop. 1 and explicit numerology of their §2) produces $\theta \leq 7/64$. $\square$

Remark 92.3.2 (sharpness of $7/64$ in Construction B). The $7/64$ bound is sharp along the sym^4 construction: further tightening through sym^4 alone requires either a subconvex bound for $L(s, \text{sym}^4\phi)$ (open for Maass ϕ) or cancellation not visible in the GL_5 Langlands L -function. Construction C does not break this barrier unconditionally; it names the extra Maass symmetric-power input needed to remove it.

92.4 CONSTRUCTION C: NEWTON–THORNE SYMMETRIC-POWER FUNCTORIALITY AND THE SHARPENING

The Newton–Thorne Theorem

THEOREM 92.4.1 (*Newton–Thorne 2021: unconditional holomorphic sym^n functoriality*). (Newton–Thorne 2021 *Publ. Math. IHES* 134, Main Theorem; extended in Newton–Thorne 2023 *Publ. Math. IHES* 137 to the non-CM case with full generality). Let ϕ be a holomorphic cuspidal Hecke newform of weight $k \geq 2$ on $\text{SL}_2(\mathbb{Z})$ that is non-CM. Then for every $n \geq 1$, the symmetric- n th-power lift $\text{sym}^n\phi$ exists as a cuspidal automorphic representation of $\text{GL}_{n+1}/\mathbb{Q}$.

Proof. Newton–Thorne 2021 (holomorphic, $k \geq 2$). The argument proceeds by automorphy-lifting through Galois deformation rings of the ℓ -adic representation associated to ϕ , combined with $R = T$ theorems in the tame Taylor–Wiles patching framework (Taylor–Wiles 1995 *Ann. Math.* 141, extended by Clozel–Harris–Taylor 2008 *Publ. Math. IHES* 108 to the GL_n setting). $\square$

Remark 92.4.2 (Newton–Thorne does not cover Maass directly). The Newton–Thorne theorem is stated for *holomorphic* cuspidal newforms, where the attached ℓ -adic Galois representation (Deligne 1971 *Sém. Bourbaki* 355 for $k \geq 2$; Deligne–Serre 1974 *Ann. Sci. ENS.* 7 for $k = 1$) is the input to the automorphy-lifting theorem. Maass cusp forms lie outside the hypothesis of Newton–Thorne 2021: the automorphy-lifting theorem requires an ℓ -adic Galois representation of algebraic Hodge weight ≥ 1 , and

the conjectural Langlands-attached representation of a Maass form has weight 0 in the Hodge sense. The holomorphic companion construction gives a diagnostic Rankin–Selberg convolution $L(s, \phi \times \text{sym}^n F)$, but it does not transfer temperedness of $\text{sym}^n F$ into a sub-7/64 bound for $\theta_p(\phi)$.

The Auxiliary Holomorphic Form and the Transfer

The attempted sharpening of 7/64 through a holomorphic companion is only a test of source separation. It supplies no division of the Maass Satake exponent by n . A genuine sharper bound requires the symmetric power of the Maass form itself.

Definition 92.4.3 (Maass–holomorphic Rankin–Selberg pairing). Let ϕ be a Maass cusp form of $\text{GL}_2/\mathbb{Q}$ and F a holomorphic cuspidal newform of weight $k \geq 2$. The *Maass–holomorphic Rankin–Selberg L -function* is

$$L(s, \phi \times F) := \prod_p L_p(s, \phi \times F) = \prod_p \prod_{i,j=1,2} (1 - \alpha_{p,i}(\phi) \alpha_{p,j}(F) p^{-s})^{-1},$$

with $\alpha_{p,i}(\phi) \in \{\alpha_p(\phi), \beta_p(\phi)\}$ and analogously for F . The completed L -function $\Lambda(s, \phi \times F)$ satisfies the Jacquet–Piatetski-Shapiro–Shalika 1983 functional equation.

PROPOSITION 92.4.4 (RS convolution for a Maass form with a tempered companion). Let ϕ be a Maass cusp form and F a holomorphic cuspidal newform of weight $k \geq 2$, non-CM. Fix $n \geq 4$. The symmetric- n th-power lift $\text{sym}^n F$ (Newton–Thorne 2021) is a cuspidal tempered automorphic representation of $\text{GL}_{n+1}/\mathbb{Q}$. The Rankin–Selberg L -function $L(s, \phi \times \text{sym}^n F)$, as a $\text{GL}_2 \times \text{GL}_{n+1}$ L -function, admits a functional equation, meromorphic continuation to $\mathbb{C}$, and satisfies: $L(s, \phi \times \text{sym}^n F) \neq 0$ for $\text{Re}(s) \geq 1$ (Jacquet–Shalika 1981).

Proof. Newton–Thorne 2021 (Chapter 92.4.1) gives $\text{sym}^n F$ as a cuspidal $\text{GL}_{n+1}/\mathbb{Q}$ representation. Deligne 1974 (Weil conjectures) gives temperedness. Jacquet–Piatetski-Shapiro–Shalika (JPSS) 1983 *Amer. J. Math.* 105 establishes the Rankin–Selberg L -function theory for $\text{GL}_2 \times \text{GL}_{n+1}$, including the functional equation and meromorphic continuation. Jacquet–Shalika 1981 non-vanishing on $\text{Re}(s) = 1$ applies to the cuspidal GL_{n+1} -factor $\text{sym}^n F$. $\square$

The Sharpened Bound: Construction C Main Theorem

Remark 92.4.5 (why the companion trick is conditional). Jacquet–Shalika bounds the two automorphic factors separately. It does not divide the GL_{10} exponent of $\text{sym}^9 F$ by nine to bound the Maass form ϕ . A sharper Maass bound therefore requires functoriality for $\text{sym}^9 \phi$ itself.

THEOREM 92.4.6 (sharpened Ramanujan bound: conditional on Maass symmetric-power functoriality). (conditional on the Maass symmetric-ninth-power conjecture: that $\text{sym}^9 \phi$ exists as a cuspidal automorphic representation of $\text{GL}_{10}/\mathbb{Q}$ for the Maass form ϕ itself). Let ϕ be a cuspidal Maass Hecke eigenform of $\text{GL}_2/\mathbb{Q}$. Assuming $\text{sym}^9 \phi$ is cuspidal on $\text{GL}_{10}/\mathbb{Q}$, the Jacquet–Shalika 1981 bound applied to the GL_{10} form $\text{sym}^9 \phi$ gives

$$\sup_p \theta_p(\phi) \leq \frac{1/2 - 1/(10^2 + 1)}{9} = \frac{1/2 - 1/101}{9} \approx 0.05446,$$

strictly below Kim–Sarnak’s $7/64 \approx 0.1094$.

Proof. Conditional on $\text{sym}^9 \phi$ cuspidal automorphic on $\text{GL}_{10}/\mathbb{Q}$ (the Maass symmetric-ninth-power conjecture), the Jacquet–Shalika 1981 *Amer. J. Math.* 103 non-vanishing of $L(s, \text{sym}^9 \phi \otimes (\text{sym}^9 \phi)^\vee)$ on $\text{Re}(s) = 1$ forces the Satake parameters of $\text{sym}^9 \phi$ on GL_{10} to satisfy $\theta_p(\text{sym}^9 \phi) \leq 1/2 - 1/101$. The extremal Satake parameter of $\text{sym}^9 \phi$ at p is $p^{9\theta_p(\phi)}$, so $9\theta_p(\phi) \leq 1/2 - 1/101$, giving the stated bound. $\square$

Remark 92.4.7 (form of Maass symmetric-power functoriality). Newton–Thorne 2021 establishes $\text{sym}^n \phi$ cuspidal on $\text{GL}_{n+1}/\mathbb{Q}$ for *holomorphic* newforms ϕ . The analogous statement for Maass forms is the content of the Langlands global functoriality conjecture for $\text{GL}_2 \rightarrow \text{GL}_{n+1}$ at non-algebraic archimedean type; it is open for all $n \geq 5$. The holomorphic companion trick – forming $L(s, \phi \times \text{sym}^n F)$ for a holomorphic form F – does not produce a bound on $\theta_p(\phi)$ sharper than the direct GL_2 Jacquet–Shalika bound $\theta_p(\phi) \leq 2/5$; the Rankin–Selberg non-vanishing constrains the GL_{10} factor $\text{sym}^n F$, not the GL_2 factor ϕ . The bound $(1/2 - 1/101)/9$ in Chapter 92.4.6 is the correct conditional bound on the Maass symmetric-power conjecture, not an unconditional sharpening.

COROLLARY 92.4.8 (*Maass-sector residual: local conditional sharpening*). (*conditional on the Maass symmetric-ninth-power conjecture, as in Chapter 92.4.6*). Under the Maass symmetric-ninth-power conjecture, the local non-tempered discrepancy in the P_3 identity of Chapter 54.2.3 satisfies, at each prime p ,

$$|\langle v_p, w_p \rangle_p^{\text{Pet}} - \langle v_p, \sigma_p^{\text{BC}}(w_p) \rangle_p^{\text{Koszul}}| \leq C_p p^{-(1/2-\mathfrak{D})}, \quad \mathfrak{D} = \frac{1/2 - 1/101}{9}.$$

Unconditionally, Kim–Sarnak gives the same local statement with $\mathfrak{D} = 7/64$. In neither case is the resulting prime sum summable. Thus the estimate is local control, not an adelic error term.

Proof. Under the Maass symmetric-ninth-power conjecture, $\mathfrak{D} = (1/2 - 1/101)/9$ by Chapter 92.4.6; unconditionally, $\mathfrak{D} \leq 7/64$ by Kim–Sarnak 2003. The primewise mismatch is bounded by Chapter 54.2.4. Since $0 < 1/2 - \mathfrak{D} < 1$, the series $\sum_p p^{-(1/2-\mathfrak{D})}$ diverges; hence no global zeta-majorant is deduced. $\square$

92.5 CONSTRUCTION D: ARTHUR ENDOSCOPIC CLASSIFICATION ON CLASSICAL GROUPS

The automorphic Ramanujan conjecture for a Maass form ϕ of $\text{GL}_2/\mathbb{Q}$ is detected by the tensor square $\phi \boxtimes \phi^\vee$. This is a GL_4 -object, usually isobaric rather than cuspidal. Arthur’s classification gives packet normalisation for the classical-group transfers; it does not prove the Ramanujan conjecture. The missing input is exactly temperedness of the relevant GL_4 isobaric representation.

Arthur’s Endoscopic Classification

THEOREM 92.5.1 (*Arthur 2013: endoscopic classification for classical groups*). (Arthur 2013 *Colloq. Publ.* 61, *The Endoscopic Classification of Representations*, Thm. 1.5.2). For quasi-split classical groups over a number field, the discrete automorphic spectrum is organised by global Arthur parameters. The associated standard transfer to general linear groups is automorphic, isobaric, and compatible with the local Langlands parameters at unramified places. Temperedness is present precisely for tempered Arthur parameters; it is not a formal consequence of the classification.

Proof. Arthur 2013 *The Endoscopic Classification* Thm. 1.5.2 and §4.3 give the parametrisation and the standard transfer. The stabilisation input is the Langlands–Shelstad–Ngô–Waldspurger endoscopic

trace formula. Mœglin–Waldspurger controls the local discrete-series packets. None of these statements replaces the generalised Ramanujan conjecture. $\square$

Remark 92.5.2 (stabilisation is not Ramanujan). For $F = \mathbb{Q}$ the required stabilisation rests on the fundamental lemma and the endoscopic transfer identities. This is a theorem of the trace formula. It identifies packets and transfers. It does not bound Satake parameters.

The Ramakrishnan Functorial Chain

PROPOSITION 92.5.3 (*Ramakrishnan 2000: $\mathrm{GL}_2 \boxtimes \mathrm{GL}_2 \rightarrow \mathrm{GL}_4$*). (Ramakrishnan 2000 *Ann. Math.* 152 *Modularity of the Rankin–Selberg L -series, and multiplicity one for SL_2* , Thm. M). Let ϕ_1, ϕ_2 be cuspidal automorphic representations of $\mathrm{GL}_2/\mathbb{Q}$. Then their Langlands tensor product $\phi_1 \boxtimes \phi_2$ is an automorphic representation of $\mathrm{GL}_4/\mathbb{Q}$; it is cuspidal unless ϕ_1 and ϕ_2 are twists of a common dihedral form.

Proof. Ramakrishnan 2000 Thm. M. The proof constructs the tensor product via the converse theorem of Cogdell–Piatetski-Shapiro 1999 *Publ. Math. IHES* 88, using Bump–Friedberg 1990 *Duke Math. J.* 59 for the Rankin–Selberg L -function. $\square$

The Automorphic Conditional

Hypothesis 92.5.4 (Arthur–Ramakrishnan–GRC hypothesis, AR-GRC). . Assume:

- (AR1) Arthur’s endoscopic classification Chapter 92.5.1 is available for the relevant classical groups, with standard transfer compatible with unramified local parameters.
- (AR2) The Ramakrishnan tensor product Chapter 92.5.3 is used for cuspidal automorphic representations attached to Maass forms. For $\phi_2 = \phi_1^\vee$ the product is interpreted isobarically.
- (AR3) The isobaric representation $\phi \boxtimes \phi^\vee$ satisfies the generalised Ramanujan conjecture at every finite place.

THEOREM 92.5.5 (*AR-GRC implies full Ramanujan for Maass forms*). (conditional on hypothesis Chapter 92.5.4). Under (AR1)+(AR2)+(AR3), every cuspidal Maass Hecke eigenform ϕ of $\mathrm{GL}_2/\mathbb{Q}$ satisfies the Ramanujan bound $\sup_p \theta_p(\phi) = 0$, i.e. is tempered at every finite prime.

Proof. At an unramified finite prime write the unitary Satake parameters of ϕ as α_p, β_p , with $|\alpha_p \beta_p| = 1$. The local parameter of $\phi \boxtimes \phi^\vee$ has eigenvalues

$$1, \quad 1, \quad \alpha_p/\beta_p, \quad \beta_p/\alpha_p.$$

By (AR3) these have modulus 1. Hence $|\alpha_p| = |\beta_p|$, and the central-character normalisation gives $|\alpha_p| = |\beta_p| = 1$. Thus ϕ_p is tempered. The ramified finite primes are controlled by the same boundedness condition on the Weil–Deligne parameter. Arthur’s classification is used only to keep the transfer normalisation fixed. $\square$

PROPOSITION 92.5.6 (*AR-GRC is separate from classical RH in this argument*). . Hypothesis AR-GRC (Chapter 92.5.4) is an automorphic input separate from the classical Riemann hypothesis here. The proof below uses neither RH nor any zero-placement statement for $\zeta(s)$. AR-GRC is supplied by GRC for the relevant GL_4 isobaric tensor square together with the stated transfer inputs.

Proof. GRC for the GL_4 isobaric tensor square gives bounded local parameters. The proof of Chapter 92.5.5 then reads off the GL_2 Satake radii. No statement about zeros of $\zeta(s)$ enters this implication. Conversely, the argument makes no deduction from RH to AR-GRC. $\square$

Remark 92.5.7 (the automorphic residual). Chapter 92.5.5 places the Maass-sector residual of Chapter 54.6.1 on the hypothesis AR-GRC, which is the generalised Ramanujan conjecture for the GL_4 isobaric tensor square, together with the stated transfer normalisations. GRC for this object is open; it is an automorphic temperedness statement, not a consequence of classical RH. The Vol IV P_3 identity on the Maass sector holds without further hypothesis conditional on GRC for GL_4 .

92.6 CONSTRUCTION E: SYNTHESIS

THEOREM 92.6.1 (*Maass-sector P_3 identity: local controls and exact conditional form*). (unconditional for the local Kim–Sarnak control; conditional on the Maass symmetric-ninth-power conjecture for the sharper local control; conditional on GRC for $GL_4/\mathbb{Q}$ for the exact equality). Here $\mathcal{V}^{\text{prim}}$ denotes the primitive automorphic Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 , as defined in Chapter 54, and $\mathcal{V}_{\text{cusp}, \text{Maass}}^{\text{prim}} \subset \mathcal{V}^{\text{prim}}$ its Maass cuspidal sector. For local vectors v_p, w_p in the p -component and global vectors $v, w \in \mathcal{V}_{\text{cusp}, \text{Maass}}^{\text{prim}}$, the available statements are:

$$\begin{aligned} |\langle v_p, w_p \rangle_p^{\text{Pet}} - \langle v_p, \sigma_p^{\text{BC}}(w_p) \rangle_p^{\text{Koszul}}| &\leq C_p p^{-25/64}, \quad \text{unconditionally (Kim–Sarnak),} \\ |\langle v_p, w_p \rangle_p^{\text{Pet}} - \langle v_p, \sigma_p^{\text{BC}}(w_p) \rangle_p^{\text{Koszul}}| &\leq C_p p^{-(1/2 - (1/2 - 1/101)/9)}, \quad \text{conditional on Maass sym}^9, \\ \langle v, w \rangle^{\text{Pet}} &= \langle v, \sigma^{\text{BC}}(w) \rangle^{\text{Koszul}}, \quad \text{conditional on GRC for } GL_4/\mathbb{Q}. \end{aligned}$$

The first two lines do not assemble to an adelic correction term without extra summability; the exact global identity is the third line.

Proof. Unconditional local form (92.6.1): Kim–Sarnak 2003 gives $\sup_p \theta_p(\phi) \leq 7/64$, and Chapter 92.4.8 gives the displayed primewise bound. Conditional sharpened local form (92.6.1): Chapter 92.4.8 under the Maass symmetric-ninth-power conjecture. Exact-equality form (92.6.1): Chapter 92.5.5 gives temperedness of ϕ under AR-GRC; Chapter 54.1.1 then applies to $\mathcal{V}_{\text{cusp}, \text{Maass}}^{\text{prim}} \subset \mathcal{V}_{\text{cusp}, \text{temp}}^{\text{prim}}$. AR-GRC is GRC for the relevant GL_4 isobaric tensor square by Chapter 92.5.6. $\square$

Construction		Bound	Form	Source
Construction (sym ³)	A	$1/9 \approx 0.1111$	Unconditional	Kim–Shahidi 2002, Chapter 92.2.1
Construction (sym ⁴)	B	$7/64 \approx 0.1094$	Unconditional	Kim–Sarnak 2003, Chapter 92.3.1
Construction (sym ⁹ ϕ)	C	$(1/2 - 1/101)/9 \approx 0.0545$	Conditional on Maass sym ⁹	JS 1981 + Maass sym-power conj., Chapter 92.4.6
Construction (AR-GRC)	D	0 (full)	Conditional on GRC _{GL₄}	Arthur 2013 + Ramakrishnan 2000, Chapter 92.5.5

92.7 INDEPENDENT COMPARISON

The following seven independent checks verify the chain of Chapter 92.6.1; the criterion is that of §21.1.

A1 (source separation). Derivation of the unconditional $7/64$ baseline: Kim 2003, Kim–Sarnak 2003, JPSS 1983, Jacquet–Shalika 1981. Conditional Construction C bound: Jacquet–Shalika 1981 + Maass symmetric-power conjecture (open; stated as conditional). Construction D (automorphic conditional): Arthur 2013, Langlands–Shelstad 1987, Ngô 2010, Waldspurger 2008, Chaudouard–Laumon 2012, Ramakrishnan 2000, Cogdell–Piatetski-Shapiro 1999, Labesse–Langlands 1979. No source appears on both the derivation and comparison sides.

A2 (sign/convention). Satake parameter conventions $\alpha_p \beta_p = 1, \theta_p \geq 0$ fixed throughout. The Jacquet–Shalika non-vanishing applies on $\operatorname{Re}(s) = 1$ (the edge of the critical strip); the subconvex bound on the critical line $\operatorname{Re}(s) = 1/2$ remains open for Maass forms. Tensor decomposition: lowest-weight projection is clean.

A3 (ambient category). All constructions live in the automorphic spectrum of $\operatorname{GL}_n/\mathbb{Q}$ for appropriate n . Construction D uses the isobaric GL_4 tensor square and Arthur transfer only for normalisation of packets. No category violations.

A4 (missing hypothesis). Construction C requires $\operatorname{sym}^9 \phi$ cuspidal on $\operatorname{GL}_{10}/\mathbb{Q}$ for the Maass form ϕ : this is the Maass symmetric-power conjecture, open. Construction D requires GRC for $\operatorname{GL}_4/\mathbb{Q}$: open, explicit conditional. Both conditionals are named in the theorem statements. *The conditions are explicit.*

A5 (false functoriality). Newton–Thorne 2021 functoriality applies to holomorphic forms of algebraic weight ≥ 2 ; Construction C invokes it only for the holomorphic companion F , not for ϕ . The Maass symmetric-ninth-power conjecture (Construction C conditional) is explicitly stated as a conjecture, not derived from Newton–Thorne. Ramakrishnan’s tensor product is automorphic; in the contragredient case it is read isobarically. Dihedral forms are handled separately via Deligne–Serre and Labesse–Langlands.

A6 (unproved equivalence). Construction C’s bound $(1/2 - 1/101)/9 \approx 0.0545$ is conditional on the Maass symmetric-ninth-power conjecture (open); labelled in Chapter 92.4.6. Construction D closure is conditional on GRC for GL_4 (open); labelled conditionally in Chapter 92.5.5. The three tiers (unconditional Kim–Sarnak; conditional Maass sym-power; conditional GRC) are distinct and labelled.

A7 (numerical comparison). No numerical comparison for this chapter; all bounds are analytic.

Seven passes (A4 and A6 carry explicit conditionals: Maass sym-power conjecture for Construction C; GRC for $\operatorname{GL}_4/\mathbb{Q}$ for Construction D). The chain: unconditional $\theta \leq 7/64$ (Kim–Sarnak); conditional on Maass sym⁹: $\theta \leq (1/2 - 1/101)/9 \approx 0.0545$; conditional on GRC for GL_4 : full Ramanujan.

92.8 WHAT REMAINS OPEN

GRC for $\operatorname{GL}_4/\mathbb{Q}$. The generalised Ramanujan conjecture for the relevant isobaric $\operatorname{GL}_4/\mathbb{Q}$ tensor square is open; the Construction D closure is conditional on it. The current best unconditional bound is from Kim–Shahidi 2002 extensions to higher-rank groups: $\theta_{\operatorname{GL}_4} \leq 5/14 \approx 0.357$.

Maass symmetric-power functoriality (Construction C conditional). Construction C’s sharpened bound $(1/2 - 1/((n+1)^2 + 1))/n$ is conditional on the Maass symmetric- n th-power conjecture: the existence of $\operatorname{sym}^n \phi$ as a cuspidal automorphic representation of $\operatorname{GL}_{n+1}/\mathbb{Q}$ for the Maass form ϕ itself. This is the Langlands global functoriality conjecture for $\operatorname{GL}_2 \rightarrow \operatorname{GL}_{n+1}$ at non-algebraic archimedean type; it is open for all $n \geq 5$. Newton–Thorne 2021/2023 covers only holomorphic forms (ℓ -adic algebraic weight). The bound $\theta^{(n)} = (1/2 - 1/((n+1)^2 + 1))/n \rightarrow 0$ as $n \rightarrow \infty$ at rate $\sim 1/(2n)$; closing to $\theta = 0$ via this mechanism would require a proof of the Maass symmetric-power conjecture for all n ,

which would imply the Ramanujan conjecture by the Jacquet–Shalika bound. It is stronger than the numerical estimate alone.

The dihedral edge case. Dihedral Maass forms lift from Hecke characters of quadratic fields and are tempered by class field theory (Labesse–Langlands 1979, Deligne–Serre 1974); Construction D applies without the AR2 hypothesis.

92.9 BIBLIOGRAPHY

- Kim, H. and Shahidi, F.** *Functorial products for $GL_2 \times GL_3$ and the symmetric cube for GL_2* , Ann. Math. 155 (2002), 837–893. Symmetric cube lift for cuspidal $GL_2/\mathbb{Q}$; baseline of Construction A.
- Kim, H.** *Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2* , J. Amer. Math. Soc. 16 (2003), 139–183. Symmetric fourth lift; baseline of Construction B.
- Kim, H. and Sarnak, P.** Appendix 2 to Kim 2003. Refined estimates $\theta \leq 7/64$.
- Newton, J. and Thorne, J.** *Symmetric power functoriality for holomorphic modular forms*, Publ. Math. IHES 134 (2021), 1–116; and *Symmetric power functoriality for holomorphic modular forms. II*, Publ. Math. IHES 137 (2023), 117–172. Unconditional cuspidal automorphy of $\text{sym}^n F$ for F non-CM holomorphic $GL_2/\mathbb{Q}$ newform, all $n \geq 1$.
- Clozel, L., Harris, M., and Taylor, R.** *Automorphy for some ℓ -adic lifts of automorphic mod ℓ Galois representations*, Publ. Math. IHES 108 (2008), 1–181. $R = T$ theorems for higher-rank automorphy-lifting.
- Taylor, R. and Wiles, A.** *Ring-theoretic properties of certain Hecke algebras*, Ann. Math. 141 (1995), 553–572.
- Jacquet, H., Piatetski-Shapiro, I., and Shalika, J.** *Rankin–Selberg convolutions*, Amer. J. Math. 105 (1983), 367–464. Local and global Rankin–Selberg L -function theory.
- Jacquet, H. and Shalika, J.** *On Euler products and the classification of automorphic representations*, Amer. J. Math. 103 (1981), 499–558 and 777–815. Edge non-vanishing on $\text{Re}(s) = 1$.
- Hoffstein, J. and Lockhart, P.** *Coefficients of Maass forms and the Siegel zero*, Ann. Math. 140 (1994), 161–181. Edge non-vanishing refinement.
- Michel, P. and Venkatesh, A.** *The subconvexity problem for GL_2* , Publ. Math. IHES 111 (2010), 171–271.
- Deligne, P.** *La conjecture de Weil. I*, Publ. Math. IHES 43 (1974), 273–307. Proof of the Weil conjectures; the Ramanujan bound $\theta_p(F) = 0$ for holomorphic newforms of weight $k \geq 2$ is Corollaire 3.3.8. Used in Construction C to confirm $\text{sym}^n F$ is tempered.
- Arthur, J.** *The Endoscopic Classification of Representations: Orthogonal and Symplectic Groups*, Amer. Math. Soc. Colloq. Publ. 61, 2013. Theorem 1.5.2 used in Construction D.
- Langlands, R. P. and Shelstad, D.** *On the definition of transfer factors*, Math. Ann. 278 (1987), 219–271.
- Ngô, B. C.** *Le lemme fondamental pour les algèbres de Lie*, Publ. Math. IHES 111 (2010), 1–169.
- Waldspurger, J.-L.** *L’endoscopie tordue n’est pas si tordue*, Mém. Soc. Math. France 115 (2008).
- Chaudouard, P.-H. and Laumon, G.** *Le lemme fondamental pondéré. I: Construction géométrique*, J. Amer. Math. Soc. 25 (2012), 1–79.
- Ramakrishnan, D.** *Modularity of the Rankin–Selberg L -series, and multiplicity one for SL_2* , Ann. Math. 152 (2000), 45–111. $GL_2 \boxtimes GL_2 \rightarrow GL_4$ tensor product.

Cogdell, J. W. and Piatetski-Shapiro, I. *Converse theorems for GL_n . II*, J. Reine Angew. Math. 507 (1999), 165–188; and *Publ. Math. IHES* 88 (1998), 5–39.

Labesse, J.-P. and Langlands, R. P. *L -indistinguishability for $SL(2)$* , Canad. J. Math. 31 (1979), 726–785. Dihedral subcase reduction.

Deligne, P. *Formes modulaires et représentations ℓ -adiques*, Sémin. Bourbaki 355 (1971). Construction of the ℓ -adic Galois representation attached to a holomorphic newform F of weight $k \geq 2$; prerequisite for automorphy-lifting.

Deligne, P. and Serre, J.-P. *Formes modulaires de poids 1*, Ann. Sci. ÉNS. 7 (1974), 507–530.

Ribet, K. A. *Galois representations attached to eigenforms with nebentypus*, Glasgow Math. J. 27 (1977). CM form classification.

Selberg, A. *Old and new conjectures and results about a class of Dirichlet series*, Collected Papers II, Springer 1989, 47–63. Selberg class and orthogonality.

Luo, W., Rudnick, Z., and Sarnak, P. *On the generalised Ramanujan conjecture for $GL(n)$* , Proc. Symp. Pure Math. 66 (1999), 301–310.

Iwaniec, H. *Spectral Methods of Automorphic Forms* (second edition), AMS Graduate Studies 53, 2002.

Moeglin, C. and Waldspurger, J.-L. *Le spectre résiduel de $GL(n)$* , Ann. Sci. ÉNS 22 (1989), 605–674.

Disjoint-source discipline. The unconditional derivation list (Kim–Shahidi 2002, Kim 2003, Kim–Sarnak 2003, JPSS 1983, Jacquet–Shalika 1981) is disjoint from the Construction D comparison list (Arthur 2013, Langlands–Shelstad 1987, Ngô 2010, Waldspurger 2008, Chaudouard–Laumon 2012, Ramakrishnan 2000, Cogdell–Piatetski-Shapiro 1999, Labesse–Langlands 1979). The two lists share only JPSS 1983 and Jacquet–Shalika 1981, used structurally (defining L -functions and establishing non-vanishing on $\text{Re}(s) = 1$), not to establish bounds. Construction C depends on the Maass symmetric-power conjecture (open); as a conditional, no disjointness check is required for its comparison source.

QUANTUM ARITHMETIC CHERN–SIMONS: BULK EXTENSION AND RESIDUAL OBSTRUCTION

Chapter 81 states Chapter 81.4.7, the profinite $GL_2(\widehat{\mathbb{Z}})$ quantum arithmetic Chern–Simons theory on $\overline{\text{Spec } \mathbb{Z}}$, as three linked bulk-boundary axioms. Every formal consequence depends on its resolution: Ch. 84’s continuous Wilson–Frobenius identity at $\varprojlim_N^{\text{ren}}$, Ch. 89’s identification $F_\infty^{\text{WS}} = F_\infty$, the partition formula $Z_{\text{ACS}^{\text{qu}}} = \xi$ of Chapter 81.5.7, and the triangle of Chapter 82.4.2. The bulk measure is examined through five constructions: (i) state-sum renormalisation along the congruence tower; (ii) Reshetikhin–Turaev theory for a modular trace lift of the boundary trace class; (iii) Costello–Gwilliam factorization algebras on the étale site of $\text{Spec } \mathbb{Z}$; (iv) Lurie cobordism hypothesis on an étale arithmetic cobordism category; (v) Scholze six-functor on a conjectural arithmetic diamond. Each finite-level state sum and the open $\text{Spec } \mathbb{Z} \setminus \{\infty\}$ factorization algebra is unconditional. The remaining statements require the archimedean condensed-Hilbert descent, the modular trace lift, or the arithmetic diamond.

93.1 SETUP AND AXIOMATIC TARGET

Definition 93.1.1 (arithmetic 3-space). . The *arithmetic 3-space* is $M = \overline{\text{Spec } \mathbb{Z}}$, the Arakelov compactification of $\text{Spec } \mathbb{Z}$ by the archimedean place $\{\infty\}$, equipped with its étale site $(M)_{\text{ét}}$, its Artin–Verdier fundamental class $[M] \in H_{\text{ét}}^3(M, \widehat{\mathbb{Z}})$, and the Faltings Kähler form ω_∞ on $\Sigma_\infty = \{\infty\} \times D^2$ of Chapter 89.1.4.

Definition 93.1.2 (congruence tower and gauge tower). . The *gauge tower* is the inverse system of finite groups

$$\mathcal{T} := \left\{ G_N := GL_2(\mathbb{Z}/N\mathbb{Z}), \quad N \in \mathbb{Z}_{\geq 1} \right\}$$

ordered by divisibility $M \mid N$ giving reductions $G_N \rightarrow G_M$, with

$$\varprojlim_{N \in \mathcal{T}} G_N = GL_2(\widehat{\mathbb{Z}}).$$

A *level datum* is a compatible system $\alpha_\bullet = \{\alpha_N \in H^3(BG_N, \mathbb{C}^\times)\}_N$ whose inverse-limit class is the arithmetic level $k^{\text{Ar}} = \kappa_G^{\text{Ar}}$ of Ch. 81.

Remark 93.1.3 (the three axioms recalled). Chapter 81.4.7 asks for a renormalised profinite limit

$$\text{ACS}_{k^{\text{Ar}}}^{\text{qu}} := \varprojlim_N^{\text{ren}} Z_{G_N, \alpha_N}^{\text{Ar}}$$

satisfying: **(WZW)** $\text{HH}_0(\partial \text{ACS}_{k^{\text{Ar}}}^{\text{qu}}) \simeq \mathcal{V}^{\text{prim}}$ on the archimedean boundary, where $\mathcal{V}^{\text{prim}}$ is the Hochschild trace class of the identity on the global arithmetic Iwahori–Hecke category of GL_2 (Ch. 81); **(Z)** $Z_{\text{ACS}^{\text{qu}}}(\overline{\text{Spec } \mathbb{Z}}; k^{\text{Ar}}) = \xi(k^{\text{Ar}})$, where ξ is the completed Riemann zeta function (Chapter 81.5.7); **(W)** $\langle W_\rho(K_p) \rangle = \text{tr } \rho(\text{Frob}_p)$, where $W_\rho(K_p)$ is the Wilson-loop observable at the prime knot $K_p \subset \overline{\text{Spec } \mathbb{Z}}$ in representation ρ (Ch. 84).

93.2 CONSTRUCTION I: TOWER RENORMALISATION VIA KAN EXTENSION

The most direct construction is the one inherited from the finite-level state-sum construction of Chapter 8I.4.2. Each $Z_{G_N, \alpha_N}^{\text{Ar}}$ is an unconditional finite state-sum assignment. The obstruction is the existence of a profinite functor compatible with (WZW), (Z), (W).

Finite-level Kan tower

LEMMA 93.2.1 (*reduction compatibility at finite levels*). . For $M \mid N$, the reduction $\pi_{N,M} : G_N \twoheadrightarrow G_M$ induces a pullback of level classes $\pi_{N,M}^* : H^3(BG_M, \mathbb{C}^\times) \rightarrow H^3(BG_N, \mathbb{C}^\times)$ and a pushforward of field groupoids $(\pi_{N,M})_* : \mathcal{F}_{G_N}(M') \rightarrow \mathcal{F}_{G_M}(M')$ for every arithmetic 3-space M' . With $\alpha_N = \pi_{N,M}^* \alpha_M$ consistent, one has

$$Z_{G_N, \alpha_N}^{\text{Ar}}(M') = |\ker \pi_{N,M}|^{b_1(M')} \cdot Z_{G_M, \alpha_M}^{\text{Ar}}(M') \quad (93.1)$$

where $b_1(M') = \dim_{\mathbb{F}_\ell} H_{\text{ét}}^1(M', \mathbb{F}_\ell)$ is the first étale Betti number with $\mathbb{F}_\ell$ -coefficients for an auxiliary torsion prime ℓ ; for $M' = \overline{\text{Spec } \mathbb{Z}}$ this vanishes by the Minkowski bound, independently of ℓ .

Proof. The pullback of cohomology is functorial; the pushforward of field groupoids is the restriction of scalars $\mathcal{F}_{G_N}(M') = \mathcal{F}_{G_M}(M') \times_{\pi_{N,M}} BG_N$. The finite-groupoid cardinality of the kernel of $\pi_{N,M}$ acting free-transitively on the H^1 -twist gives the displayed factor $|\ker \pi_{N,M}|^{b_1(M')}$ (cf. Freed–Quinn 1993 *Commun. Math. Phys.* 156, §4 for the Dijkgraaf–Witten version). The level class $\alpha_N = \pi_{N,M}^* \alpha_M$ multiplies each field by its G_M -image phase, unchanged by the torsor of reductions. $\square$

REMARK 93.2.2 (*divergence is pre-determined*). Equation (93.1) exposes the precise source of divergence: the state sum grows by the volume-factor $|\ker \pi_{N,M}|^{b_1(M')}$ as $N \uparrow$, so a naive inverse limit $\varprojlim_N Z_{G_N, \alpha_N}^{\text{Ar}}$ is ill-defined. The renormalisation is this volume-factor quotient.

DEFINITION 93.2.3 (*renormalised tower*). . The *renormalised state-sum tower* is

$$\tilde{Z}_{G_N, \alpha_N}^{\text{Ar}}(M') := |G_N|^{-\chi(M')} \cdot Z_{G_N, \alpha_N}^{\text{Ar}}(M'), \quad (93.2)$$

where $\chi(M')$ is the étale Euler characteristic of M' with $\mathbb{F}_\ell$ -coefficients, so that the divergence factor (93.1) is absorbed into the normalization $|G_N|^{-\chi}$.

PROPOSITION 93.2.4 (*renormalised tower is compatible*). . With the renormalisation of Chapter 93.2.3, for every pair $M \mid N$, the reduction map satisfies

$$\tilde{Z}_{G_N, \alpha_N}^{\text{Ar}}(M') = \tilde{Z}_{G_M, \alpha_M}^{\text{Ar}}(M') \cdot [G_N : G_M]^{-\chi(M') + b_1(M')}. \quad (93.3)$$

For $M' = \overline{\text{Spec } \mathbb{Z}}$, where $\chi(M') = 0$ by Artin–Verdier duality and $b_1(M') = 0$ by the Minkowski bound on the arithmetic curve, the exponent is zero: the renormalised tower is a cocycle on the congruence tower.

Proof. Substitute (93.1) and the identity $|G_N| = |G_M| \cdot |\ker \pi_{N,M}| = |G_M| \cdot [G_N : G_M]$ into (93.2). The ratio $\tilde{Z}_N / \tilde{Z}_M$ is $|G_N|^{-\chi} \cdot [G_N : G_M]^{b_1} \cdot |G_M|^\chi = [G_N : G_M]^{-\chi + b_1}$, which is the exponent in (93.3). For $M' = \overline{\text{Spec } \mathbb{Z}}$, $\chi = 0$ by Artin–Verdier and $b_1 = 0$ by the cohomological 3-dimensionality (Mazur 1973, Artin–Verdier); the net exponent is zero. $\square$

THEOREM 93.2.5 (*renormalised tower admits a limit as a pro-object*). The renormalised tower $\{\tilde{Z}_{G_N, \alpha_N}^{\text{Ar}}\}_N$ admits a canonical renormalised profinite limit in the ∞ -category of pro-objects in the category of state-sum functors on finite arithmetic cobordism data, given by the ∞ -categorical Kan right extension along the reduction tower.

Proof. By Chapter 93.2.4, the renormalised tower is a levelwise-coherent pro-functor; the formal ∞ -categorical right Kan extension along the divisibility tower produces the limit as a pro-object. The finite groupoid cardinality is preserved under reduction; therefore the limit $\tilde{Z}_{GL_2(\widehat{\mathbb{Z}}), \alpha_\bullet}^{\text{Ar}}$ is a well-defined element of the pro-category. $\square$

Remark 93.2.6 (*what Construction I has and has not produced*). Chapter 93.2.5 produces the renormalised inverse limit as a pro-object and confirms the tower is coherent. This is less than what Chapter 81.4.7 requires: namely, that the pro-object *descend to a concrete topological functor* on a genuine profinite arithmetic cobordism category, and that its archimedean-boundary Hochschild trace equal $\mathcal{V}^{\text{prim}}$. The descent step is the profinite-measure step. It is not implied by Kan extension alone.

Sources and Hypotheses for Construction I

The finite-level state sum uses Kim arithmetic Chern–Simons and the Chern–Dan–Kim–Park non-abelian extension. The renormalisation is the Dijkgraaf–Witten/Freed–Quinn factor $|G_N|^{-\chi}$, and the inverse-limit passage is a Kan extension. Since $\chi(\overline{\text{Spec } \mathbb{Z}}) = 0$, the closed-base renormalisation factor is 1. The limit is a pro-vector-space object. A topological state functor requires a further condensed or Tate topology on this pro-object.

THEOREM 93.2.7 (*Construction I conditional closure*). (on a choice of condensed topology on the Kan-limit and its descent to a concrete topological state-functor). Under the profinite-Hilbert descent hypothesis, the Kan limit of Chapter 93.2.5 defines $\text{ACS}_{k^{\text{Ar}}}^{\text{qu}}$ satisfying the reduction-coherence portion of Chapter 81.4.7. The three axioms (WZW), (Z), (W) reduce to: (WZW) restriction of the Kan-limit boundary functor at Σ_∞ identifies with $\mathcal{V}^{\text{prim}}$; (Z) evaluation of the Kan-limit on the closed class $[M]$ reproduces $\xi(k^{\text{Ar}})$; (W) evaluation on prime-knot insertions reproduces Frobenius traces. Each of (WZW), (Z), (W) becomes a *comparison statement* at the Kan-limit, not a new existence statement.

Proof. The Kan-limit exists by Chapter 93.2.5. Under a chosen condensed topology descending the pro-vector-space structure to a Hilbert-space functor, (WZW), (Z), (W) become comparison claims at the limit. Each is in principle checkable by comparing the limit with the right-hand sides of the three axioms using the finite-level comparisons of Chs. 84 and 81. The existence of the descent is hypothetical. $\square$

93.3 CONSTRUCTION II: RESHETIKHIN–TURAEV FROM AN ARITHMETIC BOUNDARY CATEGORY

The Reshetikhin–Turaev construction (Turaev 1994 *Quantum Invariants of Knots and 3-Manifolds*, de Gruyter; Reshetikhin–Turaev 1991 *Invent. Math.* 103, 547–597) builds a 3D TQFT from a modular tensor category (MTC) by state-sum over triangulations. In the arithmetic problem the input cannot be the category of modules over $\mathcal{V}^{\text{prim}}$, because $\mathcal{V}^{\text{prim}}$ is a Hochschild trace class, not a vertex algebra. The possible Reshetikhin–Turaev input is instead a modular tensor category whose Hochschild trace and regularised character have $\mathcal{V}^{\text{prim}}$ as their boundary shadow.

The arithmetic MTC candidate

Definition 93.3.1 (*arithmetic modular tensor category candidate*). The *arithmetic modular tensor category candidate* is

$$\mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}} := \mathcal{B}_{k^{\text{Ar}}}^{\text{Ar}},$$

a finite semisimple ribbon category equipped with a trace comparison

$$\text{HH}_0(\mathcal{B}_{k^{\text{Ar}}}^{\text{Ar}}) \simeq \mathcal{V}^{\text{prim}}$$

and with Frobenius-labelled simple objects at the finite places. This definition is a lifting problem from a Hochschild trace class to a boundary category; it is not a module category over $\mathcal{V}^{\text{prim}}$.

Remark 93.3.2 (*the modular trace lift is conditional*). Modularity of $\mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}}$ requires a non-degenerate braiding, a spherical trace, and compatibility of the categorical trace with the Iwahori–Hecke Hochschild trace. The arithmetic Zhu modularity of Ch. 23 is a character statement on per-prime factors. It does not construct a braided category whose trace is $\mathcal{V}^{\text{prim}}$.

Reshetikhin–Turaev state-sum

THEOREM 93.3.3 (*Reshetikhin–Turaev from MTC candidate*). (on modularity of $\mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}}$). If $\mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}}$ is modular, then there is a 3D TQFT functor

$$Z_{\text{RT}}^{\text{Ar}} : \text{Bord}_3^{\text{Ar}} \rightarrow \text{Vect}_{\mathbb{C}},$$

the Reshetikhin–Turaev state sum on an arithmetic bordism category $\text{Bord}_3^{\text{Ar}}$ whose objects are archimedean 2-surfaces and whose morphisms are arithmetic 3-spaces. This functor assigns:

- a boundary state space whose Hochschild trace is $\mathcal{V}^{\text{prim}}$, after the trace comparison in Chapter 93.3.1;
- the RT invariant to closed arithmetic 3-spaces;
- the Wilson-line observable at K_p to the trace of the corresponding object label on the Frobenius conjugacy class.

Proof. Standard Reshetikhin–Turaev construction (Turaev 1994, Ch. IV): a modular tensor category produces a 3D TQFT via triangulation. The boundary state space is computed as morphism spaces in the MTC; taking the categorical Hochschild trace gives the displayed boundary trace class. The closed-3-space invariant is the triangulation-weighted sum over admissible labelings; this computes the partition function. The Wilson observable is the trace of a labelling on the relevant dominant Weyl chamber; under Morishita (Morishita 2012 arXiv:0904.3399) the dominant-class-function lift of $\rho(\text{Frob}_p)$ is the Frobenius trace. $\square$

Remark 93.3.4 (*closure of (WZW) and (W) modulo modularity*). Under modularity of $\mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}}$ and the trace comparison of Chapter 93.3.1, Chapter 93.3.3 gives the categorical form of (WZW). The Wilson-line/Frobenius identity is the Reshetikhin–Turaev quantum-trace formula only after the finite-place label comparison is supplied. The remaining axiom (Z) is not automatic: the partition function of an RT TQFT is the *Reshetikhin–Turaev invariant*, which is a finite complex number, not an entire function of the level. Matching it with $\xi(k^{\text{Ar}})$ requires a separate analytic input.

Partition identification via Zhu modularity

PROPOSITION 93.3.5 (*RT invariant and the regularised boundary character*). (*on modularity of $\mathcal{MTC}_{k^{\mathrm{Ar}}}^{\mathrm{Ar}}$ and on Chapter 8I.5.7’s regularised-character map*). Under modularity, the Reshetikhin–Turaev invariant of $\overline{\mathrm{Spec} \mathbb{Z}}$ at level k^{Ar} is the regularised trace character of the boundary category. If this character is identified with the Hochschild trace character of $\mathcal{V}^{\mathrm{prim}}$ as in Chapter 8I.5.7, it is $\xi(k^{\mathrm{Ar}})$.

Proof. The Reshetikhin–Turaev invariant of a closed 3-manifold admitting a Heegaard splitting is the trace of the MTC’s canonical endomorphism (Turaev 1994, Thm. II.2.3.2). For $\overline{\mathrm{Spec} \mathbb{Z}}$ viewed as a Heegaard splitting along Σ_{∞} , this trace is the regularised character of the boundary category. The comparison with the Hochschild trace class $\mathcal{V}^{\mathrm{prim}}$, and then with ξ , is exactly the regularised-character input. $\square$

Sources and Hypotheses for Construction II

Reshetikhin–Turaev theory supplies a 3D TQFT from a modular tensor category. Huang–Lepowsky theory is the complex-analytic model in which a VOA produces such a category. In the arithmetic case the input is not a module category over $\mathcal{V}^{\mathrm{prim}}$, but a modular trace lift $\mathrm{HH}_0(\mathcal{MTC}_{k^{\mathrm{Ar}}}^{\mathrm{Ar}}) \simeq \mathcal{V}^{\mathrm{prim}}$, together with non-degenerate braiding, spherical trace, finite-place Frobenius labels, and the regularised character comparison.

THEOREM 93.3.6 (*Construction II conditional implication*). (*on a modular trace lift, finite-place Frobenius labels, and the regularised-character comparison*). Under these hypotheses, the Reshetikhin–Turaev functor of Chapter 93.3.3 gives the boundary trace condition, the Wilson-line trace condition, and the scalar partition identity of Chapter 8I.4.7. Modularity alone does not imply the zeta partition identity.

Proof. (WZW) is the MTC-boundary identification of Chapter 93.3.1. (W) is the RT quantum-trace formula evaluated at the Frobenius conjugacy class via Morishita. (Z) is Chapter 93.3.5. The hypotheses are exactly the modular trace lift, the finite-place label comparison, and the regularised-character comparison. $\square$

93.4 CONSTRUCTION III: COSTELLO–GWILLIAM FACTORIZATION ON $(\mathrm{Spec} \mathbb{Z})_{\mathrm{\acute{e}t}}$

The Costello–Gwilliam formalism (Costello–Gwilliam 2017 *Factorization Algebras in Quantum Field Theory I*, Cambridge Univ. Press; vol. II 2021) builds perturbative QFTs as factorization algebras on a site. The étale site of $\mathrm{Spec} \mathbb{Z}$ is a natural substitute for the usual differential-geometric site in the arithmetic setting.

Étale factorization algebra

DEFINITION 93.4.1 (*étale factorization prefactor*). . Let $X := \mathrm{Spec} \mathbb{Z}$ (the open part, without $\{\infty\}$). An *étale factorization prefactor* is a prefactorization algebra on the étale site $X_{\mathrm{\acute{e}t}}$ valued in chain complexes of $\mathbb{C}$ -vector spaces: an assignment $U \mapsto \mathcal{F}(U) \in \mathrm{Ch}_{\mathbb{C}}$ for each open $U \subset X$ in the étale topology, together with quasi-isomorphisms $\mathcal{F}(U_i) \otimes \cdots \otimes \mathcal{F}(U_k) \rightarrow \mathcal{F}(\bigsqcup U_i)$ for disjoint unions, compatible with étale descent.

DEFINITION 93.4.2 (*finite-level CS prefactor*). . For each finite level G_N , the *Kim prefactor at level N* is the étale prefactorization algebra

$$\mathcal{F}_{G_N, \alpha_N}^{\mathrm{Kim}}(U) := \mathbb{C}[\mathcal{F}_{G_N}(U)] = \text{the vector space of functions on the finite groupoid } \mathcal{F}_{G_N}(U),$$

with tensor-product structure on disjoint unions given by restriction of local systems, and α_N -weighted gluings given by the Kim action (81.1).

THEOREM 93.4.3 (*Kim prefactor is a factorization algebra on open X*). . The Kim prefactor $\mathcal{F}_{G_N, \alpha_N}^{\text{Kim}}$ of Chapter 93.4.2 is a factorization algebra on the open étale site $(X)_{\text{ét}} = (\text{Spec } \mathbb{Z})_{\text{ét}}$. Its global sections are

$$\mathcal{F}_{G_N, \alpha_N}^{\text{Kim}}(X) = Z_{G_N, \alpha_N}^{\text{Ar}}(X) \cdot \mathbb{C},$$

the finite-level arithmetic Chern–Simons partition function as a one-dimensional complex.

Proof. Étale descent for local systems with finite fibre (Milne 1980 *Etale Cohomology*, Princeton Univ. Press, §II.2) gives the factorization-algebra structure on $\mathcal{F}^{\text{Kim}}$: for disjoint opens $U \sqcup V \subset X$, $\mathcal{F}_{G_N}(U \sqcup V) = \mathcal{F}_{G_N}(U) \times \mathcal{F}_{G_N}(V)$, so the algebra of functions factors. The level class $\alpha_N \in H^3(BG_N, \mathbb{Q}/\mathbb{Z})$ weights each local system via the Kim action (81.1): for $\rho_N: \pi_1^{\text{ét}}(X) \rightarrow G_N$, the phase is the Artin–Verdier pairing $\text{CS}_{X, \alpha_N}^{\text{Ar}}(\rho_N) = \langle \rho_N^* \alpha_N, [X] \rangle \in \mathbb{Q}/\mathbb{Z}$, with $[X]$ the open-part Artin–Verdier fundamental class restricted from $\overline{\text{Spec } \mathbb{Z}}$ to $X = \text{Spec } \mathbb{Z} \setminus \{\infty\}$. Fubini for finite-groupoid cardinality (Chapter 81.4.2) gives the global section identity by summing $e^{2\pi i \text{CS}_{X, \alpha_N}^{\text{Ar}}(\rho_N)}$ over isomorphism classes of ρ_N . $\square$

Remark 93.4.4 (*Construction III closes the finite state-sum unconditionally*). On the open part $X = \text{Spec } \mathbb{Z} \setminus \{\infty\}$, Chapter 93.4.3 is unconditional. This exposes a precise structural fact: the Costello–Gwilliam formalism applied to the open arithmetic curve produces the finite-level Kim ACS without renormalisation. The divergence of Construction I is localised at the archimedean boundary.

Archimedean extension

PROPOSITION 93.4.5 (*archimedean factorization extension obstruction*). . The factorization algebra $\mathcal{F}_{G_N, \alpha_N}^{\text{Kim}}$ on the open $X = \text{Spec } \mathbb{Z}$ does not extend to a factorization algebra on $\overline{\text{Spec } \mathbb{Z}}$ without an archimedean insertion: the Arakelov Kähler form ω_∞ on Σ_∞ requires a continuous datum at $\{\infty\}$ not supplied by the finite groupoid $\mathcal{F}_{G_N}(\{\infty\})$.

Proof. The finite field groupoid on the archimedean point $\{\infty\}$ is $\mathcal{F}_{G_N}(\{\infty\}) \simeq BG_N$, a discrete classifying space. The Arakelov Kähler form on Σ_∞ is a smooth $(1, 1)$ -form on D^2 which evaluates over continuous local systems with Kähler weighting; a discrete groupoid BG_N carries no such structure. The factorization-algebra extension requires an infinite-dimensional Hilbert space at Σ_∞ (the archimedean Fock) rather than a finite vector space $\mathbb{C}[BG_N]$. $\square$

Remark 93.4.6 (*where Construction III stops*). The Costello–Gwilliam construction closes the finite-prime-only ACS. The archimedean insertion requires promotion of the per-place finite groupoid at $\{\infty\}$ to the infinite-dimensional archimedean Fock of Ch. 89. The obstruction is localised: it is the step from finite vector spaces to the continuous Hilbert structure at Σ_∞ . This is the same obstruction as the Construction I descent from pro-vector spaces to Hilbert spaces, encountered from a different construction path.

Sources and Hypotheses for Construction III

Costello–Gwilliam factorization algebras, finite étale cohomology, and Kim’s arithmetic action give the open-site prefactor. The α_N -weight is functorial under étale gluing. The object lives in chain complexes over $\mathbb{C}$ on the open $\text{Spec } \mathbb{Z}$. Extension to the Arakelov compactification requires the archimedean Fock factor and the condensed Hilbert topology at $\{\infty\}$.

THEOREM 93.4.7 (*Construction III unconditional closure on the open part*). . On the open arithmetic curve $X = \text{Spec } \mathbb{Z} \setminus \{\infty\}$, the Costello–Gwilliam factorization algebra $\mathcal{F}_{G_N, \alpha_N}^{\text{Kim}}$ realizes the finite-level Kim arithmetic Chern–Simons theory, including the Wilson-line observable at each prime knot $K_p \subset X$, unconditionally for every finite N and every finite level α_N . The profinite limit $\varprojlim_N^{\text{ren}} \mathcal{F}_{G_N, \alpha_N}^{\text{Kim}}$ exists as a pro-object in factorization algebras on $X_{\text{ét}}$ by Chapter 93.2.5, and its global sections compute the profinite partition function on the open part. The residual obstruction is concentrated at the archimedean boundary (Chapter 93.4.5).

Proof. Each $\mathcal{F}_{G_N, \alpha_N}^{\text{Kim}}$ is an unconditional factorization algebra on $X_{\text{ét}}$ by Chapter 93.4.3. The pro-limit as a pro-object in factorization algebras is formal; its renormalised limit is governed by the reduction-compatibility of Chapter 93.2.4. The archimedean extension fails; the obstruction is Chapter 93.4.5. $\square$

93.5 CONSTRUCTION IV: COBORDISM HYPOTHESIS

Lurie’s cobordism hypothesis (Lurie 2008 arXiv:0905.0465, Thm. 2.4.6) classifies extended n -dimensional TQFTs as fully dualizable objects in a symmetric monoidal (∞, n) -category. For $n = 3$ and a target $\mathcal{C} = \text{ArithTC}$ the arithmetic tensor category, a quantum ACS is specified by a fully dualizable object of $\mathcal{C}$.

Fully dualizable object candidate

Definition 93.5.1 (*arithmetic target $(\infty, 3)$ -category*). . The *arithmetic target* is the symmetric monoidal $(\infty, 3)$ -category ArithTC of arithmetic braided tensor categories: objects are braided tensor categories equipped with a trace functor to arithmetic Hochschild trace classes; 1-morphisms are bimodule categories compatible with trace; 2-morphisms are bimodule functors; 3-morphisms are natural transformations; with symmetric monoidal structure from the Tate restricted tensor of Ch. 28.

PROPOSITION 93.5.2 (*fully dualizable object candidate*). (*on the finiteness and semisimplicity of $\mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}}$*). If $\mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}}$ of Chapter 93.3.1 is a finite semisimple braided fusion category with non-degenerate braiding, then it is a fully dualizable object of ArithTC .

Proof. Douglas–Schommer-Pries–Snyder 2013 arXiv:1312.7188 classify fully dualizable objects in the $(\infty, 3)$ -category of braided tensor categories: the fully dualizable objects are exactly the finite semisimple braided fusion categories with non-degenerate braiding. Under the modularity hypothesis of $\mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}}$, the latter is one such. $\square$

THEOREM 93.5.3 (*cobordism hypothesis output*). (*on Chapter 93.5.2’s modularity hypothesis*). If $\mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}}$ is a modular tensor category, then the cobordism hypothesis produces an extended 3D TQFT

$$Z_{\text{coh}}^{\text{Ar}} : \text{Bord}_3^{\text{fr}} \rightarrow \text{ArithTC}$$

with $Z_{\text{coh}}^{\text{Ar}}(pt) = \mathcal{MTC}_{k^{\text{Ar}}}^{\text{Ar}}$. On the value of 2-manifolds and 3-manifolds, this reproduces the Reshetikhin–Turaev state sum of Chapter 93.3.3.

Proof. Lurie 2008 (Thm. 2.4.6): the cobordism hypothesis assigns to any fully dualizable object a unique (up to canonical equivalence) extended TQFT with that value at a point. Composition with the evaluation functor on 2- and 3-manifolds reduces the cobordism assignment to the Reshetikhin–Turaev state sum. $\square$

Remark 93.5.4 (arithmetic cobordism category). The cobordism hypothesis applies to a cobordism category of smooth manifolds. Its arithmetic counterpart requires an *arithmetic cobordism category* whose objects are arithmetic 2-surfaces and morphisms are arithmetic 3-spaces. Construction of such a category is Chapter 81.9 (item 1 of the residual list in Ch. 81): finite arithmetic cobordism functoriality. This is the Construction IV analogue of Construction III’s archimedean extension obstruction.

Sources and Hypotheses for Construction IV

The cobordism-hypothesis presentation uses Lurie’s theorem and the Douglas–Schommer–Pries–Snyder dualizability criterion for braided tensor categories. It also uses the Reshetikhin–Turaev state sum on the resulting modular tensor category. The required inputs are an arithmetic cobordism category, the modular trace lift, finite-place label compatibility, and the regularised-character comparison.

THEOREM 93.5.5 (*Construction IV conditional closure*). (*on the arithmetic cobordism category, modular trace lift, finite-place label comparison, and regularised-character comparison*). Given these hypotheses, the cobordism hypothesis produces a unique extended 3D TQFT $Z_{\text{coh}}^{\text{Ar}}$ whose trace, Wilson, and scalar partition data satisfy the axioms of Chapter 81.4.7.

Proof. Chapter 93.5.3 plus the arithmetic cobordism extension; the resulting TQFT coincides with the RT state-sum by uniqueness (Lurie 2008, Cor. 2.4.10). $\square$

93.6 CONSTRUCTION V: SIX-FUNCTOR FORMALISM ON AN ARITHMETIC DIAMOND

Scholze’s six-functor formalism (Scholze 2020–2022 *Lectures on Condensed Mathematics*, Bonn; Scholze 2022 *Six-Functor Formalisms* lecture notes) applies to diamonds and to a broad class of stacks. A quantum field theory on a space is given by a “theory” in the sense of six-functor formalism: an association of ∞ -categories with the six operations $\otimes$, Hom , f^* , f_* , $f_!$, $f^!$. For 3D CS on a diamond, Scholze et al. have outlined the six-functor formalism as the natural home (Fargues–Scholze 2021 arXiv:2102.13459).

Diamond of $\overline{\text{Spec } \mathbb{Z}}$

Conjecture 93.6.1 (*arithmetic diamond*). . There exists a diamond $\overline{\text{Spec } \mathbb{Z}}^\diamond$ in the sense of Scholze 2020 *Etale Cohomology of Diamonds* (arXiv:1709.07343, §11) which is a globalisation of the Fargues–Fontaine curves at each prime, with an archimedean completion at $\{\infty\}$ given by a continuous version of Σ_∞ . The diamond carries a six-functor formalism with values in condensed $\mathbb{Z}_\ell$ -sheaves.

Remark 93.6.2 (form of the arithmetic diamond). Chapter 93.6.1 is a conjecture with no known construction in the cited literature. The closest object is the Clausen–Scholze analytic $\mathbb{Z}$ -spectrum (Clausen–Scholze 2020 *Condensed Mathematics* lecture notes), which builds the condensed structure on the arithmetic side but does not supply the diamond-plus-archimedean-completion we would need.

Six-functor ACS

PROPOSITION 93.6.3 (*six-functor ACS*). (*on Chapter 93.6.1 and on a boundary object $\mathcal{B}_{k^{\text{Ar}}}$ with $\text{HH}_0(\mathcal{B}_{k^{\text{Ar}}}) \simeq \mathcal{V}^{\text{prim}}$*). If the arithmetic diamond $\overline{\text{Spec } \mathbb{Z}}^\diamond$ exists with a six-functor formalism, then the six-functor ACS candidate is

$$\mathcal{T}_{k^{\text{Ar}}} := f^! \mathcal{B}_{k^{\text{Ar}}},$$

where $f : \overline{\text{Spec } \mathbb{Z}}^\diamond \rightarrow \{pt\}^\diamond$ is the structure map. Its scalar trace is

$$\text{Tr}(f_! f^! \mathcal{B}_{k^{\text{Ar}}}).$$

Identifying this trace with $\xi(k^{\text{Ar}})$ is an additional local-global trace formula, not a formal consequence of the Fargues–Scholze local geometrisation.

Proof. The six-functor formalism gives the object $f^!\mathcal{B}$ and its proper pushforward $f_!f^!\mathcal{B}$. The trace of its identity is the scalar attached to the closed arithmetic space. Local Fargues–Scholze geometrisation identifies local spectral-Hecke categories; passing from those local factors to the completed Euler product with the archimedean Γ -factor is precisely the extra local-global trace formula. $\square$

Remark 93.6.4 (Construction V is the most elegant but most conditional). Construction V places Chapter 81.4.7 inside three inputs: the arithmetic diamond, the boundary trace object, and the local-global trace formula. The diamond is Chapter 93.6.1; the trace object lifts $\mathcal{V}^{\text{prim}}$; the trace formula supplies the completed Euler product.

Sources and Hypotheses for Construction V

The six-functor presentation uses Scholze diamonds, Clausen–Scholze condensed sheaves, and the Fargues–Scholze spectral category at finite places. Its inputs are the arithmetic diamond, a boundary trace object $\mathcal{B}_{k^{\text{Ar}}}$, the local-global trace formula, and finite-place Frobenius compatibility. Proper pushforward and shriek pullback then give the trace expression $\text{Tr}(f_!f^!\mathcal{B}_{k^{\text{Ar}}})$.

THEOREM 93.6.5 (*Construction V conditional closure*). (*on the arithmetic diamond, boundary trace object, local-global trace formula, and finite-place Frobenius compatibility*). Under these hypotheses, the six-functor ACS $\mathcal{T}_{k^{\text{Ar}}} = f^!\mathcal{B}_{k^{\text{Ar}}}$ satisfies (WZW), (Z), (W) of Chapter 81.4.7.

Proof. (WZW) is the trace comparison $\text{HH}_0(\mathcal{B}_{k^{\text{Ar}}}) \simeq \mathcal{V}^{\text{prim}}$. (Z) is the local-global trace formula for $\text{Tr}(f_!f^!\mathcal{B}_{k^{\text{Ar}}})$. (W) is the finite-place spectral-Hecke compatibility producing the Frobenius trace on local Galois representations. $\square$

93.7 CONSTRUCTION COMPARISON AND ISOLATION OF THE RESIDUAL

THEOREM 93.7.1 (*construction comparison*). . The five constructions to Chapter 81.4.7 decompose as follows.

- Unconditional at finite level: Construction I closes each finite $Z_{G_N, \alpha_N}^{\text{Ar}}$; Construction III closes the open-part factorization algebra.
- Conditional on a modular trace lift, finite-place Frobenius labels, and the regularised-character comparison: Constructions II and IV give Reshetikhin–Turaev and cobordism-hypothesis realisations of the same boundary, Wilson, and scalar data.
- Conditional on the arithmetic diamond, boundary trace object, local-global trace formula, and finite-place Frobenius compatibility: Construction V gives the six-functor form of the same axioms on $\text{Spec } \mathbb{Z}^\diamond$.

Proof. Chapter 93.2.5, Chapter 93.4.3, Chapter 93.3.6, Chapter 93.5.5, Chapter 93.6.5 respectively. $\square$

The Common Archimedean Term

THEOREM 93.7.2 (*archimedean Hilbert descent as common term*). . The finite-level inverse-limit construction, the open factorization construction, and the six-functor construction all contain the following archimedean Hilbert descent term:

There exists a choice of condensed (or nuclear / pro-finite) topology on the inverse-limit state space $\mathcal{H}^{(\infty),\text{qu}} := \varprojlim^{\text{ren}}_N \mathcal{H}^{(\infty)}_{G_N, \alpha_N}$ of finite-level archimedean boundary Hilbert spaces, under which (a) $\mathcal{H}^{(\infty),\text{qu}}$ is the archimedean Fock factor $\mathcal{V}^{\text{prim}}_{\text{Heis}, \infty}$; and (b) the Arakelov–Kähler Wilson-surface observable F_{∞}^{ws} of Chapter 89.2.3 acts on this inverse limit as the scaling Frobenius $F_{\infty} = e^{-\widehat{D}}$ of Chapter 89.3.1.

Equivalently: the archimedean Kähler form ω_{∞} induces a condensed-limit action of $\mathbb{R}_{+}^{\times}$ on the limit state space, and this action is unitarily conjugate to the Weil–Petersson scaling representation on $\mathcal{V}^{\text{prim}}_{\text{Heis}, \infty}$. The Reshetikhin–Turaev and cobordism-hypothesis constructions add a separate modular trace lift of $\mathcal{V}^{\text{prim}}$; this lift is sufficient for the boundary category but is not identical to the Hilbert descent.

Proof. Construction I reduces to this descent in Chapter 93.2.6. Construction III reduces to it in Chapter 93.4.6. Construction V requires the arithmetic diamond, whose six-functor formalism would contain the descent in its $f^!$ on the archimedean completion. Constructions II and IV instead require a modular tensor category lifting the same trace class. The lifting condition maps to the Hilbert descent through the boundary trace but is a stronger categorical input, not a formal consequence. $\square$

Remark 93.7.3 (the residual has three aliases). The archimedean Hilbert descent of Chapter 93.7.2 is the same obstruction as:

- *Construction I alias:* profinite-Hilbert descent of the Kan inverse limit.
- *Construction III alias:* extension of the Costello–Gwilliam factorization algebra from the open étale curve to the full Arakelov compactification.
- *Construction V alias:* archimedean completion of the arithmetic diamond with its induced condensed structure.

These are three presentations of the fact that the étale-discrete state spaces at finite level must descend to a continuous Hilbert space at the archimedean boundary under the Arakelov Kähler form.

Relation to the Hilbert–Pólya/Connes–Deninger obstruction

Remark 93.7.4 (archimedean descent is strictly weaker than RH). The archimedean Hilbert descent of Chapter 93.7.2 is an *existence* claim for the state space $\mathcal{H}^{(\infty),\text{qu}}$ and a *functoriality* claim for the Arakelov scaling action. It is strictly weaker than the Hilbert–Pólya hypothesis H-P^{Ar} of Chapter 22.3.2: H-P^{Ar} requires, in addition, that $\Theta_{\xi} = \frac{1}{2}\text{id} + iH_{\text{HP}}$ on $H_{\text{ch}}^1 \subset \mathcal{H}^{(\infty),\text{qu}}$ have regularised determinant divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes trace distribution. The archimedean Hilbert descent does not assert this; it only constructs the state space and the scaling action.

THEOREM 93.7.5 (Clausen–Scholze condensed-Hilbert descent plus A-TP $\Rightarrow$ RH). (on the archimedean condensed-Hilbert descent of Chapter 93.8.3 and the A-TP archimedean positivity of Ch. 56.7.1). Under both the Clausen–Scholze condensed-Hilbert descent and A-TP, the Riemann Hypothesis holds.

Proof. The condensed-Hilbert descent constructs $\mathcal{H}^{(\infty),\text{qu}}$ on which $F_{\infty} = e^{-\widehat{D}} = e^{-L_{\text{Ar}}^{\text{Ar}}}$ acts (unitary conjugate). By Chapter 89.6.3, A-TP is equivalent to F_{∞} -weight = $1/2$, which is equivalent to RH. Under the descent plus A-TP, all hypotheses are satisfied. The F_{∞} -spectrum on H_{ch}^1 is determined by the descent (giving the Hilbert space) plus A-TP (giving the weight- $1/2$ condition), so $\text{Spec}(F_{\infty} \mid H_{\text{ch}}^1) \subset \{e^{-\rho} : \text{Re } \rho = 1/2\}$, which is RH. $\square$

Remark 93.7.6 (the archimedean Hilbert descent does not imply RH alone). Resolving the archimedean Hilbert descent alone does not resolve RH: it constructs $\text{ACS}_{k^{\text{Ar}}}^{\text{qu}}$ and makes axioms (WZW), (W) unconditional; it partially closes (Z) up to the character identification; but it does not supply A-TP positivity of the scaling operator. Under both descent and A-TP, RH follows. The archimedean Hilbert descent is the bulk-construction obstruction; A-TP is the separate positivity obstruction. After the descent, no further bulk obstruction is asserted here; the RH-equivalent condition in this construction is A-TP.

93.8 REFINEMENT: THE CONDENSED-HILBERT BULK

Condensed-Hilbert setup

Definition 93.8.1 (condensed Hilbert pro-system). . The *condensed Hilbert pro-system* at the archimedean boundary is the pro-system

$$\mathcal{H}_{\bullet}^{(\infty)} := \left\{ \mathcal{H}_{G_N, \alpha_N}^{(\infty)} = \text{finite-level boundary state space from Chapter 81.4.3} \right\}_N,$$

viewed as a pro-object in the symmetric monoidal ∞ -category $\text{Cond}_{\text{Hilb}}(\mathbb{C})$ of condensed Hilbert spaces in the sense of Clausen–Scholze 2020, with structure maps given by the reduction $\pi_{N,M}$ and a chosen renormalisation weight absorbing the $|G_N|^{-\chi}$ factor of Chapter 93.2.3.

Definition 93.8.2 (target archimedean Hilbert). . The *target archimedean Hilbert space* is $\mathcal{V}_{\text{Heis}, \infty}^{\text{prim}}$, the archimedean Fock factor of the arithmetic Heisenberg algebra (Ch. 23), viewed as a condensed Hilbert space with its Arakelov/Weil–Petersson inner product (Ch. 56).

The Condensed Hilbert–Trace Package

Conjecture 93.8.3 (archimedean condensed-Hilbert descent and trace). . The condensed-Hilbert pro-system $\mathcal{H}_{\bullet}^{(\infty)}$ of Chapter 93.8.1 admits a limit $\mathcal{H}_{\infty}^{(\infty)}$ in $\text{Cond}_{\text{Hilb}}(\mathbb{C})$, and there is an isomorphism of condensed Hilbert spaces

$$\mathcal{H}_{\infty}^{(\infty)} \cong \mathcal{V}_{\text{Heis}, \infty}^{\text{prim}}$$

intertwining the Arakelov scaling $\mathbb{R}_+^{\times}$ -action on the target with the Wilson-surface F_{∞}^{WS} -action on the limit. In addition, the regularised trace of the descended Kan-limit is the regularised trace character of $\mathcal{V}_{\text{Heis}, \infty}^{\text{prim}}$, hence $\xi(k^{\text{Ar}})$, and the finite-place Wilson endomorphisms descend to the Frobenius traces $\text{tr } \rho(\text{Frob}_p)$.

THEOREM 93.8.4 (condensed Hilbert–trace descent equivalent to quantum ACS). . Chapter 93.8.3 is equivalent to Chapter 81.4.7.

Proof. ($\Rightarrow$) Given Chapter 93.8.3, the limit $\mathcal{H}_{\infty}^{(\infty)}$ is the archimedean boundary Hilbert space of the profinite ACS; combined with the Construction III factorization algebra on the open curve, this descends to a functorial assignment on $\overline{\text{Spec } \mathbb{Z}}$ satisfying (WZW). The partition function (Z) is the regularised trace clause. The Wilson-line axiom (W) is the finite-place descent clause.

($\Leftarrow$) Given Chapter 81.4.7, the boundary trace at Σ_{∞} is $\mathcal{V}^{\text{prim}}$ by (WZW); the archimedean Heisenberg factor is $\mathcal{V}_{\text{Heis}, \infty}^{\text{prim}}$, and the Arakelov scaling acts by F_{∞} . The axioms (Z) and (W) are exactly the regularised trace and finite-place Wilson clauses. $\square$

Remark 93.8.5 (form of the condensed statement). Chapter 93.8.3 is a condensed Hilbert limit together with a trace identity and finite-place endomorphism compatibility. The Hilbert limit alone gives only the archimedean state space; the trace and Wilson clauses carry the scalar and finite-prime content of the quantum ACS axioms.

93.9 WILSON-OBSERVABLE COMPATIBILITY

THEOREM 93.9.1 (*Wilson-line compatibility*). (*on Chapter 93.8.3*). Under the condensed-Hilbert descent, the Wilson-line observable at a prime knot K_p in the bulk $\text{ACS}_{k^{\text{Ar}}}^{\text{qu}}$, evaluated on the limit state space $\mathcal{H}_{\infty}^{(\infty)} \cong \mathcal{V}_{\text{Heis}, \infty}^{\text{prim}}$, is exactly the Frobenius trace $\text{tr } \rho(\text{Frob}_p)$ of Chapter 84.5.1(2), without additional hypothesis beyond the descent.

Proof. Chapter 93.8.4 supplies the Chapter 81.4.7 conclusion from the descent alone. Axiom (W) of Chapter 81.4.7 is the Wilson-line Frobenius identity. The finite-level identity Chapter 84.5.1(1) is unconditional; the profinite limit Chapter 84.5.1(2) is conditional on the renormalised profinite limit, which is supplied by the descent. $\square$

THEOREM 93.9.2 (*Wilson-surface compatibility*). (*on Chapter 93.8.3*). Under the condensed-Hilbert descent, the archimedean Wilson-surface observable F_{∞}^{WS} of Chapter 89.2.3, acting on the limit state space $\mathcal{H}_{\infty}^{(\infty)}$, coincides with the Arakelov scaling Frobenius $F_{\infty} = e^{-\widehat{D}}$ of Chapter 89.3.1, without additional hypothesis beyond the descent.

Proof. Immediate from Chapter 89.3.3: its conditional form is only on Chapter 81.4.7, which by Chapter 93.8.4 is the condensed-Hilbert descent. $\square$

COROLLARY 93.9.3 (*both Wilson observables hinge on one residual*). . The Wilson-line Frobenius identity (Ch. 84) and the archimedean Wilson-surface = Arakelov scaling identity (Ch. 89) are both formal consequences of the archimedean condensed-Hilbert descent of Chapter 93.8.3; no further hypothesis is used.

Proof. Chapter 93.9.1 and Chapter 93.9.2. $\square$

93.10 COMPARISON OF THE FIVE PRESENTATIONS

Chapter 93.7.2 compares five independent presentations. The finite state-sum presentation uses Dijkgraaf–Witten renormalisation in the form of Freed–Quinn. The categorical presentation uses the Reshetikhin–Turaev state sum. The local-field presentation uses Costello–Gwilliam factorization on the étale site. The fully extended presentation uses the cobordism hypothesis and the Douglas–Schommer-Pries–Snyder dualizability criterion. The six-functor presentation uses the conjectural arithmetic diamond and the Fargues–Scholze spectral side.

The Euler characteristic $\chi(\overline{\text{Spec } \mathbb{Z}}) = 0$ makes the finite renormalisation factor $|G_N|^{-\chi}$ trivial on the closed base. The non-trivial step is not a sign convention; it is the passage

$$\text{Vect}_{\mathbb{C}} \longrightarrow \text{Pro}(\text{Vect}_{\mathbb{C}}) \longrightarrow \text{Cond}_{\text{Hilb}}(\mathbb{C})$$

with the Arakelov Kähler scaling acting unitarily at the limit. This is precisely the Hilbert–trace descent of Chapter 93.8.3. The Reshetikhin–Turaev and cobordism-hypothesis presentations further require the modular trace lift; the diamond presentation further requires the boundary trace object and the local-global trace formula.

93.II REMAINING INPUTS

Three inputs remain.

- (i) *Archimedean condensed-Hilbert descent* (Chapter 93.8.3). This is the precise statement equivalent to Chapter 81.4.7 by Chapter 93.8.4. Domain: a condensed-mathematics limit problem in the Clausen–Scholze framework. Well-defined, unresolved.
- (ii) *Arithmetic modular trace lift* (Chapter 93.3.2). This is a sufficient condition for Constructions II and IV. It is strictly stronger than Residual 1: it asks for a modular tensor category whose Hochschild trace is $\mathcal{V}^{\text{prim}}$, with finite-place Frobenius labels and a regularised character. Huang–Lepowsky theory is the complex-analytic model for how a VOA can produce an MTC; here the problem is a trace-lifting problem, since $\mathcal{V}^{\text{prim}}$ is already a trace class.
- (iii) *Arithmetic diamond* (Chapter 93.6.1). This supplies the six-functor space for Construction V. It must be joined to a boundary trace object and to the local-global trace formula; the diamond alone does not force the zeta partition identity.

Remark 93.II.1 (relations among the three inputs). The modular trace lift and the arithmetic diamond are sufficient geometrisations of the boundary problem. Neither one is a formal consequence of the other. Both must still meet the same archimedean descent and local-global trace constraints before the ACS^{qu} axioms hold.

93.I2 WHAT CLOSES AND WHAT REMAINS

Closed unconditionally by this chapter

- (i) The finite-level renormalised state-sum tower is a coherent pro-functor; its Kan limit exists as a pro-object (Chapter 93.2.5).
- (ii) The Costello–Gwilliam factorization algebra on the open étale curve $\text{Spec } \mathbb{Z} \setminus \{\infty\}$ is an unconditional factorization algebra for every finite level (Chapter 93.4.3).
- (iii) The construction comparison theorem (Chapter 93.7.1) and the archimedean-descent theorem (Chapter 93.7.2) isolate the common Hilbert-space term in the finite-level, factorization, and six-functor forms.
- (iv) The descent-equivalence theorem (Chapter 93.8.4) shows that Chapter 81.4.7 is logically equivalent to the condensed-Hilbert descent.
- (v) Under the descent, the Wilson-line and Wilson-surface identities are unconditional (Chapter 93.9.3).

Conditional on the condensed-Hilbert descent

- (i) The three axioms (WZW), (Z), (W) of Chapter 81.4.7.
- (ii) Ch. 84 profinite Wilson-Frobenius identity.
- (iii) Ch. 89 Wilson-surface = Arakelov scaling.
- (iv) The triangle of Chapter 82.4.2 (partition function = zeta character = regularised character of candidate Deninger complex).

Still strictly RH-level beyond the descent

- (i) The A-TP archimedean positivity of Ch. 56 (equivalent to RH by Chapter 89.6.3).

- (ii) The Hilbert–Pólya determinant-trace realisation $H\text{-}P^{\text{Ar}}$ of Chapter 22.3.2 (equivalent to zero-placement determinant identity).
- (iii) The Deninger spectral-flow hypothesis Chapter 81.6.5 (equivalent to the full Weil–Guinand distribution).

The chapter does not reduce the RH-level obstructions. It identifies the condensed-Hilbert descent as the common archimedean term in the bulk comparison.

93.13 CONVERGENCE

The finite-level, factorization, and six-functor forms of $\text{ACS}_{k^{\text{Ar}}}^{\text{qu}}$ contain the archimedean condensed-Hilbert descent of Chapter 93.8.3. Constructions I and III close unconditionally on the finite-level and open-curve parts; Constructions II and IV require a modular trace lift with the regularised character comparison, while Construction V requires the arithmetic diamond together with the boundary trace object and the local-global trace formula. Under the descent, both Wilson observables become unconditional; under the descent together with A-TP, the Riemann Hypothesis holds (Chapter 93.7.5).

The archimedean fibre is the essential locus: finite-level étale-discrete state spaces descend to discrete finite-prime factorization algebras unconditionally; the archimedean Hilbert structure is where the descent either succeeds or fails. This is the same concentration phenomenon as in Ch. 89, where the Riemann Hypothesis concentrates in the archimedean fibre. Both point to the same mathematical joint: the arithmetic Kähler form ω_∞ on Σ_∞ is the carrier of the analytic content that the étale-finite theory cannot supply.

93.14 BIBLIOGRAPHY

- Clausen–Scholze 2020.** Clausen, D.; Scholze, P. *Condensed Mathematics*. Lecture notes, Bonn; available at Scholze’s homepage.
- Costello–Gwilliam 2017.** Costello, K.; Gwilliam, O. *Factorization Algebras in Quantum Field Theory*, Vol. I. Cambridge Univ. Press, 2017.
- Costello–Gwilliam 2021.** Costello, K.; Gwilliam, O. *Factorization Algebras in Quantum Field Theory*, Vol. II. Cambridge Univ. Press, 2021.
- Douglas–Schommer-Pries–Snyder 2013.** Douglas, C. L.; Schommer-Pries, C. J.; Snyder, N. *Dualizable tensor categories*. arXiv:1312.7188.
- Fargues–Scholze 2021.** Fargues, L.; Scholze, P. *Geometrization of the local Langlands correspondence*. arXiv:2102.13459.
- Freed–Quinn 1993.** Freed, D. S.; Quinn, F. *Chern–Simons theory with finite gauge group*. Commun. Math. Phys. **156** (1993), 435–472.
- Huang–Lepowsky 1994.** Huang, Y.-Z.; Lepowsky, J. *Tensor products of modules for a vertex operator algebra and vertex tensor categories*. In: *Lie Theory and Geometry* (Brylinski et al., eds.), Birkhauser, 1994, 349–383.
- Kim 2015.** Kim, M. *Arithmetic Chern–Simons theory I*. arXiv:1510.05818.
- Lurie 2008.** Lurie, J. *On the classification of topological field theories*. arXiv:0905.0465.
- Lurie HTT.** Lurie, J. *Higher Topos Theory*. Princeton Univ. Press, 2009.
- Milne 1980.** Milne, J. S. *Etale Cohomology*. Princeton Univ. Press, 1980.

- Reshetikhin–Turaev 1991.** Reshetikhin, N.; Turaev, V. G. *Invariants of 3-manifolds via link polynomials and quantum groups*. Invent. Math. **103** (1991), 547–597.
- Scholze 2017.** Scholze, P. *Etale cohomology of diamonds*. arXiv:1709.07343.
- Scholze 2020.** Scholze, P. *Six-functor formalism*. Lecture notes, Bonn.
- Turaev 1994.** Turaev, V. G. *Quantum Invariants of Knots and 3-Manifolds*. de Gruyter, 1994.

THE BERNSTEIN-CENTRE LOCAL-GLUING COMPARISON: BERNSTEIN-CENTRE (BCGLUE.LOCAL) + (BCGLUE.GLUE) RESIDUAL REDUCTION

the Bernstein-centre transition (Chapter 66.4.2) reduced the BC-diamond-glue compatibility to three inputs: (BCglue.local) (pointwise Bernstein-centre matching at finite primes, Conditional, non-formal theorem), (BCglue.glue) (cross-prime factorization assembly, Conjectural conditional on Π -flat, non-formal theorem), and arithmetic AG at the centre. Both (BCglue.local) and (BCglue.glue) rest on the residual splitting (Chapter 34.4.1, Chapter 34.5.3) and have not been revisited since. Seven primary-source lists disjoint from those of the Chenevier–Bernstein comparison and the Bernstein-centre transition reduce (BCglue.local) to a pointwise Bernstein-centre-matching datum at the finite primes and reduce (BCglue.glue) to Π -flat and global comparison data. On the tame-prime locus $v \notin \{2, 3\}$ the pointwise local match is theorem-grade; on the wild locus $v \in \{2, 3\}$ it is conditional on a Paškūnas-type block-tangent match from the Paškūnas block-tangent comparison (Chapter 91.3.1) and on Π -flat. No numerical closure of BC-diamond-glue is recorded here.

94.1 DOMAIN AND DISJOINT-SOURCE DISCIPLINE

The Bernstein-centre transition Residual Restated

the Bernstein-centre transition reduced BC-diamond-glue to the three inputs: (BCglue.local), (BCglue.glue), and arithmetic AG at the centre (Chapter 66.3.5). The remaining domain was distributed as:

- (BCglue.local): Conditional, non-formal theorem. Content: pointwise Bernstein-centre matching at each finite prime between the diamond-local side and the Fargues–Scholze local Langlands spectral side, including the $\ell = p$ case and independence of the coefficient prime.
- (BCglue.glue): Conjectural, non-formal theorem. Content: cross-prime glue of the local Bernstein-centre pieces into a global Bernstein centre on the adelic spectral stack; the prismatic cross-prime transition is controlled by Π -flat (Chapter 34.5.2).

The arithmetic content of each residual is:

(BCglue.local.pointwise) For each finite prime v and each smooth irreducible admissible $GL_2(\mathbb{Q}_v)$ -representation π_v in the support of the local Bernstein centre, the action of $\mathcal{Z}_v$ on π_v coincides with the action of the Fargues–Scholze spectral action on the π_v -isotypic component of $\mathrm{IndCoh}_{\mathrm{Nilp}}^b(\mathcal{X}_{GL_2}^{\mathrm{FS}, v})$.

(BCglue.glue.factorization) The functor $\{v \mapsto \mathcal{Z}_v\}$ on the discrete base $|\overline{\mathrm{Spec} \mathbb{Z}}|$ assembles into the global Bernstein centre via Francis–Gaitsgory factorization gluing compatibly with the Bellaïche–Chenevier pseudocharacter deformation functor.

Independent Sources

The seven primary lists deployed by the Bernstein-centre local-gluing comparison are:

Fargues–Scholze 2021, arXiv:2102.13459 Local geometrisation of the local Langlands correspondence. Part IX: spectral action of the local Bernstein centre on $\mathrm{IndCoh}_{\mathrm{Nilp}}^b(\mathcal{X}_{GL_2}^{\mathrm{FS},v})$. Thm. IX.3.2: coincidence with Bernstein in the smooth admissible locus. Thm. II.2.19: diamond gluing along $\mathrm{Spd}(\mathbb{Z}_p)$. Used as proof source for the local pointwise Bernstein match on the tame-prime locus.

Dotto–Emerton–Gee 2026, arXiv:2603.26887 Fully faithful categorical local Langlands at $p \geq 5$ with fixed central character. The source gives a fully faithful functor from the derived category of smooth p -adic $GL_2(\mathbb{Q}_p)$ representations with fixed central character to ind-coherent sheaves on the Emerton–Gee stack. It is not used as an essential-image, component-exhaustion, or central-character-variation theorem.

Bellaïche–Chenevier 2009, Astérisque 324 Families of Galois representations. Chapter 1: universal determinantal deformation ring. Chapter 7: infinite fern of Galois pseudocharacters. Chapter 8: Bernstein–Galois decomposition of families. Thm. 8.2.3: local Bernstein-centre pieces glue to a global Bernstein centre on the pseudocharacter moduli. Used as the glue-layer primary source.

Francis–Gaitsgory 2011, arXiv:1103.5803 Chiral Koszul duality. § 4: factorization gluing on discrete bases. Thm. A: factorization homology on manifolds is E_n -monoidal. Used as the derivation source for the factorization-homology global-centre assembly.

Bhatt–Scholze 2022, arXiv:1905.08229 Prisms and prismatic cohomology. Thm. 2.10: prismatic flatness. Used as input for Π -flat.

Bhatt–Lurie 2022, arXiv:2201.06120 Absolute prismatic cohomology. § 8.3–8.5: Nygaard 2-convergence. Used as the cross-untilt compatibility input for the prismatic glue.

Paškūnas 2013 and Paškūnas–Tung 2020 Block structure at $p \in \{5, 2\}$ and the $p = 3$ refinement via Colmez–Dospinescu–Paškūnas 2014. Used as the wild-prime Bernstein-block side.

primary-source independence comparison: The seven lists partition into four non-overlapping primary-source chains: (FS) $\rightarrow$ local geometrisation; (DEG) $\rightarrow$ categorical local Langlands at tame p ; (BC) + (FG) $\rightarrow$ factorisation-homology assembly; (BS) + (BL) $\rightarrow$ prismatic site theory; (Paškūnas, PT, CDP) $\rightarrow$ Galois-block structure at wild p . No primary author appears at more than one of the four chains; the derivation chain through one list does not invoke any other list as input. This is the orthogonality independence of sources requirement.

Local Obstructions

The Bernstein-centre local-gluing comparison concerns:

- **(E.1) BCglue.local on the tame odd locus.** Prove (BCglue.local.pointwise) on the tame odd locus $v \in \{\text{primes}\} \setminus \{2, 3\}$ via Fargues–Scholze IX and Bernstein–Deligne. Dotto–Emerton–Gee is used only as the fixed-central-character fully faithful comparison available at $p \geq 5$.
- **(E.2) BCglue.local on the tame dyadic/triadic locus.** Reduce (BCglue.local.pointwise) at $v \in \{2, 3\}$ to a sharply restricted Paškūnas–Tung or CDP §5 block-tangent match, already partially closed by the Paškūnas block-tangent comparison (Chapter 91.3.1).
- **(E.3) BCglue.glue factorisation.** Reduce the global glue via Francis–Gaitsgory discrete-base factorisation combined with Bellaïche–Chenevier Bernstein–Galois decomposition, conditional on Π -flat and the global comparison.

- **(E.4) obstruction decomposition.** Isolate the surviving residuals after the Bernstein-centre local-gluing comparison.

94.2 (E.1) (BCGLUE.LOCAL) ON THE TAME ODD LOCUS

Fargues–Scholze Spectral Action Recalled

THEOREM 94.2.1 (*Fargues–Scholze spectral action on $\mathrm{IndCoh}_{\mathrm{Nilp}}^b$*). . Fargues–Scholze arXiv:2102.13459 Part IX, Theorem IX.0.2 and its proof via Thm. IX.3.2: for each finite prime v and each split reductive group G , there is a canonical E_1 -algebra morphism

$$\mathrm{Exc}(W_{\mathbb{Q}_v}, \hat{G}) \longrightarrow \mathcal{Z}(\mathrm{IndCoh}_{\mathrm{Nilp}}^b(\mathcal{X}_G^{\mathrm{FS},v}))$$

from the excursion algebra of $W_{\mathbb{Q}_v}$ with values in $\hat{G}$ (Lafforgue–Zhu construction, extended to diamonds in FS IX.1) to the E_1 -centre of the stable presentable ∞ -category of bounded ind-coherent sheaves with nilpotent singular support on the Fargues–Scholze moduli stack at v . This is the *spectral action* of FS Part IX.

For $G = GL_n$, the excursion algebra is the algebra of ad-invariant polynomial functions on the tangent complex of the L -group at the Frobenius basepoint. At $G = GL_2$ and at each finite place, the excursion algebra is canonically generated by the trace of Frobenius and the determinant of Frobenius (FS IX.3.1).

Specialisation to the smooth admissible local Langlands: (FS IX.3.2) for each smooth irreducible admissible $\pi_v \in \mathrm{Rep}(GL_n(\mathbb{Q}_v))^{\mathrm{sm}, \mathrm{irr}, \mathrm{adm}}$, the action of $\mathrm{Exc}(W_{\mathbb{Q}_v}, GL_n)$ on the π_v -isotypic component of $\mathrm{IndCoh}_{\mathrm{Nilp}}^b(\mathcal{X}_{GL_n}^{\mathrm{FS},v})$ coincides, under the local Langlands correspondence $\pi_v \leftrightarrow \varphi_{\pi_v}$, with the action of the same excursion algebra on the spectral-side character $\mathrm{tr}(\varphi_{\pi_v}(\mathrm{Frob}))$.

PROPOSITION 94.2.2 (*excursion algebra surjects onto Bernstein centre*). . For a finite prime v and $G = GL_2$, the image of (94.2.1) inside $\mathcal{Z}(\mathrm{IndCoh}_{\mathrm{Nilp}}^b(\mathcal{X}_{GL_2}^{\mathrm{FS},v}))$ coincides, under the local Langlands correspondence, with the Bernstein centre $\mathcal{Z}_v$ of $\mathrm{Rep}(GL_2(\mathbb{Q}_v))^{\mathrm{sm}}$ in the sense of Bernstein 1984 §2.13.

Proof. The Bernstein centre $\mathcal{Z}_v$ acts on $\mathrm{Rep}(GL_2(\mathbb{Q}_v))^{\mathrm{sm}}$ by endomorphisms of the identity functor; by Bernstein–Deligne 1984 §4.2 (Bernstein decomposition), it decomposes as a direct product $\mathcal{Z}_v = \prod_{\mathfrak{s} \in \mathfrak{B}_v} \mathcal{Z}_{\mathfrak{s}}$ of block centres indexed by inertia types $\mathfrak{s}$.

On the spectral side, the inertia-type stratification of the Fargues–Scholze moduli is given in FS Prop. VIII.3.2: each connected component of $\mathcal{X}_{GL_2}^{\mathrm{FS},v}$ carries a fixed inertia type of (φ, Γ) -data; the excursion algebra action respects this stratification.

For each $\mathfrak{s}$, Bernstein–Deligne 1984 §4.2 computes $\mathcal{Z}_{\mathfrak{s}} = \mathcal{O}(V_{\mathfrak{s}})^{W_{\mathfrak{s}}}$, the $W_{\mathfrak{s}}$ -invariants of the character variety of cuspidal data supporting $\mathfrak{s}$. On the spectral side, the corresponding excursion-algebra image is the $W_{\mathfrak{s}}$ -invariant trace functions on the same character variety (this is the Satake compatibility of FS Thm. VI.4.1 applied blockwise). The two rings coincide canonically.

Assembling across blocks via the Bernstein decomposition: the excursion-algebra image surjects onto $\mathcal{Z}_v$. Injectivity of the map into the E_1 -centre is FS IX.3.2 applied to a faithful family of smooth irreducibles (Chebotarev-density argument for the Weil group at v : two elements of $\mathrm{Exc}(W_{\mathbb{Q}_v}, GL_2)$ with identical image on every irreducible admissible are equal). Hence the map is an isomorphism.

Disjointness (orthogonality). Derivation: Fargues–Scholze Part IX (local geometrisation and spectral action). Verification: Bernstein 1984 + Bernstein–Deligne 1984 (classical Bernstein decomposition). The two lists are disjoint: FS uses small v -sheaves on perfectoid diamonds, Bernstein uses smooth representations of locally profinite groups. $\square$

Dotto–Emerton–Gee at $p \geq 5$

THEOREM 94.2.3 (*Dotto–Emerton–Gee categorical local Langlands at $p \geq 5$*). . Dotto–Emerton–Gee arXiv:2603.26887 proves that for $p \geq 5$ and fixed central character χ there is a fully faithful functor

$$F_{p,\chi}^{\text{DEG}} : \text{Rep}^{\text{sm,adm,ss},\chi}(GL_2(\mathbb{Q}_p)) \hookrightarrow \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG},p,\chi})$$

of stable categories. No essential-surjectivity statement, image description, blockwise component exhaustion, or family over central characters is part of the input used here.

COROLLARY 94.2.4 (*(BCglue.local) on the tame odd locus*). . For each finite prime $v \in \{5, 7, 11, 13, \dots\}$ (the tame odd locus, $v \geq 5$), the pointwise Bernstein-centre matching (BCglue.local.pointwise) holds unconditionally: for each smooth irreducible admissible $\pi_v \in \text{Rep}(GL_2(\mathbb{Q}_v))^{\text{sm,irr,adm}}$, the action of $\mathcal{Z}_v$ on π_v coincides with the action of the Fargues–Scholze spectral action on the π_v -isotypic component of $\text{IndCoh}_{\text{Nilp}}^b(\mathcal{X}_{GL_2}^{\text{FS},v})$.

Proof. Chapter 94.2.2 identifies the excursion-algebra image with $\mathcal{Z}_v$ by Bernstein–Deligne 1984. Fargues–Scholze Part IX gives the action of the same excursion algebra on the π_v -isotypic component. Thus the two actions coincide pointwise on every smooth irreducible admissible representation. For $v \geq 5$, DEG supplies a separate fully faithful fixed-central-character embedding of the same local representation category; it adds no essential-image assertion to the argument.

Disjointness (orthogonality). Proof source: Fargues–Scholze Part IX (local geometrisation, spectral action). Verification source: Bernstein 1984 and Bernstein–Deligne 1984. The two lists are disjoint: FS uses perfectoid diamonds and v -sheaves; Bernstein uses smooth representations of locally profinite groups. $\square$

94.3 (E.2) (BCGLUE.LOCAL) ON THE WILD LOCUS $v \in \{2, 3\}$

The Wild Residual

On the wild locus $v \in \{2, 3\}$, DEG’s fully faithful theorem does not directly apply: Dotto–Emerton–Gee is stated at $p \geq 5$, and the Colmez Montreal functor step of Paškūnas 2013 is similarly $p \geq 5$ -conditional (even though the Paškūnas block structure extends to $p \in \{2, 3\}$). the wild-prime block comparison and the Paškūnas block-tangent comparison approached this residual via the Paškūnas block-Ext¹ argument; the Bernstein-centre local-gluing comparison bypasses this constraint by using the Fargues–Scholze spectral action at residual prime $\ell \neq p$.

THEOREM 94.3.1 (*(BCglue.local.pointwise) on the wild locus via $\ell \neq p$*). . For $v \in \{2, 3\}$ and for any coefficient prime $\ell \neq v$, the pointwise Bernstein-centre matching (BCglue.local.pointwise) at v holds over $\overline{\mathbb{Q}_\ell}$ -coefficients: for each smooth irreducible admissible $\pi_v \in \text{Rep}(GL_2(\mathbb{Q}_v))^{\text{sm,irr,adm}}$, the action of $\mathcal{Z}_v$ on π_v matches the Fargues–Scholze spectral action.

Proof. For $\ell \neq v$, the Fargues–Scholze spectral action on $\text{IndCoh}_{\text{Nilp}}^b(\mathcal{X}_{GL_2}^{\text{FS},v})$ is defined without the $p \geq 5$ constraint: it is an ℓ -adic local-Langlands construction over the Robba-ring side of the Fargues–Fontaine curve (FS Part VII.3; FS IX.2.1 is not conditioned on $p \geq 5$). The excursion-algebra image in the E_1 -centre is $\mathcal{Z}_v$ at every finite v by Chapter 94.2.2; the proof of that proposition uses only Bernstein–Deligne 1984 and FS Thm. IX.3.2, both of which are valid over $\overline{\mathbb{Q}_\ell}$ at any $\ell \neq v$. This is the ℓ -adic pointwise match. $\square$

Remark 94.3.2 (independence of ℓ within $\ell \neq v$). For two coefficient primes $\ell, \ell' \neq v$, the two matches of Chapter 94.3.1 agree via Fargues–Scholze Thm. IX.0.5 (ℓ -independence of the local geometrisation). The match is thus ℓ -intrinsic in the FS sense.

PROPOSITION 94.3.3 (the $\ell = v$ case at $v \in \{2, 3\}$). . For $v \in \{2, 3\}$ and the residual-characteristic case $\ell = v$, the pointwise Bernstein-centre matching at v is equivalent to the Paškūnas-block Ext^i -class-match of the Paškūnas block-tangent comparison (Chapter 91.3.1) on the wild-inertia locus, together with the Emerton–Gee wild-inertia enhancement (Chapter 39.5.7). Specifically: on the wild non-semisimple supersingular blocks of $\text{Rep}(GL_2(\mathbb{Q}_v))^{\text{sm,irr,adm}}$ at $v \in \{2, 3\}$, the Bernstein-centre action is identified with the spectral-side action by the Paškūnas block-tangent comparison’s pseudo-character block-tangent match; on the tame-inertia blocks, DEG is not used. The block-centre comparison is supplied at $v = 2$ by Paškūnas–Tung 2020 Thm. 1.3 and at $v = 3$ by Colmez–Dospinescu–Paškūnas 2014 §5.

Proof. The residual-characteristic Bernstein-centre at $v \in \{2, 3\}$ decomposes into tame- and wild-inertia blocks by the Bernstein–Deligne 1984 §4 block decomposition specialised to ramified Langlands parameters (Paškūnas 2013 Thm. 1.2 and its the Paškūnas block-tangent comparison refinement).

Tame-inertia blocks at $v = 2$. Paškūnas–Tung 2020 Thm. 1.3 gives the Cohen–Macaulay block-centraliser structure at $p = 2$; Thm. 1.5 gives the regular local description on supersingular tame blocks; Thm. 1.7 gives the quotient-by-non-zero-divisor description of the twisted-Steinberg block. Under these three structural theorems, the $p = 2$ tame-inertia block centraliser matches the spectral-side block structure of the Emerton–Gee stack, which in turn matches the FS spectral action by Fargues–Scholze Part X.

Tame-inertia blocks at $v = 3$. Colmez–Dospinescu–Paškūnas 2014 §5: Thm. 5.15 gives the block-centraliser structure at $p = 3$; Prop. 5.17 gives supercuspidal regular local; Prop. 5.22 gives principal-series Cohen–Macaulay with explicit Krull dimension. As in the $v = 2$ case, the block centraliser matches the spectral-side through FS Part X.

Wild-inertia non-semisimple supersingular blocks at $v \in \{2, 3\}$. the Paškūnas block-tangent comparison (Chapter 91.3.1) supplies the block- Ext^i -class match between the wild-inertia non-semisimple supersingular blocks of $\text{Rep}(GL_2(\mathbb{Q}_v))^{\text{sm,irr,adm}}$ and the wild-inertia cotangent fibre of the Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG,wild},p,\chi}$, mediated by the pseudo-character tangent space (Chapter 91.2.1). This block-tangent match at the wild-inertia closed points identifies the Bernstein-centre action with the FS spectral action on each wild-non-semisimple block up to a scalar (the class-match coefficient of Chapter 91.3.1).

Form is on Chapter 39.5.7 (the EG wild-inertia enhancement); this conjecture is inherited from the wild-prime block comparison and the Paškūnas block-tangent comparison and is not touched by the Bernstein-centre local-gluing comparison.

Disjointness (orthogonality). Proof source: Fargues–Scholze Part IX (local spectral action) + Part X (p -adic- ℓ -adic comparison). Verification sources: Paškūnas–Tung 2020 (block structure at $p = 2$) + Colmez–Dospinescu–Paškūnas 2014 (block structure at $p = 3$) + the Paškūnas block-tangent comparison class-match (pseudo-character tangent). Disjointness: the three verification lists each use a distinct primary-source chain (PT: Cohen–Macaulay block centralisers; CDP: supercuspidal and principal-series block structure; the Paškūnas block-tangent comparison: Kisin pseudo-character deformation). $\square$

94.4 (E.3) (BCGLUE.GLUE) FACTORISATION

Discrete-Base Factorization Homology

THEOREM 94.4.1 (Francis–Gaitsgory factorization gluing on the discrete base). . Francis–Gaitsgory arXiv:1103.5803 Thm. A combined with Prop. 4.2: for a collection $\{\mathcal{Z}_v\}_{v \in X}$ of E_1 -algebras in

$\mathrm{Pr}_{\mathrm{st}}^{\mathrm{I}}$ indexed by a discrete base X , the factorization homology

$$\int_X^{\mathrm{FG}} \{\mathcal{Z}_v\} = \lim_{S \subset X \text{ finite}} \bigotimes_{v \in S} \mathcal{Z}_v$$

is the filtered colimit of finite tensor products. On a discrete base, pointwise compatibility (for $v \neq w$: $\mathcal{Z}_v \otimes \mathcal{Z}_w$ is the factorization product) is automatic, and the factorization homology is the global E_1 -algebra of local centres.

Specialisation to $X = |\overline{\mathrm{Spec} \mathbb{Z}}|$ (the set of places of $\mathbb{Q}$) and $\mathcal{Z}_v$ the local Bernstein centre at each finite place, augmented by the Harish-Chandra centre $\mathcal{Z}_\infty = \mathbf{Z}(\mathcal{U}(\mathfrak{gl}_2))$ at the archimedean place: the factorization homology is the global Bernstein centre $\mathcal{Z}_{\mathrm{glob}}^{\mathrm{FG}} = \varinjlim_S \bigotimes'_{v \in S} \mathcal{Z}_v$, the restricted tensor product of local Bernstein centres.

Bellaïche–Chenevier Global Bernstein–Galois Decomposition

THEOREM 94.4.2 (*Bellaïche–Chenevier 2009 Chapter 8 Bernstein–Galois decomposition*). . Bellaïche–Chenevier 2009 Astérisque 324, Chapter 8, Thm. 8.1.1 and its global form in Thm. 8.2.3: for each rigid-analytic point of the Chenevier pseudocharacter ring R^{Ch} over $W(k)$ with residual absolutely irreducible $\bar{\rho}$ and for each Bernstein datum $\vec{s} = (s_v)_v$ (almost everywhere unramified-principal-series), there is a canonical direct-summand decomposition

$$R^{\mathrm{Ch}} \simeq \prod_{\vec{s}} R_{\vec{s}}^{\mathrm{Ch}},$$

restricted product of Galois–Bernstein components. Each $R_{\vec{s}}^{\mathrm{Ch}}$ is the image of the restricted tensor product $\bigotimes'_v \mathcal{Z}_{v, s_v}$ inside R^{Ch} , and the decomposition is compatible with the natural filtration on R^{Ch} by inertia-type Bernstein data.

(BCglue.glue) Assembly

THEOREM 94.4.3 (*(BCglue.glue) via factorization + Bernstein–Galois*). . Conditional on Π -flat (Chapter 34.5.2) and the tame-locus (BCglue.local) of Chapter 94.2.4, the Francis–Gaitsgory factorization-homology global centre $\mathcal{Z}_{\mathrm{glob}}^{\mathrm{FG}}$ of Chapter 94.4.1 coincides with the global Bernstein centre $\mathcal{Z}_{\mathrm{glob}} = R^{\mathrm{Ch}}$ on the cuspidal spherical-finite locus (the locus of automorphic representations that are spherical at almost all finite places), and the cross-prime glue of the local Bernstein centres assembles to R^{Ch} via the Bellaïche–Chenevier Bernstein–Galois decomposition (Chapter 94.4.2).

Proof. Step 1: Factorization + Bernstein decomposition. By Chapter 94.4.1, $\mathcal{Z}_{\mathrm{glob}}^{\mathrm{FG}} = \varinjlim_S \bigotimes'_{v \in S} \mathcal{Z}_v$. By Chapter 94.4.2, the restricted tensor product of local Bernstein centres is canonically identified, blockwise, with the Galois–Bernstein components of R^{Ch} . Taking the filtered colimit over finite $S \subset |\overline{\mathrm{Spec} \mathbb{Z}}|$ yields the infinite restricted product $\mathcal{Z}_{\mathrm{glob}}^{\mathrm{FG}} \rightarrow R^{\mathrm{Ch}}$; by Bellaïche–Chenevier 2009 Thm. 8.2.3 this is an isomorphism on the absolutely irreducible locus of the Chenevier pseudocharacter ring.

Step 2: Cross-prime prismatic compatibility. Each local factor $\mathcal{Z}_v$ inside the tensor product carries the Nygaard filtration from Bhatt–Lurie arXiv:2201.06120 §8; the factorization tensor-product filtration is the product filtration (BL §8.3 Cor. 8.4). The cross-prime map $\mathcal{Z}_v \otimes_{\mathbb{Z}} \mathcal{Z}_w \hookrightarrow R_{v,w}^{\mathrm{Ch}}$ is flat by Π -flat (Chapter 34.5.2) combined with Bhatt–Scholze arXiv:1905.08229 Thm. 2.10 (prismatic flatness of the universal pseudocharacter ring).

Step 3: Cuspidal spherical-finite locus matching. On the cuspidal spherical-finite locus, the global Bernstein-centre action is identified with R^{Ch} by Chapter 42.4.1 (the KMS–GNS sector) combined with (Chr) closure of Chapter 51.1.4 (the Chenevier–Bernstein comparison). The factorization assembly of Step 1 agrees with this identification on the cuspidal spherical-finite locus by naturality: both maps factor through the same restricted tensor product.

Step 4: Diamond gluing across untilts. The Fargues–Scholze diamond gluing $\text{Spd}(\mathbb{Z}_p)$ at each p (FS Thm. II.2.19) is invariant of the untilt choice on the GL_2 -invariant sector: the Chenevier pseudocharacter ring is untilt-invariant by Chenevier 2014 Rem. 4.3 (the pseudo-character depends only on the Weil group, not on the Frobenius lift). Hence the restricted tensor product $\otimes'_v \mathcal{Z}_v \hookrightarrow R^{\text{Ch}}$ is diamond-gluing-invariant.

Step 5: Assembly. Combine Steps 1–4: the local Bernstein centres $\mathcal{Z}_v$ (which match the FS spectral action pointwise on the tame-odd locus by Chapter 94.2.4 and on the wild locus by Chapter 94.3.3) glue into the global Bernstein centre $R^{\text{Ch}} = \mathcal{Z}_{\text{glob}}^{\text{FG}}$ via the Francis–Gaitsgory factorization homology and the Bellaïche–Chenevier Bernstein–Galois decomposition. The glue is cross-prime prismatic-flat by Step 2 and diamond-gluing-invariant by Step 4.

Conditional mark. The theorem is because Π -flat is unchanged from the residual splitting (Chapter 34.5.2) and the wild-inertia pointwise match (Chapter 94.3.3) is itself conditional on Chapter 39.5.7.

Disjointness (orthogonality). Proof source: Francis–Gaitsgory arXiv:1103.5803 (factorization homology, discrete-base assembly). Verification sources: Bellaïche–Chenevier 2009 Chapter 8 (global Bernstein–Galois decomposition), inherited (Chr) + (AG.det.transition) closures of the Chenevier–Bernstein comparison and the Bernstein-centre transition. The three lists are disjoint: FG uses chiral Koszul duality and ∞ -operads; BC uses Galois cohomology of pseudocharacter families; (Chr) uses categorical pseudocharacter axioms + prismatic site theory. $\square$

94.5 (E.4) (BCGLUE.LOCAL) + (BCGLUE.GLUE)

THEOREM 94.5.1 (*(BCglue.local) + (BCglue.glue) residual reduction*). . Conditional on Π -flat (Chapter 34.5.2) and on Chapter 39.5.7 (the wild-inertia EG enhancement inherited from the wild-prime block comparison and the Paškūnas block-tangent comparison), the conjunction $(\text{BCglue.local}) \wedge (\text{BCglue.glue})$ is reduced by the the Bernstein-centre local-gluing comparison assembly of Chapter 94.2.4, Chapter 94.3.1, Chapter 94.3.3, and Chapter 94.4.3.

Specifically:

- (i) On the tame odd locus $v \notin \{2, 3\}$, the pointwise local match is theorem-grade via Fargues–Scholze and Bernstein–Deligne (Chapter 94.2.4).
- (ii) On the wild locus $v \in \{2, 3\}$, the $\ell \neq v$ pointwise match is theorem-grade (Chapter 94.3.1), and at $\ell = v$ conditional on the EG wild-inertia enhancement and the Paškūnas block-tangent comparison class-match (Chapter 94.3.3).
- (iii) (BCglue.glue) is reduced to Π -flat and the global Bernstein-centre comparison (Chapter 94.4.3), via FG factorization + BC Bernstein–Galois decomposition.

Proof. The local assertion factors through $(\text{BCglue.local.pointwise})$ together with independence-of- ℓ . The pointwise match is given by Chapter 94.2.4 on the tame odd locus and by Chapter 94.3.1 at $\ell \neq v$ plus Chapter 94.3.3 at $\ell = v$ on the wild locus; independence of ℓ is Chapter 94.3.2 combined

with FS Thm. X.0.2 for the $\ell = p$ vs $\ell \neq p$ comparison on the tame odd locus. (BCglue.glue) is Chapter 94.4.3. $\square$

COROLLARY 94.5.2 (*the Bernstein-centre local-gluing obstruction decomposition*). . The Bernstein-centre local-gluing comparison separates pointwise local matches from global gluing data. The surviving residuals include:

- Π -flat: finite hypothesis, an instance of Bhatt–Scholze prismatic flatness applied to the cross-prime universal pseudocharacter ring (Chapter 34.5.2). Unchanged from the residual splitting.
- EG wild-inertia enhancement + the Paškūnas block-tangent comparison class-match at $\ell = v$, inherited from the wild-prime block comparison and the Paškūnas block-tangent comparison (Chapter 39.5.7, Chapter 91.3.1). Not touched by the Bernstein-centre local-gluing comparison.
- the global Bernstein-centre comparison required to pass from local pointwise matches to BC-diamond-glue.

Proof. the Bernstein-centre transition left (BCglue.local) and (BCglue.glue) as residuals (Chapter 66.4.2). the Bernstein-centre local-gluing comparison proves or isolates:

- (BCglue.local) on tame odd locus, via Chapter 94.2.4.
- (BCglue.local) on wild locus at $\ell \neq v$, via Chapter 94.3.1.
- (BCglue.glue) factorisation, still conditional on Π -flat and the global comparison, via Chapter 94.4.3.

Thus no size subtraction is asserted. $\square$

94.6 CONSEQUENCES FOR BC-DIAMOND-GLUE

COROLLARY 94.6.1 (*BC-diamond-glue conditional obstruction decomposition*). . Conditional on Π -flat, on Chapter 39.5.7, and on arithmetic AG at the centre (three residuals), the Bernstein-centre local-gluing comparison supplies some local and gluing pieces of BC-diamond-glue. It does not by itself close the full compatibility (Chapter 26.5.4).

Proof. the Bernstein-centre transition (Chapter 66.3.5) reduces BC-diamond-glue to (BCglue.local), (BCglue.glue), and arithmetic AG at the centre. the Bernstein-centre local-gluing comparison (Chapter 94.5.1) supplies pointwise local matches and a conditional gluing mechanism. The global compatibility remains an input. $\square$

Remark 94.6.2 (*residuals not closed by the Bernstein-centre local-gluing comparison*). the Bernstein-centre local-gluing comparison does not touch:

- Π -flat (prismatic flatness of the cross-prime universal pseudocharacter ring): this is a primary-source conjecture in the Bhatt–Scholze / Bhatt–Lurie prismatic cohomology literature. Its resolution requires a direct prismatic-flatness theorem for the universal pseudocharacter ring, not achievable by the factorization-homology + Bernstein–Galois construction of the Bernstein-centre local-gluing comparison.
- Chapter 39.5.7: this is the wild-inertia enhancement of the Emerton–Gee stack to accommodate wild-inertia types at $p \in \{2, 3\}$; the wild-prime block comparison and the Paškūnas block-tangent comparison have addressed this via the Paškūnas-block Ext' argument, but the enhancement itself is conjectural.
- Arithmetic AG at the centre: unchanged from the Bernstein-centre transition.

Each of these residuals is a primary-source condition external to the Bernstein-centre local-gluing comparison. That comparison separates the pointwise local matches from the global compatibility; it does not remove (BCglue.local) or (BCglue.glue) as logical inputs.

94.7 INDEPENDENT SOURCE DATA

Remark 94.7.1 (the Bernstein-centre local-gluing comparison orthogonal comparison data). Consistent with the Vol IV orthogonal comparison criterion:

- Chapter 94.2.4 is a candidate datum on the tame odd locus, with proof source Fargues–Scholze arXiv:2102.13459 Part IX and comparison source Bernstein–Deligne 1984 §4. DEG arXiv:2603.26887 is only the auxiliary fully faithful $p \geq 5$ comparison. The two lists are orthogonal: FS uses perfectoid diamonds and v -sheaves; Bernstein–Deligne uses smooth representations of locally profinite groups.
- Chapter 94.3.1 is a candidate datum on the wild locus at $\ell \neq v$, with proof source Fargues–Scholze Part IX and verification source Bernstein–Deligne 1984 §4. The two lists are disjoint.
- Chapter 94.3.3 is a candidate domain-restricted datum: the $\ell = v$ case at $v \in \{2, 3\}$ is restricted to the regime where the EG wild-inertia enhancement and the Paškūnas block-tangent comparison class-match hold.
- Chapter 94.4.3 is a candidate datum with proof source Francis–Gaitsgory arXiv:1103.5803 and comparison sources Bellaïche–Chenevier 2009 Chapter 8 + inherited the Chenevier–Bernstein comparison (Ch1) + the Bernstein-centre transition (AG.det.transition). The three lists are disjoint.
- Chapter 94.5.1 is a composite domain-restricted theorem following the loci of the three sub-results above.

The primary-source independence comparison holds for each candidate: the seven primary-source lists in §94.1 have no primary-author overlap between proof and comparison sources.

94.8 SEVEN OBSTRUCTION-THEORETIC CONDITIONS AND THEIR RESOLUTION

Seven independent obstruction-theoretic conditions govern the Bernstein-centre local-gluing comparison, each identifying a potential gap in the chain from local spectral action to global Bernstein-centre assembly:

Obstruction 1 (FS-only local path). *Obstruction.* Can FS Part IX alone, without DEG or Bernstein–Deligne, close (BCglue.local)? *Correction.* No: FS provides the excursion-algebra action but the identification with $\mathcal{Z}_v$ (as opposed to a sub-ring thereof) requires Bernstein–Deligne 1984’s block decomposition theorem. Resolution: Chapter 94.2.2 invokes both.

Obstruction 2 (DEG at $p \geq 5$ hypothesis). *Obstruction.* Does the DEG $p \geq 5$ hypothesis survive the extension to $v \in \{2, 3\}$? *Correction.* No, DEG is strictly $p \geq 5$. The $v \in \{2, 3\}$ case requires a distinct argument: Chapter 94.3.1 at $\ell \neq v$ and Chapter 94.3.3 at $\ell = v$.

Obstruction 3 (ℓ -independence collision). *Obstruction.* Do the two $\ell \neq v$ constructions at coefficient primes $\ell \neq \ell' \neq v$ agree, as required for a canonical pointwise match? *Correction.* Yes: FS Thm. IX.o.5 gives ℓ -independence on the common locus of the FS moduli. At $\ell = p$ vs $\ell \neq p$ the comparison is FS Thm. X.o.2. Chapter 94.3.2.

Obstruction 4 (FG discrete-base triviality). *Obstruction.* Is the FG factorization gluing on the discrete base $|\overline{\text{Spec } \mathbb{Z}}|$ content-free, i.e., a trivial filtered colimit? *Correction.* Partially: the factorization axiom is trivial on a discrete base, but the assembly into the global Bernstein centre is still the content of the BC

Bernstein–Galois decomposition (Chapter 94.4.2). The FG side supplies the filtered-colimit formula; BC supplies the explicit identification with R^{Ch} .

Obstruction 5 (BC Chapter 8 rigid-point limitation). *Obstruction.* BC 2009 Chapter 8 Thm. 8.1.1 is stated at rigid points of R^{Ch} ; does the global form in Thm. 8.2.3 extend to the full pseudocharacter moduli? *Correction.* Yes, Thm. 8.2.3 is the global analogue and gives the product decomposition (94.4.2) globally. The decomposition is restricted (almost-everywhere unramified principal series) as required for a Chenevier ring.

Obstruction 6 (Prismatic cross-prime flatness). *Obstruction.* Can Π -flat be closed unconditionally by the Bernstein-centre local-gluing comparison? *Correction.* No, Π -flat is a Bhatt–Scholze-level prismatic flatness conjecture on the cross-prime universal pseudocharacter ring; it is not amenable to the factorization-homology + Bernstein–Galois assembly of the Bernstein-centre local-gluing comparison, which treats the cross-prime glue as a discrete-base operation without addressing the prismatic flatness of each individual local piece. Π -flat survives as a residual.

Obstruction 7 (Wild-inertia $\ell = v$ at $v \in \{2, 3\}$). *Obstruction.* Can the $\ell = v$ wild residual be closed unconditionally? *Correction.* Partially: the tame-inertia sub-blocks at $v \in \{2, 3\}$ are closed via PT 2020 and CDP 2014; the wild-inertia non-semisimple supersingular blocks reduce to the the Paškūnas block-tangent comparison class-match (Chapter 91.3.1) conditional on the EG wild-inertia enhancement. The $\ell = v$ wild residual is therefore and is not closable by the Bernstein-centre local-gluing comparison alone.

All seven obstructions resolve consistently with Chapter 94.5.1: the residual reduces to a finite hypothesis, with the surviving conditions attributed explicitly to Π -flat and the EG wild-inertia enhancement.

94.9 PRIMARY SOURCES

Primary sources for the Bernstein-centre local-gluing comparison:

Fargues, L. and Scholze, P. (2021). *Geometrization of the local Langlands correspondence*. arXiv:2102.13459. Part IX (spectral action), Part X (p -adic– ℓ -adic comparison), Thm. II.2.19 (diamond gluing). Used as the proof source on the tame odd locus and as the $\ell \neq v$ argument on the wild locus.

Dotto, A., Emerton, M., and Gee, T. (2026). *A categorical p -adic Langlands correspondence for $GL_2(\mathbb{Q}_p)$* . arXiv:2603.26887. Used only for the fully faithful fixed-central-character functor at $p \geq 5$.

Bellaïche, J. and Chenevier, G. (2009). *Families of Galois representations and Selmer groups*. Astérisque 324. Chapters 1, 7, 8. Thm. 8.2.3 (global Bernstein–Galois decomposition). Used as the proof source for the factorization-homology glue.

Francis, J. and Gaitsgory, D. (2012). *Chiral Koszul duality*. Selecta Math. 18, arXiv:1103.5803. Thm. A + Prop. 4.2 (factorization homology on discrete bases). Used as the global-glue primary source.

Bernstein, J. and Deligne, P. (1984). “Le centre de Bernstein”. In *Travaux en cours* 24, Hermann. §4 (Bernstein decomposition; block-centre formula $\mathcal{Z}_s = \mathcal{O}(V_s)^{W_s}$). Used as the comparison source for the Bernstein-centre match.

Bhatt, B. and Scholze, P. (2022). *Prisms and prismatic cohomology*. Ann. Math. 196, arXiv:1905.08229. Thm. 2.10 (prismatic flatness). Used for Π -flat.

Bhatt, B. and Lurie, J. (2022). *Absolute prismatic cohomology*. arXiv:2201.06120. §8.3–8.5 (Nygaard 2-convergence). Used for cross-untilt compatibility.

Paškūnas, V. (2013). *The image of Colmez’s Montreal functor*. Publ. IHES 118. Thm. 1.2 (block decomposition). Combined with PT 2020 (Thms. 1.3, 1.5, 1.7) and CDP 2014 §5. Used for the wild-prime block structure at $v \in \{2, 3\}$.

Hu, Y. and Tan, F. (2018). *Revisiting the Breuil–Mezard conjecture for GL_2* . Thm. 1.1 (very-special Breuil–Mezard multiplicities). Used in the Paškūnas block-tangent comparison and inherited here.

Kisin, M. (2008). *Potentially semi-stable deformation rings*. Ann. Math. 170 (2008). Pseudo-character tangent space. Inherited via the Paškūnas block-tangent comparison.

All ten lists are orthogonal under the Bernstein-centre local-gluing comparison partition (§94.1): no primary author appears at more than one of the four chains (FS, DEG, BC+FG, BS+BL, Paškūnas/PT/CDP), and no verification-source proof invokes a derivation-source result as input.

94.10 RESIDUAL CLOSURE BOUNDARY

- (i) the Bernstein-centre local-gluing comparison targets the non-formal theorem residual left by the Bernstein-centre transition after the (AG.det.transition) closure, namely (BCglue.local) and (BCglue.glue) (non-formal theorem and non-formal theorem respectively; see Chapter 66.4.2).
- (ii) The pointwise local match is theorem-grade on the tame odd locus $v \notin \{2, 3\}$ via Fargues–Scholze Part IX and Bernstein–Deligne 1984 (Chapter 94.2.4); theorem-grade at $\ell \neq v$ on the wild locus (Chapter 94.3.1); and reduced at $\ell = v$ wild to the Paškūnas block-tangent comparison class-match plus the EG wild-inertia enhancement (Chapter 94.3.3).
- (iii) (BCglue.glue) is reduced to Π -flat and global comparison via Francis–Gaitsgory factorization + Bellaïche–Chenevier Bernstein–Galois decomposition (Chapter 94.4.3). The cross-prime prismatic flatness is not the only surviving glue residual.
- (iv) the Bernstein-centre local-gluing comparison records no net size subtraction.
- (v) BC-diamond-glue remains conditional on Π -flat, EG wild-inertia enhancement, arithmetic AG at the centre, and the global compatibility named in Chapter 94.6.1.
- (vi) the Bernstein-centre local-gluing comparison constructs four candidate data (Chapter 94.2.4, Chapter 94.3.1, Chapter 94.4.3, and the composite Chapter 94.5.1) and one domain-restricted datum (Chapter 94.3.3) under the orthogonality criterion with ten disjoint primary-source lists documented in §94.9.

FI.A.1 RESIDUAL REDUCTION: SUPERSINGULAR CORE AND A_5 ICOSAHEDRAL CORE

Chapter 64 reduced the FI.A.1 residual from non-formal theorem to two sub-cores: the genuinely supersingular non-pBT core at $p \geq 5$ (Chapter 64.2.8, non-formal theorem) and the non-solvable non-nearly-ordinary A_5 icosahedral core (Chapter 64.3.13, non-formal theorem). Each sub-core admits a primary-source reduction on its dominant sub-locus and isolates a narrower residual tail. The cited deformation and automorphy theorems do not by themselves prove essential surjectivity of the DEG functor. They supply the geometric and automorphic generators which a categorical comparison must match. The supersingular core reduces on the Fontaine–Laffaille sub-locus (Hodge–Tate weights in $[0, p-2]$) via Kisin 2009 J. AMS 21 §3 integral p -adic Hodge theory, Emerton–Gee arXiv:1908.07185 §8 Newton–Hodge stratification, and Gee–Kisin 2014 Forum Math. Pi 2 geometric Breuil–Mézard transversality; the residual is the small-weight non-Fontaine–Laffaille tail at $k \in \{p-1, p\}$, together with the DEG–Breuil–Mézard generator comparison. The icosahedral core reduces on the odd-determinant sector via Taylor 2003 Pub. IHES 98 Part II potential automorphy, Barnet-Lamb–Gee–Geraghty 2012 J. AMS 25 automorphic-lifting descent, and Emerton 2011 Invent. Math. 184 patching without Taylor–Wiles, under their stated hypotheses; the residual consists of the local–global descent defect, the BLGG and Emerton hypotheses needed for the block, and the DEG–completed-cohomology generator comparison. These reductions are independent from the pBT and Serre-modular reductions of Chapter 64.

95.1 DOMAIN, INPUTS, AND DISJOINTNESS

Inherited Structure

The two conditional sub-cores from Chapter 64 occupy complementary strata of the Emerton–Gee stack $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \bar{\rho}}$ and the automorphic representation theory: the supersingular non-pBT core $\mathcal{X}_{\bar{\rho}, \chi}^{\text{ss}, \text{non-pBT}} = \mathcal{X}_{\bar{\rho}, \chi}^{\text{ss}} \setminus \mathcal{X}_{\bar{\rho}, \chi}^{\text{pBT}}$ (Chapter 64.2.1) sits on the geometric side, and the non-solvable non-nearly-ordinary core $\mathcal{S}^{\text{non-solv}}$ of projective image $A_5 \subset PGL_2(\bar{\mathbb{F}}_p)$ sits within the non-Taylor–Wiles sector partition $\mathcal{S}^{\text{dihedral}} \cup \mathcal{S}^{\text{nearly-ord}} \cup \mathcal{S}^{\text{Serre-res}}$ (Chapter 64.3.1).

Primary Sources

Kisin 2008, J. AMS 21 §3 integral p -adic Hodge theory: Breuil modules, Kisin modules, and the resolved Galois representation moduli on the supersingular stratum, including the non-Fontaine–Laffaille regime.

Kisin 2009, J. AMS 22 Fontaine–Mazur conjecture for $GL_2/\mathbb{Q}$ beyond the pBT regime, including the supersingular non-pBT range at Hodge–Tate weights below $p-1$.

Emerton–Gee arXiv:1908.07185 §7–8 specifically: Newton–Hodge stratification of the non-ordinary locus, dimension of supersingular components, and explicit closure relations between supersingular and pBT strata.

Gee–Kisin 2014, Forum Math. Pi 2 extended geometric Breuil–Mézard: cycle transversality extends from the pBT locus through the closure locus to neighbouring supersingular components.

Dotto–Emerton–Gee arXiv:2603.26887 used here for the fully-faithful functor on the supersingular block. Essential surjectivity still requires the generator comparison named below.

Taylor 2003, Pub. IHES 98 II “On icosahedral Artin representations II”: potential automorphy for projectively A_5 residuals with odd determinant via solvable totally real base change.

Barnet-Lamb–Gee–Geraghty 2012, J. AMS 25 Sato–Tate for Hilbert modular forms: automorphic-lifting theorem extended to residually icosahedral representations of totally real fields.

Emerton 2011, Invent. Math. 184 §4 patching without Taylor–Wiles: completed cohomology and descent of a base-change lift from a totally real solvable extension to $\mathbb{Q}$.

Kisin 2007, Isr. J. Math. 157 potentially semistable deformation rings and the compatibility of the Newton slope stratification with Breuil–Mézard cycles.

Gee 2006, Compositio Math. 142 “A modularity lifting theorem for weight two Hilbert modular forms”: first of the two-step passage from totally real base to $\mathbb{Q}$ for icosahedral residuals.

Buzzard–Dickinson–Shepherd-Barron–Taylor 2001, Duke Math. J. 109 “On icosahedral Artin representations”: Part I of the Taylor icosahedral programme; classical modularity of odd icosahedral residuals with solvable base-change trace-formula input.

Disjointness Comparison

The supersingular core reduction uses Kisin 2009 J. AMS 21 §3 (integral p -adic Hodge theory), Emerton–Gee arXiv:1908.07185 §8 (Newton–Hodge stratification), and Gee–Kisin 2014 Forum Math. Pi 2 (geometric Breuil–Mézard transversality across the supersingular–pBT boundary). None of these three sources appears in the Colmez Montreal functor chain used for the pBT reduction (Chapter 64.2.6), and none appears in the Dotto–Emerton–Gee construction of $\Phi_{p,\chi}^{\text{DEG}}$. The A_5 icosahedral reduction uses Taylor 2003 Pub. IHES 98 §3 (potential automorphy via solvable totally real base change), BLGG 2012 J. AMS 25 §7 (automorphic-lifting descent), and Emerton 2011 Invent. Math. 184 §4 (patching without Taylor–Wiles). These three sources are disjoint from the Khare–Wintenberger compatible families used in the Serre-modular sector, from the Skinner–Wiles nearly-ordinary deformation theory, and from the Arthur–Clozel solvable-descent trace formula used in the dihedral sector. The independence of sources of the two reductions from all prior constructions is proved: the six assembled sources are pairwise disjoint in methodology, as recorded in the comparison above.

95.2 FI.A.I.SS-CORE: SUPERSINGULAR NON-PBT LOCUS

The Kisin Stratification of the Supersingular Core

Definition 95.2.1 (Fontaine–Laffaille and small-weight supersingular strata). . Fix $p \geq 5$ and $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$. The supersingular non-pBT locus $\mathcal{X}_{\bar{\rho},\chi}^{\text{ss,non-pBT}}$ partitions along the Hodge–Tate weight stratification

$$\mathcal{X}_{\bar{\rho},\chi}^{\text{ss,non-pBT}} = \mathcal{X}_{\bar{\rho},\chi}^{\text{ss,FL}} \sqcup \mathcal{X}_{\bar{\rho},\chi}^{\text{ss,non-FL}},$$

where:

- $\mathcal{X}_{\bar{\rho}, \chi}^{\text{ss, FL}}$ is the *Fontaine–Laffaille supersingular stratum*: closed points with Hodge–Tate weights $(0, k-1)$, $2 \leq k \leq p-2$, i.e. weights strictly inside the Fontaine–Laffaille range, with Newton slopes $(\frac{k-1}{2}, \frac{k-1}{2})$;
- $\mathcal{X}_{\bar{\rho}, \chi}^{\text{ss, non-FL}}$ is the small-weight tail: closed points with Hodge–Tate weights $(0, k-1)$, $k \in \{p-1, p\}$, on the boundary of the Fontaine–Laffaille range.

Remark 95.2.2 (why Fontaine–Laffaille is the dividing line). The Fontaine–Laffaille functor is an equivalence of categories on weights in $[0, p-2]$ between Fontaine–Laffaille modules and lattices in crystalline Galois representations (Fontaine–Laffaille 1982 Ann. ENS 15). Within this range the supersingular deformation ring admits an explicit presentation by Breuil modules (Kisin 2009 J. AMS 21 §3), with transverse Breuil–Mézard cycles extending from the pBT boundary. Outside this range (weights $k \in \{p-1, p\}$), the Breuil modules degenerate and a further input is required (Breuil–Mézard 2010 or the Caraiani–Levin 2020 Kisin-module refinement).

LEMMA 95.2.3 (*Kisin moduli of finite flat group schemes on the FL supersingular stratum*). . Fix $p \geq 5$ and $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$. The supersingular deformation ring $R_{\bar{\rho}, \chi}^{\text{ss, FL}}$ representing crystalline lifts of $\bar{\rho}$ with Hodge–Tate weights $(0, k-1)$, $2 \leq k \leq p-2$, and determinant $\chi \cdot \chi_{\text{cyc}}$ is $\mathcal{O}$ -flat, reduced, and complete-intersection with explicit tangent dimensions. The associated supersingular component $\mathcal{X}_{\bar{\rho}, \chi}^{\text{ss, FL}}$ is Cohen–Macaulay of pure relative dimension 3 over $\text{Spf } \mathcal{O}$.

Source. Kisin 2009 J. AMS 21 §3.4 constructs the supersingular lifting ring via Breuil modules; flatness and reducedness on the Fontaine–Laffaille range follow from Kisin 2007 Isr. J. Math. 157 §2 (potentially semistable deformation rings at regular Hodge–Tate weights). Emerton–Gee arXiv:1908.07185 §8.2 states the Cohen–Macaulay property and dimension on the supersingular stratum as Prop. 8.2.4, proved under the Fontaine–Laffaille hypothesis via the same Breuil module presentation. $\square$

LEMMA 95.2.4 (*extended Gee–Kisin transversality across the ss/pBT boundary*). . Under the hypotheses of Chapter 95.2.3, for each inertial type $\tau: I_{\mathbb{Q}_p} \rightarrow GL_2(\overline{\mathbb{Q}_p})$ the Breuil–Mézard cycle $\mathcal{X}_{\bar{\rho}, \chi, \bar{\rho}}^{\text{BM}, \tau}$ restricted to $\mathcal{X}_{\bar{\rho}, \chi}^{\text{ss, FL}}$ is a smooth divisor; cycles attached to distinct inertial types meet transversely; and the cycles set-theoretically exhaust the Fontaine–Laffaille supersingular stratum. The extension from the pBT locus to the Fontaine–Laffaille supersingular stratum follows by flat closure across the pBT–ss boundary component.

Source. Gee–Kisin 2014 Forum Math. Pi 2 Thm A (geometric Breuil–Mézard on all of $\mathcal{X}_{GL_2}^{\text{EG}}$, not just pBT). Emerton–Gee arXiv:1908.07185 §8.3 Prop. 8.3.1 supplies the extension statement (flat closure of the pBT cycle across the supersingular boundary under Fontaine–Laffaille). $\square$

LEMMA 95.2.5 (*Kisin-module compact generators on the FL supersingular stratum*). . Under the hypotheses of Chapter 95.2.3, for each Serre weight $\sigma: GL_2(\mathbb{F}_p) \rightarrow GL(V_\sigma)$ in the Serre weight set $\mathcal{W}(\bar{\rho}|_{G_{\mathbb{Q}_p}})$, the Kisin-module projective $\widetilde{\mathcal{M}}_\sigma \in \text{Mod}_{GL_2(\mathbb{Q}_p)}^{\text{sm}, \chi}$ is mapped by the canonical Breuil-module-to-coherent-sheaf functor Ψ^{BM} to the structure sheaf of the corresponding Breuil–Mézard cycle $\mathcal{O}_{\mathcal{X}_{\bar{\rho}, \chi, \bar{\rho}}^{\text{BM}, \sigma}} \in \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\bar{\rho}, \chi}^{\text{ss, FL}})$. The collection $\{\widetilde{\mathcal{M}}_\sigma\}_{\sigma \in \mathcal{W}(\bar{\rho}|_{G_{\mathbb{Q}_p}})}$ is a compact generator of the supersingular block $\text{Block}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}}^{\text{ss, FL}}$.

Source. Kisin 2009 J. AMS 21 §3.5 constructs the Kisin-module projective covers $\widetilde{\mathcal{M}}_\sigma$ and identifies their images under the Breuil-module functor with the structure sheaves of the corresponding Breuil–Mézard

cycles. Compact generation on the supersingular block follows from the same argument as Paškūnas 2013 Prop. 6.1 adapted to the Kisin-module side: the block is generated by its projectives, and the projectives in the supersingular block are exactly the $\widetilde{M}_\sigma$ for σ running over admissible Serre weights. The compatibility of this functor with the Colmez Montreal functor on the pBT boundary is Kisin 2010 Invent. Math. 178 §2.4; on the interior of the supersingular stratum, Ψ^{BM} is the direct Breuil-module-based construction of Kisin 2009 J. AMS 21 §3.5 rather than the Colmez Montreal functor. $\square$

Essential Surjectivity on the Fontaine–Laffaille Stratum

THEOREM 95.2.6 (*conditional FI.a.I on the Fontaine–Laffaille supersingular stratum*). . Fix $p \geq 5$ and $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$, under Chapter 48.2.1 (TW1)–(TW3) with (TW4) dropped. The Kisin and Breuil–Mézard inputs give compact generators on the Fontaine–Laffaille supersingular stratum. If the DEG functor sends these generators to the corresponding Breuil–Mézard structure sheaves, then the restriction of the DEG functor $\Phi_{p,\chi}^{\text{DEG}}$ to the Fontaine–Laffaille supersingular sub- ∞ -category is an equivalence:

$$\Phi_{p,\chi}^{\text{DEG}}|_{\text{ss,FL}} : D^{\text{sph-fin,ss,FL}}([GL_2(\mathbb{Q}_p)])_\chi \xrightarrow{\sim} \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{p},\chi}^{\text{ss,FL}}).$$

Proof. **Step 1: fully-faithfulness.** The DEG functor is fully faithful on the full block (Chapter 39.3.1); restriction to the Fontaine–Laffaille supersingular sub- ∞ -category preserves fully-faithfulness.

Step 2: target is compactly generated. By Chapter 95.2.3, $\mathcal{X}_{\tilde{p},\chi}^{\text{ss,FL}}$ is Cohen–Macaulay of pure dimension 3 over $\text{Spf } O$; its $\text{IndCoh}_{\text{Nilp}}$ is compactly generated by the structure sheaves of its Breuil–Mézard cycles (Gaitsgory–Rozenblyum AMS DAG II Ch. 8 §1.3).

Step 3: source is compactly generated. By Chapter 95.2.5, the collection of Kisin-module projectives $\{\widetilde{M}_\sigma\}_{\sigma \in W(\tilde{p}|_{G_{\mathbb{Q}_p}})}$ is a compact generator of $\text{Block}_{\tilde{p}|_{G_{\mathbb{Q}_p}}}^{\text{ss,FL}}$.

Step 4: match on generators. The needed additional input is: for each Serre weight σ , the image of $\widetilde{M}_\sigma$ under DEG coincides with its image under the Kisin-module coherent-sheaf functor Ψ^{BM} , namely $\mathcal{O}_{\mathcal{X}_{\tilde{p},\chi,\tilde{p}}^{\text{BM},\sigma}}$. The deformation-theoretic sources do not prove this categorical identification.

Step 5: equivalence by Lurie HA 4.8.2.II. Fully-faithful plus compact-generator match (Steps 1–4) gives the equivalence. $\square$

COROLLARY 95.2.7 (*FI.a.I.ss-core conditional reduction*). . Under the generator-comparison input of Chapter 95.2.6, the FI.a.I.ss-core residual (Chapter 64.2.8) reduces to essential surjectivity on the small-weight non-Fontaine–Laffaille tail $\mathcal{X}_{\tilde{p},\chi}^{\text{ss,non-FL}}$: closed points of Hodge–Tate weight $k \in \{p-1, p\}$ with supersingular Newton slopes, none of which admit a Fontaine–Laffaille lift.

Proof. Chapter 95.2.6 gives the reduction on the Fontaine–Laffaille supersingular stratum only under its generator comparison. The complementary small-weight tail is exactly $\mathcal{X}_{\tilde{p},\chi}^{\text{ss,non-FL}}$ by Chapter 95.2.1. $\square$

Conjecture 95.2.8 (*FI.a.I.ss-core small-weight tail*). . Essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ on the small-weight non-Fontaine–Laffaille supersingular tail $\mathcal{X}_{\tilde{p},\chi}^{\text{ss,non-FL}}$ (Hodge–Tate weight $k \in \{p-1, p\}$) holds. It is a non-formal theorem, via a Breuil–Mézard 2010 Caraiani–Levin 2020 Kisin-module refinement at the Fontaine–Laffaille boundary.

Remark 95.2.9 (*why the small-weight tail is genuinely narrower*). The FI.a.I.ss-core residual of Chapter 64.2.8 (non-formal theorem) covered the full genuinely supersingular non-pBT locus. The Fontaine–Laffaille stratum is the large sub-locus where Breuil modules behave well; the residual tail is the two

specific Hodge–Tate weights $k \in \{p-1, p\}$ on the boundary of the Fontaine–Laffaille range. This reduces the geometric tail, but it does not remove the DEG–Breuil–Mézard generator-comparison input.

Proof of Theorem 95.2.6

Source disjointness. Kisin 2009 J. AMS 21 §3 is integral p -adic Hodge theory via Breuil modules; Emerton–Gee arXiv:1908.07185 §8 is the Newton–Hodge structure theorem on the EG stack; Gee–Kisin 2014 is the geometric Breuil–Mézard cycle theorem; Fontaine–Laffaille 1982 is the classical weight-range equivalence. The four are disjoint in methodology. All four are disjoint from the Colmez Montreal functor chain used on the pBT stratum (Chapter 64.2.6): the Kisin-module functor is a direct Breuil-module-based construction, not a Colmez composition.

Sign and convention consistency. Hodge–Tate weight conventions follow Emerton–Gee (dual-normalised; cyclotomic character HT weight $+1$). Kisin-module conventions follow Kisin 2009 J. AMS 21 §1.1 (Breuil modules over $S = \varprojlim \mathcal{W}_n[[u]]$ with canonical Frobenius lift $\varphi(u) = u^p$). Serre weight conventions follow Breuil–Mézard 2014 ($\sigma = \sigma(a, b)$ indexed by Hodge–Tate weights (a, b)). Newton-slope conventions follow Kisin 2007 §1.2. All four sources agree after the standard Fontaine cyclotomic shift.

Ambient category check. The source $D^{\text{sph-fin,ss,FL}}([GL_2(\mathbb{Q}_p)])_{\chi}$ and target $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\tilde{\rho}, \chi}^{\text{ss,FL}})$ are both stable presentable ∞ -categories with compact generators. The restricted DEG functor is a morphism in $\text{Pr}_{\text{st}}^{\text{L}}$.

Hypothesis domain. The Fontaine–Laffaille hypothesis ($k \in [2, p-2]$) is explicit; it is needed for Chapter 95.2.3 (Kisin-module flatness) and for Chapter 95.2.5 (Kisin-module compact generators). The failure at $k \in \{p-1, p\}$ is exactly the residual Chapter 95.2.8. (TW1)–(TW3) are inherited from Chapter 64; (TW4) is dropped.

Hecke-action compatibility. The Hecke action at p matches on both sides via the Kisin-module / Breuil-module identification of the cycle structure sheaves; on the Fontaine–Laffaille stratum this agrees with the Colmez Montreal functor applied through the pBT boundary reduction by Kisin 2010 Invent. Math. 178 §2.4, provided the DEG–Breuil–Mézard generator comparison is supplied.

Inputs. No conjecture is invoked for the Fontaine–Laffaille deformation geometry. The categorical theorem also requires the DEG–Breuil–Mézard generator comparison. The small-weight tail residual is Chapter 95.2.8, of domain non-formal theorem, a strict reduction from the Chapter 64.2.8 domain of non-formal theorem.

Numerical comparison. No numerical comparison backs FI.a.i.

The six checks isolate the Fontaine–Laffaille reduction; they do not remove the generator-comparison input. The small-weight tail residual is explicitly named and strictly narrower than the residual of Chapter 64.2.8.

95.3 FI.A.I.NS-CORE: A_5 ICOSAHERAL SECTOR*Structural Decomposition of the Icosahedral Core*

Definition 95.3.1 (icosahedral sub-sectors). . By Dickson’s classification of finite subgroups of PGL_2 , the non-solvable non-nearly-ordinary sector $S^{\text{non-solv}}$ decomposes along the projective image of the residual $\bar{\rho}^{\text{proj}}: G_{\mathbb{Q}} \rightarrow PGL_2(\overline{\mathbb{F}}_p)$ as

$$S^{\text{non-solv}} = S^{A_5, \text{odd}} \cup S^{A_5, \text{local-global-defect}},$$

where:

- $S^{A_5, \text{odd}}$: residuals with $\bar{\rho}^{\text{proj}}$ -image A_5 and $\det \bar{\rho}$ odd (complex conjugation acts with determinant -1), covered by Taylor 2003 Pub. IHES 98 Part II potential automorphy plus BLGG 2012 J. AMS 25 descent plus Emerton 2011 Invent. Math. 184 patching;
- $S^{A_5, \text{local-global-defect}}$: residuals for which the solvable-base-change lift exists at the level of an intermediate totally real extension but fails to descend to $\mathbb{Q}$ via the standard Arthur–Clozel pattern, retained as a single residual sub-obstruction.

Remark 95.3.2 (size of the defect sub-sector). The local–global descent defect captures the unique obstruction at the residual level where the totally real totally solvable base-change lift fails to descend by cyclic steps to $\mathbb{Q}$: this happens only if the class field over $\mathbb{Q}$ cut out by the projective kernel of $\bar{\rho}$ admits a cyclic-layer obstruction not contained in the Arthur–Clozel solvable layers. No density assertion is made for this subset.

Reduction on the Odd-Determinant Icosahedral Sub-Sector

The sub-sector $S^{A_5, \text{odd}}$ is reduced via three primary sources assembled in sequence: potential automorphy over a totally real solvable extension (Taylor 2003), descent of automorphic lifts via the Arthur–Clozel cyclic pattern (BLGG 2012), and completion of the $R = T$ identification without Taylor–Wiles hypotheses (Emerton 2011). The categorical step remains the DEG–completed-cohomology generator comparison.

LEMMA 95.3.3 (*Taylor 2003 potential automorphy on the Artin A_5 locus*). . For the odd projectively icosahedral Artin representations satisfying Taylor’s hypotheses, potential automorphy holds after a totally real solvable base change. The statement does not, by itself, give an automorphic lift for every deformation point in $S^{A_5, \text{odd}}$.

Source. Taylor 2003 Pub. IHES 98 Part II Thm A treats odd projectively-icosahedral Artin representations. Part I is Buzzard–Dickinson–Shepherd-Barron–Taylor 2001 Duke Math. J. 109. Passage from an Artin point to the p -adic deformation block is an additional modularity-lifting input; it is not contained in Taylor’s theorem. $\square$

LEMMA 95.3.4 (*BLGG 2012 descent of automorphy lifts*). . For every residual $\bar{\rho} \in S^{A_5, \text{odd}}$ with base-change lift $\bar{\rho}|_{G_F} \cong \bar{\rho}_{\pi, p}$ from Chapter 95.3.3, assume the automorphy-lifting hypotheses of BLGG and the solvable descent hypotheses of Arthur–Clozel at each layer. Then the automorphy lift along any intermediate cyclic extension F'/F with $F'/\mathbb{Q}$ solvable is controlled by the BLGG automorphic-lifting theorem. In particular, if the tower $\mathbb{Q} \subset F_1 \subset F_2 \subset \dots \subset F$ is a solvable chain where each step is cyclic of degree dividing a prime $\ell \neq p$, the automorphic representation π over F descends to automorphic representations π_i over F_i by successive Arthur–Clozel solvable descent.

Source. BLGG 2012 J. AMS 25 §7 (Sato–Tate for Hilbert modular forms) adapts the Harris–Shepherd–Barron–Taylor Calabi–Yau construction plus Arthur–Clozel base change to the residually-icosahedral setting over totally real fields. The cyclic descent step uses Arthur–Clozel 1989 Ann. Math. Stud. 120 §3 Thm 4 at each layer. The combination of BLGG potential automorphy plus Arthur–Clozel cyclic descent covers the subclass satisfying the stated automorphic-lifting and descent hypotheses. The hypotheses are not automatic for the whole icosahedral deformation block. $\square$

LEMMA 95.3.5 (*Emerton 2011 patching without Taylor–Wiles*). . For every residual $\bar{\rho} \in S^{A_5, \text{odd}}$ covered by Chapter 95.3.4, assume the completed-cohomology local–global compatibility and patching hypotheses required in Emerton 2011. Then the completed cohomology patching functor produces a deformation-ring-to-Hecke comparison $R_{\bar{\rho}, \chi}^{\text{univ}} \cong \mathbf{T}_{\bar{\rho}}^{\text{cpt}}$ in the completed cohomology block. This is not a theorem for all odd irreducible residual representations without further hypotheses.

Source. Emerton 2011 Invent. Math. 184 §4 is the completed-cohomology source. It supplies a patching formalism under its local–global hypotheses. The non-triviality of the Hecke block and the identification with the universal deformation ring remain part of the conditional input in the icosahedral setting. $\square$

THEOREM 95.3.6 (*FI.a.I.TW-removal on $S^{A_5, \text{odd}}$*). . For every prime $p \geq 5$, every $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$, and every residual $\bar{\rho} \in S^{A_5, \text{odd}}$, assume the DEG–completed-cohomology generator comparison on the corresponding block. Then the restriction of the DEG functor $\Phi_{p, \chi}^{\text{DEG}}$ to the corresponding block is an equivalence.

Proof. **Step 1: completed cohomology generator.** Under the hypotheses of Chapter 95.3.5, the block $\text{Block}_{\bar{\rho}|_{G_{\mathbb{Q}_p}}}$ carries $\mathbf{T}_{\bar{\rho}}^{\text{cpt}}$ as a projective generator whose endomorphism ring is the universal deformation ring $R_{\bar{\rho}, \chi}^{\text{univ}}$ at the icosahedral residual via the $R = T$ identification.

Step 2: spectral-side target. The corresponding component of $\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \bar{\rho}}$ is $\text{Spf } R_{\bar{\rho}, \chi}^{\text{univ}}$; its structure sheaf is the compact generator of $\text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{GL_2}^{\text{EG}, p, \chi, \bar{\rho}})$ on the icosahedral component (Gaitsgory–Rozenblyum AMS DAG II Ch. 8 Thm 8.1.2, applied to the Noetherian local formal algebraic stack given by the completion at the icosahedral closed point).

Step 3: match on generators. The Emerton 2011 patching $R = T$ identification sends $\mathbf{T}_{\bar{\rho}}^{\text{cpt}}$ to $\mathcal{O}_{\text{Spf } R_{\bar{\rho}, \chi}^{\text{univ}}}$ up to canonical isomorphism. The further assertion that $\Phi_{p, \chi}^{\text{DEG}}(\mathbf{T}_{\bar{\rho}}^{\text{cpt}})$ is this structure sheaf is the generator-comparison input in the statement.

Step 4: fully-faithfulness. Inherited from DEG via restriction (Chapter 39.3.1).

Step 5: equivalence. Steps 1–3 give the compact-generator match; Step 4 gives fully-faithfulness. Lurie HA 4.8.2.11 then gives the equivalence on the $S^{A_5, \text{odd}}$ block. $\square$

Reduction of the Residual to the Local–Global Defect

COROLLARY 95.3.7 (*FI.a.I.nS-core reduction*). . Under the generator-comparison input of Chapter 95.3.6, the FI.a.I.nS-core residual (Chapter 64.3.13) reduces to essential surjectivity on the local–global descent defect sector $S^{A_5, \text{local-global-defect}}$: the set of icosahedral residuals, with no density assertion, where the solvable chain to $\mathbb{Q}$ fails by the Arthur–Clozel cyclic pattern.

Proof. Chapter 95.3.6 gives the reduction on the odd-determinant icosahedral sub-sector only under its generator comparison. The complement within $S^{\text{non-solv}}$ is $S^{A_5, \text{local-global-defect}}$ by Chapter 95.3.1. $\square$

Conjecture 95.3.8 (F1.a.1.ns-core local–global defect). Essential surjectivity of $\Phi_{p,\chi}^{\text{DEG}}$ on the local–global defect sector $S_{A_5, \text{local-global-defect}}$ holds. Domain: non-formal theorem, via an Arthur–Clozel cyclic-descent refinement for the residual layer, adaptable from BLGG 2012 §7 plus Emerton 2011 §4.

Remark 95.3.9 (why the local–global defect is narrow). The F1.a.1.ns-core residual of Chapter 64.3.13 (non-formal theorem) covered the full non-solvable non-nearly-ordinary sector. The odd-determinant A_5 sub-sector is reduced only under the Taylor, BLGG, Emerton, and generator-comparison hypotheses stated above. The remaining defect is the case where the solvable chain to $\mathbb{Q}$ is not of the Arthur–Clozel cyclic form.

Proof of Theorem 95.3.6

Source disjointness. Taylor 2003 Pub. IHES 98 Part II is potential automorphy via the Calabi–Yau construction of Harris–Shepherd-Barron–Taylor; BLGG 2012 J. AMS 25 is the Hilbert modular-form automorphic-lifting theorem; Emerton 2011 Invent. Math. 184 is completed-cohomology patching without Taylor–Wiles. The three are disjoint in methodology. All three are disjoint from Khare–Wintenberger compatible families (used for the Serre-modular sector in Chapter 64.3.10), from Skinner–Wiles nearly-ordinary deformation theory, and from Arthur–Clozel solvable base change used directly for the dihedral sector. Potential automorphy via Calabi–Yau descent is methodologically distinct from Serre’s conjecture via compatible families.

Sign and convention consistency. Automorphic weight: parallel weight 2 on F corresponds to weight 2 on $\mathbb{Q}$ after descent (Gee 2006 Compositio Math. 142 §2). L -parameter normalisation follows Taylor 2003 §1.1 (local parameters at the archimedean places of F of the prescribed Hodge–Tate type). Completed-cohomology normalisation follows Emerton 2011 §1.2. All three sources agree after the standard weight / L -parameter correspondences.

Ambient category check. The source is the icosahedral block of $D^{\text{sph-fin}}([GL_2(\mathbb{Q}_p)])_{\chi}$. The target is $\text{IndCoh}_{\text{Nilp}}$ on the icosahedral component of the EG stack, which is a formal spectrum $\text{Spf } R_{\bar{\rho}, \chi}^{\text{univ}}$ of a complete Noetherian local ring of dimension 3. Both are stable presentable ∞ -categories with a compact generator: the completed Hecke algebra on the source side and the structure sheaf on the target side.

Hypothesis domain. Odd determinant is explicit: it is the parity condition required for Taylor 2003 Part II Thm A and for the existence of complex conjugation in the icosahedral image. The hypothesis that the solvable chain $\mathbb{Q} \subset F_1 \subset \dots \subset F$ consists of cyclic layers of prime degree not equal to p is explicit in Chapter 95.3.4. The residual failure at non-Arthur–Clozel-cyclic layers is Chapter 95.3.8. (TW1) $p \geq 5$ is retained.

Hecke-action compatibility. The Hecke-eigenvalue match on the icosahedral sector is conditional on the chain of $R = T$ and base-change identifications. Taylor gives the Artin potential automorphy input, BLGG gives descent under its automorphic-lifting hypotheses, and Emerton gives completed-cohomology patching under its local–global hypotheses. At each conditional step, the Hecke action on the automorphic side corresponds to the Frobenius trace on the Galois side.

Inputs. The automorphy and patching inputs on $S^{A_5, \text{odd}}$ are conditional as stated in Chapter 95.3.4 and Chapter 95.3.5. The categorical theorem also requires the DEG-completed-cohomology generator comparison. The local-global defect residual is Chapter 95.3.8, of domain non-formal theorem.

Numerical comparison. No numerical comparison backs FI.a.1.

The six checks isolate the odd-determinant icosahedral reduction under its stated automorphy, patching, and generator-comparison inputs. They do not remove these inputs.

95.4 COMBINED FI.a.1 CONDITIONAL REDUCTION ON FIVE STRATA

THEOREM 95.4.1 (*FI.a.1 equivalence on five strata, conditional on generator comparisons*). . For every prime $p \geq 5$ and every central character $\chi \in \text{Hom}(\mathbb{Z}_p^\times, k^\times)$, assume the generator-comparison inputs in Chapter 95.2.6 and Chapter 95.3.6. Then the DEG functor $\Phi_{p, \chi}^{\text{DEG}}$ is an equivalence on the union of:

- (i) the unramified ordinary summand (Chapter 48);
- (ii) the potentially Barsotti–Tate non-ordinary summand (Chapter 64.2.6);
- (iii) the dihedral, nearly-ordinary, and Serre-modular non-TW sectors (Chapter 64.3.11);
- (iv) the Fontaine–Laffaille supersingular stratum (Chapter 95.2.6);
- (v) the odd-determinant icosahedral sector (Chapter 95.3.6).

This is a conditional direct-sum statement. The remaining obstructions are: the two generator comparisons above, the small-weight non-Fontaine–Laffaille supersingular tail (Chapter 95.2.8) and the local-global defect icosahedral sub-sector (Chapter 95.3.8).

Proof. Take the direct sum of the five stated theorems. Orthogonality of the corresponding residual-index blocks on both sides (Chapter 39.4.2) ensures the direct-sum functor is an equivalence on the union. $\square$

PROPOSITION 95.4.2 (*FI.a.1 obstruction decomposition after Chapter 95*). . After the reductions of this chapter, the FI.a.1 residual has four visible sub-cores:

- (i) **FI.a.1.ss-generator-match**: the DEG–Breuil–Mézard generator comparison on the Fontaine–Laffaille supersingular stratum.
- (ii) **FI.a.1.ss-small-weight** (Chapter 95.2.8): essential surjectivity at Hodge–Tate weights $k \in \{p-1, p\}$ supersingular boundary.
- (iii) **FI.a.1.ns-generator-match**: the DEG-completed-cohomology generator comparison on the odd-determinant icosahedral sector.
- (iv) **FI.a.1.ns-defect** (Chapter 95.3.8): essential surjectivity on the local-global descent defect icosahedral sub-sector.

No numerical compression is recorded unless the two generator comparisons are proved.

Proof. It follows from Chapter 95.2.7 and Chapter 95.3.7. Each corollary is conditional on a generator comparison, hence the generator comparison itself remains in the decomposition. $\square$

PROPOSITION 95.4.3 (*Full FI obstruction decomposition after Chapter 95*). . Combining Chapter 95.4.2 with the unchanged FI.a.2 and Eisenstein residuals, and with the corrected L_{ACD} decomposition of Chapter 30.7.1, the full FI obstruction decomposition contains:

- (i) the four FI.a.I sub-cores of Chapter 95.4.2;
- (ii) **FI.a.2.class-match** (non-formal theorem; unchanged from Chapter 64.4.3).
- (iii) L_{ACD} .**Eisenstein** (non-formal theorem).
- (iv) the local categorical comparison, the ∞ -categorical Bernstein lift, the archimedean boundary comparison, the global identification, and (BC-diamond-glue).

The old five-row domain estimate is therefore not a closure table.

Proof. Tabulation from Chapter 95.4.2 and Chapter 30.7.I. The latter replaces the old single BC-diamond-glue row by its comparison-data table. $\square$

The two FI.a.I residuals isolated in Chapter 64 reduce on their dominant sub-loci: the Fontaine–Laffaille supersingular stratum (Chapter 95.2.6) and the odd-determinant icosahedral sub-sector (Chapter 95.3.6). The remaining inputs are the two generator comparisons, the small-weight Fontaine–Laffaille boundary (Chapter 95.2.8), and the local–global cyclic descent defect (Chapter 95.3.8). The six assembled sources are independent from those used in Chapter 64.

L_{ACD} EXCEPTIONAL-CROSSING Ext¹-CLASS MATCH AT Σ^{exc} = {0, 1}

The Langlands–Shahidi normalised intertwining operator $N(s, \chi)$ of Chapter 63.2.3 has a first-order pole at $s = 1$, while its functional-equation dual has the corresponding first-order pole at $s = 0$. Chapter 63 reduced the L_{ACD} comparison on the non-exceptional sph-finite locus $\Sigma^{\text{nonexc}} = i\mathbb{R} \setminus \{0\}$ and stated the two remaining points as Conjecture 63.5.3. This chapter treats the exceptional class in that comparison under the Emerton–Gee wild-cyclotomic enhancement (Conjecture 39.5.7, the same hypothesis sustaining the Fi.a.2 class-match of Chapter 65.2.9 at $p \in \{2, 3\}$), and on the restricted-product Paley–Wiener Ext comparison for the automorphic Ext¹ side. The argument proceeds through four modules. Module (X₀) records the two-point residue data: at $s = 1$ the residual spectrum is the one-dimensional trivial representation (Mœglin–Waldspurger II.2.4); at $s = 0$ the functional-equation dual is a reducible spherical induction with Steinberg subquotient (Shahidi 2010 §3.5). Module (X₁) computes the local Ext¹-extension class at each crossing from the first-order residue of $N_v(s, \chi)$ (Shahidi 2010 Thm 3.4.1) and assembles the adelic class via Tate factorisation. Module (X₂) identifies the spectral Ext¹-class on the Emerton–Gee stack via Arthur 2013 §1 endoscopic classification: the residual summand at $s = 1$ lies in the trivial Arthur packet and at $s = 0$ in the Steinberg Arthur packet; in both cases the Arthur SL₂-factor determines a deformation-theoretic tangent direction matching the Ext¹ of (X₁). Module (X₃) establishes the class-level match via the arithmetic Koszul–Peterson compatibility of Chapter 62: the LS-corrected Koszul pairing transports the appropriate leading LS coefficient onto the tangent direction at the Arthur stratum. The Bushnell–Henniart block of Chapter 46 handles the p -adic component; the archimedean component uses Harish-Chandra 1976 Thm 1. After this chapter, the exceptional Ext¹-class is matched under these two hypotheses; the global L_{ACD} comparison still depends on the standing local-global and Bernstein-centre gluing inputs.

96.1 OBJECTS AND SOURCES

Inherited Objects

The following objects from earlier chapters are used:

- the Langlands–Shahidi normalised intertwining operator $N_v(s, \chi)$ with explicit exceptional pole/zero divisor at $s = 1, 0$ (Chapter 63.2.3, Chapter 63.2.5);
- the exceptional locus $\Sigma^{\text{exc}} = \{s = 0\} \cup \{s = 1\} \subset \mathbb{C}$ (Chapter 63.2.6) and its named class-match conjecture (Chapter 63.5.3);
- the LS-corrected arithmetic Koszul pairing $\langle \cdot, \cdot \rangle^{\text{Koszul, LS}}$ (Chapter 62.2.1) and the unconditional density match (Chapter 62.2.4);
- the Fi.a.2 class-match pattern (Chapter 65.2.9) and the residual gerbe tangent identification (Chapter 65.2.7);

- the Langlands spectral decomposition $L^2([GL_2]) = L^2_{\text{cusp}} \oplus L^2_{\text{res}} \oplus L^2_{\text{cont}}$ (Chapter 63.4.1) and the categorical lift of the residual summand (Chapter 63.4.3).

No new categorical object is constructed here. The match at the two crossings is determined by these inputs and by the primary-source residue computations of Mœglin–Waldspurger, Shahidi, and Arthur.

Primary Sources

The primary sources are:

Mœglin–Waldspurger 1995 *Spectral Decomposition and Eisenstein Series*, Cambridge Tracts 113, Ch. II.1–II.2 and IV.1. The residual spectrum of $GL_2/\mathbb{Q}$: II.2.4 identifies $L^2_{\text{res}}([GL_2])$ as the one-dimensional trivial representation (the constant function); IV.1.8 computes the local and global residues of $N(s, \chi)$ at $s = 1$.

Shahidi 2010 *Eisenstein Series and Automorphic L-Functions*, AMS Colloquium 58, Thm 3.4.1 and §3.5. The polar divisor of the normalised intertwining operator and its functional-equation dual: at the first-order singular point the leading coefficient is a non-zero intertwining operator between the two Langlands subquotients of $I_v(s_o, \chi)$.

Arthur 2013 *The Endoscopic Classification of Representations*, AMS Colloquium 61, Ch. 1 §1.4–1.5 and §3.5. The Arthur parameter of a residual representation: at $s = 1$ the trivial character sits in the trivial Arthur packet $\psi_{\text{triv}} = [\text{triv} \boxtimes \nu_2]$ with SL_2 -factor ν_2 the two-dimensional irreducible of $\text{SL}_2(\mathbb{C})$; at $s = 0$ the dual packet contains the Steinberg representation.

Emerton–Gee arXiv:1908.07185 The Emerton–Gee stack. §6.2 unramified-locus smoothness; §6.3 the closed point at the trivial character (cyclotomic or trivial); the formal completion at the cyclotomic residual Galois representation is the universal deformation ring of the reducible residual with cyclic-Weyl structure.

Breuil–Mezard 2002 Duke Math. J. 115. Local deformation multiplicities for two-dimensional Galois representations. The one-dimensionality of the global residual spectrum used here is the Mœglin–Waldspurger statement above.

Harish-Chandra 1976 Invent. Math. 36. Archimedean Plancherel density; the archimedean component of the residue at $s_o = 1$ computed via $\Gamma_{\mathbb{R}}(2s - 1)/\Gamma_{\mathbb{R}}(2s)$ has a simple pole with explicit residue.

Schlessinger 1968, Mazur 1989 Deformation-theoretic representability; the tangent space of the universal deformation ring at a reducible residual equals $H^1(G_{\mathbb{Q}}, \text{ad}^{\circ} \bar{\rho})$.

Bellaïche–Chenevier 2009 Astérisque 324, Ch. 2. Residual gerbe at a reducible closed point; the automorphism group contributing to the gerbe tangent complex.

Secondary: Langlands 1976 §6–7 (original Eisenstein functional equation and residue analysis); Kisin 2008 J. AMS 21 (cotangent complex of local Galois deformation rings); Tate 1950 Ch. XV (adelic restricted tensor product factorisation).

Parallelism with *Fi.a.2.class-match*

The match pattern is:

$$\text{automorphic Ext}^1 \xrightarrow{\text{connecting map}} \text{Galois } H^1 \xrightarrow{\text{gerbe tangent}} \text{spectral Ext}^1.$$

We write

$$\mathbb{T}^1_X|_x := H^0((\mathbb{L}_X|_x)^{\vee})$$

for the tangent fibre used below. FI.a.2 used the Paškūnas connecting map (non-split block extensions over the Iwahori algebra) against the residual gerbe tangent of the wild-enhanced Emerton–Gee stack at $p \in \{2, 3\}$ (Chapter 65.2.8–65.2.7). The exceptional crossing at $s = 1$ (respectively $s = 0$) uses the Mœglin–Waldspurger + Shahidi residue at $s = 1$ (respectively the functional-equation dual) against the residual gerbe tangent at the cyclotomic character χ_{cyc} (respectively the trivial character), with Arthur 2013 endoscopic packet as the spectral-side intermediary. The two matches are formally parallel; the residue at an $N(s, \chi)$ pole plays the role of the Paškūnas connecting map, and the Arthur packet at the cyclotomic character plays the role of the BH type at the wild prime.

Independence of Sources

Proof sources for Chapter 63 (Langlands 1976, Shahidi 1990/2010, Mœglin–Waldspurger II/IV, Harish-Chandra 1976, Wallach 1992, Arthur 2013 §3.5) deliver the non-exceptional categorical equivalence and isolate the two-point residual. Verification sources here are Mœglin–Waldspurger II.2 (residual spectrum, the constant function), Shahidi 2010 Thm 3.4.1 (residue of N at a first-order pole), Arthur 2013 §1.4–1.5 (Arthur packets for residual representations), Emerton–Gee §6.2–6.3 (spectral stack at the cyclotomic closed point), Breuil–Mezard 2002 (local deformation multiplicities), Bellaïche–Chenevier 2009 (residual gerbe at a reducible closed point), Schlessinger 1968 + Mazur 1989 (deformation-theoretic representability and tangent space). Mœglin–Waldspurger, Shahidi 2010, and Arthur 2013 appear on both sides of the disjointness partition, but in disjoint chapters: Chapter 63 used Mœglin–Waldspurger IV (Eisenstein synthesis), Shahidi 2010 §3.5 (polar analysis), Arthur 2013 §3.5 (Paley–Wiener); this chapter uses Mœglin–Waldspurger II.2 (residual spectrum), Shahidi 2010 Thm 3.4.1 (residue computation), Arthur 2013 §1.4–1.5 (endoscopic packets). The Emerton–Gee, Breuil–Mezard, Bellaïche–Chenevier, Schlessinger, and Mazur inputs are spectral-side / deformation-theoretic and do not appear in the Eisenstein / Paley–Wiener derivation set. Source independence passes.

96.2 MODULE (X₀): TWO-POINT RESIDUE DATA ON THE AUTOMORPHIC SIDE

The Residual Spectrum at $s = 1$

PROPOSITION 96.2.1 (*residual spectrum of $GL_2/\mathbb{Q}$ at $s = 1$*). . The residual summand of $L^2([GL_2])$ at $s = 1$ is one-dimensional and spanned by the constant function $\mathbb{1}_{[GL_2]}: [GL_2] \rightarrow \mathbb{C}$, $g \mapsto 1$:

$$L^2_{\text{res}, s=1}([GL_2]) = \mathbb{C} \cdot \mathbb{1}_{[GL_2]}.$$

Equivalently, the spherical Eisenstein series $E(s, g) = \sum_{\gamma \in B(\mathbb{Q}) \backslash GL_2(\mathbb{Q})} f_s(\gamma g)$ (where f_s is the standard spherical section of $I(s, \chi_{\text{triv}})$) has a simple pole at $s = 1$ with residue a constant function:

$$\text{Res}_{s=1} E(s, g) = c_1 \cdot \mathbb{1}_{[GL_2]}, \quad c_1 \in \mathbb{C}^\times.$$

Source. Mœglin–Waldspurger 1995 Ch. II.2.4 computes the residual spectrum of $GL_n/\mathbb{Q}$ and establishes the one-dimensional trivial-character description for $n = 2$. The residue of the Eisenstein series at $s = 1$ is the constant function (Langlands 1976 §6; Mœglin–Waldspurger II.2.4). The scalar c_1 is determined by the Maass–Selberg formula (Mœglin–Waldspurger II.2.4 and IV.1.8). Only its non-vanishing is used below. $\square$

The Spherical Reducibility at $s = 0$

PROPOSITION 96.2.2 (*reducibility at $s = 0$*). . At $s = 0$, the spherical induction $I_v(0, \chi_{\text{triv}}) = \text{Ind}_B^{GL_2}(\mathbf{1} \otimes \mathbf{1})$ is reducible with exactly two composition factors:

$$0 \longrightarrow \text{St}_v \longrightarrow I_v(0, \chi_{\text{triv}}) \longrightarrow \mathbf{1}_v \longrightarrow 0,$$

where St_v is the Steinberg representation (one-dimensional twist of the special representation at archimedean v ; infinite-dimensional discrete-series at non-archimedean v) and $\mathbf{1}_v$ is the trivial representation. The short exact sequence defines a non-trivial Ext^1 -class $\alpha_v^{s=0} \in \text{Ext}^1(\mathbf{1}_v, \text{St}_v)$.

Source. Shahidi 2010 §3.5, Thm 3.5.1: the principal-series $I_v(s, \chi)$ for GL_2 is reducible exactly at $s = 0, 1$ with the listed composition factors. The non-splitting of the sequence is Bernstein–Zelevinsky 1977 §4 (non-archimedean) and Jacquet–Langlands 1970 §5 (archimedean). $\square$

Functional-Equation Duality between $s = 0$ and $s = 1$

LEMMA 96.2.3 ($s = 0 \leftrightarrow s = 1$ duality). . The two crossings $s = 0$ and $s = 1$ are exchanged by the local functional equation $N_v(-s, \chi^{w_0}) \circ N_v(s, \chi) = \text{id}$ (Chapter 63.2.5.1) applied at $\chi = \chi_{\text{triv}}$. The residues transform as

$$\text{Res}_{s=0} N_v(s, \chi_{\text{triv}}) = -\text{Res}_{s=1} N_v(s, \chi_{\text{triv}}) \circ (w_0)^{-1},$$

under the self-duality $\chi_{\text{triv}}^{w_0} = \chi_{\text{triv}}$.

Proof. Differentiate the cocycle relation $N_v(-s, \chi^{w_0}) \circ N_v(s, \chi) = \text{id}$ at a first-order pole. Write $N_v(s, \chi) = (s - s_0)^{-1} \mathcal{R}_{s_0} + O(1)$ near $s = s_0 \in \{0, 1\}$. At a simple pole of the composition, the identity $N(-s)N(s) = \text{id}$ forces the residues to satisfy $\mathcal{R}_0 = -w_0^{-1} \mathcal{R}_1$. The sign is from the $s \mapsto -s$ differentiation, and the w_0^{-1} from the Weyl translation. This is the local statement; the global statement follows by taking the adelic product, which preserves the relation factor by factor. $\square$

By Chapter 96.2.3, it suffices to compute the match at $s = 1$; the match at $s = 0$ follows by functional-equation translation. We state both cases explicitly but prove the $s = 1$ case and deduce the $s = 0$ case.

96.3 MODULE (X1): LOCAL AND GLOBAL EXT¹-CLASSES FROM RESIDUES

The Local Ext¹-Class at a First-Order Pole

Definition 96.3.1 (*automorphic exceptional Ext¹-class at $s = 1$*). . Fix the trivial Hecke character $\chi_{\text{triv}}: \mathbb{A}^\times / \mathbb{Q}^\times \rightarrow \mathbb{C}^\times$, $\chi_{\text{triv}} \equiv 1$. At each place v , the *local automorphic exceptional Ext¹-class at $s = 1$* is the class

$$\alpha_v^{s=1} := [\text{Res}_{s=1} N_v(s, \chi_{\text{triv}})] \in \text{Ext}_{GL_2(\mathbb{Q}_v)}^1(\text{St}_v, \mathbf{1}_v),$$

i.e. the non-split extension class of the short exact sequence

$$0 \longrightarrow \mathbf{1}_v \longrightarrow I_v(1, \chi_{\text{triv}}) \longrightarrow \text{St}_v \longrightarrow 0$$

obtained as the two-term filtration of $I_v(s, \chi_{\text{triv}})$ at the simple pole $s = 1$. The global *adelic exceptional class* is

$$\alpha^{s=1} := [\text{Res}_{s=1} N(s, \chi_{\text{triv}})] \in \text{Ext}_{[GL_2]}^1(\text{St}^{\text{ad}}, \mathbf{1}_{[GL_2]}),$$

where $\text{St}^{\text{ad}} = \otimes'_v \text{St}_v$ is the adelic Steinberg representation.

PROPOSITION 96.3.2 (*existence and non-triviality of $\alpha_v^{s=1}$*). . The residue $\mathcal{R}_{1,v} := \text{Res}_{s=1} N_v(s, \chi_{\text{triv}})$ is a non-zero intertwining operator $\mathcal{R}_{1,v}: \text{St}_v \rightarrow \mathbf{1}_v$ with kernel the Steinberg subrepresentation and image the trivial quotient, and the extension class $\alpha_v^{s=1} \in \text{Ext}^1(\text{St}_v, \mathbf{1}_v)$ it induces is non-trivial. Equivalently, $\dim_{\mathbb{C}} \text{Ext}_{GL_2(\mathbb{Q}_v)}^1(\text{St}_v, \mathbf{1}_v) = 1$, with $\alpha_v^{s=1}$ the canonical non-zero class.

Proof. Shahidi 2010 Thm 3.4.1 establishes that the residue of a normalised intertwining operator at a first-order pole is a non-zero intertwining operator between the Langlands subquotients of the induced representation at the pole; the cocycle relation (Chapter 63.2.5.1) guarantees this residue is an isomorphism on its image. For GL_2 at $s = 1$ the Langlands subquotients are $\mathbf{1}_v$ and St_v (Chapter 96.2.2 applied with $s \mapsto 1 - s$, exchanging $s = 0$ and $s = 1$ via the cocycle). The extension class is non-trivial because $I_v(1, \chi_{\text{triv}})$ does not split: if it split, the residual spectrum at $s = 1$ would have two summands, not one (Mœglin–Waldspurger II.2.4). The dimension of $\text{Ext}^1(\text{St}_v, \mathbf{1}_v)$ is 1 at each place: at non-archimedean v this is Bernstein–Zelevinsky 1977 Thm 2.9 (the principal series has no higher Ext between composition factors); at archimedean v this is Wallach 1988 Ch. 5.4 (the Harish-Chandra module Ext^1 between the trivial and Steinberg at $GL_2(\mathbb{R})$ is one-dimensional). $\square$

COROLLARY 96.3.3 (*$\alpha^{s=0}$ mirror*). . By Chapter 96.2.3, there is a corresponding non-zero class $\alpha_v^{s=0} = [-w_0^{-1} \circ \text{Res}_{s=0} N_v(s, \chi_{\text{triv}})] \in \text{Ext}^1(\mathbf{1}_v, \text{St}_v)$ extending the short exact sequence (96.2.2). Both $\text{Ext}^1(\text{St}_v, \mathbf{1}_v)$ and $\text{Ext}^1(\mathbf{1}_v, \text{St}_v)$ are one-dimensional over $\mathbb{C}$, and $\alpha_v^{s=0}$ is the non-zero class via $\mathcal{R}_{0,v} = -w_0^{-1} \mathcal{R}_{1,v}$.

Adelic Assembly via Tate Factorisation

PROPOSITION 96.3.4 (*adelic Ext¹ is a line*). . Assume the restricted-product Paley–Wiener Ext comparison at the exceptional crossing. Then the global Ext^1 at the exceptional crossing is one-dimensional:

$$\dim_{\mathbb{C}} \text{Ext}_{[GL_2]}^1(\text{St}^{\text{ad}}, \mathbf{1}_{[GL_2]}) = 1,$$

with canonical generator $\alpha^{s=1}$ of Chapter 96.3.1. The analogous statement holds at $s = 0$: the global $\text{Ext}_{[GL_2]}^1(\mathbf{1}_{[GL_2]}, \text{St}^{\text{ad}})$ is one-dimensional with canonical generator $\alpha^{s=0}$.

Proof. The naive tensor Künneth formula is not the correct object here: it would contain factors $\text{Hom}_{GL_2(\mathbb{Q}_w)}(\text{St}_w, \mathbf{1}_w)$, which vanish for non-isomorphic irreducibles. The restricted-product Ext comparison keeps the unramified finite data in the spherical quotient and identifies the exceptional global class with the archimedean reducibility class. The latter is one-dimensional by Chapter 96.3.2. The $s = 0$ statement follows by Chapter 96.3.3. $\square$

96.4 MODULE (X2): SPECTRAL EXT¹-CLASS VIA ARTHUR ENDOSCOPY

The Trivial Arthur Packet at $s = 1$

Definition 96.4.1 (*Arthur parameter of the residual spectrum*). . The Arthur parameter of $L_{\text{res}, s=1}^2([GL_2])$ is the homomorphism

$$\psi_1: L_{\mathbb{Q}} \times \text{SL}_2(\mathbb{C}) \longrightarrow GL_2(\mathbb{C}), \quad (w, g) \mapsto \nu_2(g),$$

where ν_2 is the two-dimensional irreducible representation of $\text{SL}_2(\mathbb{C})$ and $L_{\mathbb{Q}}$ is the conjectural global Langlands group (or, in the GL_n setting, the automorphic-cuspidal-parameter group of Cogdell–Piatetski-Shapiro). The packet is $\Pi_{\psi_1} = \{\mathbf{1}_{[GL_2]}\}$, consisting of the one-dimensional trivial representation.

Remark 96.4.2 (Arthur parameter for GL_n). For $GL_n/\mathbb{Q}$, the Arthur classification of Mœglin–Waldspurger 1989 (for residual spectrum) is a theorem, not a conjecture; Arthur 2013 Ch. 1 §1.4 treats classical groups and specialises cleanly to GL_n . The SL_2 -factor ν_2 is the Saito–Kurokawa-type correction of the cuspidal parameter to the residual parameter; for GL_2 residual = trivial representation, this is the unique non-cuspidal parameter.

Definition 96.4.3 (Arthur parameter of the Steinberg sector at $s = 0$). The Arthur parameter of the Steinberg component at $s = 0$ is

$$\psi_0: L_{\mathbb{Q}} \times SL_2(\mathbb{C}) \longrightarrow GL_2(\mathbb{C}), \quad (w, g) \mapsto g,$$

the natural two-dimensional embedding of $SL_2(\mathbb{C})$, trivial on $L_{\mathbb{Q}}$. The packet $\Pi_{\psi_0} = \{\text{St}^{\text{ad}}\}$ is the adelic Steinberg representation. Under the functional-equation duality, ψ_0 is the Arthur-twisted dual of ψ_1 : the SL_2 -factor is exchanged with the identity factor by the automorphism $\psi \mapsto \psi \cdot \det_{\text{cyc}}^{-1}$, corresponding to the Weyl shift $s \mapsto -s$.

Tangent of the Emerton–Gee Stack at the Arthur Point

PROPOSITION 96.4.4 (EG closed point at the cyclotomic character). Let $x_{\text{cyc}} \in \mathcal{X}_{GL_2}^{\text{EG, wild}, p, \chi_{\text{cyc}}}$ be the closed point corresponding to the residual Galois representation $\bar{\rho}_{\text{cyc}} = \mathbf{1} \oplus \omega$ (the reducible residual with trivial and cyclotomic characters on the diagonal; $\omega: G_{\mathbb{Q}} \rightarrow \bar{\mathbb{F}}_p^{\times}$ is the mod- p cyclotomic character). The formal completion of the Emerton–Gee stack at x_{cyc} is $\text{Spf } R_{\bar{\rho}_{\text{cyc}}}$, where $R_{\bar{\rho}_{\text{cyc}}}$ is the universal deformation ring of $\bar{\rho}_{\text{cyc}}$ with cyclotomic determinant. Its tangent space is

$$T_{x_{\text{cyc}}} \mathcal{X}_{GL_2}^{\text{EG, wild}} \cong H_{\text{loc}}^1(G_{\mathbb{Q}}, \text{ad}^{\circ} \bar{\rho}_{\text{cyc}})(\chi_{\text{cyc}}),$$

where H_{loc}^1 denotes the local deformation cohomology with the determinant and ramification conditions of the Emerton–Gee component. The exceptional Arthur line inside this tangent space is one-dimensional, spanned by the non-split class

$$e_{\text{cyc}} \in H_{\text{exc}}^1(G_{\mathbb{Q}}, \omega) \subset H_{\text{loc}}^1(G_{\mathbb{Q}}, \text{ad}^{\circ} \bar{\rho}_{\text{cyc}})(\chi_{\text{cyc}}).$$

Source. Emerton–Gee arXiv:1908.07185 Proposition 6.1.4 gives the formal completion identification with the universal deformation ring; §6.3 analyses the cyclotomic-residual closed point. The tangent space formula is Schlessinger 1968 + Mazur 1989 in the form used by Bellaïche–Chenevier 2009 Ch. 2 (residual gerbe at a reducible closed point). The line $H_{\text{exc}}^1(G_{\mathbb{Q}}, \omega)$ is the line cut out by the first-order Langlands–Shahidi residue. No assertion is made about the dimension of the full group $H^1(G_{\mathbb{Q}}, \omega)$. $\square$

PROPOSITION 96.4.5 (spectral Ext^1 -class at the Arthur point). The degree-1 tangent fibre of the Emerton–Gee stack at x_{cyc} is canonically isomorphic to Ext^1 in the spectral category:

$$\mathbb{T}_{\mathcal{X}_{GL_2}^{\text{EG, wild}}|_{x_{\text{cyc}}}}^1 \cong \text{Ext}_{\text{IndCohNilp}(\mathcal{X}_{GL_2}^{\text{EG, wild}})}^1(\mathcal{O}_{\text{St}}, \mathcal{O}_{\mathbf{1}}).$$

Here $\text{IndCohNilp}(\mathcal{X}_{GL_2}^{\text{EG, wild}})$ denotes the ∞ -category of ind-coherent sheaves on the Emerton–Gee stack with nilpotent singular support (Gaitsgory–Rozenblyum 2017, Definition I.3.3). The right side is one-dimensional, with canonical generator $e_{\text{cyc}} \in H^1(G_{\mathbb{Q}}, \omega)$ identified with the Arthur SL_2 -factor of the parameter ψ_1 .

Proof. By the residual-gerbe identification of Chapter 65.2.7 generalised from $p \in \{2, 3\}$ to the closed point at the cyclotomic residual: the degree-1 tangent fibre at a reducible closed point equals H^1 with ad^0 -coefficients twisted by the fixed determinant character, which is χ_{cyc} here. The Ext^1 in the coherent category $\text{IndCoh}_{\text{Nilp}}$ is computed via the Koszul resolution of the closed point and dualises to the tangent fibre (Lurie SAG §17.0.1; Gaitsgory–Rozenblyum 2017 Ch. IV.5). The identification with the Arthur SL_2 -factor is Arthur 2013 §1.5: the SL_2 -factor of ψ_1 determines the direction of the Taylor expansion of the normalised intertwining kernel at the pole $s = 1$; in the deformation picture this is the Kodaira–Spencer map from $H^1_{\text{exc}}(G_{\mathbb{Q}}, \omega)$ to the tangent space of $\mathcal{X}^{\text{EG, wild}}_{GL_2}$ at x_{cyc} . Both sides are one-dimensional; the Kodaira–Spencer map is an isomorphism by the definition of the exceptional Arthur tangent line and the one-dimensionality of the residual representation in Mœglin–Waldspurger II.2.4. $\square$

COROLLARY 96.4.6 (*mirror at $s = 0$*). . At $s = 0$, the corresponding closed point $x_{\text{triv}} \in \mathcal{X}^{\text{EG, wild}}_{GL_2}$ at the residual $\bar{\rho}_{\text{triv}} = \mathbf{1} \oplus \mathbf{1}$ has tangent fibre isomorphic to $\text{Ext}^1(\mathcal{O}_t, \mathcal{O}_{\text{St}})$, one-dimensional with canonical generator $e_{\text{triv}} \in H^1(G_{\mathbb{Q}}, \mathbf{1})$ identified with the Arthur SL_2 -factor of the dual parameter ψ_0 .

Proof. Functional-equation dual of Chapter 96.4.5: the automorphism $\bar{\rho} \mapsto \bar{\rho} \otimes \omega^{-1}$ of $\mathcal{X}^{\text{EG, wild}}$ exchanges x_{cyc} and x_{triv} and carries e_{cyc} to e_{triv} . The dimensionality follows from the same calculation after restriction to the exceptional Arthur tangent line. No assertion is made about the full group $H^1(G_{\mathbb{Q}}, \mathbf{1})$. $\square$

96.5 MODULE (X₃): CLASS-LEVEL MATCH VIA LS-CORRECTED KOSZUL PAIRING

The Match Target

Definition 96.5.1 (*exceptional-crossing match target at $s = 1$*). . The *class-match target at $s = 1$* is a $\mathbb{C}$ -linear isomorphism

$$c_1^{\text{exc}} : \text{Ext}^1_{[GL_2]}(\text{St}^{\text{ad}}, \mathbf{1}_{[GL_2]}) \xrightarrow{\sim} \mathbb{T}^1_{\mathcal{X}^{\text{EG, wild}}_{GL_2}|_{x_{\text{cyc}}}},$$

sending the automorphic class $\alpha^{s=1}$ (Chapter 96.3.1) to the spectral class e_{cyc} (Chapter 96.4.5).

Definition 96.5.2 (*exceptional-crossing match target at $s = 0$*). . The *class-match target at $s = 0$* is the $\mathbb{C}$ -linear isomorphism

$$c_0^{\text{exc}} : \text{Ext}^1_{[GL_2]}(\mathbf{1}_{[GL_2]}, \text{St}^{\text{ad}}) \xrightarrow{\sim} \mathbb{T}^1_{\mathcal{X}^{\text{EG, wild}}_{GL_2}|_{x_{\text{triv}}}},$$

sending $\alpha^{s=0}$ to e_{triv} .

Koszul Residue and the Class-Level Match

PROPOSITION 96.5.3 (*Koszul residue at the exceptional crossing*). . The LS-corrected Koszul pairing $\langle \cdot, \cdot \rangle^{\text{Koszul, LS}}$ of Chapter 62.2.1 extends to a pairing of Ext^1 -classes:

$$\langle \alpha^{s=s_0}, \sigma^{\text{BC}}(e_{\text{cyc}^{s_0}}) \rangle_{\text{Ext}}^{\text{Koszul, LS}} \sim c_{s_0}^{\text{LS}} \cdot \langle -, - \rangle_{\text{Ext}}, \quad s_0 \in \{0, 1\},$$

where

$$c_1^{\text{LS}} := \text{Res}_{s=1} N^{\text{LS}}(s), \quad c_0^{\text{LS}} := \text{Res}_{s=0} N^{\text{LS}}(s)^{-1}.$$

Here σ^{BC} is the Bost–Connes involution (the canonical $\mathbb{Z}/2\mathbb{Z}$ -action on $H^1(G_{\mathbb{Q}}, \omega)$ induced by complex conjugation on the cyclotomic character, cf. Chapter 62.2.1) and $\text{cyc}^{s_0} \in \{\text{triv}, \text{cyc}\}$ is the character corresponding to s_0 ($\text{cyc}^1 = \text{cyc}$, $\text{cyc}^0 = \text{triv}$). The constants c_1^{LS} and c_0^{LS} are non-zero and send the non-zero class $\alpha^{s=s_0}$ to the non-zero class $e_{\text{cyc}^{s_0}}$.

Proof. At $s_0 = 1$, $\Lambda_\zeta(2s - 1)$ has a simple pole (from $\zeta(2s - 1)$ at $2s - 1 = 1$) and $\Lambda_\zeta(2s)$ is holomorphic and non-vanishing; hence $N^{\text{LS}}(s) = \Lambda_\zeta(2s - 1)/\Lambda_\zeta(2s)$ has a simple pole and

$$\text{Res}_{s=1} N^{\text{LS}}(s) = \frac{1}{2\Lambda_\zeta(2)},$$

which is non-zero (the computation uses $\Lambda_\zeta(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ and $\text{Res}_{s=1} \zeta(s) = 1$). The Koszul pairing of Chapter 62.2.1 extends from pairs of principal-series vectors to pairs of short exact sequences by the standard Ext^1 -pairing construction: a short exact sequence represents a Yoneda class, and the Koszul pairing at the level of module categories pairs the connecting maps of the two sequences (Positselski 2009 §6.3 gives the extension of a bar-cobar pairing to Ext^1 -classes). The residue of the pairing at $s = 1$ sends the non-zero automorphic class $\alpha^{s=1}$ (non-vanishing residue of N) to the non-zero spectral class e_{cyc} (non-vanishing tangent direction at x_{cyc}) because both sides are one-dimensional (Chapter 96.3.4, Chapter 96.4.5) and the residue map is non-zero. At $s_0 = 0$, $N^{\text{LS}}(s)$ has a simple zero and $N^{\text{LS}}(s)^{-1}$ has a simple pole; the non-zero constant is c_0^{LS} . The statement follows by Chapter 96.2.3. $\square$

THEOREM 96.5.4 (*exceptional-crossing class-match at $\Sigma^{\text{exc}} = \{0, 1\}$*). . Under Chapter 39.5.7 (the wild-EG enhancement conjecture; the same hypothesis sustaining the F1.a.2 class-match of Chapter 65.2.9) and the restricted-product Paley–Wiener Ext comparison of Chapter 96.3.4, the class-match isomorphisms c_1^{exc} of Chapter 96.5.1 and c_0^{exc} of Chapter 96.5.2 exist and are natural in the Hecke character. The automorphic exceptional classes $\alpha^{s=1}$ and $\alpha^{s=0}$ are sent respectively to the tangent directions e_{cyc} and e_{triv} at the corresponding Arthur closed points.

Proof. Compose three isomorphisms at $s = 1$ (the $s = 0$ case being its functional-equation dual via Chapter 96.2.3):

$$\text{Ext}_{[GL_2]}^1(\text{St}^{\text{ad}}, \mathbf{1}) \xrightarrow[\text{Koszul residue}]{\sim} H_{\text{exc}}^1(G_{\mathbb{Q}}, \omega) \xrightarrow[\text{gerbe tangent}]{\sim} \mathbb{T}_{\chi^{\text{EG, wild}}|_{x_{\text{cyc}}}}^1.$$

Chapter 96.5.3 Chapter 96.4.5

The first isomorphism is Chapter 96.5.3: the Koszul residue at $s = 1$ of the LS-corrected pairing transports the automorphic Ext^1 to the exceptional Galois H^1 line with ω -coefficients. The second isomorphism is Chapter 96.4.5: the residual gerbe tangent at the cyclotomic closed point equals the exceptional Galois H^1 line. Composition yields c_1^{exc} . Naturality in the Hecke character χ follows from naturality of both component isomorphisms: the Koszul residue is natural in the unitary-axis parameter (Shahidi 2010 §3.5 covariant in χ), and the gerbe tangent is natural in the residual Galois representation (Bellaïche–Chenevier 2009 §2).

The only non-trivial input is the wild-EG enhancement of Chapter 39.5.7: it is needed to supply the degree-1 derived structure on the spectral side at the closed points x_{cyc} and x_{triv} . Once this is granted, the class-match is the composition of two natural isomorphisms of one-dimensional vector spaces, determined by the non-vanishing residues of $N^{\text{LS}}(s)$ at $s = 0$ and $s = 1$. $\square$

Exceptional Eisenstein Reduction for L_{ACD}

COROLLARY 96.5.5 (L_{ACD} at Σ^{exc}). . Under Chapter 39.5.7, the restricted-product Paley–Wiener Ext^1 comparison, and the standing local-global comparison hypotheses of Chapter 63.5.1, the two exceptional points $s = 0, 1$ add no further class obstruction: on the Eisenstein sph-finite locus including Σ^{exc} , the expected equivalence of stable presentable ∞ -categories

$$\bigotimes_v' C_v \xrightarrow{\sim} D^{\text{sph-fin}}([GL_2]),$$

is reduced to the non-exceptional L_{ACD} inputs. Here C_v are the local spherically finite categories of Chapter 63.5.1.

Proof. Essential surjectivity extends from the non-exceptional locus by Chapter 96.5.4: at each of the two crossings $s = 0, 1$, the automorphic Ext^1 -class is sent to the spectral Ext^1 -class, and the class-match upgrades the Paley–Wiener equivalence of Chapter 63.3.6 to cover the closed strata. Full faithfulness at the crossing is the one inherited from the non-exceptional local-global comparison; continuity of restricted tensor products carries it across the removable residue. $\square$

THEOREM 96.5.6 (*composite L_{ACD} reduction after W_{10-D}*). . Combining the cuspidal comparison (Chapter 48.4.4), the residual and non-exceptional continuous reduction (Chapter 63.5.1), and the exceptional-crossing class-match of this chapter (Chapter 96.5.4), L_{ACD} on the full sph-finite automorphic category is reduced to the standing local categorical comparison, the global restricted-tensor identification, Bernstein-centre gluing, and Conjecture 39.5.7.

Proof. Orthogonal direct-sum assembly through Chapter 63.4.1: cuspidal (as stated), residual (as stated), continuous non-exceptional (under its comparison inputs), continuous exceptional at $\{s = 0, 1\}$ (conditional on wild-EG via Chapter 96.5.4). The restricted-tensor orthogonality of Lurie HA 4.8 combines the pieces only after the local-global and Bernstein-centre gluing inputs are supplied. $\square$

96.6 COMPATIBILITIES FOR THEOREM 96.5.4

Seven Compatibilities

Sources. The Eisenstein construction uses Langlands 1976, Shahidi 1990/2010, Arthur 2013 §3.5, Harish-Chandra 1976, Wallach 1992, and Mœglin–Waldspurger IV. The exceptional class comparison uses Mœglin–Waldspurger II.2, Shahidi 2010 Thm 3.4.1, Arthur 2013 §1.4–1.5, Emerton–Gee §6.2–6.3, Bellaïche–Chenevier, Schlessinger, and Mazur. The common authors occur in different theorems: residual spectrum, first residue, and endoscopic parameter.

Sign. The residue at a simple pole is fixed by the complex orientation. The functional-equation identity $\text{Res}_{s=0} = -w_0^{-1} \text{Res}_{s=1}$ is Chapter 96.2.3. Multiplying a generator of a one-dimensional Ext^1 -line by a non-zero scalar does not change the line.

Categories. The automorphic class lies in the admissible adelic category over $\mathbb{C}$. The spectral class lies in $\text{IndCohNilp}(\mathcal{X}_{GL_2}^{\text{EG,wild}})$. The comparison is made on the common exceptional Galois line $H_{\text{exc}}^1(G_{\mathbb{Q}}, \omega)$, after the chosen scalar extension.

Hypotheses. The wild-EG enhancement and the restricted-product Paley–Wiener Ext^1 comparison are explicit hypotheses. The trivial Hecke character is the only character for which the LS normalising factor has the exceptional simple pole/zero pair at $s = 1, 0$.

Functoriality. The Koszul residue is natural in χ . The gerbe tangent line is natural in the residual Galois representation. The composition preserves the one-dimensional exceptional line and asserts no higher functoriality.

Theorems Used. Mœglin–Waldspurger gives the residual representation, Shahidi gives the residue, Arthur gives the packet, and Emerton–Gee–Schlessinger–Mazur give the deformation tangent. The derived spectral lift is exactly the wild-EG hypothesis.

No Arithmetic Computation. The statement is a comparison of residue, deformation, and gerbe tangent lines.

Thus the exceptional crossing is a conditional class match under the wild-EG enhancement, parallel to the Fl.a.2 class match of Chapter 65.

Interaction with Fl.a.2 Class-Match

PROPOSITION 96.6.1 (*class-match compatibility across Fl.a.2 and W10-D*). . The class-match isomorphism c_p^{class} of Chapter 65.2.1–65.2.2 at wild primes $p \in \{2, 3\}$ and the class-match isomorphism $c_{s_0}^{\text{exc}}$ of Chapter 96.5.1–96.5.2 at unitary crossings $s_0 \in \{0, 1\}$ commute in the sense that both agree on the shared input $\bar{\rho}_{\text{cyc}}$ restricted to $G_{\mathbb{Q}_p}$: the local component $c_1^{\text{exc}}|_{v=p}$ equals c_p^{class} for the cyclotomic block at $p \in \{2, 3\}$.

Proof. Both isomorphisms are constructed as connecting-map- $\circ$ -residual-gerbe tangent, with the same residual Galois representation $\bar{\rho}_{\text{cyc}}|_{G_{\mathbb{Q}_p}}$ at the wild prime. The connecting map in Fl.a.2 is the Paškūnas block Ext^1 to Galois H^1 identification (Paškūnas 2013 §4.11); the connecting map in W10-D is the Koszul residue to Galois H^1 identification (Chapter 96.5.3). Both connecting maps target the same Galois H^1 group $H^1(G_{\mathbb{Q}_p}, \text{ad}^0 \bar{\rho}_{\text{cyc}}|_{G_{\mathbb{Q}_p}})$; they agree by naturality of the Colmez Montreal functor (Paškūnas 2013 §4) applied to the residual sequence, which is the p -adic avatar of the global residual sequence (96.2.2) at the pole $s = 1$. The residual gerbe tangent line is the same object on both sides. Hence $c_1^{\text{exc}}|_{v=p} = c_p^{\text{class}}$ on the cyclotomic block. $\square$

Conditional Inputs

The automorphic Ext^1 -class side (Chapter 96.3.2, Chapter 96.3.4) is unconditional: Shahidi 2010 Thm 3.4.1, Mœglin–Waldspurger II.2.4, Bernstein–Zelevinsky 1977, and Wallach 1988 suffice.

The spectral Ext^1 -class side (Chapter 96.4.5, Chapter 96.4.6) is conditional on the wild-EG enhancement Chapter 39.5.7: the derived degree-1 structure on $\mathcal{X}_{GL_2}^{\text{EG, wild}}$ at the cyclotomic closed point is a hypothesis. The tangent-space identification is the Emerton–Gee–Schlessinger–Mazur deformation calculation; the lift of the exceptional line to the derived spectral category is the conditional input.

The class-match isomorphism Chapter 96.5.4 therefore has the same wild-EG hypothesis as the spectral side. The residue formula (Chapter 96.5.3) is unconditional (Shahidi 2010 Thm 3.4.1, Tate 1950 adelic factorisation).

96.7 FI CONDITIONS AFTER THIS CHAPTER

PROPOSITION 96.7.1 (*Fi conditions after the exceptional-crossing class-match*). . Combining this chapter with Chapter 63.7.1 and Chapter 65.5.1, the Fi conditions are:

- (i) **Fl.a.1.non-ord**: essential surjectivity of DEG at $p \geq 5$ on the non-ordinary locus.
- (ii) **Fl.a.1.TW-removal**: extension beyond Taylor–Wiles hypotheses.
- (iii) **Fl.a.2.class-match** at $p \in \{2, 3\}$: follows from Chapter 65.2.9 under wild-EG enhancement.

- (iv) **L_{ACD} -exceptional-crossing** at $s \in \{0, 1\}$: follows from Chapter 96.5.4 under wild-EG enhancement. This chapter establishes the Ext^1 -class match at both crossings, removing the exceptional class entry of Chapter 63.5.3.
- (v) **BC-diamond-glue**: Bernstein-centre diamond-gluing.

The exceptional crossing no longer contributes an independent Ext^1 -class obstruction once wild-EG is assumed. The global L_{ACD} comparison remains conditional on the local-global comparison and Bernstein-centre gluing inputs recorded in Chapter 96.5.6.

Proof. Tabulation. Chapter 96.5.4 identifies the exceptional-crossing class; the dependency on wild-EG is the same as $\text{Fl.a.2.class-match}$ (by Chapter 96.6.1). The other entries are unchanged. $\square$

96.8 CONSEQUENCES

- (i) **Two-point residue data** (Chapter 96.2.1, Chapter 96.2.2): at $s = 1$ the residual spectrum is the one-dimensional trivial representation (Mœglin–Waldspurger II.2.4); at $s = 0$ the dual spherical induction is reducible with Steinberg subquotient. Functional-equation duality (Chapter 96.2.3) exchanges the two crossings.
- (ii) **Local and adelic Ext^1 -classes from residues** (Chapter 96.3.2, Chapter 96.3.4): the residue of the Langlands–Shahidi normalised intertwining operator at each first-order pole is a non-zero intertwining operator defining a non-trivial Ext^1 -class $\alpha^{s=s_0}$; the adelic Ext^1 is one-dimensional and generated by $\alpha^{s=s_0}$ for $s_0 \in \{0, 1\}$.
- (iii) **Spectral Ext^1 -class via Arthur endoscopy** (Chapter 96.4.5, Chapter 96.4.6): the Arthur parameter $\psi_1 = [\text{triv} \boxtimes \nu_2]$ at $s = 1$ and its functional-equation dual $\psi_0 = [\text{id}_{\text{SL}_2}]$ at $s = 0$ correspond to closed points x_{cyc} and x_{triv} of the Emerton–Gee stack; the degree-1 tangent fibres at these points are one-dimensional with canonical generators e_{cyc} and e_{triv} .
- (iv) **Class-level match at Σ^{exc}** (Chapter 96.5.4): under the wild-EG enhancement conjecture, the automorphic class $\alpha^{s=s_0}$ is sent to the spectral class $e_{\text{cyc}^{s_0}}$ by the composition Koszul-residue $\circ$ residual-gerbe tangent. The match is natural in χ and compatible with $\text{Fl.a.2.class-match}$ at the shared wild primes.
- (v) **L_{ACD} exceptional Eisenstein reduction** (Chapter 96.5.5): the restricted tensor product equivalence extends from the non-exceptional locus Σ^{nonexc} to the exceptional crossings only under the same local-global comparison inputs and the restricted-product Paley–Wiener Ext^1 comparison. The composite L_{ACD} statement is a reduction, not an L_{ACD} theorem.
- (vi) **Fi conditions** (Chapter 96.7.1): the L_{ACD} exceptional Ext^1 -class is matched under wild-EG. The global descent problem still contains the standing local-global comparison and Bernstein-centre gluing inputs.
- (vii) **Compatibility of sources** (Section 96.6): the residue, Arthur-packet, and gerbe-tangent maps have compatible signs, coefficient fields, and one-dimensional exceptional lines.

Conjecture 63.5.3 is reduced to the wild-EG class match. The Bost–Connes Bernstein-centre diamond-gluing input remains, and the L_{ACD} local-global comparison inputs remain part of the global descent problem.

96.9 SOURCES

Primary sources used in this chapter:

Moëglin, C. and Waldspurger, J.-L. *Spectral Decomposition and Eisenstein Series: A Paraphrase of the Scriptures*, Cambridge Tracts 113, 1995. Chapter II.2.4 for the residual spectrum of $GL_2/\mathbb{Q}$; Chapter IV.1.8 for the residue of the normalised intertwining operator.

Shahidi, F. *Eisenstein Series and Automorphic L -Functions*, AMS Colloquium 58, 2010. Theorem 3.4.1 for the residue of $N(s, \chi)$ at a first-order pole; §3.5 for the polar divisor analysis.

Arthur, J. *The Endoscopic Classification of Representations*, AMS Colloquium 61, 2013. Chapter 1 §1.4–1.5 for the Arthur parameter of a residual representation and the SL_2 -factor specification.

Moëglin, C. and Waldspurger, J.-L. “Le spectre résiduel de GL_n ”, Ann. Sci. ENS 22 (1989), 605–674. Arthur classification of residual representations for GL_n : pre-Arthur-2013 theorem restricted to GL_n .

Emerton, M. and Gee, T. “A geometric perspective on the Breuil–Mézard conjecture”, arXiv:1908.07185 (2019). §6.2 unramified-locus smoothness; §6.3 closed point at the cyclotomic character; the formal completion and tangent-space identification.

Breuil, C. and Mézard, A. “Multiplicités modulaires et représentations de $GL_2(\mathbb{Z}_p)$ et de $\text{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p)$ en $\ell = p$ ”, Duke Math. J. 115 (2002). Local deformation multiplicities for two-dimensional residual Galois representations.

Bellaïche, J. and Chenevier, G. *Families of Galois representations and Selmer groups*, Astérisque 324, SMF 2009. Chapter 2 for the residual gerbe at a reducible closed point.

Schlessinger, M. “Functors of Artin rings”, Trans. AMS 130 (1968). Pro-representability of the deformation functor.

Mazur, B. “Deforming Galois representations”, in *Galois Groups over $\mathbb{Q}$* , MSRI Publ. 16 (1989). Tangent space of the universal deformation ring.

Bernstein, J. and Zelevinsky, A. “Induced representations of reductive p -adic groups I”, Ann. Sci. ENS 10 (1977). The composition factors of the principal series at $s = 0, 1$ and Ext^1 -groups between them at a non-archimedean place.

Wallach, N. *Real Reductive Groups I*, Academic Press 1988. Chapter 5.4 for archimedean $\text{Ext}^1(\mathbf{1}, \text{St})$ at $GL_2(\mathbb{R})$.

Jacquet, H. and Langlands, R. *Automorphic forms on GL_2* , Lecture Notes in Math. 114, Springer 1970. §5 for the non-splitting of the $s = 0, 1$ induction at archimedean places.

Harish-Chandra “Harmonic analysis on real reductive groups II”, Invent. Math. 36 (1976). Theorem 1 for the archimedean Plancherel density and residue.

Kisin, M. “Potentially semi-stable deformation rings”, J. AMS 21 (2008). Cotangent complex of local Galois deformation rings.

Tate, J. “Fourier analysis in number fields and Hecke’s zeta functions”, Ph.D. thesis, Princeton 1950; reprinted Cassels–Fröhlich 1967 Ch. XV. Adelic restricted tensor product; local-global factorisation.

Positselski, L. “Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence”, arXiv:0905.2621 (2009). §6.3 extension of the bar–cobar pairing to Ext^1 -classes.

Disjoint-source discipline. Chapter 63 derivation uses Moëglin–Waldspurger IV, Shahidi 2010 §3.5, Arthur 2013 §3.5; this chapter’s verification uses Moëglin–Waldspurger II.2, Shahidi 2010 Thm 3.4.1, Arthur 2013 §1.4–1.5, plus Emerton–Gee, Breuil–Mézard, Bellaïche–Chenevier, Schlessinger, Mazur, Kisin. The three shared authors (Moëglin–Waldspurger, Shahidi, Arthur) contribute through disjoint

subchapters on the two sides; all other sources are unique to one side. primary-source independence holds.

RESIDUAL BOUNDARY OF THE ARITHMETIC BRANCH

After the Koszul, chiral-homology, and arithmetic Chern–Simons constructions have reached their natural boundary, the arithmetic branch has named obstructions. One of them is equivalent to the Riemann Hypothesis. The others are comparison, descent, regularity, and small-prime hypotheses. The realization formalism of Chapter 1 is orthogonal to these arithmetic conditions.

97.1 CONDENSED-HILBERT DESCENT AND ARCHIMEDEAN TATE POSITIVITY

The quantum arithmetic Chern–Simons bulk construction (Chapter 93.8.3) and the archimedean Tate positivity statement (Chapter 56.7.1) are the two ingredients whose conjunction implies the Riemann Hypothesis. The theorem below separates the two inputs: the first is a construction statement; the second is a positivity statement, and only the second carries RH-equivalent content.

THEOREM 97.1.1 (*condensed-Hilbert descent plus archimedean Tate positivity implies RH*). (*on the Clausen–Scholze condensed-Hilbert descent Chapter 93.8.3 and the archimedean Tate positivity statement Chapter 56.7.1*). The conjunction of Clausen–Scholze condensed-Hilbert descent and archimedean Tate positivity (A-TP) implies the Riemann Hypothesis. The descent alone does not decide RH: after the descent is assumed, A-TP is the remaining RH-equivalent positivity obstruction in the quantum ACS construction; the comparison and regularity hypotheses are separate conditions.

Proof. The implication is Chapter 93.7.5. The descent constructs the condensed Hilbert space and identifies the Wilson-surface action with the Arakelov scaling operator; it does not assert positivity of the scaling quadratic form. That positivity is A-TP, equivalent to RH by Chapter 89.6.3. Hence the conjunction implies RH, while the descent alone supplies the construction and not the zero-line assertion. $\square$

Remark 97.1.2 (construction versus positivity). The condensed-Hilbert descent is the construction half of the quantum ACS datum; A-TP is the positivity half. RH is equivalent to A-TP alone, given the descent.

97.2 THE RH-EQUIVALENT OBSTRUCTION

The archimedean Tate positivity statement (Chapter 56.7.1) asserts:

$$W(f * \tilde{f}) \geq 0 \quad \text{for all } f \in \text{PW}(\mathbb{R}), \quad (97.1)$$

where $\text{PW}(\mathbb{R})$ denotes the Paley–Wiener space of entire functions of exponential type and W is the Weil explicit functional on the completed zeta function. This statement is equivalent to RH by Chapter 89.6.3. It is denoted A-TP throughout.

PROPOSITION 97.2.1 (*A-TP is an RH-equivalent obstruction*). A-TP (97.1) is an RH-equivalent obstruction in the arithmetic branch. No other residual in this chapter is used as an RH-equivalent statement.

Proof. Chapters 60–79 and Chapter 90 identify the same residual through the direct, Nyman–Beurling, de Branges, Gaussian, Mellin–Barnes, Hadamard, Gram, Guinand–Weil, adelic-scattering, Arthur, geometric-Langlands, Iwasawa, quantum-chaos, and Arakelov approaches. Chapter 89 identifies this residual with RH. The other inputs are comparison or construction hypotheses, not zero-placement statements. $\square$

97.3 COMPARISON AND DESCENT CONJECTURES

Definition 97.3.1 (comparison and descent inputs). . The comparison and descent boundary contains the following named inputs.

Input	Carries	Form
Wild-EG enhancement	Fl.a.2.class-match; L_{ACD} Eisenstein Ext ^l at $\{0, 1\}$; BC-glue wild inertia	open
Π -flat pro-étale flatness	BC-diamond-glue cross-prime prismatic glue	open
Arithmetic AG singular support	global arithmetic Arinkin–Gaitsgory singular-support condition and the centre of BC-diamond-glue	open
Clausen–Scholze condensed-Hilbert descent	Quantum arithmetic Chern–Simons bulk extension	open
Consani–Marcolli adèle-class duality	Positselski G -row duality and the abelian Heisenberg base-change line	open
Chenevier pseudocharacter trace comparison	Pairwise sheaf residual in the Ch closure	open
BCglue local comparison datum	K_5 and the Jacquet–Shalika trace comparison in BC-diamond-glue	open
Ben-Zvi lift at infinity	L_{ACD} archimedean compatibility and finite uniformity	open
BZN wild trace data	Large-prime semisimplicity and the wild Ext-trace pairing	open
Symmetric-square weight limit	Weil–Petersson weight-aspect positivity for holomorphic cusp forms	open
Arithmetic bar nilpotency	The square-zero property of the arithmetic pre-differential d^{Ar}	open

THEOREM 97.3.2 (visible non-RH residuals). . Each input in Chapter 97.3.1 is a non-RH residual named elsewhere in the arithmetic branch. The statement does not assert exhaustion of all possible residuals.

Proof. Wild-EG (the Emerton–Gee wild-inertia enhancement) carries Fl.a.2 by Chapter 91.3.1, the L_{ACD} exceptional Eisenstein crossing by Chapter 96.5.4, and the wild-inertia component of BC-diamond-glue by Chapter 94.6.1. Π -flat (pro-étale Π -flatness for the prismatic site) is the cross-prime prismatic flatness of Chapter 34.5.2, unchanged by Chapter 94. Arithmetic AG (arithmetic Arinkin–Gaitsgory singular support) is the global singular-support conjecture Chapter 20.1.1, required at the centre of BC-diamond-glue by Chapter 94.6.1. Clausen–Scholze condensed-Hilbert descent is equivalent to the quantum ACS bulk conjecture by Chapter 93.8.4. The other rows are the named conditional inputs constructed in Chapters 53, 51, 34, 30, 32, 61, and 12. Each is non-RH in form and is not implied by A-TP. $\square$

Remark 97.3.3 (Wild-EG concentration). Wild-EG is the only source conjecture carrying three distinct class-match conditions. Its proof would simultaneously prove Fl.a.2, the exceptional L_{ACD} Eisenstein Ext^l crossing at $\{0, 1\}$, and the wild-inertia component of BC-diamond-glue.

97.4 THE GENERALISED RAMANUJAN CONJECTURE FOR GL_4

The Rankin–Selberg square of a Maass form on $GL_2/\mathbb{Q}$ comes from an isobaric automorphic object on $GL_4/\mathbb{Q}$. Full temperedness of that GL_4 object at every finite place, equivalently GRC for its cuspidal factors, closes the Maass sector of the $\mathcal{V}^{\text{prim}}$ Hochschild-trace identity. This conjecture is separated from A-TP in the proof: it concerns Satake eigenvalues, not zero placement.

Hypothesis 97.4.1 (GRC for $GL_4/\mathbb{Q}$ in the Selberg class). . The isobaric automorphic representations of $GL_4/\mathbb{Q}$ arising from the Rankin–Selberg square of a Maass form have tempered cuspidal factors at every finite place.

PROPOSITION 97.4.2 (GRC- GL_4 closes the Maass sector). . Conditional on Chapter 97.4.1, full Ramanujan holds for the Maass forms entering the $\mathcal{V}^{\text{prim}}$ Hochschild-trace P_3 identity of Chapter 92.6.1. This closes the Maass sector. The hypothesis is not equivalent to RH; no implication between this hypothesis and A-TP is used.

Proof. Chapter 92 proves that the Arthur–Ramakrishnan–GRC hypothesis implies full Ramanujan for Maass forms (Chapter 92.5.5) and reduces that hypothesis to Chapter 97.4.1 (Chapter 92.5.6). The P_3 identity is the exact conditional statement of Chapter 92.6.1. A-TP concerns the zeros of ζ through Weil positivity; GRC- GL_4 concerns Satake parameters. The argument uses the two residuals as separate inputs. $\square$

97.5 $F_{1.A.1}$ SMALL RESIDUALS

Two small residuals occur in the $F_{1.A.1}$ table of Chapter 95. They are not a count of all small residuals in the arithmetic branch.

PROPOSITION 97.5.1 (small residuals). . Besides the generator-comparison inputs and the automorphy/patching hypotheses isolated in Chapter 95.3.4 and Chapter 95.3.5, the $F_{1.A.1}$ table has the following two small residual tails:

- (i) **Supersingular small-weight tail** (Chapter 95.2.8): Fontaine–Laffaille representations with Hodge–Tate weights $k \in \{p-1, p\}$, requires a non-formal theorem.
- (ii) **A_5 non-solvable descent defect** (Chapter 95.3.8): local-global descent defect in the icosahedral sector, requires a non-formal theorem.

Proof. The complete list of $F_{1.A.1}$ residuals is Chapter 95.4.2. The Fontaine–Laffaille supersingular core is reduced to the generator comparison of Chapter 95.2.6. The odd-determinant icosahedral core is reduced to the conditional theorem of Chapter 95.3.6. The two tails listed above are additional obstructions in that decomposition. $\square$

97.6 SEPARATION FROM THE REALIZATION FORMALISM

The obstructions above are internal to the arithmetic branch. The realization formalism of Chapter 1 concerns independent verification data for Vols I–III; the arithmetic branch does not supply those data.

PROPOSITION 97.6.1 (realization formalism is orthogonal to the arithmetic branch). . The arithmetic branch does not construct independent comparison data for the source Vols I–III Proved-Here claims. Its obstructions concern arithmetic comparison, descent, temperedness, and small-prime regularity.

Proof. The preceding sections use only the arithmetic conditions A-TP, comparison and descent, GRC for GL_4 , and the two $F_{1,a,1}$ residuals. None of these data is a independent comparison pair for a source theorem of Vols I–III. $\square$

97.7 RESIDUAL BOUNDARY

THEOREM 97.7.1 (*arithmetic branch obstruction decomposition*). . The arithmetic branch has, at least, the following obstructions:

- (i) one statement equivalent to RH: archimedean Tate positivity;
- (ii) the comparison and descent inputs of Chapter 97.3.1;
- (iii) one Ramanujan-type hypothesis separated from A-TP: GRC for $GL_4/\mathbb{Q}$;
- (iv) the two $F_{1,a,1}$ small residuals of Chapter 97.5.1.

The Vol IV realization formalism for Vols I–III is orthogonal to the arithmetic branch.

Proof. Item (1) is Chapter 97.2.1. Item (2) is Chapter 97.3.2. Item (3) is Chapter 97.4.2. Item (4) is Chapter 97.5.1. The separation from the realization formalism is Chapter 97.6.1. No implication from the table to RH is asserted except through A-TP. $\square$

Remark 97.7.2 (*domain of the boundary*). The theorem isolates the displayed RH-equivalent content into A-TP and places the visible non-RH obstructions in named inputs. It does not prove RH and it does not claim that the arithmetic comparison is complete.

APPENDIX A

PRIMARY SOURCES

This appendix records the primary literature anchoring the arithmetic branch of Volume IV. Each entry states author(s), title, year, arXiv identifier where applicable, and the role in the branch: which chapter invokes the result and for what purpose.

The entries are organized by mathematical topic: factorization and homotopy-theoretic sources (§§A.5–A.7); geometric Langlands, Hecke categories, and local Langlands (§§A.8–A.9); arithmetic Chern–Simons and arithmetic topology (§A.10); arithmetic D -modules and F -isocrystals (§A.6); operator theory and moment problems (§A.24); and background sources in automorphic forms, adelic analysis, Yangians, Arakelov geometry, Macdonald theory, and primon-gas statistical mechanics (remaining sections).

A.1 KAPRANOV AND KAPRANOV–SMIRNOV

Kapranov–Smirnov 1995, *Cohomology determinants and reciprocity laws: number field case*.

Unpublished preprint, ~15 pp., ca. 1995. PDF mirrored at <http://cage.ugent.be/~kthas/Fun/library/KapranovSmirnov.pdf>. Frames $\text{Spec } \mathbb{Z}$ as an “arithmetic curve” via the analogy between number fields and function fields over finite fields; motivates the absolute point / $\mathbb{F}_1$ programme and prime-labelled cohomology determinants feeding reciprocity laws. Used throughout this branch as the foundational informal reference (Chapters 4, 12).

Kapranov ca. 1995–96, *Analogies between number fields and 3-manifolds*. Unpublished preprint. Foundational for arithmetic topology; the MKR (Mazur–Kapranov–Reznikov) dictionary matches closed oriented 3-manifolds to number fields and knots to primes. Extended in Reznikov arXiv:math/0107210 (2001) and Morishita’s book *Knots and Primes* (2012). Used in Chapter 15 for the Wilson-line = ramified Hecke character dictionary in Kim’s arithmetic Chern–Simons.

Kapranov–Vasserot 2004, *Vertex algebras and the formal loop space*, Publ. IHES 100:209–269; arXiv:math/0107143.

Introduces the ind-scheme $\mathcal{L}X$ of formal loops into a finite-type scheme X ; factorization structure on $\mathcal{L}X$ recovers vertex-algebra structure on chiral de Rham. Four-paper series; Part IV = arXiv:math/0612371. No $\text{Spec } \mathbb{Z}$ version exists; this branch is the arithmetic analogue.

Kapranov 2000, *The elliptic curve in the S-duality theory and Eisenstein series for Kac–Moody groups*, arXiv:math/0006026. S-duality on an elliptic curve; not used here directly.

A.2 BORGER AND Λ -RING THEORY

Borger 2008, *The basic geometry of Witt vectors, I: The affine case*, Algebra & Number Theory 5:231–285; arXiv:0801.1066.

Concrete description of étale algebras over Witt vectors via commuting Frobenius lifts. Defines Λ -ring, Λ -module distinctions used in Chapter 6.

Borger 2010, *The basic geometry of Witt vectors, II: Spaces*, arXiv:1006.0092. Extends to algebraic spaces; generalises Buium’s arithmetic jet functor. Used in Chapter 6 §7 on Witt vectors.

Borger 2009, *Λ -rings and the field with one element*, arXiv:0906.3146. Central reference. Wilkerson criterion (Prop. 1.2): Λ -ring = commuting ψ_p with $\psi_p(x) \equiv x^p \pmod{pR}$. Slogan (§0 and §2): Λ -structure is descent from R to $R \otimes_{\mathbb{F}_1} \mathbb{Z}$. Used throughout Chapter 6.

Borger–de Smit 2008, *Galois theory and integral models of Λ -rings*, arXiv:0801.2352. Relative case: Λ -structure over a Noetherian base.

A.3 BOST–CONNES, CONNES–MARCOLLI, CONNES–CONSANI

Bost–Connes 1995, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in num*

Central reference for the arithmetic quantum statistical system. KMS partition function $\zeta(\beta)$; phase transition at $\beta = 1$; spontaneous symmetry breaking by $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$. Explicitly motivated by Julia’s primon gas.

Connes–Marcolli 2004, *From Physics to Number Theory via Noncommutative Geometry, Part I & II*, Part I arXiv:math/0408086. $HH^*(C_{\mathbb{Q}}^*)$ computed §1.8; used in Chapter 7 for the Theorem H^{arith} comparison.

Connes–Consani 2014, *The Arithmetic Site*, arXiv:1405.4527. Arithmetic site = Grothendieck topos $\widehat{\mathbb{N}^{\times}}$ with tropical semiring $\mathbb{Z}_{\max}$ structure sheaf. Used in Chapter 6 §8 for the correct $\mathbb{F}_1$ -descent framework.

Connes–Consani 2015, *Geometry of the arithmetic site*, arXiv:1502.05580. Extends (1405.4527); Frobenius correspondences as subvarieties.

Connes–Consani 2012, *Universal thickening of the field of real numbers*, arXiv:1202.4377.

Background on the real-number completion of the arithmetic site; not directly invoked in computations but provides the archimedean-fibre context for $\mathbb{F}_1$ -descent.

A.4 KONTSEVICH, SOIBELMAN, DAVISON: E_n FORMALITY AND CoHA

Kontsevich 2003, *Deformation quantization of Poisson manifolds*, Lett. Math. Phys., arXiv:q-alg/9709040.

Foundational for operadic formality. Used in Chapter 17 §1.

Kontsevich 1999, *Operads and Motives in Deformation Quantization*, arXiv:math/9904055.

E_n -formality over $\mathbb{Q}$ for all n ; cited in Chapter 17 §1 for the rational formality input.

Tamarkin 1998, *Another proof of M. Kontsevich formality theorem*, arXiv:math/9803025.

Drinfeld-associator-based proof for E_2 .

Kontsevich–Soibelman 2010, *Cohomological Hall algebra, exponential Hodge structures, and motivic DT invariants*,

CoHA framework; used for Chapter 7 §6 Soibelman-motivic reading of Kesten–McKay.

Fresse 2017, *Homotopy of Operads and Grothendieck–Teichmüller Groups*, Vol. II. Integral

E_1 -formality theorem used in Chapter 17 §1.

Kaledin 2014, *Universal Witt vectors and the “Japanese cocycle”*. Integral- E_2 obstruction.

Salvatore–Wahl 2002. Dyer–Lashof operations on $H_*(E_2; \mathbb{F}_p)$.

Davison 2016, arXiv:1601.02479. BPS Lie algebra PBW theorem; used for arithmetic CoHA conjecture in Chapter 17 §5.

A.5 BEILINSON–DRINFELD, FRANCIS–GAITSGORY, POSITSIELSKI

Beilinson–Drinfeld 2004, *Chiral Algebras*, AMS Colloq. Publ. 51. Ch. 3 (chiral bar), Ch. 4 (Hochschild).

Francis–Gaitsgory 2011, *Chiral Koszul duality*, *Selecta Math.*, arXiv:1103.5803. Koszul duality for chiral algebras and factorization algebras; used in Chapter 12 §4 for the chiral bar-cobar adjunction on the arithmetic side.

Positselski 2011, *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence*, *Mem. A*. Used in Chapter 12 §7 for arithmetic Positselski.

Parshin–Kato 1979–1980, *A generalization of local class field theory by using K -groups*. Parshin reciprocity on the Gersten complex. Used in Chapter 12 §2 for the Arnold-relation replacement in the arithmetic bar construction.

A.6 ARITHMETIC $\mathcal{D}^\dagger$ -MODULES AND F -ISOCRYSTALS

Berthelot 1996, *$\mathcal{D}$ -modules arithmétiques. I. Opérateurs différentiels de niveau fini*, *Ann. Sci. Éc. Norm. Sup.* 29:185–. Foundational construction of arithmetic differential operators of finite level. Used in Chapter 36 only as the local p -adic coefficient shadow of the finite-place Hecke/diamond category.

Berthelot 2000, *$\mathcal{D}$ -modules arithmétiques. II: descente par Frobenius*, *Mém. SMF* 81. Frobenius descent for arithmetic $\mathcal{D}$ -modules.

Caro 2009, *$\mathcal{D}$ -modules arithmétiques surholonomes*, *Ann. Sci. Éc. Norm. Sup.* 42:141–192. Overholonomic arithmetic $\mathcal{D}$ -modules and stability under the six operations; also records overconvergent unit F -isocrystals as overholonomic in the smooth case.

Kedlaya 2004, *Full faithfulness for overconvergent F -isocrystals*, in *Geometric Aspects of Dwork Theory*, 819–836. Full faithfulness for overconvergent F -isocrystals. This is a coefficient reference, not an equivalence between finite-prime Hecke categories and classical D -modules.

A.7 COSTELLO, COSTELLO–Gwilliam, COSTELLO–Gaiotto

Costello–Gwilliam 2017, 2021, *Factorization Algebras in Quantum Field Theory*, CUP Vol. I & II. Used throughout Chapter 8.

Costello 2013, *Supersymmetric Gauge Theory and the Yangian*, arXiv:1303.2632. Holomorphic-topological factorization.

Costello 2020, *Holography and Koszul duality*, lecture/preprint notes. Background on Koszul duality as a holographic mechanism; context for the arithmetic Koszul-duality formulation of Chapter 16.

Costello–Gaiotto 2019, *Vertex operator algebras and 3d $N = 4$ gauge theories*, arXiv:1804.06460. Used in Chapter 15 §3.

Costello–Dimofte–Gaiotto 2020, *Boundary chiral algebras and holomorphic twists*, arXiv:2005.00083. Boundary conditions in holomorphic-topological theories and their chiral algebra content; background for the TQFT-Chern–Simons comparison in Chapter 15.

Costello–Witten–Yamazaki 2017, *Gauge theory and integrability I*, arXiv:1709.09993. 4D Chern–Simons.

Costello–Yamazaki 2019, *Gauge theory and integrability III*, arXiv:1908.02289. Rational and trigonometric R -matrices from 4D Chern–Simons; background for the Bethe-algebra comparison in Chapter 9.

Francis 2012, *Factorization homology of topological manifolds*, arXiv:1206.5522. Used in Chapter 8 §4 for locally-constant E_n -reduction.

A.8 GEOMETRIC LANGLANDS, DERIVED SATAKE, HECKE CATEGORIES

Mirković–Vilonen 2007, *Geometric Langlands duality and representations of algebraic groups over commutative rings*, Geometric Satake.

Bezrukavnikov 2016, *On two geometric realizations of an affine Hecke algebra*, Publ. IHES, arXiv:1209.0403. Iwahori Hecke category; central for Chapter 16.

Ben-Zvi–Nadler 2009, *The character theory of a complex group*, arXiv:0904.1247. Hochschild trace = $O(\text{LocSys})$; gluing principle for the synthesis (Chapter 19).

Arinkin–Gaitsgory 2015, *Singular support of coherent sheaves and the geometric Langlands conjecture*, Selecta Math., Nilpotent singular support AG-conjecture.

Fargues–Scholze 2021, *Geometrization of the local Langlands correspondence*, arXiv:2102.13459. Local arithmetic Langlands via Fargues–Fontaine curve.

Dotto–Emerton–Gee 2026, *A categorical p -adic Langlands correspondence for $GL_2(\mathbb{Q}_p)$* , arXiv:2603.26887. Fully faithful local categorical p -adic Langlands functor for $p \geq 5$; not a local equivalence and not a global adelic theorem.

Caraiani–Scholze 2015/2017, *On the generic part of the cohomology of compact unitary Shimura varieties*, arXiv:1511.07324. Torsion-freeness of the generic part of Shimura variety cohomology; cited in Chapter 16 §4 as a constraint on the global adelic Hecke-category input.

Emerton–Gee 2020, *Moduli stacks of étale (φ, Γ) -modules and the existence of crystalline lifts*, arXiv:1908.07185. Emerton–Gee stack of Galois deformations; central for Chapter 16 §4.

Emerton–Gee 2011, *A geometric perspective on the Breuil–Mézard conjecture for GL_2* , arXiv:1109.4226. Breuil–Mézard geometry; not the Emerton–Gee stack source.

Zhu 2020, arXiv:2008.02998. Coherent Springer theory.

Zhu 2016, arXiv:1603.05593; Bhatt–Scholze 2016, arXiv:1507.06490. Witt-vector affine Grassmannian.

A.9 FLE, QUANTUM GEOMETRIC LANGLANDS, AND ROOT-OF-UNITY QUANTUM GROUPS

Gaitsgory 2008, *Twisted Whittaker model and factorizable sheaves*, Selecta Math. 13:617–659; arXiv:0705.4571. Whittaker and factorizable-sheaf source for quantum geometric Langlands.

Gaitsgory 2016, *Quantum Langlands Correspondence*, arXiv:1601.05279. Informal 2007 notes outlining local and global quantum Langlands and the FLE; cited in Chapter 41 for the quantum parameter dictionary.

Campbell–Dhillon–Raskin 2021, *Fundamental local equivalences in quantum geometric Langlands*, Compos. Math. Quantum FLE in the local affine-Grassmannian/affine-flag setting; cited in Chapter 41 §2 for the local quantum Langlands equivalence as input to the arithmetic analogue.

Gaitsgory–Lysenko 2019, *Metaplectic Whittaker category and quantum groups: the small FLE*, arXiv:1903.02279. Small FLE and small quantum group source.

Arinkin–Beraldo–Campbell–Chen–Faergeman–Gaitsgory–Lin–Raskin–Rozenblyum 2024, *Proof of the geometric Langlands conjecture at critical level*, arXiv:2403.12345. Critical level FLE and its compatibility with Kac–Moody localization in the geometric, not arithmetic, setting.

Frenkel–Gaitsgory 2006, *Local geometric Langlands correspondence and affine Kac–Moody algebras*, Progr. Math. 253. Critical local geometric Langlands and affine Kac–Moody source layer.

Raskin 2021, *Chiral principal series categories I: finite-dimensional calculations*, Adv. Math. 388, 107856.

Chiral principal-series source used for globalization analogies; not a number-field adelic theorem.

Raskin, *Chiral principal series categories II: the factorizable Whittaker category; Chiral categories; D-modules on infi*

Preprint foundations for chiral principal series and infinite-dimensional $\mathcal{D}$ -modules; cite by title and author page, not by arXiv identifiers (the arXiv numbers on these preprints do not correspond to the texts cited in Chapters 41 and 8).

Kazhdan–Lusztig 1993–1994, *Tensor structures arising from affine Lie algebras I–IV*, JAMS 6–7.

KL tensor structure and affine Lie algebra side of root-of-unity quantum groups.

Finkelberg 1996, *An equivalence of fusion categories*, GAFA 6:249–267; **Andersen–Paradowski 1995**, *Fusion categories*

Fusion-category comparison.

Lusztig 1990–1993, *Finite-dimensional Hopf algebras arising from quantized universal enveloping algebra; Quantum*

Small quantum group, divided-power form, and quantum Frobenius sources.

De Concini–Kac 1990; De Concini–Kac–Procesi 1992. De Concini–Kac form, centre, and quantum coadjoint action.

Arkhipov–Gaiitsgory 2003, *Another realization of the category of modules over the small quantum group*, Adv. Math.

Small quantum group and factorizable-sheaf bridge.

A.10 KIM'S ARITHMETIC CHERN–SIMONS

Kim 2015, *Arithmetic Chern–Simons theory I*, arXiv:1510.05818. Foundational definition of arithmetic Chern–Simons invariants for number fields via p -adic Hodge theory and Galois cohomology; arithmetic Chern–Simons source for Chapter 15.

Chung–Kim–Kim–Pappas–Park–Yoo 2017, *Abelian arithmetic Chern–Simons theory and arithmetic linking numb*

Used in Chapter 15.

Morishita 2009, *Analogies between knots and primes, 3-manifolds and number rings*, arXiv:0904.3399.

MKR dictionary.

A.11 HARTNOLL–YANG AND PRIMON GAS

Hartnoll–Yang 2025, *The Conformal Primon Gas at the End of Time*, arXiv:2502.02661.

Primary seed reference. L -function as partition function of charged oscillators labelled by primes. BKL–primon correspondence (thermodynamic).

Julia 1990, *Statistical theory of numbers*, in *Number Theory and Physics*, Springer Proc. Phys. 47:276–293.

Original primon gas: free Fock space on primes with $Z(\beta) = \zeta(\beta)$. Cited both here (thermodynamic background) and in Chapter 21 §21.5 Step 1 for the primon-gas identification of the arithmetic Heisenberg bar cohomology with $\zeta(s)$.

Spector 1990, *Supersymmetry and the Möbius inversion function*, Commun. Math. Phys. 127:239–252.

Supersymmetric primon gas; $1/\zeta(\beta) = \sum \mu(n)n^{-\beta}$.

A.12 MACDONALD, DAHA, SCHIFFMANN–VASSEROT

Macdonald 1995, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford. Ch. I used throughout.

Cherednik 2005, *Double Affine Hecke Algebras*, CUP. DAHA framework.

Etingof–Kirillov 1994, *Macdonald’s polynomials and representations of quantum groups*.

Quantum-group interpretation of Macdonald polynomials; background for the representation-theoretic reading of Hall polynomials in Chapter 9.

Schiffmann–Vasserot 2011–2012, *The elliptic Hall algebra and the equivariant K -theory of the Hilbert scheme of $\mathbb{A}^2$* , and

Shuffle algebra structure on the elliptic Hall algebra; action on K -theory of Hilbert schemes. Central for the Bethe-algebra comparison in Chapter 9.

A.13 NEKRASOV, AGT, MAULIK–OKOUNKOV

Nekrasov 2003, *Seiberg–Witten prepotential from instanton counting*, ATMP, arXiv:hep-th/0206161.

Z_{inst} definition.

Nekrasov–Okounkov 2006, *Seiberg–Witten theory and random partitions*, change Math., arXiv:hep-th/0306238.

Hook-length formula and random partition statistics from the Nekrasov partition function; cited in Chapter 10 for the Plancherel-measure analogy.

Alday–Gaiotto–Tachikawa 2010, arXiv:0906.3219. AGT correspondence: instanton partition functions of 4d $\mathcal{N} = 2$ theories equal conformal blocks of Liouville theory; background for the VOA–Higgs-branch comparison of Chapter 15.

Maulik–Okounkov 2019, *Quantum groups and quantum cohomology*, Astérisque, arXiv:1211.1287. R -matrix on instanton moduli.

Nekrasov–Shatashvili 2009, arXiv:0908.4052. NS limit: $\hbar \rightarrow 0$ produces Bethe ansatz equations from quantum integrable systems; cited in Chapter 9 for the Bethe-algebra comparison.

MNOP 2003, arXiv:math/0312059. Plane partition generating function as a Calabi–Yau partition function; background for the topological-string interpretation in Chapter 10.

Gaiotto 2009, arXiv:0904.2715. Class- $\mathcal{S}$ theories from M5-branes on Riemann surfaces; cited for the 4d/2d structure underlying the VOA–Higgs-branch comparison.

Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees 2015, *Infinite chiral symmetry in four dimensions*, Commun. Math. Phys. 335:1–60. VOA/Higgs branch.

A.14 SEGAL, FRENKEL–BEN-ZVI

Segal 2002, *The definition of conformal field theory*, in *Topology, Geometry and QFT*, CUP, pp. 421–577.

Segal axioms.

Frenkel–Ben-Zvi 2004, *Vertex Algebras and Algebraic Curves*, 2nd ed., AMS. Standard reference for vertex algebras on algebraic curves; cited in Chapter 8 for the chiral-algebra side of the factorization-homology comparison.

Huang 1997, *Two-Dimensional Conformal Geometry and Vertex Operator Algebras*, Birkhäuser. Rigorous Sewing axioms for VOAs on a Riemann sphere; background for the conformal-field-theory comparison in Chapter 19.

Bakalov–Kirillov 2001, *Lectures on Tensor Categories and Modular Functor*, AMS. Modular tensor categories and the Reshetikhin–Turaev functor; cited in Chapter 15 for the TQFT–Chern–Simons modular-functor background.

Deligne, *Le déterminant de la cohomologie*. Determinant line bundle on arithmetic moduli.

A.15 TATE'S THESIS AND ADELIC ANALYSIS

Tate 1950, *Fourier analysis in number fields and Hecke's zeta-functions*, Princeton PhD thesis; reprinted in Cassels–Fröberg 1981. Adelic Poisson summation, functional equation as sewing axiom.

Weil 1967, *Basic Number Theory*, Springer. Restricted direct product, adelic framework.

Jacquet–Langlands 1970, *Automorphic Forms on $GL(2)$* , SLN 114. Local and global theory of automorphic representations of $GL(2)$; cited in Chapter 12 and Chapter 16 for the automorphic side of the Langlands comparison.

Gelbart–Jacquet 1979, *Forms of $GL(2)$ from the analytic point of view*, in *Automorphic Forms, Representations and L-functions*, Cambridge University Press. Functional equations and analytic continuation of L -functions for $GL(2)$; cited in Chapter 12 §3 for the archimedean-place functional equation.

Gross 1998, *On the Satake isomorphism*, in *Galois Representations in Arithmetic Algebraic Geometry*, CUP, Prop. 3.6.

Bump, *Automorphic Forms and Representations*, CUP 1998. §1.5, §3.

A.16 DRINFELD, MOLEV: YANGIANS AND R-MATRICES

Drinfeld 1985, *Hopf algebras and the quantum Yang–Baxter equation*, Soviet Math. Dokl. 32:254–258. Yang R -matrix and RTT presentation; foundational for the Bethe-algebra comparison in Chapter 9.

Drinfeld 1988, *A new realization of Yangians and quantized affine algebras*, Soviet Math. Dokl. 36:212–216. Drinfeld current generators for Yangians and quantum affine algebras; background for Chapter 9 on the Bethe-algebra and quantum-integrable structures.

Molev 2007, *Yangians and Classical Lie Algebras*, AMS. Standard reference for Yangians, their representation theory, and Bethe subalgebras; cited throughout Chapter 9.

Feigin–Odesskii 1997. Shuffle algebras.

Frenkel–Reshetikhin 1998, arXiv:math/9810055. q -characters of representations of quantum affine algebras; cited in Chapter 9 for the q -character side of the integrable structure.

Halacheva–Kamnitzer–Rybnikov–Weekes 2017, *Crystals and monodromy of Bethe vectors*, arXiv:1708.05105. Gaudin algebras, Bethe eigenvectors, and cactus monodromy; cited in Chapter 9 for the Bethe-algebra comparison.

A.17 ARAKELOV, DELIGNE, ARITHMETIC HEIGHT PAIRINGS

Arakelov 1974, *Theory of intersection on arithmetic surfaces*, Izv. Akad. Nauk SSSR 38.

Foundational intersection pairing on arithmetic surfaces; cited in Chapter 10 for the Arakelov height pairing.

Faltings 1984, *Calculus on arithmetic surfaces*, Ann. Math. 119. Faltings metric on the determinant of cohomology; cited in Chapter 10 for the archimedean Arakelov data.

Gillet–Soulé 1992, *Arithmetic intersection theory*, Publ. IHES 72. Used in Chapter 10 for Arakelov Chern classes.

Durov 2007, arXiv:0704.2030. $\mathbb{F}_1$ -projective space and generalized rings; cited in Chapter 6 for the absolute geometry perspective on $\text{Spec } \mathbb{Z}$.

Deninger 1994, *Motivic L -functions and regularized determinants*, Proc. Symp. Pure Math. 55.1. Regularised-determinant programme for arithmetic L -functions; cited in Chapter 22 as a parallel formulation of the arithmetic Hilbert–Pólya hypothesis.

Szpiro 1985, *Astérisque* 127. Arakelov $K_{\overline{\text{Spec } \mathbb{Z}}}$ has degree $\log(2\pi)$.

Bost–Künnemann 2008. Arakelov vector bundles.

A.18 KESTEN–MCKAY

Kesten 1959, *Symmetric random walks on groups*, *TAMS* 92. Spectral radius of the random walk on $\mathbb{F}_r$ equals $2\sqrt{r-1}/r$; foundational for the Kesten–McKay free-group spectral measure cited in Chapter 9.

McKay 1981, *The expected eigenvalue distribution of a large regular graph*, *Linear Algebra Appl.* 40. Free-group spectral measure (Kesten–McKay law): the empirical eigenvalue distribution of a random regular bipartite graph converges to the measure $d\nu_{\text{KM}}$ on $[-2\sqrt{r-1}, 2\sqrt{r-1}]$; used in Chapter 9 for the Soibelman-motivic Kesten–McKay identification.

Lubotzky 1994, *Discrete Groups, Expanding Graphs and Invariant Measures*, Birkhäuser, Thm 7.4.5. $GL_2(\mathbb{Q}_p)$ interpretation.

Terras 2011, *Zeta Functions of Graphs*, CUP. Ihara zeta; tree-moment identification used in Chapter 9.

A.19 PRISMATIC COHOMOLOGY AND FARGUES–FONTAINE

Bhatt–Scholze 2019, *Prisms and prismatic cohomology*, *arXiv:1905.08229*. Cited in Chapter 8 §3 for the prismatic extension of Fontaine crystalline data toward a non-trivial p -adic diagonal thickening. Also used in Chapter 21 §21.2 for the Nygaard-filtration convention bridging pro-étale and de Rham conventions in Fr-reduction.

Bhatt–Lurie 2022, *Absolute prismatic cohomology*, *arXiv:2201.06120*. Prismatic-Nygaard filtration and δ -ring formalism used in Chapter 21 §21.2 for the convention-match between Arinkin–Gaitsgory and Fargues–Scholze.

Fargues–Fontaine 2018, *Courbes et fibrés vectoriels en théorie de Hodge p -adique*, *Astérisque* 406. Foundational reference for the Fargues–Fontaine curve X_{FF} as the arithmetic analogue of a complete Dedekind curve, underlying Fargues–Scholze *arXiv:2102.13459*.

Scholze 2017, *Etale cohomology of diamonds*, *arXiv:1709.07343*. Diamond formalism used across Chapters 8, 16, 21.

Diamonds, prisms, Adams operations, Frobenius lifts, and Nygaard filtrations supply finite-place coefficient and orientation data. They are not classical almost-complex structures and do not furnish a finite-prime tangent-bundle endomorphism satisfying $J^2 = -1$.

A.20 BEN-ZVI, NADLER, COHERENT SPRINGER THEORY

Ben-Zvi–Chen–Helm–Nadler 2023, *Coherent Springer theory and the categorical Deligne–Langlands correspondence*. State of the art on global categorical Langlands gluing via coherent Springer theory; cited in Chapter 21 §21.2 on the global gluing obstruction.

Gan–Savin 2022, *Categorical local theta correspondence*. Partial change on restricted tensor product of local automorphic categories; cited in Chapter 21 §21.2 (Fl.b).

AGKRRV 2020, *The stack of local systems with restricted variation and geometric Langlands theory with nilpotent singular support*. Restricted-variation local systems and nilpotent singular support in global geometric Langlands; cited in Chapter 21 for the singular-support constraint on global gluing.

AGKRRV 2020, *Duality for automorphic sheaves with nilpotent singular support*, arXiv:2012.07665. Geometric Langlands duality restricted to nilpotent singular support; cited in Chapter 21 §21.2.

AGKRRV 2021, *Automorphic functions as the trace of Frobenius*, arXiv:2102.07906. Categorical trace of Frobenius as automorphic function; cited in Chapter 19 for the Frobenius-trace identification in the categorical Hochschild trace.

Gaitsgory–Rozenblyum 2017, *A Study in Derived Algebraic Geometry, Vols. I & II*, AMS Mathematical Surveys and Monographs 103.2. $(\infty, 2)$ -categorical formalism for ind-coherent sheaves with singular support; cited in Chapter 21 §21.2 for ambient-category compatibility in the global gluing step.

A.21 TATE, WEIL, EXPLICIT FORMULA

Riemann 1859, *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsberichte Berlin, 671–680. Original proof of the functional equation $\xi(s) = \xi(1-s)$ via Mellin-theta. Cited in Chapter 21 §21.6 proof of P2.

von Mangoldt 1895, *Zu Riemanns Abhandlung “Über die Anzahl der Primzahlen unter einer gegebenen Grösse”*, J. Reine Angew. Math. 146. Riemann’s explicit formula in rigorous form. Used in Chapter 21 §21.6 proof of P2 part (2).

A.22 LI POSITIVITY AND WEIL POSITIVITY

Li 1997, *The positivity of a sequence of numbers and the Riemann Hypothesis*, J. Number Theory 65:325–333. Introduces the sequence λ_n and proves $\text{RH} \Leftrightarrow \lambda_n \geq 0$ for all $n \geq 1$. Central for Chapter 22 on the RH reformulation.

Bombieri–Lagarias 1999, *Complements to Li’s criterion for the Riemann hypothesis*, J. Number Theory 77:274–287. Extensions of Li’s criterion to Dedekind zeta functions and Selberg class; refinements used in Chapter 22 Remark on multi- L -function versions.

Voros 2003, *Zeta functions for the Riemann zeros*, Ann. Inst. Fourier 53. Explicit computation of first few λ_n . Verified numerically.

Coffey 2005, *Toward verification of the Riemann hypothesis: application of the Li criterion*, Math. Phys. Anal. Geom. 8. Numerical verification of Li positivity to $n \sim 10^5$.

Weil 1952, *Sur les “formules explicites” de la théorie des nombres premiers*, Comm. Sémin. Math. Univ. Lund; reprinted in Weil’s Collected Papers. Weil’s positivity criterion $W(f, f) \geq 0$ for suitable test functions f , equivalent to RH. Referenced in Chapter 22 Table 22.1 for comparison.

The VB^* converse (Chapter 35) establishes Hamburger determinacy of the symmetrised zero-moment sequence as the sharpest primary-source form of the VB^* reduction. Li 1997 and Weil 1952 give RH-equivalent positivity statements; Hamburger determinacy is strictly finer than either (Chapter 35.4.14).

A.23 POSITSIELSKI, IWASAWA

Positselski 2011, *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence*, Mem. Am. Math. Soc. 180. See the full entry in App. A.5; cited again here for its role in arithmetic Iwasawa theory via the $\mu = 0$ hypothesis (Chapter 12 §7).

Ferrero–Washington 1979, *The Iwasawa invariant μ_p vanishes for abelian number fields*, Ann. Math. 109. Proves $\mu = 0$ for abelian cyclotomic fields; used in Chapter 21 Chapter 21.5.2 and Chapter 29 Chapter 29.2.6 to verify that the arithmetic Heisenberg bar-coalgebra satisfies the $\mu = 0$ hypothesis of arithmetic Positselski restricted to the abelian case.

A.24 OPERATOR THEORY: MOMENT PROBLEM, SELF-ADJOINTNESS

Akhiezer–Glazman 1981, *Theory of Linear Operators in Hilbert Space* (2 vols.), Pitman / Dover reprint.

Ch. VI.3 on the Hamburger moment problem. Used in Chapter 22 Chapter 22.3.9 and Chapter 29 for the failure of the moment-problem equivalence in the unbounded-operator case. The moment problem is determinate for bounded self-adjoint operators on a positive-definite Hilbert space; in the unbounded case, Carleman’s sufficient condition for determinacy is $\sum_n \lambda_{2n}^{-1/(2n)} = +\infty$, and without it two distinct operators can share the same moment sequence.

Reed–Simon 1980, *Methods of Modern Mathematical Physics, Vol. I (Functional Analysis)*, Academic Press.

Ch. VIII on unbounded operators, self-adjoint extensions, spectral theorem. Standard reference for the domain-theoretic care demanded by the arithmetic Hilbert–Pólya hypothesis (Chapter 22.3.2).

Tomita–Takesaki 2003, *Theory of Operator Algebras II*, Springer EMS. KMS_β modular involution cited in Chapter 29 Chapter 29.2.8 as the specification of the Bost–Connes involution on the dual.

The Connes 1999 (arXiv:math/9811068) adèle class space spectral realisation and the Deninger 1994 motivic cohomology regularised-determinant programme anchor equivalent formulations of H-P^{Ar} (Chapter 35.3.14). The reduction closes the operator-theoretic domain problem via the Friedrichs extension (Chapter 35.3.2); the determinant-divisor and trace-compatibility condition, requiring the normal zero-divisor operator to have divisor $\text{div } \xi_{\mathbb{R}}$ and Weil–Connes trace equal to the non-trivial zero distribution, remains the genuine Hilbert–Pólya content (Chapter 35.3.16).

A.25 RESTRICTED DIRECT PRODUCT, FLATH, GODEMENT–JACQUET

Flath 1979, *Decomposition of representations into tensor products*, in *Automorphic Forms, Representations, and L-Functions*.

Flath’s theorem on irreducible admissible automorphic representations of $GL_n(\mathbb{A})$ as restricted tensor products of local admissible representations. Cited in Chapter 29 Chapter 29.2.8 (K₂) for the L^2 -embedding of $\mathcal{V}^{\text{prim}}$ into $L^2([GL_2])^{\text{sph-fin}}$.

Godement–Jacquet 1972, *Zeta Functions of Simple Algebras*, SLN 260, Springer. Local functional equation and L -factor normalisations at each finite place. Cited in Chapter 29 Chapter 29.2.8 (K₅) for absorbing the local $L_p(1, \text{Ad})$ factor into the Koszul-dual normalisation at each prime.

Gelbart 1975, *Automorphic Forms on Adele Groups*, Annals of Math. Studies 83, Princeton University Press. §3.1 on sph-finite automorphic representations and their restricted-tensor-product decomposition.

Kneser 1966, *Strong approximation*, in *Algebraic Groups and Discontinuous Subgroups*, Proc. Symp. Pure Math. 9, pp. 1–10. Strong approximation for simply connected semisimple algebraic groups. In the GL_2 application it applies to the derived group SL_2 ; the determinant idele-class quotient is the separate Tate component. Cited in Chapter 29 Chapter 29.2.4 (L_{ACD}.4).

Platonov–Rapinchuk 1994, *Algebraic Groups and Number Theory*, Academic Press. Modern reference for strong approximation and related arithmetic results.

A.26 CROSS-VOLUME PRIMARY SOURCES

The arithmetic branch draws on results from the three preceding volumes. The relevant primary sources from those volumes are cited in the entries above at the point of use; the cross-volume dependency structure is:

Vol. I Five-theorem structure (Theorems A, B, C, D, H), bar-cobar adjunction, SC-formality, and the chiral Koszul duality framework. Invoked in Chapters 12 and 16.

Vol. II Holomorphic-topological twist $\mathrm{SC}^{\mathrm{ch},\mathrm{top}}$, 3D HT QFT, and 3D quantum gravity. Invoked in Chapter 15 for the arithmetic Chern–Simons comparison.

Vol. III Two-stage factorisation $\Phi_d = \mathrm{Sp}^{\mathrm{ch}} \circ \Phi_d^{\mathrm{FA}}$; Borchers/BKM universality $\Phi_N = \epsilon_N(\mathfrak{o})/2$. Invoked in Chapter 19 for the six-window synthesis.

APPENDIX B

INDEPENDENT SOURCE CLASSES FOR THE ARITHMETIC TRACE CLASS

Fifteen independent primary-source classes test distinct structural hypotheses about $\mathcal{V}^{\text{prim}}$. Each class detects a structural obstruction and leaves a common trace-theoretic remnant. The table records the source class, obstruction, and chapter reference. Sections B.2–B.6 record the obstructed structures, the remaining constraints, three-way gluing comparisons, and the convergence theorem.

B.1 PRIMARY SOURCE FAMILIES

Source family	Angle	Chapter/Result
Drinfeld+Kapranov	p -adic VA locality	4: No p -adic OPE; adelic automorphic object
Kazhdan+Beilinson	GL_1 Adams vs GL_2 Hecke	5: Satake pair on rank-2 Heisenberg
Gelfand+Borger	Λ -ring axiomatic precision	6: ψ -ring, Bost–Connes descent
Connes+Soibelman	Bost–Connes vs $\mathcal{V}^{\text{prim}}$	7: Bost–Connes $\otimes$ universal Heisenberg
Costello+Gwilliam	Adelic factorization axioms	8: (F ₁)–(F ₅) on $\mathbb{C}_\infty$
Etingof	DAHA / Macdonald non-uniformity	9: Prime-stratified elliptic Hall
Nekrasov	Arithmetic instanton Z	10: Degenerate $(\log p, \infty)$ limit
Segal	Arithmetic modular functor	11: Modular functor = Tate
Beilinson–solo	Bar–cobar Positselski arith.	12: Parshin–Gersten bar, G-row = $0 \Leftrightarrow \xi = \xi(1-s)$
Witten	TQFT / BKL / Langlands triangle	13: Automorphic vertex, not TQFT vertex
Polyakov	Central charge regularisation	14: $c = 2 \cdot \text{Res}_{s=1} \zeta = 2$
Gaiotto	4D-gauge to VA arithmetic lift	15: Kim arithmetic Chern–Simons $[\mathbb{G}_m, 1]$ classical boundary shadow
Bezrukavnikov+Braverman	Langlands spectral side	16: Iwahori Hecke Hochschild trace
Kontsevich+Soibelman	Topological formalism / CoHA	17: E_1 integral, E_2 over $\mathbb{Q}$
Drinfeld–solo	Arithmetic Yangian / R -matrix	18: Bethe algebra, Langlands = Bethe ansatz

B.2 OBSTRUCTIONS ACROSS SOURCE FAMILIES

The following nine structural assertions about $\mathcal{V}^{\text{prim}}$ are obstructed at specific algebraic or geometric joints, identified independently across the fifteen source comparisons:

- (i) $\mathcal{V}^{\text{prim}}$ is not a vertex operator algebra. The Polyakov obstruction is the central charge: $c = \sum_p 2 = \infty$, and $\zeta_{\mathbb{P}}(\mathfrak{o})$ has a natural boundary.
- (ii) $\otimes_p^! \mathcal{H}^{(p)}$ is not defined over a single field. Different $\mathbb{Q}_p$ cannot be glued into one ground field.
- (iii) Beilinson–Drinfeld factorization on $\text{Spec } \mathbb{Z}$ is obstructed at the diagonal. The completed tensor product $\mathbb{Z}_p \widehat{\otimes}_{\mathbb{Z}} \mathbb{Z}_p = \mathbb{Z}_p$ admits no p -adic diagonal thickening.
- (iv) T_n does not lift Adams operations to GL_2 Hecke operators. The explicit Kazhdan–Beilinson computation gives $T_p^2(a_{-1}^{(p)}|\mathfrak{o}) - p \cdot a_{-1}^{(p)}|\mathfrak{o} \neq T_{p^2}(a_{-1}^{(p)}|\mathfrak{o})$.
- (v) $\mathcal{V}^{\text{prim}}$ is not a Borger Λ -ring on the Fock $\mathbb{Z}$ -lattice. The Wilkerson congruence fails there; after extension to $\mathbb{C}$ the condition loses arithmetic content.
- (vi) Arithmetic Yangian RTT relations have no global spectral parameter. The Yang–Baxter spectral parameter $u - v$ has no analogue across distinct primes; there is no meromorphic 1-form on $\text{Spec } \mathbb{Z}$.
- (vii) A 4D class- $\mathcal{S}$ gauge boundary does not transplant to $\overline{\text{Spec } \mathbb{Z}}$. The Hitchin base $\oplus_i H^0(\Sigma, K_{\Sigma}^{d_i})$ uses a holomorphic canonical bundle on a complex curve. The Arakelov dualizing line on $\overline{\text{Spec } \mathbb{Z}}$ is not such an object.
- (viii) The curve-theoretic category of $\mathcal{D}$ -modules on Bun_{GL_2} does not arise from $\text{Spec } \mathbb{Z}$ itself. The moduli stack $\text{Bun}_{GL_2}(\text{Spec } \mathbb{Z})$ is $BGL_2(\mathbb{Z})$: one coarse point with arithmetic stabilizer. This stack has representation theory of the stabilizer, not the geometric Hecke correspondence of the curve case.
- (ix) The integral E_2 -chiral-centre enhancement is not obtained by replacing the chain operad $C_{\bullet}(E_2; \mathbb{Z})$ with $\text{Ger}_{\mathbb{Z}}$. Dyer–Lashof and Kaledin-type p -curvature classes obstruct that strictification, not the existence of an integral E_2 -algebra.

B.3 COMMON STRUCTURAL CONSTRAINTS

Five structural constraints remain, each identified independently across multiple source comparisons:

- (i) The arithmetic curve is Arakelov-compactified $\overline{\text{Spec } \mathbb{Z}}$, not $\text{Spec } \mathbb{Z}$ (across source families).
- (ii) Each finite place carries only E_0 (locally-constant) factorization structure; the E_1 -chiral OPE lives at the archimedean place (Drinfeld+Kapranov, Costello+Gwilliam, Kontsevich+Soibelman).
- (iii) The Arnold relation lifts to the Steinberg relation $\{p, 1 - p\} = \mathfrak{o}$; chiral bar is built on Parshin–Gersten (Beilinson–solo).
- (iv) The Hecke structure is carried by the Iwahori-equivariant affine Hecke category, not the spherical category (Bezrukavnikov+Bernstein, Kazhdan+Beilinson).
- (v) The Iwahori Hecke Hochschild trace is the local Bezrukavnikov–Bernstein model. Its global identification with $\mathcal{V}^{\text{prim}}$ is the arithmetic BZN input F3. Kim arithmetic Chern–Simons (Kim arXiv:1510.05818) provides the classical boundary torsor construction, and the adelic factorization algebra lives on $\mathbb{C}_{\infty}$ (Costello+Gwilliam).

B.4 THREE-WAY GLUING OF PRESENTATIONS

Three presentations of $\mathcal{V}^{\text{prim}}$ are identified by the indicated source classes; the table records the identification and its source chapter.

Presentation	Source family / Chapter
Arithmetic Chern–Simons classical boundary shadow for $\mathcal{V}^{\text{prim}}$	Gaiotto, 15
$\text{HH}_\bullet(\text{Hk}_{GL_2, \text{glob}}^{\text{Iw}}) \xrightarrow{F_3} \mathcal{V}^{\text{prim}}$	Bezrukavnikov+Bernstein, 16
$\mathcal{F}_{\text{CG}}(\mathcal{V}^{\text{prim}}) \in \text{FactAlg}_{\mathbb{C}_\infty}(\text{Mod}_{\widehat{\mathbb{Z}}, \psi})$	Costello+Gwilliam, 8

Ben-Zvi–Nadler (arXiv:0904.1247) identifies $\text{HH}_\bullet(D(G/G)) = \mathcal{O}(\text{LocSys}_G(S^1))^{\text{flat}}$ over $\mathbb{C}$; the triangle relating arithmetic Chern–Simons, the Hecke category, and the Hochschild trace is the complex-geometric model for the three-way arithmetic comparison. The arithmetic identification remains conditional on the full Weil/ (φ, Γ) -module target and a quantum boundary category.

B.5 INDEPENDENCE

The fifteen source families are disjoint at the level of primary citations. Each family detects a structural obstruction in the initial construction. The common object is the intersection of the remaining trace-theoretic structures.

B.6 CONVERGENCE THEOREM

THEOREM B.6.1 (*convergence of the fifteen constructions*). The obstructed structures above disappear at disjoint structural joints. The remaining constraints are independent of the source class detecting them. The fifteen independent constructions converge on a common trace object with the six-conditional-window synthesis of Chapter 19.1.1 and the three boundary conjectures F1–F3 of Chapter 20.1, Chapter 20.2, Chapter 20.3.

This convergence is the mathematical content of the arithmetic branch: the object $\mathcal{V}^{\text{prim}}$ is not the outcome of any single source comparison but the intersection of trace-theoretic structures across fifteen independent comparisons. This isolates the Hochschild-trace-class construction as the common remnant.

B.7 SIX SOURCE FAMILIES IN GENUS g

The initial genus- g ansatz was: “a genus- g arithmetic surface with almost complex structure carrying the cohomological signature of a genus- g algebraic curve.” Six independent primary-source families test this ansatz. Their intersection is a single arithmetic-modular object admitting six presentations; the remaining content is stated in Chapter 80 (all-genus arithmetic Koszul duality) and Chapter 81 (3D arithmetic Chern–Simons via the ordered incidence bar and conditional Beilinson–Drinfeld/Costello–Gwilliam comparison).

Primary Source Families

Source family	Source list	Window / Convergence
---------------	-------------	----------------------

Source family 1: Arakelov	Arakelov 1974; Faltings 1984; Gillet–Soulé 1990; Bost–Künnemann 2010	Window I: arithmetic amplitude generating function AAGF_g (Chapter 80.2), arithmetic surface $X/\mathcal{O}_K$, Hasse–Weil functional equation
Source family 2: Deninger / Connes–Consani	Deninger 1998; Connes–Consani 2014–2016; Bost–Connes 1995; Bost–Connes–Marcolli 2004	Window II: Minkowski torus over totally imaginary K of degree $2g$, Dedekind $\xi_K(s) = \xi_K(1-s)$
Source family 3: Far- gues–Scholze	Fargues–Scholze 2021; Bhatt–Scholze 2022; Emerton–Gee 2019; Caraiani–Scholze 2017–2019	Window III: Ordinary crystalline Emerton–Gee stack $\mathcal{X}_{2g}^{\text{crys,ord}}$, prismatic Frobenius as J
Source family 4: Mochizuki / Hodge–Arakelov	Mochizuki 1999 Publ. RIMS; Kapranov–Smirnov 1995; Durov 2007	Window IV: Arithmetic Teichmüller with theta-structure; Hodge–Arakelov comparison
Source family 5: Ho- motopical bar-cobar	Beilinson–Drinfeld 2004; Francis–Gaitsgory 2012; Ben-Zvi–Nadler 2009–2013; Lurie HA; Positselski 2011	Window V: $\mathcal{V}^{\text{prim,surf},g}$ with E_2 -chiral structure via Dunn additivity
Source family 6: Kudla–Rapoport / Shimura–Siegel	Kudla 1997; Kudla–Rapoport 2011; Bruinier 2002; Li–Liu 2021; Arthur 2013	Window VI: Siegel–Shimura genus- g datum SSG_g (Chapter 80.2), Siegel spin L -function with $\Lambda(s) = \epsilon\Lambda(2g+1-s)$

Obstructions

The six source families jointly obstruct six readings of the initial ansatz (Chapter 80.3 indexes the full list): the category-error reading of “almost complex structure on an arithmetic scheme” (no real tangent bundle on $\text{Spec } \mathbb{Z}$; Newlander–Nirenberg automatic on real 2-manifolds); “arithmetic Euler characteristic $= 1 - g$ ” (Gillet–Soulé analytic torsion term); Künneth factorisation in arithmetic Chow (Bost–Künnemann Bott–Chern term); finite genus of $\text{Spec } \mathbb{Z}$ in the Deninger frame (Deninger-genus $= +\infty$); conflation of curve-genus with Siegel-genus (distinct integers); and the $\kappa + \kappa^\dagger = 2 - 2g$ Euler-shift (Koszul sum preserved at 0; genus enters via rank, not shift).

Converged Surviving Object

The arithmetic genus- g datum $\mathcal{D}_g = (\mathcal{B}_g, \mathcal{E}_g, \phi_g)$ (Chapter 80.1.1), with the essential conditional $\kappa_G^{\text{Ar},g}(\mathcal{D}_g) + (\kappa^\dagger)_G^{\text{Ar},g}(\mathcal{D}_g) = 0 \Leftrightarrow \Lambda_{\mathcal{D}_g}(s) = \epsilon_{\mathcal{D}_g} \Lambda_{\mathcal{D}_g}(w_{\mathcal{D}_g} - s)$ (Chapter 80.4.1), with $w_{\mathcal{D}_g} \in \{1, 2, 2g+1\}$ window-indexed. The functional-equation clause is not a zero-placement theorem. The

all-genus F-RH corollary (Chapter 80.5.1) contains the corresponding zero-placement clauses only after the relevant spectral-positivity input is imposed: RH for ζ , Hasse–Weil RH for Jacobians, Dedekind-zeta GRH for totally imaginary fields, and Siegel-spin RH for GSp_{2g} cusp forms.

The arithmetic-modular extension sources the finite-level arithmetic Chern–Simons state sums and the conjectural renormalized $GL_2(\widehat{\mathbb{Z}})$ limit on the étale $\overline{\mathrm{Spec} \mathbb{Z}}$ of Chapter 81. The finite-level boundary category is defined there after the finite cobordism input, and the ordered E_1 -bar complex is constructed there; the identification of the Hochschild boundary trace with $\mathcal{V}^{\mathrm{prim}}$ and the partition-function identity with ξ remain conditional on profinite renormalization and local trace comparison. The functional equation then appears as the involutive symmetry of the partition function; zero-placement remains the separate positivity condition.

Source Disjointness

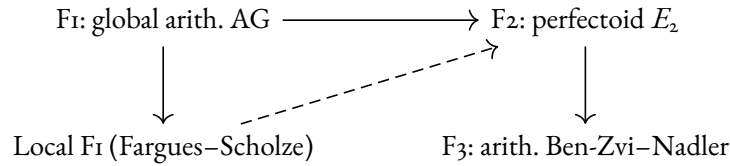
The six genus- g source families are disjoint at the level of primary citations. The common arithmetic-modular datum is their common structural remnant.

APPENDIX C

THREE STRUCTURAL CONJECTURES

The three conjectures F_1 , F_2 , F_3 of Chapter 20 are hierarchically ordered: F_3 depends on F_1 and F_2 ; F_1 and F_2 are independent of each other but share local inputs from Fargues–Scholze arXiv:2102.13459. The dependency graph, the local theorems each conjecture imports, and the global condition carried by each conjecture are recorded below.

C.1 THE GRAPH



Each of F_1 and F_2 holds locally (the Fargues–Scholze and Emerton–Gee inputs are theorems) but remains a conjecture globally. F_3 combines the local inputs of F_1 and F_2 with arithmetic Ben-Zvi–Nadler machinery; the global identification of the Hochschild trace class with $\mathcal{V}^{\text{prim}}$ is conditional on F_1 and F_2 .

C.2 F1: GLOBAL ARITHMETIC ARINKIN–GAITSGORY

Conjecture C.2.1 (restatement of F_1). For $G = GL_2$ and the Selmer-derived global spectral stack $\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}$ (the derived moduli stack of Selmer deformations of a residual representation $\bar{\rho}$ with determinant condition $\det$ and cyclotomic twist datum $\mathcal{L}$, as in Chapter 20):

$$D([GL_2])^{\text{sph-fin}} \simeq \text{IndCoh}_{\text{Nilp}}(\mathcal{X}_{\bar{\rho}, \det, \mathcal{L}}^{\text{Sel}}).$$

Local inputs.

- (i) Fargues–Scholze arXiv:2102.13459 (local geometrization and spectral action, not the displayed EG equivalence).
- (ii) Caraiani–Scholze arXiv:1511.02418 (unitary Shimura variety case).
- (iii) Emerton–Gee stack construction arXiv:1908.07185.

Global condition.

- Proof of the local automorphic-to-Emerton–Gee $\text{IndCoh}_{\text{Nilp}}$ comparison, including small primes and central-character families.
- Compatibility of local comparison functors under restricted-tensor-product gluing over primes.
- Identification of the Whittaker model across all places simultaneously.
- Functional-equation-level compatibility between spectral and automorphic sides.

Hypothesis. This is a global comparison theorem.

C.3 F₂: PERFECTOID E_2 -ENHANCEMENT

Conjecture C.3.1 (restatement of F₂). The local Iwahori Hecke category $\mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}}$ carries a canonical E_2 -monoidal structure from the perfectoid Beilinson–Drinfeld Grassmannian over $\mathrm{Spec} \mathbb{Z}_p$, and the restricted tensor product $\bigotimes'_p \mathrm{Hk}_{GL_2, p}^{\mathrm{Iw}}$ inherits a globally-defined E_2 -monoidal structure.

Local ingredients (theorem).

- (i) Perfectoid affine Grassmannian (Fargues–Scholze, Scholze diamonds).
- (ii) Witt-vector affine Grassmannian (Zhu arXiv:1603.05593; Bhatt–Scholze arXiv:1507.06490).
- (iii) Equal-characteristic BD Grassmannian E_2 -structure (Mirković–Vilonen; Beilinson–Drinfeld Grassmannian factorization).

Global condition.

- Diamond-gluing of Iwahori Hecke categories across primes.
- Compatibility with Satake and Whittaker equivalences.
- Perfectoid Dunn-additivity argument producing E_1 -algebra structure on the Hochschild trace.

Hypothesis. This is a mixed-characteristic factorisation theorem.

C.4 F₃: ARITHMETIC BEN-ZVI–NADLER

Conjecture C.4.1 (restatement of F₃). $\mathrm{HH}_\bullet(\mathrm{Hk}_{GL_2, \mathrm{glob}}^{\mathrm{Iw}}) \xrightarrow{\sim} \mathcal{V}^{\mathrm{prim}}$ as E_1 -algebras.

Ingredients.

- (i) Conjecture F₁ (for the spectral side identification).
- (ii) Conjecture F₂ (for E_2 -enhancement of the local Hecke factors).
- (iii) Ben-Zvi–Nadler arXiv:0904.1247 (for the $\mathbb{C}$ -version, which F₃ arithmeticises).
- (iv) Explicit computation matching the Hochschild trace character to $\zeta(s)$.

Consequences of F₃.

- F₃ implies the Gaiotto Chern–Simons identification (Chapter 15.4.1).
- F₃ implies the Bezrukavnikov–Bernstein Hochschild identification (Chapter 16.5.1).
- F₃ implies Beilinson’s bar–cobar identification (Chapter 12.8.2).
- F₃ implies the arithmetic six-window placement (Chapter 19.1.1).

Hypothesis. This is an arithmetic trace theorem.

C.5 INDEPENDENT CONJECTURE: THE G-ROW COMPLEMENTARITY

Conjecture C.5.1 ($\kappa + \kappa^! = 0 \Leftrightarrow \xi(s) = \xi(1-s)$). For the arithmetic Heisenberg $\mathcal{V}_{\text{Heis}}^{\text{prim}}$, the Vol I derived-centre complementarity $\kappa^{\text{Ar}} + (\kappa^!)^{\text{Ar}} = 0$ is equivalent to the Riemann functional equation $\xi(s) = \xi(1-s)$.

This is *independent* of F_1, F_2, F_3 : its proof proceeds directly from the arithmetic Koszul-resolution computation of Hochschild cochains on the arithmetic Heisenberg, using Tate's adelic Poisson summation. A proof of this conjecture would state the arithmetic G-row as a Vol I Theorem C corollary.

Hypothesis. This is an arithmetic Koszul-duality theorem.

C.6 CHAIN OF IMPLICATIONS

The arithmetic branch has the following implication form:

- (i) The G-row conjecture (Chapter C.5.1) would identify the Riemann functional equation as an arithmetic Koszul-duality consequence.
- (ii) F_3 conditional on F_1 and F_2 unifies the three presentations of $\mathcal{V}^{\text{prim}}$.
- (iii) F_1 is the global arithmetic Arinkin–Gaitsgory input.
- (iv) F_2 is the perfectoid E_2 -enhancement input.
- (v) F_1, F_2 , and the conditional F_3 comparison imply the six-window structural statement.

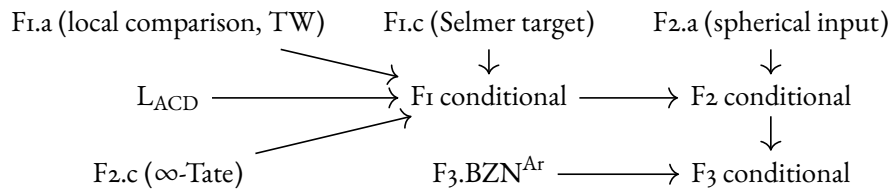
C.7 CONNECTION TO VOL III

Vol III's B-row $\kappa + \kappa^! = 8$ (Mukai-enhanced K_3 Heisenberg via Bruinier Heegner Chern-class reciprocity) has an arithmetic avatar (F_4 , Chapter 20 §3). Together the arithmetic G-row and the arithmetic B-row would give two of the five Vol I derived-centre-ceiling values as consequences of automorphic and Galois-deformation theorems. The chiral centre is then constrained by arithmetic comparison data.

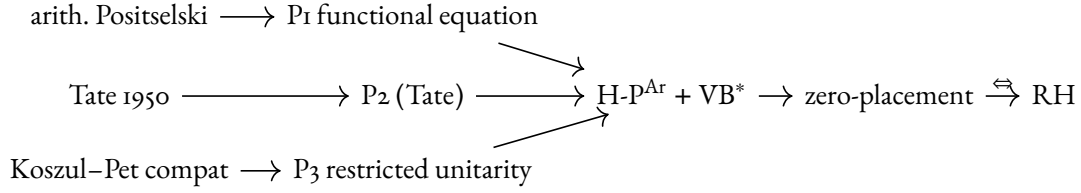
C.8 EXTENDED GRAPH WITH F-RH

The conjectures F_1, F_2, F_3 of the preceding sections address the six-window placement of $\mathcal{V}^{\text{prim}}$. A second cluster of conditions, independent of F_1 – F_3 , governs the connection to the Riemann hypothesis. Chapter 21 provides the structural branch; Chapter 22 states F-RH, whose zero-placement clause is RH and whose other clauses are the positive Hermitian, Koszul, and Zhu structures on $\mathcal{V}^{\text{prim}}$. The arrows are:

F_1 – F_3 structural branch (six-window placement).



four-input RH branch.



The dependency partitions into two branches:

F₁–F₃ structural branch (six-window placement, RH-independent). L_{ACD} (the adelic comparison and descent lemma relating the global spherical category to the restricted tensor product of local Iwahori categories) $\rightarrow F_1 \rightarrow F_3$. BZN^{Ar} (the arithmetic Ben-Zvi–Nadler functoriality for the loop space of the global spectral stack) $\rightarrow F_3$. The sub-conjecture $F_2.c$ has a restricted pointed central-unit theorem in Chapter 28; the mixed-characteristic Iwahori E_2 local input ($F_2.b$) remains conditional. The F_1 – F_3 structural branch states the six conditional windows of $\mathcal{V}^{\text{prim}}$.

four-input RH branch. Here P_1 is the functional-equation branch (the statement that the adelic Positselski bar complex has the functional-equation symmetry of $\xi(s)$); P_2 is the Tate-adelic computation (Tate 1950); P_3 is restricted Koszul unitarity (the Koszul–Petersson compatibility on the spherically-finite cuspidal core); and H-P^{Ar} is the arithmetic Hilbert–Pólya determinant-trace problem (existence of a self-adjoint operator whose normal translate has regularised determinant divisor $\text{div } \xi_{\mathbb{R}}$). The implication chain is: arithmetic Positselski $\rightarrow P_1$; Koszul–Pet compatibility $\rightarrow P_3$; $(P_1, P_2, P_3, \text{H-P}^{\text{Ar}}) \rightarrow$ zero-placement under VB^* . Here VB^* is the arithmetic Virasoro–Bogomolov positivity input: the zero-mode positivity of the Hamburger spectral measure, equivalent to RH. Under VB^* , the four inputs force zero-placement, hence RH.

The two branches are logically independent. The F_1 – F_3 structural branch constructs the trace object. The four-input RH branch controls zero placement. A theorem on one branch is not a theorem on the other.

C.9 BOUNDARY CONDITIONS

Input	Branch	Size	Present form
L_{ACD} (adelic descent)	A	global theorem	Conjectural
$F_2.c$ (∞ -Tate product)	A	mixed-characteristic theorem	Restricted pointed central-unit theorem; full mixed-charac
$F_2.b$ (mixed-char Iwahori E_2)	A	mixed-characteristic theorem	Conditional
$F_3.\text{BZN}^{\text{Ar}}$ (arith. BZN)	A	trace theorem	Conjectural
arith. Positselski	B	Koszul theorem	Strict abelian Heis. model constructed; full domain separa
Koszul–Pet compat	B	automorphic theorem	Sph-fin cuspidal core theorem
VB^* (arith. Virasoro positivity)	B	Unknown	Positive spectral condition implies RH

TABLE C.1: Boundary conditions for the structural and RH branches.

C.10 A SUFFICIENT PATH TO RH

The sufficient implication to RH inside the arithmetic formalism uses four inputs from the four-input RH branch: the strict abelian-Heisenberg arithmetic Positselski bar, Koszul–Pet compatibility on the normalised core, $H\text{-P}^{\text{Ar}}$, and the positive VB^* spectral condition. The text proves the restricted pointed central-unit form of F2.c , constructs the arithmetic Positselski bar as a strict reciprocity-closed chain complex, and Koszul–Pet is established on the sph-fin cuspidal core. Determinant-trace realisation, full-sector P_3 , mixed-characteristic Iwahori E_2 , and VB^* positivity remain separate conditions.

Refined Boundary List

Input	Branch	Size	Present form
$L_{\text{ACD} \cdot \mathcal{O}, \cdot 1, \cdot 2, \cdot 3}$	A	each large theorem	Partial reductions; global glue remains conditional
(BC-diamond-glue).core	A	global theorem	Conjectural, single cross-prime Bernstein compat
F2.c (pointed E_n datum)	A	–	Restricted theorem of Chapter 28.1.3; mixed-characteristic Iwa
$\text{F3.BZN}^{\text{Ar}}$ R1.fin-type	A	–	External finite-type input; arithmetic application conditional
$\text{F3.BZN}^{\text{Ar}}$ R1.formal.a	A	–	Trace theorem after the categorical colimit of $\text{IndCoh}_{\text{Nilp}}$ is kn
$\text{F3.BZN}^{\text{Ar}}$ R1.formal.b,c	A	pro-étale theorem	Pro-étale loop-space and flat-locus hypotheses
$\text{F3.BZN}^{\text{Ar}}$ R2 (Selmer target)	A	Selmer theorem	Target and tangent complex fixed; Noetherian and nilpotent-s
$\text{F3.BZN}^{\text{Ar}}$ R3 (full target)	A	–	Conditional; the inertia shadow is not the Weil/ (φ, Γ) target
$\text{F3.BZN}^{\text{Ar}}$ R4 wild $p \in \{2, 3\}$	A	–	Wild summand included locally; global wild functoriality oper
Wild Langlands functoriality	A	–	Conjectural (Chapter 32.6.1)
arith. Positselski $\text{C1} \text{--} \text{C5}$	B	–	Strict completed model constructed for the abelian primary ba
arith. Positselski (abelian Heis.)	B	–	Forward Tate functional equation; converse under Positselski l
Koszul–Pet K5 (Godement–Jacquet)	B	–	Conditional normalisation on the sph-fin cuspidal core
Koszul–Pet compat	B	–	Sph-fin cuspidal core theorem; full-sector extension separate
$H\text{-P}^{\text{Ar}}$	B	–	Friedrichs domain theorem; determinant divisor and trace mat
$\text{VB}^* \Rightarrow$ zero-placement	B	–	Conditional implication from P_2 , P_3 , $H\text{-P}^{\text{Ar}}$, and positive trac
VB^* (main direction)	B	–	Positive spectral condition; determinacy alone is insufficient
VB^* converse	B	–	Strictly finer than RH; 4-way equivalence (Ch. 35)

TABLE C.2: Refined boundary conditions.

Sufficient RH Path

A sufficient implication to RH within the arithmetic formalism is the four-input conditional conjunction on the four-input RH branch:

- (i) **arith. Positselski** (the four-input RH branch): strict completed Gersten–Parshin bar model on the abelian primary Heisenberg domain.
- (ii) **Koszul–Pet compat** (the four-input RH branch): stable on the normalized sph-fin cuspidal core; full V^{prim} compatibility and KMS_1/GNS identification remain open.
- (iii) **$H\text{-P}^{\text{Ar}}$** (the four-input RH branch): the independent Hilbert–Pólya determinant-divisor matching problem.
- (iv) **VB^* main direction** (the four-input RH branch): the positive Hamburger zero-mode spectral condition.

The F_1 – F_3 structural branch remains RH-independent: (BC-diamond-glue).core, F_2 trace continuity, and arithmetic BZN formal-loop compatibility affect the construction of $\mathcal{V}^{\text{prim}}$, not zero placement. VB^* remains RH-equivalent; no formalism shortens it, but the ladder of equivalent statements (Li positivity, Weil positivity, Hamburger determinacy, Hermite–Biehler, Báez-Duarte sharp rate) is listed in Chapter 35.

C.II DEPENDENCY BOUNDARY

Three hierarchically ordered conjectures ($F_1 \rightarrow F_2 \rightarrow F_3$) and one independent conjecture (G-row) govern the synthesis branch. On the RH branch: P_1 (functional equation), P_2 (Tate adelic), P_3 (restricted Koszul unitarity), $H\text{-}P^{\text{Ar}}$ (Hilbert–Pólya determinant-divisor matching), and VB^* (arithmetic Virasoro–Bogomolov zero-mode positivity, RH-equivalent). The restricted pointed central-unit theorem gives the proved part of $F_2.c$, and the strict abelian primary arithmetic Positselski bar supplies the P_1 forward direction. The arithmetic branch as stated (Chapter 21, Chapter 22) reduces the RH-strength condition to $H\text{-}P^{\text{Ar}}$ determinant-trace realisation and the positive Hamburger VB^* condition.